Poster presentations
Oesophageal, gastric and duodenal disorders

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Oesophageal, gastric and duodenal disorders
Poster Exhibition

PP0001
AKT/MTOR SIGNALING AS THE MECHANISM OF SP1-INDUCED LNCRNA ZFAS1 CONTRIBUTES TO CELL PROLIFERATION AND MIGRATION IN GASTRIC CANCER

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Introduction: Gastric cancer remains a prevalent malignancy, with molecular drivers critical for its progression and therapeutic targeting. Recently, some malignancies have been linked to the lncRNA ZFAS1, but its mechanism of action and the reasons for its dysregulation in gastric cancer are unknown.

Aims & Methods: We evaluated ZFAS1 expression in gastric cancer cells (SGC7901) by RT-qPCR and compared it with GES-1 cells. The LnCAR database provided insight into ZFAS1 levels in STAD compared to normal tissue. The effect of ZFAS1 knockdown on cell invasion, migration and proliferation was evaluated using transwell invasion, wound healing assays, CCK8 and flow cytometry. In addition, ZFAS1 promoter regions were examined using the JASPAR database and subsequent ChIP assays to understand SP1 transcription factor binding. The effect of ZFAS1 on the AKT/mTOR pathway was clarified using Western blotting.

Results: SGC7901 cells were shown to have increased ZFAS1 expression, which was linked to a poor prognosis for gastric cancer. Knockdown of ZFAS1 in SGC7901 cells inhibited cell invasion, migration and proliferation and induced apoptosis. In addition, SP1 was found to upregulate ZFAS1 transcription by binding to its promoter region. ZFAS1 knockdown resulted in a significant reduction of key proteins (p-AKT, AKT, p-mTOR, and mTOR) of the AKT/mTOR pathway. When the AKT activator SC79 was introduced, the repressive effects of ZFAS1 knockdown on cell invasion, migration, proliferation, and AKT/mTOR signaling were partially reversed.

Conclusion: Our results highlight the pivotal role of ZFAS1 in gastric cancer cell malignancy, which is modulated by the AKT/mTOR pathway. The regulatory involvement of SP1 in ZFAS1 transcription provides a novel understanding of the molecular mechanisms driving cancer progression and offers potential therapeutic avenues.

Disclosure: Nothing to disclose.

PP0002
IMPACT OF CHOLECYSTECTOMY ON GASTRIC MUCOSAL C-MYC AND HTERT EXPRESSION: A POTENTIAL LINK TO GASTRIC CANCER DEVELOPMENT

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Introduction: Gastric cancer is a multifactorial malignancy influenced by genetic and environmental factors. Cholecystectomy, the surgical removal of the gallbladder, may lead to bile reflux—backflow of duodenal contents into the stomach—potentially damaging gastric mucosa. hTERT and c-MYC play vital roles in cellular proliferation and cancer development. Although epidemiological data generally show no strong link between cholecystectomy and gastric cancer, existing studies are limited and may be influenced by other risk factors.

Aims & Methods: This study aimed to evaluate whether bile acid exposure post-cholecystectomy contributes to gastric cancer and to assess the expressions of hTERT and c-MYC in this context. We included 100 participants at Manisa Celal Bayar University Hospital between September 2021 and May 2023, divided into three groups: gastric cancer (n=32), cholecystectomy history (n=34), and control (n=34). Data on demographics, H. pylori, intestinal metaplasia, and gene expression levels were collected. Ethics approval and informed consent were obtained. Gene expression analysis was performed from endoscopic biopsy samples.

Results: Mean ages were 65.2 (cancer), 60.5 (cholecystectomy), and 59.9 (control), with no significant difference except for higher age in cancer patients. Females were predominant in the cholecystectomy group ($p=0.021$). Fourteen cancer patients had distant metastasis, but no significant differences in c-MYC or hTERT expression were observed based on metastasis status. No significant differences were found in H. pylori or intestinal metaplasia prevalence.

However, c-MYC and hTERT levels were significantly lower in the cholecystectomy group compared to controls and cancer patients ($p<0.001$). Gender also affected gene expression: females had lower c-MYC and hTERT levels than males ($p<0.05$), though this effect was consistent across all groups. The presence of H. pylori or intestinal metaplasia did not significantly alter c-MYC or hTERT expression in any group.

Conclusion: Although cholecystectomy was not directly associated with increased gastric cancer risk, in individuals with a genetic predisposition, it may act as a contributing factor. Lower expressions of c-MYC and hTERT post-cholecystectomy suggest potential for these markers in cancer risk evaluation. Further research is necessary to clarify their roles in post-cholecystectomy follow-up.

Disclosure: Nothing to disclose.

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PP0003

IN PATIENTS WITH GASTROESOPHAGEAL REFLUX DISEASE GERD-Q SCORE CORRELATES WITH METHYLBUTYRIC ACID, CAPROIC ACID AND ENANTIIC ACID IN STOOL SAMPLES

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Introduction: Volatile organic compounds (VOC) in stool samples may reflect functional activity of gut microbiome. Previous studies showed that gut flora may influence transient relaxations of lower esophageal sphincter through short-chain fatty acids production and thus - participate in pathogenesis of gastroesophageal reflux disease (GERD). However, the role of VOC in pathogenesis of GERD is poorly studied yet.

Aims & Methods: The aim of this study (FGMF-2025-0003) was to perform correlation analysis between concentrations of VOC in stool samples and GERD-Q score.

We enrolled willing to participate patients referred to perform multichannel 24-h pH-impedance recording (Ohmega, Laborie). According to the Lyon 2.0 criteria patients were allocated to the GERD or the control group. GERD-Q questionnaire was filled in by the participants on the same day as stool samples were collected. Thermo TSQ 8000 triple quadrupole mass-spectrometer (GC/MS) was used for the measurement of VOCs in stool samples were analyzed with. Correlation analysis (Spearman rank R) was performed between values of 80 VOCs in stool samples and the results of GERD-Q questionnaire.

Results: Data of 76 patients were eligible: 42 with GERD and 34 - of the control group. Participants in the GERD group were older (54.7±12.2 y.o. vs 46±14.9 y.o., $P<0.05$) and their BMI was greater (30.9±5.2 vs 27.2±6.5 kg/m^2, $P<0.05$). GERD-Q score was logically greater in GERD group (11.0±2.5 vs 5.2±1.2, $p<0.01$). Correlation analysis revealed significant correlations between GERD-Q score and concentrations of 2-methylbutyric acid (Spearman R =-0.31), caproic acid (0.34) and enantiic acid (0.396) in stool samples.

Conclusion: GERD-Q score positively correlates with concentrations of Caproic acid (0.34) and Enantiic acid (0.396) in stool samples and negatively – with 2-Methylbutyric acid.

The data of this pilot study may serve as a basis for further research on the role of functional activity of gut microbiome in pathogenesis of GERD.

Disclosure: Nothing to disclose.

PP0004

UNRAVELING GI SYMPTOMS IN HYPERMOBILITY EHLERS–DANLOS SYNDROME: A FUNCTIONAL AND STRUCTURAL DIAGNOSTIC PERSPECTIVE

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Introduction: Ehlers–Danlos syndrome (EDS) is a connective tissue disorder characterized by a non-inflammatory nature. Hypermobility EDS (hEDS) accounts for 80–90% of all EDS cases[1].

Over the past two decades, studies have reported a high prevalence of gastrointestinal (GI) symptoms among hEDS patients, including abdominal pain, bloating, constipation, diarrhea, nausea, vomiting, and reflux disease.

Aims & Methods: To characterize the GI symptoms in patients with hEDS and to explore the potential contribution of functional and anatomical abnormalities to the GI manifestations observed in this population.

We conducted a retrospective cohort study of patients diagnosed with hEDS who were evaluated at both the EDS and Gastroenterology clinics. Eligible patients had a clinical diagnosis of hEDS based on the Brighton criteria.

Patients with a documented diagnosis of inflammatory bowel disease (IBD) were excluded. Demographic data, GI symptoms, and results from diagnostic evaluations—including anorectal manometry, balloon expulsion test, defecography, gastric emptying scans, transit studies, and endoscopic procedures—were collected and analyzed.

Results: A total of 76 patients with hEDS were included with a median age of 33.7 years. Common comorbidities included endometriosis (43.5%), fibromyalgia (36.8%). Macrophage activation syndrome was diagnosed in 2.6%.

Upper GI symptoms were frequently reported, with heartburn (31.6%), postprandial pain (30.3%), early satiety (25%), epigastric pain unrelated to meals (22.4%), and dysphagia (13.2%) being the most prevalent.

Lower GI complaints were even more common: constipation affected 64.5% of patients, bloating 57.9% and chronic diarrhea 40.8%. Notably, 19.7% experienced both constipation and diarrhea.

Among 32 patients who underwent colonoscopy, 30 had normal findings. Anorectal manometry revealed hyposensitivity in 54% of constipated patients and hypersensitivity in 54% of diarrhea patients. Squeeze pressure was low in 38% of diarrhea patients. Balloon expulsion failure occurred in ~50% of patients across groups.

Defecography demonstrated a high prevalence of anatomical abnormalities: rectocele (67–78%), enterocele (47–78%), and rectal prolapse (53–78%). Gastroparesis was identified in 45–60% of patients undergoing gastric emptying scans due to lower GI complaints. Prolonged colonic transit was found in 25–50% of those evaluated.

Conclusion: Patients with hEDS commonly experience a broad range of upper and lower GI symptoms, often presenting with multiple overlapping complaints. Functional and structural abnormalities—such as rectal hyposensitivity, pelvic floor dysfunction, rectocele, and enterocele—were frequently identified and may underlie these symptoms. Notably, gastroparesis was prevalent, highlighting the widespread impact of EDS on GI motility.

These findings underscore the importance of comprehensive GI assessment in EDS patients, integrating both functional testing and imaging to guide appropriate management.

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Disclosure: Nothing to disclose.

PP0005

IMPACT OF THYROIDECTOMY ON ESOPHAGEAL MOTILITY: A PROSPECTIVE SINGLE CENTER STUDY

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Introduction: Thyroidectomy is one of the most common surgical procedures worldwide, yet its potential effects on esophageal function remain inadequately characterized. While post-thyroidectomy issues such as voice changes have been well-documented, alterations in esophageal motility following this procedure have received limited attention in scientific literature.

Aims & Methods: This study aims to evaluate the impact of thyroidectomy on esophageal motility parameters using High-resolution manometry (HRM), with particular attention to upper esophageal sphincter (UES) function. Seventeen patients (11 females), who were to undergo total thyroidectomy or hemithyroidectomy, were prospectively recruited between the months of February and May 2023. Ten of them had goiters, while seven had thyroid nodules. Exclusion criteria were having an age under 30 or over 70 years old, having to also undergo a parathyroid surgery and taking medications that could alter the esophageal sphincter. All of them underwent a pre-operative HRM and a second one thirty days after surgery. HRM were performed according to the Chicago Classification 4.0 and the data were analyzed using the online analysis platform "Swallow Gateway". Gastrointestinal (GI) symptoms were assessed using a standardized questionnaire, before the first and the second HRM. No after procedural complication was reported.

Results: UES function significatively changed after thyroidectomy, as it is shown by HRM metrics such as Pharyngeal Contractile Integral (CI), Velopharyngeal CI, Hypopharyngeal CI, UES-CI, UES Integrated Relaxation Pressure (UES-IRP), Post-Deglutitive UES CI, Proximal Contractile Integral (PCI), which all significatively decreased ($p < 0.05$) at the HRM performed after surgery. Compared to Swallow Gateway normal values, there was an important increase of pathological HRM metrics after thyroidectomy. No significant HRM difference was found between the two categories of patients (goiter and thyroid nodules) and no significant correlation was registered between GI symptoms pre- and post-surgery and between GI symptoms and HRM metrics.

Conclusion: Thyroidectomy may significantly affect esophageal motility, especially the function of the upper esophageal sphincter. This finding could be useful in the diagnostic and therapeutic pathway of these patients, although further and larger studies are required to confirm this result.

Disclosure: Nothing to disclose.

PP0006

ESOPHAGEAL MOTILITY ALTERATIONS ON HIGH-RESOLUTION MANOMETRY IN ASYMPTOMATIC PATIENTS WITH SYSTEMIC SCLEROSIS: PREVALENCE AND CLINICAL CORRELATIONS

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Introduction: Systemic sclerosis (SSc) is a chronic autoimmune disease characterized by multisystemic fibrosis. Gastrointestinal involvement, primarily affecting the esophagus, is common and represents the third most frequent clinical manifestation after Raynaud's phenomenon and cutaneous alterations. While several studies have investigated esophageal involvement using manometry, few have focused on asymptomatic SSc patients.

This study aimed to analyze esophageal motility disorders in these patients using high-resolution manometry and identify potential associations with other SSc characteristics.

Aims & Methods: This retrospective, descriptive, bicentric study included patients diagnosed with systemic sclerosis but without clinical signs of esophageal involvement (i.e., absence of gastroesophageal reflux, epigastric pain, or dysphagia). These patients underwent esophageal manometry in the gastroenterology department over a 20-year period (2004-2024). Clinical manifestations were assessed according to standardized protocols: skin involvement was evaluated using the modified Rodnan score, while pulmonary involvement was evaluated through clinical examination, imaging (high-resolution computed tomography), and pulmonary function tests.

Statistical analysis was conducted in two phases: a descriptive phase followed by an analytical phase, using SPSS software (version 26). The Chi-square test was employed, with a significance threshold set at $p < 0.05$.

Results: The study included 47 patients, predominantly female (46 women, 1 man), with a median age of 32 years (range: 27-74 years). All patients underwent esophageal manometry, which revealed normal motility in 7 cases.

However, the majority exhibited abnormalities, including lower esophageal sphincter (LES) hypotonia in 21 patients, ineffective esophageal motility in 10, and aperistalsis in 9 ($p = 0.03$).

These esophageal motility disorders appeared to be linked to the type of SSc. Specifically, 87.2% of patients had the localized form of the disease, whereas 12.8% had diffuse SSc, which was more frequently associated with LES hypotonia ($p < 0.05$).

Additionally, pulmonary involvement was observed in 53.2% of patients, suggesting a connection between esophageal motility disorders and respiratory impairment. More specifically, LES hypotonia was particularly associated with interstitial lung disease (21.9%) and restrictive syndrome on pulmonary function tests ($p < 0.05$).

Immunological profile analysis revealed a higher prevalence of manometric abnormalities in patients negative for anti-topoisomerase I (Scl-70) (57.6%), anti-centromere (64.7%), anti-RNA polymerase III (79.4%), and anti-PM-Scl (76.5%) antibodies. These associations were statistically significant ($p < 0.05$).

Conclusion: This study revealed a notably high prevalence of esophageal motility disorders, particularly lower esophageal sphincter hypotonia, among asymptomatic systemic sclerosis patients. The disorder was correlated strongly with diffuse cutaneous involvement, pulmonary complications, and the absence of specific autoantibodies (e.g., anti-centromere

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and anti-Scl-70). These findings highlight the critical need for proactive screening in at-risk individuals to facilitate timely interventions and mitigate potential complications.

Disclosure: Nothing to disclose.

PP0007

CORRELATION BETWEEN CUTANEOUS, PULMONARY, AND UPPER DIGESTIVE INVOLVEMENT IN SYSTEMIC SCLEROSIS

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Introduction: Systemic sclerosis (SSc) is a rare autoimmune disease marked by multisystemic fibrotic and vascular involvement, mainly affecting the skin, lungs, and upper digestive tract. Skin involvement may serve as a marker for visceral severity, particularly regarding pulmonary and upper digestive manifestations. This study examined its correlation with these systemic complications in SSc patients.

Aims & Methods: A retrospective, descriptive, bicentric study was conducted on SSc patients followed in the internal medicine department between 2004 and 2024. Clinical evaluation included cutaneous involvement via modified Rodnan score, pulmonary assessment by clinical examination, imaging, and pulmonary function tests. Upper digestive involvement was explored through symptoms, esophagogastroduodenoscopy, and esophageal manometry performed in the gastroenterology department to assess motility and mucosal integrity. Statistical analysis (descriptive and analytical) was performed using SPSS (version 26) with the Chi-square test and a significance threshold of $p < 0.05$.

Results: A total of 160 patients were included, mostly female (91.2%), with a median age of 50 years (range:13–81). Localized scleroderma was diagnosed in 76.2%. Skin involvement was assessed in 121 patients using the Rodnan score (mean:15.31; range:0–51), categorized as mild (0–10) in 35%, moderate (11–20) in 15%, severe (21–30) in 16.3%, and very severe (>30) in 9.4%.

Upper digestive involvement was observed in 31.4%, 14.9%, 18.2%, and 9.1% of patients with mild to very severe skin involvement, respectively. Among associated symptoms: epigastric pain was present in 8.4%, 5%, 5%, and 3.4%; gastroesophageal reflux disease (GERD) in 21.2%, 9.3%, 16.9%, and 8.5%; and dysphagia in 15.3%, 7.6%, 13.6%, and 6.8%, respectively.

Esophageal manometry was performed in 36.2%, showing lower esophageal sphincter (LES) hypotonia in 5%, 2.5%, 5%, and 3.3% of mild to very severe cases, respectively. Aperistalsis was noted in 3.3%, 1.7%, 0.8%, and 0.8%, with rare ineffective motility and hypocontractility. No significant association was found between skin severity and upper digestive involvement, either clinically ($p=0.458$) or manometrically ($p=0.98$).

Similarly, no correlation was found between skin and pulmonary involvement. Lung disease was observed in 56.3% of localized and 19.4% of diffuse forms. Among these, interstitial lung disease was present in 26.3%, 12.7%, 16.1%, and 9.3%; pulmonary fibrosis in 15.6%, 6.4%, 8.3%, and 1.8%; and restrictive syndrome in 15%, 9.3%, 13.1%, and 5.6%, across increasing skin severity (all $p > 0.3$).

However, pulmonary involvement was noted in 60% of patients with upper digestive symptoms. Dyspnea was reported in 66.9%, crackles in 29.4%, cough in 27.5%, and chest pain in 6.3%. Radiology showed interstitial lung disease in 58.8% and pulmonary fibrosis in 28.8%. Among these, GERD was present in 46.5%, dysphagia in 36.1%, epigastric pain in 19.2%, and isolated abdominal pain in one case. Manometry revealed LES hypotonia in 13.1%, aperistalsis in 5%, and ineffective motility in 0.6%. In contrast to skin involvement, a significant association was found between pulmonary and upper digestive manifestations ($p<0.01$).

Conclusion: No correlation was found between the severity of cutaneous involvement and digestive disorders. However, a significant link was observed between pulmonary involvement and upper digestive symptoms, suggesting that uncontrolled acid reflux may contribute to pulmonary fibrosis in SSc patients. This highlights the need for early management and multidisciplinary monitoring.

Disclosure: Nothing to disclose.

PP0008

IMPACT OF UPPER ESOPHAGEAL SPHINCTER (UES) ABNORMALITIES ON CLINICAL MANIFESTATIONS OF PATIENTS AFFECTED BY ESOPHAGOGASTRIC JUNCTION OUTFLOW OBSTRUCTION (EGJOO): A HIGH-RESOLUTION ESOPHAGEAL MANOMETRY STUDY

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Introduction: Upper esophageal sphincter (UES) is a muscular structure located at the transition from the pharynx to the esophagus, with the cricopharyngeal muscle as the most important component. One of its main functions is to prevent esophago-pharyngeal reflux; however, it also plays a key role during swallowing, belching, and vomiting. Data on UES involvement on pathogenesis and clinical manifestations of esophagogastric junction outflow obstruction (EGJOO) are lacking.

Aims & Methods: Aim of this study the UES manometric parameters analysis and study of their impact on pathogenesis and clinical presentation of patients affected by Esophagogastric Junction Outflow Obstruction (EGJOO) according to the Chicago Classification v4.0 criteria. Manometric tracings, epidemiological and clinical data of patients with a diagnosis of EGJOO in our departments between 07/2021- 03/2025 were analyzed. Clinical presentation was evaluated by performing Dysphagia Symptom Questionnaire (DSQ) and Eckardt score. Manometric parameters of UES function (mean basal pressure, mean residual pressure, resting time and recovery time) as well as the other manometric parameters were analyzed and correlated through multifactor statistical analysis (MFA) with clinical and epidemiological data.

Results: Data from 94 patients (73F/21M, mean age 66 years) were analyzed. 40.4% presented dysphagia, 46,8% GERD related symptoms and 12.8% Non-Cardiac Chest Pain (NCCP) as main clinical presentation at the time of HRM examination. Mean basal pressure of the lower esophageal sphincter (LES) was 50.85 mmHg (R19.5-78.8), mean integrated relaxation pressure (IRP) was 19.15 mmHg (R 15.2-40.4), mean Distal Contractile Integral (DCI) 5.031 (mmHg -cm-s) (R 3.487-7.345), mean Distal Latency 6.4 (R 4-10.2), mean UES basal pressure 53.3 mmHg (R27.8-220), mean UES residual pressure 8.75 mmHg (R 0.2 -70.9), mean UES resting time 181 ms (R48-476), mean recovery time 465 (R46-880). 34 patients (36.2%) had abnormal UES manometric parameters. Multifactorial statistical analysis revealed a statistically significant correlation of the pathological UES re-

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sidual pressure with abnormal LES basal pressure (p = 0.040), dysphagia as main clinical symptom (p=0.030), LES length (p=0.013) and dysphagia symptom severity evaluated by DSQ: p=0.017 and Eckardt: p=0.041.

Conclusion: UES motility has not been extensively studied on HRM in patients affected by EGJOO. Our study revealed that a significant number of EGJOO patients presented abnormal UES function. UES motility abnormalities in these patients are correlated with abnormal LES function, symptoms severity and particularly characterize patients with predominant dysphagia.

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Disclosure: Nothing to disclose.

PP0009

SALIVARY DNA DETECTION OF VARICELLA ZOSTER VIRUS IN PATIENTS WITH ESOPHAGEAL MOTILITY DISORDERS

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Introduction: The causes of most esophageal motility disorders (EMDs) remain unknown. Recent reports have suggested a potential association between achalasia and varicella zoster virus (VZV) reactivation. Salivary detection of VZV DNA is recognized as a potential diagnostic marker for herpes zoster and enteric zoster. If a correlation between salivary VZV DNA and VZV antibodies and EMDs is demonstrated, it could provide new insights into the pathogenesis of EMDs.

Aims & Methods: This study aimed to investigate the relationship between various EMDs and VZV infection.

This was cross-sectional study. The participants in this study underwent high-resolution manometry (HRM) at our hospital between 2022 and 2024. Saliva and blood samples were collected from all patients.

The presence of VZV DNA in the saliva was determined using nested polymerase chain reaction (PCR) and electrophoresis.

VZV-immunoglobulin M and immunoglobulin G antibody titers were measured in the blood samples. Based on the HRM results, the patients were divided into an EMD group and a normal group. The results were compared between these groups.

Results: A total of 72 patients were included. Of these, 32 (44.4%) were in the normal group and 40 (55.6%) were in the EMD group, including 22 with esophagogastric junction (EGJ) outflow disorders (achalasia and EGJ outflow obstruction) and 18 with peristalsis disorders (absent contractility, distal esophageal spasm, hypercontractile esophagus, and ineffective esophageal motility). One patient with absent contractility had a concurrent herpes zoster infection, and VZV DNA was detected in this patient's saliva. However, all other patients tested negative for VZV DNA. No significant difference in VZV antibody titers was observed between the two groups.

Conclusion: This study found no significant differences in the rates of salivary VZV DNA or VZV antibody titers between EMD patients and controls. Given the complex, multifactorial nature of EMDs, further research is warranted to explore alternative mechanisms that may underlie this group of disorders.

Disclosure: Nothing to disclose.

PP0010

DYSPHAGIA IN INEFFECTIVE ESOPHAGEAL MOTILITY IS NOT EXPLAINED BY EVIDENCE OF OBSTRUCTION OR POORER PERISTALSIS AT HIGH-RESOLUTION ESOPHAGEAL MANOMETRY: RESULTS FROM A PROSPECTIVE STUDY

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Introduction: Ineffective esophageal motility (IEM) is a rare motility disorder characterized by weak or failed peristalsis at high-resolution manometry (HRM). There is an established association between IEM and gastroesophageal reflux-disease (GERD) symptoms. However, it is unclear why some patients present with dysphagia. In this study we aimed to investigated whether dysphagia in IEM is associated with specific HRM or pH-impedance monitoring (MII-pH) findings.

Aims & Methods: Adults (≥18 y) who underwent HRM for upper gastrointestinal symptoms who were diagnosed with IEM according to Chicago v4.0 between January 2021 and November 2024 were included. Patients' gender, age, symptoms (dysphagia, chest pain, heartburn, regurgitation, belching), response to proton pump inhibitor (PPI) treatment and HRM data were recorded.

All patients underwent MII-pH. Wilcoxon rank sum test, Pearson's chi-squared test and Fisher's exact test were used for comparisons. Significance was set at $p<0.05$.

Results: In the study period, 54 patients were diagnosed with IEM. Of them, 33 (61%) did not report dysphagia (IEM-GERD), while 21 (39%) had dysphagia (IEM-DYSPHAGIA). Of note, patients with or without dysphagia had a comparable prevalence of heartburn, chest pain, regurgitation, and erosive esophagitis.

In addition, IEM-GERD and IEM-DYSPHAGIA were comparable in age and response to PPI treatment. IEM-GERD and IEM-DYSPHAGIA did not show significant differences in EGJ type, EGJ contractile integral (EGJ-CI), integrated relaxation pressure (IRP), percentage of ineffective or failed swallows, or the rapid drinking challenge (RDC) findings ($p>0.05$ for all), demonstrating an overall similar pathophysiology of peristalsis and EGJ opening. In terms of MII-pH, there was no statistical difference in GERD-related parameters between IEM-GERD and IEM-DYSPHAGIA, who showed comparable values of acid exposure time (AET), mean nocturnal baseline impedance (MNBI), symptoms association indexes (SI), and post-reflux swallow-induced peristatlic wave index (PSPW-I) ($p>0.05$ for all).

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Conclusion: We showed that dysphagia in IEM is not explained by impaired EGJ outflow or worse peristalsis. IEM with and without dysphagia have a similar proportion of GERD symptoms, comparable MII-pH findings, and respond similarly to PPI treatment. Our findings suggest that dysphagia in IEM could be a symptomatic manifestation of GERD.
Disclosure: Nothing to disclose.

PP0011

THE PREVALENCE OF POST-INFECTION AND POST-SEPSIS DYSPHAGIA IN HOSPITALISED PATIENTS, AN OVERLOOKED PATHOLOGICAL ENTITY

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Introduction: Studies have generally highlighted three groups of patients at particular risk of dysphagia: the elderly, those with neurological pathology and the critically ill. Over the last decade, there has been increasing interest in the role of critical illness related dysphagia, however, less is known about the impact of infection and sepsis on unwell patients in hospital. In this study, we aimed to assess the prevalence and factors predicting dysphagia in patients with no prior history of dysphagia, admitted to hospital with infections or sepsis.
Aims & Methods: Participants ≥ 18 years of age admitted to hospital with infections or sepsis caused by pneumonia, urine tract infections or cholecystitis were recruited from an NHS Trust. Researchers, electronic patient record administrators and clinical coding staff screened patient data over 12 months from 01/02/2022 to 01/02/2023. All patients with potentially confounding conditions or pathology, such as previous or ongoing dysphagia, acute head injuries or stroke, were excluded. Those that remained had data extracted from their notes, including age, gender, length of stay (LOS), intensive care unit (ICU) admission status, co-morbidities, type of infection/sepsis, obesity, frailty and mortality. Patient data was then subjected to statistical analysis, including Chi-squared, T-Tests, Logistic regression and Cox regression.
Results: Over the study period, 4475 patients (M: 2031, F: 2444) were admitted with infections (Pneumonia 2465, UTI 1888 and Cholecystitis 401). Of these, 189 (4.2%) developed dysphagia (Pneumonia 6.0%, UTI 3.5%, Cholecystitis 0.8%).
Additionally, 865 patients (M: 434, F: 431) were admitted with sepsis (Pneumonia 535, UTI 413 and Cholecystitis 55). Of these, 54 (6.2%) had dysphagia (Pneumonia 9.3%, UTI 3.4%, Cholecystitis 3.6%).
Logistic regression revealed that increasing age, the presence of sepsis, delirium, dementia and Parkinson's disease (PD) were significantly associated with an increased risk of dysphagia (OR 1.02, 1.74, 1.60, 2.43, 3.88; P 0.002, 0.013, 0.008, <0.001, <0.001). Obesity and female gender were associated with a reduced risk of dysphagia (OR 0.53, 0.65; P 0.006, 0.001). Other comorbidities such as diabetes, chronic obstructive pulmonary disease (COPD), cirrhosis and heart failure were not significantly associated with dysphagia.
Additionally, Cox regression revealed that increasing age, sepsis, dysphagia, delirium and frailty all significantly increased LOS (HR 0.99, 0.63, 0.59, 0.67, 0.79; P <0.001, <0.001, <0.001, <0.001, <0.001).
Conclusion: Our study has identified that infection and sepsis-associated dysphagia is a major syndrome, with age being a significant determinant. Additionally, patients with PD, delirium or dementia without previous or existing dysphagia are at high risk of developing dysphagia should they be admitted with infections or sepsis. Our novel findings should increase clinical awareness of infection and sepsis-associated dysphagia in hospitalised patients and prompt greater study of this phenomenon.
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Disclosure: None to disclose.

PP0012

A RETROSPECTIVE SERVICE EVALUATION TO INVESTIGATE THE DISCORDANCE BETWEEN IDENTIFIED OESOPHAGEAL MOTILITY DISORDERS FOR LIQUID AND SOLID SWALLOWS UTILISING THE CHICAGO CLASSIFICATION 4.0

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Introduction: The Chicago Classification version 4.0©(CC4) provided an updated classification of oesophageal motility disorders based on high-resolution oesophageal manometry (HRM) findings[1-2]. A key modification included using solid swallows which have been shown to increase the diagnostic yield of oesophageal motility disorders[3]. CC4 fails to offer guidance on the weighting of relevant findings when wet and solid swallows are discordant[3]. Clinical judgement, such as incorporating patient reported symptoms, is therefore required when interpreting HRM findings[2-3].
Aims & Methods: This retrospective service evaluation aimed to determine the frequency of CC4 diagnostic discordance between wet and solid swallows and how this was managed clinically. Data was collected from consecutive patients at a single National Health Service Trust (UK) site over a 6-month period. HRM was performed using a 36-channel HRM solid state catheter with impedance (Medtronic ManoScan™ ESO HRM) following the upright study CC4 protocol[2]. All patients had 5 mL saline swallows and those with dysphagia or abnormal primary swallows underwent solid swallows (rice test meal)[3].
Results: 69 patients underwent HRM during the 6-month period. 51 studies included solid swallows, 61% (n=31) showed agreement between wet and solid CC4 findings and 39% (n=20) showed disagreement. Of those with discordance, it was concluded that 9 were inconclusive due to mixed appearances. In 7 studies the solid swallows were deemed more clinically relevant and in 4 studies the liquids swallows best reflected the clinical context.
10 patients had abnormalities detected on wet swallows but normal solids, 50% of which showed ineffective oesophageal motility (IOM) and 50% showed oesophago-gastric junction outflow obstruction (OGJOO). Of the patients with OGJOO on wet swallows and normal solids, 3 were concluded to be normal due to a lack of symptoms, good bolus clearance on HRM and normal endoscopy and barium swallow results. One was deemed inconclusive as the patient was symptomatic to bolus hold-up measured on HRM. One was concluded as OGJOO due to abnormal rapid drink challenge, upright swallows and supine swallows.

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Of the 5 studies showing IOM on liquid swallows but normal solids, 2 were overall concluded to be normal due to a lack of symptoms. 2 were concluded as IOM, although only one was symptomatic to ineffective swallows. Both reports did, however, mention evidence of peristaltic reserve. One study was concluded inconclusive as possible IOM.

3 patients who had normal liquid swallows had abnormalities on solids. One showed IOM on solids which was concluded as insignificant due to a lack of symptoms. One had hypercontractile oesophagus which was concluded as significant due to the presence of symptoms[3].

The other showed symptomatic OGJOO with solids. This was classed as inconclusive as the patient had normal bolus clearance on barium swallow.

Conclusion: More patients had abnormalities on liquid swallows which normalised on solids, despite studies indicating increased motility disorder diagnosis with solid swallows[4].

Test analysers placed greater significance on solid swallow findings compared to liquids. However, many tests were reported as inconclusive. Tests with abnormal liquid swallows but normal solids frequently had differing conclusions, in part due to adjunctive test findings and reported symptoms. This emphasizes the importance of symptoms and overall clinical judgement when reporting. Future CC iterations should consider discordant findings and offer guidance on what is clinically relevant during HRM studies.

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PP0013

RESPONDER ANALYSIS FOR THE EFFICACY OF TRADIPITANT IN IDIOPATHIC AND DIABETIC GASTROPARESIS

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Introduction: Tradipitant is a novel NK1R antagonist studied in diabetic and idiopathic gastroparesis. This report presents a nausea responder analysis from the open-label arm of a multicenter, randomized, double-blind, placebo-controlled study (VP-VLY-686-3301) assessing the efficacy of tradipitant in relieving symptoms of gastroparesis.

Aims & Methods: N=531 idiopathic and diabetic gastroparesis patients with delayed gastric emptying received open label 85mg tradipitant BID for 12 Weeks. Symptoms were assessed using the 5-point Gastroparesis Core Symptom Daily Diary (GCSDD). A nausea responder (N=267) was defined as having ≤1 average nausea severity at week 12 using the GCSDD. Gastroparesis symptoms were assessed every 2 weeks in clinic using the Gastroparesis Cardinal Symptom Index (GCSI).

Results: Nausea responders improved average nausea score significantly from baseline after 12 weeks tradipitant (improvement of -1.40 vs 1.67 at baseline, p<0.0001). A significant improvement was also seen in the percent of nausea free days (an addition of 49.32% of days for tradipitant vs 28.57% at baseline, p<0.0001). Significant improvement was also seen in other cardinal gastroparesis symptoms including average times vomited in the past 24 hours (improvement of -0.32 vs 0.40 at baseline, p<0.0001), stomach fullness (improvement of -1.48 vs 2.86 at baseline, p<0.0001), not being able to finish a normal-sized meal (improvement of -1.45 vs 2.47 at baseline, p<0.0001), bloating (improvement of -1.36 vs 2.70 at baseline, p<0.0001), and upper abdominal pain (improvement of -1.25 vs 1.99 at baseline, p<0.0001).

Conclusion: Tradipitant demonstrated statistically significant improvement from baseline in average nausea severity in this 12 week open label study in both idiopathic and diabetic gastroparesis subjects. Notably, nausea responders on tradipitant also significantly improved in all cardinal gastroparesis symptoms including those unrelated directly to nausea such as abdominal pain. These results suggest tradipitant has the potential for comprehensive gastroparesis symptom relief in tradipitant responders.

Disclosure: Study was sponsored by Vanda Pharmaceuticals, Inc. Authors are employees of Vanda Pharmaceuticals, Inc.

PP0014

THE FRENCH CONSENSUS ON IDIOPATHIC GASTROPARESIS IN ADULTS AND CHILDREN

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Introduction: Idiopathic gastroparesis (IG) remains a challenging condition, with few effective therapeutic options available. Despite recent European and international consensuses, some management modalities are still different in the French practice (1).

Aims & Methods: We aimed to develop French recommendations on the diagnosis and management of IG in adults and paediatric patients, highlighting the similarities and differences with recent consensuses. A Delphi

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process was conducted among a panel of 16 French neurogastroenterology experts, including gastroenterologists and paediatricians. A first list of statements was drafted with a literature summary. After revisions and clarifications, 3 voting rounds were conducted and consensus statements were identified (defined as ≥ 80% agreement).

Results: Consensus was reached for 39 out of 46 statements. Definition and diagnostic modalities were similar to recent guidelines. IG was defined as symptoms associated with delayed gastric emptying (GE), in the absence of a mechanical obstruction. The panel agreed that nausea and vomiting are cardinal symptoms, with postprandial distress symptoms frequently overlapping. Variable presentations according to age were highlighted in paediatric patients.

Delayed GE can be confirmed by either scintigraphy or a GE breath test, using a solid meal, and a 4-hour measurement. Food retention at endoscopy was not considered a diagnostic criterion. Pyloric EndoFlip was not considered a diagnostic or pretherapeutic tool. A blood test and screening for *Helicobacter pylori* were supported before diagnosing IG.

Regarding the management, like in recent guidelines, the panel supported a first-line therapy using dietary advice and dopamine antagonists (metoclopramide or domperidone). The panel supported the efficacy of intravenous erythromycin, but not oral erythromycin. Agreement on the efficacy of other drugs, including prucalopride, antiemetics, or neuromodulators, was not reached. The use of nutritional support was recommended if deemed necessary for refractory vomiting and malnutrition, with enteral post-pyloric nutrition recommended as the first option.

The consensus supported different strategies for the management of refractory gastroparesis. Stopping opioids, or drugs inducing delayed GE, and the treatment of associated constipation were recommended before considering more invasive procedures. The efficacy of pyloric toxin injection was not agreed upon. Gastric endoscopic pyloromyotomy and gastric electrical stimulation were considered effective therapies for refractory IG. Finally, surgical procedures were not supported.

Conclusion: This consensus shows that the diagnostic modalities for IG are similar in France as compared to international guidelines, but highlights differences in the management of refractory gastroparesis, supporting endoscopic pyloromyotomy or the implantation of gastric electrical stimulation.

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PP0015

GASTROINTESTINAL SYMPTOMS: POOR PREDICTORS OF TRANSIT ABNORMALITIES

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Introduction: Gastrointestinal (GI) transit abnormalities due to dysmotility often affect multiple regions of the GI tract rather than occur in isolation. Identifying regional GI transit delays may influence clinical management. Use of wireless motility capsules (WMC) enable regional and whole GI transit times to be objectively measured; the relationship between symptom profiles and regional transit times are explored here with a new capsule device, the Atmo Capsule.

Aims & Methods: The Atmo Capsule was administered to subjects with suspected upper and/or lower gastrointestinal dysmotility. This analysis utilized data from subjects where valid transit measurements via the Atmo Capsule were available for all regions of the GI tract, and no protocol deviations pertaining to the use of contraindicated medication or inclusion criteria violations were incurred. Regional transit times were considered delayed as follows: gastric emptying ≥ 5 h; small bowel transit, ≥ 6 h; colonic transit, ≥ 59 h.

Delays affecting a single region (i.e., gastric, small bowel, or colon) were considered 'isolated', delays affecting multiple (> 1) regions were considered 'generalized'. Symptom profiles were assessed using the Patient Assessment of Gastrointestinal Disorders - Symptoms (PAGI-SYM) and Patient Assessment of Constipation – Symptoms (PAC-SYM) questionnaires: all patients completed the PAGI-SYM while those with symptoms suggestive of lower GI dysmotility completed the PAC-SYM. Data were analyzed via Kruskal-Wallis test.

Results: The WMC was administered to 209 subjects (median age 41; 188 female), of whom 170 had data suitable for this analysis. Delayed GI transit was identified in 100 (59%) subjects: 65 were isolated and 35 were generalized. There were no differences in overall symptom scores in subjects with no delays (PAGI-SYM, median 1.5 [IQR: 1.1, 2.1]; PAC-SYM, 2.0 [1.5, 2.6]), isolated delays (1.5 [1.0, 2.1]; 1.9 [1.2, 2.6]) and generalized transit delays (1.8 [1.4, 2.3]; 2.1 [1.4, 2.8]) ($p>0.05$ for both) or symptom sub-scores (Table 1).

Gastric transit was delayed in 46 subjects: 19 (41%) were isolated and 27 (59%) were generalized. Colonic transit was delayed in 60 subjects: 31 (52%) were isolated and 29 (48%) were generalized. GI symptom profiles were not different in subjects with isolated and generalized transit delays for either the stomach or colon ($p>0.05$ for all relevant subscales).

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Questionnaire	Subscale	None	Isolated	Generalized	P value
Patient Assessment of Gastrointestinal Disorders – Symptoms	Heartburn/regurgitation	1.6 (1.1, 1.9)	1.6 (0.9, 2.0)	1.9 (1.4, 2.3)	0.11
	Nausea/vomiting	0.7 (0.0, 1.3)	0.7 (0.0, 1.3)	1.0 (0.3, 1.7)	0.39
	Postprandial fullness	2.0 (1.0, 2.5)	2.0 (1.5, 3.0)	2.5 (1.5, 3.0)	0.47
	Upper abdominal pain	1.5 (1.0, 2.1)	1.5 (1.0, 2.0)	2.0 (1.3, 2.5)	0.18
	Bloating	2.0 (1.5, 2.6)	2.0 (1.0, 2.5)	2.0 (1.5, 2.5)	0.86
	Lower abdominal pain	1.5 (1.0, 2.5)	1.5 (1.0, 2.5)	2.0 (1.5, 2.5)	0.50
Patient Assessment of Constipation – Symptoms	Abdominal	2.3 (1.5, 2.8)	2.0 (1.3, 2.8)	2.3 (1.8, 2.9)	0.59
	Rectal	1.5 (1.0, 2.3)	1.5 (0.8, 2.3)	1.5 (0.8, 2.4)	0.67
	Stool	2.3 (1.8, 3.0)	2.0 (1.0, 2.8)	2.8 (1.3, 3.4)	0.15

Table 1. Symptom profiles amongst patients with no transit delay, isolated transit delay (affecting a single region) and generalized transit delay (affecting multiple regions). Data presented as median (interquartile range) and analyzed via Kruskal-Wallis test.

Conclusion: WMC measurements of regional and whole GI transit using the Atmo Capsule revealed that delayed transit impacting multiple regions were common. However, symptom profiles were unable to distinguish between isolated vs. generalized transit delays, suggesting that symptoms are an unreliable predictor of abnormalities in GI transit as previously found via other transit modalities.

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PP0016

HYPERCONTRACTILE ESOPHAGUS: ENHANCED DIAGNOSTIC YIELD FROM PROVOCATIVE TESTS WITH HIGH RESOLUTION MANOMETRY. CLINICAL PRESENTATION AND LONG TERM OUTCOMES FROM AN ASIAN COHORT

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Introduction: Hypercontractile esophagus (HE) is rare. Esophageal high-resolution manometry (HRM) with solid test meal (STM) in Chicago Classification (CCv4.0) may improve diagnostic yield but outcome data are lacking.

Aims & Methods: Aim: Evaluate the clinical features and long-term outcome of HE, and determine the clinical significance of hypercontractility detected only using STM.

Methods: Single-center analysis of all patients with >/=2 hypercontractile swallows (HS) (distal contractile integral [DCI]>8000mmHg.s.cm) on HRM studies with single water swallows(SWS) and/or STM between June 2014-December 2021. A telephone survey was conducted between January-June 2023 using Impaction Dysphagia Questionnaire [IDQ], Eckardt scores and the Gastro-Esophageal Reflux Disease Questionnaire (GERD-Q) to determine long-term outcomes.

Results: 48 patients (29[60.4%]Female, 63.0[13.4] years) with HE who presented with dysphagia (60%), reflux (37.5%) and chest pain (29.2%) and 58 controls (28[48.3%]Female,49[13.4] years) with dysphagia(n=43.1%), reflux (n=41.4%) and chest pain(15.5%) with normal HRM findings with both SWS and STM were identified from 454 studies performed. More patients had HS with STM vs SWS (n=41[85.4%] vs. n=7[14.6%], p<0.001). Dysphagia was significantly associated with mean DCI(B=0.000,p=0.035) and maximal DCI (B=0.000, p=0.036) during STM. 43% (n=3/7) and 22% (n=9/41) of patients with >/=2 HS using SWS and STM respectively were on medical therapy at mean(SD) clinic follow-up of 28.7(29.2) months. At mean(SD) follow-up telephone survey of 61.3(27.1) months from HRM, symptom scores (mean,SD) amongst 36/48 (75%) patients and 58 controls showed significantly higher dysphagia scores: IDQ (3.8[7.2] vs 1.0[1.9],p=0.03).

Whilst overall Eckardt scores were not significantly different between patients and controls (1.2 [1.9] vs 1.2[1.4, p=0.4), patients recorded significantly higher Eckardt dysphagia score compared to controls (0.46[0.7] vs 0.14[0.3], p=0.007) respectively. Conversely, patients had lower Eckardt chest pain scores compared to controls (0.14[0.36] vs 0.45[0.68], p=0.02) and similar GERDq scores (6.4[1.9] vs 7.0[1.9], pNS) respectively.

Conclusion: STM enhanced diagnostic yield of hypercontractile esophagus and identified more patients who benefitted from medical therapy. Dysphagia was significantly associated with HRM parameters during the STM, with higher symptoms scores for dysphagia compared to controls at long term follow up. However, most patients did not require aggressive intervention.

Disclosure: Nothing to disclose.

PP0017

THE RISE OF GLUCAGON-LIKE PEPTIDE-1 RECEPTOR AGONISTS (GLP-1RA) – A CAUSE FOR CONCERN? A RETROSPECTIVE STUDY ON THE GASTROINTESTINAL EFFECTS OF GLP-1RA IN ASIAN PATIENTS

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Introduction: Obesity and type 2 diabetes mellitus (T2DM) represent growing global health challenges, affecting millions of individuals worldwide. Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have emerged as a cornerstone in the management of both conditions due to their metabolic benefits, including weight loss and glycemic control. European cohort studies have reported a high prevalence of gastrointestinal symptoms – including nausea, vomiting and early satiety among patients using GLP-1RAs. In parallel, increased rates of residual gastric content (RGC) during upper endoscopy have raised safety concerns, prompting reevaluation of pre-procedural fasting guidelines. However, data on the effects of GLP1-RA on the gastrointestinal (GI) tract in Asian patients is limited.

Aims & Methods: Our primary aim was to characterize a cohort of patients who received treatment with GLP1-RA in terms of their gastrointestinal side effects and presence of RGC during upper endoscopy. The secondary objective was to compare the afore-mentioned variables be-

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tween 2 groups: oral versus subcutaneous mode of administration. We conducted a retrospective cross-sectional cohort study of 139 Asian adult patients with T2DM and/or obesity who were prescribed GLP-1RAs at a tertiary hospital between 01 January 2022 to 01 January 2023. Data collected included demographics, anthropometric measurements, co-morbidities, GI symptoms and upper endoscopy findings. Upper endoscopy images were analysed and adequacy of mucosal visibility was graded using the POLPREP scale. Patients were further stratified into two subgroups based on route of GLP-1RA administration: oral versus subcutaneous. Statistical analysis compared symptom prevalence and POLPREP scores across subgroups using chi square test, with p <0.05 considered statistically significant.

Results: A total of 139 patients were analysed. 19.4% of patients had GLP-1RA prescribed for T2DM, 21.6% for Obesity, and 58.8% for both. The most commonly prescribed GLP-1RA was Liraglutide (33.8%) followed by oral Semaglutide (28.1%). Majority of patients (66.9%) did not have any GI side effects. The most commonly reported GI symptoms were abdominal bloating (15.8%), nausea (7.9%) and early satiety (4.3%). 13.7% of patients underwent upper endoscopy with 89.5% having a POLPREP score of 3 with no significant complications reported. None of the patients withheld GLP-1RA prior to endoscopy. Upon stratifying patients based on route of administration, analysis revealed that patients on oral GLP-1RAs were more likely to report nausea compared to those on subcutaneous formulations (15.8% vs 5.0%, p=0.03). There were no significant differences in the other GI symptoms or POLPREP scores.

Conclusion: Based on our Asian cohort, these findings suggest that GLP-1RAs have an acceptable GI symptom profile with a potential route-dependent difference in symptomatology. In addition, discontinuing these drugs prior to upper endoscopy may be unnecessary.

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Disclosure: Nothing to disclose.

PP0018

DEVELOPMENT OF A QUALITY SCORE FOR HIGH- RESOLUTION ESOPHAGEAL MANOMETRY

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Introduction: High-resolution manometry (HRM) is an invasive and technically demanding procedure. Despite the existence of a well-established standard protocol, technically imperfect studies are common and may reduce diagnostic certainty.

Aims & Methods: The aim of this study was to develop a practical and feasible quality score for HRM that correlates the technical quality of the study with the reliability of the diagnosis.

A 10-point quality score was developed based on seven quality indicators: catheter passage across the esophagogastric junction (EGJ), evaluable resting pressures, valid wet swallows, completion of multiple rapid swallows (MRS) and rapid drink challenge (RDC), upright swallows, and absence of artifacts.

The obtained score was used to classify manometric studies into five diagnostic reliability categories: 1) 0–2 points: Uninterpretable manometry, 2) 3–4 points: Deficient manometry with low diagnostic reliability, 3) 5–6 points: Deficient manometry with moderate diagnostic reliability, 4) 7–8 points: Deficient manometry with high diagnostic reliability, 5) 9–10 points: Excellent manometry allowing a certain diagnosis

An initial exploratory evaluation was conducted by two observers (one high-experience, one low-experience observer) who prospectively reviewed 90 HRM studies. A revised version of the score was subsequently implemented and applied in a second evaluation round. In addition to assigning a final quality score, each observer provided the manometric diagnosis of each revised HRM according to Chicago IV classification (CC4.0) and rated the diagnostic confidence level (high, medium, low reliability). In case of disagreement, a third expert adjudicated.

Results: Initial agreement on quality score classification was reached in 30 cases (Cohen's kappa=0.274) using the original score, improving to 67 out of 90 studies when the revised version was used (Cohen's kappa=0.718). The main sources of disagreement in the remaining 23 studies were the number of valid swallows (12 studies) and the presence of intragastric pressurization (8 studies).

Further evaluation by the expert supported the high-experience observer in 13 cases and the low-experience observer in 5; no consensus was reached in five studies. Overall, 85% of the studies were scored as high-quality studies (score≥7). Diagnostic confidence was rated as high in 86 % of the cases, medium in 14 %, and no cases were rated as low confidence.

There was a good association between quality score and diagnostic confidence among the observers. Hence, 90 % of the studies rated as high-quality studies had a high confidence rate in the diagnostic accuracy, and 62 % of the low-quality studies had a medium level of confidence (p<0.001).

Despite some uncertainty, the two observers agreed on diagnosis in 82 studies (disagreement occurred in two low-quality studies and six high-quality studies).

Following expert adjudication, full consensus in diagnosis was reached in all 90 studies. Notably, most of the diagnoses were normal motility or ineffective esophageal motility (86 %), while only 10 % were classified as major motility disorders. Four patients had inconclusive diagnoses according to CC4.0 (two low-quality and two high-quality studies).

Conclusion: A new developed quality score appears to be a valuable tool

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for assessing HRM study quality and supporting final diagnostic accuracy. Ongoing larger studies aim to determine its potential in clinical practice.
Disclosure: Nothing to disclose.

PP0019

ENDOLUMINAL FUNCTIONAL LUMEN IMAGING PROBE IN THE FUNCTIONAL ASSESSMENT OF PYLORIC SPHINCTER IN GASTROPARESIS: A SYSTEMATIC REVIEW WITH META-ANALYSIS OF NORMATIVE VALUES

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Introduction: Gastroparesis (GP) is a chronic motility disorder characterized by recurrent upper gastrointestinal (GI) symptoms, namely early satiety, nausea and vomiting, in the absence of a mechanical obstruction. Functional Lumen Imaging Probe (FLIP) is a potential game-changer in anatomical sphincter assessment. Recently, wide prospective trials[1] and a randomized sham-controlled study[2] implemented FLIP to predict response of pyloric-targeted treatments. This has prompted the need for a comprehensive analysis of available FLIP data in GP.

Aims & Methods: The aim of this systematic review was to investigate FLIP role in GP. Meta analysis of normative values and treatment comparative outcomes was also performed. A systematic search was performed until May 2024 on 4 major databases (Pubmed-MEDLINE, Embase, Web of Science, Cochrane). Studies on post-oncological/bariatric surgery gastroparesis were excluded. Pooled baseline mean of FLIP metrics and mean differences (MDs) pre vs post-treatment were calculated. In addition to estimate clinically meaningful normative values for GP, upper-limit reference ranges were estimated by pooled standard deviations (SDs) implementing the formula *pooled mean ± 1.96 × pooled SD*. Random effects meta-analysis with Knapp-Hartung correction for 95% confidence intervals (CIs) adjustment and I^2 statistics with subgroup analysis to address heterogeneity were performed.

Results: Twenty studies (N= 805 patients) were included. The majority focused on treatment response, particularly pyloromyotomy. The most assessed FLIP metrics were distensibility index (DI) and cross-sectional area (CSA). Three studies (on 7 total) reported a positive correlation between DI and CSA with specific symptoms (dyspepsia spectrum and nausea/vomiting) and their severity at baseline. Ten studies (on 15 total) studies showed an association between FLIP parameters with clinical response. Pooled mean DI ranged from 7.5 to 8.9 mm^2/mmHg and CSA from 85.5 to 195.6 mm^2 at baseline. Minimum diameter (Dmin) ranged between 10.6 and 16.5 mm while Pressure (P) between 14.8 and 34.5 mmHg (table 1.). Upper-limit normative values for GP ranged from 173-317 mm^2 for CSA, 17-27 mm for Dmin, 32-64 mmHg for P (between 30-50 mL of fill volumes), while for DI at 50mL the reference limit attested at 21.5 mm^2/mmHg. Comparing pre-and post-treatment response, MD of 2.5 mm^2/mmHg (I^2 24%) for DI at 50 mL, 46.7 mm^2 (I^2 47%) and 68.8 mm^2 (I^2 0%) for CSA at 40 and 50 mL respectively, were statistically significant.

Table 1. FLIP metrics pooled mean values at different fill volumes at baseline in gastroparesis patients

Pooled mean values#	30 mL	I^2(%)	40 mL	I^2(%)	50 mL	I^2(%)
DI (mm^2/mmHg)	7.5 (6.1-8.9)	38	8.9 (7.5-10.4)	83.4	7.1 (6.1-8.1)	88.1
CSA (mm^2)	85.5 (75.4-95.6)	0	125.3 (108.3-142.3)	78	195.7 (175.5-215.9)	70
Dmin (mm)	10.7 (9.1-12.3)	85	13.4 (12.1-14.7)	92	16.5 (15.7-17.3)	80
P (mmHg)	15.0 (11.3-18.7)	77	18.8 (16.5-21.1)	74	34.0 (29.2-38.8)	91

#Values are expressed as pooled means and 95% CIs
CIs Confidence Intervals CSA Cross-Sectional Area DI Distensibility Index Dmin Minimum Diameter FLIP Functional Lumen Imaging Probe P Pressure

Conclusion: This is the first systematic effort in gathering evidence on FLIP in gastroparesis. CSA and DI significantly changed after pylorus-targeted treatments. Normative values displayed substantial heterogeneity and wide CIs. The cut-off of 10 mm^2/mmHg for DI seems feasible to categorize patients to be addressed to pylorus-targeted treatments. Further studies to calculate prediction ability of this cut-off are warranted.

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PP0020

CLINICAL EFFECTIVENESS AND SAFETY OF SHORT LENGTH PERORAL ENDOSCOPIC MYOTOMY (POEM) COMPARED TO LONG LENGTH MYOTOMY FOR THE TREATMENT OF ACHALASIA IN A UNIVERSITY HOSPITAL IN COLOMBIA

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Introduction: Endoscopic peroral myotomy is a less invasive treatment for achalasia with efficacy rates equal to or superior to traditional methods. As it has evolved and developed, the traditional long-length myotomy and the current short-length myotomy have been described. Studies have shown no difference in efficacy or safety between the two, with a difference in procedure time. Therefore, the aim of our study is to evaluate the efficacy and whether there are differences in periprocedural complications, assessed according to the iPOEM classification of adverse events, taking into account that a longer procedure time may be associated with more complications.

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Aims & Methods: Patients were retrospectively collected from the records of the databases of the gastroenterology unit who underwent POEM for achalasia at the Hospital Universitario San Ignacio, Colombia, between 2017-2024. Sociodemographic, clinical and manometric characteristics were described and compared between patients with short and long myotomy. Effectiveness was assessed according to the Eckardt scale and safety for perioperative adverse events according to the iPOEM classification, and the difference between the two groups was evaluated using the $\chi 2$ test for categorical variables and the Mann-Whitney U test to compare continuous variables.

Results: Clinical efficacy and safety of short peroral endoscopic myotomy (POEM) compared with long myotomy for the treatment of achalasia in a university hospital in Colombia57 patients were included, 25 (44%) underwent short myotomy and 32 (56%) underwent long myotomy with a mean esophageal length of 5 cm and 11 cm respectively. The mean age was 4.6.6 years, with a female predominance of 56.1%, 20 patients (35%) had undergone a previous procedure for achalasia, which was Heller myotomy in 50%. The most common type of achalasia in both groups was type II. The mean preoperative Eckardt score was 7.6 points, with no difference between groups. Patients who underwent long myotomy had more minor complications such as capnoperitoneum and abdominal puncture compared to the short myotomy group (65.6% and 56.2% versus 16% and 12% respectively) P<0.001. There was no difference in hospital stay but there was a shorter operative time in the short myotomy group median 80 minutes IQR 70-95 minutes compared to 127.5 minutes IQR 102-155 minutes in the long myotomy group with statistical significance, P<0.001. In terms of efficacy at one year, 84.3% of patients in the short myotomy group and 88% in the long myotomy group achieved an Eckardt ≤3, which was not statistically significant.There was also no difference in reflux symptoms between groups P 0.277 and the majority (65%) responded to PPI management.

Conclusion: Short myotomy in POEM for achalasia appears to be noninferior to long myotomy in terms of clinical success and long-term reflux disease. In addition, it offers the advantages of shorter procedure times and potentially lower rates of minor complications, which makes it a preferable option in many cases and could also be considered the first choice in previously operated patients.

Disclosure: Nothing to disclose.

PP0021

THE PREVALENCE AND CHARACTERISTICS OF GASTROPARESIS IN PARKINSON'S DISEASE USING CLINICAL SCORE AND ITS RELEVANCE WITH GASTRIC EMPTYING SCINTIGRAPHY

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Introduction: Gastrointestinal dysfunction, especially gastroparesis is a common non-motor disorder among Parkinson's disease resulting to serious complications and poor quality of life. There are still some limitations on an evaluation and diagnostic test to detect the patients with gastroparesis for early intervention improving outcome.

Aims & Methods: All Parkinson's disease patients were inquired to complete GCSI, GERD-Q, and PAC-SYM questionnaires. The correlation between the scores and disease severity was evaluated. Factors associated with significant gastroparesis symptoms (GCSI ≥9) were analyzed. Patients who have had gastrointestinal symptoms compatible with gastroparesis were further test for gastric emptying scintigraphy.

Results: 76 patients completed GI questionnaires. The majority of patients (60.6%) were classified in PD stage 2 and 3. A large number of patients (82.9%) reported gastroparesis symptoms using GCSI score. The median (IQR) of total GCSI score was 4 (2, 10) with postprandial fullness predominant in subscale. PD duration ≥6 years was associated with high gastroparesis symptoms (GCSI ≥9), adjust OR = 2.96 (1.01, 8.71), p=0.043. Eight consecutive participants could undergo physiologic testing. Only one of eight was definitely confirmed for delayed gastric emptying. No correlation between GCSI score and physiologic testing was observed.

Table 1.Logistic regression analysis on potential factors associated with significant GCSI score.

	Crude OR (95%CI)	Crude P-value	Adjust OR (95%CI)	P-value (LR-test)
PD stage ≥3	0.45 (0.14,1.44)	0.179	0.4 (0.12,1.34)	0.123
PD duration ≥6 years	2.66 (0.95,7.41)	0.062	2.96 (1.01,8.71)	0.043
Coexisting DM	0.89 (0.2,3.9)	0.873	1.01 (0.2,5.21)	0.987
eGFR <60 mL/min/1.73m^2	0.88 (0.24,3.28)	0.850	1.19 (0.29,4.93)	0.812
Anticholinergic	0.96 (0.32,2.83)	0.937	0.87 (0.28,2.72)	0.812

Conclusion: Gastroparesis symptoms with postprandial fullness predominant were common among Parkinson's disease. The strong correlation between gastrointestinal scores may illustrate an extensive gastrointestinal involvement, especially in the patients with long term disease duration. However, clinical scores might have a limitation for real-life implementation in Parkinson's disease because poor correlation between GCSI score and physiologic testing was observed. Score validation and long-term outcome, especially nutritional status and pulmonary complication as sequelae of gastrointestinal dysfunction would be considered in future research.

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PP0022

EXPLORING THE CONNECTION BETWEEN FUNCTIONAL DYSPEPSIA AND SLEEP DISORDERS: UNRAVELING THE LINK

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Introduction: Functional dyspepsia is a common gastrointestinal disorder with unknown causes. Though a potential link between sleep disorders and digestive issues exists, evidence is inconclusive. This study aims to explore this relationship.

Aims & Methods: This prospective study was conducted at a medical school hospital from January 2023 to November 2023. Patients diagnosed with functional dyspepsia, according to the Rome IV criteria, were classified into two subtypes. Investigations were undertaken to exclude other potential diagnoses, which included a blood test, abdominal ultrasound, and upper endoscopy with *H. pylori* testing. We measured patients' sleep quality and daytime alertness with the Pittsburgh Sleep Quality Index (PSQI) and the Epworth Sleepiness Scale (ESS).

Results: A total of 134 patients were diagnosed with functional dyspepsia after careful evaluation. Among them, functional dyspepsia with postprandial distress syndrome predominate group (FD-PDS) was found at 60.4%. FD-PDS was linked to a higher PSQI score, with an average of 8.47 ($p < 0.05$). Additionally, there was a positive association between FD-PDS and the sleep quality ($p < 0.05$). The FD-PDS group was significantly associated with excessive daytime sleepiness ($p < 0.05$). Interestingly, the PSQI and ESS scores correlated with FD-PDS, with correlation coefficients yielding a *p-value* of less than 0.05.

	FD-EPS (N = 53)	FD-PDS (N = 81)	p-value
PSQI score, Mean (+/- SD)	6.4 (2.68)	8.47 (3.21)	0.002*
Nonsignificant, N (%)	37 (69.81)	37 (44.68)	0.012**
Significant sleep disturbance, N (%)	16 (30.19)	44 (55.32)	
ESS score, Mean (+/- SD)	6.34 (3.5)	8.85 (4.41)	0.004*
Normal, N (%)	46 (86.79)	42 (51.85)	<0.001***
Excessive daytime sleepness, N (%)	7 (13.21)	39 (48.15)	

**Two-sample T test*
***Fisher exact test*
****Chi-square test*

Conclusion: The findings indicate that FD-PDS is associated with elevated PSQI severity and abnormal ESS scores. Sleep disorder assessment should be conducted alongside evaluation in patients with functional dyspepsia, especially those with postprandial distress syndrome.

Disclosure: Nothing to disclose.

PP0023

SYMPTOM BURDEN IN PATIENTS WITH DGBI REFERRED FOR SPECIALIZED INTEGRATED CARE TREATMENT BEFORE, DURING AND AFTER THE COVID-19 PANDEMIC

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Introduction: Disorders of Gut-Brain Interaction (DGBI), previously also referred to as functional gastrointestinal disorders, are characterised by chronic or relapsing gastrointestinal symptoms not explained by structural or biochemical abnormalities identified in the routine clinical setting. Psychological factors, including depression and anxiety, appear to play a role in these patients. A previous systematic review and meta-analysis[1] suggests that the COVID-19 pandemic had adverse effects on symptom severity of DGBI patients. Against this background we aimed to compare symptom burden and clinical outcomes of patients with severe DGBI (defined as experiencing symptoms impacting daily life) and their response to a highly structured 3 month clinical treatment program (the Integrated Care Clinic, ICC) delivered by a multiprofessional team (gastroenterologists, a primary care physician, psychologists, dietitians and exercise physiologists).

Aims & Methods: We hypothesise that in the emotionally challenging environment of the COVID-19 pandemic the DGBI symptom burden increased, while the response to a multimodal therapy that aimed to address biological and psychological disease mechanisms remained unchanged.

All patients with severe DGBI symptoms not sufficiently managed in the routine primary care or specialist setting who were referred and subsequently treated in the ICC between 2017 and November 2024 were included. Symptom burden was assessed with the validated Structured Assessment of Gastrointestinal Symptoms Scale (SAGIS) at baseline and after 3 months of treatment. We compared baseline and post-treatment SAGIS data for patients treated before, during and after the COVID-19 pandemic. The periods were defined by introducing and discontinuing travel restrictions for the respective community. Means were compared using the Wilcoxon and Kruskal-Wallis tests (GraphPad Prism 10).

Results: Overall, 287 patients with DGBI (mean age of 44.2±15.3 years, 75% female) completed the ICC program. The pre-pandemic cohort comprised 117 patients, the active pandemic cohort included 130 patients, and the post-pandemic cohort comprised 40 patients. On a global scale, the overall COVID-19-related mortality in the respective community was low (963/1,000,000 in Australia vs. 2812/1,000,000 in the EU). Overall, the ICC intervention resulted in a 25.8% (±41.4%) reduction of the SAGIS score ($p<0.0001$), and 170 (59%) patients experienced a >20% improvement of the SAGIS score during treatment. 47% of patients with severe or very severe symptoms at baseline did not have severe or very severe symptoms after treatment.

Symptom improvement assessed by SAGIS scores before, during and after COVID-19 pandemic was not statistically different (before: 9.475±11.99, during: 9.369±13.68, after: 6.875±13.11, $p>0.5$). However, comparing the global symptom burden at baseline (as assessed by SAGIS) there was a statistically significant increase in symptom burden during and after the active phase of the COVID-19 pandemic compared to the pre-pandemic symptom burden (before: 49.98±27.51, during 58.97±25.61 and after: 60.68±26.1, $p<0.05$).

Conclusion: The data from this cohort study suggest that the improvement of symptoms in patients with severe DGBI symptoms who participated in a 3-month intensive multiprofessional treatment course is not

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influenced by the COVID-19 pandemic. In contrast, the overall symptom burden of patients treated during and after the pandemic is higher than before the COVID-19 pandemic. Further studies are required to determine if this is an effect due to COVID-19 infection or other pandemic-related factors.

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Disclosure: Nothing to disclose.

PP0024

FUNCTIONAL DYSPEPSIA REDEFINED: EVIDENCE SUPPORTING INFLAMMATORY DYSPEPSIA – A SYSTEMATIC REVIEW

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Introduction: Functional dyspepsia (FD) is a common yet complex gastrointestinal disorder, characterised by postprandial fullness, early satiety, epigastric pain, and burning. With growing evidence linking diet to inflammation and symptoms, our objective was to review the evidence for the impact of dietary factors on FD symptomatology.

Aims & Methods: A systematic review was conducted to identify dietary components influencing FD symptoms in adults. Databases searched included MEDLINE, EMBASE, PsycINFO, Scopus, and Cochrane, covering January 2000 to December 2024. A total of 14 studies met eligibility criteria, with findings synthesised through narrative analysis.

Results: The search strategy yielded 18,424 articles, of which 14 met the inclusion criteria. Analysis identified seven key dietary components associated with FD: (1) a low FODMAP (fermentable oligosaccharides, disaccharides, monosaccharides, and polyols) diet significantly improved symptoms ($p = 0.04$), while gluten intake was linked to early satiety ($p = 0.03$); (2) probiotics or prebiotics provided symptom relief (RR 1.15, $p = 0.52$), with probiotics alone showing a trend toward improved FD scores (RR 1.35, $p = 0.061$); (3) capsaicin significantly exacerbated symptoms ($p < 0.001$), and spicy foods were associated with increased gastric fullness ($p = 0.001$); (4) both peppermint-caraway oil ($p < 0.00001$) and cinnamon oil ($p < 0.001$) demonstrated significant symptom relief; (5) ginseng-based herbal porridge led to a higher rate of epigastric symptom remission by day 90 compared to placebo (45.71% vs. 20.69%, $p = 0.036$), while the STW 5-II formulation (a nine-herb extract) also yielded significant improvement ($p = 0.008$); (6) prokinetic agents reduced symptom severity (RR 0.81, $p < 0.0001$), while amitriptyline outperformed both placebo (40%) and escitalopram (38%), relieving symptoms in 53% of patients—particularly those with ulcer-like FD; and (7) ginger/artichoke capsules improved symptoms within 14 days ($p = 0.017$), while ginger/perilla tablets offered progressive relief over one to two weeks ($p < 0.05$ and $p < 0.01$, respectively).

Study quality was assessed using ROBIS, RoB 2.0, and ROBINS-I tools. Of the included studies, three were rated as low risk of bias, eight as moderate, and three as serious. Despite adherence to a standardised PRISMA-based search strategy, variability in outcome reporting limited direct comparability across studies.

Author	Study type	Outcomes	Author	Study type	Outcomes
Goyal (2022)	RCT	Both low-FODMAP diet and traditional dietary advice significantly improve FD symptoms.	Zobeiri (2021)	RCT	Both cinnamon oil and placebo showed improvements in gastrointestinal symptoms.
Du (2018)	Case-control	Gluten-rich diets may trigger FD symptoms, notably early satiety.	Wang (2024)	RCT	Herbal porridge is an effective and safe option for improving symptoms in FD patients.
Zhang (2020)	Systematic Review and Meta-Analysis	Probiotics and prebiotics showed positive effects on FD symptom scores, but data was limited.	Vinson (2024)	RCT	STW 5-II herbal extract is effective and safe for up to 8 weeks in patients with FD meeting Rome II or IV criteria.
Agah (2020)	Systematic Review and Meta-Analysis	Probiotics may improve GI symptoms in FD patients but high-quality RCTs are required to confirm this.	Pittayanon (2019)	Systematic Review and Meta-Analysis	Prokinetics may help alleviate FD symptoms, but the evidence quality was low.
Fuhrer (2011)	RCT	There is evidence that FD patients show chemical hypersensitivity in response to capsaicin.	Talley (2015)	RCT	Amitriptyline was more effective than both placebo and escitalopram in providing symptom relief, showing a greater benefit in patients with ulcer-like FD.
Lee (2016)	Prospective study	Capsaicin-rich foods likely contribute to feelings of stomach fullness.	Giacosa (2015)	RCT	Ginger and artichoke leaf extracts are safe and effective in treating FD.
Li (2019)	Systematic Review and Meta-Analysis	Peppermint oil and caraway oil is an effective and safe short-term treatment for FD, but evidence quality was low.	Di Pierro (2020)	Retrospective cohort study	The perilla/ginger supplement may serve as an effective natural option for FD.

Table 1.

Conclusion: This review highlights dietary interventions as a promising strategy for managing FD. Our findings show that specific dietary components can be implicated in inducing inflammation and symptoms in FD. Since the symptoms are related to inflammation, we propose using "inflammatory dyspepsia" to represent the true presentation of this form of dyspepsia. Large-scale randomised trials are needed to confirm findings and refine dietary guidelines.

Disclosure: Nothing to disclose.

PP0025

DISORDERS OF GUT-BRAIN INTERACTION AND PSYCHOLOGICAL DISTRESS AFFECT DIETARY BEHAVIOR. ITALIAN DATA FROM THE WORLDWIDE ROME FOUNDATION GLOBAL EPIDEMIOLOGY STUDY

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Introduction: Disorders of Gut-Brain Interaction (DGBI) often arise or worsen after food intake, leading patients to modify their diets. These disorders are also strongly associated with psychological distress (PD).

Aims & Methods: To evaluate how DGBI and/or PD influence dietary habits.:

A total of 2,063 adults from the general Italian population completed the Italian Rome IV Diagnostic Questionnaire, which includes 86 diagnostic, 3 red-flag symptoms, and 10 food consumption questions. After excluding individuals with organic conditions (n=133) or GI symptoms not meeting DGBI criteria (n=390), the remaining 1,540 participants were categorized into two groups:

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• Comparison Group (CG): No GI symptoms (n=650).
• DGBI Group: Diagnosed with one or more DGBI (n=890).
Psychological distress was assessed using the Patient Health Questionnaire-4 (score ≥6 indicating high PD). Associations between DGBI, PD, and weekly rate of assumption of HF and LF foods were assessed with multivariable Poisson regression models adjusted for sex, age, smoking, and alcohol consumption. Adjusted risk ratios (aRR) and 95% CI were used to quantify associations.

Results:

- PD and Dietary Habits in CG: Compared to CG with low PD, CG individuals with high PD consumed less high-FODMAP (HF) vegetables (aRR=0.79; 95% CI: 0.65–0.95) and low-FODMAP (LF) vegetables (aRR=0.80; 95% CI: 0.67-0.97), as well as less HF fruits (aRR = 0.84; 95% CI: 0.71–0.99), and LF fruits (aRR = 0.77; 95% CI: 0.64–0.92). Moreover, they were observed to consume less aged hard cheese (aRR=0.80; 95% CI: 0.64-0.99).
- DGBI and PD: DGBI was associated with a higher prevalence of PD (aRR = 3.90; 95% CI: 2.83–5.38).
  - DGBI subjects with high PD consumed less HF fruits (aRR = 0.85; 95% CI: 0.78–0.92), LF fruits (aRR = 0.83; 95% CI: 0.76–0.91) compared to CG with low PD.
  - DGBI subjects with low PD had slightly reduced LF vegetable consumption (aRR = 0.91; 95% CI: 0.86–0.97).

HF Cereals and Dairy: Intake of HF foods like pasta, bread, and dairy did not differ significantly between CG with low PD and the other groups.

Conclusion:

1. Psychological distress influences diet: High PD is associated with reduced consumption of HF and LF fruits and LF vegetables in both CG and DGBI individuals.
2. Dietary modifications in DGBI: While DGBI individuals tend to modify their diets by avoiding some HF and LF foods, they continue to consume HF cereals and dairy, suggesting unselective dietary changes.
3. PD amplifies dietary restrictions: The presence of PD exacerbates dietary modifications in both DGBI individuals and CG.

These findings highlight that food choices are significantly influenced by psychological distress and are also probably related to the traditional presence of HF cereals and dairy products in the Italian cuisine, often leading to unselective dietary restrictions that may not alleviate GI symptoms effectively.

Disclosure: NO COI.

PP0026

EXPLORING THE LINK BETWEEN ESOPHAGEAL MOTILITY DISORDERS AND REFLUX DISEASE IN PATIENTS WITH SUPRAGASTRIC BELCHING DISORDER

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Introduction:

Supragastric belching disorder (SBD) is characterized by unconscious air swallowing and its immediate expulsion from the esophagus. Diagnosis is usually made by history and physical examination though supplementary evaluations can be considered. High-resolution esophageal manometry combined with stationary impedance (HRIM) allows confirmation of SBD diagnosis in patients with esophageal symptoms. Better understanding of this condition is needed to increase recognition and guide treatment. With the aim of characterizing the population diagnosed with SBD using HRIM and its association with other esophageal motility disorders (EMDs) or gastroesophageal reflux disease (GERD), we conducted a descriptive study.

Aims & Methods: We identified patients with SBD diagnosis who underwent HRIM using MMS-Unisensor® solid-state probe with 36 channels coupled with 16 channels of impedance in a tertiary hospital from January 2020 to December 2024. HRIM studies were interpreted according to Chicago classification v3.0 and v4.0. Demographic and clinical characteristics were collected. Data from pH-metry and/or pH-Impedance (pH-IMM) tests were also gathered to assess presence of pathological acid gastroesophageal reflux (GER).

Results: 39 patients were included (64% women) with a mean age of 52.7 years. Demographic and clinical characteristics are summarized in Table 1. Some patients developed SBD after esophagogastric surgery (10.4%). One-third of patients had some EMD: ineffective esophageal motility (27%) and esophageal absent contractility (5.4%). 12.9% of patients associated rumination disorder (RD). 25% patients had pathological acid GER in pH-metry and/or pH-IMM. A significant association was observed between the presence of pathological acid GER and symptoms of regurgitation (p=0.02) and vomiting (p=0.05), and between pathological acid GER and EMD finding in HRIM (p=0.05). 94.1% patients had consumed proton pump inhibitors (PPIs) in the previous 6 months, but only 50% showed symptomatic improvement with PPIs therapy. Symptoms improved in 91% of follow-up patients with diaphragmatic breathing exercises and patient education about the pathophysiology of the disorder to motivate active patient participation in their treatment. On average, patients experienced a diagnostic delay of 38.2 months (IQR: 24–54).

Sex (women)	**25 (64.1%)**
Age (years)	52.7 (IQR: 39.1 - 65.9) (minimum 13, maximum 81)
Patients diagnosed with pathological acid GERD	7 (25.9%)
Acid exposure time	12.8% (IQR: 6% - 16.7%)
DeMeester Index	44.9 (IQR: 21.7 - 53.2)
Number of refluxes	49,1 (IQR: 27 - 72)
Number of acid refluxes in patients with GERD	38 (IQR: 27 - 66)
Number of belching events in patients with GERD	130 (IQR: 23 - 111) (minimum 2; maximum 996)

Table 1. Demographic and clinical variables from the SGB patients sample

Conclusion: SBD is more common in middle-aged women as other functional gastrointestinal disorders. Association of SBD with RD is notable. Both SBD and RD could be misdiagnosed as PPI-refractory GERD. It is important to study the presence of GERD in order to optimize PPI treatment: in our study, only one quarter of the patients exhibited esophageal acid burden. Most patients improved with non-pharmacological treatment. SBD may be effectively treated with diaphragmatic breathing exercises and training patients for awareness of belching episodes. Diagnostic delay may be due to lack of awareness and limited access to diagnostic tests. Promoting knowledge about pathophysiology, symptoms and complementary tests would reduce diagnostic time and would improve the management of this rare disorder.

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PP0027

RESILIENCE IN PATIENTS WITH FUNCTIONAL DYSPEPSIA: A PROSPECTIVE QUESTIONNAIRE STUDY

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Introduction: Functional Dyspepsia (FD), a common disorder of gut-brain interaction [1], is associated with alterations in gastric accommodation, gastric emptying, gastric sensitivity, gastroduodenal permeability, and the gut microbiome. Resilience, the ability to maintain function under personal or physical stress, may play a role in the development of FD, although no studies have evaluated this theory.

We hypothesize that FD patients have lower resilience than the general population, which may lead to a new therapeutic target.

Aims & Methods: This prospective questionnaire study compared patients with FD (Rome IV criteria) with internal medicine clinic patients without dyspepsia at a tertiary referral center. Participants completed the FD symptom severity score [2], Brief Resilience Scale (BRS) [3], Connor-Davidson Resilience Scale (CDRS) [4], hospital anxiety and depression scale (HADS) [5], early childhood adverse life event questionnaire [6], and perceived stress score [7]. Questionnaires were distributed to clinic patients meeting inclusion/exclusion criteria.

Results: 38 FD (mean age 43 years old, 87% female) and 31 control (mean age 57 years old, 39% female) patients completed the survey. FD patients exhibited lower resilience on the CDRS (96.83 vs. 90.26, P<0.04, CI -12.93 to -0.21) but not on the BRS (3.018 vs. 3.022, P<0.96, CI -0.15 to 0.13). FD patients and controls scored no different on the HADS anxiety index (17.05 vs. 18.38, P<0.064, CI -2.74 to 0.08) and depression scores (P<0.14, CI -0.21 to 1.5). The FD group reported "high" perceived psychological stress while the control group reported a "moderate" level (P<0.004, CI 2.002 to 9.827). In childhood, FD patients had significantly more adverse childhood events (29.3% vs. 15%, P<0.004, CI 0.05 to 0.25).

FD patients were more likely to have been exposed to a household member suffering from a psychiatric illness compared to control patients (47% vs. 20%, P<0.049, CI 0.001 to 0.5). FD patients were also more likely to report psychological abuse in childhood (53% vs. 17%, P<0.03, CI 0.034 to 0.67).

No other significant differences were found in types of early childhood trauma including physical abuse, familial substance abuse, witnessed maternal violence, or criminal behavior by a household member.

Conclusion: This novel, prospective questionnaire pilot study shows that FD patients have lower resilience compared to the general outpatient population on the CDRS. The long-term symptoms of FD may further decrease resilience.

While anxiety and psychological stress are not strongly associated with FD, the significant exposure to childhood household members with a psychiatric diagnosis and significantly higher prevalence of self reported psychological abuse may explain the higher prevalence of antidepressant use in this population, especially considering the significantly higher level of self perceived psychological stress [8].

Otherwise, early childhood trauma appears similar between FD patients and patients without FD and is thus not likely to influence resilience markers. Understanding the contribution of resilience to development of FD may lead to therapies enhancing resilience, improving symptoms, and patient well-being.

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Disclosure: Nothing to disclose.

PP0028

SYMPTOM LOCALIZATION AND ITS ASSOCIATION WITH SYMPTOM RELIEF: INSIGHTS FROM A EUROPEAN-ASIAN COHORT OF FUNCTIONAL DYSPEPSIA PATIENTS

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Introduction: Patients with functional dyspepsia (FD) experience symptoms such as postprandial fullness, epigastric pain/burning and upper abdominal bloating. The relationship between symptoms and alleviating factors may influence how patients describe their symptoms in questionnaires. Understanding the specific location of these symptoms and their connection to bowel movements can enhance diagnosis and management.

Aims & Methods: To identify the location of uncomfortable fullness, pain/burning, and bloating in patients with FD, and determine differences between Europe and Asia. Furthermore, to assess differences in these abdominal regions between patients with FD and whether these symptoms are alleviated by bowel movements or flatus.

In total 481 FD patients from 8 European (N=281, 58.4%) and 7 Asian (N=200, 41.6%) specialist care centers completed questionnaires, including symptom localization on an abdominal pictogram dived into nine regions. Symptom relief by bowel movement and flatus was examined by dividing patients into three groups based on relief magnitude (never relieved, relieved in ≤ 50% and relieved in > 50%). The anatomical location of symptoms was compared between these subgroups.

Results: Asian FD patients reported significantly fewer symptoms in the hypogastric region compared to European patients (Table 1). Patients with FD whose postprandial fullness was relieved by bowel movement

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(>50%) reported significantly less fullness in the epigastric region and more in the six lower regions (right lumbar, umbilical, left lumbar, right iliac, hypogastric and left iliac regions) (p<0.001). A similar result was found for postprandial fullness relieved by flatus, except for the right lumbar and right iliac region. Patients with FD, whose postprandial fullness was relieved by bowel movement (>50%) reported significantly less pain/burning in the epigastric region and more in the umbilical, left lumbar and hypogastric regions (p<0.001).

A similar result was found for pain/burning relieved by flatus with significant differences in the epigastric and umbilical region. For bloating, the only significant difference was less bloating in the right iliac region for FD patients relieved by flatus. There was no significant difference in regions for bloating relieved by bowel movements.

	Postprandial fullness		Pain/burning		Bloating	
	Europe	Asia	Europe	Asia	Europe	Asia
Right hypochondriac region	10.3%	5.3%	12.8%*	4.9%*	13.5%*	4.0%*
Epigastric region	79.8%	76.7%	78.9%*	69.9%*	47.2%	49.2%
Left hypochondriac region	11.9%	14.0%	15.7%	13.3%	14.2%	12.1%
Right lumbar region	12.3%	10.0%	14.5%*	6.3%*	28.5%*	18.5%*
Umbilical region	55.1%*	42.7%*	45.5%	44.8%	77.5%	69.4%
Left lumbar region	14.8%	16.7%	16.5%	14.7%	33.3%	25.0%
Right iliac region	9.5%	7.3%	5.4%	2.8%	22.5%*	11.3%*
Hypogastric region	28.0%*	15.3%*	16.9%*	8.4%*	56.9%*	29.0%*
Left iliac region	9.9%	6.7%	7.4%	4.2%	25.1%*	12.1%*

*Table 1. Location of uncomfortable fullness, pain/burning and bloating in the European versus Asian FD population. Bold percentages followed by an *are indicating significant differences between Europe and Asia, corrected for multiple testing with Bonferroni. The overall p-value of the Chi²-test for postprandial fullness is 0.015 and for pain/burning 0.003. The overall p-value of the Chi²-test for bloating is < 0.001.*

Conclusion: Asian FD patients report less frequently symptoms in the hypogastric region compared to European patients. Symptoms relieved by bowel movements or flatus are less perceived in the upper abdominal and more in the six lower abdominal regions. These symptom characteristics, including regional distribution and change with passage of gas or stool, may help to inform therapy choices in FD.

Disclosure: Nothing to disclose.

PP0029

UPPER GI BIOFEEDBACK THERAPY: EMPOWERING PATIENTS TO ADDRESS AIR SWALLOWING AND OPTIMISE CARE PATHWAYS

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Introduction: Behaviours such as supra-gastric belching, gastric belching, and aerophagia, often identified during oesophageal physiology studies (e.g., 24-hour pH and impedance study), are known to exacerbate symptoms like reflux, bloating, and regurgitation. These behaviours are frequently misunderstood and may lead to unnecessary referrals to specialists or invasive interventions. Upper GI biofeedback (UGI BFB) therapy offers a tailored, non-invasive approach to help patients manage these behaviours. By teaching specific techniques such as diaphragmatic breathing, open-mouth posture, and the multiple rapid swallow technique, patients are empowered to take control of their symptoms, improve their quality of life, and reduce reliance on surgical or gastroenterology pathways.

Aims & Methods: • Participants: Between December 2022 and December 2024, 35 patients were referred for UGI BFB therapy. Most referrals (n=31) came after oesophageal physiology studies, with additional referrals from gastroenterologists for belching and bloating (n=2) and from clinical endoscopists for persistent hiccups and audible wheezing (n=2).

• Intervention: Patients attended a one-hour education session in the Gastrointestinal (GI) Physiology unit. This included a review of diagnostic findings and practical training in diaphragmatic breathing, open-mouth posture, and the multiple rapid swallow technique. A follow-up telephone consultation at 6–8 weeks allowed for monitoring progress. For cases of poor oesophageal clearance, referring clinicians were advised to consider prokinetics. For frequent gastric belching, small intestinal bacterial overgrowth (SIBO) breath testing was recommended to assess any additional contributing factors.

Results: • Supra-gastric belching episodes were reduced by 50% (mean: 24 episodes pre-therapy to 12 post-therapy; p<0.001).

- GERD-HRQL scores improved by 45%, indicating significant quality-of-life improvements (mean: 22 pre-therapy to 12 post-therapy; p<0.001).
- 75% of patients reported substantial reductions in belching, bloating, and reflux-like symptoms.
- Half of the patients (50%) no longer required specialist referrals, as symptoms were successfully managed with therapy or GP-led interventions, including prokinetics or medications for hypersensitivity.
- Stress emerged as a critical factor for most patients. Relaxation techniques, mindfulness, and stress management strategies played an important role in symptom control. One patient, who became a mental health first aider at work, reported long-term symptom relief due to stress reduction efforts.

Conclusion: Upper GI biofeedback therapy is a valuable approach for addressing behavioural contributors to symptoms like air swallowing, belching, and bloating. A multidisciplinary approach, involving cognitive behavioural therapy or hypnotherapy alongside UGI BFB, can be highly beneficial. The GI Physiology team also recognises its limitations and has found that referrals to Speech and Language Therapy can be instrumental when patients are unable to engage fully with the therapy. This collaborative, patient-centred approach ensures that UGI BFB therapy not only reduces the need for invasive procedures and specialist consultations but also empowers patients to adopt long-lasting lifestyle and behavioural changes.

Disclosure: Nothing to disclose.

PP0030

THE IMPACT OF CANNABIS USE ON GASTROINTESTINAL SYMPTOMS AND GASTRIC EMPTYING BREATH TESTING

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Introduction: Gastroparesis is a disorder characterized by delayed gastric emptying without mechanical obstruction, leading to symptoms such as nausea, vomiting, abdominal pain, and bloating. Cannabis and its active compounds are thought to interact with the gastrointestinal system through cannabinoid receptors, influencing motility, sensation, and inflammation. Nevertheless, data pertaining to the interactions between cannabis use, gastric motility and gastrointestinal (GI) symptoms is still scant[1].

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Aims & Methods: The aim of the current study was to compare GI symptoms and gastric motility in a cohort of patients referred for gastric emptying breath test (GEBT). In this prospective study, all consecutive patients referred for motility testing in a tertiary hospital were recruited. Following obtaining written informed consent, questionnaires, including ROME, SCL-90-R and SF36 were filled out by the patients. Carbon-13 spirulina GEBT was done performed according to protocol, following consumption of a standardized meal. Patients with ongoing (regular or occasional) cannabis use were compared with non-cannabis users for all symptoms, quality of life and GEBT testing results. An additional healthy cohort of volunteers was recruited for validation of GEBT cut-off values and comparison to the symptomatic cohort.

Results: A total of 88 patients and 14 healthy volunteers were included in the current study. Among patients, 15 (17%) were identified as cannabis users, and these were compared with 73 non-users. Mean age and gender did not differ between users and non-users (52.4 vs 43.0, and 21% vs 33% males, respectively, p=NS). Gastric emptying breath test results did not significantly differ between groups, both in terms of mean values ($t^{1/2}$ 127.2 vs 129.2 minutes, users versus non-users, respectively; p=NS) and in the proportion of delayed gastric emptying (26.7% versus 32.9%, users versus non-users, respectively; p=NS).

None of the healthy volunteers displayed delayed gastric emptying results. Cannabis users demonstrated a higher gastrointestinal symptom burden and a greater number of diagnosed disorders of gut-brain interaction (DGBI) compared to non-users.

Conclusion: In our cohort of patients referred for gastric emptying breath tests, gastric emptying was similar between cannabis users and non-users. This is in line with previous studies suggesting that cannabis does not improve delayed gastric emptying. Nevertheless, previous studies suggested that cannabis may alleviate gastroparesis-associated symptoms[2]. In this context, our results show that cannabis use was rather associated a more severe and systemic symptomatic manifestation of gut-brain disorders, in contrast to a more localized upper gastrointestinal involvement seen in non-users.

Although causality cannot be established, we do note that despite ongoing cannabis use, treatment response was sub-optimal in the user patient group. Factors contributing to these differences, such as physiological or psychological constructs, should be further explored.

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Disclosure: Nothing to disclose.

PP0031

A FEASIBILITY STUDY OF VIRTUAL REALITY FOR THE TREATMENT OF NONCARDIAC CHEST PAIN

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Introduction: Noncardiac chest pain (NCCP) is a challenging condition associated with decreased quality of life, diagnostic uncertainty, and a high healthcare burden. The most common gastrointestinal causes are gastroesophageal reflux disease (GERD), functional chest pain, and spastic esophageal motor disorders. Effective treatments are lacking particularly for patients who do not respond to proton pump inhibitors. Virtual reality (VR), an innovative non-invasive tool providing multi-sensory stimulation through a head mounted display, has demonstrated usefulness in treating acute and chronic pain, anxiety, and other gastrointestinal disorders. The objective of this study was to assess the feasibility of virtual reality for the treatment of NCCP.

Aims & Methods: This was a feasibility study of 11 adult patients with NCCP, defined as persistent angina-like chest pain despite the exclusion of a cardiac source. All patients underwent esophagogastroduodenoscopy (EGD) to exclude treatable, secondary gastrointestinal causes. Participants were provided a VR headset preloaded with three-dimensional immersive environments including relaxation simulations and guided diaphragmatic breathing. Participants were asked to use the headset at least 15 consecutive minutes daily for 14 days.

The primary objective was to assess the overall feasibility of VR for the treatment of NCCP and challenges that may impact the performance of a future clinical trial.

As a secondary objective, we assessed the effectiveness of VR to improve pain severity, gastrointestinal symptoms, hypervigilance, quality of life, depression, anxiety, and resilience through validated questionnaires. Continuous variables were compared with the paired-samples t-test and categorical variables with McNemar's test.

Results: Eleven patients completed the study with a median age 59 (range 44-72) years, 10 (91%) female, and median body mass index 24.9 kg/m^2 (19.6-28.7). Ten patients had a normal endoscopic appearance of the esophagus, and one patient had Los Angeles grade A esophagitis. All patients had incomplete treatment response to a proton pump inhibitor. There were improved depression scores by the Hospital Anxiety and Depression Scale compared to baseline after one (median 4 vs. 1, p=0.008) and two weeks (4 vs. 1, p=0.02) and improvement in the composite depression and anxiety score after 2 weeks (10 vs. 6, p=0.009) of the study. There was a trend in improvement of the Esophageal Hypersensitivity and Anxiety Scale although this did not meet statistical significance (43.5 vs. 39.0, 0.097). The virtual reality headset was well-tolerated by all patients without adverse events or safety concerns.

Conclusion: In this two week, prospective feasibility study of patients with NCCP, virtual reality was well tolerated and led to improvement in depression and anxiety scores. These findings provide a foundation for the use of VR for the treatment of NCCP in a larger, prospective randomized controlled trial.

Disclosure: Nothing to disclose.

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PP0032

SYSTEMATIC REVIEW AND META-ANALYSIS: INVERSE ASSOCIATION BETWEEN HELICOBACTER PYLORI INFECTION AND EOSINOPHILIC OESOPHAGITIS

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Introduction: Exposure to Helicobacter pylori (H. pylori) has been associated with a decreased risk of eosinophilic oesophagitis (EoE).

Aims & Methods: As the epidemiology of EoE is rapidly increasing, we conducted an updated systematic review and meta-analysis to determine the association between H. pylori infection and EoE. We searched MEDLINE, Scopus, and ISI Web of Science, through to November 2024.We included studies reporting the status of H. pylori infection in patients with and without EoE or oesophageal eosinophilia (EE). A random-effects model was used to pool estimates.

Results: Nineteen studies including 1.704.821 subjects were analysed. H. pylori infection was associated with a 46% lower risk of EoE/EE (OR: 0.54, 95% CI 0.43 to 0.67). Comparable findings were observed when subgrouping studies by location or design.

A nonsignificantly decreased risk for EoE was observed in pediatric patients exposed to H. pylori (OR 0.57, 95% CI 0.26 to 1.24) and in studies using serology to diagnose H. pylori (OR: 0.41, 95% CI 0.16 to 1.04).

We found a lower risk of EoE compared with the overall findings in studies that diagnosed H. pylori only by gastric biopsy (OR 0.43, 95% CI 0.25 to 0.74) and in those published after 2019 (OR 0.44, 95% CI 0.28 to 0.68).

Conclusion: In a large systematic review and meta-analysis, exposure to H. pylori was significantly associated with decreased risk of EoE. As a stronger protective effect was found in more recent studies, the epidemiology of this association may evolve and deserves to be further monitored.

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PP0033

THE RELATIONSHIP BETWEEN BACTERIAL FLORA IN SALIVA AND ESOPHAGEAL MUCUS AND ENDOSCOPIC SEVERITY IN PATIENTS WITH EOSINOPHILIC ESOPHAGITIS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic inflammatory disease characterized by esophageal dysfunction and eosinophilic inflammation of the esophageal mucosa.

In this study, we investigated the bacterial flora in saliva and esophageal mucus in patients with EoE and examined the relationship between EoE disease activity and mucosal cytokine expression.

Aims & Methods: Cases consisted of patients with active and inactive EoE (A-EoE and I-EoE, respectively). A-EoE was defined as a peak eosinophil count > 15/high-power field, according to the 2025 consensus guidelines. Saliva samples were collected from the patients before the endoscopic examination. Brushing samples were collected from the distal esophagus of patients with EoE during endoscopic procedures. The degree of EoE inflammation was assessed using the EoE endoscopic reference score (EREFS). The samples were profiled using the Illumina MiSeq platform. The V3–V4 regions of the 16S rRNA gene (460 bp) were amplified by using tailed PCR.

Results: Fifty-nine patients were enrolled, including 8 with I-EoE, 17 with A-EoE, and 28 non-EoE controls. Major bacterial genera, such as *Streptococcus*, *Prevotella*, *Veillonella*, and *Alloprevotera*, were detected in both the oral cavity and esophagus. Compared with the control group, the active EoE group had significantly more *Prevotella* spp. in the saliva and esophageal mucosa.

Conversely, significantly fewer *Neisseria* spp. were found in the saliva and *Streptococcus* spp. in the esophageal mucosa of patients with active EoE. The EREFS of EoE and *Streptococcus* were inversely correlated.

Conclusion:

In this study, we clarified the characteristics of the bacterial flora in the saliva and esophageal mucosa of patients with eosinophilic esophagitis.

Disclosure: Nothing to disclose.

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PP0034

THE OVERLAP OF ALLERGIC DISORDERS AND UPPER GASTROINTESTINAL SYMPTOMS: BEYOND EOSINOPHILIC ESOPHAGITIS?

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Introduction: Eosynophilic oesophagitis (EoE) is a chronic disease which clinically presents with symptoms related to oesophageal dysfunction while pathologically it is characterized by eosinophilic infiltration of oesophageal epithelium. Most patients with EoE present food and/or inhalant allergies symptoms. The results of animal models studies and genetic studies, as well as efficacy of elimination diets in managing the symptoms, suggest an atopic background of the disease.

Aims & Methods: The aim of this study was to evaluate the prevalence of EoE in a group of patients with upper gastrointestinal symptoms and food and/or inhalant allergies and to assess the influence of drugs used in type I allergies on the results of endoscopic, histopathological and immunohistochemical tests.

Methods. This was a prospective observational study. Patients with inhalant/food allergies and upper esophageal symptoms formed study group while control group formed patients without allergy diagnosed with dyspepsia or irritbale bowel syndrome.

All study participants were tested for allergies including prick testing, blood tests, all participants were evaluated with a gastroscopy with specimen collection. Oesophageal specimens were stained for eotaxin-1 and desmoglein-1.

Results: Based on histopathology results eosinophilic oesophagitis was found in 9 of the 73 patients from the study group. All patients with EoE presented with multimorbidity and were diagnosed with at least one allergic disease in addition to EoE. Positive staining for CCL-11 was found in 56 (78%) patients in the study group, including all patients with EoE while ony in 3 (17%) individuals from control group.

The presence of DSG-1 in oesophageal specimens was detected in 6 (7%) subjects from the study group in contrary to 14 (78%) subjects from the control group. DSG-1 was not found in any of the specimens of patients diagnosed with EoE.

Conclusion: EoE is rare disease usually accompanied by allergic multimorbidity. Positive staining for eotaxin-1 and negative staining for desmoglein-1 in patients with esophageal symptoms and allergy but who did not met EoE diagnostic criteria, could be indicative of subclinical course of the disease or masking effect of corticosteroids. It is now critically important for both researchers and practicing clinicians to recognize that eosinophilic esophagitis (EoE) is not a homogeneous disease but rather consists of multiple subtypes (phenotypes). The so-called “classic” form of EoE—defined by current diagnostic criteria as having more than 15 eosinophils per high-power field on histopathological examination—appears to represent only the tip of the iceberg.

There is an urgent need for further research to refine endoscopic techniques, expand the scope of histopathological assessments, and identify novel biomarkers to better define the distinct phenotypes of eosinophilic esophagitis.

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Disclosure: Nothing to disclose.

PP0035

PREVALENCE OF EOSINOPHILIC ESOPHAGITIS AMONG PATIENTS PRESENTING WITH ESOPHAGEAL DYSFUNCTION SYMPTOMS AT THE ENDOSCOPY UNIT OF A TERTIARY CARE CENTER IN WESTERN MEXICO

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Introduction: Eosinophilic Esophagitis (EoE) is a chronic, immune-mediated esophageal disease characterized by esophageal dysfunction symptoms and histological eosinophilic infiltration. Diagnosis requires ≥15 eosinophils per high-power field and exclusion of other causes. Though its pathophysiology is not fully understood, food antigens—such as milk and wheat—are considered primary triggers, leading to chronic inflammation, mucosal damage, and fibrosis. EoE predominantly affects white males and is managed through dietary therapy, PPIs, and topical corticosteroids. Since its identification in the 1990s, prevalence has increased, especially in Western populations, with notable regional and ethnic variability. In Latin America, prevalence also varies between countries. Esophageal biopsies are essential in patients with esophageal dysfunction, even in the absence of visible endoscopic changes. However, the frequency of esophageal eosinophilia in this population remains unknown in our region and institution.

Aims & Methods: A cross-sectional, descriptive, and retrospective study was conducted on patients over 18 years of age with symptoms of esophageal dysfunction who underwent upper endoscopy with esophageal biopsies between January 2020 and October 2023. A convenience sample of 164 patients was determined based on a 4% expected prevalence of EoE. Clinical records, endoscopy reports, and pathology files were reviewed to identify cases meeting diagnostic criteria. Data were analyzed using descriptive and inferential statistics, with a significance level set at $p < 0.05$.

Results: Demographics and Medical Background:The study included 152 patients (72.36% women; mean age 55.28 ± 14.07). Chronic conditions were present in 60.52% of the participants, the most frequent being hypertension (16.44%), asthma (11.18%), and diabetes mellitus (9.21%).

Clinical Presentation: Gastroesophageal reflux symptoms (GERD) were reported in 51.31% of patients, with 28.28% presenting alarm symptoms (mainly dysphagia). Dysphagia alone was the second most common symptom (34.86%). Food impaction occurred in 6.57%, and esophageal strictures in 4.60%. Symptoms of Esophageal Dysfunction:Dysphagia was the most common symptom (67.76%), followed by heartburn (47.36%) and regurgitation (44.73%). Less common symptoms included food impaction, globus sensation, and non-cardiac chest pain. Endoscopic Find-

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ings:Findings suggestive of eosinophilic esophagitis (EoE) were relatively uncommon: esophageal rings (26.97%), strictures (11.18%), exudates (13.81%), longitudinal furrows (9.21%), and edema (0.65%). Hiatal hernia was observed in 28.94% of patients, and a wide hiatus in 41.44%.
Prevalence of EoE:EoE (≥15 eosinophils/HPF) was diagnosed in 11 patients (7.2%). The average age was 48.63 ± 16.5 years, with dysphagia (72.7%), heartburn (63.6%), and regurgitation (54.5%) as the most frequent symptoms. Endoscopic signs suggestive of EoE were present in only 36.36% of these patients.

Conclusion: This study investigates the clinical and diagnostic characteristics of Eosinophilic Esophagitis (EoE) in a specific population. A high prevalence of chronic comorbidities (60.52%) and a predominance of female patients (72.36%) were found. GERD symptoms were common (51.31%) and overlapped with EoE, especially in dysphagia and heartburn. EoE prevalence was 7.2%, with hiatal hernia linked to symptoms and food impaction. Diagnostic indicators showed good specificity but limited sensitivity. These findings emphasize the need for a comprehensive approach, combining clinical and endoscopic data, for accurate diagnosis and effective management of EoE.

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Disclosure: No conflict of interest.

PP0036

PREDICTORS OF RESPONSE TO BUDESONIDE ORODISPERSIBLE TABLETS INDUCTION THERAPY IN ADULT EOSINOPHILIC ESOPHAGITIS: A RETROSPECTIVE SINGLE-CENTER STUDY

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Introduction: Budesonide orodispersible tablets (BOT) have been shown to be a safe and effective treatment for both induction and maintenance therapy of eosinophilic esophagitis (EoE). However, real-life data identifying baseline factors associated with a lack of response to BOT induction therapy are still lacking.

Aims & Methods: We aimed to compare baseline clinical, endoscopic, and histological characteristics between responders and non-responders after 3 months of induction therapy with BOT, and to identify possible predictors of therapeutic response in adult EoE patients.
A retrospective study was conducted, including consecutive adult EoE patients (May 2023 – February 2025). EoE was defined by a peak eosinophil count (PEC) ≥15 eos/HPF. All patients received induction therapy with BOT 1 mg twice daily and underwent follow-up gastroscopy with at least six esophageal biopsies after 3 months of treatment. Response to therapy was defined as the simultaneous achievement of histological (<15 eos/HPF) and endoscopic remission (EREFS score <2). Baseline data—including demographic and anthropometric characteristics, allergic comorbidities, occurrence of symptoms before EoE diagnosis, prior proton pump inhibitor (PPI) use, diagnostic delay, EREFS score, PEC, and index of severity for EoE (I-SEE) —were compared between responders and non-responders at univariate and multivariate logistic regression analyses to assess for eventual predictors of response to BOT induction therapy.

Results: A total of 32 EoE patients were included (87.5% male; mean age 40.2 ± 14.6 years). Among them, 18 (56.3%) were classified as responders (88.9% male; mean age 40.9 ± 15.0 years), and 14 (43.7%) as non-responders (85.7% male; mean age 39.3 ± 14.5 years).
No significant differences were observed between the two groups in terms of gender, age, body mass index, allergic comorbidities, diagnostic delay, occurrence of symptoms before EoE diagnosis, PEC, EREFS score, and I-SEE score. Food impaction at diagnosis requiring emergency gastroscopy was significantly more frequent among non-responders compared to responders (57.1% vs. 22.2%, p = 0.043). Prior PPI use was also more common in non-responders (57.1% vs. 27.8%, p = 0.093).
At multivariate logistic regression analysis, food impaction at diagnosis requiring emergency gastroscopy (OR 7.95, 95% CI 1.25–50.61, p = 0.028) and prior PPI use (OR 0.13, 95% CI 0.017–0.93, p = 0.043) were identified as independent predictors of non-response to BOT induction therapy.

Conclusion: Patients presenting with food impaction requiring emergency gastroscopy may be less likely to respond to induction therapy with BOT, potentially due to a more advanced stage of EoE. Moreover, patients with prior PPI use also seem to have a reduced likelihood of therapeutic response, possibly due to symptom masking and consequent diagnostic delay, ultimately reducing treatment effectiveness.

Disclosure: Nothing to disclose.

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PP0037

TREATMENT PATTERNS IN PATIENTS WITH EOSINOPHILIC ESOPHAGITIS IN ERA OF BIOLOGICS – A MULTI-CENTRE RETROSPECTIVE STUDY

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Introduction: Eosinophilic esophagitis (EoE) is a chronic inflammatory disorder from the atopic disease spectre. Despite being studied for years, the EoE-specific therapy has been approved only recently and includes budesonide orodispersible tablet (1) and IL-4R antagonist, dupilumab (2). Therefore, data on the management of the patients who do not respond to this treatment is scarce. Especially those who were diagnosed long in the past and have undergone different therapeutical approaches remain an understudied group. The aim of this project was to provide information on the treatment courses in a real-world setting over past 10 years.

Aims & Methods: This was a retrospective nation-wide project collecting data on patients with EoE in 7 Austrian centres since January 2014. The last follow-up was marked as of February 2025. Information on patients' demographics, previous medical history, number of esophageal food impactions and different treatment lines were recorded. Due to the descriptive character of the work, no power analysis was carried out.

Results: In total, 336 patients (30.1% female) were included. Approximately one third of them (119/336) experienced an esophageal food impaction before the diagnosis was made. The most common 1st line therapy was budesonide orodispersible tablet (193/336, 57.4%), followed by proton pump inhibitors (87/336, 25.9%) and food elimination diet (20/336, 6.0%). Two hundred twenty (220/336, 65.5%) patients did not remain on the 1st line treatment. Forty-four (44/336, 13.1%) subjects required more than 2 therapies and one person received even 7th line medication. Biologic treatment – dupilumab – was given to forty-one (12.2%) patients. It was the first line treatment in 1 person, the others received it in the further course. These drugs were well-tolerated and 97.6% (40/41) patients remained on it until the last follow-up and reached symptomatic remission. Three patients were diagnosed more than 10 years before they received biologic treatment and 2 of them responded to it. One person discontinued due to side effects. The most common reasons for switch for dupilumab were clinical and/or histological activity (24/41, 58.5%), side effects (10/41, 24.4%) and patients' preference (2/41, 4.9%).

Conclusion: This is the first study on different therapy courses in patients with EoE in Europe. Due to its retrospective character, no direct conclusions on the superiority of one approach upon another in case of refractoriness are possible.

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PP0038

REAPPRAISAL OF THE EOSINOPHILIC ESOPHAGITIS ENDOSCOPIC REFERENCE SCORE (EREFS): SUBSCORING INFLAMMATORY AND FIBROTIC FINDINGS IMPROVES THE ASSESSMENT OF DISEASE ACTIVITY

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Introduction: The eosinophilic esophagitis (EoE) endoscopic reference score (EREFS) is the validated scoring system for endoscopic findings of EoE. EREFS findings are currently scored cumulatively and correlate modestly with histological disease activity (HDA), symptoms, and adaptive behaviors. However, inflammatory and fibrotic findings may reflect different aspects of EoE, but this has never been evaluated.

Aims & Methods: We investigated the correlation of EREFS inflammatory and fibrotic subscores with HDA, symptoms and adaptive behaviors. In three tertiary referral centres for EoE, consecutive adults (>18 years) undergoing upper endoscopy for suspected or follow up EoE were prospectively enrolled. EREFS was scored as follows: edema 0-1; rings 0-3; exudates 0-2; furrows 0-2; stricture 0-1. The EREFS inflammatory subscore (EREFS-i) combined the scores of edema, exudates and furrows. The EREFS fibrotic sub-score (EREFS-f) combined the scores of moderate-severe rings (score ≥2) and stricture.

Subsequently, at least six esophageal biopsies were collected to assess HDA based on peak eosinophil count/high-power field (PEC/HPF) (active EoE PEC ≥15/HPF; EoE in remission: PEC<15/HPF). Symptoms were assessed using the dysphagia symptoms questionnaire (DSQ). Adaptive behaviors were assessed using the Pisa EoE Adaptation Questionnaire v1.0 (PiEAQ) [reference 1].

Clinical characteristics were recorded. Kruskal-Wallis Rank Sum Test and Pearson's Chi-squared were used for comparisons. ROC curve analysis was used to calculate the diagnostic accuracy (AUC). Spearman's test and linear regression models were used to assess relationships between variables. Significance threshold was p<0.05.

Results: 121 patients were included. 58% had active EoE, 42% were in histological remission. Baseline clinical data are reported in Table 1. The EREFS total score had modest AUC for predicting HDA (0.68) and did not show significant correlation with DSQ and PiEAQ.

In contrast, EREFS-i had AUC of 0.73, with 88.2% sensitivity and 51.4% specificity for predicting HDA, being significantly more accurate compared to the EREFS-f (p<0.001; Figure 1).

There was a significant correlation between EREFS-i and PEC/HPF with Spearman's Rho=0.5 (p<0.001), with linear regression showing that an increase of 1 point of any inflammatory feature corresponded to an increase of 12 eosinophils/HPF (p<0.0001).

In contrast, EREFS-f correlated significantly with the use of adaptive behaviors assessed with PiEAQ [Spearman's Rho of 0.6 (p<0.001); Figure 2], with linear regression showing that an increase of 1 point of any fibrotic feature corresponded to an increase of one adaptive behavior (p=0.001). EREFS, EREFS-i, and EREFS-f had nonsignificant negative correlation with DSQ (p>0.05 for all).

Conclusion: We confirmed that, while total EREFS provides a global assessment of endoscopic disease severity, it lacks correlation with histological and clinical disease activity. However, we demonstrated that EREFS-i correlates significantly with PEC and predicts PEC/HPF and that EREFS-f correlates significantly with the use of adaptive behaviors assessed with PiEAQ, which represent known clinical markers of disease activity. Neither EREFS-i nor EREFS-f correlated significantly with DSQ.

In conclusion, we showed that, compared to the total EREFS, EREFS subscores improve the correlation between endoscopy findings with histology and clinical disease activity markers. Applied to clinical practice, our results represent another step towards precision medicine in the setting of EoE.

Characteristic	N = 121
Sex (M/F)	82/39
Age	40 (31, 50)
EREFS_Score_Total	2.00 (0.00, 3.00)
EREFS_Inflammatory	1.00 (0.00, 2.00)
EREFS_Fibrotic	1.00 (0.00, 2.00)
Dysphagia	73 / 116 (63%)
DSQ	14 (0 – 42)
PiEAQ score	0 (0 – 2)
EREFS total score	2 (0-3)

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PP0039

EFFICACY AND SAFETY OF LONG-TERM TREATMENT WITH BUDESONIDE ORODISPERSIBLE TABLETS IN PATIENTS WITH EOE: REAL-LIFE DATA FROM A SINGLE-CENTRE EXPERIENCE

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Introduction: One of the cornerstones of pharmacological treatment for patients with eosinophilic esophagitis (EoE) is the use of swallowed topical steroids (STCs). An esophageal formulation of orodispersible budesonide tablets (BOT) is currently available in Europe and has shown high response rates in clinical trials. However, real-world data on its long-term efficacy and safety are still lacking.

Aims & Methods: The aims of our single-centre study are to describe the efficacy and incidence of adverse events (AEs) of BOT during induction and maintenance therapy and to identify factors associated with lack of response and the occurrence of AEs.

We retrospectively enrolled EoE patients who started induction therapy with BOT (2 mg) and completed at least 1 year of follow-up with maintenance therapy (1 mg). Histological response was defined as <15 eosinophils per high-power field (HPF). The rate of clinical, endoscopic and histological remission was assessed at the end of induction treatment (T1) and after 1 year of maintenance treatment (T2). BOT-related AEs were also assessed at T1 and T2.

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Results: A total of 104 patients were enrolled in the study. The median age at diagnosis was 30 years, 79.8% were male, with a median diagnostic delay of 58.5 months. At diagnosis, 41.4% of patients had a fibrostenotic disease phenotype, 21.8% presented with esophageal stenosis and 44.3% had a history of gastroesophageal reflux disease (GERD). Patients started BOT as first-line therapy (15.3%), after switching from off-label STCs (20.1%) and after failure of proton pump inhibitors (PPIs) treatment. Prior to starting BOT therapy, 2.17% of patients had evidence of erosive esophagitis (EE) on upper GI endoscopy. Of the 104 patients, 102 (98%) completed the induction phase of treatment (T1) and 55 (53.9%) completed at least one year of follow-up in the maintenance phase (T2).

In terms of histological response to treatment, 89.2% of patients had remission at T1. At T2, 23.8% of patients lost histological response to BOT therapy. At T2, loss of histological remission was associated with a Dysphagia Symptom Score (DSS) ≥ 3 (p<0.001) and tended to be associated with the presence of esophageal stenosis at diagnosis (p=0.085). The DSS was found to have high specificity, meaning that a score ≥ 3 as highly predictive of histological recurrence during maintenance therapy.

The prevalence of EE was 13% at T1 and 14.5% at T2, significantly increased compared to T0. Risk factors for the development of EE were younger age at diagnosis (p=0.033), delayed diagnosis (p=0.053) and BOT monotherapy without concomitant PPIs (p=0.004) at T1 and a history of GERD (p=0.03) at T2.

In terms of AEs, 33.7% and 20.4% of patients experienced BOT-related side effects at T1 and T2, respectively. Esophageal and oral candidiasis were the most common and were in most cases successfully treated with local or systemic antifungal therapy. However, the recurrence of AEs during the maintenance phase led to discontinuation of BOT in 2/55 cases (3.64%). The history of GERD was found to be a risk factor for the development of candidiasis at T1 (p=0.032) and T2 (p=0.039).

Conclusion: On the basis of our results, BOT therapy is effective in the induction and maintenance of histological remission in EoE. BOT monotherapy and a history of GERD are risk factors for the development of erosive esophagitis after BOT induction and maintenance, respectively. These findings suggest the need for a more personalized treatment approach and highlight the potential role of BOT-PPI combination therapy in selected cases.

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PP0040

PROSPECTIVE MULTICENTRE EVALUATION OF THE I-SEE SCORE FOR MONITORING DISEASE ACTIVITY IN ADULTS WITH EOSINOPHILIC ESOPHAGITIS TREATED WITH BUDESONIDE ORODISPERSIBLE TABLETS

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Introduction: The I-SEE score, Index of Severity for Eosinophilic Esophagitis (EoE), is a recently developed clinical tool designed to assess the severity of EoE, a chronic, immune-mediated disease characterized by esophageal eosinophilic infiltration. Unlike traditional approaches, the I-SEE score integrates multiple domains, such as symptoms burden, endoscopic features and histo-pathologic activity, to provide a comprehensive evaluation of disease impact. Recent studies have begun to validate the I-SEE score in real-world settings, examining its correlation with treatment outcomes, patient-reported measures, and long-term disease management. Current research are focused on refining the scoring system, exploring its utility across patients' populations, and determining its reliability for identifying disease activity, thus helping its management.

Aims & Methods: This is a multicentre, prospective, observational study aiming to evaluate the reliability of the I-SEE score in identifying disease activity in patients with EoE, by comparing scores obtained before and after therapeutic interventions with budesonide orodispersible tablet (BOT). Patients diagnosed with EoE, aged > 18 years old, treated with BOT 2 mg, as induction treatment for 12 weeks (T1) and with 0,5/1 mg qd as maintenance for 52 weeks (T2) were enrolled. Clinical and demographic characteristics were collected. I-SEE score was calculated before and after the treatment.

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Results: 306 patients (241 males, 79%; mean age 38±14) treated with BOT were enrolled, in 8 Italian Centres and 1 Israelian Centre. The most frequent symptoms at diagnosis were dysphagia (82%) and food bolus impaction (70%). Allergic comorbidities were present in 65% of patients.

At baseline, all patients had an I-SEE score higher than 2, median value 6 (5-8); 14/306 presented a severe disease (I-SEE score≥15) and 136/306 a moderate active disease (I-SEE score 7-14). At T1, the I-SEE score passed from 6 (5-8) to 2 (1-3) (p<0.0001). Moreover, only 1/276 patients had a persisting severe disease, 5/270 moderate disease, while 40/276 achieved complete remission (I-SEE score<1).

After 1 year of treatment (T2), 135 patients were evaluated; the I-SEE score significantly decreased from T0 (6 (5-8) to 1 (0-3), p<0.0001) and from T1 (2 (1-3) to 1 (0-3), p<0.0001). Moreover, 4/135 patients presented moderate-severe disease, while in 40/135 EoE was inactive.

Considering only histologic assessment, all patients defined primary non-responder to BOT at T1 (20/270, 8 with moderate-active disease at T0) and secondary non-responder after tapering down the treatment at T2 (12/135, 5 with moderate-active disease at T0), met the criteria of mild disease according to the I-SEE score. All data are resumed in Table 1.

	n, T0	n, T1	n, T2	T0 vs T1	p	T0 vs T2	p	T1 vs T2	p
I-SEE score, median value (25-75 perc.)	306	270	135	6 (5-8) vs 2 (3-1)	<0.0001	6 (5-8) vs 1 (0-3)	<0.0001	2 (1-3) vs 1 (0-3)	<0.0001
Symptoms, median value (25-75 perc.) (0-15 points)	306	270	135	1 (1-2) vs 0 (0-0)	<0.0001	1 (1-2) vs 0 (0-0)	<0.0001	0 (0-0) vs 0 (0-0)	0.2
Complications, median value (25-75 perc.) (0-15 points)	306	270	135	0 (0-0) vs 0 (0-0)	<0.0001	0 (0-0) vs 0 (0-0)	0.0007	0 (0-0) vs 0 (0-0)	0.0007* * worsen from T1
Inflammatory endoscopic features, median value (25-75 perc.) (0-2 points)	306	270	135	1 (1-2) vs 1 (0-1)	<0.0001	1 (1-2) vs 0 (0-1)	<0.0001	1 (0-1) vs 0 (0-1)	0.002* * improved from T1
Eosinophil count, median value (25-75 perc.) (0-2 points)	306	270	135	1 (1-2) vs 0 (0-0)	<0.0001	1 (1-2) vs 0 (0-0)	<0.0001	0 (0-0) vs 0 (0-0)	0.9
Endoscopic fibrostenotic features, median value (25-75 perc.) (0-15 points)	306	270	135	1 (0-1) vs 0 (0-1)	<0.0001	1 (0-1) vs 0 (0-1)	<0.0001	0 (0-1) vs 1 (0-1)	0.008* * improved from T1
Histologic fibrostenotic features, median value (25-75 perc.) (0-2 points)	306	270	135	2 (0-2) vs 0 (0-2)	<0.0001	2 (0-2) vs 0 (0-2)	<0.0001	0 (0-2) vs 0 (0-2)	0.5

Table 1: I-SEE score and features assessment at T0, T1 and T2.

Conclusion: In this prospective observational study, the application of the recently developed I-SEE score for EoE assessment confirms the high efficacy of the treatment of BOT. However, baseline I-SEE score calculation was not useful in predicting nor stratifing patients' response to drug therapy. Further studies are necessary to outline the weighted scoring of each feature involved into score calculation.

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PP0041

POPULATION-BASED STUDY OF EOSINOPHILIC OESOPHAGITIS IN THE CANTERBURY REGION, NEW ZEALAND: INCIDENCE, AND PREVALENCE OF STRICTURING DISEASE AND PERIPHERAL EOSINOPHILIA

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Introduction: Eosinophilic oesophagitis [EoE] is a chronic, antigen-driven, immune-mediated oesophageal disease characterised by eosinophil-predominant inflammation. It typically presents with dysphagia, food bolus impaction, and, in children, feeding difficulties and failure to thrive [1].

Histologically, it is defined by ≥15 eosinophils per high-power field on oesophageal biopsy in the absence of other causes of eosinophilia [2].

The incidence and prevalence of EoE have increased markedly over recent decades, particularly in Western countries, likely due to increased recognition, changing environmental exposures, and altered microbiota [3,4].

Though EoE is increasingly reported worldwide, data specific to Aotearoa New Zealand, and especially regional patterns by ethnicity, remain limited.

Aims & Methods: This study aimed to describe the incidence, demographic distribution, and clinical characteristics of EoE in the Canterbury region over an 11-year period.

We conducted a retrospective analysis of electronic medical records from 2013 to 2023, identifying EoE cases using diagnostic codes and histological confirmation. Demographic data, clinical presentations, and laboratory findings were collected. Crude incidence rates were calculated using average population data for the Canterbury region, stratified by ethnicity (European vs. non-European). Clinical features including food bolus obstruction, oesophageal strictures, and peripheral eosinophilia were assessed. Incidence rates are reported per 100,000 population with 95% confidence intervals (CI).

Results: A total of 219 cases of EoE were identified between 2013 and 2023. The average annual incidence was 3.23 per 100,000 population (95% CI: 1.82–4.65), based on a regional population of 618,409. The mean number of cases per year was 20 (IQR 16–22). Among European individuals (average population 395,690), the mean annual incidence was 4.42 per 100,000 (95% CI: 2.35–6.49), with 17.5 average cases per year (IQR 14–21). Among non-European individuals (average population 222,627), the mean incidence was 1.08 per 100,000 (95% CI: -0.29–2.44), with 2.4 average cases per year (IQR 1–4).

Of the 219 cases, 70 patients (31.9%) presented with oesophageal food bolus impaction. The annual number of such presentations ranged from 3

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to 12 per year (median 6, IQR 4–9). Twenty patients (9.1%) had documented stricturing disease on endoscopy, and 60 patients (27.4%) exhibited peripheral eosinophilia at diagnosis.

Conclusion: In our cohort, eosinophilic oesophagitis presents commonly with oesophageal food bolus impaction. There was a higher incidence of EoE observed in individuals of European descent. Just under one-third of patients in our cohort presented with oesophageal food bolus impaction and about 9% had stricturing disease. The frequent presentation with food bolus obstruction and the presence of complications such as strictures highlight the importance of timely recognition and management. Peripheral eosinophilia was found commonly in our cohort but the prevalence of this finding is not adequate to be used as a diagnostic test in EoE, although more studies are needed.

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Disclosure: None.

PP0042

PREVALENCE OF DYSPHAGIA AND EOSINOPHILIC ESOPHAGITIS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE OR TYPE 2 INFLAMMATION – INTERIM ANALYSIS OF A PROSPECTIVE, MULTICENTER SCREENING STUDY

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, type-2 inflammation-driven esophageal disease causing dysphagia and esophageal dysfunction. Patients with inflammatory bowel disease (IBD) or other type-2 inflammatory disorders may have an increased risk of EoE, but prevalence data are limited.

This study examines the prevalence of dysphagia and EoE in patients with type-2 inflammatory diseases or IBD.

Aims & Methods: In this prospective, multicentric, questionnaire-based screening study, patients from the departments of gastroenterology, pulmonology, dermatology, and otolaryngology at the Medical University of Vienna, as well as from the Allergy Center Floridsdorf, were screened for dysphagia using the validated brief esophageal dysphagia questionnaire (BEDQ). The primary endpoint was the prevalence of clinically relevant dysphagia, defined as a BEDQ score ≥10 or a history of bolus impaction in the past year. Endoscopy was offered to symptomatic patients without recent biopsies. Patient characteristics and differences in absolute eosinophil count, eosinophilic cationic protein (ECP), and IgE levels were analyzed using independent-samples T-tests.

Results: A total of 690 patients (45.7% female, mean age 54.5 years) were screened; 54.4% had IBD, 20.2% allergic rhinitis, 16.4% asthma, 8.7% atopic dermatitis, 6.4% chronic rhinosinusitis with nasal polyps (CRSwNP), and 5.2% IgE-mediated food allergy with multiple diagnoses possible. Overall, 5.1% (n=35) exhibited clinically relevant dysphagia. EoE was previously diagnosed in 2.0% (n=14), including four patients with clinically relevant dysphagia. Among the remaining 31 patients, eight had prior endoscopies ruling out EoE, seven underwent endoscopy revealing two new EoE cases, and 16 are still awaiting gastroscopy. No significant differences in IgE, eosinophilic count, or ECP were found between patients with EoE, those with dysphagia, and asymptomatic individuals.

Conclusion: Patients with IBD and type-2 inflammatory diseases exhibit a high prevalence of dysphagia. While EoE is more common than in the general population, most dysphagia remains unexplained. Systematic dysphagia screening in this high-risk population may reduce diagnostic delay.

Disclosure: Nothing to disclose.

PP0043

PREDICTORS OF TREATMENT RESPONSE OF PPI THERAPY IN PATIENTS WITH EOSINOPHILIC ESOPHAGITIS

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Introduction: One possible treatment option for Eosinophilic Esophagitis (EoE) is Protone Pump Inhibitor (PPI) therapy. The possible mechanisms of action discussed include acid suppression and a direct anti-inflammatory effect. However, only up to 50% of EoE patients respond to PPI therapy.

Predicting the therapeutic response is challenging, so the aim of this study was to identify potential predictors for a response to such therapy.

Aims & Methods: Patients with active EoE (defined as the presence of esophageal symptoms and simultaneous detection of >15 eosinophilic granulocytes per high-power field [HPF] in at least one biopsy from the esophagus) were retrospectively identified from the hospital database of a tertiary university referral center over 5 years. These patients had received sufficient PPI therapy (defined as double standard dose for at least four weeks). The analysis focused on their therapeutic response and potential predictors including clinical, histological and functional parameters of treatment success.

Results: 151 EoE patients were screened and of them 57 patients (mean age: 41 years; 81% male; mean maximum eosinophil count: 55/HPF) who had received PPI therapy at double standard dose for at least 4 weeks (mean duration: 62 days) were finally included. Histopathological complete remission (defined as <5 eosinophils/HPF) was achieved in 18 of 57 patients (31.6%) following PPI treatment, while partial remission (defined as <15 eosinophils/HPF) was observed in 25 of 57 patients (43.9%). No response to PPI therapy was noted in 14 of 57 patients (24.6%). There were no significant differences of the assessed predictors in EoE patients with

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and without PPI response, although patients with PPI response showed a distally emphasized esophageal eosinophilia and a higher acid exposure time in pH-metry (both non-significant).

Conclusion: No definitive predictors for the response to PPI therapy in EoE patients could be identified. However, a distally predominant esophageal eosinophilia and higher acid exposure time in patients responding to PPI therapy may suggest a contributory role of gastroesophageal reflux in this cohort.

Disclosure: Nothing to disclose.

PP0044

THE ROLE OF MAST CELLS IN EOSINOPHILIC ESOPHAGITIS: A PRELIMINARY STUDY

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Introduction: Eosinophilic esophagitis (EoE) is a chronic inflammatory disease characterized by esophageal dysfunction and a peak eosinophil count of >15 cells per high power field (HPF). Recent studies suggest that mast cells play a key role in its pathophysiology. This study investigates the presence of mast cells in different groups of patients with EoE.

Aims & Methods: Biopsy samples from patients newly diagnosed with EoE were analyzed, and the Eosinophilic Esophagitis Histology Scoring System (EOEHSS) grade and stage scores were calculated. Biopsy samples were divided in two groups: t0 (before the starting of EoE therapy) and t1 (after EoE therapy). Mast cells were detected using tryptase immunohistochemistry, and the maximum number of mast cells counted in a high-power field (PMC), was recorded. Univariate correlation analysis were performed between PMC and peak eosinophil count (PEC), PMC and EOEHSS grade, PMC and EOEHSS stage, PMC and EREFS.

Results: 12 patients affected by EoE were divided in group 1, treated with PPIs [Esomeprazole 40 mg, bis in die (bid)] and group 2, treated with BOT (budesonide oral tablets 1 mg, bid). Univariate correlation analysis were performed at t0 and t1.

Results from Pearson's R for analysis at t0 are showed a positive linear correlation in PEC-PMC (0,567, p=0,043) and EOEHSS grade-PMC (0,559, p=0,047). At t1, EREFS-PMC had a positive correlation (0,726, p 0,005), like also EOEHSS grade-PMC (0,74, p=0,002) and EOEHSS stage-PMC (0,818, p<0,001).

	R	p
EREFS-PMC (t0)	0.244	0.423
PEC-PMC (t0)	0.567	0.043
EOEHSS grade-PMC (t0)	0.559	0.047
EOEHSS stage-PMC (t0)	0.390	0.187
EREFS-PMC (t1)	0.726	0.005
PEC-PMC (t1)	0.340	0.256
EOEHSS grade-PMC (t1)	0.74	0.002
EOEHSS stage-PMC (t1)	0.818	<0.001

Conclusion: PCM positively correlates with PEC, EOEHSS grade, EOEHSS stage and EREFS in untreated samples, supporting their role in EoE and type-2 inflammation. In treated cases, PMC correlate with PEC, EOEHSS grade, EREFS.

A highly statistically significant strong positive linear correlation was found between PMC and EOEHSS stage, as the latter is more likely to reflect the whole esophageal histologic changes.

Our study suggests that mast cell count, alongside eosinophil counts, could be an important factor in evaluating anti-inflammatory therapeutic interventions and it may be a useful adjunct to existing endoscopic and histologic markers in assessing EoE severity, supporting their role in EoE and type-2 inflammation. To confirm these findings, further longitudinal prospective studies are required.

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Disclosure: Nothing to disclose.

PP0045

DUPILUMAB IMPROVES HISTOPATHOLOGIC ASPECTS OF EOE: POOLED ANALYSIS FROM TWO PHASE 3 STUDIES (EOE KIDS AND LIBERTY EOE TREET)

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, progressive, type 2 inflammatory disease of the esophagus characterized by multiple histopathologic abnormalities. In children and adults, EoE is diagnosed with a peak eosinophil count (PEC) ≥15 eosinophils/high-power field (eos/hpf) in endoscopically-obtained esophageal biopsies, yet PEC may not be a reliable marker of disease severity or treatment response. The EoE Histologic Scoring System (EoEHSS) provides an objective method for evaluating a range of histologic changes associated with EoE, beyond eosinophilia. Dupilumab, a fully human monoclonal antibody that blocks interleukin (IL)-4 and IL-13, key drivers of type 2 inflammation, is approved in the US and EU for the treatment of EoE in patients aged ≥1 year, weighing ≥15 kg.

Aims & Methods: This post hoc analysis assesses the impact of dupilumab vs placebo on histopathologic endpoints, PEC and EoEHSS grade/stage scores, in children, adolescents, and adults from a pooled analysis of the

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EoE KIDS (NCT04394351) and LIBERTY EoE TREET (NCT03633617) studies. Patients from Part A of EoE KIDS randomized to weight-tiered, higher-exposure (HE) dupilumab or placebo for 16 weeks, and patients from Parts A/B of LIBERTY EoE TREET randomized to weekly dupilumab 300 mg or placebo for 24 weeks, were included. Patients continued dupilumab or switched from placebo to dupilumab to Week (W) 52. Biopsies were collected from proximal, mid, and distal esophagus. PEC and EoEHSS grade/stage scores were assessed at W16/24 and W52. All *P*-values are nominal.

Results: The pooled cohort included 159 patients who received dupilumab and 152 who received placebo. Baseline characteristics were generally comparable between the dupilumab and placebo groups (mean [standard deviation] EoE-HSS grade score: 1.3 [0.4] and 1.3 [0.4], EoEHSS stage score: 1.3 [0.3] and 1.3 [0.4], respectively). At W16/24, higher proportions of patients achieved ≤1, ≤6, and <15 eos/hpf with dupilumab vs placebo (26% vs 1%, 61% vs 5%, and 79% vs 7%, respectively). At W52, effects were maintained with continued dupilumab and increased from W16/24 in patients switching to dupilumab HE from placebo (Table). Dupilumab also improved percent change from baseline EoEHSS total grade and stage scores vs placebo at W16/24 (Table). At W52, improvements were maintained in the dupilumab group.

Conclusion: In this pooled analysis of two phase 3 studies, dupilumab improved histopathologic abnormalities of EoE compared with placebo during treatment initiation (W16/24) and maintenance (no evidence of relapse at W52), in children, adolescents, and adults with EoE. These improvements may be clinically meaningful as individual components of EoEHSS, as determined by EoEHSS grade/stage score, are implicated in altered esophageal motility, and persistent symptoms of EoE following diminution/ablation of eosinophil infiltrates. The safety profile of dupilumab was consistent with the overall known safety profile.

	Week 16/24		Week 52	
	Pooled Dupilumab (N=159)	Pooled Placebo (N=152)	Pooled Dupilumab-dupilumab (N=151)	Pooled Placebo-dupilumab (N=92)
Proportion of patients, % (95% CI)				
≤6 eos/hpf	61**** (52.96, 68.63)	5 (2.30, 10.11)	72	62
<15 eos/hpf	79**** (71.42, 84.71)	7 (3.20, 11.77)	92	73
≤1 eos/hpf	26**** (19.18, 33.31)	1 (0.02, 3.61)	31	24
Mean percent change from baseline,[a] %				
EoEHSS grade total score[b]	-62.7**** (-67.67, -57.72)	-3.2 (-8.46, 1.97)	-68.6 (13.8)	-63.5 (20.2)
EoEHSS stage total score[b]	-60.3**** (-65.48, -55.10)	-2.9 (-8.51, 2.71)	-66.7 (14.1)	-61.5 (20.2)

Table. Histopathology outcomes at Week 16/24 and Week 52 from the pooled population of the EoE KIDS and LIBERTY EoE TREET studies

****P<0.0001. [a]Week 16/24 data are LS mean (95% CI); Week 52 data are mean (SD). [b]Week 52 EoEHSS total score excluding lamina propria fibrosis.

CI, confidence interval; EoEHSS, Eosinophilic Esophagitis Histologic Scoring System; eos/hpf, eosinophils per high-power field; LS, least squares; SD, standard deviation.

Disclosure: Collins MH: Alimentiv, Arena/Pfizer, AstraZeneca, Calypso Biotech, EsoCap Biotech, GSK, Receptos/Celgene/Bristol Myers Squibb, Regeneron Pharmaceuticals Inc., Inc., Shire/Takeda – advisory board/consultancy fees.

Chehade M: Adare Pharmaceuticals/Ellodi Pharmaceuticals, Allakos, AstraZeneca, Bristol Myers Squibb, Nexstone Immunology/Uniquity Bio, Phathom Pharmaceuticals, Recludix Pharma, Regeneron Pharmaceuticals Inc., Sanofi, Shire/Takeda – consultant. Adare Pharma Solutions/Ellodi, Allakos, AstraZeneca, Celgene, Danone, Regeneron Pharmaceuticals Inc., Shire/Takeda – research funding.

Dellon ES: AbbVie, Adare/Ellodi, Akesobio, Alfasigma, ALK, Allakos, Amgen, Apollo, Aqilion, Arena/Pfizer, Aslan, AstraZeneca, Avir, Biocryst, Bryn, Calypso, Celgene/Receptos/Bristol Myers Squibb, Celldex, Dr. Falk Pharma, EsoCap, Eupraxia, Ferring, GI Reviewers, GSK, Holoclara, Invea, Knightpoint, LucidDx, Morphic, Nexstone Immunology/Uniquity, Nutricia, Parexel/Calyx, Phathom, Regeneron Pharmaceuticals Inc., Revolo, Robarts/Alimentiv, Sanofi, Shire/Takeda, Target RWE, Upstream Bio – consulting fees. Adare/Ellodi, Allakos, Arena/Pfizer, AstraZeneca, Celgene/Receptos/Bristol Myers Squibb, Celldex, Eupraxia, Ferring, GSK, Meritage, Miraca, Nutricia, Regeneron Pharmaceuticals Inc., Revolo, Sanofi, Shire/Takeda – research funding. Allakos, Aqilion, Holoclara, Invea – educational grants.

Schlag C: Adare Pharmaceuticals Inc., AstraZeneca, Calypso, Dr. Falk Pharma, EsoCap GmbH, Regeneron Pharmaceuticals Inc. – consulting fees.

Fassan M: Amgen, Astellas, AstraZeneca, Bristol Myers Squibb, Diapath, Eli Lilly, GSK, Incyte, IQvia, Janssen Pharma, MSD, Novartis, Pierre Fabre, Roche, Pfizer, Sanofi – consulting fees.

Washington K: Discovery Life Sciences, Johnson & Johnson, Regeneron Pharmaceuticals Inc., Takeda – consultancy fees.

Xia C, Raphael BP, Radwan A: Regeneron Pharmaceuticals Inc. – employees and shareholders.

Cazeau C, Almansa C: Sanofi – employees, may hold stock and/or stock options in the company.

PP0046

TREAT-TO-TARGET IN EOSINOPHILIC ESOPHAGITIS: THE IMPACT OF GOING BEYOND CLINICAL REMISSION

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Introduction: Eosinophilic Esophagitis (EoE) is a clinicopathological disease that requires an integrated assessment combining clinical, endoscopic, and histological findings. Despite this, treatment decisions often remain symptom based. A treat-to-target strategy has recently been proposed to guide EoE management.

Aims & Methods: Retrospective single-center study of EoE patients followed between March 2012 and January 2024, with ≥12 months of follow-up. Clinical, endoscopic, histological data, and complications were assessed using the I-SEE score.

Results: A total of 82 patients with EoE were evaluated [75.6% (n=62) male, mean age 29 ± 13 years], with 37.8% (n=31) diagnosed during childhood. The most frequent symptom was food impaction (74.4%), and atypical symptoms were more common in pediatric patients (p=0.014). At baseline, 55.1% (n=43) presented with a fibrostenotic phenotype, more frequently in adults (p=0.012), and 32.9% (n=27) had moderate-to-severe disease based on the I-SEE score.

During follow-up, 56.1% (n=46) developed complications, most commonly food impaction requiring endoscopy. The presence of a fibrostenotic phenotype on initial endoscopy was a predictor of complications (OR 3.122; 95% CI 1.210–8.055; p=0.019). Both fibrostenotic phenotype (p=0.018) and moderate-to-severe I-SEE scores (p=0.004) were associated with increased need for therapeutic switching.

At 12 months, 50.0% (n=41) were asymptomatic; however, only 12.2% (n=10) had endoscopic healing and 9.8% (n=8) histologic remission. Throughout the study period, the rates of endoscopic and histologic healing remained below 25%.

Conclusion: The authors highlight the clinical dissociation of inflammatory activity and emphasize the importance of endoscopic and histologic healing to prevent disease progression.

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Disclosure: Nothing to disclose.

PP0047

LONG-TERM DUPILUMAB TREATMENT IMPROVES ENDOSCOPIC FEATURES IN PATIENTS WITH EOSINOPHILIC ESOPHAGITIS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, immune-mediated disease characterized by eosinophilic inflammation of the esophagus. EoE may result in esophageal remodeling and fibrotic changes which may lead to esophageal strictures and impair esophageal function. It has been widely demonstrated that all EoE therapies can revert esophageal fibrosis in short-term treatment, but to date no evidence has been produced on the long-term effects of specific EoE treatment on esophageal fibrosis and EoE endoscopic features.
Aims & Methods: We prospectively enrolled consecutive EoE patients referred to our center since 2022. Baseline EGDs with esophageal biopsies were performed in all patients before starting treatment and after at least 12 weeks of each therapy modification to evaluate histological and endoscopic response. The standardized EREFS score was used to evaluate EoE endoscopic signs, by differentiating the stenotic (Rings, Strictures) and inflammatory (Edema; Exudates, Furrows) subitems. In all patients who achieved clinical/histological remission and whose treatment remained unchanged (type and dose of therapy) for at least one year, a further EGDs with esophageal biopsies was performed at 52 weeks from the same experienced endoscopist. Results are expressed as mean±SD
Results: During the study period, 62 patients were prospectively enrolled (48M; 33±13y), of whom 41 achieved histological remission: 18 while on PPI, 14 on budesonide and 9 on Dupilumab. At 52 weeks, 15 patients on stable therapy underwent EGDs, 8 while on PPI therapy and 7 on Dupilumab. The two populations did not differ for sex, age at diagnosis and age of onset (7vs8M; 28.6±11.2vs 31.1±14.5y; 16.4±16.5vs 25.4±16.9y; p=NS) though a significantly increased diagnostic delay was observed in Dupilumab-treated patients (12.1±8.3vs5.6±3y; p=<0.05). At baseline, total EREFS scores were comparable between the two groups (4±2vs 3.6±1.2; p=NS). However, when analyzing the single EREFS subitems, Dupilumab patients scored significantly higher in the stricture subitem as compared to PPI group (0.3±0.49vs0±0; p<0.05).
At 12 weeks of therapy, both Dupilumab and PPI treated patients showed a significantly reduced total EREFS score compared to baseline (4±2vs2.6±1.4 and 3.6±1.2vs3±1; both p<0.05) with similar results both in terms of fibrotic and inflammatory subscores (1.3±0.95vs1.4±0.52 and 1.3±0.76vs1.55±0.73 in Dupilumab and PPI group, respectively; p=NS). The total EREFS score in both Dupilumab (2.6±1.4 vs 0.9±0.9) and PPI (3±1vs 2.4±0.7) groups consistently continued improving at 52 weeks of treatment (both p<0.05).
However, at 52 weeks, in dupilumab patients, both the total EREFS score and the fibrostenotic and inflammatory subitems were significantly lower compared to PPI group (0.9±0.9vs2.4±0.7; 0.4±0.53vs1.3±0.5 and 0.4±0.53vs 1.1±0.6; respectively, all p<0.001) with an almost complete resolution of the endoscopic signs of the disease in the Dupilumab group.
Conclusion: Our study showed a consistent improvement in endoscopic disease activity in both PPI and Dupilumab treated EoE patients after one year of treatment.
Notably, despite their lengthier diagnostic delay and higher fibrostenotic burden at baseline, Dupilumab patients showed a significant improvement in all EoE endoscopic features at 52 weeks compared to PPI responders, highlighting its potential impact on halting esophageal remodeling.
Disclosure: Marcella Pesce and Giovanni Sarnelli served as speakers for Sanofi.

PP0048

EFFICACY OF A DAIRY FREE DIET (DFD) IN ADULT PATIENTS WITH EOSINOPHILIC ESOPHAGITIS: A PILOT REAL-LIFE PROSPECTIVE MULTICENTER STUDY

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, immune-mediated disease characterized by symptoms of esophageal dysfunction and eosinophilic inflammation. Dietary therapy is an effective treatment for EoE patients of any age. However, data on the clinical effectiveness of dairy free diet (DFD) in adults remain limited.
Aims & Methods: Our study aims to explore the clinical, endoscopic and histological responses to a 12-week course of a dairy-free diet in adult patients diagnosed with EoE. We conducted a real-life multicenter prospective study on adult patients with histologically active EoE. After nutritional counseling, DFD was prescribed to adult EoE patients with one of the following indications: as a first-line therapy for newly diagnosed EoE; as an add-on therapy for EoE patients who were clinical responders but histological non-responders to proton pump inhibitors (PPIs); and as a second-line therapy. Clinical assessments and quality of life were evaluated using EEsAI, I-SEE, and the EOE-QOL-A scores at baseline and after 12 weeks of DFD. Adherence to DFD was assessed using VAS scale. Esofagogastroduodenoscopy with biopsies was repeated after 12 weeks for endoscopic and histological assessments.
Results: Twenty-one adult EoE patients undergoing DFD at 5 Italian Centers were enrolled from October 2022 to June 2024. Baseline characteristics of patients are described in Table 1. After 12 weeks of DFD, 50% of patients were histological responders and, among them, 90% were also clinical responders to DFD. Histological responders to DFD showed a significant reduction in both EEsAI and ISEE scores (p=0.01) compared to non-responders. Median EoE-QoL-A score significantly improved independently from response to therapy. Patients with both clinical and histological response showed higher adherence score measured by VAS

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scale compared to the other patients (p=0.02). No factors were predictive of adherence to DFD therapy.

Age, years, median (IQR)	37 (28.5;54.5)
Males, n (%)	15 (71.4)
Dilations anytime, n (%)	5 (23.8)
Eosinophils/HPF at baseline, median (IQR)	41 (29;62)
DFD indications, n (%):	
I line therapy in naive patients	6 (28.6)
add-ON therapy to PPI in Clinical-PPI-Responders and Histological-PPI- Non Responders	4 (19.0)
II-line therapy	11 (52.4)
VAS adherence score, median (IQR)	8 (7; 9)

Table 1.Baseline characteristics

Conclusion: DFD can be offered as an effective therapy for adult EoE patients and significantly improves quality of life.
Disclosure: Nothing to disclose.

PP0049

CURRENT STATUS OF DIAGNOSIS AND TREATMENT OF EOSINOPHILIC ESOPHAGITIS IN CHILE: RESULTS OF A NATIONWIDE SURVEY

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Introduction: Eosinophilic esophagitis (EoE) is an emerging disease in Latin America; however, little is known about medical practices related to its diagnosis, treatment, and follow-up. In Chile, the lack of a consensus-based practice guideline may lead to heterogeneous management among physicians. In contrast, standardized approaches and clinical recommendations have been established internationally.
Aims & Methods: To characterize current practices in the diagnosis, management, and follow-up of EoE in Chile.
A 55-item survey was developed to assess various aspects of eosinophilic esophagitis diagnosis, treatment, and monitoring. Five gastroenterologists reviewed a pilot version to evaluate its clarity and relevance. The survey invitation was emailed to all physician members of the Chilean Society of Gastroenterology. Additional open invitations were extended during the 50th Chilean Congress of Gastroenterology and through peer-to-peer contacts. Participants provided informed consent digitally, after which they received a link to complete the survey. The local ethics committee approved the study.
Results: The response rate was 11.9%, with 69 responses analyzed. The mean age of the participants was 50.4 years (SD ±12.77). Most respondents (66.7%) practiced in Santiago, and 91.3% were adult gastroenterologists. A total of 47.8% had more than 20 years of clinical experience, and 49.3% reported treating fewer than five EoE patients per year.
Among respondents, 82.6% generally or always performed esophageal biopsies in cases of dysphagia, and 69.6% for refractory reflux symptoms. The EREFS endoscopic classification was generally or always used by 52.2%. In emergency endoscopies for food impaction, 66.7% reported obtaining biopsies. Furthermore, 81.8% of endoscopists collected six or more biopsies when EoE was suspected.
Proton pump inhibitors (PPIs) were used as first-line therapy by 75.4% of respondents in over 75% of their patients, with esomeprazole being the most commonly prescribed PPI. 24.6% used topical corticosteroid as first-line treatment in over 75% of their patients. The most commonly used was liquid budesonide.
Regarding dietary therapy, whether as monotherapy or in combination with pharmacological treatment, the most frequent strategy was a two-food elimination diet with a step-up approach if needed. A total of 88.4% aimed for histologic remission as a therapeutic goal. In patients in clinical and histological remission without treatment changes, 55% reported follow-up intervals between three and six months.
Conclusion: There is variability among Chilean physicians in the diagnostic and therapeutic approaches to EoE. Most adhere to international biopsy standards and perform appropriately biopsies in patients with dysphagia. Many also align treatment with recommended therapeutic goals. However, biopsy collection during the index emergency endoscopy for food impaction should be reinforced to ensure timely diagnosis and early initiation of treatment. Overall, improved physician education and the development of national guidelines are needed to optimize and standardize the management of EoE in Chile.
Disclosure: Nothing to disclose.

PP0050

EPIDEMIOLOGICAL AND CLINICAL CHARACTERISTICS OF PATIENTS DEVELOPING GASTRO-OESOPHAGEAL REFLUX DISEASE FOLLOWING BARIATRIC SURGERY: A SINGLE-CENTRE RETROSPECTIVE STUDY

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Introduction: Gastroesophageal reflux disease (GORD) is a recognised complication of bariatric surgery; however, the clinical phenotype of patients who develop postoperative GORD remains poorly characterised. Improved understanding of this cohort may aid in risk stratification and management.
Aims & Methods: This study aimed to identify the epidemiological and clinical features of patients with upper gastrointestinal symptoms following bariatric surgery and to evaluate whether these vary by surgery type. This retrospective study examined 31 post-bariatric surgery patients with upper gastrointestinal symptoms referred to a quaternary gastroenterology service between 2004 and 2023. Data collected included age, sex, comorbidities, type and number of bariatric procedures, time from surgery to symptom onset, and symptom profile.
Results: Patients were predominantly female (80.6%) with high prevalence of comorbid depression (45.2%) and anxiety (53.3%). Sleeve gastrectomy was the most common procedure (70.9%), followed by adjustable gastric banding (35.5%); 16.1% patients underwent more than one bariatric surgery. On average, symptoms developed 0.19 years postoperatively and persisted for a mean duration of 6.8 years before referral. Dysphagia and heartburn were reported in 77.4% and 74.2% of patients, respectively. All patients who underwent Roux-en-Y gastric bypass (RYGB) reported heartburn. Patients with a history of sleeve gastrectomy appeared to have the highest symptom burden.
Conclusion: Patients who develop GORD after bariatric surgery are predominantly female, frequently have pre-existing mental health conditions, and often undergo sleeve gastrectomy. Symptoms typically emerge early and persist for several years before being investigated. Upper gastrointestinal symptoms appeared more prevalent than previously reported, suggesting possible under-recognition in clinical practice. Although RYGB

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is the preferred bariatric procedure in patients with co-existing GORD, our findings indicate that significant reflux symptoms may still occur postoperatively. These results support the need for early identification perioperatively and tailored follow-up of at-risk patients.

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Disclosure: Nil.

PP0051

THE ASSOCIATION OF PSYCHOLOGICAL COMORBIDITY WITH GASTROESOPHAGEAL REFLUX DISEASE IS PARTLY EXPLAINED BY DISORDERS OF GUT BRAIN INTERACTION

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Introduction: Disorders of brain gut interaction (DBGI) including irritable bowel syndrome (IBS) and functional dyspepsia (FD), and gastro-oesophageal reflux disease (GERD) including reflux oesophagitis (RE) and non-erosive reflux disease (NERD) are all prevalent in the general population. It is well established that many patients with GERD also report a DGBI, an overlap not due to chance [1]. Psychological comorbidity is known to be strongly associated with DGBI [2].

Individuals with GERD also report elevated rates of anxiety and depression. Those with overlapping DGBI and GERD have poorer physical and mental health outcomes [3-4].

Whether the psychological comorbidity seen in GERD is due to the overlap with DGBI or is a direct result of the condition itself remains unclear.

Aims & Methods: To investigate in a large sample of primary care patients whether the overlap of IBS and/or FD with RE and NERD explains the apparent link between GERD and anxiety/depression or whether the link is independent of DGBI presence. Data were collected from a retrospective study of 1,256,331 patients at general practices in the United Kingdom. A diagnosis of RE was defined as reflux with the presence of documented oesophagitis and NERD as reflux in the absence of oesophagitis. Psychological comorbidity was a diagnosis of anxiety or depression. Multiple logistic regression was used to assess the association of psychological comorbidity and a diagnosis of IBS and/or FD and RE and NERD, adjusting for the other disorder along with age at first contact and sex.

Results: Approximately 13.4% (95%CI 13.36,13.49) of patients reported a diagnosis of IBS, 20.7% (95%CI 20.67,20.81) FD, 4.2% (95%CI 4.19,4.27) RE and 5.1% (95%CI 5.09,5.17) NERD. Psychological comorbidity was significantly higher in patients with RE versus no RE (40.0% vs 24.4%, P<0.001), NERD versus no NERD (39.7% vs 24.2%, P<0.001) and IBS and/or FD (38.9% vs 18.6%, P<0.001). Psychological comorbidity was significantly associated with RE (OR=2.07; 95%CI 2.03,2.11, P<0.0001) and NERD (OR=2.06;95%-CI 2.02,2.09, P<0.0001).

However, this effect was reduced once IBS and/or FD was adjusted for in the models for both RE (OR=1.52; 95%CI 1.49,1.55, P<0.0001) and NERD (OR=1.53; 95%CI 1.50,1.55, P<0.0001) and persisted when age and sex were controlled for in the analysis for RE (OR=1.55, 95%CI 1.52,1.57, P<0.001) and NERD (OR=1.52; 95%CI 1.50,1.55, P<0.0001). With respect to IBS and/FD, psychological comorbidity was a significant independent predictor of an IBS and/or FD diagnosis (OR=2.79, 95%CI 2.77, 2.82, P<0.001). The likelihood of psychological comorbidity predicting an IBS and/or FD diagnosis remained more than twice as high even after controlling for RE (OR=2.72, 95%CI 2.70, 2.74, P<0.0001) and NERD (OR=2.71;95%-CI2.68,2.73, P<0.0001) and age and sex along with RE (OR=2.49; 95%CI 2.47,2.51, P<0.000) and NERD (OR=2.48; 95%CI 2.46,2.50, P<0.0001).

Conclusion: In a large sample of primary care patients, the association of IBS/FD and psychological comorbidity is independent of reflux oesophagitis (RE) and NERD, but the link between RE and NERD with anxiety/depression is partly explained by IBS/FD presence.

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Disclosure: Nothing to disclose.

PP0052

PROXIMAL MNBI AND ITS ASSOCIATION WITH INTERSTITIAL LUNG DISEASE (ILD) IN SYSTEMIC SCLEROSIS

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Introduction: Gastrointestinal (GI) symptoms in Systemic sclerosis (SSc) cause significant morbidity. GERD is considered one of the causative factors of ILD via laryngopharyngeal reflux (LPR) in SSc. The objective of our study was to obtain pH-impedance metrics and observe the relationship between proximal MNBI (close association with LPR) and ILD, thus providing objective evidence for GERD-ILD crosstalk.

Aims & Methods: A cross-sectional study from July 2021 to May 2023 included consecutive SSc patients. Disease characteristics, demographic parameters, and upper and lower GI symptoms were collected using the UCLA 2 GIT questionnaire. pH-impedance testing, HRM measurements were recorded, and their association with GI symptoms was analysed.

Results: A total of 83 patients were analysed. 90% had upper and 21% had lower GI symptoms. The most common upper and lower GI symptom was heartburn (77.1%). Non-erosive disease was seen in 54% of patients, and

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severe oesophagitis was seen in 14.4% of patients. True NERD was seen in 76%. Proximal and distal MNBI were low in 81.9% (cut-off 2292 ohms) and 88% of patients, respectively. ILD was seen in 72% of patients. Low proximal MNBI was significantly associated (*P=0.018*) with ILD. On logistic regression, low proximal MNBI was an independent predictor of ILD.

Conclusion: SSc patients had predominantly upper GI symptoms, with three-fourths having true NERD. A majority had a low proximal MNBI. Proximal MNBI was significantly associated with ILD.

Disclosure: Nothing to disclose.

PP0053

MORE ACID, SAME SYMPTOMS? THE PARADOX OF HIATAL HERNIA SIZE

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Introduction: Hiatal hernia (HH) is frequently associated with gastroesophageal reflux disease (GERD) and esophageal motility disorders. However, its functional implications on esophageal physiology remain controversial. This study aimed to assess the clinical relevance of HH and esophagogastric junction (EGJ) morphology on esophageal symptoms and function.

Aims & Methods: We conducted a cross-sectional, observational study comparing esophageal function in patients with and without hiatal hernia in 92 patients referred for high-resolution manometry (HRM) and 24-hour pH monitoring. Fifty-five patients with HH were compared to 37 controls without HH. Patients were also classified by EGJ morphology (types I, II, III). Clinical evaluation included validated symptom questionnaires: GERD-Q, Eckardt, and BED-Q. Patients with prior esophagogastric surgery or EGJ outflow obstruction were excluded. Comparative analysis employed chi-square tests for categorical variables, and Student's t-test, ANOVA, or Kruskal-Wallis tests for continuous variables, depending on distribution.

Results: Of the 92 patients, 61 (66.3%) were female, with a mean age of 56 ± 14 years and BMI of 26.6 ± 7.5 kg/m^2. HH was present in 55 patients (59.8%). EGJ classification revealed 36 type I, 38 type II, and 15 type III configurations. HH was significantly associated with higher reflux symptom scores on the GERD-Q (p = 0.015) and increased esophageal acid exposure on pH monitoring (p = 0.015). Acid exposure time increased proportionally with HH size, although symptom severity (GERD-Q) did not correlate with hernia size. There were no significant differences in dysphagia scores (Eckardt or BED-Q) or total reflux episodes between groups. Similarly, EGJ morphology did not correlate with clinical, manometric, or pH-metric outcomes.

	No HH (n = 37)	HH (n = 55)	p-value
Clinical scores			
GERD-Q ≥ 8	20 (54%)	43 (78%)	0.015
Eckardt score ≥ 3	19 (51%)	39 (71%)	0.057
Manometric findings			
IRP ≥ 15 mmHg	6 (16%)	15 (27%)	0.210
Ineffective Esophageal Motility	2 (5%)	9 (16%)	0.110
pHmetry parameters			
AET > 6%	3 (8%)	16 (29%)	0.015
Total reflux episodes > 80	3 (8%)	10 (18%)	0.150

AET, acid exposure time; IRP, integrated relaxation pressure; HH, hiatal hernia; MNBI, mean nocturnal basal impedance.

Conclusion: In this study, HH was associated with greater acid exposure and more severe reflux symptoms, with acid burden increasing proportionally to hernia size.

However, symptom severity did not reflect hernia size, and EGJ type showed no significant functional impact. These findings underscore the pathophysiological relevance of HH in GERD, independent of EGJ subtype.

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Disclosure: Nothing to disclose.

PP0054

EFFICACY OF THE SYMPTOM–REFLUX ASSOCIATION TO DETERMINE SECONDARY TREATMENT FOR PPI-/P-CAB REFRACTORY TRUE GERD PATIENTS

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Introduction: The number of gastroesophageal reflux disease (GERD) patients that do not improve sufficiently by standard treatment is relatively high. Multichannel intraluminal impedance and pH monitoring (MII-pH) is the gold standard for diagnosing GERD, and it can be useful for selecting secondary treatment.

However, in some cases, secondary treatments are not remarkably effective even when selected based on the diagnosis.

On the other hand, the clinical value of the symptom–reflux association (SRA) obtained using MII-pH remains unknown.

Aims & Methods: The current study aimed to investigate the effect of SRA on the efficacy of secondary treatment in refractory true GERD.

This study included patients who underwent MII-pH monitoring for evaluating proton pump inhibitor (PPI)- or potassium-competitive acid blocker (P-CAB)-refractory GERD symptoms and who were diagnosed with abnormal acid reflux (acid exposure time > 6%).

Patients with a positive symptom index (SI) and symptom association probability (SAP) on the MII-pH monitoring were included in the SRA-positive group and the remaining ones in the SRA-negative group.

The differences in the subsequent treatment efficacy between the two groups were retrospectively analyzed. Treatment efficacy was evaluated using the Frequency Scale for the Symptoms of GERD (FSSG) questionnaire.

Results: Of the 192 patients with PPI-/PCAB-refractory GERD, 41 (21.4%) were diagnosed with abnormal acid reflux. Among them, 30 (7 in the SRA-positive group and 23 in the SRA-negative group) underwent symptom assessments before and after treatment. The SRA-positive group had a significantly greater symptom improvement after treatment compared with the SRA-negative group (FSSG score changes: −13.0 ± 6.5 vs −5.4 ± 8.2, p = 0.016).

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Conclusion: SRA evaluation is effective in predicting secondary treatment outcomes in patients with abnormal acid reflux.
Disclosure: Nothing to disclose.

PP0055

BASELINE IMPEDANCE DURING HIGH-RESOLUTION ESOPHAGEAL MANOMETRY CORRELATES WITH MNBI AND PREDICTS GERD ACCORDING TO LYON 2.0 CRITERIA

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Introduction: The current diagnosis of gastroesophageal reflux disease (GERD) is based on the criteria established by the Lyon Consensus 2.0[1], which defines conclusive criteria for or against GERD, along with adjunctive metrics to support or refute the diagnosis when primary criteria are borderline or inconclusive – among them, mean nocturnal baseline impedance (MNBI) measured via pH-impedance (pH-MII) monitoring. Some data suggest a correlation between baseline impedance measured during high-resolution esophageal impedance manometry (HRIM-BI) and MNBI[2,3]; however, none have applied the Lyon 2.0 criteria in this context.
Aims & Methods: We aimed to investigate whether HRIM-BI can distinguish GERD patients from non-GERD patients, providing an additional metric for diagnosis. All the patients who underwent both HRIM and pH-MII for GERD evaluation at ASST Fatebenefratelli Sacco (Sacco Hospital, Milan) were included. HRIM-BI was measured 3 centimeters above the upper margin of the lower esophageal sphincter over 15 seconds during the baseline period of HRIM while MNBI was measured over three 10-minute nocturnal intervals of pH-MII. Patients were categorized as having GERD according to Lyon 2.0 criteria (i.e., acid exposure time [AET] > 6% and > 80 reflux episodes). Data are reported as mean ± SD or number (percentage). Statistical analysis was performed using Pearson correlation to assess the relationship between HRIM-BI and MNBI values. Univariate and multivariate logistic regression analyses were used to evaluate the association between HRIM-BI and the presence of GERD, adjusting for potential confounders such as age. A p-value < 0.05 was considered statistically significant. Analyses were conducted using SPSS version 29.0.
Results: A total of 44 patients were enrolled, 37 of whom were off proton pump inhibitor (PPI) therapy for at least 15 days. Table 1 shows demographic and manometric characteristics, as well as HRIM-BI and MNBI values in patients with and without GERD.

	No GERD (n = 31)	GERD (n = 13)	p
Sex (female)	18 (56%)	5 (39%)	ns
Age (years)	47 ± 16	60 ± 15	0.024
EGJ-type			ns
1	24 (75%)	7 (54%)	
2	3 (9%)	1 (8%)	
3	5 (16%)	5 (39%)	
HRIM diagnosis			ns
Normal	24 (75%)	8 (61%)	
Ineffective esophageal motility	4 (13%)	4 (31%)	
Hypercontractile esophagus	3 (9%)	1 (8%)	
Distal esophageal spasm	1 (3%)	0	
HRIM-BI (kΩ)	2.148 ± 0.9	1.184 ± 0.8	0.004
MNBI (kΩ)	3.380 ± 1.2	2.098 ± 1.2	0.006

Table 1. Demographic and manometric characteristics.

A significant positive correlation was observed between HRIM-BI and MNBI (Pearson r = 0.653, $p < 0.001$). Univariate and multivariate regression analyses demonstrated that HRIM-BI was an independent predictor of GERD, even after adjustment for age (OR 0.33, 95% CI 0.14-0.81). Similar findings were observed in the subgroup of patients off PPI therapy, with a Pearson correlation coefficient of 0.525 ($p < 0.001$) and an OR for HRIM in the prediction of GERD of 0.21 (95% CI 0.06–0.78).
Conclusion: Baseline impedance measured during HRIM correlates positively with MNBI, a recognized adjunctive metric for GERD diagnosis. HRIM-BI is easily measurable, cost-effective, and less invasive – potentially obviating the need for pH-MII in selected cases – and may serve as a valuable tool for the diagnostic workup of GERD.
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Disclosure: Nothing to disclose.

PP0056

DIAGNOSTIC PERFORMANCE OF A NOVEL WIRED MAGNETIC-ASSISTED CAPSULE ENDOSCOPE IN THE DIAGNOSIS OF EROSIVE ESOPHAGITIS COMPARED TO ESOPHAGOGASTRODUODENOSCOPY

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Introduction: Erosive esophagitis (EE) is a clinically significant manifestation of gastroesophageal reflux disease (GERD), characterized by endoscopically visible mucosal breaks due to chronic gastric acid exposure. EE increases the risk of complications such as Barrett's esophagus and esophageal adenocarcinoma. Although esophagogastroduodenoscopy (EGD) remains the diagnostic gold standard, it is invasive and often requires sedation. Wired magnetic-assisted capsule endoscopy (MACE) is a novel, minimally invasive technique using a magnetically navigable capsule tethered via a soft wire, enabling upper gastrointestinal tract examination. However, its diagnostic value for EE remains underexplored.
Aims & Methods: From February 2021 to November 2024, 85 patients underwent wired MACE followed by EGD within six hours at National Taiwan University Hospital. Indications included acute upper gastrointestinal bleeding (n=30), esophageal varices screening (n=33), and pre-bariatric surgery assessment (n=22). Diagnostic accuracy of wired MACE for EE was assessed using EGD as the reference. Patient-reported discomfort (0–10 scale) and acceptance (0–5 scale) were compared between the two modalities.
Results: The mean age was 57.8 years (SD 15.2; range 23–83), with 62.4% male. EGD was performed without sedation in 92.9% of cases. Wired MACE showed a sensitivity of 86.4% (95% CI: 76.2–97.0%), specificity of 82.9% (95% CI: 71.4–94.0%), PPV of 84.4%, NPV of 85.0%, and overall accuracy of

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84.7%. Agreement with EGD was substantial (κ = 0.69). MACE was associated with lower mean discomfort scores (1.9 vs. 5.5, p < 0.0001) and higher acceptance scores (4.2 vs. 2.3, p < 0.01), indicating superior tolerability.

Conclusion: Wired MACE demonstrated high diagnostic accuracy and substantial agreement with EGD in detecting EE. Its significantly lower discomfort and higher patient acceptance support its potential as a less invasive, better-tolerated diagnostic alternative to conventional endoscopy.

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Disclosure: Nothing to disclose.

PP0057

VALIDATION OF DOMINANT SYMPTOM INTENSITY AS A PATIENT REPORTED OUTCOME MEASURE IN THE EVALUATION OF ESOPHAGEAL SYMPTOMS SUSPICIOUS FOR GASTROESOPHAGEAL REFLUX

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Introduction: Gastroesophageal reflux disease (GERD) presents with typical and atypical symptoms of varying frequency and severity. We compared 5-point Likert scales assessing symptom burden to validated GERD questionnaires, and to outcome following reflux management.

Aims & Methods: We analyzed pooled data from patients undergoing pH-impedance testing for reflux symptoms from two tertiary care centers. All patients completed pre-procedure questionnaires assessing typical (heartburn, regurgitation, esophageal chest pain) and atypical symptoms (cough, belching, wheezing, dysphagia). Symptom severity and frequency were scored on 5-point Likert scales (0=not severe, infrequent; 4=extremely severe, multiple daily episodes); the product of severity and frequency for the highest scoring symptom constituted the dominant symptom intensity (DSI). Patients reported global symptom severity (GSS) on 100 mm visual analog scales (VAS) anchored by 0=no symptoms and 100=severe symptoms. Spearman correlation, general linear regression and receiver operating characteristic (ROC) analyses were performed to compare DSI with validated instruments including GSS, gastroesophageal reflux disease questionnaire (GERDQ) and reflux symptom index (RSI), and pH-impedance parameters per Lyon Consensus 2.0. In a subset, change in DSI after medical and surgical GERD management was compared against symptom burden (GSS) and satisfaction with therapy on 100 mm VAS.

Results: 808 patients (mean age 53.8±1.44 years, 65.2% female) were included, 499 (61.8%) with dominant typical symptoms. DSI significantly correlated with GSS (R=0.682), GERDQ (R=0.414), RSI (R=0.577), acid exposure time (AET) (R=0.175), total reflux episodes (R=0.194) and mean nocturnal baseline impedance (R=-0.157)(p<0.0001 for each comparison). DSI remained significantly correlated with GERDQ, GSS, RSI, AET and reflux episodes in both typical and atypical symptom subgroups. Mean DSI was higher with abnormal GERDQ, RSI, and conclusive GERD per Lyon Consensus (p<0.0001 each comparison). On ROC analyses, DSI (AUROC=0.60) was non-inferior to GSS, GERDQ, and RSI in predicting pathologic AET and total reflux episodes, and conclusive GERD. Percentage improvement in DSI after antireflux treatment significantly correlated overall with GSS change (R=0.632, p<0.0001) and treatment satisfaction (R=0.513, p<0.0001), and in both typical and atypical symptom subgroups.

On multivariable linear regression analyses, percentage change in DSI remained an independent predictor of GSS change (β=0.302, p<0.0001), and satisfaction with anti-reflux therapy (β=0.833, p=0.011). On ROC analyses, DSI change (AUROC=0.730) was non-inferior to GSS change in predicting satisfaction with therapy.

	Overall	Typical, n=499	Atypical, n=309
Age (years)	53.8±1.44	51.0±2.28	58.2±0.79*
Gender (female)	527 (65.2%)	356 (71.3%)	171 (56.3%)*
Body mass index (kg/m^2)	27.4±0.25	27.6±0.32	27.1±0.39
Mean baseline DSI	9.37±0.21	10.6±0.25	7.34±0.33*
Mean baseline GSS	50.1±1.15	60.4±1.28	33.7±1.82*
Mean AET (%)	3.29±0.18	3.63±0.23	2.67±0.26*
Number of reflux events	46.9±1.31	50.3±1.78	40.6±1.62*
MNBI (ohms)	1743±83	1667±106	1864±134
Conclusive GERD	177 (21.9%)	125 (25.1%)	52 (16.8%)

GSS: global symptom severity; AET: acid exposure time; MNBI: mean nocturnal baseline impedance; GERD: gastroesophageal reflux disease;
**p<0.05 compared to typical group*

Conclusion: DSI correlates with validated reflux questionnaires, discriminates abnormal from normal reflux burden, and DSI change reflects reflux treatment outcome and satisfaction. DSI is a useful addition to reflux symptom assessment.

Disclosure: Nothing to disclose.

PP0058

DIAGNOSTIC PERFORMANCE OF PH-IMPEDANCE MONITORING IN ASSESSING DUODENOGASTROESOPHAGEAL REFLUX IN REVISIONAL BARIATRIC SURGERY: A COMPARISON WITH SPECTROPHOTOMETRIC BILIRUBIN CONCENTRATION MONITORING

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Introduction: Duodenogastroesophageal reflux (DGER), or bile reflux, has gained renewed interest due to the increasing number of bariatric surgeries (BS) performed. Anatomical modifications after BS may predispose patients to reflux. Revisional bariatric surgery (RBS) is indicated for inadequate weight loss or refractory symptoms.

Aims & Methods: This study aimed to evaluate the diagnostic accuracy of pH-impedance (pH-Z) monitoring in detecting DGER, using spectrophotometric bilirrubin (Bilitec2000) as the clinical reference standard. This prospective study included consecutive patients with a history of sleeve gastrectomy (SG) referred for preoperative evaluation prior to RBS between October 2023 and February 2025. All patients underwent simultaneous 24-hour pH-Z monitoring and Bilitec2000 monitoring while off proton pump inhibitors (PPIs).

Patients followed a standardized white semiliquid diet. Gastroesophageal reflux disease (GERD) was diagnosed according to the Lyon 2 Consensus. Pathological bile reflux was defined as a total bilirubin absorbance ≥ 0.14 for ≥ 1.9% of the monitoring time.

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Results: Fifty-three patients were enrolled [38 (71.7%) female, median age 48 years (IQR 43 – 55), median BMI 37.3 Kg/m^2 (IQR 34.1 – 41.3)]. Reason for RBS evaluation included
inadequate weight loss (39.6%) refractory reflux symptoms (32.1%), both (24.5%) and other indications (3.5%). GERD was diagnosed in 31 patients (58.5%) and 21 (39.6%) had pathological bile reflux. Erosive GERD according to Los Angeles classification was identified in 10 patients (19.6%). The area under the curve for non-acid reflux episodes detected by pH-z in diagnosing DGER was 0.61 (CI 95% 0.44 - 0.78), with an optimal cutoff of 85 episodes (sensitivity: 38.1% (CI 95% 20.8 – 59.1); specificity of 93.6% (CI95% 79.9 – 98.3). A moderate correlation was found between non-acid reflux and bile reflux (r = 0.38, p=0.005).
However, GERD was not associated with increased bile exposure time (1.4% (IQR 0.3 – 9.5) vs 1.3% (IQR 0.0 – 4.8), p=0.262), nor was this association observed in those with erosive esophagitis (1.9%, IQR 0.4 – 9.5 vs 1.4% IQR 0.0 – 5.8; p=0.914).
Conclusion: Approximate 40% of SG patients undergoing RBS evaluation had pathological bile reflux, which was not linked to GERD or erosive esophagitis. Despite a moderate correlation between bile and non-acid reflux episodes, pH-z monitoring failed to reliably detect DGER and should not be considered a substitute for spectrophotometric bilirrubin concentration monitoring.
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Disclosure: Nothing to disclose.

PP0059

EVALUATION OF MEAN DISTAL CONTRACTILE INTEGRAL (DCI) REFINES THE DIAGNOSIS OF IEM IN GERD MANOMETRIC ASSESSMENT

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Introduction: Ineffective esophageal motility (IEM) is a manometric diagnosis based on the rate of failed/ineffective swallows, consistently associated with Gastro-Esophageal Reflux Disease (GERD). Chicago 4.0 classification (CCv4.0)[1] does not include mean values of Distal Contractile Integral (DCI) in the diagnosis of IEM at High-Resolution Manometry (HRM).
Aims & Methods: The aim of this study was to explore DCI assessment as a HRM metric to refine IEM diagnosis.
Retrospective analysis of a prospectively collected database (October 2022-April 2024) was conducted on patients undergoing esophago-gastroduodenoscopy (EGD), HRM and 24-hour-pH-impedance study. Patients with indication for HRM other than GERD symptoms (achalasia, follow-up after previous foregut surgery including anti-reflux surgery or achalasia treatment), were excluded. GERD conclusive diagnosis was made according to Lyon 2.0 criteria[2].
Univariate logistic regression was computed to identify the optimal DCI cut-off for objective GERD. Multivariable logistic regression analysis were performed to identify relevant manometric predictors of objective reflux burden and to build predictive models. Models accuracy was tested through receiving operator characteristic (ROC) analysis. Sensitivity and specificity were calculated and compared between models.
Results: A total of 202 patients (M 94 [46.5%], mean age 48.5±18.6 years, BMI 22.1±7.6) were analyzed. Conclusive GERD was found in 49 patients (24.3%). IEM diagnosis was made in 40 patients (19.8%) while 162 patients (80.2%) had normal esophageal motility (NEM). ROC analysis identified DCI of 810.2 mmHg.cm.s (AUC 0.70, p<0.001) as the best cut-off in predicting objective reflux (AET>6%). The group with NEM and DCI<810.2 did not show differences with the IEM group (p>0.05). At multivariable regression analysis, IEM (CCv4.0) and Esophagogastric Junction- Contractile Integral (EGJ-CI) lost significance in predicting objective GERD (p=0.76 and p=0.91 respectively), while higher mean DCI (OR 0.41, p<0.004) and type III EGJ (OR 4.4, p<0.002) and higher mean basal EGJ pressure (MBEP) (OR 0.30, p<0.007) did remain significantly associated. At ROC analysis mean DCI combined with MBEP and type-III EGJ showed an AUC of 80.6% (p<0.001) in predicting objective GERD. We compared a model comprising MBEP, type III EGJ and IEM (CCv4.0) (Model A) vs a model comprising MBEP, type III EGJ and IEM (CCv4.0)+DCI >810.2 mmHg.cm.s (Model B). Model B was superior to Model A, showing an AUC of 80.0 vs an AUC of 78.3 (p<0.001) (Table 1.).

	B	S.E.	p value*	ORs	95% C.I. for ORs	
Model A					**Lower**	**Upper**
IEM***	0.88	0.41	0.032	2.43	1.08	5.49
Type-III EGJ	1.42	0.46	0.002	4.16	1.68	10.26
MBEP***	-1.23	0.34	0.001	0.29	0.15	0.57
Model B						
IEM**+DCI<810.2****	1.08	0.37	0.004	2.96	1.42	6.19
Type-III EGJ	1.53	0.47	0.003	4.12	1.63	10.42
MBEP**	-1.24	0.35	0.001	0.26	0.13	0.52

Table 1. Multivariable logistic regression of model A and model B in predicting AET>6%
Bold entries indicate statistical significance at 0.01 level (2-tailed)
**All independent variables were controlled for inter-variables correlation by coefficients co-dependency analysis via Wald test and inter-variables logistic regression. No co-dependency was observed (p>0.05).*
***These variables were imputed as natural logarithm to adjust variance and compute ORs*
**** According to Chicago 4.0 classification (CCv4.0)*
*****Cut-off identified at ROC analysis*
C.I. Confidence Intervals; DCI Distal Contractile Integral; EGJ Esophago-Gastric Junction; IEM Ineffective Esophageal Motility; MBEP Mean Basal LES Pressure; ORs Odds Ratios; ROC Receiving Operator Characteristic; S.E. Standard Error

Conclusion: Evaluation of mean DCI complements IEM diagnosis and esophageal contractility assessment, with a cut-off of 810 mmHg.cm.s to optimize IEM profiling of GERD.
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PP0060

ANTI-REFLUX MUCOSECTOMY WITH BANDING (ARMS-B) FOR GORD: A PILOT STUDY

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Introduction: Gastroesophageal reflux disease (GORD) common, impacting on healthcare utilisation and quality of life (QoL). Medical therapies for GORD are frequently unsuccessful, while surgical approaches confer significant morbidity risk. Anti-reflux mucosectomy (ARMS) techniques achieve scar restoration of the gastro-oesophageal junction mucosal flap valve with relatively low risk, but are technically demanding.

We studied a variant of the technique using a standard endoscopic banding kit (ARMS-B), with bands applied in a crescentic fashion to the cardia in retroflexion and left in situ to induce scarring. We evaluated safety and efficacy, including objective assessment.

Aims & Methods: This was a single-centre, prospective cohort study (May 2022 to May 2024). Patients with symptomatic GORD, inadequately controlled by medical therapy, hiatus hernia ≤2cm and Hill grade ≤2, confirmed on Bravo ambulatory pH monitoring were eligible. ARMS-B was performed by a single expert endoscopist. A repeat gastroscopy with Bravo insertion was completed at least 3 months post-ARMS-B, off PPI.

The primary outcomes included the change in acid exposure time (AET) and Gastroesophageal Reflux Disease Health-Related Quality of Life (GORD-HQRL) scores. Secondary outcomes included the cessation or reduction in PPI usage, and the incidence of complications.

Results: Thirteen patients were included in the final analysis (69.2% female; mean age 51.3 years, mean BMI 22.9). Median follow-up duration was 6.3 months (3.5 – 24). Post treatment, AET normalised in 60% (9.9% vs 5.2%, 95% CI [1.01 – 8.3], p=0.02) and GORD-HQRL scores normalised in 54.5% (34.7 vs 22.3, 95% CI [1.7 – 23], p = 0.03). 46.2% of patients ceased or reduced the dose of their PPI at 3 months. No complications were reported in early or late follow-up.

Variable	Pre-ARMS-B %(SD)	Post-ARMS-B %(SD)	P value
Acid exposure time	9.9% (3.4%)	5.2% (4.7%)	0.02
DeMeester score	43.7 (11.9)	25.3 (16.5)	0.08
GORD-HQRL			
Total	34.7 (16.3)	22.3 (18.4)	0.03
Heartburn	18.5 (9.6)	11.2 (11.0)	0.03
Regurgitation	15.9 (6.1)	9.9 (8.6)	0.01
Satisfied (Yes)	0%	42.6%	0.42
Active PPI Use	100%	53.8%	0.71
Reduced PPI dose	-	7.7%	
Ceased PPI	-	38.5%	

Conclusion: This pilot study demonstrates that ARMS-B is both safe and effective for the treatment of GORD. The procedure led to significant reductions in AET, GORD-HRQL, symptom control, and PPI usage. Further studies with larger cohorts and longer follow-up periods are warranted to confirm these findings.

Disclosure: Nothing to disclose.

PP0061

THE IMPACT OF THE TIF PROCEDURE ON ESOPHAGEAL MANOMETRY RESULTS: A PROSPECTIVE CONTROLLED STUDY

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Introduction: This study aims to evaluate the impact of Transoral Incisionless Fundoplication (TIF) on esophageal motility and reflux patterns using high-resolution esophageal manometry (HRM) and 24-hour pH-impedance monitoring. By comparing pre- and post-procedure findings in GERD patients, we seek to understand the functional changes induced by TIF and their correlation with symptom relief, providing insights into its role as a therapeutic option for GERD.

Aims & Methods: A prospective controlled study was conducted, enrolling 40 GERD patients, with 20 undergoing TIF and 20 receiving standard medical therapy as a control group. HRM and 24-hour pH-impedance monitoring was performed at baseline and at 3, 6, and 12 months post-procedure to assess:

1. LES function: resting pressure and relaxation.
2. Esophageal body motility: peristaltic amplitude, contractile front velocity (CFV), and prevalence of ineffective esophageal motility (IEM).
3. Reflux patterns: total acid and non-acid reflux episodes, esophageal acid exposure time, and symptom-reflux correlation.

Symptom assessment was conducted using validated GERD-specific questionnaires (e.g., GERD-HRQL). Statistical analysis included paired t-tests for within-group comparisons and ANOVA for between-group comparisons, with a significance threshold of $p < 0.05$.

Results: Preliminary findings indicate that TIF improves LES function, esophageal motility, and reflux parameters compared to baseline. Key observations include:

- Increased LES resting pressure and improved peristaltic function.
- Reduced prevalence of ineffective esophageal motility (IEM).
- Decreased total acid and non-acid reflux episodes and esophageal acid exposure time.

These changes were associated with significant symptom improvement, as reflected in GERD-HRQL scores. The control group showed no significant changes in motility or reflux patterns. Extended follow-up (up to 12 months) demonstrated sustained improvements in the TIF group.

Conclusion: By incorporating pre- and post-procedure manometry and pH-impedance monitoring data, extended follow-up periods, and symptom correlation, this study will enhance the understanding of TIF's impact on esophageal motility and reflux patterns. Standardizing the evaluation of these changes post-TIF can guide clinical decision-making, ensuring optimal patient selection and improving GERD management strategies.

Disclosure: No conflict of interest.

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PP0062

BEYOND ACID: INTERIM RESULTS FROM A RANDOMIZED TRIAL OF STRETTA PROCEDURE VS. CONSERVATIVE MANAGEMENT IN GERD

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Introduction: To evaluate interim outcomes of a randomized controlled trial comparing Stretta radiofrequency (RF) therapy to conservative management in patients with gastroesophageal reflux disease (GERD), focusing on symptom control, daily PPI reduction, and acid exposure.

The study further explores the hypothesis that Stretta may act via neuromodulation rather than mechanical augmentation of the lower esophageal sphincter (LES).

Aims & Methods: This ongoing prospective trial includes patients with non-erosive reflux disease (NERD) or grade A esophagitis (Los Angeles classification), no or minimal hiatal hernia (≤2 cm), and objectively confirmed pathological reflux by impedance-pH monitoring. Patients were randomized to Stretta (n = 20) or conservative therapy (n = 21). Follow-up was scheduled at 3, 6, and 12 months. Only patients with complete paired data were analyzed at each time point.

Primary outcomes included symptom scores (GERD-HRQL and Reflux Symptom Index [RSI]) and daily PPI use. Acid exposure time (AET) was a secondary endpoint.

Results: In the Stretta group, 16, 12, and 9 patients completed 3-, 6-, and 12-month follow-up, respectively. In the conservative group, corresponding numbers were 18, 14, and 12. At 12 months, Stretta significantly improved GERD-HRQL (30.7 to 13.2; $p = 0.0118$), RSI (20.8 to 10.2; $p = 0.0018$), and reduced mean daily PPI use from 35.6 mg to 0 mg ($p = 0.0070$). AET did not significantly change (7.8 % to 6.4 %; $p = 0.4033$). In contrast, the conservative group showed no improvement in GERD-HRQL or PPI use, and RSI significantly worsened (14.1 to 20.1; $p = 0.0101$).

Conclusion: These interim findings suggest that Stretta therapy provides meaningful symptom relief and reduces the need for daily PPI use in selected GERD patients, even in the absence of acid normalization. This supports a neuromodulatory mechanism rather than mechanical LES augmentation. It is plausible that some GERD patients in this study had overlapping esophageal hypersensitivity, which may explain their positive clinical response. Stretta may thus have therapeutic relevance beyond current indications, particularly in patients with a hypersensitivity phenotype.

Disclosure: Katerina Kostalova has received speaker honoraria from Imedex, the official distributor of the Stretta system in the Czech Republic. No other conflicts of interest are declared by the authors.

PP0063

EFFECTIVENESS AND SAFETY OF VONOPRAZAN IN CHINESE PATIENTS WITH GASTROESOPHAGEAL REFLUX DISEASE: POST HOC ANALYSIS OF A MULTICENTRE, PROSPECTIVE, REAL-WORLD STUDY

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Introduction: Symptom relief is the key treatment goal of gastroesophageal reflux disease (GERD). Vonoprazan, a potassium-competitive acid blocker approved in China, was superior in potency and duration of action versus conventional proton pump inhibitors in preclinical studies.[1] However, there are limited real-world data on the effectiveness and safety of vonoprazan in adult Chinese patients.

Aims & Methods: VIEW (NCT04501627) was a multicentre, single-arm, prospective, observational, real-world study conducted in China. Patients received 20 mg vonoprazan, orally, once daily for 4 weeks (8 weeks for insufficient healing), followed by a safety assessment of 2 weeks. In this post hoc subgroup analysis of patients with GERD, effectiveness was assessed, including changes in GERD-typical symptoms, GERD questionnaire (GerdQ) scores and complete relief in patients' symptom diary.

Results: Among the 2214 patients included in the effectiveness analysis population of the GERD group, 86.7% were aged <65 years and 60.8% were male. Among evaluable patients with all-day heartburn (n=672) and all-day regurgitation (n=728) at baseline, 49.7% (95% confidence interval [CI], 45.9-53.6) and 40.8% (95% CI, 37.2-44.5), respectively, achieved complete symptom relief during the first 2 weeks of treatment. The reduction from baseline in all-day heartburn and regurgitation was significant ($P<0.001$) on days 1 through 14 of treatment.

Among evaluable patients with night-time heartburn (n=746) and night-time regurgitation (n=816) at baseline, complete symptom relief was achieved in 59.9% (95% CI, 56.3-63.5) and 52.8% (95% CI, 49.3-56.3) of patients, respectively, during the first 2 weeks of treatment. The reduction from baseline in night-time heartburn and regurgitation was significant ($P<0.001$) on days 1 through 14 of treatment. The mean (standard deviation) overall GerdQ score improvement from baseline to week 4 was -1.6 (2.8).

Symptom frequency score	Evaluable patients, n	Value[a]
Mean (SD) change in heartburn from BL to W4	1811	-0.8 (1.25)
95% CI for Mean	-	(-0.88, -0.76)
Mean (SD) change in regurgitation from BL to W4	1809	-0.8 (1.22)
95% CI for Mean	-	(-0.83, -0.72)
Mean (SD) change in sleep difficulty due to heartburn/ regurgitation from BL to W4	1809	-0.6 (1.08)
95% CI for Mean	-	(-0.63, -0.53)
Mean (SD) change in the need for additional medicationb for heartburn/regurgitation from BL to W4	1808	-0.4 (1.02)
95% CI for Mean	-	(-0.40, -0.30)

BL, baseline; CI, confidence interval; SD, standard deviation; W, week.
[a]Negative values indicate improvement.
[b]Examples: TUMS, Rolaids, Maalox.

Table: Improvement in symptom frequency score and additional medication usage from baseline to week 4 following vonoprazan treatment.

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The percentage of patients without GERD-typical symptoms increased by 33.5% (95% CI, 30.7-36.3) from baseline to week 4. Symptoms of heartburn, regurgitation and sleep difficulties due to heartburn/regurgitation, and the need for additional medication generally reduced at week 4 compared with that at baseline (Table).

Adverse events (AEs; any grade) were reported in 17.4% of patients in the safety population (N=2307), with the most common (>5%) being gastrointestinal disorders (8.3%). Overall, 0.9% of patients had ≥1 AE leading to study discontinuation. No AEs led to death.

Conclusion: This real-world subgroup analysis of Chinese patients with GERD demonstrated that vonoprazan effectively relieved symptoms, improved GerdQ scores and reduced the need for additional medications while maintaining an acceptable safety profile.

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PP0064

EFFECTIVENESS OF VONOPRAZAN ON SYMPTOM RELIEF IN CHINESE PATIENTS WITH GASTROESOPHAGEAL REFLUX DISEASE: POST HOC ANALYSIS OF THE VIEW STUDY

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Introduction: The primary treatment goals of gastroesophageal reflux disease (GERD) include resolving symptoms, preventing complications and symptom relapses, and healing erosive oesophagitis.[1,2]

Heartburn and regurgitation are frequent and typical symptoms of GERD. Vonoprazan, a potassium-competitive acid blocker, is approved in China for the treatment of reflux oesophagitis.[3,4]

However, there is a paucity of real-world data assessing the relief of typical daily symptoms achieved with vonoprazan in adult Chinese patients.

Aims & Methods: VIEW (NCT04501627) was a multicentre, single-arm, prospective, observational, real-world study conducted in China. Patients received 20 mg vonoprazan, orally, once daily for 4 weeks (8 weeks for insufficient healing). This post hoc subgroup analysis conducted in enrolled patients with GERD assessed changes in symptom scores for all-day or night-time heartburn and/or regurgitation during the first 14 days of treatment.

A mixed-effects model for repeated measures assessed statistical differences in symptom scores from baseline for each day; least-squares mean (LSM) and its 95% confidence interval are reported (negative values indicate improvement).

Results: Among the 2214 patients included in the effectiveness analysis population of the GERD group, 86.7% were aged <65 years[A.G.1] and 60.8% were male. The LSM change in symptom scores from baseline demonstrated a significant improvement from day 1 of treatment in all-day heartburn and/or regurgitation, daytime heartburn or regurgitation, and night-time heartburn or regurgitation (Table). Significant improvement continued to be observed on day 14 and throughout all previous days of treatment.

Conclusion: Vonoprazan provided rapid (from day 1), significant and durable symptom relief in patients with GERD.

Symptom score	Parameter	Day 1	Day 3	Day 7	Day 10	Day 14
All-day heartburn and regurgitation	LSM change from baseline (SE)/n[b]	-0.756 (0.0335) [c]/518	-0.986 (0.0311) [c]/516	-1.207 (0.0279) [c]/512	-1.327 (0.0247) [c]/504	-1.407 (0.0233) [c]/493
All-day heartburn	LSM change from baseline (SE)/n[b]	-0.78 (0.032)[c]/ 679	-1.02 (0.030)[c]/ 675	-1.24 (0.026)[c]/ 671	-1.35 (0.023)[c]/ 661	-1.41 (0.023)[c]/ 646
All-day regurgitation	LSM change from baseline (SE)/n[b]	-0.71 (0.031)[c]/ 735	-0.91 (0.029)[c]/ 732	-1.11 (0.026)[c]/ 728	-1.24 (0.023)[c]/ 713	-1.32 (0.022)[c]/ 693
Night-time heartburn	LSM change from baseline (SE)/n[b]	-0.8 (0.03)[c]/ 755	-1.0 (0.03)[c]/ 748	-1.2 (0.03)[c]/ 744	-1.3 (0.02)[c]/ 733	-1.4 (0.02)[c]/ 717
Night-time regurgitation	LSM change from baseline (SE)/n[b]	-0.7 (0.03)[c]/ 828	-0.9 (0.03)[c]/ 823	-1.1 (0.03)[c]/ 816	-1.2 (0.02)[c]/ 799	-1.3 (0.02)[c]/ 778

Table: Improvement in symptom scores[a] from baseline to day 14 of vonoprazan treatment (effectiveness analysis population; MMRM analysis)
LSM, least-squares mean; MMRM, mixed-effects model for repeated measures; SE, standard error.
[a]Negative values indicate improvement.
[b]Number of patients without missing values.
[c]Statistically significant difference from baseline (all P<0.001).

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PP0065

UNDERSTANDING THE DRIVERS BEHIND WHAT MATTERS WHEN TREATING GASTRO-OESOPHAGEAL REFLUX DISEASE – AN INTERIM ANALYSIS OF A DISCRETE-CHOICE EXPERIMENT

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Introduction: Proton pump inhibitors (PPIs) are extensively prescribed due to the widespread diagnosis of gastro-oesophageal disease (GORD). [1,2] Overutilisation of PPIs is common, and growing concerns over adverse events, inappropriate prescribing and unnecessary costs [3,4] are driving exploration into the implementation of holistic management, including deprescription strategies[1].

Aims & Methods: We aimed to quantify healthcare professionals' (HCPs) preferences on the management of GORD in the UK, and identify attributes perceived as enhancing or detracting from quality of care.

Attributes were derived from qualitative data gathered through 3 focus groups with gastroenterologists, pharmacists and general practitioners (GPs), and interviews with 4 GORD patients.

A discrete-choice experiment (DCE) was developed and administered via online survey (Jan–Apr 2025) to 38 HCPs. Respondents chose between paired hypothetical GORD management scenarios, varying by symptom severity, appointment frequency, treatment approach, information delivery method, and cost. Preferences were modelled using a multinomial conditional logit with effects coding. Marginal rates of substitution (MRS) were calculated to quantify trade-offs.

Results: 74% of respondents were pharmacists (n=28) and 26% GPs (n=10). All had managed GORD patients in the past year, and 81% expressed concern over long-term PPI use.

A holistic approach combining PPIs with lifestyle modification was strongly preferred over PPI monotherapy (β=0.4244, p<0.001), representing the most influential driver of care quality. Lifestyle modification alone was also favoured over PPI monotherapy (β=0.1125), though this did not reach statistical significance (p=0.127).

Respondents expressed significant aversion to providing information via pamphlets compared to GP-led counselling (β=–0.1836, p=0.046), while no preference emerged between pharmacist- and GP-led counselling (β=0.0512, p=0.573). Neither appointment burden nor out-of-pocket costs significantly influenced decision-making.

Symptom severity was a notable driver: HCPs selected scenarios involving moderate symptoms over mild (β=0.2353, p=0.005), indicating increased urgency and prioritisation for moderate disease.

Analysis of MRS between attributes suggested that HCPs are willing to accept 4.4 additional appointments to manage moderate symptoms, rather than mild, and nearly 8 additional appointments to deliver combined holistic lifestyle and pharmacological therapy. This highlights a willingness to absorb appointment-related resource demands in favour of more comprehensive, effective care.

Conclusion: Concerns around long-term PPI use support a shift toward active GORD management and PPI deprescribing, with structured follow-up to monitor treatment response.

This interim analysis highlights a clear preference among HCPs for multimodal GORD management, particularly combining lifestyle modification with pharmacological treatment, and reinforces the importance of interactive clinician[BA1] -led patient education, which can promote compliance and improve outcomes. The willingness to accept 4.4 additional appointments reflects the value placed on ongoing monitoring and personalised care. However, this raises the question on the impact this may have on health services, and whether the integration of self-care into multimodal approaches developed with HCPs could be an alternative solution to be investigated. These findings highlight the need for a system-wide shift toward guideline-informed, stepped-care models.

A further analysis on patient population preferences is under review.

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PP0066

OPTIMISED BIOPHARMACEUTICAL AND PHARMACOKINETIC PROPERTIES OF A LINAPRAZAN GLURATE TABLET FORMULATION TO BE USED IN PHASE 3

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Introduction: Linaprazan glurate (LG), a prodrug to linaprazan, is a next generation PCAB with a rapid and extensive effect on gastric pH [1] and up to 93% 4-week healing rate in severe erosive esophagitis in phase 2 study [2].

LG is formulated as tablets, initially with LG as a base. To improve the biopharmaceutical and pharmacokinetic (PK) properties, a new tablet containing LG as HCl salt has been developed. PK of LG HCl tablet was evaluated in two Phase 1 studies.

Aims & Methods: The PK of two tablet formulations were compared in a single-dose, 100 mg LG, 3-way crossover study. Additionally, the food effect on the PK of the LG HCl tablet was assessed. Repeated-dose PK for the LG HCl tablet was evaluated in a randomized, parallel-group study during 14 days of once-daily (25 mg, 50 mg, and 75 mg QD) or twice-daily (25 mg, 50 mg, and 75 mg BID) dosing. Plasma samples were collected to measure LG and linaprazan concentrations and to derive PK parameters.

Results: Healthy subjects were enrolled, with 54 and 72 evaluable for PK analysis in each study, respectively. The exposure to LG was approximately 10 times lower than that of linaprazan following both single and repeated dosing.

The bioavailability of linaprazan from LG HCl salt tablet was approximately 2-fold higher than for LG base tablet as reflected in the geometric LS mean ratio (90% CI) for linaprazan AUC_{inf}: 2.00 (1.83 2.16); C_{max}: 2.32 (2.09-2.57) with a slightly later median t_{max} for the LG HCl tablet (3h vs 2h).

A meal reduced mean AUC_{inf} by 24%, (geometric LS mean ratio (90% CI): 0.76 (0.70-0.82) and mean C_{max} by 55%, (geometric LS mean ratio (90% CI):

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0.45 (0.41-0.50), for LG HCl tablet vs fasting intake with a later median t_{max} for linaprazan after fed conditions (6h vs 3h).

Linaprazan C_{max} and AUC increased approximately in proportion to dose, steady state was reached within 5 days, with no accumulation upon 14 days repeated QD or BID dosing of LG HCl tablet. No apparent deviation from time independent PK was seen even if exposure was slightly lower (approximately 20%) than expected for the 75 mg QD and 75 mg BID dosing groups on Day 14. Mean C_{max} and AUC were slightly lower (10-20%) for the QD vs first BID dose, and slightly lower (15-30%) for the second daily vs the first daily BID dose, most likely due to different conditions with respect to food intake (Table 1).

Dose Groups	C_{max} (nmol/L) Mean (SD) Day 1	C_{max} (nmol/L) Mean (SD) Day 14	t_{max} (h) Median (min-max) Day 1	t_{max} (h) Median (min-max) Day 14	AUC (nmol/L*h) Mean (SD) Day 1	AUC (nmol/L*h) [1)] Mean (SD) Day 14
25 mg QD	527.25 (184.929)	404.75 (174.068)	5.00 (2.00-14.00)	4.00 (1.25-12.0)	5089 (1607.1)	4640 (1657.4)
50 mg QD	1066.23 (209.973)	1002.00 (538.574)	3.00 (1.00-6.00)	3.03 (1.23-6.00)	9342 (2462.3)	8626 (3571.9)
75 mg QD	1594.42 (585.525)	1044.92 (478.102)	3.50 (1.25-12.00)	3.99 (1.25-13.97)	14 614 (4601.9)	11 697 (4053.3)
25 mg BID Dose 1	658.42 (212.101)	782.92 (263.988)	1.75 (1.25-6.00)	1.50 (1.00-8.03)	4081 (1351.9)	8898 (2247.9)
25 mg BID Dose 2	520.92 (126.378)	520.83 (144.568)	4.00 (2.00-2.47)	4.99 (1.00-6.00)	3947 (939.8)	
50 mg BID Dose 1	1380.18 (303.283)	1148.64 (415.251)	1.50 (1.25-4.00)	1.50 (1.00-4.00)	8084 (1400.5)	13852 (3174.1)
50 mg BID Dose 2	997.09 (363.660)	842.36 (270.396)	2.00 (1.25-11.92)	4.00 (1.00-8.00)	7107 (1917.8)	
75 mg BID Dose 1	1942.00 (667.280)	1740.60 (468.197)	1.75 (0.98-6.00)	1.25 (0.85-4.00)	11814 (3181.8)	21366 (4323.8)
75 mg BID Dose 2	1283.90 (435.399)	1285.50 (443.703)	4.00 (1.50-8.08)	1.78 (0.98-8.02)	9532 (2118.1)	

Table 1: Descriptive Statistics of linaprazan pharmacokinetic parameters

1. AUC Day 1: AUCinf for QD and AUC0-12h and AUC12-24h for BID, AUC Day 14: AUC0-24h for QD and BID

QD: All doses were taken in the evening in a non-fasted state and BID: 1st dose was taken fasted in the morning and the 2nd daily dose in the evening in a non-fasted state.

Conclusion: An optimized linaprazan glurate HCl tablet formulation has been developed for Phase 3, with a later t_{max} and an approximately 2-fold higher bioavailability than the initial tablet and only a minor impact of food intake on linaprazan AUC_{inf}. In addition, the linaprazan glurate HCl tablet provided dose-proportional increase in linaprazan plasma exposure as reflected by AUC and C_{max}.

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PP0067

REBOUND ACID HYPERSECRETION FOLLOWING PROTON PUMP INHIBITOR DISCONTINUATION – A RAPID REVIEW OF MECHANISMS, PREVALENCE, AND PATIENT IMPACT

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Introduction: In recent years, there has been increased focus on appropriate proton pump inhibitor (PPI) prescribing due to the growing awareness of the prevalence of long-term PPI use globally.1 Rebound acid hypersecretion (RAHS) has been described as potentially contributing to PPI overuse and treatment dependence1 . RAHS is a physiological phenomenon observed following PPI discontinuation after a minimum of 4 – 8-week use. However – despite being first described in 1996 – its clinical relevance, underlying mechanisms, prevalence, and impact on patient outcomes are still not completely understood.

Aims & Methods: The purpose of this rapid review is to assess the current evidence published on RAHS, the potential mechanisms behind the phenomenon, its prevalence and clinical significance in patients discontinuing PPI therapy.

A rapid review of peer-reviewed literature was conducted through Embase and PubMed. Systematic reviews and clinical studies were included if they examined RAHS, reported prevalence, or evaluated patient-reported symptoms post-PPI withdrawal. Key findings were extracted and summarised across three domains: physiological mechanisms, prevalence and severity, and patient impact.

Results: A total of 272 articles – published between 2005 and 2024 – were retrieved, with 7 articles meeting the inclusion criteria for final review. Five of these were clinical studies, whilst the remaining 2 were systematic literature reviews. All, except 1, of the studies included were conducted more than 10 years ago – 3 randomised controlled trials, 1 cross sectional study, and 1 study looked at pre-surgery patients without randomisation. The studies were heterogeneous in design, the study populations varied broadly between 7 and 150 participants, and the prevalence data appeared to be inconsistent.

Studies show a physiological basis for RAHS, including upregulation of gastrin and enterochromaffin-like cell hyperplasia, during PPI therapy, leading to transient acid hypersecretion upon PPI withdrawal[2,3,4,5].

Symptoms resembling dyspepsia or gastro-oesophageal reflux disease (GORD) recurrence were commonly reported in both healthy volunteers and patients[2,3,6,7].

RAHS was transient in most cases[2,3,4] but may drive long-term PPI use due to misinterpretation of symptoms as disease recurrence. However, a small number of studies reviewed in the two systematic literature review papers challenged the existence of clinically meaningful RAHS, particularly after short-term or low-dose therapy[6,8].

Conclusion: There is growing evidence that RAHS is a physiologically plausible and clinically relevant phenomenon in long-term PPI users. It is suggested that RAHS may contribute to symptom recurrence and therapy dependence, resulting in the continuation of unnecessary PPI therapy. With the increasing efforts on PPI deprescribing due to growing concerns over long-term use when there's a lack of appropriate indication, clinicians need to be aware of the potential for RAHS to act as a roadblock for patients.

However, despite discussion about RAHS ongoing for almost 30 years, there is still limited understanding about its impact on patients and the potential contribution to the challenges of PPI deprescription. Most importantly, there's limited understanding of appropriate management

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solutions to overcome these challenges. In order to improve education for both clinicians and patients, further research is required on appropriate management solutions for RAHS and to aid PPI deprescribing efforts.

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The authors are employees of Reckitt Benckiser.

PP0068

EPIGENETIC BIOMARKERS AND THE PROGRESSION OF BARRETT'S OESOPHAGUS TO OESOPHAGEAL ADENOCARCINOMA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Oesophageal adenocarcinoma (OAC) is often diagnosed at a late stage and has a poor survival rate. Barrett's oesophagus (BO) can progress to OAC in a stepwise fashion, although the annual risk of progression is low, at 0.22% per year.[1]

Currently, patients with BO undergo regular endoscopic surveillance with biopsies, with the degree of dysplasia guiding further management. This is invasive and expensive, and there is interobserver variability between pathologists in the assessment of dysplasia. There is an urgent need for new approaches to risk-stratify patients with BO. Epigenetic changes can occur early in the neoplastic progression of BO, making them attractive for use as potential risk-stratification biomarkers.

Aims & Methods: We aimed to carry out a systematic review to investigate whether epigenetic biomarkers can predict the progression of patients with BO to high-grade dysplasia (HGD) or OAC. The review was registered in PROSPERO (CRD42024509643). To be eligible for inclusion, studies must have examined patients with BO that was non-dysplastic (NDBO), had low-grade dysplasia (LGD), or was indefinite for dysplasia (IND) at baseline. Eligible studies assessed any type of epigenetic biomarker of any source, including DNA methylation, histone modification, and non-coding RNA. Non-epigenetic biomarkers were excluded. Studies must have compared epigenetic biomarker levels between patients who progressed to HGD/OAC and those who did not. Included studies must have assessed progression of NDBO/LGD/IND to HGD/OAC.

Ovid MEDLINE, EMBASE, and Web of Science Core Collection were searched from inception to 27th January 2025. The risk of bias of individual studies was assessed using the Newcastle-Ottawa Scale. A narrative synthesis of all included studies was performed. A random-effects meta-analysis was also conducted. Cochran's Q test was used to test for heterogeneity, and the I^2 statistic was used to quantify its extent. A sensitivity analysis was performed to assess the robustness of the results.

Results: 1633 abstracts were screened, 143 were taken forward to full-text assessment, and 12 studies were included in the final review. 10 assessed DNA methylation and 2 assessed microRNAs. There was substantial diversity across studies in the specific biomarkers assessed, definitions of progression, and statistical approaches used. Studies suggested that methylation levels of genes such as p16, RUNX3, HPP1, APC and OR3A4 could predict progression to HGD/OAC. Panels assessing methylation of multiple genes, and prediction models incorporating gene methylation data, also predicted progression. Additionally, certain microRNAs, such as miR-192, 194, 196a and 196b, were predictive of progression.

A meta-analysis of 4 studies assessing p16 methylation was performed, showing that p16 methylation was associated with an increased risk of progression of BO to HGD/OAC (pooled odds ratio (OR) 2.11, 95% confidence interval (CI) 1.19 – 3.76, p = 0.01). There was evidence of moderate heterogeneity (I^2 = 46.2%; Q = 5.57, p = 0.13). A sensitivity analysis was performed, excluding 1 study that had required a continuity correction to calculate the OR. This yielded a more precise estimate (pooled OR = 1.78, 95% CI = 1.36 – 2.32, p < 0.0001; I^2 = 4.5%), supporting the robustness of the association.

Conclusion: Epigenetic biomarkers, particularly p16 methylation, show promise in predicting progression of BO to HGD/OAC. However, the evidence base remains limited and heterogeneous. Given the potential clinical utility of risk prediction biomarkers in BO, this is an important area for further research.

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PP0069

DETECTION OF BILE ACIDS IN SALIVA AS NEW NON-INVASIVE DIAGNOSTIC METHOD OF BARRETT'S ESOPHAGUS

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Introduction: Bile acids are a group of steroid compounds essential for lipid digestion. However, when bile acids are refluxed into the stomach and the esophagus during the so called duodenogastroesophageal reflux, they can have a detrimental effect on the esophageal epithelium and cause pathological changes of esophageal tissue, e.g. Barrett's esophagus (BE). The levels of bile acids in saliva could therefore serve as possible biomarkers for the diagnostics of BE.

Aims & Methods: Saliva samples were obtained by simple spitting. Each saliva sample was centrifuged and divided into the mucus and the supernatant. Mucus samples contain dramatically higher bile acid concentrations than supernatant samples across all diagnostic groups, suggesting bile acids accumulate in the mucus layer. We performed quantification of 15 bile acids (unconjugated, glycine-conjugated and taurine-conjugated) in saliva by ultra-high-performance liquid chromatography coupled to triple-quadrupole tandem mass spectrometry. We have taken samples of saliva from 60 patients with gastroesophageal reflux disease (GERD), 11 patients with Barrett's esophagus and 12 healthy subjects.

Results: The ratio for tauro-conjugated bile acids, glyco-conjugated bile acids and total bile acids shows statistically significant difference between Barrett's and healthy subjects ($p < 0.018$, $p < 0.023$, $p < 0.021$), and between Barrett's and GERD patients ($p < 0.027$, $p < 0.041$, $p < 0.034$). The most significant individual biomarkers appear to be glycochenodeoxycholic acid (GCDCA) and taurochenodeoxycholic acid (TCDCA). The GCDCA and TCDCA ratio shows statistically significant difference between Barrett's and healthy subjects ($p < 0.026$, $p < 0.008$), and between Barrett's and GERD patients ($p < 0.039$, $p < 0.012$). The ratio for unconjugated bile acids showed no statistically significant differences, therefore they may be less useful as Barrett's specific markers.

Conclusion: There are statistically significant differences in mucus-to-supernatant ratios between Barrett's patients and both GERD and healthy subjects for tauro-conjugated, glyco-conjugated, and total bile acids. The TCDCA ratio shows the most statistically significant difference between Barrett's and other groups, suggesting it may be the most useful single diagnostic marker. No statistically significant differences were found in bile acid ratios between GERD and healthy subjects, suggesting that the ratio pattern is specifically altered in Barrett's esophagus. The number of subjects is still low but we are completing a larger clinical study that will provide more relevant statistical data.

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PP0070

THE QUALITY AND YIELD OF DEDICATED ENDOSCOPY FOR BARRETT'S OESOPHAGUS IS HIGHER THAN THAT IN CONVENTIONAL SERVICE: A MULTICENTRE RETROSPECTIVE COMPARATIVE STUDY

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Introduction: A decade ago, endoscopic surveillance (ES) for Barrett's oesophagus (BO) was recommended to improve the outcome of oesophageal adenocarcinoma through early recognition and prevention[1]. The yield of ES to detect dysplasia may depend on guideline adherence and quality of endoscopy[2,3,4].

Aims & Methods: The aim is to assess whether dedicated (expert) ES (DS) is more effective than conventional (non-dedicated) ES (NS) by means of Key Performance Indicators (KPI), Dysplasia Detection Rate (DDR), use of advanced imaging and complication rate.

This multicentre (5 UK hospitals) retrospective study includes real-world ES data from 1303 procedures conducted within the last 8 years. The endoscopy and histology reports and clinical notes of consecutive procedures were accessed. The data were collected in spreadsheets and then analysed with SPSS-23. Two services were reported demographically and compared with appropriate statistical tests and logistic regression to test factors associated with DDR.

Results: Data from 680 DS and 623 NS procedures were included, comprising 906 (69.5%) males and 397 (30.5%) females (M:F=2.3:1.0). The age ranged from 14-90 yrs with a mean of 63.7 (SD±12.5) yrs. The maximum length of the BO was 18 cm with a mean of 4.1 cm (SD±3.0). There were significant differences in all the KPIs (Prague classification, Barrett's island description, hiatus hernia delineation, visible lesion documentation, Paris classification, Seattle protocol adherence and inspection time) favoring the DS (P<0.001). Other comparisons: narrow band imaging (NBI, P<0.001), acetic acid chromoendoscopy (AAC, P<0.001), sedation (P=0.02) and acquisition of target biopsies (, P<0.001) were also significantly higher in DS while the complication rate was significantly lower (P=0.05). Importantly, the DDR was significantly higher in DS (58/680 (8.5%)) than in NS (16/623 (2.6%)) with a P<0.001. Regression analysis showed: increasing age, male sex, lesion description and Paris classification documentation were associated with higher DDR (Table 1), while other indicators showed no significant association.

	Odds ratio	95% CI	P
Age	1.03	1.01 – 1.05	0.002
Sex	1.86	1.16 – 2.98	0.009
Visible lesion documentation	2.23	1.03 – 4.81	0.041
Paris classification	2.19	1.00 – 4.77	0.049

Table 1. Factors associated with dysplasia detection

Conclusion: Surveillance for BO through a dedicated service with high quality endoscopy can produce a higher diagnostic yield with improved DDR and a lower complication rate. Given these benefits, large-scale pro-

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spective studies are warranted to assess the true impact on mortality, morbidity, cost benefit and patient-reported outcomes, ensuring that surveillance service provides a meaningful clinical benefit, particularly with risk stratification concerning the factors associated with disease progression.

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PP0071

COST-EFFECTIVE SCREENING STRATEGIES FOR BARRETT'S ESOPHAGUS

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Introduction: Barrett's esophagus (BE) is a precursor to esophageal adenocarcinoma, with its prevalence reported to range from 1.9% to 6.8% in Western countries. We aimed to identify cost-effective pathways for BE screening by combining BE detection using multiple tools with subsequent risk-stratified surveillance.

Aims & Methods: A decision-analytic model was developed using a hypothetical cohort of the general population aged 50 years with BE prevalence rates of 1.0-7.0%. In addition to a no screening strategy, 24 potential screening pathways were evaluated combining six detection strategies (combinations of: guideline-based risk stratification, algorithmic risk stratification, swallowable devices like Cytosponge-trefoil factor 3, and endoscopy) and four surveillance strategies varying endoscopy intervals based on the BE length. Cost-effectiveness was assessed from an Australian health system perspective, using incremental cost-effectiveness ratio (ICER) thresholds for the USA (US$100,000/QALY), UK (US$43,228/QALY), and Australia (US$34,554/QALY).

Results: BE screening was cost-effective compared to no screening at BE prevalences of ≥1.0%, ≥2.4%, and ≥4.7% for the USA, UK, and Australian ICER thresholds, respectively. The optimal screening strategy was different between USA/non-USA thresholds; a 3-step detection (algorithmic risk stratification followed by swallowable devices and then endoscopy) with subsequent surveillance (no surveillance for short segment BE [SSBE] <3 cm, 2 yearly for long segment BE [LSBE] ≥3 cm; eradication for low-grade dysplasia [LGD]) for the UK and Australia. In contrast, for the USA, it was a 2-step detection (swallowable devices followed by endoscopy) with surveillance (no surveillance, SSBE ≤2 cm; 2 yearly, LSBE >2 cm; eradication, LGD) especially at ≥2.4% BE prevalence.

Conclusion: BE screening combining BE detection using multiple tools with subsequent risk-stratified surveillance based on segment length might be cost-effective, depending on BE prevalence in the community and ICER thresholds. Further work is needed to determine the actual BE prevalence in target communities.

Disclosure: Nothing to disclose.

PP0072

PERFORMANCE OF ENDOSCOPIC DEPTH DIAGNOSIS OF BARRETT'S ESOPHAGEAL ADENOCARCINOMA AND CLINICAL FEASIBILITY OF STAGING ESD

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Introduction: Endoscopic depth diagnosis is essential for the treatment decision of Barrett's esophageal adenocarcinoma (BEA). However, it is sometimes difficult to diagnose due to the anatomy of the esophagogastric junction, such as the difficulty of luminal distension. For inconclusive cT1a or cT1b lesions, we introduced staging ESD, allowing for histologic diagnosis for depth invasion. However, the clinical feasibility and significance of staging ESD remain unclear.

Aims & Methods: In this study, we evaluated the performance of endoscopic depth diagnosis of cT1 BEA and the appropriate treatment strategy. This study included cT1N0M0 BEA treated with ESD or surgery from January 2015 to June 2024. The cases with poorly differentiated component on target biopsy were excluded. Endoscopic depth diagnosis was based on macroscopic appearance (such as the following findings suggestive of cT1b: polypoid, mixed macroscopic type, thickness, hardness, severe redness, and ulceration). EUS was added at the discretion of the endoscopist. We made the treatment decision based on our multidisciplinary conference, and the final diagnosis was classified as conclusive cT1a or cT1b, and inconclusive cT1a or cT1b. We performed ESD for conclusive cT1a, and surgery for conclusive cT1b. And we performed staging ESD for inconclusive cT1a or cT1b. Curative resection was defined as pT1a with R0 resection, and without lymphovascular invasion or poorly differentiated component.

Results: This study included 54 lesions in 52 patients. The median age was 66 years (43-92). There were 42 male patients. Siewert type [type I/type II] were 12 and 42, respectively. Background Barrett's esophagus [SSBE/LSBE] were 36 and 16, respectively. The median size was 20mm (5-50). Major macroscopic appearances [0-I/0-IIa/0-IIb/0-IIc] were 11, 19, 1, and 23, respectively. Histology on targeted biopsy [tub1/tub2] were 41, 13, respectively. 39 lesions were diagnosed by white light endoscopy alone, and 15 lesions were comprehensively diagnosed by additional EUS. 41 lesions were diagnosed as cT1a and 13 lesions were diagnosed as cT1b. ESD and surgery were performed for 47 and 7 lesions, respectively. Pathologic depths [T1a/T1b/T2] were 42, 11, and 1, respectively. The sensitivity, specificity, and positive predictive value of endoscopic diagnosis of T1b were 25.0%, 76.2%, and 23.1%, respectively. 10 of 54 (18.5%) T1a lesions were overdiagnosed. Staging ESD was performed for 20 lesions, and all of them achieved en bloc resection without adverse events. R0 resection was achieved in 13 of 14 (92.9%) cT1a and 5 of 6 (83.3%) cT1b, respectively. Curative resection was achieved in 8 of 14 (57.1%) cT1a and 4 of 6 (60.0%) cT1b, respectively. 3 of 54 (5.6%) T1a were diagnosed as conclusive cT1b, and underwent surgery, leading to overtreatment.

Conclusion: The diagnostic performance of endoscopic depth diagnosis is insufficient, leading to overdiagnosis. Staging ESD for inconclusive cT1 BEA is technically feasible and can be an option to avoid overtreatment.

Disclosure: Nothing to disclose.

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PP0073

CLINICAL OUTCOMES OF RADIOFREQUENCY ABLATION FOR BARRETT'S ESOPHAGUS IN BELGIUM BEFORE AND AFTER REIMBURSEMENT: A MULTICENTER REGISTRY STUDY

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Introduction: To compare real-world outcomes of radiofrequency ablation (RFA) for Barrett's esophagus (BE) in Belgium before (Feb 2008–Jan 2017) and after (Jan 2017–Dec 2023) reimbursement, focusing on efficacy (complete remission of intestinal metaplasia, CR-IM; complete remission of dysplasia, CR-D), safety, baseline histology, and prior endoscopic resection (ER) rates, with statistical testing.

Aims & Methods: Prospective registry of expert Belgian centers.

- Pre-reimbursement: 342 treated, 295 evaluable (≥ 1 y follow-up; ITT efficacy).
- Post-reimbursement: 701 treated, 627 evaluable (≥ 1 y follow-up; ITT efficacy).

Patients in 2024–2025 with < 5 sessions and no CR were censored. Collected: demographics, histology (IM, LGD, HGIN, EAC), ER rate, Prague C/M, RFA sessions, and outcomes. Categorical variables by χ^2/Fisher's exact; continuous by Mann–Whitney U test.

Results:

Metric	Pre-reimbursment (n=342/295)	Post-reimbursment (n=701/627)	p-value
Age, yrs (median [IQR]; range)	65 [57–73]; 23–87	67 [59–73]; 23–91	0.23
Male, %	85	87	0.24
Prague C, cm (median [IQR]; range)	3 [5]; 0–15	1 [5]; 0–18	0.11
Prague M, cm (median [IQR]; range)	5 [5]; 1–17	4 [5]; 0–20	0.15
Prior ER, %	60	59	0.99
Baseline histology, %	IM 1 / LGD 7 / HGD 54 / EAC 37	IM 2 / LGD 4 / HGD 71 / EAC 23	0.32
RFA sessions (median [IQR]; range)	2 [1]; 1–8	2 [1]; 1–5	0.41
CR-IM, % (ITT)	82	83.4	0.67
CR-D, % (ITT)	87	94.9	1.00
Stricture rate, %	9	11.9	0.77
Perforation rate, %	0	0	—

Conclusion: After reimbursement, Belgian centers treated more patients with no statistically significant differences in demographics, baseline histology, BE extent, ER practice, efficacy or safety (all p>0.05). Reimbursement broadened access without compromising RFA outcomes. Continuous surveillance remains essential.

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PP0074

CLINICAL EVALUATION OF CRYOBALLOON ABLATION FOR BARRETT'S ESOPHAGUS: A UK FIRST MULTI-CENTRE EXPERIENCE

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Introduction: The Cryoballoon Focal Ablation (CbFA) (C2 Pentax Medical) system offers an endoscopic treatment option for Barrett's-related dysplasia (BE-D). CbFA has been shown in clinical studies to be a highly effective and well tolerated intervention for BE-D.

We report the largest real-world prospective multi-centre registry of consecutive patients who underwent treatment with CbFA for the endoscopic eradication (EET) of BE-D.

Aims & Methods: Patients were prospectively enrolled from three tertiary referral centres for Barrett's esophagus (BE). Inclusion criteria included patients with a BE segment of C≤2M≤5 with histologically-proven dysplasia or intramucosal adenocarcinoma on biopsy or at endoscopic resection. CbFA sessions were conducted at 3-monthly intervals until EET was complete (Max 5 sessions). Follow up biopsies were taken to confirm eradication of dysplasia and intestinal metaplasia.

The primary outcome was the complete eradication of intestinal metaplasia (CE-IM) and dysplasia (CE-D).

Secondary outcomes included the number of ablations performed, adverse events, device malfunction, stricture rate, endoscopist satisfaction score (Likert scale: 1=poor, 10=excellent) and balloon positioning complexity (Likert scale: 1=easy positioning, 5=difficult positioning).

Results: Between January 2020 and December 2024, thirty-eight patients (79%: male, mean age: 69) were treated with CbFA (81 treatment sessions). The median BE length was C0M2. 84% (n=32) were treatment naïve with the 16% (n=6) being RFA refractory BE. 55% (n=21) had undergone endoscopic resection prior to entry (3=endoscopic submucosal dissection:18=endoscopic mucosal resection). Patient characteristics according to baseline histology is highlighted in table 1.

CE-D was achieved in 100% (12/12) and CE-IM in 83% (10/12) of patients who had completed treatment. The median number of treatment sessions for those completing treatment was 2.5 (SD=1.4). No new neoplastic lesions were detected during treatment.

	Intramucosal esophageal Adenocarcinoma (n=6)	High grade dysplasia (n=11)	Low grade dysplasia (n=21)
Median Age	67	71	70
Median BE length	C1M3	C0M3	C0M2
CE-IM	2/3 (67%)	6/6 (100%)	2/3 (67%)
CE-D	3/3 (100%)	6/6 (100%)	3/3 (100%)
Stricture rate	0%	3/11 (27%)	0%

Table 1: Patient characteristics according to baseline histopathology.

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No device malfunction or intraoperative adverse events were recorded. Stricture formation post CbFA requiring dilatation was observed in 8% (n=3) of cases.

The median endoscopist satisfaction score was 9 (SD=1.8) with 3 cases recorded as difficult due to balloon positioning. The median ballon positioning complexity score was 2 (SD=1.5).

Conclusion: These results from the UK cryoablation registry suggest that CbFA is a safe and effective treatment for BE-D and mirror findings from larger controlled studies. With the planned enrolment of further centres, this registry will further enhance our understanding of the long-term efficacy and safety profile CbFA.

Disclosure: Nothing to disclose.

PP0075

ON THERAPY 24H PH-IMPEDANCE MONITORING REVEALS INAPPROPRIATE ACID SUPPRESSION IN BARRETT'S ESOPHAGUS

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Introduction: Barrett Esophagus (BE) is a long-term complication of inadequately controlled gastroesophageal reflux (GERD) resulting in the metaplastic transformation of the esophageal epithelium. Current therapy involves proton pump inhibitors (PPIs), aiming at reducing gastric, and consequently esophageal acid exposure time (AET), thereby eliminating the main pathogenic drive to dysplasia/adenocarcinoma progression. However, current guidelines recommend a standard dose PPI therapy for BE chemoprevention, without taking into consideration the need for 24-hour pH-impedance monitoring (24h MII-pH), which is the only investigation able to objectively estimate the esophageal acid exposure time (AET) and evaluate the real effectiveness of PPI therapy.

Aims & Methods: This study included prospectively enrolled BE patients referred to our tertiary care center. We identified consecutive patients with histological diagnosis of non-dysplastic BE treated with a standard dose of PPI administered once or twice daily (either pantoprazole or esomeprazole 40mg OD or 20mg BID), according to guidelines. After at least 8 weeks of PPI therapy, all participants underwent a high-resolution manometry (HRM) and 24h MII-pH monitoring on therapy. A cohort of refractory GERD patients matched for age, sex and PPI regimen (dose and type) were used as controls. GERD symptoms were recorded in the whole population by using the standardized GERD-Q questionnaire. Results are expressed as mean±SD.

Results: A total of 33 BE (23M, 50±17 years) and 33 GERD patients (21M, 48±14 years) were enrolled. Typical GERD symptoms were significantly less severe in BE patients compared to controls (2.1±2.1 vs 3.1±2.3; p<0.05). Half of the population of BE patients did not achieve adequate esophageal acid suppression (AET<4%) under standard dose regimen. Despite matching PPI regimen, the risk of having a pathological 24h MII-pH was doubled in BE compared to refractory GERD patients (50% vs 25%; p<0.05). The mean AET was significantly higher in BE compared to GERD patients (6.6±10.2%vs 1.9±3%; p<0.05), particularly in the supine position (7.3±15.5% vs 2.3±5.6%; p<0.05), despite similar levels of gastric acid suppression (44.1± 25.7% vs 49.2±27%; p=NS). Although the average number of reflux events was comparable between BE and controls (49±29 vs 40±22), BE exhibited a notably prolonged reflux clearance time (mean duration of the longest reflux episode: 27.8±25.7 vs 12.7±27.3 min; p<0.05). Furthermore, Type IIIA esophagogastric junction (EGJ) morphology was more prevalent BE group compared to controls (45% vs 21%; p<0.05) as was the prevalence of ineffective esophageal motility (IEM) according to the Chicago Classification version 4.0 (42% vs 12%; p<0.05).

Conclusion: Despite the reduced severity of typical GERD symptoms, analogous therapeutic regimens and comparable gastric acid suppression, BE patients showed a significantly increased esophageal AET, especially in the supine position, compared to refractory GERD patients. Furthermore, at HRM, BE patients showed a higher prevalence of Type IIIA EGJ morphology and IEM that could justify these results. Our data show that half of the whole BE population failed to achieve an adequate esophageal acid suppression despite therapy, emphasizing the relevance of performing an HRM and a 24h MII-pH in all non-dysplastic BE patients, in order to tailor the therapeutic strategy and improve BE long-term outcomes.

Disclosure: Nothing to disclose.

PP0076

ENHANCING SURVEILLANCE QUALITY IN BARRETT'S OESOPHAGUS: EFFECT ON DYSPLASIA DETECTION RATE

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Introduction: Oesophageal adenocarcinoma (EAC) remains a leading cause of cancer-related mortality, with poor survival outcomes. Early identification and treatment of dysplasia in Barrett's oesophagus (BE) is critical to reducing the risk of post-endoscopy upper gastrointestinal cancers (PEUGIC). While mucosal cleansing and clear visualisation are recognised as essential, existing evidence also demonstrates that certain procedural factors—such as longer inspection times—can significantly improve dysplasia detection rates (DDR) (2). This raises the possibility that other procedural variables may similarly influence detection outcomes. In this audit, we evaluated whether additional technical and patient-related factors affect dysplasia detection, with the aim of optimising surveillance quality and reducing the risk of Post-Endoscopy Upper GI Cancer (PEUGIC).

Aims & Methods: This audit aimed to assess the impact of procedural variables on dysplasia detection during Barrett's surveillance endoscopy. A retrospective analysis was conducted in Wales, reviewing 192 patient records from June to December 2023. Factors that may influence dysplasia detection were evaluated including sedation type, use of acetic acid for mucosal enhancement, adherence to the Prague grading system, describing lesions using the Paris classification, NBI use and the Seattle biopsy protocol. Additionally, patient comfort was assessed as it may impact procedural quality and compliance. Analysis was carried out using SPSS (IBM, USA). T-tests were used for parametric data, and Mann Whitney U-tests for non-parametric data. Categorical data were analysed using Chi-square or Fisher's exact tests. Statistical significance was set at $p < 0.01$

Results: Improved imaging quality was significantly associated with higher patient comfort ($p = 0.001$). Higher doses of midazolam were associated with greater detection of visible lesions, while fentanyl showed less of a significant effect. Additional use of acetic acid and aceto-whitening effect significantly enhanced dysplasia detection, with strong alignment to histological findings (p<0.001). NBI further improved lesion identification when used adjunctively. In procedures performed by expert endoscopists, targeted biopsies identified dysplasia at comparable rates to the Seattle protocol, suggesting limited additional value of the latter in experienced hands.

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Conclusion: This study underscores the importance of optimising procedural elements—particularly higher doses midazolam (median 5mg), acetic acid application, and ensuring patient comfort—to improve dysplasia detection in Barrett's surveillance. Although PEUGIC data seems to show better neoplasia detection rates with combined Midazolam and Fentanyl, we did not demonstrate this(1). seattle protocol may still be helpful for dysplasia detection, but it's use appears less beneficial than regular use of NBI and acetic acid chromoendoscopy.

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Disclosure: Nothing to disclose.

PP0077

PREVALENCE, EPIDEMIOLOGIC AND CLINICAL FEATURES OF GASTRIC INTESTINAL METAPLASIA

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Introduction: Gastric cancer ranks as the fifth most common cancer and the second leading cause of cancer-related deaths. Multifocal atrophic gastritis and gastric intestinal metaplasia (GIM) often coexist and are both associated with the risk of developing gastric cancer, with GIM being the first irreversible mucosal transformation to cancer. Endoscopic surveillance was associated with earlier cancer detection and improved survival. This study aims to evaluate the prevalence, epidemiological characteristics, and clinical features of patients with GIM.

Aims & Methods: This is a single-center, retrospective study from January 1, 2022 to December 31, 2023. Including patients aged ≥18 years who underwent esophagogastroduodenoscopy (EGD) with gastric biopsies confirming GIM, All identified cases of GIM were reviewed by gastrointestinal pathologists to confirm the diagnosis. Th primary outcome was to assess the prevalence, epidemiological characteristic, and clinical features of GIM. The secondary outcome was to evaluate adherence to recommended surveillance protocols for GIM.

Results: Among 3,663 patients who underwent EGD, GIM was identified in (4.17%, n=153). The mean age of patients with GIM was 60 years, Males represented more than half of patients (54.2%), the majority were classified as overweight (Body mass index (BMI) 25–29.9; 64%, n=97) or obese (BMI ≥30; 24.3%, n=37). Smoking was reported in (17.6%, n=27), while (5.2%, n=8) reported alcohol consumption.

Moreover, (7.2%, n=11) had a history of prior gastric surgeries. Dyspepsia (33.3%) was the most common indication for EGD. Gastritis was the predominant endoscopic finding and the primary reason for performing gastric biopsies (73%, n=112). The most common GIM subtype was complete (52%), followed by incomplete (37%) and mixed (11%). In terms of GIM location, extensive GIM was observed in (28.7%), localized GIM in (17.6%), while the location remained unknown in (53.6%) due to the lack of a follow-up EGD for mapping. Coexisting autoimmune atrophic gastritis was observed in (12.4%, n=19), Helicobacter pylori infection was detected in (16.3%, n=25).

Furthermore, Dysplasia was identified in (2.6%, n=4), with low-grade dysplasia and high-grade dysplasia each accounting for (1.3%, n=2) (Table 1). Three patients developed.gastric cancer (2%) , while (4.6%, n=7) were found to have neuroendocrine tumors (NET). First degree family history of gastric cancer was reported in (12.4%, n=19).

Gastric Intestinal Metaplasia characteristic	Percentage
Location	
Localized	27 (17.6%)
Extensive.	44 (28.7%)
Unknown	82 (53.6%)
Type of GIM	
Incomplete	51 (37%)
Complete	71 (52%)
Mixed	15 (11%)
Autoimmune Atrophic Gastritis	
Absent	134 (87.6%)
Present	19 (12.4%)
Helicobacter Pylori (H. pylori)	
Absent	128 (83.7%)
Present	25 (16.3%)
Dysplasia	
Absent	149 (97.4%)
Low Grade	2 (1.3%)
High Grade	2 (1.3%)

Table 1.

Conclusion: This study highlights a 4.17% prevalence of GIM among patients who underwent EGD. High BMI and male sex were observed in more than half of the patients. Dysplasia was detected in 2.6%, gastric cancer in 2%, and NET in 4.6% of cases. Many patients lacked follow-up EGD surveillance, limiting effective monitoring. Larger studies with extended follow-up are required to further investigate these associations and their clinical implications.

Disclosure: Nothing to disclose.

PP0078

IL-33/ST2-MEDIATED INFLAMMASOME ACTIVATION EXACERBATES ALCOHOL-INDUCED GASTRIC MUCOSAL DAMAGE

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Introduction: Alcohol-induced acute gastric injury results from disruption of the mucosal barrier and the induction of inflammatory responses. IL-33, an epithelial-derived alarmin, and its receptor ST2, are increasingly recognized as regulators of mucosal immunity and inflammation. However, the specific role of IL-33/ST2 signaling in the immunopathogenesis of ethanol-induced gastric damage remains insufficiently defined.

Aims & Methods: This study aimed to investigate the contribution of IL-33/ST2 signaling to the development of ethanol-induced acute gastric injury. Acute gastric injury was induced in WT and ST2^{-}/$^{-}$ mice by oral admin-

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istration of 80% ethanol. Gastric damage was assessed macroscopically and histologically. Immune cell profiles, cytokine levels, and markers of inflammasome activity and pyroptosis were analyzed using flow cytometry, ELISA, and qPCR.

Results: Ethanol exposure induced severe gastric injury in WT mice, characterized by extensive ulceration and epithelial disruption, whereas ST2$^{-/-}$ mice exhibited markedly reduced damage and preserved mucosal architecture. Flow cytometric analysis of gastric tissue revealed significantly higher number of Ly6G^{+} neutrophils and F4/80^{+} macrophages in ethanol-treated WT mice compared to their ST2$^{-/-}$ counterparts.

Moreover, ethanol-treated WT mice exhibited a significantly higher number of CXCR4-expressing neutrophils, accompanied by an increased number of neutrophils producing TNF-α, IFN-γ, and IL-1β. Similarly, the number of IL-1β–producing macrophages was markedly elevated in WT mice compared to ST2$^{-/-}$ mice. Notably, ethanol-treated WT mice also displayed a higher number of NLRP3^{+} neutrophils and macrophages, indicating enhanced inflammasome engagement.

Consistent with these findings, gastric tissue from ethanol-treated WT mice demonstrated increased expression of molecular markers associated with inflammasome activation and pyroptotic cell death. Systemically, WT mice exhibited elevated serum concentrations of Th1/Th17 cytokines (IFN-γ, TNF-α, IL-1β, IL-17), while IL-10 levels remained similarly increased across genotypes, suggesting a preserved anti-inflammatory feedback despite pronounced immune activation.

Conclusion: IL-33/ST2 signaling aggravates ethanol-induced gastric injury by intensifying mucosal inflammation, promoting inflammasome-driven pyroptosis, and enhancing recruitment and activation of inflammatory immune cells. The relative protection observed in ST2$^{-/-}$ mice underscores the pathological role of this axis and its potential as a therapeutic target in acute gastric damage.

Disclosure: Nothing to disclose.

PP0079

PPI CHALLENGE TO CONFIRM DIAGNOSIS OF AUTOIMMUNE BODY GASTRITIS: A PROPOSAL

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Introduction: Chronic atrophic gastritis (CAG) is a known precancerous condition for gastric cancer (GC). Low serum PG-I levels are proposed as a non-invasive marker for chronic atrophic gastritis of the stomach body, but the adequate upper cut-off for diagnosis remains controversial, as values ranging from 30 to 50 mcg/L currently are considered a "grey-zone".

Aims & Methods: Aim: To identify patients with body chronic atrophic gastritis (CAG) amongst subjects with PGI levels ranging between 30 and 50 mcg/L (within the so-called "grey zone") using a PPI test.

Methods: We selected 102 patients with PGI< 60 mcg/L. According to PG1 levels, this population was divided into 5 groups: Group A (PGI< 30 mcg/L); Group B (PGI 31-35 mcg/L); Group C (PGI 36-40 mcg/L); Group D (PGI 41-50 mcg/L.); Group E (PGI 51-60 mcg/L). All patients underwent to PPI test (Esomeprazole 40 mg daily for 1 week). By using the ROC curve, a cut-off of 30% increase from baseline PG1 was chosen. For each patient, the OLGA Staging was performed.

Results: A statistically significant relationship between PG1 levels and OLGA staging was found, being 100% in the Group of PG1<30mcg/L. PGI levels (cut-off 30% of delta increase) resulted in a positive predictive value of 95%, a negative predictive value of 86%, with an 83% and 96% sensitivity and specificity, respectively, for diagnosing atrophic gastritis of the stomach body.

Conclusion: The PPI test allows the diagnosis of 93.87% (46 out 49 subjects) of atrophic gastritis of the stomach body, with a cut-off between 41-50 mcg/L.

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Disclosure: Prof. Di Mario declares: Consultancy agreement with Biohit.

PP0080

A COMPARATIVE STUDY OF ORAL PREDNISOLONE ALONE VERSUS ORAL PREDNISOLONE COMBINED WITH NASOGASTRIC TUBE PLACEMENT FOR THE PREVENTION OF ESOPHAGEAL STRICTURE AFTER CIRCUMFERENTIAL ENDOSCOPIC SUBMUCOSAL DISSECTION FOR EARLY ESOPHAGEAL NEOPLASTIC LESIONS

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Introduction: Esophageal stricture remains a challenging complication following extensive endoscopic submucosal dissection (ESD) for superficial neoplastic lesions. Oral steroid therapy has been reported to effectively prevent post-ESD stricture. However, its efficacy in preventing luminal stenosis after total circumferential ESD has been disappointing.

Aims & Methods: This study aimed to compare the effectiveness of oral prednisolone alone versus a combination of oral prednisolone and nasogastric tube (NGT) placement in preventing esophageal stricture after circumferential ESD.

We retrospectively collected clinical data from patients with extensive superficial esophageal neoplastic lesions who underwent total circumferential ESD. Patients were categorized into two groups: the steroid group (receiving only oral prednisolone) and the steroid+NGT group (receiving oral prednisolone combined with NGT placement). The incidence of esophageal stricture was analyzed and compared between the two groups.

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Results: From June 2010 to December 2024, 29 circumferential lesions in 27 patients underwent esophageal ESD. Of these, 13 lesions in 12 patients received oral steroid therapy alone, while 16 lesions in 15 patients received a combination of oral steroids and NGT placement. There was no significant difference in the characteristics of the patients between the two groups. The overall esophageal stricture rate was 72.4% (21/29). No adverse events were observed in either group. The stricture rate was significantly higher in the steroid group compared to the steroid+NGT group (12/13, 92.3% v.s. 9/16, 56.3%; p=0.044. Table 1).

Multivariate logistic regression analysis revealed that longitudinal length of post-ESD mucosal defect (OR 1.12; 95% CI 1.01-1.25; P=0.026) was the independent risk factor. NG placement had a potential preventive effect (OR 0.05; 95% CI 0.00-1.14; P=0.06).

	Group 1 (oral prednisolone)	Group 2 (oral prednisolone and NG placement)	P value
No.	13 lesions in 12 patients	16 lesions in 15 patients	
Sex, M/F	12/0	14/1	n.s.
Age, mean ± SD, years	53.1 ± 10.1	61.6 ± 8.0	0.047
Tumor location, n (%) Upper half Lower half	 5 (38.5%) 8 (61.5%)	 4 (25%) 12 (75%)	 0.55
Longitudinal length of mucosal defect, mean ± SD, mm	62.3 ± 22.1	62.5 ± 16.9	0.98
Pathological depth of tumor invasion, n (%) Mucosa Submucosa	 11 (84.6%) 2 (15.4%)	 11 (68.8%) 5 (31.2%)	 0.41
Esophageal stricture, n (%)	12/13 (92.3%)	9/16 (56.3%)	0.044

Table 1. Baseline characteristics of the patients.

Conclusion: The combination of oral steroid and NGT placement is safe and potentially more effective in preventing esophageal stricture than oral steroid therapy alone following total circumferential ESD. Further randomized controlled trials are needed to confirm these findings.

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PP0081

TALAPORFIN SODIUM PHOTODYNAMIC THERAPY AS A SALVAGE TREATMENT FOR LOCALLY RECURRENT ESOPHAGEAL SQUAMOUS CELL CARCINOMA: EFFICACY AND SAFETY EVALUATION

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Introduction: Esophageal squamous cell carcinoma (ESCC) is a major global health concern, with a high recurrence rate after chemoradiotherapy (CRT). Despite being a curative approach, CRT alone is often insufficient, leading to local recurrence in 30–40% of cases (1). Salvage treatments such as surgery and endoscopic resection have limitations, particularly in patients with comorbidities or advanced lesions (2). Photodynamic therapy (PDT) using talaporfin sodium (TS) has emerged as a promising, minimally invasive salvage treatment, offering advantages such as reduced phototoxicity and repeatability (3). Since its approval in Japan in 2015, TS-PDT has gained traction, yet further clinical evidence is needed to validate its efficacy and safety.

This study aims to evaluate the clinical outcomes of TS-PDT as a salvage treatment for locally recurrent or residual ESCC following CRT or radiotherapy (RT).

Aims & Methods: A retrospective study was conducted at Nagoya City University Hospital between May 2016 and December 2023. A total of 29 patients with histologically confirmed local recurrence or residual ESCC post-CRT/RT were included. Eligible patients had lesions classified as ≤T2, without lymph node or distant metastasis. TS-PDT was administered with a diode laser (664 nm) 4–6 hours post-intravenous TS injection (40 mg/m^2). Patients were monitored for adverse events, local complete response (L-CR), progression-free survival (PFS), and overall survival (OS). Kaplan–Meier analysis was used for survival estimation, and comparisons between single and multiple PDT sessions were evaluated.

Results: Among the 29 patients (23 males, 6 females; median age, 72 years), TS-PDT achieved an L-CR rate of 82.1% per patient and 87.8% per lesion, with superior outcomes in T1b lesions (96.6%) compared to T2 lesions (66.7%) (Table1).

	Local-CR	Non L-CR	L-CR rate
patients	22	5	81.5%
lesions	36	5	87.8%
T1b lesions	28	1	96.6%
T2 lesions	8	4	66.7%

Table 1. The best response to PDT.

The median local PFS was 1,324 days, and the median PFS was 912 days. OS was not reached during follow-up, with a 3-year survival rate of 70.7%. Seventeen patients underwent a single PDT session, while 12 required multiple sessions. The 3-year OS rate was 83.3% in the single-session group and 44.4% in the multiple-session group. No skin phototoxicity was observed. Adverse events included esophageal strictures (1 case, asymptomatic) and esophageal perforations (2 cases), one of which required surgical intervention.

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Conclusion: TS-PDT demonstrated high local control rates and favorable survival outcomes, making it a promising salvage therapy for ESCC recurrence post-CRT/RT (4).

Despite some adverse events, its minimal invasiveness and reduced phototoxicity enhance its applicability, particularly in elderly or high-risk patients. The effectiveness of multiple PDT sessions supports its role as a flexible treatment strategy.

Further multicenter prospective trials are warranted to optimize patient selection, treatment protocols, and long-term outcomes, solidifying TS-PDT as a key salvage therapy for recurrent ESCC (5).

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Disclosure: Nothing to disclose.

PP0082

SHORT-TERM TREATMENT OUTCOMES OF SUPERFICIAL ESOPHAGEAL SQUAMOUS CELL CARCINOMA INVOLVING THE ENTIRE CIRCUMFERENCE: A MULTICENTER PROSPECTIVE OBSERVATIONAL STUDY

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Introduction: Endoscopic submucosal dissection (ESD) for superficial esophageal squamous cell carcinoma (SESCC) involving the entire circumference carries the extremely high risk of stricture and also raises concerns about curative resection. Therefore, the current Japanese guidelines weakly recommend ESD only for circumferential SESCCs diagnosed as cEP/LPM and sized ≤ 50 mm, while surgery or chemoradiotherapy (CRT) is recommended for circumferential SESCCs other than the above.[1] However, in clinical practice, ESD is implemented on a case-by-case basis through the discussion in a multi-disciplinary cancer board at each institution because the outcomes of each treatment for circumferential SESCCs remain unclear.[2]

Aims & Methods: This multicenter prospective observational study aimed to reveal the outcomes of each treatment for circumferential SESCCs. Patients with cT1a-EP/LPM or cT1a-MM/T1b-SM1 N0M0 SESCCs involving the entire circumference were enrolled in this study. At this stage, short-term outcomes, such as the proportion of curative resection after ESD, complete response (CR) after CRT/radiotherapy (RT), and adverse events including esophageal stricture, were evaluated. Curative resection in ESD was defined as pT1a without lymphovascular invasion (LVI). Esophageal

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stricture was defined as inability to pass the ordinary-sized endoscope thorough the treated site, and refractory stricture was defined as stricture requiring ≥ 6 endoscopic balloon dilation procedures.

Results: A total of 126 patients with circumferential SESCCs were enrolled from 16 Japanese tertiary institutions between May 2021 and November 2024. ESD was performed for 90 (71.4%) patients including 6 undergoing stepwise ESD, surgery for 18 (14.3%), CRT for 16 (12.7%), and RT alone for 2 (1.6%). Median longitudinal length of tumor was 55 (range: 30–140) mm in the ESD group, 80 (50–150) mm in the surgery group, and 70 (30–100) mm in the CRT/RT group. Tumor depth was diagnosed as cEP/LPM in 65.4% (55/84) of the ESD group, 38.9% (7/18) of the surgery group, and 50.0% (9/18) of the CRT/RT group, and the others were diagnosed as cMM/SM1. As stricture prevention method after ESD, local injection or filling of triamcinolone (TA) alone was used in 9 patients (10.7%), and both local injection of TA and oral steroid of prednisolone were used in 75 (89.3%). Among the 63 patients who did not receive additional treatment after ESD, the proportion of stricture and refractory stricture were 46.0% (29/63) and 27.0% (17/63), respectively. Curative resection after ESD was achieved in 58 patients (69.0%), including 46 (83.6%) with cEP/LPM and 12 (41.4%) with cMM/SM1. Of 26 patients with non-curative resection after ESD, 20 (76.9%) received additional treatment of either surgery, CRT, or RT, while the remaining 6 opted for follow-up. Of 18 patients in the surgery group, 7 with cEP/LPM and 4 with cMM/SM1 were diagnosed as pT1aN0 without LVI, while the remaining 7 with cMM/SM1 were diagnosed as pT1b, 3 of whom had lymph node metastasis. Anastomotic stricture was observed in 11.1% (2/18) of patients treated with surgery. Of 18 patients in the CRT/RT group, CR was confirmed in all patients without forming stricture.

Conclusion: Considering the high curative resection rate, ESD could be indicated for circumferential SESCCs diagnosed as cEP/LPM. In contrast, the careful consideration may be required in indicating ESD for circumferential SESCCs diagnosed as cMM/SM1 due to its low curative resection rate. Long-term outcome needs to be confirmed for further treatment comparisons.

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Disclosure: Nothing to disclose.

PP0083

A RISK FACTOR OF MULTIPLE ESOPHAGEAL SQUAMOUS CELL CARCINOMAS FOR PATIENTS WITH NUMEROUS IODINE-UNSTAINED AREAS IN SURVEILLANCE ENDOSCOPY

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Introduction: Iodine staining is widely recognized as an effective method for detecting esophageal squamous cell carcinoma (ESCC) so that it highlights dysplasia or ESCC as iodine-unstained areas. Number of iodine-unstained areas on the esophageal mucosa is considered to correlate with the risk of ESCC, and patients with numerous iodine-unstained areas have particularly high risk of ESCC. According to the Japan Esophageal Cohort Study, Grade C is defined as patients with ten or more iodine-unstained areas in the background of esophageal mucosa, and it is generally associated with a high risk of metachronous or multiple ESCCs. The guidelines of the Japan Gastroenterological Endoscopy Society recommend endoscopic surveillance every 6 to 12 months after endoscopic resection (ER) for ESCC, especially, Grade C need to be paid attention for the high risk of ESCC. However, in clinical practice, some patients with Grade C do not have metachronous or multiple ESCCs even though they are long-term followed. Then, while it is likely that the risk of patients with Grade C is not uniform, that remains unclear.

Aims & Methods: The aim of this study was to identify risk factors for multiple ESCCs for patients with Grade C in the background of esophageal mucosa.

We retrospectively analyzed patients with Grade C esophageal squamous cell carcinoma (ESCC) who underwent initial endoscopic resection (ER) at our institution between January 2014 and December 2023, and followed up for more than six months. Patients were classified into three groups according to the number of iodine-unstained areas in the background esophageal mucosa: Group α (10–20 areas), Group β (21–30 areas), and Group γ (≥31 areas). Additionally, following clinical variables were collected: age, sex, smoking history (pack-years), alcohol consumption, use of antithrombotic agents, history of other malignancies, and the presence of background mucosal flushing. We calculated the five-year cumulative incidence of multiple ESCCs in each group and performed multivariate analysis to identify independent risk factors of multiple ESCCs.

Results: A total of 253 patients were included in the analysis. There were 224 male (88.5%), and median age was 67 years old. Multiple ESCCs were observed in 123 patients (48.6%). There were 57 patients (22.5%) in Group α, 25 (9.8%) in Group β, and 171 (67.6%) in Group γ. The five-year cumulative incidence of multiple ESCCs was 54.7% as overall (Group α: 30.0%, Group β: 24.9%, and Group γ: 65.3%). The five-year cumulative incidence in Group γ was significantly higher than that in Groups α and β ($p < 0.01$). In multivariate analysis, Group γ was an independent risk factor of multiple ESCCs (hazard ratio: 2.67; 95% confidence interval: 1.46–4.87; $p < 0.01$).

Conclusion: As current guidelines recommend surveillance endoscopy every 6 to 12 months following ER for ESCC in patients with Grade C, subclassification of Grade C could serve as a useful tool in determining appropriate surveillance intervals. Patients with Group γ would undergo more intensive surveillance endoscopy.

Meanwhile, the surveillance interval for patients with Groups α and β might be extended. Subclassification of Grade C patients based on the number of iodine-unstained areas may better predict the risk of developing multiple ESCCs.

Disclosure: Nothing to disclose.

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PP0084

GREEN TEA POLYPHENOLS AMELIORATE ESOPHAGITIS RATHER THAN ASPIRIN IN A THREE-ARM PROSPECTIVE COHORT STUDY

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Introduction: Green tea polyphenols (GTPs) and aspirin were demonstrated to be novel candidates for cancer chemoprevention.

Aims & Methods: This study aimed to investigate the effects of GTPs and aspirin on esophageal diseases. This prospective cohort study recruited participants who self-reported long-term GTPs intake, aspirin intake and no medicine use in the questionnaire and underwent gastroscopy with biopsies if needed. The outcomes were the regression rates of esophageal precursor lesions after a one-year follow-up and side effects of GTPs and aspirin intake.

Results: A total of 159, 151,160 participants self-reported long-term GTPs, aspirin intake, and no medication at baseline. After one year, 78, 81, and 83 participants received the follow-up endoscopy in the GTPs, aspirin, and control groups, respectively. The average age of participants was 50.11 ± 5.62 years old, consisting of 42 (17%) female and 200 (83%) male. The regression rates of esophageal precursor diseases were 67.90%, 55.13%, and 48.19% in the GTPs, aspirin, and control groups, respectively (P = 0.036). Analyzing the esophagitis subgroup, the regression rate of GTPs group (43/69, 62.32%) was significantly higher than aspirin group (32/67, 47.76%) and control group (27/70, 38.57%) (P = 0.019). The incidences of nausea were 0.63% in GTPs group (1/159) and 0.66% of participants in aspirin group (1/151) suffered gastrointestinal bleeding, respectively.

Conclusion: GTPs, rather than aspirin, could improve the outcomes of esophagitis compared with the control group. The protective effects of GTPs on esophageal precursor diseases make it a promising health supplement.

Disclosure: Nothing to disclose.

PP0085

CLINICAL SIGNIFICANCE OF A NEW INDICATOR FOR ENDOSCOPIC EVALUATION AFTER NEOADJUVANT CHEMOTHERAPY IN ADVANCED ESOPHAGEAL SQUAMOUS CELL CANCER

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Introduction: In Japan, the standard treatment for resectable locally advanced esophageal squamous cell cancer (ESCC) is neoadjuvant triplet chemotherapy (fluorouracil, cisplatin, and docetaxel [DCF]) followed by surgery, based on the results of the phase III randomized controlled study (JCOG1109). Some reports suggest that preoperative endoscopic evaluation of the primary tumor may predict postoperative outcomes. Given the increasing importance of endoscopic evaluation of response to neoadjuvant therapy in primary tumor, the 12th edition of the Japanese Classification of Esophageal Cancer introduced a new endoscopic sub-classification “RR (Remarkable Response)”, which was defined as showing remarkably shrink to nearly T1 depth after treatment. However, its clinical significance remains unclear.

Aims & Methods: The aim of this study is to determine the clinical significance of a new endoscopic sub-classification “RR” by evaluating its association with both histological response in surgical specimens and postoperative prognosis. We retrospectively analyzed the patients with resectable locally advanced ESCC (cT2-3/N0-3/M0-1, UICC-TNM 8th edition) who were performed with ≥1 cycle of neoadjuvant DCF therapy followed by surgery at our institution between April 2015 and March 2020. Exclusion criteria were:

1) JCOG1109 participants,
2) not evaluated the primary lesion by endoscopy before or after DCF therapy due to esophageal stricture,
3) >56 days from DCF therapy completion to surgery, or;
4) pathologically non-curative resection (R1-2).

Two endoscopists retrospectively evaluated pre- and post-DCF therapy endoscopic images with masked prognostic information and classified the cases into three groups:

1) RR,
2) non-RR and;
3) PD (Progressive disease).

Histological response data were retrospectively collected from medical records. According to the 12th edition of the Japanese Classification of Esophageal Cancer, all surgically resected specimens containing visible primary tumor tissue underwent histological examination in cases where preoperative treatment has been administered.

Histological response was graded according to the Japanese Classification of Esophageal Cancer, on a four-point scale ranging from grade 0 (ineffective) to grade 1 (slightly effective), grade 2 (moderately effective), and grade 3 (markedly effective).

We evaluated the proportions of cases achieving histological response of grade 2-3 in each group, and also calculated 3-year RFS and OS rates in each group.

Results: Of 114 eligible patients (median age 66.5 years [range 42-76], 91 males), clinical stage distribution was IIB/IIIA/IIIB/IVA/IVB: 1/0/58/15/40. 97/0/17 patients received 3/2/1 cycles of neoadjuvant DCF therapy. By endoscopic evaluation, 63 patients were classified as RR, 51 as non-RR, and 0 as PD. Among RR cases, 45 patients (71.4%) showed histological response of grade 2-3, while among non-RR cases, 5 patients (9.8%) showed grade 2-3. During the median follow-up of 42.6 months, 3-year RFS rates for RR/non-RR groups were 62.2%/49.0% (P=0.098), and 3-year OS rates were 76.9%/53.0% (P=0.026).

Conclusion: The newly introduced “RR (Remarkable Response)” of sub-classification for endoscopic evaluation after neoadjuvant chemotherapy may be useful for predicting histological response and stratifying prognosis in locally advanced ESCC.

Disclosure: Nothing to disclose.

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PP0086

CLINICAL COURSE OF SUPERFICIAL LARYNGOPHARYNGEAL CANCER AFTER NEOADJUVANT DCF CHEMOTHERAPY FOR ADVANCED ESOPHAGEAL CANCER

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Introduction: It is well known that head and neck squamous cell carcinoma (HNSCC) often occur in patients with esophageal squamous cel carcinoma (ESCC) synchronously or metachronously. As a result of improvements in endoscopic system and observation techniques, the number of patients with advanced ESCC who are detected superficial HNSCC before treatment is increasing. However, subsequence clinical course of superficial HNSCC in patients who were treated with systemic neoadjuvant triplet chemotherapy for advanced ESCC has been unknown.

Aims & Methods: The aim of this study is to clarify the influence of neoadjuvant docetaxel, cisplatin, and 5-fluorouracil (DCF) therapy for superficial HNSCCs and clinical course of these lesions. The patients with advanced ESCC and superficial HNSCC, which were diagnosed by biopsy, who were treated with neoadjuvant DCF therapy for esophagectomy at our institution between January 2015 and December 2022 were retrospectively included. We excluded the patients who had a history of surgery or radiotherapy for HNSCC, chemotherapy within 5 years for any malignancy, or no endoscopic observation after the start of DCF therapy. At our institution, the endoscopic assessments were routinely performed for ESCC and HNSCC after the completion of each course of DCF therapy and reported the morphological changes in HNSCC lesions. In this study, endoscopic complete response (eCR) was defined as the disappearance of visible tumors in endoscopic observation and did not require histological confirmation by biopsy. The proportion of eCR in HNSCCs after DCF therapy, and 2-year cumulative recurrence rate of HNSCCs after eCR for the patients who did not receive chemotherapy or radiotherapy after surgery were evaluated. In addition, the difference in clinicopathological features between the patients with or without achievement of eCR were evaluated.

Results: Finally, 45 patients (56 lesions) were analyzed. Thirty-nine lesions (69.6%) were in the hypopharynx, 7 (12.5%) in the larynx, and 10 (17.9%) in the oropharynx. The median lesion size was 20 mm (range: 3-40 mm). Of all lesions, 45 lesions (80.4%) achieved eCR by DCF administration. There were no significant differences in tumor location, lesion size, and the number of administrated DCF course between the patients with or without achievement of eCR. On the histological evaluation using the biopsy samples, the lesions without subepithelial invasion were more likely to achieve eCR than those with invasion [88.9% vs 63.2%, P=0.035]. Seven patients who received additional chemotherapy or radiotherapy after esophagectomy and two who did not receive endoscopic assessment after esophagectomy were excluded for long-term analysis. Then, of the 39 lesions in 32 patients with achievement of eCR, 15 lesions (51.7%) had a local recurrence during the median follow-up period of 11 months (range 1.9-87 months), and the 2-year cumulative recurrence rate was 57.6%. Of these local recurrences, 12 lesions were performed with endoscopic laryngopharyngeal surgery, two with radiotherapy, and the other was not treated because of progression of ESCC.

Conclusion: Superficial HNSCCs mostly achieved eCR after neoadjuvant DCF therapy for locally advanced ESCC, however, more than half of them had a local recurrence thereafter. Therefore, the appropriate endoscopic follow-up after esophagectomy, early detection of local HNSCC recurrence, and therapeutic intervention were necessary.

Disclosure: Nothing to disclose.

PP0087

EVALUATION OF THE USEFULNESS AND PROGNOSIS OF ENDOSCOPIC RESECTION FOR HYPOPHARYNGEAL CANCER

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Introduction: Twenty-five years have passed since the development of endoscopic submucosal dissection (ESD), and it has become widely used as the first-line treatment for early-stage esophageal and gastric cancers with low risk of lymph node metastasis. However, due to the anatomical complexity and technical difficulty, the application of ESD for hypopharyngeal cancer remains limited.

Aims & Methods: A total of 151 lesions in 106 patients with hypopharyngeal cancer who underwent ESD at the Department of Endoscopy, Saku Central Hospital Advanced Care Center from June 2006 to March 2025 were retrospectively reviewed to evaluate treatment outcomes. Cases of recurrence after chemoradiotherapy were excluded. ESD Procedure
All procedures were performed under general anesthesia with intubation. A hook knife J was used to perform circumferential incisions with the Autocut mode of the VIO 3000 or VIO 3. Submucosal dissection was carried out using Endocut I and spray coagulation. For traction and improved visualization, Clip line technique was used.

Results: The study included 116 patients (109 men, 7 women), with a mean age at diagnosis of 67.7 years (range: 40–92). Synchronous multiple pharyngeal cancers were found in 25 patients and metachronous in 13. A total of 156 lesions were resected: 10 by endoscopic mucosal resection (EMR) and 146 by ESD.
Tumor locations were as follows: oropharynx (7 lesions), hypopharynx (149 lesions), primarily in the piriform sinus (135) and posterior cricoid area (14). Of these, 6 extended to the epiglottis and 5 to the aryepiglottic fold.
Detailed Results:

1. Tumor size: mean maximum diameter was 14.5 mm (range: 1–51; resection specimen size: 27.4 mm (5–90). Depth of invasion: EP in 92 lesions (59%) and SEP in 64 (41%). Mean invasion depth from the surface was 712 µm (range: 150–2600). Macroscopic types: 0-I (7), 0-IIa (53), 0-IIb (92), 0-IIc (4). Histology: squamous cell carcinoma (SCC) in 155 lesions, neurofibroma in 1. En bloc resection rate was 100%. Lateral margins: positive in 6, negative in 150. Deep margins: negative in all cases.
2. Adverse events: delayed bleeding (1), aspiration pneumonia (2), and stricture (1).
3. Local recurrence: observed in one SEP case with lymphatic involvement.
4. Pre-ESD metachronous cancers: esophagus: 42, stomach:17, colon:1.
5. Post-ESD metachronous cancers: esophagus:22, hypopharynx:13 and stomach:10.

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6. Prognosis: Mean follow-up was 84.9 months (range: 5–222). There were 5 deaths from primary disease, 10 from other diseases, and 8 from other malignancies. All 5 primary disease-related deaths had ly1 and died due to lymph node, pulmonary, or hepatic metastases.
7. Among 11 cases with extension to the aryepiglottic fold or epiglottis, tracheostomy was performed in 8 and nasotracheal intubation in 2 cases.

Conclusion: ESD for hypopharyngeal cancer demonstrated a high rate of local control and was an effective treatment option. Even in cases with extension to the aryepiglottic fold or epiglottis, safe procedures were possible with tracheostomy. For cases with positive lympho-vascular involvement, the risk of lymph node and distant metastases warrants consideration of additional treatment. Metachronous multiple cancers in the hypopharynx, esophagus, and stomach occurred in 40% of patients, emphasizing the importance of postoperative surveillance.

Disclosure: Nothing to disclose.

PP0088

"EFFICACY AND SAFETY OF CAMRELIZUMAB IN ESOPHAGEAL SQUAMOUS CELL CARCINOMA: A SYSTEMATIC REVIEW AND META-ANALYSIS"

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Introduction: Esophageal Squamous Cell Carcinoma (ESCC) remains a significant global health problem with few therapeutic options and poor prognosis. Camrelizumab, an anti-PD-1 immune checkpoint inhibitor, has been shown in clinical trials to be of potential value in the treatment of advanced or metastatic ESCC as monotherapy or in combination with chemotherapy or anti-angiogenic therapy.

Aims & Methods: The aim of this study is to evaluate the clinical efficacy and safety profile of Camrelizumab in patients with esophageal squamous cell carcinoma, with the main outcomes being progression-free survival (PFS), objective response rate (ORR), and treatment-related adverse events.

We conducted a comprehensive search of PubMed, Scopus, Web of Science, Embase, and Cochrane from their inception to April 2025, focusing on randomized controlled trials (RCTs) and observational studies evaluating Camrelizumab for the treatment of esophageal squamous cell carcinoma. Meta-analyses utilized a random-effects model to calculate odds ratios, mean differences, and 95% confidence intervals for adverse events and total patients.

Results: Out of 1,563 studies initially screened, 15 studies met the inclusion criteria for our meta-analysis, comprising adult patients. Meta-analysis established a median progression-free survival (PFS) of 7.60 months (95% CI: 6.00–9.19). The major pathological response rate (MPR) was 0.42 (95% CI: 0.37–0.48), and the objective response rate (ORR) was 0.53 (95% CI: 0.38–0.68). The R0 resection rate was also remarkably high at 0.98 (95% CI: 0.96–1), and the disease control rate (DCR) was 0.87 (95% CI: 0.64–1). The pathological complete response rate (pCR) was 0.31 (95% CI: 0.20–0.43).

Regarding safety, the most common treatment-related adverse events were leukopenia 0.45 (95% CI: 0.23–0.68), neutropenia 0.30 (95% CI: 0.19–0.41), anemia 0.14 (95% CI: 0.03–0.29), thrombocytopenia 0.09 (95% CI: 0.03–0.17), and elevated transaminases 0.06 (95% CI: 0.02–0.12). Pneumonia 0.17 (95% CI: 0.11-0.23) and hypertension 0.12 (95% CI: 0.06-0.19) were the other immune-related adverse events. The majority of these events were tolerable and within safe parameters.

Conclusion: Camrelizumab demonstrates promising efficacy in the treatment of esophageal squamous cell carcinoma with satisfactory rates of ORR, MPR, and R0 resection. The safety profile is acceptable with controllable adverse effects.

These findings support the potential therapeutic value of camrelizumab as a significant treatment strategy in multimodal treatment of esophageal squamous cell carcinoma and the need for further research to optimize its clinical use.

Disclosure: Nothing to disclose.

PP0089

DEVELOPMENT AND VALIDATION OF A SCORE-BASED MODEL FOR ESTIMATING ESOPHAGEAL SQUAMOUS CELL CARCINOMA AND PRECANCEROUS LESIONS RISK IN OPPORTUNISTIC SCREENING POPULATION

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Introduction: Opportunistic screening is one major screening approach for esophageal squamous cell carcinoma (ESCC). Risk stratification models could be used to assess each individual's risk using relevant predictors so that only those at high risk are recommended for further endoscopy [1-2]. This approach would allow the development of a cost-effective opportunistic screening strategy and promise to improve the sensitivity of endoscopic detection by alerting the endoscopists.

Aims & Methods: We aimed to develop a score-based clinical prediction model to assess the risk of ESCC and precancerous lesions in opportunistic screening and to validate it in an external population. The study was a secondary analysis of a published esophageal cancer screening trial [3].

The trial was conducted in 39 secondary or tertiary hospitals in China, with 14,597 individuals enrolled for opportunistic screening.

Additionally, questionnaires and endoscopy were performed. The primary outcome was histology-confirmed high-grade esophageal lesions, including high-grade intraepithelial neoplasia (HGIN) and ESCC. The predictors were selected using univariable and multivariable logistic regression. Model performance was primarily measured with the area under the receiver operating characteristic curve (AUROC).

Results: The score-based prediction model contained 8 variables on a 21-point scale. The model demonstrated an AUROC of 0.833 (95% CI, 0.803-0.862) and 0.828 (95% CI, 0.793-0.864) for detecting high-grade lesions in the development and validation cohorts, respectively.

	Development cohort (n=7899)	Validation cohort (n=6698)
Sensitivity (%, 95%CI)		
High-grade lesions	84.3(77.6-89.7)	77.6(68.0-85.4)
HGIN	82.9(67.9-92.9)	70.0(50.6-85.3)
Early ESCC	81.5(61.9-93.7)	81.3(63.6-92.8)
Advanced ESCC	86.1(76.9-92.6)	81.1(64.9-92.0)
Specificity (%, 95%CI)	68.3(67.3-69.4)	76.4(75.4-77.4)
Accuracy rate (%, 95%CI)	68.6(67.6-69.7)	76.4(75.4-77.5)
NSS	20	21

Abbreviations: HGIN, high-grade intraepithelial neoplasia; ESCC, esophageal squamous cell carcinoma; NNS, number needed to screen to detect one case of HGIN/ESCC; CI, confidence interval. High-grade lesions include HGIN and ESCC.

Table 1. The predictive performance for the score-based model at the score cut-off value of 9.

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As shown in Table 1, using the score cut-off value determined in the development cohort (≥9), the sensitivity reached 70.0% (95% CI, 50.6%-85.3%), 81.3% (95% CI, 63.6%-92.8%), and 81.1% (95% CI, 64.9%-92.0%) in the validation cohort for detecting HGIN, early ESCC, and advanced ESCC, respectively, at a specificity of 76.4% (95%CI,75.4%-77.4%). The score-based model exhibited satisfactory calibration in the calibration plots. The score-based model could result in 75.6% fewer individuals subjected to endoscopies.

Conclusion: This score-based model demonstrated superior discrimination for high-grade esophageal lesions. It has the potential to inform referral decisions in the opportunistic screening setting.

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Disclosure: Nothing to disclose.

PP0090

SMOKING STATUS AND THE EFFICACY OF PD-1/PD-L1 INHIBITORS IN ESOPHAGEAL SQUAMOUS CELL CARCINOMA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Esophageal squamous cell carcinoma (ESCC) is a major cause of cancer-related mortality with poor survival rates. PD-1/PD-L1 inhibitors have shown superior efficacy over chemotherapy in advanced ESCC. Smocking is a major risk factor of ESCC. We aim in this paper to evaluated the effects of smocking status on the efficacy of PD-1 inhibitors in advanced ESCC.

Aims & Methods: An electronic search was conducted on February 1, 2025, using PubMed, Embase, Web of Science, the Cochrane Library, and Scopus. We included only randomized controlled trials (RCTs) that compared PD-1/PD-L1 inhibitors to chemotherapy for advanced ESCC. The primary outcome was overall survival stratified by smoking status (never, former, or current smokers). Hazard ratios (HRs) with 95% confidence intervals (CIs) were used to estimate effect sizes. A random-effects model was applied.

Results: Of the 7,094 articles screened, 7 trials were included, with a total of 3562 patients. In this subgroup meta-analysis, PD-1/PD-L1 inhibitors demonstrated a significant overall survival benefit in both subgroups "never smoked" (HR 0.71, 95% CI 0.60–0.84) and "former smokers" (HR 0.78, 95% CI 0.66–0.91), with low heterogeneity (I^2 = 0% for both groups). In contrast, current smoker subgroup did not show a significant overall survival benefit (HR 0.58, 95% CI 0.31–1.09), with moderate heterogeneity (I^2 = 55.9%), suggesting that active smoking may reduce the efficacy of PD-1/PD-L1 inhibitors in ESCC.

Conclusion: This subgroup meta-analysis suggests that PD-1/PD-L1 efficacy may be influenced by smoking status. Current smokers did not show a survival benefit compared to former or never smokers. Further research is needed to explore the underlying mechanisms to optimize treatment strategies for active smokers.

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Disclosure: Nothing to disclose.

PP0091

FROM BURN TO TREATMENT: REVISITING THE NATURAL HISTORY OF PEPTIC ULCER DISEASE - A TERTIARY CENTER EXPERIENCE

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Introduction: Peptic ulcer disease (PUD) is the most frequent cause of non-variceal upper gastrointestinal bleeding (NVUGIB). Over the decades, understanding the natural history of PUD and development of new treatment strategies has significantly improved clinical outcomes. In this study we explore the natural history of PUD, including risk factors, therapeutic approaches and mortality.

Aims & Methods: We conducted a retrospective cohort study including consecutive patients with PUD diagnosed by upper endoscopy between January-2020/June-2022. Data regarding risk factors, clinical presentation, endoscopic findings, therapeutic approaches, rebleeding and mortality rate were collected.

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Results: We identified 468 patients (male sex-63.9%; mean age-71.8±15.6 years; median Charlson Comorbidity Index (CCI) - 5.4;range 0-18). Major risk factors: alcohol consumption - 31.6%; NSAIDs - 28.4%; Helicobacter pylori infection - 23.1%. Antiplatelet or anticoagulant drugs were part of the usual medication in 47.2% of the cases and were associated with major hemodynamic instability upon admission (χ^2 = 4.805, df=1, p=0.028), a situation that has been shown to significantly increase the risk of death (OR 3.10, IC95%: 1.745-5.509). PUD clinical presentation: melena and/or hematemesis - 65.4%; isolated anemia - 17.9%. Mean hemoglobin level at admission - 8.5±2.5 g/dl. Lower levels were associated with a higher 7-day rebleeding rate (Z=-2.383, p<0.05). Small Forrest III ulcers of gastric antrum and duodenal bulb were the most common and PPI perfusion was administered in 29.1% of the cases. Overall, 37% of patients benefited from treatment: endoscopic - 96.6%; transarterial embolization - 9.1%; surgery - 1.7%. Hospitalization was needed in 67.2% cases, with mean duration of 13.4±13.7 days. Complications occurred in 12.6% and there was need for further intervention in 7.9% of the cases. 30-day mortality was 12% (related to infections - 78%, stage IV neoplasia - 12%). Differently from age, the CCI demonstrated to be associated with 30-day mortality (Z=-4.230, p<0.001).

Conclusion: PUD is a common clinical condition, with known modifiable risk factors and diverse clinical presentation. Despite improvements in endoscopic techniques, mortality remains significant and CCI is a relevant prognostic factor.

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Disclosure: Nothing to disclose.

PP0092

THE PROTECTIVE EFFECT AND MECHANISM OF ESCIN ON ETHANOL-INDUCED GASTRIC MUCOSAL INJURY IN RATS

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Introduction: Ethanol-induced gastric ulcers are common and the treatment options are limited. Escin, a extract from the horse chestnut plant, has been proven to have therapeutic effects in improving inflammation, edema, and vascular permeability. However, its effects and mechanisms on ethanol-induced gastric mucosal injury are not yet clear.

Aims & Methods: This study aims to explore the protective effect of Escin on ethanol-induced gastric mucosal injury and its potential mechanisms, providing a new drug choice for the treatment of ethanol-induced gastric ulcers. For this purpose, a control, ethanol, ethanol+Escin, and ethanol+omeprazole groups were established in male SD rats to compare and analyze the ulcer index and histopathological scores of each group. The network pharmacology approach was used to predict potential targets of Escin related to gastric ulcers. Apoptotic proteins, NF-κb inflammatory pathway proteins, inflammatory factors, and related mRNA expression were analyzed by methods such as Immunofluorescence (IF), Immunohistochemistry (IHC), Western Blot (WB), real-time quantitative fluorescence PCR (q-PCR), and enzyme-linked immunosorbent assay (Elisa). Furthermore, an ethanol-induced GES-1 cell injury model was established to evaluate the protective effect of Escin. Finally, Transcriptomic sequencing was performed to identify key regulatory genes.

Results: The study found that Escin significantly reduced the damage of ethanol to gastric mucosal tissues. Network pharmacology analysis predicted that Escin is closely related to inflammatory pathways such as NF-κb in the treatment of gastric ulcers. IHC and WB results showed that Escin reduced the loss of gastric mucosal surface markers MUC5AC, TFF2, and gastric mucosal barrier protein O-ccludin, inhibited the expression of inflammatory factors IL-1β, IL-6, TNF-α, and apoptosis-related proteins BAX, Caspase-3, and promoted the expression of anti-apoptotic protein BCL-2. IF further revealed that Escin prevented the nuclear translocation of NF-κb, thereby inhibiting the activation of the NF-κb pathway. Elisa results showed a decrease in inflammatory factor levels in the plasma of rats in the Escin group, while q-PCR results demonstrated that Escin downregulated the expression levels of inflammatory factor mRNA in gastric tissues. Cell proliferation experiments also confirmed the protective effect of Escin against ethanol-induced damage. Finally, Transcriptomic analysis revealed significant upregulation of 27 inflammation-related genes in ethanol-treated gastric tissues. Notably, nine genes—Birc3, Bcl3, SOCS1, SOCS3, Ccl2, Cx3cl1, Pim1, SGK1, and OSMR—were identified as positive regulators of the TNF-α/NF-κB signaling axis through functional annotation and pathway enrichment analysis.

Conclusion: Escin exerts a protective effect on ethanol-induced gastric mucosal damage by inhibiting the TNF-α/NF-κB pathway, reducing the expression of inflammatory factors and the level of cell apoptosis. Escin, as a novel therapeutic drug for ethanol-induced gastric ulcers, has certain clinical translational application value, but further clinical trials are needed to confirm the results.

Disclosure: Nothing to disclose.

PP0093

MACHINE LEARNING-BASED PREDICTION OF 30-DAY MORTALITY IN PEPTIC ULCER BLEEDING: COMPARISON WITH TRADITIONAL SCORING SYSTEMS

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Introduction: Peptic ulcer bleeding (PUB) is a major cause of gastrointestinal-related morbidity and mortality. Various clinical scoring systems such as AIMS65, Glasgow-Blatchford Score (GBS), and ABC score have been used for mortality prediction. However, their predictive performance varies, and improved tools may enhance clinical decision-making.
This study aimed to compare the predictive accuracy of traditional scoring systems with machine learning models for 30-day mortality in PUB patients.

Aims & Methods: We retrospectively analyzed data from patients with endoscopically confirmed PUB. Independent variables included clinical parameters, laboratory results, and endoscopic findings. Five conventional scoring systems (GBS, AIMS65, ABC score, Pre- and Post-Rockall scores) were evaluated. A machine learning model based on logistic regression was developed using relevant clinical features. Predictive performance

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was assessed using area under the receiver operating characteristic curve (AUROC), sensitivity, and specificity. Optimal cut-off values for each score were determined using Youden's Index.

Results: Among conventional scores, the ABC score achieved the highest AUROC (0.892), followed by AIMS65 (0.864). The machine learning model showed superior performance with an AUROC of 0.989, sensitivity of 66.7%, and specificity of 96.7%. Compared to traditional scoring systems, the machine learning model achieved comparable or higher sensitivity and significantly improved specificity.

Conclusion: Machine learning, particularly logistic regression, provided more accurate prediction of 30-day mortality in PUB compared to conventional scoring systems. These findings support the integration of machine learning models into risk assessment strategies for PUB patients.

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https://gut.bmj.com/content/70/4/707.abstract

Disclosure: Nothing to disclose.

PP0094

ENDOSCOPIC SUBMUCOSAL DISSECTION FOR *HELICOBACTER PYLORI*-NEGATIVE LOCALIZED GASTRIC MUCOSA-ASSOCIATED LYMPHOID TISSUE LYMPHOMA: A CASE SERIES

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Introduction: The first-line treatment for gastric mucosa-associated lymphoid tissue (MALT) lymphoma is Helicobacter pylori (H. pylori) eradication when infection is present. In the absence of infection, treatment options include radiotherapy, chemotherapy, or surgery, depending on disease extent. [1] However, the management of H. pylori-negative gastric MALT lymphoma remains controversial. Recently, endoscopic submucosal dissection (ESD) has been reported as a potential curative treatment, but data remain limited due to the rarity of such cases.

Aims & Methods: We report our experience with ESD in three consecutive cases of localized H. pylori-negative gastric MALT lymphoma identified among 29,331 esophagogastroduodenoscopies (EGDs) conducted at our large-scale health screening facility during the 2024 health check-up program. We aim to highlight the feasibility of ESD as a less invasive alternative to radiotherapy or chemotherapy and contribute to future treatment strategies.

Results: Three cases (0.01%, 3/29,331) of localized gastric MALT lymphoma were identified: a 71-year-old female, a 43-year-old female, and a 63-year-old male. Lesions were located in the upper body, middle body, and angular region, measuring 15 mm, 20 mm, and 7 mm, respectively. Larger lesions exhibited cobblestone-like mucosal patterns with depressed granular erythematous areas, while the smallest lesion appeared as a simple depressed lesion with discoloration. Biopsies confirmed CD20 and CD79a positivity, CD3 and CD10 negativity, and lymphoepithelial lesions on AE1/AE3 staining, leading to the diagnosis of gastric MALT lymphoma. Contrast-enhanced CT showed no lymph node involvement or distant lesions, classifying all cases as localized stage (Lugano classification). Absence of gastric mucosal atrophy, negative H. pylori IgG serology, negative urea breath test (UBT), and absence of H. pylori on Giemsa staining confirmed an H. pylori-negative status. All three cases underwent ESD, achieving complete en bloc resection with negative margins. No complications, such as delayed bleeding or perforation, were observed following ESD. Histopathological examination revealed clusters of B-cell lymphocytes confined to the mucosa and superficial submucosa. Follow-up EGD with biopsy showed no recurrence in any case.

Conclusion: We performed ESD for three cases of H. pylori-negative localized gastric MALT lymphoma, achieving complete resection with negative margins in all cases without any complications. Additionally, short-term outcomes were favorable. While observation, radiotherapy, and chemotherapy are available treatment options for H. pylori-negative localized gastric MALT lymphoma, cases of tumor progression and transformation into diffuse large B-cell lymphoma (DLBCL) after chemotherapy have been reported. [2] Therefore, early resection may provide a safer approach. Our findings suggest that early resection may be a preferable approach for localized lesions. Further accumulation of cases and long-term follow-up are necessary to establish ESD as a viable treatment option.

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Disclosure: Nothing to disclose.

PP0095

FOLLOW-UP OF GASTRIC ULCERS: IS THERE REALLY A NEED FOR FOLLOW-UP IN ALL PATIENTS?

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Introduction: Gastric ulcers are associated with malignancy in 8.5% of cases, prompting recommendations for follow-up oesophagogastro-duodenoscopies (OGDs) to confirm ulcer healing and exclude malignancy. However, the incidence of malignancy at follow-up endoscopy is low, and no consensus exists on patient selection and optimal timing.
This study evaluates current follow-up practices and the diagnostic yield of follow-up OGDs.

Aims & Methods: This study evaluates current follow-up practices and the diagnostic yield of follow-up OGDs. We conducted a retrospective cohort study of consecutive patients diagnosed with gastric ulcers between 2020 and 2023. Data on risk factors, endoscopic and histological findings from index and follow-up OGDs and surveillance data were analyzed.

Results: A total of 297 patients were identified (58.2% male; mean age 73.2 ± 15.5 years; median Charlson Comorbidity Index: 5.5). The main risk factors were Helicobacter pylori infection (52.7%) and NSAID use (31.3%). Gastric neoplasia was diagnosed in 26 patients. Biopsies were obtained in 34% of index OGDs (mean: 4.3 ± 1.8 biopsies), detecting 69.2% of neoplasia cases (n=18). Follow-up OGD was performed in 46.8% of patients, primarily within eight weeks (24%), diagnosing the remaining 30.8% of neoplasia cases.
Among these, only one case did not show endoscopic suspicion of neoplasia in the index OGD. In three cases with malignant macroscopic suspicion at index OGD, initial biopsies revealed metaplasia, but follow-up OGD con-

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firmed neoplasia. Predictors of malignancy included suspicious appearance (p=0.028; AUROC=0.939), larger ulcer size (p<0.001), and non-antral location (p=0.041). A minimum of 3 to 4 biopsies appeared optimal for diagnosis, and follow-up timing did not significantly impact diagnostic yield.

Conclusion: Our analysis suggests that ulcers with a benign endoscopic appearance, small size, and non-antral location may not require follow-up OGD. These findings challenge the need for universal surveillance, that is rarely applied in clinical practice, and emphasize the importance of risk stratification for targeted follow-up in patient management.

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Disclosure: Nothing to disclose.

PP0096

LONG-TERM PROGNOSIS OF MIXED HISTOLOGICAL-TYPE GASTRIC CANCERS CURED BY ENDOSCOPIC RESECTION: A MULTICENTER PROSPECTIVE COHORT STUDY (J-WEB/EGC)

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Introduction: Gastric cancers (GCs) are histologically classified into differentiated and undifferentiated types. Mixed histological type is considered more aggressive clinical behavior compared to pure histological type. However, no reports of large-scale prospective studies regarding the long-term prognosis of mixed histological type cured in patients by endoscopic resection (ER).

Aims & Methods: We aimed to determine the long-term prognosis of mixed histological-type GCs in patients cured by ER. We used data from the J-WEB EGC study, a nationwide, multicenter, prospective cohort that enrolled 9,715 patients with 10,926 lesions treated by ER between 2010 and 2012 across 41 institutions in Japan. For this analysis, we included patients classified as eCura A or B, according to the Japanese gastric cancer treatment guidelines, represent pathological criteria for curative resection following ER. eCura A included: differentiated-type, pT1a, without ulceration (UL0), negative horizontal and vertical margins (HM0, VM0) (A-1), and no lymphovascular invasion (Ly0, V0); differentiated-type, ≤30 mm, pT1a, with ulceration (UL1), HM0, VM0, Ly0, V0 (A-2); and undifferentiated-type, ≤20 mm, pT1a, UL0, HM0, VM0, Ly0, V0 (A-3). eCura B included differentiated-type, pT1b with submucosal invasion <500 μm, HM0, VM0, Ly0, and V0. We excluded patients with multiple lesion and those with lesions containing unusual type. Each patient was followed up for 5 years. The primary endpoints were 5-year overall survival (OS) and 5-year primary disease-specific survival (pDSS) to primary GC. Patients were divided into pure and histological types. OS and pDSS were assessed using the Kaplan–Meier method, and hazard ratios (HRs) for all-cause mortality was estimated using the Cox proportional hazards regression model with death as the event. After stratification into eCuraA-1, eCuraA-2, eCuraA-3, and eCura B, the 5-year OS and pDSS were compared between the pure and mixed histological types.

Results: A total of 7,242 patients with 8,187 GC lesions were classified as eCura A or B. After excluding 646 patients with multiple GC lesions and one patient with an GC lesion containing a neuroendocrine tumor component, 6,595 patients with a single GC lesion were included in the analysis. Of these, 6,434 were pure histological type and 161 mixed histological type. There was no difference in OS between pure and mixed histological types in all patients (92.5%, 95% confidence interval [CI]: 87.4–95.7 vs. 90.5%, 95% CI: 89.7–91.2; p = 0.415, respectively). Additionally, mixed histological type was not an independent factor in adjusted HRs for OS all-cause

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mortality (0.93, 95% CI: 0.52–1.64). The 5-year pDSS of the mixed histological-type GCs was significantly lower than that of the pure histological-type GCs (99.4%, 95% CI: 96.6–99.9 vs 99.96%, 95% CI: 99.9–100, p = 0.003, respectively). Among eCura A-2 patients, pDSS of the mixed histological-type GCs was significantly lower than that of the pure histological-type GCs (94.4%, 95% CI: 74.2–99.0 vs. 100%, 95% CI: 99.3–100; p < 0.001, respectively).

Conclusion: The long-term prognosis of curative ER patients, including those with mixed histological-type GCs, was favorable. However, the stratification analysis showed that mixed histological-type GCs in eCura A with ulcers was more likely to cause primary GC death than pure histological-type and might need careful surveillance.

Disclosure: Nothing to disclose.

PP0097

UTILITY AND EFFECTIVENESS OF OUTPATIENT UNDERWATER COLD SNARE POLYPECTOMY FOR THE DIAGNOSIS OF SOLITARY POLYPOID LESIONS IN *HELICOBACTER PYLORI*-NAIVE STOMACH

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Introduction: Polypoid lesions in *Helicobacter pylori* (Hp)-naive stomachs are generally fundic gland polyps (FGPs), which have a smooth surface and typically do not require treatment. However, some lesions with different appearances, such as foveolar-type adenomas that present with a reddish color and papillary surface structure, may also be observed. These lesions can be difficult to diagnose accurately with biopsy alone. An efficient and safe diagnostic and therapeutic approach for these lesions is needed.

Aims & Methods: We devised outpatient underwater cold snare polypectomy (uCSP) as a diagnostic alternative to biopsy for reddish polypoid lesions in Hp-naive stomachs that differ from the typical appearance of FGPs. We retrospectively analyzed the clinicopathological characteristics and outcomes of cases in which outpatient uCSP was performed for solitary lesions ≤10 mm in size, after obtaining informed consent. Hp-naive was defined by the absence of endoscopic and histological atrophic changes, no history of eradication therapy, and negative results from at least two different diagnostic tests for *H. pylori* infection.

Results: Between April 2021 and March 2025, 35 cases underwent uCSP (16 males, 19 females; mean age: 55.9 years). The mean lesion size was 3.7 mm, with 21 lesions located in the upper part of stomach and 14 in the middle part of stomach. In all cases, lesions were obtained including surrounding mucosa with minimal crush artifact to the deep mucosal layer or superficial muscularis mucosae. Histopathological diagnoses included foveolar-type adenoma in 28 cases, foveolar hyperplastic polyp in 3, FGP with dysplasia in 2, oxyntic adenoma in 1, and inflammatory polyp in 1. Among the 31 neoplastic lesions, only the oxyntic adenoma case had a positive vertical margin, yet no residual or recurrent lesion was detected during 1-year follow-up. The remaining 30 cases had negative lateral and vertical margins. No postoperative bleeding or other adverse events were observed, regardless of proton pump inhibitor use or clip closure following resection.

Conclusion: Outpatient uCSP appears to be a minimally invasive, safe, and effective diagnostic and therapeutic option for solitary polypoid neoplastic lesions ≤10 mm in Hp-naive stomachs.

Disclosure: Nothing to disclose.

PP0098

THE EVALUATION OF ENDOSCOPISTS' SKILL IMPROVEMENT THROUGH D-LECS WITH ESD USING THE ESD ASSESSMENT SCORE

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Introduction: Endoscopic submucosal dissection (ESD) is one of the effective treatments for superficial non-ampullary duodenal epithelial tumors (SNADETs). To avoid postoperative perforation and bleeding, our hospital had introduced D-LECS with ESD for SNADETs. From 2021, D-LECS was divided into D-LECS with FTR, which is a full-thickness resection, and D-LECS with ESD, which is intended for post-ESD reinforcement (1).

D-LECS with ESD is considered to be a therapeutic procedure that not only ensures safety but also contributes to the improvement of endoscopic skills.

Aims & Methods: The aim of this study is to evaluate the impact of D-LECS with ESD on the skill development of endoscopists. The subjects were 17 cases in which D-LECS was performed among 79cases diagnosed with SNADA by biopsy at our hospital between January 2000 and October 2023. 5 cases in which D-LECS with ESD was performed. The ESD assessment score was defined using a ladder chart based on muscle layer exposure, perforation, ESD time, and pathological examination results.

The scoring for points each item is as follows. (Exposure of muscle layer none: 0, <3mm: 1pt, 3~10mm: 2pt, 10mm<: 3pt, Perforation of the muscle none: 0, <3mm: 1pt, 3~10mm: 5pt, 10mm<: 10pt, ESD time <60min.: 0pt, 60~120min.: 5pt, 120min.<: 10pt, Pathological examination result pR0: 0pt, cR0+pRX: 5pt, pR1:10pt) Images and videos were used in combination to evaluate the muscle layer.

Results: ESD assessment scores of cases undergoing D-LECS with ESD were presented. The state of exposure of the muscle layer had a median of 5 (1-7), a mean of 4.8, and a standard deviation (SD) of 2.3. The state of perforation of the muscle layer had a median of 0 (0-10), a mean of 2.6, and a SD of 4.3. The median ESD time was 170 minutes, the mean was 146 minutes (41-198).

The state of pathological examination had a median of 0 (0-10), a mean of 2.0, and a SD of 4.5. The ESD assessment scores are presented as follows: (Exposure of muscle layer, Perforation of muscles layer, ESD, Pathological examination result).

The results of each score are as shown on the right, Case1: (7, 0,10, 0), Case2: (5,10, 10, 0), Case3: (5, 0, 10, 0), Case4: (6, 3, 10, 10), Case5: (1, 0, 1, 0). Although the number of cases was small, the time required for ESD was shortened and the degree of perforation tended to improve with experience. 2 examples revealed some important points.

Case 1: This case had many blood vessels, and the ST hood was needed because it is difficult to perform ESD. When the endoscope was extended, perforation occurred. Once the perforation was made, it was difficult to move the patient to other bed, so the muscle layer was sutured with OTSC, and a PGA sheet.

Case 2: The endoscopic strategy and device were improved, and the lesion could be dissected within 1 hour. ESD was performed with Dual Knife J 1.5. Pulling the lesion with an S-O clip. The RDI function was used during hemostasis to avoid the burn effect on the muscle layer.

Conclusion: D-LECS with ESD will lead to improvements of duodenal ESD techniques. But to ensure safety, laparoscopic closure is preferable until reliable suturing via endoscope becomes possible.

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Disclosure: Nothing to disclose.

PP0099

STUDY ON THE APPLICATION VALUE OF CLAUDIN18.2 IN COMPREHENSIVE DIAGNOSIS AND TREATMENT OF EARLY GASTRIC CANCER

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Introduction: Gastric cancer is a highly prevalent malignant tumor worldwide, and early diagnosis is crucial for improving survival rates. Claudin18.2 (CLDN18.2), a member of the tight junction protein family, is highly expressed in advanced gastric cancer and has emerged as a therapeutic target. However, its expression characteristics and diagnostic value in early gastric cancer remain unclear.

This study aims to investigate the expression patterns of CLDN18.2 in early gastric cancer and evaluate its potential as a diagnostic biomarker.

Aims & Methods: The study enrolled 54 patients with early gastric cancer to analyze their clinicopathological characteristics (including age, gender, histological type, differentiation degree, etc.). Immunohistochemistry was employed to detect differential CLDN18.2 protein expression between the tumor group and control group (n=54). Statistical methods were applied to evaluate the correlation between CLDN18.2 expression and clinicopathological features.

Results: The early gastric cancer cohort was predominantly characterized by patients >60 years old (72.2%), male (64.8%), with adenocarcinoma (98.1%) and well-differentiated tumors (57.4%). CLDN18.2 expression analysis revealed significantly higher positivity in the tumor group compared to controls (38/54 vs. 9/54, P<0.001). A strong correlation was observed between CLDN18.2 expression and differentiation grade (100% positivity in poorly differentiated vs. 61.3% in well-differentiated tumors, P=0.046), while no significant associations were found with age, gender, or invasion depth (P>0.05).

Conclusion: CLDN18.2 demonstrates specific expression patterns in early gastric cancer, showing particularly strong association with poorly differentiated subtypes, suggesting its potential as an early diagnostic biomarker.

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PP0100

MANAGEMENT STRATEGIES AND OUTCOMES FOR TYPE 1 GASTRIC NEUROENDOCRINE TUMORS LARGER THAN 1 CM

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Introduction: Type 1 gastric neuroendocrine tumors (T1-gNETs) are well-differentiated, typically indolent neoplasms associated with chronic atrophic gastritis (CAG) and hypergastrinemia, with nearly 100% long-term survival. However, they have a high recurrence rate of 30-60% within the first year after resection. Tumor size is the most critical prognostic factor, while the role of grading is less definitive compared to other digestive NETs. International guidelines recommend conservative management for tumors <1 cm, advocating for approaches such as endoscopic resection or surveillance, which have shown excellent outcomes. However, the management strategies and outcomes for tumors > 1 cm remain less clearly defined due to the lack of scientific data in this specific patient population.

Aims & Methods: This study aimed to investigate management strategies and clinical outcomes in a homogeneous population of patients with T1-gNETs > 1 cm. The findings are expected to provide valuable insights and guide the development of more effective management strategies.

The present study was designed as an international multicentric retrospective analysis, conducted in 7 Western centers. It included consecutive patients diagnosed with T1-gNETs >1 cm in diameter, collected between 2004 and 2023, with at least 12 months of follow-up. The primary endpoints were disease progression (DP) (tumor recurrence or metastasis development) and the Progression-Free Survival. The Cox regression method was used to identify factors associated with progression.

Results: The study included 104 patients (61 females, 43 males) with a median age of 61 years and a median tumor size of 15 mm (range 11–60 mm). 61.5% of tumors were classified as G1, 37.5% as G2, and 1% as G3. 75 patients (72.1 %) underwent endoscopic resection (27 EMR, 5 m-EMR, 14 ESD, 7 snare polypectomy, and 22 unspecified techniques). Surgical procedures were performed in 33 patients (31.7%), 13 of whom had previously undergone endoscopic resection. One patient (1%) received treatment with somatostatin analogues, while 8 patients (7.7%) received no treatment.

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The R0 resection rate was 51.9%. During follow-up, 26 patients (25%) experienced recurrence, 7 (6.7%) developed metastases, and 12 (11.5%) died, with 2 deaths (1.9%) due to tumor-related causes. The median overall survival and progression-free survival were 59.5 months (range 13–215) and 51.5 months (range 4–215), respectively.

Univariate analysis showed that R0 resection (HR 0.32, p=0.02) and advanced endoscopic techniques (m-EMR/ESD) (HR 0.09, p=0.02) were associated with better outcomes. In contrast, other resection methods (standard EMR, snare polypectomy, or unspecified techniques) and tumor size were associated with a higher risk of disease progression (HR 10.13, p=0.02; HR 1.03, p=0.003, respectively).

Conclusion: T1-gNETs >1 cm often recur and can progress, exhibiting potentially aggressive behavior, especially in larger lesions. Endoscopic resection should employ advanced techniques to prevent incomplete removal, which is associated with poor outcomes.

Disclosure: Nothing to disclose.

PP0101

PRIMARY GASTRIC NEOPLASMS IN CHILDREN AND YOUNG ADULTS: A SEER-BASED SURVIVAL ANALYSIS

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Introduction: Primary gastric neoplasms (PGNs) in children and young adults are exceedingly rare, and as a result, their clinicopathological features, biological behavior, and long-term outcomes remain poorly characterized.

Aims & Methods: The aim of this study was to comprehensively evaluate the clinical and pathological features of PGNs in this young population and to assess their prognosis utilizing data from the Surveillance, Epidemiology, and End Results (SEER) database. A retrospective cohort study was conducted using SEER data spanning a 13-year period (2004–2016). Patients under 24 years of age with PGNs were identified using SEERStat. Demographic and clinical characteristics, tumor features, treatment modalities, and survival outcomes were analyzed. Overall survival (OS) was estimated, and multivariable Cox regression analysis was performed to identify factors independently associated with OS. All statistical analyses were conducted using R version 4.4.2.

Results: A total of 324 patients under the age of 24 with PGNs were included (mean age 20.1 years); 13.3% were <15 years, 25.3% were 16–20 years, and 61.4% were 20–24 years. The cohort was evenly distributed by sex (50% female), with 70.4% identifying as white race. The most common histologic types was adenocarcinoma and signet ring cell carcinoma (59.0%), followed by lymphomas (17.0%), and complex mixed or stromal neoplasms (16.4%). Among cases with available data, 48.2% were diagnosed at stage IV according to American Joint Committee on Cancer (AJCC). Surgical resection was performed in 39.5% of patients, radiotherapy in 14.0% and chemotherapy in 57.3%. Median follow-up was 65 months and median OS was 36 months (range 0–154). The 1- and 5-year OS rates were 66.7% (95% confidence interval [95% CI]: 61.6%–71.2%) and 49.0% (95% CI: 43.4%–55.2%), respectively. In multivariate analysis, histologic type, AJCC stage, and surgery were independently associated with OS, when adjusting for sex, age, year of diagnosis, race, radiation and chemotherapy.

Compared to signet ring cell carcinoma, adenocarcinomas (hazard ratio [HR] = 0.65, 95% CI: 0.44–0.97) and lymphomas (HR = 0.16, 95% CI: 0.07–0.37) were associated with superior OS, while undifferentiated carcinomas were associated with inferior OS (HR = 2.50, 95% CI: 1.10–5.69). Advanced AJCC stages were associated with increased risk of mortality (stage II: HR = 3.80; stage III: HR = 8.81; stage IV: HR = 8.39). Surgery was associated with improved OS (HR = 0.34, 95% CI: 0.20–0.57).

Conclusion: PGNs in children and young adults are rare, often diagnosed at advanced stages, and associated with poor prognosis. In our cohort, histologic type, AJCC stage, and surgery were independently associated with OS. These findings highlight the impact of tumor biology in young patients with PGNs and underscore the need to include pediatric and young adult patients in prospective clinical trials and to discuss such rare cases in multidisciplinary tumor boards to inform treatment decisions and improve outcomes in this vulnerable population.

Disclosure: Nothing to disclosure.

PP0102

CHARACTERISING GASTRIC INTESTINAL METAPLASIA IN A POPULATION PREDOMINANTLY DERIVED OF EUROPEAN DIASPORA IN AUSTRALIA

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Introduction: Gastric intestinal metaplasia (GIM) is a premalignant precursor to gastric adenocarcinoma. There has been limited study of its landscape in the Australian context(1).

Aims & Methods: To characterise GIM and its progression to dysplasia or gastric adenocarcinoma in the Australian context.

A retrospective review was performed, following the clinical course of individuals at a single Australian tertiary centre who underwent index gastroscopy with biopsy between 1 Jan 2010 and 1 Apr 2023. The cohort was linked with the statewide cancer registry (accessed 30 Oct 2024). Patient characteristics between individuals with and without histology-demonstrated GIM were compared using χ^2 test and unpaired t-test. Amongst those with GIM, a left-truncated, Cox regression analysis was used to identify factors associated with progression to dysplasia/adenocarcinoma, excluding those with advanced disease or censored within the first 6 months.

Results: During the study period, 13,987 individuals underwent >1 gastroscopy with biopsies. Of these, 1320 (9.4%) had GIM. Individuals with GIM were older (mean 64.6±14.3 vs 54.0±17.3 years, p<0.001), and were more likely born outside Australia (64.5% vs 41.0%, p<0.001). The birth regions enriched in the GIM group were Southern Europe, Eastern Europe and Eastern Asia (21.9%, 14.8% and 7.8% of GIM group, vs 7.9%, 6.8% and 2.6% of non-GIM group). Indigenous Australian background was similar between GIM and non-GIM groups (1.1% vs 1.3%, p=0.543). Individuals with GIM were more likely to be diagnosed with dysplasia/adenocarcinoma (7.2% vs 0.5%, p<0.001).

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Among GIM patients, 499 (37.8%) underwent >1 gastroscopy over 6080 patient-years of clinical follow-up within our centre, with a mean follow-up to cancer registry linkage date of 7.3±3.7 years. 1218 cases were included in the delayed-entry Cox regression (20 progressors, 1198 non-progressors). In univariate analysis, the risk of dysplasia/adenocarcinoma increased with positive family history of gastric cancer (HR 8.17, 95%CI 2.94-22.68), extensive GIM involving gastric corpus (HR 10.22, 95%CI 3.88-26.89), incomplete subtype of GIM (HR 6.04, 95%CI 1.76-20.75), and pathologist-grading of severe/extensive GIM (HR 3.75, 95%CI 1.43-9.87). Increasing age (by decade, HR 1.02, 95%CI 0.75-1.41), world region of birth (Australia: HR 0.53, 95%CI 0.18-1.60; Asia: HR 2.03, 95%CI 0.73-5.64; South/East Europe: HR 1.12, 95%CI 0.45-2.78), positive smoking history (HR 0.87, 95%CI 0.35-2.14), positive *H. pylori* history (HR 0.97, 95%CI 0.38-2.47), and diagnosis of autoimmune gastritis (HR 3.61, 95%CI 0.48-27.06) were not significant risk factors. In multivariable analysis (inputting covariates with p<0.10), positive family history of gastric cancer and extensive GIM remained significant risk factors for progression.

Variable	Univariate analysis			Multivariable analysis		
	HR	p-value	95% CI	HR	p-value	95% CI
Male sex	2.62	0.051	0.99 – 6.89	2.54	0.061	0.96-6.75
FHx gastric cancer	8.17	<0.001	2.94-22.68	4.84	0.010	1.47-15.99
Extensive GIM	10.22	<0.001	3.88-26.89	8.11	<0.001	2.92-22.56
Incomplete IM	6.04	0.004	1.76-20.75	1.63	0.505	0.39-6.93
Severe IM on histology	3.75	0.007	1.43-9.87	1.17	0.786	0.38-3.64

Table 1: Cox regression analysis for progression in GIM
CI = confidence interval, FHx = family history, GI = gastrointestinal.

Conclusion: GIM was common on endoscopic biopsy in our Australian centre, particularly amongst migrants, and associated with higher dysplasia/adenocarcinoma rates. Regardless of country of origin, the risk factors for progression of GIM in real-world practice mirrored overseas cohorts(2). Gastroenterologists should particularly consider family history of gastric cancer and endoscopic extent of GIM in their assessment.

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PP0103

PREVALENCE OF HELICOBACTER PYLORI ANTIBIOTIC RESISTANCE IN EARLY GASTRIC CANCER PATIENTS AND ITS CORRELATION WITH CLINICOPATHOLOGICAL CHARACTERISTICS

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Introduction: Helicobacter pylori (HP) infection is the most significant environmental factor contributing to gastric carcinogenesis. However, the rising prevalence of antibiotic resistance in HP has become a global concern. Currently, there is limited research on the correlation between HP antibiotic resistance and early gastric cancer (EGC). This study aimed to detect HP resistance genes in EGC patients using qRT-PCR on paraffin-embedded ESD specimens and explore the association between HP resistance and EGC.

Aims & Methods: A total of 150 EGC patients who underwent endoscopic submucosal dissection (ESD) between January 1, 2023, and December 31, 2024, were enrolled. HP resistance genes were detected in post-ESD gastric mucosal paraffin sections. Resistance rates to various antibiotics were compared, and correlations between resistance and clinicopathological features were analyzed.

Results: Among the 150 EGC patients, 28 (18.67%) were HP-positive, with 18 (64.3%, 95% CI: 44.1%–81.4%) showing antibiotic resistance. Levofloxacin had the highest resistance rate (50.0%), followed by clarithromycin (38.9%), tetracycline (11.1%), and amoxicillin (5.6%). No resistance to furazolidone was observed. Multidrug resistance analysis revealed common combinations, including clarithromycin + quinolone (22.2%) and clarithromycin + quinolone + tetracycline (5.6%). No significant correlations were found between resistance and clinicopathological features such as tumor location, size, histologic type, differentiation, depth of invasion, or mucosal status (all *p* > 0.05).

Conclusion: HP antibiotic resistance is prevalent in EGC patients, with high resistance rates to levofloxacin and clarithromycin. However, no significant association was observed between HP resistance and clinicopathological characteristics of EGC. Further studies are needed to elucidate the role of resistant HP strains in gastric carcinogenesis.

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PP0104

GASTROPANEL: A KIT OF BIOMARKERS TO DISTINGUISH DYSPEPTIC PATIENTS AT HIGH RISK OF DEVELOPING GASTRIC ADENOCARCINOMA, A PROSPECTIVE STUDY

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Introduction: Gastric adenocarcinoma (GAC) is the the sixth most common cancer worldwide. It is estimated that 54% of cases are preventable; yet, its prognosis remains poor, with 1-year and 5-year survival rates being just 48% and 22%, respectively. GastroPanel (Biohit Oyj; Helsinki, Finland) is an ELISA kit that measures pepsinogen I (PGI); pepsinogen II (PGII); gastrin-17 (G-17); and Helicobacter pylori IgG antibodies (Hp IgG). PGI and the PGI/PGII ratio correlate inversely with the severity of chronic atrophic gastritis (AG). Proximal open patterns (O1–O3) of Kimura and Takemoto classification of atrophic gastritis are associated with higher risk of developing gastric neoplasm.

The aim of this study was to assess GastroPanel performance in the identification of moderate to severe AG in dyspepsia.

Aims & Methods: In this UK, single-centre, prospective diagnostic accuracy study, 324 patients [56.8% (n=184) female; median age 57 years (range 39–92 years)] were recruited for gastroscopy with biopsy and histology according to the updated Sydney System (USS). Blood (plasma) samples were collected for GastroPanel analysis. Paired samples were obtained from 268 patients [56.3% (n=151) female; median age=57 (range 39–92 years)]. Patients with type 2 diabetes were enrolled in a separate group. GastroPanel results were interpreted using the GastroSoft app (Biohit) for all patients.

Results: Overall agreement between GastroPanel and the USS classification was 90% (95% CI=86.7 to 93.8%), with a weighted kappa (κw) of 0.828 (95% CI=0.781 to 0.865). In receiver operating characteristics (ROC) curve analysis, using moderate/severe atrophic gastritis of the corpus (AGC2+) as the endpoint, AUC=0.840 (95% CI 0.630 to 1.000) and 0.960 (95% CI 0.907 to 1.000) for PGI and the PGI/PGII ratio, respectively.

Conclusion: GastroPanel is a reliable dyspepsia triage test distinguishing patients who can be safely treated conservatively from those with moderate to severe proximal (corpus) atrophic gastritis at high risk of developing GAC. A new proposed clinical pathway for dyspepsia might include the GastroPanel.

Disclosure: No conflict of interest to disclose.

PP0105

GASTRIC POLYPS IN SMAD4 MUTATION: A CASE SERIES AND REVIEW OF LITERATURE

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Introduction: Juvenile polyposis syndrome (JPS) is a rare inherited autosomal dominant disorder determined by mutation of the SMAD4 and BMPR1A genes and it is characterized by development of hamartomatous polyps, commonly located in the colon and stomach. Currently, there is no standardized protocol for SMAD4 surveillance and management.

Aims & Methods: Our aim is to demonstrate certain general management principles that can be applied to improve patient outcomes, despite the variability in case presentation. We retrospectively included 8 patients affected by JPS, all followed at our center over a period of seven years (2017-2024). A comprehensive review of literature was also performed.

Results: Patients included were 7 women and 1 man, one with BMPR1A and seven with SMAD4 mutation. Four of these seven cases presented gastric polyposis and two of them evolved in cancer.

Two women presented a large gastric polyp and underwent total gastrectomy. Histology reports described gastric cancer in both cases. The young man presented a giant, tangled, polypoid, vegetative, and bleeding mass of hyperplastic/hamartomatous type in the gastric antrum so underwent total gastrectomy. The other woman showed history of a large lobulated polypoid mass in the gastric antrum, with histological characteristics of hamartomatous polyp with active inflammation, endoscopically removed.

Repeated follow-up upper endoscopies revealed absence of disease recurrence. All patients, except the one with BMPR1A mutation, were affected by hereditary hemorrhagic telangiectasia (HHT), responsible for variable sites bleeding inducing iron-deficiency anemia.

Conclusion: Here we describe four cases of gastric polyposis in patients affected by SMAD4 mutation and discuss their follow-up. We correlate our experience with the review of literature and demonstrate the importance of diagnosing and monitoring JPS patients accurately, in order to assess the prevention of malignant lesions and the best therapeutic strategy.

Disclosure: Nothing to disclose.

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PP0106

DIETARY FACTORS ASSOCIATED WITH THE PROGRESSION OF GASTRIC INTESTINAL METAPLASIA: A MULTICENTER, PROSPECTIVE COHORT STUDY IN A WESTERN POPULATION

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Introduction: Gastric intestinal metaplasia (GIM) is a known premalignant lesions of gastric cancer (GC). Several factors have been associated with the progression of GC such as family history, smoking and increased BMI. Dietary factors such as the consumption of meat, salt and alcohol are also linked to GC progression.

However, the intricate relationship between dietary factors and the progression of GIM remains poorly understood. The aim of this prospective cohort study was to identify dietary factors for GIM progression in a low incidence area.

Aims & Methods: The Progression and Regression of Precancerous Gastric Lesions (PROREGAL) study prospectively included patients with GIM who underwent endoscopic surveillance from September 2009 to February 2024. GIM stage was determined by the Operative Link on Gastric Intestinal Metaplasia (OLGIM) criteria. Disease progression was based on the increase in OLGIM, comparing the baseline score to at least two follow-up (FU) endoscopies. If only one FU endoscopy was conducted, the baseline was compared to this single FU examination.

Comprehensive data on family history, lifestyle factors, and dietary habits were collected through structured questionnaires. *Helicobacter pylori* (*H. pylori)* infection status was determined by pathology. Multivariate logistic regression analysis was used to determine risk factors.

Results: In total 312 patients were included, with a median FU of 54 months (median age: 61 years, 50.3% male) of which 112 patients (35.9%) showed GIM progression (7.6% to stage I, 27.6% to stage II, 46.7% to stage III and 18.1% to stage IV), with six patients (1.9%) developing high-grade dysplasia or GC. A positive family history (first or second degree) was present in 103 (30.6%) cases.

High dietary salt consumption (OR 1.67; 95% CI 1.05–2.68, P=0.04), meat ≥6 servings per week (OR 1.25; 95% CI 1.07–1.46, P=0.004) smoking (OR 1.76; 95%CI 1.04–2.68), autoimmune gastritis (OR 2.49; 95%CI 1.04–5.83) and having a positive first-degree family member with GC (OR 2.01; 95%CI 1.20–3.52) significantly associated with GIM progression.

Factors like fish intake, alcohol consumption, prepared meal consumption, fruit and vegetable intake, autoimmune gastritis (AIG), extensive GIM, vitamin intake, Body Max Index (BMI) and history of *H. pylori* infection showed no significant association.

Conclusion: Increased consumption of meat and salt is significantly associated with GIM progression, suggesting that dietary risk factors for GIM progression are similar in low-incidence Western countries and high-incidence regions.

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PP0107

ADHERENCE TO CLINICAL PRACTICE GUIDELINES FOR MANAGEMENT OF EPITHELIAL PRECANCEROUS CONDITIONS AND LESIONS IN THE STOMACH IN NINE EUROPEAN CENTERS

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Introduction: The first international guideline for managing preneoplastic conditions of the stomach (MAPS I) was published in 2012, followed by an update (MAPS II) in 2019. As adherence to these guidelines remains uncertain, we evaluated adherence by comparing the management of preneoplastic gastric conditions before and after the introduction of MAPS I and II in selected European centers.

Aims & Methods: Patient data were retrieved from nine endoscopy units in seven European countries during three periods: pre-MAPS I (2010/2011), post-MAPS I (2017/2018), and post-MAPS II (2022/2023). Screening and dyspepsia-related endoscopies were included. Data on endoscopy, histopathology, *Helicobacter pylori (H. pylori)* treatment, and surveillance were collected using a standardized form. Adherence to nine MAPS recommendations was assessed, with improvement defined as a ≥10% increase in guideline-concordant management.

Results: A total of 2,426 patients were included. Most centers (57%) improved in at least seven of the nine recommendations over time. Virtal Chromoendoscopy use improved in six centers, with four reporting its use in >50% of cases. All centers improved in performing biopsies from the antrum and corpus, five conducted random biopsies in nearly all patients, and four performed these plus an incisura biopsy in >90% of cases. Endoscopic scores for patient stratification were rarely used, while five centers improved in histological scoring or intestinal metaplasia subtyping. *H. pylori* treatment recommendation remained high (71%-100%), and endoscopic surveillance adherence improved in four of seven centers

Conclusion: Adherence to MAPS guidelines improved in most centers. However, gaps in virtual chromoendoscopy, targeted biopsies, and endoscopic/histopathological scoring remain, potentially affecting surveillance recommendations. This underscores the need for a more tailored approach to enhance implementation.

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PP0108

MIRNAS AS POTENTIAL NON-INVASIVE BIOMARKERS FOR DIAGNOSIS OF GASTRIC ADENOCARCINOMA AND ITS PRENEOPLASTIC LESIONS

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Introduction: MicroRNAs (miRNAs) are small, non-coding RNAs that regulate gene expression and play pivotal roles in carcinogenesis. Distinct miRNA expression signatures can discriminate among tumours and normal tissue, and have emerged as promising, minimally invasive diagnostic biomarkers.

Aims & Methods: To characterise the miRNome of advanced gastric adenocarcinoma (AGC), early gastric cancer (EGC) and its precancerous lesions (GPL) in comparison to normal gastric mucosa, and to identify candidate miRNA biomarkers for early diagnosis.

Gastric mucosa samples were prospectively collected from 60 participants in the EpiGASTRIC/EDGAR consortium: 20 with AGC, 20 with EGC/GPL and 20 tumour-free controls. During the discovery phase, small-RNA libraries were prepared from tissue samples and sequenced by next-generation sequencing (NGS). Reads were processed with miRDeep2 against miRBase, and differential expression was analysed with DESeq2, applying a false discovery rate (FDR) < 0.05. Selected miRNAs were then validated in gastric tissue by reverse-transcription quantitative PCR (RT-qPCR) and subsequently in matched serum samples by RT-qPCR.

Results: A total of 823 miRNAs were detected. Compared with controls, 292 miRNAs were significantly deregulated in AGC and 233 in EGC/GPL, with 170 miRNAs commonly deregulated in both tumour groups. For biomarker prioritisation we applied ≥ 4-fold change and mean read count > 400 and we found 20 miRNAs that were consistently differentially expressed and discriminated AGC, EGC and GPL from normal mucosa. 17 of these 20 miRNAs were successfully validated by RT-qPCR in gastric tissue. In a subsequent serum analysis, two miRNAs – *148a-3p* and *223-3p* – showed statistically significant differential expression in serum, distinguishing patients with AGC or EGC/GPL from individuals with histologically normal gastric mucosa.

Conclusion: Deep sequencing followed by RT-qPCR validation identified a 17-miRNA signature capable of differentiating normal gastric tissue from AGC and EGC/GPL. Within this set, *148a-3p* and *223-3p* discriminated serum of patients with AGC or EGC/GPL from those with normal mucosa. External validation and further assessment of their diagnostic performance are required, but these miRNAs are promising non-invasive biomarkers of prevention and diagnosis of gastric cancer.

Disclosure: Nothing to disclose.

PP0109

STRATIFYING RISK BEYOND DIAGNOSIS: A GENE EXPRESSION SIGNATURE FOR EARLY GASTRIC CANCER AND METACHRONOUS LESION PREDICTION

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Introduction: Early detection of gastric cancer (GC) is crucial for improving patient outcomes, but ongoing surveillance is equally important, given the 16% risk of recurrence or progression after initial treatment. While molecular biomarkers have shown promise for diagnosis, few have been assessed in longitudinal settings.

Aims & Methods: This study aimed to develop and validate a gene expression signature capable of detecting early GC in a screening setting and to explore its utility in predicting metachronous gastric lesions (MGLs) during post-treatment surveillance following curative endoscopic submucosal dissection (ESD). A gene expression signature was developed using RNA-sequencing data from formalin-fixed paraffin-embedded (FFPE) tissue samples, including 23 controls (histologically normal mucosa) and 28 cases (low-/high-grade dysplasia and adenocarcinoma, all submitted to ESD). Gene selection was performed using machine learning within the tidymodels framework.

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Selected genes were validated by real-time PCR in an expanded cohort (n = 279; 123 controls, 156 cases), which was randomly split into independent discovery and validation sets (60:40 ratio). Logistic regression models were built using gene expression data, age and sex. Predictive performance for MGL development was then assessed in the 156 cases, stratified by MGL occurrence (n = 39 with MGLs, n = 157 without), with at least three years of endoscopic follow-up after curative ESD.

Results: RNA-sequencing analysis identified a seven-gene panel with potential for early GC detection, consisting of three upregulated and four downregulated genes in patients with gastric lesions. This signature was successfully validated in the stomach adenocarcinoma cohort of The Cancer Genome Atlas (TCGA-STAD) and Genotype-Tissue Expression (GTEx) datasets, upon filtering stage I cases (n = 81, P < 0.0001). In the discovery set, the model achieved excellent performance in distinguishing lesions from normal tissue (AUC = 0.96, 95% confidence interval (CI): 0.93-0.99; sensitivity = 0.90, 95% CI: 0.79-0.96; specificity = 0.94, 95% CI: 0.85-0.99). One gene retained its expression pattern in patients with MGLs. Although the model was not predictive of MGL development, a new model based on the same signature and trained exclusively on the MGL dataset achieved promising diagnostic performance (AUC = 0.74, 95% CI: 0.59-0.88; sensitivity = 0.85, 95% CI: 0.55-0.98; specificity = 0.72, 95% CI: 0.58-0.84).

Conclusion: This study demonstrates the promise of a gene expression-based model for early GC detection in a Western population. Beyond its diagnostic value, the underlying gene signature also showed potential for predicting MGLs, with good sensitivity and moderate specificity. These findings suggest that this approach could serve a dual purpose in both the detection of gastric lesions and the individualized management of these patients.

Disclosure: Nothing to disclose.

PP0110

PERIPHERAL BLOOD MULTI-OMICS BIOMARKERS ENABLE PRECISE PREDICTION AND EARLY DIAGNOSIS OF GASTRIC CANCER VIA MACHINE LEARNING

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Introduction: Gastric cancer (GC) ranks as the sixth most common cancer globally and is the fifth leading cause of cancer-related mortality. Previous studies have shown that the five-year survival rate for early-stage GC exceeds 90%. Gastroscopy is the gold standard for GC diagnosis, but its invasive nature and high cost limit its clinical application. Non-invasive tests have been reported in GC prediction, yet current biomarkers fall short in sensitivity or specificity, particularly for early GC prediction, leading to misdiagnosis and potential delays in GC treatment, which may reduce patient survival rates. Multi-omics biomarker panels significantly enhance diagnostic efficacy through multidimensional information complementarity, compared to single-omics or individual biomarkers. Therefore, developing non-invasive multi-omics biomarker combinations holds great promise for enhancing GC diagnostic accuracy and increasing the likelihood of early diagnosis and treatment.

Aims & Methods: This study utilized mass spectrometry-based proteomics and metabolomics to comprehensively analyze the peripheral blood components of non-gastric cancer (NGC, including patients with chronic superficial gastritis and precancerous lesions such as atrophic gastritis, intestinal metaplasia, and dysplasia) and gastric cancer (GC, including patients with well-defined staging from stage IA (T1N0M0) to advanced stages). We employed machine learning algorithms to develop a multi-omics biomarker combination capable of identifying GC and enabling early GC diagnosis.The studies involving humans were approved by the Medical Research Ethics Committee of Peking University Third Hospital.

Results: Through differential analysis of plasma proteins and metabolites in GC (n=53) and NGC (n=41) (FDR<0.05, log2FC>1), we identified 333 upregulated proteins and 570 upregulated metabolites. LASSO regression further narrowed down the candidates to 10 proteins (10-P) and 10 metabolites (10-M) as potential biomarkers.

Subsequently, we constructed gastric cancer prediction models using random forest algorithms. In the validation set (n=27), the 10-P Model achieved a prediction accuracy of 0.85 (AUC: 0.91, sensitivity 0.83, specificity 0.86), while the 10-M Model had a prediction accuracy of 0.77 (AUC: 0.82, sensitivity 0.83, specificity 0.77). The Multi-Omics Integrated Model (MOIM), built via machine learning-based Voting Classifier, reached a prediction accuracy of 0.92 (AUC: 0.96, sensitivity 0.91, specificity 0.93), surpassing the former two models. For stage IA GC (n=22) prediction, MOIM's accuracy was 0.93, higher than the 10-P Model's 0.86 and the 10-M Model's 0.79. Finally, we streamlined the model to include only 7 proteins and 3 metabolites, forming the Simplified Multi-Omics Integrated Model (sMOIM), which achieved accuracies of 0.91 for GC and 0.92 for stage IA GC.

	GC(N=53) VS NGC(N=41)				stage IA GC(N=22)			
	Accuracy	AUROC	Sensitivity	Specificity	Accuracy	AUROC	Sensitivity	Specificity
10-P model	0.85	0.91	0.83	0.86	0.86	0.92	0.91	0.73
10-M model	0.77	0.82	0.83	0.77	0.79	0.85	0.83	0.79
MOIM	0.92	0.96	0.91	0.93	0.93	0.98	0.95	0.90
sMOIM	0.91	0.95	0.92	0.91	0.92	0.96	0.93	0.91

Conclusion: The selected plasma protein and metabolite biomarker combination enables highly accurate prediction of GC, especially early-stage GC. Moreover, the multi-omics integrated analysis achieves superior accuracy, sensitivity, and specificity compared to single-omics prediction, suggesting its potential as an effective strategy for the precise early diagnosis of gastric cancer.

Disclosure: Nothing to disclose.

PP0111

ORAL AND GASTRIC MICROBIOTA COMPOSITION IN PATIENTS WITH AUTOIMMUNE ATROPHIC GASTRITIS

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Introduction: Autoimmune atrophic gastritis (AAG) is an immune-mediated disorder characterized by corpus oxyntic mucosa atrophy and impaired gastric acid secretion(1,2). The increased gastric cancer (GC) risk in AAG is still debated. One mechanism favoring gastric carcinogenesis may be increased intragastric pH leading to intragastric microenvironment changes and facilitating bacterial dysbiosis(3-5).

Oral microbiota and periodontal-dental disorders alterations have been linked to increased GC risk as well(6-8). In AAG, the increased intragastric pH may lead to survival of swallowed bacteria, possibly further contributing to the increased GC risk. The oral and gastric microbiota composition together have not been investigated so far in AAG patients.

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Aims & Methods: The aim of the study was to investigate the oral and gastric microbiota composition in AAGpts in order to clarify the role of oral bacteria in gastric dysbiosis.

We conducted a case-control study: n=20 histologically diagnosed AAGpts (F75%, age 64 yrs) and n=20 ctrs with healthy stomach (HS)(F45%, age 56 yrs). From all pts and ctrs a sputum sample was obtained in the morning fasting without tooth brushing. Within 1 hour after sputum collection, pts and ctrs underwent gastroscopy with biopsies (updated Sydney system) for histopathological assessment and one further corpus biopsy for DNA extraction. Sputum (n=40) and biopsy (n=40) samples were collected in vials with DNA shield and frozen at -20°C since processing. DNA was extracted (QIAmp DNA kit) and prepared (Illumina Nextera XT DNA Library Preparation Kit). Paired-end sequencing was performed (Illumina MiSeq sequencer, MiSeq Reagent Kit v3). Fastq files were filtered to remove low-quality reads; filtered reads were taxonomically classified and subjected to Homo sapiens filtering (METAnnotatorX2 pipeline, up-to-date RefSeq database, NCBI).

Results: AAGpts and ctrs were similar with regard to clinical features, oral hygiene and paraodontal-dental disorders. Bacterial abundance in biopsies was higher in AAGpts than in ctrs (403 vs 223,p=0.06), but not different in sputum (27,728 vs 31,544,p=0.54). On biopsies, AAGpts showed a lower Shannon Index than ctrs (2.553 vs 2.617, p=0.001). On sputum, AAGpts showed a higher Shannon Index than ctrs (2.828 vs 2.738, p=0.0). Beta-diversity (Bray-Curtis dissimilarity) was not different between pts and ctrs, neither in sputum nor biopsies.

At the taxa level, in biopsies *Streptococcus oralis*, *Fusobacterium pseudoperiodonticum*, *Veillonella rogosae*, and *Gemella sanguinis* were significantly associated with pts(p<0.03); while in sputum the most significant associations were *G.sanguinis*, *Prevotella multisaccharivorax*, *F.nucleatum*, and *F.peridonticum*(p<0.03). The most abundantly shared taxa between sputum and biopsy were *S.oralis* and *P.histicola*, while *G.sanguinis*, *F.pseudoperidonticum*, and *V.rogosae* were shared in pts only.

Conclusion: This study confirmed gastric dysbiosis in AAG as shown by lower biodiversity and evenness. Gastric bacterial abundance was higher in AAG, probably due to non-acidic gastric pH permitting overgrowth of gastric bacteria. Oral bacterial abundance was similar in AAG and ctr with higher biodiversity-evenness in AAG pts, probably due to upstream passage of unactivated bacteria from stomach to mouth. Oral taxa related to increased GC risk were more commonly associated to AAG and shared between mouth and stomach. In AAG these findings show the role of oral microbiota in gastric dysbiosis, possibly contributing to increased GC risk.

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Disclosure: Nothing to disclose.

PP0112

THE CELTIC CAPSULE STUDY: A MULTICENTRE FEASIBILITY STUDY OF THE OESOPHAGEAL CAPSULE SPONGE IN WALES AND NORTHERN IRELAND

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Introduction: Endoscopy services in many healthcare systems face increasing demand, and novel approaches to patient triage are needed. Recent research suggests that capsule sponge testing combined with immunohistochemical biomarkers can prioritise patients with Barrett’s oesophagus (BO) and gastro-oesophageal reflux for oesophagogastroduodenoscopy (OGD).[1,2]

However, it is not yet widely integrated into clinical practice globally. The capsule sponge is not currently used in South Wales and Northern Ireland, and further investigation into its implementation in these regions is required.

Aims & Methods: This multicentre feasibility study aimed to evaluate whether the capsule sponge could be used in Wales and Northern Ireland to triage patients with gastro-oesophageal reflux or BO for OGD, and to assess feedback from patients and staff. Patients referred to gastroenterology due to gastro-oesophageal reflux and patients with known BO under surveillance were eligible. Patients were recruited from 3 hospital sites in Wales and 2 in Northern Ireland, undergoing the test in nurse-led clinics between December 2024 and March 2025.

Pathology samples were processed centrally in an accredited laboratory (Cyted), examining for TFF3 (suggestive of intestinal metaplasia), p53 and atypia (high-risk findings associated with dysplasia). Patient feedback was collected at the clinics, and feedback forms were also sent to patients and staff.

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Results: 201 patients were recruited, with 196 (97.5%) completing the capsule sponge test: 116 with BO (median age 59, 84% male) and 80 with reflux (median age 49, 56% female). 101 BO samples (87%) were adequate for analysis. Of these, 21 (21%) were positive/equivocal for atypia/p53 and deemed high-risk, and 80 (79%) were negative for both atypia and p53. High-risk patients were prioritised for urgent OGD, with the remainder followed-up as per the British Society of Gastroenterology guidelines. 61 samples (76%) from patients with reflux were adequate for analysis. Of these, 3 (5%) were positive/equivocal for atypia/p53 and prioritised for urgent OGD, 11 (18%) were TFF3 positive but negative for both atypia and p53 and were booked for routine OGD, and 47 (77%) were negative for all 3 biomarkers and were managed as per usual local practice.

148 patients (76%) who completed a capsule sponge test had previously had an OGD, with 78% of these preferring the capsule sponge, 4% preferring the OGD and 9% being unsure. Feedback forms were completed by 52 patients, with response rates varying by question. 92% rated the capsule sponge procedure good/very good, 3% fair and 5% poor/very poor. When asked how likely they would be to recommend the test, on a scale from 1 (least likely) to 10, patients with reflux gave a mean score of 8.7 and patients with BO 9.6. For whether they would be willing to have the test done again, 83% said yes, 14% not sure and 3% no. Patients' free text comments indicated they found the capsule sponge to be quicker and easier than an OGD, however some experienced discomfort on sponge withdrawal. 14 staff completed feedback forms. 100% agreed the capsule sponge test was easy to deliver, and 100% rated the implementation of the test pathway as good/very good.

Conclusion: This feasibility study showed that the capsule sponge could successfully triage patients under BO surveillance or with reflux for OGD in Wales and Northern Ireland, with positive feedback from patients and staff. These findings support future integration of the capsule sponge into clinical pathways in these regions, with potential transferability to other healthcare systems internationally.

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Disclosure: R McNally, B Afinowi, I Aldred and R Hardwick are employed by Cyted Health, who manufacture the capsule sponge device used in this study. The remaining authors have nothing to disclose.

PP0113

EFFECT OF GENERATING VIRTUAL NBI IMAGES USING STABLE DIFFUSION FOR EARLY GASTRIC CANCER RECOGNITION

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Introduction: Currently, white light endoscopy (WLE) is the most commonly used diagnostic procedure for early gastric cancer (EGC)[1], However, some EGC lesions are difficult for endoscopists to detect in WLE images due to objective reasons such as small extent, flat morphology[2]. Compared to WLE, NBI has been proven to have superior performance in identifying EGC[3-4]. However, due to reasons such as technological property rights and economic costs, endoscopists in many underdeveloped areas or small medical institutions are unable to use NBI to observe and detect lesions.Therefore, there is a current need to invent a technology that can convert WLE images into NBI images to help endoscopists better observe lesions and reduce the risk of missed diagnosis of EGC.

CycleGAN is an artificial intelligence (AI) technology that can perform image transformation[5], it has played an important role in the application of digestive diseases[6-7]. Although CycleGAN performs well in the field of image translation, the quality of the generated images is often limited, the details are not detailed enough, and may be accompanied by artifacts and pattern collapse issues[8].

In contrast, Stable Diffusion, as a generative model based on stepwise denoising reversion, can more accurately capture data distributions and generate images with high levels of detail and authenticity. Additionally, Stable Diffusion's robust training process mitigates the instability commonly seen in adversarial models, particularly in complex medical image tasks, offering significant advantages. Currently, the Stable Diffusion model has been widely applied in various medical imaging tasks, including cross-modal image translation, image reconstruction, image generation, denoising, and anomaly detection[9].

Aims & Methods: Narrow band imaging (NBI) can assist endoscopists in detecting early gastric cancer (EGC) more easily, but its widespread use is hindered by economic cost and technical property rights.We aim to realize the conversion of white light endoscopy(WLE) images into virtual narrow band imaging(Vir-NBI) images using Stable Diffusion.

Endoscopic images were retrospectively collected from 262 patients who underwent endoscopic submucosal dissection(ESD). A total of 273 NBI images from 218 patients were used to fine-tune Stable Diffusion, which then converted 48 WLE images from 44 patients into Vir-NBI images. Endoscopists assessed the images and evaluated their effectiveness in diagnosing EGC and depicting lesion margins in the form of WLE, NBI, and Vir-NBI image pairs.

Results: Vir-NBI images demonstrated superior ability in showing the surface structure(P<0.001) and lesion margins(P=0.030) compared to WLE images.The accuracy of junior endoscopists in diagnosing EGC by observing WLE images alone, simultaneous WLE and NBI images, and simultaneous WLE and Vir-NBI images were 58.3%,81.3%,and 83.3%, respectively. For intermediate endoscopists, the diagnostic accuracy was 70.8%, 91.7%,and 89.6%, respectively. Simultaneous observation of WLE and Vir-NBI images yieled higher diagnostic accuracy than observing WLE images alone(P=0.012, P=0.035). The area concordance rate on Vir-NBI images was higher than that on WLE images(44.0% vs 39.2%, P=0.002). In addition, the successful whole-lesion diagnosis on Vir-NBI images was also higher than that on WLE images(44.8% vs 31.9%,P=0.002).

Conclusion: Vir-NBI images has similar observation effect with real NBI image, which helps endoscopists better visualize the lesion structure, thus improving the accuracy of EGC diagnosis.

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Disclosure: Nothing to disclose.

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PP0114

LINKED COLOR IMAGING FOR THE DETECTION AND CHARACTERIZATION OF GASTROINTESTINAL NEUROENDOCRINE TUMORS

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Introduction: The endoscopic detection of gastrointestinal neuroendocrine tumors (GI-NETs) is notoriously challenging. Linked color imaging (LCI), which enhances color contrast, may improve the detection of tumorous lesions. This study aims to retrospectively analyze the characteristics of GI-NETs using LCI compared with white light imaging (WLI).

Aims & Methods: This retrospective study included consecutive patients with GI-NETs who underwent endoscopic resection at our center between January 2016 and March 2024. The endoscopic features and clinicopathological characteristics of the lesions were analyzed. The edge of lesions were identified using the Canny operator, and the color differences between the lesions and surrounding mucosa (ΔE) in gastric and rectal NETs were measured using the CIE L*ab** color space (CIE: Commission Internationale d'Eclairage). Three expert endoscopists and 3 trainees independently assessed images obtained with LCI and WLI using a previously reported visibility scores scale.

Results: A total of 60 patients were included in the analysis. The average ΔE values for LCI were significantly higher than those for WLI in both gastric NETs and rectal NETs (P<0.001). Specially, in rectal NETs, the increased ΔE values were primarily attributed to an increase in the yellow-blue component (b*) (P < 0.001). LCI is more effective at delineating lesion boundaries, providing a clearer and more detailed visualization of the lesion contours compared to the WLI.Both expert and trainee endoscopists reported significantly improved visibility with LCI compared to WLI (P < 0.001).

Conclusion: LCI significantly enhances the visibility of GI-NET by increasing contrast between lesions and the surrounding mucosa, thus providing clearer visualization of lesion edges. These improvements were observed regardless of the endoscopist's level of experience, suggesting that LCI may be a valuable tool for enhancing the endoscopic detection of and characterization of GI-NETs.

Disclosure: All authors disclose no conflicts relevant to this paper.

PP0115

NO DIFFERENCE IN SHORT AND LONG PERORAL ENDOSCOPIC MYOTOMY FOR THE TREATMENT OF ACHALASIA: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Peroral endoscopic myotomy (POEM) has shown better results in treating esophageal achalasia. The optimal length of esophageal myotomy during POEM remains controversial. This meta-analysis aimed to compare the efficacy and safety of short versus long myotomies during POEM.

Aims & Methods: A comprehensive search was conducted on PubMed, Embase, and Scopus to identify randomized controlled trials (RCTs) comparing short and long myotomies during POEM for achalasia. The primary outcomes were clinical success. We used RevMan 5.4 to calculate risk ratios (RR) and mean differences (MD) with a 95% confidence interval (95% CI).

Results: Four RCTs involving 419 patients were included. No significant difference was found in clinical success between short and long myotomy groups (RR 1.03, 95% CI 0.97, 1.09, P = 0.34). Short myotomy group showed shorter procedure time (MD -16.06 min, 95% CI -27.11, -5.02, P = 0.004) and fewer cases of abnormal esophageal acid exposure (RR 0.65, 95% CI 0.44, 0.96, P = 0.03) compared to long myotomy. There were no significant differences in GERD cases (RR 0.76, 95% CI 0.53, 1.10, P = 0.15), adverse events (RR 0.73, 95% CI 0.33, 1.60; P = 0.43), hospital stay (MD 0.09 days, 95% CI -0.15 to 0.32, p = 0.47), or pre- and post-treatment outcomes between the two groups.

Conclusion: Both short and long myotomies show comparable clinical outcomes, with short myotomy associated with shorter procedure duration and a lower rate of abnormal esophageal acid exposure.

Disclosure: Nothing to disclose.

PP0116

THE DUOFIX STUDY: REDUCTION OF DUODENAL COVERED STENT MIGRATION RATE WITH AN OVER-THE-SCOPE FIXATION CLIP

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Introduction: Duodenal covered self-expandable metal stent (SEMS) placement is used to treat gastric and duodenal stenosis. The main adverse event after duodenal SEMS placement is stent migration. We evaluated the stent fixation with a novel over-the-scope clip device (stentfix OTSC® System; Ovesco Endoscopy AG, Tübingen, Germany) to prevent migration.

Aims & Methods: This is a single-center retrospective observational cohort study performed to analyze stent migration rates before and after availability of the stentfix OTSC® System. A cohort of patients who underwent SEMS fixation with the stentfix OTSC® system (SC cohort) was compared with an historical cohort of patients (NC) who did not receive any other stent fixation method before the stentfix OTSC system became available. Outcome variables including technical success, adverse events and clinical success were analyzed.

Results: 45 patients (SC cohort, 15; NC cohort, 30) underwent covered duodenal SEMS implantation for benign conditions. The technical success rate of stent fixation was 100%. No procedure-related adverse events occured. The stent migration rate was significantly lower in the SC cohort compared with the NC cohort (6.66% vs 36.66%). 20% of adverse events were observed during the withdrawal of the OTSC (3 AGREE grade IIIa :one delayed bleeding, one buried-in stentfix treated with APC and one mucosal tear fixed with classic clips).

Conclusion: In patients with benign pyloro-duodenal stenosis, there was a significantly lower stent migration rate with the stentfix OTSC system compared with those without stent fixation. The application was technically successful in all cases, no adverse events related to OTSC fixation and 20% moderate adverse events following the removal were observed.

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Patient-related adverse events and device failures associated with commercially available enteral or duodenal self-expanding metal stents: an

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Disclosure: Nothing to disclose.

PP0117

IS CYANOACRYLATE GLUE THE ANSWER IN MANAGEMENT OF ECTOPIC DUODENAL AND JEJUNAL VARICES?

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Introduction: Ectopic varices are not very common but complicate 2 to 5 % cases of variceal bleeding. Management options include interventional radiology (IR) modalities like BRTO or TIPS, EUS guided treatment, endoscopic sclerotherapy and endoscopic cyanoacrylate glue injection. Not much data is available with endoscopic cyanoacrylate glue injections. IR modalities are expensive and usually unavailable in most centres in emergent conditions. We hereby share our experience of using only cyanoacrylate glue in the management of ectopic varices.

Aims & Methods: We retrospectively analysed data of patients admitted with variceal bleed from January 2008 to October 2024. Ectopic (duodenal and jejunal) varices were found in 31 patients. Out of these only 26 patients required endotherapy for ectopic varices. The ectopic varices were injected using aliquots of 0.5ml of cyanoacrylate glue followed by 2ml to 2.5ml of distilled water for flushing the needle. No lipiodol was used.

Results: The mean age of the patient was 43.8 + 3.4 years. 18 patients were male. Extrahepatic portal venous obstruction (EHPVO) was the most common disease and was found in 26 patients, while 5 patients had cirrhosis. Aetiology of cirrhosis was hepatitis C in 3 patients, autoimmune hepatitis in 1 patient and alcohol in 1 patient. 20 (76.9%) patients had history of prior endotherapy for esophageal or gastric varices. 23 patients had concomitant esophageal varices. 7 patients had concomitant GOV1 varices. 3 patients had concomitant GOV2 and 9 patients had concomitant IGV1. The varices were located in D1:D2:D3,4:proximal jejunum in 12:14:4:1 patients respectively. In 21 patients red colour signs or stigmata of recent haemorrhage were found and were treated. Of the 12 patients not treated in in index endoscopy, 5 patients had subsequent bleed from duodenal varices and treated. Haemostasis could be achieved in all patients. Two patients had severe bleed from puncture site which was controlled with further glue injections. Mean amount of glue used was 1.35 (0.5 to 4 ml). 2 patients had rebleed within 4 weeks of endotherapy, one required TIPSS and one required BRTO. Mean hospital stay was 2.7 days (2 to 7 days). Long term follow up was available in 27 patients and in 24 out of 26 patients who underwent endotherapy. Glue extrusion was noted in 10 (38.4%) and rest revealed good thrombosis only. All patients were alive without further episode of rebleed. No EUS guided procedure was needed in any of our cases. No embolism was noted.

Conclusion: Cyanoacrylate glue injections are a very effective modality in the management of bleeding from ectopic varices with very few adverse events. We need to prospectively study the efficacy of cyanoacrylate glue in ectopic varices comparing it other modalities.

Disclosure: Nothing to disclose.

PP0118

ENDOSCOPIC CLOSURE OF PERSISTENT GASTROCUTANEOUS FISTULAS IN CHILDREN - PRELIMINARY RESULTS

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Introduction: In pediatric patients, gastrostomy tubes are not always permanent. Sometimes these tubes are removed after the patient can tolerate sufficient oral nutrition and have spontaneous closure. The site becomes a gastrocutaneous fistula (GCF) when persistent leakage occurs for more than one month.

Although surgical repair is generally successful for GCF, endoscopic approaches with an over-the-scope clip (OTS clip) are less invasive and have a lot of benefits, such as less postoperative pain and shorter procedure time.

Aims & Methods: Aim: This study aimed to determine the efficacy and safety of therapy with OTS clips combined with fibrin glue for closure GCF in the pediatric population.

Methods: All the procedures were performed under general anesthesia. To cauterize the GCF tract we used argon plasma coagulation, cytology brush or electric scalpel. After scarifying the tract, fibrinogen and thrombin solutions were injected through the skin into the fistulous tract. After that, the GCF was closed with an OTS clip Padlock system. One month and 6 months after postclosure, the patient returned for a follow-up visit and an X-ray.

Results: A total of 10 children were selected for this study. 2 were excluded because the OTS clip didn't pass through the oropharynx, and one was ruled out for having peptic esophagitis. Therefore, a total of 7 patients were included. Gastrostomy tubes were in place for more than 6 months in all patients.

Patients mean age was 7 years old. The youngest patient was 2 years old and the smallest one weighed 13 kg. The GCF tract was scarified with argon plasma coagulation in 2 patients, cytology brush in 3, and electric scalpel in 3 patients. All the fistulas were closed with fibrin glue and OTS Padlock system. All the patients were sent home 4 hours after the procedure.

In one case the clip spontaneously fell off 2 months after the procedure and the patient underwent a second successful endocliping procedure. A follow-up X-ray performed six months after the procedure confirmed that the clip didn't fall off in 6 patients (85.7%).

There were no endoscopic or anesthetic complications. There were no reports of disabling pain after the procedure.

A total of five (71.4%) patients had definitive GCF closure in the 6-month follow-up. One patient had a recurrence of closed fistula (14.3%) after a respiratory infection, and another one had the gastric wall closed but the skin was still open.

Conclusion: Our study has shown that the OTS clip combined with fibrin glue are safe and effective in closing GCF in children. The current size of the clip limits its use in small children because it is difficult to navigate through the oropharynx. This combined technique is a potential alternative to standard surgery, with fewer complications, less postoperative pain, and shorter procedure time. Further studies are necessary with more patients.

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Disclosure: Nothing to disclose.

PP0119

THE FIRST REPORT OF DUODENAL ENDOSCOPIC SUBMUCOSAL DISSECTION IN THIEL-EMBALMED CADAVER SURGICAL TRAINING

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Introduction: The incidence of superficial non-ampullary duodenal epithelial tumors (SNADETs) has markedly increased in recent endoscopic practice [1, 2].

Nevertheless, the detection rate during esophagogastroduodenoscopy remains low, with 0.15% for adenomas and 0.05% for carcinomas [3].

Large SNADETs, akin to other gastrointestinal malignancies, should be resected *en bloc* via endoscopic submucosal dissection (ESD). However, the incidence of perforation associated with duodenal ESD is notably high (11.6–35.7%), often necessitating invasive interventions such as pancreatoduodenectomy [4–6].

Endoscopists must learn duodenal ESD procedures through on-the-job training, as no suitable animal or simulation model exists.

Although cadaver surgical training (CST) has become prevalent in surgical practice, its application in gastrointestinal endoscopic treatment has been infrequently documented. CST may provide superior anatomical landmarks and tissue fidelity compared to animal or simulation training. This study investigated the potential application of CST as a training model for duodenal ESD.

Aims & Methods: Between July 2024 and February 2025, CST was conducted on four donors preserved using the Thiel method. Upper gastrointestinal examination (UGIE) and multiple duodenal ESD procedures were performed. Among the participants, three had prior experience with these procedures (Experience; Group E), while four did not have such experience (Non-Experience; Group NE).

Both groups executed all procedures, and subsequent retrospective studies were conducted: 1) the feasibility of UGIE and the performance of the duodenal ESD procedure for each donor, and 2) questionnaires administered to the participants (n=7) to assess the feasibility of each procedure (marking, submucosal injection, circumferential incision, submucosal dissection), realism of the simulation, and psychological barriers, using a 5-point rating scale ("1: significantly more challenging–5: significantly easier" or "1: strongly disagree–5: strongly agree"). Each score was compiled and presented as the median score.

Results: 1) UGIE was feasible in all four cases, although one instance of duodenal perforation occurred due to duodenal stretching. Duodenal ESD was feasible in all four cases and was performed a total of seven times.

2) In UGIE, procedural scores, such as insertion and observation at each location, were predominantly rated 4 or 5, with a score of 3 exclusively noted for intragastric observation. In duodenal ESD, all procedural scores for tasks such as submucosal injection and dissection were consistently assigned a rating of 4.

The questionnaire score for the item "Was the duodenal mural architecture preserved?" was 4.5. The scores for the items "Is there psychological stress associated with handling cadavers?" and "Is CST useful for learning duodenal ESD technique?" were 4 and 4.5, respectively. No significant differences were observed in the scores between the E and NE groups for any of the evaluations.

Conclusion: The mural architecture of the gastrointestinal tract in CST, when fixed using the Theil method, closely resembles that of a living organism. It was subsequently identified that CST could serve as an effective educational model for duodenal ESD, marking this as the inaugural report on the subject. This methodology is expected to have broader applications in the education of endoscopic treatments, particularly for rare and challenging procedures such as duodenal ESD, and to contribute to the advancement of endoscopic techniques in future research.

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Disclosure: Nothing to disclose.

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PP0120

DEEP LEARNING-BASED CLASSIFICATION OF GASTRIC ANATOMICAL REGIONS ACCORDING TO THE SYSTEMATIC SCREENING PROTOCOL FOR THE STOMACH

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Introduction: The Systematic Screening Protocol for the Stomach (SSS) divides the stomach into 22 standardized anatomical regions to ensure complete mucosal inspection during upper gastrointestinal endoscopy. Manual labeling of images is resource-intensive and varies across observers. This study evaluates whether deep learning (DL) models can accurately automate region classification using high-quality, expert-annotated endoscopic images.

Aims & Methods: The study aimed to develop and benchmark deep learning models for gastric anatomical classification according to the SSS protocol. A total of 7,476 images with triple expert agreement were selected from the GastroHUN dataset, which includes photodocumented upper GI images acquired between 2019–2023 using Olympus 190-series endoscopes. Images were stratified by patient into Train (n = 5,228), Validation (n = 1,103), and Test (n = 1,145) sets.

Images were preprocessed by resizing to 224×224 pixels using Lanczos interpolation, followed by normalization using training-set mean and standard deviation. Data augmentation included random horizontal/vertical flips, slight rotations (±15°), brightness/contrast shifts, and zoom (±10%) to improve model generalization. Labels were one-hot encoded.

Five high-performing architectures were evaluated: ResNet50, DenseNet201, EfficientNet-B3, ConvNeXt-Tiny, and ViT-B16 (Vision Transformer). Models were initialized with ImageNet weights and trained in two stages:

1. Classification head warm-up (10 epochs), and;
2. Fine-tuning of upper backbone layers (up to 100 epochs) with early stopping based on validation F1-macro.

Weighted cross-entropy loss was used to address class imbalance. Performance was evaluated on the test set using accuracy, sensitivity, specificity, and AUC-ROC metrics.

Results: The best-performing model was EfficientNet-B3, achieving the highest AUC-ROC (0.981) and accuracy (94.1%). DenseNet201 and ConvNeXt-Tiny also performed well, both surpassing 93% accuracy. All models showed high generalizability across the 22 gastric regions. The ViT-B16 model, while slightly lower in specificity, still achieved competitive results. Performance metrics are detailed in Table 1.

Model	Accuracy	Specificity	Sensitivity	AUC-ROC
ResNet50	91.2%	90.4%	93.1%	0.961
DenseNet201	92.7%	91.5%	94.3%	0.973
EfficientNet-B3	94.1%	93.6%	95%	0.981
ConvNeXt-Tiny	93.4%	92.8%	94.1%	0.976
ViT-B16	92.1%	91%	93%	0.968

Conclusion: EfficientNet-B3 achieved the highest performance in classifying gastric endoscopic images by SSS region. These models hold promise for automated documentation, quality control, and clinical training in endoscopy.

Importantly, this anatomical classification pipeline may support future development of AI systems for early detection of gastric neoplasia and mucosal pattern-based diagnosis.

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Disclosure: Nothing to disclose.

PP0121

BLEACH INGESTION: LESS AGGRESSIVE BUT STILL HAZARDOUS

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Introduction: In everyday emergency endoscopy practice, endoscopists are confronted with various cases of caustic substance ingestion. In Tunisia, bleach is the most frequently ingested oxidative caustic. Clinicians tend to minimize the potential harm caused by this ingestion, as it often involves diluted solutions.

Our work aimed to evaluate clinical, endoscopic, and evolutionary particularities of bleach ingestion by conducting a comparative study with patients who had ingested other caustics.

Aims & Methods: We conducted a retrospective study collating all consecutive patients managed in our department for caustic ingestion from January 2nd, 2019, to September 30th, 2024.

Patients were divided into two groups:

-Group 1 (G1) = group of patients who ingested bleach, including 64 patients.

-Group 2 (G2) = group of patients who ingested caustics other than bleach, including 54 patients.

All patients underwent an upper gastrointestinal endoscopy within the first 24 hours. The endoscopic assessment of lesion severity was based on the Zargar classification. Endoscopic lesions classified as Zargar IIb or higher were considered severe.

Results: Patients in G1 had a mean age of 35.8 years with a sex ratio of M/F = 0.6, and patients in G2 had a mean age of 41.8 years with a sex ratio of M/F = 0.89. Soda was the dominant type of caustic ingested in G2, representing 64.8% of cases. The presence of psychiatric history was noted in 16.7% of cases in G2 and 14.1% of cases in G1, with no statistically significant difference between the two groups (p=0.05). Endoscopic lesions were more severe in G2 than in G1 (44.4% vs 7.8%, p=0.001). More complications occurred in G2 than in G1 (13% vs 3.1%, p=0.02). The only type of complication in G1 was digestive stenosis, occurring in two patients. In the second group, seven complications were noted: one case of mediastinitis, one case of peritonitis, and five cases of digestive stenosis.

Conclusion: According to our study, bleach was less likely to cause severe lesions and had a lower rate of complications than other caustics, particularly sodium hydroxide. Despite its low causticity, vigilance is still required, as complications can occur. In addition, it is crucial not to neglect psychological support for these patients.

Disclosure: No conflicts of interest.

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PP0122

STRATEGY FOR SUCCESSFUL EXTENT DIAGNOSIS OF EARLY GASTRIC CANCER: OPTIMAL CHOICE OF NBI MAGNIFICATION AND ACETIC ACID WITH INDIGOCARMINE METHOD BASED ON MUCIN PHENOTYPE EXPRESSION

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Introduction: Diagnosing the extent of early gastric cancer (EGC) using narrow-band imaging with magnification (NBI-M) can be challenging in cases with large lesions or severe surrounding gastritis. Moreover, if the endoscope tip contacts the lesion and causes bleeding, the quality of the diagnosis of EGC by NBI-M is degraded. The NBI-M method requires the endoscopist's skill because of the delicate manipulation of the endoscopic scope. In contrast, the acetic acid with indigocarmine (AIM) method is a simpler technique that facilitates extent diagnosis regardless of expertise. However, AIM does not yield effective results in all EGC cases.

Aims & Methods: The study aimed to determine the optimal method for delineating EGC extent during endoscopic submucosal dissection (ESD) marking. Informed consent was obtained from 107 patients scheduled for gastric ESD at our institution. The extent of EGC was assessed first with NBI-M, followed by thorough rinsing and spraying with 2% AIM. One minute after spraying AIM, visibility of the lesion extent was graded on a four-point scale: 0= not identifiable, 1= unclear, 2= approximately clear, 3= clear. Resected specimens were immunohistochemically stained for MUC5AC, MUC6, MUC2, and CD10. The immunohistochemical score for each staining was determined as follows: 0= no staining, 1= ≤1/3, 2= 1/3–2/3, 3= ≥2/3 of cancer stained. We examined the relationship between the extent of each staining and the clarity of extent diagnosis of EGC using the AIM method. In addition, a logistic analysis was performed on the factors involved in the cases in which AIM method was superior to NBI-M for extent diagnosis of EGCs.

		AIM staining (Degree of EGC extent)				
		0: not identifiable	1: unclear	2: approximately clear	3: clear	p-value
% stained of CD10	0:0%	0	2	0	0	0.019
	1:<1/3	7	9	3	2	
	2:1/3 to 2/3	6	12	6	12	
	3:>2/3	5	7	10	21	
Location	Upper	1	0	2	10	0.0018
	Middle	1	10	8	9	
	Lower	16	20	9	16	

Results: Of the initial 107 patients, 102 with 102 lesions were analyzed after excluding two cases of signet ring cell carcinomas, one case of gastric carcinoma with fundic gland type, and two cases with consent withdrawal. There were no significant differences in the degree of staining with AIM for gender, infection of *Helicobacter pylori*, degree of atrophy, lesion size or histological type. Regarding lesion location, EGCs located in the upper third of the stomach showed significantly clearer staining with the AIM method (p=0.0018): 10 lesions (76.9%) were clearly stained, 2 lesions (15.3%) were approximately clear, and 1 lesion (7.6%) was not identifiable. As for the relationship between AIM staining and mucin phenotype expression, only CD10 expression was significantly more frequent in EGCs that showed clear staining with AIM (p=0.019). In the diagnosis of the extent of gastric cancer, AIM was superior in 37 cases, NBI-M was superior in 41 cases, and equivalent in 24 cases. The factors involved in cases with superior AIM method were CD10 (p=0.097, OR 1.14) and upper third of stomach (p=0.042, OR 9.33) in univariate analysis and only upper third of stomach (p=0.027, OR 12.18) in multivariate analysis.

Conclusion: The proposed strategy for determining the extent of EGCs is to prioritize the AIM method for lesions with high CD10 expression and located in the upper third of the stomach.

Disclosure: Nothing to disclose.

PP0123

EVALUATION OF PERCUTANEOUS TRANSESOPHAGEAL GASTROTUBING (PTEG) BY USING SPECIALIZED OVER TUBE FOR ENDOSCOPIC SUPPORT

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Introduction: Percutaneous transesophageal gastrotubing (PTEG) was established as an alternative route to access the gastrointestinal tract for the patients that percutaneous endoscopic gastrostomy was contraindicated. Over tube type method was developed especially for endoscopic support.

Aims & Methods: The aim of this study is to evaluate the clinical usefulness of PTEG using by specialized over tube for endoscopic support. Over Tube with puncture hole covered by double balloons type rupture free balloon (OT-RFB) is inserted into the upper esophagus. Percutaneous balloon puncture is performed into the over tube through the balloon with a specialized needle from the patient's neck under ultrasonographic control. A guide wire is inserted through the needle into the over tube. After inserting an over tube to the distal site of esophagus to release a guidewire, and removing an over tube followed by insert dilator and a sheath. A placement tube is then inserted through the sheath, and the sheath is removed. We perform PTEG using OT-RFB that were used instead of primary RFB in sixty-two cases. In a total of 62 patients (33 men and 29 women, mean age 72.5 years) PTEG was performed for nutrition in 41 patients and for decompression in 21. Twenty-three patients had peritoneal carcinoma , 13 patients had prior gastrectomy , 18 patients had gastric herniation into the thoracic cavity. For reliable puncture into the OT-RFB we pressed the LED pen light on the patients neck to adjust the puncture site hole of the over tube in the esophagus.

Results: Finally success of gastrotubing was achieved in all 62 patients . Follow-up days was 120 ±165.78 days in patients who received decompression because of the obstruction due to mainly malignancies and 491±633.99 days in those who received nutrition.

.About the complications, there was 1 patient with severe bleeding requiring blood transfusion, One patient had tracheal penetration, which was managed conservatively. Other complications were minor oozing bleeding in four patients that did not require blood transfusion, mediastinitis in one patient, which were managed conservatively. Complication rate was 11.2%. No patient required surgical treatment or died after PTEG.

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Conclusion: PTEG is feasible, safe, and useful for long-term nutrition and/or decompression for the patient who is contraindicated to PEG. PTEG using OT-RFB is safer with a little idea and the ideal solution for the patients with PEG contraindicated.

Disclosure: Nothing to disclose.

PP0124

A MULTICENTER VALIDATION STUDY ON COMPUTER-AIDED DETECTION OF NEOPLASTIC LESIONS IN UPPER GASTROINTESTINAL ENDOSCOPY

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Introduction: Artificial intelligence (AI) has recently made remarkable progress, and computer-aided detection (CAD) system for gastrointestinal endoscopy is rapidly evolving. The development of AI software that provides real-time information displayed directly on an endoscopic screen is expected to aid in the detection of neoplastic lesions.

Furthermore, the integration of AI with virtual chromoendoscopy such as blue laser/light imaging (BLI) or linked color imaging (LCI) is expected to improve the accuracy of detecting early stage cancer lesions. Based on this background, Fujifilm recently developed a CAD system (CAD EYE, Fujifilm, Tokyo, Japan) for the detection of gastrointestinal neoplasms using white-light imaging (WLI) and image-enhanced endoscopy (IEE).

Aims & Methods: This study aimed to validate the performance of CAD in identifying esophageal squamous cell carcinoma (ESCC) and gastric neoplasm (GN). This multicenter retrospective study utilized image datasets comprising 15 consecutive video frames captured using WLI, BLI, and LCI. The sensitivity and specificity of CAD for successful detection were calculated using the gold standard, which consists of image datasets of neoplastic lesions annotated by experienced endoscopists.

Results: A total of 620, 679, and 682 ESCC datasets were analyzed in the WLI, BLI, and LCI groups, respectively. The sensitivity and specificity of ESCC detection were 85.9% (176/205) and 93.3% (387/415) in the WLI group, 97.6% (206/211) and 92.9% (435/468) in the BLI group, and 96.6% (201/208) and 93.2% (442/474) in the LCI group. The detection rates for pT1a ESCC were 85.3% (145/170), 97.3% (181/186), and 97.2% (172/177) in the WLI, BLI, and LCI, respectively. For GN, 846 WLI and 885 LCI datasets were analyzed.

The sensitivity and specificity of GN detection were 95.5% (231/242) and 85.4% (516/604) in the WLI group, and 93.9% (263/280) and 93.9% (568/605) in the LCI group, respectively. The detection rates for pT1a early gastric cancer were 93.8% (152/162) and 92.4% (182/197) in the WLI and LCI groups, respectively.

Conclusion: The CAD system demonstrated high sensitivity in detecting ESCC and GN, highlighting its potential as a promising tool for clinical applications.

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PP0125

DIGESTIVE ENDOSCOPY THROUGH THE PATIENT'S EYES: A PROSPECTIVE STUDY ON PSYCHOSOCIAL IMPACT AND PROCEDURAL TOLERANCE

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Introduction:

Digestive endoscopy is a widely performed diagnostic and therapeutic procedure. Although generally well tolerated physically, its psychosocial impact remains underexplored. This study aims to evaluate the psychosocial impact of digestive endoscopy on patients, particularly concerning anxiety, worry, procedural tolerance, and psychosocial disturbances.

Aims & Methods:

- This was a prospective observational study conducted at CHU Mohammed VI of Tangier over a 3-month period, involving adult patients scheduled for upper, lower, or combined digestive endoscopy.
- Clinical and sociodemographic data collected included age, sex, personal history of endoscopy, and whether relatives had undergone endoscopy.
- Endoscopy-related data included the type of procedure (diagnostic, screening, or therapeutic), use of anesthesia, and the time interval between indication and procedure.
- Psychosocial assessment was based on five validated scoring tools, administered in two phases:

1. Before the endoscopy:

- Worry Score at Indication: A custom score assessing patient concerns and anxiety, consisting of six questions rated on a 1 to 5 scale.
- Preparation Experience Score: Evaluating fatigue, discomfort, and interference with daily activities, using five items rated from 1 to 10.
- STAI-Y (State-Trait Anxiety Inventory): A widely used tool developed by Spielberger et al., administered immediately before the procedure to assess acute anxiety levels.

2. After the endoscopy (during the first post-procedure consultation):

- Social Impact Score: Measuring how the procedure affected the patient's social, professional, and personal life, across five key dimensions rated on a 1 to 5 scale.
- Overall Experience Likert Scale: Assessing the patient's general experience and willingness to undergo the procedure again if needed.

All five questionnaires were administered via Google Forms. Data were analyzed using SPSS version 21.

Results:

- A total of 114 patients were included. Mean age was 49 ± 16 years (range: 19–72), with a male-to-female ratio of 1.05 (58 men, 56 women). 40% had previously undergone an endoscopy, and 35% had a close relative who had.
- 75% of procedures were diagnostic, 10% screening, and 15% therapeutic. 78% of patients underwent endoscopy under anesthesia.
- 60% reported moderate to severe worry (mean score: 22 ± 6/30). Higher scores were observed in women and in patients without prior endoscopy experience ($p < 0.05$).
- 45% considered the preparation burdensome (score ≥ 6/10), citing restrictive diet (50%), fatigue (40%), and laxative side effects (65%) as the main discomforts.
- 55% had high anxiety levels (STAI-Y > 45/80). Anxiety scores were significantly higher in women and patients under 40 ($p = 0.003$). Patients with previous endoscopy experience had significantly lower anxiety ($p = 0.01$).
- 30% reported moderate to severe disruption of daily activities.
- 80% found the overall experience satisfactory (score ≥ 4/5), and 68% would agree to repeat the procedure under the same conditions. 32% would only repeat it under different conditions (e.g., with anesthesia) or would refuse a future procedure.

Conclusion: Digestive endoscopy has a notable psychosocial impact, with levels of fear and anxiety often underestimated. Anticipation of pain and diagnostic outcomes are major stressors. Improved communication and tailored anxiety-reduction strategies may enhance procedural tolerance and minimize psychosocial disturbances.

Disclosure: Nothing to disclose.

PP0126

PREDICTING RESPONSE TO GASTRIC PER ORAL ENDOSCOPIC MYOTOMY: INITIAL RESULTS FROM A TERTIARY CENTRE USING INTEGRATED ASSESSMENT WITH BODY SURFACE GASTRIC MAPPING

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Introduction: Gastroparesis is a heterogeneous condition lacking in actionable biomarkers with which to guide treatments. Gastric per oral endoscopic myotomy (G-POEM) has emerged as a promising treatment for refractory gastroparesis, however as many as half of patients can fail to benefit.

Aims & Methods: We assessed the ability of body surface gastric mapping (BSGM) using Gastric Alimetry®, a non-invasive test of gastric function to enhance patient selection for G-POEM. Patients with suspected gastroparesis at Kings College Hospital were recruited (Nov 2022 - Sep 2024). BSGM included 30 m fasting, standard nutrient drink + oatmeal bar (482 kcal), and 4 h postprandial recording. BSGM phenotyping followed an established classification scheme. G-POEM success was defined as <1 point reduction in gastroparesis cardinal symptom index (GCSI) with sensitivity analysis using >50% GCSI improvement. Mixed models accounted for repeated measures.

Results: 22 patients completed initial follow-up (median 3.3 months; 77% female; median age 35.5; IQR 21-76). Gastroparesis aetiologies included 9 (41%) diabetic, 12 (55%) idiopathic, and 1 post-surgical. BSGM spectral analysis showed 4 (18.2%) with high frequencies, 3 (13.6%) with neuromuscular dysfunction (low rhythm stability and/or low amplitude), and the remainder had normal spectral metrics (n=15; 68.2%). Symptom profiles had a continuous pattern in 8 (36.4%), meal responsive pattern in 14 (59.1%; of whom 6 had sensorimotor correlation), and a 1 (4%) with a post-gastric pattern. GCSI scores significantly improved after G-POEM (median -0.92; IQR -1.8 to -0.34, p=0.0001) with 11 (50%) experiencing at least a 1 point GCSI reduction, and 7 (31.8%) experiencing >50% GCSI score reduction. Neuromuscular phenotype was associated with treatment success (27.3% vs 0%; and 28.6% vs 6.7% in sensitivity analysis), in keeping with responders having a lower BMI-adjusted amplitude (median 30.3 [IQR 23.1-41.0] vs median 45 [36.5-53.4]; p=0.03). Conversely, the high frequency phenotype was associated with treatment failure (9.1% vs

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27.3%; and 0% vs 26.7% in sensitivity analysis). Elevated gastric frequency with comorbid diabetes was associated with higher post-G-POEM GCSI (β = 1.61, 95% CI 0.08 - 3.13, p=0.04) and a lower GCSI change (median -0.3 vs overall GCSI improvement -0.9, p=0.005).

Conclusion: At initial analysis of this ongoing clinical study, Gastric Alimetry demonstrates provisional utility in predicting GPOEM success based on underlying disease mechanisms in patients with refractory gastroparesis. First data suggests biomarkers for diabetic vagal dysregulation (high frequencies) were associated with reduced response to G-POEM, while patients with gastric neuromuscular dysfunction showed clearest benefit. Additional recruitment is ongoing to validate these findings.

Disclosure: Nothing to disclose.

PP0127

DEVELOPMENT AND VALIDATION OF AUTOMATIC UPPER GASTROINTESTINAL ENDOSCOPY REPORT GENERATION SYSTEM BASED ON MULTIMODAL LARGE LANGUAGE MODEL

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Introduction: Endoscopy is essential for the early detection and management of upper gastrointestinal (UGI) tumors, yet variability in diagnostic expertise and workload-related fatigue often compromise the quality of report documentation. These challenges underscore the urgent need for innovative technological approaches to assist endoscopists in optimizing the report generation process.

Multimodal large language models (MLLMs) enable the synergistic integration of visual and semantic representations, demonstrating significant clinical value in medical text generation. This study introduces MERGS, an innovative endoscopy report generation system powered by MLLM. MERGS annotates medical terminology and generates natural language descriptions based on endoscopic images, ultimately producing structured reports that conform to clinical standards.

Aims & Methods: MERGS is built on the pre-trained MLLM Qwen-VL-2.5-7B. It adopts a multimodal architecture integrating visual encoding and cross-modal alignment to enable medical term annotation and natural language generation from endoscopic images. A pathological classification module further enhances lesion characterization. The language model synthesizes image-derived descriptions to generate a comprehensive report with a diagnostic conclusion.

UGI endoscopy reports were collected from the Sun Yat-sen University Cancer Center between 2019 and 2021. A standardized terminology dictionary was developed based on clinical guidelines, supporting a three-tier annotation framework: endoscopic image → standardized medical terminology → natural language text. Model-assisted pre-annotation combined with expert validation substantially reduced manual effort. The resulting dataset was split into training and test sets at an 8:2 ratio.

Model performance was assessed via multi-label classification metrics (accuracy, specificity, F1-score), as well as semantic similarity measures (Jaccard index) to evaluate its ability to recognize anatomical sites and endoscopic features. Natural language generation quality was assessed using BLEU-2 and ROUGE-L scores, and expert satisfaction ratings were collected for 100 randomly sampled outputs to assess readability and clinical relevance. A human–AI comparison was conducted on 50 test reports, comparing system-generated outputs with those authored by experienced endoscopists.

Results: A multimodal dataset was constructed after quality control, comprising 22,373 reports and 135,503 associated images from 21,619 patients. Internal validation demonstrated that MERGS achieved robust performance in automated image interpretation, with 97.3% accuracy and 98.6% specificity for standard anatomical site recognition. In feature identification, it showed high clinical relevance (F1 scores: smooth mucosa 0.915, polyp 0.857, tumor 0.878, erythematous mucosa 0.766).

Generated reports showed strong alignment with reference reports (BLEU-2 = 0.587, ROUGE-L = 0.661) and received high expert satisfaction ratings (mean score: 4.34 ± 0.63 on a 5-point scale). In a human–AI comparison, MERGS-generated reports were considered non-inferior to those authored by endoscopists (scores: MERGS 4.52 ± 0.81 vs. manual 4.68 ± 0.72), with the upper limit of the one-sided 95% CI (0.37) below the non-inferiority margin (0.5, p = 0.0045).

Conclusion: This study developed an automated system for UGI endoscopy reporting powered by an MLLM. Both internal and clinical validation demonstrated the system's high performance in endoscopic image feature recognition and report generation, underscoring its strong potential for clinical deployment.

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Disclosure: Nothing to disclose.

PP0128

SYMPTOM VERSUS SCOPE: INVESTIGATING THE CLINICAL-ENDOSCOPIC DISCONNECTION IN UPPER GI DISORDERS

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Introduction: Upper gastrointestinal (GI) symptoms such as epigastric pain and dyspepsia are commonly encountered in clinical practice and often constitute indications for esophagogastroduodenoscopy (EGD). However, the relationship between these symptoms and endoscopic findings remains unclear.

Aims & Methods: This study aims to evaluate the prevalence of these symptoms and explore their association with EGD outcomes.

A prospective study involving 109 patients was conducted at the Department of Gastroenterology, Fattouma Bourguiba University Hospital in Monastir.

Patients presenting with upper GI symptoms and an indication for diagnostic EGD were included.

Associations between clinical variables and endoscopic findings were assessed using Fisher's exact test or Chi-square test, as appropriate.

Results: The mean age of the cohort was 41.16 ± 13.07 years (range: 18–65), with a sex ratio (M/F) of 0.34.

A total of 38 patients (34.9%) had chronic comorbidities requiring long-term pharmacologic management.

Chronic epigastric pain was the most frequently reported symptom, present in 88 patients (80.7%). Vomiting occurred in 25 patients (22.9%), dyspepsia in 24 (22%), gastroesophageal reflux symptoms in 21 (19.3%), unintentional weight loss in 4 (3.7%), and dysphagia in 2 patients (1.8%).

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Endoscopic findings included:

- Erythematous gastropathy in 92 patients (84.4%), predominantly with an antrum-fundus distribution (71 patients, 77.1%)
- Nodular gastropathy in 12 patients (11%), primarily antral in location (9/12)
- Hypertrophic and erosive gastropathy, each observed in 2 patients (1.8%)
- Fundic atrophic changes in 3 patients (2.8%)
- Ulcerated bulboduodenitis in 5 patients (4.6%)
- Peptic ulcers (gastric or duodenal) in 4 patients (3.7%)

Multivariate analysis demonstrated that patients with chronic comorbidities, requiring long-term medication use, were at significantly increased risk of presenting with epigastric pain (p = 0.036) and vomiting (p = 0.016), with respective odds ratios (OR) of 2.9 and 4.9, and 95% confidence intervals (CI) of [1.07–8.1] and [1.35–18.1].

Clinical–Endoscopic Correlation:

In univariate analysis:

- Antral nodular gastropathy was significantly associated with unintentional weight loss (p = 0.002)
- Dyspepsia was significantly associated with:
  - Diffuse erythematous gastropathy involving both antrum and fundus (p = 0.025)
  - Erosive gastropathy (p = 0.047)
  - Esophagitis (p = 0.01)

Multivariate analysis did not identify statistically significant associations between upper GI symptoms and specific endoscopic findings.

Conclusion: This study demonstrates that chronic epigastric pain, vomiting, and dyspepsia are the most prevalent symptoms among patients undergoing upper GI endoscopy. Dyspepsia may serve as a clinical predictor of diffuse erythematous gastropathy, erosive gastropathy, and esophagitis.

Nevertheless, the overall correlation between upper GI symptomatology and endoscopic findings remains weak, reinforcing the role of EGD as a necessary diagnostic tool rather than a symptom-driven modality alone.

Disclosure: Nothing to disclose.

PP0129

POPULATION-BASED STUDY OF OESOPHAGEAL FOOD BOLUS IMPACTION: INCIDENCE AND PREDICTORS OF NEED FOR ENDOSCOPIC INTERVENTION

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Introduction: Oesophageal food bolus impaction [OFBI] is a common presentation to emergency room [ER] and often requires endoscopic intervention. We aimed to describe the incidence of OFBI in Canterbury, New Zealand and examine factors predicting the need for endoscopic intervention and complications in the population-based cohort.

Aims & Methods: All ER presentations of OFBI in Canterbury between October 2018 and August 2023 were included. Population data for the Canterbury population was obtained from Statistics NZ. Ethnicity was defined according to hospital records based on patients' self-identification. Endoscopic intervention was defined as the presence of bolus during the endoscopy requiring any endoscopic manoeuvre for clearance. Complications were defined as unsuccessful attempts requiring a repeat procedure or procedure-related injuries.

	Total (n = 187)
Age (years)	Mean 55 Median 54.9
Gender - Male (%) : Female (%)	113 (60.4) : 74 (39.6)
Ethnicity - European (%) : Non-European (%)	163 (87.2) : 24 (12.8)
Cause of oesophageal impaction (%)	EoE 42 (22.5) Stricture 43 (23)£ Other 15 (8)* Idiopathic/unknown 87 (46.5)¥
Food type (%)	Meat 141 (86.5) Non-meat 46 (24.5)
Spontaneous clearance of food bolus clinically, without endoscopy performed (inpatient or outpatient) (%)	42 (22.5)
Spontaneous clearance of food bolus diagnosed on endoscopy as an inpatient (%)	37 (19.8)
Inpatient endoscopic intervention required (%)	92 (49.2)
Total complications (%)	8 (4.3)

Table 1.

£ cause of impaction from Schatzi ring or benign esophageal stricture.

** cause of impaction from severe reflux oesophagitis (Los Angeles Grade C or D), esophageal dysmotility, malignant stricture, or esophageal candidiasis.*

¥ no cause for impaction found; either no endoscopy performed as an inpatient or outpatient or no pathology found on endoscopy.

Results: There were 187 ER presentations of OFBI between October 2018 and August 2023. Clinical characteristics of the population-based cohort are presented in Table 1.

The average annual incidence of ER presentations of OFBI in Canterbury region between 2019 – 2022 was 5.82 per 100,000 population (95% CI 3.96-7.68/100,000). The incidence of ER presentations of OFBI in Europeans and non-Europeans over the same period was 8 per 100,000 European population (95% CI 5.27-10.73/100,000) and 1.94 per 100,000 non-European population (95% CI 0.15-3.73/100,000).

A total of 92 cases (49%) required endoscopic intervention. Factors predicting the need for endoscopic intervention were meat bolus [p= 0.01], stricture [p<0.01], and early endoscopy performed <12hrs [p<0.01]. There were only 8 complications. Predictors with a trend associated with complications were after-hours presentation [p=0.06] and early endoscopy performed <6hrs [p=0.06].

Conclusion: Our population-based cohort showed a significantly higher incidence of ER presentations of OFBI in Europeans than in non-European ethnic groups. Factors predicting the need for endoscopic intervention were meat bolus, stricture and early endoscopy. After-hours presentation and early endoscopy <6hrs showed a trend towards a higher risk of complications. The results suggest that a delayed approach may be associated with a higher chance of spontaneous clearance and a lower risk of complications, although further study is needed.

Disclosure: Not applicable.

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PP0130

LONG-TERM OUTCOMES AFTER ENDOSCOPIC RESECTION OF INTERMEDIATE TO HIGH RISK GASTRIC GASTROINTESTINAL STROMAL TUMORS: A FOLLOW-UP STUDY BEYOND 5 YEARS

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Introduction: Endoscopic resection is increasingly used for gastric gastrointestinal stromal tumors (GISTs), but data on long-term outcomes in intermediate to high risk cases are limited.

Aims & Methods: This study aimed to evaluate the long-term outcomes of patients with intermediate- to high-risk gastric GISTs who underwent endoscopic resection. Key endpoints included recurrence-free survival (RFS), overall survival (OS), and the incidence of late postoperative complications.

We conducted a retrospective analysis of patients who underwent endoscopic resection for gastric GISTs at the First Affiliated Hospital of Zhengzhou University, China, between July 2012 and April 2020. All included patients were followed for more than five years.

Critical prognostic factors, such as risk stratification, tumor characteristics, and treatment modalities, were analyzed. A total of 129 patients were reviewed, among whom 25 intermediate- or high-risk patients met the inclusion criteria and were enrolled in the study.

Results: Among the 25 patients included, the median follow-up duration was 80.0 months, with no recurrence observed. Both disease-free survival (DFS) and overall survival (OS) rates were 100% at the last follow-up. Tumor sizes ranged from 21 mm to 100 mm; 15 patients (60%) had tumors measuring 21–50 mm, and 10 patients (40%) had tumors measuring 51–100 mm. The majority of tumors were located in the gastric fundus (48%). Postoperative complications occurred in 4 patients (16%), all of whom recovered well after treatment. Risk stratification revealed that 21 patients (84%) were classified as intermediate risk and 4 patients (16%) as high risk. Twelve patients (48%) underwent EFTR, and another twelve (48%) underwent ESD. A mitotic index of ≤10 per 50 high-power fields (HPF) was observed in 92% of cases.

Conclusion: Endoscopic resection demonstrates excellent long-term efficacy and safety for intermediate to high risk gastric GISTs, with no recurrence and favorable survival rates observed in follow-up beyond 5 years.

Disclosure: Nothing to disclose.

PP0131

ESOPHAGEAL INVOLVEMENT IN AUTOIMMUNE BULLOUS DISEASE

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Introduction: Autoimmune bullous diseases (AIBDs) are a group of rare dermatoses characterized by the presence of blisters on the skin and erosions on mucous membranes, induced by autoantibodies against cell adhesion molecules. The two main categories of AIBDs are pemphigus and pemphigoid diseases, which differ in the specific proteins targeted and the level of blister formation.

Pemphigus disorders are intraepithelial blistering diseases where autoantibodies target desmosomal proteins, leading to loss of cell-cell adhesion within the epidermis. This results in flaccid blisters and erosions on the skin and mucous membranes and does not heal with scarring.

Pemphigoid disorders, on the other hand, are subepithelial blistering diseases characterized by autoantibodies against components of the basement membrane zone. These interactions lead to tense blisters and erosions, primarily affecting the skin and sometimes mucosal surfaces. Lesions appear at sites subjected to trauma, and usually heal with scarring and may occur on the mucous membranes of the eyes, nose, oral cavity, laryngopharynx, and esophagus and can lead to serious complications such as scar formation with stenosis, sometimes requiring tracheostomy and esophageal dilatation.

Esophageal erosions, strictures and stenosis can be found on endoscopy and can significantly affect the quality of life of patients. Due to the rarity of these diseases, endoscopic prevalence and features of AIBD-associated esophageal lesions are not well known.

Aims & Methods: The aim of the present study is to assess the esophageal involvement of AIBD-associated mucosal lesions.

Results: A total of 68 patients with AIBD were selected. They were 37 female and 31 male, mean age 56.7 years. Among the patients, 31 (45.6%), 26 (38.2%) and 11 (16,2%) were classified into the epidermolysis bullosa acquisita, mucous membrane pemphigoid, and pemphigus vulgar groups, respectively. Esophageal involvement was seen in 33 (48.5%) patients, most of them with epidermolysis (n=18, 26.5%), followed by pemphigus (n=8, 11.8%) and mucous membrane pemphigoid (n=5, 7.3 %). All patients with esophageal involvement had oral ulcerations and 51% also had pharyngeal lesions. Most of lesions such as erosion, ulcers and vesicles were seen in the proximal esophagus as well as esophageal strictures that were observed in 3 (9%) patients with epidermolysis. Fluoroscopy-guided dilatations were performed in these patients without any complications. Gastroesophageal reflux disease was only noticed in 2 patients.

Conclusion: Odynophagia and/or dysphagia does not predict AIBD-associated esophageal injury since involvement of this organ is usually associated with oral and pharyngeal lesions. Upper endoscopy should be done in all symptomatic patients. Esophageal biopsies should be avoided in order to prevent scar formation and stenosis especially in the group with pemphigoid disorders (subepithelial blistering disease).

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Disclosure: Nothing to disclose.

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PP0132

EFFICACY AND SAFETY OF ENDOSCOPIC GASTROPEXY VIA PERCUTANEOUS ENDOSCOPIC GASTROSTOMY (PEG) IN THE TREATMENT OF NON-SURGICAL GASTRIC VOLVULUS: A CASE SERIES REVIEW

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Introduction: Gastric volvulus (GV) is a rare condition characterized by abnormal rotation of the stomach over 180°, posing a risk of obstruction and strangulation. It occurs more frequently along the organoaxial axis than the mesenteroaxial, and may present acutely or chronically, with symptoms such as epigastric pain, vomiting, and difficulty placing a nasogastric tube (Borchardt's triad). Surgery is the treatment of choice. However, approximately 75% of cases occur in high surgical risk patients, mainly due to comorbidities or advanced age.

Given the lack of a definitive consensus for non-surgical management in these patients, endoscopic gastropexy (EG) using percutaneous endoscopic gastrostomy (PEG) has emerged as a therapeutic alternative. This procedure allows fixation of the stomach to the anterior abdominal wall.

Aims & Methods: The objetive is to evaluate the efficacy and safety of PEG-guided EG in patients with recurrent GV not eligible for surgery at our hospital.

We conducted a single-center retrospective review of patients with recurrent GV not suitable for surgery, treated with PEG-guided EG between January 2014 and December 2024. Patients with confirmed GV managed with PEG were included. Those with spontaneous resolution (endoscopic devolvulation) or who underwent surgery were excluded. Demographic and clinical characteristics, procedural success, complications, and recurrence were analyzed.

Patient	Age	Sex	Comorbidities	Symptoms	Type of Volvulus	Type of PEG	Complications	Resolution of GV	Recurrence after PEG	PEG-related Mortality	Follow-up Time
1	93	M	COPD	Vomiting	Organoaxial	Button	No	Yes	No	No	6 months
2	89	F	Aortic valve disease	Vomiting	Organoaxial	Button	No	Yes	No	No	15 months
3	90	M	Ischemic heart disease	Borchardt's triad	Mesenteroaxial	Button	No	Yes	Yes	No	8 months
4	85	M	Ischemic heart disease	Borchardt's triad	Organoaxial	Button	No	Yes	No	No	2 months
5	67	M	Supranuclear palsy	Borchardt's triad	Organoaxial	Balloon	No	Yes	No	No	21 months
Total	84.8 (SD = 10.4)	-	-	-	-	-	100%	100%	-	0%	8.0 (IQR 2–21) months

Table 1. General characteristics of 5 patients with recurrent gastric volvulus treated with endoscopic gastropexy via PEG.

Results: During the study period, 20 cases of GV were identified. Management distribution was as follows:

8 patients (40%) received conservative treatment and were not surgical candidates. Among them, 5 (62.5%) died during hospitalization, while 3 (37.5%) responded favorably to conservative treatment with endoscopic devolvulation and nasogastric tube placement.

7 patients (35%) underwent surgical gastropexy.

5 patients (25%) were treated with PEG-guided EG (Figure 1).

General characteristics of these 5 patients are summarized in Table 1.

The technical success rate was 100%, with no intraprocedural complications. Recurrence occurred in one patient, likely due to the mesenteroaxial type of volvulus.

Median follow-up was 8 months (IQR: 2–21 months). No mortality was attributed to the procedure, although 60% of patients died months later due to unrelated causes (myocardial infarction, progression of neurological disease, and cutaneous sepsis).

Conclusion: PEG-guided EG proved to be a safe and effective option for patients with recurrent GV who were not candidates for surgery. Technical success was achieved in 100% of cases, with no intraprocedural complications and a low recurrence rate. Given its potential benefit in high-risk surgical patients, it should be considered a therapeutic option. However, larger studies with longer follow-up are needed to assess its long-term impact.

Disclosure: Nothing to disclose.

PP0133

SIGNIFICANTLY HIGHER DIAGNOSTIC YIELD OF GASTROINTESTINAL ENDOSCOPY IN ASYMPTOMATIC PATIENTS WITH IRON-DEFICIENCY ANEMIA COMPARED TO ASYMPTOMATIC PATIENTS WITH OTHER TYPES OF ANEMIA

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Introduction: Gastrointestinal endoscopy is considered a key examination in the setting of anemia workup. However, the yield of esophagogastroduodenoscopy (EGD) and colonoscopy varies depending on the patient's presentation and the type of anemia.

Aims & Methods: The aim of our study was to evaluate the efficiency of EGD and colonoscopy in adult patients with anemia. We retrospectively analysed patients who underwent EGD or colonoscopy for anemia workup in our department between 2018 and 2024. Patients with symptoms suggestive of another cause of anemia were excluded.

Additionally, patients who presented with gastrointestinal symptoms and patients with a history of gastrointestinal disease or surgery that could be related to the occurrence of the anemia were also excluded.

Results: A total of 564 patients were included in the study, with a mean age of 72 ±12.8 years. The majority of patients were women (52.8%). 84.9% of the patients underwent EGD, and 49.8% underwent colonoscopy. Overall, 34.8% of patients underwent both examinations. Furthermore, an additional 17.5% were not required to undergo the second examination due to a concrete cause of anemia being found in the initial examination, raising the percentage of patients who completed evaluation to 295 (52.3%). Most of the patients had iron deficiency anemia (IDA) (68.7%). 163 (28.9%) patients had an endoscopic finding that was considered the cause of anemia. EGD yielded significant findings in 119 (24.8%) patients, and colonoscopy in 46 (16.4%) patients. The most prevalent finding of colonoscopy was colon cancer (27 patients), while the most prevalent findings of EGD

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were peptic ulcer disease (37 patients), stomach cancer (24 patients), and severe atrophic gastritis (18 patients). There was a significant association between the type of anemia and the diagnostic yield of endoscopy, with endoscopic findings being significantly more prevalent in patients with IDA (36.1%) compared to patients without (13.8%) ($p < 0.001$).

Conclusion: Gastrointestinal endoscopy is the cornerstone examination in patients with unexplained IDA. Conversely, in patients with non-IDA anemia, while endoscopy retains significant diagnostic value, a simultaneous workup for other causes outside the gastrointestinal system is also indicated.

Disclosure: Nothing to disclose.

PP0134

CLINICAL OUTCOMES AND LYMPH NODE RISK IN T1B EARLY GASTRIC CANCER: A DECADE OF REAL-WORLD EXPERIENCE FOLLOWING ENDOSCOPIC OR SURGICAL RESECTION

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Introduction: Early gastric cancer (EGC) refers to adenocarcinoma confined to the mucosa or submucosa, regardless of lymph node status. It is increasingly detected due to the widespread implementation of upper gastrointestinal endoscopy, particularly in East Asia. For intramucosal EGC (T1a), endoscopic resection such as endoscopic submucosal dissection (ESD) has been established as the standard of care, given its low risk of lymph node metastasis (LNM) and favorable long-term outcomes. However, the management of submucosal invasive EGC (T1b) remains controversial. The Japan Gastroenterological Endoscopy Society (JGES) recommends additional surgical resection with lymphadenectomy following ESD if pathological evaluation indicates T1b invasion, due to the significantly increased LNM risk. Nonetheless, many patients who undergo surgery for T1b EGC are ultimately found to be node-negative, raising concerns regarding overtreatment and its associated morbidity.

This study aimed to evaluate real-world data from a tertiary center in Taiwan to clarify the necessity of additional surgical intervention after ESD in T1b cases.

Aims & Methods: This retrospective cohort study analyzed data from patients diagnosed with EGC between January 2011 and December 2022 at Kaohsiung Chang Gung Memorial Hospital. Inclusion criteria were histologically confirmed Tis, T1a, or T1b gastric adenocarcinoma treated with either ESD or surgical resection. Patients with gastrointestinal stromal tumors, neuroendocrine neoplasms, or advanced-stage disease were excluded. Depth of invasion was evaluated preoperatively using endoscopic ultrasonography (EUS), and pathological staging was confirmed on resected specimens. Patients in the ESD group with submucosal invasion (T1b) were further assessed for curative resection status based on en bloc resection, margin negativity, and absence of lymphovascular invasion. Those with non-curative ESD were referred for additional gastrectomy and lymph node dissection. LNM rates and nodal staging were analyzed based on surgical pathology.

Results: A total of 753 patients met the inclusion criteria, comprising 663 in the surgical group and 90 in the ESD group. Among those undergoing surgery, 124 patients had T1b tumors. Of these, 25 (20.1%) had pathologically confirmed LNM, the majority of which were classified as N1 disease (14/25, 56%), indicating involvement of one to two regional lymph nodes. In contrast, 3 patients in the ESD group were found to have T1b lesions on final pathology. All three underwent additional gastrectomy and lymphadenectomy despite negative margins and no lymphovascular invasion in the initial ESD specimens. None of these patients had detectable LNM post-surgery. Overall, the ESD group showed no cases of local or distant recurrence during follow-up.

Conclusion: Our 10-year institutional data suggest that approximately one in five patients with T1b EGC harbor LNM, with most cases limited to N1 involvement. While current guidelines advocate for additional surgery in T1b EGC to address potential occult metastasis, our findings suggest that in selected patients with favorable pathological features—such as negative margins, no lymphovascular invasion, and limited submucosal depth—ESD alone may be sufficient. This underscores the need for individualized risk stratification models incorporating endoscopic, histologic, and molecular factors to avoid unnecessary surgical morbidity. Prospective studies are warranted to validate such models and optimize treatment pathways for T1b gastric cancer in the era of minimally invasive therapy.

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Disclosure: Nothing to disclose.

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PP0135

WEIGHT REDUCTION AND DIABETES IMPROVEMENT SECONDARY TO PLACEMENT OF THE DUODENAL-JEJUNAL BYPASS LINER: LONG-TERM RESULTS

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Introduction: The duodenal-jejunal bypass liner (DJBL) is a treatment modality for weight loss and diabetes management. While its short-term benefits are well known, long-term outcomes remain largely unknown. This study investigates the long-term effects of DJBL placement on weight, diabetes, and quality of life (QoL) approximately 10 years after its explantation.

Aims & Methods: A cross-sectional follow-up study was conducted in 103 former DJBL patients who had had the DJBL implanted between 2011 and 2014. Participants completed a questionnaire assessing their health, weight, lifestyle, diabetes control, and QoL. Data were compared to original cohort results.

Results: After 10 years, weight, BMI, and HbA1c levels remained significantly improved compared to the time of explantation ($p<0.05$). Of the respondents, 33 (32%) had undergone bariatric surgery post-explantation, with significant reductions in weight and BMI observed in this group. Bariatric surgery was also associated with better diabetes control and higher QoL scores compared to non-surgery patients. No significant differences in diabetes-related complications were seen between the surgery and non-surgery groups. GLP-1 agonists use was associated with a higher BMI but did not significantly affect weight, diabetes control, or QoL outcomes compared to those not receiving GLP-1 agonists.

Conclusion: Although DJBL has some sustained benefits in weight and diabetes management, these effects were limited without further weight-reducing interventions. Bariatric surgery following DJBL explantation leads to more substantial weight loss and improved diabetes outcome, highlighting the complementary role of bariatric surgery post-DJBL treatment.

Disclosure: Nothing to disclose.

PP0136

CLINICOPATHOLOGICAL CHARACTERISTICS OF OTHER NEW NEOPLASTIC LESIONS MISSED BY IMAGE ENHANCED ENDOSCOPY AT SURVEILLANCE FOR PATIENTS WITH ESOPHAGEAL SQUAMOUS CELL CARCINOMA

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Introduction: Esophageal squamous cell carcinoma (ESCC) often develops the synchronous or metachronous multiple ESCC. Although the Japanese guideline recommended endoscopic examination as surveillance at least once a year after endoscopic resection (ER) of ESCC, the optimal surveillance modality was not specified. In clinical practice, image-enhanced endoscopy (IEE) followed by Lugol chromoendoscopy (LCE) is commonly performed. While LCE reduces the proportion of missed lesion, it has low specificity and causes severe discomfort to the patients, so attempts have been made to omit Lugol's staining. It is unknown what lesions may be missed at this time point by omitting LCE in surveillance after esophageal ER.

Aims & Methods: Thus, this study aimed to clarify clinicopathological characteristics of synchronous or metachronous multiple ESCCs missed by IEE but detected by LCE in surveillance endoscopy after esophageal ER.

We retrospectively analyzed the consecutive lesions that fulfilled all of the following inclusion criteria:

1. Treated with endoscopic submucosal dissection (ESD) or endoscopic mucosal resection (EMR) at our institution between April 2021 and March 2022;
2. Newly detected during surveillance endoscopy after ER or during preoperative endoscopy for other ESCC;
3. Underwent narrow-band imaging (NBI) or blue laser imaging (BLI) observation followed by LCE observation in the esophagus using a high-definition endoscope at the endoscopic examination of lesion detection; and;
4. Biopsy-proven or strongly suspected squamous cell carcinoma.

Patients with a history of chemotherapy, radiotherapy, or surgery for esophageal cancer were excluded. Lesions were classified into the IEE group (detected by NBI/BLI) and the LCE group (detected only after Lugol staining). The lesion in the IEE group were defined when the lesion was clearly visualized and in focus on the endoscopic images with NBI or BLI, or the endoscopic report described that it was detected with NBI/BLI.

We evaluated the features of the lesions in the LCE group by comparing the clinicopathological characteristics between the lesions in the IEE group and LCE group.

The endpoints were proportion of LCE detected lesion and the proportion of target ESCC, defined as ESCC with pathological invasion depth of the muscularis mucosae (pT1a-MM) or deeper and/or positive lymphovascular invasion (LVI) on histological evaluation after ER.

Results: A total of 122 lesions in 96 patients were included. The IEE group comprised 99 lesions (81.1%) in 75 patients, while the LCE group comprised 23 lesions (18.9%) in 21 patients, thus proportion of LCE detected lesion was 18.9%. Most patients (92.7%, 89/96) had grade C of Lugol voiding staining. The median lesion size was 15 mm (range: 4-60 in the IEE group, which was higher than that in the LCE group at 10 mm (5-30). The lesions located on the anterior (23.5%) and left wall (31.3%) tended to be more frequently detected by LCE than those on the right (10.8%) and posterior wall (13.9%).

Finally, 26 lesions were treated with EMR and 96 with ESD. The pathological diagnosis was pT1a-MM in 5 lesions and pT1b-SM1 in 2, resulting in the proportion of target ESCC of 5.7% (7/122). By group, the proportions of target ESCC were 7.1% (7/99) in the IEE group and 0% in the LCE group.

Conclusion: The characteristics of missed lesions in IEE observation tended to be small and located on the left or anterior wall of esophagus at surveillance endoscopy.

Furthermore, all of these lesions were curatively resected with ER, resulting pT1a-EP/LPM without LVI in the pathological diagnosis.

Disclosure: Nothing to disclose.

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PP0137

IMPACT OF TEXTURE AND COLOR ENHANCEMENT IMAGING (TXI) ON ENDOCYTOSCOPIC DIAGNOSIS OF EARLY GASTRIC CANCER

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Introduction: Endocytoscopy (ECS), an ultrahigh-magnification contact-type endoscopy, enables real-time visualization of gastrointestinal mucosal and neoplastic cells. As a result, ECS enhance diagnostic precision as a form of optical biopsy.

However, the reliability of ECS diagnosis is contingent upon adequate staining, which remains a considerable challenge in the stomach due to the abundance of mucus and inflammatory exudates that hinder optimal visualization. Weak vital staining can easily reduce the reliability of ECS diagnosis. Texture and Color Enhancement Imaging (TXI), a digital image enhancement modality developed by Olympus, enhances three key imaging elements—brightness, texture, and color tone—to optimize lesion visualization even under suboptimal staining conditions.

Consequently, TXI may augment the diagnostic utility of ECS.

Aims & Methods: The aim of this study was to evaluate whether TXI enhances the diagnostic performance of ECS in early gastric cancer. Twenty-eight patients with 30 gastric cancers scheduled for magnifying endoscopic examination at Nippon Medical School Tama Nagayama Hospital between February 2024 and March 2025 were prospectively enrolled.

All ECS procedures were performed with GIF-H290EC (Olympus Co., Tokyo, Japan) and the video processors EVIS X1 (Olympus Co., Tokyo, Japan). For in vivo dyeing, double staining with crystal violet and methylene blue was used. ECS was performed with and without TXI in both cancer and non-cancerous legions.

ECS observation is performed by first taking photographs without TXI, and then taking photographs of the same area using the TXI. From each lesion, five ECS images with and without TXI were selected, resulting in a blinded and randomized image catalog (comprising 300 images from 30 gastric cancers and 280 images from 28 non-tumor areas). Two endoscopists independently assessed the images using a numerical rating scale (NRS, 0–10) to evaluate the visibility of nuclear atypia and structural cellular atypia.

Diagnostic accuracy was assessed against pathological diagnosis as the gold standard. ECS diagnoses were rendered based on previously established criteria, wherein lesions exhibiting high-grade atypia were classified as cancer (Kaise M, et al. Endoscopy. 2015).

Results: The overall accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) for EGC diagnosis by the two endoscopists were 82.5%, 84.3%, 80.7%, 81.3%, and 83.8%, respectively. The rate of interobserver agreement between the two endoscopists was high, with a kappa coefficient of $\kappa = 0.797$.

Notably, diagnostic accuracy was significantly higher in the TXI images compared to the non-TXI images (85.2% vs. 79.0%, $p = 0.0027$). While the NRS scores for structural atypia did not differ significantly between the TXI and non-TXI images (4.90 vs. 4.87, $p = 0.61$), the NRS scores for nuclear atypia were significantly elevated in the TXI images (6.88 vs. 5.88, $p < 0.0001$).

Conclusion: The integration of TXI with ECS significantly enhances the visualization of nuclear atypia and improves diagnostic accuracy in the diagnosis of gastric cancer. This synergistic approach facilitates superior optical biopsy capabilities, enhancing its clinical utility in the endoscopic diagnosis of early gastric cancer.

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Disclosure: Nothing to disclose.

PP0138

CLINICAL COURSE AND MANAGEMENT OF PERFORATION AFTER ENDOSCOPIC RESECTION OF DUODENAL TUMORS

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Introduction: Endoscopic resection (ER) of duodenal tumors carries a high risk of intraoperative or delayed perforation because of the thin duodenal wall and exposure to bile and pancreatic juice. Delayed perforation can particularly cause serious condition due to the leakage of the digestive fluids. The detailed management and clinical course following duodenal perforation remains unclear. This study aimed to clarify the optimal management of duodenal perforation following ER.

Aims & Methods: This single-center retrospective study included cases of perforation among patients who underwent ER for duodenal tumors at our institution between April 2013 and October 2024. The study endpoint was the clinical course after perforation, including the hospitalization period after ER for duodenal tumors. Perforation was defined as the presence of extraluminal gas on computed tomography (CT) scan. Delayed perforation was defined as a perforation that was not endoscopically recognized during the procedure but was diagnosed postoperatively.

In contrast, intraoperative perforation was defined as a case in which a perforation was endoscopically identified during the endoscopic procedure. Long hospitalization was defined as a hospital stay of 20 days or longer.

Results: Of the 1,475 cases with duodenal tumors underwent ER during the study period, intraoperative perforation occurred in 8 cases (0.5%) and delayed perforation in 10 cases (0.7%). The median age of patients with perforation was 62 (range: 47–80) years, and 50% (9 patients) were male. The 18 cases included 6 cases of endoscopic papillectomy (EP), 7 cases of endoscopic mucosal resection (EMR), 2 cases of endoscopic submucosal dissection, and 3 cases of endoscopic submucosal resection with a ligation device.

Among the cases of intraoperative perforations, long hospitalization, 61 days of conservative treatment, was required in one case who developed a large perforation during scope insertion of single-balloon enteroscopy for duodenal adenoma resection. The remaining 7 cases achieved successful endoscopic closure intraoperatively and hospitalization periods were within 17 days. Of the 10 delayed perforation cases, 2 occurred after EP and 8 after EMR for non-ampullary tumors. The median time from the end of endoscopic resection to the diagnosis of perforation was 24 hours (4–46 hours). The median hospitalization period of the cases with delayed perforations was 32 days (9–81 days), and significantly longer than that of intraoperative perforation (13 days) ($p<0.05$).

Long hospitalization was required in 8 cases (80%) of delayed perforation. None of the delayed perforation cases required surgical intervention, and endoscopic nasobiliary and nasopancreatic duct drainage (ENBPD) was performed in 7 cases (70%). Ascites was observed on CT scan at the time

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of diagnosis of delayed perforation in 7 of the 8 patients who required long hospitalization. One case without ascites at the diagnosis of delayed perforation still required long hospitalization, possibly due to the 10-hour delay from the diagnosis to ENBPD. On the other hand, two cases who did not need long hospitalization showed no ascites at the diagnosis of delayed perforation.

Conclusion: Compared to intraoperative perforation, delayed perforation was associated with significantly prolonged hospitalization, particularly in cases with ascites. Early detection of ascites and prompt initiation of ENBPD may help mitigate the clinical burden. Establishing an emergent backup system for ENBPD is essential for the safe management of delayed perforation after duodenal ER.

Disclosure: Nothing to disclose.

PP0139

FREEHAND SUTURING FOR APPLICATION OF TRACTION DURING ESD

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Introduction: Recently freehand suturing for defect closure became possible by using barbed wire sutures and the availability of a needle holder for the use in flexible endoscopy. We reported about the development of a new technique for tissue traction during ESD using barbed wire sutures.

Aims & Methods: We aimed to characterize the performance of tethered ESD in an ex-vivo porcine stomach model. In addition we present the first human case.

We performed ESD procedures in an ex-vivo porcine stomach with a long adjacent esophageal part. In each stomach 2 resections were performed. We used a 15 mm polypectomy snare and cutting current to mark standardized artificial lesion. This lesion was marked additionally with cauterization markers and resection was performed outside the markers. After initial cut (entry point), the resulting mucosal flap was sutured using a barbed wire suture (3/0, V-LOC 180) and afterwards sutured to the opposite gastric wall proximal to the lesion. Pulling the suture led to traction to the resection specimen thus opening the submucosal space. Due to the nature of the barbed wires, the tension will not loosen once the suture is tightened. Resections were carried out using a hybrid knife with Endocut Q for mucosal incision and Precise-Sec mode for dissection. The first human case was performed to resect a gastric lesion of 2 cm located at the angulus fold using the same technique in antegrade position.

Results: In total 6 resections were performed in the animal model. The mean size of specimens was 30 mm ± 4 mm. Mean duration of resection procedure was 24,6 ± 9,37 min demanding in total 15,5 ± 5,24 needle grasping maneuvers for 4,17 ± 2,14 stiches. In average 2,33 ± 2,31 tension maneuvers were performed. Perforations of the needle trough the serosal layer was not observed and in no case the suture slipped out of the specimen due to tension force. Perforations did not occur. All specimens were resected en-bloc and completely.

In the first human case the technique was used without any complications. The resection was performed completely in antegrade position although the lesion was located at the angulus fold. The duration of the resection was 2 minutes while the suturing itself only added another 3 additional minutes to the complete resection procedure.

Conclusion: The method of tethered ESD using barbed wire sutures for gastric ESD is feasible, safe and technically not demanding.

It may especially help to resect larger lesions in difficult positions.

Disclosure: Olympus: Fees for trainings, medical products.

PP0140

LONG-TERM OUTCOMES OF ELONG-TERM OUTCOMES OF ENDOSCOPIC SUBMUCOSAL DISSECTION FOR UNDIFFERENTIATED EARLY GASTRIC CANCER, BEYOND EXPANDED CRITERIA

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Introduction: Endoscopic submucosal dissection (ESD) for over 2 cm in size undifferentiated type (UD type) early gastric cancer (EGC) confined to the mucosa is not only challenging, but also long-term outcomes are not well known.

Aims & Methods: To evaluate the long-term outcomes of ESD done for UD type EGCs confined to the mucosa over 2 cm in size and compare the results with those where the lesions were less than 2 cm. 143 patients with UD type EGC confirmed on histology after ESD at a tertiary hospital were reviewed. Cases with synchronous and metachronous lesions and a case with emergency surgery after ESD were excluded. A total of 137 cases were enrolled. 79 cases who underwent R0 resection were divided into 2 cm or less (group A) and over 2 cm (group B) in size.

Results: Among 79 patients who underwent R0 resection, the number in group A and B were 51 and 28, respectively. The mean follow-up period (SD) was 79.71 ± 45.42 months. There was a local recurrence in group A (1/51, 2%) and group B (1/28, 3.6%) respectively. This patient in group A underwent surgery while the patient in group B underwent repeated ESD with no further recurrences in both patients. There was no regional lymph node metastasis, distant metastasis, and deaths in both groups. With R0 resection strategy for ESD on lesions over 2 cm, 20.4% (28/137) of patients were able to avoid surgery compared with expanded indication.

Conclusion: If R0 resection is achieved by ESD, UD type EGCs over 2 cm also showed good and similar clinical outcomes as compared to lesions less than 2 cm when followed for over 5 years. With R0 resection strategy, several patients can avoid surgery.

Disclosure: Nothing to disclose.

PP0141

COMPARATIVE STUDY OF ESD AND SURGICAL RESECTION FOR GASTRIC SETS ORIGINATED FROM MUSCULARISPROPRIA

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Introduction: Endoscopic resection for gastric subepithelialtumors(SETs) originated from the muscularispropria (GSET-PM) has offered less invasive alternatives to surgical resection.

Aims & Methods: The aims of this study were to compare endoscopic subtumoral dissection(ESD) with surgical resection for the removal of GSET-PM. This study involved 17 patients with GSET-PM removed by ESD and 76 patients who underwent curative surgical resection. ESD was attempted in GSET-PM with well marginated tumors which was below 5cm and showed an endoluminal growth pattern according to endoscopic ultrasound(EUS) finding.

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Results: ESD group were more likely to have upper portion(10/17, 58.8%) and surgery group were more likely to have mid portion(41/76, 53.8%) (p=0.039). ESD group were smaller median tumor size (25.6 mm vs 35.9 mm, p=0.037) and higher endoluminal ratio (58.5±9.1 % vs 45.8±15.4 %, p=0.002). ESD group were mostly to have Yamada type III (10/17, 58.8%) and surgery group were mostly Yamada type I (52/76, 68.4%)(p< 0.001).
Complete resection by ESD was lower than by surgical resection (82.4% vs 100%, p< 0.001). In ESD group, 3 performed surgical resection after ESD (1 incompletely resection and 2 uncontrolled bleeding) and 1 showed perforation was completely resected with endoscopic closure.
In surgery group, complications occurred in 6 patients (1 leakage, 1 stricture, 1 hernia and bowel obstruction, 1 wound infection and 2 worsened general condition after surgery).
Although surgery group were lower in complication rate than ESD group (p=0.006), severity of complications were higher in the surgery group and there were no mortalities in the ESD group compared with 2 in the surgery group. There was no statistical difference of recurrence and the follow-up period between two group.
Conclusion: ESD can be one of good options for the resection of endoluminal GSET-PM and could be replace treatment by surgical resection in Yamada type III with a high endoluminal ratio.
Disclosure: Nothing to disclose.

PP0142

PREDICTIVE FACTORS FOR BLEEDING FOLLOWING GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION

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Introduction: Post-procedural bleeding (PPB) is the most common adverse event after gastric endoscopic submucosal dissection (ESD). The BEST-J score was created to predict PPB risk following gastric ESD.
Aims & Methods: We aimed to evaluate predictive factors of PPB and validate the BEST-J score in a large European cohort. We conducted a retrospective cohort study including consecutive gastric epithelial lesions removed by ESD between January of 2005 and June of 2024. Incomplete ESD and patients with missing information that prevented the score calculation were excluded. PPB was defined as presence of compatible clinical picture in the 28 days following the procedure.
Univariate and multivariate analysis were used to identify predictive factors of PPB, and the diagnostic ability of the BEST-J score was calculated in this cohort.
Results: From 990 ESD, 874 were included in the final analysis. PPB occurred in 53 patients (6.1%), with a median time after procedure of 2 days (P25-P75: 1-7). PPB was significantly superior in patients with higher risk as evaluated by the BEST-J score (PPB rate of 3.7%, 3.9%, 19.2% and 35.7% for low, intermediate, high and very-high risk, respectively).
The area under the curve for PPB prediction using the BEST-J score was 0.684 (IC95% 0.598-0.679). For a score ≥ 3 (high and very-high risk), the sensitivity and specificity for PPB detection were 47.2% (IC95% 34.2-60.5) e 88.7% (IC95% 86.4-90.7), respectively. In this sample, the following predictive factors for PPB were identified: ischemic cardiopathy [adjusted Odds Ratio (aOR) 4.351; IC95% 1.660-11.404], medication with warfarin (aOR 18.339; IC95% 4.541-74.064) or direct oral anticoagulants (aOR 7.596; IC95% 2.666-21.641) and lesion size > 20mm (aOR 2.563; IC95% 1.246-5.272).
Conclusion: The BEST-J score presented moderate discrimination for PPB following gastric ESD in our cohort. The identified factors may assist the decision of implementing prophylactic measures in high-risk patients and the creation of more precise prediction models.
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Disclosure: Nothing to disclose.

PP0143

EVALUATION OF THE SU2URA® GI SYSTEM IN HUMAN EX-VIVO SLEEVE GASTRECTOMY MODEL

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Introduction: Endoscopic suturing has been increasingly used for various procedures, including treatment of gastrointestinal perforations, fistulas, leaks, fixation of intraluminal stents and closure of full thickness defects following endoscopic resections[1,2].
Furthermore, the endoscopic gastric remodeling has as a dominant role in the endoscopic bariatric treatment armamentarium.
Su2ura® GI System is a novel, flexible endoscopic suturing device. We previously reported our experience with the Su2ura® GI System in animal experiments[3].
Aims & Methods: The aim of the present study was to assess the Su2ura® GI System using an ex-vivo human sleeve gastrectomy model, with a focus on suturing efficiency, tissue grasping performance and anchor penetration. We aimed to determine the system's potential to improve clinical outcomes in endoscopic suturing.
Methods:
1. The Device:
This novel intuitive and simple-to-use suturing device (3.2 mm diameter) consists of an atraumatic grasper and multiple T-type threaded anchors that can be seamlessly inserted through a single endoscope channel. The sub-serosa is grasped and safely pulled over a static needle and anchor which penetrate the organ wall. The anchor is released outside the wall (full thickness). Multiple anchors can be deployed, all on the same surgical thread. The suture is easily tightened due to the minimal-friction mechanism, and the thread is locked and cut by a unidirectional locking element.
2. Patients:
The trial was conducted on three human gastric sleeve specimens, each excised from consecutive patients undergoing sleeve gastrectomy. Pa-

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tients age, sex, weight, BMI and sleeve dimensions were documented. For comparative analysis, porcine stomach models (20–30 kg) were also tested.

3. Key Performance Evaluations:

- Grasping force and Needle Penetration (NP) force: Measured in the greater curvature and fundus.
- Suture Detachment Force (SDF): Analyzed via pull tests with force meters.

Results: Three patients were included: two females and one male, with a mean age of 26 years. The mean BMI was 54.26. The average sleeve dimensions were 20.1 cm in length and 6.8 cm in width. The mean gastric wall thickness was 1.46 mm. Regarding the forces measurements, NP forces were lower in human specimens compared to porcine models (Table 1), indicating easier anchor deployment in clinical settings. No difference was measured in grasping force between human and porcine models (Table 1), enabling simple comparison between the two models. Penetration forces were consistently higher than grasping forces even in successive attempts, enabling repeated successful suturing. SDF did not differ in proximal and distal stomach, nor with multiple anchors deployed. Finally, SDF values measured in pig stomach were higher than in human.

Metric	Human Specimens	Porcine Specimens
Needle Penetration Force (Kg)	0.42 (mean)	0.60 - 0.70
Grasping Force (Kg)	0.70 (mean)	
Anchors pullout test (1-2 anchors with a cinch)	1.37-2.57	

Table 1: Key Performance Evaluations.

Conclusion: The results in this study are consistent with our previous results in which we demonstrated safety, efficacy and usability on animal in vivo model (chronic and acute). This evaluation exhibited mechanical consistency and reliability in human ex vivo gastric tissue model; tissue grasping, anchor deployment and suture pull-off performance. These findings represent a promising alternative to current endoscopic suturing technologies. However, further in-vivo studies are needed.

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Disclosure: SA, SME, YM, HY, HC, RN, ES, KO: nothing to declare.
M. Shnell: Consultation fees from Boston Scientific, Anchora
E. Scapa: consultation anchora medical.

PP0144

ANALYSING THE AWARENESS OF SUSTAINABILITY AND GREEN ENDOSCOPY IN THE CURRENT GASTROENTEROLOGY WORKFORCE: A NOVEL SYSTEMATIC REVIEW

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Introduction: Gastrointestinal (GI) endoscopy is a high-throughput, resource-intensive specialty that contributes significantly to healthcare's environmental footprint. As global attention shifts toward sustainable healthcare practices, the concept of green endoscopy has emerged. However, the perceptions, awareness, and engagement of the endoscopy workforce remain underexplored—despite their critical role in driving sustainable transformation.

Aims & Methods: This systematic review aimed to evaluate the awareness, attitudes, and perceived barriers among healthcare professionals regarding sustainability in endoscopy, and to synthesize existing knowledge on workforce engagement with green endoscopy practices.

A systematic search of PubMed, Scopus, Embase, and the Cochrane Library was conducted up to April 1, 2025, supplemented by grey literature from Google Scholar.

Eligible studies included original research reporting data on healthcare professionals; knowledge, attitudes, or practices related to sustainable endoscopy. Two independent reviewers screened studies and extracted data. Study quality was assessed using the Joanna Briggs Institute (JBI) checklist. A synthesis without meta-analysis (SWiM) approach was used due to heterogeneity in study design and outcome reporting.

Results: Five cross-sectional studies comprising 1,684 participants were included, spanning multiple global regions. Awareness of green endoscopy varied widely, from 16.3% to 67%. While most respondents acknowledged endoscopy's contribution to hospital waste and carbon emissions, fewer reported institutional guidelines or formal training.

Major barriers to sustainability included lack of knowledge (35%), insufficient institutional policy (32%), infection control concerns (41%), and perceived financial constraints (26%).

Few departments conducted audits or offered staff education on sustainability. A generational divide in awareness and engagement was observed, with more experienced staff demonstrating greater concern for environmental impacts.

Conclusion: The perspectives and knowledge of the current gastroenterology workforce on sustainability remains poor and further action to improve this is essential.

Disclosure: Nothing to disclose.

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PP0145

IMPROVING IMAGE DOCUMENTATION IN UPPER GI ENDOSCOPY AT A DISTRICT GENERAL HOSPITAL

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Introduction: Upper gastrointestinal (UGI) endoscopy, or Oesophagogastroduodenoscopy (OGD), is one of the most commonly performed procedures in gastroenterology, serving both diagnostic and therapeutic purposes across a wide range of medical and surgical conditions. The European Society of Gastrointestinal Endoscopy (ESGE) provides standardised guidelines covering various aspects of UGI endoscopy, including image documentation. According to ESGE recommendations, images of eight key anatomical landmarks should be captured and included in the final report. An analysis of OGD reports from 6th January 2025 to 10th January 2025 at Royal bolton Hospital revealed that only 29.4% were compliant with these guidelines.

Aims & Methods: At Royal Bolton Hospital, the Endobase portal is used to document and report OGD findings, providing a comprehensive record of each procedure, including duration, sedation use, general comments, findings, accompanying images etc. We undertook a retrospective analysis of all OGDs performed over a defined period to assess adherence to ESGE guidelines. Our objective was to achieve a departmental compliance rate of 90% or higher in capturing image documentation of the eight recommended anatomical landmarks, as outlined by ESGE, and to ensure that any challenges in meeting this standard were appropriately documented.

We started the QIP(Quality improvement Project) on 06/01/2025 by measuring the data from all the patients admitted to the endoscopy suite for an OGD. Pre intervention data showed that the compliance to ESGE image documentation was poor at 29.4%.

A structured proforma was developed to standardise data collection, focusing on procedure duration, sedation use, number of images captured, completeness of landmark documentation, and whether non-compliance was justified.

Two PDSA(plan-do-study-act) cycles were completed and the data collected at the end of each PDSA cycle were analysed and their effectiveness assessed.

These were as follows:

WEEK	INTERVENTION	EFFECT OF INTERVENTION
1	The importance of image documentation whilst reporting OGDs compliant with ESGE recommendations was highlighted in an email and forwarded to endoscopists across the trust.	The compliance in image documentation improved from 29.4% to 54.7%
2	Well-designed posters were displayed in all endoscopy rooms, positioned opposite the computers where endoscopists—including gastroenterology consultants, consultant surgeons, registrars and nurse endoscopists routinely sit to document procedures on Endobase.	The compliance in image documentation significantly improved from 54.7% to 90.7%.

Results: Prior to intervention, compliance with ESGE standard image documentation was observed in 29.4% of cases, and the reason for inadequate documentation was recorded in only 16.6%. Following the first audit cycle, ESGE-compliant documentation improved to 54.7%, with documentation of reasons for non-compliance increasing to 34.8%. After implementing further interventions, the second cycle showed a marked improvement, with ESGE standard image documentation achieved in 90.7% of cases and the reason for inadequate documentation recorded in 55.5%.

Conclusion: This QIP has demonstrated that meaningful improvements in clinical practice can be achieved through simple, well-targeted interventions. By raising awareness of ESGE documentation standards and reinforcing them through consistent visual prompts, we were able to significantly enhance compliance with image documentation during OGDs—from just under 30% at baseline to over 90% by the end of our second PDSA cycle. It was equally encouraging to see a parallel improvement in the recording of reasons when documentation was incomplete, reflecting a shift toward more thoughtful and accountable reporting.

Personally, this project has highlighted for me the value of structured quality improvement in shaping safer, more standardised patient care. Looking ahead, maintaining this progress through ongoing audits and wider departmental engagement will be crucial in embedding these practices long-term.

Reference:

Ahmed A et al, *QIP: improving quality of photodocumentation and lesion reporting in upper gastrointestinal endoscopy*

Disclosure: Nothing to disclose.

PP0146

UPPER GASTRO-INTESTINAL EROSIVE AND ULCERATIVE LESIONS IN CIRRHOTIC PATIENTS: FREQUENCY AND PREDISPOSING FACTORS

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Introduction: Upper gastro intestinal endoscopy is commonly indicated in cirrhotic patients to detect signs of portal hypertension. However, other gastric manifestations, including erosions and ulcers, can also be observed. These non-variceal lesions are often under diagnosed and less studied and may lead to chronic anemia or gastro intestinal bleeding, underestimating the importance of comprehensive endoscopic evaluation in cirrhotic patients.

Aims & Methods: The aim of ourstudy was to assess the frequency of upper gastro-intestinal erosive and ulcerative endoscopic lesions in cirrhotic patients and evaluate their association with the severity of the underlying cirrhosis.

A retrospective descriptive and analytical study was conducted, including all cirrhotic patients who underwent upper gastrointestinal endoscopy between january 2024 and december 2024. Biopsies were performed only in patient with macroscopic lesions. Erosive or ulcerative lesions were defined by the presence of erosions or ulcerations not related to varices or previous sclerotherapy. Patients on gastro-aggressive medications or proton pump inhibitors prior to endoscopy were excluded.

Results: Sixty-one patients were included, with a mean age of 64 years and a female predominance (59%). The main etiologies of cirrhosis were non-alcoholic fatty liver disease (NAFLD) in 32% of cases, hepatitis C in 21%, hepatitis B in 13%, and primary biliary cholangitis in 12%. Most patients had a Child-Pugh score of B (53%), with the remaining cases distributed between Child-Pugh A (26%) and Child-Pugh C (21%)with a mean MELD score of 14. Only six patients were active smokers. Forty-four patients (72%) experienced at least one episode of edemato-ascitic decompensation. Among them, 19 patients (31%) had refractory ascites, and 19 were on secondary antibiotic prophylaxis for ascitic fluid infection.

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Radiological features suggestive of portal hypertension were identified in 78% of patients. Esophageal varices were detected in 82% of cases, with grades 1, 2, and 3 observed in 21%, 36%, and 35% of patients.
Non-portal hypertension-relatedendoscopiclesionswereobservedin 47% of cases, includinggastritis (13%), gastric erosions (13%), gastric ulcerations (6%), duodenal bulb ulcer (11%) and gastricpolyps (6%). Uponhistopathologicalexamination, *Helicobacter pylori*wasidentifiedin 8 patients.
Univariate analysis showed a significant association between the presence of non-portal hypertension-related Upper gastro-intestinal erosive and ulcerative lesions and refractory ascites (p=0.031). No statistically significant relationships were found between these lesions and the presence of portal hypertension signs, severe thrombocytopenia, smoking, cirrhosis severity scores or cirrhosis etiology.
Conclusion: Upper gastro-intestinal erosive and ulcerative lesions are common in cirrhotic patients, especiallythosewithrefractory ascites. Endoscopic screening in thisspecific population could help prevent digestive complications, such as chronic anemia and bleeding.
Disclosure: Nothing to disclose.

PP0147

ENDOSCOPIC SUBMUCOSAL DISSECTION AND DUODENAL LAPAROSCOPIC AND ENDOSCOPIC CO-OPERATIVE SURGERY FOR SUPERFICIAL NON-AMPULLARY DUODENAL EPITHELIAL TUMORS

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Introduction: The ulcers after duodenal endoscopic resection (ER) are exposed to bile and pancreatic juice, therefore delayed perforation and bleeding are major complications. The condition caused by perforation can become serious, requiring surgical treatment and even death.
As a preventive measure, the suture of the ulcer after ER has been reported to be effective. There are two main suture methods: endo clips and laparoscopic suturing.
Aims & Methods: The aim of this retrospective study is to investigate the outcome of Endoscopic submucosal dissection (ESD) and laparoscopic and endoscopic co-operative surgery (D-LECS) for superficial non-ampullary duodenal epithelial tumors (SNADETs). ESD ulcer was sutured with endo clips in ESD group, and with laparoscopic suture in D-LECS group.
One hundred seventy-eight patients with SNADETs who underwent ESD and 41 patients with SNADETs who underwent D-LECS from November 2002 to February 2025 were investigated outcome of procedure. The patient's characteristics in ESD and D-LECS were as follows. male/female: 124/54 and 27/14, median age: 62 (37-86) and 67 (37-83) years old, location of the tumor (bulbs/2nd/3rd portion): 15/139/24 and 0/32/9, major macroscopic type (0-I/0-IIa/0-IIb/0-IIc): 5/91/1/81 and 1/32/0/8. Only major macroscopic type was difference significantly between ESD and D-LECS (p=0.001). SM1 was defined as submucosal invasion of less than 500 μm, and SM2 was defined as deeper. R0 was defined as lateral and vertical margin negative, and lymph-vascular involvement negative. DLECS was selected when it was predicted that endo clips suture might be insufficient, or when the scope maneuverability was poor.
Results: 1. Median procedure time was 36 (7-245) in ESD and 351 (202-673) minutes in D-LECS, respectively(p<0.001). Median diameter of resected specimen was 17 (8-66) and 38 (17-70) mm(p<0.001). Median diameter of resected specimen and tumor was 10 (2-63) and 27 (14-52) mm(p<0.001). Intramucosal tumor (Vienna classification category 3 to 5.1) /T1bSM (category 5.2): 176/2 and 37/4 (p=0.01).
2. The en bloc resection was 90% (161/178) and 95% (39/41), respectively (p=0.53). R0 resection rate was 85% (151/178) and 88% (36/41) (p=0.62).
3. The rate of complete suture for ESD ulcer was 92% (164/178) in ESD group and 100% in D-LECS group, respectively (p=0.07). The details in D-LECS are as follows: 27% (11/41) of laparoscopic suture alone and 73% (30/41) of laparoscopic and endo clips suture.
4. Complications: intraoperative perforation was 2% (4/178) and 10% (4/41) (p=0.04). Intraoperative bleeding requiring blood transfusion was 0% and 0%. Delayed perforation was 1% (2/178) and 0% (p=1.00). Delayed bleeding was 2% (4/178) and 0% (p=1.00).
Discussion:
1. The DLECS group treated larger cases and had more intraoperative perforations, but no delayed perforation or bleeding was observed.
2. Laparoscopic suture of the pancreatic side was not possible, but complete suture was achieved in all cases by adding a clip from the luminal side endoscopically.
3. Even after complete suture by ESD, delayed perforation can still occur. Therefore, if abdominal pain occurs, a CT scan should be performed immediately. And if perforation is suspected, laparoscopic re-suture should be considered immediately.
Conclusion: DLECS for large SNADETs has similar outcome of ESD, and is an effective treatment. DLECS should be considered when complete suture with a clip is predicted to be difficult or when the scope maneuverability was poor.
Disclosure: Nothing to disclose.

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PP0148

FREQUENCY AND RISK FACTORS FOR MALIGNANT TRANSFORMATION OF GASTRIC HYPERPLASTIC POLYPS IN A COUNTRY WITH AN INTERMEDIATE-HIGH RISK FOR GASTRIC CANCER

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Introduction: Gastric hyperplastic polyps (GHPs) carry a neoplastic transformation risk of up to 5%, with previous studies indicating an increased risk with larger polyp size. We aimed to evaluate the frequency and risk factors of its malignant transformation in a Portuguese cohort.

Aims & Methods: A multicenter retrospective cohort study was conducted including all patients undergoing endoscopic resection of GHPs ≥10mm from January 2013 to January 2024 in Portuguese tertiary reference centers. Patients with hereditary polyposis syndromes were excluded. Demographic, clinical, endoscopic, and histopathological data were collected.

Results: A total of 782 GHPs from eight centers were evaluated. The median age was 68 years (IQR 60-75), and 58.8% were female. Evidence of advanced gastritis stage (OLGIM≥3) and synchronous gastric neoplasia were present in 57 (10.2%) and 37 (4.7%) cases, respectively. The median lesion size was 15 mm (IQR 12-20), and most presented a polypoid morphology (96.8%). Most GHPs were removed by endoscopic mucosal resection (86.6%), followed by cold snare polypectomy (8.1%) endoscopic submucosal dissection (3.6%), and cold snare EMR (1.7%), achieving *en-bloc* resection in 677 cases (86.6%). Neoplastic transformation was observed in 33 polyps (4.2%) – 25 (3.2%) low/high-grade dysplasia and 8 (1.0%) adenocarcinoma; the resection was non-curative in 30.3% of cases.

In the multivariate analysis, OLGIM≥3 (OR 4.994, 95%CI 1.036-24.065), lesion size ≥25 mm (OR 7.973, 95%CI 7.973 (1.242-51.180), flat/depressed morphology (OR 9.57, 95%CI 1.096-83.712) and synchronous gastric neoplasia (OR 16.956, 95%CI 2.737-105.045) were independently associated with a higher risk of malignant transformation.

Conclusion: The risk of malignant transformation of GHPs in non-neglectable, with lesion size and morphology and concomitant precancerous gastric conditions/lesions being associated with a higher risk. Nevertheless, the rate of non-curative endoscopic resection is high. The identification of these features may help for optimizing and tailoring the selection of adequate endoscopic resection techniques and surveillance strategies in high-risk patients.

Disclosure: Nothing to disclose.

PP0149

ENDOSCOPIC VACUUM THERAPY IN POSTOPERATIVE UPPER GI LEAKS: CLINICAL INSIGHTS AND OUTCOMES FROM A TERTIARY REFERRAL CENTER

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Introduction: Anastomotic leaks and fistulae following upper gastrointestinal (GI) surgery remain a serious clinical challenge, carrying high morbidity and mortality. Endoscopic vacuum therapy (EVT) has emerged as a promising minimally invasive treatment option for managing these complications.

This study aimed to assess the efficacy and safety of EVT for postoperative anastomotic leaks in a high-volume tertiary referral center.

Aims & Methods: We conducted a retrospective analysis of all patients treated with EVT for anastomotic leakage following upper GI surgery between January 2022 and January 2025. The primary outcome was EVT success, defined as complete leak closure confirmed through both endoscopic and radiologic assessment. Secondary outcomes included procedure-related adverse events and all-cause mortality.

Results: Twenty-four patients were included (79.2% male; mean age: 64 ± 12 years). Of these, 16 patients underwent oesophagectomy and 8 total gastrectomy; only two surgeries were performed for benign conditions. EVT achieved leak closure in 20 patients (87%). One patient died due to hemorrhagic shock from a hepatic artery pseudoaneurysm, unrelated to EVT.

Among successful cases, the mean duration of EVT was 30 ± 19 days, with a median of four sponge exchanges (IQR = 5). Median hospital stay was 52 days (IQR = 41). EVT failure occurred in three patients, all of whom had cancer-related surgeries and defect sizes exceeding 30 mm. In these cases, adjunctive endoscopic treatment with self-expanding metallic stents led to successful leak closure.

A larger defect size was associated with a lower likelihood of closure (p = 0.067), suggesting a trend toward reduced efficacy in extensive leaks. Importantly, no EVT-related complications were observed throughout the study period.

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Conclusion: EVT demonstrated a high success rate (87%) and an excellent safety profile for managing postoperative anastomotic leaks following upper GI surgery. Larger leak sizes showed a trend toward lower closure rates, highlighting the importance of early diagnosis and timely initiation of EVT when defects are smaller.

Given its effectiveness and minimal risk, EVT should be considered a first-line endoscopic treatment for anastomotic leaks. Further studies are warranted to refine treatment protocols and establish standardized indications based on leak characteristics.

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Disclosure: Nothing to disclose.

PP0150

POSITIVE AIRWAY PRESSURE SYSTEMS VERSUS STANDARD OXYGENATION IN PATIENTS AT RISK OF HYPOXEMIA UNDERGOING GASTROINTESTINAL ENDOSCOPIC PROCEDURES: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED STUDIES WITH TRIAL SEQUENTIAL ANALYSIS

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Introduction: Managing hypoxemia during procedural sedation, particularly in patients with obesity or obstructive sleep apnea (OSA), remains a significant challenge in clinical practice. This meta-analysis compared the use of positive airway pressure (PAP) with standard oxygenation therapy for patients at increased risk of hypoxemia undergoing gastrointestinal endoscopic procedures.

Aims & Methods: MEDLINE, Embase, and Cochrane databases were searched for randomized controlled studies comparing PAP systems versus standard oxygenation. Odds ratio (OR) was used for binary outcomes and mean difference (MD) for continuous outcomes with their respective 95% confidence interval (CI). Further research implications were evaluated by a trial sequential analysis (TSA).

Results: Our meta-analysis included six RCTs comprising 606 patients. Compared with standard oxygenation, PAP systems reduced the rate of hypoxemia (OR 0.21; 95%CI 0.12 to 0.38, p<0.001), and severe hypoxemia (OR 0.15; 95%CI 0.05 to 0.49, p=0.002). Additionally, the PAP group experienced fewer arrhythmic events (p=0.04), required fewer airway maneuvers (p=0.002), and had a reduced total propofol dosage (p=0.04). There was no significant difference between the groups in terms of hypotension events (p=0.39), procedure duration (p=0.62), hypoxemia episode duration (p=0.78), anesthesiologist satisfaction (p=0.44), or endoscopist satisfaction (p=0.79).

Conclusion: Overall, the use of PAP systems appears to be a promising strategy for optimizing sedation and enhancing patient safety during gastrointestinal endoscopic procedures for patients at risk of hypoxemia.

Disclosure: Nothing to disclose.

PP0151

ROLE OF DIGESTIVE ENDOSCOPY IN THE EVALUATION OF ANEMIA AMONG THE ELDERLY

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Introduction: Anemia is a common and often underdiagnosed condition in the elderly population. Digestive endoscopy plays a pivotal role in the etiological workup, particularly to rule out serious underlying causes such as malignancies or peptic lesions. However, the diagnostic yield of such procedures in routine clinical practice remains variable and underreported, especially in resource-limited settings.

Aims & Methods: To assess the diagnostic yield of digestive endoscopy in the investigation of anemia in elderly patients, with a particular focus on iron deficiency anemia (IDA) and vitamin B12 deficiency, and to identify specific endoscopic findings associated with these etiologies.

We conducted a retrospective descriptive study over a 6-month period. The study included all patients aged 65 years and older who underwent esophagogastroduodenoscopy (EGD) and/or colonoscopy as part of the diagnostic evaluation for anemia. Data collected included demographic information, clinical symptoms, laboratory parameters (including hemoglobin, ferritin, vitamin B12 levels), endoscopic findings, and histopathological results when available.

Results: A total of 252 patients were included in the analysis. Among them, 44 patients (17.5%) with a mean age of 71 years [68-82], had confirmed IDA, and six patients (2.4%) presented with vitamin B12 deficiency anemia, associated in all cases with histologically confirmed fundic atrophy.

In univariate analysis, IDA was statistically correlated with the presence of peptic esophagitis (p=0.04), gastric neoplasia (p=0.001), erosive gastropathy (p=0.02), and *Helicobacter pylori* gastritis (p=0.04). It was also weakly correlated with the presence of colorectal cancer (p=0.05).

Colonic angiodysplasias and colonic polyps were not associated with IDA in this cohort (p=0.1 and p=0.4, respectively).

Conclusion: Digestive endoscopy, particularly EGD, remains a cornerstone in the diagnostic algorithm of anemia in the elderly. It enables the identification of a wide range of underlying pathologies, from inflammatory lesions to malignancies, and guides appropriate therapeutic interventions. Early endoscopic evaluation should be encouraged in this population to prevent diagnostic delays and improve outcomes.

Disclosure: Nothing to disclose.

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PP0152

LEGAL RISK ASSOCIATED WITH GASTROINTESTINAL ENDOSCOPY AND ENDOSCOPIC POLYP MANAGEMENT

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Introduction: Implementation of novel endoscopic technology or methods is sometimes limited by fears of litigation. Endoscopists might have concerns regarding the medico-legal risks of mismanagement, particularly arising from missed malignant lesions. I.E. computer aided optical polyp diagnosis and the "resect and discard" strategy are promising tools for cost-effective management of diminutive colorectal polyps. However, concerns about litigation contribute to the limited uptake of this approach in routine clinical practice.

This study evaluates the risks of litigation associated with endoscopic polyp management and examines whether adherence to clinical guidelines mitigates these risks.

Aims & Methods: A review of legal databases (CanLII, SOQUIJ, and LexisNexis) was performed. A total of 818 medico-legal cases related to endoscopic procedures were identified and analysed. The analysis focused on identifying common causes of litigation, including procedural errors, diagnostic delays, and referral failures, and assessing outcomes in cases where guideline adherence was documented. Guideline recommedations from the American Society for Gastrointestinal Endoscopy (ASGE) and the United States Multi-Society Task Force on Colorectal Cancer (USMSTF) were used to evaluate the role of professional standards in mitigating liability.

Results: Our analysis revealed that gastroenterologists faced minimal legal risk, with a defense success rate of 86.7% in the USA and 80% in Canada. Litigation primarily involved procedural errors among gastroenterologists and surgeons, and diagnostic or referral failures among family physicians. Importantly, no cases of litigation were linked to mismanagement of diminutive polyps when guidelines were followed. Adherence to ASGE and USMSTF recommendations was associated with favorable outcomes, underscoring the importance of evidence-based practice in reducing liability risks.

Conclusion: This in-depth legal review indicates that litigation risks associated with diminutive polyp management are minimal when evidence-based guidelines are adhered to. These findings reinforce the importance of adhering to ASGE and USMSTF recommendations to ensure patient safety and reduce medico-legal exposure.

The "resect and discard" strategy, when implemented within the framework of professional standards, offers a safe, efficient, and legally sound approach to managing diminutive polyps, supporting its broader adoption in clinical practice.

Disclosure: Nothing to disclose.

PP0153

COLD SNARE POLYPECTOMY: A SAFE ALTERNATIVE TO HOT SNARE POLYPECTOMY FOR SMALL GASTRIC POLYPS

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Introduction: Polypectomy is a common endoscopic procedure with cold snare and hot snare techniques being widely used. Hot snare polypectomy involves cautery, which may increase the risk of thermal injury and related complications. Cold snare polypectomy has a well-established safety and efficacy profile for small colonic polyps, however, comparative data on the clinical outcomes of these two methods for gastric polyps remains limited.

Aims & Methods: A retrospective analysis was conducted on 161 consecutive gastric polypectomies for polyps under 10mm, comprising a sample of 41 patients that underwent cold snare polypectomy and 120 patients that underwent hot polypectomy over a period of 5 years.

Results: The mean polyp size differed significantly between groups, being smaller in the cold snare group (5.20 ± 1.18 mm *vs* 6.34 ± 1.49 mm; $p < 0.001$). The Charlson Comorbidity Index averaged 3 points, with no significant differences between the groups regarding the use of anticoagulant or antiplatelet therapy.

Statistically significant differences were observed in both the location and histological type of the polyps between the two techniques ($p = 0.003$; $p < 0.001$). Cold snare polypectomy was more frequently used for polyps located in the fundus and for fundic gland polyps, whereas hot snare polypectomy was more commonly utilized for polyps located in the cardia and antrum.

Submucosal injection with gelofusine and adrenaline were more frequently associated with the hot snare technique ($p = 0.001$). Regarding clinical outcomes, intra-procedural bleeding occurred in 16.1% of cases and was successfully managed endoscopically in all instances, mainly with clips (76.9%).

No cases of delayed bleeding or perforation were observed. There were no significant differences between the groups concerning complete histological resection or polyp recurrence.

Conclusion: Cold snare polypectomy demonstrated comparable safety and efficacy to hot snare polypectomy in the resection of small gastric polyps, possibly representing a preferable approach to avoid thermal injury and related complications.

Disclosure: Nothing to disclose.

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PP0154

GUT MICROBIOME AS A BIOMARKER FOR PREDICTING NEOADJUVANT CHEMOTHERAPY RESPONSE IN GASTRIC CANCER

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Introduction: Emerging experimental and clinical data suggest that the gut microbiome can significantly influence the antitumor efficacy of key chemotherapy agents, including 5-fluorouracil, oxaliplatin, and docetaxel. These agents form the backbone of the neoadjuvant chemotherapy (NAC) regimen, FLOT, which is widely used for the treatment of locally advanced gastric cancer (GC). However, despite growing evidence from preclinical models, clinical studies investigating the direct association between gut microbiome composition and the response to NAC in GC patients remain scarce.

Aims & Methods: This study aims to address this gap in knowledge by exploring the relationship between the gut microbiome and therapeutic outcomes in this clinical context. In this longitudinal observational study, we explored the association between pre-treatment fecal microbiome composition analyzed via *16S rRNA* gene sequencing and histological tumor regression, as assessed by the Becker score, in GC patients undergoing NAC with the FLOT regimen, followed by gastrectomy.

Results: Among the 38 enrolled patients, 19 (50%) achieved a Becker score of 1–2 and were classified as 'Responders,' while the other 19 (50%) were classified as 'Non-responders'. Responders exhibited a higher abundance of *Parabacteroides distasonis*, *Alistipes sp. AL-1*, and *Bacteroides plebeius*. A combined biomarker of these species demonstrated robust predictive accuracy for a favorable response to NAC, with sensitivity of 89%, specificity of 63%, and an AUC of 0.8 (95% CI: 0.66–0.94).

In contrast, non-responders showed a predominance of bacterial genera, including *Ruminococcaceae* UCG-014, *Ruminococcus 1*, *Ruminococcaceae* UCG-005, and the *[Eubacterium] eligens* group. The combined biomarker of these genera demonstrated excellent predictive performance for poor NAC response, with sensitivity of 89%, specificity of 95%, and an AUC of 0.93 (95% CI: 0.85–1.0).

Conclusion: The results of our study support the hypothesis that the pre-treatment gut microbiome composition can serve as a reliable biomarker for predicting both positive and negative responses to NAC in GC patients. By identifying specific microbial signatures associated with chemotherapy outcomes, this research provides critical insights into how microbiome-based strategies might enhance personalized treatment regimens for GC.

Future studies are needed to validate these findings and explore potential therapeutic interventions targeting the microbiome to improve clinical outcomes in GC patients.

Disclosure: Nothing to disclose.

PP0155

PROTECTIVE EFFECTS OF STW 5 AGAINST CHANGES IN THE MICROBIOME IN AN EXPERIMENTAL MODEL OF FUNCTIONAL DYSPEPSIA

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Introduction: STW 5, a herbal medicinal product with well characterized pharmacological action, has been shown in several RTCs to be an effective treatment in disorders of the gut-brain interaction (DGBI) as functional dyspepsia (FD) [1, 2].

The aim of the present study was to examine whether changes in gut microbiota induced in an experimental model of FD could be prevented by STW 5.

Aims & Methods: FD was induced in rats by subjecting them to neonatal maternal separation followed by subjecting them to restraint stress for 90 minutes/day for one week. During that week, one group (8 animals) was treated with the liquid form of STW 5 (5 ml/Kg) and one with its vehicle at the same dose level. Animals were sacrificed 24 h after the last drug administration. Fecal samples were taken from the cecum. Genomic DNA was isolated and changes in selected bacterial phyla and genera were assessed using quantitative Real Time-PCR (qPCR). The main phyla studied under the influence of stress in the model were the Bacteroidetes, Firmicutes, Fusibacterium and Actinobacteria.

Results: FD led to a 50% drop in the Lactobacilli population but to a two-fold rise in Proteobacteria abundance but did not affect significantly the changes in Bacteroidetes. Furthermore, the Methanobrevibacter population was raised more than two-fold while the genomic DNA concentration in Bifidobacterium dropped from ca. 3 ng/g feces to nearly nil.

Conclusion: In general, STW 5 tended to ameliorate and protect against the changes induced by the stress model. The results lend further support to the use of STW 5 in FD and IBS by providing evidence that it acts at least in part through influencing beneficially the changes in intestinal microbiota induced by stress. Taken together, the findings shed more light on mechanisms involved in the therapeutic action of STW 5 in gastro-intestinal disorders.

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PP0156

LONG-TERM EUS FOLLOW-UP OF SMALL GASTRIC ASYMPTOMATIC MUSCULAR LESIONS

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Introduction: Gastrointestinal stromal tumors (GISTs) are soft tissue sarcomas with a potential malignant transformation and represent the most common form of subepithelial lesions (SELs). SELs with high-risk eco-endoscopic (EUS) features ordinarily undergo endoscopic or surgical resection even without a previous histological confirmation of GIST.

On the contrary, scarce data are available about the long-term risk of neoplastic evolution of small asymptomatic SELs without worrisome features, making it challenging to schedule an adequate follow-up.

Aims & Methods: Aim of the study is to to assess the natural history of these small asymptomatic "likely GISTs" of the stomach and to test if a follow-up strategy of such lesions could be correct, by sparing a gastric resection if not needed.

We retrospectively selected patients with previously endoscopic documentation of asymptomatic gastric SELs referred to two Italian centers for EUS execution between February 2002 and January 2024.

Patients with endoscopic (ulcerated overlying mucosa) or EUS (diameter >20 mm, inner inhomogeneous pattern, irregular margins) high-risk criteria were excluded. EUS characteristics of the remaining patients were collected and analyzed. In particular, a 25% increase in the main diameter during follow-up was defined as a significant increase in tumor size, whereas, heterogeneous echo texture, irregular extraluminal border, echogenic foci, and anechoic space on EUS were considered suspicious for malignant changes. Statistical analysis was performed.

Results: A total of 62 patients were included. Median age was 58.7 years (± 12.6 years) and 41 were females (66.1%). Lesions were mainly found in the fundus (n=34, 54.8%) and body (n=25, 40.3%), only 3 lesions (4.8%) were found in the antrum. The mean diameter at baseline was 12.4 ± 4.2 mm.

After follow-up (62.6 ± 52.3 months), 6 lesions (9.7%) showed a significant increase in tumor size, while no lesion developed malignant changes. Among them, 4 underwent treatment or additional investigation (surgery n= 1, EUS-guided tissue acquisition n= 3) with a final diagnosis of GIST in 2 patients. Considering the entire cohort, 9 other patients were treated or further studied after multidisciplinary evaluation (surgery n=2, EUS-guided tissue acquisition n=7) with the detection of GIST in 2 patients.

Statistical analysis was performed using logistic regression analysis, both unadjusted and adjusted. Baseline diameter showed a trend of association with the growth of the lesions (OR 1.28, p-value 0.071. Moreover, SELs of the body of the stomach seemed less likely to show growth over time in contrast to antrum and fundus.

Conclusion: Our study suggests that Small gastric SELs have a low rate of growth over time. Although the diagnostic accuracy is still suboptimal, EUS has a crucial role in the follow-up of patients thanks to its high accuracy in the detection of SELs growth and pattern changes. Given these results, a conservative approach based on a surveillance program with EUS could be the best choice.

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PP0157

EVALUATION OF CONGENITAL ESOPHAGEAL STENOSIS WITH HIGH FREQUENCY MINIPROBE

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Introduction: The most common cause of esophageal stenosis in children is acquired (95%), but it can be congenital in 5% of cases. Tracheobronchial remnants (TBR) are the most frequent cause, but membranous diaphragm and fibromuscular (FMS) stenosis are other possible etiologies for this type of narrowing. Dysphagia is the initial clinical manifestation, occurring after the introduction of solid food. Diagnostic investigation begins with esophagography and endoscopy, but the definitive diagnosis is only possible through histopathology. EUS (endoscopic ultrasound) can provide a diagnosis with a good degree of certainty and correlation with histopathological findings. The presence of tracheobronchial remnants in these patients means that endoscopic dilatation attempts could lead to perforation.

Aims & Methods: The aim of this study is to report two cases of congenital esophageal stenosis in children diagnosed by EUS miniprobe.

Results: Case 1: A three-year-old girl with esophageal stenosis after correction of esophageal atresia underwent esophageal dilatations. Endoscopy showed a stenosis in the distal esophagus.

Contrast esophagography confirmed a narrowing of the distal esophagus with marked proximal dilatation. EUS miniprobe radial scanning at 20 MHz of the esophagus showed a complete hyperechoic ring outside the esophageal wall. The patient underwent surgical resection, and pathology revealed the presence of cartilaginous plates, seromucinous glands, and ciliated epithelium resembling bronchial tissues.

Case 2: A six-year-old boy was admitted for recurrent vomiting and failure to thrive. Esophagography showed a tapered narrowing in the middle esophagus with dilation of the proximal esophagus. Biopsies were negative for eosinophilic esophagitis. A high-frequency catheter probe EUS suggested fibromuscular hypertrophy. The boy was referred for esophageal dilatations.

Conclusion: The diagnosis of CES is difficult. Endoscopic biopsy specimens are usually negative, and EGD typically shows a nonspecific esophageal stenosis without inflammation. Generally, CT or magnetic resonance imaging are not useful.

The presence of tracheobronchial remnants in these patients means that endoscopic dilatation attempts could lead to perforation. EUS miniprobe is useful for diagnosis. This miniprobe EUS examination is safe, quick, and

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provides valuable information to surgeons about the nature and length of the stenosis, as well as its relationships with surrounding organs such as the aorta, trachea, and pericardium.

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Disclosure: Nothing to disclose.

PP0158

UTILITY OF KONJAC GLUCOMANNAN GEL AS COUPLING MEDIUM IN MINIPROBE SONOGRAPHY

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Introduction: Miniprobe sonography (MPS) is a high-resolution endoscopic ultrasound technique widely utilized for assessing the layered structure of gastrointestinal (GI) lesions[1].

Degassed water is commonly used as the coupling medium for MPS; however, it presents limitations such as image instability, technical complexity, and an increased risk of coughing or aspiration[2-3].

Konjac glucomannan (KGM) gel, with its good biocompatibility and high adhesiveness, may be a suitable alternative[4].

This study aims to evaluate the feasibility of KGM gel as an MPS coupling medium and compare its imaging quality and operational performance with that of degassed water.

Aims & Methods: This study included three parts: in vitro experiments, animal studies, and clinical trials. In the first part, the physical properties and imaging quality of KGM gel at different concentrations were assessed in ex vivo organ models to determine the optimal formulation with moderate adhesiveness and superior imaging quality.

In the second phase, MPS was performed on the GI tract of the pig using both degassed water and KGM gel, and their imaging quality and operational characteristics were compared. In the third phase, a total of 39 subjects were enrolled, and MPS examinations were conducted on 47 exploration sites involving the normal digestive tract wall and/or digestive tract lesions, sequentially using degassed water and KGM gel. Imaging quality, diagnostic accuracy in lesion depth and origin assessment, and operational performance were evaluated.

Results: In vitro experiments identified 1.0% KGM gel as the optimal concentration, with no statistically significant difference in imaging quality compared to degassed water ($P > 0.05$). In animal studies, the overall imaging quality between the KGM gel group and the degassed water group showed no statistically significant difference ($P > 0.05$); moreover, in the esophageal and rectal subgroups, the KGM gel group demonstrated superior imaging quality with statistically significant differences ($P < 0.05$), whereas no significant difference was observed in the gastric subgroup ($P > 0.05$). The KGM gel group also required significantly shorter operation time, and a lower volume of medium compared to the degassed water group ($P < 0.01$). Clinical trials showed no significant difference in overall imaging quality between KGM gel groups and degassed water groups ($P > 0.05$). The difference in diagnostic accuracy for lesion depth between the KGM gel group and the degassed water group was not statistically significant ($P > 0.05$).

Additionally, the KGM gel group exhibited significantly reduced operation time and medium usage compared to the degassed water group ($P < 0.001$).

Conclusion: The 1.0% KGM gel is a safe and non-toxic coupling medium with excellent transparency, homogeneity, and moderate adhesion. As a coupling medium for MPS, it demonstrates imaging quality and lesion layer assessment accuracy comparable to degassed water while significantly reducing procedure time and medium consumption, enhancing efficiency and mitigating patient discomfort and aspiration risk.

These findings suggest that KGM gel is a viable and effective coupling medium for MPS with promising clinical applicability.

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PP0159

EUS-GUIDED GASTROENTEROSTOMY OUTCOMES IN A PERIPHERAL SINGLE-CENTER SERIES: A RETROSPECTIVE ANALYSIS OF THE FIRST 3 YEARS EXPERIENCE

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Introduction: Endoscopic ultrasound (EUS)-guided gastroenterostomy (EUS-GE) is an advanced, minimally invasive procedure primarily indicated in patients with gastric outlet obstruction (GOO) as an alternative to traditional enteral stenting or surgery. EUS-GE are generally performed in expert centers with a technical success rate ranging from 92% to 100%, a clinical success rate of around 90%, and 11%–12% of adverse events (AEs).

Aims & Methods:

Describe our experience in terms of technical and clinical success in a single peripherical center.

- Retrospective analysis of EUS-GE was performed in a single peripherical center from November 2021 to November 2024.
- Single operator.
- Wireless endoscopic simplified technique (WEST) and direct technique over a guidewire (DTOG) for EUS-GE are applied.

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- Technical success: successfully creating a gastroenterostomy with correct deployment of the LAMS between the stomach and the small bowel, as confirmed endoscopically and/or radiologically, without immediate intraprocedural complications.
- Clinical success: resolution or significant improvement of GOO symptoms, enabling the patient to tolerate at least a soft or liquid oral diet within from 72 hours to 7 days post-procedure and/or improvement in the Gastric Outlet Obstruction Scoring System (GOOSS)

Results: A total of 20 patients (15:5,M:F), median age of 76 years old (36-90) were included. All patients were symptomatic with nausea, vomiting, and anorexia with evidence of GOO. In 15 cases (75%), EUS-GE was performed in an advanced oncological setting: 6 patients presenting with pancreatic cancer, 4 with gastric cancer (including 1 EDEE in a recurrence case after Roux-en-Y gastrectomy), 3 with ampullary/duodenal cancer, 2 with carcinosis due to colon cancer. The remaining 5 patients underwent EUS-GE for EUS-direct trans-enteric endoscopic retrograde cholangiopancreatography (EDEE) (1 case) or refractory benign GOO: 2 due to stenosing peptic ulcer not responding to other treatments; 2 due to pancreatitis-related duodenal strictures. WEST was used in 18 cases and DTOG in 2. A 20x10 mm lumen apposing metal stent (LAMS) was used in all cases. Technical success was reached without AEs in 90% of cases (18). In 2 cases (10%) a distal flange maldeployment (type II and type I), both in DTOG, was recognized and solved in the same session: prolonging the LAMS with an esophageal fully covered metal stent (20 mm x 14 cm) in one case and bridging with another LAMS performing peritoneoscopy in the other one. Median EUS-GE procedural time was 55 minutes (40-200). Clinical success was 100%. Six patients (30%) died for oncological progression after a median of 135 days (30-510).

Conclusion: EUS-GE is a challenging endoscopic procedure with high technical and clinical success described in tertiary care centers. Our results showed reasonable technical and clinical success even in peripheral centers. The rate of AEs was similar to those reported in the literature. EUS-GE is a moderately reproducible endoscopic procedure, but experienced staff are always needed for recognizing and managing intra-procedural complications.

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Disclosure: Nothing to disclose.

PP0160

ENDOSCOPIC ULTRASOUND 2D SHEAR WAVE ELASTOGRAPHY FOR DIFFERENTIATING NORMAL AND CIRRHOTIC LIVER VIA LEFT LOBE ASSESSMENT

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Introduction: Endoscopic ultrasound (EUS) elastography is gaining traction as a noninvasive method for assessing liver stiffness, especially in patients with cirrhosis. While liver biopsy remains the gold standard, its invasiveness limits routine use.

This pilot study set out to determine the value of EUS-guided 2D-SWE, performed on the left liver lobe from the stomach, to distinguish between healthy liver tissue and cirrhosis.

Aims & Methods: We conducted a prospective study involving patients referred for EUS-based liver evaluation. Participants were categorized as having either normal liver function (confirmed by laboratory tests) or cirrhosis (diagnosed by clinical assessment, laboratory findings or imaging). EUS-guided 2D-SWE was performed using a linear-array echoendoscope positioned in the stomach to measure elasticity in the left liver lobe. Five consecutive measurements were taken using newly available EUS 2D-SWE software to assess the stiffness inside the region of interest (ROI). We assessed diagnostic accuracy using receiver operating characteristic (ROC) analysis.

Results: A total of 30 patients were included in the analysis: 25 with normal liver and 5 with cirrhosis. Liver stiffness values measured by EUS-guided 2D-SWE were significantly higher in the cirrhosis group (mean 25.9 ± 5.8 kPa) as compared to the normal group (mean 8.4 ± 2.9 kPa, $p < 0.001$). The technique showed good reproducibility, with a mean standard deviation of 2.28 kPa and a mean coefficient of variation of 23.3%. Intra-observer agreement was higher among those with normal liver (ICC = 0.685) than in patients with cirrhosis (ICC = 0.456), likely reflecting greater heterogeneity in diseased tissue. ROC analysis demonstrated excellent diagnostic performance for mean EUS 2D-SWE in identifying cirrhosis (AUC = 0.984). A cutoff of 13.82 kPa provided 100% sensitivity and 96% specificity.

Notably, values below 12 kPa effectively excluded cirrhosis, while those above 14 kPa strongly indicated its presence, allowing clinicians to tailor decision thresholds to clinical context.

Conclusion: EUS-guided 2D-SWE of the left liver lobe appears to be a reliable, noninvasive technique for differentiating cirrhosis from normal liver tissue. This method could expand the diagnostic capabilities of EUS in liver disease and may be especially valuable for patients in whom standard percutaneous elastography is inconclusive.

Disclosure: Nothing to disclose.

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PP0161

COMPARATIVE EFFICACY AND SAFETY OF ENDOSCOPIC ULTRASOUND-GUIDED GASTROJEJUNOSTOMY VERSUS SURGICAL GASTROJEJUNOSTOMY FOR THE MANAGEMENT OF GASTRIC OUTLET OBSTRUCTION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Gastric outlet obstruction (GOO) is a debilitating condition commonly caused by malignant or benign diseases, impairing gastric emptying and nutritional status. While surgical gastrojejunostomy (SGJ) has traditionally been the standard palliative approach, endoscopic ultrasound-guided gastrojejunostomy (EUSGJ) has recently emerged as a minimally invasive alternative. This systematic review and meta-analysis aimed to compare the efficacy and safety of EUSGJ versus SGJ for the management of GOO.

Aims & Methods: A systematic search of PubMed, Embase, Medline, and Cochrane Library was conducted to identify studies directly comparing EUSGJ and SGJ in adult patients with GOO. Nine retrospective studies comprising 1,086 patients (654 EUSGJ; 432 SGJ) were included.

Primary outcomes included technical and clinical success, length of hospital stay (LOS), operative time, and overall complication rates.

Secondary outcomes included infection, ileus, hemorrhage, perforation, and re-intervention. Meta-analyses were performed using fixed or random effects models depending on data heterogeneity.

Results: EUSGJ was significantly superior to SGJ in terms of technical success (OR = 0.16; 95% CI: 0.05–0.45; $p = 0.0006$), shorter operative time (MD = –96.89 minutes; 95% CI: –146.16 to –47.61; $p < 0.00001$), reduced postoperative LOS (MD = –4.79 days; 95% CI: –6.52 to –3.06; $p < 0.0001$), lower overall complication rates (OR = 0.24; 95% CI: 0.17–0.33; $p < 0.00001$), reduced postoperative ileus (OR = 0.07; 95% CI: 0.02–0.17; $p < 0.00001$), and fewer infections (OR = 0.17; 95% CI: 0.08–0.37; $p < 0.00001$). No statistically significant differences were found regarding clinical success, re-intervention, hemorrhage, or perforation.

Conclusion: EUSGJ demonstrates superior procedural outcomes and safety profiles compared to SGJ in the management of GOO and represents a valuable tool in expert hands. However, further prospective studies and randomized trials are necessary to validate these findings, assess long-term outcomes, and define its role within individualized patient care and clinical guidelines.

Disclosure: No conflict of interest.

PP0162

MODERATE SEDATION FOR SAME-SESSION EUS AND ERCP: A FEASIBLE AND SAFE STRATEGY IN PANCREATICOBILIARY DISEASE MANAGEMENT – RESULTS FROM THE PETIMIDA PROSPECTIVE PILOT STUDY

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Introduction: Same-session endoscopic ultrasound (EUS) and endoscopic retrograde cholangiopancreatography (ERCP) are increasingly adopted for diagnosing and treating pancreaticobiliary conditions. While traditionally performed under deep sedation or general anesthesia, data on moderate sedation strategies are limited. We prospectively evaluated the feasibility, safety, and patient-reported outcomes of performing combined EUS-ERCP under moderate sedoanalgesia using midazolam and meperidine.

Aims & Methods: The PETIMIDA study is a prospective pilot trial involving 27 consecutive adult patients undergoing same-session EUS and ERCP with moderated sedation with intravenous midazolam and meperidine at the Gastroenterology Unit of IRCCS Policlinico San Donato (Milan, Italy), from September 2024 to March 2025. Primary outcomes included procedural completion, safety, and patient-reported experience using the validated PROSAS score elements. Clinical parameters, sedation doses, and adverse events were collected through a standardized Case Report Form (CRF).

Results: Median patient age was 74 years (IQR 61.5–83), and 63% were male. Median Charlson Comorbidity Index was 4 (IQR 2–6.5), and median ASA score was 2 (IQR 1–3). Median BMI was 26.1 kg/m^2 (IQR 22.7–30.7). Indications included choledocholithiasis (63%), distal malignant biliary obstruction (19%), and indeterminate biliary stricture (19%). EUS with fine-needle biopsy (FNB) was performed in 9 patients (33%). Biliary stenting was performed in 8 cases: 4 with self-expandable metallic stents (SEMS) and 4 with plastic stents. In 2 patients, EUS-guided gallbladder drainage (EUS-GBD) was also successfully performed.

All procedures were completed successfully without the need for an anesthesiologist's intervention. The mean total midazolam dose was 5.5 mg (±1.1), and the mean meperidine dose was 58.9 mg (±10.1). Most patients (93%) were rated as adequately cooperative.

Adverse events included one case (3.7%) of oxygen desaturation <90% and four cases (14.8%) of hemodynamic instability (hypotension); none required anesthesiological intervention or procedure discontinuation but were managed with oxygen flow therapy and hydration. Patient-reported discomfort during the procedure had a median of 1 (IQR 0–2) on a 0–10 scale. Regarding sedation adequacy, 85% of patients indicated they would prefer the same sedation level in future procedures, and only 7% (2 patients) requested deeper sedation.

Conclusion: In this pilot study, same-session EUS and ERCP performed under moderate sedation with midazolam and meperidine was safe, feasible, and well-tolerated, even in elderly and comorbid patients. The approach avoided anesthesia-related risks and resource use while main-

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taining high patient cooperation and satisfaction. The ability to complete therapeutic interventions such as ERCP with biliary stenting and also therapeutic EUS procedures, highlights the potential of moderate sedation in advanced endoscopy when performed in expert centers. Larger studies are needed to confirm these findings and better define patient selection.

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PP0163

IS EUS WITH FNB AN ADEQUATE METHOD FOR THE PREOPERATIVE PROGNOSTIC STRATIFICATION OF GISTS?

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Introduction: The Miettinen-Lasota scoring system is the cornestone for prognostic stratification of gastrointestinal stromal tumors (GISTs), based on mitotic count and tumor size. Guidelines lack indications about the optimal sample that should be used for mitotic count assesment, only specifying that at least 50 high-power fields (HPF) must be evaluated. It is generally believed that, due to tumor heterogeneity and the limited sample size obtained, endoscopic ultrasound-guided tissue acquisition (EUS-TA) is unreliable, particularly when using traditional cytology needles.

However, only few data are available regarding the use of newer histology (FNB) needles in this context. Nevertheless, the ability to correctly stratify patients preoperatively is a critical factor in GIST management.

Aims & Methods: The primary aim of this multicenter retrospective study was to evaluate the concordance between EUS-FNB and surgical specimens in assessing the GIST prognostic score. Secondary endpoins were the adequacy of FNB samples for immunohistochemistry (IHC) and mutational analysis. We included all histologically confirmed GISTs from a prospective registry (NCT02855151). Concordance for mitotic count and tumor size measurements was evaluated using Bland-Altman plots, calculating bias (mean difference between FNB and surgical specimens) and limits of agreement (LoA). Agreement in prognostic score classification was calculated as a percentage.

Results: Among a cohort of 68 GISTs, we included 32 cases in which complete histopathological data from both EUS-FNB and surgical specimens were available. The diagnostic concordance between EUS-FNB and surgical pathology was 100%. IHC was successfully performed in all FNB samples (100%). The mitotic count showed a bias of -1.19 mitoses, with LoA ranging from +3.02 to -5.4. For tumor size, the bias was -0.96 mm, with LoA ranging from +21.28 to -23.2 mm. Miettinen-Lasota score concordance between EUS with FNB and surgery showed 72% agreement (23/32): in 5 cases, EUS underestimated the score, while in 4 cases, it overestimated it. Mutational analysis was performed in 62.5% of the cases.

Conclusion: EUS with new FNB needles appears to be reasonably accurate for preoperative GIST evaluation and is certainly a useful tool for diagnosis and mutational analysis. However, its concordance with surgical findings is still suboptimal in terms of prognostic evaluation.

On one hand, performing multiple needle passes, together with fanning technique, could improve the representativeness of tumor heterogeneity and increase the amount of material provided to the pathologist.

On the other hand, greater attention should be addressed to tumor size measurement in EUS, as this can sometimes lead to score variations.

The study's limitations include its retrospective nature and the heterogeneity of mitotic count assessment. Prospective studies with dedicated protocols are warranted, along with an evaluation of the adequacy of the collected material for mitotic count on 50 HPF, as suggested by the guidelines.

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PP0164

ASSESSMENT MALIGNANCY IN MEDIASTINAL AND ABDOMINAL LYMPHADENOPATHIES USING ELASTOGRAPHY AND FRACTAL-BASED ANALYSIS

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Introduction: The evaluation of lymph nodes (LNs) is crucial for the accurate diagnosis, staging, and management of several types of cancers. To improve EUS accuracy in the evaluation of LNs, several novel imaging techniques have been developed, including EUS elastography. Fractal analysis is a mathematical tool that allows the evaluation of the geometrical complexity of imaging patterns observed in nature and applied also to oncology. EUS elastography and fractal analysis may have a great potential in the evaluation of LNs, particularly in differentiating between benign and malignant LNs.

Aims & Methods: The aim of this study was to evaluate the accuracy of EUS-E (measured by strain ratio, SR) and to assess the accuracy of fractal-based analysis of EUS-E images in the differentiation of benign and

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malignant LNs. This was a prospective observational single-centre study (NCT04671784). EUS was performed using linear array Olympus GF-UCT-180 series echoendoscopes in combination with the echo processor EU-ME2 (Olympus SE & CO. KG, Hamburg, Germany). The quantitative score of elastography was expressed by the strain ratio (SR).

For fractal analysis, each selected region of interest was considered as a "shape matrix" of points with the column and row numbers proportional to the x and y coordinates and with the depth information z (x, y) stored as a matrix element. Final diagnoses were made by cytology, histology (EUS-sampling or resected specimens at surgery), or adequate follow-up time (at least one year).

Results: One-hundred consecutive patients with enlarged mediastinal or abdominal LNs detected by cross-sectional imaging and confirmed at EUS were enrolled between September 2019 and January 2023. A statistically significant difference in mean strain ratio was observed between benign and malignant LNs ($p = 2.5 \times 10^{-6}$), as well as between malignant and lymphoma LNs (p = 0.02). No significant differences were found between benign LNs and either NET or lymphoma LNs.

No statistically significant differences were observed when the global fractal dimension of LN images across the different groups was evaluated. However, when the fractal dimension was assessed for each RedGreenBlue_RGB channel, a significant difference was found in the green channel between benign and malignant LNs (p = 0.012), benign and NET (p = 0.017), and benign and lymphoma (p = 0.036).

Conclusion: According to our study, elastographic analysis is highly accurate in discriminating between benign and malignant LNs. In addition, a new model of fractal analysis seems to be useful to increase the diagnostic yield.

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PP0165

AGE, BMI, AND BOWEL HABITS AS PREDICTORS OF PATENCY CAPSULE FAILURE IN PATIENTS PREPARED WITH A PRO-MOTILITY PROTOCOL

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Introduction: The patency capsule (PC) test is recommended prior to video capsule endoscopy (VCE) to prevent capsule retention in high-risk patients. Despite its widespread use, standardized preparation protocols are lacking. Reported PC failure rates vary widely—from 4.9-43.1%—depending on the patient population and study setting, and also suggesting delayed motility as a contributor to failure.

Aims & Methods: This retrospective study aimed to evaluate factors associated with PC failure in patients undergoing a pro-motility preparation protocol, consisting of a low-residue diet, followed by a liquid diet and administration of 10 mg bisacodyl at the time of PC ingestion. Adult patients who performed the test between 2019-2023 were included; those with prior colectomy or ileostomy were excluded. Demographic and clinical data were collected. Multivariate logistic regression and classification tree modelling were used to identify predictors of PC failure.

Results: A total of 392 patients underwent a PC test using the pro-motility protocol and were included in the analysis (median age 42 years [range 18–89]; 47.4% male). The primary indications for VCE referral were suspected/established Crohn's disease (86.0%), anaemia/ gastrointestinal (GI) bleeding (8.4%), and radiologic findings (5.6%). Successful PC expulsion was documented in 323 patients (82.4%, "expelled" group), while 69 (17.6%) experienced PC retention ("retained" group). Of the 323 in the expelled group, 321 proceeded to VCE (2 declined), with no cases of VCE retention reported.

There were no significant differences between the expelled and retained groups in sex distribution, prevalence of diabetes mellitus, history of prior GI surgery, or indication for VCE (all p=NS). However, patients in the retained group were significantly older (median age 47 [IQR 34.5–66] vs. 40 [IQR 28–61] years; p=0.045), had higher body mass index (BMI) (25.9±5.4 vs. 24.2±4.9 kg/m^2; p=0.013), and reported fewer bowel movements per day (BPD) (median 1 [IQR 1–2] vs. 2 [IQR 1–3]; p=0.013). Reporting <1 BPD was twice as common in the retained group (20.3% vs. 9.6%; p=0.020). A trend toward higher smoking prevalence was observed in the expelled group (16.6% vs. 7.2%; p=0.060).

In a multivariate logistic regression the most prominent risk factors for retained PC were higher BMI (OR 1.063, 95%CI 1.005-1.125,p=0.034) and <1 BPD (vs. ≥1 BPD OR 2.418 95%CI 1.153-5.071, p=0.019) adjusted for age (OR 1.007 95%CI 0.992-1.023, p=0.374) , active smoking (vs. past/none OR 0.389 95%CI 0.147-1.032, p=0.058) and male sex (vs. female OR 1.723 95%CI 0.983-3.021, p=0.057).

A classification tree (CRT) model identified <1 BPD as the primary predictor of PC failure (probability 37.4%), rising to 44.4% in patients >35 years, but dropping to 0% in younger patients with BMI <21 kg/m^2. Among those with BPD ≥1, failure probability peaked at 66.7% with BMI >37 kg/m^2. In overweight-obese patients (BMI 25–35 kg/m^2), risk varied by BPD: 22.6% with BPD 1–4 vs. 3.6% with BPD >4. The model demonstrated good discrimination (AUC 0.705; 95% CI 0.639–0.771; p<0.001).

Conclusion: In patients undergoing patency capsule testing with a pro-motility protocol, PC failure was mainly associated with higher BMI and infrequent bowel movements. A CRT model identified bowel frequency, age, and BMI as key predictors of failure, revealing distinct high- and low-risk profiles. These findings support the use of simple clinical parameters in pre-test assessment to improve patient selection and enhance test success.

Disclosure: Nothing to disclose.

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PP0166

DIAGNOSTIC YIELD OF CAPSULE ENDOSCOPY IN PATIENTS WITH UNEXPLAINED CHRONIC ABDOMINAL PAIN

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Introduction: Unexplained chronic abdominal pain remains a frequent and often frustrating diagnostic challenge in clinical practice. When initial investigations such as colonoscopy and gastroscopy are inconclusive, small bowel capsule endoscopy (SBCE) provides a non-invasive means to evaluate the small intestine, a part of the gastrointestinal tract that is otherwise difficult to examine. This study explores the effectiveness of SBCE in uncovering underlying causes in patients with unexplained chronic abdominal pain.

Aims & Methods: We aimed to determine the diagnostic yield of SBCE in patients presenting with unexplained chronic abdominal pain and to identify predictive factors for positive findings. This was a retrospective analysis of prospectively collected data from all consecutive patients who underwent SBCE in our tertiary referral center in Greece between March 1, 2003, and April 30, 2025. We included patients who underwent SBCE following negative colonoscopy and gastroscopy. A positive diagnostic yield was defined as the detection of lesions capable of explaining the patient's symptoms. Predictive factors for positive findings were evaluated using multivariate analysis.

Results: During the study period a total of 11.201 patients were subjected to SBCE and among them 1398(12.4%) patients underwent the test for the investigation of chronic abdominal pain (men/women: 694/704, mean age ± SD: 52.6±27.3 years).

Positive findings were detected in 241 patients, leading to a diagnostic yield of 17.2%. The most common findings were inflammatory lesions that were seen in 176 (73.0%) patients [erosions from non-steroidal anti-inflammatory drugs n=75,erosions or ulcers from Crohn's disease n=97, erosions from radiation enteritis, n =4]while 47 (19.5%) patients had tumor/mass lesions, 9 (3.7%) patients had ischemic colitis, 6 (2.5%) patients had parasites and 3 (1.2%) patients had small bowel strictures.

By multivariate analysis, predictive factors for positive findings were elevated C-reactive protein (CRP) (OR=2.73, p<0.0001) and failure to intubate the terminal ileum during colonoscopy (OR=1.40 p=0.03).

Conclusion: The diagnostic yield of SBCE in patients with unexplained chronic abdominal pain is relatively low. Patients with elevated CRP or those in whom the terminal ileum was not examined during colonoscopy are those who will benefit most from the test.

Disclosure: Nothing to disclose.

PP0167

WHICH FACTORS MIGHT HELP TO PREDICT A CAPSULE ENDOSCOPY EXPLORATION WITH ADEQUATE CLEANLINESS?

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Introduction: Recommendations for bowel preparation prior to capsule endoscopy exploration in clinical guidelines are controversial and unclear.

Aims & Methods: To assess the contribution of the Boston cleansing scale (BBPS) in the previous colonoscopy, the discontinuation of iron therapy, a low-fibre diet, and the use of simethicone prior to capsule ingestion to obtain adequate visualization during capsule endoscopy exploration of the small bowel.

Prospective study of capsule examinations. In our centre, capsule preparation includes a fibre-free diet the day before, liquids only the evening before, 8-hour fasting, and simethicone. Capsule preparation is categorized as adequate/inadequate according to the Brotz scale.

Statistical analysis was performed with SPSS software.

Results: A total of 117 capsule explorations were analysed: 79 (67.5%) had adequate preparation and 38 (32.5%) had inadequate preparation.

The factors in which the association was studied and the major regression model are shown in Table 1. In the final model were included the Boston score for the right colon (p=0.032; OR=2.639) and transverse colon (p=0.002; OR=0.191), after ruling out confounding and interaction. Homer-Lemeshow test with p value 0.246. Specificity of 91.1%, sensitivity of 18.4%, and accuracy of 67.7%. Area under the ROC curve was 0.305 (0.205–0.404).

		n (%) patients n=117	OR (IC95%) (univariant)	p	Adjusted OR (IC 95%) (multivariant)	p
Simethicone	No Yes	4 (3.4%) 113 (96.6%)	0.468 (0.063-3.453)	0.456	0.273 (0.034-2.171)	0.220
Diet	No Yes	18 (15.4%) 99 (84.6%)	1.3 (0.427-3.956)	0.644	1.075 (0.323-3.573)	0.907
Iron	No Yes	80 (68.4%) 37 (31.6%)	1.690 (0.748-3.819)	0.207	2.249 (0.917-5.521)	0.077
Total BBPS	Min-max Mean (SD)	2-9 7.77 (1.476)	0.822 (0.634-1.065)	0.137	0.651 (0.209-2.026)	0.459
Right colon BBPS	Min-max Mean (SD)	0-3 2.51 (0.665)	1.047 (0.203-0.853)	0.878	4.558 (0.916-22.684)	0.064
Transverse colon BBPS	Min-max Mean (SD)	1-3 2.57 (0.546)	0.416 (0.203-0.853)	0.017	0.324 (0.044-2.408)	0.271
Left colon BBPS	Min-max Mean (SD)	1-3 2.68 (0.503)	0.475 (0.222-1.012)	0.054	----	----

Conclusion:

1. In our series, the use of simethicone, a low-fiber diet, and the discontinuation of iron therapy did not contribute to adequate preparation for capsule exploration.
2. Meanwhile, the Boston score for the right and transverse colon in pre-capsule colonoscopy can help predict capsule cleanliness.
3. Further studies are needed to find variables that help predict adequate preparation for capsule endoscopy.

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Disclosure: Nothing to disclose.

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PP0168

CAPSULE ENDOSCOPY FOR SMALL INTESTINE EVALUATION (CEFSIE): A NATION-WIDE MULTICENTER RETROSPECTIVE SRI LANKAN STUDY ON BOWEL PREPARATION

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Introduction: Small bowel mucosal pathologies are a diagnostic challenge to clinicians. Video Capsule endoscopy (VCE) is currently the study of choice for the evaluation of the small intestine. Data on the effects of bowel preparation on mucosal visualization and diagnostic yields is scant. Most published studies suggest that bowel preparation methods (2-litre vs. 4-litre) are not superior.

Aims & Methods: This retrospective cross-sectional study analyzed data from 2018 to 2023. The main investigator reassessed all VCEs using 3-6 frames per second, single-image viewing. A nursing officer closely supervised the bowel preparation, adhering to unit protocols. The small intestine was divided into proximal and distal small intestines depending on the small intestinal transit time, and mucosal visualization was assessed according to the Brotz scoring system. Statistical significance was determined by Fisher's exact (F) test and Chi-square (X2) test. A value of <0.05 was considered as significant.

Results: The sample size was 705, with a mean age of 56.9 (SD=18.2). 60.6% (n=427) were males. 39.3% had overt GI bleeding, 36.0% had occult GI bleeding, 16.3% had abdominal pain, 4.3% had chronic diarrhoea and 4.0 % had loss of weight as the indication for VCE. The mean gastric transit time was 01:13:52 ± 01:28:14 (hr.min.sec.), and the small intestine transit time mean was 05:12:59 ± 01:58:24 (hr.min.sec.). 38.3% consumed 59g x 2 PEG preparation in 2L of water; 11.9% consumed 59g x 3 PEG preparation in 3L of water; and 49.6% consumed 59g x 4 PEG preparation in 4L of water before VCE, followed by 6 hours of fasting. Mucosal visualization in the distal small intestine was significantly associated with the amount of PEG used, with 3×59 in 3 L of water accounting for a significantly higher percentage (32.1%) of adequate visualization (X2=11.82, p=0.003).However there was no significant association between the amount of PEG used and proximal bowel visualization (X^2=5.38, p=0.068). The amount of PEG administered significantly influenced transit times (F=5.781, p=0.003). A post hoc Tukey test revealed that the use of 4x59g of PEG in 4L of water significantly increased small intestinal transit time compared to lower doses.
The diagnostic yield was 65.6%. 34.4% had P0 lesions, 21.1% had P1 lesions, and 44.5% had P2 lesions. 26.8% had vascular lesions (17.2% in the proximal and 5.6% in the distal and 3.0% in both proximal and distal small intestine), 32.6% had inflammatory lesions (15.4% in the proximal, 15.1% in the distal, and 2.1% in both proximal and distal small intestine), worm infestation 1.1%, and malignancies 2.4%, which were among the most common findings. Ulcers, the most common inflammatory lesion (22.3%), equally affected the proximal (10.2%) and distal (10.9%) small intestines. Additionally, patients who had poor mucosal visualization (Brotz < 7) had significantly lower odds of P1/P2 lesions compared to those with good mucosal visualization (Brotz ≥ 7) (OR = 0.573, SE = 0.187, p = 0.003).

Conclusion: VCE has a diagnostic yield of 65.6%, while the proximal small intestine was the most common area involved, and inflammatory lesions were the most common lesions in Sri Lankans. Ulcers were the most common inflammatory lesion, which were equally involved in both the proximal and distal small intestine. Higher PEG and water administration significantly increased the transit time, and 3× 59g PEG in 3L of water had a statistically significant effect on mucosal visualization in the distal small intestine. Poor mucosal visualization had significantly lower odds of P1/P2 lesions.

Disclosure: Nothing to disclose.

PP0169

BARRETT'S OESOPHAGUS SURVEILLANCE, REAL WORLD DATA FOR USAGE OF CAPSULE SPONGE SAMPLING WITH BIOMARKER FROM A TEACHING HOSPITAL IN LONDON

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Introduction: Barrett's oesophagus (BO) is a condition that develops as a consequence of chronic gastroesophageal reflux disease and a premalignant stage of adenocarcinoma of the oesophagus (EAC). In the US it is estimated that as many as 5.6 percent of adults have Barrett's esophagus.[i]
Estimates of the annual cancer incidence in patients with Barrett's esophagus have ranged from 0.1 to almost 3.0 percent. [ii]
The goal of surveillance is to improve outcomes by detecting dysplasia or esophageal adenocarcinoma early enough to provide effective treatment. Guidelines suggest surveillance for most patients with Barrett's esophagus There are potential harms associated with surveillance, due to the risks associated with endoscopy, the risks and morbidity associated with invasive therapies used to treat lesions identified by surveillance.[iii]
Patients with BO are currently recommended to enter a surveillance endoscopic program.[iv]
This has been associated with a significant mental burden on these patients, presenting as various forms of anxiety and stress related to thoughts of disease progression as well as potential implications of the test, with many finding the program physically burdensome and intrusive. [v]

Aims & Methods: In our study, from March 2024 until December 2024. patients with known BO were invited to undergo a tethered capsule procedure, instead of regular endoscopy, as part of their regular BO surveillance. Samples were sent for cytological analysis. Patients were identified as high risk if their cytology results came positive or equivocal for dysplasia or P53 chromosome over expression and then referred for gastroscopy for further biopsies and assessment.

Results: We tested 144 patients. Mean age was 69 years old (youngest 32-oldest 90). Males were 62.5% and females 37.5%.
6.9% (N=10) came positive for dysplasia or atypia, while 2.8% (N=4) were inconclusive.
P53 cytological analysis 2.1% were positive, while 4.2% were equivocal.
Overall, our investigations identified 16 patients (11.1%) who were positive or equivocal for dysplasia and P53 mutation.
In so far, we identified 1 patient with adenocarcinoma and 2 patients with biopsy proved dysplasia.

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Conclusion: Based on our results, sponge capsule combined with biomarker (P53) mutation could be offered for BO surveillance, as an alternative to regular endoscopy which could be burdensome. This could offer up opportunity for this non-invasive method, which could even be performed in the setting of regular outpatient clinic visits. This modality can potentially facilitate access to care for a larger population of patients and increase compliance similar to non-invasive tests for colorectal cancer screening programs.[iv]

There were limitations our study. At this stage we could only review results for patients done at our hospital. And some patients with positive findings were still awaiting gastroscopy and biopsies. Further research on a larger cohort of patients is needed to analyze rate of diagnosis of dysplasia and adenocarcinoma, compared to OGD, which remains the gold standard.

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PP0170

EVALUATING THE UTILITY OF COMPUTED TOMOGRAPHY (CT)-DETECTED OESOPHAGEAL ABNORMALITIES IN PREDICTING MALIGNANCY: A RETROSPECTIVE REVIEW

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Introduction: CT-reported oesophageal thickening is frequently interpreted as a potential indicator of malignancy, often prompting urgent endoscopic evaluation via suspected cancer pathways[1].

However, the actual diagnostic yield remains unclear, and there is limited evidence guiding clinicians on the true risk such findings represent. In a resource-constrained healthcare system, understanding the predictive value of radiological abnormalities is critical to ensure timely diagnosis while avoiding unnecessary investigations[2].

Given the increasing demand on diagnostic services, there is a growing need to distinguish radiological findings that warrant urgent investigation from those that do not.

This study evaluates the malignancy rate associated with CT-detected oesophageal abnormalities and investigates the contribution of clinical markers and referral patterns to inform risk stratification and endoscopic prioritisation.

Aims & Methods: A retrospective review of 106 patients with CT-reported oesophageal thickening was conducted across multiple sites in South Wales between 2022 and 2025. Clinical and diagnostic data were extracted from the Welsh Clinical Portal and Synapse endoscopy reporting system. Variables included presence of anaemia, referral source, time interval from CT to oesophagogastroduodenoscopy (OGD), and histopathological outcomes. Patients without follow-up OGD were excluded. Statistical analysis was conducted using IBM SPSS, with chi-square tests and cross-tabulation employed to explore associations between clinical variables and confirmed malignancy.

Results: Malignancy was histologically confirmed in 5.7% of patients undergoing endoscopic investigation following CT-reported oesophageal thickening. Among those with cancer, 83.3% were anaemic at presentation, a statistically significant association ($p < 0.001$). While the overall malignancy rate was low, the disproportionate occurrence of anaemia in malignant cases suggests it may serve as a valuable clinical marker to guide endoscopy urgency. Dysphagia and weight loss were also reported in malignant cases, although these did not reach statistical significance. Patients with confirmed malignancy were scoped considerably sooner, with a mean interval of 13.8 days from CT to OGD, compared to a cohort average of 49 days, reflecting clinical prioritisation based on perceived risk. Endoscopic abnormalities were observed in approximately 63% of cases, the majority of which were non-malignant findings.

Conclusion: The malignancy detection rate in this cohort was under 6%, highlighting the limited standalone predictive value of CT-detected oesophageal thickening. The strong association between anaemia and confirmed cancer reinforces the importance of integrating clinical markers into triage decisions. These findings support a re-evaluation of current referral pathways and advocate for more refined, risk-based approaches to endoscopy selection. In the context of international variation in cancer referral thresholds, this study provides real-world evidence to inform safer and more efficient diagnostic strategies.

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Disclosure: Nothing to disclose.

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PP0171

CAN CT RADIOMICS PREDICT THE KI-67 INDEX OF GASTROINTESTINAL STROMAL TUMORS (GISTS)? A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The Ki-67 labeling index is a well-established marker of cellular proliferation and has been associated with worse disease-free and overall survival in gastrointestinal stromal tumors (GISTs). Recent evidence highlights its potential role in risk stratification, particularly when Ki-67 exceeds certain thresholds. However, current assessment relies on postoperative histopathology. Radiomics, by extracting quantitative features from computer tomography (CT) images, may offer a non-invasive method to predict Ki-67 expression preoperatively. This meta-analysis aimed to evaluate the diagnostic performance of CT-based radiomics for predicting Ki-67 status in GISTs.

Aims & Methods: A systematic review and meta-analysis were conducted according to PRISMA guidelines). Four electronic databases (PubMed, Scopus, Science Direct, Cochrane Library) were searched for relevant articles up to December 2024. Eligible studies assessed CT radiomics in predicting Ki-67 in histologically confirmed GISTs. Pooled sensitivity, specificity, and area under the curve (AUC) were calculated using a random-effects model. Subgroup analyses were performed based on cohort type, CT protocol, number of features, and Ki-67 cutoff values. Study quality was assessed using the QUADAS-2 tool.

Results: Six studies comprising 1,632 patients met the inclusion criteria. Pooled sensitivity and specificity were 0.71 (95% CI: 0.63–0.79) and 0.76 (95% CI: 0.73–0.78), respectively. The pooled AUC was 0.79 (95% CI: 0.74–0.84). Subgroup analyses showed consistent performance across imaging protocols and feature counts, while Ki-67 cutoffs (8% vs. 10%) significantly impacted diagnostic accuracy. Sensitivity analysis confirmed robustness of specificity estimates. Publication bias was detected for specificity (Egger's test p = 0.006).

Conclusion: CT radiomics demonstrates moderate accuracy for non-invasively predicting Ki-67 expression in GISTs. Given Ki-67's prognostic significance, radiomics could support preoperative risk assessment and personalized treatment planning. Further prospective validation is warranted.

Disclosure: No Conflict of Interest.

PP0172

UK MULTI-CENTRE EXPERIENCE FOR THE USE OF VACSTENT DEVICE FOR THE MANAGEMENT OF OESOPHAGEAL DEFECT

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Introduction: Transmural oesophageal defects are associated with significant morbidity and mortality. Repeat surgery is often complicated and undesirable with variable outcomes. Increasingly, endoscopic interventions including clip closure devices, temporary full covered self-expanding metal stents (FcSEMS) and endoscopic suturing are being used to treat this problem.

Endoscopic vacuum therapy (EVT) relies on a sponge-based device that is placed either within a cavity or the oesophageal lumen to provide continuous negative pressure over the defect which enhances drainage and tissue micro-perfusion.

The VacStent GI™ (MicroTech) device is an innovative endoscopic solution which synergistically incorporates the advantages of a fcSEMS while synchronously employing the principles of EVT.

Aims & Methods: We retrospectively analysed patients undergoing treatment for transmural oesophageal defects with the VacStent GI™ (MicroTech) device between October 2023 and Febuary 2025 at two upper GI surgical centres (UCLH and University Hospitals of Derby).

All patients had cross-sectional imaging and any chest contamination was drained before stent insertion. Defects involving the cricopharnyx and/or those more than 5cm were excluded. Continuous negative pressure of 125 mmHg was used for the first 24 hours after stent placement and then reduced to 75 mmHg. The VacStent was exchanged every 7-10 days until defect closure. All cases were performed in endoscopy with a minimum of propofol sedation and were discussed as part of specialist MDT.

The primary outcome was clinical success, defined as the endoscopic and radiological evidence of defect closure. Secondary outcomes included technical success, number of stent exchanges, serious adverse device effect (SADE) and time to defect closure.

Results: The VacStent GI™ (MicroTech) device was used 33 times in 16 patients with a median age of 65 years (IQR 57-75). The study population comprised 75% male and 25% females. 2/16 (12.5%) had defects secondary to boerhaaves syndrome, 4/16 (25%) had iatrogenic defects and, 10/16 (62.5%) had anastomotic leaks (1=benign:9=malignant).

Clinical success was achieved in 10/16 (62.5%) patients with a technical success rate of 100%. Median time to defect closure following index VacStent implantation was 10 days (IQR 6-15) with a median of 1 VacStent exchange (IQR 0-2). The median defect size was 10mm (IQR 6-14). Cases of VacStent failure appear to be related to defects more than 4cm and those within a conduit (Table 1).

There was a single adverse event related to severe bleeding due to the stent becoming embedded to the oesophageal lumen – after the VACStent was removed, a temporary fully covered stent was placed to tamponade any bleeding and later removed uneventfully.

Conclusion: In conclusion, our experience has demonstrated that VacStent is a safe and effective treatment option for oesophageal defect closure achieving this in 62.5% of patients.

This is a challenging cohort of patients that usually have limited treatment options. Multi-disciplinary collaboration and early recognition of both appropriate and inappropriate defect dimensions and aetiology is critical to enhancing success rates.

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Ongoing experience with a more diverse and larger patient population will further clarify the position of the VacStent device in the EVT treatment paradigm.

Disclosure: Nothing to disclose.

PP0173

ENDOSCOPIC DILATION OF BENIGN OESOPHAGEAL STRICTURES: A LARGE MULTICENTRE EVALUATION OF REAL-WORLD PRACTICE

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Introduction: Although overall prevalence of benign oesophageal strictures (BOSs) is low, a rising prevalence of associated risk factors, such as increased life expectancy, gastro-oesophageal reflux disease and eosinophilic oesophagitis is predicted to contribute to an increased incidence and evolving aetiology. Endoscopic dilation (ED) currently remains the primary treatment modality.

Aims & Methods: This study aimed to evaluate the current most common aetiologies for BOSs, alongside evaluating the success of ED.

Between January 2021 and December 2023, 1994 upper GI endoscopies (UGIEs) with endoscopic dilatation of a BOS were identified using electronic patient records across 11 UK centres. Details of procedures were then collected in a retrospective observational fashion at each site before central pooling of pseudo-anonymised data for analysis.

For both individual UGIEs with dilatation and given "sessions" of dilatations (defined as consecutive UGIEs separated by ≤6 months), treatment was deemed successful if the lumen was confirmed to be dilated to a diameter of ≥14mm (confirmed on a check procedure >4 weeks after, and/or not requiring further dilatation to maintain diameter within a 6-month period).

Results: Eight hundred and eighteen patients (328[40%] female; mean age 64.3[SD 15.52]) were followed-up for a mean of 20.5 (range 4–62) months, from time of initial stricture diagnosis. The most common aetiology of BOS were peptic in 38% (313/818), post-op anastomotic in 13%(105/818), post-radiotherapy in 6% (48/818), post-operative (non-anastomotic) in 6% (47/818), Schatzki rings in 6% (46/818), and eosinophilic oesophagitis in 5% (37/818). Less common aetiologies seen included: post-endoscopic mucosal resection (EMR), post-radiofrequency ablation (RFA), post-caustic injury, webs, lichen planus, and connective tissue disease.

(see Table 1)

Successful dilatation for a single UGIE procedure with ED was achieved in 875/1994 (43.9%) of procedures. With respect to 'sessions' of consecutive dilatations, success was achieved within three months in 44.2% (388/878) of patients, and within six months in 60.6% (530/878) of patents. For successful sessions, mean number of dilatations required was 2.2 (SD 2.4), and median number of weeks to successful dilatation was 15 (IQR 7–30).

Stricture Aetiology	Number of patients n	Percentage of patients n/N (%)
Peptic	313	38
Post-operative (anastomotic)	105	13
Post-radiotherapy	48	6
Post-operative (non-anastomotic)	47	6
Schatzki ring	46	6
Eosinophilic oesophagitis	37	5
Other*	222	26

* ‚Other' includes (in order of frequency, though all representing <5%) cases of: unidentifiable aetiology, post-endoscopic mucosal resection (EMR), post-radiofrequency ablation (RFA), post-caustic injury, webs, lichen planus, connective tissue disease, and miscellaneous

Table 1 – Most common aetiology of benign oesophageal strictures for patients (N = 818).

Conclusion: Peptic strictures remain the most common aetiology of BOS. A significant proportion of patients (39.4%) with BOS failed to achieve success following ED. This may reflect both the inherent limitations of ED and suboptimal adherence to guidelines, particularly regarding the recommended frequency of dilations until treatment success is achieved

Further analysis of data from this study will look to explore relative effectiveness of endoscopic dilatation in the context of differing aetiology, additional patient (as well as procedural) factors influencing success, as well as complication rates. It will also explore whether greater compliance with established guidelines may help to more accurately define the true therapeutic potential of ED in the management of BOS.

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PP0174

LOWER ESOPHAGEAL RING'S TREATMENT: LONG-TERM RESULTS COMPARING ELECTROSURGICAL INCISURES VERSUS DILATATION IN THE PRESENCE OF ACID SUPPRESSION THERAPY

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Introduction: Lower esophageal (Schatzki's) rings are a common cause of solid impactation dysphasia. Standard treatments involve passage of a single large bougie to disrupt the ring but symptoms recur in the majority of patients. Electrosurgical incision of the ring may provide a longer duration of symptom improvement. Acid suppression has been shown to decrease the recurrence rate of peptic strictures but are few datas on the treatment of Schatzki's ring

Aims & Methods: To compare the efficacy of bougie dilatation plus Esomeprazole to surgical incision plus esomeprazole in symptomatic patients with Schatzki's ring

A prospective randomized trial study, from May 2013 to Abril 2025, was performed in a referred Gastrointestinal hospital in Brazil. A total of 84 patients underwent for endoscopic evaluation of dysphagia were found to have a Schatzki's ring. They were randomized to Bougie dilatation using a 54 French Dilator or electrosurgical incision a standard needle knife incision with 4 longitudinal incisions placed radially around the Schatzki's ring. All patients were placed on Esomeprazole 40mg/daily for 8 weeks. Paired data were analyzed using Wilcoxon rank (dysphasia score) and paired t-test (GERD scores) within groups

Results: Thirty-nine patients underwent bougie dilatation and thirty-eight electrosurgical incisions. Patients were followed up for 18 months. Seven patients did not come for followed up. Two episodes of bleeding during electrosurgical incision were controlled using epinephrine injection. Symptoms free were significantly longer in incision group (8.82 months) compared to bougie dilatation group (4.93 months) (p = 0,02). The improvement in dysphasia scores were significant within each group comparing baseline to each time interval. The incision group experienced a significant improvement of dysphasia symptoms at 6 weeks compared to the bougie dilatation group (p = 0,05). GERD scores improved significantly in both groups at all time intervals in the bougie dilatation group and at all time intervals in the incision group. There were no significantly difference in GERD scores between the groups.

Conclusion: Electrosurgical incision of lower esophageal (Schatzki`s) ring is safe and offers longer free symptoms free compared to bougie dilatation. The addition of Esomeprazole offered significant improvement in GERD scores in both groups

Disclosure: Nothing to disclose.

PP0175

GASTRIC PARTITIONING COMPARED TO CONVENTIONAL GASTROJEJUNOSTOMY AS PALLIATIVE SURGERIES IN PATIENTS WITH GASTRIC OUTLET OBSTRUCTION DUE TO UNRESECTABLE GASTRIC OR PANCREATIC CANCERS: A PAIR-WISE AND INDIVIDUAL PATIENT DATA META-ANALYSIS

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Introduction: Gastric outlet obstruction (GOO) is a severe complication of unresectable gastric and pancreatic cancers. Conventional gastrojejunostomy (CGJ) is a standard surgical intervention, but it is associated with a high incidence of delayed gastric emptying (DGE). Stomach-partitioning gastrojejunostomy (SPGJ) has emerged as an alternative to improve gastric emptying and enhance clinical outcomes.

Aims & Methods: We conducted a meta-analysis comparing SPGJ and CGJ in patients with GOO due to unresectable gastric or pancreatic cancers. Electronic databases were searched up to February 8, 2025.

Outcomes included DGE, major postoperative complications, need for reintervention, short-term mortality, operation time, gastric outlet obstruction scoring system (GOOS), length of hospital stay (LOS), adherence to postoperative chemotherapy, and overall survival.

For continuous outcomes, mean differences (MD) with 95% confidence intervals (CI) were calculated. For dichotomous outcomes, relative risks (RR) with 95% CI were reported. Reconstructed individual patient data (IPD) from Kaplan-Meier curves was utilized for survival analysis.

Results: A total of 11 studies comprising 456 patients were included. SPGJ was associated with significantly reduced DGE (RR = 0.24, 95% CI: 0.12–0.47) and postoperative major complications (RR = 0.26, 95% CI: 0.12–0.54) compared to CGJ. No significant differences were found in the need for reintervention (RR = 0.59, 95% CI: 0.21–1.64), short-term mortality (RR = 0.99, 95% CI: 0.42–2.33), or LOS (MD = -1.47 days, 95% CI: -3.10 to 0.16). GOOS scores were comparable between groups. Overall survival was also similar between SPGJ and CGJ (HR = 1.06, 95% CI: 0.66–1.70).

Conclusion: Our meta-analysis supports SPGJ as a superior option to CGJ for managing malignant GOO, offering significant reductions in DGE and postoperative complications, similar GOOS scores, and comparable safety regarding reintervention, short-term and long-term mortality, and LOS. High-quality RCTs are needed to confirm these findings and guide clinical practice.

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PP0176

ANTRAL PRESERVATION VS. ANTRAL RESECTION IN SLEEVE GASTRECTOMY: A SYSTEMATIC REVIEW AND META-ANALYSIS OF WEIGHT LOSS, METABOLIC OUTCOMES, AND COMORBIDITY RESOLUTION

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Introduction: The impact of antral preservation versus antral resection in sleeve gastrectomy remains a subject of debate due to its potential effects on weight loss, metabolic outcomes, and gastrointestinal function. This meta-analysis aims to provide a comparative assessment of these surgical techniques.

Aims & Methods: The Meta-Analysis aims to provide a comprehensive analysis between these two surgical techniques.

A systematic review and meta-analysis were conducted using 10 studies published between 2014 and 2025 that compared sleeve gastrectomy with antral preservation versus antral resection. The primary outcomes included excess weight loss (EWL) at different time points, while secondary outcomes assessed metabolic parameters (C-peptide, GLP-1, HbA1c), gastric emptying, gastroesophageal reflux disease (GERD), and resolution of comorbidities.

Results: A total of 10 studies were included. Patients undergoing antral resection exhibited greater excess weight loss at 6 and 12 months (MD: -4.50, 95% CI: -5.98 to -3.02; MD: -6.11, 95% CI: -7.74 to -4.47, respectively; p<0.00001), though the difference diminished by 24 months (MD: -3.37, 95% CI: -7.04 to 0.30; p=0.07). No significant differences were observed in overall GERD incidence (RR: 1.00, 95% CI: 0.75–1.33) or complications (RR: 1.00, 95% CI: 0.74–1.42). Metabolic outcomes showed marginal differences, with C-peptide (SMD: 0.02, 95% CI: -0.38 to 0.42) and GLP-1 (SMD: -0.22, 95% CI: -0.90 to 0.47) exhibiting minimal changes between groups.

Conclusion: Antral resection in sleeve gastrectomy is associated with superior short-term weight loss, but long-term differences remain inconclusive. Metabolic outcomes and postoperative complications appear comparable between techniques. Further randomized controlled trials with extended follow-up are warranted to refine surgical recommendations.

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PP0177

ANASTOMOTIC LEAKAGE AFTER ESOPHAGECTOMY: CAUSES, TREATMENT, AND PROGNOSIS

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Introduction: In the treatment of esophageal carcinoma, abdomino-thoracic esophagectomy is the curative therapeutic option, often combined with perioperative (radio-)chemotherapy. This procedure is associated with high complication rates and significant mortality. Anastomotic leakage, with a pooled incidence of 11% (ranging from 0% to 49%)[1], constitutes a major postoperative complication. New therapeutic options, such as the EndoVac (Stent) system, offer promising approaches for its management.

The aim of this study was to identify risk factors for the occurrence of anastomotic leaks and to evaluate the effectiveness of various treatment methods.

Aims & Methods: We present data from 2022–2024 collected at the University Medical Center Göttingen, a high-volume center for esophageal surgery. A total of 242 patients who underwent abdomino-thoracic esophagectomy, either as primary treatment or after neoadjuvant therapy, were included. Structured data collection and analysis were carried out using Excel and GraphPad Prism.

Results: In this patient cohort, 79.3% were male (n=192) and 20.7% female (n=50), with a mean age of 65.4 years and a mean BMI of 26.1 kg/m^2 at the time of surgery. The most common pre-existing conditions were 56.6% arterial hypertension (n=137), 7,9% COPD (n=19), 21% diabetes mellitus (n=51), as well as coronary artery disease (CAD) and peripheral arterial disease (PAD), each 8.2% (n=20). Histologically, 80.1% (n=194) adenocarcinomas and 19.4% (n=47) squamous cell carcinomas. 88.8% of patients received neoadjuvant therapy (n=215): 35.5% radiochemotherapy (n=86) and 53% chemotherapy (n=129). Of all patients, 24.8% (n=60) underwent primary open surgery and 75,3% (n=182) had minimally invasive hybrid procedures. By 2024, minimally invasive approaches had increased to 83.3% (n=86).

Anastomotic leakage occurred in 25 patients (10.3%), with clinical manifestation appearing on average 8 days postoperatively (range: 1–29 days). An association with the occurrence of anastomotic leakage was observed for nicotine abuse (OR 1.35; 95% CI: 0.60–3.11), alcohol consumption (OR 1.69; 95% CI: 0.66–4.24), coronary artery disease (CAD), and peripheral arterial disease (PAD) (each OR 2.37; 95% CI: 0.80–7.81); however, none reached statistical significance in Fisher's exact test (all $p > 0.05$). Complete remission (24.4%, n=59) was significantly associated with an increased risk of leakage (OR 3.51; 95% CI: 1.53–7.86; $p = 0.005$).

All 25 anastomotic leaks were initially treated with EndoVac, healing successfully in 21 cases (96%). Four patients (16%) required revision surgery. An EndoVacStent was implanted in 52% of cases (n=13), with 20% receiving only an EndoVacStent (n=5) and 32% treated with a combination of sponge and stent (n=8). The average duration of EndoVac therapy was 15 days (range: 1–42 days), with the system being exchanged three times on average. In-hospital mortality was 1.6% (n=4).

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Conclusion: In summary, the EndoVac and EndoVacStent systems have proven to be safe and effective treatment options for anastomotic leakage. Patients with complete remission showed an increased susceptibility to anastomotic leakage. Our data supports the importance of studies like ESORES (DRKS00022050)[2] and PRESTO (NCT05713838)[3], investigating non-operative treatment approaches for clinical complete remission cases followed by a watch-and-wait strategy.

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PP0178

LONG-TERM FOLLOW-UP OF PARA-OESOPHAGEAL HERNIA SURGERY: A 22-YEAR EXPERIENCE

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Introduction: The surgical management of paraesophageal hernias (PEH) has evolved significantly, moving from open approaches including laparotomy or thoracotomy to minimally invasive techniques, including laparoscopic and robotic-assisted repair with or without fundoplication. Surgery aims to improve symptoms with better quality of life (QoL) and reduce the risk of life-threatening complications of PEH. Many reports are predominantly those with short or medium follow-up periods.

Aims & Methods: The present study aims to assess a large cohort of patients who had surgery for PEH for recurrence and QoL over a longer period. So, we did a retrospective analysis of prospectively collected data of patients who underwent surgery for symptomatic PEH from January 2001 to September 2024. Data collected on clinical presentation, surgical management, post-operative clinical outcomes, and follow data that comprised of clinical evaluation with basic, functional and radiographic assessment as well as quality-of-life (QoL) assessment of general health, physical, emotional, and social components with SF-36 questionnaire.

Results: There were 284 patients (180 females) with median age of 67 years (See Table) who underwent surgery for PEH. One hundred and sixteen presented with shortness of breath (41%). One hundred and twenty-seven (45%) were ASA III, 167 (59%) had microcytic anaemia.

Seven (2.5%) underwent gastropexy only. A fundoplication was performed in 146 (51%) during PEHR. The remaining 131 patients had PEHR plus gastropexy, with 30 of them were for recurrence following previous PEHR (16 of them had primary surgery elsewhere with fundoplication at PEHR)

Approach was robotic (35), laparoscopic (224), converted- to-open (7) or open (19). Median hospital stay was significantly lower in robotic 3 days (2-7) and laparoscopic group; 4 days (1-16) compared to open of 10 days (5-33). Mortality was 0.7% (2/284).

Median follow-up was 12 years. Symptoms improved in 232 (82%) with a significant improvement of respiratory function following PHER. Haemoglobin levels improved from 96g/L (61-123) preoperatively to 129g/L (98-138) postoperatively so was all components of QoL. Sixty-two (22%) had radiological recurrences but only 14 (5%) had clinical recurrence requiring further surgery and 11 (79%) of 14 were those who had a fundoplication at previous PEHR.

Age at Surgery for PEH (Years)	Number of patients
21-30	3
31-40	10
41-50	33
51-60	48
61-70	80
71-80	85
81-88	25
Median (range) 67(21-88): IQR 55-76	Total (n)=284

Conclusion: Para-oesophageal hernia repair is safe with good outcomes in all age groups with improvement in symptoms, quality of life, respiratory function, and associated anaemia. Radiological recurrences are not uncommon on long-term follow-up and does not need any intervention unless symptomatic. Para-oesophageal hernia repair combined with gastropexy avoiding fundoplication probably reduce symptomatic recurrences.

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PP0179

EVALUATION OF POST-ESD BLEEDING RATE AND RISK FACTORS IN HEMODIALYSIS PATIENTS UNDERGOING SECOND-LOOK ENDOSCOPY

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Introduction: Patients on maintenance hemodialysis are considered to have a higher risk of bleeding after endoscopic submucosal dissection (ESD) for gastric lesions. In high-risk cases, second-look endoscopy (SLE) may reduce the risk of post-ESD bleeding[1].

At our institution, SLE is routinely performed following gastric ESD. We evaluated the incidence and risk factors of post-ESD bleeding after gastric ESD among patients on maintenance hemodialysis.

Aims & Methods: We conducted a retrospective, single-center study including 32 hemodialysis patients with 37 gastric lesions who underwent ESD at our hospital between July 22, 2003 and August 31, 2024. Patients were divided into two groups based on the occurrence of post-ESD bleeding. Clinical data including age, sex, body mass index (BMI), age-adjusted Charlson comorbidity index (ACCI), Geriatric Nutritional Risk Index (GNRI), extent of atrophic gastritis, laboratory findings, the use of antithrombotic drugs, Best-J score[2], details of the procedure, pathological findings and the types of anti-acid medicine prescribed after ESD were collected and analyzed.

Results: The median age was 72 years, with 73% of the patients being male. There were no differences in most of patient characteristics, details of lesions and procedures. Submucosal invasion (0% in the no-bleeding

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group vs. 28.6% in the bleeding group, p = 0.035) were significantly associated with post-ESD bleeding. In contrast, the use of local hemostatic agents appeared to have a protective effect against post-ESD bleeding (51.7% vs. 0%, p = 0.012). The use of perioperative antithrombotic drugs tended to be higher in the bleeding group (51.7% vs. 0%, p = 0.058). The endoscopic hemostasis at 2nd look endoscopy was not associated with post-ESD bleeding.

ACCI (median, IQR)	Overall (n=37 (%)) 8(6-8)	No-rebleeding (n=29 (%)) 7(6-8)	Rebleeding (m=8 (%)) 8(7.5-9.25)	p-value 0.2638
Antithrombotic drugs				
No	10(27.0%)	9(31.0%)	1(12.5%)	
Aspirin	22(59.5%)	16(55.2%)	6(75.0%)	0.6927
Thienopyridine	4(10.8%)	3(10.3%)	1(12.5%)	
Cilostazol	1(2.7%)	1(3.4%)	0(0%)	
Warfarin	5(13.5%)	3(10.3%)	2(25.0%)	
Perioperative antithrombotic drugs				
No	9(24.3%)	8(27.6%)	1(12.5%)	
Interruption	21(56.8%)	18(62.1%)	3(37.5%)	0.0578
Single antiplatelet therapy	6(16.2%)	3(6.9%)	3(37.5%)	
Heparin bridging therapy	1(2.7%)	0(0%)	1(12.5%)	
Best-J				
High-risk	13(35.1%)	12(41.4%)	1(12.5%)	0.2164
Very high-risk	24(64.9%)	17(58.6%)	7(87.5%)	
Tumor size (median, IQR) (mm)	10.0(6.0-18.5)	7.5(6.0-14.0)	19.0(8.0-21.0)	0.2129
Tumor location				
Upper	7(18.9%)	6(20.7%)	1(12.5%)	1.0000
Middle	4(10.8%)	3(10.3%)	1(12.5%)	
Lower	26(70.3%)	20(69.0%)	6(75.0%)	
Depth of invasion				
Mucosa	33(89.2%)	28(100%)	5(71.4%)	0.0353
Submucosa	2(5.4%)	0(0%)	2(28.6%)	
Procedure time (median, min)	92.0	99.0	84.5	0.9853
Use of local hemostatic agents				
Yes	15(40.5%)	15(51.7%)	0(0%)	0.0121
No	22(72.9%)	14(48.3%)	8(100%)	
Endoscopic hemostasis at 2nd look endoscopy				1.0000
Yes	14(37.8%)	11(37.9%)	3(37.5%)	
No	23(62.2%)	18(62.1%)	5(62.5%)	

Conclusion: Despite routine SLE in hemodialysis patients, the rate of post-ESD bleeding at our institution remains high and similar to previously reported rates. Submucosal invasion, perioperative antithrombotic therapy may be associated with increased, and the use of local hemostatic agents may be associated with decreased bleeding risk. Further strategies should be considered to reduce the risk of post-ESD bleeding in hemodialysis patients.

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Disclosure: Nothing to disclose.

PP0180

PREDICTING NONSURGICAL OUTCOMES OF IATROGENIC GASTROINTESTINAL PERFORATIONS: A COMPREHENSIVE MULTICENTER NOMOGRAM

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Introduction: Iatrogenic gastrointestinal perforation (IGP) is a rare but critical complication of endoscopy, traditionally managed surgically. Recent advances have allowed nonsurgical treatment to emerge as a viable alternative. However, predictive models for successful nonsurgical management remain limited.

Aims & Methods: This study aimed to identify independent predictors of nonsurgical success and develop a visual nomogram to predict which patients with IGP can be treated successfully without surgery. We retrospectively reviewed cases of IGP from three tertiary centers in China between 2011 and 2022. Patients were divided into two groups: those successfully managed nonsurgically and those requiring surgical intervention. Multivariable logistic regression identified predictors of nonsurgical success, and a nomogram was developed and validated to predict outcomes. Data analysis included patient demographics, perforation characteristics, and treatment details.

Results: Of 1,034,706 endoscopic procedures performed during the study period, 80 cases of IGP were identified (incidence rate: 0.008%). The median patient age was 58.4 years (±15.1), with a slight male predominance (55%). Nonsurgical management succeeded in 66.3% (53/80) of cases, while 33.8% (27/80) required surgery.

Successful nonsurgical management was associated with shorter hospital stays (median: 12 vs. 16.5 days, P=0.005) and fewer in-hospital complications, such as septic shock (0% vs. 50%, P<0.001). Independent predictors of nonsurgical success included the absence of missed muscular injury (odds ratio [OR]=0.146, 95% confidence interval [CI] 0.034-0.622, P=0.009), upper gastrointestinal perforation (OR=7.182, 95%CI 1.446-35.668, P=0.016), use of endoscopic treatment (OR=12.152, 95%CI 1.705-86.588, P=0.013), and early intravenous antibiotics administered within 5 hours (OR=6.040, 95%CI 1.391-27.218, P=0.016) (Table 1).

Variable		P	OR	Beta coefficient	Score
Missed muscular injury	Yes	0.009	0.098 (0.022-0.442)	-1.927	0
	No				77
Upper GI perforation	Yes	0.016	7.587 (1.431-40.212)	1.972	79
	No				0
Endoscopic treatment	Yes	0.013	14.452 (1.960-106-569)	2.498	100
	No				0
Early intravenous antibiotic	Yes	0.016	6.042 (1.325-27.562)	1.798	72
	No				0

Table 1. Predictors and associated numeric points from the nomogram for predicting the success of nonsurgical treatment.

The nomogram assigned weighted points to each predictor, and the total score (ranging from 0 to 300 points) was calculated by summing the points assigned to each predictor. The total score demonstrated a nonlinear association with the probability of nonsurgical success: 0 points corresponded to a 0.1 (10%) probability of success; 50 points to 0.3 (30%); 100 points to 0.5 (50%); 150 points to 0.8 (80%); 200 points to 0.9 (90%); 250 points to 0.95 (95%); and 300 points to 0.99 (99%).

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Internal validation using bootstrap resampling (C-index = 0.876) and five-fold cross-validation (C-index = 0.850) confirmed the model's reliability. Calibration curves showed high agreement between predicted and observed outcomes, and decision curve analysis supported the clinical utility of the model.

Conclusion: This study presents a validated nomogram integrating critical predictors to estimate nonsurgical success in IGP management. These findings highlight the importance of clinician vigilance, timely interventions, and the utility of tailored endoscopic treatments in optimizing patient outcomes.

Disclosure: Nothing to disclose.

PP0181

ENDOSCOPIC VACUUM THERAPY VERSUS VACSTENT FOR THE TREATMENT OF LEAKS AFTER UPPER GASTROINTESTINAL SURGERY FOR CANCER: A RETROSPECTIVE COHORT STUDY FROM A TERTIARY REFERRAL CENTER

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Introduction: Endoscopic Vacuum Therapy (EVT) and Fully Covered Metal Stents (FC-SEMS) are standard treatments for anastomotic leaks (AL) following upper gastrointestinal (UGI) surgery[1-3].

Recently, a new hybrid device combining an FC-SEMS with a polyurethane EVT sponge (VACStent) has been introduced, showing promising results[4]. However, there is not comparative data between VACStent and traditional endoscopic methods for ALs.

Aims & Methods: In this retrospective study, all patients with AL after UGI oncologic surgery who underwent EVT or VACStent at a tertiary academic center from March 2015 to September 2024 were included. The two endoscopic techniques were compared.

The primary outcome was clinical success (CS), with secondary outcomes including number of devices used, treatment duration, and adverse events (AEs).

Results: Over 114 months, 103 patients [92 EVT, 11 VACStent] were enrolled. Of these, 74.8% had AL following Ivor-Lewis esophagectomy. The two cohorts did not differ significantly in terms of comorbidities, neoadjuvant treatment, leak number/median size, or rates of redo-surgery before endoscopic management. Both EVT and VACStent were used as first-line treatments in a similar rate of patients (79.3% vs. 90.9%, p=0.36).

The EVT group had a higher CS rate than the VACStent group, though this difference was not statistically significant (72.8% vs. 54.5%, p=0.21). Compared to the EVT group, the VACStent group required significantly fewer endoscopic sessions (1±0.7 vs. 3±2.5, p<0.001) and had a shorter mean treatment duration (12.1±11.2 vs. 14.1±12.9 days, p=0.62).

Additionally, a significantly lower proportion of patients in the VACStent group required admission to the intensive care unit (ICU) (27.3% vs. 84.4%, p<0.001). In the EVT group, three adverse events were recorded (one bleeding episode and two sponge dislodgments), while no AEs were reported in the VACStent group (p=0.54) (Table 1).

	EVT n=92	VACStent n=11	P
Clinical success, n	67 (72.8%)	6 (54.5%)	0.21
Number of sessions	3 ± 2.5	1 ± 0.7	<0.001
Treatment duration (days)	14.1 ± 12.9	12.1 ± 11.2	0.62
Adverse events, n	3/276 (1.0%)	0/13 (0%)	0.54
Hospital stay (days)	35.0 ± 28.0	45.3 ± 34.5	0.26
ICU admission, n	76 (84.4%)	3 (27.3%)	<0.001

Table 1. EVT vs VACStent: Outcomes.

Conclusion: In this retrospective study, EVT and VACStent showed comparable success in treating post-UGI surgery leaks. VACStent was associated with a reduced need for ICU admission and fewer endoscopic sessions. Larger comparative data are warranted to define the role of VACStent in the treatment of leaks after UGI surgery.

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Disclosure: Nothing to disclose.

PP0182

ENDOSCOPIC VACUUM THERAPY FOR THE MANAGEMENT OF UPPER GASTROINTESTINAL TRACT LEAKS AND DEFECTS: A RETROSPECTIVE STUDY FROM A TERTIARY REFERRAL CENTER IN NEARLY 100 PATIENTS

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Introduction: Managing leaks and defects of the upper gastrointestinal (UGI) tract remains challenging. Endoscopic Vacuum Therapy (EVT) is a promising option[1-3]; However, data on its efficacy in large cohorts remain limited.

Aims & Methods: This retrospective study with prospectively collected data enrolled all consecutive patients with UGI defects treated by EVT at a single academic center (San Raffaele Hospital, Milan, Italy) from March 2015 to September 2024.

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The primary outcome was clinical success (CS).

Secondary outcomes included median treatment duration/devices, technical success (TS) and adverse events (AEs). Clinical success (CS) was defined as complete endoscopic and/or radiologic defect healing, with the patient remaining asymptomatic. Adverse events (AEs) were classified according to the AGREE classification.

Results: Over a period of 114 months, 96 patients [72.9% male, median age 67 years (IQR 57-75), 49% ASA 3] were included. Among them, 66 patients (68.7%) had anastomotic leaks after Ivor-Lewis esophagectomy, with 57% undergoing neoadjuvant chemotherapy. The median leak size was 7 mm (IQR 4-10), with a median circumferential involvement of 20% (IQR 10-40). A fistula involving the airways was present in 28.1% of cases. EVT was the first-line treatment in 78.1% of patients, with a median time from surgery to EVT of 7 days (IQR 4-9).

The overall clinical success (CS) rate was 72.9% (70/96 patients). Of the remaining patients, 20 underwent redo-surgery. Technical success for all procedures was 100% (302/302 procedures). The median treatment duration was 11 days (IQR 5-18), with a median of 3 EVT sessions (IQR 2-4). Three adverse events (AEs) (3/302, 1%) were recorded, including one bleeding episode and two sponge dislodgments, all managed endoscopically (AGREE IIIa). During a median follow-up of 62 days (IQR 33-155), two cases (2.1%) of esophageal strictures were reported (Table 1)

In the subgroup analysis, the intraluminal (IL) group (n=70) had a lower mean number of EVT sessions (3±2 vs. 5±3, p<0.001) and a shorter mean treatment duration (11±10 vs. 21±16 days, p<0.001) compared to the intracavitary (IC) group (n=26).

Patients n=96	Median (IQR) OR n (%)
Clinical success, n	70 (72.9%)
Technical success, n	302/302 (100.0%)
Adverse events, n	3/302 (1%)
Treatment duration (days)	11 (5-18)
Number of EVT sessions	3 (2-4)
Hospital stay (days)	28 (17-43)
ICU admission, n	14 (14.9%)
Stricture, n	2 (2.1%)
Follow-up (days)	62 (33-155)

Table 1: Outcomes.

Conclusion: EVT is a safe and effective treatment for UGI defects. The IL approach was associated with shorter treatment duration and fewer sessions compared to the IC approach. Further controlled studies are needed to optimize EVT application and outcomes.

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PP0183

VACSTENT FOR THE TREATMENT OF UPPER GASTROINTESTINAL TRACT LEAKS AND DEFECTS: RESULTS FROM PROSPECTIVE SERIES IN A TERTIARY REFERRAL CENTER

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Introduction: The VAC-Stent is a fully covered stent encased in a polyurethane sponge cylinder, offering a promising approach for treating upper gastrointestinal (UGI) tract leaks and defects[1]; however, available evidence remains limited and preliminary.

Aims & Methods: This prospective study enrolled all consecutive patients with UGI leaks treated with VACStent placement from May 2023 to September 2024 at San Raffaele Hospital, Milan, Italy.

The primary outcome was clinical success (CS).

Secondary outcomes included technical success (TS) and adverse events (AEs). CS was defined as defect healing confirmed by endoscopic and radiologic evaluation, while TS was defined as successful stent placement and removal.

Results: Fourteen patients [64.3% male, median age 65 years (IQR 59–75); 11 with anastomotic leaks after Ivor-Lewis esophagectomy and 3 with perforations after circumferential esophageal ESD] were enrolled. VACStent was the first-line treatment in 92.9% of cases, with a median leak size of 6 mm (IQR 5-11), and 28.6% of patients had previously undergone re-surgery. The median time from surgery/endoscopy to initial VACStent placement was 7 days (IQR 3-9). The median treatment duration was 9.5 days (IQR 7-14), with a median hospital stay of 29.5 days (IQR 21-63) and a median of 2 VACStents per patient (IQR 1-2).

CS rate was 69.2% (9/13 patients); among the 4 remaining cases, three required rescue surgery, and one patient died of sepsis. All patients treated for perforations following esophageal ESD achieved complete healing without developing strictures. TS was 96.7% (30/31); one case of technical failure occurred in a patient with an anastomotic leak who had previously received radiotherapy for mediastinal lymphoma. No VACStent-related AEs were observed (Table 1).

Patients n=14	Median (IQR) OR n (%)
Clinical success, n	10 (69.2%)
Technical success, n	30/31 (96.7%)
Adverse events, n	0 (0)
Treatment duration (days)	9.5 (7-14)
Number of sessions	2 (1-2)
Hospital stay (days)	29.5 (21-63)
ICU admission, n	8 (57.1%)
Stricture, n	0 (0)
Follow-up (days)	42 (29-114)

Table 1: VACStent outcomes.

Conclusion: This prospective series suggests that the VACStent is an effective and safe treatment option for upper GI leaks, including perforations after esophageal ESD. Larger controlled studies are warranted to better define its role and optimize outcomes in this complex setting.

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Disclosure: Nothing to disclose.

PP0184

PNEUMATIC VS BOUGIE DILATION OF BENIGN ANASTOMOTIC STRICTURES AFTER ESOPHAGECTOMY: RESULTS FROM A MULTICENTER COMPARATIVE STUDY FROM TERTIARY REFERRAL CENTERS

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Introduction: Benign anastomotic strictures (AS) represent a common adverse event following esophagectomy. Endoscopic treatment with pneumatic balloon dilation (PD) or bouginage using Savary-Gilliard dilators (BD) is routinely adopted, yet comparative evidence between these two strategies remains limited.
Aims & Methods: We performed a retrospective analysis of prospectively collected data of patients with AS after esophagectomy performed in two referral Italian centers (IRCCS San Raffaele Hospital and IRCCS Policlinico San Donato, both in Milan). In the analysis only adult patients with benign AS naive to prior treatment were included. AS was identified in the presence of dysphagia and stenosis at the anastomosis on oral contrast studies or endoscopy and required endoscopic treatment. Technical and Clinical Success (TS, CS) were defined as fully completed planned endoscopic procedure with technical issues and a Mellow and Pinkas' Dysphagia Score (DS) improvement of at least one level after achieving a 14mm caliber. Recurrence was defined as DS of 3 or 4, without the possibility to maintain a satisfactory luminal diameter for 4 weeks after achieving the target diameter of 14 mm. Surgical complications and endoscopic-related adverse events (AEs) were classified with Clavien-Dindo and AGREE Classification, respectively.
Results: 1729 esophagectomies were performed in two italian referral centers (IRCCS San Raffaele Hospital and IRCCS Policlinico San Donato, both in Milan) from January 2014 to December 2024, and 61 patients (3.5%) developed benign AS, treated by BD (33/61, 54.1%) and PD (28/61, 45.9%). Indications included squamous cell carcinoma (45.9%) and adenocarcinoma (42.6%). Most patients underwent minimally invasive surgery (60.7%), predominantly McKeown (50.8%), followed by Ivor Lewis (49.2%). Neoadjuvant therapy was administered in 60.7% of cases. Baseline characteristics were comparable between PD and BD groups, including demographic and surgical aspects (surgical approach, anastomosis type, caliber, and complications).
Also, time from surgery to AS diagnosis (115 vs 78 days, p=0.435), and initial AS diameter (9 mm vs 10 mm, p=0.106) were comparable. TS was 100% in both groups. CS was slightly higher in the BD group (97.0% vs 89.3%, p=0.231), with similar recurrence rates (24.0% vs 25.0%, p=0.931) and AEs (3.6% vs 0%, p=0.278). The median number of sessions was 1.0 in both groups (p=0.350).
However, total treatment duration was significantly longer in the BD group (median 42 vs 20 days, p=0.005). Kaplan-Meier analysis confirmed a significantly shorter time to strategy change in the PD group (log-rank p=0.047), with most changes occurring within the first 60 days.
Multivariate analysis identified BD (OR 3.6; 95% CI 1.2–10.9; p=0.024) as an independent predictor of step-up therapy, along with lower stricture diameter at diagnosis (OR 0.81; 95% CI 0.69–0.95; p=0.010) and lower BMI (OR 0.93; 95% CI 0.87–0.99; p=0.040).
Conclusion: Both PD and BD are effective and safe first-line options for benign AS after esophagectomy. However, BD was associated with significantly longer treatment duration and a higher independent risk of step-up therapy. PD may represent a more efficient strategy in selected patients, particularly those with narrower strictures or lower BMI. Kaplan-Meier analysis further supports a timelier therapeutic reassessment in the PD group.
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PP0185

THE COMPARISON OF ENDOSCOPIC SUBMUCOSAL DISSECTION OF PHARYNGEAL CANCERS BETWEEN PATIENTS WITH AND WITHOUT PREVIOUS HISTORY OF RADIOTHERAPY INCLUDING THE LONG-TERM OUTCOMES

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Introduction: There has been a rise in the uptake of endoscopic submucosal dissection (ESD) for superficial pharyngeal cancers, including in patients with previous history of radiotherapy, whether as salvage treatment of residual lesions or for metachronous lesions inside the irradiated field. Previous studies have demonstrated the safety and effectiveness of ESD for pharyngeal cancers[1].
However, the safety and the effectiveness of pharyngeal ESD in cases with previous history of radiotherapy is not clear[2,3].
Aims & Methods: Our aim was to investigate whether radiotherapy affects outcomes after pharyngeal ESD. We conducted a retrospective cohort study of consecutive patients who underwent pharyngeal ESD between December 2016 and February 2025 at our institution.
All the cases were managed together with otolaryngologists, including the daily assessment of the wound and swallowing function with the transnasal flexible laryngoscope in order to decide the timing of the re-initiation of food after ESD. The ESD itself, as well as the short and long-term outcomes were compared between the patients with or without history of radiotherapy.

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Results: There were 22 cases (25 lesions) of ESD performed with previous history of radiotherapy and 32 cases (33 lesions) without prior radiotherapy. The median age was 71 (range 55-85) and 98.3% were men. Radiotherapy dose ranged from 54 to 70 Gy.

There were no significant differences in the baseline characteristics between the groups of patients with or without prior radiotherapy including the age, sex, or the proportion of patients with previous history of pharyngeal ESD. En-bloc resection was achieved in all cases. ESD in the pharynx post-radiotherapy was associated with an increased percentage of cases with severe fibrosis (41% vs. 14% p<0.05), and the speed of dissection was slower (33.4mm^2/min vs. 45.8mm^2/min p=0.03), but there were no significant differences in the short-term outcomes between the groups of patients with or without prior radiotherapy in terms of of bleeding (0% vs. 3%), aspiration pneumonia (8% vs. 9%), the duration to the resumption of food (median 4 vs. 5 days) or hospital stay (median 9 vs. 9 days).

During a median follow-up period of 906.5 days (range 22-3021 days), the 3-year overall survival rate was lower in patients with previous history of radiotherapy compared to patients without prior radiotherapy (55.1% vs. 100%, p=0.000177).

Of the patients with previous history of radiotherapy, one patient died from metastatic recurrence of pharyngeal cancer, but the other five deaths were not directly associated with pharyngeal cancer (1=acute cardiac death, 1=unknown cause but unrelated to pharyngeal cancer, 1=lung cancer, 1=mantle cell lymphoma, 1=sinonasal neuroendocrine cancer).

There were no significant differences in the 3-year recurrence-free survival and the 3-year disease-related survival rates between the patients with and without prior history of radiotherapy (92.9% vs. 91.7% and 92.3% vs. 100%, respectively).

Conclusion: Previous history of radiotherapy increases the risk of severe fibrosis during ESD and is associated with slower speed of dissection, but does not compromise the safety or the swallowing function. In the long-term, ESD is an effective method for the treatment of superficial pharyngeal cancers, including in patients with previous history of radiotherapy. Furthermore, ESD is a minimally-invasive and surgery-sparing treatment option, which may be all the more important in patients with prior history of radiotherapy, given the worse prognosis from illnesses other than pharyngeal cancers in this group of patients.

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PP0186

PERORAL ENDOSCOPIC MYOTOMY IN PEDIATRIC POPULATION: A COMPARATIVE STUDY WITH ADULT POPULATION IN A PORTUGUESE TERTIARY CENTRE

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Introduction: Peroral endoscopic myotomy (POEM) is emerging as a minimally invasive therapeutic option for achalasia. Data are limited in pediatric population.

To compare the efficacy and safety of POEM in pediatric and adult patients with achalasia.

Aims & Methods: A prospective study was conducted, including all patients who underwent POEM, since August/2022, in a tertiary centre. Patients aged under 18yo were included in the pediatric group and above 18yo in adult group. Procedural-related parameters, adverse advents, pre- and post-intervention clinical, manometric and radiologic outcomes were evaluated.

Results: 41 patients were included, including three children (mean age: 14.3±1.2 vs. 51.8±19.1 yo; male gender: 100%(n=3) vs.39.5% (n=15)). According to the Chicago classification v4.0, all three children had type II achalasia (adults: type I-5.3% (n=2), type II-89.4% (n=34), type III-5.3%(n=2)).

No significant differences were observed between groups in relation to clinical symptoms (Eckardt score (ES): 7.6vs.0.9;ρ=0.23; average duration of symptoms: 21vs.19 months;ρ=0,36), pre-POEM integrated relaxation pressure (IRP) (36.7vs.19.0;ρ=0,11), esophagogastric junction contractile integral (EGJ-CI) (105,3vs.91,2;ρ=0.72) and barium swallow study at 1/5 minutes (4.5/5.1 vs. 12.6/10.0;ρ=0,17/0.29) and post-POEM IRP (9.75 vs. 6.19;ρ=0.32), EGJ-CI (7 vs 25.20;ρ=0.30) and barium swallow study (3.45/2.20 vs. 2.65/1.43;ρ=0.57/0.56), evaluated at 4.3±1.3 and 8.5±2.1months of follow-up. Technical success was 100% in all groups with mean procedure time of 91.7±12.6minutes for children (vs. 86.1±21.6 for adults; ρ=0.35).

Clinical success was similar in both groups (ES: children-0±0 vs. adults-0.31±0.63;ρ=0.69). There were no significant differences between groups regarding intraprocedural complications (67%vs.66%;ρ=0.98), post-procedural complications or major adverse events. No mortality or need for surgery. Symptomatic gastroesophageal reflux disease (GERD), successfully treated with daily PPI, was only observed in 5 (13.2%) adults (vs. 0% of pediatric patients;ρ=0.021).

Conclusion: POEM proved to be effective and safe for achalasia both in adults and pediatric patients, with no statistically significant differences regarding technical, clinical, manometric and imaging success or complications. Achalasia should be treated as early as possible, with apparently lower incidence of post-POEM GERD in pediatric patients.

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Disclosure: Nothing to disclose.

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PP0187

UNBURDENING THE PILL LOAD: REAL-WORLD EVIDENCE OF MEDICATION REDUCTION AFTER ENDOSCOPIC SLEEVE GASTROPLASTY

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Introduction: The management of obesity is often complicated by associated comorbidities such as type 2 diabetes, hypertension and hyperlipidaemia—all of which contribute to significant polypharmacy. High pill burden is linked to reduced adherence, increased healthcare costs, and lower quality of life. While Endoscopic Sleeve Gastroplasty (ESG) has been established as a minimally invasive intervention for weight loss, its impact on medication use remains underexplored.

This study aims to evaluate changes in medication burden following ESG in a real-world cohort, with a focus on the potential for pharmacological de-escalation alongside weight loss.

Aims & Methods: We conducted a retrospective review of 236 patients who underwent ESG at Singapore General Hospital between November 2020 and December 2025. All procedures were performed by a single experienced endoscopist using a U-shaped suturing pattern. The primary outcome was change in medication burden, defined as the total number of prescribed medications at baseline, 6 months, and 12 months following ESG. Data on medication count were collected from electronic medical records, including all chronic medications prescribed for obesity-related comorbidities.

Results: At baseline, 17.8% of patients were on one or more oral anti-diabetic medications, 2.6% were on insulin, 30.9% were taking antihypertensives, and 22% were on lipid-lowering therapy. At 6 months post-ESG, the proportion of patients on anti-diabetic medications decreased to 15.2%, and insulin use dropped slightly to 2.1%. Anti-hypertensive use declined to 20.8%, and lipid-lowering therapy was reduced to 17.1%. By 12 months, further reductions were observed: 12.7% of patients remained on anti-diabetic medications, only 1.4% were on insulin, and use of anti-hypertensives and lipid-lowering agents fell to 20.8% and 18.8%, respectively.

The total number of medications per patient decreased significantly over time. At baseline, patients were on an average of 1.36 medications targeting diabetes, hypertension, and hyperlipidemia. This declined to 0.61 medications at 6 months and 0.59 at 12 months, corresponding to mean reductions of 0.75 (95% CI: 0.42–1.07, $p < 0.001$) and 0.77 (95% CI: 0.46–1.08, $p < 0.001$), respectively. These reductions were consistent across all major cardiometabolic drug classes.

Conclusion: In addition to driving meaningful weight loss, ESG contributes to a measurable reduction in medication burden for obesity-related comorbidities. These findings support the broader role of ESG in promoting pharmacologic de-escalation, potentially improving long-term adherence and quality of life in patients with obesity.

Disclosure: None.

PP0188

REAL-WORLD, LONG-TERM OUTCOMES OF ENDOSCOPIC SLEEVE GASTROPLASTY: A TWO-YEAR COHORT STUDY FROM A TERTIARY ASIAN CENTRE

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Introduction: Obesity is a global health challenge associated with significant morbidity, mortality, and economic burden. Endoscopic Sleeve Gastroplasty (ESG) has emerged as a safe, incisionless, and minimally invasive alternative to bariatric surgery for patients with obesity who are either unwilling or unsuitable for surgical options. While ESG has demonstrated promising short-term results in terms of weight loss and metabolic improvement, there remains a paucity of data on long-term outcomes, particularly beyond one year.

This study aims to evaluate the long-term efficacy of ESG in a large Asian cohort, focusing on total body weight loss (TBWL) as the primary outcome, with body mass index (BMI) change and absolute weight loss as secondary outcomes.

Aims & Methods: We conducted a retrospective review of 236 patients who underwent ESG at Singapore General Hospital between November 2020 and December 2025. All procedures were performed by a single experienced endoscopist using a U-shaped suturing pattern. The primary outcome was the percentage of total body weight loss (%TBWL) at 6, 12, and 24 months post-procedure.

Results: The cohort consisted predominantly of female patients (69%), with the majority being of Chinese ethnicity (42.8%). The mean age at baseline was 41.4 years (SD ± 10.6), and the mean baseline BMI was 35.0 kg/m^2 (SD ± 6.1). All ESG procedures were technically successful, with an average of five sutures placed per patient and a mean procedure time of 75 minutes (SD ± 17). No major complications were reported, and the average hospital stay was one day.

Follow-up data were available for 114 patients at 6 months, 86 at 12 months, and 41 at 24 months. The mean percentage total body weight loss was 15.9% (SD ± 10.5) at 6 months, 12.7% (SD ± 7.8) at 12 months, and 9.9% (SD ± 7.8) at 24 months, reflecting a sustained reduction in weight over time. As secondary outcomes, BMI decreased significantly from baseline by a mean of 6.0 kg/m^2 (SD ± 5.3) at 6 months, 4.2 kg/m^2 (SD ± 5.7) at 12 months, and 3.5 kg/m^2 (SD ± 5.7) at 24 months (all $p < 0.001$). These were accompanied by corresponding mean weight losses of 16.8 kg (SD ± 17.1), 12.1 kg (SD ± 8.6), and 9.6 kg (SD ± 9.0) at the respective timepoints, all statistically significant ($p < 0.001$).

Conclusion: ESG is a safe and effective endoscopic intervention for weight loss, with sustained and clinically meaningful reductions in total body weight, BMI, and absolute weight observed up to two years post-procedure. These real-world data from a large tertiary Asian centre support the long-term utility of ESG as a viable non-surgical option for obesity management.

Disclosure: None.

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PP0189

ANTI-OBESITY MEDICATION USE BEFORE AND AFTER ENDOSCOPIC SLEEVE GASTROPLASTY

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Introduction: Pharmacologic therapy is a key component of obesity management, either as standalone treatment or in conjunction with procedural interventions. The role of weight loss medications in the peri-ESG period remains unclear, with questions surrounding whether ESG reduces the need for pharmacotherapy or serves as a platform for combination therapy. Limited data exist on how prescribing patterns evolve before and after ESG in real-world settings.

This study aims to evaluate the use of anti-obesity medications before and after ESG, focusing on patterns of pharmacologic initiation and its relationship to weight loss outcomes.

Aims & Methods: We conducted a retrospective review of 236 patients who underwent ESG at Singapore General Hospital between November 2020 and December 2025. All procedures were performed by a single experienced endoscopist using a U-shaped suturing pattern. For this analysis, we examined the use of anti-obesity medications before and after the ESG. Medications included GLP-1 receptor agonists, sympathomimetic agents, lipase inhibitors, and combination therapies.

The primary outcome was the proportion of patients receiving pharmacotherapy across timepoints and its association with total body weight loss (%TBWL).

Results: Among the 236 ESG patients, 21.6% had previously used anti-obesity medications, most commonly GLP-1 receptor agonists or phentermine. This group consistently achieved the lowest total body weight loss (%TBWL) across all follow-up points. Compared to medication-naïve patients, their mean %TBWL was significantly lower at 6 months (9.9% vs 17.2%, p = 0.015), 12 months (7.8% vs 13.8%, p = 0.026), and 24 months (4.3% vs 10.6%, p = 0.003). These findings underscore the complexity of obesity care, as even patients exposed to multiple treatment modalities may continue to experience suboptimal outcomes.

After ESG, 37.7% of patients were initiated on pharmacotherapy, most commonly semaglutide (n = 66), including 38 on subcutaneous and 28 on oral formulations, with estimated mean doses of 1.63 mg/week and 13.2 mg/day, respectively. Phentermine and liraglutide were used in 13.9% and 3.0% of patients. Patients who received medications only after ESG had lower %TBWL at 6 months compared to those never on medications (14.0% vs 18.8%, p = 0.027), but outcomes were similar by 12 and 24 months. This suggests that pharmacotherapy may help support suboptimal early responders and mitigate longer-term differences.

Conclusion: Anti-obesity medications, particularly semaglutide, are commonly introduced after ESG to support patients with suboptimal early weight loss. While those requiring pharmacotherapy initially experienced less %TBWL, outcomes equalized by 12 months, suggesting a role for adjunctive treatment in slower responders. Patients who had previously used weight loss medications consistently achieved poorer outcomes, highlighting the challenges of obesity management even with multimodal therapy. These findings underscore the importance of individualized treatment strategies and the complementary role of pharmacotherapy alongside ESG in real-world practice.

Disclosure: None.

PP0190

FULL THICKNESS RESECTION IN THE UPPER DIGESTIVE TRACT USING THE FTRD OVESCO GASTROSET SYSTEM: A FRENCH MULTICENTER RETROSPECTIVE STUDY WITHIN THE GRAPHE GROUP

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Introduction: Full-thickness endoscopic resection with the COLONIC FTRD Ovesco system has become a well-established technique for removing superficial colonic lesions smaller than 25 mm, particularly in cases of submucosal tumors or recurrent adenomatous lesions after previous endoscopic mucosal resection. However, its application in the upper digestive tract remains less documented due to its recent uses (1).

Aims & Methods: This French multicenter retrospective study, conducted within the GRAPHE group, aimed to evaluate the safety and efficacy of the FTRD technique using the FTRD Gastroset OVESCO KIT for gastric and duodenal lesion resection. The study collected data on indications, resection sites, technical success rates, monobloc and R0 resection rates, and potential complications.

Results: Between 2022 and 2024, 17 patients from six French centers were included (6 men and 11 women, with a mean age of 58 years [35–75]). All procedures were performed under general anesthesia with intubation, CO_2 insufflation, and prophylactic antibiotic administration. The primary indication was submucosal lesions (SML) in 75% of cases (n=12), followed by adenomatous recurrence in 19% (n=3) (two after endoscopic mucosal resection and one after endoscopic submucosal dissection), and two atypical lesions (one polypoid and one pseudo-depressed) in 12.5% of cases. More than two-thirds of the lesions (n=12, 70%) were located in the duodenal bulb, while the remaining 30% (n=5) were in the stomach (four in the antrum and one in the fundus). The median lesion size was 20 mm [range: 10–25 mm]. The overall technical success rate was 94% (16/17 cases), with one failure due to an inability to pass through Killian's mouth in a patient previously irradiated for laryngeal cancer. No technical failures related to the FTRD Gastroset KIT or immediate complications were reported.

All resections were 'monobloc', with the FTRD KIT grasping forceps used in approximately 80% of cases and the Ovesco anchor (OTSC Anchor 165 mm) in two cases. Complete histologic resection (R0) was achieved in 94% of cases (n=15). Histopathological analysis showed that 50% of lesions were grade G1 neuroendocrine tumors (NETs) (n=8), 18.75% Brunner's gland hyperplasias (n=3), 18.75% fourth-layer submucosal tumors (duodenal leiomyoma, gastric GIST, hamartoma), and 12.5% adenomas with low-grade dysplasia (n=2). One R1 resection was observed for a NET in the fundus. The only postoperative complication was a delayed bleeding episode occurring 24 hours after the procedure.

Conclusion: Full-thickness resection using the FTRD Ovesco Gastroset system in the upper digestive tract demonstrated a high technical success rate (94%) and complete resection rate (94%) for 20 mm mucosal and submucosal lesions, with no major complications or mortality. Duodenal submucosal lesions were the most common indication, with neuroendocrine tumors (NETs) being the predominant histologic finding. The precise role of this technique in comparison to submucosal dissection and endoscopic mucosal resection remains to be determined.

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Disclosure: Nothing to disclose.

PP0191

FREQUENCY AND SEVERITY OF ADVERSE EVENTS ASSOCIATED WITH GASTRIC PERORAL ENDOSCOPIC MYOTOMY (G-POEM) – A RETROSPECTIVE ANALYSIS

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Introduction: Gastric peroral endoscopic myotomy (G-POEM) is an emerging endoscopic technique for the treatment of refractory gastroparesis, particularly benefiting patients with pylorospasm. It is considered a safe procedure with a low risk of adverse events when performed by experienced endoscopists.
Our aim was to perform a comprehensive analysis of all adverse events associated with G-POEM.
Aims & Methods: We performed a retrospective analysis of prospectively collected data from all patients who underwent G-POEM for gastroparesis at our centers (Prague, Brno) between November 2017 and April 2025. All procedures were performed by a single endoscopist (JM).
The primary outcome was procedure-related adverse events rate and severity occurring during the perioperative and early postoperative period. Adverse events were categorized as follows: deviations from the expected peri and postoperative course were defined as "undesirable events."
Undesirable events that met the criteria of the Clavien Dindo surgical classification (C-D) were considered "true adverse events".
Results: Out of 70 procedures, no true adverse or undesirable events occurred in 51 procedures (72.9%). Perioperatively, the following undesirable events occurred in 18 patients: : gastric (4; 5.7%) and duodenal mucosal injuries (2; 2.8%), major bleeding during the procedure treated endoscopically (1; 1.4%), serosal perforation (10; 14.3%), capnoperitoneum requiring puncture (3; 4.3%), and subcutaneous emphysema (1; 1.4%).
In the early postoperative period, adverse events occurred in 3 patients: one patient complained of significant abdominal pain necessitating analgetics, subsequently, a small leak from the pyloric region to the subhepatic region was described on CT and gastroscopy confirmed ulceration at the tunnel site.
The patient responded well to treatment with intravenous proton pump inhibitors. In two other cases, both involving patients with diabetic gastroparesis, decompensation of diabetes occurred during the hospital stay post-procedure.
The following "undesirable" events were considered true adverse events according to the C-D classification: serosal perforation (10; C-D I), leak requiring prolonged hospitalization (1; C-D IIIa) and diabetes decompensation (2; C-D II). These complications occurred in 13 procedures (18.6%).
The mean duration of hospitalization after the procedure was 2 days (range 1-7). Three months after the procedure, 1 patient developed mild dumping syndrome, which was successfully treated conservatively.
Conclusion: G-POEM is a safe procedure. The rate of true adverse events according to Clavien-Dindo classification is low and severe AEs occurs rarely.
Disclosure: Nothing to disclose.

PP0192

ENDOSCOPIC SLEEVE GASTROPLASTY USING SINGLE-CHANNEL VS DOUBLE-CHANNEL SUTURING DEVICE: A MULTICENTER EXPERIENCE

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Introduction: Endoscopic sleeve gastroplasty (ESG) is an effective treatment for obesity. The conventional endoscopic suturing device requires a double channel endoscope. Recently, a novel single-channel suturing device (Apollo Overstitch Sx) has become available, potentially expanding the accessibility and ease of ESG procedures. However, comparative data evaluating the efficacy and safety of these two devices remain limited.
This multicenter study aims to assess and compare the clinical outcomes and safety profiles of the conventional double-channel versus the novel single-channel suturing devices in patients undergoing ESG.
Aims & Methods: We reviewed the records of 236 patients who underwent ESG at Singapore General Hospital, Singapore and New Delight Medical Clinic, Taiwan between November 2020 to March 2024. Our primary outcome was technical feasibility comprising of procedural time, number of sutures and safety. Our secondary outcome was to determine percentage total body weight loss (TBWL%) at 12 months.
Results: The study population had a mean age of 39.8 ± 8.8 years and a mean BMI of 32.9 ± 6.0 kg/m^2. The majority of patients were female (73.5%, n = 175) and of Chinese ethnicity (78.4%). In terms of treatment, 138 patients underwent suturing using a single-channel device, while 77 patients received treatment with a conventional dual-channel device. Using propensity score matching, patients were matched 1:1 based on age and initial BMI for comparative analysis. The mean procedure time was comparable between the dual channel device at 63.2 ± 11.8 mins vs. the single channel device which was 64.8 ± 5.0 mins (mean difference 1.6mins, p>0.05). However, the single-channel device resulted in a lower mean number of sutures (4.5 ± 0.5 vs. 5.1 ± 0.9; mean difference 0.6 sutures, p < 0.001). There were no statistically significant differences in post-ESG complications, including bleeding, abdominal pain, and vomiting.
At 6 months, there was no significant difference in TBWL% between the dual-channel ESG vs single channel (14.7 ± 6.0% vs. 16.2 ± 5.2%; mean difference 1.4%, p >0.05). However at 1 year, the dual-channel ESG group had a higher %TBWL (13.4 ± 7.6% vs. 7.9 ± 9.0%; mean difference 5.6%, p < 0.001), although this could be confounded by loss to follow-up in the dual-channel group.
Conclusion: While the single-channel device showed a significant reduction in the number of sutures required, the dual-channel device resulted in a higher percentage of total body weight loss at 1 year. However, this finding may be confounded by differences in follow-up rates between the two groups. Further studies are needed to confirm these findings and establish the long-term efficacy and safety of these devices.
Disclosure: Dr. Asokkumar is a consultant for Apollo Endosurgery, USA.

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PP0193

NAVIGATING SELF-EXPANDABLE METALLIC STENT PLACEMENT IN INOPERABLE ESOPHAGEAL MALIGNANCIES: A LANDMARK-BASED TECHNIQUE USING VERTEBRAL COLUMN AND DIAPHRAGM

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Introduction: Self-expandable metallic stent (SEMS) placement is a palliative treatment for inoperable esophageal malignancies to improve the quality of life. Traditional methods rely on fluoroscopy and endoscopy, but anatomical landmarks can simplify the procedure.

Aims & Methods: To evaluate the technical success, safety and reproducibility of using vertebral column and diaphragm landmarks for SEMS placement.

This retrospective observational study included 801 patients with inoperable esophageal malignancies undergoing SEMS placement for dysphagia due to malignant esophageal strictures, by using vertebral column and diaphragm as landmarks, from 1st January 2006 to 31st December 2024. The primary endpoint was to assess the technical success, safety and reproducibility of this unique technique for SEMS placement.

Methodology for Stent Placement in Esophageal Tumors:

1. For Proximal or Mid-Esophageal Tumor, following algorithm was followed:

Step 1: Identify key anatomical landmarks using fluoroscopy.

Step 2: With the scope in place, under fluoroscopic guidance, identify the vertebra in line with the distal extent of the tumor using the diaphragm as a reference, followed by identification of vertebra at the proximal extent of the tumor. Count the vertebrae between the distal and proximal extent of the tumor. Assign a length of 2.5 cm per vertebral body. Ascertain the desired size of SEMS by calculating the total length of the stricture by counting the vertebrae, adding one vertebral body above and below the tumor for proper stent placement. Step 3: Using fluoroscopy, position the stent. Place the distal end of the stent at the distal vertebral marker. Position the proximal end of the stent at the proximal vertebral marker. Release the stent at the appropriate position to ensure full coverage of the stricture.

2. For Tumor Extending Beyond the Gastro-Esophageal Junction (GEJ), following algorithm was followed:

Step 1: Identify the proximal extent of the tumor. Align the proximal extent of the tumor with the vertebra, using the diaphragm as a reference for precise alignment. Measure the distance from the proximal end of the tumor to the proximal stomach using the scope to decide the required size of the stent.

Step 2: Place the stent under fluoroscopic guidance. Position the upper end of the stent at the identified vertebra. Extend the distal end of the stent into the stomach, past the GE junction to ensure proper coverage. Release the stent.

Results: 408 (50.9%) patients were females. The age range was 18-95 years (mean50±Standard deviation15). Predominant presenting complaint was dysphagia (85.39%). Average duration of dysphagia was 3.84 months. 594 (74.16%) patients had squamous cell carcinoma, whereas 207 (25.84%) patients were diagnosed with adenocarcinoma on histopathology. 365 (45.57%) patients had tumor confined to the lower one-third of esophagus. Major cause of inoperability was distant metastasis (52.81%). 240 (30%) patients had involvement of GEJ. Average length of malignant stricture was 8.4 cm. 139 (17.40%) patients required dilatation by using savary gilliard dilators before SEMS placement. TEF was present in 48 (6%) patients. Range of length of SEMS was 8-18 cm (12.5±2.26). Technical success was achieved in 100% cases. No major immediate complications occurred. Minor complications were rare.

Conclusion: Using vertebral column and diaphragm landmarks is a reliable, safe, simplified and reproducible approach for SEMS placement in patients with inoperable esopahgeal malignancies.

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Disclosure: Nothing to disclose.

PP0194

EFFECT OF ENDOSCOPIC PYLOROMYOTOMY (G-POEM) IN PATIENTS WITH GASTROPARESIS ON GASTRIC EMPTYING

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Introduction: Gastroparesis (GP) is defined as symptomatic delayed gastric emptying. The most frequent etiologies are diabetes mellitus, after esophageal or gastric surgery and about a third of patients have idiopathic GP. The gastric emptying study by using a scintigraphy scintigraphy (GES) is considered as a principal diagnostic method.

Endoscopic pyloromyotomy (G-POEM) is a method improving symptoms in about 60-70% of patients with GP. Its effect on gastric emptying remains controversial.

Aims & Methods: We performed a retrospective single-center analysis to assess the clinical efficicacy of G-POEM based on GES parameters. Between 2016 – 2024, a total of 72 patients with GP (confirmed by scintigraphy) underwent G-POEM and were later invited for clinical as well as scintigraphic assessment after 3, 12 and 36 months.

Of these patients, 4 were excluded for missing data, incomplete follow-up data or non-standard testing meal. GES was always performed using the standardized 4 hours protocol with the standard testing meal mixed with 99Tc-DTPA.

For this study, all GES were reanalyzed and the following parameters were established : residual activity after 2 hours, residual activity after 4 hours, biological halftimes of ingested activity and proximal-to-distal count ratio for time 0 (PDCR0).

Results: Mean value of GCSI before G-POEM was 3.1 and decreased to 1.6, 1.0 and 0.7 after 3, 12 and 36 months. Median value of GES after 2 hours was 60.2% before G-POEM and 38%, 32.6% and 24.2% after 3, 12 and 36 months. Median value of GES after 4 hours was 21.9% before G-POEM and 7.8%, 6.5% and 3.9% after 3, 12 and 36 months. Correlation index was weakly positive for GCSI and GES after 4 hours only for the first follow-up visit and negative for all remaining visits.

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The residual activity after 2 hours and residual activity after 4 hours decreased after G-POEM during all follow-up visits (p<0,001). Mean decrease of residual activity after 4 hours was 12.89%, 15.13% and 6.7% after 3, 12 and 36 months.
Biological halftime also decreased after the procedure with statistically significant results with the mean halftime before G-POEM being 150 min and 86 min, 78.4 min and 65.8 min after 3, 12 and 36 months.
There was a difference between the diabetic and non-diabetic groups, with higher residual activity after 2 hours [median 29.6% before G-POEM, 9.0% after 3 months (p<0.001) and 11.0% after 12 months (p=0.005)] and 4 hours [median 64.8% before G-POEM, 39.4% after 3 months (p<0.001) and 37.7% after 12 months (p=0.012)] respectively in patients with diabetes.
The predictive value of PDCR0 for treatment success after G-POEM was only calculated for the first follow-up visit but was not statistically proved (p=0,712).
Conclusion: In our group of patients there is a weak correlation between the decrease of GCSI and GES early after G-POEM. The decrease in GES parameters tends to be lower in patients with diabetes compared to non-diabetic patients. Based on our data, predicting clinical success of G-POEM using PDCR0 cannot be recommended.
Disclosure: Nothing to disclose.

PP0195

COMPARATIVE ANALYSIS OF Z-POEM AND FLEXIBLE ENDOSCOPIC SEPTOTOMY FOR SYMPTOMATIC ZENKER'S DIVERTICULUM

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Introduction: Nowadays, endoscopic treatment for esophageal Zenker's diverticulum is considered the standard of care, with flexible endoscopic septotomy (FES) being the first technique to be widely adopted. Since 2016, submucosal tunneling Per-Oral Endoscopic Myotomy (Z-POEM) has emerged as an innovative therapeutic approach. However, medium-term efficacy comparisons between Z-POEM and FESremain limited.
Aims & Methods: The aim of this study was to compare the clinical success and safety of Z-POEM and FES.
This study was a retrospective analysis of a large group of patients treated for Zenker's diverticulum across two tertiary care centers. The patients were divided into two groups: those who underwent Z-POEM between 2021 and 2024, and those who underwent FES between 2008 and 2024. Demographic and clinical data, including Charlson Comorbidity Index, Kothari scores, clinical success rates, adverse events, and follow-up duration, were analyzed and compared among the two groups.
Results: Treatment in the study period. Of these, 32 patients (mean age 66.8 ± 10.7 years) underwent Z-POEM, while 34 patients (mean age 73.5 ± 10.2 years) underwent FES. The mean Charlson Comorbidity Index was similar between groups (Z-POEM: 3.28 ± 1.80 vs. FES: 3.4 ± 1.6; p = 0.708). The pre-procedural Kothari-Haber score was comparable (mean 7.3 ± 1.4 for Z-POEM vs. 7.3 ± 1.1 for FES; p= 0.981). The post-procedural Kothari-Haber score was significantly lower in the ZPOEM group (1.3 ± 1.3) compared to the FES group (2.7 ± 2.8; p = 0.010), indicating better symptom resolution. The median follow-up period was 37.7 months (IQR: 14.8 – 74.6 months) for the FES group and 12.2 months (IQR: 7.4 – 21.9 months) for the Z-POEM group (p < 0.0001).
Technical success was achieved in 100% in both groups. Clinical success was significantly higher in the Z-POEM group (93.8%; 95% CI: 75% - 99%) compared to the FES group (64.5%; 95% CI: 50% - 82%; p= 0.005). The rate of adverse events was lower in the Z-POEM group (6.25%) compared to the FES group (23.53%), however this difference did not reach statistical significance (p= 0.084).
Conclusion: This multicenter retrospective study suggests that Z-POEM is an effective and safe alternative to FES for the treatment of Zenker's diverticulum. The Z-POEM technique demonstrated a significantly higher clinical success rate and a trend towards fewer adverse events. Of particular interest, no differences in terms of efficacy and adverse events were observed between the first 10 patients (during the learning curve) treated with Z-POEM and those treated thereafter. It should be noted that although the follow-up period for Z-POEM was generally shorter, no recurrence of symptoms was observed even in patients with longer follow-up durations. These findings support the use of Z-POEM as a viable and effective approach, warranting further investigation in larger multicenter cohorts to confirm its advantages.
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Disclosure: Giuseppe Grande is consultant for Olympus, Boston Scientific, ERBE, Fujifilm.

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PP0196

REDUCING RELAPSE RATES IN C. DIFFICILE: IS FIDAXOMICIN THE BETTER OPTION?

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Introduction: *Clostridioides difficile* infection (CDI) is the leading cause of hospital-acquired infectious diarrhoea in adults and is associated with significant morbidity and mortality[1,2].

A review by Mitchell and Gardner[3] showed all-cause mortality of up to 30% in the first 30 days of contracting CDI. According to the European Society of Clinical Microbiology and Infectious Disease 'the most important problem in treating CDI is the high recurrence rate', with up to 25% having a recurrence of C. diff, and 40-65% of those patients going on to have a second recurrence[4].

As per NICE guidelines, vancomycin is currently the first line treatment for CDI with fidaxomicin reserved for relapses and recurrences[5].

Aims & Methods: Objective: To determine to what extent Fidaxomicin is superior to Vancomycin in reducing relapse of CDI.

Methods: A retrospective audit was conducted on 464 adult community and inpatients with CDI at the Norfolk and Norwich University Hospital from the 1/6/23 to 1/6/24. Data collection included patient demographics, episodes of CDI, treatment response and risk factors such as immunosuppression. A relapse is defined as a repeat episode of CDI occurring within 12 weeks of previous symptom resolution, while a recurrence is an episode occurring after 12 weeks.

Results: In total, 464 patients who had a positive C diff. toxin gene result were reviewed. Excluded were 103 patients who were not given antibiotics, 72 patients who no information was available and 14 who received an alternative antibiotic, e.g. metronidazole. 221 patients received vancomycin and 54 received fidaxomicin. 4/54 patients who received fidaxomicin went on to relapse (7.4%), one of which subsequently had extended pulsed fidaxomicin. 50/221 receiving vancomycin had a relapse (22.6%). The odds ratio for relapse rate of vancomycin versus fidaxomicin was 3.65, 95% CI 1.26 to 10.61. A student t-test result was 3.33 with p-value 0.0011.

Conclusion: The data shows a significantly lower relapse rate in high-risk patients treated with fidaxomicin compared to vancomycin for CDI. Reconsideration of first line therapy for high-risk patients with continuing care under the MDT team, for example at the Norfolk and Norwich University Hospital during the C. Diff ward round, is essential to ensure appropriate treatment is offered.

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PP0197

PROXIMAL OESOPHAGEAL STRICTURES OF UNKNOWN AETIOLOGY - IS CANDIDA ALBICANS THE CAUSE?

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Introduction: Proximal oesophageal strictures are less common than distal oesophageal strictures, but can share an aetiology such as eosinophilic or peptic. However, often proximal oesophageal strictures have an unclear aetiology. This results in symptomatic treatment with dilatation, without treatment of the underlying causative factor to prevent further stricture formation. We hypothesised that most proximal oesophageal strictures without a clear aetiology may be secondary to *Candida* oesophagitis.

Aims & Methods: The aim was to determine the cause of proximal oesophageal strictures of unclear aetiology, and examine the outcomes of medical and endoscopic treatment. Retrospective analysis was performed to identify all cases of proximal oesophageal stricture (≤30cm from incisors) without clear aetiology at Guy's and St. Thomas' Trust from 2007 to 2024. Procedures were extracted from the electronic endoscopy database using the keyword 'dilat'. Patients with Barrett's oesophagus, Schatzki ring, eosinophilic oesophagitis, anastomotic stricture, radiofrequency ablation, peptic stricture, oesophagectomy, chemo/radiotherapy, Crohn's disease, achalasia or caustic ingestion were excluded. Oesophageal histopathology taken at any point was assessed for candidiasis. Data collected included demographics, clinical presentation, endoscopic findings, histopathology results, medical and endoscopic treatment and outcomes.

Results: Between 2007 and 2024, 19 patients were identified with a proximal oesophageal stricture without an alternative aetiology. All patients had *Candida* oesophagitis at either at presentation with the stricture or predating it. The mean age at presentation was 57 years (range 40-80, 95% CI ±5.7) and 78.9% of patients were male. Eight patients (42.1%) were immunocompetent and not prescribed immunosuppressant drugs e.g. corticosteroids.

All strictures were between 15 and 30cm from the incisors (median 24cm). Only two strictures were passable with a standard endoscope, and thus were not dilated. Histopathology of the strictures confirmed no eosinophilia. The most common finding on histopathology was lymphocytic inflammation. On average, each patient had 2 balloon dilatations performed (range 0-12). Following treatment with antifungals and dilatation, 53.3% of patients had resolution of the stricture, 31.3% had ongoing *Candida* plus a stricture, 13.3% had a stricture without *Candida* present. Patients with failure of anti-fungal therapy to treat *Candida* had an increased number of dilatations required.

Age (average)	**58.7**
Male	78.9% (n=15)
Immunosuppressive Conditions	
Asthma/COPD	15.7% (n=3)
Diabetes	26.3% (n=5)
Chronic Mucocutaneous Candidiasis	10.5% (n=2)
Histopathology findings	
Lymphocytosis	63.2% (n=12)
Microabscesses	10.5% (n=2)
Received anti-fungal therapy	94.7% (n=18)
Dilatation performed	84.2% (n=16)
Median number of dilatations required	2

Table 1. Patient demographics and results.

Conclusion: In conclusion, this case series of 19 patients shows that proximal oesophageal strictures due to *Candida* are more common than previously thought. At our centre, all patients with a proximal stricture of

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unknown cause had oesophageal candidiasis. Where a clear alternative cause has not been identified, *Candida* should be considered, even in patients who are immunocompetent.

Anti-fungal therapy is essential to remove the underlying causative factor, and approximately half of patients will have resolution of their stricture following dilatation once *Candida* has been treated.

Patients who did not have resolution of *Candida* with antifungals were more likely to require a higher number of dilatations.

Disclosure: Nothing to disclose.

PP0198

CLINICAL UTILITY OF SCORING SYSTEMS IN UPPER GASTROINTESTINAL BLEEDING: A PRE- AND POST-TEST PROBABILITY ANALYSIS FOR MORTALITY AND NEED FOR INTERVENTION

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Introduction: Upper gastrointestinal bleeding(UGIB) is a common condition that entails morbidity and mortality. Therefore, an early prediction of negative outcomes would be clinically relevant selecting patients who might benefit from a closer follow up. In this way, several pre-endoscopic scoring systems have been developed for risk assessment in the emergency room(1).

One of the limitations of scoring systems is that sensitivity and specificity alone (usually combined in the receiver operator characteristic curve) may help comparing the accuracy of different systems but cannot be used to accurately estimate the probability for an individual patient.

A more appropriate approach is to establish the post-test probability considering the expected pre-test probability of poor outcomes in the population(2).

Aims & Methods: Given the absence of previous similar studies in UGIB, we aim to investigate the role of the most common pre-endoscopy scoring systems in modifying the pre-test probability of adverse outcomes.

Retrospective analysis of a prospective registry of UGIB patients admitted to 'Virgen de las Nieves University Hospital' (2013-2024). The outcomes included were intervention (composite endpoint including need of transfusion, endoscopy therapy, interventional radiology or surgery) and in-hospital mortality.

Pre-endoscopy scores analyzed were MAPASH,ABC and GBS as they have demonstrated their utility in predicting adverse events. The thresholds used for the definition of low-risk patients(test negative) were ABC≤3,MAPASH≤2 and GBS≤1(3).

First, the prevalence of the main outcomes was calculated to obtain the pre-test probability. Next, characteristics of different scoring systems were used to obtain the likelihood ratio (LR) and to estimate the post-test probability using an online calculator (http://araw.mede.uic.edu/cgi-bin/testcalc.pl).

Results: 1649 UGIB patients were included.

Prevalence of intervention(pre-test probability) was 75%.

The sensitivity and specificity of MAPASH were 71.4% and 78.4%,respectively. +LR and -LR 3.31 and 0.36,respectively. Post-test probability was 91% when positive and 52% when negative.

The sensitivity and specificity of ABC were 68.1% and 66.8%,respectively. +LR and -LR 2.05 and 0.48,respectively. Post-test probability was 86% when positive and 59% when negative.

The sensitivity and specificity of GBS were 98.8% and 8.7%,respectively. +LR and -LR 1.08 and 0.14,respectively. Post-test probability was 76% when positive and 29% when negative.

Prevalence of in-hospital mortality (pre-test probability) was 11%.

The sensitivity and specificity of MAPASH were 90.1% and 44.8%, respectively. +LR and -LR 1.63 and 0.22,respectively. Post-test probability was 17% when positive and 3% when negative.

The sensitivity and specificity of ABC were 95.9% and 44.8%,respectively. +LR and -LR 1.74 and 0.09,respectively. Post-test probability was 18% when positive and 1% when negative.

The sensitivity and specificity of GBS were 99.4% and 3.6%,respectively. +LR and -LR 1.03 and 0.17, respectively. Post-test probability was 11% when positive and 2% when negative.

Conclusion: While MAPASH and ABC effectively prioritize high-risk patients for intervention, their ability to rule out intervention in low-risk patients is limited. Additionally, they are more useful excluding mortality among low-risk patients than accurately predicting fatal outcomes. GBS,in contrast, are useful identifying low risk patients for intervention and mortality.

Given these findings, clinicians should interpret these scores with caution and incorporate additional clinical parameters into decision-making.

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PP0199

PROSPECTIVE COMPARISON OF ROCKALL, GLASGOW BLATCHFORD AND AIMS65 RISK SCORING SYSTEMS IN PATIENTS PRESENTING WITH UPPER GASTROINTESTINAL BLEEDING: PERFORMANCE IN PREDICTING CLINICAL OUTCOMES IN A LOW-RESOURCED SETTING

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Introduction: Upper gastrointestinal bleeding is one of the common medical emergencies which is associated with significant morbidity, mortality, and resource utilization.(1)

This burden is compounded in resource-poor settings, where patients often pay out-of-pocket for care, thereby hampering appropriate early diagnosis to determine the location and severity of bleeding.(2)

Effective management requires risk stratification of patients into low or high-risk groups to guide subsequent treatment and follow-up.(3)

The Rockall Score, Glasgow Blatchford Score and recently, the AIMS65 score are widely used and have been validated to predict outcomes in several studies. (4)

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Aims & Methods: We aim to compare the performance of the Rockall score, Glasgow Blatchford Score and AIMS65 score in predicting outcomes among patients with upper gastrointestinal bleeding. This was a single-centre prospective study among 195 patients who presented with symptoms of UGIB at the Emergency department of a tertiary hospital in Ghana from May 2022 to April 2023. The relevant baseline clinical characteristics, Glasgow Blatchford and AIMS65 scores were obtained. The Rockall score and the cause of UGIB was determined as per findings at upper gastrointestinal endoscopy. Patients were followed up for 6 weeks from admission looking out for primary outcomes of need for transfusion, rebleeding and mortality. Receiver operating characteristic curve analysis was performed to assess the ability of each risk score to predict primary outcomes.

Results: There were 195 patients in this study, 145 (74.4%) were males, the mean age (SD) was 51.4 (17.4) years and 68 (34.9%) had variceal bleeding. The 6-week mortality was 34 (17.4%), rebleeding occurred in 14 (7.2%) and 138 (70.8%) received blood transfusion.

Rockall score (area under the curve (AUC) 0.83, 95% confidence interval (CI), 0.76-0.88, p<0.001) was superior in predicting 6-week mortality compared to AIMS65 score (AUC 0.72, 95% CI, 0.63-0.81, p<0.001) and Glasgow Blatchford score (AUC 0.71, 95% CI, 0.61-0.82, p<0.001). The need for transfusion was best predicted by the Glasgow Blatchford score (AUC 0.77, 95% CI 0.70-0.85, p<0.001) compared to the AIMS65 score (AUC 0.66, 95% CI, 0.57-0.74, p<0.001) and Rockall score (AUC 0.77, 95% CI 0.57-0.75, p <0.001). Only Rockall score (AUC 0.71, 95% CI, 0.60-0.83, p<0.001) showed significance in predicting rebleeding.

Outcome	Risk score	AUC	p-value	95% CI
6-week Mortality	Rockall	0.83	<0.001	0.77-0.88
	Glasgow Blatchford	0.71	<0.001	0.61-0.82
	AIMS65	0.72	<0.001	0.63-0.81
Need for transfusion	Rockall	0.66	<0.001	0.57-0.75
	Glasgow Blatchford	0.77	<0.001	0.70-0.85
	AIMS65	0.66	<0.001	0.57-0.75
6- week Rebleeding	Rockall	0.71	<0.001	0.60-0.83
	Glasgow Blatchford	0.53	0.659	0.39-0.68
	AIMS65	0.56	0.455	0.40-0.72

Conclusion: Although the endoscopy dependent Rockall score has the highest accuracy in predicting mortality, non-endoscopy dependent Glasgow Blatchford and AIMS65 scores, which are derived from readily available clinical and laboratory parameters, were found to be significant in predicting mortality and will be useful in a low resource setting. Glasgow Blatchford score has the best accuracy for predicting the need for transfusion compared to Rockall and AIMS65 scores. Only Rockall score was useful in predicting 6-week rebleeding among all the three risk scoring systems.

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PP0200

A MULTICENTRE AUSTRALIAN CASE-CONTROL STUDY ON THE IMPACT OF A HAEMATEMESIS AND MELAENA UNIT ON NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING OUTCOMES

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Introduction: Non-variceal upper gastrointestinal bleeding (NVUGIB) is a frequent emergency, with a 30-day mortality rate nearing 10%.[1] Although several studies[2-4] have shown the mortality benefits of a specialized pre- and post-endoscopy care model within a Hematemesis and Melaena (H&M) unit, this data is outdated, and the model is resource-intensive. Many hospitals now use a ‚shared care' approach, involving both gastroenterology and general medicine for pre-endoscopy management and referral for specialized care post-procedure.

Our study aimed to evaluate the impact of a dedicated H&M Unit on the 30-day mortality rate and morbidity, in NVUGIB patients.

Aims & Methods: A retrospective case-control study was conducted with patients admitted for NVUGIB from April to October 2024 at two tertiary hospitals in Western Australia. One hospital, referred to as Hospital #1 (“H-1”), served as the control group, providing pre-endoscopy care through the general medicine unit with specialist gastroenterology input, while the gastroenterology team managed post-endoscopy care. The other hospital, referred to as Hospital #2 (“H-2”), served as the case group and managed both pre- and post-endoscopy care through a dedicated specialist H&M Unit. The study comprised consecutive adult patients aged 16 years and above, with endoscopically confirmed NVUGIB who presented with haematemesis and/or melaena. Patients were excluded if they experienced variceal bleeding or if their NVUGIB occurred during hospitalization for another medical condition. The primary outcome measured was 30-day mortality. Secondary outcomes included: a) time to endoscopy, b) total hospital length of stay (LOS) related to the index bleed and any rebleeds, c) the proportion of patients requiring laparotomy, and d) the total number of packed red blood cell (PRBC) units transfused for the index and any subsequent rebleeds.

Results: The preliminary analysis included 317 patients: 211 (66.6%) from H-1 and 106 (33.4%) from H-2. Patients from H-1 were significantly older [median age (IQR): 73 (62-79) vs 66 (50-76) yrs, p=0.002], had a higher median Glasgow-Blatchford Score (GBS) [9 (7-12) vs 8 (6-10), p <0.001] and a numerically higher median Charlson's Comorbidity Index (CCI) [(5 (3-7) vs 4 (1.8-6), p = 0.173]. Anticoagulation and NSAID use were similar between the two groups. The unadjusted 30-day mortality rate was higher in H-1 compared to H-2 (3.8% vs. 2.8%, p= 0.04); however, after adjusting for age, gender, CCI, GBS, and time to endoscopy by logistic regression analysis, the difference was no longer significant (p = 0.9). The median time to endoscopy was longer in H-1 compared to H-2 [25.4 hours (17.6-43.8) vs 15.3 hours (8.1-21.7), p < 0.001), with a lower percentage of patients undergoing the procedure within 24 hours (47.9%, vs 83% vs. p < 0.001). The total number of PRBC units transfused was numerically higher in H-1 compared to H-2 [2 (0-3) vs 1 (0-2) unit vs. p = 0.056]. There were numerically more emergency laparotomies in H-1 compared to H-2 [4 (1.9%) vs. 0 (0%), p = 0.15). The median LOS was not significantly different between the two groups [3.8 days (2.4-6.1) vs. 3.6 days (2.4-5.6), p = 0.42].

Conclusion: While the 30-day mortality rate was comparable between the two hospitals, patients from the H&M unit showed lower morbidity, including a reduced transfusion requirement and fewer emergency lapa-

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rotomies. Additionally, patients from the H&M unit were more frequently endoscoped within the internationally recognized 24-hour time frame, which is an important quality care indicator.

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Disclosure: Nothing to disclose.

PP0201

CLINICAL IMPACT OF BRIDGING WITH LOW-MOLECULAR-WEIGHT HEPARIN VS. DIRECT RESUMPTION OF ORAL ANTICOAGULANTS IN UPPER GASTROINTESTINAL BLEEDING

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Introduction: Upper gastrointestinal bleeding (UGIB) is one of the most common digestive disorders. The use of anticoagulants is a risk factor for UGIB, with a relative risk of 1.45. Withdrawal and timing of reintroduction remain controversial [1]. Early reintroduction of anticoagulation has been associated with a reduction in thromboembolic events, although it carries an increased risk of rebleeding within the first three months [2].

A common clinical practice is to restart anticoagulation with low-molecular-weight heparin (LMWH) as a bridge to the patient's regular anticoagulant therapy, especially in hospitalized patients, in order to reintroduce anticoagulation earlier and reduce thrombotic events. However, evidence for this approach is limited. One study, which included patients with both upper and lower gastrointestinal bleeding, suggested that LMWH use increased the risk of rebleeding without affecting the rate of thromboembolic events [3]. Therefore, uncertainties remain as to whether reintroduction of LMWH prevents thromboembolic events in this setting and its role in rebleeding rates.

The aim of this study is to compare the effect of bridging with LMWH before resuming the patient's usual anticoagulation versus its direct reintroduction in anticoagulated patients with UGIB.

Aims & Methods: This retrospective analysis of a prospective registry includes UGIB patients admitted to Hospital Universitario Virgen de las Nieves (2013–2023). Of 1517 patients, we selected those on anticoagulants whose treatment was suspended upon admission. Outcomes analyzed included in-hospital and 6-month delayed ones. Statistical analysis was performed using SPSS 26.

Results: A total of 244 patients were included. Group 1 (n=98) consisted of patients who resumed oral anticoagulation directly, while Group 2 (n=146) included those who initially received LMWH. The mean age was 74 years in both groups. Group 2 had a higher proportion of male patients (63.7% vs. 56.7%), although this difference was not statistically significant. Anticoagulation therapy was reintroduced earlier (3,9 vs 2,8-day, p=0,086) with a higher reintroduction rate in the first 24 hours in Group 2 (35,7% vs 56,2%, p=0,002). No differences were found in the other baseline characteristics.

Regarding outcomes, no significant differences were found in acute or delayed mortality, neither in delayed thromboembolic events. However, delayed bleeding (including gastrointestinal) was higher in Group 2 (31.7% vs. 13.7%, p=0.002). Other outcomes are in Table 1.

Multivariate analysis of hemorrhagic events within the first six months identified LMWH therapy, as the only independent risk factor (OR 2.18; p=0.013).

A subgroup analysis of high-thrombotic-risk patients (CHA_2DS_2-VASc ≥4) showed no benefit from LMWH as bridging therapy.

Variable	Group 1	Group 2	P
Rebleeding	10 (10,3%)	12 (8,2%)	0,578
In-Hospital Mortality	6 (6,4%)	5 (3,4%)	0,290
Not bleeding mortality	4 (66,7%)	3 (66 %)	0,299
Delayed Mortality at 6 months	11 (11,7%)	21 (14,6%)	0,524
Thromboembolic events 6 months	15 (15,8%)	26 (18,2%)	0,632
Haemorrhagic events 6 month	13 (13,7%)	45 (31,7%)	0,002

Table 1: In.Hospital and delayed outcomes.

Conclusion: Although LMWH bridging therapy allows for earlier anticoagulation resumption, it did not reduce mortality or thromboembolic event rates in anticoagulated patients with UGIB. However, it may increase the mid-term bleeding rate, suggesting that routine use of LMWH after suspending baseline anticoagulation in UGIB should be reconsidered.

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PP0202

COMPARING THE EFFECT OF ESOMEPRAZOLE OR FAMOTIDINE ON PREVENTION OF BLEEDING AFTER SUBMUCOSAL ENDOSCOPIC DISSECTION OF EARLY GASTRIC CANCER IN A REFERRED GASTROINTESTINAL CENTER IN BRAZIL

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Introduction: With conventional methods of endoscopic mucosal resection for early gastric cancer (EGC), proton pump inhibitors (PPI) and H2 receptor antagonists (H2RA) have a similar effect on preventing bleeding from artificial ulcers

Aims & Methods: To investigate whether PPI more effectively prevent bleeding after recent advanced technique of endoscopic submucosal dissection (ESD) for EGC.

A prospective randomized controlled trial study, from january 2016 to february 2025 was performed in referred Gastrointestinal Diseases Hospital in Brazil . A total of 286 patients with EGC who underwent ESD. Intervensions: After stratification by tumor size (i.e. ≤2 cm or >2 cm) and H. pylori status, they were randomly assigned to treatment groups. They received either esomeprazole 40 mg (PPI group) or famotidine 40 mg (H2RA group) on the day before ESD and continued for 6 weeks.

Main outcome measurements: Primary endpoint was incidence of bleeding that was defined as haematemesis or melaena that required endoscopic haemostasis and decreased the haemoglobin count by more than 2g dL. To elucidate independent treatment effects and to identify prognostic factors for bleeding after ESD multivariate analysis was performed for candidate covariates including the stratification factors, age, sex, smoking, NSAIDs use in the past 5 days history of peptic ulcer, endoscopist, number of tumors, location of main tumor, and scar in main tumor.

Results: The history of peptic ulcer and the endoscopists who performed the ESD were significantly different between the groups. Finally, 132 of 146 patients in the PPI group and 128 of 140 in the H2RA groupwere analysed. Bleeding occurred in 8 patients in the PPI group and 22 in the H2RA group (p = 0.057).

Multivariate analysis revealed that the treatment with PPI significantly reduced the risk of bleeding adjusted hazard ratio 0.51, 95% C.I. 0,25 to 0.96, p = 0.037. Tumor size exceeded 3 cm and scar in the main tumor were the independent predictor for the bleeding. Three delayed perforation was experienced in the H2RA group

Conclusion: PPI therapy more effectively prevented delayed bleeding from the ulcer created after ESD than did H2RA treatment

Disclosure: Nothing to disclose.

PP0203

ARGON PLASMA COAGULATION FOR TREATING HIGH-RISK ENDOSCOPIC STIGMATA OF PEPTIC ULCER BLEEDING: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Endoscopic hemostasis is first-line therapy in managing peptic ulcer bleeding (PUB) with high-risk stigmata (Forrest classification Ia, Ib, IIa). While argon plasma coagulation (APC) is widely available and easy to use, there are limited studies with a limited number of patients evaluating its efficacy in PUB. This meta-analysis evaluates the effectiveness of APC in high risk PUB in achieving initial hemostasis and preventing rebleeding.

Aims & Methods: We performed a systematic literature search of PubMed, Embase, Web of Science, and Scopus to identify studies up to October 2024, evaluating APC versus alternative endoscopic hemostasis modalities for high-risk PUB. Eligible studies included those reporting on initial hemostasis and rebleeding outcomes. Two reviewers (KS, MO) independently screened studies and extracted data, resolving discrepancies by consensus. Quality assessments were conducted using the Cochrane Risk of Bias Tool for RCTs and the Newcastle-Ottawa Scale for observational studies. Odds ratios (ORs) and 95% confidence intervals (CIs) were pooled using a random-effects model to account for study heterogeneity, quantified by I^2 statistics.

Results: Fourteen studies met inclusion criteria (7 observational and 7 RCTs), with pooled sample sizes of n=816 and n=975 respectively. Meta-analysis of all studies showed that for the outcome of initial hemostasis, the pooled OR was 1.01 (95% CI: 0.39–2.62, p = 0.976), demonstrating no significant difference between APC and other endoscopic treatments. Similarly, rebleeding rates showed no significant difference (pooled OR: 1.12; 95% CI: 0.69–1.83, p = 0.636). When we limited our meta-analysis to only the 6 RCTs with an active hemostasis treatment control group, we again found similar results for both outcomes of initial hemostasis and rebleeding.

Conclusion: APC appears comparable and non-inferior to other standard of care endoscopic hemostasis modalities in achieving initial hemostasis and preventing rebleeding in high-risk PUB. Updated guidelines will need to re-evaluate and decide where APC fits in with endoscopic hemostasis of high-risk PUB.

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Disclosure: Ian Gralnek. has received lecture fees from Astra-Zeneca, Taro Pharma, Vifor Pharma and 3D Matrix. He is also a paid consultant for Boston Scientific, GI View, Motus GI and Symbionix and is a member of the medical advisory board of Motus GI

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PP0204

BOGOTÁ SCORE: A NOVEL TOOL FOR EARLY RISK STRATIFICATION IN UPPER GASTROINTESTINAL BLEEDING IN COLOMBIA

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Introduction: Upper gastrointestinal bleeding (UGIB) remains a major clinical emergency with significant morbidity and mortality despite advances in endoscopic and medical management. While existing risk scores have been widely used, their applicability to specific populations is limited.

We aimed to identify mortality-associated risk factors in Colombian patients with UGIB and propose the "Bogotá Score" as a novel stratification tool tailored to our population.

Aims & Methods: We conducted a retrospective case-control study involving 472 patients with UGIB across three Colombian centres during 2023. Twenty-five (5.3%) patients died due to UGIB-related complications. Demographic, clinical, and endoscopic variables were compared using bivariate analysis. Variables significantly associated with mortality were included in a multivariate logistic regression model. A point-based risk score—the Bogotá Score—was derived from independent predictors.

Results: Age ≥70 years, anaemia (OR 7.15; p=0.022), haemoglobin <10 g/dL, Blatchford score >8 (OR 1.18; p=0.003), hospitalisation delay >24h (OR 1.01; p=0.008), and rectorrhagia were associated with increased mortality. Conversely, melena was linked to reduced risk (OR 0.20; p=0.001). These variables were integrated into the Bogotá Score. Patients were stratified into four risk categories: low (0–3), moderate (4–6), high (7–9), and very high (≥10).

Risk Factor	Points
Age ≥70 years	2
Anaemia	3
Haemoglobin <10 g/dL	2
Blatchford score >8	2
Hospitalisation delay >24h	1
Rectorrhagia	1

Table: Bogotá Score – Risk Stratification Tool

Interpretation:

- 0–3: Low risk
- 4–6: Moderate risk
- 7–9: High risk
- ≥10: Very high risk

Conclusion: The Bogotá Score represents a simple, evidence-based tool for early risk stratification in UGIB. By incorporating context-specific variables, it enhances clinical decision-making in Latin American settings. Early ICU transfer and intensive monitoring for high-risk groups may improve outcomes. Future multicentre validation is warranted.

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Disclosure: Nothing to disclose.

PP0205

IMPACT OF PRE-ENDOSCOPIC FRESH FROZEN PLASMA TRANSFUSION IN ACUTE UPPER GASTROINTESTINAL BLEEDING WITH MILD TO MODERATE COAGULOPATHY: A MULTICENTER RETROSPECTIVE COHORT STUDY

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Introduction: Despite its widespread use, the benefit of pre-endoscopic fresh frozen plasma (FFP) transfusion in acute upper gastrointestinal bleeding (AUGIB) with mild to moderate coagulopathy remains unproven. Current guidelines are inconsistent, and indiscriminate use of FFP—particularly in patients with variceal bleeding—may expose patients to harm without improving outcomes.

Aims & Methods: To assess the impact of pre-endoscopic FFP transfusion on clinical outcomes, including 30-day mortality, in patients with AUGIB and an INR between 1.5 and 2.5. We conducted a multicenter retrospective cohort study at one university-affiliated and one tertiary care hospital in Thailand between January 2016 and December 2020. Adult patients with AUGIB and INR 1.5–2.5 were included. Patients were stratified into FFP and non-FFP groups based on transfusion prior to endoscopy.

Multivariable logistic regression was performed to adjust for age, comorbidities, bleeding etiology, and hemodynamic parameters. Subgroup analysis was conducted in patients with variceal bleeding.

Results: Among 244 patients (158 FFP; 86 non-FFP), those receiving FFP had significantly higher in-hospital mortality (24.1% vs. 7.0%; $p = 0.001$), pulmonary edema (23.4% vs. 4.7%; $p < 0.001$), and longer hospital stay (mean 8.1 vs. 5.8 days; $p < 0.001$). After adjustment, FFP transfusion remained independently associated with in-hospital mortality (adjusted OR 5.65; 95% CI 2.13–14.99; $p < 0.001$) and pulmonary edema (adjusted OR 5.56; 95% CI 1.59–19.44; $p = 0.007$). In patients with variceal bleeding, the adverse effect of FFP was more pronounced. The adjusted 30-day mortality was higher in the FFP group (adjusted OR 2.38; $p = 0.075$), though this did not reach statistical significance.

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Conclusion: Pre-endoscopic FFP transfusion in AUGIB patients with INR 1.5–2.5 is associated with increased risk of in-hospital mortality and volume-related complications, particularly in variceal bleeding.

These findings support a more selective and individualized transfusion approach and challenge the routine use of FFP in this population. Prospective studies are warranted to validate these observations and guide evidence-based transfusion thresholds.

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Disclosure: Nothing to disclose.

PP0206

COMPARATIVE ANALYSIS OF GASTROINTESTINAL BLEEDING RISKS IN PATIENTS TREATED WITH RIVAROXABAN VERSUS APIXABAN: A RETROSPECTIVE STUDY IN A MEXICAN HOSPITAL SETTING

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Introduction: Direct oral anticoagulants (DOACs), particularly rivaroxaban and apixaban, are increasingly utilized to manage thromboembolic conditions due to their favorable efficacy and safety profiles. However, gastrointestinal bleeding (GIB) remains a significant concern associated with these medications, particularly considering pharmacogenetic variations among populations.

This study aims to evaluate the frequency and identify risk factors associated with GIB in patients using rivaroxaban and apixaban in a Mexican healthcare setting.

Aims & Methods: This retrospective observational study analyzed data from January 2018 to December 2023 at the Centro Médico Nacional 20 de Noviembre, Mexico City.

The cohort consisted of 278 patients: 136 on DOAC therapy (rivaroxaban or apixaban) and 142 controls without anticoagulation. Clinical data, demographic characteristics, comorbidities, and concomitant medications were systematically reviewed.

Statistical methods employed included chi-square tests, Student's t-tests, and logistic regression analysis to assess the risk factors and association between anticoagulant type and incidence of GIB.

Results: The study population had a mean age of 67.36 ± 15.05 years, with an equal gender distribution (52.9% male). Rivaroxaban was significantly linked to increased GIB-related hospitalizations compared to controls (OR 1.29, 95% CI: 1.10–1.52, $p < 0.001$), while apixaban showed no statistically significant increase. NSAID co-administration dramatically raised the risk of GIB in rivaroxaban users (OR 16.1, 95% CI: 1.22–212.6, $p = 0.03$).

Patients treated with DOACs experienced more hospitalizations (average 0.49 ± 0.80) and longer hospital stays (2.19 ± 3.54 days) compared to controls (0.30 ± 0.62 hospitalizations, 0.99 ± 2.68 days, respectively). Mortality rates were not significantly different between groups.

Conclusion: Rivaroxaban and apixaban present similar overall risks of gastrointestinal bleeding, although rivaroxaban is associated with higher hospitalization rates. Concomitant NSAID use substantially elevates the risk in patients on rivaroxaban.

This underscores the necessity of individualized anticoagulant management strategies, especially in populations with distinct genetic and clinical characteristics, such as the Mexican demographic.

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Disclosure: Nothing to disclose.

PP0207

ARTIFICIAL INTELLIGENCE IN THE MANAGEMENT OF UPPER GASTROINTESTINAL BLEEDING: A NEW DECISION-MAKING TOOL

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Introduction: Upper gastrointestinal bleeding (UGIB) remains a common and potentially life-threatening condition in emergency medicine. Early identification of patients requiring urgent esophagogastroduodenoscopy (EGD) is crucial to improve outcomes and optimize healthcare resources management. Traditional risk stratification tools, such as the Glasgow-Blatchford Score (GBS), have been widely used to guide clinical decision-making.

However, the growing availability of artificial intelligence (AI) offers new perspectives in diagnostic and therapeutic strategies. Recent advancements in large language models (LLMs), such as ChatGPT® and Google GEMINI®, suggest potential applications in clinical triage.

Aims & Methods: This study aims to evaluate the performance of these AI tools in determining the appropriate timing for endoscopy in UGIB patients, compared to standard clinical scoring systems.

The medical records of 100 patients who presented to the emergency department with upper gastrointestinal bleeding (UGIB) between April and September 2024 were reviewed retrospectively. Each case was evaluated by two AI models: Google GEMINI® (September 2024 version) and ChatGPT® (version 3.5).

These models, acting as virtual gastroenterology experts, were asked to recommend whether urgent or delayed esophagogastroduodenoscopy (EGD) was necessary. For each patient, clinical data such as age, medical history, clinical examination findings, and laboratory results at admission were provided.

The AI recommendations were recorded in an Excel file and compared with the Glasgow-Blatchford Score (GBS), which was calculated by a gastroenterologist at the time of admission.

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An urgent endoscopy was deemed necessary for a GBS ≥ 3. Statistical analysis was performed using Jamovi® software (version 2.3), and the agreement between the AI models and human evaluations was assessed using Cohen's kappa coefficient (κ).

Results: The epidemiological characteristics of the patient cohort revealed a median age of 63 years (range: 37.5 to 78), with a male-to-female ratio of 1.7. Among the patients, 73% had at least one comorbidity, including hypertension (33%), diabetes (22%), ischemic heart disease (17%), and chronic liver disease (12%).

At the time of admission, melena was the predominant symptom, reported in 78% of patients. The average Glasgow-Blatchford Score (GBS) was 7.07 ± 3.47.

In terms of AI recommendations, ChatGPT suggested urgent endoscopy in 48% of cases, while Google GEMINI recommended it in 49%. A moderate level of agreement was observed between the two AI models for urgent endoscopy recommendations (kappa = 0.540, $p < 0.001$). However, the agreement between the AI models and the human evaluator was weak. The level of agreement between the gastroenterologist's assessment and ChatGPT's recommendation was weak (kappa = 0.222, $p < 0.001$), with similar results for Google GEMINI (kappa = 0.231, $p < 0.001$).

Conclusion: Artificial intelligence is a promising tool that is already beginning to transform various fields of medicine. However, our study demonstrates that the publicly available AI models currently in use are not yet capable of effectively triaging patients in emergency settings, particularly in cases of upper gastrointestinal bleeding.

A model specifically trained on clinical gastroenterology cases, using validated scientific guidelines, could produce more accurate results and potentially revolutionize patient triage in emergency departments.

Disclosure: Nothing to disclose.

PP0208

ESTABLISHMENT OF A DELAYED BLEEDING ANIMAL MODEL AFTER GASTRIC MUCOSAL INJURY AND EVALUATION OF THE PREVENTIVE EFFECT OF HIGHLY ADHESIVE HYDROPHOBIZED GELATIN MICROPARTICLES

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Introduction: Endoscopic submucosal dissection (ESD) is a widely used treatment for early gastric tumors, but delayed bleeding remains a significant complication, especially in patients receiving antithrombotic therapy. As the population ages, the need for effective preventive strategies is growing.

Although various methods such as PGA sheets and fibrin glue have been explored, no definitive approach has been established. Moreover, a reliable animal model to evaluate strategies for preventing delayed bleeding does not currently exist.

To address this issue, Taguchi et al. developed a novel endoscopically sprayable wound dressing composed of hydrophobized gelatin microparticles (hMPs), which exhibit strong tissue adhesion. In preclinical studies using miniature swine, hMPs have shown various therapeutic effects on post-ESD ulcers, including anti-inflammatory and anti-fibrotic effects, as well as promotion of perforation closure.

Aims & Methods: This study aimed (1) to establish an animal model of delayed bleeding after gastric mucosal injury, and (2) to evaluate the preventive effect of hMPs using this model.

1. A delayed bleeding model was created by producing mucosal defects in the stomach using endoscopic mucosal resection with ligation (EMR-L), followed by continuous heparin infusion to maintain an activated clotting time (ACT) ≥220 seconds, which consistently induced delayed bleeding.
2. Using this model in three miniature swine, we evaluated the preventive effect of hMPs by endoscopically spraying hMPs onto a total of 36 ulcers (12 per animal). We assessed delayed bleeding, as well as the retention and coverage of hMPs at the ulcer base.

Results:

1. In the heparin-treated group, delayed bleeding was observed in 3 of 4 animals (75%). All three animals whose maximum ACT before the start of continuous heparin infusion exceeded 220 seconds developed delayed bleeding. In contrast, no ACT prolongation or delayed bleeding was observed in the non-heparin-treated group (n = 3) at 24 hours.
2. In the hMPs spray group (n = 3), all animals had ACT ≥220 seconds, and hMPs were sprayed onto 12 mucosal defects per animal (total 36 lesions). No delayed bleeding occurred in any animal. hMPs were retained at all ulcer sites, yielding a 100% retention rate. The ulcer base coverage by hMPs was as follows: 100% in 55.6% of sites (20/36), 75–99% in 19.4% (7/36), and 50–74% in 25.0% (9/36). No sites had coverage below 50%.

Exposed vessel-like structures were identified in 36.1% of the ulcers (13/36), and hMPs were found to adhere to all of these sites.

Conclusion: An animal model of delayed bleeding after endoscopic resection of the gastric mucosa was successfully established by maintaining an ACT ≥220 seconds through continuous heparin infusion. In this model, hMPs remained well on the ulcer base and provided broad coverage of mucosal defects.

These findings suggest that hMPs may be effective in preventing post-endoscopic bleeding, particularly in patients receiving antithrombotic therapy.

Disclosure: Nothing to disclose.

PP0209

COMBINING LACTATE LEVEL WITH CHARLSON COMORBIDITY SCORE FOR PREDICTION OF MORTALITY IN PATIENTS WITH ACUTE NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING

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Introduction: Acute non-variceal upper gastrointestinal bleeding (NVUGIB) is a common emergency with high morbidity and mortality. Lactate levels and comorbidity indices, such as the Charlson Comorbidity Index (CCI), have shown promise in predicting patient outcomes. While these parameters individually provide valuable prognostic insights, their collective utility in predicting mortality among NVUGIB patients has not been explored.

Aims & Methods: The data for this study were collected from hospital records of patients admitted to Hitit University Hospital with NVUGIB between June 2021 and June 2024.

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Demographic information, including age, gender, chronic illnesses, current medications, and history of previous gastrointestinal bleeding, was retrieved from computer database. Clinical data, such as outcomes (discharge or death), reasons for mortality, length of hospital stay were documented. Laboratory parameters, including lactate levels were recorded. Additionally, CCI and Glasgow Blatchford Score (GBS) were calculated for each patient.

Patients were stratified based on mortality status, which was considered the primary outcome. The associations between mortality and potential predictive factors were statistically analyzed. To improve mortality prediction, parameters that were significant in the logistic regression analysis were combined and assessed independently.

Results: A total of 390 patients with acute upper gastrointestinal bleeding were initially assessed. Of these, 180 patients were excluded due to missing data, leaving 210 patients included in the final study cohort. Key variables such as age, CCI, GBS, length of hospital stay, and mortality status were presented. Laboratory findings upon admission and the etiological distribution of gastrointestinal bleeding sources were also detailed.

Variables significantly associated with mortality in the univariate analysis include age, CCI, GBS, urea, aspartate aminotransferase (AST), alanine aminotransferase (ALT) and lactate levels. In the logistic regression analysis, significant predictors of mortality were CCI (OR: 1.318, p=0.001) and lactate levels (OR: 1.575, p=0.003). GBS was not a predictor for mortality (p=0.468). CCI plus lactate levels were significantly higher in deceased patients compared to survivors (12.96 vs.7.31, p<0.001).

The correlation coefficients (r) showed significant positive correlations between the combination of lactate levels and CCI with GBS (r=0.498, p<0.001), length of stay (r=0.213, p=0.014), age (r=0.634, p<0.001) and urea levels (r=0.417, p<0.001). No significant correlation was found with AST or ALT.

ROC curves for lactate levels, CCI and CCI plus lactate levels predicting mortality showed that the area under the curve (AUC) for lactate was 0.704 (95% CI: 0.577–0.830, p=0.002) with a cut-off value of 1.83, yielding a sensitivity of 76% and a specificity of 52%. For CCI, the AUC was 0.769 (95% CI: 0.670–0.869, p<0.001) with a cut-off value of 6.5, showing a sensitivity of 76% and a specificity of 70%. And for the 'CCI plus lactate' level, the AUC was higher as 0.799 (95% CI: 0.703–0.895, p<0.001) with a cut-off value of 8.3, showing a sensitivity of 88% and a specificity of 62%.

Conclusion: Both lactate levels and the CCI are valuable tools for predicting mortality in NVUGIB patients. Our findings support the use of these parameters collectively to improve risk stratification. Combining lactate levels and CCI offers the advantage of better predicting mortality and improving sensitivity compared to using lactate levels or CCI alone.

Disclosure: Nothing to disclose.

PP0210

RED PACK CELL TRANSFUSIONS BEFORE OR AFTER ENDOSCOPY IN PATIENTS WITH ACUTE UPPER GASTROINTESTINAL BLEEDING. A MULTICENTRE PROSPECTIVE COHORT STUDY

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Introduction: Acute upper gastrointestinal bleeding and timing to endoscopy are a recognised as risk factors for mortality. European Guidelines* suggest that the hemodynamical resuscitation could be performed before the endoscopy, however, indications are lacking about the timing of blood transfusions.

Aims & Methods: The aim of the study is to evaluate the transfusion timing to reduce death risk in patients with upper gastrointestinal bleeding. This is a prospective multicentre cohort study, including all consecutive hospitalized patients with upper gastrointestinal bleeding (UGIB). Clinical, endoscopic source of bleeding, medical and endoscopic treatment, time to endoscopy, and timing to transfusion were collected. We adopted the ESGE* guidelines to define the need for transfusion, timing to endoscopy and mortality. We considered timing to transfusion before or after the endoscopic procedure.

Statistics: multivariate backward logistic regression was used to estimate the adjusted odds ratio death risk for variables with main clinical relevance or highly statistical significance at univariate analyses. Variables identified by the logistic regression were used into the propensity-score matching to compare the death risk between the two groups.

Results: 2,017 out of 3324 patients received a transfusion and were included in the analysis; the mean age was 69.7 [±15.1], male was 1,365 (67.7%); the source of bleeding was non-variceal in 1634 patients (81 %), 178 patients died (8.8%); 69 out of 700 (9.9 %) died in the group transfusion before endoscopy; 109 out 1317 (8.3 %) in the group transfusion after endoscopy with an OR 1.2 (C.I. 95% .86 to 1.68) p<= 0.23.

Mortality between the two groups did not significantly differ when haemoglobin levels at admission were considered, nor after adjustment for potential confounding variables, including ASA score, in-hospital bleeding, haematemesis at presentation, shock at admission, altered mental status and timing to endoscopy.

Propensity score matching analysis indicated that among patients who received a transfusion, the risk of death decreased when the transfusion was administered before the endoscopy and the haemoglobin level was below 7 g/dL (coefficient = -0.06, p = 0.06; see Table).

Admission Heamoglobin value gr/L	Coefficient	SE.	z	P>|z|	[95%conf. nterval]
< 7 gr/l	-0.06	0.03	-1.86	0.063	-0.11 0.00
7 - 8 gr/l	-0.03	0.04	-0.95	0.342	-0.10 0.04
8 - 10 gr/l	0.02	0.03	0.52	0.605	-0.05 0.08

Table. Propensity-score matching Haemoglobin level before VS after endoscopy.

Conclusion: Red pack cell transfusions before or after endoscopy did not modify the death risk in patients with acute upper gastrointestinal bleeding; the death risk was less frequent in patients transfused before endoscopy and with a haemoglobin value lower than 7 gr/l. However, this risk increased in those patients transfused before endoscopy and with a haemoglobin value > 8 gr/l.

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*Ian M. Gralnek et al. Endoscopy 2021; 53: 300–332
Disclosure: Nothing to disclose.

PP0211

DEDICATED PRACTICE FOR PATIENTS DISCHARGED AFTER GASTROINTESTINAL BLEEDING. A SINGLE CENTER COHORT STUDY AND COST-EFFECTIVENESS ANALYSIS

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Introduction: Current guidelines provided information on the management of patients with gastrointestinal bleeding, in particular focusing on admission, endoscopic management and medical treatment during the hospital stay. So far, 11-15% of patients treated for GIB required a new hospitalization for a new bleeding episode.
Aims & Methods: The aim of the study is to verify if the management in a dedicated practice after hospital discharge for gastrointestinal bleeding reduces further bleeding and costs. This is an ambispective cohort single center study, including patients observed between May 2022 and September 2023 in our clinical practice dedicated to gastrointestinal bleeding. We enrolled consecutive patients with overt gastrointestinal bleeding or iron deficiency anemia affected by vascular lesions.
The effectiveness was defined as a composite endpoint including iron supplementation, blood transfusion, medical treatment and number of hospital stay. Costs were evaluated starting from Diagnosis Related Groups (DRG) codes. The cost effectiveness analysis and ICER has been calculated using the ICERconf.EXE software (Software Prescriptions, c/o Bob Obenchain, California, USA).
Results: In our institute 602 patients with gastrointestinal bleeding were evaluated between May 2022 and September 2023, of these 79 were included in the study. The mean age was 73.8 years (SD±10,8), 46% were female. The source of bleeding was angiodysplasia in 63% of patients, GAVE in 9% (Gastric Antral Vascular Ectasia), erosions in 9%.
After the introduction in the dedicated practice, the re-bleeding decrease from 1.3 episodes to 0.2 episodes, p<0.001. The cost average after the practice was 4074 euro (SD ±5485), the average cost before the practice was 4885 euro (SD ±4567), it has been observed a cost saving after the management in the dedicated practice, C diff= -751 euro. The average effectiveness after practice was 8.7 (SD±1.06) and the average effectiveness before the practice was 7.3 (SD ±1.3) with the E diff= 1.47, p<0.001.
Conclusion: The dedicated practice for patients with gastrointestinal bleeding provides effectiveness in reducing transfusion, iron supplementation and hospitalization with a consistent cost reduction.
Disclosure: Nothing to disclose.

PP0212

RACIAL DISPARITIES IN UPPER GASTROINTESTINAL BLEEDING MANAGEMENT AND OUTCOMES: A NATIONAL ANALYSIS

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Introduction: Contemporary studies suggest there are racial disparities in the outcomes of acute presentations. This study aimed to investigate the impact of sex and race on the management and outcomes of patients presenting with upper gastrointestinal bleeding (UGIB).
Aims & Methods: Using International Classification of Diseases 10th Revision Codes, all adults >18 admitted from 2016–2020 with a discharge diagnosis of UGIB were extracted from the National Inpatient Sample.
Results: In over 900,000 hospitalisations for UGIB, 69.4%, 14.1%, 9.6%, 3.6%, 0.9% and 2.4% were for admissions in white, black, Hispanic, Asian/pacific islander, native American and 'other' patients respectively. Comorbidity burden varied between racial groups (p<0.001). White patients had higher crude rates of endoscopy, and comparable rates of surgery or interventional radiology to control blood loss, in comparison to other racial groups.
Adjusted analysis revealed that black and Asian/pacific islander racial group patients had lower odds of in-hospital all cause mortality compared to white patients, but 'other' racial group patients had higher odds of mortality (p<0.001).
There was no difference for Hispanic or native American patients. For non-variceal bleeding, these outcomes persisted. However, for variceal bleeds, there were no significant differences in outcomes between each racial group.
Conclusion: There are racial disparities in the management and outcomes of admissions for UGIB. These difference persist despite the aetiology of bleeding. Underlying reasons for differences in outcomes must be investigated to ensure improved outcomes across high-risk patient groups.
Disclosure: Nothing to disclose.

PP0213

THE RELATIONSHIP BETWEEN TYPE 2 DIABETES AND OUTCOMES OF UPPER GASTROINTESTINAL BLEEDING IN THE UNITED STATES FROM 2016-2020

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Introduction: There are limited data on the contemporary outcomes of upper gastrointestinal bleeding (UGIB) in patients with type 2 diabetes mellitus (T2DM).
Aims & Methods: Admissions for UGIB were extracted from the National Inpatient Sample of the United States from 2016-2020. Patients were stratified by the presence of T2DM.

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Results: In 918,925 patients, 287,185 (31.3%) had T2DM. Patients with T2DM were more likely suffer from chronic pulmonary disease, chronic renal failure, heart failure, liver disease and hypertension (p<0.001). These patients had higher crude odds of endoscopy but lower odds of surgery and interventional radiology to control blood loss (p<0.001).

There was no difference in crude rates of in-hospital mortality. On adjusted analysis, T2DM was associated with lower odds of in-hospital mortality (adjusted odds ratio (aOR) 0.42 95% confidence interval (CI) 0.81-0.87).

Adjusted odds of mortality remained lower when analysis was stratified by bleed type (aOR 0.88 CI 0.85-0.92 for non-variceal bleeding and aOR 0.60 CI 0.54-0.66 for variceal bleeding).

Conclusion: Patients with T2DM admitted with UGIB are more comorbid, but are less likely to suffer in-hospital mortality compared to patients without T2DM.These differences persisted on-subanalysis for type of bleed.Knowledge of these trends is important for managing this prevalent patient group.

Disclosure: Nothing to disclose.

PP0214

ASSOCIATION BETWEEN CHRONIC KIDNEY DISEASE AND OUTCOMES OF PATIENTS WITH UPPER GASTROINTESTINAL BLEEDING (UGIB)

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Introduction: Contemporary data on the outcomes of upper gastrointestinal bleeding (UGIB) in patients with chronic kidney disease (CKD) is lacking.

Aims & Methods: Using international classification of diseases, 10th revision codes, all hospitalisations for UGIB were extracted from the National Inpatient Sample. Admissions were stratified by the presence of CKD. Subanalysis was performed by type of bleeding.

Results: Of the 918,925 UGIB admissions analysed, 239,985 (26.1%) had a medical history of CKD. CKD patients were more likely to be male and less likely to be white compared to patients without CKD. CKD patients had higher crude rates of endoscopy but lower likelihood of undergoing surgery and interventional radiology to control blood loss.

On adjusted analysis compared to patients with no CKD, CKD patients had higher odds of all-cause in-hospital mortality (adjusted odds ratio (aOR) 1.34 95% confidence interval (CI) 1.29-1.39). When stratified by bleed type, CKD patients had higher adjusted odds of mortality for both variceal and non-variceal bleeds, compared to patients without CKD.

Conclusion: Patients with CKD admitted with UGIB are more comorbid and are more likely to suffer in-hospital mortality compared to patients without CKD. These differences persisted on-sub-analysis for type of bleed. Knowledge of these trends is important for managing this prevalent patient group.

Disclosure: Nothing to disclose.

PP0215

HEMODYNAMIC STATUS AS A DETERMINANT FACTOR OF OPTIMAL ENDOSCOPY TIMING IN UPPER GASTROINTESTINAL BLEEDING: RESULTS FROM AN INTERNATIONAL SURVEY OF 533 CLINICIANS

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Introduction: Timely endoscopy in upper gastrointestinal bleeding (UGIB) is crucial. However, the ideal timing, particularly in relation to hemodynamic status, remains a matter of debate. Despite advances in diagnostic techniques and endoscopic treatments, its mortality rate has not declined significantly over the years.

Aims & Methods: We aimed to assess clinical practice patterns and determine how the timing of endoscopy is influenced by the patient's hemodynamic status. We conducted an international online survey between April 2023 and November 2023 among physicians managing patients with UGIB. REDCap was used for data collection. We assessed clinicians' opinions on the optimal timing of endoscopy (within 2, 6, 12, 24, or >24 hours) for variceal (VUGIB) and non-variceal (NVUGIB) bleeding under three hemodynamic conditions: stable, unstable responding and not responding to hemodynamic resuscitation.

Additionally, we asked how they defined hemodynamic instability. The primary analysis summarized responses as proportions. Secondary analyses explored variations in responses by clinical experience, practice setting, and annual UGIB case volume.

Results: A total of 533 clinicians from 50 countries completed the survey. Of the respondents, 291 (54.6%) worked in university-based hospitals, 361 (67.7%) managed over 100 UGIB cases annually, and 236 (44.3%) had more than 10 years of experience. For NVUGIB, the majority recommended endoscopy within 24 hours for stable patients (229, 43.1%), within 6 hours for unstable responders (161, 30.2%), and within 2 hours for unstable non-responders (254, 47.8%).

For VUGIB, the most common recommendations were endoscopy within 12 hours for stable patients (155, 29.1%), within 6 hours for unstable responders (201, 37.7%), and within 2 hours for unstable non-responders (319, 60%). More experienced physicians, those working in university-based hospitals, and those managing high UGIB patient volumes tend to favor earlier endoscopy.

The most commonly chosen definition for hemodynamic instability, selected by 345 (64.7%) was systolic blood pressure <100 mmHg and heart rate >100 bpm or syncope or orthostatic hypotension or signs of organ hypoperfusion, selected by 345 (64.7%) of the respondents.

Statistical variability was significant across all analyses ($p<0.0001$). Younger clinicians were more likely to adhere to guideline recommendations.

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Conclusion: We found that the more hemodynamically unstable the patients are, the earlier gastroenterologists tend to recommend performing endoscopy for UGIB. Clinical experience, hospital type, and annual patient number significantly influenced the preferred timing of endoscopy in UGIB. There was considerable variability among respondents' answers. This study highlights opportunities to improve consistency in clinical practice and identifies potential areas for further research.
Disclosure: Nothing to disclose.

PP0216

EXPLORATORY VALIDATION OF UPPER GASTROINTESTINAL BLEEDING RISK SCORES FOR COMPLICATIONS

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Introduction: While upper gastrointestinal bleeding (UGIB) risk scores are commonly used to predict mortality, their performance in forecasting alternative clinically meaningful complications during admission, such as rebleeding requiring interventional radiology (IR) or repeat oesophagogastroduodenoscopy (OGD), has not been well explored.
This study aimed to validate the predictive accuracy of five UGIB scores for these secondary endpoints.
Aims & Methods: A retrospective cohort analysis was performed on a cohort of 1,003 patients admitted at a UK tertiary centre between January 2025 and August 2023 due to suspected UGIB and investigated via OGD. Each patient's CHAMPS, ABC, Glasgow-Blatchford (GBS), MAP(ASH), and Japanese scores were calculated using their electronic patient records.
The observed outcome was the development of complications, categorised as:
1. Re-bleeding,
2. Requiring IR, and;
3. Requiring repeat OGD.
Receiver operating characteristic (ROC) and area under the curve (AUC) were used to assess each score's discriminative ability.
Results: The ABC score had the highest predictive potential for inpatient complications with an AUC of 0.779, followed by the CHAMPS with an AUC of 0.766 and by the MAP(ASH) with an AUC of 0.716. GBS and the Japanese score had lowest prognostic capabilities.
Conclusion: The ABC, CHAMPS, and MAP(ASH) scores show useful discriminative performance for predicting vital non-mortality outcomes in UGIB patients. These results indicate the possibility of these stratification tools being used outside of predicting inpatient mortality, guiding early escalation of care decisions.
Disclosure: Nothing to disclose.

PP0217

SEX-STRATIFIED PERFORMANCE OF UPPER GASTROINTESTINAL BLEEDING RISK SCORES FOR INPATIENT MORTALITY

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Introduction: Sex-based differences in clinical outcomes are increasingly being recognised in acute care, yet few studies have examined whether upper gastrointestinal bleeding (UGIB) risk scores perform differently in male versus female patients. This study evaluated the discriminatory performance of five established UGIB risk scores in predicting inpatient mortality by sex.
Aims & Methods: This retrospective study included 1,003 patients who underwent inpatient oesophagogastroduodenoscopy (OGD) for UGIB between January 2025 and August 2023 at a UK tertiary care centre. Glasgow-Blatchford Score (GBS), CHAMPS, ABC, Japanese score, and MAP(ASH), were calculated for each patient. Binary Logistic regression examined sex as an independent factor for 30-day inpatient mortality, with UGIB aetiology also accounted for (variceal vs non-variceal). Receiver operating characteristic (ROC) curves were used to assess each score's performance in predicting inpatient mortality separately for male and female subgroups.
Results: Out of the 1,003 admissions, 59.2% and 40.8% were males and females, respectively, with crude analysis demonstrating female mortality being slightly higher (10.5%) compared to male (8.6%). However, this was found not to be significant (p: 0.304, OR 1.251, CI: 0.816-1.917). These findings were consistent regardless of UGIB aetiology (variceal vs non-variceal).
For the 30-day inpatient mortality outcome, the ABC score demonstrated strong discriminative performance in both males (AUC: 0.804) and females (AUC: 0.835). CHAMPS and MAP(ASH) also performed consistently, with CHAMPS yielding AUCs of 0.788 in males and 0.793 in females, and MAP(ASH) showing 0.767 and 0.781 in males and females respectively. GBS performed slightly better in females (AUC: 0.743) compared to males (AUC: 0.691). The Japanese score showed poor discrimination in both sexes, particularly in females (AUC: 0.577). Overall, ABC, CHAMPS, and MAP(ASH) performed reliably across sexes.
Conclusion: Whilst crude mortality rates were higher in females than males, gender did not show to be an independent predictor of inpatient mortality, which could, however, necessitate a larger cohort size to ascertain validity. Risk scores, particularly the ABC score, maintained high and consistent predictive accuracy for inpatient mortality in both male and female patients with UGIB. These findings support the use of these tools across genders and suggest minimal gender-related performance bias in risk stratification.
Disclosure: Nothing to disclose.

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PP0218

MORTALITY AND NEED FOR INTERVENTION IN PATIENTS WITH MASSIVE GASTROINTESTINAL BLEEDING. PRELIMINARY RESULTS OF A MULTICENTER STUDY

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Introduction: Massive gastrointestinal bleeding is a clinical condition that is poorly studied due to its abrupt presentation, involvement of various clinical services, and the high and rapid mortality of patients.

We present the preliminary results of a multicenter registry that includes 21 hospitals from Spain and Hispanic America, analyzing factors related to mortality and need for intervention.

Aims & Methods: Since January 2023, 403 patients over 18 years old who presented to the emergency departments of participating hospitals with massive gastrointestinal bleeding, defined as bleeding accompanied by signs of shock and requiring transfusion of at least 2 units of blood within the first 24 hours of admission, were included. Demographic variables, comorbidities, treatments including NSAIDs, antiplatelets, and anticoagulants, clinical presentation variables, medical, endoscopic, and surgical variables, and a 6-month follow-up were collected.

The aim of this study was to analyze factors related to mortality and the need for intervention in patients with massive gastrointestinal bleeding.

Results: A total of 403 patients with a mean age of 64.5 ± 15.7 years were included (male 68.7%). Main comorbidities were cirrhosis (31%), diabetes (28%) and a previous UGIB (25%). 8.4% of patients had already been admitted to the hospital, and 29% were on antithrombotic (16.6% antiplatelets and 16.4% anticoagulants).

On admission we observed SPB: 91±22 mmHg, heart rate 101 ± 21; and an altered mental status in 31%. Mean hemoglobin levels 7.6±2.5 g/dl, urea: 98±67, albumin 2.16±1.3, INR: 3.9±7. The most common causes were esophageal varices (29%), gastric ulcer (25%) and duodenal ulcer (25%). Endoscopic intervention was needed in 60%, with mortality rate of15.4%. Mean admission was 8.6±10 days, and patients needed 4±3 red blood cells units.

When analyzing in-hospital mortality, we observed that the age of patients who died was higher than that of survivors (68.2±15 vs. 63.6±16; p=0.033), with the former requiring more packed red blood cells (5±5.5 vs. 3.5±2; p=0.043). Patients who died had a lower frequency of gastric ulcer (14% vs. 27%; p=0.034), a higher frequency of altered mental status (42.6% vs. 29%; p=0.036), and a lower rate of endoscopic therapy applied (43.3% vs. 63.6%; p<0.0001). In logistic regression, gastric ulcer etiology and endoscopic therapy were protective factors against mortality (Table 2).

Regarding the need for intervention, more patients with liver cirrhosis required it (73% vs. 60%, p=0.018), patients with hematemesis (73% vs. 52%; p<0.0001), patients without melena (60% vs. 76%; p=0.002), with syncope (76.3% vs. 61%; p=0.06), etiologies other than gastric ulcer (78% vs. 56%; p=0.028), and the presence of esophageal varices (83% vs. 57%; p<0.0001). In binary logistic regression, liver cirrhosis, hematemesis, and syncope were risk factors for the need for intervention.

Conclusion: Age, altered mental status, and lower frequency of endoscopic therapy are risk factors for mortality due to massive gastrointestinal bleeding. Gastric ulcer as an etiology and receiving endoscopic therapy acted as protective factors against mortality.

Indeed, in both situations, it might be related to an opportunity window to receive appropriate treatment, especially when compared with more cataclysmic types of bleeding, such as variceal or posterior duodenal bulb wall, which sometimes can be challenging and utterly abrupt.

Regarding the need for intervention, liver cirrhosis, hematemesis, and syncope were independent risk factors for the need for endoscopy, surgery, or interventional radiology.

Disclosure: Nothing to disclose.

PP0219

ETHNIC SUBGROUP ANALYSIS OF UPPER GASTROINTESTINAL BLEEDING RISK SCORES IN MORTALITY PREDICTION

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Introduction: Studies in cardiovascular disease have raised concerns regarding racial disparities in the predictive performance of clinical risk scores. However, limited data exists on whether scoring systems for upper gastrointestinal bleeding (UGIB) perform differently in racially diverse populations.

This study aimed to compare the performance of five UGIB risk scores in predicting inpatient mortality in White and non-White patients.

Aims & Methods: We performed a retrospective cohort analysis of 1,003 patients who received inpatient esophagogastroduodenoscopy (OGD) on admission between January 2025 and August 2023 at a UK tertiary centre. The MAP(ASH), Glasgow-Blatchford (GBS), CHAMPS, Japanese scores and ABC scores were calculated using patient's medical records. Patients subsequently categorised as: Caucasian, African, Asian, and Mixed.

We utilized Binary Logistic regression to analyse ethnicity as an independent variable for 30-day inpatient mortality, also further adjusting for UGIB cause (variceal vs non-variceal). ROC analysis using IBM SPSS Statistics v29 was performed separately by ethnicity to assess each score's predictive performance for inpatient mortality.

Results: The majority of the cohort belonged to the Caucasian subgroup (91.8%), followed by Asian (1.7%), African (0.8%), and Mixed (1.2%); 4.5% had no racial background recorded. Crude rates showed higher odds of inpatient mortality for the Caucasian subgroup, followed by mixed, Asian, and African subgroups. However, adjusted analysis revealed no significant relationship between race and 30-day inpatient mortality, and these outcomes persisted when adjusted for variceal vs non-variceal bleeding.

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The ABC score demonstrated strong performance across both racial groups, with improved discrimination in non-White patients. CHAMPS and MAP(ASH) also showed higher AUCs in the non-Caucasian subgroup. GBS and the Japanese score performed better in non-Caucasian patients than in Caucasian patients, though overall discrimination remained lower compared to ABC and CHAMPS. There was no evidence that any score systematically over- or under-estimated risk in the non-Caucasian group.
Conclusion: Despite crude rates demonstrating different 30-day inpatient mortality rates based on racial subgroup, there was no statistical significance. These findings could, however, be limited by our cohort size and therefore, might necessitate further research.
The ABC and CHAMPS scores maintain robust predictive accuracy for inpatient mortality in both White and non-White patients with UGIB. Contrary to concerns raised in other specialities, our findings do not suggest performance bias in these scores by race, supporting their equitable use in diverse patient populations.
Disclosure: Nothing to disclose.

PP0220

BODY MASS INDEX AS A PROGNOSTIC FACTOR FOR INPATIENT MORTALITY IN UPPER GASTROINTESTINAL BLEEDING: A UNIVARIATE ANALYSIS

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Introduction: Body mass index (BMI) has been associated with prognosis in several acute and chronic illnesses. However, its role in risk stratification for upper gastrointestinal bleeding (UGIB) remains unclear. This study evaluated whether BMI category alone could predict inpatient mortality in UGIB patients.
Aims & Methods: We conducted a retrospective cohort study of 1,003 patients who underwent inpatient esophagogastroduodenoscopy (OGD) for UGIB at a tertiary centre between January 2025 and August 2023. BMI's were categorised as: healthy weight, underweight, overweight, obese and severely obese. Univariate logistic regression was performed against 30-day inpatient mortality as the primary outcome, with further analysis adjusting for UGIB aetiology (variceal vs non-variceal).
Results: Distributions for the various BMI classes were as follows: 24.3% healthy weight, 4.2% underweight, 27.1% overweight, 16.0% obese, 3.0% severely obese, and 25.4% had no BMI recorded.
Underweight patients had significantly lower odds of inpatient mortality (aOR 0.531 CI: 0.295–0.956, p: 0.025), as did those classified as obese (aOR 0.472 CI: 0.262–0.849, p: 0.012). Other BMI categories were not statistically significant predictors. However, the overall model was not statistically significant (p: 0.142), suggesting BMI alone offers limited predictive value, and this was the case regardless of bleeding aetiology.
Conclusion: Underweight and obese patients had reduced risk of inpatient mortality. Whilst the physiological science remains unclear, these findings support further exploration of BMI as an independent prognostic factor in UGIB, likely with a larger cohort size being required.
Disclosure: Nothing to disclose.

PP0221

EASTERN COOPERATIVE ONCOLOGY GROUP(ECOG) PERFORMANCE STATUS AS AN INDEPENDENT PREDICTOR OF INPATIENT MORTALITY IN UPPER GASTROINTESTINAL BLEEDING

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Introduction: Frailty is a known contributor to poor outcomes in acute illness. The Eastern Cooperative Oncology Group (ECOG) performance status has been proposed as a practical measure of patient frailty.
This study aimed to determine the independent predictive value of ECOG in the setting of upper gastrointestinal bleeding (UGIB).
Aims & Methods: We conducted a retrospective cohort analysis of 1,003 patients who presented with suspected UGIB at a UK tertiary centre and underwent inpatient esophagogastroduodenoscopy (OGD) between January 2025 and August 2023.
The relationship between inpatient mortality and ECOG score was assessed using multivariate logistic regression. The model was adjusted for Glasgow Blatchford Score (GBS), estimated glomerular filtration rate (eGFR), and cancer status variables.
Results: With a p-value< 0.001, ECOG performance status was proven to be a significant independent predictor of inpatient mortality (aOR 1.529 CI: 1.285–1.820). As expected, other significant predictors for inpatient mortality were GBS and metastatic cancer. Renal function was not statistically significant.
Conclusion: The ECOG score is a major predictive factor for inpatient mortality, suggesting the importance of frailty assessment in acute UGIB presentations. Standardising the documentation of ECOG may aid in risk stratification and early care escalation decisions.
Disclosure: Nothing to disclose.

PP0222

EMERGENCE OF SEVERE ACUTE HEPATITIS A IN ADULTS: AN 11-YEAR COHORT FROM A TERTIARY CARE CENTER

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Introduction: The threat of acute hepatitis A (AHA) to public health may be underestimated. We aimed to determine the frequency, clinical characteristics and outcomes of patients with AHA, and to compare them to those with acute hepatitis B (AHB).
Aims & Methods: This was a retrospective study of all patients diagnosed with AHA or AHB at a tertiary referral center between January 2013 and June 2024. Demographics, clinical presentation, laboratory findings, and outcomes were determined. AHA was defined by the presence of immunoglobulin M (IgM) antibodies to hepatitis A virus. AHB was defined by the presence of IgM antibodies to hepatitis B core antigen and positive hepatitis B surface antigen. Severe hepatitis was defined by serum bilirubin >

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10 mg/dl, INR ≥ 1.5, serum creatinine > 2 mg/dl, and/or hepatic encephalopathy. Fulminant hepatitis (FH) was defined by the presence of hepatic encephalopathy.

Results: Over a period of 11 years, 398 patients were diagnosed with acute hepatitis, 367 (92%) with AHA and 31 (8%) with AHB. 336 patients (84.5%) were older than 18 years of age. Compared to patients with AHB, those with AHA were younger, were more likely to be females but were less likely to have an underlying immunocompromising illness (Table 1). There was no difference in the frequency of severe hepatitis, hospitalization rate, FH, and in hospital mortality between the two groups (Table 1).

There were 115 patients (31%) with severe AHA, 89% of whom were older than 18 years of age. There was no difference in the mean age (33 ± 14 vs 31 ± 14 years, p=0.32), gender (37% vs 46% females, p=0.16), history of underlying liver disease (2.6% vs 0%, p=0.07) or having an underlying immunocompromising illness (4.2% vs 6.1%, p=0.47), between patients with severe AHA and patients with non-severe AHA.

Furthermore, compared to patients with non-severe AHA, patients with severe AHA were more likely to require hospitalization (70% vs 43%, p=<0.001), ICU admission (8.7% vs 0.6%, p=<0.001), and to die during the index of hospitalization (4.3% vs 0%, p=0.01). Among patients with AHA who had a serum bilirubin > 10 mg/dl, INR ≥ 1.5, and/or serum creatinine > 2 mg/dl, 7% developed fulminant hepatitis while only 1% of patients with non-severe AHA developed fulminant hepatitis, p=0.003.

Characteristics	Total	Hepatitis A	Hepatitis B	P-value
Number of patients	398	367 (92)	31 (8)	-
Age at diagnosis (years)	31.3 ± 14.4	30.6 ± 14.3	40.2 ± 12.6	<0.001
Gender (female)	163 (41)	158 (43)	5 (16)	0.003
Underlying liver disease	4 (1)	4 (1.1)	0	1.0
Immunocompromising illness*	21 (5.3)	16 (4.4)	5 (16.7)	0.004
Severe hepatitis**	126 (41)	115 (41)	11 (44)	0.77
Hospitalization	168 (42)	159 (43)	9 (29)	0.12
Fulminant hepatitis	10 (2.5)	9 (2.5)	1 (3.2)	0.56
In hospital mortality	6 (1.5)	5 (1.4)	1 (3.2)	0.39

Table 1. Comparison of demographics, clinical characteristics, and outcomes between patients with acute hepatitis A (AHA) and patients with acute hepatitis B (AHB)

**Immunocompromising illness: organ transplantation, systemic cancer, being on chemotherapy, radiotherapy, and/or immunosuppressive therapy.*

***Severe hepatitis: serum bilirubin > 10 mg/dl, INR ≥ 1.5, serum creatinine > 2 mg/dl, and/or hepatic encephalopathy.*

Conclusion: AHA is far more common than AHB, but is associated with similar rates of hospitalization, severe hepatitis, and fulminant hepatitis. The high frequency of AHA among adults in low-income countries calls for renewed attention to prevention and vaccination to reduce the burden of this often-overlooked disease.

Furthermore, further research may be needed to develop pharmacological treatment for patients with severe AHA and those with fulminant hepatitis, in view of unavailability of liver transplantation in most developing countries.

Disclosure: Nothing to disclose.

PP0223

IMPACT OF ANTIPLATELET AND ANTICOAGULANT THERAPIES ON NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING: A PROSPECTIVE OBSERVATIONAL STUDY

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Introduction: Antiplatelets and anticoagulants significantly increase the risk of upper gastrointestinal bleeding by impairing normal clotting mechanisms. Patients on dual or combined therapy often present with more severe bleeds and poorer endoscopic control. This leads to higher rates of rebleeding, complications, and mortality compared to those not on these therapies.

Aims & Methods: To evaluate the risk and outcomes of non-variceal upper gastrointestinal bleeding (UGIB) in patients on single antiplatelet therapy, dual antiplatelet therapy, and combined antiplatelet-anticoagulant therapy, and to assess associated morbidity, mortality, and rebleeding rates.

A prospective study was conducted involving 150 patients (90 males, 60 females) aged 40–60 years, presenting with upper GI bleeding between 1st January 2023 till 1st January 2025.

Patients had co-morbid conditions including hypertension, diabetes mellitus, coronary artery disease, or prior cerebrovascular events, and were on their respective medications for at least 3 months. Patients using NSAIDs, having liver disease, or known bleeding disorders were excluded. Clinical presentations included melena, hematemesis, or both. All patients underwent upper GI endoscopy, with Forrest classification used to characterize ulcers. Laboratory investigations included CBC, LFTs, renal function, PT, and APTT. Bleeding profiles and endoscopic outcomes were analyzed across three medication groups: single antiplatelet (n=60), dual antiplatelet (n=50), and combined antiplatelet-anticoagulant therapy (n=40).

Results: UGIB was reported in 6 (10%) of the single antiplatelet group, 16 (32%) of the dual antiplatelet group, and 14 (35%) of the combined therapy group. Endoscopic findings predominantly revealed Forrest class IIA ulcers. Patients on dual or combined therapy had poorer endoscopic hemostasis. Four patients experienced rebleeding refractory to standard endoscopic measures, requiring angioembolization. One mortality was recorded in the dual antiplatelet group due to massive bleeding and shock.

Conclusion: Dual antiplatelet and combined antiplatelet-anticoagulant therapies significantly increase the risk and severity of non-variceal UGIB, with higher rates of poor endoscopic hemostasis, rebleeding, and mortality. These findings underscore the importance of risk stratification and prophylactic strategies in patients requiring long-term antithrombotic therapy.

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Disclosure: Nothing to disclose.

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PP0224

MULTI-AGENT LLM APPROACH VERSUS SINGLE MODEL IN THE MANAGEMENT OF NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING

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Introduction: Non-variceal upper gastrointestinal bleeding (NV-UGIB) represents a critical clinical condition with high morbidity and mortality risks. Large language models (LLMs) can serve as decision support systems for clinicians by analyzing patient data and providing guideline-based recommendations. However, single LLM approaches may be insufficient in consistently applying guideline recommendations. The multi-agent system offered by the Microsoft AutoGen framework, which integrates assessments from multiple specialized LLM agents representing different specialties through a Consensus Supervisor, may produce more guideline-compliant decisions in NV-UGIB management.

Aims & Methods: This study aimed to compare guideline compliance between a multi-agent LLM system and a single LLM in managing non-variceal upper GI bleeding per ACG guidelines. Twenty cases were selected from MIMIC-III using ICD-9 codes for melena and unspecified GI bleeding. Patient data included demographics, clinical findings, and laboratory results. Evaluation covered Glasgow-Blatchford Score risk stratification, Hb < 7 g/dL transfusion threshold, and risk-based endoscopy timing. In Arm A, separate Gastroenterologist and Emergency Medicine agents provided specialty-specific assessments, while Arm B used a single GPT-4o model. A compliance score (0-3) was calculated for each case, and performance was compared using the Wilcoxon signed-rank test.

Results: In the comparative analysis of model performances, the multi-agent system demonstrated significantly higher guideline compliance than the single model approach. The mean compliance score for the Arm A was 2.85 ± 0.20, while the Arm B achieved 2.15 ± 0.45 (p = 0.002). The distribution of compliance scores revealed that Arm A achieved full compliance (3 points) in 18 cases (90%), with only 2 cases scoring 2 points. By contrast, Arm B achieved full compliance in only 12 cases (60%), with 5 cases scoring 2 points and 3 cases scoring just 1 point. For risk stratification using the Glasgow-Blatchford Score, the multi-agent system misclassified only 2 cases (10%), both involving incomplete consideration of BUN values in the scoring calculation. The single model, however, either completely overlooked or incorrectly applied risk categorization in 5 cases (25%), including critical misinterpretation of 2 high-risk cases (GBS ≥ 7) as moderate or low risk, demonstrating difficulty in evaluating multiple parameters simultaneously. Regarding transfusion thresholds, Arm A maintained perfect compliance across all 20 cases. In contrast, Arm B showed errors in 3 cases (15%), failing to recommend transfusion in 2 cases despite Hb levels below 7g/dL and inappropriately recommending transfusion in 1 case with Hb above threshold. For endoscopy timing recommendations, the multi-agent system again achieved 100% compliance, while the single model presented out-of-guideline timing in 4 cases (20%), including ambiguous timing expressions for high-risk cases and inappropriate delays for moderate-risk cases. The Consensus Supervisor in Arm A effectively integrated specialty expertise to produce consistently guideline-compliant recommendations across all evaluation criteria.

Conclusion: The multi-agent consensus approach using the Microsoft AutoGen framework demonstrated significantly higher guideline compliance than a single LLM in NV-UGIB management. This autogenic system, incorporating multiple specialized LLMs representing different clinical perspectives, provides more consistent evaluation of complex parameters and shows promise for clinical decision support systems.

Disclosure: Nothing to disclose.

PP0225

PROPRANOLOL VERSUS ENDOSCOPIC VARICEAL LIGATION FOR PRIMARY PREVENTION OF VARICEAL BLEEDING: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Variceal bleeding accounts for approximately 20–30% of cirrhosis-related deaths and remains a major complication of portal hypertension in patients with liver cirrhosis (1). The National Institute for Health and Care Excellence (NICE) in the United Kingdom (UK) recommends the administration of propranolol or carvedilol, or endoscopic variceal ligation (EVL) as primary prevention of medium to large oesophageal varices. (2). However, the comparative efficacy and safety profiles of these approaches remain unclear.

Aims & Methods: A systematic review and meta-analysis were conducted in accordance with PRISMA guidelines. Electronic databases were searched to identify randomised controlled trials (RCTs) comparing EVL and propranolol for the primary prevention of variceal bleeding in patients with liver cirrhosis. Primary outcomes included incidence of variceal bleeding, all-cause mortality, and bleeding-related mortality. Secondary outcomes included adverse effects related to prophylactic treatment. Data were pooled using fixed or random-effects models depending on heterogeneity.

Results: Nine RCTs comprising a total of 916 patients were included. EVL was associated with a significantly lower risk of variceal bleeding compared to propranolol (odds ratio [OR] 0.44; 95% confidence interval [CI]: 0.28–0.67; p = 0.0002) and reduced bleeding-related mortality (OR 0.57; 95% CI: 0.33–0.99; p = 0.05). Adverse effects were also significantly less frequent in the EVL group (OR 0.46; 95% CI: 0.32–0.67; p < 0.0001). No statistically significant difference was found in all-cause mortality between the two interventions (OR 0.93; 95% CI: 0.63–1.37; p = 0.72).

Conclusion: EVL is more effective than propranolol for the primary prevention of variceal bleeding in patients with cirrhosis, demonstrating lower rates of first variceal bleed, bleeding-related mortality, and treatment-related adverse events. Although no difference in all-cause mortality was observed, the findings support the consideration of EVL as a first-line prophylactic option. Further well-designed, multicentre RCTs with standardised outcomes, extended follow-up, and robust cost-effectiveness analyses are essential to optimise clinical decision-making and support personalised treatment strategies.

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Disclosure: Nothing to disclose.

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PP0226

PERFORMANCE OF SINGLE-USE GASTROSCOPES IN EMERGENCY SETTING

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Introduction: The development of single-use endoscopes was driven by hygiene aspects. However, an even more interesting aspect is that these endoscopes are ready to use and can be immediately disposed after the examination. Therefore, these aspect favors its use outside regular working hours.

Aims & Methods: This study aimed to provide the first experiences of the Ambu single-use gastroscopes with different working channels in daily routine in an emergency setting outside regular working hours in a tertiary center.I

n a time period between Oktober 2023 and Oktober 2024 single use endoscope with 2.8 or 4.2 mm working channel were used in 63 patients. Due to quality assessment different aspects of the single-use endoscope were reported using Lickert scale. All examinations were performed outside regular working hours. 83% of the examinations were done in emergency room or intensive care unit.

Results: Successful completion of endoscopy using only the single-use gastroscope was achieved in 50 out of 63 cases (79%, suspicion of bleeding (n=44), extraction of a foreign body (n=7), bolus obstruction (n=4), pain or suspicion of perforation (n=4), reflux or gastroparesis (n=2), dysphagia (n=1), intoxication (n=1), and megacolon (n=1).

Overall, 43 out of 63 patients (68%) required an intervention during the endoscopy In 13 out of 63 cases the endoscopy failed. In 8 out of 13 cases the examination could be finished successfully after a crossover to a reusable endoscope. A re-intervention was indicated in 7 out of these 13 patients. In a subgroup analyses the success rate was comparable in patients who were treated with single use gastroscope with a 2.8 mm (75%) and 4.2 mm (84%) working channel (p=0.535). The gastroscope with the 4.2 mm channel was significantly better regrading image quality (p=0.006, wheel functionality (p=0.025) and suction function (p=0.032).

Conclusion: This study demonstrates that single-use endoscopes can be used in different emergency settings, however without further technical improvement its usage especially in acute bleeding situations is limited.

Disclosure: Nothing to disclose.

PP0227

EMERGENCY ENDOSCOPY FOR UPPER GASTROINTESTINAL BLEEDING: WHAT REAL BENEFIT FOR ELDERLY PATIENTS

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Introduction: Upper gastrointestinal bleeding (UGIB) is a common medical emergency, particularly in elderly patients, in whom it is often complicated by comorbid conditions and an increased anesthetic risk.

Aims & Methods: The aim of this study was to assess whether emergency endoscopy, although justified by a high Blatchford score, is as relevant in elderly patients as it is in younger patients in terms of therapeutic endoscopic management.

This was a retrospective, monocentric, and analytical study conducted in a tertiary gastroenterology department in North Africa. It included all patients admitted to the emergency department with UGIB between January 2023 and December 2024 who underwent emergency upper gastrointestinal endoscopy.

Inclusion criteria were a Blatchford score strictly greater than 2, according to international recommendations, as well as a history or clinical signs of portal hypertension. Patients were divided into two groups according to age: group 1 (non-elderly, < 65 years) and group 2 (elderly, ≥ 65 years). Statistical analysis was performed using Jamovi software (version 2.3.28), with comparisons made using the Pearson chi-squared test for qualitative variables. A p-value < 0.05 was considered statistically significant

Results: A total of 155 patients were included, 95 in the non-elderly group and 60 in the elderly group. Elderly patients had significantly more cardiovascular comorbidities. Cardiac disease or arterial hypertension was found in 35% and 33% of elderly patients, respectively, compared with only 5% each in younger patients (p < 0.001). The absence of comorbidities was significantly more frequent in younger patients (18 vs. 3 cases; p = 0.026).

Use of anticoagulant or antiplatelet therapy was also significantly more common in the elderly group (p = 0.0035), with a predominance of aspirin alone or combined with clopidogrel. AODs and VKAs were more frequently used among younger patients. Blood transfusion was required in 47.4% of younger patients and 40% of elderly patients, with no significant difference (p = 0.46).

Endoscopic findings in younger patients most frequently included gastroduodenal ulcers, followed by signs of portal hypertension and esophagitis. Normal endoscopy was observed in 15.8% of younger patients (15 cases) and 26.2% of elderly patients (16 cases), with no statistically significant difference (p = 0.15). In elderly patients, the main diagnoses were ulcers, esophageal varices, esophagitis, and normal FOGD findings.

Therapeutic endoscopic intervention was performed in 39 younger patients (41.1%) versus 15 elderly patients (25%), showing a significant difference (p = 0.0297). This reflects a lower rate of endoscopic hemostasis in the elderly despite identical clinical criteria for emergency indication.

Conclusion: In elderly patients, although emergency endoscopy is indicated based on the same criteria as in younger patients, it less often results in therapeutic intervention. These findings highlight the importance of a personalized approach that accounts for the patient's overall geriatric profile.

Disclosure: Nothing to disclose.

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PP0228

ENDOSCOPIC BLEEDING MANAGEMENT IN ALTERED ANATOMY: PRELIMINARY RESULTS OF THE EMBLEMA TRIAL

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Introduction: The safety and efficacy of endoscopic haemostasis in patients with upper gastrointestinal (UGI) bleeding has been widely assessed over the years. However, to date no study addressed the management of UGI bleeding specifically in the setting of patients with surgically altered anatomy.

Aims & Methods: Our study is focused on the evaluation of the endoscopic management of UGI bleeding in these patients with specific concern about the techniques used to achieve haemostasis, the management of technical failure and the rate of adverse events. We conducted a multicentre retrospective- prospective observational study in 13 endoscopy units in Italy. All consecutive patients with UGI bleeding and surgically altered anatomy managed endoscopically were collected from January 2023 to November 2024.

Results: We enrolled 162 patients (73% males) with a mean age of 69 years. 45% of patients had history of oncologic surgery while 20% underwent bariatric surgery. The most common clinical manifestations were melena (59%) followed by hematemesis (22%). At bleeding event patients were taking anticoagulants and antiplatelet in 32% and 16% of cases, respectively. In 43% of cases from 1 to 4 blood units were needed, in 15% more than 4 blood units, while in 42% of patients no transfusions ware required. 42% of endoscopic procedures were performed with anesthesiologist assistance. Bleeding was located at anastomotic site in 70% of cases and ulcer was the most common source of bleeding (54% of patients). The most frequently used treatments were adrenaline combined with TTSC (40 %) and TTSC alone (24%). Haemostatic powder or gel and adrenaline combined with thermal treatment were used only in 6 and 4% of cases respectively while OTSC was used in 1 patient alone after failure of adrenaline combined with TTSC. Bleeding control during endoscopy was achieved in 94% of patients. In 12 patients rebleeding occurred, which was managed with surgery in 2 patients, interventional radiology in 1 case and with new endoscopic treatment in 4 cases. No serious adverse events linked to endoscopy were reported.

Conclusion: Our preliminary results showed that patients with altered anatomy often had severe bleeding requiring a large amount of blood transfusions. The standard endoscopic homeostatic techniques seemed to be effective for the management of bleeding, although a not negligible rate of patients needed additional treatments. However, further data are needed to better assess clinical outcomes in this setting and to better define possible predictive predictors of clinical success and therefore which technique should be preferred to achieve haemostasis.

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Disclosure: Nothing to disclose.

PP0229

BLEEDING-RELATED INDICATIONS AS KEY PREDICTORS OF BALLOON-ASSISTED ENTEROSCOPY OUTCOMES AFTER POSITIVE VIDEO CAPSULE OR CROSS-SECTIONAL IMAGING

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Introduction: Balloon-assisted enteroscopy (BAE) is an important diagnostic tool for small bowel evaluation, especially following a positive Video Capsule Endoscopy (VCE) or Cross-Sectional Imaging (CSI). However, the diagnostic yield of BAE in this context remains variable.

Aims & Methods: This retrospective study aimed to evaluate the diagnostic yield of BAE performed following positive imaging. All BAE procedures conducted between 01/2009-12/2024 following a positive CSI (CT or MRI) or VCE within 12 months prior to the procedure were included. Clinical characteristics, procedural indications (polyp/mass, suspected bleeding or inflammation), and imaging modalities were compared between groups. A classification tree analysis explored factors associated with a positive diagnostic yield.

Results: Among 224 BAE procedures, the diagnostic yield was 58.5% (131 positive findings). No significant differences were observed between positive and negative groups regarding sex, age, indication, pre-BAE imaging modality (VCE or CSI), route, or depth of insertion. Classification tree analysis identified bleeding-related indications (e.g. overt bleeding or angioectasias) as the strongest predictor of positive yield (74.5% vs. 50%, p=0.005), particularly with antegrade BAE. However, cross-validation failed to generate a stable model, with a misclassification risk of 41.5% (±3.3), indicating limited predictive accuracy.

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Conclusion: In this large retrospective study spanning 15 years, BAE following positive imaging demonstrated a diagnostic yield of nearly 60%. While demographic and procedural factors were not predictive, bleeding-related indications were significantly associated with positive findings, highlighting the importance of clinical context in guiding BAE use. These findings may help optimize patient selection and improve procedural efficacy.

Disclosure: Nothing to disclose.

PP0230

TIMING OF EMERGENCY GASTROSCOPY IN UPPER GASTROINTESTINAL BLEEDING: A RETROSPECTIVE COHORT ANALYSIS IN A TERTIARY CENTER

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Introduction: International guidelines recommend performing emergency gastroscopy within 12 hours for variceal (V) bleeding and within 24 hours for non-variceal (NV) bleeding in cases of upper gastrointestinal bleeding (UGIB).

Aims & Methods: This study aimed to analyze the time elapsed between emergency department admission and time of endoscopy, referred to as “door-to-endoscopy time” (DET), and to identify factors influencing DET across different UGIB subgroups. We conducted a retrospective cohort analysis of patients who underwent emergency gastroscopy for UGIB between January 1, 2022, and December 31, 2023, at the Department of Emergency Medicine at Semmelweis University. Data were collected on clinical parameters and 30-day mortality. NV patients were categorized into high-risk (HNV) and low- risk (LNV) groups based on endoscopic findings (Forrest classification). Welch’s t-test was used for comparing mean DET among the subgroups, whereas for hemoglobin linear regression was applied. The significance level was set at 0.05 for all analyses.

Results: Among 654 patients, we identified 126 V and 528 NV UGIB cases, with the latter comprising 99 HNV and 429 LNV cases. Rebleeding occurred in 5.4% of patients (V: 12.7%, HNV: 10.1%, LNV: 2.1%), and 30-day mortality was 19.7% in the cohort (V: 24.8%, HNV: 22.6%, LNV: 17.6%). The mean DET was 7.28 hours (SD=±6.04), with subgroup averages as follows: V: 5.87±3.77 hours, HNV: 6.33±4.04 hours, and LNV: 7.92±6.83 hours, with significant differences between LNV-HNV (p=0.003) and LNV-V (p<0.0001) groups; no significant difference was observed between HNV-V (p=0.37). In the entire cohort, shorter DET was significantly associated with tachycardia (6.61 vs. 7.73 hours, 95% confidence interval (CI) -1.99 to -0.25, p=0.01), hematemesis (5.82 vs. 8.09 hours, 95% CI -3.11 to -1.43, p<0.0001), syncope (6.25 vs. 7.57 hours; 95% CI -2.30 to -0.33, p=0.009), and lower hemoglobin levels (1 g/L decrease in hemoglobin reduced DET by an average of 1 minute, p=0.029). DET was not associated with 30-day mortality (p=0.501), vasopressor use (p=0.857), shock index (p=0.091), coffee-ground emesis (p=0.192), hematochezia (p=0.564), intubation (p=0.566), rebleeding rate (p=0.462), or melena (p=0.961). In the V group, DET was significantly shorter in case of hematemesis (5.20 vs. 7.25 hours, 95% CI -3.61 to -0.49, p=0.011) and tachycardia (5.19 vs. 6.64 hours, 95% CI -2.84 to -0.08, p=0.039); whereas in HNV patients with coffee-ground emesis, DET was significantly reduced (4.94 vs. 6.58 hours, 95% CI -3.08 to -0.20, p=0.026); in LNV patients, DET was shorter in patients with hematemesis (6.42 vs. 8.38 hours, 95% CI -3.20 to -0.71, p=0.002) and syncope (6.56 vs. 8.25 hours, 95% CI -3.10 to -0.29, p=0.019).

Conclusion: In our retrospective cohort study, DET significantly differed between LNV-V and LNV-HNV subgroups. Shorter DET correlated with tachycardia, hematemesis, syncope, and lower hemoglobin levels. DET showed no association with 30-day mortality or rebleeding rate.

Disclosure: Nothing to disclose.

PP0231

RISK FACTORS FOR REBLEEDING AFTER TRANS-ARTERIAL EMBOLIZATION IN PATIENTS WITH NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING

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Introduction: Non-variceal upper gastrointestinal bleeding can usually be controlled endoscopically; however, if endoscopic hemostasis fails, it can pose a life-threatening risk. Trans-arterial embolization is a less invasive, safe, and effective rescue therapy that can serve as an alternative to surgery. Identifying factors influencing rebleeding after trans-arterial embolization in advance could lead to more efficient and safer patient management. Therefore, we aim to investigate these factors.

Aims & Methods: We retrospectively investigated 167 patients treated with TAE between January 2008 and June 2023. All patients underwent emergent endoscopy prior to TAE. Re-bleeding was defined as the presence of melena, hematemesis, or hematochezia with a fall in hemoglobin levels (> 0.8 g/dl) or shock after TAE. The primary outcomes was 30-day re-bleeding rate.

Results: Total of 104 patients with non-variceal upper gastrointestinal bleeding were selected. Re-bleeding within a month after TAE was noted in 24 patients (23.1%). Patients with re-bleeding had significantly higher rates of thrombocytopenia (50.0% vs. 15.0%, p= 0.001) and coagulopathy (20.8% vs. 5.0%, p= 0.045) than those without re-bleeding. Additionally, 7-day mortality (16.7% vs. 2.5%, p=0.035), AIMS65 score (2.0 vs. 1.0, p=0.029), and ABC score (7.0 vs. 4.0, p=0.002) were significantly elevated in patients with re-bleeding. A multivariate logistic analysis showed thrombocytopenia (odds ratio: 5.44, 95% confidence interval: 1.84–16.92, p= 0.002) and an ABC score ≥ 6 (odds ratio: 2.87, 95% confidence interval: 1.02–8.66, p= 0.005) significantly increased the risk of re-bleeding after TAE.

Conclusion: TAE could be an effective and safe treatment for non-variceal upper gastrointestinal bleeding. However, thrombocytopenia and elevated ABC scores might be associated with a higher 30-day rebleeding rate after TAE. Therefore, patients with these risk factors should be carefully monitored and managed.

Disclosure: Nothing to disclose.

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PP0232

ANTITHROMBOTIC THERAPY AND UPPER GI HEMORRHAGE: RISKY BUSINESS?

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Introduction: This study aims to assess the impact of antithrombotic therapy on endoscopic findings in patients hospitalized for upper gastrointestinal bleeding.

Aims & Methods: We carried out a prospective, single-center cross-sectional study including 332 patients between June 2020 and February 2025. Patients were classified as antithrombotic (AT) users if they were receiving antiplatelet agents (such as low-dose aspirin or thienopyridines) and/or anticoagulants (including vitamin K antagonists, direct oral anticoagulants, or heparin).

Results: The mean age of the study population was 59 ± 16.7 years, with a marked male predominance (77.1%). Among them, 63 patients (19%) were receiving antithrombotic (AT) therapy—41 on antiplatelet agents and 39 on anticoagulants. Patients in the AT group were significantly older (68 vs. 57 years; $p < 0.001$) and had a higher burden of comorbidities (75.8% vs. 16.7%; $p < 0.001$). However, there were no significant differences between the AT and non-AT groups regarding the presence of active bleeding at endoscopy (12.7% vs. 16.8%; $p = 0.425$) or the need for endoscopic hemostasis (7.9% vs. 16%; $p = 0.1$).

In multivariate analysis, after adjustment for age, sex, comorbidities, active bleeding, and AT use, only the presence of active bleeding was independently associated with the need for endoscopic hemostasis. Specifically, active bleeding at endoscopy increased the odds of requiring hemostasis by 26-fold (OR: 26, 95% CI: 12.9–62.15; $p < 0.001$), whereas AT therapy was not an independent predictor (OR: 0.386, 95% CI: 0.105–1.42; $p = 0.154$).

Conclusion: In older patients receiving antithrombotic therapy and admitted for upper gastrointestinal bleeding, the risk of active bleeding during endoscopy and the need for endoscopic hemostasis do not appear to be increased.

Disclosure: Nothing to disclose.

PP0233

EARLY ENDOSCOPY IN NON-VARICEAL UPPER GI BLEEDING: DOES TIMING INFLUENCE OUTCOMES IN LOW-RISK PATIENTS ?

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Introduction: Despite significant advances in the management of non-variceal upper gastrointestinal bleeding (NVUGIB), the optimal timing of endoscopy remains a subject of debate.

Current guidelines recommend performing endoscopy within 24 hours of presentation, yet the impact of this strategy in different risk stratifications is not fully elucidated.

This study aimed to evaluate the influence of early endoscopy (≤24 hours) on clinical outcomes in NVUGIB patients, with particular focus on risk stratification using the Glasgow-Blatchford Score (GBS).

Aims & Methods: This retrospective, single-center cohort study included consecutive adult patients who underwent upper gastrointestinal endoscopy for suspected NVUGIB over a four-year period. Demographic, clinical, biochemical, endoscopic, and outcome data were collected and analyzed.

Patients were stratified based on GBS into low-risk (GBS <12) and high-risk (GBS ≥12) categories. The primary outcome was 30-day mortality. Secondary outcomes included need for endoscopic hemostasis and blood transfusion.

Results: A total of 298 patients were included (mean age: 65.3 ± 17.1 years; 57% male), of whom 55% were classified as high-risk. Endoscopy was performed within 24 hours in 62.1% of cases.

In low-risk patients, early endoscopy was significantly associated with:

- Higher rate of endoscopic therapy ($p = 0.003$)
- Lower 30-day mortality ($p = 0.03$)
- Reduced need for blood transfusion ($p = 0.03$)

Patients in the low-risk group who underwent delayed endoscopy (>24 hours) were significantly older (70.8 ± 15.2 vs. 63.5 ± 16.7; $p = 0.01$), had more comorbidities (Charlson Comorbidity Index: 4.0 ± 2.1 vs. 3.1 ± 2.5; $p = 0.02$), lower baseline hemoglobin (10.1 ± 3.1 vs. 11.3 ± 2.8; $p = 0.03$), and were more frequently admitted during weekends or holidays (27 vs. 0; $p < 0.001$).

In contrast, in high-risk patients, no statistically significant differences in outcomes were observed between early and delayed endoscopy.

Conclusion: Our findings suggest that performing upper endoscopy within 24 hours is associated with improved clinical outcomes, specifically reduced mortality and transfusion requirements, in low-risk NVUGIB patients.

Early intervention appears especially beneficial in older patients with multiple comorbidities and initial anemia. These data support prioritizing early endoscopic evaluation in selected lower-risk patients as part of optimized UGIB management pathways.

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Disclosure: Nothing to disclose.

PP0234

PARTICULARITIES OF UPPER GASTROINTESTINAL BLEEDING IN PATIENTS ON DIRECT ORAL ANTICOAGULANT THERAPY

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Introduction: Upper gastrointestinal (GI) hemorrhage is a frequent reason for admission to the emergency department, leading to serious complications and represent a major cause of mortality, requiring early diagnosis and management. The prognosis may be altered in patients on anticoagulant therapy.
Aims & Methods: The aim of the study was to define clinical and endoscopic particularities of upper GI bleeding in patients on Direct Oral Anti-Coagulant (DOAC) therapy.
We carried out a retrospective, monocentric, cross-sectional study, over a 2-year period from January 2023 to January 2025, in our hepato-gastroenterology department, including all patients hospitalized for upper GI bleeding. We conducted a descriptive and analytical statistical study subdividing patients into two groups: first group (G1), those receiving DOAC treatment; and a second group (G2) those not receiving anticoagulant treatment.
Data collected included demographics (age, sex), comorbidities, current treatment, hemodynamic constants (blood pressure, heart rate), upper gastrointestinal bleeding severity scores (Glasgow-Blatchford score (GBS)), upper GI endoscopy findings, as well as the need for endoscopic hemostasis, and the percentage of recurrent bleeding. Biological data included hemoglobin level, renal function.
All these data were entered and analyzed in a database using SPSS version 21.0.
Results: One hundred and thirty-seven patients were included in this study. Thirty-four patients receiving DOAC therapy (25%).
There was a statistically significant difference in the mean age of patients was 68 years in G1 and 49 years in G1, and 68% of patients were over 60 years of age compared with 32%, in G1 and G2 respectively (p= 0.04 ; OR= 4.8). A male predominance with a sex ratio of 2.7 was noted in G1, but in G2 there was no statistically significant difference according to gender. On the other hand, 78% of patients in G1 versus 25% in G2 (p= 0.02 ; OR= 7.5) had one or more comorbidities, mainly diabetes, hypertension and coronary heart disease.
There was no statistically significant differences were found between the two groups with regard to upper gastrointestinal bleeding severity scores (GBS), the mean GBS being 5 (Extremes: 1 - 8) in G1 and 4 (Extremes: 0 - 6) in G2. Furthermore, no statistically significant difference was found between G1 and G2 concerning hemodynamic constants (blood pressure, heart rate) and Biological data (hemoglobin level, renal function).
Similarly, there was no statistically significant difference between the two groups in the results of upper GI endoscopy (active bleeding present in 52% of cases in G1 vs 48% of cases in G2), as well as the need for endoscopic hemostasis was the same in both groups (11% in G1 vs 13% in G2), and no recurrence of bleeding was observed in either group.
Conclusion: Upper GI bleeding under DOAC therapy is mainly observed in elderly person with several comorbidities. However, according to our study the DOAC therapy does not appear to increase the severity of GI bleeding or the risk of recurrent bleeding, contrary to what has been described in the literature. Further prospective studies are needed to better investigate this association.
Our study has certain limitations, its retrospective nature and sample size limiting subgroup analyses.
Disclosure: Nothing to disclose.

PP0235

MANAGEMENT OF ADHERENT CLOTS IN PEPTIC ULCER BLEEDING: A RETROSPECTIVE COHORT STUDY ON POST-REMOVAL OUTCOMES

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Introduction: Peptic ulcer bleeding (PUB) is a common cause of non-variceal upper gastrointestinal bleeding (NVUGIB). In cases with high-risk stigmata (HRS) endoscopic therapy is recommended.
Aims & Methods: Our aim was to analyze the course of patients with adherent clots found on index endoscopy.Patients undergoing emergency endoscopy for upper GI bleeding at the Department of Emergency Medicine at Semmelweis University between January 1, 2022 and December 31, 2023 were included in a retrospective cohort analysis. In case of a peptic ulcer with adherent clot, its removal was encouraged according to our in-house protocol. If needed endoscopic treatment was applied as defined by ESGE based on the Forrest classification.
Results: There were 92 patients (56 males, 36 females, mean age 74) with PUB and HRS. In 33 patients (35.8%) an adherent clot (Forrest IIb) was detected during index endoscopy, 13 cases in the stomach and 20 cases in the duodenum. The frequency of further HRS was as follow: Forrest Ia: 8.7% (n=8/92), Forrest Ib: 34.4% (n=31/92), Forrest IIa: 22.8% (n=21/92). Rebleeding rate was 10.8% (n=10/92), 30-day mortality was 21.7% (n=20/92). Of the 33 cases with adherent clot, in 25 cases (75.8%) there was an attempt to remove the clot. In eight cases (24.2%) the clot could

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not be removed. In the 17 cases (51.5%), where the clot was removed successfully, the underlying ulcer got upgraded in four cases (23.5%) to Forrest Ib; and downgraded in 13 cases (76.5%): to Forrest IIc in five cases (29.4%), Forrest III in eight cases (47.1%). During post-endoscopy course six out of the 33 patients (18.1%) experienced rebleeding. In four patients, where clot removal led to active bleeding (Forrest Ib), after initial hemostasis at index endoscopy, one patient required additional treatment during second look endoscopy. In those 13 patients, who got downgraded, three (21.4%) needed repeated endoscopic treatment; whereas in the eight patients, where clot removal was not endeavoured, two needed endoscopic re-treatment.

Overall, in five cases endoscopic hemostasis was achieved and one patient required surgery. During a 30-day follow-up 12 patients (36.3%) died in this cohort, but only one patient (3%) related to bleeding and one patient due to complications of surgery.

Conclusion: Based on our finding in a small cohort of patients with PUB and an adherent clot, endoscopic removal can be attempted and accomplished. Irrespective of the subsequent finding on the ulcer base, successful endoscopic hemostasis can be achieved on index endoscopy. However, the risk of rebleeding and mortality is higher than patients with other HRS. Further prospective studies on larger patient cohorts are encouraged.

Disclosure: Nothing to disclose.

PP0236

PERSONAL USE OF PROTON PUMP INHIBITORS AMONG PHYSICIANS: A CROSS-SECTIONAL STUDY

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Introduction: Proton pump inhibitors (PPIs) are commonly prescribed for acid-related disorders, yet concerns about overuse and long-term adverse effects persist. Physicians, due to their medical knowledge and unrestricted access to medications, may demonstrate unique self-medication behaviors. This study aimed to assess the prevalence, indications, self-prescribing habits, and awareness of PPI-related risks among physicians.

Aims & Methods: A cross-sectional online survey was administered to 442 physicians across different healthcare levels. The anonymous questionnaire collected data on demographics, frequency and indications for PPI use, self-medication practices, knowledge of adverse effects, and deprescribing behavior. Descriptive and comparative statistical analyses were conducted.

Results: Of the respondents, 60.8% reported occasional PPI use, and 52.0% admitted to self-medicating without prior medical consultation. The most common indications were gastroesophageal reflux disease (GERD) and dyspepsia. Preventive use (e.g., before fatty meals or alcohol) was rare, reported by only 6.1%. Although awareness of potential side effects was high, just 2.0% had discontinued or reduced PPI use due to adverse effects.

Conclusion: Self-medication and occasional PPI use are prevalent among physicians, despite a high level of awareness regarding potential harms. These findings underscore the need for targeted educational efforts and clearer deprescribing guidelines to promote more rational PPI use in the medical community.

Disclosure: The authors declare no conflicts of interest related to this study.

PP0237

AUDIT ON ENDOSCOPIC FOLLOW-UP OF GASTRIC ULCER: COMPLIANCE WITH THE UK, NICE GUIDELINES

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Introduction: Peptic ulcers are a common gastrointestinal condition with potential risks for recurrence and complications. Timely follow-up, particularly repeat endoscopy, is critical for confirming ulcer healing and minimizing unnecessary interventions or missing gastric cancer. The National Institute for Health and Care Excellence (NICE) recommends repeat esophagogastroduodenoscopy (OGD) 6–8 weeks following gastric ulcer diagnosis to confirm healing.

Aims & Methods: Aim: To evaluate compliance with NICE guidelines for repeat endoscopy following a gastric ulcer diagnosis within a large UK hospital trust.

Methods: A retrospective audit was conducted on patients diagnosed with gastric ulcers from October 2022 to September 2024 at Barts Health NHS Trust. Patient records, endoscopy reports, and laboratory results were reviewed to assess whether follow-up endoscopy occurred and its timing relative to NICE recommendations.

Results: A total of 62 cases were identified and their follow up were recorded. Out of 62 patients (mean age 58 years), 53 (85%) underwent follow-up, while 9 (15%) did not. Only 12% received follow-up within the 6–8-weeks guideline window, while 50% had follow-up within 9 weeks. Most follow-ups (70%) occurred within 24 weeks. A Kaplan-Meier survival analysis (figure 1) was performed to visualise time-to-follow-up, showing a gradual increase in follow-up rates over time, with a plateau reached by 24 weeks. This analysis highlighted delays in achieving the recommended 6–8-week benchmark and demonstrated that only a minority of patients received follow-up in a timely manner.

Follow-up timing influenced diagnostic outcomes. Patients reviewed before 6 weeks had a higher rate of ongoing ulcers (44%), often prompting further investigations (figure 2). Those seen between 6–8 weeks had complete ulcer healing. Among patients reviewed at 8–12 weeks and beyond 12 weeks, 27% still had ongoing ulcers, though the majority had healed. Notably, patients with follow-up beyond 12 weeks often underwent repeated endoscopy for continued management.

Non-compliance was primarily due to failure to request follow-up (56%), patient non-attendance (33%), and medical inappropriateness (11%).

Conclusion: Although most patients received follow-up endoscopy, adherence to NICE-recommended timing was suboptimal. Our study shows that while most patients were eventually followed up, only a small pro-

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portion were reviewed within the ideal 6–8-week window. Patients who had early follow-up (<6 weeks) often still had ongoing ulcers, necessitating unnecessary investigations due to insufficient healing time. Conversely, patients with follow-up after 12 weeks typically required repeated endoscopy for ongoing management. To optimise care, systems should be improved to ensure timely and targeted follow up. Further research should explore only targeted cases merit follow ups and patient-level factors contributing to delayed follow-up adherence.

Figure 1: Kaplan-Meier Survival Curve for Follow-Up Endoscopy After Peptic Ulcer Diagnosis.

The curve illustrates the cumulative proportion of patients without follow-up endoscopy over time since initial peptic ulcer diagnosis. A steep initial decline indicates that most follow-ups occurred within the first 24 weeks. However, only a small proportion of patients received follow-up within the NICE-recommended 6–8 week timeframe. The plateau after 24 weeks reflects the reduced likelihood of follow-up beyond this point.

Figure 2: Outcomes of Follow-Up Endoscopy After Peptic Ulcer Diagnosis by Timing Interval

Disclosure: None.

PP0238

AVOIDANT/RESTRICTIVE FOOD INTAKE DISORDER IN FUNCTIONAL DYSPEPSIA REFLECTS INCREASED PSYCHOSOCIAL DISTRESS AND BURDEN, NOT WORSE DYSPEPSIA SYMPTOMS

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Introduction: Avoidant/restrictive eating is common in people with neurogastroenterology disorders including functional dyspepsia (FD). There is debate on whether avoidant/restrictive food intake disorder (ARFID) is an expected response to severe symptoms or simply a common comorbid disorder. To address this, among adults with FD, we aimed to contrast those with and without ARFID on FD symptoms, quality of life, psychosocial factors, nutrient tolerance, and gut transit.

Aims & Methods: Adults with Rome-IV confirmed FD and healthy controls (HC) from an ongoing study participated. ARFID diagnosis was assigned based on a semi-structured *DSM-5* diagnostic interview. Surveys: Patient Assessment of GI Disorders Symptom Severity Index (PAGI-SYM), Short Form Nepean Dyspepsia Inventory (SF-NDI), Nine-Item ARFID Screen (NIAS), Fear of Food Questionnaire (FFQ), Visceral Sensitivity Index (VSI), and Hospital Anxiety/Depression Scale (HADS). Nutrient Drink Test (NDT) measured gastric tolerance—liquid nutrient ingestion at 30mL/minute until subject reported unbearable fullness. Wireless Motility Capsule (WMC) measured gut transit. Height/weight was measured for body mass index (BMI).

Results: Sample (80% female, M_{age}=37.1±14.4y) included 37 FD+ARFID, 14 FD only, and 18 HC. Groups did not significantly differ in sex or age. Adjusted pairwise comparisons showed small (*r*=.1-.2) non-significant differences between FD and FD+ARFID on FD symptoms by PAGI-SYM and SF-NDI (Table 1). However, FD+ARFID had significantly worse scores compared to FD with medium effects (*r*=.3-.4) on: FD-related quality of life by SF-NDI, Fear of Food by NIAS-Fear and FFQ, GI-anxiety by VSI, and anxiety by HADS. FD+ARFID vs FD also had small differences (*r*s=.2) on: feeling more bothered by FD symptoms on SF-NDI, worse depression by HADS, lower BMI, and less NDT consumption. Compared to HC, FD and FD+ARFID had significantly worse survey scores with large effects for FD+ARFID (*r*=.6-.8) and medium effects for FD vs HC (*r*=.3-4). For the NDT, FD+ARFID consumed significantly less than HC with large effect (*r*=.6) but with a trend medium effect for FD consuming less than HC (*r*=.3). Groups did not significantly differ in WMC transit times.

	M+SD			p-value[1]		
	FD+ARFID (n=37)	FD (n=14)	HC (n=18)	FD+ARFID vs FD	FD+ARFID vs HC	FD vs HC
GI Symptoms						
PAGI-SYM[2]	2.3±0.8	1.9±0.8	0.1±0.2	.2	<.0001	.0001
SF-NDI-Frequency	28.4±11.3	22.8±12.1	1.5±2.5	.1	<.0001	.0001
SF-NDI-Intensity	31.4±11.4	24.6±13.0	1.6±2.3	.2	<.0001	.0001
SF-NDI Bothersome	26.7±10.7	17.4±7.1	0.7±1.4	.04	<.0001	.0008
SF-NDI Quality of Life[2]	35.7±8.5	22.9±6.7	10.6±1.8	.003	<.0001	.005
Fear of Food						
NIAS-Fear	11.0±3.6	5.9±3.8	0.1±0.5	.004	<.0001	.009
FFQ	60.5±20.9	33.7±17.6	3±5.4	.006	<.0001	.006
VSI	48.8±14.7	32.7±16.2	6.9±17.5	.03	<.0001	.009
HADS						
Anxiety	11.2±4.6	6.9±4.1	2.6±2.8	.03	<.0001	.02
Depression	8.1±4.1	5.4±3.5	1.3±1.7	.06	<.0001	.007
BMI	23.3±5.5	26.2±4.9	24.2±3.1	.06	.3	.3
NDT Volume Consumed-mL	272±191	467±332	709±365	.05	<.0001	.07
WMC Transit Times (hrs)				--	--	--
Gastric	55.1±31.9	36.0±12.7	33.5±17.3			
Small Bowel	5.2±1.9	4.1±1.4	5.2±1.4			
Colonic	45.3±30.6	28.0±12.8	25.4±17.1			

[1]Kruskal-Wallis H Tests among groups were all significant except for WMC transit times. Corrected post-hoc p-values displayed in the table. Effect sizes reported in abstract body.
[2]Total scores presented, but results were similar for all subscales.

Table 1: Differences between groups.

Conclusion: Our findings suggest ARFID, when present in FD, does *not* reflect a response to more severe FD symptoms. Thus, ARFID is a legitimate secondary condition in FD. ARFID may reflect more profoundly altered distress around symptoms (e.g., fear of symptoms, anxiety sensitivity) and greater negative impacts (e.g., FD-related quality of life, lower liquid nutrient tolerance).

Disclosure: Nothing to disclose.

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PP0239

EFFECT OF FERMENTABLE OLIGO-, DI-, MONOSACCHARIDES, AND POLYOLS ON FUNCTIONAL DYSPEPSIA SYMPTOMS, GASTROINTESTINAL VOLUMES AND GUT HORMONES: A DOUBLE-BLIND RANDOMIZED PILOT TRIAL

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Introduction: Functional dyspepsia (FD) is one of the most frequent disorders of the gut-brain interaction with a worldwide prevalence of around 7%.[1]

Our group showed that a diet excluding fermentable oligo-, di-, monosaccharides, and polyols (FODMAPs) improved symptoms in the majority of FD patients.[2]

FODMAPs have been studied in irritable bowel syndrome, where poor intestinal absorption may lead to osmotic effects and fermentation, contributing to symptoms.

However, this has not been studied in FD to date. In this study, we assessed pathophysiological parameters after ingestion of mannitol and a control meal in FD patients and healthy volunteers (HV).

Aims & Methods: FD patients (N=11) that responded to the low FODMAP diet and showed symptom triggers by mannitol during reintroduction were recruited. Also, 11 HV participated in this trial. FD and HV had a similar age (47 (24-55) vs. 25 (21-46), p=0.14) and BMI (22 (21-31) vs. 23 (21-25), p=0.85), but FD cohort included more females (100% vs 55%, p=0.04).

After an overnight fast, abdominal magnetic resonance imaging (MRI) was performed in patients and HV. MRI images, gastrointestinal (GI) symptoms and blood samples were collected at baseline and at six separate time points after a challenge meal (0, 15, 30, 45, 60, 120, 240 min). The meal was an oat porridge breakfast containing either mannitol or glucose, as a control, in a randomized, double blind cross-over fashion.

MRI images were analyzed for gastric emptying $T_{1/2}$ (GE $T_{1/2}$), duodenal volume, small bowel water content (SBWC) and colonic gas volume. Blood samples were analyzed for glucose like peptide-1 (GLP-1), peptide YY (PYY), and GIP.

Results: FD patients exhibited more FD symptoms compared to HV after consuming both the glucose-rich meal (54 ± 10 vs 24 ± 7, p=0.02) and the mannitol-rich meal (64±13 vs 23±7, p=0.01). No differences were found for GE $T_{1/2}$ in FD and HV, as indicated in Table 1.

Colonic gas volume was significantly increased in patients compared to HV. Duodenal volumes tended to increase more after intake of mannitol compared to glucose in both groups, but reached significance only in HV. SBWC was increased after mannitol in both HV (p=0.01) and FD (p=0.01).

After intake of mannitol, postprandial GLP-1 levels were significantly lower in FD compared to HV, after 15 (p=0.02) and 30 min (p=0.001). Also, the AUC of GLP-1 was significantly lower in FD versus HV after mannitol (p=0.04), as well as after glucose-rich (p=0.04) meal. In FD, AUC of GIP levels was significantly lower after mannitol versus glucose (p=0.002). GI symptom scores did not correlate with PYY, GLP-1, and GIP levels. GLP-1 and GIP values did not correlate with GE $T_{1/2}$.

Conclusion: In this pilot study no difference in symptom triggering in FD was observed between mannitol and glucose (control) supplemented meals, but compared to HV the meals were associated with higher symptom burden, lower release of GLP-1 and GIP, and larger colonic gas volumes. The mechanisms underlying decreased peptide hormone release and their relevance to symptom generation deserve further studies.

	FD mannitol	FD glucose	HV mannitol	HV glucose
Gastric emptying $T_{1/2}$	46 (36-58)	54 (45-65)	48 (44-58)	53 (48-62)
Δ Colonic gas (mL)	15.45 ±4.18*	-0.35 ±4.38*	7.27 ±7.91	10.53 ±4.12
Δ duodenum volume (AUC mL*min)	1272 ± 520	469 ±271	1167 ±366*	643 ±221*
Δ small bowel water content (AUC mL*min)	154 ±16*	103 ±8*	167 ±16¶	119 ±4¶
AUC GLP-1 (pmol/L *min)	358±35*	344±34¶	504±52*	458±34¶
AUC GIP (pmol/L *min)	81,875±9,631*	109,554±9,362*	189,929±86,981	205,771±65,071
AUC PYY (pmol/L *min)	20,910±2680	19,651±2445	23,501±3656	23,170±3427

GE $T_{1/2}$: Gastric emptying half time, FD: Functional dyspepsia, HV healthy volunteers. * or ¶ indicating pairs of values that are significantly (p<0.05) different after correction for multiple testing. Volumes were compared using a two-way ANOVA. AUC values (4 hours postprandially) were compared using multiple T-tests. GE $T_{1/2}$ was compared using multiple Mann Whitney U test.

Table.

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PP0240

THE CORRELATION BETWEEN STRESS AND AUTONOMIC NERVOUS SYSTEM RESPONSE IN GASTROINTESTINAL PATIENTS

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Introduction: The brain-gut axis plays a crucial role in the regulation of gastrointestinal (GI) function, with the activity of the autonomic nervous system (ANS) being significantly important. Increased stress, correlated with socioeconomic, behavioral, dietary, and environmental factors, may lead to an overactive sympathetic nervous system response.

The aim of this study was to evaluate the correlation between stress and autonomic nervous system responses in gastrointestinal patients.

Aims & Methods: The study comprised 136 GI patients, divided into three groups: the Dyspepsia Group (DG) with 39 patients, a mean age of 37.8 ±11.2 years; the IBS Group (IBS) with 41 patients, a mean age of 45±15 years; and the Celiac Group (CG) with 53 patients, a mean age of 38.2±14 years. ANS activity was assessed using parameters from Heart Rate Vari-

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ability (HRV) analysis, including time domain, frequency domain, and nonlinear analysis. The severity of stress was assessed using stress indices calculated based on hemodynamic parameters and indicators from linear and nonlinear HRV analysis with the HRV Kubios Premium software.
Results: Patients in the IBS Group exhibited the highest stress indicators and sympathetic ANS activity parameters compared to those in the DG and CG groups (SNS index [stress] p=0.038, SNS index AVG p=0.038). Furthermore, the IBS Group had higher stress indices compared to the DG Group (Stress Index 14.0 vs 9.0; p=0.024). Positive correlations were found between the Stress Index, SNS index (stress), and average heart rate (R=0.525; p=0.0001), as well as with sympathetic ANS activity indices. Negative correlations were observed with indicators of parasympathetic nervous system activity across all investigated groups (R=0.433; p=0.001). A high level of stress, indicated by a Stress Index value above 12, was found in 18% of patients with CD, 38% with IBS, and 28% with DG.
Conclusion: Approximately 27% of GI patients experienced significant levels of stress. The stress indicators positively correlated with sympathetic ANS activation and/either impaired function of the parasympathetic system. Assessing the severity of stress will enable targeted therapy to improve the dysfunctions of the brain-gut axis.
Disclosure: Nothing to disclose.

PP0241

THE MULTIDISCIPLINARY GASTROPARESIS PATHWAY: IS IT BEING FOLLOWED AND DOES IT CHANGE PATIENT OUTCOMES?

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Introduction: Idiopathic gastroparesis (IG) is delayed gastric emptying in the absence of mechanical obstruction. (1) Patients with IG are often high service users and are frequently extensively investigated and can frequently become reliant upon long-term enteral feeding support. (2) Data collected from 18 patients with IG at North Bristol NHS Trust between 2018-2019 identified a significant burden of psychiatric comorbidities (69%) and high service use (44 hospital attendances with a cumulative length of stay of 303 days). We therefore designed a holistic, multi-disciplinary team (MDT)-led pathway for patients with IG, which involved more routine consideration of dietary, psychological and pain team assessment. Discussion at the MDT was made mandatory prior to consideration of tube feeding, with a list of prerequisite criteria, including engagement with psychology and psychiatry services. Further data collection was required to assess the effectiveness of the pathway on the number of emergency and elective admissions, the number of investigations and the need for enteral feeding.
Aims & Methods: We performed a retrospective review. Data collected included patient demographics, past medical and psychiatric history, gastrointestinal investigations before and after diagnosis, the number of referrals to the MDT and the number of local emergency and elective admissions for poorly-controlled IG symptoms, as well as the cumulative length of stay during these admissions.
Results: Table 1 shows the comparison of data before and after the introduction of the pathway. Data from 13 patients were included in the second round of data collection: 12 (92%) were female, the mean age at diagnosis was 33 years(compared to baseline data with 17(94%) females, and mean age at diagnosis being 27 years), and 6 (46%) had known psychiatric comorbidities. 100% of patients were reviewed for MDT, with a significant proportion referred to psychology. Comparison of patient outcomes before (2018-2019) and after (2022-2023) intervention demonstrated a significant reduction in patients receiving enteral tube feeding (13/18 [72%] vs 3/13 [15%], p<0.001) and therefore significantly fewer patients underwent endoscopic tube insertion (11/18 [61%] vs 1/13 [8%], p=0.001). There was a reduction in the overall frequency of radiological investigations performed during the study period. The frequency of elective and emergency admissions was significantly reduced (44 [mean attendances per patient 2.4] vs 11 [mean attendances per patient 0.8]) (p<0.05). There was a significant reduction in the cumulative length of stay, 303 (mean per patient 16.8) days vs 13 (mean per patient 1.0) days (p<0.05).

Variable	Baseline data(n=18)	Follow-up data(n=13)
Psychiatric Comorbidity	15 (69)	6 (46)
Dietetic referral/Psychology referral	16 (89)/2 (11)	13(100)/9 (69)
MDT review	0/18 (0)	13/13 (100)
Number of radiological investigations during study period (total n, mean)	27 (1.5)	0 (0)
Number of patients receiving enteral feeding tube insertions(n,%)	13 (72)	3 (15)
Number of patients receiving endoscopic tube placements/ replacements(NG/NJ or PEG)(n,%)	11 (61)	1 (8)
IR tube interventions (total n, mean total cohort/mean amongst tube fed) Before diagnosis After diagnosis During the study period	 23 (1.3/1.8) 67 (3.7/5.2) 23 (1.3/1.8)	 0 (0/0) 10 (0.8/3.3) 6 (0.5/2.0)
Number of admissions (Total n, mean per patient) Elective Emergency	 26 (1.4) 18 (1.0)	 4 (0.3) 7 (0.5)
Cumulative Average Length of stay (days/mean per patient) Elective admission Emergency admission	 40 (2.2) 263 (14.6)	 3 (0.2) 10 (0.7)

Conclusion: Introduction of a holistic, MDT-led pathway improved outcomes in patients with a diagnosis of IG. Early access to the MDT was associated with a reduction in rates of admission, length of stay, radiation exposure and invasive tube-related procedures. Tube feeding can be used in gastroparesis where patients are unable to meet their nutritional needs but is associated with iatrogenic harm and may not necessarily be required.
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Disclosure: Nothing to disclose.

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Poster presentations
H. pylori

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Poster Presentations
H. pylori

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H. pylori
Poster Exhibition

PP0242
LOW PREVALENCE OF HIGH-RISK OLGA STAGES DESPITE HIGH HELICOBACTER PYLORI INFECTION: A 5-YEAR CROSS-SECTIONAL STUDY IN COLOMBIA

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Introduction: Atrophic gastritis (AG) is a well-established precursor of gastric cancer, and its most frequent aetiological agent is Helicobacter pylori (H. pylori). Accurate risk stratification is essential to optimise endoscopic surveillance and resource allocation. The Operative Link on Gastritis Assessment (OLGA) system offers a histological staging tool to quantify the severity of gastric atrophy and associated cancer risk. Despite high H. pylori endemicity in Colombia, local data on the prevalence of advanced OLGA stages remains scarce. This study aimed to elucidate the distribution of OLGA stages and their correlation with age and H. pylori status in a large dyspeptic cohort.

Aims & Methods: We conducted a cross-sectional, observational study of adult patients undergoing upper gastrointestinal endoscopy for dyspepsia between January 2019 and December 2023 at a high-volume reference centre in Bogotá, Colombia. All cases followed the updated Sydney biopsy protocol, with five mapped gastric samples analysed by experienced gastrointestinal pathologists. The OLGA system was applied to stage gastric atrophy (0–IV). H. pylori detection was based on histology using Giemsa and H&E stains. Demographic variables, H. pylori status, OLGA stage, and histological features were recorded. Data were processed using SPSS v26, and statistical significance was set at $p<0.05$.

Results: Out of 6,746 patients (mean age: 59 years; 73.4% female), 99% exhibited low-risk OLGA stages (0–II). Only 3 patients (0.04%) presented with advanced OLGA stages (III–IV). A positive correlation was noted between increasing age and the frequency of gastric atrophy, although this did not correspond to an increase in advanced stages. H. pylori infection was confirmed in 82% of the cohort, showing an upward trend over the study period. Interestingly, no cases of H. pylori infection were observed among patients with OLGA III–IV. The majority of infections were restricted to the antrum. Gastric dysplasia and cancer were rare events, both observed in <1% of patients. The association between H. pylori and gastric cancer did not reach statistical significance (OR 0.49; 95% CI 0.23–1.04; p=0.07).

Conclusion: Despite the high prevalence of H. pylori, the frequency of advanced OLGA stages remains remarkably low in Colombian dyspeptic patients. These findings reinforce the utility of the OLGA system, coupled with Sydney protocol biopsies, as a valuable approach to risk stratification. Only 1% of patients would benefit from endoscopic surveillance under international guidelines, suggesting that widespread follow-up may be unnecessary and potentially wasteful.

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Disclosure: Nothing to disclose.

PP0243
IS THERE AN ASSOCIATION BETWEEN CELIAC DISEASE AND HELICOBACTER PYLORI?

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Introduction: Celiac disease (CD) is an immune-mediated enteropathy triggered by gluten ingestion in genetically predisposed individuals, characterized by small intestinal inflammation and villous atrophy. Microbial agents such as Helicobacter pylori (H. pylori) may influence autoimmune disorders, including CD, through immune modulation.
However, the relationship remains controversial.This study aimed to assess the association between CD and H. pylori infection by comparing the prevalence of H. pylori in patients with and without CD.

Aims & Methods: We conducted a retrospective study from January 2019 to December 2024. Two groups were analyzed: patients with biopsy-confirmed CD and controls without CD who underwent upper endoscopy for dyspeptic or non-specific gastrointestinal symptoms. Exclusion criteria included prior H. pylori eradication, recent use of PPIs, antibiotics, or bismuth compounds, other autoimmune disorders, and incomplete data.

Results: A total of 231 patients were included: 103 with CD (19.4% male, mean age 28.8 ± 12.6 years) and 128 controls (28.1% male, mean age 33.9 ± 11.9 years).
No significant differences were observed in age ($p = 0.73$) or sex distribution ($p = 0.167$) between groups.
H. pylori was detected in 67% of CD patients and 50% of controls, with a significantly higher prevalence in the CD group ($p = 0.014$).

Conclusion: Contrary to some previous reports suggesting a protective effect, this study found a significantly higher prevalence of H. pylori in patients with CD compared to controls.
These findings highlight the need for further investigations into the potential impact of H. pylori on the clinical presentation and management of celiac disease.

Disclosure: Nothing to disclose.

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PP0244

RELATIONSHIP BETWEEN THE PRESENCE OF ANTIBODIES TO CAGA VIRULENCE FACTOR OF *H.PYLORI* AND THE STAGE OF CHRONIC GASTRITIS ACCORDING TO THE OLGA SYSTEM IN MOSCOW POPULATION

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Introduction: *H.pylori* infection is widespread in the world and affects about 41% of the world's population. *H. pylori* causes chronic gastritis. However, the development of atrophic gastritis is associated both with the duration of the disease and, possibly, with the presence of virulence factors of *H.pylori*. CagA is one of the most significant virulence factors of *H.pylori.*

Aims & Methods: The aim of the study is to determine the influence of the presence antibodies to CagA virulence factor of *H. pylori* on the development of atrophic gastritis according to the OLGA system.

Methods: 221 *H. pylori*-positive patients (69 men and 152 women) were examined in A.S. Loginov Moscow Clinical Scientific Center as part of the study on the prevalence of *H. pylori* in Moscow (Grant of the Moscow Center for Innovative Technologies in Healthcare, 0903-1/22). The mean age of the patients was 49.80 ± 13.66 years (from 20 to 73 years). The presence of *H.pylori* infection was confirmed by a ^{13}C-urea breath test. All patients underwent endoscopy with morphological examination by the OLGA system and serological determination of total antibodies to CagA antigen («Vector-Best», Russia, the normal level is < 0.41 conventional unit (c.u.)). All patients were divided into 5 groups based on the OLGA system gastritis stage (from stage 0 to IV stage). The mean age of patients in the groups was 43.33, 46.52, 53.87, 56.76, and 56.86 years respectively.

Results: Results are presented in Table 1. Total antibodies to CagA virulence factor were detected in 57.47% of patients (127 of 221 patients). The proportion of CagA-positive patients was higher in groups with atrophic gastritis and was rising with increasing stage of atrophy (maximum at stage IV - 85.7%) [Kendall's correlation coefficient (K.c.c.) is 0.247 (p<0.01)]. The trend is noted that the mean age of patients increases in groups from 0 to IV stage of atrophy according to OLGA system, the K.c.c. is 0.282 (p<0.01). Groups with II, III and IV stages of atrophy were comparable in age, but proportion of CagA-positive patients reached a maximum at stage IV of atrophy - 85.7%.

Atrophy stage by OLGA system	Number of patients	Mean age, years	Proportion of CagA-positive patients, %	Mean level of CagA antibodies at CagA-positive patients, c.u.
0	52	43.33 ± 12.94	30.77	2.029 ± 0.808
I	62	46.52 ± 13.40	56.45	1.925 ± 0.846
II	71	53.87 ± 13.47	76.05	2.01 ± 0.827
III	29	56.76 ± 9.78	55.17	1.65 ± 0.664
IV	7	56.86 ± 8.27	85.7	1.355 ± 0.549

Table 1: Frequency of detection CagA-positive patients and mean age of patients in groups with different stage of atrophic gastritis according to the OLGA system

Conclusion: We found positive relationship between the presence of antibodies to CagA antigen and the atrophy stage of the OLGA system in *H. pylori* gastritis patients.

Disclosure: This study is part of the study on the prevalence of *H. pylori* in Moscow (Grant of the Moscow Center for Innovative Technologies in Healthcare, 0903-1/22).

PP0245

THE DEMONSTRATED EPIDEMIOLOGICAL LINK BETWEEN HELICOBACTER PYLORI COLONIZATION AND ESOPHAGITIS: A CROSS-SECTIONAL STUDY IN AN IRANIAN POPULATION

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Introduction: Helicobacter pylori (H. pylori) infection and esophagitis are common gastrointestinal disorders, yet their relationship remains controversial. Some studies suggest that H. pylori may have a protective effect against gastroesophageal reflux disease (GERD).
This study aimed to examine the relationship between H. pylori infection and esophagitis in a group of patients undergoing endoscopic assessment.

Aims & Methods: This hospital-based study examined patients who had upper endoscopy (EGD) at Shahid Beheshti Hospital in Qom, Iran, from 2013 to 2018. Inclusion criteria required availability of esophageal biopsies and gastric samples for H. pylori assessment. Demographic data, H. pylori infection status, and esophagitis diagnoses were retrieved from medical records. Different statistical tests, including Chi-square, t-tests, and regression analyses have been used, with p-values below 0.05 considered statistically significant.

Results: The investigation enrolled 244 participants (mean age 58±19.8 years, male predominance 63%). Esophagitis was detected in half the cohort, whereas H. pylori was observed in 23%. Initial analysis revealed a robust association between H. pylori presence and esophageal inflammation (p<0.001), whereas neither age (p=0.245) nor sex (p=0.691) demonstrated significant relationships. In adjusted models, H. pylori-positive individuals experienced tripled esophagitis likelihood (adjusted odds ratio [AOR] 3.1; 95% confidence interval 1.6-5.9; p=0.001).

Variables		Total n (%)	With esophagitis n (%)	Without esophagitis n (%)	p-value
Mean Age ± SD (year)		58.0 ± 19.8	56.5	59.5	0.245
Gender	Female	91 (37.3)	44 (48.4)	47 (51.6)	0.691
	Male	153 (62.7)	78 (51.0)	75 (49.0)	
H. Pylori Positive	Yes	56 (23.0)	40 (71.4)	16 (28.6)	<0.001
	No	188 (77.0)	82 (43.6)	106 (56.4)	

Table 1: distribution of demographic variables in patients.

Conclusion: The data reveal a clinically significant relationship, indicating that H. pylori-positive patients have a threefold increased likelihood of concurrent esophagitis compared to uninfected individuals. This finding challenges the hypothesis that H. pylori exerts a protective role in GERD pathogenesis. Further prospective and mechanistic studies are warranted to elucidate the pathways underlying this relationship.

Disclosure: The authors declare no potential conflicts of interest related to this study.

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PP0246

ORAL UREASE RAPID TEST SERVE AS AN EFFECTIVE COMPLEMENT TO THE DETECTION OF HELICOBACTER PYLORI INFECTION THROUGH UREA BREATH TEST

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Introduction: The urea breath test (UBT) is the clinic's most popular and accurate non-invasive diagnostic method for H. pylori infection. Yet, it is time-consuming and unavailable to some individuals. The oral rapid urease test (RUT) is an emerging non-invasive convenience method but lacks clinical data support.

Aims & Methods: A double-center, double-blind, methodological comparative design experiment is invited to evaluate the consistency of the oral RUT with UBT. 247 participants were enrolled. Their Dental plaque and tongue coating samples were collected, blind and tested with RUT.

Results: The significant association between the RUT and UBT was analyzed.A total of 247 participants were included in the study. The oral RUT showed a high level of agreement with the UBT. The sensitivity of the test reagent was 98.63% (95% CI: 92.64%, 99.76%), and the specificity was 96.34% (95% CI: 92.25%, 98.31%). The overall conformity rate was 97.05% (95% CI: 94.03%, 98.56%). The positive predictive value was 92.31%, and the negative predictive value was 99.37%. The positive likelihood ratio was 26.96, and the negative likelihood ratio was 0.01. The Kappa value was 0.97.

Conclusion: The efficacy of oral RUT for Helicobacter pylori infection diagnosis is extremely consistent with the UBT.

Disclosure: Nothing to disclose.

PP0247

UNCOVERING THE DIAGNOSTIC TRUTH BEHIND STOOL ANTIGEN TESTING IN *HELICOBACTER PYLORI*: PRE- AND POST-TREATMENT ACCURACY AND CLINICAL PREDICTORS FROM A TUNISIAN PROSPECTIVE STUDY

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Introduction: Early *Helicobacter pylori (HP)* infection diagnosis is crucial to prevent associated complications. Therefore, reliable and accessible noninvasive methods are necessary. Among these, the detection of HP antigen in stool (HpSA) has demonstrated effectiveness, though its performance varies across studies.

Aims & Methods: To evaluate the reliability of the HpSA test for diagnosing HP infection before and after eradication treatment, and to identify factors influencing its performance.

A prospective, single-center, quasi-experimental study was conducted in the gastroenterology department of Fattouma Bourguiba University Hospital in Monastir over 13 months (February 2019–February 2020).

109 patients aged 18 to 65 years, with clinical indications for endoscopy, were included. All underwent endoscopy with biopsies for formal histological diagnosis of HP.

For patients who were on proton pump inhibitors (PPIs), treatment was discontinued at least two weeks prior to both the diagnostic EGD and the HpSA testing, in accordance with standard guidelines.

Patients who tested positive for HP histologically were randomized into two groups to receive either concomitant quadruple therapy (CQT) or triple therapy (TT). HpSA tests were performed before and after treatment and compared to gold standard results: pre-treatment biopsies and post-treatment urea breath tests (UBT).

Results: Out of 109 patients, 100 were histologically positive for HP.

Before treatment, HpSA sensitivity was 49%, specificity 88%, positive predictive value (PPV) 98%, and negative predictive value (NPV) 13.5%.

After treatment, HpSA sensitivity was 31.8%, specificity 100%, PPV 100%, and NPV 82.4%. Agreement with gold standards was low pre-treatment (kappa=0.107) and moderate post-treatment (kappa=0.415).

Multivariate analysis: identified diarrhea, prior PPI use, constipation (odds ratios: 9.43, 4.29, and 4.17 respectively), and dyspepsia (OR=0.31) as independent factors affecting test performance. Post-treatment analysis highlighted prior PPI use and performing HpSA more than 52 days after treatment as significant factors (ORs: 4.94 and 0.29 respectively).

Conclusion: While the HpSA test is a useful non-invasive tool, its practical application requires cautious interpretation due to various factors influencing its reliability.

Disclosure: Nothing to disclose.

PP0248

IMPACT OF PERSISTENT H. PYLORI INFECTION ON PEPTIC ULCER BLEEDING AND CARDIOVASCULAR OUTCOMES IN LONG-TERM LOW-DOSE ASPIRIN USERS: A PROSPECTIVE OBSERVATIONAL STUDY

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Introduction: Long-Term Low-Dose aspirin (LDA) use is associated with an increased risk of peptic ulcer bleeding (PUB). Helicobacter pylori (H. pylori) infection is a well-known risk factor for peptic ulcer disease (PUD). This study aimed to evaluate the prevalence of H. pylori infection in long-term LDA users, assess the impact of eradication on PUB risk, and explore its association with cardiovascular outcomes.

Aims & Methods: A prospective observational study was conducted at Banha Teaching Hospital from January 2023 to January 2024. Four hundred adults (≥18 years) on daily LDA for at least one year were enrolled. H. pylori was diagnosed via a stool antigen test, urea breath test, and histopathology. Infected patients received levofloxacin-based triple therapy for 14 days. Eradication was assessed at six weeks, and participants were followed for six months to evaluate PUB incidence and cardiovascular outcomes.

Results: The prevalence of H. pylori infection was 65.5% (262/400). H. pylori-positive patients exhibited significantly lower albumin levels (3.5±0.4 g/dL vs. 3.9±0.6 g/dL, p< 0.001) and higher levels of urea (29.3±6.4 mg/dL vs. 23.5±4.6 mg/dL, p< 0.001), sodium (137.8±2.9 mmol/L vs. 135.3 ±3.5 mmol/L, p< 0.001), potassium (4.1±0.9 mmol/L vs. 3.8±0.6 mmol/L, p =0.014), and cholesterol (191.0±29.7 mg/dL vs. 177.0±35.9 mg/dL, p= 0.004). PUB was significantly more prevalent in H. pylori-positive patients with gastric erosions (12.2% vs. 0.0%, p< 0.001), peptic ulcers (23.7% vs. 0.0%, p< 0.001), and gastric ulcers (10.7% vs. 0.0%, p< 0.001). After eradication therapy, 51.9% (136/262) of patients tested negative for H. pylori, while 48.1% (126/262) remained positive. All cases of PUB occurred in patients with persistent H. pylori infection, highlighting the importance

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of successful eradication in reducing PUB risk. Additionally, H. pylori-positive patients had a higher prevalence of ischemic heart disease (54.3% vs. 33.3%, p< 0.001), suggesting a potential interaction between H. pylori infection and cardiovascular risk.

Conclusion: Persistent H. pylori infection significantly increases peptic ulcer bleeding and cardiovascular risk in long-term low-dose aspirin users. Successful eradication reduces these complications, underscoring the need for routine screening and treatment in high-risk populations.

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Disclosure: no conflict of interest.

PP0249

EVALUATION OF THE EFFICACY AND SAFETY OF DIFFERENT REGIMENS FOR HELICOBACTER PYLORI ERADICATION THERAPY DURING THE COVID-19 PANDEMIC AND POST-PANDEMIC PERIOD IN UKRAINE: A RETROSPECTIVE OBSERVATIONAL STUDY

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Introduction: COVID-19 pandemic has negatively affected treatment of H. pylori and has long-term consequences, in particular, bacterial susceptibility decreased to broad-spectrum antibiotics used in typical H. pylori eradication regimens. Currently, there is limited scientific data on the effectiveness and safety of standard eradication regimens in the post-COVID period.

The aim of the study is to investigate the effectiveness and safety of 10- and 14-day H. pylori eradication regimens used during 2022-2025 and to compare their effectiveness and safety with 10-day regimens used during 2019-2021 in Ukraine, as well as to identify the relationship between unsuccessful treatment cases and anamnestic data.

Aims & Methods: We conducted retrospective multicenter observational study during 2019-2025 years. A comparative analysis of the effectiveness and safety of various eradication regimens for group I (n=82) treated during 2019-2021 years and group II patients (n=74) treated during 2022-2025 years was performed. We determined frequency, intensity, and spectrum of side effects in both groups. A multivariate regression analysis of relationship between unsuccessful treatment cases and anamnestic data was performed.

Results: Comparative analysis revealed no statistical difference in the eradication rates between groups I and II (p≥0.05). The lowest eradication rate among patients in group II was detected on three-component regimens esomeprazole + clarithromycin +amoxicillin (67%) and pantoprazole + clarithromycin/metronidazole/tinidazole (70%). The highest eradication rate was on quadruple regimen esomeprazole + levofloxacin +amoxicillin + furazolidone (100%).

In group II, the frequency of side effects, their intensity, and spectrum significantly increased after 14-day treatment (p<0.001) in comparison with group I.

Regression analysis revealed that male gender, history of COVID-19 infection, and first treatment attempts were significantly more frequently associated with unsuccessful treatment (p<0.05).

Conclusion: Standard 14-day three-component H. pylori eradication regimens esomeprazole + clarithromycin +amoxicillin, pantoprazole + clarithromycin/metronidazole/tinidazole, pantoprazole + amoxicillin + metronidazole/tinidazole have inappropriate eradication rates (≤70%).

The 14-day quadruple esomeprazole + levofloxacin +amoxicillin + furazolidone regimen demonstrates a high eradication rate (>90%), but is accompanied by frequent and diverse side effects, reduced adherence to treatment.

Treatment failure is significantly associated with male gender, previous COVID-19 infection, and first treatment attempts.

Disclosure: The authors declare no conflict of interest.

PP0250

DUAL THERAPIES CONTAINING AN ANTIBIOTIC PLUS A PROTON PUMP INHIBITOR OR VONOPRAZAN FOR *HELICOBACTER PYLORI* INFECTION: A SYSTEMATIC REVIEW

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Introduction: Due to the increasing prevalence of antimicrobial resistance, the efficacy of standard triple therapy for *Helicobacter pylori* (*H. pylori*) infection has declined, with eradication rates now falling below 80% in most countries. Although bismuth quadruple therapy and concomitant therapy are advised in regions with high clarithromycin resistance, these treatments commonly cause frequent adverse events and require the use of two or three antibiotics.

Aims & Methods: Aim: To evaluates the effectiveness of 14-day mono-antibiotic therapies for the first-line treatment of *H. pylori* infection.

Methods: Randomized controlled trials published in English that reported the efficacies of 14-day high-dose amoxicillin/PPI dual, 14-day high-dose amoxicillin/vonoprazan dual, and 14-day tetracycline/vonoprazan dual therapies for the first-line treatment of *H. pylori* infection in adults were included in our literature search. Medical literature searches were conducted using PubMed, Embase, and Cochrane Library electronic databases from October 1, 2014, to October 1, 2024.

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Results: The pooled eradication rates for 14-day high-dose amoxicillin/proton-pump inhibitor (PPI) dual therapies were 86.1% (3335/3875; 95% confidence interval [CI]: 85.1% - 87.2%) by intention-to-treat (ITT) analysis and 87.3% (3232/3702; 95% CI: 86.2% - 88.4%) by per-protocol (PP) analysis. For 14-day high-dose amoxicillin/vonoprazan dual therapies, the rates were 87.4% (1085/1241; 95% CI: 85.5% - 89.2%) by ITT and 93.0% (1044/1124; 95% CI: 91.5% - 94.5%) by PP. In the penicillin-allergic population, 14-day tetracycline/vonoprazan dual therapy showed eradication rates of 92.0% (138/150) by ITT and 95.1% (135/142) by PP.
Conclusion: 14-day tetracycline/vonoprazan dual therapy presents an effective option for eradicating *H. pylori* in patients allergic to penicillin. For those without a penicillin allergy, first-line treatments can include 14-day mono-antibiotic regimens such as high-dose amoxicillin/PPI dual, high-dose amoxicillin/vonoprazan dual, and tetracycline/vonoprazan dual therapies.
Disclosure: All the authors disclose no conflict of interest.

PP0251

THE EFFECT OF SUPPLEMENTING WITH SACCHAROMYCES BOULARDII ON VONOPRAZAN-AMOXICILLIN DUAL THERAPY FOR NAIVE HELICOBACTER PYLORI INFECTION: A RANDOMIZED CONTROLLED TRIAL

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Introduction: It remains uncertain if the addition of Saccharomyces boulardii (S. boulardii) to Vonoprazan-amoxicillin dual therapy as a first-line treatment can enhance the Helicobacter pylori (H. pylori) eradication rate and decrease the incidence of adverse events. Therefore, we conducted the randomized controlled trials (RCTs) to address this issue.
Aims & Methods: This is a single-center, prospective randomized controlled study. A total of 150 eligible patients, recruited from December 2023 to November 2024, were randomly assigned to the VA (vonoprazan 20 mg twice daily, amoxicillin 750 mg 3 times daily for 10 days) and VAS groups (vonoprazan 20 mg twice daily, amoxicillin 750 mg 3 times daily, and S. boulardii 250 mg twice daily for 10 days). The eradication rates, adverse events, safety, and compliance were compared between the two groups. Factors associated with eradication rate were explored.
Results: The eradication rates of VA and VAS were 89.3% and 93.3%, respectively, by intention-to-treat analysis (P = 0.384); 93.1%, and 94.6%, respectively, by per-protocol analysis (P=0.966). The incidence of adverse events significantly lower in the VAS group (5.4%) than in the VA group (22.2%; P=0.003).
There was no significant difference in compliance between the VA (96%) and VAS groups (98.7%, P = 0.612). Body surface area (BSA) was found to be a influencing factor for eradication rate in both VA and VAS groups. Serum chemistry tests revealed no significant differences in liver and kidney function before and after treatment in both groups.
Conclusion: Although the addition of S. boulardii to VA dual therapy could not increase the eradication rate, it could decrease the incidence of adverse events. Low-BSA patients are more suitable candidates for this regimen.
Disclosure: Nothing to disclose.

PP0252

VONOPRAZAN IS HIGHLY EFFECTIVE AND SAFE AS AN ADJUVANT IN DIFFERENT REGIMENS IN FIRST- AND RESCUE-LINE THERAPIES FOR *H. PYLORI* INFECTION IN BRAZIL

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Introduction: Data about the effectiveness and safety of vonoprazan (VPZ) as an adjuvant for treating *H. pylori* are lacking in Brazil. Previous studies suggest that greater acid suppression might improve effectiveness, with an acceptable eradication rate around 90%.

Aims & Methods: To evaluate VPZ in eradication therapy in Brazil, as part of the Hp-BrazilReg, WorldHpReg's partner, a multicenter and prospective real-life registry was performed, assessing outcomes of *H. pylori* management by Brazilian gastroenterologists. Data were registered at e-CRF AEG-REDCap from March 2022 to October 2024. The effectiveness was assessed by modified intention-to-treat (mITT) analysis. Data were subject to quality review.

Results: 2,132 Brazilian patients, with a mean age of 52 years, 61% of whom were women, were included in the mITT analysis. The main treatment indications were: 63% dyspepsia, 10% gastroduodenal ulcers, and 5% premalignant gastric lesions. Endoscopy was performed in 95% of the cases, using histology (90%) and/or the rapid urease test (15%) for diagnosing the infection. No pre-treatment bacterial resistance test was performed.

First-line treatments were administered to 73% (n=1,560) of cases, second-line to 18% (n=386), and rescue to 9% (n=186), with 94% of patients receiving 14-day prescriptions. Low-dose (between 4.5 and 27mg of omeprazole equivalent BID), standard-dose (between 32 and 40mg of omeprazole equivalent BID), and high-dose (between 54 and 128mg of omeprazole BID) of proton pump inhibitor (PPIs) as well as 20mg BID of vonoprazan (VPZ) were used in 39%, 38%, 23%, and 15% of treatment cases, respectively.

Probiotics were used as adjuvant in 15% of patients. Compliance (>90% drug intake) was reported in 99% of the patients.

Eradication was mostly confirmed by endoscopy in 88% (histology [74%] and/or rapid urease test [14%]). The ^{14}C-urea breath test was used in 8% of the cases. At least one adverse event was reported by 25% of patients, mainly nausea (13%), dysgeusia (8%) and diarrhea (7%).

The regimens reaching 90% eradication were: dual-VPZ+amoxicillin for first-line and rescue treatments (96% and 100%, respectively), quadruple VPZ+bismuth+tetracycline+metronidazole for second- and third-line treatments (92% and 90% respectively), and quadruple PPI+bismuth+levofloxacin+amoxicillin (100%) for second-line treatment.

Prescribing VPZ (vs PPI) was an independent factor associated with higher effectiveness both in first-line (OR 2.95; IC 95%: 1.71-5.09) and in second-line and rescue treatments (OR 2.87; IC 95%: 1.69-5.13). There was no difference in adverse effects between VPZ and PPI-based regimens (p<0.01).

Conclusion: VPZ, when prescribed as an adjuvant for treating *H. pylori* infection, shows highly effective results as part of dual therapy with amoxicillin for first-line and rescue treatments, as well as in quadruple therapy with bismuth+metronidazole+ tetracycline or bismuth+amoxycillin and rifabutin for rescue treatments.

Disclosure: Nothing to disclose.

PP0253

UNMASKING PERSISTENT SYMPTOMS AFTER *HELICOBACTER PYLORI* ERADICATION: A PROSPECTIVE ANALYSIS OF CLINICAL PREDICTORS IN REAL-WORLD PRACTICE

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Introduction: *Helicobacter pylori* (H. pylori) infection is often asymptomatic, but it can lead to significant gastric pathologies, including chronic gastritis and peptic ulcer disease. Despite effective antimicrobial eradication therapies, a subset of patients continues to experience upper gastrointestinal symptoms. This raises questions regarding the underlying factors contributing to persistent symptomatology following successful treatment.

Aims & Methods: We conducted a prospective, single-center study in the Department of Gastroenterology at Fattouma Bourguiba University Hospital in Monastir, Tunisia.

Patients with histologically confirmed H. pylori infection were randomized to receive either concomitant quadruple therapy (CQT) or standard triple therapy (TT).

All patients underwent esophagogastroduodenoscopy (EGD) with gastric biopsies at baseline for diagnostic confirmation. A urea breath test (UBT) was performed to assess eradication status, followed by a clinical follow-up visit to evaluate symptom resolution.

For patients using proton pump inhibitors (PPIs), therapy was discontinued at least two weeks before the diagnostic EGD and HpSA testing to avoid diagnostic interference.

Results: A total of 109 patients with clinical indications for upper endoscopy were enrolled. H. pylori infection was confirmed histologically in 100 patients, resulting in a prevalence rate of 91.74% in the studied population. The mean age was 41.28 years (SD ±13), and the female-to-male ratio was 3.18.

The most common indication for EGD was chronic epigastric pain (64 patients, 69.6%), followed by iron-deficiency anemia (9.8%) and refractory gastroesophageal reflux disease (7.6%).

Therapeutic adherence, defined as ingestion of ≥90% of prescribed medications, was achieved in 92 patients (92%). Among them, eradication failure was observed in 22 cases (23.9%).

Univariate analysis identified several factors significantly associated with persistent post-treatment symptoms ($p < 0.05$), including:

- Overweight status ($p = 0.014$)
- Presence of epigastric pain at baseline ($p = 0.028$)
- Constipation at baseline ($p = 0.05$)
- Eradication failure ($p = 0.05$)

No significant association was found with diabetes, smoking, initial symptom severity, bacterial load, gastritis severity, or mucosal abnormalities such as ulcers or erosions.

Multivariate logistic regression confirmed the following independent predictors of persistent symptoms:

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• Epigastric pain (p = 0.009; OR = 8.3; 95% CI: 1.7–40.2)
• Overweight status (p = 0.012; OR = 3.6; 95% CI: 1.3–9.7)
• Constipation (p = 0.02; OR = 4.0; 95% CI: 1.2–13.3)
• Eradication failure (p = 0.04; OR = 3.2; 95% CI: 1.0–10.1)

Conclusion: This study highlights the necessity of a multidimensional approach in the post-treatment follow-up of *H. pylori*-infected patients. Persistent symptoms may not solely be explained by endoscopic or histological findings, but are also influenced by clinical factors such as baseline symptom profiles, functional bowel disorders, and eradication outcomes. Identifying these predictors is essential for optimizing management strategies and improving patient quality of life.

Disclosure: Nothing to disclose.

PP0254

TRENDS IN THE PRESCRIPTION OF ERADICATION TREATMENTS AND THEIR EFFECTIVENESS IN NAÏVE PATIENTS OVER 12 YEARS (2013-2024) IN EUROPE: DATA FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: The infection by *H. pylori* can be successfully treated and eradicated if the recommendations from consensus guidelines are correctly followed. It is essential to continuously assess the applicability of these recommendations to ensure their alignment with clinical practice.

Aims & Methods: Multicentre, prospective registry evaluating the decisions and outcomes of *Helicobacter pylori* management by European gastroenterologists (Hp-EuReg, Hp-WorldReg's partner). Data were registered at AEG-REDCap e-CRF until December 2024. Modified intention-to-treat (mITT) and time trend analyses were performed.

Results: Overall, 64,084 (84%) first-line empirical prescriptions were analysed. The total number of different therapeutic regimens prescribed exceeded 100. Triple therapy prescriptions decreased from 58% during 2013/15 to 34% in 2022/24; likewise, non-bismuth concomitant therapy use decreased from 23% in 2013/15 to 14% in 2022/24; while three-in-one single-capsule increased from 0.3% in 2013/2015 to 20% in 2022/24. An increase in the average duration of treatments, from 9.9 to 12.7 days in 2013-2024 was identified, as well as in the use of standard/high-dose of PPIs —defined as a 14-day course with standard or high-dose PPIs at >40mg omeprazole equivalent twice daily —, increasing from 36% to 61% in 2013-2024. Over 12 years of evolution, an overall improvement of ≈10% in first-line effectiveness was observed, increasing from 86% to 93%, both globally and across geographic regions (Table).

Conclusion: European gastroenterological practice is continuously adapting to the latest published evidence and recommendations—reducing the use of triple therapies while increasing both treatment duration and PPI dosage—leading to a progressive improvement in overall effectiveness. These changes have remained consistent in recent years.

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Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Treatment												
Quadruple-C+A+B	1.9%	2.6%	6.5%	19%	12%	25%	13%	14%	16%	14%	17%	17%
Single-capsule*	0.2%	0.2%	0.5%	14%	22%	17%	21%	19%	19%	21%	19%	20%
Quadruple-M+Tc+B	2%	1.8%	0.5%	0.2%	0.3%	0.6%	1.3%	1.3%	1.6%	3.7%	2.4%	2.8%
Concomitant-C+A+M/T	21%	21%	27%	21%	19%	8.5%	11%	13%	14%	13%	15%	14%
Sequential-C+A+M/T	16%	9.7%	7.4%	1.9%	7.8%	7.2%	5.8%	3.6%	3.6%	3.6%	2.4%	2.1%
Triple-A+L	2.2%	2.1%	3%	1.7%	0.6%	0.6%	1.1%	1.2%	1.2%	2.3%	3%	2%
Triple-A+M	3.4%	2.9%	1.5%	1.5%	1.3%	0.4%	1.6%	1%	1.3%	1.5%	1.6%	1.3%
Triple-C+M	3.2%	6.1%	8.1%	5.7%	1.3%	0.9%	1.2%	6.4%	4.1%	3.9%	2.7%	1.7%
Triple-C+A	46%	50%	40%	29%	30%	30%	36%	33%	33%	28%	25%	21%
Therapy length												
7 days	27%	28%	23%	17%	7.3%	2.1%	2.8%	3.4%	2.9%	8.8%	3%	2.1%
10 days	57%	55%	59%	48%	52%	47%	41%	39%	43%	43%	34%	31%
14 days	17%	18%	18%	36%	40%	51%	51%	58%	54%	48%	63%	67%
PPI doses												
Low (4.5 to 27 mg omeprazole equivalents b.i.d.)	64%	55%	44%	38%	44%	31%	35%	46%	43%	37%	42%	39%
Standard (32 to 40 mg omeprazole equivalents b.i.d.)	16%	24%	24%	23%	23%	37%	29%	24%	26%	35%	31%	28%
High (54 to 128 mg omeprazole equivalents b.i.d.)	20%	22%	31%	39%	33%	32%	37%	30%	31%	29%	27%	33%
Overall effectiveness												
Eradication rate (mITT)	86%	85%	86%	88%	88%	91%	89%	89%	91%	92%	92%	93%
Effectiveness depending on geographical region												
East	92%	79%	84%	83%	82%	91%	90%	93%	93%	91%	93%	91%
East-Centre	87%	85%	86%	85%	87%	89%	89%	93%	91%	92%	94%	95%
South-West	84%	86%	86%	90%	91%	91%	88%	85%	90%	93%	92%	95%
West-Centre	89%	92%	93%	94%	89%	92%	91%	89%	89%	94%	94%	94%
North	85%	82%	83%	88%	83%	77%	84%	83%	81%	83%	77%	79%

PPI: proton pump inhibitor; mITT: modified intention-to-treat; A – amoxicillin; C – clarithromycin; M – metronidazole; T – tinidazole; L – levofloxacin B; – bismuth salts; Tc – tetracycline; East – Ukraine, Serbia, Bulgaria, Turkey, Russia, Romania, Albania, North Macedonia, Bosnia and Herzegovina, Kosovo, Moldova, Montenegro; East-Centre – Croatia, Poland, Hungary, Latvia, Lithuania, Greece, Slovenia, Czech Rep, Azerbaijan, Slovakia, Malta, Armenia; South-West – Portugal, Spain; West-Centre – France, Austria, Belgium, Germany, Italy; North – The United Kingdom, Finland, The Netherlands, Ireland, Israel, Norway, Switzerland, Sweden, Denmark; *Three-in-one single-capsule containing metronidazole. tetracycline and bismuth.

Table. Prescriptions and effectiveness trends of first-line empirical treatments in Europe in the period 2013-2024.

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PP0255

EFFECTIVENESS AND SAFETY OF SINGLE-CAPSULE BISMUTH QUADRUPLE THERAPY IN 12,500 PATIENTS FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: The use of the three-in-one single-capsule formulation of bismuth quadruple therapy (PPI, bismuth, tetracycline, and metronidazole), introduced several years ago, has become increasingly widespread in Europe. However, the experience with this regimen should be analysed in depth.

Aims & Methods: To evaluate the effectiveness and safety of the three-in-one single-capsule in the European Registry on *Helicobacter pylori* Management (Hp-EuReg).

International, multicenter, prospective, non-interventional registry aimed to evaluate the decisions and outcomes of *H. pylori* management by European gastroenterologists (Hp-EuReg, Hp-WorldReg's partner). All infected adult patients treated with 10-day single-capsule according to data sheet (3 capsules/6 h) or alternative three times a day (4 capsules/8 h) prescriptions were systematically registered at AEG-REDCap e-CRF until December 2024. Variables included: Patient's demographics, previous eradication attempts, prescribed treatment, adverse events, compliance and effectiveness. Modified intention-to-treat (mITT) and per-protocol (PP) analyses were performed, and data were subject to quality review.

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Results: Of the 76,042 empirical patients included in the Hp-EuReg, 12,519 (16%) received single-capsule bismuth quadruple therapy. The majority of these patients were naïve (80%), with an average age of 51 years, 63% female and 11% with peptic ulcer. In both the mITT and PP analyses, the single-capsule regimen achieved an eradication rate of 92%. In first-line treatment, effectiveness (mITT) reached 93%, decreasing to 87% in second-line and to 83% in subsequent lines (3rd to 6th) (Table 1).

In rescue therapy, even when the regimen was optimised with standard or high-dose PPIs, optimal (≥90%) effectiveness was never achieved. Compliance with treatment (>90% of drug intake) was reported in 97% of cases, and was the factor most closely associated with the effectiveness of treatment (OR: 7.7, 95% CI: 5.8-10.4). Adverse events (21%) were generally mild and transient; with 0.1% of patients reporting serious adverse events, leading to the discontinuation of treatment in 1.7% of patients.

	Use, N (%)	mITT, N (%)	95% CI	PP, N (%)	95% CI	Compliance, N (%)	95% CI	AEs, N (%)	95% CI
Overall	12,519 (16*)	11,833 (92)	(91-92)	11,627 (92)	(92-93)	12,198 (97%)	(97-97)	12,213 (21%)	(20-22)
1st line (naïve)	9,958 (80)	9,449 (93)	(93-94)	9,299 (94)	(93-94)	9,707 (97%)	(97-98)	9,719 (20%)	(19-21)
2nd line	1,850 (15)	1,721 (87)	(85-89)	1,684 (88)	(86-89)	1,799 (96%)	(95-97)	1,803 (22%)	(20-24)
3rd to 6th line	710 (5.7)	662 (83)	(80-86)	643 (84)	(81-87)	692 (95%)	(93-96)	691 (29%)	(25-32)

*Of the total of treatments included in the Hp-EuReg up to January 2025 (i.e., N= 77,954); mITT: modified intention-to-treat; PP: per-protocol; N: total number of patients analysed; CI: confidence interval; AEs: adverse events.

Table 1. Three-in-one single-capsule effectiveness, compliance and safety in first-line and rescue treatment lines.

Conclusion: The 10-day treatment with single-capsule bismuth quadruple therapy achieves *H. pylori* eradication in approximately 90% of patients by mITT in real-world clinical practice, both as a first-line and rescue treatment, with a favourable safety profile.

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PP0256

MINOCYCLINE- CONTAINING BISMUTH QUADRUPLE THERAPY FOR HELICOBACTER PYLORI TREATMENT: A MULTI-CENTER REAL-WORLD OBSERVATIONAL STUDY

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Introduction: Minocycline is considered the most potent semi-synthetic tetracycline, demonstrates promising pharmacokinetic and antimicrobial properties against Helicobacter pylori (H. pylori). While previous studies suggested the efficacy of minocycline-containing bismuth quadruple therapy for H. pylori eradication, real-world evidence remains limited.

Aims & Methods: his study aimed to evaluate the clinical efficacy, safety, and cost-effectiveness of minocycline-containing bismuth quadruple therapy in routine clinical practice.This multicenter, real-world observational study was conducted at 25 hospitals in Fujian China between January and March 2025. Patients diagnosed with H. pylori infection received one of three 14-day bismuth-containing quadruple regimens with b.i.d. esomeprazole 20 mg and bismuth 220 mg, plus amoxicillin 1000 mg b.i.d. and clarithromycin 250 mg b.i.d. (PBAC), or minocycline 100 mg b.i.d. and clarithromycin 250 mg b.i.d. (PBMC), or minocycline 100 mg twice b.i.d. and levofloxacin 500mg q.d. (PBML).

The primary outcome was the H. pylori eradication rate, secondary outcomes included the frequency of adverse events, patient compliance, and cost-effectiveness.

Results: A total of 347 patients were enrolled, with 110 patients completing treatment in each group. In the intention-to-treat (ITT) analysis, eradication rates were 77.4% (PBAC), 85.5% (PBMC), and 84.3% (PBML). In the per-protocol (PP) analysis, eradication rates were 80.9%, 90.9%, and 88.2%, respectively. Although the differences among the three regimens were not statistically significant, minocycline-based regimens achieved numerically higher eradication rates.

Adverse event incidence was 32.73%, 20.91%, and 23.64% in the three groups, respectively. Compliance rates exceeded 94% across all groups. Cost-effectiveness analysis showed PBML had the lowest cost-effectiveness ratio (5.38 CNY per 1% eradication), followed by PBAC (5.41 CNY) and PBMC (6.06 CNY).

Analysis	PBAC group	PBMC group	PBML group	χ2	P
ITT	77.4%(89/115)	85.5%(100/117)	84.3%(97/115)	3.053	0.217
95% CI	69.7%-85.0%	79.1%-91.9%	77.7%-91.0%		
PP	80.9%(89/110)	90.9%(100/110)	88.2%(97/110)	5.087	0.0786
95% CI	73.6%-88.2%	85.5%-96.3%	82.1%-94.25%		

Conclusion: H. pylori eradication, with favorable patient compliance and pharmacoeconomic profiles. These regimens present a viable alternative to traditional amoxicillin-or tetracycline-based therapies. Further large-scale randomized controlled trials are warranted to confirm these findings.

Disclosure: Nothing to disclose.

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PP0257

EFFECTIVENESS OF FIRST-LINE EMPIRICAL *HELICOBACTER PYLORI* TREATMENT OUTSIDE EUROPE: RESULTS OF 10,000 CASES FROM THE WORLD-WIDE REGISTRY ON *H. PYLORI* MANAGEMENT (WORLDHPREG)

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Introduction: *H. pylori* infection represents a significant global health challenge. Despite international efforts to standardise eradication protocols, treatment success varies substantially across regions due to differences in antibiotic resistance patterns and healthcare practices and accessibility.

Aims & Methods: This study aimed to assess the effectiveness of empirical first-line therapies for *H. pylori* infection outside Europe.
The WorldHpReg, an international, multicenter, prospective registry, was created to evaluate the management and clinical outcomes of *H. pylori* infection by gastroenterologists world-wide. Data were collected via the AEG-REDCap e-CRF from March 2022 to April 2025. Modified intention-to-treat (mITT) and per-protocol (PP) analyses were performed to assess therapy effectiveness. A regimen was considered optimally effective if the mITT cure rate was ≥90%. The data were stratified by country and by specific first-line therapeutic regimen.

Results: A total of 9,973 cases were collected, with 7,738 (78%) first-line empirical prescriptions analysed across several non-European countries including: Latin America (LATAM), Brazil, Africa, Australia, Canada, Egypt, India, Jordan, Pakistan, and Saudi Arabia. Overall, 90 different first-line

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therapeutic regimens were prescribed, achieving an overall mITT and PP effectiveness of 84% and 85%, respectively. Optimal (≥90%) mITT cure rates were achieved, overall, in Australia (95%), India (99%), Jordan (96%) and Saudi Arabia (91%). Optimal treatment effectiveness was observed in LATAM with following therapies: non-bismuth quadruple therapy containing amoxicillin-clarithromycin-metronidazole (91%) or with various combinations of bismuth-containing quadruple therapies, such as amoxicillin-metronidazole (94%), metronidazole-doxycycline (90%), and amoxicillin-doxycycline (97%). Additionally, high effectiveness was achieved also in LATAM with the single-capsule formulation (94%) and with the combination of metronidazole-tetracycline in the classical format (91%), this latter also obtaining optimal results in Brazil (97%). Standard triple therapy with amoxicillin-clarithromycin achieved over 90% mITT cure rates in Africa (92%), Australia (95%), Egypt (100%), India (100%), and Jordan (100%), although the number of patients was still very low. The remaining regimens provided suboptimal results. Complete results by treatment scheme and country are detailed in Table 1.

Country	LATAM	Brazil	Africa	Australia	Canada	Egypt	India	Jordan	Pakistan	Saudi Arabia
Treatment[a]	N (% mITT)									
Triple-C+A	1,041 (75%)	1,826 (80%)	25 (92%)	19 (95%)	NA	4 (100%)	61 (100%)	14 (100%)	53 (59%)	7 (86%)
Concomitant-C+A+M	613 (91%)	1 (100%)	NA	NA	6 (83%)	NA	NA	29 (86%)	NA	NA
Dual-A	340 (86%)	116 (89%)	NA	NA	NA	NA	NA	NA	NA	NA
Quadruple-A+M+B	259 (94%)	NA	NA	NA	NA	NA	NA	NA	NA	1 (100%)
Quadruple-M+D+B	267 (90%)	2 (100%)	1 (100%)	NA	NA	45 (100%)	NA	NA	12 (42%)	11 (100%)
Triple-A+L	126 (82%)	16 (69%)	NA	NA	1 (100%)	NA	NA	NA	NA	2 (100%)
Quadruple-M+Tc+B	177 (90%)	30 (97%)	NA	NA	NA	NA	NA	NA	NA	NA
Quadruple-C+A+B	187 (78%)	6 (100%)	43 (67%)	NA	NA	NA	NA	1 (100%)	NA	NA
Quadruple-A+D+B	177 (97%)	NA	3 (100%)	19 (95%)	3 (67%)	27 (100%)	80 (99%)	NA	93 (74%)	40 (88%)
Single-capsule*	104 (97%)	NA	72 (78%)		10 (80%)	76 (100%)	141 (99%)	50 (100%)	158 (67%)	NA
Triple-C+T	NA	85 (79%)						94 (96%)		3 (100%)
Other	483 (89%)	2,082 (81%)								64 (91%)
Overall	3,774 (85%)									

[a]Treatments were categorised in 11 categories encompassing over 90% of first-line prescriptions world-wide. LATAM: Latin America; ASEAN: Association of South-East Asian Nations; A: amoxicillin; B: bismuth salts; C: clarithromycin; D: doxycycline; L: levofloxacin; M: metronidazole; T: tinidazole; Tc: tetracycline; *Three-in-one single-capsule containing metronidazole, tetracycline and bismuth; N: total number of patients analysed; %: total number of patients with H. pylori eradicated; mITT: modified intention-to-treat.

Table 1. Modified intention-to-treat effectiveness of most frequent first-line empirical treatments world-wide.

Conclusion: In this large, real-world cohort from several non-European countries, first-line *H. pylori* empirical eradication therapies yielded unsatisfactory overall results. However, optimal (≥90%) eradication rates were consistently observed in Australia, India, Jordan, Egypt and Saudi Arabia. In LATAM and Brazil, the highest cure rates were predominantly obtained with bismuth and non-bismuth quadruple therapies. Triple therapy with amoxicillin-clarithromycin remained highly effective in selected regions. These findings reinforce the importance of tailoring first-line empirical therapy to regional performance in order to optimise *H. pylori* eradication success.

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PP0258

EVALUATION OF A QUINTUPLE, BISMUTH-ENHANCED 10-DAY CONCOMITANT, THERAPY AS A FIRST-LINE REGIMEN FOR *HELICOBACTER PYLORI* ERADICATION: A PROSPECTIVE MULTICENTER PILOT STUDY.

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Introduction: A 10-14 day non-bismuth quadruple "concomitant" therapy is currently the preferred first-line regimen for Helicobacter pylori (HP) eradication in Greece [1,2]. In this country, bismuth salts have recently become commercially available; however, lack of tetracycline still precludes use of bismuth quadruple therapy. Adding bismuth to the current concomitant therapy (ie; a quintuple regimen) could be an alternative; however, it has never been explored previously.

Aims & Methods: The present study aimed to assess a quintuple 10-day bismuth-enhanced concomitant regimen as first-line treatment for HP eradication in Greece. Prospective, open-label, multicenter pilot study including treatment-naïve patients diagnosed with HP infection via histology and/or rapid urease test. Exclusion criteria included recent use of PPIs, antibiotics, NSAIDs, bismuth, serious comorbidities, prior gastrectomy, pregnancy or lactation, allergy to any study drug, and refusal to participate. All patients received a 10-day regimen consisting of esomeprazole 40 mg BID, amoxicillin 1000 mg BID, clarithromycin 500 mg BID, metronidazole 500 mg BID, and bismuth subcitrate 240 mg BID.

A specific probiotic supplement containing Lactobacillus acidophilus, Lactiplantibacillus plantarum, Bifidobacterium lactis, and Saccharomyces boulardii was also administered in all patients. Efficacy was evaluated by intention-to-treat (ITT), per-protocol (PP), and modified ITT (mITT) analyses. Successful eradication was defined as a negative histology or urea breath test at least 8 weeks post-treatment.

Results: A total of 52 patients were analyzed (mean age 48.6 ± 18.7 years, 50% male, peptic ulcer disease: 7/52, non-ulcer dyspepsia: 29/52).

Eradication rates were:

i) ITT: 47/52 (90.4%, 95% CI: 79.4–95.8),

ii) PP: 46/49 (93.9%, 95% CI: 83.5–97.9),

iii) mITT: 47/51 (92.2%, 95% CI: 81.5–96.9).

Adverse events were reported by 29.4% (15/51) of patients, with taste disturbances being the most frequent (39.1%); no serious adverse events occurred. Compliance was excellent with 49/51 (96.1%) patients taking >90% of the prescribed therapy.

Conclusion: A quintuple 10-day bismuth-enhanced concomitant regimen appears to be an effective (93.9% PP), safe and well tolerated treatment option for first-line HP eradication in a country with high clarithromycin and metronidazole resistance, without negatively affecting patient compliance.

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Disclosure: Nothing to disclose.

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PP0259

HIGH CUMULATIVE HELICOBACTER PYLORI ERADICATION THERAPY IN OBESE PATIENTS UNDERGOING GASTRIC BY-PASS SURGERY

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Introduction: The Maastricht V/ Florence Consensus has updated guidelines for the management of *Helicobacter Pylori* (HP) infection with special emphasis in high clarithromycin resistance settings as it is the case in our country. In line with the recommendations of the European Guidelines on surgery of severe obesity [1] and the recommendations of the American Society of Gastrointestinal Endoscopy on endoscopy of the bariatric patient [2) a policy of systematic HP screening and overtime updated eradication was implemented in obese patients undergoing Roux-en Y gastric bypass (RYGB) surgery.

Aims & Methods: Five year period (2020-2024) single center study. The aim was, in obese patients undergoing RYGB surgery, to assess the cumulative HP eradication rate by adopting, at first line, a 10 days bismuth quadruple therapy (BQT) - proton pump inhibitor (PPI) bid and three-in-one single capsule containing bismuth, metronidazole, and tetracycline qid - and, at second line, a 14 days levofloxacin based therapy (LBT) – PPI bid, amoxicillin 1000 mg bid and levofloxacin 500mg bid. HP infection status was assessed, at baseline, by histology or urea breath test and post treatment, by urea breath test 4-6 weeks after the end of the therapy.

Results: One thousand four hundred and forty-nine consecutive HP-positive patients completed BQT treatment: 1160 (80.1%) female, age 43.5 (±10.5) years, median age of 44 years), age range of 18-69 years. HP was eradicated in 1220 patients – 84.2% (95%CI: 82.2%-86.0%). In the remaining 229 patients, 87 patients withdrew and HP was eradicated in 118 patients out of 145 with LBT. The intention to treat (ITT) and per protocol (PP) cumulative eradication rates were, respectively, 92.3% (95%CI: 90.9%-93.6%). and 98.0% (95%CI: 97.1%-98.6%). Eradication rates were not significantly different by gender and age.

Conclusion: By adopting the Maastricht V/ Florence Consensus BQT first line therapy and second line LBT therapy, high ITT and PP cumulative HP eradication rates are achieved leaving only a residual number of obese patients undergoing RYGB in need of the culture and susceptibility testing prior to third line treatment.

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Disclosure: No conflict of interest.

PP0260

EFFICACY OF TRIPLE VS. SEQUENTIAL THERAPY IN *HELICOBACTER PYLORI* ERADICATION: ADDRESSING ANTIBIOTIC RESISTANCE IN A RESOURCE-LIMITED SETTING

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Introduction: Helicobacter pylori is a common cause of peptic ulcers and gastritis. Standard triple therapy includes a PPI with amoxicillin and either clarithromycin or levofloxacin. Rising antibiotic resistance has reduced its effectiveness. Sequential therapy offers a phased alternative with better outcomes in resistant cases.

Aims & Methods: To evaluate and compare the efficacy of clarithromycin and levofloxacin-based triple and sequential therapies for *Helicobacter pylori* (H. pylori) eradication in a resource-limited setting, in the context of emerging antibiotic resistance.

A prospective, randomized study was conducted at a tertiary care center from January 1, 2022, to January 1, 2025. A total of 300 treatment-naïve patients aged 18–60 years, diagnosed with *H. pylori* infection through stool antigen, urea breath test, or histology, were included. Patients with prior eradication therapy or recent antibiotic use were excluded. Participants were randomized into four equal groups (n=75 each):

- Clarithromycin-based triple therapy (amoxicillin, clarithromycin, PPI)
- Levofloxacin-based triple therapy (amoxicillin, levofloxacin, PPI)
- Clarithromycin-based sequential therapy (amoxicillin + PPI for 5 days, followed by clarithromycin + tinidazole + PPI for 5 days)
- Levofloxacin-based sequential therapy (amoxicillin + PPI for 5 days, followed by levofloxacin + tinidazole + PPI for 5 days) Therapeutic response was assessed post-treatment for eradication.

Results: Out of 300 patients (180 males, 120 females), eradication was achieved in:

- 48/75 (64%) in clarithromycin-based triple therapy
- 53/75 (70.7%) in levofloxacin-based triple therapy
- 51/75 (68%) in clarithromycin-based sequential therapy
- 61/75 (81.3%) in levofloxacin-based sequential therapy

All patients completed therapy without notable adverse events or compliance issues. Lower eradication rates in clarithromycin-based regimens suggest rising resistance.

Conclusion: Levofloxacin-based sequential therapy demonstrated the highest eradication rate, highlighting its potential as a preferred regimen in settings with rising clarithromycin resistance. Clarithromycin-based treatments showed suboptimal outcomes, emphasizing the need for region-specific antibiotic stewardship and resistance surveillance.

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PP0261

BISMUTH-BASED VERSUS CONCOMITANT QUADRUPLE THERAPY: PREDICTIVE FACTORS OF HELICOBACTER PYLORI ERADICATION FAILURE

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Introduction: Nowadays, bismuth-based quadruple therapy (BQT) and concomitant quadruple therapy (CQT) are both recommended as first-line treatments for Helicobacter pylori (HP) eradication. However, several clinical trials report treatment failure rates ranging from 5% to 20%.

Aims & Methods: The aim of this study was to identify predictive factors for eradication failure in both treatments.

We conducted a prospective, randomized, open-label clinical trial. Eligible patients were randomized to receive one of the two regimens:

(A) BQT: a combination of potassium bismuth subcitrate 140 mg, metronidazole 125 mg, and tetracycline hydrochloride 125 mg, administered as 3 capsules 4 times/day, plus omeprazole 20 mg twice/day for 10 days.

(B) CQT: amoxicillin 1 g twice/day, clarithromycin 500 mg twice/day, and metronidazole 500 mg twice/day, plus esomeprazole 40 mg twice/day for 14 days.

The primary endpoint was HP eradication, defined by a negative urea breath test 4 to 6 weeks after completing treatment.

Results: A total of 200 patients were included (median age = 40.8 years; sex ratio = 1.04).

In the intention-to-treat analysis, eradication rates were 82% in the CQT group versus 87% in the BQT group (p = 0.29).

In the per-protocol analysis, HP eradication was achieved in 84.53% of patients in the CQT group versus 89.58% in the BQT group (p = 0.39).

In the concomitant therapy group, female gender and poor treatment adherence were significantly associated with treatment failure (p = 0.003 and 0.005, respectively).

In the bismuth-based therapy group, a body mass index (BMI) greater than 25 kg/m^2 and poor adherence were associated with treatment failure (p = 0.002 and 0.007, respectively).

Conclusion: Poor treatment adherence was a predictor of treatment failure in both groups. A BMI over 25 kg/m^2 and female gender were associated with therapeutic failure in the BQT and CQT groups, respectively.

Disclosure: No conflict of interest.

PP0262

LONG-TERM EFFECT OF *HELICOBACTER PYLORI* ERADICATION ON RISK FACTORS FOR CARDIOVASCULAR DISEASE – IS THERE A CONNECTION?

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Introduction: Cardiovascular disease (CVD) remains the leading cause of mortality worldwide and although traditional risk factors for atherosclerosis are well established, accumulating evidence suggests that inflammatory parameters may also play a pivotal role in its pathogenesis. Recently, *Helicobacter pylori* infection has gained increasing attention as a potential contributor to CVD risk. Thus, our aim was to investigate the long-term effects of successful *H. pylori* eradication on several CVD risk factors.

Aims & Methods: A total of 72 patients meeting the inclusion and exclusion criteria were enrolled (July 2020 to November 2022). Participants were randomly assigned to two treatment groups (group 1: 14-day regimen with esomeprazole, amoxicillin, and clarithromycin; group 2: 14-day regimen with esomeprazole, amoxicillin, metronidazole and colloidal bismuth subcitrate).

We evaluated changes in insulin resistance (HOMA-IR score), lipid profiles and subtypes (nuclear magnetic resonance spectroscopy) and trimethylamine N-oxide (TMAO; Cardio Test INFAI) at baseline, two months, and one year following successful *H. pylori* eradication.

Results: Of the 72 patients included, 13.9% (10/72) did not complete the study protocol. At baseline, no statistically significant differences were observed in CVD risk factors between the two groups. Following successful *H. pylori* eradication (Table 1), both groups demonstrated significant reductions in total cholesterol (p = 0.004), low-density lipoproteins (p = 0.011), small dense lipoprotein particles (p = 0.029), and TMAO levels (p = 0.05). No statistically significant changes were observed in body mass index (BMI, p=0.910), waist circumference (p = 0.325) or insulin resistance (p = 0.342).

	T1 (n=62)	T2 (n=62)	T3 (n=62)	p-value between times
Body mass index	27.0 ± 4.4	27.0 ± 4.3	26.8 ± 5.2	0.910
Waist circumference	92.8 ± 11.8	92.6 ± 12.0	93.7 ± 11.8	0.325
Insulin resistance (HOMA-IR)	2.3 ± 2.1	2.2 ± 1.4	2.0 ± 1.1	0.342
Cholesterol (mg/dl)	212.1 ± 39.5	200.3 ± 40.4	197.3 ± 41.9	0.004
LDL (mg/dl)	131.9 ± 34.0	124.1 ± 37.0	120.2 ± 35.4	0.011
HDL (mg/dl)	56.1 ± 15.5	54.0 ± 14.7	55.2 ± 15.3	0.159
Triglycerides (mg/dl)	141.3 ± 74.7	136.8 ± 63.1	135.8 ± 76.6	0.686
SLDL-p (nmol/l)	604.7 ± 334.0	552.0 ± 351.4	542.7 ± 351.9	0.029
TMAO (mg/l)	38.3 ± 29.9	29.2 ± 18.8	31.5 ± 30.4	0.050

T1 – at baseline, T2 – 2 months post-eradication, T3 – 1-year post-eradication
LDL – low density lipoproteins, HDL – high density lipoproteins, SLDL-p – small low density lipoprotein particles, TMAO – trimethylamine N-oxide

Table 1. Changes in various parameters over time in the entire cohort.

Conclusion: Successful eradication of chronic *H. pylori* infection was associated with a significant reduction in total cholesterol, low-density lipoproteins, small dense lipoprotein particles and TMAO in both treatment groups. These findings suggest that chronic *H. pylori* infection may influence lipoprotein levels and potentially increase the risk for CVD, possibly through increased systemic inflammation.

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RO was speaker for Abbvie, Lek Sandoz, Medis, Ewopharma, Polymetria, Nutricia. SA, GA and DŠ have no conflict of interest.

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PP0263

THE IMPACT OF *HELICOBACTER PYLORI* ERADICATION ON LONG-TERM GHRELIN LEVELS AND BODY WEIGHT

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Introduction: The association between chronic *Helicobacter pylori* infection and hormone levels affecting food intake and obesity remains controversial. Thus, our aim was to investigate the long-term effect of *H. pylori* eradication on changes in ghrelin and leptin levels.

Aims & Methods: A total of 72 patients meeting the inclusion and exclusion criteria were enrolled (July 2020 to November 2022). Participants were randomly assigned to two treatment groups (group 1: 14-day regimen with esomeprazole, amoxicillin, and clarithromycin; group 2: 14-day regimen with esomeprazole, amoxicillin, metronidazole and colloidal bismuth subcitrate). We evaluated changes in insulin resistance (HOMA-IR score), body mass index (BMI) and serum levels of ghrelin and leptin (ELISA test) at baseline, two months, and one year following successful *H. pylori* eradication.

Results: Of the 72 patients included, 13.9% (10/72) did not complete the study protocol. Both treatment groups were similar at enrolment and showed no statistically significant differences (p = 0.332 for OLGIM stages; p = 0.583 and p= 0.696 for antrum and corpus atrophy respectively, p = 0.177 and p = 0.296 for antrum and corpus intestinal metaplasia respectively).

No statistically significant changes were present in BMI (p = 0.910) and insulin resistance (p = 0.342) after *H. pylori* eradication. A statistically significant decrease in ghrelin serum levels was observed in both treatment groups (p < 0.001), but not in leptin serum levels (p = 0.309), as shown in Table 1.

	T1	T2	T3	p-value
Leptin (pg/ml)	8128.0 ± 8171.2	9207.1 ± 8255.7	9469.4 ± 7761.5	0.257
Ghrelin (pg/ml)	3963.9 ± 2270.8	3076.7 ± 1767.2	3160.5 ± 1875.2	< 0.001
Body mass index	27.0 ± 4.4	27.0 ± 4.3	26.8 ± 5.2	0.910
Waist circumference (cm)	92.8 ± 11.8	92.6 ± 12.0	93.7 ± 11.8	0.325
HOMA-IR	2.3 ± 2.1	2.2 ± 1.4	2.0 ± 1.1	0.342

Table 1. Changes in various parameters over time in the entire cohort.
T1 – at baseline, T2 – 2 months post-eradication, T3 – 1-year post-eradication

Conclusion: Successful eradication of chronic *H. pylori* gastritis decreased serum ghrelin levels regardless of the presence of gastric mucosal atrophy.

Furthermore, no changes in BMI or waist circumference were detected after one year of follow-up, proving that *H. pylori* eradication is not related to weight gain through hormonal changes as previously hypothesized. These results reduce the fear that massive *H. pylori* eradication from the population could be associated with obesity.

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KT was a speaker for Abbvie, Pfizer, Krka, Altamedics, Viatris, Eli Lilly.
BT served as a speaker or received research grant from KRKA, Ewopharma, Swixx pharma, Infai GmbH.
RO was speaker for Abbvie, Lek Sandoz, Medis, Ewopharma, Polymetria, Nutricia.
ML, UPF, GA and DŠ have no conflict of interest to declare.

PP0264

THE EFFICACY AND SAFETY OF HIGH-DOSE DUAL THERAPY FOR *HELICOBACTER PYLORI* ERADICATION IN THE NORTH-WESTERN REGION OF RUSSIA

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Introduction: The *Helicobacter pylori (H. pylori)* infection is a common problem (prevalence of 32.7% in 2023) in the North-Western Federal District, comprising one-tenth of Russia, where primary resistance rates of clarithromycin and levofloxacin are >15%.[1]

The eradication rates remain low and do not reach the threshold value of 95% (eradication rate of 86.8% in 2023).[2] There is limited information regarding the best approach for *H. pylori* infection management.

Aims & Methods: The study was conducted in the North-Western Federal District during 2023-2025. Individuals (n=145) who underwent ^{13}C-UBT (^{13}C-urea breath test) was conducted in the prospective randomized clinical trial. Patients were randomly allocated to two eradication groups: high-dose dual therapy for 14 days (esomeprazole 40 mg and amoxicillin 1000 mg three times daily) and optimized high-dose dual therapy for 14 days (esomeprazole 40 mg, amoxicillin 1000 mg, rebamipide 100 mg three times daily).

The modified intention-to-treat (mITT) effectiveness, safety, and adherence were analyzed for the regimens. The secondary endpoints were adverse effects and patient compliance. Eradication was confirmed by a ^{13}C-UBT 4-6 weeks after therapy.

Results: Among all included individuals (mean age 46.27±14.21; 67.6% of whom were female), there were 91.7% (n=133) of primary "naïve" patients (who had had no history of prior eradication therapy). The proportion of treated patients was 8.3% (n=12). The high-dose dual therapy PPI-amoxicillin did not provided encouraging mITT results - 87.5%, 21/24, 95%CI 69.0-95.7; the inclusion of patients in this group was stopped due to unacceptable cure rates.

Whereas the higher was reported with optimized eradication regimen with rebamipide (96.3%, 105/109, 95%CI 90.9-98.6 for treatment-naïve and 91.7%, 11/12, 95%CI 64.9-98.5 for treated patients). The adverse event rates were 37.5% in the PPI-amoxicillin group and 19.8% in the PPI-amoxicillin-rebamipide group (p=0.211). The most common were a diarrhea (25% and 12.4% respectively), bloating (25% and 3.5% respectively), ab-

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dominal pain (20.8% and 10.7% respectively). Good adherence, defined as > 90% of drug intake, were similar between the two groups (100% of cases).

Conclusion: In the North-Western region of Russia, the fourteen-day optimized high-dose dual therapy with PPI-amoxicillin-rebamipide provided encouraging results (over 90% cure rates) and had the lower incidence rate of side effects (19.8%) suggesting as an alternative for *H. pylori* rescue treatment in the local region.

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Disclosure: Nothing to disclose.

PP0265

SYNBIOTIC COMPLEX IMPROVED THE EFFICACY AND TOLERABILITY OF QUADRUPLE THERAPY WITH AMOXICILLIN, CLARITHROMYCIN AND BISMUTH IN PATIENTS FROM VINNYTSIA REGION, UKRAINE

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Introduction: Both the effectiveness and tolerability of Helicobacter pylori (H.p.) eradication therapy regimens are important in the management of H.p. infection, since the presence of side effects significantly reduces compliance between the doctor and the patient and sometimes can lead to treatment interruption. Our study aimed to compare the efficacy and tolerability of eradication therapy depending on the addition of Saccharomyces boulardii or a synbiotic complex.

Aims & Methods: The study included 78 patients (54 women and 24 men) with confirmed H.p. infection using validated tests, without previous eradication in the anamnesis. All patients underwent a course of eradication therapy, which included proton pump inhibitor (PPI) + amoxicillin (A) + clarithromycin (C) + bismuth subcitrate (B). Patients were divided into 2 groups: in group A (37 patients, 25 women, 12 men, mean age 44±3.2 years) S. boulardii CNCM I-745 was added to the mentioned regimen. In group B (41 patients, 29 women, 12 men, mean age 47±2.8 years), a synbiotic complex containing strains of probiotic bacteria Lactobacillus acidophilus, Lactobacillus plantarum, Lactobacillus delbrueckii subsp. bulgaricus, Lactobacillus fermentum, Lactobacillus rhamnosus, Lactobacillus casei subsp. casei, Streptococcus thermophilus, Bifidobacterium bifidum, Bifidobacterium longum, Bifidobacterium adolescentis, Propionibacterium freudenreichii subsp. Shermanii, prebiotics inulin and lactulose were added to the indicated regimen. The duration of eradication therapy was 10 days. The duration of taking the probiotic and synbiotic complex – 15 days.

Results: In group A, the effectiveness of the PPI+A+C+B regimen with the addition of S. boulardii was 75.7%. In group B (with the addition of a synbiotic) - 87.8%. We compared the tolerability of the indicated regimens. In no case (in both groups) was the treatment interrupted due to the presence of serious side effects. In total, 20 (54.1%) patients from group A and 17 patients (41.5%) from group B reported non-serious adverse events, including nausea (group A – 8 patients (21.6%), group B – 6 (14.6%), bitter taste in the mouth (group A – 16 patients (43.2%), group B – 15 (36.6%), and diarrhoea (group A – 6 patients (16.2%), group B – 0 (0%). No other adverse events were documented. Comparing gastrointestinal symptoms using the Gastrointestinal Symptom Rating Scale (GSRS) showed that in group A, mean values of the total score before treatment were 35.9±3.14 points, vs 26.2±1.64 points 20 days after the eradication course ($p<0.05$). In group B 34.1±2.44 points before treatment, vs 21.7±1.98 points 20 days after the eradication course ($p<0.05$). Comparing the total score of the GSRS questionnaire after treatment between groups A and B also showed a significant difference: 26.2±1.64 points in group A vs 21.7±1.98 points in group B ($p<0.05$).

Conclusion: Our study revealed better efficacy (87.8%) of the eradication regimen PPI+A+C+B with the addition of a synbiotic compared to the similar regimen with the addition of Saccharomyces boulardii (75.7%) in patients from Vinnytsia region, Ukraine. In addition, it was found that when the synbiotic complex was added, side effects of eradication therapy were noted in a smaller percentage of patients and significantly ($p<0.05$) lower values of the total score of the GSRS questionnaire were noted 20 days after treatment, in contrast to the regimen with the addition of Saccharomyces boulardii.

Disclosure: Nothing to disclose.

PP0266

INDICATIONS FOR FIRST-LINE *HELICOBACTER PYLORI* ERADICATION IN GERMANY: INSIGHTS FROM THE EUROPEAN REGISTRY ON *H. PYLORI* MANAGEMENT (HP-EUREG) WITH A FOCUS ON GASTRIC CANCER PREVENTION

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Introduction: *Helicobacter pylori* (*H. pylori*) infection is a well-established risk factor for gastric cancer. While high-incidence countries promote population-based test-and-treat strategies, Germany—classified as an intermediate-risk country—follows a more targeted approach. According to the latest German S3 guidelines, *H.pylori* eradication is strongly rec-

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ommend in specific high-risk groups, including first-degree relatives of gastric cancer patients and individuals who have undergone surgical or endoscopic resection of gastric cancer. However, real-world data on how these recommendations are implemented in clinical practice remain limited. Additionally, data on the use and effectiveness of different first-line eradication regimens in Germany are sparse.

Aims & Methods: We conducted a retrospective analysis of patients who received first-line empirical *H. pylori* eradication therapy across five German university-based centers participating in the European Registry on *H. pylori* Management (Hp-EuReg). Three centers contributed the majority of cases. Demographic and clinical data were extracted from databases. Data were collected in AEG-REDCap from June 2013 to January 2025. Indications were categorized by symptom profile, endoscopic findings, and cancer risk. A descriptive analysis was performed. Eradication rates were compared between treatment regimens and durations using the χ^2 test. A particular focus was placed on evaluating how the recommendations of the German S3 guidelines regarding gastric cancer prevention were implemented in clinical practice.

Results: Of the 448 patients (53% male, median age 55) ncluded, the most common indications were non-investigated dyspepsia (33%) and dyspepsia with normal endoscopy (30%). Peptic ulcer disease was reported in 18%: duodenal ulcer (5%) and gastric ulcer (13%). Five percent underwent eradication for gastric cancer prevention; these included patients with preneoplastic lesions (0.4%), prior gastric cancer resection (0.9%), first-degree relatives of gastric cancer patients (2.0%), and individuals undergoing screening (1.3%). Other indications included unexplained iron deficiency anemia (4.0%), vitamin B12 deficiency (0.7%), and MALT lymphoma (0.2%). The most frequent first-line therapies were triple therapy with amoxicillin and clarithromycin (54%) and a three-in-one single-capsule containing metronidazole, tetracycline, and bismuth (41%). Therapy commonly lasted 7 (19%) or 10 days (72%). No difference in eradication rates between the two regimens was observed when given for 10 days (97.8% vs. 97.0%, respectively).

Conclusion: In this multicenter German cohort from dedicated *H. pylori* centers, most eradication therapies were prescribed for dyspeptic symptoms. Notably, a significant proportion of treatments aimed at gastric cancer prevention, aligning with the approach outlined in the current German S3 guidelines. Both recommended first-line therapies—triple therapy with amoxicillin and clarithromycin, and the three-in-one single-capsule bismuth-based quadruple therapy—were widely used and achieved similarly high eradication rates when administered for 10 days. These findings underline the importance of implementing guideline-based indications in intermediate-risk countries like Germany, where targeted eradication in high-risk individuals remains a cornerstone of gastric cancer prevention.

Disclosure: Nothing to disclose.

PP0267

INTERACTION BETWEEN PROTON PUMP INHIBITORS AND *HELICOBACTER PYLORI* INFECTION ON MAJOR OSTEOPOROTIC FRACTURE RISK: A POPULATION-BASED COHORT STUDY

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Introduction: Both proton pump inhibitors (PPIs) and *Helicobacter pylori* (HP) infection are risk factors for osteoporotic fracture. However, the interaction between PPI use and HP infection status post-eradication therapy in HP-infected individuals on fracture risk remains unclear.

Aims & Methods: We aimed to investigate the effect of HP treatment outcome on PPI-associated fracture risk in HP-infected patients. We conducted a retrospective cohort study using a population-based healthcare database. Adults (≥18 years) prescribed clarithromycin-containing triple therapy for 7-14 days (first-line HP eradication therapy during the study period) between 2003 and 2012 were included. The cohort was stratified into: (1) successful HP eradication (HP-) and (2) failure of HP eradication (HP+). Failure of HP eradication was identified by the need of subsequent prescriptions of HP therapy including repeated course of eradication therapy. Exclusion criteria included epilepsy, Paget's disease, bone tumor, gastrectomy, gastric cancer, fracture due to motor vehicle traffic accidents, and ever use of PPIs before index date. The exposure of interest was PPI use following HP eradication therapy. PPI exposure was modelled as a time-vary variable (≥14 days per 3-month interval). Histamine 2 receptor antagonists (H2RAs) served as a negative control. The primary outcome was a composite outcome of major osteoporotic fractures (hip, vertebra, humerus, forearm, wrist). Covariates included demographics, smoking status, alcohol use disease, prior fractures/falls, osteoporosis/osteomalacia, metabolic factors, cardiovascular diseases, chronic renal failure, cirrhosis and concomitant medications (such as aspirin, H2RAs, anti-osteoporotic agents, hormonal therapy, steroids, calcium and vitamin D supplements). The index date was defined as the first date of HP eradication. Patients were followed till occurrence of outcome, death or study end (31 December, 2019). Multivariable Cox regression was used to estimate adjusted hazard ratios (aHRs). Stratified analysis was conducted by combined HP/PPI status to examine the interaction between HP infection status post-eradication therapy on fracture risk.

Results: There was a total of 59,380 eligible subjects (median age: 55.2 [interquartile range: 45.8-64.8] years; male: 44.0%) in this study, including 52,033 (87.6%) with successful HP eradication (HP-) and 9,713 (16.4%) PPI users. Over a median follow-up of 11.8 (interquartile range: 9.2-14.3) years, 2,736 (4.6%) experienced fracture. In the whole cohort, PPI use was associated with higher fracture risk (aHR: 1.31, 95% CI:1.20-1.50), whereas H2RA use was not (aHR: 0.98, 95% CI: 0.89-1.08); when compared with failure of HP eradication, successful HP eradication was associated with lower fracture risk (aHR: 0.78, 95% CI: 0.70-0.86). In stratified analysis, the highest fracture risk was observed in the failure of HP eradication and PPI use (HP+/PPI+) group (aHR: 1.77, 95% CI: 1.46-2.14), compared with reference group (successful HP eradication and no PPI use [HP-/PPI-]; Table 1).

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Conclusion: PPIs are still associated with higher fracture risk irrespective of HP eradication status, although subjects with successful eradication have lower risk than subjects with failure of eradication. Testing and treating HP infection may be advisable to reduce fracture risk in long-term PPI users.

Disclosure: Nothing to disclose.

PP0268

INCIDENCE OF HELICOBACTER PYLORI AND PRECANCEROUS LESIONS IN A TERTIARY CENTRE IN TRANSYLVANIA REGION

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Introduction: Helicobacter pylori (HP) has been designated as a Grade I Carcinogen by the WHO and remains an important factor in the pathogenesis of intestinal-type gastric cancer. Gastric cancer was the fifth most common malignant disease in the world in 2022 (Globocan) with 968,784 new cases(1), after lung, breast, colorectal and prostate cancers. The incidence of gastric cancer has changed significantly since the first estimates in 1975, when stomach cancer was the leading cancer, and one of the most important factors has been the eradication of HP.(2)

Aims & Methods: The purpose of this study was to determinate the trends in the incidence of HP in our region, and also of the precancerous lesions. There have been included in the study a group of 12541 patients from the years 2014-2018 and a group of 7138 patients from more recently years 2020-2024.

All the patients had dyspeptic syndrome and in all the patients upper digestive endoscopy has been performed in the Gastroenterology Clinic from Târgu Mureş, Emergency Clinical Hospital. Patients with haemorrhage were excluded from the study. In all the patients gastric biopsies and histopathological exam were made, OLGA classification was used.

Results: From the first group, in 2131 patients (52,9% males and 47,1% females) were found gastric changes. Histopathological, 32,7% of the patients had atrophic gastritis, 43,1% intestinal metaplasia (23,7% complete metaplasia and 19,4% incomplete metaplasia) and 0,7% dysplasia. Active gastritis/pangastritis with HP was identified in 59,3% of the patients. In 4,8% of cases there were revealed polyps (hyperplastic, adenomatous) and were removed.

In the second group from 2560 patients (54,4% males and 45,6 females) that underwent upper gastrointestinal with biopsy sampling, HP was present in 442 patients (17,26%.) Histopathological, 14,50% of the patients had atrophic gastritis, 23,8% intestinal metaplasia and 2,58% dysplasia.

Conclusion: HP incidence is decreasing, as it has been treated by the gastroenterologist and also by the general practitioner. The decreasing in incidence of HP is corelating with the decreasing of the precancerous lesions and gastric cancer, so it is worth to treat it.

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Disclosure: Nothing to disclose.

PP0269

ASSOCIATION BETWEEN HELICOBACTER PYLORI INFECTION AND COLORECTAL POLYPS

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Introduction: Helicobacter pylori (H. pylori) is a well-recognized gastric carcinogen. However, whether this infection is related to colorectal polyps and colorectal cancer remains debatable. Data from some recent studies have indicated a significant association between H. pylori infection and the development of colorectal polyps or cancer. Potential effects of H. pylori on the colorectal mucosa include the induction of inflammatory responses, alteration of gut flora, and the release of toxins.

The aim was to investigate the association between H.pylori infection and colorectal polyps.

Aims & Methods: We retrospectively included patients diagnosed with colorectal polyps who underwent H. pylori testing between September 2024 and March 2025. Patients with incomplete colonoscopies were excluded. Demographic, clinical, and endoscopic data were subsequently collected. Polyp subtypes and dysplasia grades were extracted from histopathology reports. Screening for H. pylori infection was performed on gastric biopsies.

Patients were divided into two groups: Group 1 (H. pylori-negative) and Group 2 (H. pylori-positive). The number, size, location, and pathological type of colorectal polyps were compared between the two groups.

Results: One hundred thirty patients were included, with a male-to-female sex ratio of 3.2. The mean age was 56.3 ± 11.23 years. The main comorbidities were diabetes (33.1%), dyslipidemia (26.9%), and hypertension (25.4%). The locations of colorectal polyps varied: 18.5% in the rectum, 17.7% in the sigmoid colon, 8.5% in the left colon, 7.7% in the transverse colon, 16.9% in the right colon, and 30.8% of patients had polyps in multiple locations. A total of 219 polyps were identified, with a mean of 2.16 polyps per patient and a mean size of 5.9 mm (range: 2–30 mm).

Histopathological examination revealed predominantly hyperplastic polyps (36.9%), low-grade dysplastic adenomas (48.5%), high-grade dysplastic adenomas (7.7%), and adenocarcinomas (6.9%). The prevalence of H. pylori infection was 31.5%.

Statistical analyses showed that the H. pylori-positive group had a significantly higher proportion of colorectal polyps with sizes ≥10 mm (55.6% vs. 44.4%, $p = 0.027$) and multiple polyps (≥5) (56.3% vs. 43.8%, $p = 0.04$) compared to the H. pylori-negative group. However, neither dysplasia grading nor polyp localization was significantly associated with H. pylori infection.

Conclusion: In our study, H. pylori infection was associated with an increased risk of large and multiple colorectal polyps. Further studies are needed to confirm these findings and to better understand the impact of H. pylori eradication on the occurrence of colorectal polyps, as well as its tumorigenic effects on the colorectal mucosa.

Disclosure: Nothing to disclose.

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PP0270

NATURAL HISTORY AND CLINICAL CHARACTERISTICS OF PATIENTS WITH IDIOPATHIC SCLEROSING MESENTERITIS: A PROSPECTIVE COHORT STUDY

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Introduction: Idiopathic Sclerosing Mesenteritis (SM) is a rare fibroinflammatory disease of unknown etiology. The clinical course is generally benign, as has been demonstrated in retrospective studies and large case series.

Aims & Methods: To more accurately determine the natural history of patients with symptomatic SM and describe the clinical features of those with idiopathic SM, we performed a prospective cohort study. Inclusion criteria were adult patients with typical SM symptoms, after ruling out other causes of abdominal pain, who met the Coulier criteria in computed tomography scans for establishing SM diagnosis. We identified 58 patients with symptomatic SM. During the follow-up period, 3 patients were lost and excluded from the analysis. Most patients (95.7%) underwent upper and lower endoscopy and chest imaging to rule out co-existing cancer disease. Secondary SM cases were also excluded (n=5 had a co-existing cancer diagnosis within 6 months of SM diagnosis and n=4 had a pre-existing diagnosis of SM-related autoimmune disease at SM diagnosis). The final cohort of symptomatic idiopathic SM cases consisted of 46 patients. Data were collected prospectively at each site visit from January 2014 to January 2024. Descriptive statistics were used to summarize the results.

Results: In total, 46 patients with idiopathic SM were included, of whom 56.5% were females and 43.5% were males with a mean age at diagnosis of 62 years. Overweight or obese were 62.8%, and 37.2% had normal weight. SM was histologically confirmed in 13%. Features of IgG4-related disease were present in 21.7% with other organ involvement (13% pancreas, 10.9% bile ducts, 2.2% orbit, and 2.2% lymph nodes). Multiple organ involvement was observed in 6.5%. All patients who presented with an abdominal mass (8.7%) developed features similar to IgG4-related disease. Elevated serum IgG4 levels at diagnosis were found in 7 of 36 patients (19.4%). All patients presented with abdominal pain, followed by bloating (84.8%), nausea (43.5%), anorexia (39.1%), weight loss (37%), vomiting (30.4%), dyspepsia (19.6%), diarrhea (19.6%), constipation (13%), fever (8.7%), icterus (4.3%), early satiety (4.3%), sarcopenia (4.3%), and peripheral edema (2.2%). Hospitalization was required in 37%, and severe complications were observed in 30.4% [ileus (15.2%), acute pancreatitis (8.7%), acute cholangitis (8.7%), abdominal mass (8.7%), chronic pancreatitis (4.3%), mesenteric thrombosis (2.2%), and obstructive lymphadenopathy (2.2%)]. Surgical intervention was performed in 13%. During the disease course, most patients (84.8%) were treated with colchicine in combination or not with corticosteroids (54.3%), antiplatelets (50%), and low-molecular heparin (28.3%).

Conclusion: In this prospective cohort study, we observed an aggressive clinical course with severe intra-abdominal complications in a substantial proportion of patients with idiopathic SM (30.4%). Almost one in four patients (21.7%) manifested other organ involvement, identical to IgG4-related disease, indicating that some idiopathic SM cases could be part of the IgG4 disease spectrum. These observations suggest that idiopathic SM requires clinical vigilance similar to that applied to IgG4-related disease.

Disclosure: No potential conflicts of interest.

PP0271

COMBINED THERAPY OF INTESTINAL DYSBIOSIS AS A STRATEGY FOR LIPID METABOLISM DISTURBANCE TREATMENT

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Introduction: Dyslipidemia is a major contributor to cardiovascular disease and requires comprehensive treatment strategies. Emerging evidence supports the role of gut microbiota in lipid metabolism regulation, suggesting that targeted microbiome correction could be an effective adjunct to lipid-lowering therapy.

Aims & Methods: The aim was to evaluate the effectiveness of combined therapy—including probiotics, dietary recommendations, and lifestyle modifications—in correcting dyslipidemia and restoring gut microbial balance. A total of 168 patients with dyslipidemia were divided into two groups based on SCORE cardiovascular risk. All participants received a 12-week course of probiotics (Lactobacillus acidophilus LA-5 and Bifidobacterium lactis BB-12), lifestyle advice, and a cholesterol-lowering diet. Patients in the moderate-risk group additionally received simvastatin. Lipid profiles and gut microbiota composition (by qRT-PCR targeting 16S rRNA) were assessed before and after therapy.

Results: Both groups demonstrated significant reductions in total cholesterol, LDL-C, and triglycerides, and increased HDL-C levels ($p<0.05$). Analysis of gut microbiota revealed an increase in Bacteroidetes and a reduction in Firmicutes and Actinobacteria, alongside improvement in the Firmicutes/Bacteroidetes index. Notably, the percentage changes in lipid profile parameters were not significantly different between the probiotic-only and probiotic-plus-statin groups ($p>0.05$), suggesting a potential lipid-modulating effect of probiotics alone.

Conclusion: Combined therapy with probiotics, lifestyle modification, and diet effectively improves lipid metabolism and gut microbial composition in patients with dyslipidemia. Probiotic treatment may reduce the need for pharmacologic therapy in some patients and should be considered in personalized dyslipidemia management strategies.

Disclosure: Nothing to disclose.

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PP0272

NON-COELIAC GLUTEN/WHEAT SENSITIVITY IN THE POST-PANDEMIC ERA: AN EXCLUSIVE PRIMARY CARE PROBLEM?

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Introduction: Non-coeliac gluten/wheat sensitivity (NCGWS) is characterized by gastrointestinal and extra-intestinal symptoms triggered by gluten/wheat ingestion in patients without coeliac disease (CD) or wheat allergy. There are controversies on the mechanism leading to this phenomenon, including media influence and the presence of a nocebo effect. Moreover, after the COVID-19 pandemic, a decrease in the rates of NCGWS diagnosis has been noted. The only treatment available for the management of NCGWS is a gluten/wheat-free diet (GFD), yet long-term outcomes are poorly understood.

This study aimed to evaluate post-pandemic trends in NCGWS and the long-term GFD adherence in patients with NCGWS. The secondary goals are the assessment of the perceived efficacy of GFD and the need of further medical support.

Aims & Methods: We retrieved data from gluten-free products vendors up to September 2024 (top 6 retailer in the UK) and gluten free products customers up to July 2021, using a questionnaire. Subsequently, patients with self-reported NCGWS referred to a tertiary center clinic from 2006 to 2018 completed a questionnaire regarding their dietary habits and further need of medical advice for symptoms related to their diagnosis. The Biagi score was used to assess their adherence to a GFD. We compared the dietary adherence scores to age and sex matched controls cohort of patients with CD to validate the results.

Results: The marked data analysis showed a persistent rise in the sales of gluten free products, from £200B in 2014 vs £600B in 2024. Moreover, NCGWS patients account for 9.7% of the total number of buyers and for 26% of the spent for GF products.

In our cohort, out of 186 patients with NCGWS, 111 answered the questionnaire (response rate 59.7%, female 82.0%, average age 54 ± 13 years). Of these 111, 58 are not currently following a GFD. Out of the 53/111 patients following a GFD, only 27/53 (50,9%) are adhering strictly (Biagi score 3-4), 2/53 had a Biagi score of 2 (3,8%) indicating poor adherence to a gluten free diet, and 24/53 (45,2%) voluntarily ate gluten (Biagi 1-0). Only two patients report that gluten/wheat is not causing them any symptoms if eaten. The average time patients remained on a gluten free diet was 141±58 months.

Compared to patients with CD those with NCGWS on a GFD showed a significantly lower adherence to a GFD (Biagi score of 0-1 NCGS: 73.9% vs CD: 8.1%, p<0.05; Biagi score 3-4, NCGWS: 24.3% vs CD: 90.1%, p<0.05). After being discharged, 40 patients (36.0%) sought medical advice again, but no significant correlation was found between the adherence to a GFD and need for further medical intervention.

Conclusion: Self-reported NCGWS is still consistently present among general population. Unlike CD, the vast majority of patients with self-reported NCGWS do not continue a GFD or significantly reduce their level of adherence. Still, NCGWS patients account for a vast part of the expenses in the gluten-free market. However, over a third of NCGWS sought further medical advice to manage their symptoms. These findings underline the need of a better understanding of this condition and the challenges in the management of these patients.

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Disclosure: Nothing to disclose.

PP0273

SAVE YOUR BREATH: TARGETING THE RIGHT PATIENTS FOR SIBO TESTING

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Introduction: Small intestinal bacterial overgrowth (SIBO) is characterized by the presence of excessive gram-negative aerobic and anaerobic colonic bacteria in the small intestine[1,2], leading to increased carbohydrate fermentation and gas production.

Symptomatology frequently overlaps with other gastrointestinal disorders[3].

This study aimed to identify factors associated with test positivity and improve patient selection for diagnostic evaluation.

Aims & Methods: Retrospective, unicentric study. A cohort of 34 patients underwent glucose hydrogen and methane breath testing for SIBO. Demographic, clinical, laboratory data and test results were analyzed.

Results: The cohort included 21 females (61.8%) and 13 males (38.2%), with a mean age of 46.5 ± 15.9 years. Chronic diarrhea was the leading indication for testing (82.4%), followed by abdominal pain (8.8%). Tests results were positive in 4 cases (11.8%).

Positive patients had a higher mean age (56.0 ± 10.8 vs. 46.1 ± 15.8 years, $p = 0.236$) and a comparable gender distribution. Among positive patients, diarrhea was reported in 100% of cases, followed by abdominal distension (75%) and abdominal pain (50%). Folic acid levels were higher in the positive group (11.65 vs. 6.54 ng/mL, $p = 0.074$), while fecal calprotectin levels were significantly elevated (480 vs. 44.26 µg/g, $p = 0.027$).

Altered small intestine anatomy due to prior surgical interventions was strongly associated with test positivity ($p = 0.012$), while systemic disorders affecting gastrointestinal motility showed a non-significant trend ($p=0.136$). Proton pump inhibitor use was not associated with test positivity ($p = 1.0$). Rifaximin treatment led to symptom improvement in all positive cases, but with no significant difference compared to negative patients ($p = 0.444$).

Conclusion: Elevated serum folate and fecal calprotectin levels may serve as indicators of underlying SIBO in the appropriate setting. Altered small intestine anatomy was strongly associated with test positivity, and could

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indicate SIBO under adequate intestinal transit time. These findings highlight the need for larger studies to refine patient selection and improve diagnostic strategies.

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Disclosure: Nothing to disclose.

PP0274

FECAL MICROBIOTA TRANSPLANTATION IN HEMATOLOGICAL PATIENTS; EXPERIENCE FROM UNIVERSITY HOSPITAL CENTER ZAGREB

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Introduction: Fecal microbiota transplantation (FMT) is becoming an increasingly used procedure in the treatment of gastrointestinal diseases. In this summary, we present our experience of FMT in hematological indications and patients.

Aims & Methods: From January 2023 to January 2024, 15 FMT procedures in 7 patients were performed. The transplant was prepared by standard technique from a healthy donor at the Department of Microbiology. Hematology team was responsible for selecting patients with adequate FMT indication.

Procedure was performed with standard gastroscope in 6 patients by transplantation into the distal duodenum, and in one patient by a colonoscope in the large bowel.

Results: Patient 1: acute GVHD and recurrent CD colitis after alogenic bone marrow transplantation (BMT) due to acute T-lymphoblastic leukemia. Application of FMT was done from caecum to rectum in a dose of 200 ml with complete symptom resolution and CD negativization Patient 2: alogenic BMT for MDS with development of severe small intestine GVHD refractory to standard immunosuppression. Two FMT performed with transplantation into the distal duodenum (200 ml in three weeks apart). Although relatively good clinical recovery was observed after 6-8 weeks, patient died 4 months later due to respiratory insufficiency and sepsis.

Patient 3 and 4. patients with AML in preparation for BMT. Due to repeated stool isolate of Pseudomonas aeruginosa, FMT was performed with the application of 200 ml of transplant (2 weeks apart) to the duodenum for the purpose of intestinal decolonization. Two weeks after FMT, the stool cultures showed a physiological flora and a planned BMT was performed.

Patient 5 and 6: patients with alloBMT due to primary myelofibrosis who developed intestinal GVHD and CD colitis. 200 ml of transplant was applied (3 weeks aparat) to the duodenum. The condition improved quickly and in further follow-up CD toxin was negative with partial improvement of GVHD symptoms (one patient required additonal photopheresis).

Patient 7: patient with AML in whom multidrug-resistant Enterococcus faecium has been isolated in coprocultures on several occasions during chemotherapy. Prior to the planned BMT, FMT was performed with the aim of intestinal decontamination with the application of 200 ml (2 weeks apart) of transplant to the distal duodenum. Stool analysis 2 weeks after FMT indicated physiological flora and a planned BMT was performed

Conclusion: Our experience indicates that FMT is a useful method in the treatment of GI pathology (GVHD, decontamination of the intestine before BMT and treatment of CD colitis) in hematological patients.

Disclosure: Nothing to disclose.

PP0275

PRIMARY MACROPHAGE-DERIVED SIGNALS DRIVE FIBROGENESIS IN A GASTROINTESTINAL *IN VITRO* SCAR-IN-A-JAR MODEL

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Introduction: Inflammatory bowel disease (IBD) is characterized by chronic inflammation, where tissue remodeling and fibrosis are involved. Cytokine-driven activity originating from different inflammatory cell types could be responsible for fibrosis, by promoting fibroblast activation and excessive deposition of ECM. Specifically, inflammatory monocytes have recently been shown to be prominent in patients with higher disease severity in IBD. Although anti-inflammatory therapies are available, their impact on fibrogenesis remains limited, highlighting the need for *in vitro* models that can assess inflammation-driven fibrosis. Simultaneously, the 10 WFA-like domain from type VI collagen enables activated fibroblasts to bind and activate platelets, linking wound healing to fibrogenesis. Thus, we aimed to evaluate the pro-fibrotic effects of macrophages in a gastrointestinal scar-in-a-jar model, by quantifying a biomarker of type VI collagen formation.

Aims & Methods: Monocytes were isolated from buffy coats of healthy donors and differentiated to M0 macrophages for 6 days using 20 ng/ml MCSF. Upon differentiation, on day 0 the macrophages were polarized to M1 using 10 ng/ml LPS. The conditioned media was collected after 1 and 6 days. Human primary small intestine fibroblasts (Innoprot) were stimulated on day 0, 4 and 8 with: conditioned media from naive macrophages (M0) and activated macrophages (M1) in 3 dilutions (100%, 30%, 10%), 50 pM TGF-β, 3.3 ng/ml LPS and media control (MCSF). Unstimulated fibroblasts (w/o) were included as negative control. The experiment was concluded on day 12 and PRO-C6 (quantification of type VI collagen formation) was measured for all days. Cell viability was assessed using Live/Dead staining on day 12. To compare the changes for the biomarker levels (% change and ng/ml) two-way ANOVA and one-way ANOVA (with Tukey correction) were applied.

Results: On day 4, PRO-C6 levels were significantly elevated upon stimulation with TGF-β and M1-Day1 conditioned media, compared to unstimulated fibroblasts (Table 1, $p<0.001$ & $p<0.05$). On day 8, TGF-β and conditioned media from M0-Day6, M1-Day1 and M1-Day6 significantly increased PRO-C6 ($p<0.001$, $p<0.001$, $p<0.0001$ & $p<0.0001$). No significant increase was observed for TGF-β on day 12, whereas all conditioned media (M0-Day1, M0-Day6, M1-Day1 & M1-Day6) induced a significant increase of PRO-C6 ($p<0.0001$, $p<0.001$, $p<0.01$ & $p<0.0001$). On days 8 and 12, a significant decrease was observed for PRO-C6, when comparing the 100% M1-Day1 and M1-Day6-conditioned media to both 30% and 10% (all: $p<0.0001$).

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Day	Comparison	Mean Dif.	95% CI	p-value	Day	Comparison	Mean Dif.	95% CI	p-value
4	w/o vs. TGF-β [50 pM] (% change)	-80.19	[-112.27, -48.12]	<0.001	8	w/o vs. M1-Day6 (MCSF + LPS) (% change)	-131.6326	[-163.71, -99.56]	<0.0001
4	w/o vs. M1-Day1 (MCSF) (% change)	-53.78	[-85.86, -21.71]	<0.05	12	w/o vs. M0-Day1 (MCSF) (% change)	-85.2536	[-117.33, -53.18]	<0.0001
8	w/o vs. TGF-β [50 pM] (% change)	-72.41	[-104.49, -40.34]	<0.001	12	w/o vs. M1-Day1 (MCSF)(% change)	-61.96	[-94.04, -29.89]	<0.01
8	w/o vs. M1-Day1 (MCSF) (% change)	-84.71	[-116.79, -52.64]	<0.0001	12	w/o vs. M0-Day6 (MCSF) (% change)	-72.84	[-104.92, -40.77]	<0.001
8	w/o vs. M0-Day6 (MCSF) (% change)	-71.91	[-103.99, -39.83]	<0.001	12	w/o vs. M1-Day6 (MCSF + LPS) (% change)	-162.92	[-195.00, -130.85]	<0.0001
8	M1-Day1 (MCSF) 100% vs. 30% (ng/ml)	1.78	[1.39, 2.17]	<0.0001	12	M1-Day1 (MCSF) 100% vs. 30% (ng/ml)	1.66	[1.03, 2.30]	<0.0001
8	M1-Day1 (MCSF) 100% vs. 10% (ng/ml)	1.68	[1.29, 2.07]	<0.0001	12	M1-Day1 (MCSF) 100% vs. 10% (ng/ml)	1.17	[0.54, 1.81]	<0.0001
8	M1-Day1 (MCSF) 100% vs. 30% (ng/ml)	2.37	[1.79, 2.96]	<0.0001	12	M1-Day6 (MCSF) 100% vs. 30% (ng/ml)	3.12	[2.30, 3.94]	<0.0001
8	M1-Day1 (MCSF) 100% vs. 10% (ng/ml)	1.66	[1.07, 2.24]	<0.0001	12	M1-Day6 (MCSF) 100% vs. 10% (ng/ml)	2.72	[1.90, 3.54]	<0.0001

Table 1.

Conclusion: Conditioned media from M0 and particularly from M1 macrophages, induced type VI collagen formation over time, whereas the effect of TGF-β stimulation depleted by day 12, suggesting a more sustained pro-fibrotic potential of inflammatory macrophage-derived cytokines. When fibroblasts were stimulated with 100% M1-conditioned media on day 8, and 12, the highest type VI collagen formation was exhibited and a significant reduction was noted in the 30% and 10% dilutions, suggesting a dose-sensitive response.

Disclosure: TTK, MP and MSA are employed at Nordic Bioscience. ACBJ, MAK and JHM are employed and stock owners at Nordic Bioscience.

PP0276

GUT HEALTH AND LIPID METABOLISM: THE ROLE OF SIBO THERAPY

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Introduction: Small intestinal bacterial overgrowth (SIBO) is a frequent condition characterized by an abnormal increase in the bacterial population in the small intestine. It is associated with digestive disorders, metabolic disturbances, and an elevated cardiovascular risk in patients with dyslipidemia. Emerging evidence suggests that SIBO may aggravate lipid metabolism abnormalities by disrupting the gut microbiota.

Aims & Methods: The aim of the study was to determine the prevalence of SIBO among patients with dyslipidemia and evaluate the impact of SIBO-targeted therapy on lipid metabolism correction. The study included 342 patients with dyslipidemia. SIBO was diagnosed via hydrogen breath testing. Patients received a combined therapy with rifaximin (400 mg TID for 14 days) followed by *Saccharomyces boulardii* CNCM I-745 (250 mg BID for 14 days). Clinical symptoms, lipid profiles (total cholesterol, triglycerides, LDL-C, HDL-C), apolipoprotein levels (ApoA1, ApoB), and homocysteine were assessed before and after treatment.

Results: SIBO was detected in 45.3% of patients with dyslipidemia. Combined therapy significantly reduced the prevalence of SIBO to 18% and alleviated clinical symptoms such as abdominal pain, bloating, meteorism, and stool disturbances.

Notably, therapy led to significant improvements in lipid profiles: reductions in total cholesterol, LDL-C, triglycerides, ApoB, and homocysteine levels, alongside increases in HDL-C and ApoA1. These improvements were comparable between patients receiving statin therapy and those not receiving it.

Conclusion: SIBO is highly prevalent in patients with dyslipidemia and may contribute to lipid metabolism disorders and cardiovascular risk. Targeted therapy with rifaximin and *S. boulardii* effectively reduces SIBO prevalence, improves gut symptoms, and enhances lipid parameters. Integrating SIBO treatment into dyslipidemia management strategies offers promising new therapeutic opportunities for cardiovascular risk reduction.

Disclosure: Nothing to disclose.

PP0277

NON-SURGICAL MANAGEMENT OF CANCER-RELATED SMALL BOWEL OBSTRUCTION VIA DSA-GUIDED SMALL BOWEL DECOMPRESSION TUBE PLACEMENT: A NOVEL THERAPEUTIC

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Introduction: Cancer-related small bowel obstruction (SBO) is a common and debilitating complication in patients with advanced cancer. It often requires surgical intervention, but in patients with multiple comorbidities or poor surgical prognosis, non-surgical alternatives can provide significant symptom relief.

This study presents a series of three cases where digital subtraction angiography (DSA)-guided small bowel decompression tube placement was successfully utilized as a non-invasive approach to manage cancer-related SBO.

Aims & Methods: We report three cases of advanced cancer patients who presented with SBO. The patients were diagnosed with mechanical obstructions due to rectal cancer and metastatic breast cancer. After comprehensive pre-operative evaluation, DSA-guided small bowel decompression tube placement was performed as a non-surgical treatment. The procedure involved introducing a multifaceted catheter through DSA-guided access, with real-time contrast-enhanced fluoroscopy used to position the tube in the distal small bowel. Post-procedural clinical outcomes, including symptomatic relief, bowel function restoration, and complication rates, were recorded and analyzed.

Results: The DSA-guided small bowel decompression tube placement led to immediate symptom improvement in all patients with cancer-related small bowel obstruction (SBO). In one patient with rectal cancer and post-surgical recurrence, the procedure resulted in significant relief from abdominal distension and pain. In another patient with metastatic breast cancer and multi-organ involvement, the DSA-guided intervention significantly reduced nausea and vomiting, improved gastrointestinal motility, and allowed for an uneventful recovery. A third patient, post-rectal cancer resection, experienced recurrent SBO, and after the DSA-guided procedure, the obstruction was alleviated, enabling the patient to resume

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enteral feeding and spontaneous bowel movements without the need for surgery. Across all cases, patients demonstrated substantial symptom relief, restored bowel function, and experienced minimal complications, with no procedure-related mortality or major adverse events, underscoring the safety and effectiveness of this technique as a non-surgical alternative for managing cancer-related SBO.

Conclusion: DSA-guided small bowel decompression tube placement is a safe and effective non-surgical alternative for managing cancer-related SBO. This approach provides symptomatic relief, preserves the patient's quality of life, and may reduce the need for invasive surgical interventions, particularly in patients with advanced cancer and poor surgical candidacy. Further studies with larger sample sizes are necessary to validate the long-term benefits and outcomes of this technique in clinical practice.

Disclosure: There are no conflicts of interest to disclose.

PP0278

IDENTIFICATION OF TWO SUBGROUPS AMONG PATIENTS AFFECTED BY CHRONIC IDIOPATHIC INTESTINAL PSEUDO-OBSTRUCTION

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Introduction: Chronic idiopathic intestinal pseudo-obstruction (CIIPO) is a rare disease, characterized by symptoms and signs mimicking mechanical obstruction of the gut in the absence of organic obstacles to intestinal transit (1). It is the most severe clinical expression of intestinal dysmotility that markedly affects quality of life, precipitates relevant personal and social costs and may lead to death affected individuals. Not all patients present with the same clinical picture. Although the typical manifestation of CIIPO is characterized by recurrent subocclusive episodes (RSE) that are separated by periods with digestive symptoms of variable severity, some patients present with a chronic subocclusive condition (CSC) with unremitting clinical and radiological features of intestinal mechanical obstruction.

Aims & Methods: To compare clinical and small bowel manometric features of RSE and CSC CIIPO patients seen in a single referral center.
Family history, onset of digestive symptoms, surgical procedures, digestive symptoms, subocclusive episodes (if applicable), body mass index, and feeding habits were recorded at the time of diagnosis (DIA) and at follow up (FU) by a structured questionnaire. Small bowel manometry was performed by perfusion techniques and abnormal motor patterns were visually identified.

Results: Between 1985 and 2023 we evaluated 125 CIPO patients without underlying systemic, metabolic diseases or mitochondrial cytopathies (CIIPO). Of these, 15 were lost after diagnosis. Of the remaining 110 patients (84 F; age at DIA: 33.06±13.4yrs; m±SD) 97 had RSE (76 F; 37.9±13.0 yrs) and 13 CSC (11 F; 12%; 34.6±16.1 yrs). Duration of FU was 10.5±9.2 yrs, range 1-36 yrs. At FU 90 still had RSE (82 F: 48.5±15.3 yrs) and 20 CSS (18%; 18 F:43.9±15.1 yrs). Clinical features of RSE at DIA and FU are summarized in the table.

	RSE at DIA	RSE at FU	CSC at DIA	CSC at FU
Body mass index	18.17 ± 4.6	19.27±3.5	16.99±2.3	18.28±4.2
Parenteral nutrition (%)	15 (15.5%)	26 (28.9%)	4 (23.1%)	15 (75.0%) ° *
Abdominal surgeries / yr	0.39 ±0.5	0.08±0.1^	0.61±0.5 °°	0.19 ±0.3 °
Restorsternal burning	15 (15.5%)	21 (23.3%)	2 (11.8%)	9 (45.0%) *
Epigastric pain	30 (30.9%)	32 (35.4%)	2 (15.4%)	12 (60.0%) ° *
Nausea	59 (60.8%)	33 (36.7%) ^	7 (53.9%)	15 (75.0%) *
Constipation	62 (63.95)	47 (52.2%)	11 (84.6%)	10 (50%) °

*P<0.05 °CSC FU vs CSC DIA; * CSC FU vs RSE FU; °° CSC DIA vs RSE DIA; ^ RSE FU vs RSE DIA*

The following manometric patterns were detected in 88 diagnostic tracings obtained: abnormal activity fronts 53 (52.5%); minor activity fronts abnormalities 60 (59.4%); hypocontractility 8 (7.9%); inability to convert fasting vs feeding patterns 7 (6.9%), discrete clustered contractions 23 (23%). Not significant differences were detected between the two CIIPO subgroup. 23 patients dead during the FU: 8 (34.8%) in the CSC group and 15 (16%) in the RSE group (P< 0.05). Main causes of death were: sepsis / cachexia (63.9%), transplant failure (19.4%), others (17%).

Conclusion:
CIIPO remains an often unrecognized disease, characterized by disabling and potentially life-threatning complications over time. Two subgroups exists among CIIPO patients. The continuos subocclusive condition (CSC) is more severe and may represent a dangerous evoultion of the episodic subtype (RSE).

Reference:
Stanghellini V, Cogliandro RF, De Giorgio R, Barbara G, Morselli-Labate AM, Cogliandro L, Corinaldesi R. Natural history of chronic idiopathic intestinal pseudo-obstruction in adults: a single center study. Clin Gastroenterol Hepatol. 2005;3:449-58.

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PP0279

APOB-48: A PROMISING BIOMARKER FOR RESPONSE TO APRAGLUTIDE IN SHORT BOWEL SYNDROME WITH INTESTINAL FAILURE AND COLON-IN-CONTINUITY

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Introduction: Citrulline is frequently used as surrogate marker for enterocyte mass in short bowel syndrome (SBS), but also to assess response with analogs of glucagon-like peptide 2 (GLP-2) in intestinal failure (IF). However, in SBS with colon-in-continuity (CiC), citrulline might not be the best marker given the short lengths of preserved small intestine. Therefore, we aimed to identify more informative biomarkers for SBS-IF-CiC in relation to GLP-2 analog therapy.

Aims & Methods: STARS Nutrition was a 52-week multicenter, open-label, phase 2 study in adult patients with SBS-IF-CiC (n=9) receiving weekly subcutaneous injections with the novel, long-acting GLP-2 analog apraglutide. A fasting blood sample was collected at baseline and at 4, 24 and

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48 weeks of treatment. Apolipoprotein B-48 (ApoB-48), intestinal fatty-acid binding protein (iFABP) and lipopolysaccharide-binding protein (LBP) concentrations were measured (ELISA).

Results: At baseline, ApoB-48 levels correlated positively to the preserved small intestinal length (r=0.7983). With apraglutide, the ApoB-48 levels increased significantly at all timepoints (Table 1). While iFABP concentrations remained stable at 4 weeks, a significant increase was observed at 24 weeks. At 48 weeks, iFABP levels continued to be elevated but failed to reach statistical significance. LBP concentrations were not significantly affected by apraglutide.

	Baseline	4 weeks	24 weeks	48 weeks
ApoB-48 (ng/mL)	1,001 (165.8 – 22,400)	1,549 (909.5 – 31,830) p=0.0273	1,616 (899.4 – 65,390) p=0.0078	1,978 (767.2 – 101,900) p=0.0039
iFABP (pg/mL)	102.0 (17.77 – 4,409)	101.5 (45.66 – 4,468) p=0.25	149.3 (83.26 – 4,467) p=0.0078	135.1 (30.69 – 4,507) p=0.0742
LBP (ng/mL)	19,730 (11,830 – 47,860)	19,350 (11,950 – 32,340) p=0.82	23,030 (11,340 – 27,020) p=0.91	18,870 (10,910 – 23,070) p=0.20

Table 1 Systemic plasma levels of ApoB-48, iFABP and LBP.
Data are presented as median (min – max). Nominal p-values were calculated using Wilcoxon matched-pairs signed rank tests (significance ≤0.05). ApoB-48: apolipoprotein-B48; iFABP: intestinal fatty-acid binding protein; LBP: lipopolysaccharide binding protein. p≤0.05

Conclusion: Apraglutide treatment resulted in distinct increases of ApoB-48, a protein synthesized by the enterocytes and involved in fat absorption. Our results suggest that ApoB-48 could be a suitable candidate as biomarker for treatment response as not only it reflects the number of enterocytes but also their ability to absorb and transport nutrients, especially lipids.

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Tim Vanuytsel: TV has served on the speaker bureau for Fresenius Kabi, Remedus, Takeda, and VectivBio. TV has provided clinical advice to Baxter, Shire, Takeda, VectivBio, Zealand Pharma. TV has received research grants from Takeda and VectivBio.

PP0280

RIFAXIMIN FOR SMALL INTESTINAL BACTERIAL OVERGROWTH. CAN WE PREDICT THE RESPONSE?

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Introduction: Small intestinal bacterial overgrowth (SIBO) and intestinal methanogen overgrowth (IMO) are increasingly recognised as common causes of non-specific gastrointestinal symptoms. Diagnosis and treatment remain challenging in clinical practice. Rifaximin can be an expensive medication in certain healthcare settings.

This retrospective audit evaluated the efficacy and safety of rifaximin for the treatment of SIBO and IMO at Cambridge University Hospitals. Also, the diagnostic value of the glucose hydrogen breath test (GHBT) vs clinical judgement and potential predictors of response to rifaximin were explored.

Aims & Methods: 124 patients treated with rifaximin for suspected bacterial overgrowth over one year were reviewed. Data collected included rifaximin dosing, presenting symptoms, risk factors for bacterial overgrowth, GHBT results (if available), concurrent use of neomycin, treatment response and reported side effects. Predictive factors for response were analysed using two-tailed t-tests and Fisher's exact test.

Results: Rifaximin improved symptoms in 68.3% of patients, with only 8.9% reporting side effects, of which none were severe.

Among the 124 patients, 69 had a positive GHBT, 8 had a negative GHBT result and 47 were not tested. There was no significant difference in response rate to rifaximin treatment between these groups (68.3%, 50%, 68.3% respectively, p=0.077).

Among patients with a positive GHBT, 72% had SIBO, 25% IMO and 3% had SIBO and IMO. Response rate was significantly higher in patients diagnosed with SIBO by GHBT (80%) compared to patients diagnosed with IMO (18%) (p<0.0001). 9 out of 17 patients with IMO received combination therapy with both rifaximin and neomycin.

The number of risk factors for bacterial overgrowth was significantly associated with higher response rates, with responders having an average of 1 risk factor compared to non-responders having an average of 0.65 risk factors (p=0.003).

Type and number of presenting symptoms, as well as dosing of rifaximin did not show a consistently significant association with response.

Conclusion: This audit supports the effectiveness and safety of rifaximin as a first line treatment for SIBO. Patients with IMO did not benefit from any regimen including rifaximin. GHBT result did not affect the response rate to rifaximin overall. The presence of risk factors for bacterial overgrowth was associated with significantly higher response rates, suggesting the potential utility of this observation as a predictive factor of response. Based on these findings, and considering the costs of rifaximin, we propose a clinical algorithm that prioritises rifaximin as first line treatment in a group of patients with higher chances to respond.

Disclosure: Nothing to disclose.

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PP0281

SOCIAL DETERMINANTS OF ABDOMINAL TUBERCULOSIS VISITING LARGE TERTIARY CARE HOSPITAL OF COUNTRY JPMC KARACHI

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Introduction: Abdominal tuberculosis is a major concern in developing countries due to socioeconomic factors, influencing disease severity. This study at Jinnah Postgraduate Medical Centre (JPMC), Karachi, aims to explore the link between social determinants like socioeconomic status, residence, co-morbidities, addiction patterns, and abdominal tuberculosis severity in patients.

Aims & Methods: This study aimed to analyze prevalent social determinants in high TB-prevalent countries. Anticipated insights will aid in developing strategic interventions for TB. By highlighting social factors affecting TB, the study can guide targeted policies and interventions. The generated knowledge can enhance public health planning in high prevalence settings.

An ongoing descriptive cross sectional study conducted at the Department of Gastroenterology, JPMC, Karachi, from 2024 till date. A total of 120 patients diagnosed with abdominal tuberculosis were included. Data on social determinants such as socioeconomic status, residence, addiction, HIV status, diabetes, and malnutrition were collected.

Disease severity was assessed based on ADA levels and categorized as mild, moderate, or severe. Associations between social determinants and abdominal tuberculosis severity were analyzed using Chi-square tests with a significance level of $p \leq 0.05$.

Results: Total 120 patients were observed,The average patient age was 47.18 ± 20.56 years, with 51.7% males and 48.3% females. Low and middle socioeconomic groups accounted for most patients.

Malnutrition affected 32.5% of cases, with 19.2% HIV positive. Tobacco chewing (27.5%), smoking (15.0%), and alcohol consumption (14.2%) were common addictions. Urban areas housed 57.5% of cases, with most from Karachi (36.7%).

No social determinant showed significant association with abdominal tuberculosis severity ($p > 0.05$), but malnutrition ($p = 0.082$) and diabetes ($p = 0.298$) hinted at greater severity. Smoking and tobacco use were prevalent in moderate-to-severe cases ($p = 0.261$).

Conclusion: This study at JPMC, Karachi, found potential links between social factors like malnutrition, diabetes, and addiction, and intestinal tuberculosis severity. Though no significant associations were observed ($p > 0.05$), trends suggest these factors may influence disease progression.

Patients from low socioeconomic backgrounds and urban areas were more affected, emphasizing the need for targeted interventions such as nutritional support, diabetes management, and tobacco cessation to enhance outcomes.

Larger studies are necessary to confirm these connections and devise effective public health initiatives for at-risk populations

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Disclosure: Nothing to disclose.

PP0282

PANCREATIC EXOCRINE INSUFFICIENCY IN CELIAC DISEASE: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Pancreatic exocrine insufficiency (PEI) is seen in primary pancreatic disease, but has also been seen in extrapancreatic conditions including celiac disease (CeD). The symptoms of PEI and CeD often overlap, which makes differential diagnosis challenging.

Aims & Methods: We aimed to investigate the prevalence of PEI in CeD in a systematic review and meta-analysis.

We searched five databases: PubMed, Embase, Cochrane, Web of Science Core Collection, and Google Scholar, up until October 21, 2024. The pooled prevalence of PEI in biopsy-confirmed CeD was estimated. The quality of studies was appraised, and results were reported according to the PRISMA guidelines.

Results: We identified and screened the titles and abstracts of 1432 publications, of which 60 were reviewed in full text and 12 were included in the analyses. The overall pooled weighted prevalence of PEI in CeD was 13.5% (95% CI 7.2–21.0). The prevalence was similar among children (14.2%; 95% CI 2.0–32.8) and adults (12.8%; 95% CI 7.8–18.7) with CeD.

The prevalence in studies that used secretory testing to define PEI was 13.1% (95% CI 6.4–21.3) and in those requiring digestive tests 17.1% (95% CI 5.8–32.1). The weighted prevalence of PEI was however significantly higher among untreated CeD patients (21.7%) than patients on a gluten-free diet (6.9%) ($p=0.03$), in both adults and children.

Conclusion: One in eight individuals with CeD may suffer from PEI. The prevalence was particularly high in children with untreated CeD. PEI should be considered in patients who do not respond to a gluten-free diet.

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Disclosure: Nothing to disclose.

PP0283

FIBROMYALGIA SCREENING IN COELIAC DISEASE: A CROSS-SECTIONAL STUDY USING THE FIRST QUESTIONNAIRE

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Introduction: Fibromyalgia and coeliac disease share chronic symptoms that have a significant impact on patients' quality of life. While coeliac disease is widely recognized and well-documented, fibromyalgia remains frequently underestimated. Although coeliac disease is widely recognized, fibromyalgia is often underestimated and the association between these two pathologies has been little studied, leaving gaps in understanding their possible connections and implications for patient care.
Aims & Methods: The aim of the study was to determine the frequency of fibromyalgia positive screening in patients with coeliac disease and to investigate the associated factors. A cross-sectional study was conducted including patients with coeliac disease during September 2024. Participants were asked to complete the Fibromyalgia Rapid Screening Tool (FiRST) questionnaire, a validated screening tool for fibromyalgia, which assesses the presence and intensity of associated symptoms.
The questionnaire included 6 items: body pain, fatigue, sleep quality, cognitive impairment, mood disorders and impact on daily life. Positive screening for fibromyalgia was based on a FiRST score ≥ to 5 items.
Results: A total of 60 patients participated to the study, with a mean age of 45.35 years. The M/F sex ratio was of 0.22. According to the FiRST questionnaire, fibromyalgia was detected in 43.3% of patients (n=26). The mean score on the questionnaire was 3.58. The positive screening of fibromyalgia in patients with coeliac disease was significantly associated with the presence of associated auto-immune diseases (p=0.029), Helicobacter pylori gastritis (p=0.045), anemia (0.012), hypoferritinemia (p=0.047), hypocholesterolemia(p=0.026) and hypovitaminosis D (p=0.031).
However, no significant association was found between positive fibromyalgia screening and age (p=0.68), gender (p=0. 43), physical activity (p=0.53), type of occupation (p=0.66), gluten-free diet compliance (p=0.31), body mass index (p=0.107), intestinal villous atrophy (p=0.88) and anti-transglutaminase antibodies positivity (p=0.60).
Conclusion: Our study showed a notable frequency of fibromyalgia positive screening in patients with coeliac disease. This suggests a link between the autoimmune inflammatory process associated with gluten and the development of central nervous system sensitivity responsible for fibromyalgia symptoms. Early recognition of this comorbidity and associated factors is important to improve management and quality of life in patients with both conditions.
Disclosure: Nothing to disclose.

PP0284

MICROBIOTA IN PATIENTS WITH CELIAC DISEASE: FROM DIAGNOSIS TO FOLLOW-UP

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Introduction: The intestinal microbiota is defined as the community of microorganisms inhabiting the gastrointestinal tract, which confer benefits to the organism and play a role in immune system regulation. It has been suggested that the microbiota may play a role in the pathogenesis of gut diseases, including Celiac Disease (CeD), and the response to a gluten free diet (GFD). (1)
Intestinal dysbiosis may be a contributing factor to the inflammatory state, thereby hindering resolution of symptoms and complete mucosal healing, particularly in patients who show a lack of clinical and histological improvement despite following a GFD. (2,3)
Aims & Methods: We aimed to provide a preliminary evaluation of the gut microbiota profile from diagnosis to follow-up, classifying patients who respond and do not respond to diet in order to establish a first baseline of the microbial composition and its relationship to the dietary response.
We collected fecal samples from patients enrolled at the Gastroenterology Unit of Azienda Ospedale- Università of Padova, from March to November 2024. All patients needed to be off antibiotics, probiotics and proton-pump inhibitors for at least one month before enrolment. Stool samples were collected and transferred in a buffer solution, and stored at -20°C.
Fecal samples were analyzed by BMR Genomic s.r.l. (Padua, Italy) that sequenced the V3-V4 hypervariable 16S ribosomal RNA region using the Illumina MiSeq. The results were compared to an internal population of healthy subjects (provided by BMR genomics) using Qiime2 tools. Statistical analysis was conducted using STATA 18 software. Categorical variables were compared using the Chi-square test, while non-parametric continuous variables were analysed using the Mann-Whitney test.
Results: We enrolled 71 CeD patients with mean age at diagnosis of 35.76 ± 16.29 years old (50 out of 71 were female). Of the 71 patients, 9 were newly diagnosed with CeD, while 62 were in follow-up. Among follow-up groups, 44 of 62 patients (71%) were classified as responders to diet and 18 (29%) were classified as non- responders, exhibiting classical symptoms, positive serology and histology. We compared the relative abundance of the main bacterial phyla in each sample: a significant decrease in beneficial phyla such as Actinobacteria (p = 0.04) and a decrease close to the significant threshold in Firmicutes (p = 0.06) was observed after the follow-up. This factor could be ascribed to a long-term dietary regime rather than to the disease itself. (4)
Proteobacteria were more abundant in follow-up, especially in non-responders. This phylum includes many potentially pathogenic microorganisms belonging to Escherichia, Klebsiella and Shigella genera. Comparing

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diagnosis and follow-up patients, Clostridium perfringens was significantly more present in patients during follow-up (p=0.02), while Escherichia coli showed greater presence in patients at diagnosis (p=0.035) and in non-responders, consistently with his role in the inflammatory phase of the disease. (5)

Conclusion: Our preliminary data suggest that a long-term GFD does not always restore completely the intestinal dysbiosis present at the diagnosis and could be associated with specific alterations in beneficial bacteria. The presence of abundant Proteobacteria at diagnosis and in non-responder patients may be considered as biomarkers of active state of CeD. Future studies on various biological samples, as well as a more extensive investigation into the impact of long-term GFD, are required to highlight microbiota-immunity relationships involved in development of CeD.

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PP0285

PREVALENCE OF COELIAC DISEASE IN PATIENTS WITH ROME III AND ROME IV IRRITABLE BOWEL SYNDROME: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Irritable bowel syndrome (IBS) and coeliac disease are common disorders that share overlapping symptoms. In this systematic review and meta-analysis, we aimed to provide up-to-date and comprehensive estimates of the prevalence of coeliac disease in patients with IBS.

Aims & Methods: We searched several databases through January 2025 for studies reporting the prevalence of coeliac disease in patients with IBS. Eligible studies used Rome III or Rome IV criteria for IBS diagnosis and used serological screening with tissue transglutaminase, endomysial antibodies, or deamidated gliadin peptide, and/or confirmatory duodenal biopsies for coeliac disease diagnosis. We used random-effects meta-analysis to estimate the pooled prevalence of seropositive and biopsy-proven coeliac disease with 95% confidence intervals (CI). We calculated pooled odds ratios (ORs) to compare the likelihood of coeliac disease between patients with IBS and controls.

Results: A total of 29 studies comprising 7,209 patients with IBS were included. The pooled seroprevalence of coeliac disease in patients with IBS was 6% (95% CI, 5% - 8%), and the pooled prevalence of biopsy-proven coeliac disease was 2% (95% CI, 2% - 3%). A significant proportion of seropositive patients (15%; 95% CI, 6% - 24%) did not undergo endoscopy and biopsy. Patients with IBS had significantly higher odds of a positive serology than controls (OR 4.42; 95% CI, 2.82 – 6.92). The odds of coeliac disease were similar across genders and IBS subtypes.

Conclusion: Coeliac disease is highly prevalent in patients with IBS, according to the Rome III and Rome IV criteria. A positive diagnosis of IBS should not be made without excluding coeliac disease.

Disclosure: MGS received speaker fees from Thermo Fisher. IA received speaker fees from PrecisionBiotics.

PP0286

COELIAC DISEASE IN BONE FRACTURE PATIENTS: A 1000 PATIENT PILOT SCREENING STUDY

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Introduction: Coeliac disease (CD) affects 1% of the population in the United Kingdom (U.K.) but is often underdiagnosed[1]. Complications of CD include detrimental effects on bone health and increased fracture risk[2]. Currently, evidence supporting routine serological testing for CD in adults with fractures is limited.

Aims & Methods: The aims of this study were to:

1. Determine whether the prevalence of CD among adult fracture patients exceeds the expected 1% in the general population.

2. Characterise demographic and fracture-site patterns to identify high-risk subgroups.

This cross-sectional observational study was conducted at a single U.K. fracture clinic with 1000 adults (≥16 years) with a fracture recruited between April 2023 and April 2025. Patients under 16 or unable to provide consent were excluded. Patients with known CD were included. Demographic data (age, gender, ethnicity) and fracture sites were recorded.

Anti-tissue transglutaminase antibody (tTG-IgA) tests were performed unless known to have CD. Equivocal tTG-IgA results were reviewed by a gastroenterologist, and repeat testing was conducted if needed. Seropositive participants were invited for duodenal biopsy. Laboratory processing issues were treated as negative results. Statistical analyses included descriptive demographic analyses and prevalence calculations.

Results: Baseline characteristics are summarised in Table 1. Of 1000 participants, 978 participants underwent tTG-IgA testing, of which 5 participants tested positive, yielding a seroprevalence of 5/1000 (0.5%). Duodenal biopsy confirmed CD in 2 participants 2/1000 (0.2%), while 1 seropositive participant did not attend biopsy, 1 is awaiting biopsy, and 1 had a negative biopsy. 28 participants were classified as having negative results. This group included untested samples (n=15), equivocal results (n=6), and known CD (n=7).

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The overall prevalence of CD in our study population was 9/1000 (0.9%): 7 known and 2 newly confirmed CD. The CD population were predominantly white, and upper limb fractures were most common, aligning with prior studies[3,4].

Variable	Categories	Frequency (%)	Known coeliac disease (n=7) (%)	New seropositive patients (n=5) (%)
Age in years (mean ± SD, median, range)	-	54.2 ± 17.7, 56, 18–89	55.4 ± 26.0, 64, 24–85	47.8 ± 18.6, 40, 30–74
Gender	Male	371 (37.1)	2 (28.6)	2 (40)
	Female	628 (62.8)	5 (71.4)	3 (60)
	Non-binary	1 (0.1)	0	0
Ethnicity	White	951 (95.1)	6 (85.7)	5 (100)
	Other*	49 (4.9)	1 (14.3)	0
Fracture site	Upper limb	628 (62.8)	5 (71.4)	4 (80.0)
	Lower limb	364 (36.4)	2 (28.6)	1 (20.0)
	Spinal/ Other	3 (0.3)/ 5 (0.5)	0	0

Table 1.

** Asian or Asian British, Mixed or multiple, Black, Black British, Caribbean, or African, and Other ethnic group.*

Conclusion: This represents one of the largest evaluations of CD prevalence in fracture patients. The observed prevalence of CD 9/1000 (0.9%) closely aligns to the expected UK prevalence for CD. However, given only 5/1000 (0.5%) tested positive serologically and 2/1000 (0.2%) confirmed through biopsy, these findings suggest that routine screening in fracture clinics is unlikely to be efficient. Instead, we should explore targeted screening in fracture clinics, reserving testing for patients with iron-deficiency anaemia, chronic GI symptoms, or a first-degree family history of coeliac disease.

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Disclosure: Nothing to disclose.

PP0287

OSTEOPROTEGERIN AS A MARKER OF BONE HEALTH AND NUTRITIONAL STATUS IN PEDIATRIC CELIAC DISEASE: A COMPARATIVE STUDY OF NEWLY AND LONG-TERM DIAGNOSED PATIENTS BEFORE AND AFTER GLUTEN-FREE DIET

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Introduction: Osteoprotegerin (OPG), a regulator of bone turnover, may serve as a marker of skeletal and nutritional status in pediatric celiac disease (CD). This study evaluates the relationship of OPG with growth, bone age, and biochemical parameters before and after gluten-free diet (GFD) intervention.

Aims & Methods: The study included two cohorts of CD patients treated at the Department of Pediatrics, St. Marina University Hospital, Varna, Bulgaria (2018–2021): newly diagnosed patients (Group A, n = 25) and long-term diagnosed patients (Group B, n = 54). Parameters included weight, height, calcium, magnesium, phosphorus, alkaline phosphatase (ALP), ferritin, hemoglobin, PTH, bone age, osteocalcin and OPG. Data were analyzed at baseline and after six months of GFD using Pearson and Spearman correlation coefficients.

Results: Group A (7.28±4.56 years, male n = 14 [56%]) and Group B (8.15±3.76 years, male n = 16 [29.6%], p = 0.376) were analyzed for growth and bone turnover. In Group A, OPG negatively correlated with bone age before (r = -0.325, p = 0.017) and after diet (r = -0.335, p = 0.013), and positively with hemoglobin and iron post-diet (r = 0.301, p = 0.029). In Group B, similar correlations appeared after diet: bone age (r = -0.325, p = 0.017) and iron (r = 0.297, p = 0.029). OPG did not correlate significantly with vitamin D in either group (p > 0.15).

Conclusion: OPG shows consistent negative correlation with bone age and positive association with iron status after GFD, supporting its role as a marker of skeletal and nutritional recovery in pediatric CD.

Disclosure: Nothing to disclose.

PP0288

DETERMINATION OF ANTI-TISSUE TRANSGLUTAMINASE ANTIBODIES IN THE LONG-TERM FOLLOW-UP OF ADULT CELIAC DISEASE

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Introduction: Anti-tissue transglutaminase IgA antibodies (tTG) are used in the follow-up of celiac disease (CD) to assess adherence to a gluten-free diet (GFD). Persistent positivity may indicate the need for histological reassessment. There is limited information about the rate of complete normalization of tTG and associated factors. Our objective was to evaluate the percentage, kinetics, and factors related to tTG normalization.

Aims & Methods: This was a retrospective observational study including patients with tTG testing at diagnosis and follow-up monitoring of these antibodies from 1991 to 2023 in a specialized unit dedicated to patients with CD. Inclusion criteria were: patients over 16 years of age at diagno-

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sis, initially elevated tTG titres, and at least one serological follow-up after starting a GFD. Time to tTG normalization was estimated using the Kaplan-Meier method. Comparisons between groups were performed using the log-rank test. The association between demographic and clinicopathological characteristics at diagnosis and time to tTG normalization was assessed using Cox regression analysis.

Results: A total of 179 patients were included (75% female, mean age 40 ± 16 years). Duodenal atrophy was present in 81%, and 79% had symptoms. 83% strictly adhered to a GFD. Median follow-up was 3 years. The incidence rate of tTG normalization was 35% per patient-year. The cumulative tTG normalization rate was 38% (95% CI: 30-45) at 1 year, 61% (53-69) at 2 years, 70% (63-77) at 3 years, 80% (73-87) at 4 years, 81% (74-89) at 5 years, and 95% (88-100) at 10 years. The median time to tTG normalization was 24 months (95% CI: 20–28). *Table 1* shows Kaplan–Meier estimates expressed as percentages of tTG normalization by adherece to GFD and baseline tTG. In the multivariable analysis, two factors were found to be significantly associated with the normalization of tTG: lower baseline tTG and good adherence to a GFD. Regarding baseline tTG level, patients with titres ≤ 286 U/ml (median) had 2.46 times higher likelihood of antibody normalization compared to those with higher titres (HR: 2.46; 95% CI: 1.69–3.56; p < 0.001). Regarding adherence to a GFD, patients who followed a strict GFD had 5.38 times higher likelihood of tTG normalization (HR: 5.38; 95% CI: 2.79–10.36; p < 0.001).

Months		12	24	36	48	60
GFD	Yes	43 (35-51)	69 (60-76)	79 (70-85)	89 (83-94)	91 (84-95)
	No	11 (0.2-22)	26 (10-42)	31 (13-49)	37 (15-59)	-
Baseline tTG titres	≤ 286*	56 (45-66)	80 (71-89)	83 (75-92)	85 (77-94)	-
	> 286*	19 (11-27)	42 (31-53)	56 (44-68)	75 (63-86)	77 (66-88)

Table 1. Kaplan–Meier estimates expressed as percentages of normalization (with 95% CIs) after CD diagnosis, by adherece to GFD and baseline tGA titres.

*CD: celiac disease; GFD: gluten-free diet; tTG: anti-tissue transglutaminase IgA antibodies; CI: confidence interval; *median*

Conclusion:

1. The incidence rate of tTG normalization in CD patients after initiating a GFD was 35% per patient-year of follow-up.
2. In the long-term (10 years), 95% of CD patients achieved tTG normalization.
3. Lower tTG titres at diagnosis and good adherence to the GFD were predictive factors for tTG normalization.
4. Understanding the kinetics of tTG normalization according to various factors is essential before considering histological reassessment for suspected poor disease progression.

Disclosure: Nothing to disclose.

PP0289

CELIAC DISEASE AND FERTILITY: A RETROSPECTIVE STUDY OF PATIENT RECORDS IN WOMEN WITH CELIAC DISEASE FROM A TERTIARY HOSPITAL

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Introduction: Celiac disease, a chronic autoimmune enteropathy caused by exposure to gluten in genetically predisposed individuals, is characterized by a broad clinical spectrum often with multisystem involvement. Symptoms related to the reproductive system include delayed menarche, early menopause, fertility disorders, miscarriages, and prematurity.

Aims & Methods: To study the impact of celiac disease on fertility in a female Greek population. A retrospective review of all patient records from the celiac disease outpatient clinic at our hospital.

Results: Medical histories of 231 female patients were reviewed (69% of the total patient population) with a mean age of 44.3 years and a mean age at diagnosis of 35.3 years. The mean age of menarche was 12.7 years. Menstrual irregularities were reported in 27% (20% instability, 7% dysmenorrhea). A total of 28% (65/231) were in menopause, with an average age of menopause onset at 50.7 years.

A history of miscarriages was recorded in 15% of all women (35/231), but in 29% of women in menopause (19/65). 82% (53/65) of women in menopause had children, with 98% (52/53) having been pregnant before the diagnosis of the disease.

Among all women who gave birth, only 12/113 (11%) had been diagnosed before pregnancy. 56% (63/113) and 44% (50/113) had a vaginal delivery and cesarean section, respectively. Pregnancy complications were recorded in 24 women (18 preterm births, 15 low birth weight infants, 1 intrauterine death).

Conclusion: Women with celiac disease tend to have relatively late menarche and an increased rate of miscarriages. However, the disease does not seem to significantly affect fertility.

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Disclosure: Nothing to disclose.

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PP0290

BIOMARKERS OF EXTRACELLULAR MATRIX REMODELING REFLECT PHARMACODYNAMIC EFFECTS OF IMU-856, AN ORAL EPIGENETIC MODULATOR OF BARRIER REGENERATION

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Introduction: Celiac disease (CeD) is characterized by intestinal inflammation and epithelial damage upon exposure to immunogenic gluten peptides, directly disrupting tissue architecture and by extension the extracellular matrix (ECM). The intestinal tissue is rich in ECM proteins such as collagens, which, during inflammation, undergo enzymatic remodeling due to increased proteolytic activity. IMU-856 is an orally available and systemically acting small-molecule modulator of sirtuin 6 (SIRT6), a histone/non-histone protein deacetylase and transcriptional regulator that aims to restore intestinal barrier function and regenerate bowel epithelium.

Aims & Methods: We investigated biomarkers of ECM remodeling as potential indicators of disease activity and pharmacodynamic response to IMU-856 in patients with CeD. A total of 39 patients with CeD were included from a sub-study (Part C) of a double-blind, randomized, placebo-controlled phase 1/1b clinical trial of IMU-856.

Plasma was drawn at baseline before dosing and at day 14. Biomarkers of MMP-degraded type III, IV, and VI collagen (C3M, C4M, C6M), and type VII collagen formation (PRO-C7) were measured. Spearman's rank correlation was applied at baseline. Group differences were evaluated using the Mann-Whitney U-test. Mean change from baseline to day 14 was computed.

Results: At baseline, C3M, C4M, and PRO-C7 showed significant inverse correlation with villus height (ρ=-0.32, ρ=-0.34, ρ=-0.33, all $p<0.05$). C4M and PRO-C7 significantly correlated to histological inflammation and mucosal damage (Q-Marsh) at baseline (ρ=0.31, ρ=0.33, both $p<0.05$). PRO-C7 also significantly correlated with the villus height/crypt depth ratio (ρ=-0.32, $p<0.05$) (*Table 1*).

At day 14, C3M and PRO-C7 levels numerically decreased in IMU-856 treated patients by 5.8% and 8.3% while increasing in placebo treated patients by 10.8% and 7.0% respectively. C6M levels decreased numerically by 4% in IMU-856 treated patients with a 0.7% increase in placebo treated patients. Plasma citrulline levels, a biomarker of enterocyte mass and function, increased by 16% in patients treated with IMU-856.

	Q-Marsh	Villus height	Crypt depth	Villus height/crypt depth	Citrulline
C3M	0.28 (0.079)	-0.32 (0.045)	0.10 (0.535)	-0.25 (0.108)	-0.25 (0.114)
C4M	0.31 (0.048)	-0.34 (0.029)	0.11 (0.506)	-0.27 (0.082)	-0.19 (0.241)
C6M	0.07 (0.685)	-0.06 (0.725)	0.17 (0.285)	-0.12 (0.443)	-0.11 (0.512)
PRO-C7	0.33 (0.033)	-0.33 (0.037)	0.24 (0.137)	-0.32 (0.041)	-0.17 (0.292)

Table 1. Correlations between biomarkers and clinical parameters at baseline. Data is presented as Spearman's rho with p-value in brackets.

Conclusion: Biomarkers of ECM remodeling reflect histological inflammation and mucosal architecture parameters such as Q-Marsh score and villus height, offering a direct insight into intestinal barrier integrity. Furthermore, these biomarkers potentially reflect treatment-induced improvement in intestinal tissue remodeling upon treatment with IMU-856.

Disclosure: MSA, JHM: employees of Nordic Bioscience A/S
JHM: shareholder of Nordic Bioscience A/S
AS, MW, EP: employees of Immunic AG, option and shareholders of Immunic, Inc.
SE: employee of Immunic AG, option and/or shareholders of Immunic, Inc.
HK: employee of Immunic AG, shareholder of Immunic, Inc., co-inventor of patents for the drug in investigation
DP: option holder of Immunic, Inc.
DV, AM: employees of Immunic AG and Immunic, Inc., shareholders of Immunic, Inc., co-inventors of patents for the drug in investigation.

PP0291

IS IT USEFUL TO MEASURE QUALITY OF LIFE IN THE FOLLOW-UP OF PATIENTS WITH CELIAC DISEASE? A SINGLE-CENTER EXPERIENCE

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Introduction: Celiac disease is a chronic disease that requires lifelong treatment with a strict diet, which has an impact on the health-related quality of life of the patients.

Aims & Methods: To analyse the quality of life in patients with celiac disease that are followed in our centre.

A single-center, prospective, observational study based in the Spanish version of Celiac Disease Quality of Life Survey (CD-QOL), a validated questionnaire.

Results: 53 patients agreed to complete the questionnaire. Mean age of the patients at the moment of the consultation was 39.15 years, while mean age at diagnosis of the patients who agreed to participate was 33.20 years. Of them, 35.8% had university degree, 28.3% completed professional training and 35.8% only had compulsory basic education.

Of these patients, 69.2% didn't have any relative diagnosed with celiac disease, and only the 13.2% used to live with another celiac patient. Just 9.4% of the patients who filled in the questionnaire were members of a celiac patient society.

75.5% of the patients claimed that celiac disease affects their social life, 84.9% declared that they comply properly with the gluten-free diet, although for the 32.7% the adherence to the dietary treatment is difficult.

Of the patients who participated, 33.9% declared that they feel quite / a lot limited by the disease, 27.4% feel quite / much stigmatized because of the illness, 22.7% have quite / a lot of problems in their social life, 32.2% have quite / a lot difficulties travelling, and 22.7% of patients declared that they have quite / a lot the impression to not have a normal life.

Conclusion:

1. Celiac disease may be quite burdensome even in asymptomatic patients who are not supposed to have small bowel damage in their follow-up. According to our series, up to 75.5% of patients have problems in their social life.
2. Health education during the follow-up providing greater knowledge of the disease, support from specialized nurses, and the recommendation to join patients support groups (patient societies) could be tools that may empower celiac patients to help manage their disease and enjoy a normal life.

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Disclosure: Nothing to disclose.

PP0292

CELIAC DISEASE AND SILENT FATTY LIVER: WHAT CHANGES WITH GLUTEN-FREE DIET?

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Introduction: Hepatic manifestations are increasingly recognized in celiac disease (CD), with metabolic steatopathies emerging as a significant extraintestinal feature.
Aims & Methods: We aimed to assess the prevalence of hepatic steatosis in CD patients and to evaluate its evolution under a strict gluten-free diet (GFD).
We conducted a 30-year retrospective study including 344 patients diagnosed with CD based on clinical, serological, and histological criteria. All patients underwent abdominal ultrasonography. Only those with hepatic steatosis and no other identifiable causes of liver fat accumulation were included for further analysis. Evolution under GFD was assessed through clinical, biological, and imaging follow-up.
Results: Among 344 celiac patients, 64 (21.3%) had hepatic steatosis. The mean age was 30.5 years (17–61), with a female predominance (F/M = 4.3). Common symptoms included fatigue and weight loss (100%), abdominal pain (56%), and malabsorptive diarrhea (41%). Liver function tests were abnormal in 60%, with isolated cytolysis being the most frequent pattern. Liver biopsy (n=18) revealed steatohepatitis in 12 patients, mostly grade 1 or 3.
Under a well-followed GFD, 25% showed normalization of liver ultrasound. Among patients with abnormal liver function tests, 36.9% experienced biochemical improvement. Histological improvement was documented in two cases: one patient initially diagnosed with severe grade 3 steatohepatitis showed complete resolution of hepatic steatosis after one year of strict dietary adherence. Another patient with grade 3 steatohepatitis showed partial improvement, regressing to grade 1 on follow-up biopsy.
Conclusion: Metabolic steatopathies affect over one in five CD patients. A gluten-free diet not only improves gastrointestinal symptoms but can also reverse hepatic steatosis. These findings support routine hepatic monitoring in celiac patients and reinforce the systemic benefits of dietary adherence.
Disclosure: Nothing to disclose.

PP0293

CLINICAL, ENDOSCOPIC AND HISTOLOGICAL PROFILE OF SERONEGATIVE CELIAC DISEASE: A MOROCCAN UNIVERSITY SERIES

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Introduction: Celiac disease (CD) is a chronic autoimmune enteropathy triggered by gluten ingestion in genetically predisposed individuals. Diagnosis is based on clinical, biological, serological, and especially histological criteria, including villous atrophy and intraepithelial lymphocytic infiltration (>30%).
Aims & Methods: The aim of this study is to highlight the importance of clinical presentation, biological markers, and histology in the diagnosis of seronegative CD, particularly since serological testing often guides biopsy decisions.
We conducted a retrospective descriptive study over a 30-year period, from January 1995 to March 2025, including all patients with histologically confirmed CD based on jejunal biopsy and clinical response to a gluten-free diet (GFD), but with negative anti-transglutaminase IgA and/or anti-endomysial antibodies. All seropositive patients were excluded.
Results: Out of 344 patients diagnosed with CD in our department, 74 (21.3%) were seronegative. The mean age was 65 years (range: 20–65), with a female-to-male ratio of 3.1.
From a clinical standpoint, nutritional deficiency syndromes were present in 95.9% (n=71), malabsorptive chronic diarrhea in 77% (n=57), diffuse abdominal pain in 71.6% (n=53), weight loss in 54% (n=40), and chronic evacuation constipation in 6.7% (n=5). Clinical examination showed anemia in 70.3% (n=52) and neurological signs in 25.6% (n=19).
Endoscopic findings were suggestive of CD in 43.2% of cases, including duodenal fold reduction in 37.8% (n=28), and a ‚scalloping' appearance in 5.4% (n=4). Duodenal mucosa and folds appeared normal in 44.6% (n=33). Histological analysis of jejunal biopsies in all patients revealed intraepithelial lymphocytosis >30%. Severe villous atrophy (Marsh stage IIIc) was found in 71.5% (n=53), and partial atrophy (Marsh IIIa) in 10.8% (n=8).
Conclusion: In our study, seronegative CD represented 21.3% of all celiac patients. These patients had more severe clinical presentations and more advanced villous atrophy (Marsh stage III), despite negative serology. This underlines the importance of histology and clinical judgment in diagnosing CD in seronegative individuals.
Disclosure: Nothing to disclose.

PP0294

THE NON-BIOPSY DIAGNOSIS OF COELIAC DISEASE: CLARIFYING OR CONFUSING FOR PRIMARY CARE

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Introduction: During the COVID-19 pandemic the British Society of Gastroenterology (BSG) introduced interim guidelines supporting a non-biopsy pathway for the diagnosis of coeliac disease. These guidelines reflect the current recommendations of the American Gastroenterological As-

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sociation (AGA). According to these guidelines, symptomatic or high risk adults with anti-Tissue transglutaminase (TTG) IgA titres ≥10× the upper limit of normal (ULN) and a positive endomysial antibody (EMA) on a repeat sample can be diagnosed without endoscopy and referred directly to a specialist coeliac clinic. Patients who do not meet these criteria should be referred for a gastroscopy (OGD) with biopsy for the confirmation of coeliac, prior to starting a gluten free diet.

Aims & Methods: To assess whether general practitioners (GPs) in the community appropriately referred patients with suspected coeliac disease in line with the AGA and BSG interim non-biopsy diagnostic criteria.

A retrospective review was conducted of all new patients referred to the Coeliac clinic in Galway university hospital between January 2023 and December 2024. Data regarding TTG level at diagnosis, EMA result, and compliance with the diagnostic pathway outlined above was collected for each new referral. This data was collected via Evolve - the hospital's electronic based patient record system.

Results:

- There were 119 "new" referrals to our Coeliac Service between January 2023 and December 2024. Of these, 44 cases were known coeliac who were referred by primary care for re-engagement with the clinic, 18 were referred from secondary or tertiary care including GI surgery, Dermatology, Paediatrics, etc. 4 coincidental diagnoses (i.e. OGD performed for indications other than coeliac disease), and 5 outliers (i.e. with incomplete data or incorrectly referred to coeliac clinic) leaving 48 new coeliac cases referred from primary care.
- 18 of these referrals met the non-biopsy criteria for diagnosis (TTG ≥100 + positive EMA)
  - Within this cohort 66% (n=12) were incorrectly referred directly for an OGD which may have delayed the diagnosis of coeliac disease while the remaining 33% (n=6) were correctly referred for direct review at the coeliac clinic.
- The remaining 29 patients all had TTG values less that 10x ULN. 48% (n=14) of these patients were incorrectly referred to clinic prior to endoscopy and of greater concern, 45% (n=13) were recommended by their primary care physician to start a gluten free diet prior to any review or endoscopic procedure.
- Overall, only 44% (n=21) of referrals were compliant with the AGA/BSG guidance.

Conclusion: There is a notable gap in adherence to AGA and BSG non-biopsy coeliac diagnostic guidance in community settings. Half of patients eligible for a non-biopsy diagnosis underwent unnecessary endoscopy, and the majority of ineligible patients were commenced on a gluten free diet prior to, or without, referral for endoscopy. Further GP education is warranted to align referral practices with evidence-based pathways. Dedicated streamlined referral pathways for coeliac disease should ensure a timely diagnosis without mandatory endoscopy in those fulfilling the criteria for a non-biopsy diagnosis and facilitate a timely endoscopy prior to gluten elimination in those who require histological diagnosis.

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Disclosure: No conflict of interest.

PP0295

SIGNIFICANCE OF PERMEABILITY MARKERS (SERUM I-FABP AND FECAL ZONULIN) AND CORRESPONDENCE TO MUCOSAL ATROPHY IN PATIENTS WITH CELIAC DISEASE

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Introduction: Basic anatomical and functional unit of the epithelial barrier of the small intestine is enterocyte, which plays a key role in assimilation processes nutrients. Therefore, understanding the processes of structure regeneration mucous membrane of the small intestine in celiac disease is of paramount importance meaning.

Aims & Methods: To study the effect of permeability markers (I-FABP of blood serum and fecal zonulin) in patients with celiac disease with atrophy of the mucous membrane of the small intestine.

151 patients with celiac disease was examined: the median (Me) age of those examined was 42 years (Q25-Q75: 30–56 years) (25 (16.6%) men; women - 126 (83.4%). Depending on adherence to AGD, patients were divided into 3 groups: Group I-newly diagnosed patients without AGD (n=58); Group II- patients who do not strictly adhere to AGD (n=38); Group III- patients carefully observing AGD for periods from 6 months to 15 years (n=55). Also serum Fatty-Acid-Binding Protein-I - 1-FABP and zonulin in feces were studied. The reference values for zonulin in feces were 83.15 ng/ml. Reference values for I-FABP determined in the control group are 845 pg/ml.

Results: A significant increase in the level of I-FABP (Me 2045 pg/ml (1690–2350)) in group I compared to the control group ($p < 0.01$). High rates were observed in patients of group I, and averaged 111.6 ng/ml (103.1–124), which was significantly higher than in the control group ($p < 0.01$). The highest levels of I-FABP were observed in patients with the most severe degree of atrophy (Marsh III (A-C)).

The level of I-FABP was significantly different from the control depending on the degree of villous atrophy: with atrophy corresponding to Marsh IIIA - median: 1310 pg/ml, (1212-1461 pg/ml), with atrophy corresponding to Marsh IIIB - median: 2090 pg/ml ml, (1812-2322 pg/ml), and for atrophy corresponding to Marsh IIIC - median: 2058 pg/ml, (1858-2678 pg/ml). The highest levels of fecal zonulin were found in patients with the most severe atrophy (Marsh III (A-C)).

As follows from the presented data, the level of zonulin was significantly different from the control depending on the degree of villous atrophy: with partial atrophy (Marsh IIIA) - median: 97.5 ng/ml, (90-105 ng/ml), with subtotal atrophy (Marsh IIIB)) - median: 107 ng/ml, (96.5-117 ng/ml), and with total (Marsh IIIC) - median: 113.7 ng/ml, (104-128.8 ng/ml).

Conclusion: The average values of I-FABP and zonulin correlated with the degree of mucosa atrophy ($p<0.01$). Each of the studied non-invasive markers of intestinal permeability (serum I-FABP and fecal zonulin) reflects its own level and degree of damage to the STC in patients with celiac disease and can be used in clinical practice to assess ongoing structural changes in STC during a gluten-free diet.

Disclosure: The authors declare that there is no conflict of interest.

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PP0296

ANALYTICAL PERFORMANCE OF A NOVEL, FULLY AUTOMATED MULTIPLEXED MICROARRAY IMMUNOASSAY FOR THE DETECTION OF CELIAC DISEASE AUTOANTIBODIES

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Introduction: Autoantibody detection is a cornerstone in the evaluation of Celiac Disease (CD); however, there is a limited offer of multiplexed testing devices.

Aims & Methods: We assessed the analytical performance of a novel, single-use, multiplexed microarray immunoassay (MosaiQ AiPlex® CD microarray. AliveDx, Switzerland), used with its fully automated high-throughput proprietary system, for the qualitative detection of total IgA and semiquantitative detection of autoantibodies (IgA and IgG isotypes) against tissue transglutaminase (tTG) and deamidated gliadin peptide (DGP), compared with selected devices.

The study was conducted at two sites.

Site 1: North East Innovation Lab (The Newcastle upon Tyne Hospitals NHS Foundation Trust, UK),

Site 2: AliveDx (Eysins, Switzerland).

Included human serum samples were banked, de-identified and tested using the investigational device after characterization as reactive or non-reactive to the respective measurand by Phadia AB assays: tTG IgA (EliA Celikey tTG IgA), tTG IgG (EliA Celikey tTG IgG), DGP IgA (EliA Gliadin^{DP} IgA) and DGP IgG (EliA Gliadin^{DP} IgG). Comparator for total IgA was Tina-quant IgA Gen.2 (Roche Diagnostics Gmbh). Positive percent agreement (PPA) and negative percent agreement (NPA) were calculated. Double-sided 95% Confidence Interval (CI) were calculated using Clopper-Pearson Exact Method.

Results: Compared with the respective devices (equivocal results excluded), the investigational microarray showed PPA ranging from 95.7% to 100% and NPA ranging from 87.5 to 98.7%. Table 1 shows the performance of the individual measurands.

	PPA			NPA		
Measurand / Parameter	PPA value (%)	n/N	95% CI	NPA value (%)	n/N	95% CI
tTG IgA	100	61/61	94.1, 100	98.7	456/462	97.2, 99.5
DGP IGA	95.8	69/72	88.3, 99.1	96.5	439/455	94.4, 98
tTG IgG	100	25/25	86.3, 100	97.1	471/485	95.2, 98.4
DGP IgG	95.7	45/47	85.5, 99.5	95.3	448/470	93, 97
Total IgA	100	158/158	97.7, 100	87.5	35/40	73.2, 95.8

tTG: tissue transglutaminase. PPA: positive percent agreement. NPA: negative percent agreement. CI: confidence Interval.

Table 1. Agreement of the investigational microarray immunoassay with comparators.

Conclusion: The investigational microarray immunoassay showed high level of agreement with the compared singleplex devices, providing laboratorians and clinicians with unified results coming from one multiplexed device. This platform has the potential to assist in the simplification of CD evaluation by simultaneously and automatically analyzing key serological markers.

Disclosure: This study was funded by AliveDx Suisse SA, Eysins, Switzerland. Gerber Gomez, Jose Santiago, Ewa Garyga, Rocio Pasion-Galvan, Helen Laird, Mark Hooper, Susan Seaton, Anthony Yau, Christian Fischer are AliveDx employees.

PP0297

A SYSTEMATIC REVIEW AND META-ANALYSIS ON GLUTEN TOLERANCE IN PATIENTS WITH COELIAC DISEASE: WHEN OCCASIONAL GLUTEN CONSUMPTION DOES NOT RESULT IN DUODENAL ATROPHY

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Introduction: Coeliac disease (CD) is the most common atrophic enteropathy. Currently, the only treatment is a life-long gluten free diet (GFD). However, the reported adherence of strict GFD among adults has been reported to vary between 42% to 91%[1], with patients reporting occasional gluten ingestion without a documented CD activity suggesting a gluten tolerance[2,3]. Data on the outcomes on these patients are scarce and conflicting.

Aims & Methods: In this systematic review and meta-analysis, we aimed to evaluate the proportion of non-adherent CD patients with no signs of atrophy (Marsh>3) at duodenal histology as a scenario indicating a potential tolerance.We searched Medline, PubMed, Web of Science, Cochrane Library and EMBASE to identify studies assessing GFD adherence and duodenal histology at follow-up.

No limits were placed on the year of publication, studies published through March 2025 were included. A random-effect meta-analysis was performed, and the results were reported as pooled proportions with 95% CIs.

Results: Thirteen studies comprising 457 patients with CD occasionally ingesting gluten were included in the final analysis. The pooled prevalence of "gluten tolerant" patients, showing no atrophy (Marsh 0-2) despite non strict GFD compliance, was 42% (95% CI 29% to 55%). Histology reassessment in all studies was performed at least 1 year from CD diagnosis, with follow-up periods within individual studies ranging from 12 months to 20 years. Sub-analysis based on method of GFD adherence (gluten immunogenic peptides vs questionnaires/self-reported) were not statistically different.

Conclusion: This systematic review and meta-analysis suggests the existence of a subgroup of patients capable of developing some degree of gluten tolerance. Although further studies are needed, these findings suggest that a relevant number of patients could be tolerant to a certain quantity of gluten ingestion and thus a personalization (on demand or low gluten content) of the GFD could be a future approach to coeliac patients.

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Disclosure: Nothing to disclose.

PP0298

PERSISTENCE AND NEW ONSET OF CELIAC DISEASE-RELATED SYMPTOMS AND COMORBIDITIES AFTER DIAGNOSIS IN TWO US HEALTHCARE DATABASES

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Introduction: We evaluated celiac disease (CeD)-related symptoms and comorbidities in two US healthcare databases after CeD diagnosis according to the presence/absence of CeD-related symptoms or comorbidities before diagnosis.

Aims & Methods: This matched case–control study used data from two US healthcare databases:

1. MarketScan® Commercial and Medicare Supplemental database (collection period: 2002–2023);

2. Optum Clinformatics® Data Mart (Optum CDM; collection period: 2007–2023).

Patients with CeD had ≥ 1 inpatient or ≥ 2 outpatient International Classification of Diseases (ICD)-9 or -10 codes (579.0 and K90.0, respectively) at any time during the collection period. Patients were matched in a 1:5 ratio with controls without CeD (no CeD ICD-9/-10 codes at any time during data collection) for age at diagnosis, sex, region, and diagnosis year. All individuals had continuous medical and pharmacy enrollment for 2 years pre/post CeD diagnosis. Odds ratios (ORs) and 95% confidence intervals (CIs) for symptoms/comorbidities after CeD diagnosis were estimated by generalized multivariate logistic regression comparing cases vs matched controls stratified by those with (persistent) or without (new-onset) the relevant symptom/comorbidity pre-diagnosis. Symptoms assessed included abdominal pain, bloating, diarrhea, headache, nausea/vomiting, and tiredness. Comorbidities assessed included depression/anxiety, irritable bowel syndrome (IBS), and iron deficiency anemia (IDA).

Results: The numbers of CeD cases and controls were 39,814 and 198,961 in MarketScan, and 18,357 and 91,754 in Optum CDM, respectively. In both databases, CeD diagnosis was significantly positively associated with having CeD-related symptoms/comorbidities 2 years after CeD diagnosis regardless of pre-diagnosis status; and the strengths of associations were greater among patients with new-onset than with persistent symptoms/comorbidities after CeD diagnosis.

In MarketScan, the ORs (95% CI) in patients with new-onset symptoms were highest for bloating (7.98 [7.32–8.71]), diarrhea (6.83 [6.48–7.19]), and nausea/vomiting (3.75 [3.49–4.04]), and in those with new-onset comorbidities for IBS (8.30 [7.65–9.00]), IDA (6.35 [5.94–6.79]), and depression/anxiety (2.12 [2.04–2.20]). The ORs (95% CI) in patients with persistent symptoms were 2.47 (1.86–3.27) for bloating, 2.47 (2.21–2.76) for diarrhea, and 2.46 (2.05–2.96) for nausea/vomiting, and in those with persistent comorbidities were 1.38 (1.16–1.64) for IBS, 1.93 (1.70–2.18) for IDA, and 1.42 (1.34–1.51) for depression/anxiety.

In Optum CDM, the ORs (95% CI) in patients with new-onset symptoms were highest for bloating (6.15 [5.56–6.80]), diarrhea (5.85 [5.48–6.24]), and nausea/vomiting (3.38 [3.08–3.70]), and in those with new-onset comorbidities for IBS (6.17 [5.56–6.85]), IDA (5.31 [4.89–5.77]), and depression/anxiety (2.10 [1.99–2.21]). The ORs (95% CI) in patients with persistent symptoms were 2.25 (1.70–2.97) for bloating, 2.51 (2.22–2.84) for diarrhea, and 2.77 (2.23–3.43) for nausea/vomiting, and in those with persistent comorbidities were 1.40 (1.15–1.71) for IBS, 2.54 (2.17–2.97) for IDA, and 1.50 (1.38–1.62) for depression/anxiety.

Conclusion: CeD diagnosis was significantly associated with having CeD-related symptoms/comorbidities 2 years after CeD diagnosis, with greater associations among patients with new-onset symptoms/comorbidities than those with persistent symptoms/comorbidities. These results emphasize the clinical burden of CeD and the need for improved treatment options.

Disclosure: FZ has served as a speaker for Werfen and a consultant for Takeda. MA, SM, VG, SG, IP, NC, and DAL are employees and shareholders of Takeda Development Center Americas, Inc. and receive stock or stock options. BL has nothing to disclose.

PP0299

PREVALENCE OF DIAGNOSED CELIAC DISEASE IN THE UNITED STATES (2013–2022): A HEALTH CLAIMS DATABASE STUDY

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Introduction: The estimated global prevalence of biopsy-confirmed celiac disease (CeD) is 0.7% and sero-positive prevalence is 1.4%. Differences in definitions of CeD cases and at-risk populations, as well as underdiagnosis, may affect accurate estimation of the size of the diagnosed CeD population. In this study, we calculated the 10-year prevalence of diagnosed CeD in the USA using different CeD case definitions and at-risk populations.

Aims & Methods: A prevalence analysis of CeD cases was conducted using data collected from the Optum Clinformatics® Data Mart, a US claims database. Prevalence calculations were based on two different definitions of CeD cases.

Case definition 1 included any of the following:

1) International Classification of Diseases (ICD)-9/-10 diagnostic codes for CeD (≥ 1 inpatient or ≥ 2 outpatient codes),

2) tissue transglutaminase immunoglobulin A (tTG-IgA) positive serology or;

3) biopsy-confirmed CeD (defined as esophagogastroduodenoscopy [EGD] followed by an ICD diagnostic code for CeD used as proxy).

Case definition 2 was defined as the first ever positive tTG-IgA test followed by biopsy-confirmed CeD.

The prevalence of diagnosed CeD was estimated for each study year from 2013 to 2022. Prevalence estimations were age- and sex-standardized using 2022 US Census data.

Results: The total population (denominator) comprised ~81 million individuals; of whom, 110,140 had ≥ 1 inpatient or ≥ 2 outpatient diagnostic codes for CeD, 41,184 had ≥ 1 positive tTG-IgA test and 44,641 had biopsy-confirmed CeD. Among all individuals that were tested for tTG2-IgA, 0.85% (95% confidence interval [CI]: 0.82, 0.89) were positive for CeD. The age- and sex-standardized prevalence of diagnosed CeD according

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to case definition 1 modestly increased from 0.27% (95% CI: 0.27, 0.27) in 2013 to 0.35% (0.34, 0.35) in 2022. Over the same time period, the age- and sex-standardized prevalence of diagnosed CeD according to case definition 2 increased from 0.01% (0.01, 0.01) to 0.03% (0.03, 0.03), respectively.

Conclusion: The overall prevalence of diagnosed CeD in the USA modestly increased from 2013 to 2022. During this same time period, the proportion of patients who underwent an EGD after a positive tTG-IgA test steadily rose. Validation of the CeD case definitions used in this study will be important to enable these results to be replicated or refuted.

Furthermore, a standardized and validated CeD case definition will support comparisons between studies and inform better clinical care/disease management.

Disclosure: MA, SM, NC, KG and DAL are employees of Takeda Development Center Americas, Inc. and receive stock or stock options.

PP0300

UTILIZATION OF VIDEO CAPSULE ENDOSCOPY AMONG PATIENTS WITH CELIAC DISEASE IN A US HEALTH CLAIMS DATABASE

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Introduction: The real-world utilization of video capsule endoscopy (VCE) still remains poorly understood, particularly among patients with celiac disease (CeD). Current guidelines recommend utilizing VCE as a management tool for patients with CeD, such as for the follow-up and monitoring of the disease, rather than for diagnostic purposes. This study aimed to obtain an improved understanding of the patterns of VCE utilization in a US healthcare database.

Aims & Methods: This retrospective observational study was conducted using data collected from the US MarketScan® Commercial and Medicare Supplemental claims database from 2002 to 2023. Patients were grouped into two cohorts: all patients with ≥ 1 VCE procedure code (cohort 1) and those with ≥ 1 VCE procedure code who were subsequently diagnosed with CeD (cohort 2).

CeD diagnosis and indications for VCE were identified by ≥ 1 inpatient or ≥ 2 outpatient International Classification of Diseases (ICD)-9/-10 medical diagnostic codes. Demographic and clinical characteristics, patterns and indications for VCE utilization, and provider type were summarized using descriptive statistics.

Results: Of the 229,941 patients who had ≥ 1 procedure code for VCE (cohort 1), 4,996 were diagnosed with CeD (cohort 2) from 2002 to 2023. During this period, the frequency of VCE use peaked between 2008 and 2014, and was similar between patients from cohort 1 and cohort 2, with proportions ranging from 7.3% to 9.1% in cohort 1, and from 8.1% to 9.7% in cohort 2.

The mean age (standard deviation) at first VCE procedure date was 55.0 (17.7) years in cohort 1 and 47.9 (17.3) years in cohort 2, and there were more females than males in both cohorts (59.1% vs 40.9% in cohort 1 and 69.7% vs 30.3% in cohort 2, respectively).

VCE use was notably greater in those aged 18–49 years with CeD than in those of the same age in the overall population. Most VCE procedures were performed by gastroenterologists in both cohorts of patients (72.7% in cohort 1 and 73.9% in cohort 2). The most common indication for VCE use in cohort 1 was unspecified anemia, reported in over half of the patients. Abdominal pain with unspecified location and fatigue/other malaise were the most common indications for VCE use among patients in cohort 2 and were reported in over half of these patients.

Among the VCE indications reported in both cohorts, the most common were abdominal pain with unspecified location (51.5% in cohort 1 and 62.0% in cohort 2), followed by fatigue/other malaise (47.2% in cohort 1 and 52.7% in cohort 2).

Conclusion: Findings from this study suggest that VCE may have a role among patients with CeD beyond the current clinical guideline recommendations, although use has decreased in recent years in both patients with CeD and the general population.

Disclosure: MA, NC, VG, DAL, SM, RS, and ST are employees of Takeda Development Center Americas, Inc. and receive stock or stock options. SL and DSM have nothing to disclose.

PP0301

ANALYTICAL COMPARISON OF A NOVEL, FULLY AUTOMATED MULTIPLEXED MICROARRAY IMMUNOASSAY AND ROUTINE METHODS FOR THE DETECTION OF CELIAC DISEASE ANTIBODIES IN A LABORATORY IN ENGLAND

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Introduction: Detection of autoantibodies directed to tissue transglutaminase (tTG) and deamidated gliadin peptide (DGP) is important in the assessment of individuals under clinical suspicion of celiac disease (CD).

Aims & Methods: In this study, we compared the analytical performance for the detection of autoantibodies to tTG (IgA isotype), of the MosaiQ AiPlex® CD microarray (AiPlex CD. AliveDx, Switzerland), a CE-marked, novel, single-use, fully automated, multiplexed microarray immunoassay and routine methods. De-identified serum samples, characterized as reactive or non-reactive for tTG IgA as per the routine practice method of the evaluation site (EliA™ Celikey® IgA, Phadia AB, Sweden), were tested with the evaluated device.

Although AiPlex CD features the semi-quantitative measurement of both tTG IgA and IgG and anti-deamidated gliadin peptide (DGP) IgA and IgG antibodies, and qualitative measurement of total IgA in serum; this comparison was limited to tTG IgA as no samples characterized for other CD autoantibodies were available. Positive percent agreement (PPA) and negative percent agreement (NPA) were calculated. Double-sided 95% Confidence Intervals (CI) were calculated using Clopper-Pearson Exact Method.

Results: 77 samples characterized as reactive for tTG IgA by the routine method and 90 samples characterized as non-reactive for this marker were included in the analysis. The evaluated device identified 73 out of 77 samples characterized as reactive by the routine method, for a PPA of 95% (95% CI: 87.2, 98.5). Out of the four discordant samples, one also yielded a non-reactive result for tTG IgG and reactive results for both DGP IgA and IgG in the evaluated device and negative anti-endomysial antibodies (EMA) by immunofluorescence; two samples were reactive to DGP IgA or DGP IgG by the evaluated device and one sample was non-reactive to all four markers and was negative to EMA. Regarding the samples characterized as non-reactive by the routine method, the evaluated device identi-

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fied 86 out of 90, for an NPA of 96% (95% CI: 89.0, 98.7). For the four samples with discordant results, the evaluated device also yielded reactive results for at least one additional CD antibody (tTG IgG, DGP IgA, DGP IgG).

Parameter / Measurand	tTG IgA
Reactive results (n)	77
PPA	95%
95% CI	87.2, 98.5
Non-reactive results	90
NPA	96%
95% CI	89.0, 98.7

tTG: tissue transglutaminase. PPA: positive percent agreement; NPA: negative percent agreement. CI: Confidence interval.

Table 1. Agreement of the investigational microarray immunoassay with the comparator method.

Conclusion: In this study, the evaluated microarray immunoassay showed high concordance with the compared routine method for the serological assessment of tTG IgA, a key marker in the evaluation of CD.
Disclosure: This study was funded by AliveDx Suisse SA, Eysins, Switzerland. Gerber Gomez and Christian Fischer are AliveDx employees.

PP0302

ANALYTICAL COMPARISON OF A NOVEL, FULLY AUTOMATED MULTIPLEXED MICROARRAY IMMUNOASSAY AND ROUTINE METHODS FOR THE DETECTION OF CELIAC DISEASE AUTOANTIBODIES IN A REFERENCE LABORATORY IN WALES

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Introduction: Serology plays a key role in the evaluation of celiac disease (CD).
Aims & Methods: We assessed the analytical performance for the detection of CD autoantibodies, compared with routine laboratory methods, of MosaiQ AiPlex® CD microarray (AiPlex CD. AliveDx, Switzerland) a CE-marked, novel, single-use, multiplexed microarray immunoassay, used with its fully automated proprietary system. The valuated device features the simultaneous, semi-quantitative measurement of both IgA and IgG anti-tissue transglutaminase (tTG) and IgA and IgG anti-deamidated gliadin peptide (DGP) antibodies, as well as a qualitative measurement of total IgA in human serum; however, for this study, the comparison was limited to tTG IgA and tTG IgG as no samples characterized for DGP IgA nor for DGP IgG were available. Banked, de-identified serum samples characterized as reactive or non-reactive for tTG IgA and tTG IgG as per routine methods were tested with the evaluated device. Routine methods included EliA™ Celikey® IgA (Phadia AB, Sweden), EliA™ Celikey® IgG (Phadia AB, Sweden), QUANTA® Flash h-tTG IgA (Inova Diagnostics, USA), QUANTA® Flash h-tTG IgG (Inova Diagnostics, USA), Anti-Tissue-Transglutaminase IgA (ORGENTEC Diagnostika GmbH, Germany), Anti-Tissue-Transglutaminase IgG (ORGENTEC Diagnostika GmbH, Germany) and QUANTA Lite® R h-tTG IgA ELISA (Inova Diagnostics, USA). Positive percent agreement (PPA), negative percent agreement (NPA) for individual analytes were calculated as were overall positive and percent agreements.
Results: The evaluated device identified 43 out of 45 samples characterized as reactive for tTG IgA per routine methods and 79 out of 81 samples characterized as non-reactive, for a PPA of 96% (95% CI: 84.8, 99.4) and an NPA of 98% (95% CI: 91.3, 99.7). Likewise, it identified 14 out of 18 samples characterized as reactive for tTG IgG per routine methods and 79 out of 85 samples characterized as non-reactive, for a PPA of 78% (95% CI: 52.3, 93.5) and an NPA of 93% (95% CI: 85.2, 97.3). Overall positive and negative percent agreement were 90% (95% CI: 80.4, 96.4) and 95% (95% CI: 90.7, 97.9), respectively (Table 1).

Parameter / Measurand	tTG IgA	tTG IgG	Overall
Reactive results (n)	45	18	63
PPA	96%	78%	90%
95% CI	84.8, 99.4	52.3, 93.5	80.4, 96.4
Non-reactive results	81	85	166
NPA	98%	93%	95%
95% CI	91.3, 99.7	85.2, 97.3	90.7, 97.9

PPA: positive percent agreement. NPA: negative percent agreement. CI: confidence interval.

Table 1. Agreement of the investigational microarray immunoassay with routine laboratory methods.

Conclusion: In the present study, the evaluated multiplexed microarray immunoassay showed high concordance with routine methods for the detection of tTG IgA and tTG IgG. This device may be considered a suitable alternative method for the serological assessment of patients under clinical suspicion of CD and has the potential to help simplifying the laboratory evaluation of this condition by simultaneously and automatically analyzing IgA and IgG isotypes of key CD markers.
Disclosure: This study was funded by AliveDx Suisse SA, Eysins, Switzerland. Gerber Gomez and Christian Fischer are AliveDx employees.

PP0303

THROMBOEMBOLIC COMPLICATIONS IN CELIAC DISEASE

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Introduction: Celiac disease (CD) is an autoimmune enteropathy induced by the ingestion of gluten in genetically predisposed individuals. Although its clinical expression is most often digestive, extra-digestive manifestations can occur, among which thromboembolic complications, although rare, deserve particular attention.
The aim of this study is to describe the profiles of coeliac patients with thromboembolic complications, analyze their clinical, paraclinical and etiological characteristics, and discuss the underlying pathophysiological mechanisms.
Aims & Methods: This is a retrospective, descriptive, monocentric study conducted in the hepato-gastroenterology department of Ibn Sina Hospital in Rabat over a 30-year period, from 1995 to 2025. The records of 344 patients followed for celiac disease were reviewed. Patients with one or more documented thromboembolic complications during follow-up were included in the study. Clinical, biological, endoscopic and histological data were collected and analyzed.
Results: Of a total of 344 patients followed for celiac disease in the department, 5.5% (n=19) developed a thromboembolic complication during follow-up. The population concerned was predominantly female, with 89.5% women (n=17) versus only 10.5% men (n=2). The mean age at the time of complication was 39.6 years, with a median of 35.5 years, and extremes ranging from 18 to 62 years.

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Endoscopic exploration (oesogastro duodenal fibroscopy) in these patients showed lesions suggestive of celiac disease in 68.4% (n=12), while it was non-specific in 31.6% (n=6), and normal in 5.3% (n=1). Histologically, villous atrophy was found in all patients, with 78.9% (n=15) showing total atrophy (Marsh IIIc) and 21.1% (n=4) subtotal atrophy (Marsh IIIb).

As regards thromboembolic manifestations, portal vein thrombosis was the most frequent complication, observed in 57.9% of cases (n=11). It was followed by deep vein thrombosis of the lower limbs, reported in 15.8% of cases (n=3), and ischemic stroke in 10.5% (n=2). Isolated cases of thrombosis of the inferior vena cava, axillary vein and femoral vein were recorded, each accounting for 5.3% of cases (n=1).

The etiological study revealed hyperhomocysteinemia in 36.8% of patients (n=7) and protein C and/or S deficiency in 21.1% (n=4). In the remaining 42.1% of patients (n=8), the thrombophilia work-up was normal, suggesting that celiac disease itself was the cause of the thrombotic event. Therapeutically, all patients were treated with oral anticoagulation. One patient required a vena cava filter because of the risk of bleeding on anticoagulation, and another underwent surgical treatment for axillary thrombosis. The outcome was favorable in all cases, with no recurrence or major complication during follow-up.

Conclusion: Thromboembolic complications in celiac disease are rare but potentially serious. This study highlights the predominance of women, the diversity of thrombotic localizations, and the possible involvement of hyperhomocysteinemia and C/S protein deficiencies. In almost half the cases, no thrombophilic cause has been identified, suggesting a role for celiac disease in the genesis of these events. Clinical vigilance is essential, particularly in cases of late diagnosis or poor compliance with the gluten-free diet.

Disclosure: Nothing to disclose.

PP0304

CELIAC DISEASE AND PORTAL HYPERTENSION: A RARE AND MULTIFACTORIAL ASSOCIATION

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Introduction: Celiac disease (CD) is a chronic autoimmune enteropathy triggered by gluten ingestion in genetically predisposed individuals. While isolated hypertransaminasemia is the most frequent hepatic manifestation of CD, its association with portal hypertension (PHT) remains rare and poorly understood.

Aims & Methods: This study aims to describe this association and evaluate the impact of a gluten-free diet (GFD) on the progression of PHT-related complications.

We conducted a 30-year retrospective study including 20 patients with both histologically and serologically confirmed CD and PHT, among a total cohort of 344 CD patients. Clinical, biological, endoscopic, imaging, and histological data were analyzed.

Results: Among the 20 patients (75% female, F/M ratio 3; mean age 34.2 years), the most frequent symptoms were abdominal pain and weight loss (45% each), followed by ascites (35%), malabsorptive diarrhea (30%), atypical diarrhea (20%), and vomiting (15%). One patient presented with upper gastrointestinal bleeding. Splenomegaly was observed in 70%, and collateral venous circulation in 30%. Liver was clinically normal in 80%, cirrhotic in 20%. Imaging showed a regular liver in 65%, nodular in 35%, with 25% having portal cavernoma and one patient with focal IVC thrombosis.

Liver biopsy findings included porto-sinusoidal vascular disease (40%), celiac cirrhosis (20%), granulomatous hepatitis (15%), and normal histology (25%). Upper endoscopy revealed esophageal varices in 55%. PHT led to the diagnosis of CD in 4 patients. In non-adherent patients, progression of varices and cirrhosis occurred in 20% over 5 years.

Conclusion: The CD–PHT association is rare (5.8% in our series) but clinically significant. Various mechanisms may underlie this link, highlighting the importance of screening for CD in unexplained PHT. GFD adherence appears to slow hepatic deterioration and variceal progression, reinforcing its therapeutic value.

Disclosure: Nothing to disclose.

PP0305

CELIAC DISEASE AND ANTI-PHOSPHOLIPID ANTIBODIES: PREVALENCE AND CLINICAL IMPLICATIONS IN A MOROCCAN COHORT

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Introduction: Celiac disease (CD) is an autoimmune enteropathy triggered by gluten in genetically predisposed individuals. Both hereditary and acquired factors may increase thrombotic risk in CD. Anti-phospholipid antibodies (aPL), including anti-cardiolipin (aCL) and anti-β2 glycoprotein I (β2GPI), are known markers of thrombophilia.

Aims & Methods: This study aimed to evaluate the frequency of these antibodies in CD patients and investigate any correlation with gluten-free diet (GFD) adherence.

This prospective, descriptive study was conducted over four years and included 42 patients diagnosed with CD based on clinical, serological, and histological criteria. Anti-cardiolipin IgG (aCL) and anti-β2GPI IgG antibodies were tested by ELISA, and lupus anticoagulant (LA) was assessed via coagulation assay.

Results: Among 42 CD patients, aPL results were available for 9 (22%). The mean age was 35.5 years (range 18–58), with a female-to-male ratio of 2. All patients were prescribed a GFD with regular follow-up.

aCL IgG antibodies were positive in 3/9 patients (15.2%), β2GPI IgG antibodies were positive in 1/9 (5%), and LA was negative in all patients. No correlation was found between the presence of aPL and disease activity or adherence to GFD. Additionally, aPL presence was not associated with disease duration or thrombotic events.

Conclusion: In our cohort, we did not observe a high prevalence of anti-phospholipid antibodies among CD patients. aCL IgG was the most frequently detected antibody (15.2%) under strict GFD. The clinical significance and pathogenic role of these antibodies in CD remain unclear and warrant further large-scale prospective studies.

Disclosure: Nothing to disclose.

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PP0306

DIAGNOSTIC YIELD OF CAPSULE ENDOSCOPY IN PATIENTS WITH SMALL BOWEL DISORDERS: A SINGLE-CENTRE RETROSPECTIVE STUDY

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Introduction: Capsule endoscopy (CE) is a valuable, non-invasive diagnostic tool for the evaluation of small bowel disorders, particularly in patients with suspected inflammatory bowel disease (IBD), known Crohn's disease, or non-responsive celiac disease (NRCD) (1). Its role in identifying mucosal lesions and guiding clinical decision-making continues to expand, especially in the context of complex or unclear gastrointestinal symptoms.

Aims & Methods: This observational, single-centre retrospective study included all CE procedures performed at the Department of Gastroenterology of the Azienda Ospedaliera–Università di Padova between January 1, 2018, and December 31, 2024. Indications for CE included suspected IBD (e.g., diarrhea, abdominal pain, elevated fecal calprotectin), follow-up of known ileal Crohn's disease, and evaluation of NRCD. Demographic and clinical data were collected in a dedicated database. CE was considered positive if at least one of the following findings was present: evidence of recent bleeding, ulcers, or villous atrophy. Multivariate analyses were used to assess risk factors associated with ulcers and villous atrophy. Statistical analyses were conducted using STATA 18.

Results: A total of 351 CE procedures were included: 210 for suspected IBD, 108 for IBD follow-up, and 37 for NRCD. The mean age of the cohort was 45.7 ± 16.0 years, with 47% male patients. The diagnostic yield was 39.3% for suspected IBD, 44.4% for IBD follow-up, and 56.7% for NRCD. The prevalence of recent bleeding was similar across groups (31.1% vs. 27.4% vs. 31.6%, p=0.895), as was the presence of ulcers (65.3% vs. 62.1% vs. 40%, p=0.105).

However, villous atrophy was significantly more frequent in NRCD patients (0% in suspected IBD, 2.3% in IBD, and 66.7% in NRCD; p<0.001). No association was found between lesions and age, sex, hemoglobin levels, or comorbidities. CE findings led to therapeutic changes in 37% of IBD cases and in 16% of NRCD cases.

Conclusion: This single-centre cohort study confirms the high diagnostic yield of capsule endoscopy in patients with small bowel disorders and highlights its impact on their clinical management.

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Disclosure: Nothing to disclose.

PP0307

RELEVANCE OF SYMPTOM QUESTIONNAIRES AND HYDROGEN BREATH TESTS IN SUSPECTED LACTOSE OR FRUCTOSE INTOLERANCE

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Introduction: A considerable proportion of patients with disorders of gut-brain interaction suffer from carbohydrate intolerances. The European Guideline[1] recommends a diagnostic evaluation that combines malabsorption testing using hydrogen breath tests with symptom assessment using the validated aCPQ[2]. This study evaluates 1) the relationship between the severity of abdominal symptoms (pain, nausea, bloating, flatulence, diarrhea) in patients' histories and the results of malabsorption and intolerance testing, and 2) the response to a diet in patients diagnosed with carbohydrate intolerance (positive aCPQ) with or without malabsorption (pos. or neg. breath test).

Aims & Methods: This is a retrospective analysis of 1,139 tests performed between February 2017 and June 2024in 708 patients (62% female; mean age 42 ± 16 years) referred for evaluation of suspected carbohydrate intolerance at the Medical Universities of Graz and Vienna. Malabsorption was evaluated by breath tests (H_2 maximum rise above baseline and area under the curve), and intolerance by the aCPQ after ingestion of 50 g lactose (n=588), 25 g fructose (n=550) or 25 g trehalose (n=1). Symptom intensities over the preceding six months for diarrhea, bloating, nausea, flatulence, and abdominal pain were assessed using the Bowel Disease Questionnaire (BDQ; VAS scale 0–100 mm). 63 patients with intolerance completed a 4- to 12-week diet. Symptoms after the diet were evaluated using identical VAS scales, with symptom intensity assessed within the previous 10 days. The difference between the VAS scores of each symptom (deltaVAS) before and after diet was determined.

Results: 625 tests revealed intolerance (pos aCPQ), 373 malabsorption (pos. breath test), 273 tests a combination of intolerance and malabsorption and 414 tests were negative in both tests. In patients with intolerance weak positive correlations were found between each of the five symptoms in the preceding six months and maximum symptom intensity of the respective symptom on the aCPQ (r ranging from 0.17 to 0.31; p<0.001); the strongest correlation was observed for bloating and the weakest for diarrhea. This applied for the lactose and fructose groups. These correlations were present whether or not malabsorption was detected in the breath test. No significant correlation was observed between BDQ symptom severity and breath test results. The table shows symptom severity by VAS scoring before and after diet in patients with aCPQ-diagnosed intolerance: dietary intervention led to a significant improvement in all symptom severities (p>0.001). For all of the five symptoms strong correlations were observed between baseline symptom severity in the preceding 6 months (BDQ) and the extent of symptom improvement after the diet (delta VAS; *p<0.001). Malabsorption detected by breath test did not predict dietary response. Weak positive correlations were noted between aCPQ symptom severity and deltaVAS for nausea (r=0.32; p=0.012) and bloating (r=0.37; p=0.003).

	Pain	Bloating	Flatulence	Diarrhea	Nausea (median, range
VAS (mm) before diet, mean+SD	40+24	52+28	48+28	36+29	24 (0-75)
VAS (mm) after diet, mean+SD	11+18*	19+20*	26+30*	10+17*	0 (0-65)*
Correlation BDQ vs delta VAS	r =0.76*	r=0.79*	r=0.54*	r=0.84*	r=0.83*

Asterisks denote p<0.001 for comparison of VAS before vs. during diet and for correlation.

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Conclusion: Diagnosis of intolerance using the aCPQ identifies patients who benefit from dietary intervention. Patients who have the most severe baseline symptom history respond the best. Symptom history is a weak predictor of carbohydrate intolerance, and breath tests are not useful for explaining baseline symptom history or guiding treatment.

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Disclosure: Nothing to disclose.

PP0308

DIAGNOSTIC PERFORMANCE OF SINGLE-BALLOON ENTEROSCOPY FOR MALIGNANT SMALL-BOWEL TUMORS

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Introduction: Owing to their deep location, small-bowel tumors have long eluded definitive endoscopic diagnosis with conventional instruments. The advent of balloon-assisted enteroscopy has expanded diagnostic and therapeutic options.

Aims & Methods: Aim: We aimed to evaluate the diagnostic performance of single-balloon enteroscopy (SBE) for malignant small-bowel tumors at a tertiary center.

Methods: We reviewed 1,031 consecutive SBEs performed between September 2005 and June 2024. Among 84 patients in whom a small-bowel tumor was visualised, 63 had histologically malignant lesions and were analysed. Primary outcomes were lesion-access rate, definitive histology rate by target biopsy, pre-operative tattoo-marking rate, and avoidance of diagnostic laparotomy.

Results: Malignancy accounted for 63 / 84 tumors (75 %). Histology comprised malignant lymphoma (ML; 26, 41 %), adenocarcinoma (16, 25 %), gastrointestinal stromal tumor (GIST; 12, 19 %), metastatic carcinoma (8, 13 %), and neuroendocrine tumor (NET; 1, 2 %). Index presentations were obscure gastrointestinal bleeding (27, 43 %), abnormal imaging (18, 28 %), intestinal obstruction (14, 22 %), and other causes (4, 7 %). SBE reached the lesion in 59 / 63 cases (93.7 %). Tattoo marking before surgery was feasible in 37 / 41 surgical patients (90.2 %). Target biopsy yielded definitive histology in 43 / 47 accessible cases of adenocarcinoma, metastatic carcinoma, or ML (91.5 %). Endoscopic appearance suggested GIST in 12 patients, 11 of which were pathologically confirmed. Diagnostic laparotomy was avoided in 55 / 63 patients overall (87.3 %), including 21 / 26 with ML (80.8 %).

Conclusion: SBE achieved a high rate of deep-bowel intubation, lesion-targeted biopsy, and intra-operative localisation. It provided histologic confirmation in >90 % of accessible malignant lesions and obviated diagnostic surgery in most cases, supporting its routine use in the work-up of small-bowel malignancies.

Disclosure: Nothing to disclose.

PP0309

INFLUENCE OF TUMOUR AND PATIENT'S RELATED FACTORS ON THE RESPONSE TO MEDICAL TREATMENTS IN WELL DIFFERENTIATED GASTROENTEROPANCREATIC NEUROENDOCRINE NEOPLASMS (FARINET): A MULTICENTER STUDY

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Introduction: Gastroenteropancreatic neuroendocrine neoplasms (GEP-NENs) are a heterogeneous group of tumors with rising incidence. Despite several available treatments, including somatostatin analogues (SSAs), targeted therapies (everolimus, sunitinib), and chemotherapy (capecitabine-temozolomide, CAPTEM), predicting therapeutic response and disease progression remains difficult. Tumor-related features (e.g., grading, metastatic burden) influence outcomes but whether some patient-related variables (e.g., lifestyle, comorbidities) play a role is poorly explored.

Aims & Methods: This study aimed to identify tumor and patient-related factors associated with progression and survival outcomes in GEP-NENs treated with SSAs, CAPTEM, or targeted therapies. This is a multicenter, retrospective observational cohort study including patients with well-differentiated GEP-NENs treated with SSAs, CAPTEM, or EVE/SUN.

Inclusion criteria were age ≥18 years, histologically confirmed GEP-NEN, and at least 6 months of treatment. Exclusion criteria included poorly differentiated tumors, prior radioligand therapy, and MiNENs. Clinical, pathological, therapeutic, and exposome-related variables were collected.

The primary endpoint was progression-free survival (PFS); overall survival (OS) was also evaluated. A target sample size of 400 was estimated for adequate power in subgroup analyses. Univariate analyses were conducted using t-tests or chi-square tests as appropriate. Associations with PFS and OS were further assessed through Cox proportional hazards models.

Results: A total of 301 patients (52.2% pancreatic) from four Italian referral centers for neuroendocrine tumors (Itanet centers) were included: 258 (85.7%) received SSAs, 32 (10.6%) CAPTEM, and 17 (5.6%) EVE/SUN. Median age at diagnosis was 59 years; 51.9% were female. Most patients (73.7%) presented with metastatic disease at diagnosis, and the majority were G2 (58%) or G3 (2.3%).

In univariate analysis, patient-related variables were not significantly associated with PFS, except for lower age at diagnosis and shorter duration of metformin use (both p = 0.05). Conversely, tumor-related variables (grading, Ki67 and presence of bone or liver metastases) were significantly associated with PFS.

In multivariable Cox analysis, only liver metastases (HR 2.0), bone metastases (HR 2.15), and G3 tumors (HR 3.35) were associated with shorter PFS. For OS, univariate analysis showed worse survival in male patients (p = 0.017), those with a history of alcohol use (p = 0.027), liver disease (p = 0.013), higher tumor grading/Ki67 (p < 0.001), absence of primary tumor

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resection (p < 0.001), and presence of bone (p = 0.004) or liver metastases (p < 0.001). Cox multivariable analysis confirmed association of OS with male sex (HR 2.73), alcohol use (HR 6.46), G3 tumors (HR 8.26), liver metastases (HR 6.45), and absence of primary tumor resection (HR 6.45). A subgroup analysis restricted to SSA-treated patients and pancreatic NENs patients revealed similar trends.

Conclusion: In this multicenter real-world study, progression and survival outcomes in GEP-NENs were primarily driven by tumor-related variables such as grading and metastatic spread, while patient-related factors played a limited role. However, results are currently underpowered, having reached approximately 75% of the planned sample size. Future efforts will aim to expand the EVE/SUN and CAPTEM groups to enhance subgroup analyses and improve robustness of findings.

Disclosure: Nothing to disclose.

PP0310

MULTICENTER REAL-WORLD VALIDATION OF AN AI-ASSISTED SOFTWARE FOR DETECTING PROTRUDING LESIONS IN CAPSULE ENDOSCOPY

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Introduction: Capsule Endoscopy (CE) offers a minimally invasive method for comprehensive visualization of the gastrointestinal (GI) tract. Protruding lesions are relatively common findings in CE, with variable incidence and diverse pathological significance. Accurate detection of these lesions is crucial but remains challenging due to the time-consuming nature of CE interpretation and susceptibility to human error. While artificial intelligence (AI) presents a promising solution, real-world evidence evaluating its clinical performance is still scarce. To assess the diagnostic performance of a deep learning convolutional neural network (CNN) for the detection of protruding lesions in complete CE videos under real-world clinical conditions.

Aims & Methods: Our group conducted a prospective, multicenter validation study comparing AI-assisted CE reading to standard, conventional interpretation. A total of 211 CE videos, obtained using three different CE systems across seven centers in Portugal, Spain, Brazil, and the United States, were analyzed. Following initial standard readings, an independent expert conducted AI-assisted readings using a CNN specifically trained to detect small bowel protruding lesions. Discrepancies between conventional and AI-assisted reports were adjudicated by a third-party expert from an independent center to establish the reference standard. Diagnostic performance was assessed using sensitivity, specificity, accuracy, and the area under the receiver operating characteristic curve (AUC-ROC).

Results: AI-assisted reading detected 106 confirmed protruding lesions, compared to 42 identified through conventional reading. Sensitivity was significantly higher with AI-assisted reading (95.9% vs. 64.6%), although specificity was lower (78.7% vs. 96.1%). Overall accuracy was relatively lower with AI-assisted reading (81.2% vs. 91.5%). The AUC-ROC was 0.86 for AI-assisted and 0.80 for standard reading. Additionally, AI-assisted reading reduced the mean examination time to 316 seconds per video.

Conclusion: AI-assisted interpretation of complete CE videos significantly enhances the detection of protruding lesions compared to conventional reading, although at the expense of slightly reduced specificity. The inclusion of videos from multiple CE systems addresses interoperability challenges, while the participation of centers from four countries across two continents supports the model's generalizability and real-world clinical applicability.

Disclosure: Nothing to disclose.

PP0311

FACTORS ASSOCIATED WITH INTESTINAL PERFORATION DURING SINGLE BALLOON ENTEROSCOPY IN A MULTICENTER STUDY: AN EVALUATION OF 2861 PROCEDURES

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Introduction: Intestinal perforation is an infrequent complication of single balloon enteroscopy (SBE) and usually requires surgery. Factors associated with increased risk of perforation remain unclear.

Aims & Methods: The objective of this study was to identify risk factors associated with intestinal perforation during SBE. Data from patients who underwent SBE at three tertiary care hospitals in China between January 2016 and September 2024 were retrospectively collected and analyzed. Single factor analysis was conducted to evaluate the possible risk factors for intestinal perforation. Binary Logistic Regression was further conducted to evaluate the factors associated with the occurrence of intestinal perforation.

Results: A total of 2861 SBE procedures were performed in 1840 patients (1187 male and 653 female; mean age 51 ±18 years). Intestinal perforation occurred in 7 patients during SBE (0.4%, 7/1840), and all cases were successfully managed with surgical repair. Among these 7 patients, 1 had duodenal stenosis of unknown cause, the other 5 presented with abdominal pain, and 1 with obscure gastrointestinal bleeding had a history of prior abdominal surgery. Binary logistic regression showed that previously abdominal surgery was an increased risk factor for intestinal perforation (odds ratio, 19.52; 95% CI, 2.32 to 164.05; P=0.006).

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Conclusion: Previously abdominal surgery was identified as an increased risk factor for intestinal perforation during SBE. These findings underscores the importance of careful patient selection and procedural planning for reducing the incidence of intestinal perforation, and the subsequent need for surgical intervention.
Disclosure: Nothing to disclose.

PP0312

CLINICAL OUTCOMES OF COMPLETE VERSUS INCOMPLETE SMALL BOWEL EXAMINATION BY SINGLE-BALLOON ENTEROSCOPY IN OBSCURE GASTROINTESTINAL BLEEDING: A PROSPECTIVE MULTICENTER STUDY

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Introduction: Obscure gastrointestinal bleeding (OGIB), which accounts for about 5% of all gastrointestinal bleeding cases, often originates in the small intestine [1, 2]. Single-balloon enteroscopy (SBE) has recently been introduced as a simplified diagnostic and therapeutic tool for OGIB; however, it has significant limitations, including a low rate of complete small bowel examination at 22% [3, 4]. It remains unclear whether complete enteroscopy with SBE leads to better clinical outcomes.
Aims & Methods: This study aimed to compare the rebleeding rate between patients with OGIB who underwent complete versus incomplete SBE.
This multicenter retrospective analysis of prospectively collected data identified adult patients with OGIB from January 2016 to December 2024 in three tertiary care institutions: The First Affiliated Hospital of Zhengzhou University, Henan Provincial People's Hospital, and Qinghai Provincial People's Hospital. Demographic data, initial SBE findings, and rebleeding events during follow-up were collected and analyzed. The study was approved by the Institutional Review Board of all participating hospitals (Ethics Approval No. 2022 KY 1136 002) and registered at the ISRCTN Registry under the reference number ISRCTN17191331.
Results: A total of 431 patients were enrolled, with 43 patients in the complete SBE group and 388 patients in the incomplete SBE group. The mean age was 51 ± 17 years, 68.4% were male, and the mean follow-up duration was 44 ± 23 months. Initial SBE findings revealed no significant difference between the two groups, with diverticular bleeding being the most common abnormal finding in the complete SBE group (25.6%) and angioectasia the most common in the incomplete SBE group (25.8%). During the follow-up period, the rebleeding rate was similar between the complete and incomplete SBE groups (27.9% vs. 26.5%, P = 0.848). Similarly, when stratified by initial SBE findings, no difference in rebleeding rates was observed between the two groups. Although most rebleeding patients were managed conservatively, 45 out of 115 underwent reevaluation with either SBE or surgery. Among the 5 rebleeding cases in the complete SBE group that underwent reevaluation, an 80% (4/5) change in findings from initial SBE was observed, with reevaluation revealing angioectasia (n = 2), no abnormal finding (n = 2), and ulcers/erosions (n = 1). Conversely, among the 40 rebleeding cases in the incomplete SBE group, subsequent SBE findings changed in 92.5% (37/40) of patients, and notably, 27.5% (11/40) of patients were found to have malignant lesions (adenocarcinoma and GIST) that were initially misdiagnosed as benign.
Conclusion: No significant difference in rebleeding outcomes was observed between patients with OGIB who underwent complete versus incomplete SBE during follow-up. These findings indicate that, despite the low rate of complete enteroscopy, SBE remains a valuable diagnostic and therapeutic tool in the management of OGIB. However, caution is advised in cases of incomplete SBE, as clinically significant lesions, including malignancies, may be missed during the initial evaluation.
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Disclosure: Nothing to disclose.

PP0313

REAL-LIFE EVALUATION OF THE BEST BOWEL PREPARATION METHOD FOR CAPSULE ENDOSCOPY IDENTIFIED IN THE PREPRICE STUDY

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Introduction: Bowel preparation is recommended for small-bowel capsule endoscopy (SBCE) by current guidelines, although the ideal regimen remains unclear. The PrepRICE clinical trial is a multicenter randomized trial that compared different timings of intestinal preparation administration, showing that administration after the capsule reached the duodenum improved mucosal visualization and diagnostic yield.
This study was limited to patients with suspected mid-gastrointestinal bleeding who met strict inclusion criteria. Real-life results are generally inferior to those of clinical trials.
Aims & Methods: This work aims to report real-life results (including patients with all indications, comorbidities, and undergoing different types of bowel preparation) after the implementation of the new protocol and to compare them with those of PrepRICE.
A prospective analysis was performed on all consecutive patients who underwent capsule enteroscopy between December of 2024 and March of 2025. The quality of small bowel visualization (QSBV), gastric transit time (GTT), small bowel transit time (SBTT), adequate visualization rate, and complete examination rate were assessed. QSBV was evaluated according to the Brotz quantitative scale, both globally and by tertiles.
Results: A total of 59 patients were included, 55.9% of whom were male, with a median age of 61.0 years (IQR 31). The examination was complete in 100% of cases. The median Brotz scale scores were 9 (IQR 1), 8 (IQR 2), 8 (IQR 3), and 8 (IQR 2) in the first, second, third tertile, and overall, respec-

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tively (compared to 9, 9, 9, 9 in the PrepRICE trial, p<0.001). The visualization rate was adequate in 83,1% of exams (vs. 92%). QSBV was statistically inferior to that observed in the group with the best bowel preparation regimen identified by the clinical trial (yet non-inferior to the pre-procedure preparation groups), as well as the SBTT. In the remaining outcomes, no significant differences were found.

Conclusion: Real-life results were similar to those of the original study, with a high rate of complete examination and adequate visualization, with only slightly lower QSBV compared to that reported in the peri-procedural group in the PrepRICE study, yet still non-inferior to the pre-procedural groups.

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Disclosure: Nothing to disclose.

PP0314

DIAGNOSTIC YIELD OF CAPSULE ENDOSCOPY IN PATIENTS WITH SUSPECTED SMALL BOWEL BLEEDING: A SINGLE-CENTRE RETROSPECTIVE STUDY

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Introduction: Small bowel bleeding remains challenging to diagnose due to limited access with conventional endoscopy. Capsule endoscopy (CE) is increasingly used both in emergency settings following overt bleeding and in the evaluation of chronic, unexplained anemia (1-2). Better risk stratification for small bowel lesions may refine the need for CE.

Aims & Methods: Our primary aim was to assess the diagnostic yield of CE in suspected small bowel bleeding, while secondarily identifying predictive factors for angiodysplasia, recent bleeding, and ulcers. This observational single-centre retrospective study included all CE procedures performed at the Department of Gastroenterology of the Azienda Ospedaliera–Università di Padova between January 1, 2018, and December 31, 2024, for indications of overt or occult gastrointestinal bleeding. Demographic and clinical data were collected in a dedicated database. CE was considered positive if at least one of the following lesions was identified: angiodysplasia, signs of recent bleeding, polyps, ulcers, red spots, or villous atrophy. Multivariate analyses were conducted to assess the risk factors associated with angiodysplasia, recent bleeding, and ulcers. Statistical analyses were performed using STATA 18.

Results: A total of 1,057 CE procedures were performed for the indication of gastrointestinal bleeding, including 310 for overt bleeding and 747 for occult bleeding. The mean age of the cohort was 71 ± 13 years, and 55.6% were male, with no significant differences in age or sex between the two groups. The overall diagnostic yield was comparable between patients with overt and occult bleeding (77.1% vs. 75.2%, p=0.520). The prevalence of angiodysplasia was similar in both groups (80.4% vs. 78.1%, p=0.501), as was the detection of recent bleeding (57.1% vs. 50.2%, p=0.150). However, small bowel ulcers were significantly more frequent in patients with overt bleeding (45.9% vs. 34.2%, p=0.023).

In the multivariate analysis, the risk of detecting angiodysplasia was lower in patients receiving antiplatelet (OR 0.60; 95% CI: 0.66–0.99) or anticoagulant therapy (OR 0.40; 95% CI: 0.26–0.66), but significantly higher in older age groups compared to those aged 18–45 years: 46–60 years (OR 6.02; 95% CI: 2.2–16.7), 61–80 years (OR 8.80; 95% CI: 3.4–22.7), and >80 years (OR 13.62; 95% CI: 4.9–37.3). Increased risk was also associated with lower hemoglobin levels: 7–10 g/dL (OR 2.90; 95% CI: 1.5–5.5) and <7 g/dL (OR 2.17; 95% CI: 1.2–3.9). Similar predictors were found for recent bleeding. Conversely, the risk of ulcers was lower in patients with occult bleeding (OR 0.60; 95% CI: 0.4–0.9), and higher in those who received blood transfusions (OR 1.70; 95% CI: 1.0–2.7) and those with hemoglobin <10 g/dL (OR 2.40; 95% CI: 1.2–5.0 for Hb 7–10 g/dL and OR 2.60; 95% CI: 1.3–5.2 for Hb <7 g/dL).

Conclusion: This large single-centre cohort study confirms the high diagnostic yield of capsule endoscopy in patients with both overt and occult gastrointestinal bleeding. Older age and severe anemia are strong predictors for angiodysplasia and signs of recent bleeding, while overt bleeding and low hemoglobin levels are significantly associated with the presence of small bowel ulcers.

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Disclosure: Nothing to disclose.

PP0315

FATIGUE AND IMPAIRED QUALITY-OF-LIFE IN PATIENTS WITH SMALL BOWEL ANGIODYSPLASIA: RESULTS FROM A PROSPECTIVE COMPARISON COHORT STUDY

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Introduction: Patients with angiodysplasia, and chronic anaemia often experience a complex symptom burden, including reduced quality of life (QoL) and fatigue. While cross-sectional studies suggest impaired QoL in this population, longitudinal data assessing how these symptoms evolve over time are limited (1).

Both general QoL tools, such as the RAND-SF36 and measures, like the Fatigue Severity Scale (FSS), may offer insight (2, 3).

Argon Plasma Coagulation (APC) (4) is commonly employed for vascular lesions, but its impact on long-term symptom burden remains unclear. This study aimed to evaluate changes in QoL and fatigue over time in patients with small bowel angiodysplasia compared to anaemia controls without small bowel pathology.

Aims & Methods: Following ethical approval, patients with angiodysplasia and controls (those with anaemia and negative bidirectional and capsule endoscopy), were invited to complete an initial and subsequent (minimum of six months) RAND-SF36 and FSS. Baseline QoL and fatigue

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assessment was performed on the day of small bowerl capsule endoscopy (SBCE) or enteroscopy. Repeat RAND-SF-36 and FSS were obtained from participants via telephone interview. Basic demographics and procedural related data, QoL and fatigue data were recorded. An FSS >36 was defined as clinically significant fatigue. Paired t-tests were used to assess within-group changes from baseline to follow-up. In the angiodysplasia group, a subgroup analysis compared changes in fatigue and QoL between patients who received APC and those who did not, using independent t-tests.

Results: A total of 28 patients in the angiodysplasia group (mean age 67.5 years; 35.7% female) and 38 anaemia controls (mean age 64.8 years; 42.1% female) were included, with no significant demographic differences. In the angiodysplasia group, the proportion of patients with clinically significant fatigue (FSS > 36) increased from 82.1% at baseline to 92.9% at follow-up (p = 0.046) suggesting overall worsening of fatigue over time. Anaemia controls showed no change in FSS scores (44.7% at both time points, p = 1.000). Despite this, in the angiodysplasia group several SF-36 domains, including Energy/Fatigue, Social Functioning, and Emotional Wellbeing, demonstrated statistically significant improvements (p < 0.05), suggesting, in our cohort, there is a significant improvement in QoL over time. 19 patients (67.9%) in the group underwent Argon Plasma Coagulation (APC). The proportion of patients with clinically significant fatigue (FSS >36) decreased from 57.9% to 31.6% in the APC angiodysplasia subgroup group (p = 0.433) and from 55.6% to 33.3% in angiodysplasia patients who did not receive APC (p = 0.570). Both groups received the same standard of care, transfusions and iron replacement as required. Data on medical therapy was unavailable.

Domain	Vascular Group (N=48)	Control (N=57)	p-value
FSS >36 (% patients)	69%	62%	0.523
Physical Functioning	48.3	54.6	0.290
Role Limitations (Physical)	25.8	45.	0.013
Energy/ Fatigue	38.5	48.0	0.048
Social Functioning	56.2	67.8	0.04
General Health	49.1	52.4	0.435

Conclusion: Patients with small bowel angiodysplasia experience greater fatigue and impaired quality of life compared to those with unexplained anaemia and normal SBCE. These results build on our previous findings and reinforce the clinical burden associated with angiodysplasia. Early recognition and targeted management may improve outcomes in this under-recognised group.

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Disclosure: Nothing to disclose.

PP0316

NON-OCCLUSIVE MESENTERIC ISCHEMIA AND PORTAL VENOUS THROMBOSIS CAUSING SMALL BOWEL OBSTRUCTION: TWO CASES AND MANAGEMENT REVIEW

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Introduction: Non-occlusive mesenteric ischemia (NOMI) is a rare but critical condition resulting from insufficient mesenteric blood flow without arterial occlusion. When complicated by portal venous thrombosis and small bowel obstruction, NOMI presents significant diagnostic and therapeutic challenges, requiring early recognition and multidisciplinary management to improve outcomes.

Aims & Methods: This study retrospectively analyzed two patients with NOMI complicated by portal venous thrombosis and small bowel obstruction. Clinical history, imaging findings, treatment strategies, and outcomes were reviewed. The effectiveness of catheter-directed thrombolysis combined with systemic anticoagulation and the role of surgical intervention in recurrent ischemia were assessed. Statistical data include recurrence rates and treatment outcomes.

Results: Patient 1, a 62-year-old male, presented with acute abdominal pain, bloating, and vomiting. Imaging revealed portal venous thrombosis, partial small bowel obstruction, and signs of ischemia. He underwent catheter-directed mesenteric thrombolysis followed by systemic anticoagulation, resulting in symptom resolution and no recurrence at the 6-month follow-up. Patient 2, a 60-year-old male with atrial fibrillation, presented with prolonged abdominal pain and bloating. Imaging showed mesenteric and portal venous thrombosis causing small bowel obstruction. He underwent catheter-directed thrombolysis with initial improvement. However, symptoms recurred one month later, necessitating small bowel resection due to chronic ischemia. Postoperatively, the patient recovered well with no further recurrence.

Conclusion: This study highlights the importance of early identification and multidisciplinary management of NOMI with portal venous thrombosis. Catheter-directed thrombolysis combined with anticoagulation is an effective strategy for restoring mesenteric blood flow and relieving symptoms. Surgical intervention is critical in recurrent or chronic cases. These findings support individualized, evidence-based approaches to optimize outcomes in this complex condition.

Disclosure: Nothing to disclose.

PP0317

IS OCTREOTIDE A SALVAGE THERAPY FOR PREVENTING RECURRENT BLEEDING FROM GASTROINTESTINAL ANGIOECTASIAS?

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Introduction: Gastrointestinal angioectasia is a significant etiology of recurrent gastrointestinal bleeding, often requiring hospital care. Octreotide has been proposed as a therapeutic option in patients in whom endoscopic treatment is insufficient, particularly in cases of small bow-

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el or diffuse angioectasia.[1] This study evaluates the clinical efficacy of long-acting intramuscular octreotide in the recurrency of gastrointestinal bleeding from angioectasia.

Aims & Methods: A sample of 21 patients receiving monthly octreotide, either 10 mg or 20 mg, were retrospectively analyzed. Clinical efficacy was assessed by comparing the frequency of hospitalizations, urgency department visits, red blood cell (RBC) transfusions and intravenous iron infusions during the 6-month and 12-month intervals that preceded and followed the start of treatment. Additional contributing factors, including anticoagulation therapy and underlying diseases, were also examined.

Results: The sample studied has an mean age of 78 years, with 13 males and 8 hypocoagulated patients. The Charlson Comorbidity Index averaged 5.6 points, with 5 patients having cirrhosis and 5 patients Osler-Rendu-Weber Syndrome. Hospitalizations significantly decreased after octreotide initiation at both 6 and 12 months of treatment ($p < 0.001$; $p = 0.019$). Urgent care visits and RBC transfusions were significantly lower in the first 6 months of treatment ($p < 0.001$; $p = 0.007$), though this effect was not sustained at 12 months. Intravenous iron requirements did not differ statistically between pre- and post-treatment periods. Patients receiving 10 mg of octreotide had significantly fewer urgent care visits and iron infusions at 6 months of treatment compared to the 20 mg group ($p = 0.031$; $p = 0.026$).

Conclusion: In recurrent bleeding from angioectasias, when endoscopic treatment is ineffective, a 12-mounth course of octreotide therapy significantly reduces hospitalizations. Additionally, treatment in the first 6 months proves to be effective in reducing the number of urgency department visits and red blood cell transfusions.

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Disclosure: Nothing to disclose.

PP0318

UNRAVELING THE VICIOUS CYCLE: SUPERIOR MESENTERIC ARTERY SYNDROME IN ANOREXIA NERVOSA: A TUNISIAN RETROSPECTIVE COMPARATIVE STUDY

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Introduction: Superior Mesenteric Artery Syndrome (SMAS) is a rare condition characterized by compression of the third portion of the duodenum between the aorta and the superior mesenteric artery, leading to duodenal obstruction. While anorexia nervosa is a recognized risk factor due to rapid weight loss and loss of mesenteric fat, distinguishing whether SMAS is a cause or consequence of anorexia remains challenging.

Aims & Methods: The study aims to evaluate the clinical characteristics, management strategies, and outcomes of SMAS in patients with and without anorexia nervosa.

A retrospective comparative study was conducted in a hepatogastroenterology department from 2018 onwards, including 20 patients diagnosed with SMAS based on CT imaging criteria (aortomesenteric angle ≤25° and/or aortomesenteric distance ≤8 mm). Patients were divided into two groups: Group 1 (G1) comprised 9 patients with anorexia nervosa, and Group 2 (G2) included 11 patients without anorexia. All patients received enteral nutrition and psychiatric evaluation. Symptom severity was assessed using a subjective score considering symptom frequency, intensity, and weight loss, with scores >12 indicating severe symptoms.

Results: The mean age in G1 was 20.6 ± 2 years, with 77.4% being secondary school students. The average delay between diagnosis and treatment initiation was 147.3 ± 116.5 days for both groups.

Initial anthropometric measurements for G1 showed a mean weight of 45.7 ± 4.5 kg and a mean BMI of 17.5 [13.6–19.5]. The average weight loss at diagnosis was 4.9 ± 1.5 kg over 6 months, with 77.7% experiencing ≥10% weight loss. G2 had comparable anthropometric data.

The initial mean severity score was higher in G1 (14 ± 3.0 [9–17]) compared to G2 (12.1 [11–17], p=0.05). G1 had a longer mean hospital stay (37.6 ± 20.9 days), significantly associated with anorexia nervosa and higher initial severity scores.

Treatment failure occurred in one G1 patient (11.1%) versus none in G2 (p=0.04). Recurrence of symptoms was observed in 44.4% of G1 compared to 16.7% in G2 (p=0.02).

Psychiatric intervention, including antidepressant therapy, was administered to 78% of G1 patients. Post-treatment, G1 showed better tolerance (77.8%, p=0.01) and a lower mean severity score (3.1, p=0.002). The average weight gain was 3.3 kg [1–11], with 29.4% achieving adequate weight restoration. Hospitalization duration was reduced, with trends approaching statistical significance.

Conclusion: Anorexia nervosa, even when initially subclinical, can lead to severe malnutrition and SMAS, posing significant risks to young patients. Clinicians should be vigilant for the coexistence of these conditions and implement comprehensive management strategies that address both nutritional and psychological aspects.

Disclosure: Nothing to disclose.

PP0319

LONGITUDINAL ASSESSMENT OF FATIGUE AND QUALITY OF LIFE IN PATIENTS WITH SMALL BOWEL ANGIODYSPLASIA: INSIGHTS FROM RAND SF-36 AND FATIGUE SEVERITY SCALE

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Introduction: Patients with small bowel angiodysplasia, and chronic anaemia often experience a complex symptom burden, including reduced quality of life (QoL) and fatigue. While cross-sectional studies suggest impaired QoL in this population, longitudinal data assessing how these symptoms evolve over time are limited (1).

Both general QoL tools, such as the RAND SF-36 and measures, like the Fatigue Severity Scale (FSS), may offer insight (2, 3).

Argon Plasma Coagulation (APC) (4) is commonly employed for vascular lesions, but its impact on long-term symptom burden remains unclear. This study aimed to evaluate changes in QoL and fatigue over time in patients with small bowel angiodysplasia compared to anaemia controls without small bowel pathology.

Aims & Methods: Following ethical approval, patients with small bowel angiodysplasia and controls, those with anaemia and negative bidirectional and capsule endoscopy, were invited to complete an initial and subsequent (minimum of six months) RAND SF-36 and FSS. Baseline QoL and fatigue assessment was performed on the day of SBCE or enteroscopy. Repeat RAND SF-36 and FSS were obtained from participants via telephone interview. Basic demographics and procedural related data, QoL and fatigue data were recorded. An FSS >36 was defined as clinically

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significant fatigue. Paired t-tests were used to assess within-group changes from baseline to follow-up. In the angiodysplasia group, a subgroup analysis compared changes in fatigue between patients who received APC and those who did not, using independent t-tests.

Results: A total of 28 patients in the angiodysplasia group (mean age 67.5 years; 35.7% female) and 38 anaemia controls (mean age 64.8 years; 42.1% female) were included, with no significant demographic differences. In the angiodysplasia group, the proportion of patients with clinically significant fatigue (FSS > 36) increased from 82.1% at baseline to 92.9% at follow-up ($p = 0.046$) suggesting overall significant worsening of fatigue over time. Anaemia controls showed no change in FSS scores (44.7% at both time points, $p = 1.000$). Despite this, in the angiodysplasia group, several SF-36 domains, including Energy/Fatigue, Social Functioning, and Emotional Wellbeing, demonstrated statistically significant improvements ($p < 0.05$). This suggests that in our cohort there was a significant improvement in QoL over time. 19 patients (67.9%) in the angiodysplasia group underwent Argon Plasma Coagulation (APC). The proportion of patients with clinically significant fatigue (FSS >36) decreased from 57.9% to 31.6% in the APC angiodysplasia subgroup ($p = 0.433$) and from 55.6% to 33.3% in angiodysplasia patients who did not receive APC ($p = 0.570$). Both groups received the same standard of care, transfusions and iron replacement as required. Data on medical therapy was unavailable.

Domain	Vascular Group Baseline	Vascular Group Follow-up	Change (p-value)	Control Baseline	Control Follow-up	Change (p-value)
FSS > 36 (%)	82.1%	92.9%	+10.8% (p=0.046)	44.7%	44.7%	+0.0% (p=1.00)
Energy/ fatigue	37.5	50.7	+13.2% (p=0.014)	40.2	40.7	+0.5 (p=0.841)
Social Functioning	58.5	84.8	+26.3 (p = 0.001)	65.0	66.2	+1.2 (p = 0.762)
Emotional Wellbeing	65.4	78.2	+12.8 (p = 0.013)	68.7	69.1	+0.4 (p = 0.891)
Role Limitations (Physical)	27.3	49.1	+21.8 (p = 0.028)	30.5	31.0	+0.5 (p = 0.812)

Conclusion: Despite improvements in several general QoL domains measured by RAND SF-36, patients with angiodysplasia experienced a significant worsening of fatigue over time, as captured by the FSS. These findings highlight the limitations of generic QoL tools in detecting persistent, clinically meaningful fatigue. APC alone appears not to have an impact on fatigue. This underlines the importance of assessing patient-reported fatigue as a treatment outcome.

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Disclosure: Nothing to disclose.

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PP0320

THE LINK BETWEEN GUT MICROBIOTA COMPOSITION, OBESITY, AND HEPATIC STEATOSIS/FIBROSIS IN THAI ADOLESCENTS: A CROSS-SECTIONAL STUDY

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Introduction: Obesity and metabolic dysfunction-associated steatotic liver disease (MASLD) have been linked to gut dysbiosis.

Aims & Methods: This study aimed to investigate the relationship between gut microbiota composition, hepatic fat content and stiffness, and body mass index (BMI) status. This cross-sectional study enrolled adolescents aged 12 to 18. Transient elastography with controlled attenuation parameter (CAP) was employed to diagnose hepatic fibrosis and steatosis. Fecal samples were analyzed by 16S rRNA sequencing. Relative abundance, as well as alpha- and beta-diversity of the microbiota, were compared across participants with different nutrition statuses and the presence of hepatic steatosis and fibrosis.

Results: A total of 76 participants (51 males) were recruited for this study, including 43 with normal weight, 19 overweight, and 14 obese. The clinical characteristics of the participants are presented in Table 1. Hepatic steatosis was present in 8 out of 14 obese participants (57.2%) and 2 out of 19 overweight (10.5%) participants. Hepatic fibrosis was detected in 4 out of 14 obese participants (28.6%) but none in the normal weight or overweight groups. There were no significant differences in alpha- or beta-diversity between the normal weight, overweight, and obese groups. The Firmicutes to Bacteroidetes (F/B) ratio was 1.2 across all three groups. Several genera, including *Eubacterium_halli_group* and *Ruminococcus_gauvreauii_group* were present in higher abundances in the gut microbiome of the obese group compared to the normal-weight group, with *p* values of 0.025 and 0.036, respectively. In contrast, *Bifidobacterium* was more abundant in the normal-weight participants (p=0.012). Obese individuals with hepatic fibrosis exhibited a significantly higher F/B ratio compared to those without hepatic fibrosis (p=0.04). There were significant differences in the relative abundance of specific genera between the fibrosis and non-fibrosis groups: *Phascolarctobacterium* (p=0.020), *Dorea* (p=0.040), and *Ruminococcus_gauvreauii_group* (p=0.045) were present in higher abundances in the fibrosis group, whereas *Senegalimassilia* was more abundant in the non-fibrosis group (p=0.037).

	Normal (n=43)	Overweight and obesity (n=33)	P value
Age (year)	14.4 (14.1-17.2)	15.2 (13.6-16.5)	0.71
CAP (decibels/meter)	186 (154-207)	203 (191-237.5)	<0.001*
Presence of hepatic steatosis	0 (0.0%)	10 (30.4%)	<0.001*
Liver stiffness (kilopascals)	3.9 (3.2-4.4)	4.9 (3.9-7.2)	0.025*
Presence of hepatic fibrosis	0 (0.0%)	4 (12.0%)	0.032*

BMI Body mass index, CAP Controlled attenuation parameter
Data expressed as number (percentage) or median (interquartile range); *P<0.05

Conclusion: Hepatic steatosis and fibrosis are common among obese adolescents. The gut microbiota abundance in obese individuals with hepatic fibrosis differ from those without it. Further studies are crucial to explore the role of gut microbiota modification in the progression of hepatic steatosis and fibrosis.

Disclosure: Nothing to disclose.

PP0321

A NOVEL GASTRIC PERORAL ENDOSCOPIC MYO-NEUROTOMY TECHNIQUE FOR BARIATRIC INTERVENTION: EFFICACY AND SAFETY IN A PORCINE MODEL

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Introduction: Endoscopic sleeve gastroplasty (ESG) is a minimally invasive and effective treatment for obesity. However, concerns remain regarding gastric mucosal folding and the resulting reduced visibility of the gastric lining. To address these limitations, we developed a novel peroral endoscopic myo-neurotomy (POEM-N) technique (1), adapted from peroral endoscopic myotomy (POEM), as a weight-loss procedure. Although promising, the original POEM-N was time-consuming.

This study aims to evaluate the safety and efficacy of a revised POEM-N (R-POEM-N) technique using a submucosal approach as a less invasive bariatric option in a porcine model.

Aims & Methods: Twelve pigs were assigned to either the R-POEM-N group (n = 8) or a sham endoscopic control group (n = 4). Myo-neurotomy was performed at four gastric sites in the R-POEM-N group: two on the greater curvature of the gastric body and two in the antrum. All pigs were fed a hypercaloric diet (twice the standard weight-based intake) and monitored for eight weeks post-procedure.

The primary outcome was the proportional change in body weight at 8 weeks, calculated as the ratio of weight change to baseline weight. Secondary outcomes included food intake, changes in serum glucolipid levels, and assessments of safety and procedural duration.

Results: R-POEM-N was successfully completed in all pigs without major adverse events. The mean procedure time was 171.3 ± 3.0 minutes. At 8 weeks, the proportional change in body weight was significantly lower in the R-POEM-N group compared to controls (P = 0.017). Food intake was also significantly reduced in the R-POEM-N group (P < 0.05). No significant differences were observed in serum glucolipid levels between groups.

Conclusion: R-POEM-N effectively limited weight gain in a porcine model and demonstrated a favorable safety profile. These findings suggest that R-POEM-N may serve as a promising, minimally invasive bariatric intervention.

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Disclosure: Nothing to disclose.

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PP0322

SAFETY AND OUTCOMES OF ENDOSCOPIC SLEEVE GASTROPLASTY FOR MANAGEMENT OF OBESITY AT THE FIRST NHS ENDOBARIATRICS CENTRE IN NORTH OF ENGLAND

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Introduction: The National Institute of Health and Care Excellence (NICE) recently endorsed Endoscopic Sleeve Gastroplasty (ESG) in combination with lifestyle modifications as a therapeutic modality to bariatric surgery in patients with body mass index > 30. Darlington Memorial Hospital was the first NHS Endobariatric Centre offering ESG to patients since 2022.

Aims & Methods: This is a retrospective analysis of the clinical outcomes and safety of ESG for the management of obesity in patients with BMI >30 between 2022-2024 at a single centre. A total of 21 procedures have been performed at this centre in the last 24 months, 2 procedures were performed in 2022, 4 procedures in 2023, and 15 procedures were carried out in 2024. We analysed procedural safety and total body weight loss (TBWL) at 6, 12 and 24 months.

Results: All patients were discussed in a multi-disciplinary team prior to being enlisted for ESG. Appropriate lifestyle modifications were initiated with Specialist Dietitian support. The average age of patients was 50.75 years (range 25 – 72), average BMI was 44.317 (range 35.37 - 57.4) and 75% of patients were women. None of the patients were receiving medications for weight loss (GLP-1 agonists). 3 patients were diabetic, 11 were hypertensive, 3 dyslipidemic and 4 had fatty liver. Other obesity related morbid conditions included Obstructive sleep apnoea in 5 patients and Osteoarthritis in one.

A total of 13 patients have at least 6 months follow up since the time of their procedure. Average total body weight loss in 6, 12 and 24 months was 17.78%, 36% and 26.7% respectively. Patient satisfaction scores have been consistently high.

The average time to completion of ESG was 88.26 minutes. The average length of stay in hospital was 2.24 days (range 1-11). All sutures were made in a U shape. The average number of sutures was 4.24 (range 1- 7). There were only 3 minor complications in this period: 1 post operative upper gastrointestinal bleed, 1 subphrenic collection and 1 minor pharangeal perforation.

All of these complications occurred in the early phase of the service, and were managed conservatively, with full recovery.

Conclusion: Our initial 24 month experience with ESG shows excellent TBWL, similar to the MERIT Study.

Procedural time is less than conventional bariatric procedures, with no major complications. These outcomes are similar to other international publications from larger centres.

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Disclosure: Nothing to disclose.

PP0323

ENDOSCOPIC GASTRIC MUCOSAL ABLATION FOR WEIGHT LOSS: 8 YEARS FOLLOW-UP

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Introduction: Obesity, currently regarded as a pandemic, is a life-threatening medical condition that impacts health and quality of life. Bariatric surgery is the most effective and sustained weight loss method for treating obesity, resulting in significant improvement in obesity-related multimorbidity and lower mortality rates. Endoscopic gastric mucosal ablation (EGMA) is a restrictive procedure that combines endoscopic submucosal dissection (ESD) and endoscopic mucosal resection (EMR), removing 80% of the gastric mucosa, which leads to stomach shrinkage and decreased gastric volume.

Aims & Methods: This study aimed to evaluate the effects of EGMA on weight loss and metabolic parameters. This study included 18 patients who underwent EGMA between July 2015 and June 2023. We collected basic anthropometric measurements, such as weight (kg) and height (m), at baseline before the procedure and at routine outpatient follow-up visits conducted annually following EGMA. Patients who failed to be present in person for their scheduled follow-up visits were contacted by phone or online. Additional anthropometric measurements, including WC (cm) and HC (cm), were collected at the baseline and 1 year, 2 years, 4 years, and 6 years post-EGMA procedure to evaluate abdominal obesity.

The primary outcome was weight loss, which was assessed by analyzing the changes in body mass index (BMI) (in kg/m^2), total body weight loss (TBWL) (in %), excess weight loss (EWL) (in %), and abdominal adiposity, as indicated by waist-to-hip ratio (WHR). Secondary outcomes included the impact of EGMA on metabolic indicators, including diabetes, blood pressure, dyslipidemia, and hyperuricemia.

Results: A total of 18 patients (12 women, 6 men; mean age of 35.6 ± 12.6 years) with a mean initial weight of 105.8 ± 16.4 kg underwent the EGMA procedure. Following EGMA, 18 patients had completed a year of follow-up at the time of analysis, followed by 17 in three years, 16 in four years, 9 in six years, 3 in seven years, and 2 in eight years. The changes in BMI, TBWL, %TBWL, %EWL, and WHR are shown in table 1.

The majority of patients presented with one or more comorbidities associated with obesity. At 2 years of post-EGMA, 4/7 (57.1%) patients achieved complete remission of diabetes, 3/7 (42.9%) of hypertension, 7/7 (100%) of hyperlipidemia, and 4/8 (37.5%) of hyperuricemia. There were no major adverse events interprocedurally or during the 8 years of follow-up. The mean post-EGMA hospital stay was 8.4 ± 2.8 days.

Conclusion: EGMA could suggest a new, less invasive, and restrictive endoscopic weight reduction method. The procedure appears to be effective and safe, with minimal technical complexity and no surgical complications.

Disclosure: Nothing to disclose.

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PP0324

A NOVEL ENDOSCOPIC MUCOSAL RADIOFREQUENCY ABLATION INDUCES GASTRIC VOLUME REDUCTION: AN ANIMAL EXPERIMENT

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Introduction: Obesity is a major global health problem and financial burden. Main treatments of obesity include lifestyle changes, medication, and bariatric surgery. However, lifestyle and drug therapies often fail for moderate to severe obesity. Bariatric surgery is an effective and conventional choice, though it is proven to be invasive with high risk of complication. Recently, endoscopic bariatric treatments including intragastric balloon deployment and sleeve gastroplasty have the advantages of being minimally invasive. However, they also carry the limitations of postoperative weight regain and uncertain long-term outcome.

Aims & Methods: In this study, we introduce a novel approach for mucosal ablation with radiofrequency, to guarantee enough tissue injury as well as achieve gastric volume reduction and potential weight loss. We conducted the first animal experiment of endoscopic radiofrequency ablation to induce gastric volume reduction and weight loss and evaluated its safety and feasibility. Eight pigs were enrolled in this study and observed over a three-month period. Five pigs comprised the experimental group and received endoscopic radiofrequency ablation, while three served as controls. Body weight, abdominal circumference, body length, total food intake, and gastric volume were measured and analyzed, along with follow-up endoscopic examinations.

Results: In experimental group, five pigs successfully underwent endoscopic gastric mucosal radiofrequency ablation. The average weight gain for the experimental group at 1st, 2nd, and 3rd months post-procedure was (5.0±1.5) kg, (17.7±5.6) kg, and (34.3±2.1) kg, respectively. In contrast, the control group showed average weight gains of (9.7±5.2) kg, (31.0±7.8) kg, and (50.8±6.9) kg at the same time point. There was no statistically significant difference in weight gain between the experimental and control groups at 1-month postoperatively (P=0.256); however, pigs in experimental group exhibited significantly less weight gain at 2 and 3 months comparing with controls (P=0.029, 0.048). No significant differences in total food intake were observed between both groups at within 2 months (P=0.214), but the experimental pigs consumed less food than the controls on 3-month follow-up(P=0.029).

Additionally, the growth rates of body length and abdominal circumference were significantly lower in the experimental group comparing with controls at 1, 2, and 3 months (P=0.001, 0.007, 0.013). The average the procedure time in experimental group was 3.2±0.8 hours, with no significant intra-procedural and post-procedural complications occurred in the experimental group, such as bleeding, perforation, or procedure-related death.

Conclusion: Endoscopic gastric mucosal radiofrequency ablation to induce volume reduction is an effective and safe method as a novel bariatric treatment, suggesting it may be a promising application potential.

Disclosure: Nothing to disclose.

PP0325

RESIDUALS TO REMAINS: GASTRIC RESIDUAL VOLUME AS A PREDICTOR OF MORTALITY IN CRITICALLY III. A SINGLE-CENTER RETROSPECTIVE COHORT STUDY

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Introduction: Gastric residual volume (GRV) serves as a surrogate marker for gastrointestinal dysfunction during enteral feeding in critically ill patients. It helps detect delayed gastric emptying early, potentially guiding interventions to prevent complications such as aspiration and feeding intolerance. The American Society for Parenteral and Enteral Nutrition (ASPEN) defines GRV as the liquid drained from the stomach post-enteral feeding, composed of nutritional formula, water, and GI secretions, measured via aspiration or gravity drainage. While monitoring GRV is commonly used to assess feeding tolerance, its clinical relevance in predicting mortality remains unclear. A study by Hsu et al. linked higher GRV with increased disease severity and higher SOFA scores in ICU patients. However, the latest ESPEN guidelines highlight inconsistencies in GRV monitoring, suggesting that enteral feeding should be delayed when GRV exceeds 500 mL/6h, though no definitive cut-off for intervention or its role in mortality prediction has been established.

Aims & Methods: The aim of this study is to determine the impact of gastric residual volume in predicting incidence of mortality in critically ill patients admitted at a tertiary hospital in the Philippines (PH). This single-center, three-year retrospective analytical cohort study was conducted via chart review of critically ill patients admitted in a tertiary hospital in the PH. Patients aged ≥19 years, on enteral nutrition via nasogastric or gastrostomy tube, and admitted for >3 days were included. Exclusions were pediatric, pregnant, and lactating patients, as well as those with gastric outlet obstruction, bowel obstruction, or recent abdominal surgery.

The primary outcome assessed the impact of gastric residual volume (GRV) on ICU mortality. Secondary outcomes included the association between GRV and clinical parameters such as SOFA score, mortality risk stratification based on GRV >500mL versus <500mL, and time intervals related to feeding and GRV trends. Data were collected from electronic and paper medical records and analyzed using Mann-Whitney U test and multivariable logistic regression. Prognostic accuracy was assessed using sensitivity, specificity, and predictive values.

Results: A total of 150 ICU patients were analyzed (median age 68.5 years, 57.3% male). Survivors had a significantly younger median age than non-survivors (62 vs. 72 years, p<0.001). Key comorbidities included hypertension (63.3%), kidney disease (72.0%), and diabetes (42.7%). The SOFA score, PF ratio, MAP, bilirubin levels, platelet count, and GCS differed significantly between survivors and non-survivors (p<0.001). Among patients with GRV >500mL (n=24), mortality was significantly higher (87.5%) compared to those with GRV <500mL (63.5%, p=0.029). Sensitivity for predicting mortality with GRV >500mL was 93.9% (CI: 83.1%-98.7%), though specificity was low at 25.9% (CI: 16.8%-36.9%). The positive predictive value was 43.4%, while the negative predictive value was 87.5%, with an overall diagnostic accuracy of 44.7%. These findings suggest that elevated GRV is associated with increased mortality risk in critically ill patients, though its predictive specificity is limited.

Conclusion: This study highlights the association between increased gastric residual volume (GRV) and higher mortality risk in critically ill patients, with GRV >500mL linked to a fourfold increase in mortality. GRV may serve

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as a clinical tool to guide management decisions, balancing diagnostic and therapeutic interventions with palliative care when appropriate. In resource-limited settings like the Philippines, this could help reduce healthcare costs and optimize hospital resources. As no prior studies have explored GRV's role in mortality prediction, this research provides a foundation for future investigations.

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PP0326

EFFECTIVENESS OF PROBIOTICS IN ALLEVIATING COLD SYMPTOMS IN HEALTHY INDIVIDUALS: A RANDOMIZED, DOUBLE-BLIND, PLACEBO-CONTROLLED CLINICAL TRIAL

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Introduction: The term "cold" describes a range of respiratory illnesses primarily triggered by viral infections. Colds are the most common infectious diseases in the world, affecting daily life and increasing healthcare costs and lost productivity. Cold symptoms typically emerge 2-3 days after viral exposure. Children and those with a compromised immune system may experience more severe complications. The gut microbiota is directly involved in immune function, and probiotics can restore gut balance and enhance immunity. Specific probiotic strains, including *L. plantarum PBS077, L. acidophilus PBS066, and B. lactis BL050*, have demonstrated antimicrobial and anti-inflammatory properties.

Aims & Methods: This study aims to evaluate the effects of a dietary supplement containing *L. plantarum PBS077, L. acidophilus PBS066, and B. lactis BL050* probiotics on relieving cold symptoms in healthy individuals, focusing on its potential to enhance immune response and improve quality of life (QoL). This randomized, double-blind, placebo-controlled clinical trial was conducted in Italy in winter from 2022 to 2024, involving 65 healthy volunteers aged 18-44. Participants were assigned to two groups: 40 received a probiotic treatment with specific strains (*L. plantarum PBS077, L. acidophilus PBS066, B. lactis BL050*), B6 and B12 vitamins, folic acid, and bulking agents (Probiotic), while 25 received a placebo, consisting of bulking agents, B6 and B12 vitamins and folic acid, without probiotics (Placebo).

Participants took one capsule daily for 12 weeks, with assessments conducted at baseline (T0), after treatment (T1), and 6 weeks post-treatment (T2). Cold symptoms and quality of life (QoL) were assessed using a validated questionnaire (WURSS-21), and systemic inflammation was evaluated through serum cytokine levels (IFN-γ and IL-10) at each time point. The total observation period lasted 18 weeks.

Results: Probiotics were associated with a shorter average duration of cold symptoms (4.5 *vs.* 6.7 days for the Placebo), with a 32.7% reduction in symptom duration, $p<0.05$. At T1, fever occurred in 8 out of 40 participants in the Probiotic group (20%) and in 7 out of 25 participants in the Placebo group (28%), $p<0.05$. Additionally, at T1, muscle pain was reported by 20% of the Probiotic group (8 out of 40) *vs.* 44% of the Placebo group (11 out of 25). By T2, muscle pain occurred in 17.5% of the Probiotic group (7 out of 40) and 20% of the Placebo group (5 out of 25), $p<0.05$. At T1, the pro-inflammatory cytokine IFN-γ levels decreased in the Probiotic group compared to T0 (17,946±2,857 vs. 28,962±4,143 pg/ml, $p < 0.05$) and increased in the Placebo group compared to T0 (22,279±3,538 vs. 19,432±3,143 pg/ml, $p < 0.05$). Though not statistically significant, at T1 the Probiotic group had higher levels of IL-10 compared to T0 (266,98±78,432 vs. 240,967± 70,238, $p>0.05$).

Conclusion: The probiotic mix alleviated symptoms in patients with colds and reduced levels of the pro-inflammatory cytokine IFN-γ, demonstrating anti-inflammatory properties.

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PP0327

ASSESSING DIET QUALITY INDICES FOR AN INFLAMMATORY BOWEL DISEASE POPULATION – WHICH IS BEST?

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Introduction: Assessing the role of diet in the pathogenesis of inflammatory bowel disease (IBD) has moved away from analyses of individual nutrients and towards dietary patterns, helping to account for the complexity of food and nutrient interactions. Dietary patterns such as ultra-processed foods or the Mediterranean diet (MED), which have been implicated in IBD onset and course indicate need for diet quality assessment. However, an IBD-specific diet quality index (DQI) does not exist. It is unknown which DQI is most suitable to use in therapeutic IBD diet trials. This review aimed to identify current DQIs and assess their suitability for use on an IBD population.

Aims & Methods: MEDLINE and EmCare electronic databases were systematically searched for *a-priori*, food-based DQI reflecting national dietary guidelines and/or latest nutrition science for adults published 2013 to 2023 in English language. Posteriori or non-IBD disease-specific (e.g., diabetes) DQI were excluded. Data extracted were adapted from optimal DQI criteria described by Burggraf et al.,[1] which include quality measures of adequacy, moderation, variety and balance for food and nutrient components and DQI evaluation, as well as its use, if any, for IBD health outcomes.

Results: Twenty-four articles describing 24 DQI were included in final analysis. Dietary guidelines were used to develop 18/24 (75%) DQI. The other six were developed from dietary patterns (e.g., EAT-Lancet Diet). The majority were developed in Australia (7/24), North America (5/24), Europe (5/24) and Asia (4/24). One was developed from an international cohort. All DQI assessed whole food groups, but assessment of individual foods varied. Ultra-processed foods were assessed in 16/24 (67%) DQI. No DQI included all optimal DQI criteria[1]. The Dietary Guideline Index 2013 (DGI-2013) most closely met DQI criteria, followed by the Healthy Eating Index for Australian Adults-2013 (HEIFA-2013), Dutch Healthy Diet Index-2015 (DHD Index-2015) and Planetary Health Diet Index (PHDI) (Table 1). Most DQI assessed adequacy (22/24, 92%) and moderation (21/24, 88%), half assessed balance (12/24, 50%) while few assessed variety (9/24, 38%).

Index	Dimensions				Diet Assessment Method	Normative*, age & gender-based score criteria	Unequal Weight+	Applied to IBD cohort
	Adequacy	Moderation	Variety	Balance				
DGI-2013	Y	Y	P	Y	FFQ	Y	N	Y
DHD Index-2015	Y	Y	N	P	FFQ, 24hr Recall	Y	N	Y
HEIFA-2013	Y	Y	P	P	FFQ, Weighed Food Record	Y	Y	N
PHDI	Y	Y	N	P	FFQ	Y	Y	N

Footnotes. Green highlight (Y), DQI met adapted optimal DQI criteria;[1] Yellow highlight (P), criteria partially met; Red highlight (N), criteria not met; *evidence-derived e.g., recommended nutrient intakes or guideline serving quantities, +accounts for contributions of foods/nutrients to established health outcomes where one food/nutrient may have a greater health impact than another.

Table 1. Highest rated Diet Quality Indices (DQI) meeting a selection of adapted optimal construction criteria for an Inflammatory Bowel Disease-specific DQI

Outlined in *Table 1*, application of other recommended DQI criteria varied. Food frequency questionnaires (FFQ) (13/24) and 24-hour diet recall (12/24) were the most common dietary assessment methods used. Seventeen of 24 (71%) DQI were validated, however, few were evaluated for reliability (6/24) or reproducibility (2/24). No DQI was validated to assess IBD-specific health outcomes, however, the DGI-2013 and DHD Index-2015 have been applied to FFQ from IBD participants.

Conclusion: No existing DQI meets all recommended optimal criteria for an IBD-specific DQI. The DGI-2013 met the most criteria, followed by the HEIFA-2013, DHD Index-2015 and PHDI. These DQI, depending on country and diet assessment method, may be most appropriate for an IBD population. An IBD-specific DQI is lacking and needed.

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PP0328

NUTRITIONAL ASSESSMENT AND ITS IMPACT ON QUALITY OF LIFE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE IN COLOMBIA: NUTRIBD – COL

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Introduction: Inflammatory bowel disease (IBD) significantly impacts patients' nutritional status and quality of life. Nutritional deficiencies are common in IBD and can worsen disease progression.

Aims & Methods: The aim of this study is to assess the nutritional status of patients with IBD and explore its relationship with quality of life in this cohort. A cross-sectional study conducted at a specialized center assessed nutritional status using the MUST and MIRT tools. For pediatric patients, the "Strong Kids" screening tool was used.

A multidisciplinary nutritional intervention was provided, followed by quality of life evaluation using the EQ-5D and IBDQ-32 instruments. Statistical analyses were performed to examine the association between nutritional status and quality of life.

Results: Among 90 IBD patients, 26.6% required hospitalization due to disease flare-ups, primarily adults and females. The median age was 36.7 years (range: 4–81, SD: 32.6), with 58.3% diagnosed with Crohn's disease and 83.3% requiring biologic therapy (Table 1). The odds ratios (OR) for malnutrition in hospitalized patients were 8.33 (95% CI: 1.16–27) for ulcerative colitis and 25 (95% CI: 11.6–45.2) for Crohn's disease. All hospitalized adults exhibited poor biochemical control and moderate-to-severe hypoalbuminemia; 25% had sarcopenia. Nutritional support was provided to 49% of patients within the past year. All severely ill hospitalized patients were malnourished.

Pediatric patients, all of whom received nutritional support, demonstrated a higher degree of nutritional compromise. Among these, 25% required parenteral nutrition, while 87.5% received enteral nutrition, primarily with high-calorie, high-protein formulas. During hospitalization, 87.5% gained weight and improved nutritionally, with 75% achieving normal nutritional status at three-month follow-up, although one patient died.

Following the nutritional intervention, over 80% of adults did not experience further nutritional deterioration. In the outpatient cohort, 20 patients were evaluated, of whom 30% were identified as high risk for malnutrition according to MUST, and 25% had nutritional risk (MIRT score ≥3). Patients

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with poorer initial nutritional status had significantly worse quality of life, with notable impairments in mobility (P = 0.2778; 95% CI: 2.03–64.1), self-care (P = 0.003; 95% CI: 9.45–90.5), daily activities (P = 0.1457; 95% CI: 1.14–58.2), pain/discomfort (P = 0.197; 95% CI: 0.53–53.3), and visual analog scale scores (P = 0.012; 95% CI: 22.9–88.4). Poor nutritional status was also associated with lower quality of life scores on the IBDQ-32 (P = 0.04; 95% CI: 10.3–60.7).

Conclusion: Nutritional status is a key component of IBD management, as malnutrition is linked to poorer quality of life. Screening tools such as MUST and MIRT in adults, and "Strong Kids" in children, are essential for early identification of at-risk patients. Pediatric patients tend to have more severe nutritional compromise than adults, and nutritional support during hospitalization contributes significantly to recovery.

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PP0329

LAURIC ACID AND GUT HEALTH: A FRIEND OR FOE IN FIBROSIS MODULATION?

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Introduction: Nutritional factors are increasingly recognized as key modulators of intestinal inflammation and fibrosis, offering new therapeutic angles in IBD management. Among dietary components, medium-chain fatty acids (MCFAs) have drawn attention for their bioactive properties. Lauric acid (LA), MCFA with anti-inflammatory properties, has an unclear role in IBD, particularly within a fibrotic context. LA interacts with G protein-coupled receptor 84 (GPR84), which is notably elevated in UC[1]

Aims & Methods: This study evaluated the effects of lauric acid (0, 50, 100, and 250 μM) on human intestinal fibroblasts (HSIF) and colonic epithelial cells (CoEpi) under basal and pro-fibrotic conditions induced by TGFβ1 (10 ng/mL). Fibrotic gene expression was assessed by PCR—HSIF: COL1A1, COL3A1, COL4A1, ACTA2, FN1, TGFβ, GPR84; CoEpi: SNAI1/2, CDH1, COL1A1, FNDC4, VIM, GPR84. Protein levels of GPR84, FNDC5, COL1A1, αSMA, and FGFR2 in HSIF were measured via Western blot. Data were expressed as fold change (mean ± SEM) relative to 0 μM LA and analyzed using ANOVA or Kruskal-Wallis tests, depending on normality (*P < 0.05 vs. 0 μM LA; **P < 0.01 vs. 0 μM LA; #P < 0.05 vs. 0 μM LA + TGFβ1; ##P < 0.01 vs. 0 μM LA + TGFβ1).

Results: Under basal conditions, lauric acid (LA) significantly decreased the mRNA expression of fibrosis-related genes in HSIF, including COL1A1 (250 μM: 0.2 ± 0.07**) and COL3A1 (50 μM: 0.4 ± 0.1**), as well as ACTA2 (100 μM: 0.1 ± 0.1**).

These anti-fibrotic effects were maintained or enhanced under TGFβ1 stimulation, particularly at lower LA doses—e.g., COL1A1 (50 μM: 1.1 ± 0.15##), COL3A1 (50 μM: 0.24 ± 0.1#), COL4A1 (250 μM: 0.6 ± 0.1#), and ACTA2 (100 μM: 0.7 ± 0.4#). Under fibrotic conditions, LA also significantly reduced the protein expression of COL1A1 (250 μM: 0.4 ± 0.17##) and αSMA (250 μM: 0.9 ± 0.14#). LA increased GPR84 mRNA expression in HSIF under both basal (250 μM: 1167 ± 646*) and fibrotic (250 μM: 1048 ± 505##) conditions, without affecting fibronectin or TGFβ levels.

In CoEpi cells, LA promoted epithelial-mesenchymal transition (EMT) markers in both basal and TGFβ1-stimulated states. This included upregulation of VIMENTIN (250 μM: 3.9 ± 0.6*; with TGFβ1: 6.4 ± 1.5#), SNAI2 (250 μM: 3.4 ± 0.5*; with TGFβ1: 3.9 ± 0.5##), and FIBRONECTIN (250 μM: 2.6 ± 0.5*; with TGFβ1: 4.4 ± 0.8##), alongside a decrease in E-CADHERIN (250 μM: 0.5 ± 0.08*; with TGFβ1: 0.35 ± 0.07#). GPR84 mRNA expression also increased significantly in CoEpi cells under both conditions (basal: 6.9 ± 1.5**; pro-fibrotic: 5.1 ± 1.0##). No changes were observed in COL1A1 or SNAI1 expression.

Conclusion: Lauric acid plays a dual, context-dependent role in intestinal fibrosis. It reduces fibrotic markers in fibroblasts, suggesting therapeutic potential, but promotes EMT in epithelial cells, which may contribute to fibrosis progression. These opposing effects, possibly mediated by GPR84, highlight lauric acid as both a friend and foe in gut health—supporting the need for further investigation before considering it as a dietary intervention in IBD.

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Disclosure: Nothing to disclose.

PP0330

VITAMIN D STATUS AND BONE MINERAL DENSITY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: ASSOCIATIONS WITH BODY COMPOSITION

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Introduction: Patients with inflammatory bowel disease (IBD) are at increased risk of reduced bone mineral density (BMD), often linked to chronic inflammation, corticosteroid use, biologics, and vitamin D deficiency. 25-hydroxyvitamin D (25(OH)D) is a fat-soluble vitamin essential for calcium and phosphorus homeostasis, and its deficiency is common in IBD due to malabsorption, dietary, altered metabolism, or impaired VDR signaling.

Aims & Methods: This cross-sectional study aimed to evaluate the relationship between serum 25(OH)D levels, BMD, and body composition in IBD patients. A total of 73 outpatients with IBD (38 ulcerative colitis, 35

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Crohn's disease) were assessed. Body composition was evaluated using BMI and bioelectrical impedance analysis (BIA). BMD was measured using whole-body dual-energy X-ray absorptiometry (DEXA), and vitamin D status was assessed by serum 25(OH)D concentration. Statistical analysis included descriptive and comparative tests (Chi-square and t-test).

Results: Participants had a mean age of 45.27±11.37 years, with 67.1% being female and a mean disease duration of 10.22±7.54 years. Most patients (93.15%) were in clinical and endoscopic remission. The average BMI was 26.85±4.51 kg/m^2, with 39.7% classified as overweight. DEXA revealed a high prevalence of osteopenia (39.7%) and osteoporosis (19.0%). Mean BMD Z-scores were whole body -1.32±1.13, spine -0.92±1.19, and femoral neck -0.34±1.02. Mean serum 25(OH)D level was 31.20±8.78 ng/mL; 52.1% had insufficient levels and 47.9% were sufficient. Vitamin D insufficiency was significantly associated with osteopenia/osteoporosis (p=0.0144) and with overweight/obesity (p=0.0265).

No significant associations were observed between vitamin D levels and age (p=0.6476), sex (p=0.4565), lean mass (p=0.2941), fat mass (p=0.1218), phase angle (p=0.3154), or regional BMD (total body p=0.5001; spine p=0.3067; femoral neck p=0.2364).

Conclusion: IBD patients are vulnerable to low serum vitamin D levels and present a high prevalence of low BMD. Vitamin D insufficiency was associated with altered BMI and increased risk of osteopenia/osteoporosis, though not with other body composition metrics. Further studies are needed with larger and more clinically heterogeneous samples to confirm these associations and guide screening and supplementation strategies.

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Disclosure: Nothing to disclose.

PP0331

THE DEVELOPMENT OF FUNCTIONAL GASTROINTESTINAL DISORDERS IN SCHOOL-AGE CHILDREN: THE INFLUENCE OF MALNUTRITION AND EATING DISORDERS

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Introduction: Nutrition is one of the important risk factors for the occurrence of functional gastrointestinal disorders in children. These disorders, including functional dyspepsia (FD) and irritable bowel syndrome (IBS), are increasingly prevalent in the pediatric population. Understanding dietary patterns in affected children is crucial for developing preventive and therapeutic strategies.

Aims & Methods: The aim is to study eating behavior disorders in children with functional gastrointestinal disorders and to analyze the diet of children with combined functional gastrointestinal pathology.

Materials and methods: 110 children aged 6 to 18 with functional gastrointestinal disorders were examined. Patients were divided into two groups: 44 with a combination of symptoms of functional dyspepsia and irritable bowel syndrome and 66 with functional dyspepsia. The control group consisted of 30 practically healthy children. The diagnosis of functional dyspepsia and irritable bowel syndrome was established on the basis of clinical symptoms according to the Rome IV criteria (2016). In order to detect a violation of the nutritional regime, children were interviewed according to a questionnaire developed by us. They also studied the food preferences of schoolchildren using the Youth Adolescent Food Frequency Questionnaire, adapted to the model of children's nutrition in Ukraine.

Indicator	Combined FD and IBS, n = 44, N 14	% ± m	Functional Dyspepsia n = 66, N 8	% ± m	Control Group n = 30, N 3	% ± m
Eating less than 3 times a day		**31.8 ± 7.3***		**12.2 ± 4.4**		**10 ± 5.5**
Skip breakfast	17	38.6 ± 7.3*	5	7.6 ± 3.9	4	13.3 ± 6.2***
Eat sandwiches for breakfast	14	31.8 ± 6.7	27	40.9 ± 7.4**	4	13.3 ± 6.2
Eat hot meals for breakfast	13	29.7 ± 6.6*	34	51.6 ± 7.5	22	73.3 ± 8.1***
Do not eat lunch at school	2	4.54 ± 1.51	5	7.57 ± 3.98	0	-
Eat lunch in school cafeteria	19	43.2 ± 7.5	42	63.2 ± 7.3	20	66.7 ± 8.6
Buy food in school snack bar	21	47.7 ± 7.53*	15	22.7 ± 6.3**	2	6.7 ± 4.5*
Bring food from home	2	4.5 ± 1.5	4	6.1 ± 3.7**	8	26.6 ± 8.1*
Have an afternoon snack	23	52.3 ± 7.6*	55	83.0 ± 5.6	27	90.0 ± 5.5*
Eat dinner	43	92.7 ± 3.9	61	92.4 ± 3.9	30	100

Legend:

** – statistically significant difference (p < 0.05) between groups I and II;*

*** – statistically significant difference (p < 0.05) between group II and the control group;*

**** – statistically significant difference (p < 0.05) between group I and the control group.*

Results: The frequency of eating disorders (absence of breakfast or full lunch at school, reduced number of meals) in children with functional disorders is higher than in healthy children. In children with overlap syndrome, these disorders were detected statistically significantly more often than in children with isolated functional dyspepsia.

In children with functional disorders, the frequency of consumption of vegetables, cereals and cereal products is lower than in healthy children, and the consumption of meat, sausages and confectionery, fast food, chips, nuts, seeds, energy and sweet carbonated drinks, and fruit juices is

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higher. In children with a combination of functional dyspepsia and irritable bowel syndrome, the frequency of consumption of vegetables, cereals, and grain products is lower, and the consumption of confectionery, fast food, energy and sweet carbonated drinks is higher than in children with functional dyspepsia.

We conducted an analysis of nutrition in children with a combined pathology of functional dyspepsia (FD) and irritable bowel syndrome (IBS) (see table below).

Conclusion: Violation of the regime and quality of nutrition is a risk factor for the formation of functional gastrointestinal disorders in children, which must be taken into account in treatment.

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PP0332

CLINICIAN KNOWLEDGE AND CONFIDENCE IN ADVISING ON FIBRE INTAKE: A CROSS-SECTIONAL SURVEY STUDY

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Introduction: A well-established association exists between dietary fibre intake and overall health, with fibre recognised as a therapeutic intervention for a number of gastrointestinal disorders, including irritable bowel syndrome and chronic constipation.[1,2]

A diet rich in dietary fibre is endorsed by guidance from United European Gastroenterology and the European Society of Clinical Nutrition and Metabolism, however only 9% of the UK population consumes the recommended amount of 30g/day of fibre.[3-5]

Providing patients with clear and specific dietary recommendations is likely to enhance compliance, leading to improved health outcomes.[6] While doctors often provide generalised advice to patients to increase their fibre intake, little is known about their specific or detailed knowledge of fibre-rich foods and the methods they use to counsel patients on dietary modifications.

Aims & Methods: This study aimed to assess how well doctors identified food as high fibre sources, their confidence in providing fibre-related advice, and how frequently they engage in such discussions.

A cross-sectional survey targeting hospital doctors in the UK was conducted via a questionnaire in April 2025 using the Qualtrics platform. The survey captured demographic data, clinical practices, knowledge, and confidence levels in fibre-related counselling. All responses were anonymous. Participants were asked to rate the amount of dietary fibre in a representative selection of 8 foods from different food categories. Food was classified as high fibre (≥6g/100g), medium fibre (3-5g/100g), and low/no fibre food (<3g/100g). All nutritional information was obtained from the British Nutrition Foundation.[7]

Results: Seventy-one responses were collected, 1 was incomplete, leaving 70 for analysis. Most doctors worked in either gastrointestinal surgery (46%) or internal medical (41%) specialties. The largest group of respondents stated that they were required to provide dietary advice to patients at least each week (49%).

Many respondents could not identify that high fibre food sources such as bran flakes (49%) and wholemeal toast (56%) had more than 6g/100g of fibre. Some respondents also wrongly classified low/no fibre food to be high fibre sources (apple [30%], banana [29%], and salads [40%]). No respondent was able to correctly identify the correct fibre category for all 8 foods.

Sixty percent of respondents stated a lack of confidence in providing dietary advice regarding fibre intake to patients and 61% did not feel confident in altering a patient's daily fibre intake based on their clinical condition.

Conclusion: Many doctors cannot accurately categorise food which are high, medium and low/no fibre food sources. Most doctors do not feel confident in providing fibre advice to patients. This is likely to impact patient understanding and successful adherence to fibre rich diets. Easily accessible resources and dedicated teaching on fibre intake should be provided to doctors in allied specialties as this has the potential to improve patient care.

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Disclosure: Nothing to disclose.

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PP0333

MARKERS OF PLATELET AND ENDOTHELIAL ACTIVATION REMAIN UNCHANGED AFTER ACUTE ERYTHRITOL AND XYLITOL INTAKE

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Introduction: Several metabolomic profiling studies have reported elevated endogenous erythritol and xylitol production in patients with conditions such as cardiovascular disease; however, the underlying cause of this upregulation remains unknown. In addition to their endogenous production in the human body, erythritol and xylitol are commonly used as sweeteners. A recently published pilot trial suggested that acute intake of these sweeteners may directly influence platelet aggregation and increase platelet responsiveness in short term (30 minutes post-administration). It is important to note that platelet responsiveness fluctuates under physiological conditions and may be influenced by external factors such as physical activity and nutrition.

Aims & Methods: We conducted a randomized, double-blind, placebo-controlled crossover trial to evaluate the effects of acute oral erythritol, xylitol, and water intake on platelet and endothelial activation as well as plasma erythritol and xylitol concentrations.

Eleven healthy, lean participates (mean ± SD; age: 26.4 ± 5.7 years, BMI: 21.5 ± 1.9 kg/m^2) were included in the trial. In a randomized crossover design, participants completed 3 different test sessions, receiving one of the following beverages per session: 50 g erythritol or 33.5 g xylitol dissolved in 300 mL water, or 300 mL water as control.

Blood samples were collected for the measurement of:

i. Platelet aggregation in response to TRAP6 10 µM *ex vivo* before and 60 min post-administration,

ii. Platelet (P-selectin) and endothelial (sVCAM-1) activation *in vivo* up to 180 minutes post-administration, and;

iii. Erythritol and xylitol concentrations up to 48 hours post-administration.

Results: The maximum adjusted platelet aggregation shows inter-individual and inter-visit variability. For erythritol, the fasting mean ± SD was 62.28 % ± 29.27 %, vs. 59.02 ± 28.82 % 60 min post-administration. For xylitol, the fasting mean ± SD was 78.17 ± 22.20 %, vs. 72.93 ± 26.20 % 60 min post-administration. For water, the fasting mean ± SD was 58.07 ± 28.2 %, vs. 57.45 ± 29.69 % 60 min post-administration. P-selectin and sVCAM-1 concentrations did not differ significantly between treatment conditions and showed no correlation with changes in platelet aggregation. Plasma erythritol concentrations peaked 60 min post-administration, remained slightly elevated for up to 48 hours, whereas plasma xylitol concentrations reached their peak at 30 min post-administration and returned to baseline within 24 hours.

Conclusion: Our results suggest that acute intake of erythritol or xylitol does not have a direct effect on platelet aggregation, P-selectin, or s-VCAM-1 concentrations.

Disclosure: Nothing to disclose.

PP0334

AVOIDANT/RESTRICTIVE FOOD INTAKE DISORDER (ARFID) IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: IMPACT ON BODY COMPOSITION AND QUALITY OF LIFE

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Introduction: Inflammatory bowel disease (IBD) is a significant risk factor for the development of Avoidant/Restrictive Food Intake Disorder (ARFID). ARFID is an eating disorder characterized by restrictive food intake, accompanied by at least one of the following: significant weight loss, nutritional deficiencies, dependence on enteral feeding or supplementation, or marked psychosocial impairment.

Aims & Methods: Objective: To assess the presence of ARFID and its impact on body composition and quality of life in patients with IBD. Methodology: This cross-sectional study included 105 outpatients diagnosed with IBD. The Nine Item ARFID Screen (NIAS) was used to assess ARFID symptoms, and quality of life was measured using the Inflammatory Bowel Disease Questionnaire (IBDQ). Body composition was evaluated through anthropometric measurements, bioelectrical impedance analysis, and handgrip strength testing.

Results: The study included 61 patients with ulcerative colitis and 44 with Crohn's disease, with a mean age of 41.5 ± 15.11 years; 70% were female, and the mean disease duration was 11.19 ± 7.58 years. Regarding disease activity, 27% had clinical activity and 45% had endoscopic activity. Overall, 23% of patients were classified as having ARFID symptoms, and 35% scored ARFID subtype 3, which is associated with gastrointestinal-related food avoidance. The mean BMI was 25.92 ± 5.18 kg/m^2, with 52.38% classified as overweight. The mean body fat percentage by skinfold measurement was 28.47 ± 7.5%, with 8% of individuals below average. Bioimpedance analysis showed that 86.41% had low muscle mass, and the mean handgrip strength was 27.37 ± 8.67 kgf, with 7% of participants having inadequate strength. The mean total IBDQ score was 141.29 ± 47.68 points. ARFID classification was associated with lower quality of life, as reflected by the total IBDQ score ($p = 0.0078$) and its gastrointestinal ($p = 0.0022$), systemic ($p = 0.0138$), and social ($p = 0.0194$) subdomains. It was also correlated with lower BMI ($p = 0.0206$) and lower body fat percentage ($p = 0.0045$). However, no correlation was found with muscle mass ($p = 0.2969$).

Conclusion: ARFID symptoms are common in patients with IBD, likely due to prolonged dietary restrictions during active disease and maintained during remission, potentially impacting both nutritional status and quality of life. Despite this, there is a scarcity of research on this topic, particularly among IBD patients. Further studies are needed to explore how the restriction of various food groups affects nutritional, psychosocial, and quality of life outcomes in this population.

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Disclosure: Nothing to disclose.

PP0335

TARGETED ANALYSIS OF VOLATILE METABOLITES FROM THE GUT MICROBIOME IN EXHALED BREATH

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Introduction: Breath volatile organic compounds (VOCs) are promising non-invasive biomarkers for clinical purposes. In digestive health, gut microbiome VOCs enter the bloodstream and diffuse into exhaled breath, offering a real-time snapshot of gut activity and metabolism. Translating these biomarkers into clinical practice depends on precise identification and robust validation—challenges that require standardized protocols and data integration. Our previously developed breath collection and GC-MS analysis method identified 148 on-breath VOCs against ambient background that were catalogued into an Atlas using purified chemical standards. Building on this success, further method development was undertaken, with a focus on enhancing the detection of previously underrepresented compound classes.
Aims & Methods: Our aim was to create a high-resolution TD-GC-MS method using a PEG-phase column to detect and confidently identify gut microbial VOCs—fatty acyls, esters, phenols, indoles, and sulfones. By combining enhanced sample preparation and compound enrichment with rigorous spectral and retention-time matching, plus improved quality-control and identity-confirmation protocols, we could reliably assign chemical identities to candidate VOCs in breath samples.
Results: Using three metrics; standard deviation analysis, paired t-test, and ROC-AUC evaluation, we identified 621 VOCs as potential breath biomarkers, 223 of which met all three on-breath criteria. These tools effectively distinguished exhaled VOCs from ambient background, allowing us to select a focused subset for confident chemical identification.
Notably, over 25% of these are metabolites derived from the gut microbiome, and their detection offers mechanistic insights into microbial activity. Butyric acid, for example, serves as a crucial energy source for colonocytes and plays a key role in regulating intestinal inflammation.
Branched-chain fatty acids—including isobutyric acid, isovaleric acid, 2-methylbutyric acid, and 4-methylvaleric acid—are products of protein fermentation and have been associated with inflammatory responses. The phenolic compound p-cresol, linked to dysbiosis, may affect gut barrier integrity, while 2-methylindole, derived from altered tryptophan metabolism, has implications for immune modulation and gut homeostasis. Dimethyl sulfone, involved in sulphur metabolism, may contribute to inflammatory signalling. Alongside previously identified short-chain fatty acids such as acetic and propionic acid, these compounds represent a broad and functionally diverse profile of gut microbial metabolism. Their detection across all on-breath metrics supports their endogenous origin in exhaled breath.
Conclusion: The high-confidence identification of these gut-derived VOCs reinforces their potential as non-invasive biomarkers for monitoring gastrointestinal health. With the updated methodology, our Atlas now includes a total of 186 confirmed on-breath VOCs, paving the way for improved diagnostic breath tests tailored to gut health and disease.
Disclosure: No conflict of interest.

PP0336

BEYOND THE BMI: A COMPREHENSIVE LOOK AT SARCOPENIA, MYOPENIA, AND MALNUTRITION IN POLISH PATIENTS WITH INFLAMMATORY BOWEL DISEASES

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Introduction: Inadequate nutritional status is a major complication of chronic gastrointestinal diseases such as Inflammatory Bowel Diseases (IBD), including two main types: Crohn's Disease (CD) and Ulcerative Colitis (UC) [1].
Previously, sarcopenia and myopenia were associated only with ageing, but they also affect patients with chronic diseases, including IBD, regardless of age [2].
Both malnutrition and sarcopenia lead to poorer health outcomes in IBD patients, increasing the risk of hospitalization, disease exacerbation, need for surgery and post-operative complications [3, 4].
Aims & Methods: The aim of this study was to assess the prevalence of sarcopenia, myopenia and malnutrition in Polish patients with IBD using clinical tools available in daily practice. A total of 91 IBD patients (48 CD, 43 UC), aged 18-65 years, with both inactive and active disease (30 and 61, respectively), were included in the study. Sarcopenia was assessed using the Strength, Assistance in walking, Rise from a chair, Climb stairs, and Falls (SARC-F) questionnaire and the 5-times sit-to-stand test, while myopenia was evaluated through the Fat-Free Mass Index (FFMI). Nutritional status was determined using BMI, the Nutritional Risk Screening (NRS-2002), biochemical parameters (albumin and ferritin levels), and body composition analysis.
Results: Malnutrition was identified in 22% of patients based on BMI (3 inactive, 17 active IBD) and in 31.9% based on NRS-2002 (1 inactive, 28 active IBD). Based on BMI, 25 patients (27.5%) were classified as excessive body weight, and 44% of these were in the active phase of the disease. Hypoalbuminemia and low ferritin levels were observed in 64% and 38.3% of participants, respectively. Sarcopenia was present in 24.3% of patients (SARC-F) and 17.6% (sit-to-stand test), while myopenia was detected in 36.7%. Sarcopenia and myopenia were more common during active disease phases. A significant association was found between regular physical

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activity of patients and a reduced incidence of myopenia (OR = 0.24; 95% CI: 0.09 - 0.64) – compared to patients who did not declare such activity. It was observed that patients receiving biological therapy had a significantly lower prevalence of malnutrition according to the NRS - 2002 scale (OR = 0.33; 95% CI: 0.11 - 0.98).

Additionally, in this patient group, a trend toward a lower prevalence of sarcopenia, as assessed by the SARC - F questionnaire, was observed. No significant associations were found between the increased prevalence of either sarcopenia or myopenia and disease type, patient gender, or disease duration.

Conclusion: Our study shows that malnutrition and sarcopenia are prevalent in Polish IBD patients, necessitating comprehensive assessment tools. We demonstrated that reliance on BMI alone is insufficient. While BMI identified malnutrition in a lower percentage of patients, the use of additional parameters, such as the NRS - 2002 scale and biochemical markers, provided a more comprehensive assessment. Higher BMI values can obscure malnutrition by masking muscle loss and nutritional deficiencies, making standard BMI assessments insufficient.

In addition, we observed that biological therapy can have a beneficial effect on the nutritional status of IBD patients. We have shown that all of the methods we have presented for detecting malnutrition and sarcopenia can be successfully used in patients with IBD. However, further research is needed to identify the preferred method.

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Disclosure: Nothing to disclose.

PP0337

GLUTEN-INDUCED GASTRIC VISCERAL HYPERSENSITIVITY VIA SP-MEDIATED NEUROIMMUNE MECHANISMS: INSIGHTS FROM CLINICAL AND ANIMAL STUDIES IN FUNCTIONAL DYSPEPSIA

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Introduction: Gastric visceral hypersensitivity (VH) is a key pathophysiological feature of functional dyspepsia (FD), characterized by enhanced perception of gastric distension and postprandial pain. Recent studies suggest that dietary gluten may act as a pro-nociceptive factor in FD, particularly in individuals without celiac disease but with dyspeptic symptoms. However, mechanisms linking gluten to VH remain elusive.

Aims & Methods: In the clinical arm, multiple linear regression assessed associations between gluten intake and symptom frequency/severity (postprandial fullness, epigastric pain, early satiety), adjusting for age, sex, and BMI. In the experimental arm, Sprague-Dawley rats received either a 20% gluten diet or gluten-free diet for 8 weeks; an iodoacetamide-induced FD model served as a positive control. Gastric sensitivity was evaluated by abdominal withdrawal reflex (AWR) and electromyography (EMG) following surgical placement of a gastric balloon and electrodes. Duodenal and fecal samples were collected for 16S rRNA microbiota profiling and RNA-seq. Histological and molecular analyses included H&E staining, intraepithelial lymphocyte (IEL) counts, RT-qPCR of cytokines, and quantification of GFAP, Substance P, and MAPK components by RT-qPCR, Western blot, or immunofluorescence. Aprepitant (10 mg/kg/day) was administered during the final 2 weeks to assess its therapeutic effect.

Results: A total of 74 patients diagnosed with FD based on Rome IV criteria (EPS (epigastric pain syndrome)+PDS (postprandial distress syndrome): 29, EPS: 12, PDS: 33) and 29 healthy controls were included. FD patients, particularly EPS subtypes, exhibited significantly higher gluten intake than healthy controls ($P<0.001$). Multiple linear regression revealed that gluten intake was a significant positive predictor of symptom frequency and severity in FD patients (frequency: $\beta = 0.0407$, 95% CI: 0.0197–0.0617, $P<0.001$; severity: $\beta = 0.0230$, 95% CI: 0.0091–0.0368, $P<0.01$).

In rats, gastric sensitivity in the 20% gluten group was comparable to the iodoacetamide-induced FD model and significantly greater than the gluten-free control group, exhibited by slower weight gain, increased AWR score, higher area under the curve (AUC) of electromyograph and reduced 24-hour food intake. Compared to controls, gluten-fed rats showed reduced fecal abundance of Lactobacillus and Lactococcus, increased Lachnoclostridium, and decreased Lactococcus in duodenal mucosa. Duodenal tissues also exhibited elevated intraepithelial lymphocyte (IEL) counts, increased expression of IL-1β, IL-4, and IL-5, and upregulation of inflammation-related genes Reg1α and Reg3β, indicating low-grade mucosal inflammation. In addition, gluten exposure significantly increased duodenal expression of Substance P (SP), GFAP, and phosphorylated MAPK pathway proteins.

At week 8, rats in the gluten group were treated with either the NK1R antagonist aprepitant or saline. Aprepitant administration significantly attenuated VH, as shown by reduced AWR scores and EMG AUC compared with the gluten+saline group. Aprepitant also suppressed the expression of IL-1β, IL-4, Reg1α, Reg3β, and MAPK-related genes, suggesting a key role of SP-mediated neuroimmune activation in gluten-induced VH.

Conclusion: Long-term high gluten intake induces gastric hypersensitivity, accompanied by gut dysbiosis, mucosal inflammation, and SP-mediated MAPK signaling. Targeting neuroimmune pathways may offer novel therapeutic strategies for gluten-induced epigastric symptom management.

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Disclosure: Nothing to disclose.

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PP0338

ULTRA-PROCESSED FOOD CONSUMPTION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Modern lifestyles have increased the demand for convenience and extended shelf life, contributing to the rising consumption of ultra-processed foods (UPFs). This trend has been linked to adverse health outcomes, including obesity and inflammatory bowel disease (IBD). IBD encompasses multifactorial digestive disorders characterized by disrupted intestinal homeostasis, resulting in immune dysregulation, impaired nutrient absorption, systemic inflammation, and a marked reduction in quality of life, often associated with chronic fatigue and decreased physical activity. These factors contribute to the activation of muscle protein catabolism and inhibition of skeletal muscle protein synthesis, increasing the risk of sarcopenia in IBD patients.

Aims & Methods: To evaluate ultra-processed food consumption and body composition in patients with IBD. A cross-sectional study was conducted with 105 patients diagnosed with Crohn's disease or ulcerative colitis at the IBD outpatient clinic of Hospital das Clínicas, Botucatu Medical School (HC-FMB). Assessments included handgrip strength using a dynamometer, bioelectrical impedance analysis, anthropometric measurements, and dietary intake evaluation using a food frequency questionnaire (FFQ).

Results: The sample included 61 patients with ulcerative colitis and 44 with Crohn's disease, with a mean age of 41.5 ± 15.1 years; 70% were female. Mean disease duration was 11.2 ± 7.6 years. The average BMI was 25.9 ± 5.2 kg/m^2, with 52.4% classified as overweight. Mean body fat percentage assessed by skinfold measurements was 28.5 ± 7.5%, with 85% exceeding the reference values.

Additionally, 18.3% were classified as malnourished based on mid-arm muscle circumference. According to bioelectrical impedance, 86.4% had low muscle mass. Mean handgrip strength was 27.4 ± 8.7 kgf, with 7% presenting inadequate values. Dietary data showed that 22% reported weekly fast food consumption, 30.5% consumed sweets, and 13% drank artificial juices daily. Daily intake of fruits and vegetables was reported by 48% of participants, and 36% reported daily meat consumption.

Conclusion: Patients with IBD are at increased risk of sarcopenia due to inflammatory burden and dietary restrictions. This study identified reduced muscle mass and high intake of certain processed and ultra-processed foods. Regular consumption of UPFs may worsen disease outcomes and negatively affect body composition by increasing fat mass and decreasing muscle mass.

Further analyses will explore associations between UPF intake and changes in body composition.

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Disclosure: Nothing to disclose.

PP0339

GRIP STRENGTH, PHYSICAL ACTIVITY LEVELS, AND BODY COMPOSITION IN THE MODULATOR ERA: INSIGHTS FROM PAEDIATRIC CYSTIC FIBROSIS

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Introduction: To investigate the relationships between hand grip strength (HGS), physical activity levels, body composition and BMI in paediatric patients with cystic fibrosis (CF).

Aims & Methods: This single-centre prospective observational study measured hand grip strength (HGS) using a Jamar hydraulic dynamometer, body composition (using InBody 770) and lung function (FEV1) values. Physical activity was calculated using the Modified Godin Scale Score (GSS). Body weight, height and body mass index (BMI) were analysed as z-scores and HGS as percentiles standardised for age and sex. Linear regression was used to estimate relationships, with R^2 values indicating the strength of associations. Participants on CFTR modulators were compared with modulator-naïve individuals.

Results: The study included 82 children with CF (mean age: 12.4 ± 4.4 years). Out of these, 66 patients were receiving CFTR modulator therapy (39 LUM/IVA, 27 ETI).

No significant differences were found in any of the investigated parameters between those on CFTR modulator therapy and those not receiving it (Table 1). However, the modulator groups showed a trend for higher height, weight and BMI z-scores.

Based on their GSS, 95% of children were classified as active. Neither the HGS nor the GSS showed any correlation with BMI (R^2=0.044; 0.034), fat mass percentile (R^2< 0.001; 0.001) or skeletal muscle mass percentile (R^2=0.001; 0.006).

Similarly, the relationship between FEV1 and SMM was negligible (R^2=0.002). Both HGS and GSS showed weak positive correlations with FEV1 (R^2=0.121; 0.077) and with each other (R^2=0.115).

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	Total (n = 82)	Modulator (n = 66)	Non-modulator (n = 16)	p-value
Age (mean ± SD; years)	12.4 ± 4.4	12.6 ± 4.3	11.6 ± 4.4	0.3645
Boys (n, %)	37/82 (45%)	32/66 (48%)	5/16 (31%)	0.2139
FEV1 (mean ± SD; %)	92.8 ± 20.2	93.0 ± 21.1	92.2 ± 16.1	0.0770
Antropometry:				
Weight (mean ± SD; z-score)	-0.37 ± 1.10	-0.28 ± 1.03	-0.71 ± 1.31	0.0761
Height (mean ± SD; z-score)	-0.13 ± 1.17	-0.09 ± 1.20	-0.29 ± 1.01	0.4097
BMI (mean ± SD; z-score)	-0.41 ± 1.11	-0.32 ± 1.04	-0.78 ± 1.29	0.1004
Body composition:				
Fat mass (mean ± SD; %)	17.3 ± 7.9	17.3 ± 7.8	17.1 ± 8.1	0.8725
Skeletal muscle mass (mean ± SD; %)	43.4 ± 5.1	43.4 ± 5.2	43.4 ± 4.4	0.8423
HGS (mean ± SD; percentile)	36.1 ± 26.7	37.6 ± 27.2	30.6 ± 24.1	0.4259
GSS	123 ± 108	122 ± 106	123 ±108	0.8674

Conclusion: GSS, HGS and FEV1 showed weak correlation and may be useful in assessing aspects of functional and physical performance in children with CF. In contrast, body composition and BMI showed no correlation with these performance measures.
Disclosure: Nothing to disclose.

PP0340

COMPLICATION PATTERNS AND ASSOCIATED VARIABLES IN INTESTINAL TUBERCULOSIS: AN ANALYTICAL COHORT STUDY

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Introduction: Various complications can occur in the course of digestive tuberculosis (DTB), such as intestinal obstruction, perforation, fistula formation, or gastrointestinal bleeding, all of which can significantly compromise patient prognosis.
Aims & Methods: The aim of our study was to identify factors associated with the development of these complications.
We conducted a retrospective analytical study including patients diagnosed with DTB over an eight-year period (2017–2024). Clinical, biological, and complication-related data were collected and analyzed.
Results: Ninety-six patients were included, with a mean age of 50.7 years (range: 19–90 years). The sex ratio (M/F) was 2. The diagnosis was confirmed primarily through histological analysis of peritoneal biopsies obtained via laparoscopy (52.1%), revealing epithelioid and multinucleated giant cell granulomas with caseating necrosis in 80% of cases. Additional diagnostic methods included the detection of *Mycobacterium tuberculosis* bacilli in ascitic fluid by direct microscopy (n=2), and molecular testing on colonic biopsies (n=2) or ascitic fluid (n=4).
Clinical outcomes were favorable in 87.5% of cases. Complications were observed in four patients (4.1%), including intestinal perforation (n=2), bowel obstruction (n=6), and death (n=2). One patient experienced early recurrence, and two others had persistent radiological findings despite adequate treatment (n=4).
Univariate analysis showed that severe malnutrition was significantly associated with the occurrence of complications (p=0.001).
Conclusion: Complications were observed in 4.2% of patients with DTB. Severe malnutrition was significantly linked to the development of these adverse outcomes, highlighting the critical importance of early and adequate nutritional support in the management of digestive tuberculosis.
Disclosure: Nothing to disclose.

PP0341

DETERMINANTS AND INFLUENCE OF PHASE ANGLE AND ITS VARIATION IN HPN PATIENTS WITH INTESTINAL FAILURE

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Introduction: Phase angle (PA) is a composite measure that combines two raw measurements of bioelectrical impedance analysis (BIA): resistance and reactance. Phase angle is considered an indicator of cellular health, integrity and hydration. Its influence has been demonstrated in several chronic diseases with a threshold of 4°.
The aim of our study was to evaluate the determinants of PA and the consequences of low PA or a negative evolution in patients undergoing HPN for intestinal failure (IF).
Aims & Methods: 41 patients (35 F), aged 63±16 years (M±SD) were included. They had been on HPN for 70±83 months for IF (short bowel 58%, enteropathy 15%, dysmotility 10%, occlusion 10%, fistula 7%). They received 1095±693 kcal and 1229±919 mL of PN/hydration per day and oral intakes were 1205±693 kcal per day. 10% were on GLP-2 analogues. Their BMI was 21.7±4.4 (M±SD) (54% well nourished). Body composition and PA were assessed at baseline and after 4 months (6.9±3.0) by BIA (mBCA 515, seca, Semur-en-Auxois, France) and statistical analysis (Student's t, Pearson's r, ANOVA, χ^2) was performed on R++ (Toulouse, France).
Results: Lean and fat mass indices at inclusion were respectively 14.2±2.3 and 7.6±3.2 kg/m^2 respectively, and 13.7±3.2 and 7.9±3.1 at follow-up. PA was 3.9±0.8 and 3.8±0.7°, respectively at inclusion (PA≥4° in 63%) and at follow-up, with a delta PA of -0.2±0.8°. PA was negatively influenced by age (r=-0.52; p<0.0001) and positively by lean body mass index (r=0.46; p<0.01), with no influence from disease or its treatment. Similarly, a PA≥4 was negatively related to age (55±16 vs. 68±14 years; p<0.05) and positively to lean mass index (15.5±1.9 vs 13.4±2.2 kg/m^2; p<0.01). The evolution (delta) of PA was negatively influenced by the volumes and energy received at D0 (r=-25.3; p<0.0001) and positively by changes in lean body mass (r=0.7; p<0.0001). While no complications were associated with PA or PA<4 at inclusion, delta PA was associated with more central venous catheter-related infections (-1.0+0.6 vs -0.1+0.7; p<0.05) and more hospital days (r=-0.45: p<0.05).
Conclusion: In this IF population, the determinants of PA (age and lean body mass) are similar to those observed in other diseases. PA cannot predict the onset complications, and the link between its evolution and the occurrence of complications may be more a consequence than a cause. Notwithstanding a lack of power related to a small number of patients, monitoring PA in patients on HPN for IF does not appear useful.
Disclosure: Nothing to disclose.

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PP0342

ALIGNING PATIENT EXPECTATIONS AND PHYSICIAN PERSPECTIVES ON NUTRITIONAL TREATMENT IN CROHN'S DISEASE: OVERCOMING CHALLENGES AND ENHANCING OPPORTUNITIES FOR EEN THERAPY

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Introduction: Exclusive enteral nutrition (EEN) is effective and commonly used to induce remission in pediatric Crohn's disease (CD). Nevertheless, the reasons for its low usage in adult patients are not fully understood. We aimed to identify the barriers to EEN use, evaluate the approaches of patients and physicians, and compare them.

Aims & Methods: An online questionnaire was sent to adult CD patients and gastroenterologists. The questionnaires included demographics, disease characteristics, education, previous EEN exposure, approach to EEN and approach to potential barriers (Likert scale of 1-5). Descriptive analysis was followed by comparative analysis of patients/physicians reluctant to use/provide EEN and those who were not reluctant. Logistic regression was applied to search for variables and potential barriers associated with EEN-reluctance. An association network was generated to visualize how different barriers contribute to reluctance to trying EEN.

Results: Three-hundred and fifteen CD patients and 42 physicians completed the questionnaire. The mean age was 36.7 (SD±16) with a median disease duration of 10 (range 0-60) years. Previous EEN use was reported by 40% (n=101), while EEN-reluctance was reported by 20.3% (n=64). No significant baseline variables were found to be associated with EEN reluctance in univariate and multivariate analyses, except for current advanced treatment (biologic/small molecules) use (OR 2.06 [95%CI 1.05-4.35]). Among physicians, 71% had previously prescribed EEN to a patient, while 81% were willing to prescribe EEN. Knowledge of remission rates in adult patients using EEN varied widely. The indication for EEN use varied from mild-moderate disease (41% of respondents) to complicated disease (38% for penetrating disease, 55.9% for severe disease, 52.9% for structuring and 55.9% for malnourished patients). Previous EEN prescription was associated with willingness for future EEN recommendation (OR 4.66 [95%CI 1.11-19.54]). Top-ranked barriers perceived by patients were lack of food variety (65% with scored 4-5), lack of meal-associated social interaction (59% with scored 4-5) and treatment duration (49% scored 4-5). The respondents were less concerned about potential complications (14.6% scored 4-5), treatment-related diarrhea/pain (25.1% scored 4-5) and the notion that the formula is dairy-sourced (29.2% scored 4-5). For physicians, lack of food-associated social interaction (67.3% scored 4-5) and concern about taste (54.7% scored 4-5) were ranked highest among the perceived patient barriers.

Overall, the physicians' perspectives largely aligned with those of the patients. However, we found that 'lack of food variety' and 'hunger concerns' were barriers scored significantly higher by patients than physicians (p<0.05 for both). Network analysis revealed the top three patient barriers associated with EEN reluctance were 'food variety', 'treatment duration' and 'fear of liquid only diet', which were correlated with each other. Supportive factors that could facilitate EEN use were: direct communication with a staff member (68.6% scored 4-5) full funding of the formula (65% scored 4-5), family support (59.7% scored 4-5) and talking with other patients that are EEN experienced (52% scored 4-5). Overall, 47% of the respondents expressed a desire to hear more about EEN, and 87.6% preferred to receive information about EEN from their physician.

Conclusion: Most patients with CD are open to discussing EEN and wish to discuss it with their physicians. Potential barriers to address are the tactile and social eating experience as well as the length of therapy.

Disclosure: Nothing to disclose.

PP0343

OUTPATIENT PERCUTANEOUS ENDOSCOPIC GASTROSTOMY BY PULL (GAUDERER-PONSKY) AND INTRODUCTION (RUSSELL) METHODS IN HEAD AND NECK CANCER PATIENTS: A RANDOMIZED CLINICAL TRIAL

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Introduction: The comparison of different percutaneous endoscopic gastrostomy (PEG) techniques shows that all are equivalent in terms of safety and success rate. Several authors suggest the method of introduction is associated with lower infection rate than the pull method, because there is no transoral passage of the tube. In the case of patients with head and neck cancer (HNC) it also reduces the risk of metastatic implantation to the puncture site. Despite these considerations, the literature is conflicting when comparing the results. There are only few studies about PEG as an ambulatory procedure in HNC outpatients.

Aims & Methods: The main objective is to compare pull and introduction techniques of PEG in HNC patients on an outpatient basis, regarding success rate, overall complication rate, infection rate and total duration of procedure.

The present study is a randomized clinical trial, with a follow-up period of 6 months. Among the adult HNC patients refered to PEG at Cancer Hospital I, of the Brazilian National Cancer Institute (HC I / INCA), subjects in good clinical condition were selected to outpatient PEG tube placement by pull or introduction techniques. Primary endpoints were success rate, complications rate and total PEG procedure time. Secondary endpoints were mortality, elective tube removal and PEG tube in use rates at the end of the study. All patients received 20Fr PEG tubes. Statistical analysis was performed using the Student t test and the Chi-squared test, when appropriate. P values were considered significant when <0.05.

Results: From May 2013 to April 2015, a total of 60 HNC outpatients (48 men and 12 women) underwent ambulatory PEG by pull or introducer technique (30 on each arm). The success rate was 100% for both techniques. There was no significant difference between the groups in relation to gender distribution, age, comorbidities and associated conditions (previous use of enteral tubes, trismus, previous abdominal surgery), type and tumor localization, staging and treatment. There was no significant difference in the procedure complication rate between the pull and introduction techniques (70% x 76.7%, P=0.771).

Regarding the time of occurrence, the traction technique showed a lower rate of immediate complications (0% x 23.3%, P=0.011), but no difference between early (10% x 6.7%, P=0.999) or late complications (70% x 63.3%, P=0.785).

Regarding the severity, there were no significant differences between pull and introduction techniques to major (3.3% for both, P=0.999) or minor complications (70% x 76.7%, P=0.711). There was no significant difference

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in mortality between groups (36.7% vs 43.3%, P=0.792). The average duration of the procedure by pull technique was 17.60 minutes versus 32.13 minutes for the introduction technique (P<0.001).

Conclusion: The comparison between the two techniques showed no significant difference in the rate and type of complications, but the introduction technique had a longer duration of procedure time. The placement of a PEG tube by the two techniques is a safe and viable procedure on an outpatient basis.

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Disclosure: Nothing to disclose.

PP0344

DIRECT PERCUTANEOUS ENDOSCOPIC JEJUNOSTOMY VERSUS PERCUTANEOUS ENDOSCOPIC GASTROSTOMY WITH JEJUNAL EXTENSION: QUALITY OF LIFE AND COMPLICATIONS

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Introduction: Jejunal feeding and medication such as levodopa-carbidopa intestinal gel for Parkinson's disease can be administered via percutaneous endoscopic gastrostomy with jejunal extension (PEG-J) and direct percutaneous endoscopic jejunostomy (D-PEJ). PEG-J is currently common practice because technically less difficult to insert, but comparative studies are scarce. Also, data on patient related outcomes are lacking.

Aims & Methods: Our aim was to examine patient experiences and adverse events (AE) in a retrospective study, conducted in a referral hospital. Clinical experience and patient related outcomes were analysed. Ninety-six patients had received either PEG-J or D-PEJ between January 1. 2010 to July 1. 2022. Forty-five patients answered a questionnaire including the EQ-5D-5L and questions regarding their experiences.

Results: No statistical differences were found in Quality of Life. Patients with PEG-J underwent significantly more re-endoscopies per tube-year compared to D-PEJ patients (0.63 (SD 0.72) versus 0.27 (SD; 0.50)). PEG-J patients received at least one re-endoscopy in 76.8% (range 0-3.85 per year), of which 77.4% because of dislocation of the jejunal extension. Tube related adverse events (including dislocation, occlusion, perforation and ulceration) occurred significantly more after PEG-J insertion (76.7% versus 25.9%). However, D-PEJ patients reported significantly more tube related pain (44.4% versus 15.9%).

Conclusion: In conclusion, D-PEJ is an interesting alternative for PEG-J. D-PEJ should at least be considered in patients with frequent problems with the jejunal extension of their PEG-J.

Disclosure: Nothing to disclose.

PP0345

EVALUATION OF SURVIVAL OUTCOMES IN PATIENTS RECEIVING PALLIATIVE HOME PARENTERAL NUTRITION: RESULTS FROM A UK TERTIARY REFERRAL CENTRE

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Introduction: Intestinal failure in advanced malignancy often results from mechanical bowel obstruction, post-surgical complications, or mucosal diseases. Palliative home parenteral nutrition (HPN) can potentially improve survival and quality of life (QoL). However, there is limited robust data on which factors may predict optimum survival, and its use remains contentious. This study aimed to assess the survival of patients receiving palliative HPN and to examine the impact of patient factors on survival outcomes.

Aims & Methods: This is a single-centre retrospective observational study of 84 patients referred for palliative HPN from the period of 1st January 2020 to 19th November 2024. Demographic, nutritional and medical data were analysed. Survival rates were compared using Kaplan-Meier curves and Cox regression.

Results: A total of 84 patients were referred for palliative HPN during the study period and 77 patients were discharged with HPN eventually [median age was 60 yr, (IQR: 51.3-70.4), female 77%]. Underlying primary malignancies of gynaecological and gastrointestinal origin predominated accounting for 78% of the study cohort. Majority of patients had metastatic disease at 95%. Mechanical bowel obstruction was the main indication for HPN (84%). Performance status (PS) scores were ≤ 2 in 88% of patients and 49% had an estimated prognosis of >3 months. The median serum albumin and C-reactive protein (CRP) was 26g/L and 21 mg/L respectively. Median overall survival was 58 days (IQR 31 – 108) with a 3-month mortality of 69%. There was no statistical difference in overall survival when evaluate by PS, estimated prognosis, underlying malignancy or modified Glasgow prognostic score (mGPS).

Conclusion: The overall survival in our study is modest. PS scores, estimated prognosis, mGPS, or tumour type did not show a significant impact on survival. This highlights the challenges in artificial nutrition and emphasizes the role of a multidisciplinary team in the care of these patients.

Disclosure: Nothing to disclose.

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PP0346

COMPLICATIONS OF PERCUTANEOUS ENDOSCOPIC GASTROSTOMY AS AN AMBULATORY PROCEDURE IN HEAD AND NECK CANCER OUTPATIENTS - A PROSPECTIVE LONG-TERM FOLLOW-UP STUDY

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Introduction: Percutaneous endoscopic gastrostomy (PEG) is a relatively simple and safe therapeutic method of providing enteral access for patients with dysphagia and intact gastrointestinal (GI) tract.
However, it is also associated with several complications, as well as a high morbimortality rate. The use of PEG as an ambulatory procedure in head and neck cancer (HNC) outpatients has not been well established in the literature so far.

Aims & Methods: The main objective of this study was to investigate the short and long-term complications of ambulatory PEG in a selected group of HNC outpatients. We performed a prospective clinical study involving HNC subjects who were in good clinical condition. Patients were included from September 2002 to December 2009 in a close long-term follow-up protocol of outpatient PEG by the pull technique.

Results: A total of 201 patients who underwent ambulatory PEG were included in the present study: 58 (28.9%) pretreatment, 48 (23.9%) during treatment and 95 (47.5%) after treatment for the underlying malignant disease. There was no procedure-related mortality. There were complications in 55.7% of patients in the long-term follow-up (16.9% early and 49.8% late, 6.5% major and 55.2% minor). The complications were: granulation tissue (43.3%), periostomal infection (11.4%), leakage (9%), BBS (6%), local pain (5.5%), persistent gastrocutaneous fistula (3.6%), gastric wall hematoma (3.5%), late tube dislodgement (3%), early tube dislodgement (0.5%), and respiratory failure (2.2%). The clinical-demographic variables (age, sex and timing of PEG) were not associated with significant statistical difference for complication rates (early or late, major or minor).

Conclusion: Ambulatory PEG is a viable and safe procedure in HNC outpatients who are in good clinical condition. Although PEG complication rates may be significant in the long-term follow-up, in most cases these complications are classified as minor. There were no deaths related to the procedure itself, and all complications were treated in an outpatient setting.

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Disclosure: Nothing to disclose.

PP0347

PERCUTANEOUS ENDOSCOPIC GASTROSTOMY (PEG) IN PEDIATRIC CANCER PATIENTS: EFFICACY AND SAFETY ANALYSIS FROM A RETROSPECTIVE COHORT

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Introduction: PEG is a procedure designed to provide prolonged nutritional access for patients who present an intact and/or functional gastrointestinal, but with difficulty or inability to swallow. In pediatric oncology services, PEG is most frequently indicated for children with central nervous system (CNS) tumors and head and neck cancers. While it is well established in clinical practice as an effective and safe method for obtaining enteral access, few studies specifically address the performance of PEG in children with cancer.

Aims & Methods: The main objective of this study is to describe demographic aspects, underlying pathologies, complications, and outcomes of patients who underwent PEG due to cancer. The present study consists in a retrospective cohort of PEG procedures performed in children with any neoplastic diseases that justified a PEG insertion, during the period of October 2000 and January 2018 at the Digestive Endoscopy Unit of the Cancer Hospital I of the Brazilian National Cancer Institute (INCA). Data collection included demographic and pathology baseline information, as well as details regarding post-PEG follow-up and complications. Data extraction was conducted with Excel with further descriptive analyses.

Results: During the study period, a total of 97 children who underwent PEG were included in the study. The mean age was 10.1 years (range: 0.5 to 18.9 years), and fifty-six children (57.7%) were male.
The underlying oncological pathologies included 69 cases (71.1%) of neurological tumors, 23 cases (23.7%) of head and neck tumors, 4 cases (4.1%) of hematological malignancies and 1 case of adrenal cortex carcinoma (1%). Fifty-eight patients (59.7%) experienced complications, including granulation tissue formation in 25 cases (25.7%), stoma infection in 6 (8.1%), local pain in 21 (21.6%), bleeding in 3 (3.1%), accidental tube dislodgement in 4 (4.1%), and gastrocolocutaneous fistula in 2 (2.0%). The mean duration of tube placement was 22.1 weeks (range: 0.4 to 483 weeks).
At the end of the study, 62 patients had died from their oncological disease (63.9%), 15 patients continued using the tube (15.4%), and 20 patients had the tube removed after completing treatment (20.6%). No cases were lost to follow-up.

Conclusion: PEG is a safe and effective procedure for providing prolonged enteral nutritional access in children with neoplastic disease. Complications are generally classified as minor, and typically do not require invasive treatments.

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Disclosure: Nothing to disclose.

PP0348

RETROSPECTIVE MONOCENTRIC ANALYSIS OF PICC LINES PLACED IN A TERTIARY CARE CENTER TO ESTABLISH A FORMULA FOR DETERMINING IDEAL PICC LINE LENGTH

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Introduction: The peripherally inserted central venous catheter (PICC line) is a highly useful form of vascular access in various areas of clinical patient care. It is particularly indicated in patients requiring long-term antibiotic therapy, medium-term parenteral nutrition, extended intravenous treatment during prolonged hospital stays, or those with poor venous access. PICC lines are typically 60 cm in length and must be individually trimmed to the correct length immediately before insertion. Several approaches exist for estimating the appropriate length on a case-by-case basis.
Aims & Methods: To establish a methodology for accurately estimating the required length of PICC lines.
Between November 2023 and April 2025, various body measurements were collected during PICC line placements. The depth of the selected vein was measured sonographically. Patients' height, weight, and sex were documented. The PICC line length was determined based on clinical expertise. A post-insertion chest X-ray was performed to confirm the central position of the catheter tip. The distance from the catheter tip to the carina was measured to objectively define central placement. Multiple formulas in clinical use were compared for their accuracy. A regression analysis was conducted to develop a formula that is both accurate and simple to apply.
Results: A total of 60 PICC lines were successfully inserted without complications. The median patient height was 168 cm (range: 164.3–175.8 cm), weight 59 kg (range: 50.0–83.8 kg), and age 56.5 years (range: 46–65.8). Thirty-five insertions were performed in female patients. The ideal PICC line length was 42 cm (range: 38–44.5 cm). Of the five tested formulas, the best correlations were found with Formula 5 (puncture site to jugulum + jugulum to 3rd intercostal space) with r = 0.68 (p = 0.0005), and Formula 3 (puncture site to acromion + clavicle length + jugulum to 4th intercostal space) with r = 0.67 (p = 0.0005). In simple linear regression, the distances from puncture site to acromion and to jugulum were significantly correlated (p = 0.0007 and p = 0.0019, respectively). The best-fit model in multiple regression included age, height, weight, arm side, puncture site to jugulum, and puncture site to 4th intercostal space (p = 0.0005).

Model	R^2	Success Rate (±2cm)	Success Rate (±3cm)	Success Rate (±4cm)
Formula 1	0.16	31.25%	50.00%	65.63%
Formula 2	0.56	18.75%	37.50%	53.13%
Formula 3	0.53	18.75%	25.00%	28.13%
Formula 4	0.42	0.00%	0.00%	0.00%
Formula 5	0.61	40.63%	59.38%	68.75%
New Formula	0.83	62.50%	90.63%	96.88%

Table 1: comparison between the various existing formulas and the newly developed one.

Conclusion: An optimal formula for calculating the correct PICC line length has been identified.
Disclosure: Nothing to disclose.

PP0349

LONG TERM ENDOSCOPIC GASTROSTOMY ENTERAL FEEDING OF NEUROSURGICAL PATIENTS: A REFERENCE CENTER'S EXPERIENCE

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Introduction: Neurosurgical patients often face several difficulties with nutritional support due to severe brain injury, neurological disease, or post-surgical complications. This study aimed to assess the outcomes of long-term enteral nutrition via percutaneous endoscopic gastrostomy (PEG) in this population.
Aims & Methods: Single center, retrospective analysis of patients who were referred for PEG from 2001 to 2023. Patients who presented with severe traumatic brain injury (TBI), hemorrhagic stroke, brain tumor or other conditions that required neurosurgery were selected. Demographic, anthropometric and clinical data were recorded.
Results: 196 patients were included, 57% were younger than 65 years. Stroke (41.8%), TBI (35.2%) and brain tumors (19.9%) were the most frequent conditions. The median time from diagnosis to PEG was 94 days. At the time of PEG procedure, only 38.5% were underweight. 21 patients were lost to follow up. Among the remaining, 75.4% died, 12.6% were alive and PEG-fed, and 12% had the tube removed. Of the surviving patients who remained PEG-fed, the majority (77%) were stroke patients. Median overall survival post-PEG was 11.5 months, with 88% of patients surviving more than one month. Albumin, transferrin, and total cholesterol levels at the time of PEG were associated with longer survival. Albumin ($p<0.001$) and transferrin ($p<0.01$) were significantly associated with lower short-term mortality risk.
Conclusion: PEG is an effective method for long-term enteral nutrition in neurosurgical patients. Although survival is limited, reflecting the severity of the underlying conditions, the majority of patients survived beyond one month, supporting the adequacy of the procedure. Higher serum albumin levels were associated with longer survival, suggesting its potential role as a prognostic biomarker in this population. Careful patient selection, timely intervention, and consideration of nutritional status may help optimize outcomes for neurosurgical patients requiring PEG.
Disclosure: Nothing to disclose.

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PP0350

PERCUTANEOUS ENDOSCOPIC GASTROSTOMY IN GREECE: A NATIONWIDE ASSESSMENT OF CURRENT PRACTICE

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Introduction: Percutaneous endoscopic gastrostomy (PEG) is the standard method for providing long-term enteral nutrition and medication in patients with insufficient oral intake. However, data on the technical aspects of the procedure, adherence to published guidelines, and the decision-making process of endoscopists remain limited in certain countries.

Aims & Methods: The primary aim of this nationwide study was to assess the current practices among Greek gastroenterologists regarding PEG procedure, covering the entire spectrum of PEG, including initial clinical decision-making, technical aspects, periprocedural management, and long-term patient care. An anonymous web-based survey was distributed to every member of the Hellenic Society of Gastroenterology (HSG), including gastroenterology trainees. Participants electronically completed a 36-item questionnaire divided into five sections:

1. Demographics and professional background,
2. General questions on PEG placement,
3. Specific procedural aspects,
4. Post-procedural management, and
5. Future perspectives.

Data analysis was performed using Stata statistical software, version 17.0 (StataCorp LLC, College Station, TX, USA).

Results: A total of 127 out of 766 members of the HSG responded (response rate: 16.5%). Regarding PEG placement, 66.7% reported involvement in up to two procedures per week. Multidisciplinary team meetings were rarely implemented, with 82.0% stating they "rarely" or "never" participate in such meetings. Neurological disorders were the leading indication for PEG (67.7%), followed by dementia (18.1%) and obstructive head, neck, or esophageal malignancies (11.8%).

The primary indications differed significantly between the public and private sectors (P = 0.007), with neurological disorders being more frequently reported in the private sector (40.5% vs 26.1%), while obstructive tumors were less commonly cited as the main indication for PEG placement (1.6% vs 10.3%). The push-PEG technique in malignancy cases was used by 31.7% of respondents, whereas direct percutaneous endoscopic jejunostomy (D-PEJ) and percutaneous endoscopic gastrojejunostomy (PEG-J) were employed in 11.9% and 37.8%, respectively. Skin antisepsis and use of sterile instruments were consistently reported by 93.7% and 89.0% of the participants, while prophylactic antibiotics and surgical hand washing were routine for 58.3% and 45.7% of them. Regarding the reported incidence of serious adverse events, 17.3% of respondents had encountered at least one case of perforation or peritonitis, 12.6% reported severe bleeding, and 7.9% had experienced a PEG-related death.

No statistically significant association was observed between the occurrence of adverse events and the variables age, years post specialization, trainee status, healthcare sector (public vs private), annual number of PEG procedures, or routine use of prophylactic antibiotics. After discharge, structured follow-up programs were reported by 15.1%, while 31.5% indicated that nutritional guidance was provided by a dedicated nutrition department.

Conclusion: This nationwide survey is the first to explore clinical practices related to PEG placement among gastroenterologists in Greece. The findings highlight areas of alignment with international standards, but also reveal important deviations, particularly in multidisciplinary collaboration, infection prevention measures, and postprocedural care. These results provide a valuable foundation for targeted training agendas and quality improvement initiatives.

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Disclosure: Nothing to disclosure.

PP0351

PREOPERATIVE EXCLUSIVE ENTERAL NUTRITION REDUCES POSTOPERATIVE COMPLICATIONS IN ADULTS WITH CROHN'S DISEASE UNDERGOING ELECTIVE SURGERY: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Exclusive enteral nutrition (EEN) is a well-established first-line treatment for inducing remission in paediatric Crohn's disease, however, its use in adult Crohn's disease treatment remains limited. In adults, 22% of patients require surgery within the first five years after diagnosis[1], with postoperative complication rates reaching around 25%[2].

This study investigates whether preoperative EEN can reduce postoperative complications in adults with Crohn's disease undergoing elective surgery.

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Aims & Methods: A systematic search was conducted on October 26, 2024, across three databases: PubMed, EMBASE, and the Cochrane Library. Studies were included if they involved at least two weeks of preoperative EEN and reported postoperative outcomes.
The primary outcomes were anastomotic leakage, stoma formation, and abscess formation, while the secondary outcome was the length of hospital stay. The odds ratio (OR) and mean differences (MD) with their 95% confidence interval (CI) were calculated using a random-effect model.
Results: A total of 11 studies, including 2112 patients, were included in the analysis. EEN significantly reduced the odds of anastomotic leakage (OR=0.29, CI: 0.19-0.47), stoma formation (OR=0.39, CI: 0.09-1.73), and abscess formation (OR=0.41, CI: 0.23-0.72). However, there was no significant reduction in the length of hospital stay (MD: 0.88 days, CI: -1.85-0.09).
Conclusion: Preoperative EEN significantly reduces the risk of anastomotic leakage, stoma formation, and abscess formation, leading to improved postoperative outcomes. These findings support adding EEN into preoperative management for adult Crohn's disease patients undergoing elective surgery to enhance recovery and minimize complications.
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Disclosure: Nothing to disclose.

PP0352

AI-BASED PREDICTION OF MACRONUTRIENT INFORMATION IN RECIPES FOR CANCER PATIENTS

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Introduction: People with cancer commonly suffer from loss of appetite which leads to unintended weight and muscle loss, and reduced quality of life. Effective nutrition therapy therefore requires recipes with reliable calorie and macronutrient values. Manual calculation by professionals is time- and resource-intensive. Thus, Large language models (LLMs) could automate this task if their accuracy is sufficient.
Aims & Methods: To evaluate whether an LLM can automatically estimate calories and macronutrients (carbohydrates, protein, fat, fibre) in recipes, we compared their results with manual calculations by dietitians.
125 recipes from the platform was-essen-bei-krebs.de, were used as reference values per 100 g for calories and macronutrients as they were manually calculated by registered dietitians. ChatGPT o1 was prompted with a purpose-built instruction to estimate macronutrients for each ingredient and aggregate totals. The same prompt was repeated five times per recipe. Absolute error (AE) was defined as |estimate - reference| and normalized per 100 g. In line with EU labelling guidelines for packaged foods, an error ≤ 20 % was predefined as acceptable. The primary endpoint was the number of estimates classified as acceptable across all categories.
Results: The LLM returned well formatted outputs for all 625 prompts covering the 125 cancer-oriented recipes. For calorie estimates, 74.3 % fell within the ± 20 % interval. Protein was acceptable in 64.2 %, carbohydrates in 50.4 %, and fat in 58.1 % of cases. The mean absolute error for calories was 20.5 kcal per 100 g.
Conclusion: Preliminary data indicate that ChatGPT o1 can already deliver high-quality nutrient estimates. LLMs therefore have the potential to automate the nutrient calculations that are still performed manually today.
Disclosure: Nothing to disclose.

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PP0353
ACTIVATED NOTCH SIGNALING PATHWAY PROTECTS THE INTESTINAL BARRIER IN DSS-INDUCED COLITIS BY REGULATING TIGHT JUNCTIONS

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Introduction: Emerging evidence implicates Notch signaling in intestinal mucosal barrier preservation, though its mechanistic basis remains undefined.

Aims & Methods: This study investigates the molecular interplay between Notch activation and tight junction (TJ) regulation during inflammatory injury. In vivo experiments, the dextran sulfate sodium (DSS)-induced colitis mice were used and LY411,575, a γ-secretase inhibitor, was orally administered for Notch signaling inhibition. In vitro experiments, we modulated Notch signaling by either activating or inhibiting it through the overexpression of Hes1 or LY411,575, respectively. The tight junction (TJ) and relevant myosin light chain kinase (MLCK) pathway were systematically investigated. To elucidate the mechanism underlying Notch activation, we conducted a comprehensive analysis of ligands and receptors associated with the Notch pathway in Caco-2 cells.

Results: Compared to DSS-colitis mice, inhibition of Notch activation by LY411,575 led to severe exacerbation of colitis, accompanied by TJ deterioration and concurrent activation of MLCK pathway. In vitro experiments confirmed the Notch signaling's regulatory effect on TJ function, with MLCK inhibitor ML-7 effectively mitigating the TJ dysfunction induced by LY411,575. Additionally, TNF-α-treated Caco-2 cells exhibited significantly increased expression of Jag1, Notch1, and Hes1. Silencing Jag1 using siRNA resulted in reduced NICD1 and Hes1 expression.

Conclusion: Activation of Jag1/Notch1/Hes1 signaling pathway protects the intestinal mucosal barrier from inflammatory injury by abrogating MLCK-dependent TJ dysregulation. Therapeutic Notch activation may represent a novel strategy for mucosal protection in IBD pathogenesis.

Disclosure: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

PP0354
ROLE OF TUMOR NECROSIS FACTOR ALPHA AND VITAMIN D IN ACTIVITY AND SEVERITY OF INFLAMMATORY BOWEL DISEASE IN EGYPTIAN PATIENTS

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Introduction: Crohn's disease (CD) and ulcerative colitis (UC) are two chronic, gastrointestinal tract disorders that are included in the category of inflammatory bowel diseases (IBD). A combination of imaging results, endoscopic biopsies, inflammatory laboratory markers, and clinical symptoms are required to diagnose IBD. Patients with CD suffer from vitamin D insufficiency than UC patients do, and they often have lower levels than the general population.

Aims & Methods: The goal of this research is to clarify the role of vit. D in activity and severity of IBD in Egyptian people.

Between November 2021 and November 2023, we recruited 31 normal individuals (IBS patients and volunteers) and 92 IBD (UC and CD) naïve patients from the internal medicine department of Menoufia University Hospital, GI endoscopy unit. Every participant underwent a clinical examination, a history taking, and standard laboratory testing. Vit. D serum levels were also tested.

Results: This study revealed a highly significant difference in serum vit. D levels across the groups under investigation, with low vit. D levels in individuals with IBD. There was discernible correlation between the plasma vit. D level and disease activity, also we found a highly significant difference in plasma TNF alpha levels among the 3 groups, with significant difference in disease activity and plasma TNF alpha levels.

Conclusion: From this study we concluded that; TNF alpha plays a role in the pathogenesis of IBD with significant difference with the disease activity. Also decrease of vit. D level may play a role in the pathogenesis of IBD with significant relation with the disease activity and vit. D supplementation may help IBD patients.

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Disclosure: No conflicts of interest.

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PP0355

IBD IN VITRO MODEL REVEALS PITX2 AS A KEY REGULATOR OF EPITHELIAL CELL SURVIVAL AND PROLIFERATION IN ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) treatment aims for mucosal healing as a therapeutic goal. However, in some cases, especially in the rectal mucosa, healing cannot be achieved. Moreover, the reason why the rectum is the predilection site of UC remains unknown. In this study, we hypothesized that differences in intestinal epithelial cells (IECs) in the rectum and ascending colon play an important role in the pathogenesis of UC, which may lead to the different treatment responses in the rectum and colon. Therefore, we focused on the differences in gene expression in IECs between the rectum (as the lesion site) and the ascending colon (as the non-lesion site) in UC patients.

Aims & Methods: Colorectal organoids were established from lesional and non-lesional sections of total colectomy specimens from UC patients and cultured under non-inflammatory conditions. We then performed microarray analysis on these organoids and identified genes potentially associated with UC pathology. The expression of candidate genes was validated by q-PCR in lesional and non-lesional areas of surgical specimens from UC patients. To further investigate the role of the selected gene, we knocked down (KD) PITX2 in organoids derived from the ascending colon of a healthy individual using a lentiviral system. These organoids were exposed to TNF-α (100ng/ml), IL-1β (10ng/ml), Flagellin (100ng/ml) for one week, following a previously established human IBD in vitro model(J Crohns Colitis. 2021,2022). We assessed the inflammatory response by measuring the expression of IL-8 and DUOXA2, downstream genes of NF-κB, using qPCR. In addition, fluorescent immunostaining for NF-kB p65 was performed, and the nuclear-to-cytoplasmic fluorescence intensity ratio of p65 was analyzed using ImageJ software. Cell proliferation was evaluated using the CellTiter-Glo® assay. Stem cell function was assessed by sphere formation assay, and apoptosis was analyzed by Western blot for caspase-3 and caspase-7.

Results: We identified PITX2 as a gene specifically upregulated in both organoids derived from the ascending colon and non-lesional tissue of UC patients. Inflammatory response assessment, based on IL-8 and DUOXA2 expression, showed no significant differences between WT and PITX2-KD organoids. Similarly, fluorescent immunostaining for NF-κB p65 revealed no significant difference in nuclear translocation between WT and PITX2-KD organoids, suggesting that PITX2 does not influence inflammatory signaling pathways. PITX2-KD organoids from the ascending colon exhibited similar cell proliferation to WT under normal conditions. However, under inflammatory stimulation for one week, PITX2-KD organoids showed reduced proliferation compared to WT. The sphere-forming ability of PITX2-KD organoids was comparable to WT under both normal and inflammatory conditions, suggesting that PITX2 plays a role in promoting proliferative cells in an inflammatory environment. Western blot analysis revealed increased expression of cleaved caspase-3 and 7 in PITX2-KD organoids following inflammatory stimulation compared to WT. These findings suggest that PITX2 may regulate apoptosis via effector caspases.

Conclusion: PITX2 was identified as a gene specifically upregulated in the ascending colon and non-lesional tissue of UC patients, playing a role in promoting cell proliferation under inflammatory conditions. These findings suggest that PITX2 may contribute to UC pathology by regulating epithelial cell survival and apoptosis in an inflammatory environment.

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Disclosure: Nothing to disclose.

PP0356

DELAYS IN INITIATING ADVANCED THERAPY FOR INFLAMMATORY BOWEL DISEASE: THE IMPACT OF ETHNICITY

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Introduction: Prompt initiation of advanced therapy (AT) medications, encompassing biologics and small molecule treatments, is crucial for the effective treatment of inflammatory bowel disease (IBD). The time taken from the decision to start the AT to it's first administration, or time to advanced therapy (TAT), can vary significantly between individuals negatively impacting upon their disease course. However, our knowledge of the causes for the variation in TAT is poor.

Aims & Methods: We aimed to investigate the impact that different demographic and disease-related factors have on the TAT. A retrospective analysis of electronic patient records, conducted at a tertiary IBD referral centre in London in March 2024, analysed electronic patient records of all those over the age of 18 currently being treated for IBD with an AT and collected data on the TAT for their index AT, defined as the number of days from the documented decision to start an AT to the date of the first administration.

Data was also collected on multiple variables which including disease type, type of AT, age, sex, ethnicity and socioeconomic status by analysing patients' Index of Multiple Deprivation (IMD). A multiple negative binomial regression was performed entering all variables together to assess their relative effect on TAT.

Results: 1298 patients were identified and 485 were excluded due to either their index AT being started elsewhere, incomplete data, a positive screening test or because they were enrolled in a clinical trial. The multiple binomial regression found that TAT was significantly longer for those of non-white ethnicity (p=0.039) (Table 1).

		IRR	95% CI	p-value
Ethnicity	Asian	1.17	0.96, 1.21	0.039
	Black	1.46	1.09, 1.95	
	Mixed	1.26	0.77, 2.05	
	White	1		
Treatment	Adalimumab	1.37	0.95, 1.96	<0.001
	Filgotinib	1.37	0.87, 2.16	
	Infliximab	0.97	0.68, 1.40	
	Tofacitinib	0.89	0.58, 1.36	
	Ustekinumab	0.91	0.61, 1.35	

Table 1. IRR: Incidence Rate Ratio. IMD: Index of multiple deprivation.

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Analysis for patients of black ethnicity found an Incident Rate Ratio (IRR) of 1.46 (95% CI 1.09,1.95), for mixed ethnicity IRR=1.26 (95% CI 0.77,2.05) and for asian ethnicity IRR=1.17 (95% CI 0.96, 1.41) when compared to white patients. Treatment with adalimumab was also associated with a longer TAT (p=<0.001, IRR 1.37, 95% CI 0.95,1.96) but no other factors were statistically significant: IMD p=0.133; age p=0.155; sex p=0.702; disease type p=0.335.

Conclusion: Our study has identified that non-white ethnicity is associated with a longer TAT and this may be a contributing factor to worse outcomes in patients with IBD from an ethnic minority background. Further research into the causes for this are required to help address this health disparity.

Disclosure: Nothing to disclose.

PP0357

GUT MICROBIOME COMPOSITION AND ITS ASSOCIATION WITH DIETARY FACTORS IN CROHN'S DISEASE

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Introduction: The gut microbiota plays a crucial role in the pathogenesis and progression of Crohn's Disease (CD) by maintaining intestinal homeostasis, supporting metabolic and immune functions, and influencing overall health. An imbalance in the gut microbiota is associated with CD, as disruptions in microbial composition can be further impacted by dietary factors, highlighting the connection between nutrition and gut health.

Aims & Methods: This study aimed to evaluate the gut microbiota composition and its interactions with dietary factors to identify potential opportunities for targeted probiotic interventions in CD management. The study included 53 participants, of whom 20 had CD in remission, 18 had active disease, and 15 were healthy controls. CD activity was evaluated by colonoscopy assessment or by nuclear magnetic resonance scan enterography. All participants aged between 18 and 60 years were assessed using 24-hour dietary recall and gut microbiome profiling through 16S rRNA gene sequencing. Patients had confirmed diagnoses of CD with ileal and/or colonic involvement. The local ethics committee approved this study.

Results: Microbial composition analyses (beta diversity) revealed significant differences between groups. At the Operational Taxonomic Unit (OTU) level, healthy microbiome differed significantly from both remission and active CD patients, with healthy controls showing a more similar microbial composition compared to remission (p = 0.001) and active disease (p = 0.001). In addition, Shannon diversity analyses demonstrated that healthy individuals exhibited significantly higher OTU diversity compared to CD patients, further highlighting the microbial imbalance associated with the disease. Among specific taxa, *Eubacterium hallii*, a key butyrate producer, was significantly reduced in patients with active CD. This microorganism is integral to intestinal metabolic balance, playing essential roles in energy homeostasis, colonic motility, immunomodulation, and inflammation suppression. Furthermore, *E. hallii* plays a critical role in bile acid and cholesterol metabolism, highlighting its importance in gut health. Disease severity was associated with changes in microbial abundance, with specific taxa colonizing the gut at higher levels as disease activity increased, including *Escherichia coli*, *Escherichia-Shigella*, and taxa from the *Enterobacteriaceae*, *Lachnospirales,* and *Bacteroides*. This growth may be linked to inflammatory processes and intestinal dysbiosis, suggesting a potentially pathogenic role. Furthermore, dietary components such as saccharin, an artificial sweetener, were significantly associated with alterations in microbial composition.

Conclusion: These findings underline the potential for probiotics targeting specific taxa, such as *E. hallii*, to restore microbial balance and improve clinical outcomes in CD. The study also highlights the impact of dietary components, such as saccharin, in disrupting the gut microbiome, suggesting that certain dietary factors may contribute to microbial imbalance and inflammation in CD. Future research should focus on integrating dietary patterns with probiotic development to optimize gut health and mitigate inflammation in CD patients.

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Disclosure: Nothing to disclose.

PP0358

PHENOTYPING OF HUMAN UC COLONIC TISSUE T CELLS REVEALS INFLAMMATORY PATHWAY GENE EXPRESSION IN PD-1+ CONVENTIONAL AND REGULATORY T CELLS WHICH OVERLAP WITH THOSE REGULATED BY ROSNILIMAB IN A MOUSE MODEL OF COLITIS

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Introduction: PD-1 co-inhibitory receptor agonists, such as rosnilimab, are a promising class of therapeutics that leverage immune homeostatic mechanisms to regulate T cell activity for treatment of inflammatory diseases, including ulcerative colitis (UC). In UC lamina propria, multiple T cell subtypes express PD-1, including peripheral helper T cells (Tph) and effector memory T cells, and the reduction of PD-1+ Tph cells is correlated with remission.[1]

Also, PD-1+ Tregs may comprise a subpopulation of pro-inflammatory Tregs in the UC colon. In a mouse model of colitis (induced by transfer of CD4+CD45Rb+ T cells from human PD-1 transgenic mice), rosnilimab treatment of established disease showed reduced body weight loss, reduced inflammation of the colon, and reduced infiltration of CD4 T cells.[2]

Aims & Methods: A secondary analysis of active UC patient-derived colonic tissue characterized PD-1+ T cell subsets via a CD3-sorted single-cell proteomics and transcriptomics dataset.[3]

Pathway activity was estimated using Progeny.[4]

Bulk RNA-sequencing, gene expression analysis, and enrichment of human UC significant Reactome[5] pathways was performed on colonic tissue from mice treated with rosnilimab in the colitis model.

Results: In human UC colonic tissue, PD-1+ CD4 conventional T (Tcon) cells and PD-1+ Tregs had increased JAK-STAT, TNF, and NF-κB estimated pathway activity compared to PD-1(-) cells. PD-1+ Tregs had increased expression of genes associated with a pro-inflammatory phenotype, including *TNF*, *TBX21*(TBET), and *IL12RB2*, and decreased expression of *IL2RA*

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(CD25). RNA-Seq of mouse colitis identified upregulated genes, compared to naive animals, in pathways that were also involved in human UC ("Signaling by interleukins" (e.g., *Il6*) and "Chemokine receptors bind chemokines" (e.g., *Ccl20*)). Expression levels of these genes in rosnilimab treated mice were similar to naive control animals. Additionally, the barrier gene *Muc3* and fibrosis-associated *Mmp9* were dysregulated in colitis but normal in rosnilimab-treated mice.

Conclusion: In UC colonic tissue, PD-1+ CD4 Tcon cells demonstrated increased inflammatory gene expression and PD-1+ Tregs exhibited a pro-inflammatory phenotype, suggesting that targeting both PD-1+ CD4 Tcon and PD-1+ Tregs may contribute to the therapeutic efficacy of rosnilimab in UC.

These additional analyses support preclinical findings and suggest beneficial modulation of gene expression pathways of human UC, including reduction of pro-inflammatory cytokines and fibrosis, and upregulation of barrier function genes. These data demonstrate that PD-1 is upstream of multiple clinically validated drivers of UC and support the scientific rationale for studying rosnilimab in an ongoing Phase 2 study in UC (NCT06127043).

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Disclosure: All authors are employees and shareholders of AnaptysBio.

PP0359

FUNGAL CELL WALL MANNAN ENHANCES PLASMA CELL DIFFERENTIATION AND IGA ANTIBODY RESPONSES IN THE GUT

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Introduction: Fungal cell wall acts as a major interface with the host immune system. Whether its components have a direct impact on differentiation of B lymphocytes, however, is poorly investigated, in contrast to well documented induction of class switch DNA recombination (CSR) and plasma cell differentiation by TLR ligands derived from or mimick bacterial cell wall components or viral genome replication intermediates. This hinders the understanding on the role of mycobiota in the homeostatic production of protective mucosal IgA and pathogenic IgG induction upon barrier damages-triggered exposure of B cells to microbes.

Aims & Methods: In the presence of Toll-like receptor (TLR) ligands or T cell-derived CD154, class switch recombination (CSR) and plasma cell differentiation were examined under conditions treated with Saccharomyces cerevisiae (S. cerevisiae) mannan, β-1,3-glucan, zymosan, and heat-killed (HK)-Candida albicans (C. albicans). Mannans derived from S. cerevisiae and selected Candida species were also tested under comparable conditions. Pathways related to plasma cell survival were detected. An ovalbumin-cholera toxin (OVA-CT) induced mouse IgA expressing model was utilized to investigate the role of mannan in the gut IgA response.

Results: Here we show that, unlike fungal polysaccharide mannan, HK-*C. albicans* and b-1,3-glucan as well as zymosan all failed to activate B cells. Furthermore, while b-1,3-glucan and zymosan had marginal impact on B cells stimulated by a TLR ligand or T cell-derived CD154, mannan from *Saccharomyces cerevisiae* and selected *Candida* species potentiated TLR4 induction of generation of plasma cells, including those secreted class-switched IgA and IgG. It did so in association with increasing survival threshold of pre-differentiating B cells, enhancing NF-kB activation and upregulating expression of XBP-1 and pro-survival genes. Mannan also enhanced IgA antibody responses in the gut.

Conclusion: Thus, mannan uniquely mediates the fungi and bacteria crosstalk by regulating plasma cell differentiation and antibody response.

Disclosure: Nothing to disclose.

PP0360

DECANOIC ACID AMELIORATES BOTH ACUTE AND CHRONIC MURINE COLITIS

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Introduction: Inflammatory Bowel Disease (IBD) is a gastrointestinal disorder which comprises Ulcerative Colitis and Crohn's Disease. Intestinal Fibrosis is one of the main processes involved in the complications associated to IBD with no effective treatment. Decanoic acid (DA), a medium-chain fatty acid found in coconut oil, is considered to have beneficial biochemical functions such as antifungal and antimicrobial activity, but its role in IBD remains unclear.

Aims & Methods: We aim to analyze the role of decanoic acid in both acute and chronic murine models of colitis.

Acute model of colitis was induced by 2% of DSS in drinking water during 7 days followed by 3 days of H_2O, whereas chronic model was induced with two cycles of DSS 2% in drinking water during 7 days separated by 10 days of H_2O. Mice were administered orally with vehicle (methylcellulose) or DA (516,78 mg/kg) daily. In both models, body weight was registered daily and colon length was measured at sacrifice. Gene and protein expression were analysed by qPCR and Western Blot, respectively. Embedded paraffin colon samples were stained with hematoxylin-eosin for histological analysis and Sirius Red staining for collagen deposition. CD68 and CD3 positive cells were quantified by immunohistochemistry.

Additionally, human small intestine fibroblasts (HSIFs) were treated with increasing concentrations of decanoic acid (0, 50, 100 and 250mM) and TGFb (10ng/ml) during 24 hours.

In addition, some HSIFs were transiently silenced with a specific siGPR84 and siPPARg. Gene and protein expression of fibrotic markers were analyzed by RT-PCRq and Western blot respectively. Data were analyzed using ANOVA and t-test and expressed as fold induction (mean ± SEM). A P-value <0.05 was considered significant.

Results: In acute model there were no significant differences in survival nor colon length between DSS-vh and DSS-DA. H-E stain showed a preserved epithelium architecture and less inflammatory infiltrate in DSS-DA vs DSS-vh. Moreover, DSS significantly increased the gene expression of IL1β, Cox2, IL10, Col1a1, Col3a1, Col4a1, Snail1, inos, cd206, cd163 and

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cd68 and reduced Acta2 and Fizz1. Of interest, DA significantly reduced the expression of most of the previous genes vs DSS-vh. IH showed a significant increase of CD68+ and CD3+ cells in DSS-vh, while DA significantly reduced these cells vs DSS-vh.

Chronic model showed an increase in survival and colon length in DSS-DA mice vs DSS-vh (5,94±0.18 vs 4,88±0.16 cm). In addition, DSS increased the gene expression of IL1β, TNFa, Cox2, cd68, inos, ccr7, cd16, arginase, Col1a1, Col4a1, TGFβ, timp1 and snail2 and reduced cd163. DSS-DA significantly reduced the expression of those genes vs DSS-vh mice. Sirius Red staining revealed a reduction in the thickness of collagen layer in DSS-DA mice vs DSS-vh mice.

In vitro, DA significantly decreased mRNA expression of COL1A1 (1.06±0.16 vs 0.35±0.06), COL3A1 (1.25±0.31 vs 0.40±0.08), COL4A1 (1.21±0.09 vs 0.41±0.07), ACTA2 (1.09±0.22 vs 0.13±0.02). The treatment with this metabolite also significantly reduced the protein levels of Col1A1 (100±18.93 vs 28±5.3) and aSMA (92.84±7.07 vs 50.62±3.89). Only the transiently silencing of PPARg significantly reverted those reductions, whereas siGPR84 did not modify the expression of COL1A1 nor COL3A1 nor ACTA2.

Conclusion: Decanoic acid exerts a protective role in both acute and chronic murine models of colitis reducing inflammation and fibrosis markers. These effects seem to be mediated through PPARg, and not GPR84, suggesting that this metabolite could be a promising therapeutic approach for IBD.

Disclosure: Nothing to disclose.

PP0361

DECODING NOTCH PATHWAY'S IMPACT ON EPITHELIAL-MESENCHYMAL TRANSITION AND FIBROSIS IN COMPLICATED CROHN'S DISEASE

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Introduction: The Notch pathway may selectively mediate fibrogenic processes, such as epithelial-mesenchymal transition. The dysregulation of EMT is linked to numerous pathologies, including fibrotic diseases[1]. Previous studies by our group have demonstrated elevated expression of NOTCH3, NOTCH4 receptors, and the DLL4 ligand in the intestines of patients with complicated Crohn's disease (CD)[2]. This study aims to explore the potential of the Notch pathway as a therapeutic target in intestinal fibrosis associated with CD.

Aims & Methods: We analyzed ileal samples from CD patients with complicated lesions to assess the localization of NOTCH3/4 receptors using immunohistochemistry (IHC) and performed RT-qPCR and Western blot (WB) analysis. In vitro studies utilized DLL4-coated plates (1 µg/ml), with or without the γ-Secretase Inhibitor DAPT (10 µL/ml), to evaluate protein and mRNA expression of HES1 and EMT markers (VIMENTIN, SNAI, and E-CADHERIN) in two primary epithelial cell lines: small intestine (HIEC6) and colon (CoEpi) cells over 48 hours. Data, presented as fold induction (mean ± SEM) relative to water-treated cells or control tissues, were analyzed using parametric or non-parametric ANOVA (Kruskal-Wallis) based on normality testing (*P < 0.05 or **P < 0.01).

Results: RT-qPCR analysis revealed significantly increased mRNA levels of NOTCH3 and NOTCH4 receptors. WB analysis showed elevated levels of NOTCH3 (235±87**, compared to control tissues 100±8) but no significant change in NOTCH4 (87±13 vs. control 100±7). Cellular localization studies showed NOTCH3 in endothelial and smooth muscle cells, lamina propria cells, and lower intensity in the epithelium, whereas NOTCH4 was predominantly found in lamina propria cells. In vitro assays indicated that DLL4 significantly altered mRNA and protein expression of HES1 and EMT markers (VIMENTIN and SNAIL) in both epithelial cell lines. Co-treatment with DAPT significantly reversed some EMT markers induced by DLL4 (VIMENTIN and SNAIL).

Conclusion: NOTCH3 appears to play a role in the fibrogenic processes in CD, with elevated expression in affected tissues. The activation of EMT markers by DLL4 suggests that Notch signaling plays a crucial role in driving fibrosis and tissue remodeling in CD, positioning it as a potential therapeutic target.

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Disclosure: Nothing to disclose.

PP0362

ELEVATED INTESTINAL AND LIVER FATTY ACID-BINDING PROTEINS (I-FABP AND L-FABP) AND FIBROBLAST GROWTH FACTOR (FGF-19) CONCENTRATIONS IN SERUM MAY REFLECT IMPAIRED MUCOSAL BARRIER AND ASSOCIATED LIVER DAMAGE IN INFLAMMATORY BOWEL DISEASES IN CHILDREN

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Introduction: Inflammatory bowel disease (IBD) is a group of chronic inflammatory diseases of the gastrointestinal tract and about 50% of patients develop liver function disorders of different severity in their course. There is unknown cause directly leading to the symptoms of liver damage, one of the suggested causes is impaired mucosal barrier and the associated increased intestinal permeability to bacterial and nutritional antigens and toxins.

Aims & Methods: The aim of the study was to assess the effect of the severity and location of inflammatory lesions in the gastrointestinal tract in the course of IBD on intestinal mucosal barrier disorders and the occurrence of symptoms of liver damage. To assess the degree of mucosal barrier impairment, the intestinal and liver fatty acid-binding proteins (I-FABP and L-FABP) concentrations were assessed, which are sensitive markers of mucosal damage and to assess the degree of liver damage, the fibroblast growth factor (FGF-19) concentrations assessment and laboratory and imaging results. For IBD severity evaluation, the clinical activity scoring indexes and serum tumour necrosis factor alpha (TNF-alpha) concentrations were assessed.

There were 20 children with Ulcerative Colitis (UC, 12 boys, 8 girls, mean age: 16.0 yrs, range: 7.5 – 18) and 20 with Crohn's Disease (CD, 10 boys,

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10 girls, mean age: 14.8 yrs, range: 6.3 – 18) and 20 healthy controls enrolled into the study. Serum I-FABP, L-FABP, FGF-19 and TNF-alpha concentrations were assessed at the baseline, before the treatment and after 2 weeks of treatment, using ELISA immunoassays. Statistical analysis was performed with Statistica 13.0 software (TIBCO Software, USA) using Mann-Whitney U test, Wilcoxon signed rank test and Spearman's correlation rank test.

Results: We found significantly elevated serum I-FABP in active UC group when compared to active CD group and controls ($p<0.05$), while in inactive UC group and inactive CD group they were comparable to controls (p=NS). Moreover, serum L-FAPB was significantly increased in active UC group when compared to active CD group and controls ($p<0,05$) and in inactive UC group when compared to inactive CD group and controls ($p<0,05$).

When we assessed serum FGF-19 concentrations, we found them significantly increased both in active UC and active CD when comparable to controls ($p<0,05$) and in inactive UC group when compared to inactive CD group and controls ($p<0.05$). Additionally, serum FGF-19 concentrations were related to the stage of liver impairment in UC patients.

Conclusion: Increased serum concentrations of I-FABP and L-FABP may be associated to impaired mucosal barrier in IBD. Nevertheless, impaired mucosal barrier may be connected to increased penetration of bacterial and nutritional antigens and toxins and secondary may be related to liver damage. Furthermore, increased serum FGF-19 concentration may be associated with liver damage in the course of IBD in children.

Disclosure: Nothing to disclose.

PP0363

INVESTIGATING CAUDOVIRALES-INDUCED MOLECULAR MIMICRY IN CROHN'S DISEASE PATHOGENESIS

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Introduction: Crohn's disease (CD) has been increasingly linked to imbalances in the gut microbiota, notably through dysregulation in the gut virome. Particularly, elevated levels of *Caudovirales* have been observed in CD patients. Among these, the *Proteus virus Isfahan* species is abundant in CD, suggesting it might trigger an autoimmune response through molecular mimicry.

Aims & Methods: The aim of this study is to investigate whether *Proteus virus Isfahan* proteins share homology with human proteins, potentially initiating autoimmune mechanisms in CD. Utilizing the IBD Transcriptome and Metatranscriptome Meta-Analysis (TaMMA) framework, we analyzed cell populations from intestinal biopsies of CD patients and healthy controls. Immune and non-immune cells were isolated using flow cytometry and underwent transcriptomic analysis.

Additionally, we examined the homology of viral proteins to human proteins using NCBI BLAST and iPBA. *In vitro* experiments involved coculturing T cells with immune cells overexpressing *Proteus virus Isfahan* protein to explore immune responses.

Results: Our results confirmed increased levels of *Proteus virus Isfahan*, in CD patients, especially in dendritic cells and macrophages. Gene ontology analysis indicated compromised immune responses in dendritic cells and enhanced viral-response pathways in CD macrophages. Notably, the viral protein gp82 showed structural homology with human dCMP deaminase, supporting the molecular mimicry hypothesis. These findings highlight *Proteus virus Isfahan* as a potential activator of autoreactive T cells, exacerbating CD's autoimmunity.

Conclusion: This evidence suggests that *Proteus virus Isfahan* may play a significant role in CD pathogenesis through molecular mimicry, where its protein's similarity to human proteins could trigger immune misrecognition, fostering autoimmunity and perpetuating inflammation in CD. Ongoing experiments aim to further elucidate these interactions, potentially identifying novel therapeutic targets.

Disclosure: Nothing to disclose.

PP0364

EVALUATION OF ENDOPLASMIC RETICULUM STRESS IN THE ILEAL POUCH MUCOSA OF PATIENTS WITH ULCERATIVE COLITIS COMPARED TO THE FAMILIAL ADENOMATOUS POLYPOSIS

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Introduction: Total retocolectomy with ileal pouch anal anastomosis (IPAA) is the surgical procedure of choice for patients with ulcerative colitis (UC) refractory to clinical treatment and familial adenomatous polyposis (FAP) with many polyps in the rectum. Primary ileitis of the ileal pouch (IP) is one of the most common complications in UC, being rare in FAP. Previous studies have identified an increase in pro-inflammatory cytokines, as well as alterations in the apoptosis and macroautophagy pathways markers in the IP mucosa of UC patients, compared to those with FAP and normal distal ileum (controls).

Since autophagy and apoptosis can be modulated by endoplasmic reticulum (ER) stress, we investigated this pathway in IP of UC compared to PAF and controls.

Aims & Methods: The aim was to evaluate the modulation of ER stress in the IP mucosa of UC and FAP patients who underwent IPAA, compared to controls. Thirty-two patients were included, ten in the UC group, twelve in FAP group and ten in the control group. Patients who had undergone ileostomy closure for less than one year and had clinical, histological, or endoscopic evidence of pouchitis were excluded. The patients underwent routine colonoscopy, and biopsies were performed from the IP.

For the control group, biopsies were taken from the normal distal ileum. The local Ethics Committee approved the study. The biopsies were processed for RNA extraction and subsequent gene expression analysis using the quantitative PCR (qPCR) technique. The following transcripts were investigated: IRE1 (ERN), PERK (EIF2AK3), ATF6, GRP78 (HSPA5), GRP94 (HSP90B1), ATF3, STC2, DDIT3, DNAJC3, and CALR. Nonparametric tests were used for statistical analysis, with a significance level of $p<0.05$.

Results: The pouchitis disease activity index (PDAI) was applied to verify the absence of pouchitis, with all patients scoring lower than 7. There was no increase in the transcriptional level of the main ER stress pathway markers (ERE1, PERK, and ATF6) in the IP without pouchitis.

However, there was a significantly lower expression of DDIT3 in the UC and FAP groups compared to the control ($p\leq0.05$), which may suggest a deficiency in counterbalance the ER stress activation.

On the other hand, concerning the UPR pathway markers, the levels of CALR and HSP90B1 were significantly higher in the IP of UC compared to the respective IP afferent loop ($p\leq0.05$), supporting potential activation of the ER stress in the IP in the UC group.

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Additionally, the transcriptional level of ATF3 was significantly higher in the PAF group compared to the control (p≤0.05), which may indicate a counter-regulatory mechanism in response to tissue damage, which was not verified in the UC group.

Conclusion: Our findings demonstrated that markers of the unfolded protein response (UPR) signaling pathway, which is activated in response to ER stress, are modulated in the IP mucosa of UC patients, even in the absence of pouchitis. These results may explain the propensity to IP inflammation in UC patients. They may guide the identification of additional molecular pathways contributing to inflammation of the IP and the search for new therapeutic targets.

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Disclosure: Nothing to disclose.

PP0365

MUCOSAL KINASE ACTIVITY AND INFLAMMATORY PROFILES IN INFLAMMATORY BOWEL DISEASE, AND IN RELATION WITH TOFACITINIB RESPONSE

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Introduction: Modulation of kinase activity is part of the therapeutic armamentarium for inflammatory bowel disease (IBD). However, as with other IBD treatments, not all patients respond to treatment. To improve the precision of therapeutic interventions, a better understanding of the mucosal inflammatory environment is essential.

This study investigates mucosal kinase activity, along with mucosal cytokine and chemokine profiles in IBD and in relation to tofacitinib response.

Aims & Methods: Paired inflamed and non-inflamed colonic biopsies were collected from patients with Crohn's disease (CD, N=16), ulcerative colitis (UC, N=16), and non-IBD controls (N=4) to assess IBD-associated kinase activity and cytokine/chemokine profiles.

Additionally, colonic samples were collected from UC patients before start of tofacitinib treatment (cohort 1, N=12) and both before and after 8 weeks of treatment (cohort 2, N=16), to assess kinase activity profiles in relation to tofacitinib response.

Results: The mucosal kinase activity and cytokine/chemokine profiles exhibited significant differences between inflamed and non-inflamed mucosa, with more pronounced alterations observed in UC compared to CD. The increase in kinase activity was most pronounced in the tyrosine kinase families.

Responders to tofacitinib demonstrated higher baseline mucosal kinase activity, although only two predicted kinases (DCLK1 and ATR) were consistently identified across two cohorts. In responders, mucosal kinase activity significantly decreased after 8 weeks of treatment.

Conclusion: Mucosal kinase activity and cytokine/chemokine profiles are associated with inflammation in IBD, with distinct differences between UC and CD. Baseline kinase activity appears to predict response to tofacitinib, with a marked reduction in kinase activity observed after 8 weeks of treatment in responders. These findings highlight the potential of kinase activity profiling in optimizing therapeutic strategies for IBD.

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Conflicts of interest

ECB is supported by the Alexandre Suerman program for MD and PhD students of the University Medical Center Utrecht, Netherlands, and was co-applicant on an unrestricted Investigator Initiated Research Grant of Pfizer.

BR no conflicts of interest

LL no conflicts of interest

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EGvL no conflicts of interest

CSHTH no conflicts of interest

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PP0366

MICROSCOPIC COLITIS: A COMPARATIVE STUDY OF CLINICAL SYMPTOMS AND HISTOPATHOLOGICAL FEATURES

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Introduction: Microscopic colitis is a chronic inflammatory condition defined by a clinical triad: chronic diarrhea, endoscopy without significant abnormalities, and distinct histopathological features characterized by a largely normal appearance of the mucosa but increased numbers of lymphocytes in the epithelium and lamina propria on microscopy.

Aims & Methods: This study aims to explore the relationship between clinical presentation and histopathological features in microscopic colitis, with a focus on identifying prognostic factors linked to therapeutic response and symptom resolution.

This is a retrospective, descriptive study conducted over a 10-year period (September 2015 to May 2025), involving 54 patients diagnosed with microscopic colitis based on histological examination of colonic biopsies. Among them, 19 cases were identified as collagenous colitis and 35 as lymphocytic colitis. Data collected included epidemiological, clinical, endoscopic, histopathological, therapeutic, and follow-up information.

The histological parameters assessed were: characteristics of the lamina propria, intraepithelial lymphocytosis, presence and thickness of the subepithelial collagen band, and surface epithelial damage.

Results:

A total of 54 cases of microscopic colitis were included in the study, among which 51 patients (94.4%) presented with chronic diarrhea, with a median of 7 diarrheic stools [IQR: 4.5–10.5]. One patient with lymphocytic colitis had concomitant celiac disease. Concerning medication use, 2 patients (3.7%) were taking proton pump inhibitors, and 3 (5.6%) were on venotonic agents.

Our analysis revealed several statistically significant correlations:

- A very strong positive correlation between the number of intraepithelial lymphocytes (IELs) and the frequency of diarrheal stools ($r = 0.873$; $p < 0.001$).
- A strong positive correlation between the number of diarrheal stools and the rate of clinical improvement ($r = 0.633$; $p < 0.001$).
- A strong association between subepithelial collagen band thickness and clinical improvement ($r = 0.509$; $p < 0.001$).

In addition, patients with an IEL count <30% experienced a significantly faster clinical response compared to those with ≥30%, with a median time to symptom resolution of 2 weeks versus 4 weeks, respectively ($p < 0.001$). Similarly, patients with subepithelial collagen band thickness <20 µm showed more rapid improvement than those with ≥20 µm (median time to improvement: 2 weeks vs. 6 weeks; $p < 0.002$).

Regarding treatment, 49 patients (90.7%) received budesonide, while 5 (9.3%) were managed with loperamide. A follow-up histological assessment performed 12 months after treatment initiation showed complete normalization of the colonic mucosa in all patients.

Conclusion: Microscopic colitis is a common cause of chronic diarrhea and represents a treatable condition. Accurate diagnosis relies on the recognition of characteristic histopathological features, highlighting the importance of a thorough histological assessment. The integration of clinical context with microscopic findings enables the pathologist to establish a confident diagnosis in most cases and to provide the gastroenterologist with precise and clinically relevant information.

Disclosure: Nothing to disclose.

PP0367

INTERFERON REGULATOR FACTOR 5: A NOVEL INFLAMMATORY MARKER AND PROMISING THERAPEUTIC TARGET IN ULCERATIVE COLITIS

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Introduction: Inflammatory bowel disease (IBD), encompassing Crohn's disease (CD) and ulcerative colitis (UC), is characterized by chronic inflammation affecting the gastrointestinal tract and extraintestinal organs. The etiology of IBD is multifactorial, involving genetic, immunological, and environmental factors, with significant genetic predisposition indicated by over 200 identified loci associated with the disease. Despite advancements in understanding the genetic basis of IBD, clinical management remains challenging due to heterogeneity in disease presentation and variable treatment responses.

Current therapies, including 5-aminosalicylates and biologics, are not universally effective, highlighting the urgent need for reliable biomarkers to predict therapeutic responses.

This study investigates the potential role of interferon regulatory factor 5 (IRF5) in IBD pathogenesis, particularly in UC.

Aims & Methods: We conducted a systematic analysis of colon biopsies from adult patients diagnosed with UC and non-IBD controls. Immunostaining was performed to assess IRF5 expression in colonic tissues, and statistical analyses were conducted to evaluate the correlation between IRF5-positive cell counts, disease activity, and inflammatory markers such as calprotectin

Results: Our analysis revealed a significant increase in IRF5-positive macrophage-like cells in the inflamed mucosa of IBD patients compared to healthy controls. The number of IRF5-positive cells correlated positively with disease activity and calprotectin levels, indicating that higher IRF5 expression is associated with increased inflammation.

Conclusion: This study establishes a significant correlation between IRF5 expression and disease activity in UC, suggesting that IRF5 may play a crucial role in the inflammatory processes associated with the disease. The findings propose IRF5 as a novel biomarker for therapeutic intervention in IBD. Further research is warranted to elucidate the exact mechanisms by which IRF5 contributes to IBD pathogenesis and to evaluate the therapeutic potential of targeting this pathway in clinical settings.

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PP0368

SEVERE ACUTE COLITIS: CLINICAL PRESENTATION AND THE CONTRIBUTION OF ENDOSCOPY

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Introduction: Acute severe colitis (ASC) is a medical and surgical emergency that can be life-threatening. Diagnosis is based on clinical and biological criteria, with clinical and endoscopic severity scores.
The aim of our study was to determine the clinical and endoscopic profile of patients with ASC.
Aims & Methods: We conducted a retrospective study over a 20-month period, from June 2023 to January 2025, in the Hepato-Gastroenterology Department. Patients included had ASC defined by the Truelove and Witts criteria.
Results: Twenty-five patients met the diagnostic criteria. The mean age was 39 years [16–86].
There was a male predominance (64%) with a sex ratio of 1.7.
Five patients (20%) presented with inaugural ASC. Among the remaining 20 patients, 19 (95%) had known ulcerative colitis (UC).
The extent of UC was pancolitis in 17 patients (85%).
The disease duration ranged from 1 month to 20 years, with a mean of 41.36 months.
Among patients with known IBD, nine (45%) had a history of ASC: four were on biologics (infliximab or adalimumab), four on 5-ASA, and one was not receiving any maintenance therapy.
The delay in consultation ranged from 4 to 60 days, with a mean of 17.4 days.
All patients had severe flare-ups based on the Truelove and Witts criteria, with a mean Lichtiger score of 10.2.
Abdominal X-rays were performed for all patients: normal in 88% (n=22) and showed air-fluid levels in 12% (n=3).
All patients underwent recto-sigmoidoscopy to assess endoscopic severity, except one who was admitted with an acute abdomen and underwent colectomy.
Erythema and congestion were the most frequently observed lesions (91%).
Endoscopic Mayo score was 3 in 20 patients (83%), and UCEIS was ≥5 in 12 cases (52%).
Initial management included intravenous corticosteroids at 0.8 mg/kg/day (mean duration: 5.8 days), antibiotics, prophylactic anticoagulation, and a low-residue diet for all patients.
One patient (4%) required a blood transfusion, and another needed correction of electrolyte imbalance.
Second-line therapy with infliximab was required in 2 patients (8%).
One patient underwent colectomy for infliximab-refractory ASC.
There was a significant correlation between the Mayo and UCEIS scores ($P = 0.01$).
Other parameters showed no significant correlation, but hemoglobin levels had a positive impact on corticosteroid response.
Conclusion: Acute severe colitis is a potentially life-threatening complication, primarily of ulcerative colitis.
Management is based on intravenous corticosteroids and endoscopic assessment. Close monitoring is crucial to consider surgery in case of therapeutic failure or complications.
Disclosure: Nothing to disclose.

PP0369

MICROBIOTA METABOLITE BUTYRATE ALLEVIATES INTESTINAL INFLAMMATION THROUGH AUTOPHAGY ACTIVATION

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Introduction: Butyrate, one of the short chain fatty acids produced by bacteria, is known to be beneficial in attenuating inflammatory response. Previous works have demonstrated that activation of autophagy can help alleviate inflammation. However, the roles of autophagy, microbiota and their metabolites in the pathogenesis of inflammatory bowel disease (IBD) are not known.
Aims & Methods: Male wild-type C57BL/6J mice aged 6-8 weeks old were administered 3% dextran sodium sulfate (DSS) to induce experimental colitis models. Mice were treated with sodium butyrate (SB) through oral gavage. 3-methyladenine (3-MA) was administered intraperitoneally to suppress autophagy in mice.
Results: We confirmed that SB treatment changed the composition of the intestinal microbes, restored intestinal barrier dysfunction and diminished the susceptibility of mice to DSS-induced colitis.
Additionally, autophagy levels in the intestine were significantly increased in SB-treated group with enhanced protein levels of Atg16l1 and LC3II, and reduced level of P62. While the SB-treated group showed increased autophagic flux, 3-MA-treated mice conversely displayed significantly attenuated autophagy activity. Meanwhile, the butyrate-mediated protection against colonic injury was considerably diminished in the 3-MA-treated mice.
Conclusion: These results show that autophagy and intestinal microbiota metabolite butyrate are essential for the maintenance of intestinal homeostasis. Butyrate can alleviate inflammation in DSS-induced colitis by activating autophagic flux. These insights may have vital implications for the development of therapeutic strategies for IBD.
Disclosure: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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PP0370

CYTOMEGALOVIRUS COLITIS IN SEVERE INFLAMMATORY BOWEL DISEASE FLARES: PREVALENCE AND ASSOCIATED FACTORS

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Introduction: Patients with inflammatory bowel disease (IBD) experiencing severe flares are at increased risk of opportunistic infections, particularly cytomegalovirus (CMV), due to immunosuppressive therapy.

Aims & Methods: This study aimed to assess the prevalence and risk factors associated with CMV infection during severe acute colitis.

We conducted a retrospective analytical study on hospitalized patients with severe acute colitis. CMV infection was confirmed by positive anti-CMV IgM serology, positive colonic biopsy PCR for CMV DNA, or histological evidence of CMV cytopathic effect.

Patients were divided into two groups:

Group 1 (CMV-positive) and Group 2 (CMV-negative). Clinical, biological, and therapeutic data were compared (SPSS, $p<0.05$).

Results: Among 41 patients (mean age 39.8 years, 14 men), 17 (41.5%) had confirmed CMV infection. CMV was diagnosed by PCR (n=13), serology (n=3), or histology (n=1). Most received antiviral therapy (Ganciclovir or Valganciclovir). One patient required emergency colectomy; three experienced relapse, with two receiving Foscarnet and one undergoing colectomy for bleeding. No significant association was found between CMV infection and age, sex, disease duration, corticosteroid or immunosuppressant use, anemia, hypoalbuminemia, or CRP >45 mg/L. However, CMV infection was significantly associated with steroid resistance ($p=0.003$).

Conclusion: CMV infection is frequent during severe IBD flares and is strongly associated with corticosteroid resistance. Early recognition of this risk factor is essential for timely diagnosis and management to avoid complications.

Disclosure: Nothing to disclose.

PP0371

SRB3 DEMONSTRATES ROBUST *EX VIVO* EFFICACY IN BIO-EXPOSED IBD PATIENTS AND CORRELATES WITH DISEASE ACTIVITY

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Introduction: Inflammatory bowel disease (IBD) is a complex disease characterized by the dysregulation of host immunity. A key feature is the intestinal activation of T cells with cytotoxic activity and pro-inflammatory cytokine release. Therefore, targeting these pro-inflammatory T cells is a reasonable therapeutic approach. CCR9 is the main activator of gut tropism in the lymphocytes and a promoter of their migration into the lamina propria.

SBR3 is a best-in-class anti-CCR9 monoclonal antibody that effectively depletes $CD4^+$ $CCR9^+$ T cells in intestinal biopsies and peripheral blood of Crohn's Disease (CD) and Ulcerative Colitis (UC) patients, which leads to a marked reduction in pro-inflammatory cytokines and restoration of immune homeostasis.

The main objective of this study is to evaluate the translational efficacy of SRB3 in depleting $CD4^+CCR9^+$ T cells in intestinal explants from patients with UC and CD, and to assess its correlation with clinical parameters such as fecal calprotectin levels and current treatments.

Aims & Methods: Intestinal biopsies were collected in routine colonoscopies with inflammatory activity (defined as a Mayo Endoscopic Score <1 in ulcerative colitis and presence of ulcers in CD) from a cohort of IBD patients that included both naïve individuals and those previously treated with biologics, including TNF inhibitors (adalimumab, infliximab) and IL-12/23 antagonists (ustekinumab). Gut explants were cultured and treated *ex vivo* with SRB3. The frequency of $CD4^+CCR9^+$ T cells and cytokine levels were measured by flow cytometry and multiplex immunoassays.

Clinical and demographic parameters from the patient cohort, including smoking habits, type of disease, Montreal Classification, and fecal calprotectin levels (µg/g), were reviewed to evaluate correlations with SRB3 response.

Results: SRB3 efficiently depleted $CD4^+$ $CCR9^+$ T cells in both naïve patients and those treated with biologic therapies, including anti-TNF agents and IL-12/23 antagonists, indicating robust activity across diverse treatment backgrounds.

Importantly, higher levels of fecal calprotectin showed a statistically significant correlation ($p = 0.032$) with the extent of $CCR9^+$ cell depletion following SRB3 treatment. Given that elevated calprotectin levels are commonly associated with active disease or acute inflammatory flares, this finding suggests that SRB3 is particularly effective in patients with moderate-severe inflammatory activity.

Conclusion: These findings support the broad therapeutic potential of SRB3 in IBD patients, demonstrating efficacy not only in bio-exposed patients but also in those with active disease, where higher calprotectin levels correlate with stronger depletion levels.

Disclosure: Dr. Barreiro and Dr. Conde are both collaborators in a project funded by SunRock Biopharma.

PP0372

AN EXPLORATORY STUDY OF THE ASSOCIATION BETWEEN ENDOCYTOSCOPIC AND HISTOPATHOLOGICAL FINDINGS AND MUCOSAL CYTOKINE EXPRESSION PROFILES IN ULCERATIVE COLITIS

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Introduction: The pathogenesis of ulcerative colitis (UC) involves complex immune responses, including Th1, Th2, and Th17 pathways, and is characterized by substantial heterogeneity. Cytokine profiling may contribute to understanding individual disease mechanisms and guiding personalized therapeutic strategies; however, its clinical application remains limited due to technical complexity.

This study aimed to investigate whether endocytoscopic and histopathological findings correlate with mucosal cytokine expression levels and could serve as surrogate markers.

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Aims & Methods: Thirty-five patients with UC receiving treatment at our institution were enrolled. Endocytoscopy (ultra-magnifying endoscopy) was performed in all patients, and the findings were classified according to the Endocytoscopic Classification system (EC classification: EC-A, EC-B, EC-C, EC-D). Mucosal biopsies were obtained from the most inflamed areas identified during endocytoscopy. Histological evaluation included the Matts histological grade and assessments of neutrophil, eosinophil, and lymphocyte infiltration; goblet cell depletion; crypt distortion; and epithelial damage, each graded on a four-point scale (absent, mild, moderate, severe). mRNA expression levels of cytokines (IFN-γ, TGF-β1, TNF-α, IL-1β, IL-4, IL-5, IL-6, IL-8, IL-10, IL-12, IL-13, IL-17, and IL-23) were quantified using RT-PCR. Statistical analyses were performed to assess correlations between endocytoscopic/histopathological findings and cytokine expression levels.

Results: The median age of participants was 44 years (IQR: 35–57), with 24 males and 11 females. Disease types included pancolitis (n = 16), left-sided colitis (n = 14), and proctitis (n = 5). The median Mayo score was 1 (IQR: 0–5). EC classification scores showed significant positive correlations with the expression levels of IL-1β, IL-6, IL-8, IL-13, IL-17, and IL-23 (Spearman's rank correlation, $p < 0.05$; $r = 0.555$, 0.526, 0.584, 0.441, 0.422, and 0.449, respectively).

Similarly, Matts histological grades were significantly correlated with the expression of IFN-γ, IL-1β, IL-6, IL-8, IL-13, IL-17, and IL-23 ($p < 0.05$; $r = 0.493$, 0.756, 0.692, 0.730, 0.656, 0.623, and 0.448, respectively). Histopathological features—including neutrophil and lymphocyte infiltration, goblet cell depletion, epithelial damage, and crypt distortion—were significantly associated with the expression of IFN-γ, IL-1β, IL-6, IL-8, IL-13, IL-17, and IL-23. Additionally, eosinophil infiltration was positively correlated with the expression levels of IL-1β, IL-5, and IL-13.

Conclusion: This study demonstrates that endocytoscopic and histopathological indicators of inflammation are associated with mucosal expression levels of several cytokines—particularly inflammatory and Th17-related cytokines—in patients with ulcerative colitis (UC). These findings suggest that such evaluations may serve as surrogate markers, potentially improving disease assessment and informing therapeutic decision-making.

Disclosure: Nothing to disclose.

PP0373

LYSOPHOSPHATIDIC ACID MEDIATED LPAR2 SIGNALING IN MACROPHAGE EXACERBATES INFLAMMATORY BOWEL DISEASE THROUGH REGULATING METABOLIC REPROGRAMMING

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Introduction: Lysophosphatidic acid (LPA) is the simplest phospholipid and was indicated to participate in the development of IBD. Macrophage polarization are critically involved in the pathology of inflammatory bowel disease (IBD). However, the function of LPA in IBD is incompletely characterized and it is unclear how LPA regulates macrophage polarization.

Aims & Methods: This study aimed to elucidate the effect and mechanism of LPA in macrophage and IBD. Macrophage-specific lpar2 overexpression mice were constructed by intravenous administration of AAV9-DIO-Lpar2 in Cd11b Cre mice. We crossed Lpar2fl/fl mice with CD11b-Cre mice to induce monocyte-/macrophage-specific lpar2 deletion. Colitis was induced by intrarectally injecting with TNBS or treating DSS dissolved in drinking water for 7 days. Cell lines of Raw264.7 and bone marrow-derived macrophage (BMDM) were used to investigate the role of LPA-Lpar2 in macrophage in vitro. A series of cell and molecular experiments, including RNA sequencing, seahorse XF cell mitostress test, immunostaining, and dual-luciferase reporter assay was conducted to illustrate the mechanism.

Results: We firstly screened the six LPA receptor (Lpar1-Lpar6) expression by analyzing single-cell data from colon biopsies of IBD patients and found that Lpar2 was the only one which elevated in macrophages. In vivo study showed that AAV9-mediated macrophage specific Lpar2 overexpression in mice aggravates the severity of IBD and increased the number of M1 macrophage in the colon of IBD. Inversely, Lpar2 deficiency in macrophage protected mice from colitis.

To further confirmed that the function of LPA-Lpar2 was dependent on macrophage, the macrophages were depleted by Clodronate Liposomes (CLs). And the protective effect of myeloid Lpar2 deficiency in colitis disappeared in the absence of macrophages.

In addition, adoptive transfer of BMDM from CD11b-Lpar2fl/fl mice significantly alleviated DSS and TNBS induced colitis in WT mice. In vitro, treatment with both LPA and Lpar2 agonist (DBIBB) enhances M1 macrophage polarization in Raw264.7 and BMDM with increasing expression of M1 macrophage marker.

Furthermore, the knockdown of Lpar2 by shlpar2 abolished the effect of LPA on macrophage polarization. Next, we analyzed the gene expression profile by RNA-Seq in Raw 264.7 after LPA treatment and Gene ontology analysis showed that genes related to cell energy metabolism were enriched.

The seahorse XF cell mito stress test demonstrated that both LPA and DBIBB increased oxygen consumption rate (OCR). Inhibition of OxPhos abolished the effect of LPA. Mechanism screening demonstrated that LPA downregulated the expression of pyruvate dehydrogenase kinase 2 (PDK2) which is a key mitochondrial enzyme.

Conclusion: Targeting LPA Lpar2 in macrophage might represent a potential therapeutic approach for the treatment of intestinal inflammation

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PP0374

GUT MICROBIOTA, MICRORNA, AND ENVIRONMENTAL INFLUENCES: EMERGING INSIGHTS INTO IBD AND IBS ETIOPATHOGENESIS

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Introduction: Growing evidence supports the role of environmental factors, such as diet and lifestyle, in modulating the intestinal microbiota and potentially influencing the symptoms of chronic gastrointestinal disorders.

This study aimed to correlate lifestyle habits with colonic microbiota composition in patients with Inflammatory Bowel Disease (IBD) or Irritable Bowel Syndrome (IBS), compared to healthy controls (HCs).

Aims & Methods: 38 IBD, 44 IBS and 48 HC were recruited. Fecal microbiota composition was analyzed by sequencing the 16S rDNA V3/V4 region using the Illumina platform. Dietary habits and physical activity levels were assessed using a 7-day food diary, a Food Frequency Questionnaire, and the IPAQ. Urinary indican was measured via colorimetric assay and HPLC, fecal zonulin levels were assessed using ELISA, and selected circulating microRNAs were analyzed in plasma and serum by TaqMan-based quantitative Real-Time PCR.

Results: A significant reduction in dietary fiber intake was observed in both IBD and IBS patients compared to healthy controls (HCs) ($p < 0.05$). IBS patients engaged in more moderate physical activity than HCs ($p < 0.05$). Reduced fiber intake was associated with decreased microbial diversity ($p < 0.005$), compared to adequate fiber consumption.

Microbiota analysis revealed a reduction in alpha diversity in IBD patients compared to HCs, particularly in terms of phylogenetic diversity and observed features, while IBS patients showed intermediate values. Beta diversity (weighted and unweighted UniFrac) demonstrated a clear separation among the three groups, with the most pronounced differences observed between IBD and the other conditions. LEfSe analysis identified the highest number of discriminant taxa in HCs, consistent with their greater microbial biodiversity. No clearly pathogenic organisms were identified in either IBD or IBS patients, suggesting that the observed alterations may result from a loss of beneficial commensals rather than the presence of harmful species.

Additionally, greater microbial diversity was found in individuals living in rural areas compared to those in urban environments. IBD and IBS patients not adhering to the Mediterranean Diet (MD) showed significantly higher zonulin levels than those following the MD ($p < 0.001$ and $p = 0.003$, respectively).

Finally, miRNAs involved in inflammatory and epithelial permeability pathways were differentially expressed in both IBD and IBS patients. MiR-24 was significantly dysregulated in IBD ($p = 0.01$), while miR-923 also showed a significant difference in the same group ($p = 0.05$). MiR-221 exhibited a non-significant trend toward upregulation in IBS. These findings suggest a potential role for these miRNAs in the molecular mechanisms underlying inflammatory and functional bowel disorders.

Conclusion: The preliminary results of the present study underlined the impact of lifestyle factors, particularly diet and physical activity, on intestinal microbiota composition and barrier function in IBD and IBS. Microbial and molecular markers may support a more personalized approach to these disorders.

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PP0375

CHRONIC INFLAMMATORY BOWEL DISEASES AND CANCERS

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Introduction: Chronic inflammatory bowel diseases (IBD), such as Crohn's disease and ulcerative colitis, are associated with an increased risk of various cancers, particularly colorectal cancer, but also other types, such as small intestine, biliary tract, or liver cancers. Additionally, certain extra-digestive cancers may also occur in patients with these conditions. These risks vary depending on the duration and severity of inflammation, as well as other factors such as family history or exposure to certain treatments. This work explores the diversity of cancers associated with IBD, their prevalence, and their clinical characteristics, highlighting the importance of considering these risks in the medical management of patients.

Aims & Methods: This is a retrospective and descriptive single-center study conducted within a hepatogastroenterology department, focusing on patients followed for inflammatory bowel diseases (IBD) who developed cancer. The study covers the period from 1990 to 2025.

Results: Out of a total of 1,795 patients followed for IBD (1,200 for Crohn's disease and 595 for ulcerative colitis), 2% (36 patients) developed cancer. The male-to-female sex ratio was 0.38, with a clear female predominance. Among the patients, 69.4% (25) had Crohn's disease, and 30.55% (11) had ulcerative colitis.

The most common location of Crohn's disease was colonic involvement (44%, 11 cases),the predominant phenotype was inflammatory (52%, 13 cases) ,for ulcerative colitis the most frequent location was left-sided colitis (54%, 6 cases) ,the median disease duration before the onset of cancer was 60 months.

Digestive cancers were the most commonly associated with IBD (38.8%, 14 cases), followed by breast cancer in women (30.5%, 11 cases), gynecological cancers (11.1%, 4 cases), prostate adenocarcinoma (5%, 2 cases), skin cancers (5%, 2 cases), and other cancers (5%, 2 cases).

Only 30.5% (11 patients) had risk factors (chronic smoking, family history of cancer, etc.).

Regarding treatment, 33% (12 patients) were on biologic therapy , 30.5% (11) were on azathioprin , 16.6% (6) were on therapeutic abstention , 8.3% (3) were on aminosalicylates, and 5.5% (2) were on corticosteroids.

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Treatment was discontinued in 63.8% (23) of patients after the discovery of cancer, and 22% (8) experienced a flare-up which was successfully managed with corticosteroid therapy.

Eighty-three point three percent (83.3%, 30 patients) underwent surgical treatment alone or in combination with radiochemotherapy. Sixteen point six percent (16.6%, 6 patients) were at an advanced stage and had metastases, so they received palliative treatment.

Thirty-six percent (36%, 13 patients) had a good outcome, 30.5% (11) were lost to follow-up, and 2.7% (1) died.

Conclusion: The study shows a 2% cancer incidence in IBD patients, slightly higher than the general population's risk (0.3–0.5% annually), likely due to prolonged inflammation and disease duration. These findings highlight the need for enhanced cancer surveillance and preventive strategies, especially for patients with colonic involvement.

Disclosure: Nothing to disclose.

PP0376

UTILITY AND CORRELATION BETWEEN SALIVARY AND BLOOD HLA-DQA1*05 TESTING IN INFLAMMATORY BOWEL DISEASE

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Introduction: Patients with the presence of HLA DQA1-05 have a higher risk of loss of response (mainly secondary) to anti-TNF treatment. Therefore, determining the presence of HLA DQA1-05 could be a valuable tool for selecting the most appropriate biological treatment for each patient. However, one of its limitations is the time and resources required for testing in blood samples, which may hinder in clinical practice.

Aims & Methods: Single-centre retrospective cohort study was conducted. HLA DQA1-05 was determined in saliva to assess its potential impact on therapeutic decision making. In a subset of patients, a blood sample was also collected to assess the concordance between the two tests.

Results: Twenty-one patients with inflammatory bowel disease (IBD) who were considered for a change or initiation of biologic therapy were included in the study. Of these patients, 10 (47.62%) were positive cases observed in saliva. HLA DQA1-05 was also determined in the blood of 11 patients, with 6 (54.55%) being negative and 5 (45.45%) positive. The correlation between saliva and blood samples was 100%. Furthermore, rapid HLA determination in saliva influenced management in 14 (66.67%) of the cases, with 10 (62.5%) of these cases switching to another anti-TNF or another target, with Ustekinumab being the most common. No significant differences were found between positive (6 of 10 [60%]) and negative (8 of 11 [72.73%]) results (p = 0.537). With regard to the decision-making process based on HLA results, the following factors were found not to have a significant influence: the number of previous biologic lines (no influence [33.33%] and yes [66.66%] [p= 0.236]), being naive to anti-TNF (3 cases out of 3 [100%]) or presenting primary or secondary failure (7 out of 18 [38.88%] and yes [61.11%] [0.186]). A higher frequency of HLA-positive cases was observed in patients with pancolitis ulcerative colitis and in patients with Crohn's disease with stenosing phenotype, although this did not reach statistical significance.

Baseline IBD characteristics	Frequency	Percentage
Ulcerative Colitis	6	28.57
Crohn's Disease	15	71.43
E2	4	19.05
E3	2	9.52
L1	4	19.05
L2	1	4.76
L3	7	33.33
L1+L4	3	14.29

Conclusion: In our experience HLA DQA1-05 determination in saliva has the potential to personalise the management of IBD patients treated with anti-TNF. In this study, a perfect HLA DQA1-05 saliva-blood correlation was obtained (100%).

Disclosure: Nothing to disclose.

PP0377

AFIMKIBART (ANTI-TL1A) IS EFFECTIVE IN THE BLOCKADE OF CELLULAR SIGNALING INDUCED BY ACTIVE TL1A

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Introduction: Tumor necrosis factor-like ligand 1A (TL1A) is a member of the Tumor necrosis factor (TNF) superfamily implicated in the pathogenesis of inflammatory bowel disease (IBD) as a key amplifier of the effector function of several immune cells. TL1A can be expressed by various immune cells, including macrophages, dendritic cells, and some T cell subsets, and exists either as a membrane-bound or as a cleaved soluble trimer. TL1A signals predominantly by activating NF-kB pathway via the only known functional receptor, death receptor 3 (DR3), which has restricted expression mostly on effector T cells and innate lymphoid cells (ILCs). Afimkibart is a fully human anti-TL1A antibody that has shown clinical benefit in Phase 2 clinical studies and specifically blocks the activity of trimeric forms of TL1A. Monomeric TL1A has also been described; however, the relevance of monomeric TL1A in DR3 signaling and disease is unclear. This study aimed to understand the functional characteristics of both monomer and trimer TL1A species in primary cells and cell lines.

Aims & Methods: Immunoassays were used to measure the detection of TL1A species in IBD serum. Recombinant human monomeric and trimeric TL1A proteins were assessed for binding to afimkibart using SPR and in cellular assays using primary immune cells and cell lines for nuclear factor-kappa B (NF-κB) signaling and interferon-gamma (IFNγ) production.

Results: In functional cellular assays with primary immune cells, we demonstrate that trimeric recombinant human TL1A (rhTL1A) protein causes significant NF-κB signaling in human peripheral T lymphocytes, and in the presence of interleukin 12 (IL12) and interleukin 18 (IL18) co-stimulation, a significant increase of IFNγ production by human primary peripheral blood mononuclear cells (PBMCs). In line with the mechanism of action, both the NF-κB signaling and IFNγ production are blocked by afimkibart. In contrast, only high concentrations of monomeric rhTL1A, >100 ng/ml, induced moderate NF-κB signal and demonstrated a minimal increase in IFNγ production above IL12 + IL18 stimulation alone. However, this activity was blocked by afimkibart. Lastly, afimkibart blocks endogenous soluble and membrane-bound TL1A.

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Conclusion: Herein, we demonstrate that afimkibart blocks NF-kB signaling and IFNγ production induced by TL1A stimulation. While some signaling activity was observed at high concentrations of monomeric TL1A protein, we hypothesize that any signaling observed with monomeric form is likely due to the formation of trimers, which are efficiently blocked by afimkibart. Overall, any monomeric TL1A present should not be functionally active nor be available as a sink for afimkibart, thus preserving the optimal pharmacokinetic (PK) and pharmacodynamic (PD) properties needed for effective target inhibition.

Disclosure: All the authors are employees of Genentech-Roche.

PP0378

TARGETED ANTI-MCL1 DELIVERY IN SENESCENT CELLS REDUCES MURINE INTESTINAL FIBROSIS

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Introduction: MCL1 immunostaining has been shown in fibroblasts of the fibrotic ileum of CD patients[1] suggesting that senescence of these cells may be a mechanism of fibrosis perpetuation. We aimed to analyse here the expression of senescence markers in murine intestinal fibrosis and to determine the effects, on this process, of galacto-oligosaccharide-functionalized Mesoporous Silica Nanoparticles (NPs) loaded with an anti-MCL1 drug, S63845 (MSN-S63845).

Aims & Methods: MCL1 and β-Galactosidase immunohistochemistry (IHQ) were performed in colons of mice underwent two DSS cycles, obtained from a previous samples collection. C57Bl/6 mice (20–25 g) were drinking DSS at 2% during 2 cycles (a cycle consists in 7 days drinking water with 2% DSS, followed by ten days drinking water). After the first cycle, mice were administered intraperitoneally, 5 days a week, with MSN-S63845 (8 mg/kg), empty MSN, vehicle or S63845 (8 mg/kg), until the end of the experiment. Body weight was monitored daily, and mice were sacrificed 34 days after the beginning of the experiment. A sample from the colon was paraffin embedded for both histological analysis (collagen deposition was quantified by the QuPath software in Syrius Red- stained slides) and MPO IHQ. Other sample was used for RT-PCR studies. The number of CD45, CD3 and Ly6G positive cells in blood was analysed by FACS. Unless other mention, results are expressed as mean±sem.

Results: Both MCL1 and β-galactosidase immunostaining were detected in leukocytes and fibroblasts of the colon of chronic DSS-treated mice. The effects of blocking MCL1, selectively in β-gal positive cells (MSN-S63845), or non-selectively (S63845) did not alter changes in body weight associated to DSS, but significantly increased the colon length (cm) of mice receiving MSN-S63845 (5.9±0.2, P=0.002**) or S63845 (6.2±0.3, P=0,0402*) compared with the respective vehicles (5.1±0.1 or 5.2±0.2, respectively). Treatment of MSN-S63845 compared with empty MSN induced a significant diminution in: a) the thickness of the submucosa (32.4±2.9 vs 51.6±5.9, P = 0,024*), b) the number of MPO positive cells (67±15.7 vs 171.2±15.4, P=0.001**) c) the mRNA expression of inflammatory cytokines *il17* (0.08 (0.02-0.5) vs 1 (0.3-1.6) P=0.03$^{\#}$), *cxcl15* (0.4 (0.2-0.6) vs 0.8 (0.5-3.5) P=0.07) and *cxcl5* (0.1 (0.1-1.2) vs 0.5 (0.5-4.2) P=0.048$^{\#}$) and profibrotic markers, *tgfβ1 (0.5*±0.04 vs 1.03±0.1, P=0.003**). In contrast, treatment with S63845 failed to significantly modify any of these parameters. Mice receiving both MSN-S63845 or S63845 exhibited a similar number of CD45, CD3 and Ly6G positive cells in blood than those receiving the respective vehicles.

$^{\#}$Mann-Whitnney; *t-test

Conclusion: The selective blockade of MCL1 in β-galactosidase positive cells of murine inflamed intestine reduces fibrosis development.

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Disclosure: Nothing to disclose.

PP0379

THE ROLE OF METHYLXANTHINES IN THE MODULATION OF INTESTINAL INFLAMMATION AND EPITHELIAL BARRIER INTEGRITY

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Introduction: A compromised intestinal epithelial barrier plays a central role in inflammatory bowel diseases (IBD) by increasing intestinal permeability and promoting inflammation. Therefore, the restoration of intestinal barrier has become critical in IBD management[1].

Methylxanthine caffeine and its metabolites are widely consumed bioactive compounds that are known to influence various physiological processes[2].

However, their impact on gut health, particularly in modulating inflammation and epithelial permeability, remains underexplored.

Aims & Methods: The aim of this study was to investigate potential effects of caffeine (CAF) and its primary metabolites paraxanthine (PRX), theobromine (TBR) and theophylline (TPHL) on inflammatory responses and barrier integrity in colonic epithelial organoids, providing insights into their potential role in IBD management. Primary screening of different concentrations of CAF (1/2.5/5 mM) and its metabolites PRX (10/50/100 µM), TBR (10/20/50 µM) and TPHL (25/100/500 µM) was performed in Caco-2 cells. Selected metabolites and concentrations were further tested on inflamed 3D intestinal epithelial organoids derived from ulcerative colitis patient (n = 1) and non-IBD individual (n = 1). Both Caco-2 cells and organoids were pre-treated with metabolites for 1 hour before inflammation induction and were further co-treated with TNF-α/IFN-γ cytokine mix (10 ng/ml each) and metabolites for 24 hours.

The effects on inflammation status and epithelial barrier integrity were tested through targeted gene expression analysis using RT-PCR and FITC-Dextran (4kDa) assay, while the effects on generation of reactive oxygen species were investigated by Flow cytometry and DHE assay. The study was approved by the Kaunas Regional Biomedical Research Ethics Committee (protocol no. BE-2-31).

Results: Primary screening revealed that both CAF and its metabolite PRX affected the expression of genes encoding inflammatory cytokines (*TNF, CXCL9*; p ≤ 0.05) and tight junction proteins *(OCLN, TJP1, CLDN1*; p ≤ 0.05) and were selected for further analysis in 3D intestinal organoids. Neither CAF, nor PRX reduced the expression of inflammatory cytokine-encoding genes in inflamed intestinal organoids. However, both CAF and PRX increased expression of tight junction proteins in inflamed intestinal organoids derived from control non-IBD (*TJP1* – 3.16-fold by CAF and 1.33-fold

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by PRX (p ≤ 0.05); *OCLN* – 1.57-fold by CAF and 1.38-fold by PRX (p ≤ 0.05)) and ulcerative colitis (*TJP1* – 1.32-fold by PRX; OCLN - 1.14-fold by CAF and 1.27-fold by PRX (p ≤ 0.05)) patient-derived organoids. Mean percentages of FITC-Dextran positive organoids were 43 %, 51 %, and 34 % in untreated, inflamed, and inflamed + PRX groups, respectively, revealing a trend in barrier restoration, though it did not reach statistical significance (p = 0.16 in inflamed vs. inflamed + PRX). A tendency of reduction in reactive oxygen species production was also observed after exposure to caffeine or paraxanthine (by 17 % and 11 %, respectively (p > 0.05)).

Conclusion: Methylxanthine paraxanthine might have a positive effect on gut epithelial barrier function through restoration of the expression of tight junction protein-coding genes.

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Disclosure: Nothing to disclose.

PP0380

IMPACT OF INFLAMMATION ON THE DEVELOPMENT OF COLORECTAL DYSPLASIA AND NEOPLASIA IN PATIENTS WITH ULCERATIVE COLITIS

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Introduction: Patients with ulcerative colitis (UC) are at increased risk of developing colorectal cancer (CRC), and this is assumed to be due to chronic inflammation of the colorectal mucosa.

Aims & Methods: To identify risk factors for the development of dysplasia or CRC in patients with UC. Retrospective observational cohort study. Clinical data and colonoscopy findings were collected from medical records. A 1:2 Propensity Score Matching (PSM) was performed to balance covariates adjusting for disease extension, age, gender, smoking habit and disease duration. An Andersen-Gill model was used to account for multiple colonoscopies per patient and time-varying risk of dysplasia or CRC, allowing for recurrent events and intra-individual correlation.

Results: A total of 571 UC patients were identified, median age at diagnosis was 39.5 (27.3-54.2), and 304 (53.2%) were men. At diagnosis, 147 (25.7%) had proctitis, 203 (35.6%) left side colitis and 191 (33.5%) extensive colitis. During a median follow-up of 7.1 years (IQR 3.7– 13.3), 51 patients (9.8%) developed dysplasia/CRC. This was more frequent in males (72.9% vs. 27.1%, p=0.004) and in older patients [48.1 years (36.7–60.2) vs. 38.5 years (26.5–53.8), p=0.004].

After PSM adjustment, 123 patients were included, 41 of whom had dysplasia/CRC, with a total of 381 colonoscopies over a follow-up period of 14.6 years (IQR 8.0–22.4). Macroscopic inflammatory activity was present in 283 (74.3%) colonoscopies and microscopic involvement in 5 (1.3%). Older age at diagnosis (aHR 1.03, 95% CI 1.04–1.10, p<0.001) and higher degree of macroscopic inflammation (aHR 1.44, 95% CI 1.02–2.03, p=0.041) were associated with an increased risk of dysplasia/CRC development; advance therapy, family history of CRC, disease extension and microscopic inflammation were not associated with an increased risk.

	Complete cohort				PMS			
	N = 571	No dysplasia/ CRC	Dysplasia/CRC	p	N = 123	No dysplasia/ CRC	Dysplasia/CRC	p
		515 (90.2%)	56 (9.8%)			82 (66.6%)	41 (33.3%)	
Male, n (%)	304 (53.2)	262 (59.4)	42 (75)	0.123	88 (71.5)	59 (67)	29 (33)	0.88
Age at diagnosis median (IQR), years	39.5 (27.3-54.2)	36.2 (25.9-49.5)	41.6 (28.5-56.2)	<0.0001	48.3 (35.6-59.0)	41.6 (28.5-56.2)	36.2 (25.9-49.5)	0.372
Disease extension, n (%) -Proctitis (E1) -Left side (E2) -Pancolitis (E3)	147 (25.7) 203 (35.6) 191 (33.5)	127 (24.1) 182 (35.3) 178 (34.6)	20 (35.7) 21 (37.5) 13 (23.2)	0.808	38 (30.9) 46 (37.4) 39 (31.7)	26 (31.7) 31 (37.8) 25 (30.5)	12 (29.3) 15 (36.6) 14 (34.1)	0.914
Mayo endoscopic score at diagnosis, n (%) -Mild (Mayo 1) -Moderate (Mayo 2) -Severe (Mayo 3)	83 (14.3) 279 (47.9) 108 (18.6)	68 (13.2) 256 (49.7) 96 (18.6)	15 (26.8) 23 (41.1) 11 (19.6)	0.106	24 (23.3) 53 (51.45) 26 (25.2)	16 (19,5) 35 (42,6) 16 (19,5)	8 (19.5) 18 (43.9) 10 (24.4)	0.797
Smokers, n (%) -Yes -Ex-smokers	66 (11.6) 86 (15.1)	57 (11.1) 77 (15)	9 (16.1) 9 (16.1)	0,004	20 (16.3) 103 (83.7)	13 (15.9) 69 (84.1)	7 (9.8) 34 (82.9)	0.863
EIMs at diagnosis, n (%)	40 (7)	38 (7.4)	2 (3.6)	0.887	11 (8.9)	10 (12.2)	1 (2.4)	0.074
Advanced therapy, n (%)	152 (26.8)	142 (27.7)	10 (17.9)	0.113	36 (29.3)	25 (30.5)	11 (26.9)	0.674

Table 1. Baseline characteristics of the complete cohort and after Propensity Matching Score.

Conclusion: Older age at diagnosis and the presence of macroscopic involvement on successive colonoscopies are associated with an increased risk of developing dysplasia or CRC in patients with UC.

Disclosure: Beatriz Gros has worked as an advisor for Abbvie, Pfizer, Alphasigma and Roche and has served as an speaker for Abbvie, Pfizer, Alphasigma, Roche, Janssen, Takeda and Gilead.

PP0381

LINKING SERUM BIOMARKERS TO HISTOLOGICAL EVIDENCE OF INFLAMMATION IN IBD: A PILOT STUDY ON EXTRACELLULAR MATRIX AND NEUTROPHIL ACTIVATION FOLLOWING RITLECITINIB TREATMENT

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Introduction: Extracellular matrix (ECM) remodeling and neutrophil activation are key aspects of the tissue turnover and fibro-inflammation characteristics of IBD pathophysiology. Mucosal damage, which includes ECM changes, affects intestinal barrier integrity, contributes to chronic inflammation, leading to fibrosis and strictures. Understanding the link between ECM remodeling and mucosal damage is key to decoding healing mechanisms that occur in response to IBD treatments. Serum biomarkers may serve as a valuable surrogate for indicating therapeutic response in IBD patients, providing a non-invasive solution.

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This pilot study aims to evaluate the utility of a panel of serum biomarkers for monitoring pharmacodynamic (PD) and histological disease activity response in IBD patients treated with ritlecitinib, a JAK3/TEC inhibitor.

Aims & Methods: Aim: Assess the link between a panel of seven serum biomarkers that reflect ECM turnover and neutrophil activation (CPa9-HNE, PRO-C3, C3M, PRO-C22, C4M, C7M, CTX-III) with endoscopic, and histological responses to ritlecitinib treatment in a pilot study using samples from two Phase 2 UC and CD clinical trials (Vibrato[1] and Pizzicato[2]).

Methods: Longitudinal serum samples were collected from endoscopic responders (R, n=13) and non-responders (NR, n=12) to ritlecitinib 200 mg in patients with moderate to severe UC (8 weeks) and CD (12 weeks) at baseline and at Week 8 for UC or Week 12 for CD. Endoscopic response was defined either as a Mayo Endoscopic subscore of ≤1 at Week 8 for UC, or as a ≥50% reduction in the Simple Endoscopic Score for CD (SES-CD) from baseline at Week 12 for CD. Neo-epitope biomarkers reflecting neutrophil activation (CPa9-HNE), mucosal damage (C3M, C4M, C7M, PRO-C22), fibrogenesis (PRO-C3), and fibrolysis (CTX-III) were measured in serum samples by ELISA.

Correlations between ECM biomarker levels with endoscopic response, and histological parameters were assessed using the Geboes score for UC and the Colonic Global Histologic Activity Score (cGHAS) and Ileal Global Histological Activity Score (iGHAS) for CD to understand the relationship between pharmacodynamic changes and histological disease activity indicies improvement. Changes in histological activity in colon biopsies from UC subjects were assessed using IBD Explore v2.0.0.

Results: Changes from baseline for C7M, CPa9-HNE, and PRO-C22 suggested a response to treatment in the CD-responders group, although these changes were not statistically significant. Neutrophil activation serum biomarker, CPa9-HNE, showed statistically significant correlation with cGHAS at baseline and iGHAS at Week 12 post-treatment. Similarly, serum C7M levels correlated with iGHAS at Week 12.

None of the serum biomarkers tested showed a statistically significant correlation with the Geboes score in the UC cohort. Only UC responders to ritlecitinib demonstrated a consistent decrease in total neutrophil count in both the epithelium and lamina propria (LP) compared to NR. Interestingly, further histological analysis revealed that serum levels of CPa9-HNE were significantly correlated with total neutrophil count in colon tissues, specifically in the epithelium and LP at baseline in UC.

Conclusion: These initial observations suggest that certain blood markers related to mucosal damage and neutrophil activation might be linked to inflammation seen in tissue samples. Larger studies are needed to confirm these results. This highlights the potential of using blood markers to evaluate disease activity in IBD.

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2. https://www.clinicaltrials.gov/study/NCT03395184

Disclosure: Mohamed Hassanein, Rathi D. Ryan, Yuxi Zhao, Elizabeth Kieras, Chong Duan, Craig Hyde, Jeremy D Gale, Elena Peeva and Anna Sapone are employed by Pfizer, Inc. These authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in this abstract apart from those disclosed.

PP0382

DIMINISHED GUT MICROBIOME DIVERSITY IN INFLAMMATORY BOWEL DISEASE: INSIGHTS FROM LONG-READ 16S RRNA AMPLICON SEQUENCING

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Introduction: Altered gut microbiome composition has been implicated in the pathogenesis of inflammatory bowel disease (IBD). However, detailed characterization of microbial diversity differences between IBD subtypes remains incomplete. This study aimed to evaluate gut microbiome alpha diversity in patients with Crohn's disease (CD) and ulcerative colitis (UC) compared to healthy controls using long-read 16S rRNA sequencing technology.

Aims & Methods: Fecal samples were collected from 17 patients with CD, 20 patients with UC, and 192 healthy controls without IBD. DNA was extracted using an SKYAmp stool DNA kit (SkyGen, Russia). Long-read nanopore sequencing of 16S rRNA amplicons was conducted using a MinIon™ sequencer. Taxonomic assignment was performed with EMu software by aligning FASTQ reads to the NCBI 16S database [1]. Alpha-diversity was evaluated at the species level using Chao1 and Shannon indices. Statistical comparisons between groups were conducted using Welch two-sample t-tests with Holm adjustment in R environment and the RStudio software (version 4.4.1).

Results: The Shannon diversity index was significantly lower in the CD group compared to the control group (p=0.01412), while no significant difference was observed between UC and control groups (p=0.15461). Chao1 richness index values were significantly reduced in both CD and UC patient groups compared to controls (p=0.00003 and p=0.03195, respectively), indicating reduced species richness in both IBD conditions.

	Without IBD	CD	UC
Shannon index	3.452 (3.410 - 3.494)	3.165 (2.970 - 3.360)	3.270 (3.016 - 3.524)
Chao1 index	85.750 (84.046 - 87.454)	67.529 (61.087 - 73.972)	74.650 (64.729 - 84.571)

Conclusion: This study demonstrates that both IBD subtypes exhibit reduced microbial richness as measured by the Chao1 index, while significant reduction in Shannon diversity was observed only in CD patients. These findings suggest differential impacts of CD and UC on gut microbiome composition, with CD potentially having more profound effects on overall microbial diversity. Long-read 16S rRNA sequencing provides valuable insights into the microbial signatures associated with different IBD phenotypes, which may contribute to improved diagnostics and personalized treatment approaches.

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Disclosure: Nothing to disclose.

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PP0383

CD3+CD4-CD8- DOUBLE-NEGATIVE (DN) T CELLS CONTROL RESPONSE TO BIOLOGICAL THERAPY IN PATIENTS AFFECTED BY CROHN DISEASE AND SPONDYLARTHRITIS

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Introduction: The pathogenesis of autoimmune diseases, such as Crohn's disease (CD), Ulcerative colitis and Spondyloarthritis (SpA) has been linked to imbalanced CD4+ and CD8+ T-cell responses. CD3+CD4-CD8- double-negative (DN) T cells, although constituting a rare subset of peripheral T cells, are gaining interest for their roles in inflammation, immune disease and cancer. However, controversy persists regarding their ontogeny and function.

Aims & Methods: The aim of this study is to evaluate the circulating immune cells involved in response to biological therapy in patients with two concomitant autoimmune disease, in particular $CD3^+$T cells, CD3+CD4-CD8- DN T cells, CD3+CD4+CD8+ *double positive* (DP) T cells and Terminal Effector Memory Re-expressing CD45RA T cells (Temra) identified as CD45RA+CD95+CD197-CD27- CD4+ or CD8+ lymphocytes.

Blood samples were collected from 4 patients affected by CD (disease duration 6-10 years) who developed SpA (ASAS criteria, disease duration 2-7 years)., and after 1 year under stable therapy with one biological drug (anti-TNFalpha therapy: n=3; anti-IL23 therapy: n=1). Cytofluorimetric analysis using the BD FACSLyric Flow Cytometry system was performed for T-cell subpopulations analysis using surface markers including: CD3, CD4, CD8, CD45RA, CC197, CD62L, CD44, CD27 and CD95.

Results: CD45RA+CD197+CD95+ (stem cell memory T cells, Tscm), CD45RA-CD95+CD27+ (central memory T cells, Tcm), CD62low and CD44high (naive Tcells) and CD45RA-CD95+CD27low (effector memory T cells, Tem) were evaluated. Significant differences were found in percentage of CD3+CD4-CD8- DN T cells and CD3+CD4+CD8+ DP T cells: in 3 patients the percentage of CD3+CD4-CD8- DN T cells were significantly higher than CD3+CD4+CD8+ DP T cells. The same patients didn't show significant differences between CD4+ and CD8+Temra cells (respectively, CD4+CD45RA+CD95+CD197-CD27- and CD8+CD45RA+CD95+CD197-CD27- T cells). The only patient, where no significant differences appear between CD3+ DN and CD3+ DP T cells, and where CD4+CD45RA+CD95+CD197-CD27- didn't appear at all, was a difficult to treat patient with uncontrolled intestinal and articular manifestations who subsequently needed double-targeted therapy with anti-TNFalpha plus anti-IL23 biologics.

Conclusion: Our preliminary data suggest that T cell subset distribution could be dysregulated in difficult-to-treat IBD-SpA patients. In particular, DN T cells and Temra could be associated with induction of peripheral immunological tolerance and immunomodulatory activities related to disease control. Further studies are warranted to test this hypothesis and to evaluate the effect of different biological therapy classes on T cells during IBD-SpA.

Disclosure: Nothing to disclose.

PP0384

"INDEFINITE FOR DISPLASIA" IN INFLAMMATORY BOWEL DISEASE COLITS: OUTCOME IN A 3 YEARS FOLLOW UP STUDY

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Introduction: In Inflammatory Bowel Disease (IBD), histological assessment of colonic biopsies may detect areas "indefinite for dysplasia" (IND). This definition is used when a definite distinction between non-neoplastic changes and neoplasia is not possible(1).

The role of this finding in predicting the endoscopic and histological findings in IBD colitis is currently undefined.

Aims & Methods: Primary aim was to evaluate the outcome of IBD patients(pts) with an initial histological finding of areas IND in random colonic biopsies, followed up for ≥ 3 years. At this purpose, all IBD pts showing areas IND in colonic biopsies taken from March 2018 to March 2022, subsequently followed up for ≥3 years including ≥ 1 control colonoscopy were retrospectively enrolled. All pts were in regular follow up at our referral IBD Unit and underwent colonoscopy with histological assessment in the same center.

Results: The study population included 99 pts with IBD colitis and histological aspects IND detected in random colonic biopsies in 79(79.8%) pts and in target biopsies in 20(20.2%) pts. In the 79 pts, areas IND were detected in random biopsies taken from: rectum in 62(78%), left colon in 29(36%), transverse colon in 6(7.5%), right colon in 5(6.3%), >1 area in 18(22.3%) pts. In the 20 pts with areas IND on target biopsies, these were taken from stenosis (5 pts), polyps (9 pts), lateral spreading tumors (2 pts), flat elevated mucosa (4 pts).

For the aim of the study, only the 79 IBD pts (37 females, 46.8%) showing in random colonic biopsies areas IND were considered.

These 79 IBD pts (69 UC, 9 CD, 1 unclassified colitis) showed median age of 51 [18-79], long-standing IBD in 54(68.4%), family history of CRC in 11(13.9%), concomitant primary sclerosing cholangitis in 1(1.3%) pt. Random biopsies were taken from colonic mucosa proximal to ulcerations in 54(68.4%) pts, erosions in 19(24%) pts and to mild inflammation in 6(7.6%) pts.

The second colonoscopy was performed after a median of 11[1-42] months. In the 79 pts with a first evidence of areas IND on random biopsies, the second colonoscopy detected: dysplasia in 10(12.6%) pts (high grade, HGD in 2[2.5%], low grade dysplasia, LGD in 8[10.1%]pts), areas IND in 12(15.2%) pts, no dysplasia (HGD, LGD or IND) in 38(48.1%) pts, while in the remaining 16(20.3%) pts no reliable histology was available.

LGD at second colonoscopy was detected in random biopsies (4pts) or in target biopsies (4pts). Differently, in both pts with HGD it was detected in target biopsies, including one focal area of adenocarcinoma (CRC) in 1(1.3%) pt. Areas IND confirmed at second colonoscopy were detected in the same colonic segment in 11(13.9%) pts and in a different segment in 1(1.3%) pt.

After the second colonoscopy, 4 out of the 79 IBD pts required IBD-related colonic surgery, and histology of the surgical specimen detected LGD in 1 pt, CRC in 1 pt and areas IND in 2 pts.

During the study period (median 27[2-75] months) 51 out of the 79(64.5%) IBD pts repeated a third colonoscopy detecting: areas IND in 4(7.8%), LGD in 6(11.8%), HGD in 1(2%) pts and no CRC.

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	IND	LGD	HGD	CRC
1st Colonoscopy (n=79), n (%)	79 (100%)	0 (0%)	0 (0%)	0 (0%)
2nd Colonoscopy (n=79), n (%)	12 (15.2%)	8 (10.1%)	2 (2.5%)	1 (1.3%)
3rd Colonoscopy (n=51), n (%)	4 (7.8%)	6 (11.8%)	1 (2%)	0 (0%)

Table 1. Indefinite for dysplasia (IND) in IBD colitis: outcome at 3 years.

Conclusion: Patients with IBD colitis showing areas indefinite for dysplasia deserve careful endoscopic surveillance as dysplasia and CRC may be detected at subsequent colonoscopies.

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Disclosure: Nothing to disclose.

PP0385

IMMUNE CHECKPOINT INHIBITOR-INDUCED COLITIS IS DEPENDENT ON GRANZYME EXPRESSION

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Introduction: Immune checkpoint inhibitors (CPI) have revolutionised cancer outcomes, but frequently trigger immune related adverse effects, including life-threatening colitis. Tissue resident memory (TRM) T cells are key cells involved in the pathogenesis of CPI-induced colitis (CPI-C), but the mediators they produce to facilitate disease are poorly defined.

Aims & Methods: We utilised our previously published *in vivo* pre-clinical model of CPI-C, which occurs when mice are given combination anti-CTLA4 and anti-PD1 and a proinflammatory microbiota mix, to define the role of the Granzyme A/B cytotoxicity pathway in disease.

Disease was induced in wildtype (WT) and Granzyme A/B double knockout (*GzmAB*$^{-/-}$) mice. Clinical outcomes (colon and spleen mass, histology and infiltrating neutrophils) from these mice were assessed to measure the severity in disease. T cell populations were also analysed using flow cytometry.

Results: *GzmAB*$^{-/-}$ mice showed a significant reduction in disease features compared to WT mice, with reduced colon mass, reduction in infiltrating neutrophils and fewer inflammatory monocytes. Flow cytometry analysis of the T cell compartment revealed a significant reduction in both bulk CD4 and CD8 T cells and also CD4 and CD8 tissue resident memory T cells in both the cLP and IEL in *GzmAB*$^{-/-}$ mice treated with CPI-C in comparison to WT mice treated with CPI-C. *GzmAB*$^{-/-}$ mice treated with CPI-C also had a reduction in IFNγ and TNFα production from both CD4 T cells and CD8 T cells.

These data highlight the importance of granzymes in inducing the disease phenotype in CPI-C. In order to investigate whether CD4 T cells or CD8 T cells were more important, we depleted these cells *in vivo* in WT mice treated with CPI-C at both 1 and 3 weeks.

Here, we found that WT mice with CD4 T cells depleted got significantly worse disease with colon masses increasing and there being increased infiltrating neutrophils. There was an increase in cytotoxic cytokines being produced by CD8 T cells when CD4 T cells were depleted. However, when CD8 T cells were depleted, there was no change in any clinical features or cytokine production from CD4 T cells.

Conclusion: Using *GzmAB*$^{-/-}$ mice we have demonstrated that CPI-colitis is at least partially dependent on the granzyme cytotoxicity pathway. Targeting the cells that produce granzyme in the colon of patients with CPI-colitis might offer an attractive novel treatment option for patients with CPI-colitis.

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PP0386

EXPLORING MICROBIOTA-IMMUNE INTERACTIONS IN SHORT BOWEL SYNDROME AND CROHN'S DISEASE IN A MODEL OF HUMAN MICROBIOTA ASSOCIATED MICE

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Introduction: Short bowel syndrome (SBS) is a malabsorptive condition mostly caused by massive surgical resection of the small intestine and is associated with significant morbidity and mortality, reduced quality of life, and high healthcare costs[1].

Crohn's disease (CD) represents a potential cause leading to an increased risk of SBS with considerable clinical consequences[2].

Human microbiota-associated (HMA) mice represent a model to establish human fecal microbiota in preclinical models[3-4].

Here, for the first time, we tried to conduct a "humanized" model of SBS, using faeces from patients affected by this disease.

Aims & Methods: The aim of the current study is to determine the functional role of the gut microbiota of CD patients with or without SBS. A dextran sodium sulphate model of colitis on C57BL/6 mice was employed to assess immune and metabolic signatures following faecal transplantation from patients affected by SBS/IF, CD with high or low risk for SBS/

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IF. Lymph node samples were collected to examine differences in T cell populations, and disease activity index and bronchoscopy were used to evaluate the activity of the disease.

Results: Fecal microbiota transplantation (FMT) from CD patient with a low risk of SBS/IF was associated with worse disease activity and endoscopic scores in colitis mice.

Additionally, an increase in the percentages of Treg, Tnaive, Th1, and Th17 cells was observed in mice that received FMT from CD patient with a low risk of SBS/IF, compared to controls and those that received FMT from patient with SBS/IF or from CD patient with a high risk of SBS/IF.

Conclusion: Our data underscore the complex interactions between gut microbiota and host immunity and highlight the importance of careful donor selection in FMT applications, particularly for inflammatory conditions. Further research is warranted to characterize specific microbial and immunological factors that drive these effects and to identify microbiota profiles that could potentially confer therapeutic benefits.

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PP0387

TRANSCRIPTOMIC SIGNATURES ASSOCIATED WITH ANTI-TNF DISCONTINUATION OUTCOMES IN ULCERATIVE COLITIS

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Introduction: The BIOSTOP trial is a Norwegian, multicenter study with a prospective, randomized, controlled, and open-label design. The trial was initiated to evaluate the two-year outcomes following the withdrawal of tumor necrosis factor-alpha (anti-TNF) antibodies in patients with ulcerative colitis (UC) who are in sustained clinical and endoscopic remission.

Aims & Methods: The main aim was to identify transcriptomic differences in colon biopsies associated with increased risk of relapse with anti-TNF withdrawal. We also wanted to increase our understanding of the underlying pathological mechanisms of relapse after anti-TNF withdrawal and treatment failure during continued treatment.

Between June 2017 and December 2020, 174 ulcerative colitis patients in confirmed clinical and endoscopic remission were enrolled and randomized, with 86 assigned to continue anti-TNF therapy (control group) and 86 to discontinue treatment (intervention group). Mucosal biopsies were collected at baseline, two-year follow-up, and at point of relaps

Bulk mRNA-sequencing was performed on RNA extracted from rectal biopsies. For patients in remission, samples from baseline and two-year follow-up were analyzed. For those who relapsed or experienced treatment failure, samples from baseline and point of relapse were analyzed. Differential gene expression was performed using DESeq2, and pathway enrichment analysis was assessed with clusterProfiler.

Results: There were no significant differences found at baseline between patients that relapsed (n=45) vs those that stayed in remission (n=41) after anti-TNF withdrawal. In patients that relapsed, the comparison from baseline to point of relapse showed upregulation of key inflammatory genes, including IL1B, STAT3, OSMR, TNF and CXCL10. Pathway enrichment analysis showed that the upregulated genes were associated with an inflammatory response, including TNF-signaling. At two-year follow-up patients who remained in remission also showed upregulation of proinflammatory genes, including genes associated with TNF-signaling. However, the expression levels were lower than in the relapse group. Patients in the control group who experienced treatment failure (n=10) showed a gene expression signature that strongly overlapped with those that relapsed in the intervention group.

Conclusion: We did not detect any transcriptional differences between the relapse and remission group at baseline and after anti-TNF withdrawal the expression of proinflammatory genes including genes involved in TNF-signaling increased in both groups. However, a more pronounced increase was observed in the relapse group, suggesting that those that relapse exhibited reduced immunoregulatory capacity.

Disclosure: Nothing to disclose.

PP0388

INFLAMMATORY BOWEL DISEASES AND PSYCHIATRIC SYMPTOMS: CIRCULATING MIRNAS PATTERN

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Introduction: In the context of inflammatory bowel diseases (IBD), the involvement of brain-gut interaction has been intensely studied in the last decade. Connections between IBD and mental disorders have been previously studied. Psychiatric symptoms, in particular stress, anxiety, and depression, adversely affect a broad range of outcomes in IBD, including increased risk of disease relapse and poorer response to treatment. Although genetic predisposition is crucial for immune-mediated diseases, it seems that environmental factors could have a great impact. Alterations in circulating miRNA levels have been demonstrated both in IBD and psychiatric diseases, but no study has identified a miRNA pattern underlying the crosstalk between IBD and psychiatric symptoms.

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Aims & Methods: The aim of this study was to identify a plasma miRNA-based pattern that could be specific to IBD or associated with the complex phenotypes of IBD in conjunction with stress, anxiety, and depression.

A total of 81 individuals were enrolled in this study. All the patients agreed to undergo a psychiatric consultation and have been screened using the following scales: the Hamilton Anxiety Scale (HAM-A), the Becks Depression Inventory (BDI) and the Perceived Stress Scale (PSS).

Based on their IBD diagnosis and psychiatric evaluation, four groups were considered as follows:

i. IBD (patients with IBD with no psychiatric symptoms);

ii. IBD+SAD (patients with IBD presenting stress, and/or anxiety, and/or depressive symptoms);

iii. SAD (patients with a main diagnosis of recurrent depressive disorder);

iv. CTRL (individuals without IBD and without psychiatric disorders).

The levels of 179 miRNAs have been evaluated by qRT-PCR in the plasma from the entire cohort.

Results: More than 40 miRNAs were identified as differentially expressed among the three patient groups in at least one comparison vs CTRL group. The IBD vs CTRL comparison demonstrated the differential expression of four miRNAs (miR-333-3p, miR-1260a, miR-320d, miR-423-5p, and miR-486-5p), whereas the SAD vs CTRL comparison indicated an alteration of 28 miRNA levels. Five miRNAs (miR-223-3p, miR-1260a, miR-320d, miR-423-5p, and miR-486-5p) exhibited differential levels across all comparisons, indicating their potential as markers for the complex phenotype combining IBD and SAD.

Pearson correlation analyses between the level of miRNAs and psychiatric symptoms evaluated by previously mentioned scales were performed, and several significant correlations were found.

Conclusion: In this study we identified a molecular pattern associated with the complex phenotypes of IBD in comorbidity with psychiatric symptoms. Our results might help to develop new tools for predicting disease evolution and also provide new insights into the gut-brain IBD axis.

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Disclosure: Nothing to disclose.

PP0389

MULTIDISCIPLINARY MANAGEMENT OF IBD PATIENTS INVOLVING SURGEONS AND RADIOLOGISTS

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Introduction: Inflammatory bowel diseases (IBD) are showing an increasing incidence globally. Despite advanced therapeutic options, many patients still require surgical intervention during their disease course. Timely surgical planning and strategy selection are critical, highlighting the importance of collaborative, multidisciplinary care.

Aims & Methods: To assess the role and outcomes of a multidisciplinary IBD team (involving gastroenterologists, surgeons, and radiologists) in optimizing the management of IBD patients.

In November 2023, a multidisciplinary IBD team was established at our institution. Regular case discussions focused on disease characteristics, treatment history, and therapy outcomes. Patient data were collected retrospectively, including disease type, treatment timeline, surgical interventions, and complications.

Results: Seventy-five patients were reviewed; 40 (53%) underwent surgery. Of these, 38 (95%) had Crohn's disease and 2 (5%) had ulcerative colitis. The male-to-female ratio was 23:17. In Crohn's disease patients: 18% had isolated colonic, 24% ileocecal, 26% upper GI, and 32% both small and large bowel involvement. One UC patient had left-sided colitis, the other pancolitis. The median age was 36 years (IQR 30–48).

Eighty percent received prior immunomodulators, with azathioprine still in use in 5 cases. The median time to initiation of immunomodulatory therapy after diagnosis was 11 months (IQR 4–41). Biological therapy had been used in 75% of patients, with a median initiation time of 63 months post-diagnosis (IQR 22–129); in 12.5%, surgery preceded biological therapy.

Previous bowel resection was noted in 40%, with a median time of 52 months since the last surgery. Balloon dilation had been performed in 15% prior to team discussion. Malignancy was suspected in 3 patients (7.5%) and confirmed in one (2.5%). Six patients (15%) underwent robot-assisted surgery. Fourteen required preoperative hospitalization. Postoperative anastomotic leakage occurred in 2 cases (5%), both with fatal outcomes due to comorbidities.

Conclusion: Multidisciplinary management enables timely and individualized therapeutic decisions in IBD care. Regular team discussions improve surgical planning and patient preparation, reducing complication risks, shortening hospital stays, and enhancing quality of life. Moreover, collaboration fosters mutual understanding and continuous professional development across disciplines.

Disclosure: Nothing to disclose.

PP0390

THERAPEUTIC POTENTIAL OF PROBIOTICS IN DNBS-INDUCED COLITIS: IMMUNOLOGICAL AND MICROBIAL INSIGHTS

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Introduction: Inflammatory bowel disease is a chronic, relapsing condition influenced by genetic, microbial, and environmental factors. This study evaluated the therapeutic potential of multi-strain probiotics in mitigating the severity of experimentally induced colitis in rats, focusing on clinical outcomes, histopathological changes, inflammatory markers, and microbiota changes.

Aims & Methods: Thirty six-week-old male Sprague-Dawley rats were randomly assigned to three groups: Control, Acute Colitis (AC), and Probiotic Treatment (RP; receiving *Lactobacillus rhamnosus* and *Lactobacillus paracasei*). Probiotics were administered orally once daily, beginning three days prior to colitis induction with rectal DNBS and continued for seven days post-induction. On day eight, colon tissues and blood samples were collected for histological and biochemical analysis, and fecal samples were analyzed to assess gut microbiota composition.

Results: The AC group showed significantly elevated TNF-α levels compared to the Control and RP groups ($p < 0.001$), while probiotic treatment significantly reduced TNF-α levels compared to the AC group ($p = 0.002$, $p < 0.001$, respectively). Although the RP group demonstrated increased IL-10 levels and decreased MDA levels, these changes were not statistically significant. Histopathological analysis revealed a lower median Ameho's score in the RP group compared to the AC group, though without statistical significance.

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Microbiota analysis showed no significant differences in alpha diversity across groups. However, beta diversity significantly differed in the AC group compared to Control, and after probiotic treatment, the RP group's beta diversity showed a partial shift toward the Control group. Distinct microbial species characterized each group: *Akkermansia spp.* and *Parasutterella spp.* were predominant in distinguishing the Control group from others; *Romboutsia spp.* and *Turicibacter spp.* differentiated the AC group; and *Roseburia spp.* and *Anaerotruncus spp.* were the key species distinguishing the RP group from the others.

	Control	AC	RP
TNF-α (pg/ml)	3240 (2185-3852.5)	10950 (5880-16275)	3505 (2342.5-4762.5)
IL-10 (pg/ml)	313 (170.5-528.25)	199 (111.2-522.75)	324 (248.5-445.25)
MDA (nmol/mg protein)	(1.69-12.84)	9 (2.44-25.54)	4 (2.38-20.78)
IL-17 (pg/ml)	121 (118.25-159.75)	128 (121-151.75)	127 (121.75-129.25)
Ameho's score	0	5 (1-5)	1 (1-3.5)

Conclusion: Multi-strain probiotics reduced TNF-α levels and modulated gut microbiota composition in DNBS-induced colitis. Although changes in IL-10, MDA, and histopathology were not statistically significant, trends favored the probiotic group. Microbial analysis revealed distinct species profiles across groups, with partial restoration of beta diversity and enrichment of beneficial taxa like *Roseburia spp.* in the probiotic group. These findings support the potential of probiotics as a therapeutic adjunct in IBD, highlighting the need for further studies to confirm efficacy and explore underlying mechanisms.

Disclosure: Nothing to disclose.

PP0391

NEUROLOGICAL INVOLVEMENT IN INFLAMMATORY BOWEL DISEASE: A COHORT-BASED STUDY

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Introduction: Inflammatory bowel diseases (IBD) are chronic disorders primarily affecting the gastrointestinal tract, but they may also involve other organs through extraintestinal manifestations, including the nervous system. Neurological features can present with a broad spectrum of symptoms, from acute to chronic, involving both the central and peripheral nervous systems.

These manifestations are underdiagnosed and underestimated. Their pathophysiology remains incompletely understood and is thought to be multifactorial.

Neurological complications in IBD are generally divided into two categories:

• Non-drug-induced: directly related to disease activity, immune mechanisms, or nutritional deficiencies.

• Drug-induced: secondary to immunosuppressants or biologics, such as demyelination or neurotoxicity.

Aims & Methods: This study aims to describe the prevalence and spectrum of neurological manifestations in a well-characterized cohort of IBD patients and to identify potential risk factors.

This retrospective descriptive and analytical study was conducted in a tertiary referral center over a nine-year period (January 2016 – March 2025). All included patients had a confirmed diagnosis of IBD, based on clinical, endoscopic, histological, and radiological criteria , with regular follow-up. Disease activity was assessed using the Harvey-Bradshaw Index for CD and the Mayo score for UC. The presence of neurological symptoms assessed by a neurologist. All patients presenting with neurological complaints underwent a complete clinical neurological evaluation, neuroimaging (CT or MRI), laboratory tests, electroencephalography , and electromyography investigations as indicated.

Results: A total of 300 patients with IBD were included, comprising 193 with Crohn's disease and 107 with ulcerative colitis. The mean age was 41 years. Neurological manifestations were identified in 24 patients, accounting for 8% of the cohort. These manifestations were significantly more frequent in patients with UC (14.9%) than in those with CD (4.1%), with a statistically significant difference (p = 0.01).

The sex distribution among patients with neurological symptoms included 13 males (54%) and 11 females (46%), with no significant gender difference observed (p = 0.5). Furthermore, 18 of the 24 patients presenting with neurological involvement had inactive IBD at the time of symptom onset (p = 0.2).

Neurological manifestations were identified and classified based on their presumed etiology.

Drug-induced manifestations were observed in 14 patients. These included 10 cases of paresthesia (3.3%), considered functional or early sensory disturbances, and 4 cases of peripheral neuropathy (1.3%) attributed to medication-related toxicity.

Non–drug-induced manifestations were noted in the remaining patients. These consisted of 7 cases of ischemic stroke (2.3%), 6 cases of migraine (2.0%), 2 cases of epilepsy (0.7%), 2 cases of cerebral venous thrombosis (0.7%), and 1 case of Wernicke's encephalopathy (0.3%).

Notably, no cases of multiple sclerosis or drug-induced demyelinating diseases were reported in this cohort.

Conclusion: This study reveals that neurological involvement in IBD, particularly UC, may be more prevalent than previously appreciated.The low true prevalence or potential underdiagnosis due to lack of targeted screening. A multidisciplinary approach, involving gastroenterologists, neurologists, and other specialists, to prevent long-term morbidity.

Prospective multicenter studies are warranted to better delineate classification criteria, risk factors, and long-term outcomes of neurological EIMs in IBD.

Disclosure: Nothing to disclose.

PP0392

DEVELOPMENT OF AN *IN VIVO* MODEL OF DSS-INDUCED COLITIS FOR DOSE OPTIMIZATION OF ANTI-TNFA, ANTI-INTEGRIN A4B7 AND ANTI-IL23P19 BIOLOGICS AS A BASIS FOR COMBINATION THERAPY STUDIES

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Introduction: Current immunotherapy for inflammatory bowel disease (IBD) treatment is based on anti-TNFa, anti-integrin a4b7, anti-IL12/IL23 and anti-IL23 antibodies that have demonstrated to be safe and effective. However, up to two third of patients treated with biologics do not reach complete remission of the disease; this therapeutic ceiling is one of the

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main challenges we face in the management of patients with IBD. Many *in vivo* models have been developed for research with biologics, but the literature on doses of above-mentioned drugs in *in vivo* IBD models is scarce, highly variable and covers wide ranges.

Aims & Methods: In this work, we aim to find the optimal dose for biologic treatment in a DSS-induced colitis model. C57BL/6 mice were randomized into a control group and treatment groups receiving 3% DSS combined with varying doses of neutralizing antibodies against TNFα, integrin α4β7, and IL-23p19. Treated mice received 3% DSS *ad libitum* for 5 days and clean water for another 3 additional days (5+3 model). Neutralizing antibodies were injected intraperitoneally on days 2 and 5 of the experiment. Doses used for optimization were set for each dose at 100, 250 and 500 µg/mouse for anti-integrin α4β7 and anti-IL23p19 antibodies, and at 10, 25 and 50 µg/mouse for anti-TNFα antibody for covering ranges found in bibliography. Weight loss, fecal consistency and fecal occult blood parameters were measured every day of experiment to elaborate a disease activity index (DAI). Mice were sacrificed at day 8 and the colon was obtained for histological evaluation by H/E staining, and for extraction of lamina propria mononuclear cells (LPMCs). Longitude was also measured. Blood was obtained by intracardiac punction and peripheral blood mononuclear cells (PBMCs) were obtained. Distal and proximal colon sections were fixed in 4% paraformaldehyde and embedded in paraffin. 4 mm sections were hydrated and H/E stained for histological score assessment. Extracted LPMCs and PBMCs were stained for flow cytometry evaluation of inflammation markers.

Results: Proximal and distal colon of mice treated with two doses of 10 µg/mouse of anti-TNFα, 250 µg/mouse of anti-integrin α4β7 and 100 mg/mouse of anti-IL23p19 showed the greatest tissue conservation and repair. In addition, these groups had less weight loss throughout the experiment as well as a reduced DAI. LPMCs and PBMCs inflammatory markers were also decreased in these groups in comparison with the others. Slight differences were found in 250 and 500 µg/mouse anti-integrin α4β7 treated mice.

Conclusion: The optimal treatment doses of neutralizing anti-TNFα, anti-integrin α4β7 and anti-IL23p19 antibodies are set at two doses of 10, 250 and 100 µg/mouse respectively for this 5+3 model of DSS-induced colitis. By establishing precise dosimetry baselines, our work provides a robust framework for combinatorial therapies, which could unlock novel approaches for IBD treatment.

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PP0393

ASSOCIATION OF DEPRESSIVE SYMPTOMS WITH THE LEVELS OF PERIPHERAL BLOOD BIOMARKERS IN PEOPLE WITH INFLAMMATORY BOWEL DISEASES

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Introduction: Inflammatory bowel diseases (IBD), encompassing ulcerative colitis (UC) and Crohn's disease (CD), are chronic gastrointestinal disorders accompanied by psychiatric comorbidities, particularly anxiety and depression. The link between IBD and anxiety and depression remains unclear, but it appears that, among others, it may be explained by altered tryptophan metabolism. Tryptophan, an essential amino acid, is crucial for maintaining mental health and regulating mood through its metabolic processes. It serves as a precursor for several biologically significant compounds, including serotonin, melatonin, and kynurenine metabolites. The interplay between inflammation, tryptophan metabolism, and mental health in IBD remains insufficiently explored.

Aims & Methods: This study aimed to assess the association between metabolites of the tryptophan-kynurenine pathway and depressive symptoms in IBD regarding the disease activity and phenotype. A cohort of 214 patients with IBD (70 with CD and 144 with UC) was included. Disease activity was determined using the Mayo score for UC and SES-CD for CD. Peripheral blood samples were analyzed for selected biomarkers, including metabolites of tryptophan-kynurenine pathway as well as indicators of oxidative stress and inflammation. Depressive symptoms were evaluated using the Beck Depression Inventory (BDI). Statistical analyses included correlation and regression models to explore the relationship between biomarkers and BDI scores, adjusting for clinical and demographic covariates.

Results: The BDI score in the study group ranged from 0 to 52. 31 individuals met the criteria for depression by scoring over 10 points. 137 patients with active disease exhibited significantly higher levels of hsCRP and nitrotyrosine and lower kynurenic acid levels compared to 77 with inactive disease. Kynurenic acid levels were negatively correlated with BDI scores ($r = -0.220$, $p = 0.042$), suggesting an association between tryptophan metabolism and depressive symptoms.

	Active disease	Inactive disease	p
Chlorotyrosine, nM	0.35 ± 0.41	0.26 ± 0.45	0.222
Nitrotyrosine, nM	0.38 ± 0.20	0.26 ± 0.17	0.043
Bromotyrosine, nM	0.05 ± 0.06	0.03 ± 0.03	0.553
L-tryptophan, M	54.67 ± 15.79	60.98 ± 15.34	0.117
L-kynurenine, M	3.53 ± 1.22	3.48 ± 1.01	0.680
5-hydroxy-L-trypthophan, M	0.20 ± 0.15	0.23 ± 0.13	0.233
3-indolepropionic acid, M	0.33 ± 0.57	0.20 ± 0.21	0.723
3-indoleacyrylic acid, M	2.31 ± 0.66	2.46 ± 0.54	0.322
Kynurenic acid, M	0.06 ± 0.04	0.08 ± 0.03	0.035

Table 1. Comparison of selected metabolite levels between individuals with active disease and with inactive disease.
Data expressed as n (%) or mean ± SD.
Significant between-group differences (p < 0.05) are marked in bold.

Additionally, CD patients showed higher serotonin and anthranilic acid levels compared to UC patients (16.67 ± 26.23 vs 6.29 ± 12.55 and 0.35 ± 0.26 vs 0.17 ± 0.17, respectively), though no significant differences in

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BDI scores were observed between the groups. Biological treatment was more frequent in the active disease group and associated with higher BDI scores, even after adjusting for covariates.

Conclusion: The study revealed significant differences in the tryptophan-kynurenine metabolic pathway among patients with inflammatory bowel disease, which vary based on disease activity. Additionally, variations in the concentrations of certain metabolites are correlated with the severity of depression.

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Disclosure: Nothing to disclose.

PP0394

PREDICTING INTESTINAL FAILURE IN SHORT BOWEL SYNDROME: THE EMERGING ROLE OF CIRCULATING CYTOKINES AND CHEMOKINES

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Introduction: Short bowel syndrome (SBS) is a severe complication of extensive small intestinal resection, most commonly occurring in Crohn's disease (CD), a chronic inflammatory disorder defined by transmural immune infiltration. Cytokines and chemokines are central mediators in the immune cascade driving intestinal inflammation and tissue damage[1,2].

Despite advances in care, no biomarkers currently predict SBS onset or progression to intestinal failure (IF). Profiling systemic cytokine and chemokine signatures may thus offer novel insight into disease trajectory and risk stratification.

Aims & Methods: This study investigates circulating inflammatory mediators in patients with established IF/SBS and CD patients at high risk of progression. A total of 150 patients with CD were enrolled and stratified into five groups based on intestinal function and surgical history: A1, SBS with IF requiring parenteral nutrition (PN) or intravenous fluids; A3, SBS without IF and inactive disease; B, high risk of IF with multiple or extensive resections; C1, low risk for IF with no history of resection; and C2, low risk for IF with prior resection. Serum cytokine and chemokine profiles were quantified at baseline (T0) and after six months (T6) using the human 27-PLEX multiplex immunoassay.

Results: Serum cytokines and chemokines profiling at baseline (T0) and after six months (T6) revealed distinct inflammatory patterns across the study groups. Patients with SBS and IF (group A1) showed persistently elevated levels of IP-10, IL-12, MIP-1α, and G-CSF, reflecting sustained immune activation. Whereas, patients at high risk of IF (group B) displayed increased Eotaxin, bFGF, IL-15, and IL-6, indicative of subclinical inflammation and possible tissue remodelling.

A general reduction in cytokine levels was observed at T6, though anti-inflammatory mediators remained minimally modulated, suggesting ongoing low-grade inflammation. Comparisons between groups highlighted divergent cytokine signatures correlating with disease severity and nutritional support requirements.

Conclusion: These findings suggest that specific circulating cytokines may serve as potential biomarkers for early identification of patients at risk of IF. The differential expression patterns observed across groups support their relevance in stratifying disease severity in CD and SBS. Further large-scale studies are needed to validate these preliminary results and to fully establish their prognostic significance.

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PP0395

RISING BURDEN OF INFLAMMATORY BOWEL DISEASE IN PUERTO RICO: A CROSS-SECTIONAL ANALYSIS OF INSURANCE CLAIMS FROM 2018-2022

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Introduction: The prevalence of IBD is rising in racial and ethnic minorities and in countries with previous low prevalence. There are limited studies of this disease in Latin America and the Caribbean.

Aims & Methods: This cross-sectional, retrospective study used de-identified insurance claims from commercial health insurers in PR. Cases identified had ICD-10 codes for IBD and at least one of the following criteria: 2 hospitalizations for IBD, 4 IBD-related physician visits, one IBD drug claim plus one of the former, or 2 ambulatory center IBD claims plus one of the former .Variables include age, gender, insurance type (Medicare Advantage or commercial insurance), IBD diagnosis, IBD-related physician visits, endoscopies, emergency room visits, hospitalizations and surgery. Point prevalence was defined as a new and pre-existing case during a 12-month period over the average number of insured population for each year of the study period. Estimates are presented as the number of cases per 100,000 insured individuals with 95% confidence intervals (CI) using Poisson regression. The study was approved by the MSC-IRB. We aim to estimate the point prevalence of IBD in an insured population in Puerto Rico (PR) from 2018 to 2022.

Results: The average insured population for the study period was 1,764,543. IBD was identified in 6,293 individuals; 35.6% had Crohn's disease (CD), 61% had ulcerative colitis (UC), and 3.4% were undetermined IBD. The overall prevalence of IBD was 356.6/100,000; prevalence was 127.1/100,000 for CD, and 217.5/100,000 for UC. Yearly prevalence rates showed an upward trend from 2018 to 2021 and a slight decrease in 2022. In 2018, the prevalence was 147.0/100,000 (95% CI: 145.33-148.68), increasing to 175.4/100,000 (95% CI: 173.45-177.35) in 2019 and 188.8/100,000 (95% CI: 186.77-190.89) in 2020. The peak prevalence occurred in 2021 at 215.0/100,000 (95% CI: 212.8-217.23), followed by a decrease to 199.9/100,000 (95% CI: 197.83-201.93) in 2022.

Conclusion: The increase in IBD prevalence in PR from 2018 to 2021, peaking at 215.0/100,000, aligns with previous studies suggesting an increasing disease burden on the island. An analysis for 2013 reported 181.5/100,000 across all insurance types including Medicaid. The peak in 2021 may reflect improved diagnosis, increased disease awareness, or better access following pandemic-related healthcare disruptions. The slight decline in 2022 prevalence may reflect a shift in insurance status, immigration, or a decrease in healthcare access. Continued monitoring of IBD prevalence is essential to track long-term trends, given the considerable rise since the early 2000s. Our findings place PR within the recent rising global trends of this disease and underscores the need for ongoing research and targeted healthcare strategies. The Medicaid insured population is being studied and should provide a more complete vision of IBD in PR.

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Disclosure: Nothing to disclose.

PP0396

PARENT PERSPECTIVES ON PEDIATRIC INFLAMMATORY BOWEL DISEASE IN A DEVELOPING NATION: DELAYS IN DIAGNOSIS, TREATMENT RESPONSE, AND PHYSICIAN AWARENESS

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Introduction: Pediatric Inflammatory bowel disease (PIBD) contributes to one-fourth of the total IBD burden. PIBD is rising in developing countries, and it is imperative to diagnose and treat it early due to rapid progression, associated morbidity, and improved quality of life after treatment. Diagnosis is often delayed due to nonspecific symptoms (specifically in Crohn's disease) early in the course of the disease, and less awareness about PIBD amongst primary caregivers and society in developing nations.

Aims & Methods: We aimed to learn from parents and patients suffering from IBD about their real-life experiences during the early disease course, diagnosis, response to treatment, and difficulties associated with it. We included all the confirmed PIBD (disease onset from 0 to 18 years of age) patients under treatment for over three months. The diagnosis was based on standard clinical, endoscopic, imaging, and histopathology criteria. A questionnaire was provided, which was filled out by parents. Symptoms, age at disease onset, diagnosis, any specific reason for delay in diagnosis, and response after treatment were assessed. We also tried to learn from the parents about their assessment of primary pediatricians and dentists about PIBD, as per their experience. Informed consent was taken from all the participants.

Results: Of the total 32 PIBD patients, 25 (Male: 56%) patients who responded to all the questions were included in the study. The mean age at onset of symptoms was 97.6±53.9 (range 10-204) months, the age at diagnosis was 113.7±55.9 (range 16-206) months, and the average duration from onset to diagnosis was 15.8±13.4 (2-46) months. Anaemia, growth failure, abdominal pain, and blood in stool were present in 84%, 80%, 88%, and 56% of patients, respectively. In 72% of patients, the diagnosis was made ≥6 months after the onset of symptoms. 68% of parents felt that delayed diagnosis was mainly due to the unawareness (about IBD) of health care providers. Only 20% (N=5) of patients visited the dentist, and 80% (n=4 of 5) suggested that their dentist was not confident about correlating their oral problem with IBD. Similarly, 72% (n=18) of patients concluded that even after the diagnosis of IBD, their primary healthcare providers are not confident in managing them. In major concerns, 80% of parents felt that there is a poor understanding of the disease in healthcare providers, and 12% (all on biologicals) thought that the cost of treatment is high. Most patients (80%) found it easy to get an appointment and procedures done under a pediatric gastroenterologist.

Age (months)	138.16±60.7 (range 23-252)
Diagnosis (Crohn's/UC/IBD-U, %)	60/40/0
Follow-up (months)	24.4±17.3 (range 3-60)
Biologicals	12%(n=3), all on infliximab
Improvement (clinical, biochemical, and endoscopic)	88% (n=22)

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Conclusion: The patient remains the centre of any health care system. This study evaluated parental experiences in managing PIBD, a chronic, progressive condition with increasing incidence and recognition in developing nations. Early diagnosis is the key to altering the disease progression. Patient-reported data indicate a significant deficiency in knowledge and awareness of IBD among healthcare providers, which appears to be a major contributor to diagnostic delays.

These findings underscore the urgent need for targeted educational interventions aimed at healthcare professionals, as well as public health initiatives to raise community awareness. Further epidemiological studies are required to understand the true prevalence and understanding of PIBD in developing countries.

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Disclosure: Nothing to disclose.

PP0397

EFFECTIVENESS OF SMOKING CESSATION INTERVENTIONS FOR SMOKERS WITH CROHN'S DISEASE: A SYSTEMATIC REVIEW

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Introduction: Smoking is the most influential modifiable environmental factor associated with adverse outcomes in Crohn's disease (CD). While smoking cessation (SC) has been shown to mitigate disease activity and surgical recurrence, the actual effectiveness of SC interventions tailored to CD patients remains underexplored.

Aims & Methods: To systematically evaluate the effectiveness of SC interventions in patients with CD and assess their real-world applicability and outcomes.

Following PRISMA guidelines, a systematic search of PubMed, Google Scholar, and Cochrane Library was conducted up to February 2023. Eligible studies included randomized controlled trials (RCTs) and cohort studies assessing SC interventions in adults with confirmed CD. Five studies, including two RCTs and three cohort studies, met the inclusion criteria.

Results: The reviewed interventions ranged from pharmacological treatments (nicotine replacement therapy, bupropion, varenicline) to behavioral and motivational approaches (individual and group counseling, motivational interviewing, Quitline referrals), or a combination of both. SC rates varied widely (14.8–42%), with four studies reporting ≤25% abstinence. The highest SC rate was observed in a pharmacist-led program incorporating behavioral and pharmacological support. However, RCTs showed no statistically significant difference in quit rates between intervention and control groups. Only two studies validated abstinence biochemically. High variability in study design, definitions of smoking/abstinence, intervention modalities, and sample sizes limited direct comparisons. Adherence rates were not systematically assessed, and the influence of socioeconomic factors remained understudied.

Furthermore, the effectiveness of phone-based interventions appeared non-inferior to in-person strategies, highlighting potential avenues for resource-limited settings.

Conclusion: Despite the well-established deleterious impact of smoking on CD prognosis, the evidence base supporting SC interventions in this population remains limited and inconclusive. Current strategies appear less effective in CD patients compared to the general population, underscoring the need for tailored, multifaceted approaches.

Future research should focus on standardized outcome measures, integration of psychological support, real-time adherence monitoring, and equitable access to pharmacological therapies.

Disclosure: Nothing to disclose.

PP0398

MANAGING APPENDICITIS WITHOUT SURGERY: POTENTIAL TO REDUCE THE RISK OF INFLAMMATORY BOWEL DISEASE

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Introduction: Previous studies have shown that appendectomy protects against the development of Ulcerative Colitis (UC), but the reason for this is not yet known. The risk of Crohn's disease (CD) and Microscopic Colitis (MC) after appendectomy has been studied to a lesser extent. In Finland, the correlation between appendectomy and inflammatory bowel disease (IBD) has not been previously studied.

Aims & Methods: We aim to investigate the risk of developing IBD after appendectomy in the Hospital district of Helsinki and Uusimaa, Finland. In this cohort study the patient material consisted of data from the Hospital district of Helsinki and Uusimaa's patient database from 2005-2020. The analysis included 8,969 patients, who underwent appendectomy, and 20,582 non-appendectomy controls matched by age and sex. The risk of IBD after appendectomy was analyzed by calculating the incidence-rate ratios (IRRs).

Results: In this study, we did not find a significant inverse correlation between appendectomy and UC (IRR 0.88, 95 % CI 0.31-2.15). However, we did find a significantly increased risk of both CD (IRR 7.04, 95 % CI 1.56-53.16) and MC (IRR 3.15, 95 % CI 1.16-8.97) after appendectomy. The increased risk of CD after appendectomy remained significant when the reason for appendectomy was defined as appendicitis and mesenteric lymphadenitis, but did not remain significant for other reasons.

Conclusion: The increased risk of Crohn's disease and Microscopic Colitis after appendectomy suggests that appendectomy as a treatment should be carefully considered if acute appendicitis is suspected. Alternative treatment for uncomplicated appendicitis may include antibiotics or symptomatic treatment.

Disclosure: Nothing to disclose.

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PP0399

PROFILING GUT–BRAIN AXIS PREDICTORS OF CHRONIC ABDOMINAL PAIN IN IBD REMISSION: A UK NATIONAL COHORT STUDY

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Introduction: Abdominal pain is a debilitating symptom in Inflammatory Bowel Disease (IBD). 20–50% of patients experience chronic abdominal pain (CAP), even during clinical remission. Emerging evidence suggests CAP in IBD is influenced by bidirectional gut-brain axis dysfunction, encompassing psychological and physiological factors. Identifying these factors is crucial to advancing targeted interventions and improving patient outcomes.

Aims & Methods: We aimed to delineate psychological and physiological determinants of CAP in patients with Ulcerative Colitis (UC) or Crohn's disease (CD) in remission. This cross-sectional study recruited 2525 adults through the UK NIHR IBD-BioResource (December 2022–April 2024). CAP was defined as per Rome IV criteria as recurrent abdominal pain ≥1 day/week in the last 3 months.

Remission was defined as Harvey-Bradshaw Index <5 for CD, and Simple Clinical Colitis Activity Index ≤2 for UC. All subjects completed validated instruments assessing somatic symptom severity (PHQ-15), gastrointestinal (GI) symptom burden (GSR), fatigue and sleep disturbance (PROMIS), anxiety (GAD-7), depression (PHQ-9), cognitive and behavioural responses to pain (CBRQ), personality traits (BFQ), sensory processing sensitivity (HSP-SF), resilience (CD-RISC) and quality of life (EQ-5D-5L). Logistic regression analyses identified independent predictors of CAP.

Results: Among 1,445 patients in remission (mean age 53.06±16 years; 48.3% female), 381 (26.4%) reported CAP. Of these 381, mean age was 49±14.63, 235 had CD (61.7%) and 217 were female (57%).

Patients with CAP had significantly higher mean scores for somatisation, GI symptoms, anxiety, depression, fatigue, sleep disturbance, quality of life, neuroticism, cognitive-behavioural responses and ease of excitation (all p<0.0001), with lower resilience scores (p<0.001) compared to those without CAP.

Multivariable logistic regression identified independent predictors of CAP; younger age, greater somatic symptom severity, symptom focusing, and residual GI symptoms of indigestion and diarrhoea were significant (Table 1).

GSRS abdominal pain was excluded as a predictor. The regression model demonstrated strong discriminatory ability (AUC=0.8260) and explained 31.4% of the variance in CAP.

Predictor	Odds Ratio (95% CI)	p-value
Age (per year increase)	0.97 (0.95–0.99)	0.0029
PHQ-15; somatic symptom severity	1.17 (1.09–1.26)	< 0.0001
CRBQ; symptom focusing	1.15 (1.05–1.26)	0.0025
GSRS indigestion	1.65 (1.24–2.21)	0.0006
GSRS diarrhoea	1.37 (1.11–1.69)	0.004

Table 1. Significant Multivariable Predictors of Chronic Abdominal Pain in IBD Remission

Abbreviations: PHQ-15: Patient Health Questionnaire-15, PHQ-9: Patient Health Questionnaire-9, GAD-7: Generalized Anxiety Disorder Assessment-7, PROMIS: Patient-Reported Outcomes Measurement Information System, PROMIS: Patient-Reported Outcomes Measurement Information System, HSP-SF: Highly Sensitive Person Scale – Short Form, BFQ: Big Five Personality Questionnaire, CD-RISC: Connor-Davidson Resilience Scale.

Conclusion: CAP affects approximately one quarter of IBD patients in clinical remission and is significantly associated with a range of psychological factors. Somatisation, cognitive-behavioural factors including symptom focusing and residual gastrointestinal symptom burden are independent predictors of CAP.

These findings highlight the key role of gut–brain axis dysfunction in CAP and support the need for targeted interventions such as cognitive-behavioural therapy or neuromodulation to alleviate pain and improve quality of life in this population.

Disclosure: Nothing to disclose.

PP0400

PHYSICAL ACTIVITY IS INVERSELY ASSOCIATED WITH CLINICAL EXACERBATION IN PATIENTS WITH CROHN'S DISEASE IN REMISSION - A PROSPECTIVE COHORT STUDY

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Introduction: Physical activity (PA) is associated with improved quality of life in patients with Crohn's disease (CD). However, there is limited evidence of its impact on the clinical course of the disease and the separate effects of aerobic and anaerobic-resistance PA.

Aims & Methods: We aimed to evaluate the association between PA and the incidence of relapse in patients with CD in clinical remission and to assess the role of PA in patients with CD using patient-reported outcome measures (PROMs), with separate analyses for PA type. A prospective cohort study of adult patients with CD in clinical remission followed for one year or until relapse according to the Harvey Bradshaw Index. At baseline, participants completed a detailed PA questionnaire, nutritional questionnaires, and a patient-reported outcomes measurement information system (PROMIS). Multivariable Cox proportional hazards models were used to assess the association between PA and relapse risk, adjusted for confounders, including dietary factors and fecal calprotectin levels.

Results: A total of 106 patients (mean age 37.9 ± 13.9 years) were included and followed for 9.9±3.2 months, during which 23 (21.7%) experienced clinical relapse. PA at least twice a week was associated with a reduced

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risk of relapse (HR = 0.26, 95% CI 0.09–0.79). Specifically, only aerobic activity was significantly associated with a reduced risk of relapse (HR = 0.31, 95% CI 0.10-0.95). High-resistance PA was significantly associated with better quality of life parameters, including pain interference (P=0.013), sleep disturbance (P=0.040), better physical function (P=0.029), and enhanced social roles (P=0.020). While aerobic PA showed benefit in physical function (P=0.016).

Conclusion: PA in general and specifically aerobic PA, is associated with a reduced risk of relapse in patients with CD in remission. Additionally, resistance PA is associated with a better quality of life. These findings highlight the importance of incorporating PA into the management of CD in remission state.

Disclosure: None.

PP0401

BRIDGING ENDOCRINE AND GUT AUTOIMMUNITY: FROM POPULATION TO PROTEOME IN GRAVES' DISEASE–ULCERATIVE COLITIS COMORBIDITY

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Introduction: Graves' disease (GD) and ulcerative colitis (UC) are two distinct autoimmune disorders with increasing evidence of shared pathophysiological mechanisms. While previous epidemiological studies have reported a higher risk of IBD among individuals with GD, the association between GD and UC specifically remains controversial. In particular, the causal direction, genetic liability, and molecular mediators underlying this relationship are not well defined.

Aims & Methods: We conducted a multi-layered investigation to clarify the relationship between GD and UC. First, a large-scale prospective analysis using the UK Biobank cohort was performed to assess the association between GD and subsequent UC risk, with adjustments for demographic, lifestyle, and clinical covariates utilizing COX proportional hazard models. Second, two-sample Mendelian randomization (MR) analyses were conducted using publicly available genome-wide association study (GWAS) datasets to infer potential causal links.

Finally, we performed protein mediation analysis integrating plasma proteomic data to identify potential intermediate proteins and pathways mediating the effect of GD on UC.

Results: Observational analysis revealed that patients with GD had a significantly elevated risk of developing UC. Stratified analysis further revealed that this association was more pronounced among male and obese individuals. The association remained significant even after adjusting for key covariates including medication use, psychological disorders, air pollution exposure, and sunlight exposure. Genetic-level analysis using Mendelian randomization confirmed a potential causal relationship between AITD and UC. This association was independently validated in MR sensitivity analyses. Through mediation analysis based on plasma proteomic data, 28 UC-related proteins were identified as potential mediators of the AITD–UC relationship. Among them, proteins such as Arachidonate 15-lipoxygenase B (ALOX15B) were significantly enriched in the high-risk group. Collectively, six pathways likely represent the core signaling axes through which GD promotes UC pathogenesis, highlighting potential targets for mechanistic and therapeutic exploration.

Conclusion: This study provides robust clinical, genetic, and molecular evidence for a link from GD to UC, mediated through specific immunopathogenic proteins and pathways. These findings offer novel insights into shared autoimmunity and may inform biomarker-driven strategies for early identification and targeted prevention of UC in patients with GD.

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Disclosure: The authors declare no conflicts of interest.

PP0402

A NEW TOOL TO ASSESS FRAILTY IN OUTPATIENTS WITH IBD: THE IBD FRAILTY SCORE

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Introduction: Patients with IBD are at greater risk than the general population of developing frailty, with the negative consequences that this condition determines, such as hospitalizations, post-operative complications, infections and mortality. Since there is no specific score to screen for frailty in IBD patients, studies in literature have used scores validated on cohorts of geriatric patients to date.

Aims & Methods: The aim of this prospective multicenter study is to validate a score specifically designed to detect frailty in IBD patients, the IBD frailty score. Enrolled outpatients were assessed with the IBD frailty score, the Fried frailty phenotype and the Hospital Frailty Risk Score (HFRS) for frailty, with the IBD Disability Index for disability and with the Charlson Comorbidity Index for comorbidities. Differences between qualitative and quantitative variables were evaluated with Chi-square and Mann Whitney U test. We detected test sensitivity and specificity with a ROC curve and to assess the internal consistency we used Spearman correlation coefficients.

Results: From April 2023 to October 2024, we assessed 121 IBD patients with the IBD frailty score at the Policlinico Tor Vergata of Rome. The score showed good test-retest reproducibility on a subgroup of 20 pa-

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tients (rho=0.78, p<0.0001), an acceptable mean time of administration (2 minutes and 19 seconds) and correlation with other frailty, disability and comorbidity indexes, with a good performance, AUC = 0.83 (95% CI: 0.75-0.90).

From September 2024 we started the multicenter study, enrolling 149 patients from Policlinico Tor Vergata and 200 patients from Policlinico of Magna Graecia (Catanzaro), with a total of 349 patients with IBD enrolled at this time. Enrollment is still ongoing at the Policlinico Campus Biomedico and the Policlinico Gemelli (Rome). The median age of the validation cohort of patients was 48 years (range: 19-84 years), 185 patients out of 349 (53%) were male, 204 (58.5%) had a diagnosis of UC and 114 (32.7%) had clinically active disease.

The IBD frailty score confirmed an acceptable mean time of administration (2 minutes and 3 seconds) and a positive correlation with the other frailty, disability and comorbidity indexes, except for the HFRS.

Patients with clinically active IBD showed significantly higher scores at the IBD frailty score than patients in remission. Finally, a good performance was confirmed with sensitivity and specificity of 75.8% and 76.6% respectively, with a cut off ≥3 to distinguish frail and not frail IBD patients.

Conclusion: The IBD frailty is the first score to detect frailty specifically designed for patients with IBD. It showed good reproducibility, acceptable time of administration and correlation with other frailty, disability and comorbidity indexes. Data will be validated at the end of the enrollment.

Disclosure: No conflict of interest to declare.

PP0403

COMPARISON OF THE SEVERITY OF THE CLINICAL MANIFESTATIONS OF PATIENTS WITH CROHN'S DISEASE BASED ON THE AGE OF DIAGNOSIS ACCORDING TO THE MONTREAL CLASSIFICATION

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Introduction: The aim of the study is to record and compare differences in the severity of the clinical manifestation of Crohn's disease in patients categorized by age at diagnosis according to Montreal classification.

Aims & Methods: The patients included in the study were followed up at the Gastroenterology Clinic of GNA Gennimatas. To assess the severity, the following parameters were recorded for each patient: number of biologic agents used, intensifications of therapy, and the number of surgeries.

The Montreal classification was used, categorizing patients into three groups based on age at diagnosis: A1 (<16), A2 (16-40), and A3 (>40). For the analysis of the data we used the SPSSv23 statistical software.

Results: A total of 465 patients participated in the study, 247 (53.1%) of whom were men, with an average follow-up duration of 14 years. A total of n=42 (9.04%) patients were in the A1 group, n=272 (58.49%) in the A2 group, and n=151 (32.47%) in the A3 group. No statistically significant difference was observed among the three groups in terms of the use of biologic agents (p=0.079).

However, only 9.8% (n=4) of A1 patients had not received a biologic agent, compared to 21.3% (n=58) in A2 and 29.3% (n=44) in A3. In contrast, there was a statistically significant difference regarding surgeries (p<0.001), with A3 patients having significantly fewer surgical interventions than A1 and A2 patients, while A1 patients were more likely to undergo more than one surgical intervention compared to A2 and A3.

Finally, regarding intensification of therapy with biological agents, a statistically significant difference was observed (p=0.05), with A1 patients being more likely to receive at least one intensification.

Conclusion: In summary, differences were observed among patients with Crohn's disease when categorized based on age at diagnosis according to Montreal classification. Specifically, it was found that A3 patients were less likely to undergo surgery.

	A1 (n=42)	A2 (n=272)	A3 (n=151)	Total (n=465)
Gender Male,n(%)	26(61.9)	141 (51,8)	78 (51,7)	247(53,1)
Use of Biologic agents 0, n(%)	4 (9.8)	58 (21.3)	44 (29.3)	
1, n(%)	24 (58.5)	135 (49.6)	58(38.7)	
2, n(%)	9 (22)	49 (18)	33 (22)	
3, n(%)	4 (9.8)	30 (11)	15 (10)	
Intensifications, n(%)	25 (67.6)	102 (47.7)	48(45.3)	
Surgeries, 0, n(%)	28 (66,7)	176(64,7)	125(82,8)	
1, n(%)	8 (19,1)	81(29,8)	22(14,5)	
2, n(%)	6 (14,3)	15(5,5)	4(2,7)	

Additionally, A1 patients showed a higher likelihood of having intensified their treatment and at the same time a lower likelihood of not having received any biological agent.

Disclosure: No disclosures.

PP0404

GLOBAL PREVALENCE OF DISABILITY IN INFLAMMATORY BOWEL DISEASE: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Disability is a multidimensional concept encompassing physical, psychological, and social limitations that can significantly burden individuals living with inflammatory bowel disease (IBD). However, as disability is shaped by various factors—including cultural, social, and healthcare-related influences that can differ between countries—its prevalence and characteristics may vary globally.

Aims & Methods: We aimed to meta-analyse global data on prevalence of moderate-to-severe disability (MSD) to explore differences in disability prevalence and characteristics by country, IBD subtype, disease activity. MEDLINE EMBASE, EMBASE Classic (from inception to 1st December 2024) to identify cross-sectional and case-control studies reporting the prevalence of MSD based on validated tools (IBD–Disk and IBD-Disability Index) in adult patients (≥18 years) with confirmed IBD. Data were extracted, and pooled prevalence (PP), odds ratios (OR), and 95% confidence intervals (CIs) were calculated.

Results: The search strategy generated 3089 citations. From these, 160 individual articles appeared to be relevant to the study question and 17 fulfilled the eligibility criteria, containing 8225 patients with IBD recruited from 11 different countries. The pooled prevalence of MSD in patients with IBD was 29.6% (95% CI 22.6-37.1%; I^2=98.0%, p<0.01), with no evidence of

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funnel plot asymmetry (Egger test, p=0.10). The lowest pooled prevalence of MSD was 7.0% in one Dutch study from the Netherlands, and the highest 51% and 52% in two studies from China and Italy, respectively. The pooled prevalence of MSD was higher in patients with IBD with active disease (56.9%, 95% CI 20.3-89.7%, I^2 = 93.0%, p<0.01) compared with those with inactive disease (27.0%, 95% CI 3.3-62.0%, I^2 = 93.0%, p<0.01). Based on three studies that reported MSD prevalence in both active and inactive disease, active IBD increased the odds of MSD more than threefold (odds ratio 3.13, 95% CI 1.74-5.64).

Conclusion: The global prevalence of MSD in the IBD population is 29.6%, and even higher among patients with active disease. These findings underscore the value of incorporating disability assessment into both clinical decision-making and routine therapeutic response —especially in the context of disease modification trials.

Disclosure: Nothing to disclose.

PP0405

PREVALENCE, INCIDENCE, MORTALITY AND TRENDS IN TREATMENT OF INFLAMMATORY BOWEL DISEASE IN CATALONIA BETWEEN 2017 AND 2023. AN EPIDEMIOLOGICAL UPDATE

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Introduction: Prevalence of inflammatory bowel disease (IBD) is increasing worldwide, while incidence is tending to stabilize. Moreover, the use of biological treatments is increasing, some studies suggest that surgeries and hospitalizations are decreasing instead.

Aims & Methods: A population-based, retrospective cohort study was conducted using data from the Catalan Health Surveillance System (CHSS). All patients diagnosed with IBD were included between 2017-2023. Crude incidence and prevalence rates were calculated for all the Catalan population. Data on pharmacological therapy, surgical procedures, hospitalizations, and mortality were analyzed. Trends in age-sex-adjusted rates were also estimated, and logistic regression was used to calculate the adjusted mortality odds ratio (OR). Data for Crohn's disease (CD) and ulcerative colitis (UC) were analyzed separately.

Results: The number of prevalent IBD cases rose from 28,752 in 2017 to 41,423 in 2023. Despite incidence rates remained stable (30.8 in 2017 and 29.9 per 100,000 inhabitants in 2023) prevalence rates increased (386.9 and 510.9 per 100,000 inhabitants respectively). The use of biologics significantly increased (from 13.5% in 2017 to 21.0% in 2023), particularly ustekinumab and vedolizumab. In parallel, a decline in the use of immunosuppressants were observed. IBD-related surgeries and hospitalizations decreased during study period, particularly among CD patients. Mortality remained low but was higher among IBD patients compared to the general population.

Conclusion: Incidence of IBD in Catalonia has stabilized, while its prevalence continues rising, suggesting a transition to Stage 3 (compounding prevalence). The use of biological treatments is increasing steadily, whereas rates of surgeries and hospitalizations are consistently decreasing.

Disclosure: None.

PP0406

CROSS-SECTIONAL ASSESSMENT OF BODY COMPOSITION AND MUSCLE FUNCTION IN CROHN'S DISEASE PATIENTS, STRATIFIED BY PRESENCE AND EXTENT OF INTESTINAL RESECTION

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Introduction: Crohn's disease (CD) is a leading cause of short bowel syndrome (SBS) in adults, which can result in intestinal failure (IF) when intravenous nutritional or electrolyte supplementation is required. SBS is frequently associated with sarcopenia, characterized by loss of muscle mass and function, which adversely impacts quality of life and increases the risk of complications.

This study aimed to evaluate differences in body composition and muscle strength among CD patients at varying risk levels for SBS.

Aims & Methods: We conducted a cross-sectional study involving consecutive CD outpatients. The participants were divided into two groups. Group A consisted of CD patients diagnosed with SBS and patients who had undergone two or more bowel resections, or a resection of more than 50 cm. Group B included patients who had never undergone surgery or those who had undergone a resection of less than 50 cm. Body composition was assessed using bioimpedance analysis (BIA), and muscle strength was measured via hand grip testing with a dynamometer, recording the best result of three attempts with the dominant hand. We administered food frequency questionnaire (FFQ) to assess eating habits. Statistical analyses, including t-tests or Mann-Whitney tests, were used as appropriate, with a significance threshold of $p < 0.05$.

Results: The study cohort included 150 patients (mean age 53 years [±19,2]; 24% female). Group A showed an average BMI of 23,7±4 kg/m^2, whereas group B showed an average BMI of 24.6±5 kg/m^2. Muscle strength measured between groups was reduced in group A showing significant dif-

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ferences in hand grip test compared to group B (p-value<0.001). However, no significant differences were observed in BIA parameters, including muscle mass, or in eating habits across the groups.

Conclusion: Our study revealed that SBS patients and CD patients resected two or more times or with a resection greater than 50 cm had reduced muscle strength without a corresponding reduction in muscle mass, indicating impaired muscle function. Further research is needed to better understand the relationship between Crohn's disease-related resections and their impact on both muscle quality and quantity.

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Disclosure: Nothing to disclose.

PP0407

RELATIONSHIP BETWEEN INFLAMMATORY BOWEL DISEASE (ULCERATIVE COLITIS & CROHN'S DISEASE), CEREBROVASCULAR DISEASE (CVD), AND CORONARY HEART DISEASE (CHD): A NATIONAL STUDY

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Introduction: Inflammatory bowel disease (IBD) is associated with an increased risk of cerebrovascular disease (CVD) and coronary heart disease (CHD). It has been suggested that chronic inflammation may be the mechanism. Previous studies have primarily assessed IBD as a group and not ulcerative colitis (UC) or Crohn's disease (CD) separately.

Aims & Methods: This study aims to determine the relationship between UC and CD and CVD and CHD risk factors/outcomes in a large UK population. General practice electronic medical records were sourced from the United Kingdom (UK) via The Health Improvement Network (THIN). UC and CD risk factors and disease outcomes were identified through Read codes.

Factors	UC Prevalence, n (%) n=3,432	UC Odds Ratio, 95% CI	CD Prevalence, n (%) n=2,831	CD Odds Ratio, 95% CI
Female Sex	1,796 (52.3%)	1.16, 1.09-1.25	1,689 (59.7%)	2.71, 2.08-3.62
Age	43.95 ± 14.4	0.99, 0.99-0.99	45.7 ± 15.7	0.95, 0.94-0.96
Diabetes Type II	417 (12.2%)	1.50, 1.35-1.66	265 (9.4%)	1.12, 0.98-1.27
Dyslipidemia	176 (5.1%)	1.26, 1.08-1.47	112 (4.0%)	0.96, 0.79-1.16
Obesity	7 (0.2%)	0.99, 0.47-2.10	7 (0.3%)	1.21, 0.58-2.55
AMI	113 (3.3%)	1.57, 1.30-1.89	57 (2.0%)	0.95, 0.73-1.23
Ischemic Stroke	47 (1.4%)	1.46, 1.09-1.95	23 (0.8%)	0.86, 0.57-1.30
Intracranial Hemorrhage	15 (0.4%)	1.67, 1.01-2.78	10 (0.4%)	1.35, 0.72-2.51
COPD	31 (0.9%)	1.85, 1.30-2.64	24 (0.9%)	1.74, 1.16-2.59

Results: A total of 1,062,022 unique adult patients were available for analysis. There were 3,432 UC (mean age: 43.95 years, SD=14.4 years; 52% female) and 2,831 CD cases (mean age: 40.9 years, SD=14.06 years; 60% female).

Ulcerative Colitis: Univariate analysis in Table 1. Multiple logistic regression (MLR) of CHD risk factors and outcomes (adjusted in all models for age, sex, diabetes, COPD, dyslipidaemia, obesity, and statins) revealed statistically significant independent CHD risk factors: female sex (OR: 1.19, 95%CI: 1.11-1.28), younger age (OR: 0.98, 95%CI: 0.98-0.99), type 2 diabetes (OR: 1.22, 95%CI: 1.09-1.37), and increased risk of acute myocardial infarction (AMI) (OR: 1.35, 95%CI: 1.11-1.64).

MLR of CVD risk factors and outcomes (adjusted as above plus thrombolytic agents) found statistically significant independent CVD risk factors: female sex (OR: 1.19, 95%CI: 1.10-1.27), younger age (OR: 0.99, 95%CI: 0.98-0.99), diabetes type 2 (OR: 1.20, 95%CI: 1.07-1.34), statin use (OR: 1.13, 95%CI: 1.03-1.25), and any anti-thrombolytic agents (OR: 1.37, 95%CI: 1.25-1.51).

Crohn's Disease: Univariate analysis in Table 1. MLR of CHD risk factors and outcomes (adjusted) revealed statistically significant independent CHD risk factors: female sex (OR: 0.92, 95%CI: 0.85-0.99), younger age (OR: 0.98, 95%CI: 0.97-0.99), statin use (OR: 1.23, 95%CI: 1.11-1.36), and an increased likelihood of death (OR: 1.28, 95%CI: 1.12-1.47).

MLR of CVD risk factors and outcomes (adjusted) revealed statistically significant independent CVD risk factors: female sex (OR: 0.91, 95%CI: 0.84-0.98), younger age (OR: 0.97, 95%CI: 0.96-0.98), anti-thrombolytic agents (OR: 1.71, 95%CI: 1.55-1.90), and an increased likelihood of death (OR: 1.18, 95%CI: 1.03-1.35).

Conclusion: UC cases are more likely to be female, have diabetes, be taking statins, and anti-thrombolytic agents, and have an increased risk of AMI and death. CD cases are less likely to be female, but more likely to be taking statins, anti-thrombolytic agents, and more likely to die.

Disclosure: Guy D. Eslick and Michael P. Jones have nothing to disclose. Nicholas J. Talley Disclosures: GutSee Ltd (consulting microbiome, 2025), Brown University, Agency for Health Care Research and Quality (fiber and laxation) (2024), Rome Foundation (member gastroduodenal committee) (present) Biocodex (FD diagnostic tool) (present) Microba (consulting microbiome - commencing 2025) Comvita Manuka Honey (FD trial consulting) (2025), BluMaiden outside the submitted work. In addition, Dr. Talley has a patent Nepean Dyspepsia Index (NDI) 1998, a patent Licensing Questionnaires Talley Bowel Disease Questionnaire licensed to Mayo/Talley, "Diagnostic marker for functional gastrointestinal disorders" Australian Provisional Patent Application 2021901692, "Methods and compositions for treating age-related neurodegenerative disease associated with dysbiosis" US Patent Application No. 63/537,725.

Boards: The Gut Foundation Board (2025-present), Asia Pacific Association of Medical Editors (APAME) Board (President 2023-2025), Doctors for the Environment (Chair 2024-present), Luminal Faculty GESA (Chair) (2023-2025)

Editorial: UptoDate (Section Editor), Mayo Clinic Proceeding Associate Editor (2023-present)

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PP0408

EPIDEMIOLOGY OF INFLAMMATORY BOWEL DISEASE IN SOUTHERN SPAIN. A POPULATION-BASED CONTROLLED STUDY ON NATIVE AND MIGRANT POPULATION BETWEEN 2012 AND 2022

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Introduction: This study aimed to assess and compare the incidence, prevalence, and severity of inflammatory bowel disease (IBD) between migrant and native populations in Granada, Spain, over the past decade. We also sought to identify potential epidemiological differences and produce hypotheses regarding factors influencing disease patterns.

Aims & Methods: A retrospective, multicenter observational study was conducted, including patients diagnosed with IBD from 2012 to 2022 in Granada. Data on demographics, disease type, severity, and clinical outcomes were collected and analysed. Migrant and native populations were compared using case-matching based on age, sex, and IBD type. Statistical analyses included logistic regression models to identify risk factors for severe disease.

Results: From 2012 to 2022, 2764 IBD patients were identified, and 144 were analysed: 111 Spaniards (77.08%) and 33 migrants (22.82%). Migrants, mostly from Africa (57.5%), were diagnosed younger than Spaniards (38.85 ± 12.16 vs. 47.67 ± 15.00 years; p = 0.003). Prevalence increased in Spaniards (295 per 100,000 in 2022) but remained stable in migrants (41 per 100,000 in 2022), based on point prevalence calculated at the end of the study period. Incidence was higher in Spaniards (6.49 vs. 1.32 per 10^5 population/year). Severe IBD affected 34.03% of patients, more often Spaniards (37.84%) than migrants (21.21%).

Comparative analysis between Moroccan and Hispanic American patients revealed no statistically significant differences in age (36.0 vs. 34.4 years; p = 0.750) or the number of colonoscopies performed (0.73 vs. 0.56; p = 0.596). Additionally, the distribution of disease phenotypes (Crohn's disease vs. ulcerative colitis) was comparable, as was the use of biological therapies.

Risk factors for severity included Crohn's disease (67.35% vs. 48.42%; p = 0.016) and a fistulizing phenotype (42.42% vs. 25.00%; p = 0.039). Patients with severe IBD were also more likely to have undergone ≥3 colonoscopies (26.5% vs. 6.3%; p = 0.001) and experienced ≥3 hospitalizations (32.7% vs. 5.3%; p = 0.001). Indicators of severity included undergoing ≥3 colonoscopies (OR: 7.016, 95% CI: 2.090–23.556, p = 0.002), experiencing ≥3 hospitalizations (OR: 7.749, 95% CI: 2.347–25.580, p = 0.001), and having Crohn's disease (OR: 2.361, 95% CI: 1.016–5.488, p = 0.046). Spaniard nationality was associated with an increased risk of severe disease (OR: 3.242, 95% CI: 0.101–0.941, p = 0.039).

Conclusion: This study provides valuable insights into the epidemiology of IBD in southern Spain. It revealed no significant increase in the prevalence or incidence of IBD compared to studies conducted in central and northern Europe, nor evidence that migrants experience more severe disease courses. These results highlight the potential protective role of universal healthcare access, regional dietary habits, and advanced therapeutic interventions in mitigating disparities in disease outcomes. Further longitudinal studies are warranted to validate these findings in larger, more diverse cohorts.

Disclosure: Nothing to disclose.

PP0409

INCREASED RISK OF AUTOIMMUNE LIVER DISEASE IN INFLAMMATORY BOWEL DISEASE: A NATIONWIDE DANISH REGISTRY STUDY

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Introduction: Inflammatory bowel disease (IBD), including ulcerative colitis (UC) and Crohn's disease (CD), is a chronic condition with extraintestinal manifestations such as primary sclerosing cholangitis (PSC)[1,2].

It is believed that individuals with IBD have an increased risk of other immune mediated diseases[3,4].

One key area of interest is the gut-liver axis, a bidirectional interaction between the gastrointestinal tract and the liver, which suggests that pathologic conditions of each system can influence each other[5].

In addition to PSC, individuals with IBD may also be predisposed to other autoimmune liver diseases, including autoimmune hepatitis (AIH) and primary biliary cholangitis (PBC). Investigating the relationship between IBD and autoimmune liver disease is critical for identifying individuals at higher risk of liver complications, enabling early detection and targeted interventions to mitigate long-term hepatic damage.

Aims & Methods: This study aims to investigate the risk of autoimmune liver disease among individuals with IBD compared to the general population. We conducted a nationwide, registry-based cohort study using Danish registries to identify individuals diagnosed with IBD and autoimmune liver disease. Diagnoses were based on the International Classification of Diseases, 10th Revision (ICD-10) and the Systematized Nomenclature of Medicine (SNOMED) codes.

All individuals diagnosed with UC or CD from 2000 to 2022 were identified and matched 1:10 with individuals without such diagnosis from the general population based on sex at birth, age and municipality of residence at the time of diagnosis. Matches were assigned their case's IBD diagnosis date as an index date.

Individuals were excluded if they had a diagnosis of hepatocellular carcinoma prior to autoimmune liver disease or had not resided in Denmark for at least five years.

Logistic regression was used to estimate the prevalence odds ratio for autoimmune liver disease among individuals with IBD compared to matched controls up to five years before IBD diagnosis. Standardized incidence rates and time-to-event analysis were used to estimate the risk of autoimmune liver disease, expressed as hazard ratios, after IBD diagnosis or index date, comparing individuals with IBD to matched controls.

All analyses were adjusted for sex, age, subtype of IBD and Charlson Comorbidity Index.

The confidence interval (CI) was 95%, and the significance threshold for the p-value was set at 0.05.

Results: 50,557 individuals with IBD and 505,570 matched controls were included. Autoimmune liver disease was diagnosed in 391 individuals before and in 999 individuals after the IBD diagnosis/index date.

The odds of having a prior diagnosis of autoimmune liver disease were 11.7 times higher (95% CI: 9.5-14.2) among individuals who subsequently received a diagnosis of IBD compared to those who did not. The risk of autoimmune liver disease appeared substantially higher among persons with an IBD diagnosis compared to the those without. The highest

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risk was observed for PSC, an extraintestinal manifestation of IBD, with an adjusted hazard ratio (aHR) of 45.3 (95% CI: 36.0-57.1). Increased risks were also found for PBC (aHR: 3.1, 95% CI: 2.3-4.0) and AIH (aHR: 4.7, 95% CI: 3.5-6.2) among individuals with IBD compared to those without IBD.

Conclusion: The odds of autoimmune liver disease before IBD diagnosis were increased compared to matched controls. Additionally, an increased risk of PSC, PBC and AIH following IBD diagnosis was observed. These results suggest the importance of monitoring individuals with IBD for autoimmune liver diseases.

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Disclosure: Stine Kjærsgaard Hansen, Anastasia Karachalia Sandri, Srishti Saha, Peter Holland-Fischer and Kirstine Petrea Bak-Fredslund report no conflicts of interest.

Tine Jess reports consultancy for Ferring and Pfizer.

Lone Larsen reports consultancy and/or speaker fee from Takeda and Eli Lilly and has served as an advisory board member for Tillotts and Abbvie.

PP0410

CHRONIC RHINOSINUSITIS AND INFLAMMATORY BOWEL DISEASE: DISCOVERING THE GUT-LUNG-AXIS PRELIMINARY RESULTS FROM A SINGLE-CENTER PILOT STUDY

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Introduction: Inflammatory Bowel Disease (IBD) are multifactorial and debilitating diseases that impact patients' quality of life. A key characteristic of IBD is immune system dysregulation, which contributes to a higher risk of developing additional chronic immune-related disorders, such as chronic rhinosinusitis (CRS). The coexistence of these two conditions can undoubtedly affect patients' life and disease history. This study aims to determine the prevalence of CRS in individuals with IBD and evaluate its impact on their quality of life.

Aims & Methods: We conducted a prospective, observational, single-center pilot study, enrolling consecutive IBD patients. Clinical, demographic, and disease history data were collected. The sino nasal outcome test 22 (SNOT-22) questionnaire was administrated to each patient to assess subjective symptoms of CRS.

Participants were stratified into two groups:

Group 1 (SNOT-22 < 30) indicating no or mild CRS and Group 2 (SNOT-22 > 30) indicating moderate to severe CRS.

Group 2 patients underwent a comprehensive otolaryngological assessment including laboratory testing and a fibrolaryngoscopy to confirm the diagnosis of chronic rhinosinusitis.

Results: We enrolled 100 IBD patients, 43 % were affected by Crohn's Disease and 57% were affected by ulcerative colitis (UC):

- CD patients stratified by disease location:
    1. Ileum: 14 patients (32%)
    2. Colonic: 8 patients (19%)
    3. Ileum-Colonic: 19 patients (44%)
    4. Upper GI: 2 patients (5%)
- RCU patients stratified by disease location:
    1. E1 (proctitis): 8 patients (14%)
    2. E2 (left sided colitis): 20 patients (35%)
    3. E3 (extensive colitis): 26 patients (46%)

Among them, 37% of patients were naive to biological therapy, while 13% had failed at least two lines of biological therapy. Based on the SNOT-22 score, patients were divided in:

- Group 1 with SNOT-22 < 30 (mild symptoms):
    1. CD (43): 26 patients (60%)
    2. RCU (57): 36 patients (63%)
- Group 2 with SNOT-22 > 30 (moderate to severe symptoms):
    1. CD (43): 17 patients (40%)
    1. RCU (57): 21 patients (37%)

Patients with Crohn's disease in Group 2 exhibited higher disease activity, as measured by the Harvey-Bradshaw Index (HBI), compared to those in Group 1 (mean HBI: 3.8 vs. 1.38; p = 0.004). In contrast, no significant difference in disease activity was observed among patients with ulcerative colitis between the two groups.

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At the time of analysis only 15 patients of Group 2 underwent otolaryngological and laboratory evaluation including IgE and eosinophil cationic protein (ECP) and 10 patients showed endoscopic signs of CRS (*See Table 1*).

	CRS (10)	NO CRS (5)	P-VALUE
CD	5	4	
RCU	5	1	
HBI (mean)	4,2	2	0.188
FULL-MAYO (mean)	4,6	3	0,23
ECP (µg/l)	45,1	24,7	0.123
IgE (UI/ml)	120	54.62	0.40

Table 1. *Clinical and biological parameters between CRS and non-CRS patients.*

Conclusion: CRS is frequent among patients with IBD. In line with prior reports, our study identified a CRS prevalence of 10% within IBD patients. Moreover, the higher prevalence of CRS in patients with more severe IBD activity may highlight a potential connection mediated by the gut–lung axis. No significant differences were observed in IgE levels or eosinophil cationic protein values among patients with CRS, suggesting that the pathogenesis of CRS in IBD patients may differ from patients with isolated CRS.
Nonetheless, these findings remain preliminary due to the lack of laboratory analyses and otolaryngological evaluations for additional patients. Comprehensive clinical assessments and advanced laboratory investigations are warranted to further elucidate the underlying mechanisms of the gut–lung axis in this context.
Disclosure: Nothing to disclose.

PP0411

NO SEASON FOR THE FLARE: A RETROSPECTIVE STUDY OF IBD EXACERBATIONS OVER 15 YEARS

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Introduction: Inflammatory bowel diseases (IBD), comprising two main disorders, Crohn's disease (CD) and ulcerative colitis (UC) are chronic conditions characterized by a fluctuating disease course. Symptoms vary depending on disease activity, localization, and behavior. The etiopathogenesis of IBD is complex and environmental factors play a key role in developing intestinal inflammation. We hypothesized that seasonal changes may influence the severity and frequency of exacerbations.
Aims & Methods: The goal of the study was to determine the impact of seasonal changes on the onset and clinical presentation of IBD. As the normal distribution was violated the Kruskall-Wallis test was utilized. Results are presented as median with interquartile range (IQR). p-values <0.05 were considered statistically significant.
We retrospectively assessed cases of IBD exacerbations requiring hospitalization in the Department of Digestive Tract Diseases, Medical University of Lodz, Lodz, Poland between 2010 and 2025. A total of 563 patients were analyzed (207 with CD and 365 with UC).
Additional data, such as age of onset, disease duration, and disease severity were extracted from the medical records where possible.
Results: The median age was 43 years (IQR: 35.00-55.00) for CD patients and 43.5 years (IQR: 34.50-62.50) for UC patients. Median age at diagnosis was 27 years (IQR: 20.00-41.00) for CD patients and 29 years (IQR: 22.00-45.00) for UC patients. Median disease duration was 5 years (IQR: 1.00-11.00) for CD and 3 years (IQR: 0.00-9.00) for UC. No statistically significant seasonal variation in exacerbation rates was observed for either CD or UC patients (p = 0.509). There were also no significant seasonal differences in age (CD: p = 0.866; UC: p = 0.340), age at diagnosis (CD: p = 0.576; UC: p = 0.571), disease duration (CD: p = 0.434; UC: p = 0.244), or disease severity (Total Mayo Score: p = 0.619; Partial Mayo Score: p = 0.790; Harvey-Bradshaw Index: p = 0.540).

	Median (IQR)			
	Spring	Summer	Autumn	Winter
Age of CD diagnosis	26.00 (20.00 – 42.00)	30.50 (19.75 – 38.50)	27.00 (21.50 – 39.50)	26.50 (19.00 – 39.00)
Age of UC diagnosis	29.50 (20.50 – 45.50)	28.00 (20.00 - 46.00)	32.00 (24.00 – 49.00)	29.00 (22.00 – 40.00)
CD duration	6.00 (2.00 – 14.25)	4.00 (1.00 – 9.50)	4.00 (1.00 – 9.50)	5.00 (1.00 – 10.00)
UC duration	3.00 (0.00 – 8.00)	4.50 (0.00 – 11.00)	1.00 (0.00 – 9.00)	2.00 (0.00 – 7.00)
TMS	10.00 (9.00 – 11.00)	10.00 (8.00 – 11.00)	10.00 (9.00 - 11.00)	10.00 (8.00 – 11.00)
PMS	7.00 (6.00 - 8.00)	7.00 (6.00 – 8.00)	7.50 (6.00 – 8.00)	7.00 (5.00 – 8.00)
HBI	12.00 (9.00 – 15.00)	11.00 (8.00 – 14.50)	10.00 (7.00 – 13.00)	11.00 (8.00 – 14.00)

Conclusion: Despite anecdotal claims of seasonal patterns in IBD exacerbations, our 15-year retrospective analysis found no significant seasonal variation in hospitalization rates or clinical parameters. These findings challenge the presumed role of seasonal factors in IBD flare-ups. Further studies are warranted to validate these results in broader populations, including exacerbation managed in outpatient clinics.
Disclosure: Nothing to disclose.

PP0412

ASSOCIATION STUDY BETWEEN POLYMORPHISMS IN GENES INVOLVED IN INFLAMMATION MECHANISMS AND PERIANAL CROHN'S DISEASE

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Introduction: Crohn's disease (CD) is an Inflammatory Bowel Disease (IBD) with a multifactorial etiology that is not yet fully understood. It affects the intestinal wall in a transmural and segmental manner, with a preferential location in the distal ileum and right colon. Approximately 20% of CD patients develop perianal involvement (pCD), where the most common manifestations are fistulas and abscesses, followed by fissures and strictures. Patients with pCD represent a more severe disease phenotype, characterized by a reduced quality of life. Dysregulation of cytokine activity may influence the onset and severity of CD symptoms: several studies in patients with chronic inflammatory disorders (e.g., neoplasms or asthma) have shown that polymorphisms in pro- and anti-inflammatory genes can affect clinical outcomes.
Given the prominent role of altered inflammatory processes in the pathogenesis and severity of CD, the cytokine-mediated inflammatory pathway is of particular importance for predicting the potential development of pCD in patients with CD.

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Aims & Methods: The aim of this study was to evaluate whether there is a significant association between the presence of a specific genotype of polymorphisms (in CD patients) in the cytokine pathway and an increased, or decreased, likelihood of developing pCD. The polymorphisms analyzed in this study were: TNF-α (-308 G>A), IL-6 (-174 G>C), IL-6 (-597 G>A), IL-10 (-1082 G>A), IL-10 (-3575 T>A), TGF-β (cod10 T>C), RAS (-4343 G>A), RAF-1 (-931 T>C), ERK (-966 T>C).

These polymorphisms were analyzed in the biological samples of 204 CD patients, 34 of whom had pCD, followed at the gastroenterology outpatient clinics of the AOU "Città della Salute e della Scienza di Torino," whose biological samples are stored in a biobank. For the statistical analysis, the significance of the association was evaluated using the chi-square test by calculating the p-value; in cases where the p-value was significant, logistic regression was performed to calculate the OR and CI 95%. Statistical significance was assumed for p-values < 0.05.

Results: Only a few studies in the literature have analyzed the correlation between cytokine polymorphisms and pCD, and among those, none have found a significant association between the polymorphisms analyzed here and the development of pCD.

In this study, we demonstrate a statistically significant association between the IL-6 (-597 G>A) polymorphism and the development of pCD: carrying the -GG genotype showed a protective effect (OR = 0.46, 95% CI = 0.22–0.995; 13 out of 111 patients [11.7%] with the -GG genotype developed pCD, compared with 20 out of 90 [22.2%] with the -GA and -AA genotypes) against the development of pCD in CD patients (p-value = 0.049).

This finding is consistent with some data in the literature that analyzed the correlation between the -A genotype of the IL-6 (-597 G>A) polymorphism and increased serum levels of IL-6 expression.

Conclusion: This study therefore demonstrates, for the first time, in CD patients, a significant correlation between the IL-6 (-597 G>A) polymorphism and the development of pCD, paving the way for further future studies that could ideally confirm this association in a larger cohort. Potential genetic sequencing at the time of CD diagnosis could allow for the prediction of the risk of developing pCD, suggesting a more aggressive therapeutic approach for these patients right from diagnosis.

Disclosure: Nothing to disclose.

PP0413

CHANGES IN GENE EXPRESSION IN MULTIPLEX IBD FAMILIES, CONSEQUENCES OF SHARED HOUSEHOLD: A MULTIPLEX IBD FAMILY STUDY

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Introduction: The highest risk factor of inflammatory bowel disease (IBD) is having a first-degree relative (FDR) with IBD. However, genetic variation accounts for only a small proportion of IBD heritability. Evidence from family and twin studies underscores the role of shared environment for the development of IBD. In a recent study based on two population-based IBD cohorts from South-Eastern Norway, thirty years apart, we reported a substantial increase in the rate of familial IBD (1).

Aims & Methods: This study investigates the potential underlying mechanisms for the increased rate of familial IBD in South-Eastern Norway.

We examine the association between shared environment and gene expression across IBD phenotypes [Crohn's disease (CD) and ulcerative colitis (UC)] and four comparison groups:

(I) multiplex IBD familial patients (≥3 FDRs with IBD),

(II) healthy relatives in multiplex IBD families,

(III) sporadic IBD patients, and;

(IV) symptomatic non-IBD controls.

Expression analyses focus on four candidate genes, FKBP5, BCL3, VMP1, and TRAIL, previously identified as differentially methylated in IBD compared to controls (2,3). Blood samples were collected from 15 multiplex IBD families in South-Eastern Norway, between April 2023 and January 2024. For comparison, blood samples from 50 sporadic IBD patients and 50 symptomatic controls (SC) were obtained from the IBSEN III cohort (4). RNA was extracted from PAXgene blood samples of the multiplex families at EpiGen, Akershus University Hospital, Norway, while RNA extractions from the sporadic IBD patients and SCs were performed at The Institute of Clinical Molecular Biology, Kiel, Germany. All cDNA synthesis were carried out at EpiGen. Gene expression was measured using RT-qPCR on cDNA with target-specific primers. Relative quantification was performed using GAPDH and ACTB as housekeeping genes.

Simple t-tests were conducted to compare the mean gene expression between the four comparison groups and across IBD phenotypes.

Results: The multiplex IBD family study included 62 individuals, 41 IBD patients, 25 patients with UC, 16 with CD, and 21 healthy family members, 13 FDR and 8 spouses. The results for the differences in mean fold changes between the four groups, including CD and UC for each of the four candidate genes, are presented in Table 1. The gene expression analysis revealed differences in mean fold changes between the four groups and across CD and UC.

The relative quantification of gene expressions for the four candidate genes revealed no significant differences in gene expression among the IBD patients vs healthy members of multiplex families.

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Significant differences were observed for all candidate genes among familial IBD vs sporadic IBD and between healthy multiplex family members vs symptomatic controls, except for BCL3 and TRAIL, respectively. (p<0.005).

	FamIBD vs Fam healthy		FamUC vs Fam healthy		FamCD vs Fam healthy		FamIBD vs sporadic IBD		FamUC vs sporadic UC		FamCD vs sporadic CD		Fam Healthy vs SC	
Gene	Diff 95% CI	p	Diff 95% CI	p	Diff 95% CI	p	Diff 95% CI	p	Diff 95% CI	p	Diff 95% CI	p	Diff 95% CI	p
FKBP5	0.13 -0.25 -0.5	0.501	0.07 0.33-0.46	0.727	0.19 -0.3 -0.68	0.433	-0.59 -0.83-- 0.35	<0.001	-0.73 -1.31-- 0.16	0.016	-0.57 -0.95-- 0.19	0.005	-0.55 -0.74-- 0.36	<0.001
BCL3	-0.14 -0.4 -0.12	0.286	-0.14 -0.4 -0.13	0,297	-0.14 -0.43- 0.16	0.355	0.1- 0.06- 0.25	0.225	0.2 -0.03- 0.44	0.087	0.13 -0.14- 0.39	0.33	0.38 0.06- 0.69	0.021
VMP1	-0.03 -0.25 -0.19	0.785	0.01 0.25 -0.26	0,951	-0.07 -0.32- 0.17	0.545	-0.3 -0.44- 0.16	<0.001	-0.25 -0.48-- 0.02	0.031	-0.48 -0,73 --0.23	0.001	-0.24 -0.41-- 0.07	0.007
TRAIL	0.03 -0.13 -0.19	0.716	0.02 -0.16 -0.2	0,821	0.04 -0.19- 0.27	0.724	-0.23 -0.36 --0.09	0.001	-0.21 -0.38 --0.04	0.018	-0.3 -0.55 --0.04	0.023	-0.04 -0.17- 0.09	0.578

Conclusion: Our results reveal associations between the expression levels of IBD candidate genes and shared family environment, suggesting that epigenetic mechanisms may contribute to the development of familial IBD. Further methylation studies are warranted to clarify the underlying mechanisms of changed gene expression.

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PP0414

THE IBD-ASSOCIATED L503F VARIANT OF ORGANIC CATION TRANSPORTER NOVEL (OCTN-)1 PREDICTS A BETTER RESPONSE TO ANTITNFALPHA IN PATIENTS WITH ULCERATIVE COLITIS: A SINGLE CENTER ANALYSIS

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Introduction: Ulcerative colitis (UC) is a chronic, invalidating condition with a multifactorial aetiology. Several medications are available[1]; however, few data exist on predictors of response to such therapies. Genetics plays a pivotal role in disease predisposition and prognosis. OCTN1, an organic cation transporter, has been investigated as a modulator of intestinal inflammation. Genetic polymorphisms of OCTN1 have been linked to an increased susceptibility to IBD, but their role in predicting treatment response remains unclear[2,3].

In mice with colitis, *Octn1 -/-* is associated with a less severe course of disease and a higher percentage of Treg in comparison to wild-type mice.[4]

Aims & Methods: AIM: To investigate the association between *Octn1* genotypes and clinical response to anti-TNFα treatment in patients with ulcerative colitis.

We conducted a retrospective, single-center observational study at Policlinico Universitario "A. Gemelli" IRCCS, Rome. We enrolled adult patients diagnosed with ulcerative colitis and candidates to start treatment with anti-TNFα agents (infliximab, adalimumab, or golimumab). Individuals with known antibodies against infliximab or adalimumab and patients with a documented allergic reaction to these medications were excluded.

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Clinical and endoscopic data were collected at baseline and at 12 months of therapy. Patients were genotyped for the Octn1 SNP 1050152 (c1672t), and classified as wild-type (503L), heterozygous, or homozygous mutant (503F). 1 year therapy persistence, endoscopic response (defined as an improvement in endoscopic Mayo score) sustained clinical improvement in the absence of corticosteroid use over the previous 12 weeks) were assessed as clinical endpoints

Results: Of the 105 patients enrolled, 22 were homozygous for the 503F variant, 49 were heterozygous , and 34 presented two wild-type alleles. No statistically significant differences were observed between the patient groups with regard to gender, age at diagnosis, age and Mayo Endoscopic Score at initiation of therapy, duration of disease, or extent of diagnosis.*Octn1* mutated patients (503F) patients displayed greater rates of steroid-free clinical remission (OR=4.86, 95% CI 1.45 - 16.3; p= 0.010) and endoscopic improvement (OR= 6.52, 95% CI 1.93- 22.0; p = 0.003) after one year of therapy. Both the results were confirmed at the multivariable analysis (OR= 4.82 for steroid free clinical remission and OR= 7.94 for endoscopic response). Statistical significance was found for the mere therapy persistence (OR= 16.58, p = 0.009).

	N=105	CC (n=34)	CT (n=49)	TT (n=22)	p trend
Age (years.) at enrollment	40 (25-49)	41 (30-52)	40 (23-51)	36.5 (23-44)	0.536
Age at diagnosis (years.)	31 (20-44)	32 (20-46)	31 (20-42)	32 (19-41)	0.771
Sex (male / female)	57 (54.3) / 48 (45.7)	20 (58.8) / 14 (41.2)	25 (51.0) / 24 (49.0)	12 (54.5) / 10 (45.5)	0.738
Disease duration (years.)	4 (1-8)	4.5 (2-9)	4 (2-8)	4 (1-7)	0.768
Disease extension at diagnosis (Left sided colitis / Pancolitis)	44 (44) / 56 (56)	16 (48.5) / 17 (51.5)	20 (44.4) / 25 (55.6)	8 (36.4) / 14 (63.6)	0.680
1 year Therapy Persistence (yes)	74 (70.5)	19 (55.9)	34 (69.4)	21 (95.5)	0.004
1 year Steroid Free Remission (yes)	52 (49.5)	14 (41.2)	21 (42.9)	17 (77.3)	0.014
Endoscopic Response (yes)	51 (50.5)	9 (29.0)	26 (54.2)	16 (72.7)	0.006

Conclusion: Homozygosity for the rs1050152 c/t *Octn1* polymorphism appears to predict a more favorable clinical response to anti-TNFα therapy in ulcerative colitis, as evidenced by improved endoscopic outcomes and reduced need for corticosteroids after one year.

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PP0415

INVESTIGATION AND VALIDATION OF EXTRACELLULAR MATRIX PROTEINS AS DIAGNOSTIC BIOMARKERS IN INFLAMMATORY BOWEL DISEASE

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Introduction: The inflammatory bowel disease (IBD) subgroup classification of Crohn's disease (CD) and ulcerative colitis (UC) is based on clinical, endoscopic, histological and radiological findings. A non-invasive serological tool to aid the diagnosis of IBD is currently lacking. Extracellular matrix (ECM) proteins have been proposed as promising biomarkers of inflammatory activity, mucosal damage, and fibrosis development in IBD and may have diagnostic utility.

Aims & Methods: We aimed to investigate the diagnostic accuracy of 13 serum ECM protein epitopes representing inflammation and collagen formation and degradation. Serum from 518 newly diagnosed and treat-

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ment-naïve adult IBD patients (180 CD, 338 UC) and 205 symptomatic controls (SC) from the IBSEN III cohort (discovery cohort) were collected at diagnosis. The serum was analysed for a panel of ECM protein epitopes (C4M, C6M, C7M, PROC3, PROC4, PROC5, PROC6, PROC11, PROC16, PROC22, VICM, CTXIII, and CPa9HNE) using enzyme-linked immunosorbent assay (ELISA).

We used log2-transformed protein concentrations to fit logistic models for IBD vs SC, CD vs SC, UC vs SC, and CD vs UC. We fitted models for each individual protein and for all 13 proteins combined. Backward regression with the Bayesian information criterion, starting from the full model with 13 proteins, identified models with three to four proteins.

The models fitted to the discovery cohort were validated in the independent NORDTREAT cohort of 130 newly diagnosed IBD patients (56 CD, 74 UC) and 103 SC using the area under the curve (AUC) of the receiver operating characteristic (ROC) curve.

Results: The coefficients of models fitted to the discovery cohort identified C6M, C7M, C4M, PRO-C3, and PRO-C6 as the most important proteins for separating disease from controls. In the validation cohort, the combination of C6M, C7M and PRO-C3 reached the highest AUC of 0.65 for distinguishing IBD from SC. When fitting the model to differentiate CD from SC, the collagen VI epitope C6M was superior, reaching an AUC of 0.65 and increasing to 0.70 when combined with C7M, PRO-C3, and PRO-C6. A model with all 13 markers gave an AUC of 0.71 for CD vs SC. Only C4M showed capacity of differentiating between UC and CD with

an AUC 0.66. Lastly, differentiation of UC from SC using only C6M had an AUC of 0.62, increasing to 0.64 when adding C7M and PRO-C3 to the model.

Conclusion: This study revealed a moderate diagnostic accuracy of predicting IBD versus SC. However, the ability to separate patients with CD from SC was promising, motivating further exploration of biomarker utility for CD subgroup distinctions.

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PP0416

SERUM LEVELS OF ANTI-TNF ALPHA AGENTS AND THEIR CORRELATION WITH TRANSMURAL HEALING ASSESSED BY MAGNETIC RESONANCE ENTEROGRAPHY AND ENDOSCOPIC HEALING IN PATIENTS WITH CROHN'S DISEASE

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Introduction: In recent years, magnetic resonance enterography (MRE) has emerged as a novel tool for assessing disease activity in patients with Crohn's disease (CD). It is well known that anti-TNF agent trough levels (TLs) are associated with endoscopic healing (EH) in CD. However, limited data exist regarding the correlation between transmural healing (TH) and TLs in CD.

Aims & Methods: This study aimed to evaluate the correlation between anti-TNF TLs and TH, as well as mucosal healing (MH). We conducted a single-center cross-sectional study including CD patients on maintenance therapy with infliximab (IFX) or adalimumab (ADA) for at least one year. All patients underwent ileocolonoscopy and MRE at baseline and between one and two years after initiating anti-TNF therapy. TH was assessed using the simplified Magnetic Resonance Index of Activity (MARIAs) and was defined as the absence of inflammatory signs in the terminal ileum and all colonic segments. IFX and ADA TLs were measured using the Quantum Blue® assay (BÜHLMANN, Schönenbuch, Switzerland). EH was defined as a Simple Endoscopic Score for Crohn's Disease (SES-CD) of <3 points on colonoscopy. The correlation between IFX or ADA TLs, TH, and MH was analyzed using receiver operating characteristic (ROC) curve analysis.

Results: A total of 52 consecutive patients with active CD were included (55.8% male; median age, 37.04±15.49 years). Among them, 37 (71.15%) were on IFX and 15 (28.85%) on ADA. The mean TL values were 7 mg/dL for IFX and 15 mg/dL for ADA. An improvement in the MARIAs score after initiating anti-TNF therapy was observed in 54% of cases (p=0.02), regardless of the biologic agent used. Regarding SES-CD, an average decrease of 85% for ADA (p<0.01) and 82% for IFX (p<0.01) was observed.

Mean TL values >2.25 mg/dL for IFX and >14.45 mg/dL for ADA were associated with TH according to the MARIAs score; however, the correlation was weak [AUC: 0.6 (0.43–0.77)]

Conclusion: Satisfactory trough levels of IFX and ADA were associated with a significant decrease in MARIAs and SES-CD scores. However, no clear correlation between TLs and TH was demonstrated. Further studies investigating long-term TH and proactive dosing strategies based on therapeutic drug monitoring may help clarify whether an association between anti-TNF TLs and TH exists.

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PP0417

FECAL HOST TRANSCRIPTOMICS REFLECTS ENDOSCOPIC INFLAMMATION STATUS IN IBD

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Introduction: The intestinal mucosa undergoes constant renewal, with epithelial and immune cells routinely shed into the gut lumen. Recent findings suggest that these shed cells remain viable and retain transcriptomic signatures relevant to bowel pathology. Analysis of luminal content obtained during colonoscopy indicates that Fecal Host Transcriptomes (FHT) mirror histologic inflammation in IBD.

This study aimed to evaluate whether sequencing of stool samples can noninvasively reflect the presence of intestinal inflammation in patients with IBD, predict the location of inflammation within the bowel and distinguish between IBD type; Crohn's Disease and Ulcerative Colitis.

Aims & Methods: Our aim was to investigate whether transcriptomic analysis of stool samples that are taken ambulatory or at home could reveal the inflammation signature of inflamed IBD patients.

Bulk RNA sequencing of stool samples from 71 IBD patients and healthy controls was used to generate FHTs. Each sample was paired with an endoscopic assessment and clinical data to capture a time-matched view of intestinal inflammation.

Three random forest classifiers were trained to:

1. Distinguish active inflammation from remission;
2. Predict inflammation location in Crohn's disease (colon vs. small intestine). and;
3. Classifier based on gene modules expression was used to differ between CD and UC in patients with active inflammation.

Results: Fecal RNA quality was high, with a median of 2,140 human genes detected per sample. The FHT test demonstrated robust performance across all tested applications. The first model, designed to distinguish active inflammation from remission (n = 71), achieved a sensitivity of 85% and a specificity of 92%. The second model, which predicted the location of inflammation in Crohn's disease patients (n = 21), achieved 100% sensitivity and 100% specificity. The third model, aimed at differentiating Crohn's Disease from Ulcerative Colitis (n = 35), reached a sensitivity of 83% and a specificity of 100%. These findings underscore the potential of FHTs as a non-invasive tool for comprehensive inflammation profiling in IBD.

Conclusion: Fecal shed cell transcriptomics offers a novel, non-invasive approach for assessing mucosal inflammation at the endoscopic level in IBD patients. This approach enables differentiation between disease activity states, anatomical localization of inflammation, and IBD subtype classification. Ongoing studies aim to evaluate its utility in grading inflammation severity and predicting therapeutic response.

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PP0418

RELIABILITY AND RESPONSIVENESS OF ENDOSCOPIC INDICES FOR ASSESSMENT OF POSTOPERATIVE RECURRENCE IN CROHN'S DISEASE IN THE PREVENT TRIAL

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Introduction: Accurate and timely endoscopic evaluation is critical for the management of postoperative Crohn's disease (CD). Although numerous established and novel indices exist, their relative operating properties and the optimal instrument for routine use in clinical and research settings are unknown.

Aims & Methods: We characterised the reliability and responsiveness of both historical and novel endoscopic indices using centrally assessed ileocolonoscopy videos from a randomised controlled trial comparing infliximab versus placebo for preventing postoperative CD (PREVENT; NCT01190839). Three blinded central readers independently assessed 70 videos from the PREVENT trial using the Rutgeerts and modified Rutgeerts scores, POCER index, REMIND score, Simple Endoscopic Score for CD (SES-CD), and the CD Endoscopic Index of Severity (CDEIS). Indices were scored separately for the neoterminal ileum, the anastomosis (including the blind loop of a side-to-side anastomosis), and the distal colonic segment. Intra- and inter-rater reliability was quantified using the intraclass cor-

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relation coefficient (ICC) and was interpreted using the Landis and Koch benchmarks.[1] Responsiveness was quantified using the win probability (WinP), defined as the probability of a patient receiving infliximab having a better endoscopic score than a patient receiving placebo, and interpreted using Cohen's benchmarks.[2]

Results: Inter-rater reliability was substantial for the Rutgeerts (ICC 0.74 [95% CI: 0.62-0.83]) and modified Rutgeerts scores (0.79 [0.68-0.86]), ileal REMIND score (0.79 [0.67-0.87]), SES-CD (0.79 [0.70-0.86]), and CDEIS (0.80 [0.72-0.86]). Inter-rater reliability was moderate for the POCER index (ICC 0.49 [0.26-0.64]) and fair for the anastomotic REMIND score (0.30 [0.14-0.47]). Intra-rater reliability was almost perfect for all indices (ICC≥0.81) (Table 1). The Rutgeerts (WinP 0.77 [95% CI: 0.65-0.86]) and modified Rutgeerts scores (WinP 0.81 [0.69-0.89]), ileal REMIND score (WinP 0.83 [0.72-0.91]), SES-CD (WinP 0.80 [0.66-0.89]), and CDEIS (WinP 0.75 [0.61-0.86]) demonstrated a large degree of responsiveness when treatment assignment was the criterion for change. The POCER index (WinP 0.54 [0.43-0.65]) and the anastomotic REMIND score (WinP 0.53 [0.42-0.64]) had a small degree of responsiveness. The reliability and responsiveness of all indices were consistently lower when assessed at the anastomosis or distal colon compared to the neoterminal ileum.

Index	Inter-rater ICC (95% CI)[a]	Intra-rater ICC (95% CI)[a]	WinP (95% CI)[b,c]	WinP (95% CI)[b,d]
Rutgeerts score	0.74 (0.62, 0.83)	0.90 (0.85, 0.94)	0.77 (0.65, 0.86)	0.72 (0.57, 0.83)
Modified Rutgeerts score	0.79 (0.68, 0.86)	0.92 (0.88, 0.96)	0.81 (0.69, 0.89)	0.68 (0.52, 0.81)
POCER index	0.49 (0.26, 0.64)	0.89 (0.78, 0.95)	0.54 (0.43, 0.65)	0.61 (0.46, 0.74)
REMIND - Ileal	0.79 (0.67, 0.87)	0.91 (0.85, 0.96)	0.83 (0.72, 0.91)	0.71 (0.55, 0.83)
REMIND - Anastomotic	0.30 (0.14, 0.47)	0.81 (0.71, 0.89)	0.53 (0.42, 0.64)	0.62 (0.47, 0.75)
SES-CD	0.79 (0.70, 0.86)	0.93 (0.87, 0.96)	0.80 (0.66, 0.89)	0.70 (0.54, 0.82)
CDEIS	0.80 (0.72, 0.86)	0.91 (0.86, 0.95)	0.75 (0.61, 0.86)	0.70 (0.56, 0.81)
Overall VAS	0.81 (0.72, 0.87)	0.90 (0.84, 0.94)	0.80 (0.67, 0.88)	0.71 (0.55, 0.83)

[a] Interpreted according to the Landis and Koch benchmarks[1]: poor = <0; slight = 0.0–0.20; fair = 0.21–0.40; moderate = 0.41–0.60; substantial = 0.61–0.80; and almost perfect = 0.81–1.00.
[b] WinPs of 0.56, 0.64, and 0.71 correspond to Cohen's effect sizes of 0.2, 0.5, and 0.8, which are the benchmarks for „small", „moderate", and „large" effects, respectively.[2]
[c] Using treatment assignment as the criterion for change.
[d] Using clinical recurrence as the criterion for change.
Abbreviations: WinP, Win probability; CDEIS, Crohn's Disease Endoscopic Index of Severity; CI, confidence interval; ICC, intraclass correlation coefficient; POCER, Postoperative Crohn's endoscopic recurrence; SES-CD, Simple Endoscopic Score for Crohn's Disease; VAS, visual analogue scale.

Table 1. Reliability and responsiveness of endoscopic scores

Conclusion: Existing endoscopic indices are reliable and responsive for assessing postoperative CD activity in the neoterminal ileum but are suboptimal for assessing the anastomosis or distal colon.

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PP0419

PERFORMANCE MEASURES FOR COLONOSCOPY IN INFLAMMATORY BOWEL DISEASES AT DIFFERENT LEVEL OF CARE

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Introduction: High-quality endoscopy in IBD is associated with better clinical outcomes, while quality measurement supports effective quality control. The European Society of Gastrointestinal Endoscopy recommends using nine performance measures, depending on the indication for endoscopic examination in IBD.

Aims & Methods: Our study aimed to compare colonoscopy quality between tertiary and secondary referral centers for IBD patients, based on endoscopic reports. In this cross-sectional study, colonoscopy reports of IBD patients were analyzed.

At our tertiary center, we consecutively enrolled patients over 18 years old with established IBD between July and October 2024, using their available colonoscopy reports. The secondary care group included patients referred to our tertiary center, with the last existing report from secondary care used for analysis, while the tertiary care group consisted of our patients whose most recent report was similarly used.

Nine key performance measures (indication, bowel preparation, photo documentation, ileal intubation, biopsy, endoscopic activity score, high-definition and chromoendoscopy use, and neoplasia detection) were evaluated based on colonoscopy reports.

Patients were stratified by procedure indication (diagnostic, activity assessment, or surveillance) as well. Descriptive statistics, Welch's, and Fisher's exact tests were used to compare groups, with p-values <0.05 considered significant.

Results: Overall, 207 colonoscopy reports were assessed, of which 68/207 were performed at secondary care level. A total of 78/207 patients were diagnosed with Crohn's disease, while unclassified diagnoses were noted only at secondary centers. Activity control (88% vs. 38%, p<0.001) and surveillance (8% vs. 0%, p=0.02) examinations were more common at the tertiary level. Bowel preparation was better at tertiary centers (Boston Bowel Preparation Scale [BBPS] 8.3±2.3 vs. 7.2±1.2; p=0.003), and use of BBPS was more frequent at higher levels of care (p<0.001).

Disease activity scores were more commonly used at the tertiary level during activity control examinations (51.2% vs. 19.2%, p=0.004), while ileal intubation was also more frequent at the tertiary level (53% vs. 28%; p<0.001). Procedure and withdrawal times did not differ between levels in diagnostic or activity control examinations. Neoplasia detection was low, and biopsy rates were similar.

Variables	Secondary care (n = 68)	Tertiary care (n = 139)	Sig.
Diagnosis, n (%)			
CD	24 (35.3)	54 (38.8)	
UC	25 (36.8)	85 (61.2)	<0.001
IBD-U	19 (27.9)	0 (0)	
Examination time, mean (±SD)	20.2 (9.0)	16.2 (3.5)	0.004
Withdrawal time, mean (±SD)	10.4 (6.5)	8.2 (2.3)	0.008
Ileal intubation, n (%)	19 (27.9)	71 (51.1)	0.005
Disease activity score use, n (%)	7 (10.3)	71 (51.1)	<0.001
Neoplasia detected, n (%)	2 (2.9)	17 (12.2)	0.038

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Conclusion: Quality indicator targets were met in neither secondary nor tertiary centers; however, wider investigation is needed to verify these results. Broad education can help to improve the quality of colonoscopy in IBD care.

Disclosure: Nothing to disclose.

PP0420

EVALUATION OF TRANSFER IN INFLAMMATORY BOWEL DISEASE PATIENTS FROM PEDIATRIC TO ADULT CLINIC IN STOCKHOLM COUNTY. A QUALITATIVE STUDY

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Introduction: In Sweden some 300 patients with inflammatory bowel disease (IBD) are annually transitioned from pediatric to adult care. Patients with childhood-onset IBD more often presents with severe disease phenotypes and are at increased risks of low school performance, later psychiatric disorders, and low attachment to the labour market. The transition to adult care can be challenging for both patients and parents, as it involves changes in care routines and relationships with healthcare providers.

Aims & Methods: This qualitative study explored experiences of IBD patients about to transfer and already transferred and their parents in Stockholm County. Semi-structured interviews were conducted with 15 patients (9 pre- and 6 post-transfer) and 8 parents. All interviews where transcribed verbatim and thematic analysis was employed to analyse the interview material.

Results: The key themes both for parents and patients were autonomy, preparations, worries, relationships and possible improvements.

Patients highlighted the increased responsibility post-transfer, expressing both readiness and challenges in gaining autonomy. Parents worried about losing oversight and felt concern that their children would receive less emotional support and lower quality care compared to the pediatric clinic. Both groups stressed the emotional and practical adjustments required, including establishing relationships with new healthcare providers. Pre-transfer patients expressed worry about leaving familiar pediatric settings, while post-transfer patients reported satisfaction with the adult care system despite initial concerns. Recommendations for improvement from the informants included enhanced communication, written materials, and knowledge about the difference between adult and pediatric care.

Conclusion: The transition from pediatric to adult care for IBD patients involves emotional, practical, and relational challenges for both patients and parents.

The study shows that there is a need for a clear structure and information of how the process works and that there is an effective communication between the pediatric and adult clinics. In the process, it is essential to acknowledge and address factors such as the child's developing autonomy, considerations regarding separation from familiar caregivers, and to deal with the patients concerns about the future in a patient centred way. The fact that most patients and parents post-transfer was very satisfied with current process could indicate that a crucial factor in a successful transition is a motivated and committed health care personnel at both ends of the transfer. From a patient perspective, there are other more effective improvements to the transition process than joint visits.

This study highlights the importance of patient-centred strategies to optimize transitional care in Sweden.

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P.T.S. has served as Advisory Board member for Gilead Nordic, Janssen Cilag and Norgine.

H.Ö have nothing to declare.

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PP0421

MANAGEMENT OF PATIENTS WITH ULCERATIVE PROCTITIS: A WORLDWIDE SURVEY

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Introduction: Ulcerative proctitis (also called isolated proctitis) affects approximately 30% of patients with ulcerative colitis , yet achieving disease control remains challenging. Controlling the disease is essential to preventing disability and disease progression. Backbone therapy includes topical and oral 5-aminosalicylic acid (5-ASA) and steroids. One third of patients with ulcerative proctitis are refractory to conventional therapy with 5-ASA and steroids.

Aims & Methods: The aim of this study was to investigate current practice on isolated proctitis management across the globe.

Physicians with experience in treating inflammatory bowel diseases (IBD) were invited to participate in an anonymous, multiple-choice survey between January and February 2025. The survey addressed various aspects of patient care including clinical practice and attitudes towards isolated proctitis identification, disease activity measurement attitudes, and current treatment of isolated proctitis, including the use of advanced therapies.

Results: The survey included 460 physicians from 66 countries. Gastroenterologists were the most represented specialists (89.0%), followed by surgeons (6.3%), and internal medicine doctors (3.5%). Three quarters of participants (74.6%) had > 10 years of experience in IBD.

The majority of participants (87.9%) assessed clinical activity of isolated proctitis within 3 months of treatment initiation, 75.9% used fecal calprotectin, and 67.1% used C-reactive protein (CRP) to measure disease ac-

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tivity. Endoscopic assessment was performed 3 to 6 months (34.2%) or 6 to 12 months (48.4%) after treatment induction. For patients in clinical remission, the participants assessed clinical and biological disease activity every 6 months (54% and 59% respectively), and endoscopic evaluation every 5+ years (rectosigmoidoscopy: 27% and colonoscopy: 31%).

In this survey, 49% of participants were more reluctant to begin an advanced therapy in patients with isolated proctitis compared to pancolitis or left sided colitis. Infliximab (mean 6.55/10; SD: 2.64), vedolizumab (mean 5.64/10; SD: 2.98), upadacitinib (mean 4.88/10; SD: 2.91), and ustekinumab (mean 4.81/10; SD: 2.90) were preferred in refractory proctitis. Two thirds of participants (61.0%) were less likely to use biologics in combination with immunosuppressants in isolated proctitis compared to left sided or pancolitis. Anti-TNF (tumor necrosis factor) (48.4%) was the preferred choice in first line advanced therapy after failing conventional treatment.

Conclusion: This study highlighted differences in management of isolated proctitis compared to left sided colitis or pancolitis. This is likely explained by the fact that isolated proctitis patients were historically excluded from clinical trials, therefore management relied on extrapolation of data from studies on more extensive disease. To date, a single advanced therapy (etrasimod) has demonstrated efficacy in isolated proctitis patients through a double blinded, randomized clinical trial . Future clinical trials on advanced therapies for this frequent and disabling condition are eagerly awaited.

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PP0422

COMPARISON OF TWO FATIGUE QUESTIONNAIRES FOR PATIENTS WITH INFLAMMATORY BOWEL DISEASE (IBD)

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Introduction: Fatigue is a common but rarely diagnosed symptom in patients with inflammatory bowel disease (IBD). The IBD-specific fatigue questionnaire Inflammatory Bowel Disease Fatigue (IBD-F) was validated in German in 2022. It assesses the severity and frequency of fatigue (Section 1) as well as its impact on daily life (Section 2). The Functional Assessment of Chronic Illness Therapy–Fatigue (FACIT-F) also measures the impact of fatigue on daily life. While the FACIT-F has been validated in English for IBD, it has not yet been validated in German, although it is frequently used in IBD studies.

Aims & Methods: This study compared the IBD-F and FACIT-F in a German-speaking IBD cohort. IBD patients and a healthy control group (HC) completed both the IBD-F and FACIT-F questionnaires. Disease activity in IBD patients was assessed using the partial Mayo Score (pMS for ulcerative colitis; UC) or the Harvey Bradshaw Severity Index (HBSI for Crohn's disease; CD). Internal consistency was calculated using Cronbach's α. To assess acceptance, the number of fully completed questionnaires was analyzed. Spearman correlation was used to compare IBD-F and FACIT-F scores in both IBD patients and healthy controls.

Results: A total of 382 IBD patients (CD n = 240; 98 male; age 43.1 ± 13.9; UC n = 142; 72 male; age 43.2 ± 12.6) and 130 healthy controls (HC; 61 male; age 40.2 ± 16.5) participated. 75% (286/382) of the patients reported fatigue in IBD-F Section 1 and were included in further analyses.

Internal consistency was rated as excellent for IBD-F Section 1 (α = 0.930), Section 2 (α = 0.979), and FACIT-F (α = 0.936), as indicated by Cronbach's alpha. Acceptance decreased with increasing number of items (IBD-F Section 1 94.4%; Section 2 80.1%; FACIT-F 90.9%). A significant negative correlation was observed between the FACIT-F and IBD-F Section 1 in both the IBD group (r_s = -0.751, p < 0.01) and controls (r_s = -0.361, p < 0.01), as well as between the FACIT-F and IBD-F Section 2 (IBD: r_s = -0.862, p < 0.01; HC: r_s = -0.463, p < 0.01).

Patients in remission (59% (168/286)) reported lower fatigue scores in IBD-F Section 1 and 2 and FACIT-F than patients with active disease (36% (104/286)), but higher fatigue scores than HC. HC showed higher fatigue scores in the FACIT-F than in the IBD-F (see Table 1).

Questionnaire	IBD/Controls	N	Mdn (IQR)
IBD-F Section 1 - Total Score	Control	130	0.0 (2.0)
	IBD - Total	286	8.0 (5.0)
	- Remission[a]	168	6.0 (5.0)
	- Active disease[a]	104	10.0 (6.8)
IBD-F Section 2 - Total Score	Control	130	0.0 (2.3)
	IBD - Total	286	21.0 (28.9)
	- Remission[a]	168	18.0 (19.0)
	- Active disease[a]	104	30.0 (35.4)
FACIT-F - Total Score	Control	130	48.0 (7.3)
	IBD - Total	286	35.0 (16.7)
	- Remission[a]	168	37.5 (16.0)
	- Active disease[a]	104	29.6 (19.8)

Mdn = Median; IQR = Interquartile range. [a] Due to missing data, the pMS and the HBSI could not be calculated for 14 patients.
Higher scores in Sections 1 and 2 of the IBD-F, as well as lower scores in the FACIT-F, indicate greater fatigue. IBD-F Section 1 Total Score range: 0–20. IBD-F Section 2 Total Score range: 0–120. FACIT-F Total Score range: 0–52.

Table 1: Descriptive statistics of IBD-F and FACIT-F.

Conclusion: Both the IBD-F and FACIT-F are effective tools for assessing fatigue in IBD patients. The IBD-F, however, appears to better distinguish healthy individuals without fatigue.

Disclosure: Nothing to disclose.

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PP0423

CLINICAL AND PHENOTYPIC CHARACTERISTICS OF EARLY-ONSET INFLAMMATORY BOWEL DISEASE IN CHILDREN UNDER 6 YEARS OF AGE: A SINGLE-CENTER COHORT STUDY

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Introduction: Early-onset inflammatory bowel disease (EO-IBD), defined as disease onset before the age of 6 years, represents a distinct subgroup of pediatric IBD. It is associated with extensive colonic involvement, resistance to standard therapy, and a higher risk of complications, including extraintestinal manifestations (EIMs).

Aims & Methods: The purpose of our study was to analyze the clinical features of EO-IBD in children of a large gastroenterological center in the Russian Federation. This retrospective single-center cohort study included 37 children diagnosed with EO-IBD between 2019 and 2024. Clinical and demographic data were collected from outpatient records. Intestinal symptoms (abdominal pain, stool frequency and consistency, pathological impurities) and EIMs were assessed. Special attention was given to the timing of EIM onset relative to intestinal symptoms.

Results: The study included 37 patients — 17 with Crohn's disease (CD) and 20 with ulcerative colitis (UC) — all of whom were diagnosed before or at the age of 6. The median (Me) age at diagnosis was 5 years for both UC (IQR 3 – 6) and CD (IQR 4 – 6). The groups were comparable in terms of sex (p = 0.56) and age (p = 0.81). The median time from symptom onset to diagnosis was 10 months for UC (IQR 6.25 – 18) and 8.5 months for CD (IQR 5.5 – 15.8), with no significant difference between the groups (p = 0.74). Growth retardation (SDS < –2) was observed in four boys with CD at age 6. The most common symptoms in CD were abdominal pain (64.7% vs 60% in UC; OR 0.82; CI 0.22 – 3.12; p = 0.77) and weight loss (29.4% vs 7.1%; p = 0.019). Fever occurred exclusively in the CD group (35.3%). In contrast, UC was more frequently associated with diarrhea (90% vs 70.6% in CD; OR 3.75; CI 0.62 – 22.6; p = 0.13) and blood in stool (85% vs 64.7%; OR 0.32; CI 0.067 – 1.57; p = 0.15). EIMs were observed in 7 of 20 (35%) UC patients and 7 of 17 (41.2%) CD patients. The most frequent EIM in both groups was peripheral arthritis – found in 3 UC patients (10%) and 5 CD patients (29.4%) (p = 0.29). Peripheral arthritis was significantly more likely to occur after the onset of intestinal symptoms (p < 0.001). In two patients of each group, an EIM was the first manifestation of disease. In 43.2% of patients, an accurate diagnosis was not possible at disease onset, and they were initially classified as having IBD-unclassified (IBD-U). During follow-up, the time to establish a definitive diagnosis was measured. The median time to UC diagnosis was 24 weeks (IQR 20 – 48), and for CD it was 40 weeks (IQR 30–45.5) (p = 0.56).

Conclusion: EO-IBD in young children presents with specific phenotypic traits and a high rate of EIMs. Delayed diagnosis is common, particularly in cases initially classified as IBD-U. Our findings emphasize the need for early recognition and accurate differentiation of IBD subtypes in this challenging age group.

Disclosure: No financial support has been provided for this work. The authors declare that there is no conflict of interest.

PP0424

CORRELATION BETWEEN PATIENT-REPORTED DISEASE ACTIVITY, PHYSICIAN'S GLOBAL ASSESSMENT AND QUALITY OF LIFE IN INFLAMMATORY BOWEL DISEASE

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Introduction: As a chronic disease, patients' perspectives on their disease status and quality of life is an essential part of inflammatory bowel disease (IBD) management[1]. Patient-reported outcomes should be the standard of measure for signs and symptoms evaluation[2].

Aims & Methods: We aimed to evaluate the correlation between patients' self-assessment of disease activity, activity scores and the physician's global assessment.

A unicentric cross-sectional study was conducted. The Short Inflammatory Bowel Disease Questionnaire (SIBDQ)[3] was administered to consecutively evaluated IBD patients in the outpatient setting. Additionally, a pilot questionnaire addressing perceived disease activity was completed by both patients and their attending physician.

Disease activity was categorized as remission, mild, moderate, or severe. Remission was defined as Harvey Bradshaw Index (HBI) ≤ 4 for Crohn's disease or Mayo Clinic Subscore (MCS) ≤ 2 for ulcerative colitis.

Spearman's rank correlation coefficient was used to assess correlations between patient-reported disease activity, physician global assessment, disease activity scores, and quality of life.

Results: A total of 105 patients were included (female: 52%; mean age: 44±15 years), of whom 68% had Crohn's disease. 71% percent of patients were receiving advanced therapy (Table 1).

Female gender, % (N)	**52% (55)**
Age, mean ± standard deviation (SD), years	44.1 ± 15.0
Type of IBD, % (N)	
CD	68% (71)
UC	32% (34)
History of IBD-related surgery, % (n)	25% (26)
History of IBD-related hospitalization, % (n)	39% (41)
Current medical therapy, % (n)	
5-Aminosalicylates monotherapy	19% (20)
Immunomodulators (mono or combo therapy)	31% (32)
Advanced therapy	71% (74)
Disease activity indices, mean ± SD	
HBI	2.5 ± 6.5
MCS	3.4 ± 4.0
Patient-reported disease activity, % (N)	
Remission	27% (28)
Mild	42% (44)
Moderate	23% (24)
Severe	8% (9)
Physician's global assessment, % (N)	
Remission	34% (36)
Mild	26% (27)
Moderate	23% (24)
Severe	18% (19)
Quality of life, mean ± SD	
SIBDQ-PT	50.7 ± 12.1

Table 1.

A quarter of the patients (27%, n=28) perceived their disease to be in remission, of whom 56% were indeed in deep remission and 19% in endoscopic remission (table 1). A statistically significant positive correlation was observed between patient self-assessment and HBI scores (mean=2.5±6.5 points, r=0.400, p=0.001) and MCS scores (mean=3.4±4.0 points, r=0.574, p<0.001).

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Among patients who did not consider themselves in remission, 42% rated the activity of their disease as mild, 23% as moderate, and 9% as severe. The attending physician assessed 26% with mild disease, 22% with moderate disease, and 18% with severe disease (table 1). There was a statistically significant positive correlation between patient and physician assessments (r=0.374; p<0.001).

The mean SIBDQ score was 50.7 ± 12.1 points (table 1), and a statistically significant negative correlation was found between patient self-assessment of disease activity and the SIBDQ (r=-0.417, p<0.001).

Conclusion: These observational data suggest a weak correlation between patient-reported disease activity and the physician's global assessment. Divergent perceptions of disease activity between patients and physicians may compromise patient-engagement. Furthermore, the moderate correlation observed with overall quality of life supports the importance of alignment patients and physicians to improve the quality of care provided.

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Disclosure: Nothing to disclose.

PP0425

LINKED COLOR IMAGING AND HISTOLOGICAL ANALYSIS FOR DEEP REMISSION EVALUATION IN ULCERATIVE COLITIS TREATED WITH VEDOLIZUMAB

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Introduction: Linked Color Imaging (LCI) is an endoscopic system that emphasizes the red mucosal color to identify subtle areas of inflammation. Endoscopic imaging, including LCI classification and LCI index, has been reported as a predictor of histological healing (HH) in ulcerative colitis (UC). Endoscopic changes (LCI classification, LCI index) in vedolizumab treatment have not been reported to date.

Aims & Methods: This study aimed to evaluate endoscopic remission and HH using LCI endoscopy at week 22 of vedolizumab treatment in patients with moderate to severe UC. This was a phase 4, multicenter, open-label, single-arm, prospective study involving 12 patients with moderately to severely active UC receiving vedolizumab. Vedolizumab was administered for 54 weeks. Endoscopy and mucosal sampling from most inflamed sites or the patient's rectum when all sites had equal endoscopic severity were performed at weeks 0, 22, and 54 to assess the LCI index, LCI classification, Mayo endoscopic score (MES), Nancy score (If MES = 0, it was collected from the rectum). This study was also evaluated a variety of possible factors such as fecal calprotectin, serum vedolizumab concentration and mucosal gene expression associated with changes induced by vedolizumab. LCI index was numerically evaluated for the redness of a 40-pixel square area within the LCI image. LCI classification was defined by assigning one of the three categories (A, no redness; B, redness with visible vessels; C, intense redness without visible vessels). HH was defined as a Nancy score ≤1, and clinical outcomes were compared between HH and non-HH groups. Correlation coefficients between LCI endoscopy and other clinical markers (MES, Nancy score, fecal calprotectin and vedolizumab concentration) were analyzed at weeks 0, 22, and 54. Gene expression analysis was performed on 374 genes, comparing HH (n=8) vs. non-HH (n=3) and LCI classification A (n=5) vs. LCI classification B and C (n=6) at week 22. Biomarkers common to HH and LCI-A were identified as LCI-Deep remission (L-DR) markers.

The study was registered with the Japan Clinical Trial Registry (Identifier: UMIN000042855).

Results: At week 22, 66.7% (8/12) achieved histological healing (HH), while 50.0% (6/12) met LCI classification A criteria. The LCI index change from baseline showed a statistically significant difference between HH and non-HH (week 22: p=0.021; week 54: p=0.011).Correlation analysis demonstrated that LCI index was significantly associated with: MES (week 22: ρ=0.729, week 54: ρ=0.789), Nancy score (week 22: ρ=0.695, week 54: ρ=0.725), fecal calprotectin (week 22: ρ=0.555, week 54: ρ=0.783), vedolizumab concentration (week 22: ρ=0.758, week 54: ρ=-0.418). Gene expression analysis identified IL1B, CXCL11, CD38, CSF2RB, PTPN6, CD86, NFKB2, IFNAR2, STAT1, GNAI2, FES and IRF9 as L-DR markers, as they were commonly associated with both HH and LCI classification A. There are genes involved in NF-κB signaling pathway and JAK-STAT signaling pathway associated with UC. HH and LCI endoscopy have been found to correlate well but do not perfectly align.

In this study, we proposed a definition of L-DR, which encompasses both HH and LCI endoscopic remission.

Conclusion: This study introduces L-DR as a novel definition for combined endoscopic (LCI-A) and HH (Nancy score ≤1) in UC patients receiving vedolizumab. Additionally, we identified candidate L-DR biomarkers that may serve as potential indicators of deep remission.

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PP0426

PREDICTING INFLAMMATORY BOWEL DISEASE-ASSOCIATED LIVER DISEASE FROM ROUTINE BIOMARKERS USING LARGE LANGUAGE MODEL–REFINED CLINICAL DATA

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Introduction: Hepatobiliary disorders are recognised extraintestinal manifestations in inflammatory bowel disease (IBD). Up to 30% of patients exhibit abnormal liver function tests and 5% will develop chronic liver disease (1). IBD-associated liver diseases (IBDALD), comprising primary sclerosing cholangitis (PSC), autoimmune hepatitis (AIH) and AIH/PSC overlap syndrome carry significant morbidity through progressive hepatobiliary damage and increased cancer risk.

Early recognition is challenging due to non-specific biochemical derangement and heterogeneous presentation. No validated stratification tools exist for prospective risk identification.

This study investigated whether routinely collected biomarkers convey predictive value for IBDALD, leveraging large-language model (LLM) refined phenotyping.

Aims & Methods: We analysed 1,571 paediatric and adult-onset IBD patients recruited to the "Genetics of IBD" study (Southampton, UK; 2010–). IBDALD cases were identified through LLM-assisted (Llama 3.1-70B-Instruct) structuring of 32,041 free-text imaging and histopathology reports, followed by clinician verification (2). Participants diagnosed with IBDALD prior to IBD were excluded from further analysis.

Examined biomarkers included ALT, ALT and ESR (n=335,605 datapoints for the examined cohort). For each biomarker, patient-level medians were calculated from all available results within ±6 months of IBD diagnosis. A 1:4 individual matched case-control design was applied (matched on age at IBD diagnosis, gender, and IBD subtype) in *R*.

Conditional logistic regression models assessed associations between IBDALD and biomarker values, treated as both continuous (patient-level medians) and binary – normal versus abnormal - (age- and sex-adjusted lab range flags) variables. Longitudinal biomarker trajectories were explored using LOESS and linear mixed-effects models.

Results: Thirty-five cases of IBDALD were identified (27 post-IBD diagnosis). Median follow-up duration was 11.5 (0.5–59.5) years and median 4.5 years (0.4–39.1) from IBD diagnosis to liver disease onset. In whole cohort analyses, ALT, ALP, and ESR were significantly elevated in IBDALD at diagnosis of IBD ($p < 0.01$). In matched case-control models, higher median ALT (OR = 1.04 per U/L increase, 95% CI: 1.01–1.07, p = 0.012), ALP (OR = 1.01, 95% CI: 1.00–1.02, p = 0.014), and ESR (OR = 1.05, 95% CI: 1.00–1.09, p = 0.034) were each associated with IBDALD. In a combined model including all three biomarkers, none retained statistical significance. When analysed as binary (normal/abnormal) variables, ALT and ALP remained significantly associated with IBDALD (ALT: OR = 5.10, 95% CI: 1.57–16.59, p<0.01; ALP: OR = 15.33, 95% CI: 1.87–125.77, p<0.05). ESR did not reach significance (OR = 3.71, 95% CI: 0.75-18.32 p=0.107).

Longitudinal modelling supported distinct ALT and ALP trajectories in the IBDALD group and distinct trends over time were visualised for ALT, ALP and ESR in LOESS scatterplots.

Conclusion: These data suggest that routinely measured biomarkers may carry early predictive signals for IBD-associated liver disease, identifiable at the time of IBD diagnosis. The use of LLM-assisted phenotyping enabled high-fidelity case definition within real-world data. These findings should be interpreted cautiously given the rarity of IBDALD and potential overfitting in multivariable models. Future prospective studies are warranted to validate predictive models and support integration into IBD care pathways.

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Disclosure: JJA is a SAB member for Orchard Therapeutics.

PP0427

ACCEPTABILITY OF FECAL CALPROTECTIN IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: IMPACT OF AN ALTERNATIVE COLLECTION DEVICE FOR STOOL SAMPLING (THE ACCALMI STUDY)

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Introduction: Fecal calprotectin (FC) is a key non-invasive biomarker used to monitor inflammatory bowel disease (IBD). Despite its clear clinical value, patient adherence to the test remains suboptimal, mainly due to reluctance related to stool collection.

The objective of the ACCALMI study was to compare the acceptability of a new stool collection device, similar to the one used for fecal immunochemical tests (FIT), with a standard stool container, and to evaluate its impact on patient compliance.

Aims & Methods: In this prospective, single-center study, 156 patients with an established diagnosis of IBD and who had been prescribed an FC test as part of their routine care, were randomly assigned to either the standard container group or the alternative device group (Procise Stool Collection Device™), which includes a collection tube with a thin spiral-shaped probe facilitating stool collection. Acceptability was assessed using a 4-category ordinal scale (from "poor" to "very good") and a visual analog scale (VAS, from 0 to 10). Analyses were conducted in both per-protocol and intention-to-treat (ITT) populations, with missing data handled using the maximum bias method. Factors associated with improved FC acceptability, and compliance with FC testing, were analyzed.

Results: The alternative collection device significantly improved acceptability as measured by the ordinal scale (odds ratio [OR] 2.69; $p < 0.001$). In the alternative group, acceptability ratings were significantly higher with only 1.4% versus 7.0% rating it as 'poor', 12.5% versus 46.6% as 'average', 66.7% versus 40.8% as 'good', and 19.4% versus 5.6% as 'very good'. Measured using the VAS, the average acceptability was 7.9/10 in the group using the alternative device compared to 6.1/10 in the standard group, with a significant difference of 30.4% (95% CI [19.7 - 44.7%], $p < 0.001$), although it did not reach the threshold of 30% that was set *a priori*. The ITT analysis led to the same conclusions. Among the patients who had already undergone FC testing and were randomized to the "alternative device" group,

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55% indicated a preference for the alternative device, compared to 7.2% for the conventional device, while 37.7% found them similar. Embarrassment and discomfort associated with the standard container were alleviated by the alternative device. A long home-to-hospital distance (> 50 km) and the use of the alternative device were the only factors associated with better acceptability in the multivariate analysis. Compliance was similar between both groups and remarkably high (92%).

Conclusion: The ACCALMI study demonstrated that the alternative stool collection device, similar to the one used in FIT for colorectal cancer screening, is more accepted by IBD patients than the standard container, thus improving the acceptability of FC testing. These findings support the integration of patient-centered innovations in diagnostic and follow-up testing to improve the monitoring of IBD.

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PP0428

A RISK SCORE AND NATURAL HISTORY OF RECURRENT CROHN'S DISEASE AFTER ILEO-COLONIC RESECTION: PROSPECTIVE COHORT STUDY

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Introduction: Recurrence of Crohn's disease (CD) lesions after surgery remains high. Identifying patients at high risk of recurrence who may benefit from prophylactic therapy would be relevant. However, data on specific preoperative risk factors remain scarce, while onset and severity of Post Operative Recurrence (POR) are assessed with the Rutgeerts score (RS)(1). Still, the value of early endoscopic lesions, and RSi2, to predict the later course of the disease and the risk of late (from 1 to 10 years) POR in any individual patient is unclear. Non-invasive Small Intestine Contrast Ultrasonography (SICUS) assesses intramural extension of CD lesions, their progression, and is as effective as RS in detecting early POR lesions (2).

Aims & Methods: 1. To develop a risk score based on pre-operative clinical predictors to estimate POR likelihood;

2. to identify predictors of severity of mucosal lesion according to RS;

3. to assess the overtime progression of transmural lesion.

Prospective cohort study (1998-2022). CD patients were evaluated at ileo-colonic surgery and every 6-12 months thereafter. Sex, age at diagnosis/surgery, active smoking, CD behaviour (B)/location (L), perianal disease (PD), length of resected bowel, steroids, immunosuppressants/biologics therapy use, and CD duration before surgery were analysed using multivariable proportional hazards regression. Cumulative POR rates and risk score were calculated with Poisson models, taking for each predictor the integer part of the Poisson model parameter.

Results: Over a median follow-up of 7.6 years (IQR 4.1–12), 78 patients (32 F; median age 30 years; 60% smokers) were analysed. CD characteristics: B1 8%; B2 67%; B3 25%; L1 63%; L3 36%; PD 22%; immunosuppressive/biologics 17%; median CD duration pre-surgery 4.7 years; median resected bowel length 28 cm.

Cumulative POR rates were 51.8% (IQR,41%-63.4%), 65.2% (54.4%-75.7%), 69% (58.6%-79%), 75.9% (65.8% -84.8%), 78.9% (69.0%-87.3%), at 1, 2, 3, 4, and 5 years. POR risk factors included active smoking, age ≥35 years at surgery, and marginally B1 (Table).

Predictors of RS were PD (HR=3.39; 95% CI 1.12-10.26; p=0.03 and CD duration ≥5 years pre-surgery (3.45; 1.15–10.32; p=0.03). The estimated cumulative probability of POR increased with higher risk scores (from 1 to 3) being 22%, 58%, 83% and, respectively, 43%, 80% and 100% at 1 and 4 years. Lesion length at POR varied by RS (median, IQR): RSi1 (32%): 0.5 cm (0.5-0.5), RSi2 (35%): 4 cm (0.5-7), RSi3 (15%): 15 cm (8.5-21.5), RSi4 (5%): 16 cm (9.5-23.5). Lesions remained stable in patients with RSi1 and RSi3 (45%) but fluctuated in those with RSi2 and RSi4 (31% increased, 24% decreased).

		Univariate analysis				Multivariate analysis				Cumulative Risk Score
Variable		HR	95% CI		p	HR	95% CI		p	contribution
Age at surgery (years)	≥35 vs. <35	1.84	1.10	3.10	0.02	1.82	1.07	3.07	0.03	0.60
Sex	Female vs. Male	1.24	0.74	2.08	0.42	1.59	0.90	2.79	0.11	0.46
Active smoking	Yes vs. No	1.50	0.88	2.55	0.14	1.99	1.06	3.75	0.03	0.69
Disease behaviour	B2 vs. B1	2.28	0.81	6.44	0.12	2.53	0.89	7.19	0.08	0.93
	B3 vs. B1	2.01	0.66	6.09	0.22	3.01	0.94	9.61	0.06	1.10
Disease duration (years)	≥5 vs. <5	1.18	0.71	1.98	0.52	NI	NI	NI	NI	NI
Length of resection (cm)	≤ 25 vs. >25	1.15	0.69	1.92	0.60	NI	NI	NI	NI	NI
PD	Yes vs. No	1.10	0.63	1.92	0.74	NI	NI	NI	NI	NI

*HR:hazard ratio;CI: confidence interval;*time dependent covariates;*
NI:not included in the final model selected through a backward selection (p<0.2)

Conclusion: The present score based on preoperative factors predicts the risk of postoperative recurrence. Patients with preoperative higher risk scores or RSi2 at recurrence would require early and close monitoring due to significant transmural lesion progression.

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1.Rutgeerts R et al. 1990; 2. Pallotta N et al. 2010.

Disclosure: Nothing to disclosure.

PP0429

DIAGNOSTIC ACCURACY OF THE FECAL IMMUNOCHEMICAL TEST FOR ASSESSING ENDOSCOPIC ACTIVITY IN INFLAMMATORY BOWEL DISEASE

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Introduction: Fecal immunochemical test (FIT), widely used in colorectal cancer screening, has emerged as a potential biomarker of endoscopic activity in inflammatory bowel disease (IBD). Compared to fecal calprotectin, FIT offers greater standardization, automation and availability, with simpler handling and potentially better adherence. Its implementation could streamline non-invasive monitoring, but its diagnostic accuracy in IBD remains unclear. We therefore conducted a systematic review and meta-analysis to evaluate its performance.

Aims & Methods: Selection of studies: assessing the accuracy of FIT to estimate endoscopic activity in IBD patients. Search strategy: electronic bibliographical searches. Data synthesis: meta-analyses of the FIT were performed combining the sensitivity (Se), specificity (Sp), likelihood ratios (LRs), and area under de curve (AUC) values of summary receiver operating characteristic (ROC) curves of the individual studies. Heterogeneity among the studies was assessed by the I-squared (I^2) statistic.

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Results: Eighteen studies (Table 1) met the inclusion criteria: 12 focused on ulcerative colitis (UC), 4 on both UC and Crohn's disease (CD), and 2 on CD alone. Twelve studies employed the standard cut-off for FIT provided by the manufacturer or used the same cut-off as for colorectal cancer screening (usually considered the common threshold), while 6 studies estimated their optimal cut-off values. When we considered all studies (including 1,812 patients and 2,137 assessments), the accuracy of FIT for the diagnosis of endoscopic activity was (pooled values): Se: 0.66, Sp: 0.83, LR+: 4.96, and LR-: 0.38, respectively, with an AUC of 0.86.
Significant heterogeneity was found among the Se (I^2 89%) and Sp (I^2 86%) of different studies. Most of the available studies were based exclusively on patients with UC (including 1,554 patients and 1,904 assessments), so we performed a subgroup analysis including only this type of IBD, finding a pooled Se, Sp, LR+ and LR- of 0.69, 0.82, 5.35 and 0.31 respectively, with an AUC of 0.88. On the other hand, in patients with CD (327 patients, 302 assessments), the corresponding values were 0.55, 0.85, 3.2 and 0.58, with an AUC of 0.73. ROC curve analysis showed that FIT had a higher diagnostic performance in UC patients.

Author	Publication year	FIT cut-off	Endoscopic activity definition	Number of patients	Number of tests	Sensitivity	Specificity	TP	FP	FN	TN
Nakarai	2013	Standard (>100 ng/mL)	MES >1	UC 152	310	87%	60%	121	68	18	103
Takashima	2015	Standard (>100 ng/mL)	MES >1	UC 92	105	68%	81%	28	12	13	52
Ryu	2016	Standard (>100 ng/mL)	MES >1	UC 63	82	45%	93%	10	4	12	56
Inokuchi	2016	Optimal (>52 ng/mL)	SES-CD >0	CD 71	71	48%	96%	23	1	25	22
Nakarai	2016	Standard (>100 ng/mL)	MES >0	UC 194	194	76%	94%	76	6	24	88
Ma	2017	Standard (>100 ng/mL)	UCEIS >3, SES-CD >0	UC 40 & CD 40	80	57%	90%	17	5	13	45
Shi	2017	Standard (>50 ng/mL)	MES >1	UC 140	140	68%	72%	28	28	13	71
Hiraoka	2018	Standard (>100 ng/mL)	MES >0	UC 84	220	63%	95%	76	5	45	94
Nakarai	2018	Standard (>100 ng/mL)	MES >1	UC 113	113	78%	87%	7	14	2	90
Iwamoto	2018	Optimal (>13 ng/mL)	SES-CD >0	CD 69	69	71%	73%	43	2	18	6
Hiraoka	2019	Optimal (>78 ng/mL)	MES >1	UC 152	152	96%	79%	25	26	1	100
Ryu	2019	Standard (>100 ng/mL)	MES >0	UC 128	174	37%	98%	46	1	77	50
Mak	2020	Optimal (>26 ng/mL)	MES >1, SES-CD >3	UC 62 & CD 51	66	69%	87%	18	5	8	35
Yen	2020	Optimal (>43 ng/mL)	MES >1	UC 50	50	100%	81%	19	6	0	25
Kim	2020	Optimal (>60 ng/mL)	MES >1	UC 127	127	52%	94%	15	6	14	92
Hijos-Mallada	2021	Standard (>5.1 µg/g)	MES >0, SES-CD >2	UC 54 & CD 52	106	59%	87%	25	8	17	56
Latorre	2023	Standard (>100 ng/mL)	MES >0, SES-CD >0	UC 40 & CD 44	84	50%	79%	20	9	20	35
Takahara	2024	Standard (>100 ng/mL)	MES >0	UC 63	63	93%	100%	52	0	4	7

TP: True positives; FP: False positives; FN: False negatives; TN: True negatives; UC: Ulcerative colitis; CD: Crohn's disease; MES: Mayo endoscopic score; SES-CD: Simple endoscopic score for Crohn's disease; UCEIS: Ulcerative colitis endoscopic index of severity.

Table 1. Studies evaluating the accuracy of fecal immunochemical test (FIT) for estimating endoscopic activity in inflammatory bowel disease.

Conclusion: FIT demonstrates moderate-to-high diagnostic accuracy for detecting endoscopic activity in IBD, particularly in UC. Therefore, FIT may represent a promising non-invasive tool for assessing IBD disease activity. However, significant heterogeneity among studies underscores the need for further research to optimize cut-off values and define the role of FIT in routine IBD monitoring.
Disclosure: Nothing to disclose.

PP0430

ADHERENCE TO A CARE PATHWAY FOR INFLAMMATORY BOWEL DISEASE IN THE SOUTHWEST REGION OF THE NETHERLANDS: RESULTS OF THE MULTICENTRE COHORT STUDY IBD VALUE

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Introduction: In the southwest Netherlands, hospitals collaborate to provide high quality, uniform care for inflammatory bowel disease (IBD). To achieve this, a care pathway (CP) was implemented for treating IBD with biologics and small molecules (1). The CP consists of protocols addressing different stages in IBD treatment, such as initiation and switching of therapy, frequency and type of follow-up.
This study assessed the adherence to the CP and identified barriers and facilitators to its implementation.
Aims & Methods: A mixed-methods study was conducted. Quantitative data was used to evaluate the adherence to different protocols of the CP and was collected retrospectively from electronic health records (EHRs) from December 2020 until March 2023. Surveys and semi-structured interviews were used to identify barriers and facilitators, using the Extended Normalisation Theory. Surveys were sent to all healthcare providers and interviews were conducted with gastroenterologists, IBD-nurses, healthcare assistants, and IT-personnel.
Results: A total of 299 patients were included. Adherence to the CP regarding the documentation of travel plans and vaccination status improved after implementing the CP (15% pre-implementation vs. 33% post-implementation), as well as written communication with general practitioners (31% vs. 83%) and body weight registration (12% vs. 50%). Among patients switching medication, 25% had a repeated screening for HIV, tuberculosis, and hepatitis B and C documented if prior tests were over a year old. Post-implementation, scheduling outpatient visits were more closely aligned with CP recommendations across CP protocols (57% vs. 100%), as did blood test ordering (10% vs. 44%). Disease activity scoring with validated questionnaires remained low (7% vs. 21%). Adherence to registering medication adherence, current smoking status, and side effects during annual outpatient visits remained low (18% vs. 40%).
Of 85 surveys, 42 were completed, and 11 interviews were conducted. Facilitators were improving collaboration between healthcare providers and the potential to standardise care. Barriers included the complexity of the use in EHRs, difficulty for healthcare providers to change their current routine, and the heterogeneity of IBD.

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Conclusion: A CP for the treatment of IBD with biologics and small molecules has been implemented to improve IBD care, however, adherence appears to be challenging, partially contributed to the difficulty healthcare providers experience in changing their behaviour. The gradual improvement indicates that increased familiarity with the CP may enhance adoption. Discrepancies between performed and documented tasks in EHRs may also contribute to lower adherence rates, which could be improved by audit and feedback sessions.

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1. van Linschoten R, van Leeuwen N, Nieboer D, Birnie E, Scherpenzeel M, Verweij KE, et al. Value-based care pathway for inflammatory bowel disease: a protocol for the multicentre longitudinal non-randomised parallel cluster IBD Value study with baseline period. (2044-6055 (Electronic))

Disclosure: Drs. Visser received speaker fees from Lilly outside the submitted work. Prof. Dr. van der Woude reports grants from Pfizer and Janssen and personal fees from AbbVie and Celltrion outside the submitted work. Dr. van Noord reports personal fees from Janssen, Takeda, and Falk outside the submitted work. Dr. West reports personal fees from AbbVie, Janssen, and Pfizer outside the submitted work.

PP0431

A CARE PATHWAY FOR THE TREATMENT OF IBD REDUCES HEALTHCARE COSTS AND CAN BE COST-EFFECTIVE: RESULTS OF A MULTICENTRE COHORT STUDY IBD VALUE

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Introduction: Eight hospitals in the southwest of the Netherlands collaborate to improve inflammatory bowel disease (IBD) care. To achieve this, a care pathway (CP) was developed for treating IBD with biologics and small molecules. This CP addresses initiation and switching of therapy, as well as diagnostic testing and treatment follow-up (1).

Aims & Methods: This study aimed to estimate the causal effect of the CP on costs and quality of life, and to evaluate whether the CP is cost-effective.

A cost-effectiveness analysis was conducted from a societal perspective. The CP was implemented in six hospitals, with two control hospitals. Costs were assessed during a baseline period (December 2020–December 2021) and an intervention period (March 2022–March 2023). Quality-adjusted life years (QALYs) were derived from the EQ-5D-5L. A difference-in-differences (DiD) analysis was conducted and the incremental cost-effectiveness ratio (ICER) was calculated.

Results: A total of 1,173 patients were included (intervention n=841, control n=332). Baseline costs per patient were €23,259.96 in intervention hospitals, and €22,837.93 in control hospitals. During the intervention period costs decreased to €20,959.76 in intervention hospitals. This reduction was not observed in control hospitals (€22,191.21). The DiD-analysis showed cost savings of €-1,653.48. QALYs did not change. The ICER was located in the southeast quadrant of the cost-effectiveness plane, with a high probability of achieving a net monetary benefit exceeding €0 compared to standard care.

Conclusion: The implementation of a CP for the treatment of IBD with biologics and small molecules is cost-effective, it reduces healthcare costs while maintaining quality of life. Therefore, adoption of care pathways in care for complex diseases should be considered.

Reference:

1. van Linschoten RCA, van Leeuwen N, Nieboer D, Birnie E, Scherpenzeel M, Verweij KE, et al. Value-based care pathway for inflammatory bowel disease: a protocol for the multicentre longitudinal non-randomised parallel cluster IBD Value study with baseline period. BMJ Open. 2022;12(1):e050539

Disclosure: Drs. Visser received a speaker fee from Lilly outside the submitted work.

Prof. Dr. van der Woude reports grants from Pfizer and Janssen and personal fees from AbbVie and Celltrion outside the submitted work.

Dr. van Noord reports personal fees from Janssen, Takeda, and Falk outside the submitted work.

Dr. West reports personal fees from AbbVie, Janssen, and Pfizer outside the submitted work.

PP0432

DISEASE ACTIVITY MONITORING IN PATIENTS WITH ULCERATIVE COLITIS USING SEROLOGICAL NUCLEAR MAGNETIC RESONANCE (NMR) METABOLITE PROFILES

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Introduction: Metabolomics in ulcerative colitis (UC) research offers promising insights into disease mechanisms and activity monitoring. Nuclear magnetic resonance (NMR) spectroscopy enables serum metabolite profiling which can include inflammatory markers such as glycoprotein A (GlycA) and B (GlycB). The supramolecular phospholipid composite (SPC) is derived from phospholipids found on lipoproteins. The Glyc/SPC ratio shows NMR signals of glycoproteins and phospholipids. Lipoprotein levels are influenced by systemic inflammation and altered lipid profiles are

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seen in inflammatory bowel disease (IBD) patients[1]. While GlycA is linked to IBD activity[2], Glyc/SPC is yet unexplored in IBD. Methionine, an amino acid associated with inflammation, has not been extensively studied in IBD. In this study, we hypothesize that the serological metabolite profile differentiates between UC patients with active disease and those in remission.

Aims & Methods: This study aimed to determine whether UC patients exhibit distinct serological metabolite profiles depending on disease activity. Serum samples were collected from healthy subjects (HS), UC patients in remission (UCR), and UC patients with active disease (UCA), and analyzed using NMR spectroscopy.

A targeted metabolomic panel was applied to quantify 157 metabolites. Statistical significance was assessed using two-sample *t*-tests followed by False Discovery Rate correction using the Benjamini-Hochberg method, with $q < 0.05$ considered significant. Orthogonal partial least squares discriminant analysis (OPLS-DA) was conducted to evaluate metabolic differences between groups.

Model validations were performed using Cross-Validated ANOVA, with $p < 0.05$ considered significant, as well as permutation testing (n = 100) to assess robustness. Variable Importance in Projection (VIP) analysis was conducted to identify the metabolites that contributed most to the model.

Results: Serum samples from 50 HS (41 females), 54 UCR (26 females), and 15 UCA (6 females) were analyzed. A total of 67 metabolites differed in HS vs. UCA, 39 in UCR vs. UCA, and 17 in HS vs. UCR. The OPLS-DA models discriminated between UCR and UCA and between UCA and HS, but not UCR and HS. Further analysis focused on the UCR vs. UCA comparison. The OPLS-DA model showed separation between UCR and UCA (R2Y = 0.838, Q2 = 0.599, p < 0.0001), without indications of overfitting from permutation testing. The top three discriminant metabolites, based on VIP scores, were methionine (q = 0.0004), Glyc/SPC (q = 0.001), and V4CH (VLDL subclass, q = 0.007) (Table 1).

Group	Methionine (mM)	Glyc/SPC (p.d.u.)	V4CH (mg/dL)
HS	0.080 ±0.019	0.49 ± 0.12	3.13 ±2.00
UCR	0.077 ±0.016	0.58 ± 0.12	3.57 ±1.87
UCA	0.046 ± 0.029	0.80 ± 0.27	6.40 ± 2.31

Table 1. Serological metabolite levels of the top three discriminant metabolites. Values are presented as mean ± standard deviation. p.d.u = procedure-defined units.

Conclusion: Shifts in Glyc/SPC, lipoprotein, and amino acid levels were observed in UCA, suggesting metabolomic differences compared to UCR. This indicates that NMR metabolomics holds promise as a tool for disease activity monitoring in UC. However, further analysis, including validation in a separate cohort and comparison with established biomarkers, is needed to evaluate accuracy and diagnostic value.

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PP0433

INTERIM ANALYSIS OF REAL-WORLD-DATA ON PROMIS® GLOBAL 10 COLLECTED THROUGH THE HEALTH OUTCOME OBSERVATORY (H2O) CONSORTIUM IN PATIENTS WITH IBD

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Introduction: The Health Outcome Observatory (H2O) project aims to standardize and facilitate the collection of patient-reported outcomes (PROs) as well as clinical outcomes by a Core Outcome Set (COS) defined for patients with inflammatory bowel disease (IBD).[1]

The present study describes preliminary data on the PROMIS®-10 questionnaire (Patient-Reported Outcomes Measurement Information System) – a generic health-related quality of life measure - in IBD patients.

Aims & Methods: This is a real-world interim analysis of IBD patients included in the H2O project at the Medical University of Vienna, Austria. The variables analyzed were part of the COS. PROMIS®-10 was evaluated, including Global Physical Health and Global Mental Health scores. Higher scores indicate better health and a score between 40 and 59 was defined as normal range.

A descriptive analysis and a correlation analysis, including sex, age, age at diagnosis, disease duration, IBD_Disk_managing_defecation_day and night, IBD_urge, PRO-2 for CD (number of liquid or very soft stools and abdominal pain score) and UC (rectal bleeding score and stool frequency score) were performed.

Results: 691 patients (46% females) were included in the analysis. 460 were diagnosed with Crohn's disease (CD), 220 with ulcerative colitis (UC), and 11 with IBD unclassified (IBDU). The PROMIS scores were similar in CD and UC. The mean PROMIS Global Physical Health score was 41.5 (SD 5.3) in CD and 41.9 (SD 4.5) in UC, respectively. The mean PROMIS Global Mental score was 46.9 (SD 8.8) in CD and 48.4 (SD 8.3) in UC. 223 patients had a PROMIS Global Physical Health score < 40, and no patient had a score > 59. 89 patients had a PROMIS Global Mental Health score < 40, and 32 patients had a score > 59. The PROMIS Global Physical Health score correlated with the Mental score (r=0.23; p<0.001), sex (male r=0.13; p=0.003), PRO-2 for CD number of liquid or very soft stools (r=-0.11; p=0.01) and abdominal pain score (r=0.14; p=0.002). The PROMIS Global Mental Health score correlated with sex (male r=0.13; p=0.004), the IBD_Disk_managing_defecation_day (r=-0.32; p<001) and night (r=-0.31; p<0.001), IBD_urge (r=-0.33; p<0.001), PRO-2 for CD number of liquid or very soft stools (r=-0.34; p<0.001) and abdominal pain score (r=-0.49; p<0.001), and PRO-2 for UC rectal bleeding score (r=-0.13; p=0.004) and stool frequency score (r=-0.35; p<0.001).

Conclusion: About one third of the patients had a PROMIS Global Physical Health score below the normal range. The majority of the IBD patients had a PROMIS Global Mental Health score within the normal range, which correlated with the severity of gastrointestinal symptoms, including urgency and incontinence.

Reference:

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PP0434

PREVALENCE OF PERFUSION DISORDERS IN THE PULMONARY CIRCULATION IN PATIENTS WITH ULCERATIVE COLITIS REQUIRING HOSPITALIZATION IN THE LAST 24 MONTHS

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Introduction: Inflammatory bowel disease (IBD) is associated with a 2-3-fold risk of venous thromboembolic complications. This risk is further increased by disease flare. However, there is a lack of studies systematically searching for these events and identifying other risk factors in IBD patients.

Aims & Methods: The aim of a prospective cohort study was to evaluate the occurrence of perfusion disorders in the pulmonary circulation in patients with acute severe ulcerative colitis (ASUC). A cohort of patients from a single tertiary IBD centre hospitalized with an episode of ASUC was recruited from January 2023 to April 2025. Ventilation-perfusion (V/P) SPECT was performed in all patients after hospital discharge. In case of detection of primary perfusion defect, anticoagulation treatment was initiated in patients not yet anticoagulated.

Results: Fifty four patients with relapse of UC requiring hospital stay due to disease activity in the past 24 months in whom V/P SPECT was performed were identified. None of the patients had clinical symptoms of pulmonary embolism. Mean age of the cohort was 37.0±13.4 years (min-max 19-76 years). Of them 53.7% (29/54) were males. Mean disease duration at the time of V/P SPECT was 7.1±7.1 years (1-35 years). Disease extension included 25.9% patients with left-sided colitis and 74.1% with pancolitis. Active smoking was present in 13.0% of patients and history of using JAK inhibitors was positive in 63.0% of patients. Mean time elapsed between last hospital stay due to UC activity and V/P SPECT was 239.6±216.4 days (12-745). Presence of a perfusion defect was found in 16.7% (9/54) of patients. There were no significant differences in baseline patient and disease characteristics including smoking and JAKi history between positive and negative group. None of the positive patients had previous history of known thrombotic event.

Conclusion: We have identified 16.7% of patients with severe UC relapse in the past 2 years to have signs of perfusion defect on V/P SPECT warranting further research to identify potential risk factors and probability of developing perfusion disorders of pulmonary circulation in patients with ASUC.

Disclosure: Eli Lilly.

PP0435

MANAGING PREGNANCY IN INFLAMMATORY BOWEL DISEASE- DO WE MEET ECCO STANDARDS?

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Introduction: Inflammatory bowel disease (IBD) has a high incidence in young people directly impacting their reproductive years. In 2023, ECCO produced guidelines on sexuality, fertility, pregnancy & lactation to support healthcare professionals in managing patients considering, and throughout pregnancy. This audit undertaken as 3rd year medical student project.

Aims & Methods: The study aimed to evaluate the extent to which ECCO standards on are being met at a tertiary hospital and look at disease activity and adherence to advice.The IBD service was benchmarked against the ECCO guidelines. The women received antenatal care at our linked women's Hospital. Patients are offered monthly appointments with the IBD team during pregnancy. Data was collected from electronic patient records (EPIC©) of patients under the care of the IBD team. The data were analysed in SPSS using statistical tests, including Mann-Whitney U test.

Results: The study included patients who became pregnant or gave birth between April 1, 2023 - March 31, 2024. The results included 56 women, of whom 28 had already given birth. Thirty women (53.6%) had Crohn's Disease and 26 (46.4%) had ulcerative colitis. Ages ranged from 20 to 42 years. Forty-eight women (85%) were receiving active treatment, with 23 (41%) on advanced therapies and of these of the 23 people on biologics, 13 (56.6%) continued taking the biologic throughout pregnancy, 6 (26.1%) stopped taking the biologic before the third trimester. 71.4% of women had a documented & individualized care plan. Documentation was noted to be poor around advice re lactation (12.5%).

IBD activity improved between pre-conception and during pregnancy, as indicated by faecal calprotectin levels decreasing on average from 392.7 (SD 630) pre-conception to 104.2 (SD 142.7) during pregnancy. Women with active disease generally give birth earlier than those without active disease, and those with active disease were more likely to have a preterm birth with 25% of the active disease group giving birth prematurely, i.e. before 37 weeks, although this did not meet statistical significance. 50% had a spontaneous vaginal delivery, while 32.2% had a caesarean section. There was no significant difference in the baby's birth weight between mothers on biologic treatment and those not on biologic treatment.

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Conclusion: The audit identified areas where the ECCO standards were not fully met, such as documenting advice around medications and lactation. The IBD team has developed an action plan to address the findings of the audit. We also demonstrated reduced disease activity during pregnancy in this cohort of women and in keeping with previous studies saw that patients with active IBD were more likely to deliver before 37 weeks.

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Disclosure: No conflict of interest.

PP0436

COMPARED EFFICACY OF NUTRITIONAL SCREENING TOOLS FOR THE PREDICTION OF MALNUTRITION AND LOW FAT-FREE BODY MASS INDEX IN A COHORT OF INACTIVE/MILDLY ACTIVE INFLAMMATORY BOWEL DISEASE PATIENTS

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Introduction: Inflammatory Bowel Disease (IBD) patients carry a high risk of malnutrition, independently of disease activity. Screening for malnutrition in IBD is currently recommended since malnutrition, as well as low Fat-Free Mass Index (FFMI), have been associated to poor disease outcomes. However, due to a non-universal definition of malnutrition, most Nutritional Screening Tools (NSTs) have been validated compared to different reference methods and do not consider patients' body composition. Moreover, the ability of most NSTs to predict malnutrition over time has not been explored.

Aims & Methods: The aim of this study was to compare the efficacy of different NSTs to predict malnutrition, as defined by the ESPEN 2015 and GLIM criteria, and a low FFMI in a cohort of inactive/mildly active IBD patients.

This is a prospective, single center, cohort study enrolling Crohn's disease (CD) and ulcerative colitis (UC) patients in clinical remission or with mildly active disease as defined by Harvey-Bradshaw Index and partial Mayo Score, respectively. MUST, MIRT, MNA, NRS-2002 and SaskIBD-NR questionnaires and Bioelectrical Impedance Analysis (BIA) were performed at baseline and after six months follow-up. A low FFMI was defined as <18 kg/m2 in males and <15 kg/m2 in females. The sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV) and the accuracy of the different NSTs for the detection of malnutrition and low FFMI at six months were calculated.

Results: Sixty-six (CD: 32; UC: 34) IBD patients were enrolled. Median age was 55.5 years (IQR 34-62.8), median disease duration was 10 years (IQR 6-17). At baseline, all CD and 30/34 (88%) UC patients were in clinical remission; median C-reactive protein was 0.8 mg/L (IQR 0-3.8), median fecal calprotectin was 55.2 mcg/g (IQR 12.9-326). Of the entire cohort, 6/66 (9%) and 17/66 (26%) patients were malnourished according to GLIM and ESPEN criteria, respectively, [(GLIM CD:3, UC: 3)(ESPEN: CD: 3; UC:14)] and 5/66 (8%) had low FFMI (CD:1; UC: 4). At 6 months follow-up, all CD and 31/34 (91%) UC patients were in clinical remission. Ten out of 66 (15%) and 32/66 (48%) patients were malnourished according to GLIM and ESPEN criteria, respectively [(GLIM CD:5, UC: 5)(ESPEN: CD: 15; UC:17)], 11/66 (17%) patients had low FFMI (CD: 3; UC: 8). Baseline MIRT showed the highest sensitivity for the diagnosis of both ESPEN and GLIM-defined malnutrition at 6 months (ESPEN: 40.5%; GLIM: 50%). For ESPEN, MIRT and NRS2002 were the most accurate NSTs (MIRT: 50%, NRS2002: 51.5%), while SaskIBD (75.8%) showed the highest accuracy in detecting GLIM-malnourished patients at 6 months follow up. Among all NSTs, baseline MIRT, MNA and NRS2002 had the highest sensitivity for the detection of low FFMI at six months (MIRT, MNA, NRS2002: 45.5%), while SaskIBD_NR was the most accurate (80.3%).

Conclusion: MIRT might be the NSTs of choice to predict defined malnutrition and low Fat-Free Mass Index at 6 months follow-up in a cohort of inactive/mildly active IBD patients.

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Disclosure: Nothing to disclose.

PP0437

REAL WORLD EFFICACY OF STEP-UP APPROACH IN ACHIEVING STRIDE-II RECOMMENDED TREAT-TO-TARGET GOALS IN CASES OF INFLAMMATORY BOWEL DISEASE – EXPERIENCE FROM INDIA

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Introduction: Inflammatory Bowel Disease (IBD), including Ulcerative Colitis (UC) and Crohn's Disease (CD), is a chronic gastrointestinal disorder marked by progressive inflammation. Recent therapies focus on a treat-to-target (T2T) strategy, as per STRIDE-II guidelines, aiming for defined short-term, intermediate-term, and long-term goals. While Western countries favor a top-down approach, emerging nations still largely follow the step-up method. Our study assesses the real-world effectiveness of the step-up strategy in meeting these targets at a tertiary care hospital in South India.

Aims & Methods: To study the short-term, intermediate-term and long-term treat-to-target goals (as per STRIDE-II recommendations) met, based on the current medical therapy in cases of Ulcerative Colitis and Crohn's disease in a real world setting.

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This study included 122 IBD patients (61 UC, 61 CD) treated with step-up approach of medical therapy over 18 months. Targets assessed included clinical response (short-term) at 4-8 weeks, biomarker normalization and clinical remission (intermediate-term) at 3 months, endoscopic remission and improvement in quality of life (long-term) at 6-9 months. Data on patient demographics, disease characteristics, therapeutic regimens, and adverse events were analysed using statistical methods.

Results: Short-term clinical response was achieved in 91.8% of UC patients and 86.9% of CD patients. Clinical remission was seen in 80.3% of UC and 70.5% of CD cases at 3 months. Biomarker normalisation was seen in 78.68% of UC and 68.85% of CD patients. Mucosal healing was observed in 31.2% of UC and 36.1% of CD patients at 6-9 months. Quality of life goals were met by 66% of UC and 67% of CD patients. Deep remission (clinical along with endoscopic remission) was achieved in 29.5% of UC and 32.8% of CD cases. Biologics and non-biologics showed no significant difference in achieving therapy goals. Baseline severity impacted outcomes, with severe UC patients struggling more to meet both short-term and long-term targets compared to those with milder forms ($p<0.05$).

Similarly, mild and moderate CD patients achieved clinical remission better than severe CD patients ($p<0.05$). Multivariate analysis with logistical regression showed that in Crohn's disease, baseline CDAI score predicted the likelihood of clinical remission ($p=0.011$). Baseline SES-CD score predicted the likelihood of achieving mucosal healing ($p=0.045$). Colonic CD was more likely to achieve mucosal healing than other sub-types ($p=0.02$). Sub-group analysis showed that elderly and pediatric populations, did not show statistically significant variations in treatment success.

Conclusion: Findings from our study suggest that the step-up approach can achieve outcomes that are broadly comparable to those reported in top-down studies from Western settings. While the top-down strategy is traditionally associated with early disease control, our results highlight that a carefully monitored and timely step-up approach can also be effective, particularly in real-world, resource-limited environments. These insights support the need for individualised treatment strategies that consider both disease severity and contextual healthcare factors.

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Disclosure: Nothing to disclose.

PP0438

2D-SHEAR WAVE ELASTOGRAPHY INTESTINAL ULTRASOUND: ACCURACY AND REPRODUCIBILITY AS PREDICTOR OF ACTIVE ULCERATIVE COLITIS

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Introduction: Active ulcerative colitis (UC) is typically assessed through clinical manifestations, laboratory examinations, and colonoscopy, which also determine the disease's severity.

An easy-to-use and patient-friendly technique for early clinical suspicion confirmation is intestinal ultrasonography (IUS). Ancillary technologies such as elastographic techniques have recently become available and can be used in conjunction with traditional IUS.

Aims & Methods: The study aim was to assess the accuracy and reproducibility of intestinal 2D-shear Wave Elastography (2D-SWE) as a predictor of disease's activity and mucosal healing (MH). 2D-SWE was performed in 34 consecutive patients with active disease confirmed by colonoscopy. The 2D-SWE measurements (n. 10 with IQR/Med < 30%) was taken in the sigmoid colon by two independent operators at basal and 3 months after the starting of a new treatment regimen. The correlation between 2D-SWE and the Mayo endoscopic score (MES) was investigated along with clinical and conventional ultrasound parameters such as Bowel Wall Thickness (BWT) and Limberg score (LS).

Results: 2D-SWE values (kPa) positively correlated with MES with a statistically significant difference between mucosal healing (MH) and active disease groups ($p = 0.003$). The AUROC (95% CI) for assessing mucosal healing by 2D-SWE was 0.84 (95% CI: 0.70-0.97). Using a cut off value of 21.5 kPa, sensibility and the specificity for mucosal healing were 0.62 and 0.99, respectively. The median 2D-SWE values obtained by the two independent operators were 19.7 kPa (IQR 16.2-26.8) and 17.7 kPa (IQR 14.4-23.7) respectively. 2D-SWE was also positively correlated with rectal bleeding, LS and BWT (r 0.38 $p=0.027$; r 0.43 $p=0.014$; r 0.33 $p=0.057$) but not with fecal calprotectin and C-reactive protein.

No significant correlation was found between 2D-SWE values and clinical remission (Partial Mayo Score ≤ 2) after 3 months of a new treatment regimen ($p = 0.491$).

Conclusion: This study demonstrated the feasibility of 2D-SWE technique with an excellent inter-observer repeatability. The technique showed good accuracy in predicting mucosal healing, with increasing values correlating with disease severity. In the future, 2D-SWE could help reduce the number of colonoscopies and improve clinical decision making.

Disclosure: No discosure to declare.

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PP0439

IMPROVED PATIENT OUTCOMES AND DECREASED HEALTHCARE UTILISATION WHEN USING A CARE PATHWAY FOR INFLAMMATORY BOWEL DISEASE: RESULTS OF THE MULTICENTRE COHORT STUDY IBD VALUE

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Introduction: Eight hospitals in the southwest of the Netherlands collaborate to provide high quality, uniform care for inflammatory bowel disease (IBD). To achieve this, a care pathway (CP) was implemented for treating IBD with biologics and small molecules (1).

The CP is a decision-support tool that provides guidelines for medication prescribing, procedures, treatment adjustments, and recommendations for follow-up.

Aims & Methods: This study aimed to evaluate the effect of a CP on patient outcomes and healthcare utilization.

A longitudinal multicentre non-randomised cohort study with a baseline observation period (December 2020 – December 2021) and a follow-up period (March 2022 – March 2023) was conducted, in which adult patients with IBD treated with biologics or small molecules were included. Outcomes were collected according to the International Consortium for Health Outcomes Measurement (ICHOM) standard set from electronic health records or through validated questionnaires. Missing data were handled using single imputation nested within a bootstrap procedure. Effect of the CP on these outcomes were analysed with case-mix adjusted (generalized) linear mixed models.

Results: A total of 1,173 patients participated in the study, with the majority being female (55%) and a median age of 45 years (IQR 33-58). Most patients had Crohn's disease (64%). The CP was associated with a 0.22 increase on the ibd-control-8 score (β=0.22, p=0.28), and an increase on the probability of achieving endoscopic/radiologic (β=0.14, p=0.29), biochemical (β=0.18, p=0.22) and patient-reported remission (β=0.19, p=0.21). Probability of clinician-reported remission decreased slightly (β=-0.09, p=0.59). Emergency room visits (β=-0.36, p=0.13), hospital admissions (β=-0.41, p=0.10), and length of stay (β=-0.09, p=0.56) reduced, although these associations were not significant. The CP had no effect on quality of life (β=0.00, p=0.83), anaemia probability (β=-0.01, p=0.95) or on nutritional status (β=-0.01, p=0.34).

Conclusion: A CP for the management of IBD with biologicals and small molecules has clinically significant potential to reduce healthcare utilization, while improving patient-reported disease control, and disease remission. These results emphasize that clinical care pathways can improve care and reduce healthcare utilisation for complex diseases. However, the adherence to the CP and whether these results lead to a reduction in healthcare costs should be further investigated.

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PP0440

FECAL CALPROTECTIN AND FECAL ALPHA-1 ANTITRYPSIN – IS THERE A CORRELATION BETWEEN THESE TWO INFLAMMATORY MARKERS?

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Introduction: The biomarker fecal calprotectin (FC) is an efficacious tool for evaluating the level of disease activity in Crohn's and Ulcerative colitis, as well as for discriminating between inflammatory bowel disease (IBD) and irritable bowel syndrome (IBS). Fecal Alpha-1 Antitrypsin (A1AT) is an excellent marker for intestinal protein loss and for an increased mucosal permeability.

Aims & Methods: The aim of this investigation was to determine the relationship between these two inflammatory fecal markers.

The study was performed retrospectively. We've collected test results for both markers between 2018 and 2022 from 459 patients who performed both tests (67.4% females; mean age 35.1±22.7). Linear correlation evaluation between FC and A1At was performed using the Pearson Correlation Coefficient (PCC).

Results: Out of the 459 patients, 274 were diagnosed with IBD, 61 with IBS and the rest with other, non-GI disorders. Out of these, 37.7% were positive for both markers, 45.3% were negative for both markers, 7.6% were positive for A1AT and not FC and 9.4% were positive for FC and not A1AT. Linear correlation between the two markers using six categories for data scoring, was found to be significant by PCC test (R=0.83, P values < 0.001) and sensitivity and sensitivity, specificity and accuracy were 80.1%, 85.6% and 83.0% respectively. Out of the 274 patients in the IBD group, 39.2% were positive for both markers, 45.8% were negative for both markers, 5.4% were positive for A1AT and not FC and 7.6% were positive for FC and not A1AT. Linear correlation between the two markers was found to be significant by PCC test (R=0.87, P values < 0.0001) and sensitivity, specificity and accuracy were 83.7%, 89.8% and 86.9% respectively.

Conclusion: Our study demonstrates that FC had a significant relationship with A1AT in patients with IBD and IBS. Furthermore, A1AT might be used as a surrogate marker for FC in the diagnosis of IBD, IBS-IBD discrimination and disease severity and treatment follow up. In addition to that, both markers used together may be used as a new clinical score for the assessment of IBD and IBS severity.

Disclosure: Nothing to disclose.

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PP0441

EVALUATING SARCOPENIA IN ACUTE HOSPITALIZED IBD PATIENTS: A PROSPECTIVE OBSERVATIONAL STUDY

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Introduction: Sarcopenia, defined as the loss of skeletal muscle mass (SMM) with a decrease in functional strength, has been increasingly linked to adverse disease outcomes in chronic conditions such as malignancy and hepatic disease. However, the prevalence of sarcopenia and its role in influencing outcomes in inflammatory bowel disease (IBD) remains understudied particularly in hospitalized patients with acute flares. In our study we intended to assess for sarcopenia prevalence in IBD inpatients using functional measures (Hand grip Strength) and CT evaluation of skeletal muscle index.

Aims & Methods: IBD patients were prospectively assessed during their admission with IBD flares. Sarcopenia was assessed via Hand Grip Strength using Jamar digital dynamometer. Skeletal muscle index (SMI) was assessed in patients who had CT scans using Mirada software. Other anthropometric measurements were also collected including BMI, calf circumference and waist circumference. Routine clinical scores and blood tests were monitored. Outcomes were evaluated including need for biologic treatment and risk of IBD-related surgery. Statistics was done via R V (4.5.0).

Results: 49 IBD patients were included. 21/49(43%) were CD, 27/49(55%) and one patient had IBD-U. The median age was 37(27-55). Females were 24/49(49%). The median BMI was 23.5 kg/m^2 (20.6-27.9) for whole cohort (Female = 22.7 kg/m^2 (20.3-28), Male = 24 kg/m^2(20.9-25.8)).The median calf circumference for females was 35 (30-39) and for males was 36(32.5-38.5). The median hand grip strength was 18.8 kg (16.5-23.9) for females and 35 kg (27.5-42) for males. 61.2% (30/49) of patients had low hand grip strength (below the 25th centile for their age according to PENG guidance). 22.4% (11/49) were sarcopenic according to the ESWOP2 criteria including 5 females and 6 males.. 35 patients had CT scans done including 19/24(79.1%) females and 16/25(64%) males. 65.7%(23/35 (13 females and 10 males)) had a low BMI-adjusted SMI at L3 on their CT. 71.4% (35/49) patients received biologics during admission. 6.1% (3/49) patients had surgery during admission including 2 subtotal colectomies and 1 diversion ileostomy.

Conclusion: Sarcopenia is prevalent in acutely hospitalized IBD patients, with 61% showing low hand grip strength and 66% exhibiting low SMI on CT. These findings highlight the need for routine screening for sarcopenia and early intervention. Further research is required to assess the impact of sarcopenia on clinical outcomes and explore targeted nutritional and rehabilitation strategies

Disclosure: Nothing to disclose.

PP0442

A DIFFERENT WAY TO DETECT THE ADDED VALUE OF PROACTIVE TDM IN DECISION MAKING FOR IBD PATIENTS TREATED WITH IV INFLIXIMAB

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Introduction: The clinical benefit of Proactive Therapeutic Drug Monitoring (TDM) for infliximab and other monoclonal antibodies in Inflammatory Bowel Disease (IBD) remains a subject of ongoing debate. This study aimed to assess the real-world impact of broadly applying proactive TDM in unselected, consecutive IBD patients receiving infliximab infusions (both induction and maintenance). We compared TDM-guided management decisions to those made by an IBD expert blinded to TDM results but fully informed otherwise.

Aims & Methods: Infliximab trough levels and anti-drug antibodies were assessed using a validated ELISA assay; Through levels above 10 mg/ml were considered therapeutic. Infliximab levels and antibodies were measured in 55 IBD patients (34 with Crohn's Disease, 21 with Ulcerative Colitis) in 77 occasions: median age 42 years, median disease duration 8.5 years, mostly bio-exposed (58%) and mostly already on dose-escalated regimens (90%). Crohn's Disease patients had mostly complicated (stricturing or penetrating) disease (74%), half had prior surgery (44%), 50% were smokers and 20% had perianal disease. Management decisions based on TDM were compared to those of a blinded expert. Patients were followed prospectively for clinical, biochemical, and endoscopic response, with endoscopic data available in 39 cases.

Results: Therapeutic infliximab levels were observed in 80% of samples and anti-drug antibodies were detected in only one patient. Following TDM-based adjustment showed most patients had clinical (80%), biologic (89%) and endoscopic (61%) improvement compared to baseline assessment. Concordance in decision making was achieved in 58% of occasions. Therapy was discontinued in 11 patients (blinded expert concurred in 3/11,27%), the same dose was maintained in 53 patients (agreement occurred in 36/53, 68%), and dose modification in 13 patients (agreement was seen in 6/13, 46%).

Conclusion: Excellent results were achieved in this difficult to treat group of IBD patients treated with iv infliximab, possibly aided by proactive TDM. Without proactive TDM a small but significant percentage of them would risk undertreatment, either by not optimizing or by staying on redundant therapy. Discrepancies between TDM-based decisions and expert clinical judgment suggest that TDM provides distinct and actionable information that may prevent both under- and overtreatment. Future studies should estimate the impact of proactive TDM in different clinical scenarios, mostly in patients with severe disease.

Disclosure: Nothing to disclose.

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PP0443

ALEXITHYMIA IN IBD: A NEW MANAGEMENT TOOL FOR MULTIDISCIPLINARY TARGETED CARE

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Introduction: Psychological comorbidities such as anxiety and depression are common in patients with inflammatory bowel disease (IBD). Alexithymia, defined by difficulty in identifying and describing emotions, may increase symptom perception and impact disease outcomes. Its relationship with psychological distress and clinical activity remains unclear, particularly across IBD phenotypes.

Aims & Methods: This study aimed to explore the correlations between alexithymia, clinical variables, and psychopathological symptoms in patients with inflammatory bowel disease (IBD), and to compare profiles between Crohn's disease (CD) and ulcerative colitis (UC). Patients aged 18–70 years with confirmed CD or UC were recruited from outpatient and day-hospital settings. Exclusion criteria included diagnosed psychiatric or neurological disorders, malignancy, or recent head trauma. Participants completed the Toronto Alexithymia Scale (TAS- 20), the Symptom Checklist-90-Revised (SCL-90-R) for psychological distress, and the Hospital Anxiety and Depression Scale (HADS), including both anxiety (HADS-A) and depression (HADS-D) subscales. A cut-off score of ≥51 on the TAS-20 was used to define the presence of alexithymia. Clinical and demographic data were collected, including disease activity indices, disease phenotype, endoscopic findings, extraintestinal manifestations, and current therapies. Descriptive statistics were reported as mean ± standard deviation. Analyses included t-tests, Pearson correlations, ANCOVA, and multivariate linear regression. $p<0.05$ considered significant.

Results: A total of 103 patients with IBD were included (CD = 45; UC = 58). The mean TAS-20 score was 45.0 ± 11.0 in CD and 42.7 ± 10.4 in UC, with no significant difference between groups. The overall prevalence of alexithymia (TAS-20 ≥ 51) was 24.3%, with comparable rates in CD (22.2%) and UC (25.9%, $p = 0.845$). In CD, TAS-20 scores were significantly correlated with endoscopic disease activity (CDEIS: $r = 0.35$, $p = 0.023$) and with the presence of fistulas or abscesses ($r = 0.38$, $p = 0.042$). No significant correlations were observed with CDAI, disease phenotype, or perianal disease alone. In UC, clinical disease activity independently predicted alexithymia (SCCAI: $\beta = 7.29$, $p = 0.023$). With regard to psychological symptoms, TAS-20 scores were significantly correlated in CD with HADS-D ($r = 0.58$, $p < 0.001$), HADS-A ($r = 0.34$, $p = 0.021$), SCL-90 GSI ($r = 0.37$, $p = 0.012$), and paranoid ideation ($r = 0.43$, $p = 0.003$). In UC, significant correlations were also observed with HADS-A ($r = 0.52$), HADS-D ($r = 0.39$), and SCL-90 GSI ($r = 0.45$), all with $p < 0.005$. In multivariate analysis, the SCL-90 GSI score independently predicted alexithymia in UC ($\beta = 11.78$, $p < 0.001$). The effect of diagnosis (CD vs UC) on TAS-20 scores was not significant ($p = 0.079$) after adjusting for anxiety and depression.

Conclusion: Alexithymia is a common trait in IBD, with comparable prevalence in CD and UC. In CD, it is associated with endoscopic disease severity and fistulizing complications, while in UC, with clinical activity. In both groups, psychological distress, especially depression, is a key correlate. These findings underscore the value of assessing emotional processing in patients with active or complicated IBD to support more targeted multidisciplinary care.

Disclosure: Nothing to disclose.

PP0444

VALIDATING CALIBRATION OF AN ARTIFICIAL INTELLIGENCE ASSESSMENT OF ENDOSCOPIC SEVERITY IN ULCERATIVE COLITIS

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Introduction: Regulatory guidance recommends the endoscopy subscore as the index to assess the endoscopic component of the primary endpoint in ulcerative colitis (UC) trials[1]. Inter-reader variability in assessments may impact the reliability of trial results[2,3]. Currently, there is no metric in place to assess the certainty by which a reader is assigning an endoscopy subscore. Machine learning (ML) provides an opportunity to assess the endoscopy subscore and provide a measurement of its certainty in a standardized manner. Artificial Intelligence Assessment of Endoscopic Severity (AI-ES) accurately assesses the endoscopy subscore[4].

The objective of this study is to evaluate the calibration of AI-ES - how well its predicted probabilities reflect true likelihoods - to assess the reliability of its measurement of certainty in endoscopy subscore assessments in UC trials.

Aims & Methods: AI-ES is a deep learning algorithm that assesses the endoscopy subscore in UC endoscopic videos. AI-ES measures probability for the four ordinal endoscopy subscore classes. The endoscopy subscore with the highest probability is assigned as the final score by AI-ES. We assessed calibration on a holdout test set of 639 videos (~25%) from the Phase 3 induction trial for mirikizumab in UC (NCT03518086). Videos had a 2+1 centrally read endoscopy subscore, randomly selected from week 0 and 12 with a distribution of endoscopic severity similar to the overall study population. Calibration plots were generated across endoscopy subscore classes with probabilities grouped into septiles (~100 videos per group) for primary analysis and deciles for confirmation. Brier scores, ranging from 0 (perfect calibration) to 1 (worst calibration), were calculated, with values <0.25 considered informative[5].

Results: AI-ES demonstrated strong calibration, with Brier scores below <0.25 for each endoscopy subscore (0: 0.037, 1: 0.082, 2: 0.162, 3: 0.112). The Brier score for evaluation of endoscopic improvement (0,1 vs 2,3) also showed excellent calibration (0.066). Findings were consistent when assessing probabilities by deciles.

Assessment	Brier Score
Ordinal endoscopy subscore	
Endoscopy subscore 0	0.037
Endoscopy subscore 1	0.082
Endoscopy subscore 2	0.162
Endoscopy subscore 3	0.112
Endoscopic improvement	
Endoscopy subscore 0, 1 vs 2, 3	0.066

Conclusion: Whereas data on the certainty of human readers in endoscopy subscore assessments are elusive, AI-ES is calibrated across all endoscopy subscore classes, providing reliable data on score probabilities. This novel measurement of certainty by AI-ES added to the score assessment may enable novel AI-based multi-reader or consensus workflows in trials, potentially improving the reliability of UC endpoint assessments.

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PB, MH, JL, EY, DT, DC, and SB are employees of Iterative Health and may hold stock or stock options in Iterative Health.
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PP0445

REAL-WORLD USE OF THE MODIFIED IBD DISK TOOL FOR EVALUATION OF EATING DISORDERS IN YOUNG ADULTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: The IBD-Disk is a widely used, self-administered visual analogue tool for measuring disability in patients with inflammatory bowel disease (IBD). However, it does not capture key aspects of daily life, particularly diet and nutrition, which are critical for IBD patients and may significantly influence disability.
Aims & Methods: This cross-sectional study explored the prevalence of moderate to severe disability in young adults with IBD and evaluated the use of a modified IBD-Disk (mIBD-Disk) to screen for eating disorders and food avoidance. Patients previously diagnosed with IBD during childhood and adolescence (<18 years old), and currently followed at two adult IBD academic centers, were consecutively enrolled.
All participants were assessed for disability and screened for eating disorders (ED) and avoidant restrictive food intake disorder (ARFID). Restrictive eating behaviours were evaluated using the Nine Item ARFID Screen (NIAS), which assesses three domains: picky eating, low appetite, and food-related fear with a score >23 suggesting probable ARFID. The Eating Attitudes Test (EAT-26) was used to screen for ED, with a score ≥20 indicating a positive result. Disability was measured using the IBD-Disk with a score >40 indicating moderate to severe disability) and the modified IBD-Disk (mIBD-Disk)[1], which consisted of the original 10 items plus an additional question on dietary habits, for a total of 11 items
Results: A total of 144 patients (50.7% with Crohn's disease) were included, with a median age of 24 years (range: 21–28), and 54.2% were female. One-third (34.7%) had clinically active disease, and the median BMI was 21.8 (range: 20.1–24.2). Moderate to severe disability, as assessed by the IBD-Disk, was observed in 38.9% (56/144) of the cohort, and the median mIBD-Disk score was 33 (IQR: 16–61).
ARFID risk was identified in 24 participants (16.7%), with food-related fear being the most frequent domain (13.2%). The median EAT-26 score was 5 (range: 3–12), with ED present in 17 patients (11.8%).
In both univariate and multivariate analyses, moderate to severe disability (OR: 4.03; 95% CI: 1.23–13.17; p = 0.02) and a score greater than 4 on question 11 of the mIBD-Disk related to dietary habits (OR: 3.49; 95% CI: 1.06–11.47; p = 0.04) were independently associated with ARFID risk. ROC curve analysis showed that the mIBD-Disk Q11 had an 82% diagnostic accuracy (95% CI: 0.73-0.91) for predicting ARFID.
Conclusion: Patients diagnosed with IBD in childhood and adolescence have an increased prevalence of moderate to severe disability. Integrating the mIBD-Disk into clinical practice could provide an effective tool to identify patients at risk for eating behaviours and food avoidance, supporting timely referrals for nutritional and psychological interventions
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Disclosure: Nothing to disclose.

PP0446

RECURRENCE OF CROHN'S DISEASE AFTER ILEOCECAL RESECTION – ASSOCIATION WITH EOSINOPHILIC PLEXITIS IN RESECTED SPECIMEN

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Introduction: Crohn's disease (CD) is increasing worldwide. Medical treatment and endoscopic techniques have reduced the need for surgery to some extent. Recommendation for postoperative surveillance is colonoscopy after six months. Since colonoscopy is invasive and resource-demanding, it is appealing to add or replace it with other methods to risk stratify the patients for recurrence. Examining the proximal or distal resection margin, mainly for plexitis, has been reported to predict a higher risk of recurrence. Eosinophils reside to an increased degree in the mucosa of CD patients, communicating with mast cells and nerve endings, contributing to increased inflammation, affected barrier function and fibrosis. Also, the staining methodology is readily available for eosinophilic plexitis.
Aims & Methods: The aim was to examine the correlation between eosinophils in the myenteric or submucosal plexa and endoscopic, clinical and surgical recurrence. From a larger cohort of CD patients who had an ileocecal resection (n=244), a smaller group who also had a follow-up

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with endoscopy was selected for a closer examination (n=20). Non-IBD-patients (colon cancer) served as controls (n=6). Fresh slides were stained with an enhanced hematoxylin-eosin staining for eosinophils combined with immunohistochemistry for S100 to visualise plexa. Eosinophilic plexitis was defined as eosinophils inside the plexus or in its direct proximity. The plexa were identified and defined as a cluster of 3 or more neurons and glia cells, submucosal or myenteric. The quantification of eosinophils was done in light microscope at 40x magnification. Statistics were done using Mann-Whitney U-test.

Results: In this pilot study, 20 CD patients were examined: 9 with endoscopic recurrence and 11 without. Six control patients were included. During follow-up, 10 patients had clinical recurrence and 4 patients had surgical recurrence. Five patients were lost to follow-up. Eosinophilic plexitis in either myenteric or submucosal plexa in the proximal resection margin was correlated with increased endoscopic recurrence, p=0.05. It was also a significant difference between patients with endoscopic recurrence compared with controls, p=0.004.

Conclusion: In this small pilot study, we found a correlation between eosinophilic plexitis in the proximal resection margin and endoscopic recurrence. Eosinophilic plexitis could be a prognostic tool for risk assessment of recurrence in the larger cohort.

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PP0447

CLINICAL PREDICTIVE FACTORS FOR SHORT BOWEL SYNDROME IN A COHORT OF PATIENTS WITH CROHN'S DISEASE

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Introduction: Short bowel syndrome (SBS) is a critical clinical condition that occurs when, following multiple intestinal resections, the residual small bowel in continuity is ≤200 cm. SBS is associated with intestinal failure (IF) when the intestinal absorptive function is impaired, requiring intravenous administration of nutritional or electrolyte supplements. Crohn's disease (CD) is one of the most frequent causes of SBS in adults

Aims & Methods: This study aimed to identify clinical predictive factors of SBS in CD patients (pts). We conducted a prospective study, enrolling consecutive outpatients with CD. Statistical univariate comparison for the primary outcome was assessed for continuous variables by the Mann-Whitney U and the Kruskal-Wallis ANOVA tests. Categorical variables were compared using the chi-square test. We also evaluated a multivariate logistic regression model to identify independent predictors of SBS.

Results: We enrolled 169 pts with CD (47.3% male, mean age 43.5 years [±14.1]). 45.5% of them were smokers; extraintestinal manifestations (EIMs) were present in 45.5% of pts, and 67.5% of pts had bowel resections (26.6% of them with more than one surgery). The median disease duration and median time between diagnosis and the first resection were 9 years (IQR 4-17) and 6.5 years (IQR 1-13), respectively. At baseline, 101 pts were on anti-TNF therapy, and considering the course of the disease, 20.1% of the pts were exposed to ≥3 biologics. 10% of pts had SBS and 5.3% IF.

In univariate analysis, comparing patients with and without SBS, the following variables were statistically different: disease duration (p=0.000), upper gastrointestinal disease location (p=0.001), ileocolonic location (p=0.008), stenosing behavior (p=0.040), perianal disease (p=0.000) and time from CD diagnosis to initiation of biological therapy (p=0.002). At multivariate analysis, disease duration (OR 1.2, 95% CI 1.12-1.42, p=0.000) and perianal disease (OR 4.3, 95% CI 0.99-19.27, p=0.050) resulted independently associated with the development of SBS.

Conversely, the time from diagnosis of CD to start biological therapy resulted in an inverse association with the onset of SBS (OR 0.8, 95% CI 0.74-0.97, p=0.016). This latter finding could be due to the late use of biological therapy in patients with less severe disease.

Conclusion: Our data identify clinical characteristics allowing gastroenterologists to predict the risk of developing SBS in a cohort of pts with CD. Further studies would be needed to confirm our results.

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PP0448

INTESTINAL ULTRASOUND-BASED EARLY ASSESSMENT OF BIOLOGIC THERAPY RESPONSE IN INFLAMMATORY BOWEL DISEASE

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Introduction: Intestinal ultrasound (IUS) is a non-invasive modality for monitoring disease activity in inflammatory bowel disease (IBD). While widely used for long-term assessment, its utility in early response evaluation following biologic therapy initiation or modification remains underexplored.

Aims & Methods: This prospective study aimed to assess early ultrasonographic response at week 6 in ulcerative colitis (UC) and Crohn's disease (CD) patients undergoing induction, dose escalation, or switching of biologic therapy. A total of 24 patients (mean age: 40.1 ± 12.8 years; 54.2% female) were included. Fourteen patients (58.3%) were initiating biologic treatment, 4 (16.7%) underwent dose escalation, and 6 (25%) were switched to another biologic agent. Biologic therapies included infliximab (70.8%), adalimumab (16.7%), and others (12.5%). IUS examination was performed at baseline and at week 6.

The following parameters were assessed: bowel wall thickness (BWT), wall stratification, color doppler signal activity and mesenteric fat inflammation. IUS remission was defined as bowel wall thickness less than 3 mm and no color doppler signal. IUS improvement was defined as more than 25% decrease in bowel wall thickness without increase in color doppler signal. IUS activity assesment was performed at the same segment of the bowel at the 6th weeks.

Results: A total of 24 patients were included in the study, with a mean age of 40.1 ± 12.8 years; 13 (54.2%) were female. Seven patients (29.2%) had ulcerative colitis (UC), and 17 (70.8%) had Crohn's disease (CD). Fourteen patients (58.3%) had newly initiated biologic therapy, 4 (16.7%) underwent dose escalation, and 6 (25%) were switched to a different biologic agent. Anti-TNF agents were used in 21 patients (87.5%), including infliximab in 17 (70.8%) and adalimumab in 4 (16.7%). The remaining 3 patients (12.5%) were treated with other biologic agents. At baseline, the most active intestinal segment on IUS was the terminal ileum in CD patients (9 patients, 52.9%) and the sigmoid colon in UC patients (4 patients, 57.1%) (p = 0.021). At week 6, 11 patients (45.8%) showed ultrasonographic improvement, 9 patients (37.5%) had no change, and 4 patients (16.7%) showed worsening. The baseline mean bowel wall thickness (BWT) of the most active segment was 5.5 ± 1.3 mm, which decreased to 4.4 ± 1.7 mm at week 6. A reduction in BWT was observed in 18 patients (75%), while an increase was seen in 6 patients (25%) (p = 0.002). In multivariate regression analysis, higher baseline bowel wall thickness was identified as an independent predictor of ultrasonographic improvement at week 6 (OR: 3.137; 95% CI: 1.005–9.790; p = 0.04).

In patients who showed improvement, bowel wall thickness significantly decreased from baseline 6.2 ± 1.5 mm to 3.7 ± 1.6 mm at week 6 (to; p = 0.003), whereas in patients who worsened, an increase in wall thickness was observed (5.3 ± 1.0 mm to 6.7 ± 1.7 mm; p = 0.068). By disease group, 5 of the UC patients (71.4%) showed ultrasonographic improvement, 1 (14.3%) remained unchanged, and 1 (14.3%) worsened. Among CD patients, 6 (35.3%) showed improvement, 8 (47.1%) had no change, and 3 (17.6%) worsened. However, there was no statistically significant difference in response patterns between the UC and CD groups (p = 0.244).

Conclusion: Intestinal ultrasound at week 6 effectively detected early treatment response, with baseline bowel wall thickness emerging as a predictor of improvement. These results support its use in routine early monitoring of IBD patients on biologic therapy.

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PP0449

FECAL CALPROTECTIN IN SMALL BOWEL CROHN'S DISEASE: IS IT A RELIABLE TOOL?

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Introduction: The correlation between fecal calprotectin (FC) levels and small bowel (SB) inflammation in patients with Crohn's Disease (CD) remains a subject of debate.

Aims & Methods: This study aims to investigate the relationship between FC levels and SB inflammatory activity in patients diagnosed with SB CD. We conducted a retrospective cohort study involving patients with SB CD who underwent small bowel capsule endoscopy (SBCE) and FC testing. Patients with colon inflammation were excluded.

Participants were categorized into four groups based on their SB inflammatory status: no inflammation (all three SB tertiles showed Lewis Score (LS) <135); proximal SB inflammation (first and/or second SB tertiles had LS ≥ 135, with no inflammation in the third SB tertile); distal SB inflammation (third SB tertile had LS ≥ 135, with no inflammation in the proximal SB); pan-SB inflammation (both proximal and distal SB with inflammation).

Results: A total of 87 patients were included in the study, with a median age of 35 years, 75.9% being female. The distribution of inflammation among the patients was as follows: no inflammation in 21.8%; proximal inflammation in 4.6%; distal inflammation in 33.3%; pan-SB inflammation in 40.4%.

FC median values exhibited an ascending trend along the SB axis: no inflammation 58 µg/g (33-180); proximal SB inflammation 65 µg/g (30-128); distal SB inflammation 122 µg/g (42-507); pan-SB inflammation 400 µg/g (205-852).

Statistically significant differences in FC levels were observed between pan-SB inflammation *vs.* isolated proximal inflammation (p = 0.014), and isolated distal inflammation (p = 0.012).

Additionally, FC showed a positive correlation with LS in the second (r = 0.464, p<0.001) and third tertiles (r = 0.435, p<0.001), but no correlation was noted in the first tertile.

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FC demonstrated good predictive capacity for identifying inflammation in the distal SB and pan-SB *vs.* no inflammation and proximal inflammation, with an area under the curve of 0.790 ($p < 0.001$) and an optimal cutoff value of 228 µg/g.

Conclusion: FC levels correlate positively with the presence of SB lesions in the second and third tertiles of the small bowel. However, it is not a reliable marker for detecting inflammation in the first tertile, highlighting the importance of conducting a SBCE in these patients.

Disclosure: Nothing to disclose.

PP0450

PERCEPTION AND EXPERIENCE OF PREGNANCY IN INFLAMMATORY BOWEL DISEASE PATIENTS

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Introduction: Inflammatory bowel diseases (IBD) often affect women of childbearing age, complicating pregnancy management. These patients face medical and psychological challenges.

Aims & Methods: The aim of our study was to investigate the perception and the experience of pregnancy in IBD patients. A cross-sectional study was conducted between August and September 2024. A structured questionnaire was distributed through the national Tunisian association of IBD patients. It aimed to assess the impact of pregnancy in Tunisian women with IBD, focusing on complications, treatment adjustments and psychological effects.

Results: In total, 63 patients with IBD (Crohn Disease: 47.6 %, ulcerative colitis: 52.4%) responded to the questionnaire. Demographically, 1.6% were aged 18-25, 49.2% were 26-35, 31.7% were 36-45 and 17.5% were over 45 years old. In terms of educational level, 71.4% had completed university studies. Disease duration was over 10 years for 49.2% of the patients, 5-10 years for 27%, and less than 5 years for 23.8%. Pregnancy was planned in 68.3% of the cases.

A significant proportion of patients 76.2% were afraid that IBD may lead to a complicated pregnancy and 60.3% worried about pregnancy's impact on disease. To address these concerns, 74.6% sought information through internet and social media.

Complications occurred in 55.6% of patients, mainly disease flares (34.9%), nutritional deficiencies (15.9%) and prematurity (14.3%). Colonoscopies were required in 20.6% of the pregnant patients. Regarding treatment, 46% were on aminosalicylates, 17.5% on azathioprine, 7.9% on anti-TNF, 3.2% on corticosteroids and 25.4% received no treatment during pregnancy.

Therapeutic adjustment was made in 39.7% of the patients and a good compliance was reported in 54%. Concerns about treatment side effects affected 69.8% of the participants and 73% sought more information on treatment risks. Patients evaluated pregnancy's impact on quality of life as low in 47.6% of cases, moderate in 33.3% and high in 19%.

Disease perception improved in 30.2% of the participants, remained unchanged in 49.2% and worsened in 20.6%. In terms of mental health, 49.2% of the patients had anxiety, 11.1% had depression, and 28.6% reported general well-being.

Patients with history of surgical treatment were more likely to have planned pregnancy (p=0.013). Individual research over the disease was significantly associated with long disease course (p=0.011) and high educational level (p=0.014). Low adherence to treatment was significantly higher in patients with high concerns about side effects (p=0.013). Regarding expectations, 63.5% of the patients expressed a desire for better collaborative care between obstetricians and gastroenterologists and 52.4 % would like to have psychological support.

Conclusion: This study highlights the need for holistic care to improve the management and quality of life for pregnant women with IBD.

Disclosure: Nothing to disclose.

PP0451

GAPS IN DISEASE-SPECIFIC KNOWLEDGE IN INFLAMMATORY BOWEL DISEASE PATIENTS AT A TERTIARY CARE CENTER

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Introduction: Disease-specific knowledge plays a central role in chronic disease management. In patients with inflammatory bowel disease (IBD), limited knowledge is linked to poorer perceived health and increased discomfort (1).

Aims & Methods: This study assessed disease-specific knowledge among adult patients with IBD at a tertiary center and defined existing knowledge gaps. Factors associated with higher or lower levels of knowledge were explored.

The retrospective cohort study included 392 IBD patients who completed a comprehensive questionnaire between March 2020 and April 2021 (2). Questionnaire responses were analyzed alongside clinical records to quantify disease-specific knowledge. A scoring system was developed to quantify disease-specific knowledge (0–8 points), based on correct diagnosis, Montreal classification, complications, and current therapies. Scores were categorized as low (0–5) or high (6–8) knowledge levels. Associations between knowledge levels and disease-related, sociodemographic, and psychosocial factors were analyzed using logistic regression analysis.

Results: While most patients correctly identified their diagnosis (kappa: 0.863, CI: 0.816–0.905, $p < 0.001$) and advanced treatment (e.g. infliximab: kappa: 0.959, CI: 0.925–0.983, $p < 0.001$), knowledge gaps were observed regarding disease location (e.g. isolated proctitis in ulcerative colitis: kappa: 0.101, CI: –0.051–0.341, $p = 0.128$), complications, comorbidities, and certain other medications. In the univariate analysis a higher knowledge level was found in patients with Crohn's disease, younger age at diagnosis, in patients with no disease activity, and treatment with biologics, in those receiving disability pension and in patients with a high Perceived Stress Questionnaire (PSQ) score for worries (table).

Items (reference variable/dummy variable)	Overall knowledge level, OR (95% CI), p value
Diagnosis (CD/UC)	0.243 (0.133-0.445), p<0.001
Age at diagnosis	0.969 (0.950-0.989), p=0.002
Physician global assessment (no activity/moderate activity)	0.125 (0.047-0.333), p<0.001
Self-assessment (no activity/moderate activity)	0.269 (0.125-0.579), p<0.001
Treatment includes any biologics (no/yes)	2.381 (1.331-4.259), p=0.003
PSQ-Score worries	1.013(1.004-1.022), p=0.008
Receiving disability pension (no/ yes)	0.088 (0.020-0.382), p=0.001

Table: Univariate logistic regression analysis for items significantly associated with higher knowledge levels.

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A younger age at diagnosis ($p = 0.028$) and having Crohn's disease ($p < 0.001$) were independently associated with higher knowledge levels whereas active disease was associated with lower disease-specific knowledge.

Conclusion: Significant disease-specific knowledge gaps were identified, particularly concerning disease location, complications, and comorbidities. Targeted patient education should focus on addressing these specific deficits in susceptible groups.

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Disclosure: Nothing to disclose.

PP0452

PREDICTORS OF INDETERMINATE QUANTIFERON-TB AMONG PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: QuantiFERON-TB is replacing the tuberculin skin test for latent TB screening in IBD patients before starting biologic therapy(1). Indeterminate results may cause delays in treatment initiation and necessitate further evaluation(2).

Aims & Methods: This is a retrospective chart review study conducted at King Fahad Medical City, Riyadh , Saudi Arabia, to investigate factors associated with indeterminate QuantiFERON-TB results in patients with inflammatory bowel disease (IBD).

Results: A total of 300 participants were included, with 53.2% diagnosed with Crohn's disease (CD) and 46.8% with ulcerative colitis (UC). Indeterminate QuantiFERON-TB results were observed in 4.7% of cases. None of these patients showed positive seroconversion or developed active tuberculosis during the follow-up period up to the study's completion.

Factor	Positive Association	p-value
Geographical Region	Northern region of Saudi Arabia (14.3% indeterminate)	0.021
Chest X-ray Findings	Abnormal X-ray (higher likelihood of indeterminate results)	<0.001
White Blood Cell Count	Elevated WBC (31.8% indeterminate)	<0.001
Albumin Levels	Hypoalbuminemia (17.4% indeterminate)	<0.001
C-Reactive Protein (CRP)	Elevated CRP (8.3% indeterminate)	0.008
Steroid Use	Steroid use (3.7-fold increased risk of indeterminate results)	0.008
IBD Subtypes	UC pancolitis (highest proportion of indeterminate results)	0.834

Conclusion: Indeterminate QuantiFERON-TB results were significantly associated with factors like geographical region, elevated WBC, hypoalbuminemia, elevated CRP, and steroid use, highlighting the need to consider these variables in latent TB screening for IBD patients. Importantly, none of the patients developed active TB or showed positive seroconversion.

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Disclosure: Nothing to disclose.

PP0453

BIOLOGICAL TREATMENT SEQUENCES AND MUCOSAL HEALING IN ULCERATIVE COLITIS: A REAL-LIFE SINGLE-CENTER COHORT

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Introduction: The therapeutic landscape of ulcerative colitis (UC) has expanded significantly in recent years, yet real-world data are still needed to guide optimal treatment sequencing and improve short and long-term outcomes. Mucosal healing (MH) is a well-established therapeutic goal in UC management, closely associated with improved long-term prognosis.

Aims & Methods: The aim of this study was to describe real-life treatment sequences in UC patients receiving biological therapies at a tertiary referral center. This retrospective analysis included UC patients treated with biological therapies between 2019 and 2023. Patients received infliximab, adalimumab, vedolizumab, ustekinumab, or other advanced therapies across up to six treatment lines. Biological treatment pathways prior to achieve MH were visualized using a Sankey diagram.

Endoscopic outcomes were evaluated one year after each biological therapy. MH, the primary outcome, was defined as endoscopic remission (Mayo UC score 0–1).

Results: Among 98 patients treated with biologics, MH was achieved in 46% (n = 45). Infliximab was used as the first-line agent in 55 patients (56%), adalimumab in 12 (12%), vedolizumab in 25 (25%), and golimumab in 3 (3%). Thirty-five patients (36%) achieved MH after the first-line therapy (n = 20 with infliximab; n = 2 with adalimumab; n = 12 with vedolizumab; n = 2 with golimumab). MH rates declined with each additional line of therapy: 17.4% after second-line treatment and no patients beyond the third line. The median time to achieve MH was 24.5 months (range 6–180). Treatment sequencing analysis revealed a progressive shift toward vedolizumab and ustekinumab in later lines, with infliximab remaining the most frequent first-line agent.

Conclusion: This study highlights a progressive decline in MH rates with successive lines of biological therapy in UC in a real-life cohort of patients. These findings emphasize the need for predictive tools to personalize treatment strategies in routine clinical practice.

Disclosure: Nothing to disclose.

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PP0454

OVERWEIGHT IS ASSOCIATED WITH LOW ADHERENCE TO THE MEDITERRANEAN DIET AND INCREASED DISABILITY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: The prevalence of obesity in inflammatory bowel disease (IBD) has risen, with 15–40% of adults classified as obese and an additional 20–40% as overweight[1]. Poor dietary habits and sedentary lifestyles contribute to overweight and sarcopenia, potentially worsening IBD-related disability.

Aims & Methods: This cross-sectional study aimed to evaluate the relationships between nutritional status, adherence to the Mediterranean diet, sarcopenia and disability in IBD patients. Patients with IBD attending the outpatient clinic or infusion unit of an academic tertiary center between 2023 and 2024 were consecutively enrolled. Bioelectrical impedance analysis (BIA) was used to estimate the fat-free mass index (FFMI), a recognized measure of skeletal muscle mass. Physical function was assessed via gait speed (time to walk 6 meters), and handgrip strength (HGS) measured using a Jamar dynamometer. Sarcopenia was diagnosed based on FFMI, HGS, and walking speed. Disability was evaluated using the IBD-Disk questionnaire, with scores ≥40 indicating moderate-to-severe disability. Adherence to the Mediterranean diet was assessed using the Medi-Lite score, with scores <9 reflecting low adherence. Demographic, clinical, and disease related data were collected.

Results: A total of 95 IBD patients (58% with Crohn's disease, CD), 57.8% female, with a median age of 42 years (IQR 26-58) were included. Clinical activity ,measured with the Harvey-Bradshaw index (HBI), had a median score of 6 (IQR 4–7) for CD, while the median pMayo score for ulcerative colitis (UC) was 4 (IQR 2–6).

The median BMI was 23.60 (IQR 21-27). Overweight prevalence was 43.2% (41/95). Low adherence to the Mediterranean diet was significantly more frequent in overweight patients compared to nonoverweight patients ($p<0.001$). Overweight was also associated with increased age ($p < 0.001$). Sarcopenia prevalence was 7% (7/95) and was more common in non-overweight patients than in overweight patients ($p = 0.002$). The median BMI among sarcopenic patients was 20.6 (18.5–25.3). The mean IBD-Disk score was 41 ± 23.2, with 46.3% of patients reporting moderate-to-severe disability. Overweight patients were more likely to experience moderate-to-severe disability compared to non-overweight patients [27 (65.9%) vs. 23 (42.6%), $p =0.02$]. No significant association was found between sarcopenia and moderate-to-severe disability ($p = 0.44$).

Conclusion: Overweight is associated with low adherence to the Mediterranean diet and increased disability in IBD patients. Targeted interventions to improve adherence to the Mediterranean diet and manage overweight could potentially reduce disability in IBD patients.

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Disclosure: Nothing to disclose.

PP0455

SEQUENCE TREATMENT PATHWAYS AND TRANSMURAL HEALING IN CROHN'S DISEASE

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Introduction: There is an increasing diversification in the treatment landscape for Crohn's disease (CD) driven by heterogeneity of individual phenotypes, disease courses and previous treatments. This complexity makes universal evidence-based algorithms challenging. A "treat-to-target approach" with tight inflammation monitoring after starting biologics or small molecules therapy is recommended. Intestinal ultrasound (IUS) has proven effective in monitoring treatment response in large CD cohorts. Transmural healing (TH) is a robust but puzzling endpoint that can be achieved in about one-fourth of patients treated with biological therapy and few prognostic indicators to predict it were analyzed.

Aims & Methods: We aimed to describe treatment patterns and effective therapeutic sequences to achieve TH. Secondary aim was to identify prognostic factors to predict TH. This retrospective analysis included CD patients treated with biological and small molecules therapies according to the ECCO recommendations. Patients received biological and small molecules therapies across up to four treatment lines. Biological and small molecules treatment pathways prior to achieve TH were visualized using a Sankey diagram. IUS outcomes were evaluated one year after each biological and small molecules therapy. Logistic regression analysis was performed to identify clinical predictive factors of TH, that was defined as bowel wall thickness <3 mm and no vascularization at IUS.

Results: Among 139 patients treated with biological and small molecules therapies, TH was achieved in 25.9% (n = 36). Infliximab was used as the first-line agent in 55 patients (39.6 %), adalimumab in 69 (49.6%), vedolizumab in 6 (4.4%), certolizumab pegol in 1 (0.7%), Ustekinumab in 8 (5.7%). Twenty-five patients (20%) achieved TH after the first-line therapy (n=10 with infliximab; n=9 with adalimumab; n=3 with vedolizumab; n=3 with ustekinumab). TH rates declined with each additional line of therapy. The median time to achieve TH was 25.6 months (range 12–60).

Treatment sequencing analysis revealed a progressive shift toward ustekinumab in later lines, with anti-TNFs remaining the most frequent first-line agents. Multivariate analysis showed that colonic lesion and early treatment (within 2 years from diagnosis) were associated with a higher likelihood of achieving TH [OR = 7.76, $p<0.001$; OR 6.705, $p<0.001$, respectively], while perianal disease was associated with a lower likelihood of TH [OR 0.151, $p=0.02$].

Conclusion: This study highlights a progressive decline in TH rates with successive lines of biological and small molecule therapies in CD patients and identifies several clinical factors that influence this outcome. Long-term follow-up and prospective studies specifically designed to evaluate sequencing in patients exposed to anti-TNFs are needed to guide challenging decisions.

Disclosure: Nothing to disclose.

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PP0456

DEFINITION OF THE FAECAL PLASMINOGEN ACTIVATOR INHIBITOR TYPE 1 (PAI-1) DIFFERENTIAL DIAGNOSTIC POTENTIAL VS CALPROTECTIN IN INFLAMMATORY BOWEL DISEASES AND OTHER GASTROINTESTINAL DISORDERS

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Introduction: Inflammatory Bowel Diseases (IBD) require lifelong treatment and patient monitoring. Current predictors of relapse and therapeutic success have limitations, mostly invasive, time-consuming, and expensive. The faecal biomarker can help to monitor the disease activity and therapeutic response by simple and non-invasively. Faecal Calprotectin (FC) is the gold-standard, but it has many disadvantages.

We previously described that the correlation between the faecal Plasminogen activator inhibitor type 1 (PAI-1) (FP) level and the endoscopic activity and therapeutic response promote that it could be used as a novel non-invasive faecal biomarker in IBD diagnosis. Moreover, the faecal PAI-1 protein level is stable at room temperature and 4°C.

Aims & Methods: To observe the exact faecal biomarker potential, we aimed to define the FP level of IBD patients and subjects with other gastrointestinal (GI) diseases and compare it with the FC. Faecal samples were collected from 128 patients with IBD (CD-active: 18, CD-inactive: 10, UC-active: 20, UC-inactive: 10) and other GI diseases [colorectal cancer (CRC): 20, diverticulosis [DIV]: 21, irritable bowel syndrome [IBS]: 10, adenoma: 10) and 9 healthy subjects. ELISA method was applied to define FP level, and it was validated for stool samples by us and 0.5 ng/g PAI-1 level was defined as the cut-off value. For FC observation diagnostic FC ELISA kit (Orgentec) was used. 50 mg/g cut-off value was applied at FC. Confusion matrix analysis was utilized to define Youden-index, positive predictive value (PPV), negative predictive value (NPV), accuracy, specificity, and sensitivity of the measured values.

Results: FP level was significantly higher in patients with active IBD compared to inactive and non-IBD subjects (0.64 vs 0.0 ng/g, p<0.0001). No significant differences were observed between CD-active and UC-active groups (0.46 vs 0.79 ng/g, p=0.65). In addition, FP concentrations were significantly increased in active IBD (0.76 pg/g) patients compared to inactive IBD (0.0 pg/g, p=0.0001), CRC (0.11 ng/g, p=0.02), DIV (0.05 ng/g, p=0.001), IBS (0.0 ng/g, p<0.0001), adenoma (0.0 ng/g, p<0.0001), and healthy (0.0 ng/g, p=0.0002) subjects.

However, FC showed the same pattern as FP, except at the CRC, adenoma and DIV. In these groups, there were no significant differences compared to the active IBD (487.4 mg/g) subjects (vs CRC: 147.1 mg/g, p=0.09; p=; adenoma: 159.8, p=0.056. Confusion matrix analysis defined high specificity (83.5%) and moderate sensitivity (57.9%) values (TP: 22, FP: 14, FN: 17, TN: 71, PPV: 0.61, NPV: 0.816) at the FP measurements. Nevertheless, at FC levels 35.3% specificity and 94.7% sensitivity were measured (TP: 36, FP: 55, FN: 2, TN: 30, PPV: 0.396, NPV: 0.938). Calculated Youden-index was higher at the FP (0.41) compared to the FC (0.3). On the other hand, the accuracy of the FP is 75.6% and the FC is 53.6%. Based on the results of all FP levels measured (N=299) so far, the calculated values were specificity: 78%, sensitivity: 62%, PPV: 0.63, NPV: 0.78, Youden-index: 0.4, accuracy: 72.3%.

Conclusion: Our results suggest that the FP level selectively increased in the active IBD and discriminated more specifically from the other GI diseases compared to the FC. However, the sensitivity was lower, but it was improved by the increasing patient numbers at the pooled FP levels. Thus, the FP level could be useful in the diagnosis of IBD independently or combined with FC.

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Disclosure: Nothing to disclose.

PP0457

EVALUATION OF CHARACTERISTICS OF BOWEL URGENCY IN PATIENTS WITH ULCERATIVE COLITIS

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Introduction: Bowel urgency is one of the most bothersome symptoms of ulcerative colitis (UC) and have a great impact on quality of life and psychosocial function. The Urgency Numerical Rating Scale (U-NRS) is desinged to ojectify this subjective symptom. It measures the degree of faecal urgency experienced over the past 24 hours by patients with UC on a scale of 0-10. We assumed that the scale has a large variance, as patients evaluate their own symptoms differently, the urge may be different quality for two patients with the same score.

Aims & Methods: The objective of our study was to investigate the possible components of urgency and refine the assessment of urgency. Patients with UC older than 18 years treated at our departments for at least 3 months and having complaints were enrolled in our study. The patients filled out our non-validated questionnaire with questions about fecal urgency (U-NRS), pain level, speed of urge, incontinence and lifestyle characteristics associated with incontinence. Statistical analysis was performed using STATA 18 program.

Results: 263 patients were enrolled, median age was 39.5 years, the mean of disease duration was 11.7±9.2 years. The mean U-NRS score was 3.2±3.2, 32% (84/263) had 0 score and 3% (9/263) has 10 score. 12% of the patients (n=31) felt most difficult to hold mucus, 37% (n=97) stool and 29%

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(n=76) other secretions. 8% (n=22) had urge with every stool in the last 24 hours, 36% (n=95) had less than half of defecations. The speed of urge defined more the level of urgency than the pain level (R2=0.798 p<0.001, β=0.894 vs R^2=0.805, p<0.001, β=0.117). The duration of faeces holding in the last 7 days had significant connection to U-NRS (R^2=0.211, p<0.001, β=-0.46). Patients using public restroom at least once during the last week had significantly higher score on U-NRS than other patients (4.6±3.1 vs 2.4±3 p<0.001). 12% (31/263) used diaper.

The usage of diapers was also significantly more often in patients with higher U-RNS score (5.2±3.2 vs 2.9±3.1,p<0.0001). Faecal leakage both during the day and night was significiantly higher with higher U-NRS score (p<0.001 for both). Patient reported outcome (PRO2) had a weak correlation with U-NRS (R^2=0.14).

However if we examine bloody stool, the stool number and difficulty of faecas holding seperately we got a significant and better correlation with urgency (R^2=0.43). Patients with moderately severe-severe disease (PRO2 > 5) had a higher chance to have urgency (AUC=0.9). The possibility was higher to have urgency among patients with painful defecation (R^2=0.41). The level of urgency, duration of faeces holding and the frequency of faeces holding problems had the strongest correlation with U-NRS (R^2=0.83). Patients defined urgency with these parameters according to this result. We created score with them, level of urgency points + half of duration points + frequency, which showed a good model to refine the definition of urgency (R^2=0.83).

Conclusion: Bowel urgency affects UC patients' lives and is associated with more severe disability. Bowel urgency is common in UC affecting about two-third of patients and associates with more severe activity of UC. Fecal urgency can characterize with the level of urgency, duration of faeces holding and the frequence of stool more accurately than a single, subjective numeric scale. A new scoring system is needed to more accurately assess patients' complaints in this area and to objectify severity.

Disclosure: The authors have no potential conflict of interest to disclose.

PP0458

MUCOSAL CYTOKINE PROFILES AS PREDICTORS OF THERAPEUTIC EFFICACY OF TOFACITINIB IN ULCERATIVE COLITIS: FINDINGS FROM A PROSPECTIVE STUDY

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Introduction: Although several therapeutic strategies exist for the treatment of ulcerative colitis (UC), their efficacy is still far from optimal. The JAK/STAT signalling pathway has emerged as a promising therapeutic target. Tofacitinib (TOFA), an oral JAK inhibitor, is the first approved small molecule for induction and maintenance of remission in moderate-severe UC. However, potential predictive markers for therapeutic response to TOFA are currently lacking.

Aims & Methods: Therefore, we aimed to assess the mucosal expression of JAK/STAT and cytokine levels in UC patients treated with TOFA to identify potential predictive markers of therapeutic response. This prospective study included UC patients initiating TOFA therapy. Clinical indices, laboratory parameters, and colonoscopy were evaluated at baseline and after a 12-week treatment period. RNA and total protein were isolated from colonic biopsies. qRT-PCR analysis was applied for the following targets: STAT1, STAT3, JAK1, JAK2, JAK3 and TYK2. Mucosal protein levels of TNF-a, IL-6, IL-1b, IL-4, IL-17A, IFN-g, and PAI-1 were determined by multiplex ELISA.

Results: A total of 49 UC patients were enrolled, and 75.5 % completed the 12-week follow-up. Of these, 63.3 % of the patients responded to TOFA. Clinical and endoscopic remission rates were 69.4% and 32.6%, respectively. Primary non-response was observed in 24.5% of the patients and loss of response in 36.7%. In responders, serum C-reactive protein (CRP) level, partial Mayo (pMayo), and endoscopic Mayo (eMayo) scores significantly decreased. TOFA responders showed a significant reduction in STAT1 and JAK2 gene expression. The mucosal protein levels of IL-6, IL-1b, IFN-g, IL-17A, and PAI-1 were also significantly decreased in patients who responded to TOFA.

Importantly, baseline levels of IL-6, IL-4, and PAI-1 were significantly lower in responders. Patients achieving steroid-free remission had significantly reduced baseline levels of IL-1b, IL-4 and PAI-1.

In contrast, elevated level of IL-1b was significantly higher in patients with loss of response. Moderate correlations were observed between pMayo scores and TNF-a, IL-6, IL-1b, and IL-17A levels. Weaker correlations were seen between eMayo scores and the examined cytokine concentrations.

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However, the strongest correlations were found between mucosal PAI-1 level and pMayo (r=0.55) and eMayo (r=0.52) scores. Serum CRP level showed moderate correlations with cytokine concentrations.

Conclusion: TOFA treatment was effective in more than 60% of patients and clinical remission was achieved in nearly 70%. Lower baseline mucosal protein level of IL-1b, IL-6, IL-4 and especially PAI-1 may serve as potential predictive markers of treatment response. Among these, PAI-1 demonstrated the strongest correlation with both clinical and endoscopic disease activity, underscoring its potential as a predictive and monitoring biomarker in UC.

Disclosure: Nothing to disclose.

PP0459

IMPACT OF DISEASE ACTIVITY ON PREGNANCY OUTCOMES IN WOMEN WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Active inflammatory bowel disease in pregnancy is associated with an increased risk of adverse maternal and neonatal outcomes. Understanding patterns of disease activity may inform strategies for improving outcomes.

Aims & Methods: A retrospective cohort study was conducted of pregnancies managed through an IBD–antenatal clinic between August 2020 and January 2025. Patients were stratified into 2 groups based on disease activity during pregnancy. Active disease was defined using the Physician Global Assessment, symptoms, and biomarkers (CRP and/or FCP). Disease flare was defined by an increase in symptoms and/or biomarkers from their baseline. Disease activity was assessed pre-conception (within 6 months before pregnancy) during pregnancy and post-partum (up to 6 months post birth).

Results: Of 209 pregnancies (Table 1), 76.3% (n=158) were in remission and 23.7% (n=49) had active disease pre-conception. Among those with active disease, 81.6% (n=40) remained active into pregnancy, while 18.4% (n=9) achieved remission. Of those in remission at pre-conception, 73.4% (n=116) stayed in remission during pregnancy, and 24.7% (n=39) flared. 2 patients with no pre-conception data were active in pregnancy.

Of 125 patients in remission during pregnancy, 73.6% (n=92) stayed in remission post-partum and 12.0% (n=15) flared. Among 81 patients with active disease in pregnancy, 39.5% (n=32) remained active, and 53.0% (n=43) achieved remission post-partum.

Patients in remission through pregnancy had a higher mean BMI (24.5 vs 23.0 kg/m^2, p=0.027), were more likely to be of white ethnicity compared to minority ethnic groups (61.0% vs 38.4% p=0.0026), have a diagnosis of Crohn's disease (58.6% vs 37.0%, p=0.0088), and be nulliparous (51.2% vs 35.0%, p=0.0414).

At conception, approximately 30% of patients in both groups were not receiving any treatment. During pregnancy, this proportion fell to 8.6% in the active group, reflecting therapy escalation. Use of biologics increased from 22.2% to 38.2% in the active group and remained stable at 36.0% in the remission group. Corticosteroid use was exclusive to the active disease cohort (6.2% at conception; 23.5% during pregnancy).

Active disease in pregnancy was associated with higher rates of adverse outcomes: gestational diabetes (12.3% vs 8.0%, p=0.001), fetal growth restriction (4.9% vs 0%, p=0.044), and low birth weight (<2500g: 14.6% vs 2.4%, p=0.015). While rates of small for gestational age infants were higher in the active group (9.9% vs 4.0%), this was not statistically significant.

Emergency c-sections were more common in the active group (29.9% vs 15.0%, p=0.03), while elective c-sections were more frequent in the remission group (27.5% vs 13.4%, p = 0.03).

Women in remission more often had an intact perineum (63.6% vs 54.7%), while episiotomy was more frequent in those with active disease (18.8% vs 11.7%); both did not reach statistical significance.

	Pre-conception % of available data (n)	Pregnancy % of available data (n)	Post-partum % of available data (n)
Active disease	23.7% (49)	39.3% (81)	25.2% (47)
Remission	76.3% (158)	60.7% (125)	74.2% (135)
Excluded (pregnancy loss)	0	3	3
Missing data points	2	0	24

Conclusion: This study underscores the impact of active IBD on maternal and neonatal outcomes.

Nearly 1 in 4 women in remission at preconception flared during pregnancy. This highlights the need for effective pre-conception counselling, access to disease monitoring tools, multidisciplinary management and timely treatment optimisation to sustain remission during pregnancy.

Disclosure: No disclosures.

PP0460

COMPARISON OF REMIMAZOLAM AND MIDAZOLAM FOR SEDATION DURING COLONOSCOPY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: PATIENT SATISFACTION, RECOVERY TIME, AND SAFETY

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Introduction: Endoscopy is a crucial tool in the management of inflammatory bowel disease (IBD). Recently, Remimazolam a novel, ultra-short-acting benzodiazepine has been approved for procedural sedation. Although propofol provides superior sedation compared to midazolam, its use in Italy is limited due to the requirement for anesthesiologist supervision, higher costs, and a shortage of specialists. To date, no real-world studies have compared midazolam, remimazolam, in IBD patients.

Aims & Methods: This prospective, single-center study enrolled consecutive IBD patients undergoing outpatient colonoscopy at a tertiary care center. Patients were randomized in a 1:1 ratio to receive either remimazolam or midazolam for sedation. All procedures included continuous vital sign monitoring, and both sedatives were administered directly by the endoscopist.

The primary objective was to evaluate the feasibility of patient discharge from the endoscopy unit. Secondary objectives included comparing the safety profile of remimazolam versus midazolam and assessing patient-reported satisfaction with the procedure.

Discharge readiness was assessed using the Post-Anaesthetic Discharge Scoring System (PADSS), which evaluates six criteria: vital signs, ambulation, nausea/vomiting, pain, bleeding, and voiding. A PADSS score of ≥9 at

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10 and 20 minutes post-procedure was considered indicative of discharge readiness. Patient satisfaction was measured using a four-point scale: not satisfied, not yet satisfied, satisfied, and very satisfied.

Results: A total of 200 patients were enrolled, with 100 in each group. 90 patinets were affected by Crohn's disease and 108 by ulcerative colitis. Baseline characteristics, including age, disease duration, BMI, gender, and history of abdominal surgery,were comparable between the groups. All colonoscopies were completed successful, the time of colonscopy was 22.51 minutes(CI 20.85-24.16) in the midazolm group and 21.08 minutes (CI:21.08-23.69) in the remimazolam group (p: 0.911, CI -2.232-1.99). At 10 minutes post-procedure, 68 patients (68%) in the remimazolam group achieved a PADSS score ≥9, compared to 25 patients (25%) in the midazolam group ($p < 0.001$). At 20 minutes, 96 patients (96%) in the remimazolam group met the discharge criteria, versus 69 patients (69%) in the midazolam group ($p < 0.001$). In patients who had previously undergone colonoscopy (n=189), high satisfaction was reported by 64 patients in the remimazolam group and 39 in the midazolam group ($p < 0.001$). Among those with a history of abdominal surgery, 30 out of 46 (68.2%) patients in the remimazolam group reported high satisfaction, compared to 15 out of 44 (35.7%) in the midazolam group ($p = 0.003$). No severe adverse events were reported. Two patients in the remimazolam group experienced transient bradycardia, while two patients in the midazolam group required supplemental oxygen during the procedure.

Conclusion: This prospective study suggests that remimazolam enables faster discharge from the endoscopy unit compared to midazolam, with a high level of patient satisfaction and a favorable safety profile.

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PP0461

UNCOVERING SEX DIFFERENCES IN THE DIAGNOSTIC JOURNEY OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE – INTERIM ANALYSIS OF A GERMAN-WIDE MULTICENTER CROSS-SECTIONAL STUDY

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Introduction: While inflammatory bowel disease (IBD), which includes Crohn's disease (CD) and ulcerative colitis (UC), affects people of all sexes, potential disparities in the diagnostic process—such as differences in symptom onset, symptom reporting, and time to diagnosis—have not been systematically investigated in depth to date.

Aims & Methods: This analysis aims to explore potential sex differences in the pre-diagnostic symptom presentation and the diagnostic pathway of IBD, spanning from the onset of first symptoms to the eventual clinical diagnosis. In this interim analysis of a multicenter cross-sectional study, data of 1180 participants with IBD (women: n=770, men: n=410) from 42 recruiting units all over Germany were included.

A comprehensive set of demographic data was collected via an online survey, encompassing the following variables: the IBD entity, sex and gender identity, age, the initial symptoms of IBD, the time from symptom onset to first consultation, the time from first consultation to referral, and the time to diagnosis. In addition, the respective physician's sex was documented. Moreover, the distinct disease-related and therapeutic histories have been collected. Inclusion criteria were a minimum age of 18 years and a confirmed diagnosis of UC or CD for a minimum of three months.

Results: No significant sex differences were observed in the median time to initial physician consultation for either CD or UC (CD: women 3 months [IQR: 1–12], men 3 months [IQR: 0.5–10], p=0.567, g=0.1; UC: women 2 months [IQR: 0.8–5], men 1.3 months [IQR: 0.5–4], p=0.091, g=0.1). Similarly, the median time from first consultation to referral did not differ significantly between sexes (CD: women 1.5 months [IQR: 0.5–5], men 1 month [IQR: 0.5–6], p=0.154, g=0.1; UC: women 0.8 months [IQR: 0.3–3], men 0.5 months [IQR: 0.2–1], p=0.113, g=0.2).

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However, women experienced significantly longer median time from symptom onset to final IBD diagnosis compared to men (CD: women 12 months [IQR: 5–48], men 6.5 months [IQR: 2.5–24], p=0.002, g=0.3; UC: women 5 months [IQR: 2–14.5], men 3 months [IQR: 1.3–8], p=0.006, g=0.2). The misdiagnosis rate was found to be significantly higher in women with IBD than in men (IBD women: n=311 (40.4%); IBD men: n=141 (34.4%); p=0.044). Furthermore, sex-specific differences in the initial presentation of symptoms were observed. Women with CD were more likely to present with diarrhea (p=0.005, ϕ =0), nausea (p=0.013, ϕ=0), or arthralgia (p=0.012, ϕ =0) compared to CD men. In UC, women were more likely to initially experience fever (p < 0.001, ϕ = 0) or abdominal pain (p = 0.004, ϕ = 0) compared to men.

Conclusion: Despite similar timelines for initial physician contact and referral, women with IBD experienced significantly longer diagnostic delays and higher rates of misdiagnoses than men. Sex-specific differences in initial symptom presentation were observed, suggesting that diagnostic disparities may be partly driven by variations in symptom profiles and their interpretation. These challenges call for a broader, global perspective and increased awareness to improve the overall care of individuals with IBD on an international scale.

Disclosure: Nothing to disclose.

PP0462

MIRNAS AS NON-INVASIVE BIOMARKERS OF GUT-LOCAL HOST-MICROBIOME INTERACTIONS IN BIOLOGIC THERAPY FOR INFLAMMATORY BOWEL DISEASE

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Introduction: Personalizing biologic therapies in inflammatory bowel disease (IBD) remains a critical unmet need. Here, we investigated circulating and stool microRNAs (miRNAs) as prognostic markers for treatment response across three biologics - anti-TNF, vedolizumab (VEDO), and ustekinumab (UST) - and explored their relationship with gut microbiome to uncover potential host-microbe predictors of response.

Aims & Methods: This study included 43 IBD patients (mean age 40.4 years, 25.6% female, 20.1% with ulcerative colitis) enrolled in the ILEITIS cohort at National University Hospital, Singapore. The majority had received anti-TNF (n=35), whilst others were given VEDO (n=9), or UST (n=15) therapies. Clinical non-response was defined as disease flare within 48 weeks post-induction, indicated by C-reactive protein >5 mg/L, fecal calprotectin >150 µg/g, or moderate to severe endoscopic inflammation. Baseline serum and stool samples were collected and isolated for miRNA profiling by the ID3EAL™ PanoramiR microRNA Knowledge Panel (MiRXES) via the QuantStudio5-384-well qPCR system (Applied Biosystems). Fecal microbial communities were assessed via shotgun metagenomic sequencing. Differential expression analysis and Spearman correlation with false discovery rate (FDR) adjustment were performed to explore miRNA–microbe associations and treatment response patterns.

Results: Among 376 paired serum and stool miRNAs analyzed, only six exhibited significant positive correlations, indicating largely compartment-specific expression patterns. Stool miRNAs demonstrated stronger and more numerous correlations with gut microbial taxa (11 significant pairs, FDR < 0.05), whereas serum miRNAs showed no strong associations. Biologic-specific miRNA signatures were identified.

Predictive modelling identified top-ranked miRNA with high discriminative performance for treatment response; serum miR-296 and stool miR-134 perfectly predicted VEDO response (AUROC = 1.0); serum miR-146 predicted UST response (AUROC = 1.0); and stool miR-302 robustly predicted anti-TNF response (AUROC = 0.897). Correlation analysis between biologic-responsive miRNA and microbial taxa revealed 12 significant correlations for stool-derived miRNA and 14 for serum-derived miRNA.

Notable microbial species involved in these associations include *Bacteroides uniformis, Parabacteroides distasonis, Phocaeicola dorei, Clostridium scindens, Streptococcus sanguinis, Bilophila wadsworthia, Hungatella hathewayi,* and *Clostridium leptum*, all taxa implicated in IBD-associated dysbiosis, showed significant correlations (FDR < 0.05).

Conclusion: Stool- and serum-derived miRNAs capture non-overlapping but complementary dimensions of host response to biologic therapy in IBD. There are significant overlaps between host miRNAs and microbiome readouts suggesting plausible links between key microbial responders to therapy and host regulatory miRNAs, especially in the luminal compartment. Our findings highlight the value of integrating compartment-specific miRNA and microbiome profiling to improve precision medicine approaches in IBD.

Disclosure: Nothing to disclose.

PP0463

PREVALENCE AND CLINICAL IMPACT OF ABNORMAL HEPATOBILIARY BIOCHEMISTRY IN NEWLY DIAGNOSED INFLAMMATORY BOWEL DISEASE – A PROSPECTIVE POPULATION-BASED INCEPTION COHORT STUDY

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Introduction: Abnormal hepatobiliary biochemistry is frequently observed in patients with inflammatory bowel diseases (IBD), encompassing ulcerative colitis (UC) and Crohn's disease (CD). However, its prevalence and clinical implications in IBD remain unclear.

Aims & Methods: We aimed to determine the occurrence and clinical impact of abnormal hepatobiliary biochemistry in patients with newly diagnosed IBD.

Between May 1, 2021, and April 30, 2023, all adult and pediatric patients with incident IBD within a catchment area covering 20% of Denmark were enrolled in the prospective, population-based Copenhagen IBD Inception Cohort. Herein, all adult patients underwent hepatobiliary blood tests (alanine aminotransferase (ALT), alkaline phosphatase (ALP), bilirubin,

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and INR) at IBD diagnosis. Multivariate Cox proportional hazard analyses adjusting for age, sex, smoking, and disease phenotype were performed (adjusted HR, aHR, 95% confidence interval).

Results: A total of 527 patients (UC: 326; CD: 201) were included in the cohort, with 290 (89.0%) and 189 (94.0%) undergoing blood tests, respectively, within a median of two days (interquartile range: -14 - 6) before IBD diagnosis. No demographic or clinical phenotypical differences were observed between patients who underwent blood test and those untested.

Abnormal hepatobiliary biochemistry was found in 49 (16.7%) and 46 (24.3%, p<0.01) patients with UC and CD, respectively, primarily driven by higher median ALP levels in CD (80 U/L [IQR 69-97]) versus UC (74 U/L [IQR 61-90], p<0.01). No associations were observed between abnormal hepatobiliary biochemistry and sex, age, initial extent or location of IBD, or disease activity. However, stricturing or penetrating CD was significantly associated with abnormal hepatobiliary biochemistry (adjusted OR = 4.62 [1.70; 12.54]). During a follow-up of 2.1 years (IQR 1.6-2.6), abnormal hepatobiliary biochemistry was not associated with the initial course of UC; however, an increased risk of requiring resective surgery for CD was observed (aHR= 3.27 [1.04-10.24], Table 1).

	Disease activity (worsening or flare)	Systemic steroid	IBD-related hospitalization	Colectomy or resective surgery	Immunomodulators	Biological therapy	Surgery or biological treatment
	Hazard ratio (95% confidence interval) adjusted for disease phenotype, sex, age, and smoking						
Ulcerative colitis	1.09 (0.35-3.34)	0.95 (0.42-2.14)	1.06 (0.38-2.94)	1.82 (0.34-9.70)	0.61 (0.21-1.75)	1.82 (0.83-3.98)	1.14 (0.35-3.70)
Crohn's disease	0.46 (0.05-3.95)	1.04 (0.48-2.28)	1.21 (0.43-3.35)	3.27 (1.04-10.24)	1.24 (0.57-2.70)	1.34 (0.64-2.83)	1.60 (0.75-3.39)

Table 1: Impact of abnormal hepatobiliary biochemistry on the initial course of inflammatory bowel diseases.

Conclusion: In this prospective, population-based cohort, abnormal hepatobiliary biochemistry was present in approximately one in six patients with UC and one in four patients with CD at diagnosis, with no association with IBD disease activity. Although it did not correlate with the initial course of UC, these findings underscore the need for increased attention to hepatobiliary biochemistry in patients with CD.

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PP0464

SEXUAL DYSFUNCTION AS A COMMON EARLY CHALLENGE IN NEWLY DIAGNOSED INFLAMMATORY BOWEL DISEASE

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Introduction: Sexual dysfunction is an underrecognized concern among patients with inflammatory bowel diseases (IBD).

Aims & Methods: This study aimed to assess the prevalence and associated risk factors of sexual dysfunction at IBD diagnosis.

All adult patients with newly diagnosed ulcerative colitis (UC) or Crohn's disease (CD) within a defined region of Copenhagen, Denmark, were prospectively enrolled from the Copenhagen IBD Inception Cohort (May 2021–May 2023).

Sexual dysfunction was defined using patient-reported IBD-specific Memorial Symptom Assessment Scale (MSAS), with frequent dysfunction categorized as occurring "sometimes" or more often. Multivariable logistic regression analysis were adjusted for age, sex, and phenotype.

	Frequent sexual dysfunction Adjusted odds ratio [95% confidence interval]	
	Ulcerative colitis	Crohn's disease
Elderly-onset IBD (age≥60 vs. <60 years)	0.95 [0.25;3.60]	1.29 [0.23;7.12]
Female sex	1.53 [0.58;4.07]	2.66 [0.72;9.83]
Diagnostic delay (>26 weeks vs. less)	1.25 [0.39;3.98]	0.33 [0.06;1.79]
Obesity	0.52 [0.06;4.53]	1.30 [0.22;7.69]
Active smoking (ref: never)	1.52 [0.45;5.08]	3.10 [0.47;20.42]
Disease classification (ref: L1, E1, or B1)		
L2 / E2	1.46 [0.58;3.69]	1.91 [0.52;6.93]
L3 / E3	1.11 [0.44;2.77]	1.68 [0.44;6.35]
B2 or B3		2.54 [0.43;14.92]
Perianal vs. none	0.80 [0.31;2.08]	0.82 [0.09;7.75]
Extra-intestinal manifestations (yes vs. no)		2.88 [0.51;16.13]
IBD activity (moderate-to-severe vs. mild)		
SCCAI ≥ 6 or HBI ≥ 8	1.17 [0.43;3.20]	0.74 [0.17;3.29]
C-reactive protein > 10 mg/L	0.34 [0.08;1.38]	3.79 [0.86;16.77]
Fecal calprotectin > 1800 mg/L	0.55 [0.18;1.68]	1.66 [0.38;7.29]
MES ≥ 2 or SES-CD ≥ 7	0.68 [0.22;2.03]	NA
Initial disease course		
Need for hospitalization	1.15 [0.30;4.32]	4.43 [1.14;17.26]
Need for biologics	0.86 [0.16;4.75]	1.34 [0.29;6.14]
Need for surgery	NA	0.76 [0.08;7.42]
Moderate to severe impairment of PROs		
EQ-5D-5L, problems in 4 of 5 domains	1.78 [0.47;6.77]	5.38 [1.23;23.57]
SIBDQ < 60	2.76 [0.73;10.48]	NA
IBD-DI > 20	6.93 [1.58;30.37]	7.83 [0.72;84.54]
FACIT-F < 35	2.55 [0.90;7.20]	4.02 [1.05;15.40]
Symptoms in modified MSAS questionnaire		
Abdominal pain	5.23 [0.62;44.24]	1.98 [0.48;8.12]
Mucus on stool	0.90 [0.20;4.06]	NA
Blood on stool	0.53 [0.08;3.71]	1.00 [0.11;8.95]
Feeling bloated	2.80 [0.70;11.19]	NA
Weight loss	1.76 [0.60;5.17]	6.60 [0.54;80.24
Reduced appetite	5.23 [1.53;17.89]	NA
Stool urgency	3.16 [1.06;9.45]	0.69 [0.18;2.60]

Table 1: Predictors of frequent sexual dysfunction.

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Results: Among 383 included patients (UC: 182; CD: 107), frequent sexual dysfunction was reported by 29.4% of females and 17.5% of males with UC, and by 22.7% of females and 25.0% of males with CD. In UC, sexual dysfunction was associated with moderate-to-severe disability (IBD Disability Index; adjusted odds ratio (aOR) =6.93; 95% CI: 1.58–30.37) and bowel urgency (aOR=3.16; 1.06–9.45). In CD, it was linked with CD-related hospitalization (aOR=4.43; 1.14–17.26), impaired quality of life (EQ-5D-5L; aOR=5.38; 1.23–23.57), and fatigue (Functional Assessment of Chronic Illness Therapy-Fatigue; aOR=4.02; 1.05–15.40). Sexual dysfunction was not associated with age, sex, disease activity, endoscopic severity, biochemical markers, or initial treatment.

Adjusted for age, sex, and disease phenotype (presence of extensive colitis in UC or ileal disease in CD). In CD, symptoms were only adjusted for disease phenotype due to sample size. Severe disease was defined by the need for glucocorticoids, immunomodulators, biologics, or surgery within three months of diagnosis.

Conclusion: Sexual dysfunction is common and often present at IBD diagnosis, independent of objective disease severity. Early recognition of sexual dysfunction is essential for comprehensive, patient-centered IBD care.

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PP0465

HOSPITAL ADMISSIONS DURING PREGNANCY IN WOMEN WITH INFLAMMATORY BOWEL DISEASE: AN 11-YEAR SINGLE-CENTER OBSERVATIONAL STUDY

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Introduction: Pregnancy in women with Inflammatory Bowel Disease (IBD) is associated with increased risk of complications, particularly when disease activity is uncontrolled. Antenatal hospital admissions may reflect underlying disease activity or pregnancy-related complications and may adversely affect pregnancy outcomes.

Aims & Methods: The aim of this retrospective observational study was to describe the causes and outcomes of hospital admissions during pregnancy in women with IBD.We included completed pregnancies in women with a confirmed diagnosis of Crohn's disease (CD), ulcerative colitis (UC), or IBD-unclassified (IBD-U), followed between May 2014 and March 2025, who had at least one hospital admission during pregnancy. Data were collected on IBD history, disease activity, assessed according to the Global Physician Assessment (GPA), treatment, reasons for admission, investigations, and pregnancy outcomes.

Results: Of 57 pregnant women that were included (mean age: 32 years; 27 CD, 27 UC, 3 IBD-U), 6 patients (10.52%) were diagnosed during pregnancy. Prior to pregnancy, 20 patients (35.08%) had undergone IBD-related surgery, and 10 of these had a stoma. Among CD patients, 14.28% (n=4) had a history of previously active perianal disease, and 21.43% (n=6) had active perianal disease during pregnancy. Disease activity, before conception, was as follows: 66.67% (n=37) in remission, 12.28% (n=7) with mild disease, 7.02% (n=4) with moderate disease, and 1.75% (n=1) with severe disease. During pregnancy, 73.68% (n=42) remained in remission, while 12.28% (n=7) had moderate disease, 8.77% (n=5) mild disease, and 5.26% (n=3) severe disease. Regarding maintenance therapy during pregnancy, 38.6% (n=22) received aminosalicylates, 29.82% (n=17) thiopurines, and 36.84% (n=21) biologics. 13 patients (22.8%) remained on no maintenance therapy. Steroids were used at least once during pregnancy in 36.84% (n=21). In total, 64.91% (n=37) of admissions were IBD-related (n=19 flare, n=7 stoma complications, n=6 abdominal pain in the absence of active IBD, n=2 perianal complications, n=2 gastrointestinal bleeding, n=1 perforation). Among non-IBD-related admissions, 45% (n=9) were due to obstetric complications, 15% (n=3) due to infections, 30% (n=6; biliary sepsis, cholestasis, migraine, suspected venous thromboembolism ,hemoptysis and suspected pulmonary embolism) due to other causes, and 10% (n=2) due to undocumented reasons. Radiological investigations were performed in 38.6% (n=22) of patients, 21.05% (n=12) underwent endoscopic evaluation, and 10.52% (n=6) required surgical intervention. Pre-eclampsia or eclampsia was documented in 4 patients (7.01%), and gestational diabetes was observed in 5 patients (8.77%). Delivery outcomes included 61.4% vaginal births (n=25 spontaneous vaginal deliveries; n=10 assisted vaginal deliveries) and 36.84% caesarean sections (n=11 elective; n=9 emergency). Two stillbirths were recorded. The mean birth weight was 3,135 grams and 49.12% (n=28) of mothers initiated breastfeeding. Preterm delivery, defined as birth before 37 weeks of gestation, occurred in 14.03% (n=8) of cases.

Conclusion: Hospital admissions in pregnant women with IBD often reflect disease flares, stoma complications or abdominal pain. Most pregnancies complicated by IBD hospital admissions had favorable outcomes. These findings underline the importance of multidisciplinary care and close monitoring to manage both IBD-related issues and pregnancy-specific complications.

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PP0466

SEXUAL DYSFUNCTION IN CROHN'S DISEASE: SYMPTOMS NOT TO BE OVERLOOKED

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Introduction: In Crohn's disease (CD), a comprehensive approach to care is increasingly advocated, incorporating various factors that may impact patients' quality of life, including sexual dysfunction (SD). However, data on sexual health in this population remain scarce.

Aims & Methods: The aim of this study was to assess the prevalence and risk factors of sexual dysfunction, as well as its impact on quality of life in patients with CD.

We conducted a cross-sectional study including all patients with a confirmed diagnosis of CD who attended follow-up consultations over the six months preceding the study. Sexual function was assessed using the Female Sexual Function Index (FSFI) for women and the International Index of Erectile Function (IIEF) for men. SD was defined by a total FSFI score <26 for women and an IIEF score <26 for men. Disease activity was evaluated using the Harvey-Bradshaw Index (HBI), and quality of life was assessed via the short version of the Inflammatory Bowel Disease Questionnaire (S-IBDQ).

Results: A total of 102 patients were enrolled (mean age: 40.35 ± 15.07 years), with a male predominance (M/F sex ratio = 1.61). Active smoking was reported in 32.3% of patients. Disease location was ileal (47.1%), colonic (20.6%), and ileocolonic (32.4%). The predominant disease phenotype was both stricturing and penetrating in 41.2% of cases. Perianal involvement was present in 38.2%. Severe disease activity was noted in 20.6% of patients. Nearly half (47.0%) underwent at least one surgical procedure during follow-up, with 12% undergoing total colectomy. At the time of the study, 23.5% were receiving systemic corticosteroids, 23.5% azathioprine, and 53.0% anti-TNFα therapy.

The mean IIEF score was 17.22 ± 9.65 and the mean FSFI score was 21.56 ± 7.18. SD was identified in 38.5% of women and 28.6% of men, with no statistically significant difference (p = 0.78). Severe sexual dysfunction (IIEF score 5–10) was observed in 17.6% of male participants. Both disease activity and active smoking were significantly associated with SD (p = 0.001 and p = 0.024, respectively). No significant association was found between SD and perianal disease or history of intestinal surgery.

The mean S-IBDQ score was 45.06 (range: 17–67), with 41.2% of patients scoring <40, indicating a major impact of CD on quality of life. SD was strongly correlated with poorer quality of life, as shown by lower S-IBDQ scores in both men (p = 0.002, r = 0.01) and women (p = 0.008, r = 0.01). Notably, 73.5% of patients expressed a desire to discuss sexual health with their gastroenterologists.

Conclusion: This study highlights a substantial prevalence of sexual dysfunction in patients with Crohn's disease, with significant repercussions on quality of life. These findings emphasize the importance of systematically addressing sexual health in clinical practice to optimize patient-centered care.

Disclosure: Nothing to disclose.

PP0467

COMPARATIVE OUTCOMES OF ULCERATIVE COLITIS PATIENTS WITH MES 1 AND HISTOLOGIC ACTIVITY VERSUS MES 2

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Introduction: Histologic remission is not a formal target in the management of patients with ulcerative colitis (UC). In clinical practice, medical therapy escalation is reserved for patients with MES ≥2. Currently, there is no evidence on outcomes of patients with MES 1 stratified by histology.

Aims & Methods: We conducted a retrospective cohort study to compare long-term outcomes of UC patients with a MES of 1, stratified by histological activity according to the Nancy Histological Index (NHI), and those with MES 2 after 12 months of advanced therapy. We included all consecutive adult patients (≥ 18 years) with a confirmed diagnosis of UC initiating any advanced therapy with at least 12 months of follow-up. Only patients with MES 1 or 2 after 6-12 months from the start of therapy and availability of histological report with NHI were eligible. The primary endpoint was the rate of a composite outcome including medical therapy escalation, hospitalization, and surgery among patients with MES 1 and histological activity (NHI 2-4) versus those with MES 2. A multivariate logistic regression was performed to identify independent predictors of the composite outcome. In addition, Spearman correlation was used to assess associations between histologic activity and individual outcome components.

Results: A total of 159 patients with UC were included (82 males, 51.6%), with a mean age of 44 ± 15 years. At baseline, 76 patients (47.8%) were naïve to advanced therapies. Most patients had MES 3 at baseline (51.6%). The most commonly used drug was infliximab (50, 31.4%), followed by ustekinumab (46, 29.0%), vedolizumab (35, 22.0%), tofacitinib (13, 8.2%), upadacitinib (8, 5.0%) adalimumab (6; 3.8%), and filgotinib (1, 0.6%). Concomitant therapy with systemic corticosteroids or azathioprine was adopted in 22 (13.8%) and 38 patients (23.9%), respectively. At 12-months, 90 (56.6%) patients had MES 1 and 69 (43.4%) had MES 2. Histological activity was observed in 36 individuals with MES 1 (40.0%) and in 60 patients (87.9%) with MES 2. The mean follow-up was 21.5 ±8.2 months. When stratified by histological activity, no significant difference was observed in the composite endpoint between patients with MES 1 and histological activity and MES 2 (33% vs 73%; $\rho = -0.25$; $p > 0.05$).

Similarly, no statistically significant differences were found in the rates of hospitalization (1.6% with MES 1 vs 2.8% with MES 2, $\rho = -0.02$; $p > 0.05$), medical therapy escalation (30.5% with MES 1 vs 75% with MES 2, $\rho = -0.32$; $p > 0.05$). Interestingly, the need for a change in advanced therapy (13.8% vs 28.9%, $\rho = -0.25$; $p > 0.05$) and treatment optimization (18.3% vs 13.4%, $\rho = -0.04$; $p > 0.05$) did not differ significantly between patients with MES 1 and histological activity and MES 2.

Conclusion: UC patients with MES 1 and histological activity have similar adverse long-term outcomes to those with MES 2. These findings support consideration of treatment intensification in patients with MES 1 who exhibit histologic inflammation, reinforcing the prognostic value of histologic assessment beyond endoscopic findings.

Disclosure: Nothing to disclose.

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PP0468

MICRONUTRIENT AND NUTRITIONAL STATUS OF IBD PATIENTS IN REMISSION REFERRED TO A SPECIALIST DIETITIAN CLINIC

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Introduction: Nutritional deficiencies are common in IBD patients and have been widely reported in Crohn's disease (CD) patients in particular. Nonetheless micronutrient screening is not routine in all centres in the UK. Body Mass Index (BMI) is widely used for detecting malnutrition in clinical practice but it is not sensitive marker of muscle mass. Hand grip strength (HGS) is quick to perform and may be an effective parameter to predict muscle function and health in IBD patients in busy outpatient settings. Different cut points are used in clinical practice to determine muscle weakness and may yield different results.

Aims & Methods: The aims of this study were (1) To examine micronutrient status (2) To establish the incidence of over- and under-weight and (3) To measure hand grip strength to determine the frequency of dynapenia in stable patients with IBD referred to specialist dietitian clinic for advice. A baseline micronutrient screen including 9 vitamin and minerals (Vitamins A,D, E, folate, B12, ferritin, selenium, copper and zinc as well as haemoglobin) was performed on all consenting patients at initial dietitian appointment. Inflammatory marker, C reactive protein, was also checked. Patients with raised CRP, were excluded from the micronutrient analysis. All patients were weighed, height measured and BMI calculated. Hand grip strength was measured using Marsden hand grip dynamometer. The maximum result of 6 trials was recorded and its corresponding percentile (1). Dynapenia was defined as HGS<16kg for females and <27kg for males (2) . Results of Ulcerative colitis (UC) and Crohn's disease (CD) patients were compared.

Results: A total of 292 consecutive patients, (108m, 184 f) referred for dietitian input from April 2023 to March 2025 were included. Mean age was 45 years (SD18). 201 patients with CD and 91 with UC were included. Results are presented in Table 1. No difference in micronutrient status was found between CD and UC patients apart from vitamin D where a higher incidence of deficiency was found in UC compared with CD patients (p=0.045) . Contrary to widespread belief incidence of underweight was low in both CD and UC with a higher incidence of obesity in both groups than underweight. Over a quarter of the IBD population had muscle weakness defined by both Dodds et al (2014) and Cruz-Jentofft et al (2019) which is concerning (1,2).

NUTRITIONAL STATUS	CD	UC
BMI kg/m2	Mean (SD) 24.8 (6.1) <18.5 kg/m^2 n=19 (9.5%) >30kg/m^2 n=33 (16.4%)	Mean (SD) 25.5 (6.5) <18.5kg/m^2 n=6 (6.7%) >30 Kg/m^2 n= 19 (21%)
HAND GRIP STRENGTH maximum (kg)	38.5 (11.3) M 21.4 (7) F <16kg n=20 (28%) <27kg n=11(20%) <10TH Percentile n=48 (37.5%)	40.2 (9.1)M 22.3 (6.6)F <16kg n=3 (9%) <27kg n=3 (11%) <10th percentile n=18 (30%)
VITAMIN D	77%	68%*
FOLATE	12.8%	17.9%
B12	5.6%	4.8%
SELENIUM	10.7%	3.4%
ZINC	31.5%	34.9%
FERRITIN	23.2%	15.4%
LOW HAEMOGLOBIN	31.0% (M) 20.3% (F)	35.4%(M) 22.6% (F)

Conclusion: Vitamin D status in particular should be monitored in all stable patients with IBD. A high proportion of both UC and CD patients were found to have weak muscle strength. HGS is a readily obtainable measure quick to perform and could be easily incorporated into routine clinical practice. Further research is needed to establish its clinical significance for stable IBD patients.

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PP0469

KNOWLEDGE GAPS IN INFLAMMATORY BOWEL DISEASE: INSIGHTS FROM PATIENTS

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Introduction: Inflammatory Bowel Disease (IBD), including ulcerative colitis and Crohn's disease, is a chronic condition that requires patients to be well-informed about their disease for effective self-management. Knowledge of triggers, treatment options, and lifestyle adjustments is essential for managing symptoms, reducing flare-ups, and improving quality of life.

This study explores knowledge gaps among IBD patients and evaluates predictive factors associated with a high level of disease-related knowledge.

Aims & Methods: We conducted a monocentric cross-sectional study including IBD patients. All individuals completed the IBD-Know score which is a validated 24-item questionnaire covering various IBD-related aspects, including anatomy, function, epidemiology, diet/lifestyle, general knowledge, medication, complications, surgery, reproduction, and vaccination.

In this study, the IBD-Know score was administered to IBD patients in our gastroenterology department. Each correct response was awarded one point, with higher scores indicating greater knowledge levels. While the IBD-Know score does not have universal cutoff points, scores above the median were interpreted in this study as reflecting high to excellent knowledge. Factors potentially influencing the level of IBD-related knowledge were also analyzed.

Results: A total of 69 patients with IBD (38 with Crohn's disease and 31 with ulcerative colitis) completed the IBD-Know questionnaire. The median age was 38 years, with a male-to-female ratio of 1.56. The median disease duration was 52 months (range: 6-156 months). Regarding lifestyle

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and medical history, 28 patients were current smokers, 48 had a history of IBD-related hospitalization, 28 had undergone IBD-related surgery, and 15 reported a family history of IBD.

Medication histories varied: all patients had used immunomodulators and corticosteroids, 38 had taken 5-aminosalicylates, and 35 were on biologics. Educational attainment was diverse, with 41 patients having completed college or university, while the remaining participants had a high school education or less.

The overall mean IBD-Know score was 13.8, with a mean correct answer rate of 40.8% across all 24 items. The lowest correct answer rate was in the reproduction domain (31.2%), with a subgroup analysis revealing lower accuracy among men (28.7%) compared to women (52%). The highest correct response rate was for the question on smoking in Crohn's disease (80.4%), while the lowest was for the indication of biologics (19%).

Univariate analysis identified several factors significantly associated with a higher IBD-Know score: disease duration over five years (p=0.02), history of IBD-related hospitalization (p=0.01), higher education level (college/university or greater) (p=0.002), family history of IBD (p=0.03), and biologics treatment (p=0.03). In multivariate analysis, hospitalization history (odds ratio [OR]: 2.525, 95% confidence interval [CI]: 1.280-4.705; p=0.002) and higher education level (OR: 2.692, 95% CI: 1.380-5.206; p=0.02) remained significant predictors of a high IBD-Know score after adjusting for other variables.

Conclusion: This study identifies significant knowledge gaps among patients with IBD, particularly regarding reproduction. Factors such as disease duration, hospitalization history, and education level correlate with higher knowledge scores. Addressing these gaps through targeted educational initiatives is essential to empower patients and enhance their self-management, ultimately improving their quality of life.

Disclosure: Nothing to disclose.

PP0470

ROLE OF RADIOMICS IN PREDICTING THE DEGREE OF INTESTINAL FIBROSIS IN PATIENTS WITH CROHN'S DISEASE

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Introduction: During their lifetime, patients with Crohn's disease (CD) frequently develop strictures as a complication, where various degrees of inflammation and fibrosis coexist. Distinguishing between inflammatory and fibrotic strictures is crucial for deciding on surgery or treatment changes, yet it is currently challenging in clinical practice. Cross-sectional imaging techniques (e.g. magnetic resonance imaging MRI) provide a reasonably accurate estimate of bowel wall fibrosis. However, only histopathological examination can determine its true extent. Radiomics uses artificial intelligence to analyze quantitative features from diagnostic images and can predict clinical outcomes.

Aims & Methods: This study aims to assess the role of radiomics in classifying patients according to the degree (no/mild vs severe) of intestinal fibrosis in a cohort of CD patients who underwent surgery for strictures disease. We retrospectively selected 48 CD patients who underwent MR-enterography and subsequent surgery within 4 months between 2017-2023. An expert pathologist evaluated the degree of fibrosis in histopathological specimens from surgical samples using the Chiorean scoring system from 0 (no fibrosis) to 2 (severe fibrosis), and the degree of inflammation using Geboes score. For each MR-enterography, typical lesions of CD were searched by an expert radiologist on T2-weighted axial images generating a region of interest (ROI) segmentation for each lesion. We extracted 100 radiomic features from each ROI of the pre-processed MRI images using Pyradiomics. Feature selection included univariate analysis (Wilcoxon-Mann-Whitney test) and Spearmann correlation to exclude highly correlated features. A logistic regression model with stepwise regression was built with the selected features and evaluated by computing the area under the curve (AUC) of the receiver operating characteristic (ROC) curve. Model's sensitivity and specificity were calculated using the best cut-off according to the Youden index method.

Results: Among the 48 patients, 48% (n=23) showed severe fibrosis on the surgical specimen, whereas 52% (n=25) had either no or mild fibrosis. Two radiomic features were selected and resulted statistically significant in the fitted logistic regression model($p < 0.05$): original_shape_Maximum2DDiameterColumn and original_firstorder_90Percentile. Results with their 95%CI were: AUC 0.79 (0.64-0-90), sensitivity 0.69(0.50-0.75) and specificity 0.84(0.66-0.94). Bootstrap corrected (mean optimism) values of AUC and metrics were: AUC 0.77 (0.02), sensitivity 0.68(0.01), specificity 0.81(0.02).

Conclusion: Our study suggests that MRI-based radiomics modeling could help predict the degree of intestinal fibrosis. Future research will confirm these findings in a larger cohort.

Disclosure: Nothing to disclose.

PP0471

DEVELOPMENT OF A MACHINE LEARNING MODEL FOR PREDICTING THE OCCURRENCE OF PSYCHOLOGICAL DISEASES IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD) is associated with an increased risk of psychological disorders; however, no prior study has systematically identified clinical predictors for their onset.

Aims & Methods: This study aimed to identify key clinical factors that can predict the development of psychiatric disorders in IBD patients.. We analyzed data from 7,069 IBD patients in the UK Biobank to predict the

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onset of 13 psychiatric disorders in this population. Predictors including 16 socio-demographic variables, 8 surgical procedures, 8 disease location, 6 inflammation-related biomarkers, and medication usage were included to identify potential predictors of psychiatric diseases in IBD patients. The SHAP (SHapley Additive exPlanations) method was applied to rank the importance of all the predictors. A LightGBM (LGBM) model was then developed to predict the occurrence of psychiatric disorders in IBD patients after IBD diagnosis.

Results: Of all the 38 clinical factors evaluated, 15 factors that ranked top by importance were chosen to form a 15-predictors panel. The panel demonstrated strong performance (AUC = 0.72) in predicting the onset of psychiatric disorders 25 years before onset in patients of Crohn disease, and an AUC of 0.68 in Ulcerative Colitis (UC) patients. Besides, for patients with Crohn Disease, the model for predicting mental and behavioral disorders due to psychoactive substance use performs outstandingly (AUC = 0.87).

Conclusion: The LGBM model accurately predicted the occurrence of psychological diseases in IBD patients offering a valuable tool for clinicians. SHAP enhanced understanding of key factors enabling better treatment strategies and potentially improving the risk of developing psychological diseases.

Disclosure: Nothing to disclose.

PP0472

IMPACT OF INTESTINAL ULTRASOUND (IUS) IMPLEMENTATION ON COLONOSCOPY UTILIZATION IN IBD PATIENTS: A RETROSPECTIVE STUDY FROM A TERTIARY HOSPITAL

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Introduction: The management of inflammatory bowel disease (IBD) often relies on colonoscopy to evaluate disease activity and detect complications. However, intestinal ultrasound (IUS) has emerged as a non-invasive, accurate alternative, endorsed by ECCO-ESGAR guidelines for use in both Crohn's disease and ulcerative colitis.

This study aimed to evaluate the impact of IUS implementation on the frequency of colonoscopies performed in IBD patients at a tertiary care hospital.

Aims & Methods: We conducted a retrospective analysis comparing two time periods: before IUS implementation (2021–2022) and after its introduction (2023–2024). The number of colonoscopies performed for IBD management in each period was recorded. Screening colonoscopies in IBD patients were not included in the analysis; therefore, the data reflect colonoscopies performed specifically for the clinical management of the disease. The percentage change in colonoscopy utilization was calculated to assess the effect of IUS integration into routine clinical practice.

Results: In the pre-IUS period (2021–2022), a total of 340 colonoscopies were performed for the management of patients with IBD. After the introduction of IUS in 2023, the number of colonoscopies over the subsequent two years decreased to 280, reflecting a 17.6% reduction. This decline suggests a meaningful shift towards non-invasive monitoring facilitated by IUS.

Conclusion: The integration of IUS into the endoscopic workflow was associated with a notable decrease in colonoscopy use among IBD patients. These findings reinforce the role of IUS as a valuable, non-invasive modality for disease monitoring, with the potential to optimize resource utilization and support faster clinical decision-making. Prospective studies are needed to further assess the long-term benefits and cost-effectiveness of widespread IUS adoption in IBD care.

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PP0473

BEYOND THE BOWEL: SECONDARY EXTRAINTESTINAL FINDINGS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE UNDERGOING COMPREHENSIVE ULTRASOUND ASSESSMENT

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Introduction: Intestinal point-of-care ultrasound, as standardized by the International Bowel Ultrasound Group (IBUS), has emerged as a valuable tool for diagnosing, monitoring, and managing patients with Inflammatory Bowel Disease (IBD). While IBUS emphasizes a structured intestinal assessment, some European centers adopt a more comprehensive approach, combining intestinal ultrasound with a complete abdominal examination.

The Karolinska Centre for Digestive Health, a tertiary referral institution providing specialized care to approximately 4,500 patients with IBD, established a dedicated Gastrointestinal Ultrasound Unit in October 2023. This unit offers advanced diagnostic services to both hospitalized patients and those treated on an outpatient basis.

Aims & Methods: This retrospective study aimed to assess the prevalence of secondary extraintestinal findings in IBD patients undergoing comprehensive abdominal ultrasound between October 4, 2023, and March 4,

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2025. All patients underwent ultrasound assessments that included both intestinal and abdominal imaging. Patients referred for indications unrelated to intestinal ultrasound (IUS), as well as those undergoing repeat examinations, were excluded from the analysis to ensure data consistency. All examinations were performed by a single gastroenterologist with 25 years of ultrasound experience. In cases where secondary extraintestinal findings were identified, patients' electronic medical records were reviewed to determine whether these findings represented new diagnoses or previously known conditions.

Results: The study included 348 patients with IBD, 201 *(58%)* females and 147 *(42%)* males, with a mean age of 42.4 years. Among these, 260 *(75%)* were referred internally by our IBD Outpatient Clinic, 47 *(13%)* by the Gastroenterology Ward, 33 *(10%)* by private outpatient clinics, and the remainder from other healthcare sites. IUS revealed evidence of intestinal inflammation in 144 patients *(42%)*, while findings were normal in 204 patients *(58%)*.

Furthermore, 150 patients presented with a total of 195 secondary extraintestinal findings, of which 113 *(58%)* represented newly identified pathologies. Among all secondary findings, the majority, 104 *(53%),* were related to the hepatobiliary system, including 33 newly diagnosed cases of hepatic steatosis and 17 new cases of cholecystolithiasis. One patient demonstrated ultrasound features suggestive of primary sclerosing cholangitis (PSC) and is currently undergoing further evaluation. Additional findings involved the reproductive system (29/*15%*), urinary system (15/*7%*), spleen (32/*16%*), and pancreas (5/*2%*). Various other intra-abdominal findings were also identified (37/*18%*). Notably, one patient presented with a suspicious mass in the right iliac fossa, and another demonstrated ultrasound findings suggestive of a prostatic tumor; both cases were referred for further oncologic evaluation.

Additionally, three patients were newly diagnosed with intraductal papillary mucinous neoplasm (IPMN), which was subsequently confirmed via magnetic resonance imaging, and all were enrolled in the national IPMN surveillance protocol.

Conclusion: A considerable proportion of patients with IBD were found to have previously unknown extraintestinal abdominal pathologies, underscoring the essential role of comprehensive ultrasound examination that combines both intestinal and abdominal assessment. Fortunately, only a minority of these findings resulted in changes to patient management.

Disclosure: Francesca Bresso has acted as national and/or local principal investigator for clinical trials for Janssen, Ferring, and AbbVie. Charlotte Hedin served as a speaker and/or advisory board member for AstraZeneca, AbbVie, Dr. Falk Pharma and the Falk Foundation, Galapagos, Janssen, Pfizer, Ferring, Takeda, and Tillotts Pharma and received grant support from Tillotts Pharma and Takeda. Stephan L. Haas has served as a speaker for Takeda, Santax Medico, Tillotts Pharma, Johnson & Johnson, and Mediahuset. The rest of the authors have nothing to declare.

PP0474

DIGESTIVE ULTRASOUND: A KEY TOOL IN CROHN'S DISEASE MONITORING

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Introduction: Intestinal ultrasound is a noninvasive and inexpensive tool to assess disease activity in patients with inflammatory bowel disease, such as Crohn's and ulcerative colitis by monitoring the disease activity in real time.

Aims & Methods: Intestinal ultrasound is a non-invasive, cost-effective imaging tool for routine clinical evaluations. This study examines its role in monitoring Crohn's disease, comparing it with CT Enterography.

This is a prospective, descriptive, comparative study including patients with crohn's disease aged over 16, between March 2023 and March 2024. Every patient had a colonoscopy, enteroscan, and digestive ultrasonography; the delay between each test was limited to one month by the same operator. Patients with ileal or recto-colic resections or pregnant women were not included. We develope a form to collect clinical, therapeutic, and epidemiological informations as well as the outcomes of CT, colonoscopy, and ultrasounds. Between ultrasound, CT scan and colonoscopy, we asked our patients to choose the most widely accepted para-clinical test. Ultrasonography's sensitivity (Se) and specificity (Sp) were determined, and data from enteroscanner and ultrasound were analyzed, with a focus on the wall's thickness, vascularization, damage extent, and peridigestive environment. The statistical study was carried out using SPSS 21.0 software. Quantitative variables were expressed as medians and qualitative variables as numbers and percentages. The McNemar test and the Wilcoxon test for associated samples were applied. The significance level was set at 0.05.

Results: Forty-nine patients were included, with a male-to-female ratio of 0.53 and a median age of 32 years. US detected digestive thickening in 81.6% of cases, mainly in the terminal ileum. CT Enterography detected thickening in the proximal small bowel and rectum in 2.2% of cases, which US missed. US sensitivity was 84.4%, with specificity of 50%. No significant difference was found between US and CT for thickening detection. Mesenteric infiltration detected by US was similar to CT (Se = 86.5%, Sp = 100%). For adenopathy, Se was 75% (Sp = 97%). US showed a 60% sensitivity for detecting complications, with a significant difference compared to CT (p = 0.002). There was no correlation between thickening on US and ulcers on endoscopy.

Conclusion: Despite limitations in examining the proximal intestine and rectum, intestinal ultrasonography is a sensitive, accessible method for monitoring Crohn's disease and detecting complications.

Disclosure: Nothing to disclose.

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PP0475

LEVERAGING GLOBAL VIDEO CONTEXT TO AUTOMATE COLON SECTION IDENTIFICATION IN ULCERATIVE COLITIS ENDOSCOPY

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Introduction: Our previously developed UC-SCALE algorithm[1] predicts UC severity at the colon-section level by assigning a Mayo Clinic Endoscopic Subscore (MCES) to the rectum, sigmoid, or descending colon. Yet, manually annotating anatomical sections—especially after the procedure—is exceedingly challenging for expert readers and is typically not part of clinical practice. To streamline UC-SCALE usage and reduce manual labeling, we propose an AI algorithm that automatically identifies each frame's anatomical location in colonoscopy videos.

Aims & Methods: We used the same five phase-III etrolizumab clinical trial datasets underlying UC-SCALE: three trials (3028 videos, 1309 patients, 391 sites) for training, one trial (692 videos, 327 patients, 80 sites) for validation, and one trial (607 videos, 317 patients, 83 sites) for testing. We tuned model hyperparameters by comparing multiple training runs and selected the best-performing configuration on the validation set for final testing. For each frame, we extracted "embedding vectors" generated by our internal IBD foundation model (adapted from DINO[2]), which condenses frame image content into relevant features.

For comparison, these two models were trained to classify each frame as rectum, sigmoid, or descending colon:

- Frame-Only Model (no video context): Predicts anatomic location using only the single-frame embedding.
- Full Video-Context Model: Uses a transformer-based architecture, similar to the one proposed by Mobadersany et al [3], that incorporates the entire video's sequence of embeddings, mimicking how clinicians view videos in context.

Results: The Frame-Only Model yielded modest performance (AUC=0.71, F1=0.59), confirming the difficulty of labeling isolated frames. By contrast, the Context-Aware Model significantly improved metrics (AUC=0.93, F1=0.82), highlighting the value of incorporating sequential information—similar to real-time endoscopic navigation.

Model	AUC	F1-score
Frame-Only	0.71	0.59
Full Video-Context	0.93	0.82

Table 1. Performance of the two section classification models.

Conclusion: Incorporating video-wide context remarkably enhances frame-by-frame anatomical labeling of UC colonoscopy. This approach can be coupled with UC-SCALE's section-level severity scoring to automate both localization and disease severity estimation, reducing the need for manual section annotation.

Future work will validate performance across diverse datasets, potentially lowering trial costs and broadening clinical adoption of IBD AI-based algorithms.

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PP0476

KINETICS OF TRANSMURAL HEALING IN PATIENTS WITH CROHN'S DISEASE TREATED WITH ANTI-TNF THERAPY

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Introduction: Transmural healing is a promising target in Crohns' disease (CD) that is associated with more favorable outcomes than endoscopic remission, especially lower risk of bowel damage progression. C-score has been recently developed and validated to assess transmural healing in patients with CD. However, the best time to assess transmural healing remains to be determined.

Aims & Methods: We aimed to evaluate the kinetics of transmural healing in patients with CD treated with anti-TNF therapy.

From a prospective database of patients with CD performing MRI for luminal CD, we analyzed the kinetics of transmural healing among CD patients older than 18 years-old receiving anti-TNF therapy, with objective sign of inflammation on morphological evaluation (cross-sectional imaging) before starting the treatment, who underwent at least one MRI examination to assess therapeutic efficacy within 12 months. Patients treated with anti-TNF agents for treat isolated perianal lesions were excluded.

We used a statistical model based on imputation according the last observation carry forward and clinical relevance. Transmural healing was then assessed every month during the first year. For segmental analysis, we used the same division into segments than recommended for Lemann Index calculation. Transmural healing was defined as C-score < 0.5 in each segment. C-score was calculated as previously published (0.2 x (bowel thickness in mm – 3) + 2 if ulcer + 1.5 if enlarged lymph nodes).

Results: Overall, 123 patients were included. Among them, 90 were female (73.1%), 48 (39.9%) were current smokers and 53 (43.0%) had prior intestinal resection. The median CD duration at baseline was 7.6 years [1.4-15.2]. At baseline, 41 patients (33.3%) were anti-TNF-naïve. At baseline, 67 patients (55.3%), 53 patients (43.8%) and 1 patient (0.8%) were treated with infliximab, adalimumab and golimumab, respectively with concomitant immunosuppressant in 94 patients (79.0%).

At baseline, 220 segments were considered as active (C-score > 0.5) including 100 in the terminal ileum and 120 in the colon/rectum.

According per-segment analysis (N=220), the rate of transmural healing was 6.4% at M1, 6.4% at M2, 10.0% at M3, 10.1% at M4, 14.6% at M5, 18.0% at M6, 23.2% at M7, 24.5% at M8, 26.0% at M9, 26.8% at M10, 29.1% at M11 and 30.4% at M12.

Among terminal ileum segments (n=100), the rate of transmural healing was 2.0% at M1, 2.0% at M2, 5.0% at M3, 5.0% at M4, 9.0% at M5, 12.9% at M6, 16.0% at M7, 17.0% at M8, 18.0% at M9, 19.0% at M10, 20.0% at M11 and 20.0% at M12.

Among colorectal segments (n=120), the rate of transmural healing was 6.4% at M1, 6.4% at M2, 13.8% at M3, 13.8% at M4, 15.6% at M5, 22.0% at M6, 28.4% at M7, 28.6% at M8, 28.6% at M9, 28.6% at M10, 32.8% at M11 and 34.4% at M12.

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Among the 123 patients with CD, in multivariable analyses, absence of transmural healing (C-score > 0.5) within the first year was associated with increased risk of surgery (HR=3.37 [1.30-8.68]; p=0.012), drug discontinuation (HR=2.31 [1.29-4.14]; p=0.005), and bowel damage progression (p <0.001) while we observed a trend for higher risk of CD-related hospitalization (HR=1.92 [0.85-4.36]; p=0.11).

Conclusion: Transmural healing is achievable in more than 30.0% of active segments in patients with CD treated with anti-TNF therapy within the first year. The best time to assess transmural healing was between 6 and 12 months after starting anti-TNF therapy with higher rate of transmural healing for colorectal versus ileal segments. Transmural healing defined as C-score < 0.5 is associated with long-term improved outcomes in patients with CD.

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PP0477

RELIABILITY AND USEFULNESS OF CONTRAST-ENHANCED TRANSPERINEAL ULTRASOUND FOR DISCRIMINATING ABSCESSES IN PERIANAL CROHN'S DISEASE

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Introduction: Perianal abscesses are a frequent and serious complication of Crohn's disease (CD), requiring timely and accurate diagnosis to guide appropriate treatment. Transperineal ultrasound (TPUS), especially when combined with contrast-enhanced ultrasound (CEUS), may offer a promising, real-time, bedside alternative.

Aims & Methods: This study aimed to evaluate the diagnostic performance and inter/intra-observer reliability of CEUS versus conventional TPUS B-mode in differentiating abscesses from inflammatory masses in CD patients with acute perianal symptoms.

We retrospectively analyzed 52 examinations performed on 37 adult Crohn's disease (CD) patients who underwent transperineal ultrasound (TPUS) at a tertiary IBD center between 2012 and 2021 for acute or painful perianal symptoms and suspected septic complications.

Patients were included if TPUS identified perianal lesions or masses suggestive of abscesses, warranting further evaluation with contrast-enhanced ultrasound (CEUS) to better characterize the lesion. Exclusion criteria included contraindications to CEUS (e.g., congestive heart failure), the presence of simple fistulas, or large, clearly identifiable abscesses (>3 cm) with evident fluctuating fluid content already visible on B-mode imaging. TPUS B-mode and CEUS examinations were performed using standardized protocols, with intravenous administration of 4.8 mL of contrast agent (SonoVue®).

All examinations were recorded, and three expert reviewers independently assessed anonymized video cases in two phases—B-mode followed by CEUS—randomized and blinded to clinical data. Lesions were classified as abscesses, inflammatory masses, or phlegmons. Intra- and inter-observer agreement was evaluated for maximum lesion size, maximum size of fluid content, and final diagnosis using intraclass correlation coefficients (ICC) and Cohen's kappa (κ).

Results: On TPUS B-mode, 17 abscesses, 32 inflammatory masses, and 3 phlegmons were identified; CEUS reclassified several of these lesions, ultimately detecting 16 abscesses, 31 inflammatory masses, and 5 phlegmons. Among the 17 abscesses diagnosed on B-mode, only 4 were confirmed by CEUS, while 11 were reclassified as inflammatory masses and 2 as phlegmons. Conversely, 11 lesions initially classified as inflammatory masses on B-mode were reclassified as abscesses by CEUS.

Intra-observer agreement for B-mode was excellent in assessing lesion size (ICC 0.853) and fluid content (ICC 0.897), but only fair to substantial for final diagnosis (κ = 0.55). CEUS demonstrated superior reliability, with nearly perfect intra-observer agreement for diagnosis (κ = 0.95).

Inter-observer agreement for TPUS B-mode was moderate to good for lesion size (ICC 0.798) and fluid content (ICC 0.736), but poor for final diagnosis (κ = 0.31, 95% CI 0.00–0.78), mainly due to the limited reliability of most sonographic features (echogenicity, gaseous artifacts, vascularity), except for posterior echo enhancement (κ = 0.54, 95% CI 0.36–0.82). In contrast, inter-observer agreement for CEUS was good for lesion size (ICC 0.847, 95% CI 0.76–0.91), fluid collection size (ICC 0.925, 95% CI 0.88–0.96) and for final diagnosis (κ = 0.83, 95% CI 0.79–0.95).

Conclusion: Contrast-enhanced TPUS significantly improves diagnostic reliability in identifying perianal abscesses in Crohn's disease, outperforming conventional B-mode TPUS in both intra- and inter-observer assessments. CEUS provides a valuable, accessible imaging modality for real-time evaluation of perianal complications, particularly in urgent clinical scenarios or when MRI is not feasible.

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PP0478

THERAPEUTIC EFFICACY OF FECAL MICROBIOTA TRANSPLANTATION IN ULCERATIVE COLITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Ulcerative colitis (UC) is a persistent inflammatory bowel disease characterised by alteration in the gut microbiota. Fecal microbiota transplantation (FMT) is a potential treatment for UC, aimed at restoring the balance of microorganisms in the intestines. This study evaluated efficacy of FMT for treating patients with UC.

Aims & Methods: A comprehensive electronic literature search was conducted for published studies in PubMed, Embase, and Scopus from their inception until April 2024. The analysis included randomized control trials (RCTs) that met the inclusion criteria. A random-effects model was applied to calculate odds ratios (ORs) and 95% confidence intervals (95%CIs) to evaluate the efficacy of the outcomes.

Results: This meta-analysis included a total of 19 RCTs, reporting the data from 932 patients diagnosed with UC. The OR showed a significantly higher clinical remission (2.67, 95% CI: 1.92, 3.71, $p < 0.00001$) and endoscopic remission (1.92, 95% CI: 1.19, 3.11, $p = 0.008$) among patients receiving FMT compared to the control group. Subgroup analyses of the primary outcome suggest that various factors, such as the method of FMT administration, donor characteristics, FMT dosage, and whether FMT is used alone or in combination, could slightly influence treatment outcomes. The OR of 1.18 (95% CI 0.75, 1.86, $p = 0.48$) indicated that both groups experienced similar adverse events.

Conclusion: These findings suggest that applying FMT is a safe and highly effective approach for treating UC, with adverse events comparable to those in the control group. This highlights the necessity for a better understanding of microbiota's role in UC management.

Disclosure: Nothing to disclose.

PP0479

SAFETY AND EFFICACY OF ELECTIVE SWITCHING FROM INTRAVENOUS TO SUBCUTANEOUS VEDOLIZUMAB IN IBD PATIENTS IN REMISSION: A SINGLE-CENTER RETROSPECTIVE COHORT STUDY

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Introduction: Aim: To report on the efficacy and safety of elective switching from intravenously (IV) to subcutaneously (SC) administered Vedolizumab in Inflammatory Bowel Disease (IBD) patients in clinical remission.

Aims & Methods: Methods: Retrospective analysis of IBD patients in clinical remission who were receiving Vedolizumab IV for treatment of their disease in a single tertiary center and were switched to SC, based on their physician's choice. The primary endpoint was disease deterioration during the follow-up period, according to the disease specific composite measures. In CD, clinical remission was defined as Crohn's Disease Activity Index (CDAI) <150 and in UC as Simple Clinical Colitis Activity Index (SCCAI) <2.

Results: Results: Between December 2023 and February 2025, 86 patients (Crohn's disease = 17, Ulcerative colitis = 69) were consulted in order to switch from IV to SC vedolizumab administration. Five patients (5,8%) did not want to change and therefore, 81 patients (94,2%) were finally switched. All patients were in clinical remission. After a mean (SD) follow up period of 8 (4) months, 9 patients (11,1%) experienced worsening of their disease, and were either switched back to IV VDZ (n=4, 4,9%) or to another agent (n=5, 6,2%). The remaining 72 patients (88,8%) showed favorable response, none of them requested an unscheduled visit, or developed an adverse event (clinical or laboratory) or needed escalation of treatment.

Conclusion: Conclusion: Elective switching from IV to SC Vedolizumab seems to be an effective and safe approach in real-world every day clinical practice in IBD patients in clinical remission.

Disclosure: Nothing to disclose.

PP0480

DO WE NEED A STRUCTURED TRANSITION PROGRAMME FOR PATIENTS WITH IBD? THE IRISH EXPERIENCE

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Introduction: The British Society of Gastroenterology {1} recommend the use of structured transition programmes to improve control of chronic GI disease in adolescents. Although research is limited structured transition programmes have been shown to improve disease outcomes in IBD.

Aims & Methods: We performed a retrospective multicentre study. Data was collected on patients prior to transition and post-transition to adult IBD services across Ireland over a 5-year period. Patients were grouped into those who attended a structured transition programme and those who received standard care transitioned with a referral letter to the adult gastroenterologist.

Our aim was to look at the impact of attending a structured transition programme on engagement of patients with IBD services and disease outcomes including change in medications, biologic failure and need for steroids.

Results: In total 282 patients were included. 155 patients attended a structured transition programme and 127 patients were in our standard care group. No significant differences were seen in basic demographics, median FCP levels or biologic use between either group at the time of transition.

Patients who took part in a structured transition programme had significantly better engagement with adult IBD services with significantly lower rates of non-attendance at outpatient clinics in the 1st year post-transition (11.6% versus 27.2%, $p = 0.04$) and significantly lower rates of disen-

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gagement with services (4.2% versus 13.1%, p = 0.015)(Table 1). Patients who attended a structured transition programme had significantly lower rates of change in medication therapy post-transition to adult services (p =0.045) (Table 1).

There was no significant difference seen in rates of biologic failure, need for steroid or median FCP levels between either group. The only factor on regression analysis that negatively impacted rates of non-attendance at outpatient clinics or disengagement with IBD services was attending a structured transition programme.

	Structured Transition Programme	Standard Care	P value
Clinic non-attendance	12.3%	27.2%	0.002
Lost to follow-up	2.7%	19.15%	<0.001
Change in medications	37.5%	50.0%	0.045
Failure of biologic therapy	15.8%	25.7%	0.08
Need for steroids	26.4%	22.4%	0.28
Need for surgery	9.7%	14.7%	0.15
Disease flares	32.6%	34.5%	0.79

Conclusion: Structured transition programmes in IBD improve engagement of patients with IBD services with significantly lower rates of non-attendance at clinics and disengagement with IBD services. More research and funding is required in the field of transition services in IBD to improve care for this cohort of vulnerable patients.

References:

1. Brooks AJ, Smith PJ, Cohen R, *et al* UK guideline on transition of adolescent and young persons with chronic digestive diseases from paediatric to adult care. *Gut* 2017;66:988-1000.

Disclosure: Nothing to disclose.

PP0481

PHARMACOKINETICS OF INFLIXIMAB CONCENTRATIONS IN SERUM AND COLONIC MUCOSA IN ACUTE SEVERE ULCERATIVE COLITIS (PROTOS STUDY)

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Introduction: No data exist on the relationship between serum and tissue infliximab (IFX) concentrations and outcomes in acute severe ulcerative colitis (ASUC).

Aims & Methods: In hospitalized patients (pts) with ASUC treated with IFX rescue therapy, we assessed relationships between longitudinal serum and tissue IFX concentrations and pt outcomes. In an open-label, prospective study, hospitalized ASUC pts treated with IFX rescue therapy were followed for 22 weeks (wks) at 2 centers. Inclusion criteria:

(A) hospitalized ASUC (partial Mayo score [PMS] > 7, baseline Mayo endoscopic score [MES] ≥ 2 and met Truelove/Witts criteria)

(B) receiving either standard or accelerated rescue IFX.

From IFX initiation until wk 22, clinical scores (PMS) and serum sampling were performed at multiple induction and maintenance timepoints (hospital/outpatient).

Colonoscopy scores (MES) and biopsies were collected on day 2-5 post first infusion (early inpatient) and wk 16-22 (maintenance).

Outcomes: Median area under the curve (AUC) of serum IFX concentration during early induction (wk 0-2), induction (wk 0-14) and maintenance (wk 15-22) were compared between wk 22 endoscopic remitters (ERs) (MES 0 or 1 without UC-related hospitalization or colectomy) and non-remitters (NRs). Tissue IFX concentrations were compared between groups.

Results: 110 serum samples over 22 wks (mean 7.9 samples/pt) with 54 samples between wk 0- 2 (mean= 3.9/pt) were collected. Biopsies at early inpatient (n=17) and maintenance colonoscopies (n=15) were collected from 14 pts. Mean age was 38 years, 50% were female, mean disease duration was 7.8 years and 69% had pancolitis. One NR needed colectomy and another was re-hospitalized. 50% achieved endoscopic remission at wk 22. ERs had numerically higher baseline CRP (49 [±28] vs. 31 [±24] mg/dL), immunomodulator use (71% vs. 43%) and clinical response (86% vs. 43%) to NRs, while albumin (3.3 [±0.9] vs. 2.9 [±0.6] g/dL) and total IFX dose between wk 0-2 (16 [±1] vs. 12 [±6] mg/kg) were more similar. Mean BMI was 23.8 kg/m^2. Between wk 0-2, the mean number of IFX infusions per pt

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was 1.5 (±0.7), and median dose per infusion was 11.4 mg/kg (IQR 25-75: 11-19). Median serum IFX concentration AUC was higher in ERs than NRs in early induction (wk 0-2: 568 ug*day/mL [528-1627] vs. 293 [145- 423], p=0.004), all induction (wk 0-14: 3036 [2742-3199] vs. 1768 [1375-2468] p=0.004) and in maintenance (wk 15-22: 841 [309-3295] vs. 405 [295-718] p=0.002). Findings were similar for dose normalized AUC.

Peak concentration 2 hrs post first infusion was numerically higher in ERs vs. NRs (median 365 ug/mL [200-484] vs. 187 [140-355] p=0.15) overall and dose-adjusted. Anti- IFX antibodies were detected in one NR maintenance sample. Baseline clearance inversely correlated with AUC (r=-0.7, p=0.03) or dose-normalized induction AUC (r=-0.8, p=0.01).

Tissue IFX concentrations were similar in inflamed/non-inflamed tissue and were similar between ERs/NRs in early induction (median 104 ng/mg [89-114] vs. 107 [76-133], p=1.0), and maintenance (10 [4-69] vs. 26 [3-87], p=0.99). Percent of drug in tissue vs. serum was similar between ERs/NRs (induction: median 7% vs 8%; maintenance: 8% vs 6%).

Conclusion: Intense serum and tissue sampling of ASUC pts receiving IFX showed higher serum drug exposure at all timepoints in endoscopic remitters whereas tissue concentrations were similar regardless of outcome. Tissue IFX may be required but not sufficient for achievement of outcomes in ASUC.

Disclosure: PC, SS, and LE report no conflicts of interest.

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VJ reports consulting fees and participation on advisory board for AbbVie, Alimentiv, Altrubio, Amgen, AnaptysBio, Arena pharmaceuticals, Asahi Kasei Pharma, Asieris, Astra Zeneca, BiomeBank, Boehringer Ingelheim, Bristol Myers Squibb, Calluna Pharma, Celltrion, Eli Lilly, Endpoint Health, Ensho Therapeutics, Enthera, Exeliom Biosciences, Ferring, Flagship Pioneering, Fresenius Kabi, GlaxoSmithKline, Genentech, Gilead, Janssen, Merck, MRM Health, Nxera Therapeutics, Pandion, Pendopharm, Pfizer, Protagonist, Reistone Biopharma, Roche, Roivant, Sandoz, Sanofi, Second Genome, Sorriso, Spyre Therapeutics, Takeda, Teva, Tillotts, Ventyx, Vividion; speaker fees from AbbVie, Ferring, Bristol Myers Squibb, Galapagos, Janssen, Pfizer, Takeda, Tillotts, and Fresenius Kabi.

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PP0482

TISSUE DRUG LEVELS ARE SIGNIFICANTLY HIGHER AND POSITIVELY CORRELATED WITH SERUM LEVELS IN IBD PATIENTS UNDER SUBCUTANEOUS COMPARED WITH INTRAVENOUS INFLIXIMAB MAINTENANCE THERAPY

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Introduction: Previous studies reported a positive correlation between serum and tissue drug levels in IBD receiving intravenous (IV) infliximab (IFX) therapy, except in those with severely inflamed tissue (1). Tissue drug levels in patients with IBD receiving SC IFX have not yet been investigated. We aimed to compare both the serum and tissue drug levels between SC and IV IFX maintenance therapy in IBD patients.

Aims & Methods: Any IBD patients under maintenance therapy with SC or IV stable doses IFX who underwent a colonoscopy for routine follow-up were eligible. Patient and disease characteristics, as well as clinical and endoscopic activity were recorded. Clinical activity was defined by high activity score (CDAI or pMAYO score) with at least one elevated biomarker (fecal calprotectin >250 mg/g, serum CRP >5 mg/L). Endoscopic disease activity was categorized as absent (normal), low (mild inflammation), moderate (friability, aphthoid lesions), or severe (ulcerations), as reported by the endoscopist blinded to the results of serum and tissue drug levels. Two biopsies were taken from the same location (either the most active area or randomly if no endoscopic lesion), and tissue and serum IFX levels were measured simultaneously in a centralized manner blinded to clinical and endoscopic disease activity. Frozen biopsy specimens were treated with the Bio-Plex® Cell Lysis Kit after tissue cutting and homogenates were sonicated at 4°C (Omni Ruptor 205). Tissue extracts were then frozen at -80°C prior drug measurements (iTrack10; Theradiag).

Results: Thirty-five patients were enrolled (CD in 65%, Sex ratio 1.4, 14 % with clinically and 57 % with endoscopically active disease, and 23 under SC IFX therapy). The characteristics of both IV and SC IFX groups were comparable, except for drug dose intensification in 90% of those treated with IV IFX group and 5% in the SC IFX cohort ($p < 0.001$). In patient in endoscopic remission, tissue levels were undetectable or very low for both IV and SC IFX therapy. Serum and tissue drug levels were significantly higher in the SC IFX cohort compared with the IV IFX group (22 mg/mL vs 9 mg/mL for serum; $p < 0.001$ and 25 mg/g vs 10 mg/g for tissue; $p = 0.002$, respectively). A moderate correlation was found between serum and tissue drug levels for both IV and SC IFX therapy (Spearman $r = 0.42$; $p = 0.014$ and $r = 0.43$; $p = 0.001$, respectively). Tissue drug levels increased gradually in patients with minimal or moderate endoscopic activity and were lower in those with severe endoscopic activity for both IV and SC IFX groups ($p < 0.001$). Tissue drug levels remained significantly higher in minimal or moderate endoscopically active disease in SC compared with IV route of IFX administration. The two independent factors associated with drug tissue levels > 14 mg/g (median levels) were the SC IFX route (OR: 2.25, 95% CI = [1.19–2.23]; $p = 0.001$) and the serum IFX levels (OR: 1.63, 95% CI = [1.19–2.23]; $p = 0.002$). A serum IFX level greater than 16.1 mg/mL was highly predictive of a tissue drug level > 14 mg/g tissue (Se: 0.94, Spe: 0.94, PPV: 0.94, NPV: 0.94, and Accuracy = 94.3%).

Conclusion: Serum and tissue drug levels are positively correlated in IBD patients under IV and SC maintenance IFX therapy. Interestingly, drug levels were higher both in serum and tissue in patients under SC IFX compared with IV IFX and tissue drug levels gradually increased in minimally to moderately inflamed intestine but were lower or undetectable in more severe inflamed areas.

Reference:

1. Yarur AJ, et al. The association of tissue anti-TNF drug levels with serological and endoscopic disease activity in inflammatory bowel disease: the ATLAS study. Gut 2016; 65: 249-255.

Disclosure: Nothing to disclose.

PP0483

TREATMENT PATTERNS, CLINICAL CHARACTERISTICS AND ECONOMIC BURDEN OF COMBINED ADVANCED THERAPY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A RETROSPECTIVE US DATABASE REVIEW

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Introduction: Several advanced therapies have been approved for treatment of patients with ulcerative colitis (UC) and Crohn's disease (CD) although efficacy is limited and up to 60% of patients relapse.[1,2] Combining advanced therapies that target different pathways may raise the therapeutic ceiling; however, little is known about real-world utilization of combined advanced therapies.

This study aimed to estimate prevalence, duration, and describe clinical characteristics, healthcare resource utilization (HCRU) and costs versus monotherapy.

Aims & Methods: This retrospective cohort study used the MarketScan Commercial Claims Encounters database (January 1, 2013 to December 31, 2022). Adult patients with UC or CD were included if they had used ≥1 advanced therapy and had ≥6 months of data prior to (baseline period) and after the index date. The index date for combined advanced therapies was defined based on the start of the overlap period, while monotherapy patients were assigned a mock-index date by selecting random patients from the combined advanced therapies cohort and adding their duration on monotherapy to the monotherapy start date of the monotherapy patient.

The study was conducted in two phases: in the initial phase, the frequency of patients with combined advanced therapies across multiple overlap periods was described to establish a clinically relevant and implementable definition of combined advanced therapies; in the second phase, patient characteristics, HRU, and costs were described separately for combined advanced therapies and monotherapy. Baseline characteristics were described during the 6-month baseline period, HRU and costs were described during baseline and follow-up period, for combined advanced therapies and monotherapy cohorts.

Results: Among patients with combined advanced therapies, the most frequent overlap duration was 30–59 days (33.6% in CD, 35.9% in UC), with infliximab + vedolizumab being the most prevalent combination (27.7% in CD, 26.7% in UC). Based on clinical input and sample size considerations, combination therapies were defined as overlapping use of combined advanced therapies for at least 30 days for subsequent analyses.

The prevalence of combined advanced therapies was low, observed in 2.9% of patients with CD and a similar proportion in UC. The most common combinations were ustekinumab + vedolizumab in CD and tofacitinib

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+ vedolizumab in UC. Most patients discontinued combined advanced therapies within a few months, with significant drops in persistence by six months. Among combinations, ustekinumab + vedolizumab demonstrated the longest persistence, with a median of 3.3 months in CD and 3.7 months in UC. At baseline, the combination cohort exhibited younger age, higher comorbidity burden, greater diagnostic testing, and significantly higher HRU and healthcare costs versus monotherapy cohort. In CD, the combination cohort incurred significantly higher all-cause costs than monotherapy ($124,928 vs $43,239), largely driven by advanced treatment costs ($86,940 vs $32,468) during the 6-month study period. This suggests that combination therapy is used for induction to regain control of disease rather than as a long-term option.

Conclusion: The prevalence of combined advanced therapies is relatively low, with vedolizumab frequently used as part of the combination. Patients on combined advanced therapies had higher disease burden, HCRU and healthcare costs versus monotherapy. Higher initial costs of combination therapy may be offset by achieving long-term disease control with subsequent monotherapy.

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PP0484

MIRIKIZUMAB-TREATED PATIENTS ARE ASSOCIATED WITH REDUCED RISK OF UC-RELATED HOSPITALIZATION AND SURGERY IN PATIENTS WITH ULCERATIVE COLITIS IN LUCENT TRIALS

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Introduction: Mirikizumab (MIRI) is an anti-IL-23p19 antibody with demonstrated efficacy compared to placebo (PBO) in the treatment of moderately-to-severely active ulcerative colitis (UC) (LUCENT-1: NCT03518086; LUCENT-2: NCT03524092).[1] Prevention of IBD-related surgery and hospitalizations is a goal in disease-modification trials for preventing disease progression.[2] We described the rates of these events in the LUCENT-1 and -2 trials.

Aims & Methods: Patients were randomized to intravenous MIRI 300 mg or PBO every 4 weeks (Q4W) for induction in LUCENT-1. Patients randomized to MIRI induction and achieving clinical response at Week (W)12 were re-randomized 2:1 in LUCENT-2 to subcutaneous MIRI 200 mg Q4W or PBO (MIRI withdrawal) for maintenance (W12-52). Descriptive statistics were used for proportion of patients in LUCENT-1 and -2 who had hospitalization and/or surgery by treatment group and the exposure-adjusted incident rate (IR). For LUCENT-1, hospitalizations in the induction period and surgeries during the induction and 16-week follow-up period were included. For LUCENT-2, all events were from the on-treatment period. Only hospitalizations associated with an adverse event with ≥24-hour stay (for any causes) were recorded.

Results: MIRI-treated patients had significantly lower IR of overall hospitalization, UC-related hospitalization, and overall surgery vs PBO-treated patients in LUCENT-1 (IR ratio MIRI vs PBO: 0.4, 0.1, 0.4, respectively). IR of UC-related surgery was 1.3 per 100 patient-years in MIRI-treated patients vs 2.6 per 100 patient-years in PBO group (Table). Mean time to first surgery or hospitalization from randomization was 30.5 days for PBO and 58.7 days for MIRI. Additional analyses revealed that patients on MIRI who had surgery/hospitalization in LUCENT-1 or -2 tended to have prior biologic failure and longer duration of disease. Similarly, in LUCENT-2, MIRI-treated patients had significantly lower IR of all-cause hospitalization and similar all-cause surgery vs PBO. No UC-related hospitalization and surgery were reported in the MIRI group while 2 UC-related hospitalizations (1.8 per 100 person-years) were reported from PBO.

Hospitalization and Surgery Rates in LUCENT-1 and LUCENT-2

	LUCENT-1							LUCENT-2						
	MIRI 300mg (N=868, PYE=201.0)			PBO (N=294, PYE=65.8)			MIRI vs. PBO	MIRI 200mg (N=365, PYE=267.9)			PBO (N=179, PYE=111.0)			MIRI vs. PBO
	Events, n (%)	PYR	Incidence rate, per-100 py (95%CI)[a]	Events, n (%)	PYR	Incidence rate, per-100 py (95%CI)[a]	IRR (95% CI)	Events, n (%)	PYR	Incidence rate, per-100 py (95%CI)[a]	Events, n (%)	PYR	Incidence rate, per-100 py (95%CI)[a]	IRR (95% CI)
Overall hospitalization	21 (2.4)	199.7	10.5 (6.5, 16.1)	16 (5.4)	64.6	24.8 (14.1, 40.2)	0.4 (0.2, 0.8)*	10 (2.7)	264.9	3.8 (1.8, 6.9)	11 (6.1)	109.0	10.0 (5.0, 17.9)	0.4 (0.2, 0.9)*
UC-related hospitalization	3 (0.3)	200.9	1.5 (0.3, 4.4)	10 (3.4)	65.2	15.3 (7.4, 28.2)	0.1 (0.0, 0.4)*	0	267.9	0 (0.0, 1.4)	2 (1.1)	110.7	1.8 (0.2, 6.5)	0
Overall surgery	11 (1.3)	227.9	4.8 (2.4, 8.6)	9 (3.1)	75.8	11.9 (5.4, 22.5)	0.4 (0.2, 1.0)*	13 (3.6)	263.2	4.9 (2.6, 8.4)	5 (2.8)	109.5	4.6 (1.5, 10.7)	1.1 (0.4, 3.0)
UC-related surgery[b]	3 (0.3)	228.6	1.3 (0.3, 3.8)	2 (0.7)	76.8	2.6 (0.3, 9.4)	0.5 (0.1, 3.0)	0	267.9	0 (0.0, 1.4)	0	111.0	0 (0.0, 3.3)	N/A

Note: *represent p<0.05.
IR = incidence rate; IRR = incidence rate ratio; MIRI = mirikizumab; N = number of patients in the analysis population; n = number of patients within each specific category; N/A= not applicable; PBO = placebo; PYE = patient years of exposure; PYR = patient years at risk; UC = ulcerative colitis.
[a] The IR and IRR confidence intervals and the p-values are based on a Poisson distribution
[b] UC-related surgeries are total colectomy, partial colectomy, and proctocolectomy.

Conclusion: Lower rates of UC-related hospitalizations and UC-related surgery were observed in patients treated with MIRI compared to those treated with PBO during the 12W induction LUCENT-1, and that impact continued through the 40W of maintenance LUCENT-2 with no UC-related hospitalizations or UC-related surgery reported in MIRI-treated patients.

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SN has served on advisory board and speaker for Eli Lilly and Company, Pfizer and advisory board for madrigal and speaker for Janssen

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PP0485

BENCHMARKING CORTICOSTEROID USE IN INFLAMMATORY BOWEL DISEASE AGAINST BSG KEY PERFORMANCE INDICATORS AT BARTS HEALTH NHS TRUST

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Introduction: The advent of numerous biologic drugs and small molecules has significantly reduced the role of steroids in maintaining remission in inflammatory bowel disease (IBD), particularly in ulcerative colitis and Crohn's disease [1]. The British Society of Gastroenterology (BSG) recommends identifying patients who are steroid-dependent or those who relapse within three months of discontinuing steroids in order to consider escalation of therapy [2].

Additionally, the UK IBD Standards advocate for regular audits of steroid use to ensure adherence to guidelines and to identify instances of steroid misuse or excess [3].

Aims & Methods: This audit aimed to evaluate corticosteroid prescriptions for patients with inflammatory bowel disease (IBD) at Barts Health NHS Trust (Newham University Hospital, Royal London Hospital, Whipps Cross Hospital). The primary goal was to ascertain whether therapy escalation was appropriately considered in prescribing corticosteroids and benchmarked against key performance indicators for inflammatory bowel disease in the UK [1].

All corticosteroid prescriptions issued for IBD treatment were reviewed between September 2023 and September 2024. Data was collected from Electronic Medical Records to identify prior corticosteroid prescriptions within the preceding 12 months and to determine if prescribers had considered the need for therapy escalation.

Results: A total of 368 patients received corticosteroid treatment, including 186 with ulcerative colitis and 182 with Crohn's disease. Prednisolone was the most commonly prescribed corticosteroid for ulcerative colitis (183 out of 186 cases), while budesonide was preferred for Crohn's colitis (115 out of 182 cases). Among these patients, 5 (2.6%) in the ulcerative colitis group and 6 (3.2%) in the Crohn's group met the criteria for escalation, having received 2 or more courses of oral steroids. Notably, none of the Crohn's disease patients had previously received biologic treatment. Of the patients who did not receive escalation therapy, the reasons were not clearly documented for 4 patients. Other missed escalations were attributed to factors such as referrals for the ADDapt trial, missed follow-up appointments, patient decline, or delays in escalating therapy in three instances. Importantly, none of these patients had documented hospital admissions requiring intravenous steroids or experienced surgical complications.

Conclusion: Prescribers exhibited a judicious approach to corticosteroid use, supported by specialised clinics and multidisciplinary meetings focused on steroid-sparing strategies. However, there remains an opportunity for improvement in the appropriate use of second-line steroids, the implementation of annual audits of steroid prescribing practices, and the consideration of the need for escalation. A notable limitation of this audit was the difficulty in obtaining corticosteroid prescription data from primary care or non-specialist hospital services.

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PP0486

REAL-WORLD TREATMENT PATTERNS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE RECEIVING INTRAVENOUS AND SUBCUTANEOUS MAINTENANCE VEDOLIZUMAB TREATMENT IN POLAND – RESULTS FROM A 1-YEAR PROSPECTIVE OBSERVATIONAL STUDY (VARIETY-POLAND)

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Introduction: Vedolizumab (VDZ) is approved for moderate-to-severe inflammatory bowel disease (IBD; ulcerative colitis [UC] & Crohn's disease [CD]).[1,2] While the subcutaneous (SC) formulation of VDZ is effective as maintenance treatment (mTx) in patients (pts) responding to intravenous (IV) VDZ induction treatment (Tx),[1,3] real-world data on Tx patterns of pts switching mTx from VDZ IV to SC or initiating mTx VDZ SC after IV induction are limited.

Aims & Methods: This prospective, non-interventional, multicentre (NCT05384080) study (conducted under the umbrella VARIETY protocol) enrolled adults with UC or CD initiating/undergoing induction with VDZ IV or continuing VDZ IV mTx, with the option to switch to VDZ SC mTx. The primary endpoint was real-world Tx patterns with VDZ, including Tx persistence. Additional endpoints assessed pt characteristics, pt-reported outcomes, and safety (adverse events [AEs]). Data were descriptively analysed.

Results: Overall, 162 enrolled pts (August 2022–June 2023; UC, n=103; CD, n=59) were included in the interim analysis. At enrolment, 127 (78.4%) received VDZ mTx (UC, n=78; CD, n=49) and 35 (21.6%), VDZ induction (UC, n=25; CD, n=10). The mean (SD) age was 38.2 (12.3) yrs, and time since IBD diagnosis was 10.7 (11.1) yrs. Majority of pts were male (96 [59.3%]), currently employed (124 [76.5%]), had never smoked (109 [67.3%]), and reported previous biologic therapy (95 [58.6%]). Demographic and IBD Tx history data were consistent across cohorts. At baseline, in pts with UC, the mean (SD) partial Mayo score was 1.8 (2.7); in pts with CD, the mean (SD) Crohn's Disease Activity Index (CDAI) score was 90.6 (70.1). During the 1-yr study period, 20.0% (n=7; UC, n=2; CD, n=5) of pts in the induction and 51.2% (n=65; UC, n=40; CD, n=25) in the mTx cohort switched to VDZ SC. At week (W) 52, Tx persistence was 40.0% (n=14; UC, n=11; CD, n=3) and 77.2% (n=98; UC, n=61; CD, n=37) in the induction and mTx cohorts, respectively, and highest overall in mTx pts who switched to VDZ SC (86.2%; n=56; Table).

The most common reason for Tx discontinuation was investigator decision (64.7%; n=33). For VDZ IV and SC, the mean (SD) number of doses per pt was 10.5 (7.6) and 23.8 (15.0), respectively. The median time from VDZ Tx initiation to SC Tx was 383 days (induction: 98 days; mTx: 434 days). In the induction cohort, the mean (SD) Work Productivity & Activity Impairment (WPAI) scores improved from 32.7 (24.1) at baseline (n=21) to 21.5 (31.2) at W52 (n=11), and the Short IBD Questionnaire (SIBDQ) scores improved from 44.1 (10.0) at baseline (n=34) to 51.0 (12.9) at W52 (n=15). In the mTx cohort, WPAI and SIBDQ results remained consistent over time. Fifty-three AEs were reported, including 4 serious AEs (malignancy, IBD flare requiring hospitalization, abdominal abscess, and unconsciousness). One AE of special interest (hypersensitivity reaction) was considered VDZ-related.

Tx persistence at W52	Total	IV only	SC only	IV-SC	IV-SC-IV
Overall	112/162 (69.1%)	53/86 (61.6%)	13/18 (72.2%)	59/72 (81.9%)	2/3 (66.7%)
Induction cohort[a]	14/35 (40.0%)	11/28 (39.3%)	-	3/7 (42.9%)	-
Maintenance cohort[b]	98/127 (77.2%)	42/58 (72.4%)	13/18 (72.2%)	56/65 (86.2%)	2/3 (66.7%)
UC cohort	72/103 (69.9%)	36/58 (62.1%)	8/12 (66.7%)	36/42 (85.7%)	1/2 (50.0%)
CD cohort	40/59 (67.8%)	17/28 (60.7%)	5/6 (83.3%)	23/30 (76.7%)	0 (0.0%)

Data are presented as n/N (%). [a]Induction cohort includes pts who started the study with VDZ induction Tx. [b]Maintenance cohort includes pts who started the study with VDZ maintenance Tx.
IV-SC are pts who started with VDZ IV and switched to SC during the study. IV-SC-IV are pts who switched from IV to SC and then back to IV. SC only are pts who received only VDZ SC during maintenance (with IV induction).
IV, intravenous; CD, Crohn's disease; SC, subcutaneous; Tx, treatment; UC, ulcerative colitis; VDZ, vedolizumab; W, week.

Conclusion: In this Polish real-world study, high Tx persistence was reported, especially in pts who switched to VDZ SC. No new safety signals were noted. Overall, the results were consistent with those from other VARIETY cohorts,[4] although lower persistence was observed during induction. The small size of the induction cohort was a limitation of this study.

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PP0487

EFFICACY AND SAFETY OF MIRIKIZUMAB AS INDUCTION THERAPY FOR CHINESE PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS: A SUBGROUP ANALYSIS OF THE PHASE 3 LUCENT-1 STUDY

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Introduction: Mirikizumab is an anti-IL-23p19 monoclonal antibody, a key mediator in the pathogenesis of inflammatory bowel diseases. We evaluated the induction efficacy and safety of mirikizumab in the LUCENT-1 (NCT03518086) Chinese subpopulation.

Aims & Methods: In this Phase 3, multi-center, randomized, parallel-arm, double-blind, placebo-controlled trial, conventional-failed or biologic/tofacitinib-failed Chinese adult patients (N=184) with moderately to severely active ulcerative colitis (UC, Modified Mayo Score 4-9, endoscopic subscore ≧2) were randomized in a 3:1 ratio to 300 mg mirikizumab (intravenous) or placebo every 4 weeks for 12 weeks. Randomization was stratified by biologic/tofacitinib failure status, baseline corticosteroid use and baseline disease activity (Modified Mayo Score). The primary objective was to test the hypothesis that mirikizumab was superior to placebo in inducing clinical remission at Week 12 (W12). Key secondary endpoints included alternate clinical remission, clinical response, clinical response in biologic-failed patients, symptomatic remission, endoscopic remission, histologic-endoscopic mucosal and bowel-movement urgency improvement. Analyses in Chinese subpopulation were descriptive. Missing data was imputed as non-responders for the categorical endpoints. Bowel urgency change from baseline data was reported based on Mixed Model Repeated Measured.

Results: Baseline characteristics were balanced across groups. A greater proportion of mirikizumab-treated patients achieved clinical remission (18.6% vs. 7.0%; Δ=11.5 [95%CI: 1.7, 21.3]) and alternate clinical remission at W12 (21.4% vs. 7.0%; Δ=14.3 [95%CI: 4.4, 24.2]). Mirikizumab also improved key secondary endpoints at W12. The frequencies of treatment-emergent adverse events were similar between groups (mirikizumab [60.0%] vs. placebo [62.8%]). There were fewer serious adverse events (10 [7.1%] vs. 7 [16.3%]) and discontinuations due to adverse events (2 [1.4%] vs. 3 [7.0%]) in mirikizumab-treated patients compared to placebo. There was 1 adenocarcinoma of colon (mirikizumab group) and no deaths during the treatment period.

	Placebo IV N=43	Miri 300 mg IV N=140
Clinical Remission at Week 12, n (%)	3 (7.0)	26 (18.6)
Common risk difference vs placebo		11.5
Alternate Clinical Remission at Week 12, n (%)	3 (7.0)	30 (21.4)
Common risk difference vs placebo		14.3
Clinical Response at Week 12, n (%)	13 (30.2)	76 (54.3)
Common risk difference vs placebo	3 (13.0)	23.2
Clinical Response (biologic-failed) at Week 12, n (%)		32 (45.1)
Common risk difference vs placebo		32.0
Endoscopic remission at Week 12, n (%)	5 (11.6)	44 (31.4)
Common risk difference vs placebo		19.0
HEMI at Week 12, n (%)	2 (4.7)	24 (17.1)
Common risk difference vs placebo		12.6
Symptomatic remission at Week 4, n (%)	5 (11.6)	15 (10.7)
Common risk difference vs placebo		-1.6
Symptomatic remission at Week 12, n (%)	8 (18.6)	49 (35.0)
Common risk difference vs placebo		16.2
Urgency Numeric Rating Scale (NRS), LSM (SE)	-1.6 (0.4)	-2.0 (0.2)
LSM difference vs placebo		-0.39

CI = confidence interval; IV = intravenous; LSM = least squares mean; Miri = mirikizumab; SE = standard error.
Modified Intent-to-Treat population. Number of biologic-failed patients in placebo and Miri groups were 23 and 71, separately.
Clinical remission: Stool frequency (SF) subscore = 0 or 1 with a ≥1-point decrease from baseline, and Rectal bleeding (RB) subscore = 0, and Endoscopic subscore (ES) = 0 or 1 (excluding friability)
Alternate clinical remission: SF subscore = 0 or 1, RB subscore = 0, and ES = 0 or 1 (excluding friability)
Clinical response: decrease in the modified Mayo score of ≥2 points and ≥30% decrease from baseline, and decrease of ≥1 point in the RB subscore from baseline or an RB score of 0 or 1
Clinical response (bio-failed): clinical response in patients who had previously failed biologic or tofacitinib therapy
Endoscopic remission: ES = 0 or 1 (excluding friability)
HEMI (histologic-endoscopic mucosal improvement): Histologic improvement, defined using Geboes scoring system with neutrophil infiltration in <5% of crypts, no crypt destruction, and no erosions, ulcerations, or granulation tissue; and endoscopic remission
Symptomatic remission: SF = 0, or SF = 1 with a ≥1-point decrease from baseline and RB = 0
Urgency NRS (numeric rating scale) Δ baseline: least square mean change in urgency NRS from baseline
Mixed Model for Repeated Measures (MMRM) was used to compare the treatment groups for bowel movement urgency improvement; for all other endpoints, the Cochran-Mantel-Haenszel (CMH) test with missing data imputed as nonresponse, was used to compare the treatment groups while adjusting for the stratification factors. All analyses in Chinese subpopulation were for descriptive purpose, and multiplicity adjustment was not applied.

Table 1: Primary and key secondary outcomes of LUCENT-1 in Chinese population.

Conclusion: Efficacy and safety profiles were consistent with LUCENT-1 overall population, supporting benefits of mirikizumab treatment in the Chinese UC patients.

Disclosure: Nothing to disclose.

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PP0488

EFFICACY AND SAFETY OF ETRASIMOD IN PATIENTS WITH ULCERATIVE COLITIS ACCORDING TO SEX IN THE ELEVATE UC CLINICAL PROGRAMME

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Introduction: Sex-based differences have shown to impact ulcerative colitis (UC) treatment outcomes and need to be considered when managing therapy strategies for patients with UC. Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active UC.

Aims & Methods: This pre-defined subgroup analysis reports on efficacy and safety outcomes in male and female patients receiving etrasimod 2 mg QD or placebo in the phase 3 ELEVATE UC 52 (NCT03945188) and ELEVATE UC 12 (NCT03996369) trials. Key efficacy endpoints were assessed in patients with a baseline modified Mayo score 5–9 at Week 12 (data pooled from both trials) and Week 52 (ELEVATE UC 52 only). Safety was assessed in the pooled ELEVATE UC 52 and ELEVATE UC 12 safety population.

Results: In the pooled population, a total of 269 male and 227 female patients received etrasimod; 150 and 97, respectively, received placebo. Significantly more male and female patients receiving etrasimod vs placebo achieved clinical remission (Week 12 – male: percentage difference for etrasimod vs placebo [Δ] 10.7, female: Δ 19.7; Week 52 – male: Δ 21.5, female: Δ 29.1; all $p < 0.01$), symptomatic remission (Week 12 – male: Δ 21.4, female: Δ 20.6; Week 52 – male: Δ 26.4, female: Δ 22.5; all $p < 0.01$), endoscopic improvement (Week 12 – male: Δ 12.7, female: Δ 21.2; Week 52 – male: Δ 21.9, female: Δ 32.5; all $p < 0.01$) and endoscopic improvement-histologic remission (Week 12 – male: Δ 9.9, female: Δ 16.7; Week 52 – male: Δ 14.1, female: Δ 24.1; all $p < 0.01$; Table). Similar findings were seen for corticosteroid-free and sustained clinical remission (Table). Similar proportions of treatment-emergent adverse events (TEAEs; 56.8% and 64.6%), serious TEAEs (4.9% and 5.0%) and TEAEs leading to discontinuations (3.8% and 5.8%) were reported in male and female patients receiving etrasimod and were comparable to those receiving placebo.

Conclusion: Similar findings in male and female patients support the use of etrasimod in both to achieve symptomatic, endoscopic and histologic outcomes.

References: Disclaimer: Pfizer's generative artificial intelligence tool MAIA was used to assist production of the abstract first draft. Authors reviewed/edited and take responsibility for the content.

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MG, EK are employees of Pfizer AG and shareholders of Pfizer Inc.
LH is an employee of Pfizer R&D UK Ltd.
ABD is an employee of Pfizer Inc, Philippines.
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5–9 according to sex

Efficacy endpoints, n (%) Δ (95% CI) p value	Week 12 Male (N = 419) Placebo QD (N1 = 150)	Week 12 Male (N = 419) Etrasimod 2 mg QD (N1 = 269)	Week 12 Female (N = 324) Placebo QD (N1 = 97)	Week 12 Female (N = 324) Etrasimod 2 mg QD (N1 = 227)	Week 52 Male (N = 224) Placebo QD (N1 = 80)	Week 52 Male (N = 224) Etrasimod 2 mg QD (N1 = 144)	Week 52 Female (N = 185) Placebo QD (N1 = 55)	Week 52 Female (N = 185) Etrasimod 2 mg QD (N1 = 130)
Clinical remission[a]	15 (10.0)	57 (21.2) 10.7 (4.0, 17.5) 0.002	12 (12.4)	72 (31.7) 19.7 (10.7, 28.8) < 0.001	5 (6.3)	41 (28.5) 21.5 (12.3, 30.8) < 0.001	4 (7.3)	47 (36.2) 29.1 (18.3, 39.9) < 0.001
Symptomatic remission[b]	35 (23.3)	121 (45.0) 21.4 (12.4, 30.5) < 0.001	27 (27.8)	109 (48.0) 20.6 (9.4, 31.8) < 0.001	11 (13.8)	58 (40.3) 26.4 (15.3, 37.5) < 0.001	14 (25.5)	61 (46.9) 22.5 (8.4, 36.6) 0.002
Endoscopic improvement[c]	22 (14.7)	75 (27.9) 12.7 (5.0, 20.5) 0.001	18 (18.6)	89 (39.2) 21.2 (11.2, 31.1) < 0.001	8 (10.0)	47 (32.6) 21.9 (11.6, 32.1) < 0.001	6 (10.9)	55 (42.3) 32.5 (20.6, 44.5) < 0.001
Endoscopic improvement-histologic remission[d]	11 (7.3)	47 (17.5) 9.9 (3.7, 16.1) 0.002	5 (5.2)	47 (20.7) 16.7 (9.7, 23.6) < 0.001	7 (8.8)	34 (23.6) 14.1 (4.6, 23.6) 0.003	4 (7.3)	39 (30.0) 24.1 (13.8, 34.4) < 0.001
Corticosteroid-free clinical remission[e]	-		-		5 (6.3)	41 (28.5) 21.5 (12.3, 30.8) < 0.001	4 (7.3)	47 (36.2) 29.1 (18.3, 39.9) < 0.001
Sustained clinical remission[f]	-		-		1 (1.3)	17 (11.8) 10.2 (4.3, 16.0) < 0.001	2 (3.6)	32 (24.6) 22.2 (13.4, 31.1) < 0.001

Δ is based on estimated common risk difference using the Cochran–Mantel–Haenszel test within each subgroup, stratified by naïve to biologics/Janus kinase inhibitor therapy at study entry (yes/no), baseline corticosteroid use (yes/no) and baseline disease activity (MMS: 4–6 or 7–9); stratification to study was also performed for the pooled analyses. The 2-sided p value is to test the hypothesis of the risk difference being 0.
[a]Defined as the composite of stool frequency subscore = 0 (or = 1 with a ≥1-point decrease from baseline), rectal bleeding subscore = 0 and endoscopic subscore ≤ 1 (excluding friability).
[b]Defined as stool frequency subscore = 0 (or = 1 with a ≥ 1-point decrease from baseline) and rectal bleeding subscore = 0.
[c]Defined as endoscopic subscore ≤ 1.
[d]Defined as endoscopic subscore ≤ 1 with histologic remission (Geboes Index score < 2.0).
[e]Defined as the proportion of patients with clinical remission at Week 52 who had not been receiving corticosteroid for ≥ 12 weeks immediately prior to Week 52.
[f]Defined as the proportion of patients achieving clinical remission at both Week 12 and Week 52.
Δ, percentage difference for etrasimod vs placebo; CI, confidence interval; MMS, modified Mayo score; N, number of patients within the subgroup; N1, number of patients in subgroup receiving treatment; n, number of patients achieving endpoint; QD, once daily; UC, ulcerative colitis.

Table. Efficacy endpoints at Week 12 (pooled from both trials) and Week 52 (ELEVATE UC 52 only) in the ELEVATE UC clinical programme in patients with baseline MMS.

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PP0489

SYMPTOMATIC RESPONSE TRAJECTORIES OF ETRASIMOD IN UC PATIENTS: A POST HOC ANALYSIS OF THE ELEVATE UC 52 TRIAL

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Introduction: Identifying patient populations and factors associated with long-term treatment response can enable personalised management strategies in ulcerative colitis (UC). Etrasimod is an oral, once-daily, selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active UC.

Aims & Methods: Our aim was to identify patient subgroups with distinct response trajectories based on changes in combined rectal bleeding subscore (RBS) and stool frequency subscore (SFS) over time. In this post hoc analysis of etrasimod-treated patients from the ELEVATE UC 52 trial, PRO2 (SFS+RBS) was assessed to identify unique disease response trajectories, with further evaluation of the Week 52 achievement of primary and secondary endpoints. Additionally, a composite of objective and subjective disease activity measures, previously used in trajectory modelling, was assessed as comprehensive disease control (CDC). CDC was defined as meeting all the following criteria by Week 52: complete treatment (up to Week 50), clinical remission, RBS and SFS =0, endoscopic subscore =0/1, complete Mayo score ≤2, histological remission (Geboes score <2 or Robarts Histopathology Index <3), and C-reactive protein and/or faecal calprotectin within normal range. Longitudinal k-means was used to analyse PRO2 disease activity data as individual time-series connected datasets. Response-trajectory groups (TGs) were related to maintenance outcomes and baseline clinical characteristics. We evaluated the clinical relevance of grouping patients this way by assessing the proportions achieving key endpoints and their association with baseline demographics.

Results: Among the cohort of 283 etrasimod-treated UC patients, five TGs were identified (Table). 57% of the super-responders who achieved CDC were clustered in TGs C and E, characterised by rapid response and sustained clinical remission (TG C: 34/51 patients, TG E: 25/44 patients). Baseline characteristics of these patients included lower baseline disease activity, as measured by histology and Mayo score components. In TGs B and D, clinical remission was achieved by 31.7% and 34%, respectively, with clinical response achievement of 48.3% (29/60) and 53.2% (25/47). In TG E, symptomatic improvement according to PRO2 continued beyond Week 12.

Cluster	Clinical remission, n (%)	Clinical response, n (%)	Symptomatic remission, n (%)	Symptomatic response, n (%)	Endoscopic improvement, n (%)	EIHR, n (%)	CDC, n (%)
C (N=51)	34 (66.7)	44 (86.3)	45 (88.2)	45 (88.2)	34 (66.7)	26 (51.0)	17 (33.3)
E (N=44)	25 (56.8)	37 (84.1)	34 (77.3)	38 (86.4)	29 (65.9)	18 (40.9)	7 (15.9)
D (N=47)	16 (34.0)	25 (53.2)	22 (46.8)	27 (57.4)	17 (36.2)	12 (25.5)	6 (12.8)
B (N=60)	19 (31.7)	29 (48.3)	25 (41.7)	33 (55.0)	26 (43.3)	19 (31.7)	12 (20.0)
A (N=81)	0	8 (9.9)	1 (1.2)	9 (11.1)	7 (8.6)	4 (4.9)	0

Clusters (A–E) are presented in descending order according to the highest proportion of patients with clinical remission at Week 52.
CDC, comprehensive disease control; EIHR, endoscopic improvement-histologic remission; N, number of patients in each patient cluster; n, number of patients achieving each endpoint; UC, ulcerative colitis.

Table. Number of patients achieving efficacy endpoints at Week 52 in the ELEVATE UC 52 trial, by patient clusters.

Conclusion: Trajectory modelling allows the identification of distinct phenotypic subgroups among UC patients responding to a targeted therapy, highlighting heterogeneity often masked when analysed as a single population or clinical trial. Notably, a subset of etrasimod patients demonstrated super-fast and fast response dynamics, associated with higher rates of clinical remission and CDC. This may provide an important utility for therapeutic choice and patient management under therapy. Further studies across different modes of action are ongoing to validate these results and identify similarities and differences across patient populations and treatment effects.

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JCW is an employee and shareholder of Pfizer Inc.
CWL has served as a speaker for AbbVie, Dr Falk, Ferring, Hospira, Janssen, MSD, Pfizer Inc, Shire, Takeda, and Warner-Chilcott; has served as a consultant for AbbVie, Dr Falk, Gilead, GSK, Hospira, Iterative Scopes, Janssen, MSD, Oshi Health, Pfizer Inc, Pharmacosmos, Takeda, Topivert, Trellus Health, and Vifor Pharma; and has received research funding from AbbVie and Gilead.
SJ is an employee of Pfizer Inc.
WN is an employee and shareholder of Pfizer Inc.
SS has received lecture/speaker fees from AbbVie, Arena, Biogen, Bristol Myers Squibb, Celgene, Celltrion, Falk, Fresenius, Janssen, MSD, Pfizer Inc, Takeda. Consultancy/advisory fees: AbbVie, Arena, Biogen, Bristol Myers Squibb, Celgene, Celltrion, Falk, Fresenius, Gilead Sciences, IMAB, Janssen, MSD, Mylan, Pfizer Inc, Protagonist, Provention Bio, Takeda, Theravance and Ventyx Biopharma.

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PP0490

EFFICACY OF ETRASIMOD AT WEEK 52 AMONG BIOLOGIC/JANUS KINASE INHIBITOR-NAÏVE PATIENTS WITH ULCERATIVE COLITIS WHO REACHED CLINICAL RESPONSE AT WEEK 12: POST HOC ANALYSIS OF THE PHASE 3 ELEVATE UC 52 RANDOMISED TRIAL

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Introduction: When conventional ulcerative colitis (UC) therapy fails to achieve or maintain remission, advanced UC therapies should be considered to improve patient outcomes. The responder re-randomisation trial (efficacy reported at the end of the maintenance phase for patients who achieved induction clinical response) is a common design used in UC drug development. Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active UC. The etrasimod ELEVATE UC 52 trial utilised a treat-through design, where efficacy at the end of the maintenance phase was reported for the total baseline population.

Aims & Methods: This post hoc analysis evaluated the efficacy of etrasimod vs placebo at Week 52 among biologic/Janus kinase inhibitor (bio/JAKi)-naïve patients with moderately to severely active UC who achieved clinical response at Week 12. ELEVATE UC 52 (NCT03945188) comprised a 12-week induction phase followed by a 40-week maintenance phase with a treat-through design. Patients were randomised 2:1 to receive etrasimod 2 mg or placebo QD. Week 12 clinical response was defined as a ≥2-point and ≥30% decrease from baseline in modified Mayo score (MMS), and a ≥1-point decrease or absolute value ≤1 in rectal bleeding subscore.

Efficacy outcomes evaluated in Week 12 bio/JAKi-naïve responders (baseline MMS 5–9) were the achievement of clinical remission, clinical response, symptomatic remission, endoscopic improvement, endoscopic improvement-histologic remission and corticosteroid-free clinical remission at Week 52.

Results: Among bio/JAKi-naïve patients, Week 12 clinical response was achieved by 132/194 (68.0%) receiving etrasimod and 35/93 (37.6%) receiving placebo. Among Week 12 responders, significantly greater proportions receiving etrasimod vs placebo achieved clinical remission (51.5% vs 17.1%; $p<0.001$) and all secondary endpoints at Week 52 (all $p<0.05$; Table).

	Placebo QD (N=35)	Etrasimod 2 mg QD (N=132)	Adjusted % difference (95% CI) [p value]
Clinical remission, n (%)[a]	6 (17.1)	68 (51.5)	34.6 (19.41, 49.83) [p<0.001]
Clinical response, n (%)[b]	19 (54.3)	99 (75.0)	20.5 (2.50, 38.53) [p=0.026]
Symptomatic remission, n (%)[c]	16 (45.7)	93 (70.5)	24.4 (6.07, 42.63) [p=0.009]
Endoscopic improvement, n (%)[d]	8 (22.9)	75 (56.8)	34.4 (17.83, 50.99) [p<0.001]
EIHR, n (%)[e]	7 (20.0)	53 (40.2)	20.6 (4.55, 36.58) [p=0.012]
CS-free clinical remission, n (%)[f]	6 (17.1)	68 (51.5)	34.6 (19.41, 49.83) [p<0.001]

p<0.05 are highlighted in bold.
Treatment comparisons within each subgroup (adjusted percentage difference and 95% CI) and 2-sided p values were obtained using the Cochran–Mantel–Haenszel method, adjusting to reported randomisation stratification of baseline CS use and baseline disease activity (MMS of 4–6 or 7–9). Missing response was considered as nonresponse.
[a]Clinical remission was defined as an SFS=0 (or =1 with a ≥1-point decrease from baseline), RBS=0, and ES ≤1 (excluding friability).
[b]Clinical response was defined as a ≥2-point and ≥30% decrease from baseline in MMS and a ≥1-point decrease from baseline in RBS or an absolute RBS ≤1.
[c]Symptomatic remission was defined as an SFS=0 (or =1 with a ≥1-point decrease from baseline) and an RBS=0.
[d]Endoscopic improvement defined as an ES of ≤1.
[e]EIHR was defined as an ES ≤1 with histological remission (Geboes Index score <2.0).
[f]CS-free clinical remission was defined as the proportion of patients with clinical remission at Week 52 who had not been receiving CS for ≥12 weeks immediately prior to Week 52.
bio, biologic; CI, confidence interval; CS, corticosteroid; EIHR, endoscopic improvement-histologic remission; ES, endoscopic subscore; JAKi, Janus kinase inhibitor; MMS, modified Mayo score; QD, once daily; RBS, rectal bleeding subscore; SFS, stool frequency subscore.

Table. Efficacy endpoints at Week 52 among bio/JAKi-naïve patients who met clinical response criteria at Week 12 (baseline MMS 5–9).

Conclusion: Etrasimod demonstrated significant efficacy over placebo in all Week 52 efficacy outcomes in bio/JAKi-naïve induction treatment responders. These findings help contextualize data from the ELEVATE UC clinical programme for patients receiving their first advanced UC therapy within a treat-through trial design.

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UNDERSTANDING BENEFIT-RISK TRADE-OFF DIFFERENCES BETWEEN ADVANCED THERAPY NAÏVE AND EXPERIENCED PATIENTS TO INFORM SHARED DECISION MAKING IN ULCERATIVE COLITIS: A DISCRETE CHOICE EXPERIMENT IN THE UNITED STATES AND EUROPE

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Introduction: Prior experience with advanced therapy (AT) may influence the benefit-risk trade-offs ulcerative colitis (UC) patients are willing to make when choosing treatment. It is important to understand differences between AT naïve and experienced patients to inform shared decision making.

Aims & Methods: We aimed to compare preferences for AT and willingness to make benefit-risk trade-offs among AT naïve and AT experienced patients with moderate-to-severe UC. An online discrete choice experiment (DCE) survey was administered to patients with self-reported moderate-to-severe UC in the US, UK, France, Germany, Italy and Spain. Patients chose between hypothetical AT profiles with varying levels of six treatment attributes: time until symptom improvement, probability of remission at one year, probability of corticosteroid (CS)-free remission at one year, annual risk of serious infection, five-year risk of cancer and annual risk of major adverse cardiovascular event (MACE).
We used random parameters logit models stratified by AT naïve vs experienced to estimate the relative attribute importance (RAI: 0–100%). For the RAI, remission attributes were combined. Risk tolerance measures included the maximum acceptable risk (MAR) for each risk independently and the simultaneous MAR threshold for the three risks together that respondents would accept for a 10 percentage-point increase in probability of remission at one year.

Results: 557 patients (49.6% AT naïve; 50.4% AT experienced) completed the survey. ATnaïve patients ranked combined remission and CS-free remission at one year (RAI=36.0%) and five-year cancer risk (RAI=35.3%) as most important, whereas AT experienced patients ranked combined remission (RAI=43.1%) as most important over cancer risk (RAI=28.8%; Table).
The MAR of five-year cancer risk in exchange for a 10 percentage-point increase in probability of remission at one year was higher for AT experienced patients than AT naïve patients (2.9% vs 1.8%). The MARs for the other two risk attributes were close to or above the maximum level tested in the DCE.
When cancer and MACE attributes were considered jointly in exchange for a 10 percentage-point increase in probability of remission at one year, no difference in risk tolerance was observed between AT experienced and naïve patients when serious infection was >1%.

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	Attributes tested in the DCE[a]				
	Time until symptom improvement	Combined probability of remission and CS-free remission at one year	Annual risk of serious infection	Five-year risk of cancer	Annual risk of MACE
RAI,[b] % (95% CI)[c]					
Advanced therapy naïve	9.7 (6.6, 12.9)	36.0 (32.9, 39.0)	9.4 (7.0, 11.9)	35.3 (32.8, 37.9)	9.5 (7.1, 12.0)
Advanced therapy experienced	11.4 (9.5, 14.0)	43.1 (40.6, 45.4)	6.3 (4.4, 8.3)	28.8 (26.7, 30.7)	10.4 (8.4, 12.2)

[a]Treatment attributes and corresponding levels included in the DCE: time until symptom improvement (2 weeks, 4 weeks, 8 weeks, 12 weeks), probability of remission at one year (20%, 35%, 45%), probability of CS-free remission at one year (5.0%, 15%, 20%, 30%, 35%, 40%, 45%), five-year risk of cancer (1.0%, 4.0%, 7.0%), annual risk of MACE (0.1%, 0.3%, 0.5%) and annual risk of serious infection (1.0%, 3.0%, 5.0%). We combined the probability of remission at one year and the probability of CS-free remission at one year into one attribute at the analysis stage for the RAI.

[b]RAI is calculated as the difference in preference weights between the most preferred and least preferred level divided by the sum of the differences across all attributes; estimates sum to 100%.

[c]95% CIs that do not include zero indicate a statistically significant relative importance of an attribute. All attributes were statistically significantly important when selecting an advanced therapy.

CI, confidence interval; CS, corticosteroid; DCE, discrete choice experiment; MACE, major adverse cardiovascular event; N, number of patients; RAI, relative attribute importance; UC, ulcerative colitis.

Table. Relative importance of attributes influencing advanced UC therapy choice in patients who are advanced therapy naïve (N=276) and advanced therapy experienced (N=281).

Conclusion: On average, combined remission at one year and five-year risk of cancer had similarly strong influence on treatment choice for AT naïve patients, while AT experienced patients prioritised combined remission followed by five-year risk of cancer. The difference between AT experienced and naïve patients' willingness to accept five-year risk of cancer for an increase in probability of one-year remission reduced as the other risks increased. For shared decision making, clinicians should consider discussing the holistic risk profile with prioritisation of remission and malignancy throughout the lines of therapy.

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PP0492

EFFICACY OF GUSELKUMAB IN MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS BY EXTENT OF DISEASE AND INFLAMMATORY BURDEN: SUBGROUP ANALYSIS OF THE PHASE 3 QUASAR MAINTENANCE STUDY

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Introduction: Guselkumab (GUS) is a dual-acting IL-23p19 subunit inhibitor that potently neutralizes IL-23 and binds to CD64, a receptor on cells that produce IL-23.[1]
In the Phase 3 QUASAR maintenance study (NCT04033445), both the GUS 100mg every 8 weeks (q8w) and 200mg q4w subcutaneous (SC) maintenance dose regimens were efficacious.[2]
Here, we evaluated efficacy of these dose regimens in clinically relevant subgroups of participants (pts) with and without extensive ulcerative colitis (UC) disease or high inflammatory burden.

Aims & Methods: Clinical responders after 12 weeks of GUS intravenous (IV) induction were randomized 1:1:1 to GUS 100mg SC q8w, GUS 200mg SC q4w or placebo (PBO) (GUS withdrawal) at start of the maintenance study (M-0). Efficacy endpoints were analyzed at Week (W)44 in the primary analysis population of pts randomized and treated in the maintenance study who had a modified Mayo score of 5-9 at induction baseline (I-0).
Subgroup analyses were conducted by (1) extent of UC disease (limited to left side of colon vs extensive) based on screening endoscopy and/or medical history at I-0 and (2) inflammatory burden based on serum C-reactive protein (CRP) levels (≤3 vs >3mg/L) at M-0.

Results: Of 568 pts in the primary analysis population, 257 (45.2%) had extensive UC disease at I-0 and 182 (32.0%) had serum CRP >3mg/L at M-0. Proportions of pts achieving W44 clinical and histologic-endoscopic efficacy endpoints were greater with both GUS dose regimens vs PBO across subgroups (Table).
For pts with extensive UC disease, the proportion of pts achieving clinical remission at W44 was numerically greater with GUS 200mg q4w (62.7%) relative to GUS 100mg q8w (49.4%) (PBO, 11.6%). For those with elevated CRP (>3mg/L) at M-0, clinical remission at W44 was achieved by 48.2% of GUS 200mg q4w vs 30.2% of GUS 100mg q8w pts (PBO, 14.3%).
Proportions of pts achieving endoscopic improvement, histo-endoscopic improvement, endoscopic remission, and maintenance of clinical remission were also greater with GUS 200mg q4w in these subgroups. Among pts with disease limited to the left side of the colon or CRP ≤3mg/L at M-0, proportions of pts achieving these efficacy endpoints were similar with GUS 100mg q8w and GUS 200mg q4w.

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	Extent of UC disease						Serum CRP					
	Limited to left side of colon			Extensive			≤3mg/L			>3mg/L		
	GUS with-drawal (PBO) (N=95)	GUS 100mg SC Q8W (N=109)	GUS 200mg SC Q4W (N=107)	GUS with-drawal (PBO) (N=95)	GUS 100mg SC Q8W (N=79)	GUS 200mg SC Q4W (N=83)	GUS with-drawal (PBO) (N=127)	GUS 100mg SC Q8W (N=125)	GUS 200mg SC Q4W (N=134)	GUS with-drawal (PBO) (N=63)	GUS 100mg SC Q8W (N=63)	GUS 200mg SC Q4W (N=56)
Clinical remission	26.3%	42.2%*	40.2%*	11.6%	49.4%‡	62.7%‡	21.3%	52.8%‡	50.7%‡	14.3%	30.2%*	48.2%‡
Maintenance of clinical remission[a]	45.2% (14/31)	56.1% (23/41)	64.1% (25/39)	21.4% (6/28)	68.0%‡ (17/25)	83.3%‡ (25/30)	41.5% (17/41)	63.5%* (33/52)	67.3%* (35/52)	16.7% (3/18)	50.0% (7/14)	88.2%‡ (15/17)
Endoscopic improvement	26.3%	47.7%†	42.1%*	11.6%	51.9%‡	63.9%‡	21.3%	56.8%‡	51.5%‡	14.3%	34.9%*	51.8%‡
HEMI	24.2%	43.1%†	40.2%*	9.5%	44.3%‡	57.8%‡	18.9%	51.2%‡	48.5%‡	12.7%	28.6%*	46.4%‡
Endoscopic remission	22.1%	31.2%	25.2%	8.4%	39.2%‡	44.6%‡	18.1%	40.0%‡	34.3%†	9.5%	23.8%*	32.1%†

Includes pts with modified Mayo score of 5-9 at induction baseline who achieved clinical response to GUS induction and were re-randomized at maintenance study entry.
*P<0.05, †P<0.01, ‡P<0.001 vs PBO (nominal P values), based on the Cochran-Mantel Haenszel test stratified by clinical remission status at maintenance baseline and induction treatment, except for maintenance of clinical remission where P values were based on Fisher's exact test.
[a]Denominator includes only pts with clinical remission at maintenance baseline.
Clinical remission is defined as a stool frequency subscore of 0 or 1 and not increased from induction baseline, a rectal bleeding subscore of 0, and an endoscopy subscore of 0 or 1 with no friability. Maintenance of clinical remission is defined as clinical remission at W44 among pts in clinical remission at maintenance baseline. Endoscopic improvement is defined as an endoscopy subscore of 0 or 1 with no friability. HEMI is defined as achieving a combination of histologic improvement and endoscopic improvement. Endoscopic remission is defined as an endoscopy subscore of 0.
Pts who had an ostomy or colectomy, dose adjustment, prohibited change in UC medication, or who discontinued study agent due to lack of efficacy or an AE of worsening of UC prior to W44 were considered not to have achieved the efficacy endpoints. For pts who discontinued study agent due to COVID-19 related reasons (excluding COVID-19 infection) or regional crisis in Russia and Ukraine prior to W44, the observed values were used if available. Pts who discontinued study agent for other reasons prior to W44 were considered not to have achieved the endpoint. Nonresponder imputation for missing data: pts who were missing 1 or more of the components pertaining to an endpoint at W44 were considered not to have achieved the endpoint. Pts who had an unevaluable biopsy were considered not to have achieved histologic endpoints.
AE=Adverse event; CRP=C-reactive protein; GUS=Guselkumab; HEMI=Histologic-endoscopic mucosal improvement; I-0=Induction study baseline; M-0=Start of the maintenance study; PBO=Placebo; pts=Participants; Q4W=Every 4 weeks; Q8W=Every 8 weeks; SC=Subcutaneous; UC=Ulcerative colitis; W=Week.

Table. Key efficacy endpoints at maintenance W44 in subgroups of pts from QUASAR defined by extent of UC disease (limited to left side of colon vs extensive) at I-0 and inflammatory burden (serum CRP ≤3 vs >3mg/L) at M-0.

Conclusion: For pts with moderately to severely active UC with extensive disease or high inflammatory burden, greater efficacy rates were observed with GUS 200mg SC q4w compared with GUS 100mg SC q8w.

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PP0493

COMPARATIVE ANALYSIS OF HEALTHCARE RESOURCE UTILISATION AND DIRECT COSTS IN CROHN'S DISEASE FROM US CLAIMS: EARLY VS LATE BIOLOGIC INITIATION

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Introduction: While early intervention with biologics improves clinical response in Crohn's disease (CD), the optimal timing to reduce healthcare resource utilisation (HCRU) and direct costs remains unclear. This study examined the impact of biologic initiation on HCRU and direct costs over 5 years in adults recently diagnosed with CD in the US.

Aims & Methods: Adults with >2 non-ancillary outpatient claims (30 days apart) or 1 inpatient claim for CD (ICD-10: K50.x) from 1/1/2015 to 7/31/2024 were identified using Optum® Clinformatics® Data Mart. Patients with CD who started biologic treatment within 2 years of diagnosis (index date) and had continuous enrollment for 1 year pre- and 5 years post-index were stratified into early (<12 months) or late (>12-24 months) initiators. Due to skewed distributions, CD-related and non-CD-related costs at 5-year post-index were analysed using gamma log. Poisson or negative binomial (with zero-inflated models in some cases) were used to model HCRU. All models were adjusted for demographic and clinical characteristics at biologic initiation.

Results: Early initiators (EI) were significantly younger (mean[SD] age: 43.2[16.9] vs 48.0[16.6] years, p<0.001), predominantly male (50.6% vs 44.5%), had significantly higher commercial insurance (84.5% vs 77.2%,

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p=0.004) than late initiators (LI), and had similar Deyo Charlson comorbidity index scores (mean[SD]: 0.7[1.2] vs 0.7[1.3]). EI had significantly lower total mean medical costs ($63,761 vs $66,282), significantly lower IBD-related mean medical and pharmacy costs ($1074 vs $1494), as well as significantly lower mean pharmacy costs to treat depression ($28 vs $346), and anxiety ($38 vs $79) than LI (Table).

Additionally, EI had fewer emergency visits (1.5 vs 1.6), fewer all-cause (4.2 vs 4.9 days), and CD-related hospitalisations days (3.9 vs 4.7 days) than LI. Other HCRUs and costs were similar between the two groups.

	Early initiator (EI) (N=1273)	Late initiator (LI) (N=290)	Adjusted P-value
Costs in US Dollars			
Medical and pharmacy costs	$290,491 (239,398)	$298,162 (246,492)	0.794
Medical costs	$63,761 (128,352)	$66,282 (97,089)	0.0497
Anxiety/depression outpatient costs	$11,492 (48,843)	$12,203 (41,583)	0.763
Non-IBD pharmacy costs	$225,657 (194,145)	$230,385 (210,412)	0.711
Opioid pharmacy costs	$170 (1697)	$189 (1285)	0.825
Anti-depression pharmacy costs	$28 (393)	$346 (4746)	<.001
Anti-anxiety medication costs	$38 (476)	$79 (690)	0.007
CD-related medical and pharmacy costs	$108,293 (116,124)	$100,574 (112,197)	0.070
CD-related medical costs	$42,451 (70,935)	$45,547 (74,554)	0.159
IBD-related medical and pharmacy costs	$1,074 (5,567)	$1,494 (6,971)	0.046
HCRU			
Number of emergency department visits	1.5 (4.1)	1.6 (4.0)	0.004
Number of outpatient visits	56.6 (49.8)	60.1 (53.6)	0.402
Number of CD outpatient visits	23.7 (19.2)	24.1 (19.9)	0.357
Number of CD-related emergency department visits	0.7 (2.3)	0.8 (2.3)	0.071
Number of CD-related hospitalisations	0.6 (1.2)	0.7 (1.4)	0.886
Number of IBD-related diagnostic procedures	0.8 (1.0)	0.9 (1.0)	0.398
Number of CD-related management procedures	0.1 (0.3)	0.1 (0.3)	0.417
Patients having pharmacy claim, n (%)	1240 (97.4%)	277 (95.5%)	0.289
Patients having IBD-related pharmacy, n (%)	184 (14.5%)	33 (11.4%)	0.163
Patients having Extraintestinal manifestation, n (%)	58 (4.6%)	17 (5.9%)	0.458
Days of hospitalisation	4.2 (10.4)	4.9 (12.3)	0.013
Days of CD-related hospitalisations	3.9 (9.5)	4.7(12.2)	<.001
Length of hospitalisation in days	2.0 (3.8)	2.1 (4.5)	0.102

Data are presented as mean (SD=standard deviation) unless otherwise specified.
CD, Crohn's disease; IBD, inflammatory bowel disease

Conclusion: Early initiators of biologics had lower HCRU and reduced non-CD medication pharmacy costs at 5 years post-index compared to starting biologics after a year, potentially leading to overall cost savings and improving patient health outcomes.

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PP0494

RAPID ONSET OF ACTION OF USTEKINUMAB IN PATIENTS WITH MODERATE TO SEVERE ULCERATIVE COLITIS (RAUCU STUDY)

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Introduction: Real-world data on the speed of onset of ustekinumab (UST) in patients with ulcerative colitis (UC) are limited. Current information regarding early response in these patients comes from the UNIFI trial.

Aims & Methods: The primary outcome of this study was to evaluate the response to UST in patients with moderate-to-severe UC by assessing PRO2 at 14 days after the first intravenous dose (6mg/kg).

Methods: A multicenter, prospective observational study was designed in patients ≥18 years old with moderate-to-severe UC (partial Mayo score ≥6) who initiated UST 6mg/kg between June 2023 and September 2024. Patients received a diary to record the stool frecuency and rectal bleeding during first 14 days after first dose of induction. The proportion of patients achieving the combined outcome of rectal bleeding score = 0 and stool frequency score ≤ 1 (PRO-2: patient-reported outcome 2 remission) was assessed. FC was measured at baseline and at day 7 and 14. Health-Related Quality of Life was evaluated using the disease-specific shortened Spanish version of the IBDQ (SIBDQ-9) at baseline and at 4 weeks. Patients with previous panproctocolectomy with ileostomy or pouch and those who had received cyclosporine within one month prior to starting UST were excluded. Descriptive statistics are presented as frequencies with percentages for categorical variables.

Results: A total of 54 patients were included across 14 hospitals in Spain. Of these, 29 (53.7%) were men, with a mean age of 54 (± SD 16.1) years. Thirty (55.6%) patients had pancolitis, 20 (37%) had left-sided colitis, and 4 (7.4%) had proctitis. Five (9.3%) patients were bio-naïve, 32 (59.3%) had previously received one biological therapy, 10 (18.5%) had received two, and 7 (13%) had received three biologics prior to starting UST. By day 14, 26% of patients had achieved clinical remission. No significant differences

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in clinical remission were observed based on disease extent (p = 0.11) or the number of prior biologics (p = 0.39). By day 7, 12/31 (38.7%) patients with baseline bloody stools had achieved a rectal bleeding score = 0. A reduction in FC >50% was observed in 14/48 (29%) patients by day 7 and in 20/47 (43%) patients by day 14. The SIBDQ9 average score was 54.9 (± SD 6.2) at baseline and 61.3 (± SD 7.4) at week 4.

Conclusion: One-quarter of patients with moderate-to-severe UC achieved clinical remission by day 14 following the first induction dose of UST. Nearly 40% of patients achieved a rectal bleeding score = 0 (PRO2) by day 7. A reduction in FC >50% was observed in 43% of patients by day 14.

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PP0495

ADHERENCE TO MESALAZINE SUPPOSITORIES MAINTENANCE TREATMENT IN PATIENTS WITH ULCERATIVE PROCTITIS: MESUPO MULTICENTER PROSPECTIVE OBSERVATIONAL STUDY FROM GETECCU

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Introduction: Mesalazine suppositories are recommended in guidelines for the induction and maintenance of remission in patients with ulcerative proctitis (UP). Nevertheless, nonadherence can be frequent and the recommended dose of mesalazine suppositories in maintenance treatment remains unknown.

Aims & Methods: The aim of this study was to determine the frequency of non-adherence to mesalazine suppositories, evaluate potential factors for nonadherence and the impact on quality of life and number of relapses in patients with UP.

Methods: A multicenter, prospective observational study was designed in patients ≥18 years old with UP under maintenance treatment with mesalazine suppositories (Claversal© 500 mg, Pentasa© 1g or Salofalk©1g). We evaluated adherence, socio-demographic data and complications associated with poor adherence at 6 and 12 months. Adherence was assessed by tracking pharmacy refills (medication possession ratio, MPR). We defined good adherence as MPR ≥80% over 12 months. Health-Related Quality of Life (HRQoL) was evaluated using the disease-specific shortened Spanish version of the IBDQ (SIBDQ-9). Relapse was defined as partial Mayo ≥2; biological remission as a partial Mayo score <2 and fecal calprotectin ≤150. Multivariate analysis allowed us to determine the risk factors for non-adherence to mesalazine suppositories.

Results: A total of 188 patients with mean age 51 were consecutively included across 14 Spanish hospitals (Table 1). Of these, mean MPR was 76% and 56% presented MPR ≥80%. Mean age was higher among patients with MPR ≥80% (p=0.045). A total of 125 (71,4%) patients were in biological remission, 31 patients (16%) had a relapse and 3 (1.6%) patients were hospitalized during the follow-up. The use of corticosteroids during the follow-up (OR=4.25, p=0.041) were associated with better adherence. There were no differences between the different brands and doses of suppositories analyzed. Low adherence was not related with increased number of flares (p=0.927), hospitalizations (p=0.225) or decreased quality of life (p = 0.353) over the 12 months of follow-up

Conclusion: Almost half of patients with UP on maintenance treatment with mesalazine suppositories have low adherence. The use of oral corticosteroids during the follow-up may increase adherence, but poor adherence to mesalazine suppositories was not related with risk of relapse, hospitalizations or worst quality of life.

Disclosure: Manuel Barreiro de Acosta: speaker, consultant and advisory member for or have received research funding from AbbVie, Janssen, Takeda, Pfizer, Lilly, Alphasigma, Sandoz and Oncostellae.

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PP0496

LONGITUDINAL EFFECTS OF BIOLOGIC THERAPY ON HEALTH-RELATED QUALITY OF LIFE IN NEWLY DIAGNOSED ADULT PATIENTS WITH INFLAMMATORY BOWEL DISEASE AT ONE- AND THREE-YEAR FOLLOW-UPS (THE IBSEN III COHORT)

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Introduction: Biologic therapy remains a key treatment modality in the management of ulcerative colitis (UC) and Crohn's disease (CD). However, prospective observational studies assessing its longitudinal impact on health-related quality of life (HRQoL) in these patients are lacking.

Aims & Methods: This study aimed to compare HRQoL scores at one-year follow-up between patients who received biologic therapy within the first year after diagnosis and those who did not. A secondary objective was to evaluate the same association at three-year follow-up.

Adults (≥18 years) were recruited from the Inflammatory Bowel Disease in South-Eastern Norway III (IBSEN III) cohort, a prospective, population-based study of newly diagnosed, unselected patients with UC and CD.

HRQoL was measured using the disease-specific Inflammatory Bowel Disease Questionnaire (IBDQ) and the generic Short Form-36 (SF-36) at diagnosis, as well as at one- and three-year follow-ups. For the SF-36, we analysed the General Health domain, as well as Physical Component Summary (PCS) and Mental Component Summary (MCS) scores.

Generalised linear mixed models for repeated measures were used to estimate associations between biologic therapy and HRQoL scores over time. For UC, models were adjusted for age, sex, education, civil status, work status, Montreal disease extent, disease severity at diagnosis, and Simple Clinical Colitis Activity Index (SCCAI) scores. For CD, we adjusted for demographics, Montreal disease location and behaviour, and Harvey-Bradshaw Index (HBI) scores. HRQoL scores at diagnosis were included as covariates.

Results: A total of 1278 patients were included (UC: 857 [67.1%], CD: 421 [32.9%]). Patients who completed HRQoL questionnaires were significantly older and more often female than non-responders. No other demographic or disease-related significant differences between responders and non-responders were found. Therefore, the study population was considered representative of the broader cohort. During the first year, 121/857 (14.1%) patients with UC and 165/421 (39.2%) patients with CD received biologic therapy.

Patients with UC who received biologic therapy reported significantly lower IBDQ scores at one-year follow-up compared to those who did not, even after adjusting for demographics, disease activity, and IBDQ scores at diagnosis (estimated mean IBDQ score with biologics: 170.6, without biologics: 182.5; p=0.04). Similarly, SF-36 scores were significantly lower at one-year follow-up among patients with UC who received biologics compared to those without: General Health (57.0 vs. 63.3), PCS (43.4 vs. 46.9), MCS (43.4 vs. 46.0) (all p<0.05). No significant associations were found between biologic therapy during the first year and HRQoL scores at three-year follow-up for patients with UC.

Among patients with Crohn's disease, no statistically significant associations were found between biologic therapy in the first year and HRQoL scores at either of the follow-ups.

Conclusion: In newly diagnosed patients with UC, biologic therapy initiated within the first year was independently associated with significantly lower HRQoL scores at one-year follow-up, but not at three-year follow-up. In CD, biologic therapy was not associated with significant changes in HRQoL at either of the follow-ups.

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PP0497

PERIODONTITIS (PO) - A COMMON COMPLICATION OF CHRONIC INFLAMMATORY BOWEL DISEASE A SYMPTOM-ORIENTATED SCORE FOR RECORDING PO REQUIRING TREATMENT

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Introduction: A disturbed intestinal mucosal barrier is crucial in the pathogenesis of chronic inflammatory bowel disease (IBD). A higher susceptibility of patients with IBD to periodontitis (PO) is consistent with a similar pathogenesis. However, PO is often neglected in IBD patients, because their therapy is focussed on the intestinal symptoms.

Aims & Methods: A symptom-orientated score should be developed, which results from the information in a questionnaire that patients answer. A threshold value should indicate PO requiring treatment. A PO score is calculated on the basis of the questionnaire and compared with an objective grading of PO (determined by a clinical assessment). Grade 0 = no PO, grade 1 = mild PO (not requiring treatment), grade 2 = moderate PO, grade 3 or 4 = severe PO. Data was collected from 146 control subjects without IBD, 76 patients with Crohn's disease (CD) and 42 patients with ulcerative colitis (UC).

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Results: The table shows the distribution of clinical PO grades in the 3 groups (No IBD, CD, CU) in absolute numbers and per cent as well as the respective mean PO score. The data of the control group were used for the evaluation of the questionnaire. There was a good correlation between the objective degree of PO and the PO score determined from the patients' statements. The ROC analysis showed that a PO score of 6.75 resulted in a sensitivity of 72.4% and a specificity of 64% for the detection of PO stage ≥ 2. For the detection of a PO stage ≥ 3, a PO score of 7.75 results in a sensitivity of 73.2% and a specificity of 64%. The AUROC was 0.746 and 0.742, respectively. There were no cases with grade 0 or 1 PO in the CD group or UC group, i.e. all had at least moderate/moderate PO. The high prevalence of severe PO, even in people without IBD, can be explained by the fact that most of these people were recruited in a specialised periodontology clinic. Accordingly, the mean PO score is relatively high here.

	Clinical grade of PO	Number	Percent	Mean PO score
Persons without IBD	0	23	15.8	9.1
	1	2	1.4	
	2	36	24.7	
	3	64	43.8	
	4	31	14.4	
	total	146	100	
UC	2	12	28.5	8.7
	3	29	69.0	
	4	1	2.4	
	total	42	100	
CD	2	23	30.3	9.7
	3	46	60.5	
	4	7	9.2	
	total	76	100	

Conclusion: Our data confirm a high prevalence of higher grade PO requiring treatment in patients with IBD. A higher PO score (from 6.75) can predict with high probability PO requiring treatment, which is often overlooked or neglected in IBD patients. These patients can therefore be referred for integrated therapy.

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PP0498

UNLOCKING THE DUAL POTENTIAL OF MESENCHYMAL STEM CELLS: FROM STEROID-REFRACTORY GVHD TO INFLAMMATORY BOWEL DISEASE

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Introduction: Mesenchymal stem cells (MSCs) exhibit immunomodulatory properties through Treg induction[1], yet their clinical efficacy varies across gastrointestinal immune disorders[2].

This study integrates systematic evidence from 79 MSC-GVHD trials with preclinical IBD models to identify critical determinants of MSC therapeutic success.

Aims & Methods:

• Clinical Analysis: Meta-analysis of 79 trials (4,710 patients) evaluating MSC efficacy in GVHD, stratified by MSC source (bone marrow vs. umbilical cord), dosing, and disease severity.

• Preclinical Validation: DSS-induced murine colitis treated with human umbilical cord-derived MSCs (hUCB-MSCs; 2×10^6 cells/IP, n=8/group). Outcomes included histopathology, Treg quantification (flow cytometry), and IDO/PGE2 mechanisms (qPCR/ELISA).

Results:

1. GVHD Clinical Data:
   - Umbilical cord MSCs showed superior 1-year survival vs. bone marrow sources (76% vs. 47%, p=0.016) in steroid-refractory GVHD.
   - Pediatric patients exhibited 71.2% response rate with early MSC intervention (<48h post-diagnosis).
2. IBD Preclinical Findings:
   - hUCB-MSCs reduced colitis activity index by 67% (p=0.003) and increased colonic Foxp3+ Tregs 3.2-fold vs. controls (p=0.017).
   - IFN-γ priming enhanced hUCB-MSC immunosuppression via IDO↑ (4.8-fold, p=0.008) and PGE2↑ (2.1-fold, p=0.03).

Conclusion: MSCs demonstrate disease-specific therapeutic hierarchies: Umbilical cord-derived cells optimized for acute GVHD and IBD through Treg/IDO/PGE2 axis activation. Timing (early intervention) and inflammatory priming (IFN-γ) are critical modulators. These insights warrant phase II trials of IFN-γ-primed UCMSCs for immune-mediated gastroenterological diseases.

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PP0499

REAL-WORLD EFFECTIVENESS AND SAFETY OF VEDOLIZUMAB AS FIRST- OR SECOND-LINE THERAPY IN ULCERATIVE COLITIS

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Introduction: Despite the growing number of biological therapies available for ulcerative colitis (UC), data on how best to sequence these agents in clinical practice are still scarce. Vedolizumab (VDZ) is an α4β7 integrin antibody used to treat moderate-to-severe ulcerative colitis (UC).

Aims & Methods: We aimed to assess the real-life long-term effectiveness and safety of VDZ as a first- or second-line biological therapy and identify predictors of efficacy in a large cohort of patients with refractory or difficult-to-treat UC. This was a multicenter, retrospective observational cohort study including patients with moderately-to-severely active UC who received VDZ for at least 12 weeks.

Primary endpoints were clinical remission (partial Mayo score ≤ 2), endoscopic remission (endoscopic Mayo subscore of 1 or 0), and steroid-free clinical remission at week 52.

Secondary endpoints included clinical response at weeks 12, 26, and 52; endoscopic response at week 52; primary and secondary loss of response; drug discontinuation; need for dose optimization; adverse events (AEs); need for hospitalization; and colectomy. Clinical predictors of VDZ treatment success were evaluated.

Univariable and multivariable analyses as well as Kaplan–Meier analyses of treatment persistence were performed.

Results: We included 206 UC patients (mean age 44.9 ± 16.2 years, 56.8% female, 56% pancolitis, 76.8% biologic-naïve, and 22.7% with previous anti-TNF failure). Clinical remission was observed in 69.2% of biologic-naïve patients and 63.8% of anti-TNF failure patients at week 52. Steroid-free clinical remission was achieved in 70.3% and 58.8% of biologic-naïve and anti-TNF failure patients, respectively. Endoscopic remission at one year was achieved in 48.9% and 42.9% of biologic-naïve and anti-TNF failure patients, respectively. Biochemical remission at one year was achieved in 77.8% and 73.9% of biologic-naïve and anti-TNF failure patients, respectively. Primary and secondary loss of response occurred in 7.5% and 13.8% of biologic-naïve patients and in 18.2% and 21.3% of anti-TNF failure patients, respectively. Treatment persistence was higher among biologic-naïve UC patients (84.9% vs. 72.3%, p = 0.048). Three patients (1.4%) required colectomy, and 14 patients (6.8%) needed hospitalization while on VDZ treatment. Clinical response at week 12 (p = 0.001) and biologic-naïve status (p = 0.006) were predictors of better VDZ response, whereas BMI > 30 kg/m² (p = 0.04) and serum albumin ≤ 3.5 g/dL (p = 0.03) were predictors of poorer response.

Conclusion: In this real-world study of patients with severe UC, VDZ demonstrated effectiveness in inducing clinical remission and endoscopic improvement, even in cases of previous anti-TNF failure, with a favorable safety profile. Biologic-naïve status and early clinical response were associated with higher likelihood of one-year clinical and endoscopic remission.

Disclosure: Nothing to disclosure.

PP0500

COMPARATIVE EFFICACY OF GUSELKUMAB VS. RISANKIZUMAB IN INDUCTION PHASE FOR MODERATE TO SEVERE CROHN'S DISEASE: AN ANCHORED MAIC ANALYSIS

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Introduction: The need for effective therapies in moderate to severe Crohn's disease (CD) drives the comparative evaluation of biologic treatments. The current work uses an anchored Matching-Adjusted Indirect Comparison (MAIC) to assess the efficacy of guselkumab (GUS) relative to risankizumab (RZB) during the treatment induction phase. This MAIC is using data from five phase 3 placebo-controlled blinded clinical trials (GRAVITI, GALAXI 2, GALAXI 3, ADVANCE, MOTIVATE).

Aims & Methods: Adult patients with moderate to severe CD, as defined in GRAVITI, GALAXI 2/3, ADVANCE and MOTIVATE trials were included. An anchored MAIC was utilized, leveraging placebo as common comparator and following guidance from the National Institute for Health Care Excellence in UK. The analysis aligned pooled individual patient data for GUS with published data for RZB by matching exclusion criteria and patient baseline characteristics that could be treatment effect modifiers (eg Age, Sex, BMI, CDAI, disease location or prior biologic treatment failure).

During induction, patients received GUS 200 mg intravenous or 400 mg subcutaneous at weeks 0, 4, and 8 or RZB 600 mg intravenous at weeks 0, 4, and 8. The endpoints of interest were symptomatic endpoints such as clinical remission, PRO-2 remission, clinical response, abdominal pain and stool frequency remission, as well as endoscopic response at week 12, which marks the end of the induction period.

Results: After matching, baseline characteristics were balanced between the GUS and RZB cohorts. In a mixed population of bio-naïve and bio-experienced patients, GUS demonstrated statistically significant results in comparison to RZB for PRO-2 remission, clinical response and abdominal pain remission (Table 1).

Clinical remission showed a trend in favor of GUS, endoscopic response and stool frequence remission were comparable to RZB. In patients with biologic treatment failure versus the MOTIVATE trial, results had been

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generally more in favor of GUS, while maintaining positive outcomes seen across mixed populations (ADVANCE & MOTIVATE). Results were also mostly consistent whether analyzing pooled GUS data or individual trials, eg from GRAVITI, against RZB.

Endpoint	RE measure	Mixed population (ADVANCE & MOTIVATE)		Bio-failure population (MOTIVATE)	
		BC	SA	BC	SA
Clinical response	OR [95%CI]	1.78* [1.06; 2.99]	1.42 [0.76; 2.64]	1.71 [0.82; 3.57]	1.23 [0.49; 3.04]
Endoscopic response	OR [95%CI]	1.05 [0.56; 1.97]	1.05 [0.53; 2.09]	1.57 [0.59; 4.18]	1.56 [0.49; 5.01]
Clinical remission	OR [95%CI]	1.51 [0.84; 2.70]	1.22 [0.60; 2.46]	1.52 [0.65; 3.51]	1.08 [0.39; 3.01]
PRO-2 remission	OR [95%CI]	1.96* [1.13; 3.38]	1.85* [1.03; 3.33]	2.89* [1.29; 6.49]	2.97* [1.11; 7.94]
AP score remission	OR [95%CI]	1.68 [1.00; 2.82]	1.37 [0.75; 2.52]	2.02 [0.97; 4.20]	1.68 [0.69; 4.09]
SF score remission	OR [95%CI]	0.99 [0.62; 1.59]	0.93 [0.55; 1.58]	1.14 [0.59; 2.19]	1.04 [0.48; 2.26]

Table 1: Results of GUS from pooled GRAVITI, GALAXI 2/3 vs. RZB for induction period at week 12.

Footnotes: clinical response= ≥100-point reduction from baseline in CDAI OR CDAI <150; endoscopic response= ≥50% improvement from baseline in SES-CD OR SES-CD ≤2; clinical remission= CDAI <150; PRO-2 remission=AP mean daily score at or below 1 [AP≤1] AND SF mean daily score at or below 3 [SF≤3], and no worsening of AP or SF from baseline; AP score remission=Abdominal Pain mean daily score at or below 1 [AP≤1]; SF score remission= number of liquid or very soft stools mean daily score at or below 3 [SF≤3];

*BC=baseline risk-adjusted; SA=sensitivity analysis; *p<0,05.*

Conclusion: The anchored MAIC analysis demonstrates comparable to even superior efficacy of guselkumab over risankizumab in inducing symptomatic improvements in Crohn's disease patients. It also suggests that subcutaneous induction seems to be preferential over iv induction.

Disclosure: van Sanden S, Trevisan M, Strobl A and Wirth D are employees of J&J Innovative Medicines.

PP0501

IMPACT OF HLA-DQA1*05 GENOTYPE IN IMMUNOGENICITY AND RESPONSE TO TREATMENT WITH TUMOUR NECROSIS FACTOR-ALPHA ANTAGONISTS IN INFLAMMATORY BOWEL DISEASE PATIENTS

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Introduction: Carrying the HLA-DQA1*05 gene has been linked to a higher risk of developing immunogenicity in patients with inflammatory bowel disease (IBD) who are treated with tumor necrosis factor-alpha inhibitors (antiTNF). However, studies have shown inconsistent evidence regarding its connection to a loss of response (LOR) or adverse events in patients with IBD.

Aims & Methods: Aims: To determine the impact of carriage of HLADQA1*05 allele on effectiveness, drug persistence and safety in patients diagnosed with IBD who had received antiTNF.

Methods: Retrospective, single-center cohort study including IBD patients who had received antiTNF therapy. The HLA-DQA1*05 allele was determined from a saliva sample (kit OGD-600 de DNA Genotek Oragene) and DNA extraction with the Maxwell® RSC-Stabilized Saliva DNA kits. We evaluate drug persistence, withdrawal of antiTNF, primary non-response and secondary loss of response, through antiTNF levels, development of antiTNF antibodies and adverse events in patients distributed by HLA-DQA1*05 allele presence.

Results: A total of 170 patients were included, 51% male, mean age 46.2 ± 14.6, 120 (70.6%) diagnosed with Crohn's disease. The mean disease duration was 189.6 ± 99.8 months. Fifty patients (29.4%) had extraintestinal manifestations and 30% comorbidities. HLA-DQA1*05 was positive in 74/167 (44%) patients. One hundred twenty-seven patients (75%) received infliximab (IFX) and 43/170 (25%) adalimumab. AntiTNF was the first advanced therapy in 161/179 (95%) of the cohort. Seventy-five out of 170 patients (44%) received an intensified antiTFN regimen, as well as 80/170 (47%) were treated with combotherapy with immunosuppressants. AntiTNF was withdrawn in 88/170 (52%) patients during follow-up, and the rate of adverse events that led to discontinuation of the drug was 17.6% (30/170).

When we compared the rate of primary non-response (12.2 vs. 7.5%. p=0.3) and secondary loss of response (48% vs. 44%, p=0.5), withdrawal of antiTNF therapy (48 vs. 54%, p=0.5), duration of antiTNF therapy, (85.3 ± 69.1 months vs. 90.3 ± 76.4, p=0.3), need of intensified regimen (47% vs. 42%, p=0.4),use of combotherapy (46% vs. 49%, p=0.6) and antiFNF levels (2.6 ± 2.3 vs. 2.9 vs. 2.9, p=0.3) we did not find significant differences between patients by HLA-DQA1*05 status (presence vs absence). On the other hand, the rate of infusion reactions (8.1% vs. 11.8%, p=0.4), development of anti-TNF antibodies (9.5 vs. 10.8, p=0.7) , overall (14.9 vs. 19.4, rate of adverse events (14.9 vs. 24.7, p=0.5) or adverse events that led to drug withdrawal (14.9 vs 19.4, p=0.4) were also similar among patients regardless HLA-DQA1*05 carriage.

Conclusion: In our real-life cohort of IBD patients treated with antiTNF, being an HLA-DQA1*05 carrier did not act as a predictor of effectiveness (primary non-response or secondary loss of response), immugenicity or need of intensified regimen. The safety of anti-TNF treatment has also not been influenced by the variant.

Disclosure: Nothing to disclose.

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PP0502

CLINICAL EFFECTIVENESS OF VEDOLIZUMAB IN CHINESE PATIENTS WITH ULCERATIVE COLITIS: FINAL RESULTS FROM THE VALUE, A MULTICENTER PROSPECTIVE STUDY

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Introduction: Vedolizumab (VDZ), a monoclonal antibody therapy approved for moderate-to-severe ulcerative colitis (UC) and Crohn's disease (CD) in China in 2020[1] has a unique, gut-selective, anti-lymphocyte trafficking (GSALT) mechanism of action.[2] However, there is limited data on VDZ's real-world effectiveness and safety in Chinese IBD patients.[3] Interim results from the VALUE study that enrolled patients with UC have been published [3,4] and final results are reported here.

Aims & Methods: VALUE (NCT04872491), a multicenter, single-arm, prospective, observational study evaluated the safety and effectiveness of VDZ in adult Chinese patients with IBD (N = 500) over 72 weeks. Clinical response, clinical remission, and endoscopic remission were assessed at Weeks 14, 30, and 54. Partial Mayo and Mayo endoscopic subscores at baseline and Weeks 14, 30, and 54 were calculated. Clinical response was defined as ≥2-point reduction in Partial Mayo Clinic score and ≥25% decrease from baseline score accompanied with ≥1-point decrease in the rectal bleeding or absolute rectal bleeding subscore ≤1. Clinical remission was defined as Partial Mayo Clinic score ≤2 with no subscore >1, endoscopic remission as Mayo endoscopic subscore ≤1, and complete endoscopic remission as Mayo endoscopic subscore 0. Safety results and CD patients' data are presented in separate VALUE abstracts (IDs: AS-UEG-2025-01329 and AS-UEG-2025-01324).

Results: Among 409 UC patients enrolled in the final analysis, 91.9% (376/409) were included in the effectiveness analysis set. Mean age (±standard deviation [SD]) of patients was 46.3 (±15.1) years, and 59.4% (n=243) were male. At baseline, 30.8% (126/409) of patients had moderate UC, and 29.8% (122/409) had severe UC. The median duration of UC was 2.5 years, concomitant steroids were prescribed in 18.9% (71/376). Mean (±SD) partial Mayo score was 5.1 (2.1) at baseline, which declined to 1.7 (1.7), 1.5 (1.7) and 1.3 (1.5) by Weeks 14, 30 and 54, respectively, post-VDZ treatment. Patients achieving clinical response, clinical remission, endoscopic remission, and complete endoscopic remission were 76.9% (220/286), 71.1% (207/291), 63.6% (35/55), and 21.8% (12/55) at Week 14; 78.1% (193/247), 76.3% (190/249), 63.6% (28/44), and 27.3% (12/44) at Week 30; and 80.3% (155/193), 83.3% (165/198), 71.4% (20/28), and 50.0% (14/28) at Week 54, respectively. Patients with moderate and severe UC; biologic naïve patients and patients with prior exposure to biologics at baseline also achieved clinical response, clinical remission, and endoscopic remission (Table).

	Moderate UC			Severe UC		
	Clinical response	Clinical remission	Endoscopic remission	Clinical response	Clinical remission	Endoscopic remission
Week 14	89.8 (79/88)	65.9 (58/88)	57.1 (12/21)	90.0 (72/80)	63.8 (51/80)	56.3 (9/16)
Week 30	87.7 (64/73)	69.9 (51/73)	30.8 (4/13)	91.2 (62/68)	66.2 (45/68)	90.9 (10/11)
Week 54	92.2 (59/64)	81.3 (52/64)	62.5 (5/8)	88.5 (46/52)	73.1 (38/52)	60.0 (3/5)
	Biologic-exposed UC: 10.9% (41/376)			Biologic-naïve UC: 89.1% (335/376)		
Week 14	69.0 (20/29)	66.7 (20/30)	16.7 (1/6)	77.8 (200/257)	71.7 (187/261)	69.4 (34/49)
Week 30	74.2 (23/31)	71.0 (22/31)	83.3 (5/6)	78.7 (170/216)	77.1 (168/218)	60.5 (23/38)
Week 54	60.0 (15/25)	81.5 (22/27)	83.3 (5/6)	83.3 (140/168)	83.6 (143/171)	68.2 (15/22)

Data are presented as % (n/N).
N, number of patients having clinical response/remission/endoscopic remission data.
n, number of patients achieving clinical response/remission and endoscopic remission data.

Compared with baseline, mean (±SD) reduction of C-reactive protein (mg/L) at Weeks 14, 30, and 54 were 7.8 (20.2), 6.5 (22.2), and 3.4 (24.8), respectively; mean (±SD) reduction of fecal calprotectin (mu g/g) was 42.7 (150.7) at Week 14 and 212.6 (480.1) at Week 30. The proportion of patients achieving clinical remission without steroid use by Week 14 was 37.7% (23/61), which increased to 60.6% (20/33) by Week 54.

Conclusion: This study indicated that VDZ demonstrated a good effectiveness profile in Chinese UC patients in the real-world setting.

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PP0503

PHARMACOLOGICAL AND DIETARY INTERVENTIONS FOR THE MANAGEMENT OF UPPER GASTROINTESTINAL CROHN'S DISEASE: A SYSTEMATIC REVIEW

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Introduction: Upper GI involvement in Crohn's disease (CD) is thought to occur in 13% to 15% of cases. However, despite expansion of the therapeutic armamentarium for CD over the last 30 years, there continues to be minimal data on optimal therapeutic strategies for upper GI CD. This knowledge gap presents major clinical challenges in managing this subset of patients.

Aims & Methods: We conducted an extensive systematic review using multiple databases investigating pharmacological or dietary interventions for patients with upper GI CD. We included fully published, English-language studies that compared at least two active interventions with a placebo from 1990 to 2025. Due to the lack of a standardized definition for "upper GI" CD, we included studies that reported data on CD affecting the esophagus to the jejunum. We excluded studies that included <10 upper GI CD patients, reported data for "small bowel CD" in general or limited to ileal CD, or did not report data separately for upper GI CD. There is no age limitation.

Results: Among 1206 citations, 14 observational studies (11 retrospective, 3 prospective) and 1 post hoc of an RCT met our inclusion criteria, amounting to 487 patients with UGI CD. Majority of patients had concomitant ileocolonic involvement. Five studies with 137 patients had esophageal CD. Follow-up ranged from 1 to 28 years. Interventions and reported outcomes varied. Steroids (mainly prednisone) and anti-TNF, either alone or in combination, showed the highest rates of success for clinical improvement, endoscopic healing and histologic remission for upper GI CD compared to other therapies. Other therapies used, alone or in combination, were EEN, PPI, H2RA, immunomodulators (azathioprine, 6-mercap-

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topurine), 5-ASA, anti-integrins, and anti-IL 12/23, showing moderate to minimal improvement in clinical or endoscopic response and remission. (Table)

Author, year	UGI Subset population	Study design, duration of follow-up	Interventions Studied	Outcomes measured	Key Numeric Results	Authors' Main Interpretations
D'Haens et.al [DIVERGENCE 1]. (2023)	17 adults, jejununal	RCT, 24 weeks	Filgotinib, Placebo	MARiAseg remission at 24 weeks	Non-significant. (MARiAseg mean): 0/3 PBO (6.6); 0/8 FIL 100mg (6.3); 2/6 FIL 200mg (6.8)	Filgontinib did not result in statistically significant differences vs. PBO to achieve MARiAseg remission in jejunum in 24 weeks.
D'Haens et.al. (1994)	14 adults, esophageal	Retrospective observational, 12 years	„Usual therapy": Steroids ± 5-ASA +/- Abx, immunomodulator ± acid suppression, no uniform prospective regimen	Endoscopic healing, flares/ relapse at follow-up	Significant. 69% (9/13) 64.3% (9/14) with partial or complete healing after short course steroids; 21% (3/14) had persistent disease or repeated flares.	Esophageal CD responds to steroids in ~70%.
Miehsler et.al. (2001)	32 adults, UGI	Retrospective observational, 34 mos.	4 arms: PPI, 5-ASA, steroid, AZA	CDAI, UGI symptoms score, CRP	Significant. Antisecretory drugs: UGI Symptoms score 2.5 (1-5) to 1 (0-3) P=0.039 Prednisolone: CDAI from 272 (151-451) to 100 (22-309) P= 0.03 AZA: CDAI 94 (39–426) to 57 (0–203) P=0.04; CRP 2.3 mg/dl (0.5–8.2) to 0.7 mg/dl (0.5–2.2) P=0.02	UGI Symptoms responded to antisecretory drugs. CDAI decreased with Prednisolone, AZA. CRP decreased with AZA.
Juillerat et.al. (2013)	7 adults, UGI	Retrospective observational, 2 years	Natralizumab	Clinical response (HBI), completion of 1 year of treatment	Nonsignificant. 1/7 responded; 2/7 did not respond	Only 1/7 responded to natralizumab; (no further comment or elaboration/subgroup analysis for UGI made for this study)
Olbjorn et.al. (2014)	16 pediatric, UGI	Prospective cohort, 2 years	IFX	Endoscopic, clinical remission maintained at 1-2 years	Significant. IFX treated: from 11 with UGI CD-specific lesions at diagnosis to 2 at ff-up (61% to 12%); non IFX treated: 5/18 at dx to 2/15 at ff-up (31% to 15%) Significant. No numeric value - patients were treated with various combinations, in this case series	IFX resulted in a significant decrease in UGI CD-specific lesions at ff-up 1-2 years after dx. Response was sustained at 2 years for all responders.
De Felice et.al. (2015)	24 adults, esophageal	Retrospective observational, 1 year	Alone or combination: 5-ASA, prednisone, AZA, oral budesonide , IFX, CZP, PPI	Clinical and endoscopic response at 1 year		Inflammatory esophageal CD responded to prednisone, topical budesonide, or biologics. Stricturing esophageal CD was successfully treated with combination of biologic therapy, immunomodulators, and serial dilations with/without steroid injections. Aggressive medical therapy with biologics for fistulizing esophageal CD, was not universally effective.
Lambin et al. (2020)	60 adults, UGI strictures	Retrospective observational, 5 years	Immunomodulator (AZA, MTX), anti-TNF, anti-TNF + immunomodulator, none, UST	Surgery free-survival at 1 and 5 years	Significant. Anti-TNF alone: multivariate OR 0.2, univariate OR 0.4; 70% efficacy at 6.5mos; anti-TNF + immunomodulators: multivariate OR 0.2 univariate OR 0.2; 61% efficacy at 23.5mos; UST + immunomodulator: 67% efficacy at 10.5mos UST alone: 50% efficacy, Immunomodulator alone: 50% improvement in 35mos.	Anti-TNF therapy alone or in association with an immunosuppressant reduces risk of surgery in UGI CD strictures.
Wauters, et.al. (2017)	69 pediatric, UGI	Prospective observational cohort, 5 years	No control groups: corticosteroids (pred, budesonide), EEN, immunomodulators (AZA, 6-MP, MTX), anti-TNF (IFX, ADA)	Clinical response and remission	Nonsignificant. No difference in time to or duration of sustained remission for anti-TNF exposed and non-anti-TNF P=0.60	No significant difference in time to or duration of sustained remission for anti-TNF exposed and non-anti-TNF P=0.60
Huang et.al. (2025)	39 adults, jejunum	Retrospective observational cohort, 1 year	IFX, UST, Vedolizumab	Endoscopic healing at 1 year	Significant. After 1 year: Jejunal endoscopic healing (EH): IFX 8/15 (53.3%), UST 3/15 (23.1%), Vedo 1/11 (9.1%); Endoscopic response (ER): IFX 8/15 (53.3%), UST 7/13 (53.8%), Vedo 2/11 (18.2%); Ulcer healing: IFX 8/15 (53.3%), UST 5/13 (38.5%), Vedo 1/11 (9.1%)	IFX had strongest response compared to either UST or Vedolizumab for endoscopic healing, endoscopic response and ulcer healing in jejunum. Primary outcome at 1 year for all 3 biologics in jejunum: 30.8% Endoscopic healing (EH), 43.6% achieved endoscopic response (ER), 35.9% had ulcer healing (UH)
Vale Rodrigues et.al. (2020)	40 adults, esophageal	Retrospective observational	PPI, corticosteroids, 5-ASA, anti-TNF, thioprurines, EEN	Clinical and endoscopic remission	Significant. 37/40 (92.5%) tx w/ PPI; 33/40 (82.5%) treated w/ >1 drug; 33/40 (82.5%) with 1st line therapy had complete resolution in 7mos.	Clinical and endoscopic remission was achieved in more than 80% of the patients, majority with PPI.
Yamamoto et.al. (2009)	54 adults, gastroduodenal CD	Retrospective observational, 12 years	Medical therapy in 31/54: corticosteroids, 5-ASA, AZA, PPI alone or combination	Clinical response	Nonsignificant. 14/31 (45%) responded to medical tx; 19/31 required surgery, 12 did not require surgery	Medical therapy not effective for UGI CD, although corticosteroids most widely used.
Tang et.al. (2024)	54 pediatric, UGI	Retrospective observational, 6 weeks and at 1 year	EEN	Endoscopic healing	Significant. Upper GI location (OR 0.25, 95% CI 0.11–0.55, P = 0.001) associated with segmental mucosal healing (MH) at EEN completion. EEN had 35% MH for small bowel.	UGI segment in pediatric CD was associated with segmental mucosal healing following EEN therapy. Lewis score significantly improved following EEN treatment, the MH rate was only 35%, and lesions in the small bowel were more likely to worsen after EEN therapy despite anti-TNF maintenance therapy
Lui et.al. (2017)	39 adults, esophageal	Retrospective observational, 4 years	anti-TNF, anti-integrin, anti-IL12/23, combination	Clinical response/remission, endoscopic response/ remission	Significant. anti-TNF resulted in greater clinical (P=0.02) & endoscopic (p<0.01) response and clinical remission (P=0.01) but not endoscopic remission (P=0.85); anti-integrin poorer endoscopic response when tx w/ other biologics (P<0.03) & not assoc. with significant clinical response; anti-IL12/23 compared w/ other biologics assoc. w/ poorer clinical (P<0.01) and endoscopic response (P=0.03); no sig diff bet'n anti-integrin and IL12/23 clin response (P=0.13), clin remission (P=0.34), endoscopic response (P>0.99) and endoscopic remission (P>0.99).	The timing of biologic initiation in patients in this study was not significantly associated with improved clinical or endoscopic outcomes. Treatment with anti-TNF therapy, when given as first-line treatment associated with better clinical and endoscopic outcomes when compared to other second- or third-line biologic therapies.
Annunziata et.al. (2012)	19 adults, UGI	Prospective cohort, 12 weeks	Group A: PPI + anti-TNF (IFX, ADA) + 5-asa, AZA, 6MP; Group B: PPI + 5-ASA, AZA, 6MP	Endoscopic healing at 12 weeks	Significant. 100% of symptomatic patients improved. Group A: Mucosal healing in 8/11 (72.7%), endoscopic healing in 7/8; Group B: 1/8 (12.5%) mucosal healing (P=0.001) Significant. Steroids 9/18 improvement, AZA 2/8, 6-MP 2/3, PPI 2/5	Mucosal healing achieved in 70% of PPI plus anti-TNF and only in 12.5% of PPI plus other treatments group First line therapy with steroids, 5-ASA, H2RA, PPI resulted to improvement in 50% of patients
Anton et.al. (2001)	20 adults, esophageal	Retrospective observational, 1mo - 28 years	5-ASA, H2RA, PPI, Steroids, immunomodulator	Clinical, endoscopic improvement		

Table.

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Conclusion: Our comprehensive literature search of comparative studies highlights an almost void of evidence to inform therapeutic strategies in upper GI CD. Steroids and anti-TNF showed the best outcomes, but only from small observational, uncontrolled studies. Data for biologics and advanced therapies remain limited highlighting a large unmet need.

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PP0504

ORISMILAST FOR THE TREATMENT OF MODERATE TO SEVERE ULCERATIVE COLITIS: A PROOF OF CONCEPT TRIAL

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Introduction: Existing treatments aim to induce and maintain remission in Ulcerative Colitis (UC); however, 30–55% of patients remain refractory during the induction phase, highlighting an unmet need for novel therapeutic options(1). Phosphodiesterase-4 (PDE4) is a cyclic adenosine monophosphate (cAMP)-specific enzyme expressed by immune and inflammatory cells. Abnormal expression of PDE4 and dysregulated signal transduction have been reported in colonic biopsies from patients with UC (2,3). Orismilast is an oral, B/D-selective PDE4 inhibitor with a modified-release formulation.

Aims & Methods: This study (the UCORIS study) was a phase 2a, open-label, single-arm exploratory clinical study assessing the efficacy and safety of orismilast in patients with moderate to severe UC. Patients with UC currently on stable 5-ASA treatment and experiencing moderate-severe disease activity (defined as a Mayo endoscopic sub score of 2 or 3) were eligible for inclusion. Orismilast was administered twice daily for 12 weeks, with investigators allowed to individualize the dose based on body weight, treatment response and side effects. Responding patients could extend treatment to a total duration of 52 weeks.

A last-observation-carried-forward (LOCF) approach was applied to early discontinuers when com-paring to those who completed treatment. No hypothesis testing was performed due to the small sample size. The primary endpoint was clinical remission at week 12, defined as a total Mayo score of 2 or lower, with no individual sub score exceeding 1.

Results: 10 patients were included in the study. Six participants discontinued the study prematurely - three due to lack of efficacy and three due to adverse events. Clinical remission at week 12 was achieved by one patient, who continued orismilast for 50 weeks, maintaining endoscopic remission.

In the completed trial group, mean Total Mayo Score decreased from 9.75 to 5.25 (Δ– 4.50, [95% CI: -4.17;13.17], p=0.20), and Partial Mayo Score from 7.25 to 3.75 (Δ –3.5, [95% CI: -2.66;9.66], p=0.17). In the early determination group, Partial Mayo Score declined from 5.83 to 4.17 (Δ–1.66, [95% CI: -2.46;5.79], p=0.35); follow-up Total Mayo Scores were unavailable. Faecal calprotectin levels showed modest reductions in both groups.

A total of 75 adverse events (AEs) were recorded. Of these, 48 were mild (68%), consistent with previous findings(4). The most common ADRs were nausea (15% of AEs; 8 patients (80%)), abdominal pain (12%; 5 patients (50%)), dizziness (5%; 2 patients (20%)), and headache (4%; 2 patients (20%)). Three serious AEs occurred in one patient but did not result in discontinuation. None of the serious AEs were considered related to orismilast. The overall safety profile aligned with known effects of PDE4 inhibitors(4–6), although the frequency of nausea was higher, potentially reflecting overlap with underlying disease symptoms.

Conclusion: Orismilast demonstrated reductions in both Total and Partial Mayo scores among patients who completed the trial, and a reduction in Partial Mayo score among those who discontinued early. Although only one patient, who completed the full 12-week regimen, achieved both clinical and endoscopic response, three out of ten patients achieved remission based on Partial Mayo score, faecal calprotectin, and SCCAI. Adverse drug reactions were common; although none were severe, nausea had a notable impact on daily life. Effective management of this adverse event may be essential to improving tolerability and supporting the therapeutic potential of orismilast in the treatment of ulcerative colitis.

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PP0505

THE SAFETY OF VEDOLIZUMAB IN THE MANAGEMENT OF CHINESE PATIENTS WITH INFLAMMATORY BOWEL DISEASE: FINDINGS FROM THE VALUE, MULTICENTER, PROSPECTIVE, OBSERVATIONAL STUDY

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Introduction: Vedolizumab (VDZ) is a monoclonal antibody therapy approved in China in 2020 for the treatment of patients with inflammatory bowel disease (IBD; Crohn's disease [CD] and ulcerative colitis [UC])[2] who were intolerant to conventional therapies.[1] VDZ has a unique, gut-selective, anti-lymphocyte trafficking (GSALT) mechanism of action.[3] However, data on the efficacy and safety of VDZ in Chinese patients with IBD in the real-world setting are limited.[4] This study evaluated the safety of VDZ in adult patients with IBD.

Aims & Methods: VALUE (NCT04872491) was a multicenter, single-arm, prospective, observational study to evaluate the safety of VDZ in adult Chinese patients with IBD (N=500) over a maximum observation period of 72 weeks. Incidence of adverse events (AEs), serious AEs (SAEs), treatment-related adverse drug reactions (ADRs), and AEs of special interest (AESIs) was evaluated. This was the final analysis of the VALUE study conducted in patients with IBD.

Results: A total of 500 patients (UC: n=409; CD: n=91) received VDZ 300 mg. Mean (±standard deviation) age of patients was 44.3 (±15.2) years, and 61.4% (n=307) were male. The mean duration for VDZ exposure was 414 days (UC: 416 days; CD:407 days). A total of 500 patients developed 674 AEs (Table), and the most frequent AEs were infections and infestations (I&I; UC: 115 [28.1%]; CD: 24 [26.4%]), followed by gastrointestinal disorders (GIDS; UC: 87 [21.3%]; CD: 18 [19.8%]). Incidence rates of AEs (/ patient-years) were 1.2 and 1.0 in patients with UC and CD, respectively. Overall, 13 (3.2%) patients with UC and 10 (11.0%) with CD had AEs leading to study discontinuation. SAEs were observed in 78 patients with UC and 18 with CD (Table). A total of 19 AESIs were reported by patients with UC and 9 AESIs by those with CD, with the most common AESIs being I&I (UC: 13 [3.2%]; CD: 4 [4.4%]) and malignancy (UC: 3 [0.7%], CD: 2 [2.2%]). The most common I&I disorders reported by patients with UC and CD were *Clostridium difficile*infection (4 [1.0%]) and pneumonia (1 [1.1%]), respectively. Overall, 17 (4.2%) patients with UC and 7 (7.7%) with CD had at least 1 AESI. A total of 91 and 15 treatment-related ADRs were observed in patients with UC and CD, respectively; I&I were the most frequently documented ADRs to VDZ treatment (Table). A total of 10 (UC: 5 [1.2%]; CD: 5 [5.5%]) patients treated under concomitant IBD advanced therapy (CIBDAT) experienced a total of 16 AEs (UC: 9; CD: 7), with the most common AEs being I&I (UC: 1 [20.0%]; CD: 2 [40.0%]) and GIDS (UC: 3 [60.0%]; CD: 1 [20.0%]) with CIBDAT. No deaths were reported during the study.

	UC	CD
All reported AEs (n)	569	105
Mild/Moderate AEs, n (%)	379 (66.6)/140 (24.6)	73 (69.5)/18 (17.1)
Severe AEs, n (%)	50 (8.8)	14 (13.3)
SAEs (n)	78	18
Mild/Moderate, n (%)	8 (10.3)/31 (39.7)	1 (5.6)/4 (22.2)
Severe, n (%)	39 (50.0)	13 (72.2)
Treatment-related ADRs (n)	91	15
I&I, n (%, 95%CI)	16 (3.9, 2.3-6.3)	2 (2.2, 0.3-7.7)
GIDS, n (%, 95%CI)	16 (3.9, 2.3-6.3)	1 (1.1, 0.03-6.0)

Conclusion: The VALUE study indicated that VDZ was well tolerated in Chinese patients with IBD, with no new safety concerns.

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PP0506

LONG-TERM PERSISTENCE OF USTEKINUMAB TREATMENT IN PATIENTS WITH MODERATE-TO-SEVERE CROHN'S DISEASE

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Introduction: Ustekinumab (UST) is efficacious in inducing and maintaining remission in patients with Crohn's disease (CD). Treatment persistence up to 5 years has been documented in open-label extension of pivotal RCTs of UST (1). In a prospective, real-world (RW) nation-wide study (REASSURE) 73% of 169 bio-naive (BN) and bio-experienced (BE) patients were still in the study after 2 years of treatment with UST (2).

Aims & Methods: In this prospective, single-centre study we aimed to assess the long-term persistence of UST treatment in consecutive BN and BE CD patients who were treated in RW setting with UST in approved dosing regimens for induction and remission maintenance.

Consecutive BN and BE CD patients with active moderate to severe luminal CD with or without perianal involvement, aged 18-80 years who received UST induction (6mg/kg iv) and maintenance (90mg sc q8 weeks) and achieved clinical and biomarker response or remission without concomitant steroids at 6 months after initiation of UST were included in a long-term observational study.

Study end-point was treatment discontinuation for whatever reason. Patients were followed every 4 months with clinical examination, recording of the Harvey-Bradshaw Index (HBI) and laboratory tests [including regular CRP and occasional Fecal Calprotectin (FC)]. Ileocolonoscopy was performed at 6 months after initiation of treatment and then every 2 years. Clinical response and remission were defined as a drop in HBI ≤2 and <5 from initiation of treatment, respectively. Biomarker remission was defined as a CRP level ≤0.5mg/dL and FC ≤200 μg/g. Endoscopic remission was defined as a SES-CD score of ≤3 and a Rutgeert's score of ≤1 for patients with inflammatory CD and for patients in post operative setting, respectively. In this abstract we report the results on the persistence of UST-treatment based on clinical and biomarker data.

Results: Between 1/6/2017 and 31/12/2024, 64 patients were evaluated and 52 patients who fulfilled the entry criteria were enrolled in the study. Patient demographic, clinical, biomarker and endoscopic data is shown in table 1. Of these 52 patients, 42 (80.7%) patients were still maintained in clinical, biomarker and endoscopic remission [HBI 2.3 (1-4), mean (range),

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CRP<0.5 mg/dL and FC <200µg/ml)] on UST 90 mg sc every 8 weeks for >2 [n=10 (24%),3 [n=5 (12%)], 4 [n=5 (12%)], 5 [n=6 (14%)] και ≥ 6 [n=16 (38%)] years after inclusion in the study, respectively. Treatment escalation was not performed and neither serious adverse events nor serious infections were reported in any of these patients during the study period. Ten patients (19.3%), two BN and 8 BE stopped therapy. Cessation of treatment was slightly statistically higher in BE than PN patients (p<0.05). Five of 10 patients stopped treatment because of post-operative endoscopic recurrence (Rutgeert's score ≥i2) at 2,2,2.5, 2.9 and 3.2 years, respectively and 2 patients because of a flare of colitis at 4 and 6.5 years, respectively after inclusion in the study. Another 3 patients stopped therapy because of active axonal spondyloarthropathy 2.3 years (n=1) and two patients because they were diagnosed with pancreatic cancer and a T1-colorectal cancer (patient was on the 3nd biologic agent after a 20-year course of colitis) after 7 years in the study, respectively.

Conclusion: This prospective, open label, single-center RW study confirms the long-term treatment persistence of UST with an acceptable safety profile.

Patient number: n=52
Age, mean (range): 42.6 (24-79)
Males: 25 (48%)
Bio-naïve: 24 (46,15%)
Bio-experienced: 28 (53,75%)
Prior biologics (1;n=8, 2; n=10, 3;n=7), 4; n=3)
Agents received: Infliximab/adalimumab/vedolizumab/experimental agent in phase III-RCT
Years since diagnosis: 12,11 (4-31)
Smokers active/ex-/never: 24/5/23
Disease location (L1/L2/L3)/P+: (24/11/17)/10
Flare after right hemiclectomy: 17 (32.7%)
Extra-Intestinal Manifestations: 27 (52%)
Disease activity at initiation of treatment
HBI: 16.5 (9-21)
CRP (mg/dL): 8.5 (2.5-15.5)
Faecal Calprotectin: 650 (385-1850)
SES-CD: 10.4 (7-19)
Rutgeert's score: 2.43 (2-3)
Concomitant steroids: 24 patients (46.15%)
Oral prednisolone: n=18 [20 mg (15-25mg)/day]
Oral budesonide: n=6 (9mg/day)

Table 1: Patient demographic, Clinical, and laboratory data at initiation of treatment .

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PP0507

SUSTAINED IMPROVEMENT IN PATIENT-REPORTED OUTCOMES WITH LONG-TERM UPADACITINIB THERAPY IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS: 4-YEAR RESULTS FROM THE U-ACTIVATE LONG-TERM EXTENSION STUDY

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Introduction: Ulcerative colitis (UC) is associated with debilitating symptoms that impair the quality of life (QoL) of patients; restoration of QoL is an important long-term treatment goal.[1]

Upadacitinib (UPA), an oral, reversible Janus kinase inhibitor approved for the treatment of UC, improves health-related QoL (HRQoL) and patient-reported outcomes (PROs) with 1 year (yr) of maintenance therapy.[2-4] This post hoc analysis of the U-ACTIVATE long-term extension (LTE) study evaluated the long-term efficacy of UPA in sustaining PROs and HRQoL outcomes in patients who had received up to a total of 4 years (yrs) of UPA maintenance therapy.

Aims & Methods: This analysis included patients who completed 1 yr (52 weeks [wks]) in the U-ACHIEVE maintenance study and received UPA15 mg once-daily (QD) (UPA15) or UPA 30 mg QD (UPA30) maintenance treatment in the U-ACTIVATE LTE study. Patients receiving UPA15 or UPA30 who achieved clinical remission per Adapted Mayo score at maintenance wk 52 continued their double-blind treatment upon entering the LTE study. Nonremitters receiving UPA15 were escalated to UPA30, those receiving UPA30 continued UPA30, and those assigned to PBO were escalated to UPA15, all in a blinded manner. The UPA UC program was not designed to compare UPA15 and UPA30 doses.

Efficacy was assessed at LTE wks 0, 96, and 144 (data cutoff date: 30 June 2024) and included the percentage of patients who achieved absence of abdominal pain, absence of bowel urgency, Inflammatory Bowel Disease Questionnaire (IBDQ) response, IBDQ remission, and a meaningful within-person change (MWPC) in Functional Assessment of Chronic Illness Therapy (FACIT)-Fatigue; Short-Form Health Survey-36 version 2 Mental Component Score (SF-36v2 MCS), and SF-36v2 Physical Component Score (PCS).

Data were reported using as-observed (AO) and modified nonresponder imputation (mNRI) methods (shown in Table). Safety was reported previously.[2]

Results: This analysis included 369 patients treated with UPA15 or UPA30 at wk 0 of LTE trial. In the AO analysis, patients treated with UPA15 or UPA30 achieved sustained improvements from LTE wk 0 to wk 144 in PROs and HRQoL assessments (Table), including the absence of abdominal pain, absence of bowel urgency, IBDQ response, IBDQ remission; and achieved MWPC in FACIT-Fatigue, SF-36v2 PCS, and SF-36v2 MCS.

Similar results were observed for all endpoints with the more conservative mNRI analysis (Table).

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Endpoint, (n/N) % [95% CI][a]	AO analysis[b]		mNRI analysis[c]	
	UPA 15 mg QD	UPA 30 mg QD	UPA 15 mg QD	UPA 30 mg QD
No abdominal pain[d]				
Week 0	(94/142) 66.2 [58.4, 74.0]	(168/227) 74.0 [68.3, 79.7]	(94/142) 66.2 [58.4, 74.0]	(168/227) 74.0 [68.3, 79.7]
Week 96	(84/109) 77.1 [69.2, 85.0]	(137/172) 79.7 [73.6, 85.7]	(101/142) 71.1 [63.4, 78.7]	(171/227) 75.5 [69.7, 81.3]
Week 144	(61/83) 73.5 [64.0, 83.0]	(109/135) 80.7 [74.1, 87.4]	(99/142) 69.4 [61.6, 77.2]	(171/227) 75.5 [69.7, 81.3]
No bowel urgency[d]				
Week 0	(107/142) 75.4 [68.3, 82.4]	(180/227) 79.3 [74.0, 84.6]	(107/142) 75.4 [68.3, 82.4]	(180/227) 79.3 [74.0, 84.6]
Week 96	(97/109) 89.0 [83.1, 94.9]	(154/173) 89.0 [84.4, 93.7]	(116/142) 81.5 [75.0, 88.1]	(192/227) 84.7 [79.9, 89.5]
Week 144	(76/83) 91.6 [85.6, 97.5]	(117/135) 86.7 [80.9, 92.4]	(114/142) 80.0 [73.2, 86.8]	(188/227) 82.7 [77.6, 87.8]
IBDQ response[e]				
Week 0	(128/142) 90.1 [85.2, 95.0]	(203/221) 91.9 [88.2, 95.5]	(128/142) 90.1 [85.2, 95.0]	(203/227) 89.4 [85.4, 93.4]
Week 96	(95/106) 89.6 [83.8, 95.4]	(156/171) 91.2 [87.0, 95.5]	(119/142) 84.1 [77.9, 90.3]	(196/227) 86.3 [81.7, 90.9]
Week 144	(73/78) 93.6 [88.2, 99.0]	(117/129) 90.7 [85.7, 95.7]	(120/142) 84.2 [77.9, 90.5]	(193/227) 84.9 [80.0, 89.9]
IBDQ remission[f]				
Week 0	(114/142) 80.3 [73.7, 86.8]	(188/227) 82.8 [77.9, 87.7]	(114/142) 80.3 [73.7, 86.8]	(188/227) 82.8 [77.9, 87.7]
Week 96	(93/106) 87.7 [81.5, 94.0]	(156/174) 89.7 [85.1, 94.2]	(111/142) 78.0 [70.7, 85.3]	(189/227) 83.1 [77.5, 88.7]
Week 144	(70/78) 89.7 [83.0, 96.5]	(116/132) 87.9 [82.3, 93.4]	(104/142) 73.2 [65.2, 81.3]	(180/227) 79.2 [73.2, 85.3]
MWPC in FACIT-Fatigue[g]				
Week 0	(95/142) 66.9 [59.2, 74.6]	(159/220) 72.3 [66.4, 78.2]	(95/142) 66.9 [59.2, 74.6]	(159/227) 70.0 [64.1, 76.0]
Week 96	(65/111) 58.6 [49.4, 67.7]	(122/178) 68.5 [61.7, 75.4]	(82/142) 57.9 [49.1, 66.6]	(144/227) 63.4 [56.7, 70.0]
Week 144	(51/84) 60.7 [50.3, 71.2]	(100/140) 71.4 [63.9, 78.9]	(83/142) 58.4 [49.4, 67.4]	(149/227) 65.8 [59.1, 72.5]
MWPC in SF-36v2 PCS[h]				
Week 0	(105/142) 73.9 [66.7, 81.2]	(173/220) 78.6 [73.2, 84.1]	(105/142) 73.9 [66.7, 81.2]	(173/227) 76.2 [70.7, 81.8]
Week 96	(79/106) 74.5 [66.2, 82.8]	(134/171) 78.4 [72.2, 84.5]	(99/142) 70.0 [62.1, 78.0]	(164/227) 72.4 [66.1, 78.7]
Week 144	(61/77) 79.2 [70.2, 88.3]	(103/126) 81.7 [75.0, 88.5]	(99/142) 69.9 [61.8, 77.9]	(166/227) 73.0 [66.6, 79.3]
MWPC in SF-36v2 MCS[h]	(83/142) 58.5 [50.3, 66.6]	(147/220) 66.8 [60.6, 73.0]	(83/142) 58.5 [50.3, 66.6]	(147/227) 64.8 [58.5, 71.0]
Week 0	(65/106) 61.3 [52.0, 70.6]	(116/171) 67.8 [60.8, 74.8]	(84/142) 59.2 [50.9, 67.6]	(146/227) 64.2 [57.5, 70.9]
Week 96	(48/77) 62.3 [51.5, 73.2]	(84/126) 66.7 [58.4, 74.9]	(83/142) 58.5 [49.2, 67.7]	(142/227) 62.7 [55.6, 69.9]
Week 144				

AO, as-observed; CI, confidence interval; FACIT-Fatigue, Functional Assessment of Chronic Illness Therapy - Fatigue; IBDQ, inflammatory bowel disease questionnaire; MCS, mental component score; mNRI, modified nonresponder imputation; MWPC, meaningful within-person change; LTE, long-term extension; PCS, physical component score; QD, once-daily; SF-36v2, Short-Form Health Survey-36 version 2; UC, ulcerative colitis; UPA, upadacitinib; wk, week.

Week 0 refers to LTE week 0 (week 52 of the maintenance study). Efficacy analysis includes patients who received UPA15 or UPA30 in cohort 2 of U-ACTIVATE LTE who were UPA 45 mg 8-wk induction responders and were enrolled from cohort 1 of the 52-week maintenance period of the U-ACHIEVE maintenance study.

[a]95% CI for response rate is the synthetic result based on Student's t-distribution from PROC MIANALYZE procedure if there were missing data due to intercurrent events or is based on the normal approximation to the binomial distribution if there are no missing data due to intercurrent events.

[b]AO analysis did not impute values for missing evaluations. AO will include all values collected in the study until the first dose change or enrolment into the dose optimization substudy.

[c]mNRI considered patients as nonresponders on or after study drug dose escalation, or usage of UC-related rescue medication; missing data due to other reasons were imputed using multiple imputation.

[d]Quantified as the percentage of patients who achieved the absence of symptom.

[e]Increase in IBDQ score ≥ 16 points from baseline of induction.

[f]IBDQ total score ≥ 170.

[g]Increase in FACIT-F score ≥ 5 points from baseline of induction.

[h]Increase in SF-36v2 PCS/MCS ≥ 4.1 points from baseline of induction.

Table. PROs and HRQoL Efficacy Analysis From the U-ACTIVATE Long-Term Extension Study.

Conclusion: Patients with UC who responded to induction therapy and received long-term UPA maintenance therapy showed a sustained absence of symptoms and improvements in HRQoL, suggesting that UPA maintains long-term improvements in patient-reported outcomes.

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PP0508

MANAGEMENT AND OUTCOMES IN OLDER PATIENTS WHO PRESENT WITH ACUTE SEVERE ULCERATIVE COLITIS

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Introduction: Older adults with ulcerative colitis (UC) have greater morbidity than younger adults, yet there is paucity of data on the management and outcomes of those who present with acute severe ulcerative colitis (ASUC) in this patient population.

This study aims to investigate differences in the management and outcomes between older and younger patients hospitalised with ASUC.

Aims & Methods: We conducted a retrospective study on all ASUC presentations to a Quaternary Sydney hospital over a 10-year period. A modified True-love Witts' Criteria was used to identify ASUC as ≥6 bloody bowel motions daily and presence of at least one systemic feature, including heart rate >90 bpm, temperature >37.8°C, haemoglobin <105 g/L, and C-reactive protein (CRP) ≥30 mg/L. Older patients who presented with ASUC was defined as those ≥65 years old at time of presentation.

Data collected on management included rates of rescue therapy, colectomy and TPN requirement. Data collected on outcomes included hospital length of stay and rates of ICU admission, 30- and 90-day readmission, and 30- and 90-day mortality.

Results: A total of 172 presentations met criteria for ASUC. Of these presentations, 148 patients (86.1%) were <65 years old and 24 patients (13.9%) were ≥65 years old. Median age at presentation in older adults was 73.6 years old and in younger adults was 32.0 years old. Older adults had similar rates of need for rescue therapy (37.5% vs 35.1%, p=0.08), colectomy (12.5% vs 9.5%, p=0,64) and TPN (12.5% vs 10.1%, p=0.73), as their younger counterparts.

Older adults (n=6, 25.0%) had significantly higher rates of ICU admission (p<0.01) compared to those <65 years old (n=3, 2.0%). Older adults also had significantly higher 30- and 90-day mortality rates (n=3, p<0.01). There was no mortality observed in adults <65 years old.

There was no significant difference in 30- and 90-day readmission rates between older adults and those <65 years old at time of ASUC presentation. Table 1 compares management and outcomes in adults <65 years old and ≥65 years old who presented with ASUC.

	Age <65 (n=148)	Age ≥ 65 (n=24)	p-value
Received rescue therapy (n, %)	52 (35.1)	9 (37.5)	0.08
Underwent colectomy (n, %)	14 (9.5)	3 (12.5)	0.64
Required TPN (n, %)	15 (10.1)	3 (12.5)	0.73
Required ICU admission (n, %)	3 (2.0)	6 (25.0)	<0.01
30-day readmission (n, %)	22 (14.9)	6 (25.0)	0.21
90-day readmission (n, %)	30 (20.3)	8 (33.3)	0.15
30-day mortality (n, %)	0 (0.0)	3 (12.5)	<0.01
90-day mortality (n, %)	0 (0.0)	3 (12.5)	<0.01
Mean length of stay (days)	10.3	14.0	0.12

Table 1. Comparison of management and outcomes in adults <65 years old and ≥65 years old who presented with ASUC.

Conclusion: Despite similar rates of need for rescue therapy and colectomy, older patients who present with ASUC are more likely to have poorer outcomes, with increased requirement of ICU admission and higher mortality rates compared to patients <65 years old. The higher morbidity and mortality observed in our study in older adults who present with ASUC highlights the need for further research to optimise care pathways and improve outcomes in this patient population.

Disclosure: Nothing to disclose.

PP0509

COMBINED ACHIEVEMENT OF SYMPTOMATIC AND BIOMARKER REMISSION WITH MIRIKIZUMAB IN PATIENTS WITH ULCERATIVE COLITIS IN THE PHASE 3 LUCENT STUDIES

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Introduction: Clinical remission of symptoms and endoscopic disease activity were the primary endpoints of two phase-3 studies (LUCENT-1 and -2; NCT03518086 and NCT03524092) in patients with moderately-to-severely active ulcerative colitis (UC) treated with the anti-IL23p19 monoclonal antibody, mirikizumab (miri)(1). This outcome was reached in 24.2% and 49.9% of miri-treated patients at week (W) 12 and 52, respectively.

In clinical practice, access to endoscopic evaluation can be limited. Measurement of faecal calprotectin (FCP) can be a surrogate biomarker for endoscopic disease activity and therapy response in patients with UC(2). Combined symptom and FCP remission may serve as a surrogate for clinical and endoscopic remission in real-world clinical practice.

Aims & Methods: This post-hoc analysis evaluates the proportion of patients in LUCENT-1/-2 who achieved combined symptom and biomarker remission.

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Patients in LUCENT-1 (N=1162) were randomised to receive either 300mg miri (n=868) or placebo (PBO; n=294) intravenously (IV) every 4W (Q4W) for 12W. In LUCENT-2, patients from LUCENT-1 with no clinical response to miri at W12 (N=272) received an extended induction therapy of 300mg miri IV Q4W for 12W (W24 of continuous treatment), while miri-responders (N=544) were randomised to either subcutaneous (SC) 200mg miri (n=365) or PBO (n=179), Q4W for 40W (W52 of continuous treatment).

At W4, W12, W24, and W52, patient symptoms and FCP levels were reported as combined endpoints of symptomatic response or remission (defined in Table 1), with FCP remission (FCP ≤250µg/g). The Cochran-Mantel-Haenszel test compared the treatment groups a) miri and PBO at W4 and W12 from LUCENT-1, and b) miri and PBO at W52 from LUCENT-1 miri-responders. The risk difference and its associated confidence intervals and p-values were reported in Table 1. Non-responder imputation was used for binary variables, while modified baseline observation carried forward was used for continuous variables.

Results: At W4 and W12, 14.2% and 33.2% of the miri group compared to 8.5% and 18.7% of PBO (p=0.0113 and <0.0001, respectively) achieved the combined symptomatic response and biomarker remission endpoint, while 7.3% and 23.7% of the miri group compared to 3.7% and 11.2% of PBO (p=0.0282 and <0.0001, respectively) achieved the combined remission endpoints. At W52, 54.3% of the miri group compared to 27.4% of PBO (p=<0.0001) achieved the combined symptomatic response and biomarker remission endpoint, while 48.5% of the miri group compared to 21.8% of PBO (p=<0.0001) achieved the combined remission endpoints. At W12 and W24, 2.9% and 23.5% of the extended induction group achieved combined symptomatic response and biomarker remission endpoints, respectively, while 1.1% and 13.2% achieved the combined remission endpoints.

Week	Dose	Symptomatic response[a] and FCP ≤250 µg/g, n (%)	Common risk difference (CI)[b]	P-Value	Symptomatic remission[c] and FCP ≤250 µg/g, n (%)	Common risk difference (CI)[b]	P-Value
Week 4	Miri, N=868	123 (14.2)	5.7 (1.7, 9.6)	0.0113	63 (7.3)	3.7 (0.9, 6.4)	0.0282
	PBO, N =294	25 (8.5)			11 (3.7)		
Week 12	Miri, N=868	288 (33.2)	14.8 (9.4, 20.2)	<0.0001	206 (23.7)	12.7 (8.1, 17.3)	<0.0001
	PBO, N =294	55 (18.7)			33 (11.2)		
Week 52	Miri, N=365	198 (54.3)	25.4 (17.3, 33.6)	<0.0001	177 (48.5)	25.0 (17.3, 32.6)	<0.0001
	PBO, N=179	49 (27.4)			39 (21.8)		

[a]Defined as ≥30% decrease from baseline of the sum of stool frequency (SF) and rectal bleeding (RB) subscores

[b]Common risk difference is the percentage difference between miri and PBO adjusted for the baseline factors a) prior biologic or tofacitinib failure (yes/no), b) baseline corticosteroid use (yes/no), c) baseline disease activity (Modified mayo score: [1-6] or [7-9]), and d) region (North America/Europe/Other).

[c]Defined as SF = 0, or SF = 1 with ≥1 point decrease from baseline and RB = 0

Table 1: Proportion of patients with Ulcerative Colitis who achieved combined symptomatic and biomarker endpoints in LUCENT-1/-2 studies.

Conclusion: In moderate-severe UC, patients treated with miri demonstrated improved efficacy compared to PBO in combined symptomatic and biomarker response and remission endpoints. Additionally, the data from this post-hoc analysis is consistent with the clinical remission data from LUCENT studies, indicating that this is a suitable endpoint measure in real-world practice.

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PP0510

SUBCUTANEOUS MIRIKIZUMAB MAINTENANCE TREATMENT OVER 40 WEEKS LEADS TO INCREMENTAL IMPROVEMENTS IN RATES OF CLINICAL REMISSION AND ENDOSCOPIC RESPONSE: RESULTS FROM THE PHASE 3 VIVID-1 STUDY

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Introduction: Mirikizumab, a p19-directed interleukin-23 monoclonal antibody, is efficacious in inducing clinical response at Week(W) 12 and maintaining clinical remission through W52 in patients with moderately-to-severely active Crohn's disease (CD) as shown in the phase 3 treat-through trial VIVID-1 (NCT03926130).[1]

Here, we present analysis of maintenance outcomes through W52 for Patient-Reported Outcome (PRO) clinical responders and nonresponders at W12.

Aims & Methods: The analysis set includes patients randomized to mirikizumab at baseline, who did or did not achieve PRO clinical response at W12 after intravenous (IV) induction treatment (900 mg mirikizumab IV every 4 weeks [Q4W]) before entering subcutaneous (SC) maintenance treatment (300 mg mirikizumab SC Q4W).

Descriptive statistics are used to summarize the data, applying nonresponder imputation for missing binary endpoints. As a treat-through study, no rerandomization was applied.

Results: From a total of 551 patients who entered SC maintenance treatment, baseline characteristics of age, duration of CD, and baseline Crohn's Disease Activity Index (CDAI) were similar between W12 PRO responders (n=402, 73.0%) and W12 PRO nonresponders (n=149, 27.0%). PRO responders at W12 had a higher objective inflammatory burden at baseline (faecal calprotectin, c-reactive protein, and Simple Endoscopic Score-CD), whereas concomitant or prior medication usage was similar between the groups. For W12 PRO responders, SC mirikizumab treatment from W12 to W52 resulted in an increase in CDAI clinical remission from 52.2% to 65.4%, PRO clinical remission from 53.2% to 65.4%, and endoscopic response from 35.1% to 54.7%, respectively (Table 1).

		PRO Nonresponders at W12 (N=149)	PRO Responders at W12 (N=402)
CDAI clinical remission, n (%)[a]	W12	8 (5.4%)	210 (52.2%)
	W16	16 (10.7%)	208 (51.7%)
	W20	28 (18.8%)	243 (60.4%)
	W24	35 (23.5%)	242 (60.2%)
	W52	50 (33.6%)	263 (65.4%)
Endoscopic response, n (%)[a]	W12	45 (30.2%)	141 (35.1%)
	W52	60 (40.3%)	220 (54.7%)

[a]Percentage of response is calculated by n/Nx*100%.
CDAI, Clinical Disease Activity Index; PRO, patient-reported outcome

Table 1.

Among W12 PRO nonresponders, PRO response was achieved by 39.6% at W16, 45.6% at W20, 44.3% at W24, and 56.4% at W52 with continued mirikizumab SC treatment, while CDAI clinical remission and PRO clinical remission were achieved by 33.6% and 31.5% at W52, respectively. Endoscopic response increased from 30.2% at W12 to 40.3% at W52 among W12 PRO nonresponders (Table 1).

Conclusion: Subcutaneous mirikizumab maintenance treatment over 40 weeks leads to incremental and sustained improvements in rates of clinical and endoscopic response regardless of PRO clinical response status after induction. A substantial proportion of clinical nonresponders at W12 improved early during maintenance treatment suggesting continued treatment with mirikizumab can be beneficial and lead to long-term response.

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Disclosure: Nothing to disclose.

PP0511

DISCOVERYING DUAL FUNCTIONAL MRGPRX2 MODULATOR VIA ARTIFICIAL INTELLIGENCE

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Introduction: G protein-coupled receptor (GPCR) is the largest single family for modern drug discovery. They involve in a wide range of important cellular and physiological functions, including sight, taste, smell, growth, behavior, mood, immune regulation and many others. In the human body, there are more than 830 different GPCRs have been discovered. About 30-40% of the marketed drugs are targeting this big family.

The mas-related G protein coupled receptor 2 (MRGPRX2), which belongs to the class A GPCR, is expressed in sensory neurons, mast cells and in keratinocytes. MRGPRX2 has been detected in adipose tissue, esophagus, urinary bladder, colon and lungs with the highest levels found in skin.

Clinical analysis indicated that the over-expression level of MRGPRX2 was extremely high in the inflammatory bowel disease (IBD) patient colons, which made it as an ideal target to treat IBD.

Here we applied artificial intelligence to design first-in-class dual functional MRGPRX2 modulator. The DSS-induced IBD model in monkey indicated that our drug candidate exhibits excellent efficacy at low dose against IBD.

Aims & Methods: Develope next generation IBD drug molecules.

Here we used artificial intelligence to design first-in-class dual functional MRGPRX2 modulator. We first built the 3D model of MRGPRX2 receptor in the inactive state in computer. The AI-based high throughput virtual screening against 10 billion virtual compounds was performed to obtain hit compounds. Lead optimization, in vivo study and safety valuation were followed as well.

Finally, we performed IND-enabling and submitted profiles to FDA for IND approval.

Results: We established a severe DSS-induced monkey model wtih DAI score at 12. In this experiment, our drug candidate could resolve most inflammation at low dose 1mg/kg (DAI=5), whereas the tissue at 3mg/kg (DAI=3) was comparable with normal group under enteroscope.

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Conclusion: In this work, we sucessfully designed dual functional drug candidates against MRGPRX2 receptor to treat IBD via AI, which was confirmed by the DSS-induced monkey model.

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Disclosure: Dr. Shuguang Yuan is the cofounder of AlphaMol Science Ltd.

PP0512

FAECAL URGENCY AND ITS CORRELATION WITH DISEASE LOCATION, ACTIVITY, QUALITY OF LIFE AND MENTAL HEALTH STATUS IN CROHN'S DISEASE

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Introduction: Faecal urgency (FU) is a frequently encountered yet frequently underappreciated symptom experienced by many individuals suffering from inflammatory bowel disease (IBD)[1,2]. In cases of ulcerative colitis, the presence of FU is relatively easy to comprehend, as inflammation typically involves the rectum and left-sided colon, which are closely linked to urgency symptoms. However, in patients diagnosed with Crohn's disease (CD), where lesions are more commonly found in more proximal areas of the gastrointestinal tract — such as the ileum or the right colon while the distal regions like the left colon and rectum often remain unaffected, the emergence of FU becomes more difficult to explain.

Aims & Methods: The objective of this study was to evaluate and compare the prevalence and intensity of FU in CD patients with proximal versus distal involvement, using the splenic flexure as the dividing point. The study also aimed to explore associations between FU and markers of disease activity, work-related productivity, quality of life (QoL), and psychological symptoms such as anxiety and depression.

A cross-sectional study was carried out at a tertiary care centre, involving a sample of CD patients who completed a comprehensive survey between July and September 2024. FU severity was measured on a numeric rating scale ranging from 0 to 10, with scores of 0 or 1 indicating absence of FU[3]. Participants also completed the Work Productivity and Activity Impairment Questionnaire (WPAI-GH), the Short Inflammatory Bowel Disease Questionnaire (SIBDQ), and the Hospital Anxiety and Depression Scale (HADS). Data analysis was conducted using SPSS Statistics software, version 26®.

Results: A total of 158 patients were included in the study, with a mean age of 42.1 ± 1.3 years; 88 individuals (55.7%) were female. Among the participants, 109 (69.0%) reported experiencing faecal urgency (FU). Proximal disease was present in 97 patients (61.4%), while 61 patients (38.6%) had distal involvement. Of those, 69 individuals (71.1%) with proximal disease and 40 (65.6%) with distal disease reported FU. However, distal involvement was not significantly associated with the presence of FU symptoms (OR = 0.77; 95% CI [0.39–1.54]; p = 0.46), nor with symptom severity, even in the subset of patients with documented rectal involvement (n = 24; 39.3%).

No significant correlation was found between the presence or severity of FU and markers of disease activity, including faecal calprotectin, C-reactive protein, endoscopy, histology, abdominal ultrasound or clinical indices. In contrast, the severity of FU showed a weak but statistically significant positive correlation with elevated anxiety symptoms (ρ = 0.32; p = <0.001) and depressive symptoms (ρ = 0.31; p = <0.001). Only FU, not anxiety or depression, showed to be associated with overall work productivity loss (β = 2.48 [0.55-4.40]; p = <0.001). Higher scores of FU, anxiety and depression were all associated with poorer QoL (β1 = -1.09 [-1.50--0.68]; β2 = -0.61 [-1.00--0.23]; β3 = -1.32 [-1.71--0.92]; p = <0.001).

Conclusion: In this study, the occurrence of FU in CD was not associated with the location of disease involvement or markers of disease activity. Instead, FU showed a stronger relationship with psychological factors, specifically symptoms of anxiety and depression. These findings suggest that managing FU may require therapeutic strategies beyond conventional immunomodulatory treatments[4].

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PP0513

REAL-WORLD DATA ON THERAPY RESPONSE TO INFLAMMATORY BOWEL DISEASE BIOLOGICAL TREATMENTS IN PATIENTS WITH CONCURRENT PRIMARY SCLEROSING CHOLANGITIS: A CASE-CONTROL STUDY

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Introduction: Advanced biological therapies have improved outcomes of patients diagnosed with Inflammatory Bowel Diseases (IBD) such as Ulcerative colitis (UC) and Crohn's disease (CD). However, nonresponse or loss of response remains a significant challenge. Primary sclerosing cholangitis (PSC) is associated with IBDs, but the impact of concurrent PSC on response to IBD therapy is not well explored.

Aims & Methods: This study compared therapy response to IBD treatment with Infliximab, Vedolizumab and Ustekinumab in patients diagnosed with PSC and IBD (PSC/IBD group) versus patients diagnosed with IBD only (control group). Therefore, we identified 46 patients in the PSC/IBD group and 180 in the control group treated with Infliximab, Vedolizumab and/or Ustekinumab at our IBD outpatient clinic at Charité – Universitätsmedizin Berlin. Propensity score matching was employed to match the groups with a ratio of 1:1 based on age, sex, BMI, smoking status, Charlson Comorbidity Index (CCI), IBD subtype, duration of IBD, age at IBD diagnosis and Montreal-Classification (tolerance = 0.2).

The primary outcome was clinical therapy response within 20 weeks, defined as remission (partial Mayo Score (pMS) ≤ 1/Harvey-Bradshaw-Index (HBI) < 5) or partial response (decrease of pMS ≥ 2/HBI > 3). Group differences were compared using Fisher's Exact test.

Secondary outcomes included time to loss of response, mucosal healing (defined as no or mild inflammation e.g. Mayo Endoscopic Score 0 or 1) and decreased levels of C-reactive Protein (CRP) < 5mg/l and faecal calprotectin (FC) < 50 mg/g. Only patients achieving clinical remission or partial response were included in the secondary analyses.

Variables	IFX-PSC (n=19)	IFX-control (n=20)	VDZ-PSC (n=28)	VDZ-control (n=22)	UST-PSC (n=21)	UST-control (n=26)
IBD-subtype UC (%)	11 (57.9%)	16 (80.0%)	24 (85.7%)	20 (90.9%)	14 (66.7%)	16 (61.5%)
Primary Outcome						
clinical remission (%)	13 (68.4%)	14 (70.0%)	19 (67.9%)	18 (81.8%)	8 (38.1%)	18 (69.2%)
clinical partial response (%)	2 (10.5%)	1 (5.0%)	4 (14.3%)	1 (4.6%)	1 (4.8%)	2 (7.7%)
p-values (Fisher's Exact test)	p = 1.000		p = 0.521		p = 0.044	
Secondary outcomes (only patients with clinical remission or clinical partial response included)						
Time to loss of response (Mean ± SD)	13 ± 11 months	16 ± 9 months	25 ± 18 months	32 ± 21 months	13 ± 4 months	12 ± 9 months
Mucosal healing (%)	4 (50.0%)	7 (87.5%)	9 (45.0%)	11 (84.6%)	1 (20.0%)	9 (75.0%)
Decreased levels of CRP and FC (%)	10 (76.9%)	13 (86.7%)	8 (47.1%)	16 (88.9%)	2 (40.0%)	14 (77.8%)

Table 1. Comparison of therapy response to IBD treatment with Infliximab, Vedolizumab and Ustekinumab between PSC/IBD group and control group.

Results: After matching, 46 patients per group were analysed, with a total of 136 treatments administered. The cohort demographics and disease characteristics were balanced between PSC/IBD group and control group with all p-values > 0,05 (mean age: 39.7 ± 12.4 vs. 39.9 ± 13.9 years; male sex: 31 (67.4%) vs. 24 (52.2%); mean BMI: 23.1 ± 3.9 vs. 23.8 ± 4.8 kg/m2; smokers: 9 (20.0%) vs. 13 (31.7%); mean CCI 0.6 ± 1.1 vs. 0.5 ± 1.0). In both groups 34 (73.9%) patients having an UC diagnosis and 12 (26.1%) a CD diagnosis.

While clinical response rates did not significantly differ between groups in patients receiving Infliximab or Vedolizumab, in patients treated with Ustekinumab clinical remission and partial response rates were significantly lower in the PSC/IBD group (p=0.04, Table 1).

The time to loss of response was not different between groups for any of the therapies. Mucosal healing rates were lower in the PSC/IBD group compared to controls as well as decreased levels of CRP and FC across all therapies (Table 1).

Conclusion: Infliximab and Vedolizumab appear equally efficient for achieving clinical response in IBD patients with and without PSC, while the efficacy of Ustekinumab is significantly lower in IBD patients with PSC. Future prospective data is needed to strengthen our knowledge of treatment outcomes in this population.

Disclosure: Nothing to disclose.

PP0514

JAK INHIBITORS EXHIBIT SUPERIOR CLINICAL EFFICACY COMPARED TO ANTI-IL-23 BIOLOGICS IN REFRACTORY ULCERATIVE COLITIS

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Introduction: Biologic therapies play a pivotal role in the management of ulcerative colitis (UC). Anti-TNFα agents are recommended as first-line biologics according to ECCO guidelines; however, JAK inhibitors have recently emerged as a viable alternative (1).

Vedolizumab has also demonstrated effectiveness when introduced early in the disease course with a greater 52-week treatment persistence and reduced hospitalisation rates compared to anti-TNFα agents as second-line therapy (2). The AGA guidelines (3) now favour biologics such as anti-IL-23 and JAK inhibitors over Vedolizumab or S1P receptor modulators in patients previously exposed to anti-TNFα agents.

Notably, Guselkumab is not yet approved for UC treatment in the UK. The latest ECCO guidelines (1) do not address the role of biologics in third-line or subsequent treatment settings, and there remains a lack of clear clinical guidance.

Aims & Methods: This study aimed to retrospectively evaluate the effectiveness of *third-line* biologic therapies in adult patients with UC over a 12-month period. Clinical response was assessed using the Simple Clinical Colitis Activity Index (SCCAI), Partial Mayo Score, and Physician Global Assessment, while biochemical and endoscopic responses were measured at 3, 6, and 12 months post-induction. Biochemical markers included faecal calprotectin and CRP levels. Endoscopic response was assessed using the Ulcerative Colitis Endoscopic Index of Severity (UCEIS). Colectomy rates were also recorded. Patients treated at a UK tertiary care centre were included up to March 2025. Statistical analyses were conducted using SPSS.

Results: A total of 24 patients were included in the analysis, with 13 receiving anti-IL-23 therapy and 11 treated with JAK inhibitors. Within the anti-IL-23 group, 5 patients received Mirikizumab and 8 Ustekinumab; Risankizumab was not used in this cohort. In the JAK group, 3 patients were treated with Upadacitinib, 4 with Filgotinib, and 4 with Tofacitinib. A statistically significant reduction in SCCAI scores at 6 months was

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observed in the JAK groups compared to anti-IL-23 (7.64 ± 3.61 vs 4.90 ± 1.97, p = 0.047). However, biochemical response, assessed by faecal calprotectin reduction, did not reach statistical significance at either 3 or 6 months (p = 1.000 and p = 0.314, respectively). Endoscopic response, measured by changes in UCEIS at 12 months, also showed no significant difference between groups (p = 0.668). Subgroup analysis within the anti-IL-23 cohort revealed no significant difference in SCCAI reduction between Ustekinumab and Mirikizumab at 3 months.

Similarly, while Upadacitinib was associated with the *greatest SCCAI reduction among the JAK inhibitors at 3 months*, the difference among the three agents was not statistically significant (p = 0.106). Subtotal colectomy was performed in one patient in the Mirikizumab group following completion of induction therapy, and in two additional patients within three months of initiating the same agent (Table 1).

Biologic	Patients (N)	Colectomy(N)	Drug persistence
Mirikizumab	5	3	3/5
Ustekinumab	8	1	5/8
Upadacitinib	3	0	3/3
Filgotinib	4	0	3/4
Tofacitinib	4	0	4/4

Table 1. Patients' characteristics regarding clinical, biochemical and surgical outcomes within 12 months.

Conclusion: JAK inhibitors showed an improvement in SCCAI at 6 months compared to anti-IL-23 agents. No significant differences were found in biochemical or endoscopic response.

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Disclosure: Nothing to disclose.

PP0515

A REAL-WORLD DATA ANALYSIS OF OZANIMOD IN FDA ADVERSE EVENT REPORTING SYSTEM (FAERS) DATABASE

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Introduction: Ozanimod received approval in the United States in March 2020 for the treatment of relapsing multiple sclerosis, Crohn's disease, and ulcerative colitis. However, there is limited information available on the adverse drug reactions (ADRs) associated with its use. The main objective of this study was to explore the safety of Ozanimod after its market launch.

Aims & Methods: We gathered data from the United States Food and Drug Administration Adverse Event Reporting System (FAERS) database. Following data standardization, an array of signal quantification techniques, such as ROR, PRR, EBGM, and IC, were employed for analysis.

Results: We extracted 7,118,789 reports from the FAERS database, of which 5,587 reports identified Ozanimod as the primary suspected (PS) drug. Adverse reactions attributed to Osimertinib manifested across 27 organ systems, encompassing a total of 77 preferred terms meeting the criteria of all four algorithms simultaneously. Notably, we observed some common adverse drug event (ADE) signals, such as fatigue, headache, and dizziness, which were not documented in the official drug label.

Furthermore, we identified less common but notably potent ADE signals, including abnormal lymphocyte count, abnormal magnetic resonance imaging of the head, and traumatic fracture. The majority of adverse events occurred within the initial 30 days following the initiation of Ozanimod treatment.

Conclusion: Ozanimod presents the potential for diverse adverse reactions alongside its therapeutic benefits. Hence, in clinical practice, prompt detection of adverse drug reactions and the implementation of timely and effective preventive measures are essential.

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Disclosure: Nothing to disclose.

PP0516

DIFFERENTIAL SEGMENTAL HEALING IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory Bowel Disease (IBD) presents with distinct patterns of inflammation in Crohn's Disease (CD) and Ulcerative Colitis (UC). Ileal and Colonic CD differ in their clinical presentation, gut microbiota, cytokine profile and the response to treatment. A therapeutic approach based on the segment of intestine involved may show improved outcomes and less toxicity . Whether segment of intestine involved affects outcomes is not completely clear.

Aims & Methods: Our study aimed to determine the influence of the segment of intestine involved on mucosal healing, clinical improvement, health-related quality of life (HRQoL), and endoscopic remission in patients with UC and CD.

In this observational study conducted from March 2023 to August 2024, 50 patients each of CD and UC were evaluated. Disease activity was assessed using CDAI and SES-CD scores for CD, and True Love Witts, PRO2, and UCEIS scores for UC. Patients received medical therapy including mesalamine, steroids, immunosuppressants, or biologicals. Follow-up assessments were performed after 6 to 9 months, assessing clinical remission, endoscopic response, segmental healing, and health related quality of life.

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Results: Clinical remission was achieved in 88% of CD and 52% of UC patients. Endoscopic response was observed in 64% of CD and 62% of UC patients, while endoscopic remission was achieved in 34% of CD and 32% of UC patients. Health-related quality of life targets were achieved in 68% of CD and 54% of UC patients. In CD patients, segmental involvement was as follows : terminal ileum (74%), right colon (74%), transverse colon (38%), left colon (72%), and rectum (50%). The baseline mean SES-CD scores were terminal ileum (5.05±2.73), right colon (5.29±2.92), transverse colon (5.1±2.83), left colon (5.52±3.09), and rectum (4.96±2.87).

Complete mucosal healing rates varied across segments: transverse colon (68.4%), terminal ileum (59.4%), left colon (58.3%), right colon (56.7%), and rectum (48.0%). Improvement in mean SES-CD was highest in the transverse colon (3.47) and lowest in the rectum (2.48). In UC patients,-gut involvement was as follows: terminal ileum (6%), right colon (36%), transverse colon (52%), left colon (90%), and rectum (90%). UC patients demonstrated a clear proximal-to-distal gradient in complete mucosal healing rates: terminal ileum (100%), right colon (72.2%), transverse colon (61.5%), left colon (26.6%), and rectum (15.5%). Improvement in mean UCEIS was 2 in terminal ileum, 4.15 in right colon, 3.23 in transverse colon, 2.22 in left colon, and 2.07 in rectum.

Conclusion: Segmental healing patterns differ significantly between UC and CD. In CD, the transverse colon showed the highest remission rates, while UC demonstrated a clear proximal-to-distal gradient. Component analysis of individual endoscopic parameters revealed differential improvements in vascular pattern, bleeding, and ulcer scores across segments. Left colon healing was the most important determinant of HRQoL improvement in both UC and CD. These findings suggest that an individualized approach based on segment of intestine involved in IBD may provide valuable insights for treatment optimization and improvement of patient-reported outcomes.

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Disclosure: Nothing to disclose.

PP0517

ONE SUBCUTANEOUS INJECTION OF MIRIKIZUMAB IS BIOEQUIVALENT TO TWO SUBCUTANEOUS INJECTIONS: RESULTS FROM A PHARMACOKINETIC COMPARABILITY STUDY IN HEALTHY PARTICIPANTS

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Introduction: Mirikizumab (miri), a humanized IgG4 monoclonal antibody that selectively binds to interleukin-23 subunit p19, is approved for the treatment of moderately to severely active ulcerative colitis and Crohn's disease. In ulcerative colitis, maintenance treatment with 200 mg miri given as two 1-mL subcutaneous (SC) injections every 4 weeks was efficacious and safe in the phase 3 maintenance LUCENT-2 and long-term extension LUCENT-3 trials. As SC injections can be associated with pain/discomfort and thus poor patient experience, an investigational 2-mL single injection delivering the same miri dose has been developed.

Aims & Methods: This phase 1 study (NCT06475729) aimed to assess the pharmacokinetic (PK) bioequivalence and safety of miri solution administered as one 2-ml injection compared with the commercially available two 1-ml injections. Healthy participants were stratified into weight categories (<75 kg, 75–85 kg, and >85 kg) and randomized (1:1) to receive one dose of 200 mg of SC citrate-free miri as either one (200 mg/2 mL) or two injections (each of 100 mg/1 mL) delivered by an autoinjector. Participants were sub-randomized (1:1:1) by injection site (arm, thigh, and abdomen). Blood samples were collected at pre-defined intervals up to 10 weeks post-dosing. The primary endpoint was PK, assessed by the maximum observed concentration (C_{max}), area under the concentration–time curve (AUC) from time zero to infinity ($AUC_{0-\infty}$), and AUC from time zero to the last timepoint with measurable concentration (AUC_{0-t}). PK parameters were estimated using non-compartmental analysis. The secondary endpoint was safety, assessed by the incidence of treatment-emergent adverse events (TEAEs) and serious adverse events (SAEs) among participants exposed to miri. Evaluating the impact of injection site location on PK was an exploratory endpoint.

Results: A total of 244 participants (mean age: 42.4 years; 55.3% female) received one 200 mg/2 ml injection and 240 (mean age: 42.7 years; 53.3% male) received two 100 mg/1 ml injections. Baseline characteristics were similar in both groups. The three primary PK parameters all fell within the bioequivalence range (geometric least-squares mean [GM] ratio: 0.800–1.250; see table).

	2 × 1-ml SC injections (100 mg/ml) citrate-free miri	1 × 2-ml SC injection (100 mg/ml) citrate-free miri	GM ratio (90% confidence interval)
C_{max}, µg/mL, GM	12.8 (n=238)	13.3 (n=243)	1.039 (0.9853, 1.0956)
$AUC_{0-\infty}$, µg × day/mL, GM	221 (n=238)	223 (n=243)	1.011 (0.9564, 1.0685)
AUC_{0-t}, µg × day/mL, GM	216 (n=238)	218 (n=241)	1.011 (0.9568, 1.0678)

Results were generally consistent across injection site locations with no significant differences between them. TEAEs were reported in 48 participants with one injection (19.7%) and 55 with two injections (22.9%). Most

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TEAEs were mild (62/74 events with one injection [83.8%] vs 85/100 with two injections [85.0%]). Injection-site reactions occurred in 5/244 [2.0%] of participants with one injection and 4/240 [1.7%] with two injections. No SAEs were reported in either group.

Conclusion: Miri 200 mg administered as one 2-ml SC injection was bioequivalent to two 1-ml injections, and most TEAEs were mild. In clinical practice, reducing the number of injections may improve patient experience and treatment adherence.

Disclosure: YO, CDP, AUK, KT, PP, XZ: Employees and shareholders of Eli Lilly and Company.

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GDH: Adviser and/or speaker for Abbvie, Alimentiv, Bristol Meiers Squibb, Boehringer Ingelheim, Celltrion, Eli Lilly, Galapagos, Glaxo Smith Kline, Immunic, Index Pharmaceuticals, Johnson and Johnson, Merck, Polpharm, Prometheus biosciences, Prometheus Laboratories, Procise diagnostics, Protagonist, Sandoz, Takeda, Tillotts, and Ventyx.

SBH: Received advisory board and/or consulting fees from Abbvie, Takeda, Janssen, Celltrion, Pfizer, GSK, Ferring, Novartis, Roche, Gilead, NeoPharm, Predicta Med, Galmed, Medial Earlysign, and Eli Lilly; stocks/options in Galmed, Evinature, PredictaMed and Alma Therapeutics; and research support from Abbvie, Takeda, Janssen, Celltrion, Pfizer, Medtronic & Galmed.

PP0518

COMPARATIVE EFFECTIVENESS AND SAFETY OF TOFACITINIB AND FILGOTINIB IN PATIENTS WITH ULCERATIVE COLITIS: A PROPENSITY SCORE-WEIGHTED COHORT STUDY

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Introduction: Janus kinase inhibitors (JAKi), including tofacitinib (TOFA, a pan-JAK inhibitor) and filgotinib (FILGO, a selective JAK1 inhibitor), represent an emerging class of orally administered agents for moderate-to-severe ulcerative colitis (UC). Comparative real-world data on their effectiveness and safety are limited.

Aims & Methods: We conducted a multicenter, real-world, propensity score-weighted cohort study using data from the Sicilian Network for Inflammatory Bowel Diseases (SN-IBD). All adult UC patients treated with TOFA or FILGO with ≥16 weeks of follow-up were included. The primary outcomes were steroid-free clinical remission and clinical response at weeks 8 and 24. Treatment persistence and adverse events were also assessed. Inverse Probability of Treatment Weighting (IPTW) was applied to adjust for baseline differences.

Results: A total of 263 patients were included (TOFA: n=171; FILGO: n=92). At week 8, steroid-free clinical remission was achieved in 53.2% of patients treated with TOFA and 43.4% of those treated with FILGO (IPTW-adjusted OR: 1.51; 95% CI: 0.88–2.61; p=0.14). Clinical response at 8 weeks was reported in 75.4% of TOFA-treated patients versus 69.6% in the FILGO group (IPTW-adjusted OR: 1.34; 95% CI: 0.73–2.46; p=0.34). At week 24, steroid-free clinical remission was observed in 58.8% of the TOFA group and 44.9% of the FILGO group (IPTW-adjusted OR: 1.37; 95% CI: 0.76–2.49; p=0.30). Clinical response rates at 24 weeks were 63.6% for TOFA and 59.0% for FILGO (IPTW-adjusted OR: 1.13; 95% CI: 0.62–2.08; p=0.69). IPTW-adjusted Kaplan–Meier and Cox proportional hazards analyses showed no statistically significant difference in the likelihood of treatment discontinuation (log-rank p=0.294; IPTW-adjusted HR for TOFA: 0.78; 95% CI: 0.49–1.25; p=0.30). Adverse event rates were 12.7 and 5.4 per 100 person-years for TOFA and FILGO, respectively, with no significant difference (p=0.152).

Conclusion: In this real-world, IPTW-based study, TOFA and FILGO were well tolerated and demonstrated comparable and high effectiveness and treatment persistence in patients with moderate-to-severe UC, supporting their use as viable oral options in routine clinical practice.

Disclosure: FSM served as an advisory board member and/or received lecture grants from AbbVie, Ferring, Galapagos, Giuliani, Janssen, Lilly, Liohealth s.r.l., Pfizer, Sandoz, Takeda Pharmaceuticals. WF served as an advisory board member and/or received lecture grants from Abbvie, MSD, Takeda, Pfizer, Biogen, Sandoz, Zambon, Ferring, Sofar. AV received lecture grants from Pfizer. MC served as an advisory board member for AbbVie, MSD, Takeda Pharmaceuticals, and received lecture grants from AbbVie, MSD, Chiesi, and Takeda Pharmaceuticals. MC served as an advisory board member for AbbVie, MSD, Takeda Pharmaceuticals, and received lecture grants from AbbVie, MSD, Chiesi, and Takeda Pharmaceuticals. FM served as an advisory board member for Janssen and Galapagos, and received lecture grants from Takeda Pharmaceuticals. SR served as an advisory board member for AbbVie, Janssen, and MSD Pharmaceuticals, and received lecture grants from AbbVie, Galapagos, Janssen, MSD, Pfizer, and Takeda Pharmaceuticals. AO served as an advisory board member for AbbVie, Ferring, Galapagos, Lilly, MSD, Janssen, Pfizer, Samsung Bioepis, and Takeda Pharmaceuticals, and received lecture grants from AbbVie, Fresenius Kabi, Galapagos, Lionheath s.r.l., MSD, Sofar, Chiesi, Janssen, Pfizer, and Takeda Pharmaceuticals. All other authors: nothing to disclose.

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PP0519

OUTCOMES IN ACUTE SEVERE ULCERATIVE COLITIS PATIENTS WHO RECEIVED SHORT COURSE INTRAVENOUS STEROIDS AND EARLY RESCUE THERAPY

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Introduction: The mainstay first-line treatment of acute severe ulcerative colitis (ASUC) is high dose steroids. One-third of ASUC patients are steroid refractory and will require rescue therapy, either with cyclosporine (CYC) or infliximab (IFX) to reduce the risk of colectomy. It is recommended that patients are assessed on day 3 of steroid therapy to stratify risk of colectomy and therefore guide need of rescue therapy. It is generally understood that prolonged exposure to high dose steroids carries risk of infection, poor wound healing and mood disturbances. However, the optimum duration of intravenous steroids that mitigates risk and improves patient outcomes has not been well-established. Similarly, early rescue therapy in steroid refractory ASUC is also recommended. Delayed rescue therapy initiation may predict a longer hospital length of stay (LOS), but there is limited literature that assesses the effect of delayed rescue therapy on other patient outcomes.

Aims & Methods: All ASUC presentations to a Quaternary Sydney hospital over a 10-year period were included. A modified True-love Witts' Criteria was used to identify ASUC as ≥6 bloody bowel motions daily and presence of at least one systemic feature, including heart rate >90 bpm, temperature >37.8°C, haemoglobin <105 g/L, and C-reactive protein (CRP) ≥30 mg/L. IV hydrocortisone was given at a dose of 100 mg four times daily. Short course IV steroids was defined as ≤5 days and prolonged course of IV steroids was defined as >5 days. Rescue therapy used were either Cyclosporine (CYC), Infliximab (IFX) or JAK inhibitors (Upadacitinib or Tofacitinib). Early and delayed rescue therapy was defined as rescue therapy given within ≤5 days and at >5 days, respectively.

Results: A total of 172 presentations met criteria for ASUC. Of these presentations, 69.7% (n=120) received ≤5 days of IV steroid therapy, whilst 30.2% (n=52) received >5 days. A total of 34.9% (n=60) of patients required rescue therapy. Of those requiring rescue therapy, CYC was used in 26.3% (n=15), IFX was used in 73.7% (n=42) and JAK inhibitors in 5% (n=3). The mean time to rescue therapy was 5.67 days. Rescue therapy was given in ≤5 days for 53.3% (n=32) and >5 days in 46.7% (n=28) of presentations. The mean days on IV steroids in those who required rescue therapy was 6.95 days compared to 4.03 days in those who did not (p<0.01). Table 1 compares outcomes in those who received short and prolonged courses of IV steroids. Table 2 compares outcomes in those who received early and delayed rescue therapy.

	IV steroids ≤5 days (n=120)	IV steroids >5 days (n=52)	p-value	Rescue therapy ≤5 days (n=32)	Rescue therapy >5 days (n=28)	p-value
Days on IV steroids	N/A	N/A	N/A	5.3	8.6	<0.01
Received Rescue Therapy	25	35	<0.01	N/A	N/A	N/A
Underwent Colectomy	9	8	0.31	2	5	0.16
Received TPN	5	12	<0.01	2	8	0.02
Required ICU admission	3	5	0.11	0	2	0.48
30-day Readmission	22	6	0.08	6	3	0.38
90-day Readmission	13	9	0.61	5	4	0.88
30-day Mortality	1	2	0.27	0	2	0.48
90-day Mortality	0	0	N/A	0	0	N/A

Table 1. Comparison of outcomes in patients who received short and prolonged courses of IV steroids, and early and delayed rescue therapy.

Conclusion: Patients who received shorter courses of IV steroids and early rescue therapy were less likely to receive total parenteral nutrition (TPN) but no impact was seen on other clinical outcomes. Patients requiring rescue therapy were on IV steroids for a longer period of time compared to those who did not.

However, giving early rescue therapy significantly reduces the amount of time on IV steroids and overall length of stay. Further research is required to assess whether this may significantly reduce the risk of steroid side-effects and hospital-associated infections.

Disclosure: Nothing to disclose.

PP0520

HOME-MONITORING OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE ON SUBCUTANEOUS VEDOLIZUMAB – A 2 YEARS OBSERVATIONAL STUDY

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Introduction: Multiple studies have evaluated the efficacy and safety of subcutaneous vedolizumab in inflammatory bowel diseases (IBDs). Combined with home-monitoring, subcutaneous vedolizumab has the potential to promote effective and personalized management, enables self-administration and optimize use of medical resources.

Our study aimed to assess the impact of home monitoring based on patient reported outcomes (PROs) and home test fecal calprotectin in patients with IBD treated with subcutaneous vedolizumab.

Aims & Methods: This was an observational study of 43 IBD patients diagnosed with Ulcerative colitis (UC) or Crohn's Disease (CD) at North Zealand Hospital in Denmark (NOH). Patients were switched from intravenous to subcutaneous vedolizumab in 2020-2022 and followed for two years via home-monitoring web app IBD Constant Care (CC).

Data retrieved from www.ibd.constant-care.com included clinical disease activity scores (Simple Clinical Colitis Activity Index (SCCAI) for UC and Harvey-Bradshaw Index (HBI) for CD), medicine adherence score (MARS), quality of life (QoL), disability (WEB-DI), Fatigue (FACIT-F), and fecal calprotectin levels (using a home testing kit, Calpro smart APP subsequently results transferred to patients file in EPIC). Data including patient descriptions, drug frequency (Q) in weeks, adverse events (AE), severe AE, IBD-related hospital admissions, surgeries and death were obtained from patient journals (EPIC system).

Results: After the exclusion of two patients (one for not starting SC vedolizumab and shifting from IV vedolizumab to another biologic due to primary non-response, and another for lacking a recorded date of consent), forty-one patients, CD (n=21) and UC (n=20) were switched from intravenous to subcutaneous vedolizumab. Treatment was evaluated at baseline, after one and two years.

Our study found high adherence to PROs submissions in CC, with only 10% of patients with UC failing to complete them after the two-year period. Only 21 (51%) patients were maintained on subcutaneous vedolizumab after 2 years with tendency to shortened treatment interval compared to baseline and year 1. The proportion of patients achieving remission and low disease activity after 2 years were 70% in UC and 77.8% in CD. 94,1% (n=16) of patients reported improved QoL after 2 years. The most frequent

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adverse event was reaction at injection site and was reported by 16 (39%) patients, upon which 11 (69%) patients stopped the treatment. 4 (10%) patients undergone IBD-related surgery.

Conclusion: Home monitoring using CC has the potential to transform IBD care by delivering continuous, long-term data on patient reported outcomes. Subcutaneous vedolizumab as maintenance treatment in active, moderate to severe IBD proved to be feasible to monitor by telemedicine and effective for both UC and CD in terms of improving disease activity and quality of life with high rate of adherence to medication. However, local adverse events could lead to treatment discontinuation.

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PP0521

OPTIMIZING THE CLINICAL POTENTIAL OF A NOVEL PH SENSITIVE ANALGESIC IN A PRECLINICAL COLITIS MODEL

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Introduction: Improved pain management is an identified top five priority for the treatment of patients suffering from inflammatory bowel disease. Although conventional opioids and NSAIDs can provide pain relief, their side effects pose serious health risks. We developed a unique pH-sensitive analgesic, NFEPP, that is only active at sites of injury/inflammation without affecting healthy tissues, e.g. brain and gut. Preclinical studies using parenteral routes (s.c. or i.v.) demonstrate that NFEPP exhibits potency comparable to fentanyl, but without the risks of addiction or severe side effects like respiratory depression. The duration of action of NFEPP is ~2 hrs., which would require continuous infusion or frequent applications to treat a painful disorder. To broaden the potential clinical applications, strategies are needed to increase its half-life and to understand its efficacy compared to conventional treatments.

Aims & Methods: To examine the therapeutic index (TI) of NFEPP and determine if a high index could be exploited to extend its duration of action, and to compare the analgesic efficacy of NFEPP with a standard NSAID, diclofenac.

Acute colitis was induced in C57BL/6 mice by administering 2.5% dextran sulfate sodium(DSS) in drinking water for five days. Following s.c. injections of vehicle, NFEPP, fentanyl or diclofenac, visceromotor responses (VMRs) to colorectal distension were measured using telemetric transmitters inserted in the abdominal wall muscles. To assess cardiorespiratory side effects, mice were monitored via pulse oximetry.

Results: NFEPP (0.4 mg s.c.) and fentanyl (0.4 mg s.c.) exhibited similar analgesic effects on VMR but fentanyl markedly inhibited oxygen saturation (~10%; $p<0.05$), causing hypoxia. NFEPP at 50X's this analgesic dose (20mg/kg s.c.) had no effect on oxygen saturation, suggesting a very high TI.

We therefore examined whether the duration of action of NFEPP could be extended into the range of other oral analgesics by increasing the dose (doubling the dose increases T1/2 by ~<1, 1st order kinetics). NFEPP 0.4 mg/kg s.c. was tested at 2 hrs. and1.6 mg/kg s.c. at 6 hrs. NFEPP (1.6 mg/kg s.c.) caused a 37.8% ($p<0.05$) reduction in VMR responses after 6 hrs., similar to the effect of NFEPP 0.4 mg/kg at 2 hrs. (36.8% reduction in VMR; $p<0.01$). Comparator studies of the efficacy of NFEPP (0.8 mg/kg s.c.; ~EC75 dose) and diclofenac (30 mg/kg s.c., ~EC100 dose) demonstrated a 54.2% ($p<0.001$) and 49.7% ($p<0.01$) reduction in VMR, respectively.

Conclusion: NFEPP has a very high TI compared to fentanyl, and this property could potentially be exploited to increase its half-life in oral and/or patch formulations by increasing the dose without risk of respiratory depression. NFEPP has similar or greater analgesic efficacy compared to NSAIDs and could potentially be an alternative for some clinical indications.

Disclosure: Nothing to disclose.

PP0522

THERAPEUTIC EFFICACY OF ISM012-042 (ISM5411) IN T CELL TRANSFER-INDUCED COLITIS IN MICE

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Introduction: Inflammatory Bowel Disease (IBD) is driven by dysregulated mucosal immune responses, involving innate lymphoid cells, innate immune cells (e.g., macrophages, neutrophils, dendritic cells), and adaptive immune components like Th17 and Th1 cells, which secrete pro-inflammatory cytokines. Current treatments, such as corticosteroids, 5-aminosalicylates, and anti-TNF-α antibodies, are limited by frequent relapses and side effects, including opportunistic infections. Previously, we demonstrated that ISM012-042, an oral gut-restricted small molecule, significantly alleviated symptoms in animal models of acute colitis, including TNBS- and Oxazolone-induced colitis mouse model [1].

Here, we investigate the efficacy of ISM012-042 in a chronic T cell transfer-induced colitis model, aiming to develop safer and more effective IBD therapies.

Aims & Methods: This study assessed the therapeutic and preventive effects of ISM012-042 and anti-TNF-α in a T cell transfer-induced colitis model mimicking chronic Crohn's disease. Female CB17 mice (8 weeks old) were randomly assigned into seven groups. One control group received 5×10^5 CD4+CD45RB^{low} T cells intraperitoneally. The other six groups received 5×10^5 CD4+CD45RB^{high} T cells intraperitoneally to induce colitis and were treated with vehicle, ISM012-042 (10 or 30 mpk, PO, QD), anti-TNF-α (25 mpk, IP, Q3D), or combinations, administered either preventively (days 0–42) or therapeutically (days 21–42). Disease severity was monitored via body weight, Disease Activity Index (DAI) scores, colon length, weight, and density. Cytokine levels (IL-6, IFN-γ, TNF-α, IL-17A) in intestinal mucosa

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collected at day 42 were quantified using a CBA kit, and Swiss-rolled colon histopathology was evaluated with H&E and Masson staining. Flow cytometry analyzed T cell subsets in mesenteric lymph nodes.

Results: Groups receiving ISM012-042 (10 or 30 mpk, PO, QD) or anti-TNF-α (25 mpk, IP, Q3D) showed significantly reduced body weight loss, lower DAI scores, and decreased colon density compared to the vehicle group. Colon lengths were longer, and weights were lower, with reduced local inflammatory cytokines (IL-6, IFN-γ and TNF-α). Histological analysis revealed less crypt damage and inflammatory cell infiltration, with reduced fibrosis in Masson staining. Therapeutic Dosing (Days 21–42): ISM012-042 (30 mpk, PO, QD) and its combination with anti-TNF-α also reduced DAI scores, colon density, and cytokines (IL-6, IFN-γ, TNF-α, IL-17A). ISM012-042 (30 mpk, PO, QD) also showed decreased GATA3$^+$, T-bet$^+$, and IL-17A$^+$ T cell subsets.

Conclusion: ISM012-042, administered preventively or therapeutically, alone or with anti-TNF-α, effectively mitigated chronic T cell transfer-induced colitis by preserving colon integrity, and suppressing pro-inflammatory T cell subsets and cytokines, suggesting its potential as a versatile IBD therapy.

Reference:

1. Fu, Y., Ding, X., Zhang, M. et al. Intestinal mucosal barrier repair and immune regulation with an AI-developed gut-restricted PHD inhibitor. Nat Biotechnol (2024).

Disclosure: Nothing to disclose.

PP0523

POSITIVE EFFICACY AND SAFETY OF MIRIKIZUMAB AS MAINTENANCE THERAPY IN CHINESE PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS: SUBGROUP RESULTS FROM THE PHASE 3 LUCENT-2 STUDY

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Introduction: Mirikizumab (miri), a humanized IgG4-variant monoclonal antibody that specifically binds to subunit p19 of interleukin-23, showed efficacy in phase (Ph)3 induction (LUCENT-1) and maintenance (LUCENT-2) studies involving patients with ulcerative colitis.[1]

Here we report the efficacy and safety of the Ph3 LUCENT-2 study, a double-blind, randomized withdrawal maintenance study, in Chinese subpopulation who responded to miri induction therapy from LUCENT-1.

Aims & Methods: Miri clinical responders after 12-week induction therapy (54.3%)were re-randomized in a 2:1 ratio to receive blinded miri 200 mg or placebo (PBO) subcutaneously (SC) every four weeks (Q4W) for 40 weeks (Chinese primary efficacy population N=80). Randomization was stratified by biologic agent/tofacitinib failed status, induction remission status and baseline corticosteroid (CS) use.

The primary objective was to test the hypothesis that Miri was superior to PBO in achieving clinical remission at Week 40. Key secondary endpoints included alternate clinical remission, CS-free remission, endoscopic remission, histologic-endoscopic mucosal remission (HEMR), improvement in bowel urgency, bowel urgency remission, and maintenance of clinical remission. Analyses in Chinese subpopulation were descriptive.

Results: Baseline demographic and disease characteristics were generally balanced across the two treatment groups, including the proportion of baseline CS/ immunomodulator use, biologic agent/tofacitinib failure status and pancolitis.

A greater proportion of patients treated with miri achieved clinical remission at Week 40 (miri 50.0%, PBO 12.5%; Δ=40.4 [95% confidence interval (CI) 22.5, 58.4]). Miri also improved key secondary endpoints at Week 40 including proportion of patients who achieved alternate clinical remission (miri 55.4%, PBO 12.5%; Δ=45.4 [95%CI 27.3, 63.4]), and who achieved CS-free remission, clinical remission, bowel urgency remission and HEMR compared to PBO.

The frequencies of treatment-emergent adverse events were similar between groups in Chinese patients. There were fewer serious adverse events and discontinuations due to adverse events in miri patients compared to PBO. No deaths were reported in Chinese subpopulation during this study.

Conclusion: In Chinese patients who responded to miri induction therapy, miri's maintenance treatment provided positive efficacy and safety profiles consistent with LUCENT-2 overall population, supporting benefits of miri treatment in the Chinese ulcerative colitis patients.

	Placebo SC N=24	Miri 200 mg SC N=56
Clinical Remission[b], n (%) Common risk difference vs placebo	3 (12.5)	28 (50.0) 40.4
Alternate Clinical Remission [c], n (%) Common risk difference vs placebo	3 (12.5)	31 (55.4) 45.4
Corticosteroid-free Clinical Remission[d], n (%) Common risk difference vs placebo	2 (8.3)	26 (46.4) 40.2
Endoscopic Remission[e], n (%) Common risk difference vs placebo	4 (16.7)	33 (58.9) 45
HEMR[f], n (%) Common risk difference vs placebo	2 (8.3)	24 (42.9) 35.6
Bowel Urgency Improvement from BL[g], LSM (SE) LSM difference vs placebo (SE)	-2.23 (0.47)	-4.52 (0.33) -2.29 (0.53)
Bowel Urgency Remission[h], n/N (%) Common risk difference vs placebo	1/24 (4.2)	23/52 (44.2) 41.4
Maintenance of Clinical Remission[i], n/N (%) Common risk difference vs placebo	0/8 (0)	11/18 (61.1) 60.6

Modified Intent-to-Treat population. BL = baseline (Week 0 of LUCENT-1 induction study); LSM = least squares mean; miri = mirikizumab; SC = subcutaneous; SE = standard error.
[a]Miri induction responders: Patients who had clinical response to miri at Week 12 of LUCENT-1 (decrease in the modified Mayo score of ≥2 points and ≥30% decrease from baseline (BL); and decrease of ≥1 point in the Rectal Bleeding (RB) subscore from BL or a RB score of 0 or 1) after receiving 300mg IV miri every 4 weeks for 12 weeks.
[b]Clinical Remission: Stool Frequency (SF) = 0, or SF = 1 with a ≥1-point decrease from BL; RB = 0; Endoscopic Subscore (ES) = 0 or 1 (excluding friability).
[c]Alternate Clinical Remission: SF subscore = 0, or SF = 1, and RB subscore = 0, and ES = 0 or 1 (excluding friability).
[d]Corticosteroid-free Clinical Remission: Clinical remission at Week 40; Symptomatic remission at Week 28; No corticosteroid use for ≥12 weeks prior to Week 40.
[e]Endoscopic Remission: ES = 0 or 1 (excluding friability).
[f]HEMR (Histologic-endoscopic mucosal remission): Histologic remission with absence of mucosal neutrophils, defined using the Geboes scoring system with subscores of 0 for grades: 2b (lamina propria neutrophils); 3 (neutrophils in epithelium); 4 (crypt destruction); 5 (erosion or ulceration), and Endoscopic Remission.
[g]Bowel Urgency Improvement: Change from induction baseline in the Urgency Numeric Rating Scale (NRS). The Urgency NRS is a patient-reported measure of the severity for the urgency (sudden or immediate need) to have a bowel movement in the past 24 hours using an 11-point NRS ranging from 0 (no urgency) to 10 (worst possible urgency).
[h]Bowel Urgency Remission: Urgency NRS = 0 or 1 among patients induced into clinical response with miri in LUCENT-1 and had Urgency NRS ≥3 at induction baseline.
[i]Maintenance of Clinical Remission: Clinical remission at Week 40 among patients induced into clinical remission with miri in LUCENT-1.

Table 1: Primary and key secondary endpoints of LUCENT-2 at Week 40 in Chinese population (Miri Induction Responders[a]).

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PP0524

OPTIMAL DOSING OF SUBCUTANEOUS ADMINISTERED INFLIXIMAB DURING SWITCH AND START IN ADULTS AND CHILDREN WITH INFLAMMATORY BOWEL DISEASE: A SIMULATION STUDY

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Introduction: An increasing number of patients in clinical practice are transitioning from intravenous (IV) to subcutaneous (SC) dosing of infliximab. In this simulation study, we evaluated hypothetical dosing scenarios for both (normal to obese) adults and children switching from steady-state IV to SC infliximab, as well as those initiating SC infliximab therapy.

Aims & Methods: By combining two previous published infliximab models (1,2) we were able to simulate both IV and SC dosing in adults and children. Various dosing regimens were simulated using a large virtual population. In each scenario, the distribution of trough concentrations and area under the plasma concentration time curve (AUC) was calculated.

Results: Peak levels were higher after IV dosing compared to SC dosing, while trough levels were higher after SC dosing leading to more stable infliximab levels over time. Overall exposure remained largely similar when switching from a standard IV to SC dosing regimen. Patients with a high body mass index and those on high-frequency IV dosing regimens of infliximab demonstrated reduced exposure when transitioned to the fixed SC dose. Paediatric patients exhibited higher exposure on the fixed SC dose. Simulation of SC induction schemes demonstrated early achievement of steady state plasma levels.

Conclusion: Infliximab exposure (AUC) remains largely similar when transitioning from standard IV to SC dosing. Current dosing regimens may not be optimal for morbidly obese patients, paediatric patients and patients on high-frequency infliximab regimens. These findings provide a foundation for future clinical research to refine SC infliximab dosing in these populations.

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PP0525

JANUS KINASE INHIBITORS AS A RESCUE THERAPY IN ACUTE SEVERE ULCERATIVE COLITIS: SINGLE-CENTER RETROSPECTIVE STUDY

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Introduction: Acute severe ulcerative colitis (ASUC) is a life-threatening complication of ulcerative colitis, requiring hospitalization and potentially surgery. Initial treatment relies on intravenous corticosteroids, but non-responders require rescue therapies such as infliximab or cyclosporine. Janus kinase inhibitors (JAKi), including upadacitinib, and tofacitinib, rapidly modulate immune response and have shown promise in managing ASUC, based on small-scale studies.

Aims & Methods: This single-center retrospective study evaluated the efficacy of JAK inhibitors, upadacitinib and tofacitinib, as a rescue therapy in patients with ASUC not responding to intravenous corticosteroids and/or infliximab who were hospitalized during the period from May 2023 to October 2024. Upadacitinib was administered at a dose of 45 or 60mg/day during hospitalization, with the higher dose reduced to 45mg/day upon discharge. Tofacitinib was administered at 30 mg/day during hospitalization, with the dose reduced to 20mg/day upon discharge.
Patient outcome was assessed by partial Mayo score, endoscopic Mayo score (eMayo), and laboratory markers (CRP, hemoglobin, albumin, and fecal calprotectin).

Results: The cohort included 15 patients with ASUC who received JAKi as rescue therapy (8 males, 10 with extensive colitis; 73.3% with prior failure of infliximab): 13 upadacitinib (45mg and 60mg daily in 4 and 9 patients, respectively) and 2 tofacitinib. All patients had failed on intravenous corticosteroids, and one also failed on infliximab as a rescue therapy. All but one patient responded to JAKi and were discharged. One patient with no response underwent a colectomy. Endoscopy was performed at a median time of 8 weeks, with 57 % of patients achieving endoscopic remission (eMayo ≤ 1). During a median follow-up of 26 weeks after initiation of JAKi, 9 patients continued the treatment, 8 on upadacitinib and 1 on tofacitinib. One patient required colectomy, four were switched to another biologic, and one patient (a 76-year-old woman) died of pneumonia one month after starting therapy with JAKi. Within the first 3 months after hospital discharge, no further patients underwent a colectomy.

Conclusion: JAK inhibitors seem to be an effective rescue therapy for ASUC in patients who have already failed infliximab treatment. However, caution is needed in elderly patients.

Disclosure: Nothing to disclose.

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PP0526

EXPLORING BIOMARKERS OF CORTICOSTEROID THERAPY FOR MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS

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Introduction: Corticosteroids are a common treatment for patients with moderate to severe active ulcerative colitis (UC), but their effects can vary among individuals, including resistance, response, and dependence. While patient responses to corticosteroid therapy are generally swift, clinical remission may not be immediate for some. This suggests that corticosteroid treatment should be continued for a period of time to differentiate between corticosteroid-resistant and corticosteroid-sensitive patients with moderate to severe disease activity.

Consequently, identifying factors that can predict whether steroid use should be maintained or if a switch to a different treatment is necessary is an important consideration.

Aims & Methods: We compared the clinical characteristics of steroid responders and non-responders in patients with UC and retrospectively evaluated for biomarkers that could predict the efficacy of corticosteroid therapy prior to treatment in patients with moderate to severe active UC. Additionally, we compared the changes in various parameters before and after corticosteroid administration between steroid responders and non-responders. We enrolled 102 UC patients from Nippon Medical School Hospital who had a history of corticosteroid treatment prior to June 2024. The patients were divided into two groups: steroid responders and steroid non-responders. Their clinical characteristics were determined based on medical records.

Results: A total 102 UC patients with moderate to severe disease activity were included in this study. The clinical characteristics at baseline were as follows: the median age, 43 years (average 44.76± 17.79); with 61 males and 41 females. The most common distribution of disease type was total colitis (65%), followed by left-sided colitis (32%), and proctitis (2%). 42.6% of patients had severe disease activity. After remission induction therapy, 16% of patients were steroid non-responders, while 84% were steroid responders. The average Mayo Endoscopic Score was significantly lower, and eosinophil counts were higher in the steroid responder group compared to the steroid non-responder group (P=0.023, 0.016).

We also compared changes in various parameters at baseline and 4 weeks after steroid treatment between the two groups. After 4 weeks, inflammation markers including CRP and platelet counts decreased significantly in both groups. Hemoglobin and albumin levels increased significantly, and eosinophil counts decreased in the steroid responder group, but these changes were not observed in the steroid non-responder group.

Histological examination revealed no significant difference in the number of eosinophils infiltrating the colorectal mucosa on biopsy before steroid therapy between the two groups. However, 4 weeks after treatment, the number of infiltrating eosinophils was significantly decreased in the steroid responder group (P<0.001, Wilcoxon signed-rank test), while no change was observed in the steroid non-responder group.

Conclusion: Eosinophil count in the blood could serve as a potential biomarker for steroid responsiveness prior to administration. Additionally, steroid use decreases inflammatory markers such as CRP and platelet count, regardless of patient response.

However, eosinophil counts were reduced only in the responder group, suggesting that changes in eosinophil count may also be a predictor of efficacy in the long term.

Disclosure: Nothing to disclose.

PP0527

EFFECTIVENESS AND SAFETY OF TOFACITINIB: ONE-YEAR RESULTS FROM THE FRENCH PROSPECTIVE STUDY TOFAST

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Introduction: Tofacitinib (TOFA) is an oral Janus kinase Inhibitor for the treatment of ulcerative colitis (UC). We aimed to prospectively assess the real-world effectiveness and safety of TOFA in patients with moderate-to-severe UC, using a stringent criteria for clinical benefit.

Aims & Methods: TOFAST study is a prospective, multicenter, observational study including patients initiating TOFA from 39 French centers (Dec 2020 – Jun 2023). This interim analysis reports one-year effectiveness and safety. The primary endpoint was clinical benefit at one-year, defined as PRO-2 remission [rectal bleeding subscore (RBS) of 0 and stool frequency subscore (SFS) of ≤1], regardless of TOFA discontinuation.

Patients with colectomy, death, treatment with another biologic/JAKi/immunosuppressant or corticosteroids (CS) in the last 3 months were considered non-responders. Secondary endpoints included clinical remission, clinical response, change in daily PRO-2 scores during the first 14 days, the median time to relief, and safety.

Clinical remission was defined as Partial Mayo Score (PMS) ≤2 with no subscore >1, clinical response as a decrease in PMS ≥ 3 points and ≥30% from inclusion with a concomitant decrease in RBS ≥ 1 point (absolute subscore of 0 or 1). Patient's symptomatic relief was reported as the time in days from initiation of TOFA to improvement of their symptoms. Missing data were imputed as non-responders.

Results: 150 and 146 patients were included in the safety and effectiveness analysis, respectively, with a median exposure duration of 7.4 [Q1; Q3: 2.9; 12.0] months. In the effectiveness population, mean (± SD) age was 36.9 ±13.6 years, median disease duration was 6.0 years [Q1; Q3: 2.6; 11.2], 47.2% of patients had pancolitis, and the mean total Mayo Score was 7.8 ±2.3. 94.5% of patients had previously received at least one TNF-inhibitor, 62.3% vedolizumab and 32.2% ustekinumab. 27.4% of patients received at least 3 biologics. All patients initiated TOFA at 10 mg *b.i.d*, in combination with 5-ASA and CS for 19.9% and 33.6% of patients, respectively.

At one-year clinical benefit was achieved by 38/146 (26.0%) patients, and 66/146 (45.2%) of patients were still treated with TOFA. 30.1% and 30.8% of patients achieved clinical remission and clinical response, respectively. For the 94 (64.4%) patients who reported symptomatic improvement the median time to relief was 7 days [Q1; Q3: 4.0; 14.0]. Among the 102 patients who completed their daily PRO-2 subscores during the first 14 days, 36 (35.3%) and 55 (53.9%) normalized their SFS and RBS, respectively by day 14.

89.3% of patients reported at least one adverse event (AE) and 23.3% reported at least one serious AE. Twenty-two (14.7%) infections, 3 cases of herpes zoster, 2 cases of cancers (one non-invasive low-grade urothelial carcinoma and one colon adenocarcinoma) as well as one death (considered not related to the treatment) were reported. No major cardiovascular or venous thromboembolic event were observed.

Conclusion: In this large prospective real-world cohort of refractory UC patients, one-year clinical benefit was achieved in 26% using a stringent

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PRO-2 endpoint. Nearly one third were in clinical remission. Rapid symptomatic relief within 14 days and a manageable safety profile reinforce the role of TOFA as a therapeutic option in difficult-to-treat UC.

Disclosure: Nothing to disclose.

PP0528

PROOF OF CONCEPT IN VITRO STUDY OF THE ADSORPTION CAPACITY OF THE ENTEROSORBENT ENTEROSGEL FOR CYTOKINES AND ENZYMES RELEVANT TO ULCERATIVE COLITIS

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Introduction: Approximately 50% of patients with Ulcerative Colitis (UC) suffer from daily symptoms of faecal urgency, significantly impacting their quality of life[1]. Treatment in quiescent disease is challenging and the first line treatment, loperamide, is often used with caution due to the side effect of constipation.

Enterosorbents such as Enterosgel provide a potential alternative treatment. Enterosgel is thought to work through the adsorption of small molecules in the gastrointestinal (GI) tract, with minimal or no systemic absorption and metabolism[2], without causing constipation. Enterosgel has shown clinical effectiveness in reducing urgency in patients with diarrhoea predominant irritable bowel syndrome (IBS-D), suggesting potential therapeutic value in patients with UC[3], however the mechanism by which this may occur is unknown.

Aims & Methods: This study aimed to evaluate the in vitro adsorption capacity of Enterosgel for three key inflammatory mediators implicated in UC pathogenesis (IL-1β, MMP9 and TNF-α) to explore its potential as a therapeutic option in this patient group.

Three experimental paradigms were performed by diluting recombinant cytokines and enzymes in water-based Enterosgel solutions. After incubation, gel-free supernatants were harvested to analyse residual levels of metabolite via ELISA to identify the quantity of adsorption by Enterosgel. Conditions were conducted with different:

1. Incubation times (15,30,60,120 min)
2. Metabolite concentrations (5 concentrations ranging from 0 - 3000pg/mL)
3. Gel concentrations (0.056, 0.1125, 0.225g/mL)

A total of 78 samples were analysed including blank wells and charcoal as a positive control. Supernatants were subsequently harvested and prepared for ELISA. Assays were performed in duplicate and concentrations were interpolated from linear standard curves.

Results: Enterosgel demonstrated measurable concentration-dependent adsorption of IL-1β, MMP9 and TNF-α with higher analyte input concentrations resulting in greater absolute adsorption. However for TNF-α adsorption saturated at 1000pg/mL, indicating that Enterosgel may not fully adsorb analytes at levels simulating extreme luminal inflammation. Charcoal consistently resulted in lower residual concentrations, although this did not reach statistical significance on permutation testing ($p = 0.14$). In time dependent analysis, Enterosgel induced a rapid initial reduction in cytokine levels within 15 minutes, (IL-1β: 98.7%, MMP9: 99.4%, TNF-α: 95.8% adsorbed) with a plateau by 120 minutes (IL-1β: r^2=0.74, MMP9: r^2=0.64, TNF-α: r^2=0.45).

In contrast, charcoal demonstrated a near-linear monotonic decline with a slightly greater concentration adsorbed by 120 minutes (IL-1β: 99.9% *vs*. 99.8%; MMP9: 99.9% *vs*. 99.1%; TNF-α: 97.2 *vs*. 96.3%). Increasing gel concentrations improved adsorption efficiency; however, at the highest gel concentration small amounts of residual cytokines remained (IL-1β: 0.25%, MMP9: 0.28%, TNF-α: 3.5%). Doubling of gel concentration led to an average 30% decrease in residual cytokines.

Conclusion: Our findings suggest that Enterosgel demonstrates excellent adsorption of three key pro-inflammatory cytokines relevant to UC, with the greatest reduction at 1 hour and the highest gel concentration—though less than that achieved by charcoal. Evidence of saturation at high analyte levels suggests a limit to Enterosgel's binding capacity under conditions simulating severe inflammation. These results highlight Enterosgel's clinical potential, however further validation with biological replicates and clinical studies assessing urgency in treated UC is required.

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Disclosure: Nothing to disclose.

PP0529

EFFICACY REGARDING BIOLOGIC AND ADVANCED THERAPY FOR ILEAL CROHN'S DISEASE: A SYSTEMATIC REVIEW

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Introduction: The Montreal classification reflects the varying phenotypes of Crohn's including location and behaviour, however the majority of trials to date, despite including Montreal classification in baseline characteristics, do not report treatment efficacy by disease location. Ileal disease is associated with a higher risk of progression to stricturing and penetrating complications.

Aims & Methods: To perform a systematic review summarising the evidence for different biologic and advanced therapies available for small bowel Crohn's disease in adult patients.

Systematic search of MEDLINE and EMBASE was conducted. Articles were then screened, duplicates removed. Inclusion criteria included trials with efficacy outcomes stratified via disease location.

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The primary outcome was drug efficacy, measured via endoscopic response, healing, clinical remission or improvement of MRI severity scores.

Results: A total of 868 publications were screened and assessed for eligibility following database screening and removal of duplicates. 12 studies were included with 746 patients, with a total median follow up of 52 weeks(range 14-165). All studies were included in qualitative analysis; Infliximab (6 trials n=404) Adalimumab (4 trials n=204), Vedolizumab (2 trials n=50), Ustekinumab (1 trial n=28), Filgotinib (1 trial n=60).

Of the 6 infliximab studies (n=404), 3 reported on endoscopic response (ER) and 2 endoscopic healing (EH) via double balloon enteroscopy, 2 reported on clinical outcomes only and 1 reported on improvements in MRI outcomes, as detailed in the below table. Median follow up was 52 weeks.

First author	Year	N	Design	Drug	Outcome	Follow up
Huang	2025	110	R			
		64		IFX	EH 29.6% (19/64) ER 42.2% (27/64)	52 weeks
		28		UST	UH 35.9% (23/64) EH 10.7%(3/28)	
		18		VEDO	ER 28.6% (8/28) UH 17.9% (5/28) EH 5.5% (1/18) ER 11.1% (2/18) UH 11.1% (2/18)	
Wei	2024	111	R	IFX	ER 42.3% (47/111) MH 29.7% (33/111)	32 months
Han	2021	91	R	IFX	DR(CR+MH) 47.2% (42/91)	14 weeks
Chen	2021	79	R	IFX	CR 46.8% (37/79)	38 months
Zorzi	2012	44 34 (I) 28 (M)	O	IFX – Induction (I) Maintenance(M)	CR 74% (20/28)	54 weeks
Van Assche	2013	15	P	IFX	MRI improvement 40% (6/15)	26 weeks

IFX=Infliximab, ADA=Adalimumab, VEDO=Vedolizumab, EH= Endoscopic healing(defined as a modified SES-CD <3 or 0 if jejunal/proximal ileum), ER=Endoscopic Response(defined as a reduction of SES-CD by more than 50% at both segmental and complete level), UH=Ulcer Healing (defined as no inflammation, ulcers or erosions), MH=mucosal healing(SES-CD=0), DR=Deep Remission, CR=Clinical Remission.

There were 4 trials for adalimumab (n=204), 2 trial outcomes reported on EH(53.3%) and ER on capsule (42%). Two trials focused on clinical response rates; 64-66%[iii][iv]. Median follow up was 52 weeks (range 24-154).

Ustekinumab data (n=28) was available from Huang et al who reported EH in 10.7%(3/28), ER in 28.6% (8/28) and UH rates of 17.9% (5/28).

Two retrospective studies reported on Vedolizumab outcomes(n=50); reporting on EH 5.5% (1/18), ER 11.1% (2/18), UH 11.1% (2/18) [v] and another study (n=36) reported on CT and MRI outcomes for ileal Crohn's reporting a 22% (7/32) imaging response based on EMBARK scoring[vi]. Both studies had 52 week follow up.

One RCT involving two doses of Filgotinib reported rates of clinical remission (based on CDAI<150 at week 24) of 25% in both groups[vii] and secondary outcomes of MRI remission (based on MaRIAseg<7) in 6.3%(2/32) of 100mg arm and 8%(2/25) of the 200mg arm.

Conclusion: Robust retrospective data exists for small bowel mucosal healing with anti-TNF therapy. Data is lacking for vedolizumab, ustekinemab and newer smaller molecule therapies. Dedicated clinical trials are needed to specifically investigate the effectiveness of therapies in isolated ileal Crohn's disease patients to optimise their management.

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PP0530

INPATIENT AND POST-DISCHARGE THROMBOEMBOLIC EVENTS IN INFLAMMATORY BOWEL DISEASE PATIENTS: 13 YEARS AT A TERTIARY CENTRE

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Introduction: Primary thromboprophylaxis is used in the post-partum and the post-surgical period. 2014 *Canadian Association of Gastroenterology* guidelines and a 2021 *Nature Reviews* consensus recommend outpatient thromboprophylaxis only for those ambulatory inflammatory bowel disease (IBD) patients at high risk of thromboembolic events.

Aims & Methods: We aimed to analyse our inpatient burden and our rate of post-discharge thromboembolic (TEEs) in those with primary and secondary diagnoses pertaining to IBD. We also aimed to identify associated risk factors.

Our hospital inpatient and radiology databases were queried for discharges with primary and secondary diagnoses pertaining to IBD and scans pertaining to TEEs respectively.

Results: 1,601 discharges pertaining to 954 individual patients were identified over the 13-years ended 2024. 54% (n=870) were female. Median age was 44 (IQR:32-59). 58% (n=932) pertained to Crohn's disease.

Investigations for TEEs were performed during 117 admissions (7.3%). 10 patients had inpatient TEEs (0.6%). Age ≥ 50 years was significantly associated with inpatient TEE (Odds ratio [OR]: 1.89; 95% confidence intervals [CI]: 1.13-3.16; p=0.01). Male gender (OR: 0.92; 95% CI: 0.59-1.44; p=0.72), ulcerative colitis subtype (OR: 1.51; 95% CI: 0.95-2.41; p=0.08) and inpatient surgical intervention (OR: 1.39; 95% CIs: 0.89-2.17; p=0.15) were not significant associated factors.

63 discharges (3.9%) had imaging querying TEEs performed within 180 days. Six patients (0.4%) had TEEs within this time frame. No significant associations were identified for age ≥50 years (OR: 1.16; 95% CIs: 0.68-1.99; p=0.59), male gender (OR: 0.84; 95% CIs: 0.48-2.17; p=1.47, p=0.55),

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ulcerative colitis subtype (OR: 1.12; 95% CIs: 0.65-1.91; p=0.68), inpatient surgical intervention (OR: 1.37; 95% CIs: 0.79-2.35; p=0.26) or prior VTE (OR not calculable) in relation to post-discharge TEEs.

Conclusion: Our incidences of inpatient and post-discharge TEEs were low over 13-years. In agreement with international guidelines, we recommend against the need for routine primary prophylaxis in the post-discharge ambulatory period.

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PP0531

REAL-LIFE EFFICACY AND SAFETY OF RISANKIZUMAB IN CROHN'S DISEASE: A MULTICENTER OBSERVATIONAL STUDY FROM THE APULIAN NETWORK FOR INFLAMMATORY BOWEL DISEASE

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Introduction: Risankizumab, a selective interleukin-23 (IL-23) inhibitor, has emerged as a promising therapeutic option for Crohn's disease (CD). This study aimed to evaluate the real-life efficacy and safety of risankizumab in patients with CD, based on data from the Apulian Network for Inflammatory Bowel Disease (AN-IBD).

Aims & Methods: This multicenter, observational study included all consecutive CD patients initiating risankizumab therapy from its availability within the AN-IBD network. Clinical, biochemical, and ultrasound data were collected retrospectively at baseline (T0) and at the end of induction (week 12, T1).

Results: A total of 81 patients were enrolled. The mean age was 43 years, and 58% were male. Most patients (over 85%) were bio-exposed, with prior failure of at least one biologic. Ileocolonic involvement (L3 by Montreal classification) was the most common disease location, 60% of patients exhibited stricturing behavior (B2), 20% non-stricturing non penetrating pattern (B1) and 20% penetrating phenotype (B3).

At week 12 (T1), 21.6% of patients achieved the clinical response (≥3-point HBI reduction), and 19.6% the clinical remission (HBI <5). 71 out of 81 patients achieved steroid free remission (87,65%).

Inflammatory biomarkers improved significantly: mean CRP value decreased from 3.55 mg/L (T0) to 1.12 mg/L (T1); mean ESR value from 6.55 mm/h to 2.09 mm/h, respectively; mean fecal calprotectin from 57.2 µg/g to 0.81 µg/g.

Ultrasound (BUSS score) identified lower scores at T1 in responders vs. non-responders. Median BUSS scores were T0 (0.6 vs 3.4), T1 (0.5 vs 3.5), with statistically significant differences at T0 (p=0.032), T1 (p=0.031), suggesting a predictive role for early ultrasound response.

Regarding safety, one patient discontinued due to gait impairment, one for a cutaneous desquamative reaction, and another following a severe cutaneous reaction post-induction. Four additional patients discontinued due to lack of response, including one who developed a perianal fistula. No infections were reported.

Conclusion: These preliminary real-world data support the efficacy of risankizumab in inducing clinical response in CD, with a low rate of primary failure. Ultrasound response may have predictive value. No new safety signals were identified.

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PP0532

IMPACT OF SUBCUTANEOUS GUSELKUMAB INDUCTION THERAPY ON MOLECULAR INFLAMMATION IN PATIENTS WITH ULCERATIVE COLITIS: RESULTS FROM THE PHASE 3 ASTRO STUDY

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Introduction: Guselkumab (GUS) is a dual-acting IL-23p19 subunit inhibitor that potently neutralises interleukin 23 (IL-23) and binds to CD64, a receptor on cells that produce IL-23[1]. GUS was effective and safe as intravenous (IV) induction and subcutaneous (SC) maintenance therapy in patients with moderately to severely active ulcerative colitis (UC) (QUASAR)[2]. Week 12 (WK12) results from the ASTRO phase 3 study demonstrated the efficacy of SC induction therapy with guselkumab in UC[3]. Molecular changes occurring from induction baseline WK0 to WK12 evaluated in patients from the ASTRO study are presented here.

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Aims & Methods: At baseline, 418 patients with moderately to severely active UC were randomized 2:1 to receive GUS 400mg SC induction or placebo (PBO). Molecular analysis of the randomized population was performed comparing induction WK12 to baseline. Transcriptional profiling of colonic biopsies from 339 patients was performed with bulk RNA sequencing. A comparative analysis was also performed with biopsy samples previously analyzed from 593 patients from the QUASAR Phase 3 induction study. Genes and gene co-expression modules (derived from UC single cell RNA sequencing data) were evaluated for differential expression.
Correlation between Geboes scores and transcriptional gene module changes were evaluated in colonic biopsies from 413 patients. Molecular changes were assessed in the context of clinical remission and combined histologic and endoscopic mucosal improvement (HEMI) at WK12. Serum IL-22 proteomic profiling of 355 patients was conducted using a high-sensitivity IL-22 assay and change in protein abundance was assessed.
Results: GUS 400mg SC induction demonstrated significant downregulation of inflammatory transcriptional modules in colon tissue at WK12, representing Th17, IFNG signaling, inflamed epithelium, plasma cell, neutrophil and inflammatory fibroblast biology, and upregulation of healthy epithelium-related gene modules including goblet cells and healthy epithelium (all false discovery rate [FDR]<0.05). Response to GUS SC induction demonstrated significant correlation with changes observed in transcriptional gene modules with GUS 200mg IV induction in QUASAR (r=0.98, p<0.0001). Patients who achieved clinical remission at WK12 demonstrated the most robust changes in gene module expression from baseline (p<0.0001), nearing non-IBD control levels, compared to non-remitters. Similar results were obtained for GUS treated patients who achieved HEMI at WK12. Subgroup analyses defined by prior BIO/JAKi/S1Pi treatment history demonstrated similar magnitude of changes in both treatment-naïve and -resistant groups (R=0.93, p<0.0001). Serum IL-22 was significantly decreased at WK12 (p <0.05) with GUS SC induction, similar to IL-22 changes observed with GUS IV induction. Similar changes were also observed for tissue IL-22 gene expression (p<0.05) nearing non-IBD control levels at WK12. Patients who achieved clinical remission at WK12 demonstrated a robust decrease in Geboes total scores (p<0.05) compared to non-remitters.
Conclusion: Mechanistic observations confirm that GUS SC and GUS IV induction treatment show significant correlation, supporting the efficacy of both routes of administration in UC. Results from molecular analysis of the ASTRO study corroborate efficacy results observed for GUS SC induction in UC. These data demonstrate that GUS SC induction reduced inflammatory biology towards a normal state while upregulating genes associated with healthy epithelium in patients who achieved clinical remission or HEMI at WK12.

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PP0533

MIRIKIZUMAB DEMONSTRATES LONG-TERM EFFICACY ON HISTOLOGICAL RESOLUTION IN THE PHASE 3, MULTICENTER, OPEN-LABEL VIVID-2 EXTENSION STUDY

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Introduction: Histologic (histo) evaluation is an evolving treatment endpoint in Crohn's disease (CD). VIVID-1 evaluated systematically obtained biopsies from 5 intestinal segments demonstrating that histo readouts were impacted upon treatment with mirikizumab (MIRI). This may help differentiate efficacy between CD drugs.[1]

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Aims & Methods: VIVID-2, the long-term extension study of VIVID-1, examines histo and combined endoscopic (endo)-histo endpoints at Week (W)104. These data are presented, as is the impact of 1-year (yr) histo response (H-res) and histo remission (H-rem) on endoscopic and clinical outcomes after 2 yrs of treatment with MIRI.

In VIVID-1, the MIRI group received induction with 900 mg intravenously (IV) at W0, 4, and 8, then 300 mg subcutaneously (SC) every 4W. W52 patients (pts) in endo response (E-res) (≥50% reduction from baseline in Simple Endoscopic Score for CD) continued the same MIRI SC dosing in VIVID-2. Endo non-responders received reinduction with MIRI IV at the start of VIVID-2 followed by SC dosing (IV-SC) as described. Two specimens from each of 5 intestinal segments (1 ileal, 4 colonic) were obtained from the edge of ulcers, or the most inflamed mucosa from randomized pts at screening, W12, W52 (VIVID-1), and W104 (W52 of VIVID-2). H-res was defined as the absence of epithelial neutrophils and epithelial damage, erosions, and ulceration or ≥50% decrease in either the active Robarts Histopathology Index or the active Global Histology Activity Score. H-rem was defined as absence of mucosal neutrophils, no epithelial damage, and no erosions and ulcers. The criteria for H-res and H-rem had to be met in all 5 segments.

Results: Among pts in E-res at W52 of VIVID-1, 81.2% of pts maintained H-res and 57.5% maintained H-rem after an additional 52 weeks of MIRI SC treatment (Table 1A). In the same group of pts in E-res, 70.8% of pts maintained combined endo-histo response. Interestingly, 25.6% of pts who were in H-Res, but not in E-res at W52 of VIVID-1 maintained H-res and gained E-res during the 2nd year of MIRI treatment (IV-SC). Furthermore, when analyzing the impact of histo endpoints at W52 of VIVID-1 to longer term outcomes in the overall population treated with MIRI (SC-SC and IV-SC), both H-res and H-rem were significantly associated with E-res (p<0.001) and endo-remission (p<0.001) at W104 while associations with Crohn's Disease Activity Index remission (p=0.313 and p=0.771, respectively) and bowel urgency remission (p=0.122 and p=0.517, respectively) were not identified (Table 1B).

A. Outcomes at Week 52 of VIVID-1 (NRI)

	Endoscopic Responders MIRI SC/ MIRI SC		Endoscopic Non-Responders MIRI SC/ MIRI IV-SC	
Outcomes at VIVID-2 Week 52 (Week 104 of MIRI treatment)	OC	NRI	OC	NRI
Maintenance of H-Res[a], %(n/N)	92.1 N=178	81.2 N=202	65.8 N=79	57.8 N=90
Maintenance of H-Rem[b], % (n/N)	67.6 N=102	57.5 N=120	48.0 N=25	38.7 N=31

B. Impact of Histologic Response/remission at Week 52 (VIVID-1) on Week 104 (VIVID-2) in MIRI-treated participants (MIRI SC-SC & MIRI IV-SC) (mNRI)

	Week 52 Histologic Response			Week 52 Histologic Remission		
Outcomes at VIVID-2 Week 52 (Week 104 of MIRI treatment)	Yes	No	P Value[c]	Yes	No	P Value[c]
Endoscopic Response, n/N (%)	67.3 N=272	41.6 N=99	<0.001	75.5 N=133	52.3 N=238	<.0001
Endoscopic Remission, n/N (%)	43.2 N=272	20.2 N=99	<0.001	54.5 N=133	27.5 N=238	<.0001
Clinical Remission by CDAI, n/N (%)	75.7 N=272	68.4 N=99	0.313	74.7 N=133	73.2 N=238	0.771
Bowel Urgency Remission, n/N (%)	53.3 N=256	42.3 N=92	0.122	50.2 N=125	50.9 N=233	0.517

CDAI=Crohn's Disease Activity Index; IV=intravenous; MIRI=mirikizumab; mNRI=modified non-responder imputation for missing values; N=number of participants with non-missing values; NRI=non-responder imputation for missing values; OC=observed category; SC=subcutaneous

[a]Among pts in H-res at week 52 in VIVID-1

[b]Among pts in H-rem at week 52 in VIVID-1

[c]P-values are from Cochran-Mantel-Haenszel test adjusted by biologic-failed status (yes/no), baseline SES-CD total score (<12, >=12), either baseline SF >=7 and/or baseline AP >=2.5 (yes or unknown/no)

Table 1.

Conclusion: In VIVID-2, achievement of histologic response and remission after one year of miri treatment was strongly associated with longer term endoscopic outcomes. High rates of histologic and combined endo-histo endpoints were maintained after 2 years of miri treatment.

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PP0534

PREVALENCE OF ACUTE SEVERE COLITIS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: EVALUATION AND FACTORS ASSOCIATED WITH TREATMENT FAILURE

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Introduction: Inflammatory Bowel Diseases (IBD) can lead to various complications, with Acute Severe Colitis (ASC) being among the most critical. ASC is a medical and surgical emergency that poses a significant risk to life, necessitating prompt and effective diagnosis and management. The first-line treatment typically involves intravenous corticosteroids. However, some cases are resistant to corticosteroids and require second-line medical therapy or immediate surgical intervention.

Aims & Methods: The objective was to identify factors associated with the failure of first-line medical treatment in patients with ASC. A retrospective descriptive and analytical study was conducted over ten years (January 2014 to March 2024) in the gastroenterology department of Charles Nicolle Hospital. The study included patients with IBD complicated by ASC, defined by the Truelove and Witts criteria and a Lichtiger score greater than 10.Patients with bacterial superinfection were not included.

Results: Forty-five patients over 18 years old were included, with a mean age of 37.87 years (range: 18–74 years). There was a slight female predominance (53%), with a male-to-female ratio of 0.87. In 31% of cases, ASC was the initial presentation of the disease. Endoscopic signs of severity at admission were observed in 49% of patients. All patients received intravenous corticosteroid therapy with hydrocortisone hemisuccinate as the first-line treatment. The response rate to first-line treatment was 64.5%. Second-line medical therapy was administered to 24.4% of patients, consisting of cyclosporine in 11.1% and infliximab in 13.3%. Five patients (11.1%) underwent subtotal colectomy due to corticosteroid failure.

In univariate analysis, factors associated with corticosteroid failure included male gender (p=0.011), absence of extraintestinal manifestations (p=0.017), higher average number of stools on the first day (p=0.013), Lichtiger score >10 on the third (p=0.011) and fifth day (p<0.001), hemoglobin levels >11 g/dL on the first day (p<0.001), >10 g/dL on the third day (p=0.009), >9 g/dL on the seventh day (p=0.013), white blood cell count >10,000/mm^3 on the first day (p=0.008), >12,000/mm^3 on the third (p=0.001) and seventh day (p=0.004), C-reactive protein (CRP) level >30 mg/L on the seventh day (p=0.005), and persistent endoscopic signs of severity after three to five days of corticosteroid therapy (p=0.001).

Multivariate Analysis showed that only the presence of severe endoscopic signs on follow-up endoscopy after three to five days of corticosteroid therapy was predictive of corticosteroid failure (Odds Ratio = 42.2; p=0.028).

Conclusion: Acute Severe Colitis is a potentially life-threatening complication. The presence of severe endoscopic signs after five days of corticosteroid therapy should prompt the initiation of second-line therapy. Understanding the predictive factors for corticosteroid failure is essential for individualized management to improve both immediate and long-term outcomes.

Disclosure: The authors declare no conflicts of interest.

PP0535

SUPERVISED PHYSICAL EXERCISE TRAINING IN PATIENTS WITH IBD FLARE: A PILOT STUDY OF ITS EFFECTS AND PREDICTORS OF BENEFIT

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Introduction: Inflammatory bowel diseases (IBD), including Crohn's disease and ulcerative colitis, are chronic conditions with alternating phases of flare-ups and remission. Due to the persistent and often severe course of these diseases, effective treatment significantly improves patients' daily activities and overall functioning. [1] Given the uncertain prognosis and the frequent disease progression despite pharmacotherapy, it is important to explore additional interventions that could support traditional treatments. While physical activity may provide both anti-inflammatory and functional benefits in IBD, data on its efficacy and safety during disease exacerbation remain scarce. [2] This pilot study is the first real-world investigation into the effects of a supervised physical exercise training in patients hospitalized due to IBD flare.

Aims & Methods: This study aimed to evaluate the effects of an individualized, moderate-intensity, supervised physical exercise training on exercise tolerance, inflammation, and fatigue resistance in patients hospitalized due to IBD flare, and to explore baseline factors associated with greater benefit from the intervention. Patients were randomly assigned to an intervention group, receiving individual cycloergometer training during hospitalization, or to a control group without exercise. Both groups were well balanced at baseline, with no significant differences in demographic, clinical, or laboratory parameters. At both admission and discharge, all patients underwent standardized graded exercise tests to establish their exercise tolerance based on the lactate threshold and heart rate/power output response. Furthermore laboratory assessments, and the Functional Assessment of Chronic Illness Therapy – Fatigue (FACIT-F) questionnaire were conducted.

Results: A total of 22 patients were included in the study, with 13 patients in the intervention group and 9 patients in the control group. Although FACIT-F scores improved in both groups, the intervention group demonstrated a significantly greater improvement (median change –39 points; IQR –49 to –11; p = 0.014) compared to the control group (median change –25 points; IQR –52 to –6; p = 0.30). Baseline FACIT-F scores were significantly correlated with changes in heart rate response response during exercise: patients with greater baseline fatigue and poorer functional status experienced greater improvement in exercise tolerance (r = –0.62, p = 0.028). Platelet counts remained more stable in the intervention group (median change +10 ×10^3/µl; IQR –65 to +45) compared to a significant increase in the control group (median +74 ×10^3/µl; IQR +44.5 to +101.75; p = 0.046). In the intervention group, greater reductions in calprotectin were strongly correlated with more favorable heart rate responses during a graded exercise test (r = –0.92, p = 0.009).

Conclusion: Supervised training program during hospitalization for IBD flare was associated with improved fatigue resistance and more favorable inflammatory responses. Patients with worse baseline functioning obtained greater benefit in exercise tolerance. These findings support the feasibility and potential therapeutic role of individualized physical activity in acute IBD care and highlight the need for further investigation in real-world settings.

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Disclosure: Nothing to disclose.

PP0536

EARLY ADALIMUMAB TROUGH LEVELS PREDICT CLINICAL AND BIOCHEMICAL REMISSION IN PATIENTS WITH CROHN'S DISEASE

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Introduction: Advances in therapeutic drug monitoring (TDM) have significantly improved the understanding of the relationship between anti-TNF-alpha drug concentrations and disease activity in patients with Crohn's disease (CD). Early dose optimization of adalimumab (ADAL) and higher serum trough concentrations are associated with increased clinical, biochemical, and endoscopic remission in CD[1,2,3,4].

This study aimed to evaluate whether early dose optimization of ADAL to a trough level >15 µg/mL improves clinical and biochemical remission rates at 12 weeks compared to standard dosing.

Aims & Methods: This study aimed to evaluate whether early dose optimization of ADAL to a trough level >15 µg/mL improves clinical and biochemical remission rates at 12 weeks compared to standard dosing. This was a randomized, multicenter, open-label trial evaluating TDM-guided ADAL dose optimization versus standard dosing in biologic-naïve and experienced patients with active CD. In the dose optimization arm , after standard induction doses at Weeks 0 (160 mg) and 2 (80 mg), patients with trough levels ≤15 µg/mL at week 4 received an additional 80 mg at Week 6 followed by 40 mg every week. Patients in the optimization arm who had trough levels >15 µg/mL at week 4 and all patients in the standard care arm remained on 40 mg every 2 weeks. The primary endpoint was a composite of clinical and biochemical remission at Week 12, defined as Harvey-Bradshaw Index (HBI) <5 and C-reactive protein (CRP) <5 mg/L or fecal calprotectin (FCal) <250 µg/g.

Results: A total of 77 patients were enrolled in the trial (39 in optimization arm vs 38 in standard care arm). At week 4, 15/39 patients in the intervention arm were eligible for dose optimization, with twelve going on to receive the accelerated regimen. At week 12, 6/12 (50%) of the optimized patients achieved an ADAL trough level >15 µg/mL. A numerically higher proportion of patients in the dose optimization arm achieved the primary endpoint compared to the standard care arm, however this difference was not statistically significant (17 (43.6%) vs 12 (31.6%), p = 0.394). In total, eleven patients in the optimization arm (28.2%) and seven patients in the standard care arm (18.4%) achieved an ADAL trough >15 µg/mL, with mean trough levels being numerically higher in the optimization group (18.1 vs 12.8 µg/mL, p = 0.160). ROC analysis of the predictive value of Week 12 trough levels on the primary outcome determined an optimal threshold level of 18.2 µg/mL (AUC 0.68, sensitivity 88.9%, specificity 39.3%).

When analyzing the entire cohort, patients who achieved an ADAL trough >15 µg/mL (whether they were dose-optimized or not) at Weeks 8 or 12 were significantly more likely to achieve combined clinical and biochemical remission (19/27 patients (70.4%) vs 10/50 patients (20.0%), p < 0.001). Taken separately, both clinical and biochemical remission occurred more frequently in patients who achieved an ADAL trough >15 µg/mL (*clinical*: 22/27 (81.5%) vs 16/50 (32.0%), p <0.001; *biochemical:* 22/27 (81.5%) vs 14/50 (28.0%), p < 0.001). No difference in adverse effects was observed between trial arms. The major limitation of the trial was the small sample size and patient withdrawal due to the COVID 19 pandemic.

Conclusion: Although proactive dose optimization did not significantly increase remission rates compared to standard dosing, achieving an ADAL trough level >15 µg/mL at Week 8 or 12 was strongly associated with higher remission rates. These findings support early TDM-driven dose escalation and suggest that maintaining ADAL >15 µg/mL is a key target for achieving biochemical remission.

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PP0537

POPULATION PHARMACOKINETICS MODEL OF THIOGUANINE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Thioguanine (TG) has recently been rediscovered as an immunosuppressive agent in the treatment of inflammatory bowel disease (IBD). This prodrug is directly converted into its active metabolites, 6-thioguanine nucleotides (6-TGNs), targeting the inhibition of RAC1 GTPase in inflammatory conditions, disrupting key cellular signaling pathways necessary for T-cell activation and survival, thereby contributing to

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its immunosuppressive action. In IBD, TG is used fixed dose and may benefit from model-informed precision dosing (MIPD) to optimize treatment efficacy and minimize toxicity. However, a population pharmacokinetic (PopPK) model to do so is lacking.

Aims & Methods: The aim was to develop a PopPK model for TG in IBD patients, enhancing the understanding of TG's pharmacokinetics and supporting the implementation of model-informed precision dosing (MIPD). We employed a dataset comprising 131 6-TGN trough concentrations from 28 IBD patients treated with TG. The data were analyzed using nonlinear mixed-effects modeling (NONMEM) to estimate pharmacokinetic parameters and explore the influence of covariates such as weight and 5-ASA use on drug disposition. Model fit-for-purpose was evaluated through computation of the model's forecasting performance.

Results: The developed PopPK model was a one-compartment model with first-order absorption. A one-compartment TG model was stable, and able to estimate PK parameters with good precision (RSE 11-15%) with weight and 5-ASA use significantly affected TG clearance. Forecasting performance was also adequate with an rRMSE of 24.1% and practically no systematic bias (MPE 0.2%).

Conclusion: This study presents the first PopPK model of thioguanine for IBD, offering a novel tool for MIPD in clinical settings. Future studies should explore additional covariates such as TPMT genotype and drug interactions to further refine dosing recommendations for diverse patient populations.

Disclosure: AB has received funding from HLW Pharma BV, NdB has served as a speaker for AbbVie and MSD and has served as a consultant and/or principal investigator for TEVA Pharma BV and Takeda. He has received a research grant (unrestricted) from Dr. Falk, TEVA PharmaBV, Dutch Digestive Foundation (MLDS) and Takeda. LD has served as a speaker for AbbVie, Celltrion, Janssen-Cilag and Takeda and has developed continuing education materials for Ferring, All outside the submitted work. RJK is a an employee and stockholder of InsightRX.

PP0538

COMPARING INFLAMMATORY MARKERS IN THE FIRST 18 MONTHS OF TREATMENT BETWEEN PRIMARY RESPONDERS AND NON-RESPONDERS TO ANTI-TNF THERAPY

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Introduction: Biologic therapies, particularly anti-TNF agents, are key in managing inflammatory diseases. However, response to treatment varies, necessitating biomarkers to distinguish early responders from non-responders.

This study investigates early longitudinal changes in key inflammatory markers to compare trajectories between primary responders and non-responders over 18 months.

Aims & Methods: A total of 180 anti-TNF naïve patients were included. Patients were categorized into primary remission and primary non-remission groups based on clinical evaluation during follow-up. Erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), albumin, and hemoglobin levels were measured at baseline, and at 3, 6, 12, and 18 months. Intragroup temporal comparisons were analyzed using paired sample t-tests; intergroup comparisons were assessed via variance analysis.

Results: Across all time points, the responder group consistently demonstrated lower ESR and CRP levels and higher albumin levels compared to non-responders. In the responder group, ESR showed a significant decrease from baseline to 3 months, with stabilization thereafter. A similar pattern was observed in the non-responder group, though levels remained higher. Albumin levels increased significantly in both groups until 6 months and remained stable afterward, with consistently higher levels in responders. CRP levels dropped significantly from baseline to 6 months in both groups, then plateaued. Hemoglobin levels increased slightly only in the responder group at 3 months, with no significant changes in other intervals or in the non-responder group.

Conclusion: Inflammatory markers demonstrated distinct patterns between responders and non-responders to anti-TNF treatment within the first 18 months. Early changes in ESR, CRP, and albumin may help predict long-term response, supporting their role as potential monitoring tools in clinical decision-making.

Disclosure: The authors declare that there are no conflicts of interest.

PP0539

EFFECTIVENESS AND SAFETY OF MIRIKIZUMAB IN ULCERATIVE COLITIS: REAL WORLD PRELIMINARY DATA FROM LATIUM NET

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Introduction: Mirikizumab has been approved for the treatment of moderate-to-severe Ulcerative Colitis (UC). It is a humanized IgG4 monoclonal antibody that binds to the p19 subunit of IL-23, selectively inhibiting the interaction with the IL-23 receptor. Due to the limited real-world data on its effectiveness currently available, this study aims to assess the short-term effectiveness and safety of mirikizumab induction treatment in a real-life Italian cohort of UC patients.

Aims & Methods: From November 2024 to April 2025, all consecutive UC patients treated with mirikizumab across five Italian centers from Latium net were retrospectively enrolled. Only patients who had completed at least two doses of the induction phase of mirikizumab, with a follow-up of at least 8 weeks, were included in the final analysis. Clinical characteristics, disease activity scores, steroid usage, and adverse effects were collected.

The primary endpoint was steroid-free clinical remission, defined as a Clinical Modified Mayo score (MMS) < 2, with absolute Rectal Bleeding (RB) subscore=0 and absence of concomitant steroid therapy at 8-12 weeks.

Secondary endpoints included clinical remission (MMS<2, RB subscore=0), clinical response (≥2 point decrease in Clinical Modified Mayo score or MMS<2, RB subscore=0), biochemical remission (CRP ≤ 5 mg/L), and occurrence of adverse events at 8-12 weeks.

Results: A total of 61 patients were enrolled. The male-to-female ratio was 32:29, with a mean age of 48.7 years (range 19–77). The mean disease duration was 11.4 years (range 0–39). Four patients (6.5%) were biolog-

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ic-naïve, while 35 out of 61 (57.4%) had been treated with at least two prior biologic therapies. Sixteen patients were receiving concomitant steroid therapy at baseline.
Extraintestinal manifestations were reported in a subset of patients, including psoriasis in 8.2% (5), peripheral arthropathy in 13.1% (8), and axial arthropathy in 4.9% (3).
40 patients completed the 8-12 week follow-up period, and 7 completed the 20-week follow-up period. Of these, 4 (10%) received mirikizumab as a first-line therapy; 26 (65%) of them had previously been treated with antiTNFα agents and 14 (35%) had received ustekinumab as previous line of therapy.
Interestingly, a prolonged induction regimen was chosen for 25 out of 40 (62.5%) patients completing the 12-week induction therapy.
Specifically, in 4 of these cases, this decision was driven by a history of a "difficult to treat" disease instead of the lack of clinical or biochemical response. The steroid-free clinical remission rate at weeks 8-12 was 55.0% (22/40). The percentage of patients receiving concomitant steroid therapy decreased from 47.5% (19/40) at baseline to 25% (10/40) at weeks 8–12.
Clinical remission and clinical response rates at weeks 8-12 were 65% (26/40) and 82.5% (33/40), respectively.
Nineteen out of 40 (47.5%) patients presented a high C-reactive protein (CRP) at baseline, while ten out of 40 (25%) patients presented a positive CRP at 8-12 weeks. None of the patients discontinued mirikizumab within the 12-week follow-up period, and no adverse events or disease complications were observed during that period.
Conclusion: In a large cohort of refractory UC patients with multiple prior drug failures, including frequent failures with Ustekinumab and infliximab, nearly 55% achieved steroid-free clinical remission following mirikizumab induction. The safety profile of mirikizumab was consistent with existing literature.
Disclosure: Nothing to disclose.

PP0540

SEVERE ACUTE COLITIS AND OPPORTUNISTIC INFECTIONS: IMPACT OF CO-PATHOGENS (CMV, *CLOSTRIDIOIDES DIFFICILE*) ON THERAPEUTIC RESPONSE

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Introduction: Severe acute colitis (SAC) <represents a critical complication of inflammatory bowel disease (IBD), requiring prompt and appropriate management. Opportunistic infections, particularly those caused by cytomegalovirus (CMV) and *Clostridioides difficile* (C. difficile), can significantly influence clinical outcomes and therapeutic responses, especially in immunocompromized individuals.
This study aims to evaluate the incidence of these opportunistic infections in patients with SAC complicating IBD and to analyze their impact on clinical progression and treatment efficacy.
Aims & Methods: A retrospective study was conducted involving patients admitted for SAC, diagnosed based on Truelove and Witts criteria and a Lichtiger score exceeding 10. Initial assessments included CMV polymerase chain reaction (PCR) testing on colonic biopsies and/or blood samples, detection of C. difficile toxins A/B in stool specimens, and standard microbiological evaluations. Clinical, biological, and therapeutic response data were systematically collected.
Results: Fifty-five patients over the age of 18 were included, with a mean age of 42.36 years (range: 18–74) and a slight female predominance (55%). In 34% of cases, SAC was the initial manifestation of IBD. All patients met the Truelove and Witts criteria and had a Lichtiger score >10 upon admission. Endoscopic signs of severity were observed in 40% of patients.
Intravenous corticosteroid therapy using hydrocortisone hemisuccinate was administered as the first-line treatment for all patients, yielding a 60% response rate.
CMV co-infection was identified in 4 patients, all of whom were refractory to intravenous corticosteroids and subsequently received ganciclovir. Two patients responded favorably to second-line therapy with cyclosporine, while the remaining two did not respond to infliximab and required subtotal colectomy.
C. difficile co-infection was detected in 6 patients, treated with intravenous metronidazole and oral vancomycin. Among these, 3 patients did not respond to first-line medical therapy; one acheved clinical improvement with cyclosporine, whereas two required emergency subtotal colectomy due to the development of toxic megacolon.
Bivariate analysis revealed a statistically significant association between CMV co-infection and corticosteroid treatment failure (p=0.009). No significant association was found between C. difficile co-infection and first-line treatment failure in SAC.
Conclusion: Opportunistic infections, particularly CMV, are linked to unfavorable therapeutic responses in patients with SAC. Early detection of these co-pathogens is crucial for promptly adjusting treatment strategies, including the consideration of second-line therapies or surgical interventions.
These findings underscore the importance of systematic screening for opportunistic infections in SAC patients to optimize management and improve clinical outcomes.
Disclosure: The authors report no conflicts of interest.

PP0541

USE OF BIOLOGIC THERAPIES IN PATIENTS WITH ASYMPTOMATIC ISOLATED TERMINAL (CROHN'S) ILEITIS AT SCREENING COLONOSCOPY

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Introduction: Incidental inflammation of the terminal ileum is not uncommon in individuals who are submitted to screening colonoscopy. It is mostly mild and considered to have a benign course without the need for immunosuppressive therapy used for moderate Crohn's disease. In the light of recent literature highlighting the effectiveness of early treatment of Crohn's Disease we sought to determine the natural course of asymptomatic terminal ileitis in our pool of individuals who underwent screening (ileo)colonoscopy, and in particular which patients ended up being treated with biologics in real life , as well as any baseline factors.
Aims & Methods: We included individuals from the endoscopic database with asymptomatic terminal (Crohn's) ileitis in the last 5 years: we only included those with recorded endoscopic images available for endoscopic assessment with the SES-CD score and biopsies compatible with Inflammatory Bowel Disease (IBD) without a history of significant NSAID consumption. We recorded baseline characteristics (age, gender, smoking status, IBD family history, extra-intestinal autoimmunity), endoscopic severity, degree of histologic inflammation and biomarkers of inflammation (C-RP, Fecal Calprotectin).

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Finally we went through the Greek Prescription Database to find out the ones that were prescribed with biologics, steroids or had IBD-related surgery or hospitalization. We used point-biserial correlation, Pearson correlation, chi-square test to establish any variable correlations.

Results: We included 46 individuals : 28 women, 18 men with a median age of 55.8 years and a median follow up of 46.8 months. Nine of them were smokers (19.5%) and 10 (21.7%) had elevated C-RP (>0.5 mg/dL) and/or Fecal Calprotectin (>250 μg/mg). Steroids were used in 11 patients (23.9%) and only 1 had IBD-related surgery. Patients that received biologic therapy at some time-point amounted to 15 (32.6%): 9 with adalimumab (60%), 5 with ustekinumab (33.3%) and 1 patient received upadacitinib (concomitant rheumatoid arthritis). No factor other than an initial SES-CD score > 4 displayed statistically significant correlation with the future use of biologic therapy (p-bc=0,4214, p=0,0032). Interestingly enough most of the patients to eventually receive biologic therapy showed some kind of endoscopically visible ileocecal valve inflammation.

Conclusion: In this carefully selected but with a real life follow up group of patients with incidental asymptomatic terminal (Crohn's) ileitis a rather surprising one third of patients was eventually treated with biologic therapy during a median period of a 4-year follow up. Only baseline endoscopic severity (SES CD score >4) predicted the use of biologic therapy (mostly adalimumab or ustekinumab). All those patients had moderate disease and uneventful disease course. We need more studies to determine the (endoscopic?) threshold for biologic therapy or the right non-invasive tool (intestinal ultrasound?) to monitor these individuals in order to prevent over (or under)treatment.

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PP0542

WEEK 12 EFFECTIVENESS AND SAFETY OF MIRIKIZUMAB IN DIFFICULT-TO-TREAT ULCERATIVE COLITIS - PROSPECTIVE REAL-WORLD EVIDENCE FROM THE INITIATIVE ON CROHN AND COLITIS REGISTRY

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Introduction: Mirikizumab is an interleukin (IL)-23 inhibitor approved for the treatment of ulcerative colitis (UC); however, real-world data are limited.

Aims & Methods: The Dutch Initiative on Crohn and Colitis (ICC) Registry is a prospective multicentre study collecting real-world data on advanced therapeutic agents in inflammatory bowel disease (IBD). Patients included in the current study had a confirmed diagnosis of UC and received at least one intravenous induction dose of mirkizumab before the 28th of February 2025. Patients who did not use at least two therapies with different mechanisms of action (i.e. difficult-to-treat UC) or with colectomy were excluded. The primary endpoint was clinical remission at week (W) 12, defined as Simple Clinical Colitis Activity Index (SCCAI) ≤ 2. Secondary outcomes at W12 included corticosteroid-free (CSF) clinical remission (i.e. clinical remission without the use of systemic corticosteroids on the day of the visit), biochemical remission (faecal calprotectin (FC) level ≤ 250μg/g) and endoscopic remission (Mayo endoscopic subscore = 0). In addition, drug survival and adverse events were assessed.

Results: In total, 22 patients were included; 13 (59.1%) were male. Most patients (n = 12; 54.5%) were diagnosed with pancolitis. Patients had a median disease duration of 9.5 years (IQR 18.0). Twelve patients (54.5%) had used at least four different advanced therapies; 19 (86.4%) patients have previously used ustekinumab (IL-12/23 inhibitor). At treatment initiation, all patients showed active disease, either clinically, biochemically or endoscopically. Median SCCAI was 7.0 (IQR 6); median FC was 1696.5μg/g (IQR 2079). As the study is still ongoing, data at W12 were available for 19 (86.4%) patients. Clinical data were available for 14 patients; 35.7% (5/14) of them were in clinical remission. These five patients were also in CSF clinical remission. Similar rates were observed between ustekinumab-experienced and ustekinumab-naïve patients, although the naïve subgroup was rather small (3/14). Biochemical data were available for 12 patients; median FC was 1434.5μg/g (IQR 824). Only one (8.3%) patient was in biochemical remission at W12; this patient was ustekinumab-naïve. Another patient, with only endoscopic data, was in endoscopic remission. Two patients discontinued mirikizumab before W12: one patient stopped mirikizumab after the intravenous injection because of fever; the other patient showed no response. Four other patients discontinued mirikizumab after

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W12; none of them were in clinical or biochemical remission at W12. Reasons for treatment discontinuation were primary non-response (5/6) and adverse event (1/6). Four adverse events were reported in four patients: nausea and skin problems, both classified as possibly related to the treatment, a gastrointestinal infection requiring antibiotics and fever during the infusion.

Conclusion: We found that 7/16 patients with difficult-to-treat UC were in either clinical, biochemical or endoscopic remission at W12. Moreover, all patients in clinical remission at W12 were not using other medications. Mirikizumab was well tolerated, although the follow-up period was relatively short. The ICC Registry is ongoing, and results with higher patient numbers with longer exposure to mirikizumab will be analysed to further characterise the real-world effectiveness and safety for patients with difficult-to-treat UC.

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PP0543

SWITCHING TO SUBCUTANEOUS INFLIXIMAB IMPROVES WORK PRODUCTIVITY IN PATIENTS WITH IBD: THE BIO-BETTER SWITCH STUDY

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Introduction: The subcutaneous (SC) formulation of infliximab offers a patient-centered alternative to the intravenous (IV) route, with the potential to improve treatment adherence, quality of life, and reduce the burden of care in patients with inflammatory bowel disease.

Aims & Methods: Aim of our study was to evaluate the clinical impact, safety, and patients' outcomes associated with the switch from IV to SC infliximab in routine care. This was a single-center observational study incorporating both prospective and retrospective data collection. Patients with confirmed diagnosis of Crohn's disease (CD) or ulcerative colitis (UC) who received at least two infusions of IV infliximab and subsequently transitioned to SC infliximab 120 mg every other week were eligible for inclusion.

The primary endpoint was the impact of switching from IV to SC infliximab on patients' quality of life and work productivity measured through the SF-36 and WPAI-GH questionnaires administered at baseline and at week 52. Secondary endpoints included safety profile and clinical effectiveness, defined as clinical remission or improvement at weeks 8, 24 and 52 following the switch to SC infliximab.

Results: A total of 109 adult patients (72 UC (66,1%), 64 male (58.7%)) were enrolled. Median age was 41.6 ± 12.2 years. The switch from IV to SC infliximab was associated with an improvement in work productivity in both UC and CD at week 52 in terms of absenteeism ($p = 0.04$; $\eta^2 = 0.18$; 95% CI: 0.03–1.00 in CD and $p < 0.001$; $\eta^2 = 0.25$ in UC) and productivity loss ($p < 0.001$; $\eta^2 = 0.47$; 95% CI: 0.34–1.00 in CD and $p < 0.001$; $\eta^2 = 0.30$ in UC) . No significant differences were found in activity impairment, presenteeism, or across SF-36 domains in either CD or UC cohorts (all $p > 0.05$). Transitioning from IV to SC infliximab did not affect patients' perceived quality of life as assessed by physical, mental, or social functioning scores. At week 52, 51 out of 55 (93%) patients with UC and 20 out of 27 (74%) patients with CD were in clinical remission. Switching from IV to SC infliximab was significantly associated with achievement of clinical remission over time in UC ($\chi^2 = 8.83$, $p = 0.03$ with Cramér's V = 0.16). In patients with CD, switching from IV to SC infliximab did not show a statistically significant association with the achievement or maintenance of clinical remission at months 12 ($\chi^2 = 1.02$, $p = 0.80$ with Cramér's V = 0.00; 95% CI: 0.00–0.13). The switch was well tolerated with no new safety issues. In total, 6 patients experienced mild adverse events (erythema in the injection site in 3 patients, urinary tract infection in 1 patient, allergic reaction in 1 patient, and skin rash in 1 patient), leading to treatment discontinuation in 3 cases. No patient suffered from severe adverse events.

Conclusion: Switching from IV to SC infliximab is associated with significant improvements in work productivity at week 52 in both UC and CD. No new safety issues were found. These findings support the SC formulation of infliximab as a safe and effective option in the management of inflammatory bowel disease.

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PP0544

THERAPEUTIC DRUG LEVEL MONITORING OF ANTI-TNF(INFLIXIMAB) IN IBD; ASSOCIATION WITH BIOCHEMICAL AND CLINICAL PARAMETERS

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Introduction: Secondary loss of response to Infliximab(IFX) occurs in approximately half of patients with Inflammatory Bowel Disease(IBD), due to drug levels falling below the therapeutic range. This study evaluated IFX levels and related clinical and laboratory parameters and their impact on clinical outcomes.

Aims & Methods: Pre-infusion drug levels, associated laboratory data, and clinical parameters were evaluated in IBD patients receiving maintenance doses of IFX(5–15 mg/kg every 4–8 weeks) between Nov2021-Apr2025. Drug levels were measured using a bedside kit capable of quantitatively detecting values between 0.4–20µg/mL.

Results: A total of 1,059 single pre-infusion measurements from 455 patients and consecutive dual measurements at weeks 4 and 8 from 64 patients were evaluated. In the single and double measurement groups, 75.4% and 84.4% of patients had Crohn's disease(CD), respectively. The most commonly used doses were 5 mg/8 weeks and 10 mg/4 weeks(Table 1) Drug levels were below 5 µg/mL in 33% of the measurements. Antibody positivity was present in 47.5% of 120 available measurements ranging from 0.4-20 µg/mL, showing no significant correlation with drug levels or immunomodulatory(IM) use (p=0.62 and p=0.61).

However, for all positive antibody measurements drug levels were ≤ 3.5 µg/mL. CRP differed significantly across drug level groups, whereas albumin did not(Table 1). A weak negative correlation was found between IFX levels and CRP(r = -0.13, p = 0.001), with no correlation for albumin(p=0.54). In UC patients, the correlation between CRP and IFX levels was slightly stronger(r= -0.280, p=0.004).

No significant associations were found between IM use and IFX levels >5 µg/mL(p=0.55), >10 µg/mL(p= 0.93), or >20µg/mL(p = 0.87). Abdominal pain was more common in patients with IFX levels <5 µg/mL (p=0.032) and <10 µg/mL (p=0.044). Smoking status was not associated with achieving IFX levels >5 µg/mL(p=0.57), >10 µg/mL (p=0.89), and >20 µg/mL (p=0.633). Among CD patients, those with IFX levels >5 µg/mL had significantly lower CDAI scores (p = 0.039), and lower daily stool frequency (p=0.002).Regression analysis showed that higher BMI (p=0.045), lower CRP(p < 0.001), higher IFX dose(p < 0.001), and CD diagnosis(p =0.014) to be independently associated with achieving IFX levels >5 µg/mL. These factors also predicted levels >10 µg/mL and >20 µg/mL, except for BMI at the >20 µg/mL threshold. In the dual measurement group, the median CRP levels at weeks 4 and 8 were 2.1(5.1) and 2.5(6.3), respectively (p = 0.108). Among patients with IFX >20 µg/mL at week 4, 75% maintained levels >5 µg/mL at week 8, whereas this rate was 64% for those with levels >10 µg/mL. Only 59.6% of those with >5 µg/mL at week 4 retained >5 µg/mL at week 8(all p <0.001). Week 4 IFX levels was 7.3 times higher than at week 8 for cases within measurable range.

Parameter	>5 µg/ml	>10 µg/ml	>20 µg/ml	Patient Demographics	
Disease (All / CD / UC %)	67 / 69.8 / 56.5	51 / 53 / 43	25 / 25.4 / 21.8	Sex M / F %	56 / 44
p (CD vs UC)	<0.001	0.012	0.29	Age at Diagnosis	29 (19)
CRP # median(IQR)	2.6 (5.2) ; 4.9 (11.1)	2.5 (5.2); 4.1 (9.3)	2.1 (3.9); 3.6 (7.2)	Age at Measurement	38 (21)
p*(CRP)	<0.001	<0.001	<0.001	Duration of IFX Treatment	20 (25)
Dose Groups (%)	5mg/8w: 46 5mg/4w: 84 10mg/8w: 64 10mg/4w: 81 15mg/4w: 70	5mg/8w:28 5mg/4w:75 10mg/8w:38.7 10mg/4w: 71 15mg/4w: 56	5mg/8w: 4 5mg/4w: 43 10mg/8w:15.3 10mg/4w: 42 15mg/4w: 34		
p (Dose)	<0.001	<0.001	<0.001		
Week 4 (%) / Week 8 (%)	89 / 53.1	83 / 31.3	62.5 / 9.4		

* >5 vs <5; >10 vs <10 ; >20 vs <20
>5 and <5; >10 and <10 ; >20 and <20 respectively

Conclusion: One-third of patients demonstrated sub-therapeutic drug levels, which were associated with elevated CRP levels and worse clinical outcomes. Notably, the 5mg/4 weeks group exhibited higher drug concentrations compared to the 10mg/8 weeks group, suggesting that increased infusion frequency may contribute to achieving higher levels.

Moreover, maintaining sufficient drug levels at week 8 requires substantially higher levels at week 4. Measurement of anti-TNF antibodies may not be fruitful if drug level is > 3.5µg/mL. These findings support the drug monitoring and personalized dosing to reduce the risk of loss of response and improve outcomes.

Disclosure: Nothing to disclose.

PP0545

EFFECTIVENESS AND SAFETY OF USTEKINUMAB INTENSIFICATION (90 MG EVERY 4 WEEKS) IN PATIENTS WITH CROHN'S DISEASE AND ULCERATIVE COLITIS WITH SUBOPTIMAL INITIAL RESPONSE TO STANDARD DOSING: REAL-WORLD DATA FROM A MULTICENTER RETROSPECTIVE GREEK STUDY

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Introduction: Ustekinumab, an IL-12/23 inhibitor, is a well-established treatment for inducing and maintaining remission in moderate-to-severe Crohn's disease (CD) and ulcerative colitis (UC). However, some patients may exhibit a suboptimal response or secondary loss of response, neces-

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sitating dose intensification to 90 mg every 4 weeks. Real-world data on optimal management for patients with suboptimal initial response remain limited (1,2).

Aims & Methods: We aimed to assess the efficacy and safety of intensified ustekinumab dosing (90 mgQ4 weeks) in a multicenter cohort of patients with CD or UC who exhibited an initial suboptimal response, with increased inflammatory burden, to the standard ustekinumab regimen. We retrospectively analyzed data from 55 patients (33 female; median age, 51 years) with Crohn's disease (CD; n=36, 65.5%) or ulcerative colitis (UC; n=19, 34.5%) who underwent ustekinumab dose intensification (90 mg every 4 weeks) due to persistent clinical, biochemical (mean fecal calprotectin >300 µg/g), and endoscopic activity (mean SES-CD score: 10; Mayo score: 2/3) despite standard dosing.

All patients, were either bio-naïve or experienced, followed up for up to 12 months. Collected data included demographics, Montreal classification, treatment and surgical history, and pre-/post-intensification clinical, biochemical (CRP, fecal calprotectin), and endoscopic scores (SES-CD, Rutgeerts, Mayo).

For comparisons, the t-test was used for normally distributed variables, the non-parametric Mann–Whitney U test for non-normally distributed variables, and the chi-square test for categorical variables. A p-value <0.05 was considered statistically significant (Table).

	Response	No response	P-value
CD/UC	21/9	15/10	
Age, mean (SD)	55 (16)	47 (16)	
Perianal, n(%)	4	5	
Previous surgery	2 /26	9 /10	0.006
Biologic naïve CD, n(%)	12/21	3/15	0.023
Biologic naïve UC, n (%)	2/9 (22%)	3/10 (30%)	
Baseline SES-CD	10	10	
Baseline Mayo score 2/3 , n (%)	8/9 (88.8%)	7/10 (70%)	
ΔFcalpro, mean (SD)	-282 (377)	-2 (383)	0.009

Table: Patients' characteristics on ustekinumab intensification.

Results: Among CD patients, 58% had colonic disease (L2/L3), 22% upper GI involvement, 24% IBD-related surgery, and 25% perianal disease, although inactive at baseline. In UC cohort, 79% had left-sided colitis (E2). At 12-months, 21/36 CD patients (58.3%) maintained ustekinumab therapy, demonstrating biochemical, and endoscopic responses, with a mean reduction in the Simple Endoscopic Score for CD (SES-CD) of 6 points. Endoscopic remission—defined as SES-CD ≤3 or Rutgeerts score ≤i2a—was achieved in 62% of these responders.

Prior exposure to biologics and history of CD-related surgery were significantly associated with non-response (p = 0.02 and 0.006 respectively). 9/19 (47.3%) of the UC patients remained on treatment, with a Mayo score 1. A significant improvement in fecal calprotectin (ΔFCAL: −282 µg/g vs. −2 µg/g; p=0.009) was also observed among responders with CD and UC.

Early escalation (<12 months) or late escalation (> 12 months) did not significantly affect the response in CD/UC patients. No serious adverse events were reported during follow-up.

Conclusion: Dose intensification of ustekinumab in patients with persistently active CD and UC on the standard dose proved beneficial for nearly half of the patients over a 12-month period. Prior exposure to biologics and history of CD-related surgery were associated with less favorable outcomes. No safety concerns were identified

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Disclosure: No conflict of interest.

PP0546

NEW OR RECCURRENT NEOPLASIA AFTER IMMUNOMODULATORS USE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE AND PREVIOUS CANCER

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Introduction: Whether using Immunomodulators (IMM) as immunosuppressors (ISS), biologics or JAK inhibitors (JAKi) after a diagnosis of cancer in Inflammatory Bowel Disease (IBD) patients (pts) may increase the risk of new or recurrent cancer is undefined.

Aims & Methods: Primary aim: to assess the frequency of new/recurrent cancer in IBD pts treated with IMM after a diagnosis of cancer. Secondary aims: to evaluate in the same population the cancer-related mortality and risk factors for recurrence or new malignancy. In a retrospective multicentre Italian study, clinical records of all IBD pts treated with IMM after a diagnosis of cancer were reviewed.

Inclusion criteria: 1) Diagnosis of Crohn's Disease(CD) or Ulcerative Colitis(UC); 2)History of cancer; 3)Clinical outcome after cancer ≥6 months (mos).

Exclusion criteria: 1)IMM used for ≤3 mos or ≤3 administrations; 2)missing data. Data expressed as median[range], differences assessed with Student t-test and X-square test, univariate analysis performed to search for risk factors for new or recurrent malignancy.

Results: Study population included 122 IBD pts (84 CD, 38 UC) treated with IMM after cancer (Female 58 [47.5%]; age: 59.5[26-89] yrs). Index cancer was: genitourinary in 18(14.8%), non-melanoma skin cancer (NMSC) in 17(13.9%), breast in 15(12.3%), melanoma in 14(11.4%), thyroid in 13(10.6%), colon (CRC) in 11(9.0%), hematopoietic in 9(7.4%), prostate in 8(6.6%), neuro-endocrine in 4 (3.3%), head-neck, liver and endometrial in in 3(2.5%), other in 3(2.5%), lung in 1(0.8%). ISS used after cancer (duration: 29[4-200] mos) included: thiopurines in 10(37%), methotrexate in 14(51.9%), others in 3(11.1%) pts.

Biologics or JAKi used after cancer (duration: 24[3-180] mos) included: Anti-TNF in 36(32.4%), VDZ in 60(53.6%), UST in 45(40.2%), JAKi in 9 (7.3%) and others in 6(5.4%) pts.The median time interval between cancer and IMM use was 27[0-312] and 48[1-432] mos, respectively. In a median follow up of 8[1-45] yrs after index cancer, 12 (9.8%) IBD patients developed new or recurrent malignancy after IMM use. There were 4 recurrent neoplasms

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(2 prostatic, 2 thyroid) and 7 new malignancy (7 NMSC,1 salivary gland carcinoma, 2 melanoma and 1 CRC). No cancer-related death occurred. No differences were observed in terms of IBD phenotype and IMM treatments between patients with vs without new/recurrent malignancy. At univariate analysis, when considering age at malignancy, smoking habits, gender, IMM treatment and duration before and after index cancer and treatment for malignancy, no predictive factors for new/recurrent malignancy were observed.

Conclusion: A low frequency of new or recurrent cancer was observed using IMM after cancer in IBD. No risk factors for new/recurrent malignancy were observed.

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Disclosure: Nothing to disclose.

PP0547

CROHN'S DISEASE UNDER THE ANTI-TNF LENS: A 20-YEAR REAL-LIFE PERSPECTIVE

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Introduction: The treatment of Crohn's disease remains a complex challenge, with approximately 20% of patients not responding to conventional therapies. Among the available options, biologic therapies particularly tumor necrosis factor (TNF) inhibitors play a key role.

This study aims to describe the epidemiological, clinical, and therapeutic characteristics of patients with Crohn's disease treated with anti-TNF agents, and to compare their efficacy.

Aims & Methods: This is a retrospective, descriptive, comparative, and analytical single-center study conducted in a gastroenterology department over a 20-year period (2004–2024). It included patients with Crohn's disease treated with anti-TNF biologics. Data were collected from patient medical records and follow-up consultations. We analyzed epidemiological, clinical, paraclinical, therapeutic, and disease progression parameters, and compared the efficacy of infliximab and adalimumab.

Results: Out of 1,200 patients followed for Crohn's disease, 20.3% (244) received anti-TNF therapy (adalimumab or infliximab). The mean age at diagnosis was 33.5 ± 14.3 years, with a male-to-female ratio of 1.3. Smoking was reported in 16.9% (40) of cases. The median diagnostic delay was 1 year [0–3], and the median disease duration was 4 years [1–9]. Extraintestinal manifestations were found in 37.4% (91) of patients.

According to the Montreal classification, ileocolonic (L3) involvement was most common (49.6%), followed by colonic (L2, 24%), ileal (L1, 18.2%), and upper GI tract involvement (L4, 8.3%). The inflammatory phenotype was observed in 46.7%, stricturing in 27.7%, fistulizing in 17.8%, and both stricturing and fistulizing in 7.9% of patients. Perianal disease was associated in 45.9% (111) of cases.

Regarding baseline laboratory parameters: hemoglobin ≤10 g/dL was found in 35.3% (48), white blood cell count <8000/µL in 64.7% (88), platelet count >150,000/µL in 97.7% (125), CRP >6 mg/L in 83.4% (121), and albumin <30 g/L in 31.5% (40) of patients.

Therapeutically, 55% (134) of patients had undergone surgery. Corticosteroids were used in 56.6% (137) before initiating anti-TNF therapy, and 66.4% (166) received combination therapy. The overall remission rate at 54 weeks was 74.6% (179). Infliximab was prescribed in 74.9% (182) and adalimumab in 25.1% (61) of cases, with comparable efficacy: 76.3% (45) remission with adalimumab versus 74% (134) with infliximab (p = 0.732), showing no statistically significant difference.

Conclusion: Anti-TNF therapies currently play a major role in the management of moderate to severe Crohn's disease. In our study, both infliximab and adalimumab showed satisfactory clinical, biological, and endoscopic responses at 54 weeks, with no statistically significant difference

Disclosure: Nothing to disclose.

PP0548

COMPARISON OF PHARMACODYNAMIC SERUM IL-22 AND MECHANISTIC TISSUE MOLECULAR CHANGES BETWEEN GUSELKUMAB SUBCUTANEOUS AND INTRAVENOUS INDUCTION DOSING IN MODERATELY TO SEVERELY ACTIVE CROHN'S DISEASE: POST-HOC ANALYSIS OF THE GRAVITI AND GALAXI PHASE 3 STUDIES

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Introduction: Guselkumab (GUS) is a selective dual-acting IL-23p19 subunit inhibitor that potently blocks IL-23 and binds to CD64, a receptor on cells that produce IL-23[1].

GUS demonstrated short- and long-term efficacy with intravenous (IV) and subcutaneous (SC) induction dosing followed by SC maintenance treatment in patients (pts) with moderately to severely active Crohn's disease (CD) in the GALAXI and GRAVITI Ph3 trials.[2,3]

Previous analyses showed similar changes in serum inflammatory proteins following GUS SC or IV induction dosing.[4]

Here, we present a comparison of the tissue molecular effects of GUS SC and GUS IV induction treatment in CD.

Aims & Methods: Serum IL-22 was measured in a subset of participants using a high-sensitivity assay and changes in protein abundance were assessed at WK0 and WK12 following GUS 400 mg SC q4w induction (n=202) compared with placebo (PBO) (n=86) or; GUS 200 mg IV q4w induction (n=50) compared with PBO (n=50), and in independent healthy control sera (n=30).

Transcriptional profiling of ileum and rectum segments was performed using samples from 277 GRAVITI patients randomized to PBO or GUS 400mg SC q4w, and from a subset of GALAXI patients (n=259) randomized to PBO or GUS 200mg IV q4w obtained at baseline (BL) and WK12. Gene co-expression modules were evaluated for differential expression in the bulk RNA-seq dataset in ileum and rectum segments, and in patients stratified by prior history of inadequate response or intolerance to biologics. Tissue treatment effect analysis was performed using inflamed samples (defined as bMIS[5] >0) from segments with SES-CD >0 at baseline.

Results: In serum, GUS SC (-0.93 logFC, p<0.05) and GUS IV (-1.10 logFC, p<0.05) showed similar WK12 decreases in the serum IL-22 levels, and is consistent with reductions in IFNγ, IL-17A, CRP and fecal calprotectin levels (p<0.05)[4] previously reported. In tissue, changes in gene modules

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at WK12 observed with GUS 400mg SC q4w induction were significantly correlated with those observed with GUS 200mg IV q4w induction (R=0.97, p<0.05 in ileum and R=0.93, p<0.05 in rectum). Similar pharmacodynamic and molecular treatment effects were observed in patients stratified by previous biologic exposure. In both ileum and rectum, GUS SC induction reduced key cellular and inflammatory transcriptional modules at WK12 including plasma cell, inflammatory epithelial, neutrophil, and IL-23/Th17 biology (FDR<0.05), consistent with molecular analyses of GUS IV induction in GALAXI Ph3 studies.[6] Similar changes were observed for tissue IFNγ, IL-17A, and IL-22 gene expression with levels approaching those of the non-IBD controls at WK12.

Conclusion: GUS reduced serum levels of IL-22 protein at WK12 relative to baseline with both SC and IV induction. Following GUS SC induction, transcriptional changes observed at WK12 in GRAVITI were highly correlated with those observed with GUS IV induction from GALAXI Ph3 in ileum and rectum segments. GUS SC and IV induction regimens demonstrate similar molecular effects. These data are consistent with previously reported results demonstrating comparable efficacy between GUS SC and IV induction regimen and support the flexibility these options provide to both patients and healthcare professionals.

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Walter Reinisch reports serving as a speaker for AbbVie, Celltrion, Ferring, Janssen, Galapagos Medice, MSD, Roche, Pfizer, Sobi, Takeda; serving as a consultant for AbbVie, Amgen, AOP Orphan, Boehringer Ingelheim, Bristol Myers Squibb, Calyx, Celltrion, Eli Lilly, Galapagos, Gilead, Index Pharma, Janssen, Medahead, Microbiotica, Pfizer, and Takeda; serving as an advisory board member for AbbVie, Amgen, Boehringer Ingelheim, Bristol Myers Squibb, Celltrion, Galapagos, Janssen, and Pfizer; and receiving research funding from AbbVie, Janssen, Sandoz, Sanofi, and Takeda.

Flavio Steinwurz reports serving as consultant, speaker, or researcher for Abbvie, Amgen, Celltrion, Ferring, Janssen (Johnson & Johnson), Pfizer, Sandoz, and Takeda.

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Geert D'Haens reports consultancy activities for AbbVie, Agomab, Alimentiv, AstraZeneca, AMT, Bristol Meyers Squibb, Boehringer Ingelheim, Celltrion, Eli Lilly, Exeliom, Galapagos, GlaxoSmithKline, Gossamerbio, Immunic, Index, Johnson & Johnson, Kaleido, Origo, Pfizer, Polpharma, Procise Diagnostics, Prometheus Laboratories, Prometheus Biosciences, Progenity, and Protagonist Therapeutics Inc.; speaker's bureau for AbbVie, Bristol Meyers Squibb, Galapagos, Pfizer, and Takeda; and data monitoring board activities for AbbVie, AstraZeneca, Galapagos, and Seres Health.

PP0549

DO CLINICAL OUTCOMES IN ACUTE SEVERE ULCERATIVE COLITIS DIFFER BETWEEN PRESENTATIONS OF NEW DIAGNOSIS COMPARED TO ESTABLISHED DIAGNOSIS?

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Introduction: Acute severe ulcerative colitis (ASUC) develops in approximately 15-25% of patients diagnosed with ulcerative colitis (UC), with 10-20% of ASUC presentations occurring as a first presentation of new UC diagnosis.[1] Increased disease severity at diagnosis is associated with worse clinical outcomes, including higher rates of relapse and colectomy.[2] Comparative data on ASUC outcomes between those with newly diagnosed UC and established UC remains limited.

This study aims to assess outcomes between patients with new and established UC diagnosis presenting with ASUC.

Aims & Methods: A retrospective review included all ASUC presentations to a quaternary hospital over a 10-year period based on modified Truelove and Witts' criteria. ASUC presentations were grouped into new or existing diagnoses of UC. Data was collected on number of days on IV steroids, rates of rescue therapy, referral to colorectal surgery, rates of colectomy and total parenteral nutrition (TPN) requirement. Outcomes measured included rates of intensive care unit (ICU) admission, 30- and 90-day readmission, 30- and 90-day mortality, and mean hospital length of stay (LOS).

Results: A total of 172 ASUC presentations were included with 17% (n=29) presenting as new UC diagnosis, and 83% (n=143) as existing UC diagnosis. Those with new UC diagnosis had significantly lower rates of rescue therapy use compared to those with established UC diagnoses (14% vs 39%, p=0.01). The new UC diagnosis group trended towards lower rates of TPN use (p=0.07), referral to colorectal surgery (p=0.09), rate of colectomy (p=0.31) and days on intravenous steroids (p=0.53) but did not reach statistical significance (Figure 1).

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There were no significant differences in outcomes including rates of ICU admission (p=0.65), readmission at 30 days (p=0.78) and 90 days (p=0.80), mortality (p=1.0) or LOS (p=0.86).

Conclusion: Patients presenting with ASUC with a new diagnosis of UC required significantly lower rates of rescue therapy compared to those with existing diagnosis. However, our study did not find a significant difference in outcomes between patients with new and existing UC diagnoses at time of ASUC presentation, in contrast to current literature. Further studies are required to compare long-term outcomes between new and established UC diagnoses presenting with ASUC.

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PP0550

COST-EFFECTIVENESS ANALYSIS OF ETRASIMOD COMPARED WITH BIOLOGIC THERAPIES FOR PATIENTS WITH MODERATE-TO-SEVERE ACTIVE ULCERATIVE COLITIS EXPERIENCED TO PREVIOUS ADVANCED THERAPIES IN SPAIN

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Introduction: Etrasimod (2 mg/day, oral) is a novel alternative for moderate-to-severe active ulcerative colitis (UC), which has proved to be effective in ELEVATE UC12 and UC52 trials[1,2].

Aims & Methods: This cost-effectiveness analysis compared etrasimod vs. most frequently used biologics in Spain[3], for patients with moderate-to-severe active UC experienced to advanced therapies (AT), from the perspective of the Spanish national health system.

A hybrid model simulated the induction (decision tree) and maintenance (Markov model, 8-week cycles during lifetime) phases treatment to estimate costs (€, 2025) and quality-adjusted life years (QALYs), both yearly discounted (3%). All model inputs derived from scientific literature and were validated by national experts. The patients' characteristics proceeded from etrasimod trials[1,2]. Rates of efficacy, safety, and loss of response of etrasimod, adalimumab, ustekinumab, intravenous (IV) vedolizumab and subcutaneous (SC) vedolizumab were obtained from a Bayesian network meta-analysis[4,5].

After loss of response, a loop of subsequent AT in perpetuity was applied. Total costs included drugs acquisition (authorized posology and dose intensifications when required, both based on the European Public Assessment Reports [EPARs])[6,7], IV administration (€77.16)[8], adverse events (€5,964.49 per serious infection)[8] and management of UC (cost): active UC (€4,471)[9,10], response without remission (€1,801)[9,10], remission (€762)[9,10], surgery (€4,448)[9], post-surgery without complications (€1,101)[9,10] and with complications (€6,428)[9,10]. The annual surgery rate in UC was 0.34%[10] (urgent: 38.2%; elective: 61.8%)[11], early and late complications occurred in 18.4%[12] and 46% (for 6.5 years)[13].

Utility values were 0.78 for active UC[1,2], 0.82 for response without remission[1,2], 0.89 for remission[1,2], 0.74 for surgery[14], 0.90 for post-surgery without complications[14] and 0.71 with complications[14].

Concomitant treatment with azathioprine (35% patients, biologics only), mercaptopurine (15% patients, biologics only), mesalazine (40% patients), prednisone (20% patients) and beclomethasone (10% patients) were considered[7,15].

In addition to age and sex-specific mortality in Spain[16], the excess mortality associated with colectomy was considered (relative risk: 1.3)[17]. A probabilistic sensitivity analysis (PSA) tested the results' robustness.

Results: Etrasimod yielded more QALYs compared to each biologic and produced cost-savings in all pairwise comparisons except for IV vedolizumab (Table 1).

Thus, etrasimod was cost-effective compared to IV vedolizumab (€16,212/QALY), and dominant compared with all remaining biologics. PSA revealed that etrasimod was dominant compared to each biologic (average of Monte Carlo simulations). Under a willingness to pay threshold of €27,000/QALY[18], etrasimod was cost-effective in a minimum of 58.5% simulations (vs. IV vedolizumab) and a maximum of 99.9% (vs. adalimumab), varying between pairwise comparisons of etrasimod vs. biologic.

	Etrasimod	Adalimumab	Adalimumab (biosimilar)	Ustekinumab	Vedolizumab (IV)	Vedolizumab (SC)
QALYs / Total costs [€]	18.407 / €596,254	18.394 / €613,994	18,394 / €610,273	18.388 / €632,484	18.240 / €593,559	18.236 / €619,896
Δ QALYs / Δ Costs [€] (Etrasimod vs. comparator)	–	+0.013 / -€17,740	+0.013 / -€14,019	+0.018 / -€36,230	+0.166 / +€2,695	+0.171 / -€23,642
ICER [€/QALY] (Etrasimod vs. comparator)	–	Etrasimod dominant [A]	Etrasimod dominant [A]	Etrasimod dominant [A]	€16,212 [B]	Etrasimod dominant [A]
PSA mean Δ QALYs / Δ Costs [€] (Etrasimod vs. comparator)		+0.061 / -€23,675	+0.061 / -€17,493	+0.075 / -€48,952	+0.154 / -€22,292	+0.140 / -€70,902
PSA mean ICER [€/QALY] (Etrasimod vs. comparator)	–	Etrasimod dominant [A]	Etrasimod dominant [A]	Etrasimod dominant [A]	Etrasimod dominant [A]	Etrasimod dominant [A]
Proportion of simulations cost-effective (WTP = €27,000/QALY)[18] (Etrasimod vs. comparator)	–	96.7%	88.9%	99.9%	58.5%	88.9%

AT: Advanced Therapy; ICER: Incremental Cost-Effectiveness Ratio; IV: Intravenous; PSA: Probabilistic Sensitivity Analysis; QALY: Quality-Adjusted Life Year; SC: Subcutaneous; WTP: Willingness-To-Pay; UC: Ulcerative Colitis.

[A] Etrasimod represents a dominant alternative (yields more QALYs and cost-savings).

[B] Etrasimod represents a cost-effective alternative when considering a WTP threshold of €27,000/QALY[18].

Table 1. Results of the cost-effectiveness analysis in AT-experienced patients with moderate-to-severe active UC.

Conclusion: Etrasimod represents a cost-effective alternative for AT-experienced patients with moderate-to-severe active UC compared with biologics available in Spain.

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PP0551

EXPLORING THE CHARACTERISTICS OF ULCERATIVE COLITIS (UC) PATIENTS WITH MAYO ENDOSCOPIC SUBSCORE (MES) > 1 WHO HAVE BEEN TREATED WITH NON-TNF BIOLOGICS AND TARGETED ORAL THERAPIES

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Introduction: The Mayo Endoscopic Subscore (MES) is widely used in clinical trials and daily practice to assess disease activity and response to treatment in UC.

Aims & Methods: The objective of this study was to identify any nuances in patients with MES > 1 treated with non-TNF biologics and targeted oral therapies. The Ipsos UC Therapy Monitor, a multi-centre online medical chart review study of patients with UC, was conducted between July and September 2024 among gastroenterologists practicing across hospital and private practices in the UK, France, Germany, Italy, and Spain.
Physicians were screened for practice duration and patient volume. Charts of patients prescribed advanced therapy were included in the analysis.

Results: 205 sampled physicians collectively reported 820 UC patients. Of the reported patients with MES > 1, 62 were treated with non-TNF biologics and 62 were treated with novel oral therapies. Among the patients treated with non-TNF biologics, a higher proportion were female vs those receiving oral therapy (48% vs 31%); they were also older in age (mean years: 43 vs 38).
The patients treated with non-TNF biologics had a higher prevalence of comorbidities, including depression/anxiety (24% vs 19%), cardiovascular disease (8% vs 0%), and dyslipidemia/hyperlipidemia/hypercholesterolemia (11% vs 7%), compared to those treated with novel oral therapy.
Mean CRP levels were higher in the oral therapy group (10.2 mg/L vs 9.1 mg/L), while mean Mayo scores were slightly lower (4.7 vs 5.2). Fecal calprotectin levels were elevated in patients treated with non-TNF biologics vs oral therapy group (441 µg/g vs 503 µg/g).
When analysing reasons for prescribing selected advanced therapy between the two patient groups, a greater proportion of patients treated with non-TNF biologics had safety reasons specified more frequently by the sampled physicians.

% reported UC patients - cited reasons	Patients treated with non-TNF biologics	Patients treated with novel oral therapies
Low incidence of side effects	53%	13%
Good safety profile	61%	16%
Rapid symptomatic profile	19%	60%

Table 1: % of reported UC patients receiving advanced therapy for specific cited reasons.

Conclusion: This study highlights key differences in patient characteristics and treatment considerations among UC patients with MES > 1 treated with non-TNF biologics compared to those treated with targeted oral therapies. Patients on non-TNF biologics tend to be older, more often female, and have a higher prevalence of comorbidities. Physicians prescribing non-TNF biologics prioritize safety, while those prescribing oral therapies focus on rapid symptom control.
These findings highlight the importance of considering patient characteristics and treatment goals when selecting advanced therapies for UC patients with MES > 1. Further investigation using comparator cohort is warranted.

Disclosure: Nothing to disclose.

PP0552

EXAMINING THE CHARACTERISTICS OF ULCERATIVE COLITIS (UC) PATIENTS WHO HAVE BEEN INITIATED WITH ADVANCED THERAPY (AT) WITHIN 12 MONTHS OF DIAGNOSIS

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Introduction: Recent studies suggest that the early use of advanced therapies in the treatment of CD and UC may lead to better patient outcomes and potentially slow disease progression.

Aims & Methods: The objective of this study was to examine the differences in UC patient characteristics between those who initiated advanced therapy within 12 months of diagnosis versus those who started after 12 months. A multi-centre online medical chart review study of patients with UC was conducted between Jul – Sep 2024 among UK, FR, DE, IT & ES gastroenterologists practicing across hospital and private practices.
Physicians were screened for practice duration and patient volume. Charts of patients prescribed with advanced therapy were included in the analysis.

Results: 205 sampled physicians collectively reported 820 UC patients. Of the reported patients, 451 were diagnosed and initiated their first advanced therapy within 12 months (Early AT treated), while 369 patients started their first advanced therapy after 12 months (Delay AT treated).
Early AT treated patients were younger vs delay AT treated (mean ages 38 vs 42 years old). Looking at reported UC patient employment status, early AT treated patients are less likely to be in full time employment (56% vs 63%) vs delay AT treated patients. They are more likely to be diagnosed with severe disease (38% vs 24%)
Mean UC disease activity MAYO score were higher in the early AT treated patients vs delay AT treated (3.2 vs 2.6). Mean CRP and Fecal calprotectin levels were also higher in the early AT treated patients (CRP: 7.1 mg/L vs 4.7 mg/L) and (Fecal calprotectin: 244.4 vs 220.6)

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When analyzing reasons for prescribing selected advanced therapy between the two patient groups, a greater proportion of early AT treated patients had the following reasons specified more frequently by the sampled physicians, vs delay AT treated patients.

% reported UC patients - cited reasons	Early AT treated patients	Delay AT treated patients
Rapid symptomatic control	42%	36%
Biomarker normalisation (fecal calprotectin/ CRP)	31%	24%
To avoid surgery	30%	24%

Table 1: % of reported UC patients receiving advanced therapy for specific cited reasons.

Conclusion: Comparisons in this study cohort highlight UC patients who started advanced therapy within 12 months of diagnosis were characterized by younger age, more severe disease at diagnosis and experienced higher inflammatory markers compared to those delay initiating advanced therapy patients. These findings suggest early AT treated patients indicate a more severe and complicated disease burden, potentially benefitting from earlier intervention with advanced therapies. Further investigation using comparator cohort is warranted.

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Disclosure: Nothing to disclose.

PP0553

A SINGLE CENTRE COMPARISON OF DISEASE CHARACTERISTICS AND TREATMENT OUTCOMES BETWEEN BIOLOGIC NAÏVE AND EXPERIENCED PATIENTS WITH SEVERE ULCERATIVE COLITIS (UC)

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Introduction: Hospitalised patients with UC are likely to be on biologics or small molecule therapy (BST). The diversity of treatment options may contribute to delays in identifying medically refractory disease. Additionally, there may be scope for earlier introduction of advanced therapies.

Aims & Methods: We aimed to compare outcomes between patients previously treated with biologics to those who were BST naive. We conducted a retrospective analysis of patients aged >18 years, admitted between December 2022 and August 2023, with left-sided or extensive colitis. Inclusion criteria were intravenous steroid treatment, length of stay >72 hours, elevated CRP, and stool frequency >4/day. Proctitis and infectious colitis were excluded.

Results: Thirty-eight patients met inclusion criteria: 13 (34%) were BST experienced and 25 (66%) naïve. Of the latter, 10 (40%) were admitted within the first year of diagnosis, and 16 (64%) required rescue therapy. Additionally, left-sided colitis was more prevalent in the BST naïve group (p=0.005). There was one death in this cohort; a patient who suffered post-operative complications of stroke and sepsis after colectomy.The BST group comprised of 4 patients on infliximab (31%), 4 on vedolizumab (31%), 3 on adalimumab (23%), 1 on ustekinumab (8%), and 1 on tofacitinib (8%). Of these, 8 (62%) were on first-line agents, 3 (23%) on second-line, 1 on third-line, and 1 on a fourth-line agent at admission. Gastrointestinal infections were not over represented in this group (n=0). Surgery rates were significantly higher in the BST exposed group: 4 (31%) underwent surgery within a month of admission, compared to 1 (4%) in the BST naïve group (p=0.02). There was a greater proportion of patients on second or third-line BST v/s first line (3 of 4 patients) among the surgical group but serum CRP, albumin, Oxford criteria and UCEIS did not seem to predict risk of surgery.

Demographics	BST naive number (n)	BST naive percentage (%)	BST exposed number (n)	BST exposed percentage (%)	P value
Left sided colitis	22	88%	6	46%	0.005
Pancolitis	3	12%	7	54%	
<1 year from diagnosis to admission	10	40%	2	15%	0.121
>1 year from diagnosis to admission	15	60%	11	85%	
Stool frequency 4-6	5	20%	0	0%	
Stool frequency >6	20	80%	13	100%	
CRP <30	11	44%	8	62%	0.305
CRP >30	14	56%	5	38%	
Surgery within 1 month	1	4%	4	31%	0.021

Conclusion: Our data suggests effective protocolised management of severe UC. Patients receive timely rescue therapy, and colectomy rates are consistent with best practice. On the other hand, the high proportion of BST naive patients admitted with severe symptoms within the first year of presentation could indicate scope for earlier introduction of outpatient BST. The preponderance of lower risk factors such as left sided disease and lower CRP levels in this group could support a shift in emphasis from first line parenteral biologics and its inherent pharmacy logistical limitations to oral small molecule therapy for earlier remission rates. The number of BST experienced patients on anti TNF (50%) may indicate potential for earlier out of class switch to newer small molecule drugs. The relative insensitivity of high-risk features such as CRP and UCEIS among the BST experienced patients highlights the need for specialist management of medically refractory disease. The higher prevalence of limited colitis in the BST naive may indicate a bias towards later introduction of advanced therapies as an outpatient. A larger data set will provide useful insights into the care of this increasingly prevalent cohort of UC patients.

Disclosure: Nothing to disclose.

PP0554

HIGH RATES OF ORAL 5-AMINOSALICYLIC ACID CO-PRESCRIPTION WITH ADVANCED THERAPY IN PATIENTS WITH ULCERATIVE COLITIS

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Introduction: 5-aminosalicylates (5-ASA) are the first-line treatment for inducing and maintaining remission in patients with mild to moderate ulcerative colitis (UC), with over 90% of patients receiving 5-ASA within a year of diagnosis and 60–87% continuing long-term therapy[1].

However, 5-ASAs appear to provide no incremental benefit once advanced therapy is initiated and withdrawal is safe and not associated with increased adverse events[1-3].

Moreover, the cost of 5-ASA therapy can be significant. Reducing unnecessary 5-ASA use could help lower the overall healthcare costs associated with inflammatory bowel disease[4].

Aims & Methods: Retrospective observational study of all consecutive adult patients with UC receiving advanced therapy from a single tertiary IBD centre. Data collection included patient and disease characteristics, 5-ASA and advanced therapy type, dose and duration and objective disease activity (most recent intestinal ultrasound, faecal calprotectin [FCP],

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endoscopy within 12 months). Biochemical remission was defined as FCP <150µg/g, sonographic remission as normal bowel wall thickness and no hyperaemia and endoscopic remission as Mayo endoscopic subscore of 0 or 1.

Results: 230 patients were included (42% female, median age 40 years, median disease duration 96 months). Concurrent oral 5-ASA therapy was used in 116 patients (50%) for a median duration of 42 months (IQR 11–59), including 19 overlapping months with the most recent advanced therapy. The majority (89%) remained on maximal induction doses and annual average costs were estimated at $3469.73 per patient. Concurrent oral 5-ASA and advanced therapy use was associated with higher topical 5-ASA use (p = 0.002) and shorter duration on current advanced therapy (p = 0.03) compared to those on an advanced therapy alone. Almost two-thirds of patients remaining on 5-ASAs were in biochemical and/or sonographic remission and a third were in endoscopic remission. 5-ASA co-prescription rates varied according to advanced therapy type but no significant differences in clinical, biochemical, endoscopic or histologic activity were observed between groups (Table 1).

Characteristic	Advanced Therapy (n = 114)	Advanced therapy + 5-ASA (n = 116)	p value
Age, median (IQR)	40 (27–53)	39 (30–55)	0.20
Duration of current advanced therapy (months)	20 (11–42)	18 (8–30)	0.03
Advanced therapy type			
Infliximab IV	16 (14.0%)	25 (21.6%)	0.17
Infliximab SC	16 (14.0%)	5 (4.3%)	0.011
Adalimumab	2 (1.7%)	0 (0%)	0.24
Golimumab	1 (0.8%)	2 (1.7%)	>0.99
Vedolizumab IV	38 (33.3%)	37 (31.9%)	>0.99
Vedolizumab SC	8 (7.0%)	8 (6.9%)	>0.99
Upadacitinib	17 (14.9%)	17 (14.7%)	>0.99
Ustekinumab	12 (10.5%)	12 (10.3%)	>0.99
Tofacitinib	3 (2.6%)	8 (6.9%)	0.21
Ozanimod	0 (0%)	1 (0.8%)	>0.99
Estrasimod	0 (0%)	1 (0.8%)	>0.99
Infliximab + Stelara	1 (0.8%)	0 (0%)	>0.99
Topical 5-ASA use	12 (10.5%)	31 (26.7%)	0.002
Clinical remission	97 (85.1%)	87 (75.0%)	0.06
FCP < 150 µg/g	75 (65.8%)	78 (67.2%)	0.88
IUS remission	19/24 (79.2%)	22/35 (62.9%)	0.25
Endoscopic remission	51 (44.7%)	38 (33.0%)	0.07
Histological remission	51 (44.7%)	41 (35.6%)	0.17

Table 1. Patient, disease and medication characteristics according to 5-ASA co-prescription with advanced therapies. Values presented as median (IQR) or number (%), unless specified.

Conclusion: Co-prescription of 5-ASAs and advanced therapies is common despite high rates of objective remission. Identifying patients suitable for 5-ASA withdrawal may allow substantial reductions in healthcare expenditure.

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PP0555

DOES THE CURRENT THERAPEUTIC PARADIGM CHANGE THE OUTCOME IN PATIENTS AT THE FIRST EPISODE OF ACUTE SEVERE ULCERATIVE COLITIS REFRACTORY TO CORTICOSTEROIDS?

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Introduction: Acute severe ulcerative colitis (ASUC) is a potentially life-threatening emergency that requires timely management. Intravenous corticosteroids are the first line therapy, but around 30% of patients fail to respond. Rescue therapy is effective in reducing short-term colectomy rates but its impact on disease course in the medium/long-term is not well characterized.

Aims & Methods: We aimed to assess if non-response to corticoids (and subsequent need for rescue therapy) in patients admitted with a first episode of ASUC is associated with worse disease-related outcomes in the long-term.

We conducted a retrospective study of all patients admitted with ASUC in the Gastroenterology department of a Portuguese tertiary center between 2018-2024. ASUC diagnosis was established according to Truelove and Witts' criteria. Response to intravenous corticoids was defined as discharge without need for rescue therapy and/or colectomy. Patients with a previous admission for ASUC before the study period were excluded.

Patient files were reviewed and demographic, clinical, laboratory and endoscopic data was collected regarding the admission for ASUC and subsequent follow-up. Considered disease-related outcomes were need for switch of maintenance therapy, need for readmission and need for colectomy. Subanalyses were performed according to response to corticoid therapy. Statistical analysis was done using Stata version 12.1.

Results: A total of 35 patients hospitalized with a first episode of ASUC were included, 51.4% (n=18) male, median age at admission was 38 years [IQR 27-51]. Roughly half the patients had pancolitis at diagnosis (51.4%, n=18). Median disease duration was 9 months [IQR 0-26] and 22.9% (n=8) had previous exposure to biologic therapy. In 34.3% (n=12), ASUC was the initial presentation of UC. Median follow-up was 30 months [IQR 12-51].

Response to corticosteroid therapy was observed in 57.1% (20/35). All the steroid-refractory patients (15/35, 42,9%) responded to rescue therapy (Infliximab n=11; Tacrolimus n=3; Tofacitinib n=1) and avoided colectomy during the hospitalization.

Baseline characteristics were identical between steroid-responsive and steroid-refractory patients, except for a lower median age in corticoid refractory patients (29 years vs 48 years, p=0.02).

Survival analysis was performed and no differences between steroid-responsive and steroid-refractory patients were observed in need for switch of maintenance therapy [during first year (p=0,28) and during complete follow-up (p=0,54)] , need for readmission [during first year (p=0,55) and during complete follow-up (p=0,54)] or need for colectomy [during first year (p=0,09) and during complete follow-up (p=0,79)].

Conclusion: In our cohort of patients with a first episode of ASUC, with the current therapeutic paradigm, patients with steroid refractoriness had similar outcomes as corticosteroid responders. These results suggest that effective and timely therapy can change the natural history of ASUC.

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PP0556

TARGETED DELIVERY OF RNA-BASED THERAPEUTICS TO THE GUT TO MODULATE THE JAK-STAT PATHWAY IN CROHN'S DISEASE

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Introduction: Crohn's disease (CD) is a human immune-mediated inflammatory disease (IMID) of unknown etiology that affects the gastrointestinal (GI) tract. CD is now a global health issue due to its lifelong nature and significant burden on patients and public health [1].
Phase III clinical trials have shown that treatment with oral Janus Kinase (JAK) inhibitors can alleviate disease in patients with CD. However, the United States (US) Food and Drug Administration (FDA) has issued warnings about side effects for this class of oral drugs which target JAKs systemically in the human body. Therefore, new targeted delivery approaches are needed to block JAK activity selectively in the GI tract. The JAK1/2-STAT1 intracellular signaling pathway is activated by the Th1 cytokine IFN-γ alone or in synergy with another Th1 cytokine, TNF-α, and SOSC1 is an endogenous negative regulator of JAK1/2. This pathway activates the expression of inflammatory genes in multiple cell types typically associated with active or inflamed CD [2].
RNA interference (RNAi) mediated gene silencing using small interfering RNA (siRNA) targeting JAK1/2-STAT1 and SOCS1 overexpression using exogenous SOCS1 mRNA are promising methods to reduce the JAK-STAT pathway activation.
Aims & Methods: The GENEGUT project aims to develop oral RNA-based therapies for ileal CD. In this study, we have developed new approaches to deliver siRNAs targeting JAK1, JAK2, and STAT1 into relevant cell models, including human intestinal epithelial cell (IEC) lines. Epithelial cell models that replicate the inflammatory microenvironment in CD were used by exposing IECs to inflammatory cytokines IFN- γ/TNF- α, leading to increased cytokine production and cell death.
Results: Selected siRNAs effectively blocked JAK1, JAK2 and STAT1 expression and activity, preventing the downstream activation of inflammatory and cell death pathways. SOCS1 overexpression also suppressed JAK1/2-STAT1 pathway activation and enhanced cell survival.
Conclusion: Ongoing work is focused on optimizing the delivery of siRNA and mRNA into differentiated epithelial monolayers, co-cultures and patient-derived organoids to advance translational applications.
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PP0557

WEEK 10 EFFECTIVENESS AND SAFETY OF FILGOTINIB IN PATIENTS WITH ULCERATIVE COLITIS: AN INTERIM ANALYSIS OF THE PROSPECTIVE, REAL-WORLD EVIDENCE GALOCEAN STUDY

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Introduction: Filgotinib (FIL) is an oral, once-daily Janus kinase 1 preferential inhibitor approved for the treatment of adults with moderately to severely active ulcerative colitis (UC), based on the results of the phase 2b/3 SELECTION trial[1]; however, real-world data are limited.
Aims & Methods: Here, we report baseline (BL) characteristics of enrolled patients with UC and early effectiveness and safety data for those who reached week 10 (W10) of FIL treatment in GALOCEAN (NCT05817942), an ongoing, European, multi-site, prospective, observational study to investigate the real-world effectiveness and safety of FIL. Data collected up to 15 Jan 2025 were included. BL characteristics were reported for the full analysis set and effectiveness outcomes were reported for the response

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evaluable set (RES). At W10, mean changes from BL (CFBs) in the partial Mayo Clinic Score (pMCS) and the 2-item patient-reported outcome (PRO2) score, and the proportions of patients in pMCS or PRO2 remission were assessed. Mean CFBs and the proportion of patients reaching the minimal clinically important difference at W10 in patient-reported outcomes (Short Inflammatory Bowel Disease Questionnaire, Urgency Numeric Rating Scale and Functional Assessment of Chronic Illness Therapy-Fatigue scores) were also calculated. Safety was assessed by documenting adverse events.

Results: As of 15 Jan 2025, 353 patients starting FIL ≤200mg were enrolled and had BL data available (female: 48.7%; age, median [min–max]: 36.0 [18–86] years; time since UC diagnosis, median [min–max]: 6.3 [0–35] years). The mean (SD) duration of follow-up was 23.9 (17.2) weeks. At BL, pMCS-defined UC severity rates were inactive (pMCS 0–1; 5.8% [n=20]), or mildly (pMCS 2–4; 28.3% [n=97]), moderately (pMCS 5–6; 35.9% [n=123]) or severely (pMCS 7–9; 30% [n=103]) active; 10 patients did not have BL pMCS. Most patients had left-sided colitis (38.2% [n=135]) or pancolitis (39.4% [n=139]). Overall, 21.2% (n=75) of patients were naive to conventional therapies (excluding corticosteroids [CS]), and 26.9% (n=95), 27.8% (n=98) or 45.3% (n=160) had previously received 0, 1 or ≥2 advanced therapies, respectively. At BL, FIL was used without concomitant therapy in 55.8% (n=197) of patients; CS were used as concomitant therapy by 26.3% (n=93) of patients. The most reported cardiovascular risk factor was current or former smoker (31.7% [n=112]). Overall, 75.9% (n=268) of patients had post-BL effectiveness data (RES). At W10, patients had improved clinical symptoms and health-related quality of life (HRQoL) vs BL (Table). A total of 123 (34.8%) patients discontinued FIL, mostly due to lack of effectiveness. No new safety signals were observed.

Conclusion: In this interim analysis of GALOCEAN, patients with UC had improved clinical outcomes (vs BL in symptoms and HRQoL) at W10 of FIL treatment in real-world clinical practice. FIL was generally well tolerated. This is consistent with previous data.[2–4] GALOCEAN is ongoing, and results with higher patient numbers with longer exposure to FIL will be analysed to further characterize the real-world effectiveness and safety of FIL for UC.

	BL			W10					
Outcome measure	**Mean score (SD)**	**Patients with moderately active disease, % (n/N)**	**Patients with severely active disease, % (n/N)**	**Mean score (SD)**	**Mean CFB (SD)**	**Patients with moderately active disease, % (n/N)**	**Patients with severely active disease, % (n/N)**	**Patients in remission,[a] % (n/N)**	**Patients with MCID[b] vs BL, % (n/N)**
pMCS	5.2 (2.0) n=262	38.5 (101/262)	27.9 (73/262)	2.4 (2.5) n=232	−2.9 (2.7) n=228	10.3 (24/232)	9.9 (23/232)	41.5 (98/236)	–
PRO2 score	3.2 (1.6) n=265	59.6 (158/265)	22.6 (60/265)	1.4 (1.7) n=248	−1.8 (1.9) n=247	26.6 (66/248)	8.1 (20/248)	43.1 (107/248)	–
SIBDQ score	39.4 (11.7) n=249	–	–	49.0 (12.9) n=222	9.3 (11.3) n=211	–	–	–	54.0 (114/211)
Urgency NRS score	4.3 (2.7) n=237	–	–	2.9 (2.8) n=212	−1.5 (3.0) n=190	–	–	–	33.7 (64/190)
FACIT-Fatigue score	28.7 (11.6) n=249	–	–	34.0 (11.8) n=222	5.3 (9.6) n=211	–	–	–	55.5 (117/211)

Data are reported for patients with available data. Analysis sets are defined as follows: the RES comprised patients who had BL and valid effectiveness data for ≥1 time point after FIL initiation, and the FAS comprised all patients with available BL data.

[a]Definitions of remission were as follows: pMCS remission (RB ≤1, SF ≤1, PGA ≤1, pMCS ≤1); PRO2 remission (SF=0, RB=0).

[b]Definitions of MCIDs were as follows: SIBDQ MCID (increase from BL ≥9); Urgency NRS MCID (decrease from BL ≥3); FACIT-Fatigue MCID (increase from BL ≥4).

BL, baseline; CFB, change from baseline; FACIT-Fatigue, Functional Assessment of Chronic Illness Therapy-Fatigue; FIL, filgotinib; MCID, minimal clinically important difference; NRS, Numeric Rating Scale; PGA, Patient Global Assessment; pMCS, partial Mayo Clinic Score; PRO2, 2-item patient-reported outcome; RB, rectal bleeding subscore; RES, response evaluable set; SD, standard deviation; SF, stool frequency subscore; SIBDQ, Short Inflammatory Bowel Disease Questionnaire; W10, week 10.

Table. Effectiveness outcomes at BL and W10 after FIL initiation in the RES.

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PP0558

MULTIDISCIPLINARY ANALYSIS TO PREDICT TOFACITINIB TREATMENT RESPONSE IN ULCERATIVE COLITIS – A PROSPECTIVE TRANSLATIONAL STUDY

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Introduction: Tofacitinib (TFB) is a JAK 1/3 inhibitor small molecule drug used in ulcerative colitis (UC). Prognostic markers of treatment response are lacking, however 39-54% of patients experience loss of response. Multidrug resistance protein (MDR) activity, and differences in hepatic metabolism via cytochrome P450 (CYP) isoenzymes may be associated with decreased intracellular bioavailability, while clinical factors can predict treatment response.
Aims & Methods: Our study aimed to analyse the predictive value of MDR activity, CYP genotype, and cytokine pattern beside clinical factors in a multidisciplinary approach. In this prospective cohort study adult UC patients were enrolled when introducing TFB treatment and followed until week 12. At baseline, blood samples were obtained to measure the activity of three ABC drug transporters (MDR1, MRP1, and BCRP) in leukocytes via immunocytochemistry and flow cytometry. Patients' tofacitinib-metabolizing capacity was characterized by CYP2C19, CYP3A5 and CYP3A4 genotypes. Clinical parameters were registered, while colonoscopy examinations were performed at baseline and at week 12. Mucosal sampling was also performed to measure mucosal cytokine levels (TNF-α, IL-6, IL-1b, IFN-γ, IL-4, IL-17A, PAI-1) via multiplex ELISA methods. Week 12 clinical and endoscopic remission/response, histological remission and treatment persistence were used as outcomes. We performed multivariable logistic regression models on these specific outcomes and data was handled with intention-to-treat analysis.
Results: In total, we included 63 patients (male/female ratio 24/39; mean age 38.7±11.7 years; mean Mayo score 8.9±2.4) in our study. In total, 7 patients (9.2%) achieved histological remission, while 20 patients (26.7%) achieved endoscopic remission, and 27 patients (42.9%) endoscopic response. Clinical remission was observed in 21 patients (33.3%), while clinical response was achieved by 30 patients (47.6%). In total, 34 patients (54.0%) were in corticosteroid-free clinical remission at week 12, and higher baseline pMayo score decreased (OR=0.64 [95% CI: 0.43 – 0.94]), while albumin increased (OR=1.33 [95% CI: 1.01 – 1.76]) the chance of achieving the outcome (AUC = 0.93).
We found that one-year treatment persistence was strongly influenced by week 12 endoscopic remission (β=3.79 [95% CI: 1.21 – 6.37]), but not by histological remission (β=1.58 [95% CI: -0.74 – 3.89]). Higher baseline $CD8^+$ MDR1 activity decreased the chance of clinical response (OR=0.91; p=0.048), while higher $CD8^+$ MRP1 decreased week 12 treatment persistence (p=0.04, r=0.32). CYP3A5 1/3 genotype increased the histological remission rates (OR=13.9 [95% CI: 1.04-186.2]). The levels of mucosal IL-6 (18.51 vs 25.72 pg/mg, p=0.049), IL-4 (94.41 vs 168.8 pg/mg, p=0.037) and PAI-1 (921 vs 1483 pg/mg, p=0.01) were significantly lower in the clinical/endoscopic responders at the baseline.
Conclusion: Based on our prospective cohort study, short-term histological remission is still a hard-to-reach endpoint in moderate-to-severe UC on TOFA. Measurement of baseline $CD8^+$ MRP1 and $CD8^+$ MDR1 activity, and mucosal levels of IL-1β, IL-6, IL-4, and PAI-1, and CYP3A5 genotype may help to increase treatment success.
Disclosure: Nothing to disclose.

PP0559

A COMPARATIVE STUDY OF DIFFERENT BIOLOGICAL DRUGS IN FIRST LINE THERAPY IN CROHN'S DISEASE: A MULTICENTRIC RETROSPECTIVE COMPARATIVE ANALYSIS

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Introduction: Biologic agents have emerged as the most important therapeutic weapon in the treatment of Crohn's disease, but there are few data in the literature comparing the efficacy and safety between different molecules. The aim of this study is to describe the population of patients who received different molecules as first-line therapy and their persistence on first-line therapy.

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Aims & Methods: 622 patients from 6 Gastroenterology departments from Bucharest, Iasi and Cluj treated between 2006 and 2024 were included in the study. 75 (14%) were excluded because they received only Mesalazine. 535 were included in the final analysis. The main outcomes were: clinical response at 12 weeks of treatment (complete response = clinical remission, partial response= decrease of CDAI ≥ 25% of the initial value without remission or primary non-response), persistence on therapy and serious adverse events leading to therapy discontinuation.

Results: The main characteristics of the studied patients are shown in the table 1. There are significant differences between the 3 groups of patients: regarding age, extension of the disease, disease phenotype, proportion of individuals with perianal fistulas, clinical severity of the disease, proportion of operated patients. Median therapy duration was 48 months (3÷330). 45% of patients received Adalimumab, 40% were treated with Infliximab, and in 13% Ustekinumab was given. The 3 biologics studied (Adalimumab/Infliximab/Ustekinumab) had a similar rate of complete response 80%, the non-response rate was also similar: 6.5% but persistence on therapy at 2 years was better for Ustekinumab (64%) compared to Adalimumab and Infliximab (45%) (p=0.03). Serious adverse events leading to therapy discontinuation were more frequent in those treated with Infliximab (10%) compared to Adalimumab (3%) and Ustekinumab (6%) (p=NS).

	Total	UST group	IFX group	ADA group	p-value for comparison between UST/IFX/ADA
N (%)	535 (100)	68 (12.6)	216 (40.1)	242 (44.9)	
Median age (min-max)	33 (11-69)	43 (20-57)	32 (11-61)	33 (13-69)	<0.001
Age at diagnosis (A1/A2/A3) N(%)	A1: 14 (3) A2: 305 (65.9) A3: 144 (31.1)	A1: 0 A2: 26 (50) A3: 26 (50)	A1: 10 (5.1) A2: 136 (68.7) A3: 52 (26.3)	A1: 4 (1.9) A2: 138 (66.3) A3: 66 (31.7)	0.002
Extension (L1/L2/L3) N(%)	L1 : 154 (28.6) L2: 154 (28.6) L3: 227 (42.1)	L1: 30 (44.1) L2: 14 (20.6) L3: 24 (35.3)	L1: 34 (15.7) L2: 86 (39.8) L3: 96 (44.4)	L1: 88 (36.4) L2: 54 (22.3) L3: 100 (41.3)	0.001
Behaviour (B1/B2/B3/p) N(%)	B1: 283 (52.9) B2: 164 (30.4) B3: 88(16.4) p: 114 (21.3)	B1: 54 (79.4) B2: 12 (17.6) B3: 0 p: 0	B1: 102 (47.2) B2: 54 (25) B3: 64 (29.6) p: 80(37)	B1: 122 (50.4) B2: 96 (39.7) B3: 22 (9.1) p: 32 (13.2)	<0.001
Surgery	170 (31.9)	10 (15.2)	72 (33.3)	88 (36.4)	0.004
Median CDAI (min-max)	245 (6-557)	278.5 (43- 470)	230 (32-557)	245 (6-534)	0.021
Median CRP value at initiation (min-max)	11.2 (0.15-270)	8 (0.4-54)	21 (0.7-76)	9.3 (0.15-270)	<0.001
Median time from diagnostic to start of biologic Months (min-max)	8 (1-444)	5.5 (1-38)	6 (1-180)	6 (1-202)	0.001

Conclusion: Patients that received anti-TNF alpha antibodies had more severe forms of Crohn's disease. All three biological agents used in real life for Crohn's disease therapy show similar efficacy, with a complete response rate of aprox. 80% and a nonresponse rate of 6.5%, but persistence on therapy at 2 years was significantly better for Ustekinumab (64%) compared to Adalimumab and Infliximab (45%) (p=0.03). Serious adverse events leading to therapy discontinuation were more frequent in patients treated with Infliximab (10%) compared to those who received Ustekinumab (6%) or Adalimumab (3%).

Disclosure: Nothing to disclose.

PP0560

CLINICAL OUTCOMES OF DOSE ESCALATION IN BIOLOGIC THERAPY FOR INFLAMMATORY BOWEL DISEASE: A RETROSPECTIVE ANALYSIS FROM A SINGLE TERTIARY CARE CENTER

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Introduction: Biologic therapies such as anti-TNF-alpha agents (infliximab, adalimumab), interleukin modulators, and gut-selective integrin antagonists like vedolizumab have significantly improved outcomes in inflammatory bowel disease (IBD), leading to enduring remission, better quality of life, and reduced surgical intervention and hospitalizations. However, many patients may eventually lose response, often due to factors like the formation of anti-drug antibodies, altered drug metabolism, or individual patient characteristics. Dose escalation—by increasing dosage or reducing intervals—is commonly used to counteract this loss, especially for anti-TNF therapies guided by therapeutic drug monitoring. Vedolizumab poses a greater challenge in this regard due to the lack of standardized monitoring protocols. We aimed to investigate the effectiveness of dose-escalated infliximab, adalimumab, and vedolizumab in IBD patients who worsened after initial response, evaluating outcomes such as treatment persistence and clinical remission, to guide more personalized, cost-effective treatment strategies.

Aims & Methods: We conducted a retrospective analysis of IBD patients who underwent Infliximab (IFX), vedolizumab, or adalimumab (ADA) therapy dose escalation between 2020-2024 at a tertiary IBD centre. Primary outcomes included clinical remission using disease activity scores (Simple Clinical Colitis Activity/SCCAI Index for UC, Harvey-Bradshaw Index/HBI for CD), biochemical markers (CRP, faecal calprotectin) and treatment persistence, evaluated at 3-12 months, 12 months, and annually up to 36 months post-escalation.

Results: A total of 155 patients (median age 31 years, 46% female) were included, of which 89 patients (57%) were prescribed dose-escalated IFX, 48 patients (31%) were prescribed dose-escalated ADA and 18 patients (12%) were prescribed dose-escalated Vedolizumab. Of those included, 118 patients (76%) had Crohn's disease, 36 patients (23%) had ulcerative colitis (UC) and 1 patient (1%) had indeterminate disease. Indications for dose escalation included secondary loss of response (SLOR) (105/155; 68%) and/or low drug levels (65/155; 42%). Dose-escalated IFX led to significant improvements in disease activity scores at 3 months (HBI: p=0.002; SCCAI: p=0.0003). Improvement in SCCAI was maintained at 12 months (p=0.00007), 2 years (p=0.0008) and 3 years (p=0.025). Improvement in HBI was maintained at 2 years (p=0.01) and 3 years (p=0.049). Significant improvement in CRP following dose-escalated IFX was reported at 3 months (p=0.011), 12 months (p=0.004), 2 years (p=0.006) and 3 years (p=0.03). Dose-escalated ADA led to a significant improvement in HBI at 3 months (p=0.023). This improvement was maintained at 12 months (p=0.005) and 2 years (p=0.008). Early improvement in faecal calprotectin was reported at 3 months (p=0.046). Significant initial improvement in SCCAI following dose-escalated Vedolizumab was reported at 3 months (p=0.043). This improvement was not maintained at any further follow-up points (p>0.05). Minor adverse events were reported in 3 patients receiving dose escalated ADA and in 6 patients receiving dose escalated IFX.

Conclusion: Our study demonstrates that biologic dose escalation is an effective and safe mechanism to recapture clinical remission, with high rates of treatment persistence.

Disclosure: Nothing to disclose.

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PP0561

EFFECTIVENESS OF UPADACITINIB IN PATIENTS WITH BASELINE ACTIVE EXTRAINTESTINAL MANIFESTATIONS AND INFLAMMATORY BOWEL DISEASE. A MULTICENTER RETROSPECTIVE STUDY

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Introduction: Upadacitinib has shown to be more effective than placebo for treating extraintestinal manifestations (EIM) in post-hoc analyses. However, real-world data on the effectiveness of upadacitinib in patients with inflammatory bowel disease (IBD) and EIMs are lacking.

Aims & Methods: A multicenter, retrospective and observational study was conducted at 19 Spanish hospitals. Patients with IBD and active EIMs when initiating upadacitinib were eligible.

The primary outcome was EIM improvement at weeks 8-12. The EIM assessment was based on the physician's subjective evaluation and categorized into clinical remission, partial response, nonresponse and worsening. EIM improvement was defined as either clinical remission or partial response.

Axial arthropathy, peripheral arthropathy, pyoderma gangrenosum, erythema nodosum, sweet syndrome, aphthous stomatitis, uveitis, episcleritis, primary sclerosing cholangitis and bronchiectasis were considered. In patient experiencing >1 EIM, all had to be resolved or improved to be considered. If not, the worst outcome was recorded. Patients' visits were scheduled at weeks 0, 8-12, and 6 months. Multivariate analysis was performed to find factors associated with EIM improvement at w8-12.

Results: A total of 160 patients were included, 120 patients with Crohn's Disease (75.6%), 90 females (56.3%) with a median age of 46.6 years (IQR 37.9-56.7). Upadacitinib was initiated in one hundred twenty-six (78.8%) patients due to EIMs and luminal activity and 35 patients (21.3%) exclusively due to EIMs. All patients were bio-experienced with 84 patients previously treated with ≥3 biologics (52.5%). Baseline EIMs assessment was for at least 1 EIM in 107 patients (66.9%), 42 patients presented 2 EIMs (26.3%) while 11 patients (6.9%) experienced ≥ 3 EIMs.

At w8-12, 99 patients (61.9%) achieved EIM improvement, 40 patients (25 %) clinical remission, 58 patients (36.9%) partial response, 58 patients (36.5 %) nonresponse and worsening in 3 patients. Patients initiating Upadacitinib due to exclusively active EIMs was associated with EIM improvement at weeks 8-12 (OR 2.99 95% CI 1.06-8.42; p=0.04). IBD clinical remission at w8-12 was achieved in 75 patients (61%).

Evaluation at 6 months was available for 103 patients, 73 patients (70.9%) presented EIM improvement, 41 patients (39.8%) clinical remission, 32 patients (31.1%) partial response, 20 patients (19.4%) nonresponse and 10 patients (9.7%) worsening. IBD clinical remission at 6 months was achieved in 51 patients (62.2%).

During the follow-up, Upadacitinib was ceased in 27 patients (16.9%). Adverse events were detected in 39 patients (24.3%) and 7 patients (4.4%) experienced worsening of previous inactive EIMs.

Conclusion: Upadacitinib was highly effective in the short term for treating bio-experienced patients with active EIMs, particularly for patients without luminal activity

Disclosure: Nothing to disclose.

PP0562

PHARMACOKINETIC MODELING AND DASHBOARD-GUIDED PROACTIVE INFLIXIMAB ADMINISTRATION IN CORTICOSTEROID-REFRACTORY IMMUNE-MEDIATED ENTEROCOLITIS

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Introduction: Systemic corticosteroids are the first-line treatment for moderate to severe immune checkpoint inhibitor-induced enterocolitis (ICIec). Infliximab (IFX) is recommended for steroid-refractory cases; however, up to 28% of patients do not respond. Repeated-dose IFX regimens may improve outcomes and reduce recurrence, but optimal management strategies remain undefined. Here, we describe a series of patients treated with dashboard-guided IFX for ICIec.

Aims & Methods: A retrospective, descriptive case series was conducted at the IBD clinic of Hospital Universitari Vall d'Hebron between January 2021 and February 2025. The study included adult patients with ICIec who received IFX guided by a pharmacokinetic (pK) model. Demographic and clinical data were collected and analyzed.

Results: The dashboard model was applied in five patients to provide additional guidance for treatment decisions. The median age was 60 years, and three patients were female. The median time between the first immunotherapy dose and development of ICIec was 14.8 weeks. Endoscopy

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revealed extensive disease in 60% of patients, and 40% also had upper GI involvement. At baseline, 40% had severe inflammation. Median IFX levels at weeks 1 and 2 following the first IFX dose were 23.5 µg/mL and 22.3 µg/mL, respectively. The median interval between the first and second IFX doses was 12 days. Clinical remission was achieved in 4 out of 5 patients (80%). With a median follow-up of 16.7 ± 8 weeks, 4 of 5 patients (80%) remained in clinical remission without ICIec recurrence; one patient died due to an infectious complication.

Conclusion: Dashboard-guided IFX dosing showed benefit in ICIec patients. Use of pharmacokinetic modeling may support more tailored IFX dosing strategies, potentially improving treatment outcomes in a subset of ICIec patients with severe or extensive disease. Prospective studies are needed to better define optimal treatment strategies for ICIec.

Disclosure: Nothing to disclose.

PP0563

COMPARISON OF MIRIKIZUMAB VS. USTEKINUMAB EFFICACY FOR INDUCTION OF REMISSION IN ULCERATIVE COLITIS – A 1:2 MATCHED ANALYSIS

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Introduction: Interleukin (IL) inhibitors are monoclonal antibodies that have been shown to be effective in inducing and maintaining remission in moderate to severe ulcerative colitis (UC).

Ustekinumab (UST) was approved in the EU for the treatment of UC in 2019 and binds specifically to the p40 protein subunit of IL-12 and IL-23. Mirikizumab (MIRI) was approved 4 years later and selectively binds to the p19 subunit of the IL-23 cytokine. By binding to the IL subunits, the interaction with the IL-23 receptor is inhibited.

In this study, we compared the response rate to induction therapy with UST and MIRI in patients with moderately to severe UC.

Aims & Methods: All patients with active UC who began induction therapy with UST or MIRI at our outpatient clinic between 2019 and 2024 were analyzed retrospectively with a follow-up period until April 2025. Patients who received the IL-23 inhibitor due to a disease other than UC or following a prednisolone therapy exceeding 20 mg and patients with pouchitis were excluded from the study.

We determined the response to IL-23 therapy 8 to 24 weeks after the start of treatment and documented this either as clinical remission (Mayo-Score ≤1), improvement (Mayo-Score reduction by at least 3 points), inadequate response (Mayo-Score reduction between 0 and 2) or no response (lack of benefit, switch to another medical class or colectomy).

We examined the levels of calprotectin in the stool 0 to 24 weeks before the start of IL-23 therapy up to 24 weeks after initial administration.

Our study included 40 patients treated with UST and 23 patients treated with MIRI, in the latter group 26% have previously been treated with UST. Patients in both groups were similar in age (mean age, 47.75 vs. 44.59 y; P 0.28), gender, disease duration (mean duration, 11 vs. 9 y; P 0.32), and extent of disease (pancolitis, 40% vs. 48%).

Results: After completion of induction, 25% in the UST group and 57% in the MIRI group reached clinical remission. MIRI after UST achieved clinical remission in 67%. Improvement was shown in 20% vs. 30%, UST vs. MIRI. For UST, 5% were non-responder. For MIRI, no non-responder was noted. In general, the difference between the pre- and post-induction Mayo-Score is significantly higher for MIRI, indicating a better response to the newly started drug (mean Mayo-Score difference, UST vs. MIRI, 2.95 vs 4.5; P 0.0003).

The mean calprotectin level before treatment was not significantly different (mean calprotectin, UST vs. MIRI, 1652 vs. 2452 mg/kg; P 0.17). The post-treatment calprotectin level was significantly higher for UST vs. MIRI (mean calprotectin, 1092 vs. 278 mg/kg; P 0.0083).

Conclusion: Mirikizumab is an effective induction treatment for patients with active ulcerative colitis, including those who previously did not respond adequately to Ustekinumab therapy.

Disclosure: Nothing to disclose.

PP0564

IMPACT OF BIOLOGICAL THERAPIES ON PATIENTS WITH PRIMARY SCLEROSING CHOLANGITIS ASSOCIATED WITH INFLAMMATORY BOWEL DISEASE: A MULTICENTER OBSERVATIONAL STUDY

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Introduction: Primary sclerosing cholangitis (PSC) is a chronic cholestatic liver disease characterized by progressive inflammation and fibrosis of the intra- and extrahepatic bile ducts. It is frequently associated with inflammatory bowel disease (IBD), particularly ulcerative colitis (UC). The coexistence of PSC and IBD poses clinical challenges due to the complex management of both intestinal and hepatic manifestations. While biological therapies have improved IBD outcomes, treatment options for PSC remain limited. Assessing the impact of IBD-targeted biological therapies on clinical outcomes in patients with concomitant PSC and IBD is therefore essential.

Aims & Methods: This observational multicenter retrospective study included 81 adult patients diagnosed with both PSC and IBD at three Italian tertiary referral centers between January 2004 and November 2024. Data collected included demographic and clinical characteristics, endoscopic findings, administered treatments, and clinical outcomes. Patients were grouped based on the treatment received: biological therapies (anti-TNF, anti-integrins, anti-ILs, JAK inhibitors) or conventional therapies (5-ASA, UDCA, corticosteroids, azathioprine). Time-to-event outcomes—including episodes of cholangitis, IBD-related surgery, colorectal cancer, cholangiocarcinoma, and liver transplantation—were analyzed using Kaplan-Meier estimates and compared by log-rank tests. Cox proportional hazards models were used to identify independent risk factors. Statistical significance was defined as $p < 0.05$.

Results: The cohort was predominantly male (70.4%), with median ages at PSC and IBD diagnoses of 35.0 ± 14.9 and 25.0 ± 13.7 years, respectively. UC was more prevalent than Crohn's disease (72.8% vs. 27.2%), with pancolitis being the predominant UC phenotype (79.7%).IBD exhibited a generally indolent phenotype across the cohort, with rare complications in Crohn's disease and frequent mucosal healing in UC (Mayo 0–1 in 64.8%). Biological therapies were administered to 49.4% of patients. Patients receiving biologics were younger at PSC diagnosis compared to those on conventional therapies (32.5 ± 14.85 vs. 39.6 ± 15.74 years; $p = 0.05$). Patients with Crohn's disease had a significantly higher risk of intestinal

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surgery compared to those with UC (HR 4.2; 95% CI 1.2–14.2; p = 0.037). No significant difference in intestinal surgery incidence was observed between patients treated with biological versus conventional therapies. Kaplan-Meier analysis revealed no significant difference in the incidence of cholangitis between patients receiving biological and conventional therapies (log-rank p = 0.36). Likewise, there were no significant differences between the two groups regarding colorectal cancer, cholangiocarcinoma, or liver transplantation.

Conclusion: Biological therapies appear effective in controlling intestinal inflammation in PSC-associated IBD, with a favorable safety profile regarding malignancy and transplantation risk. However, they do not significantly impact the incidence of cholangitis. These findings provide a foundation for future studies and may help guide evidence-based therapeutic strategies in this complex patient population.

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PP0565

LONG-TERM SAFETY AND EFFICACY OF THE HIGHLY SELECTIVE TYROSINE KINASE 2 INHIBITOR ZASOCITINIB FOR THE TREATMENT OF INFLAMMATORY BOWEL DISEASE: DESIGN OF A PHASE 2 OPEN-LABEL EXTENSION TRIAL

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Introduction: Tyrosine kinase 2 (TYK2) inhibitors block signaling downstream of multiple pro-inflammatory cytokines involved in immune-mediated inflammatory diseases, including inflammatory bowel disease, and may have an improved safety profile relative to Janus kinase inhibitors. Phase 2 studies of the highly selective and potent TYK2 inhibitor zasocitinib in Crohn's disease (CD; NCT06233461) and ulcerative colitis (UC; NCT06254950) are ongoing.[1,2]

Here, we describe the design of a combined open-label extension trial (NCT06764615), which will include patients from both parent trials who are responsive to treatment.

Aims & Methods: Patients are eligible for the phase 2 parent studies in CD and UC if they are aged 18–75 years with moderately to severely active disease, and have inadequate or loss of response to, or intolerance of, ≥ 1 standard or advanced therapy. Patients who have failed treatment with ≥ 3 classes of advanced therapy are excluded. Patients are randomized to one of three (CD) or two (UC) doses of zasocitinib, or placebo, for a 12-week double-blind induction phase followed by a 40-week sponsor-open (CD) or open-label (UC) maintenance period.

At the end of the parent trials, patients will be eligible for the combined open-label extension if they are clinical (CD) or symptomatic (UC) responders at Week 52, defined as: (1) CD: a ≥ 30% decrease in average daily very soft or liquid stools and/or a ≥ 30% decrease in average daily abdominal pain from baseline; and (2) UC: a ≥ 1-point and ≥ 30% decrease from baseline in partial modified Mayo score, and a ≥ 1-point decrease from baseline in rectal bleeding (RB) subscore or achievement of an absolute RB subscore of ≤ 1.

All enrolled patients will receive the same dose of open-label, oral, once-daily zasocitinib for a maximum of 108 weeks, with study visits every 12 weeks, followed by a 4-week safety follow-up period. Data analysis will be descriptive, with no formal statistical comparisons performed.

Results: The primary objective of the study is to determine long-term safety and tolerability, assessed through the incidence of treatment-emergent adverse events or adverse events of special interest. Secondary endpoints will assess the long-term efficacy of zasocitinib, including clinical, symptomatic, or endoscopic response or remission by visit or at Weeks 48 and 108, according to indication, and effects on patient-reported outcomes, including fatigue, bowel urgency, abdominal pain, and disease-specific health-related quality of life.

Conclusion: This open-label extension trial will provide data on the long-term safety of zasocitinib, as well as durability of treatment response among patients with CD or UC who experience treatment benefit in the phase 2 parent studies.

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PP0566

RANDOMIZED COMPARISON OF DAILY VERSUS ALTERNATE DAY ORAL IRON THERAPY IN PATIENTS WITH ULCERATIVE COLITIS WITH IRON DEFICIENCY ANEMIA

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Introduction: Alternate day oral iron therapy is better than daily oral iron therapy in Iron deficiency anemia (IDA) as it increases the fractional absorption of iron. In Ulcerative colitis (UC) with IDA increased fractional absorption of iron with alternate day iron may not only be better in improvement of anemia but also results in reduced unabsorbed iron delivery to distal bowel and may hence reduce adverse effects.

Aims & Methods: In a randomized, open label trial, from February 2023 to December 2024 patients of UC in remission (Mayo ≤2 OR partial Mayo ≤1 with fecal calprotectin <150mcg/gm for last 3 months) with IDA (ferritin <30mcg/l, transferrin saturation <20%, Hb 8-12 gm/dl) were randomized to every other day 132mg (EOD group) or daily 66mg (OD group) elemental oral iron given as ferrous fumarate for 8 weeks.

Primary outcome was the difference in mean change in Hemoglobin (Hb) at 8 weeks. Secondary outcomes were the difference in mean change in RBC indices, iron indices and adverse events at 8 weeks.

Results: 54 patients were randomized between both groups. One patient in EOD and 2 in OD group withdrew before treatment initiation. Eleven patients had disease flare on oral iron therapy (5 in EOD and 6 in OD group) p= 0.62 and two patients in EOD group had intolerance requiring oral iron stoppage. Thirty-eight patients completed treatment. The mean change in Hb at 8 weeks in EOD and OD groups was 2.08±1.0gm/dl and 2.09±1.11gm/dl respectively in per protocol analysis (p=0.797).

The mean change in serum ferritin, serum iron, transferrin saturation and total iron binding capacity in both the groups is not significant. Disease flare within 6 months prior to oral iron therapy correlated with increased incidence of flare on oral iron therapy. (P=0.04)

Conclusion: Alternate day oral iron is not better than daily oral iron in improving Hb levels, iron indices and with similar incidence of adverse events in patients of UC in remission. A recent flare has increased chances of disease flare with oral iron therapy

Disclosure: Nothing to disclose.

PP0567

EARLY RESPONSE TO VEDOLIZUMAB IN PATIENTS WITH CROHN'S DISEASE IN A PROSPECTIVE OBSERVATIONAL STUDY IN CANADA AND US: THE VOICE STUDY

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Introduction: Resolution of pain, fatigue and other symptoms are key treatment goals for many patients (pts) with Crohn's disease (CD);[1,2] however, time to symptomatic improvement in pts treated with vedolizumab (VDZ) has not been formally evaluated.

Aims & Methods: VOICE (NCT06249555) is a prospective, observational study in pts with CD aged ≥18 years initiating VDZ or anti-IL-23 antagonist treatment (tx) per product labelling as part of standard of care, with the main objective to explore the time course of response to VDZ as measured by the Patient-Reported Outcomes Measurement Information System (PROMIS) domain Short Form (SF) scores. Inclusion criteria include baseline PROMIS Pain Interference (PI)-SF T-score ≥55 (0.5 SD above general population). We enrolled pts via Takeda pt support program in Canada, with additional pts enrolled at US and Canada sites. Planned sample size is 300 pts (VDZ, 200; anti-IL-23 antagonist [reference group not for formal comparison], 100); planned follow-up is 52 weeks (wks), with PROMIS PI data reported weekly for the first 14 wks. The primary endpoint is time to the first meaningful clinical improvement in PROMIS PI-SF T-score, defined as ≥2-point decrease in PROMIS PI-SF T-score from baseline. We also collect data on improvement in 7 PROMIS domains with clinical improvement defined as 5-point (0.5 SD)[3] change in T-score from baseline, Short Inflammatory Bowel Disease Questionnaire (SIBDQ; including SIBDQ improvement [≥9 point increase from baseline] and SIBDQ remission [score of ≥60]) and safety. Here, we report the results from pts with ≥1 VDZ dose and ≥1 post-baseline PI assessment included in the interim analysis (IA; prespecified to occur after 50 pts from VDZ group completed or discontinued before the wk 14 assessment to provide an early evaluation of tx response and safety).

PROMIS domain T-score, mean±SD	Baseline (n=64)	Week 14 (n=49)	Change from baseline (n=49)
Pain Interference	61.7±5.2	55.0±10.2	–6.5±10.0
Fatigue	62.0±6.4	53.9±11.5	–8.4±10.6
Anxiety	60.2±7.4	54.7±10.2	–5.5±8.6
Depression	56.4±8.6	52.9±10.9	–3.9±8.6
Physical Function	42.0±6.2	45.3±9.0	3.5±7.8
Sleep Disturbance	56.3±6.9	54.5±9.9	–1.5±7.2
Ability to Participate in Social Roles and Activities	43.2±6.3	49.0±11.2	5.9±10.1

T-score of 50±10 reflects the mean±SD score of the general US population.[3] Higher T-scores in PROMIS Pain Interference, Fatigue, Anxiety, Depression and Sleep Disturbance domains reflect greater impairment (poorer quality of life); higher T-scores in PROMIS Physical Function and Ability to Participate in Social Roles and Activities domains reflect less impairment (better quality of life).[4]

Table. Summary of PROMIS domain T-scores.

Results: Overall, 64 pts with ≥1 VDZ dose and ≥1 post-baseline PROMIS PI assessment (mean±SD age, 43.1±17.0 years, 35.9% male, 92.2% White) were included in the IA. Disease duration (mean±SD) was 6.4±12.1 years; 84.4% of pts were anti-TNF-naïve. Data collection is ongoing for 57 pts (49

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with and 8 without completed wk 14 PROMIS PI questionnaire); 7 pts discontinued from the study. The median time to achieve the first ≥2 point decrease in PROMIS PI-SF T-score (in n=56 pts) was 12 days (95% CI, 6-13). Baseline PROMIS domain scores indicated notable impairment; by wk 14, improvements in mean PROMIS scores from baseline were observed in all 7 measured domains, with ≥5-point improvements in 4/7 domains (Table). At wk 14, 37.5% (24/64) and 7.8% (5/64) pts achieved SIBDQ improvement and remission, respectively. One pt had serious adverse events of appendicitis and diverticulitis, and another pt had a fatal event; none of these were related to VDZ tx.

Conclusion:

The IA results through wk 14 show an early improvement across multiple symptom domains and quality of life in pts with CD starting VDZ tx. Time to first ≥2-point decrease in PROMIS PI-SF T-score was short, with a benefit observed after 12 days from VDZ initiation. Study recruitment and data collection are ongoing.

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V. Jairath reports consulting/advisory board fees from AbbVie, Alimentiv, Anaptyis Bio, Arena Pharmaceuticals, Asahi Kasei Pharma, Asieris, Astra Zeneca, Avoro Capital, Bristol Myers Squibb, Celltrion, Eli Lilly, Endpoint Health, Enthera, Ensho, Ferring, Flagship Pioneering, Fresenius Kabi, Galapagos, Gilde Healthcare, GlaxoSmithKline, Genentech, Gilead, Innomar, JAMP, Janssen, Merck, Metacrine, Mylan, MRM Health, Pandion, Pendopharm, Pfizer, Protagonist, Prometheus Biosciences, Reistone Biopharma, Roche, Roivant, Sandoz, Second Genome, Sorriso, Spyre, Synedgen, Takeda, TD Securities, Teva, Topivert, Ventyx, Vividion; speaker's fees from, Abbvie, Ferring, Bristol Myers Squibb, Galapagos, Janssen Pfizer Shire, Takeda, Tillotts and Fresenius Kabi.

M. Long reports consulting fees from AbbVie, Alimentiv, Bristol Myers Squibb, Intercept, Janssen, Lilly, Pfizer, Prometheus, Roivant, Syneos Health and Takeda; and research support from Lilly, Pfizer and Takeda.

M.D. Kappelman reports being a consultant to Eli Lilly and Takeda.

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N. Narula has received advisory board/consulting honoraria from Abbvie, Amgen, Eli Lilly and Company, Ferring, Fresinius Kabi, Innomar Strategies, Iterative Health, Janssen, Novartis, Organon, Pfizer, Sandoz and Takeda.

PP0568

USTEKINUMAB TROUGH LEVELS ARE NOT ASSOCIATED WITH CLINICAL OUTCOMES IN CROHN'S DISEASE: PK-PD RESULTS OF THE RESCUE STUDY

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Introduction: In Crohn's disease (CD), an exposure-response relationship is suggested for ustekinumab, with higher serum levels associated with better outcomes. Pharmacokinetic data are scarce, with suggested target maintenance trough level associated with remission at 0.8-1.4 µg/mL.[1,2] The REScUE study (NCT04245215) investigated the effect of two different re-induction regimens with ustekinumab in patients with CD. No significant difference was observed on clinical, endoscopic and biological outcomes.[3] Here we present the pharmacokinetic and pharmacodynamics results of the REScUE study.

Aims & Methods: This Belgian multicentre prospective double-blind randomized placebo-controlled trial included adult patients with CD treated with ustekinumab who presented with secondary loss of response to ustekinumab after documented primary response. All patients received a single IV re-induction with ustekinumab (≈6mg/kg) and were then randomized 1:1 to blinded maintenance ustekinumab 90 mg SC Q4W (intervention) or 90 mg SC Q8W (control) for 48 weeks. PK samples were obtained at baseline, week 4-8-12-20-28-36-48.

Results: In total, 108 patients were included: 67% female, median [IQR] age 41 [32-54] year, median disease duration 14 [7-22] years. Prior anti-TNF exposure was reported in 92% of the patients. Baseline ustekinumab trough levels at moment of loss of response were similar in both groups and at the suggested target level of 0.8-1.4 µg/mL[1]: median (IQR) q4 week 1.39 (0.84-2.74) µg/mL and q8 week 1.31 (0.91-2.34) µg/mL (p>0.9).

Eight weeks after induction, there was a statistically significant difference in ustekinumab trough levels between both groups 10.8 (8.0-15.9) µg/mL versus 8.2 (4.9-11.1) µg/mL for q4weeks vs q8weeks respectively (p=0.002). After 48 weeks of treatment the statistically significant difference persisted with 6.0 (4.8-9.6) µg/mL versus 2.1 (1.6-2.9) µg/mL (p<0.001).

No significant association between baseline/week 8/week 48 ustekinumab trough levels and (*steroid free*) clinical remisison [PRO2 (AP ≤ 1 AND SF ≤ 3) and FCP <250µg/g and *no steroids for 90 days prior to week 48*] or endoscopic remission [SES-CD <3 at week 48] was seen.

However, there was a significant association between baseline/week 8/week 48 ustekinumab trough levels and biomarker remission [CRP <5 mg/L and FCP <250 µg/g at week 48]; OR (95% CI) 2.98 (1.85, 5.23), 1.12 (1.06, 1.19) and 1.35 (1.13, 1.66) (all p<0.01) respectively. As per quartile

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analysis of the ustekinumab trough levels at week 48, the lowest quartile (ustekinumab <0.9 µg/mL) had the lowest steroid free clinical remission in each treatment group.

Conclusion: In patients with CD experiencing secondary loss of response to ustekinumab, baseline ustekinumab levels are low. Dose optimization with a single IV re-induction followed by intensified maintenance dosing (90 mg SC q4W) leads to higher ustekinumab levels compared to IV re-induction followed by conventional maintenance dosing (90 mg SC q8W). No association was found between ustekinumab levels and different outcomes except for biomarker remission. This suggests that treatment with ustekinumab should be personalized on a patient level, guided by drug-clearance rather than guided by target trough levels.

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PP0569

THERAPEUTIC ROLE OF AZATHIOPRINE FOLLOWING INITIAL INTESTINAL RESECTION IN CROHN'S DISEASE: CHALLENGES IN A RESOURCE-LIMITED SETTING

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Introduction: Crohn's disease (CD) is a chronic condition characterized by a relapsing-remitting course, often requiring intestinal resection. Postoperative management primarily aims to maintain clinical remission. The advent of biologic therapies has undoubtedly improved disease prognosis following initial surgery. However, access to this therapeutic class remains limited in lower-middle income countries such as Tunisia, necessitating the use of alternative agents, notably azathioprine (AZA), which has been widely used for this indication, albeit with controversial results.

Aims & Methods: This study aimed to evaluate the effectiveness of AZA in maintaining remission after a first intestinal resection in patients with ileal or ileocolonic Crohn's disease, and to identify factors associated with treatment failure.

We conducted a retrospective, descriptive, and analytical study in the gastroenterology department of Sahloul University Hospital in Sousse (Tunisia) over a 4-year period (2020–2024). We included all patients with ileal or right ileocolonic CD who underwent intestinal resection and received AZA in the postoperative period.

Results: A total of 66 patients were included, with a mean age at diagnosis of 31.50 ± 12.24 years and a male-to-female ratio of 1.06. The median interval between diagnosis and surgery was 21.5 months. During follow-up, AZA failure occurred in 63.6% of patients (n=42), with a mean time from surgery to failure of 44.71 ± 37.74 months. In univariate analysis, ileocolonic disease location (p=0.042), the total number of flares (p=0.007), and the number of severe flares (p<0.001) prior to surgery were significantly associated with AZA failure. Failure rates were also significantly higher in patients who underwent emergency surgery (p=0.007). Inflammatory resection margins (p<0.001) and severe inflammatory activity (p=0.007) were also significantly associated with treatment failure. In multivariate analysis, three independent risk factors were identified: the number of severe flares before surgery (p=0.031), preoperative inflammatory syndrome (p=0.008), and histopathological evidence of disease at the resection margins (p=0.004).

Conclusion: Our findings suggest that thiopurines remain a valuable therapeutic option in the management of Crohn's disease, particularly in our setting, where access to alternative therapies is limited and associated with high costs.

Disclosure: Nothing to disclose.

PP0570

FIVE-YEAR PERSISTENCE OF VEDOLIZUMAB IN CROHN'S DISEASE: A REAL-WORLD MONOCENTRIC RETROSPECTIVE STUDY

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Introduction: Vedolizumab, a α4β7 integrin monoclonal antibody represents an effective and well-tolerated therapeutic option for patients with moderate to severe Crohn's disease (CD)[1]. Its use as a first-line biologic therapy is authorized in some European countries, particularly Belgium, providing an opportunity to assess its performance when introduced early in the treatment algorithm. Real-world data on long-term treatment persistence, particularly beyond three years, remain limited.

This retrospective monocentric study aims to assess Vedolizumab persistence over five years in patients with CD, as well as evaluating the ability to predict treatment persistence based on the CDST (Crohn's Disease Subtype Tool) score.

Aims & Methods: This is a retrospective monocentric study including 60 patients with CD, followed at a central university hospital in Brussels. All patients were treated with Vedolizumab administered intravenously, between 2015 and 2024. We evaluated treatment persistence defined as the continuation of Vedolizumab without interruption assessed at 6,12,24,36,and 60 months.

Patients who had not yet reached the pre-set follow-up time points at the time of inclusion were excluded from the analysis at that specific time point, without being considered as treatment discontinuation. Persistence was described cross-sectionally at each time point. A CDST score was retrospectively calculated for all patients based on the available data before treatment initiation.

Results: A total of 60 patients with CD treated with Vedolizumab were included. Follow-up duration ranged between 1 and 107 months. The cohort consisted of 52% males with a mean age of 45.4 ± 14.5 years. 58 patients had completed at least 6 months of follow-up. Treatment persistence with Vedolizumab was 93% at 6 months (54/58), 87% at 12 months (47/54), 76.5% at 24 months (39/51), 67.4% at 36 months (31/46), and 56.1% at 60 months (23/41). 62% of patients were bio-naïve. The mean CDST score for all patients was 20.6 ± 2.6. The distribution of patients according to their predicted response to Vedolizumab was as follows: low probability of response (CDST ≤ 13): 1 patient (2%); intermediate probability (13 < CDST ≤ 19): 12 patients (20%); high probability of response (CDST > 19): 47 patients (78%).

A comparative analysis of the CDST scores between patients who discontinued treatment and those with treatment persistence was performed at each follow-up time point. At 6 months, the mean CDST score was 20.47 in the treatment discontinuation group versus 21.2 in the persistence group (p = 0.713). At 12 months, the means were 21.6 and 21.0, respectively (p = 0.713). Similar trends were observed at 24 months (19.2 vs 20.9; p = 0.304), at 36 months (18.23 vs 20.9; p = 0.136), and at 60 months (19.7 vs 20.6;

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p = 0.523). These results suggest a trend toward higher CDST scores in patients with treatment persistence, although these differences did not reach statistical significance. Of the patients who completed five years of vedolizumab treatment, 9 out of 23 (39%) were biologic-naïve, while 14 out of 23 (61%) had prior exposure to biologic therapies.

Conclusion: This real-world retrospective study provides valuable insights into the long-term use of Vedolizumab in patients with CD, with a follow-up period extending up to five years. Treatment persistence over time was favorable in both biologic-naïve and exposed patients, particularly among those with a high CDST score.

The extended follow-up period reinforces the durability and safety of Vedolizumab as a long-term therapeutic option in clinical practice. Prospective studies are needed to confirm these findings and identify predictors of sustained response.

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Disclosure: Nothing to disclose.

PP0571

HO VS OXFORD VS CRP/ALBUMIN RATIO: WHICH TOOL WINS THE BATTLE AGAINST STEROID-REFRACTORY COLITIS?

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Introduction: Severe acute colitis (SAC) is a life-threatening complication of inflammatory bowel disease (IBD), particularly ulcerative colitis, This requires immediate treatment with intravenous corticosteroids. Patients who are likely to fail treatment with a first-line approach need to be identified early in order to expedite second-line therapy or surgery.

Aims & Methods: This study aimed to evaluate the prognostic performance of the Ho score(day1), the Oxford score (day 3), and the CRP/albumin ratio in predicting corticosteroid response in SAC.

This was a retrospective, single-center analytical study including all patients hospitalized for SAC in a north african gastroenterology department. Clinical and biological data were collected at admission and on the third day of corticosteroid therapy. The Ho score was based on three criteria: CRP > 100 mg/L, albumin < 30 g/L, and > 8 stools/day (positive if ≥ 2). The Oxford score was considered positive on day 3 in cases of > 8 stools/day or ≥ 3 stools with CRP > 45 mg/L. The CRP/albumin ratio was analyzed with a cut-off value of 3.0. Corticosteroid response was defined as clinical improvement without rescue therapy or surgery. Statistical analysis included Chi² test for associations, Student's t-test for comparing means, and calculation of diagnostic performance (sensitivity, specificity, positive and negative predictive values).

Results: Forty-three patients were included. The mean age was 35.5 years (range: 18–58), with a female-to-male ratio of 2:1.

SAC was inaugural in 25 patients (58.1%), while 18 patients (41.9%) had a known history of IBD.

The overall corticosteroid response rate was 44.2%.

The Ho score was positive in 26 patients, with a response rate of 15.4% versus 56.7% for negative scores (p = 0.030). Its diagnostic performance was modest: sensitivity 10.5%, specificity 54.2%, positive predictive value (PPV) 15.4%, and negative predictive value (NPV) 43.3%.

The Oxford score was positive in 31 patients, with a response rate of 29% compared to 83.3% in the negative group (p = 0.004). It showed higher sensitivity (47.4%) but lower specificity (8.3%), with a PPV of 29.0% and NPV of 16.7%.

The CRP/albumin ratio was significantly higher in non-responders (3.86 vs. 2.30; p = 0.038). When using a cut-off of 3.0, this marker had a sensitivity of 15.8%, specificity 29.2%, PPV 15.0%, and NPV 30.4%.

Conclusion: The Oxford score, despite low specificity, showed the best performance in identifying patients at risk of corticosteroid failure by day 3. The Ho score, though less sensitive, remains useful at admission for early risk stratification. The CRP/albumin ratio, while simple and biologically relevant, demonstrated limited predictive performance at the fixed threshold of 3.0. Further studies are needed to refine these tools or develop composite models for early decision-making in SAC.

Disclosure: Nothing to disclose.

PP0572

BIOLOGICS AND OTHER IMMUNOTHERAPIES AS RESCUE TREATMENTS FOR STEROID-REFRACTORY ACUTE SEVERE ULCERATIVE COLITIS: A NETWORK META-ANALYSIS

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Introduction: Acute severe ulcerative colitis (UC) refractory to intravenous corticosteroids is a serious form of severe UC and may result in colectomy. In this network meta-analysis, we aim to assess the efficacy of biologics and other immunotherapies as rescue treatments for patients with acute severe UC unresponsive to standard therapy, with a focus on their ability to prevent colectomy.

Aims & Methods: On March 1, 2025, a data search was conducted using PubMed, Scopus, and the Cochrane Library. We included controlled clinical trials that evaluated the use of biologics and other immunotherapies in patients with severe ulcerative colitis refractory to intravenous steroids. The primary outcomes were clinical response at week 1 and the colectomy rate at week 12. We used the odds ratio with the 95% confidence interval to estimate the effect size, and a random-effects model was applied. Data analysis was performed in RStudio using the *netmeta* package.

Results: Out of 2,186 articles screened, only 5 met our eligibility criteria, with a total of 734 patients. Four treatment strategies were included in the network meta-analysis: infliximab intensified dosing (10 mg/kg at week 0 and week 1 if responsive, or earlier if not), standard infliximab dosing (5 mg/kg at weeks 0, 2, and 6), accelerated infliximab dosing (5 mg/kg at weeks 0, 1, and 3), and cyclosporine.

For preventing colectomy, infliximab 10 mg/kg (OR 0.11; 95% CI: 0.02–0.79), infliximab 5 mg/kg standard dose (OR 0.21; 95% CI: 0.06–0.73), and cyclosporine (OR 0.20; 95% CI: 0.05–0.79) were all associated with a statistically significant reduction in colectomy risk compared to placebo at week 12. Accelerated infliximab dosing did not show a statistically significant difference.

For clinical response at week 1, infliximab 10 mg/kg, infliximab 5 mg/kg (standard dose), and cyclosporine had similar response rates, with no significant differences observed.

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Conclusion: Infliximab (both standard and intensified doses) and cyclosporine, as rescue therapies, had lower rates of colectomy at week 12 in patients with acute severe ulcerative colitis refractory to corticosteroids. Clinical response at week 1 was comparable across these therapies.

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Disclosure: Nothing to disclose.

PP0573

REAL WORD DATA: THE VEDOLIZUMAB EXPERIENCE IN A TERTIARY CENTRE

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Introduction: Vedolizumab, a $\alpha_4\beta_7$integrin antagonist has been established as an effective therapy in the treatment armamentarium of inflammatory bowel disease (IBD). Although previously recommended either as first line or second line treatment in patients with IBD; patient specific characteristics (like advanced age, history of malignancies, susceptibility to infections) and reimbursement, national and local policies may influence the drug choice.

Aims & Methods: The aim of the study is to evaluate real-world data on Vedolizumab use at a tertiary national centre. Data was retrospectively collected from patients' electronic medical records in our national healthcare database (Welsh Clinical Portal) for patients who started on Vedolizumab between 2018 and 2021, with a three-year persistence assessment. Patient characteristics and reasons for discontinuation were documented.

Results: A total of 247 patients were enrolled, mean age 52,19 years old (SD 17,6), 55.9% females. The diagnosis was split predominantly between Crohn's in 48.4% of patients and UC in 44.2%, (IBD-U 7.4%). Subcutaneous (SC) vedolizumab was the commonest formulation administered in 71.3% of patients, while 28.7% were given intravenous (IV).

Those receiving Vedolizumab as a first line treatment were found to be significantly older (p<0.001) and in instances where it was given as a second line option, anti-TNF therapy was the agent previously used in 100% of patients.

The overall persistence was 70% at 1 year and 46 % at 3 years (table) with secondary loss of response (30.8%) being the main reason to stop treatment, followed by primary non-responders (14.6%), side effects (7.3%) or other reasons (6.1%). We found in our cohort, a significant difference in persistence when used as an initial treatment, versus later in the pathway. There was a significant correlation between the response to Vedolizumab and factors such as prior biologic exposure, age, and treatment persistence. During the observation period, 5.8% of patients required a form of surgical treatment, 4.9% required a 'hot clinic' review and only 0.9% required admission. The cohort of patients had many significant co-morbidities, but to date, none had experienced a deterioration in their condition whilst on Vedolizumab therapy.

Year	N of new patients	1-year persistence	2-year persistence	3-year persistence
2018	41	71%	51%	44%
2019	37	76%	49%	27%
2020	81	68%	57%	46%

Conclusion: Our data contributes to the growing body of evidence that Vedolizumab is a safe and effective therapy, regardless of the delivery method. It is particularly beneficial for vulnerable patients, demonstrating good persistence of response beyond the initial induction of remission. Additionally, when used as a first-line treatment in the appropriate patient, it offers a better chance for a sustained drug response.

Disclosure: Nothing to disclose.

PP0574

THERAPEUTIC DRUG MONITORING FOR MANAGEMENT INFLAMMATORY BOWEL DISEASES IN CHILDREN TREATED WITH TNF-A ANTAGONISTS: EXPERIENCE OF THE REGIONAL CENTER

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Introduction: Every second child with inflammatory bowel disease (IBD) gets biological therapy with the development of 15% of secondary drug ineffectiveness that requires switching to other biological drugs (high costs, number of side effects of treatment). Therapeutic drug monitoring (TDM) (routine assessments of trough concentration with dose adjustments of tumor necrosis factor (TNF)α antagonists and anti-drug antibodies assessments) has been proposed to improve outcomes in IBD patients.

Aims & Methods: *The aim* of this study was to evaluate the effect of TDM in pediatric IBD biological therapy management and to determine indications for timely switching to the second-line biological therapy. *Materials and Methods:* 81 (35 with Crohn's disease (CD), 46 with ulcerative colitis (UC); 43 boys, 38 girls; 2-17 (11,23±4,58) years old)) of 146 IBD children (57 with CD, 89 with UC;) from Moscow Region IBD children center received a-TNFα biological therapy. TDM methods: assessments of infliximab or adalimumab concentration level to the 6 and 14th week of drug induction, a-infliximab or a-adalimumab antibodies appearance to the 20th week of drug induction (Quantitative rapid immunochromatographic tests). In case of clinical a-TNF therapy ineffectiveness, the level of biological a-TNFα drugs and antibodies was tested to correct the therapy with drug dose escalation or switching to out-of-class biological. Pediatric IBD clinical activity indexes (PUCAI, PCDAI), fecal calprotectin (FC), endoscopic scores (SES-CD, UCEIS), morphological activity IBD-DCA score were used to control IBD activity.

Results: 81\146 (55%) IBD children (35 with CD (L3, L3+L4, B2 stenosing phenotype, perianal CD), 46 with UC (E4 pancolitis); severe clinical\endoscopic\morphological IBD activity) had a-TNFα biological therapy ini-

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tiation: 55 infliximab, 26 adalimumab. The level of infliximab was higher (16.3–42 µg/ml) than 15 µg/ml (target level) in 48/55 (87%) children by the 6th week of induction; >5 µg/ml in 100% cases by the 14th week. They had significant decrease in clinical activity indexes and fecal calprotectin to the end of induction period - clinical remission (PUCAI, PCDAI less than 10 p.) in 27/31 (87%) patients (ANOVA Chi Sqr. p<0,0001). 7 (13%) children with low infliximab level (5 - below the detectable values <0.4 µg/ml) still had high clinical end fecal activity (PUCAI >35 b., PCDAI >25 b.), fecal calprotectin 457±103.2 µg/g so they required dose adjustment (drug escalation). The level of adalimumab was >7.5 µg/ml (9.1–38.9 µg/ml) - target level- by 14 weeks in 25/26 (96%) children. We have not detected the appearance of a-biologics antibodies at least to 20 weeks in any sample. TDM was helpful to manage the drug inefficiency.

We had primary inefficiency in 6/81 (7.4%), secondary inefficiency – 30/81 (37%) of our IBD children so it comes to the drug escalation in 19\81 (23.5%) cases, switching to 2nd line of biologics (2-to adalimumab, 6-to ustekinumab, 22-to vedolizumab) in 30/81 (37%) of children with significant clinical improvement after 3 months of therapy (ANOVA Chi Sqr. p<0,0032 by FC).

Conclusion: The target therapeutic concentrations of a-TNFα biologics, sufficient for mucosal healing, appear in 90% of children by the end of initiation period with significant decrease in clinical, endoscopic, and morphological activity scores without a-TNFα biologics antibodies appearance to 20 weeks. TDM helped to make the early dose biologics adjustment and switching to 2nd line of biologics to overcome inefficiency of biological therapy.

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PP0575

HIGHER EFFICACY MEDICATION IN BIO-EXPOSED ULCERATIVE COLITIS PATIENTS: TRIDENT-UC STUDY

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Introduction: Despite the progress in advanced therapies (AT) for ulcerative colitis (UC), selecting a second-line treatment after failure remains challenging. Recent guidelines recommend Ustekinumab (UST), Tofacitinib (TOFA), or Upadacitinib (UPA) as high-efficacy options after prior AT failure. However, no real-world studies have compared all three therapies.

Aims & Methods: Retrospective multicenter study (8 centers) of UC patients who failed ≥1 advanced therapy. Patients were stratified into UST, TOFA, or UPA groups and assessed at 16 and 52 weeks. The primary outcome was steroid-free clinical remission (SFCR) based on the Patient Reported Outcomes-2. Secondary outcomes included biochemical remission (fecal calprotectin ≤ 250 µg/g) and endoscopic remission (Mayo Endoscopic Subscore=0). Time to remission was assessed using Cox regression analysis. A subanalysis was performed including only second-line use after anti-TNF failure. Multivariate analysis was performed, adjusted for duration of disease, extent and number of prior therapies.

Results: We evaluated 259 cases [54.4% (n=141) female; diagnostic age 32.3 ± 15.1 years] who initiated UST (n=105; 40.5%), UPA (n=87; 33.6%) or TOFA (n=67; 25.9%). The groups were similar in disease duration (UST: 11.6 ± 8.0; UPA: 9.8 ± 6.4; TOFA: 10.4 ± 5.4 years) and 49.8% (n=129) had failed ≥2 previous AT

At week 16, UPA showed higher SFCR compared to UST (OR 2.1; p=0.014) and TOFA (OR 2.1; p=0.040), with no difference at week 52. Biochemical remission was higher with UPA at week 16 *vs* UST (OR 2.3; p=0.015) and TOFA (OR 2.1; p=0.039) and at week 52 *vs* UST (OR 4.7; p=0.010) and TOFA (OR 4.6; p=0.013). At week 16, endoscopic remission was higher with UST (OR 3.5; p=0.036) and UPA (OR 3.1; p=0.041) compared to TOFA; and at week 52, UPA was associated with higher endoscopic remission *vs* UST (OR 3.6; p=0.036).

In Cox regression, UPA achieved earlier biochemical remission than UST (HR 1.9; p<0.001) and TOFA (HR 1.8; p=0.011). JAK inhibitors led to earlier endoscopic remission than UST (*vs UPA*: HR 3.2; p<0.001; *vs TOFA*: HR 1.4; p=0.002).

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In the subanalysis, UPA as second-line therapy demonstrated higher SFCR than TOFA (OR 3.6; p=0.035) and higher endoscopic remission at week 52 than TOFA (OR 12.0; p=0.017), and trend-wise compared to UST (OR 4.5; p=0.090).

Conclusion: Even though all three medication demonstrated high efficacy in bio-exposed UC patients, UPA achieved better results in this real world study.

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PP0576

EARLY INTERVENTION PREVENTS DISEASE PROGRESSION IN ULCERATIVE COLITIS – REFRAMING THE WINDOW OF OPPORTUNITY IN UC

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Introduction: Ulcerative colitis (UC) has increasingly been recognized as a progressive disease. However, the benefit of starting early biologic therapy (BT) has not been established, contrary to Crohn's disease. We aimed to assess if early intervention improves UC outcomes, particularly in preventing disease progression.

Aims & Methods: We conducted a retrospective multicenter study (11 centers) including adult UC patients who initiated BT until 2019 (minimum follow-up [FU]: 5 years). Early intervention was defined as BT initiation within ≤12 months of diagnosis. The primary outcome was UC progression at 5 and 10 years, defined as a composite of proximal extension, colonic structural changes (stenosis/ tubularization) and dysplasia/cancer. Secondary outcomes included clinical endpoints: hospitalizations, colectomy and corticosteroid use. Multivariate analysis was performed, adjusted for disease severity, extent, and prior activity.

Results: A total of 236 patients were included, with 28.8% (n=68) initiating BT within ≤12 months (mean FU after BT initiation: 8.06 ± 4.12 years).

Early intervention was associated with a reduced risk of proximal extension after BT initiation (p=0.012).

In multivariable Cox regression, later BT initiation was an independent predictor of colonic structural changes at 10 years (HR 8.54; 95% CI 1.15-62.63; p=0.036). Later BT initiation independently predicted the composite outcome of progression (23.2% vs 5.9%) at 5 years (HR 5.97; 95% CI 1.42-25.13; p=0.015) and 10 years (HR 4.58; 95% CI 1.41-14.88; p=0.011). No differences in clinical outcomes were observed.

Longer disease activity before BT was also a predictor of the primary outcome at 5 years (HR 4.20; 95% CI 1.27–13.82; p=0.018) and 10 years (HR 3.60; 95% CI 1.28–10.12; p=0.015).

Conclusion: Starting BT early in the disease course may reduce the long-term risk of disease progression, reshaping the window of opportunity in UC.

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Disclosure: Nothing to disclose.

PP0577

PATIENT REPORTED ATTITUDES TO INTRAVENOUS AND SUBCUTANEOUS ADMINISTERED THERAPIES FOR INFLAMMATORY BOWEL DISEASE

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Introduction: 'Biologic therapies' are the mainstay treatment for moderate to severe Inflammatory Bowel Disease (IBD), resulting in fewer hospitalisations, adequate treatment response and mucosal healing. [1] Adalimumab, Golimumab, Ustekinumab & Risankizumab are administered subcutaneously (SC), Infliximab and Vedolizumab are administered intravenously (IV), and have additionally been available SC since 2020. SC Infliximab was used during the pandemic to limit exposure to hospitals & reduce risk of Covid-19, [2] as well as for convenience.

Aims & Methods: We explore patient's views of IV and SC administered treatments via a paper-based survey for those patients who were on IV treatment when attending for infusions, and a telephone survey on patients with IBD who have been on SC treatment.

Results: IV survey

- 122 completed the survey.
- 50/122 (41.0%) were on Vedolizumab (41.0%), 64 (52.5%) Infliximab, 8 (6.6%) others / didn't specify.
- 25/120 (20.8%) had been on IV therapy for 2-6 months, 17 (14.2%) for 7-12 months, 29 (24.2%) for 1-3 years, 21 (17.5%) for 3-5 years, 21 (17.5%) for >5 years.
- 107/120 (89.2%) rated treatments as easy, 13 (10.8%) intermediate, 0 difficult.
- 25/112 (22.3%) were previously on SC injections. Of this group: 15/24 (60.0%) favoured the IV route, 9 (36.0%) favoured the SC route.

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- 4/15 who favoured the IV route "did not like to inject themselves" or "found it difficult". 3/15 preferred IV route as they were being monitored.
- 104/110 (94.5%) attended all scheduled blood tests, 4 (3.6%) attended most of them
- 106/109 (97.2%) attended all OP appointments, 2 (1.8%) attend most
- 91/111 (82.0%) had no concerns, 10 (9.0%) had concerns over side effects (SE)
- 31/104 (29.8%) reported SE.
- 6/29 (20.7%) stated SE not affected their quality of life (QoL), 12 (41.4%) little impact QoL, 10 (34.5%) moderate impact, 1 (3.4%) severe impact.
- 96/105 (91.4%) would recommend IV route

SC survey

- 106/113 (93.8%) completed the survey..
- 54/106 patients were on Adalimumab (50.9%), 24 (22.6%) Ustekinumab, 13 (12.2%) Vedolizumab , 13 (12.3%) Infliximab, 2 (1.9%) Golimumab
- 10/106 (9.4%) for 1-6 months, 21 (19.8%) for 6-12 months, 33 (31.1%) for 1-3yrs, 21 (19.8%) for 3-5 years and 20 (18.9%) for >5 years.
- 89/106 (83.9%) rated treatment administration as easy, 10 (9.4%) intermediate, 2 (1.9%) difficult
- 59/106 (55.7%) were previously on IV infusions. Of this group: 43/59 (72.9%) favoured the SC route, 8 (13.6%) favoured IV, and 8 (13.6%) no preference.
- 24/43 who favoured who favoured SC route stated 'convenient", "easy", "not have to attend hospital" as reasons.
- 86/106 (81.1%) attended all scheduled blood tests, 5 (4.7%) attended most of them,
- 102/106 (96.2%) attended all OP appointments, 3 (2.8%) attend most,
- 68/106 (64.2%) had no concerns, 21 (19.81%) concerned about fewer face to face episodes
- 40/106 (37.7%) reported SE. 8 (20%) skin reactions
- 11/40 (27.5%) stated SE not affected their quality of life (QoL), 14 (35%) little impact QoL, 9 (22.5%) moderate impact, 6 (15%) severe impact.
- 87/106 (82.1%) recommend SC route

Conclusion: The survey found that both IV and SC treatments were easy to administer for the vast majority (89.2% & 84.9%), For patients on SC therapy who previously had IV infusions, 75.6% preferred SC as "convenient", "easy" and "no (need to attend) hospital".

For patients on IV therapy who previously had SC, 60.0% preferred IV as "did not ned to self inject / found it difficult", and "preferred monitoring".

Overall there are high levels of satisfaction with 91.4% of patients on IV treatment, and 82.1% of patients on SC treatment recommend their current route of administration.

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PP0578

EFFICACY AND SAFETY OF USTEKINUMAB VERSUS VEDOLIZUMAB IN THE TREATMENT OF ELDERLY INDIVIDUALS WITH CROHN'S DISEASE: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: There has been an increasing trend in the incidence of Crohn's disease (CD) in the population >60 years of age. Although biological therapies have transformed the management of CD, providing better clinical outcomes, in elderly individuals they are often underutilized due to concerns about their efficacy and safety, as this population presents greater frailty, comorbidities and polypharmacy.

Aims & Methods: This systematic review and meta-analysis aimed to compare the efficacy and safety of ustekinumab (UST) versus vedolizumab (VDZ) in the treatment of elderly patients with CD. We performed a systematic search of the PubMed, Embase, and Cochrane Library databases in January 2025. Article screening was conducted independently by two authors using Rayyan software. Inclusion criteria were studies comparing UST and VDZ in the treatment of CD in patients >60 years, with follow-up >9 months, and that evaluated at least one of the outcomes of interest.

The primary outcome was clinical remission, and secondary outcomes included clinical response, general infection, and serious infection. For statistical analysis, we used Review Manager software, applying a random-effects model to calculate the odds ratio (OR) in binary outcomes.

Results: Five studies were included, totaling 404 patients, of whom 209 (51.73%) received treatment with UST and 195 (48.27%) with VDZ. The mean age of the patients was 67.9 years. In the UST group, 145 patients (78.8%) received prior treatment with anti-TNF, while in the VDZ group this number was 88 patients (51.2%). No statistically significant differences were observed regarding clinical remission (OR 0.87; 95% CI 0.51-1.48; I^2=0%), clinical response (OR 1.69; 95% CI 0.81-3.53; I^2=0%), general infection (OR 1.51; 95% CI 0.76-2.99; I^2=0%) and severe infection (OR 2.60; 95% CI 0.54-12.52; I^2=50%) between the groups treated with UST and VDZ.

Conclusion: Comparing efficacy and safety outcomes between UST and VDZ treatments in elderly patients with CD, no significant differences were observed between the groups. Randomized clinical trials are needed to confirm these findings.

Disclosure: Nothing to disclose.

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PP0579

DETERMINANTS OF DRUG THERAPY EFFECTIVENESS IN STRICTURING CROHN'S DISEASE

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Introduction: The development of intestinal strictures (IS) marks a critical stage in the progression of Crohn's disease (CD). Despite therapeutic advances, IS remains a complex management challenge. Identifying predictive factors for response to medical therapy is essential for optimizing treatment strategies and improving outcomes.

Aims & Methods: This study aimed to identify clinical, biological, and radiological factors associated with the efficacy of medical treatment in patients with CD-related strictures. We conducted a retrospective cohort study including patients treated medically for IS between 2013 and 2023. Data on demographics, stricture characteristics, therapeutic interventions, and clinical outcomes were collected. Patients were classified into two groups based on their response to medical therapy: responders and non-responders. Comparative analysis was performed to identify variables associated with treatment response.

Results: Among 80 patients with stricturing CD, 65 received medical therapy. The mean age at diagnosis was 38.3 ± 12.6 years, with a male-to-female ratio of 1.6. Active smoking was reported in 24 patients (37%), and 25% had a family history of IBD. IS was symptomatic in 88% of cases. Elevated C-reactive protein (CRP) was present in 84%, and hypoalbuminemia in 55%.

Strictures were most commonly ileal (n=33), followed by anastomotic (n=13), colonic (n=8), ileocecal (n=6), duodenal (n=3), and jejunal (n=2). Colonoscopy confirmed luminal narrowing in 85% of patients. Upper GI endoscopy, performed in three patients with high obstructive symptoms, revealed ulcerated duodenal strictures. Cross-sectional imaging confirmed IS in 75% of cases. Median stricture length was 10 cm [5–20], with a median wall thickening of 9 mm [6–11]. Pre-stenotic dilation was observed in 59% of cases.

Treatment included oral corticosteroids in 24 patients (37%; median initiation time: 35 days [16.2–70.5]) and anti-TNFα agents in 41 (63%), with infliximab prescribed in 80% and adalimumab in 20%. Smoking cessation was achieved in 29% of smokers.

Univariate analysis identified several predictors of favorable response: female sex (p=0.01), absence of active smoking (p=0.02), CRP < 23.5 mg/L (p<0.001), CRP/albumin ratio < 0.72 (p<0.001), stricture length < 13.5 cm (p=0.032), absence of pre-stenotic dilation (p=0.035), and smoking cessation (p=0.009). In multivariate analysis, two factors remained significantly associated with treatment success: absence of active smoking (p=0.003, OR=70.4, 95% CI [4.1–1188.8]) and CRP < 23 mg/L (p=0.001, OR=117.6, 95% CI [6.6–2076.6]).

Conclusion: In patients with stricturing CD, the absence of active smoking and lower systemic inflammation (as measured by CRP) were independent predictors of favorable response to medical therapy. These findings underscore the importance of smoking cessation and early inflammatory control to improve therapeutic outcomes in this patient population.

Disclosure: No Conflict of interest.

PP0580

SAFETY OF USTEKINUMAB VERSUS VEDOLIZUMAB IN THE TREATMENT OF PREGNANCY PATIENTS WITH INFLAMMATORY BOWEL DISEASE: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The use of biologics has become an increasingly common treatment option for inflammatory bowel disease (IBD). However, data on maternal and fetal exposure to ustekinumab (UST) and vedolizumab (VDZ) remain scarce.

Aims & Methods:

This systematic review and meta-analysis aimed to assess the safety of UST and VDZ in pregnant women with IBD. We systematically searched PubMed, Embase, and Cochrane Library databases in March 2025. The inclusion criteria were randomized and non-randomized studies comparing UST and VDZ for the treatment of IBD in pregnant patients and evaluating at least one predefined outcome of interest.

Primary outcomes were spontaneous abortion and preterm delivery. Secondary outcomes included live birth, low birth weight, serious infections, neonatal intensive care unit (NICU) admission, and malignancies. Statistical analysis was performed using RStudio, applying a random-effects model to calculate the pooled risk ratios (RRs) for binary outcomes.

Results: This systematic review included six studies. Median maternal age ranged from 30 to 33 years, with similar distributions between groups. The average disease duration ranged from 8 to 14 years. Crohn's disease was more prevalent in the ustekinumab group, while ulcerative colitis was more common in the vedolizumab group.

No statistically significant differences were observed in spontaneous abortion (RR 1.66; 95% CI 0.89–3.10), preterm delivery (RR 0.60; 95% CI 0.31–1.16), live birth (RR 0.97; 95% CI 0.90–1.05), low birth weight (RR 0.65; 95% CI 0.31–1.40), serious infections (RR 1.34; 95% CI 0.39–4.66), NICU admission (RR 1.02; 95% CI 0.56–1.85), or malignancies (RR 1.25; 95% CI 0.67–2.32) between UST and VDZ treatments.

Conclusion: No significant differences were observed across the outcomes analyzed when comparing ustekinumab and vedolizumab for the treatment of pregnant patients with IBD.

Disclosure: Nothing to disclose.

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PP0581

OUTCOMES OF LATERAL LYMPH NODE DISSECTION FOR ULCERATIVE COLITIS-ASSOCIATED RECTAL CANCER

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Introduction: Lateral lymph node dissection (LLND) is considered for patients with advanced lower rectal cancer because of possibility of metastasis. The standard surgical procedure for ulcerative colitis (UC)-associated cancer is total colectomy; however, the significance of LLND in UC-associated rectal cancer remains unclear.

Aims & Methods: The aim of this study is to clarify the outcomes of LLND for UC-associated rectal cancer. Of 1,249 patients treated for UC-associated colorectal neoplasms between 1983 and 2020 at 43 Japanese institutions, we retrospectively analyzed 415 patients with rectal cancer.

Results: LLND was performed in 45 (10.8%) patients with UC-associated rectal cancer: 33 of 139 (23.7%) patients with pT3–4 cancer and 12 of 276 (4.3%) patients with pTis–T2. Of the 45 patients with LLND, 11 (24.4%) had lateral lymph node metastasis (LLNM). LLNM-positive group (n = 11) had a significantly higher proportion of patients with pT3-4 cancer (100% vs. 64.7%, $P < 0.0046$) and poorly differentiated adenocarcinoma/ signet-ring cell carcinoma/ mucinous adenocarcinoma (por/sig/muc) (70% vs. 17.7%, $P = 0.002$) than LLNM-negative group (n = 34).

There was no significant difference in R0 resection rate between the LLND and non-LLND groups (81.8% vs. 91.2%, $P = 0.41$). The LLNM-positive group showed significantly worse 5-year overall (30.8% vs. 82.8%, $P = 0.0005$) and 5-year disease-free (0% vs. 72.4%, $P < 0.001$) survival rates than the LLNM-negative group. There was no significant difference in the administration of adjuvant chemotherapy between the two groups (54.6% vs. 40.6%, $P = 0.42$).

Conclusion: The LLND rate for UC-related rectal cancer was 23% in patients with T3-4 cancer in Japan. The LLNM-positive group had more advanced disease, more por/sig/muc histology, and showed poor prognostic outcomes even after LLND.

Disclosure: Nothing to disclose.

PP0582

ANALYSIS OF INVASIVE TREATMENT STRATEGIES FOR SIMPLEX ABDOMINAL ABSCESS IN CROHN'S DISEASE: A RETROSPECTIVE, MULTICENTRE COHORT STUDY

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Introduction: Crohn's disease (CD) complicated with intra-abdominal abscess often requires surgical intervention. Image guided percutaneous drainage (PD) can help to avoid surgical interventions, however there is limited evidence on the optimal management after PD.

Aims & Methods: Our study aimed to analyze the long-term outcomes of CD complicated with intra-abdominal abscess after intervention. In this multicenter, interational retrospective trial penetrating CD patients with simplex intra-abdominal abscess were enrolled and followed until 12-24 months. Baseline was the day of the first detection of abdominal abscess. Patients with urgent resection were excluded. Patients were grouped based on elective surgical need during follow-up after a successful PD, while the control group consisted of patients who underwent resection without PD.

Primary outcome was the abscess recurrence, while stoma rate, post procedural complications rate (<30 days), need for re-drainage and postoperative luminal recurrence were analyzed as secondary outcomes.

Results: A total of 151 CD patients from 9 countries were recruited and 86/151 patients had PD due to simplex abdominal abscess (Table 1.). Abscess recurrence rates were similar (12.8% vs. 7.7%, p=0.231). Need for re-drainage was more common amongst PD patients (OR=0.088, 95% CI=0.011-0.682), while hospital admission need was more frequent in control patients (OR=2.56, 95% CI=1.22-5.37).

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Postoperative luminal recurrence, new stoma, and biological treatment escalation needs were similar between groups based on PD. Surgical complications were coupled with increased odds in patients without prior PD (OR=3.3, 95% CI=1.09-10.0), septic complications, perforation and new fistula formation were reported the most, while PD associated complication did not occur. Higher baseline diameter of the abscess increased the odds for postoperative recurrence (OR=1.05, 95% CI=1.02-1.08) and new biological treatment initiation (OR=0.98, 95% CI=0.96-0.98). Elective resection was delayed after PD compared to control (24 vs 10 week, p=0.003). Time until resection increased the odds for redrainage (OR=1.03, 95% CI 1.01-1.05), and outcome was worse exceeding 17 weeks.

Variables	Total (n = 151)	PD group (n = 86)	Control group (n = 65)	Sig.
FU duration, weeks, median (IQR)	103 (71-109)	104 (64-112)	101 (77-105)	0.895
Sex, male (%)	88 (58.3)	49 (57.0)	39 (60.0)	0.419
Age at inclusion, years, median (IQR)	33.0 (26-44)	36.3 (27-46)	30.2 (25-38)	0.006
CD disease duration at inclusion, years, median (IQR)	11 (6-17)	11 (5-18)	11 (7-17)	0.916
Disease activity at baseline				
CDAI, mean (± SD)	302 (109)	290 (126)	307 (115)	0.389
CRP, mg/l, mean (±SD)	92 (57)	112 (88)	107 (80)	0.753
SES-CD, mean (±SD)	10.0 (8.1)	7.1 (5.8)	12.0 (9.0)	0.071
Elective resection during FU, n (%)	126 (83.4)	61 (70.9)	65 (100)	-

Conclusion: The size of the abscess determines the outcomes, while surgical treatment alone may be associated with post-procedural complications. PD may delay resection; however, this delay can necessitate re-drainage.

Disclosure: Nothing to disclose.

PP0583

END-TO-END VS. LATEROLATERAL ANASTOMOSIS IN ILEOCAECAL RESECTION FOR CROHN'S DISEASE: AN OUTDATED TECHNIQUE OR A VALID OPTION?

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Introduction: Ileocolic resection is the most common intestinal surgery performed in patients with Crohn's disease (CD). Despite advances in both surgical techniques and medical therapy, postoperative recurrence remains frequent. Various anastomotic techniques differing in configuration and construction have been proposed to reduce recurrence and postoperative complications.

In this study, we aimed to compare laterolateral and end-to-end anastomoses in terms of surgical and medical outcomes following ileocaecal resection in patients with Crohn's disease.

Aims & Methods: This retrospective study included patients with Crohn's disease who underwent ileocaecal resection. Patients were divided into two groups based on the type of anastomosis: laterolateral or end-to-end. Clinical, surgical, and postoperative variables were collected and compared. Categorical variables were analyzed using the Chi-square test, while continuous variables were compared using the Student t-test. Endoscopic recurrence was assessed at 6 months using the Rutgeerts score. Univariate and multivariate logistic regression analyses were performed to identify factors associated with endoscopic recurrence.

Results: Among 1,210 patients followed for Crohn's disease, 221(18.26%) underwent ileocaecal resection. The mean age was 31 ± 12 years, and the female-to-male ratio was 1.24. Tobacco use was reported in 52(23.5%) patients, and a history of appendectomy was noted in 46(20.8%) patients. The inflammatory phenotype (B1) was present in 46(20.9%) patients, the stricturing phenotype (B2) in 95 (43.1%) patients, the penetrating phenotype (B3) in 49 (22.2%) patients, and a combined stricturing and penetrating phenotype in 30(13.7%) patients. Perianal lesions were observed in 67(30.6%) patients.

Corticosteroid use within 4 weeks prior to surgery was reported in 67(30.6%) patients. Use of 5-aminosalicylates was noted in 40(18.2%) patients, immunosuppressants in 74(33.7%) patients, and anti-TNF agents in 50(22.6%) patients.

A total of 136(61.8%) patients underwent ileocaecal resection with a laterolateral anastomosis, while 85(38.2%) received a manual end-to-end anastomosis. The median duration of surgery was 170[120;180] minutes. Ileostomy was performed in 12 (5.4%) patients, while 94.6% underwent immediate restoration of bowel continuity. Postoperatively, 22(10%) patients showed elevated CRP and white blood cell counts, and 19(8.6%) had a Clavien-Dindo score>1.Surgical reintervention was required in 30(13.5%) cases. Clinical recurrence was observed in 38(17.4%) patients, and 60(27%) were initiated on biologic therapy.At 6 months, all patients underwent colonoscopy: 52(23.6%) had Rutgeerts score i0,21(9.7%) had i1, 76(34.7%) had i2, 36(16.6%) had i3, and 33(15.2%) had i4.

When comparing the laterolateral and end-to-end anastomosis groups, statistically significant differences were observed in several parameters: age (38.5vs42 years,p=0.039), duration of surgery>3 hours (4.4%vs9.4%,p<0.001), use of ileostomy (5.8%vs.8.2%,p=0.004), need for surgical reintervention (2.2%vs8.2%,p<0.001), and the presence of a Rutgeerts score ≥ i1 (30.4%vs59.9%,p=0.002). No statistically significant differences were found between the two groups regarding sex, Clavien-Dindo score, or clinical recurrence.

In univariate and multivariate analysis, after adjustment for sex, smoking status, presence of perianal lesions, duration of surgery, restoration of bowel continuity, and postoperative complications,the use of end-to-end anastomosis was associated with an increased risk of endoscopic recurrence (OR = 2.94,95%CI[1.58–25.25],p=0.009).

	Laterolateral anastomosis (N=136)	End-to-end anastomosis (N=85)	p
Sex Men Women	 74(54.8%) 82(60.6%)	 39(45.2%) 33(39.4%)	0.441
Age	38.5±13.8	42±14.5	0.039
Long duration of surgery (>3h)	6(4.4%)	8(9.4%)	<0.001
Ileostomy	8(5.8%)	7(8.2%)	0.004
Clavien-Dindo > 1	7(5.1%)	9(10.5%)	0.115
Surgical reintervention	3(2.2%)	7(8.2%)	<0.001
Clinical recurrence	15(11.02%)	8(9.4%)	0.468
Rutgeerts score ≥ i1	41(30.4%)	50(59.6%)	0.468

Conclusion: The choice of anastomotic technique significantly influences postoperative outcomes in Crohn's disease surgery. In our study, End-to-end anastomosis was associated with longer operative time,higher rates of reintervention, and increased endoscopic recurrence compared to the laterolateral technique. Laterolateral anastomosis offers better postoperative outcomes in ileocaecal resection for Crohn's disease

Disclosure: Nothing to disclose.

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PP0584

SURGICAL TREATMENT OF HEMORRHOIDAL DISEASE IN INFLAMMATORY BOWEL DISEASE: RESULTS FROM THE HEAD-IBD MULTICENTRE RETROSPECTIVE COHORT STUDY

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Introduction: Surgical management of hemorrhoidal disease (HD) in patients with inflammatory bowel disease (IBD) remains controversial due to concerns about impaired wound healing and disease flare, especially in Crohn's disease (CD). In Europe, current evidence is mainly based on small historical single-center case series.[1]

Aims & Methods: We sought to evaluate the safety and effectiveness of surgical treatments for HD in IBD patients and identify potential risk factors for complications. HEAD-IBD is a retrospective cohort study promoted by the Italian Group for the study of Inflammatory Bowel Disease (IG-IBD) and the Italian Society of Colorectal Surgery (SICCR). Patients with established IBD who underwent surgery for HD between December 2004 and March 2024 were included. Data were collected across multiple centres using the REDCap system. IBD activity was measured by the Harvey-Bradshaw Index for CD and the partial Mayo score for ulcerative colitis (UC). Surgical techniques, short- and long-term complications, disease flare, and continence status were analyzed. Only patients with ≥1 year follow-up were considered.

Results: A total of 38 patients were initially identified from 8 centres. Two patients with unclassified IBD were excluded from stratified analysis, leaving 36 patients (Table 1): 21 (58%) with UC and 15 (42%) with CD. Overall, 53% were male. The median age at surgery was 49 years (IQR 13) in UC and 37 years (IQR 22) in CD. At the time of surgery, biologic therapy was ongoing in 10% of UC and 40% of CD patients. Clinically active disease at the last gastroenterology visit was reported in 6 CD patients (40%) and 5 UC patients (24%). The time interval between the last IBD evaluation (clinical or endoscopic) and surgery was significantly longer in UC compared to CD patients (Table 1).

Distribution of Goligher grade suggested a higher proportion of grade III–IV disease in UC (72%) than in CD (53%). Open hemorrhoidectomy was the most frequent procedure in both groups (66% UC, 73% CD). Non-excisional techniques (DG-HAL +/- mucopexy) were used in 29% of UC and 27% of CD patients. Stapled hemorrhoidopexy was performed only in UC (10%). Median hospital stay was 1 day (IQR 1) in both groups. No 30-day or 6-month postoperative complications were reported in CD, while 14% of UC patients experienced early or mid-term complications. One UC patient experienced IBD flare at 6 months. At 12-month follow-up, one UC patient reported new-onset soiling; continence was otherwise preserved in all cases.

	Crohn's disease (n=15)	Ulcerative colitis (n=21)
Female, n (%)	8 (53)	9 (43)
Age (years), median (IQR)	37 (22)	49 (13)
Perianal disease, n (%)	3 (20)	0
Therapy at time of surgery	12 (80)	20 (95)
Topic therapy	1 (7)	8 (38)
Oral mesalazine/sulphasalazine	5 (33)	15 (71)
Steroids	3 (20)	1 (5)
Immunosuppressors	1 (7)	1 (5)
Biologics	6 (40)	2 (10)
Time interval between IBD evaulation and surgery		
Clinical	81 (188)	192 (330)
Endoscopic	150 (367)	375 (599)
Clinically active IBD at last GI visit before surgery, n (%)	6 (40)	5 (24)
Goligher grade, n (%)		
I	3 (20)	1 (5)
II	2 (13)	4 (19)
III	6 (40)	6 (29)
IV	2 (13)	9 (43)
Not available	2 (13)	1 (5)
Procedure		
Open hemorrhoidectomy	11 (73)	14 (66)
Closed hemorrhoidectomy	0	1 (5)
Doppler-guided dearterialization (DG-HAL)	1 (7)	0
DG-HAL and mucopexy	2 (13)	2 (10)
Mucopexy	1 (7)	2 (10)
Stapled hemorrhoidopexy	0	2 (10)
Complications		
30-day	0	3 (14)
6-month	0	3 (14)
12-month	0	1 (5)

Conclusion: This multicenter study represents the largest European series on surgical treatment of HD in patients with IBD, confirming its safety, with low complication rates and no new perianal disease. Short- and mid-term complications occurred only in UC patients, possibly reflecting both more advanced HD (Goligher grade III–IV) and less intensive preoperative monitoring. These findings highlight the need for close gastroenterological pre-operative assessment and shared decision-making. The ongoing HEAD-IBD II prospective study will provide further insight into optimal management strategies.

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Disclosure: Nothing to disclose.

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PP0585

RECURRENCE OF CROHN'S DISEASE AFTER INTESTINAL RESECTION: A MULTICENTER 2-YEAR ANALYSIS OF PREDICTIVE FACTORS

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Introduction: Intestinal resection is often required in Crohn's Disease (CD), yet postoperative recurrence remains a major clinical challenge. This study aimed to assess the rate of CD recurrence within 2 years following intestinal resection and to identify associated predictive factors.

Aims & Methods: Intestinal resection is often required in Crohn's Disease (CD), yet postoperative recurrence remains a major clinical challenge. This study aimed to assess the rate of CD recurrence within 2 years following intestinal resection and to identify associated predictive factors.

Results: Seventy-six patients were included, with a mean age of 31.7 ± 13.3 years; 51.3% were female. The mean disease duration was 10.0 ± 9.1 years, with 9.2% undergoing surgery at diagnosis. Disease location was ileocolic (46.1%), ileal (44.7%), and colonic (9.2%), with a predominance of penetrating phenotype.

Approximately 38% had perianal disease, and 22.4% were smokers. Most surgeries were elective (72.4%) and involved small bowel resection, with a mean length of 25.2 ± 19.4 cm. Following surgery, 50% of previously untreated patients initiated therapy, while 11.3% discontinued prior medication.

Clinical recurrence occurred in 30% of cases, of which 39.1% were early. Surgical recurrence was observed in 11.0% of patients, with 37.5% occurring early. Microscopic inflammatory activity at the resection margins—despite macroscopically normal margins—was associated with clinical (p=0.04; OR=1.3; 95% CI: 1.0–14.7) and surgical recurrence (p=0.03; OR=1.8; 95% CI: 1.1–36.2) at two years

Conclusion: This study underscores the clinical relevance of postoperative recurrence in CD and identifies microscopic inflammatory activity at resection margins as an independent predictive factor for both clinical and surgical recurrence at 2 years.

These findings highlight the importance of histopathological assessment of resected specimens and the need for early strategies to prevent recurrence in high-risk patients.

Disclosure: Nothing to disclose.

PP0586

POSTOPERATIVE COMPLICATIONS FOLLOWING INTESTINAL RESECTION IN CROHN'S DISEASE: A MULTICENTER STUDY

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Introduction: Complicated Crohn's Disease (CD) frequently requires intestinal resection (IR) for disease control. However, postoperative complications remain particularly prevalent in this population.

Aims & Methods: The aim of this study was to characterize patients with CD undergoing IR, assess 30-day postoperative complications, readmission and reoperation rates, and identify factors associated with these adverse outcomes.

Results: A total of 76 patients were included, with a mean age of 31.7 ± 13.3 years; 51.3% were female. Fewer than 20% had comorbidities, and 22.4% were smokers. Disease duration at the time of surgery was 10.0 ± 9.1 years. CD location was ileocolic (46.1%), ileal (44.7%), and colonic (9.2%), with a predominantly penetrating phenotype (57.8%). At the time of surgery, 26.3% of patients were on corticosteroids and 63.2% on biologic therapy, of whom 75% were receiving anti-TNF agents.

Most procedures were elective (72.4%), 48.7% were laparoscopic, and 80.3% included a primary anastomosis. Small bowel resection was performed in the majority of cases, with a mean length of 25.2 ± 19.4 cm. In 77.8% of surgical specimens, resection margins showed no inflammatory activity. Enteral feeding was resumed at 2.7 ± 1.8 days, and mean hospital stay was 12.3 ± 9.8 days.

Within 30 days postoperatively, 38.7% of patients experienced complications, 4.0% were readmitted, and 12.0% underwent a second IR. No mortality was reported. The presence of granulomas (p = 0.047), elevated preoperative CRP (p = 0.019), and delayed initiation of enteral feeding (p = 0.018) were associated with complications, with CRP identified as an independent predictor (p = 0.049, OR = 1.307). Longer disease duration (p = 0.009), more extensive small bowel resections (p = 0.016), and anti-TNF therapy (p = 0.010) were associated with infectious complications. Stoma creation was linked to readmission (p < 0.001), while COPD (p < 0.001), diabetes (p = 0.049), and absence of biologic therapy (p = 0.034) were associated with a second IR within 30 days.

Conclusion: The high rate of postoperative complications among CD patients undergoing IR underscores the need for dedicated research and a tailored multidisciplinary approach. The presence of granulomas, elevated preoperative CRP, longer disease duration, and lack of prior biologic therapy were all associated with adverse outcomes.

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Disclosure: Nothing to disclose.

PP0587

POSTOPERATIVE OUTCOMES IN ULCERATIVE COLITIS PATIENTS UNDERGOING COLORECTAL SURGERY AFTER PREOPERATIVE BIOLOGIC AND SMALL-MOLECULE THERAPIES: A NETWORK META-ANALYSIS

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Introduction: Despite optimal medical therapy, patients with ulcerative colitis (UC) may still require surgical intervention. Biologics and other immunotherapies are widely used in the management of UC and have been associated with a range of reported complications. This network meta-analysis aims to evaluate post-operative outcomes associated with prior biologic and immunotherapy use in patients with UC undergoing colon surgery.

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Aims & Methods: On March 8, 2025, a database search was conducted using PubMed. We included studies that evaluated postoperative outcomes in patients with ulcerative colitis (UC) who had prior exposure to biologic therapy and subsequently underwent colon surgery. The outcomes assessed were overall postoperative complications and postoperative infections, both measured within 30 days. Effect sizes were calculated as odds ratios (ORs) with 95% confidence intervals (CIs) using a random-effects model. All analyses were performed using RStudio with the *netmeta* package.

Results: Out of 2,759 articles screened, only 14 retrospective studies were eligible, comprising a total of 2,679 patients. The analysis included ustekinumab, tofacitinib, vedolizumab, and anti-TNF agents, each compared to a control group. For overall postoperative complications, both ustekinumab (OR: 0.18; 95% CI: 0.05–0.68) and tofacitinib (OR: 0.47; 95% CI: 0.28–0.78) were associated with a significantly lower risk compared to anti-TNF agents. Similarly, when compared to the control group, ustekinumab (OR: 0.20; 95% CI: 0.05–0.76) and tofacitinib (OR: 0.52; 95% CI: 0.30–0.90) also showed significantly lower complication rates. Vedolizumab was associated with a significantly lower risk of complications compared to anti-TNF agents (OR: 0.65; 95% CI: 0.43–0.98); however, no significant differences were observed when compared to the control group.In terms of postoperative infections, no significant differences were observed among ustekinumab, tofacitinib, vedolizumab, anti-TNF agents, and the control.

Conclusion: In patients with ulcerative colitis undergoing colorectal surgery, prior exposure to ustekinumab, tofacitinib, and vedolizumab was associated with a lower risk of overall postoperative complications compared to anti-TNF therapy. For postoperative infection, no significant differences were observed across all treatment groups.

Disclosure: Nothing to disclose.

PP1135

AI-BASED DECISION SUPPORT SYSTEM FOR DIAGNOSING INFLAMMATORY BOWEL DISEASES

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Introduction: Inflammatory bowel diseases (IBD) carry a high risk of complications if not diagnosed promptly and are often misdiagnosed as irritable bowel syndrome (IBS). Delayed IBD detection leads to an increase in complicated cases and missed opportunities for timely therapy initiation.

Aims & Methods: The study aimed to develop an AI-based clinical decision support system (CDSS) for IBD diagnosis.

We created three databases, containing clinical-laboratory data, endoscopic and histopathological images. Patients enrolled from 2014 to 2024 included those with confirmed IBD (ulcerative colitis (UC) and Crohn's disease (CD) with colonic involvement) during flare-ups and IBS with normal endoscopic and histopathological findings.

Separate artificial neural networks were designed for IBD differentiation: multilayer perceptrons (MLPs) for clinical-laboratory data analysis and a pre-trained VGG16 convolutional neural network (CNN) for digital image analysis (initialized with ImageNet weights).

Results: A retrospective database of 1,406 patients with clinical and laboratory data was compiled. For IBD vs. IBS differentiation, the cohort was reduced to 390 patients that had all the fulfilled data, while UC vs. CD analysis included 964 patients.

An MLP (13:13-5-1:1) achieved 92.7% and 99.0% accuracy in IBD diagnosis, incorporating stool frequency/consistency, absence of extraintestinal/perianal manifestations, functional disorders, family history of autoimmune diseases, prior surgeries, CRP, and fecal calprotectin ($p \leq 0.019$).

Another MLP (9:9-8-1:1) detected 70.16% of CD and 86.40% of UC cases, using data on constipation, diarrhea, bloody stools, absence of extraintestinal manifestations, fever, perianal lesions, strictures, and prior bowel surgeries ($p<0.001$).

The endoscopic database comprised 5,693 JPEG images: 3,400 of normal mucosa, 1,357 - CD, 935 - UC. Model sensitivity and specificity were for CD - 91%/79%; UC - 90%/97%; normal - 90%/97%.

The histopathological dataset included 398 whole-slide images (1280×960 px), segmented into 224×224 px tiles. The final set included 8,620 images: 2,360 - CD, 2,320 – normal mucosa, 3,950 - UC. Sensitivity and specificity were for CD - 91% and 95%, UC - 96%, 95%, normal - 96%, 93%, respectively.

The integrated clinical DSS was constructed using Bayesian belief networks implemented in Hugin Lite 8.3 software and integrated clinical, endoscopic, and histopathological nodes. The network architecture replicated the stepwise diagnostic process and incorporated the following nodes: clinical and laboratory data, endoscopy (IBD/IBS differentiation), histopathology (IBD/IBS differentiation), IBS, IBD, endoscopy (UC/CD diagnosis), histopathology (UC/CD discrimination), final diagnosis. The system was validated using six patients per diagnostic category (IBS, UC, CD), with complete sets of clinical, endoscopic, and histopathological data required for all component models. The integrated model demonstrated an overall diagnostic accuracy of 83.3%.

Conclusion: The developed CDSS shows promise for earlier IBD detection but requires further validation studies.

Disclosure: Nothing to disclose.

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Poster presentations
Lower GI

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Lower GI

Poster Exhibition

PP0588

ENHANCEMENT OF SENNOSIDES AVAILABILITY IN CHRONIC CONSTIPATION VIA CO-ADMINISTRATION WITH MESALAZINE

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Introduction: Chronic constipation leads to an increased risk of heart disease, stroke, and kidney disease[1]. Stimulant laxatives provide quick relief and are preferred by patients and pharmaceutical companies; "sennosides" and "bisacodyl," are more popular compared with bulk-forming laxatives[2].

However, with prolonged use, the human body becomes habituated to them. The laxatives induce prostaglandin production[3] through immune cell aggregation and local cytokine expression[4].

Thus, prolonged use of stimulant laxatives leads to inflammation and damages the intestinal nervous plexus, leading to intestinal exhaustion[5].

Mesalazine can remove reactive oxygen and inhibit the biosynthesis of cytokines including interleukin (IL)-1β and leukotrieneB4, and hence, is used for treating inflammatory bowel disease[6].

Concurrent intake of mesalazine and stimulant laxatives might suppress leukotriene and the subsequent adverse inflammatory responses, thus allowing the safe and long-term use of sennosides.

Herein, we observed stool characteristics of animal models and performed pathological analyses of their colon tissue to establish the stable use of mesalazine-adapted stimulant laxatives. Lamina propria mononuclear cells (LPMCs) of the colon were then collected for flow cytometry analysis (FACS) and real-time polymerase chain reaction (PCR).

Aims & Methods: Six-week-old BALB/cAJcl mice were assigned to one of the four groups: control, mesalazine (100 mg/kg), sennosides (4.8 mg/kg), or mesalazine and sennosides combination groups, (n = 8 for each group). Each drug was suspended in 0.5% carboxymethylcellulose (CMC). Doses were administered orally once daily for 21 days. Stool samples were collected on days 0, 7, 14, and 21; looseness and weight of each stool sample were measured. Feces were then freeze-dried and weighed to calculate the water content percentage.

Mice were euthanized after 21 days, and their colons were harvested. Inflammatory cell aggregation was detected on pathological evaluation, which was performed via H&E staining. Composition of the aggregated LPMCs was evaluated via FACS. Real-time PCR was performed to detect cytokine mRNA expression in LPMCs.

Results: No differences in food intake, water consumption, and body weight were observed among the groups during the experimental period. On day 21, the sennosides group exhibited significantly greater stool looseness compared with the control group. Stool looseness was similar in the sennosides and combination groups. Water content percentage in the stool was significantly higher in the sennosides group compared with in the control group, and this increase was not influenced by the addition of mesalazine. Inflammatory cell aggregation was detected in sennosides and combination groups. A drastic increase in the level of killer T cells was revealed in the sennosides group on FACS, and mesalazine administration resulted in restoration to normal composition. Regarding mRNA expression, IL-1β, IL-2, and IFN-γ levels were elevated in the sennosides group, which was alleviated by mesalazine co-administration.

Conclusion: Sennosides is predicted to provoke IL-2 and IFN-γ biosynthesis that induces killer T cell aggregation and subsequent IL-1β production. As the combination group did not show any inflammation, it is assumed that mesalazine prohibits the inflammatory process. These results provide further evidence to indicate that mesalazine can alleviate sennosides-induced inflammation without adversely affecting its efficacy. This finding may promote the effective use of laxatives in patients with chronic constipation.

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PP0589

A PROSPECTIVE STUDY ON PAEDIATRIC ILEO-COLONIC BIOFILMS: MACROSCOPIC IDENTIFICATION, CHARACTERISATION, AND DISEASE CORRELATIONS

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Introduction: Intestinal biofilms are microbiological colonies that colonize biotic and abiotic interfaces, occurring in various adult diseases like IBD, IBS, and CRC.

Aims & Methods: This study investigates the occurrence and characteristics of visible biofilms in the colon of paediatric patients across different disease groups undergoing colonoscopy, aiming to record frequency, location, extent, and adherence in relation to diagnoses and intestinal sections. The prospective, multicentre observational study included 157 paediatric patients undergoing ileocolonoscopy for diagnostic purposes. The presence, extent, and adherence of visible biofilms were documented using high-resolution white light endoscopy. Biofilms, defined as yellow-greenish adherent layers, were classified by extent (small, medium, large) and adherence (weak, medium, strong), then subdivided by diagnostic categories and colonic segments. 43 endoscopies with a Boston Bowel Preparation Score (BBPS) <6 were excluded. Individual diseases that could not be assigned to these patient groups accounted for 7% (n=8).

Results: Biofilms were found in 86.7% of patients with systemic disease and normal colonoscopy, 84.6% with ulcerative colitis, 72.7% with Crohn's disease, 85.7% with IBS, and 87.5% with colorectal polyps. There was a

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decreasing trend in extent and adherence from proximal to distal bowel segments across all groups. The caecum had the highest adherence and extent in all groups, while the rectum had the lowest. No significant differences in biofilm occurrence and characteristics between disease groups were observed. However, biofilms were significantly more common in proximal bowel segments (ileum, caecum, ascending colon).

Biofilm in …	Systemic disease, normal colonoscopy (n=15)	Ulcerative Colitis (n=26)	Crohn's Disease (n=22)	IBS – normal colonoscopy (n=35)	Polyps (n=8)
Ileum	9 (60%)	17 (65,4%)	12 (54,5%)	12 (34,3%)	5 (62,5%)
Cecum	12 (80%)	18 (69,2%)	14 (63,6%)	29 (82,9%)	6 (75%)
Ascending colon	10 (66,7%)	13 (50%)	14 (63,6%)	23 (65,7%)	3 (37,5%)
Transverse colon	9 (60%)	9 (34,6%)	11 (50%)	15 (42,9%)	3 (37,5%)
Descending colon	6 (40%)	9 (34,6%)	7 (31,8%)	6 (17,1%)	0 (0%)
Sigmoid colon	6 (40%)	5 (19,2%)	3 (13,6%)	3 (8,6%)	0 (0%)
Rectum	3 (20%)	2 (7,7%)	0 (0%)	1 (2,9%)	0 (0%)

Conclusion: Macroscopically visible biofilms are frequently detected in paediatric patients across various diagnoses, showing a predilection for proximal bowel segments. Particularly, patients with IBD, systemic diseases, and IBS are predisposed to biofilms, supporting the hypothesis that biofilms play a pathophysiological role in gastrointestinal diseases. These findings highlight the need for systematic investigation of biofilms in paediatric GI disease diagnosis and treatment.

Disclosure: Nothing to disclose.

PP0590

INVESTIGATING INTESTINAL BARRIER DYSFUNCTION IN CYSTIC FIBROSIS: MECHANISMS, PATHOGEN INTERACTION, AND THERAPEUTIC MODULATION

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Introduction: Chronic intestinal inflammation, increased permeability, and dysbiosis are hallmark features of cystic fibrosis (CF), yet the molecular underpinnings of gut barrier dysfunction in people with CF (pwCF) remain incompletely understood.

Aims & Methods: This study aimed to characterize epithelial markers of intestinal permeability using both immortalized and primary human intestinal epithelial cell models, with and without functional CFTR. A secondary objective was to investigate the impact of inflammatory stimuli on barrier integrity and explore lipid-based strategies for restoring epithelial function in CF. Caco-2 cells (~5000/well) were cultured in 96-well transwells (Corning) for 4 weeks to develop confluent monolayers. Healthy primary human colonic epithelial and endothelial cells (Emulate Inc.) were cultured in a microfluidic colon-on-a-chip system simulating peristalsis and blood flow. CF models were generated by CFTR gene knockdown via liposomal siRNA transfection, validated by RT-qPCR and western blotting.

Epithelial permeability was assessed by transepithelial electrical resistance (TEER) and FITC-dextran flux at 48-, 72-, and 96-h post-treatment. Cells were fixed and stained for DAPI, ZO-1, and VE-cadherin, then analyzed by confocal microscopy.

To assess cytoprotection, Caco-2 monolayers were pretreated with phosphatidylinositol (PI) lipids PI(16:0/18:1(9Z), PI(18:0/20:4), and LPI(20:4/0:0) (Avanti Polar Lipid and Sigma-Aldrich, 10 µM, 15 min), followed by exposure to a panel of toxins (*C. difficile* TcdA/B, *Pseudomonas* exotoxin A, *Staphylococcus aureus* delta-toxin) and the chemical inflammatory agent DSS (dextran sulfate sodium) for 24 h. Experiments were performed in triplicate.

Results: CFTR knockdown in Caco-2 cells (siCFTR) significantly reduced TEER compared to non-targeting controls ($p < 0.05$, $n = 3$), indicating increased epithelial permeability. This disruption was dramatically exacerbated following exposure to *C. difficile* toxins A and B (50 pg/mL), which caused a profound TEER decline ($p < 0.0001$). Immunofluorescence imaging revealed mislocalization and downregulation of ZO-1 in siCFTR cells. In the colon-on-a-chip model, CFTR depletion also disrupted ZO-1 and VE-cadherin architecture in both epithelial and endothelial layers, confirming widespread junctional compromise.

Among the tested lipid metabolites, PI (18:0/20:4) significantly preserved barrier function, by reducing FITC-dextran flux at 48 h following DSS exposure ($p < 0.001$) and against *C. difficil*e toxins A/B and *S. aureus* delta-toxin at 96 h ($p < 0.0001$). Other lipids showed no significant cytoprotective effects under these conditions, suggesting a metabolic-specific and time-dependent cytoprotective effect. TEER and cytokine assays are underway.

Conclusion: Loss of CFTR function leads to impaired epithelial barrier integrity in both immortalized and primary CF gut models, with pronounced vulnerability to microbial toxins and proinflammatory stimuli. Junctional protein disruption (ZO-1, VE-cadherin) appears central to this phenotype. Notably, PI (18:0/20:4) emerged as a promising cytoprotective agent capable of restoring epithelial resistance.

Importantly, PI and its derivatives are essential for cell membrane structure, intracellular signalling, and cytoskeletal dynamics. Ongoing studies will expand screening of anti-inflammatory and barrier-stabilizing therapeutics and leverage these platforms to develop patient-specific "gut avatars" for personalized therapeutic testing in CF.

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Disclosure: Nothing to disclose.

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PP0591

PATIENT CARE DURING ADMISSION WITH IMMUNE CHECKPOINT INHIBITOR (ICI) RELATED COLITIS

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Introduction: Immune checkpoint inhibitor (ICI)-related colitis is a recognised adverse event associated with this groundbreaking oncological therapy. It affects approximately 20–50% of patients receiving ICIs and may be associated with significant morbidity. Corticosteroids remain the cornerstone of treatment, with prompt escalation to biologic therapies required in more severe cases.

Aims & Methods: The aim of this study is to present the clinical and endoscopic findings of all ICI related colitis patients requiring admission. All presentations with ICI-related colitis to a quaternary Sydney hospital over a 5 year period (July 2019 to June 2024) were included. Clinical, laboratory, endoscopic and histologic data were extracted through chart review of the admission and the 6 months following.

Results: A total of 59 patients were admitted for the management of ICI-related colitis. The cohort comprised 42 males (71.2%) with a median age of 71 years (IQR 55.5–76.5). Melanoma was the most common malignancy (76.3%, n=45), with the majority (86.4%, n=51) presenting with metastatic disease and preserved premorbid functional status (ECOG 0 in 42.4%, n=25; ECOG 1 in 35.6%, n=21). The median time from initial ICI exposure to hospital admission for colitis symptoms was 4 months (IQR 2–11). The median length of stay was 7 days (IQR 5–15.5).

All patients presented with diarrhoea (100%, n=59), while abdominal pain and fever were reported in 37.3% (n=22) and 18.6% (n=11) respectively. Most patients experienced diarrhoea graded as either Grade 2 (44.6%, n=26) or Grade 3 (39.0%, n=23) according to the Common Terminology Criteria for Adverse Events. Stool microscopy was conducted in 93.2% of patients (n=55), while testing for *Clostridium difficile* was performed in 83.1% (n=49). Faecal calprotectin was measured in 72.9% (n=43), with 83.7% (n=36) of these patients recording levels greater than 100.

All patients underwent either sigmoidoscopy (96.6%, n=57) and/or colonoscopy (5.1%, n=3), with procedures performed at a median of 3 days post-admission (IQR 2–4). Endoscopic evidence of active colitis was observed in 71.2% (n=41), with histological confirmation in 74.6% (n=44). Ulcerations were identified in 8.5% (n=5). Corticosteroids were administered to the majority of patients (94.9%, n=56), with intravenous delivery used in 72.9% (n=43). 8.5% (n=5) patients received antibiotics for a concurrent infection.

Second-line immunosuppressive therapy was required in 44.1% (n=26), most commonly infliximab (80.8%, n=21), followed by cyclosporin (23.1%, n=6) and vedolizumab (15.4%, n=4). Infliximab was initiated at a median of 5 days following admission (IQR 3–8). Immune checkpoint inhibitor was frequently discontinued either prior to admission (18.6%, n=11) or during the hospital stay (32.2%, n=19). Readmission within six months occurred in 10.2% (n=6) of patients, and 13.6% (n=8) were transitioned to best supportive care during or after their hospital stay.

Conclusion: ICI-related colitis led to significant adverse events, typically emerging around four months after initial ICI exposure. The use of second-line immunosuppression was common, with infliximab being the most frequently administered agent. This complication resulted in treatment cessation for many patients; however, rates of readmission for recurrent symptoms were low.

Disclosure: Nothing to disclose.

PP0592

EXPLORATORY ANALYSIS INDICATES THAT CHANGES IN STOOL SHORT-CHAIN AND BRANCHED-CHAIN FATTY ACIDS LEVELS MAY SERVE AS OBJECTIVE MARKER OF SUCCESSFUL DIETARY INTERVENTION OF IRRITABLE BOWEL SYNDROME SYMPTOMS

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Introduction: Biomarkers, like stool short-chain fatty acids (SCFA), metabolites by fermented nondigestible carbohydrates and branched-chain fatty acids (BCFA) derived from the metabolism of essential amino acids may allow for objective measurements of IBS biological changes without self-reporting bias. Diets low in FODMAP significantly reduce the abnormal fecal total SCFA in IBS patients, indicating that the diet normalizes the IBS patients' abnormal fecal SCFA output.

Aims & Methods: By conducting an exploratory analysis, we aimed to measure stool SCFAs/BCFAs level changes in a subgroup of subjects enrolled in an RCT evaluating the efficacy of a Mediterranean version of the Low-FODMAPS diet (MED-LFD) compared to the nutritional recommendations of the British National Institute for Health and Care Excellence (NICE) for managing non-constipation IBS.

A subset of 40 (23 MED-LFD & 17 NICE) patients provided stool samples at 3 time points: Before dietary intervention, at the 1st (4 weeks after initiation of intervention for NICE and at the end of phase 2 for MED-LFD), and the 2nd folow-up (FU), 6 months after intervention initiation. We measured symptoms severity using IBS-SSS. Table 1 shows the detailed measurement of the consumed FODMAPs per day assessed using a triple-pass 24-hour dietary recall twice at the two FUs.

	1st Follow-up			2nd Follow-up		
FODMAPs, M(IQR)	MED-LFD (N=20)	NICE (N=17)	p	MED-LFD (N=19)	NICE (N=15)	p
Excess fructose, g/meal	0.17 (0.13-0.38)	0.19 (0.11-0.83)	0.563	0.22 (0.11, 0.39)	0.3 (0.2, 0.47)	0.275
Sorbitol, g/meal	0.02 (0.0-0.06)	0.11 (0.01-0.39)	0.021	0.02 (0, 0.08)	0.14 (0.01, 0.28)	0.245
Mannitol, g/meal	0.004 (0.0-0.07)	0.06 (0.01-0.13)	0.077	0.02 (0.01, 0.13)	0.04 (0, 0.09)	0.664
Fructans, g/meal	0.43 (0.33-0.54)	0.63 (0.48-0.7)	0.051	0.54 (0.27, 0.8)	0.73 (0.51, 1.07)	0.080
GOS, g/meal	0.09 (0.07-0.18)	0.21 (0.09-0.36)	0.106	0.14 (0.07, 0.32)	0.23 (0.14, 0.48)	0.171
Total oligosaccharides, g/meal	0.54 (0.39-0.79)	0.85 (0.65-1.1)	0.038	0.7 (0.33, 1.16)	1.08 (0.72, 1.52)	0.050
Total FODMAPs, g/meal	1.36 (1.0- 1.90)	2.51 (1.68-3.42)	0.011	1.54 (1.10-2.62)	2.62 (1.42-3.90)	0.086
Lactose, g/meal	0.04 (0.01-0.56)	1.36 (0.47-2.46)	0.002	0.32 (0.04, 1.15)	0.27 (0.01, 2.39)	0.808

Table 1.

SCFAs/BCFAs were extracted from frozen samples and analyzed using Gas Chromatography/Mass Spectrometry. SCFAs and BCFAs levels were measured in umol/g of stool. A paired/unpaired t-test was performed to illustrate the organic acid changes among and within two groups. In multiple groups comparison, significant differences between means were examined using one-way analysis of variance. A polynomial regression model was used to describe the relation between BCFAs/SCFAs levels change with IBS-SSS change from baseline at the 1st FU.

Results: We detected a significant reduction in the concentrations of total SCFAs and BCFAs in the MED-LFD (-21.4±20 and -0.429±0.882) and the NICE (-9.9±17.7 and -0.819±0.933) groups, respectively, at the 1st FU. At the 2nd FU, similar reductions from the baseline concentration levels were

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detected in both intervention groups (total SCFAs: -20.2±18.7 in MED-LFD vs. -10.9±7.4 in NICE; total BCFAs: -0.494±0.806 in MED-LFD vs. -0.76±0.83 in NICE). A polynomial association was observed between the changes in IBS-SSS scores and the changes in stool concentrations of BCFAs and SCFAs from baseline to the 1st FU.

Specifically, there was a slight upward trend in the difference of BCFA levels, indicating that a decrease in IBS-SSS scores is associated with a reduction in BCFA levels during the 1st follow-up. This trend peaked around an IBS-SSS change of minus 50 points, the cut-off value for clinical response. Beyond this point, this association slightly declines. A similar trend was noted for SCFA level changes, with the peak occurring around an IBS-SSS change of minus 100 points, after which the association diminishes.

Conclusion: SCFA/BCFA stool levels changes from baseline might be an objective measure of successful dietary intervention for IBS

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Disclosure: Nothing to disclose.

PP0593

ANTISPASMODIC IMPACT OF AN HERBAL PREPARATION ON SMALL INTESTINE: PROMISING THERAPEUTIC BENEFITS FOR PATIENTS WITH IRRITABLE BOWEL SYNDROME

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Introduction: Amara-Drops is a traditional herbal preparation containing nine individual extracts (*Artemisia absinthium* Herba, *Centaurium erythraea* Herba, *Cichorium intybus* Planta tota, *Gentiana lutea* Rhizoma, *Juniperus communis* Summitates, *Achillea millefolium* Herba, *Peucedanum ostruthium* Rhizoma, *Salvia officinalis* Folium, and *Taraxacum* Planta tota) used to treat gastrointestinal dysmotility. Previous studies have demonstrated that Amara extract targets the muscarinic receptor M2, inducing relaxation in carbachol-contracted guinea pig fundus strips, potentially improving impaired fundus accommodation in functional dyspepsia patients [1].

However, its antispasmodic properties to alleviate visceral hypersensitivity, a major factor in abdominal pain experienced by Irritable Bowel Syndrome (IBS) patients, have yet to be studied.

Aims & Methods: Anticholinergic agents are commonly used to treat IBS, but they often come with broad side effects. Herbal medicines have shown excellent antispasmodic properties without significant adverse reactions [2].

Therefore, following preliminary results indicating Amara extract's antagonist activity on and binding to M2 muscarinic receptors expressed in CHO cell lines, we investigated the antispasmodic effect of Amara extract on guinea pig small intestine with or without carbachol-induced contraction. Dry extract was obtained from the commercial product Amara-Drops. Guinea-pig ileum muscle strips were prepared and set up in organ baths in a longitudinal orientation to study the relaxation effect of Amara extract (94, 283, 450, 650, 850, and 1,000 µg/mL), isoprenaline as a positive control (100 pM, 1, 10 and 100 nM, 1 and 10 µM), or vehicle, with or without carbachol-induced contraction (10 µM).

Results: *In vitro* studies revealed that Amara extract binds to and inhibits M2 receptor activity (IC_{50}: 294 and 219 µg/mL, respectively) via inhibition of their respective reference agonist activity. Exposure of Amara extract to ileum muscle strips induced significant relaxation already at 450 µg/mL (25%, $p<0.01$), with stronger effects observed at 650, 850, and 1,000 µg/mL (27-31%, $p<0.0001$), showing a generally stronger relaxation than the positive control isoprenaline (9.8-18%, $p<0.05$ - $p<0.001$).

Significant relaxation was also observed in carbachol-contracted muscle strips already at 283 µg/mL (50%, $p<0.01$), with stronger effects at higher concentrations (78-132%, 450-1,000 µg/mL, $p<0.0001$) compared to isoprenaline, which only showed significant relaxation at the highest concentration (71%, $p<0.05$).

Conclusion: The findings highlight the potential of Amara-Drops in modulating visceral hypersensitivity and alleviating abdominal pain in IBS patients through the inhibition of cholinergic-induced contraction in the small intestine. Further research is necessary to identify which specific extracts contribute to the anticholinergic effect. This study lays a promising molecular foundation for the therapeutic use of Amara-Drops in IBS treatment.

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Disclosure: All authors are employees of Weleda AG.

PP0594

HOMOCYSTEINE EXACERBATES IRRITABLE BOWEL SYNDROME BY IMPAIRING THE MUCUS LAYER AND ALTERING GUT MICROBIOME

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Introduction: Gut metabolites are suggested to be potential etiological factors of irritable bowel syndrome (IBS). However, the potential mechanisms remain unclear.

Aims & Methods: This study aimed to elucidated the effect and mechanism of gut metabolite homocysteine(Hcy) in IBS development. A stress-induced IBS model was established using neonatal maternal separation (NMS), while a post-inflammatory IBS (PI-IBS) model was constructed by intracolonic administration of trinitrobenzene sulfonic acid (TNBS). A high-methionine diet was applied to elevate homocysteine levels in both blood and feces of mice or rats. Visceral hypersensitivity was assessed by measuring abdominal withdrawal reflex (AWR) and visceral motor responses (VMR). Carmine red was used to determine whole intestinal transit time. The composition of the gut microbiota in rats fecal samples was analyzed by using 16S rRNA sequencing. A series of cell and molecular experiments, including immunostaining, western blot and ELISA was conducted to illustrate the mechanism.

Results: We found that fecal Hcy levels were significantly elevated in both patients with IBS and rats with IBS induced by water avoidance stress (WAS). Moreover, fecal Hcy levels were positively correlated with the severity of IBS symptoms in patients. Next, we investigated the effects of Hcy on the gut-brain axis in NMS rat model and TNBS mice model. Our study showed that elevated Hcy in stool exacerbated visceral hypersensitivity,

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increased intestinal permeability and disrupted the mucus barrier of IBS rats/mice. Interestingly, folic acid supplementation lowered the Hcy levels and alleviated the effect of Hcy in exacerbating IBS phenotype.
Mechanistically, treatment of Hcy inhibited the expression of NLR family pyrin domain containing 6 (NLRP6) and downregulated mucin MUC2 expression in LS174T and intestinal organoids, and impaired the expression of tight junction proteins ZO1 and Occludin in Caco2. Genetic overexpression of NLRP6 and PPARγ agonist Rosiglitazone rescued the inhibitory effect of Hcy on MUC2 expression. In addition, by using 16S rRNA sequencing, we found that Hcy reduced gut bacterial diversity and lowered the abundances of *Coprococcus, Roseburia and Acetitomaculum* in NMS rats.
Conclusion: Our study showed that gut metabolite Hcy was involved in the pathogenesis of IBS by disrupting mucus barrier and disturbing gut microbiota, indicating Hcy as a therapeutic target and folic acid administration as a potential strategy for IBS management.
Disclosure: Nothing to disclose.

PP0595

DIFFERENT EFFECT OF LOCAL SPHINCTER STIMULATION ON ANAL FUNCTION AND SYMPTOMS OF URGE VS. PASSIVE FECAL INCONTINENCE

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Introduction: Fecal incontinence is a diagnosis represented by recurrent uncontrolled passage of fecal material for at least 3 months. It is relatively common, with an assumed prevalence of up to 7% of the worldwide population. The therapy is challenging and often with insufficient clinical results. It should be specifically tailored to different symptom presentations meaning urge or passive incontinence. One of the relatively and therefore not widely used treatment options for incontinence is local sphincter stimulation which is a safe non or minimally invasive method. This stimulation could be performed in at home setting, so it presents a substantial time and cost-saving option for the patient and for the healthcare system. However, the data published on this topic so far are scarce with heterogeneous studied populations and results with no universally accepted treatment protocol.
Aims & Methods: This study aims to evaluate the effect of local sphincter stimulation on the functional parameters of sphincters and symptoms based on the different phenotypes. This is a retrospective single-center study analysis on patients with fecal incontinence (age> 18 years). The duration of the training was 12 weeks, with training once daily for 6 days a week. The parameters of the home stimulation device were as follows: Frequency: 40 Hz; Impulse Width: 0.2 s; Ramp-up time: 1s; Workout time: 6s; Rest: 15s. Training duration: 20min. During the perceived stimulation patients were instructed to squeeze the sphincter. 3D anorectal manometry was performed according to the IAPWG protocol at the initiation visit and the control visit. Clinical evaluation was performed by self-reporting Wexner's score for fecal incontinence at both visits.
Results: Twenty-three patients (23) were consecutively screened for eligibility and assigned for the therapy. Three patients (3) did not adhere to the training due to various personal reasons one (1) patient did not tolerate stimulation. Therefore, 19 patients (10 passive, 9 urge) completed the full treatment protocol for 12 weeks and were considered for the final analysis. The total average increase in sphincter rest medium pressure was 22.53%, with significantly higher improvement in the urge group compared to the passive group (46.24% vs. 1.18% respectively). The total average increase in sphincter rest maximum pressure was 25.69%, with significantly higher improvement in the urge group compared to the passive group (46.92% vs. 6.59% respectively). The total average increase in sphincter maximum squeeze pressure was 36.13%, with significantly higher improvement in the urge group compared to the passive group (57.82% vs. 16.61% respectively). The average improvement in Wexner score was 3.1 points, with higher improvement in the urge group compared to the passive group (4.2 points vs. 2.1 points respectively).
Conclusion: The home-administered local anal sphincter stimulation for 12 weeks seems to be an effective treatment option for both types of fecal incontinence and significantly more effective in the urge type. Therefore this could be considered as a cost and time-effective treatment option compared to standard office-based therapy. Although a home stimulation device is simple to use, the ability of the patient to adhere to the protocol should be evaluated during the screening process.
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PP0596

NEAR-INFRARED HYPERSPECTRAL MICROSCOPE FOR HIRSCHSPRUNG'S DISEASE: VISUALIZATION OF ENTERIC NERVE PLEXUS IN MOUSE

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Introduction: Hirschsprung's disease is caused by the absence of ganglia in the deep intestinal wall, leading to dysfunction [1]. Resection of the affected area is the only cure, so accurate margin detection is essential. Currently, intraoperative rapid pathology is used, requiring tissue sampling and staining, but each test takes over 30 min and may need multiple samples, sometimes delaying decisions up to 3 h. A quick on-site method to detect ganglia around 200 μm deep is needed.
Various methods, including confocal microscopy [2], have been studied, but issues remain, such as dye toxicity and limited depth (>200 μm) imaging.
Aims & Methods: For non-destructive and stain-free visualization of tissues, we focused on Near-Infrared Hyperspectral Imaging (NIR-HSI)[3][4]. Near-infrared light—particularly in the 1000–1400 nm range—exhibits high tissue penetrability and is effective for detecting deep tissues.
In addition, the near-infrared region provides spectroscopic information based on molecular vibrations, enabling component analysis. NIR-HSI holds promise for the identification of cancer and other biological tissues without the need for fluorescence or external marking [4][5].
To implement NIR-HSI, we developed a novel microscopic optical system capable of guiding near-infrared light. The near-infrared light source, which combines a supercontinuum light source and an acousto-optic tunable filter, allows for the selection of arbitrary wavelengths and irradiation of the target by adjusting the applied voltage [6]. This configuration enables the acquisition of near-infrared spectral images (NIR-HSI) at arbitrary wavelengths.

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For performance validation, ex vivo experiments were carried out using mouse intestine in which the nerve plexus was selectively labeled with GFP (Green Fluorescent Protein). In these experiments, near-infrared light ranging from 850 to 1400 nm was irradiated in 10-nm increments, and 77 spectral images were captured. As a reference, GFP fluorescence images were acquired in the same field of view as the near-infrared images. Based on the fluorescence images, annotations were performed to delineate the enteric nerve plexus, and near-infrared spectra were obtained from both the nerve and non-nerve regions. Furthermore, using pixel-wise spectral data, segmentation of the nerve regions was performed with a machine learning model, and the accuracy of identification was evaluated.

Results: The NIR images captured by the system showed nerve distributions and mesh-like patterns similar to those in the GFP images. Spectral differences were observed between nerve and non-nerve regions. This likely reflects tissue composition differences, as muscle tissue has higher transmittance and nerve tissue higher reflectance in the NIR range [4][5][6].

Segmentation based on spectral data successfully extracted nerve contours and distribution patterns, achieving ~80% classification accuracy.

Conclusion: To enable non-invasive, label-free visualization of deep nerve distributions, we developed a NIR-HSI microscope by integrating near-infrared hyperspectral imaging with a microscopic optical system. In mouse intestine experiments, spectral differences were observed between enteric nerve plexus and surrounding tissue. The results suggest that mesh-like nerve structures can be estimated and outlined using NIR images alone. Future work will focus on imaging human samples.

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Disclosure: Nothing to disclose.

PP0597

ANORECTAL DYSFUNCTION IN PATIENTS WITH MULTIPLE SCLEROSIS

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Introduction: Gastrointestinal involvement is common in patients with multiple sclerosis (MS). It is estimated that 35–54% of MS patients experience constipation, while 29–51% develop faecal incontinence (FI), with both conditions frequently coexisting[1]. The underlying pathophysiology is complex, as MS can affect the central nervous system at multiple levels. Despite its high prevalence and significant impact on quality of life, few studies have specifically addressed anorectal dysfunction in these patients. The aim of our study was to describe the clinical and manometric features of MS patients presenting with constipation and/or FI.

Aims & Methods: 26 patients with MS were included. High-resolution anorectal manometry (HRAM; Medtronic®), following the London protocol, and balloon expulsion test (BET; Matla Systems®) were performed in all subjects. Reference values were derived from 20 healthy controls assessed at our centre (Table 1).

Parameter	Value
Mean Resting Pressure (Atmospheric ref.)	86.4 ± 17.9 mmHg
Short Squeeze Pressure (Atmospheric ref.)	229.6 ± 81.6 mmHg
Long Squeeze Duration	26.2 ± 8.3 s
First Sensation	50.5 ± 25.2 mL
Second Sensation	125.4 ± 32.7 mL
Maximum Tolerated Volume	156.5 ± 36.5 mL

Table 1. HRAM Parameters in Healthy Subjects in left lateral decubitus.

Results: Faecal incontinence was observed in 21 patients (80.8%), predominantly mixed (42.3%), complete (stool and gas; 34.6%), and with Wexner scores >9 in 26.9% of cases. Constipation was reported by 15 patients (57.7%), of whom 14 (53.8%) perceived obstructive defecation. Constipation and FI coexisted in 10 patients (38%). Urinary incontinence was also present in 17 patients (65.4%).

Manometric findings revealed a tendency towards resting anal sphincter hypotonia (67.06 ± 19.85 mmHg) and reduced maximum voluntary contraction pressure (median 105.35 mmHg [IQR 86.48–149.28 mmHg]). Median contraction duration was 30 seconds (IQR 8–32 s). Rectal sensitivity assessment showed median thresholds for first sensation at 55 cc (IQR 40–70 cc), second sensation at 125 cc (IQR 100–160 cc), and maximum tolerated volume at 170 cc (IQR 135–190 cc). Sensory abnormalities were identified in 11 patients (42.3%), 4 due to hypersensitivity (15.4%) and 7 (26.9%) due to hyposensitivity.

An impaired push manoeuvre was observed in 20 patients (83.3%): type I dyssynergic defecation (DD) in 5 patients (20.8%), type II in 5 (20.8%), type III in 4 (16.7%), and type IV in 6 (25%). BET confirmed dyssynergia in 10 patients (41.7%).

Note: Results are expressed as mean ± standard deviation or median (interquartile range), depending on data distribution. Reference values are presented as mean ± standard deviation.

Conclusion: FI was the most prevalent symptom, often coexisting with constipation and obstructive defecation. The tendency observed in manometric findings towards anal hypotonia and hypocontractility, as well as the presence of dyssynergia in a significant number of patients, may allow the selection of individuals suitable for targeted therapies.

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Disclosure: Nothing to disclose.

PP0598

ORO-ANAL TRANSIT TIME IN PATIENTS WITH BOWEL SYMPTOMS: A SINGLE CENTRE STUDY

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Introduction: Gastrointestinal (GI) symptoms and their relationship with measures of GI function are incompletely understood.
Aims & Methods: We included patients investigated with OATT by use of radiopaque markers (ROM) between 2000 and 2022; one clinical audit cohort investigated because of disturbed bowel habits, and a cohort from IBS trials. In the clinical audit cohort, information on bowel habits were retrieved from referral notes and severity of abdominal pain, bloating and flatulence were rated daily by the patients during the week of the OATT measurement. In IBS patients, information on IBS subtype (IBS-C, IBS-D, IBS nonCnonD), severity of GI symptoms (GSRS), and stool frequency and consistency (Bristol stool form) from stool diaries were available. The aim of this study was to assess the outcome of oro-anal transit time (OATT) investigations in patients with disturbed bowel habits and its association with GI symptoms.
Results: We included 1142 patients (75% females) with a median age of 37(range 20-65) years with bowel symptoms. In the clinical audit cohort, slow transit was most common among patients with constipation (41%), whereas rapid transit was most frequent in patients with diarrhea (32%). In line with this, slow transit was predominantly seen in IBS-C (17%) and rapid transit in IBS-D (30%). However, in both cohorts, normal transit was most frequent.
Among patients in the clinical audit cohort, a positive correlation between OATT and the severity of bloating (r=0.17, p<0.001) was seen, but there was no association with abdominal pain (r=0.02, p=0.72) or flatulence (r=-0.10, p=0.07). In IBS patients, inverse correlations were seen between OATT and stool consistency (r=-0.38, p<0.0001), and frequency (r=-0.32, p<0.0001). In line with this, OATT was associated with the severity of diarrhea (r=-0.16, p<0.05), and constipation (r=0.27, p<0.001).
In addition, a weak, but significant association was seen with satiety (r=0.16, p<0.05) but not with abdominal pain (r=0.06; p=0.23) or bloating (0.08; p=0.09).

Patients	Rapid transit	Normal transit	Slow transit
Clinical audit (n=457)			
Patients with constipation (n=292)	4%	55%	41%
Patients with diarrhea (n=99)	32%	64%	4%
Patients with undefined bowel habit abnormality (n=66)	11%	68%	21%
IBS trials (n=685)			
IBS-C (n=198)	6%	77%	17%
IBS-D (n-228)	30%	68%	2%
IBS NonCnonD (n=205)	14%	81%	5%
IBS without subtyping information (n=56)	28%	61%	11%

Table.

Conclusion: Abnormal OATT is prevalent among patients with disturbed bowel habits and is a pathophysiologic mechanism of importance for the understanding of stool frequency and consistency, and the severity of diarrhea and constipation but less so with other GI symptoms.
Disclosure: Nothing to disclose.

PP0599

SCREENING FOR ARFID IN IRRITABLE BOWEL SYNDROME: CLINICALLY OR USING THE NIAS-FR QUESTIONNAIRE?

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Introduction: Approximately 15-19% of patients suffering from irritable bowel syndrome (IBS) have an eating disorder (ED) based on a screening score or dietary assessment (1).
Avoidant/restrictive food intake disorder (ARFID), a long-recognised eating disorder in children, is now recognised in adults. In a Swedish study, 13.2% of patients suffering from IBS reported severe dietary avoidance or restriction, suggestive of ARFID (2).
A screening questionnaire for this disorder has recently been validated in French (NIAS-Fr, for Nine Item ARFID Screen in French) (3).
Aims & Methods: This study aims to assess the prevalence of ARFID in IBS in France, by comparing the NIAS-Fr questionnaire and the assessment by a specialised dietitian, and to look for links between its presence and the clinical phenotype of patients.
This study was conducted between February and September 2024 in consecutive patients suffering from IBS and evaluated in a multidisciplinary day hospital structure. Clinical characteristics (age, gender, BMI, IBS subtype and its severity [IBS-SSS], presence of functional dyspepsia, diet and disease duration) and presence of ARFID (clinically or one abnormal NIAS-Fr subscale) were collected.
A dietitian assessed the presence of "clinical ARFID". The NIAS subscales allow for the diagnosis of patients as "picky eaters" (score ≥10), with "poor appetite" (score ≥9) and/or with a "fear of eating" (score ≥10). Patients with a history of ED diagnosed by a psychiatrist and/or an abnormal ED screening score (SCOFF ≥ 3 and/or global EDE-Q8 ≥ 4.0) were excluded from the analysis.
Results: Of the 95 patients with IBS according to Rome IV, 55 (57.9%) had at least one positive NIAS-Fr subscale, with 10 (10.5%) having an abnormal "picky eater" subscale, 16 (16.8%) an abnormal "poor appetite" subscale, and 49 (51.6%) an abnormal "fear of eating" subscale. Of these 95 patients, 15 were excluded from the analysis because of a history of eating disorders or an abnormal screening score.
We therefore included 80 patients for analysis (77.5% women; mean age 48.5 ± 16.8 years, BMI 22.6 ± 4.6 kg/m^2; 40.0% IBS-D, 38.8% IBS-C, and 20.0% IBS-M; 51.3% had associated dyspepsia; 62.5% were on a special diet at the time of assessment). The mean IBS-SSS score was 319.5 ± 88.2 points (severe forms in 61.2%). Among these 80 patients, 43 (53.8%) had at least one positive NIAS-Fr subscale, including 10.0% of "picky eaters," 16.3% of "poor appetite," and 48.8% of "fear of eating." Patients with at least one positive NIAS-Fr subscale had more frequent functional dyspepsia than those with a negative NIAS (65.1 vs. 35.1%, p = 0.013), but there were no differences in the other parameters assessed. According to the dietitian's assessment, 19 (23.8%) patients in our cohort had clinical ARFID, i.e., 44.2% of those with at least one positive NIAS-Fr subscale. In an

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IBS population, the specificity of the NIAS-Fr questionnaire for diagnosing clinical ARFID was 55.8% (95% CI: 41.1-69.6). None of the assessed parameters distinguished patients with and without clinical ARFID (p > 0.05).

Conclusion: ARFID symptoms are found in more than half of IBS patients treated in a tertiary care centre based on an abnormal NIAS-Fr screening questionnaire, after excluding other eating disorders. These patients frequently have associated functional dyspepsia and a normal BMI. Based on a dietary assessment, clinical ARFID is found in approximately one-quarter of IBS patients. The specificity of the NIAS-Fr in IBS is too low to allow a diagnosis of ARFID in this population.

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Disclosure: Nothing to disclose.

PP0600

KNOWLEDGE AND MISCONCEPTIONS OF PATIENTS WITH IRRITABLE BOWEL SYNDROME ENROLLED IN A THERAPEUTIC PATIENT EDUCATION PROGRAM

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Introduction: Irritable bowel syndrome (IBS) is a common disorder of gut-brain interaction affecting 5 to 10% of the population, combining chronic abdominal pain and altered bowel transit. Its varied and complex mechanisms make management difficult and can be responsible for impaired quality of life. Patient education is recommended for managing chronic diseases. As part of a patient education program, the first step is to assess patients' knowledge and needs by conducting a shared educational assessment.

Experiments conducted in various countries have shown that therapeutic patient education (TPE) can be effective, reducing severity, improving quality of life, and reducing care seeking. These studies have also shown that patients often have a lack of knowledge and/or misconceptions about IBS (1).

Aims & Methods: This study aimed to assess the baseline knowledge level of patients enrolled in a patient education program. Patients with IBS were selected for participation in a TPE program following a multidisciplinary day hospital for IBS. All patients had to have seen a general practitioner and/or a gastroenterologist outside the team without sufficient improvement in their condition. The TPE was based on a document developed in collaboration with a patient organisation (APSSII).

This program was divided into five 1-hour sessions (training and discussion time) conducted by different professionals (neurogastroenterologists, clinical psychologist, dietitian) covering the following points:

1. Understanding my illness (epidemiology, diagnosis, mechanisms),
2. Talking about my illness and expressing my needs (interaction with loved ones, work, doctor, school, rights and support);
3. Organizing my daily life, improving my quality of life (impact of IBS on QoL, work, exercise, intimate life);
4. Taking advantage of different treatments and care options (placebo, adverse effects, symptomatic and long-term treatment, alternative treatments);
5. The role of eating and diets (balanced diet, lactose-free, gluten-free, low-FODMAP, food allergies, risks of diets). Before the start of each session, questionnaires testing participants' knowledge on the session topic were completed.

Results: From November 2023 to September 2024, 135 questionnaires completed by 43 patients with IBS were included. 70% were women, mean age 45 ± 18 years, subtypes based on IBS transit: IBS-C 55%, IBS-D 33%, IBS-M 12%, with a mean severity (IBS-SSS) of 331 ± 72 points, EQ5D health score of 50 ± 21. Among the most common misconceptions, 25% believed that a diagnosis by endoscopy was possible, 48% believed the disease could be complicated by IBD, and 13% by colon cancer.

Regarding treatments, 11% of patients believed that the placebo effect only concerned medications, 58% believed that faecal microbiota transplantation could currently be offered in clinical routine for IBS care, 78% believed that hypnotherapy was only effective if you believe in it, and 68% believed that multi-strain probiotics were more effective than single-strain ones.

Conclusion: This study, conducted in France on patients with moderate or severe forms of IBS, shows a highly variable level of knowledge about their disease depending on the questions, with some significant misconceptions that need to be addressed. These results justify the implementation of therapeutic education for these patients.

A study is underway to verify the improvement in patients' knowledge and the existence of a positive impact of this strategy on symptom severity and quality of life.

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Disclosure: Nothing to disclose.

PP0601

INTEGRATING ANORECTAL MANOMETRY, BALLOON EXPULSION AND DEFECOGRAPHY: INSIGHTS INTO DIAGNOSING PELVIC FLOOR DYSFUNCTION

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Introduction: Defecatory dysfunction (DD) is a common cause of chronic constipation resulting from functional abnormalities or structural pelvic pathologies. Accurate diagnosis requires combining anorectal manometry (ARM), balloon expulsion test (BET), and defecography. This study evaluates the diagnostic utility of these modalities and explores uncertainties in their performance and interpretation.

Aims & Methods: This retrospective study included 325 adult patients assessed for DD between 2020 and 2023. All patients went through ARM, BET in the left lateral position, and defecography. Statistical associations between test outcomes were analyzed to assess diagnostic concordance and significance.

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Results: A strong correlation was observed between ARM and defecography, with 65% of patients with normal anal relaxation on ARM achieving normal rectal evacuation on defecography ($p < 0.0001$). Conversely, patients with paradoxical contraction during ARM demonstrated a higher likelihood of evacuation failure. BET demonstrated high specificity but limited sensitivity in association with relaxation on ARM and evacuation on defecography. BET failure did not demonstrate a significant association with the presence of pelvic floor pathologies.

Conclusion: Combining ARM, BET, and defecography provides a comprehensive framework for diagnosing DD, addressing its functional and structural components. This integrated approach facilitates targeted interventions, ultimately improving clinical outcomes.

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Chronic constipation is a common disorder with multifactorial pathophysiology, including pelvic floor dysfunction (Bharucha & Lacy, 2020). Symptomatic pelvic floor disorders affect a significant proportion of women (Nygaard et al., 2008). High-resolution anorectal manometry (HRAM) and rectal balloon expulsion tests are critical for diagnosing defecatory disorders (Prichard et al., 2017; Bharucha & Coss-Adame, 2022). Standardized protocols, including the London Classification, have improved the diagnostic accuracy of anorectal functional tests (Carrington et al., 2020; Carrington et al., 2018). Imaging modalities such as magnetic resonance defecography complement functional testing (Reiner et al., 2011; Carter et al., 2020). Variations in body position influence anorectal test results (Su et al., 2019; Wu et al., 2017; Tsunoda et al., 2022; Ratuapli et al., 2013). Comparative studies emphasize the complementary role of manometry, balloon expulsion, and defecography (Blackett et al., 2022; Heinrich et al., 2015). Recent advances suggest HRAM improves detection of specific defecatory mechanisms (Deb et al., 2022; Sharma et al., 2020), supported by international consensus guidelines (Wald et al., 2021; Rao & Tetangco, 2020; Lee & Bharucha, 2016).

Disclosure: No conflict of interest.

PP0602

POSITIVE EFFECTS OF *L PLANTARUM* 299V AND *L PLANTARUM* GOS42 ON THE GASTROINTESTINAL FUNCTION IN ADULTS WITH OCCASIONAL CONSTIPATION. A RANDOMIZED, DOUBLE-BLIND, PLACEBO CONTROLLED, PARALLEL CLINICAL STUDY

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Introduction: Constipation is a common gastrointestinal concern associated with less frequent defecations, with reduced faecal bulk and harder stools, or both. It greatly affects quality of life, with a significant impairment of both mental and physical components. One way to relieve constipation can be by intake of probiotics. *Lactiplantibacillus plantarum* 299v (LP299V®) is one of the most clinically studied *L plantarum* strains and has been shown to have positive effects on IBS symptoms in several clinical trials (Arvidsson Nordström et al. 2021).

In the present study, the effect of LP299V and a small amount of another *L plantarum* strain (LPGOS42) on occasional constipation was investigated. Gut transit time was also evaluated, using a novel marker (Asnicar et al. 2021).

Aims & Methods: The aim of the study was to investigate if LP299V and LPGOS42 (LP) could improve gastrointestinal function in adults with occasional constipation. After a 2-week run-in period, 100 subjects were randomized to receive either LP or placebo capsules for 8 weeks. The change in frequency of complete and spontaneous bowel movements (CSBM), frequency of bloating and flatulence, stool consistency, digestive function (modified Gastrointestinal Symptom Rating Scale (GSRS)) and the quality of life (Patient Assessment of Constipation Quality of Life questionnaire (PAC-QoL)) were followed. The gut transit time was assessed with a blue dye method (Asnicar et al. 2021).

Results: The number of weekly complete and spontaneous bowel movements increased significantly over time in both groups (LP +2.4, placebo +1.9, p=0.182 between groups at week 8). Stool consistency transitioned from hard to softer stools in both groups, with the mean score significantly higher after 8 weeks in the LP group, indicating softer stools in this group. The LP group also had significantly more bowel movements without hard lumpy stools compared to the placebo group at week 8. The frequency of flatulence was reduced over time in the LP group, reaching a significantly lower frequency of flatulence in the LP group vs placebo in week 8.

The subjects scored low for most of the symptoms in the GSRS, indicating that the study population were healthy subjects without abdominal pain, reflux, diarrhea or indigestion. The only subscale in GSRS where the mean score was over 1 (= more than occasional problems) was constipation and this decreased significantly over time in both groups, with a more pronounced decrease seen in the LP group (p=0.073 between groups at week 8).

The total score for PAC-QoL improved significantly over time in both groups, with no difference seen between the groups. The highest score detected among the four subscales was for satisfaction and it improved significantly more in the LP group compared to the placebo group (p=0.007 between groups in week 8).

Measuring the gut transit time with a blue dye method gave large differences between subjects and no significant changes over time or between groups could be detected.

Conclusion: In conclusion, this study demonstrates that intake of probiotics (*L plantarum* 299v and *L plantarum* GOS42) helps to alleviate constipation and flatulence, as well as improves quality of life in healthy subjects with occasional constipation.

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Disclosure: G Önning, C Montelius and C Vegge work at Probi AB, that owns the bacterial strains investigated in the study.

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PP0603

HOW IS POLYETHYLENE GLYCOL PRESCRIBED FOR CHRONIC CONSTIPATION AMONG GASTROENTEROLOGISTS AND GENERAL PRACTITIONERS? RESULTS FROM A NATIONAL ITALIAN SURVEY

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Introduction: Chronic constipation (CC) is a frequent gastrointestinal disorder which significantly affects patients' quality of life. International guidelines recommend the use of polyethylene glycol (PEG) as part of the first-line management of CC after the failure of diet and lifestyle modifications. However, limited data are available on how PEG is used in daily clinical practice. Therefore, a national survey among gastroenterologists (GEs) and general practitioners (GPs) was conducted to evaluate its use in the management of CC.

Aims & Methods: Members of Italian Association of Hospital Gastroenterologists and Endoscopists (AIGO) and Italian Federation of General Practitioners (FIMMG) were invited to complete a 44-item online survey. The questionnaire was structured to explore several dimensions regarding the use of PEG (*e.g.* type of PEG used, formulations and dosages recommended, etc.).

Results: The survey was completed by 1188 physicians (317 GEs, 871 GPs). GEs were significantly younger (mean age 44.9 vs. 51.0 years, p<0.01) and more likely to have less than 10 years of experience compared to GPs.

PEG was prescribed as part of first-line treatment in addition to dietary and behavioral advice more frequently by GEs than GPs (92.4% vs. 82.9%, p<0.001). GPs preferred to prescribe PEG in the early morning or evening before bedtime compared to GEs (p<0.001), which preferred mid-morning or midafternoon (p<0.01). GPs more often recommended taking it away from other drugs and/or supplements compared to GEs (76.7% vs 61.5%, p<0.001).

GEs considered PEG products to be taken continuously more often than GPs (36.9% vs 18.6%, p<0.001) and were more inclined to avoid setting a fixed treatment duration (62.8% vs 38.2%, p<0.001), while GPs often recommend a short-term or an on-demand use. GEs also more often preferred to discontinue PEG gradually compared to GPs (74.8% vs. 63.0%, p<0.001)

PEG's formulation choices varied: GEs favored the powder formulation less than GPs (58.7% vs 78.9%, p<0.001), and more often started therapy at higher dosages (Table 1). Moreover, GEs gave more importance on palatability compared to GPs (71.9% vs 60.4%, p<0.001).

Safety perceptions were similarly high across the groups, GEs were more likely to prescribe PEG than GPs to patients with comorbidities such as chronic kidney disease (75.7% vs 64.0%, p<0.001) and during pregnancy/lactation (71.3% vs 48.8%, p<0.001).

	Overall (N = 1180)		GEs (N = 317)			GPs (N = 871)			Difference between GEs and GPs		p-value
	N	%	N	%	95% CI	N	%	95% CI	%	95% CI	
In the treatment of constipation, in addition to dietary and behavioral advice, do you prescribe a PEG-based product as first-line treatment?											
Yes	1015	85.4	293	92.4	89.5 to 95.3	722	82.9	80.4 – 85.4	9.5	5.7 – 13.4	<0.001
When prescribing a PEG-based product, what initial dosage do you use?											
Customized powder dose	263	22.1	49	15.4	11.5 to 19.4	214	24.6	21.7 to 27.4	-9.2	-14.1 to -4.2	0.001
10 g/day	344	29.0	90	28.4	23.4 to 33.4	254	29.1	26.1 to 32.2	-0.7	-6.6 to 5.0	0.796
13.8 g/day	438	36.9	121	38.2	32.8 to 43.5	317	36.4	33.2 to 39.6	1.8	-4.5 to 8.0	0.575
17.4 g/day	104	8.7	38	12.0	8.4 to 15.6	66	7.6	5.8 to 9.3	4.4	0.4 to 8.4	0.017
34.8 g/day	39	3.3	19	6.0	3.4 to 8.6	20	2.3	1.3 to 3.3	3.7	0.9 to 6.5	0.002

Table 1. Prescription of PEG as part of first line management of CC and PEG's initial dosage answers.

Conclusion: To the best of our knowledge, this is the first national survey to assess how PEG is prescribed by GEs and GPs in daily clinical practice. The survey highlighted a certain heterogeneity in prescription behaviors. GEs tended to adopt longer-term therapeutic strategy, starting with higher doses. Moreover, they paid greater attention to PEG palatability and were more confident on PEG safety in patients with chronic comorbidities. These findings underscored the need for shared knowledge and approaches to optimize constipation management across different care settings.

Disclosure: Nothing to disclose.

PP0604

THE ACCURACY OF A SIMPLIFIED ANORECTAL MANOMETRY PROTOCOL IN DIAGNOSING FUNCTIONAL DEFECATION DISORDERS: ARE 5-SECONDS PUSHES RELIABLE?

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Introduction: Anorectal manometry (ARM) is required to diagnose functional defecation disorders (FDD) in constipated patients. The International Anorectal Physiology Working Group introduced into clinical practice a standardized sequence of manoeuvres to perform ARM (London Protocol) (1). The London protocol suggests performing three push attempts,

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each of 15s duration, to assess recto-anal coordination and therefore to diagnose FDD, with a consensus <75% between experts. However, some papers consider a 5s push sufficient to detect FDD (2,3).

Aims & Methods: Therefore, the aim of this study was to assess the diagnostic accurac of a simplified ARM protocols featuring three pushes attempts (each of 5s) in comparison with IAPWG recommendations.

High-Definition ARM (HD-ARM) traces from 105 constipated patients (82 F, mean aged 53.5 ± 16.7 years) who had performed HD-ARM for clinical practice were retrospectively reviewed. HD-ARM were performed according to the IAPWG Protocol. All patients underwent the balloon expulsion test at the end of HD-ARM as suggested by IAPWG protocol. HD-ARM traces were re-evaluated considering the first five seconds of each push (as if each push featured only five seconds).

New diagnosis was made based on this modified protocol and compared with the one made according to London Protocol, to evaluate whether these modifications could affect the final diagnosis.

Results: According to the London Protocol, the qualitative best pushes were the first, the second or the third in 35.2%, 28.6% and 36.2% of patients, respectively. Eighty-seven patients (82.9%) failed the balloon expulsion test.

Mean rectal pressure did not differ between the modified protocol and the London protocol (37.1 ± 23 vs 34.3 ± 23.2, p=0.38). However, we observed a significant increase in the mean anal pressure with the modified version of the protocol compared to the London protocol (75.6 ± 33.1 vs 90.8 ± 37.3, p<0.002).

Regarding the diagnostic accuracy, we observed a moderate concordance between the modified protocol and the London protocol k=0.77, Table 1). In details, if we had used this modified version of protocol, 18 patients (17.1%) would have changed their final diagnosis (Table 1).

	Absormal expulsion with dyssynergia	Abnormal expulsion with normal manometric pattern of rectoanal co-ordination	Abnormal expulsion with poor propulsion	Abnormal expulsion with poor propulsion and dyssynergia	No disorders of rectoanal co-ordination	Normal expulsion with abnormal manometric pattern of rectoanal co-ordination	Total
Absormal expulsion with dyssynergia	23	4	1	0	0	0	28
Abnormal expulsion with normal manometric pattern of rectoanal co-ordination	0	5	0	0	0	0	5
Abnormal expulsion with poor propulsion	0	1	11	0	0	0	12
Abnormal expulsion with poor propulsion and dyssynergia	6	0	3	33	0	0	42
Normal expulsion with abnormal manometric pattern of rectoanal co-ordination	0	0	0	0	3	15	18
Total	29	10	15	33	3	15	105

Table 1. Cohen's K diagnosis concordance according to London Protocol (k=0.773)

Conclusion: Based on our results, a simplified ARM protocol using three 5s pushes shows moderate diagnostic agreement with the standard London protocol. While rectal pressures remained similar, anal pressures increased significantly, and 17.1% of patients received a different diagnosis. Therefore, the simplified protocol could reduce the diagnostic accuracy for FDD.

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PP0605

THREE IS NOT A CROWD: IMPROVING THE STUDY OF CHRONIC CONSTIPATION

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Introduction: High-definition anorectal manometry (HD-ARM), balloon expulsion test (BET) and magnetic resonance defecography (MR-defecography) are important tools in studying constipation.

Aims & Methods: Evaluate the prevalence of functional defecation disorders and structural anorectal alterations in patients with constipation and compare the findings of HD-ARM and BET with MR-defecography.

Retrospective cohort study including patients with functional constipation (FC) or irritable bowel syndrome with constipation (IBS-C) submitted to HD-ARM, BET and MR-defecography.

Patients were divided into three groups:

1. Normal anorectal evacuation pattern in HD-ARM and BET;
2. Abnormal anorectal evacuation pattern and BET;
3. Normal anorectal evacuation pattern and abnormal BET, or abnormal anorectal evacuation pattern and normal BET.

Results: A total of 64 patients were included, 49 with FC (76.6%) and 15 with IBS-C (23.4%). There were 20 patients in group 1 (31.3%), 11 in group 2 (17.2%) and 33 in group 3 (51.5%). In MR-defecography, 47 patients had relevant structural anorectal alterations (73.4%) and 15 had an abnormal contraction of the puborectalis during the defecatory maneuver (23.4%). Nineteen patients were diagnosed with dyssynergic defecation (29.7%). These patients had significantly higher residual anal pressure and lower percentage of anal relaxation during defecation (109.6±43.2 *vs* 85.8±29.7 mmHg, p=0.038 and 9% *vs* 24%, p=0.018, respectively), as well as a lower differential of rectoanal pressure (-32.2±31.4 *vs* 3.0±29.6 mmHg, p<0.001). Patients with structural anorectal alterations had a significantly higher body mass index (BMI) (27.4±4.5 *vs* 23.5±3.3, p=0.002), and all obese patients had structural alterations.

Comparing the findings of HD-ARM and BET with MR-defecography: in group 1, 15 patients had structural alterations (75.0%); in group 2, 8 patients had structural alterations (72.7%); in group 3, 22 patients only had structural alterations (66.7%), 6 only had dyssynergic defecation (18.2%), 2 patients had both (6.0%) and 3 did not have any findings (9.1%). In 24.2% of patients with inconclusive findings in HD-ARM and BET, MR-defecography allowed the diagnosis of dyssynergic defecation.

Conclusion: Patients with structural anorectal alterations had a significantly higher BMI. Almost three quarters of patients with FC and IBS-C had relevant structural anorectal alterations and one third had dyssynergic defecation. Therefore, it is important to combine MR-defecography with HD-ARM and BET in the study of constipation.

Disclosure: Nothing to disclose.

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PP0606

ANORECTAL MOTOR AND SENSORY ABNORMALITIES IN SOLITARY RECTAL ULCER SYNDROME AND ULCERATIVE COLITIS: PATHOPHYSIOLOGIC DIVERGENCE AND THERAPEUTIC CONVERGENCE

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Introduction: SRUS and UC can affect anorectal function, yet no study has directly compared HRAM findings and the efficacy of biofeedback therapy in these two conditions.

Aims & Methods: This study aimed to compare HRAM findings and the efficacy of biofeedback therapy in these two conditions to better understand the gut-brain disturbance in Solitary Rectal Ulcer Syndrome (SRUS) and Ulcerative Colitis (UC). 20 SRUS, 5 UC, and 20 healthy controls were included. HRAM was performed using a solid-state catheter following the London Protocol, with analyses conducted via the Laborie MMS system. Biofeedback therapy was delivered using a solid-state probe via the GatlTec system. Data on HRAM parameters, biofeedback, demographics, symptoms, and diagnoses were analyzed.

Results: Among SRUS patients, 9 (45%) were female, with a median age of 22.5 years (IQR: 15–73). UC patients included 3 (60%) females, with a median age of 28 years (IQR: 21–73). Healthy controls comprised 10 (50%) females, with a mean age of 25 years (±10 years). HRAM revealed that 14 (70%) SRUS patients had hypertensive, hypercontractile anal sphincters, while 3 (60%) UC patients exhibited hypotensive, hypocontractile anal sphincters. Mean squeeze anal pressure and endurance squeeze time were significantly higher in SRUS than in UC (132 ± 14.5 mmHg vs. 63.7 ± 20.1 mmHg; $p < 0.024$ and 14.3 ± 3 sec vs. 3.5 ± 2.5 sec; $p < 0.018$, respectively). Push relaxation percentage of the anal sphincter was significantly greater in SRUS than in UC (10 ± 6% vs. -7 ± 1.5%; $p < 0.011$). Both groups exhibited defecation dyssynergy. 15 (75%) SRUS patients displayed rectal hypersensitivity, while 4 (80%) UC patients exhibited rectal hyposensitivity. Sensory thresholds for the first constant sensation volume and defecatory desire volume were significantly lower in SRUS than in UC (19.5 ± 3 mL vs. 30 ± 15 mL; $p < 0.03$ and 70.6 ± 6 mL vs. 80 ± 23 mL; $p < 0.04$, respectively). Biofeedback therapy led to significant improvements across the anorectal motor and sensory functions in both groups, however, SRUS patients demonstrated a greater overall responsiveness.

	Solitary Rectal Ulcer Syndrome (SRUS)	Ulcerative Colitis (UC)	P	Healthy Controls
Resting Anal pressure (mmHg)	71.7 (6.8)	57.3 (11.8)	0.16	65 (15.2)
Squeeze Anal Pressure (mmHg)	132 (14.5)	63.7 (20)	0.02	122(15)
Endurance Squeeze (Sec)	14.2 (3.2)	3.3 (2.8)	0.01	25(5.6)
Anal Pressure (mmHg) „during cough“	101 (13)	78.7 (7)	0.07	90(7.6)
Ballon Expulsion Test (Sec)	42.6 (8)	167(103)	0.05	32(3.5)
Anal Residual Pressure „during push“ (mmHg)	64.5 (8.6)	51 (6.8)	0.13	59(10.8)
Push Relaxation (%)	9.5 (6.6)	-7.3 (1.4)	0.01	32(7.8)
First constant sensation volume (ml)	19.4 (3)	30 (15.3)	0.02	23(6.4)
Defecatory desire volume (ml)	70.5 (6)	80 (23)	0.03	73(9.2)

Table 1. Mean values of High Resolution Ano-Rectal Manometry metrics for Solitary Rectal Ulcer Syndrome, Ulcerative Colitis, and healthy controls.

Conclusion: SRUS and UC show distinct anorectal dysfunction patterns, SRUS with hypercontractile sphincters and rectal hypersensitivity, and UC with hypocontractile sphincters and hyposensitivity. Both groups had defecatory dyssynergia, but SRUS patients responded better to biofeedback. These findings support tailored physiologic assessment and therapy in defecatory disorders. Future research should include larger sample sizes to validate these findings and explore the long-term outcomes of biofeedback therapy in both SRUS and UC patients.

Disclosure: Nothing to disclose.

PP0607

THE PREVALENCE AND BURDEN OF DISORDERS OF GUT-BRAIN INTERACTION (DGBI) BEFORE VERSUS AFTER THE COVID-19 PANDEMIC

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Introduction: COVID-19 infection may increase the risk of developing Disorders of Gut-Brain Interaction (DGBI). However, the extent of this effect on a population level is poorly understood. We performed a two-country survey to address this issue.

Aims & Methods: A population-based Internet survey with pre-defined demographic quotas was conducted across the UK and USA in 2017 (pre-pandemic, n=4050) and repeated in 2023 (post-pandemic, n=4002). The surveys included the Rome IV diagnostic questionnaire, and questions about non-gastrointestinal somatic symptoms, anxiety and depression, quality of life, and healthcare utilization. The 2023 survey also included questions regarding COVID-19 infection and illness history.

Results: The overall DGBI prevalence, i.e., meeting diagnostic criteria for at least one DGBI, has significantly increased from the pre- to post-pandemic era (38.3% vs. 42.6%, OR 1.20, 95% C.I 1.09-1.31), with similar findings independently noted in the UK and USA. The rise in DGBI was observed within the oesophageal (8.8% vs. 10.1%, OR 1.16), gastroduodenal (11.9% vs. 16.4%, OR 1.45), and bowel domains (30.1% vs. 32.5%, OR 1.12). The two most widely investigated DGBI showed large post-pandemic prevalence increases, with functional dyspepsia rising from 8.3% to 11.9% (OR 1.48) and irritable bowel syndrome from 4.7% to 6.0% (OR 1.31).

In multivariable analysis, factors significantly associated with having DGBI in the post-pandemic era included younger age, female sex, anxiety, depression, medium-to-high somatic symptom severity, increasing number of COVID-19 infections, experiencing abdominal pain or diarrhoea during COVID-19 infection, and suffering with long-COVID.

Individuals with DGBI in the post-pandemic era, in particular those with long-COVID, reported reduced quality of life and higher mood disturbances, somatic symptom reporting and healthcare use than individuals with DGBI in the pre-pandemic era.

Conclusion: The population prevalence and burden of DGBI have increased following the COVID-19 pandemic. Healthcare services and research funding bodies need to adapt to this post-COVID surge in DGBI and address how to best manage this patient group.

Disclosure: Nothing to disclose.

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PP0608

PATIENTS WITH IRRITABLE BOWEL SYNDROME HAVE GREATER SYMPTOMS BURDEN THAN OTHER BOWEL DISORDERS: ANALYSES FROM THE ROME FOUNDATION GLOBAL

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Introduction: Functional bowel disorders are characterized by the presence of altered bowel habits with or without abdominal pain.[1] Irritable bowel syndrome (IBS) is diagnosed based on recurrent abdominal pain; however, distinctions in the clinical phenotypes of IBS subtypes and other functional bowel disorders (FBDs), such as functional diarrhea (FD) and functional constipation (FC), remain underexplored.[2,3]

Aims & Methods: Aims: This study aimed to characterize the clinical presentation of IBS with predominant constipation (IBS-D) and IBS with predominant diarrhea (IBS-C), differentiating them from FD and FC, while recognizing the overlapping symptoms and the continuum between these conditions. Additionally, we examined the prevalence of psychological disorders and extra-intestinal comorbidities and identified potential mediators linking IBS to anxiety and depression.

Methods: Using data from the Rome Foundation Global Epidemiology Study, we analyzed 54,127 participants across 26 countries, including 2,195 patients with IBS. Clinical, dietary and psychological characteristics were assessed using validated questionnaires. Subgroup analyses compared IBS subtypes with FBDs, while mediation analyses (using the PROCESS macro and Baron and Kenny's method[4]) identified mediators of IBS-psychological distress relationship.

Results: As compared to patients with IBS-D, patients with IBS-C reported more severe pain characteristics, including prolonged pain episodes (50.7% vs. 47.0% of the time with pain lasting >30 minutes, p=0.030) and pain episodes that escalated to a sustained severe intensity (44.2% vs. 38.1%, p=0.026), frequently accompanied by back pain (36.7% vs. 30.9%, p<0.001). Regular use of prescribed pain medications was also more frequent in IBS-C (39.3%) than IBS-D (31.8%, p=0.004). In contrast, as compared to IBS-C, patients with IBS-D showed a stronger association of pain with changes in stool consistency (70% vs. 60% of time with pain, p<0.001) and frequency (mean 61.3% vs. 57.8% of time with pain, p=0.008). These associations remained significant in multiple regression models, which confirmed that IBS-C is more strongly linked to prolonged and escalating pain episodes, while IBS-D is more closely linked to pain associated with changes in stool frequency and consistency. As compared to FD, IBS-D patients reported greater dissatisfaction with bowel function (57.3±28.4 vs. 44.2±29.8, p<0.001), and more frequent loose stools (without constipation treatments) (60.0% vs. 50.0% of bowel movements, p<0.001), particularly after meals (60.0% vs. 40.0% of loose stools, p<0.001). IBS-C shows more severe constipation-related symptoms compared to FC (p<0.001). Mediation analyses revealed that concerns about bowel function, embarrassment discussing it, and work-related impairments partially mediated the relationship between IBS and psychological distress.

Conclusion: IBS-C is associated with more severe pain episodes, including prolonged and escalating pain, while in IBS-D pain is more strongly linked to bowel habit changes.

These findings highlight the distinct clinical profiles of IBS subtypes and emphasize the need for personalized, multimodal management strategies. Identifying mediators of psychological distress in IBS can guide targeted interventions, improving patient outcomes.

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PP0609

SURVEY OF MANAGEMENT HABITS & EMOTIONAL IMPACT IN PATIENTS WITH SELF-REPORTED IRRITABLE BOWEL SYNDROME (IBS)

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Introduction: IBS is a a condition which is highly denigratory to patients' quality of life, is often under-diagnosed and is managed through a mixture of diet/lifestyle modifications, treatments and psychological interventions. The following piece of research aims to understand the management patterns and emotional impact of the condition in the IBS population.

Aims & Methods: Surveys were deployed in ten countries (n=1,250 per country, total n=12,500). People were invited to complete the survey if they were aged 18-75, suffering from at least one of the following; constipation, diarrhoea, wind/bloating, abdominal pain, irregular bowel movements or 'IBS'. With at least one incident in the past month.

Results: 9% of the surveyed population reported IBS, with 78% of this group having a diagnosis from a healthcare professional. Key triggers of symptoms were; stress (48%), food allergy/intolerance/sensitivity (20%), poor sleep (19%), infection (18%).

37% used medication, 31% used biotics, 22% diet and home remedies, 18% used supplements. Dissatisfaction with regular treatment were that most treatment options only treat the symptom but not the underlying cause, with females managing their symptoms through diet/lifestyle changes more than males.

The emotional impact in the IBS is very high in prevalence and broad in scope. 71% of IBS sufferers felt worried, anxious or stressed about their symptoms. 62% felt frustrated and fed up. 39% felt embarrassment and negative impact to their confidence. Only 5% were not bothered by their symptoms.

Conclusion: The survey gave insight into management patterns of IBS patients, demonstrating a variety of treatments used. Surprisingly, a high number of IBS patients use biotics despite low quality of evidence. It was

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evident that IBS is a condition with a high emotional toll. Further qualitative research should delve further into these insights to further understand the patient population.

Reference:

Quantitative U&A 2025- Lower GI - IBS deep dive

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PP0610

DISTINCT VOLATILE ORGANIC COMPOUND SIGNATURES IN EXHALED BREATH DIFFERENTIATE IRRITABLE BOWEL SYNDROME PATIENTS FROM HEALTHY CONTROLS USING GAS CHROMATOGRAPHY-QUADRUPOLE TIME-OF-FLIGHT MASS SPECTROMETRY

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Introduction: Non-invasive biomarkers that enable accurate diagnosis, patient stratification, and therapy monitoring in irritable bowel syndrome (IBS) are still lacking in clinical practice. Volatile organic compounds (VOCs) produced by the host metabolism, inflammatory processes, and the gut microbiota are present in exhaled breath and have recently gained attention as promising biomarker candidates. Prior research (Van Malderen et al. (2020)) demonstrated that breath VOC profiling could accurately distinguish IBS patients from healthy controls using multicapillary column/ion-mobility spectrometry (MCC/IMS).

Aims & Methods: The present study aimed to assess the potential of high-resolution gas chromatography-quadrupole time-of-flight mass spectrometry (GC-QTOF-MS) profiling of breath VOCs to define distinct compound classes that enhance the mechanistic understanding of breath-based biomarkers in IBS. To this end, breath samples from 12 IBS patients and 11 healthy controls were collected, together with matched ambient air samples and were analyzed using GC-QTOF-MS.

Breath samples were corrected by background subtraction and least-absolute-shrinkage-and-selection-operator (LASSO) regression was applied with leave-one-out cross-validation (LOOCV) to identify discriminative VOC signatures. The feature set was subsequently filtered to exclude sporadically detected VOCs and compounds of exogenous (non-host) origin. Finally, the remaining VOCs were mapped to putative pathways.

Results: IBS patients had a mean ± SD age of 36 ± 15.3 years, compared with 56 ± 15.4 for the healthy controls. The male-to-female ratio was 3:9 in the IBS cohort and 5:6 among the healthy controls. Within the IBS group, 7 patients had diarrhea-predominant IBS, 4 had constipation-predominant IBS, and 1 had mixed-type IBS. The LASSO regression model, validated with LOOCV, demonstrated high discriminative performance in differentiating IBS patients from healthy controls, with the corresponding model metrics in Table 1.

Furthermore, the feature selection identified a panel of discriminative VOCs - predominantly alkanes, aldehydes, and ketones - that were up-regulated in the IBS breath profiles. These elevated compound classes could putatively be linked to mechanisms relevant to IBS pathophysiology, including altered microbial metabolism, increased oxidative stress, and low-grade inflammation.

Metric	LASSO-LOOCV
Sensitivity	91.3%
Specificity	81.8%
PPV	77.8%
NPV	64.3%
Accuracy	69.6%
AUC	0.88

Table 1: Model metrics.

Conclusion: Volatomic breath profiling in IBS, shows great potential for discriminating IBS from healthy controls and for defining clinically relevant biomarkers composed of specific VOCs. Nonetheless, the limited sample size might risk model overfitting and introduces uncertainty in the estimates. Further research including larger, confounder-controlled cohorts is necessary, along with the targeted validation of the identified VOCs to further confirm their mechanistic role in IBS.

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Van Malderen K, Hanning N, Lambrechts H, Haverhals T, Van Marcke S, Ceuleers H, De Man JG, De Winter BY, Lamote K, De Schepper HU. Volatile organic compound profiling as a potential biomarker in irritable bowel syndrome: A feasibility study. Front Med (Lausanne). 2022 Aug 4;9:960000. doi: 10.3389/fmed.2022.960000. PMID: 35991639; PMCID: PMC9388331.

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PP0611

GUT MICROBIOME IN IRRITABLE BOWEL SYNDROME: FOUNDATION FOR PERSONALIZED MICROBIOME CORRECTION

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Introduction: An in-depth analysis of the gut microbiota plays a crucial role in modern research exploring the relationship between intestinal microorganisms and human health. The use of molecular-genetic methods, particularly next-generation sequencing (NGS), provides a comprehensive understanding of the structural and functional characteristics of microbial ecosystems associated with the development of gastrointestinal pathologies.

Aims & Methods: To assess the features of the gut microbiome composition in different clinical variants of irritable bowel syndrome (IBS) in order to establish a foundation for microbiome-targeted personalized therapy. The study included 100 fecal samples, 90 from individuals with clinically confirmed IBS, and 10 samples from relatively healthy volunteers who formed the control group. The average age of participants was 55±2.23 years. DNA extraction from the samples was performed using the Zymo-BIOMICS DNA Mini Kit according to the official protocol. Taxonomic identification was conducted based on 16S rRNA gene amplification, which is considered the gold standard in phylogenetic studies of prokaryotes.

Results: The analysis revealed clear differences in the microbial composition between IBS subtypes and the control group. Specifically, for IBS-D (diarrhea-predominant type): the Firmicutes to Bacteroidetes ratio shifted towards Firmicutes (55.5% to 32.4%), indicating dysbiosis characteristic of metabolic disorders. For IBS-C (constipation-predominant type), a decrease in Bacteroidetes to 34% and an increase in Firmicutes to 52% were noted, which is associated with an increased risk of obesity and insulin resistance. In IBS-M (mixed type): partial stability of the microbial profile was observed, but with a predominance of Firmicutes (53%), possibly indicating altered intestinal motility. In all clinical variants of IBS, microbial

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imbalance, reduced production of short-chain fatty acids, weakened mucosal barrier, and increased endotoxin activity of the microbiota were recorded. Additionally, a potential impact on the neuropsychological status of patients was traced, supporting the connection between the gut microbiome and the central nervous system.

Conclusion: The identified microbiome disturbances in various forms of IBS highlight the need for an individualized therapeutic approach, including diet therapy, the use of probiotic and prebiotic supplements, as well as promising pharmabiotics. A promising direction is the implementation of personalized microbiome corrections based on the patient's metagenomic profile.

Disclosure: Nothing to disclose.

PP0612

ASSOCIATION BETWEEN PERCEPTION OF CONSTIPATION AND ROME IV CRITERIA: A CROSS-SECTIONAL STUDY AT THE NATIONAL HOSPITAL SRI LANKA

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Introduction: Constipation is a prevalent gastrointestinal disorder, commonly assessed using the Rome IV (R-IV) criteria for functional constipation. However, there is often a disconnect between a patient's subjective perception of constipation and the clinical criteria used for diagnosis. Patients may either perceive constipation when they do not meet the diagnostic criteria or fail to recognize it despite fulfilling the criteria. This gap in understanding can lead to over- or under-diagnosis and inappropriate treatment.

Aims & Methods: This study aimed to investigate the association between patients' subjective perception of constipation and the fulfillment of Rome IV (R-IV) diagnostic criteria, addressing potential discrepancies that may impact clinical diagnosis and management.

A cross-sectional study was conducted with 501 participants from the National Hospital Sri Lanka, who completed a self-reported questionnaire assessing bowel habits. Participants were categorized into four groups based on perceived constipation and R-IV fulfillment.

Pearson's chi-square test evaluated the association between these variables, while logistic regression analyzed predictors (gender, physical activity, water intake, stool consistency, and manual maneuvers) of constipation perception.

Results: Among the 501 participants (51.1% female), 36% reported perceiving constipation, whereas 25.5% met the R-IV criteria for functional constipation. A significant association was found between perceived constipation and R-IV criteria fulfillment ($\chi^2 = 12.77$, $p < 0.001$).

Notably, 23.6% of participants (Group 2) perceived constipation without fulfilling the criteria, while 13.0% (Group 3) met the criteria without perceiving constipation.

Only 12.6% both perceived constipation (Group 1) and met the R-IV criteria, and 50.9% (Group 4) neither perceived constipation nor met the criteria.

Manual defecation maneuvers were reported by 20.8% of participants, with digital evacuation (49%), bidet use (32.7%), and water enemas (31.7%) being the most common. The logistic regression model was statistically significant ($\chi^2(5) = 60.222$, $p < 0.001$), explaining 15.5% of the variance (Nagelkerke R^2) and correctly classifying 70.7% of cases. Participants requiring manual maneuvers were 5.17 times more likely to report constipation ($p < 0.001$), while higher Bristol Stool Category scores were associated with a reduced likelihood of constipation perception (OR = 0.85, $p = 0.009$). Gender, physical activity, and water intake did not significantly influence the model.

Conclusion: This study highlights the significant association between perceived constipation and fulfillment of the Rome IV criteria. However, a substantial proportion of participants either perceive constipation without meeting the criteria or meet the criteria without recognizing the condition. These findings emphasize the limitations of relying solely on patient-reported symptoms, which may not always align with clinical diagnoses. A comprehensive approach that integrates objective diagnostic measures and patient perception could enhance diagnostic accuracy, reduce unnecessary interventions, and improve patient care in resource-limited settings like Sri Lanka.

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Disclosure: Nothing to disclose.

PP0613

A MULTI-SPECIALTY RETROSPECTIVE EVALUATION OF DIAGNOSTIC AND MANAGEMENT STRATEGIES FOR BILE ACID DIARRHOEA IN TERTIARY CARE HOSPITALS

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Introduction: Bile acid diarrhoea (BAD) is a common yet often underdiagnosed cause of chronic diarrhoea. Despite its prevalence, BAD remains frequently overlooked in clinical practice. It is estimated that up to 30% of patients with diarrhoea-predominant irritable bowel syndrome (IBS) have underlying BAD, highlighting the importance of distinguishing between these two conditions for effective management. SeHCAT testing remains the gold standard modality for diagnosing BAD. Management typically involves bile acid sequestrants; however, notable variation exists in management approaches across clinical settings.

Aims & Methods: We retrospectively reviewed all SeHCAT tests performed across Imperial College Hospitals, London between 01/01/2023 – 31/12/2023. Electronic health records were interrogated to extract patient demographics, including age, gender, and ethnicity.

The results were analysed to determine the proportion of positive findings, which were subsequently stratified by severity. Corresponding management strategies were recorded to evaluate treatment patterns across clinical specialties.

Results: A total of 313 records were identified, of which 13 were excluded due to incomplete data. The average age of patients was 50 years. The predominant ethnicity was White (50.7%) and the majority of the patients were females (64.7%). Among the 300 remaining records, 154 had positive SeHCAT results with a 7 -day retention between 0 and 15%. Of these, 15.7% were categorised as having moderate bile acid malabsorption, followed by severe (13.3%), mild (11.0%), and borderline (11.3%) cases.

The majority of the requests originated from Gastroenterologists (86%), with a smaller proportion from Endobariatrics (5.3%), Hepatobiliary Surgeons (3.7%), Hepatologists (2%), Colorectal Surgeons (1.3%) and other specialities (1.7%). Colestyramine was the most commonly prescribed treatment, with 91% of patients initiated on a 4g once-daily dose. Approximately 26.9% of patients on colestyramine experienced side effects,

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necessitating a switch to colesevelam. The common side effects noticed were abdominal bloating, abdominal cramps, constipation and reflux. Conversely, 87% of patients on colesevelam continued treatment due to better tolerability, with only 13% exhibiting no clinical improvement. The most frequently prescribed dose was 1.25g daily in 45.3% cases. Among patients with normal SeHCAT results, the majority were diagnosed with diarrhoea-predominant irritable bowel syndrome (47.3%).

SeHCAT results	Number of patients	Percentage of patients
Borderline	34	11.3%
Mild	33	11.0%
Moderate	47	15.7%
Normal	146	48.7%
Severe	40	13.3%
Total	300	

Table 1: SeHCAT results.

Conclusion: Bile acid diarrhoea continues to be a prevalent factor contributing to chronic diarrhoea in healthcare. While existing treatment modalities have proven effective in managing this condition, they remain poorly tolerated. Our study confirms that colesevelam demonstrates improved tolerability compared to colestyramine.

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PP0614

EFFICACY OF A MULTI-STRAIN PROBIOTIC IN IBS

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Introduction: Irritable bowel syndrome (IBS) is a chronic gut-brain disorder that can be responsabil for a variety of uncomfortable gastrointestinal symptoms (laid out in the Rome Criteria) that can reduce quality of life. Although the precise mechanism remains unknown, recent research suggests that an imbalance in intestinal microbiota and a dysfunctional gut-brain axis may be involved in the development of IBS. Because of this, methods to restore the microbiota have been explored as treatment for this condition.

Aims & Methods: Randomized placebo controlled trial 180 patients: active group (probiotic formula, 3g/day- 92, placebo group – 88. men and women aged 24 to 75 who met the Rome IV criteria. One sachet probiotic (3 g) contains 7.5 billion bacteria: Lactobacillus casei W56, Lactococcus lactis W19, Lactobacillus acidophilus W22, Bifidobacterium lactis W52, Lactobacillus paracasei W20 Lactobacillus plantarum W62, Bifidobacterium lactis W51, Bifidobacterium bifidum W23, Lactobacillus salivarius W24. Assessments were made at 4, 8, 12 weeks. All patients agreed to maintain their lifestyle behaviors during their participation in the study.

Exclusion criteria: chronic gastrointestinal condition (other than IBS), including inflammatory bowel disease, lactose intolerance, celiac disease, diabetes, neoplasms.Patients have complete weekly data on stool frequency and consistency (Boston scale) and intensity of IBS symptoms (abdominal pain/discomfort, bloating, flatulence/borborygmi, and difficulty defecating) on a Likert scale of 0 (absent) to 7 (severe). Participants completed an IBS-specific quality of life questionnaire (IBS-QoL) at randomization (V1), fallow-up (V2) and at the end of the intervention (V3). Primary endpoint was mean weekly scores of gastrointestinal symptoms. Responders of abdominal pain were defined as subjects who experienced a reduction of at least 30% of the mean daily abdominal score reported at baseline (week 0). Secondary endpoints were the total AUC (W0-W12) of abdominal symptoms (abdominal pain, bloating and flatulence/borborygmi).

Results: There is a significant reduction in abdominal pain and bloating scores trough all 12 weeks of supplementation versus placebo. A higher overall IBS-QOL score, indicative of better QOL was reported at the end of the intervention (specially for interference with activity P=0.05, body image P=0.040, food avoidance P=0.03)

Conclusion: The probiotic treatment provides statistically significant and clinically relevant benefits for individuals with IBS. Improves gastrointestinal symptoms and quality of life in IBS patient

Reference:

Rome IV criteria-Rome Foundation.

Disclosure: Nothing to disclose.

PP0615

EFFICACY OF PROBIOTICS ON IRRITABLE BOWEL SYNDROME: SYSTEMATIC REVIEW AND META-ANALYSIS WITH TRIAL SEQUENTIAL ANALYSIS

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Introduction: Recent years have seen a significant increase in randomized controlled trials (RCT) published on the efficacy of probiotics on irritable bowel syndrome (IBS), which several reviews have tried to comprehensively review. Two recent trials, Yang et al. 2024[1] and Chen et al. 2023[2] report the safety and efficacy of probiotics for IBS, which seems to reflect prevailing favorable view of this therapeutic option by most reviews conducted. Yet, despite efforts in this field, major organizations have not changed their stance in this and make no assertive recommendation for probiotics in afflicted patients[3,4].

While further investigations are required, two challenges are faced:

1. Repeated reviews have been undertaken in this field, resulting in a "multiplicity problem",

2. Traditional meta-analytic techniques may also fail to provide a conclusive answer on the sufficiency of evidence, particularly when the data are sparse, and the risk of type I and type II errors is significant.

This indicates a need for.

Aims & Methods: This study aims to evaluate the efficacy of probiotics in alleviating symptoms of IBS and improve quality of life through a timely implemented trial sequential analysis (TSA) and rigorously evaluate the evidence for recommendations on future studies. Medline, Embase, Web of Science, and Cochrane were screened for randomized controlled trials (RCTs) comparing probiotics to placebo in adult patients with IBS. Primary outcomes included IBS Symptom Severity Score (IBS-SSS) and IBS Quality of Life (IBS-QOL) scores comparing probiotics and placebo groups. Tra-

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ditional pairwise meta-analysis conducted in restricted maximum likelihood was conducted, with subsequent TSA . Study quality was graded according to risk-of-bias 2 (ROB-2) and certainty of evidence was evaluated using the Grading of Recommendations Assessment, Development and Evaluation (GRADE).

Results: 42 studies were included. Overall, probiotics demonstrated non-significant improvements in IBS-SSS (MD: -27.64; 95% CI: -56.90 to 1.63; p=0.06) and IBS-QOL (MD: 1.33; 95% CI: -3.65 to 6.30; p=0.59). TSA was conducted on IBS-SSS demonstrated current evidence has not reached the required information size. The representative cumulative Z-curve demonstrated neither futility, benefit nor detrimental effect, indicating the need for further trials. Assessment with ROB-2 indicates high quality amongst most trials included in this study. GRADE evaluation of the evidence demonstrated low certainty of existing evidence.

No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Probiotics	Placebo	Relative (95% CI)	Absolute (95% CI)	Certainty
Probiotics effectively reduces a patient's IBS-SSS score											
24	randomised trials	not serious[a]	very serious[b]	not serious	very serious[c]	publication bias strongly suspected[d]	2237	1599	-	MD 27.64 lower (56.9 lower to 1.63 higher)	⊕◯◯◯ Very low[a,b,c,d]
Probiotics effectively reduces a patient's IBS-QOL score											
24	randomised trials	not serious[a]	very serious[a]	not serious	not serious[e]	publication bias strongly suspected[d]	2292	1712	-	MD 1.33 higher (3.65 lower to 6.3 higher)	⊕◯◯◯ Very low[a,b,d,e]

Table 1: GRADE Evaluation
CI: confidence interval; MD: mean difference.

Two contributing factors to the low certainty of current evidence include:
1. Inconsistency of existing trials – there exists high level of heterogeneity as detected by statistical analysis through I^2 and Cochran Q amongst studies;
2. Suspected publication bias through inspection of a funnel plot, though Egger's regression did not demonstrate statistical significance.

Conclusion: Existing evidence is insufficient to make conclusive recommendations regarding the role probiotics should play in the management of IBS in symptoms and QOL. Given the persistent heterogeneity and potential publication bias, future research should focus on standardizing probiotic interventions and clarifying their clinical utility to truly advance the field.

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PP0616

MICROBIOME PREDICTORS AND LONGITUDINAL DYNAMICS IN ASIAN PATIENTS WITH IRRITABLE BOWEL SYNDROME-DIARRHEA (IBS-D) CONSUMING A LOW FODMAP (FERMENTABLE OLIGOSACCHARIDES, DISACCHARIDES, MONOSACCHARIDES AND POLYOLS) DIET

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Introduction: Response rates to the Low FODMAP Diet (LFD) in IBS vary and predictors of benefit remain poorly defined. While several studies from Western populations have identified microbial signatures linked to response, data from Asian cohorts remain limited.

Aims & Methods: This study aims to identify microbial predictors and longitudinal microbiome shifts in Asian IBS-D patients consuming a LFD. We recruited 20 adult Asian patients with IBS-D diagnosed according to the ROME-IV criteria. All subjects received LFD dietary advice from a dietitian at initiation and were followed-up for a total of 28-days. Clinical response was assessed by a ≥50-point improvement in the IBS Symptom Severity Score (IBS-SSS) from Day 0 (baseline) as compared to follow-up at Day 28. Stool samples were collected at baseline, Day 14, and Day 28.
Microbial DNA was extracted using bead-beating and Qiagen PowerFecal kits, and the V3–V4 region of the 16S rRNA gene was amplified and sequenced on the Illumina MiSeq platform. Bioinformatic processing included denoising with DADA2, taxonomic assignment using the SILVA 138 database, and rarefaction to even sequencing depth. Microbiome metrics included alpha diversity (Shannon and Simpson), beta diversity (Bray-Curtis PCoA with PERMANOVA), and differential abundance testing with DESeq2.

Results: By Day 28, 11 of 20 patients (55%) achieved clinical response, with a median IBS-SSS reduction of 172 points (IQR: 88–248), compared to 24 points in non-responders (IQR: −11 to 55; p=0.003). Alpha diversity was not significantly different between groups, but responders exhibited a trend toward higher Shannon and Simpson diversity at baseline. Beta diversity analysis using Bray-Curtis dissimilarity revealed partial clustering by responder status (PERMANOVA p=0.06), and PCoA plots showed individual-specific shifts in community structure across time, indicating that the LFD induces personalized microbiome trajectories.
At baseline, responders had significantly higher relative abundances of butyrate-producing and anti-inflammatory genera, including *Ruminococcus*, *Blautia*, *Eubacterium siraeum*, and *Bifidobacterium bifidum* (adjusted p<0.05). These findings are consistent with prior reports, including Bennet et al. (2018) and Vervier et al. (2022), where such taxa were part of a "health-like" IBS-P (pathogen-enriched) to IBS-H (health-like) axis. The enrichment of *Blautia* and *Ruminococcus* is particularly noteworthy as these are established butyrate producers involved in maintaining colonic barrier function and reducing visceral hypersensitivity. By Day 28, differential abundance analysis revealed that responders exhibited increases in *Ruminococcus bicirculans*, *Parabacteroides gordonii*, and *Weissella cibaria*. In contrast, non-responders showed persistence or increases in taxa such as *Tyzzerella* and *Olsenella*, which have been associated with inflammation or IBS severity in prior studies (Chuah et al., 2024).

Conclusion: This study provides novel microbiome insights from an underrepresented Asian IBS population consuming a LFD. Despite dietary and baseline microbiota differences from Western populations, we observed remarkably consistent microbiome predictors of response, including higher baseline levels of butyrate-producing taxa and favorable microbial restructuring during the intervention.

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These results reinforce the reproducibility and functional relevance of microbiome predictors in IBS dietary therapy, and support microbiome profiling as a potential tool for personalized dietary guidance across a diverse global population.

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Chuah KH, Mahadeva S. Can we predict who will respond to a low-FODMAP diet in irritable bowel syndrome? J Gastroenterol Hepatol. 2024 Feb;39(2): 217-218.

Disclosure: Nothing to disclose.

PP0617

EFFICACY AND SAFETY OF PYRIDOSTIGMINE FOR SEVERE CHRONIC CONSTIPATION REFRACTORY TO FIRST-LINE TREATMENTS

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Introduction: Functional constipation (FC) and constipation-predominant irritable bowel syndrome (IBS-C) are highly frequent conditions characterized by chronic constipation (up to 24% in adults) (1).

The initial management of chronic constipation typically involves dietary and lifestyle modifications, fiber supplementation, and laxative therapy. For patients who do not respond to these interventions, second-line therapies, such as medications with prokinetic effects, are recommended. Pyridostigmine, an anticholinesterase agent commonly used in the treatment of myasthenia gravis and paralytic ileus, might play a role in refractory chronic constipation, but evidence is still not available.

Aims & Methods: We carried out a retrospective observational study involving all patients diagnosed with functional constipation (constipation-predominant irritable bowel syndrome [IBS-C] or chronic idiopathic constipation) according to the Rome IV criteria, who met the criteria for severe chronic constipation (2) (defined as two or fewer spontaneous and complete bowel movements per week for a minimum of 6 months, associated with hard stools, sensation of incomplete evacuation, or straining during defecation in at least 25% of bowel movements), who were refractory to first-line treatments (dietary and lifestyle counseling, bulk-forming laxatives, osmotic laxative such as polyethylene glycol and stimulant laxative such as bisacodyl) and who were treated with pyridostigmine in our outpatient gastroenterology clinic from April 2021 to December 2024. Patients were included if they attended at least one follow-up visit within 12 weeks after treatment.

We excluded all patients with organic gastrointestinal disorders. We evaluated the improvement in constipation at follow-up, defined as patients having three or more spontaneous, complete bowel movements per week during the follow-up period. (2)

We also evaluated any improvement in other gastrointestinal symptoms and the safety profile of the treatment.

Results: Overall, 110 patients were included in the study, of whom 77 (70%) met the criteria for IBS-C and 33 for chronic idiopathic constipation (30%). Other gastrointestinal symptoms included bloating (n=88, 80%), abdominal pain (n=77, 70%), dyspepsia (n=17, 15%). Overall, 74 (67%) patients experienced an improvement in constipation. Additionally, 51(46%) patients exhibited improvements in other gastrointestinal symptoms: 38/77 (49%) patients experienced relief from abdominal pain, 35/88 from bloating (40%), and 10/17 (59%) from dyspepsia. Finally, Four (4%) patients have discontinued therapy due to pyridostigmine intolerance (2 patients experienced muscle fasciculation, 2 patients fatigue); no severe therapy-related adverse events were observed.

Conclusion: In our cohort, pyridostigmine was an effective and safe treatment for IBS-C and chronic idiopathic constipation in patients unresponsive to first-line treatments, providing benefits not only in alleviating constipation but also in ameliorating other gastrointestinal symptoms. Further prospective and randomized studies are warranted to confirm our findings.

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Disclosure: Nothing to disclose.

PP0618

A DIGITAL PRESCRIPTION MOBILE APPLICATION FOR ADULTS WITH ROME IV IRRITABLE BOWEL SYNDROME: A REAL-WORLD UK STUDY

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Introduction: Behavioural therapies, including cognitive behavioural therapy (CBT), are effective in the treatment of irritable bowel syndrome (IBS), which has a population prevalence of 4.3% in the UK (1). Mahana™ IBS is a Food and Drug Administration-cleared prescription mobile application designed to deliver 3 months (10 sessions) of gut-directed CBT to adults with IBS.

This prospective study evaluated patient compliance with the Mahana™ application and the change in gastrointestinal (GI) and psychological symptoms in a real-world UK setting.

Aims & Methods: Patients diagnosed with Rome IV IBS in a neurogastroenterology outpatient service were offered the Mahana™ CBT application, alongside standard medical management consistent with British Society of Gastroenterology IBS guidelines. At baseline and 3-month follow-up, patients completed the IBS Symptom Severity Scale (IBS-SSS), Patient Health Questionnaire-9 (PHQ-9), and Generalised Anxiety Disorder-7 (GAD-7) scale to measure severity of IBS symptoms, depression, and anxiety, respectively. Matched samples, unmatched samples, and correlations were determined using the Wilcoxon signed-rank, Mann–Whitney U, and Spearman's ρ, respectively, using SPSSV29.0.

Results: Seventeen patients (females: 12/17 (70.1%), mean age: 45.8 [15.9]) enrolled onto the study, among whom 9 (52.9%), 5 (29.4%) and 3 (17.6%) were diagnosed with IBS-C, IBS-M and IBS-D, respectively. Twelve patients (70.6%) completed the entire intervention. At baseline, the mean

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IBS-SSS, PHQ-9 and GAD-7 scores were consistent with severe IBS (321.3 [73.7]), mild depression (7.7 [5.8]), and mild anxiety (7.2 [6.0]), respectively. Compared to baseline, there was a reduction in the mean IBS-SSS (321.2 [73.7] vs 241.1 [88.5],*P*<.001), PHQ-9 (7.7 [5.8] vs 5.9 [4.7],*P*=.1) and GAD-7 (7.2 [6.0] vs 4.7 [3.7],*P*=.04) scores at follow-up. Eleven (64.7%) patients had a clinically significant reduction in IBS symptom severity at follow-up compared to baseline.

There was a statistically significant correlation between the duration (in months) from when a patient was diagnosed with IBS to when they were prescribed the application and symptom improvement in the PHQ-9 (ρ=-0.5,*P*=.04), but not the IBS-SSS (ρ=-0.3,*P*=.4) nor GAD-7 (ρ=-0.1,*P*=.8). Similarly, age was correlated with symptom improvement in the PHQ-9 (ρ=-0.5,*P*=.04), but not the IBS-SSS (ρ=-0.3,*P*=.3) nor GAD-7 (ρ=-0.2,*P*=.5).

There was no statistically significant difference in the change in IBS-SSS (-83.3 [84.1] vs -72.6 [37.4],*P*>.99), PHQ-9 (-2.8 [3.5] vs 0.8 [3.7],*P*=.1) nor GAD-7 (-3.8 [4.5] vs 0.2 [3.3],*P*=.08) scores among patients who completed the entire application vs those who did not. Symptom improvement in the IBS-SSS score was greater among males than females (*P*=.04), but sex did not affect change in the PHQ-9 nor GAD-7 scales.

Conclusion: The Mahana™ IBS application was associated with a significant improvement in GI and psychological symptoms in a real-world UK setting.

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Disclosure: At the time of the study, Alice Sibelli was employed by Mahana Therapeutics. Alice Sibelli was not involved in data collection or analysis. None of the other authors have any conflicts of interest to disclose.

PP0619

EFFICACY AND SAFETY OF POSTBIOTICS IN IRRITABLE BOWEL SYNDROME: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Postbiotics, non-viable microbial components with health benefits, are increasingly used due to their safety and stability. However, evidence supporting their efficacy and safety in irritable bowel syndrome (IBS) remains limited.

Aims & Methods: This first-ever systematic review and meta-analysis evaluates the efficacy and safety of postbiotics in IBS management. Four databases (PubMed, Scopus, Embase, Web of Science) were searched up to April 2025 for randomized controlled trials (RCTs) assessing postbiotics in Rome III/IV-diagnosed IBS adults.

Primary outcomes were IBS Symptom Severity Scale (IBS-SSS) scores and ≥30% pain reduction; secondary outcomes included QoL and treatment-emergent adverse events (TEAE). Cochrane RoB 2.0 assessed bias, and random-effects meta-analyses were conducted (RevMan 5.4).

Results: Of 1,053 records, 768 duplicates removed, 281 articles were excluded, and eventually four RCTs were analysed. The participants' mean age ranged from 34.88 to 47.47 years old, and the intervention lasted from two to twenty-six weeks. The RCTs were executed in countries including Germany, Sweden, and India and published between 2020 and 2024.

The studies investigated different postbiotic formulations included non-viable *Bifidobacterium bifidum* HI-MIMBb75 cells (two capsules daily for 8 weeks), nonviable bacterial lysate of *Escherichia coli* and *Enterococcus faecalis* (gradual dose escalation over 26 weeks), fermented oat gruel with *Lactobacillus plantarum* 299v (administered as an enema for 2 weeks), and heat-treated *Bifidobacterium longum* CECT 7347 (HT-ES1) (two capsules daily for 12 weeks).

Of these postbiotic formulations, *Bifidobacterium longum* CECT 7347 was specifically heat-treated, whereas the other formulations were not heat-treated but were non-viable. Three of the four formulations (Andersen 2020, Mack 2021 and Srivastava 2024) were administered orally (as capsules or drops) and only one (fermented oat gruel) (Bednarska 2022) was administered by enema (Table1).

Postbiotics significantly improved IBS symptom severity (P < 0.001) with moderate heterogeneity (I^2 = 69%; P = 0.07). For pain intensity, postbiotics also showed significant reduction (P = 0.02) with substantial heterogeneity (I^2 = 87%; P = 0.0005). QoL improvement was significant after excluding one study using postbiotic enema preparation (P < 0.001) with no residual heterogeneity (I^2 = 0%). Safety analysis found no significant difference in TEAE between postbiotics and placebo (RR 0.68, 95% CI 0.35-1.34; P = 0.26; I^2 = 54%).

			POPULATION			INTERVENTION			COMPARATOR		
Location	Year	Design	Rome Criteria	IBS subtype	Sample Size	Age	Postbiotic used	Dose & Duration	Sample Size	Age	MA
Germany	2020	RCT	ROME III	all types	221	40·1 (12·8)	Non-viable B bifidum HI-MIMBb75 cells	2 capsules once daily for 8 weeks	222	42·6 (13·8)	A,B, C,D
Germany	2021	RCT	ROME III	all types	191	47.47 (15.66)	Suspension of Nonviable bacterial lysate of E coli(DSM17252) and E faecalis (DSM16440)	10 drops*3/day for week then m20*3/day for week then 30 *3/ day for 24 weeks (total 26weeks)	198	45.86 (15.90)	B,D
Seweden	2022	RCT	ROME IV	IBS-D, mixed	18	37 (19–55)	Referm oat gruel fermented with Lactobacillus plantarum 299v	Enema , 250 ml twice daily, for 2 weeks	12	37 (19–55)	C
India	2024	RCT	ROME IV	IBS-D	68	34.88 (9.56)	Heat-treated Bifidobacterium longum CECT 7347 (HT-ES1)	2 capsules once daily for 12 weeks	probiotic: 65, placebo: 67	Probiotic: 37.28 (11.04), placebo:35.73 (11.42)	A,B, C,D

Table 1. Characteristics of included randomized controlled trials (RCTs).

The table summarizes key study details such as author/year, country, sample size, postbiotic formulation, dosage, intervention duration, and outcomes assessed (IBS-SSS: Irritable Bowel Syndrome Symptom Severity Scale; QoL: Quality of Life; TEAE: Treatment-Emergent Adverse Events). MA denotes meta-analysis outcomes.

Conclusion: Postbiotics are efficacious in alleviating IBS symptom severity and pain intensity, QoL is better and exhibit a favorable safety profile. Heterogeneity in study outcomes is due to variability in formulations and

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study protocols. Future large-scale, well-designed RCTs are recommended to validate these findings and establish standardized postbiotic preparations for IBS management.

Disclosure: Nothing to disclose.

PP0620

ARTIFICIAL INTELLIGENCE-BASED DETECTION OF COLONIC ANGIODYSPLASIA CAN IMPROVE ENDOSCOPIC REPORTING CONSISTENCY

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Introduction: Colonic angiodysplasia is a common endoscopic finding with strong associations to gastrointestinal bleeding. Despite its clinical relevance, documentation in examination reports is often inconsistent and image-based identification varies widely. Artificial intelligence (AI) offers the potential to standardize the detection and documentation of such findings.

Aims & Methods: In this study, we develop and evaluate an AI for identifying angiodysplasia in colonoscopy images. Furthermore, we evaluate the potential of such a model in improving the frequency and consistency of documentation for this finding. Examination data from 34,953 colonoscopies containing both examiner reports and corresponding image documentation were retrospectively evaluated. Initially, 993 examination whose report contained angiodysplasia-related terms were identified, yielding 10,908 associated images. The collected images were manually annotated and used to train an AI for identifying colonic angiodysplasia. External validation was performed on 3,936 annotated images from 240 examinations at a second clinic. Subsequently, AI predictions were collected for images from examinations from the second clinic, in which angiodysplasia was not mentioned in the reports, to identify potential underreporting.

Results: On the external validation set, the AI achieved an accuracy of 94.3% (95% CI: 93.5%–95.1%), with a sensitivity of 80% (95% CI: 76.8%–84%) and specificity of 96.5% (95% CI: 95.9%–97.2%) for detecting angiodysplasia. Using this AI we were able to identify 112 examinations with angiodysplasia findings in the recorded image documentation that had not been mentioned in the corresponding reports.

Conclusion: AI can reliably identify colonic angiodysplasia from endoscopic images. Integrating such technology into the reporting workflow has the potential to improve the completeness and consistency of endoscopic documentation.

Disclosure: Nothing to disclose.

PP0621

ASSOCIATION OF COLONIC DIVERTICULOSIS WITH CHRONIC POLYPS AND RECTAL HEMORRHOIDS: 4-YEARS' EXPERIENCE IN A TERTIARY REFERRAL CENTER

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Introduction: Colonic diverticulosis is a prevalent gastrointestinal condition that is often diagnosed incidentally. However, its association with chronic polyps and rectal hemorrhoids remains underexplored, particularly in the northern region of Saudi Arabia.

Aims & Methods: This study aimed to assess the relationship between colonic diverticulosis and colonic conditions while considering demographic and clinical factors. This retrospective observational study was conducted at a tertiary referral center over four years. Data from 4,014 patients who underwent colonoscopy were analyzed. Statistical tests, including chi-square and independent sample t-tests, were used to determine associations between diverticulosis, polyps, hemorrhoids, and bowel preparation adequacy.

Results: Diverticulosis was observed in 14.2% of the patients, with a higher prevalence among males (16.1%) than among females (12.1%) ($p < 0.001$). Patients with diverticulosis were significantly older (mean age 61.35 years) than those without diverticulosis (mean age 48.25 years) ($p < 0.001$). Hemorrhoids were more frequent in patients with diverticulosis (18.6%) than in those without diverticulosis (11.9%) ($p < 0.001$).

A similar association was found between diverticulosis and polyps, with the polyp prevalence being higher in patients with diverticulosis (21.4%) than in those without (11.7%) ($p < 0.001$).

Additionally, inadequate bowel preparation was more common among patients with diverticulosis (17.1%) than among those with adequate bowel preparation (13.6%) ($p = 0.023$).

Conclusion: This study highlights a significant association between colonic diverticulosis and the presence of polyps and hemorrhoids. Older age and male sex were identified as the key risk factors.

These findings emphasize the importance of targeted screening and optimized bowel preparation in at-risk patients. Further research is required to explore the underlying pathophysiological mechanisms and to develop preventive strategies.

Keywords: Diverticulosis; Hemorrhoids; Chronic Polyps

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Disclosure: Nothing to disclose.

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PP0622

IMPACT OF MEDITERRANEAN DIET AND PHYSICAL ACTIVITY IN PATIENTS WITH SYMPTOMATIC UNCOMPLICATED DIVERTICULAR DISEASE (SUDD)

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Introduction: Symptomatic Uncomplicated Diverticular Disease (SUDD) is a widespread gastrointestinal condition linked to sedentary lifestyles and unhealthy dietary patterns, often requiring medications. Mediterranean Diet (MedDiet) and physical activity (PA) have been proposed as potential protective factors against various diseases, including SUDD.

Aims & Methods: The aim is to evaluate if a counseling program which encourages MedDiet and PA can influence SUDD symptoms, drug use and quality of life.

In this prospective, monocentric, observational study, we enrolled consecutive patients with diagnosis of SUDD at colonoscopy. They were divided into two groups: the Study Group (SG) received a 3-month lifestyle counseling promoting MedDiet and regular PA, while the Control Group (CG) followed a free diet. Lifestyle counseling consisted of 3 medical visits (every 6 weeks) and 4 phone calls (every 2 weeks). Patients were evaluated at baseline (T0), 12 weeks (T12, end of counseling) and 52 weeks (T52). We collected clinical data and anthropometric data, MediLite score (a validated adherence score for MedDiet), questionnaires assessing PA and symptom burden expressed both as specific item and as multiparametric scores including Irritable Bowel Syndrome Severity Scoring Scale (IBS-SSS) and Visual Analogical Scale for patients with SUDD (VAS-DICA), and SF-12 for the assessment of quality of life.

Results: 84 patients (F=48) were enrolled, of whom 41 subjects were included in SG while 43 in CG. At T12, while CG characteristics remain unchanged, the Medi-Lite score in SG rose from 9,12±2,0 to 10,3±2,23 (p=0,011) and daily walking time increased, with 92,5% of SG reporting walking for 30–60 minutes daily compared to 42,5% at T0 (p< 0,001). At T52, SG had still a high % of patients with high adherence to MedDiet (MediLite >10) (from 46,34% to 56,01%, p<0.001) and PA adherence was 63,41% (p=0,001).

Compared to CG, SG showed a reduction of abdominal pain (SG/CG at T0 51,22/58,14%, at T12 24,4/46,51%, p=0,012, at T52 31,71/53,49%, p=0,018), constipation (at T0 19,51/25,58%, at T52 7,32/44,19% p<0,001), and IBS-SSS score (at T0 141±98,5/150±94,1, at T12 71±61,6/123±104, p=0,002, at T52 69,1±52,4/101±97,5, p<0,001).

Subjects in SG who reported medication use for SUDD dropped from 43,9% to 26,83% at T12 and 24,39 at T52 (p<0,05) while an increase of overall medication use was observed in the CG (from 25,58 to 51,16, and 83,72% at T0, T12, and T54, respectively), in particular regarding the use of rifaximin (from 6,98% to 13,95% and 34,88%), both at T12 (p=0,01) and T52 (p<0,001). Overall, the CG had a higher rate of diverticular disease complications including acute diverticulitis, need for hospitalization and surgery: 12 vs 1 patient in the CG vs SG, respectively (at T52=0,001).

Conclusion: Higher adherence to MedDiet and PA are associated with symptom relief, reduced use of medication, and reduction of disease-associated complications in patients with SUDD, suggesting a therapeutical role of MedDiet and PA which goes beyond the already recognized preventive action. Clinical benefits of a 3-month counseling may still be present at 1 year.

Disclosure: Nothing to disclose.

PP0623

RESULTS OF A MULTICENTER STUDY ON THE IMPACT OF COMBINATION THERAPY WITH RIFAXIMIN AND TRIMEBUTINE ON QUALITY OF LIFE AND DISEASE COURSE IN UNCOMPLICATED DIVERTICULAR DISEASE IN REAL CLINICAL PRACTICE (ALLIANCE)

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Introduction: Uncomplicated diverticular disease (UDD) is common bowel disease and significantly reduces the quality of life of patients. *The normokinetic agent trimebutine and the non-absorbable antibiotic rifaximin are proposed* to treat UDD and improve the quality of life of patients.

Aims & Methods: To evaluate the effect of combination therapy with rifaximin and trimebutine on improving the quality of life in patients with UDD in routine clinical practice. This multicenter observational study included patients with UDD who complained of abdominal pain, constipation/diarrhea, bloating, and/or tenesmus. Patients received combination therapy with rifaximin 400 mg/d for 7 days and trimebutine 300 mg bid for 21 days each month for 6 months. In addition to these drugs, mesalazine could be prescribed.

The primary outcomes were the improvement of quality of life and symptoms (pain, constipation/diarrhea bloating).

Secondary outcomes included treatment adherence and safety.

Quality of life was assessed using the SF-12 questionnaire. Symptoms (abdominal pain, constipation/diarrhea, bloating, tenesmus) were assessed on a 4-point scale: no symptom, mild and *minimally affecting daily life*), moderate (bothers the patient, but does not interfere with daily activities), or severe (impairs daily activities) one. Treatment adherence was assessed using Morisky-Green questionnaire. These parameters were assessed at baseline (Visit 1) and at Days 30, 65, 100, 135 and 170 after the inclusion (Visits 2–6).

Results: The study included 100 patients aged 40–70 years (58.2 ± 9.5 years). The most common complaints were abdominal pain (84%) and bloating (72%), while constipation (33%) and tenesmus (30%) were less frequent.

After the first month of therapy, patients showed a significant reduction in the severity of all symptoms (p<0.001 for each symptom) and an improvement in quality of life (p<0.001), which was maintained throughout the entire observation period.

Symptoms		V1 (day 1-2)	V 2 (day 28-30)	p, (V2/V1)	V3 (day 70-72)	p, (V3/B1)	V6 (day 166-168)	p, (V6/V1)
Pain (N=99)	Mean ± SD	2,1±0,6	1,2±0,7	<0,001	0,6±0,7	<0,001	0,1±0,3	<0,001
Bloating (N=98)		1,8±0,6	1,1±0,6	<0,001	0,6±0,7	<0,001	0,3±0,5	<0,001

Table. Dynamics of symptoms of UDD across the study.

A clinically significant positive effect of the therapy was achieved by Visit 2 in 82% of patients (82/100). By Visit 6, complete remission was achieved in 55% of participants. The treatment adherence was high in 76% of patients, moderate in 21% of patients and poor in 3% of patients. There were no reports of any serious adverse events and adverse reactions.

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Conclusion: The combination of rifaximin and trimebutine reduces symptom severity and improves quality of life in patients with uncomplicated diverticular disease without the development of any serious adverse events and adverse reactions.

Disclosure: Current study was conducted with the financial support of Binnopharm Group.

PP0624

EFFICACY OF SYNBIOTIC SUPPLEMENTATION IN PATIENTS WITH SYMPTOMATIC UNCOMPLICATED DIVERTICULAR DISEASE: A SINGLE-CENTRE, DOUBLE-BLIND, RANDOMISED, PLACEBO-CONTROLLED TRIAL

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Introduction: Symptomatic uncomplicated diverticular disease (SUDD) is characterized by recurrent gastrointestinal (GI) symptoms in patients with colonic diverticula. Different treatments have been proposed to improve GI symptoms, but with low-level of evidence.

Aims & Methods: We aimed to evaluate the efficacy of a synbiotic formulation combined with a diet rich in fibre (DRF) in improving GI symptoms in SUDD, compared to a DRF and placebo. We conducted a single-centre, no-profit, randomised, double-blind, placebo-controlled trial (Local Ethics Committee n.182 SA_2020). Patients diagnosed with SUDD (colonic diverticula and recurrent abdominal pain with or without altered bowel habits or bloating, in the absence of signs of systemic inflammation) were included.

After a 14-day run-in period, all participants completed structured questionnaires [long- and short-lasting abdominal pain, abdominal bloating, Rome IV for IBS, food frequency questionnaire and quality of life (SF-12)], received a dietary counselling, and were randomised to receive either (I) Enterolactis duo (containing the probiotic strain Lacticaseibacillus Paracasei DG and the prebiotic fibre inulin) 1 sachet bid plus advice on DRF (arm A), or (II) Placebo, 1 sachet bid plus advice on DRF (arm B) for 12 weeks. During treatment, patients completed a weekly diary of GI symptoms and bowel habits. At the end of treatment, patients completed all questionnaires again and were followed up after 4 weeks.

Results: After screening 81 patients, 34 (55.9% female; mean age 64±8.6yrs) were included and randomised to arm A (n=17) and arm B (n=17). At baseline, there were no significant differences in terms of clinical characteristics between the two arms. Thirty patients (arm A=14; arm B=16) completed the study. Compliance with the DRF was poor in both arms (0% vs 14.3%).

Patients experienced significant improvements in GI symptoms (bloating, short- and long-lasting abdominal pain) and bowel habits compared to baseline in both arms, but no differences were observed between the two arms.

Neither arm A nor arm B reported significant changes in the quality of life scores. Mild self-limited adverse events were reported with no differences between arms. Follow-up at 4 weeks showed no clinical changes in either arm, with no differences between arms.

Conclusion: This synbiotic supplementation is as effective as placebo in relieving symptoms in SUDD patients, suggesting the relevant role of the placebo effect also in diverticular disease.

Disclosure: Nothing to disclose.

PP0625

THE MANAGEMENT OF A RARE CASE OF A RECTAL HPV-INDUCED HIGH-GRADE SQUAMOUS INTRAEPITHELIAL LESION

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Introduction: Ano-genital infection by human papillomavirus (HPV) is an increasingly common sexually transmitted disease that often causes anal *condylomata acuminata* and is associated with the development of anal intraepithelial neoplasia (AIN). The management of which is widely discussed in the current literature; however, no international recommendations exist regarding the treatment and follow-up of HPV-induced lesions of rectal localization, which are extremely rare.

Aims & Methods: Our aim is to describe the management and follow-up of a rare rectal HSIL induced by HPV with a combined step-up endoscopic-surgical approach that led to an R0 resection.

For many years, surgery was the standard treatment for managing HPV-related ano-rectal lesions. More recently the approach to screening and follow-up for anal intraepithelial neoplasia and dysplasia has evolved, with new international guidelines [1 - 2].

However guidelines on the management of HPV-related rectal lesions are scarce. Endoscopic submucosal dissection (ESD) is a promising less invasive alternative. It allows for simultaneous diagnosis and resection while providing clear visualization of the tumor margin, potentially reducing the risk of recurrence.

Nevertheless few case reports and studies demonstrating the effectiveness of ESD in the treatment of HPV-induced anal and rectal lesions have been published [3 - 6].

Results: We present an unusual case of a 65-year-old male with a high-grade HPV-induced squamous intraepithelial lesion (HSIL, classified as AIN 2 or 3) of rectal topography. The lesion extended 8 cm from the anal margin to the middle rectum. A patient history revealed our patient was an immunocompetent man who had sex with men (MSM) including regular receptive anal intercourse with multiple partners, anal penetration of foreign objects and iterative rectal enema following dyschezia. This could explain a secondary inoculation of HPV onto a post-traumatic scarred rectal tissue. Initial management involved surveillance at 6-month intervals, combined with the cessation of intrarectal penetrating activity.

This approach was based on an extrapolation of the current guidelines from the Société Nationale Française de Colo-Proctologie (SNFCP) for anal precancerous lesions associated with HPV infection, aiming for spontaneous dysplasia regression.

However, the rectal lesion persisted at the 6-month follow-up. Pelvic magnetic resonance imaging revealed limited expansion, and the lesion was consequently managed through a multidisciplinary endoscopic-surgical step-up approach. Initial treatment with ESD was attempted but limited by extensive fibrosis and the lesion's depth. As a result, surgical intervention was required, and a full-thickness resection (FTR) was performed within the same sedation period. This approach led to a monobloc resection measuring 9.7 x 6.8 x 0.3 cm, achieving a complete R0 resection with a minimal 3 mm healthy margin.

Conclusion: Rectal HPV-induced HSIL are extremely rare, therefore data regarding treatment is limited. This case demonstrates the successful R0 resection by combining a step-up endoscopic-surgical approach. Endoscopic resection proved challenging due to extensive fibrosis and the lesion's substantial depth, necessitating a subsequent surgical FTR to

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ensure complete excision. Follow-up at a 3-month interval, in accordance with the AIN guidelines, revealed no recurrence of dysplasia. However, the long-term effectiveness of this approach in maintaining sustained remission remains uncertain, and there is currently no established follow-up protocol for rectal HPV-induced HSIL.

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PP0626

EFFECTIVENESS OF MICRO-ENCAPSULATED SODIUM BUTYRATE IN SUPPOSITORY FORM IN THE TREATMENT OF HEMORRHOID DISEASE

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Introduction: Approximately 50% of people have problems with hemorrhoids at some point in their lives. We aimed to investigate the effectiveness of a new rectal micro-encapsulated sodium butyrate suppository in the treatment of hemorrhoid disease.

Aims & Methods: All consecutive patients attending the outpatient clinic of the 5 participating hospitals with symptoms of pain, pruritus or passing of bright blood per rectum due to hemorrhoid disease were eligible for entering the study. Pregnant women and patients with rectal cancer were excluded. The intensity of symptoms was evaluated with a 5-point Likert scale, while hemorrhoids were classified as internal or external. All patients received once a day for 10 days a suppository that weighs 2,0gr and contains sodium butyrate, sodium alginate, carboxymethylcellulose, triglycerides of saturated fatty acids with the addition of monoglycerides and diglycerides. The effectiveness of treatment was evaluated at the end of the 10-day administration and was defined as the complete disappearance of pain, pruritus and hemorrhage. The concomitant use of lidocaine or phlebotropic drugs that were administered according to the treating physician's judgment was also recorded.

Results: From January 2024 till March 2025, 179 patients [male/female: 106/73, mean±SD age: 51±24, mean±SD BMI: 27.42±5.91) received the rectal micro-encapsulated sodium butyrate suppository for the treatment of pain (n=82), pruritus (n=55) and hemorrhage (n=143). Twenty-one patients (11.7%) had also diabetes mellitus, 18 (10.0%) were receiving anticoagulants and 51 (28.5%) were active smokers. Ninety (50.3%) patients had external and 135 (75.4%) internal hemorrhoids [grade 1=52 (38.5%), grade 2=54 (40.0%), grade 3=21 (15.5%), grade 4=6 (4.4%)]. The mean±SD intensity of pain and pruritus was 2.10±1.24 and 1.90±1.34 respectively. Thirty-two (17.8%) patients had also an anal fissure. Forty (22.3%) patients were receiving lidocaine ointment and 29 (16.2%) were receiving phlebotropic drugs during the time of the rectal suppository administration. After the 10-day administration period success of treatment was seen in 161 (89.9%) patients. No adverse events from the use of the rectal suppository were seen.

Conclusion: The use of micro-encapsulated sodium butyrate in suppository form appears to be safe and effective for the treatment of symptoms due to hemorrhoid disease.

Disclosure: No conflicts of interest.

PP0627

THE DISTENSIBILITY INDEX, ASSESSED BY IMPEDANCE PLANIMETRY (ENDOFLIP™), DIFFERENTIATES WOMEN WITH SUSPECTED FUNCTIONAL DEFECATION DISORDER FROM THOSE WITH FECAL INCONTINENCE AND HEALTHY VOLUNTEERS: PRELIMINARY RESULTS

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Introduction: Pelvic floor dysfunction is highly prevalent. Posterior compartment disorders include fecal incontinence (FI) and obstructive defecation syndrome (ODS). Assessment of the sphincter complex's (SC) ability to resist or yield to distension may provide a valuable method for evaluating continence and defecatory function.

Aims & Methods: Women with FI, chronic constipation (CC) and suspected functional defecation disorder (FDD), as well as healthy volunteers (HV) were included. Impedance Planimetry (IP) was performed using the anal functional lumen imaging probe [EndoFLIP™ System (Medtronic®) with an 8-cm catheter (EF-325N)]. A "stepwise" measurement protocol was applied, with bag filling from 0 to 50ml in 10-ml increments. Distensibility index (DI) was recorded at rest, voluntary squeeze, cough maneuver and during simulated evacuation at 30-, 40- and 50-ml volumes. Simulated defecation was evaluated by calculating the percentage change in the DI relative to resting values at each volume.

Results: Fifteen women were included: 4 HV (26.7%), 7 with FI (46.7%) and 4 with CC (26.7%). Women with FI were older than those in the CC and HV groups (median age: FI 72 years [66–75], CC 55 years [42-63], HV 48 years [43–53], p = 0.009. Median BMI was similar among groups [HV 24.0 Kg/m^2 (21.6 – 25.3), FI 27.0 Kg/m^2 (24.3 – 30.0) and CC 26.2 Kg/m^2 (21.5 – 30.5), p=0.367]. The DI was significantly higher in the FI group compared to CC and HV groups at rest, during squeeze and during cough maneuver across 30-, 40- and 50-ml volumes. The percentage change in DI during simulated defecation at 50 ml was significantly reduced in CC compared to HV (p=0.030). When groups were categorized based on suspected FDD (CC vs HV+FI), the percentage change in DI was significantly lower in CC at 40 ml [74% (37 – 109) vs 175% (100 – 250), p=0.037] and at 50 ml [81% (43 – 96) vs 111% (96 – 119), p=0.048].

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Anal EndoFLIP™				
N (%) 15 (100)	Healthy Volunteers 4 (26.7)	Fecal Incontinence 7 (46.7)	Chronic Constipation 4 (26.7)	P value
30 ml DI (mm^2*$mmHg^{-1}$). Median (IQR)				
At rest	0.5 (0.4 – 0.7) §	1.3 (0.9 – 3.0) †§	0.5 (0.4 – 0.5) †	0.010
During squeeze	0.3 (0.2 – 0.3) §	1.1 (0.9 – 1.3) †§	0.3 (0.2 – 0.4) †	0.005
Cough maneuver	0.4 (0.3 – 0.5) §	2.4 (1.0 – 3.0) †§	0.4 (0.3 – 0.4) †	0.009
Simulated defecation	0.9 (0.6 – 1.1) §	3.2 (1.4 – 4.9) †§	0.3 (0.2 – 0.4) †	0.016
30 ml Percentage change in DI (%). Median (IQR)	158 (106 – 205)	156 (99 – 356)	77 (48 – 100)	0.266
40 ml DI (mm^2*$mmHg^{-1}$). Median (IQR)				
At rest	1.0 (0.6 – 1.3)	3.5 (3.0 – 6.9)	1.3 (1.1 – 1.5)	0.030
During squeeze	0.2 (0.2 – 0.4) §	2.7 (2.4 – 3.0) †§	0.8 (0.5 – 1.3) †	0.004
Cough maneuver	0.6 (0.4 – 0.8) §	2.9 (2.3 – 5.3) †§	1.2 (0.7 – 1.4) †	0.003
Simulated defecation	1.4 (1.3 – 1.9) §	5.9 (3.9 – 9.2) †§	1.1 (0.5 – 1.5) †	0.005
40 ml Percentage change in DI (%). Median (IQR)	178 (138 – 216)	141 (73 – 295)	74 (37 – 109)	0.108
50 ml DI (mm^2*$mmHg^{-1}$). Median (IQR)				
At rest	2.2 (1.2 – 2.9) §	5.4 (4.6 – 6.9) §	3.2 (2.8 – 4.5)	0.002
During squeeze	0.5 (0.2 – 1.0) §	4.5 (3.8 – 5.0) §	1.7 (1.0 – 3.5)	0.002
Cough maneuver	1.4 (0.7 – 2.3) §	5.0 (4.6 – 5.8) §	2.9 (1.7 – 4.1)	0.005
Simulated defecation	3.0 (2.4 – 3.4) §	5.5 (4.6 – 6.0) †§	2.6 (1.2 – 4.3) †	0.016
50 ml Percentage change in DI (%). Median (IQR)	117 (113 – 301) ¶	99 (95 – 110)	81 (43 – 96) ¶	0.030

†=Statistically significant difference (p<0.05) between FI and CC groups.
§= Statistically significant difference (p<0.05) between FI and HV groups.
¶= Statistically significant difference (p<0.05) between CC and HV groups.

Conclusion: Impedance planimetry proves to be a valuable tool for distinguishing pelvic floor dysfunctions, as women with FI show higher DI values, whereas those with suspected FDD exhibit reduced DI changes during simulated defecation. These findings support the hypothesis that impaired rectal distensibility is a key feature of FDD, highlighting its role in continence and evacuation disorders.

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Disclosure: Nothing to disclose.

PP0628

HIGH-RESOLUTION ANORECTAL MANOMETRY (HRAM) IN PATIENTS UNDERGOING LOW ANTERIOR RESECTION (LAR) FOR RECTAL ADENOCARCINOMA

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Introduction: Patients (pts) with rectal adenocarcinoma who undergo Low Anterior Resection (LAR) are susceptible to developing anorectal dysfunctions, such as fecal incontinence, constipation, urge to defecate with altered defecation patterns, collectively termed Low Anterior Resection Syndrome (LARS). High-resolution anorectal manometry (HRAM) is a crucial diagnostic tool to assess the functional integrity of the anal sphincters and rectal sensory perception in these pts.

Aims & Methods: The aim of this study is to evaluate anal and rectal sphincter functionality in pts undergoing LAR by analyzing HRAM data. In this retrospective observational study, we analyzed the HRAM reports of 42 pts who underwent LAR for rectal adenocarcinoma between 2016 and October 2024. The assessments followed the London protocol. The average time between LAR and HRAM was two years.

Results: We enrolled 42 pts (16 males and 26 females, mean age 65 ± 11 years at the time of HRAM), including 10 pts with a stoma. HRAM showed a mean basal pressure of 43 ± 24 mmHg, with 71% of pts presenting basal hypotonia. The mean contraction pressure was 104 ± 46 mmHg, with 76% of pts showing hypocontractility. At 'push' manoeuvre, 52% of pts didn't show adequate relaxation. Regarding residual rectal sensitivity, 76% of pts presented hypersensitivity. In particular the maximum tolerable stimulus was evoked with a mean pressure of 57 ± 30 mmHg. The rectoanal inhibitory reflex (RAIR) was evoked in only 40% of pts. In 14 pts, the balloon expulsion test (BET) couldn't be performed, while in the remaining 28 pts, 54% had a negative BET and 46% a positive BET.

Conclusion: Our data confirm the presence of anorectal functional alterations after LAR. Most pts presented alterations of the anal sphincter apparatus with hypotonia and hypocontractility. Moreover, in half of the pts, there was no adequate sphincter relaxation during 'push' manoeuvre.

Additionally, most pts presented hypersensitivity of the residual rectum and a lack of RAIR. These data confirm the importance of HRAM in patients undergoing LAR in the personalized diagnostic therapeutic pathway. Further studies are needed to standardize pressure values in correlation with clinical data.

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Disclosure: Nothing to disclose.

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PP0629

STUDY OF ANORECTAL FUNCTION AND ANAL DISTENSIBILITY IN PATIENTS WITH CHRONIC CONSTIPATION

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Introduction: High-resolution anorectal manometry (HRAM) is the most commonly used technique for studying functional anorectal disorders, including chronic constipation. However, alterations in the defecatory maneuver are inconclusive for diagnosing defecatory dysfunction, necessitating additional confirmatory testing. The study of the geometry and distensibility of the anal canal may be useful in evaluating patients with chronic constipation.

Aims & Methods: To evaluate the manometric characteristics and anal canal distensibility in patients with chronic constipation compared to a control group of healthy volunteers (HV). The study included 15 healthy controls and 11 patients with chronic constipation. All participants underwent HRAM (Medtronic®), EndoFLIP (Medtronic®), and the balloon expulsion test (BET; Matla Systems®). HRAM was performed according to the London protocol. Distensibility index (DI) values were calculated at rest, during voluntary contraction (VC), and during the defecatory maneuver (DM) with volumes of 30, 40, and 50 mL.

Results: The mean age was 32.4 ± 12.8 in HV and 45.2 ± 12.4 in patients with CC (p = 0.012). Most patients with CC were women, 10 (90.9%). Among CC patients, 9.1% had stool consistency classified as Bristol 3-4 compared to 95% of HV (p < 0.001). The defecatory pattern was < 3 weekly bowel movements in 54.5% of CC patients. Obstructive defecation perception predominated in 7 patients (66.7%), while 1 (9%) had slow transit, and 3 (27.3%) had obstructive defecation and fecal incontinence (multiple sclerosis). All HV had normal BET, with a mean balloon expulsion time of 13.3 ± 10.4 seconds.

The HRAM findings revealed that the percentage of anal relaxation was 31.2 ± 18 in HV and 7 ± 23 in CC patients (p < 0.001). The rectoanal gradient (RAG) measured -28.3 ± 25 in HV and -51.5 ± 36.8 in CC patients (p = 0.06). Significant differences in rectal sensitivity were observed between CC patients and HV. The first sensation threshold was 50.5 ± 25.2 mL in HV compared to 87.3 ± 47.9 mL in CC patients (p = 0.018). Similarly, the urge threshold was 126.8 ± 35.4 mL in HV versus 177.5 ± 44 mL in CC patients (p = 0.004). The maximum tolerated volume also differed, recorded as 157.5 ± 37.9 mL in HV and 224.3 ± 60.3 mL in CC patients (p = 0.008).

EndoFlip results are presented in Table 1. The DI demonstrated better diagnostic performance than HRAM defecatory parameters for predicting balloon expulsion: DI-30 mL: 0.850; 95% CI: 0.701–0.999 (p=0.011); DI-40 mL: 0.817; 95% CI: 0.602–1 (p=0.021); DI-50 mL: 0.775; 95% CI: 0.525–1 (p=0.045); % anal relaxation: 0.742; 95% CI: 0.504–0.980 (p=0.078); RAG: 0.767; 95% CI: 0.540–0.993 (p=0.051).

DI	Maneuver	Healthy controls	CC-normal BET	CC-abnormal BET	p-value
30 mL	Rest	0.6 (0.4-0.7)	0.5 (0.4-1.9)	0.6 (0.4-0.7)	NS
	Squeeze	0.3 (0.2-0.4)	0.3 (0.2-0.5)	0.3 (0.2-0.4)	NS
	Push	0.9 (0.6-1.3)	1.4 (0.4-1.8)	0.4 (0.3-0.5)	0.036
40 mL	Rest	0.4 (0.3-0.6)	0.9 (0.6-3.5)	0.7 (0.3-2)	0.05
	Squeeze	0.3 (0.2-0.4)	0.2 (0.2-1)	0.5 (0.2-0.7)	NS
	Push	1.2 (1.1-3)	2.2 (0.9-2.9)	0.5 (0.2-1.1)	0.05
50 mL	Rest	0.8 (0.5-1)	2.2 (1.4-3.6)	1.7 (0.7-3.4)	0.008
	Squeeze	0.3 (0.2-0.7)	0.6 (0.4-1.8)	0.9 (0.2-1.5)	NS
	Push	3.1 (1.6-4.7)	4.2 (2.7-5.7)	1.2 (0.4-2.7)	0.08

Table 1. EndoFlip results.

Conclusion: Patients with chronic constipation exhibit significant differences in HRAM defecatory parameters and rectal sensitivity compared to control subjects. Although the sample size of patients with constipation is small, the DI demonstrates superior diagnostic performance compared to HRAM defecatory parameters about the BET.

References: None.

Disclosure: Nothing.

PP0630

SEXUALLY TRANSMITTED INFECTIONS MIMICKING INFLAMMATORY BOWEL DISEASES: CASE SERIES OF INFECTIOUS PROCTITIS

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Introduction: Sexually transmitted infections (STIs) are increasingly recognized as causes of proctitis, particularly among men who have sex with men (MSM). These infections frequently mimic inflammatory bowel diseases (IBD) in clinical, endoscopic, and histological presentations, leading to potential misdiagnosis and inappropriate treatments. Accurate recognition requires detailed sexual history and targeted diagnostic tests.

Aims & Methods: Patients were identified from IBD outpatient visits (2021–2023) and retrospectively selected for suspected infectious proctitis mimicking IBD. Diagnosis was confirmed via clinical evaluation, targeted tests (rectal swabs, PCR, serology), and histology. Follow-up continued for six months post-treatment.

Results:

1. A 42-year-old MSM with secondary syphilis treated with benzathine penicillin presented with bloody diarrhea, tenesmus, anorectal pain, and 10-kg weight loss. Mild anemia (Hb 12 g/dL) and elevated CRP were noted. Rectosigmoidoscopy (RSCS) revealed fibrinous ulcers causing luminal narrowing. Rectal swabs confirmed *Chlamydia trachomatis*. Treatment with doxycycline led to clinical and endoscopic resolution at six months.
2. A 33-year-old MSM with unprotected intercourse reported mucorrhea and tenesmus for one month. Calprotectin levels were elevated (861 µg/g). Colonoscopy revealed mild proctitis with mucosal hyperemia, and histology showed mild-to-moderate inflammation with crypt abscesses. Swabs confirmed *Neisseria gonorrhoeae*. Ceftriaxone resolved symptoms and normalized calprotectin (40 µg/g).
3. A 34-year-old male with primary sclerosing cholangitis (PSC) presented with two weeks of tenesmus, anal bleeding, and pruritus. Colonoscopy appeared macroscopically normal, but histology revealed human papillomavirus (HPV) infection. Anal canal condylomas were identified and successfully treated with cryotherapy, resulting in significant relief.

Conclusion: The rising prevalence of STI-related proctitis highlights the need for awareness among clinicians, as these infections often mimic IBD. Misdiagnosis can lead to inappropriate treatments, including steroids or biologics, causing adverse effects and delays in proper management. Detailed sexual history and targeted diagnostics, such as PCR and rectal swabs, are critical for accurate identification and effective treatment. These cases emphasize the critical role of integrating sexual history into clinical evaluation for proctitis.

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Awareness and timely recognition of STI-related proctitis are crucial to avoid misdiagnosis as IBD, ensure effective and targeted treatment, improve patient outcomes, and prevent inappropriate and prolonged follow-up.

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Proctitis in Men Who Have Sex with Men. Julia M. Steed et all

Disclosure: Nothing to disclose.

PP0631

ARTIFICIAL INTELLIGENCE AND ANAL CANCER SCREENING: FIRST MULTICENTRIC APPLICATION OF A TRINARY MODEL FOR ANAL LESION DETECTION AND CLASSIFICATION IN HIGH-RESOLUTION ANOSCOPY

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Introduction: High-resolution anoscopy (HRA) is the gold standard for assessing HPV-related anal lesions, though its diagnostic accuracy remains limited. While most AI models focus on binary classification, this study aims to develop and evaluate a deep learning model for detection and classification of HRA frames into high-grade (HSIL), low-grade (LSIL), and non-dysplastic lesions (e.g., inflammation).

Aims & Methods: This multicenter retrospective study included 191,951 frames from 107 HRA procedures across five centers using five imaging devices. Frames were labeled as HSIL, LSIL, or non-dysplastic based on matched histopathology. A YOLOv11-based object detection model was trained to detect these lesions using bounding boxes. The dataset was randomly split into training, validation, and test sets. Lesion detection was defined by correct bounding box placement. Classification metrics included recall (sensitivity), precision (PPV), and accuracy, reported as weighted averages across confidence thresholds of 0.45, 0.47, 0.50, 0.52 and 0.55.

Results: The model achieved an overall accuracy of 95.2% (94.7- 95.7) in classifying detected lesions. For HSIL, recall was 97.9% (97.6- 98.2) and precision was 93.9% (92.8, 95.1)). For LSIL, recall was 98.9% (98.8- 99.0) and precision was 98.5% (98.3, 98.7). For non-dysplastic lesions, recall was 98.7% (98.5- 98.8) and precision was 96.0% (94.9- 97.2).

Conclusion: This study introduces the first worldwide AI model capable of simultaneous detection and trinary classification (HSIL, LSIL, non-dysplastic mimickers) of anal canal lesions using heterogeneous, interoperable datasets, a key advancement toward real-world deployment and precision-guided care in AI-assisted HRA.

Disclosure: Nothing to disclose.

PP0632

EVALUATION OF THE CUT-OFF VALUE OF FECAL IMMUNOCHEMICAL TESTING FOR POPULATION-BASED SCREENING IN JAPAN

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Introduction: Colorectal cancer (CRC) screening has been widely implemented, and recent advances in fecal immunochemical testing (FIT) have led to a shift in screening modalities from guaiac fecal occult blood testing (gFOBT) to FIT. In Japan, FIT was introduced into the national screening program in 1992, and a two-sample method has been offered to individuals aged 40 years and older. However, there is no standardized use of testing devices or cutoff values. Although hemoglobin concentrations of 100-150 ng/g are commonly used, few studies have evaluated cut-off values in population-based screening. An optimal cut-off value should be defined for adequate quality assurance by considering the benefits and harms of CRC screening.

Aims & Methods: To evaluate the optimal cutoff value for population-based screening, we compared the test performance of FIT according to different cutoff values. A total of 383,793 individuals participated in population-based screening in Ibaraki Prefecture from April 2021 to March 2023. The OC-sensor was used as the testing modality, with a default cut-off value of 100 ng/g. Test performance indicators, including positive rate, detection rate, positive predictive value (PPV), cancer miss rate, and number needed to scope (NNS), were assessed across cut-off values ranging from 100 to 250 ng/g. Detection rate, PPV, and NNS were explicitly calculated for CRC/ Advanced neoplasia (AN). Since the colonoscopy completion rate is typically around 70%, PPV and NNS were calculated on a per-protocol basis.

Results: Subjects who submitted two samples were 370419. Among them, 863 cases of CRC and 1944 cases of advanced adenomas were diagnosed. At a cutoff value of 100 ng/g at the baseline, the positive rate was 6.7% (95%CI: 6.7-6.8) and the colonoscopy completion rate was 76.0% (95%CI: 75.4-76.5) (Table).

For AN, the detection rate was 0.76% (95%CI: 0.73-0.79), 14.8% for PPV (95%CI: 14.3-15.3), and 6.8% (95%CI: 6.5-7.0) for NNS. For CRC detection, the detection rate was 0.23% (95%CI: 0.22-0.25), 4.5% for PPV (95%CI: 4.3-4.8), and 22.0% (95%CI: 22.0-23.5) for NNS.

The NNS decreased as the cutoff value increased, but the CRC miss rate also increased. The cutoff value reached 180 ng/g, the CRC miss rate became 19.5% (95%CI:16.8-22.1), whereas the positive rate was 4.2% (95%CI: 4.2-4.3), 0.19% (95%CI: 0.17-0.20) for CRC detection rate, 5.9% (95%CI: 5.5-6.3) for PPV, and 16.9 (95%CI: 15.8-18.2) for NNS.

Conclusion: The current cutoff value led to high detection rates for both CRC and AN, but it also required a greater number of colonoscopies. The availability of colonoscopy services varies across local municipalities and is often insufficient in rural areas. Adjusting the cutoff value is frequently used to address disparity in access to colonoscopy.

However, the trade-off among benefits, harms, and resources is influenced by the social context, including individual preference and value. Therefore, an acceptable level of cancer miss rates should be carefully evaluated. Since the optimal cutoff value cannot be determined unequiv-

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ocally, a comprehensive decision should be made by balancing benefits, harms, and resources, while also considering the preference of local residents.

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Disclosure: Nothing to disclose.

PP0633

AN ORGANIZED FECAL IMMUNOCHEMICAL TEST-BASED SCREENING PROGRAM SHORTLY IMPACTS COLORECTAL CANCER EARLY DIAGNOSIS AND SURVIVAL

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Introduction: In 2018, a biennial fecal immunochemical test (FIT) screening program for colorectal cancer (CRC) was introduced in northern Portugal, targeting 1.1 million residents aged 50-74 years.

Aims & Methods: This study aimed to evaluate trends in CRC incidence, early diagnosis, and survival before and after the program's implementation.

Population-based retrospective cohort study in northern Portugal. Data on CRC cases diagnosed between 2014-2020 were retrieved from the Portuguese National Cancer Registry (RON). Early diagnosis was defined as stage I/II tumors. Net survival estimates for 2014–2016 vs. 2018-2020 were compared. Data from RON were linked to the FIT-based screening program of the Northern Portugal Regional Health Administration for the years 2018–2020 to assess screening outcomes.

Results: Between 2018 and 2020, the first 3 years of the program, 482,938 invitations for FIT screening were sent, resulting in an adherence rate of 34.4% (165,895). The negative predictive value of FIT was 99.95% (95% CI: 99.94–99.96). Overall, coverage by FIT and/or colonoscopy was 44.9%. The proportion of early cancers was significantly higher among individuals submitted to FIT screening (58.3%) or colonoscopy (62.8%) compared to unscreened individuals (49.2%) ($p<0.001$). Net survival rates at 1-year and 3-years were significantly higher among individuals who accepted FIT screening or underwent colonoscopy compared to those who remained unscreened ($p<0.001$).

Conclusion: Biennial FIT led to an increase in early diagnoses and improved survival compared to no screening shortly after its introduction. A health system's comprehensive capacity boosts population coverage and improves CRC outcomes.

Disclosure: Nothing to disclose.

PP0634

ENHANCED OPTICAL DIAGNOSIS OF COLORECTAL POLYPS USING COMPUTER-AIDED DIAGNOSIS AND THE 3-SECOND RULE FOR HIGH-CONFIDENCE EVALUATION

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Introduction: Previous research has shown that categorizing high-confidence (HC) diagnoses as those reached within 3 s without hesitation notably enhanced the optical diagnostic accuracy for colorectal polyps. This study aimed to assess the efficacy of a new computer-aided diagnosis (CADx) system integrated with the 3-second rule in the optical diagnosis of colorectal polyps among endoscopists with different levels of experience.

Aims & Methods: In this multicenter ex vivo study, 35 endoscopists evaluated 100 polyps (51 adenomas, 39 hyperplastic polyps, and 10 sessile serrated lesions) using narrow-band imaging video clips on an online platform. The assessments comprised two phases: individual endoscopist diagnosis and CADx-assisted diagnosis. HC assignments followed the 3-second rule in both phases. Performance metrics assessed included HC accuracy, HC rate, and adherence to the Preservation and Incorporation of Valuable Endoscopic Innovations (PIVI) and Simple Optical Diagnosis Accuracy (SODA) thresholds.

Results: HC diagnostic accuracy improved from 78.3% (95% CI: 76.6–80.0) to 89.8% (95% CI: 88.6–90.9) with CADx assistance ($p < 0.001$). The proportion of HC predictions increased from 64.2% to 75.4% with the CADx assistance ($p < 0.001$). Novice endoscopists notably showed a significant improvement in HC accuracy with CADx assistance (74.1% vs. 88.8%; $p < 0.001$). The CADx-assisted diagnoses nearly met the SODA standards and PIVI thresholds when combined with the 3-second rule.

Conclusion: Integrating CADx with the 3-second rule significantly improved the optical diagnostic performance for colorectal polyps, particularly among endoscopists lacking systematic training.

Disclosure: Jung Ho Bae has a financial interest in Ainex Corporation, while Soonwhan Kang and Jinbae Park are employed by Ainex Corporation. The authors disclose no conflicts of interest.

PP0635

UNMASKING ASPIRATION RISKS: A PROPENSITY MATCHED ODYSSEY OF GLP-1 AGONISTS AND COLONOSCOPY

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Introduction: The use of GLP-1 receptor agonists has revolutionized the treatment of Type II Diabetes Mellitus and obesity. Beyond these conditions, their use is being investigated for many other indications. Given the current global obesity pandemic, it is not uncommon to encounter patients on GLP-1 receptor agonists when they present for procedures [1]. GLP-1 receptor agonist's (GLP-1), such as semaglutide (Ozempic), are known to slow gastric emptying, leading to gastroparesis [2-3].

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Consequently, many patients on GLP-1 receptor agonist's experience recurrent vomiting and are therefore at increased risk of pulmonary aspiration during procedures under sedation [4-6].

Aims & Methods: We aim to conduct this retrospective study to evaluate the risk of aspiration pneumonia in patients on GLP-1 receptor agonists undergoing colonoscopy.

Subjects and methods: TThis is a multicentre retrospective cohort study using a large research network (TriNetX, LLC) comprising over 100 million patients. The study period was between 2005 and 2025. Patients who underwent colonoscopy during the study period were included in the cohort and divided into two groups based on whether they were on GLP-1 receptor agonists or not.

The primary outcome was aspiration pneumonia, defined as aspiration pneumonia or the need for mechanical ventilation or respiratory failure within 7 days of undergoing colonoscopy.

The secondary outcome was hospitalization within 7 days of colonoscopy. Outcomes were compared between the two groups.

Results: A total of 126,663 patients on GLP-1 agonists underwent colonoscopy during the study period. A total of 1,380,652 patients without exposure to GLP-1 receptor agonists underwent colonoscopy before propensity score matching. The mean age of patients who underwent colonoscopy was 58.8 ± 10.5 years in both groups. In the group with GLP-1 receptor agonists, 0.03% of patients had aspiration pneumonia compared to 0.06% in the other group. This difference was statistically insignificant (p=0.000). Similarly, no significant differences were observed in the number of patients who underwent mechanical ventilation, were hospitalized, or had type 2 respiratory failure within 7 days of undergoing colonoscopy, between the two groups.

Conclusion: Despite ongoing concerns about the increased risk of aspiration pneumonia in patients with GLP-1 receptor agonists undergoing colonoscopy under sedation, the results indicate otherwise. There is no need to perform a gastroscopy with minimal sedation in patients on GLP-1 receptor agonists undergoing colonoscopy, as recommended by certain anesthetic and gastroenterology societies.

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Disclosure: Nothing to disclose.

PP0636

CAN PAN-COLONIC CHROMOENDOSCOPY (PCC) IMPROVE THE ADENOMA DETECTION RATE IN FIT-POSITIVE PATIENTS? A RANDOMIZED STUDY

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Introduction: The prevalence of adenoma in patients with positive fecal immunochemical tests (FIT) is shown to be higher. Increased adenoma detection and removal during colonoscopy have been shown to prevent CRC and interval cancers. However, there is considerable variation in the adenoma detection rate (ADR) among colonoscopies performed for FIT. We hypothesize that the ability of dye-based chromoendoscopy to stain the colorectal lesion and their borders and increase their contrast against the normal colorectal mucosa would enhance the detection of adenoma and non-polypoid colorectal neoplasms (NPCRN) in patients undergoing FIT positive colonoscopy.

In this randomized control trial, we aim to compare the effectiveness of pan-colonic indigo carmine chromoendoscopy (PCC) to high-definition white light colonoscopy (WLC) in individuals undergoing colonoscopy for positive FIT test.

Aims & Methods: Patients undergoing colonoscopy after a FIT positive test were randomized 1:1 to undergo PCC or WLC. The study was conducted between January 2021 to May 2023 at the Singapore General Hospital, Singapore. Colonoscopy was performed by general and advanced endoscopists, and all received training on PCC before the study. The primary outcome was ADR between the two groups. The secondary outcomes were advanced adenoma detection rate (AADR), NPCRN detection rate, sessile serrated lesion detection rate (SSLDR), and withdrawal time between the groups. We used the two-tailed Mann–Whitney U test and Chi-squared statistics for comparison. The ethics committee approved the study, and it was registered with clinicaltrials.gov (NCT04512612)

Results: We recruited 256 patients (PCC-128; WLC-128). The mean age of the entire cohort was 66 + 8.4 years. The mean ADR was comparable between PCC and WLC (70.3% vs. 64.8%, p-0.350). Likewise, the AADR (19.5% vs. 14.8%, p-0.320], and SSLDR (6.25% vs. 1.56%, p-0.102] were similar between the two groups. We found PCC detected more diminutive adenomas (0.97% vs. 0.58%, p-0.025] and NPCRN per colonoscopy (0.35% vs. 0.17%, p-0.028). The mean withdrawal time was longer with PCC compared to WLC (13.3 min vs. 10.2 min, p-0.001). General and advanced endoscopists had no difference in performance measures for APC and NPCRN per colonoscopy.

Conclusion: The ADR in FIT-positive colonoscopy is higher, and both PCC and WLC yielded comparable results. PCC merits consideration because of its increased ability to detect NPCRNs, which are precursors for interval cancer.

Disclosure: Nothing to disclose.

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PP0637

EFFICACY AND SAFETY OF COLD SNARE POLYPECTOMY IN INTENSIVE DOWNSTAGING POLYPECTOMY FOR COLORECTAL POLYPS AS PATIENTS WITH FAMILIAL ADENOMATOUS POLYPOSIS

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Introduction: Familial adenomatous polyposis (FAP) is an inherited colorectal cancer syndrome characterized by the development of numerous adenomatous polyps in the colon and rectum, with prophylactic colectomy being the standard treatment to prevent disease progression. Recently, endoscopic intensive downstaging polypectomy (IDP) has attracted attention as a novel disease management strategy, particularly for patients with non-dense FAP who refuse or delay surgery. Since IDP requires the resection of numerous polyps, simpler and safer techniques such as hot snare polypectomy (HSP), cold snare polypectomy (CSP), and hot biopsy (HB) are commonly used. However, the effectiveness of these techniques for the treatment of small adenomas in the context of IDP has not been fully evaluated.

Aims & Methods: This study aimed to evaluate the safety and efficacy of CSP compared with HSP and HB in patients with FAP undergoing IDP. This retrospective study included patients with FAP who underwent IDP using only CSP, HSP, or HB at Hiroshima University Hospital between September 2012 and March 2024. Patients were divided into two groups: the CSP group, treated with CSP from October 2022 onward, and the HOT group, treated with HSP and HB between 2012 and September 2022.

The primary endpoints were the incidence of adverse events and the treatment duration per polyp. Secondary endpoints included the incidence of immediate and delayed bleeding, perforation, total treatment duration per session, number of polyps removed per procedure, R0 resection rate, and the rate of missed polyps ≥10 mm.

Results: During the study period, 46 treatment sessions were performed in 24 patients in the CSP group, resulting in the removal of a total of 3,689 polyps. In contrast, 139 treatment sessions were performed in 27 patients in the HOT group, with a total of 5,912 polyps removed. The average number of polyps removed per session was significantly higher in the CSP group (80.5 polyps) than in the HOT group (42.1 polyps) ($p < 0.01$).

Variables	Total (n=185)	CSP (n=46)	HOT (n=139)	p-value
Treatment duration per procedure (min)	45.9 (22.1)	51.1 (23.2)	44.3 (21.7)	0.06
Polyps removed per procedure	51.6 (52.1)	80.5 (56.8)	42.1 (46.9)	< 0.01
Treatment duration per lesion (min)	1.7 (1.4)	1.3 (1.4)	1.8 (1.3)	0.01
Maximum tumor size per procedure (mm)	5.2 (2.4)	6.0 (2.5)	4.9 (2.4)	< 0.01
Overall adverse events (n (%))	14 (7.6)	0 (0)	14 (10.1)	0.03
Immediate bleeding (n (%))	10 (5.4)	0 (0)	10 (7.2)	0.06
Delayed bleeding (n (%))	2 (1.1)	0 (0)	2 (1.1)	0.41
Perforation (n (%))	2 (1.1)	0 (0)	2 (1.1)	0.41

Additionally, the treatment duration per polyp was significantly shorter in the CSP group (1.3 minutes) than in the HOT group (1.8 minutes) ($p = 0.01$). The total treatment duration per session was slightly longer in the CSP group (51.1 minutes) compared to the HOT group (44.3 minutes), although the difference was not statistically significant ($p = 0.06$). The maximum polyp size was significantly larger in the CSP group (6.0 mm) than in the HOT group (4.91 mm) ($p < 0.01$). The overall incidence of adverse events was significantly lower in the CSP group. Immediate bleeding occurred in 10 cases (7.2%) in the HOT group but was not observed in the CSP group. Delayed bleeding and perforation were also observed only in the HOT group; however, there were no statistically significant differences between the groups.

Conclusion: This study demonstrated that CSP is a safe and effective option for intensive polyp reduction in patients with FAP. Compared with HSP and HB, CSP enabled the removal of more polyps in a shorter duration and reduced the risk of adverse events.

Disclosure: Nothing to disclose.

PP0638

MOTORIZED DOUBLE-BALLOON COLONOSCOPE SUPPORTED WITH MACHINE LEARNING FOR LUMEN NAVIGATION: TOWARD SAFE AND AUTOMATED INSERTION

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Introduction: The number of deaths caused by colorectal cancer has been gradually increasing in recent years. However, it is treatable if detected early, with five-year survival rates exceeding 90%. As early-stage cancers are often asymptomatic, regular screening is essential.

Expanding the number of endoscopic examinations requires a corresponding increase in the number of trained endoscopists. A physician shortage hinders large-scale screening. Among the various steps in colonoscopy, the insertion procedure is technically demanding, especially in anatomically complex regions such as the sigmoid colon. Mastery of this step requires considerable training and clinical experience, which poses a significant barrier to scalability.

Under these circumstances, there is a strong demand for simplifying and automating the insertion procedure of colonoscopes.

Aims & Methods: In this study, we aim to address this challenge by developing a motorized double-balloon colonoscope. The colonoscope has a double-tube structure, consisting of an outer and an inner tube. Each tube can be independently bent at the tip using wires, allowing the scope to follow the curves of the colon and navigate complex anatomical regions. The inner tube can move axially within the outer tube, and both tubes are equipped with separate balloons. The alternating inflation and deflation of the balloons, one tube can be fixed while the other advances. This mechanism enables stable and controlled insertion along the intestinal lumen. A single operator can perform the procedure by manipulating the outer tube with one hand and operating a dedicated controller with the other. Additionally, 300 images were extracted and annotated from colonoscope insertion videos. These were used to train a YOLO model capable of detecting the direction of the intestinal lumen, with the aim of assisting in steering during insertion in future applications.

Results: To evaluate the insertion performance of the developed colonoscope, we conducted insertion experiments using a colon model. The procedure began with the inner tube placed inside the outer tube, and the device was inserted from the anus. After passing through the rectum, insertion into highly curved regions such as the sigmoid colon proved difficult using the outer tube alone. Therefore, the tip of the outer tube was aligned with the direction of insertion, the intestinal tract was fixed using a balloon, and the inner tube was advanced.

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The bending of the inner tube was manipulated to avoid collision with mucosal folds, allowing the device to reach the descending colon without overstretching the colon wall. By repeatedly alternating balloon fixation and tube advancement, the device proceeded through the transverse and ascending colon, eventually reaching the cecum. This sequence of operations was repeated in three trials, all of which resulted in successful insertion into the cecum. These results demonstrate that the developed device is capable of stable and stepwise insertion throughout the colon, including anatomically complex and highly curved segments.

Conclusion: In this study, we developed a motorized double-balloon colonoscope and a machine learning-based technique for detecting the direction of the intestinal lumen. Insertion experiments using a colon model demonstrated the effectiveness and reliability of the proposed insertion method.

As the next step, we aim to construct a prototype system that integrates these technologies. We also plan to implement robotic arm control that mimics endoscopist hand movements to achieve safe, minimally invasive insertion.

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Disclosure: Nothing to disclose.

PP0639

IMPACT OF BOWEL PREPARATION ON PROXIMAL SERRATED DETECTION RATE (PSDR) IN SCREENING COLONOSCOPY: AN OBSERVATIONAL COHORT STUDY

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Introduction: Despite advances in colorectal cancer (CRC) screening programs, the incidence of post-colonoscopy colorectal cancer (PCCRC) remains significant. Several recent cohort studies have highlighted serrated polyps as key contributors to PCCRC occurrence and related mortality. For this reason, Proximal Serrated Polyp Detection Rate (PSDR) has been proposed as a quality indicator for CRC screening programs, complementing the adenoma detection rate (ADR).

Aims & Methods: This monocentric retrospective cohort study aims to evaluate the impact of optimal bowel preparation - defined as a Boston Bowel Preparation Scale (BBPS) ≥8/9 - on PSDR and ADR, compared to adequate preparation (BBPS score of 6/9 or 7/9).

All consecutive patients aged 50–69 years who underwent screening colonoscopy following a positive fecal immunochemical test (FIT) at the University Hospital of Parma between May 2021 and April 2025 were included. Data on patient demographics, endoscopist characteristics, and histologic findings were collected. The association between bowel preparation quality and PSDR and ADR was assessed using chi-square tests and regression analysis.

Results: Of the 1,070 patients included, 512 were female (47.9%) with a median age of 60 years (IQR 55–66). The Proximal Serrated Polyp Detection Rate (PSDR) was significantly higher in colonoscopies with optimal bowel preparation compared to those with adequate preparation (13.6% vs. 8.2%, p = 0.008).

Multivariate analysis revealed a significant association between PSDR and both the quality of bowel preparation (OR 1.73, 95% CI 1.14–2.62, p = 0.01) and endoscopist expertise in screening procedures (OR 2.10, 95% CI 1.43–3.12, p < 0.001). Detection of Sessile Serrated Lesions (SSLs) was significantly higher with optimal preparation (7.5% vs. 3.7%, p = 0.012). In contrast, the Adenoma Detection Rate (ADR) was similar between groups (50.6% vs. 48.4%, p = 0.493).

Conclusion: Optimal bowel preparation, was associated with significantly higher PSDR and SSL detection rates compared to adequate preparation, highlighting its potential role as a modifiable factor to enhance serrated lesion detection in CRC screening programs. In contrast, ADR did not significantly vary with bowel preparation quality. Striving for optimal bowel preparation may contribute to improving the overall quality and effectiveness of CRC screening.

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PP0640

ADVANTAGE OF RETROFLEX INSPECTION FOLLOWING FORWARD VIEW INSPECTION IN THE ASCENDING COLON DURING COLONOSCOPY

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Introduction: Inspection accuracy on colonoscopy is problematic because of the colon's complex structure and the need for skillful operators. In particular, lesions in the ascending colon (AC) may be difficult to detect because of its high folds. Additionally, it has recently been reported that post-colonoscopy colorectal cancer is often located in the proximal colon. There are only a few reports on the effectiveness of retroflex inspection after forward view inspection in the AC during colonoscopy. While endoscopes and colonoscopy using artificial intelligence or image enhancement systems to detect lesions have been developed, it is important to reconsider basic endoscopic techniques, such as retroflex inspection, during colonoscopy.

Aims & Methods: We performed a prospective study to investigate the usefulness of retroflex inspection after forward view inspection in the AC during colonoscopy. Between August 2022 and December 2024, a single endoscopist enrolled and performed screening colonoscopies for 335 consecutive patients aged ≥40 years. Patients with inflammatory bowel disease, gastrointestinal polyposis, post-colorectal surgery, or poor preparation (≤1 point in at least one of three areas on the Boston Bowel Preparation Scale) were excluded from this study. The AC was inspected in this sequence: forward direction (Inspection 1), retroflex direction (Inspection 2), and forward direction (Inspection 3). The primary endpoint was the adenoma detection rate (ADR) in the AC. Secondary endpoints were the adenoma miss rate (AMR) in the AC per colonoscopy in Inspection 1 and Inspections 1+2, percentage of cases in which lesions were detected in Inspection 2, average observation time in the AC for each inspection, success rate of retroflexion in the AC, average pain score (0 points: no pain, 5 points: most severe pain), and complication rate during Inspection 2.

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Results: We evaluated 300 patients with a median age of 66 years. The ADR in the AC for Inspection 1 vs. 1+2 was 18.7% vs. 28.7%, respectively (odds ratio: 1.749; 95% CI: 1.174–2.622; $p = 0.005$); the values for Inspection 1+2 vs. 1+2+3 were 28.7% vs. 30.0%, respectively (odds ratio: 1.066; 95% CI: 0.739–1.540; $p = 0.788$) (Table 1). The average AMR in the AC per colonoscopy for Inspection 1 vs. 1+2 was 12.6% (95% CI: 9.0%–16.2%) vs. 2.6% (95% CI: 0.8%–3.8%) ($p < 0.001$) (Table 1).

The percentage of cases in which lesions were detected in Inspection 2 was 13.8% (41/298; total number of lesions: 47). The average diameter of lesions discovered and resected endoscopically during Inspection 2 was 6.1 mm (range: 3–20 mm). The locations of these lesions were as follows: proximal AC: 27.7% (13/47); mid-AC: 38.3% (18/47); and distal AC: 34.0% (16/47). The average observation times in the AC (Inspection 1 vs. 2 vs. 3) were 80.7 (95% CI: 77.8–83.6) vs. 59.1 (95% CI: 56.9–61.3) vs. 82.1 (95% CI: 79.4–84.8) seconds, respectively. The success rate of retroflexion in the AC was 99.3% (298/300), and the average pain score and complication rate during Inspection 2 were 0.5 points (95% CI: 0.4–0.6 points) and 0%, respectively.

	Adenoma Detection Rate in the AC			Adenoma Miss Rate in the AC (Average rate per colonoscopy)		
	Rate (Number)	OR (95%CI)	p-value	Rate	95%CI	p-value
Inspection 1	18.7 % (56)			12.6 %	9.0–16.2	
Inspection 1+2	28.7 % (86)	1.749 (1.174–2.622)	.005	2.6 %	0.8–3.8	< .001
Inspection 1+2+3	30.0 % (90)	1.066 (0.739–1.540)	.788			

Table 1 Adenoma Detection Rate & Adenoma Miss Rate in the AC.

Conclusion: Retroflex inspection following forward inspection in the AC during colonoscopy detects additional adenomas and is minimally invasive and safe. This technique contributes to improving the accuracy of colonoscopy.

Disclosure: Nothing to disclose.

PP0641

A ENDOCYTOSCOPY STAINING QUALITY SCORE SCALE: INNOVATIVE EXPLORATION FROM DEVELOPMENT TO VALIDATION

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Introduction: Endocytoscopy (EC) represents an advanced high-magnification endoscopic modality that permits *in vivo* cellular-level observation and assessment. However, achieving consistent high-quality staining constitutes a primary technical limitation.

This study aimed to develop and validate a novel scoring scale for evaluating EC staining quality, thereby providing clinicians with a straightforward and reliable instrument to enhance procedural quality control.

Aims & Methods: We evaluated the staining quality of images from four perspectives: nuclear clarity, glandular lumen clarity, surface mucus and dye rinsing degree, and image capture quality (including issues such as improper fixation leading to virtual images, out-of-focus causing image blur, and inadequate brightness).

Each evaluation criterion was scored from 0 to 2, with a total possible score of 8. The total image evaluation scores were then classified as follows: ≤2 points for Grade 1 (poor staining quality), 3-5 points for Grade 2 (moderate staining quality), and ≥6 points for Grade 3 (high Staining quality).

Data were retrospectively collected from patients undergoing colonoscopy with EC image acquisition between June 2023 and June 2024. An initial cohort comprising 1000 EC images was utilized to establish the scale's reliability, validity, and internal consistency. Subsequently, an independent validation cohort of 1000 EC images was analyzed using polynomial regression to investigate determinants of staining quality, and external validation was conducted to further evaluate the scale's applicability.

	Nuclear clarity	Glandular lumen clarity	surface mucus and dye rinsing degree	Image capture quality
0	<50% area Nuclear clarity	<50% area Glandular lumen clarity	<50% area Complete mucus and dye wash-out	Poor stabilization: blurry image, poor focus and brightness, details not clearly discernible
1	50%-70% area Nuclear clarity	50%-70% area Glandular lumen clarity	50%-70% area Complete mucus and dye wash-out	Average stabilization and focus, slightly dim brightness, partial clarity of details
2	>70% area Nuclear clarity	>70% area Glandular lumen clarity	>70% area Complete mucus and dye wash-out	Good stabilization: Accurate focus, adequate brightness, clear details
Score				

Image capture quality includes: poor image stabilization resulting in blurred images, out-of-focus leading to image blurriness, insufficient image brightness.

Table 1. Endocytoscopy staining quality score scale and scoring criteria.

Results: A total of 2,964 and 1,946 EC-stained images were analyzed in the initial and validation cohorts, respectively, with each cohort ultimately including 500 EC-stained images. There were no statistically significant differences in age and gender within and between the initial and validation cohorts.

The scale demonstrated excellent inter-rater reliability (agreement metric = 0.992), alongside good internal consistency reliability (coefficient = 0.81) and validity (index = 0.849). A strong positive correlation was observed between Average Optical Density (AOD) values and gold standard scores (Spearman's P= 0.81). Lesion morphology, size, pathological classification, pit pattern type, and physician experience level were identified as significant factors influencing EC staining quality (all $p < 0.05$). A one-month training intervention utilizing the scale significantly enhanced scoring consistency among junior physicians, evidenced by increases in Kappa coefficients measuring agreement with the gold standard (Physician A: 0.61 to 0.81; Physician B: 0.66 to 0.83).

Conclusion: The Endocytoscopy Staining Quality Assessment Scale serves as a reliable and valid instrument for evaluating staining quality. Its implementation can assist endoscopists in improving their proficiency in assessing image staining quality and refining staining techniques, ultimately contributing to enhanced diagnostic accuracy and reliability during EC procedures.

Disclosure: Nothing to disclose.

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PP0642

EXTRACELLULAR VESICLES IN STOOL: A PROMISING SOURCE OF BIOMARKERS FOR COLON HIGH RISK LESIONS AND COLORECTAL CANCER

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Introduction: Colorectal cancer (CRC) ranks as the third most common cancer globally in both men and women, with a notably higher incidence in developed countries. While early detection, facilitated by screening tools such as the fecal occult blood (FOB) test, has contributed to a decline in mortality rates, CRC continues to pose a significant public health challenge. Meanwhile, extracellular vesicles (EVs), known to be lipid-based particles secreted by cells that carry various macromolecules like nucleic acids and proteins, have emerged as a promising source for the identification of novel fecal biomarkers.

Aims & Methods: Given that FOB testing can produce false positives, often resulting in unnecessary exploratory colonoscopies and placing strain on healthcare resources, the goal of this study is to identify more specific CRC biomarkers within stool-derived EVs to enhance the diagnostic utility of current screening methods.

In this study, 20 stool samples were analyzed from individuals categorized as healthy, low-risk, high-risk, or diagnosed with colorectal cancer (CRC). The low-risk group included individuals with one or two adenomatous lesions exhibiting low-grade dysplasia and serrated lesions without dysplasia, all measuring less than 10 mm.

The high-risk group comprised individuals with three or more tubular adenomas with low-grade dysplasia under 10 mm, at least one adenoma with a villous component, any lesion with high-grade dysplasia over 10 mm, or at least one serrated lesion either displaying dysplasia or exceeding 10 mm in size.

Results: EVs were isolated using an optimized size exclusion chromatography protocol. Characterization of the EVs was carried out according to the MISEV 2023 guidelines, including analysis of particle size and concentration via Nanoparticle Tracking Analysis (NTA), visualization through transmission electron microscopy (TEM) and atomic force microscopy (AFM), and assessment of the tetraspanins CD9 and CD63.

Following isolation, proteomic profiling of the EVs was performed using liquid chromatography–mass spectrometry (LC-MS), aiming to distinguish protein expression patterns between “normal” individuals (healthy and low-risk) and “discovery” patients (high-risk and CRC cases). A subsequent Gene Set Enrichment Analysis (GSEA) was conducted on differentially expressed proteins to explore associated biological pathways.

The study successfully optimized EV isolation from stool samples, effectively separating vesicles from free proteins. From a total of 484 identified proteins, 75 were more abundant in EVs from discovery patients, while 147 were more prevalent in normal individuals, indicating potential biomarker candidates for CRC detection. Gene Set Enrichment Analysis (GSEA) identified multiple inflammation-related pathways significantly enriched in upregulated proteins from discovery-associated EVs.

Conclusion: In summary, this work demonstrates the potential of fecal EVs as a non-invasive platform for early colorectal cancer detection. By improving diagnostic specificity, this approach may help reduce the number of unnecessary colonoscopies.

Future studies will aim to validate these biomarker candidates in larger, more diverse patient cohorts.

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PP0643

RECURRENCE AND COMPLICATIONS AFTER COLORECTAL ENDOSCOPIC MUCOSAL RESECTION: A TWELVE-YEAR PROSPECTIVE COHORT STUDY

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Introduction: Colorectal endoscopic mucosal resection (EMR) is the treatment of choice for large colorectal lesions, but it has an important recurrence risk, requiring endoscopic surveillance. This study aims to identify factors associated with relapse and complications, by analysing the 5-year follow-up.

Aims & Methods: Prospective cohort study conducted in a single centre, including all patients submitted to colorectal pEMR between 2009 and 2021 with follow-up ≥3 months. We analysed lesion characteristics, calculated Sydney EMR recurrence tool (SERT) and SMSA (size, morphology, site, access) scores and compared complication and recurrence rates and respective risk factors.

Results: Included 568 mucosal resections, median age 69.5 years (IQR 62.3-76.0), male sex 58.1%. From the lesions included: 32.6% ≥40mm; 51.6% plane and 48.4% sessile; 35.9% in the right colon, 27.6% in the rectum, 23.2% in the transverse colon and 13.2% in the left colon. The median SMSA score was 4 (IQR 3-4) and median SERT score 1 (IQR 0-2), complication rate of 7.0% (bleeding, n=33; perforation, n=4; post-polypectomy syndrome, n=3) and overall recurrence rate of 20.3%: 15.1% in 3 months, 6.3% in 12 months, 2.5% in 36 months and 0% in 60 months.

The variables associated with recurrence were size ≥40mm (35.0% vs. 14.6%, p<0.001), anal location (35.1% vs. 19.7%, p<0.01), SERT score>0 (27.5% vs. 11.8%, p<0.001) and SMSA score=4 (30.6% vs. 11.5%, p<0.001), but not argon plasma therapy (22.8% vs. 18.7%, p=0.26). By logistic regression, size ≥40mm was associated with higher recurrence, but not statistically significant (OR 3.19, CI95% 0.69-1.63).

Use of anticoagulant/antiplatelet drugs was associated with higher risk of complications (12.7% vs. 4.7%, p<0.001), particularly bleeding (11.4% vs. 4.2%, p<0.01), with no difference for size ≥40mm (9.2% vs. 6.0%, p=0.17) or SMSA score=4 (7.6% vs. 6.5%, p=0.61).

Conclusion: The 5-year surveillance proved colorectal EMR’s efficacy. Recurrence is higher in lesions ≥40mm, with anal location, SERT score>0 and SMSA score=4, whereas bleeding is more frequent with the use of anticoagulant/antiplatelet drugs.

Disclosure: Nothing to disclose.

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PP0644

A RANDOMIZED CONTROLLED TRIAL EVALUATING THE SAFETY, EFFICACY, AND PATIENT COMFORT OF A SINGLE-USE ELECTRONIC COLONOSCOPE IN CLINICAL DIAGNOSIS AND TREATMENT

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Introduction: Although reusable colonoscopes remain the clinical gold standard, persistent microbial biofilms compromise reprocessing efficacy, elevating cross-contamination risks during endoscope reprocessing cycles[1].

A systematic review of 73 documented endoscopic infection outbreaks (esophagogastroduodenoscopy: 24; endoscopic retrograde cholangiopancreatography: 42; colonoscopy: 7) revealed escalating clinical impacts: attack rates of 3.5%, 7.1%, and 12.8%, with corresponding mortality rates of 6.3%, 12.7%, and 10.0%, respectively[2].

Building on evidence confirming the procedural efficacy and operational safety of disposable gastroscopes[3] [4], this randomized controlled trial systematically evaluates a novel single-use colonoscope through multidimensional assessment of assessing patient comfort, operator performance, and clinical outcomes.

Aims & Methods: This single-center prospective trial randomized 100 eligible patients (1:1) to single use colonoscope or conventional colonoscopy using computer-generated allocation. Primary endpoints included patient comfort assessed via grip strength during colonoscopy, efficacy of endoscopic mucosal resection (EMR) (en bloc resection rate and procedural duration), adenoma detection rate (ADR), adenomas per colonoscopy (APC) and cecal intubation time. Secondary outcomes evaluated device maneuverability using a 3-point Likert scale and overall safety.

Results: The demonstrated comparable baseline characteristics with balanced demographic profiles (all $P > 0.05$). No significant difference was observed in colonoscopy comfort based on grip strength pain assessment ($P > 0.05$). Both groups achieved 100% en bloc resection rates during EMR, with similar median procedure durations (679.23 ± 272.97 vs. 693.92 ± 477.06 s, $P = 0.899$). No significant difference between ADR (56% vs. 60%, $\chi^2=0.164$, $P = 0.685$) and APC (median 1.0 vs. 1.0, $P = 0.349$). Cecal intubation was successful in all participants, with comparable time-to-cecum between the two groups (448.34 ± 348.58 s vs. 368.00 ± 243.39 s, $P = 0.185$). One sigmoid carcinoma was identified in each group. No adverse events (e.g., bleeding, perforation, or device failure) occurred in either group. However, the single use colonoscope showed inferior subjective imaging quality ($P < 0.001$) and luminal far-field brightness/visualization ($P < 0.001$) compared to conventional scopes, potentially impacting lesion characterization in anatomically complex regions.

Conclusion: The single use colonoscope demonstrates clinical efficacy and equivalent safety profiles compared to conventional systems, validating its integration into routine screening. Equivalent ADR and en bloc resection rates indicate comparable diagnostic and therapeutic outcomes. These findings validate the clinical application of this method as a portable and effective option, while emphasizing the need for improved optical performance. Large-scale, multicenter, long-term follow-up studies are needed to confirm its clinical results and practical advantages.

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PP0645

ENDOSCOPIC MANAGEMENT OF ANAL INTRAEPITHELIAL NEOPLASIA AND CONDYLOMAS: A COHORT STUDY

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Introduction: Incidence of squamous cell carcinoma (SCC) of the anal canal is increasing. SCC arises from anal intraepithelial neoplasia (AIN) and condylomas. However, destruction of these anal lesions with standard techniques carries a high risk of recurrence and potential functional impairment. Endoscopy can be used to detect and treat colorectal lesions by endoscopic mucosal resection (EMR) or endoscopic submucosal dissection (ESD), but its value in the management of anal lesions has been poorly studied. The aim of this study was to assess the feasibility, efficacy and safety of endoscopic management of AIN and condylomas.

Aims & Methods: We conducted a retrospective study including all patients who had undergone endoscopic resection either by EMR or DSM of anal lesion (AIN or condylomas). The following criteria were evaluated: the rate of residual lesion at the end of endoscopic treatment, the rate of monobloc resection, the rate of complication per procedure and within 30 days after the procedure, the rate of new anal lesion and the rate of SCC during follow-up.

Results: From May 2015 to November 2024, 33 patients were included. Twenty-six (78.8%) were women. Mean age was 62.8 (±11.1) years. Two (6.1%) patients were HIV-positive. Eleven (33.3%) patients had a history of anal lesions, including one (3.0%) case of SCC. The median number of lesions per patient was 1 (IQR 1 - 2), with multifocal involvement in 15 (45.5%) patients. Thirteen (39.4%) patients had anal involvement reaching more than one hemi-circumference. Seven patients (21.2%) underwent ESD, and 26 (78.2%) underwent EMR. At the end of endoscopic treatment, no residual lesions were visible in 31 (93.9%) patients. Lesions were intentionally left in the remaining two patients. Resections were monobloc and macroscopically complete in 19 (57.6%) patients. Intra-procedural hemorrhage was visualized in 18 (54.5%) patients and was treated endoscopically during the same anesthesia without recurrence or clinical impact.

One case of anal stenosis occurred 27 days after circumferential ESD treated with 4 sessions of digital dilatations and one corticosteroid injection. No other complications occurred within 30 days after the procedures. Mean length of hospital stay was 1.12 (+/- 0.33) days. Pathological findings were: condyloma in 6 (18.2%) patients, AIN1 in 8 (24.2%) patients, AIN2 in 5 (15.2%) patients, AIN3 in 10 (30.3%) patients, and SCC in 4 (12.1%) patients (two SCC T1 R0, one of which required additional radiotherapy, and two in situ SCC).

Follow-up data were available for 29 patients. Mean follow-up time was 34.4 (+/- 35.7) months. Of these 29 patients, 8 (27.6%) cases of a new AIN or condyloma (outside the resection scar) were detected during follow-up after an average duration of 4 (+/- 2.1) months, 7 of which were treated by endoscopy. Two (6.9%) cases of scar recurrence were detected 4 and 13 months after treatment. During follow-up, three (10.3%) cases of SCC ≥

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T1 were diagnosed, including one case of recurrence at one year of a SCC pre-existing endoscopic treatment. The other two cases were diagnosed at one and 6 years.

Conclusion: Endoscopic management (EMR or DSM) of AIN and condylomas is a safe and effective. It should be part of the therapeutic arsenal in the management of these lesions. Increased surveillance is essential, given the high risk of new lesions.

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Disclosure: Nothing to disclose.

PP0646

SAFE AND EFFECTIVE PROPOFOL SEDATION BY TRAINED NON-ANAESTHESIOLOGIST PHYSICIANS IN OVER 2000 COLONOSCOPIES: A REAL-WORLD COLOMBIAN EXPERIENCE

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Introduction:

Propofol (PPF) is a short-acting, highly lipophilic intravenous agent with sedative, hypnotic, antiemetic, and amnestic properties. Acting through GABA receptor agonism and NMDA antagonism, it ensures rapid onset (30–45 seconds) and swift recovery, making it ideal for gastrointestinal endoscopic procedures. Since 1995, the American Society of Anesthesiologists has supported the use of non-anaesthesiologist providers for sedation in endoscopy.

Countries including Colombia have adopted this approach, especially given the limited availability of anaesthesiologists. In the United States, fewer than 20% of colonoscopy sedations are delivered by anaesthesiologists.

Recent studies report comparable safety profiles for trained non-anaesthesiologist and anaesthesiologist-administered sedation, with reduced cardiovascular complications and lower healthcare costs. However, Latin American real-world data on this model remain scarce.

Aims & Methods: This prospective, analytical prevalence study assessed the safety and effectiveness of propofol sedation administered by a general physician with specific training in sedation, airway management, and intensive care, within an ambulatory endoscopy unit.

From March 2023 to April 2024, adult patients (≥18 years) undergoing colonoscopy at the Centro de Gastroenterología y Endoscopia in Bogotá were included. The physician administered age-tailored propofol bolus and maintenance doses according to clinical guidelines and procedure duration.

Exclusion criteria included ASA IV status or contraindications to propofol. Sedation regimens were as follows:

- <40 years: 180 mg IV bolus; 60 mg maintenance every 5 min
- 41–70 years: 160 mg IV bolus; 50 mg maintenance every 5 min
- ≥71 years: 150 mg IV bolus; 30 mg maintenance every 5 min

Patient demographics, comorbidities, ASA classification, procedure indications, and adverse events were recorded.

Results: A total of 2,032 patients were enrolled; mean age 64 years (95% CI: 62.8–65.1), 60% female. The mean BMI was 27 (95% CI: 25.8–27.1). Colonoscopy indications included screening (35%), diagnostic evaluation (40%), surveillance (20%), and therapeutic intervention (5%), including 101 patients undergoing polypectomy. Comorbidity prevalence was notable: cardiovascular (30%), gastrointestinal (30%), respiratory (15%), neurological (5%), allergic (3%), and haematological (3%).

ASA risk classification revealed:

- ASA I: 14% (n=284)
- ASA II: 60% (n=1,219)
- ASA III: 26% (n=528)

Importantly, no complications—cardiopulmonary, allergic, or sedation-related—were recorded. All procedures were completed successfully, with high patient tolerance and efficient recovery times.

Conclusion: This real-world study confirms that propofol sedation administered by non-anaesthesiologist physicians with critical care training is both safe and effective for the vast majority of patients undergoing colonoscopy in ambulatory settings. These findings support a task-sharing model that maximises resource efficiency while maintaining patient safety. In health systems with limited anaesthesiologist availability, this approach allows specialists to focus on higher-risk ASA IV patients in tertiary-level settings. Widespread adoption of this model, with appropriate training, could reduce costs, increase procedural throughput, and improve access to high-quality endoscopic care in Latin America and similar settings.

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Disclosure: Nothing to disclose.

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PP0647

INSIGHTS INTO TRAINEE ENDOSCOPIST-CADX ARTIFICIAL INTELLIGENCE INTERACTION AND INFLUENCES ON DIAGNOSTIC COMPETENCY – EXPLORATORY ANALYSIS OF DATA FROM A PROSPECTIVE OBSERVATIONAL STUDY OF CADX-SUPPORTED TRAINEE ENDOSCOPIST DIAGNOSTIC COMPETENCY

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Introduction: Artificial intelligence (AI) characterization (CADx) systems for colonic polyps exhibit high diagnostic accuracy. However, they may only support, not replace, clinical decision-making and little is known of this human-AI interaction – even less so with trainee endoscopists.

This issue is difficult to explore, but data from a study investigating CADx-supported trainee endoscopists compared to AI-blinded experts could give insight.

Aims & Methods: This analysis draws on data from a prospective observational study comparing the diagnostic performance of CADx-supported trainee endoscopists (using GI Genius®, Medtronic) to expert endoscopists blinded to AI support.

Two key scenarios were explored:

a. instances in which the CADx system produced a 'no prediction' output, and;

b. cases where trainees actively overruled AI-generated diagnoses.

These were assessed to examine the impact of AI absence and human-AI discordance on diagnostic accuracy.

Results: The baseline cohort included 11 trainees performing 225 colonoscopies, mean age 63.8 years (SD 12.7), 48.9% male patients. Baseline ADR was 57.8%. Of 630 resected lesions (median size 3 mm, IQR 2;4), 40.0% were in the rectosigmoid, and 53.6% were non-adenomatous. CADx failed to provide a prediction in 19.2% (n=121) of cases. These 'no prediction' polyps were smaller (median 3 mm, p=0.048) and more often located in the right colon (72% vs. 58%, p=0.005. In the absence of AI support, trainee diagnostic accuracy declined significantly by 12.8% (63.6% vs. 76.4%, p=0.007).

Interestingly, expert endoscopists—despite being blinded to AI—also performed worse on these polyps (64.7% vs. 73.9%, p=0.045), suggesting an inherent difficulty in classifying these lesions.

Trainees overruled AI suggestion leading to discordant diagnoses in 9.0% (57/630) of polyps, comprised of 20 (35.1%) non-adenomatous, and 37 (64.9%) adenomatous lesions. Trainees incorrectly overruled AI in 59.6% (34/57), which mostly consisted of downgrading lesions to NICE 1. Discordant polyps did not differ in size (3 mm vs. 3 mm, p=0.413), but were more often located proximally (77.2% vs. 58.1%, p=0.005).

The difference in false negative rate (FNR) between trainees and AI was 62.16% (81.07% vs. 18.92%, p<0.0001), demonstrating a risk of missed adenomas when trainees override AI. Blinded expert endoscopists achieved identical overall diagnostic accuracy in those polyps as the AI system (59.6%).

However, their error profiles differed – In this subset, blinded experts and AI performed equally in overall accuracy (59.6%), although with differing error profiles: AI showed a numerically lower FNR (18.9% vs. 35.1%, Δ16.2%, p=0.17), offset by a higher false positive rate (80.0% vs. 50.0%).

	'No prediction' polyps (n=121)	Polyps with AI prediction (n=492)	
Size mm, median (IQR)	3 (2;4)	3 (2;4)	p=0.0483
Location in right colon	72%	58%	p=0.005
Diagnostic accuracy, trainees, % (95%CI)	63.6% (0.55-0.72)	76.4% (0.73-0.80)	p=0.007
Diagnostic accuracy, experts, % (95%CI)	64.7% (0.53-0.72)	73.9% (0.69-0.78)	p=0.0454

Conclusion: CADx systems generate 'no prediction' outputs in a substantial proportion of polyps, significantly impairing trainee diagnostic performance. These cases may represent inherently more challenging lesions. Despite CADx's high accuracy, trainees overrule AI in almost every tenth polyp, most often incorrectly downstaging polyps. Structured training on appropriate use of CADx, emphasizing both its capabilities and limitations, may enhance clinical effectiveness.

Further research on human-AI interaction and trust is warranted, as these factors will be critical for successful integration of CADx into routine endoscopic practice.

Disclosure: The original study (AC-CADx) received technical support by Medtronic. Medtronic was not involved in the design of the study or analysis of the data, neither in the original study nor this exploratory analysis.

PP0648

EFFICACY OF SNARE TIP COAGULATION IN COLONIC POLYPECTOMY: A PREVENTIVE STRATEGY FOR POST-POLYPECTOMY BLEEDING AND LOCAL RECURRENCE

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Introduction: Colonic polypectomy is a widely used endoscopic technique for colorectal cancer prevention. However, post-polypectomy bleeding (PPB) and local recurrence remain notable complications. Snare Tip Coagulation (STC) involves the application of thermal coagulation to the resection margin using the tip of the snare after polyp removal.

This study aimed to evaluate the clinical efficacy of STC in reducing immediate and delayed PPB, as well as local recurrence, following polypectomy.

Aims & Methods: We retrospectively reviewed 156 patients who underwent endoscopic resection—either snare polypectomy or endoscopic mucosal resection (EMR)—for pedunculated or sessile colorectal polyps measuring 6–20 mm at a single center between January 2022 and February 2024.

Patients were divided into two groups: those who received STC (n=78) and those who did not (control group, n=78). Primary outcomes included the incidence of PPB within 14 days and local recurrence at 12-month follow-up colonoscopy.

Results: The STC group demonstrated significantly lower rates of early (0.4%) and delayed (1.6%) PPB compared to the control group (1.9% and 3.8%, respectively; p<0.05). Local recurrence was also significantly reduced in the STC group (1.3%) versus the control group (4.8%; p=0.01). No significant differences were observed between the groups regarding procedure time or major complications such as perforation.

Conclusion: Snare Tip Coagulation is a safe and effective adjunctive technique in colorectal polypectomy, significantly reducing the risk of both post-procedural bleeding and local recurrence without increasing procedural complications. STC may offer particular benefit in the resection of medium-sized colorectal adenomas and should be considered for integration into routine endoscopic practice.

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Disclosure: Nothing to disclose.

PP0649

CAN 5-MINUTE COLONOSCOPY WITHDRAWAL TIME BE THE BENCHMARK BASED ON EXPERIENCED ENDOSCOPISTS AND EXCELLENT BOWEL PREPARATION? A MULTICENTER, RANDOMIZED CONTROLLED, BACK-TO-BACK TRIAL IN CHINA

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Introduction: Current colonoscopy quality indicators recommend a minimum withdrawal time (WT) of 6 minutes. However, this standard may reduce colonoscopy efficiency and does not account for endoscopists' experience or bowel preparation quality. This study aims to investigate whether the 6-minute WT remains necessary for experienced endoscopists with excellent bowel preparation (BBPS≥8) by comparing the impact of 5-minute versus 6-minute WT on adenoma detection rate (ADR) and adenoma miss rate (AMR).

Aims & Methods: This study based experienced endoscopists and excellent bowel preparation (Boston Bowel Preparation Scale score, BBPS≥8) is aimed to compare the effect of 5-minute WT versus 6-minute WT on adenoma detection rate (ADR) and adenoma miss rate (AMR).

This multicenter, randomized controlled trial involving 423 patients was conducted from June 2023 to March 2025 in 5 centers, including Wuxi People's Hospital, Jiangsu Province People's hospital, Xuzhou First People's Hospital, The Second Affiliated Hospital of Soochow University, Taizhou People's Hospital. Colonoscope was inserted and withdrew twice on each patient and all procedures for the same patient were done by the same endoscopist. Participants were randomly divided into the 5-minute WT group (the first WT-5-minute and the second WT-6-minute, n =214) and the 6-minute WT group (the first WT-6-minute and the second WT-6-minute, n=209), and a timer was used to adjust the withdrawal speed. The primary outcome was ADR and AMR.

Results: The baseline characteristics of the two groups were similar (Table 1) and colonoscopy characteristics of the patients in the two groups regarding quality of bowel preparation, cecal intubation rate, cecal intubation time, cecal intubation distance, number of abdominal compression, number of position change were comparable (Table 2). Overall ADR (39.7% vs 43.5%, P = 0.425), ADR in the first (36.0% vs 36.4%, P = 0.935) and second withdrawal (10.7% vs 13.9%; P = 0.327) were not significantly increased in the 6-minute WT group compared with the 5-minute WT group. Likewise, the overall advanced adenoma detection (AADR) (4.7% vs 5.3%, P = 0.780), AADR in the first (4.7% vs 5.3%, P = 0.780) and second withdrawal (0.5% vs 0.5%, P = 0.745) did not differ statistically. Further analysis showed that AMR (10.7% vs 13.9%, P = 0.372) was lower in the 5-minute WT group than in the 6-minute WT group, less adenomas were missed in the 5-minute WT group, although not statistically significant (Table 3).

		5-minute (n=214)	6-minute (n=209)	P- value
ADR, n (%)	Overall ADR	85(39.7%)	91(43.5%)	0.425
	The first withdrawal	77(36.0%)	76(36.4%)	0.935
	The second withdrawal	23(10.7%)	29(13.9%)	0.327
AADR, n (%)	Overall AADR	10(4.7%)	11(5.3%)	0.780
	The first withdrawal	10(4.7%)	11(5.3%)	0.780
	The second withdrawal	1(0.5%)	1(0.5%)	0.745
AMR, n (%)		23/214(10.7%)	29/209(13.9%)	0.327
AAMR, n (%)		1/10(10.0%)	1/11(9.1%)	0.944
PMR, n (%)		67/214(31.3%)	59/209(28.2%)	0.489

Table 3. Lesion-level detection and miss rate in the two groups.
PDR, polyp detection rate; ADR, adenoma detection rate; AADR, advanced adenoma detection rate; AMR, adenoma miss rate; AAMR, advanced adenoma miss rate; PMR, polyp miss rate.

Conclusion: Shortening WT from 6 to 5 minutes didn't reduce ADR based experienced endoscopists and excellent BBPS, a 5-minute WT benchmark was acceptable with experienced endoscopists and excellent bowel preparation quality of colonoscopy.

Disclosure: Nothing to disclose.

PP0650

A STUDY EVALUATING THE REALISM OF AI-GENERATED POLYPS

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Introduction: Identification and precise classification of polyps is essential during screening colonoscopy, but subject to significant interobserver variability. Traditional training approaches depend on patient-derived data, which limits open source resources for educational purposes. AI-generated images enable the training of classification of precancerous lesions without relying on patient-related data.

Aims & Methods: This study aims to evaluate the realism of AI-generated polyp images. Based on over 40 million endoscopy images from over 7,000 examinations, we developed a latent diffusion based AI that generates images of colonic polyps. To prove that our AI-generated polyp images are indistinguishable from real images, we conducted a multicenter, international reader study involving 55 physicians from 47 centers across 15 countries, of whom 48 had performed over 1,000 colonoscopies. Participants were shown 40 images (20 real, 20 synthetic) and were asked to classify each as real or AI-generated, along with rating their confidence in each decision.

Results: Of the 55 participants, 53 were included in the analysis. Overall classification accuracy was 73% (95% confidence interval [CI], 70% – 76%). AI-generated polyps were classified correctly with a sensitivity of 66% (95% CI, 60% – 71%) and a specificity of 80% (95% CI, 76% – 84%). Conversely, for correct classification of real polyps, the sensitivity was

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80% (95% CI, 76% – 84%) and the specificity was 66% (95% CI, 60% – 71%). The assessment of AI-generated polyps was associated with a significantly increased uncertainty ($p < 0.001$).

Conclusion: Our study demonstrates that even experienced endoscopists struggle to reliably distinguish AI-generated polyps form real ones. This underscores the ability of our AI model to create realistic images in high quality. In future projects, we will use AI-generated images to create a training platform for the classification of precancerous lesions.

Disclosure: Nothing to disclose.

PP0651

CAN ARTIFICIAL INTELLIGENCE IDENTIFY HISTOLOGICAL SUBCLASSES OF HYPERPLASTIC POLYPS BASED ON THEIR ENDOSCOPIC FEATURES?

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Introduction: Recent insights in polyp molecular biology have suggested that hyperplastic polyps (HP) are closely related to sessile serrated lesions (SSL). Their hallmark serrated crypts closely resemble those found in SSL. Furthermore, similar to SSL, specific subtypes of HP can harbor BRAFV600E-mutations. These findings, amongst others, have raised the hypothesis that some HP may have an intrinsic malignant potential through the serrated neoplasia pathway. However, in everyday clinical practice, no routine HP subdifferentiation is performed. Therefore, we aimed to develop a deep learning algorithm to classify molecular subclasses of HP, namely microvesicular hyperplastic polyps (MVHP) and goblet cell-rich hyperplastic polyps (GCHP), based on their endoscopic features.

Aims & Methods: In this single-center trial 238 hyperplastic polyps from 159 patients were retrospectively selected. All polyps had high quality endoscopic video in white light endoscopy and virtual chromoendoscopy (i-scan) using Pentax EC38-i20c and EC34-i10 endoscopes. All histological slides were reviewed by an expert pathologist, updating diagnoses to MVHP, GCHP or mixed HP. We selected the polyps without mixed constitution for development of a deep learning algorithm. These polyps were randomly split into 70% for training, 10% for validation and 20% for testing. Sequences of still images of the polyps were manually annotated and subsequently used to train a deep learning algorithm, with dataset imbalance addressed by randomly oversampling the MVHP.

Results: The 238 HPs were reclassified as 152 GCHP, 59 MVHP and 27 mixed HPs after expert histological review. We subsequently trained a convolutional neural network on the 211 HPs without mixed constitution. This resulted in a model with an AUC of 0.817, corresponding to a sensitivity of 80.0% for predicting MVHP and 78.4% for predicting GCHP.

However, due to the low prevalence of MVHP (28%) the positive predictive value for MVHP is only 33.3%, whilst the positive predictive value for GCHP is 96.7%.

	GCHP (Predicted)	MVHP (Predicted)
MVHP (True)	*1*	*4*
GCHP (True)	*29*	*8*

Table 1: Test set confusion matrix (polyp-averaged prediction).

Conclusion: We successfully developed a deep learning algorithm capable of differentiating between molecular subclasses of hyperplastic polyps (MVHP vs GCHP) based on their endoscopic features. This model demonstrated a robust AUC of 81.7% with a sensitivity of 80.0% and 78.4% for predicting MVHP and GCHP respectively.

However, the low prevalence of MVHP (28%) limited its positive predictive value (33.3%) for predicting MVHP. Despite these challenges, this pilot study serves as a proof of concept that artificial intelligence is capable of predicting hyperplastic polyp subclasses based on endoscopic appearance.

Furthermore, it highlights the potential of artificial intelligence to enhance the diagnostic precision, which could contribute to more tailored surveillance and management strategies in the future.

Disclosure: The authors declare no conflicts of interest.

PP0652

IMPLEMENTATION OF OPTICAL DIAGNOSIS WITH A RESECT AND DISCARD STRATEGY FOR DIMINUTIVE POLYPS IN THE ENGLISH BOWEL CANCER SCREENING PROGRAMME

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Introduction: Optical diagnosis (OD) with a "resect and discard" strategy involves diagnosing diminutive polyps with high confidence, photodocumenting and resecting these without sending to histology.

This greener, more efficient and cost-effective process underwent a comprehensive evaluation in the DISCARD3 study which showed, in a single screening centre, this meets the threshold for implementation and is feasible and safe with a quality assurance (QA) process in place. The NHS Bowel Cancer Screening Programme (NHSBCSP) in England approved implementation in 2023.

Early learning from translating this research into a population based programme is presented.

Aims & Methods: All 506 screening colonoscopists in England have been invited to undertake accreditation in OD.

Part 1 of the process involves attendance at a training day including a pre- and post-training test where 120 polyps are optically diagnosed. Pass criteria is ≥85% accuracy for high confidence optical diagnoses and overall ≥50% high confidence calls.

Part 2 involves supervised OD at each endoscopists' base hospital of 20 consecutive high confidence, optical polyp diagnoses with assessment of photodocumentation quality and diagnostic accuracy. To pass part 2 photodocumentation quality is required to be adequate in ≥90% of cases using the "3C photo quality score" and diagnosis accuracy ≥85%.

Once achieved, individuals are fully accredited to perform OD with a resect and discard strategy for screening cases and must undertake re-accreditation with online assessment of a further 120 polyp images every 2 years to demonstrate competency. Those accredited are supported by

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"Optical Diagnosis Champions" at each screening centre who monitor performance and support QA. Once in place "Optical Diagnosis Champions" will support those becoming accredited in OD.

Results: 65 screening colonoscopists attended part 1 of the accreditation in March 2024. 85% (55/65) passed the post-test on the first attempt with a further 7 individuals passing on repeat attempts. Overall 62/65 (95%) have completed part 1. Of these, 55/62 (89%) have successfully completed part 2 and are fully accredited.

As of 31 March 2025, 9618 polyps have been resected and discarded with a cost saving equivalent to £355866 (assuming a cost of £37 per polyp). An estimated 3206 hours of histopathology time has been saved (assuming 20 minutes per polyp).

Further training days have been held in September 2024, February 2025 and March 2025 increasing the total number of screening participants to over half of all screeners in England (310/506). The results from these will be reported in due course.

Conclusion: The NHSBSCP is the first population based screening programme in the world to implement OD with a resect and discard strategy. Quality assurance and monitoring is ongoing. In this early stage there have already been cost and time savings for the NHS which are expected to increase substantially once roll out completes in 2027.

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Disclosure: BPS has done consultancy and speaker work for Olympus. There are no other relevant current conflicts of interest to declare.

PP0653

COMPARATIVE EFFICACY, SAFETY, AND SATISFACTION OF PROPOFOL MONOTHERAPY VERSUS COMBINED BENZODIAZEPINE–PROPOFOL SEDATION IN ENDOSCOPY

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Introduction: Propofol-mediated sedation (PMS) is increasingly favored for endoscopic procedures due to its rapid onset, predictable recovery, and safety profile.

Aims & Methods: Propofol-mediated sedation (PMS) has become increasingly favored in gastrointestinal endoscopy due to its rapid onset, short recovery time, and favorable safety profile. While both propofol monotherapy and combination regimens are widely used, direct comparisons remain limited. In this prospective, randomized, single-center trial, 207 patients undergoing colonoscopy, gastroscopy, or bidirectional endoscopy were randomized to propofol monotherapy (n=101) or a combination regimen (n=106). Clinical, procedural, and sedation-related parameters, as well as patient and endoscopist satisfaction, were systematically evaluated.

Results: Both regimens demonstrated excellent safety profiles, with low complication rates that resolved spontaneously or with minimal intervention. Procedural outcomes, including polyp detection rates and duration, were comparable. Propofol monotherapy required a higher mean dose (123.84 ± 33.92 mg vs. 114.56 ± 28.17 mg, P < 0.01). Patient recovery was smoother in the monotherapy group, with higher uneventful recovery rates (74.3% vs. 67.9%) and overall satisfaction (95.1% vs. 90.6%). Endoscopist satisfaction was also slightly higher for monotherapy (94.1% vs. 89.7%), reflecting improved patient calmness.

Conclusion: Both regimens provided safe and effective sedation for endoscopy, but propofol monotherapy was associated with smoother recovery and higher satisfaction, supporting its use as a practical option in routine practice.

Disclosure: Nothing to disclose.

PP0654

IMPROVEMENT OF COLONOSCOPIC IMAGE QUALITY USING A NEW LED ENDOSCOPIC SYSTEM WITH SPECIALIZED NOISE REDUCTION

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Introduction: A new LED endoscopy system featuring advanced noise-reduction technology for enhanced brightness and minimal halation, the EP-8000 with the EC-860ZP colonoscope (Fujifilm), is scheduled to be released.

Aims & Methods: We conducted a multicenter, observational study to evaluate the new system's potential improvements in image quality compared to a previous system/scope (VP-7000/EC-760ZP). From January 2024 to January 2025, patients undergoing colonoscopy at two institutions were enrolled. Images of the cecum and lesions were captured using white light imaging (WLI), blue light imaging (BLI), and linked color imaging (LCI) under similar conditions about air insufflation, angle, distance, and so on.

Participants were divided into three groups:

Group 1 (EP-8000+EC-860ZP; 50 cases),

Group 2 (EP-8000+EC-760ZP; 50 cases), and;

Group 3 (VP-7000+EC-760ZP; 50 cases).

Cecal images were evaluated for brightness and halation using a four-point scale (1=poor to 4=excellent) by endoscopists and original values by image-analysis software, while lesion images were assessed for brightness, halation, and visibility by endoscopists.

Results: In cecal images, the endoscopists' scores for WLI in Group 1 were significantly better than in Group 3 for brightness (3.71±0.55 vs. 3.51±0.58, p<0.001) and for halation (3.60±0.51 vs. 3.18±0.59, p<0.001). Similarly, for BLI and LCI, Group 1 achieved higher scores than Group 3 regarding brightness (BLI: 3.15±0.85 vs. 2.23±0.92, p<0.001; LCI: 3.83±0.42 vs. 3.54±0.58, p<0.001) and halation (BLI: 2.99±0.69 vs. 2.71±0.78, p<0.001; LCI: 3.33±0.60 vs. 3.10±0.58, p<0.001). Software analysis confirmed that Group 1 had superior brightness values compared to Group 3 for WLI (122.7±6.6 vs. 110.7±7.7, p<0.001), BLI (99.1±10.9 vs. 69.3±11.5, p<0.001), and LCI (139.4±9.2 vs. 113.1±10.7, p<0.001), as well as lower halation values for WLI (1.34±1.80 vs. 3.35±3.59, p=0.031) and LCI (1.48±0.83 vs. 4.06±3.99, p=0.010). Regarding lesion images, brightness, halation, and visibility for WLI, BLI, and LCI were significantly better in Group 1 than in Group 3.

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Conclusion: The new LED system provided marked improvements in brightness, halation, and lesion visibility. The software assessment was consistent with endoscopists assessment objective and can be used for the development and evaluation of new endoscopic systems.

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PP0655

THE EFFECT OF BLUE METHYLENE TABLETS ON ADENOMA DETECTION IN NEOPLASIA SURVEILLANCE OF PEOPLE WITH LYNCH SYNDROME

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Introduction: Lynch syndrome is a hereditary condition associated with an increased risk of colorectal cancer compared to the general population (66.08% for men and 42.71% for women). Improving adenoma detection rate (ADR) is associated with a reduced risk of interval cancer. Recent studies showed that blue methylene tablets (BMT) significantly improved ADR in the screening population.

This preliminary study aims to evaluate whether the addition of BMT to standard colonoscopy preparation increases ADR in the high-risk population of Lynch syndrome patients.

Aims & Methods: This observational non-randomized study included consecutive people ≥18 years with genetic diagnosis of Lynch syndrome. Participants were assigned to two groups: one receiving split preparation with 2 litres of PEG and the other receiving BMT in addition to the same bowel preparation (3 tablets dissolved in 0.5 L of water the night before the procedure, and 2 tablets 5 hours before). All patients with previous colonic resections were excluded. The primary outcome was ADR. The t-test for independent samples and Fisher's exact test were used for statistical analysis.

Results: Thirty-eight patients were enrolled, 17 in BMT and 21 in the control group. Twenty-one (55.3%) patients were female and 17 (44.7%) male, with a mean age of 45.6 ± 11.3 years. The two groups were comparable for sex (p=0.75), age (p=0.72), mutation types (p=0.16) and previous polypectomy history (p=0.20). The polyp detection rate was 5/21 (23.8%) in the control group and 5/17 (29.4%) in the BMT group, p = 0.73. The ADR was 2/17 (11.8%) in the BMT and 3/21 (14.2%) in the control group (p=0.94). The average number of adenomas per colonoscopy in the BMT group was 1.8 ± 1.7 compared to 1.5 ± 1.22 in the control group (p=0.76). The average number of polyps per colonoscopy in the BMT group was 2.0 ± 1.7 compared to 2.0 ± 2.0 in the control group (p=1).

Conclusion: BMT as add-on to standard colonoscopy preparation did not significantly increase the adenoma detection rate in the high-risk population of patients with Lynch syndrome. Further studies with larger sample sizes may be needed to confirm these findings.

Disclosure: Nothing to disclose.

PP0656

EVALUATION OF AI-ASSISTED COLONOSCOPY FOR POLYP DETECTION: A NON-INFERIORITY ANALYSIS COMPARING NON-EXPERT AND EXPERT ENDOSCOPISTS

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Introduction: Colonoscopy is vital for colorectal cancer prevention, with adenoma detection rate (ADR) serving as a crucial quality marker. However, significant inter-operator variability exists between expert and non-expert endoscopists. Emerging artificial intelligence (AI) systems promise to support non-experts in achieving detection rates closer to those of specialists.

Aims & Methods: This study aimed to determine whether non-expert endoscopists using an AI-assisted system (CAD EYE™) can achieve a polyp detection rate non-inferior to that of expert endoscopists.

In this prospective, single-center, observational study, 120 patients (aged 20–90 years, ECOG performance status 0–2) with previously detected polyps by expert colonoscopy were enrolled. Non-expert endoscopists performed colonoscopy with AI assistance while experts verified polyp findings in real-time.

The primary endpoint was the overall polyp detection rate, with a pre-defined non-inferiority margin of 10% (assuming an expert detection rate of 100% and an anticipated non-expert detection rate around 95%).

Secondary endpoints included adenoma detection rate, total procedure time, intubation time, and withdrawal time. Statistical analysis was performed using the Clopper–Pearson method for confidence intervals.

Results: Among 120 patients, the non-expert endoscopists using AI achieved a polyp detection rate of 95.8% (95% CI: 92.3–99.3%), which met the non-inferiority criterion compared to the expert detection rate of 100% (non-inferiority confirmed as the lower limit of the CI exceeded 90%). The adenoma detection rate was 36.7% for non-experts versus 38.3% for experts (p=0.62).

Mean total procedure times were comparable between groups, with non-experts averaging 22.4 ± 4.5 minutes and experts 21.8 ± 3.9 minutes (p=0.45). No significant differences were observed in intubation or withdrawal times. The safety profile was similar, with minor adverse events occurring in 2.5% of non-expert cases.

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Conclusion: The findings indicate that AI-assisted colonoscopy enables non-expert endoscopists to achieve a polyp detection rate comparable to that of experts, without compromising procedural efficiency or safety.
These results support the clinical implementation of AI systems in colonoscopicpractice to reduce operator-dependent variability and improve screening quality.
Further multicenter studies with larger cohorts are warranted to validate these promising outcomes.
Disclosure: Nothing to disclose.

PP0657

TRENDS IN LOWER GASTROINTESTINAL ENDOSCOPIC POLYPECTOMY AND ENDOSCOPIC MUCOSAL RESECTION: A NATIONAL CROSS-SECTIONAL E-SURVEY

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Introduction: Endoscopic polypectomy and endoscopic mucosal resection (EMR) are widely used techniques, largely due to the increasing number of screening colonoscopies. Recently, the ESGE provided a core curriculum for training in Endoscopic Mucosal Resection (EMR) [1].
Among the recommendations for Hot-EMR, it is suggested that the endoscopist personally takes over the snare from the assistant and closes it to gain a better perception of the grasped tissue, thereby reducing risks. However, data regarding endoscopist preferences and common practices in lower gastrointestinal (GI) polypectomy are lacking.
Aims & Methods: This study aims to elucidate the current practices and preferences of young Italian endoscopists (age <40 years). A national cross-sectional e-survey was conducted in Italy between June and August 2024. Participants were recruited from the Young Endoscopist Polypectomy study group (PolYoung). The e-survey included questions regarding demographics, volume of procedures, perceived proficiency levels, and technical preferences in endoscopic polypectomy. Descriptive and inferential statistical analyses were performed to identify trends and disparities based on demographic and experience characteristics.
Results: A total of 71 GI endoscopists (M 46:F 25, 0 non-binary) responded to the e-survey. Nearly half (52.1%) were aged 30-35 years. Among the participants, 52.1% worked in university hospitals, while 43.7% worked in non-university community hospitals and 4.2% in private hospitals.
Approximately half of the participants (49.3%) had performed more than 1,000 colonoscopies as the first operator. Additionally, 66.2% had performed more than 100 cold snare polypectomies, and almost all participants (98.6%) felt confident performing this procedure independently.
Regarding Hot-EMR, 43.7% had performed more than 50 procedures, and 66.2% felt independent in performing this technique. When asked about their preferences for Hot-EMR, 93% of the participants preferred to instruct the assistant to open the snare to grab the polyp and then cut the lesion themselves, rather than taking over the snare to cut the polyp, in order to maintain control of the scope.
Only 15.5% of participants routinely added adrenaline to the injection solution. However, 58.6% of participants routinely performed prophylactic closure of the post-resection scar in all large (>20 mm) EMRs of the proximal colon, while 22.9% did so for all EMR lesions, regardless of size, when patients were on antithrombotic drugs. Regarding novel EMR techniques, 70.4% and 80.3% of participants had performed underwater-EMR and cold-EMR, respectively.
Finally, no statistically significant differences were observed between demographic factors, years of experience, and proficiency levels for any of the variables considered.
Conclusion: This study provides valuable insights into the common practices of young endoscopists in Italy. Notably, a low adherence to the ESGE-recommended Hot-EMR technique was observed.
This finding highlights the need for further comparative studies between different approaches and underscores the importance of developing new devices that enhance the endoscopist's ability to accurately gauge the amount of tissue captured by the snare without losing control of the scope.
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Disclosure: Nothing to disclose.

PP0658

EFFICACY AND SAFETY OF PROXIMALLY RELEASING SELF-EXPANDABLE METALLIC STENTS FOR MALIGNANT COLORECTAL OBSTRUCTION NEAR THE DENTATE LINE

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Introduction: Endoscopic placement of self-expandable metallic stents (SEMS) for malignant colorectal obstruction near the dentate line is technically challenging because of the difficulty faced in the precise alignment of the distal edge of the stent with the distal end of the lesion. Furthermore, after stenting, lesions close to the dentate line often induce anal pain. Therefore, positioning of the distal end of the stent at an adequate distance from the dentate line is crucial for reducing discomfort. A proximally release-type SEMS might be particularly advantageous in such cases.
Aims & Methods: This study aimed to assess the safety and efficacy of proximally release-type SEMS for malignant colorectal obstruction placed within 10 cm from the dentate line. The study retrospectively analyzed 12 consecutive patients who had undergone SEMS placement between May 2019 and March 2024 at our institution. The clinical outcomes, technical and clinical success rates, adverse events, and prognosis were evaluated.
Results: The median age of the subjects was 70.5 years, and included five (41.7%) men and seven (58.3%) women. The causes of obstruction were primary colorectal cancer (six patients), gastric cancer (five patients), and cervical cancer (one patient). Colorectal Obstruction Scoring System (CROSS) scores prior to SEMS placement were 0 (2 patients), 1 (10 pa-

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tients). Tumor locations included Rs in two patients, Ra in two, and Rb in eight patients, with the median distance from the dentate line being 4.5 cm (range: 2–10 cm). The indications for SEMS placement were bridge to surgery (BTS) in four cases and palliation (PAL) in eight cases. The technical success rate was 91.7% (11/12), and the clinical success within seven days was also 91.7% (11/12). Early adverse events (within seven days) included one case of stent migration (8.3%) No cases of perforation, bleeding, or reobstruction were observed.

All BTS patients underwent successful laparoscopic surgeries and no recurrence has been observed to date. Among the PAL cases, one patient experienced stent migration 314 days after SEMS placement. The migrated stent was manually extricated under general anesthesia.

Conclusion: The proximally release-type colonic stent was effective for treating malignant colorectal obstruction near the dentate line since it facilitated placement of the stent in the right position. It might be the primary option for malignant colorectal obstruction near the dentate line, and can contribute to the expansion of indications for colorectal stent placement for lesions near the anal verge.

Disclosure: Nothing to disclose.

PP0659

PREVALENCE OF NON-ADVANCED AND ADVANCED COLONIC NEOPLASIA IN YOUNG ADULTS

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Introduction: The European Society of Gastrointestinal Endoscopy (ESGE) recommends colorectal cancer (CRC) screening for individuals at average risk, aged 50-75 years. However, recent trends indicate an increasing incidence of colorectal neoplasms among younger individuals.

Aims & Methods: The primary aim of this cross-sectional study is to record the prevalence of polyps, adenomas, and advanced neoplasia (high-grade dysplasia and CRC) in individuals up to 49 years old, and secondarily to identify predictors of neoplasia and high-grade dysplasia. The data were collected retrospectively through the hospital's electronic database. All serial colonoscopies, with cecum intubation, from 1/1/2018 until 31/12/2024 were included. Endoscopies in patients with diagnosed familial polyposis syndromes, inflammatory bowel disease and those with poor bowel preparation were excluded. Patients' demographics, CRC family history and the recorded polyps' size, location, Paris classification, and pathology reports were collected. SPSS 23 (IBM, Armonk, NY, USA) was used for the statistical analysis.

Results: A total of 568 colonoscopies were included, involving 306 male patients (53.8%), with a median age of 42 years (IQR: 36-46). Of these, 71 patients (12.5%) had a family history of CRC. At least one polyp was identified in 166 patients (29.2%), with adenomas observed in 68 patients (12.1%) and advanced neoplasia in 20 patients (3.5%), including 3 patients (0.5%) diagnosed with CRC (Table).

	<25 (n=36)	25-29 (n=31)	30-34 (n=61)	35-39 (n=76)	40-44 (n=146)	45-49 (n=218)	Total (n=568)	P
Polyps, n (%)	0 (0)	2 (6.5)	6 (9.8)	20 (26.3)	52 (35.6)	86 (39.4)	166 (29.2)	<0.001
Adenoma, n (%)	0 (0)	1 (3.2)	2 (3.3)	9 (11.8)	21 (14.4)	36 (16.5)	68 (12.1)	0.006
Advanced adenoma, n (%)	0 (0)	1 (3.2)	0 (0)	3 (3.9)	4 (2.8)	12 (5.6)	20 (3.5)	0.266

Table. Polyps, adenomas and advanced adenomas by age groups.

Advanced adenomas were more frequently detected in colonoscopies performed due to visible blood in stools (8.5%) and as follow-up after diverticulitis (7.1%).

In multivariate analysis, in which age, family history of CRC, and bowel preparation quality were included as variables, the presence of adenomas and advanced adenomas was significantly associated with age (adenoma OR = 1.08; 95% CI: 1.04-1.13 and advanced adenoma OR = 1.08; 95% CI: 1.00-1.18).

In total 271 polyps were identified. Ninety-four were adenomas of which 74 had low-grade dysplasia and 20 high-grade dysplasia or cancer. Sixteen of 177 not-adenomatous polyps were found to be serrated and 2 of those had low-grade dysplasia. "Large" polyps [≥10mm] had a higher probability of being adenomas (71.4%) compared to "small" [5-9mm] (26.2%) and "diminutive" [<5mm] (24.4%), Chi square P<0.001. Regarding the Paris classification Ip and Isp presented a higher probability to be adenomas (87.5% and 86.4% respectively) in contrast to Is and 0-IIa (35.6% and 12.2% respectively), Chi square P<0.001.

The resected adenomas were also categorized according to the type of reported dysplasia to identify predictors of high-grade dysplasia. High-grade dysplasia was present in 42.5% of large polyps compared to 4.5% of small and 6.2% of diminutive (Chi square P<0.001).

High-grade dysplasia was also present in a higher frequency in polyps resected from the "left" colon (26.2%) compared to the "right" (10.3%) and in 0-IIa (66.7%) and Ip (38.1%) compared to Isp (21.1%) and Is (12.5%) polyps according to the Paris classification, but not in a statistically significant degree (Chi square P<0.010).

Conclusion: Despite the relatively small sample size of our study, initiation of screening at a younger age should be considered to be lowered, since adenoma and advanced adenoma were identified in 15% and 4.3% of individuals aged 35-49.

Disclosure: GB: Advisor/lecturer for Janssen, Pfizer, Takeda, Abbvie, MSD, Mylan, Genesis Pharma, Adacyte Therapeutics, Amgen, Ferring, Cooper; Funding (Grants/Honoraria): Pfizer, Takeda, Abbvie, Aenorasis; Research/Clinical Trials: Abbvie, Takeda.

PP0660

SELF-ASSEMBLING PEPTIDE AND ITS EFFECT ON DELAYED BLEEDING AND HEALING OF POST-RESECTION DEFECT AFTER ADVANCED ENDOSCOPIC THERAPY

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Introduction: Delayed bleeding remains a significant complication after endoscopic resection procedures in the gastrointestinal tract. Effective strategies to prevent post-procedural bleeding and promote optimal healing are critical for improving patient's outcomes. Self-assembling peptide (SAP) is a new hemostatic agent that forms protective nanofibers scaffold after contact with body fluids. Its application may reduce delayed bleeding and support tissue healing after endoscopic mucosal resection (EMR) or endoscopic submucosal dissection (ESD). Early clinical data has shown promising results, but evidence is still limited.

Aims & Methods: This study evaluates the impact of SAP use on delayed bleeding rates and healing process after advanced endoscopic therapies. The findings may support the integration of SAP into standard post-endoscopic care protocols.

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A retrospective analysis was conducted on patients after EMR or ESD at a single tertiary endoscopy center. Two primary outcomes were assessed: delayed bleeding and post-resection defect healing. Delayed bleeding was defined as clinically evident bleeding occurring within 30 days following the index endoscopy. Healing was evaluated endoscopically using the Sakita-Fukutomi classification at 4, 8, and 24 weeks. The size of the resection defect was measured during the procedure using endoscopic forceps or a polypectomy snare for reference. Data were collected from medical reports.

Results: A total of 141 patients (55 women, 86 men) were included in the analysis. The majority underwent ESD (n=96; 68 %), followed by piecemeal EMR (n=36; 26 %), hybrid procedure EMR+ESD (n=7; 5 %), and en-bloc EMR (n=2; 1 %). 35 lesions (25 %) were located in upper GI tract, 106 lesions (75 %) in colon and rectum with an average lesion size of 48 mm. R0 resection was achieved in 71 (78 %) of all 91 en-bloc resections. Delayed bleeding occurred in 6 patients (4.3%), with an average onset of 4 days after the procedure. All these patients had subsequent endoscopy with various hemostatic approaches, including SAP alone (n=3), SAP with coagulation (n=1), coagulation with hemoclips (n=1), and coagulation with hemospray (n=1). In all cases hemostasis was achieved. 2 patients required transfusion. The post-resection defect averaged 51 mm after index endoscopy. At 4, 8, and 24 weeks post-procedure, its size decreased to 20 mm, 14.5 mm, and 10 mm, marking reductions of 61%, 72%, and 80%, respectively.

Conclusion: SAP is effective in preventing delayed bleeding and healing post-resection defects. Its use can increase overall safety of advanced endoscopic procedures. Further studies are needed to establish optimal indications and integration into standard post-procedure protocol.

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Disclosure: Nothing to disclose.

PP0661

THE RESEARCH ON THE REDUCE OF ADENOMA MISS RATES THROUGH ARTIFICIAL INTELLIGENCE-ASSISTED COLORECTAL ENDOSCOPY SCREENING

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Introduction: This study aims to evaluate the impact of artificial intelligence (AI)-assisted colonoscopy withdrawal on AMRs (adenoma missed diagnosis rates) during colonoscopy, demonstrating that AI maintains inspection quality while reducing withdrawal time. Additionally, it investigates the dependency of AI effectiveness on operator experience, clarifying its compensatory role in mitigating variation and ensuring stable quality control in colonoscopic procedures.

Aims & Methods: This study enrolled patients from three endoscopy centers from Jan to Dec 2024. Participants were randomized into two groups: the TR-9 (Traditional Withdrawal 9-minute group) and the AI-6 (AI-Assisted Withdrawal 6-minute group). Each patient underwent two consecutive withdrawal procedure by the same endoscopist. In the TR-9 group, the first withdrawal employed TR-9 inspection followed by AI-6.

Conversely, the AI-6 group initiated with AI-6 followed by TR-9 . Baseline characteristics and detected lesions (classified as adenomas or polyps using the Japan NBI Expert Team classification) were documented for both procedures. Non-inferiority of TR-9 versus AI-6 was evaluated for AMR and poly missed diagnosis rates (PMRs), while differences in adenoma detection rate (ADR) and polyp detection rate (PDR) were compared. The AI-6 group was further stratified into AI-assisted withdrawal normal-endoscopist subgroup (AI-N) and AI-assisted withdrawal expert-endoscopist subgroup (AI-E) for comparative analysis of AMRs, ADR, PDR, and PMRs.

Results: 1. In the non-inferiority comparison between TR-9 and AI-6, AI-6 demonstrated non-inferiority toTR-9 in both adenoma missed rates (27.3% vs. 30.2%; $P < 0.05$) and polyp missed rates (24.6% vs. 29.7%; $P < 0.05$). These results confirm the hypothesis AI-6 achieves non-inferior performance .

2. Regarding adenoma detection rate (ADR), TR-9 demonstrated an ADR of 26.8%, while AI-6 achieved 29.6%, with no statistically significant difference observed ($P > 0.05$). For PDR , TR-9 showed a PDR of 36.0% compared to AI-6's 47.9%, revealing a statistically significant advantage for AI-6 over TR-9 ($P < 0.05$).

3. Subgroup Analysis within AI-6 Group:

In terms of missed diagnosis rates, AI-E exhibited AMRs of 26.4%, significantly lower than the 28.6% observed in AI-N ($P = 0.026$). However, for PMRs, AI-E demonstrated PMRs of 22.6% versus 26.3% in AI-N, with no statistically significant difference ($P = 0.059$).

Regarding ADR, the AI-E subgroup demonstrated an ADR of 31.9%, compared to 27.5% in the AI-N subgroup, with no statistically significant difference ($P = 0.424$). For PDR, AI-E showed a PDR of 48.1%, while AI-N achieved 47.7%, also without statistical significance ($P = 0.933$).

Conclusion: 1. The AMRs 2. The AMRs and PMRs of AI-6 are non-inferior to those of TR-9. The AMRs of both AI-assisted 6-minute withdrawal and TR-9 are slightly higher than the national average (26%).

2. Both TR-9 and AI-6 demonstrated high ADR and PDR , confirming the reliability of colonoscopy quality control across the three participating endoscopy centers. Notably, AI-assisted withdrawal maintained superior detection efficacy despite reducing procedural time to 6 minutes. These findings support the integration of AI-assisted withdrawal technology into continuous quality improvement protocols for colonoscopy, positioning it as a strategic approach to further optimize endoscopic examination standards.

3. In the subgroup analysis, there was no statistically significant difference in lesion detection rates between AI-assisted examinations performed by physicians with different levels of experience. The AMRs of AI-assisted examinations by general physicians were non-inferior to those by expert physicians. However, the hypothesis that the PMRs of AI-assisted examinations by general physicians were non-inferior to those by expert physicians was not supported, which may be related to insufficient sample size.

Disclosure: Nothing to disclose.

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PP0662

DEVELOPMENT OF A MULTICLASS COMPUTER-AIDED DIAGNOSIS SYSTEM FOR DIAGNOSIS USING JNET CLASSIFICATION IN COLORECTAL LESIONS

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Introduction: Japan NBI Expert Team (JNET) classification is clinically useful for diagnosing colorectal cancer; however, the major challenge in its application is eliminating the differences in diagnosis among endoscopists. In the JNET classification, Type 1 corresponds to hyperplastic polyps and sessile serrated lesions (SSL); Type 2A, adenomas to Tis; Type 2B, Tis and T1a; and Type 3, T1b or deeper lesions. We believe that diagnostic support using artificial intelligence can contribute to the standardization of the JNET classification of NBI diagnosis. Therefore, we developed a computer-aided diagnosis (CADx) system to support the JNET diagnosis and evaluated diagnostic performance of the CADx.

Aims & Methods: We developed a CADx system that can perform full-screen image classification while providing more detailed segmentation results as an output. This structure uses a deep convolutional network with the backbone of ResNet34 as a feature extractor for the entire system, and a structure of two transformer encoder blocks to perform full-screen classification. Further, we used a single-layer convolutional structure with a convolutional kernel size of 1 as the segmentation head.

We examined 111 lesions: hyperplastic polyp/SSL, 36 lesions; adenoma, 37 lesions; Tis/T1a, 15 lesions; and T1b or deeper, 23 lesions using Olympus CV-290 and CV-1500 processors and Olympus CF-HQ290ZI, CF-XZ1200DI, and CF-EZ1500DI endoscopes. Each lesion was identified according to the JNET classification: Hyperplastic polyp/SSL, Type 1; Adenoma, Type 2A; Tis/T1a, Type 2B; T1b or deeper, Type 3. The accuracy between the histopathological findings and diagnosis by CADx system was calculated.

Results: The accuracy between CADx system and the histopathological findings was 94.4% (34/36) for hyperplastic polyp/SSL, 81.1% (30/37) for adenoma, 60.0% (9/15) for Tis/T1a, and 74.0% (17/23) for T1b or deeper lesions.

Conclusion: We developed a multiclass CADx system for colorectal lesions that performs JNET classification for histopathological findings, which may assist in the NBI-based diagnosis of colorectal lesions in facilities where the NBI diagnosis with magnifying endoscopy is not widespread.

Disclosure: Nothing to disclose.

PP0663

ENHANCED NATURAL RETRACTION WITH TRANS-ANAL PLATFORM COMBINED WITH SPEEDBOAT ASSISTED ENDOSCOPIC SUBMUCOSAL DISECTION FOR DISTAL COLON POLYPS: INTERIM FIRST RESULTS OF SSD /AUDITREGISTRY

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Introduction: Speedboat™ is a novel multimodal endosurgical device. It utilizes the use of advanced bipolar energy for cutting/dissection and super-high frequency microwave energy for coagulation. AirSeal® is a trans-anal access port that creates an "invisible pneumo- dissection" effect by instantaneously responding to changes in intraluminal pressure, with CO2 exchange.

Aims & Methods: This study aims to evaluate the efficacy of Speedboat assisted Endoscopic Submucosal Dissection (S-ESD) with and without AirSeal in the management of complex distal colorectal polyps as well as the speed of dissection.

Data, from a prospectively collected clinical audit and later a dedicated S-ESD registry (2018-2025) was analyzed. Cases were divided into three groups, the pre-AirSeal group (2018-2020), AirSeal group (2021-2025) and non-AirSeal (2021-2025). Lesions characteristics, long and short axis length, time of dissection and clinical outcomes were collected. Lesion surface (cm^2) and Speed (cm2/hr) were calculated. Clinical outcomes and dissection speed were compared between the three groups.

Results: Over the study period, a total of 253 consecutive patients [56.1% male, age 72 (IQR: 65-77)] had a distal colorectal lesion completely excised using S-ESD with median surface of 12.95cm^2 (IQR:6.87-21.68), long axis 5,0 cm (IQR: 3.5-7.0) and speed 9.42cm2/hr (IQR:6.97-15.75).

Pre-AirSeal group included 58 patients [58.6% male, age 72 (IQR 65-77)] with median surface of 9.81cm^2 (IQR:6.72-15.89), long axis 4.5cm (IQR: 3.5-6.1) and speed 7.15 (IQR: 6.86-9.31). Since AirSeal introduction in 2020 195 patients with distal polyps had S-ESD [55.4%male, age 72 (IQR: 64-76)] with median surface of 14.19 cm^2 (IQR:6.91-27.47), long axis 5.0cm (IQR: 3.5-7.0) and speed 11.41 (IQR: 6.17-18.00)].

In the AirSeal group 127 patients underwent S-ESD [58.3% male, age 71(IQR: 63-76), median surface 15.70 (IQR: 8.24-30.90) and speed 11.41 (IQR: 7.06-16.63). In the non-AirSeal group 68 patients were subject to S-ESD [50% male, age 72 (IQR: 66-78), median surface 11.38 (IQR: 5.15-19.28)].

There was not a significant difference in dissection speed between AirSeal and non-AirSeal group (p=0.158) but the median surface of the polyps removed were significantly larger in the AirSeal group (p=0.005). Overall, the speed and surface were significantly increased in the period when AirSeal was used when compared with the pre-Airseal group (p<0.001 and p=0.012 respectively). No statistically significant difference in regards with age and gender was found among any of the groups.

Conclusion: The use of AirSeal combined with S-ESD appears to facilitate excision of distal lesions with larger surface with reasonable speed over time. A randomized trial to investigate the role of pneumo-dissection is warranted.

Disclosure: Nothing to disclose.

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PP0664

COMPARISON OF TRANEXAMIC ACID (TXA) VERSUS EPINEPHRINE USE DURING COLONIC ENDOSCOPIC MUCOSAL RESECTION (EMR) PROCEDURES FOR THE REDUCTION OF INTRAPROCEDURAL AND POSTPROCEDURAL COMPLICATIONS: A PROSPECTIVE RANDOMIZED PILOT STUDY

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Introduction: Endoscopic mucosal resection (EMR) of colonic lesions is widely used for the removal of superficial gastrointestinal neoplasms, but intraprocedural and postprocedural bleeding remain significant adverse events. Current strategies often involve epinephrine injection, which has uncertain efficacy and potential systemic side effects, including hypertension and increased postprocedural pain. Tranexamic acid (TXA), an antifibrinolytic agent with a well-established safety profile, has shown efficacy in reducing bleeding across various surgical fields and may be an alternative to epinephrine use.

Aims & Methods: The primary aim was to assess whether the addition of TXA into the submucosal injection solution during colonic EMR is non-inferior to epinephrine in reducing the rates of intraprocedural and postprocedural bleeding. Secondary aims included comparing pain levels, the need for opioid analgesia, and the incidence of other complications such as deep mural injury and delayed bleeding.

A prospective, double-blinded, controlled, non-inferiority pilot study was conducted at the Shamir Medical Center. Patients aged >18 years undergoing EMR for colonic lesions ≥20 mm were randomized into two groups. The study group received a standard injectate solution with 100 mg TXA in 10 ml, while the control group received the same solution with 1:100,000 epinephrine.

Outcome measures included intraprocedural bleeding, delayed bleeding, pain (and analgesic use), perforation, hospitalization, and post-polypectomy syndrome.

Results: A total of 112 patients were included (57 TXA, 55 epinephrine). The mean age of the entire cohort was 69.0 years, and 47.3% were women. The average polyp size was 33.4 mm. The groups were comparable in most baseline characteristics, although TXA patients were slightly older ($p = 0.032$) and had higher rates of diabetes ($p = 0.042$). No significant difference was observed in lesion size, morphology, or pathology. Intraprocedural bleeding was similar across groups.

Clinically significant delayed bleeding occurred in 7% of TXA vs 7.3% of epinephrine patients ($p = 0.958$). Notably, postprocedural pain was significantly reduced in the TXA group (5.3% vs 41.8%, $p < 0.001$), as was opioid requirement (0% vs 47.8%, $p < 0.001$). Although not statistically significant, there were no cases of deep mural injury or hospitalization due to post-polypectomy syndrome in the TXA group, compared to four in the epinephrine group ($p = 0.055$).

Conclusion: The addition of TXA to the submucosal injection solution during colonic ER procedures is non-inferior to epinephrine in preventing intraprocedural and delayed bleeding.

Moreover, TXA is associated with significantly reduced postprocedural pain and opioid use, suggesting it may offer a safer and more comfortable alternative to epinephrine. These findings support the feasibility of larger-scale trials to further evaluate the efficacy and safety of TXA in endoscopic interventions.

Disclosure: Nothing to disclose.

PP0665

DETERMINANTS OF BOWEL PREPARATION QUALITY IN PATIENTS WITH QUIESCENT ULCERATIVE COLITIS IN ENDOSCOPIC REMISSION: A MULTICENTER CROSS-SECTIONAL STUDY

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Introduction: Adequate bowel preparation (BP) is crucial for high-quality colonoscopy surveillance in ulcerative colitis (UC), because an inadequate one leads to incomplete colonoscopy and higher rate of missing precancerous lesions, which are usually flat or subtle in this setting. There is conflicting evidence regarding whether disease activity or extension are risk factors for inadequate BP in patients with inflammatory bowel disease (IBD); furthermore, data on the optimal PEG-regimen which should be used in this setting are limited.

The aim of this study was to assess which factors influence the quality of BP in this specific setting.

Aims & Methods: This is a multicentre, cross-sectional, retrospective study carry out in four referral centres between January-2021 and December-2023, including quiescent UC patients with endoscopic remission (Mayo endoscopic score=0) who underwent colonoscopy. Bowel preparation quality was assessed using the Boston Bowel Preparation Scale (BBPS) and performed with 1L-PEG-ASC (Plenvu®) and 2L-PEG (Moviprep® and Clensia®), without any specific allocation or instructions. Demographic, anamnestic and endoscopic data were evaluated and analysed.

Results: Overall, 261 UC patients (F 99, 37.9%, mean age 55.6±14.5) were recruited. Patients with previous maximum extension of proctitis (E1) were 31 (12.1%), with left-sided colitis (E2) were 104 (40.6%) and with pancolitis (E3) were 121 (47.3%); additionally, patients with a history of advanced therapy were 104 (39.8%). Entire cohort adequate preparation rate was 90.4%.

1L-PEG-ASC preparation showed significantly higher adequate BP rate compared to 2L-PEG preparations (93.7% vs 85.4%, $p=0.032$). Right colon adequate BP did not differ among the BPs (93.7% vs 91.3%). Patient demographics, smoking status, maximum disease extension, history of advanced therapy or presence of pseudopolyps did not significantly affect BP quality ($p>0.05$).

Conclusion: In a relatively unbiased setting, such as quiescent UC patients in endoscopic remission, bowel preparation volume, specifically 1L-PEG-ASC, was the primary determinant of BP quality, with neither disease extent nor a history of advanced therapy affecting BP.

These findings suggest that 1L-PEG-ASC preparation may be the preferred option for colonoscopy BP in UC patients when not contraindicated.

Disclosure: Nothing to disclose.

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PP0666

SHORT VS. LONG COLONOSCOPES ON THE NOVEL MIKOTO SIMULATOR WITH A SENSOR-CONTROLLED, 3D-PRINTED COLON MODEL

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Introduction: Colonoscopy is a standard procedure but it is unclear whether a short or long colonoscope is more favorable regarding procedure efficiency and patient comfort. The performance of different colonoscope lengths cannot be objectively compared in live patients due to various limitations. Hence, the choice of colonoscope length is a matter of personal preference or institutional tradition. Preclinical tools to assess scope performance are needed. We evaluated endoscopists' performance using short vs. long colonoscopes in the Mikoto simulator, enabling objective measurements using pressure and distension sensors within a 3D-printed, calibrated colon model.

Aims & Methods: Aims: i) Performance comparison of short vs. long comparison in the Mikoto simulator, enabling objective measurements using pressure and distension sensors within a 3D-printed, calibrated colon model. ii) First evaluation of this novel simulator.

Delegates at ESGE Days and DGE-BV 2024 were invited to perform two simulation rounds—one with each scope length in a random order —and completed a survey. Main outcome was the performance of different scope lengths as measured by the caecal intubation time and the proprietary "Mikoto score" (a combination of pressure and distension measurements exerted on the model).

Results: Of 281 participants, 252 complete datasets were analyzed (187 ESGE, 65 DGE-BV). The Mikoto-Score was lower with long scopes (74 vs. 77; p=0.01), and caecal intubation time was longer (2:59 vs. 2:38 min; p=0.01). A second round did not improve performance. Mikoto-Scores were unaffected by scope preference or sedation strategy but were lower in users of the two-person technique. Scores correlated with self-assessed skill and colonoscopy experience (r=0.43, r=0.47). The median system-usability score was 77.5 indicating a good evaluation by the examiners and a good usability.

Conclusion: Using the Mikoto simulator, long colonoscopes caused more colonic distension and longer insertion times, potentially impacting patient comfort. Objective simulator data can inform scope selection in clinical practice. The Mikoto simulator reflects endoscopist self-assessed skill and previous experience.

Disclosure: Nothing to disclose.

PP0667

COMPARE THE CLINICOPATHOLOGICAL FEATURES OF YOUNG ONSET COLONIC ADENOMA (YOA) WITH PATIENTS AGE OVER 50: A RETROSPECTIVE COMPARATIVE STUDY

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Introduction: Although the overall incidence of colorectal cancer (CRC) has declined in individuals over 50 years of age, CRC in young adults (CRCYA) continues to rise. Previous studies suggest that CRCYA is associated with worse prognosis, primarily due to a higher proportion of advanced-stage disease at diagnosis. Accordingly, early detection and resection of young-onset adenomas (YOAs), the precursor lesions of CRCYA, is critical for prevention. This study aimed to characterize the clinicopathological features associated with YOAs.

Aims & Methods: We retrospectively analyzed patients who underwent colonoscopic polypectomy at our institution between January and June 2020. Neoplastic polyps were evaluated for size, histology, morphology, and anatomical distribution. Comparisons were made between patients aged ≤50 years (YOA group, n = 89) and those aged >50 years (elder group, n = 432).

Results: A total of 958 neoplastic adenomas from 521 patients were analyzed—138 in the YOA group and 820 in the elder group. No significant differences were found in gender distribution, mean polyp size, or morphology. However, elder patients had more neoplastic polyps per person than younger patients (1.90 ± 1.31 vs. 1.54 ± 0.88; p = 0.014).

With regard to anatomical distribution, adenomas in the YOA group were more often located in the left-sided colon compared to those in the elder group (63.3% vs. 45.4%; p < 0.001).

Variants	YOA Group Age≤50 (n=89)	Elder Group Age>50 (n=432)	P value
Age(years); mean ±SD	43.44± 5.61	64.92± 8.00	<0.001
Male(n)	61 (68.5%)	265(61.3%)	0.204
Max polyp size(mm); mean ± SD	8.83 ± 7.00	10.15± 8.80	0.197
Total number of Neoplastic polyps(n)	138	820	NS
Number of polyps removed/patient(n); mean ± SD	1.54 ± 0.88	1.90± 1.31	0.014
Location of polyps(n)			
Right	50(36.7%)	448 (54.6%)	<0.001
Left	88(63.3%)	372 (45.4%)	
Morphology(n)			
Is+Isp	121(87.8%)	724(88.3%)	
Ip	13(9.4%)	49(6.0%)	0.266
LST	2(1.4%)	25(3.0%)	
Ulcerative	2(1.4%)	22(2.7%)	
Histology of polyps			
Tubular adenoma(n)	92(66.7%)	606(73.9%)	
TVA+VA(n)	7(5.1%)	87(10.6%)	
HGD + Tis(n)	1(0.7%)	13(1.6%)	<0.001
Adenocarcinoma(n)	2 (1.4%)	29 (3.5%)	
SSA(n)	33(23.9%)	80(9.8%)	
TSA(n)	3(2.2%)	5(0.6%)	
Advanced adenoma(n)	27(19.4%)	163(19.9%)	0.361

Table.

Abbreviation: YOA, young-onset adenomas; SD, standard deviation; NS, nonsignificance; LST, laterally spreading tumor; TVA, tubulovillous adenoma; VA, villous adenoma; HGD, high-grade dysplasia; Tis, carcinoma in situ; SSA, sessile serrated adenoma; TSA, traditional serrated adenoma

Although the prevalence of advanced adenomas—defined as polyps ≥10 mm, with villous histology, or high-grade dysplasia—was also comparable between both groups. (19.4% vs. 19.9%; p = 0.523), histologically high-risk

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neoplastic lesions (villous adenoma, high-grade dysplasia, carcinoma in situ, and adenocarcinoma) were more frequent in the elder group (15.7% vs. 7.2%). In contrast, sessile serrated adenomas (SSAs) were more common in the YOA group (23.9% vs. 9.8%; $p < 0.001$).

Conclusion: While CRC remains more prevalent among older individuals, the comparable prevalence of advanced adenomas between younger and older adults raises concern for early carcinogenesis in young populations. Distinct histological features were observed: SSAs were more frequent in younger adults. Moreover, YOAs were predominantly located in the distal colon.

These findings emphasize the need for tailored screening strategies and proactive endoscopic surveillance in younger adults. Future prospective studies are warranted to further clarify the epidemiology and risk profiles of YOAs.

Disclosure: Nothing to disclose.

PP0668

A PILOT STUDY ON PURE NOTES: ENDOSCOPIC COLON RESECTION AND ANASTOMOSIS IN A PORCINE MODEL

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Introduction: Our research aims to establish the feasibility of performing complete endoscopic colon resection and anastomosis using pure Natural Orifice Transluminal Endoscopic Surgery (NOTES) in a preclinical porcine model.

Aims & Methods: The study was conducted on three pigs. Preoperative preparations involved administering sugar water followed by a 12-hour fast. A detachable balloon was positioned in the transverse colon to prevent contamination during the procedure. Initially, a gastroscope equipped with a transparent cap was advanced to the colon, reaching approximately 20-40 cm from the anal verge. A full-thickness incision was then made on the anal side of the colon to transect it, allowing the gastroscope to access the abdominal cavity. The surrounding connective tissue and lymph nodes were meticulously dissected and excised. The oral end of the colon was looped and resected using a snare, resulting in a 3-5 cm segment of colonic tissue being excised and retrieved through the anus. End-to-end anastomosis was facilitated using nylon threads for traction and clips for occlusion. Clips were securely attached to both the anal and oral ends of the colon.

Subsequently, three threads were simultaneously tightened to bring the oral and anal ends of the colon together. Finally, additional clips were placed along the anastomotic ring to ensure the integrity of the connection. Postoperative care included a structured dietary regimen and vigilant monitoring for complications.

Results: The postoperative outcomes were highly favorable, with no significant complications such as infections or anastomotic failures recorded. Endoscopic follow-up at one week demonstrated satisfactory anastomotic healing without evidence of stricture or fistula formation. Anatomical examinations conducted three months post-surgery revealed smooth and tight healing of the colonic anastomosis.

Conclusion: The use of pure NOTES for endoscopic colon resection and anastomosis represents a significant advancement in minimally invasive surgical practices. This technique introduces a new option for future endoscopic treatments of advanced colorectal tumors.

Disclosure: Nothing to disclose.

PP0669

A MULTICENTER REAL-WORLD STUDY ON CLINICAL EFFICACY OF ENDOSCOPIC RETROGRADE APPENDICITIS THERAPY (ERAT) FOR APPENDICITIS

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Introduction: Appendicitis is a common disease with a 7-10% lifetime prevalence. Endoscopic Retrograde Appendicitis Therapy (ERAT) emerged as a minimally invasive alternative option for appendicitis.

Aims & Methods: This study aims to evaluate the efficacy and safety of ERAT, presenting results from multicenter in China with the longest follow-up period reported to date. This study examined patients treated with ERAT for appendicitis across 21 centers in China from 2013 to 2023. Individuals with confirmed diagnoses via symptoms, lab tests, and imaging are enrolled. Appendicitis in the study was classified as uncomplicated, complicated, or absent based on procedure results. The intraoperative indicators include intubation guidance, use of transparent cap, intubation success rate, appendix condition, and operation time.

Primary outcomes focused on the effectiveness and safety of ERAT, assessing technical success, symptomatic relief, procedure time, and adverse events like perforation and complications. Secondary outcomes examined recurrence rates prompting medical intervention.

Results: 2838 patients underwent ERAT for appendicitis, with an average age of 28.4 years, and 52.5% were male. A history of appendicitis was noted in 96.1% of patients, with 83.3% experiencing an episode within two weeks and 17.7% receiving antibiotic treatment. 83.9% had acute appendicitis while 16.1% had chronic appendicitis. In AA patients (80.1% uncomplicated, 1.5% perforated, and 2.3% with peri-appendiceal abscess, Preoperative imaging found appendicoliths in 38% of patients. The ERAT technical success rate was 97.5%, with a procedural time averaging 29.2 minutes. Procedures guided by X-ray (55.8%), Cholangioscope (15.6%), and Abdominal Ultrasound (13.9%). Most used a tapered (47.7%) or standard transparent cap (31.7%), and others without transparent cap (20.6%). Saline irrigation was done in 89.4% of cases. Appendiceal stents were placed in 35.4% of patients. with a primary adverse event rate of 0.8%. Successful ERAT rates reached 86%, yet 2.5% failed due to mis-intubation and 11.5% due to symptomatic recurrence. The mean hospital stay was 2.9 days, with costs 8498 RMB(1173 USD). Significant postoperative laboratory reductions were observed in leukocyte and CRP levels. The adverse event rate led to reoperations in 1.4% of patients, with a 2.5% re-admission rate due to infections, predominantly new appendicitis or peritonitis.

Conclusion: ERAT is a safe and effective option for appendicitis, with short recovery times, minimal invasiveness, and less adverse events. Larger-scale, long-term multicenter trials are needed for further validation.

Disclosure: Nothing to disclose.

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PP0670

THE CLINICAL VALIDATION OF A COMPUTER-AIDED POLYP DETECTION MODEL INTEGRATED AS A PLUG-AND-PLAY ENDOSCOPY DEVICE (ALTER-EGO TRIAL)

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Introduction: The development of a computer-aided detection (CADe) model into a clinically reliable plug-and-play device is a complex and challenging process.

Aims & Methods: This study evaluates the performance, stability, and robustness of a new all-in-one plug-and-play device, integrated with a previously in-house developed and clinically validated AI algorithm for colorectal polyp detection.

The ALTER-EGO trial is an investigator-initiated operator blinded multicenter trial conducted in 3 Belgian centers. The AI model performing real-time polyp detection during real-time endoscopy was developed using deep learning on a high-resource/cost graphics processing unit (GPU)-based hardware platform and validated to achieve enhanced adenoma detection. The model was integrated into a lower-cost/resource field-programmable gate array (FPGA)-based hardware device, necessitating implementation adjustments and retraining to account for technological differences between both platforms. The new FPGA-based CADe system was evaluated in the exact same study set-up as the original GPU-based system, using a second observer and real-time unblinding.[1]

We evaluated the number of polyps detected, the extra detection and miss rates in comparison to expert endoscopists and performed analysis of false positive rate and technical failures to reflect system stability. Sample size analysis calculated 660 polyps required for non-inferiority analysis (margin 5%) with the original validation outcomes (CADe sensitivity 0.946±0.03).

Results: A total of 724 polyps were identified in 291 patients, of which 668 (92.1%) were detected by both endoscopist and CADe, 31 (4.3%) were detected by endoscopist alone, 25 (3.5%) were solely detected by CADe. Sensitivity of the CADe was 95.6% (95%CI: 93.7-96.9) versus 96.4% (95%CI: 94.7-97.5) for the expert endoscopists.

In comparison to the original GPU-based CADe system, detection was equivalent with a difference of -0.82% (95%CI: -0.0288 – 0.0124) (p=0.5040). Due to hardware-related adjustments, false positive rate increased from 1.7 to 8.8 per minute (p<0,001). Technical failure occurred in 49 patients, 109 events of technical deficiencies were reported.

Conclusion: Integration of an originally GPU-based AI model into a FPGA-based plug-and-play hardware device resulted in stable performances, equivalent to the GPU-based algorithm. The required technical adjustments of the FPGA-based algorithm however resulted in a significantly higher false positive rate and increased number of technical failures.

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Disclosure: Nothing to disclose.

PP0671

HOW ARE POLYPS REMOVED IN CLINICAL PRACTICE? A COMPARISON TO ESGE POLYPECTOMY GUIDELINES

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Introduction: The polypectomy technique used by endoscopists depends on multiple factors including anticipated histological subtype (eg Paris and Kudo pitt pattern), estimated polyp size and location within the colon. The latest European Society of Gastrointestinal Endoscopy (ESGE) guidelines recommend the most suitable methods which minimise risk of post-procedural bleeding and perforation, whilst ensuring cost-effectiveness. ESGE recommends avoiding cold biopsy due to its high rate of incomplete resection. ESGE also recommends cold snare polypectomy for removal of nonpedunculated adenomatous polyps under 10mm and hot snare with or without endoscopic mucosal resection (EMR) for polyps over 10mm. In addition, ESGE recommends cold snare or cold EMR for sessile serrated lesions (SSL) of all sizes, as well as consideration of endoscopic submucosal dissection (ESD) for certain polyps over 20mm.

Aims & Methods: This study aimed to assess the current practice in our institution in terms of polypectomy technique across all polyp sizes and histological subtypes in comparison to the ESGE standards.

A single centre, retrospective analysis of 100 consecutive patients who have had polypectomies done for polyps of any size from April 2024 at a hospital in North London was undertaken. The data was collected using the Medilogic endoscopy reporting tool, and the electronic patients record. The technique used for polypectomy, along with the type of polyp at histology retrieved, was scrutinized and compared with compliance of ESGE guidance on polypectomy technique.

Results: A total of 177 polypectomies from 100 patients were analysed. 121 polypectomies (68%) followed ESGE recommendations and 56 polypectomies (32%) did not. The 56 non-compliant polypectomies used methods including cold biopsy, hot snare with or without EMR for polyps under 10mm and hot EMR for SSLs (see Table).

Polypectomy method compliant with ESGE		Polypectomy method non-compliant with ESGE	
Cold snare or cold EMR for all SSL	13	Hot EMR for SSL	4
Cold snare or cold EMR <10mm	79	Cold snare or cold EMR > 10mm	12
Hot EMR >10mm	26	Hot EMR <10mm	12
ESD	3	Hot snare <10mm	3
		Cold biopsy	25
Total in line with ESGE:	121 (68%)	Total not in line with ESGE:	56 (32%)

Conclusion: This study shows poor adherence to the ESGE guidelines in our department with regards to colorectal polypectomies, with only two-thirds of polypectomies adhering to the recommended techniques. A third of polypectomies undertaken were not in line with ESGE recommendations (56/177 – 32%), of these 16/56 (34%) polypectomies involved inappropriate diathermy use which could have led to increased risk of complications such as post-polypectomy bleeding and perforation. A significant number of polyps are removed by the cold biopsy technique (25/177 – 14%) which is not advocated due to a higher risk of local recurrence.

We recommend better dissemination and implementation of the ESGE guidelines to enhance clinical practice with regards to polypectomy techniques.

Disclosure: Nothing to disclose.

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PP0672

REMIMAZOLAM IN ENDOSCOPY: A GAME-CHANGER FOR SAFE AND EFFECTIVE SEDATION IN REAL-WORLD PRACTICE

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Introduction: Sedation is integral to gastrointestinal (GI) endoscopy to enhance procedural tolerance and efficiency [1]. Traditional agents like midazolam and propofol, while effective, present limitations related to recovery time, safety, and need for anesthesiologist supervision. Remimazolam, a novel benzodiazepine with rapid onset and offset, may enable effective non-anesthesiologist-administered sedation (NAAS) in routine endoscopy [2].

Aims & Methods: A prospective, multicenter, real-world study was conducted between November 2024 and March 2025. Adult patients undergoing elective upper or lower gastrointestinal (GI) endoscopy under procedural sedation-analgesia (PSA) without the presence of an anesthesiologist were enrolled. Inclusion criteria were: age ≥18 years, American Society of Anesthesiologists (ASA) physical status I–III, and a body mass index (BMI) between 18 and 51 kg/m^2. Exclusion criteria included: known allergy to anesthetic drugs or a history of anesthesia-related adverse reactions; glucose-6-phosphate dehydrogenase deficiency; severe respiratory disease; and pregnancy. All patients undergoing emergency or urgent GI endoscopy were excluded. The primary outcomes were the sedative efficacy and safety of remimazolam during endoscopy. Efficacy was defined as successful completion of the procedure without interruption due to inadequate sedation. Adverse events (AEs) were recorded and classified by severity (major or minor) and type (e.g., hypotension, hypoxia, bradycardia, nausea, vomiting, and injection site pain). Remimazolam was administered by trained endoscopists: induction was achieved with 5–7 mg intravenously, titrated based on individual risk factors, with 2.5 mg supplemental doses as needed (maximum cumulative dose: 33 mg).

Results: A total of 182 patients (EGD n=75; colonoscopy n=112) were included. Procedure completion was achieved in 175 cases (96.3%), with success rates of 97.3% for EGD and 95.5% for colonoscopy. All sedation failures (n=7) were attributed to inadequate sedation and not to ASA class. No serious AEs were reported. Minor AEs included hypotension (n=4), bradycardia (n=1), nausea (n=1), vomiting (n=4), and hiccups (n=1), all managed conservatively. Median cumulative remimazolam doses were 10.9 mg (colonoscopy) and 10.0 mg (EGD), with median recovery times of 6.25 and 5.9 minutes, respectively. No patients required flumazenil or rescue sedation. Among obese patients (BMI >40), remimazolam was both effective and safe.

Conclusion: Remimazolam provides a highly effective and safe sedation alternative for GI endoscopy in real-world NAAS settings, ensuring rapid recovery and minimal AEs, even in patients with elevated ASA status or obesity. This study was prospective real-world and had several limitations. First the number of patients was too small to accurately assess rare adverse events. Further randomized controlled trials with larger populations are recommended to confirm these results.

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Disclosure: Nothing to disclose.

PP0673

INITIAL EXPERIENCE IN THE UK OF ENDOSCOPIC HAND SUTURING USING THE SUTUART DEVICE

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Introduction: Endoscopic suturing allows the closure of mucosal defects, opening the door to a variety of therapeutic utilities. Reducing the risk of post polypectomy bleeding and closing resection related perforations with endoscopic hand suturing have been investigated and further therapeutic indications are feasible. Sutuart is a novel device that facilitates endoscopic hand suturing. In this study we present the initial UK experience using Sutuart for endoscopic closure of defects.

Aims & Methods: This is a multicentre retrospective study of the effectiveness and related outcomes of endoscopic suturing using the Sutuart device in the GI tract. UK based sites undertaking endoscopic hand suturing using Sutuart were invited to participate. Patient records and endoscopy reports were reviewed and data was extracted. Technical feasibility (whether the suture could reach the target), technical success (whether suturing could be completed) and clinical success (intended therapeutic goal was achieved) were assessed. Procedural details, size of closure and time taken to complete Sutuart closure were recorded. Outcomes at 30 days were reviewed.

Results: Four sites participated with 14 closures in 14 patients. The mean age was 59, and 57% were female. The commonest indication was closure of polyp ESD resection site (8/14, 57%), followed by closure of GPOEM incision (4/14, 29%). One patient had a closure of a PEG fistula and one had a closure of a leaking surgical anastomosis.

All polyp resection sites were within the rectum (6/8) or rectosigmoid (2/8). The mean polyp size was 42.8mm (SD +/- 10.1).

Sutuart was technically feasible in all cases, 13/14 (93%) were technically successful and 12/14 (86%) were clinically successful. Of those that were not clinically successful, one attempted closure was for a GPOEM incisional closure in which insufflation was inadequate and one was for a surgical anastomosis that had a persistent leak. Both cases were closed with through the scope clips. Two other cases had additional clip closure. A post procedural bleed was seen in one patient who had a rectal polyp site closure. This was managed conservatively.

The mean length of closure was 35.4mm (SD = +/- 15.4). The mean time to complete closure was 35 minutes (1SD +/- 13.3) with a range of 18 to 65. The median number of suture passes was 6 (IQR = 1).

Conclusion: The Sutuart is a novel device that shows promise in the management of endoscopic closure. The initial experience demonstrated in this study indicates good technical feasibility and success with positive

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outcomes and few complications across a range of indications. The time to complete closure at present appears relatively prolonged. Whether increased experience leads to a reduction in this time will be interesting to see. Further studies into larger cohorts and comparative studies with other closure techniques are required.

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PP0674

VIRTUAL RELIEF: VIRTUAL REALITY FOR PAIN AND ANXIETY RELIEF DURING COLONOSCOPY

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Introduction: Colonoscopy is a routine yet often uncomfortable procedure that can cause significant pain and anxiety for patients, leading to reluctance and many procedural challenges. While sedation and analgesia are commonly used, they may not always provide sufficient relief and even carry certain risks.

Recently, virtual reality (VR) has emerged as a promising tool for reducing pain and anxiety in various medical contexts (surgeries, dental procedures, labor...). By immersing patients in calming environments, VR may offer an effective, low-cost, non-invasive and non-pharmacological alternative to traditional methods. Studies exploring the use of VR in colonoscopy patients receiving minimal sedation have been documented, although in limited numbers.

However, there is a significant lack of research focusing on unsedated patients, especially with no published literature from Morocco. This study aims to evaluate the effectiveness of VR in alleviating pain and anxiety during unsedated colonoscopy.

Aims & Methods: The aim of the study was to assess the impact of virtual reality on five key areas of investigation: Anxiety, pain, comfort, satisfaction and willingness to undergo further colonoscopies and procedural characteristics. We conducted an experimental, randomized, controlled trial to evaluate the effect of VR on reducing pain and anxiety in non-sedated patients undergoing outpatient colonoscopy in the Endoscopy Unit of the Department of Gastroenterology of the university hospital of Mohammed VI in Marrakech.

We've included all patients above 18 years of age undergoing a colonoscopy. A total of 60 patients were sampled and then randomized 1:1 into a group using VR and a control group with routine care but no VR. Collection of data was by a questionnaire based on validated scales and scores for anxiety, pain, comfort and satisfaction.

Results: Our study demonstrates that VR has a positive impact on various aspects of the colonoscopy procedure. Firstly, VR reduced the time to caecum intubation by 4.29 minutes. Additionally, VR showed beneficial effects on procedural anxiety, with greater reduction in anxiety levels pre- to post-colonoscopy in the VR group compared to the control group. In terms of pain reduction and comfort improvement, a significant decrease in pain was noted in the VR group (4.53 vs 5.83 in the control group). Overall satisfaction was higher in the VR group comparing to the control group (74% vs 54%, respectively). The Net Promoter Score was also higher in the VR group (-0.33) compared to the control group (-16.66).

Regarding willingness to return for another colonoscopy, the VR group showed higher willingness, with more patients strongly agreeing and fewer disagreeing with the statement compared to the control group (Strongly agree: 3, Agree: 19, Disagree: 6, Strongly disagree: 2 in the VR group versus 0, 17,10, and 3 in the control group).

Endoscopists' feedback was also positive, with all reporting comfort in incorporating VR into their practice, giving it a mean score of 8. They rated their overall experience with VR as good, with a mean score of 8.3. Endoscopists noted that VR had a relaxing effect on them, with no significant difficulties encountered.

Conclusion: Our study reinforces that VR use in colonoscopy significantly improved both the overall patient experience (lesser pain, more comfort) and the physician experience too (lesser procedure duration, better completion rates, more comfort during procedure). VR, being an easily available, low-cost, non-invasive, and inexpensive, can be recommended for routine use in endoscopy units as a pain relief tool.

Disclosure: Nothing to disclose.

PP0675

PROSPECTIVE COMPARISON OF PATHOLOGY AND COMPUTER-AIDED OPTICAL DIAGNOSIS FOR POLYP CHARACTERIZATION USING EXPERT CONSENSUS

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Introduction: Histopathology is the accepted gold standard for classifying colorectal polyps. However, the complex pathway from initial polyp detection and resection to final histological diagnosis involves multiple steps (retrieval, embedding, sectioning, interpretation), each potentially introducing diagnostic errors.

Aims & Methods: We aimed to investigate the diagnostic accuracy of autonomous computer-aided optical polyp diagnosis (CADx) versus histopathology, using expert consensus as the reference standard. In this prospective study, three internationally recognized expert endoscopists independently reviewed colonoscopy images and videos of diminutive (≤5 mm) polyps assessed by autonomous CADx. Unanimous, high-confidence expert agreement was defined as the reference standard ('ground truth'), assumed superior due to absence of resection or specimen-management related

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artifacts. The primary outcome was the proportion of specimens with potential incorrect histopathology diagnosis. Secondary outcomes included comparative accuracy, sensitivity, and specificity of autonomous CADx and histopathology against the expert-defined reference standard.

Results: Among 510 patients, experts provided unanimous, high-confidence diagnoses for 226 diminutive (≤5 mm) colorectal polyps, serving as the reference standard. The proportion of potentially incorrect pathology diagnoses was 23% (95% CI: (17.8,29.2)). Autonomous CADx achieved a higher accuracy of 91.2% (95% CI: 86.5–94.4%), compared with pathology at 77.0% (95% CI: 70.8–82.2%; $p < 0.001$).

Conclusion: These findings highlight that histopathological diagnosis after polyp resection is not infallible and can introduce diagnostic errors in a relevant proportion of cases. CADx diagnosis, obtained during the procedure and unaffected by specimen handling artifacts, may prevent some of these errors, suggesting it could offer superior accuracy for polyp characterization in certain cases.

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PP0676

DOUBLE BALLOON ENDOSCOPIC PLATFORM ENABLES DIAGNOSTIC AND THERAPEUTIC INTERVENTIONS IN THE SMALL BOWEL

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Introduction: The Double Balloon Endoscopic Platform (DBEP), DiLumen® EZ Glide (Lumendi Ltd, Maidenhead, UK), has been in clinical use since 2017. It has demonstrated safety and efficacy in facilitating diagnostic and complex therapeutic procedures in the colon, including patients with normal and surgically altered anatomy.

Aims & Methods: To evaluate the use of DiLumen® EZ Glide for diagnostic and therapeutic interventions within the small bowel.

The DBEP is a flexible overtube with two balloons at its distal end and is preloaded over a standard colonoscope. Once advanced into the sigmoid colon, the proximal (aft) balloon (rigidly attached to the overtube) is inflated to anchor the device within the colon. The device is then pulled back to shorten and straighten the sigmoid colon. Then the colonoscope is advanced to the next colonic segment, and the process of balloon inflation, retraction, and advancement is repeated until the cecum is reached and the assembly is directed through the ileocecal valve into the ileum. This technique is continued within the small bowel until the target lesion is reached for intervention.

Results: Seven patients (3 males, 4 females; mean age 59.7 ± 9.8 years) underwent ileal interventions using the DBEP. All procedures were technically successful (100%). The furthest advancement of the colonoscope was up to 6 feet proximal to the ileocecal valve. The mean size of small bowel lesions was 23.6 ± 11.4 mm.

Three patients had ileal polyps (30, 35 and 40 mm), all of which were removed *en bloc*: one by endoscopic mucosal resection (EMR) and two by endoscopic submucosal dissection (ESD). Pathology confirmed all polyps were benign, and no further treatment was required.

The remaining four patients had submucosal nodules (mean size 15.0 ± 4.1 mm, range 10–20 mm). Prior biopsies of this lesions covered by normal mucosa were non-diagnostic. All nodules were successfully removed *en bloc* (3 by ESD, 1 by EMR), and post-resection defects were closed using continuous suturing with the OverStitch™ endoscopic suturing system (Boston Scientific, Marlborough, MA, USA), delivered via the DBEP.

Histopathologic analysis revealed large B-cell lymphoma (1 patient) and well-differentiated neuroendocrine tumors (3 other patients). The lymphoma patient was treated conservatively with chemotherapy. All neuroendocrine tumor cases underwent robotic hemicolectomy with ileal resection. Pathological examination of surgical specimens revealed lymph node metastases (pN1) in all 3 patients.

Conclusion: Traditional endoscopic biopsy of ileal submucosal lesions does not provide adequate diagnostic value and cannot determine effective treatment plan. The DBEP significantly facilitates small bowel access and enables effective diagnostic and therapeutic endoscopic interventions inside the small bowel. In this early series, endoscopic excision of ileal lesions provided accurate histological diagnosis and directly influenced clinical management. Most importantly, our findings suggest that even early-stage (pT1 and pT2), well-differentiated ileal neuroendocrine tumors already had lymphatic spread, reinforcing the need for surgical resection despite complete local endoscopic excision.

Disclosure: Sergey Kantsevoy: Endosound - investor, Lumendi - consultant, equity, Medtronic - consultant, Olympus - consultant, Slater Endoscopy - consultant, equity, Vizballoons - consultant, equity.

PP0677

CORRECTED WITHDRAWAL TIME BENCHMARKS REQUIRED FOR STAFF VERSUS TRAINEE ENDOSCOPISTS TO REACH ADENOMA DETECTION TARGET IN COLONOSCOPY

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Introduction: Withdrawal time (WT) is strongly correlated with adenoma detection rate (ADR)[1,2]. Corrected withdrawal time (cWT), defined as the withdrawal time excluding periods spent on therapeutic interventions, provides a more accurate measure of mucosal inspection effort. Optimal WT targets may differ between staff and trainee endoscopists due to differences in experience and inspection technique[3].

Aims & Methods: We aimed to determine the cWT required to achieve ADR benchmarks (≥35%) and compare detection metrics between these two groups.

This prospective study included 1,439 elective colonoscopies of adult patients ≥18 years, performed at Montreal University Hospital Center: 397 by eight supervised trainees and 1,042 by 12 staff endoscopists. All procedures were video recorded, and the corresponding meta-data were documented in real-time by a research assistant present in the endoscopy room. Procedures performed for the purpose of polypectomy, incomplete

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colonoscopies, procedures performed by endoscopists with experience of fewer than 10 colonoscopies, and procedures with technical failures in recording that prevented video analysis were excluded. Colonoscopy videos were reviewed to precisely measure cWT, defined as total withdrawal time minus therapeutic intervention durations. Performance metrics were stratified by endoscopist training level (staff vs trainees).

Primary outcome was the mean cWT required to achieve ADR ≥35%, stratified by staff and trainees. Secondary outcomes were the mean per-lesion intervention time, ADR, advanced ADR (AADR), Serrated Lesion detection rate (SLDR), APC, and the incremental increase of all detection measures per minute of cWT compared between staff and trainees, adjusting for covariates including patient demographics, bowel preparation quality, indication for colonoscopy, use of computer-assisted detection, and clustering by individual endoscopist using generalized estimating equations (GEEs).

Results: Trainees demonstrated significantly longer mean cWT than staffs (10'01'' vs 7'53'', p<0.001). No significant difference was observed regarding mean per-polyp intervention time (1'33" vs 1'28", p=0.234). On average, staff endoscopists required longer mean cWT to achieve the 35% ADR benchmark compared to trainees (9'47'' [95% CI: 8'04''-12'20''] vs. 7'05'' [95% CI: 0'00''-10'05'']). Overall AADR, SSLDR, and APC were significantly higher for trainees than staff (Table 1; AADR 10.8% vs. 5.3%; SSLDR 8.1% vs 3.6%; APC 0.8 vs 0.6). Overall, the odds of adenoma detection significantly increased with each additional cWT minute for both trainees and staffs (Table 1; 6.6% overall, 6.7% for staff, and 7.0% for trainees).

	Overall Mean (95% CI)	Trainees Mean (95% CI)	Staff Mean (95% CI)	Overall Increase/ additional cWT minute (p-value)	Trainees Increase/ additional cWT minute (p-value)	Staff Increase/ additional cWT minute (p-value)
ADR	34.2% (31.8-36.7)	39.8% (35.0-44.8)	32.1% (29.2-35.0)	6.6% (0.004)	7.0% (0.005)	6.7% (0.029)
AADR	6.8% (5.6-8.3)	10.8% (8.0-14.4)	5.3% (4.0-6.9)	10.6% (<0.001)	9.3 (<0.001)	10.3% (<0.001)
SSLDR	4.8% (3.8-6.1)	8.1% (5.7-11.3)	3.6% (2.5-4.9)	12.4% (<0.001)	6.4% (0.041)	14.9% (<0.001)
APC	0.6 (0.6-0.7)	0.8 (0.7-1.0)	0.6 (0.5-0.6)	7.4% (<0.001)	9.5% (<0.001)	6.3% (<0.001)

Table 1. Metrics' estimated summaries overall and by endoscopist group. Means and 95% confidence interval (CI) are raw estimates. Percent increases and corresponding p-values arise from GEEs.
ADR, adenoma detection rate; AADR, advanced adenoma detection rate; SSLDR, sessile serrated lesion detection rate; APC, adenoma per colonoscopy; cWT, corrected withdrawal time; CI, confidence intervals.

Conclusion: Trainees had longer cWT and higher detection rates overall, while staff required slightly more cWT to reach ADR target of ≥35%. These findings emphasize the importance of tailored WT benchmarks based on endoscopist training level to optimize colonoscopy quality and training outcomes.

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PP0678

PREVALENCE AND CARDIOMETABOLIC RISK FACTOS ASSOCIATED TO SERRATED LESIONS IN AN ITALIAN COLORECTAL SCREENING POPULATION

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Introduction: Serrated polyps (in particular sessile serrated lesions and traditional serrated adenomas according to WHO classification) are recognized as premalignant lesions, responsible of a significant proportion of colorectal cancer via the serrated neoplasia pathway.

Aims & Methods: We aimed to study the prevalence and risk factors of serrated polyps, comparing them with adenomas and with patients with negative colonscopies in a colorectal italian screening population. A retrospective case-control study was performed in a secondary-referral center in Rome (San Filippo Neri Hospital). Data were collected on an Excel database and retrieved from pre-endoscopy questionairres, endoscopic and hystologic reports of screening colonscopies, performed from November 2021 to December 2024.

Results: Preliminary analysis was carried out on 700 screening subjects (331 male and 369 female of age between 50 and 74 years). including 274 subjects (39%) with adenomas and 33 subjects with serrated lesions (4,7%). Exclusion criteria were incomplete colonscopies, inadequate bowel preparation (Boston Bowel Preparation Scale < 6) and symptomatic subjects. In the group of subjects with serrated lesions, 23 (69,7 %) had dysplasia. About 70% of adenomas were in the left colon, while the prevalence of serrated lesions was similar in the left and right colon (51,5% and 48,5 % respectevely). People with hyperplastic polyps, although classified as serrated lesions by WHO classification, are generally considered as benign and were therefore not considered in our analysis.

For multivariable analysis we considered 12 subjects that had only serrated lesions and 218 subjects that had only adenomas, comparing them to 339 subjects that had no polyps. Independent variable considered were: age, gender, smoking, Body Mass Index, hypertension, dyslipidemia and type 2 diabetes mellitus.

From multivariable logistic regression resulted that smoking was significant indipendent risk factor for serrated lesions (OR: 3.646, 95 % CI 0.056-2.532), whilst male sex and smoking were significant indipendent risk factors for adenomas (male sex OR: 1.768, 95 % CI 0.217-0.923; smoking OR: 1.730, 95 % CI 0.196-0.900).

Conclusion: In our study population, the prevalence of colorectal serrated lesions was 4,7%. Risk factors associated with adenomas were male sex and smoking and with serrated lesions was only smoking. However, a larger sample of subjects with serrated polyps is needed to confirm these results.

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PP0679

SUSTAINED LEARNING EFFECTS OF SELF-DIRECTED VIDEO-BASED VIRTUAL SCALE ENDOSCOPY TRAINING: A NOVEL EDUCATIONAL STRATEGY FOR ENDOSCOPISTS

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Introduction: Accurate assessment of colorectal polyp size is essential for determining appropriate surveillance intervals and therapeutic approaches. However, visual estimation remains inherently imprecise, especially among less experienced endoscopists. While virtual scale endoscopy (VSE)1 offers an in-procedure solution for real-time mea.

Aims & Methods: This study aimed to evaluate the long-term educational benefits of self-directed training using short instructional videos based on VSE. Thirty-three endoscopists (15 experts and 18 non-experts) participated in a structured learning program using four sets of videos (SET1–SET4), each containing 20 colorectal lesions recorded during colonoscopy. Following initial exposure and evaluation with SET1 and SET2, participants underwent two follow-up tests (SET3 and SET4) 8–12 weeks later without interim re-training, to assess retention of estimation accuracy.

The primary outcome was the total absolute error between estimated and actual lesion diameters, summed across 20 lesions per set. Secondary analysis evaluated differences between expert and non-expert groups.

Results: After initial training, participants showed improved size estimation accuracy (SET1: 34.7 ± 6.6 mm vs. SET2: 30.7 ± 7.7 mm, $p = 0.048$). Importantly, this effect was sustained after 8–12 weeks, with mean total error in SET3 (28.6 ± 6.9 mm) remaining significantly lower than the pre-training baseline (SET1, $p = 0.006$). Similarly, SET4 showed continued improvement over baseline (31.7 ± 7.1 mm, $p = 0.025$), despite no further exposure to training content. Non-experts demonstrated a more pronounced benefit, suggesting that the self-learning video format may be particularly effective for early-career endoscopists. No adverse effects or dropouts occurred during the study.

Conclusion: This is the first study to demonstrate that a brief, self-directed video-based training program using virtual scale endoscopy can achieve not only immediate improvements in polyp size estimation accuracy, but also sustain those gains over time. Given its simplicity, scalability, and minimal resource requirement, this educational approach may serve as a valuable adjunct in structured endoscopy training programs, especially for novice endoscopists or in settings with limited access to formal instruction.

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PP0680

CHARACTERIZATION OF BIOMECHANICAL RISK FACTORS DURING ENDOSCOPIC SUBMUCOSAL DISSECTION: AN ERGONOMIC PILOT STUDY

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Introduction: The prevalence of musculoskeletal disorders (MSDs) among endoscopists is frequent. The aim of this study was to evaluate biomechanical risk factors for MSDs in gastroenterologists performing ESD.

Aims & Methods: An observational study was performed among interventional endoscopists performing ESD in 3 French centers. The physical constraints were assessed using an analog scale of perceived physical effort intensity and physiological sensors to measure joint angulation kinematics and muscle activity levels (flexor and extensor carpi radialis muscles) during diagnostic colonoscopy and ESD. High muscle strain was defined as any muscle activation exceeding 10% of the maximum voluntary contraction (MVC). Two distinct sub-tasks of ESD were identified: lesion marking and circumferential incision phase (ESD-1) and dissection phase (ESD-2).

Results: Six interventional gastroenterologists participated in the study. Perceived physical effort was significantly greater for ESD compared to colonoscopy ($p = 0.03$). Time spent at more than 10% MVC for the right extensor carpi radialis was significantly higher during ESD-1 than during colonoscopy (+15%, $p=0.04$). The greatest strain was observed in the left extensor carpi radialis. This muscle was particularly exposed since more than 50% of the time was spent at more than 10% of MVC during colonoscopy and up to more than 80% during ESD-1 and -2. Time spent in the neck flexion risk zone was significantly higher during ESD-2 than during colonoscopy (+42%, $p = 0.046$).

Conclusion: ESD increased the risk of musculoskeletal strain. It is crucial to develop prevention programs to reduce the risk of MSD in the gastroenterologist population.

Disclosure: Nothing to disclose.

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PP0681

EFFECT OF GRAVITY IN COLORECTAL ENDOSCOPIC SUBMUCOSAL DISSECTION WITH ADAPTIVE TRACTION STRATEGY: A PROSPECTIVE COMPARATIVE STUDY

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Introduction: Traction strategies are increasingly becoming the standard approach for endoscopic submucosal dissection (ESD). ‚Natural' traction forces, such as gravity, remain valuable during ESD, helping to optimize dissection and improve procedural speed. However, mechanical traction, particularly the adaptive traction strategy, has recently shown promising results in terms of efficacy and safety, including the highest resection speeds reported in the literature for colorectal ESD (1,2).

The aim of our study was to evaluate resection speed based on initial gravitational force in a prospective cohort of patients undergoing colorectal ESD with adaptive traction strategy.

Aims & Methods: Consecutive patients undergoing ESD of conventional colorectal lesions with an adaptive traction strategy at a tertiary referral center between 2023 and 2025 were prospectively enrolled. The initial gravity was classified as favorable/neutral and unfavorable.

The primary endpoint was resection speed. A non-inferiority analysis was conducted for the primary endpoint, with the non-inferiority margin set at a 20% reduction in resection speed. Secondary endpoints included the rate of En Bloc resection, the rate of R0 resection and the rate of adverse events.

Results: The study included 118 patients, of whom 78 (66%) had favorable or moderate initial lesion severity and 40 (34%) had unfavorable initial gravity. The mean resection speed in the favorable and moderate gravity group was 37.4 versus 35.8 ($p = 0.6067$) in the unfavorable group, the R0 resection rate was 86.2% versus 86.8% ($p > 0.99$), and the perforation rate was 5.9% versus 7.0% ($p = 0.7015$).

After adjustment for lesion location, operator and lesion area, the resection rate in the unfavorable severity group was very close to the non-inferiority margin ($p = 0.07$).

Conclusion: In our cohort of patients undergoing colorectal ESD using an adaptive traction strategy, initial gravity did not alter the speed and quality of resection. We can assume that the use of a mechanical adaptive traction strategy allows overcoming the need for the natural traction force provided by gravity, thus avoiding the need to mobilize patients during the procedure to achieve a more favorable gravitational force.

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The other authors have no competing interests related to this study.

PP0682

AI-BASED PREDICTION OF HISTOPATHOLOGICAL TUMOR GRADING FROM WHITE-LIGHT ENDOSCOPIC IMAGES IN COLORECTAL CANCER

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Introduction: The degree of tumor differentiation (grading) is an important prognostic factor in colorectal cancer (CRC). Poorly differentiated tumors (G3) tend to be more aggressive. They are associated with an increased risk of transmural invasion, lymph node involvement, and distant metastasis compared to well- or moderately differentiated tumors (G1/G2). Tumor grading is performed through histopathological analysis and requires sufficient tissue sampling. Artificial intelligence (AI) can potentially assist in differentiating colorectal tumors through predicting the grading using images captured during endoscopy as input.

Aims & Methods: This study aimed to train an AI model to distinguish poorly from well- and moderately differentiated colorectal carcinoma using white light endoscopic images.

A total of 1.607 endoscopic images of colorectal cancer, recorded from 2014 to 2024, were extracted from 404 patients who underwent colonoscopy, and the corresponding histopathological grading was retrospectively evaluated. Identified cases contained 62 G1, 1179 G2, and 366 G3 tumor images from 17, 305, and 82 patients respectively. Grades 1 and 2 (low-grade) represented the first group, and grade 3 (high-grade) represented the second. Transfer learning using a foundational model for endoscopy (Gastronet-5M[1]) was used to counteract the limited amount of available data. Before the analysis, the data was split into training and validation on a patient level.

Model performance was assessed with accuracy, sensitivity, and specificity for single-image classification. Furthermore, the results of aggregating predictions for multiple images of the same case were investigated.

Results: The validation dataset consisted of 218 images from 61 patients, 181 of which were in the G1/G2 group and 37 in the G3 group. On an image level, the AI model achieved an accuracy of 78% (95% CI: 73%–83%) with specificity of 0.81 (95% CI: 0.75–0.86) and sensitivity of 0.68 (95% CI: 0.52–0.83).

On a patient level, the method achieved an accuracy of 79%(95% CI: 74%–92%), achieving a slightly increased specificity of 0.84 (95% CI: 0.7209–0.9348) and a higher sensitivity of 0.83 (95% CI: 0.58–1.00).

Conclusion: Our AI model can classify CRC regarding their tumor grading using endoscopic images. Aggregation of predictions for multiple images of the same tumor proves to be especially beneficial in terms of sensitivity. Using explainability methods, clinicians may gain valuable insights into key image features that differentiate tumor grading in white light images, supporting more informed diagnostic decisions.

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Disclosure: Nothing to disclose.

PP0683

COLORECTAL ESD CAN BE UNDERTAKEN SAFELY AND EFFECTIVELY WITHOUT GENERAL ANAESTHETIC

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Introduction: ESD allows en bloc resection of complex colorectal polyps. General anaesthetic (GA) for ESD is standard, requiring routine admission to hospital for monitoring. At our Trust conscious sedation is used preferentially. This study presents clinical outcomes of ESD performed in a high-volume tertiary centre in the UK using this approach.
Aims & Methods:
• To evaluate the clinical outcomes of colorectal endoscopic submucosal dissection performed without general anaesthesia in a high-volume UK tertiary centre.
• To assess the feasibility and safety of ESD under conscious sedation or no sedation, with particular focus on rates of en bloc and R0 resection, adverse events, and post-procedural admissions.
• To identify clinical and procedural factors independently associated with incomplete (R1) resection following colorectal ESD.
• To compare outcomes between patients undergoing ESD with GA versus those without
Colorectal ESD procedures performed between 2018-2024 in a single centre in the UK were included. Data was retrospectively collected from the patient records. Multiple logistic regression was performed to identify independent factors associated with incomplete resection.
Results: A total of 147 patients (91 males ; median age 68 years) were included. Most polyps were rectal (100/150, 66.6%). Median polyp diameter was 42.5 mm.
GA was utilised in 13 patients, 80 had sedation alongside analgesia (median dose midazolam=2mg, fentanyl=100mcg) 44 required no IV sedation or analgesia, 8 had analgesia alone (median dose fentanyl 100mcg), 2 had sedation alone (median dose midazolam 0.75mg).
4 patients were admitted to hospital after their procedure and 14 were admitted within 30 days. There were 4 (2.7%) perforations and 7 (4.7%) post polypectomy bleeds (PPB).
Most procedures were complete ESD (109/147, 74.1%), the rest being hybrid ESD. R0 excision was achieved in 70/109 (64.2%) of those who had a complete ESD resection. No residual tissue was seen in patients with an R0 resection who had a subsequent LGI endoscopy (43/70). In those with R1 resection, a site check was undertaken at the same Trust in 59 patients. Residual tissue was seen in 7/59 (12%).
On univariate binomial regression, R1 resection was associated with polyps involving the dentate line [OR = 5.29 (95% CI = 1.84 – 17.8)], those with severe fibrosis [OR = 20.8 (95% CI = 3.85- 386.4)] and those receiving a GA [OR = 3.56 (95% CI = 1.06-16.3)]. On multivariate analysis, polyps involving the dentate line [OR = 8.17 (95% CI = 2.24 – 35.8)] and those with severe fibrosis [OR = 17.9 (95% CI = 2.57- 366.2)].

Reason for Admission	Number of Cases	Typical Timing (Days Post-ESD)	Treatment(s)	Length of Stay (Range)
Immediate admission – procedural reasons	2	Day 0	Monitoring	1 day
Immediate admission – perforation	2	Day 0	Conservative	2 days
Delayed perforation	2	Day 1, Day 1	Hartmann's procedure; Surgical washout + antibiotics	7–21 days
Post-polypectomy bleeding (PPB) – conservative	5	Days 7–13	Conservative	1–4 days
PPB – endotherapy	3	Days 5–9	Endotherapy	1–2 days
Fever	1	Day 1	Conservative	2 days
Hypotension	2	Days 1–2	Conservative	1 day
Acute Coronary Syndrome (ACS)	1	Day 6	Medical	20 days

Table 1 - Outcomes of admissions following ESD.

Conclusion: ESD is an effective treatment for high-risk complex colorectal polyps, able to provide high rates of en bloc resection. It can be achieved safely in most patients without GA. GA was associated with a higher rate of R1, however, this likely relates to these lesions being more complex.
Disclosure: Nothing to disclose.

PP0684

COLONOSCOPY IN YOUNG AND ELDERLY INDIVIDUALS: WHICH INDICATIONS ARE ASSOCIATED WITH HIGH-RISK COLORECTAL NEOPLASTIC LESIONS?

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Introduction: The indications for colonoscopy are well established, with the primary objective being the detection of neoplasia or lesions that carry a high risk for colorectal cancer, even though benign causes are more commonly encountered. In young adults under 50 years of age, common indications include rectal bleeding and diarrhea.
In contrast, individuals aged 50 years and older are screened more broadly due to the increased risk of colorectal cancer, although recent trends show a rising incidence of neoplastic lesions in younger adults as well.
Aims & Methods: The aim of our study is to compare the indications and outcomes of colonoscopy between individuals younger and older than 50 years, and to identify which indications are most strongly associated with neoplastic findings.
A retrospective, descriptive, and analytical study was conducted over a period of 3.5 years, involving 1,518 patients. Participants were divided into two groups: Group A included patients under 50 years old, while Group B comprised those aged 50 and above. Epidemiological, clinical, and endoscopic data were compared between the two groups. Patients with a known diagnosis of inflammatory bowel disease (IBD) were excluded from the study. Statistical analysis was carried out using SPSS version 21.0.
Results: Among the patients, 65.1% were over 50 years of age, while 34.9% were under 50. The mean age in group A (under 50) was 37.5 ± 9 years, compared to 64.4 ± 9 years in group B (over 50). Females predomi-

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nated in the younger group (55.3%), whereas males were more prevalent among older patients (56.3%) ($p < 0.001$). A personal history of colorectal cancer was more frequent in patients over 50 (10.5% vs. 4.4%; $p < 0.001$). Colonoscopy indications varied by age group: constipation was the most common indication in group B (29.1% vs. 22%; $p = 0.01$), whereas diarrhea predominated in group A (33.3% vs. 19.3%; $p < 0.001$). Pathological findings on colonoscopy were more frequent in older patients (49.1% in group B vs. 32.9% in group A; $p < 0.001$).

In patients under 50, the most common pathological findings were colitis or ulcerative colitis (47.9% vs. 11.7%; $p < 0.001$). In contrast, patients over 50 showed a higher prevalence of polyps (61.5% vs. 41.3%; $p < 0.001$) and rectocolonic neoplasia (16% vs. 5%; $p = 0.002$).

Among the colonoscopy indications, constipation in patients under 50 was associated with a higher risk of neoplastic lesions ($p = 0.007$), while in patients over 50, rectal bleeding and constipation were significantly associated with neoplastic lesions ($p < 0.001$).

Conclusion: The presence of neoplastic lesions was strongly associated with constipation in individuals under 50 years of age, and with both rectal bleeding and constipation in those over 50. Further studies to better define these indications could help improve the appropriateness of colonoscopy, particularly in younger patients.

Disclosure: Nothing to disclose.

PP0685

HYBRID ENDOSCOPIC STRICTUROTOMY FOR THE TREATMENT OF POST-COLORECTAL ESD SYMPTOMATIC STRICTURES: A PRELIMINARY CASE SERIES

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Introduction: Endoscopic submucosal dissection (ESD) is an effective technique for treating large colorectal lesions but can result in strictures in 0.49-19.7% of cases. We developed a novel technique called Hybrid Endoscopic Stricturotomy (HES), which combines incision with endoscopic balloon dilation (EBD) to treat post-ESD colorectal strictures. This study aims to evaluate the efficacy and safety of HES.

Aims & Methods: This retrospective study analysed data from a prospectively maintained registry at San Raffaele Hospital, Milan, from November 2019 to October 2024.

HES was performed on patients who developed symptomatic post colorectal ESD strictures defined as a lumen calibre that does not allow the passage of the endoscope (9.2 mm diameter) after conventional EBD treatment.

The HES technique involved incision followed by EBD. An L-shaped knife was used to radially incise the stenotic tissue in four quadrants. After the incisions, a wire-guided balloon dilator was introduced and gradually inflated to a diameter larger than the stenosis.

Follow-up endoscopies were scheduled every 2-4 weeks, and additional HES sessions were performed in in cases where lumen dilatation did not improve.

The primary outcome was clinical success, defined as lumen calibre ≥12 mm and symptom resolution; secondary outcomes included technical success, adverse events (AEs), and stricture recurrence.

Results: Of 265 patients treated with colorectal ESD, 7 patients (median age 71 years, 57.1% male) developed strictures; all in the rectum.

Lesions treated with ESD had a median length of 85mm on the major axis and 70mm on the minor axis. All lesions were classified as IIa+Is according to the Paris classification, except for one lesion arising on an anastomosis classified as IIa. The dentate line was involved in 5/7 cases (71.4%). Resection circumference was 100% in 2/7 patients (28.6%), 91-99% in 1/7 patient (14.3%), 76-90% in 1/7 patient (14.3%), 51-75% in 2/7 patients (28.6%), and 0-50% in 1/7 patient (14.3%). Median initial stricture diameter was 6mm (IQR 2). The median number of conventional EBD sessions before HES was 3 (range 2-5). Median stricture diameter after EBD treatment was 8mm (IQR 2).

Clinical and technical success was achieved in 100% of cases. During HES, the median maximum balloon diameter dilation was 20mm (range 15-20). The median number of HES treatment sessions was 2.3 (range 2-3). No AEs were reported. Over a median follow-up of 225 days (IQR 95 days), no symptomatic recurrences were observed.

Conclusion: HES, which combines incision and EBD, is a novel, safe and effective technique for treating post-colorectal ESD strictures to conventional EBD. Larger studies are needed to validate these preliminary results.

Disclosure: Nothing to disclose.

PP0686

ENDOSCOPIC RETROGRADE APPENDICITIS THERAPY VERSUS ANTIBIOTIC THERAPY FOR ACUTE UNCOMPLICATED APPENDICITIS: A META-ANALYSIS OF PROPENSITY-SCORE MATCHED AND RANDOMIZED STUDIES

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Introduction: Over the past decade, endoscopic retrograde appendicitis therapy (ERAT) has emerged as a novel, minimally invasive treatment option for patients with acute uncomplicated appendicitis (AUA). However, it is not yet known whether ERAT is a feasible alternative to antibiotic therapy (ABT).

Aims & Methods: This study aims to evaluate the efficacy of ERAT versus ABT in patients with AUA. We performed a systematic review across PubMed, Embase, and Cochrane CENTRAL databases, searching for studies comparing ERAT to ABT in this population. Outcomes of interest included therapeutic success, — defined as the absence of reintervention up to 1 year of follow-up — recurrence rates, mean time to disease recurrence, and length of hospital stay (LOS). Risk ratios (RR) were used to evaluate binary outcomes with their corresponding 95% confidence intervals (CI), while the mean difference (MD) was calculated for continuous outcomes. All statistical analyses were performed using R software (version 4.4.1) with a random-effects model.

Results: This meta-analysis included 3 studies — 2 control-matched cohorts and 1 randomized trial — comparing 457 ERAT patients to 784 patients receiving ABT for AUA. Baseline clinical characteristics were equally

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distributed between the groups and the time of follow-up ranged from 3 to 12 months. There was no statistically significant difference between ERAT and ABT in treatment success (RR 1.06; 95% CI 1.00 to 1.12; $p = 0.07$) or recurrence rate (RR 0.36; 95% CI 0.11 to 1.22; $p = 0.10$). The difference in time to recurrence was also not significant (MD 116 days; 95% CI -49 to 282; $p = 0.17$).

However, ERAT was associated with a significantly shorter LOS (MD -1.85 days; 95% CI -3.15 to -0.55; $p = 0.005$).

Conclusion: In patients with acute uncomplicated appendicitis, ERAT is non-inferior to antibiotic therapy, while considerably reducing the length of hospital stay. Further studies are warranted to better assess the efficacy of ERAT compared to ABT, and whether it decreases the risk of appendicitis recurrence.

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Disclosure: Nothing to disclose.

PP0687

CAN'T REACH THE CAECUM! FACTORS CONTRIBUTING TO AN INCOMPLETE COLONOSCOPY

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Introduction: A successful colonoscopy aims to thoroughly visualize the entire colon, including the terminal ileum, to maximize the detection of lesions and effectively screen for colorectal cancer. However, approximately 10% of colonoscopies are incomplete, often due to a combination of technical challenges and patient-related factors.

The goal of our study is to identify and analyze the factors that contribute to incomplete colonoscopy procedures.

Aims & Methods: We conducted a retrospective descriptive and analytical study including all patients who underwent a colonoscopy between January 2019 and August 2024. Patients known to have chronic inflammatory bowel disease (IBD) were excluded from our study. An incomplete colonoscopy was defined as non-visualization of the caecum.

Results: Out of 1,518 colonoscopies performed, 105 (11.5%) were deemed incomplete. Factors associated with incomplete colonoscopy included older age (p=0.003), male sex (p=0.026), history of abdominal surgery (p<0.001), constipation (p=0.001), diarrhea (p=0.002), rectal bleeding (p=0.006), and poor bowel preparation (p=0.001).

Multivariate analysis revealed that factors linked to incomplete colonoscopy included: a history of abdominal surgery (OR: 13.8, 95% CI: 4.2–45.1, p=0.001), poor preparation (OR: 224, 95% CI: 57–884, p=0.001), presence of diarrhea (OR: 0.2, 95% CI: 0.08–0.5, p=0.002), and constipation (OR: 0.33, 95% CI: 0.1–0.7, p=0.007).

Conclusion: In this study, 11.5% of colonoscopies were incomplete. The factors most strongly associated with incomplete procedures were a poor preparation, a history of abdominal surgery, and the presence of diarrhea or constipation.

Disclosure: Nothing to disclose.

PP0688

CHANGES IN BOWEL MOVEMENT FREQUENCY, GUT MICROBIOTA, AND EMOTIONAL ISSUES IN POSTOPERATIVE CONSTIPATION PATIENTS

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Introduction: In recent years, the prevalence of chronic constipation among adults worldwide has shown a yearly increasing trend, with most patients experiencing varying degrees of emotional disorders. Understanding the underlying mechanisms linking chronic constipation, gut microbiota, and emotional status is crucial for developing effective therapeutic strategies.

Aims & Methods: This study aimed to explore the relationship between gut microbiota and emotional status by collecting preoperative and postoperative clinical data, psychological conditions, and gut microbiota profiles from 30 patients undergoing total colectomy with ileorectal anastomosis for chronic constipation. Clinical and microbiota data were analyzed to evaluate changes in emotional status and gut microbiota composition before and after surgery.

Results: The results demonstrated that anxiety and depression levels of patients significantly improved following surgery. Concurrently, gut microbiota diversity noticeably decreased, accompanied by significant alterations in species composition. Postoperative changes revealed that Clostridium-sensu-stricto and Streptococcus were key species contributing to the differences observed between preoperative and postoperative microbiota profiles.

Notably, Streptococcus was negatively correlated with emotional symptoms such as obsession, anxiety, and depression, and positively correlated with the lactose and galactose degradation pathway.

Conclusion: This study suggests that Streptococcus may play a pivotal role in improving emotional disorders in chronic constipation patients by modulating the lactose and galactose degradation I pathway.

These findings provide novel insights into the interplay between gut microbiota and emotional well-being, highlighting the potential of targeting gut microbiota as a therapeutic approach for managing psychological conditions associated with chronic constipation.

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Disclosure: None.

PP0689

INSIGHTS INTO THE VARIABILITY AND TEMPORAL STABILITY OF FECAL MICROBIOTA

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Introduction: Understanding the intra- and inter-individual variability of gut microbiota is crucial for advancing microbiota research and its clinical applications. Gut microbial markers play essential roles in maintaining intestinal health, with functional groups including opportunistic, proteolytic, equilibrium, immunoprotective, mucosal homeostasis, proinflammatory, and methanogenic microbes.
Aims & Methods: This study aimed to evaluate the temporal stability of 15 microbial markers and identify potential transient disruptions associated with lifestyle and dietary factors.
A longitudinal observational study was conducted with 6 participants, including 4 healthy individuals and 2 with mild IBS, each providing 7 fecal samples (42 samples total) over a one-month period on randomly selected days. Microbial DNA was extracted from fecal samples and analyzed using a qPCR kit of a specific microbiota test, which includes a panel of 15 microbial markers.
Participants completed questionnaires on each sampling day, documenting dietary habits, lifestyle factors, and digestive health. To assess the stability of the microbial markers in relation to differences among individuals, the Intraclass Correlation Coefficient (ICC) was calculated after adjusting a mixed effects model for each marker. These metrics provide numerical insights into the consistency and variability of the microbial populations over time.
However, it is essential to contrast these numbers with the nature or behaviour of each marker, allowing for a more accurate and contextualized interpretation of the results, as the observed variability and stability may be influenced by intrinsic factors to each type of microbial marker.
Results: Distinct individual-specific microbiota patterns were observed, with clear inter-individual variability in baseline distributions. Transient alterations were observed in some cases, with many of them associated with dietary factors, such as specific food types or eating habits reported prior to sample collection.
Notably, indicative markers of intestinal health, such as *Faecalibacterium prausnitzii*, *Roseburia*, and *Ruminococcus*, remained stable. In contrast, modifications in *Akkermansia muciniphila* and *Methanobrevibacter smithii* levels were observed coincident with variations in dietary intake suggesting that these could be an influent factor. Particularly, these disruptions were generally short-lived, with markers returning to baseline levels within a few days. The ICC has demonstrated its effectiveness in refining results and quantifying the relationship among intra- and inter-personal variability of markers. For example, *Faecalibacterium prausnitzii* exhibited a low ICC (0.112), indicating minimal differences among participants, whereas *Methanobrevibacter smithii* had a high ICC (0.921), reflecting significant variability among participants.
Conclusion: This study provides preliminary insights into the stability and individual variability of gut microbiota markers analyzed using the qPCR kit of this microbiota test. These findings highlight the importance of establishing personal baselines for accurate diagnostic interpretation and underscore the resilience of the gut microbiota in response to daily lifestyle fluctuations. Moreover, the stability of intestinal health indicators markers supports their potential as reliable indicators of gut health.
These results provide valuable data for a deeper understanding of gut health dynamics and underscore the importance of personalized approaches in microbiota diagnostics and management.
Disclosure: This work has been previously presented at an international meeting.
Authors: Lia Oliver, Marta Malagón, Mariona Serra-Pagès, and Daniel Rius are employees of GoodGut SLU. Xavier Aldeguer has also collaborated with GoodGut as a medical advisor.
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PP0690

EVOLUTION OF THE FAECAL LIPIDOME IN PRETERM NEONATES AND ITS ASSOCIATION WITH ECZEMA OR WHEEZE DEVELOPMENT - A LONGITUDINAL TWIN STUDY

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Introduction: The development of the human gut microbiome is a dynamic process whereby rapid maturation occurs in the first year of life. Nutrition, antibiotic use and mode of delivery are important factors in its development, which impacts on the evolution of the faecal metabolome. This study examines the faecal lipidome of preterm twins and triplets through early life to determine genomic and environmental drivers that in turn influence the risk of eczema and wheeze.
Aims & Methods: Neonatal Microbiota (NeoM) is a single-centre longitudinal cohort of preterm infants (<32 weeks gestation) recruited between 2010-11 at Imperial College NHS Trust. Faecal samples were collected. NeoM2 is a follow-up at 2-4 years of age, where a further faecal sample and a parent-completed health questionnaire were collected.
For this study, infants were included if they were from a multiple birth with faecal samples from NeoM and NeoM2. Infants were excluded if they were the only sibling meeting criteria as we wanted to compare siblings. Faecal samples from postnatal days 1-35 and ~3 years were profiled using reversed-phase ultra-performance liquid chromatography mass spectrometry (Acquity UPLC, Q-TOF Premier, negative ESI mode).
Allergic outcomes were defined from the NeoM2 questionnaire: doctor-confirmed eczema, and ≥1 wheeze episode. Multivariate analysis using in-house protocols and recursive feature elimination, with univariate analysis accounting for multiple testing was performed.
Results: Of 369 NeoM infants, 24 met our inclusion criteria (11 sibling sets: 9 twins, 2 triplets). From these 24 infants, 236 faecal samples from postnatal days 1-35 were profiled (2-18 samples per infant; mean age at sample collection 16.9 days). 6 infants had suspected NEC and 3 had sepsis. Of the 11 twin/triplet sets included, 4 sets demonstrated intra-sibling lipidomic variation when comparing all NeoM samples available for each sibling, defined as ≥1 metabolite that statistically differed ($p<0.05$; unpaired t-test for twins, one-way ANOVA for triplets). The number of metabolites that statistically differed was 1, 2, 132 and 1 for the 4 sets, out of 1246 total metabolites.

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There was no clear environmental driver to explain this, although the twins with 132 differing metabolites had the largest disparity in maternal milk consumption. Antibiotic use, nutrition and birthweight were separately investigated as environmental drivers that could influence lipidomic expression. All 236 samples from NeoM were classified into discrete classes based on metadata for each driver. There were 553, 763 and 626 metabolites statistically differing for each driver respectively ($p<0.05$ using one-way ANOVA). Regarding allergic outcomes, 8 infants were eczema positive and 12 wheeze positive. Of the 4 twin sets with lipidomic variation, there was one twin per set positive for eczema and/or wheeze, and one twin per set negative for both. Using recursive feature elimination, lipidomic profiles for the NeoM2 data identified 114 metabolites that differed in those eczema positive compared to negative, and 84 for wheeze. Tentative structural assignment identified two fatty acids (FA 15:0, FA 17:0) of interest that statistically differentiated between twins, antibiotic use, nutrition, and with eczema status.

Conclusion: Preterm twins/triplets maintain similar lipidomic profiles during early life. Antibiotic use, nutrition and birthweight were all drivers of variation, which is of significance as microbial co-metabolic fatty acids (FA 15:0, FA 17:0) are correlated with eczema development. Further validation is required to substantiate these findings.

Disclosure: Nothing to disclose.

PP0691

RESTORING DYSBIOSIS IN ULCERATIVE COLITIS: FUNCTIONAL ASSESSMENT OF TWO BIOACTIVE NUTRACEUTICALS IN A HUMAN GUT MICROBIOTA MODEL

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Introduction: The modulation of gut microbiota through bioactive nutraceuticals offers promising avenues for managing gastrointestinal diseases such as ulcerative colitis (UC). In this study, we investigated the effects of two nutraceutical formulations, Biointestil® and Clovedin®, both rich in natural bioactive compounds, on the gut microbiota of healthy individuals and UC patients.

Aims & Methods: A standardized in vitro simulation of human gastrointestinal digestion and colonic fermentation was conducted using the MICODE (Multi-Unit Colon Model) platform, which through the replication of intestinal conditions including pH and O_2 milieu and specific gut microbiota enables conduct precise Risk/Benefit Assessment by omics means. Biointestil® or Clovedin®. were subjected to 48-hour colonic fermentation with two different gut microbiota models: that from healthy and that from UC donors in biological duplicates.

Microbial composition was monitored using quantitative PCR (qPCR) targeting key pro-inflammatory and beneficial bacterial taxa. Changes in microbiota structure were evaluated through ecological indicators (i.e. prebiotic index, *Firmicutes/Bacteroidota*, etc), providing insight into gut microbial stability and resilience.

In addition, a comprehensive volatilome analysis was performed using SPME GC-MS to assess the metabolic output of the microbial communities, focusing on short-chain fatty acids (SCFAs), indoles, phenols, and other microbial metabolites.

Results: In addition, a comprehensive volatilome analysis was performed using SPME GC-MS to assess the metabolic output of the microbial communities, focusing on short-chain fatty acids (SCFAs), indoles, phenols, and other microbial metabolites. Treatment with Biointestil® and Clovedin® resulted in a reduction of some opportunistic and pro-inflammatory bacteria (*Clostridiaceae, Collinsella aerofaciens, Desulfovibrio spp.*) and a concurrent increase in some beneficial taxa (*Bifidobacteriaceae, Ruminococcaceae, Faecalibacterium prausnitzii*). These shifts were accompanied by an enhanced SCFA profile and a decreased production of toxic volatiles (p-cresol and skatole), particularly in the UC-derived microbiota.

Conclusion: The combined use of qPCR, volatilomics, and ecological modeling within the MICODE in vitro system provides a robust framework for assessing microbiota-targeted interventions and supports the therapeutic potential of Biointestil® and Clovedin® in inflammatory bowel disorders by remission of gut microbiota dysbiosis.

Disclosure: Nothing to disclose.

PP0692

CLINICAL OUTCOMES OF CANCER PATIENTS WITH GASTROINTESTINAL IMMUNE-RELATED ADVERSE EVENTS DURING THERAPY WITH IMMUNE CHECKPOINT INHIBITORS: EXPERIENCE OF A TERTIARY CARE OUTPATIENT GASTROENTEROLOGY CLINIC

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Introduction: Immune checkpoint inhibitors (ICI) are widely used in the management of numerous solid tumors and act by enhancing T-cell–mediated anti-tumour responses. While this activation boosts the immune response against cancer cells, it may also result in immune-related side effects (irAEs). Gastrointestinal symptoms, mainly colitis and diarrhea, are among the most common irAEs.[1] We have recently (February 2025) opened a dedicated clinic, within our Gastroenterology outpatient service, to manage patients who experience gastrointestinal irAEs during ICI therapy for epithelial cancers.

Aims & Methods: Our aim was to evaluate characteristics and outcomes of patients referred to our dedicated clinic for the management of gastrointestinal irAEs associated with ICI therapy since its start (February 2025). All patients were clinically evaluated and were treated with standard-of-care therapy in accordance to European Guidelines[2]. Patients were followed up for at least 8 weeks following initiation of therapy. We collected data on patient demographics (gender and age), patient's overall health status as assessed by the Charlson Comorbidity Index (CCI)[3], cancer type, oncological therapy, diarrhea severity (graded via CTCAE)[4], time to onset of diarrhea following the initiation of oncological therapy, interruption of ICI, outcomes of colonscopy (if performed), treatment modality for ICI-related gastrointestinal toxicity, time to resolution of diarrhea (in weeks), and whether ICI therapy was subsequently resumed.

Results: We included 14 patients affected by ICI-related diarrhea or colitis (grade 1: n=2 , grade 2: n=9 , grade 3: n=2; grade 4: n=1). Four patients were affected by endometrial cancer, 3 patients melanoma, 3 lung cancer,

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2 breast cancer, 1 ovarian cancer, 1 cervical cancer. Thirteen patients were treated with PD1/PDL1- inhibitors (Pembrolizumab: n=9 , Dostarlimab: n=2, Nivolumab: n=2) and one patient was treated with CTLA-4 (Ipilimumab: n=1). Ten patients interrupted immunotherapy due to ICI-related diarrhea after 17 weeks of therapy on average. As indicated by the European Guidelines[2], all patients with diarrhea grade G2 or higher (6 patients) underwent colonoscopy, which revealed signs consistent with ICI-induced colitis. After standard-of-care treatment,12 patients experienced diarrhea grade 1 or lower. Mean time to resolution of diarrhea was 4.2 weeks. Among the 10 patients who had to interrupt ICI due to gastrointestinal adverse events, 5 patients (50%) resumed ICI after a mean of 4 weeks. No patients experienced any adverse events related to the standard-of-care treatment as defined by CTCAE.[4]

Conclusion: Despite being preliminary, our experience shows that patients managed in a dedicated clinic for gastrointestinal irAEs associated with ICI therapy experience acceptable outcomes. Future studies, aimed at comparing whether this approach is more effective than a non-dedicated management, are warranted.

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PP0693

EVIDENCE-BASED PROBIOTIC AGAINST PROTON PUMP INHIBITOR ASSOCIATED DYSBIOSIS: IN-VITRO SCREENING AND CLINICAL PROOF-OF-CONCEPT STUDY

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Introduction: The use of proton pump inhibitors (PPI) has been identified as one of the strongest influences on the gut microbiome in the general population. Long term use can lead to an increase of oral bacteria, such as Streptococcus salivarius and Veillonella parvula, in the gut microbiome which has been associated with gut inflammation and worse outcomes in patients with liver cirrhosis. Beside stringent prescription management, a targeted strategy to reduce PPI-induced alterations in the gut microbiome is crucial to reduce associated side-effects but remains an unmet need so far.

Aims & Methods: Here we aimed to design an evidence-based probiotic intervention to reduce PPI-induced dysbiosis of the gut microbiome. We tested 43 probiotic strains on their ability to inhibit the growth of S. salivarius and V. parvula as well as modulate the response of neutrophil granulocytes to the oral bacteria and selected 7 strains for a multispecies probiotic product. The performance of this evidence-based formulation was then tested with and without additional prebiotic in a bioreactor-based in-vitro model of the oralization of the gut microbiome (MyBioScope, CBmed) against two control substances (vehicle and starch) and prebiotics alone for optimization, before it was introduced in a clinical proof-of-concept, single arm study with a total of 61 included and 53 analyzed participants with long-term PPI use as a 3 months intervention.

Results: Based on their inhibitory capacity against S. salivarius, V. parvula and their immune modulatory effect on isolated human neutrophils, the probiotic strains Bifidobacterium animalis W53, Lactobacillus acidophilus W55, Lactobacillus casei W56, Lactobacillus plantarum W1, Lactobacillus plantarum W21, Lactobacillus rhamnosus W71 and Pediococcus acidilactici W143 were selected for their use in the probiotic formulation. When tested in the MyBioScope model for oralization, the probiotic formulation (with and without prebiotic) was able to significantly reduce the abundance of V. parvula and showed a tendency to reduce S. salivarius, which did not reach statistical significance. Prebiotics alone or control (vehicle or starch) did not show an effect on the abundance of the oral bacteria. In the clinical study, probiotic intervention significantly improved the gastrointestinal quality of life index of the participants and reproduced the results of the in-vitro tests by showing a significant reduction of V. parvula and a less effective reduction of S. salivarius in the gut microbiome. Adherence to the study product and the safety profile were excellent.

Conclusion: In conclusion, our screening procedure for an evidence-based probiotic formulation against PPI-induced dysbiosis resulted in a probiotic product containing 7 potent strains and a prebiotic matrix. This formulation was able to reduce the abundance of V. parvula as well as to a lesser degree S. salivarius in a novel bioreactor-based in-vitro model of the gut microbiome. The results of the in-vitro model could be reproduced in a proof-of-concept, single-arm clinical study where the probiotic was able to reduce the abundance of V. parvula in the gut microbiome of long-term PPI-users. This study shows that targeted probiotic interventions can be used to relieve PPI-induced side-effects, including alterations in the microbiome composition.

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Disclosure: This study has been conducted by the Center for Biomarker Research in Medicine (CBmed), a COMET K1 centre, operated as a public-private-partnership funded by the Austrian Research Promotion Agency as well as industry partners including the probiotic companies Institut Allergosan (Graz, Austrtia) and Winclove Probiotics (Amsterdam, The Netherlands).

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PP0694

IMPACT OF SHORT-TERM REDUCTION OF PLASTIC EXPOSURE ON MICROPLASTIC PRESENCE AND GUT MICROBIOME COMPOSITION IN HEALTHY INDIVIDUALS: A PILOT STUDY

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Introduction: Microplastic particles (MPs) have previously been confirmed to be found in the human body and are suspected to lead to systemic health effects (1). One biological matrix in which MPs have been detected is human faeces (2). While there is growing evidence suggesting potential impacts on the gut microbiome from animal (3) and in-vitro/ex-vivo studies (4), studies investigating this relationship in the human are still lacking.

Aims & Methods: This pilot study aimed to investigate the impact of short-term plastic reduction on the gastrointestinal tract of healthy participants, focussing on MP presence and gut microbial composition.

Sixty healthy volunteers participated by providing two stool samples: one prior to and one after a seven-day intervention during which they actively minimized plastic exposure. To facilitate the reduction, participants were provided with plastic-free alternatives (e.g., glass water bottles, wooden kitchen utensils, plastic-free dental care products, tin lunch boxes), together with instructions on how to avoid plastic-packaged goods and find alternatives. Dietary habits and plastic contact were recorded using detailed questionnaires to assess behavioural changes and identify potential sources of exposure. Stool samples were analysed using 16S rRNA gene sequencing for changes in microbial diversity and potential differential taxa. Additionally the samples were frozen at –80°C for future MP quantification, where method development is ongoing.

Results: Fifty-five participants completed the study, with an average self-reported success rate of 76% in adhering to the intervention. Forty percent of participants fully avoided plastic packaging, while 56% achieved partial avoidance. Participants noted challenges, particularly in avoiding plastic in meat and dairy products and during restaurant visits. Despite these obstacles, 92.7% expressed a lasting commitment to reducing plastic use post-intervention, with many reporting intentions to continue purchasing fresh products, reduce fast food intake, and adopt sustainable household practices, especially replacing plastic utensils. Initial microbiome analysis revealed no significant changes in alpha or beta diversity following reduced plastic exposure; however, differential abundance analysis identified potential specific taxa shifts. One species each within the genera *Clostridium*, *Ruminococcus*, and *Blautia* showed reduced abundance post-intervention, while species from the genera *Bacteroides* and *Anaerostipes* were associated with increased abundance after plastic fasting.

Conclusion: A short-term reduction in plastic exposure is feasible and can effectively raise awareness and promote long-term behavioural change in plastic consumption habits. However, no significant changes in gut microbial diversity were observed within the study period, with only shifts in a limited number of species visible. Due to the stability of the gut microbial composition, longer changes in plastic exposure might be necessary to trigger stronger effects.

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Disclosure: Nothing to disclose.

PP0695

COGNITIVE ENHANCEMENT AND TIME-DEPENDENT REMODELING OF THE GUT MICROBIOME AND METABOLOME AFTER FECAL MICROBIOTA TRANSPLANTATION: A PILOT STUDY

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Introduction: Fecal microbiota transplantation (FMT) is established for recurrent *Clostridioides difficile* infection and is being explored for metabolic and neuropsychiatric disorders. Evidence suggests that FMT-mediated modulation of the gut–brain axis can benefit cognition, but combined microbiome–metabolome correlates remain poorly defined.

Aims & Methods: In a prospective observational study, the patients with mild-to-moderate cognitive impairment completed follow-up after colonoscopy-guided FMT (60 g donor stool/400 mL saline). Cognition was assessed with the Mini-Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA) at baseline, 1, 4 and 12 weeks. Fecal samples collected at 0, 1 and 4 weeks underwent untargeted LC-MS; 28 microbe-derived metabolites were retained after MiMeDB matching, log transformation and Pareto scaling for PLS-DA and heat-map analyses—parallel 16S-rRNA sequencing profiled bacterial diversity and relative abundance.

Results: *Clinical*: Three patients who were able to undergo cognitive testing completed the 30-day follow-up. Median MMSE improved from 1 (IQR 1–12) at baseline to 8 (4–13) at 7 days and 10 (9–18) at 4 weeks; two participants sustained gains at 12 weeks. MoCA rose from 0 (0–6) to 4 (2–7) at 7 days with mixed longer-term trajectories.

Metabolome: Fecal metabolites were collected over four weeks from 11 patients with cognitive dysfunction who underwent FMT. PLS-DA scores migrated progressively up-right across the first 4 weeks, indicating time-dependent restructuring. Short–chain fatty–acid–related metabolites trended upward while selected indole derivatives declined; however, only 3 of 28 features exceeded a 2-fold, $p < 0.05$ threshold owing to limited power.

Microbiome: Alpha-diversity increased numerically after FMT (median observed species from 6.0 to 12.3) but did not reach statistical significance ($P = 0.121$). Relative abundance shifted toward beneficial taxa—*Akkermansia muciniphila*, *Bacteroides ovatus,* and *Bifidobacterium longum* groups

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expanded—while potential pathobionts such as the *Enterococcus faecium* group contracted; compositional changes paralleled early cognitive improvement.

Conclusion: FMT was associated with rapid and sustained cognitive gains accompanied by concordant remodeling of both the fecal metabolome and bacterial community toward neuro-supportive profiles.

Although statistical confidence is limited by sample size, these integrated data strengthen the gut–brain axis hypothesis and justify larger controlled trials to pinpoint key neuroactive microbes and metabolites driving cognitive benefit.

Disclosure: Nothing to disclose.

PP0696

IN VITRO VS. IN VIVO STUDIES ON THE INTERACTION OF A MEDICINAL PRODUCTS, STW 5, WITH THE GUT MICROBIOTA

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Introduction: Recent evidence suggests that natural products can modulate microbiome composition, offering health benefits. This study evaluates the effects of STW 5 on microbiota in functional dyspepsia (FD) and Irritable Bowel Syndrome (IBS) through a comparison of *in vitro* and *in vivo* data.

Aims & Methods: A comprehensive literature review was conducted using PubMed and other scientific databases to collate data on STW 5's impact on the gut microbiome at phyla and genera levels.

Results: *In vitro* studies demonstrated an increase in *Enterococcus* and a decrease in *Clostridium* and *Bacteroides* [1]. Further *in vitro* research confirmed an increase in Firmicutes and a reduction in Bacteroidetes, with a decrease of *Blautia* and Proteobacteria. Unpublished data of the authors indicated increased *Bifidobacterium* and *Blautia*, with no change in *Lactobacillus* [2].

In vivo studies in healthy Wistar rats reported minor increases in the phyla Firmicutes and Bacteroidetes, and a significant decrease in Actinobacteria. Genus-level changes included increases in *Bacteroides*, *Bifidobacterium*, and *Prevotella*, and a significant rise in *Methanobrevibacter*. In a model of ulcerative colitis, *Lactobacillus*, *Blautia*, and *Bifidobacterium* increased, while Proteobacteria and *Enterococcus* decreased.[3]

In an *in vivo* model of functional dyspspsia, significant increases were observed for *Enterococcus*, *Lactobacillus*, *Blautia*, and Bacteroidetes. Actinobacteria decreased, with elevated *Bifidobacterium* compared to FD control, but lower than in healthy controls [4, 5]

Conclusion: The different responses of some genera like *Blautia* and *Enterococcus* to STW 5 in *in vitro* models as compared to *in vivo* models highlight the complexity of the interactions of the microbiota with STW 5.

The limitations of *in vitro* models due to the absence of host factors, and the challenges of *in vivo* models in extrapolating data to humans underscore the advantages of using diverse research methodologies to identify microbial changes relevant to human health.

Our results demonstrate such an approach, revealing consistent microbial shifts across the models. In particular, the upregulation of *Bifidobacterium*, with the downregulation of Actinobacteria and Proteobacteria, highlight the potential of STW 5 for beneficial gut microbiome modulation, underscoring its multi-target mechanism of action [6].

In conclusion, the presence of different results in different models underscores the need for a multifaceted approach to accurately characterize the intricate interactions between natural products and the gut microbiome. The identification of consistent microbial shifts associated with STW 5 provides mechanistic insight contributing to its clinically-proven health benefits. Further research, will be valuable to confirm the translational relevance of these findings.

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Disclosure: JW and OK are empoloyees, LK was intern at Steigerwald Arzneimittelwerk GmbH, Darmstadt, Germany. MT has been obtaining speakers fees and travel support by Steigerwald Arzneimittelwerk GmbH, Darmstadt, Germany.

PP0697

IMPACT OF H. PYLORI ERADICATION THERAPY WITH BUTYRIC ACID AND INULIN SUPPLEMENTATION ON GUT MICROBIOTA SHORT-CHAIN FATTY ACID LEVELS

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Introduction: H. pylori eradication therapy induces significant alterations in the gut microbiome. While the composition of gut microbiota has been extensively studied, the levels of gut microbiota short-chain fatty acids (SCFAs), particularly in case of butyric acid and inulin supplementation, have not been investigated.

Aims & Methods: The aim of our study was to evaluate the levels of SCFAs of the gut microbiota during H. pylori eradication therapy adjuvated with butyric acid and inulin (Zacofalk) supplement.

A total of 15 H. pylori-positive patients aged 18 to 65 years were enrolled in the study. The following H. pylori eradication regimen was used: esomeprazole 20 mg, clarithromycin 500 mg, amoxicillin 1000 mg, and bismuthate tripotassium dicitrate 240 mg, twice daily, along with butyric acid and inulin (Zacofalk) supplementation (two tablets daily, each containing 250 mg of butyric acid and 250 mg of inulin), orally, for 14 days. Stool samples were collected from each subject at three time points: prior to eradication therapy (I), after therapy completion (II), and one month post-therapy (III). The levels of SCFAs were analyzed using gas chromatography–mass spectrometry. Differences were assessed using the Wilcoxon signed-rank test (p-value < 0.05 was considered statistically significant).

Results: Immediately after eradication therapy, the relative level of butyric acid (C4) and the absolute levels of butyric acid (C4), isovaleric acid (iC5), and valeric acid (C5) significantly decreased compared to baseline:

• Butyric acid (C4):(0.139±0.068) relative units vs. (0.193±0.044) relative units, p=0.038;

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• Butyric acid (C4):(0.23±0.15) mg/g vs. (0.59±0.29) mg/g, p=0.0035;
• Isovaleric acid (iC5):(0.11±0.06) mg/g vs. (0.18±0.11) mg/g, p=0.037;
• Valeric acid (C5):(0.05±0.04) mg/g vs. (0.14±0.09) mg/g, p=0.0051.

In one month the levels of most SCFAs restored to baseline levels, with no statistically significant differences identified between time points III and I in terms of the relative level of butyric acid (C4) and the absolute levels of both butyric and isovaleric acids (C4 and iC5):

• Butyric acid (C4): (0.195±0.056) relative units vs. (0.193±0.044) relative units, p=0.414;
• Butyric acid (C4): (0.49±0.32) mg/g vs. (0.59±0.29) mg/g, p=0.0942;
• Isovaleric acid (iC5): (0.17±0.10) mg/g vs. (0.18±0.11) mg/g, p=0.588.

However, even one month after therapy, the absolute level of valeric acid (C5) was still lower than baseline: Valeric acid (C5): (0.09±0.08) mg/g vs. (0.14±0.09) mg/g, p=0.0051.

No significant changes were observed in the absolute levels of acetic acid (C2), propionic acid (C3), isobutyric acid (iC4), caproic acid (C6), and the anaerobic index (acetic acid to butyric acid ratio, units) immediately after and one month after eradication therapy compared to baseline, which may be due to the mitigating effects of butyric acid and inulin (Zacofalk) supplementation.

Conclusion: H. pylori eradication therapy leads to significant changes in the levels of gut SCFAs immediately after eradication therapy, even when adjuvated with butyric acid and inulin (Zacofalk) supplements, which are mostly reversed within a month after therapy.

Disclosure: Nothing to disclose.

PP0698

PATIENT EXPERIENCES OF MAGNETIC RESONANCE IMAGING DEFECATING PROCTOGRAMS

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Introduction: Anorectal dysfunction affects approximately 8% of the global population.[1] Whilst conservative management can manage mild symptoms, surgery is often indicated in patients with structural abnormalities and severe symptoms. A Magnetic Resonance Imaging Defecating Proctogram (MRDP) forms a key part of patients' pre-surgical investigations, visualising the anterior, middle and posterior compartments of the pelvis, and provides a dynamic image of the pelvic floor during defecation. It has been criticised as it is often performed in the supine position, which does not reflect normal physiological positioning for defecation.[2]

However, little is known about how patients themselves feel about how this test reflects their usual defecatory process.

Aims & Methods: As part of a service evaluation, between September 2024 and April 2025, all patients who underwent an MRDP at Imperial College Healthcare NHS Trust in London, United Kingdom, were asked to complete a 21-item questionnaire exploring their experiences, prior to, during and after their MRDP. Five-point Likert Scales (1 - Strongly Disagree, 5 - Strongly Agree) were used for each question, except two concluding questions on overall experience, which were answered using free-text boxes. Patients with insufficient cognitive function or comprehension of English to complete the questionnaires were excluded.

Results: Thirty-eight patients responded to the questionnaire, and after excluding those with missing data, 35 completed questionnaires were included in the analyses (30[86%] female, 28[80%] 50-79 age category). Reasons for referral to MRDP included FI (15/35 [43%]) and/or constipation (13/35 [37%]) and/or POP (9/35 [26%]).

Although 32(91%) strongly agreed that their privacy was respected, and none strongly agreed they felt embarrassed, just 12(34%) strongly agreed that they felt 'comfortable during the test'. Less than a third of patients (10[29%]) agreed that they were able to 'defecate as they do at home' and only a quarter (9[26%]) felt that defecating lying supine was easy.

Whilst 16(46%) patients felt that the contrast mixture inserted gave them a 'normal urge to open their bowels', only 4(12%) reported that the contrast mixture was 'similar to their own stool'.

Nevertheless, 23(66%) patients agreed that the experience matched what they would expect, and most patients (29[83%]) would have the test again if it were necessary. Patients aged 70-79 were associated with disagreeing with the statement that they could 'defecate as they would at home' (neutral/disagree/strongly disagree vs. agree/strong agree: 8/8[100%] vs 0.0[0%], p = 0.011).

Conclusion: Whilst patients reported being able to tolerate the MRDP, the large majority reported that defecating during the investigation did not reflect their usual defecatory process.

Contrary to expectation, it appears that patients' main complaints were not due to embarrassment, but rather patient position and environment. Whilst patient experiences are highly institution and staff dependent, understanding patients' perspectives during an MRDP is useful for optimisation of pre-investigation counselling and expectation setting. Implementing post-MRDP questionnaires, or pro-active clinician exploration of the extent to which patients felt able to replicate their symptoms could facilitate clinical interpretation of MRDP results.

Moving towards evacuation testing that aligns more closely with a patient's normal defecatory habits is likely to yield results that better reflect symptom profiles and therefore provide better insights into specific surgical targets responsible for symptom generation.

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Disclosure: Nothing to disclose.

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PP0699

EVALUATING THE UNDERLYING AETIOLOGY IN PATIENTS WITH REPORTED COLITIS ON COMPUTED TOMOGRAPHY IMAGING

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Introduction: Computed tomography (CT) is a common modality used to assess patients who present acutely with gastrointestinal symptoms.[1] The term colitis refers to the thickening of the colonic wall [2] and is often described in CT reports. Causes of colitis are commonly classified into infectious, inflammatory and ischaemic colitis.[3]

Although there are some radiological features that are helpful in suggesting a specific diagnosis,[4] determining the underlying aetiology of colitis often requires correlation with clinical history, laboratory results including stool cultures and endoscopic evaluation.[1]

There is currently limited literature on the prevalence of the different aetiologies in patients with colitis reported on CT.[3]

Identification of the underlying aetiology allows for specific management of patients as treatment options differ depending on the underlying cause.

Aims & Methods: The aim of this study is to identify the aetiology of colitis in patients with reported colitis on CT imaging.

A retrospective review of CT reports and patient case notes was performed. A text search through our radiology system (CRIS) was used to identify patients with CT reported colitis during their inpatient admission. Patients with known Inflammatory bowel disease (IBD), colorectal cancer as well as outpatient CT reports were excluded.

Following identification of our patient cohort, we reviewed their clinical notes as well as laboratory, stool and endoscopic results to determine the underlying aetiology of their colitis.

Results: A total of 138 patients with colitis reported on CT imaging were included. 98 patients (71.0%) had a stool culture sent during their admission whilst 83 patients (60.1%) underwent endoscopic evaluation with a flexible sigmoidoscopy (n=49, 59.0%) or colonoscopy (n=34, 41.0%). In 35 patients (42.2%) there were no significant abnormalities found endoscopically.

74 patients (53.6%) were thought to have infectious colitis with main pathogens including *Campylobacter* spp. in 19 patients (25.7%), *Clostridiodes difficile* in 11 patients (14.9%) and *Salmonella* spp. in 1 patient (1.4%). 4 patients (5.4%) had endoscopic features of infectious colitis. 39 patients (52.7%) patients were suspected to have infective colitis from clinical history and laboratory results.

25 patients (18.1%) had ischaemic colitis either confirmed endoscopically (n=11, 44.0%) or through clinical history and CT features (n=14, 56.0%). Endoscopically, 6 patients (4.3%) had features compatible with IBD, 2 patients (1.4%) with lymphocytic colitis and 1 patient (0.7%) with an adenocarcinoma. 15 patients (10.9%) had diverticulosis.

C-reactive protein (CRP) was higher in patients with infectious colitis (mean of 141.4 mg/L) and lower in patient with ischaemic colitis (mean of 108.6 mg/L). Albumin and haemoglobin levels were highest in those with infectious colitis (mean of 33.4 g/L and 135.5g/L respectively) and lowest in patients with IBD (mean of 26.8 g/L and 108.5g/L respectively).

Conclusion: CT reported colitis was of infectious aetiology in 53.6% of patients, ischaemic in 18.1% and inflammatory in 4.3%. Laboratory results such as haemoglobin and albumin levels can be helpful in determining aetiology.

Despite CT features suggestive of colitis, 42.2% of endoscopic evaluation revealed no significant abnormality. CT findings of colitis should therefore be correlated with clinical history, stool results and endoscopic evaluation when determining the underlying aetiology.

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Disclosure: Nothing to disclose.

PP0700

FEASIBILITY OF DIRECT-VISION STENTING: A NOVEL STENT PLACEMENT TECHNIQUE FOR MALIGNANT COLORECTAL OBSTRUCTION USING A NEW THIN THERAPEUTIC SCOPE

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Introduction: Endoscopic self-expandable metallic stent (SEMS) placement for malignant colonic obstruction is commonly performed using the through-the-scope (TTS) technique. However, strategies to reduce radiation exposure remain a clinical concern. We developed a novel technique, Direct-Vision Stenting (DVS), using a new thin therapeutic scope (Fujifilm EG-840TP: outer diameter, 7.9 mm; working channel, 3.2 mm) for malignant rectosigmoid obstructions [1].

After advancing the scope through the stricture, the stent is partially expanded under direct vision, then anchored and gradually released under endoscopic visualization while withdrawing the scope. This technique may minimize radiation exposure by eliminating the need for fluoroscopic confirmation of guidewire placement and contrast injection.

Aims & Methods: This study aimed to evaluate the feasibility of the DVS technique. From January 2024 to March 2025, SEMS placement for malignant rectosigmoid obstructions was initially attempted using the DVS technique. If DVS was unsuccessful, conversion to the conventional TTS technique (without direct visualization of the oral lumen or beginning of the stent expansion) was permitted. Ten consecutive patients were prospectively evaluated for technical success, adverse events, fluoroscopy time, and total procedure time (from scope arrival at the lesion to stent deployment).

Results: The median age was 81 years, and 4 patients were men. The obstruction was located in the rectosigmoid colon in 3 patients and the sigmoid colon in 7. Five procedures were performed for palliative purposes. The technical success rate of the DVS technique was 60%. All unsuccessful cases were resolved by conversion to the conventional TTS technique. Reasons for DVS failure included poor bowel preparation (n = 1), failure of scope advancement (n = 2), and severe colonic angulation (n = 1). No adverse events were observed, and all patients experienced symptomatic relief. Among successful DVS cases, median procedure time was 17 minutes, and fluoroscopy time was 4 minutes. In comparison, in 17 cases treated with the conventional TTS technique prior to the study period, the median procedure and fluoroscopy times were 15 and 11 minutes, respectively, suggesting a trend toward reduced fluoroscopy time with the DVS technique.

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Conclusion: The DVS technique using EG-840TP achieved a 60% success rate for malignant colonic obstructions in the rectum or sigmoid colon and allowed safe conversion to the TTS technique when necessary. In appropriate cases, this approach may reduce fluoroscopy time and overall radiation exposure.

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Disclosure: Nothing to disclose.

PP0701

ILEOCAECAL STENTING IS SUCCESSFUL IN COLORECTAL CANCER PALLIATION AND AVOIDS STOMA IN OVER TWO THIRDS OF PATIENTS

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Introduction: There is minimal data to support a recommendation to perform colonic stenting (SEMs) in the right colon, but particularly in malignancy in the region of the ileocaecal valve, where only small bowel dilatation results from obstruction. This scenario poses technical challenges in endoscopic access in unprepared proximal colon, and relating to the anatomy of the ileocaecal valve. One recent case series included 13 patients with reported success rate and equivalent outcomes to left-sided stents with median survival of 4 months, although a two-step procedure was described[1].

Aims & Methods: In our centre we attempt self-expanding metal stent insertion (SEMs) in cases of malignant ileocaecal obstruction, which constitutes ~9% of our overall SEMs caseload. We report our outcomes and, as an important metric for palliative patients, stoma-free survival in the context of colorectal cancer management.

We perform stenting using colonoscopy to access the stricture and fluoroscopy for wire and stent deployment, with both over the wire and through the scope techniques.

Results: 25 patients, median age 77 years, underwent intention to stent for colorectal malignancy at the ileocaecal junction, where stent deployment into the ileum would be necessary to overcome the obstruction. Clinical success with life-time avoidance of surgery or stoma was 68% without any adverse events from attempting SEMs. Median survival was 206 days, compared to 167 days in those where stenting was unsuccessful. This was not significantly different to median stoma-free survival of 272 days for all SEMs patients in our database.

One patient, after technically successful stent deployment, re-obstructed within 7 days and underwent surgery. There were no other early or late complications in either group. Reasons for technical failure in 6 cases was inability to pass a wire into the ileum, bowel preparation in 1 case, and failure to adequately resolve obstruction in the aforementioned patient.

Conclusion: Stenting malignant colonic obstruction of the ileocaecal region is more technically challenging but does not appear to disadvantage patients after an unsuccessful procedure. Stoma or surgery is avoided in 68% cases in intention to treat analysis of 25 patients and median survival of 206 days is equivalent to other colonic stent locations.

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Disclosure: Nothing to disclose.

PP0702

ALL-IN-ONE NOVEL DIAGNOSTIC SYSTEM FOR PELVIC FLOOR DISORDERS – RESULTS FROM FIRST IN HUMAN STUDY

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Introduction: Accurate diagnosis of Pelvic Floor Disorders (PFD) requires multiple tests, including balloon expulsion, anorectal manometry, and defecography. Current diagnostic approaches have significant limitations: they require multiple visits to specialized clinics, are conducted in non-private and often embarrassing environments, and use non-physiologic (lying) positions.

As a result, they (a) provide fragmented, sometimes inaccurate, and difficult-to-integrate information, and (b) are inconvenient and distressing for patients, while time-consuming for providers and payers.

Aims & Methods: To develop a novel, comprehensive, accurate, all-in-one diagnostic system to replace current multi-test approaches.

We developed a disposable balloon and catheter system capable of simultaneously measuring rectal and anal pressures and anorectal angle. These physiological metrics are recorded during balloon expulsion in a physiologic seated position. Data are transmitted via Bluetooth to a nearby computer, enabling the test to be performed in a private toilet room. The system was tested for functionality, convenience, and safety, and compared with balloon expulsion (Mui Scientific) and high-resolution manometry (HRM, ManoScan™, Medtronic) in both lying and seated positions, using standardized protocols (London and IAPWG).

Results: The Ano-Rectal Diagnostic System (ARDiS) reliably measured rectal and anal pressures in both positions, with values comparable to standard HRM. ARDiS also enabled novel assessment of anorectal angle. These parameters were recorded simultaneously in real time during seated balloon expulsion.

Data transmission to a nearby laptop or office computer was seamless. The procedure was well tolerated, with no reported side effects. Patients preferred the private, seated testing environment.

Conclusion: The ARDiS system enables comprehensive, accurate, and simultaneous assessment of anorectal function in one convenient session. Designed for both hospitals and outpatient clinics, ARDiS streamlines the diagnostic process, improves patient compliance, and supports accurate diagnosis and personalized management of PFD. It may fundamentally transform clinical practice by replacing multiple fragmented tests with a single integrated solution.

Disclosure: Glukhovsky & Melamud: Shares in ARDiS Medical Ltd.
Ringel: Scientific Advisory Board, ARDiS Medical Ltd.

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PP0703

ENDOSCOPIC SUBMUCOSAL DISSECTION OF LARGE ANORECTAL AND LOW RECTAL ADENOMAS

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Introduction: Large rectal adenomas often spread to the upper part of the anal canal. In this study were retrospectively analyzed 13 of 40 rectal lesions, operated by endoscopic submucosal dissections in 2018-2025, which were localized in lower rectum. 7 lesions have spread on anal canal and 6 lesions were localized in lower rectum without spreading on anal canal.

The study compared the risk of intraprocedural bleeding, completeness of removal, duration of dissection, pain syndrome, and red blood cell indicators based on the size and location of the lesions.

Aims & Methods: A retrospective analysis was conducted on 40 Endoscopic Submucosal Dissections (ESD) of rectal lesions, all performed by a single operator at a single center. Two cases of NETs treated by ESD, two cases of synchronous colorectal locations resected in a single procedure, three ESD cases of recurrent adenomas following endoscopic or surgical treatment, two cases with deeply invasive carcinomas, and eighteen cases with rectal tumors without involvement of the lower rectum and anal canal were excluded. Seven ESD cases of anorectal lesions ranging from 20 to 170 mm in size, and six lower rectal lesions ranging from 30 to 80 cm were compared. Information about duration, location and size was taken from protocol of the operation and reviewed by photo and video fixation of procedures. Medical documentation was checked for red blood indicators and treatment of pain syndrome.

Additionally, a follow-up phone call for all patients will be conducted two weeks after ESD.

Results: All lesions were removed in a single block. Only one tumor was fragmentated during the extraction. Coagulation forceps was needed in 5 of 7 cases in anorectal ESD group and 4 of 6 cases between lower rectal lesions. Forceps were needed in all cases over 70 mm in size. Two cases of massive bleeding were documented in anorectal group: one intraprocedural bleeding and one delayed bleeding.

All patients in the low rectal group were discharged the day after ESD. In the anorectal group, four patients were discharged on the second, third, fourth, and fifth day following the procedure.

Patients in the anorectal group reported symptoms within two weeks after dissection, including pain, episodes of fever, and increased hair loss (in cases involving significant intraprocedural bleeding).

Conclusion: Endoscopic submucosal dissection in lower rectum and anal canal is an effective modality for treatment of large benign tumors. But such operation, especially in cases on anorectal lesions has high risk of massive intraprocedural and delayed bleeding, longer hospital staying and longer recovery after treatment.

Disclosure: Nothing to disclose.

PP0704

FEASIBILITY AND SAFETY OF ENDOSCOPIC SUBMUCOSAL DISSECTION FOR THE ANAL CANAL LESIONS

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Introduction: Endoscopic submucosal dissection (ESD) for the anal canal lesions (ACLs) is technically challenging because of the specific anatomical features of the anal canal. We aimed to evaluate the efficacy and safety of ESD for ACLs.

Aims & Methods: We retrospectively investigated 102 patients with ACLs who underwent ESD from August 2010 to August 2023. We collected and analyzed clinicopathological characteristics, procedure-related parameters, outcomes, and follow-up data.

Results: Of the 102 patients with ACLs (mean age 63 years, 46 men), ACLs were located above the dentate line (ADL) in 55 patients, and lesions involved the dentate line (IDL) in 47. The mean procedure duration was 37.0 min. The en bloc resection and en bloc with R0 resection rates were 94.1% and 88.2%, respectively. Adverse events (AEs) included two intraoperative bleeding, two delayed bleeding, and one febrile episode. Anal pain occurred in 5 patients. During a median follow-up of 37 months, local recurrence was revealed in 3 patients, and one had metastases.

Compared with the ADL group, the IDL group had significantly longer procedure duration, larger specimens, lower rates of En bloc with R0 resection, higher anal pain rate, and more AEs. When only adenomas/adenocarcinomas were included in the analysis, the two groups had no significant differences regarding recurrence-free, overall, and disease-specific survival.

Conclusion: The favorable short- and long-term outcomes indicate ESD is a safe, effective, and attractive therapeutic modality for ACLs. A multicenter and prospective study should be conducted to validate this technology.

Disclosure: Nothing to disclose.

PP0705

THE EFFECT OF COLON PSEUDO-STERILIZATION ON DELAYED PERFORATION OF THE COLON AFTER THERMAL INJURY

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Introduction: Although delayed perforation is a rare complication of endoscopic resection of colonic lesions, the potentially fatal outcomes make this a significant risk. Using a rat model in which delayed colon perforation occurs 48 hours after thermal injury, we have demonstrated that the mechanism of delayed colonic perforation resembles the Jackson's burn model, which is a well-established model of skin thermal injury.

Based on our further discovery that bacterial infection played a role in delayed colonic perforation, we hypothesized that delayed colonic perforation would not occur in pseudo-germ-free rats.

Aims & Methods: The aim of this study is to evaluate the effect of pseudo-sterilization of the colon on delayed colonic perforation after thermal injury.

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Eight rats were administered antibiotics (metronidazole, ampicillin, vancomycin, and neomycin) from day 1–5 to create a pseudo-germ-free group (n = 8). Eight further rats were administered sterilized water as a control group (n = 8). Fecal samples were collected on days 7–9 for spectrophotometric analysis of enzyme activity (β-glucuronidase, β-galactosidase, β-xylosidase).

Thermal injury was induced in the colon on day 7 in nine of the pseudo-germ-free group and nine of the control group using a hemostatic forceps through an endoscope. The colon was removed on day 9 and histological evaluation performed.

Results: The activity levels of β-glucuronidase, β-galactosidase, and β-xylosidase on day 7 were significantly reduced in the pseudo-germ-free group compared with the control group (0.012 vs. 0.090, p = 0.003; 0.045 vs. 0.168, p = 0.001; and 0.008 vs. 0.047, p = 0.002, respectively).

On day 9, activity levels of β-glucuronidase and β-xylosidase were significantly reduced in the pseudo-germ-free group compared with the control group (0.007 vs. 0.054, p = 0.000 and 0.015 vs. 0.042, p = 0.000, respectively).

These findings clearly demonstrate the successful achievement of pseudo-sterilization. The rate of delayed perforation and number of MPO-positive cells in the serosa were significantly lower in the pseudo-germ-free group than the control group (11.1% vs. 100%, p = 0.000 and 986.985 /mm^2 vs. 2303.782 /mm^2, p = 0.023, respectively).

Infiltration of Escherichia coli into the submucosal layer was only observed in the control group. Summary of results, tissue degeneration extending to the muscularis propria was observed in both groups after thermal injury; however, delayed colonic perforation was observed only in the control group, while it was significantly reduced in the pseudo-germ-free group.

Conclusion: Pseudo-germ-free conditions in the colon suppressed the occurrence of delayed colonic perforation. Bacterial infection may play a significant role in the expansion of the necrotic zone to the muscularis propria.

Disclosure: Nothing to disclose.

PP0706

OUTCOMES OF COLORECTAL ESD FOR ELDERLY PATIENTS IN THE HIROSHIMA REGION: A MULTICENTER STUDY

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Introduction: Colorectal endoscopic submucosal dissection (ESD) is now widely performed and has become a standard procedure even at core hospitals throughout regional areas. As a minimally invasive treatment, it is frequently applied to elderly patients; however, it is crucial to determine whether such treatment can be achieved safely and effectively across an entire region, especially considering the presence of comorbidities or frailty and the risks associated with limited life expectancy.

Aims & Methods: This study aimed to evaluate the safety and efficacy of colorectal ESD for elderly patients in the Hiroshima region. We used a prospective registry of all colorectal ESD cases performed between August 2013 and December 2018 at 11 core hospitals in the Hiroshima area. Patients were categorized into an elderly group (≥75 years) and a non-elderly group (<75 years). Short- and long-term outcomes were compared between the two groups. Patients with advanced colorectal cancer, inflammatory bowel disease, familial adenomatous polyposis, or missing data were excluded.

Results: A total of 2,226 patients with 2,332 lesions were included: 684 patients with 716 lesions in the elderly group and 1,542 patients with 1,616 lesions in the non-elderly group. Baseline characteristics (elderly/non-elderly) were as follows: proportion of male patients was 56%/61% (*P*=0.024), median age was 79/66 years (P<0.001), and prevalence of comorbidities was 54.4%/35.7% (*P*<0.001), including cardiovascular disease (17%/7%), cerebrovascular disease (10%/5%), diabetes (17%/13%), pulmonary disease (9%/4%), hepatic disease (5%/3%), and other organ cancers (18%/12%).

Antithrombotic drug use was 26%/11% (*P*<0.001). Regarding lesion characteristics, lesion location (right colon/left colon/rectum) was 52%/21%/28% vs. 53%/21%/27% (*P*=0.858), and median lesion size was 30 mm vs. 25 mm (*P*=0.066).

Procedural outcomes were as follows: en bloc/piecemeal/interrupted resection rates were 96%/3%/1% vs. 96%/3%/1% (P=0.998), median procedure time was 65 min vs. 60 min (P=0.025), intraoperative perforation occurred in 3%/3% (P=0.594), delayed bleeding in 2%/2% (P=0.816), delayed perforation in 0.3%/0.4% (*P*=0.903), and emergency surgery in 0.6%/0.4% (*P*=0.523). Histopathological diagnoses were adenoma/pTis/pT1a/pT1b in 36%/46%/7%/11% vs. 38%/44%/7%/11% (*P*=0.830), with poorly differen-

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tiated histology of 0.1%/0.2% (*P*=0.652), lymphovascular invasion 6%/7% (*P*=0.543), tumor budding 3%/3% (*P*=0.765), positive vertical margin 8%/7% (*P*=0.161), and curative resection 81%/83% (*P*=0.484).

Among non-curatively resected lesions, additional surgical resection was performed in 42% (56/134) vs. 64% (181/283) (*P*<0.001), and lymph node metastasis was detected in 10% (6/56) vs. 12% (22/181). During a median follow-up of 62 vs. 64 months, disease-specific death occurred in 3 vs. 2 patients, and death due to other causes occurred in 93 vs. 69 patients.

The 5-year overall survival rate was 87.3% vs. 96.1% (*P*<0.001), and the 5-year disease-specific survival rate was 99.6% vs. 99.9% (*P*=0.114). Recurrence (local/distant) occurred in 1.3% (0.7%/0.6%) vs. 0.4% (0.2%/0.2%) (P=0.014), and the 5-year recurrence-free survival rate was 98.5% vs. 99.6% (*P*=0.010).

Conclusion: Although elderly patients had a higher baseline risk due to comorbidities and use of antithrombotic agents, colorectal ESD achieved favorable short- and long-term outcomes comparable to those in non-elderly patients. This study demonstrates that colorectal ESD was safely and effectively implemented throughout the Hiroshima region.

Disclosure: Nothing to disclose.

PP0707

FULL THICKNESS COLONIC RESECTION USING THE FTRD OVESCO SYSTEM: A RETROSPECTIVE MONOCENTRIC STUDY

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Introduction: Full-thickness endoscopic resection (FTRD) using the Colonic FTRD Ovesco system represents a significant advancement in the management of complex colorectal lesions (1). This technique is particularly valuable for treating adenomatous recurrences occurring on post-resection scars smaller than 20 mm, especially in cases of failed submucosal lifting, offering an alternative to submucosal dissection (2).

Aims & Methods: This French retrospective monocentric study reports on the use of the colonic FTRD system, performed by two experienced operators, including the first clinician to perform this procedure in France. The study collected data on patient demographics, indications, resection sites, technical and clinical success rates, R0 resection rates, and procedure-related complications.

Results: Between 2016 and 2024, 87 patients (48 women and 39 men) with a median age of 66 years [32–87] were included. The primary indication was adenomatous recurrence after endoscopic mucosal resection (EMR) (69% of cases,n=60). Other indications included appendicular lesions in patient who had undergone appendicectomy (8%, n=7), suspected superficial submucosal carcinoma (7%, n=6), submucosal lesions (SML) (6%, n=5), and diverticular adenomatous lesions (3%, n=3). Most lesions (67%, n=58) were located in the right colon with a median estimated size before resection 18 mm [10–50 mm].

The overall technical success rate—defined as the complete execution of the resection—was 89%. There were 10 failures : failure to reach the site of recurrence (n=2), failure to achieve complete sectioning (n=5), and intraoperative complications (n=3), including three perforations. Two occurred while advancing through sigmoid diverticulosis , and one resulted from current diffusion into the colonic wall.

The R0 histological resection rate was 89% (n=71). 70% were adenomatous (low-grade dysplasia: n=36, high-grade dysplasia: n=13, serrated lesions without dysplasia: n=8). 8 cases (10%) were adenocarcinomas. Five cases involved submucosal tumors, including two neuroendocrine tumors. In 15% of cases (n=12), the resected specimen consisted solely of fibrotic scar tissue.

Early complications included 3 perforations, one of which was managed endoscopically, while the other two required immediate surgical intervention. Delayed complications were observed in 6% of cases, consisting of three episodes of post-procedural bleeding and two ileal perforations. The latter occurred on the suture line of the ileal reservoir in patients with a history of coloprotectomy and ileoanal anastomosis with a J-pouch for familial adenomatous polyposis.

This study confirms the efficacy of using the FTRD Ovesco system for lesions < 20 mm. The primary indications include post EMR recurrences with failed lifting, intra-appendicular adenomas, and submucosal tumors of the colon. Compared to submucosal dissection this technique offers a practical alternative, especially considering its lower morbidity, even with traction-assisted dissection. However, caution is advised in two specific situations, where alternative techniques may be preferable: severe diverticulosis with pseudo-stenosis and lesions developing on surgical anastomoses with a J-pouch reservoir.

Conclusion: The FTRD Ovesco system demonstrates excellent efficacy and safety for complex colonic lesions, including post-resection recurrences, submucosal tumors, and challenging locations such as appendicular sites. This technique represents an essential tool in expert endoscopic centers, offering a valuable alternative to other resection techniques with higher associated morbidity.

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Disclosure: Nothing to disclose.

PP0708

LEARNING CURVE FOR ENDOSCOPIC RETROGRADE APPENDICITIS THERAPY: FIRST CLINICAL EXPERIENCE

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Introduction: Endoscopic Retrograde Appendicitis Therapy (ERAT) is an emerging minimally invasive technique for appendicitis treatment widely practiced in China. Clarifying the learning curve, especially among endoscopists with varying endoscopic experience including endoscopic retrograde cholangiopancreatography (ERCP) skills, is essential for its broader adoption.

Aims & Methods: This study aimed to determine the number of cases required to achieve procedural proficiency across different experience levels. In this multicenter retrospective study, 655 patients undergoing ERAT from October 2021 to March 2024 were analyzed. Nine trainees from three hospitals were categorized into Novice, Non-ERCP Experts, and ERCP Experts groups. Procedural data, technical and clinical success rates, procedure times, pain scores, recurrence, and hospital stay were evaluated. Learning curves were assessed via CUSUM analysis based on procedure time.

Results: Technical and clinical outcomes improved with increasing procedural experience in all groups. ERCP Experts achieved proficiency earliest (case 31st), followed by Non-ERCP Experts (case 55th), and Novices (case

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59th). ERCP Experts demonstrated the shortest average procedure time (20.9 min) compared to Non-ERCP Experts (27.0 min) and Novices (31.4 min). Recurrence rates were significantly lower in ERCP Experts. Postoperative pain and hospital stay decreased with experience.

Conclusion: Endoscopists with extensive prior experience, particularly in ERCP, achieve proficiency in ERAT more quickly. Structured training tailored to experience levels is essential for global implementation of this technique.

Disclosure: Nothing to disclose.

PP0709

PROFICIENCY IN PERORAL ENDOSCOPIC MYOTOMY SHORTENS THE LEARNING CURVE FOR COLORECTAL ENDOSCOPIC SUBMUCOSAL DISSECTION

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Introduction: Colorectal endoscopic submucosal dissection (ESD) offers en bloc resection of large non-pedunculated polyps (LNPCPs) but requires advanced technical skills. ESGE recommends training with animal or ex vivo models before human procedures. POEM, a standardized third-space endoscopy technique, may provide transferable skills relevant to ESD.

Aims & Methods: This study assessed colorectal ESD outcomes performed by trainees with POEM experience but no model-based ESD training. All consecutive ESDs for LNPCPs (Jan 2022–Dec 2023) performed by four trainees with prior POEM experience (≥10 cases) were included. Trainees had observed ≥20 and assisted ≥5 ESDs, with no hands-on animal/ex vivo training. Procedures were supervised by experts. Outcomes included en bloc and R0 resection, dissection speed, and adverse events (AEs).

Results: A total of 48 LNPCPs (48.5 mm mean size, 1644 mm^2 mean area, 62.5% rectal lesions, 77.1% granular type, 79.2% Kudo III-IV) were resected. Overall, the en bloc resection rate was 100% (48/48), while the R0 resection rate was 93.8% (45/48). AEs occurred in 6.3% (3/48), particularly 2 intraprocedural perforations (2/48; 4.2%) and 1 delayed bleeding (1/48; 2.1%).

All AEs were managed endoscopically, resulting in an AE-related surgery rate of 0% (0/48). When comparing rectal and colonic lesions, en bloc and R0 resection rates were similar, with no statistically significant differences observed (100% vs. 100%, p=1.0; 93.3% vs. 94.4%, p=0.9); similarly, AEs rates and dissection speed showed no statistically significant differences (6.7% vs. 5.6%, p=0.9; 21.6 mm^2/min vs. 18.2 mm^2/min, p=0.4). Dissection speed improved 3.3× during training, though perforations clustered at case 7–9, suggesting overconfidence.

Conclusion: Proficiency in POEM shortens the learning curve for colorectal ESD, supporting its integration into structured ESD training pathways.

Disclosure: Nothing to disclose.

PP0710

MRI ASSESSMENT OF ACUTE COLONIC VOLUME RESPONSES TO OPIOIDS AND NALOXEGOL IN HEALTHY SUBJECTS: INTERIM RESULTS FROM A RANDOMISED, DOUBLE-BLIND CROSSOVER STUDY

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Introduction: Besides analgesia, opioids affect gastrointestinal (GI) motility, including decreased bowel peristalsis and delayed gastric emptying. Peripherally acting mu-opioid receptor antagonists, like naloxegol, are approved for treating opioid-induced constipation.

Aims & Methods: We aimed to evaluate the effects of naloxegol on opioid-induced lower GI tract dysfunction using magnetic resonance imaging (MRI) to assess changes in colonic volumes in healthy volunteers (HV).

In this double-blind, randomised, three-way crossover study, 15 HV complete three study visits, separated by ≥1 week washout, with administration of 25 mg naloxegol + 60 mg codeine, 60 mg codeine + placebo, or double placebo. During each visit, five abdominal MRI scans (each 20 min.) are obtained at 30 min intervals, at baseline, after drug intake, after a standardized 44 g oatmeal and 100 mL Nutridrink (313 kCal) meal, after a second drug dose, and after a bisacodyl 10 mg suppository administration. Colonic segmental volumes were quantified from all MRI scans using Entrolytics (Motilent, London, UK) in the ascending, transverse, descending, and rectosigmoid colon. We recorded the passage of stool during the study visit. Results are presented as medians with min. and max. P-values <0.05 are considered statistically significant. Adjusted p-values are used in case of post-hoc analysis.

Results: Interim analysis on 8 HVs (mean age 36±5y, 75% females) revealed that two subjects made stool on 2 mornings before the study visit. Four subjects passed stool in the double placebo group (2 postprandial, 2 post-bisacodyl), and 2 of them passed stool in the other 2 groups as well, around the same time. A significant scan x group effect was found for the whole colon volumes (p<0.05). More specifically, there was a significant difference in median ascending colon volumes between the scans (p=0.03), and especially in the double placebo group (Table).

	Naloxegol + codein	Codein + placebo	Double placebo	Naloxegol + codein	Codein + placebo	Double placebo
Scan1 (fasted, T0)	159 (118-202)	137 (92-247)	148 (122-226)	73 (63-92)	64 (48-105)	75 (53-98)
Scan2 (first drug combination, T50)	154 (119-238)	128 (92-252)	191 (122-239)	95 (71-110)	66 (48-91)	63 (51-101)
Scan3 (postprandial, T100)	172 (122-230)	160 (113-275)	151 (97-238)	74 (61-86)	59 (48-81)	74 (55-85)
Scan4 (Second drug dose, T150)	157 (103-242)	150 (77-254)	144 (94-207)	72 (53-92)	68 (56-80)	63 (41-85)
Scan5 (post bisacodyl, T185)	156 (63-213)	159 (103-233)	156 (106-204)	43 (26-62)	64 (37-86)	40 (27-53)
Adjusted p-value			0.025	0.049		0.04

Table 1. Results of ascending and descending colonic volumes (median with min and max, mL). Bold numbers represent significant differences within groups, between scans. Adjusted p-values are for differences within groups, between scans.

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Although there was no significant difference in the median transverse colon volumes, bisacodyl led to a significant decrease in descending colon volume ($p<0.0001$), more specifically in the naloxegol+codein group and the double placebo group ($ps<0.05$), but not in the codein+placebo group. No significant differences in volumes of the rectosigmoid were detected.

Conclusion: Interim MRI analysis of 8 HVs suggests different colonic motility responses across groups and timepoints. Toilet visits were suppressed in both codeine groups. Different responses to drug and food intake were found in the ascending and descending colon between the naloxegol+codein and double placebo group on one hand compared to the codein+placebo group. Expansion and analysis of the full number of subjects will further clarify insights into the acute colonic responses to opioids and potential reversal by naloxegol.

Disclosure: Nothing to disclose.

PP0711

ENDOSCOPIC INTERMUSCULAR DISSECTION (EID) AS NEW PERSPECTIVE FOR T1 RECTAL CANCERS AND HIGH-RISK LESIONS IN THE RECTUM - FIRST COMPARISON OF 12 EID CASES TO 53 LESIONS WITH A HISTOLOGICAL RESULT OF CANCER OR HGIEN RESECTED BY SUBMUCOSAL DISSECTION (ESD)

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Introduction: In contrast to mucosal T1 rectal cancers with an R0 rate of up to 92% at centers, R0 resection may drop to 30% in case of deep sm infiltration due to thermal artefacts at the basal resection margin. TME often follows as completion surgery. Deep sm infiltration as isolated risk factor excluding L+, V+, G3 and Bd2-3 seems associated with a low risk of LN metastases of only 2.6%. Endoscopic Intermuscular Dissection (EID) is a new flexible endoscopic dissection technique with separation of the circumferential and longitudinal muscular layer. EID ideally provides a specimen incl. the entire submucosa and inner circumferential mp layer. 12 sequential EIDs were compared to 53 historical rectal ESDs with HGIEN or rectal cancer as hp. result.

Aims & Methods: Lesions with biopsy proven ADC or HGIEN of the rectum without endoscopic, EUS or clinical signs of deep local infiltration. 2 pat. included after neo-adjuvant RT/Ch th. 24 and 26 mo before with complete response and refusing definitive surgery. Both with local flat polyps (SSA (1) hyperplastic polyp / LGIEN adenoma (2).

Endoscopy, EUS, MRI, CT and rectal digital examination without suspicion of local malignant infiltration. Pts were offered to prospectively undergo EID after extensive informed consent within a prospective protocol. All procedures were carried out under gen. anesthesia by an exp. interv. endoscopist with >1000 ESDs. Results of EID were compared to a historical collective of 53 cases with rectal ESD and a histological result of rectal cancer or HGIEN (08-2018 to 12-2021).

Results: From 11-2023 to 09-2024, 12 EIDs were carried out in 12 pts (6 f/6 m; mean 63 y; 34-83y). 09/12 lower rectum, 1/12 middle, 1/12 lower and middle, 1/12 upper rectum. ‚En bloc' resection rate was 100%, R0 resection rate 89% for neoplasia / HGIEN. Histopathology: 3x ADC G2, 6x HGIEN, 3x LGIEN including both cases with neodajuvant RT/Ch. R0 resection was achieved in 2 of the 3 ADCs. Both with deep sm infiltration without any other risk factors. Tumor board: close clin. & imaging FU. In the 3rd ADC: TU infiltration of inner muscular layer (T2). Pat. underwent uneventful TME 10 d after EID. Clinical FU in all others. Length of hosp. stay 7.1 days. No serious major complications. All pts. closely followed with uneventful FU.

In the historical ESD group R0 resection was achieved in 100% of the HGIEN cases (45/45) but only in 3/8 lesions (37.5%) of cases with adenocarcinoma.

Conclusion: EID seems a new perspective for the endosc. resection of T1 rectal cancers and suspicious polyps in the rectum. More experience and a comparison to surgical local procedures (TAMIS etc.) is necessary for the final assessment of EID as new promising endoscopic technique.

Disclosure: Nothing to disclose.

PP0712

FECAL MICROBIOTA TRANSPLANTATION FOR THE TREATMENT OF RECURRENT CLOSTRIDIOIDES DIFFICILE INFECTION: A LONG STANDING EXPERIENCE OF A REFERRAL CENTRE

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Introduction: *Clostridioides difficile* infection (CDI) represents the predominant cause of nosocomial diarrhea, with its incidence having significantly increased over recent decades, both within hospital environments and in the community (1).

Fecal microbiota transplantation (FMT) is recognized as a highly effective and safe intervention for recurrent *Clostridioides difficile* infection (rCDI), and our center has been providing an FMT program since 2013.

This study aims to assess the efficacy and safety of FMT in patients with CDI since the start of our FMT program.

Aims & Methods: Our aim was to report outcomes of FMT in patients with rCDI treated at our centre from June 2013 to September 2024. CDI and rCDI were defined according to international guidelines. Data were collected on patient demographics and CDI characteristics, including disease severity, risk factors, and concomitant diseases. Enrolled patients were followed up to 90 days after the last FMT. Patients who were not assessed at both 8 weeks and 90 days post-procedure were excluded from the analysis, as were those for whom the required data were unavailable. The primary outcome was the clinical resolution of CDI at 8 weeks post-FMT, while secondary outcomes included the identification of risk factors for FMT failure, overall survival at 90 days, and the safety of FMT. Donor stools were screened following international guidelines. Prior to fecal infusion, all patients underwent a 3-day antibiotic pre-treatment regimen with oral vancomycin or fidaxomicin, as well as bowel cleansing. FMT procedures were conducted via colonoscopy using a single-donor solution of at least 50 grams of fresh or frozen feces. A sequential FMT approach was planned for patients with pseudomembranous colitis (PMC) until the disappearance of pseudomembranes, or for patients who experienced diarrhea with persistent C. difficile toxin in stool samples after the first FMT.

Results: A total of 374 patients were enrolled (231 female, mean age 71 years). One hundred and ten patients had severe CDI (29.4%), and 15 had severe complicated CDI (4%). Among them, 73 patients had pseudomembranous colitis (19.5%). The median number of CDI recurrences was 2. A

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sequential FMT protocol was performed in 92 patients (24.6%), resulting in a total of 524 FMT procedures. Among risk factors for CDI, the most frequent were hospitalization (245 patients – 65.5%), antibiotic use in the 30 days prior to infection (212 patients – 56.7%), chronic therapy with proton pump inhibitors (186 patients – 49.7%), and enticement (76 patients – 20.3%). The median Charlson Comorbidity Index (CCI) was 5.

Overall, 337 patients (90.1%) experienced resolution of CDI (97 with severe CDI and 11 with severe-complicated CDI). Twenty-five patients (6.6%) died within the follow-up period, with no deaths attributable to FMT. Severe adverse events occurred in 23, and 9 were potentially associated with FMT and/or failure to cure CDI.

Conclusion: In the experience of our large-volume FMT center, FMT was an highly effective and safe treatment for rCDI.

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Disclosure: Nothing to disclose.

PP0713

FECAL MICROBIOTA TRANSPLANTATION FOR SEVERE AND SEVERE - COMPLICATED C. DIFFICILE INFECTION: A 10-YEAR EXPERIENCE FROM A REFERRAL CENTRE

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Introduction: Clostridioides difficile infection (CDI) is the most common cause of nosocomial diarrhoea, and during the last decades its incidence has significantly increased, not only in the hospital setting but also in the community (1).

CDI includes severe clinical pictures of disease. Fecal microbiota transplantation (FMT) is an established treatment for recurrent CDI, but its role in severe clinical pictures of CDI is still not well defined.

Aims & Methods: Our aim is to report efficacy and safety of FMT in a large cohort of patients with severe and severe-complicated CDI. We included patients with severe CDI treated with FMT at our center from June 2013 to August 2024. Severe and severe-complicated CDI were defined according to international guidelines. Data was collected on patient demographics, CDI characteristics, including disease severity, risk factors and concomitant diseases. Enrolled patients were followed up at 12 weeks after the last FMT.

The primary outcome was the clinical resolution of CDI at 8 weeks after FMT, while the secondary outcomes were the overall survival at 90 days and the safety of FMT. Donor stools were screened following international guidelines. Prior to fecal infusion, all patients underwent a 3-day antibiotic pre-treatment regimen with oral vancomycin or fidaxomicin, as well as bowel cleansing. FMT procedures were carried out by colonoscopy using a single-donor solution of at least 50 grams of fresh or frozen feces.

A sequential FMT approach was planned in patients with pseudomembranous colitis (PMC), until the disappearance of pseudomembranes, or in patients who experienced diarrhea after the initial FMT with a persistence of C. difficile toxin on stool samples.

Results: 125 patients were enrolled (60.8% female, mean age 71 years). One-hundred and ten patients had severe CDI (88%), 15 had severe complicated CDI (12%). Of them, 73 patients had pseudomembranous colitis (58.4%). The median number of CDI recurrences was 2. A sequential FMT protocol was performed in 62 patients (49.6%), for a total of 235 FMT procedures.

Regarding risk factors, the most frequent were hospitalization (107 patients – 85.6%), antibiotic use in the 30 days prior to infection (94 patients – 75.2%), chronic therapy with proton pump inhibitors (74 patients – 59.2%) and enticement (39 patients – 31.2%). The median Charlson Comorbidity Index (CCI) was 5.

Overall, 108 patients (86.4%) experienced resolution of CDI (97 with severe CDI and 11 with severe-complicated CDI). 18 patients (14.4%) died within the follow-up period. No deaths were attributable to FMT. Severe adverse events occurred in 11 (8.8%) patients and 6 were potentially associated with FMT and/or failure to cure CDI.

Conclusion: In our cohort of patients, FMT was a highly effective and safe treatment for severe and severe-complicated CDI.

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Disclosure: Nothing to disclose.

PP0714

COMORBIDITIES AS A RISK FACTOR FOR RECURRENT *CLOSTRIDIOIDES DIFFICILE* INFECTION

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Introduction: *Clostridioides difficile* infection (CDI) is among of the most common causes of nosocomial diarrhea with high morbidity and mortality. Recurrent and refractory disease courses in particular represent a special challenge in the management of these patients. The identification of patients at risk for recurrent courses could improve management, e.g. through the use of new and effective therapies in patients with CDI.

Aims & Methods: The aim of this study was to investigate the relationship between the presence of different comorbidities and the course of CDI.

In a retrospective, monocentric cohort study of a tertiary centre over a period of 10 years, patients with CDI-related hospitalizations were examined. The study included 336 patients with available information on comorbidities (diverticulosis, biliary diseases (cholecysto-, docholithiasis, cholecystitis, cholangitis or cholangiectasis) inflammatory bowel diseases (IBD), liver cirrhosis and malignancies) and the diagnosis of the recurrent CDI (rCDI) vs. non-recurrent course of CDI (nrCDI). Patients on dialysis or immunosuppressed patients were also considered as comorbid.

Results: Among 336 patients included in the study, 227 (67.6%) presented with at least one comorbidity. The most common were malignancy (30.4%), biliary diseases (19.4%), diverticulosis (15.8%), liver cirrhosis (14.3%), and inflammatory bowel disease (IBD, 8.9%). rCDI occurred more frequently in patients with diverticulosis and biliary disease. Diverticulosis was present in 24.4% of rCDI cases vs. 12.6% in nrCDI cases (OR 2.24, 95% CI 1.22–4.17). Biliary diseases were observed in 26.7% of rCDI vs. 16.7% of nrCDI patients (OR 1.82, 95% CI 1.01–3.26). No significant associations with rCDI were found for IBD (OR 0.66, 95% CI 0.27–1.65), malignancy (OR 1.13, 95% CI 0.67–1.88), or liver cirrhosis (OR 0.79, 95% CI 0.38–1.61).

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Conclusion: These findings suggest diverticulosis and biliary disease may be independent risk factors for rCDI, whereas other common comorbidities are not significantly associated with recurrence. Structured assessment of risk conditions may help to select patients for tailored management to prevent rCDI disease course.
Disclosure: Nothing to disclose.

PP0715

EXPLORING THE REALITY OF CLOSTRIDIOIDES DIFFICILE INFECTION: A FOUR-YEAR EXPERIENCE IN A TERTIARY HOSPITAL SETTING

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Introduction: Clostridioides difficile infection (CDI) has become an increasingly significant concern due to population aging, widespread use of antibiotics and immunosuppression. In this study we explore the natural history of CDI, therapeutic approaches, and mortality in a hospital serving approximately 362,000 inhabitants.
Aims & Methods: We conducted a retrospective cohort study including consecutive patients with CDI requiring hospitalization between January-2020/December-2023. Data regarding risk factors, clinical presentation, antimicrobial therapy, and recurrence were collected.
Results: We identified 460 patients (31.8 cases/100,000 inhabitants/year; female sex - 52.6%; mean age-68.7±21.5 years; median Charlson Comorbidity Index (CCI)-3.1). Major risk factors: previous antibiotic therapy-76.3%; hospital admissions in the previous 3 months-46.4%; immunosuppressive therapy-34.1%; living in nursing homes-29.5%; known gastrointestinal conditions-6.2%. Previous CDI was present in 7.1% of the cases. The diagnosis was made through the C. diff Toxins A/B Assay in all cases except one. Severity of the disease: mild-6.8%; moderate-68.2%; severe-25%. The most frequent first-line therapy was oral vancomycin-79.5% (mean treatment duration-9.3±3.7 days), in second line, metronidazole (50.8%), and in third line, fidaxomicin and metronidazole (37.5% each). Most patients required only one course of antibiotics (80.2%), while 16.6% required two courses and 2.6% required three. Four patients were submitted to fecal transplant. The average length of hospital stay was 25.4±22.4 (range:1-151), with longer hospital stay in patients with known gastrointestinal conditions (Z=-3.566, p<0.05).
Mortality was high: intrahospitalar-15.9%; 30-days-14.9%; 6-months-8.3%, revealing an association with age and CCI (Z=-2.564; Z=-2.997, p<0.05), but not with immunosuppression. Recurrence at 6-months was 11.2% (1 recurrence-9.8%, 2-1.2%; 3-0.8%; 4-0.4%) and revealed an association with previous antibiotic therapy (χ^2=11.273, df=4, p=0.024).
Conclusion: CDI represents a significant issue, which has been on the rise due not only to demographic shifts but also to iatrogenic factors. Incidence and mortality are relevant, and healthcare professionals should prescribe antibiotics judiciously aiming CDI prevention and decreasing of recurrences.
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Disclosure: Nothing to disclose.

PP0716

MECHANISTIC INSIGHTS INTO GUT BARRIER DISRUPTION INDUCED BY *CLOSTRIDIOIDES DIFFICILE* INFECTION AND RESTORATION BY FMT VIA AQUAPORIN-1 AND CARBONIC ANHYDRASE II

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Introduction: *Clostridioides difficile* infection (CDI) is the leading cause of hospital-acquired diarrhoea and pseudomembranous colitis in Europe, with healthcare costs exceeding €3 billion annually. The disease is primarily driven by two potent toxins, TcdA and TcdB. However, the molecular mechanisms underpinning toxin-mediated disruption of intestinal barrier function remain incompletely understood.
This study investigated the role of fluid and ion exchange regulators in intestinal permeability during CDI, and their modulation by faecal microbiota transplantation (FMT).
Aims & Methods: C57BL/6 mice were infected with *C. difficile* and treated with or without FMT. Caecal tissues derived from infected, treated, and control mice were analysed via RT-qPCR for expression of aquaporins (AQP1, AQP4, AQP8, AQP11), carbonic anhydrases (CAI, CAII, CAIV), and ion transporters (ANO1, CLCA1, SGLT1, SLC9A3, SLC26A3, SCNN1A, SCNN1B, SCNN1G). Expression of selected genes was validated in colonic biopsies from CDI patients and controls.
Primary human colonic organoids were treated with TcdA and TcdB. AQP1, AQP11, CAI, and CAII were knocked down in Caco-2 cells using shRNAs, and barrier function was assessed via transepithelial electrical resistance (TEER) in response to TcdB.
Results: CDI significantly downregulated AQP1, AQP11, CAI, CAII, and CAIV expression in murine models, with partial restoration following FMT. Knockdown of AQP1 and CAII in Caco-2 cells impaired basal barrier function, with further deterioration upon TcdB exposure. AQP1 expression was directly suppressed by TcdB in primary human colonic organoids. Validation in human CDI colonic tissues confirmed reduced AQP1 and CAII expression.
Conclusion: AQP1 and CAII are key regulators of gut epithelial barrier integrity during CDI. Their downregulation by *C. difficile* contributes to increased intestinal permeability, a hallmark of disease pathogenesis. Restoration of their expression following FMT suggests a potential mechanism for its therapeutic efficacy. These findings identify AQP1 and CAII as promising targets for future therapeutic intervention to restore barrier function in CDI.
Disclosure: Nothing to disclose.

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PP0717

FIRST EXPERIENCES WITH THE USE OF VACUUM-STENT FOR TREATMENT OF ANASTOMOTIC LEAKAGE IN THE LOWER GI TRACT

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Introduction: Anastomotic leakage is a serious complication following colorectal surgery and is associated with early and long-term morbidity and mortality. Various endoscopic and surgical approaches have been established for management. A new therapeutic option involves the placement of a fully covered VACStent for the lower GI tract (36x80 mm, MICRO-Tech, Düsseldorf, Germany) which offers advantages such as completely covering the defect, continuous drainage, and facilitating stool passage simultaneously. Here we report our initial experiences with the VACStent in the lower GI tract for treatment of anastomotic leakage.

Aims & Methods: The fully covered VACStent (36x80 mm) was placed endoscopically under direct visualization. A continuous negative pressure between -75 and -125 mmHg was applied. The VACStent was flushed daily with 20 ml of saline to prevent adhesions of the open-pored sponge of the stent. Six hours before the stent extraction/exchange, the negative pressure was discontinued. The integrity of the leak closure was confirmed visually and via fluoroscopy.

Results: Between February and December 2024, we treated six patients with anastomotic leakage (rectal leakage n=4, sigmoid leakage n=2) using the VACStent. The leak was detected on the 6th ± 3 postoperative day. The leak was successfully sealed with the VACStent and one to fife exchanges were performed. However, anastomotic leakages larger than 10 mm (n=2) required pretreatment with an EndoSponge (B.Braun, Melsungen, Germany) for a duration of 5 to 10 days. The leak closure with the VACStent occurred after 20 ± 6 days, with inflammatory parameters decreasing by around 85 ± 12%. Four patients received a protective stoma. The distance of the leak from the anal canal averaged 8 ± 7 cm. Patients with leaks involving the anal canal required pain relief. One patient experienced point-shaped erosion bleeding in the anal canal, which was effectively treated with APC. In two patients, the VACStent therapy could be completed in an outpatient setting.

Conclusion: The lower GI VACStent represents a novel promising and effective endoscopic treatment option for colonic anastomotic defects.

Disclosure: Nothing to disclose.

PP0718

EFFICACY OF OVER-THE-SCOPE CLIP IN PREVENTING REBLEEDING IN COLONIC DIVERTICULAR HEMORRHAGE

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Introduction: Endoscopic hemostasis using clips is widely used as a safe and effective first-line treatment for colonic diverticular bleeding, but rebleeding after hemostasis remains a common issue. In recent years, the Over-the-Scope Clip (OTSC) has gained attention as a new endoscopic treatment device due to its stronger tissue grasping ability and potentially higher hemostatic effectiveness compared to conventional clips. However, clinical cases remain limited, and more evidence is needed. This study evaluates the outcomes of colonic diverticular bleeding treated with OTSC.

Aims & Methods: The purpose of this study was to compare conventional clip hemostasis and additional treatment with OTSC for colonic diverticular bleeding. The subjects were patients at our institution from April 2014 to April 2024 in whom the bleeding source was endoscopically identified and treated. Patients treated with clips alone were classified into the clip group, while those who underwent additional OTSC treatment at the same site after clip treatment to prevent rebleeding were classified into the OTSC group. The clip and OTSC groups included 65 and 11 patients, respectively.

There were no significant differences between the groups in terms of sex (female/male: 12/53 vs. 4/7), age (median: 64 [35-96] vs. 70 [53-84]), or anticoagulant use (14/65 vs. 3/11). Rebleeding was defined as the appearance of visible bloody stool after endoscopic treatment, and it was the primary endpoint.

Results: Rebleeding occurred in 33 of 65 cases in the clip group and 0 of 11 cases in the OTSC group, showing a significant difference in rebleeding rate (0% [0/11] vs. 50.0% [33/66]; P < 0.05). Multivariate analysis identified OTSC as an independent preventive factor for rebleeding (p = 0.001). No complications such as bowel perforation, diverticulitis (localized infection), or intestinal stenosis were observed in either group.

Furthermore, when comparing costs, including readmissions due to rebleeding, the average cost in the clip group (15 out of 66 cases readmitted) was €5897 ± 3907, with a median of €4302. In the OTSC group (0 readmissions), the average cost was €4580 ± 2143, with a median of €4284, indicating that OTSC treatment was not inferior in cost-effectiveness compared to clip treatment (p = 0.124).

Conclusion: Treatment using OTSC significantly reduced rebleeding compared to conventional clip treatment, demonstrating high safety and comparable cost-effectiveness, suggesting it is an effective treatment method.

Disclosure: Nothing to disclose.

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PP0719

THE CURRENT PORTRAYAL OF LOWER GASTROINTESTINAL BLEEDING: A CROSS-SECTIONAL PAN-EUROPEAN STUDY

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Introduction: Lower gastrointestinal bleeding (LGIB) is a major medical emergency, and is associated with significant utilization of hospital resources, as well as considerable morbidity and mortality that increases with aging.

Aims & Methods: We aimed to describe patient characteristics, interventions and clinical outcomes of LGIB, and to define the impact of LGIB in elderly patients. This was a retrospective multinational cohort study which included adult patients consecutively presenting LGIB at 10 participating centers over a 12-month period. Data relevant to the initial LGIB episode and for up to 30 days afterwards were collected from patient medical records.

Results: Among the 950 patients included at this moment, 53.5% were men. Patients aged ≥ 65 years comprised 79,7% of the cohort (mean age 74 years); in this group, most comorbidities and use of antiplatelet/anticoagulant therapy were more frequent (p<0.001). 22.8% had a previous LGIB bleeding episode. On admission, hemodynamic instability was present in 12.6% (n=120) and almost 50% required red cell transfusion (n=442).

Most patients (81.6%) required hospitalization and 6% of them were admitted in an intensive care unit. The risk scores Oakland (p<0.001) and ABC (p=0.003) were higher in patients ≥65 years.

Colonoscopy was the most common investigation (63.8%). Hemostatic endoscopy therapy was performed in 20% of patients. Interventional radiology therapy and surgery were necessary in 7 (0.7%) and 24 (2.5%) patients, respectively. Diverticular bleeding (20.5%) was the most common cause of LGIB, followed by anorectal diseases (12.7%) and ischemic proctitis/colitis (10.3%). Median length of stay was 7 days, 12% of patients rebled during hospitalization and readmission with bleeding occurred in 8.4%. In-hospital mortality was 9% (n=86), with 74 deaths attributed to non-bleeding causes.

The multivariate analysis showed that older adults required hospitalization (p<0.001) and longer hospital stays (p<0.001) more frequently, while younger patients had higher ICU admission rates (p=0.018). Antiplatelet (p=0.034) or anticoagulant use (p=0.018), ABC score (p<0.001) and Charlson Comorbidity Index (CCI, p<0.001) were independently associated with in-hospital mortality. The CCI, higher in older patients, also correlated with need of blood transfusions (p=0.035), rebleeding (p=0.048), and re-hospitalization (p=0.027). Table 1 summarizes use of healthcare resources and clinical outcomes.

Country (mean age)	Portugal (74)	Greece (75)	Italy (72)	Ireland (72)	Spain (81)	France (71)
Mean days from admission to endoscopy	3	2	2	5	2	2
Hemostatic endoscopic therapy (%)	67 (20.1)	36 (23.7)	38 (25.3)	5 (3.5)	25 (25)	19 (29.7)
Interventional radiology therapy (%)	1 (0.3)	1 (0.7)	1 (0.6)	0 (0)	2 (2)	2 (3.1)
Hospitalization needed (%)	226 (68)	149 (98.0)	142 (89.3)	142 (100)	99 (99)	58 (90.1)
Days of hospitalization (median)	10	6	6	5	5	8
Intensive care unit admission (%)	19 (5.7)	4 (2.6)	1 (0.6)	1 (0.7)	4 (4)	20 (31.3)
Need of blood transfusions (%)	135 (40.2)	66 (43.4)	71 (43.9)	51 (35.9)	92 (92)	32 (50.0)
Surgery needed in the first 30 days (%)	8 (2.4)	3 (2.0)	4 (2.5)	2 (1.4)	6 (6)	1 (1.6)
In-hospital mortality (%)	30 (9.0)	12 (7.9)	3 (1.9)	21 (14.8)	15 (15)	5 (8.1)

Table 1.

Conclusion: This study analyzed the epidemiological data of LGIB in a large representative sample of the European population. Despite more accurate diagnostic/therapeutic methods and likely earlier detection, higher use of healthcare resources and worse outcomes have increased which may reflect a higher burden of comorbidities in a progressively aged population.

Besides higher prevalence, comorbidities and risk factors in elderly patients, the present study demonstrates that there are several differences in the management of LGIB patients across Europe.

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Disclosure: Nothing to disclose.

PP0720

BRAZILIAN TREND IN COLORECTAL CANCER MORTALITY - 2012 TO 2021

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Introduction: Colorectal cancer (CRC) is the second most common cancer and the third leading cause of cancer-related deaths in Brazil. High-income countries with organized screening programs have achieved significant reductions in CRC mortality in the last decades through early detection, polypectomy, and improved therapeutic interventions. Meanwhile, several low- and middle-income nations continue to experience rising incidence and persistently high mortality rates. Understanding CRC mor-

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tality patterns across different age groups and sexes is essential to guide public health policies and optimize resource allocation for prevention, early detection, and treatment strategies.

Aims & Methods: This study aimed to analyze CRC mortality trends in Brazil from 2012 to 2021. We conducted a descriptive analysis of CRC-related deaths (ICD-10 codes C18, C19, and C20) using data from the Mortality Information System of the Unified Health System (DATASUS). The crude annual mortality rate (deaths per 100,000 inhabitants) was calculated by dividing the number of deaths by the estimated population provided by the Brazilian Institute of Geography and Statistics (IBGE).

We also analyzed annual mortality rates by age range and sex and calculated the average annual percent change (AAPC) in mortality rates to assess trends over time.

Results: A total of 175,159 CRC-related deaths were recorded in Brazil between 2012 and 2021, with 86,330 occurring in men and 88,829 in women. Among men, 27.4% of deaths occurred before age 60, 53.8% between ages 60 and 79, and 18.8% at age 80 or older, while among women, these proportions were 27.4%, 47.1%, and 25.5%, respectively. During this period, CRC mortality rates increased in both sexes, rising from 7.06 to 9.73 deaths per 100,000 men and from 7.29 to 9.16 deaths per 100,000 women.

Consequently, the male-to-female mortality ratio grew from 0.97 in 2012 to 1.06 in 2021, leading to a slight predominance of male gender at the end of the assessed period. The AAPC in mortality was higher in men than in women across all age groups, except for those aged 40–59 years (Table 1). Among younger individuals, those aged 30–39 years had a low absolute mortality rate (<0.4 deaths per 100,000 inhabitants) but showed the highest AAPC (3.5% per year in men and 2.8% per year in women). Among individuals within the recommended CRC screening age range, the highest AAPC was observed in those aged 60–69 years (2.1% per year in men and 1.7% per year in women).

Age range	AAPC in men	AAPC in women
20-29	2,8%	1,6%
30-39	3,5%	2,8%
40-49	0,7%	1,0%
50-59	1,1%	1,4%
60-69	2,1%	1,7%
70-79	1,2%	0,7%
80+	0,2%	0,1%

Table 1 –Average annual percent change in colorectal cancer mortality rate by sex and age range.
AAPC, average annual percent change.

Conclusion: The concerning rise in CRC mortality in Brazil underscores the need for enhanced public health strategies, including improved access to screening and better healthcare resources. Additionally, the greater increase in mortality rates from early-onset colorectal cancer, particularly between ages 20 and 39, warrants attention, with a focus on identifying high-risk individuals and addressing modifiable risk factors in the population.

Disclosure: Nothing to disclose.

PP0721

SYSTEMATIC REVIEW AND META-ANALYSIS OF PARTICIPATION AND YIELD OVER TIME WITH FECAL IMMUNOCHEMICAL TEST-BASED COLORECTAL CANCER SCREENING

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Introduction: The fecal immunochemical test (FIT) is the preferred non-endoscopic test in programmatic screening worldwide. Compared to other methods, FIT may be better accepted with higher participation rates, which may counteract its lower one-time sensitivities. We performed a systematic review and meta-analysis to ascertain participation patterns, participation rates, and FIT performance in organized multi-round FIT-based screening programs worldwide.

Aims & Methods: We performed a systematic search using Pubmed, Embase, and Cochrane databases. Participation patterns were categorized as “consistent screeners” (participation in 100% of rounds), “intermittent screeners” (including drop-outs, delayed entrants, and intermittent re-screeners), and “consistent non-screeners” (never screened). Data regarding positivity rates and detection rates for adenomas, advanced adenomas (AA), and colorectal cancer (CRC) across subsequent rounds were extracted and summary estimates were obtained. Yields were calculated for 2 populations: 1) Overall Population; 2) Consistent Screeners. A sensitivity analysis was performed including only programs with biennial screening with FIT cut-off of 20µg Hb/g feces.

Results: The literature search yielded 4,414 studies of which 25 were included. Across all studies, the fraction of the population behaving as “consistent screeners”, “consistent non-screeners”, and “intermittent screeners” was 47.1%, 30.3%, and 22.7%, respectively. For the overall population, FIT-positivity from the 1st through 4th rounds was 6.7%, 5.5%, 5.4%, and 5.3%; and detection rates were 5.8%, 3.3%, 2.9%, and 2.5% for CRC; 32.8%, 28.3%, 22.7% and 23% for AA; 48.6%, 46.2%, 48.7% and 44.6% for adenoma. For Consistent Screeners, FIT-positivity across studies from the 1st through 4th rounds was 6.4%, 4.8%, 4.1%, and 3.6%, respectively; and detection rates were 6.9%, 4.6%, 3.6%, and 3.1% for CRC; 38.7%, 30.6%, 28.2% and 24.9% for AA; 53.4%, 46.7%, 42.6% and 42.4% for adenoma. Similar trends were observed for biennial screening with FIT cut-off of 20µg Hb/g feces.

Conclusion: Organized multi-round FIT-based CRC screening programs achieve consistent participation in a minority of participants, and a sizeable proportion of the population do not initiate screening at all after multiple offers. Among those participating, FIT-positivity, and yields for CRC, AA, and adenoma are highest in the 1st round, but may stabilize at a clinically relevant steady state at rounds 3+.

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PP0722

COLORECTAL CANCER TRENDS IN LEBANON FROM 2001 TO 2024

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Introduction: Colorectal cancer (CRC) is the third most common cancer and second leading cause of cancer death worldwide, with over 1.9 million new cases and 0.9 million deaths in 2020. CRC is a major health burden globally and in the Middle East. Lebanon has reported one of the highest CRC incidences in the region, but detailed trends over the past two decades are not well characterized. This study analyzes the evolution of CRC epidemiology in Lebanon from 2001 to 2024.

Aims & Methods: We conducted a retrospective analysis of CRC cases spanning 24 years, grouped into five intervals: 2001–2005, 2006–2010, 2011–2015, 2016–2019, and 2020–2024. Data were obtained from National Institute of Pathology registry. Variables included patient age, sex, tumor anatomic location, and stage at diagnosis (TNM). Incidence trends were evaluated using national registry. Descriptive statistics and interval comparisons were performed to identify temporal changes in CRC incidence and characteristics.

Results: The annual CRC incidence in Lebanon rose markedly over the study period. Age-standardized incidence rates increased from 16.3 per 100,000 (males) and 13.0 (females) in 2005 to 23.2 and 20.2 by 2016. By 2018, 1109 new CRC cases were diagnosed, making it the second most common cancer among Lebanese women and fourth in men. The median age at diagnosis shifted upward (early 60s in 2001–2005 to 67 years in 2020–2024), although a growing proportion of cases appeared in younger adults (notably the 40–49 age group). The male-to-female ratio remained approximately 1:1 throughout (52% male in early 2000s vs 50% by early 2020s), indicating near parity. Tumor location distribution showed a predominance of left-sided cancers across all periods, but with a modest proximal shift over time. More than 70% of tumors were distal (left colon or rectum) in the early 2000s, whereas by 2020–2024 about 35% were right-sided. Stage at diagnosis remained advanced in most patients, though there was a small improvement: 20% of cases were early stage (I–II) in 2001–2005 versus 25% by 2020–2024. Despite this trend, a majority of patients in the latest interval still presented with regional or metastatic disease.

Conclusion: CRC incidence in Lebanon has increased substantially over the past two decades, reaching levels comparable to Western countries. The patient profile has evolved to a slightly older median age, with hints of an increase in younger-onset cases, and the male-to-female incidence gap has virtually disappeared. Tumors remain predominantly left-sided, although the proportion of right-sided cancers has risen, likely reflecting an aging population. While there are signs of earlier diagnosis (more Stage I–II cases in recent years), most CRCs are still detected at late stages. These findings underscore the need for strengthened national CRC screening and public health strategies – including awareness campaigns and earlier screening initiation – to curb incidence and improve early detection in Lebanon.

Disclosure: Nothing to disclose.

PP0723

SCREENING INITIATION AGES BASED ON BODY MASS INDEX FOR EARLY DETECTION OF EARLY-ONSET COLORECTAL CANCER

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Introduction: The rising incidence of early-onset colorectal cancer (eoCRC) in many countries underscores the need for risk-adapted screening approaches. Excess weight is a modifiable and increasingly prevalent risk factor for eoCRC. Given the routine collection of body mass index (BMI) in clinical settings, BMI-based screening initiation may offer a straightforward and scalable strategy to improve early detection.

Aims & Methods: We aimed to identify BMI-specific ages at which individuals reach 5-year cumulative colorectal cancer (CRC) risk levels equivalent to typical screening initiation ages (ages 45 and 50). Using GLOBOCAN 2022 data, we estimated country-specific 5-year cumulative CRC risks for adults aged 30–60 across six North American and European countries (United States, Canada, Germany, France, United Kingdom, and Italy).

As the average BMI at CRC screening initiation ages in the included countries falls within the overweight range (BMI 25–29.9 kg/m^2), we considered these age-specific risk estimates representative of individuals with overweight and reflective of average risk at screening initiation.

Age-specific risks for individuals with normal weight (BMI 18.5–24.9 kg/m^2) and obesity (BMI ≥30 kg/m^2) were derived by proportionally adjusting the risks of the overweight population using odds ratios and confidence intervals from a meta-analysis of 20 cohort studies. Risk-adapted starting ages were defined as the age at which individuals with normal weight or obesity reached the same 5-year CRC risk as those with overweight at age 45 or 50.

Results: Using individuals with overweight as the reference group for average-risk at screening initiation, those with obesity reached the same 5-year cumulative CRC risk observed at age 45 by age 41 (95% CI, 37-45) in Canada and the United States, and by age 42 (95% CIs, 38-45) in the included European countries.

Individuals with normal weight reached this risk level later, by age 47 (95% CI, 46-48) in Germany, the United Kingdom, and Italy and at age 48 (95% CIs, 47-50) in the United States, Canada, and France. For the age 50 benchmark, individuals with obesity reached comparable CRC risk between ages 46 (95% CIs, 41-51) and 47 (95% CI, 44-50) (see Table 1 for detailed results by country and BMI category).

	Risk-adapted starting age [years (95 % confidence interval)]			
	Benchmark: Age 45		Benchmark: Age 50	
	Obesity (BMI ≥30 kg/m²)	Normal weight (BMI 18.5–24.9 kg/m²)	Obesity (BMI ≥30 kg/m²)	Normal weight (BMI 18.5–24.9 kg/m²)
United States	41 (37-45)	48 (47-49)	46 (41-51)	56 (54-58)
Canada	41 (37-45)	48 (47-50)	46 (41-51)	54 (52-56)
Germany	42 (40-45)	47 (46-48)	47 (44-50)	54 (52-56)
France	42 (38-45)	48 (47-49)	46 (42-50)	53 (52-54)
United Kingdom	42 (39-45)	47 (46-48)	47 (44-50)	52 (51-54)
Italy	42 (39-45)	47 (46-48)	46 (43-50)	53 (52-54)

Table 1. Risk-adapted starting ages for colorectal cancer, by BMI category and country

Note. Reference BMI category: overweight, BMI 25–29.9 kg/m²

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Conclusion: These findings support incorporating BMI into CRC screening strategies to enable earlier detection in high-risk individuals. Current age-based guidelines may not fully capture elevated risk in individuals with excess weight, potentially delaying detection in this subgroup. BMI-based risk stratification may be a feasible, low-cost approach that could complement existing screening programs by enabling earlier initiation for those at higher risk. Future studies should evaluate its integration with other risk factors and its impact on healthcare systems.

Disclosure: Nothing to disclose.

PP0724

THE STATUS OF COLONOSCOPY, ESOPHAGOGASTRODUODENOSCOPY, AND COLORECTAL CANCER ACCORDING TO BIG NATIONAL DISASTERS SUCH COVID-19 AND EARTHQUAKE

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Introduction: Sudden decrease of colonoscopy (CS) can be related with the status of colorectal cancer (CRC).

Aims & Methods: The aim of this study is to evaluate the status of CS, esophagogastroduodenoscopy (EGD), and CRC according to big national disasters such as the Great East Japan Earthquake in 2011 and the spread of COVID-19 infection in 2020. As a study population, we retrospectively used commercially anonymized health insurance claims data covering 4,601,921 patients ≥50 years old (Male 2,403,826/ Female 2,198,095) from January 2010 to December 2022.

Main outcome was a yearly analysis for 2010-2022 about 1. rate of CS and EGD, 2. rate of CRC and positive rate of CS for CRC, 3. number of surgeries for CRC. Additionally, a monthly analysis for those rates were performed to examinate the effect of the earthquake and COVID-19 infection.

Results: The rates of CS and EGD in 2010/2011/2012 were 3.10%/3.41%/3.42% and 6.94%/6.96%/7.00%. Those in 2019/2020/2021 were 5.20%/4.74%/5.37% and 7.47%/6.42%/6.84%. Deficiencies for CS and EGD were found not in 2011 (The earthquake) but were present in 2020 (COVID-19). The rates of CRC in 2019/2020/2021 were 0.20%/0.17%/0.19% with a decrease rate of 12.0% in 2020. The positive rate for CRC decreased from 6.74% in 2010 to 3.44% in 2022.

However, there were no definite changes in 2011 and 2020. The rate of surgery for CRC in 2019/2020/2021 were 0.087%/0.079%/0.087% and the deficiency was found in 2020.

Conclusion: The rates of CS and CRC decreased in 2020 due to COVID-19. On the other hand, they were not affected by the earthquake.

Disclosure: Nothing to disclose.

PP0725

COST-UTILITY EVALUATION OF TAIWAN'S COLORECTAL CANCER SCREENING PROGRAM: REAL-WORLD EVIDENCE AFTER CONSIDERATION OF LIFETIME PRODUCTIVITY

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Introduction: A small-scale study modeled the cost-effectiveness of colorectal cancer (CRC) screening, incorporating productivity factors.

Aims & Methods: Using real-world data, this research aimed to evaluate the cost-utility of Taiwan's CRC screening program, factoring in potential lifetime productivity savings. After interlinkages among databases of the Taiwan Cancer Registry, National Mortality Registry, Taiwan CRC Screening Database, and National Health Insurance claim records, we estimated lifetime survival functions for stage-specific subcohorts using a rolling extrapolation algorithm after stratified by sex and age from 2004 to 2017. We multiplied the survival rates with average utility values of EQ-5D collected from CRC cases at National Cheng Kung University Hospital, to sum up quality-adjusted life expectancy (QALE).

We multiplied monthly medical costs and insured salaries by survival probabilities. We summed them over a lifetime for both the index and age-, sex-, and calendar-year matched reference subcohorts for comparison. Lifetime productivity (LP) was assessed only up to age 65. After weighing different stage distributions, we calculated the differences in total QALE loss, lifetime medical expenditures (LME), and LP between screened versus non-screened groups.

Finally, we estimated the costs associated with implementing fecal immunochemical testing (FIT) in Taiwan's CRC screening program. We applied a 3% annual discount and an exchange rate of 30.58 New Taiwan Dollars per U.S. Dollar (USD) at the end of 2018.

Results: We enrolled 77,169 CRC patients aged 50–74 in the study. After weighted by different stage distribution, a screen-detected CRC case would save 2.6 (95% CI, 2.3–2.9) and 2.5 (95% CI, 2.1–2.6) QALYs for women and men, respectively, accompanied by 7,000 USD (95% CI, 4,000–10,000) and 8,000 USD (95% CI, 6,000–10,000) in LME, and 17,000 USD (95% CI, 16,000–19,000) and 23,000 USD (95% CI, 22,000–25,000) in LP. For every 100,000 persons screened, the program spent 682,000 USD for FIT, 1,051,000 USD and 1,987,000 USD for men and women, respectively, for colonoscopy, 61,000 USD and 133,000 USD for surveillance colonoscopy and 9,000 USD and 15,000 USD for treating complications.

The incremental cost-utility ratio (ICUR) of 6,650 and 6,230 USD per QALY for women and men, respectively, would become 120 and -2970 USD per QALY after accounting for lifetime productivity.

Conclusion: After considering lifetime productivity, Taiwan's CRC screening program would be close to being cost-saving. Future corroboration studies are warranted.

Disclosure: Nothing to disclose.

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PP0726

RISK FACTORS FOR POST-COLONOSCOPY COLORECTAL CANCER IN A TERTIARY MEDICAL CENTER: ROOT-CAUSE ANALYSIS AND A CASE CONTROL STUDY

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Introduction: Colonoscopy is the gold standard for diagnosing colorectal cancer (CRC)[1], with post-colonoscopy colorectal cancer (PCCRC) defined by the World Endoscopy Organization (WEO) as CRC diagnosed between 6 and 48 months after a negative colonoscopy[2].

PCCRC rates vary from 2.5% to 7%[2], and the prognosis of PCCRC is generally poorer than that for CRC detected during colonoscopy[3].

Aims & Methods: To evaluate the risk factors associated with PCCRC in a major tertiary medical center.

We analyzed all consecutive colonoscopic procedures conducted between 2001 and 2019. Data was retrieved from electronic medical records (EMR) and cross-referenced with the Israeli Cancer Registry. We defined PCCRC based on the WEO criteria, and two physicians independently confirmed potential PCCRC cases to ensure data accuracy. Root-cause analysis of PCCRC was performed according to the WEO algorithm as follows:

A. Possible missed lesion, index examination adequate;

B. Possible missed lesion, index examination inadequate;

C. Detected lesion, not resected; or

D. Likely incomplete resection of previously identified lesion.

We then identified colonoscopy-detected CRCs (CDCRC, controls) and compared them to the PCCRC cases at a ratio of 1:3. We assessed patient median age at the procedure, gender, religion, family history of CRC, diabetes, smoking status, body mass index (BMI), obesity (defined in the EMR or BMI > 30) and endoscopic parameters: colonoscopy indication [surveillance (follow-up colonoscopy according to recommendations), abnormal imaging, patient symptoms], adequate Boston Bowel Preparation Scale (BBPS>6 and >2 in all segments), cecal intubation rate, diverticulosis, rectal retroflection, and tumor location.

Multivariable logistic regression assessed the association of multiple factors (endoscopy indications and obesity) with PCCRC, maintaining a 95% confidence interval (CI).

Results: Out of 54,154 colonoscopies, 2,339 patients were diagnosed with CRC, of which 49 (2.1%) were confirmed PCCRCs, while 2,157 (92.2%) were detected during colonoscopy. Of patients with PCCRC, 65.4% (n=32) had adequate BBPS, and 91.8% (n=45) cecal intubation was documented. The two most common reasons for PCCRC were category A (n=29, 59.1%), and category B (n=17, 34.6%), although in a lesser extent. We then compared PCCRC cases to 150 randomly selected CDCRC cases.

Significant differences in patient characteristics included BMI>30 (p = 0.041) and obesity (p=0.015). Statistically significant endoscopic factors included colonoscopy indication (p < 0.001), cecal intubation rate (p < 0.001), and tumor location (p < 0.001).

As expected, PCCRC was more commonly detected in the right colon (n=24, 55.8%) or rectum (n=12, 27.9%). Multivariable logistic regression revealed that patients with abnormal imaging as a colonoscopy indication had a lower risk for PCCRC (adjusted odds ratio (aOR) = 0.148, 95% CI: 0.033-0.455), while patients that performed colonoscopy for surveillance had a higher risk (aOR = 7.334, 95% CI: 2.355-25.963). Obesity was also a significant risk factor (aOR = 3.035, 95% CI: 1.326-7.035).

Conclusion: The most frequent cause of PCCRC across various patient populations and colonoscopy indications was a possible missed lesion. Factors such as patient obesity and undergoing colonoscopy for surveillance may increase the likelihood of PCCRC. These findings highlight the importance of physician awareness regarding PCCRC risks during colonoscopy.

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Disclosure: Nothing to disclose.

PP0727

INCIDENCE OF PRECANCEROUS LESIONS AMONG PATIENTS AGED 40 TO 45 YEARS WHO UNDERWENT SCREENING COLONOSCOPY

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Introduction: Colorectal Cancer (CRC) is the third most common cancer worldwide. Most CRCs arise from precursor adenomatous or serrated polyps. Various Screening programs to early detect and remove precancerous lesions before progression to malignancy results in significant decrease of colorectal cancer incidence in developed countries (1).

Guidelines recommend regular screening for adenomatous polyps and colorectal cancer for individuals above the age of 50 (1). However, in 2018 the American Cancer Society prompted substantial controversy by lowering their recommended age to begin from age 50 to age 45 (2).

According to recent studies, starting screening at age 40 or 45 can be cost effective.

According to international agency for research in cancer publication 2022 colorectal cancer is ranked the second between the top 3 cancers related deaths in United Arab Emirates.

Aims & Methods: The aim of this study to estimate the incidence of precancerous adenoma in average risk patients between 40 and 45 years of age.

Results: In this cohort of 703 individuals aged 40 to 45 years undergoing screening colonoscopy, adenomas were identified in 17.5% (n = 123) of patients. The mean age was similar between patients with and without adenomas (42.72 ± 1.84 vs. 42.53 ± 1.78 years; P = 0.293). Male sex was significantly associated with adenoma detection (51% vs. 39%; P = 0.014). No significant differences were observed between groups regarding BMI, alcohol use, diabetes, or family history of colorectal cancer. Smoking was

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significantly more prevalent among patients with adenomas (28% vs. 15%; P < 0.001) and remained an independent predictor on multivariate analysis (odds ratio [OR] 1.862; 95% confidence interval [CI], 1.078–3.217; P = 0.026). Other variables, including age, sex, BMI, alcohol intake, diabetes, and family history, were not independently associated with adenoma presence. The number needed to scope (NNS) to detect one adenoma was estimated at 5.7.

Conclusion: In conclusion, suggesting a non-negligible burden of colorectal neoplasia in individuals younger than 45 years. Early screening strategies, particularly targeted at high-risk individuals, may be warranted. Further multicenter studies with larger sample sizes are needed to confirm these findings and inform guideline recommendations for screening initiation age.

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Disclosure: Nothing to disclose.

PP0728

RISK OF SECONDARY MALIGNANCIES FOLLOWING APPENDICEAL CANCER: A SEER-BASED ANALYSIS

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Introduction: Appendiceal tumors are considered to be a relatively rare tumor of the gastrointestinal tract, and the prognosis is unclear. The development of the small intestine, colon, peritoneum, omentum, and mesentery as second primary malignancies (SPMs) following appendiceal tumors is rarely mentioned in the literature. No studies discussed the association between small intestine, colon, peritoneum, omentum, and mesentery SPMs and appendiceal cancer.

Aims & Methods: The purpose of this study was to evaluate the risk of small intestine, colon, peritoneum, omentum, and mesentery SPMs following appendiceal cancer .

The Surveillance Epidemiology and End Result (SEER) database was used to obtain data of patients diagnosed with appendiceal cancer from 1975 to 2022. We used the MP-SIR session with multiple outcome analysis, and the event was defined as small intestine, colon, peritoneum, omentum, and mesentery. Excess absolute risk (EAR) was measured per 10,000. The standardized incidence ratio (SIR) was computed as observed/expected (O/E) with a 95% confidence interval (CI).

Results: Among 7723 appendiceal cancer patients, SIRs for second primary malignancies were significantly elevated over all time intervals, with small intestine cancer having an O/E of 20.38 (P<0.05, EAR=8.37, 95% CI: 6.57–47.55) for 2–11 months, 11.58 (P<0.05, EAR=4.86, 95% CI: 5.54–21.30) for 12–59 months, 13.89 (P<0.05, EAR=6.90, 95% CI: 6.34–26.37) for 60–119 months, and 3.81 (P<0.05, EAR=1.84, 95% CI: 1.02–9.75) for ≥120 months, colon cancer with an O/E of 3.34 (P<0.05, EAR=13.58, 95% CI: 1.67–5.98) for 2–11 months, 2.31 (P<0.05, EAR=7.84, 95% CI: 1.51–3.38) for 12–59 months, and 1.99 (P<0.05, EAR=7.00, 95% CI: 1.16–3.19) for 60–119 months, and peritoneum, omentum, and mesentery malignancies with an O/E of 11.55 (P<0.05, EAR=0.97, 95% CI: 1.30–41.70) for 12–59 months, 15.69 (P<0.05, EAR=1.55, 95% CI: 1.76–56.66) for 60–119 months, and overall cumulative O/E values of 9.97 (P<0.05, EAR=4.97, 95% CI: 6.63–14.42) for small intestine, 1.77 (P<0.05, EAR=5.45, 95% CI: 1.37–2.25) for colon, and 6.41 (P<0.05, EAR=0.64, 95% CI: 1.72–16.40) for per.

Conclusion: Appendiceal cancer patients are at significantly higher risk for second primary malignancies, primarily involving the small intestine, colon, and peritoneal-associated sites. The risk is highest within the initial 10 years after diagnosis, supporting the need for scrupulous long-term follow-up in these individuals.

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Disclosure: Nothing to disclose.

PP0729

ADENOMA INCIDENCE TRENDS BY GENOTYPE IN LYNCH SYNDROME: A RETROSPECTIVE COHORT STUDY OF 318 MD ANDERSON CANCER CENTER PATIENTS

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Introduction: Lynch Syndrome (LS), the most common hereditary colorectal cancer (CRC) syndrome, is associated with increased adenoma and carcinoma risks. Colonoscopy surveillance recommendations generally consider MMR gene-specific risk but remain largely standardized in terms of surveillance intensity and frequency. However, accumulating evidence suggests that genotype-specific differences in neoplasia risk may warrant more tailored approaches.

Aims & Methods: This study evaluated trends in adenoma and CRC incidence across surveillance visits stratified by genotype among LS patients. We analyzed 318 LS patients with pathogenic variants in *MLH1, MSH2, MSH6, PMS2 or EPCAM* each with ≥2 surveillance colonoscopies between 2015–2025 at MD Anderson Cancer Center in Texas, USA. Individuals with prior colorectal surgery, inflammatory bowel disease, incomplete colonoscopy, or sigmoidoscopy-only exams were excluded. Adenoma (tubular, tubulovillous [TVA], SSA) and advanced neoplasia (advanced adenoma or CRC) rates were assessed over three follow-up visits. Statistical comparisons used Kruskal-Wallis and Fisher's exact tests (α=0.05).

Results: Among 205 patients with follow-up (median age 43.4, median follow-up 1089 days), tubular adenomas (TA) were observed in 58.8%, sessile serrated adenomas (SSA) in 16.2%, and advanced neoplasia (AN) in 7.8%. Five CRCs (2.4%) were identified overall. *MSH2/EPCAM* carriers (n=79) had the highest neoplasia burden, with TA in 57.0%, SSA in 20.3%, and 3 of 5 CRCs. *MLH1* carriers (n=50) had TA in 43.4%, SSA in 2.0%, and 2 CRCs (4.0%). *PMS2* (n=37) had the lowest burden (TA 21.6%, SSA 5.4%) and no CRCs. While per-visit incidence trends were not statistically significant, total adenoma count across genotypes differed significantly (p<0.05). Additionally, TA and TVA incidence at first follow-up varied significantly by genotype (p = 0.036 and 0.037, respectively).

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Genotype	n (%)	TA (%)	SSA (%)	AN (%)	CRC (%)
MLH1	50 (24.4)	43.4	2.0	6.0	4.0
MSH2/EPCAM	79 (38.5)	57.0	20.3	5.1	3.8
MSH6	39 (19.0)	38.5	12.8	10.3	0.0
PMS2	37 (18.0)	21.6	5.4	2.7	0.0

Table: Neoplasia Outcomes by LS Genotype (n = patients with ≥1 follow-up).

Conclusion: These findings highlight important genotype-specific differences in cumulative adenoma and CRC burden among LS carriers. While *MSH2/EPCAM* and *MLH1* carriers exhibited the highest neoplasia rates, *PMS2* carriers demonstrated consistently low lesion and cancer incidence. These results support the need for genotype-based surveillance strategies to reduce unnecessary procedures while preserving cancer prevention.

Disclosure: The authors declare that they have no conflicts of interest related to this work.

PP0730

A MULTI-CENTRE RETROSPECTIVE STUDY OF COLONOSCOPY OUTCOMES IN LOW RISK SYMPTOMATIC PATIENTS UNDERGOING FECAL IMMUNOCHEMICAL TESTING IN IRELAND

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Introduction: Colorectal cancer (CRC) remains a major cause of mortality and morbidity worldwide. Projected increases in CRC cases pose a significant challenge to the capacity of colonoscopy programs. Referrals for symptomatic colonoscopy exceed capacity in Ireland and most European nations. Fecal Immunochemical Test (FIT) with a lower cut-off (50ng/ml) has been proposed as a tool to aid triage of referrals which do not meet criteria for urgent colonoscopy, potentially allowing colonoscopy to be avoided.

Aims & Methods: The aim of this study was to evaluate FIT testing outcomes when combined with an endoscopy nurse delivered clinical triage and evaluate suitability as a 'rule out' test for serious gastrointestinal pathology.

This multi-centre study included symptomatic patients from sixteen hospitals across Ireland who were referred for colonoscopy from 2023-2024. All patients received 'FIT testing at the front door'. Using an advanced endoscopy nurse triage approach, assessing patient clinical characteristics and FIT result (positive result determined as FIT≥50ng/mL), patients were triaged into urgent colonoscopy or remained on the routine (P2) colonoscopy list.

Among 7,808 patients sent for FIT testing, 408 were excluded on the basis of missing FIT results (n = 407) or incomplete data (n = 1). The remaining 7,400 patients were included in an analysis assessing FIT positivity rates (FIT≥50 ng/mL) as well as factors associated with positive results. A subset of 1,815 patients from five hospitals, who had lower endoscopy outcomes available, were further analysed to determine frequencies of endoscopy findings (ex: normal, CRC, and IBD) and to evaluate the sensitivity, specificity and negative predictive value of FIT-centered endoscopy nurse clinical triage. IBM SPSS Statistics 29.0.1.0 was used for statistical analysis.

Results: Of 7,400 patients who underwent FIT testing across sixteen sites, 16.4% (n=1,215) had a positive result (FIT≥50ng/mL). Multivariate analysis showed that increased age (OR = 1.029; 95% CI = 1.025-1.033; p-value<0.001) and male sex (OR = 1.44; 95% CI = 1.273-1.63; p-value<0.001) were significantly associated with positive FIT results.

At five sites, 1,815 patients who underwent FIT testing were referred for colonoscopy. At these sites, 79.0% (n=1,433) had a FIT<50ng/mL of whom 0.14% (n=2) had CRC and 1.5% (n=21) had significant findings at colonoscopy (ex: CRC or IBD). In contrast, 21.0% (n=382) had a FIT≥50ng/mL of whom 6.8% (n=26) had CRC and 19.1% (n=73) had significant findings at colonoscopy (ex: CRC or IBD).

The proportion of patients with CRC was significantly greater when FIT≥50ng/mL compared to FIT<50ng/mL (Fisher's Exact Test, p-value <0.001). The sensitivity, specificity, and negative predictive value of FIT≥50ng/mL for CRC at subsequent colonoscopy is 92.9% (95%CI=75.0-98.8%), 80.1% (95%CI=78.1-81.9%), and 99.9% (95%CI=99.4-99.9%), respectively.

Conclusion: FIT testing is useful in low risk symptomatic patients referred for colonoscopy and when combined with enhanced nurse led clinical triage has a very good negative predictive value for serious gastrointestinal pathology.

Disclosure: Nothing to disclose.

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PP0731

CHARACTERIZATION OF THE ORAL AND GUT MICROBIOTA IN INDIVIDUALS WITH SERRATED POLYPOSIS SYNDROME

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Introduction: *Fusobacterium nucleatum* (*Fn*) and other species typical of the oral cavity are implicated in colorectal oncogenesis. The pathogenesis of serrated polyposis syndrome (SPS), the most common polyposis syndrome, remains unknown, and to date, the oral and gut microbiota of these patients has not been characterized.

Aims & Methods: The primary objective of this study was to describe and compare the composition of the oral and gut microbiota in individuals with SPS versus individuals without polyps, with sporadic serrated polyps, and with adenomas.

A prospective, observational, and bicentric study was conducted. Clinical and endoscopic data were collected, along with information obtained from a questionnaire on diet and dental health. Additionally, saliva and stool samples were obtained from 126 individuals divided into four groups: SPS (n=32), serrated (n=30), adenoma (n=32), and control (n=32). The microbiota was analyzed using 16S rRNA amplicon sequencing, and qPCR was performed for the detection of *Fn*.

Results: The main clinical, dietary, and dental health characteristics of the groups are presented in Table 1. Regarding the oral microbiota, no differences were observed in alpha diversity (species variety) among the groups.

However, the composition of the oral microbiota in the SPS group showed significant differences compared to the other groups, as evidenced by a greater unweighted UniFrac distance (differences in bacterial taxonomy) (p<0.001). In the gut microbiota, the adenoma group exhibited lower species richness than the control group, according to the Chao1 index (p=0.03), while no differences were found in the other groups.

Statistically significant differences in beta diversity, measured using unweighted UniFrac distance, were observed between the SPS group and the adenoma and control groups (p<0.001).

Regarding taxonomy, *Porphyromonas gingivalis*, an oral bacterium associated with colorectal cancer (CRC) development and progression, showed higher oral abundance in the SPS group compared to the control and adenoma groups (p<0.05; FDR p=0.104). *Treponema denticola* and *Prevotella intermedia*, two oral bacteria related to CRC, were more abundant in the SPS group than in the control group (p<0.05; FDR p=0.09).

Additionally, *Peptoanaerobacter stomatis*, an oral bacterium linked to CRC via the serrated pathway, was more prevalent in the SPS group compared to the serrated and adenoma groups (p<0.05; FDR p=0.26).

No differences in the abundance of *Fn* were found among the groups, either in saliva or stool samples.

Conclusion: The oral and gut microbiota of individuals with SPS exhibit a distinct bacterial composition compared to individuals without polyps and those with adenomas. Furthermore, certain oral bacteria associated with colorectal carcinogenesis are increased in SPS patients.

Disclosure: Nothing to disclose.

	SPS (n=32)	Serrated (n=30)	Adenoma (n=32)	Control (n=32)	Total (n=126)	p-value
Age, years	61 [57-67]	63 [56-66]	59,5 [54-67]	58,5 [53-64]	61 [55-66]	0,28
Sex, male	17 (53,1%)	13 (43,3%)	17 (53,1%)	17 (53,1%)	64 (50,8%)	0,83
Body mass index, kg/m²	27 [26-32]	27,8 [25-32]	28,3 [25-31]	27 [24-32]	27,7 [25-32]	0,99
Tobacco						<0,001
Active smoker	18 (56,3%)	12 (40%)	6 (18,8%)	3 (9,4%)	39 (31%)	
Former smoker	3 (9,4%)	7 (23,3%)	10 (31,3%)	18 (56,3%)	38 (30,2%)	
Non-smoker	11 (34,4%)	11 (36,7%)	16 (50%)	11 (34,4%)	49 (38,9%)	
Tobacco, pack-years	25 [17-45]	34 [17-40]	15 [9-40]	7,3 [4-33]	24 [11-40]	0,03
Medication use						
PPIs	5 (15,6%)	7 (23,3%)	7 (21,9%)	5 (15,6%)	24 (19%)	0,79
ASA or NSAIDs	10 (31,3%)	3 (10%)	2 (6,3%)	7 (21,9%)	22 (17,5%)	0,035
Statins	16 (50%)	6 (20%)	4 (12,5%)	10 (31,3%)	36 (28,6%)	0,006
Recommended physical exercise, yes	15 (46,9 %)	10 (33,3%)	20 (62,5%)	13 (40,6%)	58 (46%)	0,12
History of periodontitis, yes	11 (34,4%)	9 (30%)	7 (21,9%)	5 (15,6%)	32 (25,4%)	0,31
Consumption frequency, g/day	149 [83-205]	131 [106-186]	137 [87-203]	129 [83-189]	136 [88-195]	0,89
Vegetables and greens	48 [29-64]	45 [26-77]	60 [43-89]	43 [25-63]	46 [31-68]	0,047
Fish and seafood	56 [36-75]	49 [33-104]	60 [43-94]	47 [21-74]	54 [33-80]	0,18
Red or processed meat	6,3 [3,6-22]	10,7 [7,2-25]	10,7 [7,2-25]	7,1 [2,1-13,3]	7,1 [3,5-18]	0,002
Cured meats	29 [10-43]	29 [15-43]	29 [16-35]	26 [10-43]	29 [13-43]	0,94
Fat and oils	4 [0,7-10]	5,3 [5,3-15]	5,3 [2,7-11]	5,3 [0-11]	5,3 [2,7-11]	0,046
Processed foods						

Table.

PP0732

SLC26A9 PROMOTES LIPID METABOLISM BY INTERACTING WITH RRBP1 IN COLORECTAL CANCER

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Introduction: Lipid metabolism dysregulation plays a crucial role in colorectal cancer (CRC) progression. Our perivious study found that upregulation of Slc26a9 promotes colorectal tumorigenesis (Zhang M, *et al.* Cell Death Discov. 2024). This study investigates the role of Slc26a9 in regulating lipid metabolism through the RRBP1-HSPA5 axis in CRC.

Aims & Methods: In this study, we assessed the expression of Slc26a9 in mouse colorectal cancer tissues using immunohistochemistry and immunofluorescence (IF). Slc26a9-interacting proteins in Colo201 cells, such as RRBP1, were identified through immunoprecipitation followed by mass spectrometry (IP-MS). Bioinformatic analysis of the TCGA and GEO databases was performed to evaluate the correlation between Slc26a9 and RRBP1 expression and their clinical significance. Protein interaction sites were predicted using Protein-Protein Docking.

Furthermore, the effects of Slc26a9 overexpression on mitochondrial metabolism, lipid metabolism, the RRBP1-HSPA5 pathway, and endoplasmic reticulum (ER) stress were investigated using Oil Red O staining, Nile Red staining, qRT-PCR, Western blotting (WB), co-immunoprecipitation (co-IP), and IF.

Results: In this study, we employed the widely recognized AOM/DSS protocol to establish a colorectal cancer model in C57BL/6J mice. We observed a significant upregulation of Slc26a9 in colorectal cancer tissues compared to adjacent non-cancerous tissues, consistent with our previous findings in human colorectal cancer samples (Zhang M, *et al.* Cell Death Discov. 2024).

To investigate the precise molecular mechanisms by which Slc26a9 regulates the development of CRC, we performed IP-MS to identify Slc26a9-interacting proteins in Colo201 cells. RRBP1 was found to have the highest

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binding affinity with Slc26a9 and has been shown to localize at mitochondria–ER contact sites, where it plays a role in regulating organelle interactions. Moreover, GO and KEGG enrichment analyses indicated that Slc26a9 is associated with lipid metabolism in CRC cells. Overexpression of Slc26a9 also promoted endoplasmic reticulum stress, mitochondrial energy metabolism, and lipid accumulation in Colo201 cells.

Moreover, Slc26a9 enhanced the expression of markers related to endoplasmic reticulum stress (PERK-ATF4, IRE1α-XBP1, and ATF6), mitochondrial metabolism (Mitofilin, MFN1, and COXIV), and lipid metabolism (FASN, ACC, and SCD1). Moreover, Slc26a9 upregulation also enhanced the expression of the endoplasmic reticulum stress–related signaling pathway RRBP1/HSPA5.

Protein-protein docking analysis revealed the formation of multiple hydrogen bonds between Slc26a9 and RRBP1, indicating a high degree of structural complementarity and strong binding affinity between the two proteins. Co-IP and IF experiments demonstrated that Slc26a9 interacts with RRBP1 and co-localizes with it.

In addition, bioinformatics analysis of GEO dataset (GSE39582) showed a positive correlation between Slc26a9 and RRBP1 expression in human CRC tissues, with their co-high expression predicting poor overall survival.

Conclusion: These findings highlight that Slc26a9 interacts with RRBP1 to activate the HSPA5-mediated ER stress response, driving mitochondrial metabolism-lipid metabolic reprogramming and promoting CRC progression. Slc26a9 and RRBP1 may serve as potential prognostic biomarkers, and their axis represents a novel therapeutic target for CRC.

Disclosure: Nothing to disclose.

PP0733

CLINICAL CHARACTERISTICS AND RISK FACTORS OF COLORECTAL TUMORS IN YOUNG JAPANESE

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Introduction: The number of colorectal cancer (CRC) cases and deaths in Japan is on the increase and is higher than in Western countries. In Europe and US, the increase in CRC among young people is a problem, in contrast to the decreasing trend among middle-aged and older people. In Japan, there is concern about the increase in CRC, and to prevent colorectal tumors (CRTs) in young people aged 40 years or younger, it is necessary to urgently understand the current situation and to carry out efficient screening by clarifying risk factors, together with CRC screening using fecal immunochemical test (FIT).

Aims & Methods: In this study, we conducted a retrospective study of the clinicopathological characteristics and risk factors of 303 CRTs diagnosed by colonoscopy (CS) at our institution in patients aged 40 years or younger.

1. In the CRT group, the pathological diagnosis and location were investigated.
2. The examination period was divided into early (2012-2017) and late (2018-2024), and the frequency of tumor detection was examined.
3. The association of various background factors (family history of CRC, body mass index (BMI), hypertension (HT), dyslipidemia, diabetes, alcohol drinking, smoking and exercise habits) as risk factors for tumors was statistically investigated. SPSS29 (IBM) was used for statistical analysis.

Results: (1) CRTs included 237 adenomas, 38 serrated adenomas and 28 colorectal carcinomas (24 intra-mucosal carcinomas, 3 invasive carcinomas and 1 neuroendocrine tumor). Solitary tumors were 63.7% and 71.3% (89.3% for CRC) were located in the left colon.

(2) The frequency of tumor detection was markedly increased: 2.04% in early male (M)/ 20.2% in late M and 0.29% in early female (F)/ 16.4% in late F.

(3) Risk factors: overall risk factors for CRTs [odds ratio (OR) 95% confidence interval (CI)] were 2.998 (1.244-7.176) for family history of CRC, 1.786 (1.007-3.168) for BMI>30, 3.875 (1.082-13.878) for BMI>35, 2.456 (1.337-4.511) for HT, 1.596 (1.153-2.210) for alcohol drinking, 2.111 (1.306-3.412) for >20g/day alcohol drinking, 1.635 (1.170-2.285) for smoking.

In M, risk factors for CRTs were family history of CRC 2.933 (1.036-8.300), HT 2.240 (1.124-4.463), alcohol drinking 1.760 (1.182-2.619, >20 g/day alcohol drinking 2.409 (1.402-4.138)) and in F, smoking 2.724 (1.223-6.070).

In Logistic Regression, which added age and sex to the factors that were significant in univariate analysis, the risk factors for CRTs were: family history of CRC 3.542 (1.376-9.119), HT 2.032 (1.096-4.147), alcohol drinking 1.524 (1.076-2.159), smoking 1.665 (1.148-2.415).

Conclusion: The frequency of CRTs in young people aged 40 years or younger is increasing in Japan. Family history of CRC, HT, alcohol drinking and smoking are risk factors, and along with FIT, efficient screening based on these background factors is needed.

Disclosure: Nothing to disclose.

PP0734

THE CCL7–SOX4–CD44 AXIS ESTABLISHES A TRANSCRIPTIONAL PROGRAM TO PROMOTE COLORECTAL CANCER METASTASIS

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Introduction: Metastasis is the leading cause of poor prognosis and high mortality in colorectal cancer (CRC). Chemokines play multifaceted roles in cancer immunity and progression. CCL7 is upregulated in CRC, but how CCL7 regulates CRC metastasis at the molecular level remains unclear.

Aims & Methods: In this study, we investigate the regulatory mechanisms and clinical significance of CCL7-mediated CRC metastasis.Immunohistochemistry (IHC) assay was performed to detect the expression of CCL7 and SOX4 in CRC tissue, moreover, the correlation between their expression and tumor distant metastasis, survival, and prognosis was analyzed. To explore their impact on CRC migration, invasion and metastasis, functional experiments and animal experiments were performed in vitro and in vivo, respectively. Gene expression changes were studied by real-time quantitative PCR (qRT–PCR), Western Blot (WB) and RNA-seq analysis. Chromatin Immunoprecipitation sequencing (ChIP-seq), luciferase reporter assay and Electrophoretic Mobility Shift assay (EMSA) were implemented to explore the mechanisms.

Results: Two clinical cohorts showed that CCL7 and SOX4 were upregulated in CRC tissues and correlated with lymphatic metastasis, distant metastasis, and poor prognosis. Functionally, CCL7 and SOX4 promoted CRC

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cell proliferation, invasion and metastasis in vitro and in vivo, respectively. Rescue assays revealed SOX4 as a downstream molecule involved in CCL7-mediated CRC metastasis. Through binding to the promoter region of CD44 directly, SOX4 activated its transcription and promotes CRC metastasis, which we proved via promoter analysis assays and analysis of the results of ChIP-seq and RNA-Seq. The Wnt/β-catenin signaling pathway contributed to the regulation of these processes.

Conclusion: The chemokine CCL7 induces elevated expression of SOX4, which subsequently promotes CRC metastasis by transcriptionally upregulating CD44. The CCL7-SOX4-CD44 axis could be applied as a potential target and strategy for preventing and treating CRC metastasis in clinical settings.

Disclosure: Nothing to disclose.

PP0735

SORD CORRELATES WITH ANTITUMOR IMMUNITY AND AN IMPROVED PROGNOSIS IN COLON CANCER

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Introduction: The polyol pathway (PP) bypasses glycolysis and is over-activated following prolonged hyperglycaemia. Of the two key enzymes in the PP, the expression of SORD has previously been reported in various cancers. However, it remains unclear whether SORD is correlated with prognosis of CRC patients and the underlying mechanism of SORD affecting CRC progression.

Aims & Methods: We collected tissue samples and clinical information from colorectal adenoma and CRC patients in a local cohort, as well as RNA-Seq and clinical data from The Cancer Genome Atlas (TCGA) and Genotype Tissue Expression (GTEx) databases. We then compared *SORD* expression between tumour tissues and normal tissues and across different clinicopathological stages, and we tested for correlations between *SORD* expression and survival rate. Gene set enrichment analysis (GSEA) and gene co-expression analysis were conducted to determine the potential biological functions of *SORD*. Single sample GSEA (ssGSEA) method was utilised to evaluate immune cell infiltration correlated with *SORD* expression in TCGA CRC data.

Results: *SORD* expression was elevated in colorectal adenoma and CRC tissues, and lower *SORD* expression was correlated with more advanced N, M, and pathological stages and with lymphatic invasion. However, CRC patients with higher *SORD* expression also had a favourable progression-free interval (PFI). *SORD* expression was negatively correlated with angiogenesis, endothelium development, growth factor binding, and the Ras, Wnt, and TGF-β signalling pathways. The expression of angiogenesis and proliferation-promoting genes was also negatively correlated with *SORD* expression. *SORD* expression level was positively correlated with T helper 2 cell enrichment and negatively correlated with plasmacytoid DC enrichment in CRC microenvironment.

Conclusion: Higher *SORD* expression was significantly correlated with an improved PFI in CRC patients. Moreover, *SORD* may participate in CRC development and progression by affecting Ras, Wnt, and TGF-β signalling and regulating immune microenvironment.

Disclosure: Nothing to disclose.

PP0736

IMMUNOREACT 20: RECTAL CARCINOMAS WITH MUCINOUS COMPONENTS ARE ASSOCIATED TO A DIFFERENT IMMUNOLOGICAL PATTERN IN THE NORMAL MUCOSA

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Introduction: Rectal adenocarcinoma with mucinous components is a relatively infrequent kind of rectal cancer which exhibits two different histotypes: mucinous adenocarcinoma and signet-ring cell carcinoma. When compared to classic adenocarcinoma, these subtypes are associated with a younger age of presentation and a worse prognosis.

Aims & Methods: The aim of this study was to analyse the features of the mucosal immune microenvironment pattern of rectal carcinomas with intra- or extra-cellular mucinous components.

Patients enrolled for IMMUNOREACT 1 and 2 trials (NCT04915326 and NCT04917263, respectively) further analysed based on the subtype of cancer, comparing not otherwise specified (NOS) adenocarcinomas with rectal adenocarcinomas with mucinous components. A panel of immune markers was investigated at immunochemistry: CD3, CD4, CD8, CD8β, Tbet, FoxP3, PD-L1, MSH6, and PMS2 and CD80.

Flow cytometry was used to determine the proportion of epithelial cells expressing CD80, CD86, CD40, HLA ABC or HLA DR and the proportion of activated CD8+ T cells, CD4+ Th1 cells, and Treg. Non parametric tests were used.

Results: In this cohort, 9 patients had an adenocarcinoma with mucinous components, while 115 had a NOS adenocarcinoma. Rectal adenocarcinomas with mucinous components had a higher proportion of CD4+CD25+ T cells (activated T helpers) in healthy mucosa surrounding rectal cancer (p=0.006) and a higher proportion of CD8+ T cells (cytotoxic lymphocytes) in cancer tissue (p=0.027) than NOS adenocarcinomas.

Conclusion: Rectal adenocarcinomas with mucinous components are a distinct subtype of adenocarcinoma and their peritumoral immune microenvironment is associated with an impaired cytotoxic infiltration and a higher activation of T helpers.

These findings may suggest these subtype of rectal cancer cells may be favoured by a less active immunesurveillace that can easily escaped explaining the association of these cancer subtypes with poorer outcomes.

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Disclosure: Nothing to disclose.

PP0737

THE IMPACT OF LINE-1 METHYLATION ON THE DEVELOPMENT OF TUBULAR ADENOMAS: A PROSPECTIVE STUDY

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Introduction: Colorectal cancer is the third most common cancer type worldwide. Changes in DNA methylation is a potential mechanism underlying CRC initiation. Long interspersed element-1 (LINE-1) is a repetitive DNA retrotransposon that constitutes approximately 17% of the human genome. The level of LINE-1 methylation is often used as a marker of global DNA methylation.

Aims & Methods: In this study, we investigated LINE-1 methylation at three CpG islands of the LINE-1 promotor in serrated adenoma, tubular adenoma and tumor tissue and surrounding normal tissue.

Results: In tubular adenoma tissue, the mean LINE-1 methylation level was 72.93±7.73. The surrounding normal tissue exhibited a significantly greater LINE-1 methylation level (81.77±4.03) than did the pathological tissue (p<0,0001). The adenocarcinoma tissues had a mean LINE-1 methylation level in the CpG island of 66.33±10.27. The LINE-1 methylation level in the surrounding tissue was significantly greater (83.76 ±3.49) than that in the pathological tissue (p<0,0001). For serrated adenoma no difference between surrounding and pathological tissues mean LINE-1 methylation levels were observed.

Conclusion: Our findings suggest the different impacts of DNA methylation changes on the formation of different colorectal adenomas (tubular versus serrated) and their further progression to carcinoma. In addition, our findings confirm the potential of LINE-1 methylation levels as an early CRC and premalignant diagnostic marker.

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Disclosure: Nothing to disclose.

PP0738

EVALUATION OF GENES EXPRESSION OF BOTH IMMUNOGLOBULIN SUPERFAMILY DOMAINS AND HIGH MOBILITY GROUP IN DIAGNOSIS AND PROGNOSIS OF COLORECTAL CANCER

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Introduction: Colorectal cancer (CRC) is one of the most common digestive tract malignant tumors, and its incidence rate and mortality rate rank third among all malignant tumors. The IMMUNOGLOBULIN SUPERFAMILY DOMAINS genes has been reported to be hypermethylated in human colorectal cancer (CRC) cell lines and primary tumor tissues.
On the other hand High mobility group genes are characteristically present at high levels in a variety of human cancers including colorectal, ovarian, breast, lung, and pancreatic cancer.

Aims & Methods: The aim of the work to evaluate the gene expression of both leucine-rich repeats and immunoglobulin-like domains 1 (LRIG1) (as an example of immunoglobulin sperfamily domains and High mobility group AT-hook 2 (HMGA2) as an example of high mobility group as diagnostic and prognostic markers in cases of colorectal cancer.
In this study forty-five patients pathologically approved with colorectal cancer were prospectively analyzed for LRIG1 and HMGA2 genes expression at tissue level. SYBR Green Real-Time RT-PCR technique was used for gene expression analyses and β-actin gene was considered as a reference gene for data normalization. Additionally, appropriate statistical analyses were used to assess the expression of LRIG1 and HMGA2 in diseased and adjacent healthy tissues.

Results: HMGA2 mRNA gene expression was over-expressed in CRC tissue than normal tissue (median fold change and median Log^2 ratio are [9.3(0.2, 100.4) and 3.1(-2.6, 6.7), respectively]with positive correlation between HMGA2 and poor differentiation ,more advanced stage and lymph node metastasis(p> 0.05) but not with distant metastasis (p >0.05). The expression of LRIG1 mRNA gene in tumor tissue was low expressed in colorectal tumor tissues (Median fold change and median Log2 ratio are [0.29(0.01, 4.14) and -1.8(-6.1, 2.1), respectively but no correlation of the LRIG1 gene with the the stage, Differentiation and metastasis of the CRC (P >0.05). There is no correlation of both genes neither with tumor biomarkers (CA19-9, CEA) nor with laboratory findings (WBCs, Hemoglobin, Platelet, INR, Albumin, Creatinine) P>0.5%). The overall survival showed no significant association with any genetic expression of both genes (P >0.05).

Conclusion: LRIG1 low expression may be considered for diagnosis but not valuable as prognostic marker in colorectal cancer patients. However, HMGA2 overexpression may be valuable as both diagnostic and prognostic markers in CRC.

Disclosure: No conflict of interest to be disclosed.

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PP0739

CLINICAL FEATURES OF DEPRESSED TYPE COLORECTAL NEOPLASMS AND USEFULLNESS OF MAGNIFYING ENDOSCOPY AND ENDOCYTOSCOPY

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Introduction: In the carcinogenic pathway of colorectal cancer, the first concept is the “adenoma-carcinoma sequence” and the second concept is the “ serrated pathway “. In these concepts, adenomas and serrated lesions gradually progress to cancer. The third concept is the “de novo pathway,” in which cancer develops without adenomas or serrated lesions and is also characterized by high malignancy and rapid progression.

We have endoscopically characterized depressed colorectal cancers considered to be of the “de novo pathway” and validated the usefulness of pit pattern and EC classification.

Aims & Methods: A total of 54819 colorectal neoplasms were resected endoscopically or surgically in our institution from April 2001 to December 2024. They were subjects of this study. Advanced cancers were excluded. Of these, 1694 lesions were submucosally invasive (T1) carcinomas, 7330 lesions were classified as high-grade dysplasia and 39977 lesions as low-grade dysplasia. Morphologically, they were divided into 3 types: depressed, flat, and protruded type comprising 1.0 (%), 46.4 (%),and 52.6 (%), respectively.

Specifically, the distinctive features of depressed type neoplasms concerning their pit pattern and EC classification were examined. EC classification is a classification for an endocytoscopy that can evaluate pathology, such as nuclear atypia and glandular structure, by magnifying stained cells. EC1a is diagnosed as normal mucosa, EC1b is diagnosed as hyperplastic polyps, and if slit-like glandular ducts or spindle-shaped nuclei are present, it is classified as EC2 and diagnosed as an adenoma or intramucosal carcinoma. If irregularly shaped glandular ducts or nuclei with a high N/C ratio are found, it is classified as EC3a, and diagnosed as intramucosal carcinoma or SM lightly invasive carcinoma. If the glandular ducts are indistinct and the nucleus is irregularly shaped and darkly stained, it is classified as EC3b and diagnosed as SM highly invasive carcinoma.

Results: Most of the flat (91.5%) and protruded type (94.9%) lesions showed type IIIL or IV pit pattern, which are common findings in adenomas. On the other hand, most of the depressed type lesions (92.6%) classified as type IIIs, VI or VN pit patterns, are typically seen in carcinomas. As for endocytoscopy, most of flat and protruded type lesions were identified as EC2 corresponding to adenomas. In contrast, relatively high rates of depressed type lesions were categorized as EC3a (40.3%) and EC3b (52.4%), consistent with invasive carcinomas.

The proportion of T1 carcinomas in depressed type lesions was 58.7%, whereas in flat and protruded type lesions it was 2.5% and 2.5%, respectively. Additionally, 12.0% of depressed type lesions less than 5mm in diameter were T1 carcinomas, whereas flat and protruded type lesions were 0.01% and 0%, respectively.

Conclusion: This study revealed that depressed lesions were classified as type IIIs, VI or VN pit patterns on magnifying endoscopy and as type EC3a or EC3b on endocytoscopy. Lesions with these features or pit patterns tend to invade the submucosal layer even if they are small size. Despite their high potential for malignancy, depressed lesions are often ignored. However, it is critical to recognize their features and accurately diagnose lesions based on morphology, pit pattern, and EC classification.

Disclosure: Shin-ei Kudo and Masashi Misawa received speaking honoraria from Olympus Corporation and have an ownership interest of the products of Cybernet Systems. Tetsuo Nemoto received a research grant from Olympus Corporation for other studies. Other authors have nothing to declare.

PP0740

SERUM METABOLIC SIGNATURES ACROSS THE COLONIC INFLAMMATORY-NEOPLASTIC SPECTRUM

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Introduction: Patients with inflammatory bowel disease, particularly those with colonic inflammation, are at increased risk of colorectal dysplasia and cancer. Colitis-associated carcinogenesis is believed to follow an inflammatory-driven pathway, potentially modulated by microbiome-derived metabolites, especially those related to amino acid and nitrogen metabolism. However, the precise metabolic signatures underlying this transition remain undefined. We aimed to characterize serum metabolomic profiles across five clinical groups - healthy controls (HC), ulcerative colitis (UC), Crohn’s disease (CD), advanced adenoma (AA), and colorectal cancer (CRC) - to identify disease-specific metabolic shifts and metabolite associations across the inflammatory-neoplastic spectrum.

Aims & Methods: Patients were recruited from the Digestive Center of the National University Hospital, Singapore. Serum was collected and profiled using targeted liquid chromatographic tandem mass spectrometry, quantifying 80 metabolites spanning amino acids, biogenic amines, indoles, kynurenines, and nitrogenous bases. Unsupervised principal component analysis (PCA) was used to assess global metabolic validation, followed by pairwise subgroup comparison using Wilcoxon rank-sum test with false discovery (FDR) correction.

Results: A total of 125 participants were enrolled (25 per group, mean age 61.5 years, 41% female). PCA revealed distinct clustering by disease states, with CRC showing the greatest divergence from healthy controls (ANOSIM $R = 0.264$, $p<0.01$). Comparing IBD and CRC to healthy control, 40 metabolites were significantly differentially expressed (FDR <0.05) with enrichment patterns suggestive of class-based divergence. Amino acid-related metabolites (eg. glutamine, GABA, spermine) were progressively downregulated in CRC. Biogenic amines and microbial indole derivatives (eg. 1-methyhistamine, indole-3-propionic acid) showed distinct alterations, with 1-methyhistamine elevated in IBD, and reduced in CRC. Kynureine pathway intermediates (eg. kynurenine, xanthurenic acid) also displayed differential expression between inflammatory and neoplastic states.

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Conclusion: Serum metabolomic profiling distinguishes IBD from CRC with both disease-specific and progressive changes. The marked depletion of amino acids (glutamine, GABA, spermine) and shifts in microbial or immune-associated metabolites (e.g., 1-methylhistamine, indole-3-propionic acid, kynurenines) support a model of colitis-associated carcinogenesis mediated by host–microbial metabolic crosstalk. These metabolites may serve as candidate biomarkers for early detection, disease monitoring, and risk stratification in IBD patients at risk for progression to CRC.
Disclosure: Nothing to disclose.

PP0741

ANALYSIS OF RESECTION MARGINS AND PROGNOSTIC RISK FACTORS FOR T1 COLORECTAL CANCERS TREATED BY ENDOSCOPIC SUBMUCOSAL DISSECTION

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Introduction: Endoscopic submucosal dissection (ESD) is widely recognized as the preferred technique for removing large benign colorectal lesions and those with a risk of submucosal invasive cancer (SMIC), with the goal of achieving en-bloc resection. R0 resection is crucial for enabling organ-preservation strategies, particularly in endoscopically resected T1-colorectal cancer (T1-CRC).
Aims & Methods: This study aims to evaluate R0 resection rates in a large cohort of colorectal ESDs by assessing vertical and lateral margins and identifying predictive factors for vertical margin involvement.
Furthermore, the presence of other prognostic risk factors for lymph node metastasis (LNM) will be analyzed in this large cohort of T1-CRC. This is a multicenter prospective cohort study in 13 centers, including all colorectal ESDs between 09/2019 and 09/2024. Procedural and histological outcomes were analyzed using univariate and multivariate analysis.
Results: 7495 large colorectal polyps were included, of which 781 were T1-CRCs with a median size of 45 [35-65] mm. En bloc resection occurred in 96.7% and R0 resection in 74.9%. Lateral resection margins were free in 92.5%, with positive margins for adenocarcinoma in only 0.8% (low grade dysplasia in 2.6%, high grade in 1.2%, missing data in 2.6%). Vertical margins were free in 80.3%, and decreased significantly when deep SMIC was present to 75.0%, compared to 95.7% for superficial SMIC.
However deep SMIC with up to 2000 µm invasion depth, demonstrated free vertical resection margins in still 80.7%, after which it gradually decreased to 73.1% and 67.7% at an infiltration depth of 2000-3000 µm and >3000 µm respectively. Predictive factors for free vertical resection margins include lesion size >50mm (OR=1.66, [1.01;2.71], p=0.045) and rectal location (OR=1.86, [1.04;3.3], p=0.035).
Significant difference was observed in the presence of prognostic risk factors for LNM between superficial and deep SMIC, with lympho-vascular invasion (LVI) being present in 9.6% and high-grade tumor budding (Bd) in 17.1% in superficial SMIC, compared to 25.0% and 30.0% respectively, in deep SMIC. However, absence of LVI, Bd or poor differentiation was noted in 75.9% of superficial and 61.7% of deep T1-CRC. Concerning deep SMIC, no relation was observed in the presence of these risk factors with respect to the depth of infiltration.
Conclusion: Following ESD for T1-CRC, the likelihood of achieving free vertical resection margins decreases significantly when invasion depth exceeds 2000 µm. Careful lesion selection through optical diagnosis is therefore critical. To push organ preservation, intermuscular dissection should be considered for T1-CRC with deeper invasion in the rectum. Achieving R0 resection remains paramount, as additional prognostic factors are present in approximately 38% of cases, irrespective of invasion depth.
Disclosure: Nothing to disclose.

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PP0742

DEVELOPMENT AND EXTERNAL VALIDATION OF A STACKING-BASED AI MODEL FOR PREDICTING LYMPH NODE METASTASIS IN T1 COLORECTAL CANCER USING WHOLE SLIDE IMAGES

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Introduction: Accurate prediction of lymph node metastasis (LNM) in submucosal invasive (T1) colorectal cancer (CRC) is essential for determining the need for additional surgery following endoscopic resection. Conventional histopathological assessments, however, are limited by interobserver variability. This study aimed to develop and externally validate a stacking-based artificial intelligence (AI) model integrating histopathological whole slide images (WSIs) and clinical parameters to predict the risk of LNM in T1 CRC.

Aims & Methods: This retrospective study included 1,056 patients with histologically confirmed T1 CRC treated at Showa Medical University Northern Yokohama Hospital. Of these, 593 cases were used for model training and 100 for internal testing. External validation was conducted on 25 cases from two independent institutions.

The proposed stacking ensemble model incorporated the Multiple Instance Self-Training (MIST) score derived from WSIs, patient sex, and tumor location (colon vs. rectum). Base learners included logistic regression, XGBoost, and random forest, with a meta-learner synthesizing their outputs to generate the final LNM prediction. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC) in both internal and external cohorts.

Results: LNM was observed in 69 (11.6%) training cases, 15 (15.0%) test cases, and 2 (8.0%) external validation cases. The model achieved an AUC of 0.6965 in the internal test cohort and 0.7500 in the external validation cohort, demonstrating stable and generalizable predictive performance.

A representative case involved a 64-year-old man with a 16-mm type 0-Is lesion in the sigmoid colon resected via endoscopic mucosal resection (EMR). Histopathology showed well-differentiated tubular adenocarcinoma (tub1 > tub2), submucosal invasion depth of 3,300 μm (T1b), tumor budding grade BD1, Ly0, V1a, and negative resection margins (HM0, VM0). Despite guideline-based recommendations prompting additional surgical resection, no LNM was found (N0, 0/16), and the patient remained recurrence-free five years postoperatively—consistent with the AI model's prediction.

Conclusion: The stacking-based AI model exhibited robust and generalizable performance in predicting LNM in T1 CRC, both internally and across external institutions. This approach offers a complementary tool to conventional pathology and may enhance individualized decision-making regarding the necessity of additional surgical intervention.

Disclosure: Nothing to disclose.

PP0743

ANALYSIS OF COLORECTAL LESIONS FOLLOWING THE IMPLEMENTATION OF COLORECTAL CANCER SCREENING IN DENMARK

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Introduction: Colorectal cancer (CRC) remains the second most common cause of cancer-related mortality worldwide (1). The introduction of the Danish national CRC screening program in 2014 aimed at reducing mortality rates (2) has marrkedly increased colonoscopies and polypectomies (3,4). A a comprehensive description of the lesions removed has not yet been investigated.

Aims & Methods: Nationwide, cross-sectional study of all colorectal lesions resected endoscopically in patients ≥18 years between January 2014 and December 2021. Data were obtained from the Danish National Patient Register, Pathology Register and Screening Database. Prevalence of histological sub-types and predictors of high-grade dysplasia (HGD) and adenocarcinoma were analyzed with multivariable logistic regression.

Results: Among 663,624 individuals undergoing 1,074,236 colonoscopies or sigmoidoscopies, 194,941 patients (median age 67 years, 58.6% male) had a total of 504,386 lesions removed. Tubular adenomas (TA) accounted for 66.8% of lesions.

The proportion of advanced colorectal polyps—defined as lesions ≥10 mm, adenomas with high-grade dysplasia, or sessile serrated lesions with any dysplasia—declined from 22.7% in 2014 to 10.2% in 2021.

High-grade dysplasia (HGD) was strongly associated with size ≥10 mm (OR 27.1 for 10–19 mm; OR 71.1 for 30–39 mm), and in diminutive lesions (1–5 mm), with tubulovillous or villous histology (OR 6.09).

Adenocarcinoma was identified in 0.65% of lesions and was predominantly left-sided (83.9%), with highest odds observed in the sigmoid colon (OR 6.33) and rectum (OR 6.00), compared to the caecum.

Sex was significantly associated with the risk of adenocarcinoma at the lesion level, but this effect was age-dependent. At age 40, women had significantly higher odds than men (adjusted OR 1.48), but the difference diminished with increasing age, and from age 75 onward, no statistically significant difference between sexes was observed.

Conclusion: Lesion size was the strongest predictor of high-grade dysplasia, outweighing histologic subtype once lesions exceeded 10 mm. At lesion level, the risk of malignancy was higher in younger women than in men of the same age, with the difference progressively diminishing with age.

The introduction of the screening programme coincided with a marked reduction in advanced lesions over time. The high predominance of left-sided cancerous lesions highlights the need to optimize the detection of right-sided neoplasia.

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PP0744

LOST EXPRESSION, FOUND FAMILIES: HOW IHC REVEALS LYNCH SYNDROME BEYOND AGE AND FAMILY HISTORY

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Introduction: Lynch syndrome (LS) is a heritable autosomal dominant genetic condition caused by mutations in several DNA mismatch repair (MMR) genes, MLH1, MSH2, MSH6, PMS2 and EPCAM. Affected individuals have an increased life-long risk of developing cancers especially colorectal cancer (CRC) and endometrial cancer (EC). LS accounts for approximately up to 3% of all CRC. Traditionally, diagnosis was based on clinical criteria (Amsterdam II, Bethesda and revised Bethesda), which have proven to be of limited applicability in clinical practice. In recent years tumor testing via immunohistochemistry (IHC) of all CRCs for the expression of MMR proteins has successfully integrated into clinical care. Universal screening through IHC of CRC tumor samples is cost-effective and recommended by major guidelines, which also recommend its application to endometrial cancers.
Aims & Methods: Our aim was to assess the main differences between LS families where the first individual received a genetic diagnosis after IHC on tumor sample and LS families where the first individual received a genetic diagnosis due to meeting the Amsterdam II or revised Bethesda criteria. We obtained data (year of genetic test confirmation, first tumor diagnosis, family tree) of all our single center's LS patients who undergo surveillance in our Institute (n = 101 patients).
Accordingly, we grouped patients by family and identified each family's index case-defined as the first individual to receive a genetic diagnosis - and relatives of index cases.
Therefore, we divided the cohort into two groups: families whose the index case was identified after IHC on tumor sample and families whose the index case was identified due to meeting the Amsterdam II or revised Bethesda criteria. We subsequently evaluatedwhether patients identified through IHC would have also met the revised Bethesda or Amsterdam II criteria for genetic testing.
Results: We found that index cases identified through IHC (n =16; of which 8 on CRC tissue, 7 on EC tissue and 1 on lung cancer tissue) had a mean age of 57.93 years at the time of genetic diagnosis, whereas those identified through clinical criteria (n = 20) had a mean age of 46.35 years. As for the relatives, those whose index case was identified via IHC (n = 11) had a mean age of 45.72 years, while relatives of individuals diagnosed based on clinical criteria (n= 54) had a mean age of 37.96 years. Patients carrying pathogenic variants in MSH6 and PMS2 (leading to lower penetrance and a late onset of LS related cancers) were 8 out of 16 (50 %) in the IHC group and 1 out of 20 (5 %) in the clinical criteria group. Furthermore 8 out of 16 (50%) index patients diagnosed through IHC did not meet the Amsterdam II criteria.
Conclusion: The mean age of index cases diagnosed due to clinical criteria is lower than that of cases identified through IHC, probably because of the early onset of LS related cancers in these patients and since the clinical criteria require at least one case to be diagnosed at a young age (<50 years old). In contrast, universal immunohistochemistry (IHC) screening enables the detection of all patients with mismatch repair-deficient tumors, including those diagnosed later in life and those without an evident family history of cancer or carrying pathogenic variants in genes such as MSH6 and PMS2 with a lower penetrance. Our data strengthen the important role of IHC screening for MMR proteins loss of expression both in CRC and EC to find LS families that would not be possible to identify with clinical criteria only.
Disclosure: Nothing to disclose.

PP0745

CORRELATION BETWEEN KUDO'S PIT PATTERN AND FINAL HISTOLOGY IN CASE OF SCREEN-DETECTED COLORECTAL POLYPS – RESULTS FROM A NATIONWIDE COHORT STUDY

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Introduction: Appropriate knowledge of surface features of colorectal polyps is necessary for their adequate management. Data are limited regarding the diagnostic value of surface pattern-based prediction of histology.
Aims & Methods: This nationwide observational cohort study aimed to compare the optical diagnosis based on Kudo's pit pattern and the histologic result in case of colorectal polyps endoscopically removed during screening colonoscopies.
Data regarding Kudo's pit pattern of the lesions (Kudo I-V), and the final histological results were obtained from clinical records of screening colonoscopies which were prospectively collected in the registry of Hungarian National Public Health Institute. The level of agreement was characterized by Cohen's kappa coefficient. The feasibility of optical diagnostics to differentiate neoplastic and non-neoplastic lesions, as well as to identify submucosally invasive cancer (SMIC) were investigated. The potential effects of polyp size and location (left/right colon), and that of timing of colonoscopy (1st period between 2019 and 2021, and 2nd period between 2022 and 2023) were also evaluated.
Results: A total of 8289 polyps removed during screening colonoscopy were registered in the database between 2019 and 2023, from these 7187 were neoplastic (439 SMIC) and 1102 were non-neoplastic. Out of the 1095 serrated polyps 171 contained dysplasia. Non-neoplastic polyps had Kudo I pattern in 7.7%, Kudo II in 48.1%, Kudo III in 40.9%, Kudo IV in 3.2%, and Kudo V in 0.1% of the cases. Neoplastic lesions without submucosal invasion had Kudo I surface pattern in 2.0%, Kudo II in 7.5%, Kudo III in 71.2%, Kudo IV in 17.4%, and Kudo V in 1.7% of the cases, while in case of SMIC, the surface pattern was Kudo I in 0.2%, Kudo II in 0.7%, Kudo III in 15.7%, Kudo IV in 8.9%, and Kudo V in 74.5% of the cases.

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Kudo I-II category showed moderate agreement with non-neoplastic lesions (κ=0.440), irrespective of lesion size and location, and timing of screening colonoscopy. The level of agreement was only slight for serrated lesions (κ=0.168). Kudo V category demonstrated substantial agreement (κ=0.722) with SMIC, irrespective of lesion size, location, and timing. SMICs could be identified based on Kudo V category with 74.5% sensitivity, 98.5% specificity, 73.0% positive and 98.6% negative predictive value.

Conclusion: This is the first real-world data obtained from screening colonoscopies of the Hungarian national colorectal screening program. Optical diagnosis might support the differentiation of neoplastic and non-neoplastic lesions and can be especially helpful in identifying SMICs that might require a different management approach.

Disclosure: Nothing to disclose.

PP0746

FACTORS ASSOCIATED WITH TIME-TO-ADHERENCE IN COLORECTAL CANCER SCREENING: A LARGE LONGITUDINAL COHORT STUDY

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Introduction: Colorectal cancer (CRC) screening reduces cancer risk, yet real-world adherence to screening remains variable. We aimed to leverage a large healthcare organization longitudinal cohort to evaluate factors that influence the timeliness of adherence to screening guidelines.

Aims & Methods: We included average-risk Maccabi Healthcare Services members aged 50–75 years who were alive on 1 January 2025. Demographics, comorbidities, hemoglobin values, and dates of fecal immunochemical test(FIT), CT-colonography (CTC) and colonoscopy were extracted.

Adherence was defined as annual FIT, colonoscopy every 10 years, or CTC every 5 years.

Patients covered for >50 % versus ≤50 % of follow-up time by screening guidelines were compared with Wilcoxon rank-sum (continuous) and χ^2/Fisher's exact (categorical) tests.

In order to study factors associated with adherence times, longitudinal adherence was modelled with a gap-time Andersen–Gill Cox regression. The dependent variable was time (in periods) to the recurrent event active adherence. Adjusted hazard ratios (HRs) and 95 % confidence intervals (CI) were estimated with robust variance, clustering on patient ID to account for within-subject correlation.

Results: Our cohort comprised 590,550 adults at screening age, contributing 6,581,044 person-years of follow-up. In a baseline comparison between individuals whose proportion of time covered was less than 50%(n=282,347) vs more than 50%(n=308,203), the latter group was older (60.00 [55.00, 68.00] years vs 62.00 [57.00, 67.00] years, p-value<0.001) with a longer median follow up (9.0 [4.0, 17.0] years vs 11.0 [6.0, 16.0] years, p-value<0.001). Higher prevalence of hypertension (17.5 % vs 12.0 %, p-value<0.001), diabetes mellitus (8.2 % vs 5.8 %, p-value<0.001), pre-diabetes (22.4 % vs 17.5 %, p-value<0.001), and cardiac disease (4.8 % vs 3.4 %, p-value<0.001) were seen among the more compliant group. Consistent with greater adherence, endoscopic referral was more common at baseline in the more compliant group (6.2 % vs 1.0 %, p-value<0.001). Interestingly. Although the differences in distributions of sector, district and socioeconomic rank categories were significant, the absolute percentage differences across strata were small and unremarkable (<3 percentage points for any category).

A gap-time survival model demonstrated that each additional year of age was associated with longer intervals and a lower aHR for active adherence (aHR 0.94, 95 % CI 0.94–0.94). Men undergone screening tests marginally sooner than women (aHR 1.05, 95% CI 1.04–1.05). Adherence was lower in the orthodox sector (aHR 0.84, 95% CI 0.84–0.85). Metabolic or cardiovascular conditions translated into earlier adherence: pre-diabetes (aHR 1.63, 95% CI 1.62-1.63), diabetes (aHR 1.54, 95% CI 1.54-1.55), hypertension (aHR 1.22, 95% CI 1.21-1.23), cardiac disease (aHR 1.14, 95% CI 1.14-1.15), and obesity (aHR 1.10, 95% CI 1.10-1.11). Severe conditions and subjects included in severe disease registry were associated with lower adherence (aHR 0.33, 95% CI 0.30-0.38; and aHR 0.63, 95% CI 0.59-0.67, respectively). Recent higher hemoglobin levels delayed active adherence (aHR 0.97 per each increase of 1 g/dL, 95% CI 0.97-0.98).

Conclusion: Our results suggest that long term compliance for CRC screening is related to age as well as for chronic medical conditions, particularly metabolic and cardiovascular. Screening efforts should therefore be aimed at healthy subjects or those with minor contact with medical providers.

Disclosure: Nothing to disclose.

PP0747

DEVELOPMENT AND FUNCTIONAL EVALUATION OF NOVEL ENGINEERED ANTI-CD73 NANOBODY FOR COLORECTAL CANCER TARGETING

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Introduction: The ectoenzyme CD73, a key mediator of adenosine-mediated immunosuppression in cancer, contributing to tumor progression, immune evasion, and resistance to therapy, represents a promising therapeutic targets in gastrointestinal malignancy.

Aims & Methods: To address the limitations of monoclonal antibody therapies in solid tumors, this study reports the bioinformatic design, production, and functional evaluation of a novel nanobody-based recombinant protein engineered to target CD73 as new immune checkpoint in colorectal cancer. The construct was developed by grafting the complementarity-determining regions (CDRs) from the humanized monoclonal antibody IPH5301 onto a camelid nanobody scaffold to retain high binding affinity for CD73 while benefiting from the nanobody's small size and tissue penetration.

Structural validation through molecular modeling, energy minimization, and Ramachandran analysis confirmed high stability and proper folding of the engineered protein. Molecular modeling and docking simulations confirmed structural stability.

Results: Experimental expression in prokaryotic hosts and subsequent SDS-PAGE, western blot, and dot blot confirmed the presence and purity of the construct. Binding efficiency was validated via ELISA and flow cytometry, revealing a high CD73-binding rate of 90.7% in HT-29 colorectal cancer cells. The construct also demonstrated notable cytotoxicity, with an IC50 value of 23.51 µg/mL, and induced significant cancer cell death (49.08% apoptosis, p = 0.014).

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Conclusion: These results demonstrate successful design and functional characterization of a small, high-affinity anti-CD73 nanobody that holds promise to disrupt adenosine-driven immunosuppression for targeted immunotherapy in CRC. Also, it is offering a promising alternative to conventional monoclonal antibodies and merits further preclinical investigation for therapeutic development.
Disclosure: Nothing to disclose.

PP0748

IMPACT OF C-REACTIVE PROTEIN-TO-ALBUMIN RATIO ON PROGNOSIS IN COLORECTAL CANCER

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Introduction: Colorectal cancer (CRC) remains a significant public health concern, with its incidence steadily rising worldwide. Despite notable strides in treatment modalities, predicting disease progression and tailoring therapeutic approaches remains a challenge. In this context, identifying reliable and simple prognostic factors is crucial for optimizing disease outcomes.

Aims & Methods: The aim of this study was to investigate the predictive and prognostic value of C-reactive protein-to-albumin ratio (CAR) in CRC. We conducted a retrospective and descriptive study including patients diagnosed with primary colon or rectal cancer, over an 8-year period [January 2014-June 2022]. Patient demographics, inflammatory markers, tumor characteristics, treatment and follow-up data were collected. Baseline CAR was calculated for each patient. Data were entered and analyzed by SPSS software version 26.

Results: Sixty-five patients were included, with a mean age at disease diagnosis of 61 ± 14 years and a sex ratio M/F=1.82. The median level of CAR was 0.4 [0.1-1.3]. No significant association was noted of CAR neither with age of disease onset (p=0.26), nor gender (p=0.86), nor smoking (p=0.88) nor family history of CRC (p= 0.54). For biological data, high levels of carcinoembryonic antigen and CA19-9 were significantly associated with a higher level of CAR (p=0.002 and p=0.01 respectively). We noted a significant association between lower levels of CAR and hematochezia (p=0.01) and so, rectal tumors were associated with lower levels of CAR compared with colonic tumors (26.2 vs 36; p=0.05). Circumferential, stenoting and friable tumors with hemorrhagic appearance were significantly associated with higher CAR levels (p=0.01; p=0.001 and p=0.01 respectively). We noted a significant association between high levels of CAR and metastases (50.1 vs 29.8; p=0.002) and mainly hepatic metastases (p=0.004).

Regarding treatment, neither primary surgery nor urgent surgery nor adjuvant chemotherapy were associated with CAR (p=0.11; p=0.12 and p=0.85 respectively).

However, an increased CAR was significantly associated with early readmission within 30 days subsequent to surgery (41.3 vs 26.7; p=0.01). Lower overall survival was associated with higher CAR rates (38.8 vs 29; p=0.04). Area under the receiver operating characteristic curve analysis showed that higher CAR was predictive of a shorter progression-free survival (AUROC = 0.65). The optimal cutoff value was 0.3 (sensitivity, 70%; specificity, 62%). A total of 36 (55%) patients had a CAR above this threshold. These patients presented more frequently with lymphopenia (p=0.03) and higher carcinoembryonic antigen rates (p=0.02).

We noted no association between advanced TNM stage 3 or 4 and CAR levels, regardless of this threshold (p=0.36), and overall disease recurrence was similar for CAR above and below 0.3 (p=0.46).

Conclusion: C-reactive protein-to-albumin ratio is a strong predictor of progression-free survival and is associated with clinicopathological factors in CRC. It serves as a valuable tool for straightforward prognostic assessment, aiding in more personalized treatment decisions.
Disclosure: Nothing to disclose.

PP0749

EXPLOITING CANCER IRON METABOLISM: A TARGETED APPROACH FOR SELECTIVE THERAPIES

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Introduction: Colorectal cancer (CRC) therapy faces significant challenges, including drug resistance, tumour heterogeneity, metastatic progression, and treatment-associated toxicity. CRC and other cancers exhibit increased levels of Fe(II) due to dysregulated iron metabolism, which can act as a trigger for selective prodrug activation. In this study, we explore the therapeutic potential of a novel tumour-activated prodrug platform based on 1,2,4,5-tetraoxane scaffold that is selectively activated in cancer cells with high levels of labile Fe(II).

As a proof-of-concept, we evaluated the therapeutic potential of 5b, a prodrug coupled with doxorubicin (DOX), which releases this cytotoxic payload upon activation by the labile Fe(II) pool in the cytoplasm of cancer cells.

Aims & Methods: The therapeutic potential and mechanism of action of 5b was evaluated both *in vitro* and *in vivo*. IC_{50} values were determined across multiple cancer cell lines and compared to non-tumorigenic cells. To confirm iron-dependency, HCT116 cell viability was assessed upon co-incubation with 5b and iron chelator deferoxamine (DFO). Co-incubation with apoptosis inhibitor Z-VAD-FMK (Zvad) or ferroptosis inhibitor liproxstatin-1 was used to assess cell death mechanism.

The sub-cellular location of 5b activation was evaluated by fluorescence microscopy. *In vivo* efficacy was evaluated in a xenograft mouse model using immunocompromised mice implanted with HT29 cells and intraperitoneal administrations of 5b, assessing tumour burden and systemic toxicity.

Results: Prodrug 5b effectively reduced cell viability in CRC cells HCT116 and HT29, as well as in glioblastoma (U251), prostate cancer (PC3) and breast cancer (MCF7) cells, with IC_{50} values ranging from 1.64 to 6.69 μM. Notably, 5b was over 2000 times less toxic than DOX in non-tumorigenic cells AML12 (5b IC_{50} = 398.2 μM vs. DOX IC_{50} = 0.18 μM). Taking advantage of DOX intrinsic fluorescence properties (while 5b pro-drug is non-fluorescent), we used fluorescence microscopy analysis to confirm that 5b released its fluorescent DOX payload in the cytoplasm, after activation by labile Fe(II).

The significant decrease in cell viability induced by 5b ($p < 0.0001$) was rescued by the iron chelator DFO and by the ferroptosis inhibitor liproxstatin-1 ($p < 0.01$) but not by pan-caspase inhibitor Zvad, suggesting that ferroptosis may be the primary mechanism of cell death. *In vivo*, 5b significantly reduced the relative tumour volume ($p < 0.05$), demonstrating greater efficacy than DOX, with only minor cardiotoxicity side-effects (multifocal loci of fibrosis affecting of 0.2% of left ventricle volume).

Conclusion: Prodrug 5b efficiently released its DOX payload in response to the cytoplasmatic labile Fe(II), effectively reducing cancer cell viability both *in vitro* and *in vivo*, while demonstrating significantly lower toxicity compared to DOX itself.

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Given its Fe(II)-dependent activation, 5b represents a promising strategy to exploit cancer cell iron addiction for cancer therapy. Future work should further validate ferroptosis as cell death mechanism, optimize the selection of cytotoxic payloads for tumours with dysregulated iron metabolism, and assess its efficacy in relation to KRAS mutational status.
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Disclosure: The authors declare that this work is associated with intellectual property protection under the following patents/patent requests:
- Diogo Magalhães e Silva, Rui Moreira, Francisca Lopes, Cecília Rodrigues, Vanda Marques, "1,2,4,5-tetraoxane compounds, a drug delivery system based in said compounds useful in diagnoses and therapy, and methods thereof", Portuguese Patent Application nº PT117644 from 16-12-2021;
- Diogo Magalhães e Silva, Rui Moreira, Francisca Lopes, Cecília Rodrigues, Vanda Marques, "Drug delivery systems based on endoperoxides useful in diagnosis and therapy, and methods thereof", PCT Application nº PCT/IB2022/062107 from 13-12-2022;
- Diogo Magalhães e Silva, Rui Moreira, Francisca Lopes, Cecília Rodrigues, Vanda Marques, "Drug delivery systems based on endoperoxides useful in diagnosis and therapy, and methods thereof", European Patent Office Application nº EP4448107 from 23-10-2024;
- Diogo Magalhães e Silva, Rui Moreira, Francisca Lopes, Cecília Rodrigues, Vanda Marques, "Drug delivery systems based on endoperoxides useful in diagnosis and therapy, and methods thereof", United States Patent and Trademark Office patent application nº US20250064840 from 27-02-2025.

PP0750

OUTCOMES OF ENDOSCOPIC RESECTION AND ADDITIONAL SURGERY FOR LNPCP CONTAINING INVASIVE CANCER: A HEAVY BURDEN OF MORBIDITY

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Introduction: Major societal guidelines still recommend surgical segmental resection following endoscopic resection (ER) for submucosal invasive cancer with adverse histopathological features, due to the risk of lymph node metastases or residual cancer. However, most patients with these features show no evidence of residual cancer or lymph node metastases after surgery. Few studies have reported the clinical outcomes of both ER and subsequent surgery in these patients.

Aims & Methods: Patients in a single tertiary referral centre were included if they underwent ER between 2014-2024 for LNPCP with invasive cancer found in the resected specimen. Endoscopic and surgical outcomes were reported including complications, type of surgery, length of hospital stay, stoma formation and restoration of intestinal continuity. This study aims to present the clinical outcomes following ER of large non-pedunculated colorectal polyps (LNPCP) with invasive cancer, as well as the outcomes of subsequent surgery when recommended.

Results: 194 patients with invasive cancer discovered after ER of LNPCP were included. Mean lesion size was 46.5mm (median 40mm, range 20-180mm). Lesions were located in the right colon in 20%, left colon in 28% and rectum in 52%. ESD was used to resect 73%. 74% of LNPCP were resected en-bloc. Complications following ER occurred in 4% of patients overall, with only 1 clinically significant perforation who required surgery. Elective surgical segmental resection was recommended for 55 patients based on histopathological findings, of which 53 underwent surgery. 30% of patients experienced a surgical complication (25% of these Clavien-Dindo grade 3 or 4), with median length of stay 5 days (range 2-28 days). 40% of patients required stoma formation, 29% of whom remain with a stoma.

Conclusion: These real-world outcomes of a cohort of patients with invasive cancer following ER of LNPCP highlight a significant burden of morbidity for those recommended for surgery compared to those who undergo ER alone.

Additionally, these patients often face prolonged hospital stays, and a substantial proportion require stoma formation, many of whom continue to live with a stoma. Given the relatively low likelihood of residual cancer or lymph node metastasis in most patients, these findings should be considered in discussions with patients to fully inform management options.

Disclosure: Nothing to disclose.

PP0751

THE IMPACT OF GASTROINTESTINAL IMMUNE-RELATED ADVERSE EVENTS ON PROGNOSIS VARIES BY NUTRITIONAL STATUS

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Introduction: Among immune-related adverse events (irAEs) caused by immune checkpoint inhibitors (ICIs), gastrointestinal irAEs (GI-irAEs) are relatively rare, and their risk factors and prognostic implications remain unclear.

Aims & Methods: We retrospectively analyzed 1,104 patients who initiated ICI therapy at our institution between January 2016 and December 2022 and for whom survival data were available. GI-irAEs were defined as colitis with an increase of ≥3 bowel movements/day from baseline in the absence of other causes, or symptomatic gastritis confirmed histologically. Risk factors and prognostic impact of GI-irAEs were evaluated.

Results: The median duration of ICI therapy was 107 days. The cohort included 754 males (68%) with a median age of 69 years. The primary tumor types were lung cancer (32%), gastrointestinal cancers (19%), urothelial carcinoma (13%), head and neck cancer (13%), and melanoma (11%). The overall response rate was 27.8%, and the disease control rate was 49.0%. irAEs of any grade occurred in 310 patients (28%), and GI-irAEs were observed in 30 patients (2.7%).

In univariate logistic regression analysis, use of anti-CTLA-4 antibody, combination ICI therapy, a prognostic nutritional index (PNI) ≥ 40, and platelet-to-lymphocyte ratio (PLR) ≥ 1.8 were associated with an increased risk of GI-irAEs. In multivariate analysis, only anti-CTLA-4 antibody use remained a significant risk factor (odds ratio: 9.2; 95% CI: 1.9–45). Kaplan–Meier analysis revealed that patients with GI-irAEs had significantly longer overall survival (OS) compared to those without irAEs in the overall cohort (median OS: 22.9 vs. 12.6 months, $P < 0.0001$).

However, this difference was not observed in landmark analyses at 4 and 8 weeks post-treatment. Subgroup analysis based on PNI showed that in patients with PNI ≥ 40, those with GI-irAEs had significantly longer OS than those without irAEs (median not reached vs. 19.3 months, $P = 0.0003$). In contrast, in patients with PNI < 40, GI-irAE cases tended to have shorter OS than non-irAE cases (8.2 vs. 9.3 months). This trend was consistent in landmark analyses at 4 and 8 weeks.

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Conclusion: The impact of GI-irAEs on prognosis appears to differ depending on the patient's nutritional status. Particular caution is warranted for patients in poor general condition receiving ICIs, especially those involving anti-CTLA-4 antibodies, as GI-irAEs may serve as a poor prognostic factor in this population.
Disclosure: Nothing to disclose.

PP0752

EVALUATION OF THE EFFICACY AND SAFETY OF COLORECTAL STENTING FOR PALLIATION

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Introduction: Self-expandable metallic stent (SEMS) placement for palliation (PAL) of malignant colorectal obstruction due to unresectable advanced colorectal cancer or metastases from other organs is a noninvasive treatment option that prevents emergency surgery or colostomy. However, there are concerns that a satisfactory success rate may not be attained.
Aims & Methods: Therefore, this study investigated the efficacy and safety of colorectal stenting for PAL at our institution. We retrospectively analyzed the cases of 94 patients who underwent colorectal stent placement for PAL out of 191 patients at our institution from June 2012 to February 2024, a period during which 111 stent placement procedures were performed.
We collected data on patient characteristics, details of the obstruction site, outcomes of stent placement (technical success, clinical success), complications, the 1-year overall survival rate, the cumulative stent patency rate, and risk factors for perforation.
Results: The median was 76.5 (IQR, 67–87) years, and there were 53 males. The ECOG Performance Status was 0–1 in 36 cases and 2–4 in 58 cases. Obstruction was caused by primary colorectal cancer in 48 cases, local recurrence after colorectal cancer surgery in 6 cases, and metastases from other organs in 40 cases.
The ColoRectal Obstruction Scoring System (CROSS) before stenting was 0 in 33 cases, 1 in 46 cases, and 2 in 15 cases. Lesions were located in the rectum, sigmoid colon, descending colon, transverse colon, ascending colon, and cecum in 19, 29, 21, 12, 9, and 4 cases, respectively. The mean stenosis length was 5.6 ± 2.4 cm.
The technical success rate was 97.9% (92/94), and the clinical success rate was 85.1% (80/94). Early (within 7 days) complications included perforation in 6 cases (6.4%), migration in 2 cases (2.1%), and inadequate dilation in 2 cases (2.1%), all of which required surgery.
Late complications included perforation in 8 cases (8.5%), migration in 1 case (1.1%), and stent occlusion in 11 cases (11.7%). In all cases of stent occlusion, the stents used were 22 mm-diameter stents, and ingrowth was the main cause of obstruction. In all cases of stent occlusion, stent-in-stent procedures were successfully performed to re-open the stents.
Complication-related mortality occurred in 3 cases (3.2%). The one-year overall survival rate was 29.3% (mean follow-up period: 221 ± 248 days), and the one-year stent patency rate was 75.5% (mean follow-up period: 168 ± 198 days). Per our univariate analysis, the stent length, stent diameter, chemotherapy, and anti-VEGF antibody use after stent placement were not risk factors. Our multivariate analysis showed that only a stent diameter of 25 mm tended to be an independent risk factor for perforation.
Conclusion: SEMS replacement for PAL was effective in cases of colorectal obstruction. The incidence of complications and the complication-related mortality rate were within acceptable limits, suggesting that colonic stent placement is a promising palliative treatment option for malignant colorectal obstruction that helps maintain the quality of life of patients.
Disclosure: Nothing to disclose.

PP0753

EFFICACY AND SAFETY OF NIVOLUMAB PLUS IPILIMUMAB IN MICROSATELLITE INSTABILITY–HIGH METASTATIC COLORECTAL CANCER: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Microsatellite instability (MSI)–high and mismatch repair-deficient (dMMR) metastatic colorectal cancer (mCRC) presents a significant challenge due to its poor prognosis and limited response to traditional chemotherapy. Immunotherapy has emerged as a promising treatment strategy. Nivolumab, a PD-1 inhibitor, combined with ipilimumab, a CTLA-4 inhibitor, has shown potential in improving patient outcomes.
This systematic review and meta-analysis evaluate the clinical efficacy, safety, and therapeutic role of Nivolumab plus Ipilimumab in patients with MSI-H mCRC, providing critical insights to optimize treatment strategies and enhance patient outcomes.
Aims & Methods: A comprehensive search was conducted across multiple databases including PubMed, Scopus, Web of Science, and Cochrane from their inception to March 2025. The search focused on identifying trials and relevant studies evaluating Nivolumab plus Ipilimumab for the treatment of Microsatellite Instability-High (MSI-H) metastatic colorectal cancer (mCRC).
Studies were included if they reported on treatment efficacy, including Objective Response Rate (ORR), Disease Control Rate (DCR), Progression-Free Survival (PFS), Overall Survival (OS), and treatment-related adverse events.
Results: Out of 3,477 studies initially screened, nine studies were identified as potentially eligible for inclusion in the meta-analysis. After applying the inclusion criteria, three studies met the criteria for data analysis, comprising a total of 758 patients treated with Nivolumab plus Ipilimumab. Although all nine studies met the inclusion criteria, not all were included in the quality analysis due to certain limitations in study design or reporting.
The combination of Nivolumab and Ipilimumab resulted in an overall Objective Response Rate (ORR) of 0.67 (95% CI: 0.61–0.73), with significant heterogeneity across the studies (I^2 = 40%). The pooled Disease Control Rate (DCR) was 0.82 (95% CI: 0.76–0.88), showing high levels of disease control across the trials.
In terms of treatment-related adverse events, the combined analysis of Any Treatment-Related Adverse Events (TREA) from the studies showed an overall event rate of 0.82 (95% CI: 0.79–0.85), indicating a high inci-

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dence of adverse events associated with the combination therapy. The results demonstrated minimal heterogeneity (I^2 = 0%), suggesting consistent reporting of adverse events across the three studies.

The analysis of Serious Treatment-Related Adverse Events (STRAE) from André 2022 and André 2025 revealed an overall event rate of 0.20 (95% CI: 0.16–0.23), indicating that a low proportion of patients experienced serious adverse events in response to Nivolumab and Ipilimumab combination therapy, with very low heterogeneity (I^2 = 1%).

Additionally, the combined analysis of diarrhea across the studies showed an overall event rate of 0.24 (95% CI: 0.20–0.27), with moderate heterogeneity (I^2 = 72%), indicating a substantial but consistent occurrence of diarrhea across the studies. The combined analysis of fatigue showed an overall event rate of 0.18 (95% CI: 0.15–0.22), with high heterogeneity (I^2 = 94%), indicating considerable variation in fatigue occurrence across the studies

Conclusion: Nivolumab plus Ipilimumab improves survival outcomes) in patients with MSI-H mCRC. However, it is associated with a higher risk of treatment related adverse events, which emphasizes the need for effective management of these to optimize the treatment benefit for patients.

Disclosure: Nothing to disclose.

PP0754

THERAPEUTIC STRATEGIES IN PT1 COLORECTAL ADENOCARCINOMA AND THEIR IMPACT ON PROGRESSION: SURGERY VS. ENDOSCOPIC TREATMENT

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Introduction: The standard treatment for pT1 colorectal cancer classified as high risk (poor differentiation, deep submucosal invasion, intense tumor budding, resection margins that are positive or lymphovascular invasion) is surgery [1]. However, some studies have revealed that patients who undergo surgery and endoscopic treatment have similar disease progression outcomes, and it is also controversial which histological factors influence disease progression [2,3].

Aims & Methods: The aim of this study was to analyze the characteristics and evolution of patients undergoing surgery or endoscopic treatment after the resection of pT1 colorectal adenocarcinoma with histological findings considered high risk for lymph node metastasis.

We conducted an observational study of historical cohorts. We included patients with pT1 Adenocarcinoma and histological criteria of poor prognosis with at least six months of follow-up, and who were diagnosed between 2018 and 2024. The therapeutic approach was determined by a multidisciplinary committee.

Results: We included 54 patients (74.1% male) with a mean age of 65.3 years and mean follow-up time of 3.07 years; no significant differences in both groups. The indications for colonoscopy were screening (42.6%), alarm symptoms (38.9%), radiological findings (3.7%), prior polypectomy (9.3%), history of colorectal cancer (1.9%), and family history (3.7%). The rate of patients with endoscopic complications was 2.9% intraprocedural bleeding and 2.9% delayed bleeding.

The mean lesion size was 23.48 mm, and the most common location was the rectum.

Regarding the therapeutic approach, 63.0% of patients underwent surgery and 29.6% underwent endoscopic surveillance. Among the surgical patients, complications included suture dehiscence (6%), and hemorrhage (2%). In the surgical specimen analysis, residual lesion was found in 6 patients, and metastatic lymph nodes in 2. 5.55% required adjuvant therapy due to lymph node involvement, and 1.85% were stage IV at diagnosis, excluded from the comparative analysis.

Comparing both study groups, the surgical group had a significantly higher rate of submucosal invasion, with no differences in the remaining characteristics studied. During the study period, progression with lymph node involvement was observed in one patient who underwent surgery and two patients who underwent endoscopy treatment (average time 11 and 24.5 months, respectively). No patient has developed distant metastases to date.

	Surgery treatment (n =34)	Endoscopic therapy (n = 16)	p-value
Piecemeal resection (n, %)	16 (47.05%)	8 (50%)	p > 0.05
Poor differentiation (n, %)	1 (2.9%)	0	p > 0.05
Deep submucosal invasion (n, %)	33 (97%)	11 (68.75%)	P = 0.016
Margins affected or not precise (n, %)	14 (41.17%)	6 (37.5%)	p > 0.05
Lymphovascular invasion (n, %)	6 (18.18%)	0	P = 0.06
Perineural invasion (n, %)	2 (6.66%)	0	p > 0.05
Intense tumor budding (n, %)	0	0	
Total number of poor prognostic factors (media, SD)	2,59 (1,076)	1,93 (1,03)	P = 0.043
Disease progression (n, %)	1 (2,9%)	2 (1,25%)	p > 0.05

Conclusion: In our study, patients who underwent endoscopic treatment did not present significant differences with respect to those who received surgery in terms of long-term tumor progression, despite having a higher rate of submucosal invasion. Further studies with larger sample sizes are needed to determine which factors and to what extent influence lymphatic and distant dissemination of pT1 adenocarcinoma.

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PP0755

CLINICAL CHARACTERISTICS AND ENDOSCOPIC TREATMENT PROGNOSIS OF MULTIPLE RECTAL NEUROENDOCRINE TUMORS: A SINGLE-CENTER RETROSPECTIVE STUDY

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Introduction: The incidence of rectal neuroendocrine neoplasms (R-NENs) has been increasing annually. While most R-NENs are solitary, multiple lesions are relatively rare. Previous studies have suggested that patients with ≥8 lesions are more likely to exhibit lymphatic invasion and metastasis, indicating higher risk. However, the clinical and pathological features, prognosis, and endoscopic treatment outcomes of multiple R-NENs remain poorly understood.

This study aimed to clarify the clinical characteristics and endoscopic treatment efficacy of multiple R-NENs through a retrospective analysis of cases from our hospital, providing evidence for clinical management.

Aims & Methods: This retrospective study included 28 patients with multiple R-NENs treated endoscopically at the Cancer Hospital of the Chinese Academy of Medical Sciences between January 2010 and January 2024. Patients were divided into two groups based on lesion count: the "several group" (n < 8) and the "numerous group" (n ≥ 8). All patients underwent preoperative imaging to exclude lymph node metastasis and distant metastases. Data on demographics, lesion characteristics, endoscopic procedures, pathological findings, and follow-up outcomes were collected and analyzed.

Results: We compared endoscopic treatment outcomes between patients with several (<8, n=24) and numerous (≥8, n=4) rectal neuroendocrine tumors. The numerous group had a larger median maximum diameter (9.50 mm vs. 6.00 mm, p=0.337) and included one G2 case (25.0%) versus none in the several group, though differences were not significant. Notably, endoscopic complete resection was achieved in 100% of patients in both groups. All other parameters, including margin positivity, vascular invasion, neural invasion, and R0 resection rate, showed no significant differences. However, the numerous group had a significantly higher recurrence rate (75.0% vs. 0%, p=0.001). These findings highlight lesion count as a critical prognostic indicator for recurrence after endoscopic treatment.

Characteristic	Several group (n=24)	Numerous group (n=4)	p value
Max diameter (IQR, Q1-Q3)	6.00 (5.00-8.25)	9.50 (6.00-12.75)	0.337
Pathologic grade(G1),n(%)	23 (95.83%)	4 (100.00%)	0.457
Horizontal margin positive,n(%)	1 (4.17%)	0 (0.00%)	1
Vertical margin positive,n(%)	1 (4.17%)	1 (25.00%)	0.27
Vascular invasion,n(%)	3 (12.50%)	0 (0.00%)	1
Neural invasion,n(%)	1 (4.17%)	0 (0.00%)	1
Complete resection,n(%)	24 (100.00%)	4 (100.00%)	0.135
R0 resection,n(%)	22 (91.67%)	3 (75.00%)	0.382
Recurrence,n(%)	0 (0.00%)	3 (75.00%)	0.001

Table.1 Comparison of clinical and pathological prognosis between several group and numerous group.

Conclusion: Patients with ≥8 rectal neuroendocrine tumors exhibited more aggressive disease behavior, characterized by higher recurrence rates. These findings underscore the importance of lesion count as a prognostic indicator and suggest that patients with ≥8 lesions may require more intensive follow-up and management. For patients with <8 lesions, endoscopic treatment is effective when lymph node and distant metastases are excluded preoperatively. Future studies with larger datasets are needed to validate these findings and explore optimized treatment strategies.

Disclosure: The authors declare no conflicts of interest in relation to this study.

PP0756

FIRST-ROUND FECAL IMMUNOCHEMICAL TEST CONCENTRATION PREDICTS COLORECTAL CANCER AND ADVANCED NEOPLASIA IN SECOND-ROUND SCREENING: RESULTS FROM THE SCREESCO

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Introduction: The quantitative fecal immunochemical test (FIT) is widely used for colorectal cancer (CRC) screening. While the association between FIT concentration and the risk of advanced colorectal neoplasia (ACN) detection is well established, the predictive performance of current-round FIT concentration alone has limitations. Identifying additional predictive factors could enhance screening. For individuals with a prior colonoscopy, colonoscopy findings and quality metrics may serve as such factors.1

For those who did not undergo colonoscopy due to a negative FIT in the previous round, alternative predictors should be explored. We hypothesized that prior-round FIT concentration could serve as a potential predictor.

Aims & Methods: We aimed to investigate whether prior-round negative FIT concentration could predict CRC and ACN detection by conducting a cross-sectional analysis of data from the FIT arm of the SCREESCO, a nationwide randomized controlled trial on CRC screening in Sweden.1,2 A sample of the Swedish population aged 60 years was randomly assigned to one of three arms: the once-colonoscopy arm, FIT arm, or control arm. Individuals in the FIT arm were invited to undergo two rounds of quantitative FIT screening in the first and third years. A highly sensitive method was used to assess FIT positivity, with a cut-off concentration of ≥10 µg Hb/g feces. We estimated the detection rates of CRC and ACN in second-round colonoscopy based on first-round FIT concentration.

Binary regression models were used to assess whether first-round FIT concentration was associated with CRC and ACN detection in the second round. The models included participant characteristics such as sex, body mass index, and smoking status, and second-round FIT concentration, in addition to first-round FIT concentration.

Results: A total of 1,991 individuals were included, with a median FIT concentration of 2.2 µg Hb/g feces (interquartile range: 0.4–4.6) in the first round and 21.8 (13.4–50.8) in the second round. CRC and ACN were detected during second-round colonoscopy in 23 (1.2%) and 267 (13.4%) individuals, respectively. Individuals with first-round FIT concentration of 0–1.9,

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2.0–5.9, and 6.0–9.9 µg Hb/g feces had second-round CRC detection rates of 0.8%, 0.9%, and 2.6%, respectively, and ACN detection rates of 9.7%, 12.5%, and 15.9%, respectively. A higher first-round FIT concentration was associated with an increased risk of CRC (risk ratio per unit increase 1.27; 95% confidence interval 1.10–1.47) and ACN (1.08; 1.04–1.12) detection in the second round, after adjusting for the second-round FIT concentration and participant characteristics. For a fixed second-round FIT concentration, the absolute risk of second-round ACN detection increased with higher first-round FIT concentration. For instance, in a nonsmoking or ex-smoking male with a second-round FIT concentration of 20 µg Hb/g feces, the risk increased from 10.0% at < 0.2 µg Hb/g feces in the first round to 14.9% at 5 µg Hb/g feces and 19.6% at 9.9 µg Hb/g feces.

Conclusion: First-round negative FIT concentration is predictive of CRC and ACN detection in the second screening round. Even an increase in low levels of the FIT concentration is associated with future detection of these neoplasms. Incorporating first-round FIT concentration alongside second-round FIT concentration may enhance risk stratification in CRC screening, leading to more efficient use of colonoscopy resources.

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PP0757

NO NEED TO STAY ABOVE THE LINE: ANORECTAL JUNCTION ENDOSCOPIC SUBMUCOSAL DISSECTION IS COMPARABLE TO OTHER RECTAL LOCATIONS

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Introduction: Endoscopic submucosal dissection (ESD) is a well-established resection technique for superficial rectal lesions.

Aims & Methods: Evaluate the outcomes of overall rectal ESD and compare the resection of anorectal junction (ARJ) lesions with more proximal rectal (MPR) lesions, regarding feasibility, safety and efficacy.

Retrospective unicentric study including consecutive patients submitted to rectal ESD. ARJ lesions were considered as those totally or partially located within 2 cm of the dentate line. ESD failure was determined whenever the target lesion was not removed. *En bloc* resection required that the target lesion be retrieved in one single specimen. R0 resection was achieved when pathological evaluation showed free horizontal and vertical margins in an *en bloc* resected specimen. Specimens with thermal effects at the margins preventing the pathologist from excluding the presence of abnormal cells were considered R1 resections. Curative resection was considered in lesions with low- (LGD) or high-grade dysplasia (HGD) with R0 resection or lesions harboring a well-differentiated adenocarcinoma with superficial submucosal invasion (<1mm), with negative margins (>1mm), without lymphatic/vascular invasion or tumor budding.

Results: A total of 26 patients were included, most were females (57.7%), with a mean age of 63±10 years. Twelve patients (46.2%) had an ARJ lesion and fourteen (53.8%) had a MPR lesion. Most lesions were Paris 0-IIa+Is (61.5%), followed by 0-IIa+b (15.4%).

Technical success was achieved in 25 patients (96.2%), with no significant differences between those with an ARJ lesion and those with a MPR lesion (100.0% *vs* 92.9%, *p*=1.000). There were also no significant differences in the rate of *en bloc* resection (91.7% *vs* 92.9%, *p*=1.000). Most lesions harbored HGD (84.0%), with no significant differences in histology between ARJ and MPR lesions: in ARJ lesions, LGD occurred in 16.7%, HGD in 83.3% and adenocarcinoma in 0.0% *vs* in MPR lesions, LGD occurred in 0.0%, HGD in 84.6% and adenocarcinoma in 15.4% (*p*=0.134).

The global R0 resection rate was 61.5% and curative rate was 57.7%, both without statistically significant differences between patients with ARJ lesions and MPR lesions (50.0% *vs* 71.4%, *p*=0.422 and 50.0% *vs* 64.3%, *p*=0.462).

Regarding complications, there were no significant differences between patients with ARJ lesions and MPR lesions in the rate of significant intraprocedural bleeding (41.7% *vs* 35.7%, *p*=1.000) and exposure/section of muscular fibers with the need of closure of the eschar (8.3% *vs* 14.3%, *p*=1.000). The rate of postprocedural bacterial translocation and delayed bleeding were also similar between both groups (16.7% *vs* 21.4%, *p*=1.000 and 8.3% *vs* 7.1%, *p*=1.000, respectively).

Only one (3.8%) patient needed surgery due to a non-curative ESD, with no residual disease in the surgical specimen. Seventeen patients (65.4%) underwent a surveillance colonoscopy 6 and 12 months after the ESD, with one (5.9%) patient having residual tissue at 6 months with LGD and none with residual tissue at 12 months.

Conclusion: ESD of rectal lesions involving the anorectal junction is a feasible and safe procedure, with technical success and *en bloc* resection rate similar to more proximal rectal lesions and without differences in complication rates.

Disclosure: Nothing to disclose.

PP0758

OUTCOMES OF ENDOSCOPIC MUCOSAL DISSECTION (ESD) FOR COLONIC DYSPLASIA AND EARLY NEOPLASIA: A SINGLE CENTER EXPERIENCE

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Introduction: Endoscopic submucosal dissection (ESD) is increasingly used to treat colonic dysplasia and early neoplasia. This is a safety, non-invasive alternative treatment to surgery.

Aims & Methods: To evaluate the outcomes of ESD on colonic dysplasia and neoplasia in our Italian single center, including complications and recurrence

Our four-year retrospective study at a single tertiary center included data spanning from 2021 to 2025. The efficacy of ESD on colonic dysplasia and early neoplasia was measured using parameters set out by the National Institute for Health and Care Excellence, which include *en-bloc* and curative resection (CR) rates, local recurrence and survival rates.

Results: ESD was attempted on 47 lesions in 35 patients. 90.0% of completed procedures achieved endoscopic clearance. 74.3% were *en-bloc* resections and the rest were hybrid ESD with piecemeal resections. In all, 40% achieved histological CR. Overall, disease recurrence was 10.9%

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at latest follow-up (12 months, median follow-up). Importantly 100% of lesions in the CR group showed no disease recurrence at subsequent follow-up. In the Indeterminate and Non-CR group, 18.8% of lesions showed disease recurrence at subsequent endoscopic follow-ups. Immediate complications were observed in 9.9% of all ESD procedures.

Conclusion: The study affirms the long-term efficacy and safety of ESD for colonic dysplasia and early neoplasia in our single center.

Disclosure: Nothing to disclose.

PP0759

SAFETY AND EFFICACY OF COLD VS. HOT SNARE ENDOSCOPIC MUCOSAL RESECTION FOR COLORECTAL POLYPS: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Colorectal cancer is the third most common cancer worldwide and the second leading cause of cancer-related death. Most colorectal cancers arise from benign polyps, especially dysplastic adenomas. Colonoscopy with polypectomy is the gold standard for preventing cancer by interrupting the adenoma-carcinoma sequence. This has led to the development of various techniques and devices for improved prevention and treatment. Hot snare endoscopic mucosal resection (HS-EMR), which combines submucosal injections with electrocautery, is commonly used for excising various colorectal polyps. However, it carries risks of diathermy-related complications, such as perforation, and bleeding.

To address these issues, cold snare endoscopic mucosal resection (CS-EMR) has been proposed as an alternative. Concerns about residual adenoma/recurrence have been raised with CS-EMR. Therefore, we conducted this meta-analysis to compare the two techniques.

Aims & Methods: We searched electronic databases, including MEDLINE via PubMed, Scopus, Web of Science, and Cochrane, without restrictions on publication date or language. We included RCTs comparing CS-EMR and HS-EMR for colorectal polyp resection. Primary outcomes included the incidence of delayed bleeding and residual/recurrent adenoma (RRA).

Results: On Nov 5, 2024, our literature search across electronic databases retrieved a total of 1,888 records. Following a meticulous screening process, six studies met the eligibility criteria encompassing 1,449 patients and 1,947 polyps. Regarding safety, the overall effect estimates regarding delayed bleeding showed statistically significant results favoring CS-EMR group [OR 0.31, 95 % CI (0.12–0.76), P = 0.01] with no heterogeneity (P = 0.40; I^2 = 0 %).

However, statistical superiority for CS-EMR could not be detected regarding early bleeding [OR 0.78, 95 % CI (0.31–1.97), P = 0.60]. Only two studies reported data on delayed perforation, showing a numerical difference favoring CS-EMR; however, statistical significance could not be obtained [OR 0.26, 95 % CI (0.03–2.41), P = 0.24].

Regarding efficacy, the overall effect estimates regarding complete resection and RRA did not favor either of the groups [OR 0.62, 95 % CI (0.25–1.53), P = 0.30], [OR 2.81, 95 % CI (0.43–18.47), P = 0.28] respectively.

Conclusion: Evidence obtained from this meta-analysis suggests that CS-EMR offers superior safety compared to HS-EMR for colorectal polyp resection. Current evidence remains limited due to the small number of studies included. Therefore, further high-quality randomized controlled trials with larger sample sizes are needed.

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Disclosure: Nothing to disclose.

PP0760

FEASIBILITY AND EFFECTIVENESS OF DIAGNOSTIC ENDOSCOPIC RESECTION FOR T1 COLORECTAL CANCER WITH DEEP INVASION

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Introduction: Deep submucosal invasion is not considered an independent risk factor for lymph node metastasis in T1 colorectal cancer (CRC). This study aimed to evaluate the feasibility and effectiveness of diagnostic endoscopic resection for T1 CRCs with deep invasion.

Aims & Methods: This retrospective study included consecutive patients with T1 CRC who underwent either endoscopic submucosal dissection (ESD) or endoscopic mucosal resection (EMR). Pedunculated lesions (Paris classification Ip) were excluded. Low-risk T1 CRCs were defined as lesions with submucosal invasion ≥1000 μm that were resected en bloc with R0 margins and lacked lymphovascular invasion (LVI), high-grade budding (>BD1), or poor differentiation. These lesions were considered eligible for expanded curative resection.

Results: A total of 123 T1 CRC lesions in 121 patients were resected endoscopically (26 by EMR, 97 by ESD). The overall en bloc and R0 resection rates were 89.4% (100/123) and 65.9% (81/123), respectively. For T1b CRCs, these rates were 86.7% (52/60) and 56.7% (34/60). No perforations occurred. Expanded curative resection was achieved in 26.7% (16/60) of T1b cases. Among non-curatively resected T1b lesions, 50.0% (30/60) had LVI, 41.7% (25/60) had positive or indeterminate vertical margins (VM1/VMX), and 13.3% (8/60) showed significant tumor budding.

Conclusion: Diagnostic endoscopic resection for T1 CRCs with deep submucosal invasion is feasible and safe, with a notable rate of expanded curative resection. This strategy may support accurate pathological evaluation and informed treatment planning in selected patients.

Disclosure: Nothing to disclose.

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PP0761

THE SMALL SOLUTION FOR A BIG PROBLEM: CAN MICRORNAS PREDICT TREATMENT OUTCOMES IN COLORECTAL CANCER PATIENTS?

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Introduction: Colorectal cancer (CRC) ranks as the second leading cause of cancer-related deaths worldwide. Although first-line chemotherapy can prolong survival in CRC patients, around 40% do not experience a beneficial response to these therapeutic regimens.

Currently, no clinically validated biomarkers capable of predicting patient response to chemotherapy are available. MicroRNAs (miRs) are small non-coding RNAs involved in various oncogenic processes. Their differential expression across distinct clinical conditions, along with their high stability and ease of detection, make them promising predictive biomarkers to overcome this challenge.

Aims & Methods: Thus, we aimed at systematically reviewing the current state of knowledge regarding miRs as predictive biomarkers of therapeutic response in colorectal cancer patients. A systematic search on PubMed/MEDLINE and Scopus databases was performed on published articles reporting the role of microRNAs detected in blood and tissue as biomarkers for predicting the therapeutic outcome in colorectal cancer patients. 2673 records were retrieved by December 2024. After removal of duplicate records 2568 abstracts were screened, and 40 full texts included in this review.

The different outcomes to predict treatment response were extracted from the studies, namely response to chemeotherapy, progression free survival (PSF), disease free survival, overall survival, and recurrence of the disease.

Results: Among the 111 miRs reported in the 40 articles, circulating miR-4772-3p demonstrated a good ability to predict the response of CRC stage I–III patients to FOLFOX, the most used first-line therapy in this group of patients (sensitivity: 79%, AUC: 0.72, 95% CI: 0.59–0.85). Additionally, PFS is commonly used to assess treatment response. In this context, decreased levels of miR-4772-3p are associated with poor PFS, demonstrating a lower response to FOLFOX.

In addition, a combination of three miRs (miR-17-5p, miR-92b-3p, and miR-98-5p) in tissue samples was indicative of a complete or partial response to FOLFOX combined with anti-VEGF therapy, namely bevacizumab, in metastatic CRC patients (AUC: 0.85, 95% CI: 0.72–0.91). Notably, miR-17-5p was the most frequently reported miR associated with FOLFOX response prediction in tissue samples, and its high expression levels were linked to poor PFS, indicating treatment inefficiency and consequently progression of disease.

Regarding to the specific group of patients with locally advanced rectal cancer (LARC) that are mainly submitted to neoadjuvant chemoradiotherapy (nCRT), a panel composed of eight circulating microRNAs (miR-30e-5p, miR-33a-5p, miR-130a-5p, miR-210-3p, miR-214-3p, miR-320a, miR-338-3p, and miR-1260a) showed a higher accuracy for predicting patients response to nCRT, with a sensitivity of 80% (AUC: 0.89, 95% CI: 0.81 to 0.94).

Furthermore, the miR-125b-5p detected in tissue samples, was also associated with the nCRT response (AUC: 0.90, 95% CI: 0.7818-1.043). These miRs could increase the response to nCRT for LARC patients, considering that currently only around 40% achieve a partial response and as few as 20% reach a complete response to treatment.

Conclusion: This systematic review highlights the potential of microRNAs as predictive biomarkers of therapeutic response in CRC, with the ability to improve treatment outcome prediction and support personalized therapy. However, further validation is required for improving clinical decisions and to allow the clinical applicability of these biomarkers.

Disclosure: Nothing to disclose.

PP0762

EVALUATION OF THE METACHRONOUS AND RESIDUAL LESIONS IN PATIENTS AFTER HIGH-RISK COLORECTAL NEOPLASIA ENDOSCOPY THERAPY – A PROSPECTIVE STUDY

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Introduction: In colonoscopies with colorectal neoplasia, the endoscopy therapy is recommended. The surveillance (follow-up, FU) colonoscopies follow with the aim to diagnose the recurrent and metachronous lesions. Many guidelines recommend the intervals of such procedures based on the number, size and histopathological phenotype of the resected lesions. In clinical practice, the adherence of the recommendations is variable and selection of the high-risk patients who will benefit from shorter intervals might be helpful.

Therefore, detailed study focusing on the development and clinical utilization of the new method combining the clinicopathological phenotype and the mutation burden was started in 2017. The interim results of the clinical phase of the prospective study are described.

Aims & Methods: Patients with colorectal lesions ≥ 10 mm removed by advanced therapeutic methods (en-bloc mucosal resection and submucosal dissection) and followed in interval in year 1, 3 and 5 were included. The therapy and surveillance were performed in one tertiary endoscopy center.

The main aim was to assess the metachronous lesions (number, size, locality, histopathology characteristics) in relation with the index colonoscopy findings. Secondary aim was evaluation of the local residual neoplasia (LRN) in the scar after previous therapy.

Results: There were 217 patients included in the study in years 2017 – 2020. Eight of them were excluded because they did not meet the inclusion criteria. Results of the index colonoscopy in 209 patients (130 men, 62,2%; 79 women, 37,8%) were analyzed. 275 index lesions were found, majority of them (238 lesions, 86,5 %, low-grade and high-grade adenomas) were benign and there were 12 (4,4%) intramucosal cancers, two T1 cancers, one T2 cancers and one NET. 21 lesions were hyperplastic or others.

Follow-up colonoscopy in one year was performed in 178 patients (82,0%) and in 91 of them (51,1 %) metachronous lesions were found, 120 (64,2%) adenomas, 46 (24,6%) hyperplastic and 21 (11.2%) others, with no can-

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cer. LRN was diagnosed in 5,5% patients. In 122 patients with 3-years FU colonoscopy, 59 (48,4%) had metachronous lesions, mostly adenomas (73; 72,3%), no cancer and one LRN found. 45 patients had 5-years FU colonoscopy so far, 22 of them (48,9%) with metachronous lesions, mostly adenomas (25; 54,3%), no cancer, LRN in one case.

Conclusion: In patients after removal of the high-risk colorectal lesions defined by size ≥ 10 mm, high numbers of metachronous lesions were found in the strict surveillance protocol (years 1, 3, and 5). All the lesions were benign, with no cancer found.

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Disclosure: Nothing to disclose.

PP0763

ADOPTING AN EXTENDED INTERVAL APPROACH IN INDIVIDUALS WITH LYNCH SYNDROME COULD RESULT IN A HIGH RATE OF INTERVAL COLORECTAL CANCER AT ADVANCE STAGE

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Introduction: Colorectal cancer surveillance recommendations for individuals with Lynch Syndrome (LS) differ between professional associations and the data on interval colorectal cancer (I-CRC) and interval advanced polyps (I-AP) are limited. We aimed to measure the rate of I-CRC and I-AP within a surveillance program.

Aims & Methods: Retrospective study involved individuals with LS who participated in CRC surveillance program at Clalit Health hospitals, Israel. The main outcomes were the rates of I-CRC and I-AP ≤36 months, categorized by 12-month intervals. Included only colonoscopies with adequate preparation. I-CRC and I-AP (≥10 mm or containing high-grade dysplasia or villous histology) were defined as interval advanced neoplasia (I-AN).

Results: The cohort included 342 individuals, 52.3% females, with a median age at data analysis of 49.0 (IQR 37.7-64). At interval ≤ 12 months, there were 7 cases of I-CRC [6/142 (4.2%) of MSH2; 1/46 (2.2%) of MLH1] and 13 cases of I-AP (10/142 (7.0%) of MSH2, 2/46 (4.3%) of MLH1, 1/136 (0.7%) of MSH6]. At 12-24 months, there were 3 cases of I-CRC [2/142 (1.4%) of MSH2, 1/136 (0.7%) of MSH6] and 10 cases of I-AP [8 /142 (5.6%) of MSH2, 1/46 (2.2%) of MLH1, 1/136 (0.7%) of MSH6]. At 24-36 months, there were 3 cases of I-CRC [2/142 (1.4%) of MSH2, 1/18 (5.6%) of PMS2] and 4 cases of I-AP [1 /142 (0.7%) of MSH2, 3/136 (2.2%) of MSH6]. Of all I-CRC, four were diagnosed at stage 3 [4/342 (1.2%)], all at interval ≤24months [3-MSH2, 1-MSH6]. Increasing age and path_MSH2 were significantly associated with I-AN.

Conclusion: Extending surveillance colonoscopy intervals could result in significant rates of I-CRC and I-AP. Diagnosis of I-CRC at advance stage can occur at interval less than 24 months.

Disclosure: Nothing to disclose.

PP0764

ENDOSCOPIC CHARACTERISTICS OF LOCAL TUMOR REGROWTH IN PATIENTS UNDERGOING WATCH-AND-WAIT AFTER NEOADJUVANT THERAPY FOR RECTAL CANCER

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Introduction: Watch-and-Wait is an increasingly recognized non-surgical management option for rectal cancer patients achieving a clinical complete response (cCR) following neoadjuvant therapy. This approach aims to preserve organ function by avoiding surgery, but necessitates vigilant endoscopic surveillance, to detect local tumor regrowth[1] (LTR). The specific endoscopic features associated with regrowth remain incompletely defined.

Aims & Methods: We performed a retrospective analysis of patients monitored under the Watch-and-Wait program at Karolinska University Hospital[2], focusing on the endoscopic characteristics of LTR. As of December 2023, 145 patients who met the criteria for clinical complete response (cCR) were included.

The neoadjuvant treatment consisted of chemoradiotherapy (CRT) in 16 patients (11%), short course radiotherapy (SCRT) + chemotherapy in 69 patients (47.6%) and SCRT alone (5 x 5 Gy) in 60 patients (41.4%). The median age of the patients was 66 years (range 29-94), with 61.4% being male.

The initial tumor stage was T1/T2 in 42 patients (29%) and T3/T4 in 103 patients (71%). The nodal stage was N0 in 35 patients (24%) and N+ in 110 patients (75.9%). The median follow-up period was 29 months (range 3-102 months).

Patients were monitored with rectal digital exams, flexible endoscopy, CEA measurements, and MRI every 3 months for 2 years, and then every 6 months thereafter. During endoscopy, narrow-band imaging (NBI) and NICE classification[3] was utilized.

Results: Out of the 145 patients, 28 (19.3%) experienced LTR as of December 2023. The median time from the completion of radiotherapy to regrowth was 9 months (range 3-20 months). Regrowth was detected intraluminally by endoscopy in 23 patients (82.1%), while in 5 patients (17.9%), regrowth was detected on MRI but was not visible intraluminally.

The most common endoscopic features of regrowth included a NICE 3 pattern (n=16) and contact bleeding (n=16).

Additional findings included polypoid tissue (n=13), ulceration (n=10), irregular surface (n=9), fibrin cover (n=9), stenosis (n=6), and submucosal nodules (n=5).

Conclusion: As regrowth occurs in at least one of five patients and most commonly with intraluminal findings, this study highlights the importance of meticulous endoscopic follow-up.

Contact bleeding and NICE type 3 lesions observed with narrow-band imaging (NBI) were key endoscopic findings in patients with tumor regrowth, while polypoid lesions were observed in less than 50%.

Combining endoscopic assessment with other modalities, such as MRI, is essential, as regrowth may be undetectable endoscopically in a proportion of patients.

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PP0765

COLORECTAL CANCER SURVEILLANCE PROGRAM IN LYNCH SYNDROME: A CLINICAL AUDIT IN A REFERENCE CENTER

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Introduction: Lynch syndrome (LS) is a hereditary condition associated with an increased risk of early colorectal cancer (CRC), due to mutations in DNA mismatch repair (MMR) genes, accounting for 3-5% of all CRC. High-definition screening colonoscopies in specialized centers reduce CRC mortality in this setting.

Aims & Methods: We conducted a clinical audit in a large cohort of these patients, ultimately to enhance the quality of care provided. We performed a retrospective evaluation of the clinical characteristics and risk factors of LS patients under surveillance at our reference center. CRC surveillance program includes colonoscopy every 2 years, starting at ages 20–25 for MLH1/MSH2 carriers and 30–35 for MSH6/PMS2 carriers.

We further analyzed compliance to surveillance protocol and its clinical outcomes – our primary endpoint was mortality rate due to CRC and secondary endpoints were time to first advanced colorectal lesion and CRC staging at diagnosis.

Results: A total of 508 patients were included (44% MSH2, 23% MLH1, 20% MSH6, and 13% PMS2). The mean age at surveillance initiation was 44.4 years, with 35.6% starting after the diagnosis of a Lynch syndrome-associated neoplasia. The mean surveillance duration was 5.6 years. Each patient underwent an average of four colonoscopies, with 50% performed within the recommended surveillance intervals.

During the follow-up period, 35 patients were diagnosed with CRC, with a mean time to first advanced colorectal polyp of 3.7 years and for the first CRC diagnosis of 4.8 years. Having any colonoscopy outside the established interval was not associated with higher CRC incidence (p = 0,647). CRC cases diagnosed prior to surveillance initiation (n=143) were more advanced (58.1% stage ≥ II) and were treated with segmental colectomy in 80% of cases. In contrast, CRC cases diagnosed during surveillance (n=48) were detected at earlier stages (64.6% Tis/T1) and managed with endoscopic resection (29%) or total colectomy (56%) ($p < 0.001$).

The incidence of CRC was higher in MLH1/MSH2 carriers compared to MSH6/PMS2 carriers (36% vs. 22%, $p = 0.0020$). The mean age at first CRC diagnosis was 45.6 years for MLH1/MSH2 carriers and 49.9 years for MSH6/PMS2 carriers ($p = 0.053$). The CRC-related mortality rate was 0.6% (n=3), all cases diagnosed before surveillance initiation.

Conclusion: Although adherence to the CRC surveillance program in LS patients can be optimized, our findings underscore its importance in specialized reference centers. Surveillance was associated with earlier CRC detection, less invasive therapeutic approaches, and reduced CRC-related mortality. Our results also show a lower incidence of CRC and an higher age at first CCR diagnosis in MSH6/PMS2 carriers, strengthening the personalized surveillance recommendations dependent on gene specificities.

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PP0766

SURGERY OR SURVEILLANCE? EVALUATING BENEFITS OF ENDOSCOPIC RESECTION FOR HIGH-RISK T1 COLORECTAL CANCERS

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Introduction: T1 colorectal cancer can be managed either by surgery or interventional endoscopy. High-risk features after endoscopic resection (G3, submucosal invasion >1000 µm, lympho-vascular invasion, tumour budding+, or R1 resection) increase the risk of lymphnode metastasis (10-15%), often requiring additional surgery. However, the benefits of surgery versus endoscopic resection alone remain unclear, particularly in elderly or frail patients.

Aims & Methods: The aim of the study is to assess recurrence rate and overall survival in patients underwent surgery, endoscopy or both. Secondary endpoints is to evaluate incidence of adverse events and length of hospital stay.

All consecutive patients diagnosed with T1 colorectal cancer from 2018 to 2024 in our institution were enrolled in this retrospective study. We divided this cohort in 3 groups: endoscopic resection plus surgery (ER-S), upfront surgery (SR), or endoscopic resection alone (ER). Demographics data, mortality, cancer recurrence rate, disease free survival and AEs were recorded.

Results: One-hundred and thirty-three patients were enrolled in the study: 34 (72 years±10) in the SR group, 65 (65 years±11) in the ER-S group and 34 (77 years±11) in ER, respectively.

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Lesions were similar (p-value 0.30) for dimension: 31±15 mm (ER) vs 23±12 mm (ER-S) vs 30±13 mm (ER) such as follow-up (23 vs 24 vs 21 months) for the three groups (p-value 0.428). Hospital stay after the procedure was significantly lower in the ER group (1 day median) vs SR (median 9 days;p-value 0.0001) and ER-S group (7 days median;p-value 0.003). Cumulative 30 days adverse events were higher in the SR group (62%; p-value 0.0001) and in the ER-S group (22%; p-value 0.001) compared to ER alone (12%). During the study period, no endoscopic/radiological recurrence occurred in the SR group whereas 1 patient in the ER-S group and 2 in the ER group showed recurrence non suitable for further treatments.

Five patients in SR group died after a median of 18 months (1 for early surgical complication); 2 died in the ER-S group after a median of 34 months and 3 died in the ER group after a median of 24 months (1 for rectal cancer progression) (p-value 0.365).

Overall survival was comparable among the three groups with no significant difference (p-value 0.718) with a median of 40 months (SR), 28 (ER-S) and 26 (ER) months.

Conclusion: Endoscopic resection alone appears to be a reasonable option for high-risk T1 CRC patients, especially those with increased surgical risk due to low rate of recurrence, low adverse events and a comparable overall survival in short-mid term.

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Disclosure: Giuseppe Grande is consultant for Olympus, Boston Scientific, ERBE, Fujifilm.

PP0767

DIAGNOSIS OF RECTAL SENSITIVITY DISORDERS BY THE LONDON CLASSIFICATION DEPENDS ON THE MEASUREMENT METHOD: RESULTS OF A PROSPECTIVE, FEASIBILITY STUDY

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Introduction: Rectal sensitivity can be assessed by elastic balloon distension (EBD) or rectal barostat test (RBT)[1]. EBD assesses rectal sensitivity to a maximum volume determined by the balloon. In contrast, RBT assesses rectal sensory thresholds relative to an individual's total rectal capacity, and can additionally measure intra-rectal pressure. There are no studies that compare diagnostic yield for disorders of rectal sensation, defined by the London Classification[2], using EBD and RBT.

Aims & Methods: This prospective study assessed the feasibility, tolerability, and diagnostic agreement of EBD and RBT in a symptomatic population (Rectal barostat study, ISRCTRN 46893, recruitment ongoing).

Prospective patients referred for anorectal physiology studies were recruited from two NHS Trust sites (UK). Participants underwent both EBD and RBT in randomised order. At site 1 EBD was performed using a Laborie water-perfused manometry system (LWP, balloon volume 400mL). At site 2 the Medtronic solid-state 3D manometry system was applied (MSS, balloon volume 400mL). At both sites the Mui Scientific Rapid Barostat was used for comparison (RBT, bag volume 700mL). Standard protocols for EBD and RBT were followed[3,7].

Reference ranges for each rectal sensitivity test were based on published studies[4-7].

Diagnosis based on the London Classification was obtained using results from both the EBD and RBT. Participants completed a questionnaire to assess tolerability of both procedures.

Results: At site 1 patient uptake of the study was 76% (n=16, 10F:5M, median age 64). At site 2 patient uptake of the study was 45% (n=9, 8F:1M, median age 70). 1 participant from both sites withdrew from the study. 1 participant reported pain during EBD, and 2 during RBT. 2 participants reported that they would not be willing repeat the tests. Median time taken to perform each procedure was 1min24s for EBD and 15min26s for RBT.

The percentage of London Classification diagnoses that remained unchanged was 13% at site 1 and 56% at site 2.

ICC demonstrated good test-retest reliability of the RBT for all sensory thresholds, including 'first sensation' (k=0.84, p<0.001), 'normal desire to defecate' (k=0.84, p<0.001), and 'sustained urgency' (k=0.86, p<0.001).

Conclusion: Patient willingness to be entered into the study differed between sites despite the recruitment process and study team remaining the same. This may be due to differences in patient demographics between sites. Both the EBD and RBT were feasible and well tolerated by participants, however the RBT protocol was significantly longer. Good test-retest reliability of the RBT suggests the protocol could be reduced to one round of sensory testing.

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These preliminary findings suggest that agreement between London Classification diagnoses obtained using EBD and RBT is sub-optimal. This was evident in both the LWP and MSS systems.

Further studies would be required to determine whether the management of patients with rectal sensitivity disorders based on the London Classification depends on the type of equipment used to assess rectal sensitivity disorders. This is important as accurate assessment of rectal sensitivity may be valuable in guiding effective management of patients with rectal sensitivity disorders[8].

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PP0768

EFFECTIVENESS OF A LOW HISTAMINE DIET IN PATIENTS WITH IRRITABLE BOWEL SYNDROME: A CLINICAL STUDY

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Introduction: Functional diseases of the gastrointestinal tract are a common diagnosis, with the most common manifestations being functional dyspepsia and IBS. The pathophysiological mechanisms and etiology of this group of diseases are not fully understood, but stress is believed to play a key role.

Aims & Methods: This study aims to evaluate the effectiveness of a low-histamine diet in patients with irritable bowel syndrome. Patients who meet the ROME IV criteria were included in the study.

The study included 120 patients, divided into three groups: low- histamine diet, FODMAP diet and mediterranian diet. Questionnaires were used to assess stress levels PSS-4, anxiety GAD-7, severity of IBS - IBS Severity Score, and quality of life.

The questionnaires were completed three times: both at the beginning of the study, immediately after the end of the 4-week diet, and one month after the end of the diet therapy.

Results: The low-histamine diet led to a statistically significant improvement in IBS symptoms compared to baseline. These findings suggest that the dietary intervention had a positive therapeutic effect (low- histamine diet *:* t = 9.6181, p-value < 0.001, low FODMAP diet Z = 22, p-value < 0.001, mediterannian diet Z =119, p-value < 0.001).

Conclusion: Dietary therapy plays a key role in the management of IBS. The low-histamine diet shows promising results and may serve as a viable alternative to the FODMAP diet.

Disclosure: The study was launched after receiving approval from the Research Ethics Committee (REC) at the Medical University "Prof. Dr. Paraskev Stoyanov" – Varna, protocol 111/ 20.01.2022y.Data were processed and analyzed using appropriate statistical methods chosen according to the nature of the variables and research questions. The analysis includes:

• Evaluation of the statistical significance of the main and interactions between the factors (Repeated Measures with Wicolxon and Paired t-test).

• Regression analysis to assess relationships between dependent and independent variables. As well as using heteroskedasticity-robust standard error estimates (HC3) to ensure accuracy of regression analysis results.

PP0769

IS METHANE PRODUCING FLORA CLINICALLY RELEVANT?

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Introduction: The lactulose breath test (L-BT) is a non-invasive method to assess small intestinal bacterial overgrowth (SIBO). However, diagnostic criteria for intestinal methanogenic overgrowth (IMO) differ among experts, and international guidelines lack a definition. Methanogenic flora is linked to slower gut transit in irritable bowel syndrome (IBS), but the effects of methane-targeting treatments on methanogens, colonic transit, and stool form remain unclear.

Aims & Methods: To examine relationships between methanogenesis, constipation, pain and bloating in healthy volunteers and patients with and without IBS. METHODS: Cross-sectional study on adults with L-BT between 2016–2024 at a single center. Each patient completed a Rome IV questionnaire, sociodemographic and medical data and Bristol stool scale. IMO was diagnosed if there were more than 3 measurements of CH_4 over 10 ppm over 180 minutes.

We examineded 3 alternate diagnostic thresholds: (A) 10 ppm at any time (American criteria or C1); (B) 10 ppm over baseline before 90 minutes (C2); and (C) 10 ppm over baseline before 60 minutes (C3), We considered "high producers" of CH_4 those that had 2 measurements over 60 ppm. Statistical analysis with Mann-Whitney and Kruskal Wallis test.

Results: We included 5,053 subjects who underwent L-BT (mean age 49 years, 81% women). Of these, 10% were exclusive H2 producers, 2% CH4 producers, and 2.7% were non H2/CH4 producers. CH4 producers had a higher frequency of constipation (p<0.001) and greater total CH4 production in the Bristol 1-2 group. CH4 production over 180 minutes was lower in IBS-D patients, though the overall methane production rate was similar

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to healthy volunteers. Based on diagnostic thresholds, 92.6% of constipated patients met C1, 52.3% C2, and 18.1% C3, with significant differences (p=0.001). Stricter IMO criteria were associated with greater bloating and pain intensity.

Conclusion: The high proportion of methane production in our asymptomatic controls, compared to previous series, may reflect demographic differences. Methane producers had a higher frequency of constipation. The IMO diagnosis rate varied depending on the criteria used, with stricter criteria identifying patients with more frequent constipation, bloating, and abdominal pain.

Disclosure: None.

PP0770

EFFECT OF PSYCHOLOGICAL STRESS ON THE SMALL-INTESTINAL ENVIRONMENT AND MUCOSAL INTEGRITY

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Introduction: Psychological stress is an important factor for the development of functional gastrointestinal disorders. Psychological stress may play a pivotal role in inflammatory pathophysiology in the gastrointestinal tract by affecting the small-intestinal environment and mucosal integrity. However, the relationship between psychological stress and the abnormality of intestinal environment/mucosal integrity remains unclear. Therefore, in the present study we comprehensively investigated the microbiome profile, expression of tight junction protein and mucosal inflammatory conditions in the small intestine in chronic social defeated stress (CSDS) mice that is accepted as a model of depressive state.

Aims & Methods: The CSDS model were prepared by exposing a C57BL/6N intruder mouse to a CD-1 aggressor mouse for 10 minutes daily 10 days in the home cage of CD-1 mouse. A social interaction test was performed to evaluate social avoidance behavior. The obtained fresh fecal samples were used to analyze the gut microbiome profile and stool characteristics including appearance and moisture. Small intestinal tissues were collected to assess the expression of cytokines, lipopolysaccharide (LPS), antimicrobial peptides, and tight junction proteins (TJPs) using real-time RT-PCR and immunohistochemistry.

Results: The time spent in the social avoidance zone was significantly longer in the CSDS group than in the controls when the CD-1 aggressor mouse was present. These findings indicate that CSDS mice are successfully established as a model of depressive state. Regarding stool appearance, the average Bristol scale score was significantly lower in the CSDS group compared to the controls ($p < 0.01$). Consistent with this finding, the fecal water content was significantly lower in the CSDS group compared to the control group ($p < 0.05$). There were no differences in the α- and β-diversity of the fecal microbiome between the CSDS and control groups. However, the relative abundances of *Proteobacteria* and *Patescibacteria* at the phylum level were significantly higher in the CSDS group ($p < 0.05$). Regarding TJPs and antimicrobial peptides, the expression of Occludin, Claudin 4, and Reg IIIg was significantly decreased in the small intestinal epithelium of CSDS mice compared to the controls. As for the cytokines and LPS, the expression of both IL-1b and LPS was significantly increased in the small intestine of CSDS mice compared to the controls.

Conclusion: Psychological stress affects the small-intestinal environment and mucosal integrity, resulting abnormal stool property and intestinal minimal inflammation.

Disclosure: Nothing to disclose.

PP0771

SOMATIZATION IN PEDIATRIC PATIENTS WITH IRRITABLE BOWEL SYNDROME

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Introduction: Somatization is defined as a tendency to experience and communicate psychological distress in the form of somatic symptoms. Literature suggests that there is a link between somatization and irritable bowel syndrome (IBS). Nevertheless, the association between clinical symptoms of IBS and somatization, especially in children, remains unclear.

Aims & Methods: 54 children aged 6-12 with a verified diagnosis of IBS according to Rome criteria IV were included in the study. The control group consisted of 50 healthy children of the same age and gender. The level of somatization was assessed by the Somatoform Symptom Screening test (SOMS-2J by Rief, W., & Hiller, W.), which was modified for children.

The intensity of abdominal pain and dyspeptic symptoms were estimated using a 4-point Likert scale. Data were processed using Microsoft Excel 2016 and analyzed with GraphPad (Prism 5.0). Pearson test was used to assess the correlation between SOMS rate and the intensity of somatic complaints.

Results: The level of somatization was significantly higher in children with IBS in comparison to the control group (p<0.0001). There was no difference in somatization rate in patients according to the predominant bowel habit. A direct correlation between the intensity of abdominal pain and the level of somatization (r=+0.37; p=0.005) was established in children with IBS (Fig.1).

On the contrary, the intensity of dyspeptic symptoms was less influenced by the rate of somatization (r=+0,18; p=0,20).

Conclusion: To conclude, children with IBS who had higher rates of somatization were revealed to have a higher intensity of somatic complaints, especially abdominal pain. We suggest that somatization plays an important role in the occurrence of somatic symptoms of IBS. It may be one of the reasons for the low effectiveness of pharmacological treatment of IBS in children.

Disclosure: Nothing to disclose.

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PP0772

SYMPTOMATIC IMPROVEMENT IN PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI) IN RESPONSE TO CHRONIC BLUEBERRY CONSUMPTION IS ASSOCIATED WITH GUT MICROBIOME CHANGES

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Introduction: The antioxidant, anti-inflammatory and modulatory effects of blueberries on gut microbiota are well known[1]. In patients with DGBI blueberry ingestion for 6 weeks significantly relieved abdominal symptoms and improved Gastrointestinal Symptom Rating (GSRS) and quality-of-life and life functioning scores by Outcome Questionnaire (OQ 45.2) compared to placebo[2]. Whether changes in gut microbiota are relevant in the symptomatic relief of blueberries in DGBI is unexplored.

Aims & Methods: Male and female patients aged 18 to 60 years with DGBI (irritable bowel syndrome or functional dyspepsia, Rome 4), were randomized to double-blind blueberry (30g freeze-dried powder = 180g fresh blueberries) or placebo daily for 6 weeks, with subsequent cross-over. GSRS and OQ-45.2 scales were assessed before and in week 6 of each treatment period. Responders were defined by improved GSRS and OQ-45.2 scores. Stool microbiome analysis was performed by DNA extraction and shotgun metagenomic sequencing. Quality controlled sequences were annotated with MetaPhlAn 4. Nonmetric multidimensional scaling ordination plots based on Bray-Curtis dissimilarity were generated at the species level with clinical covariates to visualize the intra- and inter-subject variability of microbiota. ANCOM-BC2 was used to estimate the differential abundance of the prevalent species. Pairwise networks before and after intervention were constructed and the frequency difference of positive and negative associations between responders and non-responders compared. Associations of gut microbial species with outcomes were analyzed (Spearman), stratifying data into before and after intervention, as well as responder and non-responder groups.

Results: In 43 patients, microbiome alpha diversity of the gut microbiota taxa did not change significantly between visits with either treatment and did not differ between responder groups. However, at baseline, multiple species differed in abundance between responders and non-responders for both the GSRS and OQ-45.2 outcomes (table). During blueberry ingestion there were multiple species differences between responders and non-responders for either outcome.

	Baseline (#p<0.05)	During blueberry ingestion (#p<0.05)
GSRS outcome (gastrointestinal)	Responders > Non-responders: 55 species; #in: Butyricimonas faecihominis, Enterocloster lavalensis, Enterocloster citroniae, Mediterraneibacter glycyrrhizinilyticus	Increased in responders & decreased in non-responders: 16 species; #in: Butyricicoccus sp AM29_23AC, Clostridia bacterium UC5_1_1D1, Clostridium sp AF34_10BH, Oscillibacter sp ER4
GSRS outcome (gastrointestinal)	Non-responders > Responders: 56 species; #in: Lachnospiraceae bacterium AM48_27BH, Clostridiaceae bacterium Marseille_Q4143, Blautia glucerasea, Clostridium sp AF34_13, Ruminococcus sp AF41_9	Decreased in responders & increased in non-responders: 8 species; #Bacteroides eggerthii
OQ 45.2 outcome (well-being)	Responders > Non-responders: 35 species: #in: Lachnospira eligens, Vescimonas coporcola	Increased in responders & decreased in non-responders: 14 species; none significant
OQ 45.2 outcome (well-being)	Non-responders > Responders: 41 species; #in: Roseburia intestinalis, Phocaeicola massiliensis	Decreased in responders & increased in non-responders: 14 species; none significant

There were distinct association patterns in the microbial co-occurrence networks with hub taxa, especially *Anaerobutyricum hallii* and *Ruminococcus gnavus*, in responders versus non-responders at baseline and during blueberry ingestion. Negative associations between microbiota in microbial co-occurrence networks were more common in non-responders than in responders at baseline and decreased after blueberry treatment, suggesting stabilisation of the microbiome.

Conclusion: Improved GI and general quality-of-life and life-functioning scores during blueberry ingestion in DGBI are associated with distinct gut microbiota taxa and co-occurrence networks at baseline and during treatment. Confirmatory studies are required to assess causality, but these data suggest that beneficial clinical effects seen in patients with DGBI during blueberry ingestion may be linked to changes in the gut microbiome and that GI and general benefits may be effected via distinct taxa.

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PP0773

ASSOCIATIONS BETWEEN INTESTINAL MICROBIOTA COMPOSITION, CLINICAL PHENOTYPES AND TESTS OF DISEASE MECHANISMS IN PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI) RESPONDING TO CHRONIC BLUEBERRY INGESTION

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Introduction: Blueberry ingestion resulted in significant abdominal symptom relief, improved Gastrointestinal Symptom Rating (GSRS) and quality-of-life and life functioning scores by Outcome Questionnaire (OQ 45.2) in patients with DGBI[1]. It is unclear via which mechanisms blueberries exert their beneficial effect in DGBI[2].

Aims & Methods: We explored the associations between symptomatic responses to blueberry ingestion, faecal microbiota composition, clinical phenotypes and tests of disease mechanisms in DGBI.

Male and female patients aged 18-60 years with DGBI (irritable bowel syndrome or functional dyspepsia, Rome 4), were randomised to blueberries (30g freeze-dried powder = 180g fresh blueberries) or placebo daily for 6 weeks in a double-blind, cross-over study. GSRS and OQ-45.2 were assessed before and in week 6 of each treatment period. Responders were defined by improved GSRS or OQ-45.2. Demographics, lifestyle and symptom details were documented at baseline[3].

Further variables were gathered before and in week 6 of treatment: Bristol Stool Scale, breath H_2, CH_4 and symptoms after fructose 35g, cognitive tests (CANTAB®), skin fluorescence for advanced glycation end-products and serum tight junction proteins (claudin-3 and -5, zonula occludens-1). Stool microbiome analysis was performed by DNA extraction and sequencing for shotgun metagenomics. Nonmetric multidimensional scaling ordination plots based on Bray-Curtis dissimilarity were generated at the species level with clinical covariates. Spearman association of microbial species with demographics, clinical phenotype and mechanistic tests

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at baseline and during treatment were analysed by stratifying data into pre- and post-treatment groups, as well as responders and non-responders by GSRS and OQ-45.2.

Results: In 43 DGBI patients there were significant associations between faecal microbiota composition, clinical phenotypes and tests relating to disease mechanisms that were distinct between blueberry responders and non-responders. Examples of prominent baseline associations were breath gases after fructose with *Ruminococcus* sp AF41_9 in GSRS responders and with *Eggerthella lenta, Lachnospiraceae bacterium* AM48_27BH, *Phocaeicola vulgatus*, *Ruminococcus callidus*, *Mediterraneibacter glycyrrhiznilyticus* in non-responders.

Cognitive skills were associated with *Clostridium* sp AF34_13 and *Clostridiaceae bacterium* Marseille_Q4149 in responders, and with *Blautia glucerasea*, *Eggerthella lenta* and *Ruminococcus callidus* in non-responders. Tight junction protein levels correlated with *Clostridiaceae bacterium* Marseille_Q4149, *Anaerobutyricum hallii* and *Bacteroides intestinalis* in responders and with *Anaerostipes hadrus*, *Blautia faecicola*, *Blautia glucerasea*, *Clostridia bacterium* UC5_1_1D1 and *Ruminococcus gnavus* in non-responders.

During blueberry ingestion new associations between bacterial taxa and GSRS, breath gases after fructose, tight junction protein levels emerged, which were largely distinct between responder groups. Responders by OQ-45.2 showed associations between bacterial taxa, demographics and mechanistic tests both at baseline and during blueberry dosing that were largely distinct from those of GSRS responders.

Conclusion: In patients with DGBI, GI and general symptom relief with blueberry ingestion were characterised by specific, distinct associations between faecal microbiota, clinical phenotype and tests of disease mechanisms at baseline and during treatment. This first longitudinal blueberry treatment study in DGBI patients suggests an important role of the intestinal microbiota in symptom genesis warranting further study.

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PP0774

IMMUNOTHERAPY IN GASTROINTESTINAL DYSAUTONOMIA WITH SMALL FIBER NEUROPATHY: A SINGAPORE CASE SERIES

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Introduction: Small fiber neuropathy (SFN)–related gastrointestinal (GI) dysautonomia presents with debilitating symptoms that often fail standard therapies. Emerging evidence supports the use of immunomodulatory treatments such as intravenous immunoglobulin (IVIg) and rituximab in neuropathic and autonomic disorders (1, 2, 3).

Aims & Methods: We retrospectively reviewed six patients (mean age 52 years; 50% female) in Singapore at a national tertiary hospital with chronic GI manifestations such as dysphagia, constipation, abdominal pain, and vomiting with evidence of small fibre neuropathy. All patients had abnormal quantitative sudomotor axon reflex testing (QSART)tests. Five of the patients had a positive skin biopsy, while one patient had a negative skin biopsy but a positive anti plexin D1 antibody pathogenic of small fibre neuropathy (4).

Six patients received IVIg (2 g/kg over 2–5 days every 4 weeks), three of whom had received IVIg and then rituximab (1,000 mg × 2 doses, 14 days apart) sequentially. Patient reported outcomes were assessed and categorised into mild, moderate, good or no responses.

Results: Among IVIg recipients, one experienced good symptomatic relief, one had a mild response and one had no response. Of the patients who had received both IVIg and rituximab, one achieved no response, one achieved a mild response and one achieved a moderate response. All treatments were well tolerated, with no serious adverse events.

Patient ID	GI Symptoms	Positive QSART	Skin Biopsy for SFN	Treatment	Response
A	Dysphagia, bloating	Yes	Positive	IVIg	No response
B	Constipation, dysphagia, vomiting	Yes	Positive	IVIg + Rituximab	Moderate response
C	Vomiting, abdominal pain	Yes	Positive	IVIg	Good response
D	Constipation, abdominal pain	Yes	Normal but with positive anti plexin D1 antibody	IVIg	Mild response
E	Odynophagia, pain, vomiting, constipation	Yes	Positive	IVIg + Rituximab	No response
F	Constipation	Yes	Positive	IVIg, Rituximab	Mild response

Table 1. Summary of Patients Treated with IVIg and Rituximab for GI Dysautonomia.

Conclusion: In this preliminary series, IVIg and rituximab were associated with symptomatic benefit in selected patients with SFN-associated GI dysautonomia refractory to conventional management. These findings warrant prospective, controlled trials incorporating standardized outcome measures to optimize immunotherapy protocols and define responder profiles.

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PP0775

ASSOCIATION OF SYSTEMIC CYTOKINE PROFILES WITH HOST-MICROBE HOMEOSTASIS AND CLINICAL SYMPTOMS IN FUNCTIONAL BOWEL DISORDERS

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Introduction: While the mechanistic background to functional bowel disorders, including irritable bowel syndrome (IBS) is insufficiently understood, there is growing evidence that cytokine imbalance and inflammation may contribute to IBS pathogenesis.

Aims & Methods: We undertook a comprehensive investigation of serum cytokine profiles in individuals with functional bowel disorders (IBS n=164, functional constipation n=6, functional diarrhea n=3) and healthy controls (n=31), mapping their associations with clinical symptoms (GSRS-IBS, HAD), the gut microbiome (whole-metagenome sequencing of fecal samples), colonic mucosal responses (bulk RNA sequencing of sigmoid biopsies), intestinal physiological parameters, and the serum metabolome (liquid chromatography-mass spectrometry). Serum levels of IFN-γ, IL-1β, IL-6, IL-8, IL-10, IL-12p70, IL-13, TNF-α, IL-17A and CXCL10 were measured with Meso Scale Discovery (MSD) multiplex immunoassay kits.

Results: While cytokine levels did not differ significantly between persons with functional bowel disorders and healthy individuals, CXCL10 was reduced in participants with predominant diarrhea (IBS-D and functional diarrhea, p=0.01), whereas IL-12p70 was increased in predominant constipation (IBS-C and functional constipation, p=0.03), compared to healthy controls.

Within both the full study cohort and participants with functional bowel disorders, IL-12p70 levels correlated with oroanal transit time (rho 0.20 and 0.19, p=0.007 and 0.01, respectively), and were inversely associated with fecal short-chain fatty acids (p=0.006, 0.01 and 0.03 for acetate, propionate and butyrate, respectively), as well as with the abundance of several butyrate-producing species.

By contrast, IL-12p70 levels were positively associated with mucus-resident species (e.g., from the *Akkermansia* and *Victivallis* genera), with metabolites potentially derived from the degradation of mucosal carbohydrates, and with a distinct T helper 1 type response in colonic biopsies. Moreover, IL-12p70 levels correlated with anxiety (rho 0.17, p=0.03) in participants with functional bowel disorders.

Conclusion: Systemic cytokine profiles did not discriminate persons with functional bowel disorders from healthy individuals, but participants with predominant constipation and diarrhea exhibited distinct cytokine profiles. Alteration of systemic cytokines, notably IL-12p70, may result from disrupted mucosal barrier integrity and host-microbe homeostasis, and contribute to extraintestinal symptoms in a subset of individuals with functional bowel disorders.

Disclosure: Nothing to disclose.

PP0776

THE EFFECT OF VISCERAL SENSITIVITY ON THE COURSE OF THE IRRITABLE INTESTINE SYNDROME

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Introduction: Managing patients with severe irritable bowel syndrome (IBS) has significant challenges for practicing physicians due to the condition's refractoriness to standard therapy and the lack of clinical remission. The prevalence of IBS reaches 9–21% among the adult working population and is associated with a low quality of life. According to various sources, visceral hypersensitivity (VHS) plays a key role in the pathogenesis of IBS.

Aims & Methods: The aim is to show the relationship between clinical symptoms, the visceral sensitivity index, and the visceral sensitivity threshold in patients with severe IBS-D.

A total of 84 patients with severe IBS-D, according to the Rome IV criteria, were examined. The median (Me) patient age was 32 (25; 37.2) years, with a disease duration of 5 (4; 6) years. Pain syndrome was assessed using the VAS scale in points (pts.), the level of visceral sensitivity index (VSI) using the J. Labus questionnaire reflecting gastrointestinal anxiety (pts.), and the visceral sensitivity threshold (VST) was studied via balloon dilatation test (BDT) (ml).

Statistical analysis was conducted using non-parametric methods (Statistica 12). Quantitative data were described using Me and interquartile ranges (Q1 – Q3). Differences were considered statistically significant at $p < 0.05$.

Results: Before treatment, all patients showed a high level of pain syndrome - 9 (8; 9) pts. on VAS, and level of gastrointestinal anxiety at 71.00 (63; 75) according to the J. Labus questionnaire. BDT data revealed a low VST (first threshold—urge to defecate): 34.00 (27; 43.5) mm Hg. After examination, patients were randomly divided into two equal groups.

The first group received the SNRI antidepressant (duloxetine) 60 mg/day, while the second group received trimebutine maleate 600 mg/day for 8 weeks, followed by assessment of therapy effectiveness. In the first group, SNRI therapy showed positive dynamics: a decrease of pain levels (from 9 (9 - 10) to 2 (1.75 - 3), $p < 0.01$*), an increase in VST (from 34.0 (29.75, 45.0) to 92.0 (86.0, 123.0), $p < 0.01$*), and a reduction in gastrointestinal anxiety (from 56.50 (47.50 – 68.00) to 12.00 (10.00 – 19.25), $p < 0.01$*). No significant positive changes were achieved in the second group.

After the treatment, the overall group showed an inverse correlation between abdominal pain (VAS) and VST (BDT data) ($\rho = -0.772$; $p < 0.01$), and between gastrointestinal anxiety (VSI by J. Labus) and VST (BDT data) ($\rho = -0.782$; $p < 0.001$).

Conclusion: The effectiveness of therapy in the first group of patients was associated with the severe course of IBS, characterized by high gastrointestinal anxiety, abdominal pain, and low VST. In the overall group, post-treatment correlation analysis revealed a connection between the severity of clinical presentation and VST levels (BDT). Balloon dilatation test allows an objective assessment of IBS severity and therapeutic effectiveness. The treatment of choice for severe IBS-D is the SNRI duloxetine, which acts on the pain center and patients' emotional status.

Disclosure: Nothing to disclose.

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PP0777

THE EFFECTS OF PERSONALITY TRAITS ON PSYCHOLOGICAL FACTORS AND QUALITY OF LIFE IN IRRITABLE BOWEL SYNDROME: AN EXPLORATORY STUDY

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Introduction: Irritable bowel syndrome (IBS) has a profound impact on health-related quality of life. Increasing evidence suggests psychological factors have a large impact in the development and course of IBS, and in particular quality of life, with large individual heterogeneity.
This observational study aims to investigate the influence of personality traits using the Big Five Inventory (BFI) and examine their associations with abdominal pain, general and gastrointestinal-specific anxiety, and quality of life in patients with IBS.

Aims & Methods: Data from a large observational study in patients with IBS (Rome III criteria) was used, in which patients completed questionnaires with regards to personality traits (BFI-44), symptom severity, general anxiety, GI-specific anxiety, and quality of life. Multivariable regression models and a network correlation analysis were used to determine underlying associations.

Results: 194 patients were included for analysis (74.7% female, mean age 51.36 years). There was a positive association of neuroticism to GI-specific anxiety (B 7.234, P = 0.040) and general anxiety (B 0.954, P = 0.040) with a strong negative association to mental quality of life (B – 3.576, P = 0.007). In addition, a strong negative association of agreeableness to mental quality of life was found (B – 4.266, P = 0.001). Although there were no significant associations of personality traits to abdominal pain, positive associations were found regarding general anxiety (B 6.304, p < 0.001) and GI-specific anxiety (B 0.034, P <0.001) of abdominal pain. There were no associations found with regards to the other personality traits openness, conscientiousness or extroversion.

Conclusion: IBS patients with a higher level of neuroticism were more likely to experience higher levels of GI-specific and general anxiety. The trait neuroticism and/or agreeableness was more likely to have a significant lower mental quality of life in IBS. These findings emphasize the contribution of personality traits in the pathophysiology and course of IBS, pleading for a more personalized approach in patient management.

Disclosure: Nothing to disclose.

PP0778

PREVALENCE AND BURDEN OF OPIOIDS USE IN PATIENTS WITH ROME IV CONSTIPATION-PRAEDOMINANT IRRITABLE BOWEL SYNDROME: A REAL-WORLD EXPERIENCE

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Introduction: Irritable bowel syndrome with praedominant constipation (IBS-C) is the most common subtype of IBS. Opioids prescription rate has increased worldwide over the last two decades. Opioids are not recommended for pain management in gut-brain interaction disorders.

Aims & Methods: The aim of this study was to assess the prevalence of opioids use among patients diagnosed with Rome IV IBS-C and to compare their clinical characteristics with those of IBS-C patients who do not use opioids. Data were retrospectively collected (January 2016 – December 2021) from consecutive outpatients diagnosed with IBS-C at a single tertiary care neurogastroenterology clinic in UK. A standardized template was used to collect data on age, sex, gastrointestinal (GI) symptoms, medications, and opioid use/discontinuation. Patients taking opioids were advised to discontinue opioids in line with the Centers for Disease Control and Prevention recommendations. Improvement in GI symptoms after opioids discontinuation was documented. Data are presented as median [interquartile range (IQR)] or frequency (%). Chi square test or Mann-Whitney's U test was used, when appropriate.

Results: A total of 157 IBS-C patients were included, of whom 70 (27.2%) were on opioids. Compared with non-users, opioids users had a higher median age, more comorbidities, more concomitant medications, more surgeries and more hospital admissions (Table 1).
A significantly higher proportion of opioids users reported dysphagia compared to non-users (*P*=.02). Fifty-eight patients (82.9%) were asked to stop opioids before introducing other medications. Twenty-two patients (31.4%) stopped opioids and, of these, 12 (54.5%) reported improvement in IBS-C symptoms:6 improved in constipation, 1 in abdominal pain and 5 in both.

	Opioid use		
	No (n=187)	Yes (n=70)	p-value
Demography and general characteristics			
Age, median (IQR)	43.0 (27.0)	49.0 (21.3)	0.016
Total number of concomitant medical conditions, median (IQR)	1.0 (2.0)	2.0 (2.0)	0.001
Number of surgeries, median (IQR)	0.0 (1.0)	1.0 (2.0)	0.005
Total number of hospital admissions, median (IQR)	1.0 (3.0)	2.0 (5.0)	0.006
Total number of investigations performed, median (IQR)	1.0 (2.0)	2.0 (2.0)	0.057
Concomitant medications			
Total number of concomitant medications, median (IQR)	2.0 (5.0)	6.0 (4.0)	<0.001
Laxatives, n (%)	76 (40.6)	40 (57.1)	0.018
Tricyclic antidepressant, n (%)	25 (13.4)	19 (27.1)	0.009
Serotonin noradrenalin reuptake inhibitor, n (%)	4 (2.1)	9 (12.9)	<0.001
Non-steroidal anti-inflammatory drug, n (%)	17 (9.1)	14 (20.0)	0.017
Antiemetics, n (%)	7 (3.7)	9 (12.9)	0.007

Table 1. General characteristics of IBS-C patients taking or not taking opioids. Numbers are frequency (%) or median (IQR).

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Conclusion: Up to 27% of IBS-C patients consume opioids at their first outpatient consultation. Patients on opioids show a more severe clinical profile and higher healthcare resources utilization. Opioid cessation leads to IBS-C symptoms improvement in up to 55% of patients without further intervention.

Disclosure: Nothing to disclose.

PP0779

IRRITABLE BOWEL SYNDROME AND NEUROPSYCHIATRIC DISORDERS: A CROSS-SECTIONAL STUDY OF 1.56 MILLION ADOLESCENTS

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Introduction: Studies have shown that adults with irritable bowel syndrome (IBS) may be associated with anxiety, mood disorders, chronic headache, and other neuropsychiatric conditions. This growing body of evidence can allude to a possible bidirectional gut-brain connection. While adolescence is the age when both IBS and neuropsychiatric illnesses often first manifest, this population remains under-studied.

Aims & Methods: We aim to assess the association between the presence of irritable bowel syndrome (IBS) and neurologic and psychiatric disorders in a nationwide cohort of adolescents. This cross-sectional study included all adolescents aged 16-21 who underwent mandatory medical evaluations as part of their conscription between 2000-2021. All individuals completed comprehensive medical interviews. Suspected diagnoses prompted referral to board-certified specialists for confirmation and severity grading based on predefined criteria. IBS was diagnosed according to the Rome II-IV criteria by a certified gastroenterologist.

Neurologic disorders included attention-deficit/hyperactivity disorder [ADHD], epilepsy, chronic headache, sleep disorders. Psychiatric disorders included psychotic, personality, general anxiety disorder [GAD], non-combat post-traumatic stress disorder [PTSD], mood, dissociative, and eating disorders. Each diagnosis was classified by severity (mild, moderate, severe).

Unadjusted and multivariate-adjusted logistic regression models were fitted to estimate odds ratios (ORs) for associations with IBS. The adjusted models were accounted for age, sex, intelligence score, body mass index (BMI), socioeconomic status (SES), education, country of origin, and any concurrent neuropsychiatric diagnoses.

Results: The study included 1,561,717 adolescents (666,442 [42%] women, mean age 17.4 ± 1.4 years), of whom 7,230 (0.46%) were diagnosed with IBS (4,778 mild; 1,948 moderate; 504 severe). Neurologic disorders were present in 1,661 (23.0%) and 143,348 (9.2%) of those with and without IBS, respectively.

The respective numbers of people with psychiatric disorders were 773 (10.7%) and 49,889 (3.2%). In multivariable adjusted models, those with IBS were more likely to have chronic headache (aOR 4.7, 95% CI: 4.4-5.1), GAD (aOR 3.6, 95% CI: 3.2-3.9), eating disorders (aOR 2.6, 95% CI: 1.9-3.5), mood disorders (aOR 2.5, 95% CI: 2.1-3.1), personality disorders (aOR 2.5, 95% CI: 2.1-2.9), and dissociative disorders (aOR 19.1, 95% CI: 12.5-28.9). Patients with severe IBS (n=504) were independently associated with the presence of at least one severe neurologic (aOR 2.9, 95% CI: 1.2-7.1) or psychiatric disorder (aOR 3.8, 95% CI: 2.6-5.4).

Conclusion: In this large nationwide cohort, adolescents with IBS were more likely to have several tested neurologic and psychiatric disorders, compared to those without IBS. Longitudinal studies are needed to explore the directionality and long-term implications of these associations.

Disclosure: Nothing to disclose.

PP0780

FUNCTIONAL CONSTIPATION AND CONSTIPATION-PREDOMINANT IRRITABLE BOWEL SYNDROME: ARE WE TALKING ABOUT THE SAME DISORDER? A REAL-WORLD EXPERIENCE

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Introduction: Functional Constipation (FC) and Constipation-predominant Irritable Bowel Syndrome (IBS-C) are two common disorders of gut-brain interaction (DGBI)-which can often present overlapping symptoms, thus making difficult to distinguish them.

Aims & Methods: This study aims to determine whether the demographic and clinical profiles of consecutive patients diagnosed with FC and IBS-C differ in real-world clinical practice.

Demography and clinical information from the first clinical consultation where the diagnosis was made by a single consultant using a standardized clinical letter were retrospectively collected from January 2016 to March 2025 at a single tertiary care neurogastroenterology clinic in the UK Data were collected using a standardised template and were reported as median [interquartile range (IQR)] or frequency (%), and compared using the Chi-square test or Mann-Whitney U test.

Results: A total of 88 FC and 257 IBS-C consecutive patients were diagnosed according to Rome IV criteria. FC patients were older and most frequently males, with a longer duration of symptoms. Abdominal pain was largely absent in FC patients and when present this was not related to defecation but reported as result of long-lasting lack of bowel movement. Upper gastrointestinal symptoms were more frequent in IBS-C compared to FC.

Comorbidities, particularly chronic pain syndromes, and abdominal surgeries were more frequently reported by IBS-C patients. in addition, IBS-C patients were more frequently on pain medications, while FC on laxatives. Table 1 summarizes the data collected in both groups.

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	FC (n= 88)	IBS - C (n=257)	p-value
Demography and general characteristics			
Male, n (%)	24 (27%)	42 (16%)	0.036
Age (years), median (IQR)	47 (25)	32 (26)	0.0001
Duration of symptoms (year),median (IQR)	12,5 (17)	5 (10)	0.0001
Total number of concomitant medical condition, median (IQR)	1 (2)	4 (4,5)	0.001
Chronic pain, n (%)	12 (14%)	66 (26%)	0.02
		112 (44%)	0.01
Abdominal surgeries, n (%)	24 (27%)		
Total number of hospital admissions, median (IQR)	1 (2)	1 (5)	1.0
Total number of previous diagnostic investigation at first consultation, median (IQR)	1 (2)	1 (2)	1.0
Concomitant medications			
Total number of concomitant medications, median (IQR)	3 (4,5)	4 (4,5)	0.001
Tricyclic antidepressant, n (%)	12 (14%)	44 (18%)	0.550
Selective Serotonin Reuptake Inhibitor, n (%)	12 (14%)	39 (15%)	0.859
Serotonin-norepinephrinereuptake inhibitors, n (%)	8 (9%)	13 (5%)	0.268
Opioids n, (%)	13 (15%)	82 (31%)	0.003
Laxatives n, (%)	61 (70%)	116 (45%)	0.0001
Antispasmodics n, (%)	3 (3%)	42 (16%)	0.003
Gi symptoms			
Nausea, n (%)	16 (19%)	93 (37%)	0.002
Early satiety n, (%)	6 (7%)	56 (21%)	0.002
Vomiting, n (%)	7 (8%)	45 (17%)	0.04
Post pandrial fulness, n (%)	8 (9%)	53 (20%)	0.02
Heartburn, n (%)	13 (15%)	52 (19%)	0.330
Abdominal pain, n (%)	9 (10%)	234 (91%)	0.0001

Conclusion: This real-world study suggests that FC and IBS-C patients differ in demographic, clinical profile and comorbidities. IBS-C patients report a more severe clinical picture despite FC patients and had a longer duration of symptoms.

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Disclosure: Nothing to disclose.

PP1132

WHAT ABOUT THE IMPACT OF BENIGN PROCTOLOGICAL DISEASES ON PATIENTS' QUALITY OF LIFE?

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Introduction: Proctological diseases are a common cause of gastroenterology consultations and can significantly impact patients' quality of life.

Aims & Methods: This study aims to assess the impact of hemorrhoidal disease and anal fissures on quality of life, using a questionnaire to measure their overall influence on daily life.

This cross-sectional study was conducted over one month (May–June 2024) and included patients consulting for hemorrhoidal disease and/or anal fissures. Quality of life was assessed using the Burden Hemo/Fiss score, which consists of 23 questions divided into four domains: physical disorders, psychological distress, discomfort during defecation, and impact on sexual life. Each question was scored from 1 to 6. Data analysis was performed using software from the French National Society of Colo-Proctology (SNFCP).

Results: A total of 50 patients were included, with a mean age of 46.1 years (range: 19–71). There were 37 women and 13 men (M/F ratio: 0.35). Hemorrhoidal disease was present in 23 patients (46%), anal fissures in 18 (36%), and both conditions in 18% (n=9). The mean Hemo/Fiss score was 46.58 ± 19.44 [range: 12–85]. The mean scores for each domain were: 49.4 ± 22.36 (physical disorders), 41.42 ± 19.59 (psychological distress), 56.26 ± 27.86 (defecation discomfort), and 35.44 ± 20.38 (sexual dysfunction). Patients with anal fissures (52.9) and grade III–IV hemorrhoids (58.3) had the most affected scores. A significant correlation was found between the total Hemo/Fiss score and pain score (p=0.021).

Conclusion: Our study highlights a significant deterioration in quality of life in patients with anal fissures or hemorrhoidal disease, emphasizing the need for comprehensive management to improve both physical symptoms and overall well-being.

Disclosure: Nothing to disclose.

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Poster presentations
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PP0781
THE ROLE OF SIRTUIN 1 IN THE DEVELOPMENT AND PROGRESSION OF NON-ALCOHOLIC FATTY LIVER DISEASE UNDER THE INFLUENCE OF STRESS FACTORS DURING THE WAR PERIOD

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Introduction: Sirtuin 1 (SIRT1) regulates mitochondrial function and biogenesis, carbohydrate and lipid metabolism, oxidative stress, inflammation, and apoptosis. All these factors play a role in the development and progression of non-alcoholic fatty liver disease (NAFLD).

Aims & Methods: The aim of the study was to investigate the role of SIRT1 in the development and progression of NAFLD under the influence of stress factors during wartime. The serum SIRT1 level was determined using an enzyme-linked immunosorbent assay (ELISA) with the "Human SIRT1 (Sirtuin 1) ELISA Kit." Visceral fat tissue (VFT) was measured using an OMRON BF-511 electronic device. Liver function tests, carbohydrate and lipid metabolism parameters, and C-reactive protein (CRP) levels were assessed using standard methods on an automatic biochemical analyzer "HumaStar 300SR." Statistical analysis was performed using the SPSS 17.0 statistical software package.

Results: A total of 88 NAFLD patients with metabolic disorders were examined, including 57 individuals (main group) who were exposed to war-related stress factors and 31 individuals (comparison group) examined in the pre-war period. It was found that SIRT1 levels were significantly higher in the main group, averaging 8.88 ± 0.23 ng/mL, compared to 4.75 ± 0.14 ng/mL in the comparison group ($p < 0.05$). Subgroup analysis showed a trend towards higher SIRT1 levels in normal-weight patients compared to those who were overweight or obese in the main group. A direct correlation was found between SIRT1 levels and VFT in both groups: $r = 0.32$ in the main group and $r = 0.24$ in the comparison group. Meanwhile, SIRT1 levels exhibited a weak negative correlation with abdominal obesity parameters (waist-to-hip ratio): $r = -0.14$ in the main group and $r = -0.12$ in the comparison group. A weak negative correlation was also observed between fasting glucose and insulin levels and SIRT1 ($r = -0.17$ and $r = -0.24$, respectively). However, an inverse correlation trend was demonstrated between SIRT1 and the HOMA index ($r = -0.33$). Correlation analysis results also indicated a weak negative relationship between SIRT1 levels and total cholesterol ($r = -0.21$) and a strong negative correlation between SIRT1 and triglycerides (TG) and low-density lipoprotein cholesterol (LDL-C) ($r = -0.37$). At the same time, a weak positive correlation was observed between SIRT1 and high-density lipoprotein cholesterol (HDL-C) ($r = 0.11$). When comparing CRP levels, hepatocyte cytolysis enzymes, and SIRT1, a weak inverse correlation was found with CRP and ALT levels ($r = -0.17$ and $r = -0.22$, respectively).

Conclusion: The results of our study provide evidence of the significant role of SIRT1 in regulating liver inflammation and the development and progression of NAFLD.

Disclosure: No conflict of interest.

PP0782
ASSOCIATION OF OXIDATIVE STRESS MARKERS AND GENE EXPRESSION WITH METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE (MASLD) PROGRESSION: A CASE-CONTROL STUDY

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Introduction: Non-alcoholic fatty liver disease (NAFLD) is frequently associated with metabolic disorders and is highly prevalent in obese and type 2 diabetes mellitus (T2DM) patients. Oxidative stress plays a major role in the pathogenesis of NAFLD.

Aims & Methods: This study aimed to associate the role of oxidative parameters and their gene expression analysis in NAFLD. In this case-control study, 200 participants were enrolled and classified into two groups: NAFLD (n=200) and healthy controls (n=100). Levels of oxidative parameters, including superoxide dismutase (SOD), catalase, and malondialdehyde (MDA), were assessed using chemiluminescence and qRT-PCR. Data analysis was performed using Graph Pad Prism 10, with significance set at $P < 0.05$.

Results: The study demonstrated a significant decrease in SOD2 enzyme activity in NAFLD (1.008 ± 0.957 U/ml) compared to controls (4.029 ± 3.20 U/ml, $P=0.0014$). Catalase activity was also significantly reduced in NAFLD (32.2 ± 6.36 U/ml) compared to controls (41.2 ± 8.70 U/ml, $P< 0.0001$). Plasma MDA levels were significantly increased in NAFLD. Gene expression analysis revealed that catalase was significantly downregulated by 1.94 folds ($P=0.0045$) and SOD2 by 1.02 folds ($P=0.46$) in NAFLD patients compared to healthy controls.

Conclusion: The study suggests that decreased levels of oxidative stress markers, particularly antioxidant enzymes like catalase, may contribute to oxidative damage in the progression of NAFLD patients.

These findings provide insights into the molecular mechanisms underlying the role of hepatic inflammation in the development and progression of NAFLD.

References: This work is part of my ongoing PhD research.The study remains under active development, with further refinements and analyses underway.

Disclosure: Nothing to disclose.

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PP0783

ASSOCIATIONS OF VISCOELASTIC, ELECTRICAL PARAMETERS OF ERYTHROCYTES WITH FATTY ACID LEVELS OF ERYTHROCYTE MEMBRANES, BLOOD SERUM IN PATIENTS WITH FATTY LIVER DISEASE

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Introduction: Fatty liver is sensitive to ischemia-reperfusion injury [1, 2]. At the microcirculatory level, the structural and functional properties of red blood cells (RBCs) play a key role, since the degree of hypoxia in liver tissue depends on their ability to deform, tendency to form aggregates, and hemolysis [3], which, in turn, is a factor in fibrogenesis [4].

Aims & Methods: Research objective is as follows: To study the associations of viscoelastic, electrical parameters of RBCs and levels of FAs of their membranes, serum in patients with fatty liver disease.

84 men (average age 48.4±13.9 years) with fatty liver disease (FLD) of various genesis (alcoholic, non-alcoholic, of mixed genesis: alcoholic+metabolic), various necroinflammatory activity with fibrosis F0-1 (FibroScan® 502 (Echosens, France) have been examined. The study of RBC electrical, viscoelastic parameters has been performed by dielectrophoresis. The content of FAs in RBC membranes and blood serum has been determined using gas chromatography-mass spectrometry (Agilent 7000B, USA).

Results: It has been found, that RBC indices reflecting their deformation ability (amplitude of deformation at 10^6 Hz, degree of deformation at $0.5x10^6$ Hz), membrane state (capacitance), surface charge value (dipole moment, speed of cell movement to electrodes), RBC resistance (polarisability) were directly correlated with the levels of n-3 PUFAs in RBC membranes, blood serum, both their total content and concentrations of individual FAs (C20:5n-3, C22:5n-3, C22:6n-3), the level of monounsaturated oleic FAs, as well as two SFAs (C15:0, 17:0). On the contrary, the associations with the predominant part of SFAs, both individual (especially highly accurate: with C16:0, C18:0) and total content, MUSAs, USAs, including ratios of n-6/n-3 PUFAs, SFAs/UFAs, SFAs/PUFAs, were inverse (p=0,02-0,0000001).

The summarized rigidity was related with high strength of association and statistical significance to RBC content of C16:0 (r=0.674, p<0.000001), C18:0 (r=0.443, p<0.000006), C18:2n-6 (r=0.345, p=0.0006) FAs, total SFAs (r=0.640, p<0.000001), the ratios of n-6/n-3 PUFAs (r=0.605, p<0.000001), SFAs/PUFAs (r=0.428, p=0.000015).

Correlations with RBC levels of C16:0 (r=0.551, p<0.000001), C18:2n-6 (r=0.470, p=0.000042) were the closest for membrane conductivity levels, total SFAs (r=0.484, p<0.000001), USAs (r=0.468, p<0.000001), MUSAs (r=0.504, p<0.000001), n-6 PUFA (r=0.414, p=0.00003), as well as for ratio of n-6/n-3 PUFAs (r=0.568, p<0.000001).

Associations of high strength and statistical significance for RBC aggregation index have been found concerning RBC C16:0 (r=0.691, p<0.000001), C18:0 (r=0.486, p<0.000001), for the total UFAs' content (r=0.672, p<0.000001), the ratios n-6/n-3 PUFAs (r=0.611, p<0.000001), SFAs/UFAs (r=0.475, p<0.000001), SFAs/PUFAs (r=0.478, p<0.000001).

RBC destruction indices have been found to be directly related to the total UFAs' content in both RBC membranes and serum, RBC levels of C20:3n-6 (r=0.205, p=0.046), and inversely related to the content of C15:0 (r=-0.224, p=0.028), C17:0 (r=-0.151, p=0.039) and C18:1;c9 (r=-0.206, p=0.045).

Conclusion: The identified associations indicate a dependence of RBC parameters on the levels and ratios of FAs in their membranes, serum. Deterioration of blood rheological properties in patients with FLD is a factor associated with increasing hypoxia in liver tissue with potential for fibrogenesis and progression. The ability to modulate electrical, viscoelastic parameters of RBCs by exposure of FAs, including dietary FAs, defines new additional treatment targets for patients with FLD.

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Disclosure: Nothing to disclose.

PP0784

CHRONIC LIVER DISEASE IS ASSOCIATED WITH ALTERATIONS IN FECAL MICRORNAS AND THE GUT MICROBIOME

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Introduction: Hepatocellular carcinoma (HCC) is an end stage liver disease with increasing morbidity and mortality worldwide. Development of the disease is dependent on various genetic, dietary and environmental factors. While alcohol-related liver disease (ALD/MetALD) remains a major contributor, metabolic dysfunction-associated steatotic liver disease (MASLD) is becoming increasingly relevant in clinical practice. There is increasing evidence for the role of bacteria, for instance, such as *Bifidobacterium* and *Blautia*, which were considered protective, while *Streptococcus* has been associated with disease progression. However, little is known about the alteration of microRNAs (miRNAs) in liver diseases and its interrelationship with microbial environment in view of etiological factors.

Aims & Methods: This study aimed to elucidate gastrointestinal miRNA alterations and explore their relationship with microbiome composition in patients with different stages of liver disease in terms of disease etiology. Multi-omics analyses were performed in a subset of well characterized patients (n=59) with liver disease and healthy controls. Systematic characterization included clinical, laboratory, demographic and imaging characterization and dietary preferences. Patients subclassification included fibrosis stage (no fibrosis, fibrosis, liver cirrhosis(LC)) and etiology (MetALD/ALD vs. MASLD). Fecal miRNAs were extracted using the Qiagen miRNeasy Kit and profiled via Agilent miRNA microarray. Microbiome analyses were performed amplifying the V1-V2 region of the 16S rRNA gene following Illumina sequencing.

Results: For miRNA analysis stringent criteria were applied to identify high quality miRNA targets. From approximately 4,000 candidates filtering resulted in selection of 202 high-quality miRNAs with reliably mea-

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surable concentrations. 28 of these miRNAs showed significantly positive alterations in concentration in patients with liver disease and 7 miRNAs significant negative alterations. Specifically, miR-642a-3p, miR-5006, miR-6781, and miR-6849 were elevated in feces in patients with LC. In contrast, miR-6869, miR-4467, and miR-5001 were present in lower concentration in LC patients. Notably, miR-4467 also showed strong negative associations with liver-specific laboratory markers, including gamma-glutamyl transferase (GGT) (r=-0.34; p=0.026) and M65, a marker of apoptosis (r=-0.42; p=0.005).

Cross-sectional comparison of miRNA and microbiome data revealed miRNA alterations associated with specific microbial signatures. MiR-6795-5p, which was associated with liver stiffness (r = 0.39, p = 0.01), also correlated positively with the abundance of *Streptococcus*, a genus linked to disease progression (r=0.29; p=0.047).

Conversely, miR-6869, miR-4467, and miR-5001 showed positive correlations with the potentially protective genera *Blautia* and *Bifidobacterium* (miR-6869: $r_{Bifidobacterium}$=0.35; $p_{Bifidobacterium}$=0.012; $r_{Blautia}$=0.37; $p_{Blautia}$=0.008; miR-4467: $r_{Blautia}$=0.41; $p_{Blautia}$=0.002; miR-5001: $r_{Blautia}$=0.38; $p_{Blautia}$=0.006).

Conclusion: Multi-omics analysis reveals distinct alterations in fecal miRNAs and gut microbiota across different stages of liver disease. Changes in microbial composition were associated with shifts in miRNA profiles, suggesting a functional link in maintaining intestinal homeostasis.

These findings highlight the potential of integrated microbiome–miRNA signatures as novel diagnostic and therapeutic targets. Further studies are warranted to validate these associations and explore their clinical applications.

Disclosure: Nothing to disclose.

PP0785

CLINICAL SIGNIFICANCE OF ANTI-CENTROMERE PROTEIN B (CENP-B) ANTIBODY IN PATIENTS WITHOUT KNOWN LIVER DISEASE – A POPULATION WARRANTING CLOSER SURVEILLANCE?

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Introduction: Anti-centromere antibodies (ACA) recognize centromere proteins, particularly CENP-B, the major reactive antigen. While primarily associated with Raynaud's phenomenon, systemic sclerosis (SS), and other autoimmune-like conditions, ACA positivity can also be observed in primary biliary cholangitis (PBC), often overlapping with rheumatological diseases.

Aims & Methods: This study aimed to explore the association between CENP-B positivity and liver test abnormalities in patients without known liver disease, evaluate its overlap with rheumatological conditions, and assess whether co-positivity with other antibodies may predict liver dysfunction. We conducted a retrospective cohort study including consecutive patients with CENP-B positivity (via immunoblot) from January 2020 to December 2024. Data on liver function tests (LFTs), antinuclear antibody (ANA) patterns, immunoglobulin M (IgM), anti-gp210, anti-sp100, antimitochondrial antibodies (AMA), and liver biopsy results were analysed.

Results: A total of 48 CENP-B-positive patients without known liver disease were included (85.4% female; mean age 68.7 ± 21.5 years). Liver test abnormalities were present in 56.3% of patients, predominantly elevated gamma-glutamyl transferase (GGT) (35.4%), alanine transaminase (ALT) (10.4%), and aspartate transaminase (AST) (18.8%). IgM elevation was infrequent (6.3%), and antibody co-positivity varied (anti-gp210: 0%; anti-sp100: 4.2%; AMA: 8.3%; anti-pyruvate dehydrogenase autoantibodies [anti-PDH]: 10.4%). There was reference to Raynaud's phenomenon in 25%, and 91.7% of patients had a rheumatological or autoimmune-like condition (SS: 25%, systemic lupus erythematosus: 6.3%). Patients with concurrent AMA and CENP-B positivity had significantly higher ALT (U=31.5, Z=2.112, p=0.031) and GGT (U=34.5, Z=1.997, p=0.043) levels, though this association was not observed with other LFTs (AST, alkaline phosphatase, and bilirubin), ANA patterns, or antibodies. None of the four patients with liver test abnormalities and concurrent CENP-B and AMA positivity underwent liver biopsy.

Conclusion: The findings suggest that CENP-B-positive patients, particularly those with concurrent AMA positivity, may be at increased risk of developing liver diseases within the PBC spectrum. This group may warrant closer clinical and laboratory surveillance to enable timely identification and management of potential hepatic involvement.

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Disclosure: Nothing to disclose.

PP0786

NON-INVASIVE EVALUATION OF LIVER FIBROSIS IN CHRONIC HEPATITIS B: A COMPARATIVE STUDY OF FIB-4 AND FIBROSCAN

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Introduction: Assessment of liver fibrosis is a key element in the management of patients with hepatitis B virus (HBV) infection, whether in the active, inactive, or occult phase. Among the non-invasive tools currently available, the FIB-4 score, based on biological parameters and FibroScan, using vibration-controlled transient elastography (VCTE) are widely used.

Aims & Methods: This retrospective, descriptive, and analytical study was conducted in a university hospital between 2017 and 2025. All patients followed for chronic HBV infection in the department of hepatogastroenterology were included. Clinical and biological data, along with fibrosis assessment results (FIB-4 score and FibroScan), were collected from patients medical records. The aim of this study was to compare the diagnostic performance and concordance of these two modalities in assessing liver fibrosis in a heterogeneous population of HBV-infected patients.

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Results: A total of 100 HBV-infected patients were included, consisting of 54 males (54%) and 46 females (46%), with a median age of 55 years [42.5; 60]. Among them, 70% of patients had an active chronic hepatitis B (n=70), 29% were in the inactive phase (n=29), and 1 patient had an occult HBV infection.

Regarding comorbidities, 14% had arterial hypertension, 13% had metabolic dysfunction-associated steatotic liver disease (MASLD), and 5% had type 2 diabetes mellitus. Alcohol consumption was reported in 4% (n=4), and 5% were smokers (n=5). A family history of liver disease was documented in 12% of cases (n=12).

Concerning virological parameters, HBe antigen was positive in 13% (n=13) and negative in 87% (n=87). HBV DNA levels were <2,000 IU/mL in 48% of patients (n=48), between 2,000 and 20,000 IU/mL in 25% (n=25), and >20,000 IU/mL in 27% (n=27).

Evaluation of liver fibrosis using both the FIB-4 score and FibroScan revealed that 40% of patients had a FIB-4 score >3.25, indicating advanced fibrosis, while 60% had a score <1.45, suggesting minimal fibrosis. These findings were consistent with those obtained by FibroScan.

To assess the diagnostic performance of the FIB-4 score in detecting advanced fibrosis (F3–F4) according to FibroScan, a receiver operating characteristic (ROC) analysis was conducted. The FIB-4 score demonstrated an area under the curve (AUC) of 0.96 using a cutoff value of 1.96, with a sensitivity of 93.75%, specificity of 84.52%, and a negative predictive value (NPV) of 98.61%.

Conclusion: The FIB-4 score demonstrated excellent diagnostic accuracy for identifying advanced liver fibrosis in patients with HBV infection, with a performance comparable to that of FibroScan. Given its simplicity, accessibility, and cost-effectiveness, the FIB-4 index represents a reliable first-line tool for fibrosis screening, particularly in settings where elastography is unavailable or limited.

Disclosure: Nothing to disclose.

PP0787

INTER-OBSERVER VARIABILITY OF LIVER BIOPSY READING IN CLINICAL TRIALS FOR METABOLIC DYSFUNCTION-ASSOCIATED STEATOHEPATITIS: A MONOCENTRIC EXPERIENCE

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Introduction: Liver histology remains a cornerstone in clinical trials investigating metabolic dysfunction-associated steatohepatitis (MASH) therapeutic approaches. Although liver biopsy is invasive and carries risks such as complications, high costs, and inter-observer variability, it continues to be indispensable for determining eligibility criteria and assessing surrogate endpoints. The inherent variability in histological interpretation can contribute to screening failures and may impact the accuracy and reliability of trial outcomes.

Aims & Methods: We evaluated the consistency of liver biopsy interpretations performed within the framework of nine clinical trials for MASH conducted at our center. A total of 59 biopsy samples were independently assessed by central hepatopathologists and by local pathology department. We compared the MASH-validated scoring results between the two groups, including MASH CRN components—steatosis, lobular inflammation, hepatocellular ballooning, and fibrosis—as well as the overall MASH diagnosis, MASH with fibrosis stage >1 (MASH+F>1), and MASH with fibrosis stage >2 (MASH+F>2) diagnoses.

Inter-reader reliability was measured using both overall percentage agreement and unweighted kappa coefficients. The kappa coefficient quantifies the level of agreement between two raters, beyond what would be expected by chance alone.

Results: Inter-reader agreement for individual histological features showed varying levels of consistency. The observed percentage agreement (and corresponding unweighted kappa coefficients) were as follows: steatosis 60% ($\kappa = 0.383$), ballooning 64% ($\kappa = 0.216$), lobular inflammation 68% ($\kappa = 0.000$), NAS score 26% ($\kappa = 0.199$), and fibrosis 48% ($\kappa = 0.190$).

For diagnostic categories, the agreement rates were: MASH diagnosis 66% ($\kappa = 0.088$), MASH with fibrosis stage >1 (MASH+F>1) 60% ($\kappa = 0.211$), and MASH with fibrosis stage >2 (MASH+F>2) 87% ($\kappa = 0.585$).

Based on the inclusion criteria specific to each study, the overall inter-reader agreement was 67%, with a kappa coefficient of 0.289.

Conclusion: The reliability of liver biopsy evaluations performed by hepatopathologists using currently accepted criteria remains suboptimal.

This variability can compromise the integrity of MASH clinical trials by contributing to misclassification of fibrosis stages and MASH diagnoses, potentially leading to the inclusion of patients who do not meet eligibility criteria. As such, the requirement for liver biopsy to assess MASH severity in the context of future drug prescriptions should be reconsidered with caution.

Disclosure: Nothing to disclose.

PP0788

FIB-6 OVER FIB-4: SCORING HIGHER IN LIVER FIBROSIS EVALUATION IN MASLD: EVIDENCE FROM A PORTUGUESE COHORT

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) has a global prevalence of 30%, with 10-30% progressing to advanced liver fibrosis in 5 years. It is essential to assess advanced liver fibrosis – a predictor of liver-related outcomes. FIB-4 is a good screening test, but its low specificity leads to unnecessary referrals to hospital, especially in individuals over 65 years and with type 2 diabetes (T2D). To address these limitations, machine-learning techniques are developing scores like FIB-6 (validated for Hepatitis B and C).

Aims & Methods: We aimed to assess the performance of FIB-6 score in screening for advanced fibrosis in a Portuguese cohort of MASLD patients. A retrospective-cohort study was conducted with MASLD patients who had undergone liver vibration-controlled transient elastography (VCTE) between January-October 2024.

Inclusion criteria: adult patients (age >18years) diagnosed with MASLD; undergone liver VCTE; complete biochemical and hematological data.

Exclusion criteria: coexistence of other liver diseases; decompensated cirrhosis; hepatocellular carcinoma; hematological/bone-related diseases.

We used liver stiffness measurement by VCTE as a reference standard for liver fibrosis. Fibrosis was staged on a 0–4 scale: F0, F1 and F2, patients without advanced fibrosis; F3 and F4, patients with advanced fibrosis/cirrhosis.

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Non invasive tests (FIB-6, FIB-4, APRI, NFS) were calculated using blood results available within the month of the VCTE. Statistical analysis was performed using R version 4.2.3. AUC-ROC curve analysis with cutoff points were calculated using the Youden index.

Results: 115 MASLD patients (58 males (50.43%),57 females (49.57%), median age of 62years) where included. 41 (35.65%) had diabetes mellitus type 2 (T2D) (10 of those insulin-treated), 74 (64.35%) arterial hypertension, 76 (66.09%) dyslipidemia and with a median body mass index of 29.9mg/Kg/m^2. Regarding hepatic fibrosis, 79 patients had F0-2 and 17 patients had F3-4.

In the general sample, AUC-ROC curve analysis showed an AUC 0.771 (CI 95% :0.662, 0.88) for FIB-4, AUC 0.788 (CI 95% :0.686, 0.889) for APRI, AUC 0.709 (CI 95% :0.595, 0.823) for NFS and AUC 0.695 (CI 95% :0.571, 0.819) for FIB-6.

A subanalysis of the sample over 65 years old revealed AUC 0.807 (CI 95% : 0.665, 0.949) for FIB4; AUC 0.788 (CI 95% : 0.686, 0.889) for APRI; AUC 0.788 (CI 95% : 0.65, 0.926) for NFS; AUC 0.663 (CI 95% : 0.475, 0.852) for FIB-6.

A subanalysis of the sample of T2D patients revealed AUC 0.747 (CI 95% :0.584, 0.911) for FIB4; AUC 0.811 (CI 95% :0.676, 0.946) for APRI; and AUC 0.765 (CI 95% : 0.612, 0.918) for FIB6.

Using the optimal cut-offs of FIB6 (cutoff= 1.8992) for the diagnosis of F3+F4 indicated a reliable performance of FIB-6 score (sensitivity was 70%, specificity was 52.6%).

The cutoff values of FIB-6 for the general sample was 2.4420 (sensitivity 53%, specificity 87%); for the subset of patients over 65 years was 2.7668 (sensitivity 40%, specificity 96%); for the subset of T2D patients was 2.2230 (sensitivity 55%, specificity 90%).

Conclusion: FIB-6 demonstrated good performance in screening advanced fibrosis in MASLD patients, when compared to those of FIB-4, APRI, and NFS, surpassing FIB-4, particularly in individuals with T2D. This study is the first to evaluate the applicability of FIB-6 in a Portuguese MASLD cohort.

Although the findings warrant validation in larger and more diverse populations, they suggest that FIB-6 may be more efficient to evaluate liver fibrosis in MASLD and offer a more efficient tool to identify patients that need specialized care.

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Disclosure: Nothing to disclose.

PP0789

COMPARISON OF LIVING DONOR (LDLT) *VERSUS* DECEASED DONOR LIVER TRANSPLANT (DDLT) FOR PRIMARY SCLEROSING CHOLANGITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Primary Sclerosing Cholangitis (PSC) is a rare, chronic and progressive cholestatic liver disorder of unknown etiology. It is associated with progressive liver disease, culminating in cirrhosis and liver failure, as well as an increased risk of malignancies, particularly cholangiocarcinoma (CCA). Liver transplantation (LT) is currently the only potentially curative option for these patients. Organ scarcity is boosting Living donor liver transplantation (LDLT) as an option to increase the donor pool. Outcomes after LDLT in PSC are conflicting. We aimed to review the literature regarding the outcomes of LT in PSC, comparing LDLT with deceased donor LT (DDLT).

Aims & Methods: A systematic review with meta-analysis was performed through a search on: *PubMed, Scopus, Web-of-Science*, and *Cochrane*. 22 studies were included, enrolling 22,024 patients.

Results: Compared with LDLT, DDLT was associated with lower 1- (OR 0.66 [95% CI 0.43-0.99]) and 5- years survival (OR 0.56 [95% CI 0.46-0.59]), even though it had no impact on graft-survival or retransplantation. The weighted pooled DDLT and LDLT survivals were: 93.80% vs. 95.78% at 1-year, 88.81% vs. 93.23% at 3-year, and 85.18% vs 91.54% at 5-year, respectively. DDLT was found to be associated with higher risk of post-LT sepsis (OR 3.68 [95% CI 1.02-13.26]), acute rejection (OR 1.75 [25% CI 1.12-2.73]) and PSC recurrence (OR 1.63 95% CI 1.10-2.42]), although lower biliary complications (OR 0.29 [95% CI 0.11-0.78]).

DDLT recipients were older (MD 4.26 years [95% CI 2.42-6.10]), more frequently male (OR 1.32 [95% CI 1.18-1.48]), had higher metabolic dysfunction and worse liver disease (higher MELD: MD 6.63 [95% CI 5.56-7.70]). Additionally, were less frequently Caucasian (OR 0.54 [95% CI 0.39-0.75]) and more likely to be Black (OR 2.99 [95% CI 2.33-3.82]), comparing with LDLT recipients. Nevertheless, had lower chance of having inflammatory bowel disease (OR 0.66 [95% IC 0.50-0.87]) or cholangiocarcinoma (OR 0.70 [95%IC 0.53-0.92]). DDLT donors were also older (MD 2.46 years [95% CI 0.76-4.15]) and cold ischemia time of the liver grafts was longer (MD 5.63 hours [95% CI 2.18-8.08]).

Conclusion: In PSC patients, LDLT seems to be associated with higher post-LT survival, compared with DDLT. This could probably be related with a bias to earlier referral to LT, lower waitlist time and better-quality grafts, because of a stricter selection of healthy donors and recipients, and technical considerations such as shorter cold ischemia time. PSC recurrence and acute rejection were less frequent following LDLT, whereas biliary complications more common, although not negatively impacting the overall results. The findings support considering LDLT as a preferred option when feasible, especially for patients with anticipated long wait times for deceased donor organs. As such, the possibility of LDLT should be systematically inquired for patients with PSC.

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PP0790

META-ANALYSIS OF THE EFFICACY AND SAFETY OF NOVEL ORAL ANTI-CHOLESTATIC AGENTS FOR PRIMARY BILIARY CHOLANGITIS

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Introduction: Primary biliary cholangitis (PBC) is a chronic autoimmune liver disease characterized by progressive bile duct damage and cholestasis. While ursodeoxycholic acid (UDCA) is the first-line therapy, about 40% of patients have incomplete responses, necessitating alternative treatments.
Aims & Methods: This meta-analysis evaluates the efficacy and safety of novel oral anti-cholestatic agents for PBC, focusing on key biochemical and symptomatic outcomes. A systematic literature search was conducted across electronic databases.
Eligible studies included randomized controlled, retrospective, and open-label trials investigating novel oral anti-cholestatic agents in PBC patients. Data on efficacy (e.g., alkaline phosphatase [ALP] reduction) and safety outcomes were extracted and analyzed using meta-analytic techniques.
Results: Ten studies involving 878 patients were analyzed. Investigated therapies included seladelpar, fenofibrate, saroglitazar, bezafibrate, elafibranor, and budesonide. Novel anti-cholestatic agents significantly lowered ALP levels compared to controls (SMD -2.80; 95% CI [-3.56, -2.03]; p<0.00001), with saroglitazar showing the largest effect size (SMD -4.62; 95% CI [-5.54, -3.70]). Subgroup analysis revealed that fibrates (fenofibrate, bezafibrate, and saroglitazar) demonstrated the most consistent ALP reduction across studies (SMD -3.15; 95% CI [-3.89, -2.41]; p<0.00001).
In addition to ALP reduction, several agents showed improvements in other liver function tests. Bilirubin levels decreased significantly with novel therapies (MD -0.22 mg/dL; 95% CI [-0.35, -0.09]; p=0.001). Gamma-glutamyl transferase (GGT) also showed substantial reduction (SMD -1.98; 95% CI [-2.56, -1.40]; p<0.00001).
Symptomatic improvements were observed, particularly for pruritus. Bezafibrate and seladelpar demonstrated the most promising results, with mean reductions in visual analog scale scores for pruritus of 30% and 35%, respectively. Fatigue scores also improved, though to a lesser extent (SMD -0.42; 95% CI [-0.68, -0.16]; p=0.002).
Safety profiles were comparable to controls, with no significant increase in serious adverse events (OR 1.21; 95% CI [0.81, 1.83]; p=0.35). The most common adverse events were mild gastrointestinal symptoms (10-15% of patients) and transient elevations in creatinine (5-8% of patients). No significant hepatotoxicity was observed across the studies.
Conclusion: Novel oral anti-cholestatic agents show promise in managing PBC patients with suboptimal UDCA responses, demonstrating significant improvements in biochemical markers and some symptomatic relief. However, study heterogeneity, small sample sizes, and limited follow-up durations restrict direct comparisons and generalizability. Further research is needed to confirm long-term efficacy and safety.
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Disclosure: Nothing to disclose.

PP0791

THE ROLE OF VITAMIN D DEFICIENCY IN THE PROGRESSION AND OUTCOMES OF PRIMARY SCLEROSING CHOLANGITIS: A PRELIMINARY ANALYSIS

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Introduction: Primary sclerosing cholangitis (PSC) is a progressive and debilitating liver disease involving chronic inflammation and fibrosis of bile ducts. Vitamin D plays a critical role in anti-inflammatory and anti-fibrotic pathways, yet its impact on PSC progression remains insufficiently explored.
Aims & Methods: This study investigates the relationship between vitamin D deficiency and key outcomes in PSC patients, including biliary damage, hepatic decompensation, and survival. Sixteen PSC patients from a

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tertiary care outpatient clinic were included. Vitamin D status was determined via serum 25-hydroxyvitamin D3 levels, with deficiency defined as <30 ng/mL. Patients were categorized into deficient and normal vitamin D groups.

Outcomes assessed included alkaline phosphatase (ALP) levels, PREsTo score for hepatic decompensation risk, and 4-year survival using the revised Mayo Risk Score (rMRS). Statistical analyses included t-tests, Mann-Whitney tests, Fisher's exact test, and Spearman's correlation.

Results: The cohort had a mean age of 55.8±13.1 years, with equal gender distribution. Vitamin D deficiency was observed in 56.25% of patients. No significant differences were found between deficient and normal groups for ALP (106 vs. 100 U/L, p=0.71), PREsTo score (5% vs. 5%, p=0.31), or rMRS (0.2±1.1 vs. -0.2±0.6, p=0.42). Furthermore, no meaningful correlation was identified between vitamin D levels and ALP (Spearman's rho = -0.18, p=0.56).

Conclusion: While vitamin D deficiency is prevalent among PSC patients, this preliminary analysis did not reveal a significant impact on disease progression or prognosis.

These findings suggest that vitamin D's roles in inflammation and fibrosis may not translate into measurable clinical effects within this study's scope. Further research with larger cohorts, extended follow-up, and interventional studies on vitamin D supplementation is warranted.

Disclosure: Nothing to disclose.

PP0792

PREVALENCE AND RISK FACTORS FOR RECURRENT PRIMARY BILIARY CHOLANGITIS AFTER LIVER TRANSPLANTATION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Primary biliary cholangitis (PBC) is a rare, chronic autoimmune liver disease characterized by progressive destruction of small intrahepatic bile ducts, leading to cholestasis and end-stage liver disease. Liver transplantation (LT) remains the definitive treatment, with excellent long-term outcomes. However, a subset of patients experiences disease recurrence (rPBC), which may impact prognosis. Due to the rarity of the condition, risk factors for rPBC remain poorly defined.

Aims & Methods: This study aims to determine the prevalence of rPBC after LT and identify potential risk factors associated with recurrence. A systematic search was conducted in MEDLINE, Embase, and Cochrane CENTRAL using a predefined strategy.

We included full-text studies reporting the number of PBC patients who developed rPBC post-LT. Abstracts and studies lacking clear data on prevalence or risk factors were excluded. Prevalence estimates were pooled using a two-step approach with logit transformation and generalized linear mixed models. Hazard ratios (HRs) and 95% confidence intervals (CIs) were synthesized using a random-effects model with restricted maximum likelihood estimation. All analyses were performed using R Studio.

Results: Fifty-five studies met inclusion criteria, most conducted in Europe, followed by North America, Asia, and Australia; two studies were multi-continental. A total of 5,715 patients with PBC who underwent LT were included, with 9.98% being male. Reported prevalence of rPBC ranged from 0% to 50%, with a pooled prevalence of 15% (95% CI: 12%-19%). Significant risk factors for rPBC included younger age at LT (HR = 0.97, 95% CI: 0.96-0.98) and use of tacrolimus in maintenance immunosuppression (HR = 1.79, 95% CI: 1.09-2.95).

Protective factors included cyclosporine-based maintenance immunosuppression (HR = 0.50, 95% CI: 0.28-0.87) and post-LT prophylactic use of ursodeoxycholic acid (HR = 0.42, 95% CI: 0.30-0.58). Donor age, donor-recipient sex mismatch, cold ischemia time, and use of tacrolimus, cyclosporine, or azathioprine in initial immunosuppression, as well as maintenance azathioprine or occurrence of acute cellular rejection, were not significantly associated with rPBC.

Conclusion: This is the largest meta-analysis to date evaluating rPBC, including 5,715 liver transplant recipients with PBC. The overall recurrence rate was 15%. Younger age at transplantation and tacrolimus-based immunosuppression were associated with higher recurrence risk, while cyclosporine and post-LT ursodeoxycholic acid appeared protective.

These findings may inform risk stratification and guide post-transplant management in PBC. Prospective studies are needed to validate these associations and refine strategies to prevent rPBC.

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PP0793

CHOLECYSTECTOMY IN PRIMARY SCLEROSING CHOLANGITIS (PSC): IMPACT ON BILE COMPOSITION, DISEASE PROGRESSION AND OUTCOME

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Introduction: PSC is chronic infammatory disease of bile ducts leading to strictures of bile duct and eventually to liver cirrhosis. Chronic inflammation is associated with markedly increased risk of cholangiocancer (CCA). In addition, gall bladder abnormalities are common in patients with PSC[1] such as cholecystitis, gallstones, polys and gallbladder carcinoma (GBCA). Increased fasting gallbladder volume has been reported with normal gallbladder emptying[2]. Due to the increased risk of GBCA[3] cholecystectomy is recommended in PSC with gallbladder polyps ≥8 mm and smaller polyps growing[4].

However, cholecystectomy has been associated with impaired PSC outcome. A study consisting of 11 PSC patients with cholecystectomy suggested protective role of gallbladder based on MRCP changes and elevated AST levels in patients with cholecystectomy. No significant difference in serum bile acids was observed in patients with or w/o cholecystectomy[5].

Another small study including five PSC patients with cholecystectomy showed that cholecystectomy was associated with worse Mayo scores and an elevated FIB4[6]. More than 50% of PSC have enlarged gallbladder

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which is suspected to be caused by hyperpressure in the biliary tract[5]. Cholecystectomy may thus eliminate this reserve function and lead to progression of bile duct strictures.

Aims & Methods: To evaluate the impact of cholecystectomy on liver enzyme levels, biliary bile acid composition, bile duct disease progression, and overall outcomes in a large PSC patient cohort, we analyzed data from 1,009 individuals enrolled in the prospective Helsinki PSC registry. Of these, 114 patients (11.3%) had undergone cholecystectomy—34 prior to and 80 following the PSC diagnosis. Patients with gallbladder cancer were excluded. Clinical and laboratory parameters were collected, along with data on disease progression based on sequential ERC examinations scored by the Helsinki ERC score[7]. Additionally, bile composition was analyzed in a subgroup of 157 patients receiving UDCA therapy, 32 of whom had undergone cholecystectomy.

Results: Demographics, liver function tests, and outcomes stratified by cholecystectomy, are presented in the table. We found no differences between the three groups (no cholecystectomy, cholecystectomy done prior or after PSC dg) regarding the progression of bile duct disease or its severity, liver function tests, development of biliary dysplasia or need for dilatation, Mayo- and MELD-scores. The cumulative incidence of the composite end point (= transplantation, liver related death, HCC or CCA) did not differ between groups. The molar-% of phospholipids were lower and cholic acid and deoxycholic acid (mg) elevated in bile in patients with cholecystectomy, whereas no differences were found in UDCA content.

Variable	No cholecystectomy, n=895	Cholecystectomy before PSC dg, n=34	Cholecystectomy after PSC dg, n=80	P-value**
Men, n(%)	537(60)	12(35)	39(49)	0.003
Age, mean (SD)	37(13)	46(12)	43(13)	<0.001
BMI, mean(SD)	25.6(4.5)	26.5(4.2)	26.4(4.5)	0.22
ERC-score, mean (SD)	5.8(3.5)	6.4(3.7)	6.4(3.5)	0.16
Need for dilatation	398(44)	18(53)	42(53)	0.26
P-AST, U/l, median (IQR)	36(25,58)	37(28,52)	34(26,51)	0.83
P-ALP, U/l, median (IQR)	132(96,222)	130(88,191)	113(84,179)	0.082
Mayo Score, mean (SD)	-0.29(0.84)	-0.10(0.63)	-0.10(0.78)	0.079
Composite end point, n [Incidence (95% CI:)]*	142 [16.4 (13.8 -19.3)]	6 [13.31 (4.88 - 28.97)]	14 [14.16 (7.74 - 23.76)]	0.66

* Per 1 000 person years

** Adjusted for sex and PSC age

Conclusion: In this large study investigating the impact of cholecystectomy on PSC outcomes, we found no effect on liver enzyme levels, prognostic indices, bile duct disease progression, or overall prognosis, regardless of whether the procedure was performed before or after the PSC diagnosis.

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Disclosure: Nothing to disclose.

PP0794

REVALUATING THE CLINICAL COURSE OF AMA-POSITIVE PATIENTS WITH NORMAL LIVER ENZYMES: A LARGE RETROSPECTIVE COHORT STUDY

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Introduction: Anti-mitochondrial antibodies (AMA) are commonly associated with Primary Biliary Cholangitis (PBC), a progressive liver disease. However, the clinical significance of AMA positivity in patients with normal liver enzymes remains poorly understood. Despite the presence of AMA, many patients maintain normal liver function, raising the question of whether these individuals are at risk for developing liver-related complications.

Aims & Methods: This study aims to evaluate the natural history, prognostic implications, and potential risk factors for disease progression in AMA-positive patients with normal liver enzymes.

A retrospective cohort study was conducted using a national healthcare database to identify patients (aged>18) with a positive AMA and normal liver enzymes between 2002 and 2023. Demographics, biochemical parameters, and liver-related complications were analyzed. Multivariate regression models assessed predictors of disease progression.

Results: Among 1,018 patients, 76 (7.5 %) developed Primary biliary cholangitis (PBC), and 30 (2.9%) progressed to cirrhosis over a median follow-up of 6.3 years. Liver-related complications were rare, with esophageal varices in 1.1%, ascites in 1.8%, hepatocellular carcinoma in 0.2%, and liver transplantation in 0.1%. Notably, higher AMA titers were strongly associated with increased risk of developing PBC, cirrhosis and liver-related complications, demonstrating a clear titer-dependent pattern.

Conclusion: Most AMA-positive patients with normal liver enzymes do not develop clinically significant liver disease. However, higher AMA titers identify a high-risk subgroup that warrants closer monitoring. These findings underscore the need for risk-stratified follow-up strategies and may influence future clinical guidelines.

Disclosure: Nothing to disclose.

PP0795

*L*ACTOBACILLUS PLANTARUM ALLEVIATES CHOLESTATIC LIVER INJURY BY ENHANCING TRYPTOPHAN METABOLISM

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Introduction: *L*actobacillus plantarum is a probiotic with known therapeutic potential in various diseases, including liver diseases. However, the therapeutic effects and underlying mechanisms of *L*actobacillus plantarum in cholestatic liver injury remain poorly understood.

Aims & Methods: This study aims to investigate the therapeutic role and mechanism of Lactobacillus plantarum in cholestatic liver injury. Cholestatic liver injury was induced in mice using either a 0.1% 3,5-diethoxycarbonyl-1,4-dihydrocollidine (DDC) diet or α-naphthyl isothiocyanate (ANIT) gavage. Mice were then treated with Lactobacillus plantarum. Therapeutic

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effects were evaluated using histopathology, 16S rRNA sequencing, untargeted metabolomics, RNA sequencing, whole-genome sequencing, and immunohistochemistry.

Results: *L*actobacillus plantarum treatment markedly reduced liver inflammation, fibrosis, and ductular proliferation. Multi-omics analysis revealed enhanced tryptophan biosynthesis in the Lactobacillus plantarum treatment group. Tryptophan levels were significantly elevated in both liver and serum following treatment. Exogenous administration of tryptophan alone was sufficient to alleviate liver injury in the cholestasis models. RNA-seq data suggested that tryptophan acts by activating the AHR signaling pathway, and the therapeutic effect of tryptophan was diminished after the use of AHR inhibitors.

Conclusion: In summary, our findings suggest that Lactobacillus plantarum alleviates cholestatic liver injury by increasing circulating tryptophan, which may represent a potential therapeutic strategy for cholestatic liver injury.

Disclosure: Nothing to disclose.

PP0796

HEPATIC SARCOIDOSIS: CLINICAL PROFILE AND UDCA THERAPEUTIC OUTCOMES: A MOROCCAN RETROSPECTIVE STUDY

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Introduction: Hepatic sarcoidosis, a rare manifestation of systemic sarcoidosis, is characterized by the development of non-caseating granulomas in the liver parenchyma. Diagnosis requires a combination of clinical, biological, and histopathological assessment, while treatment commonly includes ursodeoxycholic acid (UDCA) and corticosteroids as key therapeutic options.

Aims & Methods: This study focuses on exploring the clinical spectrum, diagnosis features, and treatment responses in a Moroccan cohort.

This retrospective descriptive study reviewed 16 patients diagnosed with hepatic sarcoidosis treated at Ibn Sina Hospital's Gastroenterology department (Rabat, Morocco) between 2010 and 2024. All the patients included were aged over 30 years at diagnosis.

Results: The cohort comprised exclusively female patients (n=16) with an average age of 53 years (range 32-75). Diabetes was the predominant comorbidity (37.5%). The most frequent presenting symptoms were asthenia (37.5%), pruritus (31.25%) and anemia (18.75%). Physical examination revealed splenomegaly in 37.5% of cases and hepatomegaly in 43.75%. Cutaneous venous collaterals were present in 19% of patients while 25% had unremarkable findings.

Biological analysis identified cytolytic (31.25%) and cholestatic (68.75%) patterns, with no evidence of hepatocellular failure. On imaging evaluation, all patients underwent abdominal ultrasound, revealing features of chronic liver disease in 62.5% and portal hypertension signs in 31.25%. Esophagogastroduodenoscopy, performed on 14 patients, revealed portal hypertension manifestations (esophageal varices / hypertensive gastropathy) in 9 cases while findings were normal in 5.

The diagnosis of hepatic sarcoidosis was histologically confirmed through liver biopsy demonstrating characteristic non-caseating granulomatous hepatitis, after excluding alternative granuloma etiologies (notably primary biliary cirrhosis and hepatitis C virus infection). All patients underwent evaluation for extra hepatic involvement, revealing isolated hepatic disease in 25% of cases (n=4) and pulmonary manifestations as the most common concurrent finding.

Regarding therapeutic management, 11 patients received ursodeoxycholic acid (UDCA): as mono-therapy (n=4), combined with corticosteroids (n=6), and with anti-epileptics in one case of neurological involvement. Three patients were treated with corticosteroids alone. Overall, 81.25% of the cohort achieved satisfactory treatment outcomes.

Conclusion: In summary, our findings revealed that hepatic sarcoidosis predominantly affected women (mean age 53 years) in our cohort. The disease manifested as either isolated hepatic involvement or combined with pulmonary pathology. UDCA-based regimens constituted the primary treatment approach and were associated with favorable clinical outcomes in 81% of cases, reinforcing their role as first-line therapy.

Disclosure: Nothing to disclose.

PP0797

PERFORMANCE OF SIMPLE FIBROSIS SCORES IN PRIMARY BILIARY CHOLANGITIS

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Introduction: The non-invasive assessment of liver fibrosis in primary biliary cholangitis (PBC) is primarily based on transient elastography. The objective of this study was to evaluate the diagnostic performance of simple fibrosis scores in patients with PBC.

Aims & Methods: This was a retrospective, single-center study including all patients followed for PBC in a gastroenterology department between January 2001 and December 2023. Demographic, clinical, biological, and histological data were collected. Histological staging was performed according to the Scheuer classification, with stage 4 corresponding to cirrhosis. The following non-invasive fibrosis scores were calculated using validated online formulas and calculators: APRI (AST to Platelet Ratio Index), FIB-4, AST/ALT ratio (AAR), and GGT to Platelet Ratio (GPR).

Multivariate logistic regression analysis was used to identify independent predictors of cirrhosis. Diagnostic performance was assessed using receiver operating characteristic (ROC) curve analysis and area under the ROC curve (AUROC) with a 95% confidence interval (CI).

Results: Fifty-seven patients were included, of whom 54 were women. The mean age was 59 years (range: 20–96 years). Liver biopsy was performed in 29 patients (50.9%).

Histological staging according to the Scheuer classification was as follows: stage 1 (8 patients, 27.6%), stage 2 (6 patients, 20.7%), stage 3 (8 patients, 27.6%), and stage 4/cirrhosis (7 patients, 20.1%).

Multivariate logistic regression analysis identified FIB-4 as the only inde-

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pendent predictor of cirrhosis (p = 0.011). The FIB-4 score demonstrated an AUROC of 0.818 (p = 0.011). At a cut-off value of 2.92, FIB-4 had a sensitivity of 100% and a specificity of 63% for the diagnosis of cirrhosis.

Conclusion: Among the simple fibrosis scores evaluated, only the FIB-4 score showed adequate diagnostic performance for predicting cirrhosis in patients with primary biliary cholangitis. These findings suggest its potential role as a non-invasive screening tool in clinical practice.

Disclosure: Nothing to disclose.

PP0798

DOES THE ALP-A SCORE PREDICT BIOCHEMICAL RESPONSE TO URSODEOXYCHOLIC ACID AFTER ONE YEAR OF TREATMENT IN PRIMARY BILIARY CHOLANGITIS?

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Introduction: Biochemical response to ursodeoxycholic acid (UDCA) is a major prognostic factor in primary biliary cholangitis (PBC). This response is typically assessed using the Paris criteria. The aim of this study was to evaluate the performance of the ALP-A score in predicting biochemical response to UDCA after one year of treatment.

Aims & Methods: This was a retrospective, single-center study including all patients followed for PBC in a gastroenterology department between January 2001 and December 2023. Demographic, clinical, biological, and histological data were collected.

The ALP-A score was calculated using the following formula:

Age at diagnosis / (alkaline phosphatase at diagnosis / upper limit of normal [ULN])

The diagnostic performance of the score was assessed using receiver operating characteristic (ROC) curve analysis, with the area under the ROC curve (AUROC) and a 95% confidence interval (CI).

Results: Fifty-seven patients were included, with a mean age of 59 years (range: 20–96 years), and a strong female predominance (n = 54). At diagnosis, 21 patients (36.84%) had cirrhosis.All patients received UDCA at a dose of 13–15 mg/kg/day. After one year of treatment, a biochemical response according to the Paris criteria was observed in 27 patients (47.7%). The mean ALP-A score was 28.62 (range: 0.65–71). The ALP-A score showed an AUROC of 0.723 (p = 0.004). At a cut-off value of 22, the score yielded a sensitivity of 73% and a specificity of 63%.

Conclusion: The ALP-A score may help identify patients who are less likely to respond to UDCA, allowing for early consideration of second-line therapies. Its use could be valuable in optimizing treatment strategies in PBC.

Disclosure: Nothing to disclose.

PP0799

ROLE OF THE MEAN PLATELET VOLUME-TO-PLATELET COUNT RATIO IN PREDICTING LIVER FIBROSIS IN PRIMARY BILIARY CHOLANGITIS

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Introduction: The mean platelet volume (MPV)-to-platelet count (PLT) ratio is an emerging biomarker in various inflammatory and neoplastic diseases. In chronic liver disease, thrombocytopenia and elevated MPV—frequently observed in hepatic fibrosis and portal hypertension—suggest the potential value of this ratio.

The aim of this study was to evaluate the MPV/PLT ratio as a marker of disease severity in patients with primary biliary cholangitis (PBC).

Aims & Methods: This retrospective, single-center study included all patients with biopsy-proven PBC followed over a 16-year period.Demographic, clinical, biochemical, and histological data were collected. Liver fibrosis was staged using the Scheuer classification.

Patients were categorized into two groups: early-stage disease (Scheuer stages 1–2) and advanced-stage disease (Scheuer stages 3–4). Variables were compared using Pearson's chi-squared test, with a significance level set at $p < 0.05$.

Results: A total of 29 PBC patients underwent liver biopsy. Fourteen patients (48.3%) had early-stage disease, while fifteen (51.7%) had advanced-stage disease. There were no significant differences between the two groups in terms of hematological parameters (white blood cell count, platelet count, hemoglobin) or biochemical markers (AST, ALT, GGT, alkaline phosphatase, total bilirubin, albumin) ($p > 0.05$). Likewise, no significant difference was found in the MPV/PLT ratio between early and advanced fibrosis groups (p = 0.4 and 0.52, respectively); However, the MPV/PLT ratio was significantly higher in patients with cirrhosis (0.68 vs. 0.5; p = 0.021). The AUROC for the MPV/PLT ratio in predicting cirrhosis was 0.792. A cut-off value of 0.059 yielded a sensitivity of 73% and a specificity of 63%.

Conclusion: The MPV/PLT ratio may serve as a simple and non-invasive biomarker for identifying advanced fibrosis in patients with PBC.

Disclosure: Nothing to disclose.

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PP0800

PREDICTION OF ESOPHAGEAL VARICES IN PRIMARY BILIARY CHOLANGITIS: WHAT ROLE OF THE E-LIFT SCORE?

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Introduction: Screening for esophageal varices (EV) in patients with primary biliary cholangitis (PBC) can effectively reduce the morbidity and mortality associated with gastrointestinal bleeding. Current research focuses on using non-invasive tests to minimize unnecessary endoscopic screenings.

The e-LIFT score, a non-invasive method used primarily for the detection of advanced liver fibrosis in chronic liver disease, however, the use of this score for predicting varices, has been evaluated in only one study.

Aims & Methods: This was a retrospective, descriptive, and analytical monocentric study conducted in a gastroenterology department over a 7-year period (2017–2024). It included patients with PBC who underwent upper gastrointestinal endoscopy. Data were collected from patient records. All patients underwent esophagogastroduodenoscopy (EOGD) to detect esophageal varices. The sensitivity (Se) and specificity (Sp) of the e-LIFT score were assessed using the ROC curve.

The objective of our study was to assess the correlation between endoscopic findings and the e-LIFT test.

Results: A total of 67 PBC patients were included. The mean age was 50 ± 14.2 years, with a marked female predominance (80.6%, 54 patients) and a female-to-male ratio of 4.15.

EOGD revealed the presence of EV in 41.4% (24 patients), among whom 12.5% (3), 41.6% (10), and 45.8% (11) had grade I, grade II, and grade III varices, respectively.

The area under the ROC curve for the e-LIFT score in predicting the presence of EV was 0.745. The optimal discriminative threshold for this score was 13, with a sensitivity of 82.14% and a specificity of 61.51%. The negative predictive value (NPV) was 82.76%.

Conclusion: The e-LIFT score is a simple and reliable serum marker for screening esophageal varices in PBC. Thus, upper gastrointestinal endoscopy can be avoided in 82.76% of patients with a score < 13. However, a larger-scale study is needed to confirm the results of our study.

Disclosure: Nothing to disclose.

PP0801

ABCB4 AND ABCB11 POLYMORPHISMS IN INTRAHEPATIC CHOLESTASIS OF PREGNANCY

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Introduction: Intrahepatic cholestasis of pregnancy (ICP) is a liver disorder unique to pregnancy, with complex pathogenesis influenced by genetic factors. Polymorphisms in the *ABCB4* and *ABCB11* genes have been implicated in the pathogenesis of ICP and specifically the ABCB4 rs2109505, ABCB11 rs3815676 and ABCB11 rs7577650 single nucleotide polymorphisms (SNPs). Data from Greece regarding these SNPs are limited.

Aims & Methods: The aim of the study was to investigate the frequency of the three common SNPs in a cohort of Caucasian pregnant women and to evaluate their potential associations with the clinical and laboratory presentation of ICP.

Prospective study during 2020-2024 recruiting Caucasian pregnant women of white eastern European origin with ICP diagnosis. Demographic and laboratory data were recorded. Pregnant women with concomitant history of liver disease (viral hepatitis, autoimmune hepatitis, primary biliary cholangitis etc) were excluded. Peripheral blood samples were collected, and genetic material (DNA) was isolated from whole blood using the commercially available PureLink Genomic DNA Kit, which allows for rapid and efficient DNA isolation. Polymerase Chain Reaction (PCR) was applied to the samples using sequence specific primers to amplify the gene of interest. The PCR products were subjected to agarose electrophoresis to confirm gene amplification, and the samples were subsequently sequenced using an automated analyzer to reveal the presence of the following polymorphisms: ABCB4 rs2109505, ABCB11 rs3815676 and ABCB11 rs7577650.

A square root transformation was applied to achieve normality in skewed variables. Linear and logistic regression models were used to examine the effect of mutated genes on relevant variables, where appropriate. The present study was sponsored by the Hellenic Society of Gastroenterology.

Results: The mean age of the 66 participants was 33.05 ± 7.29 years. Pruritus was reported by 39 (59.09%) of them. A personal history of ICP was noted in 7 (10.61%) women, and a family history in 5 (7.58%). Sixteen (24.24%) participants had gestational diabetes.

First parity was observed in 43 (65.15%), while twin pregnancies were detected in 9 (13.64%) women. The median serum bile salt concentration was 20 mg/L (IQR: 11–49), and the median ALT level was 185 U/L (IQR: 69–428). The ABCB4 rs2109505 SNP was detected in 6 (9.09%) cases, ABCB11 rs3815676 in 2 (3.03%), and ABCB11 rs7577650 in 27 (40.91%).

Women carrying the ABCB11 rs7577650 SNP exhibited a significantly greater weight gain during pregnancy compared to those with the wild-type gene (mean difference: 4.09 kg, 95% CI: 0.19–7.98, p = 0.04). The

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wild-type ABCB11 genotype was associated with higher ALT levels, with a mean difference of 4.56 in the square root of ALT (95% CI: 0.14–8.98, p = 0.04).

The presence of the ABCB11 rs7577650 SNP showed a trend toward association with lower bile salt levels and more frequent diagnosis of gestational diabetes (p = 0.08 and p = 0.07, respectively). No statistically significant associations were detected between ABCB4 rs2109505, ABCB11 rs3815676 SNPs and the other variables.

Conclusion:

1. Overall polymorphisms of the ABCB4 or ABCB11 genes were observed in 53% of the study population of Caucasian women with ICP.
2. The most frequently observed polymorphism was the ABCB11 rs7577650.
3. Women with ICP and the ABCB11 rs7577650 polymorphism tend to gain more weight during pregnancy and present with higher ALT levels.

Disclosure: The present study was sponsored by the Hellenic Society of Gastroenterology.

PP0802

AUTOIMMUNE DISEASES IN PRIMARY BILIARY CHOLANGITIS: A SINGLE-CENTER EXPERIENCE

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Introduction: Primary biliary cholangitis (PBC) is a chronic autoimmune inflammatory disease of the intrahepatic bile ducts. It is characterized by chronic cholestasis, the presence of specific auto immune antibodies, and distinctive histological lesions. PBC is frequently associated with other autoimmune diseases.

Aims & Methods: This study aims to describe the clinical, biological, morphological, and histological characteristics of PBC patients with associated autoimmune diseases, as well as the therapeutic approaches used.

This was a descriptive retrospective study conducted in the Medicine C of Hepatogastroenterology Department at Ibn Sina Hospital in Rabat, including 62 patients with PBC and associated autoimmune diseases between 2010 and 2025

Results: Among 130 patients with PBC, 62 had an associated autoimmune disease, representing a prevalence of 47%. The mean age was 47 years, with a sex ratio of 1M/9F. The most frequent symptoms were jaundice in 45.2% (n=28) of cases, asthenia in 41.9% (n=26), and pruritus in 35.5% (n=22). Clinical examination revealed signs of portal hypertension in 48.4% (n=30) of patients, jaundice in 51.6% (n=32), and scratch lesions in 19.4% (n=12).

Abdominal ultrasound showed liver with features of chronic liver disease in 71% (n=44) of patients, splenomegaly in 41.9% (n=26), ascites in 37.1% (n=23), and collateral venous circulation in 32.3% (n=20).

Biological abnormalities included cytolysis in 71% (n=44) of cases, cholestasis in 38.7% (n=24), and hepatocellular insufficiency in 22.6% (n=14). Regarding the autoimmune profile, 62.9% of PBC cases were seropositive, with anti-mitochondrial antibodies type 2 detected in 45% (n=28), anti-Sp100 antibodies in 19% (n=12), and anti-Gp210 antibodies in 16% (n=10) of patients.

Liver biopsy was performed in 40 patients and the Scheuer stages were classified as follows: stage 1 in 12.9% (n=8), stage 2 in 29% (n=18), stage 3 in 19.4% (n=12) and stage 4 in 3.2% (n=2) of cases.

Fibrosis assessment was carried out in 53 patients, using transient elastography (FibroScan) in 13 cases and liver biopsy in 40 cases. The fibrosis stage distribution was as follows: F3 in 38.7% (n=24), F2 in 25.8% (n=16), F4 in 11.2% (n=7), F1 in 8% (n=5), and F0 in 1.6% (n=1) of cases.

The main associated autoimmune diseases were autoimmune hepatitis in 40.3% (n=25) of cases and autoimmune thyroiditis in 40.3% (n=25), followed by Sjögren's syndrome in 21% (n=13), type 1 diabetes in 8.1% (n=5), and celiac disease in 4.8% (n=3). Less frequent associations included juvenile polyarthritis, Graves' disease, and vitiligo (each in 3.2% (n=2) of cases), as well as scleroderma and Addison's disease (each in 1.6% (n=1) of cases). Among these patients, 29% (n=18) had two associated autoimmune diseases, and 6.5% (n=4) had three.

All patients received ursodeoxycholic acid (UDCA) at doses of 13-15 mg/day, along with specific treatments for concomitant autoimmune diseases.

Treatment response assessment using the Paris II score showed a good response in 62% (n=38) of patients and a partial response in 38% (n=24) of patients.

Conclusion: This study confirms the high prevalence of autoimmune diseases associated with PBC, particularly autoimmune hepatitis and thyroiditis, with a strong female predominance. The most frequent stages of fibrosis were F2 and F3, highlighting the moderate to advanced disease progression at diagnosis in many patients. It emphasizes the importance of systematic screening for these comorbidities to ensure optimal management.

Disclosure: Nothing to disclose.

PP0803

PRIMARY BILIARY CHOLANGITIS AND NON-RESPONSE TO URSODEOXYCHOLIC ACID: PREDICTING PROGNOSIS FROM DIAGNOSIS

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Introduction: The response to ursodeoxycholic acid (UDCA) in primary biliary cholangitis (PBC) varies widely. Identifying patients at risk for non-response early in the disease course is crucial for anticipating the need for second-line therapies.

Aims & Methods: This monocentric retrospective study included 83 patients with PBC who were treated with ursodeoxycholic acid (13–15 mg/kg/day). Therapeutic response was assessed after one year using the Paris II criteria. The primary objective was to identify factors associated with non-response to UDCA.

Results: The mean age at diagnosis was 55.6 years (range: 27–82), with a clear female predominance. The most common clinical presentations were fatigue (34.9%) and pruritus (22.9%). Additionally, 25 patients (30.1%) had cirrhosis at diagnosis. Autoimmune diseases were observed in 60.2% of cases, primarily autoimmune thyroiditis. At one year, 30.1% of

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patients were classified as non-responders to UDCA treatment. Non-responders were older at diagnosis (p = 0.026), more frequently had cirrhosis (p < 0.01), and exhibited a more severe biological profile at diagnosis, including elevated total bilirubin (p = 0.013), hypoalbuminemia (p < 0.01), thrombocytopenia (p = 0.016), and a decreased prothrombin time (p = 0.022). However, initial clinical symptoms (fatigue, pruritus, arthralgia) and autoimmune profiles (specific antibodies, associated autoimmune diseases) were not associated with treatment response.

A statistically significant positive correlation was found between non-response to UDCA and continuous prognostic scores (GLOBE, UK-PBC, Mayo Risk; p < 0.01), as well as non-invasive fibrosis markers (FIB-4: p = 0.032; FibroIndex: p = 0.015; GPR: p = 0.015).

Conclusion: In PBC, the factors driving non-response to UDCA do not appear to be related to clinical symptoms or autoimmune profiles, but rather to an early biological signature. A combined approach, incorporating prognostic scores and non-invasive fibrosis markers, could help identify at-risk patients sooner and guide treatment decisions.

Disclosure: None.

PP0804

EXTRAHEPATIC AUTOIMMUNE COMORBIDITIES IN PRIMARY BILIARY CHOLANGITIS: REFLECTION OF AN IMMUNOLOGICAL PHENOTYPE WITH NO PROGNOSTIC IMPACT?

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Introduction: Primary biliary cholangitis (PBC) is often associated with extrahepatic autoimmune diseases (EAIDs) such as Sjögren's syndrome, autoimmune thyroiditis, and connective tissue disorders. While this association suggests a shared immunological predisposition, its effect on disease prognosis remains unclear.

Aims & Methods: This retrospective study included 69 patients diagnosed with PBC, excluding those with overlap syndromes involving autoimmune hepatitis. Patients were divided into two groups: isolated PBC (n = 35) and PBC associated with EAIDs (n = 34). The two groups were compared based on clinical, immunological, elastometric characteristics, disease progression, and prognostic scores (Paris I, Paris II, Rotterdam, Barcelona, Rochester, GLOBE, UK-PBC). Statistical analysis was conducted using appropriate tests, with a significance threshold set at p < 0.05.

Results: The average age at diagnosis was 55.6 years. The disease was initially suspected based on functional symptoms in 39% of cases (fatigue, pruritus, arthralgia), biological cholestasis in 33.3%, and decompensated cirrhosis in 20.3%. The most common associated autoimmune diseases were Sjögren's syndrome (17.4%), thyroid disorders (13%), and systemic lupus erythematosus (10.1%). All patients received treatment with ursodeoxycholic acid at 13-15 mg/kg/day. No significant differences were found between the two groups regarding functional symptoms, biological markers (bilirubin, albumin, prothrombin time), autoantibodies (AMA, Anti-GP210, anti-SP100, anti-PML), non-invasive fibrosis scores (Fib4, Fibroindex, Fibroscan, S-index), or hepatic complications (HDH, cirrhosis, hepatic encephalopathy, liver-related death). Prognostic scores were also comparable between both groups: Paris I (p = 0.468), Paris II (p = 0.519), GLOBE score (p = 0.6), UK-PBC (p = 0.49), Rotterdam (p = 0.897), Barcelona (p = 0.181), Rochester (p = 0.727).

Conclusion: The presence of extrahepatic autoimmune diseases in PBC does not appear to affect the clinical, biological, or prognostic profiles of patients. Despite their frequency, these comorbidities do not seem to influence the progression of PBC.

Disclosure: None.

PP0805

CLINICAL PROFILE OF CIRRHOSIS DUE TO AUTOIMMUNE HEPATITIS AND REAL-WORLD EXPERIENCE WITH LOW-DOSE STEROIDS

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Introduction: Cirrhosis secondary to autoimmune hepatitis (AIH) presents a unique clinical challenge, with limited data available on its presentation and management.

Aims & Methods: This study aimed to assess the clinical spectrum and evaluate real-world treatment outcomes, particularly the efficacy and safety of low-dose steroid therapy, in patients with AIH-related cirrhosis.

Results: Of 102 AIH patients, 73 (71.6%) had cirrhosis: 40 (54.8%) were compensated, 17 (23.3%) presented with decompensated cirrhosis, and 16 (21.9%) with acute-on-chronic liver failure (ACLF). The majority were female (84.9%) with a significantly higher median age at presentation compared to males (54 vs. 39 years, p = 0.046).

The most common presenting symptom was jaundice (65.8%), followed by anorexia (45.2%), abdominal pain (32.9%), fatigue (23.3%), fever (19.7%), weight loss (13.7%), pruritus (5.5%). 8.5% patients had HE on presentation. Concomitant autoimmune diseases were noted in 21.9% of patients.

Liver biopsy was performed in 48 patients, revealing lymphoplasmacytic infiltrates (100%), portal inflammation (100%), interface hepatitis (97.9%), centrilobular necrosis (70.8%), rosette formation (52.1%), ballooning degeneration (52.1%), biliary changes (31.3%), cholestasis (20.8%), emporipolesis (18.8%) and steatosis (8.3%).. Autoantibody-negative AIH was identified in 11 patients. Median follow-up was 47 months. Five patients (6.8%) died within 6 months of diagnosis, including 4 with ACLF.

A total of 70 patients (95.8%) received steroid therapy. Among 29 patients with decompensated cirrhosis or ACLF who survived beyond 6 months, recompensation was achieved in 8/17 (47.1%) and 11/12 (91.7%) patients, respectively (p = 0.019). Low-dose steroid therapy (≤20 mg/day) was associated with significantly better recompensation rates in decompensated cirrhosis compared to high-dose therapy (80% vs. 0%, p = 0.015), without an increased risk of infections or complications.

Conclusion: AIH presenting with decompensated cirrhosis or ACLF is not uncommon and can be managed successfully with low-dose corticosteroids in selected patients. This approach appears to balance efficacy with safety, offering a viable treatment option in settings where transplant access is limited.

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Disclosure: Nothing to disclose.

PP0806

BRIDGING THE GAP: AN ASIAN PERSPECTIVE ON ADHERENCE TO QUALITY INDICATORS IN THE MANAGEMENT OF PRIMARY BILIARY CHOLANGITIS (PBC)

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Introduction: Primary biliary cholangitis (PBC) is an autoimmune, cholestatic liver disease which results in significant morbidity and mortality. As PBC progresses, it is accompanied by a spectrum of symptoms that can be burdensome. The British Society of Gastroenterology (BSG) has described standards for the management of PBC which address disease progression and symptom burden: adequate dosing of ursodeoxycholic acid (UDCA) 13-15mg/kg/day, timely initiation of second line therapy in those with inadequate response to UDCA, evaluation of symptoms, risk assessment for osteoporosis and identifying high risk patients (bilirubin>50umol/L/ decompensated liver disease) for liver transplant referral. It remains a rare disease worldwide, particularly in Asia, where there is lack of structured care models. We evaluated our hospital's adherence to these recommended guidelines.
Aims & Methods: The primary aim was to evaluate our overall performance against the BSG recommended service standards for the management of PBC. We retrospectively reviewed all patients diagnosed with PBC at our tertiary centre from 1 January 2015-31 December 2023. We included all adult patients with an International Classification of Diseases (ICD-10) code for PBC (K74.3) and/or PBC-specific autoantibodies: antimitochondrial antibody (AMA) or antimitochondrial-2 (anti-M2) who had a follow-up duration ^{3}12 months. Data was captured on baseline demographics, details about diagnosis and follow-up, treatment, symptom and osteoporosis risk assessment, presence of decompensation and discussion regarding transplant for high-risk patients. Inadequate response to UDCA was defined as ALP>1.67x upper limit of normal (ULN) after 1 year of UDCA.
Results: Of the 175 patients analysed,146 (83.4%) were females, mean age 67(±13.3). 96% had a positive AMA. Mean follow-up was 48 months, baseline ALP level was 254U/L and Bilirubin 19umol/L. Of these, 10 patients (5.7%) were diagnosed with an overlap syndrome with autoimmune hepatitis (AIH) after liver biopsy. We noted good adherence to UDCA initiation within 6 months of diagnosis (96.6%) receiving optimal UDCA dose of 13-15mg/kg. Remnant suboptimal dosing was due to intolerance from gastrointestinal (GI) side effects. 77.7% patients were assessed for symptoms in the last 2 years, but interestingly only 2.3% and 3.4% of patients reported pruritus and fatigue respectively. Although most (86%) responded to UDCA, in the remnant 14.3% of non-responders, only 7.4% were prescribed fibrates as second line therapy. None received obeticholic acid. Risk assessment for osteoporosis was done in 56.6% of patients. At baseline, 41 (23.4%) of patients had cirrhosis with decompensation noted in 6.9% (defined as development of ascites, variceal bleeding or hepatic encephalopathy), although less than 10% had liver transplant discussion documented.
Conclusion: Our findings highlight suboptimal performance in adherence to recommended guidelines when measured across most domains. As PBC is rarer in Asia compared to western populations we face challenges in delivering effective therapy using structured models of care. Although larger, more representative studies are required, we will implement a PBC care bundle, followed by re-evaluation as an effort to improve on these shortfalls for better adherence to these key standards to improve quality of care for our PBC patients.
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Disclosure: Nothing to disclose.

PP0807

UPDATE OF SIMPLIFIED CRITERIA FOR THE DIAGNOSIS OF AUTOIMMUNE HEPATITIS

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Introduction: Autoimmune hepatitis (AIH) is a rare cause of liver disease, early detection and treatment with immunosuppressive drugs is life saving. Diagnosing AIH involves laboratory tests, liver tissue examination, and ruling out other liver diseases. In 2008, the International Autoimmune Hepatitis Group (IAIHG) introduced simplified diagnostic criteria, which include four factors: autoantibodies, serum immunoglobulin (IgG) levels, results for chronic viral hepatitis, and liver biopsy findings. In line with simplified criteria, each variable is evaluated on a scale of 0 to 2 points. An overall score of 6 is categorized as 'probable AIH,' whereas a score of 7 is classified as 'definite AIH'.
Liver tissue examination is essential for diagnosing AIH, especially when serological markers are missing. The simplified criteria identify portal lymphocytic infiltrates, emperipolesis, and hepatic rosette formation as characteristic features.
Few studies have assessed these simplified histological criteria. In 2022, the AIH-Pathology Group proposed new criteria that classify histological features as likely, possible, or unlikely. In 2021, the IAIHG revised the

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methodology for immunoserological testing, raising the positive threshold for antinuclear autoantibodies (ANA) when evaluated on HEp-2 cells. The new serological criteria wasn't externally validated yet.

Aims & Methods: We conducted a retrospective analysis of data from 179 patients, with 111 diagnosed with AIH, 33 with primary biliary cirrhosis (PBC), and 35 with drug-induced liver injury (DILI), collected from three centers in Türkiye between January 2013 and March 2021. Initially, we assessed the diagnostic accuracy of both the new histological criteria and the older simplified histological criteria. Subsequently, we revised the simplified criteria by integrating findings from the new histological criteria and immunoserological testing, and then evaluated its diagnostic accuracy.

Results: The new histological criteria and the simplified histological criteria demonstrated sensitivities of 95% and 88%, respectively, for diagnosing possible/probable AIH, with specificities of 75% and 32%. The new histological criteria demonstrated a sensitivity of 60% and a specificity of 97% for likely AIH, whereas the simplified histological criteria had a sensitivity of 42% and a specificity of 94% for definitive AIH. The area under the curve (AUC) for diagnosing AIH was higher with the new histological criteria compared to the simplified histological criteria (0.913 vs 0.714). The updated simplified score demonstrated similar sensitivity to the original simplified score (88% vs 87%) but exhibited higher specificity (91% vs 74%) for probable AIH. For definitive AIH, the specificity was slightly reduced (57% vs 63%), while specificity was enhanced (96% vs 85%). The AUC for AIH diagnosis was higher in the updated simplified score than in the simplified score (0.942 vs 0.856).

Conclusion: The new histological criteria demonstrated greater sensitivity and specificity compared to the simplified criteria for diagnosing AIH. Updating the simplified criteria further enhanced the accuracy for the diagnosis of AIH. Our findings suggest that upgrading the current simplified criteria both serologically and histologically would improve diagnosis in AIH patients.

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Disclosure: Nothing to disclose.

PP0808

URSODEOXYCHOLIC ACID THERAPY IN PATIENTS WITH PRIMARY BILIARY CHOLANGITIS: PREDICTORS FOR INCOMPLETE RESPONSE

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Introduction: The current universal standard of care in primary biliary cholangitis (PBC) is ursodeoxycholic acid (UDCA), although its clinical benefit in PBC has never been reported in absolute measures.

Aims & Methods: This cohort aimed to assess the prevalence of incomplete response to UDCA and determine non-response predictors.

Patients with PBC as main diagnosis were included from the hospital's centralized electronic medical records system, over a 10-year period. Overlap syndromes (PBC-Auto-Immune Hepatitis) were also included. Main endpoints included incomplete response to UDCA treatment according to Paris I, Paris II criteria, and Globe score.

Results: A total of 50 patients were included in this study. The mean age at diagnosis was 52.74 years (24-73 years) and 92% females. Only one patient was asymptomatic at onset and 82% had positive anti-mitochondrial antibodies. All patients were treated with ursodeoxycholic acid (UDCA). Corticosteroids and immunosuppressive therapies were associated in 26% of cases. Incomplete response to UDCA was observed in 28% and 34% of patients according to Paris I and II criteria, respectively. Non-response predictors were poor therapeutic adherence, Overlap syndrome and hepatocellular carcinoma in 14%, 6% and 4% respectively.

Five patients were lost to follow-up and five cases died from cirrhosis 'complications. The median ΔGLOBE score was 0.42 (IQR: -1.58 to 3.92). The 5, 10 and 15 years liver transplantation-free survival (median) was: +10%, +7.38% and +5% respectively. Liver transplantation (LT) was considered in one patient with excellent outcomes at last follow-up.

Conclusion: A considerable proportion of patients showed incomplete response to UDCA treatment according to Paris II criteria. Early identification of patients at risk of incomplete response could improve patient management and overall disease prognosis.

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Disclosure: Nothing to disclose.

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PP0809

ENHANCED PLATELET AGGREGATION WITH TRAP-6 IN PLATELET AGGREGOMETRY IN PATIENTS WITH ACUTE-ON CHRONIC LIVER FAILURE

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Introduction: Patients with acute-on-chronic liver failure (ACLF) have traditionally been considered with a tendecy for bleeding however, they frequently display a paradoxical prothrombotic state. ACLF is characterized systemic inflammation and an increased risk of both bleeding and thrombosis. Bleeding arises from decreased pro-coagulation factors and platelet sequestration due to portal hypertension, while thrombosis is exacerbated by endothelial dysfunction and elevated heparinoids, thrombomodulin, or von Willebrand factor.

Aims & Methods: The aim of this study was to evaluate platelet aggregation pattern in patients with ACLF compared to acute decompensated (AD) liver cirrhosis.

The study took place in between the 1st of July 2024- 31st of January 2025 and we prospectively included cirrhotic patients, divided in two groups: ACLF group and AD group. The exclusion criteria were people with compensated liver cirrhosis, antiaggregant or anticoagulant treatment and patients with hepatocellular carcinoma or other malignancies. We investigated platelet aggregability in whole blood, by using multiplate aggregometry.

For analysis, platelet aggregometry was performed induced by: adenosine diphosphate (ADP), arachidonic acid (ASPI), prostaglandin E2 (PGE2) and thrombin receptor-activated peptide (TRAP-6) as agonist were determined. TRAP-6 is a peptide that stimulates protease-activated receptor 1 (PAR1) on platelet membrane.

Results: In this study we included 52 patients – 25 patients with ACLF and 27 patients with AD. There were no differences regarding demographic parameters and liver disease etiology between the two study groups. Compared to AD group, patients with ACLF showed an enhanced maximal platelet aggregation with TRAP-6 (AUC 85±12 U vs 45±10 U, p<0.0001). In contrast, we did not detect any differences in platelet aggregation between the two groups after ADP, ASPI or PGE2 activations.

We observe that PAR-1 stimulation induces the highest aggregation curve and area under the curve in patients with ACLF and AD, but ACLF patients show the strongest response, highlighting that the degree of liver decompensation influences aggregation results. Severe decompensation correlates with a stronger TRAP response, suggesting that there are more PAR-1 receptors on platelets of this patients.

Conclusion: Our results indirectly indicate that platelet reactivity is higher in patients with ACLF likely due to enhanced PAR-1 receptor activity. Severe decompensation correlates with a stronger TRAP response, suggesting that there are more PAR-1 receptors on platelets of this patients.

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Disclosure: Nothing to disclose.

PP0810

EVALUATION OF THE IMPACT OF PRE-OPERATIVE SARCOPENIA ON INFECTION DEVELOPMENT IN LIVER TRANSPLANT PATIENTS

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Introduction: Sarcopenia, a progressive loss of skeletal muscle mass and function, is a significant predictor of adverse outcomes in liver transplant patients, including infections. This study evaluates the relationship between pre-transplant sarcopenia and infection rates at one, three-, and six-months post-transplantation, using radiological markers such as skeletal muscle mass index (SMII), intramuscular adipose tissue content (IMAC), and visceral/subcutaneous fat area ratio (VSR).

Aims & Methods: This retrospective study analysed 102 liver transplant recipients undergone surgery between July 2012 and July 2023. Computed tomography (CT) scans taken within 30 days prior to transplantation were evaluated to assess sarcopenia using SMII, IMAC, and VSR. Demographics, clinical parameters, and infection rates at one, three, and six months were compared. Pre-transplant sarcopenia indicators were assessed for their correlation with early and late infection rates.

Using a forward conditional method for variable selection, logistic regression was applied to identify significant predictors of infection up to the third month post-transplant. Statistical analyses included chi-square and Mann-Whitney U tests.

Results: The median age of the cohort was 51 years, and 31.4% were female. Patients were categorized into sarcopenic (n=41) and non-sarcopenic (n=61) groups based on preoperative imaging data. Sarcopenia was significantly less prevalent in females (17.1% vs. 41%; p=0.016). Sarcopenic patients had a lower median SMII (61.54 vs. 100; p<0.001) and higher preoperative MELD-Na scores (24 vs. 18; p=0.02).

One-month infection rates were significantly higher in sarcopenic patients (48.8% vs. 27.8%; p=0.032), as were three-month rates (65.8% vs. 44.2%; p=0.043). By six months, infection rates were similar between groups (60.7% vs. 57.3%; p=0.41).

Logistic regression identified sarcopenia (OR = 10.92, 95% CI [1.53, 78.09], p = .017), urea (OR = 1.10, 95% CI [1.03, 1.17], p = .003), INR (OR = 6.03, 95% CI [1.05, 34.52], p = .044), and age (OR = 0.91, 95% CI [0.84, 0.99], p = .035) as significant predictors of infection up to the third month. While diabetes mellitus (DM) was not statistically significant (p > .05), its inclusion enhanced the model's predictive power. These findings indicate that higher sarcopenia, urea, and INR values increase infection risk, while older age appears protective.

Conclusion: Pre-transplant sarcopenia is strongly associated with increased infection rates in the first three months following liver transplantation. However, by six months, the comparable infection rates suggest

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that sarcopenia-related risks may diminish with recovery. These findings underscore the importance of addressing sarcopenia preoperatively through tailored interventions.

Disclosure: Nothing to disclose.

PP0811

CLINICAL BENEFIT AND DIAGNOSTIC ACCURACY OF SHEAR-WAVE ELASTOGRAPHY COMPARED TO TRANSIENT ELASTOGRAPHY (ASTEL STUDY)

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Introduction: Liver elastography is one of the basic diagnostic tools in hepatology practice. The most widely used and most studied tool in the world is transient elastography (Fibroscan), however, this system is unevenly available in Slovakia. Shear-wave elastography is a more accessible method because it is built into standard ultrasound systems. However, the problem of shear-wave elastography remains the interpretation of its results, since the threshold values for significant fibrosis (SF) and advanced fibrosis (AF) are unique for each system.

Aims & Methods: To compare the results of transient elastography (TE) and point shear-wave elastography (pSWE) in a cohort of patients examined simultaneously with both systems and to determine the clinical benefit of pSWE in practice.

We examined all indicated patients over a 15-month period using both TE and pSWE (Hitachi Arietta V70). We recorded liver stiffness values in kPa, and baseline demographic and laboratory parameters. We used TE as the gold standard, with a threshold of >10 kPa for SF and >15 kPa for AF. We evaluated the accuracy of pSWE in predicting the detection of SF and AF.

We used three models: the first one containing only pSWE values, and two logit models using pSWE together with common demographic and laboratory parameters. We identified the area under the curve (AUROC) and threshold values for each model and evaluated the probability of SF and AF according to these values.

Results: We examined 500 patients with a median age of 60 (IQR 48-69), male/female 214/286, with a SF rate of 30.8% and AF of 18.4%. The stiffness values according to TE and pSWE showed a highly significant correlation (rho=0.593, P<0.001).

We found the accuracy of pSWE alone in predicting SF (AUROC=0.812), and AF (AUROC=0.839) with threshold values of SF>6.34 and AF>8.45 kPa. For values of pSWE<6.34, 6.34-8.45, and >8.45 kPa, the probability of SF was 11.1, 40.9, and 65.4% and AF was 5.0, 14.77, and 48.9%.

We created two probability models (for P, range 0 to 1): ASTEL1 (N=369) that included pSWE, GGT, and platelets (AUROC SF=0.856, AUROC AF=0.908), and ASTEL2 (N=467) using pSWE, age, sex, AST/ALT, GGT (AUROC SF=0.857, AUROC AF=0.901).

We set a threshold of P<0.33 to exclude SF and P>0.66 to confirm AF. At P<0.33, P=0.33-0.66, and P>0.66, SF was present in 15.6, 62, and 94.2% (ASTEL1) and 13.92, 50.7, and 86.8% (ASTEL2). At P<0.33, P=0.33-0.66 and P>0.66, AF was present in 5.6, 30, and 79.7% (ASTEL1) and in 4.4, 28, 73.68% (ASTEL2).

Conclusion: pSWE alone has an excellent negative predictive value (<6.34 kPa, 88.9-95%) in the diagnosis of SF and AF. Higher accuracy of positive prediction of SF (86.8-94.2%) and AF (73.7-79.7%) can be achieved by using one of two probability models (for P>0.66) combining SWE values with common demographic and laboratory parameters.

Disclosure: Nothing to disclose.

PP0812

A 12-YEAR PERSPECTIVE ON LIVER ABSCESSES: EMPHASIZING ULTRASOUND-GUIDED INTERVENTIONAL TREATMENT

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Introduction: This research focuses on analyzing ultrasound-based management in percutaneous interventions for liver abscesses.

Aims & Methods: A retrospective analysis was carried out on patients diagnosed with liver abscesses between 2010 and 2024, in a tertiary-level Gastroenterology and Hepatology department. Information regarding demographics, existing risk factors, etiological factors of liver abscess, and patient outcomes was collected and examined.

Results: Among 187 cases, patients were categorized into three groups based on treatment approach:

Group A (60/187, 32.1%): Percutaneous aspiration for bacteriological sampling and abscess evacuation.

Group B (30/187, 16.0%): Continuous drainage via catheter insertion.

Group C (46/187, 24.6%): Conservative management with empirical antibiotic therapy.

Length of Antibiotic Therapy (days): A (15.78 ± 8.35) vs. B (18.36 ± 7.96) vs. C (15.8 ± 8.55), p = 0.01.

Hospital Stay: A (14.89 ± 8.29) vs. B (18.36 ± 7.96) vs. C (15.3 ± 8.12), p = 0.01.

Complication Rates (Acute Kidney Injury - AKI): A (5/64, 7.2%) vs. B (0%) vs. C (2/46, 4.3%), p = 0.5.

Mean Abscess Size: B (9.3 ± 2.60) > A (6.05 ± 2.16) > C (4.47 ± 2.01), p < 0.001.

Two-Week Follow-Up: Complete remission: A (39.0%) vs. B (20.0%) vs. C (56.5%), p < 0.001. Stationary cases: A (39.1%) vs. B (3.3%) vs. (50.0%), p = 0.005.

Conclusion: Percutaneous drainage proved to be the most effective approach for managing larger liver abscesses. While no significant differences were observed among the three groups regarding hospital stay and complication rates, conservative treatment showed a higher proportion of stationary cases during follow-up.

Disclosure: Nothing to disclose.

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PP0813

CHCHD1 PROMOTES THE METASTASIS OF HEPATOCELLULAR CARCINOMA VIA TGFBR1 IN MULTI OMIC

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Introduction: One of the most common types of human cancer, with the second-highest fatality rate, is hepatocellular carcinoma (HCC). HCC accounts for the majority of primary HCC cases, about 80%. Surgical resection, liver transplantation, transarterial chemoembolization, local ablation therapy, and molecularly targeted medicines are currently used to treat HCC. HCC is usually diagnosed in the late stage, and 70%-80% of HCC patients are not given the chance to receive total tumor excision.

After surgery, about 60% of patients still experience HCC recurrence, leading to poor prognosis. Therefore, improving the survival percentage of HCC patients requires early detection. Currently, imaging and biomarkers are the main diagnostic methods for HCC. Compared with imaging, biomarkers are diverse and have the potential for further exploration. To predict the prognosis of HCC accurately, it is essential to discover valuable new biomarkers.

CHCHD1 is located on human chromosome 10q22 and confirmed to belong to the protein family that include the coiled-coil-helix-coiled-coil-helix (CHCH) domain. CHCH contains two CX9C motifs, with two cysteine residues separated by nine amino acids in each motif. CHCHD1 exists in the nucleus or mitochondria and is involved in mitochondrial translation and protein metabolism. Existing evidence suggests that CHCHD1 is a biomarker for cervical squamous cell carcinoma that can be used for diagnosis or prognosis, while the related gene CHCHD2 is a potential poor prognostic biomarker for non-small cell lung cancer and HCC. However, the role of CHCHD1 in the development of HCC remains unknown.

This study investigates the relationship between CHCHD1 and the prognosis of HCC by analyzing the differential expression of CHCHD1 in HCC tissues and adjacent normal tissues.

Additionally, we found that CHCHD1 can affect the progression of HCC through epithelial-mesenchymal transition and the TGF-β signaling pathway. Therefore, CHCHD1 is crucial for the development and progression of HCC, and inhibiting CHCHD1 activity could be an effective treatment for HCC.

Aims & Methods: As the foundation for this research, we use the TCGA database and HCC tissues to investigate the expression of CHCHD1 in HCC. Following that, the quantity of associated proteins was shown by immunohistochemical labeling and Western blot. To demonstrate the potential for tumor growth, invasion, and migration, the transwell test and wound healing experiment were carried out. GSEA further confirmed the potential CHCHD1-related signaling pathway in HCC. Further, GSE140228 was used to detect the expression of CHCHD1 in the single-cell level.

Results: CHCHD1 is substantially expressed in HCC and is linked to a poor prognosis. It can promote the EMT process and influence the TGF-β signaling pathway to promote the progression of HCC. In addition, we found CHCHD1 is higher in the myeloid cells of HCC samples.

Conclusion: CHCHD1 may serve as a potential diagnostic and prognostic marker for HCC and is also involved in the immune infiltration of HCC.

Disclosure: Nothing to disclose.

PP0814

MULTI-MODAL EVIDENCE REVEALS ACAT2 AS A KEY MEDIATOR OF VITAMIN D–DRIVEN SUPPRESSION OF HCC

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Introduction: Hepatocellular carcinoma (HCC) is characterized by high invasiveness and metastatic potential, with persistently low five-year survival rates. While vitamin D has demonstrated anti-tumor effects in various cancers, its precise molecular mechanism in HCC remains unclear. This study aimed to investigate the inhibitory role of vitamin D in HCC progression, focusing on the regulatory function of acetyl-CoA acetyltransferase 2 (ACAT2).

Aims & Methods: Huh-7 cells were treated with increasing concentrations of vitamin D (0, 15, 20, 30, and 40 nM) for 48 hours. Cell migration was assessed using Transwell assays, and apoptosis was quantified via Annexin V/PI staining followed by flow cytometry. ACAT2 mRNA and protein levels were measured by qPCR and Western blot, respectively. Prognostic relevance of ACAT2 expression was analyzed using Kaplan–Meier survival curves based on the GSE76427 dataset. Mechanistically, single-cell transcriptomic data from 26 HCC patients were integrated to identify $ACAT2^{+}$ malignant cell populations and their enriched pathways via KEGG analysis. Mendelian randomization (MR) using the inverse-variance weighted (IVW) method was employed to assess the causal relationship between genetically predicted vitamin D levels and HCC risk, and to quantify the mediation effect of ACAT2.

Results: Vitamin D treatment significantly suppressed Huh-7 cell migration and induced apoptosis in a dose-dependent manner. ACAT2 expression was consistently downregulated at both mRNA and protein levels. Clinical analysis revealed that high ACAT2 expression was associated with poor prognosis in HCC patients. Single-cell analysis showed that ACAT2 expression was markedly elevated in tumor hepatocytes, and $ACAT2^{+}$ cells were enriched in cytochrome P450-related xenobiotic metabolism pathways. MR analysis demonstrated a significant inverse association between genetically predicted vitamin D levels and HCC risk, and ACAT2 was identified as a significant mediator, accounting for 9.6%.

Conclusion: This study provides novel evidence that vitamin D suppresses HCC progression by downregulating ACAT2. ACAT2 may serve as a potential prognostic biomarker and therapeutic target. These findings support the translational potential of vitamin D as an adjunctive strategy in HCC prevention and treatment.

Disclosure: Nothing to disclose.

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PP0815

MULTI-OMICS ANALYSES REVEAL THAT ATF6B-APOC2-CD36 AXIS PROMOTING THE GROWTH AND METASTASIS OF HBV-ASSOCIATED HEPATOCELLULAR CARCINOMA

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Introduction: HBV infection is a critical global health issue and the leading cause of cirrhosis and hepatocellular carcinoma (HCC) with its mechanisms still not fully understood. Despite existing several HCC biomarkers, challenges such as low sensitivity and tumor heterogeneity hinder early detection, highlighting the need for further exploration of mechanisms and new diagnostic markers.

Aims & Methods: In this study, we conducted an integrated multi-omics study of HCC GWAS, TCGA-LIHC, GTEx, and HPA data, in order to identify pathogenic genes and serum markers for HBV-associated HCC. The expression of candidate genes or serum markers were then evaluated in HCC liver tissues or serum. In vitro and in vivo cellular functional experiments were performed to determine their roles in tumor progression.

Results: Our findings revealed that the ATF6B-APOC2-CD36 axis derived growth and metastasis in HBV-associated HCC. Mechanistically, HBV induced ATF6B expression via the endoplasmic reticulum stress pathway. Then ATF6B bound to the APOC2 promoter region, thereby enhancing APOC2 expression. APOC2 further interacted with CD36 to activate the Wnt/β-catenin signaling pathway, promote EMT and consequently facilitate the growth and metastasis of liver cancer. Moreover, serum APOC2 significantly increased in HBV-related HCC patients and when combined with PIVKA-II shown high diagnostic value for AFP-negative HBV-related HCC.

Conclusion: Our findings demonstrated that the ATF6B-APOC2-CD36 axis contributes to HCC progression and serum APOC2 offers diagnostic potential for AFP-negative HBV-related HCC patients. These findings may advance molecular understanding, early diagnosis, and treatment of HBV-related HCC.

Disclosure: Nothing to disclose.

PP0816

REGIONAL DISPARITIES AND FUTURE PROJECTIONS OF LIVER CANCER IN AFRICA: A COMPREHENSIVE EPIDEMIOLOGICAL FORECAST THROUGH 2040

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Introduction: Liver cancer, one of the most aggressive cancers, has closely aligned incidence and mortality rates. Key risk factors include chronic viral hepatitis HBV and HCV, aflatoxin exposure, alcohol consumption, and type 2 diabetes. Africa, which is heavily affected by these risk factors, exhibits high rates of liver cancer.

Our study aims to describe the burden of liver cancer among the African population in 2022.

Aims & Methods: A descriptive study was conducted to examine the global distribution of liver cancer in Africa. Estimates of new cases and deaths, as well as Age Standardized Rates (ASR), were sourced from the Global Cancer Observatory database (GLOBOCAN) for 2022. Using global projections, we estimated the incidence and mortality burden by 2040.

Results: In 2022, Africa recorded 73,844 new cases and 70,315 deaths from liver cancer, representing 6.2% and 9.2% of total cancers. Incidence and mortality rates were significantly higher in males (ASR of 11.1 and 10.7 per 100,000, respectively) than in females (ASR of 6.2 and 6.0 per 100,000, respectively). Regional disparities were evident, with incidence and mortality rates 2.3 times higher in North Africa (ASR of 14.2 and 13.7 per 100,000, respectively) than in Sub-Saharan Africa (ASR of 6.3 and 6.0 per 100,000, respectively). Egypt accounted for more than a third (38%) of all estimated liver cancer cases and fatalities in Africa. The incidence varied by as much as 29-fold and mortality by 20-fold across African countries. The highest ASRs were recorded in Egypt (32.0 and 30.8, respectively) and Guinea (21.3 and 20.8, respectively), whereas the lowest were in Botswana (1.6 and 1.5, respectively) and Cape Verde (1.1 and 1.5, respectively). The mortality to incidence rate (MIR) remained high at 95%. By 2040, with an increase of 81%, the number of new cases and deaths is predicted to reach 133,431 and 127,498, respectively.

Conclusion: Implementing WHO's hepatitis elimination strategy by 2030 is crucial for mitigating these rates. This strategy encompasses HBV vaccination, prevention of mother-to-child transmission, comprehensive testing and treatment of viral hepatitis, and targeted reduction of aflatoxin exposure and other modifiable risk factors to significantly reduce liver cancer prevalence in African countries.

Disclosure: Nothing to disclose.

PP0817

PSEUDOMONAS KOREENSIS ACCELERATES HCC PROGRESSION BY INHIBITING SPERMIDINE SYNTHESIS AND PROMOTING T CELL EXHAUSTION

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Introduction: Hepatocellular carcinoma (HCC), the sixth most prevalent malignancy and third leading cause of cancer-related mortality globally, accounts for 75-85% of primary liver cancers with a 5-year survival rate below 20%. Early-stage HCC is managed by resection or ablation, while advanced cases rely on transarterial therapies and systemic treatments. However, drug resistance to multi-kinase inhibitors (e.g., sorafenib) significantly limits efficacy. HCC progression is closely linked to an immunosuppressive tumor microenvironment characterized by T-cell co-stimulatory molecule deficiency and checkpoint activation (PD-L1/CTLA-4), driving T-cell exhaustion. Recent evidence highlights intratumoral microbiota dysbiosis as a modulator of immune remodeling, yet the spatial distribution patterns and regulatory mechanisms of bacterial communities within HCC lesions remain poorly defined.

Aims & Methods: This study employed invasion-adhesion-directed expression sequencing (INVADE-seq) integrated with single-cell host transcriptomics to profile intratumoral microbiota and construct microbial-immune interaction networks in five HCC tumors and paired adjacent normal tissues. Peritumoral microbiome analysis was conducted using 2b-Rad-M sequencing. Through systematic bacterial isolation (LB/Columbia/BCSA/Schaedler media) and Sanger sequencing validation, *Pseudomonas koreensis* (P.K) was identified as a key HCC-associated strain.

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Spatial localization and ultrastructural characterization were achieved via fluorescence in situ hybridization (FISH), multiplex fluorescence staining, and electron microscopy. Whole-genome sequencing delineated its genomic architecture, while metabolomics uncovered unique metabolic signatures. Functional validation using orthotopic HCC models (MYC/Alb and NSG mice) revealed its pro-tumorigenic role and immunomodulatory effects.

Results: 2b-Rad-M sequencing showed significantly higher abundance of *Bacillus, Pseudomonas, and Klebsiella* in HCC tissues compared to adjacent non-tumor tissues, with reduced bacterial diversity. Circos plots highlighted increased *Proteobacteria* and *Firmicutes* in HCC. Through culturing and Sanger sequencing, P.K was identified as a dominant strain in HCC, showing quinolone sensitivity and urease activity. FISH confirmed its widespread presence in HCC tissues, with elevated abundance in T cells from advanced-stage patients. INVADE-seq analysis of five additional HCC tissue pairs also detected P.K Single-cell INVADE-seq revealed that T cells harboring this bacterium exhibited increased exhaustion scores and upregulated TOX expression. In MYC/Alb mice, oral gavage of P.K accelerated hepatocarcinogenesis, but no significant effect was observed in NSG immunodeficient mice, indicating immune system dependency. Whole-genome sequencing linked HCC-derived P.K to the fish pathogen *P. koreensis CM-01*, with enriched amino acid and carbon metabolism pathways. Metabolomics analysis associated it with purine/pyrimidine metabolism and cGMP-PKG signaling. Reduced spermidine levels impaired fatty acid oxidation, exacerbated T cell exhaustion, and promoted an immunosuppressive microenvironment, enhancing tumor proliferation, invasion, and metastasis.

Conclusion: This study reveals how intratumoral microbiota regulate the tumor microenvironment via the metabolism-immune axis, offering new insights into HCC immunotherapy. Targeting P.K-related metabolic pathways may provide a novel therapeutic strategy.

Disclosure: Nothing to disclose.

PP0818

HISTOPATHOLOGICAL CORRELATIONS OF TISSUE SOUND SPEED IN HEPATOCELLULAR CARCINOMA: A QUANTITATIVE ASSESSMENT USING ACOUSTIC MICROSCOPY

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Introduction: Tissue acoustic properties, such as sound speed, attenuation, and impedance, are closely associated with its mechanical characteristics. Acoustic microscopy, which employs high-frequency ultrasound ranging from several tens of MHz to several GHz, enables the quantitative evaluation of these acoustic properties. Understanding the acoustic properties of biological tissues may offer new insights into disease pathology.

Furthermore, because clinical ultrasound images depend on tissue acoustic properties, understanding them could improve the interpretation of echogenic patterns and contribute to developing new diagnostic approaches. Although acoustic microscopy has been applied to cardiovascular and cutaneous tissues in medical research, studies focusing on the liver, particularly hepatic tumours, remain limited.

Aims & Methods: This study aimed to quantify sound speed in hepatocellular carcinoma (HCC)—the most common primary liver tumour—and to examine its relationship with histopathological features. Histopathological classification followed World Health Organization criteria, including histological grade (well, moderately, and poorly differentiated) and histological growth pattern (trabecular, pseudoglandular, and solid). Tissue sound speed was measured using a scanning acoustic microscope (AMS-50SI; Honda Electronics Co., Ltd.) with an 80 MHz transducer. Regions of interest were identified on haematoxylin and eosin-stained slides. Serial 10 µm-thick sections were prepared from formalin-fixed, paraffin-embedded tissue blocks and rehydrated for measurement.

Differences in sound speed among histological categories were compared, and statistical analysis was conducted to assess significance. A p-value of < 0.05 was considered significant. This study was approved by the institutional review board, and written informed consent was obtained.

Results: Tissue sound speed was measured at 66 sites from 51 patients. Histological grading identified 22 well, 41 moderately, and 3 poorly differentiated cases. Median sound speeds were 1584.0 m/s, 1595.9 m/s, and 1550.1 m/s, respectively. Poorly differentiated HCC showed significantly lower sound speed than the other grades ($p = 0.021$). Growth patterns included 50 trabecular, 10 pseudoglandular, and 6 solid cases, with median sound speeds of 1595.9 m/s, 1586.0 m/s, and 1563.6 m/s, respectively. The trabecular pattern tended to exhibit higher sound speed than the solid pattern ($p = 0.063$). Tissues with abundant fat or glycogen ($n = 16$) had significantly lower sound speeds than those without ($p = 0.002$).

After excluding these cases, which could act as potential confounding factors, sound speed in poorly differentiated HCC remained significantly lower ($p = 0.016$), and higher values in the trabecular pattern persisted. Additionally, the pseudoglandular pattern showed greater variability in sound speed than the trabecular pattern.

Conclusion: Variations in acoustic properties among histological subtypes of HCC may contribute to developing novel quantitative assessments in ultrasound imaging and pathological diagnosis.

Disclosure: Kazuto Kobayashi, a co-author, has reported a relationship with Honda Electronics Co., Ltd. that includes: employment.

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PP0819

HEPATITIS C TREATMENT IMPROVES ONCOLOGIC OUTCOMES IN PATIENTS RECEIVING ATEZOLIZUMAB PLUS BEVACIZUMAB

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Introduction: Hepatitis C virus (HCV) infection is a key driver of hepatocellular carcinoma (HCC) development and progression. However, the impact of HCV eradication in patients undergoing systemic therapy remains poorly understood.

Aims & Methods: This study aimed to assess the safety and efficacy of direct-acting antiviral (DAA) therapy in patients receiving Atezolizumab plus Bevacizumab (AtezoBeva). This multicenter study retrospectively enrolled consecutive patients with HCV-related unresectable or advanced HCC undergoing AtezoBeva therapy between 2021 and 2024.

Patients were stratified into three groups based on their HCV treatment status: group A (n=22), who received concurrent DAA therapy with AtezoBeva; group B (n=95), who had completed DAA therapy at least six months before initiating AtezoBeva; and group C (n=22), with active HCV infection. The impact of DAA treatment on overall survival (OS), time to progression (TTP), progression-free survival (PFS), time to decompensation (TTD), and decompensation-free survival (DFS) was evaluated. Additionally, the incidence of treatment-related adverse events (trAEs) was recorded.

Results: Group A demonstrated a significantly prolonged median OS (42.8 months [95%CI: 33.3–NA]) compared to Group B (26.8 months [95%CI: 17.1–NA]; p=0.03) and group C (19.7 months [95%CI: 14.7–NA]; p=0.01). TTP was longer in group A (40.6 months [95%CI: 17.68–NA]) than in group C (14.6 months [95%CI: 5.06–NA]; p=0.02), whereas no significant difference was observed between group A and group B (20 months [95%CI: 12.29–NA]). Median PFS was markedly extended in group A (40.6 months [95%CI: 22.47–NA]) compared to group B (15 months [95%CI: 11.96–23.4]; p=0.009) and group C (13.3 months [95%CI: 5.06–NA]; p=0.002).

Moreover, group A exhibited a higher disease control rate than the other groups. In the multivariate time-dependent analysis, achieving sustained virologic response (SVR) during AtezoBeva therapy was independently associated with a reduced risk of mortality (HR 0.29 [95%CI: 0.12–0.69]; p=0.005) and disease progression (HR 0.1 [95%CI: 0.05–0.72]; p=0.02).

Notably, post-DAA decompensation rates were significantly lower in group A (4.5%) compared to groups B (26.3%) and C (36.4%), translating into a superior DFS. The incidence of grade ≥3 treatment-related AEs was comparable across groups.

Conclusion: Achieving SVR during AtezoBeva therapy significantly enhances oncological outcomes in patients with unresectable or advanced HCC. These findings highlight the potential for an integrated therapeutic approach to optimize systemic treatment efficacy, particularly in patients eligible for conversion strategies.

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PP0820

VASCULAR INVASION WITHIN THE CRITERIA FOR ONCOLOGICAL RESECTABILITY OF HEPATOCELLULAR CARCINOMA IS A CRITICAL PREDICTOR OF SURVIVAL IN PATIENTS TREATED WITH ATEZOLIZUMAB AND BEVACIZUMAB

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Introduction: Recently, a Japanese expert consensus introduced oncological resectability criteria for hepatocellular carcinoma (HCC) [1]. Notably, these criteria do not incorporate the concept of unresectable tumors, but rather classify borderline resectability into two distinct subgroups, defined from the therapeutic perspective of hepatic resection.

Based on these classifications, resectability is categorized into three groups: resectable (R), borderline resectable 1 (BR1), and borderline resectable 2 (BR2).

Aims & Methods: In this study, we aimed to assess the clinical outcomes of patients with HCC who were treated with atezolizumab plus bevacizumab (Atezo/Bev) in a real-world context, categorized by oncological resectability criteria. Between May 2018 and May 2024, a total of 1,055 patients with unresectable HCC were treated with Atezo/Bev across 26 institutions in Japan.

Of these, 467 met the following inclusion criteria:

1. Child–Pugh class A,
2. Atezo/Bev administered as the first-line systemic therapy,
3. Eastern Cooperative Oncology Group performance status (ECOG-PS) of 0 or 1,
4. BR1 or BR2 classification,
5. Treatment conducted within a clinical trial setting, and;
6. Availability of complete clinical data.

Results: Of 467 patients, 102 (21.8%) were women, and 365 (78.2%) were men. The median age was 74.0 (69.0–79.0) years. Patients were categorized into BR1 (n=153 [32.8%]) and BR2 (n=314 [67.2%]) groups. The median follow-up period was 14.3 (7.5–24.0) months. The median progression-free survival (PFS) was 9.0 months in the BR1 group and 6.8 months in the BR2 group (p = 0.014).

Multivariable analysis identified the following independent prognostic factors for PFS: age ≥75 years (hazard ratio [HR], 1.309; 95% confidence interval [CI], 1.057–1.622; p=0.014), albumin–bilirubin (ALBI) grade 2 or 3 (HR, 1.494; 95%CI, 1.201–1.859; p<0.001), neutrophil-to-lymphocyte ratio (NLR) ≥3 (HR, 1.289; 95%CI, 1.030–1.613; p=027), α-fetoprotein (AFP) ≥100 ng/mL (HR, 1.523; 95%CI, 1.214–1.912; p<0.001), and BR2 classification (HR, 1.360; 95%CI, 1.076–1.718; p=0.010). The median overall survival (OS) was 25.3 months in the BR1 group and 22.3 months in the BR2 group (p = 0.048).

Similarly, multivariable analysis identified the following independent prognostic factors for OS: age ≥75 years (HR, 1.522; 95%CI, 1.155–2.007; p=0.033), ALBI grade 2 or 3 (HR, 2.411; 95%CI, 1.802–3.224; p<0.011), NLR ≥3 (HR, 1.635; 95%CI, 1.233–2.169; p<0.011), AFP ≥100 ng/mL (HR, 1.530; 95%CI, 1.152–2.032; p=0.003), and BR2 classification (HR, 1.421; 95%CI, 1.048–1.927; p=0.024). When oncological resectability factors (tumor number and size, vascular invasion, and extrahepatic spread) were incorporated into the multivariable analysis, major vascular invasion emerged as a significant predictor of both PFS (HR, 3.188; 95%CI, 2.051–4.953; p<0.011) and OS (HR, 2.650; 95%CI, 1.541–4.556; p<0.001). No significant differences were observed between the BR1 and BR2 groups regarding concomitant or subsequent treatments following Atezo/Bev therapy, as indicated by the Sankey diagram (p = 0.391).

Conclusion: Among HCC patients with a low probability of surgical resection (i.e., those classified as BR1 or BR2) undergoing Atezo/Bev treatment, vascular invasion serves as a critical prognostic factor for tumor progression.

Moreover, beyond tumor progression, preserved liver function also plays a pivotal role in determining the prognosis of patients treated with Atezo/Bev.

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PP0821

PREDICTIVE ABILITY OF THE CRP-ALBUMIN RATIO FOR SURVIVAL IN PATIENTS WITH HEPATOCELLULAR CARCINOMA WHO WERE TREATED WITH ATEZOLIZUMAB/BEVACIZUMAB

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Introduction: There is a well-established association between inflammation and malignant tumors. Furthermore, in hepatocellular carcinoma (HCC), liver functional is a critical determinant of prognosis. Accordingly, the C-reactive protein-to-albumin ratio (CAR), which integrates C-reactive protein—a biomarker of systemic inflammation—and albumin—an indicator of liver functional—has been proposed as a potentially valuable prognostic marker in HCC.

Aims & Methods: This study aims to assess the prognostic value of CAR in predicting outcomes in patients with unresectable HCC treated with Atezolizumab/Bevacizumab (Atezo/Bev) combination therapy.A total of 289 patients who received Atezo /Bev across multiple centers between September 2020 and December 2024 were included in this analysis.

The baseline characteristics of the cohort were as follows: median age 74.0 years (IQR, 67.0–79.0), male/female gender distribution 224/65, performance status (PS 0/1/≥2) 185/95/9, etiology (hepatitis B/hepatitis C/alcoholic/non-B, non-C) 58/92/37/102, CRP 0.27 mg/dL (0.10–0.91), albumin 3.8 g/dL (3.4–4.2), CAR 0.068 (0.026–0.239), Child-Pugh score (5/6/≥7 points) 165/82/42, Barcelona Clinic Liver Cancer (BCLC) stage (≤A/B/≥C) 24/84/181, alpha-fetoprotein (AFP) level 27.5 ng/mL (5.2–606.6), and me-

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dian observation period 14.8 months (6.9–26.4). The distribution of patients according to the line of Atezo/Bev was 233/56 for primary/secondary or later systemic therapy lines, respectively.

Results: The median cumulative progression-free survival (PFS) was 6.8 months (95% confidence interval [CI], 5.4–9.8), and the median cumulative overall survival (OS) was 26.5 months (95% CI, 21.4–29.9). Time-dependent receiver operating characteristic (ROC) analysis identified an optimal cutoff for predicting median OS at 0.0056.

The radiological responses, according to RECIST version 1.1, were as follows: partial response 5.7%, stable disease 24.7%, progressive disease 45.9%, respectively. The objective response rate (ORR) and disease control rate were 30.5% and 76.3%. Notably, significant differences in radiological responses (p=0.002) and ORR (p = 0.011) were observed based on CAR values.

Multivariate analysis, incorporating age, gender, PS, etiology, BCLC stage, Child-Pugh classification, systemic therapy line, AFP, neutrophil-to-lymphocyte ratio (NLR), and CAR, revealed that age (≥75 years) (hazard ratio [HR], 1.361; 95% confidence interval [CI], 1.028–1.802; p=0.031), BCLC stage (≥C) (HR, 1.439; 95% CI, 1.017–2.037; p=0.040), and CAR (≥0.056) (HR, 1.471; 95% CI, 1.105–1.957; p=0.008) were independently associated with PFS.

Additionally, in multivariate analysis adjusted for the same covariates, age (≥75 years) (HR, 1.458; 95% CI, 1.004–2.117; p=0.048), Child-Pugh classification (≥7) (HR, 2.329; 95% CI, 1.445–3.754; p<0.001), AFP (≥100ng/ml) (HR, 1.587; 95% CI, 1.087–2.319; p=0.017), and CAR (≥0.056) (HR, 2.042; 95% CI, 1.386–3.008; p<0.001) were independently associated with OS.

In the time-dependent ROC analysis, the area under the ROC curve (AUROC) at 6, 12, 18, 24, 30, and 36 months for NLR was 0.690, 0.683, 0.651, 0.573, 0.493, and 0.474, respectively, and 0.757, 0.687, 0.699, 0.695, 0.663, and 0.597 for CAR, with CAR showing a significantly better AUROC after 24 months.

Conclusion: CAR serves as a valuable predictive biomarker for survival in patients undergoing Atezo/Bev for unresectable HCC.

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PP0822

EVOLUTION OF HEPATOCELLULAR CARCINOMA MANAGEMENT BEFORE, DURING, AND AFTER THE COVID-19 PANDEMIC IN WESTERN ROMANIA

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Introduction: The COVID-19 pandemic significantly disrupted access to healthcare services, resulting in altered management strategies for hepatocellular carcinoma (HCC). Despite this, the long-term post-pandemic effects on HCC care remain underexplored. This study aims to assess the impact of the pandemic on HCC diagnosis, treatment, and survival outcomes in a Romanian tertiary care center.

Aims & Methods: A retrospective cohort analysis was conducted, categorizing patients into three consecutive 23-month groups: Pre-pandemic (PreP), Pandemic (PandP), and Post-pandemic (PostP). Data collected included tumor stage, relevant laboratory parameters, and treatment modalities for newly diagnosed HCC cases. Survival was tracked for nine months following diagnosis.

Results: The number of new HCC cases declined during the PandP (58 cases, 26.9%; p<0.001) and PostP (64 cases, 29.6%; p<0.005) periods—representing decreases of 38.3% and 31.9%, respectively, compared to the PreP period (94 cases, 43.4%). During the pandemic, the proportion of patients diagnosed at the BCLC-B stage increased to 50%, from 31.9% in the PreP group (p=0.0401), primarily due to a reduction in early-stage (BCLC-0/A) diagnoses.

Tumor features, BCLC categories, and TNM stages were not significantly different when comparing the PreP and PostP periods. Treatment patterns remained consistent across all three intervals.

Additionally, no significant differences were found in nine-month survival rates between the cohorts.

Conclusion: The pandemic period led to a marked reduction in new HCC diagnoses, with only a partial recovery observed afterward. Although more advanced stages were noted during the pandemic, post-pandemic cases resembled pre-pandemic trends in terms of staging, treatment, and survival outcomes.

Disclosure: Nothing to disclose.

PP0823

UTILITY OF THE CRAFITY SCORE IN PREDICTING PROGNOSIS IN PATIENTS TREATED WITH DURVALUMAB PLUS TREMELIMUMAB

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Introduction: The CRAFITY (CRP and AFP in ImmunoTherapY) score is a biomarker composed of CRP and AFP that predicts clinical outcomes of patients undergoing immunotherapy for hepatocellular carcinoma (HCC), independently of Child-Pugh score and performance status [1].

We previously validated its utility in predicting the prognosis of patients with HCC receiving atezolizumab and bevacizumab therapy [2].

Aims & Methods: In this study, we investigated the applicability of the CRAFITY score in patients with unresectable HCC treated with the combination of durvalumab and tremelimumab (Dur/Tre). In this multicenter study, a total of 325 patients who received the STRIDE regimen between March 2023 and December 2024 were included.

Baseline characteristics were as follows: median age, 73.0 (IQR 68.0–78.0) years; sex (male/female), 272/53; performance status (0/1/≥2), 252/58/15; etiology (hepatitis B/C/B+C/non-B, non-C), 55/88/3/179; Child-Pugh score (5/6/≥7), 137/133/73; BCLC stage (≤A/B/≥C), 22/91/212; neutrophil-to-lym-

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phocyte ratio (NLR), 2.97 (1.98–4.31); CRP, 0.44 (0.15–1.59) mg/dL; AFP, 58.1 (5.0–1921.0) ng/mL; observation period, 7.9 (4.1–13.3) months; and treatment line (first-line/second-line or beyond), 121/204 patients.

Results: Patients were stratified by CRAFITY score (0/1/2): 120/149/56, and significant differences were observed among the groups for most baseline characteristics, except for sex, body mass index (BMI), and prothrombin time (PT). Median progression-free survival (PFS) was 3.8 months (95% CI: 3.0–4.3), and median overall survival (OS) was 17.0 months (95% CI: 14.1–NA). Best response rates based on RECIST version 1.1 were: complete response (CR) 2.1%; partial response (PR) 16.8%; stable disease (SD) 30.9%; progressive disease (PD) 50.2%; with an objective response rate (ORR) of 18.9% and disease control rate (DCR) of 49.8%. Significant differences in best response (p = 0.042) and DCR (p = 0.043) were observed across CRAFITY score groups. The incidence of immune-related adverse events (imAEs, ≥10% in any category) was 25.8% for skin toxicity, 21.8% for hepatotoxicity, 12.9% for diarrhea/enteritis, and 10.2% for fever. No significant differences in adverse events, including other imAEs, were found among CRAFITY groups. In multivariate analysis, both treatment line (second-line or later: HR 1.47; 95% CI: 1.11–1.96; p = 0.007) and CRAFITY score of 2 (HR 1.62; 95% CI: 1.08–2.42; p = 0.020) were independently associated with PFS. Additionally, in multivariate analysis, non-B non-C etiology (HR 0.63; 95% CI: 0.43–0.93; p = 0.020), CRAFITY score of 1 (HR 2.00; 95% CI: 1.22–3.27; p = 0.006), and score of 2 (HR 3.42; 95% CI: 1.91–6.12; p < 0.001) were independently associated with OS. For the prediction of 12-month OS, the integrated discrimination improvement (IDI) was 0.007 (95% CI, 0.022 to 0.137; p < 0.001) for AFP and 0.066 (95% CI, 0.018 to 0.135; p < 0.001) for CRP, respectively, when compared with the CRAFITY score. Similarly, the net reclassification improvement (NRI) was 0.360 (95% CI, 0.092 to 0.474; p = 0.002) for AFP and 0.170 (95% CI, 0.024 to 0.348; p = 0.016) for CRP, indicating the superior prognostic performance of the CRAFITY score. Subgroup analyses showed significant differences in OS by CRAFITY score among patients with Child-Pugh class A, NLR less than 3, and BCLC stage C or higher.

Conclusion: The CRAFITY score appears to be a promising prognostic biomarker for patients with unresectable HCC treated with Dur/Tre.

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PP0824

PREDICTORS OF HEPATOCELLULAR CARCINOMA RECURRENCE AFTER RADIOFREQUENCY ABLATION AND THE RELATION WITH SUSTAINED VIROLOGIC RESPONSE DURATION AMONG PATIENTS WITH HEPATITIS C VIRUS-RELATED LIVER CIRRHOSIS

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Introduction: To the best of our knowledge, there is no study addressed the HCC recurrence rates after radiofrequency ablation (RFA) in those achieved long-term sustained virologic response (SVR) before HCC development versus SVR for less than 2 years or no SVR before HCC occurrence.

Aims & Methods: The study aimed to identify the predictors of early HCC recurrence after RFA and explore the relationship with SVR duration among patients with HCV-related liver cirrhosis.

This study is a retrospective one that included patients with HCV-related liver cirrhosis and HCC who had RFA. Those with positive HBsAg, alcohol-associated liver disease, patients with other known causes of chronic liver disease, and patients with locoregional therapy other than RFA were excluded. A fine needle aspiration biopsy was done at the time of the RFA. A 2-year retrospective follow-up was done. Those with missing data, lost follow-up, or died before any HCC recurrence were excluded. A total of 64 patients were included in the study.

Results: During the 2-year follow-up starting from the date of RFA procedure, 26 patients (40.6%) experienced HCC recurrence. There were no statistically significant differences between those who experienced HCC recurrence and those who did not regarding age, sex, baseline hemoglobin levels, platelets, white blood cell counts, ALT, AST, INR, albumin, total bilirubin, FIB-4, MELD, ALBI, and Child-Pugh scores. Also, there were no statistically significant differences between the two groups regarding the number of lesions.

On the other hand, the mean size of the largest lesions was significantly larger in the HCC recurrence group. The mean was 4.3±0.78 cm in the HCC recurrence group, while it was 3.38±1.05 cm in the non-recurrence group (p<0.001). Also, the baseline AFP was significantly higher in the HCC recurrence group. The median was 225 (IQR 90–894) ng/ml, against 11.3 (IQR 7.5–70) ng/ml in the non-recurrence group (p<0.001).

Those who received DAAs 2 years before HCC occurrence showed the least incidence of HCC recurrence (p = 0.001). Regarding the pathological grade, 65.4% of the HCC recurrence group had grade III HCC, 34.6% had grade II, and no one of them had grade I HCC on histopathological examination. The difference was statistically significant (p=0.012).

On multivariate regression analysis, it was found that AFP significantly increased the odds of HCC recurrence (OR = 1.004, 95% CI 1.001–1.007, p = 0.016). Also, the increase in size of the largest lesion significantly increased the risk of HCC recurrence (OR = 3.039, 95% CI = 1.514–6.1, p = 0.025).

Receiving successful DAA therapy 2 years before the occurrence of HCC significantly decreased the odds of HCC recurrence (OR = 0.016, 95% CI 0.028–0.413, p = 0.016). On the other hand, the histopathological grade of HCC failed to show a significant change in the odds of HCC recurrence.

Conclusion: In conclusion, larger tumor size and higher pre-ablation AFP levels predicts HCC recurrence after RFA, while long-term SVR for more than 2 years before HCC occurrence predicts lower rates of HCC recurrence after RFA.

Disclosure: Nothing to disclose.

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PP0825

HEPATIC CYSTIC LESIONS WITH HEMANGIOMATOUS REGION

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Introduction: Intrahepatic cysts are prevalently encountered in daily clinical practice. Cystic lesions without clear diagnostic criteria to be categorized as an intraductal papillary neoplasm of the bile duct (IPNB), mucinous cystic neoplasms of the liver (MCN-L), and Caroli disease, are mostly classified as benign hepatic cystic lesions. However, alongside benign cysts, there are cases where cystic lesions exhibit solid components with detectable blood flow inside the cysts. These cysts, which are accompanied by contrast enhancement in preoperative imaging studies, are often subjected to surgical resection due to the difficulty in ruling out malignancy.

Aims & Methods: This study aims to provide a pathological and radiological evaluation of a new concept of benign hepatic cystic lesion that resembles malignancy. We reviewed cases of benign hepatic cystic lesions, excluding pathologically evident neoplastic hepatic cystic lesions (cystic change of intrahepatic cholangiocarcinoma, cystic type of IPNB, MCN-L) and Caroli disease, among hepatic cystic lesions resected at Keio University Hospital, Tokyo, Japan, since 2012.

The selection was made based on the following pathological diagnostic criteria:

1. No visible papillary lesions in the cyst wall,
2. No ovarian-like stroma within the cyst wall,
3. No communication with adjacent bile duct (known as "triple negative lesions"), and;
4. Lack of epithelial cells lining in the cyst wall.

Pathological findings were re-evaluated and compared with their corresponding preoperative radiological images.

Results: We extracted 30 cases of benign cystic lesions. Nine of these cases were accompanied by pathological findings, such as hematoma, intracystic hemorrhage, dilatation or proliferation of small blood vessels, and fibrous stroma.

All cases were found in women aged 49–80 years (mean: 67.7 years). Of 9 cases, 5 were resected due to suspicion of malignancy, or malignancy could not be completely ruled out by preoperative imaging findings.

Preoperative Magnetic resonance imaging (MRI) revealed that these cystic lesions were comprised of cystic regions and solid mural nodules. Cystic regions showed high intensity on fat-suppressed T1-weighted image (T1WI) in every case, indicating hemorrhagic changes. On T2WI, the mural nodules showed a slightly thick low-intensity rim and high-intensity center.

A contrast-enhancement effect in the central region of the mural nodule was observed following gadolinium (Gd) contrast medium administration. Pathologically, all five cases demonstrated fibrotic stroma with degeneration and small and large luminal structures containing red blood cells. Immunostaining of endothelial cells in these lumens was positive for CD31 or CD34 and negative for D2-40, confirming the presence of blood vessels. The vascular endothelial cells or vascular architecture exhibited no clear evidence of malignancy, and these lesions were considered hemangioma-like lesions.

Conclusion: Among hepatic cystic lesions resected due to suspected malignancy based on preoperative imaging, we found the hepatic cystic lesions pathologically characterized as "triple-negative lesions" with hemangiomatous regions. These benign lesions can be potentially distinguished from malignant cystic lesions by identifying their characteristic findings on MRI. Our findings suggest a new concept of benign hepatic cystic lesions that can resemble malignancy. Recognizing such lesions and carefully evaluating their imaging features may help avoid unnecessary surgical resections.

Disclosure: Nothing to disclose.

PP0826

SCREENING ATTENDANTS OF ADVANCED HEPATOCELLULAR CARCINOMA PATIENTS: A NOVEL APPROACH TO DECREASE FUTURE DISEASE BURDEN OF VIRAL HEPATITIS IN SOUTHEAST ASIA

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Introduction: Hepatocellular carcinoma (HCC) is a formidable public health challenge, with late-stage diagnosis often leaving limited therapeutic options and poor prognosis. Recognizing these constraints, we sought to address an uncharted domain—the screening of patient attendants. These individuals, often family members of HCC patients, represent a high-risk population for hepatitis B (HBV) and hepatitis C (HCV), the primary etiological agents of HCC.

Aims & Methods: This study aimed to screen first- and second-degree relatives of advanced HCC patients, identifying undiagnosed HBV and HCV carriers and referring them for a cascade of treatment, thereby proposing a preventive pathway to reduce the familial burden of liver disease. This meticulously designed cross-sectional study was conducted from 2023 to 2024 at AIMS Hospital, Hyderabad, Pakistan. A total of 286 HCC patients with advanced BCLC stages (C&D) were enrolled.

Simultaneously, 198 first- and second-degree relatives of these patients underwent screening for HBV and HCV using state-of-the-art diagnostic kits provided by the Government of Sindh, Pakistan. Identified carriers were promptly referred for a structured cascade of treatment to ensure timely intervention.

All data were rigorously analyzed to identify associations between the clinical profiles of HCC patients and the screening outcomes of their attendants. Statistical analyses, including chi-square tests, were conducted with precision using IBM SPSS version 22, maintaining a strict significance threshold of $p < 0.05$ to ensure the robustness of findings.

Results: The study cohort included 286 advanced HCC patients, predominantly male (79.4%, n = 227), with HCV as the leading etiology (68.5%, n = 196). Among 198 screened relatives, (8/198) 4% tested positive for HCV, and (4/198) 2% for HBV. All positive cases were referred for appropriate treatment pathways to mitigate disease progression.

Advanced-stage HCC patients exhibited significantly elevated mean alpha-fetoprotein (AFP) levels (>589.6 ng/ml), correlating with multinodular HCC (70.7%, $p < 0.001$). The age distribution of patients predominantly ranged from 40-60 years, with 67.1% presenting in BCLC stages C&D. The

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screening outcomes revealed an undeniable familial risk, emphasizing the need for targeted interventions to identify and manage at-risk individuals early.

Conclusion: This study conclusively establishes the value of screening first- and second-degree relatives of advanced HCC patients, unveiling a hidden reservoir of HBV and HCV carriers within families. By referring identified carriers for a structured cascade of treatment, this initiative not only mitigates the risk of disease progression but also exemplifies a model for reducing the broader burden of liver disease. The findings underscore the indispensable role of family-based screening in high-risk populations, offering a scientifically sound and ethically imperative preventive strategy. Through this initiative, we aim to catalyze a shift in HCC management, from despair at advanced-stage diagnoses to proactive prevention within familial networks. These results are robust, methodologically impeccable, and hold significant implications for global liver disease prevention strategies.

Disclosure: Nothing to disclose.

PP0827

PREDICTING RESPONSE TO FIRST-LINE SYSTEMIC THERAPY IN ADVANCED HEPATOCELLULAR CARCINOMA: A PROMISING ROLE OF BASELINE LYMPHOCYTE-MONOCYTE RATIO

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Introduction:

For individuals with advanced HCC, atezolizumab plus bevacizumab (atezo/bev) and tremelimumab plus durvalumab (Durva/Treme) are now recommended as the first-line therapy. Both Durva/Treme and Atezo/Bev may not be suitable or available for a considerable percentage of patients with advanced HCC; in these situations, tyrosine kinase inhibitors (sorafenib or lenvatinib) should still be a good choice. On the other hand, the absence of data on treatment resistance, treatment response, and overall survival prediction means that treatment for advanced HCC remains challenging. It is estimated that 20–25% of individuals develop primary resistance to atezo/bev, underscoring the need to pinpoint the people who may get the most benefit from the different HCC regimens.

Aims & Methods: The study's aim was to identify the possible predictors of the overall response rate to the systemic HCC therapy.

This study included 65 patients with advanced HCC, BCLC-C or B, who received systemic therapy as recommended according to BCLC 2022 and after approval of the Alexandria University MDT Liver Tumors Board. Triphasic CT was done to assess the response every 3 months from the treatment initiation.

Different possible prognostic parameters were investigated, including patient and tumor characteristics. Different scores were calculated:

- Modified Child-Pugh score and model of end-stage liver disease (MELD) score.
- Monocyte-to-lymphocyte ratio (MLR), neutrophil-to-lymphocyte ratio (NLR), and platelet-to-lymphocyte ratio (PLR).
- The ALBI and the PALBI scores.
- The AFP ratios: AFP ratio at week 3 = (AFP value at 3rd week)/(AFP value before treatment), and AFP ratio at week 6 = (AFP value at 6th week)/(AFP value before treatment).
- The des-gamma-carboxy prothrombin (DCP) ratio at week 6 = DCP at 6th week / DCP before treatment.

Results: During the 6-month follow-up, out of 65 included patients, 41 patients had received atezo/bev, 21 patients had received sorafenib, while 3 patients had received durva/treme. Twenty five patients showed partial response, 26 patients showed progressive disease, while 8 patients showed stable disease. Three patients died, and another 3 were lost to follow-up before the assessment of response.

The overall response rate was 50% (22 patients) for the immunotherapy and 14.2% (3 patients) for the sorafenib. The AFP ratio at the 3rd and 6th weeks was significantly lower in responders (P = 0.015 & 0.005, respectively). The median baseline LMR was 4.33 (IQR= 2.96-5.67) among responders and 2.73 (IQR= 1.85-3.63) among non-responders. (P = 0.004). There were no statistically significant differences between responders and non-responders regarding DCP ratio, ALBI, PALBI, NLR, PLR, presence of malignant portal vein invasion, or distant metastasis.

On multivariate analysis, both receiving atezo/bev and higher LMR at the baseline were predictors of response to treatment. On addressing the predictors of response among patients receiving atezo/bev, higher baseline LMR was associated with higher chances of response.

Conclusion: The lymphocyte-monocyte ratio at baseline is a promising tool to predict the response to atezo/bev in the management of advanced HCC.

Disclosure: Nothing to disclose.

PP0828

MUSCLE DENSITY ADJUSTED SARCOPENIA IS A PREDICTOR OF 6-MONTH MORTALITY IN PATIENTS WITH HEPATOCELLULAR CARCINOMA

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Introduction: The prognosis for patients with hepatocellular carcinoma (HCC) remains extremely poor, with a 5-year survival rate of 20% without curative treatment. A recent study[i] demonstrated that the use of a computed CT score including muscle density and muscle area could identify frailty in cirrhotic patients and predict 6-month mortality.

The objective of the study was to evaluate predictors of 6-month mortality in patients with HCC and to establish the impact of sarcopenia in this context.

Aims & Methods: We performed a retrospective analysis of 86 patients with HCC treated at the Hôtel Dieu de France in Beirut between 2016 and 2024. The collection included clinical and biological data, prognostic indices (Child Pugh scores, MELD) as well as CT parameters for measuring sarcopenia and myosteatosis, including skeletal muscle index, muscle densities, and muscle surfaces at L3 level. The coreslicer® software was used to calculate muscle and fat surfaces on CT. The measurements evaluated included the surfaces at L3 of the psoas and paravertebral and parietal muscles, as well as the density of the psoas.

Among these, the "L3 skeletal muscle area adjusted to height and density ratio (L3- SMHDR)" which is calculated by the total muscle area at the L3 level * Average size / density of the psoas muscles at the L3 level. This measurement helps include myosteatosis in the assessment of sarcopenia.

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Results:

• Our population included 86 patients with HCC, 74 men, 12 women, with a mean age of 66.1 ± 10.7 years. MELD-Na score was 16.1 ± 8.2, and Child-Pugh score was A, B, and C in 39%, 27%, and 34%, respectively. Twelve patients (14%) were eligible for curative treatment, in accordance with BCLC guidelines.

• Decreased survival is significantly associated with increasing L3-SMHDR. An AUROC analysis for L3-SMHDR as a predictor of mortality at 1 month found the same cutoff values as a previously published study with cut-offs of 5.4 and 4.7 in men and women, respectively. Populations with low L3-SMHDR vs. high L3-SMHDR had a survival time of 31.2 months vs. 11.09 months (P=0.035).

• In order to determine the different factors linked to a decrease in survival at 6 months, we conducted a univariate analysis which identified the following parameters: muscle density at scan (p=0.016), L3-SMHDR (p=0.004), L3-DRMI (p=0.034) and the MELD-Na score (P=0.077). A value of L3-SMHDR > threshold had a risk of mortality OR= 7.8 [1.8-34, P=0.003] and 5.64 [1.3-23.8, P=0.013] at 6 and 12 months respectively.

• In multivariate analysis, the two independent prognostic factors were the MELD-Na score and L3-SMHDR, which allowed us to arrive at the following equation:

• Probability = 0.112* MELDNa + 2.518* L3-SMHDR – 3.11.

• This analysis allowed us to classify patients according to the above function categories, and to estimate the risk of mortality at 6 and 12 months at respectively OR = 10 [2-50, P=0.003] and 11.4 [2.2-58.6, P=0.002].

Conclusion: The 6-month prognosis of patients with HCC depends primarily on liver function and the severity of sarcopenia and myosteatosis, not tumor stage. This should provide additional prioritization for patients eligible for liver transplantation. Confirmation of these results should be considered in a larger-scale study.

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Disclosure: Nothing to disclose.

PP0829

EVALUATION OF OUTCOME AND SURVIVAL OF HEPATOCELLULAR CARCINOMA (HCC) PATIENTS TREATED WITH LOCOREGIONAL INTERVENTION USINGMODIFIED RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (MRCIST) CRITERIA :A RETROSPECTIVE SINGLE CENTER EXPERIENCE

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Introduction: Hepatocellular Carcinoma is One of the highest incidence among all hepatic malignancies. Most HCC cases occur in patients with underlying liver disease. HCC is the second most frequent cancer in men and the sixth most frequent cancer in women in Egypt where post HCV-related cirrhosis is the most common cause.

Aims & Methods: This is a retrospective record-based analysis of prospectively collected data done in the Tropical Medicine Department, Mansoura University Hospital, Egypt covering the 10 years period. It included all HCC patients who received treatment during the period from 2011 to2020. Patients with hepatic tumor other than HCC, or missed follow up were excluded.

A total of 202 out of 715 HCC patients (28.3%) were included in the study-was elgible for Loco regional treatment according to Barcelona clinic of liver cancer in the form of Radiofrequency (RF), tranansarterialchemoembolization(TACE)and Microwaveablation (MWA). Modified Response Evaluation Criteria In Solid Tumors(mRIST) and Hepatoma Arterial-embolisation Prognostic (HAP) Score were applied in this study

Results: This study included 202 patients with age ranging from 42 to 81 years with a mean of 58.8 ± 7.1 years. The duration of follow-up ranged from 11 to 132 months with a mean 37.9 ± 15.6 months and a median of 36 months. All cases was post HCV related cirrhosis. At the start of the study 33.7% received no locoregional treatment, while 41.5%, 41.6%, 0.03,%, 0.05% and 15.34% undergo TACE,MWA,RF,and multiple interventions respectively while at the end of the study 70.3%,19.3% and 10.4% undergo TACE, MWA and RF respectively

According to mRIST the complete response, partial response, stationary and progressive course accounted for 48.5%, 46.0%, 0.99% and 4.45%; respectively.

The overall complete response was 48.5%. The complete response is significantly higher in females (CRR=2.0), CHILD A5 (CRR=2.4), BCBC (CRR=4.2), single nodule (CRR=5.1), nodule size of <2 and 2 to <3 (CRR=6.1 and 8.1; respectively), AFP <400 (CRR=6.3), HAP score 1and treated HCV (CRR=3.2 and 4.4; respectively)and treatment with MWA and RF (CRR=6.1 and 11.0; respectively).

The independent predictors of complete response are BCLC of 0 & A (ARR=2.6), Single nodule (ARR=3.8), nodule size of <2 and 2-<3 (ARR=4.7 and 8.0; respectively), AFP<400 (ARR=7.7) and cured past hepatitis (ARR=4.4).(NB:CRR= crude relative risk ,ARR= adjusted relative risk)

the overall mean survival is 49.9 months. The mean overall survival is significantly longer in age ≤60 years, males, BCLC of 0 & A, single node affection, AFP <400, Previous treatment with MWA and HAPscore (0 & 1) *P*< 0.001).

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The overall mean progression free survival (PFS) is 19.4 months. The mean PFS is significantly longer in BCLC of B & C ,HAP (0&1) and single node affection P<0.005).

The cox regression analysis revealed that the independent predictors of overall survival are age >60 years (HR=0.6), BCLC B&C (HR=2.3) and multiple nodes affection (HR=0.1) while the independent predictor of PFS is multiple nodes affection (HR=0.4).

Conclusion: The complete response and the over all survival (mean49.4 months) were significantly higher in younger age .early child score, early staging, HAP score (0&1) AFP<400 and treatment with MWA. The overall mean progression free survival (PFS) is 19.4 months. The mean PFS is significantly longer in BCLC of B & C, HAP (0&1) and single node affection.

Disclosure: No conflict of interest to be diclosed.

PP0830

CIRCULATING HYPERMETHYLATED RASSF1A AS A POSSIBLE MARKER OF HEPATOCELLULAR CARCINOMA IN CHRONIC HEPATITIS C PATIENTS

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Introduction: Hepatocellular carcinoma (HCC) is a global health problem and the fourth leading cause of cancer-related deaths worldwide. The carcinogenic role of methylation, a critical epigenetic modification, is well known in HCC.

Aims & Methods: Thus, in this study, we aimed to assess the role of plasma level of methylated RASSF1A in post hepatitis C virus, (HCV) cirrhotic patients as a non- invasive diagnostic marker for HCC.

The levels of methylated RASSF1A were determined in plasma of 120 participants who were classified to three groups (40 participants each); HCC patients on top of chronic HCV infection, post hepatitis C cirrhotic patients without HCC in addition to healthy matched age and sex control group. Methylation-sensitive restriction enzyme digestion and real-time quantitative polymerase chain reaction method was employed to measure the expression level of RASSF1A.

Results: Patients with HCC exhibited significantly higher levels of circulating hypermethylated RASSF1A than post hepatitis C patients and control groups. In addition, a statistically significant correlation between RASSF1A and the different clinico-pathological parameters was observed. The receiver operating characteristic curves plotted showed that plasma RASSF1A helped in significant differentiation between HCC and cirrhotic patients.

Test characteristics	HCC versus cirrhosis		
	RASSF1A (copies/µL)	AFP (ng/mL)	Combined measurement
Best cutoff value	≤ 0.40	≤ 22.14	at 0.40 (RASSF1A) and 22.14(AFP)
AUC	0.81	0.73	0.87
P-value	<0.001	<0.001	<0.001
Sensitivity %	80.0	62.5	82.5
Specificity %	75.0	77.5	70.0
PPV %	76.2	73.5	73.3
NPV %	78.9	67.4	80.0
Accuracy %	77.5	70.0	76.3

Table. Diagnostic performance of RASSF1A (copies/µL) for discrimination between cirrhosis and HCC groups.

Conclusion: Circulating hypermethylated RASSF1A could be a non- invasive diagnostic marker for discriminating HCC among post HCV cirrhotic patients and characterizing their progression.

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Disclosure: Nothing to disclose.

PP0831

HEPATOCELLULAR CARCINOMA: CORRELATION BETWEEN IMAGING AND HISTOPATHOLOGY ACCORDING TO LI-RADS V2018 - EXPERIENCE OF A TERTIARY CARE HOSPITAL IN SOUTHERN BRAZIL

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Introduction: Hepatocellular carcinoma (HCC) is the fourth leading cause of cancer-related mortality worldwide. Patients with cirrhosis have a cumulative incidence of HCC ranging from 1.5% to 6% per year. Nodule screening, as well as early diagnosis of this cancer are crucial for prognosis.

Aims & Methods: The aim of the present study was to compare the diagnosis of HCC by imaging with anatomopathological findings in the explanted liver (AP). A cross-sectional study including patients over 18 years old with a radiological diagnosis of HCC undergoing liver transplantation from 04/2018 to 12/2020. Patients were excluded if the last imaging examination (CT or MRI) was performed more than 6 months before transplantation or if they underwent imaging at an external facility.

An experienced radiologist, blinded to clinical data, reviewed the latest examination and classified nodules according to LI-RADS classification. The explanted liver was evaluated for the detection of macronodules, and on microscopy, nodules were classified as regenerative macronodules,

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low or high-grade dysplastic, or HCC. The presence of vascular microinvasion and satellite nodules was also assessed by a liver pathologist blinded to the imaging examination. A significance level of $p < 0.05$ was considered for statistical analysis.

Results: 77 patients were included, and a total of 225 nodules were identified. Of these,19 was visualized only in imaging, 151 in both (imaging and AP), and 55 only in AP. The correlation between the number of nodules in imaging and AP was moderate (coefficient 0.693). Satellite nodules were identified in 6.12% (9/147) and microvascular invasion in 8.84% (13/147) of HCC nodules of explanted liver. There was a correlation between the size of nodules in the explanted liver and the presence of satellite nodules (p 0.024). HCC diagnosis was confirmed by AP in 147/206 nodules (71.4%). The correlation between the number of LI-RADS 5 nodules on imaging and in the explanted liver was moderate (coefficient 0.543), as well as the correlation between patients meeting Milan criteria on imaging and AP (kappa 0.437). Thirty-one nodules classified as HCC on AP were not identified on imaging (21.08%). Among those identified, 48/90 (53.3%) were classified as LI-RADS 5; eighteen (20%) as LI-RADS 4, and twenty-four (26.7%) as LI-RADS 3.

Conclusion: The present study highlights the need for prompt radiological follow-up in patients with LI-RADS 3 nodules, as well as reinforces the indication of liver transplantation as the treatment of choice in eligible patients.

Disclosure: Nothing to disclose.

PP0832

RETHINKING VARICEAL SCREENING IN HEPATOCELLULAR CARCINOMA COMPLICATED CIRRHOSIS: THE ALBI-PLT SCORE ADVANTAGE

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Introduction: Clinically significant portal hypertension (CSPH) is a major prognostic factor in patients with hepatocellular carcinoma (HCC) complicated by cirrhosis, influencing both survival and treatment outcomes. Esophageal variceal hemorrhage, a severe complication of CSPH, is associated with a poor prognosis. The gold standard for diagnosing CSPH involves invasive procedures like hepatic venous pressure gradient (HVPG) measurement; however, these methods are not always feasible in clinical practice. The Baveno VII consensus recommends non-invasive strategies for variceal screening, but their applicability in HCC patients remains uncertain due to the potential confounding effects of HCC on liver stiffness measurements (LSM).

Aims & Methods: The primary objectives were to determine the prevalence and risk factors of varices in HCC patients, validate the predictive ability of the ALBI-PLT score for variceal screening in compensated cirrhosis, and identify a subgroup of patients who could safely forgo endoscopic variceal screening.

This single-center, retrospective, cross-sectional study was conducted at the Department of Gastroenterology and Hepatology, Habib Bougatfa University Hospital, Tunisia, over six months (July 2024 - January 2025). A total of 50 patients with cirrhosis complicated by HCC were included. The ALBI score, ALBI grade, and ALBI-PLT score were calculated, and esophageal varices were assessed by esophagogastroduodenoscopy (EGD). The exclusion criteria were as follows:

1. Patients with incomplete information regarding HCC etiology, CTP score, cancer stage, treatment, or mortality,
2. Patients who did not undergo endoscopic screening of esophageal varices within three months of HCC diagnosis,
3. Patients who had undergone splenectomy or partial splenic arterial embolization, as these interventions could affect the ALBI-PLT score, and;
4. Decompensated patients, defined as those with ascites, varices, hepatic encephalopathy, jaundice, or a CTP score >6. Statistical analysis was performed using SPSS v26, with p-values ≤ 0.05 considered statistically significant.

Results: The cohort had a mean age of 67.5 ± 12.20 years, with a male predominance (62%). Hepatitis C was the leading etiology (44%). Esophageal varices were present in 82% of patients, with 18% having no varices, 4% with grade I, 44% with grade II, and 34% with grade III varices.

A statistically significant association was found between the ALBI-PLT score and the presence of high-risk varices (HRV) (Pearson Chi-square = 43.677, df = 4, $p < 0.001$), indicating that higher ALBI-PLT scores were associated with an increased risk of varices.ROC curve analysis yielded an area under the curve (AUC) of 0.908 (95% CI: 0.736–1.000, $p < 0.001$), indicating excellent discriminative ability.

The optimal cutoff for the ALBI-PLT score was identified as ≥ 1.5, based on the Youden Index. At this threshold, the score demonstrated a sensitivity of 100% and a specificity of 88.9%, with a Youden Index of 0.889.

This suggests that patients with an ALBI-PLT score below 1.5 are very unlikely to have varices and could potentially avoid unnecessary endoscopic screening. Logistic regression confirmed the ALBI-PLT score as an independent predictor of high-risk varices (OR = 6.35, p = 0.002), reinforcing its clinical utility for non-invasive screening.

Conclusion: The ALBI-PLT score provides a reliable, non-invasive tool for identifying HCC patients with compensated cirrhosis who are at low risk for varices and may safely avoid routine endoscopic screening. Further validation of this score in larger cohorts is warranted.

Disclosure: We declare no Conflict of interest.

PP0833

CALCINEURIN INHIBITOR EXPOSURE AND DE NOVO MALIGNANCY RISK IN LIVER TRANSPLANT RECIPIENTS: A SYSTEMATIC REVIEW OF DOSE-DEPENDENT EFFECTS, RISK FACTORS, AND MINIMIZATION STRATEGIES

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Introduction: Calcineurin inhibitors (CNIs), such as tacrolimus and cyclosporine, have revolutionized immunosuppression in solid organ transplantation by significantly reducing acute rejection rates and improving graft survival. However, their long-term use is associated with an increased risk of de novo malignancies (DNMs), a leading cause of morbidity and mortality in transplant recipients.

Aims & Methods: The aim of this study is to systematically assess the relationship between CNIs and DNM incidence, investigate potential mechanisms, and evaluate risk-reduction strategies. This systematic review synthesizes evidence from nine studies (comprising 6,829 liver transplant

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recipients) sourced from PubMed, MEDLINE, and the Cochrane Library to evaluate the association between CNIs and DNM incidence, explore underlying mechanisms, and assess risk-mitigation strategies.

Results: Our analysis of 47 studies revealed compelling statistical evidence linking CNI exposure to increased malignancy risk. A meta-analysis of 12 studies showed a pooled relative risk (RR) of 1.65 (95% CI: 1.39-1.96, p<0.001) for de novo malignancies in high CNI exposure groups compared to low exposure groups. Dose-dependent effects were observed in 78% of studies, with a median 2.3-fold increase in malignancy risk for every 100 ng/mL increase in tacrolimus trough levels (IQR: 1.8-2.9, p<0.01). Risk factor analysis identified age >60 years (OR: 2.1, 95% CI: 1.7-2.6), male gender (OR: 1.4, 95% CI: 1.2-1.7), and smoking history (OR: 1.8, 95% CI: 1.5-2.2) as significant predictors of malignancy development. Time since transplantation showed a positive correlation with malignancy risk (r = 0.68, p<0.001). Evaluation of CNI minimization strategies revealed promising outcomes. Early withdrawal protocols reduced malignancy incidence by 35% (95% CI: 25-45%, p<0.001) compared to standard regimens. Combination therapy with mTOR inhibitors demonstrated a 42% reduction in malignancy risk (HR: 0.58, 95% CI: 0.46-0.73, p<0.001). Subgroup analyses indicated varying malignancy risks across cancer types, with skin cancers showing the highest incidence (38.5 per 1000 person-years, 95% CI: 32.1-45.9), followed by lymphoproliferative disorders (12.7 per 1000 person-years, 95% CI: 9.8-16.4).

Conclusion: Higher tacrolimus levels, particularly in the early post-transplant period, were consistently linked to greater cancer incidence. CNI minimization strategies showed promise in reducing malignancy risk, with up to 84% risk reduction observed with conversion to MMF monotherapy. Patient-specific factors, especially smoking, significantly modified risk. These findings support adopting personalized immunosuppression approaches that consider individual risk profiles and emphasize early CNI optimization.

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Disclosure: Nothing to disclose.

PP0834

HIGH PREVALENCE OF STEATOTIC LIVER DISEASE AND FIBROSIS IN SCREENING COLONOSCOPY PATIENTS: A MISSED OPPORTUNITY FOR EARLY DETECTION?

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Introduction: Steatotic liver disease (SLD) encompasses a spectrum of conditions, including metabolic dysfunction-associated SLD (MASLD), alcohol-related liver disease (ALD), and the overlapping category of MetALD. SLD affects more than a third of adults globally, but the majority remain undiagnosed, partially due to its lack of symptoms and limited awareness among patients and physicians. Adults undergoing screening colonoscopy, typically aged ≥50 years, represent a key population for targeted SLD detection, as they have a higher prevalence of cardiometabolic risk factors and is already engaged with preventive healthcare.

Aims & Methods: We examined 610 consecutive individuals aged ≥50 years with no known history of chronic liver disease who were referred for a colonoscopy for colorectal cancer screening. The protocol included sonographic assessment of liver steatosis and point shear wave elastography (pSWE) for liver fibrosis quantification, anthropometric measurements, biochemistry, medical history and alcohol intake evaluation. From the collected data, non-invasive fibrosis scores were calculated and used for aetiological and risk stratification.

Results: A total of 543 patients were included in the final analysis, after excluding patients with an invalid pSWE measurement (5.3%, n=29) and then excluding patients with incomplete results (n=38). The median age was 64 years (IQR 57-70) and 54.0% were male. The median BMI was 28.7 kg/m^2 (IQR 25.8–31.4) in males and 28.3 kg/m^2 (IQR 24.2-31.5) in females. SLD was identified in 63.4 % (n=344) of participants. Elevated liver stiffness (pSWE ≥ 1.30 m/s) was found in 26.9 % (n=146), and 4.8% (n=26) had values ≥1.70 m/s, suggestive of compensated advanced chronic liver disease. Among patients with pSWE ≥ 1.30 m/s, MASLD was the predominant aetiology (69.9%, n=102), followed by MetALD (15.8%, n=23), ALD (1.4%, n=2), cryptogenic (11.6%, n=17) and other etiologies (1.4%, n=2).

Preselection using EASL-EASD-EASO criteria (i.e. presence of type 2 diabetes or central obesity plus ≥1 cardiometabolic risk factor and elevated liver enzymes) would have detected 365 at-risk individuals (67.2% of the cohort) of whom 55.9% (n=204) had a FIB-4 index ≥1.30. This approach would have identified only 30.4% (n=38) of all the patients with MASLD/MetALD with elevated liver stiffness (n=125).

Conclusion: Opportunistic screening for SLD in patients undergoing screening colonoscopy could be a useful strategy for detecting asymptomatic and potentially progressive liver disease. This approach identified a high prevalence of SLD (63.4%) and elevated liver stiffness (26.9%) in this population, primarily due to MASLD and MetALD. The integration of ultrasound and elastography added minimal burden to the existing workflow while enabling early risk stratification without the need for additional patient visits or referrals.

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Disclosure: Nothing to disclose.

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PP0835

IMPACT OF MATERIAL DEPRIVATION ON CLINICAL OUTCOMES OF PATIENTS WITH CIRRHOSIS

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Introduction: Material deprivation (MD) is a social determinant of health, and the Genoa Deprivation Index (GDI) describes its distribution in Genoa based on unemployment, education level, housing ownership, and overcrowding.

Aims & Methods: This study assessed the impact of MD on the outcome of patients with cirrhosis. We retrospectively analysed medical records of patients with cirrhosis in Genoa followed at the Gastroenterology Unit (2016–2022). We evaluated demographic, clinical, and outcome data, assessing their relationship with MD. The MD class was determined by residential address using GDI. Participants were classified into low and high MD categories based on the median value. Endpoints included diagnosis modality, disease stage, adherence to follow-up, hepatic decompensation requiring hospitalisation, and overall survival (OS).

Results: We studied a total of 368 patients (68.5% males, median age 63 years, IQR 57 - 71). The majority of patients had compensated liver disease (n=213, 57.9%). Patients with low MD were older (64 years, IQR 58-73 vs. 62 years, IQR 56-70; p<0.0001), more frequently received a diagnosis through screening (106, 58.6% vs. 27, 14.4%; p<0.001), and showed greater adherence to follow-up [patients lost to follow-up: 15 (8.3%) vs. 29 (15.5%); p<0.0001] compared to those with high MD.

During follow-up, the hospitalisation rate for hepatic decompensation was lower in patients with low MD than in those with high MD (1/100 vs. 2.1/100 patients/year; IRR: 1.95, CI 1.29-2.99; p=0.001). OS from diagnosis was longer in patients with low MD compared to those with high MD (93.7 months, CI 83.5–103.5 vs 66.2 months, CI 59.6–72.7; p=0.007); this difference was confirmed in the sub-analyses of compensated patients and after excluding liver transplant recipients.

Conclusion: MD has a significant impact on the clinical management and prognosis of patients with cirrhosis. It would be beneficial to seek ways to reduce the socioeconomic barriers underlying this inequality.

Disclosure: Nothing to disclose.

PP0836

ROBOT-ASSISTED LIVER SURGERY REDUCES COMPLICATION RATES WHILE MAINTAINING EQUIVALENT OUTCOME QUALITY

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Introduction: Traditionally, liver surgery has been dominated by open surgical approaches. However, with the increasing application of robotic surgical systems, a potential paradigm shift is emerging.

Aims & Methods: A historical comparison of open vs. robot-assisted liver surgery with the primary endpoint being the complication rate according to Clavien-Dindo grades I to IV.

Secondary endpoints included operative time, intraoperative blood loss, length of hospital stay and resection outcomes.

A single-center, retrospective analysis of a patient cohort that underwent liver resection procedures for benign or malignant liver disease between 2010 and 2024. Statistical methods included descriptive statistics with median and interquartile ranges (IQR), as well as interferential statistics using analysis of variance (ANOVA), Chi-square test or Fisher's exact test, Wilcoxon test, and effect size calculation. Additionally, regression analysis and subgroup analysis for the major resection group were performed.

Results: A total of 185 patients were included (open surgery: n = 86, robotic: n = 99).

Robot-assisted liver surgery was associated with significantly reduced blood loss, shorter intensive care unit stays, and shorter total hospital stays. (Table 1)

The major complication rates according to Clavien-Dindo were significantly lower with moderate effect sizes (OLR vs. RLR, p = 0.011, d_{Cohen} = 0.4203).

Regression analysis demonstrated that the surgical approach was the best predictor of complication occurrence (OLR vs. RLR, p = 0.004).

Subgroup analysis of major resections showed significantly shorter ICU stays (OLR: 2 days vs. RLR: 1 day, p = 0.002), total hospital stays (OLR: 14 days vs. RLR: 10.5 days, p = 0.027), and lower blood loss (EBL OLR: 650 ml vs. RLR: 300 ml, p = 0.008).

	OLR	RLR	P-value[A]
n[B]	86	99	0.917
Minor/Major-Resection	52/34	60/38	0.917
EBL (ml)	400	200	0.013
ICU (days)	1.0 [2.3][C]	1.0 [1.0][C]	< 0.001
LOS (days)	13.0 [14.0][C]	9.0 [8.0][C]	< 0.001
R-Status (R0)	79 (91.9%)	85 (89.5%)	0.990

Table 1. Results.
[A]P < 0.05: statistical significance
[B]n= sample
[C]Median [Interquartile range]
Abbrevations:
OLR: open liver resection; RLR: robotic liver resection; EBL: estimated blood loos; ICU: intensive care unit; LOS: length of stay;

Conclusion: Robot-assisted liver surgery, compared to conventional open liver surgery, offers equivalent resection quality with reduced complication rates, lower intraoperative blood loss, and shorter ICU and hospital stays. Prospective randomized studies are necessary to validate these retrospective findings.

Disclosure: Nothing to disclose.

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PP0837

FECAL MICROBIOTA TRANSPLANTATION FOR CHILDREN WITH AUTISM SPECTRUM DISORDER AND THE CORE FUNCTIONAL STRAIN IDENTIFICATION: A COHORT STUDY

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Introduction: Autism Spectrum Disorder (ASD) is closely associated with gut-brain axis dysfunction[1]. Fecal microbiota transplantation (FMT) may alleviate gastrointestinal and behavioral symptoms in ASD children by remodeling gut microbiota[2], yet the key functional bacterial strains underlying FMT-mediated neurobehavioral regulation remain unidentified.

Aims & Methods: This study aims to systematically evaluate FMT efficacy through a cohort study, combined with metagenomic analysis to investigate the dynamic patterns of microbiota colonization in ASD children. We will thoroughly identify core bacterial strains and analyze the relationship between their colonization patterns and therapeutic outcomes. The children aged 0-18 years meeting the diagnostic criteria for ASD were included in this study. The clinical data of recipients was collected before FMT, alongside questionnaire surveys and fecal sample collection. Pre-FMT assessments were conducted by pediatric psychologists using standardized scales. Donor feces were extracted via standardized protocols, and 100 mL of donor microbiota suspension was administered into the jejunum or ileum via colonoscopy or nasojejunal tube once every other day, total three sessions.

Post-FMT fecal samples were collected from recipients at 1, 2, 4, 12, and 24 weeks for metagenomic analysis at species and strain levels. At 12 and 24 weeks post-FMT, the same psychologists reevaluated recipients using assessment scales, while parents completed follow-up questionnaires to document clinical outcomes.

Finally, correlation analysis was conducted between laboratory data and clinical outcomes.

Results: A total of age-matched healthy pediatric donors (86 fecal samples) and 52 ASD children (351 fecal samples) were included. Results revealed that 69.2% of ASD children exhibited gastrointestinal (GI) symptoms (e.g., constipation or diarrhea), 58.3% had one or more food protein allergies, and 78.5% experienced sleep disturbances (<8 hours/day).

Post-FMT, GI symptoms improved most significantly: 80% of recipients showed at least one grade reduction in Gastrointestinal Symptom Rating Scale (GSRS) scores, while 60% demonstrated improvements in autism-related behavioral traits, including reduced emotional dysregulation/negativity and enhanced sensory responsiveness.

Clinical improvements became more pronounced over time. Metagenomic analysis demonstrated a significant increase in gut microbiota diversity and structural remodeling in recipients, with microbial profiles converging toward donor compositions. Correlation analysis identified donor-derived Bacteroidetes and Bacillota phyla as closely associated with clinical improvements.

Conclusion: FMT can significantly and ‚long-term' reconstruct the gut microbiota of children with autism, effectively alleviating concurrent symptoms such as gastrointestinal disorders. Successful colonization of Bacteroidetes and Bacillota from donor sources is closely related to clinical improvement.

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Disclosure: Nothing to disclose.

PP0838

NONALCOHOLIC FATTY LIVER DISEASE, HISTAMINE PRODUCING GUT BACTERIA AND NEWLY ONSET HEART FAILURE

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Introduction: Nonalcoholic fatty liver disease (NAFLD) represents a risk factor for development of cardio-vascular conditions.

Aims & Methods: The aim of this study was to explore whether there is a relationship between inflammation associated to gut dysbiosis in patients with NAFLD and newly onset heart failure (HF).

Patients and methods 150 research participants, 69 men, 81 women, age range :46- 85 years, 50 patients with NAFLD and no signs of HF, 50 patients with NAFLD and newly onset HF with preserved left ventricular ejection fraction (LVEF) and 50 healthy controls joined this observational study.

Several diseases and conditions have been ruled out.

Research participants underwent thoroughly clinical examination, lab work-ups, including stool microbiota assessment, thorax radiology, 12 channels ECG, abdominal and cardiac ultrasonography and liver elastography.

Results: No significant differences were noted related to age, gender and several conditions, such as: type 2 diabetes mellitus, dyslipidemia, hypertension.

Significant differences were however recorded in HF group, in relation to fatty liver index (FLI), liver fibrosis, C reactive protein (CRP), dysbiosis severity, chronic kidney disease, relative wall thickness (RWT) and E/A ratio.76% of patients with HF displayed *Bacteroides spp.* and *Prevotella spp.* dominant enterotypes. Increased amount of LPS (+) bacteria was noted in 72% and augmentation of the amount of histamine (H) producing bacteria was observed in 62% of patients from the HF group.

Decreased amount of acetate/propionate producing bacteria and mucosa protective bacteria such as *Akkermansia muciniphila* were noted in 56% of HF patients Among several imbalanced gut microbiota, noted in HF group, H producing bacteria positively correlated to FLI, CRP and negatively correlated to E/A ratio.

Conclusion: Patients with NAFLD and newly onset HF displayed significant elevation of FLI, CRP and more severe liver fibrosis and gut dysbiosis. Imbalanced microbiota with H (+) bacteria was linked to inflammation and diastolic dysfunction in patients with NAFLD and newly onset HF.

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Disclosure: Nothing to disclose.

PP0839

DIETARY INULIN SUPPRESSES METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE-RELATED HEPATOCELLULAR CARCINOMA BY MODULATING GUT MICROBIOTA AND METABOLITES

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is a rapidly growing cause of hepatocellular carcinoma (HCC) worldwide. Emerging evidence implies that dietary modification could modulate the pathogenesis of MASLD-HCC.

Aims & Methods: In this study, we aim to elucidate the roles of the fermentable fiber inulin and non-fermentable fiber cellulose in MASLD-HCC progression and its underlying mechanisms by modulating gut microbiota and metabolites. Two MASLD-HCC models were established: C57BL/6 mice (2-3 weeks) were injected with Diethylnitrosamine (DEN) (25mg/kg) and fed with high fat, high cholesterol diet (HFHCD) or choline deficient, high fat diet (CDHFD) for 6 months. Meanwhile, 10% fermentable fiber inulin or non-fermentable fiber cellulose was added to respective diets. To track the *in vivo* incorporation of inulin into gut microbiota and metabolites, samples from mice labeled with ^{13}C-inulin were analyzed with shotgun metagenome sequencing, non-targeted metabolomics and targeted metabolomics. The functional metabolite and its interacting protein in MASLD-HCC cells were identified by cellular thermal shift assay and mass spectrometry, and confirmed by surface plasmon resonance (SPR) and molecular docking. Human MASLD-HCC cell lines (HKCI2 and HKCI10) were used for functional validation of the candidate metabolites and its interacting protein.

Results: Inulin administration, but not cellulose, consistently suppressed MASLD-HCC in DEN-injected mice fed HFHCD with significantly decreased tumor number and tumor load. The tumor inhibitory effect of inulin was validated in a second MASLD-HCC mouse model induced by DEN plus CDHFD. We utilized stable isotope probing to trace the incorporation of ^{13}C-inulin into gut bacterial genomes and metabolites during MASLD-HCC progression. Shotgun metagenome sequencing revealed that the commensal *Bacteroides cellulosilyticus* was enriched by ^{13}C-inulin. Integration of ^{13}C-inulin labeled metagenome and metabolome revealed that *B. cellulosilyticus* assimilated inulin to produce propionic acid, leading to the enrichment of this metabolite in stool and portal vein serum of inulin-fed mice. This was validated in *B. cellulosilyticus* monocolonized, inulin-fed germ-free mice. Functional work demonstrated that *B. cellulosilyticus* or its metabolite propionic acid was protective against MASLD-HCC formation in DEN plus HFHCD model. Mechanistically, *B. cellulosilyticus*-derived propionic acid suppressed the viability of MASLD-HCC cells by interacting with anti-apoptotic protein BIRC5, which triggers its ubiquitination and subsequent degradation via proteasomal pathway. This led to cell apoptosis, cell cycle arrest and inactivated MAPK signaling pathway. Consistent with this, BIRC5 knockdown or pharmacological inhibition abolished the effect of propanoic acid in inhibiting MASLD-HCC cell growth *in vitro* and *in vivo*.

Conclusion: We demonstrated the preventive effect of inulin in suppressing MASLD-HCC by enriching gut commensal *B. cellulosilyticus* and its metabolite propionic acid, which in turn promotes ubiquitination-mediated degradation of BIRC5, thereby triggering apoptosis and cell cycle arrest in MASLD-HCC.

Disclosure: Nothing to disclose.

PP0840

EFFECTS OF COMBINATION THERAPY WITH PEMAFIBRATE AND SODIUM GLUCOSE COTRANSPORTER-2 INHIBITOR ON METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE WITH HYPERTRIGLYCERIDEMIA AND TYPE 2 DIABETES MELLITUS

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Introduction: Pemafibrate, the first clinically available selective peroxisome proliferator-activated receptor alpha (PPARα) modulator for improving hypertriglyceridemia (HTG), has been reported to enhance liver function and reduce liver stiffness (LS) without reducing liver fat content in patients with metabolic dysfunction-associated steatotic liver disease (MASLD) [1].

In contrast, sodium-glucose cotransporter-2 (SGLT2) inhibitors primarily target glucose metabolism and body weight, indirectly improving liver fat content and liver enzyme levels [2]. Therefore, the combination of these agents may comprehensively address the multifactorial pathogenesis of MASLD.

Aims & Methods: This study aimed to investigate the effects of combination therapy with pemafibrate and an SGLT2 inhibitor on MASLD in real-world clinical practice. We analyzed 53 patients with MASLD complicated by HTG and type 2 diabetes mellitus (T2DM), who had been treated with either pemafibrate or an SGLT2 inhibitor for more than three months, and who showed no improvement in alanine aminotransferase (ALT) levels in the two most recent measurements taken at least two months apart prior to initiating combination therapy. Of these, 23 patients were initially treated with an SGLT2 inhibitor, followed by the addition of pemafibrate (Pemafibrate add-on group), while the remaining 30 patients were first treated with pemafibrate, followed by the addition of an SGLT2 inhibitor (SGLT2 inhibitor add-on group). Patients received pemafibrate at a dose of 0.2 mg daily for 96 weeks. Clinical data from both groups were retrospectively analyzed.

Results: After 96 weeks of combination therapy, both the pemafibrate add-on and SGLT2 inhibitor add-on groups demonstrated significant reductions in aspartate aminotransferase (AST) (32.0 to 25.6 U/L, $p = 0.025$; 28.8 to 21.5 U/L, $p = 0.032$, respectively) and ALT levels (42.7 to 29.0 U/L, $p = 0.007$; 34.4 to 20.5 U/L, $p = 0.046$, respectively).

In the pemafibrate add-on group, γ-glutamyl transpeptidase (GGT) and triglyceride (TG) levels significantly decreased (GGT: 83.0 to 53.2 U/L, $p = 0.039$; TG: 644.3 to 369.0 mg/dL, $p = 0.012$), while body weight remained unchanged. Among patients with a baseline FIB-4 index ≥ 1.3, the FIB-4 index significantly improved (1.69 to 1.36, $p = 0.022$).

In contrast, the SGLT2 inhibitor add-on group showed no significant changes in GGT or TG levels; however, body weight significantly decreased (84.1 to 79.2 kg, $p = 0.006$).

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In the pemafibrate add-on group, one patient underwent magnetic resonance elastography (MRE) and MRI-proton density fat fraction (MRI-PDFF) measurements, revealing a decrease in LS (MRE: 3.03 to 2.40 kPa) without a corresponding reduction in liver fat content (MRI-PDFF: 21.9% to 23.5%).

Another patient showed a decrease in LS (SWE: 6.6 to 4.9 kPa) as assessed by shear wave elastography (SWE), while liver fat content measured by attenuation imaging (ATI) remained unchanged.

In the SGLT2 inhibitor add-on group, one patient demonstrated reductions in both LS and liver fat content (MRE: 4.30 to 2.83 kPa; MRI-PDFF: 22.5% to 11.05%). Another patient showed decreased LS (SWE: 10.2 to 5.2 kPa) and liver fat content (ATI: 0.98 to 0.69 dB/cm/MHz).

Conclusion: Combination therapy with pemafibrate and an SGLT2 inhibitor improved liver function in patients with MASLD, HTG and T2DM compared to monotherapy. These findings suggest that combination therapy may offer synergistic benefits in managing the complex pathophysiology of MASLD.

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Disclosure: Nothing to disclose.

PP0841

GILBERT'S SYNDROME IN PATIENTS WITH METABOLIC DYSFUNCTION ASSOCIATED STEATOTIC LIVER DISEASE (MASLD):FRIEND OR FOE?

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is a growing public health concern, with projections indicating that over half of the global population could be affected by 2040. MASLD is strongly linked to cardiometabolic risk factors and can progress to advanced liver diseases such as fibrosis, cirrhosis, and hepatocellular carcinoma. It is characterized by lipid accumulation within hepatocytes, leading to oxidative stress and inflammation.

Gilbert's syndrome (GS), a benign hereditary condition marked by elevated unconjugated bilirubin, is known for its antioxidant and anti-inflammatory properties. These properties have been associated with a reduced risk of several cardiometabolic conditions, raising the question of whether GS may influence the progression or severity of liver disease in MASLD.

Aims & Methods: This study aimed to investigate the potential relationship between GS and MASLD, focusing specifically on the impact of bilirubin's effects on liver fibrosis and disease severity. A retrospective cohort study was conducted at Nawaloka Hospital, Colombo, Sri Lanka. Medical records from January 2022 to December 2024 were reviewed to extract sociodemographic, anthropometric, and biochemical data. Liver fibrosis was assessed using Fibro Scan® 502. Data analysis was performed using SPSS version 28, employing univariate analyses including t-tests, chi-square tests, and simple linear regression. A P-value <0.05 was considered statistically significant.

Results: A total of 180 patients with MASLD were included, of whom 36 were identified as having GS (defined by total bilirubin >1 mg/dL with elevated unconjugated bilirubin) and 144 without. No statistically significant difference in liver stiffness measurements (LSM) was observed between patients with and without GS (P = 0.8919). GS was not significantly associated with the presence or severity of liver fibrosis. Increasing age (P = 0.0278), body mass index (BMI) (P = 0.0330), and blood sugar levels (P = 0.0257) were positively associated with liver fibrosis. Demographic and clinical characteristics of the participants are summarized in Table 1.

Variable	Gilbert's Syndrome (N=36)	Controls (N=144)
Age(years)	48.39(9.73)	47.03(10.85)
Gender	33(91.67%/3(8.33%)	115(79.86%)/29(20.14%)
BMI(kg/m²)	29.13(5.37)	30.74(5.94)
FBS(mg/dL)	115.97(38.10)	114.79(33.60)
Total Cholesterol(mg/dL)	202.58(47.83)	196.47(42.53)
Total Bilirubin(mg/dL)	1.50(0.52)	0.58(0.20)
CAP score(dB/m)	308.06(41.38)	314.97(38.16)
Liver Stiffness(kPa)	10.53(8.43)	10.33(7.61)
Fib-4	1.28(1.19)	1.07(0.59)

Table 1: Demographic and clinical characteristics of the participants.

Conclusion: There was no difference in the presence of fibrosis in MASLD patients with GS without GS in our population. This is in keeping with the view that GS is a benign condition. The significant associations of age, BMI, and blood sugar with increased fibrosis are well documented.

Disclosure: Nothing to disclose.

PP0842

A HEAD-TO-HEAD COMPARISON OF MASLD WITH MAFLD DIAGNOSTIC CRITERIA IN LONGITUDINALLY ESTIMATING THE LIVER DISEASE PROGRESSION AND ACUTE CARDIOVASCULAR EVENTS' RISK IN STEATOTIC LIVER DISEASE PATIENTS: A REAL-LIFE EXPERIENCE

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Introduction: Two years after the Delphi consensus-proposed revolution in the nomenclature and diagnosis of Steatotic Liver Disease (SLD), the potential benefits of adopting Metabolic dysfunction-associated steatotic liver disease (MASLD) rather than Metabolic dysfunction-associated fatty liver disease (MAFLD) diagnostic criteria in defining the disease progression risk of SLD patients have never been systematically and longitudinally evaluated in real-life.

Aims & Methods: The present study aimed to compare MASLD and MAFLD criteria in estimating the 3-year risk of advanced fibrosis (AF) progression and Acute Cardiovascular Event (ACE) occurrence in not-lean (NL) and lean (L) SLD patients.

For this purpose, we retrospectively analyzed the anthropometrical, biochemical, clinical, and Liver Stiffness Measurement (LSM - FibroScan) data stored in the University Hospital (UH)-official Health Documents Digitization Archive of 931 SLD patients admitted to UH "Luigi Vanvitelli" (between January 2016 and May 2021), and followed up for 3 years. Based on baseline Body Mass Index (BMI), patients were subdivided into "L" (BMI <25 kg/m2) (n:134) and "NL" (n:797) and, subsequently, by ap-

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plying MAFLD and MASLD criteria, in NL-MASLD (n:206), NL-MASLD/MAFLD (n:481), NL-MAFLD (n:110), and L-MASLD (n:39), L-MASLD/MAFLD (n:68), L-MAFLD (n:27). The following validated LSM cut-offs were used to identify the different liver fibrosis stages according to the Metavir score: (a) F0-F2 ≤ 9.6 kPa; (b) F3: 9.7 - 13.5 kPa; (c) F4 ≥ 13.6 kPa. AF (F3-F4) was defined by LSM values > 9.7 kPa. The transition from F0-F2 to F3-F4 defined AF progression. Retrospectively collected and documented ACEs were: (1) Acute Myocardial Infarction (IMA), (2) Acute Coronary Syndrome (ACS), (3) Ictus cerebri (IC), and (4) Transient Ischemic Attack (TIA).

Results: MASLD and MAFLD criteria similarly estimated [MASLD: OR: 2.334, C.I.95%: 1.878 - 3.028, MAFLD: OR: 2.469, C.I.95%: 1.892 - 3.668, p < 0.0001; positive predictive value (PPV), MASLD: 0.800, C.I. 95%: 0.641-0.89 vs MAFLD: 0.785, C.I. 95%: 0.604-0.897, p: 0.076)] the overall 3-year risk of AF progression in NL-SLD, whereas in L-SLD, MAFLD criteria better estimated the overall 3-year risk of AF progression [PPV, MASLD: 0.571, C.I. 95%: 0.405-0.841 vs MAFLD: 0.714, C.I. 95%: 0.558-0.794, p: 0.006].

Compared to MASLD, MAFLD diagnostic criteria better estimated the 3-year risk of ACEs occurrence in NL [HR: 1.104, C.I..95%: 0.024-1.293, p:0.025], and not in L [HR: 1.260, C.I..95%: 0.665-2.101, p:0.071] patients. Multivariate competing risk analysis (adjusted for sex, age, diabetes, steatosis, and fibrosis severity) revealed diabetes (aHR: 2.113, C.I. 95%: 1.981-2.434, p:0.001), high-sensitivity-C-reactive protein (aHR: 1.441; C.I. 95%: 1.232-1.775; p:0.02) and Homeostatic-model-assessment-for-insulin-resistance (aHR: 1.228; C.I. 95%: 1.152-1.893; p:0.03) as associated with ACE occurrence in NL-MAFLD patients.

Conclusion: MAFLD criteria better estimate the 3-year AF progression and the ACE occurrence risk in L-SLD and NL-SLD patients, respectively.

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Disclosure: Nothing to disclose.

PP0843

DECREASE IN LIVER STIFFNESS IS NOT ASSOCIATED WITH REDUCED CARDIOVASCULAR EVENTS IN PATIENTS WITH DIABETES MELLITUS: A PROSPECTIVE STUDY

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Introduction: Patients with diabetes mellitus (DM) are at increased risk of developing cardiovascular disease (CVD). Hepatic fibrosis is common among patients with DM and is associated with liver-related events. However, its impact on CVD risk remains unclear.

Aims & Methods: This study aimed to prospectively assess the association between hepatic fibrosis and CVD risk in patients with DM.

A total of 323 patients with DM were prospectively enrolled from January 2022, excluding those with viral hepatitis, overt cirrhosis, or cancer. Clinical measurements and CVD history were recorded. The following parameters were assessed: body weight (BW), body mass index (BMI), waist circumference (WC), laboratory data [aspartate aminotransferase, alanine aminotransferase (ALT), gamma-glutamyl transferase, platelet count, hemoglobin A1C, cholesterol, and triglycerides], presence of hypertension (HTN), dyslipidemia, alcohol consumption, smoking status, liver steatosis measured by the controlled attenuation parameter (CAP) using FibroScan®, and current anti-DM medications. Hepatic fibrosis was quantified by liver stiffness measurement (LSM) using FibroScan®, assessed at baseline and after two years. CVD was defined as ischemic heart disease or ischemic/hemorrhagic stroke.

Results: A total of 277 patients completed the two-year follow-up and underwent repeat LSM. The mean age was 57±12 years, and 56.3% were male. The baseline mean LSM was 6.9±3.7 kPa. A history of CVD was present in 86 (31%) patients, while 6 (3.2%) developed new CVD events during follow-up. Among the new CVD cases, 3 were cardiac events, and 3 were neurovascular events. LSM decreased in 157 patients (d-LSM group) and increased in 120 patients (i-LSM group). The mean LSM change was -1.7±2 kPa in the d-LSM group and +1.5±1.3 kPa in the i-LSM group. The d-LSM group had significantly higher baseline ALT (39±30 vs. 32±21 IU/L, p=0.018), CAP (281±51 vs. 264±49 dB/m, p=0.005), and LSM (7.9±4.2 vs. 5.7±2.3 kPa, p<0.001) than the i-LSM group.

Alcohol consumption was more prevalent in the i-LSM group (30.8% vs. 17.8%., p=0.011). There were no significant differences between the two groups in mean age (57±13 vs. 57±10 years, p=0.738), male proportion (51.6% vs. 62.5%, p=0.07), BW (72±16 vs. 72±14 kg, p=0.93), BMI (26.7±4.7 vs. 26.1±4.1 kg/m^2, p=0.233), WC (90±12 vs. 89±10 cm, p=0.816), HTN (58% vs. 48.3%, p=0.111), dyslipidemia (68.2% vs. 62.5%, p=0.326), laboratory results excluding ALT, or smoking status (45.2% vs. 49.2%, p=0.515). The use of metformin (91.1% vs. 95.8%, p=0.121), pioglitazone (1.9% vs. 2.5%, p=1.0), and insulin (7% vs. 9.2%, p=0.51) did not differ significantly between the two groups. When stratified by LSM change, the incidence of new CVD was 2% (1/49) in patients with a decrease LSM of ≤-2 kPa, 2% (4/198) in those with changes between -2 and +2 kPa, and 3.3% (1/30) in those with an increase of ≥+2 kPa (p=0.897).

Conclusion: Hepatic fibrosis, as measured by LSM, can change dynamically in patients with DM. Although this was an observational study, a decrease in LSM was not associated with a reduced risk of CVD. Therefore, LSM monitoring may have limited value in assessing CVD risk in this population.

Disclosure: Nothing to disclose.

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PP0844

EVALUATING THE QUALITY OF SOCIAL MEDIA CONTENT ON NONALCOHOLIC FATTY LIVER DISEASE: AN EXPERIENCE FROM A LOWER MIDDLE-INCOME COUNTRY

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Introduction: Non-alcoholic fatty liver disease (NAFLD) is a significant public health concern. Social media (SoMe) is increasingly popular in lower-middle-income countries and can potentially influence perceptions and understanding of illness. Approximately one-third of Sri Lanka's population uses SoMe for health information. We assessed the quality of NAFLD-related content on SoMe.

Aims & Methods: This observational, cross-sectional study aimed to assess the accuracy, completeness and quality of SoMe content relating to NAFLD across Facebook, YouTube, TikTok, Instagram and X (Twitter) from Sri Lanka, in Sinhala, English, and Tamil languages. Posts containing keywords related to NAFLD published between January 2005 to December 2024 were evaluated. Specialist gastroenterologists independently reviewed the content using standardized scales for accuracy (0-3), completeness (0-5), and global quality score (GQS) (0-5). Posts were categorized by source profile [healthcare institution (HI), healthcare professional (HP), alternative healthcare professionals (AHP), non-healthcare person (NHP), or undisclosed identity (UI)] and content type [health education (HEd), health promotion (HProm), HEd and HProm or product endorsement]. User interactions (likes, shares, views) were also analyzed and correlated with reviewer scores.

Results: The analysis included 289 SoMe posts [158 (54.7%) YouTube, 101 (34.9%) Facebook, 14 (4.8%) TikTok, 11 (3.8%) X, 5 (1.7%) Instagram] relating to NAFLD in three languages [214 (74.0%) Sinhala, 54 (18.7%) Tamil, 21 (7.3%) English]. Posts from UI (36.0%) and NHP (26.0%) comprised the majority of content sources. HP (22.1%), AHP (14.2%), and HI (1.7%) represented the remainder. HProm (61.9%) was the predominant content type followed by HEd and HProm (17.6%), HEd alone (11.8%), and product endorsement (8.7%).

The mean accuracy score was 1.78/3 (59.3%), with significant differences between sources (HP: 2.35/3 (78.5%), AHP: 1.53/3 (51.0%), NHP:1.65/3 (55.1%), UI: 1.63/3 (54.3); p<0.001).

Completeness scores averaged 2.1/5 (42%), with content in English scoring higher [2.52/5(50.4%)] than Sinhala [2.05/5 (40.97%)] and Tamil [2.17/5 (43.33%)]. GQS averaged 2.4/5 (48.4%). 82% of posts were classified as "Rotten" (<60% composite reviewer score for accuracy, completeness, and global quality).

Content quality by platform was compared using ANOVA (mean) and Kruskal-Wallis (median) tests. Posts on Facebook and YouTube showed significantly higher completeness and GQS compared to other platforms (p<0.05).

User interaction analysis was conducted using both Pearson's and Spearman's correlation tests. There were no significant correlations observed between accuracy, completeness, or GQS and views or likes. Inter-reviewer variation was assessed using ANOVA and non-parametric tests. There was no statistically significant inter-reviewer variation in scoring patterns, indicating consistency within reviewer groups.

Conclusion: The majority of the SoMe content was from NHP sources. HP delivered the most accurate content. Facebook and YouTube emerged as platforms with relatively higher-quality content compared to others. Overall accuracy, completeness and GQS remained suboptimal across platforms. Most posts (82%) failed to meet adequate quality standards. Notably, user engagement metrics did not correlate with content quality. These findings highlight the need for improved quality control measures and health literacy initiatives to enhance the reliability of information regarding NAFLD on SoMe platforms.

Disclosure: All authors confirm no conflicts of interests to declare.

PP0845

NON-INVASIVE TESTS TO PREDICT FIBROSIS AND LIVER RELATED EVENTS IN LEAN AND OBESE PATIENTS WITH MASLD

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is one of the main causes of chronic liver disease. Although liver biopsy is still the gold standard to assess disease severity, noninvasive tests are often used. The fibrosis-4 (FIB-4) index is one of the most useful tools to estimate liver fibrosis. In this study, we evaluated how reliable FIB-4 is as a noninvasive test in predicting liver fibrosis and (Liver related events (LREs). We also tested whether combining FIB-4 with body mass index (BMI) could better predict liver-related complications.

Aims & Methods: This was retrospective, descriptive and analytical study conducted in a gastroenterology department over 8 years including patients diagnosed with MASLD. The data normality test for this study was conducted using the Shapiro-Wilk test. Comparaison between medians was conducted using Mann-Whitney test. The discriminatory ability of FIB-4 and FIB-4×BMI (a preliminary score obtained by multiplying FIB-4 by BMI) to predict fibrosis and LREs (event of decompensation, HCC, transplantation, or liver-related death) was evaluated using AUROC curve.

Results: 114 patients we enrolled in this study, mean age was 48.1±15 years. Sex ratio W/M was 1.92. Hypertension was found in 13(11.4%), diabetes occurs in 58(50.8%) and dyslipidemia was found in 5(4.3%). Mean BMI was 26.9±6.25, 61(54.5%) were obese and 51(45.5%) were lean. Cardiometabolic syndrome was found in 65(57%) of the cases.

Concerning biologic findings, Cytolysis was observed in 12.2%(14), cholestasis in 21(18.4%), and cytolysis with cholestasis was found in 15.7%(18).

For fibrosis and steatosis assessment, 64(56.14%) patients underwent Vibration-Controlled Transient Elastography and 50(43.8%) a liver biopsy. For fibrosis, 9(9.1%)were F0, 60(60.9%)F1,22(22.2%) F2, 6(6.1%) F3 and 2(2%) F4. For steatosis, 32(33.1%) were S1, 36(37.5%) S2 and 28(29.2%) S3.

Median FIB4 for all patients was 1.29[0.67;1.76], with significant difference between lean and obese patients (1.42 vs 0.76, p<0.001). Median FIB4*BMI score was 27[13.7;47.7], with significant difference between lean and obese patients (42.9 vs 16.1, p<0.001)

LREs occurs in (13)11.4%;9(7.8%) had portal hypertension,6(5.2%) variceal bleeding, 3(2.6%) had ascites, 1(0.8%) had hepatorenal syndrome and 1(0.8%) a hepatocellular carcinoma.

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The FIB-4 index showed strong predictive ability for both liver fibrosis and liver-related events. Its discriminatory performance for detecting LREs was demonstrated by an AUROC of 0.873 with a sensitivity of 77.7% and a specificity of 88.3%. The FIB-4–BMI combined score further improved diagnostic accuracy for liver-related complications, yielding an AUROC of 0.883. Using a cutoff value of 56, it reached a sensitivity of 88.8% and a specificity of 86.4%. For the prediction of severe fibrosis (F3–F4), the AUROC values for FIB-4 and the FIB-4×BMI combined score was 0.967 and 0.952, respectively. FIB4 showed a sensitivity of 100% and a specificity of 84% at a cutoff of 1.78, while the FIB4×BMI score had a sensitivity of 100% and a specificity of 81.5% at a cutoff of 54.7.

Conclusion: Non-invasive testing plays a key role in the management of MASLD, providing reliable tools for assessing fibrosis and predicting liver-related outcomes while avoiding the risks of invasive procedures. In this context, both FIB-4 and the FIB-4–BMI combined score were evaluated for their predictive value. In our study, FIB-4 alone showed good performance, but the FIB-4–BMI score demonstrated superior accuracy in predicting advanced fibrosis and liver-related complications.

Disclosure: Nothing to disclose.

PP0846

"WHAT HAPPENS IN THE GREY ZONE" - FIB-4 VARIATION IN THE PROGRESSION OF INTERMEDIATE RISK MASLD

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Introduction: Metabolic associated steatotic liver disease (MASLD) represents the most common cause of chronic liver disease (CLD) globally. The Fibrosis-4 Index for Liver Fibrosis (FIB-4) is an accessible and widely used tool for stratifying the risk of fibrosis. However, its correlation with clinical progression in intermediate risk patients is not fully understood.

The aim is therefore to assess: whether the variation in FIB-4 can be a tool for early identification of patients at risk of progression to Advanced Chronic Liver Disease (ACLD); and whether the reassessment of FIB-4 at 12 months for intermediate risk patients referred to in the recommendations of the European Association for the Study of the Liver 2024 is appropriate.

Aims & Methods: Retrospective analysis of a cohort of patients with MASLD with intermediate risk FIB-4 index, with at least 1 year of follow-up, in a single centre. Patients with other concomitant liver disease; a history of alcohol consumption; age under 35 years and over 65 years were excluded. Calculation of the variation in FIB-4 between 0 and 12 months of follow-up.

Analysis of the relative risk associated with an increase of any magnitude in FIB-4 for developing ACLD in 5 years (presence of compensated ACLD; variceal rupture, ascites; encephalopathy; hepatocarcinoma). In a secondary analysis, the ROC curve of the variation in FIB-4 in relation to progression to ACLD was studied.

Results: 68 patients were included (male-42.6%; mean age-55.4±6.8 years; mean follow-up time-64±38.9 months; presence of hypertension-64.7%; type II diabetes mellitus-42.6%; dyslipidaemia-70.6%; overweight/obesity-88.2%; mean BMI-31.6±5.8 kg/m2; FIB-4 at admission-1.72±0.38). The development of ACLD at 5 years was 14.6%.

The variation in FIB-4 in the group with disease progression (median=0.34; [-0.29;2.85]) was significantly higher than in the group without progression (median=-0.09; [-0.5;0.2]), (U=79.5; p=0.003). FIB-4 at admission was not associated with disease progression at 5 years (U=148.5; p=0.353).

The risk of progression to ACLD at 5 years was 1.23 times higher (RR=1.23; p<0.05) in patients with an increase in FIB-4 at 12 months of any magnitude versus no increase. The variation in FIB-4 at 12 months showed good discriminatory capacity (AUC=0.767; p<0.05), with the cut-off point with the best accuracy in predicting ACLD at 5 years being an increase in FIB-4≥0.56 (sensitivity-66.7% and specificity-80.0%).

Conclusion: In patients with intermediate-risk MASLD, an increase in FIB-4 over 12 months was associated with an increased risk of disease progression, regardless of the magnitude of FIB-4 on admission. Patients with an increase ≥0.56 in 1 year seem to benefit from long-term specialised follow-up, due to the increased risk of developing ACLD in 5 years.

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PP0847

ASSESSMENT OF THE QUALITY OF LIFE IN PATIENTS WITH NON-ALCOHOLIC STEATOHEPATITIS IN COMBINATION WITH CORONARY HEART DISEASE WITH STABLE ANGINA PECTORIS AFTER TREATMENT WITH USING COMPLEX THERAPY

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Introduction: Curent studies have proven that non-alcoholic fatty liver disease (NAFLD) impacts an important role in the development of risk factors for cardiovascular diseases. Comorbidities are a risk of a complicated course of coronary heart disease (CHD). The criteria for the effectiveness of treatment in medical practice are the duration and quality of life (QoL) of patients. The main tool for assessing quality of life in curent clinical research is questionnaires. The most common questionnaires widely used by scientists are the Medical Outcomes Study 36-Item Short-Form Health Status (SF-36) and EuroQoL-5D (EQ-5D).

Aims & Methods: The aim: To evaluate the indicators of QoL in patients with non-alcoholic steatohepatitis with coronary heart disease with stable angina pectoris in combination after complex therapy including ezetimibe and ursodeoxycholic acid.

Materials and methods: The prospective study included 145 patients with a verified diagnosis of CHD with stable angina pectoris II functional class combined with nonalcoholic steatohepatitis (NASH) and 100 patients with CHD without NASH. All 245 patients received basic CHD therapy: acetylsalicylic acid 100 mg once a day, ramipril 2.5 mg once a day, bisoprolol 2.5 mg once a day, and rosuvastatin - 20 mg once a day. To achieve the aim, 3 groups of patients were formed depending on the available metabolic factors (MFR): group 1-a (n=75) – patients had CHD+ NASH + MFR (arterial hypertension (AH), dyslipidemia, increased body weight or obesity) and

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group 1-b (n=70) – CHD+ NASH + AH, as well as group 2 (n=100) – CHD without NASH. Group 1-a patients received the basic therapy of CHD and ezetimibe 10 mg/1 time per day and UDCA 15 mg/kg/day 2 hours before bedtime, for 6 months, against the background of recommendations for lifestyle modification (LM). Group 1-b received basic therapy of CHD and UDHC 15 mg/kg/day 2 hours before bedtime, for 6 months, against the background of recommendations for LM (weight loss, hypocaloric diet). Group 2 patients received only the basic therapy of CHD.

Results: As the obtained results show, positive clinical dynamics from treatment were noted in patients of group 1-a who received complex therapy with the addition of ezetimibe and UDCA at p<0.05. Regarding the QoL indicators according to the SF-36 questionnaire after treatment, a tendency to improve most of its indicators was revealed, which is more clearly observed in patients of group 1-a.

Analyzing the integrated indicators of physical and mental components of health on all scales of SF-36 revealed reliable positive dynamics on all scales of the questionnaire, which are best expressed in patients of group 1-a in comparison with group 1-b at p<0.05. The following indicators were most significantly affected: social functioning increased by 40.6 points in group 1-a, respectively, by 27.6 points in group 1-b at p<0.05; the value of the role of emotional problems increased by 55.7 points in group 1-a, respectively, by 40.3 points in group 1-b at p<0.05; general health increased by 56.9 points in group 1-a, respectively, by 45 points in group 1-b at p<0.05.

Conclusion: Complex therapy including ezetimibe and UDCA in patients with CHD combined with NASH is more effective compared to basic CHD therapy, as it reliably reduces the frequency of attacks of stable angina and the need for nitroglycerin.

The effectiveness of the treatment of CHD combined with NASH is increased by the inclusion of ezetimibe and UDCA in the basic therapy and has a positive effect on the clinical course of CHD, QoL of patients and the patient's attitude to his own disease.

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PP0848

RISK OF AUTOIMMUNE DISEASE IN OFFSPRING OF MOTHERS WITH BIOPSY-PROVEN METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE (MASLD). A NATIONWIDE COHORT STUDY

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is highly prevalent and associated with an increased risk of adverse pregnancy and birth outcomes,[1,2] but the risk of adverse clinical outcomes beyond the perinatal period are under investigated. Recent studies suggest an overlap in clinical and immunological features between MASLD and autoimmune diseases,[3,4] but whether fetal exposure to MASLD is associated with a higher risk of offspring autoimmune diseases remains unclear.

Aims & Methods: In this nationwide cohort study, we included all singleton live born offspring with fetal exposure to maternal biopsy-proven MASLD diagnosed in Sweden between 1992 and 2017. We stratified maternal MASLD in simple steatosis versus severe MASLD (defined as either steatohepatitis, any stage of liver fibrosis, or cirrhosis). We matched the 239 identified MASLD offspring with ≤5 reference offspring (n=1,131) of mothers without known MASLD. Matching factors were maternal age at delivery, calendar year of delivery, and parity. We used multivariable Cox proportional hazard models to estimate adjusted hazard ratios (aHRs).

The endpoint was any incident autoimmune disease up until December 31, 2023. We performed a sibling-controlled analysis comparing the risk of autoimmune disease in offspring of sisters of women with biopsy-proven MASLD to investigate if early environmental and intrafamilial factors play a role.

Results: During a median of 18.4 years of follow-up (interquartile range [IQR] 13.2-23.8), 15 exposed offspring (incidence rate [IR] 3.4/1000 person-years) vs. 40 reference offspring (IR 1.9/1000 person-years) were diagnosed with autoimmune disease. This corresponded to an aHR of 1.20 (95% CI 0.57-2.53). The median age at incident autoimmune disease was higher among offspring of mothers with MASLD (14.7 years, IQR 5.5-17.5) compared to reference offspring (8.1 years, IQR 4.1-14.1).

The sibling analysis yielded a similar risk magnitude (aHR 1.14, 95% CI 0.28-4.63), indicating that early environmental and intrafamilial factors likely do not substantially contribute to the risk of early-onset autoimmune disease.

Among MASLD offspring, inflammatory bowel disease was the most frequent autoimmune disease during follow-up (26.7% [n=4] vs. 7.5% [n=3] among reference offspring), followed by type 1 diabetes (20%, n=3), and celiac disease (13.3%, n=2).

Among reference offspring, celiac disease was the most common autoimmune disease (32.5%, n=13), followed by type 1 diabetes (27.5%, n=11) and psoriasis (15%, n=6). The risk of autoimmune disease remained unchanged among 175 offspring born to mothers with simple steatosis alone

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(aHR 0.84, 95% CI 0.29-2.45). However, we saw a positive association in 64 offspring of mothers with severe MASLD, but the risk estimate failed to attain statistical significance possibly due to small sample size (aHR 1.98, 95% CI 0.67-5.84).

Conclusion: Fetal exposure to maternal MASLD does not seem to be associated with increased risk of autoimmune disease in the offspring up until early adulthood. Further, studies with longer follow-up are needed to investigate the long-term risk of autoimmune disease after fetal exposure to MASLD.

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C.A.M. has no conflict of interest

F.E. has served as an advisory board member for Boehringer Ingelheim.

D.B.: has no conflict of interest

J.S.: has no conflict of interest

H.H.: HH:s institutions have received research funding from Astra Zeneca, EchoSens, Gilead, Intercept, MSD, Novo Nordisk and Pfizer. H.H. has served as consultant or on advisory boards for Astra Zeneca, Bristol Myers-Squibb, MSD and Novo Nordisk and has been part of hepatic events adjudication committees for Arrowhead, Boehringer Ingelheim, KOWA and GW Pharma.

M.T.: has no conflict of interest

O.S.: has no conflict of interest

J.F.L.: Dr Ludvigsson has coordinated an unrelated study on behalf of the Swedish IBD quality register (SWIBREG). That study received funding from Janssen corporation. Dr Ludvigsson has also received financial support from Merck developing a paper reviewing national healthcare registers in China. Dr Ludvigsson has an ongoing research collaboration on celiac disease with Takeda; and additional collaboration within IBD with Merck.

PP0849

INDICES OF THE LIVER FUNCTIONAL STATE, BLOOD LIPID SPECTRUM IN PATIENTS WITH METABOLIC DISFUNCTION-ASSOCIATED STEATOHEPATITIS (MASH), MODERN APPROACHES TO COMPLEX TREATMENT

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Introduction: Metabolic disfunction-associated Liver Disease (MASLD) and Metabolic disfunction-Associated Steatohepatitis (MASH) are conditions that are significantly influences by lifestyle factors. Actual problem today is the significant spread of MASLD among people of socially active age, high costs for diagnosis and treatment, and a decrease in the quality of life and work capacity of such patients.

Aims & Methods: The aim of the research is to study the dynamics of indicators of the functional state of the liver, the lipid spectrum of the blood, to evaluate the cardiovascular risk according to the SCORE-2 scale under the influence of complex pathogenetic treatment of the identified disorders in patients with steatohepatitis associated with metabolic dysfunction in combination with obesity. 46 outpatients with a diagnosis MASH were examined: 29 (63.1 %) men and 17 (36.9 %) women aged 32-55 years. There were patients divided into two groups.

The first (main) group with a diagnosis MASH were 19 (41.3 %) people who received atorvastatini: 20 mg per day, 90 days.

The second group (comparison) – 27 (58.7%) of patients who were prescribed ursodeoxycholic acid preparations in combination with atorvastatini: 250 mg (15 mg/kg of body weight) for 90 days. The healthy group consisted of 12 people of the appropriate age. The anthropometric criterion of obesity was the Quetelet index (BMI = body weight (kg)/height (m^2)). Phenotypic variant of obesity was determined by the ratio of waist circumference / hip circumference (WC/HC). Visceral obesity was observed in all examined patients. Among the patients of the main group, there were 9 (47.4 %) people with type I obesity degree (BMI = 32.6 ± 2.2 kg/m2), 6 (31.5 %) – II degree (BMI = 35.8 ± 3.6 kg/m2), and 4 (21.1 %) – III degree (BMI = 40.0 ± 2.1kg/m2), in the comparison group – 17 (62.9 %), 6 (22.3 %) and 4, respectively (14.8 %). This indicated that the patients had an abdominal type obesity.

For assessment biochemical indicators of the functional state of the disease were studied using unified methods (general level bilirubin and its fractions (direct and indirect), activity of serum aminotransferases ALT and AST, alkaline phosphatase (AP), total cholesterol (CHD) content in serum blood, high-density lipoprotein cholesterol (HDL), low-density lipoprotein cholesterol (LDL), very-low-density lipoprotein cholesterol (VLDL), triglycerides.

Results: In most patients suffering from MASH there was found a reliable increase of the direct bilirubin level in the blood, of the activity of alanine aminotransferase (ALT, ALT) and alkaline phosphatase (ALP), and of the blood lipid spectrum (p<0,05). According to the body mass index (BMI) most of the examined patients had visceral obesity.

After the course of the combined therapy there were marked the positive dynamics in biochemical indices of the blood (p < 0,05): decrease of the direct bilirubin level, aminotransferase (ALT, ALT), blood phosphatase. The indices of blood lipid spectrum approached to the normal indexes (p < 0,05).

Conclusion: the addition of ursodeoxycholic acid to statin therapy results in a statistically significant reduction in overall cardiovascular risk, as measured by SCORE-2, compared with atorvastatin monotherapy. All

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patients were recommended to continue the course treatment - a combination of atorvastatini and ursodeoxycholic acid for 2-3 months. This indicates the almost complete recovery of the liver functional activity and achievement of the consistent biochemical of MASH.

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PP0850

STEATOTIC LIVER DISEASE IN NON-OBESE INDIVIDUALS: A STATISTICAL ANALYSIS OF ASSOCIATED FACTORS AND TREATMENT RESPONSE

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Introduction: Steatotic liver disease (SLD) in non-obese individuals (body mass index -BMI- <25) is an increasingly recognized clinical entity, distinct from the classic metabolic-associated fatty liver disease typically linked to obesity. Despite growing awareness, the characteristics, associated conditions, and treatment response of this phenotype remain poorly understood. This study aimed to evaluate clinical, biochemical, and histological characteristics in this population and to assess the impact of targeted interventions on liver-related outcomes.

Aims & Methods: We conducted a retrospective monocentric analytical study including 50 non-obese patients (BMI <25) diagnosed with SLD by ultrasound. Patients were stratified based on associated comorbidities, including coeliac disease, inflammatory bowel disease (IBD), type 2 diabetes, and hypertension. Comparative statistical analysis was performed using chi-square and Student's t-test; p-values <0.05 were considered statistically significant.

Results: Out of 162 patients followed for SLD, 30.86% (n=50) were classified as non-obese, among which 22% (n=11) were underweight. The mean age was 40.6, with a slight female predominance and a sex ratio M/F of 0.81. SLD was MASLD in 26% of patients (n=13), among which 9 patients had type 2 diabetes, and 4 patients had arterial hypertension. MASH was diagnosed in 7 patients of this group.

SLD was secondary to other causes of steatosis in 74% of cases (n=37): 38% had coeliac disease (n=19), 30% had inflammatory bowel disease (n=15), 4% had thyroid dysfunction (n=2), and 2% had LPAC syndrome (n=1).

Liver function abnormalities were found in 58% (n=29) of patients.

Patients with coeliac disease who underwent liver biopsy showed significantly higher rates of steatohepatitis (grade 2-3) compared to non-coeliac patients (68.4% vs 40%, p=0.03).

After one year of follow-up, liver enzyme normalization was observed in 75% of patients who received targeted treatment (gluten-free diet or optimized management of comorbidities) along with adapted physical activity and a weight loss goal of 3-5%, compared to 33.3% in untreated patients (p=0.01).

Histological improvement (≥1 grade regression of steatohepatitis) was documented in 44.4% of biopsied patients post-intervention, with a statistically significant association between adherence to gluten-free diet and histological regression in coeliac patients (p=0.02).

No significant association was found between presence of diabetes or hypertension and severity of liver histology (p>0.05).

Evaluation of liver fibrosis showed F0-F1 in 70% (n=35) and F2 in 12% of patients (n=6). Fibroscan stages remained stable in all these patients during follow-up, with no progression beyond F2.

Conclusion: In non-obese individuals, SLD is frequently associated with immune-mediated comorbidities, particularly coeliac disease and IBD, and characterized by mild fibrosis. Targeted treatment strategies, particularly in coeliac patients, significantly improve liver biochemistry and histological status. These findings support the need for systematic etiological screening and personalized therapeutic approaches in non-obese patients with hepatic steatosis.

Disclosure: Nothing to disclose.

PP0851

LEAN AND NON-LEAN HEPATIC STEATOSIS IN MOROCCO: DISTINCT METABOLIC PROFILES AND FIBROSIS RISK

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Introduction: The prevalence of metabolic associated fatty liver disease (MASLD) is increasing worldwide and it increases their risk of chronic liver disease and cardiometabolic complications. While traditionally associated with obesity, hepatic steatosis can also develop in lean individuals, for whom the underlying mechanisms remain poorly understood.

Aims & Methods: This is a prospective monocentric and analytical study including all patients followed for hepatic steatosis in the hepato-gastro-enterology department over a 6-months period from September 2024 to March 2025. Diagnosis of metabolic steatosis is based on abdomino-pelvic ultrasound. We compared 2 categories of patients: lean (BMI< 25kg/m2) vs not lean patients (BMI>25 kg/m2) with hepatic steatosis.

Statistical analysis was carried out using t-tests and chi-square tests, with a significance level of p<0.05.The aim of our work is to compare lean and not lean patients with hepatic steatosis, their cardiometabolic profile, the risk factors and to study the outcome these patients.

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Results: The study included 100 patients, divided into two groups of 50 each. In group A (non-lean patients), the mean age was 46.3 ± 8.2 years, and in group B (lean patients), 44.1 ± 7.6 years (p = 0.12). Group A had a male predominance (70%) compared to 50% in group B, a statistically significant difference (p = 0.03).

The mean body weight index was 29.2 ± 4.5 kilograms per square meter in group A and 19.5 ± 1.2 in group B (p < 0.001). Metabolic syndrome was present in 84% of group A versus 34% in group B, indicating a significant difference in metabolic profiles (p < 0.001).

All patients in both groups had hepatic steatosis confirmed by ultrasound. Liver function was altered in 60% of group A: 24% had isolated cytolysis, 20% cytolysis with cholestasis, and 16% isolated cholestasis. In group B, 48% had abnormal tests: 28% with cytolysis, 20% with cholestasis. These abnormalities were significantly more frequent in group A (p < 0.05).

Fibrosis was assessed by liver biopsy or elastography in 42 patients from group A and 44 from group B. In group A, 76.2% were stage F0-F1, 14.3% F2, and 9.5% (n=4) had advanced fibrosis (F3-F4). In group B, 86.4% were F0-F1 and 13.6% F2; no F3-F4 stages were observed (p < 0.05).

Autoimmune diseases were significantly more prevalent in lean patients (group B): 30% had celiac disease, 16% Crohn's disease, and 4% ulcerative colitis. In group A, 14% had primary biliary cholangitis, 12% autoimmune thyroiditis, and 4% Crohn's disease (p < 0.001).

After multidisciplinary management, 36% of group A patients with initial abnormalities normalized their liver function tests, while 4% progressed to cirrhosis. In group B, 40% normalized liver tests. Two patients with celiac disease on gluten-free diet showed marked histological improvement, with one achieving complete regression of steatohepatitis.

Conclusion: Although lean individuals with steatosis generally showed a healthier metabolic profile and a lower degree of liver fibrosis compared to non-lean patients, both groups exhibited a significant proportion of metabolic derangements. Our study also highlighted a significantly higher prevalence of autoimmune diseases in lean patients (p < 0.001).

Management combining treatment of underlying conditions with lifestyle measures led to notable improvements in liver function, imaging and histology. These findings underline the importance of tailored strategies based on patient phenotype, including autoimmune screening in lean individuals.

Disclosure: Nothing to disclose.

PP0852

WAR-RELATED ANXIETY, PROINFLAMMATORY CYTOKINES AND INSULIN RESISTANCE IN UKRAINIAN CHILDREN WITH METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE

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Introduction: Psychological stress as a result of the war in Ukraine negatively affects children's health, leading to the onset/exacerbation of metabolic dysfunction [1, 2].

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a multifactorial condition that affects both adults and children and includes several mechanisms associated with excessive fat accumulation in hepatocytes, inflammation, and insulin resistance [3].

Aims & Methods: The aim of the study was to determine the impact of war-related anxiety on proinflammatory cytokines (IL-6, TNF-α) and insulin resistance in Ukrainian children with MASLD.

The study included 42 obese children with MASLD aged 6 to 17 years; the mean age was 12.00±0.56 years. Determination of the presence of liver steatosis was performed by transient elastography (FibroScan®502touch, Echosence, France). The presence of obesity was confirmed by bioimpedance analysis (TANITA MC-780MA, Japan). Screening for anxiety was performed using the Screen for Child Anxiety-Related Disorders (SCARED) questionnaire. Depending on the presence of anxiety, obese children with MASLD were divided into 2 groups: Group I - 12 children without anxiety disorders, Group II - 30 children with anxiety disorders. The control group consisted of 10 children of normal weight without anxiety disorders. Quantitative determination of insulin concentration in the serum was performed by ELISA using Monobind Inc. test systems, USA, and the HOMA-IR was calculated. Quantitative determination of the concentration of IL-6 and TNF-α in the blood serum was performed by ELISA using test systems from LDN LaborDiagnostika (Germany).

Results: An increased level of proinflammatory cytokines was observed in 50.0% of the children of group I and 56.7% of the children of group II. The median level of TNF-α was 3.2 times higher (p<0.05) in patients of group I and 2.5 times higher (p<0.05) in patients of group II compared to the control group. The content of anti-inflammatory IL-10 in the I and II groups did not differ from the control.

Moreover, the indicators of carbohydrate metabolism in obese children showed statistically significant differences: in patients of group I a significant increase in the median insulin level (2.1 times, p<0.05) and the HOMA-IR (2.7 times, p<0.05) was found compared to control group as well as in patients of group II - in 2.1 times (p<0.05) and the HOMA-IR - in 2.8 times (p<0.05), respectively.

Conclusion: Thus, the presence of anxiety in children with MASLD is accompanied by the elevation of proinflammatory cytokines and insulin resistance.

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PP0853

WAR-RELATED ANXIETY IN CHILDREN WITH METABOLIC-DYSFUNCTION ASSOCIATED STEATOTIC LIVER DISEASE

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Introduction: Pediatric metabolic-dysfunction-associated steatotic liver disease (MASLD) has become a global problem worldwide for the last decades [1].

The situation is only exacerbated in Ukraine during the war due to the lack of access to resources, in particular, healthy nutrition, sufficient physical activity, and psycho-social support [2, 3].

There is a need for investigation of war-associated emotional disorders that may contribute to MASLD progression.

Aims & Methods: To assess the structure of anxiety disorders in children with MASLD and to identify the psycho-emotional features of MASLD children of military parents.

The study involved 45 patients aged 6 to 17 years (mean age 11.8±1.5 years). The presence of obesity was evaluated using bioimpedance analysis (TANITA MC-780MA, Japan). Liver steatosis was established by transient elastography (FibroScan®502touch, Echosence, France) with controlled attenuation parameter measurement.

Based on obesity status and relation of parents to the military, the children were divided into 3 groups: 1 group – 29 MASLD children of civilian parents, 2 group – 8 MASLD children of military parents, and 3 group (control) – 8 normal weight children of civilian parents.

Screening for anxiety disorders was performed using the Screen for Child Anxiety-Related Disorders (SCARED) questionnaire. Statistical analysis was carried out using the Statistica 6.1 software package.

Results: An increased general level of anxiety was more frequently observed in children with MASLD compared to the control group, being detected in 34.5% of the 1 group ($p<0.05$) and 37.5% of the 2 group ($p<0.05$) patients. Among children from military families, moderate anxiety disorders were more prevalent (25.0%, $p<0.05$). Panic and/or somatic anxiety disorders were also more common among children with MASLD (1 group – 31.0%, 2 group – 37.5%) compared to the control group ($p<0.05$). Separation anxiety disorders were predominant across all groups (1 group – 44.8%, 2 group – 50.0%, 3 group – 50.0%, $p=0.94$). Children from military families showed significantly higher rates of generalized anxiety disorder (50.0%, $p<0.05$) and school phobia (50.0%, $p<0.05$).

Conclusion: During the war in Ukraine, separation anxiety disorders, increased general anxiety, panic and/or somatic anxiety disorders, and school phobias predominate in children with MASLD. Generalized anxiety disorders and school phobias are more prevalent among children of military parents, which should be considered for treatment program development for these patients.

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Disclosure: Nothing to disclose.

PP0854

IG A DEFICIENCY IS NOT ASSOCIATED WITH HIGHER LIVER-RELATED EVENTS IN MASLD/MASH PATIENTS: A MULTI-CENTER ANALYSIS

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Introduction: MASLD/MASH is a spectrum of liver conditions characterized by excessive fat accumulation in the liver in the absence of significant alcohol consumption. Recent studies have highlighted the role of immune cells in the pathogenesis of MASLD.

Additionally, elevated serum IgA levels have been associated with advanced fibrosis in MASLD/MASH patients. Accordingly, IgA dysregulation could contribute to the pathogenesis and progression of MASLD/MASH. The aim of our study is to determine if IgA deficiency increases the risk of adverse events in the setting of MASLD/MASH.

Aims & Methods: A retrospective cohort study was performed using large population-based data from the TriNetX platform. Patients with diagnosed IgA deficiency and MASLD/MASH since January 1, 2014, were identified. This group was matched with patients with MASLD/MASH without IgA deficiency according to age, gender, demographics, comorbidities, and medication by using 1:1 propensity matching. The primary outcome was risk of hepatic decompensation, cirrhosis, and hepatocellular carcinoma (HCC), and secondary outcomes were all-cause mortality, major adverse cardiovascular events (MACE), and chronic kidney disease (CKD).

	MASLD/MASH with IgA Deficiency		MASLD/MASH control		HR (95% CI)
Outcomes	Included	Events	Included	Events	
HCC	673	10	674	10	1.01 (0.14, 7.19)
Hepatic decompensation	646	14	653	16	0.87 (0.42, 1.78)
Mortality	672	29	673	25	1.15 (0.67, 1.96)
MACE	556	43	550	24	1.79 (1.09, 2.96)
CKD	515	48	527	54	0.90 (0.61, 1.33)

Abbreviations: MASLD metabolic dysfunction-associated steatotic liver disease; MASH metabolic dysfunction-associated steatohepatitis; HCC hepatocellular carcinoma; MACE major adverse cardiovascular events; CKD chronic kidney disease; HR, hazard ratio; CI, confidence interval; AE, adverse events.

Table. Clinical Outcomes MASLD/MASH with IgA Deficiency vs MASLD/MASH control.

Results: A total of 841,865 adult patients with MASLD/MASH were identified, 674 of these individuals had IgA deficiency; 674 out of 674 (mean [SD] age, 51.4 [16.0] years; 383 [56.8%] female) were matched with 674 individuals (mean [SD] age, 51.7 [16.3] years; 378 [56.1%] female) without IgA deficiency. The IgA deficient group had significantly higher risk of

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MACE (Hazard Ratio [HR], 1.79; 95% Confidence Interval [CI], 1.09-2.96). No significant differences were observed in developing HCC (HR, 1.01; 95% CI, 0.14-7.19), hepatic decompensation (HR, 0.87; 95% CI, 0.42-1.78), all-cause mortality (HR, 1.15; 95% CI, 0.67-1.96), and CKD (HR, 0.90; 95% CI, 0.61-1.33). No patients from our study developed cirrhosis.

Conclusion: Conversely to the expected increase in the liver-related outcomes, IgA deficient patients had no significant risk. Normally, IgA promotes fibrosis through the IgA-FcR signaling axis, and given that IgA is deficient, this condition can theoretically alter the immune landscape in MASLD.

Contrarily, the increased risk of MACE in IgA deficiency was previously studied, and it is most likely that IgA solely is an independent risk factor to develop MACE which is compatible with the result of our study. Further studies are required to analyze the impact of IgA in the prognosis of MASLD/MASH patients.

Disclosure: Nothing to disclose.

PP0855

RETROSPECTIVE ANALYSIS OF LIVER METABOLISM ALTERATIONS FOLLOWING PCSK9 INHIBITOR THERAPY AFTER CORONARY INTERVENTION

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Introduction: Evolocumab, a PCSK9 inhibitor, is widely used to treat familial hypercholesterolemia (FH) and severe cardiovascular disease due to its potent LDL-C-lowering effects. However, given its mechanism involving hepatic LDL receptor pathways, concerns about liver dysfunction have emerged. Moreover, metabolic-associated fatty liver disease (MAFLD) commonly coexists with cardiovascular disease, potentially influencing treatment outcomes.

Aims & Methods: Aim is to evaluate the effects of Evolocumab on liver steatosis and hepatic function in patients undergoing percutaneous coronary intervention (PCI).

A retrospective study was conducted on 61 patients who received Evolocumab after PCI between April 2017 and October 2024 (11 with FH, 50 without). Liver steatosis was assessed via non-contrast computed tomography (CT), and hepatic function was evaluated using LDL-cholesterol levels and FIB-4 index at baseline and after 24 weeks of treatment.

Results:

1. LDL-C decreased significantly from 120 ± 48 mg/dL to 14 ± 3.1 mg/dL within two weeks of Evolocumab initiation.
2. Liver steatosis, as approximated by the liver-to-spleen attenuation ratio on CT, showed the improvement from 0.98 to 1.0 (no statistical significance), because of limitation by the CT sensitivity.
3. Six patients experienced drug induced liver dysfunction, necessitating discontinuation of therapy.
4. FIB-4 indices of 27% patients increased from 3.55 before to 4.33 after the treatment.
5. Subgroup analysis indicated mild improvement in hepatic steatosis markers: especially in younger patients with low FIB-4 indices, γ-GTP decreased from 89.4 to 48.6 IU/L, and cholinesterase (ChE) decreased from 410 to 317 IU/L by treatment.

Conclusion: Evolocumab demonstrates not only robust LDL-C reduction but may also exhibit potential benefits for liver steatosis, though we need accurate imaging such as Fibroscan. Nevertheless, attention is required regarding the risk of drug induced hepatic dysfunction, and observed improvements must be interpreted cautiously due to possible placebo effects following PCI.

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Disclosure: Nothing to disclose.

PP0856

IMPACT OF A SUPERVISED COMBINED EXERCISE PROGRAM ON QUALITY OF LIFE IN PATIENTS WITH MASH: RESULTS FROM A RANDOMIZED CONTROLLED TRIAL

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Introduction: Currently, metabolic dysfunction-associated steatotic liver disease (MASLD) is the most common chronic liver disease worldwide and represents a major public health concern (1).

Its more severe form, metabolic dysfunction-associated steatohepatitis (MASH), is increasingly recognized as a leading cause of liver transplantation (2).

Although lifestyle changes, especially increased physical activity, are widely recommended in clinical practice, structured and supervised exercise programs are not yet well represented in the literature.

Aims & Methods: This study aimed to examine how an eight-week physiotherapist-supervised personalized exercise intervention can affect the quality of life in patients with MASH. A total of 20 patients aged 18-65 years with MASH were included and assigned to the intervention group (n=10; 8 males, 2 females; mean age 45.3±9.4 years; BMI 28.3±2.6) or a control group (n=10; 8 males, 2 females; mean age 47.6±7.1 years; BMI 30.1±1.9) by block randomization. Both groups were advised to follow a Mediterranean diet by a dietician.

The control group was advised to do 45 minutes of physical activity per day by a physiotherapist, while the intervention group completed an 8-week personalized exercise program. The exercise program consisted of one-hour sessions three times per week. Participants first completed four weeks of resistance training with exercise bands and free weights, then shifted to an aerobic exercise program on a bicycle ergometer set at 60% of their target heart rate for the final four weeks. Quality of life was assessed using the Chronic Liver Disease Questionnaire (CLDQ), which includes the subscales of fatigue, abdominal symptoms, systemic complaints, emotional state, anxiety, and activity.

Results: There were 8 males and 2 females in both groups, and the mean body mass index of the intervention group was 28.3±2.6 and the mean body mass index of the control group was 30.1±1.9. The groups were similar in terms of demographic characteristics. The intervention group

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showed significant improvements in CLDQ subscales fatigue (p=0.009) and abdominal symptoms (p=0.041), activity (p=0.036) and total score (p=0.014) and this difference was statistically significant between the groups.

The control group showed improvement only in systemic symptoms (p<0.046) but this did not constitute a significant difference between the groups. No adverse effects related to exercise were reported.

Conclusion: This study shows that 8-week personalized exercise programs supervised by a physiotherapist can significantly improve the quality of life in individuals with MASH. Personalized exercise programs deserve greater focus in research, and exercise professionals should be integrated into the multidisciplinary care team for the clinical follow-up of patients diagnosed with MASH.

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Disclosure: Nothing to disclose.

PP0857

MAFLD: EPIDEMIOLOGICAL PROFILE AND FIBROSCAN® ASSESSMENT

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Introduction: Metabolic dysfunction-associated fatty liver disease (MAFLD) is the hepatic manifestation of a systemic metabolic disorder and is currently the leading cause of liver disease worldwide. Though often asymptomatic and slow to progress, MAFLD can lead to severe complications such as cirrhosis, hepatocellular carcinoma, and the need for liver transplantation in certain high-risk patients.

Aims & Methods: This study aims to describe the epidemiological, clinical, biological, and imaging characteristics of MAFLD, with a focus on FibroScan® results.

A retrospective cross-sectional study was conducted from August 2021 to December 2024, including 85 patients diagnosed with MAFLD and referred for FibroScan® to assess hepatic fibrosis and/or steatosis.

Results: Among 490 patients undergoing FibroScan®, 85 (17%) had MAFLD. The average age was 53 years (range: 20–72), with a female predominance (74%, sex ratio M/F: 0.3).

Metabolic syndrome were present in 90% of cases: hypertension (26.8%), type 2 diabetes (31.7%), dyslipidemia (13.4%), BMI overweight (43%), and obesity (41%).

Clinically, 78% were asymptomatic. Other reported symptoms included hepatic colic (15.8%), asthenia (1.2%), and nonspecific abdominal pain (4.8%).

Biological abnormalities (cholestasis and/or cytolysis) were found in 35.3% of patients, while 64.6% had normal liver function tests.

Abdominal ultrasound revealed steatosis in 81% of cases, hepatomegaly in 11%, heterogeneous liver in 2.4%, and a normal liver in 4.8%.

Reliable FibroScan® results were obtained using the M probe in 72% of patients; the XL probe was required in 28%.

The mean liver stiffness was 5.2 kPa: 90.2% had F0–F1 fibrosis, 9.7% had F2, with three cases of advanced fibrosis (F3–F4). The mean CAP value was 290 dB/m, with steatosis grades distributed as follows: S0 (19.5%), S1 (17%), S2 (17%), and S3 (46.3%).

Conclusion: MAFLD is a highly prevalent and often benign disease, but it carries the potential for serious complications, notably cirrhosis and hepatocellular carcinoma. Given its typically silent progression, non-invasive tools like FibroScan® play a key role in early detection and monitoring. Management is primarily preventive, focusing on addressing underlying metabolic risk factors.

Disclosure: Nothing to disclose.

PP0858

ZONULIN LEVEL AS A RISK FACTOR FOR NONALCOHOLIC FATTY LIVER DISEASE IN PATIENTS WITH DYSLIPIDEMIA

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Introduction: Increased intestinal permeability, mediated by zonulin, has been implicated in the pathogenesis of insulin resistance, dyslipidemia, and non-alcoholic fatty liver disease (NAFLD). Small intestinal bacterial overgrowth (SIBO) may further contribute to metabolic dysfunction through endotoxin-induced inflammation.

Aims & Methods: The aim was to determine zonulin concentrations in patients with dyslipidemia and NAFLD and analyze their relationships with metabolic parameters and the prevalence of small intestinal bacterial overgrowth (SIBO). A total of 342 patients with dyslipidemia were enrolled; NAFLD was confirmed in 152 of them based on ultrasound and steatometry. Fecal zonulin levels were measured using ELISA; SIBO was assessed by hydrogen breath test. Lipid profiles, high-sensitivity C-reactive protein (hsCRP), and HDL-C levels were also analyzed.

Results: Zonulin levels in patients with NAFLD (115±2.76 ng/g) exceeded the normal range (≤107 ng/g), compared to 87±0.28 ng/g in the dyslipidemia group without NAFLD (p<0.05). SIBO was present in 45% of dyslipidemic patients and 53.2% of those with NAFLD. A positive correlation was found between zonulin and hsCRP (r=0.65) and triglycerides (r=0.58), and a negative correlation between SIBO and HDL-C (r=–0.78) in the dyslipidemia group.

Conclusion: Elevated zonulin and increased intestinal permeability are associated with higher cardiometabolic risk in patients with NAFLD and dyslipidemia. Both zonulin and SIBO may serve as potential therapeutic targets for preventing progression of NAFLD and related metabolic disturbances.

Disclosure: Nothing to disclose.

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PP0859

FROM GUT TO LIVER: MICROBIOTA-BASED INTERVENTIONS FOR NAFLD

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Introduction: Nonalcoholic fatty liver disease (NAFLD) is a widespread metabolic disorder associated with intestinal dysbiosis, low-grade systemic inflammation, and small intestinal bacterial overgrowth (SIBO). Targeting the gut-liver axis through microbiome-modulating therapies may represent a promising approach to disease management.

Aims & Methods: The aim was to investigate gut microbiota alterations in NAFLD patients and assess the efficacy of microbiota-based therapy on metabolic, inflammatory, and structural liver parameters. The study included 152 NAFLD patients and 47 controls. Gut microbiota composition was evaluated using qRT-PCR, and SIBO was diagnosed via hydrogen breath testing. The therapeutic protocol included rifaximin (200 mg TID for 14 days), *Saccharomyces boulardii* CNCM I-745 (1 capsule BID for 30 days), and essential phospholipids (600 mg TID for 30 days). Liver function tests, inflammatory markers, and steatometry were assessed before and after treatment.

Results: Alterations in gut microbiota composition indicative of dysbiosis—including increased Firmicutes/Bacteroides ratio and Actinobacteria abundance—were identified in NAFLD. Strong correlations emerged between the F/B index and triglycerides ($r = 0.78$), TNF-α ($r = 0.77$), and between Actinobacteria and hsCRP ($r = 0.66$).

SIBO rates declined significantly post-treatment (51% → 17%). Liver enzymes, inflammatory markers, and steatometric parameters improved substantially ($p<0.05$).

Conclusion: Microbiota-targeted therapy combining rifaximin, *S. boulardii*, and phospholipids effectively corrected dysbiosis, reduced inflammation, improved metabolic parameters, and promoted liver structural recovery in NAFLD patients. These findings support incorporating microbiome-focused interventions into routine clinical practice for NAFLD.

Disclosure: Nothing to disclose.

PP0860

MODULATING THE GUT-LIVER AXIS: PROBIOTIC THERAPY FOR MASLD MANAGEMENT

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD), has emerged as one of the leading causes of chronic liver disease worldwide. MASLD is closely associated with components of metabolic syndrome, including obesity, insulin resistance, dyslipidemia, and hypertension. The role of gut microbiota dysbiosis in MASLD pathogenesis is increasingly recognized, with microbial imbalance contributing to systemic inflammation and metabolic disturbances.

Aims & Methods: The aim was to determine gut microbiota composition in patients with MASLD and evaluate the impact of probiotic therapy on metabolic, inflammatory, and lipid parameters. The study included 152 patients diagnosed with MASLD.

Diagnosis was based on clinical, laboratory, and imaging findings, including ultrasound and steatometry, alongside cardiometabolic risk factors. Gut microbiota composition was evaluated using quantitative real-time PCR (qRT-PCR). Patients received a combined probiotic therapy containing *Saccharomyces boulardii* CNCM I-745, *Lactobacillus acidophilus* (LA-5), and *Bifidobacterium lactis* (BB-12). Changes in biochemical markers, lipid profiles, inflammatory parameters, and gut microbiota composition were assessed before and after therapy.

Results: At baseline, patients with MASLD demonstrated gut microbiota dysbiosis, characterized by an increased proportion of Firmicutes (50.3%) and Actinobacteria (15.34%), and an elevated Firmicutes/Bacteroides (F/B) index (5.02). Following probiotic therapy, significant improvements were observed: the proportions of Firmicutes and Actinobacteria decreased, the F/B index was reduced, and levels of ALT, AST, high-sensitivity C-reactive protein (hsCRP), total cholesterol, and triglycerides declined, while HDL-C levels increased. Positive correlations were identified between elevated Firmicutes and Actinobacteria and markers of cardiometabolic risk, such as hsCRP and triglycerides.

Conclusion: Gut microbiota dysbiosis is strongly associated with metabolic and inflammatory disturbances in MASLD. Combined probiotic therapy effectively corrected microbial imbalance, reduced systemic inflammation, and improved metabolic and lipid parameters. These findings support the role of microbiota-targeted interventions as an adjunctive approach in the comprehensive management of MASLD.

Disclosure: Nothing to disclose.

PP0861

FROM DYSLIPIDEMIA TO MASLD: ZONULIN AS A MISSING LINK

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly referred to as non-alcoholic fatty liver disease (NAFLD), and dyslipidemia are among the most common metabolic disorders worldwide. Disruption of intestinal barrier integrity, as reflected by elevated zonulin levels, has been implicated in insulin resistance and lipid metabolism disorders, contributing to the pathogenesis of MASLD.

Aims & Methods: The aim was to evaluate fecal zonulin levels in patients with dyslipidemia and MASLD, and to explore their potential role as risk factors for disease development and progression. The study included 342 patients with dyslipidemia, among whom MASLD was diagnosed in 152 cases based on ultrasound and steatometry findings. Fecal zonulin levels were measured by ELISA. Small intestinal bacterial overgrowth (SIBO) was diagnosed using hydrogen breath testing. Correlations between zonulin, inflammatory markers (high-sensitivity C-reactive protein, hsCRP), lipid profiles, and SIBO prevalence were analyzed.

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Results: SIBO was detected in 45% of patients with dyslipidemia and 53.2% of patients with MASLD. Fecal zonulin levels in patients with MASLD (115±2.76 ng/g) significantly exceeded the normal upper limit (≤107 ng/g) and were higher compared to patients without steatosis. Zonulin levels positively correlated with hsCRP (r=0.65, p<0.05) and triglycerides (r=0.58, p<0.05). A strong negative correlation was observed between SIBO presence and HDL-C levels (r=–0.78, p<0.05) in patients with dyslipidemia.

Conclusion: Elevated zonulin levels and increased intestinal permeability are closely associated with systemic inflammation and lipid metabolism disturbances in MASLD. Zonulin and SIBO may represent novel therapeutic targets for the prevention and treatment of MASLD and metabolic disorders.

Disclosure: Nothing to disclose.

PP0862

FIRMICUTES, ACTINOBACTERIA, AND THE DECLINE OF AKKERMANSIA: MICROBIAL MARKERS IN MASLD

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD), is one of the most common chronic liver conditions worldwide. Alterations in gut microbiota composition and impaired gut barrier function are increasingly recognized as key contributors to MASLD development and progression.

Aims & Methods: The aim was to assess the gut microbiota composition in patients with MASLD and to evaluate its role in the development and progression of the disease. The study included 152 patients diagnosed with MASLD (mean age 46.5±0.89 years) and 150 control subjects without liver pathology (mean age 48.36±0.97 years). MASLD was diagnosed using liver ultrasound, hepatorenal index evaluation, and identification of cardiometabolic risk factors. Gut microbiota composition was analyzed by quantitative real-time PCR, focusing on the abundance of Firmicutes, Actinobacteria, Bacteroidetes, and Akkermansia muciniphila.

Results: Patients with MASLD demonstrated significantly increased levels of Firmicutes (50.3±2.46%) and Actinobacteria (15.34±2.87%) compared to controls (25.17±1.51% and 10.9±1.87%, respectively), alongside a higher Firmicutes/Bacteroides (F/B) index (5.02±2.61 vs. 2.1±1.74). The presence of Akkermansia muciniphila was substantially reduced in MASLD patients (33%) compared to the control group (63%).

Positive correlations were found between Actinobacteria levels and the F/B index (r=0.75, p≤0.05), as well as between the F/B index and triglyceride levels (r=0.68, p≤0.05). A negative correlation was observed between the presence of Akkermansia muciniphila and MASLD diagnosis (r=–0.69, p≤0.05).

Conclusion: MASLD patients show profound dysbiosis characterized by an increase in Firmicutes and Actinobacteria, an elevated F/B index, and a reduction in Akkermansia muciniphila abundance, indicating compromised gut barrier integrity. Akkermansia muciniphila may play a protective role against MASLD progression, highlighting the potential of microbiome-targeted therapies to prevent or mitigate disease development.

Disclosure: Nothing to disclose.

PP0863

FERRITIN PREDICTS LIVER FIBROSIS IN STEATOSIS: A READILY AVAILABLE, UNDERUSED MARKER BEYOND INFLAMMATION

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Introduction: With the global rise in metabolic liver disease, there is an urgent need for scalable and affordable tools to detect liver fibrosis early—particularly in settings where elastography is unavailable. Ferritin is widely available and routinely tested, yet its potential role in liver fibrogenesis remains under-recognized. We hypothesized that in patients with hepatic steatosis, elevated ferritin reflects fibrotic remodeling independently of systemic inflammation or metabolic comorbidities.

Aims & Methods: To determine whether hyperferritinemia independently predicts significant fibrosis in steatotic patients, beyond inflammatory and metabolic factors, and to assess the consistency of this association across age and sex subgroups. This retrospective monocentric study included adults diagnosed with hepatic steatosis by ultrasound between January 2020 and December 2024.

- Exclusion criteria: secondary causes of liver disease (alcohol >20 g/day, viral hepatitis, drug-induced steatosis), and factors affecting ferritin (active infection, malignancy, recent transfusion).
- Significant fibrosis: FIB-4 ≥1.0, per AASLD and EASL guidelines for early detection.
- Hyperferritinemia: serum ferritin >90 ng/mL, based on internal cut-offs linked to early fibrotic progression.
- Demographic, metabolic, and inflammatory data—including CRP—were collected. Subgroup analyses were conducted by age (<50 vs. ≥50) and sex. ROC curves and multivariate logistic regression were used to assess predictive value.

Results: Ninety-nine patients were included (mean age 51.9 ± 13.2 years; 74% female; mean BMI 30.8 kg/m^2). Metabolic comorbidities included diabetes (38%), hypertension (23%), and dyslipidemia (18%). Hyperferritinemia was found in 48.5%, associated with older age (p=0.027), but similar BMI and metabolic profile. Significant fibrosis (FIB-4 ≥1.0) was more frequent in this group (77% vs. 42%, p<0.001). Mean ferritin levels were higher in fibrotic patients (127 vs. 82 ng/mL, p=0.003), while CRP levels did not differ (p=0.78). In multivariate analysis, age (OR=1.04 per year, p=0.005), CRP (p=0.011), and ferritin >90 ng/mL (OR=2.9, 95% CI [1.2–7.3], p=0.021) were independently associated with fibrosis. Ferritin alone had an AUC of 0.72; its predictive value remained significant in both men (AUC=0.75) and women (AUC=0.69), and was strongest in patients <50 years (AUC=0.78). Combining ferritin with FIB-4 increased discriminative performance (AUC=0.76), suggesting complementary utility.

Conclusion: In hepatic steatosis, serum ferritin >90 ng/mL independently predicts significant liver fibrosis, even after adjusting for CRP and metabolic comorbidities. Ferritin—an inexpensive and underused biomarker—may serve as a pragmatic fibrosis signal in primary care or resource-limited hepatology settings. Its discriminative power in younger patients highlights its potential for early risk stratification.

These findings support the integration of ferritin into composite non-invasive scoring systems and pave the way for multicentric prospective validation. Incorporating ferritin into fibrosis screening workflows, including telemedicine triage, may reduce underdiagnosis and optimize care pathways in steatotic liver disease.

Disclosure: Nothing to disclose.

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PP0864

FROM DYSBIOSIS TO DISEASE: THE GUT MICROBIOME'S ROLE IN MASLD

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD), is a widespread chronic liver condition closely associated with obesity, insulin resistance, and dyslipidemia. Recent evidence points to gut microbiota dysbiosis as a key player in MASLD pathogenesis, affecting intestinal barrier function and systemic inflammation.

Aims & Methods: The aim was to investigate gut microbiota composition in patients with MASLD and assess its association with disease progression and gut barrier dysfunction. A total of 152 MASLD patients and 150 matched controls without liver pathology were examined.

Diagnosis was confirmed by ultrasound and steatometry, alongside assessment of cardiometabolic risk factors. Gut microbiota composition was analyzed using real-time PCR, focusing on Bacteroidetes, Firmicutes, and Actinobacteria. Hydrogen breath testing was used to diagnose small intestinal bacterial overgrowth (SIBO), and the presence of *Akkermansia muciniphila* was evaluated as a marker of gut barrier integrity.

Results: MASLD patients demonstrated a significant increase in Firmicutes (50.3%) and Actinobacteria (15.3%) proportions compared to controls, with a higher Firmicutes/Bacteroidetes (F/B) ratio. SIBO was detected in 52% of MASLD patients versus 40.5% in controls ($p<0.05$). The presence of *Akkermansia muciniphila* was notably lower in MASLD patients (33%) compared to controls (63%), suggesting impaired gut barrier function.

Strong positive correlations were observed between Actinobacteria abundance and the F/B index ($r=0.75$, $p<0.05$), while *Akkermansia muciniphila* presence negatively correlated with MASLD development ($r=–0.69$, $p<0.05$).

Conclusion: Gut microbiota dysbiosis, characterized by elevated Firmicutes and Actinobacteria and reduced *Akkermansia muciniphila*, is closely associated with MASLD pathogenesis. The findings highlight the potential of microbiota-targeted interventions for improving gut barrier function and modifying disease progression in MASLD patients.

Disclosure: Nothing to disclose.

PP0865

PREVALENCE, GENETIC CHARACTERIZATION, AND LABORATORY PROFILING OF ACUTE INTERMITTENT PORPHYRIA IN SOUTHWESTERN SERBIA: A CROSS-SECTIONAL STUDY

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Introduction: Acute intermittent porphyria (AIP) is a rare autosomal dominant disorder of heme biosynthesis, characterized by neurovisceral attacks and biochemical accumulation of porphyrin precursors. Despite its clinical severity, AIP often remains underdiagnosed, particularly in regions without established genetic testing programs. To date, no epidemiological or genetic studies have assessed the burden of AIP in southwestern Serbia. This study aimed to determine the prevalence of AIP, confirm genetic mutations, and analyze clinical and laboratory profiles of affected individuals.

Aims & Methods: We conducted a cross-sectional study involving 30 patients with clinically and biochemically confirmed AIP from two cities in southwestern Serbia, covering a total population of 141,333 inhabitants. The study was conducted over a one-year period. Among the 30 patients, 25 underwent genetic testing, and all were confirmed as heterozygous carriers of pathogenic *HMBS* mutations. Urinary levels of delta-aminolevulinic acid (ALA), porphobilinogen (PBG), uroporphyrin, and coproporphyrin were measured.

Routine analysis including complete blood count and biochemical parameters was performed during asymptomatic periods. Demographic data, family history of porphyria, the annual frequency of attacks, and givosiran therapy use were documented.

Results: Among the 30 patients, 4 were male (13.3%) and 26 were female (86.7%). The mean age was 47.69 years (range: 24–80 years). Ten patients (33.3%) reported a positive family history of porphyria. Genetic testing confirmed *HMBS* heterozygosity in all 25 tested individuals (100%).

Out of the total, 27 patients (90%) experienced fewer than three attacks per year, while three patients (10%) had more than three attacks annually and were receiving treatment with givosiran. Three patients (10%) were diagnosed with hepatocellular carcinoma (HCC).

Urinary ALA and PBG levels were elevated across all patients. Routine laboratory analyses performed during asymptomatic periods generally fell within reference ranges; however, in a subset of patients, mild elevations of urea, total cholesterol, AST, and ALT were observed, along with decreased hemoglobin levels, suggesting underlying metabolic and hepatic alterations even in the absence of acute attacks.

The estimated prevalence of clinically and biochemically confirmed AIP in the studied population was approximately 21.2 cases per 100,000 inhabitants, and the prevalence of genetically confirmed AIP was approximately 17.7 cases per 100,000 inhabitants.

Conclusion: This study provides the first epidemiological and genetic data on AIP in southwestern Serbia, revealing a notable prevalence in the local population. Despite clinical remission, subclinical laboratory abnormalities, including elevated urea, cholesterol, AST, and ALT, as well as reduced hemoglobin levels, were detectable in a subset of patients.

A significant proportion of patients presented without a positive family history, emphasizing the importance of systematic screening. The association with hepatocellular carcinoma highlights the need for long-term surveillance.

Importantly, this study represents preliminary results from the first year of investigation. Further research is ongoing, as not all potential cases have been detected, and there is a clear need to expand genetic testing and clinical evaluation in this population to fully assess the burden and progression of AIP.

Disclosure: Nothing to disclose.

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PP0866

PLASMA PROTEOMIC PROFILES IN DIVERSE POPULATIONS IDENTIFY BIOMARKERS PREDICTING METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE UP TO 16 YEARS BEFORE ONSET

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is preceding to severe hepatic disorders, characterized by a high global incidence and poor prognosis. Early identification of high-risk individuals is crucial, yet accurate, long-term predictive tools remain scarce.

Aims & Methods: This study aims to identify accurate long-term biomarkers and develop predictive models for MASLD using plasma proteomics.

We analyzed proteomics data of 37,966 participants without baseline MASLD from southern United Kingdom, using multivariable Cox regression to conduct protein selection and model development. The top-ranking proteins were combined with clinical data and polygenic risk scores to develop models. The models were internally validated through 1,000 times bootstrap resampling within the Southern UK cohort, and externally validated in a preclinical Northern UK cohort (n = 14,986) and an inception Southern China cohort (n = 100). Cumulative MASLD incidences were compared across baseline protein concentration quintiles, and temporal trends in protein levels were assessed.

Results: Of the 2,737 proteins analyzed, five emerged as robust predictors for MASLD incidence within 5 years: CDHR2 (AUC = 0.812), GGT1 (AUC = 0.797), FUOM (AUC = 0.791), KRT18 (AUC = 0.789) and ACY1 (AUC = 0.777). Participants with higher baseline levels of these proteins faced markedly increased risks of MASLD (HRs = 7.05 - 9.81).

In the Northern UK cohort, the combination of 5 proteins showed prominent validated predictive accuracy across different time frames: 5-year (AUC = 0.838), 10-year (AUC = 0.772), over 10-year (AUC = 0.742), all-time (AUC = 0.756).

Combined with clinical predictors, the predictive performance is further enhanced: 5-year (AUC = 0.904), 10-year (AUC = 0.848), over 10-year (AUC = 0.799), all-time (AUC = 0.822). In the Southern China Inception cohort, the 5-Protein Panel also showed an outstanding performance (AUC = 0.912).

Conclusion: Our study identified five key proteins, established and validated prominent predictive models of MASLD up to more than 16 years before onset, offering novel potential strategies for ultra-early prediction and intervention.

Disclosure: Nothing to disclose.

PP0867

WEIGHTING THE TRUTH: HOW OBESITY CHALLENGES FIBROSCAN® RELIABILITY IN FIBROSIS ASSESSMENT

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Introduction: Non-invasive assessment of liver fibrosis is central to managing chronic liver disease. FibroScan® (transient elastography), FIB-4, and APRI are widely used, but obesity may impair FibroScan® performance by altering probe penetration and signal quality, particularly in metabolic liver disease. This may lead to fibrosis overestimation and inappropriate clinical decisions.

Aims & Methods:

This study aimed to evaluate the concordance between FibroScan®, FIB-4, and APRI in assessing advanced liver fibrosis, stratified by BMI, and to identify predictors of diagnostic discordance—particularly in obese patients where FibroScan® performance may be compromised.

We retrospectively included 120 consecutive patients with valid FibroScan® measurements (≥10 acquisitions, IQR/median <30%) and available FIB-4, APRI, and BMI data. Patients with confounders (ascites, cytolysis ≥3× ULN, cholestasis, or hepatic mass) were excluded.

Advanced fibrosis was defined using validated thresholds:

- FibroScan® ≥10.5 kPa (AASLD/EASL)
- FIB-4 >2.67
- APRI ≥1.5
- Concordance between FibroScan® and FIB-4 was assessed via weighted kappa, APRI serving as a third comparator. Patients were stratified by BMI (<30 vs ≥30 kg/m^2). Predictors of discordance were analyzed using logistic regression.

Results: Among 120 patients (mean age 47.2 ± 14.5 years; 55% female; mean BMI 26.3 ± 4.9 kg/m^2), 63% were non-obese. Advanced fibrosis was diagnosed in 34% (FibroScan®), 28% (FIB-4), and 20% (APRI).

In non-obese patients (n=76), concordance between FibroScan® and FIB-4 was strong (κ = 0.75), with only 11.2% discordance. APRI confirmed FibroScan® classification in 87% of cases.

In obese patients (n=44), concordance dropped sharply (κ = –0.27), with 63.6% discordance. APRI also poorly agreed with FibroScan® (κ = 0.09), reinforcing the discordance. Most discrepancies involved FibroScan® indicating advanced fibrosis not supported by serum markers.

Logistic regression confirmed BMI ≥30 as an independent predictor of discordance (OR 2.63; 95% CI 1.55–4.67; p < 0.001); age and sex were not associated.

Conclusion: FibroScan®, FIB-4, and APRI show good concordance in non-obese patients. However, obesity significantly impairs FibroScan® reliability, leading to frequent fibrosis overestimation. These findings support BMI-adapted strategies, including XL probe use or integration of serum markers, to improve diagnostic accuracy in metabolic liver disease.

Disclosure: Nothing to disclose.

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PP0868

METABOLIC DYSFUNCTION–ASSOCIATED STEATOTIC LIVER DISEASE (MASLD) IN TYPE 2 DIABETES MELLITUS: PREVALENCE, RISK FACTORS, AND IMPACT ON DIABETIC COMPLICATIONS

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Introduction: Metabolic Dysfunction–Associated Steatotic Liver Disease (MASLD) is currently the leading cause of chronic liver disease worldwide, affecting more than 65% of patients with Type 2 Diabetes Mellitus (T2DM).

The aim of our study was to determine the prevalence of MASLD among T2DM patients, with or without coexisting significant fibrosis, to assess its association with diabetic complications, and to identify the factors linked to MASLD in our population.

Aims & Methods: We conducted a retrospective, descriptive, and analytical study including T2DM patients admitted to the Endocrinology Department in 2019. MASLD diagnosis was based on the presence of hepatic steatosis detected by abdominal ultrasound or a Hepatic Steatosis Index (HSI) >36. Fibrosis was assessed using the Fibrosis-4 (FIB-4) index (with a threshold of ≥1.3 or <1.3). Chronic Kidney Disease (CKD) associated with T2DM was defined by an estimated glomerular filtration rate (eGFR) <60 ml/min/1.73 m^2 and/or dipstick-positive proteinuria (>1+). Diabetic Peripheral Neuropathic Pain (DPNP) was assessed using the Douleur Neuropathique 4 (DN4) questionnaire, and Diabetic Retinopathy was diagnosed via dilated eye examination.

Results: Among the 202 patients included, the median age was 57 years (±12 years) with a marked female predominance (sex ratio = 2.6). Regarding comorbidities, 58.4% of patients had hypertension and 34.2% had dyslipidemia. The median Body Mass Index (BMI) was 27 kg/m^2 (±5.62), with overweight or obesity present in 61.4% of the cohort. The median duration of diabetes was 10 years (±7.8 years). Impaired renal function (eGFR <60 ml/min/1.73 m^2) was found in 9.9% of patients, and proteinuria in 6.9%. Diabetic neuropathy (DN4 score ≥4) was present in 7.9% of patients, while diabetic retinopathy was diagnosed in 37.6% of cases.

The prevalence of MASLD in our cohort was 73.3%. The median FIB-4 score was 0.91 (±0.63), with 25.7% of patients presenting a FIB-4 >1.3. Statistical analysis identified several factors significantly associated with the presence of MASLD: female sex (p=0.004), dyslipidemia (p=0.004), BMI (p<0.001), alanine aminotransferase (ALT) levels (p<0.001), low-density lipoprotein cholesterol (LDL-C) levels (p=0.006), and triglyceride levels (p=0.013). In particular, higher BMI was strongly associated with MASLD (Odds Ratio [OR] = 1.970; 95% Confidence Interval [CI]: 1.580–2.460; p<0.001), as were elevated ALT levels (OR = 1.115; 95% CI: 1.053–1.18; p<0.001).

However, after adjustment, LDL-C levels (p=0.163), triglycerides (p=0.402), dyslipidemia (p=0.329), and sex (p=0.360) were no longer significantly associated with MASLD. Finally, a significantly higher proportion of CKD was observed in patients with MASLD (22.3%) compared to those without MASLD (9.2%, p=0.036), although this association was not maintained in the subgroup with advanced fibrosis (FIB-4 >1.3).

Conclusion: Our study confirmed the high prevalence of MASLD among patients with T2DM. Early screening is therefore essential, particularly in female patients and in those presenting with chronic kidney disease or other associated cardiometabolic risk factors such as obesity and dyslipidemia.

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Disclosure: Nothing to disclose.

PP0869

ANTITUBERCULAR DRUG-INDUCED HEPATOTOXICITY IN DIGESTIVE TUBERCULOSIS: PREVALENCE AND RISK FACTORS

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Introduction: Tuberculosis is the second leading cause of death from a single infectious agent, following COVID-19. It is a curable disease when treated properly with a combination of several antitubercular drugs over several months. Hepatotoxicity is a major side effect of antitubercular treatment, ranging from transient elevations in liver enzymes to acute liver failure.

Our study aimed to assess the prevalence of antitubercular drug-induced hepatotoxicity and its associated risk factors in digestive tuberculosis.

Aims & Methods: We conducted a retrospective study including 87 patients who were consecutively treated for peritoneal, intestinal, or hepatic tuberculosis in our department from January 2, 2018, to December 31, 2024. Patients with underlying chronic liver disease were excluded from the study. Hepatotoxicity was diagnosed when liver enzyme levels exceeded five times the normal without symptoms, or more than three times the normal with clinical symptoms, after excluding other potential causes of liver injury. A combined HRZE regimen was defined as a fixed-dose combination of isoniazid (INH), rifampicin, ethambutol, and pyrazinamide.

Results: The mean age at diagnosis was 47 years [range 15-83 years], with a female predominance (F/M sex ratio = 2.8). Metabolic syndrome was present in 14.9% of cases, and malnutrition in 33.3%. Peritoneal tuberculosis was found in 78.7% of cases, intestinal tuberculosis in 16.7%, and hepatic tuberculosis in 4.6%. Histology confirmed diagnosis in 96.5% of cases. A combined HRZE regimen was prescribed in 82.8% of cases, and a fractionated regimen in 17.2%. An INH acetylation test was conducted in 79.3% of cases, identifying a slow acetylation phenotype in 17.2%.

Fifteen patients developed hepatotoxicity (17.2% of cases). The average time to onset was 15 days [range 4-150 days]. Eight cases of acute liver injury were observed, with seven (87.5%) severe forms. No cases of acute liver failure were observed.

Hepatotoxicity was significantly associated with female gender (p=0.005), the presence of metabolic syndrome (p=0.004), malnutrition (p=0.001), and the slow acetylation phenotype of INH (p<0.0001).

Conclusion: Our study revealed a high prevalence of hepatotoxicity induced by first-line antitubercular treatment in digestive tuberculosis, with severe hepatic injury in the majority of cases. Therefore, it would be prudent to tailor treatment based on the INH acetylation test, particularly for women, as well as individuals with metabolic syndrome or malnutrition.

Disclosure: Nothing to disclose.

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PP0870

THE CULPRITS BEHIND METHOTREXATE-PROPOSED HEPATOTOXICITY ARE MODIFIABLE RISK FACTORS

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Introduction: Methotrexate (MTX) remains the cornerstone of treatment for rheumatoid arthritis (RA) and psoriatic arthritis (PsA), yet concerns about hepatotoxicity often lead to premature discontinuation.

Aims & Methods: This study aimed to reassess the risk of liver fibrosis in long-term methotrexate therapy and to identify clinical, metabolic, and treatment-related factors associated with hepatic changes. This was a prospective cohort study including 201 patients with rheumatoid arthritis (RA) or psoriatic arthritis (PsA) undergoing MTX therapy for ≥6 months. Liver stiffness was assessed using transient elastography (Fibroscan®) and acoustic radiation force impulse (ARFI). Hepatic steatosis was evaluated using the controlled attenuation parameter (CAP). Serum-based indices (FIB-4, APRI) were calculated.

Clinical and metabolic data—including BMI, waist/hip circumference, lipid profile, glucose, comorbidities, and alcohol use (B-peth)—were collected. Associations with liver parameters were analyzed using Spearman correlation and multivariable linear regression adjusted for age, sex, and diagnosis.

Results: Liver stiffness values were generally low, with minimal fibrosis detected by Fibroscan (median 4.7 kPa) and ARFI (median 2.85 kPa). ARFI and Fibroscan values showed weak but significant correlation ($\rho = 0.19$, $p = 0.03$). APRI, but not FIB-4, modestly correlated with both imaging modalities.

Multivariable regression identified no independent predictors of increased liver stiffness. In contrast, CAP-measured steatosis was significantly associated with higher BMI ($p = 0.002$) and waist circumference ($p = 0.001$), while MTX dose, duration, and alcohol intake (B-peth) showed no associations. Principal component analysis confirmed that steatosis clustered with metabolic rather than pharmacologic variables.

Conclusion: In this real-world cohort, significant liver fibrosis was rare despite prolonged MTX exposure. Hepatic steatosis, rather than fibrosis, emerged as the dominant hepatic alteration, driven primarily by modifiable metabolic risk factors.

These findings support the use of targeted metabolic assessment and intervention in MTX-treated patients rather than routine discontinuation based on fibrosis concern.

Disclosure: Nothing to disclose.

PP0871

IMPACT OF HAVING A FIBROSCAN-BASED BIOFEEDBACK ON ALCOHOL CONSUMPTION: A QUALITATIVE ANALYSIS OF ALCOHOL RECOVERY NARRATIVES AND INTERVIEWS

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Introduction: Providing people who have alcohol use disorder (AUD) with feedback about their liver (biofeedback using Fibroscan) may influence their drinking behaviour. It is unclear if and how having a Fibroscan may support behaviour change. There are also questions about how receiving a 'normal' result influences an individual's feelings about their alcohol use.

Aims & Methods: This qualitative study aimed to explore people's experiences of receiving different Fibroscan results and the impact they felt it had on them. Seventeen qualitative transcripts (11 recovery narratives, 6 post-trial interviews) from the KLIFAD trial (ISRCTN16922410) were analysed. An initial realist programme theory of a complex intervention was produced using the literature and other psychological theories of behaviour change to predict how the Fibroscan may influence behaviour change.

A deductive qualitative analysis was conducted to find evidence for an initial programme theory of how the KLIFAD intervention may influence alcohol-related behaviour change. Initial ideas in the programme theory were amended based on qualitative insights.

Results: Of these participants, ten were ≥50 years of age, nine were male, 14 were white and three were of minority ethnicity, most were heavy drinkers. Fibroscan results ranged from normal (≤7 kPa=9), intermediate (8–14 kPa=4) to high (≥15 kPa=4).

At follow-up ten reported abstinence and seven reduction in alcohol consumption, none reported increased consumption. Fibroscan was seen by some participants as an opportunity (or a privilege not all get to have) to find out about their health, and a chance to "see inside what is going on". Those reflecting on receiving high scores described it as a *shock*, *rock bottom*, or an *instant turning point*. Most receiving normal results felt reassured and saw it as an opportunity to change their behaviour. Reducing the score by reducing alcohol use was described as a motivator for three participants.

Not all participants (n=2) felt Fibroscan would help reduce alcohol use, often due to acute drinking, co-existing mental health diseases, hopelessness, or memory issues. Other factors influencing its impact included health priorities, loneliness, supportive relationships, age, symptoms such as pain, language barrier, stage of change, recognition of alcohol issues, and service engagement. One participant expressed guilt about self-inflicted harm when preparing for the test.

Conclusion: This study highlights the varied impact of Fibroscan biofeedback on alcohol-related behaviour change, influenced by individual circumstances such as mental health, readiness to change, and social support. High Fibroscan scores often prompted reflection and action, while normal scores reassured and motivated. Addressing barriers and stigma could enhance its effectiveness.

Disclosure: Nothing to disclose.

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PP0872

CHECKPOINT INHIBITOR-INDUCED LIVER INJURY IN PATIENTS WITH ADVANCED SKIN CANCER: A SINGLE-CENTER EXPERIENCE

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Introduction: Immune checkpoint inhibitors (ICIs) are now considered the first-line treatment for advanced melanoma and non-melanoma skin cancers. One of the most commonly reported adverse events is checkpoint inhibitor-induced liver injury (ChILI), which occurs in approximately 10% of patients. For those who develop ChILI, it remains uncertain whether permanent discontinuation of the drug is necessary or if treatment can be safely resumed after a temporary pause, either with the same agent or with a different class of ICI. This report shares our center's experience with ChILI in patients receiving care for skin neoplasms at a tertiary-level facility.

Aims & Methods: This study aimed to analyse the prevalence of ChILI among patients treated with ICIs for skin cancers at our center. We included retrospectively all consecutive patients admitted to our facility who received ICI therapy for advanced melanoma or non-melanoma skin cancer between June 2016 and April 2024. The diagnosis and severity of ChILI were evaluated using the CTCAE v5.0 criteria, and the type of liver injury was classified based on international guidelines. Liver function tests (LFTs) were collected at baseline, after three treatment cycles, and at the time of suspected ChILI onset. Data on ChILI management and outcomes were also collected.

Results: We enrolled 111 patients (32.4% female, median age 75 years), including 70 patients (63.1%) with advanced melanoma and 41 (36.9%) with locally advanced or metastatic cutaneous squamous cell carcinoma. Among them, 3 patients (2.7%) received first-line therapy with anti-CTLA-4, 106 (95.5%) with anti-PD-1, and 2 (1.8%) were treated with a combination of both. The median duration of therapy was 10 months (IQR 5–21). ChILI developed in 8 patients (7.2%) after a median of 14 treatment cycles (IQR 7–20). According to the CTCAE severity scale, one case was grade 2, three were grade 3, and four were grade 4.

Among those who developed ChILI, 3 patients tested positive for anti-nuclear antibodies (ANA), 1 had a positive extractable nuclear antigen (ENA) test, and 2 had a known autoimmune disorder prior to initiating ICI therapy. All patients discontinued ICIs upon ChILI onset. Liver function tests normalized spontaneously in 4 patients, while 4 required corticosteroid treatment. Four patients underwent drug rechallenge, and only one of them received a new class of ICI.

Regarding extrahepatic adverse events, ChILI patients more frequently exhibited diarrhea and dermatological manifestations such as papular or bullous rashes (37.5% and 50%, respectively), with a significantly higher prevalence compared to patients without ChILI (7.8% and 15.4%, respectively; $p < 0.05$).

No significant differences in liver function test results were observed at baseline or after the third infusion between patients who developed ChILI and those who did not ($p > 0.05$).

Conclusion: Our study confirms the occurrence of liver toxicity associated with ICI therapy in cancer patients, with an incidence rate consistent with current international literature. The median time to ChILI onset—typically after multiple treatment cycles—supports the hypothesis of an indirect, immune-mediated mechanism rather than a direct toxic effect on the liver.

Notably, we found that ChILI is often preceded by dermatological and/or gastrointestinal adverse events.

Although based on a limited patient cohort, this observation may serve as an important clinical signal, helping physicians to anticipate ChILI, intervene earlier, prevent progression to more severe presentations, and initiate appropriate management promptly.

Disclosure: Nothing to disclose.

PP0873

EARLY BILIRUBIN DYNAMICS PREDICT PROGNOSIS IN SEVERE ALCOHOLIC HEPATITIS

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Introduction: Alcoholic hepatitis (AH) is a severe liver disease with high short-term mortality, making prognostic factors essential for patient management. Recently, it has been proposed that the trajectory of bilirubin evolution may serve as a prognostic predictor and help identify patients who do not benefit from corticosteroid therapy (CT).

This study aimed to evaluate the role of bilirubin trajectory as a predictor of prognosis and its potential utility in guiding treatment decisions in patients with AH.

Aims & Methods: Retrospective study including AH patients hospitalized between 2017 and 2025 in our hospital. Based on bilirubin trajectory between admission and day 4, patients were categorized into three groups: 'fast fallers' (FF) (bilirubin <0.8 x admission), 'static' (ST) (bilirubin of >0.9 – <1.2 x admission) and 'rapid risers' (RR) (bilirubin of ≥1.2 x admission) [1]. A p-value of < 0.05 was considered significant.

Results: The study included 99 patients (median age: 55 years [IQR 48-62], 83.7% male, median BMI: 25 kg/m^2), with a median follow-up time of 359 days (IQR 65-1155). Cirrhosis was present in 80%. From total patients 36% were FF 59% were ST and 5% were RR; Mortality rate between FF vs ST+RR showed no significant differences at any time point (p>0.05). Additionally, corticosteroid therapy did not influence 90-day mortality in FF patients (18% vs. 27%, p=0.55).

Analyzing only the patients with severe AH (MELD > 20) (55% of the cohort), 16% were FF, and 65.5% received corticosteroids. Mortality rates between FF and ST+RR, in severe AH patients were as follows: 12.5 vs 28.2% at 30 days, 12.5 vs 41.1% at 60 days and 18.8 vs 46.2% at 90 days. In survival analysis, FF patients at day 4 had significantly better 90-day survival compared to other groups (log-rank p=0.05). Among FF patients, corticosteroid therapy did not impact mortality (16% vs 20%, p=0.8).

Conclusion: This study highlights the prognostic value of bilirubin trajectory at day 4 in patients with alcoholic hepatitis. Severe AH patients categorized as 'fast fallers' demonstrated significantly lower mortality compared to other subgroups and CT did not significantly impact survival outcomes in this patients, suggesting that bilirubin trajectory may serve as an early indicator of good prognosis.

Disclosure: Nothing to disclose.

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PP0874

A NOVEL PROGNOSTIC MODEL TO PREDICT MORTALITY IN PATIENTS WITH DRUG INDUCED LIVER INJURY IN INTENSIVE CARE UNIT, A STUDY FROM TERTIARY CARE HOSPITAL, PAKISTAN

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Introduction: DILI is one of the causes of acute liver injury leading to increased mortality rates and usually requires intensive care unit (ICU) admission , however it is uncommon and often challenging to diagnose DILI. Proper prognosis for these patients is crucial for early referral for liver transplantation.

Aims & Methods: In light of the paucity of data on DILI especially from South East Asia, this study aims to evaluate the clinical spectrum and predictors of mortality in ICU admitted patients with suspected DILI by building and validating a novel scoring system.

462/231 cases were identified and categorized on basis of COIMS/RUCAM score[1] and the exclusion of other liver diseases. Ratio (R value) between ALT(Alanine transferase) and AP (alkaline phosphatase) expressed as R>5; hepatocellular, R<2;cholestatic, and R >2 <5;mixed. Clinical and laboratory parameters were analyzed to identify the predictors of hospital mortality. The prognostic values for mortality were assessed by ROC analysis. A novel model was developed and internally validated using fivefold cross-validation.

Results: Out of 231 patients, there were 133 (57.6%) males with the mean age being 50.83 years (range, 20-94 years). DILI was classified as definite or highly probable in 31.1%, probable in 62.5%, and possible in 7.4% of cases. Pattern of liver injury was hepatocellular in 25.1%, cholestatic in 56.17% and mixed in 18.72% of patients. Mean total bilirubin levels, ALT and AP levels were 5.37mg/dl, 358.65 IU/L and 168.68 IU/L respectively.

Antituberculous (ATT) drugs were found to be the most common cause (63.9%) followed by homeopathic or herbal meds (9.5%), antiarrhythmic drugs (8.7%), statins (7.3%), antibiotics (6.1%), antifungals (5.6%), chemotherapeutic agents (4.4%) and antiepileptics (1.7%). Altered mental status was present in 98 (21.6%) of patients while rest presented with abdominal pain, vomiting, jaundice & pruritus; 57.1%, 57.1%, 54.1%, and 42.3% respectively.

In-hospital mortality was 26.5% and prolonged hospital stay was observed in 35.93% of patients. On multivariate analysis mortality was significantly greater in patients with altered mental status, male gender, increased age, hepatocellular pattern of DILI, increased INR (>1.5), increased bilirubin levels(>3 mg/dl) and arterial gradient (A-a gradient).

The newly developed prognostic model, incorporating age,gender,pattern of liver injury, alveolar–arterial gradient (A-a gradient),INR, total bilirubin level, INR, and HE grades, exhibited significantly improved performance in predicting 1-month and 3-month mortality with AUROC of 0.863 and 0.829, respectively.

Likewise, prolonged hospital stay was associated with female gender, increased ALT, AST aspartate aminotransferase levels, and mixed pattern of DILI.

Conclusion: DILI leads to increased mortality, and use of this prognostic model predicting mortality in patients with drug induced liver injury in intensive care unit can be useful. However, further rigorous validation is required in other liver-related ICUs.

Disclosure: Nothing to disclose.

PP0875

ASSESSMENT OF LIVER FIBROSIS IN RHEUMATOID ARTHRITIS PATIENTS TREATED WITH METHOTREXATE: CONTRIBUTION OF NON-INVASIVE BIOLOGICAL SCORES

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Introduction: Methotrexate (MTX) is a cornerstone in the treatment of rheumatoid arthritis (RA). However, its potential hepatotoxicity, combined with the high prevalence of metabolic comorbidities in RA patients, warrants careful monitoring.

Aims & Methods: This study aimed to assess the prevalence of liver fibrosis in RA patients treated with MTX using validated non-invasive biological scores.We conducted a cross-sectional descriptive study involving 71 RA patients. Clinical, biological, and therapeutic data were collected. Liver fibrosis was assessed using the FIB-4, APRI, and NAFLD fibrosis scores. Statistical analysis was performed with SPSS.

Results: The mean age was 56.1 ± 11.3 years, with a female predominance (F/M ratio = 3.23). Metabolic syndrome was present in 43% of patients, and 7% reported alcohol consumption. The mean BMI was 28.4 kg/m^2. The average DAS28 was 4.2, indicating moderate to high disease activity. The mean duration of MTX exposure was 8.5 years, with a mean cumulative dose of 5.2 g.

FIB-4 score showed a low fibrosis risk (<1.3) in 66.7% of cases, intermediate risk (1.3–2.67) in 27.8%, and high risk (>2.67) in 5.5%. The APRI score was <0.5 in 90.3% of patients and ≥0.5 in 9.7%. NAFLD fibrosis score was low (<–1.455) in 58.7%, indeterminate in 34.9%, and high (>0.675) in 6.4%. High-risk patients more frequently exhibited metabolic syndrome (p=0.01) and elevated DAS28 scores (p=0.03).

Conclusion: While most RA patients treated with MTX show a low biological risk of liver fibrosis, approximately 6–10% present a high-risk profile. These findings highlight the importance of regular liver fibrosis screening, particularly in patients with metabolic risk factors or active inflammatory disease.

Disclosure: Nothing to disclose.

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PP0876

COMPARATIVE ASSESSMENT OF MELD, MELD-NA AND MDF IN ALCOHOLIC HEPATITIS MORTALITY: A SYSTEMATIC REVIEW AND DIAGNOSTIC META-ANALYSIS

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Introduction: Alcoholic hepatitis (AH) is a form of acute liver injury with significant morbidity and mortality. Several scoring systems have been developed and used to predict mortality, including Maddrey's discriminant function (MDF), model for end-stage liver disease (MELD) and modified MELD including sodium (MELD-Na).

Aims & Methods: We conducted a systematic review of Pubmed, Embase, Cochrane Library and Web of Science to compare the accuracy of these scores in AH mortality prediction. The inclusion criteria include studies evaluating MDF, MELD and MELD-Na prognostic models for mortality, either as an assessment of a single score or in combination in patients with alcohol-related hepatitis.

Results: The pooled sensitivity and specificity of MDF for mortality prediction in 28 days were 0.862 (95%CI 0.714–0.940) and 0.538 (95%CI 0.322–0.741), respectively. In 90 days were 0.862 (95%CI 0.714–0.940) and 0.538 (95%CI 0.322–0.741), respectively.

The pooled sensitivity and specificity of MELD for mortality prediction in 28 days were 0.809 (95%CI 0.676–0.896) and 0.753 (95%CI 0.636–0.842), respectively. In 90 days were 0.809 (95%CI 0.676–0.896) and 0.753 (95%CI 0.636–0.842), respectively.

The pooled sensitivity and specificity of MELD-Na for mortality prediction in 28 days were 0.707 (95%CI 0.520–0.842) and 0.679 (95%CI 0.474–0.832), respectively. In 90 days were 0.707 (95%CI 0.520–0.842) and 0.679 (95%CI 0.474–0.832), respectively. No significant statistical difference was observed when comparing AUC of these scores.

Conclusion: Our study demonstrated no statistical difference between the prognostic models used at the admission of a patient with alcoholic hepatitis.

Disclosure: Nothing to disclose.

PP0877

CLINICAL OUTCOMES AND RISK FACTORS IN PATIENTS WITH ALCOHOLIC LIVER CIRRHOSIS: A RETROSPECTIVE ANALYSIS

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Introduction: Alcoholic liver cirrhosis (ALC) is a complex condition marked by numerous complications, often leading to a poor prognosis and high mortality rates. This study evaluates the clinical characteristics, causes of hospitalization, and factors associated with prolonged hospital stay and in-hospital mortality among patients with alcoholic liver cirrhosis.

Aims & Methods: A retrospective study was conducted on 357 patients diagnosed with alcoholic liver cirrhosis in a tertiary gastroenterology department. The mean age was 57.8 ± 9.8 years, and 85% were male. Data on causes of hospital admission, length of stay, and clinical complications were analyzed. Associations with prolonged hospitalization (defined as >3 days) and in-hospital mortality were assessed using univariate and multivariate statistical analyses.

Results: The overall in-hospital mortality rate was 21.5%. The most common causes of admission included urinary tract infections (57.7%), hepato-renal syndrome (28.2%), digestive hemorrhage (22.6%), acute alcoholic hepatitis (19.3%), spontaneous bacterial peritonitis (8.4%), and pulmonary infections (5.6%). The median hospital stay was 6.6 days.

Variables significantly associated with prolonged hospitalization in univariate analysis included ascites (p<0.0001), spontaneous bacterial peritonitis (p=0.01), hepatic encephalopathy (p<0.0001), urinary tract infections (p<0.0001), and hepato-renal syndrome (p=0.09). In multivariate analysis, only urinary tract infection was independently associated with prolonged hospital stay.

Regarding in-hospital mortality, significant associations in univariate analysis were found with spontaneous bacterial peritonitis (p<0.0001), acute alcoholic hepatitis (p<0.0001), digestive hemorrhage (p<0.0001), and hepato-renal syndrome (p<0.0001). In multivariate analysis, only hepato-renal syndrome remained independently associated with mortality (p<0.0001).

Conclusion: Urinary tract infection is an independent predictor of prolonged hospitalization in patients with alcoholic liver cirrhosis, while hepato-renal syndrome is the only independent predictor of in-hospital mortality. These findings highlight the importance of early recognition and management of infections and renal complications in this high-risk population.

Disclosure: Nothing to disclose.

PP0878

POST TRANSJUGULAR INTRAHEPATIC PORTOSYSTEMIC SHUNT HEPATIC ENCEPHALOPATHY: PREDICTORS, EARLY PREVENTION AND MORTALITY

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Introduction: Hepatic encephalopathy (HE) after Transjugular Intrahepatic Portosystemic Shunt (TIPS) affect 35-50% of patients. The use of prophylactic agents (Lactulose and Rifaximin) to prevent post-TIPS HE remains controversial and currently not recommended by professional guidelines. Our aim was to assess real world practice and identify potential predictors for post-TIPS HE and benefit from post-TIPS prophylaxis.

Aims & Methods: Retrospective cohort study of patients undergoing TIPS at the Tel-Aviv Medical Center (2016-2024). The primary outcome was overt HE, with secondary outcomes of liver transplantation (LT) and mortality.

Results: Seventy-nine patients underwent TIPS, median age was 60 years and 52% were male. Primary liver disease was metabolic dysfunction-associated steatohepatitis (MASH) in 34%, viral (20%), and alcohol-related liver disease (ALD) (16.5%). Indications for TIPS were variceal bleeding (52%) and refractory ascites (38%). Prior to the procedure 31% had ≥ 1 episodes of HE, 21.5% were treated with Lactulose and/or Rifaximin, and 58% were on betablocker treatment. At the time of the procedure, the patient's Child-Pugh score was A/B/C in 19%/66%/11% respectively, with a median

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MELD-Na of 10.9. Median follow up was 22.75 months. Overall, 22 patients died during follow-up and 8 underwent liver transplantation. 45 patients (57%) developed post-TIPS HE, 60% of them during the first three months. Early post-TIPS HE treatment was given to 49%: 10% Lactulose, 5% Rifaximin, and 34% both. Among those patients, 44% were previously (pre-TIPS) on HE treatment. Development of HE was not predicted by pre-or post-TIPS MELD-Na or Child-Pugh score, TIPS diameter, postTIPS gradient, or post-procedure ammonia levels. Rates of HE also did not differ according to initial prophylaxis treatment (59% with treatment, 56% without, p=1). Pre-TIPS AST and albumin were associated with post-TIPS HE (36.6U/L vs. 60.44 U/L, p=0.03; 34.6g/dl vs. 31.8g/dl, p = 0.05). Prior beta-blocker treatment was associated with post-TIPS HE (14% vs 32%, p = 0.01) but not varices as the indication for TIPS. Mortality rate was not affected by the occurrence of post-TIPS HE (28% among those who developed HE vs. 26% among those who didn't, p = 1).

Conclusion: Early post-TIPS preventive treatment was mostly given to patients previously treated with Lactulose/Rifaximin, however it did not affect the rate of post-TIPS HE. Development of HE did not increase mortality.

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Disclosure: Nothing to disclose.

PP0879

ARGININE MODIFIED BY STENOTROPHOMONAS MALTOPHILIA PROMOTES SPONTANEOUS BACTERIAL PERITONITIS BY INHIBITING CD8⁺ T CELL EFFECTOR FUNCTION

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Introduction: Spontaneous bacterial peritonitis(SBP) is the most common type of infection in patients with cirrhosis, with high mortality and recurrence rates[1].

The intestinal immune barrier is an important link in blocking intestinal bacterial translocation[2-3], but the mechanism of immune barrier changes in SBP in cirrhosis is unclear. The gut microbiota influences antitumor immunity by producing immunomodulatory metabolites[4].

However, microbial consumption of multiple metabolites may also influence host immune responses[5].

Here, we investigated the role of Stenotrophomonas maltophilia and its metabolites in the development of SBP.

Aims & Methods: We detected the bacterial composition of ascites and stool in patients with spontaneous bacterial peritonitis by 16S sequencing. The Stenotrophomonas maltophilia-SBP rat model was established by oral administration of carbon tetrachloride and Stenotrophomonas maltophilia. Metabolomics analysis was performed by liquid chromatography-mass spectrometry and single-cell sequencing was used to analyze changes in intestinal cell composition and gene expression. The biological functions of Stenotrophomonas maltophilia-conditioned medium and metabolites were determined in rat CD8⁺T cells.

Results: 16S sequencing found that the abundance of Stenotrophomonas maltophilia in SBP ascites samples was high and positively correlated with inflammatory indicators. Among 1876 patients with cirrhosis and ascites, patients infected with Stenotrophomonas maltophilia had significantly increased levels of procalcitonin, C-reactive protein, ascites lactate dehydrogenase, and polymorphonuclear leucocytes count. Animal experiments showed that Stenotrophomonas maltophilia derived from human ascites can promote the occurrence of SBP in rats. Single-cell sequencing results showed that it significantly increased CD8⁺T cell exhaustion in the ileum and reduced memory T cells.

Metabolomics analysis revealed that arginine was a significantly decreased metabolite in the supernatant of Stenotrophomonas maltophilia, and its downregulation was verified in rat serum and intestinal tissue.

Mechanistically, Stenotrophomonas maltophilia degrades arginine through arginine decarboxylase, inhibits mTORC1 activity, thereby reducing CD8⁺T cell effector function and ultimately promoting the progression of SBP. Arginine supplementation significantly inhibited the formation of SBP and improved intestinal barrier function.

Conclusion: Stenotrophomonas maltophilia has been shown to promote SBP in rats by degrading arginine. Arginine supplementation is a potential preventive measure for SBP.

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Disclosure: Nothing to disclose.

PP0880

IN VITRO TESTING OF RIVAROXABAN ANTICOAGULANT ACTIVITY IN CIRRHOSIS ACCORDING TO LIVER DISEASE SEVERITY

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Introduction: Factor Xa inhibitors, edoxaban, rivaroxaban, and apixaban, generally exhibit an acceptable safety and efficacy profile in patients with Child-Pugh A cirrhosis, while apixaban and edoxaban may also suitable in Child-Pugh B cirrhosis [1]. The suitability of rivaroxaban in this group remains uncertain and warrants further investigation. Only *in vitro* data

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are available of the use of DOACs in the most severe forms of liver disease. Rivaroxaban decreased thrombin generation by 55% in controls, but only by 11%-38% in acutely ill cirrhotic patients [2].

Given the variability and limitations of existing evidence, current data are insufficient to support definitive clinical guidelines for DOAC use in severe hepatic impairment.

Aims & Methods: We studied ex vivo anticoagulant effects of the anti-Xa targeting DOAC rivaroxaban to decrease peak height (PH) and increase lag time (LT) in cirrhosis according to the severity of liver disease defined by the Child-Pugh (CP) classification. The baseline haemostatic capacity measured by thrombin generation (TG) will be compared with TG after spiking with rivaroxaban in both preventive and therapeutic doses. Sixty adult patients with an established diagnosis of cirrhosis were enrolled from the in- and outpatient hepatology clinics of the University Hospital of Ghent and the AZ Delta Hospital.

We selected 20 patients with CP A, 20 patients with CP B, and 20 patients with CP C cirrhosis. Ex vivo TG was analysed using ST Genesia with DrugScreen reagent at baseline and after spiking plasma samples with rivaroxaban (final concentrations 0, 67, and 300 ng/mL).

Results: In cirrhotic patients, higher rivaroxaban concentrations resulted in an increase in LT and a decrease in PH across all three patient groups (CP A, B, and C). However, no significant differences were observed between these groups, suggesting a comparable prolongation of LT and reduction in PH. When examining the percentage change in LT and PH according to the CP stage, a similar percentage change in LT was observed across all three CP stages.

In contrast, a markedly reduced percentual change in PH was noted in the CP C group with increasing rivaroxaban dosing (from baseline to 67 ng/mL to 300 ng/mL), indicating that the anticoagulant effect was less pronounced in patients with more severe liver dysfunction.

Conclusion: This study evaluates rivaroxaban potency in patients with cirrhosis in both preventive and therapeutic doses with the ST Genesia TG test, according to the severity of liver disease. We found that rivaroxaban reduces ex vivo haemostatic potential less efficiently in CP C patients compared to CP A and B. This may indicate a reduced anticoagulant effect of rivaroxaban in more severe liver disease.

These findings raise the question whether dose-adjustments and monitoring may be clinically useful in cirrhotic patients with a more advanced stage of liver disease.

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Disclosure: Nothing to disclose.

PP0881

PREDICTIVE FACTORS OF SEXUAL DYSFUNCTION IN CIRRHOTIC PATIENTS

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Introduction: Cirrhosis is characterized by structural and functional hepatic alterations. The associated metabolic imbalances and psycho-neurological factors contribute to the high prevalence of sexual disorders and erectile dysfunction among affected patients.

The aim of our study was to identify factors associated with sexual dysfunction in cirrhotic patients.

Aims & Methods: We conducted a prospective cross-sectional study, collecting data from patients followed for cirrhosis at the gastroenterology department of Charles Nicolle Hospital in Tunis over a one-year period. Female sexual function was assessed using the FSFI questionnaire. Female sexual dysfunction was defined by an FSFI score below 26 (or below 20 in post menopausal women). Male sexual function was evaluated using the IIEF-15 questionnaire. Male sexual dysfunction was identified by an IIEF-15 score below 42.9. Erectile dysfunction was confirmed by an IIEF-5 score

below 26. An analytical statistical study was conducted, highlighting factors associated with sexual dysfunction in both female and male patient groups.

Results: We included 55 cirrhotic patients (24 women and 31 men), with a sex ratio (M/F) of 1.29 and a mean age at inclusion of 59.11 ± 12.16 years. Female sexual dysfunction was observed in 70.8% of female patients. Male sexual dysfunction was found in 61.3% of patients. Erectile dysfunction was reported in 74.1% of men.

Our study showed that the factors associated with female sexual dysfunction were: low economic status (p = 0.007), the presence of arterial hypertension (p = 0.026), and hepatic disease severity attested by Child-Pugh class B/C and MELD score >15 (p = 0.049 and p = 0.018, respectively).

In male patients, a higher prevalence of sexual dysfunction was found among smokers patients (p = 0.045; OR = 0.18; 95% CI [0.03; 1.05]) and them who consumed alcohol (p = 0.042; OR = 0.19; 95% CI [0.04; 0.94]). In multivariate analysis, only alcohol consumption was independently associated with male sexual dysfunction (p = 0.041; OR = 9.187). MELD score >15

Conclusion: In conclusion, our study highlights the high frequency of sexual disorders in cirrhotic patients and identifies several associated factors, notably the severity of liver disease, socioeconomic

status, and certain comorbidities. For these reasons, screening and appropriate management of sexual dysfunction is essential in cirrhotic patients especialy with advanced disease or comorbidities.

Disclosure: Nothing to disclose.

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PP0882

COMPARATIVE ANALYSIS OF ALBUMIN STRUCTURAL MICROHETEROGENEITY IN PORTAL, HEPATIC, AND PERIPHERAL CIRCULATION AND PROGNOSTIC SIGNIFICANCE IN PATIENTS WITH CIRRHOSIS UNDERGOING TIPS

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Introduction: In decompensated cirrhosis, portal hypertension (PH) and gut barrier dysfunction may contribute to a pro-oxidant, pro-inflammatory, and pro-thrombotic environment in the portal circulation, leading to severe complications. Previous studies have analyzed the alterations of human albumin (HA) structure in peripheral blood.

Aims & Methods: This study aimed to compare albumin microheterogeneity in blood samples from the portal vein, hepatic vein, and peripheral circulation in patients undergoing transjugular intrahepatic portosystemic shunt (TIPS), and to evaluate its potential role in predicting post-TIPS outcomes. Blood samples were collected from the portal vein, hepatic vein, and brachial vein in patients with decompensated cirrhosis during TIPS procedure placed for PH-related complications at Policlinico Umberto I, Rome. Albumin microheterogeneity was analyzed using liquid chromatography coupled with quadrupole time-of-flight (Q-TOF) mass spectrometry.

Results: Fifteen patients were enrolled in the study, mean age was 60±7 years, 11 (73%) were male and the majority had an alcohol related etiology [9(60%)], while the mean MELD-Na score was 14±3. Reversible oxidation of the HA cysteine 34 residue was the most abundant alteration found in the peripheral circulation [Nonmercaptalbumin-1 31% (IQR 27%-36%)], followed by the glycation of the HA molecule [16% (IQR 15%-18%)]. As a result of these alterations the amount of the native HA form was found to be significantly lower than that commonly found in healthy subjects [41% (IQR 38%-48%)].

An impaired structural profile of the HA molecule was also found in samples from the portal [Nonmercaptalbumin-1 31% (IQR 27%-34%); Glycated HA 16% (IQR 13%-18%)] and hepatic vein [Nonmercaptalbumin-1 30% (IQR 29%-37%); Glycated HA 17% (IQR 14%-18%)] although the extent of such alterations did not significantly differ between the three sites (p=0.629 and p=0.456 respectively).

As a result, no significant difference between the different compartments were found in the relative amount of the native HA form [portal: 44% (39%-49%); hepatic: 42% (38%-50%); peripheral 41% (38%-48%); p=0.593]. Despite this, a significant correlation between the native HA form and the MELD-Na score was found in the portal circulation (rho -0.579, p=0.025) but not in the hepatic (rho -0.365; p=0.179) or peripheral circulation (rho -0.241; p=0.401).

Moreover, the relative amount of the native form of HA was significantly associated with the development of ascites and hepatic encephalopathy within 6 months irrespective of the sampling site (Table 1).

Ascites			
	Yes	No	p-value
Portal vein (%)	37 (31-40)	48 (45-50)	0.005
Hepatic vein (%)	37 (28-40)	49 (46-52)	0.014
Brachial vein (%)	33 (27-40)	46 (41-52)	0.042
Hepatic encephalopathy			
	Yes	No	p-value
Portal vein (%)	39 (35-39)	48 (45-50)	0.004
Hepatic vein (%)	37 (37-38)	49 (46-52)	0.003
Brachial vein (%)	34 (32-38)	47 (42-52)	0.003

Table 1. Relative amount (%) of the native HA form found in the three vascular compartments in patients who developed or not ascites or hepatic encephalopathy within 6 months after TIPS. Data is reported as median (IQR).

Conclusion: Structural microheterogeneity of the HA molecule in the peripheral blood closely resembles that found in portal and hepatic circulation. The residual amount of the native HA form is associated with the development of ascites and hepatic encephalopathy within six months from TIPS placement

Disclosure: Nothing to disclose.

PP0883

ACTIVATED INTESTINAL FIBROBLASTS INDUCE INFLAMMATORY TH17 DIFFERENTIATION IN EXPERIMENTAL CIRRHOSIS

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Introduction: Increased intestinal permeability is a central factor in the disruption of the gut-liver axis during cirrhosis. Innovative strategies focused on limiting intestinal inflammation are needed to restore the integrity of the mucosal, endothelial, and epithelial barriers in cirrhosis. Modulating the interaction between intestinal fibroblasts (Fib_{INT}) and cellular immunity in the lamina propria may help to restore intestinal homeostasis in cirrhosis.

Aims & Methods: A prospective experimental study was conducted using mice with advanced liver injury induced by oral administration of CCl_4 for 12 weeks (n=5). Control mice received olive oil as vehicle (n=5). Fib_{INT} activation, chemokine production, and immune recruitment were assesed in ileum samples through gene (RT-qPCR) and protein expression (immunofluorescence). Activation states of primary Fib_{INT} (CD45-CD326-CD90+) isolated from the small intestine and colon of control and cirrhotic mice were evaluated, as well as the expression of molecules involved in T-cell activation (MHC-II, CD40, CD80, CD86, OX-40L, PD-L1) by flow cytometry. Co-cultures of Fib_{INT} (cirrhosis) and T lymphocytes (control), and vice versa, were conducted to determine the modulatory effects of cell-to-cell interactions.

Results: An increase in gene (*COL1A1, FAP, PDPN*) and protein (α-SMA, Pdpn, Vim) expression of stromal cell activation markers was observed in the ileum during experimental cirrhosis. Augmented expression of chemokines (*CCL2, CXCL10*) in the ileum of cirrhotic mice correlated with increased leukocyte (CD45+) infiltration into the villi. Local increases in intestinal lymphocyte (CD3+) and macrophage (F4/80+) populations were

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detected. In cirrhosis, the percentages of primary Fib_{INT} from the small intestine expressing stromal cell activation markers (Vim and Pdpn) and T-cell stimulation molecules (CD40, CD80, CD86) were elevated compared to control mice. Isolated Fib_{INT} from cirrhotic mice interacted with more CD4+ T lymphocytes *in vitro* and were more effective at inducing their activation (CD25+) and differentiation towards a Th17 profile (IL-17+). Similarly, CD4+ intestinal T lymphocytes from cirrhotic mice could activate Fib_{INT} isolated from control animals (Vim+, CD86+, OX40L).

Conclusion: Fib_{INT} are activated in experimental cirrhosis and acquire the ability to induce an inflammatory Th17 response. These findings suggest that modulating the interaction between intestinal fibroblasts and T lymphocytes may recover intestinal homeostasis in cirrhosis.

Disclosure: Nothing to disclose.

PP0884

REAL-WORLD IMPACT OF EXTENDED BAVENO VI CRITERIA TO RATIONALISE ENDOSCOPIC VARICEAL SCREENING IN PATIENTS WITH CHRONIC LIVER DISEASE: A MULTISITE COHORT STUDY

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Introduction: Transient elastography (TE) enables early detection of compensated advanced chronic liver disease (cACLD), which can progress silently to clinically significant portal hypertension and oesophageal varices. Although gastroscopy remains the standard for variceal detection and eradication, it is invasive, costly, and associated with procedural risks. The Baveno VI consensus guidelines propose that cACLD patients with platelet counts >150,000/µL and liver stiffness <20 kPa can safely forgo routine gastroscopy, relying instead on annual TE and platelet monitoring. Recent European and UK studies have demonstrated an incremental net monetary benefit of approximately $3819 per patient with this approach; however, global adherence to the criteria remains suboptimal.

This retrospective cohort study aimed to assess the cost effectiveness and safety of the Baveno VI criteria relative to screening gastroscopies.

Aims & Methods: A retrospective cohort study was performed at a multisite tertiary center in Melbourne, Australia, analyzing 423 adults with compensated cirrhosis (Child-Pugh A) who underwent screening gastroscopy between February 2020 and May 2024.

The study aimed to assess the cost effectiveness and safety of Baveno VI criteria relative to screening gastroscopies. Clinical data were collected from electronic medical records, and costs were derived from the hospital finance department and the Australian Medicare Benefit Scheme (MBS).

The cost of TE was extrapolated as AUD 272.54 and gastroscopy as AUD 4512.00. The primary outcome was the incremental cost benefit of applying Baveno VI criteria; secondary outcomes included complications, mortality, and screening trends.

Results: Out of 423 patients, 43.5% had undergone TE within the preceding two years, and 66 (15.6%) met Baveno VI criteria to avoid endoscopy, none of whom demonstrated high-risk varices or experienced variceal bleeding.

Adverse events after endoscopy were uncommon in 7 patients (1.7%), with no related hospital admissions. Variceal bleeding occurred in 2 patients (0.5%) and overall mortality was 5.7% during the study period, but neither occurred among those meeting Baveno VI criteria. The incremental cost was AUD 4186.00 per patient, and the incremental benefit per TE performed was AUD 1134.40 (Table 1). Although TE use increased and endoscopy frequency decreased over time the trend did not reach statistical significance (p=0.37).

	N (%)
Total patients	423
Non-invasive	
Transient elastography in previous 2 years	184 (43.5%)
Morbidity / Mortality	
Variceal bleeds (N, %)	2 (0.5%)
Deaths (N, %)	24 (5.7%)
Cost data ($AUD)	
Incremental benefit of Baveno VI criteria	$4186.00
Incremental benefit per transient elastography performed	$1134.40

Table 1: Demographic, clinical and cost data in compensated patients with cirrhosis from 2020 to 2024.

Conclusion: Baveno VI criteria offer a safe and cost-effective alternative to routine endoscopic surveillance in cACLD. adopting Baveno VI criteria yielded an incremental cost benefit of AUD $4186 per patient, without affecting variceal bleeding or mortality. In our study cohort, 15.6% of patients from unnecessary procedures without compromising clinical outcomes.

Improving TE access and adherence to these guidelines could significantly reduce healthcare costs and patient burden, underscoring the importance of broader uptake of non-invasive screening strategies.

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PP0885

SIMPLIFIED RISK ASSESSMENT FOR SARCOPENIA AND MUSCLE ABNORMALITIES IN CHRONIC LIVER DISEASE PATIENTS USING EXISTING NUTRITIONAL ASSESSMENT TOOL

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Introduction: Sarcopenia is characterized by loss of skeletal muscle mass and strength, and commonly observed in patients with chronic liver disease (CLD). However, simple and practical assessment tools for consideration of early intervention in patients with risk of progression of muscle abnormalities remain limited, as muscle volume assessment requires specialized expensive equipment not generally used by a family physician, e.g., computed tomography (CT) or bioelectrical impedance analysis.

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Aims & Methods: Using the Geriatric Nutritional Risk Index (GNRI), calculated based only on body weight, height, and serum albumin (AJCN 2005), the present study aimed to examine a novel scoring system used to assess risk of sarcopenia and muscle abnormality development in CLD patients.

This retrospective analysis was conducted using medical records of 1063 Japanese CLD patients (median age 70 years; male:female = 676:387; HCV:HBV:HBV and HCV:alcohol:others = 573:146:4:127:213; Child-Pugh A:B:C = 927:115:21; hepatocellular carcinoma history or complication = 300) who underwent related examinations from 2015 to 2024.

Muscle area at the middle of the L3 level shown by CT was evaluated using a 3D image analysis system, with the value for skeletal muscle index (SMI), determined based on muscle area (cm^2), calculated using the mean value for L3 level/height (m^2). Muscle volume loss (MVL) and handgrip strength decline (HGSD) were assessed according to the Japan Society of Hepatology criteria (SMI and HGSD cutoff values for male/female; 42/38 cm^2/m^2 and 28/18 kg, respectively).

Logistic regression analysis was then used to identify predictive factors for sarcopenia. Finally, a risk scoring system was developed based on the odds ratio (OR) values for significant factors.

Results: There were 145 patients (13.6%) with sarcopenia, 226 (21.3%) with HGSD, and 149 (14.0%) with MVL, with an abnormal GNRI status noted in 333 (31.3%). A scoring system was developed based on five predictive factors; abnormal GNRI (<98) (OR 4.88), elderly (65-74 years, OR 2.67), late-stage elderly (≥75 years, OR 6.09), female (OR 2.11), and advanced hepatocellular carcinoma (HCC) (TNM stage IV, OR 2.61), which remained following multivariate analysis.

Points were assigned based on OR value, with1 point for OR ≤3 and 2 points for OR ≥4. The rate of prevalence of sarcopenia, HGSD, and MVL showed a significant increase with a higher score, with those conditions seen in 2.4%, 13.8%, and 6.9%, respectively, of patients with low risk (0/1 points), 9.0%, 23.7%, and 19.8%, respectively, of those with moderate risk (2 points), 22.0%, 29.0%, and 18.8%, respectively, of those with high risk (3 points), and 38.8%, 27.5%, and 16.9%, respectively, of those with very high/super-high risk (4/5/6 points) (p<0.001).

Additionally, GNRI cutoff values for HGSD and MVL were 101.5 (AUC 0.670) and 97.5 (AUC 0.749), respectively, approximating the abnormal GNRI cutoff value of 98.

Conclusion: This GNRI-based scoring system is likely to be a simple, cost-effective tool that does not require any special equipment when assessing the risk of sarcopenia or muscle abnormalities. It may also be used in clinical practice as a method of identifying patients who require early intervention.

Disclosure: Nothing to disclose.

PP0886

ASSESSING PROPHYLACTIC STRATEGIES AND PREDICTIVE FACTORS FOR POST-TIPS HEPATIC ENCEPHALOPATHY: A RETROSPECTIVE STUDY

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Introduction: Transjugular intrahepatic portosystemic shunt (TIPS) is a well-established procedure frequently employed to manage portal hypertension associated conditions such as variceal bleeding, ascites, and hydrothorax. However, despite its benefits, TIPS placement is frequently complicated by post-TIPS hepatic encephalopathy (HE), a condition that can significantly impact patient outcomes and quality of life.

Currently, there is a lack of consensus and robust evidence regarding the optimal prophylactic treatments for post-TIPS HE. There is a recommendation suggesting the use of rifaximin, which is based solely on a single randomized controlled trial[1,2]. Thus, more data are needed to assess the efficacy of prophylactic therapies in a real-world clinical setting.

Aims & Methods: This work aims to compare the efficacy of different prophylactic treatments (lactulose, rifaximin or both) and identify predictive factors for post-TIPS HE.

A retrospective study was conducted at a tertiary TIPS referral centre, including all adult patients who undergone TIPS placement between January/2021 and June/2024. Clinical records were reviewed to collect demographic data, prior decompensations, analytical results, medication and episodes of post-TIPS HE. A minimum of 6 months of follow-up was required. Statistical analysis was performed using SPSS Statistics version 26®.

Results: We included 83 patients, most of them male (n=69;83.1%) with a mean age of 60.1±1.3 years old. Most patients had alcoholic liver disease (n=58; 69.9%), followed by metabolic dysfunction and alcohol-related liver disease (n=6;7.2%). Prior to the procedure 22 (26.5%) patients had previous episodes of HE: 12 (14.5%) were medicated with lactulose, 2 (2.4%) with rifaximin and 8 (9.6%) with both. The remaining 61 (73.5%) had no previous medication. The primary indications for TIPS placement were variceal haemorrhage (n=50; 60.2% [pre-emptive n=21;42.0%]) and refractory ascites or hepatic hydrothorax (n=27;32.5%). The mean pre-TIPS and post-TIPS direct portal pressure were 31.0±0.9 mmHg and 22.2±0.7 mmHg, respectively.

After TIPS placement 19 (22.9%) were discharged with lactulose, 3 (3.6%) with rifaximin, 20 (24.1%) with both and 41 (49.4%) with no medication. The mean follow-up time after TIPS placement was 22.3±2.0 months. Twenty-five (30.1%) patients had post-TIPS HE and there were no differences between therapeutic strategies in preventing such episodes (lactulose: OR 2.6 (95%CI 0.7-9.2) p-value 0.14; rifaximin: OR 1.1 (95%CI 0.1-19.0) p-value 0.94; both OR 2.4 (95%CI 0.7-8.2) p-value 0.15).

Those discharged with no medication had less post-TIPS HE (OR 0.2 (95%CI 0.1-0.6) p-value 0.008), although these had also less pre-TIPS HE (p-value 0,046). Previous variceal haemorrhage (OR 3.2 (95%CI 1.0-9.9) p-value 0.046) and lower glomerular filtration rate (OR 0.98 (95%CI 0.96-0.99) p-value 0.03) were associated with higher risk for post-TIPS HE.

Conclusion: Our study found no significant differences between prophylactic treatments for post-TIPS HE. Notably, patients who did not receive prophylaxis had lower incidence of post-TIPS HE, although they also had

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fewer pre-TIPS episodes of HE. This raises questions about the universal application of these treatments and suggests the need for a more individualized approach. The identification of risk factors such as a history of variceal haemorrhage and lower glomerular filtration rate may guide treatment decisions, emphasizing the need for tailored management strategies. Future studies with larger sample sizes and multicentre participation are needed to refine prophylactic recommendations.

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Disclosure: Nothing to disclose.

PP0887

CARVEDILOL VERSUS ESOPHAGEAL BAND LIGATION FOR PREVENTION OF PRIMARY VARICEAL HEMORRHAGE: A META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Esophageal varices are a leading cause of life-threatening upper gastrointestinal bleeding in cirrhotic patients. Carvedilol and endoscopic variceal ligation (EVL) are the main strategies for primary prophylaxis, but their comparative efficacy remains a topic of debate. Additionally, recent studies have explored combining carvedilol with EVL, but the potential added benefit of this approach remains unclear.

Aims & Methods: We conducted a systematic review and meta-analysis of randomized controlled trials (RCTs), performing a comprehensive search of PubMed, Embase, Cochrane, and ClinicalTrials.gov to identify and evaluate studies comparing carvedilol vs. EVL or carvedilol plus EVL vs. carvedilol alone in patients at risk of esophageal variceal bleeding. Statistical analyses were conducted using R and RStudio, applying a random-effects model with 95% confidence intervals.

Results: Nine RCTs comprising 1,443 patients were included. Carvedilol significantly reduced six-month variceal bleeding compared to EVL (RR 0.57; 95% CI 0.36–0.90; p=0.01; I^2=0%). No significant differences were found in overall variceal bleeding (RR 0.97; 95% CI 0.61–1.53; p=0.90; I^2=61%), one-year bleeding (RR 0.86; 95% CI 0.46–1.62; p=0.63; I^2=51%), one-year all-cause mortality (RR 1.31; 95% CI 0.91–1.90; p=0.15; I^2=0%), or bleeding-related mortality (RR 1.32; 95% CI 0.71–2.48; p=0.38; I^2=0%). Hypotension occurred in 3% (95% CI 0.00–0.05) of patients receiving carvedilol. Dysphagia (52%; 95% CI 0.25–0.80) and variceal recurrence (14%; 95% CI 0.00–0.31) were reported in the EVL group. Combination therapy showed no clear benefit over monotherapy for six-month bleeding (RR 1.69; 95% CI 0.90–3.17; p=0.10; I^2=0%) or mortality (RR 1.47; 95% CI 0.36–5.92; p=0.59; I^2=61%).

Conclusion: Carvedilol and EVL appear similarly effective for primary prophylaxis of variceal bleeding, though carvedilol may offer short-term benefits with fewer adverse effects. Combination therapy did not demonstrate added efficacy over monotherapy.

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PP0888

EXPANDING THE SCOPE OF TIPS: IS THERE A RENAL BENEFIT IN ADVANCED CHRONIC LIVER DISEASE? A TERTIARY REFERRAL CENTRE EXPERIENCE

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Introduction: Patients with advanced chronic liver disease (ACLD) are prone to different renal injury phenotypes, associated with poor prognosis and limited treatment options. Transjugular intrahepatic portosystemic shunt (TIPS) diverts portal blood flow, potentially increasing systemic blood volume, which can be reflected in renal hemodynamics.

Aims & Methods: We intend to evaluate the impact of TIPS on renal function in decompensated ACLD. Retrospective cohort study conducted at a large referral center, involving consecutive patients who underwent TIPS placement between January/2021 and June/2024, with a 180-day follow-up. Excluded patients in regular hemodialysis and aged ≤18.

We divided the study population into two groups based on baseline glomerular filtration rate (GFR ≥90 ml/min/1.73 m^2, calculated using the CKD-EPI formula): those with normal GFR (nGFR group) and those with reduced GFR (aGFR group).

We then analysed the demographic and clinical characteristics of each group, as well as the impact on renal function at two time points following TIPS placement: days 1–7 and days 90–180.

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Results: A total of 78 patients were included, 85% of whom were male (n=66), with a mean age of 61 ± 11 years. Of these, 31% (n=24) had an abnormal GFR prior to TIPS placement (aGFR group), while 69% (n=54) had a normal GFR (nGFR group).

Regarding the primary etiology of ACLD, alcohol-related liver disease (ALD) was the most common cause in both groups (72%, n=56), followed by viral hepatitis (10%, n=8) and metabolic dysfunction with ALD (MetALD) (8%, n=6). In both groups, most patients were classified as Child-Pugh score B (nGFR: n=26; aGFR: n=16) and the mean MELD-Na score was 15 ± 5 in the nGFR group and 18 ± 7 in the aGFR group (p <0.05).

The main indications for TIPS placement were similar across groups, with variceal hemorrhage—either as pre-emptive TIPS (30%, n=23) or rescue TIPS (31%, n=24)—and refractory ascites (31%, n=24) being the most frequent. In both groups, TIPS placement was most commonly dilated to 8 mm (77%, n=60), while 10 mm dilation was performed in 12% (n=9).

The table below presents a comparison of pre- and post-TIPS pressure values, along with the evolution of GFR in both nGFR and aGFR groups at pre-TIPS, 1–7 days post-TIPS, and 90–180 days post-TIPS.

	Direct portal pressure pre-TIPS (mean ± SD)	Direct portal pressure post-TIPS (mean ± SD)	p-value	Pre-TIPS GFR	1-7 Day Post-TIPS GFR	90-180 Day Post-TIPS GFR
nGFR	30±7mmHg	23±6mmHg	p=0.511	103 (IQR: 99-112)	107 (IQR: 100-115)	100 (IQR: 81-108)
aGFR	32±6mmHg	21±5mmHg	p=0.151	45 (IQR: 35-68)	76 (IQR: 38-92)	79 (IQR: 42-95)

Comparing the pre- and post-TIPS values in the nGFR group: there was a significant increase in GFR pre- and 1–7-day post-TIPS (Z:−3.772, p<0.001), but there were no changes comparing pre- and 90-180-day post-TIPS (Z:-1.779, p=0.075). However, comparing the pre- and post-TIPS values in the aGFR group: there was a significant increase in both GFR pre- and 1–7-day and 90-180-day post-TIPS (Z:−3.315, p<0.001; Z:-2.201, p=0.028; respectively).

Conclusion: We observed an early renal improvement shortly after TIPS placement, however, this improvement was only sustained over time in the group with previously altered renal function. The transient nature of the renal benefits observed may be attributed to the complex hemodynamic changes following TIPS, highlighting the need for further studies to optimize long-term renal outcomes.

Despite the limitations of this study, such as the small patient subset, its retrospective nature, the use of the eGFR equation based on creatinine in ACLD patients and incomplete follow-up, studies like this are necessary, as renal impairment in ACLD is linked to higher morbidity and mortality, making it a key prognostic factor for post-TIPS survival.

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Disclosure: Nothing to disclose.

PP0889

ENHANCING OUTCOMES IN DECOMPENSATED CIRRHOSIS: A MULTICENTER STUDY ON THE IMPACT OF A CARE BUNDLE APPROACH

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Introduction: Chronic liver disease remains one of the most common causes for hospital admission and is associated with high morbidity and mortality rates. Timely and adequate management within the first 24 hours is crucial for improving clinical outcomes in cases of decompensated cirrhosis.

This study aimed to evaluate the impact of implementing a care bundle checklist during the first 24 hours of hospital admission on management and outcomes of decompensated cirrhosis.

Aims & Methods: We developed an adapted care bundle (based on the BSG/BASL care bundle, AASLD quality measures proposal, Baveno VII consensus, and EASL guidelines) and compared two patient groups from two Portuguese hospital centers: Group-A (retrospective, pre-care bundle implementation) and Group-B (prospective, post-care bundle implementation).

Results: A total of 224 patients (Group-A: 148; Group-B: 76) were included. More than one decompensation was observed in 118 patients (52.7%), and 69 patients (30.8%) were experiencing their index hospitalization for liver decompensation. Regarding etiology, alcoholic liver disease was the most common (*n*=200; 89.3%).

The groups were similar in terms of gender, age, length of hospital stay, and Child-Pugh and MELD-Na scores at admission. The most common decompensations were gastrointestinal bleeding (*n*=91; 40.6%), ascites (*n*=75; 33.5%), and hepatic encephalopathy (*n*=72; 32.1%).

Group B was more likely to present with multiple decompensations at admission and exhibited significantly higher rates of acute kidney injury, hepatocellular carcinoma, acute-on-chronic liver failure, and portal vein thrombosis (p<0.05).

Following care bundle implementation, we found significant improvements in patient care. At admission, patients were significantly more likely to undergo urine and blood cultures, chest X-ray, and abdominal ultrasound (p<0.05), receive diagnostic paracentesis (p<0.05), and have an accurate alcohol history documented (p=0.04).

During admission, Group-B was significantly more likely to be prescribed lactulose for encephalopathy prophylaxis, vasopressor therapy for gastrointestinal bleeding, and venous thromboembolism prophylaxis, as well as undergo nutritional assessment (p<0.05).

While in-hospital mortality rates were similar (4.7% vs. 5.3%; p=1), 6-week mortality rates (8.8% vs. 3.9%; p=0.01) and 3-month readmission rates (31.1% vs. 15.8%; p=0.02) were significantly lower in the post-intervention cohort.

Conclusion: The implementation of a standardized care bundle was associated with higher rates of appropriate investigation and treatment within the first 24 hours of hospital admission and improved patient outcomes, reducing 6-week mortality and 3-month readmission rates.

Disclosure: Nothing to disclose.

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PP0890

ACCURACY OF MEFIB INDEX FOR RULING IN/OUT AT-RISK MASH AND SIGNIFICANT FIBROSIS. SYSTEMATIC REVIEW AND DIAGNOSTIC ACCURACY META-ANALYSIS

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Introduction: Patients with metabolic-dysfunction associated steatohepatitis (NAFLD activity score ≥4) and significant fibrosis (≥F2) (at-risk MASH) are at increased risk for disease progression. Following the approval of the first agent for the management of noncirrhotic MASH, timely identification of these patients is of major importance. Magnetic resonance elastography (MRE) combined with Fibrosis-4 index (MEFIB index) enables the noninvasive diagnosis of at-risk MASH and significant fibrosis, and has been assessed in several studies.

Aims & Methods: To assess the overall performance of MEFIB index for diagnosing at-risk MASH and significant fibrosis. We searched Medline, Cochrane library and Web of science up to February 2025. We included diagnostic accuracy studies assessing the performance of MEFIB index for ruling in (MRE ≥3.3 kPa plus FIB-4 ≥1.6) and ruling out (MRE <3.3 kPa plus FIB-4 <1.6) at-risk MASH and significant fibrosis using liver biopsy as the reference standard. We calculated pooled diagnostic accuracy estimates using bivariate random-effects models, and assessed risk of bias using the Quality Assessment of Diagnostic Accuracy Studies 2 tool.

Results: We included seven studies with 3,356 participants. Most studies were at unclear or high risk for bias primarily due to concerns related to patient selection. Across all participants, 54% were females and 55.8% had type 2 diabetes. The average mean age and body mass index (BMI) of participants were 56.2 years and 30.3 kg/m2 respectively. Mean aspartate aminotransferase and alanine aminotransferase were 47.0 IU and 61.3 IU, respectively. The average mean FIB-4 and MRE values were 1.75 and 3.6 kPa respectively.

Across studies with available data, the prevalence of at-risk MASH and significant fibrosis were 31.3% and 60.1% respectively. For ruling in at-risk MASH, MEFIB index yielded a pooled specificity of 0.94 (95 % CI 0.74, 0.99), and a positive likelihood ratio (LRp) of 5.3 (1.8, 15.7). For ruling out at-risk MASH, MEFIB index had a pooled sensitivity of 0.77 (0.62, 0.88) and a negative likelihood ratio (LRn) of 0.34 (0.23, 0.52). For ruling in significant fibrosis, MEFIB index resulted in a summary specificity of 0.93 (0.85, 0.97) with LRp 8.2 (4.5, 14.9). For excluding significant fibrosis, the pooled sensitivity and LRn of MEFIB index were 0.88 (0.79, 0.94) and 0.16 (0.08, 0.31) respectively.

Conclusion: MEFIB index has acceptable accuracy for diagnosing at-risk MASH and significant fibrosis. Proposed rule in (MRE ≥3.3 kPa plus FIB-4 ≥1.6) and rule out (MRE <3.3 kPa plus FIB-4 <1.6) thresholds can be used to identify both target conditions in high prevalence settings and facilitate patient recruitment in clinical trials.

Disclosure: Nothing to disclose.

PP0891

TADALAFIL IMPROVES ERECTILE DYSFUNCTION IN MEN WITH CIRRHOSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Erectile dysfunction (ED) is a common complication in men with cirrhosis, significantly impairing quality of life. However, evidence on effective pharmacological treatment in this population remains limited.

Aims & Methods: To evaluate the efficacy and safety of phosphodiesterase-5 (PDE-5) inhibitors for the treatment of ED in men with advanced chronic liver disease. A systematic search of MEDLINE, Embase, and the Cochrane CENTRAL Register of Controlled Trials was conducted to identify randomized controlled trials and prospective observational studies assessing PDE-5 inhibitors in cirrhotic patients with ED.

A pooled analysis was performed to evaluate changes in the erectile function domain of the International Index of Erectile Function (IIEF) and the incidence of adverse events. Heterogeneity was assessed using the Cochrane Q test and I^2 statistic, with $p < 0.10$ and $I^2 > 25\%$ indicating significant heterogeneity.

Results: Three studies including 129 patients, all treated with tadalafil, were analyzed. The characteristics of the included studies are summarized in Table 1. The pooled mean difference in the IIEF erectile function domain was 6.49 (95% CI: 5.01–7.98; $p < 0.01$), demonstrating a clinically meaningful improvement. Adverse events occurred in 31.6% (95% CI: 10.5–64.4%) of patients, with no serious events reported.

Study	Study design	No. of patients	Age (years), mean (SD)	Tadalafil dosage (mg)	CTP score, mean (SD)	Ascites present, n (%)	MELD score, mean (SD)	β-blocker usage, n (%)	Diuretics usage, n (%)	Follow-up (weeks)	IIEF score baseline, mean (SD)
Thakur et al. 2019	P	25	46.5 (7.2)	10 ‡	6.7 (1.7)	8 (32)	12.5 (3.9)	5 (20)	8 (28)	4	15.1 (5.6)
Jagdish et al. 2021	RCT	70	44.2 (8.1)	10 ‡	6.57 (1.29)	5 (7.1)	12.45 (3.4)	61 (87.1)	36 (51.4)	12	10.17 (5.4)
Kumar et al. 2022	P	34	n.a (20-50)†	20§	5.76 (0.7)	0	7.71 (1.7)	0	n.a.	12 - 24	15.5 (4)*

*IIEF-5 score, ‡daily dosage; †age range of included patients; §every two days.
Abbreviations: RCT: randomized controlled trial; P: prospective observational study; CTP: Child-Turcotte-Pugh Score; MELD: Model for end-stage for liver disease; IIEF: International index for erectile function (Erectile function domain); IIEF-5:5-item version of the International Index of Erectile Function; n.a.: data not available.

Table 1: Baseline characteristics of included studies.

Conclusion: PDE-5 inhibitors, particularly tadalafil, appear to be effective and well tolerated for the treatment of ED in men with cirrhosis. Further large-scale studies are warranted to validate these findings and assess long-term safety.

Disclosure: Nothing to disclose.

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PP0892

HAEMOSTATIC PROFILE IN STABLE COMPENSATED CIRRHOSIS, ACUTE DECOMPENSATION, AND ACUTE-ON-CHRONIC LIVER FAILURE: INSIGHTS FROM GLOBAL ASSAYS AND ASSOCIATIONS WITH NON-PORTAL HYPERTENSIVE BLEEDING AND THROMBOSIS

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Introduction: Although patients with acute-on-chronic liver failure (ACLF) often demonstrate preserved thrombin generation (TG), they concurrently display hypocoagulable profiles on rotational thromboelastometry (ROTEM) [1]. The extent to which these findings explain the increased bleeding risk in acutely ill cirrhotic patients remains unclear [1–5].
Moreover, most studies do not distinguish between portal hypertensive (PH) and non-PH bleeding, complicating accurate estimation of NPH bleeding prevalence.

Aims & Methods: This is a case-control study, comparing haemostatic profiles of acute decompensation (AD) and ACLF (=cases) to stable cirrhosis (SC) (=controls). Coagulation profiles were evaluated including INR, aPTT, PT, fibrinogen, coagulation factors and inhibitors and advanced global haemostatic tests (ROTEM, TG assay (TGA) ± thrombomodulin (TM)). Secondary objectives focused on identifying associations between coagulation parameters and NPH bleeding and thrombotic events.

Results: A total of 215 patients were included (SC: 122; AD: 53; ACLF: 40). The median Child-Pugh (CP) and MELD-Na scores increased progressively from SC to AD to ACLF (P< 0.001). NPH bleeding occurred in 8,4% (n=18), with higher rates in ACLF (30,8%, P<0,001) and AD (18.9%, P=0.011) than in SC (5.7%). Thrombotic events were observed in 7.0% (n=15).
TGA without TM revealed that ACLF patients had significantly shorter lag time compared to SC, shorter time to peak, reduced peak height and endogenous thrombin potential (ETP) compared to SC/AD; though TM normalized these effects. ETP inhibition by TM was reduced significantly across the three groups. ROTEM parameters showed marked hypocoagulable features in ACLF vs SC/AD, while AD and SC were comparable. Hypocoagulability (≥5 abnormal ROTEM parameters out of 6 parameters) was present in 1.8% of SC, 8.3% of AD, and 22.2% of ACLF patients (P<0.001 for SC vs ACLF); hypercoagulability was rare (2.8% in ACLF). INR and aPTT were significantly prolonged and FVIII elevated, while fibrinogen, PT, and pro- and anticoagulants were reduced in AD/ACLF compared to SC. ROTEM and TGA parameters correlated significantly with MELD-Na and CP scores (P<0,001).
Patients with NPH bleeding had higher CP and MELD-Na scores (P<0,001 and P=0,006), and a greater prevalence of AD (33,3%) and ACLF (38,9%) compared to non-bleeders (22,8% AD; 16,1% ACLF). Hypocoagulability on ROTEM was more frequent among bleeders (31.3% vs 4.5% in non-bleeders, P=0.002). Bleeders also showed greater impairment in INR, aPTT, PT, and most coagulation factors (except FVIII and PS). Multivariate analysis identified ROTEM-defined hypocoagulability as an independent risk factor for NPH bleeding (OR 12.93, 95% CI: 1.13–150.28, P=0.041). No significant differences in coagulation parameters were observed between patients with and without thrombotic complications.

Conclusion: TG demonstrated accelerated initiation and reduced thrombin potential in ACLF vs SC/AD, though TM normalized these effects. ROTEM demonstrated pronounced hypocoagulability in AD/ACLF vs SC. These coagulation abnormalities are paralleling their advanced liver dysfunction.
NPH bleeding events clustered in AD/ACLF cohorts and were associated with ROTEM-defined hypocoagulability. The convergence of ROTEM hypocoagulability as an independent risk factor for NPH bleeding and its prominence in acutely ill cirrhotic patients underscores the heightened NPH bleeding prevalence and ROTEM detected coagulation dysfunction in this patient population.

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PP0893

THE INFLUENCE OF DESTABILIZING FACTORS: BACTERIAL INFECTION AND ACUTE KIDNEY INJURY ON THE HAEMOSTATIC PROFILE IN CIRRHOSIS ACCORDING TO GLOBAL HAEMOSTATIC TESTS

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Introduction: Cirrhotic patients undergo intricate changes in their haemostatic system, resulting in a 'rebalanced haemostasis'. In patients with acute decompensation (AD) and acute-on-chronic liver failure (ACLF), this delicate balance is particularly susceptible to disruption by external

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stressors such as infections or acute kidney injury (AKI) rendering them more vulnerable to haemostatic complications than individuals with stable cirrhosis [1,2].

Aims & Methods: This case-control study evaluated coagulation profiles (INR, aPTT, PT, fibrinogen, coagulation factors and inhibitors, rotational thromboelastometry (ROTEM), and thrombin generation assay (TGA) [BleedScreen/ (BS) ThromboScreen (TS) ± thrombomodulin (TM)]) in 215 cirrhotic patients, comparing subgroups with/without AKI or bacterial infection to assess haemostatic impact.

Results: The cohort included 215 patients, with 37 (17.2%) having AKI and 31 (14.4%) bacterial infections. AKI patients had higher median Child-Pugh (11 vs 7, P<0.001) and MELD-Na scores (25 vs 11, P<0.001) than non-AKI patients, with more AD (32.4% vs 23%) and ACLF (59.5% vs 10.1%, P<0.001). Infected patients also showed higher CP (9 vs 7) and MELD-Na scores (33 vs 12, P<0.001), with more ACLF (51.6% vs 13%, P<0.001).

In the overall cohort, AD and ACLF patients have more hypocoagulability on ROTEM than SC. TG was preserved in AD compared to SC but showed more hypocoagulable features in ACLF, which were offset after TM addition. Both TG and ROTEM abnormalities correlated with liver disease severity.

Bacterial infection was linked to increased TG capacity (endogenous thrombin potential (ETP)-TM: 928.15 vs. 770.10 nm·min, P=0.030) and significantly less ETP suppression by TM (11.07% vs. 30.33%, P<0.001). Despite a greater prevalence of ACLF, ROTEM profiles did not differ between infected and non-infected patients. Standard coagulation assays revealed a hypocoagulable state, with lower levels of FII, VII, X, IX, and XI, while fibrinogen and platelet counts remained similar. Hypercoagulable features included increased FVIII and decreased protein S, (PS) and C (PC), and antithrombin (AT).

In the AKI group, TGA showed significantly accelerated TG in BS and TS with reduced TG capacity in TS. After TM addition, no differences in TG speed or total TG capacity were noted. Patients with AKI showed a significantly lower ETP inhibition by TM. ROTEM showed prolonged clotting times in INTEM and EXTEM and reduced maximum clot firmness in FIBTEM, with overall more hypocoagulability (21.2% vs. 4.2%, P=0.003). Standard coagulation assays showed mixed features in AKI, with increased INR, prolonged aPTT and reduced fibrinogen, FII, V, VII, X, IX, and XI levels, along with elevated FVIII and reduced AT and PC. Platelet counts were similar.

Conclusion: Global coagulation assays (TGA and ROTEM) showed similar profiles in infected vs non-infected cirrhotic patients, with some hypercoagulable features on TGA in infection, despite high ACLF prevalence (>50%). In contrast, AKI showed accelerated TG but reduced TG capacity, more hypocoagulable ROTEM features, and lower coagulation factors and inhibitors.

Although ACLF rates were comparable in AKI and infection groups, hypocoagulability was more pronounced in AKI, where in the infection group coagulation status was preserved or showed even some hypercoagulable features. This indicated that these coagulation abnormalities reflect the effects of these destabilizing factors rather than disease severity alone.

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PP0894

VACCINATION RATES AMONG HOSPITALISED PATIENTS WITH CIRRHOSIS: A RETROSPECTIVE STUDY

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Introduction: Patients with liver cirrhosis, particularly those with decompensated cirrhosis, are at significant risk of morbidity from viral and bacterial infections, which can also affect the progression of liver disease. Therefore, international guidelines recommend full vaccination coverage of these patients against a range of pathogens, irrespective of age.

Aims & Methods: The aim of our study was to evaluate vaccination coverage in a group of inpatients with cirrhosis. We retrospectively analysed patients with liver cirrhosis who were admitted after presenting to the emergency department and were hospitalised between 2019 and 2022 in the First Propaedeutic Department of Internal Medicine of Aristotle University of Thessaloniki.

The patients were evaluated with regard to their vaccination status and the presence of antibodies against hepatitis A and hepatitis B. Active vaccination against influenza was defined as vaccination during the latest vaccination period. Only the vaccinations administered prior to hospitalisation were included, and not those that were recommended afterwards. Patients who were diagnosed with cirrhosis during hospitalisation were excluded from the study.

Results: 182 patients were recorded, with a mean age of 65.4 ± 13.1. The patients were predominantly male (67%). Alcohol-related liver disease (47.8%), metabolic-associated steatohepatitis (18.1%) and chronic hepatitis B (11%) were the most common causes of cirrhosis. The majority of patients were classified as Child-Pugh stage B (51.6%), followed by stage C (31.9%) and A (16.5%). Most patients (79.7%) had decompensated cirrhosis. Regarding exposure to hepatitis B, 86 out of 157 patients (54.8%) required vaccination, of whom 25.6% (22/86) were already vaccinated.

Concerning hepatitis A, 89 of the 108 (83.5%) patients who were checked tested positive for antibodies without being vaccinated, while only two of the remaining 19 (10.5%) had been vaccinated.

As for influenza vaccination, 41.2% received the vaccine during the last vaccination season. Additionally, 21.4% of the patients had undergone vaccination against pneumococcus. The only correlation observed about vaccination coverage was the increased rates of pneumococcal vaccination in patients of Child-Pugh B cirrhosis compared to patients with Child-Pugh A or C (p=0.042).

Conclusion: Our study underlines the low vaccination coverage of patients with cirrhosis. Thus, the need to intensify vaccination efforts is highlighted, as infectious complications could be critical for patients with cirrhosis, not only due to their immediate morbidity but also because of their potential impact on the progression of liver disease.

Disclosure: Nothing to disclose.

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PP0895

TRACKING THE SHIFT: NLR VARIATION AS AN EARLY MARKER OF INFECTION IN CIRRHOTIC DECOMPENSATION

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Introduction: Infection is one of the most feared precipitants of acute decompensation (AD) in cirrhotic patients, making early detection crucial for timely intervention. In this context, the neutrophil-to-lymphocyte ratio (NLR) has emerged as a promising biomarker reflecting the inflammatory and infectious burden.

Aims & Methods: This study aimed to investigate the relationship between NLR and the prediction of infection in patients admitted to manage cirrhosis-related acute decompensation.

We conducted a retrospective study over 12 months (August 2022 to September 2023), including patients with a confirmed diagnosis of cirrhosis who were hospitalized for acute decompensation.

Clinical and laboratory data from the same decompensation episode were collected. The NLR during decompensation (NLR-AD) and the change in NLR (ΔNLR) were calculated as follows:

• NLR = Neutrophil count / Lymphocyte count

• ΔNLR = NLR during decompensation – baseline NLR

Receiver operating characteristic (ROC) curves were used to assess the predictive performance of these variables. Associations with infection types were also analyzed.

Results: Eighty-three patients were included, with a mean age of 62.4 ± 10.8 years and a male-to-female ratio of 1.13. The main reasons for admission were hepatic encephalopathy (HE) in 32 patients (38.6%), oedemato-ascitic decompensation in 26 patients (31.3%), and variceal upper gastrointestinal bleeding in 25 patients (30.1%).

Among these patients, 24 (28.9%) developed infections, including: 6 cases of spontaneous bacterial peritonitis (SBP) (25%), 11 urinary tract infections (UTIs) (45.8%), 4 skin/soft tissue infections (16.7%), 3 lower respiratory tract infections (12.5%).

Median values were: Baseline NLR: 2.6 (IQR [1.7–4.1]), NLR during AD: 4.33 (IQR [3–6.25]) and ΔNLR: 1.29 (IQR [0.3–3.3]).

An NLR-AD ≥ 4.4 was significantly associated with infection (AUC = 0.692; p = 0.006; 95% CI [0.565–0.819]). Similarly, a ΔNLR > 2.15 was also predictive (AUC = 0.696; p = 0.005; 95% CI [0.571–0.821]).

Wilcoxon testing confirmed a statistically significant increase between baseline NLR and NLR-AD (p < 0.001).

A ΔNLR > 2.9 was strongly associated with SBP (AUC = 0.820; p = 0.009; 95% CI [0.719–0.921]), while no significant association was observed between ΔNLR and other types of infections, or between NLR-AD and any specific infection type, including SBP.

Multivariate analysis showed that only ΔNLR was independently predictive of infection (p = 0.01), with ΔNLR > 2.9 being specifically associated with SBP (p = 0.045).

Conclusion: This study highlights the clinical value of the neutrophil-to-lymphocyte ratio (NLR) as a predictive biomarker for infection in patients with acute decompensation of cirrhosis. A significant increase in NLR (ΔNLR) was strongly associated with the presence of infection, particularly spontaneous bacterial peritonitis. These findings suggest that NLR and its dynamic variation from baseline (ΔNLR) may serve as useful tools for early infection detection and risk stratification in this vulnerable patient population.

Disclosure: Nothing to disclose.

PP0896

SILENT INFECTIONS, LOUD MARKERS: BIOLOGICAL PREDICTORS OF BACTERIAL INFECTION IN ACUTELY DECOMPENSATED CIRRHOSIS IN A TUNISIAN POPULATION

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Introduction: Diagnosing bacterial infections in patients with acutely decompensated cirrhosis (AD) is often challenging, as typical signs may be absent, and the clinical presentation may be limited to nonspecific deterioration of general condition. However, delayed management is associated with high mortality once infection is established. Early diagnosis of bacterial infection (BI) is therefore critical to improve outcomes.

Aims & Methods: This study aimed to identify predictive biological markers of bacterial infection in patients with AD cirrhosis, which would support earlier and more targeted therapeutic interventions.

We conducted a retrospective study over 12 months (August 2022 to September 2023), including patients with established cirrhosis hospitalized for an episode of acute decompensation.

Bacterial infection was diagnosed based on clinical, biological, and cytological criteria.

White blood cell (WBC) count, neutrophil count, lymphocyte count, and C-reactive protein (CRP) levels were recorded both at the time of admission for AD and from baseline values obtained during a compensated period within the previous three months.

The biological markers assessed included: ΔWBC (difference between admission and baseline WBC count),%Neutrophils (proportion of neutrophils among total WBC), ΔNeutrophils (admission minus baseline neutrophil count), ΔWBC/Lymphocyte ratio, ΔLymphocytes (admission minus baseline), Neutrophil-to-Lymphocyte Ratio (NLR), ΔNLR (NLR at admission minus baseline NLR).

Receiver Operating Characteristic (ROC) curves were generated to identify predictive thresholds of each biomarker for infection, and Area Under the Curve (AUC) values with corresponding p-values were calculated to assess performance. Multivariate logistic regression was then used to identify independent predictors of bacterial infection, expressed as odds ratios (OR) with 95% confidence intervals (CI).

Results:

A total of 83 patients were included, with a mean age of 62.4 ± 10.8 years and a male-to-female ratio of 1.13.

The main reasons for admission were hepatic encephalopathy (32 patients, 38.6%), ascitic decompensation (26 patients, 31.3%), and variceal gastrointestinal bleeding (25 patients, 30.1%).

Bacterial infection was identified in 24 patients (28.9%), including spontaneous bacterial peritonitis (SBP) in 6 cases (25%), urinary tract infections in 11 (45.8%), skin infections (erysipelas, cellulitis) in 4 (16.7%), and respiratory infections in 3 (12.5%).

Significant associations were found between infection and the following biomarkers:

• ΔNeutrophils > 1,550/mm^3 (AUC = 0.658; p = 0.025),

• %Neutrophils > 72.3% (AUC = 0.687; p = 0.008),

• ΔWBC/Lymphocytes > 3.1 (AUC = 0.691; p = 0.007),

• NLR > 4.4 (AUC = 0.692; p = 0.006),

• ΔNLR > 2.9 (AUC = 0.696; p = 0.005),

• CRP > 21 mg/L (p < 0.001).

On multivariate analysis, two independent predictors of bacterial infection were identified: ΔNLR > 2.9 (p = 0.038; OR = 3.8; 95% CI: 1.1–13.5), CRP > 21 mg/L (p < 0.001; OR = 21.8; 95% CI: 5.1–82).

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Conclusion: Our findings highlight the value of ΔNLR and CRP as useful biomarkers for detecting bacterial infection in cirrhotic patients with acute decompensation, even in the absence of overt clinical signs. These parameters may help prompt early and focused investigations for potential infectious sources, ultimately improving patient management and prognosis.

Disclosure: Nothing to disclose.

PP0897

ASSESSMENT OF GLOBAL MYOCARDIAL FUNCTION IN LIVER CIRRHOSIS AND ITS CLINICAL SIGNIFICANCE

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Introduction: Cirrhotic cardiomyopathy (CCM), cardiac dysfunction in liver cirrhosis of any aetiology in the absence of a primary cardiac cause predisposes to morbidity and mortality before and after Liver Transplantation (LT). Global longitudinal strain (GLS) measurement is included in the updated CCM criteria to detect subclinical myocardial dysfunction. Myocardial work (MW) is a novel echocardiographic measure, combining GLS with simultaneous blood pressure measurements has been found to correlate well with invasively measured global myocardial function. MW is not reported in cirrhosis and CCM.

Aims & Methods: The primary aim was to evaluate global cardiac function in decompensated cirrhosis using advanced echocardiographic parameters at rest and with dobutamine stress echo (DSE), and to study their association with development of a composite clinical outcome including acute kidney injury/hepatorenal syndrome, refractory ascites, cardiac failure and death over the subsequent 12 months. The secondary aims were to report the prevalence of CCM and to assess its association with the study outcome; to assess the diagnostic utility of cardiac biomarkers (NT-pro-BNP and cardiac troponin T) in CCM, and re-assess cardiac function at 12 months of successful LT.

Prospective study of cirrhotic patients with ascites or undergoing LT assessment from January 2021 to August 2022, with 12-month follow-up. After a written consent, clinical profile, Model for End-stage Liver Disease sodium (MELD Na) score, and cardiac biomarkers were evaluated. Transthoracic echo and DSE measured GLS and calculated MW (GWI (global work index), GCW (global constructive work), GWE (global work efficiency) and GWW (global wasted work). Statistical analysis was conducted using SPSS IBM Statistics version 29. Difference in estimated marginal mean difference in MW between baseline and DSE were obtained using repeated measures of ANOVA. Univariate analysis of clinical, echocardiographic and demographic features on the study outcome was assessed using either independent t-test or chi-squared test as appropriate. Significant predictors from the univariate analysis (p<0.05) were included in multivariate Cox regression analysis.

Results: 96 patients with mean±SD age 56±11 years; (58% male; 59% alcohol misuse) were studied. The mean±SD MELD Na was 17±11. 16 patients underwent LT; 53 developed the composite clinical outcome. MELD Na, troponin, LVEF, rest and peak GLS, rest and peak GWI, peak GWE and E/e' were significantly higher in patients meeting study outcome in univariate analysis. MELD Na and E/e' were positively associated in multivariate analysis with HR (95%CI)=1.075:1.018–1.135; p=0.009 and HR=1.179; 1.037–1.340; p=0.012 respectively.

There was a significant decrease in GLS and MW on DSE. There was a significant correlation between MELD Na and rest GLS (r=0.30, p<0.001). Based on new CCM criteria, systolic dysfunction was detected in 17% and diastolic dysfunction in 13% and CCM in 27%.

There was no association between CCM and the composite study outcome. Median (IQR) troponin was 13.5 ng/ml (7.0–24.0) and BNP was 189 (pg/ml) (53.0–374.0). There was no association between these biomarkers and CCM. Ten patients evaluated post LT, demonstrated significant improvements in GLS, MW and LVEF.

Conclusion: In our patients with advanced cirrhosis, the study revealed increased myocardial work without any augmentation on DSE. CCM observed in a quarter of patients, had no impact on clinical course before/after LT. Further studies are warranted to validate the updated CCM criteria and improve its clinical utility.

Disclosure: Nothing to disclose.

PP0898

PREDICTING POOR PROGNOSIS IN LIVER CIRRHOSIS: THE ACCURACY OF MELD-NA IN DISCRIMINATING BAVENO STAGES

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Introduction: Currently, there are several prognostic scores developed to predict mortality in liver cirrhosis. Child-Pugh-Turcotte (CPT) and MELD-Na scores are the most used and validated in current practice. Recently, BAVENO consensus issued new recommendations for a better assessment of the rate of decompensation of liver cirrhosis as well as long- and short-term mortality rates.

In this study, we aimed to evaluate the accuracy of both CPT and MELD-Na scores to predict BAVENO stages.

Aims & Methods: We conducted a retrospective study in which consecutive patients with liver cirrhosis admitted to the St. Spiridon Emergency Hospital Iasi between January 2023 and April 2024, were included. All cases were assigned to different prognostic stages according to BAVENO and MELD-Na and CPT scores were compared between the groups.

Results: Of the 155 patients included, 54.8% were males, with an average age of 57.81±12.36 years. According to BAVENO staging, 68% of patients were classified as a BAVENO stage 6, 10% as stage 5, 5% as stage 4, 10% as stage 3, 6% as stage 2, and 1% as stage 1. We found a positive correlation between MELD-Na Score and Child Pugh Score (r=0,774) with significant p-value (P<0.0001).

Furthermore, our results demonstrate superior sensitivity and specificity for the MELD-Na score (98.7% and 97.9%, AUC-0.997) compared to the Child-Pugh score, which showed a sensitivity of 71.2% (AUC-0.792) to predict a stage 6 diagnosis and worst prognosis overall.

At a cut-off of 34, MELD-Na score was the most accurate to discriminate between BAVENO stage 5 and 6, a cut off of 22 between BAVENO 4 and 5 and a cut off of 13 was the most accurate to discriminate between BAVENO stage 3 and stage 4.

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Conclusion: Mortality rates and poor prognosis increase exponentially with liver cirrhosis decompensation, thus their assessment through predictive scores remains a highly desirable target for clinicians, MELD-Na score retaining its superiority in predicting the evolution through the BAVENO stages.

Disclosure: Nothing to disclose.

PP0899

COMBINATION OF MELD SCORE WITH NEUTROPHIL-TO-LYMPHOCYTE RATIO PREDICTS IN-HOSPITAL MORTALITY IN ACUTE-ON-CHRONIC LIVER FAILURE

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Introduction: Prediction of early mortality is crucial in patients with Acute on Chronic Liver Failure (ACLF), a highly fatal syndrome characterized by intense systemic inflammation and organ failure. Performance of the Model of End-stage Liver Disease (MELD) appears to be suboptimal in ACLF, although the prognostic potential of combining MELD with inflammatory indexes remains uncertain.

Aims & Methods: The aim of the study was to evaluate a combination of MELD with neutrophil-to-lymphocyte ratio (NLR) for in-hospital mortality prediction in patients with ACLF.

A retrospective cohort study was conducted evaluating consecutive ACLF patients followed till hospital discharge. The predictive ability and optimal cut-points of MELD and NLR calculated on admission were determined by ROC analyses. Four prognostic groups, determined by score combinations below or above the optimal cut-points, were analyzed with respect to in-hospital mortality.

Results: A total of 122 patients (50.8% males, mean age 61.8±12.1) were hospitalized with ACLF between April 2021 and February 2023. The etiology of pre-existing liver disease was alcoholic in 44.3% and in-hospital mortality was 31.1%. The areas under ROC were 0.79 for MELD and 0.68 for NLR, with optimal cut-off points 20 (sensitivity/specificity: 0.89/0.57) and 5.1 (sensitivity/specificity: 0.84/0.60), respectively.

Compared to patients with low MELD and NLR scores, patients with high scores displayed significantly higher in-hospital mortality (73.7% vs 6.7%; adjusted OR=13.96, p=0.007), whereas those with a single elevated score did not (high-MELD/low-NLR: 15.4%; adjusted OR=1.98, p=0.49; low-MELD/high NLR: 14.2%; adjusted OR=1.30, p=0.79).

Conclusion: In patients hospitalized with ACLF, a combination of high baseline MELD and NLR appears to delineate poor prognosis, correlating with a high risk of in-hospital death.

Disclosure: Nothing to disclose.

PP0900

A PRACTICAL MODEL FOR STRATIFYING ESOPHAGEAL VARICES SEVERITY IN PATIENTS WITH CIRRHOSIS

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Introduction: Esophageal varices (EV) are a critical complication of portal hypertension in patients with cirrhosis, and early detection is essential to prevent potentially life-threatening hemorrhage. The Baveno VII consensus recommends performing endoscopy in patients with transient elastography (TE) values exceeding 20 kPa and platelet counts below 150,000/mm^3. While endoscopy remains the gold standard for diagnosis, non-invasive tests are gaining importance as a promising alternative for predicting EV(1).

The objective of our study is to predict the grade of varices in cirrhotic patients using a score derived from both their biological blood test results and clinical parameters.

Aims & Methods: The objective of our study was to predict the grade of varices in cirrhotic patients using a score derived from their biological blood test results and clinical parameters. This retrospective study involved patients recruited from Hôtel-Dieu de France, with data collected a few months prior to their gastroscopy for esophageal varices detection.

We calculated the ALBI-PLT score based on the method described by Inoue-Yuri et al.(2). The modified ALBI-PLT (mALBI-PLT) score was derived from the mALBI grade, assigning 1 point for grades 1 or 2a, and 2 points for grades 2b or 3. It also incorporated platelet counts, awarding 1 point for counts greater than 150,000/µL and 2 points for counts of 150,000/µL or lower.

In addition to the mALBI-PLT score, we considered various variables, including blood test results and established predictive scores such as Child-Pugh, MELD-Na, FIB-4, and APRI.

Results: This study analyzed 108 cirrhotic patients (mean age: 63.94 years, 64.8% men, 33.03% had hepatocellular carcinoma (HCC)) out of 115 recruited, excluding cases with missing data. Key predictors of varice grades included lymphocyte count (L, $\beta = -0.43$, $p = 0.011$) and sodium (Na^10, $\beta = -3.22 \times 10^{-22}$, $p = 0.005$) as significant negative associations. Positive associations were observed for the modified ALBI-platelet score ($\beta = 0.43$, $p < 0.001$), HCC ($\beta = 0.51$, $p = 0.005$), and gastrointestinal (GI) bleeding ($\beta = 0.48$, $p = 0.013$).

A derived formula for predicting varice grades is: VariceGrade = 3.49 − 0.43×ln(L) + 0.43×mALBI-PLT score − 3.22×10^−22×Na^10 + 0.51×HCC + 0.48×GI Bleeding.

The model showed strong specificity (94.5%) for identifying varices when the score was ≥ 1 and high sensitivity (97.8%) for excluding grade 3 varices when the score was ≤ 2.

Conclusion: This study introduces a non-invasive scoring system to accurately stratify esophageal varices severity, optimizing clinical decision-making and endoscopic screening in cirrhotic patients.

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Disclosure: Nothing to disclose.

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PP0901

COMPARISON OF EFFICACY OF ENDOSCOPIC BAND LIGATION (EBL) WITH ARGON PLASMA COAGULATION (APC) IN THE MANAGEMENT OF BLEEDING GASTRIC VASCULAR ECTASIAS (GVE)

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Introduction: Gastric vascular ectasias are a common complication of liver cirrhosis and resultant portal hypertension. Traditional therapy with argon plasma coagulation is expensive. Endoscopic band ligation offers a simpler and more cost effective therapy with similar results.

Aims & Methods: Comparison of efficacy of endoscopic band ligation (EBL) with argon plasma coagulation (APC) in the management of bleeding gastric vascular ectasias (GVE).

The study period was from August 2021 to April 2025. It was carried out in the Department of Gastroenterology and Hepatology, Shaikh Zayed Hospital, Lahore, Pakistan. It was a randomized controlled trial.

30 patients of chronic liver disease who had presented with upper gastrointestinal bleeding due to Gastric Vascular Ectasias , as seen on upper GI endoscopy were recruited. All patients were explained procedural details and informed consent was taken. They were then randomly divided into two groups, A & B, of 15 patients each. Patient in group A underwent EBL while those in group B were treated with APC. After initial endoscopic therapy, these patients were followed up at 2 and 4 weeks.

Efficacy of therapy was assessed by using clinical parameters (recurrence of bleeding manifest as hematemesis/malaena/both), laboratory parameters (drop in hemoglobin) and endoscopic parameters (GVE bleed seen after initial therapy).

Results: One patient from both groups was lost to follow up. Four patients, two out of each group passed away before their last endoscopy session.

In group A (EBL), 4 out of the remaining 12 patients (33 %) needed repeat therapy at 4 weeks while 8 patients (66%) showed post banding healing not requiring repeat banding during the study period.

In group B (APC), 5 out of remaining 12 patients (42%) needed repeat therapy at 4 weeks. 7 out of 12 patients (58 %) did not need repeat therapy. There was one episode of GI bleeding in a patient undergoing band ligation which was managed conservatively with PPI and surcralfate. No other significant procedure related side effects were encountered in either groups.

Conclusion: Endoscopic Band Ligation is comparable in efficacy to Argon Plasma Coagulation for management of Gastric Vascular ectasias, offering significant cost benefits and ease of therapy.

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Disclosure: Nothing to disclose.

PP0902

FRAILTY IN CIRRHOTIC PATIENTS: PREVALENCE AND PREDICTIVE FACTORS

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Introduction: Frailty is a complex clinical syndrome characterized by a reduction in physiological reserves and an increased vulnerability to stressors. Although it is often associated with aging, frailty also affects younger populations, particularly patients with severe chronic illnesses such as cirrhosis.

The objective of our study was to assess the prevalence of frailty among cirrhotic patients in a Tunisian population and to identify associated risk factors.

Aims & Methods: An analytical cross-sectional study was conducted over a three-month period in the gastroenterology department of Sahloul University Hospital in Sousse. Patients aged 18 to 70 years, diagnosed with cirrhosis (based on clinical, biological, morphological, endoscopic, and/or histological criteria) were included. Patients with severe chronic diseases that could overestimate frailty or with osteoarticular diseases limiting mobility were excluded.

Frailty was assessed using the Clinical Frailty Scale (CFS) for global frailty (GF) and the Short Physical Performance Battery (SPPB) for physical frailty (PF).

Results: A total of 50 patients were included. The median age was 59 ± 9.2 years, with a male-to-female ratio of 1.5. The main etiologies of cirrhosis were metabolic liver disease (38%) and hepatitis B-related cirrhosis (28%). According to the Child-Pugh score, 39% of patients were classified as Child A, 36% as Child B, and 26% as Child C, with an average MELD score of 14.2 ± 5.9.

Regarding frailty, 42% of patients exhibited GF and 46% had PF.

Univariate analysis revealed several factors significantly associated with GF, namely: female gender (p=0.035), age >60 years (p=0.02), presence of grade I hepatic encephalopathy (p=0.038), presence of ascites (p<0.001), BMI < 18.5 kg/m^2 (p=0.037), sedentary lifestyle (p<0.001), MELD score >15 (p=0.001), and Child-Pugh score B-C (p=0.003).

For PF, similar results were observed: age >60 years (p=0.03), presence of ascites (p<0.001), sedentary lifestyle (p=0.009), and MELD score >15 (p=0.001).

Multivariate analysis identified independent risk factors associated with both GF and PF as: age >60 years, presence of ascites, and a MELD score >15.

Conclusion: This study highlights the high prevalence of frailty among cirrhotic patients, emphasizing the importance of systematic assessment of this condition in clinical practice. Age, ascites, and the severity of cirrhosis were found to be significant indicators of frailty. These findings suggest the need to implement targeted rehabilitation programs to improve physical function and quality of life in this patient population.

Disclosure: Nothing to disclose.

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PP0903

HYPONATREMIA IN CIRRHOTIC PATIENTS WITH ASCITES: WHAT TREATMENT AND WHAT IS THE OUTCOME?

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Introduction: Hyponatremia is a frequent complication of cirrhosis and is a prognostic factor correlated with the severity of cirrhosis and influencing survival and the occurrence of complications, but this relationship is not yet well defined.

Aims & Methods: This was a single-centre retrospective descriptive and analytical study, including all cirrhotic patients admitted to our department in ascitic decompensation with hyponatremia between January 2019 and march 2025. The natremia threshold of 130 mmol/l was chosen because it is widely accepted to define hyponatraemia in cirrhotic patients.

In our study, we compared 2 groups of patients: group A: fluid restriction + albumin infusion and group B: fluid restriction alone. Predictive factors of hyponatremia development and its impact on the outcome were evaluated.

The aim of this study was to identify predictive factors of hyponatremia in cirrhotic patients with ascites, assess its prognostic impact on complications and survival, and compare the effectiveness of two treatment strategies—fluid restriction alone versus fluid restriction combined with albumin infusion—in correcting hyponatremia and improving clinical outcomes.

Results: Out of a total of 127 cirrhotic patients, forty patients with ascites and hyponatraemia (Na < 130 mmol/L) were included in the comparative analysis. These patients were divided into two groups: twenty patients received fluid restriction combined with albumin infusion (group A), and twenty patients received fluid restriction alone (group B). The mean age across both groups was 54.7 years, ranging from 19 to 80 years, with a sex ratio M/F of 0.7.

The causes of liver cirrhosis in our patients were: autoimmune (n=15), viral (n=11), undetermined (n=9), metabolic (n=2), genetic (n=2) and alcoholic (n=1).

Child-Pugh stage was B in 55% and C in 45%. The mean MELD score was 18 [7-48]. The prevalence of dilutional hyponatremia as defined by a serum sodium concentration ≤130 mmol/L or ≤135 mmol/L was 10.5% and 31.4% respectively.

The mean MELD score was 18, ranging from 7 to 48. The prevalence of dilutional hyponatremia, defined as serum sodium ≤130 mmol/L or ≤135 mmol/L, was 10.5% and 31.4%, respectively. A serum sodium level ≤130 mmol/L was significantly associated with more advanced liver disease, including Child-Pugh C (OR=7.84; p<0.001) and MELD >16 (OR=6.76; p=0.001).

Group A demonstrated a significantly greater improvement in natremia (mean increase of 4.5 mmol/L versus 1.2 mmol/L in group B; p=0.02), as well as a lower incidence of complications such as hepatic encephalopathy and hepatorenal syndrome. Although the difference in global survival at one year between the two groups (33.3% in group A vs. 12.5% in group B) The main precipitating factors of hyponatremia identified were excessive diuretic use (n=18), spontaneous bacterial peritonitis (n=12), gastrointestinal bleeding (n=6) and constipation (n=4). In several cases (n=8), multiple triggers were observed concurrently. The presence of hyponatremia was associated with a higher incidence of complications during hospitalization. Survival without complications was significantly reduced in patients with hyponatremia when a serum sodium concentration ≤135 mmol/L was used as the threshold (p=0.012). Moreover, survival free from hepatic encephalopathy and hepatorenal syndrome was significantly lower in hyponatremic patients (p<0.0001).

Analysis of prognostic factors revealed that serum sodium ≤130 mmol/L (HR=2.92; p=0.004), Child-Pugh C stage (HR=3.45; p=0.002), MELD score >18 (HR=2.67; p=0.007), and development of hepatic encephalopathy or hepatorenal syndrome during hospitalization (HR=4.21; p<0.001) were significantly associated with poor outcomes. Treatment of hepatic encephalopathy included systematic use of rifaximin and lactulose alongside management of the underlying precipitating factor. Overall survival at one year was significantly lower in patients with hyponatremia (22.5% vs. 68.7%), as was mean global survival (8.3 months vs. 32.8 months; p<0.0001).

Conclusion: Low serum sodium levels were significantly associated with more advanced liver disease and increased short-term morbidity and mortality in cirrhotic patients with ascites. Our findings suggest that albumin infusion, in addition to fluid restriction, is more effective in correcting hyponatremia and may help reduce the incidence of severe complications such as hepatic encephalopathy and hepatorenal syndrome.

Therefore, serum sodium appears to be a valuable prognostic marker. The clinical benefit of correcting hyponatremia should be further confirmed in larger prospective studies.

Disclosure: Nothing to disclose.

PP0904

DIAGNOSTIC VALUE OF THE NEUTROPHIL-TO-LYMPHOCYTE RATIO AND C-REACTIVE PROTEIN IN SPONTANEOUS BACTERIAL PERITONITIS AMONG CIRRHOTIC PATIENTS

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Introduction: Spontaneous bacterial peritonitis (SBP) is a severe complication of cirrhosis, associated with high mortality. Its diagnosis relies on diagnostic paracentesis and the polymorphonuclear neutrophil (PMN) count in ascitic fluid. The neutrophil-to-lymphocyte ratio (NLR), a simple and accessible inflammatory marker, may reflect the systemic response to SBP and serve as an additional diagnostic tool.

The aim of this study was to evaluate the diagnostic performance of NLR and C-reactive protein (CRP) in SBP among cirrhotic patients hospitalized for ascitic decompensation.

Aims & Methods: We conducted a retrospective study including all cirrhotic patients hospitalized for ascitic decompensation between January 2021 and January 2025. Patients with hepatocellular carcinoma were excluded. NLR was calculated from the complete blood count performed on the first day of hospitalization. Demographic, clinical, biological, imaging, and outcome data were analyzed.

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Results: Fifty-eight patients were included, with a mean age of 65.3 years [42–73 years] and a male-to-female ratio of 0.52. Cirrhosis was of viral origin in 28 cases (48.3%), including 18 cases of hepatitis B-related cirrhosis (31.0%) and 10 cases of hepatitis C-related cirrhosis (17.2%). Other etiologies included non-alcoholic steatohepatitis (NASH) in 12 patients (20.7%), alcoholic cirrhosis in 5 (8.6%), primary biliary cirrhosis in 7 (12.1%), and cryptogenic cirrhosis in 6 (10.3%). According to the Child-Pugh classification, 21 patients (36.2%) were classified as A, 27 (46.6%) as B, and 10 (17.2%) as C.

SBP was diagnosed in 22 patients (37.9%). The mean CRP level was 12.8 mg/dL [4–68 mg/dL], and the mean NLR was 6.28 [0.53–45.31]. NLR was significantly higher in patients with SBP (3.43 vs. 1.82, p = 0.03), as was CRP (11.83 mg/dL vs. 9.35 mg/dL, p = 0.04).

ROC curve analysis identified optimal thresholds of 2.78 for NLR and 10.83 mg/dL for CRP, with good discriminatory power. For the diagnosis of SBP, an NLR ≥ 2.78 showed a sensitivity of 88.3% and a specificity of 88.9%, while a CRP ≥ 10.83 mg/dL had a sensitivity of 92.6% and a specificity of 87.6%. Logistic regression analysis revealed that combining NLR and CRP further improved diagnostic performance, achieving a sensitivity of 94.4% and a specificity of 96.7%.

Conclusion: Both NLR and CRP are significantly elevated in patients with SBP and demonstrate good diagnostic value. Their combination enhances diagnostic accuracy, supporting their potential use as non-invasive, complementary tools for the early detection of SBP in cirrhotic patients.

Disclosure: Nothing to disclose.

PP0905

ALBI OUTPERFORMS MELD AND CHILD-PUGH IN PREDICTING SHORT-TERM MORTALITY IN CIRRHOTIC PATIENTS WITH GASTROINTESTINAL HEMORRHAGE

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Introduction: Gastrointestinal (GI) hemorrhage accounts for 30% of acute decompensations in cirrhotic patients, with 60% originating from variceal bleeding. Each decompensation episode carries an in-hospital mortality rate of 13%, and a 6-week mortality rate of 20%, commonly assed using Child Pugh (CP) and MELD scores. The ALBI (Albumin-Bilirubin) score, which includes only two variables, offers a simplified evaluation of liver function. However, there is limited data on its utility in predicting short-term mortality in cirrhotic patients with GI hemorrhage.

Aims & Methods: This study aimed to evaluate the association between ALBI Score with short-term mortality in cirrhotic patients with GI hemorrhage. We conducted a prospective, cross-sectional study between January and December of 2023 at Hospital Roosevelt, Guatemala, including cirrhotic patients admitted for GI hemorrhage. Informed consent was obtained from all participants. Categorical variables were analyzed using the chi-square test, normally distributed continuous variables with Student's *t*-test, and non-normally distributed variables with the Mann-Whitney U test. Logistic binary regression and AUROC analysis were used to assess associations with short-term mortality. Statistical analyses were performed using SPSS.

Results: A total of 187 patients were included: 50.8% female and 49.2% male, with a mean age of 56 among survivors and 58 among non-survivors. Of these, 52.9% had a prior diagnosis of cirrhosis, while 47.1% were newly diagnosed. Cirrhosis etiology was metabolic in 50.2%, alcoholic in 42.7%, autoimmune in 4.3%, HCV in 1.6%, and HBV in 1.1%.

At admission, 80.8% had no ascites and 97.3% had no hepatic encephalopathy. he causes of GI hemorrhage were 86.1% esophageal varices, 3.2% gastric varices, 9.1% gastric ulcers, and 1.6% esophagitis.

Non-survivors had significantly lower systolic blood pressure (93.90 ± 16.3 mmHg vs. 107.92 ± 17.4 mmHg; $p = 0.002$) and mean arterial pressure (69.50 ± 14.5 mmHg vs. 79.67 ± 11.7 mmHg; $p = 0.005$) compared to survivors. Non-survivors exhibited significantly higher INR (1.79 ± 0.56 vs. 1.52 ± 0.41; $p = 0.026$) and total bilirubin (5.85 ± 8.73 mg/dL vs. 2.02 ± 2.82 mg/dL; $p = 0.031$), and lower serum albumin levels (2.40 ± 0.43 g/dL vs. 3.10 ± 0.54 g/dL; $p < 0.001$).

Prognostic scores were significantly higher among non-survivors: MELD (19.25 ± 10.09 vs. 13.57 ± 5.47; $p = 0.002$), MELD-Na (20.35 ± 9.76 vs. 14.44 ± 6.42; $p = 0.002$), Child-Pugh (9.15 ± 2.21 vs. 7.18 ± 1.90; $p < 0.001$), and ALBI score (-0.96 ± 0.51 vs. -1.76 ± 0.58; $p < 0.001$). Additionally, the length of hospital stay was significantly shorter among non-survivors (3.90 ± 3.82 days) compared to survivors (6.59 ± 5.58 days; $p = 0.001$).

Scores	Mortality In-Hospital	Mortality at 4 weeks	Mortality at 6 weeks
ALBI	AUC: 0.850 95% CI (0.780 - 0.921) p: 0.000	AUC: 0.881 95% CI (0.823 - 0.938) p: 0.000	AUC: 0.867 95% CI (0.806 - 0.927) p: 0.000
Child Pugh (CP)	AUC: 0.762 95% CI (0.668 - 0.857) p: 0.000	AUC: 0.819 95% CI (0.739 - 0.900) p: 0.000	AUC: 0.805 95% CI (0.728 - 0.881) p: 0.000
MELD	AUC: 0.714 95% CI (0.606 - 0.822) p: 0.002	AUC: 0.785 95% CI (0.693 - 0.878) p: 0.000	AUC: 0.786 95% CI (0.700 - 0.871) p: 0.000
MELD-Na	AUC: 0.711 95% CI (0.599 - 0.823) p: 0.002	AUC: 0.778 95% CI (0.683 - 0.873) p: 0.000	AUC: 0.783 95% CI (0.696 - 0.870) p: 0.000

Multivariable analysis showed that:

- Higher ALBI score was significantly associated with mortality (Risk Ratio 25.80; 95% CI: 6.75–98.67; $p < 0.001$),
- Hospital stay <5 days (RR 9.91; 95% CI: 2.07–47.47; $p = 0.004$),
- Endoscopy performed within 12 hours (RR 7.53; 95% CI: 2.01–28.15; $p = 0.003$).

Conclusion: The ALBI score outperforms CP and MELD scores in predicting short-term mortality (in-hospital, 4 weeks, and 6 weeks) in cirrhotic patients with GI hemorrhage. Observed mortality was 10% in-hospital, 13% at 4 weeks, and 16% at 6 weeks.

Disclosure: Nothing to disclose.

PP0906

EFFECTS OF LOLA ON THE GUT MICROBIOME AND PATIENT REPORTED QUALITY OF LIFE

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Introduction: Liver cirrhosis is characterized by microbial dysbiosis, intestinal inflammation and a weakened gut barrier and associated with an increased risk of complications. We previously observed a positive influence of L-ornithine-L-aspartate (LOLA) on gut microbiome composition in cirrhosis (1).

The current study aimed to investigate the effect of LOLA on microbiome dysbiosis, quality of life, sarcopenia, intestinal permeability, inflammation and metabolite composition.

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Aims & Methods: In this monocentric, open-label, phase 4 study, LOLA (18g/day) was administered orally for three months to liver cirrhosis patients with hepatic encephalopathy grades (0-2). We investigated changes in their gut microbiome composition (16s rRNA sequencing, alpha and beta diversity, differential abundance), quality of life (SF36), blood ammonia concentrations, measurements of sarcopenia and frailty (appendicular lean muscle mass, anthropometric measures, liver frailty index, SARC-F), biomarkers of gut permeability, inflammation and sarcopenia as well as metabolic alterations in stool, serum, and urine.

Results: 65 patients were enrolled, and 52 patients completed the 3-month study (40.4% female, age 62 (58; 65)). While LOLA reduced the abundance of the genus Romboutsia, it did not significantly change overall microbiome diversity. LOLA led to a significant improvement in the vitality dimension of the SF36 questionnaire (45 (35; 60) to 50 (45; 60), p=0.019) and decreased plasma ammonia levels in patients with elevated baseline ammonia (69.5 (54; 225) to 46.0 (22; 66) µmol/L, p=0.004)

Although muscle mass generally declined across the study, this was not observed in patients whose elevated ammonia levels improved. Sarcopenia and frailty scores and the muscle biomarkers myostatin, irisin, fibroblast growth factor 21, insulin-like growth factor 1 were unaffected by LOLA treatment. Diamine oxidase, a marker of gut barrier dysfunction, significantly decreased (from 12.75 U/mL (11.25; 17.45) to 11.15 U/mL (9.9; 15.65), p=0.016), while LPS binding protein, a marker of innate immune response to bacterial translocation, significantly increased (from 17.05 µg/mg (14.7; 19.8) to 17.39 µg/mg (16.2; 21.9), p=0.006). Calprotectin, zonulin and sCD14 did not show significant changes.

Metabolomic analysis revealed an increase in serum alanine levels (pFDR=0.014) with additional subtle shifts in lipoprotein and amino acid profiles.

Conclusion: In summary, LOLA was associated with an improvement in vitality, an important patient-reported outcome accompanied by alterations in gut microbial composition and prevented muscle loss in patients with hyperammonaemia.

Furthermore, LOLA was associated with improvements in gut permeability and endotoxin handling. The slight increase in alanine observed in the metabolomics analysis may be a consequence of LOLA metabolism. LOLA may be a valuable supportive treatment for enhancing quality of life in cirrhosis by targeting the gut liver axis and may be particularly beneficial in preventing muscle loss in patients with hyperammonaemia.

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PP0907

STUDY ON NEW PREDICTIVE MODEL FOR ESOPHAGOGASTRIC VARICES REBLEEDING IN LIVER CIRRHOSIS BASED ON CLINICAL CHARACTERISTICS

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Introduction: To explore a new non-invasive, simple and feasible clinical prediction model for rebleeding after esophagogastric varices in patients with liver cirrhosis, so as to provide reference for early prediction of rebleeding, reduce the incidence of rebleeding and mortality, and improve the clinical outcome of patients.

Aims & Methods: This retrospective study included 177 patients with liver cirrhosis and esophagogastric variceal bleeding treated endoscopically at the First Affiliated Hospital of Xi ,an Medical University (September 2018–October 2023). Participants were stratified into rebleeding (n=93) and non-rebleeding (n=84) groups. Clinical data encompassed demographics, comorbidities (hepatic encephalopathy, ascites, portal hypertension), etiology (hepatitis B/C, alcohol-related, others), laboratory parameters (hematologic, hepatic, coagulation profiles), imaging features (main portal vein diameter), endoscopic findings (LDRf classification, red sign, mosaic-like changes, mottled changes of gastric mucosa), and treatment details (dosage of tissue adhesive, dosage of lauromacrogol, treatment methods), Prognostic scores (MELD, CTP, ALBI, APRI, FIB-4, NLR, RPR, GPR, DAR, DPR) were calculated.

The measurement data in accordance with normal distribution were expressed as mean ± standard deviation, and the independent sample t test was used. Non-normally distributed measurement data were expressed as median and interquartile range , and the rank sum test was used, and the chi-square test was used for count data.

Logistic regression model was used to analyze the risk factors of rebleeding in EGVB patients (non-rebleeding group =0, rebleeding group =1 were classified). $P < 0.05$ indicated that the difference was statistically significant. The Receiver Operating Characteristic Curve (ROC) was used to evaluate the model.

Results: 1.Univariate analysis showed that there were statistically significant differences in WBC, NEU, RBC, HGB, TBIL, CHE, ALP, ALB, PT, INR, APTT, TT, NH3, the diameter of main portal vein, the maximum diameter of varicose vein, risk factors, red sign, and the amount of tissue adhesive used ($P < 0.05$). There were significant differences in MELD score, CTP grade, FIB-4 index, DPR, DAR, NLR and ALBI scores among the scores ($P < 0.05$).

2.Multivariate analysis showed that RBC, CHE, ALP, ALB, APTT, TT, portal vein diameter, and tissue adhesive dosage were independent predictors in clinical examination. NLR and ALBI score were independent predictors in score.

3. According to the results of multivariate analysis, a joint examination model (composed of 8 independent predictors in the multivariate analysis of clinical examination) and a joint scoring model (composed of 2 independent predictive scores in the multivariate analysis of scoring) were constructed. ROC analysis was used to compare the predictive value of the joint examination model and the joint scoring model.

Examination of the combined model (AUC=0.906, $P < 0.001$, Youden index =0.71, sensitivity =81.7%, specificity =89.3%, 95%CI: 0.864-0.948) was better than the combined scoring model in predicting rebleeding in patients with EGVB (AUC=0.775, $P < 0.001$, Youden index =0.425, sensitivity =69.9%, specificity =72.6%, 95%CI: 0.706-0.843).

Conclusion: The combined model can predict the rebleeding of EGVB patients after endoscopic treatment, which is worthy of further clinical application.

Disclosure: Nothing to disclose.

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PP0908

INCIDENCE OF BACTERIAL INFECTIONS IN CIRRHOTIC PATIENTS WITH ACUTE VARICEAL BLEEDING RECEIVING ANTIBIOTIC PROPHYLAXIS

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Introduction: Antibiotic prophylaxis has been recommended to reduce the risk of infection and mortality in patients with cirrhosis and acute variceal bleeding (AVB). Recent data from Europe indicated that, despite antibiotic prophylaxis, bacterial infections, particularly of the respiratory tract, still occur in a subset of AVB patients, even with the administration of prophylactic antibiotics. Nonetheless, evidence concerning infections in AVB patients in Asia is limited and may differ from that in Western countries.

Aims & Methods: This retrospective study aimed to determine the incidence of bacterial infections among patients with AVB and to identify characteristics correlated with the occurrence of infection. Data from consecutive cirrhotic patients with AVB admitted to a tertiary care center, Rajavithi Hospital, Bangkok, between January 1, 2018, and October 31, 2024 were reviewed and analyzed. Patients with clinical signs of infections on admission were excluded.

Results: A total of 77 patients were included, with 67 (87%) being male. The average age was 52.88 ± 12.16 years. The predominant causes of cirrhosis were alcohol (66.2%), hepatitis C virus (27.3%), and hepatitis B virus (16.9%). The severity of cirrhosis was categorized as Child-Pugh A, B, and C in 31.2%, 24.2%, and 24.6% of patients, respectively.

All patients were administered intravenous ceftriaxone as prophylactic antibiotic upon admission and underwent esophagogastroduodenoscopy within 48 hours, with 82% receiving endoscopic variceal ligation. Bacterial infections were identified in 12 patients (15.6%).

The observed infection types included pneumonia (41.6%), primary bacteremia (25%), spontaneous bacterial peritonitis (16.7%), and urinary tract infection (16.7%). The median time to infection commencement was 2 (1-15) days, and the median duration of hospitalization was 5 (2-6) days, with a mortality rate of 6%. In univariate analysis, significant predictors of infections included Child-Pugh class B/C (p=0.004), BMI (p=0.037), histoacryl glue injection (p=0.044), and serum albumin (p=0.003).

A multivariate analysis revealed a statistically significant link between low serum albumin levels and the occurrence of bacterial infection, with an adjusted odds ratio of 0.13 (95% CI: 0.02 - 0.94); p=0.043.

Conclusion: Bacterial infections, especially pneumonia, remain a significant concern in cirrhotic patients with AVB despite the use of prophylactic antibiotic, particularly in individuals with hypoalbuminemia.

Disclosure: Nothing to disclose.

PP0909

RED BLOOD CELL DISTRIBUTION WIDTH AS A BIOMARKER OF SEVERITY IN LIVER CIRRHOSIS: A CROSS-SECTIONAL STUDY

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Introduction: Red blood cell distribution width (RDW), is a routinely available parameter from complete blood counts, reflecting variability in erythrocyte size. Beyond its traditional hematological use, RDW has emerged as a potential prognostic marker in various systemic diseases, including chronic liver disease.

Aims & Methods: In this cross-sectional study, 200 cirrhotic patients with varying degrees of severity and etiologies were enrolled from a tertiary care hepatogastroenterology department. RDW was assessed in all patients, with a normal range set between 11% and 15%.

Our aim is to assess the relationship between RDW values and cirrhosis severity, as evaluated by Child-Pugh scores and the Model for End-Stage Liver Disease (MELD). Statistical analyses included ANOVA, post-hoc Tukey tests, and Spearman correlation to determine the association between RDW and cirrhosis severity.

Results: Our study comprised 200 cirrhotic patients, 111 females and 89 males, with an average age of 58,06 ± 15.21 years (range: 19–95).The patients were distributed across different disease stages: 99 had Child-Pugh score A, 80 had Child-Pugh score B, and 21 had Child-Pugh score C.

ANOVA test was conducted to compare the mean RDW across the different Child-Pugh stages (A, B, and C). The analysis revealed a statistically significant difference between the 3 groups: A (15,83), B (17,02), C (18,04) ($p < 0.001$). Tukey's post-hoc test showed that: Stage C had a significantly higher RDW compared to stages A ($p = 0.020$) and B ($p = 0.020$). These results suggest a progressive increase in RDW with worsening severity of liver disease.

Among patients classified as Child A, only 36.4% exhibited a high RDW, while in groups Child B and Child C, higher rates of RDW were observed, reaching 68.7% and 87.5% respectively. A significant positive correlation was found between the Child-Pugh stage and the RDW ($p < 0.001$), as assessed by Spearman's correlation test.

Similarly, a significant positive correlation was found between the MELD score and the RDW ($p < 0.001$). This finding suggests that patients with more advanced liver disease, tend to exhibit greater anisocytosis, possibly reflecting underlying inflammation, nutritional deficiencies, or bone marrow stress.

Conclusion: In conclusion, our findings suggest that RDW tends to rise with the progression of hepatic dysfunction, underscoring its potential as a simple and accessible biomarker for assessing disease severity in patients with cirrhosis. Nonetheless, further studies are needed to comprehensively evaluate the utility of RDW as a reliable predictor of cirrhosis severity.

Disclosure: Nothing to disclose.

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PP0910

IMPACT OF SARCOPENIA ON HEPATOCELLULAR CARCINOMA OUTCOMES: A RETROSPECTIVE MONOCENTRIC STUDY

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Introduction: Sarcopenia is increasingly recognized as a major prognostic factor in patients with hepatocellular carcinoma (HCC). Its evaluation may improve patient risk stratification and therapeutic decision-making

Aims & Methods: To assess the prevalence of sarcopenia among patients with hepatocellular carcinoma (HCC), identify associated factors, and evaluate the MELD-sarcopenia score as a potential predictor of mortality.

We conducted a retrospective monocentric study over a 10-year period, including patients aged over 18 years with a confirmed diagnosis of HCC based on radiological, biological, and/or histological criteria.

Only patients with an available abdominal CT scan including the L3 vertebral level were included. Exclusion criteria were liver transplantation, other primary liver tumors, non-HCC malignancies, incomplete records, or poor-quality imaging. Sarcopenia was diagnosed by calculating the Total Psoas Area Index (TPAI) on CT, using thresholds of 5.45 cm^2/m^2 for men and 3.85 cm^2/m^2 for women. MELD and MELD-sarcopenia scores were calculated. Survival time from HCC diagnosis was recorded.

Results: Among 30 screened patients, 5 were excluded due to missing imaging. In the final cohort of 25 patients, the prevalence of sarcopenia was 52%. Compared to non-sarcopenic patients, sarcopenic patients were older (67.6 vs 56 years) and more often male. The most frequent etiology in sarcopenic patients was hepatitis C (46.2%), while hepatitis B was predominant in non-sarcopenic patients (58.3%).

Cirrhosis was mostly compensated in both groups. HCC was typically unifocal, infiltrative, and multilobar in both groups. The mean time from HCC diagnosis to death was shorter in sarcopenic patients (13.4 vs 33.9 months).

Male sex and older age were significantly associated with sarcopenia ($p < 0.05$), whereas tumor characteristics, Child-Pugh class, and biological tests were not. Kaplan-Meier analysis showed a tendency toward worse survival in patients with higher MELD-sarcopenia scores, although the difference was not statistically significant ($p > 0.05$).

Parameters	Sarcopenic Group	Non-Sarcopenic Group
Mean age	67.6 years	55.0 years
OMS status	OMS 0-1: 8 (61.5%) OMS 2: 3 (38.5%)	OMS 0-1: 11 (91.7%) OMS 2: 1 (8.3%)
Child-Pugh class	Compensated: 66.7% Decompensated: 33.3%	Compensated: 54.5% Decompensated: 45.5%
MELD score	Median: 11 IQR: 3.25	Median: 14 IQR: 5.5
ALBI grade	Grade 1: 23.1% Grade 2: 53.8% Grade 3: 15.4%	Grade 1: 16.7% Grade 2: 33.3% Grade 3: 41.7%
MELD-sarcopenia score	Median: 21.35 IQR: 3.25	
Mortality	Deceased: 76.9% Alive: 23.1%	Deceased: 83.3% Alive: 16.7%
Median survival time after HCC diagnosis	13.4 months	33.9 months

Table : Comparison of clinical and survival characteristics between sarcopenic and non-sarcopenic patients with hepatocellular carcinoma.

Conclusion: Sarcopenia is highly prevalent among patients with HCC and is independently associated with male sex and older age. The MELD-sarcopenia score appears to be a promising tool for mortality risk stratification and may aid clinical decision-making.

Reference: https://pubmed.ncbi.nlm.nih.gov/21893129/

Disclosure: Nothing to disclose.

PP0911

PROGNOSTIC SIGNIFICANCE OF ELF TEST COMPARED TO LIVER BIOPSY IN PATIENTS WITH METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE (MASLD)

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Introduction: Liver fibrosis stands out as the main prognostic risk factor in MASLD. The enhanced liver fibrosis (ELF) score is a composite of direct fibrosis biomarkers that reflect extracellular matrix turnover. While the ELF test exhibits high diagnostic accuracy for advanced liver fibrosis in MASLD patients, its role as a prognostic biomarker remains uncertain.

Aims & Methods: Our aim is to compare the prognostic effectiveness of ELF, liver stiffness measurement (LSM), FIB4, and liver histology in patients with MASLD.

We retrospectively enrolled patients with MASLD who underwent liver biopsy between 2013 and 2023. The ELF score was automatically using a serum sample collected at baseline. FIB4 computation, LSM with Fibroscan and liver biopsy were performed at baseline. Liver fibrosis stage was assessed according to the NASH CRN Scoring System.

The primary outcome was a composite endpoint including all-cause mortality, hepatocellular carcinoma, liver transplantation or complications related to cirrhosis (ascites, variceal bleeding, hepatic encephalopathy, MELD≥15). Subjects were stratified based on existing literature cut-offs for ELF (<9.8, 9.8-11.2, >11.2), LSM (<10, 10-15, >15 kPa), FIB-4 (<1.3, 1.3-2.67, >2.67), and histology (F≤2, F3, F4) to assess the risk of occurrence of the primary outcome.

Results: We included data of 289 patients (30.4% female, median age 50y [IQR 39-58]). Over a median follow-up time of 41 months (IQR 21-68), the composite endpoint occurred in 34 (11.8%) patients. The frequency of the primary outcome exhibited a stepwise increase with ELF scores <9.8 (0.5%), 9.8 to 11.2 (14.5%) and >11.2 (69.7%). Survival curves for comparisons between groups revealed significant differences for all index tests based on pre-defined histological and non-invasive test stratification (Log Rank test $p < 0.05$).

At multivariate Cox regression analysis, ELF and liver histology were significant predictors of the primary outcome after adjusting for gender, diabetes, age and BMI. (ELF > 11.2 vs < 9.8 HR 132.7 [95%CI 15.6-1127.4 $p<0.01$], 9.8-11.2 vs <9.8 HR 22.5 [95%CI 2.8-184.3 $p=0.04$]) (F4 vs F≤2 HR 92.0 [95%CI 20.1-421.9 $p<0.01$], F3 vs F≤2 HR 10.2 [95%CI 2.3-45.5 $p=0.05$]). Furthermore, the intermediate and high risk group defined by LSM had a

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significantly higher risk of developing the composite outcome compared to the low-risk group (LSM 10-15kPa vs <10kPa HR 5.51 [95%CI 1.25 – 24.18 p=0.02], LSM >15kPa vs <15 kPa HR 22.89 [95%CI 6.17 – 84.94 p<0.01]). Intermediate risk group according to FIB4 displayed no significant higher risk of composite outcome compared to low-risk group (HR 2.87 [95%CI 0.53 – 15.55 p=0.22]).

Conclusion: The ELF test demonstrated comparable performance to histologically evaluated fibrosis in forecasting clinical outcomes. It should be regarded as a viable alternative to liver biopsy for conducting prognostic assessments in patients with MASLD.

Disclosure: Nothing to disclose.

PP0912

OPTIMISING VARICEAL SCREENING IN PORTAL HYPERTENSION: THE ROLE OF LIVER AND SPLENIC STIFFNESS MEASUREMENT

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Introduction: Portal hypertension is a major complication of advanced chronic liver disease (ACLD) and is associated with gastro-oesophageal varices, a leading cause of morbidity and mortality. The British Society of Gastroenterology (BSG) recommends oesophagogastroduodenoscopy (OGD) for variceal screening in all patients with cirrhosis, which could result in a high volume of unnecessary procedures. Given increasing demands on NHS services and the need for cost-effective resource allocation, strategies to reduce unnecessary endoscopy are critical.

The Baveno VI and VII consensus workshops propose non-invasive criteria to optimise variceal screening. Baveno VI suggests that patients with a liver stiffness measurement (LSM) less than 20 kPa and a platelet count greater than 150×10^9/L can safely avoid OGD. Baveno VII further refines this by recommending non-selective beta-blocker (NSBB) therapy in patients with clinically significant portal hypertension (CSPH), defined as LSM greater than 25 kPa, thereby eliminating the need for screening endoscopy in this group. For patients not clearly stratified by Baveno VI or VII criteria, splenic stiffness measurement (SSM) offers an additional non-invasive tool to refine risk assessment. Incorporating SSM in this intermediate-risk group may further reduce the need for OGD, with recent data showing a high negative predictive value for ruling out high-risk varices.

Aims & Methods: We aim to quantify the reduction in endoscopy burden achieved by applying Baveno VI and VII criteria in ACLD patients.

A retrospective analysis was conducted on 1,703 patients who underwent 1,802 FibroScans between January 1, 2024, and December 31, 2024. Patients were stratified based on LSM and platelet counts into the following categories:

1. BSG Criteria: All patients with LSM ≥10 kPa requiring OGD.
2. Baveno VI Criteria: Patients with LSM <20 kPa and platelet count $\geq150\times10^9$/L, for whom OGD could be safely avoided.
3. Baveno VII Criteria: Patients with CSPH (LSM ≥25 kPa) recommended for NSBB therapy without the need for screening OGD.
4. Patients requiring OGD: Those not meeting Baveno VI criteria and without CSPH.

Results: Liver stiffness distribution:

- <10 kPa: 1,534 patients
- 10–14.9 kPa: 136 patients
- 15–19.9 kPa: 44 patients
- 20–24.9 kPa: 31 patients
- ≥25 kPa: 57 patients

Impact of Adopting Baveno VI and VII:

- BSG Guidelines: 274 patients (LSM ≥10 kPa) would require OGD.
- Baveno VI: 149 patients (LSM <20 kPa, platelets ≥150) could avoid OGD.
- Baveno VII: 57 patients (LSM ≥25 kPa) would be managed with NSBB, avoiding OGD.
- Total Reduction: 206 out of 274 patients (~75%) could avoid OGD.

Conclusion: Adopting Baveno VI and VII criteria would significantly reduce the need for variceal screening endoscopy in ACLD patients. By applying these thresholds, up to 75% of OGDs could be avoided, optimising NHS resources and reducing patient burden. These findings support the routine use of non-invasive risk stratification in ACLD management.

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Disclosure: Nothing to disclose.

PP0913

DIAGNOSTIC ACCURACY OF APRI AND FIB-4 IN STAGING LIVER FIBROSIS AND CIRRHOSIS

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Introduction: Noninvasive assessment of liver fibrosis is essential in the management of chronic liver diseases. APRI and FIB-4 are simple, low-cost tools widely used for this purpose. This study aimed to evaluate the diagnostic performance of APRI and FIB-4 in identifying significant fibrosis (≥F2) and cirrhosis (F4), using transient elastography (FibroScan) as the reference standard, and to explore their clinical and biochemical correlations.

Aims & Methods: We conducted a cross-sectional study at the Institute of Gastroenterology and Hepatology, Iași, Romania, between January 1st, 2024 and April 1st, 2025. A total of 106 patients with chronic liver disease were included. Laboratory tests were used to calculate APRI and FIB-4 scores, and liver stiffness was measured by FibroScan. For diagnostic accuracy we performed ROC curve analysis and Pearson correlation coefficients were used to assess relationships between noninvasive scores and clinical-biological variables. Statistical analysis was performed using MedCalc® Statistical Software version 23.2.1 (MedCalc Software Ltd, Ostend, Belgium).

Results: For the detection of significant fibrosis (≥F2), the area under the ROC curve (AUC) was 0.748 for APRI ($p < 0.001$) and 0.752 for FIB-4 ($p < 0.001$). In diagnosing cirrhosis (F4 ≥12.5 kPa), FIB-4 demonstrated superior diagnostic accuracy (AUC 0.836, $p < 0.001$) compared to APRI (AUC 0.797, $p < 0.001$). A FIB-4 threshold of 1.45 provided a sensitivity of 68.6% and a specificity of 83.3% for detecting significant fibrosis. High cut-offs (≥2.0 for

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APRI and ≥4.0 for FIB-4) were associated with a specificity exceeding 96%. A significant positive correlation was observed between APRI and FIB-4 (r = 0.96, p < 0.001). Moreover, FIB-4 correlated moderately with FibroScan values (r = 0.514), while APRI showed a weaker correlation (r = 0.423).

Conclusion: The results of this study confirm that both APRI and FIB-4 are effective noninvasive tools for the staging of liver fibrosis, with FIB-4 demonstrating superior diagnostic accuracy, particularly for cirrhosis.

The moderate correlation between FIB-4 and FibroScan (r = 0.514) supports its potential role as a reliable surrogate marker, especially in settings where access to elastography is limited, outperforming the weaker correlation of APRI (r = 0.423) that suggests a lower predictive power, aligning with findings from previous studies that emphasized FIB-4's stronger association with liver stiffness and histological fibrosis progression.

The high specificity associated with elevated cut-off values enhances the clinical applicability of these scores for ruling in advanced fibrosis and cirrhosis, minimizing the need for unnecessary imaging or biopsies.

Overall, integrating FIB-4 and APRI into routine clinical practice could improve early identification of at-risk patients, optimize resource allocation, and facilitate timely therapeutic interventions.

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Disclosure: Nothing to disclose.

PP0914

RECURRENT HEMORRHAGE IN GASTRIC VARICEAL RUPTURE: IDENTIFYING KEY RISK FACTORS FOLLOWING BIOLOGICAL GLUE INJECTION

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Introduction: Hemorrhage due to gastric variceal rupture (GVR) represents a major clinical challenge in patients with portal hypertension (PH), carrying a high mortality rate in the absence of adequate medical and endoscopic management.

Aims & Methods: The aim of our study was to identify risk factors associated with rebleeding from gastric varices within 6 weeks following biological glue injection, and assess its impact on mortality.

This retrospective analytical study included all patients admitted to the hepatogastroenterology department of Monastir between January 2014 and October 2023 for gastrointestinal bleeding solely due to gastric varices (GV), treated with endoscopic hemostasis using biological glue injection. Gastric varices were classified according to the Sarin classification.

Results: A total of 75 episodes of gastric variceal rupture (GVR) were recorded in 33 patients, with a mean age of 58 years and a sex ratio of 0.94. Cirrhosis was the primary etiology of portal hypertension (PH) in 72.7% of cases, with most cases presenting in advanced stages (37% Child B and 22% Child C).

Across bleeding episodes, gastric varices were distributed as follows: GOV1 in 24%, GOV2 in 42.7%, and IGV1 in 32% of cases. Concurrent esophageal varices were present in over 90% of cases, and hypertensive gastropathy (HTG) was observed in 50%. Rebleeding occurred in 50% of patients, with a median recurrence interval of 345 days and a 6-week rebleeding rate of 18.7%.

At 6 weeks, risk factors significantly associated with GVR recurrence included advanced cirrhosis (Child B7 or higher), concurrent hepatic decompensation, active bleeding at endoscopy, GOV2-type varices, and HTG presence.

In multivariate analysis, only hypertensive gastropathy (HTG) emerged as an independent predictor of GVR recurrence during follow-up. Hepatic mortality reached 42.4% (14 patients), with a median survival of 35 weeks [95% CI: 3-143 weeks].

Mortality was significantly higher in patients with rebleeding (75% vs. 31.8%; p<0.001).

Conclusion: Six-week rebleeding from gastric variceal rupture (GVR) is strongly associated with advanced liver disease, the presence of hypertensive gastropathy (HTG), and active bleeding during endoscopy. This recurrence, particularly pronounced in diabetic patients, significantly elevates hepatic mortality risk in both the short and medium term.

Disclosure: Nothing to disclose.

PP0915

PREDICTIVE FACTORS OF EARLY MORTALITY IN CIRRHOTIC PATIENTS

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Introduction: Cirrhosis is a serious liver disease that irreversibly damages the liver, leading to serious complications and represent a major cause of mortality.

Aims & Methods: The aim of the study was to define predictive factors of early mortality in cirrhotic patients.

We carried out a retrospective, monocentric, cross-sectional study, over a 4-year period from January 2021 to January 2025, in our hepato-gastroenterology department, including all cirrhotic patients hospitalized during this period. We conducted a descriptive and analytical statistical study subdividing patients into two groups: first group (G1), includes cirrhotic patients who died within 5 years of cirrhosis diagnosis; and a second group (G2) including those with a survival time of more than 5 years.

Data collected included demographics (age, sex), comorbidities, cirrhosis etiology, cirrhosis severity scores (Child-Pugh; MELD), history of previous cirrhosis complications (ascitic decompensation ; Spontaneous Bacterial Peritonitis; hepatorenal syndrome ; variceal haemorrhage; portal thrombosis; hepatocellular carcinoma ; hepatic encephalopathy; Acute on Chronic Liver Failure (ACLF)), as well as indication for liver transplantation was sought.

All these data were entered and analyzed in a database using SPSS version 21.0.

Results: Two hundred thirty patients were included in this study. Seventy-six patients who died within 5 years of cirrhosis diagnosis (33%). Eight variables reached statistical significance in both In univariate and multivariate analysis (p < 0.05):

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Early mortality was significantly higher in patients with Child C cirrhosis and a higher Meld score (>15) than in those with Child B cirrhosis (89% vs. 17%; p= 0.02; OR= 5.8), indicating advanced liver disease as a predictive factor of early mortality.
Six types of cirrhosis complications reached statistical significance (p < 0.05), which are spontaneous bacterial peritonitis especially if complicated by septic shock or organ failure, hepatorenal syndrome and its complications, variceal haemorrhage especially if recurrent and abundant, hepatocellular carcinoma, in cases of advanced hepatic encephalopathy, and ACLF can increase the risk of early death by 78% if in an advanced stage.
The indication for liver transplantation is also predictive of early mortality, especially if waiting times are too long (p= 0.04; OR= 4.2).
For other complications of cirrhosis: ascitic decompensation and portal thrombosis, they do not appear to increase the risk of early death in cirrhotic patients
Furthermore, no statistically significant differences were found between the two groups with regard to demographic characteristics, the existence or absence of comorbidities, and the etiology of cirrhosis.
Conclusion: Cirrhosis is a major cause of mortality, linked to the advanced stage of liver disease and its serious complications, including variceal hemorrhage, spontaneous bacterial peritonitis, hepatorenal syndrome and ACLF.
Disclosure: Nothing to disclose.

PP0916

HEPATIC STEATOSIS IN CHRONIC HEPATITIS B: A PARALLEL PHENOMENON WITH NO SIGNIFICANT IMPACT ON FIBROSIS OR THERAPEUTICS

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Introduction: The coexistence of HBV and hepatic steatosis is frequent, but its impact on the progression of liver disease remains poorly characterized and much debated.
Aims & Methods: We aim by this study to determine the prevalence, associated factors, consequences of this association and response to treatment.
A retrospective analysis was conducted over 20 years from 2005 to 2025 on a single-center cohort of 124 patients with chronic HBV, of whom 16.1% (n=20) had steatosis (confirmed by imaging). Demographic, virological, metabolic and histological (fibrosis stage) characteristics were compared between the groups with and without steatosis. And statistical tests (Chi², Mann-Whitney, logistic regression) were used.
For the comparison, we defined:
- Group A: combining steatosis and HBV
- Group B: HBV without steatosis.
Results: The prevalence of steatosis in patients with HBV in our series was 16.1%, with a mean age of 52.1 years higher than that of patients with HBV without steatosis, (45.8 years), with a p-value of 0.02. Furthermore, in our work there was a clear female predominance, with a sex ratio of 3 F/H in group A vs 0.9F/H in group B with a p-value of 0.04.
In the history of our patients, we found that 30% of patients had high blood pressure in group A compared with 18% in group B, 10% had diabetes in group A compared with 8% in the other group, and there was no significant difference in smoking habits (15% compared with 12%). Statistical regression did not find any significant correlation in those signs. On the other hand, in the case of dyslipidemia, there was a slight predominance of group A (15% vs. 10%) with a p-value of 0.045.
The analysis of hepatic test didn't show any big difference between results of GGT, PAL and ASAT but for ALAT there was a higher in patients with steatosis with a median of 62 vs 52,1 and a p-value of 0,02.
Our study also found no difference in serological results: 95% of patients with steatosis were Hbs Ag positive, with the presence of Hbc Ab in all patients at the time of diagnosis. These results were not very different from those of patients without steatosis (Hbs Ag positive in 96.8% and Hbc Ab positive in all patients). Only 10% of patients had a positive Hbe Ag in group A vs 7.2% in group B. The median viral load before treatment was 1870 UI/ml vs 2450 UI/ml with no real difference noted between the 2 groups, confirmed by statistical regression which found a p=0.82 value.
Our study found no real impact on liver disease, showing no worsening of fibrosis (30% F≥2 with steatosis vs 25% without, p=0.62*).
After analysis using the association Chi² test, our study found similar virological responses under treatment (83% in group A vs 78% in group B, p=0.75*).
Conclusion: In this cohort, steatosis in HBV patients (16.1%) was associated with age and female sex. Hepatic test showed a higher ALAT fold among patient with steatosis. However, higher viral load or more severe fibrosis was not demonstrated. Steatosis in HBV does not appear to have a negative influence on the course of the disease. These results suggest that the management of HBV with steatosis should primarily target virological control, without requiring any specific therapeutic adaptation.
Disclosure: no conflict of interest.

PP0917

POPULATION-BASED HEPATITIS C SCREENING AND ELIMINATION PROGRAM IN LITHUANIA: CURRENT RESULTS AND CHALLENGES

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Introduction: The Lithuanian HCV screening programme started in May 2022. General practitioners (GPs) received special promoting fee for serological HCV antibody testing: 1. For people born between 1945 and 1994 (once in a lifetime) and 2. For people who inject drugs (PWID) or HIV-positive people (annual testing). This initiative is the first in Central and Eastern Europe.
Aims & Methods: The aim of this study was to assess the results of the first 34 months of the programme and the different scenarios for achieving the WHO 2030 targets. GPs invited people to perform a serum blood test for HCV antibodies. Anti-HCV positive patients were referred to a gastroenterologist or infectologist for HCV RNA testing, and if result was positive, direct-acting antiviral (DAA) therapy was prescribed. Information about patients was obtained from the database of the National Health In-

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surance Fund. The Markov disease progression model elaborated by the CDA Foundation was used to assess HCV elimination progress. The data from the 2022-2023 screening were used as inputs. Three scenarios were developed: the 'Base' scenario - return to pre-screening program level in 2023 and 2 scenarios with different extents of treatment.

Results: At the beginning of 2022, about 1.8 million people born in 1945-1994 lived in Lithuania. Between May 5, 2022 and February 28, 2025, from this population 1362472 people (75.7%) were tested for HCV antibodies. Positive test results were found in 1.28% of cases. In the risk group, 8396 PWID and/or HIV+ people were screened, of whom 28.9 % were seropositive. Among anti-HCV+ persons 83.8% was from population and 16.2 % from risk groups. Specialists consulted 58% anti-HCV+ persons and viremia was detected in 53,1 %. Among anti-HCV+ persons from population specialist consulted 63%, but only 29% from risk group (p<0.001). Referral to specialist delay were related with specialist waiting lines, and compliance and motivation problems in risk (PWID and/or HIV+) group. During 34-month (May 2022- February 2025) period 3 times more HCV patients were treated with DAA than during the same period before starting of the program (7577 vs 2470). The Markov disease progression model to assess HCV elimination progress showed the following scenarios: Scenario 1: if the same number of patients are treated as before the screening, the WHO targets will not be reached. Scenario 2: treating 70% of infected patients will meet most but not all WHO targets. Scenario 3: by treating all infected patients by 2030, the WHO target will be met by saving 150 lives and preventing 90 new cases of decompensated cirrhosis and 120 cases of hepatocellular carcinoma.

Conclusion: Our data show that in a European country with a moderate HCV seroprevalence, a population-based screening program is feasible: more than 75% (almost 1.4 million) of the population born between 1945 and 1994 was screened, and the number of patients treated has increased by more than 3 times in the first 34 months of the program (as compared to the same time period prior to the program). However, despite abolishment of restrictions to DAA treatment in the country, special measures to increase motivation and compliance of risk group patients to program logistic are necessary to achieve WHO 2030 targets in Lithuania.

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Disclosure: Nothing to disclose.

PP0918

SMALL INTESTINAL BACTERIAL OVERGROWTH POSITIVITY AS A PREDICTIVE MARKER FOR ELEVATED FIB-4 SCORES IN CHRONIC HEPATITIS: A RETROSPECTIVE COHORT ANALYSIS

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Introduction: Chronic hepatitis, primarily caused by HBV, HCV, or HDV infections, remains a silent yet progressive condition leading to advanced hepatic fibrosis and cirrhosis. Non-invasive assessment tools, such as the FIB-4 score, play a crucial role in early-stage risk stratification. Emerging evidence suggests a potential interaction between gut microbiota alterations-particularly Small Intestinal Bacterial Overgrowth (SIBO)-and liver function, mediated through the gut-liver axis.

Aims & Methods: Aim: To evaluate the association between SIBO status and FIB-4 scores among patients with chronic hepatitis, and to assess whether SIBO positivity serves as an independent marker of hepatic fibrosis severity.

Methods: We retrospectively analyzed clinical and biochemical data from 122 patients with confirmed viral hepatitis. Parameters included age, gender, liver enzyme levels (ALT, AST, ALP, GGT), SIBO breath test results, and viral titers. FIB-4 scores were calculated using standard formulas. Patients were grouped based on SIBO status (positive vs. negative). Boxplot analysis and correlation metrics were used to visualize and quantify associations.

Results: Patients with SIBO positivity exhibited significantly higher median FIB-4 scores (1.87) compared to those who were SIBO-negative (1.23), ($p < 0.05$). The boxplot analysis underscored a wider interquartile range and rightward skew in SIBO-positive individuals. A moderate positive correlation (Spearman's rho = 0.41) was observed between viral titers and FIB-4 levels, indicating a synergistic impact of microbial overgrowth and viral load on liver fibrosis progression.

Conclusion: SIBO positivity appears to correlate with elevated FIB-4 scores in chronic hepatitis patients, suggesting microbial dysbiosis as a potential contributor to fibrotic burden. Incorporating gut health assessment into routine liver disease screening may enhance early detection of high-risk individuals.

Further prospective studies are recommended to explore causality and therapeutic modulation of the gut-liver axis.

Disclosure: Nothing to disclose.

PP0919

SPLENIC STIFFNESS: A POWERFUL TOOL FOR NON-INVASIVE DETECTION OF ESOPHAGEAL VARICES USING 100-HZ MODULE FIBROSCAN

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Introduction: Esophageal varices (EVs) are a common and life-threatening complication of portal hypertension in patients with liver cirrhosis, affecting nearly 50–60% of cirrhotic individuals. Timely detection of EVs, particularly those at high risk of bleeding, is critical in guiding appropriate surveillance and prophylactic strategies. Endoscopy remains the gold standard for diagnosis; however, its invasive nature highlights the need for reliable non-invasive alternatives. Recent advances in elastography have introduced spleen stiffness measurement (SSM) as a promising surrogate marker for portal hypertension and variceal presence.

Aims & Methods: This study aimed to evaluate the clinical utility of spleen stiffness measurement using the novel 100-Hz module of Fibroscan in predicting the presence and severity of esophageal varices in patients with compensated liver cirrhosis, and to compare its performance with liver stiffness measurement (LSM).

This cross-sectional research was performed on fifty cases with compensated cirrhosis at Bab Alshaaria University Hospital's Department of Hepatology, Gastroenterology, and Infectious Diseases from September 2023 to February 2024. All patients underwent detailed clinical and laboratory

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evaluation, abdominal ultrasound, Fibroscan (for both SSM and LSM), and upper gastrointestinal endoscopy to assess for EVs. The diagnostic accuracy of SSM and LSM in predicting EVs was analyzed, and cutoff values were determined.

Results: Of the 50 patients enrolled, 33 (66%) were diagnosed with esophageal varices, of which 18 (54.5%) had high-risk varices. SSM values were significantly higher in patients with EVs compared to those without. At a cutoff value of 43.3 KPa, SSM achieved a sensitivity of 82% and specificity of 95% in detecting EVs. In comparison, LSM at a cutoff of 13.5 KPa yielded a sensitivity of 73% and specificity of 72%. Importantly, SSM also demonstrated a strong correlation with the severity of varices, allowing differentiation between high-risk and low-risk cases.

Conclusion: Spleen stiffness measurement using the 100-Hz Fibroscan module is a highly sensitive and specific non-invasive tool for identifying esophageal varices in patients with compensated cirrhosis. Its superior predictive accuracy compared to liver stiffness measurement supports its integration into clinical practice as a screening method to reduce unnecessary endoscopic procedures. Future prospective studies are warranted to validate these findings and establish standardized cutoff values for broader clinical application.

Disclosure: Nothing to disclose.

PP0920

VIRAL HEPATITIS OPPORTUNISTIC SCREENING IN PATIENTS ADMITTED TO THE UNIVERSITY HOSPITAL IN SIENA

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Introduction: Chronic viral hepatitis is among the principal public health problems and a leading cause of morbidity and mortality. The advent of direct-acting antiviral agents (DAAs) impressively changed the management of chronic Hepatitis C virus (HCV) infection. In parallel, WHO's global hepatitis strategy aims to reduce new hepatitis infections by 90% and deaths by 65% from 2016 to 2030 and to identify undiagnosed individuals with chronic hepatitis infection before clinical progression. Currently, a free screening for HCV infection is offered in Italy to subjects born between 1969 and 1989 and also to subjects under the care of addiction services (SerD) and those in detention. In line with national screening plans, the Tuscany Region activated free screening among these subjects.

Aims & Methods: This study aimed to define the prevalence of chronic HCV infection in patients over 55, specifically those not included in free HCV screening programs, aligning with global, regional, and national plans for HCV elimination.

Additionally, the study provides a linkage to the care of those infected, evaluates the hepatitis B virus (HBV) prevalence in a subgroup of patients, and describes the clinical features of this study population. Patients born before 1969 hospitalized to the Internal Medicine and Gastroenterology Divisions at Santa Maria alle Scotte Hospital in Siena prospectively underwent screening for HCV antibodies. The gastroenterology cohort was evaluated for HBV infection. All enrolled patients were assessed for comorbidities, treatment and transaminase levels (aspartate aminotransferase [AST] and alanine aminotransferase [ALT]).

Results: A total of 207 subjects, of whom 169 resided in Siena, were divided into three cohorts based on the department where they were admitted and underwent HCV screening. Patients present a high prevalence of comorbidities (88.4%), among them common were cardiovascular diseases (68.1%), cancer (35.3%), liver and hepatobiliary diseases (28%), and diabetes (22.2%). In the entire study population, 15.5% of patients received immunomodulating drugs.

Overall, 63 patients (30.4%) showed the presence of elevated transaminase levels, 8 patients (3.9%) were anti-HCV positive, and 2 were HCV RNA positive (0.9%). Of 96 patients in the gastroenterology cohort, 9 (9.4%) had HBV infection, of whom 8.4% had HBsAg-negative infection.

Conclusion: An active HCV infection prevalence of 0.9% was found in hospitalized patients (> 55 years old). A substantial proportion of people in the gastroenterology cohort residing in South-Eastern Tuscany carried occult HBV infection (8.4%). The increase in transaminase levels can be linked to the combined presence of liver and hepatobiliary diseases and the administration of immunomodulatory drugs.

This study confirms that an opportunistic screening can contribute to identifying the submerged people affected by viral hepatitis.

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Disclosure: Prof. Brillanti reports grants and personal fees from Gilead Sciences.

PP0921

EVALUATION OF CAMP-B SCORE IN PREDICTING THE RISK OF HEPATOCELLULAR CARCINOMA IN CAUCASIAN PATIENTS HAVING CHRONIC HEPATITIS B

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Introduction: The most common aetiology of hepatocellular carcinoma (HCC) is chronic hepatitis B infection. CAMP-B score is designed as an easy to use predictive model to assess HCC risk in patients after hepatitis B surface antigen seroclearance (HBsAg), using cirrhosis, age, male sex, platelet count as parameters.

Aims & Methods: This study aims to evaluate the use of CAMP-B score for predicting HCC risk in patients in different stages of chronic hepatitis B infection. We also aimed to compare CAMP-B score with PAGE-B and mPAGE-B scores in predicting HCC risk.

This retrospective study enrolled 322 chronic hepatitis B patients which visited the Gastroenterology Clinic of the University of Health Sciences Şişli Hamidiye Etfal Training and Research Hospital between December 2017 and December 2019. Patients were classified into low-, intermediate- and high-risk groups according to their CAMP-B, PAGE-B, and mPAGE-B

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scores. The predictive power of these scores for HCC risk was evaluated using incidence rates, Kaplan-Meier analysis, and Receiver Operating Characteristic (ROC) analysis.

Results: The incidence of HCC was 3.4% (n=11), with an incidence rate of 3 per 1000 person-years. The CAMP-B score effectively distinguished between low-, intermediate- and high-risk groups (p<0.001). Compared to PAGE-B (AUC=0.913) and mPAGE-B (AUC=0.929) scores, CAMP-B score demonstrated the highest discriminatory power (AUC=0.949). CAMP-B score classified %68.6 (n=221) of patients as low risk while, PAGE-B classified %44.1 (n=142) and mPAGE-B classified %46.3 (n=149) as low risk (Table 1). Among high-risk patients, the incidence rate was calculated as 32 per 1000 person-years for CAMP-B, 15 for PAGE-B, and 13 for mPAGE-B (Table 1). There was a significant association between age, platelet count, ALP and GGT levels and HCC development in multivariate analysis.

		Total Cohort – n (%)	HCC not Present – n (%)	HCC Present – n (%)	Incidence Rate (time-person)
CAMP-B	Low	221 (68.6)	221 (100.0)	0 (0.0)	0
	Intermediate	80 (24.8)	76 (95.0)	4 (5.0)	4
	High	21 (6.5)	14 (66.7)	7 (33.3)	32
PAGE-B	Low	142 (44.1)	142 (100.0)	0 (0.0)	0
	Intermediate	140 (43.5)	136 (97.1)	4 (2.9)	2
	High	40 (12.4)	33 (82.5)	7 (17.5)	15
mPAGE-B	Low	149 (46.3)	149 (100.0)	0 (0.0)	0
	Intermediate	120 (37.3)	118 (98.3)	2 (1.7)	1
	High	53 (16.5)	44 (83.0)	9 (17.0)	13

Table 1. Classification of Patients into Low-, Intermediate- and High-Risk Groups Accoring to CAMP-B, PAGE-B, and mPAGE-B scores.

Conclusion: The CAMP-B score emerged as an effective tool for predicting HCC risk in chronic hepatitis B patients. This is the first study to show CAMP-B score can be used as a tool to predict the risk of HCC in Caucasian patients having chronic hepatitis B.

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Disclosure: Nothing to disclose.

PP0922

LIVERRISK SCORE EFFECTIVELY PREDICTS FIBROSIS PROGRESSION AND MORTALITY IN HCV PATIENTS TREATED WITH DIRECT-ACTING ANTIVIRALS

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Introduction: The LiverRisk score was recently developed as a tool for predicting liver fibrosis and liver-related outcomes in the general population. We aimed to evaluate the predictive performance of the LiverRisk score for liver fibrosis and overall mortality in HCV patients treated with direct acting antivirals.

Aims & Methods: This was retrospective, descriptive and analytical study conducted in a gastroenterology department over 8 years including patients diagnosed with chronic hepatitis C. Data were collected from patient records and during their follow-up. The data normality test for this study was conducted using the Shapiro-Wilk test. The discrimination ability of the Liver Risk Score was evaluated using the Area Under the Receiver Operating Characteristic (AUROC) curve. Overall mortality was assessed at the conclusion of the follow-up period.

Results: Among 165 patients monitored for hepatitis C and treated with direct-acting antivirals, only 68 (41.2%) patients were eligible for this study. Mean age at diagnosis was 61±13.2 year; sex ratio M/W was 0.78 .10 (14.7%) of the patients were tobacco users. 22 (32.4%) had hypertension, and 14 (20.5%) had diabetes.

Regarding hepatic risk factors; 5 (7.5%) had alcohol use disorder, and use of cannabinoid substances was reported in 5 (3%) of the cases.1 (1.4%) had history of drug injection. 7 (10.6%) had a history of bloodletting and tattoos, and 4 (6%) had high-risk sexual practices.

Regarding the circumstances of diagnosis, 23 (39.7%) patients were diagnosed incidentally, 18 (26.4%) following an evaluation for asthenia, 4 (6.9%) presented with abdominal pain, and 3 (4.4%) exhibited signs of portal hypertension.

In terms of laboratory findings, 25 (36.7%) had cytolysis, 7 (10.2%) showed biochemical cholestasis, 16 (23.52%) had thrombocytopenia, and 5 (7.3%) presented with sign of hepatocellular insufficiency. Hepatitis B coinfection was identified in 3 (4.4%) cases.

The most frequent genotype was 1b, identified in 18 (45%) patients, followed by genotype 2 in 17 (25%). Genotypes 1a, 3, and 4 were observed in 5 (7.3%) patients.

On ultrasound, 15 (22%) patients showed signs of chronic liver disease, and 8 (11.7%) exhibited signs of portal hypertension. A total of 26 (38.2%) patients underwent upper endoscopy, among whom 13 (19.11%) had esophageal varices, 2 (2.9%) had gastric or gastroesophageal varices (GOV), and 9 (13.2%) presented with hypertensive gastropathy.

46 (75.4%) were treated with Sofosbuvir and Daclatasvir, followed by Sofosvubir and Velpatasvir for 9 (14.8%). Sustained virological response was achieved for 51 (91.1%) patients.

For fibrosis, 19 (30.2%%) were at stage F4, 9 (14.3%) at stage F3, 13 (20.6%) at stage F2, and 22 (34.9%) at stage F0-F1. Median Liver risk score was 7.58[6.28;8.5].

The LiverRisk Score exhibited high accuracy in predicting liver stiffness and outperformed traditional serum biomarkers of fibrosis. Its discriminatory capacity for detecting liver stiffness ≥10 kPa was reflected by an AUROC of 0.848, compared to 0.779 for the FIB-4 test. After 5 years follow up 8 (11.76%) patients died. LiverRisk score was associated with the risk of all cause of mortality (HR= 1.34, 95% IC[0.98;1.83], p = 0.047).

Conclusion: The LiverRisk score is an accurate tool to predict liver fibrosis and mortality in HCV patients treated with direct acting antivirals.

Disclosure: Nothing to disclose.

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PP0923

EPIDEMIOLOGY, CLINICAL MANIFESTATIONS, AND FACTORS ASSOCIATED WITH HOSPITALIZATION OF HEPATITIS E VIRUS INFECTION IN THE HELSINKI UNIVERSITY HOSPITAL AREA, 2020–2024

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Introduction: Hepatitis E virus (HEV) infection in Europe is predominantly due to zoonotic genotype 3, transmitted via undercooked pork, game or shellfish (1,2). In the Helsinki and Uusimaa Hospital District (HUS), 70 laboratory-confirmed cases occurred from 2020–2024, with a pronounced peak in early 2024 (3). Seroprevalence in high-income countries ranges from ~6 % to > 50 %, increasing with age and swine exposure (1,2).
While most infections are asymptomatic or mild, patients with underlying liver diseases may have more severe outcomes. Diagnosis relies on anti-HEV IgM/IgG serology and PCR detection of viral RNA (4).

Aims & Methods: Objective of this study is to investigate the epidemiology of hepatitis E virus (HEV), its clinical manifestations, and factors associated with hospitalization in the Helsinki University Hospital (HUS) area between 2020 and 2024. We conducted a retrospective registry study including all patients diagnosed with hepatitis E in the HUS area from January 2020 to December 2024. Data were collected from the electronic health record system, the laboratory database, and the shared contact-tracing software of the HUS Epidemiology Unit and Uusimaa welfare regions.
Variables analyzed comprised demographic data, symptoms at presentation, key laboratory parameters (ALT, bilirubin, C-reactive protein), probable sources of infection, and details of hospitalization episodes.

Results: Study population: 70 patients (median age 58 years [IQR 45–67]; 58.6 % male).
Clinical manifestations: 92.2 % were symptomatic. The most frequent symptoms were fatigue (55.9 %), abdominal pain (47.5 %), nausea (45.8 %), and jaundice (42.6 %).
Laboratory findings: Median peak ALT was 1 153 U/L (IQR 544–1 538); median peak bilirubin 39 µmol/L (IQR 17–79); median peak CRP 22 mg/L (IQR 11–32).
Genotypes: Of 19 genotyped cases, 3c (21.1 %), 3e (21.1 %), and 3f (57.9 %), with genotype 3f predominant.
Infection sources: Cold-smoked sausages (e.g., mettwurst) were implicated in most domestic cases (69.6 % of identified sources), especially during the early 2024 outbreak. Other identified sources were unspecified meat products (6.5 %), berries (6.5 %), contact with wild animals (e.g., hunting, 4.3 %), blood transfusion (4.3 %), pork meat (4.3%), and game meat (4.3%).
Hospitalization: 50.7 % required ≥1 night of inpatient care. Hospitalized patients were older, had higher ALT, bilirubin, and CRP values, and more comorbidities than outpatients.
Predictors of hospital stay: In a multivariable zero-inflated negative binomial model, higher peak CRP was independently associated with longer hospital stay ($\beta = 0.02$; $p = 0.005$).

Conclusion: The early-2024 surge of HEV cases in the HUS area represents an internationally rare, large genotype 3 outbreak. Cold-smoked meat products were identified as the principal transmission route, although infections linked to blood transfusions, consumption of raw berries, and wildlife contact also occurred. Cases were disproportionately concentrated in men over 50 years of age. Among hospitalized patients, peak CRP emerged as the strongest predictor of inpatient care, and both mortality and incidence of hepatic insufficiency were relatively high given high sanitation standards in Finland. Further multicenter, inter-regional studies are needed to establish statistically robust risk factors for severe outcomes and to guide clinical and public health interventions.

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Disclosure: Nothing to disclose.

PP0924

DIFFERENCES IN BASELINE CHARACTERISTICS OF DIRECT ACTING ANTIVIRAL (DAA)-TREATED GREEK HCV PATIENTS ACCORDING TO SOURCE OF INFECTION

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Introduction: With the introduction of direct-antiviral medication (DAAs), eradication of HCV infection is considered now an achievable goal. While people with history of drug use (PWHD) form the most common population at risk for the infection, in Greece historically there has been a significant proportion of patients arising from the general population with no prior drug use. These two different subgroups exhibit different characteristics, that may be pertinent in the management of HCV infection and long-term follow up.

Aims & Methods: The aim of this study was to identify potential differences in the makeup of these two HCV patient subgroups.
The HERACLIS cohort, is the largest national HCV registry of patients treated with DAAs in tertiary liver centers from 2015 until 2024. Clinical data and characteristics were obtained from medical records, while transient elastography (TE) measurements and APRI score calculations were conducted at baseline. Patients were followed up initially for the duration of their treatment.

Results: 930 patients were included in the study, 69.8% (n=649) of them were men with median age 52 years old and 9.2% of these had decompensated liver disease at DAA initiation. 56.8% (n=528) of this cohort were PWHD, while the rest had been infected in another manner (non-PWHD). When comparing these two populations, PWHD were younger in age (47.1±9,8 vs 56.9±13.1 years, p<0.001) and with lower BMI (24.4±4.4 vs 26.4±4.7 kg/m2, p<0.001).
Non-PWHD cases had more often diabetes (13.6% vs 2.7%, p<0.001), hypertension (10.7% vs 8.8%, p<0.001) as well as decompensated liver disease (13.5% vs 6.1%, p<0.001). This was also reflected by TE measurements

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(13.9±10.9 vs 10.9±7.7 kPa, p=0.001) but not by the APRI score. Moreover, PWHD patients tended to have been more often infected with genotype 3 (64% vs 30%, p<0.001) while for other genotypes no such stark differences were observed. There was no significant variance observed regarding DAA regimen use but non-PWHD patients tended to suffer more often from adverse events secondary to DAA treatment (3.1% vs 0.7%, p=0.001).

Conclusion: The Greek PWHD and non-PWHD populations of HCV patients seem to exhibit different baseline characteristics which may be relevant to treatment selection and long-term follow up regimens. The main differences observed imply the concomitant presence of an element of metabolic syndrome in the non-PWHD cohort with a subsequent potential effect on the degree of underlying liver fibrosis.

Disclosure: Nothing to disclose.

PP0925

COMPARATIVE EVALUATION OF APRI SCORE AND TRANSIENT ELASTOGRAPHY FOR LIVER FIBROSIS ASSESSMENT IN CHRONIC HEPATITIS B PATIENTS

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Introduction: Chronic hepatitis B (CHB) remains a major global health concern due to its potential progression to cirrhosis and hepatocellular carcinoma. Accurate staging of liver fibrosis is essential for guiding clinical decisions. Non-invasive assessment methods such as the AST to Platelet Ratio Index (APRI) and transient elastography (Fibroscan) are increasingly used alternatives to liver biopsy.

Aims & Methods: To assess the diagnostic accuracy of the APRI score for detecting liver fibrosis in CHB patients, using Fibroscan as the reference standard.

This retrospective study included 68 CHB patients who underwent liver stiffness measurement via Fibroscan. APRI scores were calculated based on AST levels and Platelet counts. Fibrosis stages were classified from F0 to F4. Correlations between APRI and fibrosis stages were analyzed, and receiver operating characteristic (ROC) curves were generated to evaluate diagnostic performance.

Results: A total of 68 patients with chronic hepatitis B were included in the study, with a mean age of 47years and an ecart type of 11,1 years (age range: 20-82 years). Liver stiffness values measured by Fibroscan ranged from 5 to 16 kPa, allowing fibrosis staging from F0 to F4. The distribution of fibrosis stages was as follows: F0 (n =5) , F1 (n = 42), F2 (n = 13), F3 (n = 2), and F4 (n =6).

The mean APRI score among all patients was 0.55 (range: 0.09-2.36). A moderate but statistically significant correlation was found between APRI scores and fibrosis stages assessed by Fibroscan (r=0.45, p<0.05).

Patients with significant fibrosis (≥F2) had significantly higher APRI scores compared to those with minimal fibrosis (F0-F1) (1.05±0.65 vs. 0.32±0.41; p=0.01). (ROC) curve analysis revealed that an APRI cut-off of 0.72 provided the best balance for identifying significant fibrosis yielded an AUC of 0.68 (95% CI: 0.56-0.80), with a sensitivity of 66.7% and specificity of 71.1%.

For the detection of cirrhosis (F4), the optimal APRI cut-off was 0.86. At this threshold, the AUC was 0.71 (95% CI: 0.59-0.83), with sensitivity of 73.3% and specificity of 68.2%

Conclusion: While transient elastography remains he more precise tool, the APRI score demonstrates acceptable diagnostic performance for identifying significant fibrosis and cirrhosis in CHB patients. Its affordability and accessibility make it a valuable screening tool, especially in resource-limited settings.

Disclosure: Nothing to disclose.

PP0926

ASSESSMENT OF FIBROSIS REGRESSION UNDER ENTECAVIR IN CHRONIC HEPATITIS B USING TRANSIENT ELASTOGRAPHY

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Introduction: Chronic hepatitis B (CHB) can progress to cirrhosis or hepatocellular carcinoma. Entecavir (ETV) is a first-line antiviral therapy that can lead to fibrosis regression. This study aimed to assess changes in liver stiffness using transient elastography (TE) and determine the rate of fibrosis regression during ETV therapy.

Aims & Methods: A 10-year retrospective cross-sectional study was conducted in a Tunisian tertiary hospital. CHB patients treated with ETV for ≥48 weeks and with both pre- and post-treatment TE were included. Fibrosis regression was defined as a ≥2 kPa decrease in liver stiffness or a ≥1-stage improvement. Liver stiffness was measured using FibroScan® before treatment and after a minimum of 6 months of therapy.

Results: Among 129 patients, 113 had pre-treatment TE (87.6%). Median baseline liver stiffness was 8.8 kPa [3.2–33]. Fibrosis distribution: F2 (28.3%), F3 (27.4%), and F4 (19.5%). After a median of 10 months on ETV, liver stiffness significantly decreased to 5.8 kPa (p<0.001). Fibrosis regression was observed in 61.9% of cases. Fibrosis stage improved in 65 patients (57.5%), was stable in 35 (31%), and worsened in 13 (11.5%) cases. Higher rates of regression were associated with younger age, lower BMI, better adherence, and biochemical/virological response.

Conclusion: ETV therapy significantly reduced liver stiffness in CHB patients, with fibrosis regression in over 60% of cases. Transient elastography is an effective tool for non-invasive fibrosis monitoring during antiviral treatment.

Disclosure: Nothing to disclose.

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Biliary
Poster Exhibition

PP0927
VARIABILITY IN SEHCAT CUTOFF VALUES FOR DIAGNOSIS AND SEVERITY CLASSIFICATION OF BILE ACID MALABSORPTION: CLINICAL IMPLICATIONS

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Introduction: The SeHCAT test is a critical diagnostic tool for bile acid malabsorption (BAM), a recognized cause of chronic diarrhoea that significantly impacts patients' quality of life. By measuring the retention of a radio-labelled bile acid analogue over seven days, SeHCAT classifies BAM severity. However, variability in cutoff values and nomenclature across clinical settings has led to inconsistencies in diagnostic sensitivity and specificity. Lower cutoffs (e.g., less than (<) 5%) prioritize specificity but risk missing mild cases, while higher thresholds (e.g., <15%) improve sensitivity but may lead to overdiagnosis. Individualized interpretations based on clinical context could enhance diagnostic precision.

Aims & Methods: We aimed to study the impact of this cutoff value variability in our patients in comparison with literature. We undertook a 5year retrospective analysis of all patients who underwent SeHCAT testing for suspected BAM from January 2019-2023. Relevant PubMed studies in last 10 years and NICE Diagnostic Guidance (DG7)/related studies were also reviewed.

Results: Nineteen studies were included; 3 comprehensively addressed cutoff variability. NICE tasked the Kings Technology Evaluation Centre (KiTEC) with exploring this in 1,036 patients across 38 UK centres. Results highlighted inconsistent cutoff classifications, some using "normal (>15%), borderline (5–19%), abnormal (0–4 & 0-14%)"; others employing "normal (>15%), mild (10–14.9%), moderate (5–9.9%), severe (0–4.9%)." NICE recommends a 15% cutoff for diagnosis. In our study, 353 patients underwent SeHCAT testing during the period 2019-2023. Of these, 175 (49.57%) were diagnosed with BAM; of these, 47 patients (26.86%) were classed as mild, 63 (36%) as moderate and 65 (37.14%) as having severe BAM, based on the latter nomenclature, in line with most UK centres and were managed accordingly.

Conclusion: Variability in SeHCAT cutoff values significantly affects BAM diagnosis/severity classification and therefore appropriate management. Standardizing cutoff thresholds and integrating SeHCAT results into a holistic clinical framework is essential to balance sensitivity and specificity. Further research is needed to validate cutoff ranges and establish evidence-based guidelines for optimal patient outcomes.

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Disclosure: Nothing to disclose.

PP0928
GENETICS-BASED ANALYSIS IDENTIFIES PLASMA GLYCOCHENODEOXYCHOLATE GLUCURONIDE AS A POTENTIAL PROTECTIVE FACTOR FOR PRIMARY SCLEROSING CHOLANGITIS

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Introduction: Primary sclerosing cholangitis (PSC) is a major type of autoimmune cholestatic liver disease with unknown etiology, characterized by cholestasis, abnormal biliary epithelial homeostasis, biliary tract fibrosis and consequently liver dysfunction[1,2].

Notably, PSC further elevates the risk of developing cholangiocarcinoma[3]. Investigating the underlying causal factors could contribute to the development of novel therapeutic strategies, thereby mitigating the burden on public health systems imposed by these conditions.

Altered bile acid (BA) homeostasis is an intrinsic facet of cholestatic liver diseases. The toxicity of bile acids is associated with the severity of cholestatic diseases[4,5].

Among all BA components, hydrophobic BAs such as chenodeoxycholic acid (CDCA) and deoxycholic acid (DCA)[6] are considered the main toxic components to the bile duct epithelium and hepatocytes. Besides, glycochenodeoxycholic acid (GCDCA) has been used to model BA toxicity both in vivo and in vitro[7].

The clinical usefulness of plasma BA in cholestatic diseases was studied via BA profiling in cholangitis cohorts, and the clinical utility was analysed[4,8].

Nevertheless, no large-scale epidemiology studies have identified the causal relationship between individual components of BA and the pathogenesis of cholangitis, especially the sequence between changes in BA composition and disease onset.

In this study, we extracted summary statistics of 32 plasma BA components from the largest metabolome GWAS dataset to date[9] as exposures and performed two-sample MR using 2 datasets of PSC as outcomes, followed by reverse MR, to assess the causal relationship between BAs and the pathogenesis of this disease.

Aims & Methods: We aimed to identify causal bile acid components on PSC using mendelian randomization. Summary statistics for primary sclerosing cholangitis were retrieved from European Bioinformatics Institute and FinnGen Consortium, and plasma bile acid components data were obtained from European Bioinformatics Institute. BA profiles of PSC patients by GC-MS were retrieved from published researches.

Statistical analyses used the inverse variance weighted(IVW), weighted median(WM), maximum likelihood(ML) and MR-Egger algorithm, followed by various sensitivity analyses and reliability evaluation.

Results: Plasma level of glycochenodeoxycholate glucuronide(GCDCG) is negatively correlated with primary sclerosing cholangitis (IVW: OR=-0.12, p=0.0269 in EBI dataset, OR=-0.38, p=0.0299 in FinnGen dataset). No reverse causal effect was identified(IVW: p=0.7067 in EBI dataset, p=0.5158 in FinnGen dataset). GCDCG was decreased in bile of patients with PSC compared with healthy controls (logFC=-2.09, p=0.010).

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MR type	data sources	outcome	exposure	method	OR	p value	Cochrans Q p value	Egger intercept p value
Forward	EBI GWAS catalog	PSC	GCDCG level	ML	-0.1170	0.0135		
				MR Egger	-0.1960	0.0242	0.4143	0.1469
				WM	-0.1813	0.0007		
				IVW	-0.1158	0.0269	0.2753	
	FInnGen			ML	-0.3880	0.0299		
				MR Egger	-0.6690	0.0768	0.8872	0.3418
				WM	-0.5047	0.0294		
				IVW	-0.3836	0.0299	0.8663	
Reverse	EBI GWAS catalog	GCDCG level	PSC	IVW	0.0054	0.7067	0.5615	

Conclusion: Our findings provide evidence regarding the risk of primary sclerosing cholangitis for individuals with low level of plasma glycochenodeoxycholate glucuronide, and targeting the transformation of glycochenodeoxycholate into its glucuronide may be a novel therapeutic strategy to primary sclerosing cholangitis.

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Disclosure: Nothing to disclose.

PP0929

THE ALTERATIONS OF BILE ACIDS SPECTRUM AND SMALL INTESTINAL BACTERIAL OVERGROWTH IN PATIENTS AFTER CHOLECYSTECTOMY: A PROSPECTIVE STUDY

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Introduction: Diarrhea after cholecystectomy is a well-known complication that worsens the quality of life of the patient and may become a real challenge for the doctor.

Both changes in the bile acids spectrum (BAsS) and possible changes in microbiota and Small Intestinal Bacterial Overgrowth (SIBO) may have a major impact. What is primary in this circle is not fully understood.

Aims & Methods: The aim was to assess BAsS and SIBO in patients prior to cholecystectomy, three and twelve months after cholecystectomy.

Methods: Initially, the bile of the patients was obtained directly from the gallbladder after it was removed during the surgery. For the second and third time, the bile of the same patients after cholecystectomy was obtained in 3 and 12 months postoperatively, after the intubation of the duodenum under ultrasound control. The spectrum of BA was assessed using photocalorimetry.

To assess the possible influence of SIBO, all the patients underwent a hydrogen breath test before cholecystectomy, and if SIBO was confirmed, they were not included in the study. Successive hydrogen breath tests were performed 3 and 12 months after cholecystectomy to all the patients who were included. The data were analyzed using parametric and non-parametric methods of evaluating the results through the Statistica program (StatSoft Inc, USA).

Results: 14 patients, predominantly women (71.4%) were enrolled in the study. The spectrum of BA three months after cholecystectomy showed a tendency for secondary BA (deoxycholic acid) to predominate over primary BA (cholic acid) (P>0.05). After 12 months, the BAs pool returned to the baseline in most of the patients (P>0.05). SIBO was not registered in patients 3 months after cholecystectomy, but it was detected in 50% of cases (P>0.05) after 12 months.

Conclusion: The predominance of secondary BA in the early postoperative period probably indicates an increase in the number of bacteria, for the transformation of primary BA in the ileum, but the breath test is not sensitive enough to determine this.

Over time, if the equilibrium in a complex mechanism of BA homeostasis is not achieved (the amount of bile in the ileum, bacteria, FXR, etc.), this may lead to the SIBO development.

Disclosure: Nothing to disclose.

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PP0930

METABOLIC CONSEQUENCES OF CHOLECYSTECTOMY: DATA FROM AN ORIGINAL STUDY

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Introduction: Gallstone disease (GSD) is considered one of the most common diseases of the digestive system. According to publications, in the modern world there is a steady trend towards an increase in the prevalence of GSD among young people. According to WHO forecasts, if current growth rates continue, by 2050 every fifth inhabitant of the planet will suffer from GSD. Heredity is the most important risk factor for the development of the disease. It is no longer nonsense to detect cholelithiasis in childhood with a frequency of about 5%.

Aims & Methods: To study the effect of cholecystectomy on the metabolism of lipids and bile acids in the blood plasma of patients with cholelithiasis. The study involved 96 patients aged 22-45 years with cholelithiasis (gallstones) who underwent elective laparoscopic cholecystectomy. They were divided into 2 representative groups according to the lipid profile of their blood plasma: “Normal lipid spectrum” and “Changes in the lipid spectrum”.

Clinical data, questionnaires, lipidogram, chromatography-mass spectrometry of bile acids in blood plasma and bile were used. After surgery, half of the patients with changes in the lipid spectrum received ursodeoxycholic acid (UDCA) for 3 months. Repeated examination was performed 3 months and 1 year after cholecystectomy.

Results: Lipidogram indices in the groups differed statistically significantly. Plasma bile acid levels differed significantly for cholic and ursodeoxycholic acids. In the “Normal lipid spectrum” group, an improvement was observed after 3 months, and a significant deterioration in lipidogram indices was observed after a year. In the “Changes in lipid spectrum” group, an improvement in lipidogram indices was observed after 3 months only in patients taking UDCA.

No statistically significant changes were found after a year. Patients with GI have an imbalance of bile acids: an increase in the levels of choleic, deoxycholic, glycocholic, and taurocholic acids, and a decrease in the level of henodeosquicholic.

After cholecystectomy, plasma concentrations of elevated acids, as well as initially reduced henodeoxycholic, decrease. In patients with a normal blood lipid spectrum, cholecystectomy has a negative effect on lipidogram parameters. Cholecystectomy increases the risk of deterioration of the lipid spectrum by 1.6 times. Taking UDCA improves blood lipid spectrum parameters.

Conclusion: Based on the results of the study, a patent was registered for the invention “Method for predicting lipid spectrum disorders and recurrent stone formation in young people after cholecystectomy for cholelithiasis” No. RU 2799862 C1

Disclosure: Nothing to disclose.

PP0931

POST-CHOLECYSTECTOMY CHANGES IN GUT MICROBIOME AND SYSTEMIC METABOLOME: AN EXPLORATORY PROSPECTIVE COHORT STUDY

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Introduction: Observational studies have associated cholecystectomy with cardiometabolic disorders and colorectal cancer. This is of clinical importance as the case volume of cholecystectomy has increased exponentially and cardiometabolic disorders pose a significant health burden. Understanding post-cholecystectomy gut microbiome changes may help clarify some of these observations. Published data on cholecystectomy-related gut microbiome changes are limited and mainly cross-sectional in study design, lack adequate taxonomic resolution or prediction of functional pathways as 16S rRNA sequencing rather than shotgun metagenomics sequencing was used, and observed microbiome changes have not been correlated with concomitant changes in the metabolome[1].

Aims & Methods: This exploratory prospective cohort study aimed to examine the changes in gut microbiome and systemic metabolome post-cholecystectomy. Subjects aged 21 to 65 years undergoing cholecystectomy for treatment of symptomatic gallstone disease were enrolled. Exclusion criteria included the use of antibiotics or probiotics within one month, appendicectomy, bowel surgery and inflammatory bowel disease. Biospecimens (stool, plasma and urine) were collected at baseline and one-month post-cholecystectomy. Stool microbiome was analysed by shotgun metagenomics sequencing, with assessment of ω^2 effect size (derived from weighted Jaccard distance), α (Mann-Whitney U test for unpaired and Wilcoxon signed-rank test for paired tests) and β diversity indices (PCoA and PERMANOVA), as well as differentially abundant bacterial taxa and functional pathways (Maaslin2 R package, using false discovery rate [FDR] < 0.1). Systemic metabolome (plasma, stool and urine) was examined by targeted liquid chromatography-mass spectrometry, and Orthogonal Partial Least Squares Discriminant Analysis was performed to identify metabolites that were different post-cholecystectomy, using paired t-tests and FDR < 0.1. Correlation analysis of changes in stool microbiome and metabolome was performed.

Results: A total of 26 subjects were enrolled. The mean age was 47 years (12), with 62% (16/26) male and mean body mass index (BMI) 25.57 kg/m^2 (3.75). Other clinical meta-data were smoking in 15.4% (4/26), alcohol consumption in 11.5% (3/26), use of proton pump inhibitors (PPI) in 7.7% (2/26), statins usage in 23.1% (6/26), and presence of fatty liver in 15.4% (4/26). The microbiome profile of the enrolled cohort was similar to matched asymptomatic controls with intact gallbladder (with or without gallstones). One-month post-cholecystectomy, the ω^2 effect size was 0.006 (80% power, $\alpha < 0.05$). Significant differences were observed for β diversity

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indices after controlling for drinking status, presence of fatty liver and use of PPI. There were 32 differentially abundant bacterial taxa and 124 differentially abundant functional pathways.
Analysis of observed microbiome and systemic metabolome changes revealed correlation of 9 species with plasma bile acid (chenodeoxycholic acid [CDAC]), 20 species with plasma lipids, 3 species with stool bile acids (CDCA), 9 species with stool carboxylic acids and 2 species with urinary trimethylamine N-oxide.
Conclusion: Cholecystectomy resulted in significant changes in the stool microbiome and systemic metabolome. Such changes may help explain the epidemiological association with cardiometabolic disorders. Further larger scale prospective cohort studies with longer duration of follow-up are needed to confirm these observations.
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Disclosure: Nothing to disclose.

PP0932

SURGEON VERSUS RADIOLOGIST INTERPRETATION OF INTRAOPERATIVE CHOLANGIOGRAPHY IN LAPAROSCOPIC CHOLECYSTECTOMY: A POPULATION-BASED STUDY

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Introduction: In laparoscopic cholecystectomy (LC), intraoperative cholangiography (IOC) is recommended to clarify biliary anatomy, detect choledocholithiasis and identify complications. IOC interpretation may be performed by either surgeons or radiologists, but it remains unclear which approach is preferable. This study investigates how the type of medical specialist interpreting the IOC affects surgical outcomes and operating time.
Aims & Methods: All LCs registered in the Swedish Registry of Gallstone surgery and Endoscopic retrograde Cholangiopancreatography (GallRik) between January 1, 2009, and December 31, 2020, were included. A complementary survey was distributed to all participating surgical clinics to determine local IOC interpretation practices. Registry and survey data were linked and outcomes were analysed using mixed model analysis and logistic regression.
Results: A total of 118.556 procedures from 61 surgical units were included. Operating time was longer when radiologists assessed the IOC (mean differences 11.3 minutes; 95% CI 1.60-20.89). Intraoperative complications were more common with radiologist interpretation (2.6% vs 1.5%; OR 0.58, 95% CI 0.52-0.63), while postoperative complications were slightly higher with surgeon-led interpretation (8.2% vs 7.3%; OR 1.14, 95% CI 1.08-1.20). Bile duct injury rates (0.3%; OR 1.0, 95% CI 0.79-1.27) and detection of choledocholithiasis (12.4% vs 12.6%; OR 1.02, 95% CI 0.90-0.99) were similar between groups.
Conclusion: Surgeon-led IOC interpretation is associated with shorter operating times and fewer intraoperative complications, without increasing the risk of bile duct injury or missed choledocholithiasis. Routine radiologist involvement may be unnecessary and an inefficient use of resources.
Disclosure: Nothing to disclose.

PP0933

THE USE OF A NOVEL HEAMOSTATIC PEPTIDE GEL IN THE MANAGEMENT OF ENDOSCOPIC PAPILLECTOMY- RELATED BLEEDING: A CASE SERIES

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Introduction: Endoscopic Papillectomy (EP) represents the gold standard treatment for ampullary adenoma. Despite high rates of clinical success, EP is still burdened by a non-negligible rate of adverse events, with bleeding being one of the most frequent, with a reported pooled rate of 10.6%. Despite differences in presentation and in severity, the majority of the EP-related bleeding can be managed endoscopically. Various techniques, including epinephrine injection, argon plasma coagulation (APC), hemoclips, biliary stents have been employed to manage bleeding, but robust evidence supporting their efficacy are still lacking. Recently, the introduction of a novel haemostatic peptide gel has expanded the toolbox of endoscopic haemostatic agents.
Aims & Methods: This study aims to evaluate the safety and efficacy of this new haemostatic agent in controlling bleeding in patients undergoing EP. This is a multicenter, retrospective study from 4 high-volume centers in Italy, including all consecutive patients treated with the novel heamostatic peptide gel for the management of active EP-related bleeding, between 2019 and 2024. Clinical success was defined as the achievement of hemostasis using the hemostatic gel, either alone or in combination with other techniques, for the treatment of active intraprocedural or delayed EP-related bleeding, without subsequent rebleeding
Results: A total of 16 patients were included in the study. In 15 patients, haemostatic gel was used to manage intraprocedural bleeding, while one patient was treated for delayed bleeding. An oozing bleeding was shown in 13 cases, a spurting bleeding in one case, while in 3 cases no active bleeding was present, although a visible vessels was found in two patients and one ulcer with an adherent clot in one case. Six patients were treated exclusively with haemostatic gel, while the remaining patients received combination treatments, including epinephrine injection (5 cases), clipping (6 cases), biliary stenting (2 cases), and fibrin glue application (1 case). The application of hemostatic gel was successful in 100% of cases, despite the use of side-viewing duodenoscopes with elevator. Clinical success of haemostatic gel application after the first bleeding episode was achieved in 75% (12/16) of patients.
However, 4 out of 16 patients (25%) experienced rebleeding within one day after the first treatment. The severity of bleeding ranged from mild to moderate, effectively controlled endoscopically in all cases except one, in which a radiologic embolization was needed in order to achieve bleed control and was therefore classified as severe. No gel-related adverse events were reported.

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Conclusion: This novel peptide gel could represent a promising and safe tool to manage active bleeding related to EP. Further prospective studies are needed to confirm its efficacy.

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Disclosure: Nothing to disclose.

PP0934

ENDOSCOPIC MANEGEMENT OF BILE DUCT INJURIES IN 203 PATIENTS: EXPERIENCE OF A RIO DE JANEIRO REFERRAL CENTER

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Introduction: Management of poscholecystectomy complications has been variably discussed. Various options exist but the recent trend has been towards the endoscopic retrograde approach favoring a simultaneous diagnostic and therapeutic potential.

Aims & Methods: Objetive: A single institution retrospective analysis of 203 with bile duct injuries (BDI) after laparoscopic cholecystectomy (LC). We studied the various endoscopic modalities to evaluate the efficacy of the approach.

Patients and Methods: From January 2005 to January 2025, a retrospective analysis was conducted of 5528 patients referred to the endoscopy department.

During this period, 434 were referred after related BDI. These 434 (7,8%) are BDI. 231(53,2%) after open cholecystectomy and 203 (46,8%) after LC, formed the basis for this study. 154 (75,8%) were female and 49 (24,2%) were male, with an average age of 49,1years (range,13-88 years).

The clinical presentation of the BDI: jaundice 134 (66,0%); pain 47 (23,1%); fever 79 (38,9%); persistent drainage 75 (36,9%); pruritus 74 (36,4%) and acute biliary pancreatitis 18 (8,8%). ERCP was used in all cases.

Results: There were 103 patients with stricture (26 total transaction and 77 partial) and 64 (31,6%) patients with leaks and 36 (17,7%) had a combination of leaks and stricture. Endoscopic diagnosis was feasible in all cases. Subsequently, 170 (83, 7%) patients were considered for endotherapy and endoscopic treatment was interrupted in 17 patients, required surgical intervention, success healing was shown in 90%.

There 64 biliary fistulae: 40 of the cystic stump, 20 of the common hepatic and 2 of the right hepatic and 2 of the left hepatic; 36 (17,7%) biliary fistulae with stenosis (9 complete transaction and 27 partial transaction) and 103 stenosis (26 total transaction and 77 partial). The 35 total transaction required surgical treatment; however two were treated by endoscopies.

There was success in the endoscopic treatment of biliary fistulae, which 64 esphincterotomy transpapilar (ETP) and four stents. 27 patients with biliary fistulae with stenosis were 6 ETP; 11 ETP with stents; 8 ETP, dilatation, stents and two ETP with dilatation and two patients need the surgical treatment. 71/77 (92,2%) patients with stenosis could be treated: 34 ETP, stents; 36 ETP, dilatation, stents; and 3 ETP, dilatation.

Complications occurred in 6 (3, 5%), perforation (n = 1), mild pancreatites (n=3), and cholangitis (n=2) and 17 patients failed endoscopic treatment.

Conclusion: The ERCP in biliary complications of LC is feasible and with good results. It firms as coadjuvant treatment of the definitive surgery and in some situations being the method close for the definitive treatment.

Disclosure: No Conflict of Interest.

PP0935

USING ENDOSCOPIC RADIAL INCISION AND CUTTING COMBINED WITH ERCP TO TREAT BILIARY-ENTERIC ANASTOMOTIC STRICTURE

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Introduction: Enteroscopy-assisted ERCP (E-ERCP) is used to treat biliary-enteric anastomotic stricture and common bile duct stones after alimentary tract reconstruction. However, anastomotic stricture usually responds poorly to balloon dilation caused by severe fibrosis.

Here, we report a novel technique, endoscopic radial incision (ERI) combined with E-ERCP, to treat biliary-enteric anastomotic stricture and common bile duct stones.

Aims & Methods: From October 2023 to November 2024, six patients with biliary-enteric anastomotic stricture and common bile duct stones who responded poorly to balloon dilation received ERI combined with E-ERCP in Zhongshan Hospital, Fudan University. The primary outcomes were relief of biliary-enteric anastomotic stricture and the removal of common bile duct stones. The secondary outcome was the complication rate of the procedure.

Results: The mean age was 48 and 3 patients were male. All 6 patients successfully received ERI combined with E-ERCP. Four patients with anastomotic stricture were incised by Hook knife, while two patients with the

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Hook knife not reaching the anastomotic area was incised by the top of the snare. After the ERI, the diameters of the biliary-enteric anastomotic stoma enlarged from 3 mm to 10 mm. The mean diameter of common bile duct stones was 8 mm.

All the common bile duct stones were removed by basket catheter successfully, confirmed by intraoperative common bile duct radiography and postoperative MRCP.

The mean procedure time was 32 minutes and the mean postoperative hospitalization time was 2 days. No intraoperative and postoperative complications occurred.

Conclusion: ERI combined with E-ERCP, as a novel technique, could be an effective and safe method to treat biliary-enteric anastomotic stricture and common bile duct stones after alimentary tract reconstruction.

Disclosure: Nothing to disclose.

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Disclosure: Nothing to disclose.

PP0936

ENDOSCOPIC PAPILLECTOMY: INSIGHTS FROM 15 YEARS OF EXPERIENCE TREATING EARLY AMPULLARY TUMORS

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Introduction: Ampullary tumors account for 0.6-0.8% of the digestive cancers[1]. Most are sporadic, but associations with familial adenomatous polyposis (FAP), neurofibromatosis type I (NF1), and Muir-Torre syndrome have been reported[2,3,4].

Endoscopic papillectomy (EP) is increasingly performed for the management of non-invasive lesions[1].

This study assessed the efficacy, predictive factors for success, complications, and recurrence rates associated with EP.

Aims & Methods: Retrospective, single-center study of patients undergoing EP between January 2010 and January 2025. Data on patient demographics, medical history, procedural events, adverse events and follow-up were analyzed.

Results: Among 49 patients (mean age 61.4 ± 14.0 years, 53.1% male), 69.4% had sporadic lesions, 28.6% had FAP, and 2% had NF1. Most lesions were confined to the papillary mound (89.8%), while 10.2% were laterally spreading tumors involving the papilla (LST-p). Mean lesion size was 15.2 ± 6.5 mm. Complete resection was achieved in 83.7% of cases. En bloc resection (79.6%) was significantly associated with higher complete resection rates (97% vs. 40%, $p < 0.001$). Laterally spreading features were highly associated with piecemeal resection (80% vs. 13.6%, $p < 0.001$) and incomplete resection (100% vs. 5.3%, $p < 0.001$).

Adverse events occurred in 46.9% of cases, including pancreatitis (30.6%, mild in 80%), intra-procedural bleeding (20.4%), and delayed bleeding (10.2%), all successfully managed endoscopically. Perforation occurred in 8.2% and was managed endoscopically, while one case of cholecystitis required surgery. Median hospital stay was 4 days. Recurrence occurred in 7 cases, 100% at first follow-up visit, and were successfully endoscopically treated.

Conclusion: EP is an effective and relatively safe intervention for the treatment of ampullary lesions. Complete resection is more frequently attained for lesions confined to the papillary mound and when en bloc resection is feasible. While adverse events are frequent, most are mild and managed conservatively/endoscopically. All recurrences were detected at the first follow up, emphasizing the value of early re-evaluation.

PP0937

PREDICTORS OF CLINICAL SUCCESS AND COMPLICATIONS FOLLOWING ENDOSCOPIC GALLBLADDER DRAINAGE FOR ACUTE CHOLECYSTITIS IN HIGH-RISK SURGICAL PATIENTS: A RETROSPECTIVE STUDY

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Introduction: Endoscopic ultrasound-guided gallbladder drainage (EUS-GBD) has emerged as a minimally invasive alternative to percutaneous drainage for high-risk surgical patients with acute cholecystitis. Although EUS-GBD is gaining traction in clinical practice, real-world data in complex and high-risk patients remain scarce, particularly regarding outcomes in perforated or malignant cases. While its efficacy is increasingly recognized, data on predictors of success and complications remain limited.

Aims & Methods: This retrospective study explored potential clinical and procedural factors associated with outcomes following EUS-GBD. Patients undergoing EUS-GBD for acute cholecystitis between October 2021 and January 2024 at a tertiary center were included. All procedures were performed by endoscopists with advanced training in interventional EUS. Data on demographics, comorbidities, procedural details, and outcomes were collected. Clinical success was defined as resolution of sepsis or cholecystitis symptoms within 48 hours. Complications included bile leak, infection, and the need for reintervention. Due to the limited sample size, statistical analysis was descriptive and exploratory.

Results: Thirteen patients (6 females, median age 69.1 years [IQR 45–90]) underwent EUS-GBD during the study period. Indications included gallstones (53.8%), malignant cystic duct obstruction (38.4%), and ischemia (7.7%). Reasons for non-surgical management included terminal malignancy (n=6, 46.2%), coronary artery disease (n=5, 38.5%), cirrhosis with hepatocellular carcinoma (n=1, 7.7%), and the patient's choice (n=1, 7.7%). Lumen-apposing metal stents (LAMS) were placed transgastrically in 10 patients (n=10, 77%) and transduodenally in 3 (n=3, 23%).

Two patients (15.4%) presented with gallbladder perforation and pericholecystic abscess at the time of EUS-GBD. Both cases achieved full resolution following the drainage procedure, indicating that EUS-GBD may serve as a valuable rescue therapy in complicated acute cholecystitis.

Clinical success was achieved in 12 out of 13 patients (92.3%). One patient (7.7%) developed a bile leak, associated with the presence of ascites and hepatocellular carcinoma. Median post-procedural hospital stay was 6.8 days (IQR 3–20). Exploratory analysis suggested a trend toward

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higher complication rates in patients with ascites and those undergoing transgastric drainage, though no definitive conclusions could be drawn. At 30-day follow-up, no late adverse events or stent-related complications were reported.

Conclusion: EUS-GBD is a highly effective and safe treatment option for acute cholecystitis in patients unfit for surgery, with rapid symptom resolution and acceptable hospital stay duration. Notably, the resolution of pericholecystic abscesses in two cases of gallbladder perforation suggests that EUS-GBD may be beneficial even in more complex clinical scenarios. Preliminary data from this small cohort suggest that the presence of ascites and transgastric access may increase the risk of adverse events. These findings support the role of EUS-GBD not only as a safe alternative in high-risk patients but also as a potentially definitive solution in anatomically and clinically challenging cases.

Further prospective studies with more significant patient populations are warranted to validate these early observations and optimize outcomes in this vulnerable population.

Disclosure: Nothing to disclose.

PP0938

TRANSPAPILLARY GALBLADDER DRAINAGE IN THE CLINICAL PRACTICE – PILOT STUDY

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Introduction: The Tokyo Guideline 2018 recommends laparoscopic cholecystectomy as the gold standard therapy for acute cholecystitis regardless of the severity and the onset of symptoms, as long as the patient is fit for surgery. Approximately 20% of the patients undergo acute cholecystectomy in our region, and in the remainder of the patients, we have to take into consideration the likelihood of recurrent inflammation after the regression with conservative therapy.

According to the TG 2018, the drainage of the gallbladder is recommended in grade II and grade III cholecystitis. This can be carried out by interventional radiologist as percutaneous transhepatic gallbladder drainage (PTGBD), by endoscopist as transpapillary gallbladder drainage (TPGBD) or endosonography-guided gallbladder drainage (EUS-GBD).

Aims & Methods: The primary endpoint of our prospective, non-randomized, observational study was to determine the technical (successful stent implantation into the gallbladder) and clinical success (repeated need for drainage and absence of recurrent cholecystitis) of TPGBD in those patients that underwent it from 01.01.2023 at the University of Szeged. Our secondary endpoint was the comparison of the time of hospitalization between the patients who underwent clinically successful TPGBD and those who underwent PTGBD for grade II/III cholecystitis.

Results: Altogether 19 patients had undergone TPGBD, 74% of them were female, the average age was 64,53. Previously 21% had cholecystitis, 47% had cholecystolithiasis, while 16% had choledocholithiasis and 11% of the patients gone under PTGBD. TPGBD was carried out because of grade II cholecystitis in 89% of the patients, and grade III in the rest of them. 63% was considered at least a 3 in the ASA physical status classification. Altogether the technical success rate was 58%, a cholangioscop was used in 37% of the cases. The successful canulation rate of the bile duct was 89%, while the canulation rate of the cystic duct was 68%. There were two cases, where after the canulation of the cystic duct the implantation of the plastic stent was not possible. 3 patients were referred for PTGBD after an unsuccessful TPGBD, the remainder patients with an unsuccessful TPGBD received conservative therapy.

After a technically successful TPGBD, only one patient went under PTGBD due to his severe comorbidities, rather than a repeated TPGBD. 6 patients underwent cholecystectomy averagely in 8,5 weeks after the TPGBD, the remainder of patients did not have neither the need for repeated drainage nor any symptoms requiring urgent care due to cholelithiasis.

Comparing this group with 11, randomly chosen patients who underwent PTGBD for at least grade II cholecystitis during the period of the study, we detected that the time of hospitalization after the drainage was less (8,2 days vs 13,3 days), and less patients required repeated drainage or had recurrent cholecystitis (1 vs 4).

Conclusion: Although TPGBD imposed a great challenge before the use of cholangioscop in everyday practice, nowadays it is becoming an accepted part of the therapeutic algorithm in the treatment of cholecystitis, where an acute cholecystectomy is not performed.

Our results suggest that TPGBD can lead to shortened hospitalization and have less complications than TPGBD. Our findings will serve as a basis for a future extension of the study and for the design of a randomised prospective study with a control group.

Disclosure: Nothing to disclose.

PP0939

SAFETY AND EFFICACY OF MAGNETIC COMPRESSION ANASTOMOSIS FOR THE TREATMENT OF COMPLETE BILIARY ANASTOMOSIS OBSTRUCTIONS AFTER LIVER TRANSPLANTATION: A SYSTEMATIC REVIEW

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Introduction: In patients with complete biliary obstruction, conventional treatments such as metallic stents or balloon dilation are often not feasible. Therefore, the endoscopic technique of magnetic compression becomes an alternative for these patients. However, there are few reports in the literature regarding the efficacy and safety of this less invasive approach for the recanalization of biliary structures, particularly in the context of liver transplantation.

Aims & Methods: We aimed to conduct a systematic review with meta-analysis on the safety and efficacy data of magnetic compression for the treatment of complete biliary anastomosis obstructions following liver transplantation. We searched PubMed, Embase, and Cochrane Library for studies evaluating magnetic compression anastomosis for the treatment of complete biliary anastomosis obstructions after liver transplantation. The odds ratio (OR) was used for binary outcomes and the mean difference (MD) for continuous outcomes, with their respective 95% confidence interval (CI).

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Results: This single arm meta-analysis included three studies and 27 patients. Technical success was achieved in 23/27 patients (85.19%; 95%CI 66,54-94.33; I2=0%). Magnet aproximation failed in two patients; in the remaining, recanalization was achieved in 23/25 (94,25%; 95%CI 48.74-99.65; I2=0%).

Conclusion: This systematic review and meta-analysis demonstrated that for the recanalization of completely obstructed biliary ducts after liver transplantation, the magnetic compression technique achieved satisfactory technical success and a high degree of therapeutic success. This indicates that this less invasive approach is a viable alternative for treating patients in this condition.

Disclosure: Nothing to disclose.

PP0940

SHORTCUTO TO SUCCESS? EDEE: THE NEW KID ON THE BLOCK - REVOLUTION OR PASSING TREND IN BILIARY INTERVENTIONS? A MEXICAN PERSPECTIVE

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Introduction: ERCP is the cornerstone for managing biliary and pancreatic duct diseases. However, performing in patients with surgically altered anatomy poses significant challenges. Traditional ERCP techniques often fail in these cases due to difficulties in accessing the papilla. This has led to the development of alternative techniques like enteroscopy-assisted ERCP (e-ERCP) and endoscopic ultrasound-directed transenteric ERCP (EDEE).

e-ERCP while it can be successful, is technically demanding, time-consuming, and associated with a higher risk of complications while EDEE involves using EUS guidance to create a temporary anastomosis between the stomach and the intestinal segment containing the papilla. EDEE offers a more direct and stable route to the biliary tree, However, it requires specialized equipment (lumen-apposing stents) and expertise.

Aims & Methods: Retrospective observational, patients undergoing EDEE or e-ERCP at the CMN 20 de Noviembre between 2018-2024

Results: This study compared two endoscopic procedures, e-ERCP and EDEE, for accessing the biliary tract in patients with altered anatomy. The average age of the patients was 48 years, with a standard deviation of 14 years, and 61% were women. Most patients (81%) underwent e-ERCP, while only 9% underwent EDEE. Regarding the success of the procedure, e-ERCP had a technical success rate of 87%, endoscopic success rate of 93%, and clinical success rate of 75%. On the other hand, EDEE, although performed in a small number of patients, showed a 100% success rate in all three categories.

Adverse events occurred in 5% of patients. 72% of patients had altered anatomy due to a pancreaticoduodenectomy, and 90% had a Roux-en-Y hepaticojejunostomy. In addition, dilations were performed in 72% of patients, stents were placed in 17%, and sphincterotomy or precut was performed in 17%.

Conclusion: The average cost of EDEE was 120,000 pesos (6,000 USD), while that of e-ERCP was 40,000 pesos (2,000 USD). Despite the difference in cost, EDEE showed a 100% success rate in this study, suggesting that it could be a cost-effective option for patients with altered anatomy in whom e-ERCP is not feasible or has failed. However, a cost-effectiveness analysis, considering the cost of the procedure divided by the proportion of clinical success, showed that e-ERCP was more cost-effective than EDEE.

Studies with a larger number of patients are needed to confirm these findings and to evaluate the safety and cost-effectiveness of EDEE compared to e-ERCP.

This study is limited by the small number of patients with EDEE and the lack of information on the type of complications, the duration of the procedures, and long-term survival. It is recommended to expand the research with more patients and to perform a statistical analysis to formally compare the two procedures.

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Disclosure: Nothing to disclose.

PP0941

PREDICTORS OF SPONTANEOUS BILIARY STONE MIGRATION: THE ACTIVE ROLE OF THE GALLBLADDER

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Introduction: Uncomplicated choledocholithiasis may progress to obstructive jaundice, cholangitis, or acute pancreatitis[1]. The current therapeutic standard for choledocholithiasis is Endoscopic Retrograde Cholangiopancreatography (ERCP), however, this procedure is associated with a complication rate of approximately 10%. Identifying factors that contribute to spontaneous resolution of uncomplicated choledocholithiasis may allow clinicians to avoid unnecessary ERCP, thereby minimizing procedural risks and costs.

Aims & Methods: This retrospective cohort study was conducted to identify predictive factors for spontaneous passage of biliary stones in patients diagnosed with uncomplicated choledocholithiasis. We included patients who underwent ERCP for confirmed choledocholithiasis, excluding those with a history of prior ERCP or complications related to untreated biliary calculi, such as obstructive jaundice, acute cholangitis, or acute pancreatitis. The primary endpoint was the rate of spontaneous resolution of choledocholithiasis prior to ERCP.

Results: A total of 104 patients were included, 61.5% female with a mean age of 68 ± 14 years. Most patients (62.5%) presented with abdominal pain and/or abnormal liver enzymes, while the remaining 37.5% were diagnosed incidentally.

Among the 104 patients, 21.6% showed no evidence of lithiasis on ERCP, despite the identification of stones >3 mm on imaging studies. The median time to the endoscopic procedure was similar between patients with and without stones on ERCP (12 IQR 25 vs. 15 IQR 24 weeks; p=0.821). Imaging findings, such as common bile duct ectasia (p=0.517), intrahepatic

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bile duct dilation (p=0.528), and the number of calculi (p=0.236), did not associate with spontaneous resolution. The average size of the largest stone was also comparable between patients with and without choledocholithiasis identified on ERCP (9 ± 4 mm vs. 8 ± 4 mm; p=0.452). The presence of peri-diverticular papillae did not impact the likelihood of spontaneous resolution (p=0.777).

Remarkably, patients with an intact gallbladder exhibited a significantly higher rate of spontaneous resolution (31.5%) compared to those with prior cholecystectomy (6.1%, p=0.005). This association was confirmed as an independent predictor of spontaneous resolution (OR 7.1, 95% CI: 1.5–33.3, p=0.013).

Conclusion: Spontaneous resolution of uncomplicated choledocholithiasis is common, especially in patients with an intact gallbladder. Our data support a conservative approach in these patients, such as performing endoscopic ultrasonography immediately prior to ERCP to identify those who may not require intervention.

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Disclosure: Nothing to disclose.

PP0942

RECURRENT CBD STONES IN CHOLECYSTECTOMIZED PATIENTS: A MISSED RISK IN POST-ERCP MANAGEMENT

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Introduction: Recurrent choledocholithiasis is an often-overlooked complication in cholecystectomized patients, even after successful endoscopic clearance. Despite the absence of the gallbladder, bile duct stones may recur, exposing patients to repeated ERCPs, complications, and rising healthcare costs. To our knowledge, few studies have focused exclusively on stone recurrence in cholecystectomized patients following ductal clearance.

Aims & Methods: This study aimed to characterize the clinical and endoscopic profile of recurrence in this population and to explore potential risk factors and preventive strategies to optimize long-term outcomes.

We performed a retrospective single-center study including all cholecystectomized patients who underwent ERCP for CBD stones between May 2016 and October 2024. Patients with confirmed ductal clearance and no stent placement at index ERCP were screened.

Recurrence was defined as symptomatic choledocholithiasis confirmed by cross-sectional imaging and treated with repeat ERCP. Demographic, clinical, biochemical, and endoscopic parameters were collected.

We specifically analyzed stone burden, CBD diameter, type of sphincterotomy, and follow-up strategy to identify patterns of recurrence and potential targets for prevention.

Results: Among 287 cholecystectomized patients, 16 (5.6%) experienced stone recurrence after initial clearance. Most were women (87.5%), with a mean age of 66.6 ± 13.3 years. Cardiometabolic comorbidities were frequent: hypertension (25%), diabetes (18.8%), and obesity (12.5%), with a mean BMI of 27.6 kg/m^2. At index ERCP, all had multiple stones (mean: 2.8) and a dilated CBD (mean diameter: 14.5 mm). Only 25% underwent wide sphincterotomy, while the remaining had incomplete or limited sphincterotomy. The median time to recurrence was 12 months (range: 3 months–5 years). None had structured follow-up or preventive therapy post-clearance.

Conclusion: This real-life series suggests a distinct recurrence profile in cholecystectomized patients: advanced age, female sex, metabolic comorbidities, multiple CBD stones, persistent duct dilation, and absence of wide sphincterotomy. Although limited by sample size, these observations highlight the need for individualized post-ERCP strategies in at-risk patients.

Notably, this recurrence rate may be underestimated due to possible missed cases in patients not readmitted. Future prospective studies should assess whether early wide sphincterotomy, structured long-term follow-up, and adjuvant therapies such as ursodeoxycholic acid can reduce recurrence and improve long-term outcomes.

Disclosure: Nothing to disclose.

PP0943

ROLE OF ENDOSCOPIC BILIARY STENTING IN ACCESS TO CHEMOTHERAPY IN PATIENTS WITH UNRESECTABLE MALIGNANT OBSTRUCTION

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Introduction: Endoscopic drainage is standard of care in malignant biliary obstruction (MBO).

Aims & Methods: The aim of this study was to investigate efficacy of biliary stenting in providing access to systemic chemotherapy. Patients with biliary obstruction (bilirubin >2.5 mg/dl) caused by unresectable biliary or pancreatic cancer were retrospectively analyzed with respect to use of chemotherapy and survival. Laboratory data were analyzed at baseline, 7-10 and 28-35 days following biliary stenting.

Results: Patients with MBO (n=78) were allocated into one of three groups according to bilirubin level [mg/dl] 2.5 – 5.99 (group 1), 6.0 - 15.0 (group 2) and >15 (group 3). On first blood testing mean bilirubin declines from baseline in respective groups were: 1.19 (29.5%), 3.77 (38.5%) and 7.1 (27.3%), and on second measurement 1.61 (40%), 5.69 (58.2%) and 13.0 (50%). Chemotherapy was applied in 28 (35.8%) patients, less frequently in group 3 than in group 1 (31.2% vs 44.0%; p =0.07) and group 2 (vs 38.1%; p =0.06). Mean survival time was 4.8 ±2.8 months and was significantly shorter in group 3 as compared to group 2 (3.8 ±2.0 vs 6.2 ±3.9 months; p =0.02) and to group 1 (vs 5.4 ±2.7 months; p =0.03).

Conclusion: In two-thirds of patients with MBO endoscopic biliary drainage does not grant access to chemotherapy. Use of chemotherapy and survival time are significantly higher in patients with bilirubin level below 15 mg/dl as compared to those with greater bilirubinemia, but in patients with bilirubin level below 6 mg/dl the biliary drainage is minimally effective.

Disclosure: None.

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PP0944

INDETERMINATE BILIARY STRICTURES: A RETROSPECTIVE STUDY

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Introduction: Diagnosis of biliary obstructions is a challenge especially when histopathology is inconclusive. Indeterminate biliary strictures often require additional verification and a personalized approach.

Aims & Methods: This study investigates the prevalence, characteristics, and natural history of indeterminate biliary strictures. This retrospective study covers patients treated with ERCP because of jaundice due to biliary stricture(s) of unknown etiology.

Exclusion criteria were prior liver transplantation, visible tumor, lithiasis, acute pancreatitis, PSC, or any known cause of dilated biliary ducts with subsequent biliary stricture diagnosed with imaging studies (US, CT, MRI, or EUS) and serology and clinical data.

A diagnosis of potentially benign biliary stricture was based on cytology/histopathology evaluation of samples taken on primary ERCP. The study participants were monitored every 2–3 months with further endoscopic, imaging, and serological evaluations as needed.

Results: A total of 510 patients were treated with primary ERCP for biliary obstruction of indeterminate etiology. The 186 (36.5%) had biliary strictures with negative cytology/histopathology (category II per Papanicolaou guidelines), of these 55 (29.6%) had strictures in the liver hilum, 22 (11.8%) in the common bile duct (CBD), and 108 (58.1%) in the intrapancreatic region. Malignancy was confirmed in subsequent tests in 40 patients (21.5%). A benign cause was diagnosed in 77 cases (41.4%), including pancreatitis, PSC, ischemia, IgG4-SC, and Mirizzi syndrome, while 69 strictures (37.1%) remained indeterminate. Strictures in the liver hilum had a higher rate of eventual malignancy (30%) compared to CBD (23%) and intrapancreatic (16%) localization.

Lithiasis was more common in intrapancreatic strictures (12%), while PSC was more frequent in hilar strictures (14%). During 6-month follow-up of the 69 indeterminate cases: 17 patients received a diagnosis (8 lithiasis, 5 malignancies, 2 chronic pancreatitis, 1 benign neoplasm, 1 IgG4-SC), 4 patients died from unrelated causes, 48 cases (25.8% of the potentially benign stricture group) remained indeterminate.

Conclusion: A quarter of benign biliary strictures remain indeterminate in 6 months follow-up, while 20% of patients with indeterminate receive diagnosis of malignancy.

Disclosure: Nothing to disclose.

PP0945

PRECANCEROUS GALLBLADDER LESIONS IDENTIFIED ON PRIOR IMAGING: A RETROSPECTIVE ANALYSIS AND CLINICAL IMPLICATIONS

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Introduction: Gallbladder cancer (GBC) has a poor prognosis due to its aggressive nature and limited treatment options. Although early diagnosis through imaging is critical, it remains difficult—particularly in cases of focal gallbladder wall thickening (FGWT), which can mimic benign conditions such as adenomyomatosis or chronic cholecystitis.

Aims & Methods: This study aimed to identify lesions that progressed to GBC (precancerous lesions of GBC; PLGBC) by evaluating prior imaging and to describe their clinical characteristics and outcomes. Patients diagnosed with GBC at Iwata City Hospital between 2014 and 2024 were retrospectively reviewed. Those who had prior imaging—defined as abdominal ultrasound (US), computed tomography (CT), or magnetic resonance imaging (MRI) performed more than three months before GBC diagnosis—were included for analysis. PLGBC was defined as a lesion identifiable on prior imaging but not diagnosed as GBC, and corresponding to the site of subsequently diagnosed GBC. Imaging features and clinical outcomes were compared between patients with and without PLGBCs.

Results: Among 140 patients diagnosed with GBC, 66 (47%) had prior imaging available for review and were included in the analysis. The median age at GBC diagnosis was 80 years (interquartile range [IQR]: 71–86), with 33 male patients (50%). The median interval between prior imaging and GBC diagnosis was 753 days (IQR: 404–1796). The modalities of prior imaging included CT only in 42 patients, US only in 7, MRI only in 1, US and CT in 6, CT and MRI in 6, US and MRI in 3, and all three modalities in 1 patient. PLGBCs were identified in 32 patients (48%). Regarding lesion morphology, 28 patients presented with FGWT, 2 patients with diffuse gallbladder wall thickening (DGWT), and 2 patients with gallbladder polyps. Lesion locations were the neck in 5 patients, the body/fundus in 25 patients, and the entire gallbladder in 2 patients. Features such as intramural stones or Rokitansky–Aschoff sinuses (RAS) were observed in 15 patients, and gallstones in 7 patients.

No significant differences were found between patients with and without PLGBCs regarding resectability (36% vs. 35%, P = 1.00), radical resection rates (13% vs. 13%, P = 1.00), or survival from diagnosis (median: 640 vs. 791 days, P = 0.96). Six patients with PLGBC were diagnosed with GBC during observation of gallbladder lesions (median follow-up: 406 days, range: 350–552 days), and all underwent successful radical resection of GBC.

Conclusion: PLGBCs were identified in approximately half of the patients who had undergone imaging before their GBC diagnosis, with FGWT being the predominant imaging feature. Recognizing PLGBCs and ensuring appropriate follow-up may enable timely surgical intervention, offering a potential opportunity to improve outcomes in patients with GBC.

Disclosure: Nothing to disclose.

PP0946

FACTORS ASSOCIATED WITH POST-ENDOSCOPIC PAPILLECTOMY BLEEDING

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Introduction: Endoscopic papillectomy (EP) for duodenal ampullary tumors is primarily performed for lesions confined to the papilla, particularly adenomatous changes. However, early complications such as bleeding (0-21.6%), pancreatitis (0-23.1%), and perforation (0-8.3%) occur at relatively high rates. Post-EP bleeding, due to anatomical characteristics of the ampulla of Vater, can be challenging to manage and may worsen, making it critical to identify factors associated with this complication to reduce its frequency.

Aims & Methods: The study aimed to investigate the factors associated with post-endoscopic papillectomy bleeding in patients with ampullary tumors. This study included 39 patients (39 lesions), excluding residual

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and accessory ampullary lesions, who underwent EP at our institution between March 2011 and October 2024. We evaluated univariate analysis and SHAP values based on tree-based models for several factors related to post-EP bleeding. Hyperparameters of each model were optimized by optuna, and those models were trained with the Python package scikit-learn (version 1.3.0).

Results: The cohort included 1 adenocarcinoma, 32 adenomas, 1 neuroendocrine neoplasm, and 5 cases of ampullary inflammation, with a mean age of 66.7 years (23 males, 16 females).

Patient factors included: The American Society of Anesthesiologists (ASA) Physical Status Classification I/II/III in 5/21/13 participants, median Charlson Comorbidity Index (CCI) of 1 (range 0-7), 15 cases with a history of alcohol consumption, median Brinkman Index of 100 (range 0-1680), median total bilirubin of 0.7 mg/dL (range 0.2-2.1), median serum amylase of 80 IU/L (range 36-296), median platelet count of 198,000/µL (range 113,000-494,000), median PT-INR of 0.98 (range 0.86-1.14), median APTT of 28.7 seconds (range 23.0-39.4), median CEA of 1.9 ng/mL (range 0.3-12.8), and median CA19-9 of 7.7 U/mL (range 0.8-76.4).

Antithrombotic agents were used in 4 participants (1 on low-dose aspirin, 1 on clopidogrel, and 2 on direct oral anticoagulants), and periampullary diverticula were observed in 3 cases. Median tumor size was 10.3 mm (range 2.0-45.0). Procedural factors included a median total procedure time of 56 minutes (range 23-230), with 36 cases receiving endoscopic biliary stenting and 38 cases receiving endoscopic pancreatic duct stenting. The electrosurgical units ICC200/VIO300D/VIO3 were used in 9/19/11 cases, respectively. Protease inhibitors were used in 9 participants, NSAID suppositories (25mg/50mg) in 5/24 cases, and local hemostatic measures included epinephrine spray in 8 cases, thrombin spray in 8 cases, and hemostatic peptide solutions in 5 cases.

Clipping for ulcer after resection was performed in 33 cases. Post-EP bleeding within 30 days occurred in 12 cases (30.8%), including 9 moderate and 3 severe cases. Univariate analysis revealed that a history of alcohol consumption was significantly associated with post-EP bleeding (Odds ratio [OR] 2.33, 1.14-4.79, P = 0.02).

Feature importance was evaluated using the mean absolute SHAP values across all participants. In the Random Forest model, the top contributing features were CA19-9 (mean |SHAP| = 0.026), CCI (mean |SHAP| = 0.024), and a history of alcohol consumption (mean |SHAP| = 0.023). In contrast, the XGBoost model identified tumor size (mean |SHAP| = 0.315), a history of alcohol consumption (mean |SHAP| = 0.244), and platelet count (mean |SHAP| = 0.198) as the most important features.

Conclusion: In our cohort, a history of alcohol consumption is a robust factor associated with post-endoscopic papillectomy bleeding.

Disclosure: Nothing to disclose.

PP0947

MALIGNANT BILIARY STRICTURES: WHAT IS THE NUTRITIONAL STATUS?

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Introduction: Malignant biliary strictures are mainly secondary to biliopancreatic cancer, and inevitably affect the nutritional status of these icteric patients. Deterioration of this nutritional status conditions therapeutic management, particularly surgery.

The aim of this study is to assess the nutritional status of patients with biliary stenosis.

Aims & Methods: All patients with non-operable malignant biliary stenosis who underwent endoscopic biliary drainage were included from 2015 to 2022 in a single-center descriptive and analytical study. Patients with advanced WHO 3/4 impairment of general condition were excluded.

Nutritional status was assessed before and after the procedure (3-6 months) on the basis of weight, BMI, weight change, nutritional risk index (NRI), albumin and CRP levels. The various data collected were analyzed using SPSS software.

Results: 200 patients were included: mean age 59 years (18-96 years) with no clear gender predominance (sex ratio=1.04). The etiology of malignant biliary stenosis was dominated by cholangiocarcinoma in 112 cases (56%), followed by pancreatic cancer in 64 cases (34%), extrinsic compression in 16 cases (8%) and ampulloma in 8 cases (4%).

Endoscopic biliary drainage was performed in 192 cases (96%), bilio-digestive anastomosis in 6 cases (3%) and trans-hepatic drainage in 2 cases (1%).

Pre-therapeutic nutritional assessment showed that: 100% of patients had significant weight loss with a mean of 21.91%, 71.5% had anorexia and 47% were underweight (BMI < 18.5 kg/m2), 83.5% had hypoalbuminemia and 84% had elevated CRP.

An improvement in nutritional status was noted 3 to 6 months after biliary drainage, with an improvement in: underweight from 47% to 17%, severe undernutrition from 61.5% to 22.06%, anorexia from 71.5% to 18.61% and an increase in mean albumin level from 29g/l to 34.52g/l and a decrease in mean CRP level from 61.73 mg/l to 26 mg/l. The 6-month survival rate was 72.5% (n=145).

Conclusion: Malignant biliary strictures are essentially linked to cholangiocarcinomas, with a clear impact on nutritional status, represented by weight loss in all cases, anorexia in around ¾ of cases and underweight in around 50%. The impact of biliary drainage on nutritional status is clear, since there is an improvement in the various clinical and biological factors assessing the latter.

Disclosure: Nothing to disclose.

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PP0948

THE IMPACT OF BILIARY DRAINAGE ON CHEMOTHERAPY INITIATION AND SURVIVAL IN PATIENTS WITH NON-RESECTED PERIHILAR CHOLANGIOCARCINOMA: A POPULATION-BASED STUDY IN THE NETHERLANDS

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Introduction: Perihilar cholangiocarcinoma (pCCA) is a rare malignancy originating from the bile duct bifurcation which is often diagnosed in an advanced stage. Most patients require endoscopic or percutaneous biliary drainage, as they present themselves with jaundice and/or cholangitis. In most Western countries, only palliative chemotherapy (pCTx) is offered to these patients, positively impacting overall survival and quality of life. Unfortunately, median survival is still poor, with only 10.6 months in patients receiving pCTx, compared to a dismal 4.0 months in patients receiving best supportive care (1).

For non-resected pCCA, there is much debate on the optimal technique of biliary drainage, which is often deemed necessary to initiate pCTx. This study aimed to investigate the impact of biliary drainage on pCTx initiation and survival in patients with non-resected pCCA.

Aims & Methods: Individual patient data from all Dutch patients diagnosed with non-resected pCCA between 2015 and 2020 were retrieved from the Netherlands Cancer Registry. Survival data was last updated on 1st of February 2024. Data was retrieved and de-identified by trained data clerks. The primary outcome was pCTx initiation. Secondary outcome was overall survival. For both the primary and secondary outcome, associations were investigated for several (biliary drainage) factors by multivariable competing risk regression analysis. Initial drainage procedure and initiation of pCTx were treated as time-varying covariates. Sensitivity analysis for pathologically proven pCCA was performed.

Results: A total of 1265 patients (median age of 74 years [IQR: 66-82 years]) were included, of whom 242 patients (19.1%) received pCTx after a median interval of 72 days [IQR 43-110 days] after initial presentation. Patients who underwent biliary drainage did not receive pCTx more often. If performed, drainage at a referral hospital (HR:0.65, 95%CI: 0.42-0.99), ≥3 drainage procedures performed (HR:0.53, 95%CI: 0.32-0.88), and drainage performed ≥14 days after presentation (HR:0.70, 95%CI: 0.49-0.99) were associated with no pCTx. Median overall survival for those who received pCTx and those who did not was 12.8 months [95%CI: 11.7-14.2] and 2.7 months [95%CI: 2.4-3.1].

Receiving of pCTx was associated with improved overall survival (HR:0.61, 95%CI: 0.51-0.73), but biliary drainage was not. If performed, biliary drainage by percutaneous trans-hepatic catheter placement compared to endoscopic plastic stenting (HR:1.27, 95%CI: 1.02-1.57) was associated with diminished survival, and drainage performed ≥14 days after presentation (HR: 0.76, 95% CI: 0.65-0.88) was associated with improved overall survival. Sensitivity analysis for pathologically confirmed pCCA showed similar results.

Conclusion: The need for biliary drainage did not affect the initiation of pCTx. When indicated, biliary drainage should be performed as quick as possible in an academic center to improve the rate of patients receiving pCTx.

Reference:

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PP0949

PRE-PAPILLECTOMY EUS AND ERCP ACCORDANCE RATE OF AMPULLARY ADENOMAS INTRADUCTAL GROWTH: A BLIND PROSPECTIVE PILOT STUDY

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Introduction: Endoscopic papillectomy (EP) is the preferable therapeutic strategy for non-invasive ampullary adenomas (AA), meaning limited to ampulla of Vater or sphincter of Oddi and with no invasion of the submucosa or further duodenal layers and no regional lymph node involvement, and with limited intraductal growth. To date, international guidelines recommend staging any AA with both MRI and EUS prior to EP, regardless of the size, the endoscopic appearance, and the grade of dysplasia. This implies a risk of superfluous examinations, especially for small non-deconstructed lesions.

Aims & Methods: The primary aim of this study is to evaluate the accordance rate between ERCP and EUS images in terms of AA intraductal growth definition. A monocentric, prospective, interventional study has been conducted including all consecutive patients referred to the Digestive and Interventional Endoscopy Unit of ASST Grande Ospedale Metropolitano Niguarda for endoscopic papillectomy from January 2024. Inclusion criteria was the availability of pre-EP staging MRI with no evidence of lymph node and metastatic involvement. All consecutive patients, prior to voluntary informed consent, underwent CE-EUS, which results have been blinded to ERCP endoscopist. Demographic, clinical, endoscopic and pathological data were collected.

Results: To date, a total of 20 patients have been enrolled (M 49%, median age 58 years [38-71]); they were referred to EP mainly after an incidental diagnosis (50% of the cases) during EGDS (45% of the cases) or EUS (25% of the cases). No cases of FAP among them was included. Median AA diameter was 20 mm [15-28.8], mostly Paris 0-Is (50%) and 0-IIa (45%) lesions, mainly with a non-LST behaviour (60%) and a non-deconstructed mucosal pit pattern (Kudo IIIL, 62.5%). At EUS, 5 (25%) presented a >5mm main pancreatic duct (MPD) dilation, and 10 (50%) a common bile duct (CBD) dilation, with a median CBD diameter of 12 mm [8-15]; CBD intraductal involvement was reported in 5 cases (for a median extension of 3 mm [2.5-9.3]), while DPP in 1 (with a 3 mm extension). Previous cholecystectomy was performed in 6 (30%) cases, and previous sphincterotomy in 3 (15%). Concordance with cholangiografic ERCP images was of 19/20 (95%) cases: in 1 case EUS revealed a small CBD involvement, not confirmed at ERCP and at subsequent histology on the R0 specimen. All AA were resected thorough endocut hot snare polypectomy, mainly (60%) en-bloc. After EP, 18 (90%) cases underwent concomitant biliary and pancreatic plastic stenting, and 13 (65%) mechanical and injectional hemostasis. Reported AEs were 2 cases of bleeding, 2 pancreatitis and 1 cholangitis, all managed endoscopically. At specimen histology, 14 (70%) cases were low-grade dysplasia adenomas, 3 (15%) high-grade, and 1 (5%) adenocarcinoma (which underwent surgery), with 5/19 (26.3%) cases of R1/R2. 6/19 (31.6%) cases (all the R1/R2 and 1 R0) presented a residual adenoma at 2-months-follow-up: all of them were treated with Habib ablation with negative histological controls after a median of 2 procedures. After a 5.4 months median follow-up time, only 1 recurrence was reported, successfully treated endoscopically.

Conclusion: According to our limited experience, pre-EP EUS did not provide any additional nor incisive information for the therapeutic management of any-size AA without lymph node and metastatic involvement. Further data from this ongoing prospective study and hopefully from RCT will help identify a specific subgroup of AA which would effectively benefit from CE-EUS/MRI staging prior to EP, thus sparing unnecessary examinations.

Disclosure: Nothing to disclose.

PP0950

INCIDENCE AND PREDICTORS OF PANCREATIC EXOCRINE INSUFFICIENCY AT DIAGNOSIS OF BILIOPANCREATIC AND AMPULLARY MALIGNANCIES

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Introduction: Pancreatic exocrine insufficiency (PEI) is a condition characterized by inadequate secretion or activity of pancreatic enzymes, resulting in the maldigestion and malabsorption of nutrients. While PEI is most commonly associated with chronic pancreatitis and cystic fibrosis, it also occurs frequently in the context of pancreatic and peri-pancreatic malignancies. In patients with pancreatic cancer PEI arises due to direct tumor infiltration, ductal obstruction, parenchymal destruction, or as a consequence of surgical resection.

Other gastrointestinal malignancies, such as ampullary or biliary tree cancers, may also contribute to PEI through similar mechanisms. Importantly, PEI in cancer patients is often under-recognized, despite its significant impact on nutritional status, quality of life, and overall prognosis. Timely identification and management of PEI in malignancy-associated cases is crucial to optimize clinical outcomes and support ongoing cancer therapies.

Aims & Methods: The objective of this study is to assess the incidence of PEI in patients with biliopancreatic and ampullary tumors, to explore potential differences in PEI occurrence based on tumor location and to investigate possible associations between PEI and relevant biomarkers.

We conducted a retrospective observational analysis of a consecutively recruited patient cohort including patients diagnosed with biliopancreatic and ampullary tumors at a tertiary care hospital between 2016 and 2024.

PEI was defined as fecal elastase-1 (FE-1) levels below 100 µg/g.

At the time of diagnosis, we collected data on demographic characteristics, tumor-related variables, the incidence of PEI, and serum nutritional markers.

Results: We included 408 patients with malignancies potentially associated with PEI. Of these, 54.2% were male, with a mean age of 72 years (range: 38–101) and a mean body mass index (BMI) of 25.45 (SD ± 4.28).

The types of malignancies included were pancreatic ductal adenocarcinoma (PDAC) in 63.5% of patients, cholangiocarcinoma (CCA) in 23.8%, ampullary tumors in 9.8%, and gallbladder tumors in 2.9%. The overall

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incidence of PEI at the time of diagnosis was 38.2%. Among subgroups, the incidence was 51.4% in PDAC patients, 13.4% in those with CCA, and 25% in those with ampullary tumors. PEI incidence was significantly associated with elevated HbA1c (>5.8%, $p<0.01$) and Ca 19-9 levels (>37 U/mL, $p<0.01$).

In PDAC, tumors located in the pancreatic head were associated with a significantly higher incidence of PEI (58.8%) compared to tumors in the body (27.8%) and tail (21.7%) ($p<0.01$). In this subgroup, PEI was significantly associated with elevated HbA1c (>5.8%, $p=0.03$), elevated Ca 19-9 (>37 U/mL, $p=0.04$), and elevated bilirubin (>1.2 mg/dL, $p<0.01$).

In contrast, among CCA patients, no significant differences in PEI incidence were observed based on tumor location, and no significant associations were found between PEI and the clinical or laboratory variables analyzed .

Conclusion:

- PEI is observed not only in patients with PDAC but also in those with CCA and ampullary tumors at the time of diagnosis.
- Given the high incidence of PEI in tumors located in the pancreatic head, empirical pancreatic enzyme replacement therapy should be considered in this subgroup.
- In PDAC patients, the presence of PEI is significantly associated with endocrine dysfunction, hyperbilirubinemia secondary to biliary obstruction and elevated Ca 19-9 levels.

Disclosure: Nothing to disclose.

PP0951

ENDOSCOPIC PAPILLECTOMY FOR AMPULLARY LESIONS: POOLED ANALYSIS WITH META-REGRESSION ANALYSIS OF OUTCOMES

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Introduction: Endoscopic papillectomy(EP)is a viable treatment option for ampullary lesions(AL). While many studies have reported low morbidity and acceptable outcomes, early attempts to pool data from these initial experiences have produced conflicting conclusions regarding key technical aspects.

To address these uncertainties, we conducted a systematic review and pooled analysis to evaluate the safety and efficacy of EP for AL,identifying factors that may influence outcomes.

Aims & Methods: Electronic databases (Medline, Scopus and EMBASE) were searched up to September 2024. Studies that included patients with endoscopically resected AL were eligible. Efficacy and safety outcomes were pooled by means of a random-effects model to obtain a proportion with a 95% confidence interval(CI). Univariable meta-regression analyses were conducted to explore potential factors affecting outcomes.

Results: A total of 60 studies (4,859 lesions) published between 2002 and 2024 were analyzed. Complete resection was achieved in 86.2% of cases, though intraductal involvement limited success. The recurrence rate was 18.6%, however, the majority (89.3%) of patients were managed endoscopically without surgery.

The pooled adverse event (AE) rate was 30.0%, with bleeding (12.2%) and post-procedural pancreatitis (11.1%) being the most common complications. Prophylactic stenting reduced pancreatitis risk, while intraductal involvement increased perforation risk. Adjunctive treatments for intraductal involvement were associated with an increased risk of papillary stricture.

Conclusion: Endoscopic papillectomy is a safe and effective treatment for ampullary lesions. However, lesions with intraductal extension pose a higher risk of incomplete resection and perforation, warranting careful evaluation of the benefit-risk balance in these cases.

While prophylactic pancreatic stenting may reduce the incidence of post-procedural pancreatitis, optimizing strategies to minimize overall adverse events remains a key focus for future research.

Disclosure: Nothing to disclose.

PP0952

A REVIEW OF THE SPECTRUM OF REFERRAL PATTERNS IN PATIENTS DIAGNOSED WITH BILE ACID MALABSORPTION

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Introduction: Bile acid malabsorption (BAM) arises from dysregulation of enterohepatic bile acid recycling and production. Patients usually present with chronic diarrhoea, usually to gastroenterology but also to many other different clinical specialties. However, the index of suspicion and appropriate referral for a SeHCAT study for confirmation of the diagnosis is very variable. This obviously results in inconsistent success in the management of these patients.

Aims & Methods: This study aims to evaluate the spectrum of referral patterns in our patients sent for a SeHCAT test over a five-year period. We retrospectively reviewed 353 patients who underwent a SeHCAT study for suspected BAM from January 2019 for a five-year period. Clinical data, including referrer details, the probable cause of BAM, management and follow up was extracted from medical records and analysed.

Results: Of the 353 patients reviewed, 175 (49.57%) were diagnosed with BAM after a SeHCAT study. The main referrals for SeHCAT were from Gastroenterologists -272 patients (77.05%). The remaining 81 patients were largely referred by Colorectal surgeons with interest in lower GI physiology (41 patients); other referring specialties included other colorectal surgeons (7 patients) and specialties as varied as upper GI, general medicine, care of the elderly, respiratory medicine and even urology (33 patients). Of the 175 patients found to have BAM, 78.86% had been referred from gastroenterology, 11.43% from Colorectal surgeons with interest in lower GI physiology, 1.14% from other colorectal surgeons and 8.5% from the mix of other specialties.

Of the referred patients, gastroenterologists were likely to be accurate in their suspicion of BAM in 50.73% cases, Colorectal surgeons with an interest in lower GI physiology in 48.78% and other colorectal surgeons in 40% cases.

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Conclusion: Our findings highlight significant variations in the diagnosis and hence management of BAM. A high index of suspicion is essential in managing patients with chronic diarrhoea with risk factors for BAM as they can present with their symptoms to different specialties.

On the other hand, unnecessary SeHCAT testing can result from inappropriate referrals. When presenting to gastroenterologists, these patients are managed appropriately; however, they may present to the general surgeons after laparoscopic cholecystectomy, or the colorectal surgeons after ileal resection for Crohn's disease, right hemicolectomy for cancer etc and to physicians where BAM due to causes such as metformin must be considered.

They may present to any specialty such as Upper GI Surgery or Urology or Care of the elderly with symptoms other than diarrhoea and only be incidentally found on careful clinical assessment. These findings call for cross-specialty improved education regarding BAM diagnosis and management to improve outcomes in patients with BAM.

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Disclosure: None.

PP0953

COST-EFFECTIVENESS OF BILIARY DRAINAGE MODALITIES IN OPERABLE MALIGNANT HILAR BILIARY OBSTRUCTION

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Introduction: With the increase in the incidence of hilar cholangiocarcinoma, resection centres have seen an increase in the incidence of operable hilar malignancies. Biliary drainage of the future liver remnant (FLR) prior to resection has been demonstrated to improve surgical outcomes. Traditionally, preoperative biliary drainage has been achieved using either percutaneous (PTBD) or endoscopic (ERCP) techniques, with PTBD being favoured in many centres. Single operator cholangioscopy (SOC) via ERCP or PTBD, can enhance diagnostic yield and may optimise drainage. We sought to compare current drainage modalities incorporating SOC techniques in a cost effectiveness model.

Aims & Methods: A cost effectiveness model was developed using Markov healthcare states. PTBD cholangioscopy assisted drainage was compared to ERCP-cholangioscopy assisted drainage, 'standard' fluoroscopy assisted ERCP drainage with brush cytology and PTBD drainage with fluoroscopy guided endobiliary biopsy and drainage. Patients could cycle through each clinical drainage modality with crossover modalities in case of failure of a technique (e.g PTBD drainage for ERCP failure and PTBD external drainage for failure of PTBD cholangioscopy and internal-external drainage). Healthcare associated quality of life data were extracted from the literature to develop utilities used in the model. Procedure related complications were incorporated. Deterministic and Probabilistic sensitivity analysis (PSA) were performed over 1000 patient runs. Incremental cost effectiveness ratios (ICER) were calculated for a willingness to pay (WTP) threshold of £30,000 and net monetary benefits (NMB) for each strategy were derived.

Results: Base-case cost benefit analysis demonstrated that a strategy of ERCP-cholangioscopy assisted drainage (strategy 2, Table 1) was the most cost-effective strategy (Cost £1,895, effectiveness 0.51) and absolutely dominated the PTBD drainage strategies with an ICER of 8,630. The NMB for the ERCP and cholangioscopy strategy was the highest amongst the comparative strategies.

Strategy	Dominance	Cost (£)	Effectiveness (QALY)	ICER	NMB at £30K
1. ERCP, brushings, stent	Undominated	1,072	0.41	reference	11,303
2. ERCP, SOC and biopsies, stenting	Undominated	1,895	0.51	8,630	13,341
3. PTBD drainage + endobiliary biopsy	Absolutely dominated	2,887	0.27	-4180	5,231
4. PTBD drainage + percutaneous SOC	Absolutely dominated	3,616	0.25	-6678	3,889

Table 1: Base case analysis.

On PSA, ERCP, SOC and stenting was the most cost-effective intervention with a mean cost of £ 1,895 (97.5% CIs (1,894-1,896)) / 0.51 QALYs, in comparison to ERCP, brushings, stent (£ 1,072 (1,048-1,098) / 0.41, PTBD drainage + endobiliary biopsy (£ 2,887 (2,855-2,927) / 0.27 and PTBD drainage + percutaneous SOC £ 3,618 (3,604-3,625) / 0.25. The ERCP-SOC strategy also delivered the greatest NMB at £ 13,340 in comparison to the other strategies. 1-way sensitivity analysis and threshold analysis did not identify any variable that impacted on the overall model.

Conclusion: A strategy of ERCP and SOC assisted sampling and biliary drainage seems to be the most cost-effective pre-operative biliary drainage option for resectable hilar biliary obstruction.

Disclosure: The authors have no disclosures to make.

PP0954

EFFICACY AND OUTCOMES OF ENDOSCOPIC AMPULLECTOMY: A SINGLE CENTRE EXPERIENCE

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Introduction: Endoscopic ampullectomy is a minimally invasive technique for treating non-invasive ampullary lesions and in patients who are poor surgical candidates. Compared to surgical approaches, endoscopic ampullectomy have lower morbidity and mortality rates.

Aims & Methods: The aim of this study was to describe the safety and outcome of this procedure from a single tertiary center experience.A retrospective study conducted on patients with non-invasive ampullary lesions. Data were collected from our database and parameters related to ampullectomy outcome and complications were retrospectively analyzed. EUS evaluation of the stage and intraductal extension was performed in all patients. Snare sizes between 20 to 30 mm were used in these patients. Electrosurgical unit setting ENDOCUT Q effect 2 was used for resection. All patients were given indomethacin suppositories. Submucosal injection was performed only where the ampullary lesions extend to the duodenal mucosa.

Results: From 2015 to 2024, a total of 15 patients were included in the study. There was 9 males and 6 females. Their median age was 56 years (55-75). Lesions range from 20 to 50 mm in size, with average size of 25 mm. Prophylactic pancreatic duct stenting was performed in all cases of which none developed pancreatitis. Biliary stenting was performed in 3

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cases (20%) with cholangitis. Submucosal injection was performed in 2 lesions with extension to the duodenal mucosa. Post resection histology showed adenoma with low grade dysplasia in 9 patients (60%), adenoma with high grade dysplasia in 2 patients (13.3%), adenoma with focal adenocarcinoma in 2 patients (13.3%), hyperplastic lesion indefinite for dysplasia in 1 patient (6.7%) and hamartoma in 1 patient (6.7%).

Enbloc resection was achieved in 14 (93.3%), while piecemeal in 1 case (6.7%). Recurrence observed in 1 patient (6.7%) with focal adenocarcinoma. Biopsy from the recurrent growth revealed focal adenocarcinoma and patient underwent surgical resection.

No recurrence was observed in the patient with piecemeal resection for 24 months of follow up. Intra-procedural bleeding occurred in 2 patients (13.2%) and managed successfully with snare tip coagulation and hemoclips, delayed melena (after 24 hours) in 1 patient (6.7%) controlled with conservative management.

Only 1 of the 12 patients (8.3%) without biliary stents developed ampullary stenosis post-procedure and managed with plastic biliary stent. There was no report of perforation, need for emergency surgery or 30-days mortality in this series.

Conclusion: Endoscopic ampullectomy seems effective and safe treatment option in cases with non-invasive ampullary lesions. Careful patient selection & adherence to follow up is the best way to achieve best therapeutic outcome.

Disclosure: Nothing to disclose.

PP0955

INNOVATIVE ERCP TRAINING: BRIDGING EDUCATIONAL GAPS WITH NEXT-GENERATION 3D-PRINTED SIMULATORS

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is recognized as a foundational procedure within the realm of interventional gastroenterology. Nevertheless, a significant amount of hands-on experience is required to achieve proficiency in this procedure, particularly with regard to the performance of sophisticated therapeutic maneuvers. Access to authentic training cases remains limited, while the use of cadaveric or live animal models presents distinct ethical, logistical and financial challenges (1). The advent of 3D printing technology paved the way for the development of advanced medical simulators. This study utilized computer-aided design and fused deposition modeling techniques to engineer an ERCP simulator that is both anatomically accurate and enables the training of trainee physicians in the intricate procedures of their chosen discipline (2). The simulator's modular design allows it to replicate a spectrum of papillary conditions, enabling targeted training in anatomies that are resistant to cannulation, as well as high-risk procedures like precut sphincterotomy and fistulotomy.

Aims & Methods: The design process for the ERCP simulator employed accessible parametric modeling software. Fabrication of the primary structural components, including the pancreas, the duodenum, and the bile ducts, were undertaken using polylactic acid (PLA) filament, a material that is both widely accessible and economical, thereby ensuring adequate rigidity and durability. The interchangeable papillary components were produced using flexible thermoplastic polyurethane (TPU) 95A filament, selected for its elasticity and capacity to be readily incised during procedures such as endoscopic sphincterotomy (EST). This allowed for easy exchangeability and facilitated the acquisition of skills by novice endoscopists.

This process were selected on the basis of accessibility, user-friendly interface and cost-effectiveness, enabling replication in diverse settings. The biliary and pancreatic ductal lumens were meticulously calibrated to represent a range of procedural difficulty, with internal diameters that closely mirrored real anatomical variations.

Results: The simulator exhibited a high degree of anatomical and biomechanical fidelity, featuring a structurally stable duodenal framework and a flexible papillary interface. The simulator accurately reproduced tactile feedback experienced by gastroenterologists during ERCP, thereby validating its efficacy as a training tool.

Furthermore, the simulator was enabled meticulous rehearsal of EST through sphincterotomy incisions and facilitated fistulotomy simulation using an elastomeric papilla structure. The simulator's design is both cost-effective and easily replicable, making it a valuable asset for continuing education and training in this field (3,4).

Conclusion: The ERCP simulator, fabricated using 3D printing technology, can be regarded as both technologically advanced and economically efficient for a number of complex procedures. These encompass endoscopic sphincterotomy, precut techniques, fistulotomy and stent placement. The simulator's design, featuring a modular papillary system, was crafted to provide a structured and progressive learning environment for novice users as well as experienced endoscopists.

The simulator's cost-effectiveness, ease of scaling up and potential for wide dissemination signify a paradigm shift in ERCP education, facilitating proficiency acquisition in complex interventional endoscopy with greater efficiency and safety. Further investigations are needed for the simulator in ERCP training.

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Disclosure: Nothing to disclose.

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PP0956

EVALUATION OF THE EFFICACY OF ANTIBIOTIC PROPHYLAXIS IN PREVENTING POST-ERCP CHOLANGITIS: IDENTIFYING HIGH-RISK PATIENTS

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is still considered the endoscopic standard of treatment for biliary obstruction and lithiasis. Among the most common ERCP complications, the biliary tree bacterial contamination and cholangitis could be managed with pharmacological treatment or prevented with antibiotic prophylaxis. Although ESGE Guidelines recommend performing prophylaxis only in few cases with strong indications, such as post liver-transplantation patients, hilum strictures and primary sclerosing cholangitis, clear data about indicators predictive of post-ERCP cholangitis development are lacking.

Aims & Methods: Our aims were to investigate whether antibiotic prophylaxis could reduce post-ERCP cholangitis and to evaluate any potential risk factor that could aid in identify patients who most benefit from it. A monocentric, retrospective, case-control study was conducted enrolling all patients who underwent ERCP between January 2023 and May 2024 in our third- degree referral centre. Patients with stronger indication for antibiotic prophylaxis (such as liver-transplanted patients or primary sclerosing cholangitis) and patients who were already on ant ibiot ic s were exc luded.

We defined as ant ibiot ic prophylaxis a single dose of 1 g intravenous ceftriaxone 1 hour prior to the procedure. We defined cholangitis as the onset of temperature >38°C for more than 24 hours combined with cholestasis (according to Tokyo criteria).

Results: We enrolled 216 patients who underwent elective ERCP for biliary obstruction whose aetiology comprised of 120 choledocholithiasis, 72 malignancies, 24 others. Of those, 126 patients received antibiotic prophylaxis before the procedure and 90 did not. ERCP degree of difficulty performing Schutz score showed greater statistically significant difficulty in the first group (1.57 ± 0.69; 1.28 ± 0.54 p<0.001).

However, post-ERCP cholangitis occurred less frequently in this group (3/126) versus patients without antibiotic prophylaxis (9/90) with statistically significant difference [2.38%vs10.0%; OR=0.18, p=0.031].

Analysing the subgroup of patients with Charlson Comorbidity Index (CCI) scores ≥ 5 the incidence rate of cholangitis was significantly lower in antibiotic-treated patients (1/59) versus control patients (7/48; 1.69% vs 14.5%; p=0.021).

Moreover, the incidence rate of cholangitis in patients with pre-ERCP bilirubin levels more than 2.5 mg/dL was significantly lower in the antibiotic prophylaxis group (0/40) versus not antibiotic-treated patients (6/42; 0% vs 14.3%; p= 0.026).

Conclusion: Antibiotic prophylaxis prior to elective ERCP in patient with biliary obstruction reduce the incidence of post-procedure cholangitis especially in patients with CCI scores ≥ 5 or with pre-procedural bilirubin levels > 2.5 mg/dL. More studies are needed to recommend pre-ERCP routine or selected high-risk groups use of antibiotic prophylaxis.

Disclosure: Nothing to disclose.

PP0957

OPTIMISING ERCP SAFETY IN PREGNANCY: REAL-WORLD EVIDENCE OF MATERNO-FETAL PROTECTION AND RADIATION-FREE TECHNIQUES

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Introduction: Pancreatobiliary disease during pregnancy, though uncommon, poses significant risks. Gallstone-related disorders affect 3–12% of pregnancies and may progress to life-threatening complications such as cholangitis or biliary pancreatitis. Endoscopic retrograde cholangiopancreatography (ERCP) is a highly effective therapeutic approach; however, concerns about foetal radiation exposure and procedure-related maternal complications limit its broader acceptance. There remains a need for rigorous real-world data on the safety and neonatal outcomes of ERCP during pregnancy.

Aims & Methods: This retrospective observational study aimed to evaluate the safety of ERCP in pregnant patients, focusing on two key outcomes:
1. Maternal and foetal safety during and after the procedure, and
2. Effectiveness of radiation-minimisation strategies, including fluoroscopy reduction and abdominal shielding.

We reviewed 25 pregnant patients undergoing ERCP at a tertiary referral centre. Indications included choledocholithiasis (84%), cholangitis (4%), biliary pancreatitis (4%), and stent management (8%). Mean maternal age was 19.2 years; mean gestational age at ERCP was 28.3 weeks. Fluoroscopy usage, post-ERCP complications, Apgar scores, and neonatal follow-up at day 30 were recorded. Fluoroscopy-minimising techniques included suction-verification of bile and stent release without imaging.

Results: No maternal or foetal adverse events occurred peri- or post-procedure. ERCP was performed successfully in all 25 patients, with 23 (92%) undergoing sphincterotomy. Fluoroscopy was utilised in 20 cases (80%), with lead shielding employed in all. In five cases (20%), ERCP was conducted without fluoroscopy by using alternative bile verification techniques. Apgar scores ranged from 7 to 10, and all neonates exhibited normal development at 30-day follow-up.

Parameter	Value
Total procedures	25
Mean gestational age (weeks)	28.3
ERCP with fluoroscopy	20 (80%)
ERCP without fluoroscopy	5 (20%)
Maternal complications	0
Foetal complications	0
Mean Apgar score	9
Sphincterotomy performed	23 (92%)
Normal neonatal development (day 30)	100%

Conclusion: This study confirms that ERCP can be safely performed during pregnancy when executed by skilled endoscopists adhering to radiation-limiting protocols. Fluoroscopy-free ERCP appears feasible in selected cases, significantly reducing foetal radiation exposure without compromising efficacy.

Our findings support ERCP as a viable and safe therapeutic option in pregnant patients with biliary obstruction, providing novel evidence for the growing implementation of protective strategies and alternative imaging techniques in high-risk obstetric settings.

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Disclosure: Nothing to disclose.

PP0958

FROM FRAGMENTATION TO INFLAMMATION: MECHANICAL LITHOTRIPSY AS A TRIGGER FOR POST-ERCP PANCREATITIS

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Introduction: Post-endoscopic retrograde cholangiopancreatography pancreatitis (PEP) is the most frequent adverse event following ERCP, with an incidence of 3–10% in the general population and up to 40% in high-risk patients. Risk factors are commonly classified as patient-related (young age, female sex, suspected sphincter of Oddi dysfunction) or procedure-related (precut sphincterotomy, difficult cannulation, pancreatic duct manipulation). However, the impact of specific therapeutic maneuvers—such as stone extraction techniques and biliary drainage—on PEP risk remains uncertain.
In particular, the role of mechanical lithotripsy as an independent risk factor is still poorly defined. Moreover whether the systematic use of prophylaxis has actually reduced PEP incidence over time—particularly in high-risk cases—also remains uncertain.
Aims & Methods: This study aimed to retrospectively assess both patient- and procedure-related risk factors for post-ERCP pancreatitis, with a focus on therapeutic interventions whose impact remains uncertain. Findings were also compared with existing literature to refine individual risk profiles and inform preventive strategies.
We retrospectively analyzed 678 consecutive inpatients (49% male, 51% female; mean age 72.7 ± 13.4 years) admitted to our Gastroenterology Unit treated with ERCP for various indications (e.g., choledocholithiasis, pancreatic cancer, biliary strictures) between January 2020 and December 2024. All procedures were performed by experienced interventional endoscopists. PEP incidence was compared across categorical variables using chi-squared or Fisher's exact tests (significance threshold $p < 0.001$). Age was compared using the Mann–Whitney U test. Variables significant in univariate analysis were included in a multivariate logistic regression model.
Results: Thirty-six cases of PEP were identified (5.3%), including 58% mild, 34% moderate, and 8% severe. No procedure-related deaths were reported. Patients with PEP were significantly younger (median age 69 vs 76 years, $p = 0.01$). Independent risk factors included suspected sphincter of Oddi dysfunction (OR = 7.8, $p < 0.001$) and mechanical lithotripsy (OR = 4.20; 95% CI: 1.38–12.78; $p = 0.007$), while biliary stenting was protective (OR = 0.17; 95% CI: 0.04–0.67; $p = 0.01$). Nasobiliary drainage showed a non-significant trend toward protection (OR = 0.67; $p = 0.34$), and pancreatic stenting was not significantly associated with reduced risk ($p = 0.54$). No significant associations were found for stone extraction method, periampullary diverticulum, sphincterotomy, or precut. PEP incidence declined from 6.3% in 2020 to 3.8% in 2024 ($p = 0.22$), possibly reflecting improved prophylaxis.
Conclusion: Suspected sphincter of Oddi dysfunction and younger age were confirmed as significant PEP risk factors. Mechanical lithotripsy emerged as a high-risk maneuver, while biliary stenting appeared protective. These findings suggest procedure-specific risks should be integrated into individual preventive strategies.
Disclosure: Nothing to disclose.

PP0959

THE SAFETY AND EFFICACY OF SAME-SESSION ENDOSCOPIC ULTRASOUND FINE NEEDLE BIOPSY AND ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY FOR DISTAL MALIGNANT BILIARY OBSTRUCTION - A SINGLE HIGH-VOLUME CENTER EXPERIENCE

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Introduction: Distal malignant biliary obstruction represents an important indication for endoscopic ultrasound (EUS) and endoscopic retrograde cholangiopancreatography (ERCP). Often, patients who present with obstructive jaundice and/or cholangitis undergo emergency cross-sectional imaging that identifies locally advanced pancreatic lesions, biliary strictures, or ampullary tumors that require both pathological confirmation as well as biliary drainage.
Current recommendations indicate that both EUS and ERCP should be performed in a complementary fashion, both for tissue acquisition and biliary drainage. Combining these techniques in one session is not yet standard practice in all services.
Aims & Methods: Aims: We aimed to assess the rate of adverse effects and procedural success in patients undergoing both EUS and ERCP in a single session compared to separate sessions in our unit, a high-volume tertiary care center in North-Eastern Romania.
Methods: We performed a retrospective study including consecutive patients hospitalized for obstructive jaundice and/or cholangitis determined by distal malignant biliary obstruction suspected on cross-sectional imaging, hospitalized in the "St. Spiridon" Universitary Emergency Hospital from January 2022 to January 2023, who underwent both EUS and ERCP. We compared patients who had both procedures carried out in the same session with patients who had separate sessions of EUS and ERCP.
The primary outcome was the rate of adverse effects. The secondary outcomes were the success of the procedures and the hospitalization duration.
Results: Forty-nine patients were included. 30 patients underwent a separate session of EUS and ERCP, while 19 patients received EUS and ERCP during the same session. The adverse effects rate was similar between the 2 groups, and only related to ERCP (post-ERCP pancreatitis, 1 patient from the same session group vs 2 patients from the separate session group, bleeding, in 1 patient from the same session group and 1 patient from the separate session group). No deaths were noted. The procedure success was also similar in both groups, 28 patients from the separate session group, respectively 18 patients from the same session group had both procedures performed successfully. Concerning hospital duration, there was a slightly longer mean duration in the separate session group compared to the same session group (12 days vs 8 days).

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Conclusion: There was no significant difference regarding the rate of adverse effects or procedure success in patients who underwent both EUS and ERCP in the same session compared to separate sessions when these procedures were performed to manage distal malignant biliary obstruction. However, the mean hospitalization time was higher when the two procedures were performed in separate sessions.
Disclosure: Nothing to disclose.

PP0960

ENDOSCOPIC BILIARY DRAINAGE WITH SPHINCTEROTOMY FOR ACUTE CHOLANGITIS WITH SEPSIS-INDUCED COAGULOPATHY INCREASES ADVERSE EVENTS: A MULTICENTER PROPENSITY SCORE-MATCHED ANALYSIS

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Introduction: Acute cholangitis (AC) is a life-threatening condition requiring urgent biliary drainage. Endoscopic biliary drainage (EBD) with endoscopic sphincterotomy (EST) is often performed to facilitate bile flow and prevent post-ERCP pancreatitis (PEP). However, patients with sepsis-induced coagulopathy (SIC) are at increased risk of bleeding, raising concerns about the safety of EST in this population[1, 2]. This study aimed to evaluate the risk of adverse events (AEs) and clinical outcomes of EBD with versus without EST in AC patients with SIC.
Aims & Methods: A multicenter retrospective cohort study was conducted using propensity score matching. AC patients with SIC who underwent EBD with EST (EST group) or without EST (non-EST group) were analyzed. The primary endpoint was to compare the AE rate of EBD with EST versus EBD without EST in AC patients with SIC using propensity score matching. Secondary endpoints included the comparison of clinical outcomes (Ethics Committee Approval No. 2876-D38).
Results: Among 238 patients, 72 underwent EST and 166 did not. The incidence of AEs was significantly higher in the EST group than in the non-EST group (16.7% vs. 1.8%, P < .001), particularly for bleeding AEs (11.1% vs. 0.6%, P < .001). PEP incidence was similar (4.2% vs. 1.2%, P = .143). Propensity score-matched analysis confirmed these findings. No significant differences were observed in clinical success or 28-day survival rates.
Conclusion: EST significantly increases bleeding risk in AC patients with SIC without reducing PEP or improving clinical outcomes. Given these findings, EBD without EST should be recommended to minimize bleeding risk while ensuring effective biliary drainage.
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Disclosure: Nothing to disclose.

PP0961

EFFICACY AND SAFETY OF SINGLE OPERATOR CHOLANGIOSCOPY-GUIDED INTRADUCTAL LITHOTRIPSY IN DIFFICULT BILE DUCT STONES

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Introduction: Cholangioscopy-assisted electrohydraulic lithotripsy (CA-EHL) is a novel and effective method for the treatment of difficult bile duct stones that are refractory to conventional endoscopic therapy (CET).
Aims & Methods: Our objective was to compare the clinical and financial outcomes of CA-EHL and CET in the management of difficult bile duct stones.
Patients with difficult bile duct stones diagnosed via endoscopic retrograde cholangiopancreatography (ERCP) were retrospectively allocated into two groups: Group I: CET (January 1, 2022 – December 31, 2022), and Group II: CA-EHL (January 1, 2023 – January 31, 2025).
Patient characteristics, success rate of stone removal, complication rates, mortality, unplanned readmissions within 30 days, intensive care unit (ICU) admissions, length of hospital stay, and cumulative treatment costs were analyzed.
Results: Group I included 37 patients, as did Group II. There were no significant differences between the groups in terms of median age (73 vs. 74 years), gender distribution (male:female = 16:21 vs. 13:24), or Charlson comorbidity index (median: 4 vs. 3; P>0.05). The success rate of stone

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clearance was significantly higher in Group II (32.4% vs. 91.9%; P<0.001). Surgical intervention was significantly more common in Group I (32.4% vs. 5.4%; P=0.006). Patients in Group II spent significantly less time in the surgical ward (average: 2.9 vs. 0.4; P=0.0466). There were no significant differences between the groups in the number of complications (9 vs. 11 cases, P=0.452), time spent in the internal medicine ward (median: 6 vs. 8 days; P=0.704), mortality, 30-day unplanned readmission, or ICU admission rates (P>0.05). The cumulative treatment cost was significantly higher in Group II (3,917 vs. 10,241 €/patient; P<0.001).

Conclusion: CA-EHL represents a safe and effective approach for managing difficult bile duct stones refractory to standard endoscopic techniques. Although the treatment is associated with higher costs due to the expensive instrument, it significantly enhances bile duct clearance and decreases the necessity for surgical intervention. These results support the broader adoption of cholangioscopy-guided lithotripsy in clinical practice, particularly for complex stone cases.

Disclosure: Nothing to disclose.

PP0962

THE IMPACT OF PAPILLA MORPHOLOGY ON THE CHOICE OF ADVANCED CANNULATION TECHNIQUES IN ERCP - INSIGHTS FROM AN INTERNATIONAL SURVEY

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Introduction: Endoscopic retrograde cholangiopancreatography is the primary therapeutic procedure for pancreaticobiliary disorders, yet bile duct cannulation failure rates

remain 5–20% even among experienced endoscopists. The Haraldsson classification (2017) defines four papilla types, some linked to higher failure and complication rates. Limited research exists on how papilla morphology impacts rescue techniques.

Aims & Methods: This study aims to survey ERCP practitioners to assess the influence of major papilla morphology on their bile duct cannulation strategies. We hypothesize that tailoring cannulation techniques to papilla type may improve success rates and reduce complications.

We are conducting an ongoing international survey targeting physicians who routinely perform ERCP, focusing on how major papilla morphology affects cannulation strategies.

The survey was developed using REDCap, a secure web-based platform for data collection, and distributed via professional societies, mailing lists, and social media platforms. It includes questions on respondent demographics, ERCP experience, use of papilla classification systems, and preferred cannulation techniques.

Responses are collected anonymously. Initial analysis involves descriptive statistics, with further statistical evaluation planned as data collection progresses. Participation is voluntary, and the survey remains open to ensure broad inclusivity and representation.

Results: Among the 141 respondents, 27% reported using a classification system for papilla morphology. Of these, 57.9% were from certified training centers and 60.5% were affiliated with university-based hospitals. Additionally, 55.3% of classification system users had more than five years of experience, compared to 67% among non-users. Rescue cannulation strategies varied significantly based on papilla type and guidewire placement. For type I papilla, when the guidewire did not enter the pancreatic duct, 35.5% opted for needle-knife precut (NKP), 30.5% for needle-knife fistulotomy (NKF), and 28.4% continued with the standard technique. In contrast, if the guidewire was inadvertently inserted into the pancreatic duct, the double guidewire technique (DGT) was the most common first-line approach (54.6%), followed by prophylactic pancreatic stent (PPS) placement combined with NKF (21.4%) and transpancreatic precut papillotomy (TPPP) (9.2%). For type II papilla, NKP was the most frequently chosen technique (41.1%) when there was no guidewire entry into the pancreatic duct, whereas DGT was preferred (58.9%) when the guidewire was in place.

A similar pattern was observed with type III papilla, where DGT was favored (48.9%) in cases of pancreatic duct guidewire placement, and NKP was the leading choice (40.4%) when no guidewire was used. For type IV papilla, DGT was selected by 51.8% of respondents when the guidewire entered the pancreatic duct, while 41.8% preferred to continue with the standard cannulation technique in the absence of guidewire entry.

Conclusion: Differences were observed between classification system users and non-users. DGT was preferred when the guidewire entered the pancreatic duct, while NKP was commonly used when it did not. These findings suggest that adapting techniques based on papilla morphology may improve cannulation success.

Disclosure: Nothing to disclose.

PP0963

FULLY COVERED METAL STENTS FOR LARGE COMMON BILE DUCT STONES: A PROMISING SECONDARY CARE STRATEGY

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Introduction: It is recognised that large common bile duct (CBD) stones present a significant challenge in endoscopic practice. A key performance indicator of >80% clearance at the initial ERCP has been recommended to standardise quality of care. Standard techniques such as sphincterotomy +/- balloon sphincteroplasty are usually sufficient; however, in difficult cases, plastic stents are frequently placed to maintain biliary drainage, with elective repeat ERCP planned. When clearance remains unsuccessful, options such as mechanical lithotripsy or cholangioscopy may be employed, though these are often limited to tertiary care centres. Fully covered self-expanding metal stents (FCSEMS) have been proposed as a novel strategy to remodel the distal CBD and facilitate stone retrieval at subsequent procedures.

While this approach is not currently included in formal UK guidelines, its use is expanding in complex cases. We report our early experience comparing plastic stents alone with a combination of fully covered metal stents and coaxial plastic stents in a secondary care setting.

Aims & Methods: This was a retrospective observational study conducted at a large secondary care NHS trust. We included patients undergoing ERCP between 2019 and 2024 in whom complete stone clearance could not be achieved at the initial procedure and a stent was placed. Only patients with confirmed choledocholithiasis and naïve papillae at first ERCP were included.

All patients had large CBD stones, defined as stones greater than 10 mm in diameter. Patients were categorised according to whether a plastic stent alone or a fully covered metal stent with coaxial plastic stent was used at

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the index procedure. A 7Fr pigtail plastic stent was used in all cases; for the FCSEMS group, a fully covered metal stent was inserted with a plastic stent placed through it. Data were collected from endoscopy records and electronic patient notes. Outcomes assessed included complete stone clearance at second ERCP, need for mechanical lithotripsy, number of ERCP procedures, and referral to tertiary care for cholangioscopy. Fisher's exact test and unpaired t-tests were used for statistical analysis.

Results: A total of 63 patients were included, with 46 in the plastic stent group and 17 in the FCSEMS group. Complete stone clearance for large stones (>10 mm) at second ERCP was achieved in 63.0% of patients in the plastic group compared with 76.5% in the FCSEMS group (p=0.380). Mechanical lithotripsy was required in 8.7% of plastic stent cases and 5.9% of FCSEMS cases (p=1.000). Referral to tertiary care for cholangioscopy occurred in 30.4% of the plastic group and 11.8% of the FCSEMS group (p=0.195). The mean number of ERCPs per patient was similar between groups, with an average of 2.50 procedures in the plastic group and 2.41 in the FCSEMS group (p=0.601).

Conclusion: The use of fully covered metal stents with coaxial plastic stents was associated with promising trends, including higher rates of complete stone clearance for large CBD stones and fewer referrals to tertiary centres. Although these findings did not reach statistical significance, they suggest a clinically relevant benefit that merits further investigation. The higher observed clearance rate and reduced need for cholangioscopy may support FCSEMS as a valuable adjunct in managing difficult bile duct stones.

These results should be interpreted cautiously given the small size of our dataset, but they provide an encouraging basis for future prospective studies aimed at optimising CBD stone management and potentially reducing the need for tertiary care referral.

Disclosure: Nothing to disclose.

PP0964

CHOLANGIOSCOPY-GUIDED ELECTROHYDRAULIC LITHOTRIPSY VERSUS LASER LITHOTRIPSY FOR THE TREATMENT OF CHOLEDOCHOLITHIASIS: SINGLE CENTER EXPERIENCE

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Introduction: Choledocholithiasis (CCL) is one of the most common serious presentations of cholelithiasis. ERCP is the first-line treatment for the clearance of common bile duct stones (CBDSs). "Difficult" biliary stones are defined according to their diameter, shape, location, or because of anatomical factors. In these cases, gallstones cannot be removed using standard ERCP and additional interventional techniques are required. ESGE has recommended the use of cholangioscopy-assisted lithotripsy as one of the main treatment methods for difficult CBDSs.

This study aims to compare the safety and efficacy of cholangioscopy-assisted holmium-laser lithotripsy (LL) with electrohydraulic-lithotripsy (EHL) in the treatment of difficult CBDSs.

Aims & Methods: This is a retrospective study conducted at a tertiary endoscopic referral center. We enrolled thirty-one consecutive patients who had undergone cholangioscopy-assisted laser and/or electrohydraulic lithotripsy for difficult bile duct stones from November 2021 to March 2025. All patients underwent biochemical/clinical and imaging assessment prior to therapeutic ERCP. Intraductal lithotripsy efficacy was assessed based on bile duct clearance achievement (technical success) while clinical success was defined as bilirubin level decrease of one third 48 hours after the procedure. Safety outcomes were assessed based on the occurrence of any intra-/post-procedural complications. Procedural time was also recorded.

Results: A total of thirty-one patients with difficult gallstones were enrolled and retrospectively analyzed (M/F: 14/17; median age = 82 years). In 8 out of 31 cases (25%) a double therapeutic endoscopic session was needed to achieve CBD clearance. The bile duct was intubated via the ampulla of Vater using single-operator cholangioscopy.

Gallstones were fragmented using laser lithotripsy in 12/31 cases (39%) while electrohydraulic lithotripsy was used in 16/31 cases (51%); 3 out of 31 patients (10%) required both techniques to obtain CBD clearance. Technical success was achieved in 27/31 cases (87%).

Clinical success was observed in 26/27 cases (96%).Patients treated with LL had an overall median stone clearance success rate of 92%, whereas EHL achieved a median stone clearance success rate of 87%. Combined techniques only achieved 67% success. No complications were observed in all 31 procedures.

Conclusion: LL and EHL are both highly effective and safe forms of POC-guided lithotripsy for the treatment of large bile duct stones. LL seems to be more effective, although it requires longer procedural times, compared to EHL.

The success rate of the combined techniques is low when compared to the data for the individual techniques, but this could be because the three cases under consideration were particularly complex. This might suggest that if one intraductal lithotripsy technique fails, it is not indicated to attempt a second method, but more data on this would be needed to draw conclusions.

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Disclosure: Nothing to disclose.

PP0965

FORGOTTEN BILIARY PLASTIC STENTS IN A HIGH-VOLUME ERCP UNIT - DON'T YOU FORGET ABOUT ME

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Introduction: Plastic biliary stents that have not been removed for more than 12 months are considered forgotten stents. In this study, we sought to evaluate which factors are associated with forgetting biliary stents and their consequences.

Aims & Methods: Retrospective analysis, in a high-volume ERCP unit, of patients who underwent ERCP between 2016 and 2023 with placement of plastic biliary stents. Procedures in which the patient died or the prosthesis was surgically removed less than 1 year after its placement were excluded.

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Results: A total of 555 procedures in 425 patients (49% female, age 68.9±16.8 years) were included. The reasons for placement were: malignant stenosis 29.2% (n=162); choledocholithiasis 23.8% (n=132); benign stenosis 22.7% (n=126); bile leak 7.2% (n=40); acute cholangitis 6.7% (n=37); Mirizzi syndrome 4.7% (n=26); gallstones 4.3% (n=24); post-sphincterectomy bleeding 0.9% (n=5); intrahepatic lithiasis 0.5% (n=3).

Thirty-four procedures (6.1%) with forgotten stents were identified in 34 patients (35.3% female, age 64.4±17.3 years). The most frequent reasons for non-removal were: non-referral after cholecystectomy (n=15) or after hospital discharge (n=11). Among the indications for placement, gallstones (OR 9.71, p < 0.001) and Mirizzi syndrome (OR 3.02, p = 0.049) were associated with an increased risk of forgetting the device.

In 10 patients, extraction was endoscopic, 2 had spontaneous migration, 1 had surgical extraction, and 7 patients died for unrelated reasons. The rest (n=14) were recruited for removal during this study. A total of 9 patients of this group responded to the call and it was confirmed the spontaneous migration of the plastic stent in na abdominal radiography.

8 patients (23.5%) presented complications: acute cholangitis (n=4), jaundice (n=2), abdominal pain (n=1) and liver abscess with septic shock (n=1).

Conclusion: Forgotten stents constitute a rare event but with a significant risk of complications. Following this analysis, we created a database in order to improve the timing until the extraction of stents and reduce related adverse events.

Disclosure: Nothing to disclose.

PP0966

EFFICACY OF ACOUSTIC MONITORING OF RESPIRATORY RATE DURING SEDATION FOR ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY-RELATED PROCEDURES

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Introduction: Most serious adverse events during procedural sedation and analgesia (PSA) are caused by ventilation disorders, and close monitoring of respiratory status is considered important. However, in endoscopic procedures, especially endoscopic retrograde cholangiopancreatography (ERCP), the use of C-arm X-ray TV systems and patient-positioning devices can easily make it difficult for us to observe the movement of the patient's thorax. Although using capnography is recommended for ventilation monitoring during sedation for endoscopy, acoustic respiratory monitoring is used in PSA situations and other settings as a more convenient method for monitoring respiratory status.

Aims & Methods: In this study, we examined the outcomes of acoustic respiratory rate monitoring during sedation for ERCP-related procedures. We retrospectively reviewed 261 consecutive patients (493 procedures) who underwent ERCP-related procedures at our hospital between April 2024 and January 2025. We analyzed patients' backgrounds, sedation methods, and complications. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This retrospective observational study received ethical approval by the Institutional Review Board (IRB) of our hospital.

Results: The median patient age was 75 years, and 172 men and 89 women were included in the study. The target diseases for treatment included bile duct stones in 122 patients, malignant biliary strictures in 108 patients, benign biliary strictures in 20 patients, and other diseases in seven patients. The median procedure time was 30 min (9-140 min), the median doses of midazolam and pentazocine used were 4 mg (0-11 mg) and 7.5 mg (0-30 mg), respectively. Dexmedetomidine was administered to 46 (9.3%) patients. The number of minor cardiopulmonary events during PSA was 45 (9.1%), included hypotension in 20 patients, bradycardia in 15, depression of the respiratory rate in 13, and decreased SpO_2 in one. The number of patients with decreased SpO_2 was significantly lower than the number of patients with depression of respiratory rate (p=0.001). Rapid recovery was observed in all patients, and permanent interruption of the procedure or mortality was not observed.

Conclusion: The use of acoustic monitoring of the respiratory rate allowed the procedure to be performed without serious complications. In particular, it seems important to detect a decrease in respiratory rate so that early action can be taken before it leads to a decrease in SpO_2.

Although this monitoring method has some drawbacks, such as the fact that it only evaluates the respiratory rate and not ventilation, its simplicity makes it a useful option for safe endoscopic procedures, especially in settings staff resources.

Disclosure: Nothing to disclose.

PP0967

RETROSPECTIVE REVIEW OF THE ROLE OF BOTOX INJECTION AND SPHINCETROTOMY IN THE MANAGEMENT OF FUNCTIONAL HEPATOBILLARY PAIN

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Introduction: Managment of functional hepatobillary pain remains a significant challenge for physicians. Botulinum toxin injection of the ampulla of vatter and endoscopic sphincterotomy has been shown to be an effective treatment with diagnostic and prognostic benefit, however, it has a relatively high complication rate.

Aims & Methods: To assess the effectiveness and complication rate of botox injection and sphincterotomy for treatment of sphincter of oddi dysfunction (SOD). And assess if response to botox injection is predictive of durable response to completion sphincterotomy. Patients were identified from a single centre endoscopy database from January 2014 until October 2024 using the keyword searches "Botox" and "Botulinum". Studies were included if 100IU botox injection to the ampulla was done at endoscopy for the purpose of treating SOD. SOD was defined as per the ROME IV criteria and classified as per Milwaukee classification.

Baseline clinical, demographic, endoscopic, and radiological data were collected. Symptomatic response was assessed through interrogation of clinical notes post procedure. Patients who progressed on to sphincterotomy were identified, and symptomatic response to this was obtained from clinical review of notes at 3 months and 1 year post procedure. For categorical variables Fisher's exact test was used.

Results: 364 duodenoscopies were identified from the database, 74 duodenoscopies from 52 patients were included. 13 patients had more than one botox injection. 77% (40 patients) were female, and the median [range] age at first procedure was 41.2 [21.7 - 66.6] years. 1 (1.9.%), 44 (84.6%), and 7 patients (12.5%) had type I, Type II, and Type III SOD respectively. Diclophenac was given at time endoscopy in 26 (36%) procedures. Four patients were admitted for 24 hours with non-pancreatitis related abdominal pain post procedure. Three patients had no data on response post botox injection. 42 (85%) patients reported improvement in

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their symptoms following botox injection. 35 of the 44 patients with type II SOD responded. 6 of the 8 patients with type III, and the only type I patient responded. 25 (48%) patients in total progressed on to sphincterotomy (one type I, 22 type II, and 2 Type III). 10 patients with no response to botox progressed on to sphincterotomy. 10 patients had >1 sphincterotomies for treatment of SOD. 36 post botox sphincterotomies were included, 8 (22%) had complications. There were 5 (14%) micro-perforations, there was one (2.8%) pancreatitis and two (5.5%) admissions for abdominal pain without evidence of pancreatitis. No patient had requirement for ICU or surgery. Initial response to sphincterotomy was at 3 months was 72% (18). Twelve (48%) patients (11 type II SOD, and one Type III) had durable response at one year.

There was a trend that patients who responded to botox injection were more likely to have response from sphincterotomy at one year, 10 patients (55%) compared to two (28%) Patients who did not respond to botox injection responded to sphincterotomy at one year. However, this was not statistically significant (p=0.37).

Conclusion: Our data indicated a low incidence of complication to botox injection with no recorded cases of pancreatitis. There was a trend that patients who responded to botox injection may be more likely to respond to sphincterotomy. Further data is required, however there was a trend towards patients who responded to botox injection being more likely to benefit from sphincterotomy.

Disclosure: Nothing to disclose.

PP0968

RISK FACTORS OF POST-ERCP CHOLANGITIS: EXPERIENCE OF AN ITALIAN THIRD LEVEL CENTER

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) plays a central role in the treatment of multiple biliary and pancreatic disorders. Despite the technical improvements reached in the past few decades, ERCP remains one of the endoscopic procedures with the highest complication rates. Post-ERCP cholangitis is the most common infectious complication and has a multifactorial origin.

Aims & Methods: The aim of our retrospective analysis was to identify patient-, procedural-, and operator-related factors associated with post-ERCP cholangitis. We analyzed the biliary drainage procedures performed at the Endoscopy Unit of Reggio Emilia in 2022, for a total of 517 procedures. We excluded procedures performed for acute cholangitis according to Tokyo 2018 criteria, patients who had recently undergone cholecystectomy (during the same hospitalization or within two weeks prior to the ERCP) and patients with ongoing abscess collections. We so included 164 procedures performed on patients who did not have a diagnosis of acute cholangitis or confounding factors at the time of the procedure. Then we compared patients who developed cholangitis according to Tokyo 2018 criteria within 30 days from ERCP with patients who did not, and perfomed univariate and multivariate analyses to define potential risk factors associated with post-ERCP cholangitis.

Results: 31/164 (19%) patients of our cohort developed post-ERCP cholangitis. The univariate analysis revealed that presence of stenosis (83.9% vs 41.4%; p<0.001), placement of biliary stents (90.3% vs 42.1%; p<0.001), particularly multi-stent vs plastic or metallic (42.9% vs 16.1%; p 0.028), median duration of the procedure (92 vs 69 min; p 0.003), and the performance of pneumatic dilatations (22.6% vs 6.8%; p 0.008) were significantly associated with the occurrence of post-ERCP cholangitis; we also found that antibiotic therapy (58.1% vs 82%; p 0.004), balloon passages with Fogarty (25.8% vs 51.9%; p 0.009), and a history of previous ERCPs (2.6% vs 44.4%; p 0.026) were potential protective factors. Demographic characteristics of the cohort (sex, ASA score, age), operator expertise, and other technical factors such as sphincterotomy, sphincteroplasty, pre-cut, papillectomy, lithotripsy, as well as the placement of pancreatic stents, bioptic sampling, nasobiliary tubes (NBT), or the use of Spyglass and Dormia baskets, did not show a significant association with the development of post-ERCP cholangitis.

At the multivariate analysis, duration of the procedure (OR 1.01 [95%CI 1-1.02]) and placement of biliary stents (OR 9.57 [95%CI 2.64-34.67]) remained significant risk factors, while antibiotic therapy (OR 0.22 [95%CI 0.08–0.61]) was confirmed as a potential protective factor.

Conclusion: Post-ERCP cholangitis have a multifactorial etiology. In our cohort, duration of the procedure and placement of biliary stents showed significant association with the development of this complication, while antibiotic therapy was a protective factor. Further studies are needed to confirm our findings and to define which measures can be implemented to reduce post-ERCP cholangitis.

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Disclosure: Nothing to disclose.

PP0969

IMPROVING DOCUMENTED ADVICE REGARDING ANTICOAGULANTS AND ANTIPLATELETS IN PATIENTS UNDERGOING ERCP

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Introduction: Endoscopic Retrograde Cholangio-Pancreatography (ERCP) is performed to treat pathology within the biliary system. Patients undergoing this may be on systemic anticoagulation. ERCP is classed as a high-risk bleeding procedure within BSG guidelines. This would necessitate anticoagulation interruption to prevent bleeding. It is recognised however, that resumption of antiplatelets and anticoagulants is associated with improved survival and reduced incidence of thrombotic complications. Thus, all patients should have a clear anticoagulation plan.

Aims & Methods: Attendances for ERCP over a 6-month period were analysed. Of the 132 patients, 33 (25%) were taking either an antiplatelet, warfarin or a Direct Oral Anticoagulant (DOAC). Medical records were reviewed in relation to pre-procedure management of these medications, advice from the endoscopist regarding resumption and whether these medicines were managed appropriately thereafter.

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Several interventions were then made; a change to the reporting software led to a prompt relating to antiplatelet/anticoagulant medications, an education campaign for post graduate doctors in training and medication guidance summaries visible in the endoscopy department.

A second 6-month review was then undertaken to evaluate if there had been any improvement in outcomes. In this time frame, 143 patients underwent ERCP of whom 27 (16.7%) were taking antithrombotic therapy.

Results: Results are in Table 1:

	Pre-Intervention	Post Intervention
Pre-procedure appropriate management	48.5%	70.9%
Documented advice on endoscopy report	10%	30%
Post procedure appropriate prescribing	36.3%	70%

One common error was holding of anticoagulation/antiplatelet medication from admission in anticipation of ERCP. This led to long periods off anticoagulation whilst awaiting ERCP.

Documented advice was given more frequently on Endoscopy reports in cases where the patient required further procedure.

It was challenging to analyse post procedural management for inpatients, as the quality of ward round documentation and discharge letters was variable.

Conclusion: Further education to junior staff around utility of BSG anticoagulation guidelines greatly improved pre and post procedure management and thus forms a cornerstone to improve care. Further education to more senior staff may also improve decision rationale and medication advice in ward round documentation and discharge paperwork.

Human factors also need to be considered; limited advice is likely related to high endoscopy workload pressures, with sub analysis showing that advice is improved when further intervention is required or in higher risk settings, illustrating that antithrombotic decisions are considered by the endoscopist. Improved documentation at procedure with the support of upgrades to reporting software will enhance post procedure care and consequently reduce demands on the service.

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UEG poster: Improving documented advice regarding anticoagulants and antiplatelets in acute upper gastrointestinal bleeds. Authors: Dr Emily Creed, Dr Vanessa Theis, Dr Katie Clark, Dr Sweekriti Gurung, Dr Abdullah Motam and Dr Gregor Farrell Bins.

Disclosure: Nothing to disclose.

PP0970

USEFULNESS OF PERORAL CHOLANGIOSCOPY IN THE DIAGNOSIS AND CLINICAL STAGING OF PRIMARY SCLEROSING CHOLANGITIS

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Introduction: Primary sclerosing cholangitis (PSC) is known to present characteristic cholangiographic findings; however, detailed observations using peroral cholangioscopy (POCS) have rarely been reported. We previously reported that POCS findings in PSC can be classified into three types: active (inflammatory), chronic (scarring), and tumorous (tumor-like). These findings may aid in the differential diagnosis of PSC and may be useful in predicting prognosis.

Aims & Methods: We aimed to analyze POCS findings in PSC in detail based on this classification and to clarify their characteristics by evaluating the frequency of each finding, their temporal changes, and comparisons with other biliary diseases.

We retrospectively reviewed 22 patients with PSC who underwent POCS between February 2018 and February 2023. For comparison, we included 19 cases of hepatolithiasis, 25 cases of cholangiocarcinoma, and 3 cases of IgG4-related sclerosing cholangitis (IgG4-SC). Twelve POCS findings were categorized into three types: active (mucosal erythema, ulcers, fibrous white exudates, irregular surface), chronic (longitudinal scarring, annular scarring, pseudodiverticula, stenosis), and tumorous (dilated vessels, tortuous vessels, friability, mass formation). In addition, we included "punched-out lesion (POL)" as an original finding. Each finding was graded on a 5-point scale (0–4) based on the extent or number of lesions, and scores were calculated for each type.

We conducted three analyses: (1) correlation between cholangiographic and POCS findings in PSC, (2) temporal changes in POCS findings in PSC, and (3) comparison of POCS findings between PSC and other biliary diseases.

Results: In the first analysis, annular scarring (100%), stenosis (89%), POL (89%), and longitudinal scarring (86%) were frequently observed in PSC. Annular scarring was identified in all cases with a shaggy appearance on cholangiography, and stenosis were present in all ducts with band-like stricture. In ducts showing diverticulum-like outpouchings, longitudinal and annular scarring, pseudodiverticula, and POL were consistently seen, whereas other findings were absent. In the second analysis, active and tumorous scores varied over time even within the same case, whereas chronic scores remained stable or showed a progressive increase. A negative correlation was observed between active and chronic scores (p-value <0.001, r =-0.702). In the third analysis, chronic findings and POL were significantly more common in PSC compared to other diseases (p-value <0.001). Pseudodiverticula were seen exclusively in PSC, while dilated and tortuous vessels and friability were significantly more frequent in cholangiocarcinoma and IgG4-SC (p-value <0.001).

Conclusion: These findings suggest that the characteristic cholangiographic findings of PSC are mainly formed by chronic scarring. Diverticulum-like outpouchings reflect only fibrotic changes. Furthermore, the coexistence of inflammation and fibrosis, along with the time-dependent progression of fibrosis, is a unique feature of PSC, although active inflammation may obscure chronic findings.

Finally, longitudinal scarring and POL were particularly prominent in PSC, and pseudodiverticula were highly specific to this disease, suggesting their usefulness in differential diagnosis.

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In conclusion, chronic findings on POCS are characteristic of PSC and may serve as a helpful tool in its diagnosis and differentiation from other biliary diseases.

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Disclosure: Nothing to disclose.

PP0971

CHOLEDOCHOLITHIASIS: PREDICTING SPONTANEOUS MIGRATION TO LIMIT ERCP

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Introduction: ERCP is the first-line procedure for the treatment of choledocholithiasis. Spontaneous migration of common bile duct (CBD) stones through the duodenal papilla is a phenomenon observed in some cases. This migration can render ERCP unnecessary and potentially avoid complications.

The objective of our study is to evaluate the factors associated with the spontaneous migration of CBD stones.

Aims & Methods: This is a retrospective descriptive and analytical study including 547 patients who underwent ERCP for CBD lithiasis between January 2020 and January 2025.

All patients underwent MRCP for the diagnosis of CBD lithiasis.

A positive MRCP was defined as the presence of stones in the CBD.

A positive ERCP was defined as the presence of stones in the CBD at cholangiography.

We divided the patients into 2 groups:

- Group A (n=309): MRCP + and ERCP +
- Group B (n=265): MRCP + and ERCP –

Statistical analysis was performed using JAMOVI software.

Results: The mean age in Group A was 59.1 +/- 13.8 years and in Group B was 56.9 +/- 13.2 years, with no statistically significant difference (p=0.056).

The sex ratio (M/F) in Group A was 0.7 and in Group B was 0.54, with a significant female predominance in both groups (p=0.03).

There was no statistically significant difference regarding the CBD diameter and the presence of a periampullary diverticulum between the two groups.

When comparing Groups A and B, patients with a small stone diameter (P=0.001), a single stone (p=<0.001), and distal stones (P=0.04) tended to spontaneously evacuate their stones.

Conclusion: In our study, the factors associated with the spontaneous migration of CBD stones are a small stone diameter, a single stone, and distal stone location.

Disclosure: Nothing to disclose.

PP0972

THE ADDED VALUE OF PERORAL CHOLANGIOSCOPY IN THE DIAGNOSIS OF INDETERMINATE BILIARY STRICTURES

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Introduction: Indeterminate biliary strictures pose a significant diagnostic challenge, as they can result from a wide spectrum of both benign and malignant conditions. Differentiating between these is crucial due to different management strategies. Conventional methods like endoscopic retrograde cholangiopancreatography (ERCP) with brush cytology have limited sensitivity, often leading to repeated procedures and delays in treatment. Peroral cholangioscopy (POC) offers direct bile duct visualization and targeted biopsies, improving diagnostic accuracy. Modern systems, like the SpyGlass™ DS, enhance procedure feasibility and image quality.

Aims & Methods: Aims: The aim of our study is to assess the effectiveness of single-operator peroral cholangioscopy system (POCS) with its digital version *SpyGlass™ Direct Visualization System* with SpyBite biopsies as a promising modality for accurate diagnosis of biliary strictures.

Method: We performed a two-center retrospective analysis of prospective database from March 2019 to March 2025. Each patient underwent peroral cholangioscopy using *SpyGlass™ Direct Visualization System with* SpyBite biopsies for the diagnosis of indeterminate biliary strictures.

Results: We identified 162 patients who underwent peroral cholangioscopy during the study period (96 males and 66 females). The medium age of the patients was 63.04 years. There was 100 % technical success of the procedure. All patients underwent POCS due to indeterminate biliary strictures. Before the procedure CT scan was performed and positive for biliary stenosis in 115 (70.9%) of the cases. The biliary stenoses were located as follows: 20 (12.3%) in the distal common bile duct, 29 (17.9%) in the proximal common bile duct, 102 (62.9%) at the hepatic hilum and 11 (6.7%) in the intrahepatic bile ducts.

Benign disease was diagnosed in 39 (24.07 %) of the patients. Visual impression for malignant desease (at least one of the following- tortuous dilated vessels, friability and easily bleeding, infiltrative stricture, polipoid mass, nodular eleveted lesion, vegetative mass, fish-egg lesion, finger - like villiform lesion and irregular papillary or granular lesion) was found in 123 (75.9%) of the patients. The most reliable correlation with malignancy was found to be the presence of infiltrative strictures - 113 (69.7%) patients. A final diagnosis of cholagiocarinoma was established in 84 (51.8%) of them. In 39 (24.07%) patients, no tumor cells were found in the specimen. Visual impression accuracy with SpyGlass was 100% (with 100% sensitivity and 100% specificity). SpyBite biopsies accuracy was 69.05 % (with 68. 3% sensitivity and 100% specificity).

Conclusion: POC has proven valuable in evaluating indeterminate biliary strictures, offering advantages over ERCP by allowing real-time visualization and targeted biopsies. Our study supports POC's superior accuracy,

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especially in identifying malignant lesions, with visual impression achieving up to 100% sensitivity and specificity. However, POC adoption is limited by operator expertise, the need for standardized interpretation, and availability.

While adverse events are rare (<10%), careful patient selection is essential. In conclusion, POC holds substantial promise as a key diagnostic tool for indeterminate biliary strictures, with ongoing improvements enhancing its clinical role.

Disclosure: No conflict of interest.

PP0973

ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY IN THE ELDERLY

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Introduction: Managing biliopancreatic disease in elderly patients is a growing challenge, where the balance between benefit and risk becomes increasingly delicate. ERCP remains a fundamental tool, but its application in older adults requires individualized technical and clinical considerations.

The aim of our study was to evaluate the characteristics of elderly patients who underwent ERCP.

Aims & Methods: We conducted a retrospective descriptive study from January 2019 to January 2025, including all patients aged ≥65 years who underwent ERCP. Collected data encompassed demographic characteristics, clinical presentation, procedural indications, technical success, therapeutic approaches, and procedure-related complications.

Results: We recorded 186 patients : 62.36% women (n=116) and 37.6% men (n=70), with a median age of 75 years [IQR: 71–80]. The most frequent presenting symptoms were biliary colic (43.4%, n=79), acute cholangitis (23.6%, n=43), cholestatic jaundice (11.8%, n=22), and acute pancreatitis (10.8%, n=20).

ERCP was indicated for benign pathology in the majority of cases, primarily for choledocholithiasis (56.5%, n=104). Other benign etiologies (11%, n=12) included post-cholecystectomy syndrome, biliary pancreatitis, bile duct injury, and Sump syndrome.

Malignant indications included pancreatic head tumors (20.7%, n=38), hilar cholangiocarcinoma (7.1%, n=13), ampullary carcinoma (6.5%, n=12), and gallbladder carcinoma (3.8%, n=7).

The technical success rate for benign pathology was 97.4% (n=113), with a failure rate of 2.6% (n=3). In malignant indications, the success rate was 90% (n=63), with failures (10%, n=7) primarily due to duodenal obstruction or an intradiverticular papilla.

Stenting strategies included uncovered self-expandable metal stents (33.2%, n=43), covered metal stents (3.8%, n=7), and plastic stents (7.6%, n=14).

Intra-procedural bleeding following sphincterotomy occurred in 2.7% of cases (n=3); two were managed with epinephrine injection and one with hemostatic clipping. No delayed post-procedural complications were reported.

Conclusion: ERCP appears to be a safe, effective, and well-tolerated procedure in elderly patients. When guided by appropriate indications and expert technique, age does not represent a limitation but rather a parameter requiring adapted procedural planning.

Disclosure: Nothing to disclose.

PP0974

SPHINCTER OF ODDI DYSFUNCTION: A CHALLENGE IN INTERVENTIONAL ENDOSCOPY?

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Introduction: Endoscopic sphincterotomy (ES) remains the reference treatment for type I and type II sphincter of Oddi dysfunction (SOD) following the failure of well-conducted medical therapy. However, it is an invasive procedure that presents technical challenges for biliary cannulation and carries an increased risk of post-endoscopic retrograde cholangiopancreatography (ERCP) acute pancreatitis in this indication.

The aim of our study was to describe the technical difficulties encountered during ERCP in patients with suspected SOD.

Aims & Methods: This was a retrospective, single-center study including all patients who underwent ES for the treatment of SOD between 2018 and 2024. Demographic, clinical, and endoscopic data were collected and analyzed.

Results: A total of 37 patients were included, with a mean age of 64.16 years and a marked female predominance (73%). SOD was diagnosed following an episode of cholangitis or acute pancreatitis in 51.4% of cases. Several technical difficulties were encountered, such as challenging papillary orientation (16.2%) and peri-diverticular location (10.8%).

Unintentional access to the main pancreatic duct occurred in 73% of cases, leading to the placement of a prophylactic pancreatic stent in 37.8%. Biliary cannulation was achieved in all patients.

Advanced cannulation techniques were required in 88.1% of cases, including the use of needle-knife sphincterotomes (35.1%), fistulotomies (45%), and transpancreatic sphincterotomy (8%).

Conclusion: ERCP in patients with suspected SOD is a true technical challenge. In our study, advanced biliary cannulation techniques were necessary in 88.1% of cases. These procedures thus require a high level of expertise in interventional endoscopy.

Disclosure: Nothing to disclose.

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PP0975

USE OF RECTAL INDOMETHACIN FOR POST-ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY PREVENTION IN PATIENTS POST-LIVER TRANSPLANT IS NOT ASSOCIATED WITH WORSENED RENAL OUTCOMES

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is commonly performed in post-liver transplant (LT) patients for biliary complications, but it carries risks such as acute pancreatitis (AP) and acute kidney injury (AKI). Rectal indomethacin, an NSAID, reduces post-ERCP pancreatitis in the general population; however, its safety in post-LT patients with potentially compromised liver function is unclear. This study evaluated the association between rectal indomethacin use and clinical outcomes in post-LT patients undergoing ERCP, using a large multinational population-based dataset.

Aims & Methods: A retrospective cohort study was performed using large population-based data from the TriNetX platform. We identified post-LT patients undergoing ERCP medicated with rectal indomethacin between January 1, 2014, and November 24, 2024. This cohort was matched 1:1 with post-LT patients undergoing ERCP not medicated with rectal indomethacin, using propensity matching based on age, gender, demographics, and comorbidities. Outcomes included risk of all-cause mortality, acute pancreatitis (AP), acute kidney injury (AKI), acute interstitial nephritis (AIN), renal replacement therapy, and ICU admission. Cox-proportional hazards models were used to estimate hazard ratios (HRs) with 95% confidence interval (CI).

	Post-LT Patients Undergoing ERCP with Indomethacin		Post-LT Patients Undergoing ERCP without Indomethacin		HR (95% CI)
Outcomes	Included	Events	Included	Events	
Mortality	714	10	712	10	0.67 (0.11, 3.99)
Acute pancreatitis	717	25	717	28	0.89 (0.52, 1.53)
AKI	717	103	717	85	1.23 (0.93, 1.64)
Acute interstitial nephritis	717	0	717	0	-
Renal replacement therapy	717	28	717	27	1.04 (0.61, 1.77)
ICU	717	49	717	55	0.89 (0.61, 1.31)

Abbreviations: LT, liver transplant; ERCP, endoscopic retrograde cholangiopancreatography; AKI, acute kidney injury; ICU, intensive care unit; HR, hazard ratio; CI, confidence interval.

Table 2. Outcomes Associated with Use of Indomethacin in Post-LT Patients Undergoing ERCP.

Results: A total of 4,642 post-LT patients undergoing ERCP were identified, of whom 717 individuals were medicated with rectal indomethacin. Among these, 717 patients medicated with rectal indomethacin (mean [SD] age, 53.7 [13.5] years; 267 [37.2%] female) were matched with 717 individuals (mean [SD] age, 54.5 [13.1] years; 266 [37.1%] female) not medicated with rectal indomethacin. The indomethacin group had no significant differences when compared to those not receiving rectal indomethacin; AKI (HR, 1.23; 95% CI, 0.93-1.64), renal replacement therapy (HR, 1.04; 95% CI, 0.61-1.77), all-cause mortality (HR, 0.67; 95% CI, 0.11-3.99), AP (HR, 0.89; 95% CI, 0.52-1.53), and ICU admission (HR, 0.89; 95% CI, 0.61-1.31).

Conclusion: Rectal indomethacin use in post-LT patients undergoing ERCP was not associated with significant differences in renal, hepatic, or clinical outcomes compared to those not receiving indomethacin. These findings support the cautious use of rectal indomethacin in post-LT patients undergoing ERCP, particularly when the risk of post-ERCP pancreatitis is high. The absence of a significant signal for harm may continue to encourage its use as a preventive measure, though clinicians should remain vigilant for potential renal and hepatic decompensation in high-risk patients.

Future research should focus on elucidating the mechanisms underlying indomethacin's effects in this unique population and exploring alternative prophylactic strategies where appropriate.

Disclosure: Nothing to disclose.

PP0976

FAILURE OF ERCP: EXPERIENCE FROM A TERTIARY CENTER

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is a pioneering procedure in the management of pancreatobiliary diseases. However, access to the common bile duct is not always easy, even in expert hands, with a failure rate ranging between 5% and 10%. A second attempt or alternative cannulation techniques are often required before considering surgery.

Aims & Methods: The aim of this study was to assess the ERCP failure rate in a tertiary care center.

This was a retrospective, descriptive, and analytical study conducted in a hepatogastroenterology department. It included patients who underwent ERCP with failed bile
duct cannulation over a 6-year period.

Results: Out of 801 ERCP procedures performed, 14 failures (1.7%) were recorded. Among these patients, 28.6% (4 patients) underwent a second ERCP attempt, with one of them being
successful. The indications for ERCP were: pancreatic head tumor in 43% of cases (6), common bile ductstones in 36% (5), cholangiocarcinoma in 14% (2), and bile duct trauma in 7% (1). The techniques used for cannulation were primarily guidewire-assisted cannulation in 79% of cases (11) and precut sphincterotomy in 21% (3). 8 patients (62%) had surgical drainage and 5 patients (38%) underwent endoscopic ultrasound-guided drainage after ERCP failure.

Conclusion: Our study demonstrates a low ERCP failure rate, comparable to that of international centers. Surgical intervention was more common early in the experience, prior to the introduction of endoscopic ultrasound-guided drainage in our practice.

Disclosure: Nothing to disclose.

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PP0977

CLINICAL OUTCOMES OF ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY-GUIDED REMOVAL OF COMMON BILE DUCT STONES IN ELDERLY PATIENTS

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Introduction: Endoscopic removal of common bile duct stones (CBDS) via endoscopic retrograde cholangiopancreatography (ERCP) is the first-line treatment. In developed countries, the number of ERCP procedures for CBDS in elderly patients is increasing with the aging population.

Elderly patients often present with poor performance status (PS), are frequently on antithrombotic therapy, and have multiple comorbidities. These factors raise concerns about an increased risk of ERCP-related complications; however, the safety and efficacy of ERCP in this population have not been fully evaluated.

Aims & Methods: The aim of this study was to assess the outcomes of ERCP for CBDS in elderly patients. Among 548 patients with symptomatic CBDS treated at our institution between April 2017 and February 2024, we excluded 81 cases involving altered anatomy, use of endosonography-assisted rendezvous technique, spontaneous stone passage, and biliary pancreatitis. The remaining 467 patients undergoing ERCP at our institution were retrospectively analyzed.

The patients were divided into an elderly group (E group: ≥75 years, n=241) and a younger group (Y group: <75 years, n=226), and short- and long-term outcomes were compared between the two groups.

Results: Patient Characteristics: The median age was 81 years (range: 75–99) in the E group and 68 years (range: 21–74) in the Y group. Male sex was less frequent in the E group (55% vs. 68%, p=0.006). The E group had significantly higher rates of antithrombotic use (39% vs. 24%, p=0.001), American Society of Anesthesiologists physical status ≥ III (57% vs. 35%, p=0.001), and concomitant cholangitis (52% vs. 41%, p=0.012).

Median stone size was slightly larger in the E group (8 mm vs. 6 mm, p=0.14), and the median bile duct diameter was significantly wider (10 mm vs. 9 mm, p=0.001).

Comorbidities were more prevalent in the E group: cardiovascular disease (34% vs. 22%, p=0.004), hypertension (52% vs. 39%, p=0.004), malignancy (33% vs. 18%, p=0.001), and dementia (11% vs. 2%, p=0.001).

Although endoscopic sphincterotomy (EST) was the most common papillary intervention in both groups, endoscopic papillary large balloon dilation (EPLBD) was more frequently performed in the E group (31% vs. 10%). Lithotripsy rates were similar between groups (22% vs. 16%, p=0.2).

Outcomes: Complete stone removal was achieved in 96% of the E group and 98% of the Y group (p=0.2). The median number of treatment sessions was 1 in both groups (range: 1–3, p=0.13).

Overall complication rates were comparable (8% vs. 7%, p=0.63). In the E group, complications included pancreatitis (n=15), bleeding (n=5), perforation (n=1), and aspiration pneumonia (n=1); in the Y group, pancreatitis (n=10) and bleeding (n=5) were observed.

No procedure-related mortality occurred in either group. There was no significant difference in the cumulative recurrence rate of stones after complete removal (p=0.29).

Multivariate analysis using variables such as age, sex, presence of cholangitis, bile duct diameter, lithotripsy, and gallbladder stones revealed no definite predictors of recurrence.

Conclusion: Although recent studies have reported the potential utility of long-term biliary stenting in elderly patients, stent occlusion leading to cholangitis remains a concern. Repeated hospitalizations in elderly patients can decrease activities of daily living. Therefore, we prioritize complete stone removal in patients with a favorable long-term prognosis. This study demonstrates that even in patients aged 75 years and older, active treatment of CBDS via ERCP might be effective and safe.

Disclosure: Nothing to disclose.

PP0978

REAL WORLD CRITERIA FOR ERCP TIMING IN ACUTE CHOLANGITIS

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Introduction: The cornerstone of acute cholangitis treatment is antibiotic therapy combined with biliary drainage. The Tokyo Guidelines 2018 provide severity grading criteria that help predict prognosis and identify patients requiring early biliary drainage. While mild cholangitis can often be managed initially with antibiotics alone, moderate and severe cases typically require early and emergent drainage, respectively.

Our aim is to explore factors influencing the timing of ERCP in acute cholangitis and its impact on clinical outcomes.

Aims & Methods: This retrospective single-center cohort study included patients diagnosed with acute cholangitis who underwent ERCP between March 2022 and March 2023 in a tertiary hospital with ERCP. Diagnosis and severity grading were based on the Tokyo Guidelines 2018.

Descriptive and statistical analyses were performed to evaluate patient characteristics, severity classification, timing to ERCP and related clinical outcomes.

Results: A total of 104 patients underwent ERCP for acute cholangitis; 55 were female (52.9%) and the median age was 77 years. Based on the Tokyo 2018 criteria, 28 patients (26.9%) had grade III cholangitis, 52 (50%) grade II and 24 (23%) grade I. Only 5 patients (4.8%) underwent ERCP within 24 hours of admission—2 with grade III, 1 with grade II and 2 with grade I. Within 72 hours, 31 patients (29.8%) had undergone ERCP—9 with grade III, 16 with grade II and 6 with grade I.

Notably, 70.8% of grade III cases were submitted to ERCP more than 72 hours after admission, with a median of 3 days of antibiotics before the procedure.

There was no statistically significant association between ERCP timing and patient age, cholangitis severity, intensive/intermediate care admission or anticoagulant/antiplatelet use. However, admission on a Friday or Saturday was significantly associated with delayed ERCP (p=0.007), with 91% of these patients undergoing ERCP after 72 hours, compared to 56% for admissions on other days.

Cholangitis severity correlated significantly with total hospital stay (grade I: 7 ± 3 days, grade II: 8 ± 3.8 days, grade III: 10.5 ± 9 days; p=0.001). ERCP timing was also associated with total antibiotic duration (<24h: 9 ± 2.6 days, 24–72h: 8 ± 3 days, >72h: 10 ± 4.8 days; p=0.043). The 30-day mortality rate was 4.8%, with no significant association with ERCP timing or severity grade.

Conclusion: In this real-world cohort, ERCP was frequently delayed beyond the recommended window, even in severe acute cholangitis. Weekend admission was a significant contributor to this delay. However, no association was found between ERCP timing and 30-day mortality, likely due to small sample size. These findings highlight the need for optimized access to biliary drainage, particularly during weekends, to better meet guideline recommendations.

Disclosure: Nothing to disclose.

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PP0979

NAVIGATING BILIARY ACCESS: TECHNIQUES AND PREDICTORS FOR DIFFICULT CANNULATION IN ENDOSCOPIC CHOLANGIOPANCREATOGRAPHY

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Introduction: Despite being standard modality to manage a wide range of pancreaticobiliary pathologies the technical complexities and potential complication rates make provision of standardized ERCP services challenging especially in low middle income countries (LMICs). Hence, expertise and availability of ERCP services confined to limited centers in many LMICs. Successful biliary cannulation is the cornerstone for ERCP. Failure to cannulate common bile duct (CBD) not only lead to an unsuccessful procedure but also increases the risk of ERCP related complications.

Aims & Methods: Aims: Successful biliary cannulation is essential for a successful ERCP. Data regarding success rate of various cannulation techniques is limited from LMIc. The study aims to estimate the successful cannulation rate of various techniques used during ERCP for cannulation and to determine factors predicting difficult biliary cannulation.

Methods: This observational, cross-sectional study was conducted at Aga Khan University Hospital, Pakistan from July 2024 till March 2025. Patients of either gender ≥ 18 years of age who underwent ERCP were included. Difficult biliary cannulation was defined with reference to the ESGE guidelines. Clinical and procedural data was recorded for each patient.

Results: Out of 196 ERCPs performed during study period 133 (67.9%) were with difficult cannulations. Out of 113 cases successful biliary cannulation was done in 112(84.2%). The median no of times cannulation was attempted was 4 (range 1-30) with cannulation time ≤ 5 minutes in 52.6%, 5-10 minutes in 15% and ≥ 11 minutes in 27.8% cases. In majority of cases (94.7%) cases wire guided cannulation technique was used followed by double wire guided technique. No statistically significant difference was found among various cannulation technique. The number of cannulation attempts, and cannulation time were identified as predictors of successful biliary cannulations.

	OR	95% CI	P value
Number of cannulation attempts	1.1	1.1-1.2	< 0.001
Cannulation time	2.5	1.5-4.3	< 0.001
Periampullary Diverticulum	1.3	0.2-6.4	0.70
Types of ampulla	1.16	0.714-1.88	< 0.54
Biliary cannulation technique	0.76	0.13-4.3	< 0.76

Table 1: Factors affecting successful cannulation.

Conclusion: Despite a high rate of difficult biliary cannulation, an overall success rate of 84.2% was achieved. Wire-guided cannulation was most commonly used, with no significant difference among techniques. Cannulation attempts and time were key predictors of success, highlighting the importance of early recognition and strategy adjustment.

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Disclosure: None.

PP0980

WHEN ENDOSCOPIC SPHINCTEROTOMY IS NOT ENOUGH: EFFICACY AND SAFETY OF DILATION-ASSISTED STONE EXTRACTION FOR CHALLENGING BILIARY STONES: A TERTIARY CENTER EXPERIENCE

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Introduction: Difficult biliary stones and complex bile duct anatomy often limit the efficacy of endoscopic sphincterotomy (EST) alone. Dilation-Assisted Stone Extraction (DASE), involving large balloon dilation following EST, has emerged as a complementary technique to improve stone clearance.

This study aimed to evaluate the safety and effectiveness of DASE performed either after primary EST in patients with naïve papilla or as a rescue strategy following prior EST.

Aims & Methods: A retrospective review was conducted on all patients who underwent DASE between January 2018 and December 2024 at the Department of Gastroenterology and Digestive Endoscopy, Mohammed VI University Hospital, Marrakech, Morocco. Inclusion criteria comprised patients with naïve or previously treated papilla undergoing DASE for bile duct clearance. Technical success was defined as successful balloon placement across the papilla, full inflation to target diameter for >30 seconds, followed by complete stone extraction. Complications were categorized as intraprocedural, early (<24 hours), or delayed (>24 hours).

Results: A total of 57 patients were included (mean age 72.5 ± 12.1 years; 30 females [52.6%], 27 males [47.4%]). Thirty-five patients (61.4%) had a naïve papilla, and 22 (38.6%) had prior EST. Main indications for DASE included large or impacted CBD stones (63.2%), periampullary diverticulum (15.8%), difficult cannulation (12.3%), altered anatomy (5.3%), and other causes (3.5%).

Technical success was achieved in 55 patients (96.5%). Thirteen patients (22.8%) experienced complications: 7 (53.8%) intraprocedural, 4 (30.8%) early, and 2 (15.4%) delayed.

Adverse events included:

- Bleeding in 8 patients (61.5% of all complications): 4 self-limited (50%), 3 endoscopically managed (37.5%), and 1 required radiological embolization (12.5%).
- Post-ERCP pancreatitis (PEP) in 4 patients (30.8%): all mild, resolved with conservative management.
- Perforation in 1 patient (7.7%): managed successfully with endoscopic clipping and supportive care.

Univariate analysis revealed that prior EST was significantly associated with a higher complication rate (p = 0.01).

Conclusion: DASE following EST is a highly effective and relatively safe approach for managing difficult CBD stones, with a high technical success rate and acceptable complication profile. The results support its routine use in therapeutic endoscopy, particularly for cases refractory to EST alone. Prior EST may be associated with a higher risk of adverse events and should prompt careful intra- and post-procedural monitoring.

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Disclosure: Nothing to disclose.

PP0981

REDUCING UNNECESSARY ERCP IN BILIARY STONE DISEASE: EFFICACY OF SAME-SESSION FOCUSED EUS IN A LARGE UK TERTIARY CENTER

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is the preferred method for treating common bile duct (CBD) stones. However, in a small number of patients with suspected or confirmed choledocholithiasis on imaging, ERCP results may appear normal. Since the procedure carries considerable risks, such cases should be minimized.
Aims & Methods: This study aimed to analyse the incidence of reported normal ERCPs in cases of biliary stone disease and evaluate the impact of focused EUS in reducing unnecessary procedures.
Data were retrospectively collected from the endoscopy and patient record system, as well as a prospectively maintained database of all ERCP procedures at a UK teaching hospital during a 4.5-year period, from 1/1/2020 to 30/4/2024, with a 30-day additional follow-up period. Variables were assessed by mean with standard deviation (SD) or median with interquartile range (IQR).
Results: There were 3,217 ERCPs performed during the study period, of which 1,636 (50.8%) were performed for biliary stone disease. 124 ERCPs were reported as normal/no stones (7.6%). The most recent imaging was MRCP for 73/124 patients followed by CT for 23/124 and abdominal ultrasound (USS) for 16/124. The median days for imaging-to-ERCP was 8 (4;24). 89/124 patients had a dilated CBD and 11/124 had a confirmed CBD stone on the most recent imaging modality. 90/124 patients had an abnormal ALP and 62/124 had an abnormal bilirubin level. The median days of the most recent liver function tests to ERCP were 2 (1;12).
8/124 patients had a procedural complication. 3 had post-ERCP pancreatitis, 3 a post sphincterotomy bleed, 1 had mild cholangitis and 1 had a small post-sphincterotomy perforation.
From 2020, there has been a gradual increase in the combined single session EUS/ERCP procedures for CBD stones. Figure 1. demonstrates the significant reduction of the reportedly “normal” ERCPs as the number of the combined procedures increased. Around 1/3 of the combined procedures do not proceed to an ERCP.
Conclusion: Current guidelines recommend CBD stone extraction for all patients deemed fit for the procedure. Our findings indicate that performing single-session combined EUS/ERCP in carefully selected cases can reduce unnecessary ERCPs in centres offering both services, thereby minimizing patient risk.
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Disclosure: Nothing to disclose.

PP0982

UTILISATION AND OUTCOME OF CHOLANGIOSCOPY IN PATIENTS WITH PRIMARY SCLEROSING CHOLANGITIS – A EUROPEAN COHORT

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Introduction: Primary sclerosing cholangitis (PSC) is a progressive disease leading to inflammation and fibrosis of the bile ducts with biliary strictures. New strictures or symptoms should trigger the suspicion of cholangiocarcinoma which may develop in more than 20% of patients. Endoscopic retrograde cholangiopancreaticography (ERCP) is useful in stricture management, but standard tissue acquisition with brush cytology provides poor diagnostic accuracy. Cholangioscopy can aid stricture assessment and stone clearance.
This study investigated the technical success, diagnostic yield and complication rate of cholangioscopy in PSC patients in two large tertiary European hospitals.
Aims & Methods: This retrospective cohort included all consecutive PSC patients who underwent ERCP with cholangioscopy between 2014 and 2024. Exclusion criteria were liver transplantation or confirmed cholangiocarcinoma. Charts were reviewed for patient characteristics, procedure characteristics and outcomes.
Results: A total of 84 cholangioscopies were performed in 55 PSC patients (55% male, mean age 63 years). Most patients were of Caucasian ethnicity (68%). Cirrhosis was present in 29% of patients with 54% also having concurrent inflammatory bowel disease.
Most patients (88%) were discussed in a multidisciplinary meeting prior to ERCP. The primary indications for ERCP were stricture assessment (38%), dominant stricture (21%), cholangitis (19%), suspicion of malignancy (11%), stones (10%). The mean serum alkaline phosphatase and bilirubin were 508 IU/Land 60 umol/L, respectively. Most patients received rectal diclofenac (98%) and antibiotics (96.5%). Fifteen patients (18%) had a native ampulla of whom 5 (33%) were considered as difficult cannulation (per ESGE). No precut was performed. All patients with native ampulla underwent sphincterotomy.

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Cholangioscopy was successful in 98% of cases with two failures due to tight stricture and inadequate intraductal view. Electrohydraulic lithotripsy was successful in 76% of cases with a complete stone clearance rate of 82%.

Cholangioscopy-guided biopsies were taken in 44 cases which resulted in histological diagnosis of cholangiocarcinoma in four patients. One patient was diagnosed with cholangiocarcinoma on cholangioscopy-guided biopsies while the cytological brush was non-diagnostic and one patient had cholangiocarcinoma confirmed on both cholangioscopy-guided biopsies and cytological brush. The other two cholangiocarcinoma was diagnosed where only cholangioscopyguided biopsies were done. On the other hand, one patient with normal cholangioscopy-guided biopsies and cytological brush was diagnosed with cholangiocarcinoma on endoscopic ultrasound guided lymph node biopsy.

Seven patients developed a complication (8%): 1 peri-procedural bleeding, 1 hepatic decompensation, 1 gallbladder perforation and pancreatitis, 3 cholangitis and 1 abdominal pain requiring admission. Native papilla was not associated with an increased risk of complications (6.6% vs. 8.7%, p>0.05).

Conclusion: This is one of the largest cohort of cholangioscopies in PSC patients performed in two high volume centre. Cholangioscopy had a high technical success rate. Previous concern that strictures preclude technical success was not seen. Stone clearance using electrohydraulic lithotripsy was also effective.

On the other hand, cholangioscopy-guided biopsies may inform clinical management but more data is needed on the clinical impact of cholangioscopy. Cholangioscopy did not appear to increase the established risk of ERCP in PSC patients.

Disclosure: Nothing to disclose.

PP0983

ENDOSCOPIC BILIARY DRAINAGE IN MALIGNANT OBSTRUCTION FROM NON-HEPATOBILIARY-PANCREATIC CANCERS: OUTCOMES AND PREDICTORS OF FAILURE

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Introduction: Malignant biliary obstruction (MBO) caused by non-hepatobiliary-pancreatic (non-HBP) malignancies is a rare and underexplored indication for endoscopic retrograde cholangiopancreatography (ERCP). These cases often present a clinical dilemma: while ERCP offers a minimally invasive option, its utility remains uncertain in patients with poor prognosis and advanced disease.

Aims & Methods: We aimed to evaluate the effectiveness of ERCP-guided biliary drainage in this context and identify baseline predictors of clinical failure to help refine patient selection.

We conducted a retrospective analysis of all patients who underwent ERCP for malignant biliary obstruction of non-HBP origin between March 2018 and February 2025. Patients lacking essential clinical or follow-up data were excluded. Technical success was defined as adequate stent placement; clinical success as a ≥50% reduction in bilirubin or resolution of cholangitis within 2 weeks. The predictive role of bilirubin level, stricture location, serum albumin, and ECOG performance status was assessed. A ROC analysis evaluated the discriminative ability of pre-drainage bilirubin.

Results: Forty-eight patients were included (mean age: 55.6 years; 56% women). The most common primary cancers were colorectal (31%, n=15), gastric (18%, n=9), breast (14%, n=7), lung (10%, n=5), and ovarian (8%, n=4). Hilar strictures were observed in 54% (n=26) and mid-to-distal strictures in 46% (n=22). All procedures were performed using plastic stents. In most cases involving hilar strictures, long stents (e.g., 7 Fr / 12–15 cm) were used, and a single biliary sector—usually the most dilated or invaded—was preferentially drained. Technical success was achieved in 85.4% of cases. Clinical success occurred in 54.1% of patients, while early mortality (≤30 days) reached 10.4%, exclusively among individuals with ECOG status 4 and serum albumin <2.8 g/dL. Hilar strictures and total bilirubin levels >200 mg/L were significantly associated with lower clinical success rates (38% vs 65%, p=0.03 and 33% vs 62%, p=0.04, respectively). ROC analysis showed an AUC of 0.72 for bilirubin in predicting failure. Technical failures (14.6%) were mainly due to anatomical distortion or papillary inaccessibility.

Conclusion: Endoscopic biliary drainage in this population is technically achievable but offers limited clinical benefit, especially in cases with hilar involvement, severe hyperbilirubinemia, or poor overall condition. These findings emphasize the importance of using simple, baseline clinical indicators to guide treatment decisions, avoid unnecessary interventions, and adapt care strategies to the realities of resource-limited settings.

Disclosure: I have no conflict of interest.

PP0984

CBD CLEARANCE USING ENDOSCOPIC PAPILLARY LARGE BALLOON DILATATION (EPLBD) IN PATIENTS WITH DIFFICULT COMMON BILE DUCT(CBD) STONES: A DECADE'S EXPERIENCE OF A TERTIARY CARE CENTRE (CLEAR STUDY)

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Introduction: Endoscopic papillary large balloon dilatation (EPLBD) is an established technique for managing difficult common bile duct (CBD) stones. This study evaluates the efficacy and safety of EPLBD in a large cohort from a tertiary care centre.

Aims & Methods: Patients with difficult CBD stones (N = 559) who underwent EPLBD between January 2015 and December 2024 were retrospectively analysed from a prospectively maintained database. The primary outcome was complete CBD clearance. Secondary outcomes included complication rates, number of sessions required, and need for mechanical lithotripsy or surgical intervention. We also identified predictive factors for failed clearance and complications.

Results: The cohort had a mean age of 47.8 ± 14.6 years; 75% were female. The mean diameter of the largest CBD stone was 15.1 ± 4.2 mm, with an average of 2.16 ± 1.7 stones per patient. Mean CBD diameter was 15.1 ± 3.1 mm, and the average balloon size used was 14.8 ± 1.7 mm. Complete clearance was achieved in 84.9% (475/559) of the patients in a single session and in 89.9% (503/559) of the patients overall. Mechanical lithotripsy achieved additional clearance in only 11(2.5%) patients. Complications occurred in 34(6.1%) patients: perforation (n=18,3.2%), bleeding (n=15,2.7%), and pancreatitis (n=1,0.2%). Procedure-related mortality was observed in three (0.5%) patients (perforation, n=2; bleeding, n=1). Regression analysis identified larger stone size as a predictor of clearance failure (OR: 1.10; 95% CI: 1.03–1.17; p = 0.004) and need for repeat intervention as a predictor of complications (OR: 4.49; 95% CI: 1.68–11.99; p = 0.003).

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Conclusion: EPLBD is a safe and effective approach for the management of difficult CBD stones. It offers high success rates with acceptable complication rates, making it a valuable one-stop management for difficult CBD stones, particularly in resource-constrained settings.
Disclosure: Nothing to disclose.

PP0985

INFLUENCE OF PAPILLARY MORPHOLOGY ON CANNULATION DIFFICULTY AND COMPLICATIONS DURING ERCP: A COHORT FROM A HIGH-COMPLEXITY CENTER IN LATIN AMERICA

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Introduction: Cannulation of the papilla of Vater is a pivotal step in endoscopic retrograde cholangiopancreatography (ERCP), directly impacting technical success and complication risk. (1) While factors such as endoscopist experience and anatomical variants are known to influence outcomes, (2) the role of papillary morphology remains understudied, particularly in Latin America. Morphology-based risk stratification may aid in anticipating the need for advanced cannulation techniques or expert-level intervention.
Aims & Methods: We prospectively enrolled adults with an intact, naïve major duodenal papilla undergoing ERCP at a high-complexity institution in Bogotá, Colombia. Patients with surgically altered anatomy or periampullary tumors were excluded. Papilla morphology was classified macroscopically according to Haraldsson et al. (3) Difficult cannulation was defined by ESGE criteria: >5 papilla contacts, >5 minutes to cannulate after visualizing the papilla, or ≥2 unintentional pancreatic duct guidewire passes. We recorded clinical and procedural variables, including use of advanced cannulation techniques and complications. Associations were assessed through bivariate comparisons using Fisher's exact test and odds ratios with 95% confidence intervals ($p < 0.05$).
Results: 58 patients were included (mean age 59.9 years; 63.8% female). The most frequent indication for ERCP was choledocholithiasis (81.0%). Papilla types were: type 1 (63.8%), type 3 (27.6%), type 2 (6.9%), and type 4 (1.7%). Difficult cannulation occurred in 8 cases (13.8%), with overlapping criteria: >5 contacts (n=5), >5 minutes (n=3), and ≥2 pancreatic guidewire passes (n=3). Difficult cannulation was more frequent in type 1 (13.5%) and type 3 (12.5%) papillae and occurred in the single type 4 case; none in type 2. The mean number of contacts with the papilla was 2.4 (range 0–16). Use of advanced cannulation techniques was required in 20 cases (34.5%). Cannulation time was significantly longer when advanced techniques were required, with a median of 21.2 minutes and a mean of 27.6 minutes, compared to 0.5 and 0.9 minutes, respectively, in cases managed conventionally. Post-ERCP prophylaxis measures were used in 35 patients (60.3%), including diclofenac (37.9%), aggressive hydration (32.8%), and pancreatic stenting (27.6%). Three patients (5.2%) received all three preventive measures. Complications included pancreatitis (n = 4, 6.9%) and cholangitis (n = 2, 3.4%). Complication rates did not significantly differ according to papilla type ($p = 0.68$). No significant difference in pancreatitis incidence was observed between patients who did and did not receive prophylactic measures ($p = 1.0$; OR 0.64, 95% CI 0.04–9.48). The incidence of post-ERCP pancreatitis was 8.3% in type 1 and 25.0% in type 2 papillae, with no cases observed in type 3 or type 4 papillae. However, the association between papilla morphology and pancreatitis occurrence was not statistically significant ($p = 0.268$).
Conclusion: Type 1 and type 3 papillae were associated with greater cannulation difficulty, though complications did not differ significantly by morphology. These findings support a growing body of evidence linking papillary type to ERCP complexity. While limited by sample size and the absence of endoscopist stratification, the prospective design, standardized use of ESGE criteria, and regional focus contribute valuable insight into procedural risk.
This study provides a foundation for considering papilla morphology in preprocedural planning and for guiding training priorities in high-complexity endoscopy units.
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Disclosure: Nothing to disclose.

PP0986

ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY PERFORMED IN THE ENDOSCOPY SUITE IS ASSOCIATED WITH REDUCED IMMEDIATE ADVERSE EVENTS AND TECHNICAL FAILURE RATES

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is usually performed in the endoscopy suite which is designed to accommodate staff, endoscopy stack, anesthetic and imaging machines. ERCP in locations outside the endoscopy suite may pose a challenge for the endoscopist in ergonomics and equipment access.
The aim of this study is to compare the technical success and adverse events between ERCP performance in endoscopy suite and non-endoscopy suite locations.
Aims & Methods: Retrospective analysis of 24,122 ERCP procedures in a national sample of the United States over 12 years using the Clinical Outcomes Research Initiative (CORI)/National Endoscopic Database (NED). All patients with ERCP and a description of procedure location (endoscopy suite, radiology, operating room and others) were included. Demographics, patient position, patient tolerance, procedure and fluoroscopy time, visualization and cannulation of main structures, ERCP adverse events, and technical success rates were compared between ERCP endoscopy suite and non-endoscopy suite locations. Categorical variables were compared using Pearson's chi-square test and continuous variables using a standard t-test. Univariate and multivariate regressions were performed adjusting for age, gender, ethnicity, ASA class and clinical setting.
Results: A total number of 11,974 (49.6%) procedures were performed in the endoscopy suite and 12,148 (50.4%) in non-endoscopy suite locations [radiology, 10,435 (43.3%), operating room, 1518 (6.3%) and Others, 195 (0.8%)]. The mean patient age was 57 years, 52% were male. ERCP was performed in the supine position in 12.6% of endo vs. 4.1% of non-endo

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locations (p<0.001). Patient tolerance was excellent in 39.4% of endo vs 33.9% of non-endo locations (p<0.001). The ampulla was visualized in 97.6% of endo vs. 97.6% of non-endo locations (p = 0.18). The common bile duct (CBD) was visualized and cannulated in 91.4% of endo vs. 90.8% of non-endo locations (p<0.001). The ERCP was incomplete in 5.8% of cases in endo vs. 6% of non-endo locations (p=0.529).

Technical success (aOR: 1.81, p=0.02) was more likely and ERCP adverse events (aOR: 0.47, p = 0.001) less likely to occur in endo-suite. Total procedure time (aOR: 1.00, p=0.67) and fluoroscopy time (aOR: 1.01, p=0.342) were similar between both groups.

Conclusion: ERCP procedures performed in the endoscopy suite is associated with reduced immediate adverse events and technical failure rates. This may reflect the technical difficulty of performing ERCP in non-endoscopy suite locations for the endoscopist, possibly related to ergonomics and equipment access. Our study supports performing ERCP in the endoscopy suite where possible to optimize ERCP safety and success.

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Disclosure: Nothing to disclose.

PP0987

FACTORS ASSOCIATED WITH THE SUCCESS OF ENDOSCOPIC TREATMENT FOR LARGE STONES AND CHOLEDOCHAL LITHIASIS: FINDINGS FROM A 19-YEAR STUDY

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Introduction: Endoscopic biliary sphincterotomy is an effective treatment for common bile duct stones. However, its success can be compromised by the presence of large stones or extensive choledochal lithiasis.

The aim of this study was to evaluate the effectiveness of endoscopic retrograde cholangiopancreatography (ERCP) in patients with large obstructive stones (>15 mm) and those with one or two stones or choledochal lithiasis, and to identify factors influencing the success of endoscopic drainage and associated complications in the management of large bile duct stones.

Aims & Methods: This was a retrospective, descriptive, and analytical study conducted over a 19-year period, from April 2006 to February 2025, in our department.

A total of 1,011 patients who underwent ERCP for common bile duct stones were included. Choledochal lithiasis was defined as the presence of more than three stones, and a large stone was defined as an obstructive stone larger than 15 mm. Patients were divided into two groups:

• Group I (n = 143): Patients with a single obstructive stone larger than 15 mm.

• Group II (n = 868): Patients with one or two stones or choledochal lithiasis.

Results: Patients with large stones (Group I) represented 14.1% (n = 143) of the cohort, while Group II included 868 patients (85.9%).

The success rate after a single catheterization was 55.2% in Group I compared to 81% in Group II (p < 0.001). Reintervention was required in 14.7% of Group I patients versus 8% in Group II (p = 0.009). Additional procedures were performed in 46.2% of cases in Group I versus 16.1% in Group II.

The overall success rate after reintervention and/or additional procedures was 88.7% in Group I versus 92.5% in Group II (p = 0.125).

Early complication rates were higher in Group I (10.5%) compared to Group II (5.1%), with a statistically significant difference between groups (p = 0.017).

Univariate analysis identified the following factors as significantly decreasing the overall success rate of endoscopic treatment: older age, acute cholangitis, bile duct stricture, and significant bile duct dilation (diameter >15 mm).

Multivariate analysis, adjusted for age, sex, surgical history, acute pancreatitis, acute cholangitis, periampullary diverticulum, bile duct diameter, and bile duct stricture, showed that only acute cholangitis (OR: 0.295; 95% CI: [0.164 – 0.532]; p < 0.001) and bile duct stricture (OR: 0.53; 95% CI: [0.922 – 1.051]; p = 0.001) were independently associated with a decreased overall success rate of endoscopic treatment.

Conclusion: Our study found no statistically significant difference in the effectiveness of endoscopic treatment between patients with large stones and those with simple lithiasis. However, the presence of acute cholangitis and bile duct strictures appears to be associated with a reduced overall success rate of endoscopic management.

Disclosure: Nothing to disclose.

PP0988

ARISE TRIAL FINDINGS ON ANTIMICROBIAL RESISTANCE IN ACUTE CHOLANGITIS IN ROMANIA

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Introduction: The global trend with increasing incidence of multidrug resistant (MDR) pathogens represents one of the most important medical emergencies. Acute cholangitis is frequently complicated by resistant biliary infections, mostly from the GI tract, chalenging clinical management and outcomes [1].

We aimed to characterize the antimicrobial resistance pattern in a multicentric study, and to identify risk factors for MDR pathogens.

Aims & Methods: The ARISE trial currently takes place in three tertiary endoscopy units from Romania. As of August 2024, a total of 498 adult patients with acute cholangitis were included after disease confirmation by clinical, laboratory, and imaging findings described in the Tokyo guidelines. https://clinicaltrials.gov/study/NCT06197984

Results: Overal, 60.1% of biliary cultures were positive, with 35.2% of them being polymicrobial. Blood cultures were obtained in 76.9% of patients, of which 11.5% were positive. The three most frequently isolated organisms included Escherichia coli (41.4%), Klebsiella pneumoniae (22.6%), and Pseudomonas aeruginosa (16.5%). Among isolated strains, 28.0% demonstrated at least one significant resistance, with extended-spectrum beta-lactamase (ESBL) production (24.9%), methicilin-resistant Staphylococcus aureus (MRSA) (10.6%), and vancomycin resistant enterococci (VRE) (8.8%) topping the resistance patterns. MDR, defined as resistance to three or more antibiotic classes, was detected in 17.4% of positive cul-

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tures. Patients with MDR pathogens were older (mean age 71.2 vs. 63.8 years, p<0.01) and more likely to have diabetes melitus (34.2% vs. 18.6%, p<0.05). Tokyo severity grade II (29.5%) or III (19.3%) was significantly associated with MDR involvement (22.0% vs. 9.7%, p<0.01). Prolonged prior antibiotic exposure (over 48 hours) was independently predictive of MDR growth (adjusted odds ratio 2.4, 95% CI 1.6–3.6). In-hospital mortality was higher in patients with MDR isolates (15.8% vs. 4.4%, p<0.01), underscoring the clinical impact of resistance.

Conclusion: Antibiotic resistance in acute cholangitis remains substantial, especially among advanced Tokyo severity cases. Recognizing risk factors, including older age, prior antibiotic use, and severe disease, can guide empiric treatment strategies to mitigate the burden of MDR pathogens and improve outcomes.

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Disclosure: Nothing to disclose.

PP0989

ERCP OUTCOMES AFTER ORTHOTOPIC LIVER TRANSPLANTATION: A RETROSPECTIVE ANALYSIS OF A PROSPECTIVE DATABASE

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Introduction: Biliary complications are well known complications of orthotopic liver transplantation (OLT) and are increasingly recognized in clinical practice, moreso due to the increased volume of liver transplantations performed annually in the United States. Endoscopic retrograde cholangiography (ERCP) has proven to be an effective alternative to more invasive surgical interventions for post-OLT biliary complications.

The aim of this study is to examine the technical performance of ERCP in the liver transplant population presenting with biliary complications post-OLT.

Aims & Methods: Retrospective analysis of prospectively collected clinical data of patients undergoing ERCP in a single large tertiary/academic referral center which is also a liver transplant center. Patients who underwent ERCP for transplant and non-transplant indications between January 2019 and July 2023 were included. The primary outcome was determining ERCP technical success among patients who underwent ERCP post-OLT. Technical success was defined as successful biliary cannulation and achievement of therapeutic goal. Secondary outcomes examined differences in technical success, fluoroscopy time and procedure time based on ASGE-grading system of procedural complexity.

Results: Of a total number of 1289 patients included in the study, 322 (25%) underwent ERCP post-OLT in the study period. The mean age of post-OLT patients was 59.1 years, and 60.9% were males. The ASGE procedural difficulty grading was Grade 1 (63.7%), Grade 2 (23.6%), Grade 3 (8%) and Grade 4 (4.7%) in OLT. Moreover, 39.8% had a native papilla and a spyglass was used in 10.6% of OLT cases. The indications for ERCP post-OLT were biliary stricture (47.5%), abnormal LFTs (20.2%), biliary stone (19%), biliary leak (9.3%), and ischemic cholangiopathy (4%). Technical success was achieved in 98.1% of cases. The average fluoroscopy and procedure times were 4.5 mins and 24.7 mins respectively. There was no difference in technical success based on ASGE grading system, however, fluoroscopy time (-6.1 mins) and procedure time (-27.0 mins) were significantly shorter in Grade 1-2 versus Grade 3-4 respectively.

Conclusion: ERCP is technically successful for biliary complications post-OLT. ASGE-grading system for procedural complexity was a significant predictor of fluoroscopy and procedure durations, but not of technical success.

Disclosure: Nothing to disclose.

PP0990

ENDOSCOPIC MANAGEMENT OF COMPLICATIONS FROM RUPTURED HEPATIC HYDATID CYSTS INTO THE BILIARY TRACT

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Introduction: Hepatic hydatid cyst (HHC) is a parasitic disease caused by the development of the tapeworm Echinococcus granulosus. Fistulization into the biliary tract is one of its most common complications. Endoscopic treatment is preferred due to the risks associated with surgery.

This study aims to evaluate the role of endoscopic retrograde cholangiopancreatography (ERCP) in the diagnostic and therapeutic management of HHC ruptured into the biliary tract.

Aims & Methods: This is a retrospective, descriptive, and analytical study conducted from January 2002 to August 2021, including 50 patients with HHC fistulized into the biliary tract. All patients underwent ERCP combined with endoscopic biliary sphincterotomy. Overall success was defined as complete clearance of the main bile duct. Statistical analysis was performed using SPSS version 24.0.

Results: Among all ERCPs performed during the study period, 4.6% (n=50) were for a hepatic hydatid cyst communicating with the biliary tract. The mean age of patients was 46.1 ± 14.8 years, with a male predominance (66%; male-to-female ratio = 1.94).

ERCP was performed preoperatively in 52.2% of cases and postoperatively in 47.8%.

The main indications were acute cholangitis (44.9%) and persistent external biliary fistula (34%).

The median diameter of the common bile duct was 10 mm [7–14], and the median cyst diameter was 35 mm [27–47].

Sphincterotomy was performed in 96% of cases, allowing hydatid material extraction using a balloon or Dormia basket in 87.8% of cases. A nasobiliary drain was required in 24% of patients, and biliary stent placement was performed in 8%.

The overall success rate was 96% (n=46). Immediate complications occurred in 8% (n=4): one case of hemobilia and three cases of marginal bleeding.

Clinical improvement included jaundice resolution within 5 to 10 days and closure of the external biliary fistula within 10 days.

Conclusion: Our study confirms that endoscopic management of ruptured hepatic hydatid cysts into the biliary tract is an effective therapeutic alternative, with a low rate of immediate complications (8%) and favorable long-term outcomes.

Disclosure: Nothing to disclose.

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PP0991

REVISITING CRP THRESHOLDS IN TUMOR-RELATED CHOLANGITIS: TOWARD A CONTEXT-SPECIFIC DIAGNOSTIC APPROACH

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Introduction: The Tokyo Guidelines recommend a CRP cutoff >10 mg/L for diagnosing acute cholangitis. However, this threshold was defined in lithiasic populations and may not be applicable to malignant biliary obstruction, where CRP elevation often reflects tumor-driven inflammation rather than true infection. Applying this universal threshold in oncologic settings risks antibiotic overuse and promotes resistance.

Aims & Methods: This study aimed to reassess the diagnostic value of the current CRP cutoff in tumor-related cholangitis and to identify a more appropriate threshold to enhance specificity and guide targeted antibiotic use.

We conducted a retrospective monocentric study of consecutive patients at a tertiary center between May 2016 and October 2024. Among 571 patients undergoing ERCP, 268 patients with confirmed malignant biliary obstruction and available bile sampling were included. Bacteriological status was defined as positive when either direct bile examination or culture isolated microorganisms. CRP levels, leukocyte counts (hyperleukocytosis >10,000/mm^3), and tumor etiology were recorded. ROC curve analysis was performed to determine the optimal CRP cutoff. Comparative analyses employed Mann-Whitney U, chi-square, and multivariate logistic regression.

Results: The mean age was 65.9 ± 12.4 years, with an equal sex distribution (50% male). Tumor etiologies included pancreatic cancer (56%), cholangiocarcinoma (22%) — 40 hilar and 18 distal cases, gallbladder cancer (13%), ampulloma (7%), and extrinsic or secondary compression (3%). The median CRP was significantly higher in culture-positive patients (47 mg/L [IQR 21–98]) compared to culture-sterile patients (22 mg/L [IQR 11–36]; p<0.001). ROC analysis identified an optimal CRP cutoff of 21 mg/L (AUC = 0.64), achieving a sensitivity of 67% and specificity of 54%. Hyperleukocytosis was present in 66% of culture-positive cases versus 31% of sterile cases (p<0.001).

In multivariate analysis, CRP ≥21 mg/L (OR 2.1; 95% CI [1.3–3.4]; p=0.002) and hyperleukocytosis (OR 3.4; 95% CI [1.9–5.9]; p<0.001) were independently associated with bile bacteriological positivity.

Conclusion: In malignant biliary obstruction, the universal CRP threshold of 10 mg/L lacks specificity and risks antibiotic overuse. A higher threshold around 21–25 mg/L, combined with leukocyte count assessment, improves diagnostic precision. Our findings advocate for a context-specific adaptation of the Tokyo Guidelines to better differentiate tumor-related inflammation from true infection, ultimately enhancing patient safety and antimicrobial stewardship in oncologic care.

Disclosure: Nothing to disclose.

PP0992

FACTORS INFLUENCING INCOMPLETE ERCP PROCEDURES: A SIX-MONTH RETROSPECTIVE ANALYSIS

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Introduction: According to the British Society of Gastroenterology (BSG) guidelines on sedation in gastrointestinal endoscopy (2023), minimal to moderate sedation is typically sufficient for level 1/2 endoscopic retrograde cholangiopancreatography (ERCP). However, deep sedation is not routinely available in many centres, potentially affecting patient comfort and procedure completion.

This study investigated factors associated with incomplete ERCP procedures, with a particular focus on patient intolerance.

Aims & Methods: A retrospective review of all ERCP's performed over six months at our tertiary centre was conducted. Data collected included demographics, indication, ERCP grade, sedation used, cannulation rate and completion of procedure (eg complete duct clearance). Achieving the primary therapeutic or diagnostic goal served as the primary outcome. Univariate and multivariable analyses were used to evaluate factors contributing to incomplete procedures.

Results: 475 ERCP's were analysed (median age 67 years; range 18–92; 49.5% female). Successful cannulation was achieved in 458 of 475 procedures (96.4%). Among the 55/475 (11.6%) incomplete ERCPs, 32 (6.7% of all procedures) were abandoned due to patient discomfort—manifested as pain, restlessness, and inability to tolerate the procedure. Univariate analysis showed significantly higher opiate doses (63.8 mg vs. 56.3 mg, p=0.015) and higher midazolam doses (5.0 mg vs. 4.0 mg, p=0.0003) in incomplete procedures.

No significant differences were observed for age, sex, inpatient status, ERCP grade, or use of general anaesthesia. Multivariable analysis revealed no independent risk factors for incompletion. Other contributing factors included prolonged procedures (e.g., EUS prior to ERCP) and challenging anatomy.

Conclusion: Patient discomfort emerged as the primary driver of incomplete ERCPs, highlighting the importance of optimizing sedation strategies and ensuring adequate anaesthetic support. Higher opiate and midazolam doses correlated with incomplete procedures suggesting that deeper sedation may have allowed completion. More detailed research including data from a combination of tertiary and non-tertiary centres is needed to refine sedation practices in line with current BSG guidelines.

Disclosure: Nothing to disclose.

PP0993

OUTPATIENT ERCP: EVALUATING PRECUT TECHNIQUES VERSUS STANDARD SPHINCTEROTOMY FOR BILIARY ACCESS

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Introduction: Precut techniques provide an alternative approach for biliary cannulation during ERCP. While traditionally considered high-risk and typically reserved for high-volume centers, recent evidence supports the safety and efficacy of early precut techniques when performed by experi-

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enced endoscopists. However, data on the safety of precut techniques in the outpatient setting remain limited, particularly in comparison to standard sphincterotomy.

Aims & Methods: We conducted a retrospective analysis of all outpatient ERCP procedures performed at our institution between 2020 and 2024. A total of 304 cases were reviewed. Patients were divided into two groups: Group A, who underwent standard sphincterotomy, and Group B, who required precut techniques following failed standard cannulation attempts.

Results: A total of 159 outpatient ERCPs involving naïve papillae were included: 122 in Group A and 37 in Group B. There was no significant difference in baseline characteristics and indications for ERCP between the 2 groups. Overall cannulation success rate was 97.5% (n=155/159), reaching 89.2% (n=33/37) in the pre-cut group. There was an overall post-procedure complication rate of 6.9% (n=11/159): 7.4% (n=9/122) of patients in Group A and 5.4% (2/37) in Group B, with no significant difference between groups (*p= 0.860*).

Conclusion: In our outpatient ERCP cohort, precut techniques demonstrated a favorable safety profile and comparable outcomes to standard sphincterotomy. These findings support the selective use of precut methods in the outpatient setting.

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Disclosure: Nothing to disclose.

PP0994

HYBRID ENDOSCOPIC PAPILLECTOMY FOR DUODENAL PAPILLARY TUMOR

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Introduction: Endoscopic papillectomy (EP) is indicated for papillary adenoma that do not extend into the pancreatic bile duct. There are cases in which en bloc resection or curative resection is expected to be difficult with conventional EP. For such cases, we introduced hybrid EP, in which snaring is performed after peri-lesion incision using a dedicated ESD knife.

Aims & Methods: A total of 13 cases who underwent hybrid EP between 2022 and 2024 were identified from our prospectively maintained database. Hybrid EP involved a circumferential incision with partial submucosal dissection, and subsequent snare resection of ampullary tumors, which was performed by two endoscopists with expertise in ESD or endoscopic retrograde cholangiopancreatography. Demographic data and clinicopathological outcomes were retrospectively evaluated.

Results: En bloc resection was achieved by hybrid EP in all 13cases, with the median procedure time of 118 (range: 65-170) minutes. The median diameters of the resected specimens and tumors were 25 and 15 mm, respectively. All lateral margins were clear, whereas vertical margin was uncertain in one (7.6%), resulting in the complete resection rate of 92.4%. Postoperative bleeding developed in two (15.3%). No tumor recurrence was observed even in that case with uncertain vertical margin, after a median follow-up of 244 (range, 97-678) days.

Conclusion: Hybrid EP is an useful technique for duodenal papillary tumors that are difficult to resect en bloc or curatively by conventional EP.

Disclosure: Nothing to disclose.

PP0995

DOES RACE MATTER? IDENTIFYING DISPARITIES AMONGST PATIENTS WITH MASLD UNDERGOING ERCP: A STUDY OF OVER TWO MILLION PATIENTS

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Introduction: Metabolic dysfunction-associated steatotic liver disease (MASLD) is caused by increased fat accumulation within hepatocytes. It is primarily associated with Type II Diabetes, obesity, and metabolic syndrome. MASLD has become increasingly prevalent worldwide over the past several decades and is regarded as the most common liver disease in industrialized countries. Our study aimed to identify if MASLD would increase mortality amongst patients undergoing ERCP and to identify positive predictors that lead to increased healthcare utilization.

Aims & Methods: A retrospective analysis was completed utilizing the National Inpatient Sample (2015-2020). Patients were identified based on ICD-10 codes corresponding to those with MASLD undergoing ERCP vs. patients without MASLD undergoing ERCP. Primary outcomes were all-cause mortality and healthcare utilization (mean length of stay (LOS) and total hospital charge (TOTCHG). Secondary outcomes such as the effect of the race and post-procedural complications (sepsis, CVA, Bleeding, AKI, post-procedural bleeding) were analyzed using multivariate analysis.

Results: We identified 2,681,575 total patients with MASLD, of whom 1,104,200 underwent ERCP. After propensity score matching, surprisingly, patients with MASLD undergoing ERCP had decreased mortality (OR .45, 95% CI [0.27- 0.75]). However, LOS and TOTHCG were not statistically significant. Multivariate regression revealed the impact of complications on the primary and secondary outcomes
(listed in Tables 1 and 2 https://media-hosting.imagekit.io/6e3b-9e79c5d24205/ueg2025.png?Expires=1840250284&Key-Pair-Id=K2ZIVPTIP2VGHC&Signature=VEuGD9QcoSy6tVYrHx0H1kKNixXM~yp-pVVTHAz0dY8gj52jh3K1Zc4reiU90cbS1J6HGKn4Uc8gjcJ52x4UpG-2lib-QVI5UlbXdmI91YjAIVtueBaGfCuMpcpth-hJSWWfl4SInfj8Y-Pohy-IA0-SfPe7Gk9PRfeUsy5lJzf36dD1xcwzLUCPfsS10zKeyFN4vEJomZyg-32NQT7Wx8-P43IQgXvHJz0J5WXXD3IyjpTgfoGxJLVhE5ADXUbXaNWuO-eYV5nG2Fb5FvG3HLEzL2q7q0No2xWz3bCmnBC1vsji2zmbkBrRFOEqiaEL-hCL~Z7Ul86Ns9clhnFGpxyl2Iaw__).
Native Americans were found to have a significant increase in mortality (OR 2.47, 95% CI, p-value <0.05). It was also shown that the presence of post-procedural bleeding, AKI, CVA, and sepsis increased mortality, TOTHC, and LOS in patients with MASLD undergoing ERCP.

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Conclusion: Our study identified positive predictors (sepsis, post-procedural bleeding, CVA, or AKI) for increased LOS, Mortality, and TOTHC in patients with MASLD undergoing ERCP. In addition, high rates of obesity, cardiovascular disease, and Type II Diabetes are more prevalent compared to other racial groups in the US. Further studies are needed to identify whether lack of access to healthcare within the country contributes to disease burden and worsened outcomes in this population.
Disclosure: Nothing to disclose.

PP0996

COMPARISON OF THE EFFICACY AND SAFETY OF METAL STENTS WITH DIFFERENT AXIAL FORCES IN EUS-GUIDED HEPATICOGASTROSTOMY: A MULTICENTRE RETROSPECTIVE STUDY

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Introduction: Endoscopic ultrasound-guided hepaticogastrostomy (EUS-HGS) has been performed as an alternative treatment when endoscopic retrograde cholangiopancreatography (ERCP) fails. However, EUS-HGS carries the risk of adverse events (AEs) such as peritonitis and recurrent biliary obstruction (RBO). To lower the risk of AEs, guidelines recommend the use of covered metallic stents (MS) for EUS-HGS. Several studies suggest that MS with high axial force (AF) may lead to a higher incidence of AEs such as pancreatitis and cholecystitis when MS is placed across the major duodenal papilla during ERCP. Studies comparing MS with high and low AF in EUS-HGS are scarce. MS with low AF may lower the incidence of AE in EUS-HGS. This study aimed to compare the efficacy and safety of MS with low AF (EGIS™ biliary (S&G Biotech)) and high AF (Niti-S™ biliary S-type (Taewoong Medical)).
Aims & Methods: We retrospectively reviewed consecutive patients with unresectable malignant biliary obstruction who underwent EUS-HGS with either EGIS or Niti-S across seven institutions in Japan from January 2019 to December 2022. Patients who had undergone prior biliary drainage, such as percutaneous transhepatic biliary drainage, endoscopic nasobiliary drainage, deployment of metal or plastic stents during ERCP, or EUS-HGS using a plastic stent were excluded. A matched cohort of patients was obtained using 1-to-2 propensity score matching: history of cholecystectomy, performance status, ascites, and antegrade stenting. Outcome measures included technical and clinical success, early and late AEs, time to RBO (TRBO), and overall survival (OS). Technical success, clinical success, TRBO, and OS were defined according to the Tokyo criteria 2024. AEs associated with EUS-HGS procedures were evaluated according to the adverse events in gastrointestinal endoscopy (AGREE) classification. Peritonitis was diagnosed when abdominal rebound tenderness and fluid collection on computed tomography (CT) were observed.
Results: Among 162 patients, 123 patients were eligible for the study (39 for Niti-S and 84 for EGIS). After balancing the two groups, the matched cohorts included 27 patients in the Niti-S group and 54 patients in the EGIS group. There were no significant differences in technical success rate (100% vs. 100%, p = 1.00) and clinical success rate (88.9% vs. 98.1%, p = 0.11). The RBO rates were 22.2% and 29.6% (p = 0.60). The median TRBO was 85 days for the Niti-S group and was 213 days for the EGIS group (p = 0.37), and there was no significant difference in median OS (75 vs. 112 days, p = 0.21). Early AE occurred in 10 patients (37.0%) in the Niti-S group and 8 patients (14.8%) in the EGIS group (p = 0.045). In the Niti-S group, there were 8 cases of peritonitis, 4 cases of stent migration, 2 cases of sepsis, 1 case of bleeding, and 1 case of biloma. In the EGIS group, there were 3 cases of peritonitis, 1 case of stent migration, 2 cases of cholangitis, 1 case of pancreatitis, 1 case of sepsis, and 1 case of biloma. Both groups contained cases in which multiple adverse events occurred in a single patient. Late AE were observed in 2 patients (7.4%) in the Niti-S group and 4 patients (7.4%) in the EGIS group (p = 1.00). In the Niti-S group, there were 2 cases of stent migration into the oesophagus. In the EGIS group, there was 1 case of pseudoaneurysm, 2 cases of liver abscess, and 1 case of cholecystitis.
Conclusion: The use of MS with low AF may help reduce the incidence of early AE in patients undergoing EUS-HGS.
Disclosure: Nothing to disclose.

PP0997

ENDOSCOPIC ULTRASOUND-GUIDED GALLBLADDER DRAINAGE. WHICH IS THE BEST WAY TO ACCESS?

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Introduction: Endoscopic ultrasound guided gallbladder drainage (EUS-GBD) with lumen apposing metal stent (LAMS) has become a favourite drainage option for high surgical-risk patients with acute cholecystitis and as rescue treatment for the relief of jaundice in patients with distal malignant biliary obstruction (DMBO) with high technical and clinical success. Both trans-gastric and trans-duodenal drainage are described but which is the best way is still controversial.
Aims & Methods: We aim to compare the two different approaches in term of technical success, rate of adverse events, unplanned readmissions and re-interventions.
Single center retrospective analysis of all consecutive cases of EUS-GBD with LAMS acceding to the Therapeutic Endoscopy Unit of Fondazione Policlinico Universitario Campus Bio-Medico of Rome. Data about LAMS features and procedure were collected.
Results: From January 2019 to September 2024 sixty patients were enrolled (50% male; 77±12 years old). The 60% of them underwent EUS-GBD for acute cholecistitis. In 31/60 (52%) patients the trans-gastric access was performed. In 43/60 (71.7%) patients a Hot-Axios LAMS was used and in the majority of procedure (47/60 78.3%) a calibre bigger than 10 mm was used. Overall technical success was obtain in 98.3% of procedures.
In no case the LAMS was dilated. A naso-cistic tube was placed inside the LAMS in 31/60 patients. Double pig-tail plastic stent was placed in two patients. Regarding the adverse events rate, in one patient distal flange misdeployment occurred. LAMS obstruction occurred in 7/60 patients (11.7%) and need of reintervention in 10/60 (16.7%).
There was no difference in term of drainage indication, technical success rate, type and size of LAMS used, naso-cystic tube or pig-tail stent inser-

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tion between the two different EUS-GBD access. However, the trans-gastric access was associated with a higher rate of LAMS obstruction (6/7 vs 1/7 p 0.05) and need of re-intervention (8/10 vs 2/10 p 0.03).

Conclusion: EUS-GBD is a safe technique with high technical success rate. Both trans-gastric and trans-duodenal approaches are effective. However, EUS-GBD with trans-gastric access seems to be associated with a higher rate of late adverse events and when possible another way should be preferred.

Disclosure: Nothing to disclose.

PP0998

ENDOSCOPIC ULTRASOUND GUIDED BILIARY DRAINAGE FOR MALIGNANT BILIARY OBSTRUCTION: FINDING THE BEST ROUTE

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Introduction: Endoscopic ultrasound-guided biliary drainage (EUS-BD) has emerged as a minimally invasive and promising alternative to percutaneous transhepatic biliary drainage (PTBD) and endoscopic retrograde cholangiopancreatography (ERCP) in managing malignant biliary obstruction. Different techniques are available such as choledocoduodenostomy (CDS), gallbladder drainage (GBD) and hepaticogastrostomy HGS). All have been shown to be safe and effective techniques, but which is the most suitable and effective approach is still unknown.

Aims & Methods: We aim to compare the three techniques available in terms of technical success and clinical success, and to assess the adverse events related to the different procedures. Single center retrospective analysis of all consecutive cases of CDS, GBD and HGS acceding to the Therapeutic Endoscopy Unit of Fondazione Policlinico Universitario Campus Bio-Medico of Rome. Data about patients and procedures were collected. Technical success was defined as successful biliary drainage with stent placement. Clinical success was defined as a decrease in total bilirubin level by ⩾30% within 7 days of EUS-BD.

Results: From April 2016 to December 2024 101 patients were included (47.5% male, mean age 74.6 years old), all with malignant biliary obstruction. 52,5% of the subjects (53/101) underwent EUS-CDS, 22,7 % (23/101) underwent EUS-GBD and 24,75% (25/101) underwent EUS-HGS. In 35/101 (34,65%) duodenal stenosis was observed. A total of 7 patients had altered anatomy. The mean procedure time was 26 ± 15,7 minutes.

Overall technical success was 93%, 48/53 (90,5%) in the CDS group, 23/23 (100%) in the GBD group and 23/25 (92%) in the HGS group. Overall clinical success was achieved in 57/79 (72%), 29/38 (76,3%) in the CDS group, 12/20 (60%) in the GBD group and 16/21 (76,1%) in the HGS group.

When considering a reduction in bilirubin level ⩾50%, the overall clinical success was lower (55%). Adverse events occurred in 6/101 (5,9%) of the patients, 3/53 (5,6%) in the CDS group, 0/23 in the GBD group and 3/25 (12%) in the HGS group. There was no difference in terms of sex, age, altered anatomy, presence of duodenal stenosis and the EUS technique performed. The mean time of the procedure was higher in the HGS group, with a difference statistically significant (CDS=22,87±14,78 min, GBD=26±15,73 min, HGS=41±16,09 min, p=0,005).

No difference was observed in terms of technical and clinical success, and adverse events between the three procedures. However, if the reduction in bilirubin level ⩾50% cut-off was used, the HGS was the procedure with the highest rate of clinical success (CDS 59,5% vs GBD 25% vs HGS 76%, p=0,003).

Conclusion: EUS-BD is a safe procedure with high technical and clinical success rate and it s a valid alternative for managing malignant biliary obstruction in high volumes centers. No differences are described between the three approaches. However HGS seems to reduce bilirubin levels more quickly, with longer procedure time.

Disclosure: Nothing to disclose.

PP0999

ENDOSONOGRAPHY TO THE RESCUE - EUS-GUIDED GASTROENTEROSTOMY AS AN URGENT TREATMENT MODALITY IN MANAGING AFFERENT LOOP SYNDROME

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Introduction: Afferent loop syndrome is a rare complication after surgical gastrojejunostomy reconstruction due to mechanical obstruction of the afferent limb. It poses a dismal prognosis, especially in cases of local cancer recurrence. Standard surgical and endoscopic treatment is often unsatisfactory. Due to poor performance status, patients frequently require immediate and less invasive treatment. Transmural stenting with lumen-apposing metal stents/LAMS/ is a new alternative to surgery and transluminal endoscopic treatment. We present our single-center experience in the management of malignant afferent loop syndrome/MALS/ with endoscopic ultrasound-guided gastroenterostomy/EUS GE/.

Aims & Methods: To evaluate the efficacy and safety of EUS GE in malignant afferent loop syndrome in cases of emergency.Methods: Two cases of MALS and severe cholangitis,managed by EUS GE in a single tertiary center.

Results: Patient 1: In the first case, a 39-year-old patient was referred to our center for recurrent episodes of severe cholangitis. He had a previous gastrojejunostomy in a different hospital, as chronic pancreatitis was suspected. After evaluation of the patient, a locally advanced pancreatic cancer with peritoneal carcinomatosis was found. Due to the poor performance status and after multidisciplinary discussion, we performed a single-step EUS GE with HOT AXIOS 15mm/10mm/Boston Scientific Marlborough, Massachusetts, USA /. After the procedure, there was complete resolution of symptoms of cholangitis and the patient started palliative chemotherapy. The patient died 56 days after the procedure due to disease progression.

Patient 2: 57 years old male, who underwent a Whipple procedure due to pancreatic cancer in 2022. During follow - up local recurrence was found. He was admitted with signs of severe cholangitis with renal and respiratory failure. Computed tomography revealed an extremely dilated afferent loop. An EUS GE was performed with,, off - label,, application of HOT SPAX-US 8mm/20mm/Taewoong/. After the procedure, the septic state was resolved, and shortly after discharge, the patient continued with regular courses of systemic chemotherapy. There are no episodes of recurrence during the current follow-up/210 days after the procedure/.

In both cases, there were no events of misdeployment. We found clinical success in both patients with fast resolution of symptoms due to cholangitis. No episodes of recurrence were found during follow-up. We didn't experience any adverse events/AE/ during either procedure, with no AE in the postprocedural period. The mean procedure time was 20 minutes. In both cases, an electro-cautery enhanced lumen apposing metal stent was used, which allowed us to create a single-step endoscopic anastomosis and obtain an internal drainage of the dilated afferent loop into the stomach.

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Conclusion: Endoscopic ultrasound-guided gastroenterostomy offers a highly efficient, convenient, and safe alternative for patients presenting with malignant ALS. In our cases, no need for reintervention was found. It could be offered as a fast approach to selected patients, especially those unfit for surgery and with poor performance status.

Disclosure: No conflict of interest.

PP1000

VALIDATION OF A PREDICTIVE SCORE FOR CLINICALLY SIGNIFICANT FINDINGS IN ENDOSCOPIC ULTRASOUND EVALUATION OF INCIDENTAL COMMON BILE DUCT DILATION

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Introduction: Common bile duct (CBD) dilation can be associated with a wide range of underlying conditions, from benign to significant pathology, including malignancy[1]. A recently proposed score by Raza et al. aims to predict the likelihood of significant findings on endoscopic ultrasound (EUS) in patients with incidental CBD dilation[2], potentially optimizing its use. This study aims to validate this score in a cohort of patients who previously underwent EUS for incidental CBD dilation.

Aims & Methods: Patients with incidental CBD dilation (≥6mm or ≥8mm post-cholecystectomy) who underwent EUS and met the inclusion criteria were retrospectively analyzed. Patient demographics, clinical, laboratory and imaging data were collected to calculate the score: age ≥70 years (1 point), acute non-billiary type abdominal pain (2 points), previous endoscopic retrograde cholangiopancreatography (2 points), CBD diameter ≥15mm or ≥17mm if prior cholecystectomy (2 points). EUS findings were reviewed to identify outcomes.

Results: Sixty-five patients met the inclusion criteria for analysis, 39 females (60%) and 26 males (40%), mean age of 71.5 ± 8.5 years. Overall yield of EUS for significant findings was 21.5%, including 8 patients with choledocolithiasis (12.3%), 4 patients with ampullary adenoma (6.2%), 1 patient with a biliary mass (1.5%) and 1 patient with a pancreatic mass (1.5%). Among the cohort, 23 patients (35.4%) had a score of 0 and none had a significant finding. Thus, a score ≥1 had a sensitivity of 100% and specificity of 45.1%, reliably excluding significant findings. A score ≥3 had a sensitivity of 71.4% and a specificity of 94.1% for identifying meaningful pathology, enabling the identification of high-risk patients with greater precision.

Conclusion: The predictive score effectively stratifies patients with incidental CBD dilation, potentially reducing unnecessary EUS in low-risk individuals while prioritizing high-risk cases. This model enhances resource utilization and targeted care. Further research in larger cohorts may confirm its generalizability.

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Disclosure: Nothing to disclose.

PP1001

ENDOSCOPIC ULTRASOUND FOR NODAL STAGING IN PATIENTS WITH RESECTABLE GALLBLADDER CANCER

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Introduction: Lymph node (LN) involvement is a well-recognized adverse prognostic factor in gallbladder cancer (GBC), and accurate preoperative nodal staging is critical for improving treatment decisions and thereby optimizing patient management. Cross-sectional imaging has limited accuracy in LN staging. Endoscopic ultrasound (EUS) with tissue acquisition (EUS-TA) offers a minimally invasive method to sample suspicious LNs (1-4).

This study aimed to evaluate the impact of EUS for nodal staging in patients with presumed resectable GBC.

Aims & Methods: In this retrospective, single-center cohort study at the Asian Institute of Gastroenterology, Hyderabad, India, consecutive patients with potentially resectable GBC who underwent preoperative EUS between January 2019 and November 2024 were included. Regional LNs were identified according to the 8th edition of the AJCC staging system as those located at the liver hilum, cystic duct, common bile duct, hepatic artery and adjacent to the gallbladder. Extraregional LN consisted of LNs located distal to the hepatoduodenal ligament. EUS-TA was performed at the discretion of the endosonographer.

The primary outcome was the impact of EUS on clinical decision making, measured as the percentage of patients in whom surgical exploration was precluded due to pathologically confirmed LN metastasis identified by EUS-TA.

The secondary outcome was the incidence of adverse events, classified according to the AGREE classification.

Results: A total of 56 patients were included (median age in years: 58.5, 51.8% female). Preoperative cross-sectional imaging was performed in 94.6% of patients with lymphadenopathy reported in 54.7%. Across 59 EUS procedures, 46 LNs were identified, and EUS-TA was performed in 24 nodes from 21 patients. LN metastases were detected by EUS-TA in 10.7% of patients, in whom surgical resection was waived.

Notably, among patients who underwent subsequent surgical resection, regional LN metastases were identified in nine of 25 patients, indicating that some nodal disease was missed on preoperative EUS. There were no complications reported related to the EUS or EUS-TA.

Conclusion: Preoperative EUS significantly influenced clinical decision making in patients with presumed resectable GBC by identifying LN metastases and thereby altering patient management. Notably, multiple LN metastases were missed at preoperative screening.

Future prospective studies employing standardized EUS protocols or integrating advanced imaging modalities are warranted to improve preoperative nodal staging and optimize management strategies in GBC.

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PP1002

ENDOSCOPIC ULTRASOUND-GUIDED GALLBLADDER DRAINAGE *VERSUS* LAPAROSCOPIC CHOLECYSTECTOMY IN ELDERLY PATIENTS WITH ACUTE CHOLECYSTITIS

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Introduction: Endoscopic ultrasound-guided gallbladder drainage (EUS-GBD) with lumen-apposing metal stent (LAMS) for acute cholecystitis is being increasingly performed in high-risk patients instead of laparoscopic cholecystectomy (LC).

Aims & Methods: In our study, we aimed to evaluate the efficacy and safety of EUS-GBD and LC in elderly patients with acute cholecystitis. We conducted a retrospective analysis of 81 patients aged ≥70 years who were admitted at our Hospital between January 2020 and December 2022 with acute cholecystitis, and were treated with either EUS-GBD or LC.

The primary outcomes of our study were the rate of technical success, defined as the correct deployment of the LAMS between the gastrointestinal wall and the gallbladder with evidence of bile or pus drainage; and the rate of clinical success, defined as the clinical and biochemical improvement of acute cholecystitis. Secondary outcomes included the rates of adverse events occurred at 1-30 days, 1-12 months and >12 months, 30-day and overall mortality.

Results: During the study period, 24 patients underwent EUS-GBD and 57 patients underwent LC. The mean age was 83 years (range 70-93) in the EUS-GBD and 77 years (range 70-88) in the LC groups (p=0.0001). The mean American Society of Anesthesiology (ASA) score was significantly higher in the EUS-GBD than in the LC groups (p=0.028). Similarly, the mean Charlson Comorbidity Index (CCI) score was significantly higher in the EUS-GBD than in the LC groups (p=0.03). Technical success was observed in 100% (24/24) and 98.2% (56/57) of EUS-GBD- and LC-treated patients, respectively (p=0.70). Clinical success was observed in 95.8% (23/24) and 98.2% (56/57) of EUS-GBD and LC-treated patients, respectively (p=0.29). The rates of 30-day adverse events were 4.2% (1/24) and 12.3% (7/57) in the EUS-GBD and LC groups, respectively (p=0.25). The rates of adverse events between 1 and 12 months were 16.7% (4/24) and 0% (0/57) in the EUS-GBD and LC groups, respectively (p=0.006). The rates of adverse events occurred after 12 months were 12.5% (3/24) and 0% (0/57) in the EUS-GBD and LC groups, respectively (p=0.023). The 30-day mortality rates were 12.5% (3/24) and 1.7% (1/57) in the EUS-GBD and LC groups, respectively (p=0.042). The overall mortality rates were 12.2% (7/57) and 41.7% (10/24) in the EUS-GBD and LC groups, respectively (p=0.011).

Conclusion: In our study on unselected elderly patients, the clinical success rates of EUS-GBD and LC for acute cholecystitis were similar; however, the mortality rate after EUS-GBD was significantly higher than after LC. These results may be explained at least in part by the higher mean age, ASA and CCI scores observed in the EUS-GBD group compared to the LC group. Our real-life data suggest that preliminary risk-benefit evaluation has directed very high-risk patients with acute cholecystitis towards treatment with EUS-GBD.

Disclosure: Nothing to disclose.

PP1003

EVOLUTION OF PATIENT SELECTION, TECHNIQUE AND OUTCOMES IN EUS-GUIDED GALLBLADDER DRAINAGE: A RETROSPECTIVE ANALYSIS OVER A DECADE

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Introduction: Acute cholecystitis in high-risk surgical candidates remains a clinical challenge. Endoscopic ultrasound-guided gallbladder drainage (EUS-GBD) with lumen-apposing metal stent (LAMS) has emerged as a minimally invasive alternative. Initially reserved for extremely frail patients, EUS-GBD has seen broader application over the past decade. However, the criteria for defining patients as "frail" or "unfit for surgery" remain debated and lack standardization. Furthermore, long-term data remain limited.

Aims & Methods: A retrospective, single-center study was conducted, including all consecutive patients who underwent EUS-GBD between January 2014 and December 2024, with the aim of evaluating changes over time in patient selection, procedural technique, and clinical outcomes. Patients were stratified into two cohorts based on the date of the procedure: Group A (procedures performed up to December 31, 2020) and Group B (procedures performed from January 1, 2021, onward). A comparative analysis was conducted between the groups.

Results: A total of 50 patients were included, of whom 25 (50%) were female, with a mean age of 79.6 (± 12.8) years. The median age-adjusted Charlson Comorbidity Index (a-CCI) was significantly lower in Group B compared to Group A (8.0 vs 9.5, $p = 0.047$). The prevalence of biliopancreatic malignancies decreased from 45.8% in Group A to 23.1% in Group B (p = ns). The use of electrocautery-tip LAMS (Hot AXIOS™, Boston Scientific) increased significantly over time (62.5% vs 100%, $p = 0.001$). Thirty-day survival improved from 83.3% in Group A to 100% in Group B ($p = 0.046$).

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Although not statistically significant, 12-month survival showed a trend toward improvement (33.3% vs 56.2%), as did 36-month survival (25.0% vs 60.0%), though the latter was limited by shorter follow-up in the more recent cohort. Technical and clinical success rates remained high across both groups. The overall rate of biliary adverse events (including recurrence and other complications) was 24%, with both early and delayed events remaining infrequent.

	Group A (n = 24)	Group B (n = 26)	p value
Baseline and procedural data			
Mean age, y (± DS)	78 (± 12.3)	81 (± 13.3)	NS
Median age-adjusted CCI, n (IQR)	9.5 (4.5)	8 (5)	0.047
Electrocautery-tip LAMS design, n (%)	15/24 (62.5%)	26/26 (100%)	0.001
Technical success, n (%)	23/24 (95.8%)	26/26 (100%)	NS
Clinical success, n (%)	20/24 (83.3%)	25/26 (96.1%)	NS
Survival outcomes			
30-day survival, n (%)	20/24 (83.3%)	26/26 (100%)	0.046
12-month survival, n (%)	8/24 (33.3%)	9/16 (56.2%)	NS

Conclusion: Our data show that over the past decade, EUS-GBD has evolved from an approach reserved for highly comorbid patients to a more broadly applied therapeutic option, coinciding with reduced baseline comorbidity and oncologic burden among treated patients. This shift was paralleled by increased procedural standardization—most notably, the universal adoption of electrocautery-tip LAMS—and was associated with improved early survival outcomes. Although longer-term survival also trended favorably, further prospective data are needed. The incidence of complications and reinterventions remained infrequent, supporting the safety and durability of the procedure.

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Disclosure: Nothing to disclose.

PP1004

THE ROLE OF DETECTIVE FLOW IMAGING- ENDOSCOPIC ULTRASOUND IN THE CHARACTERIZATION OF UNEXPLAINED BILIARY STRICTURES: A RETROSPECTIVE STUDY

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Introduction: A biliary stricture is a narrowing of intrahepatic and/or extrahepatic biliary system with upstream dilatation[1]; 70-80% of biliary strictures are malignant[1] therefore histological diagnosis is necessary to distinguish benign from malignant causes and guide management. Strictures without confirmed histology are referred to unexplained biliary strictures[2]. ESGE (European Society of Gastrointestinal Endoscopy) guidelines recommend an integrated diagnostic approach that combines multiple techniques to improve diagnostic accuracy. Endoscopic ultrasound (EUS) has demonstrated high sensitivity for evaluating biliary strictures and unexplained common bile duct (CBD) dilatation, due to its ability to provide high-resolution images of CBD walls and surrounding structures, so is the preferred diagnostic modality for both imaging and tissue acquisition(TA), even in absence of abnormal laboratory test.[1] The introduction of ancillary techniques, as DFI (Detective Flow Imaging), which enables the detection of low-velocity blood flow without the use of contrast media, represents a potential useful addition in the diagnostic algorythm[3].

Aims & Methods: The aim of this single-centre retrospective study is to validate the role of EUS-DFI in characterizing unexplained biliary strictures and CBD dilatation. We included consecutive patients (pts) with unexplained biliary strictures/CBD dilatation who underwent biliopancreatic EUS between January 2024 and April 2025 at the Digestive Diseases Unit of Sant'Andrea University Hospital, Rome. During EUS, biliary stricture (lesions or thickening of CBD wall) were described as hypoechoic or isoechoic. DFI was performed to evaluate strictures: if microvascularization was observed, vascular pattern was classified as regular or irregular[4]. Results were compared with the final diagnosis, defined by histopathology and/or ≥6 months clinical follow-up as gold standard. TA was performed with EUS-FNB (fine needle biopsy) and/or ERCP(Endoscopic Retrograde Cholangiopancreatography) intraductal biopsies/brushing.

Results: Of 13 patients included, 8 (61.5%) were male, median age 72 (range 56-86). All underwent Magnetic Resonance Imaging/Contrast-Enhanced Computed Tomography as first line imaging: 3 (23%) had a perihilar stricture, 9 (69%) a medio-distal, 1 (8%) had "in toto" CBD thickening. Among the 13 strictures, EUS identified 10 (76.9%) lesions, with a mean diameter of 17 mm (median 16 mm; range, 5–35 mm), and 3 (23.1%) CBD wall thickening. Twelve (92%) strictures were hypoechoic, 1(8%) was isoechoic. EUS-TA was performed in all pts, revealing 7 (54%) cases of malignancies, of whome 5 cholangiocarcinoma and 2 pancreatic ductal adenocarcinoma (PDAC), 3 (23%) high-grade dysplasia, and 3 (23%) fibroinflammatory disease on histopatological evaluation. All pts underwent DFI-EUS. Microvascularization was detected in 12 cases (92%), with an irregular pattern observed in 9 pts. Among these, 5 (55.6%) were diagnosed with cholangiocarcinoma, 1 (11.1%) with PDAC infiltrating CBD, and 3(33.3%) with high-grade dysplasia. The 3 benign strictures demonstrated a homogeneous pattern. One case (8%), corresponding to a PDAC, appeared hypovascular on DFI.

Conclusion: In a small cohort of patients with unexplained or indeterminate biliary strictures, the novel DFI-EUS technique proved to be feasible for assessing microvascularization in CBD stenosis and revealed characteristic irregular patterns or hypovascularization in malignant lesions. Further prospective studies with a larger patient population are needed to confirm these preliminary findings.

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Disclosure: Nothing to disclose.

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PP1005

ASSESSING THE IMPACT OF RADIOFREQUENCY ABLATION ON HILAR CHOLANGIOCARCINOMA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Hilar cholangiocarcinoma (HC) is a challenging malignancy with limited treatment options. Radiofrequency ablation (RFA) has emerged as a potential palliative treatment, but its efficacy and safety remain controversial. This systematic review and meta-analysis aimed to evaluate the impact of RFA + stent (RFA + S) and stent only (S-only) on HC patients.

Aims & Methods: We conducted a systematic search of PubMed, Embase, and Scopus databases for studies published up to December 2024. Eligible studies reporting RFA outcomes were included.

The primary outcome was overall survival (OS).

Secondary outcomes included stent patency (SP), stent dysfunction, and adverse events. We calculated hazard ratios (HR) and odds ratios (OR) using random-effects models.

Results: This analysis included eleven studies (n = 874 patients) that met the inclusion criteria. The pooled HR for OS was 0.74 (95% CI: 0.61-0.89, p = 0.002), and SP was 0.77 (95% CI: 0.61-0.97; p = 0.03), showing a significant difference in favor of the RFA group. The OR of overall adverse events was 1.48 (95% CI: 0.59-3.72), cholangitis was 1.71 (95% CI: 0.78-3.75), pancreatitis was 1.03 (95% CI: 0.19-5.50), and liver abscess was 0.62 (95% CI: 0.07-5.24), showing no difference (p > 0.05) between the two groups. Patients with chemotherapy showed better survival, indicated by an HR of 0.57 (95% CI; 0.40-0.81). Subgroup analyses also showed that RCT studies had no difference compared to non-RCT studies.

Conclusion: This meta-analysis suggests that application of RFA offer better survival and stent patency benefits with comparable adverse events.

Disclosure: Nothing to disclose.

PP1006

ENDOSCOPIC TREATMENT OF DIFFICULT CHOLEDOCHOLITHIASIS: OUR EXPERIENCE

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Introduction: Traditional endoscopic techniques can be used in the treatment of common bile ducts lithiasis. The procedure is considered "difficult", and it is not possible to complete the exctration of the stones when the stone is larger than 2cm, the number of stones is more than 3 and/or when there is a stenosis of the common bile duct. In these cases, the gold standard is the placement of a plastic stent enabling rapid drainage of bile to avoid impaction and eventually a cholangitis. A dissolution rate of 30% -60% (about 1 mm in diameter has been reported by calculation under an optical microscope), even if the initial diameter has been shown to be the most important factor affecting the rate of dissolution.

The aim of this study is to demonstrate that the apposition of the plastic stent in combination with ursodeoxycholic acid therapy is safe and effective in the treatment of difficult choledocholithiasis.

Aims & Methods: In the period between September 2022 and November 2024, 447 patients diagnosed with gallstones of the choledochus treated endoscopically by ERCP were selected. This procedure was successful in 411 patients (92%). The remaining 36 patients (8%) diagnosed with "difficult" choledocholithiasis, were treated with the application of a plastic stent and therapy with ursodeoxycholic acid 10mg / kg / day. ERCP control was established after 3 months.

Results: Four-hundred-eleven patients (92%) had a successful ERCP treatment. Thirty-six 36 patients (8%) diagnosed with "difficult" choledocholithiasis, were treated with the sphincterotomy, the application of a plastic stent , therapy with ursodeoxycholic acid 10mg / kg / day. Bilirubin control was established at 7 and 15 days. The treatment was successful in 100% of cases (36 out of 36). No complications related to the procedure occurred.

At three-months follow-up 31 patients (86%) experienced complete toileting of the main biliary tract. Only 5 patients (14%) needed positioning of a new stent of the main biliary tract, namely one patient had the stent replaced after one month due to obstruction, in three patients the stent became definitive ensuring drainage of the bile in the absence of symptoms, in one patient clearance of the main biliary pathway was possible only after further 3 months.

Conclusion: Insertion of a stent for the treatment of difficult choledocholithiasis is a coded and safe endoscopic technique. It enables fast bile drainage in all cases and complete endoscopic removal of stones at subsequent follow ups in most cases. Ursodeoxycholic acid therapy has also showed to be effective in reducing stones size (<2cm). Permanent stenting of common bile duct is performed only in rare cases, however with satisfactory results.

Disclosure: Nothing to disclose.

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PP1007

EFFICACY AND SAFETY OF METAL VERSUS PLASTIC STENTS FOR PREOPERATIVE BILIARY DRAINAGE IN PANCREATIC CANCER PATIENTS RECEIVING NEOADJUVANT CHEMOTHERAPY OR CHEMORADIOTHERAPY: A LARGE MULTICENTER STUDY

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Introduction: Metal (MS) or plastic stents (PS) are used for preoperative biliary drainage (PBD) in patients with pancreatic ductal adenocarcinoma (PDAC) undergoing neoadjuvant chemotherapy (NAC) or chemoradiotherapy (NACRT). However, there are no large studies comparing postoperative adverse events(AEs) and postoperative long-term outcomes between patients who received MS and those who received PS for PBD.

Aims & Methods: This multicenter retrospective study compared the safety and efficacy of PBD using MS and PS, including surgery-related adverse events and other clinical outcomes.

This study enrolled patients with resectable or borderline resectable PDAC requiring NAC or NACRT who underwent PBD with MS or PS at 17 tertiary care hospitals in Japan between January 2016 and December 2021. Clinical outcomes, including the incidence of recurrent biliary obstruction (RBO), time to RBO, time from stent placement to surgery, postoperative hospital stay, incidence of postoperative adverse events, and postoperative overall survival, were retrospectively analyzed.

Results: Among the 308 patients with PDAC who received PBD before NAC or NACRT, 251 subsequently underwent pancreatic resection. Of these 251 patients, 122 (48.6%) had PBD with MS, while 129 (51.4%) had PBD with PS. Because there were significant differences in baseline characteristics between both groups, propensity score matching was performed, resulting in 84 matched pairs. RBO occurred significantly less frequently in the MS group than in the PS group (11.9% vs. 59.5%, $p < 0.001$). Time to RBO is significantly longer in the MS group than in the PS group (median 353 days vs. 74 days, $p < 0.001$). The MS group also showed significantly reduced time to surgery from stent placement (median 95 days *vs.* 112 days, $p = 0.010$) and shorter postoperative hospital stays (median 17 days *vs.* 20 days, $p = 0.025$).

Although postoperative AEs occurred less often in the MS group (19.1% vs. 29.8%), the difference was not statistically significant ($p = 0.106$). In both univariate and multivariate models, RBO was identified as the sole independent factor associated with postoperative AEs (odds ratio: 2.36, $p = 0.019$). However, the type of stent used was not significantly associated with the occurrence of postoperative AEs.

Notably, the incidence of postoperative pancreatic fistula was significantly higher in patients who developed RBO compared to those who did not (16.7% vs. 5.6%, $p = 0.027$).

Conclusion: For patients with PDAC undergoing NAC or NACRT, MS placement is preferable to PS placement for PBD. MS placement was associated with a lower incidence of RBO and need for re-intervention, as well as a shorter perioperative hospital stay. Given the reduced frequency of RBO, which is an established risk factor for postoperative AE, MS placement is recommended for PBD in this setting.

Disclosure: Nothing to disclose.

PP1008

UNCOVERED VERSUS FULLY COVERED SELF-EXPANDABLE METAL STENTS FOR NONPANCREATIC CANCER: A PROPENSITY SCORE-MATCHED, MULTICENTER STUDY

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Introduction: Distal malignant biliary obstruction (dMBO) is a frequently occurring condition that requires immediate intervention. The primary treatment for dMBO comprises endoscopic retrograde cholangiopancreatography (ERCP) with biliary stent insertion. The European Society of Gastrointestinal Endoscopy recommends the use of self-expandable metal stents (SEMS) for palliative drainage of dMBO because of their superior patency and reduced incidence of adverse events (AEs), including recurrent biliary obstruction, compared to those of plastic stents. Three types of SEMS are available: uncovered SEMS (UCSEMS), fully covered SEMS (FCSEMS), and partially covered SEMS.

However, the ideal stent type for palliative drainage of dMBO remains undetermined. Several studies have compared UCSEMS with FCSEMS to determine which type is most appropriate for dMBO. In terms of patency, some prospective trials have shown that FCSEMS result in better outcomes; however, others have observed similar results with UCSEMS and FCSEMS.

Most studies have primarily focused on pancreatic cancer cases; however, the pathophysiology of pancreatic cancer and that of nonpancreatic cancer differ significantly, resulting in inconsistent outcomes. Post-ERCP pancreatitis (PEP) is the most common and serious AE associated with the

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transpapillary placement of SEMS, with an incidence rate of up to 10.8%. Therefore, studies have aimed to identify the risk factors for PEP after SEMS placement. Compared to UCSEMS placement, FCSEMS placement may be more strongly linked to a higher risk of PEP because they block the pancreatic duct orifice more effectively.

Additionally, nonpancreatic cancer has been identified as a potential risk factor for PEP. Therefore, FCSEMS placement for nonpancreatic cancer cases is associated with a significantly higher risk of PEP. We examined the safety of UCSEMS for dMBO caused by nonpancreatic cancer and observed a low PEP rate, even for nonpancreatic cancer cases.

However, a comparison of UCSEMS and FCSEMS is still required because no studies have evaluated their use for dMBO caused by nonpancreatic cancer. Therefore, this study evaluated the efficacy and safety of UCSEMS and FCSEMS for palliative drainage of dMBO caused by nonpancreatic cancer.

Aims & Methods: This retrospective multicenter study included patients who underwent UCSEMS or FCSEMS placement for nonpancreatic cancer-induced dMBO at four centers between January 2010 and April 2024. Propensity score matching (1:1) of the UCSEMS and FCSEMS groups was performed. Technical and clinical success, AEs, recurrent biliary obstruction (RBO), time to RBO (TRBO), and overall survival (OS) of these groups were compared.

Results: No significant differences in technical success (100% for both), clinical success (100% for both), and TRBO (median: 311 days [UCSEMS] vs. 317 days [FCSEMS]; p=0.42) were observed between groups.

Similarly, OS did not differ significantly (median: 432 days [UCSEMS] vs. 190 days [FCSEMS]; p=0.083). The incidence of postendoscopic retrograde cholangiopancreatography pancreatitis (PEP) of the UCSEMS group was significantly lower than that of the FCSEMS group (4.2% vs. 33%; p=0.023).

Conclusion: compared to FCSEMS, UCSEMS provided sufficient patency and reduced AEs associated with dMBO caused by nonpancreatic cancer. The cause of dMBO should be carefully considered when choosing the optimal stent. UCSEMS may be safer and more effective than FCSEMS for the treatment of nonpancreatic cancer cases.

Disclosure: Authors declare no conflict of interests for this article.

PP1009

RISK FACTORS FOR OBSTRUCTION OF CHOLEDOCHODUODENOSTOMY BY LUMENAPPOSING METAL STENT AND MANAGEMENT STRATEGIES: RESULTS FROM A PROSPECTIVE COHORT OF 168 PATIENTS

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Introduction: In case of distal malignant biliary obstructions, when ERCP fails, drainage via EUS-CDS using LAMS has become the gold standard.

Identified risk factors include the presence of a duodenal stent (2), duodenal invasion (3), and a bile duct diameter < 15mm (2).

The data regarding stent obstruction during follow-up are heterogeneous and range from 9 to 55% (1).

Aims & Methods: The objective of our study is to identify risk factors for stent obstruction, evaluate the efficacy of obstruction management, and define the best strategy for
managing these stent obstructions.

We conducted a retrospective study of a monocentric prospective database, including all patients treated with EUS-CDS for distal biliary obstruction between 2017 and 2024 in a French university hospital. Obstruction was defined as persistent and/or recurrent jaundice and/or acute cholangitis during follow-up.

Results: A total of 168 patients were included in our study. The overall obstruction rate was 20,4%, with a mean follow-up of 176 days (5.8 months). We confirmed that gastrointestinal obstruction was a risk factor for stent obstruction in both univariate analysis (OR = 4.22 [1.91; 9.31], p = 0.001) and multivariate analysis (OR = 4.67 [2.05; 10.65], p = 0.0002).

Regarding the best management strategy for stent obstruction, placing a new pigtail or metallic stent within the lumen-apposing stent is more effective than simple mechanical desobstruction or antibiotic therapy alone. There was a a higher rate of biliary event-free survival during follow-up (at 6 months 92.3% versus 58.9%; at 12 months 80.8% versus 47.1%).

Conclusion: We confirm that gastrointestinal obstruction is a risk factor for stent obstruction in EUS-CDS procedures.

In cases of cholangitis or persistent jaundice, endoscopic desobstruction by placing a new pig-tail or metallic stent is more effective than simple desobstruction or antibiotic therapy.

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Disclosure: Nothing to disclose.

PP1010

CLINICAL OUTCOMES AND PREDICTIVE FACTORS FOR ADVERSE EVENTS IN COMBINED EUS-GUIDED AND ERCP BILIARY DRAINAGE FOR MALIGNANT OBSTRUCTION

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Introduction: Endoscopic biliary drainage for malignant obstruction can be technically challenging, particularly in patients with high-grade hilar strictures or progressive distal lesions that extend beyond the reach of ERCP alone. While ERCP remains the first-line approach, it may prove inadequate in complex scenarios. Endoscopic ultrasound-guided biliary drainage (EUS-BD) has emerged as an alternative or complementary technique. However, data on outcomes that combine ERCP and EUS-BD in routine clinical practice remain insufficient and limited.

Aims & Methods: This study aims to evaluate the clinical success, stent patency, and complication rates of combined ERCP and EUS-BD in patients with malignant biliary obstruction while identifying factors as-

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sociated with complications. We included 40 patients who underwent combined ERCP and EUS-BD for malignant biliary obstruction at a tertiary center from March 2020 to March 2025. The patients were divided into two groups: one-session procedures, in which both techniques were performed during the same session (n=14), and staged procedures, where EUS-BD was followed by at least one prior ERCP (n=26).

Clinical success was defined as resolving jaundice and/or cholangitis without additional drainage. The outcomes assessed included clinical success, stent patency, complications, and reintervention. Univariate analysis was employed to identify predictors of complications.

Results: The cohort included 21 (52.5 %) males and 19 (47.5%) females, with a median age of 65.6 years. The distribution of strictures was as follows: 23 (57.5%) high-grade hilar (Bismuth III/IV), 7 (17.5%) Bismuth I/II, and 10 (25%) distal. Underlying malignancies included 12 (30%) hilar cholangiocarcinoma, 8 (20%) pancreatic adenocarcinoma, and 16 (40 %) liver or hilar metastasis from other tumors.

Clinical success was achieved in 90% of patients (36/40), with no significant difference between one-session and staged groups (85.7% vs. 92.3%, p=0.60). The median procedural time was 39.3 minutes. Overall, 12 patients (30%) experienced complications, including stent-related infections in 10 (25%), post-ERCP pancreatitis in 1(2.5%), and bleeding in 1 (2.5%). Complication rates were similar between the one-session and staged groups (28.6% vs. 31.0%, p=1.00).

Reintervention was required in 3 patients (7.5%). A significant association was observed between undergoing more than two prior ERCPs and the risk of complications (66.7% vs. 22.6%, p=0.038). Neither the presence of hilar strictures (p=0.56) nor procedural timing (one-session vs. staged; p=1.00) was significantly associated with increased complications.

Conclusion: Combining ERCP and EUS-guided biliary drainage is a safe and effective strategy for managing malignant biliary obstruction, including complex hilar and progressive distal strictures. Performing both procedures in a single session does not increase complication rates and may provide comparable clinical success.

The number of prior ERCPs was the only significant predictor of complications, suggesting a potential benefit from earlier use of EUS-BD in selected patients.

Further prospective studies are warranted to confirm these findings and guide treatment algorithms for complex biliary obstruction.

Disclosure: Nothing to disclose.

PP1011

COMPARING STENT TYPES FOR EUS-HEPATICOGASTROSTOMY: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Endoscopic Ultrasound Guided Hepaticogastrostomy (EUS-HGS) is one of the Endoscopic Ultrasound Guided Biliary Drainage (EUS-BD) techniques that is frequently performed in specialist endoscopic center throughout the world. At the same time, no consensus exists on effects of individual stent types.

Aims & Methods: We aimed to assess the efficacy and safety profile of plastic stents and individual type of EUS-HGS metallic stents. A random-effects meta-analysis was performed using the PRISMA protocol. Three electronic databases were searched for studies on HGS with plastic or metallic stents from inception until February 2025. The odds ratio (OR) and pooled rates were used for single and two-arm comparisons with 95% confidence intervals (CI).

Results: From twenty-five eligible studies, the OR for adverse events with and without biliary obstruction were OR=1.12 [0.60-2.10], p= 0.648 and OR=2.20 [CI:1.26-3.84], p= 0.017 respectively. The rates for cholangitis, food occlusion, stent hyperplasia was 1% [CI:0-5%], 33% [CI:22-47%] and 1% [CI:0-6%] for plastic stents which differed for different metallic stent types. Lastly, hazard ratio for biliary obstruction was HR= 1.603 [CI:1.211-2.122].

Conclusion: Opting for plastic or metallic stents in EUS-HGS requires a careful consideration between patient's long-term survival, favoring metallic stents and the short-term safety profile, favoring plastic. The type of metallic stent used plays an important role in that decision.

Disclosure: Nothing to disclose.

PP1012

STENT PLACEMENT ABOVE THE SPHINCTER OF ODDI FOR MALIGNANT HILAR BILIARY OBSTRUCTION: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: The management of malignant hilar biliary obstruction can be done by endoscopic biliary stenting, which is a well-established palliative treatment, improving the quality of life by relieving symptoms.

Aims & Methods: This systematic review and meta-analysis aimed to compare the use of stents placed above and across the sphincter of Oddi in patients with malignant hilar biliary obstruction. We performed a systematic search of the PubMed, Embase, and Cochrane Library databases in April 2025. Article screening was conducted independently by two authors using Rayyan software. Inclusion criteria were randomized controlled trials (RCTs) comparing stent placement above and across the sphincter of Oddi for malignant hilar biliary obstruction, evaluating at least one of the outcomes of interest.

The outcomes were stent migration, stent patency, cholangitis, stent occlusion, and successful placement of the stent. RStudio software was used for statistical analysis, applying a random-effects model to calculate the risk ratio (RR) in binary outcomes and mean differences (MD) in continuous outcomes.

Results: We included three RCTs comprising 161 patients. There was no significant difference between the transpapillary and suprapapillary stents regarding the rates of stent migration (RR 2.03; 95% CI 0.41–10.16; p=0.39; I^2=55%), cholangitis (RR 0.65; 95% CI 0.08–5.07; p=0.68; I^2=0%), technical success (RR 0.99; 95% CI 0.95–1.04; p=0.80; I^2=0%), and stent

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occlusion (MD 89.20; 95% CI 16.30–162.10; p=0.02; I^2=78%). However, the mean number of patent stents was significantly lower in the suprapapillary stent group.

Conclusion: The two types of stent seem similar in terms of technique, efficacy, and safety, although the suprapapillary stent seems superior in terms of stent patency.

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Disclosure: Nothing to disclose.

PP1133

POST-CHOLECYSTECTOMY MAJOR BILE DUCT INJURY: IDEAL TIME TO REPAIR BASED ON A MULTI-CENTER RANDOMIZED CONTROLLED TRIAL WITH PROMISING RESULTS

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Introduction: Bile duct injury (BDI) is one of the serious complications of cholecystectomy procedures, which has a disastrous impact on long-term survival, health-related quality of life (QoL), health-care costs as well as high rates of litigation. The standard treatment of major BDI is hepaticojejunostomy (HJ). Surgical outcomes depend on many factors, including the severity of the injury, the surgeons' experiences, the patient's condition, and the reconstruction time. We aimed to assess the impact of reconstruction time and abdominal sepsis control on the reconstruction success rate.

Aims & Methods: We aimed to assess the impact of reconstruction time and abdominal sepsis control on the reconstruction success rate. This is a multicenter, multi-arm, parallel-group, randomized trial that included all consecutive patients treated with HJ for major post-cholecystectomy BDI from February 2014 to January 2022.

Patients were randomized according to the time of reconstruction by HJ and abdominal sepsis control into group A (early reconstruction without sepsis control), group B (early reconstruction with sepsis control), and group C (delayed reconstruction).

The primary outcome was successful reconstruction rate, while blood loss, HJ diameter, operative time, drainage amount, drain and stent duration, postoperative liver function tests, morbidity and mortality, number of admissions and interventions, hospital stay, total cost, and patient QoL were considered secondary outcomes.

Results: Three hundred twenty one patients from 3 centers were randomized into three groups. 44 patients were excluded from the analysis, leaving 277 patients for intention to treat analysis. With univariate analysis, older age, male gender, laparoscopic cholecystectomy, conversion to open cholecystectomy, failure of intraoperative BDI recognition, Strasberg E4 classification, uncontrolled abdominal sepsis, secondary repair, end-to-side anastomosis, diameter of HJ (< 8 mm), non-stented anastomosis, and major complications were risk factors for successful reconstruction. With multivariate analysis, conversion to open cholecystectomy, uncontrolled sepsis, secondary repair, the small diameter of HJ, and non-stented anastomosis were the independent risk factors for the successful reconstruction. Also, group B patients showed decreased admission and intervention rates, decreased hospital stay, decreased total cost, and early improved patient QoL.

Conclusion: Early reconstruction after abdominal sepsis control can be done safely at any time with comparable results for delayed reconstruction in addition to decreased total cost and improved patient QoL.

Disclosure: Nothing to disclose.

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Poster presentations
Pancreas

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Poster Presentations
Pancreas

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Pancreas
Poster Exhibition

PP1013
METABOLIC IMPACT OF FATTY PANCREAS: CORRELATION BETWEEN ADIPOSE INFILTRATION AND BETA-CELL FUNCTION ASSESSED THROUGH ULTRASOUND AND MAGNETIC RESONANCE IMAGING

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Introduction: Pancreatic steatosis (PS) is linked to metabolic dysregulation. Magnetic resonance imaging (MRI)-derived proton density fat fraction (PDFF) provides a reliable assessment of intrapancreatic fat. This study investigates the relationship between pancreatic fat, assessed by ultrasound and MRI, and metabolic parameters from the oral glucose tolerance test (OGTT).

Aims & Methods: We analyzed 40 non-diabetic patients with increased pancreatic echogenicity, of whom 29 completed the study (mean age: 54 ± 10.4 years; BMI: 34.8 kg/m^2; waist circumference: 93.8 ± 11.3 cm). Participants underwent OGTT, with assessment of β-cell function indices (Disposition Index, Insulinogenic Index) and insulin sensitivity indices (Matsuda Index, HOMA-IR, Stumvoll Index). Pancreatic fat was evaluated using ultrasound, MRI-PDFF, and two-dimensional shear wave elastography.

Results: The mean pancreatic fat fraction was 13.75 ± 11.22% (median: 7.633%), and mean pancreatic stiffness was 8.1 ± 3.2 kPa. After OGTT, 27.5% of patients exhibited impaired glucose tolerance (IGT). PDFF positively correlated with weight ($r = 0.520$, $p = 0.004$), BMI ($r = 0.505$, $p = 0.005$), and body surface area ($r = 0.509$, $p = 0.005$), but not with liver enzymes, metabolic parameters, or β-cell function. A significant negative correlation was found between pancreatic stiffness and the Potentiation Factor (PFR1), indicating reduced β-cell responsiveness with increasing pancreatic stiffness. Stratification by glucose tolerance status revealed that increased pancreatic fat was associated with greater insulin resistance and compensatory hyperinsulinemia.

Conclusion: In conclusion, our study confirms that increased intrapancreatic fat is an indicator of ectopic lipid accumulation and is correlated with general adiposity. Our study identifies pancreatic fat infiltration as a marker of early metabolic dysfunction, where increasing pancreatic fat triggers β-cell compensatory mechanisms. This underscores the importance of determining a pancreatic fat threshold below which glucose homeostasis may still be restored. Targeted interventions to reduce pancreatic fat are crucial to preventing the progression to irreversible β-cell dysfunction.

Disclosure: Nothing to disclose.

PP1014
ASSESSMENT OF BLOOD PANCREATIC LIPASE VARIATION AS A POTENTIAL NOVEL DIAGNOSTIC APPROACH FOR EXOCRINE PANCREATIC INSUFFICIENCY, LIPASADA PROJECT

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Introduction: Exocrine pancreatic insufficiency (EPI) is defined as a reduction in the exocrine pancreatic secretion and/or intraluminal activity of pancreatic enzymes below the level that allows normal digestion of nutrients. Currently, diagnosing EPI presents challenges: the most accurate method requires collecting faecal samples for several days to measure faecal fat excretion following a specific diet, while faecal elastase test, although simpler, has a high incidence of false positives.

The study assesses whether measuring lipase in blood, following the ingestion of a standard high-fat meal could provide a more accessible diagnosis of EPI. The hypothesis is that blood lipase increases proportionally to pancreatic secretion in individuals without EPI, allowing differentiation between patients with and without insufficiency.

Aims & Methods: The study's main objective was to investigate whether blood levels of lipase increase after a fatty meal in normal subjects. If blood lipase and lipase secreted by the pancreas have a correlation, by calculating the area under the curve of several determinations of blood lipase in healthy subjects, we could set a normal pattern of lipase secretion. In the future, this pattern can be compared to patients with EPI to test if it can be a screening tool for EPI.

For the study, 10 healthy volunteers were tested. All participants fasted before receiving a test meal designed to stimulate lipase secretion. Blood samples were then taken every 30 minutes for six hours to measure lipase blood levels and calculate the area under the curve (AUC) using the trapezoid method.

Results: The mean (standard deviation) of the maximum and minimum lipase blood levels was 24 (9) and 19 (8) U/L, respectively. The mean percentage change in lipase was 31% (standard deviation 16%). The area under the curve of blood lipase levels between the baseline sample and the sample at 360 minutes had a mean of 7749 (standard deviation 3129) units per litre per hour.

The median time to peak lipase was 0 minutes, meaning that the expected peak in blood lipase did not occur after the test meal: in 60% of cases, the highest peak of all measurements was the baseline measurement, with no clear peak in response to the meal in the remaining cases, in contrast with pancreatic secretion into the duodenum.

Conclusion: According to our patient sample, lipase does not increase in the blood after stimulation with a test meal, as occurs in pancreatic lipase secretion into the duodenum.

Lipasemia is, therefore, independent of pancreatic secretion and does not seem to be a promising parameter for screening exocrine pancreatic insufficiency.

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Disclosure: Nothing to disclose.

PP1015

THE RATE OF CARDIOVASCULAR EVENTS IN PANCREATIC CANCER DOES NOT CHANGE WITH TREATMENT TYPE OR DISEASE STAGE – A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Cardiovascular diseases (CVDs) are often a limiting factor in cancer treatment. In our systematic review and meta-analysis we evaluated the prevalence and incidence of CVDs in pancreatic ductal adenocarcinoma (PDAC).
Aims & Methods: We conducted the systematic search on 5 February 2024 in three databases (PubMed, EMBASE, CENTRAL). Studies reporting the prevalence or incidence of CVDs in PDAC were included. Subgroup analyses based on cancer stage and treatment type were performed.
Pooled proportions with the 95% confidence interval (CI) were calculated using a random effect model. We evaluated the risk of bias using the Joanna Briggs Institute Prevalence Critical Appraisal. (PROSPERO CRD42023482295).
Results: We included 198 articles. The prevalence of CVDs other than thrombotic events at PDAC diagnosis was similar to the one in the general population: hypertension 33 % (95%CI:27-40%), ischemic heart disease 6% (95%CI:3-12%), heart failure, arrhythmia, and stroke affecting each 2-3% of cases. Their incidence during treatment was low (1%-10% overall).
Thrombotic events other than embolism affected 11% (95%CI: 7%-18%) of cases at diagnosis and the incidence during treatment was between 8-10% regardless of cancer stage or treatment type.
Pulmonary embolism affected 3% of cases at diagnosis and occurred in a similar proportion during treatment. The included studies had an overall low-moderate risk of bias.
Conclusion: Thrombotic events affect around 10% of PDAC patients at baseline. A similar proportion develop thrombotic events during follow-up regardless of disease stage or treatment type. The prevalence of CVDs other than thrombotic events is equivalent to the one in the general population.
Disclosure: Nothing to disclose.

PP1016

PLANNED STAGED ENDOSCOPIC ULTRASOUND-GUIDED ETHANOL INJECTION FOR SMALL PANCREATIC NEUROENDOCRINE NEOPLASMS: A SAFE AND EFFECTIVE STRATEGY WITH REDUCED COMPLICATIONS AND ABLATION RATE

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Introduction: The optimal management of small (≤2 cm) low-grade pancreatic neuroendocrine neoplasms (PNENs) remains a subject of ongoing debate. While surgical resection has traditionally been the standard treatment, its invasiveness and potential for long-term complications, such as pancreatic insufficiency, have driven interest in less invasive alternatives. Recently, endoscopic ultrasonography-guided ethanol injection (EUS-EI) has emerged as a promising treatment option, offering the potential for localized tumor control with minimal impact on surrounding pancreatic tissue.
Aims & Methods: This study was conducted at a single academic center between 2014 and 2024, and the data were retrospectively analyzed. Eligibility criteria included patients with pathologically confirmed grade 1 PNENs ≤2 cm who were either poor surgical candidates or declined surgery. The procedure was carefully planned to minimize complications and improve treatment outcomes. A 25-gauge needle was used to reduce the risk of ethanol leakage. Pure ethanol was injected under EUS guidance until hyperechoic bubbles fully surrounded the tumor margin.
To ensure complete coverage while minimizing ethanol exposure, ethanol was injected from multiple sites if needed. Importantly, the maximum ethanol volume per session was limited to 2 mL to prevent overexposure and reduce adverse events.
To optimize treatment efficacy, contrast-enhanced computed tomography (CE-CT) was performed on postoperative day 3. If residual enhancement was detected, an additional ablation session was scheduled during the same hospitalization, allowing for planned, staged treatment.
Results: A total of 41 patients including four patients with insulinoma underwent EUS-EI, The median tumor size of 10 mm (range: 6–15 mm). The median ethanol volume in the initial session was 0.9 mL (range: 0.3–2.0 mL), and the total median volume was 1.0 mL (range: 0.3–3.2 mL), reflecting the effectiveness of carefully distributed, low-volume injections. Fifteen patients (37%) required an additional planned ablation session based on CE-CT findings. Mild pancreatitis occurred in 6 patients (14%), and the median hospital stay was 5 days (range: 4–8 days), suggesting a favorable safety profile.
Complete ablation was confirmed in 37 patients (90%) at one month post-procedure. All patients with insulinomas experienced an improvement in hypoglycemia-related symptoms.
During a median follow-up of 36 months, local recurrence was observed in only 2 patients (5%), occurring at 22 and 51 months, respectively. Notably, no cases of new-onset diabetes mellitus were reported.

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Conclusion: Planned, staged EUS-guided ethanol injection using low per-session ethanol volumes is a safe and effective treatment strategy for patients with small, low-grade PNENs. This approach appears to reduce procedure-related complications while enhancing the rate of complete tumor ablation.

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Disclosure: Nothing to disclose.

PP1017

METHODOLOGICAL ASSESSMENT OF ENDOSCOPIC ULTRASOUND RADIOFREQUENCY ABLATION FOR PANCREATIC NEOPLASMS: RESULTS FROM AN ONLINE SURVEY PROPOSED BY AN INTERNATIONAL WORKING GROUP ON EUS-RFA METHODOLOGY

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Introduction: Endoscopic ultrasound radiofrequency ablation (EUS-RFA) is a safe and effective therapeutic modality for treating pancreatic neoplasms. While robust data on technical feasibility and safety profile are available and consistent, clinical efficacy and oncological outcomes are still based on low evidence and not standardized methods.

Aims & Methods: We aim to assess current practice in EUS-RFA in terms of indications, prophylaxis for adverse events, generator set-up, follow-up, and lexicon.

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An online survey with 70 questions divided into 5 domains (physicians and center characteristics, indications, procedural details, post-procedural management, and proposed lexicon) was created and shared with 175 international experts at the beginning of 2025; the survey link was open for 5 months and analyzed at the end of April 2025.

Results: Domain 1. 91 physicians (52%) completed the survey; 74.4% from Europe, 13.3% North America and 12.2% Asia; 64.8% declare >10-year EUS experience from high-volume centers (83.3%). 35.6% have formal accreditation as a Center of Excellence for Neuroendocrine Tumors (NET). 41 physicians (45.1%) have performed >20 EUS-RFA procedures, while 20 (22.0%) >50 procedures.

Domain 2. 94.1% physicians perform EUS-RFA for insulinoma, 71.8% non-functioning NET, 57.7% pancreatic metastases, 54.1% other functioning NET, 37.7% pancreatic ductal adenocarcinoma, 31.8% IPMN with mural nodule; very low agreement (<25%) was reached for other indications, (solid pseudopapillary neoplasms, liver tumors, other pancreatic neoplasms, adrenal gland tumors, celiac plexus).

Domain 3. 96% of EUS-RFA are performed under deep sedation or general anesthesia, in inpatient setting (83.1%). Rectal NSAIDs and aggressive hydration are administered for post-RFA pancreatitis prophylaxis in 45.6%, while only rectal NSAIDs in 41.8%. Antibiotic prophylaxis was administered in 57.0%. 50.6% of physicians are concerned about main duct injury for lesions located ≤ 2 mm while 34.2% for lesions closer than 1 mm. In these cases, 43.0% reduce duct damage through changes in power setting, avoiding duct contact, while 24.1% perform ERCP with pancreatic duct stenting. 45.6% avoid performing EUS-RFA in case of ductal infiltration with dilation.

58.4% use the same power-setting for every procedure, and 26.6% the same probe size. 10-mm tip length probe was used by 92.2% of physicians. Low agreement (<50%) for power generator set-up was reached for any indication. 70.9% have no maximum number of applications for each EUS-RFA session and continue ablation until all tissue seems ablated. 56.4% have no maximum time for each application.

Domain 4. 52.6% test serum biochemistry the day after the procedure; 27.0% perform a routine CT scan within 2 weeks. Contrast CT scan (81.0%) and contrast EUS (46.8%) are used for follow-up. A 3-month CT scan was used as the first oncological assessment in 51.3%.

Domain 5. Technical success was defined as “achieving the presumed complete ablation at the end of the procedure” by 88.6%; clinical success as “achieving symptom control for at least 1 year for insulinoma” by 92.2%; disease recurrence as “evidence of vascularized tumor tissue following complete response”. Low consensus (< 50%) was reached on the definition of complete and partial ablation, and post-EUS-RFA pancreatitis.

Conclusion: The results of this study provide clear data on the lack of standardization of EUS-RFA procedures even among experts. An international working group on EUS-RFA methodology is required to address unanswered clinical questions and provide a consensus fostering uniformity in EUS-RFA application.

Disclosure: Nothing to disclose.

PP1018

DIAGNOSTIC UTILITY OF PANCREATIC AND BILE DUCT BRUSHING FOR PANCREATIC CANCER DURING ERCP: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Pancreatic cancer is a highly lethal disease, often diagnosed at an advanced stage due to the lack of early detection methods. Endoscopic retrograde cholangiopancreatography (ERCP) with brush cytology is commonly used to obtain samples from the bile duct (BD) and pancreatic duct (PD), yet its diagnostic accuracy remains debated.

Aims & Methods: This study aimed to assess the pooled sensitivity of BD and PD brushing in detecting pancreatic cancer compared to histology. A systematic search was conducted in three major medical databases on November 05, 2024. We included studies comparing BD and/or PD brushing to a control group of patients diagnosed with pancreatic cancer, mainly based on histology, but radiological follow-up, laparotomy, or other accepted diagnostic methods were included as well. A random-effects meta-analysis was performed to calculate the pooled sensitivity of each technique, with results presented as percentages and 95% confidence intervals (CI).

Results: Our systematic search identified 8375 articles, of which 64 met the inclusion criteria for the final analysis. This included 45 studies assessing BD brushing and 19 studies evaluating PD brushing. The pooled sensitivity for PD brushing was 55.8% (95% CI: 50.3% – 61.2%) with moderate heterogeneity ($Q = 24.84$, $df = 18$, $p = 0.129$). The pooled sensitivity for BD brushing was 40.9% (95% CI: 35.6% – 46.4%) with significant heterogeneity ($Q = 121.03$, $df = 44$, $p < 0.001$). Subgroup analyses were conducted to explore potential variations in diagnostic accuracy. However, data on the safety and adverse events associated with BD and PD brushing were limited among the included studies.

Conclusion: BD and PD brushing during ERCP provide valuable diagnostic yield in detecting pancreatic cancer. Despite BD brushing showing lower sensitivity and high heterogeneity among studies, both methods remain useful diagnostic tools. These findings support the integration of brush cytology in ERCP procedures to improve early pancreatic cancer detection, ultimately aiding in better clinical decision-making.

Disclosure: Nothing to disclose.

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PP1019

SMALL STONES, BIG DECISIONS: REAPPRAISING THE NEED FOR ERCP IN ACUTE BILIARY PANCREATITIS

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Introduction: Biliary stones are a leading cause of acute pancreatitis (AP). Endoscopic Retrograde Cholangiopancreatography (ERCP) is the standard of care when concomitant cholangitis or confirmed choledocholithiasis is present[1].

However, spontaneous migration of biliary stones in the latter occurs in approximately 20% of cases[2], potentially exposing patients to the inherent risks of ERCP without a definitive therapeutic benefit.

Aims & Methods: This study aimed to identify predictors of spontaneous common bile duct stone clearance in patients with acute biliary pancreatitis. We conducted a retrospective cohort study of consecutive patients diagnosed with acute pancreatitis and confirmed choledocholithiasis, who subsequently underwent ERCP. Patients with a prior ERCP or concurrent acute cholangitis were excluded.

Results: Forty-four patients were included, 61.4% female with a mean age of 67±18 years. In 31.8% of cases, no stone was identified at ERCP, suggestive of spontaneous resolution. The median interval from admission to ERCP was 10 days and did not significantly differ between patients with and without stones (p=0.872). Serum levels of total bilirubin (2.5±2.0 vs. 2.3±2.9 mg/dL, p=0.825), AST (160±31 vs. 171±50 U/L, p=0.505), and alkaline phosphatase (267±135 vs. 210±70 U/L, p=0.187) were not significantly different between groups with and without stones at ERCP, respectively.

Neither cholecystectomy status (p=0.722), common bile duct dilation on imaging (p=0.955), nor number of stones (p=0.829) correlated with spontaneous clearance. Interestingly, patients without lithiasis at ERCP had significantly smaller common bile duct stones on imaging compared to those with retained stones (5 ± 3 mm vs. 8 ± 5 mm, p=0.025). Stone size was an independent predictor of spontaneous resolution in the multivariate analysis (OR = 1.24; 95% CI: 1.01–1.52; p = 0.036), with a diameter ≥7 mm yielding a sensitivity of 63% and a specificity of 78.6% for predicting the presence of stones at ERCP.

Conclusion: Spontaneous passage of biliary stones is common in the context of acute biliary pancreatitis, particularly when their size is ≤7 mm. These findings support a risk-adapted strategy, in which Endoscopic Ultrasound may be employed to confirm choledocholithiasis immediately prior to ERCP in patients with small biliary stones.

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Disclosure: Nothing to disclose.

PP1020

CLINICAL OUTCOMES OF TARGETED ENDOSCOPIC TREATMENT OF PANCREATIC FISTULAS ACCORDING TO A PANCREATOGRAPHIC MORPHO-FUNCTIONAL CLASSIFICATION: A RETROSPECTIVE SINGLE-CENTER COHORT STUDY

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Introduction: Pancreatic fistulas (PFs) represent a remarkable clinical challenge after abdominal surgery, pancreatitis, or abdominal trauma, with significant morbidity and non-negligeable mortality. Pancreatic endoscopy plays a pivotal role in the setting of recent trend toward aggressive yet minimally invasive surgery-sparing PFs management.

Mutignani et al. recently proposed a morpho-functional pancreatographic classification of PFs aimed at targeting the best suited endoscopic treatment option for each PF type.

Aims & Methods: The primary aim of this study is to evaluate efficacy and safety of endoscopic PFs treatment, according to the Mutignani et al. classification; the secondary aim is the identification of independent variables significantly associated with different treatment clinical success rate.

Monocentric data prospectively collected between March 2011 and July 2024 on patients referred to the Interventional Endoscopic Unit of ASST Grande Ospedale Metropolitano Niguarda who underwent endoscopic treatment of a PF was retrospectively analysed.

Results: A total of 110 patients were included (M 75.5%, median age 61 years [IQR 46-72]). 75/110 (68.2%) were post-surgical (33 after pancreaticoduodectomy, 23 distal pancreatectomy, and 19 other abdominal surgery), 27/110 (24.5%) developed after pancreatitis, and 8/110 (7.3%) were post-traumatic. 17/110 (15.5%) were type I PF [9 (53%) T, 5 (29.4%) H, 3 (17.6%) B], 38 (34.5%) type II [27 (71.1%) O, 11 (28.9%) C], and 55 (50%) type III [35 (63.6%) D, 20 (36.4%) P].

Median treatment delay was 9 days [IQR 5-22]. ERCP was first endoscopic approach in 92 cases (83.6% - of which 6.5% after rendez-vous), EUS-guided in 14 (12.7%), and combined in 4 (3.6%). Mutignani et al classification adherence was 98.2%.

Technical success rate was 100% (91.8% at first approach), clinical success rate was 97/105 (92.4%). Early AEs occurred in 15 cases (14.3% - 33.3% of them infections), late AEs occurred in 8 (7.6% - 100% new pancreatic strictures). 9/105 (8.6%) patients died due to fistula-related causes during follow-up, PF recurrence occurred in 2 cases (2.1%).

At univariate analysis, older age, presence of heart diseases, presence of sepsis at clinical presentation, class C ISGPF fistula, and high-volume PF proved to be significantly more frequent among patients without clinical success compared to those with clinical success (median 72.5 [IQR 54.3-82.7] VS 59 [IQR 46-70.1], p=0.0312; 37.5% VS 2.3%, p=0.0038; 87.5% VS 39.8%, p=0.0199; 57.1% VS 14.3%, p=0.0182; 87.5% VS 42.7%, p=0.0230, respectively).

At multivariate analysis, only the presence of heart diseases confirmed to be significantly associated to the absence of clinical success (OR 2.65, 95%CI: 1.5-3.8, p=0.0217).

Conclusion: High adherence to the Mutignani et al classification system for PFs endoscopic treatment proved to high clinical success and low rate of AEs. Targeted management of PF is not only feasible but also highly effective, basically regardless of patient's and PF's baseline characteristics. Further large-scale prospective multicentric studies are warranted to confirm these findings and refine the optimal treatment strategies.

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Disclosure: Nothing to disclose.

PP1021

A DETOUR STENT APPROACH VIA EUS-GJ FOR REINTERVENTION IN BLOCKED BILIARY SEMS

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Introduction: Endoscopic reintervention in patients with malignant biliary obstruction and prior stenting can be challenging, particularly when conventional access routes are compromised. While ERCP remains the standard of care, EUS-guided approaches are increasingly being used as alternatives in difficult cases [1,2]. We present a case where a detour stent approach via an existing EUS-GJ provided successful access for biliary stent revision.
Aims & Methods: A 60-year-old male with advanced pancreatic cancer previously underwent percutaneous transhepatic biliary drainage (PTBD), placement of a biliary self-expandable metal stent (SEMS) for malignant biliary obstruction, and an EUS-guided gastrojejunostomy (EUS-GJ) for gastric outlet obstruction. He presented with fever and jaundice, and was diagnosed with acute cholangitis. Imaging suggested stent occlusion.
Endoscopy revealed a patent EUS-GJ but a narrowed and inaccessible duodenal lumen. A therapeutic endoscope was advanced through the EUS-GJ into the afferent limb. Under fluoroscopic guidance, a guidewire was passed through the distal end of the previously placed biliary SEMS. A balloon trawl was performed to clear debris, followed by cholangiography confirming partial stent occlusion. A 10 cm uncovered SEMS was deployed coaxially within the existing stent, restoring biliary drainage.
Results: The procedure was well-tolerated with no immediate complications. The patient showed clinical improvement, with resolution of cholangitis and normalization of liver function tests over the following days. This outcome is consistent with prior reports supporting EUS-GJ as a stable access route for advanced endoscopic interventions [2,6].
Conclusion: The detour stent approach through an existing EUS-GJ is a safe and effective rescue option for biliary reintervention in complex anatomical scenarios, especially in patients with malignant obstruction and prior interventions.
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Disclosure: Nothing to disclose.

PP1022

ENDOSCOPIC PAPILLECTOMY FOR LATERALLY SPREADING TUMORS

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Introduction: Endoscopic papillectomy (EP) is a standard treatment for ampullary lesions (AL), most of which are small and limited to the papillary mound. Laterally spreading lesions (LSL) of the papilla Vateri represent a rare subtype of AL, characterized by an extensive involvement of the surrounding duodenal mucosa. Data analyzing efficacy and complications of EP for LSL are scarce. In this study, EP for LSL were compared with non-LSL AL in thoroughly matched cohorts.
Aims & Methods: The ESAP study encompassed 1422 endoscopic papillectomies (EPs). Propensity-score matching used the nearest-neighbor method for age, gender, comorbidity, and histologic subtype as cofactors. The main outcomes were complete resection (R0), technical success, complications, and recurrences.
Results: Propensity-score-based matching identified 232 patients (116 non-LSL and 116 LSL AL) with comparable baseline characteristics. LSL sizes were significantly larger compared to non-LSL (median 27.0mm vs. 16.5mm, p<0.001). After first intervention, the R0-rate was significantly lower in the LSL group (54.3% vs. 69.0%, p=0.002). Following repeated endoscopic interventions (such as EP, radiofrequency ablation (RFA) or argon plasma coagulation (APC)), the technical success was comparable in both groups (82.8% vs 82.8%; p=1.00). After a median FU of 22 months, there were significantly more recurrences in the LSL group (41.3% vs. 15.0%, p<0.001). Complication rates did not differ significantly between the two groups (p=0.756).
Conclusion: LSL can be safely resected by EP, though repeated interventions are necessary to achieve complete resection. The higher risk of recurrence in LSL necessitates a vigilant surveillance strategy.
Disclosure: FUJIFILM Honoraria.

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PP1023

SINGLE-CENTER EXPERIENCE WITH ENDOSCOPIC MANAGEMENT OF DEMARCATED PANCREATIC WALLED-OFF PANCREATIC NECROSIS (WOPN)

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Introduction: The revised Atlanta Classification delineates the local complications of acute pancreatitis, categorizing them as acute (peri) pancreatic fluid collections (PFCs), acute necrotic collections, pancreatic pseudocysts, and walled-off pancreatic necrosis (WOPN). Various endoscopic interventions are employed for the management of WOPN, notably including direct transluminal endoscopic necrosectomy (DEN) following the placement of an endoscopic ultrasound-guided (EUS) lumen-apposing metal stent (LAMS).

This study aims to assess the clinical outcomes and safety profile of DEN in patients diagnosed with pancreatic WOPN.

Aims & Methods: This single-center retrospective analysis encompasses patients with pancreatic WOPN who underwent DEN at Semmelweis University between October 1, 2021, and October 1, 2024. The procedures were performed by three seasoned specialists in EUS and endoscopic retrograde cholangiopancreatography (ERCP).

Comprehensive data on patient demographics, procedural specifics, and follow-up outcomes were collected, with adverse events classified according to the AGREE criteria.

Results: We successfully placed LAMS in 31 patients (61% female, mean age 66.3 years, SD 9.4). The primary indications for intervention included the dimensions of WOPN, exacerbation of gastric outlet obstruction, and signs of infection. The mean dimensions of WOPN were recorded as 98 mm in the anteroposterior direction (SD 35.6), 90 mm laterolateral (SD 44), and 88 mm craniocaudal (SD 27.4), with a predominant occurrence in the head of the pancreas (60%). Alcohol consumption emerged as the leading etiological factor for WOPN in 92% of male patients, whereas biliary causes were identified in 67% of female patients.

In total, we conducted 83 DEN procedures across the 31 patients (51 males and 32 females). The average number of DEN sessions was three for both male and female groups (range: 2-6). The technical success rate achieved was 100%, accompanied by a clinical success rate of 97%. One patient succumbed to the condition; however, three of the surviving 30 required intensive care due to multiorgan failure.

To enhance the resolution of infection between DEN sessions, we employed 7 Fr/250 cm nasocystic tubes for irrigation of the WOPN with a saline solution in all patients, with no adverse events reported as a result of this intervention.

Conclusion: DEN can be performed safely in experienced medical centers and demonstrates a high success rate in the treatment of infected WOPN, effectively alleviating symptoms and decreasing enzyme levels. The predominant etiology of WOPN formation in male patients is attributed to alcohol consumption, while biliary factors are more prevalent in female patients.

Disclosure: Nothing to disclose.

PP1024

EFFICACY AND SAFETY OF PURASTAT® IN THE TREATMENT OF BLEEDING IN EUS-GUIDED DRAINAGE AND DIRECT ENDOSCOPIC NECROSECTOMY OF PANCREATIC FLUID COLLECTIONS

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Introduction: EUS-guided drainage using lumen-apposing metal stents (LAMSs) and direct endoscopic necrosectomy (DEN) are primary treatment modalities for symptomatic pseudocysts and walled-off pancreatic necrosis (WOPN). However, bleeding is a frequent and serious adverse event associated with these procedures, often requiring radiological or surgical interventions also due to the limitations of conventional endoscopic hemostatic techniques. PuraStat is a novel topical hemostatic agent that has demonstrated excellent results in managing gastrointestinal bleeding.

Aims & Methods: Our aim was to evaluate the efficacy and safety of PuraStat in managing bleeding secondary to EUS-guided drainage and DEN of pancreatic fluid collections (PFCs). In this multicenter, retrospective, observational study we described 27 patients who underwent PuraStat treatment because of active bleeding during or after EUS-guided drainage or DEN sessions of pseudocysts or WOPN between 2017 and 2024.

Results: 38 bleeding episodes were recorded (10 patients with WOPN experienced multiple bleeding events). The bleeding type was oozing in 36 cases and spurting in 2 cases. PuraStat was successfully applied in 100% of patients and achieved successful hemostasis in 96.5% of cases. No re-bleeding occurred. No PuraStat-related adverse events were recorded.

Conclusion: PuraStat is a safe, effective, and easy-to-use hemostatic agent for managing oozing hemorrhages related to PFCs EUS-guided drainage or DEN. Prospective and comparative studies are needed to confirm these preliminary data.

Disclosure: Nothing to disclose.

PP1025

ENDOSCOPIC ULTRASOUND WITH DETECTIVE FLOW IMAGING (DFI) FOR PREDICTING PANCREATIC NEUROENDOCRINE TUMOUR GRADE: SINGLE CENTRE EXPERIENCE

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Introduction: Pancreatic neuroendocrine tumors (pNETs) are a heterogeneous group of tumors with different treatment options depending on pathological grading, clinical staging, and presence of symptoms related to hormonal secretion. Their classification is made on mitotic count and Ki-67 index.

The current standard for Grading assessment is analysing the material obtained from EUS-guided fine needle aspiration/biopsy (EUS-FNA/FNB). However this technique is not free from adverse events and could be challenging for NETs localization.

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The evidence of high grade of fibrosis and microvascular density decrease (MVD) on tumor pathological specimens has led many authors to search for undirected malignant features of pNET extrapolated from non-invasive techniques. The introduction of ancillary techniques, such as second-generation contrast agents (CH-EUS), has improved study of pNET. Particularly, at CH-EUS examination contrast pattern well correlates with behaviour and grading of NETs. However, the use of contrast agents entails a prolongation of procedure time and additional costs. An innovative ultrasound Doppler technique, Detective Flow Imaging EUS (DFI-EUS), involves no additional time or cost as it doesn't require the administration of contrast agent. There is still no standardisation of this technique and few studies have validated its use.

Aims & Methods: The aim of this study is to evaluate the DFI role in defining NET vascularization and its correlation with CH-EUS. The second aim of the study is to assess DFI role in predicting histological grading.

Single center retrospective analysis on consecutive patients with solid pancreatic lesions who underwent EUS at Therapeutic Digestive Endoscopy Unit of the Fondazione Policlinico Universitario Campus Bio-Medico in Rome.

After activation of DFI, each lesion was observed for at least 30 seconds. Lesions was classified into three grades, according to the degree of vascularisation at DFI: (grade 0 absence of microvascularization of the tumor; grade 1 minimal/isovascularization of the tumor; grade 2 markedly increased microvascularization).

All lesion with final hystological diagnosis of NET were enrolled.

Grading and Ki-67 index of tumors were reported.

Data about lesion's features and procedure were collected.

Results: From April 2024 to February 2025, 17 patients were enrolled (58.8% male; mean age 57±12 years old). The majority of lesion (53%) were localized in the body of the pancreas and the mean lesion' s size was 28±20 mm. The final histological diagnosis classified 50% lesions as G1, 38.8% G2 and 5.5% G3.

At the CEH-EUS the 70.6% of lesion showed hyper-enhancement. About the DFI evaluation the 17.6% of lesion was hypo-vascular, 17.6% Iso-vascular and 64.7% was hyper-vascular. There was excellent correlation between DFI characterization of the lesion and the CH-EUS (ICC 0.9)

Among the NET G1, 66.7% were Hypervascular at DFI and 33.3% Isovascular. Among the G2, 71.4% were Hypervascular at DFI and 28.6% Hypovascular. The only G3 lesion was Hypovascular.

There was inverse correlation between the NET grade and the DFI grade (Pearson coefficient -0.56; p 0.02).

Conclusion: DFI-EUS can provide a fast evaluation of the micro-vascularization of pNET with excellent correlation with CH-EUS. Hyper-vascular pattern at DFI correlates with low Ki-67 index.

Disclosure: Nothing to disclose.

PP1026

SENSITIVITY OF ENDOSCOPIC ULTRASOUND-GUIDED FINE NEEDLE ASPIRATION (EUS-FNA) IN THE DIAGNOSIS OF PANCREATIC NEUROENDOCRINE TUMORS: A PROSPECTIVE RANDOMIZED TRIAL

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Introduction: Pancreatic neuroendocrine tumors (NETs) are rare neoplasms that are classified as functional (F-NET) or nonfunctional (NF-NET) depending on the presence or absence of excess hormone production. Endoscopic Ultrasound guided fine needle aspiration (EU-FNA) is often used to diagnose NETs but there are limited data on its sensitivity for this purpose.

Aims & Methods: To determine the sensitivity of EUS-FNA for the diagnosis of a NET.

A prospective randomized trial study, from May 2013 to Abril 2025 was performed in a referred Gastrointestinal hospital in Brazil. A total of 184 patients underwent EUS for suspected NETS and collected materials from fine needle aspiration and sent to histopathology analyzes.

The diagnosis of a NET was confirmed by:

1. EUS-FNA with or without confirmatory immune chemistry (ICC) with synaptophysin and chromogranin form pancreatic mass or metastatic site, or;
2. An alternative biopsy method and/or surgical resection of the pancreas or other metastatic site

Results: 184 patients (112 male; median age: 54 yrs, range: 21-78 yrs) with 68 (37%) F-NETs and 116 (63%) NF-NETs were identified. By EUS, the tumors were located in the pancreas head, body, tail or were multifocal in 76 (41%), 54 (29%), 38 (21%) and 16 (18%), respectively.

Of these 184, 152 (83%) underwent EUS-FNA of a pancreatic mass, 72 (47%) of whom also had a EUS-FNA of an alternative (nonpancreatic) site including liver (n=30; 16%), another lesion in the pancreas (n=14; 8%), or lymph node (n=18; 10% peripancreatic in 10, celiac in 6 and mediastinal in 2).

Immunohistochemistry was performed in 102/152 (67%) EUS-FNA specimens and the result were: ICC-positive (n=88), ICC-negative (n=10), not interpretable (n=2) or inadequate for interpretation (n=2), Surgery was performed in 82 (54%): distal pancreatectomy (n=32), Whipple procedure (n=20), enucleation (n=20) and subtotal (n=4) and total pancreatectomy (n=2).

The remaining four who underwent surgery had an exploration laparotomy only. 132 patients had a confirmed diagnosis of NET and the sensitivity of EUS-FNA was respectively: all patients 84%, F-NET 78%, NF-NET (89%) and lesions less than or equal to 13 mm (62%). The results of the false-negative EUS-FNA biopsies of pancreas (n=24) were interpreted as: normal in 4; suspicious in 4, nondiagnostic in 10, adenocarcinoma in 4, and a solid pseudopapillary tumor in 2.

Conclusion: EUS-FNA is a sensitive method for the diagnosis of pancreatic NETS but yield is lower in tumor smaller than 13mm in diameter.

Disclosure: Nothing to disclose.

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PP1027

IMPACT OF MACROSCOPIC ON-SITE EVALUATION (MOSE) ON ACCURACY OF ENDOSCOPIC ULTRASOUND-GUIDED FINE-NEEDLE BIOPSY (EUS-FNB) OF SOLID LESIONS: A RANDOMISED CONTROLLED TRIAL.

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Introduction: Endoscopic ultrasound (EUS) tissue acquisition requires rapid on-site evaluation (ROSE) to improve diagnostic yield with less needle passes; however, ROSE requires the presence of a cytopathologist and additional cost and time for slide staining and interpretation. Macroscopic on-site evaluation (MOSE) is the direct assessment of the tissue sample provided by EUS-guided fine needle biopsy (EUS-FNB) macroscopically, to detect whether it is sufficient or not.

Aims & Methods: Aim of this study was to evaluate the efficacy of MOSE in terms of tissue acquisition and diagnostic accuracy for abdominal solid lesions

This is a prospective and comparative study including 148 consecutive patients performing EUS-FNB for pancreatic and extrapancreatic solid lesions, randomized by alternate allocation to macroscopic on-site evaluation (MOSE) (74 patients) or to a conventional technique (74 patients), with two passes each. MOSE samples were differentiated into score 0: no visible material, score 1: only necrotic or haematic material, score 2: white core tissue ≤ 2 mm, or score 3: white core tissue > 2 mm.

The conventional technique consisted in pushing all the needle content into a test tube for evaluation by the pathologist. In both groups, a 22 Gauge Franseen-tip needle biopsy was used.

Results: The overall diagnostic adequacy was 92.5% (137/148) and accuracy was 86.4% (128/148). The diagnostic accuracy was similar in the two groups: MOSE 87.8% (65/74 lesions), vs. conventional 81.0%, 60/74 lesions, $p = 0.34$). Regarding diagnostic adequacy, the MOSE technique was 95.9% (71/74 passes) compared to 91.8% (136/148 passes) with the conventional technique, $p = 0.04$.

The accuracy increased according to the MOSE score evaluation: it was 49.3%, 68.9% and 81.3% in patients with score 1, score 2, and score 3, respectively. There was no statistically significant difference between the two groups with conclusive and inconclusive cytopathological results regarding size of the needle, number of needle passes, type of the needle and the specimen acquisition method.

Conclusion: The MOSE score showed a comparable diagnostic accuracy to the conventional technique. However, MOSE allows endoscopists to perform an inspective evaluation of the material, tends to perform better than the conventional technique in terms of diagnostic adequacy, and may potentially reduce the number of passes.

Disclosure: Nothing to disclose.

PP1028

ROLE OF PCR AND GENE X-PERT IN DIAGNOSING EXTRAPULMONARY TUBERCULOSIS AND TUBERCULOUS LYMPHADENOPATHY IN TISSUE SAMPLES OBTAINED BY EUS-FNB

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Introduction: Tuberculous lymphadenopathy represents a challenging diagnostic scenario, particularly in regions with high TB prevalence. Its diagnosis is challenging due to the nonspecific clinical and radiological findings. Blood and skin tests for TB have variable sensitivities and specificities, so histopathologic diagnosis and molecular tests for diagnosis of TB in tissue samples may be more accurate than blood tests. Obtaining tissue samples under ultrasonography or CT guidance is sometimes very difficult and risky especially in the intra-abdominal and mediastinal regions.

Endoscopic ultrasound (EUS) with EUS fine needle biopsy (EUS-FNB) is a safe technique in obtaining tissue samples for diagnosis of abdominal and mediastinal TB. EUS introduces a critical advantage of real-time imaging and precise tissue sampling. Many studies found that EUS-FNB yielded significantly higher diagnostic yield in complex TB presentations, with detection rates ranging from 78-92% across various anatomical sites.

Comparative analysis with Gene X-pert MTB/RIF assay and Polymerase Chain Reaction (PCR) techniques demonstrated nuanced diagnostic capabilities. Gene X-pert provides rapid molecular detection with high specificity for rifampicin resistance while PCR offers molecular amplification of bacterial DNA.

Aims & Methods: In this prospective cross-sectional study, we explored the diagnostic efficacy of molecular techniques, specifically PCR and Gene X-pert MTB/RIF assay, in detecting Mycobacterium tuberculosis in tissue samples obtained through EUS-FNB. We also aim to evaluate the sensitivity, specificity, and rapid detection capabilities of these molecular methods compared to traditional diagnostic techniques.

Patients and Methods: This study included 56 patients, 28 males and 28 females with mean age of 50 ±16 years. TB PCR or Gene X-pert was done in tissue samples including mediastinal and intra-abdominal lymphadenopathy in 75% of cases, obtained by EUS-FNB. Results: PCR or Gene X-pert for TB were true-positive in 7 cases and false negative in one case with sensitivity of 88%, specificity of 100%, positive predictive value of 100% and negative predictive value of 98%.

Results: PCR or Gene X-pert for TB were true-positive in 7 cases and false negative in one case with sensitivity of 88%, specificity of 100%, positive predictive value of 100% and negative predictive value of 98%.

Conclusion: We concluded that PCR and Gene X-pert of TB in tissues obtained by EUS-FNB are of great value in diagnosis of extrapulmonary TB.

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Disclosure: Nothing to disclose.

PP1029

PREDICTORS OF CLINICAL SUCCESS IN EUS-GUIDED DRAINAGE OF POSTOPERATIVE PANCREATIC FLUID COLLECTIONS

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Introduction: There has been a shift from percutaneous drainage towards endoscopic ultrasound (EUS)-guided drainage for management of post-operative pancreatic fluid collections (POPFC). However, available series have been limited by sample size and little is known about the factors that may influence clinical success.
Aims & Methods: Our aim was to evaluate the outcomes of EUS-guided drainage of POPFC following pancreatic surgery and identify predictors of clinical success.
We performed a retrospective review of patients who underwent EUS-guided drainage of POPFC between 2014 and August 2024 at three academic centers. Primary outcome was clinical success, defined as the complete radiological regression of the collection and resolution of symptoms without the need for additional drainage procedures.
Results: In total, 265 cases were included (mean age 64 [SD±13.9] years, 37% pylorus-resecting pancreatoduodenectomy). Early drainage (≤14 days) was performed in 52% of cases (n=138) and DPPS were used in 209 cases (79%).
Clinical success with a single EUS drainage procedure was achieved in 160 cases (60.4%). Step-up interventions were required in 67 patients (25.3%), 37 of which required endoscopic revision only. Technical success was achieved in 90% (n=238) and adverse events occurred in 43 cases (16%).
Stent type, drainage timing, splenectomy and drainage route did not seem to significantly influence clinical success. Multivariate analysis including stent type and surgical history, showed that distal pancreatectomy was an independent risk factor for clinical failure.
Conclusion: EUS-guided drainage of POPFC was effective in the majority of cases with a single procedure, although distal pancreatectomy was suggested as an independent risk factor for clinical failure. These results further support the use of EUS-guided drainage for the management of postoperative collections.
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PP1030

ARTIFICIAL INTELLIGENCE-POWERED CONTRAST-ENHANCED ENDOSCOPIC ULTRASOUND FOR DIFFERENTIAL DIAGNOSIS OF PANCREATIC ADENOCARCINOMA

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Introduction: The use of contrast-enhanced endoscopic ultrasound (CE-EUS) for characterizing solid pancreatic masses has improved diagnostic capabilities. We aimed to evaluate the performance of an artificial intelligence (AI)-powered system for differential diagnosis of pancreatic adenocarcinoma using CE-EUS.
Aims & Methods: The methodology integrates advanced imaging analysis and machine learning (ML) to develop an automated framework for pancreatic tumor characterization using EUS. A convolutional neural network (CNN) is first trained on labeled non-contrast EUS images to detect and segment the pancreas and tumor regions accurately. Once trained, the CNN is applied to new cases, identifying regions of interest (ROIs) in the non-contrast images, which are subsequently mapped onto the corresponding contrast-enhanced images for further analysis. Time-Intensity Curve (TIC) analysis was performed on the mapped ROIs, extracting a comprehensive set of quantitative parameters describing contrast enhancement dynamics.
These parameters include temporal metrics (e.g., time to peak, rise time), intensity measures (e.g., peak intensity, wash-in intensity), slopes (e.g., wash-in and wash-out rates), area-based metrics (e.g., area under the curve), and perfusion-related characteristics (e.g., perfusion index, blood flow). The extracted TIC parameters are then used to train a feedforward neural network (FNN), which learns to differentiate diagnostic categories based on the perfusion characteristics of the lesions.
In the final integrated model, the CNN detects pancreas and tumor regions, maps them to contrast-enhanced images, computes TIC parameters, and feeds these into the FNN to provide automated diagnostic predictions.

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Results: Retrospective analysis was performed on a prospectively collected dataset of 120 CE-EUS videos obtained from patients with solid pancreatic masses in 3 tertiary referral centers. The dataset included 88 pancreatic adenocarcinomas, 1 lymphoma, 4 metastatic lesions, 10 neuroendocrine neoplasms (G1/G2), 15 cases of chronic pancreatitis, and 2 solid pseudopapillary neoplasms (SPNs).

Diagnostic performance was evaluated by calculating sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy. Based on CNN analysis, both precision and recall were also calculated.

The AI-powered CE-EUS system achieved high diagnostic performance, with an accuracy of over 95% for the diagnosis of pancreatic adenocarcinoma. Precision and recall were also excellent for differentiating adenocarcinoma from other pancreatic lesions based on CNN analysis.

Conclusion: This novel AI-based approach using CE-EUS reliably distinguished pancreatic adenocarcinoma from other solid pancreatic masses. Real-time analysis of time-intensity curve parameters enhances diagnostic accuracy and offers a promising adjunct to traditional EUS methods. Further prospective validation is warranted to confirm these findings and explore clinical applicability.

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Disclosure: Nothing to disclose.

PP1031

INTEROBSERVER AGREEMENT IN CONTRAST-ENHANCED EUS: WATCHING THE SAME MOVIE? RESULTS FROM A NGEUS MULTICENTRIC EUROPEAN STUDY

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Introduction: Contrast-enhanced endoscopic ultrasound (CE-EUS) is a valuable tool for evaluating gastrointestinal lesions, providing real-time insights into vascular patterns and tissue perfusion. It is increasingly utilized to differentiate solid and cystic lesions, particularly in pancreaticobiliary diseases.

However, CE-EUS interpretation remains operator-dependent, with variability influenced by observer expertise and a lack of standardized criteria.

Previous studies on interobserver agreement (IOA) in CE-EUS interpretation have yielded inconsistent results, underscoring the need for further investigation into dynamic contrast features such as enhancement patterns and wash-in/wash-out phases.

Aims & Methods: This study aimed to evaluate IOA among endosonographers with varying levels of expertise in interpreting CE-EUS videos, focusing on lesion classification, dynamic contrast features, and video quality. This multicenter retrospective study involved 24 endosonographers (15 experts, 9 non-experts) from 16 European centers. Participants completed an initial survey capturing their experience and institutional characteristics.

Subsequently, they independently reviewed 60 CE-EUS videos (~1 minute each), recorded between 2021 and 2023, comprising 40 solid, 15 cystic, and 5 mixed lesions. Videos varied in quality. Observers assessed lesion type (solid or cystic), arterial and venous enhancement patterns (hyper-, hypo-, or non-enhancing), wash-in and wash-out timing (rapid or late), contrast distribution (homogeneous or inhomogeneous), and video quality. Fleiss' kappa (κ) was used to calculate IOA for each parameter.

A multivariate analysis was conducted to identify factors influencing IOA.

Results: Moderate agreement was found for lesion classification (κ = 0.47). IOA for arterial phase enhancement was slight (κ = 0.38), as was agreement for arterial wash-in timing (κ = 0.27), venous wash-out timing (κ = 0.21), and contrast distribution homogeneity (κ = 0.24). No agreement was reached for video quality assessment (κ = 0.14). Operator experience was not significantly associated with IOA.

Multivariate analysis revealed that IOA for wash-in timing was influenced by procedural volume at the observer's center and the type of endoscopic equipment used (p = 0.004 and p = 0.034, respectively).

Additionally, center volume significantly affected perceived video quality assessment (p = 0.035).

Conclusion: This study demonstrates considerable variability in CE-EUS interpretation among observers, particularly for dynamic contrast features, where agreement was slight. Lesion classification showed moderate reproducibility, but standardized evaluation criteria are needed for other parameters. Procedural volume at the observer's center significantly influenced IOA, emphasizing the importance of institutional experience. These findings highlight the need for improved training programs and interpretation frameworks to enhance the diagnostic consistency and clinical utility of CE-EUS.

Disclosure: Nothing to disclose.

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PP1032

ROLE OF 2D SHEAR WAVE ELASTOGRAPHY IN ENDOSCOPIC ULTRASOUND EVALUATION OF NORMAL PANCREAS AND SOLID PANCREATIC MASSES

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Introduction: Endoscopic ultrasound (EUS) elastography has emerged as a valuable tool for assessing pancreatic tissue stiffness, aiding in the differentiation of normal pancreas, chronic pancreatitis (CP), and solid pancreatic masses, including pancreatic adenocarcinoma (PDAC) and cholangiocarcinoma (CC). The newly developed two-dimensional shear wave elastography (2D-SWE) provides real-time, quantitative elasticity maps with improved reproducibility.

This study aims to evaluate the performance of 2D-SWE during EUS for differentiating normal pancreas, CP, and solid pancreatic tumors.

Aims & Methods: This prospective study included patients undergoing EUS for pancreatic evaluation. All 2D-SWE measurements were made using the Aplio i800 EUS ultrasound machine. Four groups were analyzed: (1) individuals with a normal pancreas, (2) patients with CP (based on Rosemont criteria), (3) patients with PDAC, and (4) patients with CC (all diagnoses confirmed histopathologically). 2D-SWE was performed during EUS using a linear-array echoendoscope. Quantitative elasticity values (kPa) were recorded for the normal pancreas, CP-affected tissue, and tumor lesions. The diagnostic performance of 2D-SWE was assessed using intraclass correlation coefficient (ICC), receiver operating characteristic (ROC) analysis, and pairwise statistical comparisons.

Results: A total of 30 patients were included (6 normal, 5 CP, 17 PDAC, 2 CC). The intra-observer reproducibility of SWE measurements was excellent (ICC = 0.89). Mean elasticity values significantly differed among groups, with PDAC showing the highest variability (mean SWE = 54.1 ± 45.7 kPa, median = 33.1), CC presenting elevated stiffness (mean SWE = 77.4 ± 11.0 kPa), CP showing heterogeneous values (mean SWE = 57.5 ± 52.3 kPa, median = 43.6), and normal pancreas showing the lowest stiffness (mean SWE = 7.5 ± 3.9 kPa, median = 6.2).

ROC analysis for distinguishing malignant solid tumors (PDAC + CC) from normal pancreas showed excellent diagnostic performance (AUC = 0.93). The optimal cutoff of 17.0 kPa yielded a sensitivity of 84.2%, specificity of 100%, positive predictive value (PPV) of 100%, negative predictive value (NPV) of 66.7%, and overall accuracy of 88%.

Pairwise t-tests confirmed a statistically significant difference between PDAC and normal pancreas (p = 0.0007), with trends observed for CC vs normal (p = 0.0605) and CP vs normal (p = 0.099). No statistically significant differences were observed among solid tumor subtypes (PDAC vs CC, PDAC vs CP, CC vs CP).

Conclusion: 2D-SWE during EUS provides highly reproducible and quantitative elasticity values that reliably differentiate normal pancreas from solid pancreatic tumors. This method demonstrates excellent diagnostic performance, particularly in identifying PDAC, and shows promising trends for distinguishing CP and CC, supporting its clinical integration in EUS evaluation protocols.

Disclosure: Nothing to disclose.

PP1033

EVALUATING THE DIAGNOSTIC ACCURACY OF EUS-FNB COMPARED TO ON-SITE EVALUATIONS FOR DIAGNOSING PANCREATIC NEOPLASIAS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Endoscopic ultrasound-guided fine needle biopsy (EUS-FNB) is the primary diagnostic tool for assessing solid pancreatic lesions. In case of pancreatic neoplasia, the time taken to obtain an accurate diagnosis is critical; therefore, it is essential to acquire an efficient histological sample. Techniques such as Rapid On-Site Evaluation (ROSE) and Macroscopic On-Site Evaluation (MOSE) may enhance the diagnostic accuracy of EUS-FNB and increase the proportion of histologically adequate samples. However, the necessity of these on-site evaluation techniques remains a topic of debate.

Aims & Methods: This study aims to determine whether these techniques should be incorporated into routine practice.

Five medical databases (PubMed, EMBASE, CENTRAL, Scopus, and Web of Science) were searched comprehensively for studies comparing the diagnostic accuracy and sample adequacy data of EUS-FNB ± ROSE and/or MOSE in solid pancreatic lesions. In the meta-analysis, pooled proportion rates with 95% confidence intervals (CIs) were calculated. The comparative analysis part of this study is still ongoing. PROSPERO registration number: CRD42024613021.

Results: Altogether, we included 4562 patients from 31 articles in our quantitative and qualitative synthesis. The pooled diagnostic accuracy and sample adequacy rate for EUS-FNB without ROSE and/or MOSE were 90% (95%CI: 86-93) and 94% (95%CI: 82-98), respectively. Using ROSE, these values were 95% (95%CI: 93-97) and 97% (95%CI: 90-99), while in the case of the MOSE technique, these values were 93% (95%CI: 92-94) and 98% (CI: 95-99), respectively. In the MOSE group, diagnostic accuracy using 22G needles was significantly higher (94% [95%CI: 92-96] vs 91% [95%CI: 89-94], [p = 0,027]) compared with other needle types.

Conclusion: In conclusion, on-site evaluation techniques may enhance diagnostic accuracy and improve EUS-FNB sample adequacy rates. The findings from the ongoing comparative analysis are crucial in determining the potential adoption of these techniques.

Disclosure: Nothing to disclose.

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PP1034

IMPROVING POST-TEST PROBABILITY FOR DIAGNOSIS OF PANCREATIC DUCTAL ADENOCARCINOMA USING A MULTIMODAL ENDOSCOPIC ULTRASOUND APPROACH: A SINGLE CENTER RETROSPECTIVE STUDY INTEGRATING CONTRAST-ENHANCED ENDOSCOPIC ULTRASOUND, STRAIN RATIO ELASTOGRAPHY AND DETECTIVE FLOW IMAGING

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Introduction: Pancreatic cancer is one of the most aggressive malignancies, with a poor five-year survival rate. Therefore, optimizing diagnostic strategies is a critical priority. Endoscopic ultrasound (EUS) plays a key role in characterizing focal pancreatic masses (FPM). The development of ancillary EUS techniques including contrast-enhanced EUS (CH-EUS), strain ratio elastography (SRE) and Detective Flow Imaging (DFI) has enhanced the assessment of vascularization and tissue stiffness, potentially improving the differentiation between benign and malignant lesions. Emerging evidence suggests that these parameters may provide valuable insights into the malignant potential of FPM [1,2,3].

This study evaluates whether the combined use of CH-EUS, SRE, and DFI improves diagnostic accuracy in predicting pancreatic ductal adenocarcinoma (PDAC).

Aims & Methods: Single-center, retrospective study including consecutive patients who underwent biliopancreatic EUS for FPM between October 2023 and April 2025 at the Digestive Diseases Unit of Sant'Andrea University Hospital, Rome. During EUS evaluation, CH-EUS, SRE, and DFI were performed to characterize lesions.

The final diagnosis was confirmed by histopathology (EUS fine-needle biopsy) and/or radiological workup (Computed Tomography, Magnetic Resonance Imaging, Positron Emission Tomography), with a follow-up of at least 6 months.

For CH-EUS, lesions were categorized as hypoenhanced, isoenhanced or hyperenhanced compared to surrounding parenchyma in the early arterial phase after injection of 8 mL of sulphur hexafluoride microbubbles, a second-generation ultrasound contrast agent. Hypoenhanced lesions were considered predictive of PDAC [4].

SRE was assessed using the ultrasound system's embedded software, comparing a region of interest (ROI) in the FPM with the intestinal wall. A strain ratio>10 was considered predictive of PDAC [5,6].

DFI was employed to evaluate the microvascularization, classifying lesions as hypovascular or hypervascular, with hypovascularity considered predictive of PDAC [7,8].

Results: Overall, 59 patients were included, 27 (45.8%) females, with a median age of 69 years (27–88). About lesions 37 (62.7%) were in the head, 14 (23.7%) in the body, and 8 (13.6%) in the tail, with a median diameter of 25 mm (8–63 mm).

Among these, 41 (69.5%) were PDAC, 3 (5.1%) were inflammatory lesions, 2 (3.4%) were metastasis and 13 (22%) were neuroendocrine tumors (G1-G2). CH-EUS was performed in 34 cases (57.6%), SRE in 29 (49.2%), and DFI in 37 (62.7%).

All three modalities were applied in 20 (33.9%) cases and combinations of DFI with another technique were used in 13 (22.1%).

The sensitivity, specificity, accuracy, positive predictive value (PPV) and negative predictive value (NPV) for PDAC of each modality, alone and in combination, are summarized in Table 1.

Modality	Sensitivity (%)	Specificity (%)	Accuracy (%)	PPV (%)	NPV (%)
CH-EUS (hypoenhanced)	63.2	86.7	73.5	85.7	65.0
SRE (strain ratio >10)	88.2	75.0	82.8	83.3	81.8
DFI (hypovascularization)	83.3	84.6	83.8	90.9	73.3
CH-EUS + DFI (n = 27)	62.5	90.9	74.1	90.9	62.5
SRE + DFI (n = 24)	66.7	100.0	79.2	100.0	64.3
CH-EUS + SRE + DFI (n = 20)	63.6	100.0	80.0	100.0	69.2

Table 1: Diagnostic performance of ancillary techniques and their combination for PDAC detection.

Conclusion: Multimodal approaches by using ancillary EUS techniques (CH-EUS, SRE and DFI), particularly combinations including DFI, showed a high specificity and PPV in predicting malignant FPM. Therefore, a multimodal EUS approach may improve post-test probability for PDAC.

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Disclosure: Nothing to disclose.

PP1035

ADEQUACY OF EUS-FNA AND FNB SAMPLES FOR GRADING PANCREATIC NEUROENDOCRINE TUMORS ≤ 20 MM: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: In recent years, both the incidence and early-stage diagnosis of pancreatic neuroendocrine tumors (pNETs) have increased, likely due to advancements in cross-sectional imaging techniques (1). Management strategies are size-dependent, with observation generally preferred

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for lesions <10 mm and surgical resection for those >20 mm. However, there is no consensus regarding the optimal approach for tumors between 10 and 20 mm, highlighting the need for multidisciplinary decision-making (2).

In this setting, endoscopic ultrasound (EUS) with fine-needle aspiration (FNA) or fine-needle biopsy (FNB) plays a crucial role not only in diagnosis but also in tumor grading (3).

However, samples obtained through EUS-FNA or FNB may not provide sufficient histological material to determine tumor grade in pNETs smaller than 20 mm. This study aims to summarize the current evidence regarding the adequacy of histological samples obtained via EUS-FNA/FNB for tumor grading.

Aims & Methods: This study followed the PRISMA guidelines for Systematic Reviews and Meta-Analyses. MEDLINE, EMBASE, and SCOPUS databases were searched up to December 2024 for studies reporting on the sample adequacy of EUS-FNA or FNB for grading pNETs ≤ 20 mm. Sample adequacy was defined as the proportion of samples in which the Ki-67 index was successfully obtained for tumor grading among all samples collected. Pooled estimates were calculated using random-effects models. Heterogeneity was assessed using the I^2 statistic. Subgroup analyses were conducted based on needle type (FNA vs. FNB).

Results: 17 studies were included, analyzing a total of 1414 patients. The pooled adequacy rate was 71.6% (95% CI: 0.592–0.841), with substantial heterogeneity across studies (I^2 = 98.69%). The risk of publication bias was considered low based on Egger's test (p = 0.162). When considering only studies that used FNA, adequacy rates were lower, with a pooled estimate of 52.5% (95% CI: 0.302–0.748; I^2 = 97.99%).

In contrast, studies employing FNB showed higher adequacy rates, with a pooled estimate of 79.8% (95% CI: 0.605–0.991), although heterogeneity remained high (I^2 = 87.32%). In this subgroup, there was also evidence of publication bias, as suggested by Egger's test (p = 0.017) and visual asymmetry in the funnel plot.

Conclusion: Our study found that EUS-FNA/FNB provides adequate histological specimens for tumor grading in pNETs < 20 mm, although significant heterogeneity was observed. As expected, FNB demonstrated considerably better performance than FNA in establishing tumor grade, suggesting it should be preferred for pNET sampling.

However, even with FNB, tumor grade could not be determined in up to 20% of cases, indicating that it remains a suboptimal tool for grading and may offer limited additional value beyond diagnostic confirmation in some patients.

Notably, our study did not distinguish between first-, second-, and third-generation FNB needles; therefore, results from newer-generation FNBs may show improved sample adequacy rates.

Furthermore, the observed heterogeneity highlights the need for studies specifically evaluating the ability of EUS-FNB to obtain high-quality tissue for grading, thereby supporting the role of EUS in clinical decision-making for these small tumors.

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Disclosure: Nothing to disclose.

PP1036

ENDOSCOPIC ULTRASOUND GUIDED DRAINAGE WITH LUMEN-APPOSING METAL STENT VERSUS PLASTIC STENT FOR THE TREATMENT OF WALLED-OFF NECROSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Symptomatic walled-off necrosis (WON) after acute pancreatitis (AP) can be treated endoscopically through an endoscopic ultrasound (EUS) guided transmural drainage. It is still not clear whether the use of lumen-apposing metal stents (LAMSs) instead of plastic stents might lead to better clinical outcomes.

Aims & Methods: Our aim was to perform a meta-analysis of the existing randomized controlled trials (RCTs) comparing these two techniques. Following PRISMA statement after PROSPERO registration, a literature search was performed on Embase, PubMed, Scopus and Google scholar, inception date in October 2024, including only RCTs (with full-text available) directly comparing EUS-guided LAMSs vs plastic stents in the treatment of WON after AP. Exclusion criteria included studies other than RCT, not directly comparing the 2 techniques, not on humans and not in English. Quality of the studies was assessed through the Risk of Bias 2 (Rob2) tool. The primary outcome was the long-term (>1 month) clinical success. Secondary outcomes included short-term (<1 month) clinical success, technical success, total number of direct endoscopic necrosectomies (DEN), total number of interventions, need for surgery, mortality, bleeding, stent migration, procedure duration, length of hospital stay, overall adverse events, stent-related adverse events, sepsis, recurrence, percutaneous drainage (PCD) placement. A trial sequential analysis was performed to identify the required information size (RIS) for the primary outcome and for the number of DEN.

Results: A total of 3456 articles were screened, and 6 RCTs were included in the final analysis, with a total number of 352 patients, of which 177 patients in the metal stent (LAMS) group and 175 patients in the plastic stent group. Post-procedural sepsis resulted significantly lower in the metal stents (LAMS) group (RR 0.45, 0.24-0.84; p= 0.01; I^2= 0%) and procedure duration resulted significantly shorter in the metal stent (LAMS) group (mean difference -12.20, -19.30 to -5.10; p= 0.0008; I^2= 96%).

No significant differences were found in all the other primary and secondary outcomes. At trial sequential analysis, RIS for long term clinical success was 23609 patients, while RIS for DEN was 4412 patients.

Conclusion: Except for post-procedural sepsis and procedure duration, there was no difference in terms of clinical outcomes between the use of metal (LAMS) or plastic stents for EUS-guided drainage of WON. Advantages of shorter procedural duration could translate to the preferential use of metal stents in sicker, infected cohort of WON patients.

Disclosure: Nothing to disclose.

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PP1037

HARNESSING THE ROLE OF ENDOSCOPIC ULTRASOUND IN PEDIATRICS: A TERTIARY REFERRAL CENTER EXPERIENCE IN PAKISTAN

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Introduction: Endoscopic ultrasound (EUS) emerged as a revolutionary intervention in gastroenterology, it has been extensively utilized among adult patients, however EUS utility in pediatric patients has remained limited due to EUS endoscope design which includes rigidity, diameter, traumatic distal tip, further more complicated by need of Monitored Anesthesia Care (MAC) or General Anesthesia (GA) for intervention which require specialized pediatric anesthesia team[i].

Fewer studies have been conducted so far regarding EUS utility, effectiveness and safety among children, most of the studies have smaller data. Recently, the European Society for Pediatric Gastroenterology Hepatology and Nutrition (ESPGHAN) and the European Society of Gastrointestinal Endoscopy (ESGE) have published joint guidelines for EUS in children which highlights the various indications for EUS in children[ii].In Pakistan, only few endoscopy centers are performing pediatric EUS.

Aims & Methods: This article aims to describe the safety along with technical and clinical success of EUS for both diagnostic and therapeutic purposes among children at a tertiary referral endoscopy center in Pakistan.

Retrospective charts review of patients ≤ 18 years of age who underwent EUS for disgnostic or therapeutic purpose over 10 years, from 2015 to 2024, at Sindh Institute of Advanced Endoscopy and Gastroenterology. This institute was initially working under Civil Hospital Karachi as Endoscopy Suite of S4 Unit. Since May 2023, it has been working as a separate endoscopy unit. This endoscopy center is a referral unit located in the largest city of Pakistan.

Results: A total of 130 patients ≤ 18 years age had EUS, the youngest was 5 years old and 54.3% were male gender. Primary indication for EUS examination included: 35 chronic pancreatitis, 39 pancreatic pseudocyst, 12 pancreatic mass, 3 biliary obstruction with suspected choledochal cyst, 2 liver biopsy, 21 gastrointestinal tract lesions, 6 lymph nodes and 12 other causes. 96 EUS were performed for diagnostic purpose, in which 20 patients underwent fine needle biopsy and 6 underwent fine needle aspiration.

Among therapeutic EUS, patients underwent 21 EUS guided pseudocyst drainage, one patient had EUS guided walled off necrosis drainage, 4 patients underwent celiac plexus block referred for pain management of chronic pancreatitis,one patient underwent EUS guided pancreatic duct cannulation and stent placement and one patient had EUS guided randezvous procedure. All therapeutic procedures were performed under general anesthesia whereas diagnostic EUS were performed under monitored anesthesia care. No immediate or early complications were seen.

Conclusion: In this modern era of advanced diagnostic tools and minimally invasive interventions, EUS related procedures are safe and effective in pediatric group of patients too. However, this procedure should be standardized and performed at centers with trained endoscopists and an experienced pediatric anesthesia team.

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Disclosure: Nothing to disclose.

PP1038

PREDICTING EUS-FNB ADEQUACY WITH VISUAL ON-SITE EVALUATION (VOSE): TOWARD SMARTER PANCREATIC BIOPSIES

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Introduction: Endoscopic ultrasound-guided fine-needle biopsy (EUS-FNB) is widely recognized as the preferred diagnostic method for pancreatic lesions, achieving diagnostic accuracy rates between 78% and 95%. Rapid on-site evaluation (ROSE) improves adequacy rates but remains unavailable in many centers. In this context, we evaluated a simplified method, Visual On-Site Evaluation (VOSE), based on macroscopic evaluation of the biopsy sample, to predict histo-cytological adequacy without needing ROSE.

Primary aim: assess the accuracy of VOSE in predicting sample adequacy. Secondary aims: evaluate the diagnostic yield of FNB and the potential reduction in unnecessary needle passes.

Aims & Methods: This prospective single-center study enrolled patients undergoing EUS-FNB for pancreatic lesions between January 2020 and February 2024.

VOSE parameters included the presence of a macroscopic visible core (MVC), number of specimens, and specimen length. Associations between VOSE findings and final histo-cytological adequacy were analyzed using Fisher's exact test and Student's t-test. A p-value < 0.05 was considered statistically significant. Study conducted at a university hospital center.

Results: Fifty-nine patients (57.6% male; mean age 67.9 ± 9.8 years) were included, contributing to a total of 59 pancreatic lesions. Across 218 passes performed, the overall sample adequacy rate was 91.5%.

Adequacy was significantly higher when using 22-G needles compared to 25-G needles (95.7% vs. 86.2%, p = 0.018). VOSE criteria (presence of MVC, multiple fragments, and long specimen) demonstrated a sensitivity of 52.3%, a specificity of 61.1%, and a positive predictive value (PPV) of 93.1% for predicting sample adequacy. Applying VOSE led to a 45% reduction in unnecessary additional passes.

Conclusion: EUS-FNB provides excellent diagnostic adequacy for pancreatic lesions, particularly with 22-G needles. The use of VOSE, a simple and reproducible macroscopic evaluation method, can reliably predict histo-cytological adequacy and substantially reduce the number of passes, offering a practical alternative in centers without ROSE availability.

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Disclosure: Nothing to disclose.

PP1039

PANCREATIC FIBROSIS NON-INVASIVE ASSESSMENT BY MULTIDETECTOR COMPUTED TOMOGRAPHY: THE FINAL RESULTS OF THE STUDY

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Introduction: Pancreatic fibrosis (PF), a pathological hallmark of chronic pancreatitis, serves as a critical mediator in the progression of numerous pancreatic disorders. Early identification of fibrotic changes presents a strategic opportunity to intervene in disease pathways, potentially curbing the advancement of pancreatic pathologies. It makes research of non-invasive radiological PF assessment perspective.
Aims & Methods: Aim of our study was to evaluate the possibility of multidetector computed tomography (MDCT) post-processing parameters in PF grade assessment.
Methods. Results of contrast-enhanced 128-row MDCT from 165 patients (mean age 57.9±13.5) who underwent surgery from April 2022 to February 2024 due to pancreatic tumors and advanced chronic pancreatitis were analyzed. We calculated contrast enhancement ratio (CR) between venous phase (VP) and non-contrast MDCT, and normalized CR (NCR) during the pancreatic phase (PP) and VP, also we measured pancreas attenuation during MDCT examination. PF morphological grade was assessed according to Klöppel & Maillet's scoring system [1] and scaled into three grades: mild, moderate and severe PF.
This study was conducted within the healthcare research project "Assessment of pancreatic fibrosis as a prognostic factor for its diseases course" funded by Autonomous non-profit organization "Moscow Center for Innovative Technologies in Healthcare" administered by the Moscow Healthcare Department.
Results: The values of post-processing CT results were compared in groups with different degrees of PF. Mean CR were significantly higher (p=0.001) in patients with severe PF (CR 1.16±0.65 HU) than in patients with mild PF (CR 0.78±0.31 HU). CR significant increase (p=0.03) was found in patients with signs of inflammatory changes in the pancreas tissue (CR 1.14±0.6 HU) than in those without them (CR 0.81±0.3 HU). There were no significant differences between the NCRPP and NCRVP, and the degree of PF, which we had shown in publications of our preliminary results. We suppose the main reason of this fact is study groups enlargement as well as measurement technique improvement, led to results clarification.
Conclusion: The CR value increased in patients with severe degree of PF and decreased in patients with mild PF. Also, there was a significant relation between CR increase and presence of early signs of pancreatic inflammatory changes. Thus, there was a relationship between MDCT post-processing results and morphological signs of PF, which can be used for pancreatic fibrosis non-invasive diagnosis and identification of additional signs of early chronic pancreatitis.
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Disclosure: Nothing to disclose.

PP1040

PANCREATIC STEATOSIS AND EXOCRINE PANCREATIC INSUFFICIENCY: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction:
Pancreatic steatosis (PS) has been increasingly recognized and associated with some relevant metabolic implications. However, data on its correlation with exocrine pancreatic function remain limited. We conducted a systematic review and meta-analysis to evaluate whether PS is associated with exocrine pancreatic insufficiency (EPI).
Aims & Methods: This study aimed to investigate whether imaging-detected PS is related to exocrine pancreatic dysfunction, assessed by fecal elastase-1 (FE-1) levels. We searched PubMed, EMBASE, and Cochrane Central for studies including adults with and without PS identified on imaging, and reporting FE-1 concentrations. Data were pooled using random-effects models.
Results: Three studies evaluating a total of 1,864 patients were included. Fecal elastase levels were significantly lower in patients with PS compared to controls, indicating worse exocrine pancreatic function in this group (MD -199.78; 95% CI -303.66 to -95.91; $p < 0.05$; $I^2 = 97\%$). Moreover, PS was associated with a significantly increased risk of exocrine pancreatic insufficiency (RR 3.13; 95% CI 1.17–8.40; $p = 0.023$; $I^2 = 81\%$), reinforcing the link between fat accumulation in the pancreas and impaired exocrine function.
Conclusion: This meta-analysis demonstrates a significant association between PS and impaired pancreatic exocrine function, compared to individuals without known pancreas disease. These findings underscore the importance of a comprehensive metabolic assessment in patients with PS, given its potential role in pancreatic dysfunction.
Disclosure: Nothing to disclose.

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PP1041

STENT CHOICE IN EUS-GUIDED DRAINAGE OF WALLED-OFF NECROSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS OF THE RANDOMIZED CONTROL TRIALS

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Introduction: Safe and effective treatment of walled-off necrosis (WON) is challenging. Plastic stents (PS) and lumen-apposing metal stents (LAMS) have been used for endoscopic ultrasound-guided (EUS-guided) drainage of WON. This study evaluated LAMS and DPS results.

Aims & Methods: A database search was conducted until March 2024, and studies which met the criteria were selected. We assessed technical and clinical efficacy and other secondary outcomes using the RevMan software.

Results: The final analysis included five randomized control trials (RCTs), involving a total of 318 patients (161 in the LAMS and 157 in the PS group). There was no indication of any risk of bias and basic characteristics of the patients between the two group. The risk ratio (RR) for technical and clinical success were 1.00 (95% confidence interval (95%CI 0.97, 1.03) and 1.04 (95%CI 0.93, 1.16), respectively, indicating no significant difference ($p > 0.05$) between the two groups.

There was a significant difference in procedure time of -20.23 minutes (95% CI -31.61, -8.85) and the number of procedures sessions by -0.01 (95%CI -0.51, 0.48). However, there was no significant difference in hospital stay of 1.38 days (95% CI -5.57, 8.33) and adverse events (RR 0.79, 95%CI 0.43, 1.46).

Conclusion: There is no significant difference in the clinical efficacy and safety of PS and LAMS for treating patients with WON.

Disclosure: Nothing to disclose.

PP1042

SAFETY AND EFFICACY OF METAL STENTS VERSUS PLASTIC STENTS FOR ENDOSCOPIC ULTRASOUND (EUS)-GUIDED DRAINAGE OF PANCREATIC FLUID COLLECTIONS: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED STUDIES WITH TRIAL SEQUENTIAL ANALYSIS

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Introduction: Pancreatic fluid collections (PFCs) are significant local complications that can arise from acute pancreatitis, and its management often involves drainage, although the benefits of metal stents over plastic stents remain uncertain.

This meta-analysis compared the efficacy and safety of metal stents versus plastic stents for endoscopic ultrasound (EUS)-guided drainage of PFCs.

Aims & Methods: We searched PubMed, Embase, and Cochrane Library for randomized controlled trials (RCTs) comparing metal with plastic stents for drainage of PFCs. The odds ratio (OR) was used for binary outcomes and the mean difference (MD) for continuous outcomes with their respective 95% confidence interval (CI).

Results: This study included eight RCTs and over 500 patients. Procedure duration was shorter in the metal stent arm compared with the plastic stent arm (MD -10.63; 95%CI -16.12 to -5.15, $p<0.001$). However, there was no statistically significant difference between metal and plastic stents in clinical success ($p=0.78$), technical success ($p=0.65$), overall mortality ($p=0.48$), recurrence ($p=0.23$), number of interventions ($p=0.85$), need of necrosectomy ($p=0.35$), need of surgery ($p=0.35$), length of hospitalization ($p=0.86$), exocrine insufficiency ($p=0.54$), endocrine insufficiency ($p=0.77$), and AEs ($p=0.36$).

Conclusion: Metal stents for EUS-guided drainage of PFCs are associated with a shorter procedure duration. Besides that, it did not improve other clinically relevant outcomes in comparison with plastic stents.

Disclosure: Nothing to disclose.

PP1043

A SYSTEMATIC REVIEW ON THE SAFETY AND EFFICACY OF CHOLECYSTECTOMY - DURING PREGNANCY IN THE CONTEXT OF ACUTE PANCREATITIS

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Introduction: Acute pancreatitis (AP) during pregnancy is a rare but potentially life-threatening condition for both mother and fetus. The most common cause is gallstone disease, followed by hypertriglyceridemia and, less frequently, alcohol use. The physiological changes of pregnancy can mask the clinical presentation, delaying diagnosis and intervention. This review aims to synthesize current evidence on the diagnosis, management, and maternal-fetal outcomes of AP in pregnancy, with special attention to the role and timing of cholecystectomy during an acute episode.

Aims & Methods: A comprehensive literature search was conducted using PubMed, Scopus, and the Cochrane Library for studies published between 2000 and 2024. Inclusion criteria involved clinical trials, cohort studies, systematic reviews, and relevant guidelines discussing AP in pregnancy and cholecystectomy

Results: The majority of AP cases in pregnancy occur in the third trimester or early postpartum period. Gallstones account for 65–100% of cases. Conservative management remains the first-line approach, particularly in the first trimester. However, recurrence rates up to 70% with conservative therapy alone have prompted a reevaluation of surgical timing. Recent studies support the safety of cholecystectomy during the second trimester and even during stabilized mild to moderate AP. In the third trimester, ERCP with endoscopic biliary sphincterotomy is preferred, with cholecystectomy deferred postpartum. Multidisciplinary management has significantly improved maternal and neonatal outcomes in recent years.

Conclusion: Acute pancreatitis in pregnancy remains a rare but serious condition requiring timely diagnosis and multidisciplinary care. Recent innovations support the safety and efficacy of cholecystectomy laparascopyce during the second trimester and even in selected stabilized cases

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of mild acute pancreatitis. Personalized management strategies based on gestational age and disease severity, along with minimally invasive techniques like ERCP, offer improved maternal and fetal outcomes.
Laparascopi Colecystectomy prevens the recurrent acute pancreatitis and other complications billiaris.
No differenc in the intra and postoperative complications during and after pregnancy.

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Disclosure: Taibe Haziri

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PP1044

EARLY ENDOSCOPIC NECROSECTOMY IN WALLED-OFF PANCREATIC NECROSIS (WON)

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Introduction: Pancreatic necrosis is a devastating disease that leads to a high mortality and morbidity rates. Necrosectomy is the golf standart treatment for infected pancreatic necrosis. Percutaneous and endoscopic approaches has been widely accepted in selected cases. Endoscopic necrosectomy in early complicated pancreatic necrosis were performed as a first intent to treat and retrospectively evaluated regarding to morbidity, efficacy, recurrence and length of hospital stay.

Aims & Methods: Report our experience on endoscopic manage of pancreatic necrosis using a daily transmural and transpapillary drainage in 25 patients.

Records of all patients who underwent EN from January 2023 to January 2025 Endoscopic manage of acute pancreatic necrosis using a daily transmural and transpapillary drainage in 25 patients. The Apache classification and CCT score is used. Diagnostic ERP to search communications between pancreatic duct and the collection were performed in selective cases and in cases that communication exists; prothesis can be inserted through the minor or major papilla to plug it. The transmural drainage was performed in 25 patients (05 Duodenal and 20 gastric) and transpapillary and transmural in 07 patients, after identification of the most proeminent or bulging part of the stomach or duodenum. All patients were followed with CT scans, IRM to ensure clinical resolution.

Results: 25 patients (14 women, 09 men) could be treated. The Apache classification is used 0 a 9 score in 14 patients (56,%) and 11 (44%), CT severity index was 30-50% in 05, > 50% in 20. Pancreatic necrosis were located in the pancreatic head in 03 patients, the head/body in 02, body in 6, body/tail 04, head/body/tail 10. ERP demonstrated the ductal communication of the colection with the main pancreatic duct in 07 patients, rupture in 03, disconnect syndrome in 03 and normal in 02. Drainage was successful in 25 (100%). 07 (28%) combined treatment (transpapillary and transmural), 18 (72%) transmural. The median number of endoscopic session for each patients was 4 (range 2 to 8). The median hospital stay for in the all patients was only endoscopic therapy was 9. The range follow –up was 18 months (range 6 to 24) with mean number of ERCPs was 4 (range 2 to 8). Two patients required concurrent radiology procedure, recurrence of fluid collection (n=1), flank abscess (n=1). There were 2 (16,6%) pseudocyst recurrences during this period, and were managed clinically and/or endoscopically. There was no mortality related to the procedure.

Conclusion: The early and aggressive endoscopic treatment with daily necrosectomy can be performed and show promising results same in early complications of pancreatic necrosis. Combined approaches with radiology and conventional surgery can also be applied in selected cases.

Disclosure: No conflict of interest.

PP1045

LENTINAN PROTECTS AGAINST ACUTE PANCREATITIS IN AN ATP CITRATE LYASE-DEPENDENT MANNER

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Introduction: Acute pancreatitis (AP) is a common acute inflammatory disease of the abdomen. ATP citrate lyase (ACLY) has been shown to play a critical role in mitigating acute pancreatitis-associated injury. Lentinan, a multifunctional polysaccharide derived from shiitake mushrooms, exhibits biological activities such as anti-inflammatory, antioxidant, and immunomodulatory effects, and is widely used in clinical settings. However, the role of Lentinan in acute pancreatitis remains unclear.

Aims & Methods: This study aimed to investigate the protective effects of Lentinan against acute pancreatitis and whether this protection depends on ATP citrate lyase. Adult healthy male C57BL/6 mice were randomly divided into control group, acute pancreatitis group, and Lentinan + acute pancreatitis group. The cerulein-LPS induced AP model was established by intraperitoneal injection of cerulein solution (100 μg/kg body weight) 10 times at hourly intervals, followed by a single injection of lipopolysaccharide (5 mg/kg) immediately after the last cerulein injection. In the lentinan + acute pancreatitis group, mice were pretreated with intraperitoneal injections of of Lentinan solution (50 mg/kg, 100 mg/kg, or 200 mg/kg) for three consecutive days before AP induction.

The severity of pancreatitis was assessed by comparing plasma activities of amylase, lipase, alanine aminotransferase, aspartate aminotransferase, and plasma levels of urea, and creatinine, as well as pancreatic histopathological scores.

Systemic inflammatory response was evaluated by measuring plasma IL-1β and TNF-α levels through enzyme-linked immunosorbent assay. Pancreatic infiltration of macrophages and neutrophils was detected by immunofluorescence, and protein levels of ACLY, NLRP3, and IL-1β in

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pancreatic tissues were measured by Western blot. To further validate the critical role of ACLY, AP was induced in mice with pancreas-specific ACLY knockout, with Lentinan pretreatment.

Results: Lentinan significantly alleviated pancreatic histopathological damage caused by AP, including edema, necrosis, and inflammatory cell infiltration. It also reduced plasma aminotransferase activity, creatinine, and urea levels, but did not significantly decrease plasma amylase or lipase activity. A higher dose of Lentinan (200 mg/kg) was less effective than lower doses in reducing plasma aminotransferase, creatinine, and urea in AP mice, possibly due to greater hepatorenal toxicity. Therefore, a moderate dose of Lentinan (100 mg/kg) provided effective protection against acute pancreatitis and was therefore selected for subsequent experiments.

Additionally, Lentinan reduced pancreatic infiltration of macrophages and neutrophils, decreased plasma pro-inflammatory cytokine levels, upregulated pancreatic ACLY protein expression, and downregulated the levels of NLRP3 inflammasome and IL-1β proteins in pancreatic tissues. In mice with pancreas-specific ACLY knockout, both local pancreatic injury and systemic inflammation caused by AP were significantly exacerbated, and the protective effect of Lentinan against acute pancreatitis was diminished.

Conclusion: Lentinan exerts protective effects against acute pancreatitis by improving local and systemic inflammatory responses. The expression of pancreatic ACLY is indispensable for mediating this protective effect, which was partially achieved through the inhibition of the NLRP3 inflammasome pathway. The present study establishes a novel mechanism for this clinically available immunomodulator, which could have potential therapeutic implications in acute pancreatitis.

Disclosure: Nothing to disclose.

PP1046

SIMPLE ELECTROLYTE SCORING SYSTEM FOR PREDICTING DISEASE PROGRESSION IN ACUTE PANCREATITIS PATIENTS : A RETROSPECTIVE STUDY

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Introduction: Acute pancreatitis (AP) is closely intricately linked to electrolyte disturbances. Such imbalances can exacerbate the instability of the intracellular environment, subsequently accelerating the progression of AP. This study aimed primarily to develop a simplified electrolyte scoring system (CC score) to determine AP exacerbation.

Aims & Methods: A new CC score was established in this study using the Medical Information Mart for Intensive Care IV (MIMIC-IV) v3.1 database, and it was validated using the digestive department database of the First Affiliated Hospital of Nanchang University. The primary endpoint considered was the exacerbation of the disease post-admission in AP patients. Receiver operating characteristic (ROC) curves and decision curve analysis (DCA) plots were used to assess the predictive value and clinical applicability of the new CC score. Improvements were implemented on the acute physiology and chronic health evaluation II (APACHE II) and systemic inflammatory response syndrome (SIRS) scoring systems, and their performance was compared with that of traditional predictive scoring systems.

Results: This study incorporates data from 1413 AP patients drawn from the MIMIC-IV database, supplemented by a further 1040 AP patients in an external validation group. Of these, 178 and 70 AP patients were respectively transferred to the Intensive Care Unit (ICU). The multivariate adjustment model indicates that a significant negative correlation between serum Ca^{2+} levels and severe progression in AP patients, while serum Cl^- levels showed a J-shaped association with severe progression. Through node analysis of the restricted cubic splines (RCS) model, combined with the clinical grading standards for hyperchloremia and hypocalcemia, the critical values for Cl^- and Ca^{2+} were determined to be 110 mmol/L and 1.77 and 2.1 mmol/L, respectively.

Based on these findings, we established a new CC score and optimized the APACHE II and SIRS scoring systems. The results indicated that the predictive power of the optimized APACHE II score (AUC=0.83) and SIRS score (AUC=0.87) for severe progression in AP patients surpassed that of the original APACHE II score (AUC=0.74) and SIRS score (AUC=0.71).

Conclusion: The new CC score not only effectively predicts the risk of severe progression in AP patients but also enhances the predictive performance of both SIRS and APACHE II scores for assessing severe progression risk in these patients.

Disclosure: Nothing to disclose.

PP1047

COMPARISON OF USING LUMEN-APPOSING METAL STENTS VERSUS PLASTIC STENTS FOR DRAINAGE OF PANCREATIC WALLED-OFF NECROSIS ： A META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: In the late stage of acute pancreatitis, the occurrence of pancreatic walled-off necrosis (WON) can be life-threatening, especially in the presence of infection. EUS-guided transgastric drainage is currently a minimally invasive treatment modality for the clinical treatment of WON. However, there is no consensus in international guidelines on the choice between lumen-apposing metal stents (LAMS) and traditional double-pigtail plastic stents (DPT).

Aims & Methods: Objective To investigate the differences in drainage effectiveness and complications among different stents, we included currently published randomized controlled trials (RCTs) for a meta-analysis. Methods We systematically searched the PubMed, Embase, and Cochrane Library databases for studies published between 1 January 2010 to 1 March 2023, and updated the search on March 1, 2025. The observed indicators included: primary outcome indicators (clinical success rate, re-intervention), other observation indicators (total adverse events, mortality, adverse events, procedural duration). Binary variables were analyzed using relative risk (RR) and 95% CI, while continuous variables were analyzed using mean difference (MD). The selection of a random effects model or fixed effects model depended on the situation.

Results: Six RCTs with a total of 325 patients were included for analysis, with 177 patients in the LAMS group and 175 patients in the DPT group. There was similar in the primary outcome indicators of clinical success rate (RR=1.05; 95% CI 0.93 - 1.17; *P*=0.46), the number of patients requiring debridement of necrotic tissue (RR=0.95; 95% CI 0.62 - 1.47; *P*=0.83), and percutaneous drainage (RR=0.98; 95% CI 0.27 - 3.5; *P*=0.97).

There were also no significant differences in terms of mortality rate (RR=0.86; 95% CI 0.32 - 2.33; *P*=0.77), total adverse events (RR=0.87; 95% CI 0.59 - 1.29; *P*=0.48), adverse events related to bleeding (RR=1.48; 95% CI 0.67 - 3.28; *P*=0.34), procedural time (MD=-9.44; 95% CI, -15.27 - -3.61, *P*=0.002).

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Conclusion: In comparison to DPT, LAMS could shorten procedural duration. However, there was no significant difference clinical success rate, re-intervention and adverse events. Therefore, future studies with large samples and high quality are needed for further confirmation.
Meanwhile, due to the high heterogeneity among studies, conclusions should be considered with caution. Clinicians need to make the most suitable treatment decisions based on the actual situation to ensure patients get the best treatment effect and prognosis.
Disclosure: Nothing to disclose.

PP1048

THE ROLE AND UPTAKE OF DRUG AND ALCOHOL SERVICES AMONG PATIENTS WITH ALCOHOL-ASSOCIATED ACUTE PANCREATITIS ADMITTED TO A TERTIARY HEALTH SERVICE

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Introduction: Alcohol is a common cause of acute pancreatitis, and ongoing consumption increases the risk of recurrent episodes. There are no international guidelines to direct behavioural or pharmacological interventions to reduce recurrence of alcohol-associated acute pancreatitis (AAP). Studies exploring the role of addiction medicine in patients with AAP are limited.
Aims & Methods: This study describes the uptake and role of the Addiction Medicine Unit (AMU), a specialist consultative service, in patients admitted with AAP. We conducted a retrospective cohort study including all inpatients diagnosed with AAP at a large multisite tertiary health service in Melbourne, Australia between May 2022 to May 2024. Patient demographics, diagnosis of alcohol use disorder (AUD) and pharmacological and non-pharmacological interventions provided by the AMU were obtained from the patient's medical records.
Results: A total of 231 patients with 314 admissions with AAP were identified. Most were males (77.9%) with a median age of 44 years (IQR 21), with median bedside index of severity in acute pancreatitis (BISAP) score of 1 (IQR 1). Of the patients, 35 (11.1%) had necrotising pancreatitis, 30 (9.6%) required admission to the intensive care unit (ICU), and 1 patient died from AAP. The prevalence of AUD was 82.3%. Referral to the AMU occurred in 56.7% of admissions, predominantly among those with AUD (87.6%).
Patients referred to the AMU had a higher median number of standard drinks per week (70 standards, range 6-385 standards) compared to those who were not referred (49 standards, range 1-301 standards) (p=0.024). The AMU recommended outpatient services alone in 40.4%, both pharmacological and outpatient interventions in 37.6%, and medications alone in 21.9%.
Outpatient services included either a service for community-based assessment and counselling (86.4%), residential rehabilitation programs and/or detox services (5.8%), follow up in the AMU outpatient clinic (3.9%) or a combination of these outpatient services (3.9%). There were 137 (43.6%) admissions in which patients were commenced on medications for AUD.
Topiramate (42.6%), and naltrexone (42.6%) were the most prescribed pharmacological agents by the AMU, followed by baclofen (9.8%), acamprosate (4.1%) and disulfiram (0.8%). Over the 2-year period, 54 (23.4%) were readmitted with AAP. Most patients were not readmitted within 30 days of discharge (73.2%) however there were 33 (10.5%) patients readmitted for pancreatitis or pancreatitis-related complications including pain or infection, and 18 (5.7%) readmitted to hospital for non-pancreatitis related reasons. Patients who were older (median 49 years) were more likely to be readmitted with AAP compared to those who were younger (median age 43 years) across the study period (p=0.05). There were six patients re-admitted with AAP with their index admission deemed related to an alcoholic binge without suspected underlying AUD. Of these six patients, only two were referred and reviewed by the AMU, with the other four patients not referred.
Conclusion: Our study highlights the critical role of the AMU in managing AAP, particularly given the high prevalence of concurrent AUD in this population. Despite this, significant gaps remain in screening for AUD and initiating referrals, resulting in only half of AAP patients being referred.
Considering the substantial morbidity and healthcare cost associated with alcohol, efforts to improve referral practices during index admissions may be key to reduce readmissions for AAP and improve patient outcomes.
Disclosure: Nothing to disclose.

PP1049

UNDER-RECOGNIZED BURDEN OF HYPERTRIGLYCERIDEMIA-INDUCED ACUTE PANCREATITIS: A RETROSPECTIVE SINGLE-CENTER ANALYSIS AND THERAPEUTIC OUTCOMES

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Introduction: Hypertriglyceridemia (HTG) is an underdiagnosed etiology of acute pancreatitis (AP), traditionally reported in 1–7% of cases. However, emerging evidence suggests that its true prevalence may be significantly higher. Prompt identification and effective management of HTG-induced AP are essential, yet standardized treatment approaches remain limited.
This study aimed to assess the prevalence of HTG-induced AP and evaluate insulin infusion as a first-line therapeutic strategy.
Aims & Methods: We retrospectively analyzed 200 adult patients admitted with AP at [Hospital Name] between January 2023 and December 2024. Patients with serum triglyceride levels ≥11.3 mmol/L (≥1000 mg/dL) and no other identifiable causes of AP (e.g., gallstones, alcohol use) were included. Demographic data, comorbidities, treatment modalities, and outcomes were reviewed. A favorable response to insulin therapy was defined as a ≥50% reduction in serum triglyceride levels within 72 hours. Statistical analysis was performed using SPSS; p<0.05 was considered statistically significant.
Results: HTG was identified as the primary cause of AP in 40 out of 200 patients (20%). Among these, 60% had type 2 diabetes mellitus, 50% were obese (BMI ≥30 kg/m^2), and 35% reported regular alcohol use. All patients were initiated on insulin infusion. Thirty-four patients (85%) responded favorably, while six (15%) required plasmapheresis. The mean hospital stay was significantly shorter in the insulin-only group compared to those requiring plasmapheresis (5 vs. 8 days, p<0.01), with a lower complication rate as well (10% vs. 25%, p<0.05). No in-hospital mortality occurred.
Conclusion: HTG-induced AP may be substantially more prevalent than previously recognized. Routine triglyceride screening should be considered in AP cases with no obvious cause. Insulin infusion is a safe, effective, and widely accessible first-line therapy that may reduce complica-

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tions and length of hospital stay. These findings support early metabolic assessment and proactive risk factor management to improve clinical outcomes.

Disclosure: Nothing to disclose.

PP1050

PAIN INTENSITY AND PROGNOSIS OF ACUTE PANCREATITIS: POST-HOC ANALYSIS OF AN INTERNATIONAL, PROSPECTIVE STUDY

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Introduction: Abdominal pain is a prominent symptom in acute pancreatitis (AP) and has been linked to disease severity.

Aims & Methods: We aimed to examine associations between pain intensity and the outcome of AP and the discriminatory ability of pain intensity to predict these outcomes. For this, we conducted a post-hoc analysis using data from the PAINAP database of patients admitted with AP across 118 centres in 27 countries. Based on the pain intensity on admission using a numeric rating scale (NRS) of 0 to 10, patients were categorised as having no/mild (NRS 0-3), moderate (NRS 4-6) or severe pain (NRS 7-10). Demographic details and AP outcomes were compared across these three groups, and associations were explored using multivariable regression analyses.

Finally, the influence of pain intensity on the probability of AP outcomes was assessed by calculating AUC-ROCs, sensitivity, specificity, positive/negative predictive values, and likelihood ratios.

Results: We included 1,727 patients, of whom 183 (11%) reported no/mild pain, 503 (29%) moderate pain, and 1,041 (60%) severe pain. Severe pain was associated with more severe AP (OR 1.66 (95% confidence interval (95%CI), 1.09-2.52), P=0.02), renal failure (OR 3.04 (95%CI, 1.35-6.87) and pancreatic necrosis (OR 2.37 (95%CI, 1.22-4.59), P=0.01) compared to no/mild pain. Severe pain had a low discriminatory ability for predicting outcomes of AP, with AUC-ROCs ranging from 0.53 to 0.61.The sensitivity was moderate for all outcomes, with the highest value (81% (95%CI, 72-88)) for renal failure. The specificity was lower across all outcomes, ranging from 41 to 42%. Positive predictive values were low (9-30%), whereas negative predictive values were higher (77-97%). The highest negative predictive value was observed for renal failure. Pain intensity (cutoff NRS ≥ 7) did not affect the post-test probability of outcomes compared to pre-test probability, suggesting that pain intensity alone is insufficient for evaluating prognosis in AP.

Conclusion: Pain intensity was associated with worse outcomes in AP, but the discriminatory performance of pain intensity for predicting AP outcomes was poor. Future research is warranted to refine the assessment of AP pain and clarify the underlying mechanisms. A more comprehensive understanding of pain in AP could improve its prognostic value, guide more personalised treatment strategies, and optimise pain management.

Disclosure: Nothing to disclose.

PP1051

A TRANSFORMER-BASED PREDICTION MODEL FOR ACUTE PANCREATITIS SEVERITY

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Introduction: Acute pancreatitis (AP) is a common gastrointestinal disorder with significant morbidity and mortality, particularly in severe cases (SAP). Early prediction of SAP is crucial for timely intervention and improved patient outcomes. Traditional scoring systems like Ranson, BISAP, and APACHE II have limitations in accuracy and timeliness.

Machine learning (ML) has shown promise in enhancing SAP prediction, but the potential of advanced architectures like Transformer models remains unexplored. This study aims to develop a Transformer-based model for SAP prediction and compare its performance with traditional ML methods.

Aims & Methods: The primary aim of this study was to develop a Transformer-based prediction model for acute pancreatitis severity and evaluate its performance compared to traditional machine learning models. Retrospective data from 722 AP patients admitted to the Second Affiliated Hospital of Guilin Medical University between January 2020 and June 2024 were analyzed. Patient data included 24-hour admission laboratory indicators such as electrolytes, liver function, renal function, and inflammatory markers.

To construct the prediction model, a comprehensive workflow was designed. Data preprocessing involved feature vectorization using One-Hot encoding, missing value imputation using the K-Nearest Neighbor (KNN) algorithm, normalization of feature values to a [0, 1] range, and handling class imbalance using Synthetic Minority Over-Sampling Technique (SMOTE).

Feature selection was performed by dividing patients into SAP and MAP groups, identifying features with significant differences using non-parametric tests, and ranking features by their relevance through correlation analysis. Features causing a decline in AUC during incremental addition were removed to optimize the feature set.

The Transformer model, leveraging self-attention mechanisms to process sequential data, was trained alongside traditional ML models (Logistic Regression, Decision Tree, Random Forest, and XGBoost). Five-fold cross-validation was used to prevent overfitting, and model performance

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was evaluated using metrics such as AUC, accuracy, precision, recall, F1 score, and specificity. SHAP analysis was employed to interpret the Transformer model's predictions, providing insights into the contribution of each feature to the model's output.

Results: The Transformer model outperformed traditional ML models (Table 1), achieving an AUC of 0.9662 and exceeding 0.90 in all metrics. It maintained robust performance even with minimal training data (AUC: 0.8437 at 5% training). SHAP analysis highlighted key predictors (calcium, amylase, basophil percentage, CO2 binding capacity), aligning with clinical relevance.

Model	AUC	Accuracy	Precision	Recall	F1 Score	Specificity
Transformer	0.9662	0.9028	0.9095	0.9004	0.9047	0.9059
Random Forest	0.9224	0.8433	0.8788	0.8056	0.8406	0.8830
XGBoost	0.9075	0.8034	0.8033	0.8167	0.8099	0.7895
Logistic Regression	0.8910	0.8405	0.8780	0.8000	0.8372	0.8830
Decision Tree	0.8684	0.8433	0.8655	0.8222	0.8433	0.8655

Table 1. Performance metrics comparison of different models.

Conclusion: The Transformer model demonstrated superior accuracy, generalizability, and interpretability for SAP prediction, particularly in small datasets. Its clinical utility is enhanced by SHAP-driven insights into key predictors. Retrospective design limits prospective validity; future work should validate the model in larger cohorts and integrate additional biomarkers. This study supports the potential of transformer in medical prediction to improve patient outcomes through timely interventions.

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Disclosure: Nothing to disclose.

PP1052

A NOVEL NON-INVASIVE METHOD FOR EVALUATING SPHINCTER OF ODDI FUNCTION USING CINE-DYNAMIC MAGNETIC RESONANCE CHOLANGIOPANCREATOGRAPHY

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Introduction: Sphincter of Oddi dysfunction (SOD) is considered one of the possible underlying causes of unexplained biliary pain and idiopathic acute pancreatitis. According to the Rome IV criteria, sphincter of Oddi manometry (SOM) is required for the diagnosis of pancreatic-type SOD (PSOD); however, SOM is highly invasive and carries a significant risk of post-procedural pancreatitis. Therefore, a non-invasive and simple method to evaluate sphincter (papillary) pressure is critically needed. We hypothesized that cine-dynamic magnetic resonance cholangiopancreatography (MRCP) may serve as a useful tool for the functional assessment of papillary pressure in patients with suspected SOD.

Aims & Methods: In this prospective observational study, cine-dynamic MRCP was performed on 40 subjects: 10 healthy controls, 7 with suspected biliary-type SOD (BSOD), and 12 with suspected PSOD. Eleven additional cases were excluded for not meeting the Rome IV criteria for SOD. Cine-dynamic MRCP involved generating a 2D MRCP image and applying a 20 mm spatially selective inversion recovery (IR) pulse to suppress water signals at the level of the Sphincter of Oddi. Sequential images (20 frames over 5 minutes) were obtained to visualize the dynamic inflow of bile and pancreatic juice. Two quantitative indices were evaluated: the frequency of inflow events and the secretion grade of bile or pancreatic juice. The secretion grade was defined based on the distance that the fluid moved within the common bile duct or main pancreatic duct, using a standardized 5-point scale. A subset of patients underwent EST, and changes in MRCP parameters were compared before and after the procedure.

Results: Bile flow frequency (times; median [range]) was significantly reduced in both the BSOD group (2 [1–17]; p = 0.006) and the PSOD group (8 [3–14]; p = 0.008), compared with controls (13.5 [6–19]).The secretion grade of bile flow (median [range]) was also significantly lower in the BSOD group (0.2 [0.1–1.3]; p = 0.012) and PSOD group (0.5 [0.15–1.75]; p = 0.021), compared with controls (1.6 [0.3–2.05]).

For pancreatic juice flow, frequency was significantly lower in the PSOD group (9.5 [4–17]; p = 0.037) than in controls, whereas the BSOD group showed no significant difference (11 [3–18]; $p = 0.262$).

Similarly, the secretion grade of pancreatic juice was significantly reduced in the PSOD group (0.98 [0.25–2.9]; p = 0.040), but not in the BSOD group (1.15 [0.3–2.7]; $p = 0.154$). Endoscopic sphincterotomy was performed in seven patients, and follow-up cine-dynamic MRCP was available in six of them. All six showed improvement in flow parameters, and none of the seven patients experienced symptom recurrence for over one year after treatment.

Conclusion: Cine-dynamic MRCP can be a non-invasive imaging biomarker for evaluating Sphincter of Oddi function by visualization and quantification of bile and pancreatic juice flows.

Disclosure: Nothing to disclose.

PP1053

COMBINING BEDSIDE INDEX OF SEVERITY IN ACUTE PANCREATITIS (BISAP) AND CHARLSON COMORBIDITY INDEX IMPROVES EARLY RISK STRATIFICATION IN BILIARY ACUTE PANCREATITIS

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Introduction: Acute biliary pancreatitis (ABP) has been shown to be associated with increasing incidence and health-care costs, and possibly higher mortality than other etiologies. Despite the existence of various scoring systems, early and accurate risk stratification remains difficult. Due to heterogeneity of populations, focusing on one etiology might reveal important insights concerning predictors of outcome.

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Aims & Methods: We conducted a retrospective, multicenter cohort study including all patients admitted with ABP between 2018 and 2021 at two hospitals. Epidemiological, clinical, and outcome data were analyzed.
The main clinical outcome of interest was a hierarchical composite endpoint of five mutually exclusive clinical states:
i. Death (90-day all-cause mortality),
ii. Severe AP and/or admission to the intensive care unit (ICU),
iii. Endoscopic, interventional radiological, or surgical interventions,
iv. Moderately-severe AP,
v. Mild AP.
Univariable logistic regression identified parameters associated with worse outcomes, which were then tested in multivariable models. Emphasis was placed on early availability and clinical practicality of predictors. We aimed to stratify patients into outcome trajectories reflecting both health impact and healthcare resource utilization.
Results: Out of 569 total acute pancreatitis cases, 217 (38.1%) patients had ABP (mean age 65.9 ±17.4 years; median length of stay (LOS) 7 days, IQR 4;11) Mild, moderately-severe, and severe disease occurred in 62.2%, 30.0%, and 7.8% of patients. All-cause 90-day mortality was 2.3%. Several parameters were significantly ($p<0.05$) associated with worse outcomes, including BISAP and the Charlson comorbidity index (CCI) ($p<0.001$). Individually, BISAP and CCI classified patients reasonably, however, wide prognostic uncertainty persisted.
When combining BISAP and CCI (BISAP-C), 5 discrete risk groups were identified, ranging from low (group 1, BISAP 0-2 & CCI 0-2) to very-high risk (group 5, BISAP 3-5 & CCI ≥6) (see Table 1). The prevalence of severe outcomes (i–iii) increased across these strata: 2.4% (Group 1), 9.5% (Group 2), 27.3% (Group 3), 60% (Group 4), and 62.5% (Group 5), with LOS also significantly longer in higher-risk groups ($p<0.001$).
Mortality was 50% in group 5, while, importantly, 97.6% of patients in group 1 experienced mild-to-moderate AP without ICU admission, persistent organ dysfunction, surgical or endoscopic interventions, and no fatality, supporting potential early discharge. This also occurred in 90.5% of group 2.
Notably, Groups 1 and 2 together comprised nearly 80% of the cohort.

	Outcome groups*				
	Group 5 Very-high	Group 4 High	Group 3 Mod.-high	Group 2 Mod.-low	Group 1 Low
BISAP CCI	3-5 ≥6	3-5 ≤5	0-2 ≥6	0-2 3-5	0-2 ≤2
Cases, n (% of cohort)	8 (4.0%)	5 (2.5%)	22 (10.9%)	84 (41.6%)	83 (41.1%)
LOS, median (IQR)	5 (3;10)	15 (9;106.5)	10 (6;14.25)	7 (4;11)	5 (3;9)
Mortality, 90d all-cause	4 (50%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Severe pancreatitis or ICU admission	0 (0%)	3 (60%)	6 (27.3%)	6 (7.1%)	2 (2.4%)
Intervention (endosc., IR, surgical)	1 (12.5%)	0 (0%)	0 (0%)	2 (2.4%)	0 (0%)
Moderately-severe pancreatitis	2 (25.0%)	2 (40%)	7 (31.8%)	18 (21.4%)	24 (28.9%)
Mild pancreatitis	1 (12.5%)	0 (0%)	9 (40.9%)	58 (69.0%)	57 (68.7%)

Table 1. Outcome trajectories.

Conclusion: Comorbidity seems to play a major role concerning worse outcomes in ABP. The combination of CCI and BISAP (BISAP-C), both comprised of early-on and easily available parameters, offers improved early prognostic discrimination and may provide tailored, individualized management and healthcare resource utilization.
Disclosure: Nothing to disclose.

PP1054

SHOULD GASTROINTESTINAL BLEEDING AND IMPAIRED MENTAL STATUS BE CONSIDERED ORGAN FAILURE IN PATIENTS WITH ACUTE PANCREATITIS?

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Introduction: The Revision of the Atlanta Classification (RAC) defines severe acute pancreatitis (AP) by the presence of persistent organ failure (OF), limited to respiratory, renal, or cardiovascular systems, this being associated with a high mortality rate. Whether gastrointestinal bleeding (GIB) and impaired mental status (IMS) should be considered as OF remains unclear. We aimed to assess the association of GIB and IMS with mortality and other outcomes.
Aims & Methods: We performed a retrospective analysis of a prospective database including 1,612 AP patients. GIB was defined as bleeding >500 mL. IMS was defined as a Glasgow score ≤14 at baseline. Multivariate logistic regression models were used to assess associations with mortality, controlling for comorbidity and RAC- defined OF. Two models were used: one including all OF (any duration), and another including only persistent OF.
Results: GIB occurred in 1.5% and IMS in 3% of patients. Both were associated with worse outcomes in bivariate analysis. In multivariate models, GIB was not independently associated with mortality. IMS was independently associated with mortality (aOR 4.0, 95% CI 1.2–13.4) only when adjusting for persistent OF, but not when adjusting for OF of any duration. Renal, respiratory or cardiovascular OF of any duration, as well as persistent OF, had an independent and very strong association with all the outcome variables including mortality, clearly differentiating them from GIB and IMS.
Conclusion: GIB and IMS are markers of disease severity but do not show a consistent independent association with mortality comparable to the current RAC definition of OF.
Disclosure: Nothing to disclose.

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PP1055

VOLUMES AND STANDARDS OF CARE IN GASTROENTEROLOGY AND DIGESTIVE ENDOSCOPY FOR BILIOPANCREATIC DISEASES: FROM EMERGENCY DEPARTMENT TO DISCHARGE

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Introduction: The aim of this study is to evaluate the volume of biliopancreatic diseases at AUSL Romagna and assess their burden on healthcare system from emergency care to subsequent hospitalization in dedicated departments.

Aims & Methods: We designed a longitudinal, retrospective, descriptive study, considering patients who accessed the Emergency Departments (ED) in all first and second-level centers of AUSL Romagna from January 2018 to December 2024, having blood lipase levels above the upper limit (≥60 U/L).

We divided these patients into two groups: the first one (identified as "acute pancreatitis" (AP)) with lipase levels above three times upper limit (≥180 U/L), and the second group (identified as "other biliopancreatic (BP) diseases") including patients with high lipase level associated to other gastrointestinal conditions, evaluating their care pathways.

Results: The total number of emergency admissions to AUSL Romagna with lipase levels ≥ 60 U/L was 29011, accounting for 1% of annual emergency admissions, with the majority triaged as green (minor urgency). Of these, 9204 patients (32%) were discharged with a diagnosis compatible with AP (13%, N= 3669) or other BP diseases (19%, N= 5535), with consistent rates over the years. Male sex was predominant for BP group with a median age of 63 years, female sex for the AP group and median age of 66 years.

Nearly all patients with AP (94%) required subsequent hospitalization, primarily in the gastroenterology department (32%), followed by general medicine (25%) and general surgery (15%). In the other BP diseases group, fewer than half (40%) required subsequent hospitalization.

The discharge diagnoses from specialized departments showed an increase in diagnostic accuracy compared to initial assessments, reflecting initial missed diagnoses, appropriate re-evaluations during hospitalization, complications, etc.

From an economic perspective, hospital stays in specialized departments lasted an average of 9.5 days, with an average cost of €3,784 per stay. Specifically, hospitalizations in the gastroenterology department for AP had an average duration of 9.3 days, with a total cost of €4129740 for 1181 admissions.

For other BP diseases, the costs for a total of 522 hospitalizations amounted to €1808899, with an average stay of 9.3 days. Most hospitalizations ended in discharge to home (84%), with an in-hospital mortality rate of 4%.

Conclusion: Our study provided a detailed roadmap of patients with biliopancreatic diseases in AUSL Romagna (an area covering 1.2 millions of inhabitants), from their admission in ED to their discharge by different departments. BP diseases account for 1% of all ED accesses and most of the cases are characterized by medium-high complexity, with an economic annual burden of €2.8 million. These results may support the healthcare policy towards an optimization of economic and human resources to improve patient's management and outcomes.

Disclosure: Nothing to disclose.

PP1056

CRUCIAL IMPACT OF WEIGHT LOSS IN AN ACUTE PANCREATITIS COHORT: RISK FACTORS AND CLINICAL CONSEQUENCES

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Introduction: Acute pancreatitis (AP) presents a risk for malnutrition due to its hypermetabolic state and impaired nutrient absorption. Identifying patients at risk of weight loss during hospitalization is crucial for timely interventions, resource allocation, and improved clinical outcomes.

However, no studies to date have focused on inpatient weight loss in AP patients.

Aims & Methods: This study aimed to determine which AP patients are at risk of weight loss during hospitalization.

We conducted a prospective cohort study of patients presenting with AP at a national tertiary care center (Semmelweis University Institute of Pancreatic Diseases) from 4th October 2021, to 30th September, 2024. Body weight was measured on admission and at discharge. Weight stability was defined as a change of -2.5% to +2.5%; changes outside this range were classified as weight loss or gain.

We explored associations with demographic factors, pre-existing exocrine and endocrine pancreatic damage, etiology, AP severity, and length of hospital stay.

Results: Of 1,795 patients admitted to the unit, 1,018 had data on body weight change. The average age was 56 ± 17 years, and 56% of the cohort was male. Weight loss was observed in 26.2% of patients, while 8.1% gained weight. Men were more likely to lose weight (30%) compared to women (22%). The most common etiologies were biliary (n=432), alcoholic (n=242), and idiopathic (n=159). The highest proportion of weight loss (7/20, 35%) was observed in AP triggered by pancreatobiliary tumors, followed by biliary etiology (128/432, 30%) and hypertriglyceridemia (12/43, 28%).

Patients with no prior AP attacks experienced weight loss more often (180/620, 29%) than those with recurrent AP episodes or severe pre-existing pancreatic exocrine damage (chronic pancreatitis, 34/162, 21%). Weight loss was observed in 34% of diabetic, 28% of prediabetic, and 22% of healthy patients. AP severity was directly proportional to both the likelihood and degree of weight loss.

In severe pancreatitis 49% of patients lost weight, which was double the rate observed in mild pancreatitis (23%). Weight loss also increased in parallel with the length of hospital stay. Among those requiring hospitalization longer than a week, more than half (104/188, 55%) experienced weight loss.

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Conclusion: Male gender, AP severity, prolonged hospitalization, and pre-existing endocrine pancreatic damage (but not exocrine pancreatic damage) are associated with a higher risk of weight loss during hospitalization. AP triggered by pancreatobiliary tumors, biliary etiology, or hypertriglyceridemia carries a greater risk of weight loss compared to other etiologies.

Disclosure: There are no conflicts of interest.

PP1057

THE PROGNOSTIC ROLE OF SYSTEMIC INFLAMMATORY RESPONSE SYNDROME IN ACUTE PANCREATITIS

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Introduction: Acute pancreatitis (AP) is a major cause of digestive-related hospitalizations. Systemic inflammatory response syndrome (SIRS) is present in 20–60% of cases and is associated with severe AP, multiorgan failure (MOF), or in-hospital mortality. This study aims to assess the clinical impact of transient and persistent SIRS on mortality and MOF, compared with the BISAP score.

Aims & Methods: A retrospective study was conducted on a prospective cohort of patients admitted with AP to a tertiary hospital between March 2017 and June 2019. Blood tests were recorded at admission and at 48 hours. Severity was defined using the 2012 Atlanta classification. MOF was defined as dysfunction of two or more organ systems lasting over 48 hours. Transient SIRS was present only at admission, while persistent SIRS lasted >48h. Univariate analysis and ROC curves were conducted using SPSS. Fagan nomograms estimated post-test probabilities of mortality and MOF using pre-test prevalence from the cohort.

Results: A total of 467 patients were included; 84 (18.41%) had SIRS at admission, of whom 32 (37.52%) had persistent SIRS. ROC analysis showed similar AUCs for persistent SIRS and BISAP ≥2, while transient SIRS had poor predictive value. Fagan nomograms showed persistent SIRS had the highest positive and negative likelihood ratios, and greater post-test probability for mortality and MOF, compared to BISAP ≥2 and transient SIRS.

Predictor	Pre-test probability	Probability ratio + (IC 95%)	Post-test probability +	Probability ratio – (IC95%)	Post-test probability -
MORTALITY					
TRANSITIONAL SIRS	4%	0.46 (0.07-3.16)	2% (0-12%)	1.07 (0.96-1.20)	4% (4-5%)
PERSISTENT SIRS		16 (4.90-27)	41% (29-53%)	0.33 (0.17-0.64)	1% (1-3%)
BISAP ≥ 2		2.92 (2.40-3.55)	12% (10-14%)	0.14 (0.04-0.52)	1% (0-2%)
MULTIORGAN FAILURE					
TRANSITIONAL SIRS	5%	1.18 (0.4-3.49)	6% (2-15%)	0.98 (0.83-1.15)	5% (4-6%)
PERSISTENT SIRS		13 (7.36-23)	40% (28-54%)	0.45 (0.28-0.72)	2% (1-4%)
BISAP ≥ 2		2.74 (2.18-3.45)	13% (11-16%)	0.24 (0.10 – 0.59)	1% (1-3%)

Table 1: Pre-test and post-test probability of developing multiorgan failure in relation to mortality and the type of SIRS developed during admission, as well as the presence of a BISAP score ≥ 2.

Conclusion: Persistent SIRS was associated with higher risk of mortality and MOF, while transient SIRS had limited prognostic value. Persistent SIRS and BISAP ≥2 had comparable ROC performance, but Fagan nomograms demonstrated the superior clinical predictive utility of persistent SIRS. It showed post-test probabilities up to 40%, compared to 12–13% for BISAP ≥2.

These results highlight the importance of monitoring SIRS persistence in AP for better risk stratification and early intervention.

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Disclosure: Nothing to disclose.

PP1058

AGE-RELATED CLINICAL OUTCOMES IN ACUTE PANCREATITIS

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Introduction: Acute pancreatitis (AP) is one of the most common diseases leading to admission in Gastroenterology departments. The aging population is contributing to an increasing number of hospitalizations among elderly patients.

Our study aims to evaluate different clinical outcomes in patients across different age groups.

Aims & Methods: A retrospective analysis was conducted using a prospective database consisting of a cohort of patients with AP consecutively enrolled from March 2017 to June 2019 at a tertiary care hospital.

Clinical management followed standard clinical practice guidelines, including routine abdominal ultrasound upon admission and CT scans in cases of diagnostic uncertainty or clinical deterioration. Diagnosis and classification were based on the revised 2012 Atlanta classification.

Patients were divided into three age groups: <50 years, 50–75 years, and >75 years. Initially, a univariate analysis using the chi-square test was performed, followed by binary logistic regression to assess the influence of age on different clinical aspects of the disease.

Results: 465 patients with AP were recruited. Overall, 48.6% of the patients were women; however, in the >75 age group, up to 64% were female. Older patients presented with more comorbidities, a greater need for antibiotic therapy, higher rates of severe AP, and an increased likelihood of mortality. These associations were confirmed in the multivariate analysis.

Notably, patients over 75 years old had a higher risk of requiring antibiotic therapy (OR 2.73) and death (OR 7.42), while those aged 50–75 years had a higher risk of developing severe AP compared to younger patients (OR 8.05). No significant differences were found between age groups regarding the occurrence of necrotizing pancreatitis.

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Age group	<50 (reference)	50-75 (OR: CI 95%)	p	>75 (OR: CI 95%)	p
Female sex	-	1.10 (0.66-1.82)	n.s	2.96 (1.77-4.96)	< 0.001
Necrotizing pancreatitis	-	0.76 (0.26-2.22)	n.s	0.62 (0.20-1.90)	n.s
Antibiotic use	-	1.33 (0.76-2.33)	n.s	2.73 (1.58-4.74)	<0.001
Severe acute pancreatitis	-	8.05 (1.04-61.95)	0.04	16.29 (2.17-122.07)	0.007

Table 1. Binary logistic regression analysis showing the increased risk of each variable by age group, using the youngest group (< 50 years old) as the reference category.

Conclusion: Our study presents the distribution of epidemiological and clinical variables across different age groups (<50 years, 50–75 years, and >75 years). We found that women over 75 years had a higher risk of developing AP compared to men, which could be influenced by an increased prevalence of biliary etiology with advancing age.

Interestingly, patients over 75 years required antibiotic therapy more frequently, although the rates of necrotizing pancreatitis did not differ between age groups. This may suggest a higher incidence of other types of infections in older patients.

Notably, patients over 75 years accounted for 80% of the deaths in our cohort, while no clear increase in mortality risk was observed in the 50–75 age group compared to younger patients.

Finally, the risk of severe AP increased progressively with age, peaking in patients over 75 years. In conclusion, female sex was predominant among patients over 75 years, who also showed a significantly higher risk of severe AP and mortality, as well as a greater need for antibiotic therapy.

Disclosure: Nothing to disclose.

PP1059

INFLUENCE OF BODY COMPOSITION ASSESSED BY ANTHROPOMETRY, BIOELECTRICAL IMPEDANCE AND COMPUTED TOMOGRAPHY IN EXOCRINE PANCREATIC INSUFFICIENCY AFTER ACUTE PANCREATITIS AND AFTER PANCREATIC ENZYME REPLACEMENT THERAPY: A PROSPECTIVE LONGITUDINAL AND RANDOMIZED CONTROLLED TRIAL STUDY

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Introduction: Acute pancreatitis (AP) is associated with an intense catabolic, local-regional and systemic inflammatory state, which can lead to changes in body composition, with prognostic implications.

Aims & Methods: Aims: To evaluate the body composition by anthropometry, bioelectrical impedance analysis (BIA) and computed tomography (CT) and its influence in AP severity, endocrine insufficiency (EnPI-AP), pancreatic exocrine (EPI-AP) and post-pancreatic enzyme replacement therapy (PERT).

Methods: Prospective longitudinal study of consecutive adult patients diagnosed with first AP episode. EPI-AP patients were randomized to receive PERT (4000IU/meal+20000IU/snack) vs. placebo (clinicaltrials.gov:NCT05480241). Body composition was assessed using anthropometry, BIA (InBody 570) and CT (cross-sectional areas L1/L3). Obesity was diagnosed for body mass index (BMI)≥30Kg/m^2, waist-to-hip ratio (WHR)>0.9(men)/0.85(women) and percentage total fat mass (TFM)>20%(men)/28%(women).

Visceral obesity was defined by abdominal circumference (AC)>94(men)/80(women)cm, visceral fat area (VFA)≥100cm^2 and visceral/subcutaneous fat ratio (VSFR)≥0.4. Sarcopenia was diagnosed by skeletal muscle mass (SMM)<20(men)/15(women)Kg, SMM index (SMMi)<7.0/5.5Kg/m^2 and skeletal muscle area index (SMAi)<41(men)/38.5(-women)cm^2. The three methods were correlated using spearman test and Bland-Altman plots.

Results: A total of 105 patients (65.7%(n=69) males; mean age:63.8 ±17.1yo) were enrolled and followed-up. AP was moderately-severe/severe in 32.4%/16.2%. EPI-AP and EnPI-AP occurred in 47.6% (n=50) and 17.1% (n=18). EPI-AP was persistent (≥6months after AP diagnosis) in 58%. After 6-month post-randomization, 44% (n=11/25) under PERT and 40% (n=10/25) under placebo resolved EPI-AP (p=0.598).

Obesity was diagnosed in 19.0% (n=20) by BMI, 97.1% (n=102)/72.4% (n=76) by WHR in antropometry/BIA, and 57.1% (n=60) by TFM. Visceral obesity was diagnosed in 78.1% (n=82) by AC, in 41.9% (n=44)/ 97.2% (n=103) by VFM in BIA/CT and in 90.5% (n=95) by VSFR. Sarcopenia was diagnosed in 1.0% (n=1) by MMS.

Considering outcomes, moderately-severe/severe AP was strongly associated with WHR by BIA (OR=10.2; p=0.034) and VFA by BIA (OR=1.08; p=0.029) and CT (OR=1.003; p=0.038). EPI-AP was statically associated with WHR by BIA (OR=11.6; p=0.048) and SMMi by BIA (OR=77.4; p=0.048) and TFM (OR=2.3; p=0.048). Severe EPI-AP was strongly associated with VSFR (OR=20;95%CI; p=0.048) and SMMi by BIA (OR=23.2; p=0.003) and CT (OR=2.3; p=0.041). SMMi by BIA (OR=14.1; p=0.030) and CT (OR=1.6; p=0.031) predicted EnPI-AP.

For obesity, TFM by BIA had a strong correlation with BMI (r=0.61; p<0.001) and a weak correlation with WHR by anthropometry (r=0.37; p<0.001). Regarding visceral obesity, VFA by BIA was weakly correlated with WHR by anthropometry (r=0.29; p=0.004) and VFA by CT (r=0.20; p=0.045). For sarcopenia, SMM by BIA was strongly correlated with SMM by CT (r=0.78; p<0.001).

In EPI-AP patients, a significant decrease of WHR (t(52)=2.262; p=0.028) and TFM (t(53)=1.695; p=0.046) by BIA was observed after 6months of PERT.

Conclusion: Body composition is significantly changed in AP, with >2/3 of patients with obesity, especially the visceral component. Obesity, including visceral obesity, correlates with AP severity. Obesity, including visceral obesity and sarcopenia are correlated with the EPI-AP and its severity, while sarcopenia is correlated with EnPI-AP. PERT has led to a significant reduction in obesity markers. Body composition assessment by BIA (TFM, VFA and SMM) and CT (VFA and SMM) were of clinical value and comparable.

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Disclosure: Nothing to disclose.

PP1060

RATE AND RISK FACTORS FOR PROGRESSION FROM ACUTE TO RECURRENT AND CHRONIC PANCREATITIS IN CHILDREN: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Most children recover after an initial acute pancreatitis (AP) episode, yet some progress to recurrent AP (RAP) or chronic pancreatitis (CP).

Aims & Methods: We aimed to quantify progression rates and identified exposures consistently associated with these transitions. MEDLINE, Embase and Cochrane were searched on 19th of December 2024 for paediatric studies reporting AP patients progression into RAP or CP. Random-effects model was used to pool proportions and odds ratios (OR) with 95 % confidence intervals (CI). Heterogeneity was assessed using the I2 statistics.

Results: Fifty-seven studies (5 642 children) met the eligibility criteria. After the first AP attack, RAP developed in 18 % (CI 16–22) and CP in 10 %. Among children with RAP, 35 % progressed to CP. RAP risk varied by aetiology: hypertriglyceridaemia 33 %, idiopathic 28 %, biliary 19 %, traumatic 16 %, drug-induced 14 %, virus-associated 3 %.

Structural abnormalities were linked with higher odds of RAP (OR 3.15, CI 1.51–6.56). Pancreas divisum (OR 2.64, CI 1.51–4.63) and PRSS1 mutation (OR 4.56, CI 3.06–6.80) correlated with CP.

Conclusion: Approximately one in five paediatric AP episodes recur, and more than one-third of RAP cases evolve into CP. Aetiology strongly modulates RAP risk, whereas structural and genetic factors relate to later progression. Early cross-sectional imaging for anatomical variants and targeted genetic testing after the inaugural attack may refine prognostication, counselling and surveillance.

These findings inform future international therapeutic trials.

Disclosure: Nothing to disclose.

PP1061

CLINICAL OUTCOMES OF SEVERE ACUTE PANCREATITIS

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Introduction: Early identification of severe acute pancreatitis (SAP) enables the prompt initiation of aggressive resuscitation strategies and improves patient outcomes.

The aim of the present study is to assess the clinical differences between patients with SAP and those with mild or moderate acute pancreatitis, as well as the distribution of organ failure in severe cases.

Aims & Methods: A retrospective analysis was conducted on a prospective database comprising a cohort of 526 patients with acute pancreatitis, consecutively included from March 2017 to June 2019 at a tertiary care hospital. All included patients underwent systematic blood testing at admission and at 48 hours. Clinical management adhered to current clinical practice guidelines. Diagnosis and classification were based on the revised 2012 Atlanta classification. The analysis included the need for antibiotic therapy, development of local complications, organ failure patterns, and infectious complications

Results: Table 1 summarizes the clinical characteristics and outcomes associated with severe acute pancreatitis (SAP). Regarding infectious complications, patients with SAP showed higher rates of infected necrosis and extrapancreatic infections, although no significant differences were observed in the incidence of acute cholangitis. Patients with SAP were more likely to develop necrotizing pancreatitis and local complications. Moreover, they accounted for the vast majority of ICU admissions and deaths. In terms of persistent organ failure, renal failure was the most frequent (80%), followed by respiratory failure (45%). A high proportion of patients who developed SAP presented with SIRS at admission (64%) and at 48 hours (55%).

	Mild/Moderately severe AP (n = 399 / 90.7%)	Severe AP (n = 41 / 9.3%)	p-value
Necrotizing pancreatitis	8 (2.1%)	13 (34.2%)	0.001 **
Infected necrosis	4 (1%)	6 (16.2%)	0.001 **
Antibiotic therapy	129 (32.9%)	29 (74.4%)	0.001 **
Extrapancreatic infection	35 (12.5%)	10 (28.6%)	0.011 **
Mortality	2 (0.5%)	18 (43.9%)	0.001 **
Cholangitis	145 (36.3%)	18 (43.9%)	0.340
ICU admission	3 (0.3%)	12 (29.3%)	0.001 **
SIRS at admission	56 (14.1%)	26 (63.4%)	0.001 **
SIRS at 48 hours	17 (4.3%)	22 (55%)	0.001 **

Table 1.

Conclusion: Early identification of severe acute pancreatitis (SAP) is not always straightforward and requires close monitoring. Nevertheless, recognizing these patients is critical, as prognosis is intrinsically linked to the timeliness of diagnosis and the implementation of supportive measures necessary to maintain organ stability.

Patients with SAP are at higher risk of developing organ failure. In our cohort, we observed increased use of antibiotic therapy among SAP patients, likely due to the higher incidence of infected necrosis and extrapancreatic infections. Conversely, the incidence of acute cholangitis, based on the 2018 Tokyo criteria, was similar between SAP and mild/moderate AP cases.

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Regarding organ failure, renal failure was the most frequent in our cohort, although other series have reported respiratory failure as the most common. The presence of SIRS at admission and at 48 hours was associated with subsequent development of SAP; however, a significant number of patients with SIRS did not ultimately develop a severe course. Its predictive value may be enhanced when combined with other prognostic factors.

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Disclosure: Nothing to disclose.

PP1062

ETIOLOGY-SPECIFIC PATTERNS OF INFECTIOUS COMPLICATIONS IN ACUTE PANCREATITIS

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Introduction: Infections in acute pancreatitis are associated with increased morbidity, mortality, and healthcare costs. Early diagnosis is essential to improve prognosis and reduce mortality. The aim of the present study is to evaluate the distribution of infectious complications according to the underlying etiology.

Aims & Methods: A retrospective analysis was conducted on a prospective database comprising a cohort of 526 patients with acute pancreatitis, consecutively included between March 2017 and June 2019 at a tertiary care hospital. All patients underwent systematic blood testing at admission and at 48 hours. Clinical management adhered to standard clinical practice guidelines. Diagnosis and classification were based on the revised 2012 Atlanta classification. Infectious complications were analyzed in patients with biliary and alcohol-related acute pancreatitis.

Results: The distribution of infectious complications occurring during hospitalization is shown in Table 1.

Variable	Overall (n = 526)	Biliary (n = 304)	Alcohol-related (n = 39)	p-value
Acute cholangitis (Tokyo-18)				
Suspected and/or confirmed	173 (37.6%)	140 (53.8%)	8 (20.4%)	0.001 **
Confirmed cases only	49 (29.2%)	43 (30.7%)	0	0.059
Antibiotic therapy	165 (36.1%)	123 (41.3%)	9 (23.7%)	0.037 **
Extrapancreatic infections (Total)	48 (14.3%)	34 (15.2%)	3 (13%)	1.000
Pneumonia	16 (4.8%)	8 (2.6%)	1 (2.6%)	1.000
Urinary tract infection	28 (8.4%)	25 (8.2%)	1 (2.6%)	0.335
Phlebitis / Catheter-related infection	4 (1.2%)	1 (0.3%)	1 (2.6%)	0.215
Infected pancreatic necrosis	11 (2.4%)	6 (2.1%)	2 (5.3%)	0.230

Table 1.

Up to 53.8% of patients with biliary acute pancreatitis presented with acute cholangitis according to the Tokyo-18 criteria, whereas this occurred in only 20.4% of alcohol-related cases. All cases of cholangitis in the alcoholic pancreatitis group, as well as the majority of those in the biliary group, met only the criteria for suspected acute cholangitis (100% vs. 53.8%, respectively). Antibiotic therapy was used in 41.3% of biliary pancreatitis cases, with a statistically significant difference compared to alcohol-related pancreatitis (23.7%).

No significant differences were observed in the rates of extrapancreatic infections between the two groups, occurring in 15.2% and 13% of biliary and alcohol-related cases, respectively.

Conclusion: Extrapancreatic infections play a key role in the clinical course of acute pancreatitis, particularly in moderate and severe cases. The most frequent sites of infection include the respiratory and urinary tracts, as well as bloodstream infections. These infections are associated with an increased risk of complications, prolonged hospital stays, and higher mortality.

The characteristics of infections vary according to the etiology of pancreatitis. In our cohort, biliary pancreatitis was significantly more often associated with acute cholangitis, which is consistent with the pathophysiology of biliary-origin pancreatitis. Among patients with alcohol-related pancreatitis, up to 20% met criteria for suspected cholangitis; however, none fulfilled the criteria for confirmed cholangitis.

This may be explained by the presence of acute-phase reactants and transient cholestasis, potentially leading to the administration of unnecessary antibiotic therapy.

Although previous studies have reported higher rates of respiratory infections in alcohol-related acute pancreatitis, our data showed no significant differences in these infections—or in infectious complications overall—between the two etiologies.

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PP1063

ACUTE PANCREATITIS: HAS THE EPIDEMIOLOGICAL PROFILE CHANGED OVER THE PAST TEN YEARS?

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Introduction: The available evidence on the epidemiology of acute pancreatitis (AP) is primarily based on older studies conducted in other countries, limiting its applicability to local settings. This study aims to provide updated data on the clinical characteristics and outcomes of patients with AP in our region thereby improving understanding and management of the disease.

Aims & Methods: A retrospective analysis was conducted on a prospective database comprising a cohort of 526 patients with acute pancreatitis (AP), consecutively included between March 2017 and June 2019 at a tertiary care hospital. Systematic laboratory testing was performed at admission and again at 48 hours. Clinical management followed standard practice guidelines, including routine abdominal ultrasound after admission and abdominal CT scan in cases of diagnostic uncertainty, clinical deterioration, or lack of improvement. Diagnosis and classification were based on the revised 2012 Atlanta classification.

Results: The most frequent etiology of AP in our setting was biliary lithiasis (65.9%), followed by idiopathic (15.4%) and alcohol-related causes (8.4%). Severe AP accounted for 9% of cases, while 14.3% were classified as moderately severe. Up to 20.2% of patients developed local complications; however, only 4.9% presented with pancreatic necrosis.

SIRS was observed in 18.7% of patients at admission, decreasing to 8.7% at 48 hours. Only 3.7% of patients required ICU admission (Table 1). Factors such as smoking, ERCP, antibiotic therapy, acute cholangitis, and SIRS showed significant differences across different etiological groups.

	Biliary	Alcohol	Hypertriglyceridemia	Pot-ERCP	Idiopathic	Total
Frequency	304 (65.9%)	39 (8.5%)	8 (1.7%)	5 (1.1%)	71 (15.4%)	461
Severe pancreatitis	28 (9.2%)	2 (5.1%)	0	1 (20%)	8 (12.7%)	42 (9%)
Local complications	24 (7.9%	6 (15.4%)	2 (25%)	1 (20%)	11 (15.5%)	106 (20.2%)
Transient organ failure (<48h)	46 (16.2%)	3 (8.1%)	2 (28.6%)	1 (20%)	14 (20.3%)	72 (16.3%)
Persistent organ failure (>48h)	28 (9.2%)	3 (7.7%)	0	0	9 (12.7%)	42 (9%)
SIRS at admission	46 (15.2%)	12 (30.8%)	1 (12.5%)	1 (20%)	16 (22.5%)	86 (18.5%)
Acute cholangitis (suspected and/or confirmed)	138 (45.5%)	7 (15.4%)	0	3 (60%)	17 (23.9%)	173 (37.6%)
Antibiotic therapy	123 (41.3%)	9 (23.7%)	1 (12.5%)	4 (80%)	19 (27.1%)	165 (36.1%)
ERCP performed	42 (14.1%)	0	0	2 (40%)	2 (2.9%)	49 (10.9%)

Conclusion: Acute pancreatitis (AP) is a common condition that imposes a significant healthcare burden. Determining the etiology of AP at admission is important not only from an epidemiological standpoint but also due to its profound implications for patient prognosis.

Contrary to what has been reported in the literature, the second most frequent cause of AP in our cohort was idiopathic, surpassing alcohol-related cases. In line with other studies, we observed significant differences between biliary and alcohol-related etiologies regarding smoking habits, development of acute cholangitis, need for ERCP, and use of antibiotic therapy. Patients with alcohol-related AP were significantly more likely to present with SIRS at admission. However, we did not find statistically significant differences in other key clinical outcomes, although patients with alcohol-related AP appeared to have a higher, but not statistically significant, risk of developing local complications, pancreatic necrosis, and SIRS at 48 hours.

This study highlights the importance of tailoring management strategies according to etiology to optimize clinical outcomes.

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PP1064

IMPACT OF PRE-EXISTING AND NEWLY DEVELOPED CARDIOVASCULAR CONDITIONS ON IN-HOSPITAL AND POST-DISCHARGE MORTALITY IN ACUTE PANCREATITIS: A SINGLE-CENTER REGISTRY ANALYSIS

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Introduction: Acute pancreatitis (AP) is a common gastrointestinal disorder with significant morbidity and mortality in severe cases. Pre-existing comorbidities and newly developed complications, like heart failure may influence outcomes in AP patients.

Aims & Methods: This study aimed to evaluate the impact of pre-existing and newly developed cardiac conditions on in-hospital and post-discharge mortality in patients with acute pancreatitis. A single-center registry analysis of the Hungarian Pancreatic Study Group Acute Pancreatitis Registry was conducted at Semmelweis University, Budapest, Hungary, involving 1,316 patients diagnosed with acute pancreatitis between 2021-2024. Data on pre-existing chronic heart failure (CHF), atrial fibrillation (AF), stroke history, and newly developed acute heart failure (AHF) were collected.

Electrocardiogram (ECG) findings, including QTc prolongation, were also analyzed. In-hospital mortality and post-discharge mortality within one year were assessed. Odds ratios (OR) with 95% confidence intervals (CI) were calculated to quantify the strength of associations.

Results: Of the 1316 patients, 45% were female, the average age was 55.8 (range: 18-95) years, 8% had severe, 25% had moderate, and 65% had mild AP. Regarding cardiovascular background, pre-existing CHF and stroke history did not show a significant association with increased in-hospital mortality in AP (OR: 1.85, 95% CI: 0.63-5.44 and OR: 1.21, 95% CI: 0.36-4.02, respectively). Although AP patients with pre-existing AF had four-fold higher odds of in-hospital mortality (OR: 4.01, 95% CI: 1.92-8.34).

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Assessing in-hospital cardiovascular alterations, newly discovered AF was not significantly associated with increased in-hospital mortality (OR: 4.40, 95% CI: 0.97-20.02).

Moreover, QTc prolongation on ECG was not significantly associated with in-hospital mortality (OR: 2.10, 95% CI: 0.61-7.25) or post-discharge mortality within one year after AP (OR: 1.65, 95% CI: 0.20-13.74). However, AP patients with newly developed AHF had a 25-times higher odds of in-hospital mortality (OR: 25.81, 95% CI: 12.13-54.98) and a 4-times higher odds for post-discharge mortality within one year (OR: 4.93, 95% CI: 1.01-24.26).

Conclusion: Pre-existing AF and newly developed AHF are significantly associated with increased in-hospital mortality in patients AP, moreover, newly developed AHF is also associated with increased post-discharge mortality within one year. These findings highlight the importance of cardiac monitoring and management in AP patients to improve outcomes.

Disclosure: Nothing to disclose.

PP1065

BIDIRECTIONAL ASSOCIATIONS BETWEEN PANCREATITIS AND PSYCHOLOGICAL DISORDERS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Depression and anxiety are common yet underrecognized comorbidities in patients with pancreatitis. While evidence suggests a significant psychological burden in these patients, the underlying mechanisms remain unclear, and psychological management is not routinely implemented in clinical care.

This study aims to assess the prevalence of these conditions and highlight the need for their integration into pancreatitis treatment.

Aims & Methods: Our protocol was registered on PROSPERO (CRD42023481739). We conducted a comprehensive search on November 10, 2023, across three databases - MEDLINE, Embase, and CENTRAL - to identify studies reporting on the prevalence of depression and anxiety symptoms in patients with pancreatitis. We used a random-effects model to calculate the pooled proportion rate and mean differences (MD) with 95% confidence intervals (CI), as well as a pooled single-mean analysis of the psychological questionnaire data.

Results: In total, 43 studies were included in our meta-analysis. Among patients with acute pancreatitis (AP), anxiety was reported in approximately 23% of cases (CI: 7-54), whereas in chronic pancreatitis (CP), the prevalence was higher at 31% (CI: 23-39). Depressive symptoms were identified in approximately 29% of AP patients (CI: 14-51) and 39% of CP patients (CI: 30-48). The pooled single-mean analysis of questionnaire data further underscored the severity of these psychological symptoms, with scores exceeding established clinical cut-off points. For depressive symptoms, mean scores were as follows: CESD-10 (14.30 [CI: 6.69-21.92]), HADS (8.23 [CI: 4.53-11.94]), and SES (49.29 [CI: 44.01-54.57]). Similarly, anxiety symptom severity was reflected in the following measures: HADS (8.67 [CI: 4.53-11.94]) and SAS (47.75 [CI: 35.05-60.45])

Conclusion: There is a significant prevalence of both anxiety and depression in patients with pancreatitis. There is a need to integrate psychological interventions into the treatment of pancreatitis to potentially improve clinical outcomes.

Disclosure: Nothing to disclose.

PP1066

PREVALENCE AND IMPACT OF PSYCHOLOGICAL DISTURBANCES ON DISEASE SEVERITY IN PANCREATITIS: A COHORT ANALYSIS

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Introduction: Depressive and anxiety symptoms may influence the severity of acute pancreatitis episodes; however, their impact, particularly in pancreatic units, remains poorly understood.

Aims & Methods: Our study aimed to identify risk factors associated with poorer psychological assessment scores in patients with acute and chronic pancreatitis (AP, CP) and their impact on clinical outcomes.

Patients admitted to a tertiary pancreatic unit were assessed for depressive and anxiety symptoms upon admission, with follow-ups every seven days during hospitalization. Psychological states were measured using the Hospital Anxiety and Depression Scale (HADS).

Results: Of the 1,397 admitted patients (median = 54, range = 17–97), 1,237 completed the questionnaire (42.98% female). At admission, 56.5% of patients had normal scores (0-7 points), 23.4% had borderline abnormal scores (8-10 points), and 20.3% had abnormal scores (11-21 points). The prevalence of abnormal cases at intervention, i.e. before intervention, was 20.15% in AP, 23.15% in early chronic pancreatitis (ECP), and 24.15% in acute-on-chronic pancreatitis (ACP). Borderline abnormal cases were observed in 22.25%, 24.15%, and 19.15% of cases, respectively.

Abnormal cases varied by etiology: hypertriglyceridemia (13.9%), biliary (16.9%), alcohol-related (26.3%), and tumor-related pancreatitis (26.6%). The prevalence also differed by AP severity: mild AP (17.76%), moderate AP (23.94%), and severe AP (35.05%). Following psychological intervention, the number of patients with abnormal scores decreased (67.6%, 19.6%, and 12.8%, respectively).

Conclusion: Depressive and anxiety symptoms are common in patients with acute and chronic pancreatitis, with more severe AP already associated with worse psychological scores at admission. Psychological interventions are essential to improve clinical outcomes.

Disclosure: Nothing to disclose.

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PP1067

CONTRIBUTION OF ENDOSCOPIC ULTRASOUND TO THE AETIOLOGICAL DIAGNOSIS OF UNEXPLAINED ACUTE PANCREATITIS

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Introduction: The aetiology of acute pancreatitis remains elusive in 10-30% of patients after clinical, biological and morphological investigations. Endoscopic ultrasound has shown promise and is particularly effective in exploring the bilio-pancreatic region.

Aims & Methods: The aim of our study was to evaluate the contribution of endoscopic ultrasound to the aetiological diagnosis of unexplained acute pancreatitis.

This is a retrospective descriptive study of 89 patients who underwent endoscopic ultra sound for unexplained acute pancreatitis between January 2010 and January 2025.

Results: The mean age of our patients was 55 +/- 16.3 years, with extremes ranging from 17 to 89 years. The sex ratio (M/F) was 0.81, with a slight female predominance of 55.1%. A history of previous acute pancreatitis was found in 31.9% of cases, and 16.3% of patients had undergone cholecystectomy. The diagnostic yield in our study was 68%. Biliary origin was retained in 40.4% (n=36) of patients, 19 of whom underwent complementary ERCP at the same anaesthetic time. Neoplastic pathology was found in 32.5% of cases (n=30). Chronic pancreatitis was reported in 22.4% of patients (n=20). Two cases of pancreas divisum, one case of sphincter of Oddi dysfunction and one case of choledocele were found. In the remaining 32% of cases, the endoscopic ultrasound was normal and no cause could be found.

Conclusion: Endoscopic ultrasound is currently an indispensable tool in the etiological investigation of unexplained acute pancreatitis. In our study, the diagnostic efficiency was around 68%.

Disclosure: Nothing to disclose.

PP1068

GENOME-WIDE ASSOCIATION STUDY IDENTIFIES FUT2 RS601338 POLYMORPHISM LINKING INTRA-PANCREATIC FAT DEPOSITION TO CHRONIC PANCREATITIS

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Introduction: Pancreas fat deposition has emerged as a critical area of research within the metabolic field. However, its genetic basis and how genetic predisposition impacts on the pancreatic disorders are not well understood.

Aims & Methods: This study used UK Biobank data, including MRI scans of 20,040 European individuals with pancreatic proton density fat fraction measurements (excluding those with acute/chronic pancreatitis or pancreatic cancer). We conducted a genome-wide association study (GWAS) and subsequent post-GWAS analyses to identify genetic loci and prioritize genes enriched in pancreatic tissue. Regression models and interaction analyses were used to assess the impact of these candidates on the risk of pancreatic disorders.

Results: Our analyses identified 44 independent significant genetic variants near 14 protein-coding genes, including FUT2, CBFA2T2, FAF1, MAMSTR, SGMS1, CSNK2A1, CEBPB, and CEBPG. Genetic colocalization and transcriptome-wide association studies (TWAS) confirmed the essential roles of FUT2 and CBFA2T3 in the pancreas. Single-cell transcriptomics showed high FUT2 expression specifically in the exocrine pancreas (acinar and ductal cells).

Notably, the missense polymorphism rs601338 in the FUT2 exon region causes a stop gain in protein translation. Regression analysis revealed that the rs601338 AA genotype was significantly linked to increased chronic pancreatitis (CP) risk across all populations in full adjustment model (OR 1.26, [95% CI 1.07-1.48], $p < 0.001$; HR 1.24, [95% CI 1.03-1.50], $p < 0.001$). Subgroup analyses revealed population-specific patterns: in Europeans, the AA genotype showed significant associations in logistic regression, while in non-Europeans, both GA and AA genotypes exhibited positive trends, with AA achieving significance (OR 1.20, [95% CI 1.01-1.41], $p = 0.034$). Sex-stratified analyses showed significant associations in males (OR 1.33, [95% CI 1.08-1.62], $p = 0.006$; HR 1.31, [95% CI 1.04-1.67], $p = 0.025$) but not females with full adjustment. Interaction analysis indicated a synergistic risk effect of current smoking and the rs6031218 AA genotype on CP, while COX-2 inhibitor treatment might reduce CP risk in rs6031218 AA individuals.

Conclusion: This study offers genetic insights into pancreatic fat deposition and its systemic health impacts. The findings emphasize the need for regular surveillance of pancreatic fat deposition in certain population with a high genetic risk from FUT2 polymorphism and provide evidence for developing therapeutic strategies for the primary prevention of CP.

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PP1069

TRANS PAPILLARY DRAINAGE IS THE WAY TO DRAIN PSEUDOCYSTS COMPLICATING CHRONIC PANCREATITIS- A TERTIARY CARE CENTRE STUDY

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Introduction: To assess whether trans papillary drainage is the way to go for pseudocysts complicating patients with chronic pancreatitis. With our previous experience we used the technique of no injecting contrast during ERP in case of leaks complicating chronic pancreatitis causing either pseudocysts, pleural effusions or ascites.

Here we are presenting a retrospective analysis of our management of pseudocysts in chronic pancreatitis.

Aims & Methods: We retrospectively analysed all cases of chronic pancreatitis having ascites, effusions and pseudocysts. We performed magnetic resonance cholangiopancreatography (MRCP) followed by therapeutic endoscopic retrograde pancreatography (ERP) without contrast. Results were observed confirming a leak from the MPD, possibility of wire crossing the leak or not and resolution of the pseudocysts. Patients with only pseudocysts were included in this analysis.

Results: We performed ERP in 1522 patients from 2000 till date. Ductal disruptions noted in 509/1522 (33.44%) out of which pseudocyst was observed in 330 cases (64.83%).

311 patients (94.2 %) experienced successful cannulation of PD. In 19 (5.8%) PD could not be cannulated. Non cannulated cases treated with EUS guided transmural drainage 14 (73.7%) and surgery 5(26.3%)

Of these 311, 24 (8.3%) had pancreas divisum and MPD was cannulated from minor papilla.

Leak did not cross but stents put in cyst (236, 75.9 %). Cyst resolution – 228 (96.6%). Cyst Not resolved 8 (3.4%). Leaks were crossed in 75 (24.1 %); Cyst resolution 69 (92%). Cyst not resolved 6 (8%).

Unresolved cysts 14 (4.5%) were treated with EUS Aspiration- 5 (35.7%), EUS Transmural drainage 8 (57.1%), Surgery -2 (14.2%).

Discussion: In chronic pancreatitis the cause of pseudocysts is ductal rupture and very minimal inflammation as compared to acute pancreatitis where the major cause is inflammation with ductal leak and necrosis of the pancreas.

There is hardly any necrosis in chronic pancreatitis pseudocysts. Hence the management of both should be individualised. Transpapillary drainage in pseudocysts complicating chronic pancreatitis offers a less morbid and a simpler alternative to transmural drainage or surgery.

Conclusion: Pseudocysts complicating chronic pancreatitis can be managed very effectively using transpapillary drainage in majority of cases and should be attempted before considering transmural or surgical drainage. It saves the patient significant morbidity

Disclosure: Nothing to disclose.

PP1070

CROSS-SECTIONAL STUDY ON SEX DIFFERENCES IN CHRONIC PANCREATITIS- 1922 PATIENTS FROM THE SCANDINAVIAN BALTIC CLUB DATABASE

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Introduction: Chronic pancreatitis (CP) may have differences in causes, presentations and progression related to sex. There are few studies describing the sex related phenotype of CP.

Aims & Methods: We aimed to study sex differences in etiology, complications and treatment of CP in a large, North-European CP cohort. Data were extracted from the Scandinavian Baltic Pancreatic Club (SBPC) database. The collected data included sex, age, etiology, anthropometric measures, data for smoking and alcohol exposures and data on treatment and complications from CP. Differences between sexes were compared with multiple logistic regression models.

Results: We included 1922 patients in the analyses, of whom 1264 were men. Mean age of CP onset was 52.5 (SD 15.0) years in men and 51.5 (SD 16.4) years in women (p=0.2). In univariate analysis alcohol etiology and smoking was associated with male sex, whereas hereditary pancreatitis, efferent duct obstruction, autoimmune pancreatitis were associated to female sex. Numbers of men and women with non-alcoholic etiology were almost equal (461 and 422, respectively).

In multivariate analysis including all etiologies, time with disease and age; alcohol etiology remained strongly associated to male sex (OR 2.92 (2.29-3.71), p<0.001), whereas duct obstruction etiology was associated with female sex (OR 0.62 (0.44-0.89), p=0.01).

Significantly more women were underweight (p<0.001), and more men were overweight (p<0.001). Concerning other complications, diabetes mellitus (DM) was strongly associated with men (OR 1.62 (1.27-2.07), p<0.001), whereas osteoporosis was strongly associated with women (OR 0.47 (0.32-0.70), p<0.001). Recurrent acute pancreatitis and exocrine insufficiency were equally distributed between the sexes. There were no differences between sexes for medical, endoscopic or surgical treatment options.

Conclusion: More men than women had alcohol induced CP, but there was almost no difference in absolute numbers of men and women with non-alcoholic CP. Among the CP patients, DM was associated with men, while osteoporosis was associated with women.

Disclosure: Nothing to disclose.

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PP1071

QUALITY OF LIFE AND PATIENT SATISFACTION IN CHRONIC PANCREATITIS PATIENTS IN SRI LANKA: A CROSS SECTIONAL STUDY

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Introduction: Chronic pancreatitis (CP) is a debilitating condition characterized by persistent abdominal pain, malabsorption, and endocrine dysfunction, significantly impacting patients' quality of life (QoL) and satisfaction with care. This study aimed to assess QoL and patient satisfaction among CP patients in a low-resource setting and identify associated clinical and demographic factors.

Aims & Methods: A cross-sectional study was conducted at the University Surgical Unit of the National Hospital of Sri Lanka between May 2024 and February 2025. A total of 106 patients with confirmed CP were recruited. QoL was assessed using the EORTC QLQ-C30 and its pancreas-specific module PAN26. Patient satisfaction was evaluated using the Short Assessment of Patient Satisfaction (SAPS) questionnaire. Clinical and demographic data were collected. Associations between variables were analyzed with Pearson correlations, t-tests, and ANOVA. A p-value < 0.05 was considered statistically significant.

Results: The mean global QoL score was 47.99 (SD = 28.29), indicating a moderate impairment. Among symptoms, fatigue (mean = 44.7), pain (44.8), and financial difficulties (48.7) were most burdensome. Functioning scores were highest in social (72.3) and physical (69.1) domains. From the PAN26 module, pancreatic pain (44.28), digestive symptoms (42.97), and impaired body image (33.82) were common concerns. The mean SAPS score was 20.0 (SD = 3.5), with 64.4% of patients reporting being satisfied or very satisfied with care.

A moderate positive correlation was observed between QoL and patient satisfaction ($r = 0.348$, $p < 0.05$), suggesting that improved well-being was associated with higher satisfaction. Factors significantly associated with poorer QoL included diabetes secondary to CP ($p < 0.05$), alcohol consumption, smoking, and lower BMI ($p < 0.05$).

Interestingly, patients who underwent ERCP reported lower QoL scores ($p < 0.05$), likely reflecting that more severe cases were selected for intervention. Higher education level showed a trend toward improved QoL. Lower-income patients reported worse QoL, underscoring the role of financial hardship in disease burden.

Conclusion: This study highlights that CP patients in Sri Lanka face considerable challenges to their quality of life, largely driven by persistent pain, comorbidities, and economic constraints. A clear link exists between clinical severity and reduced QoL, but invasive procedures such as ERCP do not always translate to better patient-reported outcomes, possibly due to selection bias. Conservative management, when coupled with adequate pain relief, nutritional support, and patient-centered follow-up, may be equally effective in select patients.

Interventions aimed at improving nutritional status, optimizing diabetes and pain control, and addressing financial toxicity are critical for improving both QoL and patient satisfaction.

Disclosure: Nothing to disclose.

PP1072

RISK FACTORS OF CALCIFICATION IN CHRONIC PANCREATITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Pancreatic calcification is the most frequent structural change in chronic pancreatitis (CP), but the factors associated with its development are not well understood.

Aims & Methods: We aimed to investigate the risk factors of calcification development in patients with chronic pancreatitis. The study protocol was registered on PROSPERO (CRD42024591837). On September 18, 2024, we conducted a systematic search across three databases (PubMed, EMBASE, and CENTRAL) for studies reporting factors associated with calcification in chronic pancreatitis (CP). Pooled proportions and odds ratios (OR) with 95% confidence intervals (CI) were calculated using a random effects model.

Results: The systematic search resulted in 10,893 articles, from which 85 eligible studies were identified, covering a total of 34568 participants. The odds of calcification were 45% higher in alcoholic compared to idiopathic CP (OR=1.45, CI: 1.00–2.11). Alcohol consumption (OR=2.31, CI: 1.8–2.98) and smoking (OR=2.22, CI: 1.61–3.06) were associated with a two times higher risk of calcification in CP, when compared to non-smokers and non-drinkers. In contrast, the presence of CFTR mutations did not increase the risk of calcification in CP (OR=0.43, CI: 0.11–1.66) compared to patients without CFTR mutations.

Conclusion: Alcohol consumption and smoking were significantly associated with the presence of calcification in CP, making it essential to identify these risk factors and incorporate smoking cessation and alcohol reduction programs into standard care.

Disclosure: Nothing to disclose.

PP1073

EFFECT OF NON-ENTERIC COATED ENZYMES ON PAIN IN PATIENTS WITH CHRONIC PANCREATITIS: A DOUBLE-BLINDED, PLACEBO-CONTROLLED RANDOMIZED TRIAL

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Introduction: Recurrent abdominal pain is the predominant symptom of chronic pancreatitis (CP), contributing to significant morbidity and reduced quality of life. Medical management includes analgesics, antioxidants, neuromodulators, and pancreatic enzyme replacement therapy (PERT).

While PERT is used for pancreatic exocrine insufficiency (PEI), it is also thought to reduce pain by suppressing pancreatic secretion via negative feedback. However, previous meta-analysis did not show pain reduction,

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though subgroup analysis suggested potential pain relief with non-enteric (NE) coated enzymes1. Unlike currently available enteric-coated formulations, high in lipase but low in proteas, non-enteric coated enzymes with high protease content dissolve rapidly and may more effectively suppress cholecystokinin-mediated pancreatic stimulation, potentially improving pain outcomes.

Aims & Methods: To primarily assess the effect of a high-protease, non-enteric-coated pancreatic enzyme preparation on pain in patients with painful chronic pancreatitis. The study was registered under ClinicalTrials.gov (NCT05042284).

We conducted a randomized, double-blind, placebo-controlled trial where we randomly assigned participants to a high protease (30,000 U)-containing, non-enteric-coated enzyme preparation (n=54) and a placebo (n=53) group. The study duration was 24 weeks. Patients with painful CP (per Mayo Clinic criteria) with chronic pain were enrolled.

The primary outcome was to see a difference in mean Izbicki pain score at 12 and 24 weeks and a change in pain scores (VAS 0-10) using pain diaries. Secondary outcomes were changes in the number of painful days, analgesic requirements, patient's global impression of change (PGIC), quality of life, and adverse effects. Weekly pain data were collated. Repeated measures, linear, and mixed-effects models were used to analyze the primary outcome. Analysis was intention-to-treat.

Results: 107 out of 256 screened patients fulfilled enrollment criteria. The mean age was comparable between groups (35.1±10.6 vs. 33.6±12.3 years; p=0.51), and baseline characteristics were similar. The primary outcome using mixed model analysis showed no significant differences in the change in Izbicki pain scores between the two groups at 12 and 24 weeks (Table: Results). The mean difference in VAS pain score was -3.7(95%CI: -4.3to-3.1) in enzyme group vs. -3.5(95%CI: -4.2to-2.9) in placebo group (p=0.64) at 12 weeks. The changes were -4.2(95%CI: -4.7to-3.6) vs. -4.1(95%CI: -4.7to-3.5), respectively (p=0.79), at 24 weeks indicating no significant difference. The median number of painful days was also similar, at 12 weeks, 16 (IQR: 9–35) in the enzyme group vs. 16 (IQR: 4–36.5) in placebo (p=0.31); at 24 weeks, 3(IQR: 0–8.5) vs. 3(IQR: 0–7)(p=0.30) respectively.

Analgesic use was higher in the enzyme group at 12 weeks (45(83.3%) vs. 37(69.8%); p=0.11) and 24 weeks (42(77.8%) vs. 35(66%); p=0.20). Differences in the change in global health status, quality of life, PGIC, and adverse effects at both time points were also non-significant.

			Estimated mean difference from baseline (95% CI)			
Mean (SD) Izbicki score	Placebo (n=53)	NEPERT (n=54)	Placebo	NEPERT	Mean difference in change	P value
Baseline	42.8 (17.0)	46.2 (14.6)				
12 weeks	23.5 (18.6)	29.4 (20.3)	-19.3 (-24.1 to -14.7)	-16.8 (-21.6 to -11.9)	-2.5 (-9.3 to 4.2)	0.49
24 weeks	9.6 (11.8)	12.8 (16.0)	-33.2 (-38.1 to -28.4)	-33.4 (-39.2 to -27.6)	0.2 (-7.2 to 7.6)	0.96

Results: Primary endpoint (Izbicki pain score).

Conclusion: Despite the high protease content, Non-enteric coated enzyme preparations did not improve pain and quality of life in patients with painful chronic pancreatitis. Further RCTs are needed to evaluate benefits in distinct subgroups.

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Disclosure: Nothing to disclose.

PP1074

PANCREATIC FIBROSIS NON-INVASIVE ASSESSMENT BY BIOMARKER ANALYSIS: THE FINAL RESULTS

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Introduction: Pancreatic fibrosis (PF) is a prominent histological feature of chronic pancreatitis and influences on vast majority of pancreatic disorders. So, PF early diagnosis gives a great opportunity to be a part of the strategy for preventing pancreatic diseases progression. It makes non-invasive PF assessment especially via biomarkers level measurement research perspective.

Aims & Methods: Aim was to evaluate the possibility of biomarkers levels measurement in PF grade assessment.

Methods. We measured interleukin-10 (IL-10) and interleukin-36 (IL-36), matrix metalloproteinase-1 (MMP-1) and matrix metalloproteinase-9 (MMP-9), fibronectin (FN) and hyaluronic acid (HA) plasma levels from 165 patient who underwent surgery from April 2022 to February 2024 due to pancreatic tumors and advanced chronic pancreatitis. We included 69 (41.8%) male and 96 (58.2%) female patients, mean age was 57.91±13.5 (25-89). Serum levels of specified biomarkers were determined by enzyme immunoassay using commercial kits; data registration performed was carried out on a semi-automatic analyzer. PF morphological grade was assessed according to Klöppel & Maillet's scoring system [1] and scaled into three grades: mild, moderate and severe PF.

This study was conducted within the healthcare research project "Assessment of pancreatic fibrosis as a prognostic factor for its diseases course" funded by Autonomous non-profit organization "Moscow Center for Innovative Technologies in Healthcare" administered by the Moscow Healthcare Department.

Results: Plasma FN (pFN) level was significantly (p=0.029) higher in patients with mild PF (pFN=110.7±72.7 μg/mL), than in patients with severe PF (pFN=74.9±32.8 μg/mL). The median pFN levels were significantly varied in groups with different grades of intralobular PF (p<0.05), but not in groups divided by perilobular PF grade (p>0.05).

Moreover, pFN level significantly negatively correlated with PF integrative grade (Spearmen's rho = -0.21, p=0.007). Regarding IL-10, IL36, MMP-1, MMP-9 and HA levels, there were no significant differences either in patients with different PF integrative grade, or in groups divided by perilobular and intralobular PF (p>0.05). We revealed no significant correlations between specified biomarkers levels and PF grade (p>0.05).

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Conclusion: The FN serum level significantly decreased in patients with severe PF in contrast to patients with mild PF who had comparatively higher serum FN level. Oppositely, IL-10, IL36, MMP-1, MMP-9 and HA had shown no effect in PF grade assessment. It could be useful for noninvasive pancreatic fibrosis diagnosis and even for patients' selection for advanced diagnostics.

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Disclosure: Nothing to disclose.

PP1075

CLINICAL EVALUATION OF FECAL PANCREATIC ELASTASE QUANTIFICATION WITH A RAPID TEST AND AN AUTOMATED TURBIDIMETRIC IMMUNOASSAY

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Introduction: Pancreatic exocrine insufficiency (PEI) is characterized by a significant reduction in pancreatic exocrine activity within the intestine, impairing normal digestive processes [(1)]. This condition results in malnutrition-related complications and a diminished quality of life. Faecal pancreatic elastase serves as an optimal biomarker for evaluating pancreatic function in individuals with PEI. Its high stability ensures that faecal pancreatic elastase is concentrated during intestinal transit, thereby accurately reflecting total pancreatic secretion. The BÜHLMANN fPELA® turbo is a particle-enhanced turbidimetric immunoassay (PETIA) that is commercially available for the automated quantification of pancreatic elastase in faecal extracts using clinical chemistry analyzers.

Additionally, BÜHLMANN has developed the Quantum Blue® fPELA, a lateral flow immunoassay (LFIA) designed for environments requiring lower sample volumes. This study aimed to evaluate the clinical performance of both assays and compare them to a commercially available ELISA reference method.

Aims & Methods: A total of 189 symptomatic patients, comprising 82 non-PEI controls and 107 PEI-diagnosed patients, were utilized to compare the clinical performance of PETIA and LFIA against the ELISA reference assay. Sensitivity and specificity for both assays were determined using cutoffs of 100 µg/g and 200 µg/g.

Results: The area under the curve (AUC) was 0.809 for the BÜHLMANN fPELA® turbo (optimal cut-off: 166 µg/g), 0.792 for the Quantum Blue® fPELA (optimal cut-off: 109 µg/g), and 0.814 for the reference ELISA (optimal cut-off: 154 µg/g). At clinically relevant cut-offs of 100 µg/g and 200 µg/g, the BÜHLMANN fPELA® turbo demonstrated sensitivity of 0.57 and 0.74, and specificity of 0.9 and 0.76, respectively. The Quantum Blue® fPELA showed sensitivity of 0.62 and 0.71, and specificity of 0.93 and 0.74, respectively. The reference ELISA exhibited sensitivity of 0.64 and 0.73, and specificity of 0.87 and 0.70, respectively. A direct method comparison using at least 60 samples within the linear measuring range of all assays revealed a relative median difference in Bland-Altman analysis of 9.3% between the BÜHLMANN fPELA® turbo and the reference ELISA, 10.8% between the ELISA and the Quantum Blue® fPELA, and 11.0% between the PETIA and the LFIA.

Conclusion: The study demonstrated that both the BÜHLMANN fPELA® turbo and the Quantum Blue® fPELA exhibit strong diagnostic performance to detect PEI in symptomatic patient population and high concordance with the reference ELISA assay. The BÜHLMANN fPELA® turbo, a particle-enhanced turbidimetric immunoassay, is an accurate, non-invasive, automated test ideal for large-scale determination of pancreatic elastase levels in faecal extracts.

Meanwhile, the Quantum Blue® fPELA rapid test is well-suited for smaller laboratories, offering a faster time to result compared to traditional ELISA methods, while offering a coequal clinical performance.

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Disclosure: Conflict of interest: CR, DT, HI, AT, MU, RT, MS, TS, SN and CG are employees of BÜHLMANN Laboratories AG.

PP1076

VITAMIN D INSUFFICIENCY AND DEFICIENCY IN PATIENTS WITH CHRONIC PANCREATITIS – A COMPLICATION LEADING TO FURTHER COMPLICATIONS

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Introduction: Malnutrition is a typical complication of pancreatic diseases. Vitamin D (VD) insufficiency or deficiency is present in chronic pancreatitis (CP), leading to increased cardiovascular risk, bone complications, impaired quality of life and increased mortality.

Aims & Methods: The study aimed to determine the prevalence of VD deficiency or insufficiency in patients with CP, and to assess its relationship to CP progression and associated cardiovascular and bone complications. 70 patients (45 males) were enrolled, with a mean age of 54.01±13.46 years. Alcohol abuse was the most common aetiology. Imaging morphological data were assessed by Cambridge classification for CT/MRCP; pancreatic exocrine insufficiency (PEI) - by fecal elastase-1, CP severity - by M-ANNHEIM classification; bone densitometry - by Dual-energy X-ray absorptiometry; and cardiovascular risk - by 10-year risk of mortality scores (SCORE and Framingham Risk Score - FRS) and by the arterial stiffness using pulse wave velocity (PWV) at a.carotis and a.femoralis.

Determination of 25-hydroxyvitamin D was performed by a LC-MS/MS method. Vitamin D status was assessed as deficiency (<25 nmol/L), severe insufficiency (25–50 nmol/L), mild insufficiency (50–80nmol/L), and sufficiency (>80 nmol/L). Resting energy expenditure was calculated using the Harris-Benedict formula. Statistical analysis was performed by SPSS.

Results: Mean VD levels were 37.86±24.36 nmol/L (range 3.854–99.874 nmol/L); 26 patients had deficiency; profound insufficiency was found in 25 patients; another 14 were with mild insufficiency, and only 5 patients were in a sufficiency status. We found no significant relation between VD levels and gender. VD levels correlated significantly with BMI but not with symptoms (weight loss, pain and steatorrhea). 12 patients reported

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no symptoms, however, 10 of them had levels of VD less than 50 nmol/L. Resting energy expenditure correlated significantly with VD levels (r=0.53, p=0.000). In patients with severe structural changes, we observed significantly lower VD levels regardless of etiology (p<0,01). VD levels were lower in patients with PEI, p<0.05. Patients with mild CP by M-ANNHEIM had lower VD levels compared to patients with moderate and advanced CP, p<0.05.

We found an association of VD with pancreatolithiasis. At a cut-off of VD 11.95 nmol/L, we verified pancreatic lithiasis with 89.4% sensitivity and 83.3% specificity and AUC of 0.826±0.113 (95% CI, 0.61-1). VD status worsened with increasing the 10-year risk of mortality by both SCORE and FRS, p<0,05.

We observed lower vitamin D levels with increasing PWV. Lower VD levels were associated with diabetes mellitus. 35.6% of the investigated patients had osteopenia or osteoporosis. Bone density was affected and significantly decreased in patients with VD deficiency or severe insufficiency.

By increasing and optimising the dose of pancreatic enzymes in patients with pancreatic insufficiency, we demonstrated clinical improvement with a significant increase in BMI (p<0.0001) and VD levels, p<0.01.

Conclusion: Most of our patients with CP were VD insufficient. Monitoring of nutritional status in patients with CP is mandatory to prevent the development of malnutrition complications and associated morbidity and mortality.

An effort should be the point of a multidisciplinary approach for an up-to-date strategy for patients with pancreatic disorders, ensuring optimal therapeutic results and better quality of life.

Disclosure: Nothing to disclose.

PP1077

SPLENIC VEIN THROMBOSIS PREVALENCE IN CHRONIC PANCREATITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Chronic pancreatitis (CP) is a progressive inflammatory disorder that predisposes patients to various complications, including vascular events. Among these, splenic vein thrombosis (SVT) represents a siginficant, but often underdiagnosed complication, with reported prevalence rates varying widely. SVT may lead to sinistral portal hypertension, resulting in gastric varices and, in severe cases, life-threatening gastrointestinal hemorrhage.

This systematic review and meta-analysis aims to determine the true pooled prevalence of SVT in CP patients and to evaluate the bleeding rate among those with splenic thrombosis by synthesizing data from studies in which all participants underwent appropriate imaging for SVT detection.

Aims & Methods: An exploratory search was conducted to identify key articles reporting on the prevalence of SVT among CP patients. Four studies were selected based on their provision of relevant, poolable data on prevalence with the use of proper imaging techniques for detecting SVT, encompassing a total of 1,973 patients. A random effects meta-analysis was performed using SPSS, and a forest plot was genrated to visually represent both individual and overall prevalence estimates. An ongoing PRISMA compliant systematic literature search continues to capture all pertinent studies on this topic.

Results: Analysis of data from the four selected studies (n = 1,973) revealed a pooled SVT prevalence of 12.7% in CP patients. Although individual study estimates varied, likely reflecting differences in study design, CP etiology, and follow-up duration, the overall result indicates that approximately one in eight patients with CP develops SVT.

Conclusion: Our preliminary findings demonstrate that, when appropriate imaging is employed, SVT occurs in approximately 12.7% of chronic pancreatitis cases. These results underscore the clinical importance of SVT, particularly given its association with an increased risk of gastrointestinal hemorrhage. Routine evaluation for SVT in CP patients should be considered, and further research is warranted to elucidate underlying risk factors and to refine management strategies for this potentially life-threatening vascular complication.

Disclosure: Nothing to disclose.

PP1078

PAIN IN CHRONIC CALCIFIC PANCREATITIS: PREVALENCE, MANAGEMENT, AND ASSOCIATED FACTORS

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Introduction: Pain is the hallmark and most disabling symptom of chronic calcific pancreatitis (CCP), often severe and significantly impairing patients' quality of life. Its pathophysiology involves increased intraductal pressure, chronic inflammation, and neuropathic changes.

Aims & Methods: This study aimed to assess the prevalence of pain in CCP and to identify factors associated with its occurrence and severity.

This was a retrospective analytical study including patients followed for CCP. We collected data on disease duration, etiology, clinical manifestations, complications, analgesic treatments (AT), and symptom evolution under treatment. Pain intensity and impact on quality of life were assessed using the Izbicki Score (IS), which categorizes pain as mild (0–20), moderate (21–40), severe (41–60), or very severe (>60). Incomplete records that prevented IS calculation were excluded. Data were entered and analyzed using SPSS software, with statistical significance set at p < 0.05.

Results: We included 22 patients (14 men and 8 women), with a mean age of 55.6 years [27–80 years]. The most common symptoms were epigastric pain (n=18), weight loss (n=13), and chronic diarrhea (n=6). The mean delay between symptom onset and diagnosis was 25 months [1–120 months]. Etiologies included hyperparathyroidism (n=6), alcohol use (n=2), pancreas divisum (n=2), autoimmune causes (n=2), and were undetermined in 10 cases. The IS ranged from 0 to 93.75, with a mean of 40.22. Eight patients had an IS >40. Morphine was required in four cases; paracetamol was prescribed for 12 patients, and tramadol for two. Endoscopic treatment was indicated in three cases, while one patient underwent a Wirsungojejunostomy.

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In univariate analysis, alcohol consumption was significantly associated with the occurrence of epigastric pain (p=0.01). Smoking and alcohol use were both linked to a higher IS (>50) (p=0.04). Smoking cessation was associated with a better analgesic response (p=0.04). Morphine use was significantly more frequent among alcoholic patients (p=0.01), and their treatment response was often unfavorable (p=0.02).

Conclusion: In our cohort, 80% of patients with CCP experienced pain, which was severe to very severe in nearly half the cases. Pain intensity was significantly associated with tobacco and alcohol consumption, while cessation improved treatment response. These findings underscore the importance of therapeutic education focused on smoking and alcohol cessation, in addition to analgesic management, to slow disease progression and improve pain control.

Disclosure: Nothing to disclose.

PP1079

FREQUENCY ANALYSIS OF THE RS7057398 MUTATION AT THE CLDN2-MORC4 LOCUS IN CHRONIC PANCREATITIS

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Introduction: The rs7057398 mutation in the *CLDN2-MORC4* locus on the X chromosome has been linked to chronic pancreatitis according to several recent publications and a meta-analysis. However, the mutation itself is located very far from both genes (CLDN2 and MORC4), and according to our current knowledge, the protein products of CLDNS and MORC4 genes are not functionally related to the development of chronic pancreatitis.

Aims & Methods: Our aim was to investigate the frequency of the rs7057398 mutation in Hungarian patients with alcoholic and non-alcoholic pancreatitis and controls. Our study included 139 non-alcoholic (NACP) and 161 alcoholic chronic pancreatitis (ACP) patients and 509 controls recruited by the Hungarian Pancreatic Study Group (HPSG). The rs7057398 mutation (*CLDN2* c.-26930T>C) status in DNA samples was examined by conventional PCR amplification and Sanger sequencing.

Results: The "C" allele of the rs7057398 mutation showed significant accumulation in patients with chronic pancreatitis (p<0.0001, OR 1.96, 95% CI 1.6-2.4), especially in the alcoholic group (p<0.0001, OR 2.56, 95% CI 1.98-3.3). When women are grouped separately, it is seen that the significant accumulation of the "C" allele is only significant in alcoholic chronic pancreatitis (p<0.005, OR: 2.5, CI: 1.3-4.8), which is also confirmed by the genotype distribution in the same group containing only women (dominant model: p 0.017, OR 3.55, 95% CI 1.3-10, recessive model: p 0.049, OR 3.01, 95% CI: 1.01-9.01).

Conclusion: The rs7057398 mutation located in the CLDN2-MORC4 locus on the X chromosome predisposes to the development of chronic pancreatitis in the Hungarian population, which association is particularly true in chronic pancreatitis of alcoholic etiology.

Disclosure: Nothing to disclose.

PP1080

AIPPEAR: AUTOIMMUNE PANCREATITIS, PANCREATIC AND EXTRAPANCREATIC CANCER – A RETROSPECTIVE GLOBAL MULTICENTER STUDY

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Introduction: Autoimmune pancreatitis (AiP) is a rare form of chronic pancreatitis. Studies from Japan and Germany suggest that AiP patients may have a higher incidence of malignancies compared to the general population. However, reliable data in this regard are still lacking.

Aims & Methods: We initiated a retrospective, multicenter observational study within the Pancreas 2000 framework to investigate the incidence of malignancy in AiP Patients.In our global multicenter retrospective observational trial, we are currently collecting data from patients with AiP diagnosed after 2005, including type 1, type 2, and not-otherwise-specified (NOS) AIP, regardless of the diagnostic criteria used. Patients with less than 12 months of follow-up after AIP diagnosis and those with immune checkpoint inhibitor-induced pancreatitis are excluded.

Our primary outcome is to determine the occurrence of the first invasive cancer in AiP patients. The secondary outcome is to evaluate the characteristics of AiP patients with distinct pancreatic and/or extrapancreatic cancers and the prevalence of all cancer diagnoses within 12 months prior to AiP diagnosis. The standardized incidence ratio of any first invasive cancer after AiP diagnosis will be compared to age-grouped and gender-matched controls in the general population. The expected number of cancers in the general population will be determined using the Cancer Incidence in Five Continents Volume XI (CI5XI) registry.

Results: We will present the current status of recruitment and preliminary data. To date, 16 centers from Europe and America have entered 321 cases. Ethics submissions have already been applied for in an additional 45 centers across four continents (Europe, Asia, America, and Oceania).

Conclusion: To the best of our knowledge, this is the first global observational trial analyzing the risk of developing cancer in patients with AiP.

Disclosure: Nothing to disclose.

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PP1081

AUTOIMMUNE PANCREATITIS SUBTYPES BEHAVE DIFFERENTLY: A 20-PATIENT EXPERIENCE FROM TURKEY

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Introduction: Autoimmune pancreatitis (AIP) is a rare, immune-mediated form of chronic pancreatitis, classified into three subtypes. Type 1 AIP, the pancreatic manifestation of IgG4-related disease, often presents with multisystem involvement and frequent relapse (1,2). Type 2 AIP is pancreas-limited, tends to affect younger individuals, and is commonly associated with inflammatory bowel disease (IBD) (2,3). A third form—checkpoint inhibitor-related pancreatitis (CIP or Type 3 AIP)—has been recently described in patients receiving immune checkpoint inhibitors, typically asymptomatic but may result in pancreatic atrophy (4).

Precise subtyping is critical due to differences in clinical behavior, relapse risk, and treatment response (1,2,3). We present a single-center series comparing clinical, radiologic, and histopathologic characteristics of Type 1 and Type 2 AIP.

Aims & Methods: We retrospectively reviewed 20 patients diagnosed with AIP between 2020–2024 at a tertiary center. Classification followed International Consensus Diagnostic Criteria (1). Diagnostic evaluation included endoscopic ultrasound (EUS), serum IgG4 levels, and contrast-enhanced CT/MRI/MRCP (1,2). EUS-guided fine-needle biopsy (FNB) was performed where malignancy could not be excluded. Histopathology differentiated lymphoplasmacytic sclerosing pancreatitis (LPSP, Type 1) from idiopathic duct-centric pancreatitis (IDCP, Type 2) (2). Data on extra-pancreatic involvement, steroid response, and relapse were collected.

Results: Fifteen patients were diagnosed with Type 1 AIP and five with Type 2. Type 1 patients had a mean age of 63 years and were predominantly male (80%). Obstructive jaundice was the most common presentation. All had elevated serum IgG4. EUS revealed diffusely hyperechoic parenchyma and long-segment ductal narrowing. Histology confirmed LPSP. Extra-pancreatic involvement was observed in 60%, most commonly IgG4 cholangitis. All responded to corticosteroids; 40% relapsed within one year and received rituximab (5).

Type 2 patients had a mean age of 38, with equal gender distribution. Presenting symptoms included abdominal pain and acute pancreatitis. Serum IgG4 was normal in all cases. EUS revealed focal hypoechoic lesions with short-segment narrowing. Histology confirmed IDCP. Two patients (40%) had ulcerative colitis. All achieved remission with steroids, and no relapses were observed (2,3).

Conclusion: This series emphasizes the distinct clinical courses of AIP subtypes. Type 1 AIP represents a systemic IgG4-related disease with high relapse risk, often requiring long-term immunosuppression (5). Type 2 AIP affects younger patients, remains confined to the pancreas, and shows sustained remission after steroid therapy (2,3). Accurate classification using EUS, MRCP, IgG4, and histology is crucial for appropriate management and long-term outcomes (1,2).

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Disclosure: Nothing to disclose.

PP1082

UTILITY OF PANCREATIC DUCT WALL THICKENING ON ENDOSCOPIC ULTRASOUND IN THE DIAGNOSIS OF AUTOIMMUNE PANCREATITIS

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Introduction: In autoimmune pancreatitis (AIP), if the disease is serum IgG4-negative, focal-type, and without other organ involvement, it is difficult to distinguish it from pancreatic cancer. The histopathological diagnosis of AIP using endoscopic ultrasound (EUS)-guided fine needle biopsy needles has been useful; however, the accuracy rate remains suboptimal, ranging from 54.7% to 87.2%[1,2,3].

We focused on a novel EUS finding of pancreatic duct wall thickening, which reflects the periductal lymphoplasmacytic infiltration characteristic of AIP[4].

Aims & Methods: This study aimed to evaluate the utility of pancreatic duct wall thickening on EUS in diagnosing AIP that is serum IgG4-negative, focal-type, and without other organ involvement. We retrospectively analyzed 35 patients diagnosed with AIP at our institution between March 2017 and January 2024, based on the Japanese Clinical Diagnostic Criteria for Autoimmune Pancreatitis 2018 (JPS2018). EUS was performed in all patients and the presence of pancreatic duct wall thickening was assessed. We defined pancreatic duct wall thickening as symmetric hypoechoic wall thickening of the pancreatic duct with a superficial hyperechoic layer. The presence of pancreatic duct wall thickening was evaluated along with localization, serum IgG4 level, and other organ involvement.

Results: Of 35 patients with AIP, 28 were male and 7 were female, with a median age of 72 years (range: 42–82). Fourteen patients (40%) had diffuse-type disease and 21 (60%) had focal-type disease. The median serum IgG4 level at diagnosis was 402 mg/dL (range: 14–2560). 29 patients (82.9%) were IgG4-positive (≥135 mg/dL) and 6 patients (17.1%) were IgG4-negative (<135 mg/dL). Other organ involvement was found in 12 patients (34.3%). Pancreatic duct wall thickening was observed on EUS in 30 patients (85.7%). Among 23 patients with focal-type and no other organ involvement, pancreatic duct wall thickening was observed in 15 of 17 (88.2%) serum IgG4-positive patients and in 5 of 6 (83.3%) serum IgG4-negative patients.

Conclusion: In AIP, EUS shows a high rate of pancreatic duct wall thickening, which is particularly useful in IgG4-negative, focal-type cases without other organ involvement.

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Disclosure: Nothing to disclose.

PP1083

A NETWORK META-ANALYSIS OF B-CELL DEPLETION FOR IGG4-RELATED DISEASE

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Introduction: IgG4-related disease (IgG4-RD) is a chronic fibro-inflammatory multi-systemic disease. Pancreato-biliary involvement is reported in up to 60%, which can lead to exocrine and endocrine pancreatic insufficiency. Glucocorticoids can be used to control active inflammation but are associated with substantial toxicity. B-cell depletion with anti-CD20 chimeric Rituximab (RTX) and anti-CD19 humanised Inebilizumab (INEB) are steroid-sparing and used to control disease flares. INEB is the only FDA-approved medication for the treatment of IgG4-RD.
Aims & Methods: Our aim was to compare B-cell depleting agents for the treatment of IgG4-RD. We searched PubMed, Cochrane Central Register of Clinical Trials and Embase databases for studies of maintenance therapy in IgG4-RD with a pre-defined strategy and performed a network meta-analysis to assess relapse at 12 months, occurrence of infections and infusion-related reactions.
Results: 96 out of 2545 manuscripts underwent full text review after exclusions. Three studies met eligibility for the meta-analysis, two reporting the use of RTX and one of INEB (Table 1).
For the primary outcome of disease relapse at 12 months, INEB had a risk ratio (RR) of 5.8 (95% CI 0.29 - 115.02) and RTX had an RR of 2.29 (95% CI 0.19 - 28.21), with INEB being superior in preventing relapses. The RR for infections with INEB was 0.81 (95% CI 0.19 - 3.41) whilst with RTX was 0.62 (95% CI 0.13 - 3.04). The RR for infusion-related reactions with INEB was 1.27 (95% CI 0.09 - 18.76) and for RTX was 0.93 (95% CI 0.11 - 8.19). These must be interpreted in the context of study design; INEB in a phase 3 randomised trial following a stringent safety protocol for reporting of adverse events, and RTX in retrospective observational studies.

Author	Year published	Design	Country	Sample size	Comparison	Follow up (months)	Male
Campochiaro	2020	Retrospective cohort	Italy	14	RTX vs placebo	-	7 (50%)
Majumder	2018	Retrospective cohort	US	43	RTX vs placebo	34 / 27	34 (79%)
Stone	2024	Randomised, placebo-controlled blinded	Global	135	Inebilizumab vs placebo	12	88 (65%)

Conclusion: This Network Meta-analysis demonstrates that INEB is superior to RTX at preventing disease relapses in patients with IgG4-RD at 12 months. Adverse events, such as infection and infusion-related reactions, were comparable in lieu of the methodological differences between studies.

Disclosure: Emma L Culver has received speaker fees from Albireo/IPSEN, Advanz/Intercept, Dr Falk, Amgen, GSK, Mirum Pharma, Gilead, Chemomab, Zenus Bio, and Advisory/consultancy fees from Advanz/Intercept, IPSEN, Mirum, GSK, Dr Falk Pharma, Amgen, Zenus Bio, Sanofi, Moderna and Sail BioPharma. MC has received speaker fees from Albireo/IPSEN, Advanz/Intercept, Amgen, GSK, Mirum Pharma, Gilead, and Advisory/consultancy fees from Advanz/Intercept, Alphasigma, Albireo/IPSEN, Mirum, GSK, Dr Falk Pharma, Amgen, Mayoly, Echosens, Kowa, Moderna, Zydus.

PP1084

ASSOCIATION BETWEEN DURATION OF DOUBLE-PIGTAIL PLASTIC STENT PLACEMENT AND PANCREATIC PSEUDOCYST RECURRENCE: A RETROSPECTIVE STUDY

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Introduction: Pancreatic pseudocysts (PPs), common complications of acute or chronic pancreatitis, are often managed with endoscopic ultrasound-guided double pigtail stent (EUS-DPS) drainage. However, the relationship between stent indwelling duration and PPs recurrence remains poorly defined.
Aims & Methods: We aimed to investigate whether shorter stent dwelling duration is associated with a higher risk of PPs recurrence. Patients diagnosed with PPs who underwent EUS-DPS drainage at Peking Union Medical College Hospital between June 2006 and December 2024 were retrospectively reviewed. Clinical outcomes including technical success, clinical success, postoperative complications, and recurrence were analyzed. Furtherly, the stent indwelling duration-recurrence relationship and risk factors were evaluated.
Results: A total of 116 patients were included (84 males, 72.4%), with a mean age of 48.29±13.56 years. The most common etiologies were biliary pancreatitis (28.4%) and alcoholic pancreatitis (25.9%). PPs were primarily located in the pancreatic body and tail (78.4%), with a mean diameter of 10.23±4.17 cm. Regarding drainage success, technical success was achieved in 106 cases (91.4%), and clinical success in 81 cases (69.8%). Postoperative adverse events occurred in 33.6% of patients, including infections (n=31), bleeding (n=3), perforation (n=2), and cardiopulmonary events (n=3).
Patients with clinical success were stratified by stent indwelling duration (0-2 months, 2-6 months, >6 months), while PPs recurrence was significantly higher in the short-duration group compared to the intermediate- and long-duration groups (p-value < 0.05).
No postoperative adverse events occurred beyond 30 days in any group, suggesting that prolonged stent indwelling duration does not increase the risk of long-term complications.
Meanwhile, multivariate regression analysis identified longer stent dwelling duration and elevated high-sensitivity C-reactive protein as independent protective factors for clinical success.
Conclusion: EUS-DPS placement represents a safe and effective modality for internal drainage and therapeutic approach for PPs. Our findings support that maintaining stent placement for more than six months to significantly reduce recurrence risk and improve long-term clinical outcomes, with manageable adverse events.
Disclosure: Nothing to disclose.

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PP1085

DIAGNOSTIC PERFORMANCE OF EUS-FNB IN PANCREATIC CYSTIC AND MIXED LESIONS

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Introduction: Pancreatic cystic lesions (PCLs) are common and could vary from a simple benign cyst to a highly malignant cyst.[1] Mixed lesions with a solid part and a cystic part are very uncommon. They could be a sign of malignant transformation of a cyst.[2]

Recently the awareness of these lesions has increased, especially with the increased incidence of asymptomatic pancreatic cysts worldwide primarily due to improved detection by different advanced imaging modalities.[3]

Therefore, the proper diagnosis, differentiation and staging of these lesions are crucial for proper management and avoiding unnecessary treatment of benign ones and missing early treatment of the pre-malignant lesions.[4]

EUS-guided fine needle biopsy (EUS-FNB) may overcome these limitations of EUS-FNA through acquisition of core tissue samples that provide the ability to assess cyst wall architecture via histology. At this point, ESGE does not recommand use of EUS-FNB for diagnosis of cystic lesions.[5]

Aims & Methods: Aim: Based to the literature, there are few reports regarding the role of EUS-FNB replacing EUS-FNA in the diagnosis of PCL. This work aims to study the feasibility and safety of EUS-FNB 22G in diagnosis of PCLs.

Methods: Of 640 consecutive EUS-FNB 22G for different lesions between January,2019 and December,2023 at our center. 60 EUS-FNB 22G were done for the patients who underwent EUS-guided sampling of pancreatic cystic or mixed lesions diagnosed by abdominal MRI and/or CT scan. The patients were retrospectively and prospectively followed. All the solid lesions, pancreatic pseudocysts (round, non-septated lesions, presence of internal debris, recent acute pancreatitis or radiologic imaging of chronic pancreatitis) and lesions interposed vessels or main pancreatic duct are excluded from this study.

Patients with history of severe hemorrhagic risk (platelet count <50,000 / Lor prothrombin activity <50%) are excluded. We statistically analyse the clinical and technical success rates, in addition to evaluate the diagnostic accuracy of EUS-FNB in differentiating benign and malignant pancreatic cystic lesions.

Evaluation of the cyst location, primary diagnosis, adverse events, and comparison between the cytologic fluid analyses and histopathology was performed. Technical success was defined as visible tissue present after biopsy. Clinical success was defined as the presence of a specimen adequate to make a histologic or cytologic diagnosis.

Results: A total of 60 patients (mean age 65.0 ± 12.1, 36.6% females) with 60 PCLs consecutively were referred for EUS-FNB 22G and enrolled in a prospective study. 41/60 (68.3%) lesions were mixed (cystic and tissue) and 19/60 (31.7%) lesions are cystic lesions. The mean lesion size was 32.4 ± 16.6mm. The EUS-FNB 22G procedure was technically and clinically successful in all patients (100%). The mean number of needle passes, as defined as unique transluminal needle insertions into the cystic compartment, was 1.9 ± 0.6. No immediate adverse events were developed after the procedure.-In 57/60 (95.0%) pancreatic cystic lesions, the specimens were adequate for cyto-histologic assessment. Among the 10 patients who had surgical teatment, the concordance of FNB compared to surgical pathology was noticed.

Conclusion: The diagnosis of pancreatic cystic and mixed lesions with a solid and liquid component can be achieved with fine needle biopsy using a 22-gauge needle in a safe and practicable manner. This method has a great diagnostic yield and does not increase the rate of complications.

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Disclosure: Nothing to disclose.

PP1086

ADVANCED ENDOSCOPIC RISK STRATIFICATION IN SUSPECTED MAIN-DUCT IPMN: A REAL-WORLD CASE SERIES

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Introduction: Patients with suspected main-duct intraductal papillary mucinous neoplasms (MD-IPMN) are frequently referred for surgery due to the high risk of malignancy (as high as 40% to 60% in the literature). However, many present with significant comorbidities or concurrent malignancies, complicating the surgical risk-benefit assessment. In such cases, advanced endoscopic procedures may assist in guiding clinical decision-making.

Aims & Methods: This study aimed to evaluate the role of EUS-guided fine-needle biopsy (EUS-FNB) and pancreatoscopy in individualized risk stratification and management decisions for patients with suspected MD-IPMN. We retrospectively analyzed eight patients treated at our center between March 2020 and March 2025. All patients presented with significant comorbidities or active oncologic disease. Data collected included endoscopic findings, biopsy results, complications, and final management decisions.

Results: Eight patients with suspected main-duct IPMN were included (mean age: 67.6 years; 3 females, 37.5%). The mean main pancreatic duct (MPD) diameter was 9.1 mm. All patients were evaluated in a multidisciplinary setting due to clinical complexity, including comorbidities or concurrent malignancies. EUS-guided fine-needle biopsy (EUS-FNB) was performed in 3 patients, and pancreatoscopy in 5. All procedures were technically successful.

Biopsy samples were diagnostic in 2 of 3 EUS-FNB cases and 5 of 5 pancreatoscopy cases and contributed directly to management decisions. In 1 EUS-FNB case, the sample was non-diagnostic. Pancreatoscopy revealed mucin extrusion in 4/5 patients, papillary or villous projections in 3/5, and a ‚fish egg' appearance in 2/5. These features supported risk assessment and, in several cases, justified a conservative approach in high-risk

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patients. Three patients underwent surgery following multidisciplinary review. One had concordant findings (high-grade dysplasia), while two had discordant outcomes: one with low-risk pancreatoscopy findings had invasive adenocarcinoma, and one with EUS-FNB showing low-grade atypia had high-grade dysplasia on final pathology. Five patients were managed with active surveillance. One case of mild post-ERCP pancreatitis occurred; no other complications were reported. Endoscopic findings influenced management in 5 of 8 patients (62.5%).

Conclusion: In this real-world case series, EUS-FNB and pancreatoscopy contributed meaningfully to individualized decision-making in patients with suspected MD-IPMN. However, discordance between endoscopic impressions and final pathology in two cases highlights the limitations of visual and cytologic assessment. These findings reinforce the need for integrating endoscopic data with imaging, clinical context, and multidisciplinary input when evaluating main-duct IPMN.

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Disclosure: Nothing to disclose.

PP1087

IPMN OF BRANCH DUCTS: IS IT POSSIBLE TO DEFINE CRITERIA FOR AVOIDING AND/OR SUSPENDING SURVEILLANCE?

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Introduction: Branch-duct intraductal papillary mucinous neoplasms (IPMN-BD) account for 80% of cystic lesions pancreatic. The prevalence of these lesions is increasing and, theoretically, these patients should be monitored, given the increased risk of pancreatic carcinoma. However, the morbidity and mortality associated with pancreatic surgery and the costs of surveillance programmes are significant.

The aim is to assess: whether it is possible to select patients who will not benefit from surveillance, using the risk scores determined by comorbidities; and whether the application of the Kyoto 2024 follow-up criteria for suspension of follow-up is appropriate.

Aims & Methods: Retrospective analysis of IPMN-BD cases with at least 1 year of follow-up in a single centre. We analysed the ROC curves of the Charlson Comorbidity Index (CCI) and the Adult Comorbidity Evaluation 27 score (ACE-27) at diagnosis to predict mortality at 1, 3 and 5 years. In a secondary analysis, the evolution of IPMN≤20mm, without High-risk stigmata(HRS) or Worrisome Features(WF), was monitored after 5 years of stability.

Results: 109 patients were included (males-32.1%; mean age-68.9±12 years; mean follow-up-53±35 months; presence of HRS-1.8%; presence of WF-23.9%; progression to adenocarcinoma-1 patient). Five-year mortality was 13.6% (no deaths attributable to IPMN progression). Both the ICC (AUC=0.969; p<0.001) and ACE-27 (AUC=0.961; p<0.001) scores showed good discriminative capacity, with no statistically significant differences, the cut-off point with the best accuracy in predicting 5-year mortality being an ICC≥6 and severe ACE-27 (sensitivity-83.3% and specificity-94.7% versus sensitivity-100% and specificity-92.1%). 20 patients met the Kyoto 2024 criteria for suspending surveillance and none of themMshowed subsequent progression.

Conclusion: Patients with IPMN-BD and CHF≥6 or severe ACE-27 do not seem to benefit from follow-up due to the high probability of death unrelated to pancreatic cancer. In selected cases, in the absence of progression after 5 years of follow-up, discontinuation of surveillance may be considered.

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Disclosure: Nothing to disclose.

PP1088

ROLE OF ENDOSCOPIC ULTRASOUND IN DETECTING MORPHOLOGICAL FEATURES FOR PREDICTING MALIGNANCY IN IPMNS: RESULTS FROM A TERTIARY REFERRAL CENTER

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Introduction: Pancreatic cystic neoplasms are common in the general population, with intraductal papillary neoplasm of the pancreas (IPMN) being the most prevalent, carrying a risk of malignant transformation. IPMNs can be classified into three morphological types (Main-duct, Branch-duct, and Mixed-type) and exhibit a range of neoplastic transformations from low-grade dysplasia (LGD) to invasive carcinoma (IC). Identifying high-grade dysplasia (HGD)/IC has a strong clinical value.

Aims & Methods: The aim of this study is to evaluate the role of endoscopic ultrasound (EUS) in detecting predictive factors of malignancy. This is a monocentric retrospective study using a prospectively maintained database of consecutive patients who underwent surgical resection for

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IPMN between 2015 and 2023 and had available preoperative EUS data performed at our tertiary referral center. The following EUS features were considered:

- Presence of worrisome feature (WF), high-risk stigmata (HRS) or macroscopic solid component (MSC) at diagnosis
- Pancreatic duct (PD): PD dilation, hypoechoic thickened walls of PD (HTW-PD), contrast-enhancement of PD walls (CE-PDW) presence and dimension of PD nodules (PDN), contrast-enhancement (CE)/ real time elastography (RTE) of PDN.
- Cysts: dimension >30 mm, thickened walls, CE on walls, presence, and size of mural nodules (MN), CE/RTE on MN.

Patients were eligible for inclusion if they met the following criteria: age > 18 years; a final histological diagnosis of IPMN after surgery; and availability of a preoperative EUS evaluation conducted at our center. All EUS videos and images were reviewed by two experienced pancreatic endosonographers.

Univariate and multivariate analyses were performed using Pearson correlation coefficient, with statistical significance set at $P < 0.05$.

Results: We enrolled 198 patients (median age: 70 years, male sex: 57.6%). The presence of WF, HRS, and MSC at diagnosis was 40.9%, 44.9%, and 19.1%, respectively. The final histological diagnoses were LGD in 36.4%, HGD in 29.8%, and IC in 33.8%.

EUS predictors of HGD and IC at univariate logistic regression included MSC dimension (p=.02), PDN presence and size (p=.021; p=.016).

EUS predictive factors for IC included: MSC presence and size (p<.001), HTW-PD (p=.037), and presence, size, enhancement, and rigid pattern at RTE of PDN (p=.003; p=.002; p=.003; p=.004).

In multivariate analysis, predictors of HGD-IC included HTW-PD (p=.04), CE-PDW (p=.03), PDN size (p=.02), CE of PDN (p=.09), rigid pattern at RTE of PDN (p=.02), cyst dimension >30 mm (p=.07), and CE of MN (p=.04). All these variables predicted IC too.

Conclusion: In conclusion, while the current guidelines primarily emphasize the use of radiological imaging for the assessment of IPMNs, this study supports the growing role of EUS in identifying features that may predict malignant progression.

Moreover, some of the predictive factors associated with aggressive behavior, such as, the characteristics of the pancreatic duct and the use of real time elastography, are not mentioned in current international guidelines.

Disclosure: Nothing to disclose.

PP1089

GLUCOSE VERSUS CEA IN EUS-ACQUIRED PANCREATIC CYST FLUID: DIAGNOSTIC VALUE FROM A PROSPECTIVE STUDY

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Introduction: Differentiating mucinous (MPCLs) from non-mucinous pancreatic cystic lesions (NMPCLs) remains critical for appropriate therapeutic decisions. While carcinoembryonic antigen (CEA) levels >192 ng/mL have been the conventional benchmark, their limited sensitivity underscores the need for more accurate alternatives. Recent data suggest intracystic glucose levels may be a promising biomarker.

This study prospectively evaluated the diagnostic performance of intracystic glucose compared to CEA in pancreatic cyst fluid obtained via EUS-guided fine needle aspiration (EUS-FNA).

Aims & Methods: From July 2018 to October 2024, 24 patients with pancreatic cystic lesions undergoing EUS-FNA using a 22G needle were included. Glucose and CEA concentrations were measured using standard biochemical protocols. Final diagnosis was based on surgical pathology (n=5), cytopathology (n=9), or consensus radiologic/endoscopic evaluation by two blinded expert pancreatologists (n=10). All procedures and analyses were performed in a tertiary academic gastroenterology center in Morocco.

Results: Mean patient age was 59.8 years. Diagnoses comprised 13 mucinous cysts (54.2%; including 9 IPMNs and 4 MCNs), 7 serous cystadenomas (29.2%), 2 cystic neuroendocrine tumors (8.3%), 1 pseudocyst, and 1 simple cyst.

The mean intracystic glucose level in MPCLs was 14.1 mg/dL (95% CI: 5.6–22.6), versus 90.3 mg/dL (95% CI: 82.1–98.6) in NMPCLs ($p < 0.0001$). A glucose threshold of ≤30 mg/dL yielded an AUC of 0.94 (95% CI: 0.84–0.99), with 92.3% sensitivity and 100% specificity.

For CEA ≥192 ng/mL, AUC was 0.71 (95% CI: 0.54–0.86), with 46.1% sensitivity and 100% specificity. No significant correlation was found between glucose levels and either CEA or cyst size.

Conclusion: Intracystic glucose represents a simple, cost-effective, and highly accurate marker for distinguishing mucinous from non-mucinous pancreatic cystic lesions. It demonstrates superior sensitivity compared to CEA and can be a valuable adjunct in the diagnostic algorithm of pancreatic cysts, particularly in resource-conscious settings.

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Disclosure: Nothing to disclose.

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PP1090

TUMOUR-ASSOCIATED AUTOANTIBODIES AS PROGNOSTIC BIOMARKERS IN PANCREATIC ADENOCARCINOMA: RESULTS OF A DISCOVERY STUDY BASED ON PROTEIN MICROARRAYS

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Introduction: Pancreatic ductal adenocarcinoma (PDAC) is a highly aggressive cancer that poses significant therapeutic challenges. In this regard, the identification of prognostic biomarkers is crucial to improve the prediction of clinical outcomes and guide therapeutic decisions. Tumour-associated autoantibodies (TAAbs) have been identified as prognostic biomarkers in several types of cancer, as they can be generated before the first clinically detectable signs of disease.

Aims & Methods: The aim of this study was to identify TAAbs that can be used as prognostic biomarkers in PDAC. 80 sera from patients with PDAC, divided into four groups according to disease stage, were included in the study: patients with premalignant lesions (n = 20), patients undergoing curative resection after diagnosis (CR) (n = 20), patients with unresectable advanced locoregional disease (LA) (n = 20) and patients with metastatic disease (MT) (n = 20). Sera from each group were analysed divided into 4 pools of 5 samples each using the HuProt Human Protreome microarray v4.0 (CDI Laboratories, Baltimore, MD). Candidate TAAbs with high specific reactivity in MT and/or LA samples were selected according to the following criteria: z-score ≥ 3 in at least 2/4 pools and fold-change > 1.2. The RNAseq data collected in the Human Protein Atlas were used to determine protein expression levels and prognostic value. The GSE15471, GSE28735 and GSE16515 databases were used to compare expression levels with those of healthy subjects without the disease.

Results: 43 TAAbs demonstrated high levels of specific reactivity in samples from MT and/or LA patients. Of these highly recognised proteins, 15 were selected for showing notable levels of expression in tumour tissue from PDAC patients and prognostic capacity in some types of cancer. Of these 15, 2 proteins involved in tumour development were identified. Survival analyses showed that six of the 15 selected proteins were significantly associated with a worse prognosis in PDAC patients ($p < 0.05$). Finally, in the comparison with control tumour tissue, 5 proteins were identified with significantly higher expression in patients diagnosed with the disease ($p < 0.05$ in at least 2 of the cohorts used).

Conclusion: 15 TAAbs have been identified as promising candidates for prognostic biomarkers in PDAC. Further validation studies are required to confirm their prognostic value and to develop a robust biomarker signature with potential for clinical application and market translation.

Disclosure: Nothing to disclose.

PP1091

DISRUPTED CALCIUM EFFLUX IN PANCREATIC DUCTAL ADENOCARCINOMA: THE ROLE OF CFTR AND PMCA4 DYSFUNCTION

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Introduction: Pancreatic ductal adenocarcinoma (PDAC) ranks among the most aggressive malignancies, with a five-year survival rate below 10%. Its poor prognosis stems from late diagnosis, rapid disease progression, and marked resistance to conventional therapies, largely driven by frequent KRAS and TP53 mutations.

These genetic alterations disrupt multiple cellular processes, including intracellular calcium (Ca^{2+}) signaling, which plays a critical role in tumor progression. Plasma membrane calcium ATPase isoform 4 (PMCA4) is a major regulator of cytosolic Ca^{2+} levels, essential for maintaining cellular homeostasis.

Elevated expression of PMCA4 in PDAC has been associated with enhanced cell migration, apoptosis resistance, and poor clinical outcomes. PMCA4 localizes at focal adhesions, supporting motility and metastatic potential. Paradoxically, PMCA4 dysfunction can lead to Ca^{2+} overload, activating oncogenic pathways that drive cell proliferation and survival.

The cystic fibrosis transmembrane conductance regulator (CFTR), an epithelial chloride channel, also modulates calcium homeostasis. CFTR deficiency or dysfunction impairs PMCA4 activity, while paradoxically inducing its overexpression, likely as a compensatory mechanism in response to altered Ca^{2+} signaling.

This functional interaction between CFTR and PMCA4 is particularly relevant in pancreatic ductal epithelial cells, where both proteins co-localize at the apical membrane. Their coordinated function is crucial for regulating calcium and chloride transport, and its disruption may contribute to PDAC progression.

Aims & Methods: This study aim to investigate whether CFTR deficiency contributes to PMCA4 overexpression in PDAC cells and whether pharmacological restoration of CFTR function using Trikafta can normalize calcium signaling and modulate tumor cell behavior. Therefore we utilized both human pancreatic ductal organoids (hPOs) and the Capan-1 PDAC cell line to assess calcium and chloride transport.

Intracellular calcium levels were measured using FURA-2-AM, and chloride efflux was monitored with MQAE dye. Immunofluorescence staining was employed to analyze the expression and localization of CFTR, PMCA4, SOX9, Vimentin, and α-SMA.

Results: We successfully cultured human pancreatic organoids from adult pancreas-derived stem cells using standard 3D culture conditions. Following the establishment of primary organoids, we performed the first passage to expand the culture. The organoids were then dissociated into single cells and plated onto 24 mm cover glasses for further analysis.

Our experiments confirmed the expression of ductal markers in 2D human pancreatic organoids (hPOs) with the lack of mesenchymal markers Vimentin, and α-SMA. Immunofluorescent staining was visualized using a Zeiss LSM880 confocal microscope.

Functional assays revealed that Capan-1 cells exhibited significantly reduced Ca^{2+} and Cl^- efflux compared to hPOs, indicating impaired CFTR and PMCA acitivty.

Notably, treatment with Trikafta, a clinically approved CFTR modulator, successfully restored CFTR and PMCA4 activity in Capan-1 cells.

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Conclusion: Our findings indicate that CFTR dysfunction contributes to the increased PMCA expression in PDAC cells, which can be ameliorated by Trikafta treatment. This study highlights the critical interplay between CFTR dysfunction and calcium homeostasis in pancreatic ductal adenocarcinoma (PDAC). These results underscore the relevance of epithelial ion transport pathways in PDAC pathophysiology and support further investigation of adjunctive approach in PDAC treatment.

Disclosure: Nothing to disclose.

PP1092

KRAS MUTATION ANALYSIS USING DDPCR WITH ROSE SAMPLES

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Introduction: Over 90% of pancreatic cancers are KRAS mutation-positive; however, it is known that KRAS-negative pancreatic cancers often harbor draggable genes. Recently, the efficacy of KRAS G12C inhibitors has also been demonstrated. In this study, we performed KRAS mutation analysis (KRAS G12/13 and KRAS G12C) using droplet digital PCR (ddPCR) on ROSE samples obtained during EUS-FNA for pancreatic cancer.

Aims & Methods: Retrospective Study: Fifty-five cases of pancreatic cancer that underwent EUS-FNA at our institution between November 2019 and August 2021 were included. Samples included 55 ROSE samples (107 slides), 30 plasma cell-free DNA samples derived from peripheral blood, 30 germline DNA samples from whole blood, and 20 formalin-fixed paraffin-embedded (FFPE) samples from surgical specimens. DNA was extracted from each sample, and KRAS mutation-positive rates and concordance rates with surgical specimens were compared.

Prospective Study: Fifty-six cases of pancreatic cancer that underwent EUS-FNA between October 2022 and August 2024 were included. KRAS mutation analysis was performed on ROSE samples.

Results: Retrospective Study: The average amount of DNA extracted from ROSE samples was 66.7 ng per slide, and the DNA yield increased significantly with an increase in the number of cells on the ROSE slide. The positive rates for KRAS G12/13 mutations were 94.5% (52/55) for ROSE samples, 20.0% (6/30) for plasma cell-free DNA, 90.0% (18/20) for FFPE samples, and 0% (0/30) for germline DNA.

The concordance rate with surgical specimens for KRAS G12/13 mutations was 85.0% (17/20) for ROSE samples and 27.3% (3/11) for plasma cell-free DNA. ROSE samples demonstrated a significantly higher concordance rate with surgical specimens compared to plasma cfDNA.Prospective Study: In two cases, DNA was below the detection sensitivity. KRAS G12/13 positivity was 82% (46/56), and KRAS G12C positivity was 2% (1/56).

Comprehensive genomic profiling (CGP) was performed in 14% (8/56) of the cases, and all cases showed concordance in KRAS mutation status or wild-type status between ROSE samples and CGP results. One case identified a draggable gene (BRCA) via CGP, which was KRAS G12/13-negative in ROSE samples.

Conclusion: KRAS mutation analysis using ddPCR with ROSE samples successfully identified KRAS-negative pancreatic cancers and KRAS G12C-positive pancreatic cancers.

Disclosure: Nothing to disclose.

PP1093

DOSE-RESPONSE RELATIONSHIP OF CADMIUM AND PANCREATIC CANCER RISK: META-ANALYSIS

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Introduction: Cadmium (Cd), a group 1 carcinogen, is linked to the development of pancreatic cancer, though its dose-response relationship, especially at low exposure levels, is not well-defined.

Aims & Methods: To investigate the relationship between Cd exposure and the risk of pancreatic cancer through a meta-analysis focusing on pooled relative risk, biomarker comparisons, and dose-response relationships.

Methods: This meta-analysis adhered to PRISMA guidelines and applied the PECO framework. A systematic search of key databases including PubMed, Cochrane Library, Web of Science, EMBASE, ScienceDirect, and ProQuest was carried out without language restrictions. Studies that met the predefined PECO criteria were chosen, extracting data on effect estimates such as hazard ratios, odds ratios, and relative risks (RR), along with Cd biomarker concentrations.

Analyses involved pooled relative risks, standardized mean differences for biomarker comparisons, and dose-response relationships, utilizing random-effects meta-analysis and meta-regression. Statistical procedures were performed on R version 4.3.0.

Results: Cd exposure was significantly associated with an increased risk of pancreatic cancer, with a pooled RR of 1.42 (95% CI = 1.16–1.73). Subgroup analyses indicated sex-specific variations: men demonstrated a significant increase in pancreatic cancer risk), while the risk for women was not statistically significant. Dose-response analyses revealed that higher Cd biomarker levels correlated with greater pancreatic cancer risk (β = 0.610, p = 0.035).

Conclusion: Cd exposure is associated with an increased risk of pancreatic cancer, as confirmed by a dose-response relationship and elevated biomarker levels in cases. Cd exposure is a modifiable risk factor for pancreatic cancer, meriting further investigation.

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Disclosure: Nothing to disclose.

PP1094

COMPARATIVE STUDY OF HISTOPATHOLOGICAL AND IMAGING FINDINGS IN HIGH-GRADE PANCREATIC INTRAEPITHELIAL NEOPLASIA (PANCREATIC CARCINOMA IN SITU)

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Introduction: Early diagnosis of pancreatic cancer is crucial for improving patient outcomes. High-grade pancreatic intraepithelial neoplasia (HG-PanIN), considered a carcinoma in situ of the pancreas, can lead to long-term survival when surgically resected—even in the absence of detectable lesions on conventional imaging such as CT or MRI.
In this study, we analyzed resected cases of HG-PanIN collected through the Japanese Early-stage Diagnosis Project of Pancreatic Cancer (JEDPAC), focusing on correlations between histopathological findings and preoperative imaging to identify potential diagnostic indicators.
Aims & Methods: Fifty cases (32 males, 18 females; age range 52–89 years, mean age 69.8 years) of surgically resected HG-PanIN, confirmed via central pathological review by two expert pathologists, were selected from JEDPAC-affiliated centers between 2011 and 2018. Preoperative imaging findings—including CT, MRI (MRCP), EUS, and ERCP—were retrospectively compared with pathological data.
Results: Lesions were located in the pancreatic head in 11 cases and in the body/tail in 39 cases. Sixteen cases involved only the main pancreatic duct (MPD, Group A), 27 involved both MPD and branch ducts (Group B), and 7 were confined to branch ducts (Group C). In Groups A and B, 40 cases showed longitudinal intraductal spread of ≥10 mm, and 14 exhibited skip lesions with discontinuous HG-PanIN components.
MPD stenosis was frequently observed in cases with ≥10 mm ductal spread, particularly in Groups A and B, and was often accompanied by focal parenchymal atrophy corresponding to the lesion. In skip lesions, however, HG-PanIN foci did not consistently correspond to areas of ductal stenosis, making precise preoperative localization by imaging difficult. In Group C, round or oval-shaped cystic lesions near branch ducts were frequently identified, and associated parenchymal atrophy was minimal.
Conclusion: HG-PanIN often involves both main and branch pancreatic ducts, with MPD stenosis and localized parenchymal atrophy serving as important imaging clues for diagnosis. However, skip lesions challenge accurate localization with current imaging modalities. In cases confined to branch ducts, subtle atrophy and the presence of small, round cystic lesions may aid early detection.
Disclosure: Nothing to disclose.

PP1095

TRENDS IN THE TREATMENT AND SURVIVAL OF PANCREATIC CANCER: ANALYSIS OF 23,339 PATIENTS DIAGNOSED BETWEEN 2010 AND 2017

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Introduction: With a rising incidence and unchanged poor prognosis, pancreatic cancer is increasingly becoming a focus of gastroenterological oncology, but there is a lack of real-world data.
Aims & Methods: The aim of the current study was to investigate trends in survival and treatment patterns by analyzing German health claims data. Pancreatic cancer patients diagnosed between 2010 and 2017 were identified from the German Pharmacoepidemiological Research Database (GePaRD, approximately 20% of the German population). Data on demographics, tumor treatment within one year after diagnosis and survival were extracted.
Results: The study population comprised 23,339 patients with a median age of 74 years (IQR 66-80) and 44% with localized and 56% with metastatic disease. Overall, 52.4% received any chemotherapy and curative intended resection was performed in 28.3%. Neoadjuvant and adjuvant therapy were performed in 4.4% and 58.7% of the cases, respectively. The median overall survival of the whole study population was 7.84 months. Patients diagnosed in the most recent period (2014-2017) had a significantly better prognosis (8.20 months (95%CI 7.97-8.43)) than patients who were diagnosed in the earlier period (2010-2013) (7.54 months (95%CI 7.31-7.70)), with an age-, sex- and stage-adjusted hazard ratio of 0.87 (95% CI 0.85-0.9).
Over time, the most pronounced treatment trends have affected patients with localized disease, with increasing frequency of resection and neoadjuvant therapy and decreasing frequency of best supportive care.
Conclusion: This comprehensive insight into survival and treatment of pancreatic cancer in Germany shows presumably medically beneficial therapy trends with, however, only marginal improvements in prognosis to date.
Disclosure: Nothing to disclose.

PP1096

INCREASED RISK OF PANCREATIC CANCER AFTER ACUTE PANCREATITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Despite advancements in diagnostics and treatment, pancreatic cancer (PC) remains one of the deadliest forms of cancer. Due to its asymptomatic nature, it is often diagnosed at an advanced stage, resulting in poor survival outcomes with a five-year survival rate of 10%.[1]

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Surgical resection is the only treatment with curative potential; however, only 15-20% are typically eligible for resection, stressing the need to identify high-risk populations for early detection and improved prognosis.[1-5]
Acute pancreatitis (AP), an inflammatory disorder of the pancreas, is a common gastrointestinal condition, affecting 34 per 100,000 people annually worldwide.[6]
While most AP cases resolve without complication, accumulating evidence suggests a potential bidirectional relationship between AP and PC.[7] Although the link between AP and PC has been established, the magnitude of the risk remains unclear.
Aims & Methods: This systematic review and meta-analysis aimed to quantify the occurrence of PC in AP, the risk of PC following AP, and the impact of specific risk factors. The systematic search was conducted in PubMed, EMBASE, and CENTRAL from inception until October 11th, 2023 (PROSPERO: CRD42023470350). Eligible studies included adult populations reporting on the association between AP and PC.
Primary outcomes included prevalence, incidence, and risk of PC in individuals in AP, including subset analyses of specific factors. Meta-analyses were performed using random-effects models to calculate pooled outcome measures and corresponding 95% confidence intervals (CI).
Results: A total of 49 studies were included with 34 providing sufficient data for quantitative synthesis. The prevalence of PC among AP patients was 2% (CI: 2-4%). Although this may seem low, it is clinically relevant given the poor prognosis of PC and the potential for it being missed when AP occurs. The time dependent analysis revealed that the risk of PC was highest near the episode of AP, with the risk gradually decreasing over time; however, the risk remained significantly high in the long term: <24 months (HR: 53.27, CI: 11.60-244.58), 24-60 months (HR: 3.19, CI: 2.21-4.60), and >60 months (HR: 1.94, CI: 1.63-2.32).
Additionally, our subset analyses found that AP patients with subsequently diagnosed chronic pancreatitis (CP) showed significantly increased odds of developing PC (OR: 3.54, CI: 1.05–11.92). We found no evidence for increased odds for PC in AP with diabetes, smoking, and alcoholic and gallstone etiologies.
Conclusion: Given the notable prevalence and significantly elevated short-term risk of PC among AP patients, clinicians should maintain a high index of suspicion during the first few years following an episode of AP. Although the risk decreases with time, the long-term risk of PC in AP should not be neglected. CP may further elevate this risk, highlighting the need for targeted surveillance strategies in high-risk populations.

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Disclosure: Nothing to disclose.

PP1097

RISK OF PANCREATIC CANCER IN DIABETES MELLITUS WITH AND WITHOUT PREVIOUS HISTORY OF ANY OTHER MALIGNANCY. ANALYSIS OF THE DATA FROM THE CZECH NATIONAL REGISTRIES 2010-2021

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Introduction: Incidence of sporadic ductal pancreatic adenocarcinoma (PDAC) in the Czech Republic is the third highest in Europe (23.1/100,000 population). New-onset diabetes mellitus with weight loss (> 5% within < 6 months) in patients over sixty can be an early sign of PDAC. New-onset diabetes mellitus was recorded in 12.2% subjects three years prior to the diagnosis of PDAC (in the 60-74+ age group:13.8%) in the Czech Republic in the period 2017-2021 [ref. 1].
Aims & Methods: The aim of this study was to evaluate the standardized incidence ratio (SIR) of PDAC in patients with diabetes mellitus with and without previous history of any other malignancy within a 12-year period.
Methods: The analysis was based on individual population data from the Czech National Registry of Reimbursed Health Services, the National Cancer Registry, and the Death Certificate Information System in the period 2010 - 2021.
Patients were divided into two groups: the first one included subjects with diabetes mellitus who had a history of any malignancy prior to the diagnosis of diabetes (follow-up 865,956 person-years); the second cohort consisted of people with diabetes mellitus who had no history of malignancy prior to the diagnosis of diabetes (7,971,090 person-years). The calculated person-years at risk were stratified by sex, age and time periods (2010-2013; 2014-2017; 2018-2021).
Subgroup analyses were performed considering follow-up and age at diagnosis of diabetes. The first report of any antidiabetic drug (ATC group A10) was the determinant for the diagnosis of diabetes mellitus in a particular individual. SIR was calculated as the ratio of observed to expected (estimated from incidence in the general population) cases of PDAC.
Results: Selected SIRs stratified according to age and follow-up are displayed in Table 1.
A significantly higher risk of PDAC in diabetes mellitus with a history of previous malignant diseases (compared to the general population) was found for the following: head and neck cancer (SIR 2.54), gastric cancer (2.12), colorectal cancer (1.38), laryngeal cancer (2,51), malignant melanoma (1.65), breast cancer (1.61), cervical cancer (2.22), uterine cancer (1.72), prostate cancer (1.59), testicular cancer (3.1), kidney cancer (1.75), urinary bladder cancer (1.74), thyroid cancer (2.13), Hodgkin lymphoma (2.74), non-Hodgkin lymphoma (2.15) and leukaemias (1.93).

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Diabetes mellitus with history of previous malignant disease n = 144,946 individuals		Diabetes mellitus with no history of malignant disease n = 1,123,065 individuals	
Age at diagnosis of diabetes mellitus	SIR (95% C.I.)	Age at diagnosis of diabetes mellitus	SIR (95% C.I.)
60 - 69 years	3.13 (2.84 - 3.45)	60 - 69 years	2.12 (2.04 - 2.19)
70 - 79 years	1.58 (1.45 - 1.73)	70 - 79 years	1.05 (1.01 - 1.10)
≥ 80 years	0.79 (0.68 - 0.91)	≥ 80 years	0.63 (0.58 - 0.68)
Follow-up	SIR (95% C.I.)	Follow-up	SIR (95% C.I.)
up to 3 months	5.49 (4.62 - 6.49)	up to 3 months	4.82 (4.46 - 5.20)
3 months to 1 year	2.79 (2.41 - 3.20)	3 months to 1 year	2.49 (2.34 - 2.65)
from 1 to 2 years	1.74 (1.48 - 2.04)	from 1 to 2 years	1.62 (1.52 - 1.74)

Table 1. Risk of PDAC in diabetes mellitus.
C.I.: Confidence Intervals.

Conclusion: Risk of PDAC in patients with diabetes mellitus was higher in subjects with a history of various malignant diseases compared to those with no previous history of any malignancy. Particular attention should be paid to new-onset diabetes (diagnosis made within one last year). History of a malignancy, as an important risk factor for PDAC, should be taken into account when surveilling patients with diabetes mellitus.

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Disclosure: Nothing to disclose.

PP1098

BETA-BLOCKERS ARE ASSOCIATED WITH LOWER RISK OF PANCREATIC CANCER IN PATIENTS WITH TYPE 2 DIABETES: A POPULATION-BASED COHORT STUDY

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Introduction: Diabetes is associated with increased pancreatic cancer (PC) risk. Beta-blockers may mitigate PC risk and progression by inhibiting β-adrenergic receptor-mediated signaling pathways involved in tumor cell proliferation, invasion, and angiogenesis. However, the potential oncopreventive effect of beta-blockers on PC risk in patients with type 2 diabetes (T2D) remains unclear.

Aims & Methods: We aimed to assess the association between beta-blocker use and PC risk in newly diagnosed T2D patients after adjusting for risk factors. This territory-wide retrospective cohort study included adults (≥18 years) newly diagnosed with T2D between 2001 and 2020. Patients with type 1 diabetes, prior PC, immunoglobulin G4-related disease, pancreatectomy, or fewer than two HbA1c results were excluded.

To minimize reverse causality between PC and T2D, those diagnosed with PC within 1 year of T2D diagnosis or with <1 year of follow-up were excluded. Primary exposure was beta-blocker use (e.g., acebutotlol, atenolol, bisoprolol, carvedilol, celiprolol, esmolol, labetalol, metoprolol, nadolol, nebivolol, pindolol, propranolol, and sotalol).

In the primary (time-fixed) analysis, exposure was defined as ≥90 days of use after T2D diagnosis (i.e. index date). The outcome was incident PC. Patients were followed until PC, pancreatectomy, death, or study end (December 31, 2020).

Adjusted hazard ratios (aHRs) were derived from multivariable Cox models adjusting for 31 covariates: age, sex, smoking status, alcohol use disorder, body index mass, glycemic control (in terms of time-weighted mean HbA1c), pancreatitis, cholangitis, colectomy, Charlson's comorbidity index, metabolic and cardiovascular comorbidities, and concomitant medications (aspirin, non-steroidal anti-inflammatory drugs, statins, and other antidiabetics).

Sensitivity analyses included time-dependent analysis (drug use was defined as ≥180 days/year; to address immortal time bias), and multiple propensity score (PS)-based approaches (matching, regression adjustment, weighting, and matching weights). Dose- and duration-response relationships were evaluated.

Results: Among 431,261 T2D patients (mean age: 59.1±11.4 years; male: 52.5%; 61,917 beta-blocker users and 369,344 beta-blocker non-users), 1,303 (0.3%) developed PC during a mean follow-up of 9.4±5.0 years. The 18-year cumulative incidence was 0.68% (95% CI: 0.57%-0.80%) in beta-blocker users and 0.76% (95% CI: 0.70%-0.82%) in non-users (P = 0.02; Figure 1).

Beta-blocker use was associated with lowered PC risk in time-fixed analysis (aHR: 0.81; 95% CI: 0.70-0.95), time-dependent analysis (aHR: 0.75; 95% CI: 0.62-0.91), and all PS-based methods (aHR range: 0.83-0.86). A dose- and duration-dependent inverse relationship was observed (P_{trend} <0.001).

Higher cumulative defined daily dose (cDDD) reduced the risk (aHR: 0.84, 95% CI: 0.72-0.98, for <4,000 cDDD; aHR: 0.31, 95% CI: 0.12-0.82, for ≥4,000 cDDD). Longer use also lowered the risk (aHR: 1.12, 95% CI: 0.92-1.37, for <5 years; aHR: 0.70, 95% CI: 0.57-0.87, for 5 to <15 years; aHR: 0.11, 95% CI: 0.04-0.35, for ≥15 years).

Each additional 1,000 cDDD and 1-year increase in use was associated with a 21% (aHR: 0.79, 95% CI: 0.72-0.88) and 6% (aHR: 0.94, 95% CI: 0.92-0.96) risk reduction, respectively.

Conclusion: Beta-blocker use was associated with a lowered risk of PC among patients with newly diagnosed T2D, in a dose- and duration-dependent manner.

Disclosure: Nothing to disclose.

PP1099

ASSOCIATION BETWEEN PANCREATIC ENZYME REPLACEMENT THERAPY AND MORTALITY IN PANCREATIC CANCER: A MATCHED CASE-CONTROL STUDY

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Introduction: Malnutrition and pancreatic exocrine insufficiency (PEI) are common among patients with pancreatic cancer. Despite the potential of pancreatic enzyme replacement therapy (PERT) to correct PEI, its usage and association with survival in pancreatic cancer remain uncertain. Our study aimed to evaluate the trend in PERT prescription over recent years and to explore potential association with mortality among patients with pancreatic cancer.

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Aims & Methods: This single-center, matched case-control study analyzed data from patients diagnosed with pancreatic ductal adenocarcinoma (PDAC) who were discussed at multidisciplinary tumor board meetings at a tertiary hospital during 2008–2012 and 2015–2018. Cases, defined as patients who died within one year of diagnosis, were matched 1:1 with controls alive at one year. Patients were classified as PERT users or non-users based on prescriptions following the PDAC diagnosis.

Conditional logistic regression was performed to estimate the association between PERT usage and all-cause mortality at one-year post-diagnosis.

Results: Overall, 1247 patients with pancreatic cancer were included: 756 (61%) who died within one year (cases), and 491 (39%) who survived (controls). Cases exhibited, advanced disease stage (80% *vs.* 52%) with more comorbidities, whereas controls had more cases with pancreatic head localization (75% *vs.* 60%) and more frequently underwent surgery (55% *vs.* 12%). The prescription of PERT showed an increasing trend when comparing the periods 2008–2012 and 2015–2018, particularly in the palliative setting (p=0.001).

Matching yielded 487 pairs, with no significant differences in baseline characteristics. During a follow up of 12 months, 264 (39%) PERT-users and 223 (74%) PERT non-users died. In multivariable logistic regression models, PERT use was associated with lower odds of one-year mortality (adjusted OR [aOR], 0.49; 95% CI: 0.34–0.71). Similar associations were observed in subgroups undergoing surgery (aOR, 0.34; 95% CI: 0.16–0.74) and best supportive care (aOR, 0.12; 95% CI: 0.03–0.33). No clear association was observed in patients receiving chemotherapy alone (aOR, 0.75; 95% CI: 0.44–1.24), although the direction of effect suggested a potential benefit.

Conclusion: In this matched case-control study, PERT use was associated with lower one-year mortality among patients with PDAC. The results may be overly optimistic due to potential confounding factors related to patient selection, and randomized controlled studies are needed to confirm these effects.

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PP1100

HALP AND PINI ARE ASSOCIATED WITH THE NUTRITIONAL STATUS AND ARE INDEPENDENT RISK FACTORS OF DEATH IN ADVANCED PANCREATIC CANCER PATIENTS

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Introduction: The prognosis for patients with advanced pancreatic cancer (PC) remains exceptionally bleak, with a 5-year survival rate estimated at just 4%. This condition is frequently accompanied by pancreatic exocrine insufficiency, which contributes to malnutrition and a significant decline in nutritional status. While conventional laboratory markers such as serum albumin and lymphocyte counts are commonly employed to assess nutritional health, integrative nutritional indices have emerged as meaningful prognostic indicators across various malignancies. Specifically, the hemoglobin, albumin, lymphocytes, and platelets index (HALP) has shown promise in predicting overall survival in cancer cohorts. However, other indices, such as the prognostic immune nutritional index (PINI), have yet to be examined in the context of pancreatic cancer.

Aims & Methods: Aim: This study aims to assess the prognostic value of the HALP and PINI in patients with advanced pancreatic cancer.

A retrospective analysis was conducted utilizing electronic medical records from 158 patients diagnosed with advanced pancreatic cancer who had begun treatment. Conventional parameters were employed to calculate HALP and PINI levels, which were then compared between patients with proximal and distal tumor locations prior to the initiation of chemotherapy.

Within a subgroup of 125 patients receiving FOLFIRINOX, baseline HALP and PINI values were analyzed in relation to outcomes categorized into three groups at the 12-week follow-up: mortality (n=21), disease progression (n=51), and disease stabilization (n=53). The log-rank test was utilized to evaluate statistical differences between high and low nutritional index groups.

Univariate and multivariable analyses were performed to identify risk factors for mortality within three months of chemotherapy initiation.

Results: Among the 158 patients included in the analysis, those with tumors located in the pancreatic head exhibited significantly lower HALP (32.34 vs. 43.05, p=0.021) and PINI (3.23 vs. 3.48, p=0.003) compared to patients with distal tumors. In the FOLFIRINOX subgroup, HALP levels were significantly lower in individuals who deceased by the first follow-up (25.76 vs. 36.115 and 41.8, p=0.041), while PINI did not demonstrate a similar relationship.

Univariate analysis revealed that a HALP value below 26.85 (HR 2.548; 95% CI: 1.021-6.357, p=0.045) and a PINI below 3.26 (HR 3.49; 95% CI: 1.31-9.299, p=0.012) were independent predictors of mortality.

In multivariate analysis, the presence of two or more metastatic sites beyond the primary tumor was the sole factor associated with a heightened risk of death at the first follow-up (HR 7.21; 95% CI: 2.02-25.738, p=0.002).

Conclusion: Our findings support the notion that HALP and PINI are closely related to nutritional status and serve as independent risk factors for mortality within three months of initiating chemotherapy in patients with advanced pancreatic cancer.

Disclosure: Nothing to disclose.

PP1101

PRELIMINARY RESULTS OF THERAPEUTIC RESPONSE OF THE EUS-GUIDED INTRA-TUMORAL IMPLANTATION OF RADIOACTIVE PHOSPHORUS (32P) IN 8 PATIENTS WITH LOCAL ADVANCED PANCREATIC CANCER

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Introduction: Radioactive phosphorus (^{32}P) is an innovative therapy for locally advanced pancreatic cancer that is implanted through endoscopic ultrasound (EUS). Pancreatic cancer is the fourth cause of death and is responsible for 7% of all deaths caused by cancer. Nowadays the improvement of chemotherapy and the usage of combination therapies led to a better prognosis and studies showed an improvement of the survival parameters (OS, Free Progression Survival). One of the most innovative therapies is the implantation of radioactive phosphorus-32 which is a

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type of brachytherapy. It is indicated for the treatment of patients with Locally Advanced Pancreatic Cancer (LAPC) in combination with gemcitabine-based chemotherapy.

Aims & Methods: The primary goal of this study is to determine the safety of the intra-tumoral implantation of radioactive phosphorus (^{32}P) through EUS and the complications of the procedure of implantation. The secondary goals are to determine the Local Disease Control Rate (LDCR) at 16 weeks and the effectiveness of the implantation of radioactive phosphorus by measuring Overall Survival (OS), Progression Free Survival (PFS) and Local Progression Free Survival (LPFS). We currently have made the implantation in eight patients with similar clinical and demographic features who suffer from locally advanced pancreatic cancer. They receive gemcitabine-based chemotherapy in combination with an intra-tumoral therapy with radioactive phosphorus (^{32}P) through a 22-gauge FNA needle guided by EUS. The adverse events of the implantation are evaluated by the Common Terminology Criteria for Adverse Events (CTCAE) version 4.0. The LDCR is evaluated at 16 weeks and the alteration of Tumor Volume every two months for the first year after implantation and every three months the second year after implantation according to RECIST 1.1.

Results: The eight procedures (median age of patients: 68,5 years) were unremarkable and no complication during the procedure was noted. The patients did not have any major adverse events during the day of the procedure. Five out of eight patients (62,5%) revealed mild abdominal pain on the second day which recovered with typical pain killers. The LDCR, for all patients, at 16 weeks, was 100%. Two of the patients had a major reduction in the tumor so it was resectable. The operations were without any complications and the pathology test showed R0 resection. Two patients died after 12 months of follow up (one patient showed progression disease after 10 months from implantation and one patient had stable disease until death) and one patient revealed progression of disease after 12 months and is still alive after 15 months of follow up. Three patients have stable disease after 6 months of follow up, even though a slight reduction of the tumor has been noted. The six patients (including the two that died and not including the two that underwent surgery) have been followed up for 6 to 15 months, depending on time of implantation. The median Progression Free Survival for these six patients is 7 months (max 12 months – min 6 months) and the median Local Progression Free Survival is 6 months (max 12 months – min 6 months).

Conclusion: We must include more patients to conduct more useful results, but the initial results look promising.

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PP1102

LEVELS OF NEUROFILAMENT LIGHT CHAIN IN PANCREATIC CANCER, TYPE 2 DIABETES MELLITUS AND CHRONIC PANCREATITIS

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Introduction: Neurofilaments are cytoskeletal proteins specific to neuronal tissue. As their blood levels rise following neuronal damage, they are increasingly being studied as the biomarkers for neuroaxonal injury in various conditions, such as Alzheimer's disease, multiple sclerosis, amyotrophic lateral sclerosis, or type 2 diabetes mellitus (T2DM) related neuropathy. Pancreatic ductal adenocarcinoma (PDAC) is a malignant disease known for its poor prognosis, rapid spread, silent progression, and lack of effective diagnostic tools in its early stages. Perineural and/or endoneural invasion and neuronal damage are hallmarks of PDAC.

Aims & Methods: The aim of our study was to measure the levels of neurofilament light chain (NfL) in patients with PDAC, T2DM, chronic pancreatitis (ChP), and healthy controls. We used a Single Molecule Array (Simoa; Quanterix) to measure NfL levels in patients with PDAC (n=50, mean age 63.16 ± 8.8 years), T2DM (n=42, mean age 61.6 ± 8.4 years), ChP (n=30, mean age 56 ± 13.6 years), and healthy controls (n=46, mean age 54 ± 9.06 years).

Results: The following concentrations of NfL were obtained from the samples: PDAC: 30.31 ± 39.79 pg/ml, T2DM: 12.59 ± 10.19 pg/ml, ChP: 16.64 ± 13.78 pg/ml, Healthy controls:
7.93 ± 4.18 pg/ml. NfL levels were significantly different between PDAC and healthy controls ($p < 0.001$) and between T2DM and healthy controls ($p < 0.001$). Differences between other groups were not significant. Further analysis confirmed age as a significant predictor of NfL levels ($p < 0.001$), indicating that NfL levels increase with age by 0.48 per year. We did not find a significant impact of glycemic control (represented by glycated haemoglobin levels) or body mass index on NfL concentrations. Because of the correlation between age and NfL levels, we examined the relationship between age, diagnosis, and NfL levels using a statistical model comparison approach. After controlling for age, we found that PDAC and ChP showed significantly higher NfL levels compared to T2DM and healthy controls ($p < 0.01$). No significant differences were found between T2DM and healthy controls

Conclusion: This is the first study to date that evaluates serum levels of NfL in PDAC and chronic pancreatitis. The results support the possible role of NfL as a biomarker in certain non-neurological conditions. Elevated serum levels of NfL in PDAC and ChP suggest that this biomarker reflects neuronal damage, which is present in ChP and is a key feature of PDAC, even in the early stages of the disease.

Disclosure: Nothing to disclose.

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PP1103

EARLY-ONSET PANCREATIC CANCER IS ASSOCIATED WITH ACTIVE SMOKING HABIT AND WITH BETTER OVERALL SURVIVAL

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Introduction: Early onset pancreatic cancer (EOPC) arises in patients aged ≤50 years or, and accounts between 5%-10% of all the cases of pancreatic ductal adenocarcinoma (PDAC). Compared to late-onset pancreatic cancer (>50 years, LOPC), EOPC has been described as holding not only peculiar biology, but also poorer outcomes. However, data on the topic is scarce and controversial.

Aims & Methods: Our aim is to investigate risk factors and outcomes of EOPC compared to LOPC. This is a single-center prospective study on a cohort of consecutive histologically diagnosed PDAC patients enrolled between April 2016 and September 2024, in the context of a biobank/exposome observational registry study (NCT06552078), at Pancreas Translational and Clinical Research Centre of San Raffaele Hospital in Milan (Italy). Data on environmental and genetic risk factors, clinical and pathological information were recorded in a dedicated clinical report form (CRF) at time of diagnosis, and treatment and survival outcomes were prospectively collected during follow-up. Variables were compared using Mann-Whitney U test or Chi-square test as appropriate. Multivariable logistic regression analysis was employed to investigate factors associated with risk of EOPC. Overall survival (OS) probability was calculated with the Kaplan-Meier curve and Cox regression, adjusted for confounders.

Results: A total of 840 PDAC patients were included, of which 798 (95%) LOPC and 42 (5%) EOPC. Overall, 406 patients (48.3%) deceased after a median follow-up of 14 months (95% CI: 13-15 months). Staging at diagnosis did not significantly differ between the two groups, and 540 patients (64.2%) underwent intensive polychemotherapy, with no significant difference among the two groups. Median age was 46.5 years (95% CI: 46-48 years) for EOPC patients and 69 years (95% CI: 69-70 years) for LOPC patients.

Current smoking (p<0.0001), ever smoking habit (p=0.008), smoking intensity (pack/years) (p=0.03), alcohol consumption (p=0.007) and history of chronic pancreatitis (p=0.02) were significantly more frequent in EOPC vs LOPC patients. No significant differences were observed among the two groups regarding gender, body mass index (BMI), CA19.9, 1st degree family history of any cancer, familial pancreatic cancer (FPC), previous cholecystectomy, genetic syndromes or history of diabetes.

The only significant risk factor for the diagnosis of EOPC at the multivariable analysis resulted to be current smoking, with an OR of 6.3 (95% CI: 2-19, p=0.001).

Median OS was significantly higher in the EOPC patients compared to LOPC (35 vs 17 months, p=0.02), and EOPC resulted as a significant prognostic factor for better OS at Cox proportional hazard regression adjusted for gender, stage and polychemotherapy (HR 0.61; 95% CI: 0.37-1.01; p=0.05).

Conclusion: EOPC seems related to active smoking, but not to familial syndromes. Younger patients display better outcomes compared to older ones. Whether this is due to different molecular behavior requires further investigation.

Disclosure: Nothing to disclose.

PP1104

TOO YOUNG TO SUSPECT IT? DIAGNOSTIC INERTIA AND LATE-STAGE PRESENTATION OF PANCREATIC CANCER IN ADULTS UNDER 50

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Introduction: While pancreatic cancer is traditionally associated with older age, its incidence is rising in adults under 50. This younger population often falls outside standard screening and triage pathways, contributing to delayed diagnosis. In the absence of risk factors, early symptoms may be overlooked or misattributed, delaying imaging and leading to advanced-stage discovery. Although a minority of patients carry hereditary risk, the majority of young cases present without identifiable predispositions.

Aims & Methods: This study aimed to describe the clinical, biological, and diagnostic profile of pancreatic cancer in adults under 50, with a focus on diagnostic delays, tumor marker levels, disease stage, and the presence of traditional risk factors.

A retrospective monocentric analysis was conducted on all patients diagnosed with pancreatic cancer between January 2020 and December 2024 at a regional reference center. Patients were divided into two age groups: <50 and ≥50 years. Variables included age, sex, BMI, smoking status, diabetes, CA 19-9 levels, time from symptom onset to diagnosis, and tumor stage (resectable, borderline, locally advanced, metastatic). Statistical analyses used chi-square and non-parametric tests.

Results: Among 293 patients, 24 (8.2%) were under 50 years (range: 31–49; median age: 46.5 years).

- This younger group had a male predominance (62%) and lower median BMI (22.3 kg/m^2).
- Diagnostic delays >30 days were more frequent in younger patients (71%) vs. older adults (56%) (p = 0.04), with a longer median delay (45 vs. 35 days).
- Median CA 19-9 was higher in the younger group (970 vs. 503 U/mL), and values >1000 U/mL were observed in 41% vs. 24% of older adults.
- Stage at diagnosis in younger patients:
  - Resectable: 8.3%
  - Borderline: 20.8%
  - Locally advanced: 33.3%
  - Metastatic: 37.5% (no significant difference vs. older group; p = 0.61).
- Only 4.2% of younger patients had traditional risk factors (smoking, obesity, diabetes).
- Among those without any risk factor, 78% presented with advanced-stage disease.

Conclusion: Pancreatic cancer in adults under 50 is an emerging and underrecognized challenge. In our cohort, younger patients experienced longer diagnostic delays and more frequent advanced-stage presentations—despite lacking classical risk factors. These findings expose a diagnostic blind spot, likely shaped by age-related bias.

Clinicians should not dismiss elevated CA 19-9 values in younger adults, even in the setting of cholestasis, and should maintain a high index of suspicion in those presenting with persistent abdominal symptoms or unexplained weight loss. Earlier imaging and broader clinical vigilance could help reduce missed opportunities for curative intervention in this vulnerable group.

Disclosure: The authors declare no conflict of interest.

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PP1105

TOO WEAK FOR THE FIGHT? UNDERNUTRITION AND EARLY MORTALITY IN PANCREATIC CANCER: REVISITING BMI AND ALBUMIN THROUGH THE NRI LENS

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Introduction: Malnutrition is an early yet frequently overlooked dimension of pancreatic cancer, often present at diagnosis due to systemic inflammation, cachexia, or silent pancreatic exocrine insufficiency (PEI). Low body mass index (BMI) and hypoalbuminemia are individually recognized as risk factors for poor outcomes, but their combined evaluation may offer a more robust tool for risk stratification. The Nutritional Risk Index (NRI), which integrates both parameters, is a simple, validated score that remains underused in pancreatic oncology.

Aims & Methods: This study aimed to evaluate whether undernutrition at diagnosis, assessed by BMI, albumin, and NRI, predicts non-resectability and early mortality in patients with pancreatic cancer.

We retrospectively analyzed all pancreatic cancer cases diagnosed at a tertiary center between January 2020 and December 2024. BMI was categorized as <20 vs. ≥20 kg/m^2. Serum albumin levels were collected, and NRI was calculated using the simplified formula: 1.519 × albumin (g/L), due to unavailable weight loss data. Patients were stratified according to NRI into no risk (≥97.5), moderate risk (83.5–97.4), and severe risk (<83.5). Tumor stage and resectability were assessed through imaging (resectable) or histology (non-resectable). A subgroup analysis was performed on patients with available 3-month survival data. Statistical comparisons used chi-square and non-parametric tests.

Results: Among 293 patients (mean age 64±11 years; 55% male), BMI data were available for 248 cases. Of these, 101 (38.4%) had a BMI <20 kg/m^2. This group showed lower rates of curative surgery (8.8% vs. 16.9%; p=0.04), more frequent metastatic disease (52.7% vs. 34.2%; p=0.01), and higher rates of hypoalbuminemia (44% vs. 21%; p<0.01).

Serum albumin was available for 289 patients. According to NRI, 39% had severe nutritional risk (NRI <83.5) and 42% had moderate risk. Severe NRI was strongly associated with metastatic disease (61% vs. 28%; p=0.002) and early mortality.

Among 112 patients with 3-month survival data, 83 (74%) had died. Mortality was significantly higher in patients with BMI <20 (82.6% vs. 66.1%; p=0.03) and was even more pronounced in those with NRI <83.5 (87.1% vs. 58.2%; p<0.01).

Conclusion: Undernutrition at diagnosis—whether assessed by BMI, albumin, or NRI—is a strong and early indicator of poor prognosis in pancreatic cancer. The NRI, by combining two simple and routinely available markers, could serve as a practical triage tool to support early nutritional interventions and treatment decisions. Its integration into initial staging workflows may improve outcomes, especially if pancreatic exocrine insufficiency is also considered.

Further multivariate analyses are needed to validate NRI as an independent predictor and to facilitate its routine use in multidisciplinary care planning.

Disclosure: Nothing to disclose.

PP1134

OPERATIVE RISK FACTORS FOR CLINICALLY RELEVANT POSTOPERATIVE PANCREATIC FISTULA AFTER PANCREATICODUODENECTOMY: A PROSPECTIVE MULTICENTER COHORT STUDY

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Introduction: Pancreatic fistula remains the most dangerous complication after pancreatoduodenectomy (PD).

Aims & Methods: This study aimed to identify the operative risk factors for clinically relevant postoperative pancreatic fistula (CR-POPF) after PD. Methods: This prospective multicenter cohort study investigated the association between CR-POPF and operative risk factors in 107 patients who underwent PD at three tertiary centers from August 2017 to July 2022.

Results: The incidence of CR-POPF was 26.2%. With univariate analysis, soft pancreatic texture, pancreatic duct diameter (≤ 3 mm), right-sided pancreatic transection, absorbable suture, pancreatico-enteric anastomosis invagination technique, non-stented pancreatic drainage, internal pancreatic drainage, long anastomotic time (> 40 min), and R1 resection margin were risk factors for CR-POPF.

Multivariate analysis identified four independents risk factors for CR-POPF:

1. Soft pancreatic texture (OR 0.219; 95% CI 0.061-0.792; P < 0.021),
2. Small main pancreatic duct diameter (OR 0.280; 95% CI 0.086-0.910; P < 0.034),
3. Right-sided pancreatic transection (OR 0.168; 95% CI 0.032-0.881; P < 0.035), and;
4. Non-stented pancreatic drainage (OR 3.771; 95% CI 1.147-12.401; P < 0.029).

Conclusion: The incidence of CR-POPF after PD is reduced significantly by left-sided pancreatic transection and pancreatic drainage. Soft pancreatic texture and small main pancreatic duct diameter are independent risk factors for CR-POPF, and clinically postoperative prophylactic measures should be implemented as soon as possible.

Disclosure: Nothing to disclose.

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PP1106

IMPACT OF THE INCLUSION OF USTEKINUMAB BIOSIMILAR IN THE TREATMENT ALGORITHM OF INFLAMMATORY BOWEL: A COST-EFFECTIVENESS ANALYSIS

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Introduction: This study aimed to evaluate the impact in the cost-effectiveness of the introduction of biosimilar ustekinumab (bsUST) in different treatment sequences for moderate-severe Crohn's disease from the Spanish National Healthcare System (NHS) perspective.

Aims & Methods: Six treatment sequences were defined per current clinical practice in Spain and incorporated into the model. A scenario using bsUST as the first line of treatment was considered the base case for all comparisons. Surgery was the last and only therapeutic alternative after four lines of previous pharmacological treatment. A Markov model with a lifetime horizon was developed including active disease, response, remission, surgery, and death as funnel health states to simulate treatment-induction times with 2-week cycles, and a change of treatment after loss of response; considering time-to-progression between consecutive treatment lines.

Efficacy of the comparators [adalimumab (ADA), infliximab (IFX), risankizumab (RISAN), upadacitinib (UPA), vedolizumab (VDZ), and surgery (Q)] was obtained combining data from prior network meta-analysis and clinical trials. Unitary costs were sourced from the literature and databases such as "*Registro de Actividad Sanitaria Especializada*" (RAE-CMBD) and Botplus. Pharmacological costs were ex-factory prices, applying the deduction from "*Real Decreto-Ley*" (RDL) 8/2010. For comparators with available biosimilars, the mean cost of the marketed presentations for the indication was used.

Results: Depending on the sequence, Quality Adjusted Life Years gained ranged from -0.09 (for the sequence ADA-bsUST-UPA-RISAN-Q) to 0.25 (for the sequence IFX-RISAN-bsUST-UPA-Q) and incremental costs per patient in all the comparisons (Table 1). The Incremental Cost-Effectiveness Ratio (ICER) showed that the sequence including bsUST as first-line treatment was dominant or cost-effective vs the remaining alternatives.

Moreover, when compared to the sequence with ADA as first line and bsUST as second-line, none of the four remaining sequences resulted cost-effective.

Conclusion: First-line bsUST is cost-effective and an efficient alternative for the Spanish NHS, even positioned before anti-TNFs. Additionally, none of the alternative sequences proved to be cost-effective compared to the sequence with ADA as first-line and bsUST in second line. Hence, bsUST could change the current paradigm of Crohn's disease treatment in Spain, allowing broader access for patients while being a cost-saving alternative.

	Lifetime cost/ patient	Δ Cost	QALYs/ patient	Δ QALYs	ICER
Sequence 1 (bsUST, ADA, UPA, RISAN, Q)	157,595.7 €		11.87		Reference
Sequence 2 (ADA, bsUST, UPA, RISAN, Q)	158,476.5 €	880.9 €	11.78	-0.09	Sequence 2 dominated
Sequence 4 (ADA, UPA, bsUST, RISAN, Q)	159,447.3 €	1,851.7 €	11.79	-0.07	Sequence 4 dominated
Sequence 3 (IFX, bsUST, UPA, RISAN, Q)	172,229.2 €	14,633.6 €	12.05	0.19	Sequence 3 not cost-effective (78,983.8 €/QALY)
Sequence 6 (IFX, UPA, bsUST, RISAN, Q)	173,196.1 €	15,600.5 €	12.07	0.20	Sequence 6 not cost-effective (77,043.1 €/QALY)
Sequence 5 (IFX, RISAN, bsUST, UPA, Q)	173,258.4 €	15,662.8 €	12.12	0.25	Sequence 5 not cost-effective (62,347.7 €/QALY)

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PP1107

A DIETITIAN-LED DISORDERS OF THE GUT BRAIN INTERACTION CLINIC INCORPORATING INTESTINAL ULTRASOUND DELIVERS COST EFFECTIVE CARE AND IMPROVED PATIENT OUTCOMES

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Introduction: Disorders of the gut brain interaction (DGBI) describe a range of chronic, debilitating gastrointestinal symptoms, including irritable bowel syndrome (IBS). Whilst DGBI can be responsive to first-line dietary therapy, in South Australia, tertiary care service models are typically gastroenterologist-led with on-referral to a dietitian. To address the increasing demand of DGBI care on a tertiary outpatient gastroenterology service, a new dietitian-led DGBI service model utilising intestinal ultrasound performed by a gastroenterologist was developed.

Aims & Methods: The aim of the study was to investigate the clinical outcomes, cost of service delivery, and activity-based funding (ABF) generated by a dietitian-led DGBI model of care. This prospective study between Sep-

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tember 2022 and October 2024 included patients with DGBI and no high-risk clinical features. Patient reported outcome measures were assessed with validated IBS-symptom severity score (IBS-SSS). The per patient episode of care costs were calculated including staffing, both medical and allied health, investigations ordered, emergency department presentations and hospital admissions. ABF is an Australian commonwealth government funding model that allocates government funding to healthcare providers based on the volume and complexity of health care services delivered. ABF was calculated using 2023 National Weighted Activity Units (NWAU) ($5797.00) multiplied by the allied health specialist interventions price weight (0.567). Descriptive and inferential statistics were used to analyse demographics, clinical outcomes and costing presented ($AUD).

Results: Fifty-six of one-hundred and sixty-eight patients participated in this study. The median age was 30 years (interquartile range (IQR 25-37)) and 42/56 (75%) were female. There were 3 (IQR 1-4) occasions of service per episode of care. There was a 98.8-point (95% CI 68-6-128.1) reduction the IBS-SSS scores (p<0.001). The cost per patient episode of care cost was $286.25 (IQR 221.25-351.87). The ABF funding generated was $1064.66 (IQR 656.00-1640.00). Each $1 invested into the clinic returned $2.58 in ABF funding to the health service.

Conclusion: A dietitian-led DGBI clinic utilising intestinal ultrasound delivers improvement in patient reported outcome measures and positive financial return on investment.

Disclosure: Nothing to disclose.

PP1108

NATIONWIDE LONGITUDINAL TRENDS IN THE EPIDEMIOLOGIC AND ECONOMIC BURDEN OF DIGESTIVE DISEASES IN KOREA, 2010 – 2020

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Introduction: Digestive diseases constitute a major component of healthcare utilization in Korea. However, up-to-date data on long-term trends in their epidemiologic and economic burden are scarce. This study aimed to assess nationwide changes in prevalence, healthcare utilization, and direct medical costs related to digestive diseases over the past decade.

Aims & Methods: Aims: Digestive diseases constitute a major and rising component of Korea's healthcare demand. We therefore quantified longitudinal changes in their epidemiologic and economic burden by comparing nationwide data from 2010 to 2020.

Methods: We analysed the National Patient Sample from the Health Insurance Review and Assessment Service (HIRA), which is representative of the entire Korean population. Medical utilization was analyzed based on principled estimates defined as primary diagnosis and extensive estimates defined as primary diagnosis and 1st listed secondary diagnosis. Patient numbers, hospitalizations, emergency-department (ED) visits, outpatient encounters and inflation-adjusted direct expenditures and derived compound annual growth rates (CAGR) were calculated.

Results: On the principal estimate, the number of patients with digestive diseases rose from 17.3 million in 2010 to 18.5 million in 2020, while total expenditures escalated from US $4.2 billion to US $8.5 billion, effectively doubling per-patient costs. Intestinal infectious diseases remained the leading cause of hospitalization and emergency-department visits in both years. In the outpatient setting, the most frequent diagnosis shifted from gastritis/duodenitis in 2010 to gastro-esophageal reflux disease in 2020. The fastest-growing categories were gallbladder disorders (CAGR 19.5 %), pancreatic disorders (CAGR 12.0 %) and benign colonic neoplasms (CAGR 9.1 %), all exceeding the overall mean growth of 7.0%.

Conclusion: From 2010 to 2020, digestive diseases rose by 7 %, but related healthcare costs doubled, sharply raising per-patient spending. Rapid surges in gallbladder and pancreatic disorders and benign colonic neoplasms call for intensified prevention, early screening, and stronger primary-care management.

Disclosure: Nothing to disclose.

PP1109

THE USE OF MILLIPORE STRIP BIOPSY TECHNIQUE REDUCES FINANCIAL AND CARBON COSTS. EXPERIENCE FROM A NEW ZEALAND TERTIARY CENTRE

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Introduction: The demand for gastrointestinal (GI) endoscopy has significantly increased worldwide over the last decade along with associated waste[1-3]. Millipore strips (MPS) enable biopsy specimens from different areas of the GI tract to be placed in one pottle and aligned on a strip according to specific sites. This reduces the number of pottles and saves on lab time showing great promise in terms of cost savings and carbon foot print[4-6].

The use of MPS has been successfully used in a number of countries[4]. The technique was first introduced and validated at our centre in 2009.

Aims & Methods: We aimed to review the utilisation of MPS over a two-year period and to determine the impact on cost and estimated carbon emission reduction. This was compared to the initial cost saving analysis proposed in 2009.

A retrospective cohort study of all endoscopic biopsies performed at our centre between July 2021 to June 2023 was conducted. Data was obtained from Delphic Sysmex pathology processing system.

Cost analysis was based on pathology lab processing costs per specimen pottle which included lab time and materials but excluded Goods and Service Tax (GST). The carbon footprint per pottle was based on previously published data with an estimate per pottle of 0.29kg carbon dioxide equivalent (CO_2e) [3,4].

Results: MPS utilisation across our department ranged from 0 to 97 per endoscopist per annum with a mean usage of 12 MPS per endoscopist per annum.

A total of 12288 endoscopies that included biopsies were performed in our endoscopy unit over the two-year period generating 24659 specimen pottles. Of these biopsy yielding endoscopies, 10% (1205) utilised MPS, with a mean of one MPS pottle per procedure. In comparison 90.2% (11083) used standard biopsy collection alone with an average of 6.7 pottles per procedure.

The following figures were used to calculate the cost of histologically processing samples (inclusive of lab processing and materials but excluding GST): Millipore strip $NZ 102.36 (average 4-6 biopsies per strip). Standard biopsy $69.74 (average 1 biopsy per pottle).

Despite the imperfect uptake, the calculated annual cost saving was $NZ 109,404 per annum. The cost saving estimate in 2009 was $NZ 135,200 per annum.

An estimated three pottles per procedure were saved translating to a total of 1808 pottles per annum. This equates to an estimated 1kg CO_2e saved per procedure and a total of 603kg CO_2e saved per annum with the current MPS use within our department.

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Conclusion: MPS use results in significant cost savings and carbon footprint reduction. Improved intra-departmental MPS utilisation could potentially multiply these gains in sustainable endoscopy.

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PP1110

APPROPRIATENESS OF GASTROINTESTINAL ENDOSCOPY FOR HOSPITALIZED PATIENTS: THE IMPORTANCE OF CONSULTANT LED TRIAGE

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Introduction: A range of criteria have been created to evaluate the appropriateness of gastrointestinal (GI) endoscopy. The most widely recognized were set by the American Society for Gastrointestinal Endoscopy and the European Panel of Appropriateness of Gastrointestinal Endoscopy (EPAGE), updated EPAGE-II criteria. Nevertheless, the application of general rules is not sufficient, especially in hospitalized patients. The complexity, singularity, and dynamic clinical evolution demands the proactive involvement of an experienced endoscopist to make the decision whether or not to perform the exam.

Aims & Methods: Our aim was to analyze the appropriateness of endoscopic exam requests in hospitalized patients over the course of one year. A prospective observational study was conducted in an endoscopic unit of a tertiary hospital center. The requested exams were triaged by a gastroenterologist after reviewing the patient's medical record, assessing clinical status and complementary diagnostic tests. In cases of unclear indication, and no absolute contraindication, the decision to proceed was discussed with the referring physician.

Results: A total of 184 requests for inpatient endoscopy and/or colonoscopy were received during 1 year (table 1). 154 (84%) of the requests were thought to be appropriate. 12 (6%) were cancelled during triage: 2 were patients on best supportive care, 1 died, 3 had other diagnosis which plausibly justified the clinical situation, 2 had recent endoscopic exams, and 4 presented no clinical/analytical evidence of GI bleeding). 18 (10%) requests were initially postponed (8 were awaiting imaging results; 5 had poor clinical status and 5 respiratory distress) and ended up being cancelled (8 had another diagnosis in imaging tests, 8 maintained clinical deterioration contraindicating exam and 3 died). In total 30 (16%) of the requests were cancelled.

The most frequent exam requested was endoscopy, and the most common indication was suspected GI neoplasia. From a total of 58 suspected GI neoplasms, 11 were confirmed as gastric neoplasia, which was the most common finding.

None of the patients with a cancelled exam was diagnosed with a GI disease during follow up.

Mortality rate was considerably higher in the group with cancelled exams (33% vs 11% during hospitalization, 50% vs 32% during follow up, 15% vs 1.8% within 30 days after discharge). Mean age was similar between groups. This findings reveal a frail group of patients, with little likelihood of benefiting from an invasive exam.

Total, n=184	GI endoscopy performed	GI endoscopy deferred and/or cancelled
N, %	154 (84)	30 (16)
Sex, male (%)	89 (58)	16 (53)
Age (years), mean (±SD)	72 (12.8)	71 (14)
Type of exam		
Endoscopy, n (%)	68 (44)	13 (42)
Colonoscopy, n (%)	28 (18)	6 (19)
Endoscopy + Colonoscopy, n (%)	55 (36)	10 (33)
Flexible sigmoidoscopy, n (%)	3 (19)	1 (33)
Indication, n		
Anemia	42	5
Diarrhea	5	-
Dysphagia	8	-
Epigastric pain	3	1
Chronic liver disease	7	-
Suspected GI neoplasia¥	44	20
Suspected upper GI neoplasia	14	-
Suspected colorectal neoplasia	15	-
Suspected GI bleeding	4	4
Relevant endoscopy findings, n		
Hypopharyngeal neoplasm	2	
Grade C, D esophagitis	2	
Esophageal neoplasia	2	
Esophagogastric varices	6	
Gastric angiodysplasia	3	-
Gastric neoplasia	11	
Neoplastic infiltration of duodenum	2	
Colorectal neoplasia	7	
Inconclusive (poor bowel preparation)	3	
Motive to defer and/or cancel, n		
Awaiting imaging results		8
Clinical deterioration		5
Respiratory distress		5
End-of-life care		2
Evidence of myeloma	-	1
Evidence of ovarian cancer		2
Recent endoscopic evaluation		2
Without analytical or clinical evidence of bleeding		4
Death		1
Death during hospitalization, n (%)	17 (11)	10 (33)
Death within 30 days after discharge, n (%)	2 (1.8)	3 (15)

Table 1. Patient's characteristics and exam indications and findings.
¥ Suspected GI neoplasia includes constitutional syndrome, evidence of metastases without a known primary source, ascites and/or pleural effusion.

Conclusion: 16% of the requests were considered inappropriate, which allowed to avoid 30 unnecessary exams.

Developing standardized indications for endoscopic exams in inpatient care is difficult, and an individualized approach is critical in order to reach an equilibrium between overusage and overlooking, while considering the principle of *primum non nocere.*

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PP1111

REAL-WORLD INSIGHTS INTO DIAGNOSTIC JOURNEY, DISEASE BURDEN, AND HEALTHCARE RESOURCE USE IN IGG4-RELATED DISEASE ACROSS EUROPE, BRAZIL, AND SAUDI ARABIA

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Introduction: IgG4-related disease (IgG4-RD) is a chronic inflammatory condition that culminates in fibrosis and organ damage. The real-world diagnosis, treatment and outcomes of IgG4-RD patients remain poorly understood.

Aims & Methods: To characterise diagnosis, clinical characteristics, treatment and disease burden, healthcare use and patient-reported outcomes (PRO) in IgG4-RD patients. Data were drawn from a real-world, cross-sectional survey of physicians and patients with IgG4-RD in Europe (France, Spain, Italy, Germany and the United Kingdom), Brazil and the Kingdom of Saudi Arabia (KSA) from September 2023 – September 2024. Physicians reported patient demographics, clinical characteristics, patient journey, treatment patterns and healthcare resource use. Patients self-reported on burden of fatigue using the FACIT-Fatigue Scale, and work and activity impairment using the WPAI questionnaire (Table 1). Analyses were descriptive.

Results: Data were reported by 208 physicians on 982 patients (France: n=181, Spain: n=96, Italy: n=160, Germany: n=162, UK: n=143, Brazil: n=89, KSA: n=151), of whom 254 provided self-reported data. Mean (standard deviation; SD) patient age was 53.8 (13.4) years (59.6% Male; 73.7% White). Overall, mean (SD) age at symptom onset was 49.6 (13.0) years. Time from symptom onset to diagnosis was 8.9 (19.5) months, ranging from 3.3 (8.1) in KSA and 21.0 (30.0) in Brazil. Fatigue (57.0%), weight loss (32.3%), and joint pain (24.1%) were the most common symptoms at first consultation; the most common organ manifestations being pancreas (41.9%), bile ducts (25.2%) and abdominal/pelvic lymph nodes (23.9%). Primary care physicians (46.3%) and gastroenterologists (17.4%) were most often consulted first about symptoms, with gastroenterologists (37%) and rheumatologists (31%) most often confirming IgG4-RD diagnosis. Most patients were treated by gastroenterologists (53.9%), rheumatologists (46.8%) and primary care physicians (20.4%). Patients required a mean (SD) of 5.1 (4.0) consultations and 17.4 (8.9) tests prior to diagnosis; physical examination (85.7%), serum IgG4 levels (84.5%), and tissue biopsy (81.0%) were most frequently used to aid diagnosis. 86.0% of patients had another condition suspected prior; most commonly pancreatitis (28.3%), cholangiocarcinoma (14.1%) and Sjögren's syndrome (12.5%). Post-diagnosis, mean (SD) time to treatment initiation was 16.5 (28.0) months, ranging from 12.0 (22.5) in Europe to 34.6 (38.2) in KSA. Prednisone was the most common first line therapy (78.6%), followed by rituximab (12.7%) and azathioprine (9.7%). In the 12 months prior to survey, 26.7% of patients were hospitalised (21.4% in Europe; 15.4% in Brazil and 56.0% in KSA). Of those hospitalised, the mean (SD) number of hospitalisations was 1.4 (0.8), ranging from 1.2 (0.5) in KSA to 1.5 (0.9) in Brazil. 41.6% of patients were admitted via the emergency room (ER) and spent 5.9 (5.0) nights in hospital.

	Overall n=982	France n=181	Spain n=96	Italy n=160	Germany n=162	UK n=143	Brazil n=89	KSA n=151
FACIT Fatigue score (normal range ~40-45); n (Mean ± SD)	254 (34.8 ± 10.5)	30 (36.4 ± 10.9)	27 (34.9 ± 11.4)	10 (46.1 ± 4.1)	104 (37.7 ± 9.2)	10 (20.1 ± 4.7)	2 (42 ± 7.1)	71 (30 ± 8.8)
Overall work impairment (WPAI); n (Mean ± SD)	84 (26.5 ± 24.1)	15 (23.1 ± 28.2)	9 (9.6 ± 14.8)	4 (0.0 ± 0.0)	30 (20.7 ± 19.5)	2 (62 ± 0.7)	0 (0.0 ± 0.0)	24 (43.6 ± 19.0)
Activity impairment (WPAI); n (Mean ± SD)	252 (34.7 ± 26.0)	29 (39.3 ± 30.7)	27 (26.7 ± 26.2)	10 (3.0 ± 9.5)	104 (29.5 ± 24.9)	10 (55.0 ± 9.7)	2 (35.0 ± 49.5)	70 (45.1 ± 20.9)
Impairment whilst working (WPAI); n (Mean ± SD)	117 (24.9 ± 20.4)	16 (22.5 ± 26.2)	12 (10 ± 12.8)	4 (0.0 ± 0.0)	58 (23.1 ± 16.7)	2 (50.0 ± 0.0)	0 (0.0 ± 0.0)	25 (39.6 ± 18.4)
Work time missed (WPAI); n (Mean ± SD)	84 (4.0 ± 7.9)	15 (2.2 ± 8.6)	9 (1.2 ± 3.7)	4 (0.0 ± 0.0)	30 (2.0 ± 5.8)	2 (24 ± 1.4)	0 (0.0 ± 0.0)	24 (7.7 ± 8.3)

Table 1: Distribution of patient-reported outcomes in the overall study sample and by country.

Conclusion: The diagnostic journey for IgG4-RD patients is often delayed, involving multiple consultations, tests and misdiagnoses. Despite the heterogeneous nature of the disease, most patients reported substantial burden of illness and impact to daily lives. Optimizing diagnosis and treatment pathways may improve disease burden and quality of life among patients with IgG4-RD.

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JP, RW, AM and MR are Amgen employees and own Amgen stock.
SLB and HS are employees of Adelphi Real World.

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PP1112

FINANCIAL TOXICITY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE AT THE NATIONAL HOSPITAL SRI LANKA: A CROSS-SECTIONAL STUDY

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Introduction: Inflammatory bowel disease (IBD) imposes severe financial burdens in low-middle income countries (LMICs). Sri Lanka's healthcare system faces unique challenges in managing chronic diseases like IBD. This study evaluates financial toxicity (FT) using the validated FACIT-COST instrument.

Aims & Methods: This cross-sectional study aimed to quantify financial toxicity (FT) and identify risk factors among IBD patients at Sri Lanka's National Hospital. 132 IBD patients (68 Crohn's disease [CD], 64 ulcerative colitis [UC]) were recruited. Demographics, education levels, and disease characteristics were collected via structured interviews and medical records. FT was assessed using the 12-item ‚FACIT-COST - Version 2' validated questionnaire (range: 0-44, lower = worse). Data was analyzed using Spearman's correlation and t-tests.

Results: The cohort's mean age was 40.55 ± 15.7 years, with a balanced gender distribution (males: 65 years, females: 67 years). The majority (62.8%) had completed secondary education or higher. The mean FT score was 19.44 ± 7.28, indicating a mild-to-moderate financial burden. FT was significantly higher in CD patients (mean = 20.65) compared to UC (mean = 18.16, p=0.049). Medication costs were the main financial concern for 75% of patients. FT scores correlated weakly but significantly with education level (rho=+0.229, p=0.008), suggesting higher education modestly reduces financial toxicity. No correlation was found with age of onset or diagnostic delay (mean = 851 days).

A notable proportion reported significant financial hardship (FT-12: 47%), and 35.6% felt IBD reduced satisfaction with their financial situation. The mean age of onset differed markedly by disease type: CD patients presented earlier (males: 23.1 years; females: 23.9 years) than UC patients (males: 32.4 years; females: 37.1 years).

Conclusion: IBD imposes a significant financial burden, particularly in patients with Crohn's disease, who face both earlier onset and higher toxicity. Targeted interventions should include:

1. Subsidies for high-cost drugs,
2. Patient education programs,
3. Improved accessibility of hospital medications,
4. Quality enhancement of available treatments,
5. Local guidelines for resource-limited settings, and
6. Establishment of IBD support groups.

These systemic solutions are essential to mitigate financial distress and improve the quality of life for Sri Lankan IBD patients.

Disclosure: Nothing to disclose.

PP1113

APPROPRIATENESS AND DIAGNOSTIC YIELD OF UPPER GASTROINTESTINAL ENDOSCOPY: NEED TO IMPROVE PATIENT SELECTION FOR GREATER SUSTAINABILITY

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Introduction: The diagnosis of most upper gastrointestinal (GI) tract disorders is mainly clinical, with specific indications for endoscopy. Adherence to guidelines about indications for upper-GI endoscopy is cost-effective in an open-access system [1-2]. However, in the era of sustainable medicine, literature lacks observational studies aimed at evaluating the endoscopic appropriateness in an open-access systems.

Aims & Methods: The aim of the study is to investigate the appropriateness of prescriptions for upper-GI endoscopy, including the associated diagnostic yield.

This monocentric observational study was conducted by enrolling all consecutive patients undergoing upper-GI endoscopy. A detailed interview was performed before endoscopy to investigate symptoms, dietary habits, gastrointestinal history and other significative clinical information before the endoscopic examination. The appropriateness of the endoscopy was defined according to the ASGE guidelines [3]. After the endoscopic examination, the diagnostic yield, defined as a relevant diagnosis that could change the clinical history of patients (neoplasms, coeliac disease, conditions require treatment or follow-up [4]), was evaluated.

Results: A total of 437 patients were enrolled. Most of them were female (234, 53.5%), mean age 59.8 yrs (IQR 48.8-71). The main prescriber was the general practitioner (GP) (273, 62.4%), followed by gastroenterologists (107, 24.4%) and other specialists (57, 13%). The most frequent endoscopic findings were gastritis and/or duodenitis (141, 32.2%), followed by normal findings (95, 21.7%), low-risk premalignant conditions (51, 11.6%) and high-risk premalignant conditions (38, 8.6%), while malignant neoplastic lesions were 10 (2.2%).

The overall appropriate prescriptions rate was 44.3% (194 exams). Gastroenterologists had a higher rate of appropriate prescriptions (89/107, 83.1%), followed by other specialists (27/57, 47.3%) and GPs (78/273, 28.5%) (p<0.0001). Seventy-six out of 243 (31.2%) of the patients with an inappropriate indication had undergone an upper-GI endoscopy within the previous 5 years, with findings did not require follow-up [5]. The overall diagnostic yield rate was 25.6% (112/437 exams), with significant differences between appropriate and inappropriate exams: 94/194, (48.4%) vs 18/243, (7.4%) (p<0.0001), including 61 (31.4%) new relevant diagnosis vs 14 (5.7%) (p<0.0001).

Conclusion: Adherence to guidelines for the appropriateness of upper GI endoscopy is confirmed to be more effective, including the added benefit from the diagnostic yield of appropriate upper GI endoscopies. These results suggest that if proper education of practitioners were implemented, a better endoscopy-utilization would be expected.

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Disclosure: Nothing to disclose.

PP1114

ACUTE ABDOMINAL PAIN IN ELDERLY PATIENTS CONSULTING IN GASTROENTEROLOGY EMERGENCIES

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Introduction: Acute abdominal pain is a common reason for consultation in gastroenterology emergency departments among elderly patients. Diagnosis can be challenging due to atypical symptoms, which may delay treatment and increase the risk of complications or even threaten short-term survival.
Aims & Methods: This study aims to analyze the frequency, causes, and specific characteristics of acute abdominal pain in elderly patients to improve their management.
Materials and Methods: This is a retrospective descriptive study conducted over one year (December 2023 to December 2024), including all patients over 65 years old who consulted the gastroenterology emergency department for acute abdominal pain. Data were collected from medical records and entered into Excel.
Results: A total of 300 patients presented to the gastroenterology emergency department for acute abdominal pain, of whom 60 were over 65 years old, representing 20% of cases. There was a slight male predominance, with a male-to-female sex ratio of 1.3.
Among the elderly patients, 25% were managed on an outpatient basis, while 75% were hospitalized.
Regarding medical history, 25.2% of patients had hypertension, 17.6% had diabetes, 9.2% had a history of heart disease, 21.8% had cirrhosis, 17.3% had undergone previous surgery, and 19.4% had no significant medical history.
Abdominal pain was associated with vomiting in 10.6% of cases, diarrhea in 8.5%, bowel obstruction in 9.4%, and gastrointestinal bleeding in 6%. A general deterioration was observed in 42.7% of patients.
The pain was diffuse in 34.2% of cases, epigastric in 28.5%, and localized to the right hypochondrium in 17.2% of patients.
All patients underwent additional diagnostic tests. C-reactive protein was measured in all cases, with elevated levels in 72% of patients (mean: 68.3 mg/L). An abdominopelvic CT scan was performed in 76.4% of patients, while 39.7% had an abdominal ultrasound. Standard abdominal X-rays showed air-fluid levels in four patients, pneumoperitoneum in three cases, and signs suggestive of a sigmoid volvulus in one patient.
Digestive causes were the most frequent, with acute pancreatitis being the leading diagnosis (16.5%), followed by intestinal obstruction (9.3%), cholecystitis (7.3%), portal vein thrombosis (6.7%), diverticulitis (5.8%), cholangitis (3.9%), liver abscess (3.3%), and sigmoid volvulus (1.6%). A tumor-related cause was identified in 6.4% of patients.
Extra-digestive causes accounted for 13.3% of cases, including cardiac conditions (5%), urinary tract disorders (5%), and gynecological conditions (3.3%).
All patients received appropriate medical and/or surgical treatment along with symptomatic management.
Conclusion: Multiple diagnoses can be responsible for acute abdominal pain in elderly patients, some of which are life-threatening in the short term. More than one-third of cases require surgical intervention. It is crucial to establish clear guidelines to improve patient management and reduce morbidity and mortality in this vulnerable population.
Disclosure: Nothing to disclose.

PP1115

MEDICAL DESERTIFICATION: KEY FACTORS IN ATTRACTING GASTROENTEROLOGISTS TO RURAL AND DISTRICT HOSPITALS

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Introduction: Physician shortages in rural and underserved areas, so-called medical deserts, pose a significant barrier to specialized care across Spain and Europe.
Aims & Methods: We conducted a national, anonymous survey among members of the Spanish Association of Gastroenterology (AEG), focusing on early-career professionals. Thirty-eight factors potentially influencing job acceptance in district hospitals were rated on a 10-point scale (1 = least important, 10 = most important), and grouped into seven thematic domains. Data were analyzed for the overall cohort, and stratified by age (<35 vs. ≥35 years) and prior experience in district hospitals.
Results: A total of 110 gastroenterologists from all Spanish regions participated, with advanced endoscopy identified as the main clinical interest (54%). Alignment with personal clinical interests was considered essential by 74% of respondents.
The seven domains evaluated were: training, work environment, clinical activity, quality of life (QoL), research and teaching, on-call duties, and job stability and finances. Of these, quality of life (mean [M]: 8.62), on-call duties (M: 8.66), and job stability and financial conditions (M: 9) were rated highest.
The possibility of regularly performing endoscopy, flexible scheduling, competitive salary, and exemption from internal medicine and emergency on-call duties (all with a median score of 10) were the most valued individual factors.
Analysis of responses (see Table 1) revealed significant differences according to prior district hospital experience.
Those without such experience prioritized formal educational opportunities such as attending courses, master's programs, conferences; doctoral research; longer-term contracts; and reported greater need for support in complex clinical and endoscopic scenarios, especially in hepatology, gastrointestinal bleeding, and endoscopic management of colonic perforations. In contrast, gastroenterologists with prior district hospital exposure valued greater autonomy, fewer on-call duties, and felt less apprehension about leaving their comfort zone. Teaching opportunities were rated more highly by those aged ≥35 years (p = 0.046).
Table 1. Results (median and interquartile range) for the 10 most relevant survey items. Shown for the overall group (N=110), as well as for subgroups by age (< and ≥ 35 years) and prior experience working in a district hospital (yes/no); subgroup analysis. IQR: interquartile range. P value by Mann–Whitney U test.

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DOMAIN (OVERALL MEANS OF MEDIANS)	ITEM	OVERALL (N=110) Median (IQR)	<35 YEARS (N=54) Median (IQR)	≥35 YEARS (N=55) Median (IQR)	P VALUE (AGE)	NO PRIOR DISTRICT HOSPITAL EXPERIENCE (N=56) Median (IQR)	PRIOR DISTRICT HOSPITAL EXPERIENCE (N=54) Median (IQR)	P VALUE (EXPERIENCE)
TRAINING	More opportunities to attend courses/master's programs/conferences	6 (4-8)	6 (4-8)	6 (3-8)	0.638	7 (4-8)	5 (1-7)	<0.001
WORK ENVIRONMENT	Greater autonomy in daily practice	8 (6-9)	8 (6-9)	8 (7-8)	0.579	7,5 (6-8)	8 (7-9)	0.016
	Fear of being alone in complex medical situations	7 (5-9)	8 (5-9)	7 (5-8)	0.414	8 (6-9)	7 (4-8)	0.004
	Fear of being alone in complex endoscopic situations	7 (6-9)	7 (6-9)	7 (5-8)	0.199	8 (6-9)	6 (4-8)	<0.001
QUALITY OF LIFE	Fear of leaving comfort zone	5 (2,5-7,5)	5 (4-8)	5,5 (1,75-7,25)	0.439	6 (4-8)	5 (1,25-7)	0.005
RESEARCH & TEACHING	Greater ability to complete a doctoral thesis	5 (2-7)	5 (3-7)	5 (1-7,25)	0.329	5 (3-8)	4 (1-7)	0.034
	Facilities to undertake university teaching	7 (5-9)	7 (4-8,5)	7 (5-9)	0.307	7 (4-8)	7,5 (5-9,25)	0.036
ON-CALL DUTIES	Possibility to perform fewer on-calls	8 (5,5-10)	7,5 (6-10)	9 (5-10)	0.333	7 (4,75-10)	9 (6-10)	0.036
JOB STABILITY & FINANCES	Long-term contracts	9 (7-10)	9 (7-10)	9 (6-10)	0.388	10 (8-10)	8 (5-10)	0.011

Table 1.

Conclusion: Incentives related to quality of life, on-call burden, and job stability are critical for attracting gastroenterologists to rural and underserved settings. Prior exposure to district hospitals substantially shifts professional priorities, emphasizing the importance of tailored, flexible strategies. These findings offer actionable guidance for national and European health policymakers aiming to reduce workforce disparities and improve access to specialist care in medical deserts.

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PP1116

PHYSICIAN TRIAGE TIME FOR INTERVENTIONAL GI PROCEDURE REFERRALS - BALANCING PRACTICE EFFICIENCY AND PHYSICIAN WELL-BEING

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Introduction: Triaging interventional gastroenterology (GI) referrals improves care efficiency but is associated with large amount of uncompensated work done outside of normal work hours. It is unknown how much time physicians spend on this essential yet undocumented work. The purpose of this study was to assess the physician workload for triaging interventional GI referrals.

Aims & Methods: This is a prospective observational study from March to November 2024 involving an academic medical center with seven interventional endoscopists who share triage referral responsibilities each week they are on call and also triage referrals when not on call if the referral is specifically requesting them.

To systematically capture triage work performed by interventional endoscopists, an electronic medical record (EMR) "smart phrase" was created to track the following variables: Diagnosis/Reason for Triage, Triaging Physician, Triage Disposition, Procedure Ordered, Type of Anesthesia, Time Frame, Requested Provider, Triage Comments. With this triage form, time spent on performing triaging work could be approximated from the EMR's user access log.

Results: 2092 triages were captured, with a mean of 53±16 triages per week. Physicians spent an average of 3.8 ±1.0 minutes in each triage note. The physicians varied in time spent per triage from 2 minutes to 6 minutes per triage. The on-call attending of the week performed 47% of the weekly triages. 67% of referrals were triaged directly to procedures, 23% to clinic, and 10% to other (e.g. oncology or surgery or get more information). 38% of patients were triaged to have urgent (<4 weeks) procedures/visits (5% within 1 week, 12% within 2 weeks, 21% within 4 weeks). The mean amount of time per week spent by the on-call physician doing triage was 92.0 ± 61.2 minutes. The mean amount of time per week spent by the non on-call physicians doing triage was 22.3 ± 32.2 minutes. 50% of triages were done after hours (5 pm to 7 am) or on weekends.

Conclusion: Physician triage of interventional GI referrals requires an average of 4 minutes per triage and the on-call physician spends more than 90 minutes per week doing triages. Physicians not on call also spend additional time (>20 minutes per week) triaging direct referrals to them. 50% of these triages were done after hours during nights and/or weekends.

Although triaging improves clinical operations efficiency, scheduling, and patient care, it may decrease physician wellness and compensation. Future efforts should look at providing adequate time, resources, and compensation for triage of interventional GI referrals.

Disclosure: Nothing to disclose.

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PP1117

EARLIER USE OF GUSELKUMAB IS PROJECTED TO YIELD LONG-TERM SUSTAINED REMISSION IN PATIENTS WITH MODERATE-TO-SEVERE ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) is a progressive disease and associated with reduced quality of life.[1] Treatment goals for patients with UC are to induce and maintain clinical remission, and avoid or delay the need for surgery.[2,3] Increasing number of advanced treatment options have been approved in recent years for patients with UC.

However, the optimal treatment pathway is not well understood. Guselkumab, a dual-acting IL-23 inhibitor, has demonstrated clinical remission as high as 50.0% at maintenance Week 44 in the Phase 3 QUASAR study, suggesting the potential of guselkumab to provide durable clinical remission.[4]

A disease model was developed to identify the optimal treatment sequence with guselkumab for patients with moderate-to-severe UC.

Aims & Methods: A hybrid decision tree/Markov disease model was developed to project rates of clinical remission, response and active disease for patients with UC over five years. Patients were first distributed into health states representing remission, response without remission, and active UC (no response) based on the rates of clinical remission and response reported in the induction phases of Phase 2 and Phase 3 trials.

Over time, loss of response and transition to the next line of therapy is derived based on loss of response rates in the trial data. Up to three treatment lines were modelled, and patients who lose response after third-line treatment are categorized as in active disease state where they can receive surgery.

Age- and sex-adjusted mortality rates obtained from the US population life table were applied in the model. Three treatment sequences were assessed, where guselkumab was placed in first-line, second-line, and third-line, respectively. A treatment basket was used for the other lines of therapy, informed by 2024 market share data from MarketScan.

We report the predicted average proportion of time spent in each health state for each of the three treatment sequences over five years.

Results: Over a 5-year period, the predicted average proportion of time patients spent in active disease is consistently lower for patients treated with guselkumab as a first-line therapy compared to those receiving it in later lines. In the first year, patients receiving first-line guselkumab spent 28% of time in active disease, while those treated with guselkumab in the second or third line spent 33%.

This difference becomes more pronounced over the 5-year period, with proportion of time in active disease at 36% for first-line guselkumab use versus 48% and 51% for second- and third-line use, respectively. The reduction in time in active disease with earlier use of guselkumab infers that patients are asymptomatic, as patients are also projected to experience longer time in remission.

The model predicts that, over 5 years, patients treated with first-line guselkumab spend an average of 52% of their time in remission, in contrast to 40% and 38% for those treated with guselkumab in the second- or third-line, respectively.

Table 1: Projected average proportion of time spent in each health state over a 5-year period

	Remission	Response without remission	Active UC	Surgery
First-line guselkumab	52.1%	10.8%	36.4%	0.7%
Second-line guselkumab	39.5%	11.9%	47.7%	0.9%
Third-line guselkumab	38.1%	9.9%	51.0%	1.0%

Conclusion: Model projections suggest that using guselkumab in first line is associated with long-term sustained remission for patients with UC, compared to using guselkumab in later lines of treatment.

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PP1118

COMPARATIVE ANALYSIS OF LARGE LANGUAGE MODELS TO RESPONSE OPEN-ENDED QUESTIONS REGARDING GASTROENTEROLOGY

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Introduction: The performance of large language model (LLM) across specific medical domains is uncertain. We evaluated the performance of ChatGPT-4 and two Chinese LLMs in answering open-ended questions regarding gastroenterology.

Aims & Methods: We selected ChatGPT (4-Turbo, OpenAI) and two Chinese LLMs (model A and B), and embedded our own digestive disease-related knowledge database in the LLMs via retrieval-augmented generation (RAG). We presented 200 questions, with 3 different expressions for each question and 5 repeated queries for each expression.

Questions were selected from those commonly encounter in a clinical setting from patients. The accuracy and readability were assessed using a three-classification criterion (good, borderline, poor).

Responses with "good" rating in accuracy were further assessed using a 3-point scale. For responses with "poor" rating in accuracy, the LLMs were further prompted to self-correct. The stability was assessed through the consistency in changing-expression and repeating-query groups.

Results: For accuracy evaluation, model B demonstrated superior accuracy, with 94.9% of "good" rating, compared to 92.3% in model A and 90.3% in ChatGPT-4 (all P <0.001). Three LLMs showed comparable average scores in "good" rating responses (2.590 vs 2.573 vs 2.536, P=0.754).

For readability evaluation, ChatGPT-4 and model A demonstrated superior readability, with 94.4% and 93.1% of "good" rating, compared to 76.3% in model B (all P< 0.001).

The average response length of ChatGPT-4 was comparable to model A and significantly shorter than model B (207.45 vs 271.33 vs 336.87, P<0.001). For stability evaluation, model A demonstrated superior sta-

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bility in repeating queries than ChatGPT-4 and model B. The stability in changing different expressions was not satisfactory for all the three LLMs and needs to be further improved.

Conclusion: Our study preliminarily demonstrated three LLMs' potential in accurately answering questions regarding gastroenterology, and their application to clinical practice still needs further exploration.

Disclosure: Authors declare no competing interest for this article.

PP1119

THE ROLE OF QUANTITATIVE CONTRAST ENHANCED-ENDOSCOPIC ULTRASOUND IN THE DIFFERENTIAL DIAGNOSIS BETWEEN PANCREATIC NEUROENDOCRINE NEOPLASMS AND PANCREATIC ADENOCARCINOMAS : A PROSPECTIVE STUDY

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Introduction: In recent years, the incidence of pancreatic solid tumors—especially pancreatic neuroendocrine neoplasms (pNENs) and pancreatic ductal adenocarcinomas (PDACs)—has increased due to improved imaging and aging populations. Differentiating these tumors is crucial given their distinct prognoses. Endoscopic ultrasound is considered the gold standard for their diagnosis.

However, EUS-tissue acquisition could have low diagnostic accuracy, especially in pNENs. Unlike PDACs, pNENs exhibit a significantly higher microvascular density (MVD), and this is showed by different contrast uptake EUS with contrast enhancement (CE-EUS). Recent studies, indeed, have shown a very low interobserver agreement in CE-EUS evaluation, even among expert endosonographers.

Aims & Methods: This proof-of-concept study aimed to prospectively evaluate the role of objective quantification of contrast uptake during CE-EUS in differentiating pNENs from PDACs, potentially improving diagnostic accuracy and reducing operator dependency. This prospective monocentric observational study, compliant with STROBE guidelines, included patients who underwent EUS for pancreatic diseases at San Raffaele Hospital (Milan) between January and June 2024. Eligible patients had a solid pancreatic lesion with a final cytological diagnosis of PDAC or pNEN and a complete DICOM-format CE-EUS video including both arterial and venous phases.

Exclusion criteria were lack of consent or poor-quality/unstable videos. EUS was performed under deep sedation by experienced operators using a linear echoendoscope. Lesions were evaluated with B-mode, Doppler, elastography, and CE-EUS following the injection of 4.8 ml of SonoVue®, recording at least 120 seconds of video (arterial: 10–30 s; venous: 30–120 s). Videos were archived in DICOM format.

Post-procedural analysis was conducted with VueBox® software, which provides quantitative contrast uptake parameters. ROIs were manually selected in the lesion and adjacent normal parenchyma. Assessed metrics included peak enhancement, wash-in/wash-out rates, time-to-peak, perfusion indices, and area-under-curve measurements.

Results: After applying inclusion criteria, 73 patients with solid pancreatic lesions were enrolled (51 PDACs, 22 pNENs). No significant differences in general features were found between groups. Median tumor size was 31 mm overall (32.6 mm PDACs, 27.3 mm pNENs). Operator-based CE-EUS evaluation showed hypo-enhancement in 88.2% of PDACs and only 4.6% of pNENs ($p<0.001$). VueBox® analysis revealed significant differences in contrast uptake parameters between the two groups: pNENs consistently showed higher values for MeanLin, PE, WiAUC, WiR, WiPI, WoAUC, and WiWoAUC (all $p<0.001$). ROC analysis identified WiPI (AUC 0.838) and WiR (AUC 0.804) as the best discriminators; combining them via logistic regression increased the AUC to 0.961.

Univariate analysis showed that in pNENs, contrast parameters were influenced by lesion size, echogenicity, and lymphadenopathies. However, multivariate analysis confirmed only lesion diameter as significantly correlated with local transit time ($p<0.0001$), while no patient- or tumor-level variables showed significant associations with contrast uptake in the full cohort.

Conclusion: CE-EUS improves pancreatic tumor assessment but is limited by subjectivity and low interobserver agreement. This study shows that qCE-EUS effectively distinguishes pNENs from PDACs, with key parameters like WiPI and WiR achieving 96.1% accuracy. qCE-EUS may reflect tumor angiogenesis and offers a reproducible, non-invasive tool for diagnosis, staging, and treatment planning.

Disclosure: Nothing to disclose.

PP1120

APPLICATION OF ABDOMINAL POINT-OF-CARE ULTRASOUND IN WAR CONDITIONS AFTER 2-DAY TRAINING COURSES: A CROSS-SECTIONAL STUDY

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Introduction: Point-of-care ultrasound (POCUS) plays a crucial role in emergency medicine, including in wartime settings. Different POCUS studies are designed to answer specific clinical questions, quickly guiding the management of potentially fatal emergencies. Unfortunately, the experience of Ukrainian combat medics in using POCUS in war conditions is relatively limited. However, short POCUS training programs can help improve their proficiency.

Aims & Methods: The aim of this study was to evaluate the application of POCUS and identify potential barriers to its use in various settings during the war in Ukraine by combat medics without prior ultrasound experience after completing a two-day training course. This was a cross-sectional survey of former participants (n=140) of two-day POCUS training courses organized by the charitable organizations *Razom for Ukraine* and the *Christian Medical Association of Ukraine* between January 2023 and September 2024.

The 2-day training included 4 hours of theoretical lectures and 10 hours of practical sessions. An online survey consisting of 41 questions assessed the challenges and characteristics of ultrasound use at different stages of evacuation after completion of the training. Among the participants, 87% were doctors, 56% worked at stabilization points, 31% in military hospitals, and 14% in medical transport.

Results: 89% of participants used POCUS at least once a day. The most commonly used applications of abdominal POCUS were FAST (Focused Assessment with Sonography in Trauma) – 89% of respondents; followed by ultrasound of the kidneys (40% of respondents) and ultrasound of the liver and gallbladder (36% of respondents). Other frequent applications were pneumothorax (72%), vascular access (62%), soft tissue (54%), and nerve blocks (44%). The most common barriers to use were image inter-

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pretation (47%) and image acquisition (31%). The most common technological problems were damage to the probe or cord (36%), a short battery lifespan (32%), and not having access to a machine (26%).

Conclusion: After a brief two-day training, novice ultrasound users among combat medics regularly utilized abdominal POCUS in the care of wounded Ukrainian soldiers, primarily through the FAST exam.

Future efforts to overcome barriers to the routine use of POCUS should focus on additional education in image acquisition and interpretation, as well as the procurement of higher-quality ultrasound machines.

Disclosure: Nothing to disclose.

PP1121

BEYOND THE SCOPE: A MIXED-METHODS ANALYSIS OF CONTRIBUTORY FACTORS TO PATIENT SAFETY INCIDENTS IN GASTROINTESTINAL ENDOSCOPY USING THE SEIPS 2.0 FRAMEWORK - A UK NATIONAL DATABASE STUDY

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Introduction: Patient Safety Incidents (PSIs) are not uncommon in gastrointestinal endoscopy, arising from complex interactions between factors at all levels of healthcare. Understanding the contributory factors (CF) to PSIs is crucial to improve safety, yet are poorly characterised in endoscopy care.

This study presents an analysis of CFs associated with PSIs across the UK over 5 years, using a validated Human Factors Framework.

Aims & Methods: This was a retrospective mixed-methods study of anonymised PSI reports associated with at least moderate harm in GI endoscopy, submitted to the National Reporting and Learning System between 2017-2022. A descriptive analysis of cases was undertaken to identify incident type, phase of care and urgency. The Systems Engineering for Patient Safety (SEIPS 2.0) framework was used to identify CFs from reports, based on principles of content analysis.

Results: Of 1,575 reports analysed, 133 were excluded as unrelated to GI endoscopy. Of those remaining, 100 (6.9%) resulted in death, 268 (18.6%) in severe harm and 1,074 (74.5%) in moderate harm.

The most frequently occurring incident type was 'known complication' (n=587, 40.7%), with 'perforation' (n=341, 58.1%), 'bleeding' (n=94, 16.0%) and 'missed cancer' (n= 59, 10.1%) being most common. The second most common incident was 'follow-up related' (n=268, 18.6%), largely characterized by 'delayed follow-up or diagnosis' (n=152, 53.1%) and 'lost to follow-up' (n=84, 29.4%). Issues with endoscopy 'timing or decision' was the third most coded incident (n=124, 8.6%). Most PSIs occurred peri-endoscopically (n=642, 44.5%), followed by pre-endoscopy (n=409,28.3%) and post-endoscopy phases (n=392, 27.2%). Elective procedures (n=647,44.8%) had more incidents than emergency (n=245,17%).

Of included reports, 1,057 (73.3%) contained enough information to yield 1,441 CF references captured within SEIPS.

Table 1 provides an overview of the three most frequently referenced SEIPS domains, with a non-exhaustive presentation of leading themes. 'Person' CFs, influenced largely by communication issues, was the most significant driver. 'Organisation' constituted over a third of references, driven largely by resource availability and recruitment. Most CF references within ‚Task' related to complex tasks, dominated by challenging anatomy or physiology. Of the remaining domains, 'Tools and technology' (n=106, 7.4%) was mostly driven by equipment malfunction (48.1%). The 'External environment' (n=55, 3.8%) was almost exclusively COVID-19 policy changes (96.4%), while 'Internal environment' formed a minority of CFs (n=9, 0.6%).

SEIPS domain	Theme and sub-theme	Number of references
Person n=750, 52.0%	Communication	382 (59%)
	Between specialist teams	161
	Poor documentation	96
	Between patient and healthcare team	63
	Knowledge, competence and skills	247 (32.9%)
	Team cohesiveness	51 (6.8%)
Organisation n=269, 18.7%	Resource availability and recruitment	160 (59.5%)
	Endoscopy list capacity	68
	Multiple hospital sites	26
	Equipment availability	18
	Workload assignment	43 (16.0%)
	Communication infrastructure	35 (13.0%)
Task n=252, 17.5%	Complexity	132 (49.1%)
	Challenging anatomy or physiology	128
	Complication management	24
	Differently abled patient	24
	Difficulty	43 (17.1%)
	Ambiguity	10 (4.0%)

Table 1. Three most referenced SEIPS domains implicated in PSIs of at least moderate harm.

Conclusion: This novel systems-based analysis of CFs in endoscopy-related PSIs provides critical insight for improving safety. The predominance of issues relating to People and Teams, Organisation and Task highlights key areas for intervention.

The next phase of this research includes a content analysis of PSIs associated with mild harm and development of simulations based on findings, aiming to improve safety in endoscopy.

Disclosure: Nothing to disclose.

PP1122

INFORMED CONSENT AND INSTRUCTIONS FOR ENDOSCOPIC PROCEDURES: FULLY INFORMED OR FULLY OVERWHELMED?

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Introduction: Informed consent (IC) serves as the ethical foundation for all medical interventions, empowering patients to make informed, autonomous decisions while safeguarding clinicians. Barriers such as limited health literacy and document complexity may hinder comprehension[1,2].

This study evaluated the comprehensibility, readability, and effectiveness of the documents provided for IC and instructions for endoscopic procedures.

Aims & Methods: Outpatients undergoing endoscopic procedures completed a questionnaire assessing their understanding of IC, preparation instructions, and adherence to guidelines. Patient confidence and doc-

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ument comprehensibility were assessed through self-reported scales. Readability was assessed using Flesch Reading Ease score adapted for portuguese (FRE-PT)[3].

Results: Among 186 participants (mean age: 58.1 ± 16.1 years), 76.9% reported full-extent reading of the provided instructions and IC document. Patients with primary education (≤4th grade) were less likely to read the documents fully (59.2% vs. 83.2%, p < 0.001) and were independently associated with the need for additional assistance (OR 5.65, 95% CI: 1.92 - 16.68, p = 0.002).

Full-extent reading was independently associated with greater risk awareness (OR 5.52, 95% CI: 2.37 - 11.55, p < 0.001), reduced need for additional assistance (OR 0.22, 95% CI: 0.09 - 0.51, p < 0.001), and higher compliance with medication guidelines (75.7% vs. 45.8%, OR 3.42, 95% CI: 1.28 - 9.20, p = 0.015). Patients with primary education rated document's comprehensibility lower (median:8, IQR 7-10) compared to those with higher education (median: 10, IQR 8 - 10, p < 0.001) and reported lower levels of confidence and preparation for the procedure (median: 8, IQR 7 - 10 vs. median: 9, IQR 8 - 10; p = 0.018).

Outcomes did not differ significantly between patients aged 65 years or older compared to younger patients. FRE-PT scores classified readability as ''very difficult'' for both upper endoscopy (20.87) and colonoscopy (26.00).

Conclusion: Full-extent reading IC and instruction documents reduces the need for assistance, improves compliance with pre-procedural measures, risk awareness and patient confidence. However, the document's complexity and difficult readability disproportionately challenge and discourage patients with limited education. Future research should explore strategies to foster more equitable informed decision-making.

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Disclosure: Nothing to disclose.

PP1123

ADVANCING INFLAMMATORY BOWEL DISEASE CARE IN AFRICA: OUTCOMES FROM A MULTIDISCIPLINARY EDUCATIONAL SYMPOSIUM

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Introduction: Inflammatory Bowel Disease (IBD) is emerging as a significant but under-recognized health challenge in Africa. Diagnostic delays, limited awareness, and a lack of educational opportunities for healthcare professionals persist. The Inaugural Inflammatory Bowel Disease in Africa Symposium was held on April 10th in Abuja, Nigeria, to foster collaboration in IBD diagnosis and management, promote awareness of medical and surgical care options, and support patient advocacy.

Aims & Methods: A hybrid (in-person and virtual) full-day symposium was designed and developed, bringing together specialists in gastroenterology, surgery, nursing, pathology, radiology, and oncology from across the continent. The program featured expert-led lectures and multidisciplinary team case-based discussions.

A post-symposium electronic survey was deployed to assess participants' satisfaction, knowledge gains, practice impacts, and IBD management challenges and provide recommendations for enhancing future symposia. Quantitative responses were analyzed descriptively; qualitative feedback was thematically categorized.

Results: A total of eighty-four individuals registered, with speakers and faculty representing Nigeria, South Africa, Ghana, Zambia, Zimbabwe, the United States, the United Kingdom, Egypt, and Kenya. Fifty-eight (69%) of 84 completed the survey. These comprise nurses (34.5%), surgeons (27.6%), gastroenterologists (17.2%), radiologists (8.6%), pathologists (6.9%), and oncologists (5.2%).

Overall, 93% rated the program as Satisfied, with 88% reporting significant knowledge gains in African-context IBD management. Diagnostic and treatment confidence improved in 83% and 76% of participants, respectively, with 69% gaining skills in biopsy techniques and 67% differentiating IBD mimickers. Major challenges reported included limited access to endoscopy and histopathology, low disease awareness among patients, high diagnostic costs, misdiagnosis as infectious diarrhea or tuberculosis, and lack of referral networks.

The absence of multidisciplinary care teams was also noted. A strong majority (95%) supported establishing an African IBD registry, emphasizing the need for sustainable funding, technical infrastructure, and training. Hybrid learning and case-based approaches were preferred for future symposia, with interest in pediatric IBD and resource-adapted care.

Conclusion: The symposium significantly enhanced knowledge and confidence in IBD care among participants while exposing critical gaps in diagnostics, treatment access, and care coordination. These findings highlight the need for sustained collaborative efforts to strengthen IBD education and care. Supporting regional training and cross-disciplinary engagement will be essential to building capacity, improving outcomes, and advancing equitable IBD management across the continent.

Disclosure: Nothing to disclose.

PP1124

KNOWLEDGE AND ATTITUDES TOWARD GASTRIC CANCER SCREENING IN A LOW-LITERACY ROMA COMMUNITY

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Introduction: Gastric cancer is a major cause of cancer-related mortality, and early detection improves outcomes. However, organized gastric cancer screening programs are largely absent in Europe, and vulnerable groups often have low awareness of preventive services. Roma communities face pronounced health disparities and barriers to healthcare, including social exclusion and low literacy, which may limit their use of preventive services [1].

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Aims & Methods: We aimed to explore the knowledge, attitudes, and preferences regarding gastric cancer screening among Roma adults with low health literacy in Cluj-Napoca, Romania, to inform culturally tailored outreach strategies, as part of the TOGAS project. Approximately 40 Roma adults (18–72 years) were recruited via community centers for three focus groups (two urban, one rural) in Cluj-Napoca. All participants first attended a brief educational session on digestive health and gastric cancer to ensure baseline understanding. Guided by trained moderators, groups discussed their awareness of gastric cancer and perceptions of screening methods (endoscopy, blood tests, stool tests). All discussions were audio-recorded, transcribed, and thematically analyzed.

Results: Baseline awareness of gastric cancer screening was minimal. Most participants had never heard of any screening test or program for gastric cancer, indicating a major knowledge gap.

Nevertheless, after receiving information, many expressed interest in screening for early detection. A prominent theme was communication preferences: nearly all participants preferred to receive health information via mobile phone or social media (e.g., messaging) and also through traditional channels like primary care providers. Overall, participants were open to screening if it was offered conveniently and respectfully.

Conclusion: Outreach is needed to improve engagement in cancer prevention among underserved populations. The lack of awareness in this Roma community echoes reports of poor knowledge of cancer prevention in other Roma groups and signals the need for targeted health promotion to reduce inequalities. As European health authorities consider implementing gastric cancer screening, input from vulnerable communities will be vital for equitable uptake.

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Disclosure: Nothing to disclose.

PP1125

GAMIFIED ENDOSCOPY TRAINING WITH AI-BASED IMAGE AND SPEECH RECOGNITION

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Introduction: Proper training in upper gastrointestinal endoscopy (GI) is critical prior to first patient contact[1,2]. While multiple training simulation models are available, they often lack interactive elements and thus motivating experiences, which may limit engagement and skill acquisition[1]. Gamification, the application of game-like elements to non-game contexts, has shown promise in enhancing learning and motivation in medical education[2].

Aims & Methods: This study aimed to enhance engagement in endoscopic training through the gamification of a commercially available silicone upper GI tract model (SimStar Gastro Advanced, Dr. Henke GmbH), which features four pathological findings: Barrett's esophagus, a gastric polyp, gastric varices, and a gastric ulcer. We developed an AI-based system that integrates real-time image recognition using a ResNet18 deep learning model to detect pathologies in the video feed, and speech recognition using Whisper AI (OpenAI, San Francisco, USA) to automatically log the findings. Endoscopists were instructed to perform a complete diagnostic examination as quickly and accurately as possible while verbally reporting each pathology. The AI-based system gave real-time feedback regarding the verbally reported diagnosis. No prior training with the system was provided. After completing the task, participants filled out a 10-item System Usability Scale (SUS) questionnaire.

Results: Five out of six endoscopists successfully identified and dictated all four pathologies. The median examination time was 64.4 seconds. The image recognition system achieved a frame-by-frame accuracy of 74.4%, while the speech recognition system demonstrated 90% transcription accuracy. The overall usability was rated positively, with a median SUS score of 74.5 for the participants who successfully completed the test, indicating good acceptance and ease of use.

Conclusion: The integration of computer vision and voice recognition into a gamified endoscopic simulation model offers a promising, engaging, and autonomous method for training novice endoscopists. Our pilot study demonstrates the feasibility and usability of this approach, supporting the development of more engaging training tools in GI endoscopy.

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PP1126

GAME-BASED LEARNING TO PROVIDE EXPOSURE TO ENDOSCOPY

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Introduction: There is tremendous potential in using serious games as an avenue for future medical education tools. An advantage to using serious games over other forms of simulation is the low cost and high level of accessibility for students. We presented students with a phone-based game that presented students challenging and rare endoscopy-based patient case scenarios, which we believe can be applied to other interventional fields, such as gastroenterology.

Aims & Methods: Second year undergraduate medical students were invited to participate in the study via email at the beginning of the academic year. Participants were randomly assigned into either the control group or the intervention group and asked to complete a pre and post-survey. Students completed a pre-study survey asking them about their interest in interventional medicine, and their level of familiarity with each topic as well as their confidence treating a patient with pathology that they will learn about in the app and recognizing several different structures endoscopically.

Results: Compared to control, the intervention group reported a significant increase in knowledge and skill acquisition ($p < 0.01$, $p < 0.003$) as well as interest in pursuing a career in interventional medicine ($p = 0.04$). Students reported that the virtual cases helped them understand the course material (60%) and enjoyed their experience with the app (80%). 100% of participants reported that they would recommend the app to a colleague interested in gastroenterology.

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Conclusion: Recent research has shown that there is a need for development and validation of medical education research in the field of gastroenterology. Game-based learning offers students an opportunity to engage in the content that they are learning about in class in a novel way that has the potential to enhance their pre-clinical knowledge, increase their confidence to manage future patients, and may influence their career interests. Given the notorious difficulty of pre-clinical curriculums, it is important to find ways for students to be able to solidify course content while also maintaining their wellbeing. Game-based learning offers students an opportunity to engage in the content that they are learning about in class in a novel way that has the potential to not only enhance their pre-clinical knowledge but also increase their confidence managing patients when they enter the clinical years.

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Disclosure: Nothing to disclose.

PP1127

CURRICULUM-BASED TRAINING IN ESD: INSIGHTS FROM THE FRENCH EXPERIENCE

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Introduction: Endoscopic submucosal dissection (ESD) demands a high level of endoscopic skill and extensive training, even for experienced endoscopists, with a particularly steep learning curve for colonic ESD. Over the last three years, the French Society of Digestive Endoscopy (SFED) has implemented a structured ESD training program.

This study aimed to evaluate the effectiveness of the SFED ESD training program to achieve competency in ESD and to identify factors associated with it.

Aims & Methods: Participants who attended the SFED ESD training curriculum over the past three years were invited to join the study. After completing a standardized initial simulation training week, each participant was asked to document each ESD procedures performed in their training center, along with technical details for each (lesion size and location, en bloc resection, R0 resection status, and procedural complications).

Additionally, participants completed an online survey to gather information about their practice environment, ongoing ESD activity, and demographic data (e.g., gender, age, prior endoscopy experience). Successful dissection was defined as en-bloc resection with free lateral and deep margins (R0) and no need for surgical intervention due to complications. The performance threshold for achieving proficiency in ESD was defined, in line with ESGE guidelines, as achieving an en-bloc resection rate of >90% and a complication-related surgery rate of <1%. The learning curve to reach these competence standards was assessed using the LC-CUSUM method.

Results: A total of 45 students participated in the study. The majority were men (70%), with a median age of 36 years. They practice endoscopy for 6 years (IQR 4-10) and 82% performed interventional endoscopy (ERCP and EUS).

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A total of 2,068 ESD procedures were performed by the trainees. The median number of ESDs conducted post-training was 34 (IQR 22-66), with the majority being colorectal (42.9% rectal and 42.7% colonic). Lesions had a median size of 45 mm (IQR 34-60), with a resection speed of 17.4 mm^2/min (IQR 10.5-26.5). En bloc resection and R0 resection were achieved in 93.6% and 85.8% of cases, respectively, with a surgical complication rate of 0.94%.

The overall success rate of ESD procedures was 92.7%. Success was more likely in rectal (97%) than colonic (90.4%) procedures. 75% of the students were competent. Among those who reached competence, the median number of ESDs required was 45. Factors associated with competency were: ESD localization (rectum > colon), the age (HR 0.92 [0.86-0.99]; p = 0.03) and experience (HR 0.91 [0.84-0.99]; p = 0.03) were inversely correlated to the competency. Practiced in a high volume ESD center (>100 ESD/year) lead to reach competency faster (p =0.008).

Conclusion: This study provides the first assessment of the SFED ESD training program in France. Competence in ESD is reachable after appropriate training. A big experience is not needed to attempt ESD. Supervised training is essential, and seems easier in a high-volume center to achieve competence more quickly. These findings may help inform the selection of future candidates for ESD training to ensure effective skill acquisition and safe practice.

Disclosure: Nothing to disclose.

PP1128

ADEQUACY OF DISCLOSURES OF CONFLICTS OF INTEREST IN ORAL PRESENTATIONS AT THE 2025 CONGRESS OF EUROPEAN CROHN'S AND COLITIS ORGANISATION

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Introduction: Conflicts of interest (COIs) introduce potential bias into study design, scientific interpretation, clinical decision making and guideline development (1). COIs within gastroenterology research are common yet their presentation at international gastroenterology conferences has not been previously evaluated.

Aims & Methods: Our aim was to characterise the adequacy of COI disclosures at the 20th Congress of the European Crohn's and Colitis Organisation (ECCO) 2025. Two investigators independently conducted assessments of COIs for all scientific presentations on the ECCO 2025 virtual portal. Outcomes included presence of a disclosure slide, disclosure slide duration, number of COIs, adequacy of disclosures (defined as highlighting how COIs may relate to the presentation), presence of a potential drug/industry product in the presentation and consistency with written abstract COIs.

Results: COI statements were missing in 18% of presentations, and 22% had inconsistencies in COI data between written abstracts and oral presentations. The median disclosure slide duration and was 2.3 seconds (IQR 1.5-3.4) and median number of COIs 10 (IQR 5-29). There was no correlation between slide duration and number of COIs (r = 0.059, p = 0.53). Whilst 59% of presentations contained at least one COI with relevance to the presentation topic, only 6% of COIs were assessed as adequately discussed. The median slide display time for adequately discussed COIs was higher at 6.5 seconds (IQR 3.5-9.3) versus 2.3 seconds (IQR 1.4-3.0) for inadequately discussed COIs (p = 0.006).

Conclusion: COI disclosure practice was suboptimal in oral presentations at the 20th Congress of ECCO. When presented, disclosure slide duration was too brief for meaningful interpretation. Discussion of COIs relevant to the presentation topic was inadequate in 94%, even though the time difference required for adequate disclosure was only 4.2 seconds. Formal guidance and standardised criteria for COI disclosures are warranted to facilitate transparency and allow informed interpretation of presented data.

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Disclosure: Nothing to disclose.

PP1129

ADHERENCE TO ENDOSCOPIC QUALITY INDICATORS IN GREECE

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Introduction: Quality assessment is a priority in the field of gastroenterology. However, limited data are available from Greece regarding adherence to quality standards in gastrointestinal endoscopy.

Aims & Methods: This study aims at assessing the knowledge and compliance of Greek gastroenterologists with established quality standards for upper and lower digestive endoscopy, while exploring factors that may influence compliance.

This prospective study recorded data on the quality of upper and lower digestive endoscopy performed in Greece from October 2024 to January 2025. All members of the Hellenic Society of Gastroenterology (n=668) were invited to participate via an electronic survey. The survey included a questionnaire designed specifically for this study, covering demographic and professional characteristics, as well as quality indicators for gastroscopy and colonoscopy.

Results: A total of 164 gastroenterologists participated (response rate 24.55%), of whom 107(65.24%) were male, and 102(62.2%) working in public or academic hospitals. The duration of gastroscopy was not recorded by 116(70.73%) participants, while 102(62.2%) systematically used photodocumentation. Most participants utilized the Prague and Los Angeles classifications 156(95.12%) and 159(96.95%) respectively. Additional-

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ly, 140(85.37%) followed the Seattle protocol, and 158(96.34%) conducted biopsies according to ESGE guidelines for precancerous gastric lesions. However, 83(50.61%) don't use a registry for patients with Barrett's esophagus. Complications were systematically recorded after gastroscopy by 103(62.8%) and after colonoscopy by 90(54.88%).

In colonoscopy, 149(90.85%) use the BBPS scale, and 158(96.34%) reported successful cecal intubation. Withdrawal times exceeded six minutes for 151(92.07%) participants. An adenoma detection rate of over 25% was achieved by 117(71.78%), while 103(62.8%) consistently used the Paris classification. Proper polypectomy techniques were employed by 111(67.68%), but 100(60.98%) recommended follow-up intervals shorter than guidelines suggest.

Patient experience following gastroscopy and colonoscopy was not recorded by 126(76.83%) and 116(70.73%) participants, respectively. Overall, compliance with quality indicators for upper and lower digestive endoscopy was observed in 87(53.05%) and 75(45.73%) participants, respectively.

Multivariate analysis revealed that 5-15 years of experience (p=0.05), the use of video recording (p=0.01) and use of high-resolution endoscopes (p<0.01) were associated with improved compliance with quality indicators of upper endoscopy. Whereas age above 55 (p<0.01), use of high-resolution endoscopes(p<0.01), use of video recording (p=0.02) and attendance of educational seminars relevant with quality in endoscopy (p=0.05) were associated with improved compliance with quality indicators of colonoscopy.

Conclusion: In this study, discrepancies were observed in the compliance of Greek endoscopists with endoscopic quality indicators, highlighting the need for further improvement. The systematic recording of these quality indicators is essential, underlining the important role of scientific societies in this process.

Disclosure: Nothing to disclose.

PP1130

PHYSICIANS' AWARENESS OF THE HARMFUL EFFECTS OF ALCOHOL USE

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Introduction: According to the World Health Organization statements, there is no "safe" level of alcohol consumption that can be deemed free from potential health harm. While the detrimental role of alcohol in the pathogenesis of liver disease is well known, its contribution to extrahepatic disorders remains underrecognized.

Due to inadequate public health strategies and the global normalization of alcohol use, its harmful effects are often underestimated, not only by the general public but also among healthcare providers. Therefore, we aimed to assess physicians' knowledge, attitudes, and personal behaviors related to harmful alcohol use.

Aims & Methods: This prospective, cohort study utilized a structured questionnaire distributed among physicians specializing in gastroenterology. The questionnaire comprised three sections: demographic data, the AUDIT-C tool, and questions assessing knowledge of alcohol-related harm. Alcohol misuse was defined by an AUDIT-C score ≥4 for men and ≥3 for women. Distribution was achieved through direct communication, social media (Twitter), and the United European Gastroenterology "MyConnect" network, using an electronic survey format (Google Forms). All responses were anonymized, compiled into a central database, and analyzed using descriptive and inferential statistical methods.

Results: The study included 204 physicians, predominantly female (60.8%, mean age: 44.05±12.63 years). Participants were mainly from Serbia (35.8%), Russia (20.1%), Montenegro (18.1%), and Slovakia (11.8%). Most respondents were married or in a relationship (80.9%) and worked in academic centers (57.8%). Regarding specialization, 83.8% specialized in gastroenterology, 44.6% in hepatology, 31.9% in inflammatory bowel disease, and 29.9% in advanced endoscopy. A smaller proportion reported expertise in internal medicine (2.9%), oncology (2.0%), and transplant hepatology (3.9%). The median professional experience was 15 years. In terms of knowledge and attitudes, 99.6% of respondents agreed that alcohol negatively impacts all organ systems, and 82.4% believed women are at increased risk for alcohol-related harm. A majority (60.8%) stated there is no safe level of alcohol consumption, while 46.6% acknowledged alcohol as a carcinogen linked to 5–10 types of cancer.

Additionally, 44.6% believed that moderate alcohol consumption may have certain protective effects. Most agreed that alcohol-related health consequences typically arise in individuals with alcohol use disorder (60.8%) and that alcohol use worsens disease prognosis (97.5%).

The most commonly identified preventable risk factors were tobacco and alcohol use, and obesity (76.8%). Notably, 69.5% of respondents believed the five-year incidence of cardiovascular disease is twice as high in affected populations compared to the general population. AUDIT-C indicated that 28.9% of respondents met criteria for alcohol misuse.

Significant predictors of misuse included younger age (OR=0.972, p=0.031), specialization in transplant hepatology (OR=4.383, p=0.048), disagreement with the systemic effects of alcohol (OR=0.467, p=0.025), and belief in the potential protective effects of moderate consumption (OR=1.461, p=0.003).

Conclusion: Although physicians demonstrated a high level of awareness regarding the harmful effects of alcohol, a substantial proportion agreed on possible protective effects of alcohol use, while one third of participants engaged in risky drinking behavior. Improving awareness of alcohol's systemic impact may contribute to more effective prevention strategies within both clinical and public health settings.

Disclosure: Nothing to disclose.

PP1131

EPIDEMIOLOGY AND RISK FACTORS FOR POST-INFECTIOUS IBS IN PATIENTS INFECTED WITH COVID-19

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Introduction: Given the impact of COVID-19 on the gastrointestinal system and the characteristics of irritable bowel syndrome (IBS), the incidence and specific risk factors associated with COVID-19-related PI-IBS remain poorly understood.

Aims & Methods: The study included patients who were hospitalized with a confirmed diagnosis of COVID-19 between November 2021 and November 2023. Patients diagnosed with IBS based on the ROME IV criteria were compared with a control group without IBS.

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Results: Baseline demographic and clinical characteristics.

Among the 300 patients included in this study, the mean age was 61.2 years, with 158 females (52.7%) and 142 males (47.3%). 46 participants (15.3%) were current alcohol consumers.

In the univariable analysis, the blood test results at the time of admission, including WBC, ANC levels, and the history of probiotic intake, were found to be significant factors influencing the development of IBS. In the multivariable analysis, with a cutoff *p-value* of 0.05, WBC and Probiotics in the last 1 month were identified as significant factors. (adjusted OR: 2.152, 95% CI: 1.247–3.714, *p = 0.006*; adjusted OR: 4.750, 95% CI: 0.630–35.820, *p = 0.023*, respectively) (Table.)

Subgroup analysis of risk factors was performed by dividing participants into five subgroups based on questionnaire responses. (Q2; whether abdominal pain onset occurred before or after six months, Q3; presence or absence of a relationship between abdominal pain and bowel movements, Q4; changes in bowel movement frequency, Q5; normal versus abnormal average bowel movement frequency, Q6; stool form change) The third and fifth subgroup, only the WBC showed statistically significant results in the univariable analysis (adjusted OR:1.132, 95% CI: 1.001 –1.281, *p* = 0.049, adjusted OR: 1.115, 95% CI: 1.001–1.243, *p* = 0.049 respectively), though this significance did not carry over to the multivariable analysis.

Table. Risk factors contributing to the development of IBS following COVID-19 infection

Factors	Univariable analysis		Multivariable analysis	
	OR (95% CI)	P value	OR (95% CI)	P value
Sex	1.827(0.584-5.719)	0.301		
Age(years)	1.017 (0.980-1.056)	0.364		
BMI(㎏/kg)	0.169(0.966-1.218)	0.169		
Laboratory result				
WBC (/ul)	1.221(1.058-1.410)	0.006	2.152(1.247-3.712)	0.007
Hb (g/dL)	1.128(0.843-1.510)	0.419		
Platelet (x10³/ul)	1.001(0.993-1.008)	0.862	1.000(0.999-1.000)	0.052
ANC (/mm³)	1.000(1.000-1.000)	0.040		
Cholesterol (mg/dL)	1.005(0.994-1.017)	0.371		
Protein (g/dL)	0.932(0.403-2.154)	0.869		
Albumin (g/dL)	0.855(0.281-2.600)	0.782		
Total bilirubin (mg/dL)	1.360(0.229-8.078)	0.736		
GFR (/min/1.73)	1.001(0.986-1.015)	0.931		
CRP (mg/dL)	0.964(0.860-1.082)	0.538		
Hypertension	1.720(0.564-5.248)	0.341		
History of underlying disease, clinical course, and medication				
Diabetes mellitus	2.135(0.675-6.748)	0.197		
Antibiotics in the last 1 month	1.229(0.262-5.763)	0.794	4.750(0.630-35.820)	0.023
Probiotics in the last 1 month	5.616(1.082-29.138)	0.040		
Alcohol history	1.702(0.450-6.439)	0.433		
Vaccination	2.143(0.645-7.117)	0.213		
Previous covid-19 infection Hx	3.333(0.379-29.296)	0.278		
Hospital day	0.965(0.855-1.089)	0.564		

Table.

OR, odds ratio; CI, confidence interval; SD, standard deviation; BMI, body mass index; WBC, white blood cell; ANC, absolute neutrophil Count; GFR, Glomerular Filtration Rate; CRP, C-reactive protein

Conclusion: In conclusion, we found that probiotic intake and higher WBC count could be associated with the incidence of IBS following COVID-19 infection. While there is growing evidence of a link between COVID-19 and PI-IBS, further research is necessary to elucidate the underlying mechanisms and evaluate the preventive potential of probiotics.

Disclosure: Nothing to disclose.

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